

Treatment

There is no treatment for Tay-Sachs disease. Parents who are identified as carriers may want to seek **genetic counseling**. If a fetus is identified as having TSD, parents may consider termination of the **pregnancy**.

Prognosis

Children born with Tay-Sachs disease become increasingly debilitated; most die by about age four.

Wolman's disease

Causes and symptoms

Wolman's disease is caused by a genetic defect (with a recessive pattern of inheritance) that results in deficiency of an enzyme that breaks down cholesterol. This causes large amounts of fat to accumulate in body tissues. Symptoms begin in the first few weeks of life and include an enlarged liver and spleen, adrenal calcification (hardening of adrenal tissue due to deposits of calcium salts), and fatty stools.

Treatment

No treatment is currently available.

Prognosis

Death generally occurs before six months of age.

Prevention

Couples who have family histories of genetic defects can undergo genetic testing and counseling to see if they are at risk for having a child with one of the lipidoses disorders. During pregnancy, cell samples can be collected from the fetus using amniocentesis or chorionic villi sampling. The results of these test can indicate if the developing fetus has a lipidosis disorder. Termination of the pregnancy may be considered in some cases.

Resources

ORGANIZATIONS

- National Institute of Neurological Disorders and Stroke.
P.O. Box 5801, Bethesda, MD 20824. (800) 352-9424.
< <http://www.ninds.nih.gov/index.htm> >.
- National Niemann-Pick Foundation. 3734 E. Olive Ave.,
Gilbert, AZ 85234. (602) 497-6638.
- National Organization for Rare Disorders. P.O. Box 8923,
New Fairfield, CT 06812-8923. (800) 999-6673.
< <http://www.rarediseases.org> >.

KEY TERMS

Amniocentesis—A procedure where a needle is inserted through the abdomen into the uterus of a pregnant woman to remove a small amount of the fluid that surrounds the developing fetus. This test can be preformed at about week 16 of the pregnancy. Cells from the fetus can be tested for genetic defects.

Chorionic villi sampling—A procedure to remove a small tissue sample of the placenta, the sac that surrounds the developing fetus. This test can be performed as early as week 10 of the pregnancy. The tissue can be tested for genetic defects.

Lipids—Organic compounds not soluble in water, but soluble in fat solvents such as alcohol. Lipids are stored in the body as energy reserves and are also important components of cell membranes.

Recessive—Refers to an inherited characteristic or trait that is expressed only when two copies of the gene responsible for it are present.

X-linked—Refers to a gene carried on the X chromosome, one of the two sex chromosomes.

National Tay-Sachs and Allied Diseases Association. 2001
Beacon St., Suite 204, Brookline, MA 02146. (800) 906-8723. < <http://www.ntsad.org> >.

OTHER

Gaucher Disease Treatment Program. < <http://gaucher.mgh.harvard.edu> >.

Rare Genetic Diseases In Children: An Internet Resource Gateway. < <http://mcrcr2.med.nyu.edu/murphp01/homenew.htm> >.

Altha Roberts Edgren

Lipoproteins test

Definition

Lipoproteins are the “packages” in which cholesterol and triglycerides travel throughout the body. Measuring the amount of cholesterol carried by each type of lipoprotein helps determine a person's risk for cardiovascular disease (disease that affects the heart and blood vessels, also called CVD).

Purpose

Cholesterol and triglycerides are fat-like substances called lipids. Cholesterol is used to build cell membranes and hormones. The body makes cholesterol and gets it from food. Triglycerides provide a major source of energy to the body tissues. Both cholesterol and triglycerides are vital to body function, but an excess of either one, especially cholesterol, puts a person at risk of cardiovascular disease.

Because cholesterol and triglycerides can't dissolve in watery liquid, they must be transported by something that can dissolve in blood serum. Lipoproteins contain cholesterol and triglycerides at the core and an outer layer of protein, called apolipoprotein.

There are four major classes of lipoproteins: chylomicrons, very low-density lipoproteins (VLDL), low-density lipoproteins (LDL), and high-density lipoproteins (HDL). There also are less commonly measured classes such as lipoprotein(a) and subtypes of the main classes. Each lipoprotein has characteristics that make the cholesterol it carries a greater or lesser risk. Measuring each type of lipoprotein helps determine a person's risk for cardiovascular disease more accurately than cholesterol measurement alone. When a person is discovered to be at risk, treatment by diet or medication can be started and his or her response to treatment monitored by repeated testing.

Description

Chylomicrons

Chylomicrons are made in the intestines from the triglycerides in food. They contain very little cholesterol. Chylomicrons circulate in the blood, getting smaller as they deposit the triglycerides in fatty tissue. Twelve hours after a meal, they are gone from circulation. Serum collected from a person directly after eating will form a creamy layer on the top if left undisturbed and refrigerated overnight. This creamy layer is the chylomicrons.

Very low-density lipoproteins (VLDL)

VLDL are formed in the liver by the combination of cholesterol, triglycerides formed from circulating fatty acids, and apolipoprotein. This lipoprotein particle is smaller than a chylomicron, and contains less triglyceride but more cholesterol (10-15% of a person's total cholesterol). As the VLDL circulates in the blood, triglycerides are deposited and the particle gets smaller, eventually becoming a low-density lipoprotein (LDL). Serum from a person with a large amount of VLDL will be cloudy.

Low-density lipoproteins (LDL)

LDL, often called "bad" cholesterol, is formed primarily by the breakdown of VLDL. LDL contains little triglycerides and a large amount of cholesterol (60-70% of a person's total cholesterol). Although the particles are much smaller than chylomicrons and VLDL, LDL particles can vary in size and chemical structure. These variations represent subclasses within the LDL class. Serum from a person with a large amount of LDL will be clear.

LDL carries cholesterol in the blood and deposits it in body tissues and in the walls of blood vessels, a condition known as **atherosclerosis**. The amount of LDL in a person's blood is directly related to his or her risk of cardiovascular disease. The higher the LDL level, the greater the risk. LDL is the lipoprotein class most used to trigger and monitor cholesterol lowering therapy.

High-density lipoproteins (HDL)

HDL is often called "good" cholesterol. HDL removes excess cholesterol from tissues and vessel walls and carries it to the liver, where it is removed from the blood and discarded. The amount of HDL in a person's blood is inversely related to his or her risk of cardiovascular disease. The lower the HDL level, the greater the risk; the higher the level, the lower the risk. The smallest lipoprotein, it contains 20-30% of a person's total cholesterol and can be separated into two major subclasses.

Lipoprotein(a)

Lipoprotein(a) is found in lower concentrations than other lipoproteins, yet it carries a unique and significant risk for cardiovascular disease. Because of its similarity to LDL, test methods often don't measure it separately, but include it within the LDL class. Testing specifically for this class may uncover why a person is not responding to standard cholesterol-lowering treatment. High lipoprotein(a) levels may not respond to treatment aimed at high LDL.

Measurement guidelines

The Expert Panel of the National Cholesterol Education Program (NCEP) sponsored by the National Institutes of Health has published guidelines for the detection of **high cholesterol** in adults. The NCEP panel recommends that adults over the age of 20 be tested for cholesterol and HDL every five years. If the cholesterol is high, the HDL is low (below 35 mg/dl), or other risk factors are present, a complete

lipoprotein profile that includes total cholesterol, triglycerides, HDL, and calculated LDL should be done.

Measurement methods

There are a variety of methods to measure the lipoprotein classes. All require separation of the classes before they can be measured. One way to separate them is by spinning serum (the yellow, watery liquid that separates from the cells when **blood clots**) for a long time in a high-speed centrifuge (called ultracentrifugation). The most dense classes will settle toward the bottom, the least dense toward the top. Following centrifugation, the most complete measurement of all the lipoprotein classes is done using electrophoresis. This procedure measures the quantity of each lipoprotein class based on its movement in an electrical field.

In 2003, a new test called the vertical auto profile or VAP, was developed that provides detailed measurements of cholesterol subclasses. These subclasses play important roles in patients later developing heart disease. The new tests were predicted to help identify important, emerging risk factors for heart disease.

Other, less extensive procedures also are used. For example, if only HDL is to be measured, a chemical is added to the serum that will clump the other classes, leaving HDL free in the serum to be measured by a chemical method. LDL often is not measured directly but its level is calculated based on the measurements of total cholesterol, HDL, and triglycerides. The formula is called the Friedewald formula: $\text{LDL} = \text{total cholesterol} - \text{HDL} - (\text{triglycerides}/5)$. The calculated result will be inaccurate in a person with high triglycerides. Results usually are available the same or following day.

Preparation

The patient must fast for 12 hours before the test, eating nothing and drinking only water. The person should not have alcohol for 24 hours before the test. There should be a stable diet and no illnesses occurring in the preceding two weeks.

A lipoproteins test requires 5 mL (milliliters) of blood. A person's physical position while having blood collected affects the results. Values from blood drawn while a person is sitting may be different from those while the person is standing. If repeated testing is done, the person should be in the same position each time.

Aftercare

Discomfort or bruising may occur at the puncture site or the person may feel dizzy or faint. Pressure to

KEY TERMS

Atherosclerosis—Disease of blood vessels caused by deposits of cholesterol on the inside walls of the vessels.

Cardiovascular disease—Disease that affects the heart and blood vessels.

Cholesterol—A fat-like substance called a lipid. It is used to build cell membranes and hormones. The body makes cholesterol and gets it from food.

Lipoproteins—The packages in which cholesterol and triglycerides travel throughout the body.

the puncture site until the bleeding stops reduces bruising. Warm packs to the puncture site relieve discomfort.

Normal results

People with HDL levels between 45 mg/dl and 59 mg/dl carry an average risk for cardiovascular disease. People with HDL levels above 60 mg/dl have a negative risk factor and appear to be protected from cardiovascular disease.

LDL levels below 130 mg/dl are desirable.

Some people have normal variations in their lipoprotein and total cholesterol levels. Repeat testing may be necessary, especially if a value is at a borderline risk category point.

Abnormal results

People with HDL levels 36-44 mg/dl have a moderate risk of cardiovascular disease. HDL levels below 35 mg/dl are a major risk.

LDL levels 130-159 mg/dl place a person at a borderline high risk of cardiovascular disease; levels above 160 mg/dl place a person at high risk. Relative proportions between HDL and LDL are important also. Results of a large clinical trial in 2003 showed that the new VAP cholesterol tests increased lipid-lowering therapy by 59% in high-risk patients with diabetes.

Resources

PERIODICALS

"Doctors Laboratory to Offer VAP Expanded Cholesterol Test." *Heart Diseases Weekly* September 7, 2003: 35.

"Results of a Prospective, Multi-center Study Showed that the Availability of Lipoprotein Subclass Testing

(Vertical Auto profile û VAP û Cholesterol Test)
Increased Use of Lipid-lowering Therapy by 59% in
High-risk Patients with Type 2 Diabetes.” *Diagnostics &
Imaging Week* June 19, 2003: 6.

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas,
TX 75231. (214) 373-6300. <[http://
www.americanheart.org](http://www.americanheart.org)>.

National Heart, Lung and Blood Institute. P.O. Box 30105,
Bethesda, MD 20824-0105. (301) 251-1222. <[http://
www.nhlbi.nih.gov](http://www.nhlbi.nih.gov)>.

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Liposuction

Definition

Liposuction, also known as lipoplasty or suction-assisted lipectomy, is **cosmetic surgery** performed to remove unwanted deposits of fat from under the skin. The doctor sculpts and recontours the patient's body by removing excess fat deposits that have been resistant to reduction by diet or **exercise**. The fat is permanently removed from under the skin with a suction device.

Purpose

Liposuction is intended to reduce and smooth the contours of the body and improve the patient's appearance. Its goal is cosmetic improvement. It is the most commonly performed cosmetic procedure in the United States.

Liposuction does not remove large quantities of fat and is not intended as a weight reduction technique. The average amount of fat removed is about a liter, or a quart. Although liposuction is not intended to remove cellulite (lumpy fat), some doctors believe that it improves the appearance of cellulite areas (thighs, hips, buttocks, abdomen, and chin).

A new technique called liposhaving shows more promise at reducing cellulite.

Precautions

Liposuction is most successful on patients who have firm, elastic skin and concentrated pockets of fat in cellulite areas. To get good results after fat removal, the skin must contract to conform to the



“Before” photo of patient undergoing liposuction.
(Photograph by I. Richard Toranto, M.D., Custom Medical Stock
Photo. Reproduced by permission.)

new contours without sagging. Older patients have less elastic skin and therefore may not be good candidates for this procedure. Patients with generalized fat distribution, rather than localized pockets, are not good candidates.

Patients should be in good general health and free of heart or lung disease. Patients with poor circulation or who have had recent surgery at the intended site of fat reduction are not good candidates.

Description

Most liposuction procedures are performed under **local anesthesia** (loss of sensation without loss of consciousness) by the tumescent or wet technique. In this technique, large volumes of very dilute local anesthetic (a substance that produces anesthesia) are injected under the patient's skin, making the tissue swollen and firm. Epinephrine is added to the solution to



“After” photo of same patient following liposuction.
(Photograph by I. Richard Toranto, M.D., Custom Medical Stock Photo. Reproduced by permission.)

reduce bleeding, and make possible the removal of larger amounts of fat.

The doctor first numbs the skin with an injection of local anesthetic. After the skin is desensitized, the doctor makes a series of tiny incisions, usually 0.12-0.25 in (3-6 mm) in length. The area is then flooded with a larger amount of local anesthetic. Fat is then extracted with suction through a long, blunt hollow tube called a cannula. The doctor repeatedly pushes the cannula through the fat layers in a radiating pattern creating tunnels, removing fat, and recontouring the area. Large quantities of intravenous fluid (IV) is given during the procedure to replace lost body fluid. Blood transfusions are possible.

Some newer modifications to the procedure involve the use of a cutting cannula called a liposhaver, or the use of ultrasound to help break up the fat deposits. The patient is awake and comfortable during these procedures.

The length of time required to perform the procedure varies with the amount of fat that is to be removed and the number of areas to be treated. Most operations take from 30 minutes to 2 hours, but extensive procedures can take longer. The length of time required also varies with the manner in which the anesthetic is injected.

The cost of liposuction can vary depending upon the standardized fees in the region of the country where it is performed, the extent of the area being treated, and the person performing the procedure. Generally, small areas, such as the chin or knees, can be done for as little as \$500, while more extensive treatment, such as when hips, thighs, and abdomen are done simultaneously, can cost as much as \$10,000. These procedures are cosmetic and are not covered by most insurance policies.

Preparation

The doctor will do a physical exam and may order blood work to determine clotting time and hemoglobin level for transfusions should the need arise. The patient may be placed on **antibiotics** immediately prior to surgery to ward off infection.

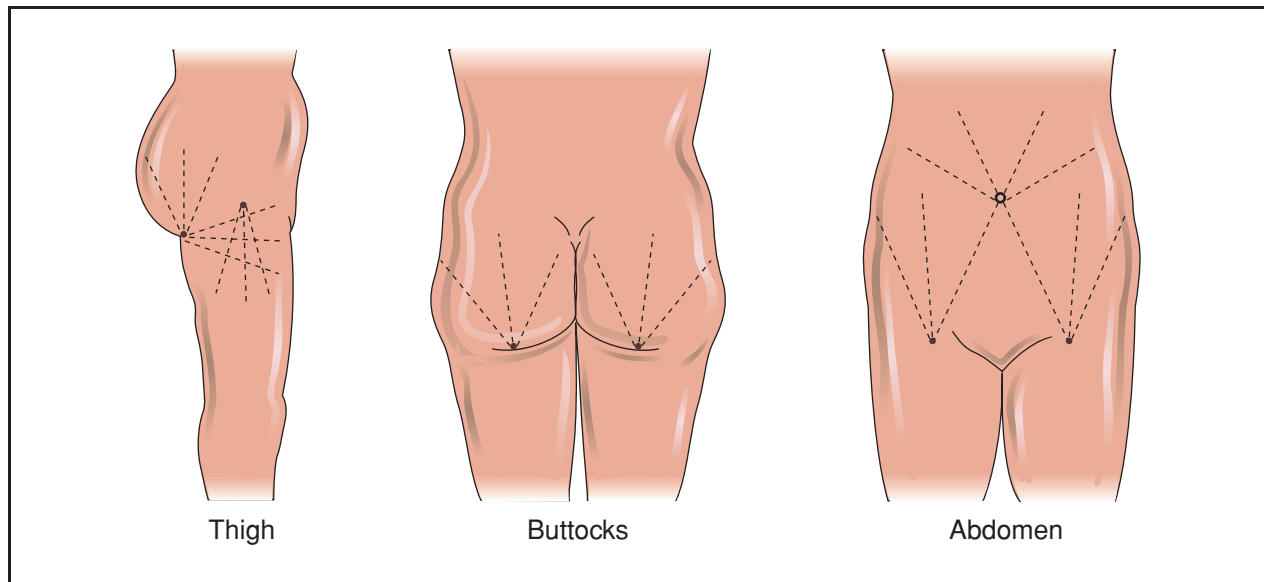
Aftercare

After the surgery, the patient will need to wear a support garment continuously for 2-3 weeks. If ankles or calves were treated, support hose will need to be worn for up to 6 weeks. The support garments can be removed during bathing 24 hours after surgery. A drainage tube, under the skin in the area of the procedure, may be inserted to prevent fluid build-up.

Mild side effects can include a burning sensation at the site of the surgery for up to one month. The patient should be prepared for swelling of the tissues below the operated site for 6-8 weeks after surgery. Wearing the special elastic garments will help reduce this swelling and help to achieve the desired final results.

The incisions involved in this procedure are tiny, but the surgeon may close them with stitches or staples. These will be removed the day after surgery. However, three out of eight doctors use no sutures. Minor bleeding or seepage through the incision site is common after this procedure. Wearing the elastic bandage or support garment helps reduce fluid loss.

This operation is virtually painless. However, for the first postoperative day, there may be some discomfort which will require light **pain** medication. Soreness or aching may persist for several days. The patient can usually return to normal activity within a week.



Common entry sites for liposuction procedures. (Illustration by Electronic Illustrators Group.)

Postoperative bruising will go away by itself within 10-14 days. Postoperative swelling begins to go down after a week. It may take 3-6 months for the final contour to be reached.

Risks

Liposuction under local anesthesia using the tumescent technique is exceptionally safe. A 1995 study of 15,336 patients showed no serious complications or deaths. Another study showed a 1% risk factor. However, as with any surgery, there are some risks and serious complications. **Death** is possible.

The main hazards associated with this surgery involve migration of a blood clot or fat globule to the heart, brain, or lungs. Such an event can cause a **heart attack**, **stroke**, or serious lung damage. However, this complication is rare and did not occur even once in the study of 15,336 patients. The risk of blood clot formation is reduced with the wearing of special girdle-like compression garments after the surgery, and with the resumption of normal mild activity soon after surgery.

Staying in bed increases the risk of clot formation, but not getting enough rest can result in increased swelling of the surgical area. Such swelling is a result of excess fluid and blood accumulation, and generally comes from not wearing the compression garments. If necessary, this excess fluid can be drained off with a needle in the doctor's office.

Infection is another complication, but this rarely occurs. If the physician is skilled and works in a sterile environment, infection should not be a concern.

If too much fat is removed, the skin may peel in that area. Smokers are at increased risk for shedding skin because their circulation is impaired. Another and more serious hazard of removing too much fat is that the patient may go into **shock**. Fat tissue has an abundant blood supply and removing too much of it at once can cause shock if the fluid is not replaced.

A rare complication is perforation or puncture of an organ. The procedure involves pushing a cannula vigorously through the fat layer. If the doctor pushes too hard or if the tissue gives way too easily under the force, the blunt hollow tube can go too far and injure internal organs.

Liposuction can damage superficial nerves. Some patients lose sensation in the area that has been suctioned, but feeling usually returns with time.

Normal results

The loss of fat cells is permanent, and the patient should have smoother, more pleasing body contours without excessive bulges. However, if the patient overeats, the remaining fat cells will grow in size. Although the patient may gain weight back, the body should retain the new proportions and the suctioned area should remain proportionally smaller.

Tiny **scars** about 0.25-0.5 in (6-12 mm) long at the site of incision are normal. The doctor usually makes the incisions in places where the scars are not likely to show.

In some instances, the skin may appear rippled, wavy, or baggy after surgery. Pigmentation spots may

KEY TERMS

Cellulite—Cellulite is dimply skin caused by uneven fat deposits beneath the surface.

Epinephrine—Epinephrine is a drug that causes blood vessels to constrict or narrow. It is used in local anesthetics to reduce bleeding.

Hemoglobin—Hemoglobin is the component of blood that carries oxygen to the tissues.

Liposhaving—Liposhaving involves removing fat that lies closer to the skin's surface by using a needle-like instrument that contains a sharp-edged shaving device.

Tumescent technique—The tumescent technique of liposuction involves swelling, or tumescing, the tissue with large volumes of dilute anesthetic.

develop. The recontoured area may be uneven. This unevenness is common, occurring in 5-20% of the cases, and can be corrected with a second procedure that is less extensive than the first.

Resources

PERIODICALS

Taylor, Mia, Lloyd A. Hoffman, and Michael Lieberman. "Intestinal Perforation after Suction Lipoplasty: A Case Report and Review of the Literature." *Annals of Plastic Surgery* 38, no. 2: 169-172.

ORGANIZATIONS

American Society of Aesthetic Plastic Surgery. (888) 272-7711.

American Society of Plastic and Reconstructive Surgeons. 44 E. Algonquin Rd., Arlington Heights, IL 60005. (847) 228-9900. <<http://www.plasticsurgery.org>>.

Lipoplasty Society of North America. (800) 848-1991.

Louann W. Murray, PhD

Listeria monocytogenes infection see

Listeriosis

Listeriosis

Definition

Listeriosis is an illness caused by the bacterium *Listeria monocytogenes* that is acquired by eating contaminated food. The organism can spread to the

blood stream and central nervous system. During **pregnancy**, listeriosis often causes **miscarriage** or **stillbirth**.

Description

Listeriosis is caused by an infection with the bacterium *Listeria monocytogenes*. This bacteria can be carried by many animals and birds, and it has been found in soil, water, sewage, and animal feed. Five out of every 100 people carry *Listeria monocytogenes* in their intestines. Listeriosis is considered a "food-borne illness" because most people are probably infected after eating food contaminated with *Listeria monocytogenes*. However, a woman can pass the bacteria to her baby during pregnancy. In addition, there have been a few cases where workers have developed *Listeria* skin infections by touching infected calves or poultry.

In the 1980s, the United States government began taking measures to decrease the occurrence of listeriosis. Processed meats and dairy products are now tested for the presence of *Listeria monocytogenes*. The Food and Drug Administration (FDA) and the Food Safety and Inspection Service (FSIS) can legally prevent food from being shipped, or order food recalls, if they detect any *Listeria* bacteria. These inspections, in combination with the public education regarding the proper handling of uncooked foods, appear to be working. In 1989, there were 1,965 cases of listeriosis with 481 deaths. In 1993, the numbers fell to 1,092 cases with 248 deaths.

In 1996, the Centers for Disease Control and Prevention (CDC) began a nationwide food-borne disease surveillance program called "FoodNet," in which seven states were participating by January 1997. Results from the program indicated that, in 1996, one person out of every 200,000 people got listeriosis. FoodNet also revealed that the hospitalization rate was higher for listeriosis (94%) than for any other food-borne illness. In addition, FoodNet found that the *Listeria* bacteria reached the blood and cerebrospinal fluid in 89% of cases, a higher percentage than in any other food-borne illness.

Persons at particular risk for listeriosis include the elderly, pregnant women, newborns, and those with a weakened immune system (called "immunocompromised"). Risk is increased when a person suffers from diseases such as **AIDS**, **cancer**, **kidney disease**, **diabetes mellitus**, or by the use of certain medications. Infection is most common in babies younger than one month old and adults over 60 years of age. Pregnant women account for 27% of the cases and immunocompromised persons account for almost

70%. Persons with AIDS are 280 times more likely to get listeriosis than others.

Causes and symptoms

As noted, persons become infected with *Listeria monocytogenes* by eating contaminated food. *Listeria* has been found on raw vegetables, fish, poultry, raw (unpasteurized) milk, fresh meat, processed meat (such as deli meat, hot dogs, and canned meat), and certain soft cheeses. Listeriosis outbreaks in the United States since the 1980s have been linked to cole slaw, milk, Mexican-style cheese, undercooked hot dogs, undercooked chicken, and delicatessen foods. Unlike most other bacteria, *Listeria monocytogenes* does not stop growing when food is in the refrigerator—its growth is merely slowed. Fortunately, typical cooking temperatures and the pasteurization process do kill this bacteria.

Listeria bacteria can pass through the wall of the intestines, and from there they can get into the blood stream. Once in the blood stream, they can be transported anywhere in the body, but are commonly found the central nervous system (brain and spinal cord); and in pregnant women they are often found in the placenta (the organ which connects the baby's umbilical cord to the uterus). *Listeria monocytogenes* live inside specific white blood cells called macrophages. Inside macrophages, the bacteria can hide from immune responses and become inaccessible to certain **antibiotics**. *Listeria* bacteria are capable of multiplying within macrophages, and then may spread to other macrophages.

After consuming food contaminated with this bacteria, symptoms of infection may appear anywhere from 11-70 days later. Most people do not get any noticeable symptoms. Scientists are unsure, but they believe that *Listeria monocytogenes* can cause upset stomach and intestinal problems just like other food-borne illnesses. Persons with listeriosis may develop flu-like symptoms such as **fever, headache, nausea and vomiting**, tiredness, and **diarrhea**.

Pregnant women experience a mild, flu-like illness with fever, muscle aches, upset stomach, and intestinal problems. They recover, but the infection can cause miscarriage, **premature labor**, early rupture of the birth sac, and stillbirth. Unfortunately, half of the newborns infected with *Listeria* will die from the illness.

There are two types of listeriosis in the newborn baby: early-onset disease and late-onset disease. Early-onset disease refers to a serious illness that is present at birth and usually causes the baby to be born prematurely. Babies infected during the

pregnancy usually have a blood infection (**sepsis**) and may have a serious, whole body infection called granulomatosis infantisepticum. When a full-term baby becomes infected with *Listeria* during **childbirth**, that situation is called late-onset disease. Commonly, symptoms of late-onset listeriosis appear about two weeks after birth. Babies with late-term disease typically have **meningitis** (inflammation of the brain and spinal tissues); yet they have a better chance of surviving than those with early-onset disease.

Immunocompromised adults are at risk for a serious infection of the blood stream and central nervous system (brain and spinal cord). Meningitis occurs in about half of the cases of adult listeriosis. Symptoms of listerial meningitis occur about four days after the flu-like symptoms and include fever, personality change, uncoordinated muscle movement, **tremors**, muscle contractions, seizures, and slipping in and out of consciousness.

Listeria monocytogenes causes **endocarditis** in about 7.5% of the cases. Endocarditis is an inflammation of heart tissue due to the bacterial infection. Listerial endocarditis causes **death** in about half of the patients. Other diseases which have been caused by *Listeria monocytogenes* include **brain abscess**, eye infection, hepatitis (**liver disease**), **peritonitis** (abdominal infection), lung infection, joint infection, arthritis, heart disease, bone infection, and gallbladder infection.

Diagnosis

Listeriosis may be diagnosed and treated by infectious disease specialists and internal medicine specialists. The diagnosis and treatment of this infection should be covered by most insurance providers.

The only way to diagnose listeriosis is to isolate *Listeria monocytogenes* from blood, cerebrospinal fluid, or stool. A sample of cerebrospinal fluid is removed from the spinal cord using a needle and syringe. This procedure is commonly called a spinal tap. The amniotic fluid (the fluid which bathes the unborn baby) may be tested in pregnant women with listeriosis. This sample is obtained by inserting a needle through the abdomen into the uterus and withdrawing fluid. *Listeria* grows well in laboratory media and test results can be available within a few days.

Treatment

Listeriosis is treated with the antibiotics ampicillin (Omnipen) or sulfamethoxazole-trimethoprim (Bactrim, Septra). Because the bacteria live within macrophage

KEY TERMS

Abscess—An accumulation of pus caused by localized infection in tissues or organs. *Listeria monocytogenes* can cause abscesses in many organs including the brain, spleen, and liver.

Immunocompromised—To have a poor immune system due to disease or medication. Immunocompromised persons are at risk for developing infections because they can't fight off microorganisms like healthy persons can.

Macrophages—White blood cells whose job is to destroy invading microorganisms. *Listeria monocytogenes* avoids being killed and can multiply within the macrophage.

Meningitis—An inflammation of the tissues that surround the brain and spinal cord. It can be caused by a bacterial infection.

Sepsis—The presence of bacteria in the blood stream, a normally sterile environment.

cells, treatment may be difficult and the treatment periods may vary. Usually, pregnant women are treated for two weeks; newborns, two to three weeks; adults with mild disease, two to four weeks; persons with meningitis, three weeks; persons with brain abscesses, six weeks; and persons with endocarditis, four to six weeks.

Patients are often hospitalized for treatment and monitoring. Other drugs may be provided to relieve **pain** and fever and to treat other reactions to the infection.

Prognosis

The overall death rate for listeriosis is 26%. This high death rate is due to the serious illness suffered by newborns, the elderly, and immunocompromised persons. Healthy adults and older children have a low death rate. Complications of *Listeria* infection include: meningitis, sepsis, miscarriage, stillbirth, **pneumonia**, **shock**, endocarditis, **abscess** (localized infection) formation, and eye inflammation.

Prevention

The United States government has already done much to prevent listeriosis. Persons at extremely high risk (pregnant women, immunocompromised persons, etc.) must use extra caution. High risk persons should: avoid soft cheeses, such as Mexican

cheese, feta, Brie, Camembert, and blue cheese (cottage cheese is safe), thoroughly cook leftovers and ready-to-eat foods (such as hot-dogs), and avoid foods from the deli.

For all people, the risk of listeriosis can be reduced by taking these precautions:

- Completely cook all meats and eggs.
- Carefully wash raw vegetables before eating.
- Keep raw meat away from raw vegetables and prepared foods. After cutting raw meat, wash the cutting board with detergent before using it for vegetables.
- Avoid drinking unpasteurized milk or foods made from such milk.
- Wash hands thoroughly after handling raw meat.
- Follow the instructions on food labels. Observe food expiration dates and storage conditions.

Resources

OTHER

“Preventing Foodborne Illness: Listeriosis.” *Centers for Disease Control*. <<http://www.cdc.gov/ncidod/diseases/foodbornelister.htm>>.

Belinda Rowland, PhD

Lithotripsy

Definition

Lithotripsy is the use of high-energy shock waves to fragment and disintegrate **kidney stones**. The shock wave, created by using a high-voltage spark or an electromagnetic impulse, is focused on the stone. This shock wave shatters the stone and this allows the fragments to pass through the urinary system. Since the shock wave is generated outside the body, the procedure is termed extracorporeal shock wave lithotripsy, or ESWL.

Purpose

ESWL is used when a kidney stone is too large to pass on its own, or when a stone becomes stuck in a ureter (a tube which carries urine from the kidney to the bladder) and will not pass. Kidney stones are extremely painful and can cause serious medical complications if not removed.

Precautions

ESWL should not be considered for patients with severe skeletal deformities, patients weighing over 300 lbs (136 kg), patients with abdominal aortic aneurysms, or patients with uncontrollable bleeding disorders. Patients who are pregnant should not be treated with ESWL. Patients with cardiac **pacemakers** should be evaluated by a cardiologist familiar with ESWL. The cardiologist should be present during the ESWL procedure in the event the pacemaker needs to be overridden.

Description

Lithotripsy uses the technique of focused shock waves to fragment a stone in the kidney or the ureter. The patient is placed in a tub of water or in contact with a water-filled cushion, and a shock wave is created which is focused on the stone. The wave shatters and fragments the stone. The resulting debris, called gravel, then passes through the remainder of the ureter, through the bladder, and through the urethra during urination. There is minimal chance of damage to skin or internal organs because biologic tissues are resilient, not brittle, and because the shock waves are not focused on them.

Preparation

Prior to the lithotripsy procedure, a complete **physical examination** is done, followed by tests to determine the number, location, and size of the stone or stones. A test called an intravenous pyelogram, or IVP, is used to locate the stones. An IVP involves injecting a dye into a vein in the arm. This dye, which shows up on x ray, travels through the bloodstream and is excreted by the kidneys. The dye then flows down the ureters and into the bladder. The dye surrounds the stones, and x rays are then used to evaluate the stones and the anatomy of the urinary system. (Some people are allergic to the dye material, so it cannot be used. For these people, focused sound waves, called ultrasound, can be used to see where the stones are located.) Blood tests are done to determine if any potential bleeding problems exist. For women of childbearing age, a **pregnancy** test is done to make sure the patient isn't pregnant; and elderly patients have an EKG done to make sure no potential heart problems exist. Some patients may have a stent placed prior to the lithotripsy procedure. A stent is a plastic tube placed in the ureter which allows the passage of gravel and urine after the ESWL procedure is completed.

KEY TERMS

Aneurysm—A dilation of the wall of an artery which causes a weak area prone to rupturing.

Bladder—Organ in which urine is stored prior to urination.

Bleeding disorder—Problems in the clotting mechanism of the blood.

Cardiologist—A physician who specializes in problems of the heart.

EKG—A tracing of the electrical activity of the heart.

ESWL (Extracorporeal shock wave lithotripsy)—The use of focused shock waves, generated outside the body, to fragment kidney stones.

Gravel—The debris which is formed from a fragmented kidney stone.

IVP (Intravenous pyelogram)—The use of a dye, injected into the veins, used to locate kidney stones. Also used to determine the anatomy of the urinary system.

Kidney stone—A hard mass that forms in the urinary tract and which can cause pain, bleeding, obstruction, or infection. Stones are primarily made up of calcium.

Stent—A plastic tube placed in the ureter prior to the ESWL procedure which facilitates the passage of gravel and urine.

Ultrasound—Sound waves used to determine the internal structures of the body.

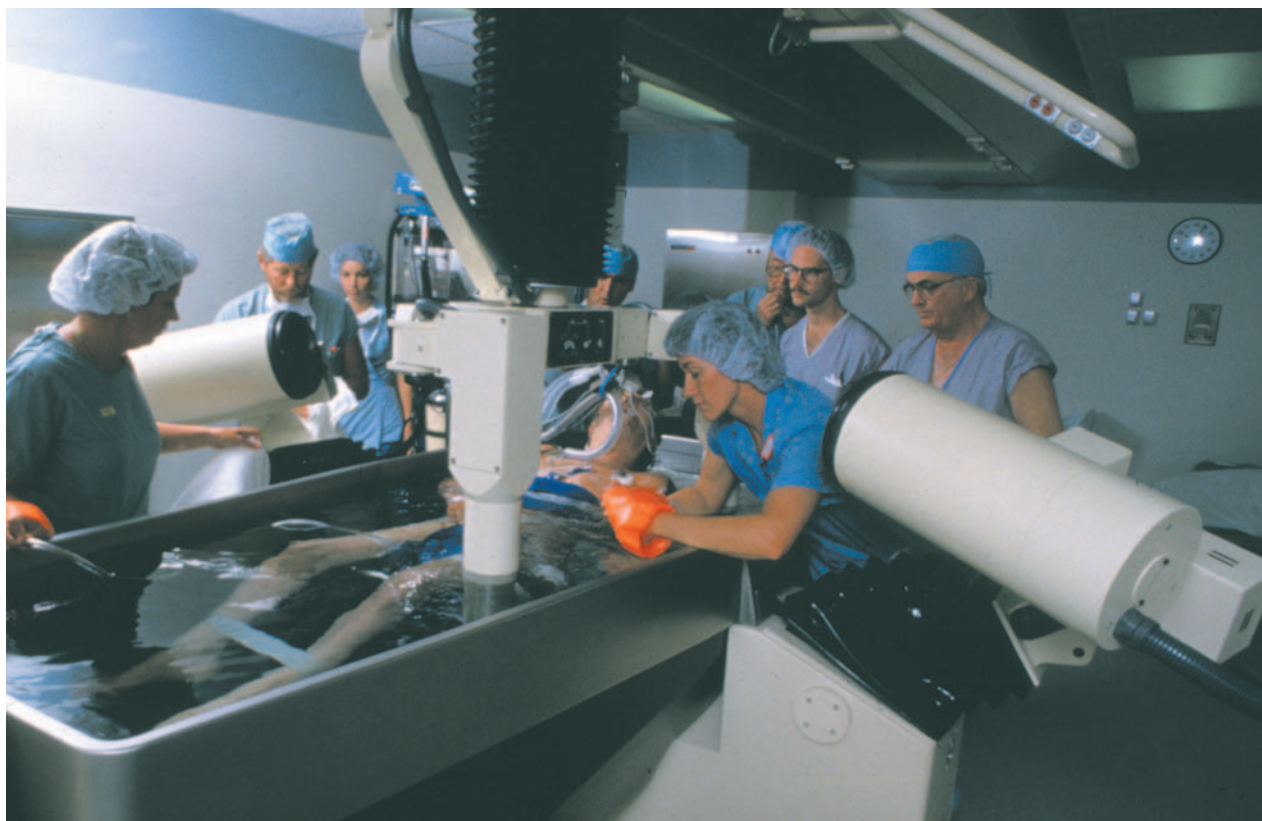
Ureter—A tube which carries urine from the kidney to the bladder.

Urethra—A tube through which urine passes during urination.

Urologist—A physician who specializes in problems of the urinary system.

Aftercare

Most patients have a lot of blood in their urine after the ESWL procedure. This is normal and should clear after several days to a week or so. Lots of fluids should be taken to encourage the flushing of any gravel remaining in the urinary system. The patient should follow up with the urologist in about two weeks to make sure that everything is going as planned. If a stent has been inserted, it is normally removed at this time. Patients may return to work whenever they feel able.



A lithotripter in use by patient in tub. This noninvasive method crushes kidney stones through shock waves. (Photo Researchers, Inc. Reproduced by permission.)

Risks

Abdominal **pain** is not uncommon after ESWL, but it is usually not cause to worry. However, persistent or severe abdominal pain may imply unexpected internal injury. Colicky renal pain is very common as gravel is still passing. Other problems may include perirenal hematomas (**blood clots** near the kidneys) in 66% of the cases; nerve palsies; **pancreatitis** (inflammation of the pancreas); and obstruction by stone fragments. Occasionally, stones may not be completely fragmented during the first ESWL treatment and further ESWL procedures may be required.

Resources

ORGANIZATIONS

American Urological Association. 1120 North Charles St., Baltimore, MD 21201-5559. (410) 727-1100. <http://www.auanet.org/index_hi.cfm>.

Joseph Knight, PA

Live cell therapy see **Cell therapy**

Liver-spleen scan see **Liver nuclear medicine scan**

Liver biopsy

Definition

A liver biopsy is a medical procedure performed to obtain a small piece of liver tissue for diagnostic testing. Liver biopsies are sometimes called percutaneous liver biopsies, because the tissue sample is obtained by going through the patient's skin.

Purpose

A liver biopsy is usually done to diagnose a tumor, or to evaluate the extent of damage that has occurred to the liver because of chronic disease. Biopsies are often performed to identify abnormalities in liver tissues after imaging studies have failed to yield clear results.

A liver biopsy may be ordered to evaluate any of the following conditions or disorders:

- Jaundice
- **Cirrhosis**
- Hemochromatosis, which is a condition of excess iron in the liver.
- Repeated abnormal results from liver function tests
- Unexplained swelling or enlargement of the liver
- Primary cancers of the liver, such as hepatomas, cholangiocarcinomas, and angiosarcomas
- Metastatic cancers of the liver.

Precautions

Some patients should not have percutaneous liver biopsies. They include patients with any of the following conditions:

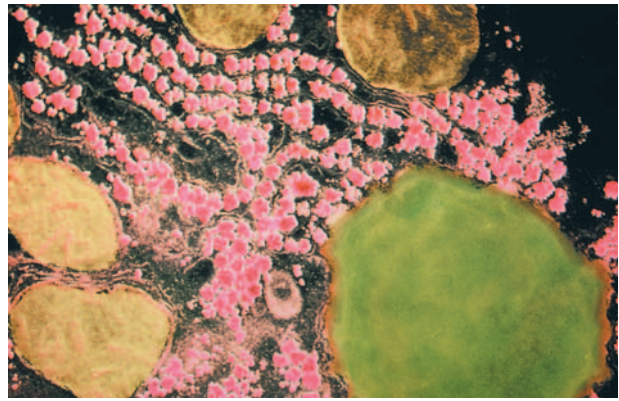
- A **platelet count** below 60,000
- A longer-than-normal **prothrombin time**
- A liver tumor that contains a large number of blood vessels
- A history of unexplained bleeding
- A watery (hydatid) cyst
- An infection in either the cavity around the lungs, or the diaphragm.

Description

Percutaneous liver biopsy is done with a special hollow needle, called a Menghini needle, attached to a suction syringe. Doctors who specialize in the digestive system or liver will sometimes perform liver biopsies. But in most cases, a radiologist (a doctor who specializes in x rays and imaging studies) performs the biopsy. The radiologist will use computed tomography scan (CT scan) or ultrasound to guide the choice of the site for the biopsy.

An hour or so before the biopsy, the patient may be given a sedative to help relaxation. He or she is then asked to lie on the back with the right elbow to the side and the right hand under the head. The patient is instructed to lie as still as possible during the procedure. He or she is warned to expect a sensation resembling a punch in the right shoulder, but to hold still in spite of the momentary feeling.

The doctor marks a spot on the skin where the needle will be inserted and thoroughly cleanses the right side of the upper abdomen with an antiseptic



A false color image of hepatocyte cells of the liver that secrete bile. (Custom Medical Stock Photo. Reproduced by permission.)

solution. The patient is then given an anesthetic at the biopsy site.

The needle with attached syringe is inserted into the patient's chest wall. The doctor then draws the plunger of the syringe back to create a vacuum. At this point the patient is asked to take a deep breath, exhale the air and hold their breath at the point of complete exhalation. The needle is inserted into the liver and withdrawn quickly, usually within two seconds or less. The negative pressure in the syringe draws or pulls a sample of liver tissue into the biopsy needle. As soon as the needle is withdrawn, the patient can breathe normally. Pressure is applied at the biopsy site to stop any bleeding, and a bandage will be placed over it. The entire procedure takes 10 to 15 minutes. Test results are usually available within a day.

Preparation

Aspirin and non-steroidal anti-inflammatory drugs (NSAIDs) such as ibuprofen are known to thin the blood and interfere with clotting. These medications should be avoided for at least a week before the biopsy. Four to eight hours before the biopsy, patients should stop eating and drinking.

The patient's blood will be tested prior to the biopsy to make sure that it is clotting normally. Tests will include a platelet count and a prothrombin time. Doctors will also ensure that the patient is not taking any other medications, such as blood thinners like Coumadin, that might affect blood clotting.

Aftercare

Liver biopsies are outpatient procedures in most hospitals. After the biopsy, patients are usually

KEY TERMS

Biopsy—A procedure where a piece of tissue is removed from a patient for diagnostic testing.

Menghini needle—A special needle used to obtain a sample of liver tissue.

Percutaneous biopsy—A biopsy in which a needle is inserted and a tissue sample removed through the skin.

Prothrombin time—A blood test that determines how quickly a person's blood will clot.

Vital signs—A person's essential body functions, usually defined as the pulse, body temperature, and breathing rate.

instructed to lie on their right side for about two hours. This provides pressure to the biopsy site and helps prevent bleeding. A nurse will check the patient's vital signs at regular intervals. If there are no complications, the patient is sent home within about four to eight hours.

Patients should arrange to have a friend or relative take them home after discharge. Bed rest for a day is recommended, followed by a week of avoiding heavy work or strenuous **exercise**. The patient can resume eating a normal diet.

Some mild soreness in the area of the biopsy is normal after the anesthetic wears off. Irritation of the muscle that lies over the liver can also cause mild discomfort in the shoulder for some patients. Tylenol can be taken for minor soreness, but aspirin and NSAIDs are best avoided. Patients should call their doctor if they have severe **pain** in the abdomen, chest or shoulder, difficulty breathing, or persistent bleeding. These signs may indicate that there has been leakage of bile into the abdominal cavity, or that air has been introduced into the cavity around the lungs.

Risks

The risks of a liver biopsy are usually very small. When complications do occur, over 90% are apparent within 24 hours after the biopsy. The most significant risk is internal bleeding. Bleeding is most likely to occur in elderly patients, in patients with cirrhosis, or in patients with a tumor that has many blood vessels. Other complications from percutaneous liver biopsies include the leakage of bile or the introduction of air into the chest cavity (**pneumothorax**). There is also a

small chance that an infection may occur, or an internal organ such as the lung, gall bladder, or kidney could be punctured.

Normal results

After the biopsy, the liver sample is sent to the pathology laboratory for study under a microscope. A normal (negative) result would find no evidence of **cancer** or other disease in the tissue sample.

Abnormal results

Changes in liver tissue that are visible under the microscope indicate abnormal results. Possible causes for the abnormality include the presence of a tumor, or a disease such as hepatitis.

Resources

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Lata Cherath, PhD

Liver cancer

Definition

Liver **cancer** is a relatively rare form of cancer but has a high mortality rate. Liver cancers can be classified into two types. They are either primary, when the cancer starts in the liver itself, or metastatic, when the cancer has spread to the liver from some other part of the body.

Description

Primary liver cancer

Primary liver cancer is a relatively rare disease in the United States, representing about 2% of all

malignancies and 4% of newly diagnosed cancers. Hepatocellular carcinoma (HCC) is the fifth most common cancer in the world as of 2004. It is much more common outside the United States, representing 10% to 50% of malignancies in Africa and parts of Asia. Rates of HCC in men are at least two to three times higher than for women. In high-risk areas (East and Southeast Asia, sub-Saharan Africa), men are even more likely to have HCC than women.

According to the American Cancer Society, 18,920 people in the United States will be diagnosed with primary liver cancer in 2004, and 14,270 persons will die from the disease. The incidence of primary liver cancer has been rising in the United States and Canada since the mid-1990s, most likely as a result of the rising rate of **hepatitis C** infections.

TYPES OF PRIMARY LIVER CANCER. In adults, most primary liver cancers belong to one of two types: hepatomas, or hepatocellular carcinomas (HCC), which start in the liver tissue itself; and cholangiomas, or cholangiocarcinomas, which are cancers that develop in the bile ducts inside the liver. About 80% to 90% of primary liver cancers are hepatomas. In the United States, about five persons in every 200,000 will develop a hepatoma (70% to 75% of cases of primary liver cancers are HCC). In Africa and Asia, over 40 persons in 200,000 will develop this form of cancer (more than 90% of cases of primary liver are HCC). Two rare types of primary liver cancer are mixed-cell tumors and Kupffer cell **sarcomas**.

One type of primary liver cancer, called a hepatoblastoma, usually occurs in children younger than four years of age and between the ages of 12 and 15. Unlike liver cancers in adults, hepatoblastomas have a good chance of being treated successfully. Approximately 70% of children with hepatoblastomas experience complete cures. If the tumor is detected early, the survival rate is over 90%.

Metastatic liver cancer

The second major category of liver cancer, metastatic liver cancer, is about 20 times as common in the United States as primary liver cancer. Because blood from all parts of the body must pass through the liver for filtration, cancer cells from other organs and tissues easily reach the liver, where they can lodge and grow into secondary tumors. Primary cancers in the colon, stomach, pancreas, rectum, esophagus, breast, lung, or skin are the most likely to metastasize (spread) to the liver. It is not unusual for the metastatic cancer in the liver to be the first noticeable sign of a cancer that started in another organ. After **cirrhosis**,

metastatic liver cancer is the most common cause of fatal **liver disease**.

Causes and symptoms

Risk factors

The exact cause of primary liver cancer is still unknown. In adults, however, certain factors are known to place some individuals at higher risk of developing liver cancer. These factors include:

- Male sex.
- Age over 60 years.
- Ethnicity. Asian Americans with cirrhosis have four times as great a chance of developing liver cancer as Caucasians with cirrhosis, and African Americans have twice the risk of Caucasians. In addition, Asians often develop liver cancer at much younger ages than either African Americans or Caucasians.
- Exposure to substances in the environment that tend to cause cancer (carcinogens). These include: a substance produced by a mold that grows on rice and peanuts (aflatoxin); thorium dioxide, which was once used as a contrast dye for x rays of the liver; vinyl chloride, used in manufacturing plastics; and cigarette smoking.
- Use of oral estrogens for birth control.
- Hereditary **hemochromatosis**. This is a disorder characterized by abnormally high levels of iron storage in the body. It often develops into cirrhosis.
- Cirrhosis. Hepatomas appear to be a frequent complication of cirrhosis of the liver. Between 30% and 70% of hepatoma patients also have cirrhosis. It is estimated that a patient with cirrhosis has 40 times the chance of developing a hepatoma than a person with a healthy liver.
- Exposure to hepatitis viruses: **Hepatitis B** (HBV), **Hepatitis C** (HCV), **Hepatitis D** (HDV), or **Hepatitis G** (HGV). It is estimated that 80% of worldwide HCC is associated with chronic HBV infection. In Africa and most of Asia, exposure to hepatitis B is an important factor; in Japan and some Western countries, exposure to hepatitis C is connected with a higher risk of developing liver cancer. In the United States, nearly 25% of patients with liver cancer show evidence of HBV infection. Hepatitis is commonly found among intravenous drug abusers. The 70% increase in HCC incidence in the United States is thought to be due to increasing rates of HBV and HCV infections due to increased sexual promiscuity and illicit drug needle sharing.

The association between HDV and HGV and HCC is unclear at this time.

Symptoms of liver cancer

The early symptoms of primary, as well as metastatic, liver cancer are often vague and not unique to liver disorders. The long period between the beginning of the tumor's growth and the first signs of illness is the major reason why the disease has such a high mortality rate. At the time of diagnosis, patients are often fatigued, with **fever**, abdominal **pain**, and loss of appetite. They may look emaciated and generally ill. As the tumor enlarges, it stretches the membrane surrounding the liver (the capsule), causing pain in the upper abdomen on the right side. The pain may extend into the back and shoulder. Some patients develop a collection of fluid, known as **ascites**, in the abdominal cavity. Others may show signs of bleeding into the digestive tract. In addition, the tumor may block the ducts of the liver or the gall bladder, leading to **jaundice**. In patients with jaundice, the whites of the eyes and the skin may turn yellow, and the urine becomes dark-colored.

Diagnosis

Physical examination

If the doctor suspects a diagnosis of liver cancer, he or she will check the patient's history for risk factors and pay close attention to the condition of the patient's abdomen during the **physical examination**. Masses or lumps in the liver and ascites can often be felt while the patient is lying flat on the examination table. The liver is usually swollen and hard in patients with liver cancer; it may be sore when the doctor presses on it. In some cases, the patient's spleen is also enlarged. The doctor may be able to hear an abnormal sound (bruit) or rubbing noise (friction rub) if he or she uses a stethoscope to listen to the blood vessels that lie near the liver. The noises are caused by the pressure of the tumor on the blood vessels.

Laboratory tests

Blood tests may be used to test liver function or to evaluate risk factors in the patient's history. Between 50% and 75% of primary liver cancer patients have abnormally high blood serum levels of a particular protein (alpha-fetoprotein or AFP). The AFP test, however, cannot be used by itself to confirm a diagnosis of liver cancer, because cirrhosis or chronic hepatitis can also produce high alpha-fetoprotein

levels. Tests for alkaline phosphatase, bilirubin, lactic dehydrogenase, and other chemicals indicate that the liver is not functioning normally. About 75% of patients with liver cancer show evidence of hepatitis infection. Again, however, abnormal liver function test results are not specific for liver cancer.

Imaging studies

Imaging studies are useful in locating specific areas of abnormal tissue in the liver. Liver tumors as small as an inch across can now be detected by ultrasound or computed tomography scan (CT scan). Imaging studies, however, cannot tell the difference between a hepatoma and other abnormal masses or lumps of tissue (nodules) in the liver. A sample of liver tissue for biopsy is needed to make the definitive diagnosis of a primary liver cancer. CT or ultrasound can be used to guide the doctor in selecting the best location for obtaining the biopsy sample.

Chest x rays may be used to see whether the liver tumor is primary or has metastasized from a primary tumor in the lungs.

Liver biopsy

Liver biopsy is considered to provide the definite diagnosis of liver cancer. A sample of the liver or tissue fluid is removed with a fine needle and is checked under a microscope for the presence of cancer cells. In about 70% of cases, the biopsy is positive for cancer. In most cases, there is little risk to the patient from the biopsy procedure. In about 0.4% of cases, however, the patient develops a fatal hemorrhage from the biopsy because some tumors are supplied with a large number of blood vessels and bleed very easily.

Laparoscopy

The doctor may also perform a **laparoscopy** to help in the diagnosis of liver cancer. First, the doctor makes a small cut in the patient's abdomen and inserts a small, lighted tube called a laparoscope to view the area. A small piece of liver tissue is removed and examined under a microscope for the presence of cancer cells.

Treatment

Treatment of liver cancer is based on several factors, including the type of cancer (primary or metastatic); stage (early or advanced); the location of other primary cancers or metastases in the patient's body; the patient's age; and other coexisting diseases, including cirrhosis. For many patients, treatment of liver

cancer is primarily intended to relieve the pain caused by the cancer but cannot cure it.

Surgery

Few liver cancers in adults can be cured by surgery because they are usually too advanced by the time they are discovered. If the cancer is contained within one lobe of the liver, and if the patient does not have either cirrhosis, jaundice, or ascites, surgery is the best treatment option. Patients who can have their entire tumor removed have the best chance for survival. Unfortunately, only about 5% of patients with metastatic cancer (from primary tumors in the colon or rectum) fall into this group. If the entire visible tumor can be removed, about 25% of patients will be cured. The operation that is performed is called a partial hepatectomy, or partial removal of the liver. The surgeon will remove either an entire lobe of the liver (a lobectomy) or cut out the area around the tumor (a wedge resection).

A newer technique that is reported to be safe and effective is laparoscopic radiofrequency ablation (RFA). RFA is a technique in which the surgeon places a special needle electrode in the tumor under guidance from MRI or CT scanning. When the electrode has been properly placed, a radiofrequency current is passed through it, heating the tumor and killing the cancer cells. RFA can be used to treat tumors that are too small or too inaccessible for removal by conventional open surgery.

Chemotherapy

Some patients with metastatic cancer of the liver can have their lives prolonged for a few months by **chemotherapy**, although cure is not possible. If the tumor cannot be removed by surgery, a tube (catheter) can be placed in the main artery of the liver and an implantable infusion pump can be installed. The pump allows much higher concentrations of the cancer drug to be carried to the tumor than is possible with chemotherapy carried through the bloodstream. The drug that is used for infusion pump therapy is usually floxuridine (FUDR), given for 14-day periods alternating with 14-day rests. Systemic chemotherapy can also be used to treat liver cancer. The medications usually used are 5-fluorouracil (Adrucil, Efudex) or methotrexate (MTX, Mexate). Systemic chemotherapy does not, however, significantly lengthen the patient's survival time.

Radiation therapy

Radiation therapy is the use of high-energy rays or x rays to kill cancer cells or to shrink tumors. Its use in

liver cancer, however, is only to give short-term relief from some of the symptoms. Liver cancers are not sensitive to radiation, and radiation therapy will not prolong the patient's life.

Liver transplantation

Removal of the entire liver (total hepatectomy) and **liver transplantation** can be used to treat liver cancer. However, there is a high risk of tumor recurrence and metastases after transplantation. In addition, most patients have cancer that is too far advanced at the time of diagnosis to benefit from liver transplantation.

Other therapies

Other therapeutic approaches include:

- Hepatic artery embolization with chemotherapy (chemoembolization).
- Alcohol ablation via ultrasound-guided percutaneous injection.
- Ultrasound-guided cryoablation.
- Immunotherapy with monoclonal antibodies tagged with cytotoxic agents.
- Gene therapy with retroviral vectors containing genes expressing cytotoxic agents.

Alternative treatment

Many patients find that alternative and complementary therapies help to reduce the **stress** associated with illness, improve immune function, and boost spirits. While there is no clinical evidence that these therapies specifically combat disease, activities such as **biofeedback**, relaxation, **therapeutic touch**, **massage therapy** and **guided imagery** have no side effects and have been reported to enhance well-being.

Several other healing therapies are sometimes used as supplemental or replacement cancer treatments, such as antineoplastons, cancell, cartilage (bovine and shark), laetrile, and mistletoe. Many of these therapies have not been the subject of safety and efficacy trials by the National Cancer Institute (NCI). The NCI has conducted trials on cancell, laetrile, and other alternative therapies and found no anticancer activity. These treatments have varying effectiveness and safety considerations. Patients using any alternative remedy should first consult their doctor in order to prevent harmful side effects or interactions with traditional cancer treatment.

KEY TERMS

Aflatoxin—A substance produced by molds that grow on rice and peanuts. Exposure to aflatoxin is thought to explain the high rates of primary liver cancer in Africa and parts of Asia.

Alpha-fetoprotein—A protein in blood serum that is found in abnormally high concentrations in most patients with primary liver cancer.

Cirrhosis—A chronic degenerative disease of the liver, in which normal cells are replaced by fibrous tissue. Cirrhosis is a major risk factor for the later development of liver cancer.

Cryoablation—A technique for removing cancerous tissue by killing it with extreme cold.

Hepatitis—A viral disease characterized by inflammation of the liver cells (hepatocytes). People infected with hepatitis B or hepatitis C virus are at an increased risk for developing liver cancer.

Metastatic cancer—A cancer that has spread to an organ or tissue from a primary cancer located elsewhere in the body.

Radiofrequency ablation—A technique for removing a tumor by heating it with a radiofrequency current passed through a needle electrode.

Prognosis

Liver cancer has a very poor prognosis because it is often not diagnosed until it has metastasized. Fewer than 10% of patients survive three years after the initial diagnosis; the overall five-year survival rate for patients with hepatomas is around 4%. Most patients with primary liver cancer die within six months of diagnosis, usually from liver failure; fewer than 5% are cured of the disease. Patients with liver cancers that metastasized from cancers in the colon live slightly longer than those whose cancers spread from cancers in the stomach or pancreas.

As of 2004, African American and Hispanic patients have much lower 5-year survival rates than Caucasian patients. It is not yet known, however, whether cultural differences as well as biological factors may be partly responsible for the variation in survival rates.

Prevention

There are no useful strategies at present for preventing metastatic cancers of the liver. Primary liver cancers, however, are 75% to 80% preventable.

Current strategies focus on widespread **vaccination** for hepatitis B, early treatment of hereditary hemochromatosis, and screening of high-risk patients with alpha-fetoprotein testing and ultrasound examinations.

Lifestyle factors that can be modified in order to prevent liver cancer include avoidance of exposure to toxic chemicals and foods harboring molds that produce aflatoxin. Most important, however, is avoidance of alcohol and drug **abuse**. Alcohol abuse is responsible for 60% to 75% of cases of cirrhosis, which is a major risk factor for eventual development of primary liver cancer. Hepatitis is a widespread disease among persons who abuse intravenous drugs.

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ORGANIZATIONS

American Cancer Society. 1599 Clifton Rd. NE, Atlanta, GA 30329. (800) 227-2345. <<http://www.cancer.org>>.

American Institute for Cancer Research (AICR). 1759 R St. NW, Washington, DC 20009. (800) 843-8114. <<http://www.aicr.org>>.

American Liver Foundation. 1425 Pompton Ave., Cedar Grove, NJ 07009. (800) 223-0179. <<http://www.liverfoundation.org>>.

- Cancer Care, Inc. 275 Seventh Ave., New York, NY 10001. (800) 813-HOPE. <<http://www.cancercare.org>>.
- Cancer Hope Network. Suite A., Two North Rd., Chester, NJ 07930. (877) HOPENET. <<http://www.cancerhopenetwork.org>>.
- Hospicelink. Hospice Education Institute, 190 Westbrook Rd., Essex, CT, 06426-1510. (800) 331-1620. <<http://www.hospiceworld.com>>.
- National Cancer Institute (National Institutes of Health). 9000 Rockville Pike, Bethesda, MD 20892. (800) 422-6237. <<http://www.nci.nih.gov>>.
- Wellness Community. Suite 412, 35 E. Seventh St., Cincinnati, OH 45202. (888) 793-9355. <<http://www.wellness-community.org>>.

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Rebecca J. Frey, PhD
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Liver cirrhosis see **Cirrhosis**

Liver disease

Definition

Liver disease is a general term for any damage that reduces the functioning of the liver.

Description

The liver is a large, solid organ located in the upper right-hand side of the abdomen. Most of the liver lies under the rib cage, which helps protect it from physical injury. The liver is made up of two main lobes and two minor lobes and has a total weight in adults of about 3.5 lb (1.6 kg).

Within the liver are tiny ducts (tubes) that collect bile, a product secreted by the liver. Bile is stored in to the gall bladder and then released into the intestines after meals to help in the digestion of fats and the absorption of certain **vitamins**. This system of bile production by the liver, transport through the bile ducts, and storage in the gall bladder is called the biliary system. Damage to this system is called biliary disease.

The liver receives blood that comes directly from the intestines. At any given time the liver contains about 13% of the blood circulating in the body. This

blood is rich in nutrients (food, vitamins, and **minerals**) that the body needs to function. Some of the most important functions of the liver are to process these nutrients.

Important functions of the liver include:

- manufacturing and regulating the production of proteins. The most important proteins made in the liver are albumin, which helps maintain blood volume, and clotting factors to help regulate blood clotting.
- making and storing fatty acids and cholesterol.
- forming and releasing bile
- processing and storing sugars in the form of glycogen, which can then be re-converted into energy
- Storing iron, an important element in blood formation
- Breaking down (detoxifying) alcohol, drugs, and environmental poisons so that they can be removed from the body.
- Processing and removing bilirubin, a product released when red blood cells break down, and ammonia, a toxic waste product of protein breakdown.
- Defending against infection by removing bacteria from the blood and making chemicals necessary to the functioning of the immune system.

Although the liver is the only organ that has the capacity to grow back, or regenerate, after injury or damage, sometimes the damage is too great for it to heal. The American Association for the Study of Liver Disease estimates that about 25 million Americans experience a liver-related disease each year. Individuals cannot live without a functioning liver. The ability to transplant livers is improving, **liver transplantation** is not nearly as common or successful as **kidney transplantation**.

Because the liver has many vital functions, there are many types of liver disease. The American Liver Foundation estimates that over 20,000 Americans die of chronic liver disease each year and another 360,000 are hospitalized. Individuals cannot live without a functioning liver.

Congenital Liver Diseases

Congenital liver diseases are disorders that are present at birth. Inherited liver diseases and disorders include:

- Alagile syndrome, a disorder that causes withering of the bile ducts. This disease occurs in less than 1 in 100,000 individuals.

- Alpha 1-antitrypsin deficiency, an inborn error in metabolism and the most common type of genetic liver disease.
- Galactosemia, a hereditary metabolic disease in which the liver is unable to break down the sugar galactose found in milk. It occurs in about one in every 20,000 births.
- Hematochromatosis, a hereditary metabolic disorder in which too much iron is absorbed from the diet and stored in the liver. This disease affects over one million Americans.
- Porphyria, a disorder in which the component of blood that contains iron is not correctly formed.
- Tyrosinemia, a rare inherited error in metabolism that causes severe liver disease in infants and children. It affects fewer than 200,000 individuals in the United States.
- Type I glycogen storage disease, a lack of the enzyme that helps regulate blood glucose (sugar) levels.
- Wilson's disease, an inherited disorder in which copper is accumulated in the liver and nervous system.

Acquired Liver Diseases

Many liver diseases are acquired from infection and exposure of the liver to toxic substances such as alcohol or drugs. In some areas of the world (although not the United States) liver parasites are a common cause of liver disease. In the United States, the most common acquired liver diseases are **hepatitis A**, **B**, and **C** and **cirrhosis**. Hepatitis A causes an acute (short-term) illness and is caused by a virus found in food or drinking water contaminated with feces. Hepatitis A infects between 125,000 and 250,000 people in the United States each year and causes about 100 deaths annually.

Hepatitis B is a viral infection spread by blood exchange and sexual contact with an infected person. It can be passed from an infected mother to her fetus. In most people hepatitis B is a short-term illness that causes mild symptoms such as **fatigue**, but in 2–6% of people, the disease lasts a long time and causes permanent liver damage. More than 75,000 people in the United States become infected with hepatitis B each year. Chinese Americans have a hepatitis B infection rate five times that of Caucasian Americans.

Hepatitis C is caused by a virus spread mainly through contact with the blood of an infected person, such as through sharing needles to inject drugs or from a mother to a fetus. Individuals infected with hepatitis C virus may not feel sick or know that they are infected for many years, but the disease can increase the likelihood of developing **liver cancer** or cirrhosis. The

American Liver Foundation estimates that 4 million Americans are infected with hepatitis C, resulting in 10,000–12,000 deaths each year. About 70% of individuals who are infected do not know that they have the virus. African Americans have the highest rate of hepatitis C infections and are twice as likely to be infected with hepatitis C as Caucasian Americans.

Cirrhosis of the liver involves the formation of permanent scar tissue in the liver and loss of liver function. It is often caused by chronic alcohol **abuse** (alcoholic liver disease), but it can also be caused by diseases such as hepatitis. Cirrhosis interferes with blood flow through the liver and can raise pressure in blood vessels supplying the liver and decrease the absorption of nutrients from the blood, leading to **malnutrition**. The liver of individuals with cirrhosis is also less effective in removing toxic wastes from the blood. Cirrhosis can be fatal.

Over 800 over-the-counter and prescription drugs, as well as illicit street drugs, can cause liver damage. One of the most common drugs to cause liver damage is **acetaminophen** (Tylenol) when taken at high doses or by individuals who already have some liver damage. Exposure to toxic chemicals, physical injury, and blockage of the bile ducts call also cause liver damage.

Liver **cancer** can either develop first in the liver (primary liver cancer) or spread there from another site (metastasized cancer). About 16,000 new cases of primary liver cancer are diagnosed each year, most commonly in middle age and older men. Although the cause of liver cancer is unclear, it appears to be associated with chronic infections of hepatitis B and C.

Causes & symptoms

The causes of liver disease are many and varied. Leading causes are viral infection, alcohol abuse, and inherited disease. A common symptoms of liver disease are **jaundice**. With jaundice, the skin and the whites of the eyes take on a yellowish color as a result of the accumulation of bilirubin and bile pigments in the blood. This is a sign that the liver or the biliary system is not functioning properly. Other symptoms of liver disease include an enlarged liver and swollen abdomen, **nausea**, **vomiting**, weight loss, and fatigue. Some infections cause flu-like symptoms of **fever**, **headache**, and weakness.

Diagnosis

Liver function tests, sometimes called a liver panel, measure various enzymes, proteins, and waste

products in the blood. These readings can tell a physician whether the liver is damaged and give an idea of how well it is functioning. Liver function tests are the most common way to diagnose liver damage. Based on the results of a liver panel, additional blood tests for infection, a **liver biopsy**, or liver scan may be done to pinpoint the reason for loss of function.

Treatment

Treatments depend on the type of liver disease an individual has. Many liver diseases are treated with altered **diets**, abstinence from alcohol, and medication. Hepatitis can be treated with antiviral medications such as interferon or ribavirin. Liver cancer is treated with **chemotherapy**, radiation, and surgery. As of 2005, over 300 clinical trials were enrolling patients with various types of liver disease in experimental treatment programs. Information on current clinical trials can be found at < <http://www.clinicaltrials.gov> > .

If the liver fails completely, a liver transplant is possible. About 5,600 liver transplants were performed in the United States in 2003. Donors and recipients are matched on the basis of blood type and must also be about the same weight. There is no machine like a **kidney dialysis** machine to perform the functions of the liver while individuals are awaiting a transplant. In 2003, 1,800 people died awaiting a liver donor, and about 18,000 more are on the waiting list awaiting a donated liver. Livers to be transplanted can come from either a living donor or a deceased donor.

Alternative treatment

A great deal of interest in alternative treatments for hepatitis C has resulted in a review of alternative and complementary treatments by the National Center for Complementary and Alternative Medicine (NCCAM), a division of the United States National Institutes of Health. Although there was in 2003 not enough solid experimental evidence to show that any herbal treatments cured hepatitis C, the most promising herbal treatment was an extract of milk thistle (*Silybum marianum*), a plant in the aster family that has been used for centuries in Europe to treat liver disease and jaundice. Some studies suggested that extracts of milk thistle promoted the growth of certain types of liver cells and acted as an anti-oxidant to protect the liver while producing few unwanted side effects. Other studies showed no protective effects.

Licorice root (*Glycyrrhiza glabra*) was also reviewed by NCCAM. Some studies suggested that

KEY TERMS

Biliary—Relating to the system that produces and transports bile.

Bile—A yellowish-green material secreted by the liver, stored by the gall bladder, and emptied into the small intestine to aid in the digestion and absorption of fats.

Bilirubin—A reddish-yellow substance that results from the breakdown of aging red blood cells. It is found in blood and bile, and if it accumulates in large quantities can cause jaundice.

Biopsy—A diagnostic procedure in which a small sample of tissue is obtained and examined under the microscope to determine they type and stage of a disease.

Congenital—Present at birth.

Feces—Waste products eliminated from the large intestine; excrement.

Jaundice—A yellowish tinge to the skin and whites of the eyes that indicates malfunction of the biliary system and/or liver and build up of bile components in the blood.

licorice root had antiviral properties, however this herb did not reduce the amount of hepatitis C virus circulating in the blood. Long term use of licorice root can have serious, health-threatening side effects.

Other alternative treatments studied by NCCAM include ginseng (*Panax quinquefolius* and *Panax ginseng*), which they concluded might possibly have a positive effect on the liver, especially in the elderly, and schisandra (*Schisandra chinensis* and *Schisandra sphenanthera*) used in Chinese medicine, which seemed to have a liver-protective effect in laboratory animals. Thymus extract and colloidal silver were found to be ineffective in treating liver disease.

Prognosis

The course of liver disease depends on the type of disease. Many individuals recover completely from infections of hepatitis A and B. However, if liver scarring occurs, the effects are irreversible. The initial success rate for liver transplants is good, with about 90% of individuals receiving a liver transplant are alive one year after the transplant operation. However, no alternative treatments produced a better outcome than traditional treatments of hepatitis C.

Prevention

Prevention is an effective way to avoid liver disease. Vaccines exist for hepatitis A and B (but not hepatitis C), although many individuals remain unvaccinated. In addition to **vaccination**, individuals can decrease the likelihood of developing liver disease by

- practicing safe sex
- avoiding sharing needles
- eating a healthy, balanced diet
- taking medications as prescribed
- avoiding drinking alcohol.

Resources

ORGANIZATIONS

American Association for the Study of Liver Disease, 1729 King Street, Suite 200, Alexandria, VA 22314. Telephone: 703-299-9766. <<http://www.aasld.org>>.

American Liver Foundation, 75 Maiden Lane, Suite 603, New York, NY 10038. Telephone: 800-465-4837 or 888-443-7872. <<http://www.liverfoundation.org>>.

OTHER

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Tish Davidson, A.M.

Description

A normally functioning liver metabolizes and detoxifies substances formed in the body during the digestive process. Impaired liver function allows substances like ammonia (formed when the body digests protein), some fatty acids, phenol, and mercaptans to escape into the bloodstream. From there, they may penetrate the blood-brain barrier, affect the central nervous system (CNS), and lead to hepatic coma.

Hepatic coma is most common in patients with chronic **liver disease**. It occurs in 50-70% of all those with **cirrhosis**.

Causes and symptoms

The cause of hepatic coma is unknown, but the condition is frequently associated with the following conditions:

- Acute or chronic liver disease
- Gastrointestinal bleeding
- Azotemia, the accumulation of nitrogen-containing compounds (such as urea) in the blood
- Inherited disorders that disrupt the process by which nitrogen is decomposed and excreted
- The use of shunts (devices implanted in the body to redirect the flow of fluid from one vessel to another)
- Electrolyte imbalances, including low levels of potassium (**hypokalemia**) and abnormally alkaline blood pH (alkalosis). These imbalances may result from the overuse of sedatives, **analgesics**, or **diuretics**; reduced levels of oxygen (hypoxia), or withdrawal of excessive amounts of body fluid (hypovolemia)
- Constipation, which may increase the body's nitrogen load
- Surgery
- Infection
- Acute liver disease.

Binge drinking and acute infection are common causes of hepatic coma in patients with long-standing liver disease.

Symptoms of hepatic encephalopathy range from almost unnoticeable changes in personality, energy levels, and thinking patterns to deep coma.

Inability to reproduce a star or other simple design (**apraxia**) and deterioration of handwriting are common symptoms of early encephalopathy. Decreased brain function can also cause inappropriate behavior, lack of interest in personal grooming, mood swings, and uncharacteristically poor judgment.

Liver encephalopathy

Definition

Liver encephalopathy is a potentially life-threatening disease in which toxic substances accumulate in the blood. Also known as hepatic encephalopathy or hepatic **coma**, this condition can cause confusion, disorientation, abnormal neurological signs, loss of consciousness, and **death**.

The patient may be less alert than usual and develop new sleep patterns. Movement and speech may be slow and labored.

As the disease progresses, patients become confused, drowsy, and disoriented. The breath and urine acquires a sweet, musky odor. The hands shake, the outstretched arms flap (asterixis or “liver flap”), and the patient may lapse into unconsciousness. As coma deepens, reflexes may be heightened (hyperreflexia). The toes sometimes splay when the sole of the foot is stroked (Babinski reflex).

Agitation occasionally occurs in children and in adults who suddenly develop severe symptoms. Seizures are uncommon.

Diagnosis

The absence of sensitive, reliable tests for encephalopathy make the physician’s personal observations and professional judgment the most valuable diagnostic tools.

Confusion, disorientation, and other indications of impaired brain function strongly suggest encephalopathy in patients known to have liver disease. CAT scans and examination of spinal fluid don’t provide diagnostic clues. Elevated arterial ammonia levels are almost always present in hepatic coma, but levels are not necessarily correlated with the severity or extent of the disease.

Magnetic resonance imaging (MRI) can show severe brain swelling that often occurs prior to coma, and **electroencephalography** (EEG) detects abnormal brain waves even in patients with early, mild symptoms. Blood and urine analyses can provide important information about the cause of encephalopathy in patients suspected of taking large quantities of sedatives or other drugs.

Treatment

This condition may disappear if the cause of symptoms is eliminated. In other cases, treatment is designed to improve liver function as much as possible; remove or relieve factors that worsen symptoms; and decrease the body’s production of poisonous substances.

All non-essential medications are discontinued. Soft restraints are recommended in place of sedatives for patients who become agitated.

Enemas or **laxatives** are used to stimulate expulsion of toxic intestinal products. All or most protein is eliminated from the diet, and supplemental feeding may be necessary to replenish lost calories. Regular doses of neomycin (Neobiotic), taken orally or

KEY TERMS

Cirrhosis—A serious disease of the liver caused by chronic damage to its cells and the eventual formation of scar tissue (fibrosis).

Coma—A condition of deep unconsciousness from which the person cannot be aroused

Electrolytes—Substances that conduct electricity when they are in solution. In the body, electrolytes in the blood and tissues enable nerve impulses to flow normally.

Encephalopathy—A dysfunction of the brain. Hepatic encephalopathy is brain dysfunction that occurs because the liver isn’t removing harmful substances from the blood.

administered to comatose patients in liquid form through a tube, may be used to decrease production of protein-digesting bacteria in the bowel.

Lactulose, a synthetic sugar, changes the characteristics of intestinal bacteria, decreases the amount of ammonia accumulated in the body, and has laxative properties. The patient is given hourly doses of lactulose syrup until **diarrhea** occurs, then dosage is adjusted to maintain regular bowel function. Lactulose and dietary-protein restrictions may be used to control chronic encephalopathy.

Prognosis

Encephalopathy may be reversible if the responsible factor is identified and removed or treated. Patients whose condition is the result of chronic liver disease may recover completely after the underlying cause is corrected.

Despite intensive treatment, encephalopathy caused by acute liver inflammation (fulminant hepatitis) is fatal for as many as 80% of patients. Those with chronic liver failure often die in hepatic coma.

Resources

ORGANIZATIONS

American Liver Foundation. 1425 Pompton Ave., Cedar Grove, NJ 07009. (800) 223-0179. < <http://www.liverfoundation.org> >.

Maureen Haggerty

Liver fluke infections see **Fluke infections**

Liver function tests

Definition

Liver function tests, or LFTs, include tests for bilirubin, a breakdown product of hemoglobin, and ammonia, a protein byproduct that is normally converted into urea by the liver before being excreted by the kidneys. LFTs also commonly include tests to measure levels of several enzymes, which are special proteins that help the body break down and use (metabolize) other substances. Enzymes that are often measured in LFTs include gamma-glutamyl transferase (GGT); alanine aminotransferase (ALT or SGPT); aspartate aminotransferase (AST or SGOT); and alkaline phosphatase (ALP). LFTs also may include **prothrombin time** (PT), a measure of how long it takes for the blood to clot.

Purpose

Liver function tests are used to aid in the differential diagnosis of **liver disease** and injury, and to help monitor response to treatment.

Precautions

Bilirubin: Drugs that may cause increased blood levels of total bilirubin include anabolic steroids, **antibiotics**, antimalarials, ascorbic acid, Diabinese, codeine, **diuretics**, epinephrine, **oral contraceptives**, and vitamin A.

Ammonia: Muscular exertion can increase ammonia levels, while cigarette **smoking** produces significant increases within one hour of inhalation. Drugs that may cause increased levels include alcohol, **barbiturates**, **narcotics**, and diuretics. Drugs that may decrease levels include broad-spectrum antibiotics, levodopa, lactobacillus, and potassium salts.

ALT: Drugs that may increase ALT levels include **acetaminophen**, ampicillin, codeine, dicumarol, indomethacin, methotrexate, oral contraceptives, **tetracyclines**, and verapamil. Previous intramuscular injections may cause elevated levels.

GGT: Drugs that may cause increased GGT levels include alcohol, phenytoin, and phenobarbital. Drugs that may cause decreased levels include oral contraceptives.

Description

The liver is one of the most important organs in the body. As the body's "chemical factory," it

regulates the levels of most of the main blood chemicals and acts with the kidneys to clear the blood of drugs and toxic substances. The liver metabolizes these products, alters their chemical structure, makes them water soluble, and excretes them in bile.

Liver function tests are used to determine if the liver has been damaged or its function impaired. Elevations of certain liver tests in relation to others aids in that determination. For example, aminotransferases (which include ALT and AST) are notably elevated in liver damage caused by liver cell disease (hepatocellular disease). However, in intrahepatic obstructive disease—which may be caused by some drugs or biliary cirrhosis—the alkaline phosphatases are most abnormal.

Alanine aminotransferase

Alanine aminotransferase (ALT), formerly called serum glutamate pyruvate transaminase, or SGPT, is an enzyme necessary for energy production. It is present in a number of tissues, including the liver, heart, and skeletal muscles, but is found in the highest concentration in the liver. Because of this, it is used in conjunction with other liver enzymes to detect liver disease, especially hepatitis or **cirrhosis** without **jaundice**. Additionally, in conjunction with the **aspartate aminotransferase test** (AST), it helps to distinguish between heart damage and liver tissue damage.

Aspartate aminotransferase

Aspartate aminotransferase (AST), formerly called serum glutamic-oxaloacetic transaminase, or SGOT, is another enzyme necessary for energy production. It, too, may be elevated in liver and heart disease. In liver disease, the AST increase is usually less than the ALT increase. However, in liver disease caused by alcohol use, the AST increase may be two or three times greater than the ALT increase.

Alkaline phosphatase

Alkaline phosphatase (ALP) levels usually include two similar enzymes (isoenzymes) that mainly come from the liver and bone and from the placenta in pregnant women. In some cases, doctors may order a test to differentiate between the alkaline phosphatase that originates in the liver and the alkaline phosphatase originating in bone. If a person has elevated ALP, does not have bone disease and is not pregnant, he or she may have a problem with the biliary tract, the system that makes and stores bile. (Bile is made in the liver, then passes through ducts to the gall bladder, where it is stored.)

Gamma-glutamyl transferase

Gamma-glutamyl transferase (GGT), sometimes called gamma-glutamyl transpeptidase (GGPT), is an enzyme that is compared with ALP levels to distinguish between skeletal disease and liver disease. Because GGT is not increased in bone disorders, as is ALP, a normal GGT with an elevated ALP would indicate bone disease. Conversely, because the GGT is more specifically related to the liver, an elevated GGT with an elevated ALP would strengthen the diagnosis of liver or bile-duct disease. The GGT has also been used as an indicator of heavy and chronic alcohol use, but its value in these situations has been questioned recently. It is also commonly elevated in patients with **infectious mononucleosis**.

Bilirubin

Bilirubin, a breakdown product of hemoglobin, is the predominant pigment in a substance produced by the liver called bile. Excess bilirubin causes yellowing of body tissues (jaundice). There are two tests for bilirubin: direct-reacting (conjugated) and indirect-reacting (unconjugated). Differentiating between the two is important diagnostically, as elevated levels of indirect bilirubin are usually caused by liver cell dysfunction (e.g. hepatitis), while elevations of direct bilirubin typically result from obstruction either within the liver (intrahepatic) or a source outside the liver (e.g. **gallstones** or a tumor blocking the bile ducts). Bilirubin measurements are especially valuable in newborns, as extremely elevated levels of unconjugated bilirubin can accumulate in the brain, causing irreparable damage.

Ammonia

Analysis of blood ammonia aids in the diagnosis of severe liver diseases and helps to monitor the course of these diseases. Together with the AST and the ALT, ammonia levels are used to confirm a diagnosis of **Reye's syndrome** (a rare disorder usually seen in children and associated with **aspirin** intake), which is characterized by brain and liver damage following an upper respiratory tract infection, **chickenpox**, or **influenza**. Ammonia levels are also helpful in the diagnosis and treatment of hepatic encephalopathy, a serious brain condition caused by the accumulated toxins that result from liver disease and liver failure.

Preparation

Preparation requirements for all these tests vary from laboratory to laboratory, so it is generally considered best that the patient be in a **fasting** state (nothing to eat or drink) after midnight the day before the test(s).

Aftercare

Because many patients with liver disease have prolonged clotting times, it is important to monitor the puncture site for bleeding after blood is drawn (venipuncture).

Risks

Risks for this test are minimal, but may include slight bleeding from the blood-drawing site, **fainting** or feeling lightheaded after venipuncture, or hematoma (blood accumulating under the puncture site).

Normal results

Reference ranges vary from laboratory to laboratory and also depend upon the method used. However, normal values can generally be found within the following ranges, unless specified differently.

- ALT: 5-35 IU/L (values for the elderly may be slightly higher, and values also may be higher in men and in African-Americans)
- AST: 0-35 IU/L
- ALP: 30-120 IU/L
- GGT: Normal values for this test vary widely, depending on the laboratory performing the test, and the age and sex of the patient. For example, females less than 45 years old have lower values than both males and females over 45 years of age. Values in the newborn can be as much as five times higher than in adults.
- Bilirubin: (Adult, elderly, and child) Total bilirubin: 0.1-1.0 mg/dL; indirect bilirubin: 0.2-0.8 mg/dL; direct bilirubin: 0.1-0.3 mg/dL. (Newborn) Total bilirubin: 1-12 mg/dL. Note: critical values for adult: greater than 1.2 mg/dL. Critical values for newborn (requiring immediate treatment): greater than 15 mg/dL.
- Ammonia: Normal values for this test vary widely, depending upon the laboratory performing the test, the age of the patient, and the type of specimen. For example, values are somewhat higher in arterial than in venous blood.
- PT: 9-12 seconds.

Abnormal results

ALT: Values are significantly increased in cases of hepatitis, and moderately increased in cirrhosis, liver tumor, obstructive jaundice, and severe **burns**. Values are mildly increased in **pancreatitis**, **heart attack**, infectious mononucleosis, and **shock**. Most useful when compared with ALP levels.

KEY TERMS

Cirrhosis—A serious disease of the liver caused by chronic damage to its cells and the eventual formation of scar tissue (fibrosis). The most common symptoms are mild jaundice, fluid collection in the tissues, mental confusion, and vomiting of blood. If left untreated, cirrhosis lead to liver failure and death.

Hemolytic disease of the newborn—Also known as erythroblastosis neonatorum, this is a condition in which a newborn's red blood cells are destroyed by antibodies that have crossed the placenta from the mother's blood. (Hemolytic disease begins in the fetus, in whom the disease is called erythroblastosis fetalis). Severe anemia caused by hemolytic disease is treated in the same way as other anemias, but when jaundice appears due to increased bilirubin, the jaundice is treated by exposing the infant to bright lights. In severe cases, exchange transfusion is required or brain damage may result.

Hepatitis—An inflammation of the liver, with accompanying liver cell damage or cell death, caused most frequently by viral infection, but also by certain drugs, chemicals, or poisons. May be either acute (of limited duration) or chronic

(continuing). Symptoms include jaundice, nausea, vomiting, loss of appetite, tenderness in the right upper abdomen, aching muscles, and joint pain. In severe cases, liver failure may result.

Hepatic encephalopathy—Also called liver encephalopathy or hepatic coma, this is a disorder in which brain function deteriorates because toxic substances, which would normally be removed by the liver, accumulate in the bloodstream due to liver damage or disease. Early symptoms include subtle changes in logical thinking, personality and behavior. As the disorder progresses, signs of drowsiness and confusion increase until eventually the patient loses consciousness and lapses into coma.

Reye's syndrome—A rare disorder characterized by brain and liver damage following an upper respiratory tract infection, chickenpox, or influenza, almost entirely confined to children under age 15, and often related to aspirin ingestion for a viral infection. Symptoms include uncontrollable vomiting, often with lethargy, memory loss, disorientation, or delirium. Swelling of the brain may cause seizures, coma, and in severe cases, death.

- **AST:** High levels may indicate liver cell damage, hepatitis, heart attack, **heart failure**, or gall stones.
- **ALP:** Elevated levels occur in diseases that impair bile formation (**cholestasis**). ALP may also be elevated in many other liver disorders, as well as some lung cancers (bronchogenic carcinoma) and Hodgkin's lymphoma. However, elevated ALP levels may also occur in otherwise healthy people, especially among older people.

GGT: Increased levels are diagnostic of hepatitis, cirrhosis, liver tumor or metastasis, as well as injury from drugs toxic to the liver. Although the causes are unclear, GGT levels may increase with alcohol ingestion, heart attack, pancreatitis, infectious mononucleosis, and Reye's syndrome.

Bilirubin: Increased *indirect* or total bilirubin levels can indicate various serious **anemias**, including hemolytic disease of the newborn and **transfusion** reaction. Increased *direct* bilirubin levels can be diagnostic of bile duct obstruction, gallstones, cirrhosis, or hepatitis. It is important to note that if total bilirubin levels in the newborn reach or exceed critical levels, exchange transfusion is necessary to avoid kernicterus, a condition that causes brain damage.

Ammonia: Increased levels are seen in primary liver cell disease, Reye's syndrome, severe heart failure, hemolytic disease of the newborn, and hepatic encephalopathy.

PT: Elevated in acute liver injury, vitamin K deficiencies, and disorders with impair the absorption of vitamin K, including cholestasis.

Resources

BOOKS

Pagana, Kathleen Deska. *Mosby's Manual of Diagnostic and Laboratory Tests*. St. Louis: Mosby, Inc., 1998.

Janis O. Flores

Liver nuclear medicine scan

Definition

A liver scan is a diagnostic procedure to evaluate the liver for suspected disease. A radioactive substance

which concentrates in the liver is injected intravenously and the image of its distribution in the body is analyzed to diagnose certain abnormalities.

Purpose

In the past, liver scans were used to evaluate the liver in a wide variety of situations. It was considered a useful study to detect abnormalities, but was often not able to establish a specific diagnosis. In the 1990s, radionuclide imaging of the liver (use of a radioactive form of cobalt or iodine) evolved into a more specialized study, used to identify individual diseases or conditions. This is accomplished by using different radioisotopes precisely designed to further evaluate a particular case. Isotopes are different forms of the same substance, such as radioactive iodine, that are injected into the body. This allows the physician to trace the process of the substance throughout the part of the body that is being tested for disease.

A liver scan is usually ordered after blood studies and other imaging procedures have shown a liver abnormality. It is most often used to further evaluate masses or tumors. These may be benign growths in the liver, or **cancer** which has developed in the liver or has spread (or metastasized) from another organ.

A liver scan may also be helpful in diagnosing specific disorders, by detecting features which are characteristic of a disorder, such as **cirrhosis** of the liver. This study may also be part of the battery of tests used to evaluate potential candidates for liver transplant.

Precautions

Women who are pregnant or breast feeding should not have this test.

Description

This test can be performed in an outpatient setting or a hospital x-ray department. The patient usually lies down while a radioactive substance (radioactive isotope) which accumulates in the liver is injected through a vein in the arm. Scanning times may vary, depending on the specific radioisotope used. It most often begins within minutes after injection. The radionuclide scanner, sometimes called a gamma camera or scintillation camera, is positioned above the upper abdomen and may lightly touch the patient. It is important for the patient to lie quietly. Position changes and brief periods

KEY TERMS

Radioisotope—A radioactive, or radiation-emitting form of an element.

Radionuclide—A substance which emits radiation that can be detected by a scanner as the substance disintegrates.

of breath holding may be required. The test usually takes approximately one hour.

A specialized liver scan used to assess blood flow is frequently used. It may be referred to as a radionuclide blood pool or volume study, a labeled red cell scintigram, or some combination of these terms. Other studies may be named for the radioisotope used. This test may also be called a liver-spleen scan.

Preparation

No physical preparation is required. A liver scan should be performed before doing any study that uses iodinated or barium-containing contrast agents, to prevent inaccurate results.

The patients should understand that there is no danger of radioactive exposure to themselves or others. Only small amounts of radionuclide are used. The total amount of radiation absorbed is often less than the dose received from ordinary x rays. The scanner does not emit any radiation, but detects and records it from the patient.

Aftercare

No special precautions are needed.

Normal results

A normal scan will show a liver of normal size, shape, and position.

Abnormal results

An abnormal liver scan may result from a mass. Depending on the radioisotope and technique used, the scan may identify particular types of tumors or certain cancers. Too much radioisotope in the spleen and bones, compared to the liver, can indicate potential **hypertension** or cirrhosis. Liver diseases such as cirrhosis or hepatitis may also cause an abnormal scan, but are rarely diagnosed from the information revealed by this study alone.

Resources

PERIODICALS

Drane, Walter E. "Scintigraphic Techniques for Hepatic Imaging." *Radiologic Clinics of North America* 36 (March 1998): 309-318.

Ellen S. Weber, MSN

Liver transplantation

Definition

Liver transplantation is a surgery that removes a diseased liver and replace it with a healthy donor liver.

Purpose

The liver is the body's principle chemical factory. It receives all nutrients, drugs, and toxins absorbed from the intestines and performs the final stages of digestion, converting food into energy and replacement parts for the body. The liver also filters the blood of all waste products, removes and detoxifies poisons and excretes many of these into the bile. It processes other chemicals for excretion by the kidneys. The liver is also an energy storage organ, changing food energy to a chemical called glycogen that can be rapidly converted to fuel.

As the liver fails, all of its functions diminish. **Nutrition** suffers, toxins build up, and waste products accumulate. Scar tissue builds up on the liver if disease is of long duration. As the liver **scars**, blood flow is progressively restricted in the portal vein, which carries blood from the stomach and abdominal organs to the liver. The resulting high blood pressure (**hypertension**) causes swelling of and bleeding from the blood vessels of the esophagus. Severe **jaundice**, fluid accumulation in the abdomen (**ascites**), and deterioration of mental function, due to the build-up of toxins in the blood (**liver encephalopathy**), eventually occur, leading to **death**.

Among the many causes of liver failure that bring patients to transplant surgery are:

- Progressive hepatitis (mostly due to virus infection) accounts for more than a third.
- Alcohol damage brings in about 20%
- Scarring or abnormality of the biliary system accounts for roughly another 20%.

National Transplant Waiting List By Organ Type (June 2000)

Organ Needed	Number Waiting
Kidney	48,349
Liver	15,987
Heart	4,139
Lung	3,695
Kidney-Pancreas	2,437
Pancreas	942
Heart-Lung	212
Intestine	137

- The remainder comes from selected cancers, other uncommon diseases, and a situation called fulminant liver failure.

Fulminant liver failure most commonly happens during acute viral hepatitis, but it is also the result of **mushroom poisoning** by *Amanita phalloides* and toxic reactions to some medicines, like an overdose of **acetaminophen**. This is a special category of candidates for liver transplant because of the speed of their disease and the immediate need of treatment.

The first human liver transplant was performed in 1963, and since then, thousands of liver transplants are done every year. Since the introduction of cyclosporine (a drug that suppresses the immune response that rejects the donor organ), success rates for liver transplantation have reached 85%.

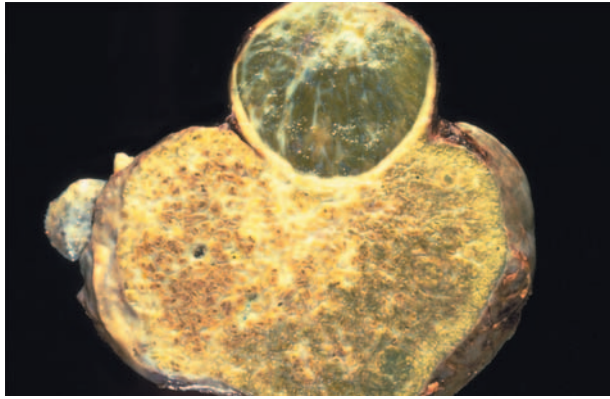
Precautions

Patients with advanced heart and lung disease, who are HIV positive, and who **abuse** drugs and alcohol are poor candidates for liver transplantation. Their ability to survive the surgery and the difficult recovery period, as well as their longterm prognosis, is hindered by their conditions.

Description

There are three types of liver transplantation methods. They include:

- Orthotopic transplantation is the replacement of a whole diseased liver with a healthy donor liver.
- Heterotopic transplantation is the addition of a donor liver at another site, while the diseased liver is left intact.
- Reduced-size liver transplantation is the replacement of a whole diseased liver with a portion of a healthy



The diseased liver of a patient ready for transplantation.
(Custom Medical Stock Photo. Reproduced by permission.)

donor liver. Reduced-size liver transplants are most often performed on children.

When an orthotopic transplantation is performed, a segment of the inferior vena cava attached to the liver is taken from the donor as well. The same parts are removed from the recipient and replaced by connecting the inferior vena cava, the hepatic artery, the portal vein and the bile ducts.

When there is a possibility that the afflicted liver may recover, a heterotopic transplantation is performed. The donor liver is placed in a different site, but it still has to have the same connections. It is usually attached very near the original liver, and if the original liver recovers, the donor shrivels away. If the original liver does not recover, it will shrivel, leaving the donor in place.

Reduced-size liver transplantation transplants part of a donor liver into a patient. It is possible to divide the liver into eight pieces, each supplied by a different set of blood vessels. Two of these pieces have been enough to save a patient in liver failure, especially if the patient is a child. It is therefore possible to transplant one liver into at least two patients and to transplant part of a liver from a living donor and have both donor and recipient survive. Liver tissue grows to accommodate its job so long as there is initially enough of the organ to use. Patients have survived with only 15-20% of their original liver, provided that 15-20% was healthy.

Availability of organs for transplant is a current crisis in the transplantation business. In October 1997, a national distribution system was established that gives priority to the sickest patients closest in location to the donor liver, but makes livers available nationally. It is now possible to preserve a liver out of the

body for 10-20 hours by flushing it with cooled solutions of special chemicals and nutrients, so it can be transported across the country.

Preparation

Before transplantation takes place, the patient is first determined to be a good candidate for transplantation by going through rigorous medical examination. A suitable candidate boosts their nutritional intake in order to ensure that they are as healthy as possible before surgery. Drugs are administered that will decrease rejection after surgery. Consultation with the patient, as well as any family, is conducted to explain the surgery and its complications. Psychological counseling is recommended.

Aftercare

In order to prevent organ rejection, immunosuppressive drugs will be taken. Hospitalization ranges from four weeks to five months, depending on the rate of recovery.

Successfully receiving a transplanted liver is only the beginning of a life-long process. Patients with transplanted livers have to stay on **immunosuppressant drugs** for the rest of their lives to prevent organ rejection. Although many can reduce the dosage after the initial few months, virtually none can discontinue drugs altogether. Prednisone, azathioprine, and tacrolimus are often combined with cyclosporine for better results. Newer immunosuppressive agents are coming that promise even better results. In spite of immunosuppressants, rejection occurs most of the time and requires additional medication. In some cases it cannot be reversed, and retransplantation becomes necessary.

Risks

Early failure of the transplant occurs once in four surgeries and has to be repeated. Some transplants never work, some succumb to infection, and some suffer immune rejection. Primary failure is apparent within one or two days. Infections happen in half the patients and often appear during the first week. Rejection usually starts at the end of the first week. The surgery itself may need revision because of narrowing, leaking, or **blood clots** at the connections.

There are potential social and economic problems, psychological problems, and a vast array of possible medical and surgical complications. Close medical surveillance must continue for the rest of the patient's life.

KEY TERMS

Acetaminophen—A common pain reliever (Tylenol).

Antigen—Any chemical that provokes an immune response.

Bile ducts—Tubes carrying bile from the liver to the intestines.

Biliary system—The tree of tubes that carries bile.

Hepatic artery—The blood vessel supplying arterial blood to the liver.

Inferior vena cava—The biggest vein in the body, returning blood to the heart from the lower half of the body.

Leukemia—A cancer of the white blood cells.

Lymphoma—A cancer of lymphatic tissue.

Portal vein—The blood vessel carrying venous blood from the abdominal organs to the liver.

Infections are a constant risk while on immunosuppressive agents, because the immune system is supposed to prevent them. A way has not yet been devised to control rejection without hampering immune defenses against infections. Not only do ordinary infections pose a threat, but because of the impaired immunity, transplant patients are susceptible to the same “opportunistic” infections that threaten **AIDS** patients—pneumocystis **pneumonia**, herpes and cytomegalovirus infections, fungi, and a host of bacteria.

Immunosuppression also hinders the body’s ability to resist **cancer**. All the drugs used to prevent rejection increase the risk of leukemias and lymphomas.

There is also a risk of the original disease returning. Hepatitis virus still inhabits the patient, as does the urge to drink alcohol. Newer **antiviral drugs** hold out promise for dealing with hepatitis, and Alcoholics Anonymous (AA) is the most effective treatment known for **alcoholism**.

Drug reactions are also a continuing threat. Every drug used to suppress the immune system has potential problems.

Resources

ORGANIZATIONS

American Liver Foundation. 1425 Pompton Ave., Cedar Grove, NJ 07009. (800) 223-0179. <<http://www.liverfoundation.org>>.

J. Ricker Polsdorfer, MD

Ller-Christi see **Histiocytosis X**

Loaiasis see **Filariasis**

Lobectomy see **Lung surgery**

Lobotomy see **Psychosurgery**

Local anesthetic see **Anesthesia, local**

Localized scratch dermatitis see **Lichen simplex chronicus**

Lockjaw see **Tetanus**

Loperamide see **Antidiarrheal drugs**

Loratadine see **Antihistamines**

Lou Gehrig’s disease see **Amyotrophic lateral sclerosis**

Louis-Bar syndrome see **Ataxia-telangiectasia**

Low potassium blood level see **Hypokalemia**

Low back pain

Definition

Low back **pain** is a common musculoskeletal symptom that may be either acute or chronic. It may be caused by a variety of diseases and disorders that affect the lumbar spine. Low back pain is often accompanied by **sciatica**, which is pain that involves the sciatic nerve and is felt in the lower back, the buttocks, and the backs of the thighs.

Description

Low back pain is a symptom that affects 80% of the general United States population at some point in life with sufficient severity to cause absence from work. It is the second most common reason for visits to primary care doctors, and is estimated to cost the American economy \$75 billion every year.

Low back pain may be experienced in several different ways:

- **Localized.** In localized pain the patient will feel soreness or discomfort when the doctor palpates, or presses on, a specific surface area of the lower back.
- **Diffuse.** Diffuse pain is spread over a larger area and comes from deep tissue layers.

- **Radicular.** The pain is caused by irritation of a nerve root. Sciatica is an example of radicular pain.
- **Referred.** The pain is perceived in the lower back but is caused by inflammation elsewhere—often in the kidneys or lower abdomen.

Causes and Symptoms

Acute pain

Acute pain in the lower back that does not extend to the leg is most commonly caused by a sprain or muscle tear, usually occurring within 24 hours of heavy lifting or overuse of the back muscles. The pain is usually localized, and there may be **muscle spasms** or soreness when the doctor touches the area. The patient usually feels better when resting.

Chronic pain

Chronic low back pain has several different possible causes:

MECHANICAL. Chronic strain on the muscles of the lower back may be caused by **obesity**; **pregnancy**; or job-related stooping, bending, or other stressful postures.

MALIGNANCY. Low back pain at night that is not relieved by lying down may be caused by a tumor in the cauda equina (the roots of the spinal nerves controlling sensation in and movement of the legs), or a **cancer** that has spread to the spine from the prostate, breasts, or lungs. The risk factors for the spread of cancer to the lower back include a history of **smoking**, sudden weight loss, and age over 50.

ANKYLOSING SPONDYLITIS. Ankylosing spondylitis is a form of arthritis that causes chronic pain in the lower back. The pain is made worse by sitting or lying down and improves when the patient gets up. It is most commonly seen in males between 16 and 35. Ankylosing spondylitis is often confused with mechanical back pain in its early stages.

HERNIATED SPINAL DISK. Disk herniation is a disorder in which a spinal disk begins to bulge outward between the vertebrae. Herniated or ruptured disks are a common cause of chronic low back pain in adults.

PSYCHOGENIC. Back pain that is out of proportion to a minor injury, or that is unusually prolonged, may be associated with a somatoform disorder or other psychiatric disturbance.

Low back pain with leg involvement

Low back pain that radiates down the leg usually indicates involvement of the sciatic nerve. The nerve

can be pinched or irritated by herniated disks, tumors of the cauda equina, abscesses in the space between the spinal cord and its covering, **spinal stenosis**, and compression **fractures**. Some patients experience **numbness** or weakness of the legs as well as pain.

Diagnosis

The diagnosis of low back pain can be complicated. Most cases are initially evaluated by primary care physicians rather than by specialists.

Initial workup

PATIENT HISTORY. The doctor will ask the patient specific questions about the location of the pain, its characteristics, its onset, and the body positions or activities that make it better or worse. If the doctor suspects that the pain is referred from other organs, he or she will ask about a history of diabetes, peptic ulcers, **kidney stones**, urinary tract infections, or **heart murmurs**.

PHYSICAL EXAMINATION. The doctor will examine the patient's back and hips to check for conditions that require surgery or emergency treatment. The examination includes several tests that involve moving the patient's legs in specific positions to test for nerve root irritation or disk herniation. The flexibility of the lumbar vertebrae may be measured to rule out ankylosing spondylitis.

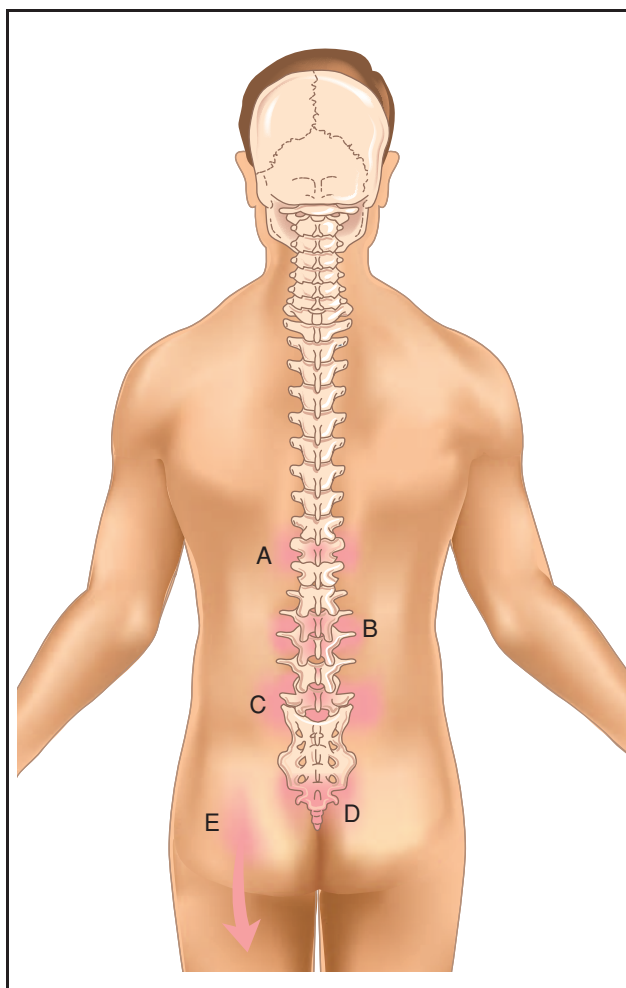
Imaging studies

Imaging studies are not usually performed on patients whose history and **physical examination** suggest routine muscle strain or overuse. X rays are ordered for patients whose symptoms suggest cancer, infection, inflammation, pelvic or abdominal disease, or bone fractures. MRIs are usually ordered only for patients with certain types of masses or tumors.

It is important to know that the appearance of some abnormalities on imaging studies of the lower back does not necessarily indicate that they cause the pain. Many patients have minor deformities that do not create symptoms. The doctor must compare the results of imaging studies very carefully with information from the patient's history and physical examination.

Treatment

All forms of treatment of low back pain are aimed either at symptom relief or to prevent interference with the processes of healing. None of these methods appear to speed up healing.



Sites of low back pain. Pain anywhere along the spine (A) can be caused by osteoarthritis. Pain along one or the other side of the spine may be (B) a kidney infection. Trauma to back muscles, joints, or disks (C) causes low back pain. Damage to the coccyx (D) can occur during a fall. Sciatica (E) can cause pain to run down from the back and buttocks area down a leg. (Illustration by Electronic Illustrators Group.)

Acute pain

Acute back pain is treated with **nonsteroidal anti-inflammatory drugs** (NSAIDs), such as ibuprofen, **muscle relaxants**, or **aspirin**. Applications of heat or cold compresses are also helpful to most patients. If the patient has not experienced some improvement after several weeks of treatment, the doctor will reinvestigate the cause of the pain.

Chronic pain

Patients with chronic back pain are treated with a combination of medications, physical therapy, and occupational or lifestyle modification. The medications given

are usually NSAIDs, although patients with hypertension, kidney problems, or stomach ulcers should not take these drugs. Patients who take NSAIDs for longer than six weeks should be monitored periodically for complications.

Physical therapy for chronic low back pain usually includes regular **exercise** for fitness and flexibility, and massage or application of heat if necessary.

Lifestyle modifications include giving up smoking, weight reduction (if necessary), and evaluation of the patient's occupation or other customary activities.

Patients with herniated disks are treated surgically if the pain does not respond to medication.

Patients with chronic low back pain sometimes benefit from **pain management** techniques, including **biofeedback**, **acupuncture**, and **chiropractic** manipulation of the spine.

Psychotherapy is recommended for patients whose back pain is associated with a somatoform, **anxiety**, or depressive disorder.

Low back pain with leg involvement

Treatment of sciatica and other disorders that involve the legs may include NSAIDs. Patients with long-standing sciatica or spinal stenosis that do not respond to NSAIDs are treated surgically. Although some doctors use cortisone injections to relieve the pain, this form of treatment is still debated.

Alternative treatment

A thorough differential diagnosis is important before any treatment is considered. There are times when alternative therapies are the most beneficial, and other times when more invasive treatments are needed.

Chiropractic

Chiropractic treats patients by manipulating or adjusting sections of the spine. It is one of the most popular forms of alternative treatment in the United States for relief of back pain caused by straining or lifting injuries. Some osteopathic physicians, physical therapists, and naturopathic physicians also use spinal manipulation to treat patients with low back pain.

Traditional Chinese medicine

Practitioners of **traditional Chinese medicine** treat low back pain with acupuncture, *tui na* (push-and-rub) massage, and the application of herbal poultices.

KEY TERMS

Ankylosing spondylitis—A type of arthritis that causes gradual loss of flexibility in the spinal column. It occurs most commonly in males between 16 and 35.

Cauda equina—The roots of the spinal nerves controlling movement and sensation in the legs. These nerve roots are located in the lower spine and resemble a horse's tail (*cauda equina* in Latin).

Chiropractic—A method of treatment based on the interactions of the spine and the nervous system. Chiropractors adjust or manipulate segments of the patient's spinal column in order to relieve pain.

Lumbar spine—The segment of the human spine above the pelvis that is involved in low back pain. There are five vertebrae, or bones, in the lumbar spine.

Radicular—Pain that is caused by the root of a nerve.

Referred pain—Pain that is experienced in one part of the body but originates in another organ or area. The pain is referred because the nerves that supply the damaged organ enter the spine in the same segment as the nerves that supply the area where the pain is felt.

Sciatica—Pain caused by irritation of the sciatic nerve. Sciatica is felt in the lower back, the buttocks, and the backs of the upper legs.

Spinal stenosis—A form of sciatica that is caused by a narrowing of the spinal canal in the lumbar vertebrae. The narrowing puts pressure on the roots of the sciatic nerve.

Herbal medicine

Herbal medicine can utilize a variety of antispasmodic herbs in combination to help relieve low back pain due to spasm. Lobelia (*Lobelia inflata*) and myrrh (*Commiphora molmol*) are two examples of antispasmodic herbs.

Homeopathy

Homeopathic treatment for acute back pain consists of applications of *Arnica* oil to the sore area or oral doses of *Arnica* or *Rhus toxicodendron*. *Bellis perennis* is recommended for deep muscle injuries. Other remedies may be recommended based on the symptoms presented by the patient.

Body work and yoga

Massage and the numerous other body work techniques can be very effective in treating low back pain. **Yoga**, practiced regularly and done properly, can be most useful in preventing future episodes of low back pain.

Prognosis

The prognosis for most patients with acute low back pain is excellent. About 80% of patients recover completely in 4-6 weeks. The prognosis for recovery from chronic pain depends on the underlying cause.

Prevention

Low back pain due to muscle strain can be prevented by lifestyle choices, including regular physical exercise and weight control, avoiding smoking, and learning the proper techniques for lifting and moving heavy objects. Exercises designed to strengthen the muscles of the lower back, and chairs or car seats with lumbar supports are also recommended.

Resources

BOOKS

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Rebecca J. Frey, PhD

Low blood magnesium see **Magnesium imbalance**

Low blood phosphate level see **Phosphorus imbalance**

Low blood pressure see **Orthostatic hypotension; Hypotension**

Low blood sugar see **Hypoglycemia**

Low calcium blood level see **Hypocalcemia**

Low sodium blood level see **Hyponatremia**

Lower esophageal ring

Definition

Lower esophageal ring is a condition in which there is a ring of tissue inside the lower part of the

esophagus (the tube connecting the throat with the stomach). This tissue causes narrowing and partial blockage of the esophagus. Lower esophageal ring can also refer to the ring itself.

Description

Lower esophageal ring (also called Schatzki's ring and B-ring) affects about 10-14% of the population. Normally, the lower part of the esophagus, near where the esophagus meets the stomach, has an inside diameter of 1.5-2 inches. The diameter of this part of the esophagus is less when lower esophageal ring is present, and diameters as small as one-eighth inch have been seen. When the inside diameter is less than about three-fourths of an inch, intermittent difficulty with swallowing can result. About 96% of people with lower esophageal ring have no symptoms.

Causes and symptoms

Causes

Lower esophageal ring seems to result from infoldings of tissue near the bottom of the esophagus, but the underlying cause is unknown. Although some specialists speculate they are due to a congenital defect, most people do not develop symptoms until they reach their forties or later. Although lower esophageal ring is generally associated with hiatal **hernia**, and sometimes with **heartburn**, the cause/effect relationship is unclear.

Symptoms

Intermittent difficulty swallowing solid food is the primary symptom of this condition. The degree of difficulty in swallowing is directly related to the degree the esophagus is narrowed. Certain foods, especially tough or fibrous foods like meat, are more likely to cause swallowing difficulties.

Diagnosis

Gastroenterologists and internists are best equipped to diagnose and treat lower esophageal ring. The diagnosis is based on the patient's history of swallowing difficulties and a barium x ray of the upper gastrointestinal tract. For a barium x ray, the patient swallows a liquid containing barium, a substance that is opaque to x rays. Subsequent x-ray photography reveals the shape of the esophagus and any narrow regions present.

The presence of a lower esophageal ring can also be shown with a test called an esophagoscopy. This

procedure visualizes the inside of the esophagus with an inserted, thin, flexible tube. However, this test is less sensitive for lower esophageal ring and costs about five times as much as barium x ray. However, if the findings of a barium x ray are not definitive, esophagoscopy should be done. Biopsies can then be done on questionable areas.

Treatment

Dietary change

Swallowing difficulties due to lower esophageal ring can often be relieved by chewing food more thoroughly. Soft foods and liquids may also be recommended.

Dilation

Lower esophageal rings can be corrected by passing a bougie (a cylindrical, mercury-filled dilator) through the esophagus. This procedure, called bougienage, is effective most of the time, but may need to be repeated every few years. Complications and adverse reactions are extremely rare.

Surgery

If bougienage is unsuccessful, lower esophageal ring tissue can be surgically removed.

Prognosis

The probability of a favorable outcome is high. Swallowing difficulties can be alleviated in almost every case, and the rate of complications from bougienage or surgery is less than 1%.

Prevention

Since the cause of lower esophageal ring is not known, there are no definitive preventive measures. Nevertheless, anyone with lower esophageal ring who also suffers from heartburn would be wise to prevent or treat the heartburn. It is possible that the stomach acid in the esophagus associated with heartburn contributes to esophageal ring.

Resources

ORGANIZATIONS

The American College of Gastroenterology (ACG). P.O. Box 3099, Alexandria, VA 22302. (800) 432-2876. <<http://www.healthtouch.com>>.

American Gastroenterological Association (AGA). 7910 Woodmont Ave., 7th Floor, Bethesda, MD 20814.

KEY TERMS

Bougie—A mercury-filled dilator in the shape of a cylinder or tapered cylinder. Bougies come in a range of different sizes.

Bougienage—The procedure of dilating tubal organs, like the esophagus, with a bougie or bougies.

Congenital—Existing at birth.

Dysphagia—Difficulty swallowing.

Esophagoscopy (also esophagoendoscopy)—Examination of the inside of the esophagus using a flexible tube that transmits video images.

Esophagus—The tube connecting the throat to the stomach, which is about ten inches long in adults. It is coated with mucus and surrounded by muscles, and pushes food to the stomach by sequential waves of contraction. It functions to transport food from the throat to the stomach and to keep the contents of the stomach in the stomach.

Heartburn—A burning sensation in the chest that can extend to the neck, throat, and face, caused by the movement of stomach acid into the esophagus.

Hiatal hernia—A condition where part of the stomach extends through the diaphragm into the chest cavity.

(310) 654-2055. <aga001@aol.com> <<http://www.gastro.org/index.html>>.

American Society for Gastrointestinal Endoscopy.

13 Elm St., Manchester, MA 01944. (508) 526-8330. <<http://www.asge.org/doc/201>>.

National Digestive Diseases Information Clearinghouse.

2 Information Way, Bethesda, MD 20892-3570. (800) 891-5389. <<http://www.niddk.nih.gov/health/digest/nddic.htm>>.

Lorraine Lica, PhD

Lower GI exam see **Barium enema**

LSD see **Lysergic acid diethylamide**

Lues see **Syphilis**

Lumbar puncture see **Cerebrospinal fluid (CSF) analysis**

Lumbar stenosis see **Spinal stenosis**

Lumbosacral radiculopathy see **Sciatica**

Lumpectomy

Definition

A lumpectomy is a type of surgery used to treat **breast cancer**. It is considered “breast-conserving” surgery because in a lumpectomy, only the malignant tumor and a surrounding margin of normal breast tissue are removed. Lymph nodes in the armpit (axilla) also may be removed. This procedure is called lymph node dissection.

Purpose

Lumpectomy is a surgical treatment for newly diagnosed breast **cancer**. It is estimated that at least 50% of women with breast cancer are good candidates for this procedure. The location, size, and type of tumor are of primary importance when considering breast cancer surgery options. The size of the breast is another factor the surgeon considers when recommending surgery. The patient’s psychological outlook, as well as her lifestyle and preferences, should also be taken into account when treatment decisions are made.

The extent and severity of a cancer is evaluated or “staged” according to a fairly complex system. Staging considers the size of the tumor and whether the cancer has spread to other areas, such as the chest wall and the lymph nodes, and/or to distant parts of the body. Women with early stage breast cancers usually are better candidates for lumpectomy. In most cases, a course of **radiation therapy** after surgery is part of the treatment. **Chemotherapy** or antiestrogens also may be prescribed.

Many studies have compared the survival rates of women who have had removal of a breast (**mastectomy**) with those who have undergone lumpectomy and radiation therapy. The data demonstrate that for women with comparable stages of breast cancer, survival rates are similar between the two groups, but the risk of the cancer recurring in the breast is slightly higher with lumpectomy. A 2003 study confirmed that younger women who have lumpectomies have a higher risk of tumor recurrence than those who have mastectomies.

In some instances, women with later stage breast cancer may be able to have lumpectomy. Chemotherapy may be administered before surgery to decrease tumor size and the chance of spread in selected cases.

Precautions

A number of factors may prevent or prohibit a breast cancer patient from having a lumpectomy. The

tumor itself may be too large or located in an area where it would be difficult to remove with good cosmetic results. Sometimes several areas of cancer are found in one breast, so the tumor cannot be removed as a single lump. A cancer that has already attached itself to nearby structures, such as the skin or the chest wall, needs more extensive surgery.

Certain medical or physical circumstances also may eliminate lumpectomy as a treatment option. Sometimes lumpectomy may be attempted, but the surgeon is unable to remove the tumor with a sufficient amount of normal tissue surrounding it. This may be termed “persistently positive margins,” or “lack of clear margins,” referring to the margin of unaffected tissue around the tumor. Lumpectomy is not used for women who have had a previous lumpectomy and have a recurrence of the breast cancer.

The need for radiation therapy after lumpectomy makes this surgery medically unacceptable for some women. For instance, radiation therapy cannot be administered to pregnant women because it may injure the fetus. If, however, delivery would be completed prior to the need for radiation, pregnant women may undergo lumpectomy. Women with collagen vascular disease, such as lupus erythematosus or **scleroderma**, would experience scarring and damage to their connective tissue if exposed to radiation treatments. A woman who has already had therapeutic radiation to the chest area for other reasons cannot have additional exposure for breast cancer therapy.

Some women may choose not to have a lumpectomy for other reasons. They may strongly fear a recurrence of breast cancer, and may consider a lumpectomy too risky. Others feel uncomfortable living with a cancerous breast and experience more peace of mind with the entire breast removed.

The need for radiation therapy also may be a barrier due to non-medical concerns. Some women simply fear this type of treatment and choose more extensive surgery so that radiation will not be required. The commitment of time, usually five days a week for six weeks, may not be acceptable for others. This may be due to financial, personal, or job-related constraints. Finally, in geographically isolated areas, a course of radiation therapy may require lengthy travel, and perhaps unacceptable amounts of time away from family and other responsibilities.

Description

Lumpectomy is an imprecise term. Any amount of tissue, from 1% to 50% of the breast, may be removed

and called a lumpectomy. Breast conservation surgery is a frequently-used synonym for lumpectomy. Partial mastectomy, quadrantectomy, segmental excision, wide excision, and tylectomy are other names for this procedure.

A lumpectomy is frequently done in a hospital setting (especially if lymph nodes are to be removed at the same time), but specialized outpatient facilities sometimes are preferred. The surgery is usually done while the patient is under **general anesthesia**. Local anesthetic with additional **sedation** may be used for some patients. The tumor and surrounding margin of tissue is removed and sent to the pathologist. The surgical site is closed.

If axillary lymph nodes were not removed in a prior biopsy, a second incision is made in the armpit. The fat pad that contains lymph nodes is removed from this area and also is sent to the pathologist for analysis. This portion of the procedure is called an axillary lymph node dissection; it is critical for determining the stage of the cancer. Typically, 10 to 15 nodes are removed, but the number may vary. Surgical drains may be left in place in either location to prevent fluid accumulation. The surgery may last from one to three hours.

The patient may stay in the hospital one or two days, or return home the same day. This generally depends on the extent of the surgery and the medical condition of the patient, as well as physician and patient preferences. A woman usually goes home with a small bandage. The inner part of the surgical site usually has dissolvable stitches. The skin may be sutured or stitched; or the skin edges may be held together with steristrips, which are special thin, clear pieces of tape.

Preparation

Routine preoperative preparations, such as having nothing to eat or drink the night before surgery, are typically ordered for a lumpectomy. Information about expected outcomes and potential complications also is part of preparation for lumpectomy, as it is for any surgical procedure. It is especially important that women know about sensations they might experience after the operation, so the sensations are not misinterpreted as signs of further cancer or poor healing.

If the tumor cannot be felt (not palpable), a preoperative localization procedure is needed. A fine wire, or other device is placed at the tumor site, using x ray or ultrasound for guidance. This is usually done in the radiology department of a hospital. The woman

is most often sitting up and awake, although some sedation may be administered.

Aftercare

After a lumpectomy, patients are usually cautioned against lifting anything that weighs more than five pounds for several days. Other activities may be restricted (especially if the axillary lymph nodes were removed) according to individual needs. **Pain** is often enough to limit inappropriate motion. Women often are instructed to wear a well-fitting support bra both day and night for approximately one week after surgery.

Pain is usually well controlled with prescribed medication. If it is not, the patient should contact the surgeon, as severe pain may be a sign of a complication that needs medical attention. A return visit to the surgeon normally is scheduled approximately 10 days to two weeks after the operation. Studies have shown that women improve their survival rates after lumpectomy if they stop **smoking**.

Radiation therapy is usually started as soon as feasible after lumpectomy. Other additional treatments, such as chemotherapy or hormone therapy, may also be prescribed. The timing of these is specific to each individual patient.

Risks

The risks are similar to those associated with any surgical procedure. Risks include bleeding, infection, asymmetry, anesthesia reaction, or unexpected scarring. A lumpectomy also may cause loss of sensation in the breast. The size and shape of the breast will be affected by the operation. Fluid can accumulate in the area where tissue was removed, requiring drainage.

If lymph node dissection is performed, there are several potential complications. A woman may experience decreased feeling in the back of her armpit. She also may experience other sensations, including **numbness**, **tingling**, or increased skin sensitivity. An inflammation of the arm vein, called phlebitis, can occur. There may be injury to the nerves controlling arm motion.

Approximately 2% to 10% of patients develop **lymphedema** (swelling of the arm) after axillary lymph node dissection. This swelling of the arm can range from mild to severe. It can be treated with elastic bandages and specialized physical therapy, but it is a chronic condition, requiring continuing care. **Lymphedema** can arise at any time, even years after surgery.

KEY TERMS

Lymph node—A small mass of tissue in the form of a knot or protuberance. Lymph nodes are the primary sources of lymph fluid, which serve in the body's defense by removing toxic fluids and bacteria.

A new technique often eliminates the need to remove many axillary lymph nodes. Sentinel lymph node mapping and biopsy is based on the idea that the condition of the first lymph node in the network, which drains the affected area, can predict whether the cancer may have spread to the rest of the nodes. It is thought that if this first, or sentinel, node is cancer-free, there is no need to look further. Many patients with early-stage breast cancers may be spared the risks and complications of axillary lymph node dissection as the use of this approach continues to increase.

Normal results

When lumpectomy is performed, it is anticipated that it will be the definitive surgical treatment for breast cancer. Other forms of therapy, especially radiation, are often prescribed as part of the total treatment plan. A 2003 study reported that radiation of the entire breast produces better results than radiation of part of the breast. The expected outcome after lumpectomy and radiation is no recurrence of the breast cancer, however, women who have had lumpectomies, particularly those who were young at the time of treatment, should continue to see their physicians for regular breast cancer check-ups, since the cancer can recur.

Abnormal results

An unforeseen outcome of lumpectomy may be recurrence of the breast cancer, either locally or distally (in a part of the body far from the original site). Recurrence may be discovered soon after lumpectomy or years after the procedure. For this reason, it is important for patients to be regularly and closely monitored by their physicians. A 2003 report showed that **magnetic resonance imaging** (MRI) is accurate in detecting any cancer left in the breast after lumpectomy. Women should continue to have regular mammograms. While the scar tissue from lumpectomy and radiation therapy can make mammograms less comfortable, a special cushion was approved by the U.S. Food and Drug Administration in 2003 that reduces

discomfort in women who have had breast conserving surgery.

Resources

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Teresa G. Odle

Lumpy breasts see **Fibrocystic condition of the breast**

Lumpy jaw see **Actinomycosis**

Lung abscess

Definition

Lung **abscess** is an acute or chronic infection of the lung, marked by a localized collection of pus, inflammation, and destruction of tissue.

Description

Lung abscess is the end result of a number of different disease processes ranging from fungal and bacterial infections to **cancer**. It can affect anyone at any age. Patients who are most vulnerable include those weakened by cancer and other chronic diseases; patients with a history of **substance abuse**, diabetes, epilepsy, or poor dental hygiene; patients who have recently had operations under anesthesia; and **stroke** patients. In children, the most vulnerable patients are those with weakened immune systems, **malnutrition**, or blunt injuries to the chest.

Causes and symptoms

The immediate cause of most lung abscesses is infection caused by bacteria. About 65% of these infections are produced by anaerobes, which are bacteria that do not need air or oxygen to live. The remaining cases are caused by a mixture of anaerobic and aerobic (air breathing) bacteria. When the bacteria arrive in the lung, they are engulfed or eaten by special cells called phagocytes. The phagocytes release chemicals that contribute to inflammation and eventual necrosis, or death, of a part of the lung tissue. There are several different ways that bacteria can get into the lung.

Aspiration

Aspiration refers to the accidental inhalation of material from the mouth or throat into the airway and lungs. It is responsible for about 50% of cases of lung abscess. The human mouth and gums contain large numbers of anaerobic bacteria; patients with **periodontal disease** or poor **oral hygiene** have higher concentrations of these organisms. Aspiration is most likely to occur in patients who are unconscious or semi-conscious due to anesthesia, seizures, alcohol and drug **abuse**, or stroke. Patients who have problems swallowing or coughing, or who have nasogastric tubes in place are also at risk of aspiration.

Bronchial obstruction

The bronchi are the two branches of the windpipe that lead into the lungs. If they are blocked by tissue swelling, cancerous tumors, or **foreign objects**, a lung abscess may form from infection trapped behind the blockage.

Spread of infection

About 20% of cases of **pneumonia** that cause the death of lung tissue (necrotizing pneumonia) will develop into lung abscess. Lung abscess can also be caused by the spread of other infections from the liver, abdominal cavity, or open chest **wounds**. Rarely, **AIDS** patients can develop lung abscess from *Pneumocystis carinii* and other organisms that take advantage of a weakened immune system.

Lung abscess is usually slow to develop. It may take about two weeks after aspiration or bronchial obstruction for an abscess to produce noticeable symptoms. The patient may be acutely ill for two weeks to three months. In the beginning, the symptoms of lung abscess are difficult to distinguish from those of severe pneumonia. Adults will usually have moderate **fever** (101-102 °F/38-39 °C), chills, chest

pain, and general weakness. Children may or may not have chest pain, but usually suffer weight loss and high fevers. As the illness progresses, about 75% of patients will **cough** up foul or musty-smelling sputum; some also cough up blood.

Lung abscess can lead to serious complications, including **emphysema**, spread of the abscess to other parts of the lung, hemorrhage, **adult respiratory distress syndrome**, rupture of the abscess, inflammation of the membrane surrounding the heart, or chronic inflammation of the lung.

Diagnosis

The diagnosis is made on the basis of the patient's medical history (especially recent operations under **general anesthesia**) and general health as well as imaging studies. Smears and cultures taken from the patient's sputum are not usually very helpful because they will be contaminated with bacteria from the mouth. The doctor will first use a bronchoscope (lighted tube inserted into the windpipe) to rule out the possibility of lung cancer. In some cases of serious infection, the doctor can use a fiberoptic bronchoscope with a protected specimen brush to take material directly from the patient's lungs, for identification of the organism. This technique is time-consuming and expensive, and requires the patient to be taken off **antibiotics** for 48 hours. It is usually used only to evaluate severely ill patients with weakened immune systems.

In most cases, the doctor will use the results of a **chest x ray** to help distinguish lung abscess from **empyema**, cancer, **tuberculosis**, or cysts. In patients with lung abscess, the x ray will show a thick-walled unified clear space or cavity surrounded by solid tissue. There is often a visible air-fluid level. The doctor may also order a CT scan of the chest, in order to have a clearer picture of the exact location of the abscess.

Blood tests cannot be used to make a diagnosis of lung abscess, but they can be useful in ruling out other conditions. Patients with lung abscess usually have abnormally high white blood cell counts (**leukocytosis**) when their blood is tested, but this condition is not unique to lung abscess.

Treatment

Lung abscess is treated with a combination of antibiotic drugs, **oxygen therapy**, and surgery. The antibiotics that are usually given for lung abscess are penicillin G, penicillin V, and clindamycin. They are given intravenously until the patient shows signs of improvement, and then continued in oral form. The

patient may need to take antibiotics for a month or longer, until the chest x ray indicates that the abscess is healing. Oxygen may be given to patients who are having trouble breathing.

Surgical treatment

Most patients with lung abscess will not need surgery. About 5% of patients—usually those who do not respond to antibiotics or are coughing up large amounts of blood (500 mL or more)—may have emergency surgery for removal of the diseased part of the lung or for insertion of a tube to drain the abscess. Antibiotic treatment is considered to have failed if fever and other symptoms continue after 10-14 days of treatment; if chest x rays indicate that the abscess is not shrinking; or if the patient has pneumonia that is spreading to other parts of the lung.

Supportive care

Because lung abscess is a serious condition, patients need quiet and bed rest. Hospital care usually includes increasing the patient's fluid intake to loosen up the secretions in the lungs, and physical therapy to strengthen the patient's breathing muscles.

Follow-up

Patients with lung abscess need careful follow-up care after the acute infection subsides. Follow-up usually includes a series of chest x rays to make sure that the infection has cleared up. Treatment with antibiotics may continue for as long as four months, to prevent recurrence.

Prognosis

About 95% of lung abscess patients can be treated successfully with antibiotics alone. Patients who need surgical treatment have a mortality rate of 10-15%.

Prevention

Some of the conditions that make people more vulnerable to lung abscess concern long-term lifestyle behaviors, such as substance abuse and lack of dental care. Others, however, are connected with chronic illness and hospitalization. Aspiration can be prevented with proper care of unconscious patients, which includes suctioning of throat secretions and positioning patients to promote drainage. Patients who are conscious can be given physical therapy to help them cough up material in their lungs and airways. Patients

KEY TERMS

Abscess—An area of injured body tissue that fills with pus, as in lung abscess.

Anaerobe—A type of bacterium that does not require air or oxygen to live. Anaerobic bacteria are frequent causes of lung abscess.

Aspiration—Inhalation of fluid or foreign bodies into the airway or lungs. Aspiration often happens after vomiting.

Bronchoscope—A lighted, flexible tube inserted into the windpipe to view the bronchi or withdraw fluid samples for testing. Bronchoscopy with a protected brush can be used in the diagnosis of lung abscess in severely ill patients.

Bronchus—One of the two large tubes connecting the windpipe and the lungs.

Leukocytosis—An increased level of white cells in the blood. Leukocytosis is a common reaction to infections, including lung abscess.

Necrotizing pneumonia—Pneumonia that causes the death of lung tissue. It often precedes the development of lung abscess.

Sputum—The substance that is brought up from the lungs and airway when a person coughs or spits. It is usually a mixture of saliva and mucus, but may contain blood or pus in patients with lung abscess or other diseases of the lungs.

with weakened immune systems can be isolated from patients with pneumonia or fungal infections.

Resources

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Rebecca J. Frey, PhD

Lung biopsy

Definition

Lung biopsy is a medical procedure performed to obtain a small piece of lung tissue for examination

under a microscope. Biopsy examinations are usually performed by pathologists, who are doctors with special training in tissue abnormalities and other signs of disease.

Purpose

Lung biopsies are useful, first of all, in confirming a diagnosis of **cancer**, especially if malignant cells are detected in the patient's sputum. A lung biopsy may be ordered to examine other abnormalities that appear on chest x rays, such as lumps (nodules). It is also helpful in diagnosing symptoms such as coughing up bloody sputum, **wheezing** in the chest, or difficult breathing. In addition to evaluating lung tumors and their associated symptoms, lung biopsies can be used in the diagnosis of lung infections, especially **tuberculosis**, drug reactions, and such chronic diseases of the lung as **sarcoidosis**.

A lung biopsy can be used for treatment as well as diagnosis. **Bronchoscopy**, which is a type of lung biopsy performed with a long slender instrument called a bronchoscope, can be used to clear a patient's air passages of secretions and to remove blockages from the airways.

Precautions

As with any other biopsy, lung biopsies should not be performed on patients who have problems with blood clotting because of low platelet counts. Platelets are small blood cells that play a role in the blood clotting process. If the patient has a **platelet count** lower than 50,000/cubic mm, he or she can be given a platelet **transfusion** as a temporary relief measure, and a biopsy can then be performed.

Description

Overview

The lungs are a pair of cone-shaped organs that lie in the chest cavity. An area known as the mediastinum separates the right and the left lungs from each other. The heart, the windpipe (trachea), the lymph nodes, and the tube that brings the food to the stomach (the esophagus) lie in this mediastinal cavity. Lung biopsies may involve entering the mediastinum, as well as the lungs themselves.

Types of lung biopsies

Lung biopsies can be performed using a variety of techniques. A bronchoscopy is ordered if a patch that looks suspicious on the x ray seems to be located deep

in the chest. If the area lies close to the chest wall, a needle biopsy is often done. If both these methods fail to diagnose the problem, an open surgical biopsy may be carried out. If there are indications that the lung cancer has spread to the lymph nodes in the mediastinum, a **mediastinoscopy** is performed.

NEEDLE BIOPSY. When a needle biopsy is to be done, the patient will be given a sedative about an hour before the procedure, to help relaxation. The patient sits in a chair with arms folded on a table in front of him or her. X rays are then taken to identify the location of the suspicious areas. Small metal markers are placed on the overlying skin to mark the biopsy site. The skin is thoroughly cleansed with an antiseptic solution, and a local anesthetic is injected to numb the area.

The doctor then makes a small cut (incision) about half an inch in length. The patient is asked to take a deep breath and hold it while the doctor inserts the special biopsy needle through the incision into the lung. When enough tissue has been obtained, the needle is withdrawn. Pressure is applied at the biopsy site and a sterile bandage is placed over the cut. The entire procedure takes between 30 and 45 minutes.

The patient may feel a brief sharp **pain** or some pressure as the biopsy needle is inserted. Most patients, however, do not experience severe pain.

OPEN BIOPSY. Open biopsies are performed in a hospital under **general anesthesia**. As with needle biopsies, patients are given sedatives before the procedure. An intravenous line is placed in the arm to give medications or fluids as necessary. A hollow tube, called an endotracheal tube, is passed through the throat, into the airway leading to the lungs. It is used to convey the general anesthetic.

Once the patient is under the influence of the anesthesia, the surgeon makes an incision over the lung area. Some lung tissue is removed and the cut closed with stitches. The entire procedure usually takes about an hour. A chest tube is sometimes placed with one end inside the lung and the other end protruding through the closed incision. Chest tube placement is done to prevent the lungs from collapsing by removing the air from the lungs. The tube is removed a few days after the biopsy.

A **chest x ray** is done following an open biopsy, to check for lung collapse. The patient may experience some grogginess for a few hours after the procedure. He or she may also experience tiredness and muscle aches for a day or two, because of the general anesthesia. The throat may be sore because of the placement of the hollow endotracheal tube. The patient may also

have some pain or discomfort at the incision site, which can be relieved by medication.

MEDIASTINOSCOPY. The preparation for a mediastinoscopy is similar to that for an open biopsy. The patient is sedated and prepared for general anesthesia. The neck and the chest will be cleansed with an antiseptic solution.

After the patient has been put to sleep, an incision about two or three inches long is made at the base of the neck. A thin, hollow, lighted tube, called a mediastinoscope, is inserted through the cut into the space between the right and the left lungs. The doctor examines the space thoroughly and removes any lymph nodes or tissues that look abnormal. The mediastinoscope is then removed, and the incision stitched up and bandaged. A mediastinoscopy takes about an hour.

Preparation

Before scheduling any lung biopsy, the doctor will check to see if the patient is taking any prescription medications, if he or she has any medication **allergies**, and if there is a history of bleeding problems. Blood tests may be performed before the procedure to check for clotting problems and blood type, in case a transfusion becomes necessary.

If an open biopsy or a mediastinoscopy is being performed, the patient will be asked to sign a consent form. Since these procedures are done under general anesthesia, the patient will be asked to refrain from eating or drinking anything for at least 12 hours before the biopsy.

Aftercare

Needle biopsy

Following a needle biopsy, the patient is allowed to rest comfortably. He or she will be checked by a nurse at two-hour intervals. If there are no complications after four hours, the patient can go home. Patients are advised to rest at home for a day or two before resuming regular activities, and to avoid strenuous activities for a week after the biopsy.

Open biopsy or mediastinoscopy

After an open biopsy or a mediastinoscopy, patients are taken to a recovery room for observation. If no other complications develop, they are taken back to the hospital room. Stitches are usually removed after seven to 14 days.

If the patient has extreme pain, light-headedness, difficulty breathing, or develops a blue tinge to the

skin after an open biopsy, the doctor should be notified immediately. The sputum may be slightly bloody for a day or two after the procedure. If, however, the bleeding is heavy or persistent, it should be brought to the attention of the doctor.

Risks

Needle biopsy

Needle biopsy is a less risky procedure than an open biopsy, because it does not involve general anesthesia. Very rarely, the lung may collapse because of air that leaks in through the hole made by the biopsy needle. If the lung collapses, a tube will have to be inserted into the chest to remove the air. Some coughing up of blood occurs in 5% of needle biopsies. Prolonged bleeding or infection may also occur, although these are very rare.

Open biopsy

Possible complications of an open biopsy include infection or lung collapse. **Death** occurs in about 1 in 3000 cases. If the patient has very severe breathing problems before the biopsy, breathing may be slightly impaired following the operation. If the person's lungs were functioning normally before the biopsy, the chances of any respiratory problems are very small.

Mediastinoscopy

Complications due to mediastinoscopy are rare; death occurs in fewer than 1 in 3000 cases. More common complications include lung collapse or bleeding caused by damage to the blood vessels near the heart. Injury to the esophagus or voice box (larynx) may sometimes occur. If the nerves leading to the larynx are injured, the patient may be left with a permanently hoarse voice. All of these complications are very rare.

Normal results

Normal results of a needle biopsy and an open biopsy include the absence of any evidence of infection in the lungs. No lumps or nodules will be detected in the lungs and the cells will not show any cancerous abnormalities. Normal results from the mediastinoscopy will show the lymph nodes to be free of cancer.

Abnormal results

Abnormal results may be associated with diseases other than cancer. Nodules in the lungs may

KEY TERMS

Bronchoscopy—A medical test that enables the doctor to see the breathing passages and the lungs through a hollow, lighted tube.

Endotracheal tube—A hollow tube that is inserted into the windpipe to administer anesthesia.

Lymph nodes—Small, bean-shaped structures scattered along the lymphatic vessels which serve as filters. Lymph nodes retain any bacteria or cancer cells that are traveling through the system.

Mediastinoscopy—A medical procedure that allows the doctor to see the organs in the mediastinal space using a thin, lighted, hollow tube (a mediastinoscope).

Mediastinum—The area between the lungs, bounded by the spine, breastbone, and diaphragm.

Sputum—Mucus or phlegm that is coughed up from the passageways (bronchial tubes) in the lungs.

be due to active infections such as tuberculosis, or may be **scars** from a previous infection. The lung cells on microscopic examination do not resemble normal cells, and show certain abnormalities that point to cancer. In a third of biopsies using a mediastinoscope, the lymph nodes that are biopsied prove to be cancerous. Abnormal results should always be considered in the context of the patient's medical history, **physical examination**, and other tests such as sputum examination, chest x rays, etc. before a final diagnosis is made.

Resources

ORGANIZATIONS

American Cancer Society. 1599 Clifton Rd., NE, Atlanta, GA 30329-4251. (800) 227-2345. <<http://www.cancer.org>>.

American Lung Association. 1740 Broadway, New York, NY 10019. (800) 586-4872. <<http://www.lungusa.org>>.

Cancer Research Institute. 681 Fifth Ave., New York, N.Y. 10022. (800) 992-2623. <<http://www.cancerresearch.org>>.

National Cancer Institute. Building 31, Room 10A31, 31 Center Drive, MSC 2580, Bethesda, MD 20892-2580. (800) 422-6237. <<http://www.nci.nih.gov>>.

Lata Cherath, PhD

Lung cancer, non-small cell

Definition

Non-small cell lung **cancer** (NSCLC) is a disease in which the cells of the lung tissues grow uncontrollably and form tumors.

Description

There are two kinds of lung cancers, primary and secondary. Primary lung cancer starts in the lung itself, and is divided into **small cell lung cancer** and non-small cell lung cancer. Small cell lung cancers are shaped like an oat and called oat-cell cancers; they are aggressive, spread rapidly, and represent 20% of lung cancers. Non-small cell lung cancer represents almost 80% of all primary lung cancers. Secondary lung cancer is cancer that starts somewhere else in the body (for example, the breast or colon) and spreads to the lungs.

The lungs

The lungs are located along with the heart in the chest cavity. The lungs are not simply hollow balloons but have a very organized structure consisting of hollow tubes, blood vessels and elastic tissue. The hollow tubes, called bronchi, are highly branched, becoming smaller and more numerous at each branching. They end in tiny, blind sacs made of elastic tissue called alveoli. These sacs are where the oxygen a person breathes in is taken up into the blood, and where carbon dioxide moves out of the blood to be breathed out.

Normal healthy lungs are continually secreting mucus that not only keeps the lungs moist, but also protects the lungs by trapping foreign particles like dust and dirt in breathed air. The inside of the lungs is covered with small hairlike structures called cilia. The cilia move in such a way that mucus is swept up out of the lungs and into the throat.

Lung cancer

Most lung cancers start in the cells that line the bronchi, and can take years to develop. As they grow larger they prevent the lungs from functioning normally. The tumor can reduce the capacity of the lungs, or block the movement of air through the bronchi in the lungs. As a result, less oxygen gets into the blood and patients feel short of breath. Tumors may also block the normal movement of mucus up into the throat. As a result, mucus builds up in the lungs and infection may develop behind the tumor. Once lung

cancer has developed it frequently spreads to other parts of the body.

The speed at which non-small cell tumors grow depends on the type of cells that make up the tumor. The following three types account for the vast majority of non-small cell tumors:

- Adenocarcinomas are the most common and often cause no symptoms. Frequently they are not found until they are advanced.
- Squamous cell carcinomas usually produce symptoms because they are centrally located and block the lungs.
- Undifferentiated large cell and giant cell carcinomas tend to grow rapidly, and spread quickly to other parts of the body.

Worldwide, lung cancer is the most common cancer in males, and the fifth most common cancer in women. The worldwide mortality rate for patients with lung cancer is 86%. In the United States, lung cancer is the leading cause of **death** from cancer among both men and women. The World Health Organization estimates that the worldwide mortality from lung cancer will increase to three million by the year 2025. Of those three million deaths, almost two and a half million will result from non-small cell lung cancer.

The American Cancer Society (ACS) estimates that 173,770 Americans will develop lung cancer in 2004, 93,110 men and 80,660 women. Of these patients, 160,440 will die of the disease.

The incidence of lung cancer is beginning to fall in developed countries. This may be a result of antismoking campaigns. In developing countries, however, rates continue to rise, which may be a consequence of both industrialization and the increasing use of tobacco products.

Causes and symptoms

Causes

Tobacco **smoking** accounts for 87% of all lung cancers. Giving up tobacco can prevent most lung cancers. Smoking **marijuana** cigarettes is considered another risk factor for cancer of the lung. Second hand smoke also contributes to the development of lung cancer among nonsmokers.

Certain hazardous materials that people may be exposed to in their jobs have been shown to cause lung cancer. These include asbestos, coal products, and radioactive substances. Air pollution may also be a contributing factor. Exposure to radon, a colorless, odorless gas that sometimes accumulates in the

basement of homes, may cause lung cancer in a tiny minority of patients. In addition, patients whose lungs are scarred from other lung conditions may have an increased risk of developing lung cancer.

Symptoms

Lung cancers tend to spread very early, and only 15% are detected in their early stages. The chances of early detection, however, can be improved by seeking medical care at once if any of the following symptoms appear:

- a **cough** that does not go away
- chest **pain**
- **shortness of breath**
- recurrent lung infections, such as **bronchitis** or pneumonia
- bloody or brown-colored spit or phlegm (sputum)
- persistent hoarseness
- significant weight loss that is not due to dieting or vigorous **exercise**; **fatigue** and loss of appetite
- unexplained fever

Although these symptoms may be caused by diseases other than lung cancer, it is important to consult a doctor to rule out the possibility of lung cancer.

If lung cancer has spread to other organs, the patient may have other symptoms such as headaches, bone **fractures**, pain, bleeding, or **blood clots**.

Diagnosis

Physical examination and diagnostic tests

The doctor will first take a detailed medical history and assess risk factors. During a complete **physical examination** the doctor will examine the patient's throat to rule out other possible causes of hoarseness or coughing, and will listen to the patient's breathing and chest sounds.

If the doctor has reason to suspect lung cancer, particularly if the patient has a history of heavy smoking or occupational exposure to irritating substances, a **chest x ray** may be ordered to see if there are any masses in the lungs. Special imaging techniques, such as computed tomography (CT) scans or **magnetic resonance imaging** (MRI), may provide more precise information about the size, shape, and location of any tumors.

Sputum analysis

Sputum analysis is a noninvasive test that involves microscopic examination of cells that are coughed up

from the lungs. This test can diagnose at least 30% of lung cancers, even if tumors are not visible on chest x rays. In addition, the test can detect cancer in its very early stages, before it spreads to other regions. The sputum test does not provide any information about the location of the tumor.

Lung biopsy

Lung biopsy is the most definitive diagnostic tool for cancer. It can be performed in three different ways. **Bronchoscopy** involves the insertion of a slender, lighted tube, called a bronchoscope, down the patient's throat and into the lungs. This test allows the doctor to see the tubes inside the lungs, and to obtain samples of lung tissue. If a needle biopsy is to be performed, the location of the tumor is first identified using a computerized tomography (CT) scan or magnetic resonance imaging (MRI). The doctor then inserts a needle through the chest wall and collects a sample of tissue from the tumor. In the third procedure, known as surgical biopsy, the chest wall is opened up and a part of the tumor, or all of it, is removed. A doctor who specializes in the study of diseased tissue (a pathologist) examines the tumor to identify the cancer's type and stage.

Treatment

Staging

Treatment for non-small cell lung cancer depends primarily on the stage of the cancer. Staging is a process that tells the doctor if the cancer has spread and the extent of its spread. The most commonly used treatments are surgery, **radiation therapy**, and **chemotherapy**.

Non-small cell lung cancer has six stages:

- Occult carcinoma. Cancer cells have been found in the sputum, but no tumor has yet been found.
- Stage 0. A small group of cancerous cells have been found in one location.
- Stage I. The cancer is only in the lung and has not spread anywhere else.
- Stage II. The cancer has spread to nearby lymph nodes.
- Stage III. The cancer has spread to more distant lymph nodes, and/or other parts of the chest like the diaphragm.
- Stage IV. The cancer has spread to other parts of the body.

Surgery

Surgery is the standard treatment for the earlier stages of non-small cell lung cancer. The surgeon will decide on the type of surgery, depending on how much of the lung is affected. There are three different types of surgical procedures:

- Wedge resection is the removal of a small part of the lung.
- Lobectomy is the removal of one lobe of the lung. (The right lung has three lobes and the left lung has two lobes.)
- Pneumonectomy is the removal of an entire lung.

Lung surgery is a major procedure and patients can expect to experience pain, weakness in the chest, and shortness of breath. Air and fluid collect in the chest after surgery. As a result, patients will need help to turn over, cough, and breathe deeply. Patients should be encouraged to perform these activities because they help get rid of the air and fluid and speed up recovery. It can take patients several months before they regain their energy and strength.

Radiotherapy

Patients whose cancer has progressed too far for surgery (Stages III and IV) may receive radiotherapy. Radiotherapy involves the use of high-energy rays to kill cancer cells. It is used either by itself or in combination with surgery or chemotherapy. The amount of radiation used depends on the size and the location of the tumor.

Radiation therapy may produce such side effects as tiredness, skin **rashes**, upset stomach, and **diarrhea**. Dry or sore throats, difficulty in swallowing, and loss of hair in the treated area are all minor side effects of radiation. These may disappear either during the course of the treatment or after the treatment is over.

Chemotherapy

Chemotherapy is also given to patients whose cancer has progressed too far for surgery. Chemotherapy is medication that is usually given intravenously to kill cancer cells. These drugs enter the bloodstream and travel to all parts of the body, killing cancer cells that have spread to different organs. Chemotherapy is used as the primary treatment for cancers that have spread beyond the lung and cannot be removed by surgery. It can also be used in addition to surgery or radiation therapy.

Chemotherapy for NSCLC has made significant advances since the early 1980s in improving the

patient's quality of life as well as length of survival. Newer cytotoxic (cell-killing) agents developed in the 1990s, such as the taxanes, are typically combined with either cisplatin or carboplatin as first-line therapy for non-small cell lung cancer.

Newer drugs for lung cancer developed since 2000 include gefinitib (Iressa) and pemetrexed (Alimta). The FDA approved gefinitib in May 2003 as a treatment for patients with NSCLC who have not responded to platinum-based or taxane chemotherapy. It is taken by mouth and works by inhibiting an enzyme involved in the growth of tumor cells. Pemetrexed, which is given by injection, was approved by the FDA in February 2004 for the treatment of **mesothelioma**, a type of lung cancer caused by exposure to asbestos fibers. However, the drug appears to be effective in treating other types of lung cancer as well.

Chemotherapy is also used as palliative treatment for non-small cell lung cancer. Palliative refers to any type of therapy that is given to relieve the symptoms of a disease but not to cure it.

Chemotherapy for non-small cell lung cancer often has severe side effects, including **nausea and vomiting**, hair loss, anemia, weakening of the immune system, and sometimes **infertility**. Most of these side effects end when the treatment is over. Other medications can be given to lessen the unpleasant side effects of chemotherapy.

Alternative treatment

Because non-small cell lung cancer has a poor prognosis with conventional medical treatment, many patients are willing to try complementary and alternative therapies. These therapies are used to try to reduce **stress**, ease side effects and symptoms, or control disease. Two treatments sometimes used are shark cartilage and mistletoe. Although shark cartilage is thought to interfere with the tumor's blood supply, clinical trials have so far been inconclusive. Mistletoe is a poisonous plant that has been shown to kill cancer cells in the laboratory. Again, however, clinical trials with cancer patients have been inconclusive.

Patients who decide to try complementary and alternative therapies should tell their doctor. Some of these therapies may interfere with conventional treatment.

Prognosis

The prognosis for non-small cell lung cancer is better if the disease is found early, and removed

KEY TERMS

Bronchi—The tubes that carry air into the lungs.

Lymph—Clear fluid containing white blood cells that is collected from the tissues of the body and flows in vessels called the lymphatic system.

Lymph node—Small oval-shaped filters in the lymphatic system that trap bacteria and other unwanted particles to ensure their removal from the body.

Palliative—Referring to any type of treatment that is given to relieve the symptoms of a disease rather than to cure it.

Respiratory distress—A condition in which patients with lung disease are not able to get enough oxygen.

surgically. For patients whose disease is caught in Stage I, the survival rate five years after surgery ranges from 60% to 80%. Up to 55% of Stage II patients are alive after five years, but only about 30% of Stage III patients make it to five years. Unfortunately, 85% of patients already have at least Stage III cancer by the time they are diagnosed. Many of these patients have disease that is too advanced for surgery. Despite treatment with radiotherapy and chemotherapy, the five-year survival for patients with inoperable disease is extremely low.

Prevention

The best way to prevent lung cancer is not to start smoking or to quit smoking. Secondhand smoke from other people's tobacco should also be avoided. Appropriate precautions should be taken when working with cancer-causing substances (carcinogens). Testing houses for the presence of radon gas, and removing asbestos from buildings have also been suggested as preventive strategies.

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ORGANIZATIONS

- Alliance for Lung Cancer Advocacy, Support and Education. PO Box 849 Vancouver, WA 98666. (800) 298-2436. <<http://www.alcase.org>>.
- American Lung Association. 1740 Broadway New York, NY 10019. (212) 315-8700. <<http://www.lungusa.org>>.
- National Cancer Institute (National Institutes of Health). 9000 Rockville Pike, Bethesda, MD 20892. (800) 422-6237. <<http://www.nci.nih.gov>>.
- National Center for Complementary and Alternative Medicine (National Institutes of Health). PO Box 8218, Silver Spring, MD 20907-8218. (888) 644-6226. <<http://nccam.nih.gov>>.

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Lung cancer, small cell

Definition

Small cell lung **cancer** is a disease in which the cells of the lung tissues grow uncontrollably and form tumors.

Description

Lung cancer is divided into two main types: small cell and non-small cell. Small cell lung cancer is the least common of the two, accounting for only about 20% of all lung cancers. In the past, the disease was called oat cell cancer because, when viewed under a microscope, the cancer cells resemble oats. This type of lung cancer grows quickly and is more likely to spread to other organs in the body.

The lungs are located along with the heart in the chest cavity. The lungs are not simply hollow balloons, but have a very organized structure consisting of hollow tubes, blood vessels, and elastic tissue. The hollow tubes, called bronchi, are multi-branched, becoming smaller and more numerous at each branching. They end in tiny, blind sacs made of elastic tissue called alveoli. These sacs are where the oxygen a person breathes in is taken up into the blood, and where carbon dioxide moves out of the blood to be breathed out.

Normal, healthy lungs are continually secreting mucus that not only keeps the lungs moist, but also protects the lungs by trapping foreign particles like dust and dirt in breathed air. The inside of the lungs is covered with small, hair-like structures called cilia. The cilia move in such a way that mucus is swept up out of the lungs and into the throat.

Small cell lung tumors usually start to develop in the central bronchi. They grow quickly and prevent the lungs from functioning at their full capacity. Tumors may block the movement of air through the bronchi in the lungs. As a result, less oxygen gets into the blood and patients feel short of breath. Tumors may also block the normal movement of mucus into the throat. As a result, mucus builds up in the lungs and infection may develop behind the tumor.

Lung cancer is a growing global epidemic. Worldwide, lung cancer is the second most common cancer among both men and women and is the leading cause of cancer **death** in both sexes. The worldwide mortality rate for patients with lung cancer is 86%. Of the 160,000 deaths from lung cancer that occur

annually in the United States, about 40,000 are caused by small cell lung cancer. Although there are differences in mortality rates between ethnic groups, this is mainly due to differences in **smoking** habits.

Causes and symptoms

Causes

Tobacco smoking accounts for nearly 90% of all lung cancers. The risk of developing lung cancer is increased for smokers who start at a young age, and for those who have smoked for a long time. The risk also increases as more cigarettes are smoked, and when cigarettes with higher tar content are smoked. Smoking **marijuana** cigarettes is also a risk factor for lung cancer. These cigarettes have a higher tar content than tobacco cigarettes.

Certain hazardous materials that people may be exposed to in their jobs have been shown to cause lung cancer. These include asbestos, coal products, and radioactive substances. Air pollution may also be a contributing factor. Exposure to radon, a colorless, odorless gas that sometimes accumulates in the basement of homes, may cause lung cancer in some patients. In addition, patients whose lungs are scarred from other lung conditions may have an increased risk of developing lung cancer.

Although the exact cause of lung cancer is not known, people with a family history of lung cancer appear to have a slightly higher risk of contracting the disease.

Symptoms

Small cell lung cancer is an aggressive disease that spreads quickly. Symptoms depend on the tumor's location within the lung, and on whether the cancer has spread to other parts of the body. More than 80% of small cell lung cancer patients have symptoms for only three months or less, and few cases are detected early. The following symptoms are the most commonly reported by small cell lung cancer patients at the time of their diagnosis:

- a persistent **cough**
- chest **pain**
- **shortness of breath** and wheezing
- persistent hoarseness
- fatigue and loss of appetite

Although some patients may experience bloody spit or phlegm, this symptom is more commonly seen in patients with other types of lung cancer.



A normal lung (left) and the lung of a cigarette smoker (right). (Photograph by A. Glauber, Photo Researchers, Inc. Reproduced by permission.)

Small cell tumors often press against a large blood vessel near the lungs called the superior vena cava (SVC), causing a condition known as SCV syndrome. This condition may cause patients to retain water, cough, and have shortness of breath. Because small cell lung cancer often spreads quickly to the bones and central nervous system, patients may also have bone pain, headaches, and seizures.

Diagnosis

If lung cancer is suspected, the doctor will take a detailed medical history that checks both symptoms and risk factors. During a complete **physical examination**, the doctor will examine the patient's throat to rule out other possible causes of hoarseness or coughing, and listen to the patient's breathing and the sounds made when the patient's chest and upper back are tapped. A **chest x ray** may be ordered to check for masses in the lungs. Special imaging techniques, such as computed tomography (CT) scans or **magnetic resonance imaging** (MRI), may provide more precise information about the size, shape, and location of any tumors.

Sputum analysis involves microscopic examination of the cells that are either coughed up from the lungs, or are collected through a special instrument called a bronchoscope. The sputum test does not, however, provide any information about the location of the tumor and must be followed by other tests.

Lung biopsy is the most definitive diagnostic tool for cancer. It can be performed in several different ways. The doctor can perform a **bronchoscopy**, which involves the insertion of a slender, lighted tube, called a bronchoscope, down the patient's throat and into the lungs. In addition to viewing the passageways of the lungs, the doctor can use the bronchoscope to obtain samples of the lung tissue. In another procedure known as a needle biopsy, the location of the tumor is first identified using a CT scan or MRI. The doctor then inserts a needle through the chest wall and collects a sample of tissue from the tumor. In the third procedure, known as surgical biopsy, the chest wall is opened up and a part of the tumor, or all of it, is removed for examination.

Treatment

Staging

Staging procedures are important in lung cancer because they tell doctors whether patients have disease only in their lungs, or whether the cancer has spread to other parts of the body. To establish the cancer stage, doctors have to perform various tests. These may include **bone marrow aspiration and biopsy**, CT scans of the chest and abdomen, MRI scans of the brain, and radionuclide bone scans. All of these tests determine the extent to which the cancer has spread. Once the stage is determined, doctors can decide on a course of treatment, and can have a better idea of the patient's prognosis.

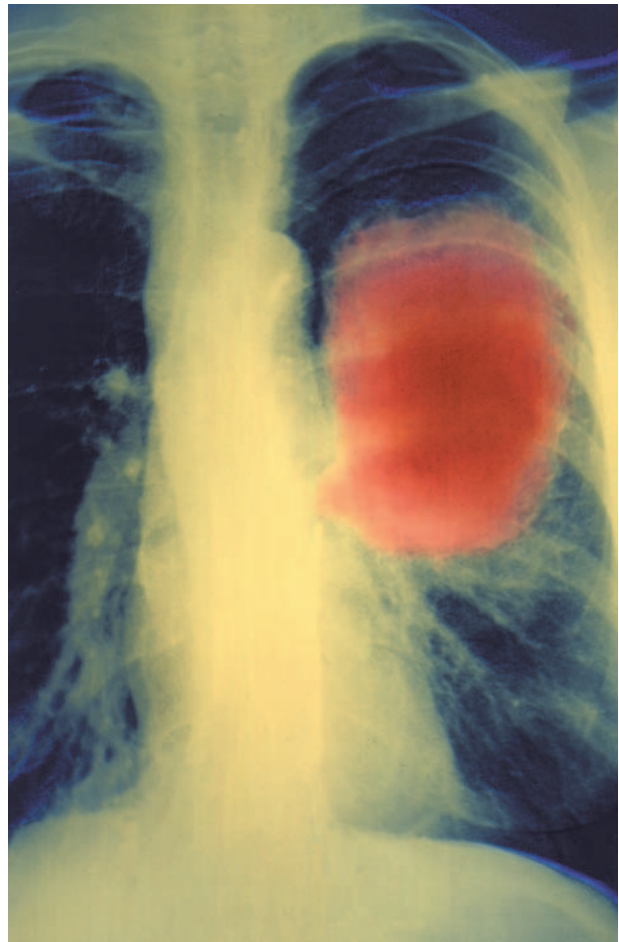
Unlike other types of lung cancer, the staging of small cell lung cancer is relatively simple. This is because approximately 70% of patients already have metastatic disease when they are diagnosed, and small differences in the amount of tumor found in the lungs do not change the prognosis. Small cell lung cancer is usually divided into three stages:

- Limited stage: The cancer is found only in one lung and in lymph nodes close to the lung.
- Extensive stage: The cancer has spread beyond the lungs to other parts of the body.
- Recurrent stage: The cancer has returned following treatment.

Without treatment, small cell lung cancer has the most aggressive clinical course of any type of pulmonary tumor, with median survival from diagnosis of only 2–4 months. Compared with other cell types of lung cancer, small cell lung cancer has a greater tendency to be widely disseminated by the time of diagnosis, but is much more responsive to **chemotherapy** and irradiation.

Treatment of small cell lung cancer depends on whether the patient has limited, extensive, or recurrent disease. Treatment usually involves radiotherapy and chemotherapy. Surgery is rarely used for this type of lung cancer because the tumor is usually too advanced.

Patients with limited-stage disease are usually treated with chemotherapy. Combinations of two or more drugs have a better effect than treatment with a single drug. Up to 90% of patients with this stage of disease will respond to chemotherapy. The chemotherapy most commonly prescribed is a combination of the drugs etoposide (Vepesid) and cisplatin (Platinol). Combining chemotherapy with chest radiotherapy and/or occasionally surgery has also prolonged survival for limited-stage patients.



An x-ray image showing an oval-shaped carcinoma in the left lung (right of image). (Custom Medical Stock Photo. Reproduced by permission.)

In addition to chest radiotherapy, some patients are also treated with **radiation therapy** to the brain, even if no cancer is found there. This treatment, called prophylactic cranial irradiation (PCI), is given to prevent tumors from forming in the brain. The combination of etoposide and cisplatin chemotherapy with chest radiation therapy and PCI has increased the two-year survival of limited-stage small cell lung cancer patients to almost 50%.

Combinations of different chemotherapy agents are also used for treating extensive-stage small cell lung cancer. However, compared with limited-stage patients, the percentage of extensive-stage patients who respond to therapy is lower. Commonly used drug combinations include cyclophosphamide (Cytoxan), doxorubicin (Adriamycin), and vincristine (Oncovin), or etoposide and cisplatin. The addition of radiation therapy to chemotherapy does not improve

KEY TERMS

Bronchi—Hollow tubes that carry air into the lungs.

PCI—A type of radiotherapy that is used to prevent tumors from growing in the brain.

Radionuclide bone scan—A test that tells if cancer has spread to the bones.

Superior vena cava (SVC) syndrome—A condition seen in lung cancer patients where the tumor presses against a large blood vessel and causes various symptoms.

survival in these patients. However, radiation therapy is used for the palliative (pain relief) treatment of symptoms of metastatic lung cancer, particularly brain and bone tumors.

Patients who have recurrent small cell lung cancer often become resistant to chemotherapy. These patients are treated with palliative radiotherapy. Their doctor may also recommend that they take part in a clinical trial of a new therapy. Patients whose relapse occurs more than six months after their initial treatment, however, may still respond to traditional chemotherapy.

Alternative treatment

Many cancer patients have tried using shark cartilage to treat their disease. Shark cartilage is thought to interfere with the tumor's blood supply. A clinical trial using this treatment in lung cancer patients is ongoing. Information on this and other alternative treatments is available on the Internet from the National Center for Complementary and Alternative Medicine.

Patients who decide to try complementary and alternative therapies should tell their doctor. Some of these therapies may interfere with conventional treatment.

Prognosis

Small cell lung cancer is a very aggressive disease. Without treatment, limited-stage patients will survive for three to six months, while extensive-stage patients will survive six to 12 weeks. However, small cell lung cancer is much more responsive to chemotherapy and radiation therapy than other types of lung cancer. Among patients treated with chemotherapy, 70–90% have a major response to treatment.

Survival in patients responding to therapy is four to five times longer than in patients without treatment. In addition, two years after the start of therapy, about 10% of patients remain free of disease. In general, women tend to have a better prognosis than men. Patients whose disease has spread to the central nervous system or liver have a much worse prognosis. Although the overall survival at five years is 5% to 10%, survival is higher in patients with limited stage disease. About 70% of patients who are disease free after two years do not relapse. After five to 10 disease-free years, relapses are rare.

Prevention

The best way to prevent lung cancer is either not start smoking, or quit smoking. Secondhand smoke from other people's tobacco should also be avoided. Appropriate precautions should be taken when working with substances that can cause cancer (carcinogens). Testing houses for the presence of radon gas, and removing asbestos from buildings have also been suggested as preventive strategies.

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Alliance for Lung Cancer Advocacy, Support, and Education. P.O. Box 849, Vancouver, WA 98666. (800) 298-2436. <<http://www.alcase.org>>.

American Lung Association. 1740 Broadway New York, NY 10019. (212) 315-8700. <<http://www.lungusa.org>>.

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National Center for Complementary and Alternative Medicine (National Institutes of Health). P.O. Box 8218, Silver Spring, MD 20907-8218. (888) 644-6226. <<http://nccam.nih.gov>>.

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Lung diseases due to gas or chemical exposure

Definition

Lung diseases due to gas or chemical exposure are conditions that can be acquired from indoor and outdoor air pollution and from ingesting tobacco smoke.

Description

The lungs are susceptible to many airborne poisons and irritants. Mucus present in the airways blocks foreign particles of a certain size, however it is unable to filter all airborne particulates. There are hundreds of substances that can pollute air and harm lungs. Harmful gases and chemicals are just one type of airborne pollutant that can adversely affect the lungs. They include:

- Vehicle exhaust
- Localized pollutants such as arsenic, asbestos, lead, and mercury
- Outdoor pollutants caused by industry and intensified by weather conditions
- Household heating, such as wood-burning stoves
- Household chemical products
- Tobacco smoke.

Lungs respond to irritants in four ways, each of which can occur separately or, more often, trigger other responses.

- **Asthma** occurs when irritation causes the smooth muscles surrounding the airways to constrict.
- Increased mucus comes from irritated mucus glands lining the airway. Excess mucus clogs the airway and prevents air from circulating.
- Constriction of the lungs results from scarring when the supporting tissues are damaged.
- **Cancer** is caused by certain irritants, such as asbestos and tobacco smoke.

The major categories that airborne irritants fall into are allergic, organic, inorganic, and poisonous, with many agents occupying more than one category.

- Allergic irritants bother only people who are sensitive to them. Cat hair, insect parts, and pollen are common allergens. Chemicals called sulfites, which are widely used as food preservatives, also cause asthma.

- There are many organic dusts that irritate the lungs. Most of them occur on the job and cause occupational lung disease. Grain dust causes silo filler's disease. Cotton and other textile dusts cause **byssinosis**. Mold spores in hay cause farmer's lung.
- Inorganic dusts and aerosolized chemicals also are found mostly on the job. Classic among them are asbestos and coal dust. Many metals (cadmium, arsenic, chromium, and phosphorus), various other fine particles (cement, mica, rock), acid fumes, ammonia, ozone, and automobile and industrial emissions are part of a very long list.
- While tobacco smoke is a culprit in many smokers, a 2003 report found that those who work in the tobacco industry experience higher incidence of lung disease from tobacco dust in their work environment.
- Most intentional poisons (cyanide, nerve gas) that enter through the lungs pass through and damage other parts of the body. Mustard gas, used during World War I and banned since, directly and immediately destroys lungs.
- Tobacco use **scars** the lungs and causes **emphysema** and lung cancer.

Causes and symptoms

Lung disease generates three major symptoms—coughing, **wheezing**, and **shortness of breath**. It also predisposes the lungs to infections such as **bronchitis** and **pneumonia**. Cancer is a late effect, requiring prolonged exposure to an irritant. In the case of tobacco, an average of a pack of cigarettes a day for forty years, or two packs a day for twenty years, will greatly increase the risk of lung cancer.

Diagnosis

A history of exposure combined with a **chest x ray** and lung function studies completes the diagnostic evaluation in most cases. Lung function measures the amount of air breathed in and out, the speed it moves, and the effectiveness of oxygen exchange with the blood. If the cause still is unclear, a **lung biopsy** aids diagnosis.

Treatment

Eliminating the offending irritant and early **antibiotics** for infection are primary. There are many techniques available to remove excess mucus from the lungs. Respiratory therapists are trained in these methods. Finally, there are several machines available to enrich the oxygen content of breathed air.

KEY TERMS

Allergen—A substance that causes an allergic reaction in those who are sensitive to it.

Asthma—Temporary airway narrowing that causes wheezing and shortness of breath due to allergies.

Bronchitis—Infection in the bronchi (breathing tubes).

Pneumonia—Infection or inflammation in the lung itself.

A surgical treatment called lung reduction volume surgery is emerging as a treatment for certain people over age 65 with severe emphysema. It promises substantial return of lung function for selected patients by cutting away diseased parts of the lungs so that healthy tissue functions better. In the fall of 2003, Medicare announced that it would begin paying for the surgery.

Prognosis

Many of these diseases are progressive, because the irritants stay in the lungs forever. Others remain stable after the offensive agents are removed from the environment. Lungs do not heal from destructive damage, but they can clean out infection and excess mucus, and function better.

Prevention

Industrial air filters, adequate ventilation, and respirators in polluted work sites now are mandatory. Tobacco smoke is the world's leading cause of lung disease and many other afflictions. **Smoking** cessation programs are widely available.

Resources

PERIODICALS

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ORGANIZATIONS

American Lung Association. 1740 Broadway, New York, NY 10019. (800) 586-4872. <<http://www.lungusa.org>>.

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Teresa G. Odle

Lung fluke infections see **Fluke infections**

Lung function tests see **Pulmonary function test**

Lung perfusion and ventilation scan

Definition

A lung perfusion scan is a nuclear medicine test that produces a picture of blood flow to the lungs. A lung ventilation scan measures the ability of the lungs to take in air and uses radiopharmaceuticals to produce a picture of how air is distributed in the lungs.

Purpose

Lung perfusion scans and lung ventilation scans are usually performed in the same session. They are done to detect pulmonary embolisms, determine how much blood is flowing to lungs, determine which areas of the lungs are capable of ventilation, and assess how well the lungs are functioning after surgery. These tests are called by different names, including perfusion lung scan, aerosol lung scan, radionucleotide ventilation lung scan, ventilation lung scan, xenon lung scan, ventilation/perfusion scanning (VPS), pulmonary scintiphotography, or, most commonly, V/Q scan.

Precautions

The amount of radioactivity a person is exposed to during these tests is very low and is not harmful. However, if the patient has had other recent radio-nuclear tests, it may be necessary to wait until other radiopharmaceuticals have been cleared from the body so that they do not interfere with these tests.

Description

In a lung perfusion scan, a small amount of the protein labeled with a radioisotope is injected into the patient's hand or arm vein. The patient is positioned under a special camera that can detect radioactive material, and a series of photographs are made of the chest. When these images are projected onto a screen (oscilloscope), they show how the radioactive protein has been distributed by the blood vessels running through the lungs.

KEY TERMS

Pulmonary embolism—A blood clot in the arteries going to the lungs.

In a lung ventilation scan, a mask is placed over the nose and mouth, and the patient is asked to inhale and exhale a combination of air and radioactive gas. Pictures are then taken that show the distribution of the gas in the lungs. Each test takes 15-30 minutes.

Preparation

There is little preparation needed for these tests. The patient may eat and drink normally before the procedure. Tests to check for **pulmonary embolism** are often performed on an emergency basis.

Aftercare

No special aftercare is needed. The patient may resume normal activities immediately.

Risks

There are practically no risks associated with these tests.

Normal results

Normal results in both tests show an even distribution of radioactive material in all parts of the lungs.

Abnormal results

In the lung perfusion scan, an absence of radioactive marker material suggests decreased blood flow to that part of the lung, and possibly a pulmonary **embolism**. However, **pneumonia**, **emphysema**, or lung tumors can create readings on the lung perfusion scan that falsely suggest a pulmonary embolism is present.

In the lung ventilation scan, absence of marker material when the lung perfusion scan for the area is normal suggests lung disease.

Certain combinations of abnormalities in lung perfusion and ventilation scans suggest pulmonary embolism.

Resources

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Tish Davidson, A.M.

Lung surgery

Definition

Lung surgery includes a variety of procedures used to diagnose or treat diseases of the lungs. Biopsies are performed to extract a small amount of tissue for diagnosis, resections remove a portion of lung tissue, and other surgeries are aimed at reducing the volume of the lungs, removing cancerous tumors, or improving lung function.

Purpose

The type of lung surgery performed will depend upon the underlying disease or condition, as well as other factors.

- Pneumonectomy usually refers to the removal of a lung, or sometimes one or more lobes (sections containing lung tissue, air sacs, ducts, and respiratory bronchiole). It is most commonly indicated in certain forms and stages of lung **cancer**.
- Thoracotomy, or surgical incision of the chest wall, is used primarily as a diagnostic tool when other procedures have failed to provide adequate diagnostic information.
- Lobectomy is the term used to describe removal of one lobe of a lung. It is most commonly indicated for lung cancer, but may also be used for **cystic fibrosis** patients if other treatments have failed.
- Other surgical procedures include segmental resection or wedge resection. A resection is the removal of a part of the lung, often in order to remove a tumor. Wedge resection is removal of a wedge-shaped portion of lung tissue.
- Volume reduction surgery is a newer surgery used to help relieve **shortness of breath** and increase tolerance for **exercise** in patients with chronic obstructive pulmonary disease, such as **emphysema**.
- Other surgeries are continuously improved upon to make biopsy less invasive and surgery more effective, such as video-assisted lobectomy. Other purposes for

lung surgery may include severe **abscess**, areas of long-term infection, or permanently enlarged or collapsed lung tissue.

Precautions

Thoracotomy should not be performed on patients whose general health status will not tolerate major surgery. Any surgery carries with it risks associated with **general anesthesia** and possibility of infection. Patients whose risk for these complications outweighs benefit may not be considered candidates for lung surgery. Each individual patient's condition will be reviewed prior to the treatment decision.

Description

Lung surgery procedures will vary depending on the underlying cause of the surgical test or intervention. A patient will be placed under general anesthesia during the surgery. An incision is made to examine the lungs. Diseased tissue is removed and may be sent for biopsy. Following the surgery, drainage tubes may be placed in the chest to drain fluids, blood, and air from the chest cavity. Tubes will most likely remain in place for one to two days, depending on the surgery and the patient's condition. The chest cavity, ribs, and skin are closed and the incision will be sutured. Hospital stay averages from three to 10 days.

Pneumonectomy consists of removal of all of one lung. It may often be indicated only when a lobectomy does not successfully remove the cancerous or damaged tissue. Thoracotomy consists of reaching the lung tissue through incision and obtaining tissue for a biopsy. The biopsy is used to diagnose or stage cancer, and thoracotomy may be avoided until other less invasive methods have failed. Volume reduction surgery involves incision and removal of those parts of the lung or lungs which are the most destroyed, in order to allow for full function of the remaining lung structure. This procedure is still being studied.

Lobectomy is performed in the same general manner as other lung surgeries, but will involve removal of an entire lobe of the lung. Most patients with Stage I or II **non-small cell lung cancer** will receive this treatment for their disease, or a less extensive resection. Lobectomy may only be performed if a wedge or segmental resection is ineffective, but is generally preferred as treatment for primary lung cancer in any patient who can tolerate the procedure. Wedge and segmental resections are still major surgery, but remove less tissue and may be the first choice for some patients, such as those with Stage I and Stage

II non-small cell lung cancer. Patients who do not have enough pulmonary function to undergo a lobectomy will receive a wedge or segmental resection instead. This may lead to a higher recurrence rate of cancer. In general, the surgery method chosen will depend on specific circumstances and consideration of benefit versus risk.

Preparation

Preparation for lung surgery is much like that for any major surgery. Patients will receive instructions from a physician concerning limit of food or water intake prior to the surgery, as well as risks and expected recovery. Patients should continue to follow treatment for the underlying condition, unless instructed otherwise by the physician, and should discuss medications and changes in condition with their physician prior to the surgery.

Aftercare

The chest tube inserted at the end of surgery will remain in place until the lung has fully expanded. Patients will be carefully monitored in the hospital for complications and infection. Deep breathing is recommended to help lessen the risk of **pneumonia** and infection. Breathing exercises will also help expand the lung. After discharge from the hospital, the patient may still receive some **pain** or infection-fighting medications and should recover within one to three months of the operation.

Risks

Risks of lung surgery follows those of any major surgery involving general anesthesia. These risks include reactions to anesthetics or medications, bleeding, infection, and problems restoring breathing. Lung surgery, in particular, offers the risk of pneumonia and **blood clots**. Thoracotomy, as a biopsy procedure, offers greater risk than most biopsy procedures.

Normal results

Outcome for any lung surgery depends on many factors and the severity of disease. In general, the predicted benefits, which justified the surgery, are normal expected results. Thoracotomy results in a definitive diagnosis in more than 90% of patients. Volume reduction surgery has been shown to result in relief of some symptoms and improvement in quality of life for selected patients with severe emphysema and have shown short-term promise.

Mortality from lung surgery improves as procedures move from the more complete pneumonectomy to lobectomy, and the lowest rate for segmental resection.

Resources

ORGANIZATIONS

American Cancer Society. 1599 Clifton Rd., NE, Atlanta, GA 30329-4251. (800) 227-2345. <<http://www.cancer.org>>.

American Lung Association. 1740 Broadway, New York, NY 10019. (800) 586-4872. <<http://www.lungusa.org>>.

National Heart, Lung and Blood Institute. P.O. Box 30105, Bethesda, MD 20824-0105. (301) 251-1222. <<http://www.nhlbi.nih.gov>>.

Teresa Odle

Lung transplantation

Definition

Lung transplantation involves removal of one or both diseased lungs from a patient and the replacement of the lungs with healthy organs from a donor. Lung transplantation may refer to single, double, or even heart-lung transplantation.

Purpose

The purpose of lung transplantation is to replace a lung that no longer functions, or is cancerous, with a healthy lung. In order to qualify for lung transplantation, a patient must suffer from severe lung disease which limits activities of daily living. There should be potential for rehabilitated breathing function. Attempts at other medical treatments should be exhausted before transplantation is considered. Many candidates for this procedure have end-stage fibrotic lung disease, are dependent on **oxygen therapy**, and are likely to die of their disease in 12-18 months.

Patients with **emphysema** or chronic obstructive pulmonary disease (COPD) should be under 60 years of age, have a life expectancy without transplantation of two years or less, progressive deterioration, and emotional stability in order to be considered for lung transplantation. Young patients with end-stage **silicosis** (a progressive lung disease) may be candidates for lung or heart-lung transplantation. Patients with Stage III or Stage IV **sarcoidosis** (a chronic lung

National Transplant Waiting List By Organ Type (June 2000)

Organ Needed	Number Waiting
Kidney	48,349
Liver	15,987
Heart	4,139
Lung	3,695
Kidney-Pancreas	2,437
Pancreas	942
Heart-Lung	212
Intestine	137

disease) with **cor pulmonale** should be considered as early as possible for lung transplantation. Other indicators of lung transplantation include pulmonary vascular disease and chronic pulmonary infection.

Precautions

Patients who have diseases or conditions which may make them more susceptible to organ rejection should not receive a lung transplant. This includes patients who are acutely ill and unstable; who have uncontrolled or untreatable pulmonary infection; significant dysfunction of other organs, particularly the liver, kidney, or central nervous system; and those with significant coronary disease or left ventricular dysfunction. Patients who actively smoke cigarettes or are dependent on drugs or alcohol may not be selected. There are a variety of protocols that are used to determine if a patient will be placed on a transplant recipient list, and criteria may vary depending on location.

Description

Once a patient has been selected as a possible organ recipient, the process of waiting for a donor organ match begins. The donor organ must meet clear requirements for tissue match in order to reduce the chance of organ rejection. It is estimated that it takes an average of one to two years to receive a suitable donor lung, and the wait is made less predictable by the necessity for tissue match. Patients on a recipient list must be available and ready to come to the hospital immediately when a donor match is found, since the life of the lungs outside the body is brief.

Single lung transplantation is performed via a standard thoracotomy (incision in the chest wall) with the patient under **general anesthesia**. Cardiopulmonary bypass (diversion of blood flow from the heart) is not always necessary for a single lung transplant. If bypass

is necessary, it involves re-routing of the blood through tubes to a heart-lung bypass machine. Double lung transplantation involves implanting the lungs as two separate lungs, and cardiopulmonary bypass is usually required. The patient's lung or lungs are removed and the donor lungs are stitched into place. Drainage tubes are inserted into the chest area to help drain fluid, blood, and air out of the chest. They may remain in place for several days. Transplantation requires a long hospital stay and recovery can last up to six months.

Heart-lung transplants always require the use of cardiopulmonary bypass. An incision is made through the middle of the sternum. The heart, lung, and supporting structures are transplanted into the recipient at the same time.

Preparation

In addition to tests and criteria for selection as a candidate for transplantation, patients will be prepared by discussing the procedure, risks, and expected prognosis at length with their doctor. Patients should continue to follow all therapies and medications for treatment of the underlying disease unless otherwise instructed by their physician. Since lung transplantation takes place under general anesthesia, normal surgical and anesthesia preparation should be taken when possible. These include no food or drink from midnight before the surgery, discussion of current medications with the physician, and informing the physician of any changes in condition while on the recipient waiting list.

Aftercare

Careful monitoring will take place in a recovery room immediately following the surgery and in the patient's hospital room. Patients must take immunosuppression, or anti-rejection, drugs to reduce the risk of rejection of the transplanted organ. The body considers the new organ an invader and will fight its presence. The **anti-rejection drugs** lower the body's immune function in order to improve acceptance of the new organs. This also makes the patient more susceptible to infection.

Frequent check-ups with a physician, including x ray and blood tests, will be necessary following surgery, probably for a period of several years.

Risks

Lung transplantation is a complicated and risky procedure, partly because of the organs and systems involved, and also because of the risk of rejection by

KEY TERMS

Pulmonary—Refers to the respiratory system, or breathing function and system.

Sarcoidosis—A chronic disease with unknown cause that involves formation of nodules in bones, skin, lymph nodes, and lungs.

Silicosis—A progressive disease that results in impairment of lung function and is caused by inhalation of dust containing silica.

the recipient's body. Acute rejection most often occurs within the first four months following surgery, but may occur years later. Infection is a substantial risk for organ recipients. An early complication of the surgery can be poor healing of the bronchial and tracheal openings created during the surgery. A late complication and risk is chronic rejection. This can result in inflammation of the bronchial tubes or in late infection from the prolonged use of **immunosuppressant drugs** to fight rejection. Overall, lung transplant recipients have demonstrated average one and two-year survival rates of more than 70%.

Normal results

The outcome of lung transplantation can be measured in survival rates, and also in improved quality of life for recipients. Studies have reported improved quality of life after lung and heart-lung transplants. One study showed that at the two-year follow-up period, 86% of studied recipients reported no limitation to their activity. Demonstration of normal results for patients may include quality of life measurements, as well as testing to ensure lack of infection and rejection.

Resources

ORGANIZATIONS

Children's Organ Transplant Association, Inc. 2501 COTA Drive, Bloomington, IN 47403. (800) 366-2682. <<http://www.cota.org>>.

Second Wind Lung Transplant Association, Inc. 9030 West Lakeview Court, Crystal River, FL 34428. (888) 222-2690. <<http://www.arthouse.com/secondwind>>.

Teresa Odle

Lupus erythematosus see **Systemic lupus erythematosus**

Luque rod see **Spinal instrumentation**

Luteinizing hormone test

Definition

The luteinizing hormone (LH) test is a test of the blood or urine to measure the level of luteinizing hormone (lutropin). This hormone level is highest immediately before a woman ovulates during her menstrual cycle.

Purpose

The LH test is frequently used to determine the timing of ovulation. Couples who are trying to become pregnant may use information about the timing of ovulation to improve their chance of conception. The LH test and other hormone tests may be used during **infertility** screening to chart a woman's menstrual cycle. It may also be used during preparation for **in vitro fertilization**, to determine when eggs are mature and ready to be removed from the ovary.

Description

Luteinizing hormone is a hormone released by the pituitary gland, a small gland at the base of the brain. The hormone stimulates the ovaries to produce and release eggs each month during the menstrual cycle. The level of LH in the blood is highest before ovulation. This increase in hormone level is sometimes called a "surge." A urine or blood sample can be analyzed by a laboratory for the level of LH present. An LH test may be used as part of an infertility screening to determine if there is a hormonal imbalance that might make it difficult to become pregnant. If fertility drugs are given to stimulate ovulation, an LH test can help determine the best time for sexual intercourse. The LH test may also be used to determine when eggs are mature enough to be surgically removed from the ovary as part of the in vitro fertilization process. LH tests may also aid in the diagnoses of polycystic ovary disease, premature ovarian failure, and **menopause**.

A urine LH detection kit is also available for use at home. These are sometimes called "ovulation tests" and are similar to home **pregnancy** test kits. A sample of the woman's first morning urine is tested with the materials provided in the kit. These home tests are often used by women who want to become pregnant. By monitoring levels of LH and watching for the "surge," they can time sexual intercourse to coincide with ovulation, increasing the chance that the egg will be fertilized.

KEY TERMS

Lutropin—Another term for luteinizing hormone, this hormone stimulates the development and release of the egg from the ovary.

Preparation

If a blood sample is taken, the skin around the vein where the needle will be inserted is swabbed with an antiseptic. No special preparation is necessary for collection of a urine sample.

Aftercare

No special aftercare is required. If the blood is tested, as with any blood sampling, the area where the needle was inserted should be kept clean.

Risks

There are no significant risks associated with either the blood or urine test for LH.

Normal results

The level of LH in the blood or urine will vary depending on when the sample was taken during the menstrual cycle. LH levels will be highest around the time of ovulation, about halfway between a woman's menstrual periods. Levels will be lower during the rest of the month. Women who have already experienced menopause will normally have lower LH levels.

Abnormal results

LH levels that remain low throughout the menstrual cycle may indicate a hormonal imbalance that could prevent ovulation. Additional testing may be required if this test is done as part of an infertility screening.

Resources

BOOKS

Zaret, Barry L., et al., editors. *The Patient's Guide to Medical Tests*. Boston: Houghton Mifflin, 1997.

Altha Roberts Edgren

Lyme borreliosis see **Lyme disease**

Lyme disease

Definition

Lyme disease is an infection transmitted by the bite of ticks carrying the spiral-shaped bacterium *Borrelia burgdorferi*. The disease was named for Lyme, Connecticut, the town where it was first diagnosed in 1975 after a puzzling outbreak of arthritis. The organism was named for its discoverer, Willy Burgdorfer. The effects of this disease can be long-term and disabling unless it is recognized and treated properly with **antibiotics**.

Description

Lyme disease, which is also called Lyme borreliosis, is a vector-borne disease. This term means that it is delivered from one host to another. It is also classified as a **zoonosis**, which means that it is a disease of animals that can be transmitted to humans under natural conditions. In this case, a tick bearing the *Borrelia burgdorferi* organism literally inserts it into a host's bloodstream when it bites the host to feed on its blood. It is important to note that neither *Borrelia burgdorferi* nor Lyme disease can be transmitted directly from one person to another, or from pets to humans.

Controversy clouds the true incidence of Lyme disease because no test is definitively diagnostic for the disease, and many of its symptoms mimic those of so many other diseases. Cases of Lyme disease have been reported in 49 of the 50 states; however, 92% of the 17,730 cases reported to the Centers for Disease Control and Prevention (CDC) in 2000 were from only nine states (Connecticut, Rhode Island, New York, Pennsylvania, Delaware, New Jersey, Maryland, Massachusetts, and Wisconsin). The disease is also found in Scandinavia, continental Europe, the countries of the former Soviet Union, Japan, and China; in addition, it is possible that it has spread to Australia.

In the United States, Lyme disease accounts for more than 90% of all reported vector-borne illnesses. It is a significant public health problem and continues to be diagnosed in increasing numbers. The Centers for Disease Control and Prevention (CDC) attributes this increase to the growing size of the deer herd and the geographical spread of infected ticks rather than to improved diagnosis. In addition, some epidemiologists believe that the actual incidence of Lyme disease in the United States may be 5–10 times greater

than that reported by the CDC. The reasons for this difference include the narrowness of the CDC's case definition as well as frequent misdiagnoses of the disease.

The risk for acquiring Lyme disease varies, depending on what stage in its life cycle a tick has reached. A tick passes through three stages of development—larva, nymph, and adult—each of which is dependent on a live host for food. In the United States, *Borrelia burgdorferi* is borne by ticks of several species in the genus *Ixodes*, which usually feed on the white-footed mouse and deer (and are often called deer ticks). In the summer, the larval ticks hatch from eggs laid in the ground and feed by attaching themselves to small animals and birds. At this stage they are not a problem for humans. It is the next stage—the nymph—that causes most cases of Lyme disease. Nymphs are very active from spring through early summer, at the height of outdoor activity for most people. Because they are still quite small (less than 2 mm), they are difficult to spot, giving them ample opportunity to transmit *Borrelia burgdorferi* while feeding. Although far more adult ticks than nymphs carry *Borrelia burgdorferi*, the adult ticks are much larger, more easily noticed, and more likely to be removed before the 24 hours or more of continuous feeding needed to transmit *Borrelia burgdorferi*.

Causes and symptoms

Lyme disease is caused by *Borrelia burgdorferi*. Once *Borrelia burgdorferi* gains entry to the body through a tick bite, it can move through the bloodstream quickly. Only 12 hours after entering the bloodstream, *Borrelia burgdorferi* can be found in cerebrospinal fluid (which means it can affect the nervous system). Treating Lyme disease early and thoroughly is important because Lyme disease can hide for long periods within the body in a clinically latent state. That ability explains why symptoms can recur in cycles and can flare up after months or years, even over decades. It is important to note, however, that not many people who are exposed to *Borrelia burgdorferi* develop the disease.

Lyme disease is usually described in terms of length of infection (time since the person was bitten by a tick infected with Lyme disease) and whether *Borrelia burgdorferi* is localized or disseminated (spread through the body by fluids and cells carrying *Borrelia burgdorferi*). Furthermore, when and how symptoms of Lyme disease appear can vary widely from patient to patient. People who experience



The first sign of Lyme disease is usually an itchy rash around the site of the tick bite. (Science Photo Library. Custom Medical Stock Photo. Reproduced by permission.)

recurrent bouts of symptoms over time are said to have chronic Lyme disease.

Early localized Lyme disease

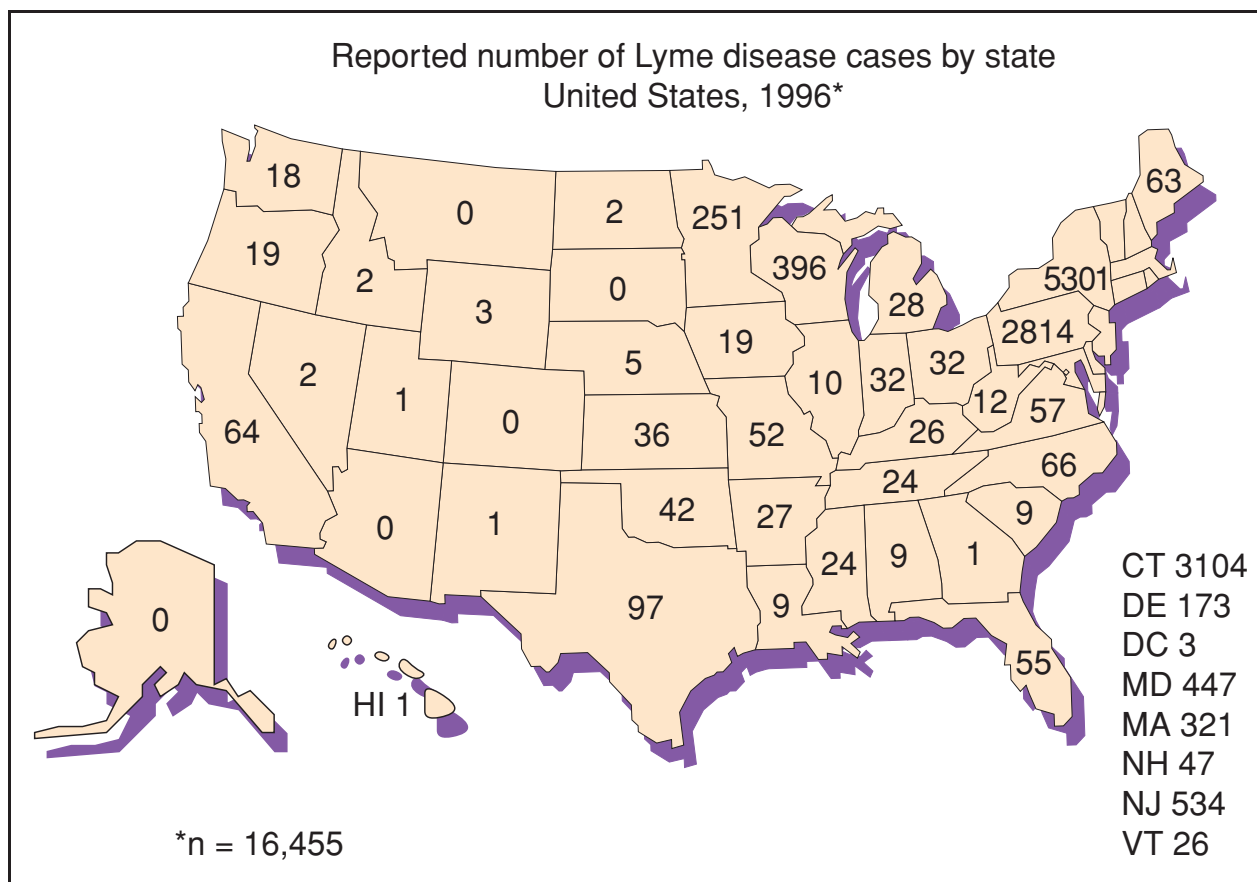
The most recognizable indicator of Lyme disease is a rash around the site of the tick bite. Often, the tick exposure has not been recognized. The eruption might be warm or itch. The rash—erythema migrans (EM)—generally develops within 3-30 days and usually begins as a round, red patch that expands outward. About 75% of patients with Lyme disease develop EM. Clearing may take place from the center out, leaving a bull's-eye effect; in some cases, the center gets redder instead of clearing. The rash may look like a bruise on people with dark skin. Of those who develop Lyme disease, about 50% notice flu-like symptoms, including **fatigue**, **headache**, chills and **fever**, muscle and joint **pain**, and lymph node swelling. However, a rash at the site can also be an allergic reaction to the tick saliva rather than an indicator of Lyme disease, particularly if the rash appears in *less* than three days and disappears only days later.

Late disseminated disease and chronic Lyme disease

Weeks, months, or even years after an untreated tick bite, symptoms can appear in several forms, including:

- Fatigue, forgetfulness, confusion, mood swings, irritability, numbness.
- Neurologic problems, such as pain (unexplained and not triggered by an injury), Bell's palsy (facial **paralysis**, usually one-sided but may be on both sides), and a mimicking of the inflammation of brain membranes known as **meningitis**; (fever, severe headache).
- Arthritis (short episodes of pain and swelling in joints) and other musculoskeletal complaints. Arthritis eventually develops in about 60% of patients with untreated Lyme disease.

Less common effects of Lyme disease are heart abnormalities (such as irregular rhythm or cardiac block) and eye abnormalities (such as swelling of the cornea, tissue, or eye muscles and nerves).



Lyme disease accounts for more than 90% of all reported vector-borne illnesses in the United States. It is caused by an infection transmitted by the bite of ticks carrying the *Borrelia burgdorferi* bacterium. (Data taken from the Centers for Disease Control. Illustration by Electronic Illustrators Group.)

Diagnosis

A clear diagnosis of Lyme disease can be difficult, and relies on information the patient provides and the doctor's clinical judgment, particularly through elimination of other possible causes of the symptoms. Lyme disease may mimic other conditions, including **chronic fatigue syndrome (CFS)**, **multiple sclerosis (MS)**, and other diseases with many symptoms involving multiple body systems. Differential diagnosis (distinguishing Lyme disease from other diseases) is based on clinical evaluation with laboratory tests used for clarification when necessary. A two-test approach is common to confirm the results. Because of the potential for misleading results (false-positive and false-negative), laboratory tests alone cannot establish the diagnosis.

In February 1999 the Food and Drug Administration (FDA) approved a new blood test for Lyme disease called PreVue. The test, which searches for antigens (substances that stimulate the production

of antibodies) produced by *Borrelia burgdorferi*, gives results within one hour in the doctor's office. A positive result from the PreVue test is confirmed by a second blood test known as the Western blot, which must be done in a laboratory.

Doctors generally know which disease-causing organisms are common in their geographic area. The most helpful piece of information is whether a tick bite or rash was noticed and whether it happened locally or while traveling. Doctors may not consider Lyme disease if it is rare locally, but will take it into account if a patient mentions vacationing in an area where the disease is commonly found.

Treatment

The treatment for Lyme disease is antibiotic therapy; however, overprescribing of antibiotics can lead to serious problems, so the decision to treat must be made with care. Disease organisms can develop

resistance to families of medications over time, rendering the drugs useless. Furthermore, testing and treatments can be expensive. If a patient has strong indications of Lyme disease (symptoms and medical history), the doctor will probably begin treatment on the presumption of this disease. The American College of Physicians recommends treatment for a patient with a rash resembling EM or who has arthritis, a history of an EM-type rash, and a previous tick bite.

The benefits of treating early must be weighed against the risks of over treatment. The longer a patient is ill with Lyme disease before treatment, the longer the course of therapy must be, and the more aggressive the treatment. The development of opportunistic organisms may produce other symptoms. For example, after long-term antibiotic therapy, patients can become more susceptible to yeast infections. Treatment may also be associated with adverse drug reactions.

For most patients, oral antibiotics (doxycycline or amoxicillin) are prescribed for 21 days. When symptoms indicate nervous system involvement or a severe episode of Lyme disease, intravenous antibiotic (ceftriaxone) may be given for 14-30 days. Some physicians consider intravenous ceftriaxone the best therapy for any late manifestation of disease, but treatments for late Lyme disease are still controversial as of 2003. **Corticosteroids** (oral) may be prescribed if eye abnormalities occur, but they should not be used without first consulting an eye doctor.

The doctor may have to adjust the treatment regimen or change medications based on the patient's response. Treatment can be difficult because *Borrelia burgdorferi* comes in several strains (some may react to different antibiotics than others) and may even have the ability to switch forms during the course of infection. Also, *Borrelia burgdorferi* can shut itself up in cell niches, allowing it to hide from antibiotics. Finally, antibiotics can kill *Borrelia burgdorferi* only while it is active rather than dormant.

Alternative treatment

Supportive therapies may minimize symptoms of LD or improve the immune response. These include vitamin and **nutritional supplements**, mostly for chronic fatigue and increased susceptibility to infection. For example, yogurt and *Lactobacillus acidophilus* preparations help fight yeast infections, which are common in people on long-term antibiotic therapy. In addition, botanical medicine and homeopathy can be considered to help bring the body's systems back to a state of health and well being. A Western herb,

spilanthes (*Spilanthes* spp.), may be effective in treating diseases like LD that are caused by spirochetes (spiral-shaped bacteria).

Prognosis

If aggressive antibiotic therapy is given early, and the patient cooperates fully and sticks to the medication schedule, recovery should be complete. Only a small percentage of Lyme disease patients fail to respond or relapse (have recurring episodes). Most long-term effects of the disease result when diagnosis and treatment is delayed or missed. Co-infection with other infectious organisms spread by ticks in the same areas as *Borrelia burgdorferi* (**babesiosis** and **ehrlichiosis**, for instance) may be responsible for treatment failures or more severe symptoms. Most fatalities reported with Lyme disease involved patients coinfecting with babesiosis.

Prevention

Update on vaccination

A vaccine for Lyme disease known as LYMERix was available from 1998 to 2002, when it was removed from the United States market. The decision was influenced by reports that LYMERix may be responsible for neurologic complications in vaccinated patients. Researchers from Cornell-New York Hospital presented a paper at the annual meeting of the American Neurological Association in October 2002 that identified nine patients with neuropathies linked to **vaccination** with LYMERix. In April 2003, the National Institute of Allergy and Infectious Diseases (NIAID) awarded a federal grant to researchers at Yale University School of Medicine to develop a new vaccine against Lyme disease. As of late 2003, the best prevention strategy is through minimizing risk of exposure to ticks and using personal protection precautions.

Minimizing risk of exposure

Precautions to avoid contact with ticks include moving leaves and brush away from living quarters. Most important are personal protection techniques when outdoors, such as:

- Spraying tick repellent on clothing and exposed skin.
- Wearing light-colored clothing to maximize ability to see ticks.
- Tucking pant legs into socks or boot top.
- Checking children and pets frequently for ticks.

KEY TERMS

Babesiosis—A disease caused by protozoa of the genus *Babesia* characterized by a malaria-like fever, anemia, vomiting, muscle pain, and enlargement of the spleen. Babesiosis, like Lyme disease, is carried by a tick.

Bell's palsy—Facial paralysis or weakness with a sudden onset, caused by swelling or inflammation of the seventh cranial nerve, which controls the facial muscles. Disseminated Lyme disease sometimes causes Bell's palsy.

Blood-brain barrier—A blockade of cells separating the circulating blood from elements of the central nervous system (CNS); it acts as a filter, preventing many substances from entering the central nervous system.

Cerebrospinal fluid—Clear fluid found around the brain and spinal cord and in the ventricles of the brain.

Disseminated—Scattered or distributed throughout the body. Lyme disease that has progressed beyond the stage of localized EM is said to be disseminated.

Erythema migrans (EM)—A red skin rash that is one of the first signs of Lyme disease in about 75% of patients.

Lyme borreliosis—Another name for Lyme disease.

Spirochete—A spiral-shaped bacterium. The bacteria that cause Lyme disease and syphilis, for example, are spirochetes.

Vector—An animal carrier that transfers an infectious organism from one host to another. The vector that transmits Lyme disease from wildlife to humans is the deer tick or black-legged tick.

Zoonosis (plural, zoonoses)—Any disease of animals that can be transmitted to humans under natural conditions. Lyme disease and babesiosis are examples of zoonoses.

In highly tick-populated areas, each individual should be inspected at the end of the day to look for ticks.

Minimizing risk of disease

The two most important factors are removing the tick quickly and carefully, and seeking a doctor's evaluation at the first sign of symptoms of Lyme disease. When in an area that may be tick-populated:

- Check for ticks, particularly in the area of the groin, underarm, behind ears, and on the scalp.
- Stay calm and grasp the tick as near to the skin as possible, using a tweezer.
- To minimize the risk of squeezing more bacteria into the bite, pull straight back steadily and slowly.
- Do not try to remove the tick by using petroleum jelly, alcohol, or a lit match.
- Place the tick in a closed container (for species identification later, should symptoms develop) or dispose of it by flushing.
- See a physician for any sort of rash or patchy discoloration that appears three to 30 days after a tick bite.

Resources

BOOKS

Beers, Mark H., MD, and Robert Berkow, MD., editors. "Bacterial Diseases Caused by Spirochetes: Lyme Disease (Lyme Borreliosis)." In *The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck Research Laboratories, 2004.

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ORGANIZATIONS

- Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.
- Lyme Disease Foundation. One Financial Plaza, Hartford, CT, 06103. (800) 886-LYME. <<http://www.lyme.org>>.
- Lyme Disease Network of NJ, Inc. 43 Winton Road, East Brunswick, NJ 08816. <<http://www.lymenet.org>>.

National Institute of Allergy and Infectious Diseases (NIAID). 31 Center Drive, Room 7A50 MSC 2520, Bethesda, MD, 20892. (301) 496-5717. <<http://www.niaid.nih.gov>>.

OTHER

Centers for Disease Control and Prevention, Division of Vector-Borne Infectious Diseases. *CDC Lyme Disease Home Page*. <<http://www.cdc.gov/ncidod/dvbid/lyme/>>.

National Institute of Neurological Disorders and Stroke (NINDS) Fact Sheet. *Bell's Palsy*. Bethesda, MD: NINDS, 2003.

NINDS Information Page. *Neurological Complications of Lyme Disease*. Bethesda, MD: NINDS, 2003.

Rebecca J. Frey, PhD

Lymph node angiogram see
Lymphangiography

Lymph node biopsy

Definition

A lymph node biopsy is a procedure in which all or part of a lymph node is removed and examined to determine if there is **cancer** within the node.

Purpose

The lymph system is the body's primary defense against infection. It consists of the spleen, tonsils, thymus, lymph nodes, lymph vessels, and the clear, slightly yellow fluid called lymph. These components produce and transport white blood cells called lymphocytes and macrophages that rid the body of infection. The lymph system is also involved in the production of antibodies. Antibodies are proteins that fight bacteria, viruses, and other foreign materials that enter the body.

The lymph vessels are similar to veins, only instead of carrying blood as veins do, they circulate lymph to most tissues in the body. Lymph nodes are about 600 small, bean-shaped collections of tissue found along the lymph vessel. They produce cells and proteins that fight infection, and clean and filter lymph. Lymph nodes are sometimes called lymph glands, although they are not true glands. When someone talks about having swollen glands, they are actually referring to lymph nodes.

Normal lymph glands are no larger than 0.5 in (1.3 cm) in diameter and are difficult to feel. However, lymph nodes can enlarge to greater than 2.5 in (6 cm) and can become sore. Most often the swelling is caused by an infection, but it can also be caused by cancer.

Cancers can metastasize (spread) through the lymph system from the site of the original tumor to distant parts of the body where secondary tumors are formed. The purpose of a lymph node biopsy is to determine the cause of the swelling and/or to see if cancer has begun to spread through the lymph system. This information is important in staging the cancer and devising a treatment plan.

Precautions

Women who are pregnant should inform their doctor before a lymph node biopsy, although **pregnancy** will not affect the results.

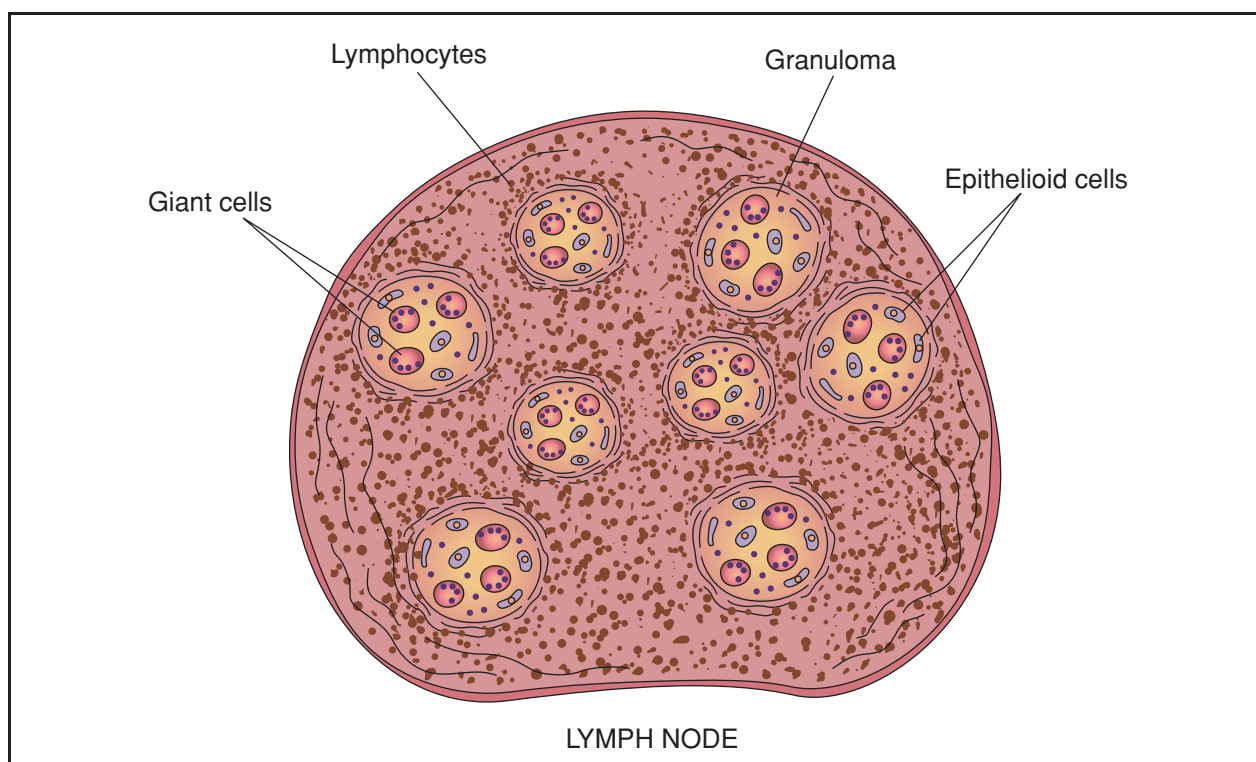
Description

There are three kinds of lymph node biopsy. Sentinel lymph node mapping and biopsy is a promising new technique that is discussed in its own entry. Fine needle aspiration (FNA) biopsy, often just called needle biopsy, is done when the lymph node of interest is near the surface of the body. A hematologist (a doctor who specializes in blood diseases) usually performs the test. In FNA biopsy, a needle is inserted through the skin and into the lymph node, and a sample of tissue is drawn out of the node. This material is preserved and sent to the laboratory for examination.

Advantages of a needle biopsy are that the test is minimally invasive. Only a local anesthetic is used, the procedure generally takes less than half an hour, and there is little **pain** afterwards. The disadvantage is that cancer may not be detected in the small sample of cells removed by the needle.

Open lymph node biopsy is a surgical procedure. It is done by a surgeon under **general anesthesia** on lymph nodes in the interior of the body and under **local anesthesia** on surface lymph nodes where FNA biopsy is considered inadequate. Once there is adequate anesthesia, the surgeon makes a small cut and removes either the entire lymph node or a slice of tissue that is then sent to the laboratory for examination. Results in both kinds of biopsies take one to three days.

Open biopsy can be advantageous in that it is easier to detect and identify the type of cancer in a large piece of tissue. Also, lymph nodes deep in the body can be sampled. Disadvantages include a longer recovery time,



Lymph node biopsy is a procedure in which a sample of lymph node tissue is removed for laboratory analysis. It is generally performed on an outpatient basis. (Illustration by Electronic Illustrators Group.)

soreness at the biopsy site for several days, and the use of deeper anesthesia, increasing the risks to the patient. The procedure is done in a hospital or outpatient surgery center and takes about an hour, with additional time to recover from general anesthesia.

Preparation

No particular preparation is necessary for a needle biopsy. For an open biopsy, patients need standard pre-operative blood tests and other tests to evaluate general health. The doctor should be informed about any medications (prescription, non-prescription, or herbal) the patient is taking, as well as past bleeding problems or **allergies** to medication or anesthesia.

Aftercare

Little aftercare is needed in a needle biopsy other than a bandage to keep the biopsy site clean. Patients who have general anesthesia for an open biopsy often feel drowsy and tired for several days following the procedure, and should not plan to drive home after biopsy. The incision site must be kept clean and dry, and a follow-up visit to check on healing is usually necessary.

Risks

There are few risks associated with lymph node biopsy. The main risks are excessive bleeding (usually only in people with blood disorders) and allergic reaction to general anesthesia (rare). Occasionally the biopsy site becomes infected.

Normal results

Normal lymph nodes are small and flat. When examined under the microscope, they show no signs of cancer or infection.

Abnormal results

Abnormal lymph nodes are usually enlarged and contain cancerous (malignant) cells and/or show signs of infection.

Resources

ORGANIZATIONS

American Cancer Society. 1599 Clifton Road NE, Atlanta, GA 30329. 800(ACS)-2345). < <http://www.cancer.org> >.

KEY TERMS

Lymph nodes—Small, bean-shaped organs located throughout the lymphatic system. The lymph nodes store special cells that can trap cancer cells or bacteria that are traveling through the body in lymph. Also called lymph glands.

Lymphocytes—Small white blood cells that bear the major responsibility for carrying out the activities of the immune system; they number about 1 trillion.

Malignant—Cancerous. Cells tend to reproduce without normal controls on growth and form tumors or invade other tissues.

Spleen—An organ located at the left side of the stomach that acts as a reservoir for blood cells and produces lymphocytes and other products involved in fighting infection.

Thymus—An organ near the base of the neck that produces cells that fight infection. It is at its largest at puberty, then declines in size and function during adult life.

Tonsils—Small masses of tissue at the back of the throat.

Cancer Information Service. National Cancer Institute. Building 31, Room 10A19, 9000 Rockville Pike, Bethesda, MD 20892. (800)4-CANCER. <<http://www.nci.nih.gov/cancerinfo>>.

OTHER

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Tish Davidson, A.M.

Lymphadenitis

Definition

Lymphadenitis is the inflammation of a lymph node. It is often a complication of a bacterial infection of a wound, although it can also be caused by viruses or other disease agents. Lymphadenitis may be either generalized, involving a number of lymph nodes; or limited to a few nodes in the area of a localized infection. Lymphadenitis is sometimes accompanied by lymphangitis, which is the inflammation of the lymphatic vessels that connect the lymph nodes.

Description

Lymphadenitis is marked by swollen lymph nodes that are painful, in most cases, when the doctor touches them. If the lymphadenitis is related to an infected wound, the skin over the nodes may be red and warm to the touch. If the lymphatic vessels are also infected, there will be red streaks extending from the wound in the direction of the lymph nodes. In most cases, the infectious organisms are hemolytic *Streptococci* or *Staphylococci*. Hemolytic means that the bacteria produce a toxin that destroys red blood cells.

The extensive network of lymphatic vessels throughout the body and their relation to the lymph nodes helps to explain why bacterial infection of the nodes can spread rapidly to or from other parts of the body. Lymphadenitis in children often occurs in the neck area because these lymph nodes are close to the ears and throat, which are frequent locations of bacterial infections in children.

Causes and symptoms

Streptococcal and staphylococcal bacteria are the most common causes of lymphadenitis, although viruses, protozoa, rickettsiae, fungi, and the **tuberculosis** bacillus can also infect the lymph nodes. Diseases or disorders that involve lymph nodes in specific areas of the body include rabbit **fever (tularemia)**, **cat-scratch disease**, **lymphogranuloma venereum**, **chan-croid**, **genital herpes**, infected **acne**, dental abscesses, and bubonic **plague**. In children, **tonsillitis** or bacterial sore throats are the most common causes of lymphadenitis in the neck area. Diseases that involve lymph nodes throughout the body include mononucleosis, **cytomegalovirus infection**, **toxoplasmosis**, and **brucellosis**.

The early symptoms of lymphadenitis are swelling of the nodes caused by a buildup of tissue fluid and an increased number of white blood cells resulting from the body's response to the infection. Further developments include fever, often as high as 101-102°F (38-39°C) together with chills, loss of appetite, heavy perspiration, a rapid pulse, and general weakness.

Diagnosis

Physical examination

The diagnosis of lymphadenitis is usually based on a combination of the patient's history, the external symptoms, and laboratory cultures. The doctor will press (palpate) the affected lymph nodes to see if they



Swollen lymph node glands in a young girl's neck. (Custom Medical Stock Photo. Reproduced by permission.)

are sore or tender. Swollen nodes without soreness are often caused by cat-scratch disease. In children, the doctor will need to rule out **mumps**, tumors in the neck region, and congenital cysts that resemble swollen lymph nodes.

Although lymphadenitis is usually diagnosed in lymph nodes in the neck, arms, or legs, it can also occur in lymph nodes in the chest or abdomen. If the patient has acutely swollen lymph nodes in the groin, the doctor will need to rule out a **hernia** in the groin that has failed to reduce (incarcerated inguinal hernia). Hernias occur in 1% of the general population; 85% of patients with hernias are male.

Laboratory tests

The most significant tests are a white blood cell count (WBC) and a **blood culture** to identify the organism. A high proportion of immature white blood cells indicates a bacterial infection. Blood cultures may be positive, most often for a species of *staphylococcus* or *streptococcus*. In some cases, the doctor may order a biopsy of the lymph node.

Treatment

Medications

The medications given for lymphadenitis vary according to the bacterium or virus that is causing it. If the patient also has lymphangitis, he or she will be

KEY TERMS

Hemolytic—Able to break down or dissolve red blood cells. The bacteria that cause lymphadenitis are hemolytic.

Hernia—The bulging of a part of the intestine or other organ through its surrounding wall of tissue. Most hernias are in the abdominal cavity. An inguinal hernia is located in the groin area.

Lymph nodes—The glandlike masses of tissue in the lymphatic system that contain lymphocytes. The lymph nodes also filter lymph, which is a clear yellowish tissue fluid that carries lymphocytes and fats throughout the body.

Lymphangitis—Inflammation of the lymphatic vessels. It often occurs together with lymphadenitis.

Septicemia—The presence of bacteria and their toxins in the bloodstream. Septicemia is sometimes called blood poisoning.

Staphylococcus—Any of several species of spherical bacteria that occur in groups of four or irregular clusters. *Staphylococci* frequently cause skin infections.

Streptococcus—Any of several species of bacteria that are spherical in shape and form pairs or chains. *Streptococci* cause scarlet fever, tonsillitis, and pneumonia, and are often involved in lymphadenitis.

treated with **antibiotics**, usually penicillin G (Pfizerpen, Pentids), nafcillin (Nafcil, Unipen), or **cephalosporins**. Erythromycin (Eryc, E-Mycin, Erythrocin) is given to patients who are allergic to penicillin.

Supportive care

Supportive care of lymphadenitis includes resting the affected limb and treating the area with hot moist compresses.

Surgery

Cellulitis associated with lymphadenitis should *not* be treated surgically because of the risk of spreading the infection. Pus is drained only if there is an **abscess** and usually after the patient has been started on antibiotic treatment. In some cases, a biopsy of an inflamed lymph node is necessary if no diagnosis has been made and no response to treatment has occurred.

Prognosis

The prognosis for recovery is good if the patient is treated promptly with antibiotics. In most cases, the infection can be brought under control in three or four days. Patients with untreated lymphadenitis may develop blood **poisoning** (septicemia), which is sometimes fatal.

Prevention

Prevention of lymphadenitis depends on prompt treatment of bacterial and viral infections.

Resources

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Rebecca J. Frey, PhD

Lymphangiography

Definition

Lymphangiography, or lymph node angiogram, is a test which utilizes x-ray technology, along with the injection of a contrast agent, to view lymphatic circulation and lymph nodes for diagnostic purposes.

Purpose

The lymphatic system is a one way circulation that channels tissue fluid back into the heart. The watery fluid called lymph seeps out of the blood into tissues, and while journeying back to the heart, it picks up germs, **cancer** cells, and some waste products. Lymph passes through the lymph nodes, which are major arsenals of immune defense that attack germs carried in the lymph. Cancer cells are also subject to attack in lymph nodes.

Cancers of the lymph system, such as **Hodgkin's disease** and non-Hodgkin's lymphomas, spread throughout the body. Treatment often depends upon finding all the disease and directing radiation to each location. Planning other kinds of treatment, such as surgery or **chemotherapy**, may also require that the full extent of the disease be known.

The lymphatic circulation may become clogged by infection, injury, or several other types of cancer that

KEY TERMS

Contrast agent—A substance that makes shadows on x rays.

Filariasis—A tropical disease caused by worms that live in lymph channels.

Hodgkin's disease—A cancer of the lymphatic system.

Lymphoma—A type of lymphatic cancer.

have spread through lymphatic channels. Swelling, sometimes massive, can result from blocked lymphatics. The most outstanding example of this is the tropical disease **filariasis**, which results in the swelling of the legs termed elephantiasis.

Lymphangiography gives precise information on the extent and location of lymph vessels and lymph nodes. Oftentimes, it is performed to evaluate the extent of a lymphatic cancer. Rarely, it is a tool, which aids surgeons attempting to reconstruct the lymphatics.

Precautions

Lymphangiography should not be performed on patients with dye or shellfish **allergies** or on patients with chronic lung disease, **kidney disease**, heart disease, or **liver disease**.

Description

A lymphangiogram begins by injecting a blue dye into a hand or foot. The lymph system picks up dye, which in turn will highlight the lymph vessels. This process may take a full day. When the lymphatic channel is clearly visible, the radiologist will insert an even tinier needle into that vessel and inject a contrast agent. X rays outline the journey of the contrast agent as it travels to the heart through lymph vessels and nodes.

Preparation

Unless a dye allergy is suspected, no special preparation is need. If an allergy is suspected, a non-ionic contrast agent can be administered instead.

Aftercare

Prior to suture removal seven to 10 days after the procedure, the patient should watch for any sign of infection around the site.

Risks

Lipid **pneumonia** can occur if the contrast agent penetrates the thoracic duct. An allergic reaction to the contrast agent is possible, causing a range of symptoms that can range from innocuous to life threatening.

Resources

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J. Ricker Polsdorfer, MD

Lymphedema

Description

Lymphedema involves blockage of the lymph vessels, with a resulting accumulation of lymphatic fluid in the interstitial tissues of the body. The lymphatic system consists of lymph vessels and lymph nodes throughout the body. The lymph vessels collect lymphatic fluid, which consists of protein, water, fats, and wastes from cells. The lymph vessels transport the fluid to the lymph nodes, where waste materials and foreign materials are filtered out from the fluid. The fluid is then returned to the blood. When the vessels are damaged or missing, the lymph fluid cannot move freely throughout the system but accumulates. This accumulation of fluid results in abnormal swelling of the arm(s) or leg(s), and occasionally swelling in other parts of the body.

Lymphedema is a very serious condition. There is no cure for lymphedema and once it develops, it can be a long-term, uncomfortable, and sometimes painful condition requiring daily treatment. When lymphedema is not treated, the protein-rich fluid continues to accumulate, leading to even more swelling and hardening (referred to as fibrosis) of the tissues. This fluid is a good culture medium for bacteria, thus resulting in reoccurring infections when there are injuries to the skin, decrease or loss of functioning of the affected limbs, and skin breakdown. Infections, referred to as lymphangitis, can affect the connective tissue under the skin. Repeated infections may result in scarring, which in turn makes the tissue susceptible to more swelling and infection. Over time, these infections result in tissue hardening (i.e., fibrosis), which is

a characteristic of advanced chronic lymphedema. In very severe cases, untreated lymphedema may even result in a rare form of lymphatic **cancer** called lymphangiosarcoma.

Lymphedema affects approximately 100 million people worldwide, including at least 3 million people in the United States.

Symptoms of lymphedema include:

- swelling of an affected limb, which may develop gradually or suddenly
- tightness of the skin and a feeling of heaviness in the affected area
- discomfort or a feeling of "pins and needles" in the affected area
- pitting **edema**, which can be identified by observing a temporary indentation in the swollen area when pressure is placed on the affected area
- aching in the adjacent shoulder or hip due to the increasing weight of the swelling limb
- tight fitting of a ring, wristwatch, or bracelet, without a gain in weight.

In 90% of the cases, lymphedema is diagnosed through observations, measurements, and symptoms. The remaining 10% require the use of more complex diagnostic tests such as lymphoscintigraphy. Lymphoscintigraphy is a technique in which a radioactive substance that concentrates in the lymphatic vessels is injected into the affected tissue and is mapped using a gamma camera, which images the location of the radioactive tracer. **Magnetic resonance imaging (MRI)**, computed tomography (CT) scanning, and duplex ultrasound are imaging techniques that are also sometimes used as diagnostic tools for lymphedema.

There are three stages of lymphedema:

- Stage 1 (spontaneously reversible) - tissue is still at the pitting stage and soft to the touch. Upon waking in the morning, the limbs or affected areas are of normal or almost normal size
- Stage 2 (spontaneously irreversible) - tissue is non-pitting and no longer soft to the touch, fibrosis begins to form, and the limbs increase in size
- Stage 3 (lymphostatic **elephantiasis**) - swelling is irreversible and the affected areas are very swollen. The skin hardens and begins to break down, fibrosis is more extensive, and patients may need surgery to remove some of the swollen tissues.

Causes

Primary lymphedema is an inherited condition, where the patient is born without lymph vessels and nodes. The swelling associated with primary lymphedema usually occurs during adolescence and affects the foot or calf. A rare form of primary lymphedema, called Milroy's Disease, occurs in **pregnancy**. However, secondary lymphedema, or acquired lymphedema, develops as a result of an injury to the lymph system. Specific causes include surgical treatments for certain types of cancers, especially those cancers that currently require the removal of lymph nodes. Radiation treatment for cancer or for some AIDS-related diseases such as Kaposi-Sarcoma may also result in lymphedema, as radiation may damage or destroy lymph nodes or cause the formation of scar tissue that can interrupt the normal flow of the lymphatic fluid. Specific cancers and their treatment that may result in lymphedema include **malignant melanoma**, breast (in both women and men), gynecological, head and neck, prostate, testicular, bladder, and **colon cancer**. Other causes of lymphedema include trauma to the lymphatic system from **burns**, **liposuction**, tattooing, injuries, surgery, radiation, **obesity**, heart or circulatory disease, and **multiple sclerosis**. Lymphedema in people at risk may not develop the condition immediately, but develop the condition weeks, months, or even years later. Aircraft travel has been linked to the development of lymphedema in patients after cancer surgery, possible due to the decreased cabin pressure.

In Western countries, one of the most common causes of lymphedema is **mastectomy** with axillary dissection (removal of the breast and underarm lymph tissue for treatment of **breast cancer**), which may result in lymphedema of the breast, underarm, or arm on the side of the surgery in 10-20% of patients. This occurs because the lymphatic drainage of the arm passes through the axilla (armpit), and tissue in the axilla is removed during the mastectomy. To reduce the risk of developing lymphedema after breast cancer treatment, there is an alternative treatment that avoids axillary lymph node dissection. Sentinel **lymph node biopsy** is a new diagnostic procedure used to determine whether the breast cancer has spread (metastasized) to axillary lymph nodes. A sentinel lymph node biopsy requires the removal of only one to three lymph nodes for close review by a pathologist. If the sentinel nodes do not contain tumor (cancer) cells, this may eliminate the need to remove additional lymph nodes in the axillary area. Early research on this technique indicates that sentinel lymph node biopsy may be associated with less

pain and fewer complications than standard axillary dissection. Because the procedure is so new, long term data are not yet available. However, there is still a risk for developing lymphedema because of follow-up radiation treatments or **chemotherapy**, which may also damage the lymph nodes.

Persons who have developed lymphedema after cancer treatment should be checked for a possible reoccurrence of cancer if they experience a sudden increase of swelling, for the tumor growth may be responsible for blocking lymphatic flow.

Treatments

Lymphedema is a chronic condition that cannot be cured, but it can be improved with treatment. There are several major components of a lymphedema treatment program, which should be administered by the health care provider in cooperation with a physical therapist trained in lymphedema treatment. Complete Decongestive Therapy (CDT; also referred to as Complete Decongestive Therapy (CDT) or as Complete Decongestive Physiotherapy (CDP)) combines manual lymph drainage (MLD) with compression techniques and with patient education on self-care needs. The goals of the treatment program are to:

- remove the stagnant lymph fluids out of the tissues
- reduce and help control swelling
- soften fibrotic tissue
- improve the overall health of the patient.

However, some lymphedema specialists feel that lymphedema patients with metastatic cancer should not be treated with CDT, to prevent the spreading of the cancer.

MLD was developed in 1932 in Denmark by a doctor and his wife. It was widely used in Europe and now is accepted as a therapy for lymphedema patients in the United States. In MLD a series of rhythmic, light strokes are made in a specific sequence along the lymphatic vessels and the adjoining tissues. These movements remove the lymph fluids from the tissues and return them to the circulatory system, thus reducing swelling in the affected area.

Compression techniques include the use of compression garments, compression aids, and compression bandages. These techniques encourage natural drainage and prevent swelling by supporting tissues in a way that aids in drainage. Compression garments are knit, stretch sleeves or stockings. Compression aids are custom-fitted sleeves, stockings, or pads made of fabric-covered foam. Bandages are an

effective and flexible means of compression. They work when the patient is active or is resting and can easily be adjusted to fit changing limb sizes. However, the bandage should be a special type of short-stretch bandage and not the long-stretch bandage that is commonly known as Ace bandages. Only persons who are trained in lymphedema therapy should tape or wrap swollen areas.

Self-care techniques are practiced by the patient or his or her caregiver at home, between visits to the therapist. Self-care techniques include self-massage, skin care to maintain healthy tissue, nutritious diet, and **exercise** to increase lymph flow, increase mobility, and to improve the patient's general health.

Exposure to extreme heat has the potential to increase lymphedema swelling, so an affected person or a person at risk of developing lymphedema should avoid hot tubs, saunas, and steam rooms.

To keep the affected extremities as healthy as possible, a person with lymphedema should keep the swollen areas clean and avoid heavy lifting and pulling as well as avoid any type of trauma, such as cuts, **bruises**, **sunburn** or other burns, injections, **sports injuries**, insect bites, or cat scratches. Some doctors and lymphedema therapists recommend that a person with lymphedema use a preventative course of **antibiotics** when having dental treatment, that is, starting antibiotics several days before the appointment and continuing several days afterwards. A person at risk of developing lymphedema (for example, a woman who has been treated for breast cancer) should also observe the same type of precautions to prevent the development of the condition.

If infections occur, then all treatments for lymphedema should be discontinued while the infection is present, and the infection treated with antibiotics.

Surgery is sometimes used to remove excess tissue ("debulking") if the swollen limb becomes so large and heavy as to interfere with movement.

Exercise is important for a person with lymphedema, but only in moderation. If the extremity starts to ache, the person should lie down and elevate the swollen limb. Recommended exercises include walking, swimming, light aerobics, bike riding, and **yoga**.

Persons with lymphedema should wear a lymphedema alert bracelet or necklace for safety during a medical emergency, explaining the risk of infections. They may also benefit from counseling and membership in support groups to deal with the psychological impact of the disease. Sometimes patients with lymphedema will be denied insurance coverage for

treatment; as a result patient advocacy groups in 2005 are attempting to get a law passed through the U.S. Congress guaranteeing insurance coverage for lymphedema.

Alternative and complementary therapies

The use of clinical **aromatherapy** in conjunction with CDT may improve the quality of life for persons with lymphedema. Clinical aromatherapy involves the use of essential oils to improve the functioning of the immune system, for the immune system is closely associated with the lymphatic system. Also a massage oil comprised of a blend of frankincense, grapefruit, hyssop, and lavender, may be used to soften scarred and fibrotic tissues. Radiation treatments can cause skin **contractures**, which can be helped by massage with a blend of cajeput, frankincense, hyssop, lavender, sage, and tea tree. Radiation can also have adverse effects on the bowel, resulting in poor bowel functioning, scarring, and activity restrictions. Massaging the abdomen with a blend of grapefruit, fennel, helichrysum, lavender, myrrh, and sage may improve intestinal functions. When compression techniques are used, the underlying skin can be treated with a blend of bay laurel, chamomile, geranium, helichrysum, lavender, patchouli, and vetiver in a combination of castor oil, safflower oil, and grapeseed oil as carrier oils. Good skin care is important in preventing infections. Body oils that contain cajeput, cypress, lavender, marjoram, and rosewood can be applied after bathing to keep the skin moist and healthy. Finger nail beds can be a portal of entry for infections, so can be kept moist with an essential oil blend of chamomile, geranium, lavender, lemon, sage, tea tree, and ylang ylang.

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KEY TERMS

Axillary nodes—Lymph nodes found in the armpit that drain the lymph channels from the breast.

Clinical aromatherapy—Aromatherapy is the therapeutic use of plant-derived, aromatic essential oils to promote physical and psychological well-being. It is sometimes used in combination with massage and other therapeutic techniques as part of a holistic treatment approach.

Debulking—General term used for surgeries in which subcutaneous tissue is removed from lymphodematous limb.

Fibrosis—Formation of fibrous tissue as a reaction or as a repair process; may occur due to treatment and/or disease. In lymphedema condition known as hardening of the limb with resulting restriction of circulatory flow, increased infection, and weeping sores.

Fibrotic—Pertaining to or characterized by fibrosis. In dermatological description, “fibrotic” would be used to describe leathery, bound-down, or thickened, scarred skin.

Interstitial fluid—The fluid between cells in tissues. Referred to as the liquid substance of the body.

Interstitial space—The fluid filled areas that surround the cells of a given tissue; also known as tissue space.

Long-stretch bandages—Specialized bandages, similar to an Ace bandage, that have 100 to 200% stretch.

Low-stretch bandage—Specialized bandages, with 30 to 90% stretch, that are used to obtain the correct compression during the treatment of lymphedema; also known as short-stretch bandages.

Lymph—The almost colourless fluid that bathes body tissues and is found in the lymphatic vessels that drain the tissues of the fluid that filters across the blood vessel walls from blood. Lymph carries antibodies and lymphocytes (white blood cells that help fight infection) that have entered the lymph nodes from the blood.

Lymph nodes—Small bean-shaped organs of the immune system, distributed widely throughout the

body and linked by lymphatic vessels. Lymph nodes are garrisons of B, T, and other immune cells.

Lymph System—When sickness or infection invades the body, the immune system is the first line of defense. A big part of that defense is the lymph system. Lymph is carried through the body by lymph vessels that have valves and muscles to help move the fluid. Along the route are lymph nodes that serve as filters for harmful substances. This network of vessels and nodes together is called the lymph system.

Lymphatic fluid—The clear fluid found outside the cells which bathes the tissues. It is collected, filtered, and transported by the lymphatic system from around the tissues to the blood circulatory system. Fluid that collects as a result of lymphedema.

Nail beds—The underlying connective tissue that nourishes the finger and toenails.

Pitting edema—When a swollen area is pressed, the pressure leaves an indentation (pit) that takes time to fill back in.

Sentinel node biopsy—A newer procedure performed in order to determine whether breast cancer has spread to auxiliary (underarm) lymph nodes. A blue radioactive tracer and/or blue dye is injected into the area of the breast tumor. The lymphatic vessels carry the dye or radioactive material, to a “sentinel node”. This sentinel node is thought to be the first lymph node receiving fluid from the tumor and the one most likely to contain cancer cells if the cancer has spread. Only if the sentinel node contains cancer cells are more lymph nodes removed.

Skin contracture—A permanent tightening of the skin that prevents normal movement of the associated body part and that can cause permanent deformity. A contracture develops when the normally elastic connective tissues are replaced by inelastic fibrous tissue. This makes the affected area resistant to stretching and prevents normal movement.

OTHER

Lymph Notes, an online information resource and support group for those with lymphedema and for the family, friends, and therapists who care for them. Web site: < www.lymphnotes.com/index.php >

National Lymphedema Network, Latham Square Building, Suite 1111, 1611 Telegraph Avenue, Oakland, CA 94612-2138. Telephone: (800) 541-3259. Fax: (510) 208-3110. Web site: < www.lymphnet.org >

Lymphedema People. Web site:

< www.lymphedemapeople.com/ >

Lymphatic Research Foundation. < <http://www.lymphaticresearch.org> >

Lymphedema Awareness Foundation. < <http://www.elymphnotes.com/> >

Judith Sims

Lymphocyte typing

Definition

Lymphocyte typing focuses on identifying the numbers and relative percentages of lymphocytes in an individual's bloodstream. Lymphocytes, primarily T cells and B cells, are types of white blood cells, the underlying supports of the immune system in the bloodstream.

Purpose

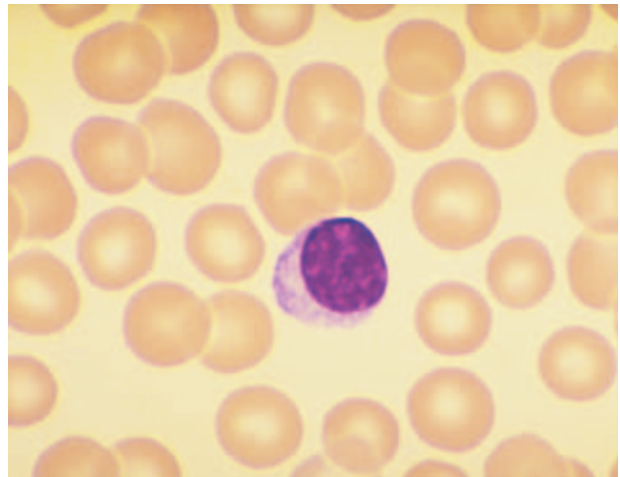
Determining the numbers and relative percentages of T cells and B cells provides information on the state of a person's immune system. By comparing these values to normal numbers and percentages, the presence of disease and the side effects of certain drugs can be revealed. Lymphocyte typing can also show whether a person has been exposed to certain poisonous substances.

Description

To do a white blood cell count, a small amount of blood is drawn from a vein. The total number of white blood cells is calculated, either through microscopic examination of a blood smear or by using automated counting equipment. For a white blood cell count with differential, 100 white blood cells are counted and the proportion of each type is calculated. Since T cells and B cells have similar appearances, a differential can only give the proportion of lymphocytes in the blood, not the proportion of specific lymphocyte types.

For more specific information on B cells and T cells, it is necessary to divide the blood into its separate components. In this procedure, a tube of blood is placed in a centrifuge, a piece of equipment that spins the tube in circles at high speed. The force generated by the spinning causes the various elements in the bloodstream to settle at different levels of the tube.

The lymphocytes are extracted from the tube and treated with special dyes, or stains. Each stain is equipped with an antibody portion that adheres to a specific type of lymphocyte, such as a B cell or a T cell. The stains make the cells visible to an automated counting machine, called a flow cytometer. Based on the number of times the machine detects a particular stain, it can calculate the number of the associated cell type. This procedure can also be used to classify T cells and B cells into their subtypes.



A lymphocyte cell. (Photograph by Lester V. Bergman, Corbis Images. Reproduced by permission.)

KEY TERMS

Immune system—The body's system of defenses against infectious diseases.

Lymphocytosis—A condition in which the number of lymphocytes increases above normal levels.

Lymphopenia—A condition in which the number of lymphocytes falls below normal levels.

White blood cell—A class of cells in the blood that form the foundation of the body's immune system.

Preparation

If possible, a person should avoid eating a heavy meal within hours of the test or engaging in strenuous exercise for the 24 hours preceding the blood test.

Normal results

In general, normal levels of white blood cells vary slightly by age and gender. Normal values are lower in children under the age of 15 and in young adults between the ages of 20 and 30. After age 30, men have slightly higher levels of white blood cells than women.

Normal adult levels of white blood cells are 4,500-11,000 cells per microliter of blood. Lymphocytes account for approximately 25-45% of the total white blood cell count; the normal range is 1,000-4,800 lymphocytes per microliter of blood. Of the total

lymphocytes, 60-80% are T cells and approximately 15% are B cells. (There are two other types of lymphocytes; natural killer and K-type; that constitute a minor proportion of the total lymphocyte numbers.)

Abnormal results

A higher-than-normal level of lymphocytes is called lymphocytosis. Lymphocytosis occurs if a person has a viral, bacterial, or other type of infection. It can also occur with certain blood disorders, such as leukemia.

Lower-than-normal levels of lymphocytes is called lymphopenia. Lymphopenia can be an indicator of certain cancers, bone marrow failure, or immune system deficiency. Medical treatments, such as **chemotherapy** and **radiation therapy**, can also deplete the body's supply of lymphocytes, as can exposure to poisonous substances.

Resources

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Julia Barrett

Lymphocytic choriomeningitis

Definition

Lymphocytic choriomeningitis (LCM) is a viral infection of the membranes surrounding the brain and spinal cord and of the cerebrospinal fluid.

Description

Lymphocytic choriomeningitis virus infection is relatively rare and recovery usually occurs spontaneously within a couple of weeks. Many cases are probably not even identified because the symptoms range from extremely mild to those resembling severe flu. A few patients develop symptoms of **meningitis**. In some rare cases, the LCM viral infection can spread throughout the central nervous system, and may even be fatal.

KEY TERMS

Prodrome—Symptom(s) experienced prior to the onset of a disease. For example, visual disturbances may precede and signal the onset of a migraine headache.

Causes and symptoms

LCM is caused by an arenavirus, which is an RNA virus and is a mild cousin in the family containing the much more threatening arenaviruses that cause hemorrhagic **fever**. Humans acquire LCM virus from infected rodents by coming in contact with the animals or their excretions. Exposure to the virus is not as unlikely to occur as it seems, because the viral hosts can be common house mice and even pets, such as hamsters and chinchillas. Most cases of LCM occur in fall and winter, when mice seek warmth inside dwellings. Food and dust can become contaminated by the excretions of rodents infected with LCM virus. In 1997, French scientists alerted physicians to suspect LCM viral infection in people who had contact with Syrian hamsters.

The symptoms of LCM occur in two phases. The first (prodrome) stage can produce fever, chills, muscle aches, **cough**, and **vomiting**. In the second phase, characteristic meningitis symptoms of **headache**, stiff neck, listlessness, and **nausea and vomiting** may occur. In adults, complications are rare and recovery may even occur before the second phase.

The virus is not spread from person to person, except through **pregnancy**. LCM virus is one of the few viruses that can cross the placenta from mother to child during pregnancy and may be an underrecognized cause of congenital infection in newborns. Infection with cytomegalovirus, *Toxoplasma gondii*, or LCM virus can appear similar enough in infants to be confused when diagnosed. In cases that have been recognized among infants, LCM viral infection has a high mortality rate (about one-third of the babies studied died).

Diagnosis

LCM can be distinguished from bacterial meningitis by the history of prodrome symptoms and the period of time before meningitis symptoms begin, which is about 15-21 days for LCM.

Treatment

No antiviral agents exist for LCM virus. Treatment consists of supporting the patient and treating the symptoms until the infection subsides, generally within a few weeks.

Jill S. Lasker

Lymphocytic leukemia, acute see

Leukemias, acute

Lymphocytic leukemia, chronic see

Leukemias, chronic

Lymphocytopenia

Definition

Lymphocytopenia is a condition marked by an abnormally low level of lymphocytes in the blood. Lymphocytes are a specific type of white blood cell with important functions in the immune system.

Description

Lymphocytes normally account for 15–40% of all white cells in the bloodstream. They help to protect the body from infections caused by viruses or fungi. They also coordinate the activities of other cells in the immune system. In addition, lymphocytes fight **cancer** and develop into antibody-producing cells that neutralize the effect of foreign substances in the blood.

Lymphocytopenia is the result of abnormalities in the way lymphocytes are produced, make their way through the bloodstream, or are lost or destroyed. These conditions can result from congenital or drug-induced decreases in the body's ability to recognize and attack invaders.

Causes and symptoms

Lymphocytopenia has a wide range of possible causes:

- **AIDS** and other viral, bacterial, and fungal infections
- Chronic failure of the right ventricle of the heart. This chamber of the heart pumps blood to the lungs.
- Hodgkin's disease and cancers of the lymphatic system
- A leak or rupture in the thoracic duct. The thoracic duct removes lymphatic fluid from the legs and abdomen.

KEY TERMS

B lymphocyte—A type of lymphocyte that circulates in the blood and lymph and produces antibodies when it encounters specific antigens. B lymphocytes are also called B cells.

Lymph—A clear yellowish fluid circulated by the lymphatic system. The lymph carries mostly lymphocytes and fats.

Lymphocyte—A specific type of white blood cell that is important in the production of antibodies.

- Leukemia
- Side effects of prescription medications
- Malnutrition. **Diets** that are low in protein and overall calorie intake may cause lymphocytopenia.
- Radiation therapy
- High **stress** levels
- Trauma.

The symptoms of lymphocytopenia vary. Lymphocytes constitute only a fraction of the body's white blood cells, and a decline in their number may not produce any symptoms. A patient who has lymphocytopenia may have symptoms of the condition responsible for the depressed level of lymphocytes.

Diagnosis

Lymphocytopenia is most often detected when blood tests are performed to diagnose other diseases.

Treatment

Treatment for lymphocytopenia is designed to identify and correct the underlying cause of the condition.

Drug-depressed lymphocyte levels usually return to normal a few days after the patient stops taking the medication.

A deficiency of B lymphocytes, which mature into antibody-producing plasma cells, can result in abnormally low lymphocyte levels. When the number of B lymphocytes is low, the patient may be treated with **antibiotics**, antifungal medications, antiviral agents, or a substance containing a high concentration of antibodies (gamma globulin) to prevent infection.

It is not usually possible to restore normal lymphocyte levels in AIDS patients. Drugs like AZT (azidothymidine, sold under the trade name Retrovir) can increase the number of helper T cells, which help other cells wipe out disease organisms.

Prognosis

Very low levels of lymphocytes make patients vulnerable to life-threatening infection. Researchers are studying the effectiveness of transplanting bone marrow and other cells to restore normal lymphocyte levels. **Gene therapy**, which uses the body's own resources or artificial substances to counter diseases or disorders, is also being evaluated as a treatment for lymphocytopenia.

Resources

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Maureen Haggerty

Lymphogranuloma venereum

Definition

Lymphogranuloma venereum (LGV) is a sexually transmitted systemic disease (STD) caused by a parasitic organism closely related to certain types of bacteria. It affects the lymph nodes and rectal area, as well as the genitals, in humans. The name comes from two Latin words that mean a swelling of granulation tissue in the lymph nodes resulting from sexual intercourse. Granulation tissue is tissue that forms during wound or ulcer healing that has a rough or lumpy surface.

Description

Although LGV is easily treated in its early stages, it can produce serious complications in its later stages. LGV is most likely to occur among people living in tropical or subtropical countries and among military personnel or tourists in countries or large cities with high rates of the disease. Prostitutes play a major role in carrying and transmitting LGV, as was documented during an outbreak in Florida in the late 1980s. There are about 1000 documented cases of LGV in the United States in an average year.

Causes and symptoms

LGV is caused by *Chlamydia trachomatis*, a globe-shaped parasitic organism that reproduces only inside of living cells. *C. trachomatis* has 17 subtypes and is responsible for a wide range of infections in both men and women; however, only subtypes L1, L2, and L3 cause lymphogranuloma venereum. The parasite has a two-part lifecycle. In the first stage, it is inert and can survive outside of cells. In its second stage, it lacks a cell wall and actively reproduces after gaining entry to a cell. As the chlamydia organism reproduces inside the cell, it pushes the nucleus aside and forms an inclusion that can be identified with tissue staining. LGV differs from other diseases caused by *C. trachomatis* in that it affects the body's lymphatic system and not just the moist tissues of the genital region. In humans, the chlamydia organism is transmitted through vaginal or anal intercourse, oral sex, or contact with fluid from open ulcers or infected tissues.

Lymphogranuloma venereum has three stages. In its primary stage, the disease is more likely to be detected in men; it may go unnoticed in women. After an incubation period of four to 30 days, a small painless ulcer or blister develops in the genital area. Second-stage LGV develops between one and six weeks later. In this stage, the infection spreads to the lymphatic system, forming buboes (swellings) in the lymph nodes of the groin area. The buboes often merge, soften, and rupture, forming sinuses and fistulas (hollow passages and ducts) that carry an infectious bloody discharge to the outside of the body. Patients with second-stage LGV may also have **fever**, **nausea**, headaches, pains in their joints, skin **rashes**, and enlargement of the spleen or liver. Third-stage LGV, which is sometimes called anogenitoretal syndrome, develops in about 25% of patients. In men, this stage is usually seen in homosexuals. Third-stage LGV is marked by rectal **pain**, **constipation**, a discharge containing pus or bloody mucus, and the development of strictures (narrowing or tightening of a body passage) in the rectum or vagina.

LGV can have a number of serious complications. *C. trachomatis* infections of any subtype are associated with long-term fertility problems in women. Strictures in the rectum can completely close off the lower bowel, producing eventual rupture of the bowel and inflammation of the abdominal cavity. The patient can develop chronic abscesses or fistulae in the anal area or in the vagina in women. Long-term blockages in the lymph nodes can produce **elephantiasis**, a condition in which the patient's upper legs and groin area become greatly enlarged. Patients with chronic LGV infection



This man suffers from lymphogranuloma venereum, a venereal disease that is caused by the bacterium *Chlamydia trachomatis*. (Photograph by Milton Reisch, M.D., Corbis Images. Reproduced by permission.)

have a higher risk of developing **cancer** in the inflamed areas.

Chronic LGV can be reactivated in patients who become infected with the **AIDS** virus. These patients develop open ulcers in the groin that are difficult to treat.

Diagnosis

The diagnosis of LGV is usually made on the basis of the patient's history, careful examination of the genital area and lymph nodes, and blood tests or cultures to confirm the diagnosis. In the early stages of the disease, the doctor will need to distinguish between LGV and such other STDs as **syphilis** and herpes. If the patient has developed buboes, the doctor will need to rule out **tuberculosis**, **cat-scratch disease**, bubonic **plague**, or **tularemia** (a disease similar to plague that is carried by rabbits and squirrels). If the patient has developed rectal strictures, the doctor will need to rule out tumors or colitis.

There are several blood tests that can be used to confirm the diagnosis of LGV. The most commonly used are the complement fixation (CF) test and the microimmunofluorescence (micro-IF) tests. Although the micro-IF test is considered more sensitive than the CF test, it is less widely available. An antibody titer (concentration) of 1:64 or greater on the CF test or

KEY TERMS

Anogenitoretal syndrome—Another name for third-stage LGV.

Aspiration—A procedure in which pus or other fluid is removed from a body cavity through a hollow needle connected to a syringe.

Bubo—An inflamed swelling inside a lymph node, characteristic of second-stage LGV.

Elephantiasis—Abnormal enlargement of the legs and groin area caused by blockage of the lymphatic system, as a complication of LGV.

Fistula—A passageway formed by a disease or injury that drains fluid from an infected area to the outside or to other parts of the body.

Lymph—A clear yellowish fluid that circulates throughout the body, carrying white blood cells and fats. The system that produces and circulates lymph is called the lymphatic system; it includes lymph vessels, lymph nodes, the thymus gland, and the spleen.

Proctitis—Inflammation of the anus and rectum.

Stricture—An abnormal narrowing or tightening of a body passage. LGV can cause strictures to form in the patient's rectum, or in the vagina of female patients.

1:512 or greater on the micro-IF test is needed to make the diagnosis of LGV. In some cases, the diagnosis can be made from culturing *C. trachomatis* taken from samples of tissue fluid from ulcers or buboes, or from a tissue sample from the patient's rectum.

Treatment

LGV is treated with oral **antibiotics**, usually tetracycline or doxycycline for 10-20 days, or erythromycin or trimethoprim sulfamethoxazole for 14 days. Pregnant women are usually treated with erythromycin rather than the **tetracyclines**, because this class of medications can harm the fetus.

Patients who have developed second- and third-stage complications may need surgical treatment. The doctor can treat buboes by withdrawing fluid from them through a hollow needle into a suction syringe. This procedure is called aspiration. Fistulas and abscesses also can be treated surgically. Patients who develop elephantiasis are usually treated by plastic surgeons. Patients with rectal strictures may need surgery to prevent bowel obstruction and rupture into the abdomen.

Prognosis

The prognosis for recovery for most patients is good, with the exception of AIDS patients. Prompt treatment of the early stages of LGV is essential to prevent transmission of the disease as well as fertility problems and other serious complications of the later stages.

Prevention

Prevention of lymphogranuloma venereum has four important aspects:

- Avoidance of casual sexual contacts, particularly with prostitutes, in countries with high rates of the disease.
- Observance of proper safeguards by health professionals. Doctors and other healthcare workers should wear gloves when touching infected areas of the patient's body or handling soiled dressings and other contaminated items. All contaminated materials and instruments should be double-bagged before disposing.
- Tracing and examination of an infected person's recent sexual contacts.
- Monitoring the patient for recurring symptoms for a period of six months after antibiotic treatment.

Resources

BOOKS

Chambers, Henry F. "Infectious Diseases: Bacterial & Chlamydial." In *Current Medical Diagnosis and Treatment*, 1998, edited by Stephen McPhee, et al., 37th ed. Stamford: Appleton & Lange, 1997.

Rebecca J. Frey, PhD

Lymphomas see **Hodgkin's disease**

Lymphopenia see **Lymphocytopenia**

Lymphosarcomas see **Malignant lymphomas**

Lysergic acid diethylamide (LSD)

Definition

Lysergic acid diethylamide (LSD), also known as "acid," belongs to a class of drugs known as hallucinogens, which distort perceptions of reality. LSD is the most potent mood- and perception-altering drug

known: doses as small as 30 micrograms can produce effects lasting six to 12 hours.

Purpose

In the United States, LSD has no accepted medical use and its manufacture is illegal.

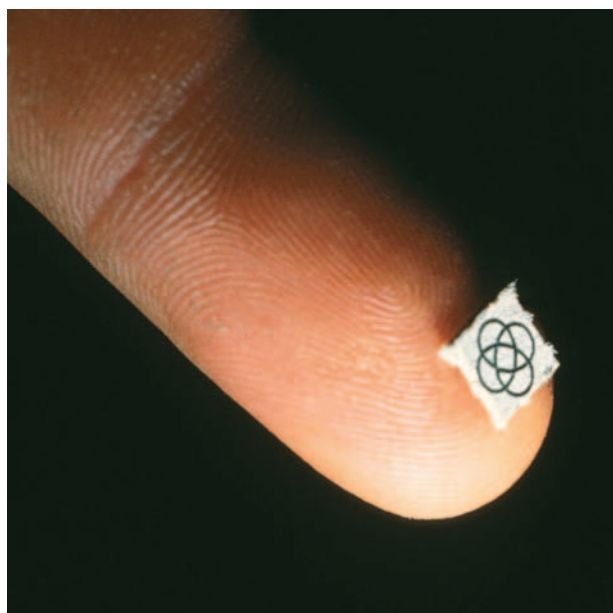
Description

LSD is produced synthetically from a fungus that grows on rye grass. This odorless, colorless, and slightly bitter-tasting chemical is generally ingested orally and absorbed from the gastrointestinal system. Manufacturers commonly distribute LSD in small squares of absorbent paper soaked with the drug, which users chew and swallow. Use of LSD and other hallucinogens by secondary school students has decreased since 1998, but has increased among older teens and young adults attending dance clubs and all-night raves, according to the National Institute on Drug Abuse.

LSD alters perceptions by disrupting the action of the neurotransmitter serotonin, although precisely how it does this is unclear. Studies suggest LSD acts on certain groups of serotonin receptors, and that its effects are most prominent in two brain regions: the cerebral cortex and the locus ceruleus. The cerebral cortex is involved in mood and perception, and the locus ceruleus receives sensory signals from all areas of the body. Natural hallucinogens resembling LSD, such as mescaline and psilocybin, have been used in social and religious rituals for thousands of years.

After its discovery in 1938, LSD was used experimentally to treat neuroses, narcotic **addiction**, **autism**, **alcoholism**, and terminally ill **cancer** patients, and to study the mechanisms of psychotic diseases like **schizophrenia**. Nearly 30 years after its discovery, manufacture, possession, sale, and use of LSD was restricted in the United States under the Drug Abuse Control Amendment of 1965.

LSD's effects generally begin within an hour of taking the drug and last for up to 12 hours. The drug is absorbed from the gastrointestinal tract, and circulated throughout the body and to the brain. It is metabolized in the liver and excreted in the urine about 24 hours after ingestion. Physical effects of LSD may include loss of appetite, sleeplessness, pupil dilation, **dry mouth**, salivation, **palpitations**, perspiration, **nausea**, **dizziness**, blurred vision, and **anxiety**, as well as increased body temperature, heartbeat, blood pressure, and blood sugar.



LSD on blotter paper. (Custom Medical Stoch Photo Inc. Reproduced by permission)

The major effects of LSD are emotional and sensory. Emotions may shift instantaneously from euphoria to confusion and despair, and users may feel as if they are experiencing several emotions simultaneously. Colors, smells, and sounds may be highly intensified, and time may appear to move very slowly. Sensory perceptions may blend in a phenomenon known as synesthesia, in which a person sees sounds, or smells colors, for example. Users may have out-of-body sensations, or may perceive their body has changed shape or merged with another person or object.

Precautions

Unlike **cocaine**, amphetamines, heroin, alcohol, and nicotine, LSD is not considered addictive, but it is considered dangerous; users are at risk for several short- and long-term side effects. LSD's effects are unpredictable and may vary with the amount ingested and the user's personality, mood, expectations, and surroundings. Users may experience enjoyable sensations on some "trips," and terrifying feelings of anxiety and despair on others. Most LSD-related deaths stem not from the LSD's physical effects on the body, but from the panicked reactions ensuing from intense LSD-triggered illusions.

KEY TERMS

Acid—Common street name for LSD.

Cerebral cortex—Brain region responsible for reasoning, mood, and perception.

Hallucinogen—A drug that distorts sensory perceptions and disturbs emotion, judgment, and memory.

Hallucinogen persisting perception disorder (HPPD)—The recurrence of LSD effects after the drug experience has ended.

Locus ceruleus—Brain region that processes sensory signals from all areas of the body.

Neurotransmitter—Chemical compound in the brain that transmits signals from one nerve cell to another.

Serotonin—A neurotransmitter that modulates the actions of other neurotransmitters in the brain.

Side Effects

Two long-term effects are associated with LSD use: **psychosis**, and hallucinogen persisting perception disorder (HPPD), also known as "flashbacks." The exact causes of these effects, including the mechanism by which LSD may cause them, is unknown. Chronic hallucinogen users or individuals with underlying personality problems are most vulnerable to these effects, but individuals with no history of psychological disorders have also experienced them. LSD-induced psychosis may include dramatic mood swings, loss of cognitive and communication skills, and **hallucinations**. Flashbacks generally involve seeing bright flashes, or halos or trails attached to moving objects after the LSD "trip" has ended. Flashbacks can last a few seconds or even several hours.

According to the Drug Abuse Warning Network (DAWN), the number of LSD-related hospital emergencies is low compared to those related to cocaine, heroin, **marijuana**, methamphetamine, and other illicit drugs. One reason for this trend may be that LSD currently sold on the black market is less potent than in the past. LSD dose strengths tend to range from 20 to 80 micrograms today, compared to 100 to 200 micrograms reported during the 1960s and early 1970s.

Interactions

LSD flashbacks can be spurred by use of drugs such as marijuana. Preliminary evidence suggests serotonin reuptake inhibitors like Prozac and Zoloft may also exacerbate the LSD flashback syndrome.

Resources

PERIODICALS

Aghajanian, G. K. and G. J. Marek. "Serotonin and Hallucinogens." *Neuropsychopharmacology* 1999: 16S-23S

ORGANIZATIONS

National Clearinghouse for Alcohol and Drug Information. 11426 Rockville Pike, Suite 200, Rockville, MD. 20852. (800) 729-6686. <<http://www.health.org>>.

National Institute on Drug Abuse. P.O. Box 30652, Bethesda, MD. 20824-0652. (888) 644-6432. <<http://www.drugabuse.gov>>.

U.S. Department of Justice, Drug Enforcement Administration. 2401 Jefferson Davis Highway, Alexandria, VA 22301. (888) 644-6432. <<http://www.usdoj.gov/dea>>.

Ann Quigley



Macular degeneration

Definition

Macular degeneration is the progressive deterioration of a critical region of the retina called the macula. The macula is a 3-5 mm area in the retina that is responsible for central vision. This disorder leads to irreversible loss of central vision, although peripheral vision is retained. In the early stages, vision may be gray, hazy, or distorted.

Description

Macular degeneration is the most common cause of legal blindness in people over 60, and accounts for approximately 11.7% of blindness in the United States. About 28% of the population over age 74 is affected by this disease.

Age-related macular degeneration (ARMD) is the most common form of macular degeneration. It is also known as age-related maculopathy (ARM), aged macular degeneration, and senile macular degeneration. Approximately 10 million Americans have some vision loss that is due to ARMD.

ARMD is subdivided into a dry (atrophic) and a wet (exudative) form. The dry form is more common and accounts for 70-90% of cases of ARMD. It progresses more slowly than the wet form and vision loss is less severe. In the dry form, the macula thins over time as part of the **aging** process and the pigmented retinal epithelium (a dark-colored cell layer at the back of the eye) is gradually lost. Words may appear blurred or hazy and colors may appear dim or gray.

In the wet form of ARMD, new blood vessels grow underneath the retina and distort the retina. These blood vessels can leak, causing scar tissue to form on the retina. The wet form may cause visual distortion and make straight lines appear wavy. A

central blind spot develops. The wet type progresses more rapidly and vision loss is more pronounced. Treatments are available for some, but not most, cases of the wet form.

Other less common forms of macular degeneration include:

- Cystoid macular degeneration. Loss of vision in the macula due to fluid-filled areas (cysts) in the macular region. This may be a result of other disorders, such as aging, inflammation, or high myopia.
- Diabetic macular degeneration. Deterioration of the macula due to diabetes.
- Senile disciform degeneration (also known as Kuhnt-Junius macular degeneration). A specific and severe type of the wet form of ARMD that involves leaking blood vessels (hemorrhaging) in the macular region. It usually occurs in people over 40 years old.

Causes and symptoms

Age-related macular degeneration is part of the aging process. There may be a hereditary component. Having a family member with ARMD increases a person's risk for developing it. There is a slightly higher incidence in females. Whites and Asians are more susceptible to developing ARMD than blacks, in whom the disorder is rare.

ARMD is thought to be caused by hardening and blocking of the arteries (arteriosclerosis) in the blood vessels supplying the retina. Some of the same things that are bad for the heart are thought to contribute to the development of macular degeneration. These risk factors include **smoking** and a diet that is rich in saturated fat. Smokers have a risk of developing ARMD that is approximately 2.4-3 times that of non-smokers. Smoking increases the risk of developing wet-type ARMD, and may increase the risk of developing dry-type as well. Dietary fat also increase the risk. In one

study of older (age 45-84) Americans, signs of early ARMD were 80% more common in the group who ate the most saturated fat compared to those who ate the least. Low consumption of antioxidants, such as foods rich in vitamin A, is associated with a higher risk for developing ARMD. Consumption of moderate amounts of red wine and foods rich in vitamin A is associated with a lower risk. It is generally believed that exposure to ultraviolet (UV) light may contribute to disease development, but this has not been proven.

The main symptom of macular degeneration is a change in central vision. The patient may notice blurred central vision or a blank spot on the page when reading. The patient may notice visual distortion such as bending of straight lines. Images may appear smaller. Some patients notice a change in color perception and some experience abnormal light sensations. These symptoms may come on suddenly and become progressively more troublesome. Sudden onset of symptoms, particularly vision distortion, is an indication for immediate evaluation by an ophthalmologist.

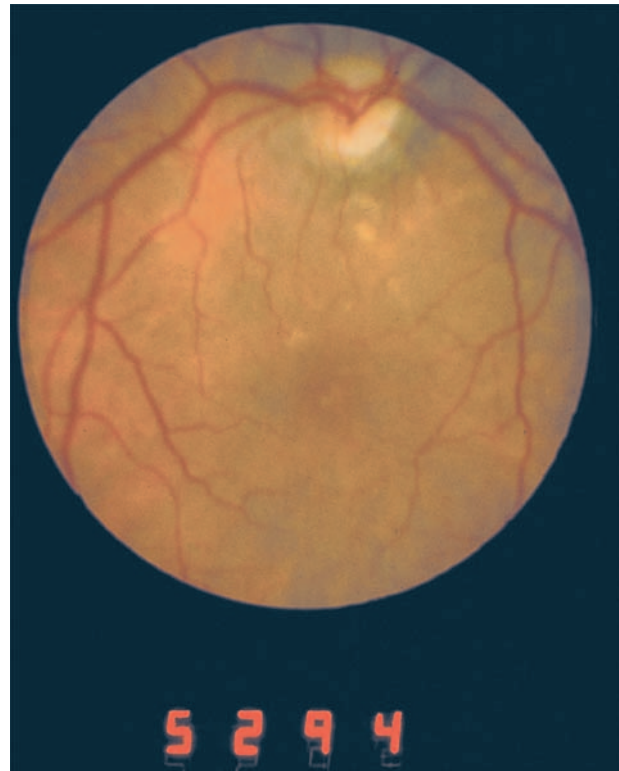
Diagnosis

To make the diagnosis of macular degeneration, the doctor dilates the pupil with eye drops and examines the interior of the eye, looking at the retina for the presence of yellow bumps called drusen and for gross changes in the macula such as thinning. The doctor also administers a visual field test, looking for blank spots in the central vision. The doctor may call for fluorescein **angiography** (intravenous injection of fluorescent dye followed by visual examination and photography of the back of the eye) to determine if blood vessels in the retina are leaking.

A central visual field test called an Amsler grid is usually given to patients who are suspected of having ARMD. It is a grid printed on a sheet of paper (so it is easy to take home). When looking at a central dot on the page, the patient should call the doctor right away if any of the lines appear to be wavy or missing. This may be an indication of fluid and the onset of wet ARMD. Patients may also be asked to come in for more frequent checkups.

Treatment

While loss of vision cannot be reversed, early detection is important because treatments are available that may halt or slow the progression of the wet form of ARMD. Treatment for the dry form is not available as of 1998, but cell transplantation studies are under study.



A slit-lamp view showing macular degeneration of the eye.
(Custom Medical Stock Photo. Reproduced by permission.)

In wet-type ARMD and in senile disciform macular degeneration, new capillaries grow in the macular region and leak. This leaking of blood and fluid causes a portion of the retina to detach. Blood vessel growth, called neovascularization, can be treated with laser photocoagulation in some cases, depending upon the location and extent of the growth. Argon or krypton lasers can destroy the new tissue and flatten the retina. This treatment is effective in about half the cases but results may be temporary. A concern with laser therapy is that the laser also destroys the photoreceptors in the treated area. If the blood vessels have grown into the fovea (a region of the macula responsible for fine vision), treatment may not be possible. Because capillaries can grow very quickly, this form of macular degeneration should be handled as an emergency and treated quickly. Patients who are experiencing visual distortion should seek help immediately.

Another form of treatment for the wet form of ARMD is **radiation therapy** with either x rays or a proton beam. Blood vessels that are proliferating (growing) are sensitive to treatment with low doses of ionizing radiation. Nerve cells in the retina are not growing and are insensitive, so they are not harmed by this treatment. External beam radiation treatment has

shown promising results at slowing progression in limited, early trials. An alternative treatment is internal beam radiation therapy. For this treatment, the patient is given a local anesthetic and an applicator containing strontium 90 is inserted into the affected eye. This brief and localized radiation therapy prevents the growth of blood vessels.

Other therapies that are under study include treatment with alpha-interferon, thalidomide, and other drugs that slow the growth of blood vessels. Subretinal surgery also has shown promise in rapid-onset cases of wet ARMD. This surgery carries the risk of **retinal detachment**, hemorrhage, and acceleration of cataract formation. Other experimental treatments include photodynamic therapy (PDT). For this treatment, a photosensitizing dye is injected, followed by irradiation of the area of new blood vessel growth with a special, low-intensity diode laser. This treatment damages the cells in the blood vessel walls and causes them to stop growing.

A controversial treatment called rheotherapy involves pumping the patient's blood through a device that removes some proteins and fats. As of 1998, this had not been proven to be safe or effective.

Alternative treatment

Consumption of a diet rich in antioxidants (beta carotene and the mixed carotenoids that are precursors of vitamin A, **vitamins** C and E, selenium, and zinc), or taking antioxidant **nutritional supplements**, may help prevent macular degeneration, particularly if started early in life. Good dietary sources of antioxidants include citrus fruits, cauliflower, broccoli, nuts, seeds, orange and yellow vegetables, cherries, blackberries, and blueberries. Research has shown that nutritional therapy can prevent ARMD or slow its progression once established. Some doctors recommend taking beta carotene and zinc as a precautionary measure. Some vitamins are marketed specifically for the eyes.

Prognosis

The dry form of ARMD is self-limiting and eventually stabilizes. The loss of vision is permanent. The vision of patients with the wet form of ARMD often stabilizes or improves even without treatment, at least temporarily. However, after a few years, patients with the wet form of ARMD are usually left with only coarse peripheral vision remaining.

Many patients with macular degeneration lose their central vision permanently and may become

KEY TERMS

Drusen—Tiny yellow dots on the retina that can be soft or hard and that usually do not interfere with vision.

Fovea—A tiny pit in the macula that is responsible for sharp vision.

Neovascularization—Growth of new capillaries.

Photoreceptors—Specialized nerve cells (rods and cones) in the retina that are responsible for vision.

Retina—The light-sensitive membrane at the back of the eye that images are focused on. The retina sends the images to the brain via the optic nerve.

legally blind. However, macular degeneration rarely causes total loss of vision. Peripheral vision is retained. The patient can compensate, to some extent, for the loss of central vision, even though macular degeneration may render them legally blind. Improved lighting and special low-vision aids may help even if sharpness of vision (visual acuity) is poor. Vision aids include special magnifiers that allow the patient to read and telescopic aids for long-distance vision. The use of these visual aids plus the retained peripheral vision usually allow the patient to remain independent. Registration as a legally blind person will enable a patient to obtain special services and considerations.

Prevention

Avoiding the risk factors for macular degeneration may help prevent it. This includes avoiding tobacco smoke and eating a diet low in saturated fat. Some other behaviors that may help reduce the risk of wet-type ARMD are eating a diet rich in green, leafy vegetables and yellow vegetables such as carrots, sweet potatoes, and winter squash; drinking moderate amounts of alcohol, such as one or two glasses of red wine a day; and taking an antioxidant vitamin supplement, especially vitamin A. Some vitamins may be toxic in large doses, so patients should speak with their doctors. Vitamins C and E have not been shown to reduce risk, nor did selenium in one large study. The use of zinc is controversial: some studies showed a benefit, others showed no benefit, and one actually showed an increased risk of ARMD with increased levels of zinc in the blood. Some doctors suggest that wearing UV-blocking sunglasses reduces risk. Use of estrogen in postmenopausal women is associated with a lower risk of developing ARMD.

Resources

ORGANIZATIONS

American Academy of Ophthalmology. 655 Beach Street, P.O. Box 7424, San Francisco, CA 94120-7424. <<http://www.eyenet.org>>.

American Optometric Association. 243 North Lindbergh Blvd., St. Louis, MO 63141. (314) 991-4100. <<http://www.aoanet.org>>.

Prevent Blindness America. 500 East Remington Road, Schaumburg, IL 60173. (800) 331-2020. <<http://www.preventblindness.org>>.

Louann W. Murray, PhD

Macule see **Skin lesions**

Mad cow disease see **Creutzfeldt-Jakob disease**

Madura foot see **Mycetoma**

Maduromycosis see **Mycetoma**

Magnesium hydroxide see **Antacids**

Magnesium imbalance

Definition

A mineral found in the fluid that surrounds cells, magnesium (Mg) is an essential component of more than 300 enzymes that regulate many body functions. Imbalances occur when the blood contains more or less magnesium than it should.

Description

Magnesium is necessary for the formation and functioning of healthy bones, teeth, muscles, and nerves. It converts food into energy, builds proteins, and is instrumental in maintaining adequate levels of calcium in the blood. Magnesium helps prevent cardiovascular disease and irregular heartbeat, reduces the risk of bone loss (**osteoporosis**), and increases an individual's chance of surviving a **heart attack**. It may also help prevent **stroke** and lessen the effects of existing osteoporosis.

Fish, dairy products, leafy green vegetables, legumes, nuts, seeds, and grains are especially good sources of magnesium, but varying amounts of this mineral are found in all foods. Some is stored in the kidneys, and excess amounts are excreted in the urine or stools.

Magnesium deficiency (hypomagnesemia) or excess (hypermagnesemia) is rare, but either condition can be serious.

Causes and symptoms

Hypomagnesemia

Magnesium deficiency most often occurs in people who have been fed intravenously for a long time, whose diet does not contain enough magnesium, or who are unable to absorb and excrete the mineral properly.

Secreting too much aldosterone (the hormone that regulates the body's salt-fluid balance), ADH (a hormone that inhibits urine production), or thyroid hormone can cause hypomagnesemia.

Other factors associated with hypomagnesemia include:

- Loss of body fluids as a result of stomach suctioning or chronic **diarrhea**
- Cisplatin (a **chemotherapy** drug)
- Long-term diuretic therapy
- Hypercalcemia (abnormally high levels of calcium in the blood)
- Diabetic acidosis (a condition in which the body's tissues have a higher-than-normal acid content)
- Complications of bowel surgery
- Chronic **alcoholism**
- Malnutrition
- Starvation
- Severe **dehydration**.

People who have hypomagnesemia usually experience loss of weight and appetite, bloating, and muscle **pain**, and they pass stools that have a high fat content. Also, they may be listless, disoriented, confused, and very irritable. Other symptoms of hypomagnesemia are:

- Nausea
- Vomiting
- Muscle weakness
- Tremor
- Irregular heart beat
- Delusions and **hallucinations**
- Leg and foot cramps
- Muscle twitches
- Changes in blood pressure.

Severe magnesium deficiency can cause seizures, especially in children.

Neonatal hypomagnesemia can occur in premature babies and in infants who have genetic parathyroid disorders or who have had blood transfusions. This condition also occurs in babies born to magnesium-deficient mothers or to women who have:

- Diabetes mellitus
- Hyperparathyroidism (overactive parathyroid glands)
- Toxemia (a pregnancy-related condition characterized by high blood pressure and fluid retention).

Hypermagnesemia

Hypermagnesemia is most common in patients whose kidneys cannot excrete the magnesium they derive from food or take as medication. This condition can also develop in patients who take magnesium salts, or in healthy people who use large quantities of magnesium-containing **antacids**, **laxatives**, or **analgesics** (pain relievers).

Magnesium **poisoning** can cause severe diarrhea in young people, and mask the symptoms of other illnesses. Very high overdoses can lead to **coma**. The risk of complications of magnesium poisoning is greatest for:

- Elderly people with inefficient kidney function
- Patients with kidney problems or intestinal disorders
- People who use **antihistamines**, **muscle relaxants**, or narcotics.

Severe dehydration or an overdose of supplements taken to counteract hypomagnesemia can also cause this condition.

People who have hypermagnesemia may feel flushed and drowsy, perspire heavily, and have diarrhea. Breathing becomes shallow, reflexes diminish, and the patient becomes unresponsive. Muscle weakness and hallucinations are common. The patient's heart beat slows dramatically and blood pressure plummets. Extreme toxicity, which can lead to coma and cardiac arrest, can be fatal.

Diagnosis

Blood tests are used to measure magnesium levels.

Treatment

The goal of treatment is to identify and correct the cause of the imbalance. Oral magnesium supplements

KEY TERMS

Hypermagnesemia—An abnormally high concentration of magnesium in the blood.

Hypomagnesemia—An abnormally low concentration of magnesium in the blood.

or injections are usually prescribed to correct mild magnesium deficiency. If the deficiency is more severe or does not respond to treatment, magnesium sulfate or magnesium chloride may be administered intravenously.

Doctors usually prescribe **diuretics** (urine-producing drugs) for patients with hypermagnesemia and advise them to drink more fluids to flush the excess mineral from the body. Patients whose magnesium levels are extremely high may need mechanical support to breathe and to circulate blood throughout their bodies.

Intravenously administered calcium gluconate may reverse damage caused by excess magnesium. Intravenous furosemide (Lasix) or ethacrynic acid (Edecrin) can increase magnesium excretion in patients who get enough fluids and whose kidneys are functioning properly.

In an emergency, dialysis can provide temporary relief for patients whose kidney function is poor or who are unable to excrete excess **minerals**.

Prognosis

Because imbalances may recur if the underlying condition is not eliminated, monitoring of magnesium levels should continue after treatment has been completed.

Prevention

Most people consume adequate amounts of magnesium in the food they eat. Dietary supplements can be used safely, but should only be used under a doctor's supervision.

Resources

OTHER

"Mineral Guide." *CNN Page*. May 2, 1998. <<http://www.cnn.com/HEALTH>>.

Maureen Haggerty

Magnetic field therapy

Definition

Magnetic therapy is the use of magnets to relieve **pain** in various areas of the body.

Purpose

Some of the benefits that magnetic therapy claims to provide include:

- pain relief
- reduction of swelling
- improved tissue alkalization
- more restful sleep
- increased tissue oxygenation
- relief of stress
- increased levels of cellular oxygen
- improved blood circulation
- anti-infective activity

Description

Origins

Magnetic therapy dates as far back as the ancient Egyptians. Magnets have long been believed to have healing powers associated with muscle pain and stiffness. Chinese healers as early as 200 B.C. were said to use magnetic lodestones on the body to correct unhealthy imbalances in the flow of *qi*, or energy. The ancient Chinese medical text known as *The Yellow Emperor's Canon of Internal Medicine* describes this procedure. The *Vedas*, or ancient Hindu scriptures, also mention the treatment of diseases with lodestones. The word “lodestone” or leading stone, came from the use of these stones as compasses. The word “magnet” probably stems from the Greek *Magnes lithos*, or “stone from Magnesia,” a region of Greece rich in magnetic stones. The Greek phrase later became *magneta* in Latin.

Sir William Gilbert's 1600 treatise, *De Magnete*, was the first scholarly attempt to explain the nature of magnetism and how it differed from the attractive force of static electricity. Gilbert allegedly used magnets to relieve the arthritic pains of Queen Elizabeth I. Contemporary American interest in magnetic therapy began in the 1990s, as several professional golfers and football players offered testimony that the devices seemed to cure their nagging aches and injuries.

Many centuries ago, the earth was surrounded by a much stronger magnetic field than it is today. Over the past 155 years, scientists have been studying the decline of this magnetic field and the effects it has had on human health. When the first cosmonauts and astronauts were going into space, physicians noted that they experienced bone calcium loss and **muscle cramps** when they were out of the Earth's magnetic field for any extended period of time. After this discovery was made, artificial magnetic fields were placed in the space capsules.

There are two theories that are used to explain magnetic therapy. One theory maintains that magnets produce a slight electrical current. When magnets are applied to a painful area of the body, the nerves in that area are stimulated, thus releasing the body's natural painkillers. The other theory maintains that when magnets are applied to a painful area of the body, all the cells in that area react to increase blood circulation, ion exchange, and oxygen flow to the area. Magnetic fields attract and repel charged particles in the bloodstream, increasing blood flow and producing heat. Increased oxygen in the tissues and blood stream is thought to make a considerable difference in the speed of healing.

Preparations

There are no special preparations for using magnetic therapy other than purchasing a product that is specific for the painful area being treated. Products available in a range of prices include necklaces and bracelets; knee, back, shoulder and wrist braces; mattress pads; gloves; shoe inserts; and more.

Precautions

The primary precaution involved with magnetic therapy is to recognize the expense of this therapy. Magnets have become big business; they can be found in mail-order catalogs and stores ranging from upscale department stores to specialty stores. As is the case with many popular self-administered therapies, many far-fetched claims are being made about the effectiveness of magnetic therapy. Consumers should adopt a “let the buyer beware” approach to magnetic therapy. Persons who are interested in this form of treatment should try out a small, inexpensive item to see if it works for them before investing in the more expensive products.

Side effects

There are very few side effects from using magnetic therapy. Generally, patients using this therapy

find that it either works for them or it does not. Patients using transcranial magnetic stimulation for the treatment of depression reported mild **headache** as their only side effect.

Research and general acceptance

Magnetic therapy is becoming more and more widely accepted as an alternative method of pain relief. Since the late 1950s, hundreds of studies have demonstrated the effectiveness of magnetic therapy. In 1997, a group of physicians at Baylor College of Medicine in Houston, Texas studied the use of magnetic therapy in 50 patients who had developed **polio** earlier in life. These patients had muscle and joint pain that standard treatments failed to manage. In this study, 29 of the patients wore a magnet taped over a trouble spot, and 21 others wore a nonmagnetic device. Neither the researchers nor the patients were told which treatment they were receiving (magnetic or nonmagnetic). As is the case with most studies involving a placebo, some of the patients responded to the nonmagnetic therapy, but 75% of those using the magnetic therapy reported feeling much better.

In another study at New York Medical College in Valhalla, New York, a neurologist tested magnetic therapy on a group of 19 men and women complaining of moderate to severe burning, **tingling**, or **numbness** in their feet. Their problems were caused by diabetes or other conditions present such as **alcoholism**. This group of patients wore a magnetic insole inside one of their socks or shoes for 24 hours a day over a two-month period, except while bathing. They wore a nonmagnetic insert in their other sock or shoe. Then for two months they wore magnetic inserts on both feet. By the end of the study, nine out of ten of the diabetic patients reported relief, while only three of nine non-diabetic patients reported relief. The neurologist in charge of the study believes that this study opens the door to additional research into magnetic therapy for diabetic patients. He plans a larger follow-up study in the near future.

As of 2000, a federally funded study is underway at the University of Virginia. This study is evaluating the effectiveness of magnetic mattress pads in easing the muscle pain, stiffness and **fatigue** associated with **fibromyalgia**.

Magnetic therapy is also being studied in the treatment of depression in patients with **bipolar disorder**. A procedure called repeated transcranial magnetic stimulation has shown promise in treating this condition. In this particular study, patients with depression had a lower relapse rate than did those

KEY TERMS

Fibromyalgia—A chronic syndrome characterized by fatigue, widespread muscular pain, and pain at specific points on the body.

Lodestone—A variety of magnetite that possesses magnetic polarity.

Transcranial magnetic stimulation—A procedure used to treat patients with depression.

using **electroconvulsive therapy**. Unlike electroconvulsive therapy, patients using magnetic therapy did not suffer from seizures, memory lapses, or impaired thinking.

Resources

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Magnetic resonance imaging

Definition

Magnetic resonance imaging (MRI) is the newest, and perhaps most versatile, medical imaging technology available. Doctors can get highly refined images of the body's interior without surgery, using MRI. By using strong magnets and pulses of radio waves to manipulate the natural magnetic properties in the body, this technique makes better images of organs and soft tissues than those of other scanning technologies. MRI is particularly useful for imaging the brain and spine, as well as the soft tissues of joints and the interior structure of bones. The entire body is visible to the technique, which poses few known health risks.

Purpose

MRI was developed in the 1980s. The latest additions to MRI technology are **angiography** (MRA) and spectroscopy (MRS). MRA was developed to

study blood flow, while MRS can identify the chemical composition of diseased tissue and produce color images of brain function. The many advantages of MRI include:

- **Detail.** MRI creates precise images of the body based on the varying proportions of magnetic elements in different tissues. Very minor fluctuations in chemical composition can be determined. MRI images have greater natural contrast than standard x rays, computed tomography scan (CT scan), or ultrasound, all of which depend on the differing physical properties of tissues. This sensitivity lets MRI distinguish fine variations in tissues deep within the body. It also is particularly useful for spotting and distinguishing diseased tissues (tumors and other lesions) early in their development. Often, doctors prescribe an MRI scan to more fully investigate earlier findings of the other imaging techniques.
- **Scope.** The entire body can be scanned, from head to toe and from the skin to the deepest recesses of the brain. Moreover, MRI scans are not obstructed by bone, gas, or body waste, which can hinder other imaging techniques. (Although the scans can be degraded by motion such as breathing, heartbeat, and normal bowel activity.) The MRI process produces cross-sectional images of the body that are as sharp in the middle as on the edges, even of the brain through the skull. A close series of these two-dimensional images can provide a three-dimensional view of a targeted area.
- **Safety.** MRI does not depend on potentially harmful ionizing radiation, as do standard x-ray and CT scans. There are no known risks specific to the procedure, other than for people who might have metal objects in their bodies.

MRI is being used increasingly during operations, particularly those involving very small structures in the head and neck, as well as for preoperative assessment and planning. Intraoperative MRIs have shown themselves to be safe as well as feasible, and to improve the surgeon's ability to remove the entire tumor or other abnormality.

Given all the advantages, doctors would undoubtedly prescribe MRI as frequently as ultrasound scanning, but the MRI process is complex and costly. The process requires large, expensive, and complicated equipment; a highly trained operator; and a doctor specializing in radiology. Generally, MRI is prescribed only when serious symptoms and/or negative results from other tests indicate a need. Many times another test is appropriate for the type of diagnosis needed.

Doctors may prescribe an MRI scan of different areas of the body.

- **Brain and head.** MRI technology was developed because of the need for brain imaging. It is one of the few imaging tools that can see through bone (the skull) and deliver high quality pictures of the brain's delicate soft tissue structures. MRI may be needed for patients with symptoms of a **brain tumor**, **stroke**, or infection (like **meningitis**). MRI also may be needed when cognitive and/or psychological symptoms suggest brain disease (like Alzheimer's or Huntington's diseases, or **multiple sclerosis**), or when developmental retardation suggests a birth defect. MRI can also provide pictures of the sinuses and other areas of the head beneath the face. Recent refinements in MRI technology may make this form of diagnostic imaging even more useful in evaluating patients with brain **cancer**, stroke, **schizophrenia**, or epilepsy. In particular, a new 3-D approach to MRI imaging known as diffusion tensor imaging, or DTI, measures the flow of water within brain tissue, allowing the radiologist to tell where the normal flow of fluid is disrupted, and to distinguish more clearly between cancerous and normal brain tissue. The introduction of DTI has led to a technique known as fiber tracking, which allows the neurosurgeon to tell whether a space-occupying brain tumor has damaged or displaced the nerve pathways in the white matter of the brain. This information in turn improves the surgeon's accuracy during the actual operation.
- **Spine.** Spinal problems can create a host of seemingly unrelated symptoms. MRI is particularly useful for identifying and evaluating degenerated or herniated spinal discs. It can also be used to determine the condition of nerve tissue within the spinal cord.
- **Joint.** MRI scanning is most commonly used to diagnose and assess joint problems. MRI can provide clear images of the bone, cartilage, ligament, and tendon that comprise a joint. MRI can be used to diagnose joint injuries due to sports, advancing age, or arthritis. MRI can also be used to diagnose shoulder problems, like a torn rotator cuff. MRI can also detect the presence of an otherwise hidden tumor or infection in a joint, and can be used to diagnose the nature of developmental joint abnormalities in children.
- **Skeleton.** The properties of MRI that allow it to see through the skull also allow it to view the inside of bones. It can be used to detect bone cancer, inspect the marrow for leukemia and other diseases, assess bone loss (**osteoporosis**), and examine complex **fractures**.

- The rest of the body. While CT and ultrasound satisfy most chest, abdominal, and general body imaging needs, MRI may be needed in certain circumstances to provide better pictures or when repeated scanning is required. The progress of some therapies, like **liver cancer** therapy, needs to be monitored, and the effect of repeated x-ray exposure is a concern.

Precautions

MRI scanning should not be used when there is the potential for an interaction between the strong MRI magnet and metal objects that might be imbedded in a patient's body. The force of magnetic attraction on certain types of metal objects (including surgical steel) could move them within the body and cause serious injury. Metal may be imbedded in a person's body for several reasons.

- Medical. People with implanted cardiac **pacemakers**, metal aneurysm clips, or who have had broken bones repaired with metal pins, screws, rods, or plates must tell their radiologist prior to having an MRI scan. In some cases (like a metal rod in a reconstructed leg) the difficulty may be overcome.
- Injury. Patients must tell their doctors if they have bullet fragments or other metal pieces in their body from old **wounds**. The suspected presence of metal, whether from an old or recent wound, should be confirmed before scanning.
- Occupational. People with significant work exposure to metal particles (working with a metal grinder, for example) should discuss this with their doctor and radiologist. The patient may need prescan testing—usually a single, regular x ray of the eyes to see if any metal is present.

Chemical agents designed to improve the picture and/or allow for the imaging of blood or other fluid flow during MRA may be injected. In rare cases, patients may be allergic to or intolerant of these agents, and these patients should not receive them. If these chemical agents are to be used, patients should discuss any concerns they have with their doctor and radiologist.

The potential side effects of magnetic and electric fields on human health remain a source of debate. In particular, the possible effects on an unborn baby are not well known. Any woman who is, or may be, pregnant should carefully discuss this issue with her doctor and radiologist before undergoing a scan.

As with all medical imaging techniques, **obesity** greatly interferes with the quality of MRI.

Description

In essence, MRI produces a map of hydrogen distribution in the body. Hydrogen is the simplest element known, the most abundant in biological tissue, and one that can be magnetized. It will align itself within a strong magnetic field, like the needle of a compass. The earth's magnetic field is not strong enough to keep a person's hydrogen atoms pointing in the same direction, but the superconducting magnet of an MRI machine can. This comprises the "magnetic" part of MRI.

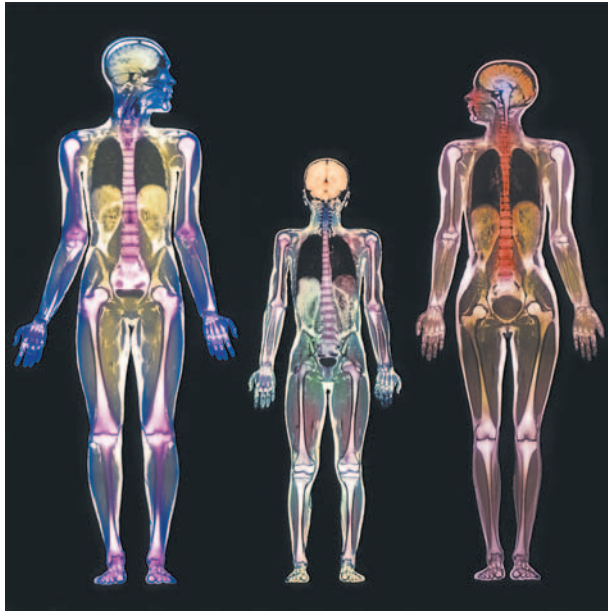
Once a patient's hydrogen atoms have been aligned in the magnet, pulses of very specific radio wave frequencies are used to knock them back out of alignment. The hydrogen atoms alternately absorb and emit radio wave energy, vibrating back and forth between their resting (magnetized) state and their agitated (radio pulse) state. This comprises the "resonance" part of MRI.

The MRI equipment records the duration, strength, and source location of the signals emitted by the atoms as they relax and translates the data into an image on a television monitor. The state of hydrogen in diseased tissue differs from healthy tissue of the same type, making MRI particularly good at identifying tumors and other lesions. In some cases, chemical agents such as gadolinium can be injected to improve the contrast between healthy and diseased tissue.

A single MRI exposure produces a two-dimensional image of a slice through the entire target area. A series of these image slices closely spaced (usually less than half an inch) makes a virtual three-dimensional view of the area.

Magnetic resonance spectroscopy (MRS) is different from MRI because MRS uses a continuous band of radio wave frequencies to excite hydrogen atoms in a variety of chemical compounds other than water. These compounds absorb and emit radio energy at characteristic frequencies, or spectra, which can be used to identify them. Generally, a color image is created by assigning a color to each distinctive spectral emission. This comprises the "spectroscopy" part of MRS. MRS is still experimental and is available in only a few research centers.

Doctors primarily use MRS to study the brain and disorders, like epilepsy, **Alzheimer's disease**, brain tumors, and the effects of drugs on brain growth and metabolism. The technique is also useful in evaluating metabolic disorders of the muscles and nervous system.



MRI body scans of a man, woman, and child. (Simon Fraser, Photo Researchers. Reproduced by permission.)

Magnetic resonance angiography (MRA) is another variation on standard MRI. MRA, like other types of angiography, looks specifically at fluid flow within the blood (vascular) system, but does so without the injection of dyes or radioactive tracers. Standard MRI cannot make a good picture of flowing blood, but MRA uses specific radio pulse sequences to capture usable signals. The technique is generally used in combination with MRI to obtain images that show both vascular structure and flow within the brain and head in cases of stroke, or when a blood clot or aneurysm is suspected.

Regardless of the exact type of MRI planned, or area of the body targeted, the procedure involved is basically the same and occurs in a special MRI suite. The patient lies back on a narrow table and is made as comfortable as possible. Transmitters are positioned on the body and the cushioned table that the patient is lying on moves into a long tube that houses the magnet. The tube is as long as an average adult lying down, and the tube is narrow and open at both ends. Once the area to be examined has been properly positioned, a radio pulse is applied. Then a two-dimensional image corresponding to one slice through the area is made. The table then moves a fraction of an inch and the next image is made. Each image exposure takes several seconds and the entire exam will last anywhere from 30-90 minutes. During this time, the patient is not allowed to move. If the patient moves during the scan, the picture will not be clear.

Depending on the area to be imaged, the radio-wave transmitters will be positioned in different locations.

- For the head and neck, a helmet-like hat is worn.
- For the spine, chest, and abdomen, the patient will be lying on the transmitters.
- For the knee, shoulder, or other joint, the transmitters will be applied directly to the joint.

Additional probes will monitor vital signs (like pulse, respiration, etc.).

The process is very noisy and confining. The patient hears a thumping sound for the duration of the procedure. Since the procedure is noisy, music supplied via earphones is often provided. Some patients get anxious or panic because they are in the small, enclosed tube. This is why vital signs are monitored and the patient and medical team can communicate between each other. If the chest or abdomen are to be imaged, the patient will be asked to hold his/her breath as each exposure is made. Other instructions may be given to the patient, as needed. In many cases, the entire examination will be performed by an MRI operator who is not a doctor. However, the supervising radiologist should be available to consult as necessary during the exam, and will view and interpret the results sometime later.

Preparation

In some cases (such as for MRI brain scanning or an MRA), a chemical designed to increase image contrast may be given by the radiologist immediately before the exam. If a patient suffers from **anxiety** or claustrophobia, drugs may be given to help the patient relax.

The patient must remove all metal objects (watches, jewelry, eye glasses, hair clips, etc.). Any magnetized objects (like credit and bank machine cards, audio tapes, etc.) should be kept far away from the MRI equipment because they can be erased. The patient cannot bring their wallet or keys into the MRI machine. The patient may be asked to wear clothing without metal snaps, buckles, or zippers, unless a medical gown is worn during the procedure. The patient may be asked to remove any hair spray, hair gel, or cosmetics that may interfere with the scan.

Aftercare

No aftercare is necessary, unless the patient received medication or had a reaction to a contrast agent. Normally, patients can immediately return to their daily activities. If the exam reveals a serious

KEY TERMS

Angiography—Any of the different methods for investigating the condition of blood vessels, usually via a combination of radiological imaging and injections of chemical tracing and contrasting agents.

Diffusion tensor imaging (DTI)—A refinement of magnetic resonance imaging that allows the doctor to measure the flow of water and track the pathways of white matter in the brain. DTI is able to detect abnormalities in the brain that do not show up on standard MRI scans.

Gadolinium—A very rare metallic element useful for its sensitivity to electromagnetic resonance, among other things. Traces of it can be injected into the body to enhance the MRI pictures.

Hydrogen—The simplest, most common element known in the universe. It is composed of a single electron (negatively charged particle) circling a nucleus consisting of a single proton (positively charged particle). It is the nuclear proton of hydrogen that makes MRI possible by reacting resonantly to radio waves while aligned in a magnetic field.

Ionizing radiation—Electromagnetic radiation that can damage living tissue by disrupting and destroying individual cells. All types of nuclear decay radiation (including x rays) are potentially ionizing. Radio waves do not damage organic tissues they pass through.

Magnetic field—The three-dimensional area surrounding a magnet, in which its force is active. During MRI, the patient's body is permeated by the force field of a superconducting magnet.

Radio waves—Electromagnetic energy of the frequency range corresponding to that used in radio communications, usually 10,000 cycles per second to 300 billion cycles per second. Radio waves are the same as visible light, x rays, and all other types of electromagnetic radiation, but are of a higher frequency.

condition that requires more testing and/or treatment, appropriate information and counseling will be needed.

Risks

MRI poses no known health risks to the patient and produces no physical side effects. Again, the

potential effects of MRI on an unborn baby are not well known. Any woman who is, or may be, pregnant, should carefully discuss this issue with her doctor and radiologist before undergoing a scan.

Normal results

A normal MRI, MRA, or MRS result is one that shows the patient's physical condition to fall within normal ranges for the target area scanned.

Abnormal results

Generally, MRI is prescribed only when serious symptoms and/or negative results from other tests indicate a need. There often exists strong evidence of a condition that the scan is designed to detect and assess. Thus, the results will often be abnormal, confirming the earlier diagnosis. At that point, further testing and appropriate medical treatment is needed. For example, if the MRI indicates the presence of a brain tumor, an MRS may be prescribed to determine the type of tumor so that aggressive treatment can begin immediately without the need for a surgical biopsy.

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ORGANIZATIONS

American College of Radiology. 1891 Preston White Drive, Reston, VA 22091. (800) 227-5463. < <http://www.acr.org> >.

American Society of Radiologic Technologists. 15000 Central Ave. SE, Albuquerque, NM 87123-3917. (505) 298-4500. < <http://www.asrt.org> >.

Center for Devices and Radiological Health. United States Food and Drug Administration. 1901 Chapman Ave., Rockville, MD 20857. (301) 443-4109. < <http://www.fda.gov/cdrh> >.

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Magnetic resonance spectroscopy see
Magnetic resonance imaging

Major depression see **Depressive disorders**

Major tranquilizers see **Antipsychotic drugs**

Malabsorption syndrome

Definition

Malabsorption syndrome is an alteration in the ability of the intestine to absorb nutrients adequately into the bloodstream. It may refer to malabsorption of one specific nutrient or for specific carbohydrates, fats, or trace elements (micronutrients).

Causes and symptoms

Protein, fats, and carbohydrates (macronutrients) normally are absorbed in the small intestine; the small bowel also absorbs about 80% of the eight to ten liters of fluid ingested daily. There are many different conditions that affect fluid and nutrient absorption by the intestine. A fault in the digestive process may result from failure of the body to produce the enzymes needed to digest certain foods. Congenital structural defects or diseases of the pancreas, gall bladder, or liver may alter the digestive process. Inflammation, infection, injury, or surgical removal of portions of the intestine may also result in absorption problems; reduced length or surface area of intestine available for fluid and nutrient absorption can result in malabsorption. **Radiation therapy** may injure the mucosal lining of the intestine, resulting in **diarrhea** that may not become evident until several years later. The use of some **antibiotics** can also affect the bacteria that normally live in the intestine and affect intestinal function.

Risk factors for malabsorption syndrome include:

- premature birth
- family history of malabsorption or **cystic fibrosis**
- use of certain drugs, such as mineral oil or other **laxatives**
- travel to foreign countries
- intestinal surgery, including bowel transplantation
- excess alcohol consumption.

The most common symptoms of malabsorption include:

- Anemia, with weakness and **fatigue** due to inadequate absorption of vitamin B₁₂, iron, and **folic acid**
- Diarrhea, steatorrhea (excessive amount of fat in the stool), and abdominal distention with cramps, bloating, and gas due to impaired water and carbohydrate absorption, and irritation from unabsorbed fatty acids. The individual may also report explosive diarrhea with greasy, foul-smelling stools.
- **Edema** (fluid retention in the body's tissues) due to decreased protein absorption
- Malnutrition and weight loss due to decreased fat, carbohydrate, and protein absorption. Weight may be 80–90% of usual weight despite increased oral intake of nutrients.
- Muscle cramping due to decreased vitamin D, calcium, and potassium levels
- Muscle wasting and atrophy due to decreased protein absorption and metabolism
- Perianal skin burning, **itching**, or soreness due to frequent loose stools.

Irregular heart rhythms may also result from inadequate levels of potassium and other electrolytes. Blood clotting disorders may occur due to a **vitamin K deficiency**. Children with malabsorption syndrome often exhibit a failure to grow and thrive.

Several disorders can lead to malabsorption syndrome, including cystic fibrosis, chronic **pancreatitis**, **lactose intolerance**, and gluten enteropathy (nontropical sprue.)

Tropical sprue is a malabsorptive disorder that is uncommon in the United States, but seen more often in people from the Caribbean, India, or southeast Asia. Although its cause is unknown, it is thought to be related to environmental factors, including infection, intestinal parasites, or possibly the consumption of certain food toxins. Symptoms often include a sore tongue, anemia, weight loss, along with diarrhea and passage of fatty stools.

Whipple's disease is a relatively rare malabsorptive disorder, affecting mostly middle-aged men. The cause is thought to be related to bacterial infection, resulting in nutritional deficiencies, chronic low-grade **fever**, diarrhea, joint **pain**, weight loss, and darkening of the skin's pigmentation. Other organs of the body may be affected, including the brain, heart, lungs, and eyes.

Short bowel syndromes—which may be present at birth (congenital) or the result of surgery—reduce the surface area of the bowel available to absorb nutrients and can also result in malabsorption syndrome. Congenital short bowel syndrome occurs in about 24 out of 100,000 live births and has a high mortality rate (about 38%).

Diagnosis

The diagnosis of malabsorption syndrome and identification of the underlying cause can require extensive diagnostic testing. The first phase involves a thorough medical history and **physical examination** by a physician, who will then determine the appropriate laboratory studies and x rays to assist in diagnosis. A 72-hour stool collection may be ordered for fecal fat measurement; increased fecal fat in the stool collected indicates malabsorption. A biopsy of the small intestine may be done to assist in differentiating between malabsorption syndrome and small bowel disease. Ultrasound, computed tomography scan (CT scan), **magnetic resonance imaging** (MRI), **barium enema**, or other x rays to identify abnormalities of the gastrointestinal tract and pancreas may also be ordered.

A newer method of obtaining diagnostic information about the small intestine was approved by the Food and Drug Administration (FDA) in 2001. Known as the M2A Imaging System, the device was developed by a company in Atlanta, Georgia. The M2A system consists of an imaging capsule, a portable belt-pack image receiver and recorder, and a specially modified computer. The patient swallows the capsule, which is the size of a large pill. A miniature lens in the capsule transmits images through an antenna/transmitter to the belt-pack receiver, which the patient wears under ordinary clothing as he or she goes about daily activities. The belt-pack recording device is returned after seven or eight hours to the doctor, who then examines the images recorded as a digital video. The capsule itself is simply allowed to pass through the digestive tract.

Preparation requires only **fasting** the night before the M2A examination and taking nothing but clear liquids for two hours after swallowing the capsule.

After four hours the patient can eat food without interfering with the test. As of the early 2000s, the M2A system is used to evaluate gastrointestinal bleeding from unknown causes, inflammatory bowel disease, some malabsorption syndromes, and to monitor surgical patients following small-bowel transplantation.

Laboratory studies of the blood may include:

- Serum cholesterol. May be low due to decreased fat absorption and digestion.
- Serum sodium, potassium, and chloride. May be low due to electrolyte losses with diarrhea.
- Serum calcium. May be low due to vitamin D and amino acid malabsorption.
- Serum protein and albumin. May be low due to protein losses.
- Serum vitamin A and carotene. May be low due to bile salt deficiency and impaired fat absorption.
- D-xylose test. Decreased excretion may indicate malabsorption.
- Schilling test. May indicate malabsorption of vitamin B₁₂.

Treatment

Fluid and nutrient monitoring and replacement is essential for any individual with malabsorption syndrome. Hospitalization may be required when severe fluid and electrolyte imbalances occur. Consultation with a dietitian to assist with nutritional support and meal planning is helpful. If the patient is able to eat, the diet and supplements should provide bulk and be rich in carbohydrates, proteins, fats, **minerals**, and **vitamins**. The patient should be encouraged to eat several small, frequent meals throughout the day, avoiding fluids and foods that promote diarrhea. Intake and output should be monitored, along with the number, color, and consistency of stools.

The individual with malabsorption syndrome must be monitored for **dehydration**, including dry tongue, mouth and skin; increased thirst; low, concentrated urine output; or feeling weak or dizzy when standing. Pulse and blood pressure should be monitored, observing for increased or irregular pulse rate, or **hypotension** (low blood pressure). The individual should also be alert for signs of nutrient, vitamin, and mineral depletion, including **nausea** or **vomiting**; fissures at corner of mouth; fatigue or weakness; dry, pluckable hair; easy bruising; **tingling** in fingers or toes; and **numbness** or burning sensation in legs or feet. Fluid volume excess, as a result of diminished

KEY TERMS

Anemia—A decrease in the number of red blood cells in the bloodstream, characterized by pallor, loss of energy, and generalized weakness.

Atrophy—A wasting away of a tissue or organ, often because of insufficient nutrition.

Biopsy—A tissue sample removed from the body for examination under the microscope.

Cystic fibrosis—A hereditary genetic disorder that occurs most often in Caucasians. Thick, sticky secretions from mucus-producing glands cause blockages in the pancreatic ducts and the airways.

Edema—From the Greek word meaning swelling, an excessive accumulation of fluid in the tissue spaces. Excessive generalized edema may also be referred to as ascites.

Gluten enteropathy—A hereditary malabsorption disorder caused by sensitivity to gluten, a protein found in wheat, rye, barley, and oats. Also called non-tropical sprue or celiac disease.

Intestines—The intestines, also known as the bowels, are divided into the large and small intestines. They extend from the stomach to the anus.

Short bowel syndrome—A condition in which the bowel is not as long as normal, either because of surgery or because of a congenital defect. Because the bowel has less surface area to absorb nutrients, it can result in malabsorption syndrome.

Steatorrhea—An excessive amount of fat in the stool.

Trace elements—A group of elements that are present in the human body in very small amounts but are nonetheless important to good health. They include chromium, copper, cobalt, iodine, iron, selenium, and zinc. Trace elements are also called micronutrients.

protein stores, may require fluid intake restrictions. The physician should also be notified of any **shortness of breath**.

Other specific medical management for malabsorption syndrome is dependent upon the cause. Treatment for tropical sprue consists of folic acid supplements and long-term antibiotics. Depending on the severity of the disorder, this treatment may be continued for six months or longer. Whipple's disease also may require long-term use of antibiotics, such as tetracycline.

Management of some individuals with malabsorption syndrome may require injections of vitamin B₁₂ and oral iron supplements. The doctor may also prescribe enzymes to replace missing intestinal enzymes, or antispasmodics to reduce abdominal cramping and associated diarrhea. People with cystic fibrosis and chronic pancreatitis require pancreatic supplements. Those with lactose intolerance or gluten enteropathy (nontropical sprue) will have to modify their **diets** to avoid foods that they cannot properly digest.

Prognosis

The expected course for the individual with malabsorption syndrome varies depending on the cause. The onset of symptoms may be slow and difficult to diagnose. Treatment may be long, complicated, and changed often for optimal effectiveness. Patience and a positive attitude are important in controlling or curing the disorder. Careful monitoring is necessary to prevent additional illnesses caused by nutritional deficiencies.

Resources

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ORGANIZATIONS

National Digestive Diseases Information Clearinghouse. 2 Information Way, Bethesda, MD 20892-3570.

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Malaria

Definition

Malaria is a serious infectious disease spread by certain mosquitoes. It is most common in tropical climates. It is characterized by recurrent symptoms of chills, **fever**, and an enlarged spleen. The disease can be treated with medication, but it often recurs. Malaria is endemic (occurs frequently in a particular locality) in many third world countries. Isolated, small outbreaks sometimes occur within the boundaries of the United States.

Description

Malaria is a growing problem in the United States. Although only about 1400 new cases were reported in the United States and its territories in 2000, many involved returning travelers. In addition, locally transmitted malaria has occurred in California, Florida, Texas, Michigan, New Jersey, and New York City. While malaria can be transmitted in blood, the American blood supply is not screened for malaria. Widespread malarial epidemics are far less likely to occur in the United States, but small localized epidemics could return to the Western world. As of late 2002, primary care physicians are being advised to screen returning travelers with fever for malaria, and a team of public health doctors in Minnesota is recommending screening immigrants, refugees, and international adoptees for the disease—particularly those from high-risk areas.

The picture is far more bleak, however, outside the territorial boundaries of the United States. A recent government panel warned that disaster looms over Africa from the disease. Malaria infects between 300 and 500 million people every year in Africa, India, southeast Asia, the Middle East, Oceania, and Central and South America. A 2002 report stated that malaria kills 2.7 million people each year, more than 75 percent of them African children under the age of five. It is predicted that within five years, malaria will kill about as many people as does **AIDS**. As many as half a billion people worldwide are left with chronic anemia

due to malaria infection. In some parts of Africa, people battle up to 40 or more separate episodes of malaria in their lifetimes. The spread of malaria is becoming even more serious as the parasites that cause malaria develop resistance to the drugs used to treat the condition. In late 2002, a group of public health researchers in Thailand reported that a combination treatment regimen involving two drugs known as dihydroartemisinin and azithromycin shows promise in treating multidrug-resistant malaria in south-east Asia.

Causes and symptoms

Human malaria is caused by four different species of a parasite belonging to genus *Plasmodium*: *Plasmodium falciparum* (the most deadly), *Plasmodium vivax*, *Plasmodium malariae*, and *Plasmodium ovale*. The last two are fairly uncommon. Many animals can get malaria, but human malaria does not spread to animals. In turn, animal malaria does not spread to humans.

A person gets malaria when bitten by a female mosquito who is looking for a blood meal and is infected with the malaria parasite. The parasites enter the blood stream and travel to the liver, where they multiply. When they re-emerge into the blood, symptoms appear. By the time a patient shows symptoms, the parasites have reproduced very rapidly, clogging blood vessels and rupturing blood cells.

Malaria cannot be casually transmitted directly from one person to another. Instead, a mosquito bites an infected person and then passes the infection on to the next human it bites. It is also possible to spread malaria via contaminated needles or in blood transfusions. This is why all blood donors are carefully screened with questionnaires for possible exposure to malaria.

It is possible to contract malaria in non-endemic areas, although such cases are rare. Nevertheless, at least 89 cases of so-called airport malaria, in which travelers contract malaria while passing through crowded airport terminals, have been identified since 1969.

The amount of time between the mosquito bite and the appearance of symptoms varies, depending on the strain of parasite involved. The incubation period is usually between 8 and 12 days for falciparum malaria, but it can be as long as a month for the other types. Symptoms from some strains of *P. vivax* may not appear until 8–10 months after the mosquito bite occurred.

The primary symptom of all types of malaria is the “malaria ague” (chills and fever). In most cases, the fever has three stages, beginning with uncontrollable shivering for an hour or two, followed by a rapid spike in temperature (as high as 106°F), which lasts three to six hours. Then, just as suddenly, the patient begins to sweat profusely, which will quickly bring down the fever. Other symptoms may include **fatigue**, severe **headache**, or **nausea and vomiting**. As the sweating subsides, the patient typically feels exhausted and falls asleep. In many cases, this cycle of chills, fever, and sweating occurs every other day, or every third day, and may last for between a week and a month. Those with the chronic form of malaria may have a relapse as long as 50 years after the initial infection.

Falciparum malaria is far more severe than other types of malaria because the parasite attacks all red blood cells, not just the young or old cells, as do other types. It causes the red blood cells to become very “sticky.” A patient with this type of malaria can die within hours of the first symptoms. The fever is prolonged. So many red blood cells are destroyed that they block the blood vessels in vital organs (especially the kidneys), and the spleen becomes enlarged. There may be brain damage, leading to **coma** and convulsions. The kidneys and liver may fail.

Malaria in **pregnancy** can lead to premature delivery, **miscarriage**, or **stillbirth**.

Certain kinds of mosquitoes (called anopheles) can pick up the parasite by biting an infected human. (The more common kinds of mosquitoes in the United States do not transmit the infection.) This is true for as long as that human has parasites in his/her blood. Since strains of malaria do not protect against each other, it is possible to be reinfected with the parasites again and again. It is also possible to develop a chronic infection without developing an effective immune response.

Diagnosis

Malaria is diagnosed by examining blood under a microscope. The parasite can be seen in the blood smears on a slide. These blood smears may need to be repeated over a 72-hour period in order to make a diagnosis. Antibody tests are not usually helpful because many people developed antibodies from past infections, and the tests may not be readily available. A new laser test to detect the presence of malaria parasites in the blood was developed in 2002, but is still under clinical study.

Two new techniques to speed the laboratory diagnosis of malaria show promise as of late 2002. The first

is acridine orange (AO), a staining agent that works much faster (3–10 min) than the traditional Giemsa stain (45–60 min) in making the malaria parasites visible under a microscope. The second is a bioassay technique that measures the amount of a substance called histidine-rich protein II (HRP2) in the patient’s blood. It allows for a very accurate estimation of parasite development. A dip strip that tests for the presence of HRP2 in blood samples appears to be more accurate in diagnosing malaria than standard microscopic analysis.

Anyone who becomes ill with chills and fever after being in an area where malaria exists must see a doctor and mention their recent travel to endemic areas. A person with the above symptoms who has been in a high-risk area should insist on a blood test for malaria. The doctor may believe the symptoms are just the common flu virus. Malaria is often misdiagnosed by North American doctors who are not used to seeing the disease. Delaying treatment of falciparum malaria can be fatal.

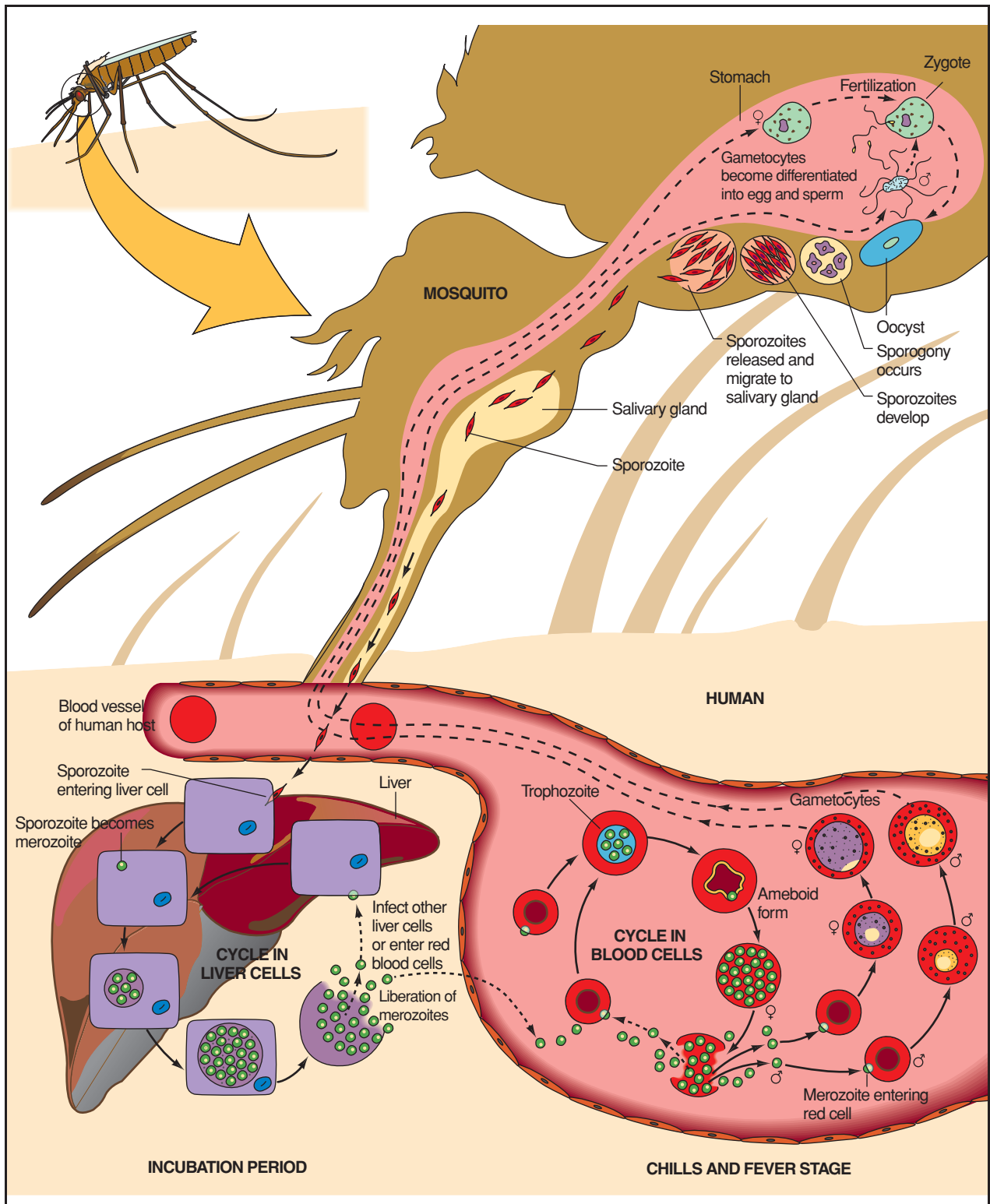
Treatment

Falciparum malaria is a medical emergency that must be treated in the hospital. The type of drugs, the method of giving them, and the length of the treatment depend on where the malaria was contracted and how sick the patient is.

For all strains except falciparum, the treatment for malaria is usually chloroquine (Aralen) by mouth for three days. Those falciparum strains suspected to be resistant to chloroquine are usually treated with a combination of quinine and tetracycline. In countries where quinine resistance is developing, other treatments may include clindamycin (Cleocin), mefloquin (Lariam), or sulfadoxone/pyrimethamine (Fansidar). Most patients receive an antibiotic for seven days. Those who are very ill may need intensive care and intravenous (IV) malaria treatment for the first three days.

Anyone who acquired falciparum malaria in the Dominican Republic, Haiti, Central America west of the Panama Canal, the Middle East, or Egypt can still be cured with chloroquine. Almost all strains of falciparum malaria in Africa, South Africa, India, and southeast Asia are now resistant to chloroquine. In Thailand and Cambodia, there are strains of falciparum malaria that have some resistance to almost all known drugs.

A patient with falciparum malaria needs to be hospitalized and given **antimalarial drugs** in different



The life cycle of *Plasmodium vivax*, the parasite that causes malaria. (Illustration by Hans & Cassady.)

combinations and doses depending on the resistance of the strain. The patient may need IV fluids, red blood cell transfusions, **kidney dialysis**, and assistance breathing.

A drug called primaquine may prevent relapses after recovery from *P. vivax* or *P. ovale*. These relapses are caused by a form of the parasite that remains in the liver and can reactivate months or years later.

Another new drug, halofantrine, is available abroad. While it is licensed in the United States, it is not marketed in this country and it is not recommended by the Centers for Disease Control and Prevention in Atlanta.

Alternative treatments

The Chinese herb qinghaosu (the Western name is artemisinin) has been used in China and southeast Asia to fight severe malaria, and became available in Europe in 1994. Because this treatment often fails, it is usually combined with another antimalarial drug (mefloquine) to boost its effectiveness. It is not available in the United States and other parts of the developed world due to fears of its toxicity, in addition to licensing and other issues.

A Western herb called wormwood (*Artemisia annua*) that is taken as a daily dose can be effective against malaria. Protecting the liver with herbs like goldenseal (*Hydrastis canadensis*), Chinese goldenthread (*Coptis chinensis*), and milk thistle (*Silybum marianum*) can be used as preventive treatment. Preventing mosquitoes from biting you while in the tropics is another possible way to avoid malaria.

As of late 2002, researchers are studying a traditional African herbal remedy against malaria. Extracts from *Microglossa pyrifolia*, a trailing shrub belonging to the daisy family (Asteraceae), show promise in treating drug-resistant strains of *P. falciparum*.

Prognosis

If treated in the early stages, malaria can be cured. Those who live in areas where malaria is epidemic, however, can contract the disease repeatedly, never fully recovering between bouts of acute infection.

Prevention

Several researchers are currently working on a malarial vaccine, but the complex life cycle of the malaria parasite makes it difficult. A parasite has much more genetic material than a virus or bacterium.

For this reason, a successful vaccine has not yet been developed.

Malaria is an especially difficult disease to prevent by **vaccination** because the parasite goes through several separate stages. One recent promising vaccine appears to have protected up to 60% of people exposed to malaria. This was evident during field trials for the drug that were conducted in South America and Africa. It is not yet commercially available.

The World Health Association (WHO) has been trying to eliminate malaria for the past 30 years by controlling mosquitoes. Their efforts were successful as long as the pesticide DDT killed mosquitoes and antimalarial drugs cured those who were infected. Today, however, the problem has returned a hundred-fold, especially in Africa. Because both the mosquito and parasite are now extremely resistant to the insecticides designed to kill them, governments are now trying to teach people to take antimalarial drugs as a preventive medicine and avoid getting bitten by mosquitoes.

A newer strategy as of late 2002 involves the development of genetically modified non-biting mosquitoes. A research team in Italy is studying the feasibility of this means of controlling malaria.

Travelers to high-risk areas should use insect repellent containing DEET for exposed skin. Because DEET is toxic in large amounts, children should not use a concentration higher than 35%. DEET should not be inhaled. It should not be rubbed onto the eye area, on any broken or irritated skin, or on children's hands. It should be thoroughly washed off after coming indoors.

Those who use the following preventive measures get fewer infections than those who do not:

- Between dusk and dawn, remain indoors in well-screened areas.
- Sleep inside pyrethrin or permethrin repellent-soaked mosquito nets.
- Wear clothes over the entire body.

Anyone visiting endemic areas should take antimalarial drugs starting a day or two before they leave the United States. The drugs used are usually chloroquine or mefloquine. This treatment is continued through at least four weeks after leaving the endemic area. However, even those who take antimalarial drugs and are careful to avoid mosquito bites can still contract malaria.

International travelers are at risk for becoming infected. Most Americans who have acquired

KEY TERMS

Artemisinins—A family of antimalarial products derived from an ancient Chinese herbal remedy. Two of the most popular varieties are artemether and artesunate, used mainly in southeast Asia in combination with mefloquine.

Chloroquine—An antimalarial drug that was first used in the 1940s, until the first evidence of quinine resistance appeared in the 1960s. It is now ineffective against *falciparum* malaria almost everywhere. However, because it is inexpensive, it is still the antimalarial drug most widely used in Africa. Native individuals with partial immunity may have better results with chloroquine than a traveler with no previous exposure.

Mefloquine—An antimalarial drug that was developed by the United States Army in the early 1980s. Today, malaria resistance to this drug has become a problem in some parts of Asia (especially Thailand and Cambodia).

Mefloquine—An antimalarial drug that was developed by the United States Army in the early 1980s. Today, malaria resistance to this drug has become a problem in some parts of Asia (especially Thailand and Cambodia).

Quinine—One of the first treatments for malaria, quinine is a natural product made from the bark of the *Cinchona* tree. It was popular until being superseded by the development of chloroquine in the 1940s. In the wake of widespread chloroquine resistance, however, it has become popular again. Quinine, or its close relative quinidine, can be given intravenously to treat severe *Falciparum* malaria.

Sulfadoxone/pyrimethamine (Fansidar)—An antimalarial drug developed in the 1960s. It is the first drug tried in some parts of the world where chloroquine resistance is widespread. It has been associated with severe allergic reactions due to its sulfa component.

falciparum malaria were visiting sub-Saharan Africa; travelers in Asia and South America are less at risk. Travelers who stay in air conditioned hotels on tourist itineraries in urban or resort areas are at lower risk than backpackers, missionaries, and Peace Corps volunteers. Some people in western cities where malaria does not usually exist may acquire the infection from a mosquito carried onto a jet. This is called airport or runway malaria.

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Centers for Disease Control Travelers Hotline. (770) 332-4559.

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Malaria Foundation Page. <<http://www.malaria.org>>.

Carol A. Turkington
Rebecca J. Frey, PhD

Malaya see **Elephantiasis**

Male breast enlargement see **Gynecomastia**

Male condom see **Condom**

Male infertility see **Infertility**

Male pattern baldness see **Alopecia**

Malignant lymphomas

Definition

Lymphomas are a group of cancers in which cells of the lymphatic system become abnormal and start to grow uncontrollably. Because there is lymph tissue in many parts of the body, lymphomas can start in almost any organ of the body.

Description

The lymph system is made up of ducts or tubules that carry lymph to all parts of the body. Lymph is a milky fluid that contains the lymphocytes or white blood cells. These are the infection-fighting cells of the blood. Small pea-shaped organs are found along the network of lymph vessels. These are called the lymph nodes, and their main function is to make and store the lymphocytes. Clusters of lymph nodes are found in the pelvic region, underarm, neck, chest, and abdomen. The spleen (an organ in the upper abdomen), the tonsils, and the thymus (a small organ found beneath the breastbone) are part of the lymphatic system.

The lymphocyte is the main cell of the lymphoid tissue. There are two main types of lymphocytes: the T lymphocyte and the B lymphocyte. Lymphomas

develop from these two cell types. B cell lymphomas are more common among adults, while among children, the incidence of T and B cell lymphomas are almost equal.

The T and the B cell perform different jobs within the immune system. When an infectious bacterium enters the body, the B cell makes proteins called "antibodies." These antibodies attach themselves to the bacteria, and flag them for destruction by other immune cells. The T cells help protect the body against viruses. When a virus enters the cell, it generally produces certain proteins that are projected on the surface of the infected cell. T cells recognize these proteins and produce certain substances (cytokines) that destroy the infected cells. Some of the cytokines made by the T cells attract other cell types, which are capable of digesting the virus-infected cell. The T cells can also destroy some types of cancerous cells.

Lymphomas can be divided into two main types: Hodgkin's lymphoma or **Hodgkin's disease**, and non-Hodgkin's lymphomas. There are at least 10 types of non-Hodgkin's lymphomas. They are grouped (staged) by how aggressively they grow; slow growing (low grade), intermediate growing, and rapidly growing (high grade); and how far they spread.

A majority of non-Hodgkin's lymphomas begin in the lymph nodes. About 20% start in other organs, such as the lungs, liver or the gastrointestinal tract. Malignant lymphocytes multiply uncontrollably and do not perform their normal functions. Hence, the body's ability to fight infections is affected. In addition, these malignant cells may crowd the bone marrow, and, depending on the stage, prevent the production of normal red blood cells, white blood cells, and platelets. A low red blood cell count causes anemia, while a reduction in the number of platelets makes the person susceptible to excessive bleeding. Cancerous cells can also invade other organs through the circulatory system of the lymph, causing those organs to malfunction.

Causes and symptoms

The exact cause of non-Hodgkin's lymphomas is not known. However, the incidence has increased significantly in the recent years. Part of the increase is due to the **AIDS** epidemic. Individuals infected with the **AIDS** virus have a higher likelihood of developing non-Hodgkin's lymphomas. In general, males are at a higher risk for having non-Hodgkin's lymphomas than are females. The risk increases with age. Though it can strike people as young as 40, people between the ages of 60 and 69 are at the highest risk.

People exposed to certain pesticides and ionizing radiation have a higher than average chance of developing this disease. For example, an increased incidence of lymphomas has been seen in survivors of the atomic bomb explosion in Hiroshima, and in people who have undergone aggressive **radiation therapy**. People who suffer from immune-deficient disorders, as well as those who have been treated with immune suppressive drugs for heart or kidney transplants, and for conditions such as **rheumatoid arthritis** and autoimmune diseases, are at an increased risk for this disease.

There have been some studies that have shown a loose association between retroviruses, such as HTLV-I, and some rare forms of lymphoma. The Epstein-Barr virus has been linked to Burkitt's lymphoma in African countries. However, a direct cause-and-effect relationship has not been established.

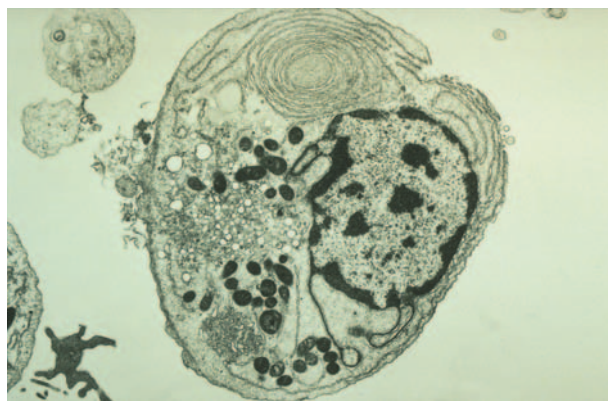
The symptoms of lymphomas are often vague and non-specific. Patients may experience loss of appetite, weight loss, **nausea**, **vomiting**, abdominal discomfort, and **indigestion**. The patient may complain of a feeling of fullness, which is a result of enlarged lymph nodes in the abdomen. Pressure or **pain** in the lower back is another symptom. In the advanced stages, the patient may have bone pain, headaches, constant coughing, and abnormal pressure and congestion in the face, neck, and upper chest. Some may have fevers and night sweats. In most cases, patients go to the doctor because of the presence of swollen glands in the neck, armpits, or groin area. Since all the symptoms are common to many other illnesses, it is essential to seek medical attention if any of the conditions persist for two weeks or more. Only a qualified physician can correctly diagnose if the symptoms are due to lymphoma or some other ailment.

Diagnosis

Like all cancers, lymphomas are best treated when found early. However, it is often difficult to diagnose lymphomas. There are no screening tests available, and, since the symptoms are non-specific, lymphomas are rarely recognized in their early stages. Detection often occurs by chance during a routine **physical examination**.

When the doctor suspects lymphoma, a complete medical history is taken, and a thorough physical examination is performed. Enlargement of the lymph nodes, liver, or spleen may suggest lymphomas. Blood tests will determine the cell counts and obtain information on how well the organs, such as the kidney and liver, are functioning.

A biopsy of the enlarged lymph node is the most definitive diagnostic tool for staging purposes. The



A malignant lymph cell. (Custom Medical Stock Photo. Reproduced by permission.)

doctor may perform a **bone marrow biopsy**. During the biopsy, a cylindrical piece of bone and marrow fluid is removed. They are generally taken out of the hipbone. These samples are sent to the laboratory for examination. In addition to diagnosis, the biopsy may also be repeated during the treatment phase of the disease to see if the lymphoma is responding to therapy.

Once the exact form of lymphoma is known, it is then staged to determine how aggressive it is, and how far it has spread. Staging is necessary to plan appropriate treatment.

Conventional imaging tests, such as x rays, **computed tomography scans** (CT scans), **magnetic resonance imaging**, and abdominal sonograms, are used to determine the extent of spread of the disease.

Lymphangiograms are x rays of the lymphatic system. In this procedure, a special dye is injected into the lymphatic channels through a small cut (incision) made in each foot. The dye is injected slowly over a period of three to four hours. This dye clearly outlines the lymphatic system and allows it to stand out. Multiple x rays are then taken and any abnormality, if present, is revealed.

Rarely, a lumbar puncture or a spinal tap is performed to check if malignant cells are present in the fluid surrounding the brain. In this test, the physician inserts a needle into the epidural space at the base of the spine and collects a small amount of spinal fluid for microscopic examination.

Treatment

Treatment options for lymphomas depend on the type of lymphoma and its present stage. In most cases,

treatment consists of **chemotherapy**, radiotherapy, or a combination of the two methods.

Chemotherapy is the use of anti-cancer drugs to kill **cancer** cells. In non-Hodgkin's lymphomas, combination therapy, which involves the use of multiple drugs, has been found more effective than single drug use. The treatment may last about six months, but in some cases may last as long as a year. The drugs may either be administered intravenously (through a vein) in the arm or given orally in the form of pills. If cancer cells have invaded the central nervous system, then chemotherapeutic drugs may be instilled, through a needle in the brain or back, into the fluid that surrounds the brain. This procedure is known as intrathecal chemotherapy.

Radiation therapy, where high-energy ionizing rays are directed at specific portions of the body, such as the upper chest, abdomen, pelvis, or neck, is often used for treatment of lymphomas. External radiation therapy, where the rays are directed from a source outside the body, is the most common mode of radiation treatment.

Bone marrow transplantation is used in cases where the lymphomas do not respond to conventional therapy, or in cases where the patient has had a relapse or suffers from recurrent lymphomas.

There are two ways of doing bone marrow transplantation. In a procedure called "allogeneic bone marrow transplant," a donor is found whose marrow matches that of the patient. The donor can be a twin (best match), a sibling, or a person who is not related at all. High-dose chemotherapy or radiation therapy is given to eradicate the lymphoma. The donor marrow is then given to replace the marrow destroyed by the therapy.

In "autologous bone marrow transplantation," some of the patient's own bone marrow is harvested, chemically purged, and frozen. High-dose chemotherapy and radiation therapy are given. The marrow that was harvested, purged, and frozen is then thawed and put back into the patient's body to replace the destroyed marrow.

A new treatment option for patients with lymphoma is known as "peripheral stem cell transplantation." In this treatment approach, cells that normally circulate in the blood are collected when the patient has normal blood counts taken, and these cells are saved via a process called "pheresis." Researchers are exploring whether these cells can be used to restore the normal function and development of blood cells, rather than using a bone marrow transplant.

KEY TERMS

Antibodies—Proteins made by the B lymphocytes in response to the presence of infectious agents such as bacteria or viruses in the body.

Biopsy—The surgical removal and microscopic examination of living tissue for diagnostic purposes.

Growth factors (cytokines)—Chemicals made by the cells that act on other cells to stimulate or inhibit their function. Cytokines that stimulate growth are called "growth factors."

Prognosis

Like all cancers, the prognosis for lymphoma depends on the stage of the cancer, and the patient's age and general health. When all the different types and stages of lymphoma are considered together, only 50% of patients survive 5 years or more after initial diagnosis. This is because some types of lymphoma are more aggressive than others.

The survival rate among children is definitely better than among older people. About 90% of the children diagnosed with early stage disease survive 5 years or more, while only 60-70% of adults diagnosed with low grade lymphomas survive for 5 years or more. The survival rate for children with the more advanced stages is about 75-85%, while among adults it is 40-60%.

Prevention

Although many cancers may be prevented by making diet and life style changes which reduce risk factors, there is currently no known way to prevent lymphomas. Protecting oneself from developing AIDS, which may be a risk factor for lymphomas, is the only preventive measure that can be practiced.

At present, there are no special tests that are available for early detection of non-Hodgkin's lymphomas. Paying prompt attention to the signs and symptoms of this disease, and seeing a doctor if the symptoms persist, are the best strategies for an early diagnosis of lymphoma. Early detection affords the best chance for a cure.

Resources

ORGANIZATIONS

American Cancer Society. 1599 Clifton Rd., NE, Atlanta, GA 30329-4251. (800) 227-2345. <<http://www.cancer.org>>.

Cancer Research Institute. 681 Fifth Ave., New York, N.Y. 10022. (800) 992-2623. <<http://www.cancerresearch.org>>.

Leukemia Society of America, Inc. 600 Third Ave., New York, NY 10016. (800) 955 4572. <<http://www.leukemia.org>>.

Lymphoma Research Foundation. 8800 Venice Boulevard, Suite 207, Los Angeles, CA 90034. (310) 204 7040.

National Cancer Institute. Building 31, Room 10A31, 31 Center Drive, MSC 2580, Bethesda, MD 20892-2580. (800) 422-6237. <<http://www.nci.nih.gov>>.

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Lata Cherath, PhD

Malignant melanoma

Definition

Malignant melanoma is a type of **cancer** arising from the melanocyte cells of the skin. Melanocytes are cells in the skin that produce a pigment called melanin. Malignant melanoma develops when the melanocytes no longer respond to normal control mechanisms of cellular growth. They may then invade nearby structures or spread to other organs in the body (metastasis), where again they invade and compromise the function of that organ.

Description

Melanocytes are derived from a structure in the human embryo called the neural crest. They are distributed in the epidermis and thus are found throughout the skin. They produce a brown pigment known as melanin and are responsible for racial variation in skin color as well as the color of **moles**. Malignant degeneration of the melanocyte gives rise to the tumor known as melanoma, which has four subtypes. These are: superficial spreading, nodular, lentigo maligna, and acral lentiginous melanomas, accounting for 70%, 15% to 30%, 4% to 10%, and 2% to 8% of cases, respectively. Malignant melanoma may develop

anywhere on the body. In men, it is most common on the trunk. In women, it is most common on the back or legs. The subtype also may influence where the tumor develops; lentigo melanoma is more common on the face while acral lentiginous melanoma is more common on the palms of the hand, soles of the feet, or in the nail beds.

The locally invasive characteristic of this tumor involves vertical penetration through the skin and into the dermis and subcutaneous (under-the-skin) tissues of the malignant melanocytes. With the exception of the nodular variety of melanoma, there is often a phase of radial or lateral growth associated with these tumors. Since it is the vertical growth that characterizes the malignancy, the nodular variant of melanoma carries the worst prognosis. Fortunately, the superficial spreading type is most common.

The primary tumor begins in the skin, often from the melanocytes of a pre-existing mole. Once it becomes invasive, it may progress beyond the site of origin to the regional lymph nodes or travel to other organ systems in the body and become systemic in nature.

Lymph is the clear, protein-rich fluid that bathes the cells throughout our body. Lymph will work its way back to the bloodstream via small channels known as lymphatics. Along the way, the lymph is filtered through cellular stations known as nodes, thus they are called lymph nodes. Nearly all organs in the body have a primary lymph node group filtering the tissue fluid, or lymph, that comes from that organ. Different areas of the skin have different primary nodal stations. For the leg, they are in the groin. For the arm, the armpit or axilla. For the face, it is the neck. Depending where on the torso the tumor develops, it may drain into one groin or armpit, or both.

Cancer, as it invades in its place of origin, may also work its way into blood vessels. If this occurs, it provides yet another route for the cancer to spread to other organs of the body. When the cancer spreads elsewhere in the body, it has become systemic in extent and the tumor growing elsewhere is known as a metastasis.

Untreated malignant melanoma follows a classic progression. It begins and grows locally, penetrating vertically. It may be carried via the lymph to the regional nodes, known as regional metastasis. It may go from the lymph to the bloodstream or penetrate blood vessels, directly allowing it a route to go elsewhere in the body. When systemic disease or distant metastasis occur, melanoma commonly involves the lung, brain, liver, or occasionally bone. The

malignancy causes **death** when its uncontrolled growth compromises vital organ function.

Of the anticipated new cases of cancer for the year 2003 in the United States, malignant melanoma will account for 5% of malignancies in men and 4% in women, being the sixth most common cancer in men and the seventh in women. It is estimated there will be 553,400 total cancer deaths in the United States in 2001. Malignant melanoma will account for 7,800 of these deaths, for an incidence of 1.5% of total deaths related to cancer.

The incidence of primary cutaneous malignant melanoma has been steadily increasing, possibly related to increase of sun exposure. Currently, the risk is about 13 per 100,000 of the population. It affects all age groups but is most commonly seen in patients between 30 and 60 years of age.

Sun exposure definitely increases the risk of developing melanoma, particularly in older males. The melanocytes are part of the integument's photoprotective mechanism; in response to sunlight, they produce melanin that has a protective role from the sun's ultraviolet rays. For Caucasians, the amount of melanin present in the skin is directly related to sun exposure. However, it is not so much the total sun exposure that seems important, rather it is the history of **sunburn**, (especially if severe or at an early age), that correlates with the increased risk. On this basis populations of fair-skinned people living in areas of high sun exposure such as the southwest United States or Australia are subject to increased risk. Malignant melanoma also affects non-Caucasians—though sun exposure probably does not play a role—at a rate of 10% that of Caucasians. The most common form of melanoma in African Americans is acral lentiginous melanoma.

Malignant melanoma may arise in the skin anywhere on the body. It is estimated that 50%–70% develop spontaneously while the remainder start in a pre-existing mole.

Causes and symptoms

The predisposing causes to the development of malignant melanoma are environmental and genetic. The environmental factor is excessive sun exposure. There are also genetically transmitted familial syndromes with alterations in the CDKN2A gene, which encodes for the tumor-suppressing proteins p16 and p19. In 2003 a group of Swedish researchers reported that 63 out of a group of 71 melanoma patients, or 89% of the group, had mutations in either the NRAS or the BRAF gene. The researchers found that these

mutations occur at an early point in the development of melanoma and remain as the tumor progresses.

As of early 2003, some researchers think there may be two pathways to malignant melanoma, one involving exposure to sunlight and the other with melanocyte proliferation triggered by other factors. This hypothesis is based on the difference in distribution of moles on the body between patients who develop melanomas on the face and neck, and those who develop melanomas on the trunk.

A small percentage of melanomas arise within burn scar tissue. As of 2003, researchers do not fully understand the relationship between deep **burns** and an increased risk of skin cancer.

As mentioned previously, melanin production in fair-skinned people is induced by sun exposure. An exposure substantial enough to result in a mild sunburn will be followed by melanin producing a tan that may last a few weeks. Both ultraviolet radiation and damaging oxygen radicals caused by sun exposure may damage cells, particularly their DNA. It is suspected that this damage induces mutations that result in the development of malignant melanoma. Though these mutations are alterations of the genome causing the melanoma, they are environmentally induced and account for sporadic or spontaneous cases of this disease.

A positive family history of one or two first-degree relatives having had melanoma substantially increases the risk on a genetic basis. A family tendency is observed in 8% to 12% of patients. There is a syndrome known as the dysplastic (atypical) nevus syndrome that is characterized by atypical moles with bothersome clinical features in children under age 10. Such individuals have to be observed closely for the development of malignant melanoma. Chromosome 9p has been identified as being involved in familial predisposition. There are mutations in up to 50% of familial melanoma patients of the tumor-suppressing gene CDKN2A. The actual number of moles increases risk, but the size of the moles needs be considered. Those with 10 larger moles of over 1 cm (0.4 in.) are at more risk than those with a higher number (50-99) of smaller moles. Finally, when a child is born with a large congenital mole, careful observation for change is appropriate because of increased risk.

An excellent way of identifying changes of significance in a mole is the ABCDE rule:

- Asymmetry
- Border irregularity
- Color variegation

- Diameter greater than 6 mm (0.24 in)
- Elevation above surrounding tissue.

Notice that three of the criteria refer to variability of the lesion (color variegation refers to areas of light color and black scattered within the mole). Thus small, uniform regular lesions have less cause for concern. It is important to realize that change in a mole or the rapid development of a new one are very important symptoms.

Another summary of important changes in a mole is the Glasgow 7-point scale. The symptoms and signs below can occur anywhere on the skin, including the palms of the hands, soles of the feet, and also the nail beds:

- Change in size
- Change in shape
- Change in color
- Inflammation
- Crusting and bleeding
- Sensory change
- Diameter greater than 7 mm (0.28 in.)

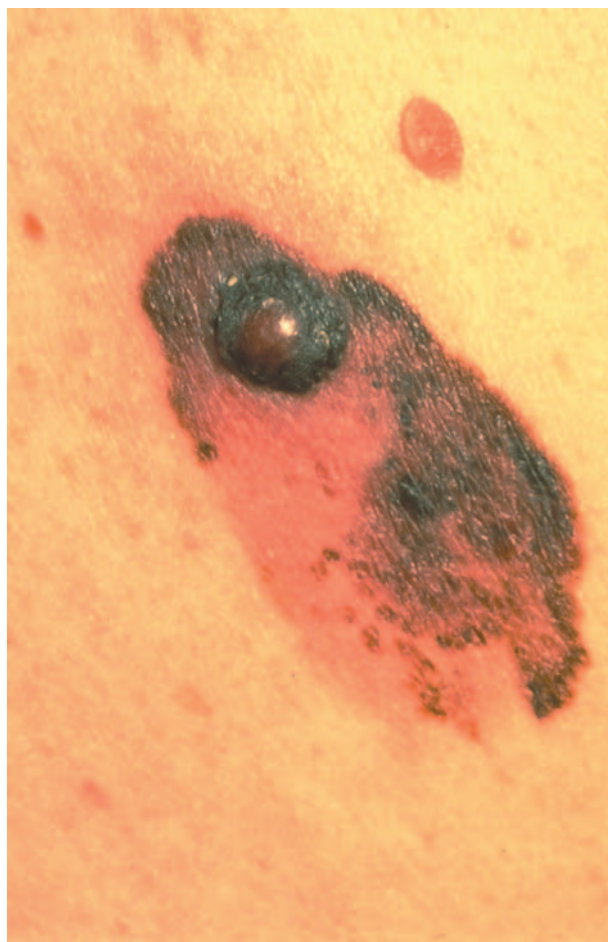
In this scheme, change is emphasized along with size. Bleeding and sensory changes are relatively late symptoms.

Symptoms related to the presence of regional disease are mostly those of nodules or lumps in the areas containing the lymph nodes draining the area. Thus nodularity can be found in the armpit, the groin, or the neck if regional nodes are involved. There is also a special type of metastasis that can occur regionally with malignant melanoma; it is known as an in-transit metastasis. If the melanoma is spreading through the lymph system, some of the tumor may grow there, resulting in a nodule part way between the primary site and the original lymph node. These in-transit metastasis are seen both at the time of original presentation or later after primary treatment has been rendered, the latter being a type of recurrence.

Finally, in those who either present with or progress to widespread or systemic disease, symptoms and signs are related to the affected organ. Thus neurologic problems, lung problems, or liver problems develop depending on the organ involved.

Diagnosis

None of the clinical signs or symptoms discussed above are absolute indications that a patient has malignant melanoma. The actual diagnosis is



A close-up image of a malignant melanoma on a patient's back. (Custom Medical Stock Photo. Reproduced by permission.)

accomplished by biopsy, a procedure that removes tissue to examine under a microscope. It is important that the signs and symptoms are used to develop a suspicion of the diagnosis because the way the biopsy is performed for melanoma may be different than for other lesions of the skin.

The doctor may also use a dermatoscope to examine the mole prior to removal. The dermatoscope, which can be used to distinguish between benign moles and melanomas, is an instrument that resembles an ophthalmoscope. An immersion oil is first applied to the mole to make the outer layers of skin transparent.

When dealing with an early malignant melanoma, it is very important to establish the exact thickness of penetration of the primary tumor. Any biopsy that does not remove the full vertical extent of the primary is inadequate. Therefore, if a skin lesion is suspicious, full thickness excisional biopsy is the approach

recommended. Shave biopsies and biopsies that remove only a portion of the suspect area are inappropriate. Often, in an early case, the excision involves just the suspicious lesion with minimal normal skin, but it should be a full vertical excision of the skin. If a melanoma is diagnosed, further treatment of this area will often be necessary but does not compromise outcome (prognosis). In some special areas of the body, minor modifications may be necessary about initial total excision, but full thickness excision should always be the goal. (See staging, below.)

Once the diagnosis is obtained, careful examination of the patient for regional lymph node involvement should be done. A careful review to uncover any symptoms of widespread disease is also appropriate.

The more common patient has an early melanoma, and extensive testing is not usually warranted. Routine testing in this situation involves a complete **blood count**, a **chest x ray**, and determinations of blood enzymes including lactic dehydrogenase and alkaline phosphatase.

If the patient has signs or symptoms of more advanced disease, or if the lesion's depth of penetration is sizeable, further imaging studies may be appropriate. These would involve CAT scans of the abdomen, the chest, or regional nodal areas, or a CT or MRI of the brain.

Treatment

The key to successful treatment is early diagnosis. Patients identified with localized, thin, small lesions (typified by superficial spreading subtype) nearly always survive. For those with advanced lesions, the outcome is poor in spite of progress in systemic therapy.

Clinical staging

Malignant melanoma is locally staged based on the depth of penetration through the skin and its appendages. There are two ways of looking at the depth of penetration. The Clarke system utilizes the layers of the dermis and the skin appendages present at that layer to identify the depth of penetration. The Breslow system uses the absolute measurement of depth. Though useful conceptually, the Clarke system is used less frequently because of the fact that skin is of different thickness in different regions of the body. The depth of penetration is much greater when the tumor reaches the subcutaneous fat when the skin involved is the back as opposed to the face. It turns out that the Breslow measurement is more

reproducible and thus more useful; therefore, for purposes here, depth of penetration by absolute measurement (Breslow) is used in local staging.

Stage I and stage II have no involvement of the regional lymph nodes and are thus localized to the site of origin. These stages are subdivided on the basis of penetration. Stage Ia is 0.75 mm or less (1 mm = 0.04 in), and Stage Ib is 0.75–1.5 mm penetration. Stage IIa is 1.5–4.0 mm and Stage IIb is over 4.0 mm or into the subcutaneous fat. In stage III and IV, there is disease beyond the primary site. Stage III is defined by the presence of in-transit or regional nodal metastasis or both. Stage IV is defined by the presence of distant metastasis.

Once the diagnosis of malignant melanoma has been established by biopsy and the stage has been identified using the results of the examination and studies, a treatment plan is developed. Melanoma is not cured unless it is diagnosed at a stage when it can be isolated and removed surgically. Considerations revolve around the extent of the local and regional nodal surgery for stages I through III. For stage IV patients, or those that are treated and then develop recurrence at distant sites, **chemotherapy** or immunotherapy is planned. Studies are in progress to improve the results from traditional chemotherapeutic regimens. Adjuvant therapy (auxiliary drug treatment used to make possibility of relapse less for those at high risk) is also considered.

Surgical therapy for the primary site is that of wide local removal of the skin including subcutaneous tissue surrounding the lesion. In the past, wide excisions were large and encompassed 2 in. of tissue in all directions wherever feasible. It has been shown that such wide local excisions are not necessary and the issue has become: how wide is enough? Studies from the World Health Organization Melanoma Group and by the Melanoma Intergroup Committee in the United States have provided general guidelines based on the depth of penetration of the melanoma. These guidelines and anatomic considerations need to be kept in mind by the surgeon.

The next issue in primary management is whether the patient should have the regional lymph nodes removed in addition to treatment of the primary tumor. The problems associated with the resection of regional lymph nodes are those of lifelong **edema** or swelling in the extremity. Though it does not occur in all patients (5% to 20%, depending on the extremity and extent of the dissection), it can be a disabling symptom. Certainly, if it could be ascertained that there was disease in the nodes, resection

(removal) would be appropriate. However, if there was no disease, the risk of edema should be avoided. In patients with no signs of regional disease, depth of penetration of the primary tumor helps guide the decision. If the tumor penetrates less than 1mm, dissection is not usually done. If it is 1-2 mm, node dissection may be done at the time of primary treatment or the patient may be observed and only undergo lymph node dissection if the area later shows signs of disease. If the patient has enlarged lymph nodes or the depth of the tumor has led to the evaluation by CAT scan showing enlarged nodes, resection of the nodes will be considered. In the latter case, more extensive imaging of the lung, liver, or brain may be appropriate to be sure the patient does not already have stage IV disease.

Questions related to which patients should have resection of regional lymph nodes have led to an intermediary procedure known as sentinel node mapping and biopsy. Intermediate thickness melanomas between 1 and 4 mm deep (0.04 and 0.16 in.) may have nodal involvement even if the examination and any other studies done are normal. If a radioisotope tracer or blue dye is injected into the area of the primary tumor, very shortly it will travel to the lymph nodes draining that area. These sentinel nodes are thus identifiable and are the most likely to harbor any regional metastatic disease. If these nodes alone are biopsied and are normal, the rest of the lymph node group can be spared. If they show microscopic deposits of tumor, then the full resection of the lymph node group may be completed. This procedure allows selection of those patients with intermediate thickness melanoma who will benefit from the regional lymph node dissection.

Patients with metastatic melanoma who do not respond well to other therapies may be candidates for treatment with aldesleukin. Aldesleukin is a form of interleukin, a specific kind of biological response modifier that promotes the development of T-cells. These cells are part of the lymphatic system and can directly interact with and fight cancer cells. Although aldesleukin is produced naturally in the body, its therapeutic form is developed via biotechnology in a laboratory setting. Treatment is considered palliative, which means that it provides comfort but does not produce a cure. Side effects, however, can be severe, and range from flu-like symptoms to whole-body infection (**sepsis**) and **coma**.

Some patients, such as those with IIb or stage III melanoma, are at high risk for the development of recurrence after treatment. Although these patients are clinically free of disease after undergoing primary

treatment, they are more likely to have some microscopic disease in the body that studies have not yet been able to identify. In an effort to decrease the rate of relapse, adjuvant therapy may be considered. Interferon alpha 2a is an agent that stimulates the immune system. This adjuvant therapy may slightly increase the duration of a patient's disease-free state and lengthen overall survival. However, interferon alpha 2a has high toxicity and patients may not tolerate the side effects.

Unfortunately, treatment for those patients who present with or go on to develop systemic disease usually fails; melanoma that has metastasized to the brain is particularly difficult to treat. The chemotherapeutic agent dacarbazine, or DTIC, seems to be the most active agent. Overall responses are noted in about 20% of patients, and they last only two to six months. Combination therapy may be an option. The regimen of DTIC + BCNU (carmustine) + cisplatin + tamoxifen delivers a response rate of 40%. Combining biologic or immunologic agents such as interferon with standard chemotherapeutic agents is under study and showing improved response rates, though toxicity is substantial and only the healthier, younger patients tolerate the treatment.

Some researchers are investigating the reasons why melanomas are so resistant to chemotherapy. One suggestion as of late 2003 is that the genes ordinarily responsible for apoptosis (cell self-destruction) do not function normally in melanomas. The development of new drugs to treat melanoma depends on a better understanding of the complex processes involved in apoptosis.

Alternative treatment

Though **radiation therapy** has a minimal role in the primary treatment of malignant melanoma, for patients who have metastatic disease, radiation may be helpful. This is true in patients who have developed tumor deposits in such areas as the brain or bone.

Prognosis

Almost all patients survive stage Ia malignant melanoma, and the survivorship for stage I overall is more than 90%. Survival drops in stage IIa to about 65% at five years and is worse yet for stage IIb at slightly over 50%. Stage III has a survival rate at 5 years of 10%–47%, depending on the size and number of regional nodes involved. Stage IV malignant melanoma is almost always a fatal disease.

KEY TERMS

Adjuvant therapy—Treatment given to patients who are at risk of having microscopic untreated disease present but have no obvious symptoms.

Dermis—The deeper portion or layer of the skin beneath the epidermis.

Dysplastic nevus syndrome—A familial syndrome characterized by the presence of multiple atypical appearing moles, often at a young age.

Epidermis—The uppermost layer of skin cells.

Genome—The genetic makeup of a cell, composed of DNA.

Immunotherapy—A form of treatment that uses biologic agents to enhance or stimulate normal immune function.

Integument—The medical name for the skin.

Lymph node dissection—Surgical removal of a group of lymph nodes.

Lymphedema—Swelling of an arm or leg following surgical removal of the lymph nodes that drain the limb.

Melanocytes—Skin cells derived from the neural crest that produce the protein pigment melanin.

Metastasis (plural, metastases)—A tumor growth or deposit that has spread via lymph or blood to an area of the body remote from the primary tumor.

Nevus (plural, nevi)—A medical term for mole.

Resection—The act of removing something surgically.

Skin appendages—Structures related to the integument such as hair follicles and sweat glands.

Variegation—Patchy variation in color.

Coping with cancer treatment

For those with familial tendencies for malignant melanoma, **genetic counseling** may be appropriate. Psychological counseling may be appropriate for anyone having trouble coping with a potentially fatal disease. Local cancer support groups may be helpful and are often identified by contacting local hospitals or the American Cancer Society.

Prevention

Though it is difficult to prove that **sunscreens** statistically reduce the frequency of malignant

melanoma at this time, most authorities recommend their use as protection from ultraviolet light (considered a major factor in the development of melanoma.) Avoidance of severe sunburns is recommended.

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American Cancer Society. 1599 Clifton Road NE, Atlanta, GA 30329. (800) ACS-2345.

National Cancer Institute (NCI). NCI Public Inquiries Office, Suite 3036A, 6116 Executive Boulevard, MSC8332, Bethesda, MD 20892-8322. (800) 4-CANCER or (800) 332-8615 (TTY). <<http://www.nci.nih.gov>>.

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Malignant plasmacytoma see **Multiple myeloma**

Malingering

Definition

In the context of medicine, malingering is the act of intentionally feigning or exaggerating physical or psychological symptoms for personal gain.

Description

People may feign physical or psychological illness for any number of reasons. Faked illness can get them out of work, military duty, or criminal prosecution. It

can also help them obtain financial compensation through insurance claims, lawsuits, or workers' compensation. Feigned symptoms may also be a way of getting the doctor to prescribe certain drugs.

According to the American Psychiatric Association, patients who malingering are different from people who invent symptoms for sympathy (factitious diseases). Patients who malingering clearly have something tangible to gain. People with factitious diseases appear to have a need to play the "sick" role. They may feign illness for attention or sympathy.

Malingering may take the form of complaints of chronic **whiplash pain** from automobile accidents. Whiplash claims are controversial. Although some people clearly do suffer from whiplash injury, others may be exaggerating the pain for insurance claims or lawsuits. Some intriguing scientific studies have shown that chronic whiplash pain after automobile accidents is almost nonexistent in Lithuania and Greece. In these countries, the legal systems do not encourage personal injury lawsuits or financial settlements. The psychological symptoms experienced by survivors of disaster (**post-traumatic stress disorder**) are also faked by malingers.

Causes and symptoms

People malingering for personal gain. The symptoms may vary. Generally malingers complain of psychological disorders such as **anxiety**. They may also complain of chronic pain for which objective tests such as x rays can find no physical cause. Because it is often impossible to determine who is malingering and who is not, it is impossible to know how frequently malingering occurs.

Diagnosis

Malingering may be suspected:

- When a patient is referred for examination by an attorney
- When the onset of illness coincides with a large financial incentive, such as a new disability policy
- When objective medical tests do not confirm the patient's complaints
- When the patient does not cooperate with the diagnostic work-up or prescribed treatment
- When the patient has antisocial attitudes and behaviors (antisocial personality).

The diagnosis of malingering is a challenge for doctors. On the one hand, the doctor does not

KEY TERMS

Antisocial personality—A personality characterized by attitudes and behaviors at odds with society's customs and moral standards, including illegal acts.

Factitious diseases—Conditions in which symptoms are deliberately manufactured by patients in order to gain attention and sympathy. Patients with factitious diseases do not fake symptoms for obvious financial gain or to evade the legal system.

Post traumatic stress disorder (PTSD)—A disorder that occurs among survivors of severe environmental stress such as a tornado, an airplane crash, or military combat. Symptoms include anxiety, insomnia, flashbacks, and nightmares. Patients with PTSD are unnecessarily vigilant; they may experience survivor guilt, and they sometimes cannot concentrate or experience joy.

want to overlook a treatable disease. On the other hand, he or she does not want to continue ordering tests and treatments if the symptoms are faked. Malingering is difficult to distinguish from certain legitimate **personality disorders**, such as factitious diseases or post-traumatic distress syndrome. In legal cases, malingering patients may be referred to a psychiatrist. Psychiatrists use certain written tests to try to determine whether the patient is faking the symptoms.

Treatment

In a sense, malingering cannot be treated because the American Psychiatric Association does not recognize it as a personality disorder. Patients who are purposefully faking symptoms for gain do not want to be cured. Often, the malingering patient fails to report any improvement with treatment, and the doctor may try many treatments without success.

Resources

ORGANIZATIONS

American Psychiatric Association. 1400 K Street NW, Washington DC 20005. (888) 357-7924.
< <http://www.psych.org> >.

Robert Scott Dinsmoor

Mallet finger

Definition

Mallet finger refers to the involuntary flexion of the distal phalanx of a finger caused by the disruption or tearing of its extensor tendon.

Description

Tendons are the strong “cables” between muscles and bones that help control movements of the body. They consist of white, glistening, fibrous cords, of various length and thickness, either round or flattened, and lacking in elasticity. In mallet finger, which often occurs as a sports-related injury, the tendon on the back of the finger becomes damaged or torn near the outermost joint. Without the support provided by the tendon, the short bone at the tip of the finger drops downward at an awkward angle. This bone, referred to as the “distal phalanx” of a finger, is the one furthest from the palm. In addition to tendon damage, mallet finger may involve a fracture of the distal phalanx. Mallet finger is sometimes called baseball finger.

Causes and symptoms

Mallet finger usually occurs while playing a sport that involves a ball—for example, reaching out to catch a hard pass in basketball or bare-handing a baseball. Instead of landing on the palm of the hand, the ball accidentally hits the tip of an extended (or partially extended) finger. This straight-on impact causes instantaneous stretch of the tendon, which may overextend or tear away. Mallet finger can also result from hitting the hand against a hard object or receiving a cut from a sharp edge such as a knife.

Symptoms of mallet finger include **pain** and swelling around the top part of the finger, near the outermost joint. These symptoms occur right after the injury. Redness and swelling develop soon afterward. The tip of the finger has an abnormal-looking downward droop, and it may be difficult to fully extend the finger.

Diagnosis

Mallet finger is usually diagnosed after a relatively brief **physical examination** conducted by an emergency care physician or by an orthopedist, the type of doctor who specializes in such injuries. The downward droop of the fingertip is the major indication of mallet finger, along with the tenderness and pain that occurs in the

KEY TERMS

Distal Phalanx—The outermost bone of any finger or toe.

Fracture—A break in bone.

Orthopedist—A doctor who specializes in disorders of the musculoskeletal system.

Phalanx—Any of the digital bones of the hand or foot. Humans have three phalanges to each finger and toe with the exception of the thumb and big toe which have only two each.

Tendon—A tough cord of dense white fibrous connective tissue that connects a muscle with some other part, especially a bone, and transmits the force which the muscle exerts.

affected area. X rays will be taken to determine if the bone at the top of the finger has been fractured. Mallet finger is typically covered by medical insurance.

Treatment

If symptoms of mallet finger appear, the affected individual should consult a physician or seek emergency care. In the meantime, ice (wrapped in a towel or cloth) can be applied to the affected area to help reduce swelling and alleviate pain.

Treatment usually involves wearing a splint around the top of the affected finger in order to keep it extended and allow the injury to heal. The splint must be worn at all times for six to eight weeks, though it may be briefly removed to wash the finger, but with extreme care so as not to allow the fingertip to bend. For the next six to eight weeks after that, the splint need only be worn during sleep or athletic activities.

If the bone at the top of the finger has sustained a large fracture, surgery may be necessary. If the tendon was damaged due to a cut, stitches may be required both to repair the tendon and to adequately close the wound.

Over-the-counter (OTC) or prescription pain medication can be used to alleviate pain.

Alternative treatment

Acupuncture, therapeutic massage, and **yoga** are believed by some practitioners of alternative medicine to have generalized pain-relieving effects. Any of these therapies may provide additional comfort while the finger heals.

Prognosis

With proper treatment, most people regain full use of the affected finger.

Prevention

Caution should be used when playing ball sports or using knives or other sharp implements.

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ORGANIZATIONS

American Academy of Orthopaedic Surgeons. 6300 North River Road, Rosemont, IL 60018-4262. (800) 346-AAOS. <<http://www.aaos.org>>.

Greg Annussek

Mallory-Weiss syndrome

Definition

Mallory-Weiss syndrome is bleeding from an arterial blood vessel in the upper gastrointestinal tract, caused by a mucosal gastric tear at or near the point where the esophagus and stomach join.

Description

Mallory-Weiss syndrome causes about 5% of all upper gastrointestinal bleeding. The condition was originally diagnosed in alcoholics and is associated with heavy alcohol use, although it can also be found in patients who are not alcoholics. Earlier episodes of heavy hiccups, **vomiting**, and retching are reported by about half the patients who are diagnosed with Mallory-Weiss syndrome. It is thought that the tear or laceration occurs when there is a sudden increase in intra-abdominal pressure. Patients with increased pressure in the vein leading into the liver (portal **hypertension**) are more likely to bleed heavily from an

KEY TERMS

Electrolytes—Salts and minerals that can conduct electrical impulses in the body. Common human electrolytes are sodium chloride, potassium, calcium, and sodium bicarbonate. Electrolytes control the fluid balance of the body and are important in muscle contraction, energy generation, and almost every major biochemical reaction in the body.

Endoscopy—A procedure in which an instrument containing a camera and a light source is inserted into the gastrointestinal tract so that the doctor can visually inspect the gastrointestinal system.

Esophageal varix—An enlarged vein of the esophagus. (Plural: esophageal varices.)

Portal hypertension—High blood pressure in the portal vein, which carries blood from the abdominal organs to the liver.

esophageal laceration than those whose blood pressure is normal.

Causes and symptoms

In Mallory-Weiss syndrome, a tear occurs in the gastric mucosa, near where the esophagus and stomach join. About 10% of the tears are in the esophagus. Most are either right at the junction of the esophagus and stomach or in the stomach just slightly below the junction.

Bleeding from the tear causes a disruption in fluid and electrolyte balance of the body. The patient often produces vomit tinged with either fresh blood or older, blackish blood. Blood loss can be considerable.

Diagnosis

A Mallory-Weiss syndrome tear is not visible on standard upper gastrointestinal x rays. A tear about one-eighth to one and one-half inches long (0.5–4 cm) is revealed by endoscopy. Endoscopy also shows that in 35% of patients there is another potential cause for gastrointestinal bleeding, such as peptic ulcer, erosive **gastritis**, or esophageal varices.

Treatment

The patient is resuscitated and stabilized with blood transfusions and intravenous fluids to restore the fluid and electrolyte balance. Most of the time,

esophageal bleeding stops spontaneously. When bleeding does not stop, patients are treated with an injection of epinephrine (adrenaline) and/or the bleeding artery is cauterized with heat. If these treatments fail, surgery is performed to stop the bleeding.

Prognosis

In 90-95% of patients whose bleeding does not stop spontaneously, cauterization without surgery will stop the bleeding. Patients at highest risk for a recurrence of bleeding are those with portal hypertension.

Prevention

Mallory-Weiss syndrome is associated with **alcoholism**. Limiting alcohol intake may help prevent the disorder.

Resources

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Tish Davidson, A.M.

Malnutrition

Definition

Malnutrition is the condition that develops when the body does not get the right amount of the **vitamins**, **minerals**, and other nutrients it needs to maintain healthy tissues and organ function.

Description

Undernutrition

Malnutrition occurs in people who are either undernourished or overnourished. Undernutrition is a consequence of consuming too few essential nutrients or using or excreting them more rapidly than they can be replaced.

Infants, young children, and teenagers need additional nutrients. So do women who are pregnant or breastfeeding. Nutrient loss can be accelerated by **diarrhea**, excessive sweating, heavy bleeding (hemorrhage), or kidney failure. Nutrient intake can be restricted by age-related illnesses and conditions,

excessive dieting, **food allergies**, severe injury, serious illness, a lengthy hospitalization, or **substance abuse**.

The leading cause of **death** in children in developing countries is **protein-energy malnutrition**. This type of malnutrition is the result of inadequate intake of calories from proteins, vitamins, and minerals. Children who are already undernourished can suffer from protein-energy malnutrition (PEM) when rapid growth, infection, or disease increases the need for protein and essential minerals. These essential minerals are known as micronutrients or trace elements.

Two types of protein-energy malnutrition have been described—kwashiorkor and marasmus. Kwashiorkor occurs with fair or adequate calorie intake but inadequate protein intake, while marasmus occurs when the diet is inadequate in both calories and protein.

About 1% of children in the United States suffer from chronic malnutrition, in comparison to 50% of children in southeast Asia. About two-thirds of all the malnourished children in the world are in Asia, with another one-fourth in Africa.

Overnutrition

In the United States, nutritional deficiencies have generally been replaced by dietary imbalances or excesses associated with many of the leading causes of death and disability. Overnutrition results from eating too much, eating too many of the wrong things, not exercising enough, or taking too many vitamins or other dietary replacements.

Risk of overnutrition is also increased by being more than 20% overweight, consuming a diet high in fat and salt, and taking high doses of:

- Nicotinic acid (niacin) to lower elevated cholesterol levels
- Vitamin B₆ to relieve **premenstrual syndrome**
- Vitamin A to clear up skin problems
- Iron or other trace minerals not prescribed by a doctor.

Nutritional disorders can affect any system in the body and the senses of sight, taste, and smell. They may also produce **anxiety**, changes in mood, and other psychiatric symptoms. Malnutrition begins with changes in nutrient levels in blood and tissues. Alterations in enzyme levels, tissue abnormalities, and organ malfunction may be followed by illness and death.

Causes and symptoms

Causes

Poverty and lack of food are the primary reasons why malnutrition occurs in the United States. Ten percent of all members of low income households do not always have enough healthful food to eat. Protein-energy malnutrition occurs in 50% of surgical patients and in 48% of all other hospital patients.

Loss of appetite associated with the **aging** process. Malnutrition affects one in four elderly Americans, in part because they may lose interest in eating. In addition, such dementing illnesses as **Alzheimer's disease** may cause elderly persons to forget to eat.

There is an increased risk of malnutrition associated with chronic diseases, especially disease of the intestinal tract, kidneys, and liver. Patients with chronic diseases like **cancer**, **AIDS**, intestinal parasites, and other gastric disorders may lose weight rapidly and become susceptible to undernourishment because they cannot absorb valuable vitamins, calories, and iron.

People with drug or alcohol dependencies are also at increased risk of malnutrition. These people tend to maintain inadequate **diets** for long periods of time and their ability to absorb nutrients is impaired by the alcohol or drug's affect on body tissues, particularly the liver, pancreas, and brain.

Eating disorders. People with anorexia or bulimia may restrict their food intake to such extremes that they become malnourished.

Food **allergies**. Some people with food allergies may find it difficult to obtain food that they can digest. In addition, people with food allergies often need additional calorie intake to maintain their weight.

Failure to absorb nutrients in food following bariatric (weight loss) surgery. **Bariatric surgery** includes such techniques as stomach stapling (gastroplasty) and various intestinal bypass procedures to help people eat less and lose weight. Malnutrition is, however, a possible side effect of bariatric surgery.

Symptoms

Unintentionally losing 10 pounds or more may be a sign of malnutrition. People who are malnourished may be skinny or bloated. Their skin is pale, thick, dry, and **bruises** easily. **Rashes** and changes in pigmentation are common.

Hair is thin, tightly curled, and pulls out easily. Joints ache and bones are soft and tender. The gums bleed easily. The tongue may be swollen or shriveled

and cracked. Visual disturbances include night blindness and increased sensitivity to light and glare.

Other symptoms of malnutrition include:

- anemia
- diarrhea
- disorientation
- night blindness
- irritability, anxiety, and attention deficits
- goiter (enlarged thyroid gland)
- loss of reflexes and lack of muscular coordination
- muscle twitches
- amenorrhea (cessation of menstrual periods)
- scaling and cracking of the lips and mouth.

Malnourished children may be short for their age, thin, listless, and have weakened immune systems.

Diagnosis

Overall appearance, behavior, body-fat distribution, and organ function can alert a family physician, internist, or **nutrition** specialist to the presence of malnutrition. Patients may be asked to record what they eat during a specific period. X rays can determine bone density and reveal gastrointestinal disturbances, and heart and lung damage.

Blood and urine tests are used to measure the patient's levels of vitamins, minerals, and waste products. Nutritional status can also be determined by:

- Comparing a patient's weight to standardized charts
- Calculating body mass index (BMI) according to a formula that divides height into weight
- Measuring skinfold thickness or the circumference of the upper arm.

Treatment

Normalizing nutritional status starts with a nutritional assessment. This process enables a clinical nutritionist or registered dietician to confirm the presence of malnutrition, assess the effects of the disorder, and formulate diets that will restore adequate nutrition.

Patients who cannot or will not eat, or who are unable to absorb nutrients taken by mouth, may be fed intravenously (parenteral nutrition) or through a tube inserted into the gastrointestinal (GI) tract (enteral nutrition).

Tube feeding is often used to provide nutrients to patients who have suffered **burns** or who have

inflammatory bowel disease. This procedure involves inserting a thin tube through the nose and carefully guiding it along the throat until it reaches the stomach or small intestine. If long-term tube feeding is necessary, the tube may be placed directly into the stomach or small intestine through an incision in the abdomen.

Tube feeding cannot always deliver adequate nutrients to patients who:

- Are severely malnourished
- Require surgery
- Are undergoing **chemotherapy** or radiation treatments
- Have been seriously burned
- Have persistent diarrhea or **vomiting**
- Whose gastrointestinal tract is paralyzed.

Intravenous feeding can supply some or all of the nutrients these patients need.

Prognosis

Up to 10% of a person's body weight can be lost without side effects, but if more than 40% is lost, the situation is almost always fatal. Death usually results from **heart failure**, electrolyte imbalance, or low body temperature. Patients with semiconsciousness, persistent diarrhea, **jaundice**, or low blood sodium levels have a poorer prognosis.

Some children with protein-energy malnutrition recover completely. Others have many health problems throughout life, including **mental retardation** and the inability to absorb nutrients through the intestinal tract. Prognosis for all patients with malnutrition seems to be dependent on the age of the patient, and the length and severity of the malnutrition, with young children and the elderly having the highest rate of long-term complications and death.

Prevention

Breastfeeding a baby for at least six months is considered the best way to prevent early-childhood malnutrition. The United States Department of Agriculture and Health and Human Service recommend that all Americans over the age of two:

- Consume plenty of fruits, grains, and vegetables
- Eat a variety of foods that are low in fats and cholesterol and contain only moderate amounts of salt, sugars, and sodium
- Engage in moderate physical activity for at least 30 minutes, at least several times a week

KEY TERMS

Anemia—Not enough red blood cells in the blood.

Anorexia nervosa—Eating disorder marked by malnutrition and weight loss commonly occurring in young women.

Bariatric—Pertaining to the study, prevention, or treatment of overweight.

Calorie—A unit of heat measurement used in nutrition to measure the energy value of foods. A calorie is the amount of heat energy needed to raise the temperature of 1 kilogram of water 1°C.

Kwashiorkor—Severe malnutrition in children primarily caused by a protein-poor diet, characterized by growth retardation.

Marasmus—Severe malnutrition in children caused by a diet lacking in calories as well as protein. Marasmus may also be caused by disease and parasitic infection.

Micronutrients—Essential dietary elements that are needed only in very small quantities. Micronutrients are also known as trace elements. They include copper, zinc, selenium, iodine, magnesium, iron, cobalt, and chromium.

- Achieve or maintain their ideal weight
- Use alcohol sparingly or avoid it altogether.

Every patient admitted to a hospital should be screened for the presence of illnesses and conditions that could lead to protein-energy malnutrition. Patients with higher-than-average risk for malnutrition should be more closely assessed and reevaluated often during long-term hospitalization or nursing-home care.

Resources

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American College of Nutrition. 722 Robert E. Lee Drive, Wilmington, NC 20412-0927. (919) 452-1222.

American Institute of Nutrition. 9650 Rockville Pike, Bethesda, MD 20814-3990. (301) 530-7050.

Food and Nutrition Information Center. 10301 Baltimore Boulevard, Room 304, Beltsville, MD 20705-2351. <<http://www.nal.usda.gov/fnic>>.

OTHER

World Health Organization (WHO) Nutrition web site. <<http://www.who.int/nut/index.htm>>.

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Malocclusion

Definition

Malocclusion is a problem in the way the upper and lower teeth fit together in biting or chewing. The word malocclusion literally means "bad bite." The condition may also be referred to as an irregular bite, crossbite, or overbite.

Description

Malocclusion may be seen as crooked, crowded, or protruding teeth. It may affect a person's appearance, speech, and/or ability to eat.

Causes and symptoms

Malocclusions are most often inherited, but may be acquired. Inherited conditions include too many or too few teeth, too much or too little space between teeth, irregular mouth and jaw size and shape, and atypical formations of the jaws and face, such as a **cleft palate**. Malocclusions may be acquired from habits like finger or thumb sucking, tongue thrusting, premature loss of teeth from an accident or dental disease, and medical conditions such as enlarged tonsils and adenoids that lead to mouth breathing.

Malocclusions may be symptomless or they may produce **pain** from increased **stress** on the oral structures. Teeth may show abnormal signs of wear on the chewing surfaces or decay in areas of tight overlap. Chewing may be difficult.

Diagnosis

Malocclusion is most often found during a routine dental examination. A dentist will check a patient's occlusion by watching how the teeth make contact when the patient bites down normally. The dentist may ask the patient to bite down with a piece of coated paper between the upper and lower teeth; this paper will leave colored marks at the points of contact. When malocclusion is suspected, photographs and x rays of the face and mouth may be taken for further study. To confirm the presence and extent of malocclusion, the dentist makes plaster, plastic, or artificial stone models of the patient's teeth from impressions. These models duplicate the fit of the teeth and are very useful in treatment planning.

Treatment

Malocclusion may be remedied by orthodontic treatment; orthodontics is a specialty of dentistry that manages the growth and correction of dental and facial structures. Braces are the most commonly used orthodontic appliances in the treatment of malocclusion. At any given time, approximately 4 million people in the United States are wearing braces, including 800,000 adults.

Braces apply constant gentle force to slowly change the position of the teeth, straightening them and properly aligning them with the opposing teeth. Braces consist of brackets cemented to the surface of each tooth and wires of stainless steel or nickel titanium alloy. When the wires are threaded through the brackets, they exert pressure against the teeth, causing them to move gradually.



Orthodontia treatments usually include the use of braces and retainers. (Photograph by Lester V. Bergman, Corbis Images. Reproduced by permission).

Braces are not removable for daily tooth brushing, so the patient must be especially diligent about keeping the mouth clean and removing food particles which become easily trapped, to prevent **tooth decay**. Foods that are crunchy should be avoided to minimize the risk of breaking the appliance. Hard fruits, vegetables, and breads must be cut into bite-sized pieces before eating. Foods that are sticky, including chewing gum, should be avoided because they may pull off the brackets or weaken the cement. Carbonated beverages may also weaken the cement, as well as contribute to tooth decay. Teeth should be brushed immediately after eating sweet foods. Special floss threaders are available to make flossing easier.

If overcrowding is creating malocclusion, one or more teeth may be extracted (surgically removed), giving the others room to move. If a tooth has not yet erupted or is prematurely lost, the orthodontist may insert an appliance called a space maintainer to keep the other teeth from moving out of their natural position. In severe cases of malocclusion, surgery may be necessary and the patient would be referred to yet another specialist, an oral or maxillofacial surgeon.

Once the teeth have been moved into their new position, the braces are removed and a retainer is worn until the teeth stabilize in that position. Retainers do not move teeth, they only hold them in place.



This patient's teeth are misarranged because of excessive thumb sucking. (Custom Medical Stock Photo. Reproduced by permission.)

Orthodontic treatment is the only effective treatment for malocclusion not requiring surgery. However, depending on the cause and severity of the condition, an orthodontist may be able to suggest other appliances as alternatives to braces.

Alternative treatment

There are some techniques of **craniosacral therapy** that can alter structure. This therapy may allow correction of some cases of malocclusion. If surgery is required, pre- and post-surgical care with homeopathic remedies, as well as vitamin and mineral supplements, can enhance recovery. Night guards are sometimes recommended to ease the strain on the jaw and to limit teeth grinding.

Prognosis

Depending on the cause and severity of the malocclusion and the appliance used in treatment, a patient may expect correction of the condition to take 2 or more years. Patients typically wear braces 18-24 months and a retainer for another year. Treatment is faster and more successful in children and teens whose teeth and bones are still developing. The length of treatment time is also affected by how well the patient follows orthodontic instructions.

KEY TERMS

Braces—An orthodontic appliance consisting of brackets cemented to the surface of each tooth and wires of stainless steel or nickel titanium alloy. Braces are used to treat malocclusion by changing the position of the teeth.

Impression—An imprint of the upper or lower teeth made in a pliable material that sets. When this material has hardened, it may be filled with plaster, plastic, or artificial stone to make an exact model of the teeth.

Occlusion—The way the upper and lower teeth fit together in biting or chewing.

Retainer—An orthodontic appliance that is worn to stabilize teeth in a new position.

Space maintainer—An orthodontic appliance that is worn to prevent adjacent teeth from moving into the space left by an unerupted or prematurely lost tooth.

Prevention

In general, malocclusion is not preventable. It may be minimized by controlling habits such as finger or thumb sucking. An initial consultation with an orthodontist before a child is 7 years old may lead to appropriate management of the growth and development of the child's dental and facial structures, circumventing many of the factors contributing to malocclusion.

Resources

ORGANIZATIONS

American Association of Oral and Maxillofacial Surgeons.
9700 West Bryn Mawr Ave., Rosemont, IL 60018-5701.
(847) 678-6200. <<http://www.aaoms.org>>.

American Association of Orthodontists. 401 North
Lindbergh Boulevard, St. Louis, MO 63141-7816. (314)
993-1700. <<http://www.aaortho.org>>.

Bethany Thivierge

MALT lymphoma

Definition

MALT lymphomas are solid tumors that originate from cancerous growth of immune cells that are

recruited to secretory tissue such as the gastrointestinal tract, salivary glands, lungs, and the thyroid gland.

Description

The digestive tract is generally not associated with lymphoid tissue, with the exception of small collections of lymphocytes such as Peyer's patches. A specific kind of white blood cell, B-lymphocytes, can accumulate in response to infections of the digestive tract and other secretory tissues, or as a result of autoimmune conditions such as Sjgren's syndrome. When the growth of these lymphocytes is maintained through continued infection or autoimmune disease, a malignant cell can arise and replace the normal lymphocytes. These lymphomas, derived from mucosa-associated lymphoid tissue (MALT), most commonly arise in the stomach. Their growth seems to be dependent upon continuous stimulation of the immune system by an infectious agent, such as *H. pylori*, or some other entity, termed an antigen, that the body recognizes as foreign. This antigen-driven growth permits these tumors to be treated by eliminating the stimulus that generated the original, normal immune response. In the stomach they are associated, in greater than 90% of all cases, with the bacteria called *Helicobacter pylori* (*H. pylori*). This bacteria is also associated with peptic stomach irritation, ulcers, and gastric **cancer**. MALT lymphomas are generally indolent, that is, they grow slowly and cause little in the way of symptoms. Those MALT lymphomas that arise in the stomach in response to *H. pylori* infections are generally successfully treated with **antibiotics**, which eliminate the bacteria.

Demographics

MALT lymphomas occur at a frequency of about 1.5 per 100,000 people per year in the United States and account for about 10% of all non-Hodgkin's lymphomas. The frequency varies among different populations. For example, in parts of Italy the frequency of MALT lymphomas is as high as 13 per 100,000 people per year. This can in part be attributed to different rates of infection with *H. pylori*. However, other hereditary, dietary, or environmental factors are almost certainly involved.

Causes and symptoms

The majority of MALT lymphomas appear to be the result of infectious agents, most commonly *H. pylori* in the stomach. It is not known if infectious agents also cause MALT lymphomas outside of the

stomach. In some cases, such as in the thyroid, MALT lymphomas seem to arise in patients who have autoimmune diseases, which make their immune systems treat their own tissue as foreign or antigenic. It is believed that there must be additional factors, in addition to infection or autoimmunity, that influence the development of MALT lymphomas. For example, in the United States, where infections with *H. pylori* are quite common, less than 1 in 30,000 people who have *H. pylori* in their stomachs develop MALT lymphomas. In addition, individuals who develop MALT lymphomas are more likely to develop other forms of cancer. This would suggest that there might be genetic factors predisposing individuals to develop MALT lymphomas or other tumors in response to environmental or infectious agents.

In general, patients have stomach **pain**, ulcers, or other localized symptoms, but rarely do they suffer from systemic complaints such as **fatigue** or **fever**.

Diagnosis

The indolent nature of most MALT lymphomas means that the majority of patients are diagnosed at early stages with relatively nonspecific symptoms. In the case of gastric MALT lymphomas, the physician will then have a gastroenterologist perform an endoscopy to examine the interior of the stomach. MALT lymphomas are then recognized as areas of inflammation or ulceration within the stomach. It is unusual for masses recognizable as tumors to be seen upon examination. Definitive diagnosis of MALT lymphoma requires a biopsy, in which a bit of tissue is removed from the stomach or other involved site. Examination of this tissue by a pathologist is the first step in distinguishing among the possible diagnoses of inflammation, indolent lymphoma, or a more aggressive form of cancer, such as gastric cancer or a rapidly growing non-Hodgkin's lymphoma. The pathologist evaluates the type of lymphoid cells that are present in the biopsy to establish the nature of the lesion. In addition, it is essential that the pathologist determine whether or not the lymphoma has grown beyond the borders of the mucosa, which lines the stomach or other gland.

Treatment

The best staging system to employ for MALT lymphomas is still the subject of discussion. However, it is standard practice that patients presenting with MALT lymphomas should be evaluated in a similar manner to individuals with nodal lymphomas, the more common type of lymphoma that originates at

sites within the lymphoid system. These procedures include a complete history and physical, blood tests, chest x rays, and **bone marrow biopsy**. This evaluation will permit the oncologist to determine if the disease is localized or if it has spread to other sites within the body.

In general, the prognosis for patients with MALT lymphomas is good, with overall five-year survival rates that are greater than 80%. The features that are most closely related to the outlook for newly diagnosed individual patients are: whether the primary site is in the stomach or is extra-gastric; if the disease has spread beyond the initial location; and whether the histologic evaluation of the initial tumor biopsies is consistent with a low-grade, slowly growing lesion, as compared to a high-grade lesion that is more rapidly growing. In general, the histologic grade is the most important feature, with high-grade lesions requiring the most aggressive treatment.

Treatment of MALT lymphomas differs from that of most lymphomas. In the most common type of MALT lymphomas—low-grade lesions originating in the stomach—treatment with antibiotics to eliminate *H. pylori* leads to complete remissions in the majority of patients. The effectiveness of this treatment is indistinguishable from surgery, **chemotherapy**, **radiation therapy**, or a combination of surgery with drugs or irradiation. Approximately one-third of patients in this group have evidence of disseminated disease, where lymphoma cells are detected at sites in addition to the gastric mucosa. The response of these patients to antibiotic treatment is not significantly different from that for individuals with localized disease. For both groups a complete remission is achieved in about 75% of patients, who remain, on average, free of disease for about 5 years.

Prognosis

Patients with MALT lymphomas arising outside of the digestive tract also have good prognoses. Effective treatment for these lymphomas has been achieved with local radiation, chemotherapy, and/or interferon. Surgery followed by chemotherapy or radiation is also effective with nongastrointestinal MALT lymphomas. Overall these patients have five-year survival rates greater than 90%.

While the outlook for patients with MALT lymphomas is good, difficulties in diagnosis and staging have left the optimal treatment a matter of continued study. This is an especially open question for those patients who fail to respond to antibiotic therapy, or

KEY TERMS

Antigen—A foreign substance that leads to an immune response, including the production of antibodies by B cells.

Autoimmune disease—A condition in which an individual's immune system reacts to their own tissues, viewing self components as if they were foreign antigens.

Bone marrow biopsy—A procedure in which cellular material is removed from the pelvis or breastbone and examined under a microscope to look for the presence of abnormal blood cells characteristic of specific forms of leukemia and lymphoma.

Indolent lymphoma (also called low-grade)—Cancerous growths of lymphoid tissue that progress slowly to more aggressive forms of cancer.

Lymphoid tissue—Sites within the body that produce cells of the immune system, including lymph nodes, bone marrow, and the thymus.

whose disease recurs. It may be the case that in these patients, the MALT lymphoma may have already progressed to a point where high-grade lesions, not observed in the original biopsies, were resistant to the initial treatment. The best treatment for these patients remains to be established. In general, these patients are treated with chemotherapy in a similar manner to patients with other types of lymphoma. Given the success of antibiotics, and the good prognosis for gastric MALT lymphomas in general, no sufficient body of evidence exists to determine the best chemotherapy for patients who fail to achieve a complete and lasting remission upon initial treatment. At present, a chemotherapeutic regime designated CHOP includes the anti-cancer drugs cyclophosphamide, doxorubicin, vincristine, and prednisone. Similar drug combinations are being used for patients whose MALT lymphomas do not respond to antibiotic treatment.

Clinical trials are underway and mostly concentrate upon optimizing treatment of gastric MALT lymphomas that involve *H. pylori*. The aspects of treatment being addressed are the most effective antibiotics and the use of **antacids** to modulate irritation in the stomach. These protocols have been designed to follow the natural history of gastric lymphomas and to establish the biological features that predict treatment response to antibiotics and duration of remission.

Prevention

There are currently no commonly accepted means to prevent MALT lymphomas. While the *H. pylori* infections are associated with this and other gastric disease, the eradication of *H. pylori* in asymptomatic individuals is not currently recommended for prevention of MALT lymphomas or gastric cancer.

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Warren Maltzman, Ph.D.

Malta fever see **Brucellosis**

Mammogram screening see **Mammography**

Mammography

Definition

Mammography is the study of the breast using x ray. The actual test is called a mammogram. There are two types of mammograms. A screening mammogram is ordered for women who have no problems with their breasts. It consists of two x-ray views of each breast. A diagnostic mammogram is for evaluation of new abnormalities or of patients with a past abnormality requiring follow-up (i.e., a woman with **breast cancer** treated with **lumpectomy**). Additional x rays from other angles or special views of certain areas are taken.

Purpose

The purpose of screening mammography is breast **cancer** detection. A screening test, by definition, is used for patients without any signs or symptoms in order to detect disease as early as possible. Many studies have shown that having regular mammograms increases a woman's chances of finding breast cancer in an early stage, when it is more likely to be curable. It

has been estimated that a mammogram may find a cancer as much as two years before it can be felt. The American Cancer Society, American College of Radiology, American College of Surgeons and American Medical Association recommend annual mammograms for every woman beginning at age 40.

Screening mammograms usually are not recommended for women under age 40 who have no special risk factors and a normal physical breast examination. Below age 40, breasts tend to be "radiographically dense," which means it is difficult to see many details. In 2003, a new technique that introduces radiographic contrast into digital mammograms was proving useful at improving visibility of breast cancer in younger women. Screening mammograms can detect cancers in their earliest stages and greatly reduce mortality, particularly among women age 40 to 69. In fact, a study in 2003 found that women age 40 and older who had annual screening mammograms had better breast cancer prognoses because their cancers were diagnosed at earlier stages than women who had mammograms less often.

Some women are at increased risk for developing breast cancer, such as those with two or more relatives who have the disease. The 2003 American Cancer Society guidelines stated that women at increased risk might benefit from earlier screening mammograms and more frequent intervals for screening. However, the society suggested that evidence was not strong enough at that time to support making specific recommendations concerning screening examinations.

Diagnostic mammography is used to evaluate an existing problem, such as a lump, discharge from the nipple, or unusual tenderness in one area. The cause of the problem may be definitively diagnosed from this study, but further investigation using other methods often is necessary. This exam also is used to evaluate findings from screening mammography.

Description

A mammogram may be offered in a variety of settings. Hospitals, outpatient clinics, physicians' offices, or other facilities may have mammography equipment. In the United States, since October 1, 1994, only places certified by the U.S. Food and Drug Administration (FDA) are legally permitted to perform, interpret, or develop mammograms under the Mammography Quality Standards Act (MQSA).

In addition to the usual paperwork, a woman will be asked to fill out a form seeking information relevant to her risk of breast cancer and special



A person undergoing a mammography. (Custom Medical Stoch Photo Inc. Reproduced by permission)

mammography needs. The woman is asked about personal and family history of cancer, details about menstruation, child bearing, birth control, **breast implants**, other breast surgery, age, and **hormone replacement therapy**. Information about Breast Self Examination (BSE) and other breast health issues usually are available at no charge.

At some centers, a technologist may perform a **physical examination** of the breasts before the mammogram. Whether or not this is done, it is essential for the patient to tell the technologist about any lumps, nipple discharge, breast **pain**, or other concerns.

Clothing from the waist up is removed and a hospital gown or similar covering is put on. The woman stands facing the mammography machine. The technologist exposes one breast and places it on a plastic or metal film holder about the size of a placemat. The breast is compressed as flat as possible between the film holder and a rectangle of plastic (called a paddle), which presses down onto the breast from above. The compression should only last a few seconds, just enough to take the x ray. Good compression can be uncomfortable, but it is necessary to ensure the clearest view of all breast tissues.

Next, the woman is positioned with her side toward the mammography unit. The film holder is tilted so the outside of the breast rests against it, and a corner touches the armpit. The paddle again holds the breast firmly as the x ray is taken. This procedure is repeated for the other breast. A total of four x rays, two of each breast, are taken for a screening mammogram. Additional x rays, using special paddles, different breast positions, or other techniques are usually taken for a diagnostic mammogram.

The mammogram may be seen and interpreted by a radiologist right away, or it may not be reviewed until later. If there are any questionable areas or an abnormality, extra x rays may be recommended. These may be taken during the same appointment. More commonly, especially for screening mammograms, the woman is called back on another day for these additional films.

A screening mammogram usually takes approximately 15 to 30 minutes. A woman having a diagnostic mammogram can expect to spend up to an hour at the mammography facility.

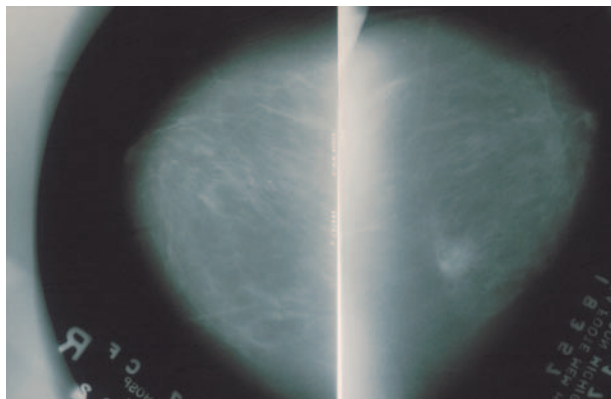
The cost of mammography varies widely. Many mammography facilities accept “self referral.” This means women can schedule themselves without a physician’s referral. However, some insurance policies require a doctor’s prescription to ensure payment. Medicare will pay for annual screening mammograms for all women with Medicare who are age 40 or older and a baseline mammogram for those age 35 to 39.

A digital mammogram is performed in the same way as a traditional exam, but the image is viewed on a computer monitor, stored as a digital file, and can be printed on film. Medicare now pays a small additional fee for digital mammography.

Preparation

The compression or squeezing of the breast for a mammogram is a concern for some women, but necessary to render a quality image. Even with concerns about pain, a 2003 study said that three-fourths of women reported the pain associated with a mammogram as four on a 10-point scale. Mammograms should be scheduled when a woman’s breasts are least likely to be tender. One week after the menstrual period is usually best. The MQSA regulates equipment compression for consistency and performance.

Women should not put deodorant, powder, or lotion on their upper body on the day the mammogram is performed. Particles from these products can get on the breast or film holder and may look like abnormalities on the mammogram film.



Comparison of two mammograms—cancerous tissue is shown on left and normal tissue on right. (Custom Medical Stock Photo. Reproduced by permission.)

Aftercare

No special aftercare is required.

Risks

The risk of radiation exposure from a mammogram is considered virtually nonexistent. Experts are unanimous that any negligible risk is far outweighed by the potential benefits of mammography.

Some breast cancers do not show up on mammograms, or “hide” in dense breast tissue. A normal (or negative) study is not a guarantee that a woman is cancer-free. Mammograms find about 85% to 90% of breast cancers.

“False positive” readings also are possible, and 5% to 10% of mammogram results indicate the need for additional testing, most of which confirm that no cancer is present.

Normal results

A mammography report describes details about the x ray appearance of the breasts. It also rates the mammogram according to standardized categories, as part of the Breast Imaging Reporting and Data System (BIRADS) created by the American College of Radiology (ACR). A normal mammogram may be rated as BIRADS 1 or negative, which means no abnormalities were seen. A normal mammogram may also be rated as BIRADS 2 or benign findings. This means that one or more abnormalities were found but are clearly benign (not cancerous), or variations of normal. Some kinds of calcification, lymph nodes, or implants in the breast might generate a BIRADS 2 rating. A BIRADS 0 rating indicates that

KEY TERMS

Breast biopsy—A procedure in which suspicious tissue is removed and examined by a pathologist for cancer or other disease. The breast tissue may be obtained by open surgery or through a needle.

Radiographically dense—Difficult to see details of breast tissue on x ray.

the mammogram is incomplete and requires further assessment.

Abnormal results

Many mammograms are considered borderline or indeterminate in their findings. BIRADS 3 means an abnormality is present and probably (but not definitely) benign. A follow-up mammogram within a short interval of six months is suggested. This helps to ensure that the abnormality is not changing, or is “stable.” This stability in the abnormality indicates that a cancer is probably not present. If the abnormality is cancerous, it will likely grow and change in the time between mammograms. Some women are uncomfortable or anxious about waiting and may want to consult with their doctor about having a biopsy. BIRADS 4 means suspicious for cancer. A biopsy is usually recommended in this case. BIRADS 5 means an abnormality is highly suggestive of cancer. The suspicious area should be biopsied.

Often, screening mammograms are followed up with additional imaging. The reasons are numerous; they may mean the radiologist suspects a cancerous lesion, only that he or she cannot make a clear diagnosis from the screening mammogram views. The most common imaging methods are additional views on the mammogram, sometimes called magnification views, and ultrasound. In recent years, some patients have received **magnetic resonance imaging** (MRI) of the breast. A new technique called dual-energy contrast enhanced digital subtraction mammography is reported to find cancers that may be missed by conventional mammography. It may be ordered in the future as a follow-up study.

Resources

BOOKS

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- "Contrast Mammography Reveals Hard-to-find Cancers." *Cancer Weekly* October 14, 2003: 34.
- "Mammography in Women Over Forty Catches Disease Earlier." *Women's Health Weekly* August 14, 2003: 14.
- "New Digital Technique Improves Mammography Results." *Women's Health Weekly* September 18, 2003: 28.
- Smith, Robert A., et al. "American Cancer Society Guidelines for Breast Cancer Screening: Update 2003." *Cancer* May-June 2003: 141-170.
- "Stress—Not Pain—Is Major Barrier to Mammography." *Contemporary OB/GYN* July 2003: 17.

ORGANIZATIONS

- American Cancer Society. 1599 Clifton Rd., Atlanta, GA 30329. (800) ACS-2345. <<http://www.cancer.org>>.
- National Cancer Institute. Office of Cancer Communications, Bldg. 31, Room 10A31, Bethesda, MD 20892. NCI/Cancer Information Service: (800) 4-CANCER. <<http://cancernet.nci.nih.gov>>.
- U.S. Food and Drug Administration. 5600 Fishers Lane, Rockville, MD 20857. (800) 532-4440. <<http://www.fda.gov>>.

Ellen S. Weber, MSN
Teresa G. Odle

Manganese excess see **Mineral toxicity**

Mania

Definition

Mania is an abnormally elated mental state, typically characterized by feelings of euphoria, lack of inhibitions, racing thoughts, diminished need for sleep, talkativeness, risk taking, and irritability. In extreme cases, mania can induce **hallucinations** and other psychotic symptoms.

Description

Mania typically occurs as a symptom of **bipolar disorder** (a mood disorder characterized by both manic and depressive episodes). Individuals experiencing a manic episode often have feelings of self-importance, elation, talkativeness, sociability, and a desire to embark on goal-oriented activities, coupled with the less desirable characteristics of irritability, impatience, impulsiveness, hyperactivity, and a decreased need for

sleep. (Note: Hypomania is a term applied to a condition resembling mania. It is characterized by persistent or elevated expansive mood, hyperactivity, inflated self esteem, etc., but of less intensity than mania.) Severe mania may have psychotic features.

Causes and symptoms

Mania can be induced by the use or **abuse** of stimulant drugs such as **cocaine** and amphetamines. It is also the predominant feature of bipolar disorder, or manic depression, an affective mental illness that causes radical emotional changes and mood swings.

The Diagnostic and Statistical Manual of Mental Disorders, Fourth Edition (*DSM-IV*), the diagnostic standard for mental health professionals in the U.S., describes a manic episode as an abnormally elevated mood lasting at least one week that is distinguished by at least three of the following symptoms: inflated self-esteem, decreased need for sleep, talkativeness, racing thoughts, distractibility, increase in goal-directed activity, or excessive involvement in pleasurable activities that have a high potential for painful consequences. If the mood of the patient is irritable and not elevated, four of these symptoms are required.

Diagnosis

Mania is usually diagnosed and treated by a psychiatrist and/or a psychologist in an outpatient setting. However, most severely manic patients require hospitalization. In addition to an interview, several clinical inventories or scales may be used to assess the patient's mental status and determine the presence and severity of mania. An assessment commonly includes the Young Mania Rating Scale (YMRS). The Mini-Mental State Examination (MMSE) may also be given to screen out other illnesses such as **dementia**.

Treatment

Mania is primarily treated with drugs. The following mood-stabilizing agents are commonly prescribed to regulate manic episodes:

- Lithium (Cibalith-S, Eskalith, Lithane) is one of the oldest and most frequently prescribed drugs available for the treatment of mania. Because the drug takes four to seven days to reach a therapeutic level in the bloodstream, it is sometimes prescribed in conjunction with neuroleptics (**antipsychotic drugs**) and/or **benzodiazepines** (tranquilizers) to provide more immediate relief of mania.
- Carbamazepine (Tegretol, Atretol) is an anticonvulsant drug usually prescribed in conjunction with other

KEY TERMS

Hypomania—A less severe form of elevated mood state that is a characteristic of bipolar type II disorder.

Mixed mania—A mental state in which symptoms of both depression and mania occur simultaneously.

mood-stabilizing agents. The drug is often used to treat bipolar patients who have not responded well to lithium therapy. As of early 1998, carbamazepine was not approved for the treatment of mania by the FDA.

- Valproate (divalproex sodium, or Depakote; valproic acid, or Depakene) is an anticonvulsant drug prescribed alone or in combination with carbamazepine and/or lithium. For patients experiencing “mixed mania,” or mania with features of depression, valproate is preferred over lithium.

Clozapine (Clozaril) is an atypical antipsychotic medication used to control manic episodes in patients who have not responded to typical mood-stabilizing agents. The drug has also been a useful preventative treatment in some bipolar patients. Other new anticonvulsants (lamotrigine, gabapentin) are being investigated for treatment of mania and bipolar disorder.

Prognosis

Patients experiencing mania as a result of bipolar disorder will require long-term care to prevent recurrence; bipolar disorder is a chronic condition that requires lifelong observation and treatment after diagnosis. Data show that almost 90% of patients who experience one manic episode will go on to have another.

Prevention

Mania as a result of bipolar disorder can only be prevented through ongoing pharmacologic treatment. Patient education in the form of therapy or self-help groups is crucial for training patients to recognize signs of mania and to take an active part in their treatment program. Psychotherapy is an important adjunctive treatment for patients with bipolar disorder.

Resources

ORGANIZATIONS

American Psychiatric Association. 1400 K Street NW, Washington, DC 20005. (888) 357-7924. <<http://www.psych.org>>.

National Alliance for the Mentally Ill (NAMI). Colonial Place Three, 2107 Wilson Blvd., Ste. 300, Arlington, VA 22201-3042. (800) 950-6264. <<http://www.nami.org>>.
National Depressive and Manic-Depressive Association (NDMDA). 730 N. Franklin St., Suite 501, Chicago, IL 60610. (800) 826-3632. <<http://www.ndmda.org>>.
National Institute of Mental Health. Mental Health Public Inquiries, 5600 Fishers Lane, Room 15C-05, Rockville, MD 20857. (888) 826-9438. <<http://www.nimh.nih.gov>>.

Paula Anne Ford-Martin

Manic depression see **Bipolar disorder**

Manic episode see **Mania**

MAO inhibitors see **Monoamine oxidase inhibitors**

Marasmus see **Protein-energy malnutrition**

Marble bones see **Osteopetroses**

Marburg virus infection see **Hemorrhagic fevers**

Marfan syndrome

Definition

Marfan syndrome is an inherited disorder of the connective tissue that causes abnormalities of the patient's eyes, cardiovascular system, and musculoskeletal system. It is named for the French pediatrician, Antoine Marfan (1858-1942), who first described it in 1896. Marfan syndrome is sometimes called arachnodactyly, which means “spider-like fingers” in Greek, since one of the characteristic signs of the disease is disproportionately long fingers and toes. It is estimated that one person in every 3000-5000 has Marfan syndrome, or about 50,000 people in the United States. Marfan syndrome is one of the more common inheritable disorders.

Description

Marfan syndrome affects three major organ systems of the body: the heart and circulatory system, the bones and muscles, and the eyes. The genetic mutation responsible for Marfan was discovered in 1991. It affects the body's production of fibrillin, which is a protein that is an important part of connective tissue. Fibrillin is the primary component of the microfibrils that allow tissues to stretch repeatedly without weakening. Because the

patient's fibrillin is abnormal, his or her connective tissues are looser than usual, which weakens or damages the support structures of the entire body.

The most common external signs associated with Marfan syndrome include excessively long arms and legs, with the patient's arm span being greater than his or her height. The fingers and toes may be long and slender, with loose joints that can be bent beyond their normal limits. This unusual flexibility is called hypermobility. The patient's face may also be long and narrow, and he or she may have a noticeable curvature of the spine. It is important to note, however, that Marfan patients vary widely in the external signs of their disorder and in their severity; even two patients from the same family may look quite different. Most of the external features of Marfan syndrome become more pronounced as the patient gets older, so that diagnosis of the disorder is often easier in adults than in children. In many cases, the patient may have few or very minor outward signs of the disorder, and the diagnosis may be missed until the patient develops vision problems or cardiac symptoms.

Marfan syndrome by itself does not affect a person's intelligence or ability to learn. There is, however, some clinical evidence that children with Marfan have a slightly higher rate of hyperactivity and attention-deficit disorder (ADD) than the general population. In addition, a child with undiagnosed nearsightedness related to Marfan may have difficulty seeing the blackboard or reading printed materials, and thus do poorly in school.

Marfan syndrome affects males and females equally, and appears to be distributed equally among all races and ethnic groups. The rate of mutation of the fibrillin gene, however, appears to be related to the age of the patient's father; older fathers are more likely to have new mutations appear in chromosome 15.

Causes and symptoms

Marfan syndrome is caused by a single gene for fibrillin on chromosome 15, which is inherited in most cases from an affected parent. Between 15 and 25% of cases result from spontaneous mutations. Mutations of the fibrillin gene (FBN1) are unique to each family affected by Marfan, which makes rapid genetic diagnosis impossible, given present technology. The syndrome is an autosomal dominant disorder, which means that someone who has it has a 50% chance of passing it on to any offspring.

Another important genetic characteristic of Marfan syndrome is variable expression. This term means that the mutated fibrillin gene can produce a

variety of symptoms of very different degrees of severity, even in members of the same family.

Cardiac and circulatory abnormalities

The most important complications of Marfan are those affecting the heart and major blood vessels; some are potentially life-threatening. About 90% of Marfan patients will develop cardiac complications.

- **Aortic enlargement.** This is the most serious potential complication of Marfan syndrome. Because of the abnormalities of the patient's fibrillin, the walls of the aorta (the large blood vessel that carries blood away from the heart) are weaker than normal and tend to stretch and bulge out of shape. This stretching increases the likelihood of an **aortic dissection**, which is a tear or separation between the layers of tissue that make up the aorta. An aortic dissection usually causes severe **pain** in the abdomen, back, or chest, depending on the section of the aorta that is affected. Rupture of the aorta is a medical emergency requiring immediate surgery and medication.
- **Aortic regurgitation.** A weakened and enlarged aorta may allow some blood to leak back into the heart during each heartbeat; this condition is called aortic regurgitation. Aortic regurgitation occasionally causes **shortness of breath** during normal activity. In serious cases, it causes the left ventricle of the heart to enlarge and may eventually lead to heart failure.
- **Mitral valve prolapse.** Between 75 and 85% of Marfan patients have loose or "floppy" mitral valves, which are the valves that separate the chambers of the heart. When these valves do not cover the opening between the chambers completely, the condition is called mitral valve prolapse. Complications of mitral valve prolapse include **heart murmurs** and **arrhythmias**. In rare cases, mitral valve prolapse can cause sudden death.
- **Infective endocarditis.** Infective endocarditis is an infection of the endothelium, the tissue that lines the heart. In patients with Marfan, it is the abnormal mitral valve that is most likely to become infected.
- **Other complications.** Some patients with Marfan develop cystic disease of the lungs or recurrent spontaneous **pneumothorax**, which is a condition in which air accumulates in the space around the lungs. Many will also eventually develop **emphysema**.

Musculoskeletal abnormalities

Marfan syndrome causes an increase in the length of the patient's bones, with decreased support from the ligaments that hold the bones together. As a result,

the patient may develop various deformities of the skeleton or disorders related to the relative looseness of the ligaments.

Disorders of the spine

- **Scoliosis.** Scoliosis, or curvature of the spine, is a disorder in which the vertebrae that make up the spine twist out of line from side to side into an S-shape or a spiral. It is caused by a combination of the rapid growth of children with Marfan, and the looseness of the ligaments that help the spine to keep its shape.
- **Kyphosis** is an abnormal outward curvature of the spine at the back, sometimes called hunch back when it occurs in the upper back. Marfan patients may develop kyphosis either in the upper (thoracic) spine or the lower (lumbar) spine.
- **Spondylolisthesis.** Spondylolisthesis is the medical term for a forward slippage of one vertebra on the one below it. It produces an ache or stiffness in the lower back.
- **Dural ectasia.** The dura is the tough, fibrous outermost membrane covering the brain and the spinal cord. The weak dura in Marfan patients swells or bulges under the pressure of the spinal fluid. This swelling is called ectasia. In most cases, dural ectasia occurs in the lower spine, producing low back ache, a burning feeling, or **numbness** or weakness in the legs.

Disorders of the chest and lower body

- **Pectus excavatum.** Pectus excavatum is a malformation of the chest in which the patient's breastbone, or sternum, is sunken inward. It can cause difficulties in breathing, especially if the heart, spine, and lung have been affected by Marfan. It also usually causes concerns about appearance.
- **Pectus carinatum.** In other patients with Marfan the sternum is pushed outward and narrowed. Although pectus carinatum does not cause breathing difficulties, it can cause embarrassment about appearance. A few patients with Marfan may have a pectus excavatum on one side of their chest and a pectus carinatum on the other.
- **Foot disorders.** Patients with Marfan are more likely to develop pes planus (flat feet) or so-called "claw" or "hammer" toes than people in the general population. They are also more likely to suffer from chronic pain in their feet.
- **Protrusio acetabulae.** The acetabulum is the socket of the hip joint. In patient's with Marfan, the

acetabulum becomes deeper than normal during growth, for reasons that are not yet understood. Although protrusio acetabulae does not cause problems during childhood and adolescence, it can lead to a painful form of arthritis in adult life.

Disorders of the eyes and face

Although the visual problems that are related to Marfan syndrome are rarely life-threatening, they are important in that they may be the patient's first indication of the disorder. Eye disorders related to the syndrome include the following:

- **Myopia (nearsightedness).** Most patients with Marfan develop nearsightedness, usually in childhood.
- **Ectopia lentis.** Ectopia lentis is the medical term for dislocation of the lens of the eye. Between 65 and 75% of Marfan patients have dislocated lenses. This condition is an important indication for diagnosis of the syndrome because there are relatively few other disorders that produce it.
- **Glaucoma.** This condition is much more prevalent in patients with Marfan syndrome than in the general population.
- **Cataracts.** Patients with Marfan are more likely to develop cataracts, and to develop them much earlier in life, sometimes as early as 40 years of age.
- **Retinal detachment.** Patients with Marfan are more vulnerable to this disorder because of the weakness of their connective tissues. Untreated retinal detachment can cause blindness. The danger of retinal detachment is an important reason for patients to avoid contact sports or other activities that could cause a blow on the head or being knocked to the ground.
- **Other facial problems.** Patients with Marfan sometimes develop dental problems related to crowding of the teeth caused by a high-arched palate and a narrow jaw.

Other disorders

- **Striae.** Striae are stretch marks in the skin caused by rapid weight gain or growth; they frequently occur in pregnant women, for example. Marfan patients often develop striae over the shoulders, hips, and lower back at an early age because of rapid bone growth. Although the patient may be self-conscious about the striae, they are not a danger to health.
- **Obstructive sleep apnea.** Obstructive sleep apnea refers to partial obstruction of the airway during sleep, causing irregular breathing and sometimes **snoring**. In patients with Marfan, obstructive sleep

apnea is caused by the unusual flexibility of the tissues lining the patient's airway. This disturbed breathing pattern increases the risk of aortic dissection.

Diagnosis

Presently, there is no objective diagnostic test for Marfan syndrome, in part because the disorder does not produce any measurable biochemical changes in the patient's blood or body fluids, or cellular changes that could be detected from a tissue sample. Although researchers in molecular biology are currently investigating the FBNI gene through a process called mutational analysis, it is presently not useful as a diagnostic test because there is evidence that there can be mutations in the fibrillin gene that do not produce Marfan. Similarly, there is no reliable prenatal test, although some physicians have used ultrasound to try to determine the length of fetal limbs in at-risk pregnancies.

The diagnosis is made by taking a family history and a thorough examination of the patient's eyes, heart, and bone structure. The examination should include an echocardiogram taken by a cardiologist, a slit-lamp **eye examination** by an ophthalmologist, and a work-up of the patient's spinal column by an orthopedic specialist. In terms of the cardiac examination, a standard electrocardiogram (EKG) is not sufficient for diagnosis; only the echocardiogram can detect possible enlargement of the aorta. The importance of the slit-lamp examination is that it allows the doctor to detect a dislocated lens, which is a significant indication of the syndrome.

The symptoms of Marfan syndrome in some patients resemble the symptoms of homocystinuria, which is an inherited disorder marked by extremely high levels of homocystine in the patient's blood and urine. This possibility can be excluded by a urine test.

In other cases, the diagnosis remains uncertain because of the mildness of the patient's symptoms, the absence of a family history of the syndrome, and other variables. These borderline conditions are sometimes referred to as marfanoid syndromes.

Treatment

The treatment and management of Marfan is tailored to the specific symptoms of each patient. Some patients find that the syndrome has little impact on their overall lifestyle; others have found their lives centered on the disorder.

Cardiovascular system

After a person has been diagnosed with Marfan, he or she should be monitored with an echocardiogram every six months until it is clear that the aorta is not growing larger. After that, the patient should have an echocardiogram once a year. If the echocardiogram does not allow the physician to visualize all portions of the aorta, CT (computed tomography) or MRI (**magnetic resonance imaging**) may be used. In cases involving a possible aortic dissection, the patient may be given a TEE (transesophageal echocardiogram).

Medications. A Marfan patient may be given drugs called beta-blockers to slow down the rate of aortic enlargement and decrease the risk of dissection by lowering the blood pressure and decreasing the forcefulness of the heartbeat. The most commonly used beta-blockers in Marfan patients are propranolol (Inderal) and atenolol (Tenormin). Patients who are allergic to beta-blockers may be given a calcium blocker such as verapamil.

Because Marfan patients are at increased risk for infective endocarditis, they must take a prophylactic dose of an antibiotic before having dental work or minor surgery, as these procedures may allow bacteria to enter the bloodstream. Penicillin and amoxicillin are the **antibiotics** most often used.

Surgical treatment. Surgery may be necessary if the width of the patient's aorta increases rapidly or reaches a critical size (about 2 inches). As of 2000, the most common surgical treatment involves replacing the patient's aortic valve and several inches of the aorta itself with a composite graft, which is a prosthetic heart valve sewn into one end of a Dacron tube. This surgery has been performed widely since about 1985; most patients who have had a composite graft have not needed additional surgery.

Patients who have had a valve replaced must take an anticoagulant medication, usually warfarin (Coumadin), in order to minimize the possibility of a clot forming on the prosthetic valve.

Musculoskeletal system

Children diagnosed with Marfan should be checked for scoliosis by their pediatricians at each annual **physical examination**. The doctor simply asks the child to bend forward while the back is examined for changes in the curvature. In addition, the child's spine should be x rayed in order to measure the extent of scoliosis or kyphosis. The curve is measured in degrees by the angle between the vertebrae as seen on the x ray. Curves of 20° or less are not likely to become

worse. Curves between 20 and 40 degrees are likely to increase in children or adolescents. Curves of 40 degrees or more are highly likely to worsen, even in an adult, because the spine is so badly imbalanced that the force of gravity will increase the curvature.

Scoliosis between 20 and 40 degrees in children is usually treated with a back brace. The child must wear this appliance about 23 hours a day until growth is complete. If the spinal curvature increases to 40 or 50 degrees, the patient may require surgery in order to prevent lung problems, back pain, and further deformity. Surgical treatment of scoliosis involves straightening the spine with metal rods and fusing the vertebrae in the straightened position.

Spondylolisthesis is treated with a brace in mild cases. If the slippage is more than 30 degree, the slipped vertebra may require surgical realignment.

Dural ectasia can be distinguished from other causes of back pain on an MRI. Mild cases are usually not treated. Medication or spinal shunting to remove some of the spinal fluid are used to treat severe cases.

Pectus excavatum and pectus carinatum can be treated by surgery. In pectus excavatum, the deformed breastbone and ribs are raised and straightened by a metal bar. After four to six months, the bar is removed in an outpatient procedure.

Protrusio acetabulae may require surgery in adult life to provide the patient with an artificial hip joint, if the arthritic pains are severe.

Pain in the feet or limbs is usually treated with a mild analgesic such as **acetaminophen**. Patients with Marfan should consider wearing shoes with low heels, special cushions, or orthotic inserts. Foot surgery is rarely necessary.

Visual and dental concerns

Patients with Marfan should have a thorough eye examination, including a slit-lamp examination, to test for dislocation of the lens as well as nearsightedness. Dislocation can be treated by a combination of special glasses and daily use of one percent atropine sulfate ophthalmic drops, or by surgery.

Because patients with Marfan are at increased risk of glaucoma, they should have the fluid pressure inside the eye measured every year as part of an eye examination. Glaucoma can be treated with medications or with surgery.

Cataracts are treated with increasing success by implant surgery. It is important, however, to seek treatment at medical centers with eye surgeons

familiar with the possible complications of **cataract surgery** in patients with Marfan syndrome.

All persons with Marfan should be taught to recognize the signs of retinal detachment (sudden blurring of vision in one eye becoming progressively worse without pain or redness) and to seek professional help immediately.

Children with Marfan should be evaluated by their dentist at each checkup for crowding of the teeth and possible misalignment, and referred to an orthodontist if necessary.

Athletic activities and occupational choice. People with Marfan should avoid sports or occupations that require heavy weight lifting, rough physical contact, or rapid changes in atmospheric pressure (e.g., scuba diving). Weight lifting increases blood pressure, which in turn may enlarge the aorta. Rough physical contact may cause retinal detachment. Sudden changes in air pressure may produce pneumothorax. Regular noncompetitive physical **exercise**, however, is beneficial for Marfan patients. Good choices include brisk walking, shooting baskets, and slow-paced tennis.

Social and lifestyle issues

Smoking. Smoking is particularly harmful for Marfan patients because it increases their risk of emphysema.

Pregnancy. Until very recently, women with Marfan were advised not to become pregnant because of the risk of aortic enlargement or dissection. The development of beta-blockers and echocardiograms, however, allows doctors now to monitor patients throughout pregnancy. It is recommended that patients have an echocardiogram during each of the three trimesters of pregnancy. Normal, vaginal delivery is not necessarily more stressful than a Caesarian section, but patients in prolonged labor may be given a Caesarian to reduce strain on the heart. A pregnant woman with Marfan should also receive **genetic counseling** regarding the 50% risk of having a child with the syndrome.

Appearance and Social Concerns. Children and adolescents with Marfan may benefit from supportive counseling regarding appearance, particularly if their symptoms are severe and causing them to withdraw from social activities. In addition, families may wish to seek counseling regarding the effects of the syndrome on relationships within the family. Many people respond with guilt, fear, or blame when a genetic disorder is diagnosed in the family, or they may overprotect the affected member. Support groups are often good sources of information about Marfan; they can

KEY TERMS

Arachnodactyly—A condition characterized by abnormally long and slender fingers and toes.

Ectopia lentis—Dislocation of the lens of the eye. It is one of the most important single indicators in diagnosing Marfan syndrome.

Fibrillin—A protein that is an important part of the structure of the body's connective tissue. In Marfan's syndrome, the gene responsible for fibrillin has mutated, causing the body to produce a defective protein.

Hypermobility—Unusual flexibility of the joints, allowing them to be bent or moved beyond their normal range of motion.

Kyphosis—An abnormal outward curvature of the spine, with a hump at the upper back.

Pectus carinatum—An abnormality of the chest in which the sternum (breastbone) is pushed outward. It is sometimes called "pigeon breast."

Pectus excavatum—An abnormality of the chest in which the sternum (breastbone) sinks inward; sometimes called "funnel chest."

Scoliosis—An abnormal, side-to-side curvature of the spine.

offer helpful suggestions about living with it as well as emotional support.

Prognosis

The prognosis for patient's with Marfan has improved markedly in recent years. As of 1995, the life expectancy of people with the syndrome has increased to 72 years, up from 48 years in 1972. This dramatic improvement is attributed to new surgical techniques, improved diagnosis, and new techniques of medical treatment.

The most important single factor in improving the patient's prognosis is early diagnosis. The earlier that a patient can benefit from the new techniques and lifestyle modifications, the more likely he or she is to have a longer life expectancy.

Resources

BOOKS

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Pyeritz, Reed E., and Cheryll Gasner. *The Marfan Syndrome*. New York: National Marfan Syndrome, 1999.

Rebecca J. Frey, PhD

Marie-Strümpell disease see **Ankylosing spondylitis**

Marijuana

Definition

Marijuana (marihuana) *Cannabis sativa* L., also known as Indian hemp, is a member of the Cannabaceae or hemp family, thought to have originated in the mountainous districts of India, north of the Himalayan mountains.

Description

The herb was referred to as "hempe" in A.D. 1000 and listed in a dictionary under that English name. Supporters of the notorious Pancho Villa first used the name marijuana in 1895 in Sonora, Mexico. They called the mood-altering herb they smoked marijuana. The term hashish, is derived from the name for the Saracen soldiers, called *hashashins*, who ingested the highly potent cannabis resin before being sent out to assassinate enemies.

Two related species of cannabis are *C. ruderalis*, and *C. indica*, a variety known as Indian hemp. Indian hemp grows to a height of about 4 ft (1.2 m) and the seed coats have a marbled appearance.

The species *C. sativa* L. has many variations, depending on the soil, temperature, and light conditions, and the origin of the parent seed. These factors also affect the relative amounts of THC (tetra-hydrocannabinol) and cannabidiol, the chemicals present in varying amounts in cannabis that determine if the plant is primarily a fiber type or an intoxicant. Generally the species grown at higher elevations and in hotter climates exudes more of the resin and is more medicinally potent.

Marijuana is a somewhat weedy plant and may grow as high as 18 ft (5.4 m). The hairy leaves are arranged opposite one another on the erect and branching stem. Leaves are palmate and compound, deeply divided into five to seven narrow, toothed and pointed leaflets. Male and female flowers are small and greenish in color and grow on separate plants.

Male flowers grow in the leaf axils in elongated clusters. The female flowers grow in spike-like clusters. The resinous blossoms have five sepals and five petals. The male and female blossoms can be distinguished at maturity. The male plant matures first, shedding its pollen and dying after flowering. Female plants die after dropping the mature seeds. Marijuana produces an abundance of quickly germinating seeds. This hardy annual is wind pollinated and has escaped from cultivation to grow wild along roadsides, trails, stream banks, and in wayside places throughout the world. The plant matures within three to five months after the seed has been sown.

History

Marijuana has been cultivated for thousands of years. Cannabis was first described for its therapeutic use in the first known Chinese pharmacopoeia, the *Pen Ts'ao*. (A pharmacopoeia is a book containing a list of medicinal drugs, and their descriptions of preparation and use.) Cannabis was called a “superior” herb by the Emperor Shen-Nung (2737-2697 B.C.), who is believed to have authored the work. Cannabis was recommended as a treatment for numerous common ailments. Around that same period in Egypt, cannabis was used as a treatment for sore eyes. The herb was used in India in cultural and religious ceremonies, and recorded in Sanskrit scriptural texts around 1,400 B.C. Cannabis was considered a holy herb and was characterized as the “soother of grief,” “the sky flyer,” and “the poor man’s heaven.” Centuries later, around 700 B.C., the Assyrian people used the herb they called *Qunnabu*, for incense. The ancient Greeks used cannabis as a remedy to treat inflammation, earache, and **edema** (swelling of a body part due to collection of fluids). Shortly after 500 B.C. the historian and geographer Herodotus recorded that the peoples known as Scythians used cannabis to produce fine linens. They called the herb *kannabis* and inhaled the “intoxicating vapor” that resulted when it was burned. By the year 100 B.C. the Chinese were using cannabis to make paper.

Cannabis use and cultivation migrated with the movement of various traders and travelers, and knowledge of the herb’s value spread throughout the Middle East, Eastern Europe, and Africa. Around 100, Dioscorides, a surgeon in the Roman Legions under the Emperor Nero, named the herb *Cannabis sativa* and recorded numerous medicinal uses. In the second century, the Chinese physician Hoa-Tho, used cannabis in surgical procedures, relying on its analgesic properties. In ancient India, around 600, Sanskrit writers recorded a recipe for “pills of gaiety,” a

combination of hemp and sugar. By 1150, Moslems were using cannabis fiber in Europe’s first paper production. This use of cannabis as a durable and renewable source of paper fiber continued for the next 750 years.

By the 1300s, government and religious authorities, concerned about the psychoactive effects on citizens consuming the herb, were placing harsh restrictions on its use. The Emir Soudon Sheikhoune of Joneima outlawed cannabis use among the poor. He destroyed the crops and ordered that offenders’ teeth be pulled out. In 1484, Pope Innocent VIII outlawed the use of hashish, a concentrated form of cannabis. Cannabis cultivation continued, however, because of its economic value. A little more than a century later, the English Queen Elizabeth I issued a decree commanding that landowners holding sixty acres or more must grow hemp or pay a fine. Commerce in hemp, which was primarily valued for the strength and versatility of its fibers, was profitable and thriving. Hemp ropes and sails were crossing the sea to North America with the explorers. By 1621, the British were growing cannabis in Virginia where cultivation of hemp was mandatory. In 1776, the Declaration of Independence was drafted on hemp paper. Both President George Washington and President Thomas Jefferson were advocates of hemp as a valuable cash crop. Jefferson urged farmers to grow the crop in lieu of tobacco. By the 1850s, hemp had become the third largest agricultural crop grown in North America. The U. S. Census of that year recorded 8,327 hemp plantations, each with 2,000 or more acres in cultivation. But the invention of the cotton gin was already bringing many changes, and cotton was becoming a prime and profitable textile fiber. More change came with the introduction of the sulfite and chlorine processes used to turn trees into paper. Restrictions on the personal use of cannabis as a mood-altering, psychoactive herb, were soon to come.

Controversy

The 1856 edition of the *Encyclopedia Britannica*, in its lengthy entry on hemp, noted that the herb “produces inebriation and **delirium** of decidedly hilarious character, inducing violent laughter, jumping and dancing.” This inebriating effect of marijuana use has fueled the controversy and led to restrictions that have surrounded marijuana use throughout history in many cultures and regions of the world. Cannabis use has been criminalized in some parts of the United States since 1915. Utah was the first state to criminalize it, then California and Texas. By 1923, Louisiana, Nevada, Oregon, and Washington had legal restrictions on the herb. New York prohibited cannabis use in 1927.

Despite the restrictions, cannabis use was woven into the cultural and social fabric in some communities, and widespread use persisted, particularly among the Mexican, Asian, and African American populations.

In 1937, the federal government passed the Marihuana Tax Act, prohibiting the cultivation and farming of marijuana. This bill was introduced to Congress by then Secretary of the Treasury Andrew Mellon, who was also a banker for the DuPont Corporation. That same year, the DuPont Chemical Company filed a patent for nylon, plastics, and a new bleaching process for paper. The 1937 Marijuana Transfer Tax Bill prohibited industrial and medical use of marijuana and classified the flowering tops as narcotic, and restrictions on the cultivation and use of cannabis continued. Marijuana was categorized as an illegal narcotic, in the company of **LSD** and heroin, **cocaine**, and morphine. Illegal use continued. The FBI publication, *Uniform Crime Reports for The United States*, 1966 reported that 641,642 Americans were arrested for marijuana offenses that year, with as many as 85% of these arrests for simple possession, rather than cultivation or commerce.

In a reversal of the state-by-state progression of criminalizing marijuana that led to the 1937 Marijuana Transfer Tax Bill, there is a movement underway, state by state, to endorse the legalized use of medical marijuana. By 1992, 35 states in the U. S. had endorsed referenda for medical marijuana. A growing body of scientific research and many thousands of years of folk use support the importance of medical marijuana in treatment of a variety of illnesses, and the economic value of hemp in the textile, paper, and cordage industries has a long history.

The controversy and misinformation persists around this relatively safe and non-toxic herb. The World Health Organization, in a 1998 study, stated that the risks from cannabis use were unlikely to seriously compare to the public health risks of the legal drugs, alcohol and tobacco. And despite thousands of years of human consumption, not one **death** has been directly attributed to cannabis use. According to Lester Grinspoon, MD, and James B. Bakalar, JD, in a 1995 *Journal of the American Medical Association* article, "Marihuana is also far less addictive and far less subject to **abuse** than many drugs now used as **muscle relaxants**, hypnotics, and **analgesics**. The chief legitimate concern is the effect of **smoking** on the lungs. Cannabis smoke carries even more tars and other particulate matter than tobacco smoke. But the amount smoked is much less, especially in medical use, and once marihuana is an openly recognized medicine, solutions may be found."

Purpose

The whole cannabis plant, including buds, leaves, seeds, and root, have all been utilized throughout the long history of this controversial herb. Despite persistent legal restrictions and severe criminal penalties for illicit use, marijuana continues to be widely used in the United States, and throughout the world, both for its mood-altering properties and its proven medicinal applications. The conflicting opinions on the safety and effectiveness of cannabis in a climate of prohibition make any discussion of its beneficial uses politically charged. Marijuana has analgesic, antiemetic, anti-inflammatory, sedative, anticonvulsive, and laxative actions. Clinical studies have demonstrated its effectiveness in relieving **nausea and vomiting** following **chemotherapy** treatments for **cancer**. The herb has also been shown to reduce intra-ocular pressure in the eye by as much as 45%, a beneficial action in the treatment for **glaucoma**. Cannabis has proven anticonvulsive action, and may be helpful in treating epilepsy. Other research has documented an in-vitro tumor inhibiting effect of THC. Marijuana also increases appetite and reduces **nausea** and has been used with **AIDS** patients to counter weight loss and "wasting" that may result from the disease. Several chemical constituents of cannabis displayed antimicrobial action and antibacterial effects in research studies. The components CBC and d-9-tetrahydrocannabinol have been shown to destroy and inhibit the growth of streptococci and staphylococci bacteria.

Cannabis contains chemical compounds known as cannabinoids. Different cannabinoids seem to exert different effects on the body after ingestion. Scientific research indicates that these substances have potential therapeutic value for **pain** relief, control of nausea and **vomiting**, and appetite stimulation. The primary active agent identified to date is 9-tetrahydrocannabinol, known as THC. This chemical may constitute as much as 12% of the active chemicals in the herb, and is said to be responsible for as much as 70–100% of the euphoric action, or "high," experienced when ingesting the herb. The predominance of this mental lightness or "euphoria" depends on the balance of other active ingredients and the freshness of the herb. THC degrades into a component known as cannabinalol, or CBN. This relatively inactive chemical predominates in marijuana that has been stored too long prior to use. Another chemical component, cannabidiol, known as CBD, has a sedative and mildly analgesic effect, and contributes to a somatic heaviness sometimes experienced by marijuana users.

Before prohibition, cannabis was recommended for treatment of **gonorrhea**, **angina** pectoris

(constricting pain in the chest due to insufficient blood to the heart), and **choking** fits. It was also used for **insomnia**, **neuralgia**, rheumatism, gastrointestinal disorders, **cholera**, **tetanus**, epilepsy, strychnine **poisoning**, **bronchitis**, **whooping cough**, and **asthma**. Other phytotherapeutic (plant-based therapeutic) uses include treatment of ulcers, cancer, **emphysema**, migraine, Lou Gehrig's disease, HIV infection, and **multiple sclerosis**.

The United States federal government policy prohibits physicians from prescribing marijuana, even for seriously ill patients because of possible adverse effects, and the disputed belief that cannabis is dangerously addictive. U. S. Attorney General Janet Reno warned that physicians in any state who prescribed marijuana could lose the privilege of writing prescriptions, be excluded from Medicare and Medicaid reimbursement, and even be prosecuted for a federal crime, according to a 1997 editorial in the *New England Journal of Medicine*.

Preparations

Cannabis extracts, prepared for medicinal application, are prohibited in the United States. Marijuana is ingested by smoking, which quickly delivers the active ingredients to the blood system. The dried herb can also be prepared for eating in cookies or other baked goods. The essential oil consists of beta caryophyllenes, humules, caryophyllene oxide, alpha-pinenes, beta-pinenes, limonene, myrcene, and betaocimene. The oil expressed from the seeds is used for massage and in making salves used to relieve muscle strain.

Precautions

Marijuana is considered a Class I narcotic and its use has been restricted by federal law since 1937. Penalties include fines and imprisonment. The National Commission on Marihuana and Drug Abuse concluded in 1972 that "A careful search of the literature and testimony of the nation's health officials has not revealed a single human fatality in the United States proven to have resulted solely from ingestion of marihuana."

Research has shown that cannabis acts to increase heart frequency by as much as 40 beats per minute. A study reported by The American Heart Association in February 2000, concluded that smoking marijuana can precipitate a **heart attack** in persons with pre-existing heart conditions. One hour after smoking marijuana, the likelihood of having a heart attack is four and one-half times greater than if the person had not smoked, according to the research.

KEY TERMS

Antiemetic—A drug or herbal preparation given to relieve nausea and vomiting. Marijuana has antiemetic properties.

Cannabinoids—The chemical compounds that are the active principles in marijuana.

Euphoria—An intense feeling of elation or well-being. Many marijuana users experience temporary euphoria.

An additional health concern is the effect that marijuana smoking has on the lungs. Cannabis smoke carries more tars and other particulate matter than tobacco smoke.

Although marijuana is less likely than some other drugs to lead to dependence, heavy users may suffer a withdrawal syndrome characterized by **anxiety**, irritability, chills, and **muscle cramps** if they stop usage abruptly.

More seriously, marijuana has been linked to the onset or worsening of certain psychiatric conditions, including **panic disorder**, **schizophrenia**, and depersonalization disorder. Persons diagnosed with or at risk for these conditions should not use marijuana.

Side effects

The *PDR For Herbal Medicine* reports, "No health hazards or side effects are known in conjunction with the proper administration of designated therapeutic dosages." Smoking the herb, however, "... leads almost at once to euphoric states (pronounced gaiety, laughing fits)," according to the PDR, while "long term usage leads to a clear increase in tolerance for most of the pharmacological effects." The ability to safely operate automobiles and machinery can be impaired for up to eight hours after ingesting the herb. Chronic abuse results in "laryngitis, bronchitis, apathy, psychic decline and disturbances of genital functions," according to the PDR.

Some people may be hypersensitive to marijuana. They may be allergic or hypersensitive to the plant. Chronic sinus fungal infections have been linked to chronic marijuana smoking.

A team of German researchers reported in early 2004 that marijuana appears to speed up the

progression of cancer. If this finding is replicated by other researchers, it would limit the usefulness of marijuana in treating pain and depression in cancer patients.

Interactions

Marijuana use may mask the perceived effects of alcohol and cocaine when the drugs are consumed together. Marijuana is said to exert a synergistic effect with other medicinal agents. When used with nitrous oxide it may enhance the effect.

Resources

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Marriage counseling

Definition

Marriage counseling is a type of psychotherapy for a married couple or established partners that tries to resolve problems in the relationship. Typically, two people attend counseling sessions together to discuss specific issues.

Purpose

Marriage counseling is based on research that shows that individuals and their problems are best handled within the context of their relationships. Marriage counselors are trained in psychotherapy and family systems, and focus on understanding their clients' symptoms and the way their interactions contribute to problems in the relationship.

Description

Marriage counseling is usually a short-term therapy that may take only a few sessions to work out problems in the relationship. Typically, marriage counselors ask questions about the couple's roles, patterns, rules, goals, and beliefs. Therapy often begins as the couple analyzes the good and bad aspects of the relationship. The marriage counselor then works with the couple to help them understand that, in most cases, both partners are contributing to problems in the relationship. When this is understood, the two can then learn to change how they interact with each other to solve problems. The partners may be encouraged to draw up a contract in which each partner describes the behavior he or she will be trying to maintain.

Marriage is not a requirement for two people to get help from a marriage counselor. Anyone person wishing to improve his or her relationships can get help with behavioral problems, relationship issues, or with mental or emotional disorders. Marriage counselors also offer treatment for couples before they get married to help them understand potential problem areas. A third type of marriage counseling involves postmarital therapy, in which divorcing couples who share children seek help in working out their differences. Couples in the midst of a divorce find that marriage therapy during separation can help them find a common ground as they negotiate interpersonal issues and child custody.

Choosing a therapist

A marriage counselor is trained to use different types of therapy in work with individuals, couples, and groups. American Association of Marriage and Family Therapy (AAMFT) training includes supervision by experienced therapists, a minimum of a master's degree (including specific training in marriage and family therapy), and specific graduate training in marriage and **family therapy**.

When looking for a marriage counselor, a couple should find out the counselor's training and educational background, professional associations, such as AAMFT, and state licensure, and whether the person has experience in treating particular kinds of problem. Also, questions should be asked concerning fees, insurance coverage, the average length of therapy, and so on.

Normal results

Marriage counseling helps couples learn to deal more effectively with problems, and can help prevent small problems from becoming serious. Research shows that marriage counseling, when effective, tends to improve a person's physical as well as mental health, in addition to improving the relationship.

Resources

ORGANIZATIONS

American Association for Marriage and Family Therapy. 1133 15th St., NW Suite 300, Washington, DC 20005-2710. (202) 452-0109. <<http://www.aamft.org>>.

American Psychological Association (APA). 750 First St. NE, Washington, DC 20002-4242. (202) 336-5700. <<http://www.apa.org>>.

Carol A. Turkington

Marshall-Marchetti-Krantz procedure

Definition

The Marshall-Marchetti-Krantz procedure surgically reinforces the bladder neck in order to prevent unintentional urine loss.

Purpose

The Marshall-Marchetti-Krantz procedure is performed to correct **stress** incontinence in women, a common result of **childbirth** and/or **menopause**. Incontinence also occurs when an individual involuntarily loses urine after pressure is placed on the abdomen (like during **exercise**, sexual activity, sneezing, coughing, laughing, or hugging).

Precautions

In some women, stress incontinence may be controlled through nonsurgical means, such as:

- Kegel exercises (exercises that tighten pelvic muscles)
- Biofeedback (monitors temperature and muscle contractions in the vagina to help incontinent patients control their pelvic muscles)
- **Bladder training** (behavioral modification program used to treat stress incontinence)
- Medication
- Inserted incontinence devices.

Each patient should undergo a full diagnostic workup to determine the best course of treatment.

Description

The Marshall-Marchetti-Krantz procedure, also known as retropubic suspension or bladder neck suspension surgery, is performed by a surgeon in a hospital setting. The patient is placed under **general anesthesia**, and a long, thin, flexible tube (catheter) is inserted into the bladder through the narrow tube (urethra) that drains the body's urine. An incision is made across the abdomen, and the bladder is exposed. The bladder is separated from surrounding tissues. Stitches (sutures) are placed in these tissues near the bladder neck and urethra. The urethra is then lifted, and the sutures are attached to the pubic bone itself, or to tissue (fascia) behind the pubic bone. The sutures support the bladder neck, helping the patient gain control over urine flow.

KEY TERMS

Biofeedback—Biofeedback training monitors temperature and muscle contractions in the vagina to help incontinent patients control their pelvic muscles.

Bladder training—A behavioral modification program used to treat stress incontinence. Bladder training involves putting the patient on a toilet schedule, and gradually increasing the time interval between urination.

Catheter—A long, thin, flexible tube. A catheter is used to drain the bladder of urine during a Marshall-Marchetti-Krantz procedure.

Kegel exercises—Exercises that tighten the pelvic floor muscles. Kegel exercises can assist some women in controlling their stress incontinence.

Urethra—The narrow tube, leading from the bladder that drains the body's urine.

risks of infection, internal bleeding, and hemorrhage. There is also a possibility of permanent damage to the bladder or urethra. The urethra may become scarred, causing a permanent narrowing, or stricture.

Normal results

Approximately 85% of women who undergo the Marshall-Marchetti-Krantz procedure are cured of their stress incontinence.

Resources

ORGANIZATIONS

American Foundation for Urologic Disease. 1128 North Charles St., Baltimore, MD 21201. (800) 242-2383. <<http://www.afud.org>>.

National Association for Continence. P.O. Box 8310, Spartanburg, SC 29305-8310. (800) 252-3337. <<http://www.nafc.org>>.

Paula Anne Ford-Martin

Preparation

A complete evaluation to determine the cause of incontinence is critical to proper treatment. A thorough medical history and general **physical examination** should be performed on candidates for the Marshall-Marchetti-Krantz procedure. Diagnostic testing may include x rays, ultrasound, urine tests, and examination of the pelvis. It may also include a series of urodynamic testing exams that measure bladder pressure and capacity, and urinary flow.

Patients undergoing a Marshall-Marchetti-Krantz procedure must not eat or drink for eight hours prior to the surgery.

Aftercare

Recovery from a Marshall-Marchetti-Krantz procedure requires two to six days of hospitalization. The catheter will be removed from the patient's bladder once normal bladder function resumes. Patients are advised to refrain from heavy lifting for four to six weeks after the procedure.

Patients should contact their physician immediately if they experience **fever**, **dizziness**, or extreme **nausea**, or if their incision site becomes swollen, red, or hard.

Risks

The Marshall-Marchetti-Krantz procedure is an invasive surgical procedure and, as such, it carries

Massage therapy

Definition

Massage therapy is the scientific manipulation of the soft tissues of the body for the purpose of normalizing those tissues and consists of manual techniques that include applying fixed or movable pressure, holding, and/or causing movement of or to the body.

Purpose

Generally, massage is known to affect the circulation of blood and the flow of blood and lymph, reduce muscular tension or flaccidity, affect the nervous system through stimulation or **sedation**, and enhance tissue healing. These effects provide a number of benefits:

- reduction of muscle tension and stiffness
- relief of muscle spasms
- greater flexibility and range of motion
- increase of the ease and efficiency of movement
- relief of **stress** and aide of relaxation
- promotion of deeper and easier breathing
- improvement of the circulation of blood and movement of lymph

- relief of tension-related conditions, such as headaches and eyestrain
- promotion of faster healing of soft tissue injuries, such as pulled muscles and sprained ligaments, and reduction in **pain** and swelling related to such injuries
- reduction in the formation of excessive scar tissue following soft tissue injuries
- enhancement in the health and nourishment of skin
- improvement in posture through changing tension patterns that affect posture
- reduction in stress and an excellent stress management tool
- creation of a feeling of well-being
- reduction in levels of **anxiety**
- increase in awareness of the mind-body connection
- promotion of a relaxed state of mental awareness

Massage therapy also has a number of documented clinical benefits. For example, massage can reduce anxiety, improve pulmonary function in young **asthma** patients, reduce psycho-emotional distress in persons suffering from chronic inflammatory bowel disease, increase weight and improve motor development in premature infants, and may enhance immune system functioning. Some medical conditions that massage therapy can help are: **allergies**, anxiety and stress, arthritis, asthma and **bronchitis**, **carpal tunnel syndrome** and other repetitive motion injuries, chronic and temporary pain, circulatory problems, depression, digestive disorders, **tension headache**, **insomnia**, myofascial pain, **sports injuries**, and temporomandibular joint dysfunction.

Description

Origins

Massage therapy is one of the oldest health care practices known to history. References to massage are found in Chinese medical texts more than 4,000 years old. Massage has been advocated in Western health care practices at least since the time of Hippocrates, the “Father of Medicine.” In the fourth century B.C. Hippocrates wrote, “The physician must be acquainted with many things and assuredly with rubbing” (the ancient Greek term for massage was rubbing).

The roots of modern, scientific massage therapy go back to Per Henrik Ling (1776–1839), a Swede, who developed an integrated system consisting of massage and active and passive exercises. Ling established the Royal Central Gymnastic Institute in Sweden in 1813 to teach his methods.

Modern, scientific massage therapy was introduced in the United States in the 1850s by two New York physicians, brothers George and Charles Taylor, who had studied in Sweden. The first clinics for massage therapy in the United States were opened by two Swedish physicians after the Civil War period. Doctor Baron Nils Posse operated the Posse Institute in Boston and Doctor Hartwig Nissen opened the Swedish Health Institute near the Capitol in Washington, D.C.

Although there were periods when massage fell out of favor, in the 1960s it made a comeback in a different way as a tool for relaxation, communication, and alternative healing. Today, massage is one of the most popular healing modalities. It is used by conventional, as well as alternative, medical communities and is now covered by some health insurance plans.

Massage therapy is the scientific manipulation of the soft tissues of the body for the purpose of normalizing those tissues and consists of a group of manual techniques that include applying fixed or movable pressure, holding, and/or causing movement of or to the body. While massage therapy is applied primarily with the hands, sometimes the forearms or elbows are used. These techniques affect the muscular, skeletal, circulatory, lymphatic, nervous, and other systems of the body. The basic philosophy of massage therapy embraces the concept of *vis Medicatrix naturae*, which is aiding the ability of the body to heal itself, and is aimed at achieving or increasing health and well-being.

Touch is the fundamental medium of massage therapy. While massage can be described in terms of the type of techniques performed, touch is not used solely in a mechanistic way in massage therapy. One could look at a diagram or photo of a massage technique that depicts where to place one’s hands and what direction the **stroke** should go, but this would not convey everything that is important for giving a good massage. Massage also has an artistic component.

Because massage usually involves applying touch with some degree of pressure and movement, the massage therapist must use touch with sensitivity in order to determine the optimal amount of pressure to use for each person. For example, using too much pressure may cause the body to tense up, while using too little may not have enough effect. Touch used with sensitivity also allows the massage therapist to receive useful information via his or her hands about the client’s body, such as locating areas of muscle tension and other soft tissue problems. Because touch is also a form of communication, sensitive touch can convey a

sense of caring—an essential element in the therapeutic relationship—to the person receiving massage.

In practice, many massage therapists use more than one technique or method in their work and sometimes combine several. Effective massage therapists ascertain each person's needs and then use the techniques that will meet those needs best.

Swedish massage uses a system of long gliding strokes, kneading, and friction techniques on the more superficial layers of muscles, generally in the direction of blood flow toward the heart, and sometimes combined with active and passive movements of the joints. It is used to promote general relaxation, improve circulation and range of motion, and relieve muscle tension. Swedish massage is the most commonly used form of massage.

Deep tissue massage is used to release chronic patterns of muscular tension using slow strokes, direct pressure, or friction directed across the grain of the muscles. It is applied with greater pressure and to deeper layers of muscle than Swedish, which is why it is called deep tissue and is effective for chronic muscular tension.

Sports massage uses techniques that are similar to Swedish and deep tissue, but are specially adapted to deal with the effects of athletic performance on the body and the needs of athletes regarding training, performing, and recovery from injury.

Neuromuscular massage is a form of deep massage that is applied to individual muscles. It is used primarily to release trigger points (intense knots of muscle tension that refer pain to other parts of the body), and also to increase blood flow. It is often used to reduce pain. Trigger point massage and myotherapy are similar forms.

Acupressure applies finger or thumb pressure to specific points located on the **acupuncture** meridians (channels of energy flow identified in Asian concepts of anatomy) in order to release blocked energy along these meridians that causes physical discomforts, and re-balance the energy flow. **Shiatsu** is a Japanese form of acupressure.

The cost of massage therapy varies according to geographic location, experience of the massage therapist, and length of the massage. In the United States, the average range is from \$35-60 for a one hour session. Massage therapy sessions at a client's home or office may cost more due to travel time for the massage therapist. Most sessions are one hour. Frequency of massage sessions can vary widely. If a person is receiving massage for a specific problem, frequency can vary

widely based on the condition, though it usually will be once a week. Some people incorporate massage into their regular personal health and fitness program. They will go for massage on a regular basis, varying from once a week to once a month.

The first appointment generally begins with information gathering, such as the reason for getting massage therapy, physical condition and medical history, and other areas. The client is asked to remove clothing to one's level of comfort. Undressing takes place in private, and a sheet or towel is provided for draping. The massage therapist will undrape only the part of the body being massaged. The client's modesty is respected at all times. The massage therapist may use an oil or cream, which will be absorbed into the skin in a short time.

To receive the most benefit from a massage, generally the person being massaged should give the therapist accurate health information, report discomfort of any kind (whether it is from the massage itself or due to the room temperature or any other distractions), and be as receptive and open to the process as possible.

Insurance coverage for massage therapy varies widely. There tends to be greater coverage in states that license massage therapy. In most cases, a physician's prescription for massage therapy is needed. Once massage therapy is prescribed, authorization from the insurer may be needed if coverage is not clearly spelled out in one's policy or plan.

Preparations

Going for a massage requires little in the way of preparation. Generally, one should be clean and should not eat just before a massage. One should not be under the influence of alcohol or non-medicinal drugs. Massage therapists generally work by appointment and usually will provide information about how to prepare for an appointment at the time of making the appointment.

Precautions

Massage is comparatively safe; however it is generally contraindicated, i.e., it should not be used, if a person has one of the following conditions: advanced heart diseases, **hypertension** (high blood pressure), phlebitis, thrombosis, **embolism**, kidney failure, **cancer** if massage would accelerate metastasis (i.e., spread a tumor) or damage tissue that is fragile due to **chemotherapy** or other treatment, infectious diseases, contagious skin conditions, acute inflammation,

infected injuries, unhealed **fractures**, **dislocations**, **frostbite**, large hernias, torn ligaments, conditions prone to hemorrhage, and **psychosis**.

Massage should not be used locally on affected areas (i.e., avoid using massage on the specific areas of the body that are affected by the condition) for the following conditions: **rheumatoid arthritis** flare up, eczema, **goiter**, and open **skin lesions**. Massage may be used on the areas of the body that are not affected by these conditions.

In some cases, precautions should be taken before using massage for the following conditions: **pregnancy**, high fevers, **osteoporosis**, diabetes, recent postoperative cases in which pain and muscular splinting (i.e., tightening as a protective reaction) would be increased, apprehension, and mental conditions that may impair communication or perception. In such cases, massage may or may not be appropriate. The decision on whether to use massage must be based on whether it may cause harm. For example, if someone has osteoporosis, the concern is whether bones are strong enough to withstand the pressure applied. If one has a health condition and has any hesitation about whether massage therapy would be appropriate, a physician should be consulted.

Side effects

Massage therapy does not have side effects. Sometimes people are concerned that massage may leave them too relaxed or too mentally unfocused. To the contrary, massage tends to leave people feeling more relaxed and alert.

Research and general acceptance

Before 1939, more than 600 research studies on massage appeared in the main journals of medicine in English. However, the pace of research was slowed by medicine's disinterest in massage therapy.

Massage therapy research picked up again in the 1980s, as the growing popularity of massage paralleled the growing interest in complementary and alternative medicine. Well designed studies have documented the benefits of massage therapy for the treatment of acute and chronic pain, acute and chronic inflammation, chronic **lymphedema**, **nausea**, muscle spasm, various soft tissue dysfunctions, anxiety, depression, insomnia, and psycho-emotional stress, which may aggravate mental illness.

Premature infants treated with daily massage therapy gain more weight and have shorter hospital stays than infants who are not massaged. A study of

40 low-birth-weight babies found that the 20 massaged babies had a 47% greater weight gain per day and stayed in the hospital an average of six days less than 20 infants who did not receive massage, resulting a cost savings of approximately \$3,000 per infant. Cocaine-exposed, preterm infants given massage three times daily for a 10 day period showed significant improvement. Results indicated that massaged infants had fewer postnatal complications and exhibited fewer stress behaviors during the 10 day period, had a 28% greater daily weight gain, and demonstrated more mature motor behaviors.

A study comparing 52 hospitalized depressed and adjustment disorder children and adolescents with a control group that viewed relaxation videotapes, found massage therapy subjects were less depressed and anxious, and had lower saliva cortisol levels (an indicator of less depression).

Another study showed massage therapy produced relaxation in 18 elderly subjects, demonstrated in measures such as decreased blood pressure and heart rate and increased skin temperature.

A combination of massage techniques for 52 subjects with traumatically induced spinal pain led to significant improvements in acute and chronic pain and increased muscle flexibility and tone. This study also found massage therapy to be extremely cost effective, with cost savings ranging from 15-50%. Massage has also been shown to stimulate the body's ability to naturally control pain by stimulating the brain to produce endorphins. **Fibromyalgia** is an example of a condition that may be favorably affected by this effect.

A pilot study of five subjects with symptoms of tension and anxiety found a significant response to massage therapy in one or more psycho-physiological parameters of heart rate, frontalis and forearm extensor electromyograms (EMGs) and skin resistance, which demonstrate relaxation of muscle tension and reduced anxiety.

Lymph drainage massage has been shown to be more effective than mechanized methods or diuretic drugs to control **lymphedema** secondary to radical **mastectomy**, consequently using massage to control lymphedema would significantly lower treatment costs. A study found that massage therapy can have a powerful effect upon psycho-emotional distress in persons suffering from chronic inflammatory bowel disease. Massage therapy was effective in reducing the frequency of episodes of pain and disability in these patients.

Massage may enhance the immune system. A study suggests an increase in cytotoxic capacity

associated with massage. A study of **chronic fatigue syndrome** subjects found that a group receiving massage therapy had lower depression, emotional distress, and somatic symptom scores, more hours of sleep, and lower epinephrine and cortisol levels than a control group.

Resources

ORGANIZATIONS

American Massage Therapy Association. 820 Davis Street, Suite 100, Evanston, IL. <<http://www.amtamassage.org>>.

Elliot Greene

Mastectomy

Definition

Mastectomy is the surgical removal of the breast for the treatment or prevention of **breast cancer**.

Purpose

Mastectomy is performed as a surgical treatment for breast **cancer**. The severity of a breast cancer is evaluated according to a complex system called staging. This takes into account the size of the tumor and whether it has spread to the lymph nodes, adjacent tissues, and/or distant parts of the body. A mastectomy usually is the recommended surgery for more advanced breast cancers. Women with earlier stage breast cancers, who might also have breast-conserving surgery (**lumpectomy**), may choose to have a mastectomy. In the United States, approximately 50,000 women a year undergo mastectomy.

The size, location, and type of tumor are important considerations when choosing the best surgery to treat breast cancer. The size of the breast also is an important factor. A woman's psychological concerns and lifestyle choices also should be considered when making a decision.

There are many factors that may make a mastectomy the treatment of choice for a patient. Large tumors are difficult to remove with good cosmetic results. This is especially true if the woman has small breasts. Sometimes multiple areas of cancer are found in one breast, making removal of the whole breast necessary. The surgeon sometimes is unable to remove the tumor with a sufficient amount, or margin, of

normal tissue surrounding it. In this situation, the entire breast needs to be removed. Recurrence of breast cancer after a lumpectomy is another indication for mastectomy.

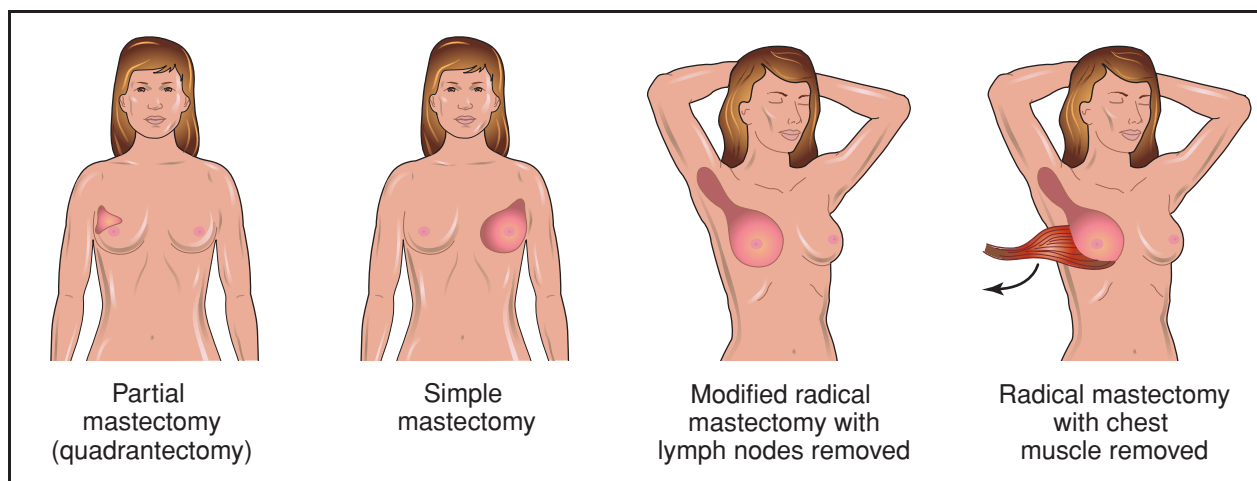
Radiation therapy is almost always recommended following a lumpectomy. If a woman is unable to have radiation, a mastectomy is the treatment of choice. Pregnant women cannot have radiation therapy for fear of harming the fetus. A woman with certain collagen vascular diseases, such as **systemic lupus erythematosus** or **scleroderma**, would experience unacceptable scarring and damage to her connective tissue from radiation exposure. Any woman who has had therapeutic radiation to the chest area for other reasons cannot tolerate additional exposure for breast cancer therapy.

The need for radiation therapy after breast conserving surgery may make mastectomy more appealing for nonmedical reasons. Some women fear radiation and choose the more extensive surgery so radiation treatment will not be required. The commitment of time, usually five days a week for six weeks, may not be acceptable for other women. This may be due to financial, personal, or job-related factors. In geographically isolated areas, a course of radiation therapy may require lengthy travel and perhaps unacceptable amounts of time away from family or other responsibilities.

Some women choose mastectomy because they strongly fear recurrence of the breast cancer, and lumpectomy seems too risky. Keeping a breast that has contained cancer may feel uncomfortable for some patients. They prefer mastectomy, so the entire breast will be removed. However, studies have shown that survival rates for women choosing mastectomy and those undergoing breast-conserving surgery have been the same.

The issue of prophylactic or preventive mastectomy, or removal of the breast to prevent future breast cancer, is controversial. Women with a strong family history of breast cancer and/or who test positive for a known cancer-causing gene may choose to have both breasts removed. Patients who have had certain types of breast cancers that are more likely to recur may elect to have the unaffected breast removed. Although there is some evidence that this procedure can decrease the chances of developing breast cancer, it is not a guarantee. It is not possible to guarantee that all breast tissue has been removed. There have been cases of breast cancers occurring after both breasts have been removed.

Studies have shown that women who choose preventive mastectomy generally are satisfied with their choice, but also believe they lacked enough information before deciding, particularly about the surgery, **genetic testing**, and **breast reconstruction**. A study



There are four types of mastectomies: partial mastectomy, or lumpectomy, in which the tumor and surrounding tissue is removed; simple mastectomy, where the entire breast and some axillary lymph nodes are removed; modified radical mastectomy, in which the entire breast and all axillary lymph nodes are removed; and the radical mastectomy, where the entire breast, axillary lymph nodes, and chest muscles are removed. (Illustration by Electronic Illustrators Group.)

released in 2003 concerning women who underwent radical mastectomy of one breast and chose surgical removal of the other breast as a preventive measure found that 83% were highly satisfied with their decision.

Precautions

The decision to have mastectomy or lumpectomy should be carefully considered. It is important that the woman be fully informed of all the potential risks and benefits of each surgical treatment before making a choice.

Description

There are several types of mastectomies. The radical mastectomy, also called the Halsted mastectomy, is rarely performed today. It was developed in the late 1800s, when it was thought that more extensive surgery was most likely to cure cancer. A radical mastectomy involves removal of the breast, all surrounding lymph nodes up to the collarbone, and the underlying chest muscle. Women often were left disfigured and disabled, with a large defect in the chest wall requiring **skin grafting**, and significantly decreased arm sensation and motion. Unfortunately, and inaccurately, it still is the operation many women picture when the word mastectomy is mentioned.

Surgery that removes breast tissue, nipple, an ellipse of skin, and some axillary or underarm lymph nodes, but leaves the chest muscle intact, usually is

called a modified radical mastectomy. This is the most common type of mastectomy performed today. The surgery leaves a woman with a more normal chest shape than the older radical mastectomy procedure, and a scar that is not visible in most clothing. It also allows for immediate or delayed breast reconstruction.

In a simple mastectomy, only the breast tissue, nipple, and a small piece of overlying skin are removed. If a few of the axillary lymph nodes closest to the breast also are taken out, the surgery may be called an extended simple mastectomy.

There are other variations on the term mastectomy. A skin-sparing mastectomy uses special techniques that preserve the patient's breast skin for use in reconstruction, although the nipple still is removed. Total mastectomy is a confusing expression, as it may be used to refer to a modified radical mastectomy or a simple mastectomy. In 2003, surgeons reported on a new technique that spared the nipple in many women with early stage breast cancer.

Many women choose to have breast reconstruction performed in conjunction with the mastectomy. The reconstruction can be done using a woman's own abdominal tissue, or using saline-filled artificial expanders, which leave the breast relatively flat but partially reconstructed. Additionally, there are psychological benefits to coming out of the surgery with the first step to a reconstructed breast. Immediate reconstruction will add time and cost to the mastectomy procedure, but the patient can avoid the physical impact of a later surgery.

A mastectomy typically is performed in a hospital setting, but specialized outpatient facilities sometimes are used. The surgery is done under **general anesthesia**. The type and location of the incision may vary according to plans for reconstruction or other factors, such as old **scars**. As much breast tissue as possible is removed. Approximately 10 to 20 axillary lymph nodes usually are removed. All tissue is sent to the pathology laboratory for analysis. If no immediate reconstruction is planned, surgical drains are left in place to prevent fluid accumulation. The skin is sutured and bandages are applied.

The surgery may take from two to five hours. Patients usually stay at least one night in the hospital, although outpatient mastectomy is increasingly performed for about 10% of all patients. Insurance usually covers the cost of mastectomy. If immediate reconstruction is performed, the length of stay, recovery period, insurance reimbursement, and fees will vary. In 1998, the Women's Health and Cancer Rights Act required insurance plans to cover the cost of breast reconstruction in conjunction with a mastectomy procedure.

Preparation

Routine preoperative preparations, such as not eating or drinking the night before surgery, typically are ordered for a mastectomy. On rare occasions, the patient also may be asked to donate blood in case a blood **transfusion** is required during surgery. The patient should advise the surgeon of any medications she is taking. Information regarding expected outcomes and potential complications also should be part of preparation for a mastectomy, as for any surgical procedure. It is especially important that women know about sensations they might experience after surgery, so they are not misinterpreted as a sign of poor wound healing or recurrent cancer.

Aftercare

In the past, women often stayed in the hospital at least several days. Now many patients go home the same day or within a day or two after their mastectomies. Visits from home care nurses can sometimes be arranged, but patients need to learn how to care for themselves before discharge from the hospital. Patients may need to learn to change bandages and/or care for the incision. The surgical drains must be attended to properly; this includes emptying the drain, measuring fluid output, moving clots through the drain, and identifying problems that need attention from the doctor or nurse. If the drain becomes

blocked, fluid or blood may collect at the surgical site. Left untreated, this accumulation may cause infection and/or delayed wound healing.

After a mastectomy, activities such as driving may be restricted according to individual needs. **Pain** is usually well controlled with prescribed medication. Severe pain may be a sign of complications, and should be reported to the physician. A return visit to the surgeon is usually scheduled 7 to 10 days after the procedure.

Exercises to maintain shoulder and arm mobility may be prescribed as early as 24 hours after surgery. These are very important in restoring strength and promoting good circulation. However, intense **exercise** should be avoided for a time after surgery in order to prevent injury. The specific exercises suggested by the physician will change as healing progresses. Physical therapy is an integral part of care after a mastectomy, aiding in the overall recovery process.

Emotional care is another important aspect of recovery from a mastectomy. A mastectomy patient may feel a range of emotions including depression, negative self-image, grief, fear and **anxiety** about possible recurrence of the cancer, anger, or guilt. Patients are advised to seek counseling and/or support groups and to express their emotions to others, whether family, friends, or therapists. Assistance in dealing with the psychological effects of the breast cancer diagnosis, as well as the surgery, can be invaluable for women.

Measures to prevent injury or infection to the affected arm should be taken, especially if axillary lymph nodes were removed. There are a number of specific instructions directed toward avoiding pressure or constriction of the arm. Extra care must be exercised to avoid injury, to treat it properly if it occurs, and to seek medical attention promptly when appropriate.

Additional treatment for breast cancer may be necessary after a mastectomy. Depending on the type of tumor, lymph node status, and other factors, **chemotherapy**, radiation therapy, and/or hormone therapy may be prescribed.

Risks

Risks that are common to any surgical procedure include bleeding, infection, anesthesia reaction, or unexpected scarring. After mastectomy and axillary lymph node dissection, a number of complications are possible. A woman may experience decreased

KEY TERMS

Axillary—Located in or near the armpit.

Lymphedema—Swelling caused by an accumulation of fluid from faulty lymph drainage.

Mastectomy, modified radical—Total mastectomy with axillary lymph node dissection, but with preservation of the pectoral muscles.

Mastectomy, radical—Removal of the breast, pectoral muscles, axillary lymph nodes, and associated skin and subcutaneous tissue.

Mastectomy, simple—Removal of only the breast tissue, nipple and a small portion of the overlying skin

feeling in the back of her armpit or other sensations including **numbness**, **tingling**, or increased skin sensitivity. Some women report phantom breast symptoms, experiencing **itching**, aching, or other sensations in the breast that has been removed. There may be scarring around where the lymph nodes were removed, resulting in decreased arm mobility and requiring more intense physical therapy.

Approximately 10% to 20% of patients develop **lymphedema** after axillary lymph node removal. This swelling of the arm, caused by faulty lymph drainage, can range from mild to severe. It can be treated with elevation, elastic bandages, and specialized physical therapy. **Lymphedema** is a chronic condition that requires continuing treatment. This complication can arise at any time, even years after surgery. A new technique called sentinel lymph node mapping and biopsy often eliminates the need for removing some or all lymph nodes by testing the first lymph node for cancer.

Normal results

A mastectomy is performed as the definitive surgical treatment for breast cancer. The goal of the procedure is that the breast cancer is completely removed and does not recur.

Abnormal results

An abnormal result of a mastectomy is the incomplete removal of the breast cancer or a recurrence of the cancer. Other abnormal results include long-lasting (chronic) pain or impairment that does not improve after several months of physical therapy.

Resources

PERIODICALS

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- “Majority Satisfied with Prophylactic Mastectomy Decision.” *AORN Journal* November 2003: 773.
- “Studies Compare Mastectomy, Lumpectomy Survival Rates.” *Clinician Reviews* January 2003: 24.

ORGANIZATIONS

- American Cancer Society. 1599 Clifton Rd., NE, Atlanta, GA 30329-4251. (800) 227-2345. <<http://www.cancer.org>>.
- National Lymphedema Network. 2211 Post St., Suite 404, San Francisco, CA 94115-3427. (800) 541-3259 or (415) 921-1306. <<http://www.wenet.net/~lymphnet/>>.
- Y-ME National Organization for Breast Cancer Information and Support. 18220 Harwood Ave., Homewood, IL 60430. 24-hour hotlines: (800) 221-2141 or (708) 799-8228.

OTHER

- ibreast.org* April 15, 2001. [cited June 12, 2001]. <<http://www.breastcancer.org>>.
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Ellen S. Weber, MSN
Teresa G. Odle

Mastitis

Definition

Mastitis is an infection of the breast. It usually only occurs in women who are breastfeeding their babies.

Description

Breastfeeding is the act of allowing a baby to suckle at the breast to drink the mother’s milk. In the process, unaccustomed to the vigorous pull and tug of the infant’s suck, the nipples may become sore, cracked, or irritated. This creates a tiny opening in the breast, through which bacteria can enter. The presence of milk, with high sugar content, gives the bacteria an excellent source of **nutrition**. Under these

conditions, the bacteria are able to multiply, until they are plentiful enough to cause an infection within the breast.

Mastitis usually begins more than two to four weeks after delivery of the baby. It is a relatively uncommon complication of breastfeeding mothers, occurring in only approximately 3% to 5% of nursing women.

Causes and symptoms

The most common bacteria causing mastitis is called *Staphylococcus aureus*. In 25-30% of people, this bacteria is present on the skin lining normal, uninfected nostrils. It is probably this bacteria, clinging to the baby's nostrils, that is available to create infection when an opportunity (crack in the nipple) presents itself.

Usually, only one breast is involved. An area of the affected breast becomes swollen, red, hard, and painful. Other symptoms of mastitis include **fever**, chills, and increased heart rate.

Diagnosis

Diagnosis involves obtaining a sample of breast milk from the infected breast. The milk is cultured, allowing colonies of bacteria to grow. The causative bacteria then can be specially prepared for identification under a microscope. At the same time, tests can be performed to determine what type of antibiotic would be most effective against that particular bacteria. Sometimes, women and their physicians confuse mastitis with breast engorgement, or the tenderness and redness that appears when milk builds up in the breasts. Mastitis often can be distinguished if symptoms are accompanied by fever.

Treatment

A number of **antibiotics** are used to treat mastitis, including cephalexin, amoxicillin, azithromycin, dicloxacillin, and clindamycin. Breastfeeding usually should be continued, because the rate of **abscess** formation (an abscess is a persistent pocket of pus) in the infected breast goes up steeply among women who stop breastfeeding during a bout with mastitis. Most practitioners allow women to take **acetaminophen** while nursing, to relieve both fever and **pain**. As always, breastfeeding women need to make sure that any medication they take is also safe for the baby, since almost all drugs they take



Mastitis is usually caused by a bacterial infection through a nipple damaged during breastfeeding. (Photograph by Dr. P. Marazzi, Photo Researchers, Inc. Reproduced by permission.)

appear in the breastmilk. Warm compresses applied to the affected breast can be soothing.

Prognosis

Prognosis for uncomplicated mastitis is excellent. About 10% of women with mastitis will end up with an abscess within the affected breast. An abscess is a collection of pus within the breast. This complication will require a surgical procedure to drain the pus.

Prevention

The most important aspect of prevention involves good handwashing to try to prevent the infant from acquiring the *Staphylococcus aureus* bacteria in the first place. Keeping the breast clean before breastfeeding also helps prevent infection. Keeping the breasts from becoming engorged may help prevent mastitis by preventing plugging of milk ducts.

Resources

PERIODICALS

Hager, W. David. "Managing Mastitis: Antibiotics Can Prove Invaluable in the Treatment of Mastitis, but Before You Prescribe Them, It's Important to Distinguish Breast Engorgement from Infectious Mastitis." *Contemporary OB/GYN* January 2004: 32-41.

ORGANIZATIONS

La Leche League International. 1400 N. Meacham Rd., Schaumburg, IL 60173-4048. (800) 525-3243. <<http://www.lalecheleague.org>>.

Rosalyn Carson-DeWitt, MD
Teresa G. Odle

Mastocytosis

Definition

Mastocytosis is a disease characterized by the presence of too many mast cells in various organs and tissues.

Description

The body has a variety of free-roaming cell populations that function as immunogenic agents. Most immunogenic cells fall into the category of white blood cells, but some remain in tissues and are not found in the blood. Mast cells are such a group.

Mast cells are found primarily in the skin and digestive system, including the liver and spleen, and produce histamine, a chemical most famous for its ability to cause **itching**. Histamine also causes acid **indigestion**, **diarrhea**, flushing, heart pounding, headaches, and can even cause the blood pressure to drop suddenly.

Mastocytosis comes in three forms. Most cases produce symptoms but do not shorten life expectancy. The three forms are:

- Mastocytoma, a benign skin tumor.
- Urticaria pigmentosa, small collections of mast cells in the skin that manifest as salmon or brown-colored patches.
- Systemic mastocytosis, the collection of mast cells in the skin, lymph nodes, liver, spleen, gastrointestinal tract, and bones.

Causes and symptoms

The cause of mastocytosis is unknown. People with systemic mastocytosis have bone and joint **pain**. Peptic ulcers are frequent because of the increased stomach acid stimulated by histamine. Many patients with systemic mastocytosis also develop urticaria pigmentosa. These **skin lesions** itch when stroked and may become fluid-filled.

Diagnosis

A biopsy of the skin patches aids diagnosis. An elevated level of histamine in the urine or blood is also indicative of mastocytosis.

Treatment

Mastocytoma usually occurs in childhood and clears-up on its own. Urticaria pigmentosa (present

KEY TERMS

Non-steroidal anti-inflammatory drugs (NSAIDs)—Aspirin, ibuprofen, naproxen, and many others.

Peptic ulcer—Ulcers in the stomach and upper duodenum (first portion of the small intestine) caused by stomach acid and a bacterium called *Helicobacter pylori*.

alone without systemic disease) also dramatically clears or improves as adolescence approaches.

Several medications are helpful in relieving symptoms of systemic mastocytosis. **Antihistamines** and drugs that reduce stomach acid are frequently needed. Headaches respond to migraine treatment. A medicine called cromolyn helps with the bowel symptoms. Several other standard and experimental medications have been used.

Prognosis

Mastocytoma and urticaria pigmentosa rarely if ever, develop into systemic mastocytosis, and both spontaneously improve over time. Systemic mastocytosis is only symptomatically treated. There is no known treatment that decreases the number of mast cells within tissue.

Resources

BOOKS

Austen, K. Frank. "Diseases of Immediate Type Hypersensitivity." In *Harrison's Principles of Internal Medicine*, edited by Anthony S. Fauci, et al. New York: McGraw-Hill, 1997.

J. Ricker Polsdorfer, MD

Mastoid tympanoplasty see **Mastoidectomy**

Mastoidectomy

Definition

Mastoidectomy is a surgical procedure to remove an infected portion of the bone behind the ear when

medical treatment is not effective. This surgery is rarely needed today because of the widespread use of **antibiotics**.

Purpose

Mastoidectomy is performed to remove infected air cells within the mastoid bone caused by **mastoiditis**, ear infection, or an inflammatory disease of the middle ear (cholesteatoma). The cells are open spaces containing air that are located throughout the mastoid bone. They are connected to a cavity in the upper part of the bone, which is in turn connected to the middle ear. As a result, infections in the middle ear can sometimes spread through the mastoid bone. When antibiotics cannot clear this infection, it may be necessary to remove the infected air cells by surgery. Mastoidectomies are also performed sometimes to repair paralyzed facial nerves.

Description

Mastoidectomy is performed less often today because of the widespread use of antibiotics to treat ear infections.

There are several different types of mastoidectomy:

- Simple (or closed). The operation is performed through the ear or through a cut (incision) behind the ear. The surgeon opens the mastoid bone and removes the infected air cells. The eardrum is cut (incised) to drain the middle ear. **Topical antibiotics** are then placed in the ear.
- Radical mastoidectomy. The eardrum and most middle ear structures are removed, but the innermost small bone (the stapes) is left behind so that a hearing aid can be used later to offset the **hearing loss**.
- Modified radical mastoidectomy. The eardrum and the middle ear structures are saved, which allows for better hearing than is possible after a radical operation.

The wound is then stitched up around a drainage tube, which is removed a day or two later. The procedure usually takes between two and three hours.

Preparation

The doctor will give the patient a thorough ear, nose, and throat examination as well as a detailed hearing test before surgery. Patients are given an injection before surgery to make them drowsy.

KEY TERMS

Cholesteatoma—A rare but chronic inflammatory disease in which skin cells and debris collect in the middle ear, usually as a result of an ear infection.

Mastoid bone—The prominent bone behind the ear that projects from the temporal bone of the skull.

Mastoiditis—An inflammation of the bone behind the ear (the mastoid bone) caused by an infection spreading from the middle ear to the cavity in the mastoid bone.

Aftercare

Painkillers are usually needed for the first day or two after the operation. The patient should drink fluids freely. After the stitches are removed, the bulky mastoid dressing can be replaced with a smaller dressing if the ear is still draining. The patient is given antibiotics for several days.

The patient should tell the doctor if any of the following symptoms occur:

- Bright red blood on the dressing.
- Stiff neck or disorientation. These may be signs of **meningitis**.
- Facial **paralysis**, drooping mouth, or problems swallowing.

Risks

Complications do not often occur, but they may include:

- Persistent ear drainage.
- Infections, including meningitis or brain abscesses.
- Hearing loss.
- Facial nerve injury. This is a rare complication.
- Temporary **dizziness**.
- Temporary loss of taste on the side of the tongue.

Resources

ORGANIZATIONS

American Academy of Otolaryngology-Head and Neck Surgery, Inc. One Prince St., Alexandria VA 22314-3357. (703) 836-4444. <<http://www.entnet.org>>.

American Hearing Research Foundation. 55 E. Washington St., Suite 2022, Chicago, IL 60602. (312) 726-9670. <<http://www.american-hearing.org/>>.

Better Hearing Institute. 515 King Street, Suite 420,
Alexandria, VA 22314. (703) 684-3391.

Carol A. Turkington

Mastoiditis

Definition

Mastoiditis is an infection of the spaces within the mastoid bone. It is almost always associated with **otitis media**, an infection of the middle ear. In the most serious cases, the bone itself becomes infected.

Description

The mastoid is a part of the side (temporal bone) of the skull. It can be felt as a bony bump just behind and slightly above the level of the earlobe. The mastoid has been described as resembling a “honeycomb” of tiny partitioned-off airspaces. The mastoid is connected with the middle ear, so that when there is a collection of fluid in the middle ear, there is usually also a slight collection of fluid within the airspaces of the mastoid.

Mastoiditis can range from a simple case of some fluid escaping into the mastoid air cells during a middle ear infection, to a more complex infection which penetrates through to the lining of the mastoid bone, to a very severe and destructive infection of the mastoid bone itself.

Causes and symptoms

Mastoiditis is caused by the same types of bacteria which cause middle ear infections (*Streptococcus pneumoniae* and *Haemophilus influenzae*), as well as by a variety of other bacteria (*Staphylococcus aureus*, *Pseudomonas aeruginosa*, *Klebsiella*, *Escherichia coli*, *Proteus*, *Prevotella*, *Fusobacterium*, *Porphyromonas*, and *Bacteroides*). Mastoiditis may occur due to the progression of an untreated, or undertreated, middle ear infection.

Symptoms of mastoiditis may at first be the same as symptoms of an early middle ear infection. With progression, however, the swollen mastoid may push the outer ear slightly forward and away from the head. The area behind the ear will appear red and swollen, and will be very sore. There may be drainage of pus from the infected ear. In some cases, the skin over the

KEY TERMS

Abscess—A pocket of infection, usually including a collection of pus.

Meningitis—Inflammation and infection of the tissues covering the brain and spinal cord (the meninges).

Otitis or oteitis—An infection of the middle ear; marked by an enlargement of bone, tenderness and dull aching pain.

mastoid may develop an opening through which pus drains. **Fever** is common.

Diagnosis

Mastoiditis is usually suspected when a severe middle ear infection is accompanied by redness, swelling, and **pain** in the mastoid area. A computed tomography scan (CT scan) will show inflammation and fluid within the airspaces of the mastoid, as well as the erosion of the little walls of bone that should separate the air spaces. If there is any fluid draining from the ear or mastoid, this can be collected and processed in a laboratory to allow identification of the causative organism. If there is no fluid available, a tiny needle can be used to obtain a sample of the fluid which has accumulated behind the eardrum.

Treatment

Identification of the causative organism guides the practitioner's choice of antibiotic. Depending on the severity of the infection, the antibiotic can be given initially through a needle in the vein (intravenously or IV), and then (as the patient improves) by mouth.

In the case of a very severe infection of the mastoid bone itself, with a collection of pus (**abscess**), an operation to remove the mastoid part of the temporal bone is often necessary (**mastoidectomy**).

Prognosis

With early identification of mastoiditis, the prognosis is very good. When symptoms are not caught early enough, however, a number of complications can occur. These include an infection of the tissues covering the brain and spinal cord (**meningitis**), a pocket of infection within the brain (abscess), or an abscess within the muscles of the neck. All of these complications have potentially more serious prognoses.

Prevention

Prevention of mastoiditis involves careful and complete treatment of any middle ear infections.

Resources

ORGANIZATIONS

American Academy of Otolaryngology-Head and Neck Surgery, Inc. One Prince St., Alexandria VA 22314-3357. (703) 836-4444. <<http://www.entnet.org>>.

Rosalyn Carson-DeWitt, MD

Maternal serum alpha-fetoprotein test see
Alpha-fetoprotein test

Maternal to fetal infections

Definition

Maternal to fetal infections are transmitted from the mother to her fetus, either across the placenta during fetal development (prenatal) or during labor and passage through the birth canal (perinatal).

Description

Antibodies in the maternal blood prevent most infections from being transmitted to the fetus. However some maternal-to-fetal infections, particularly in the first trimester of **pregnancy**, can cause **miscarriage** or severe **birth defects**. Other infections can cause preterm labor, fetal or neonatal **death**, or serious illness in newborns. Perinatal transmissions infect the fetus after its protective membranes rupture—the waters break—and during labor and delivery when the fetus is exposed to maternal blood. Perinatal transmission is more likely if the waters break prematurely.

Toxoplasmosis

Up to one-third of all people are infected with **toxoplasmosis**. The U.S. Centers for Disease Control and Prevention (CDC) estimate that 25–45% of women of childbearing age carry the parasite *Toxoplasma gondii* that causes toxoplasmosis. Very few infected people have symptoms and most pregnant women have antibodies that protect the fetus from infection. However in one-third of women who are infected for the first time during pregnancy, the parasite infects the placenta and enters the fetal circulation. Congenital (present at birth) infection occurs in

one out of every 800–1,400 infants born to infected mothers. The fetal infection rate is above 60% if maternal infection occurs during the third trimester, but the most severe fetal complications occur with first-trimester infection.

Viral respiratory infections

Cytomegalovirus (CMV) is the most common infection that can be transmitted to a fetus. From 50–80% of childbearing-age women have been infected by CMV prior to pregnancy. However about 1–3% of women have their first or primary CMV infection during pregnancy and about one-third of these infections are transmitted to fetuses. Although most infants with congenital CMV have no problems, infection in early pregnancy can cause miscarriage or birth defects and CMV is a leading cause of congenital deafness. In later pregnancy CMV infection may cause preterm labor, **stillbirth**, or serious newborn illness. In the United States about 8,000 infants annually are born with potentially fatal CMV-related birth defects.

Fifth disease, caused by the parvovirus B19, is very common among children. About one-half of all adults are susceptible. About one-third of infants whose mothers contract fifth disease during pregnancy show signs of infection at birth. Although not usually dangerous, fifth disease contracted early in pregnancy can cause miscarriage or severe fetal anemia (low **blood count**) that can lead to congestive **heart failure**.

A fetus infected from its mother by *Varicella zoster* virus may develop pocks that can cause limb deformities early in development. If a woman contracts varicella (**chickenpox**) during the first 20 weeks of pregnancy, there is a 2% chance that her newborn will have varicella syndrome. However the greatest risk from varicella is if the mother contracts the virus just before delivery when she has not yet produced antibodies to protect the newborn.

In the past **rubella** was a common cause of birth defects. However routine vaccinations have made prenatal infection rare in the developed world. Rubella infection during the first 10 weeks of pregnancy may cause fetal death and more than 50% of newborns have severe birth defects. Infections contracted later in pregnancy do not cause congenital defects, although the newborn may become seriously ill and eventually develop **diabetes mellitus**.

Bacterial infections

Invasive group B streptococcal (GBS) disease is the most common cause of life-threatening infection in newborns. Up to 20% of pregnant women carry GBS

in their vaginas during the last trimester, with the potential of infecting the fetus during birth. Although premature infants are more susceptible to GBS, 75% of infected infants are full-term. During the 1970s GBS emerged as the most common cause of newborn **sepsis**, or blood infection, and meningitis—infection of the fluid and lining surrounding the brain. GBS also is a frequent cause of newborn **pneumonia**. Maternal infection at conception or within the first two weeks of pregnancy may lead to hearing and vision loss and **mental retardation**. Between 1993 and 2002 congenital GBS infection in the United States decreased from 1.7 per 1,000 live births to 0.4 per 1,000 due to the use of **antibiotics** during delivery.

The food-borne bacterial infections listeriosis—caused by *Listeria monocytogenes*—and salmonellosis or food poisoning—caused by *Salmonella* bacteria—can be transmitted to a fetus. *Listeria monocytogenes* is ubiquitous in soil and groundwater, on plants, and in animals. Most human infections result from ingesting contaminated foods. Hormonal changes make pregnant women about 20 times more likely than other healthy adults to contract **listeriosis** and about one-third of all cases occur in pregnant women. Listeriosis can cause miscarriage, fetal or newborn death, premature delivery, or severe illness in the mother and infant.

Each year an estimated 8,000 pregnant women in the United States are infected with **syphilis** caused by the spirochete *Treponema pallidum*. Rising rates of syphilis among pregnant women are increasing the number of infants born with congenital syphilis. Congenital syphilis is a severe, disabling, and often life-threatening disease that can cause facial deformity, blindness, and deafness.

Every year in the United States an estimated 40,000 pregnant women are infected with gonorrhea—caused by *Neisseria gonorrhoeae*—and an estimated 200,000 are infected with chlamydia—caused by *Chlamydia trachomatis*. Chlamydia can cause premature membrane rupture and labor. Both infections can cause newborn conjunctivitis—a discharge of pus from the eyes.

Sexually transmitted viral infections

Each year an estimated 8,000 pregnant American women are infected with HIV, the human **immunodeficiency** virus that causes acquired immune deficiency syndrome (AIDS). About 20–25% of pregnant women with untreated HIV transmit it to their fetuses. In developed countries widespread HIV testing and anti-retroviral therapy have reduced maternal-fetal transmission dramatically.

Genital herpes are caused by herpes simplex virus (HSV) type-2 and, less frequently, by HSV type-1 that usually causes cold sores. About 25% of American adults are infected with HSV-2, affecting one in 1,800–5,000 live births. There is little risk of fetal transmission if the mother is infected before the third trimester and has no genital sores at the time of delivery. However infection during the third trimester—when the virus is likely to be active and the mother has not yet made sufficient antibodies to protect her fetus—may lead to congenital HSV infection. This can seriously damage the newborn's eyes, central nervous system, and internal organs, lead to mental retardation and, rarely, death.

Genital or venereal **warts** are caused by some types of human papillomavirus (HPV). At least 20 million Americans are infected and about 5.5 million new cases are reported annually. **Genital warts** are highly infectious and tend to grow faster during pregnancy. If vaginal warts are very large they may interfere with the infant's passage through the birth canal, necessitating a **cesarean section** (C-section).

An estimated 8,000 pregnant women are infected with **hepatitis B** in the United States every year. They are at risk for premature delivery and, if untreated, newborns may develop chronic **liver disease**.

Causes and symptoms

Toxoplasmosis

The single-celled protozoan *Toxoplasma gondii* produces eggs in cat intestines. The eggs shed in cat feces and can survive for up to 18 months in the soil. Human infection occurs from handling contaminated soil or feces or by ingesting raw or undercooked meat from infected animals.

Although the symptoms of toxoplasmosis usually are very mild or absent, infection occurring early in fetal development can cause:

- premature birth and low birth weight—under 5 lb (2.3 kg)
- slow growth
- fever
- skin rashes
- easy bruising
- anemia
- a small or large head (microcephaly or macrocephaly)
- fluid in the cavities of the brain (hydrocephaly)

- inflammation of the brain, heart, or lungs
- severe or prolonged jaundice
- an enlarged liver and spleen
- an eye inflammation called chorioretinitis which can lead to blindness
- severe illness or death shortly after birth

Symptoms of congenital toxoplasmosis may appear months or years after birth and may include:

- seizures or other neurological problems
- visual impairment
- hearing loss
- mental retardation

Viral respiratory infections

Although most CMV-infected newborns have no symptoms, 10–15% may exhibit:

- low birth weight
- rashes
- small bruises
- jaundice
- enlarged liver and spleen
- hernias in the groin
- microcephaly or hydrocephaly
- respiratory problems
- brain damage

From 0.5–15% of CMV-infected infants develop hearing, vision, or neurological problems over several years. In addition to crossing the placenta, there is a 1% risk of perinatal CMV transmission.

Symptoms of congenital fifth disease include:

- bright red rash on the cheeks
- lacy, red rash on the neck, trunk, and legs
- joint pain
- fatigue
- malaise

Varicella syndrome in a newborn is characterized by:

- abnormally small limbs and head
- scarring of the skin
- eye defects
- mental retardation

In addition to various birth defects, newborns infected with rubella early in the pregnancy may have:

- low birth weight
- bruising
- bluish-red skin lesions
- enlarged lymph nodes
- enlarged liver and spleen
- brain inflammation
- pneumonia

Bacterial infections

Although most GBS carriers have no symptoms, GBS in pregnant women may cause:

- bladder or urinary tract infections
- infection of the womb
- stillbirth

Pregnant women are more likely to transmit GBS to their fetuses if they:

- previously delivered a GBS-infected baby
- have a urinary tract infection caused by GBS
- carry GBS late in pregnancy
- begin labor or membrane rupture before 37 weeks of gestation
- have membrane rupture 18 hours or more before delivery
- experience fever during labor

Symptoms of congenital GBS infection include:

- breathing difficulties
- shock
- sepsis
- pneumonia
- meningitis

Listeriosis may cause flu-like symptoms and the infection can be transmitted prenatally even if the mother has no symptoms.

Symptoms of salmonellosis can be severe in pregnant women and newborns and may include:

- **diarrhea**
- fever
- abdominal cramps
- rarely, meningitis

Syphilis can be transmitted to a fetus either prenatally or perinatally if the mother is infected during pregnancy or was inadequately treated for a past infection. In adults syphilis usually causes genital lesions 10–90 days after exposure, with a rash developing six

weeks later. Symptoms may go unnoticed. Congenital syphilis can cause premature birth or stillbirth.

A surviving newborn with untreated congenital syphilis may have no initial symptoms but may gain little weight and, during the first month of life, develop:

- rash or small fluid-filled blisters on the palms and soles of the feet
- raised bumps around the nose, mouth, and diaper region
- cracks around the mouth
- nasal discharge of mucus, pus, or blood
- enlarged lymph nodes, liver, and spleen
- bone inflammation
- rarely, meningitis

Early-stage symptoms of congenital syphilis include:

- failure to thrive
- fever
- severe congenital pneumonia
- rash and lesions around the mouth, genitalia, and anus
- bone lesions
- nose cartilage infection or saddle nose (lacking a bridge)

Symptoms of late-stage congenital syphilis include:

- copper-colored rashes on the face, palms, and soles
- scarring around earlier lesions
- gray patches on the anus or outer vagina
- notched or peg-shaped teeth
- joint swelling
- bone pain
- abnormalities in the lower leg bones
- neurological conditions
- visual loss or blindness
- hearing loss or deafness

Both **gonorrhea** and **chlamydia** can be transmitted perinatally. **Conjunctivitis** caused by gonorrhea usually appears two to seven days after birth. Conjunctivitis caused by chlamydia usually appears 5–12 days after birth, although sometimes it takes six weeks to develop.

Symptoms of gonorrhea in women, if present, may include:

- bleeding during vaginal intercourse
- pain or burning with urination
- yellow or bloody vaginal discharge
- pelvic inflammatory disease

Sexually transmitted viral infections

HIV can be transmitted through the placenta, during labor and delivery, and through breast milk. HIV-infected infants do not have symptoms at birth, although about 15% develop serious symptoms or die within the first year. Almost one-half die by the age of 10.

Factors that may increase the risk of maternal HIV transmission include:

- the mother's viral load—the amount of HIV in her blood
- use of illicit drugs
- severe inflammation of the fetal membranes
- a prolonged period between membrane rupture and delivery

Most women carrying HSV never have recognizable symptoms; however a first episode of genital herpes during pregnancy can be passed to the fetus and may cause premature birth. Both HSV-1 and HSV-2 can be transmitted during birth if the mother has active genital sores, causing facial or genital herpes in the newborn.

Initial symptoms of congenital herpes usually appear within four weeks of birth and may be quite mild:

- blisters on the skin
- fever
- tiredness
- loss of appetite

More serious symptoms of congenital HSV infection include:

- a skin rash with small fluid-filled blisters
- chronic or recurring eye and skin infections
- cataracts
- widespread infection affecting many organs including the lungs and liver
- a life-threatening brain infection called herpes encephalitis

Nearly 50% of women infected with HPV have no symptoms although genital warts may appear weeks or months after infection. They can become larger during pregnancy causing difficulty with urination. Vaginal warts can reduce the elasticity of the vagina and cause obstruction during delivery. Symptoms of congenital HPV infection may include lung infection and obstructed air passages from warts inside the windpipe.

Although hepatitis B can be transmitted to the fetus through the placenta, most often it is transmitted perinatally. Since the virus is thought to pass through the umbilical cord, C-sections do not prevent transmission. Congenital hepatitis B can cause chronic liver infection, although symptoms usually do not become apparent until young adulthood.

Diagnosis

Diagnosis of maternal, fetal, or congenital infection can be difficult. An obstetrician may diagnose a maternal infection based on the woman's symptoms and blood tests. Sometimes a fetal infection can be diagnosed using ultrasound. Diagnosis of congenital infections in newborns may be based on a **physical examination**, symptoms, and blood or urine tests. Ultrasound scanning may be used to image the newborn's brain and **echocardiography** may be used to diagnose heart problems.

Toxoplasmosis

Prenatal tests for toxoplasmosis include:

- a blood test for maternal antibodies
- testing of the amniotic fluid and fetal blood
- fetal ultrasound

Postnatal diagnosis for congenital toxoplasmosis may include:

- antibody tests of the cord blood and cerebrospinal fluid
- an ophthalmologic examination
- neurological examinations
- a computed axial tomography (CT or CAT) scan

A 2005 study advised that all pregnant women and newborns have blood screenings for toxoplasmosis

Viral and bacterial infections

Diagnostic procedures include:

- blood tests for maternal antibodies against CMV or fifth disease

- ultrasound for fetal fifth disease
- bacterial cultures from blood, spinal fluid, skin, the vagina, or rectum for GBS
- a maternal blood test for listeriosis

Sexually transmitted infections (STIs)

The CDC recommends that all pregnant women be screened on their first prenatal visit for syphilis, gonorrhea, chlamydia, HIV, hepatitis B, and **hepatitis C**.

Infants are tested for syphilis at birth. Syphilis in an older infant may be diagnosed by a blood test, a lumbar puncture to look for signs of syphilis in the brain and central nervous system, an ophthalmologic examination, dark-field microscopy to visualize the spirochete, or **bone x rays**.

Maternal gonorrhea can be diagnosed by staining or culturing a cervical smear or testing for the bacterial DNA in a urine or cervical sample.

Women who were not screened for HIV during pregnancy may be screened during labor or delivery with a rapid test. The most common screening for HIV tests for antibodies in the blood; however most infants born to infected mothers test positive for 6–18 months because of the presence of maternal antibodies. An HIV blood test performed within 48 hours of birth detects only about 40% of infections, so testing is repeated at one and six months.

An HSV culture from an affected genital site—preferably on the first day of the outbreak—can test for herpes simplex. A blood test can show if a person has ever been infected with HSV and may distinguish between HSV-1 and HSV-2 and old or recently acquired infections. An examination or test can indicate whether a pregnant woman has active genital herpes near the time of delivery.

Genital warts are diagnosed visually. Vinegar may whiten infected areas to make them more visible. Cervical warts can be diagnosed by removing a piece of tissue for microscopic examination.

Treatment

Infants born with serious infections are treated in the neonatal care unit with intravenous drugs. Infants born to infected mothers may be treated with medications even if they show few or no signs of infection.

Toxoplasmosis

Maternal toxoplasmosis is treated with spiramycin during the first and early second trimesters of

pregnancy. Fetal toxoplasmosis may be treated by giving the mother pyrimethamine and **sulfonamides** such as sulfadiazine during the later second and third trimesters.

Newborns with symptoms of toxoplasmosis are treated with pyrimethamine and sulfadiazine for one year; leucovorin for one year to protect the bone marrow from pyrimethamine toxicity; **corticosteroids** for heart, lung, or eye inflammations; clindamycin; and a corticosteroid to reduce the inflammation of chorioretinitis.

Viral respiratory infections

There is no effective treatment for CMV, although ganciclovir may be used to treat some symptoms.

Fetal anemia caused by fifth disease may resolve on its own. If the fetus is at risk for heart failure, a fetal blood **transfusion** may be performed. The mother also may receive medication that passes through the placenta to the fetus.

Exposure to chickenpox or rubella by a non-immune pregnant woman may be treated with an injection of immune globulin to help prevent fetal transmission. Congenital chickenpox is treated immediately to prevent serious complications or death. There is no specific treatment for rubella infection.

Bacterial infections

Pregnant women with GBS in their urine are treated with penicillin. Most GBS-carriers are treated with intravenous antibiotics—from membrane rupture through labor—to prevent fetal transmission. Infants born with congenital GBS infections are treated immediately with intravenous antibiotics.

Maternal and congenital listeriosis and syphilis are treated with antibiotics.

Maternal gonorrhea may be treated with cefixime, ceftriaxone, or levofloxacin. Since women often are infected with both gonorrhea and chlamydia, a combination of antibiotics such as ceftriaxone and doxycycline or azithromycin are used to treat both infections.

An antibiotic ointment such as silver nitrate is placed under the eyelids of all newborns as preventative treatment for gonorrhea. An infant born to a gonorrhea-infected mother is treated with penicillin. Conjunctivitis caused by gonorrhea is treated with an eye ointment containing polymyxin and bacitracin, erythromycin, or tetracycline. An antibiotic such as ceftriaxone is given intravenously. Congenital

chlamydia is treated with erythromycin eye ointment and oral tablets.

Viral STIs

Women who are being treated for HIV with combination drugs may stop treatment for the first trimester of pregnancy to avoid the risk of birth defects and to avoid missing doses due to **vomiting**, which can cause the growth of drug-resistant HIV strains. Although the side effects of the anti-retroviral drugs may worsen during pregnancy, stopping treatment can worsen a woman's condition.

Zidovudine (ZDV, AZT, Retrovir) is the only drug that has been proven to help prevent fetal HIV infection. HIV-positive pregnant women usually take ZDV from 14–34 weeks of gestation. During delivery the mother receives ZDV intravenously. The newborn is given liquid ZDV every six hours for six weeks. A 2004 study of HIV-positive Thai women found that oral ZDV beginning at 28 weeks of gestation, with a single dose of nevirapine during labor, greatly reduces HIV transmission.

Pneumonia caused by *Pneumocystis carinii* often is the first AIDS-related illness to appear in HIV-infected infants and is a major cause of death during the first year. The CDC recommends that all babies born to HIV-infected mothers be treated with anti-pneumonia drugs beginning at four–six weeks and continuing until the infant is found to be HIV-negative.

Although there is no cure for genital herpes, outbreaks just prior to delivery may be prevented by acyclovir (Zovirax), famciclovir (Famvir), or valacyclovir (Valtrex). An HSV-infected newborn is treated immediately with intravenous **antiviral drugs** such as acyclovir. Eye infections are treated with trifluridine drops.

There is no cure for HPV and treatment during pregnancy often is ineffective, although it may include:

- Imiquimod cream
- 5% 5-fluorouracil cream
- trichloroacetic acid
- freezing or burning the warts with a laser
- surgical removal
- alpha interferon injected into the wart

HPV infection in newborns is treated by surgically removing the warts. If the warts obstruct breathing passages, frequent **laser surgery** is required. Interferon may be used to reduce the likelihood of recurrence.

Non-infected pregnant women may begin the hepatitis B vaccine series if they are at high-risk for infection. Infants born to mothers infected with hepatitis B are given both the first dose of hepatitis B vaccine and hepatitis B immune globulin within 12 hours of birth. The second and third doses of vaccine are given at one month and six months of age.

Prognosis

Maternal treatment with spiramycin for toxoplasmosis infection occurring within the first two weeks of pregnancy prevents transmission to the fetus. The prognosis for congenital toxoplasmosis depends on its severity.

With treatment most infants with congenital CMV survive, although almost all suffer from its effects.

A GBS-carrier's risk of delivering an infected child decreases from one in 200 to one in 4,000 if she is treated with antibiotics. GBS-infected mothers are less likely to infect their newborns if treated with antibiotics during labor. Although immediate penicillin treatment for GBS-infected newborns is very effective, about 5% of GBS-infected newborns die.

Many fetuses infected with syphilis early in gestation are stillborn. Nearly 50% of untreated fetuses die shortly before or after birth. However the fetus is at minimal risk if the mother receives adequate treatment with penicillin during pregnancy.

Pregnant women on combined antiretroviral therapy are at a 1–2% risk of transmitting HIV to the fetus. If the mother's viral load is under 1,000 and she is treated with ZDV, the risk of transmission is almost zero. Mothers with a high viral load may reduce the risk of transmission by having a C-section before labor begins and the membranes rupture. Congenital HIV infection that is treated with combination drugs, including **protease inhibitors**, may reduce the risk of death by 67%.

Women with an active HSV infection can reduce the risk of fetal transmission with a C-section. Although immediate medication for the newborn may prevent or reduce the damage from HSV, one-half of infants born with widespread HSV infections die and the other one-half may have brain damage.

Infants born to hepatitis B-infected mothers have a greater-than-95% chance of being protected against the virus if they receive the first dose of vaccine and immune globulin within 12 hours of birth.

Prevention

General advice for preventing infection during pregnancy includes:

- good hygiene—including frequent thorough hand washing and not sharing food or drinks—particularly for mothers who have or work with young children and may be at risk for CMV
- vaccinations several months before a planned pregnancy
- appropriate vaccinations after the first trimester of pregnancy
- contacting a healthcare provider immediately upon being exposed to a transmittable infection

To avoid *Taxoplasma* during pregnancy women should:

- keep cats indoors
- avoid handling cat litter without rubber gloves and wash thoroughly
- disinfect the cat box with boiling water for five minutes
- cover sandboxes
- wear gloves for gardening and wash afterwards
- avoid insects that may have been exposed to cat feces
- wash after handling cats, raw meat or poultry, soil, or sand
- avoid raw or undercooked meat and poultry, unwashed fruits and vegetables, raw eggs, and unpasteurized milk
- kill *Taxoplasma* by freezing food or cooking it thoroughly

All non-immune women of childbearing age should be vaccinated against rubella and chickenpox before pregnancy. Pregnant women should be tested for immunity to rubella at their first prenatal visit.

Women should be tested for GBS between 35 and 37 weeks of pregnancy to determine whether the bacteria are likely to be present at delivery.

Since *Listeria* can grow at temperatures below 40°F (4°C), pregnant women should handle food cautiously and avoid:

- hot dogs and luncheon and deli meats unless they are reheated to steaming
- soft cheeses
- refrigerated meat spreads
- refrigerated smoked seafood unless it is in a cooked dish
- raw unpasteurized milk

KEY TERMS

Antibody—A blood protein produced in response to a specific foreign substance including bacteria, viruses, and parasites; the antibody destroys the organism, providing protection against disease.

Cesarean section; C-section—Incision through the abdominal and uterine walls to deliver the fetus.

Conjunctivitis—An inflammation of the eye that can be caused by gonorrhea or chlamydia.

Cytomegalovirus; CMV—A common human herpes virus that is normally not harmful but may cause severe complications if transmitted to a fetus.

Fifth disease—Erythema infectiosum; a common respiratory infection among children caused by parvovirus B19 that usually is not serious but can cause fetal complications.

Group B streptococcal (GBS) disease—A common bacterial infection that is potentially life-threatening if transmitted to a fetus during early pregnancy or birth.

Herpes simplex virus; HSV—A very common sexually transmitted infection; Type-2 HSV causes genital herpes and type-1 HSV usually causes cold sores but also can cause genital herpes; congenital HSV can be transmitted to the fetus during birth if the mother has an active infection.

Human papillomavirus (HPV)—A large family of viruses, some of which cause genital warts; HPV can be transmitted to a fetus during birth.

Immune globulin—Serum containing antibodies against a specific infection.

Listeriosis—A food-borne bacterial infection caused by *Listeria monocytogenes* to which pregnant women are particularly susceptible.

Meningitis—An inflammation of the membranes covering the brain and spinal cord that can be caused by various congenital infections.

Perinatal infection—A maternal infection that is transmitted to the fetus after membrane rupture or during labor or delivery.

Placenta—The uterine organ that provides nourishment to the fetus.

Prenatal infection—A maternal infection that is transmitted to the fetus through the placenta.

Rubella—German measles; three-day measles; a viral infection that causes death or severe birth defects if transmitted to the fetus during the first 10 weeks of gestation.

Salmonellosis—Food poisoning; an infection by bacteria of the genus *Salmonella* that usually causes severe diarrhea and may be transmitted to the fetus.

Sepsis—A systemic or body-wide response to infection.

Sexually transmitted infection; STI—An infectious disease that is transmitted through sexual activity.

Ultrasound—High-frequency sound waves that are used to visualize parts of the body or a fetus in the womb.

Varicella—Chickenpox; a disease caused by the *Varicella zoster* virus—human herpes virus 3—that can cause severe birth defects if transmitted to the fetus during the first 20 weeks of pregnancy and newborn complications if it is transmitted perinatally.

Pregnant women should use precooked or ready-to-eat perishables immediately, clean the refrigerator regularly, and keep the refrigerator at or below 40°F (4°C).

Salmonellosis may be prevented by:

- cooking all meat, poultry, seafood, and eggs thoroughly
- avoiding sushi containing raw fish
- washing raw vegetables thoroughly
- avoiding unpasteurized milk, soft cheeses, and alfalfa sprouts

Prevention of STIs includes:

- abstaining from sexual contact outside of a mutually monogamous relationship

- using latex condoms correctly and consistently
- avoiding blood-contaminated needles, razors, or other items

Precautions for preventing fetal exposure to HIV-infected maternal blood include avoiding: **amniocentesis**, fetal scalp blood sampling, premature rupturing of the fetal membranes.

Prevention of maternal-to-fetal HSV transmission includes:

- abstaining from sexual activity during the last trimester of pregnancy or if there are signs of an outbreak or visible sores
- using a **condom** even if no symptoms are present
- postponing membrane rupture

- avoiding a fetal monitor that makes tiny punctures in the scalp
- avoiding vacuum or forceps deliveries which cause breaks in the infant's scalp

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ORGANIZATIONS

- American College of Obstetricians and Gynecologists. 409 12th St. SW, PO Box 96920, Washington, DC 20080-6920. 202-863-2518. < <http://www.acog.org> >.
- American Social Health Association. PO Box 13827, Research Triangle Park, NC 27709-3827. 800-783-9877. < <http://www.ashastd.org> >.
- Association of Women's Health, Obstetric and Neonatal Nurses. 2000 L Street NW, Suite 740, Washington, DC 20036. 800-673-8499. 202-261-2400. < <http://www.awhonn.org> >.
- Centers for Disease Control and Prevention. 1600 Clifton Road, Atlanta, GA 30333. 888-232-3228. < <http://www.cdc.gov> >.
- Hepatitis B Foundation. 700 East Butler Avenue, Doylestown, PA 18901-3697. 215-489-4900. < <http://www.hepb.org> >.
- March of Dimes Birth Defects Foundation. 1275 Mamaroneck Avenue, White Plains, NY 10605. < <http://www.marchofdimes.com> >.

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Mathematics disorder see **Learning disorders**

Maxillofacial trauma

Definition

Maxillofacial trauma refers to any injury to the face or jaw caused by physical force, **foreign objects**, or **burns**.

Description

Maxillofacial trauma includes injuries to any of the bony or fleshy structures of the face.

Any part of the face may be affected. Teeth may be knocked out or loosened. The eyes and their muscles, nerves, and blood vessels may be injured as well as the eye socket (orbit), which can be fractured by a forceful blow. The lower jaw (mandible) may be dislocated by force. Although anchored by strong muscles for chewing, the jaw is unstable in comparison with other bones and is easily dislocated from the temporomandibular joints that attach it to the skull. A fractured nose or jaw may affect the ability to breathe or eat. Any maxillofacial trauma may also prevent the passage of air or be severe enough to cause a **concussion** or more serious brain injury.

Athletes are particularly at risk of maxillofacial injuries. Boxers suffer repeated blows to the face and occasional knockouts (traumatic brain injury). Football, basketball, hockey, and soccer players, and many other athletes are at risk for milder forms of brain injury called concussions. There are an estimated 300,000 cases every year. Overall, there are one million new traumatic brain injuries every year, causing 50,000 deaths. Of the rest, 7–9% are left with long-term disability.

Burns to the face are also categorized as maxillofacial trauma.

Causes and symptoms

There are no reliable statistics on the incidence of maxillofacial trauma because there are so many types and many are not reported. Automobile accidents are a major cause, as well as participation in sports, fights, and other violent acts, and being hit by an object accidentally, for instance being hit by a baseball while watching a game. People most at risk are athletes, anyone who drives a vehicle or rides in one, and those who do dangerous work or engage in aggressive types of behavior.

One study reported in August 2000 that 42% of all facial **fractures** resulted from sports activity.

The major symptoms of most facial injuries are **pain**, swelling, bleeding, and bruising, although a fractured jaw also prevents the person from working his jaw properly, and symptoms of a fractured nose also include black eyes and possible blockage of the airway due to swelling and bleeding.

Symptoms of eye injury or orbital fracture can include blurred or double vision, decreased mobility of the eye, and **numbness** in the area of the eye. In severe injuries there can be temporary or permanent loss of vision.

Burn symptoms are pain, redness, and possibly blisters, **fever**, and **headache**. Extensive burns can cause the victim to go into **shock**. In that situation, he will have low blood pressure and a rapid pulse.

Symptoms of traumatic brain injury include problems with thinking, memory, and judgement as well as mood swings, and difficulty with coordination and balance. These symptoms linger for weeks or months, and in severe cases can be permanent. Double vision for months after the injury is not uncommon.

Diagnosis

Trauma is usually diagnosed in an emergency room or physician's office by **physical examination** and/or x ray. Some injuries require diagnosis by a specialist. A detailed report of how the injury occurred is also taken. In some cases, diagnosis cannot be made until swelling subsides.

Treatment

Treatment varies, depending on the type and extent of the injury.

Dislocation of the jaw can be treated by a primary care physician by exerting pressure in the proper manner. If muscle spasm prevents the jaw from moving back into alignment, a sedative is administered intravenously (IV) to relax the muscles. Afterward, the patient must avoid opening the jaw wide as he will be prone to repeat **dislocations**.

A jaw fracture may be minor enough to heal with simple limitation of movement and time. More serious fractures require complicated, multi-step treatment. The jaw must be surgically immobilized by a qualified oral or maxillofacial surgeon or an otolaryngologist. The jaw is properly aligned and secured with metal pins and wires. Proper alignment is necessary to ensure that the bite is correct. If the bite is off, the patient may develop a painful disorder called temporomandibular joint syndrome.



Face of an elderly woman suffering from maxillofacial trauma. (Photo Researchers. Reproduced by permission.)

During the weeks of healing the patient is limited to a liquid diet sipped through a straw and must be careful not to choke or vomit since he cannot open his mouth to expel the vomitus. The surgeon will prescribe pain relievers and perhaps **muscle relaxants**. Healing time varies according to the patient's overall health, but will take at least several weeks.

Another common maxillofacial fracture is a broken nose. The bones that form the bridge of the nose may be fractured, but cartilage may also be damaged, particularly the nasal septum which divides the nose. If hit from the side, the bones and cartilage are displaced to the side, but if hit from the front, they are splayed out. Severe swelling can inhibit diagnosis and treatment. Mild trauma to the nose can sometimes heal without the person being aware of the fracture unless there is obvious deformity. The nose will be tender for at least three weeks.

Either before the swelling begins or after it subsides, some 10 days after the injury, the doctor can assess the extent of the damage. Physical examination of the inside using a speculum and the outside, in addition to a detailed history of how the injury occurred will determine appropriate treatment. The doctor should be informed of any previous nasal fractures, nasal surgery, or chronic disease such as **osteoporosis**. Sometimes an x ray is useful, but it is not always required.

A primary care physician may treat a nasal fracture himself, but if there is extensive damage or the air passage is blocked, he will refer the patient to an otolaryngologist or a plastic surgeon for treatment. Initially the nose may be packed to control bleeding and hold the shape. It is reset under anesthesia. A protective shield or bandage may be placed over it while the fracture heals.

In the case of orbital fractures, there is great danger of permanent damage to vision. Double vision and decreased mobility of the eye are common complications. Surgical reconstruction may be required if the fracture changes the position of the eye or there is other facial deformity. Treatment requires a maxillofacial surgeon.

When the eyes have been exposed to chemicals, they must be washed out for 15 minutes with clear water. Contact lenses may be removed only after rinsing the eyes. The eyes should then be kept covered until the person can be evaluated by a primary care physician or ophthalmologist.

When a foreign object is lodged in the eye, the person should not rub the eye or put pressure on it which would further injure the eyeball. The eye should be covered to protect it until medical attention can be obtained.

Several kinds of traumatic injuries can occur to the mouth. A person can suffer a laceration (cut) to the lips or tongue, or loosening of teeth, or have teeth knocked out. Such injuries often accompany a jaw fracture or other facial injury. **Wounds** to the soft tissues of the mouth bleed freely, but the plentiful blood supply that leads to this heavy bleeding also helps healing. It is important to clean the wound thoroughly with salt water or hydrogen peroxide rinse to prevent infection. Large cuts may require sutures, and should be done by a maxillofacial surgeon for a good cosmetic result, particularly when the laceration is on the edge of the lip line (vermillion). The doctor will prescribe an antibiotic because there is normally a large amount of bacteria present in the mouth.

Any injury to the teeth should be evaluated by a dentist for treatment and prevention of infection. Implantation of a tooth is sometimes possible if it has been handled carefully and protected. The tooth should be held by the crown, not the root, and kept in milk, saline, or contact lens fluid. The patient's dentist can refer him to a specialist in this field.

For first degree burns, put a cold-water compress on the area or run cold water on it. Put a clean bandage on it for protection. Second and third degree burn victims must be taken to the hospital for treatment.

Fluids are replaced there through an IV. This is vital since a patient in shock will die unless those lost fluids are replaced quickly. **Antibiotics** are given to combat infection since the burns make the body vulnerable to infection.

Treatment for a **head injury** requires examination by a primary care physician unless symptoms point to

a more serious injury. In that case, the victim must seek emergency care. A concussion is treated with rest and avoidance of contact sports. Very often athletes who have suffered a concussion are allowed to play again too soon, perhaps in the mistaken impression that the injury is not so bad if the player did not lose consciousness. Anyone who has had one concussion is at increased risk of another one.

Danger signs that the injury is more serious include worsening headaches, **vomiting**, weakness, numbness, unsteadiness, change in the appearance of the eyes, seizures, slurred speech, confusion, agitation, or the victim won't wake up. These signs require immediate transport to the hospital. A neurologist will evaluate the situation, usually with a CT scan. A stay in a **rehabilitation** facility may become necessary.

Alternative treatments

Fractures, burns, and deep lacerations require treatment by a doctor but alternative treatments can help the body withstand injury and assist the healing process. Calcium, **minerals**, **vitamins**, all part of a balanced and nutrient-rich diet, as well as regular **exercise**, build strong bones that can withstand force well. After an injury, **craniosacral therapy** may help healing and ease the headaches that follow a concussion or other head trauma. A physical therapist can offer ultrasound that raises temperature to ease pain, or **biofeedback** in which the patient learns how to tense and relax muscles to relieve pain. **Hydrotherapy** may ease the **stress** of recovering from trauma. Chinese medicine seeks to reconnect the chi along the body's meridians and thus aid healing. Homeopathic physicians may prescribe natural medicines such as Arnica or Symphytum to enhance healing.

Prognosis

When appropriate treatment is obtained quickly after an injury, the prognosis can be excellent. However, if the victim of trauma has osteoporosis or a debilitating chronic disease, healing is more problematic. Healing also depends upon the extent of the injury. An automobile accident or a gunshot wound, for example, can cause severe facial trauma that may require multiple surgical procedures and a considerable amount of time to heal. Burns and lacerations cause scarring that might be improved by **plastic surgery**.

Prevention

Safety equipment is vital to preventing maxillofacial trauma from automobile accidents and sports.

KEY TERMS

Corneal abrasion—A scratch on the surface of the eyeball.

Mandible—The lower jaw, a U-shaped bone attached to the skull at the temporomandibular joints.

Maxilla—The bone of the upper jaw which serves as a foundation of the face and supports the orbits.

Orbit—The eye socket which contains the eyeball, muscles, nerves, and blood vessels that serve the eye.

Otolaryngologist—Ear, nose and throat specialist.

Shock—A reduction of blood flow in the body caused by loss of blood and/or fluids. Can be fatal if not treated quickly.

Temporomandibular joint—The mandible attaches to the temporal bone of the skull and works like a hinge.

Temporomandibular joint syndrome—TMJ Syndrome refers to an incorrect alignment of the lower jaw to the skull which causes the bite to be off line. It causes chronic headaches, nausea, and other symptoms.

Nasal septum—The cartilage which divides the nose in half.

Vermilion border—The line between the lip and the skin.

Here is a partial list of equipment people should always use:

- seatbelts
- automobile air bags
- approved child safety seats
- helmets for riding motorcycles or bicycles, skateboarding, snowboarding, and other sports
- safety glasses for the job, yard work, sports
- other approved safety equipment for sports such as mouthguards, masks, and goggles

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American Association of Oral & Maxillofacial Surgeons.
9700 W. Bryn Mawr Ave., Rosemont, IL 60018. (847)
678-6200.
Brain Injury Association, Inc. 105 N. Alfred St., Alexandria,
VA 22314. (703) 236-6000. <<http://www.biausa.org>>.

OTHER

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MCS syndrome see **Multiple chemical sensitivity**

MD see **Muscular dystrophy**

Measles

Definition

Measles is an infection caused by a virus, which causes an illness displaying a characteristic skin rash known as an exanthem. Measles is also sometimes called rubeola, 5-day measles, or hard measles.

Description

Measles infections appear all over the world. Prior to the current effective immunization program, large-scale measles outbreaks occurred on a two to three-year cycle, usually in the winter and spring. Smaller outbreaks occurred during the off-years. Babies up to about eight months of age are usually protected from contracting measles, due to immune cells they receive from their mothers in the uterus. Once someone has had measles infection, he or she can never get it again.

Causes and symptoms

Measles is caused by a type of virus called a paramyxovirus. It is an extremely contagious infection, spread through the tiny droplets that may spray into the air when an individual carrying the virus sneezes or coughs. About 85% of those people exposed to the



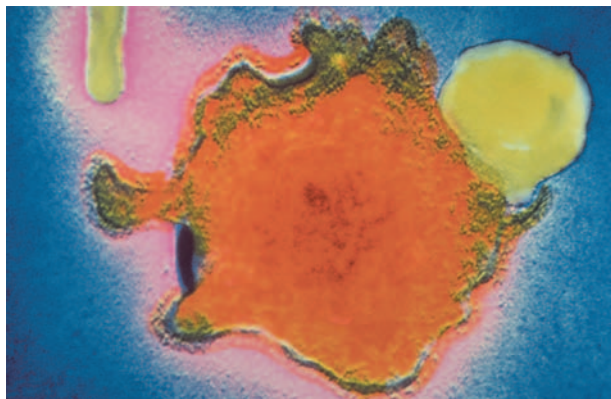
Measles on child's face. (Custom Medical Stock Photo. Reproduced by permission.)

virus will become infected with it. About 95% of those people infected with the virus will develop the illness called measles. Once someone is infected with the virus, it takes about 7–18 days before he or she actually becomes ill. The most contagious time period is the three to five days before symptoms begin through about four days after the characteristic measles rash has begun to appear.

The first signs of measles infection are **fever**, extremely runny nose, red, runny eyes, and a **cough**. A few days later, a rash appears in the mouth, particularly on the mucous membrane which lines the cheeks. This rash consists of tiny white dots (like grains of salt or sand) on a reddish bump. These are called Koplik's spots, and are unique to measles infection. The throat becomes red, swollen, and sore.

A couple of days after the appearance of the Koplik's spots, the measles rash begins. It appears in a characteristic progression, from the head, face, and neck, to the trunk, then abdomen, and next out along the arms and legs. The rash starts out as flat, red patches, but eventually develops some bumps. The rash may be somewhat itchy. When the rash begins to appear, the fever usually climbs higher, sometimes reaching as high as 105°F (40.5°C). There may be **nausea**, **vomiting**, **diarrhea**, and multiple swollen lymph nodes. The cough is usually more problematic at this point, and the patient feels awful. The rash usually lasts about five days. As it fades, it turns a brownish color, and eventually the affected skin becomes dry and flaky.

Many patients (about 5–15%) develop other complications. Bacterial infections, such as ear infections, sinus infections, and **pneumonia** are common, especially in children. Other viral infections may also strike the patient, including **croup**, **bronchitis**, **laryngitis**, or



A transmission electron microscopy (TEM) image of a single measles virion. (Custom Medical Stock Photo. Reproduced by permission.)

viral pneumonia. Inflammation of the liver, appendix, intestine, or lymph nodes within the abdomen may cause other complications. Rarely, inflammations of the heart or kidneys, a drop in **platelet count** (causing episodes of difficult-to-control bleeding), or reactivation of an old **tuberculosis** infection can occur.

An extremely serious complication of measles infection is swelling of the brain. Called **encephalitis**, this can occur up to several weeks after the basic measles symptoms have resolved. About one out of every thousand patients develops this complication, and about 10-15% of these patients die. Symptoms include fever, **headache**, sleepiness, seizures, and **coma**. Long-term problems following recovery from measles encephalitis may include seizures and **mental retardation**.

A very rare complication of measles can occur up to 10 years following the initial infection. Called **sub-acute sclerosing panencephalitis**, this is a slowly progressing, smoldering swelling and destruction of the entire brain. It is most common among people who had measles infection prior to the age of two years. Symptoms include changes in personality, decreased intelligence with accompanying school problems, decreased coordination, involuntary jerks and movements of the body. The disease progresses so that the individual becomes increasingly dependent, ultimately becoming bedridden and unaware of his or her surroundings. Blindness may develop, and the temperature may spike (rise rapidly) and fall unpredictably as the brain structures responsible for temperature regulation are affected. **Death** is inevitable.

Measles during **pregnancy** is a serious disease, leading to increased risk of a **miscarriage** or **stillbirth**.

In addition, the mother's illness may progress to pneumonia.

Diagnosis

Measles infection is almost always diagnosed based on its characteristic symptoms, including Koplik's spots, and a rash which spreads from central body structures out towards the arms and legs. If there is any doubt as to the diagnosis, then a specimen of body fluids (mucus, urine) can be collected and combined with fluorescent-tagged measles virus antibodies. Antibodies are produced by the body's immune cells that can recognize and bind to markers (antigens) on the outside of specific organisms, in this case the measles virus. Once the fluorescent antibodies have attached themselves to the measles antigens in the specimen, the specimen can be viewed under a special microscope to verify the presence of measles virus.

Treatment

There are no treatments available to stop measles infection. Treatment is primarily aimed at helping the patient to be as comfortable as possible, and watching carefully so that **antibiotics** can be started promptly if a bacterial infection develops. Fever and discomfort can be treated with **acetaminophen**. Children with measles should never be given **aspirin**, as this has caused the fatal disease **Reye's syndrome** in the past. A cool-mist vaporizer may help decrease the cough. Patients should be given a lot of liquids to drink, in order to avoid **dehydration** from the fever.

Some studies have shown that children with measles encephalitis benefit from relatively large doses of vitamin A.

Alternative treatment

Botanical immune enhancement (with **echinacea**, for example) can assist the body in working through this viral infection. Homeopathic support also can be effective throughout the course of the illness. Some specific alternative treatments to soothe patients with measles include the Chinese herbs bupleurum (*Bupleurum chinense*) and peppermint (*Mentha piperita*), as well as a preparation made from empty cicada (*Cryptotympana atrata*) shells. The itchiness of the rash can be relieved with witch hazel (*Hamamelis virginiana*), chickweed (*Stellaria media*), or oatmeal baths. The eyes can be soothed with an eyewash made from the herb eyebright (*Euphrasia officinalis*). Practitioners of **ayurvedic medicine** recommend ginger or clove tea.

KEY TERMS

Antibodies—Cells made by the immune system which have the ability to recognize foreign invaders (bacteria, viruses), and thus stimulate the immune system to kill them.

Antigens—Markers on the outside of such organisms as bacteria and viruses, which allow antibodies to recognize foreign invaders.

Encephalitis—Swelling, inflammation of the brain.

Exanthem (plural, exanthems or exanthemata)—A skin eruption regarded as a characteristic sign of such diseases as measles, German measles, and scarlet fever.

Koplik's spots—Tiny spots occurring inside the mouth, especially on the inside of the cheek. These spots consist of minuscule white dots (like grains of salt or sand) set onto a reddened bump. Unique to measles.

Prognosis

The prognosis for an otherwise healthy, well-nourished child who contracts measles is usually quite good. In developing countries, however, death rates may reach 15–25%. Adolescents and adults usually have a more difficult course. Women who contract the disease while pregnant may give birth to a baby with hearing impairment. Although only 1 in 1,000 patients with measles will develop encephalitis, 10–15% of those who do will die, and about another 25% will be left with permanent brain damage.

Prevention

Measles is a highly preventable infection. A very effective vaccine exists, made of live measles viruses which have been treated so that they cannot cause actual infection. The important markers on the viruses are intact, however, which causes an individual's immune system to react. Immune cells called antibodies are produced, which in the event of a future infection with measles virus will quickly recognize the organism, and kill it off. Measles vaccines are usually given at about 15 months of age; because prior to that age, the baby's immune system is not mature enough to initiate a reaction strong enough to insure long-term protection from the virus. A repeat injection should be given at about 10 or 11 years of age. Outbreaks on college campuses have occurred among unimmunized or incorrectly immunized students.

Measles vaccine should not be given to a pregnant woman, however, in spite of the seriousness of gestational measles. The reason for not giving this particular vaccine during pregnancy is the risk of transmitting measles to the unborn child.

Surprisingly, new cases of measles began being reported in some countries—including Great Britain—in 2001 because of parents' fears about vaccine safety. The combined vaccine for measles, **mumps**, and **rubella** (MMR) was claimed to cause **autism** or bowel disorders in some children. However, the World Health Organization (WHO) says there is no scientific merit to these claims. The United Nations expressed concern that unwarranted fear of the vaccine would begin spreading the disease in developing countries, and ultimately in developed countries as well. Parents in Britain began demanding the measles vaccine as a separate dose and scientists were exploring that option as an alternative to the combined MMR vaccine. Unfortunately, several children died during an outbreak of measles in Dublin because they had not received the vaccine. Child mortality due to measles is considered largely preventable, and making the MMR vaccine widely available in developing countries is part of WHO's strategy to reduce child mortality by two-thirds by the year 2015.

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American Academy of Pediatrics (AAP). 141 Northwest Point Boulevard, Elk Grove Village, IL 60007. (847) 434-4000. <<http://www.aap.org>>.

Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

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Mebendazole see **Antihelminthic drugs**

Mechanical debridement see **Debridement**

Mechanical ventilation see **Inhalation therapies**

Meckel's diverticulum

Definition

Meckel's diverticulum is a congenital pouch (diverticulum) approximately two inches in length and located at the lower (distal) end of the small intestine. It was named for Johann F. Meckel, a German anatomist who first described the structure.

Description

The diverticulum is most easily described as a blind pouch that is a remnant of the omphalomesenteric duct or yolk sac that nourished the early embryo. It contains all layers of the intestine and may have ectopic tissue present from either the pancreas or stomach.

The rule of 2s is the classical description. It is located about 2 ft from the end of the small intestine, is often about 2 in in length, occurs in about 2% of the population, is twice as common in males as females, and can contain two types of ectopic tissue—stomach or pancreas. Many who have a Meckel's diverticulum never have trouble but those that do present in the first two decades of life and often in the first two years.

There are three major complications that may result from the development of Meckel's diverticulum. The most common problem is inflammation or infection that mimics **appendicitis**. This diagnosis is defined at the time of surgery for suspected appendicitis.

Bleeding caused by ectopic stomach tissue that results in a bleeding ulcer is the second most frequent problem. Bleeding may be brisk or massive. The third potential complication is obstruction due to **intussusception**, or a twist around a persistent connection to the abdominal wall. This problem presents as a small bowel obstruction, however, the true cause is identified at the time of surgical exploration.

Meckel's diverticulum is a developmental defect that is present in about 2% of people, but does not always cause symptoms. Meckel's diverticula (plural of diverticulum) are found twice as frequently in men as in women. Complications occur three to five times more frequently in males.

Causes and symptoms

Meckel's diverticulum is not hereditary. It is a vestigial remnant of the omphalomesenteric duct, an embryonic structure that becomes the intestine. As such, there is no genetic defect or abnormality.

Symptoms usually occur in children under 10 years of age. There may be bleeding from the rectum, **pain** and **vomiting**, or simply tiredness and weakness from unnoticed blood loss. It is common for a Meckel's diverticulum to be mistaken for the much more common disease appendicitis. If there is obstruction, the abdomen will distend and there will be cramping pain and vomiting.

Diagnosis

The situation may be so acute that surgery is needed on an emergency basis. This is often the case with bowel obstruction. With heavy bleeding or severe pain, whatever the cause, surgery is required. The finer points of diagnosis can be accomplished when the abdomen is open for inspection during a surgical procedure. This situation is called an acute abdomen.

If there is more time (not an emergency situation), the best way to diagnose Meckel's diverticulum is with a nuclear scan. A radioactive isotope injected into the bloodstream will accumulate at sites of bleeding or in stomach tissue. If a piece of stomach tissue or a pool of blood shows up in the lower intestine, Meckel's diverticulum is indicated.

Treatment

A Meckel's diverticulum that is causing discomfort, bleeding, or obstruction must be surgically removed. This procedure is very similar to an **appendectomy**.



A close-up image of a patient's small intestine with a protruding sac. This condition, called Meckel's diverticulum, is a congenital abnormality occurring in 2% of the population, usually males. (Custom Medical Stock Photo. Reproduced by permission.)

Prognosis

The outcome after surgery is usually excellent. The source of bleeding, pain, or obstruction is removed so the symptoms also disappear. A Meckel's diverticulum will not return.

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KEY TERMS

Appendectomy—The procedure to surgically remove an appendix.

Appendicitis—Inflammation of the appendix.

Appendix—A portion of intestine attached to the cecum.

Cecum—The first part of the large bowel.

Congenital—Refers to a disorder which is present at birth.

Distal—Away from the point of origin.

Ectopic—Tissue found in an abnormal location.

Intussusception—One piece of bowel inside another, causing obstruction.

Isotope—Any of two or more species of atoms of a chemical element with the same atomic number and nearly identical chemical behavior but with differing atomic mass and physical properties.

Peptic ulcer—A wound in the bowel that can be caused by stomach acid or a bacterium called *Helicobacter pylori*.

Volvulus—A twisted loop of bowel, causing obstruction.

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- American Academy of Family Physicians. 11400 Tomahawk Creek Parkway, Leawood, KS 66211-2672. (913) 906-6000. fp@aafp.org. <<http://www.aafp.org/>>.
- American Academy of Pediatrics. 141 Northwest Point Boulevard, Elk Grove Village, IL 60007-1098. (847) 434-4000. Fax: (847) 434-8000. kidsdoc@aap.org. <<http://www.aap.org/default.htm>>.
- American College of Gastroenterology. 4900 B South 31st Street, Arlington, VA 22206. (703) 820-7400. Fax: (703) 931-4520. <<http://www.acg.gi.org>>.
- American College of Surgeons. 633 North St. Clair St., Chicago, IL 60611-32311. (312) 202-5000. Fax: (312)

202-5001. postmaster@facs.org. <http://www.facs.org/>.

American Medical Association. 515 N. State Street, Chicago, IL 60610. (312) 464-5000. <http://www.ama-assn.org/>.

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L. Fleming Fallon, Jr., MD, DrPH

Median nerve entrapment see **Carpal tunnel syndrome**

Mediastinoscopy

Definition

Mediastinoscopy is a surgical procedure that allows physicians to view areas of the mediastinum, the cavity behind the breastbone that lies between the lungs. The organs in the mediastinum include the heart and its vessels, the lymph nodes, trachea, esophagus, and thymus.

Mediastinoscopy is most commonly used to detect or stage **cancer**. It is also ordered to detect infection, and to confirm diagnosis of certain conditions and diseases of the respiratory organs. The procedure involves insertion of an endotracheal (within the trachea) tube, followed by a small incision in the chest. A mediastinoscope is inserted through the incision. The purpose of this equipment is to allow the physician to directly see the organs inside the mediastinum, and to collect tissue samples for laboratory study.

Purpose

Mediastinoscopy is often the diagnostic method of choice for detecting lymphoma, including Hodgkin’s disease. The diagnosis of **sarcoidosis** (a chronic lung disease) and the staging of lung cancer can also be accomplished through mediastinoscopy. Lung cancer staging involves the placement of the cancer’s progression into stages, or levels. These stages help a physician study cancer and provide consistent definition levels of cancer and corresponding treatments. The lymph nodes in the mediastinum are likely to show if lung cancer has spread beyond the lungs. Mediastinoscopy allows a physician to observe and

extract a sample from the nodes for further study. Involvement of these lymph nodes indicates diagnosis and stages of lung cancer.

Mediastinoscopy may also be ordered to verify a diagnosis that was not clearly confirmed by other methods, such as certain radiographic and laboratory studies. Mediastinoscopy may also aid in certain surgical biopsies of nodes or cancerous tissue in the mediastinum. In fact, the surgeon may immediately perform a surgical procedure if a malignant tumor is confirmed while the patient is undergoing mediastinoscopy, thus combining the diagnostic exam and surgical procedure into one operation when possible.

Although still performed in 2001, advancements in computed tomography (CT) and **magnetic resonance imaging** (MRI) techniques, as well as the new developments in ultrasonography, have led to a decline in the use of mediastinoscopy. In addition, better results of fine-needle aspiration (drawing out fluid by suction) and core-needle biopsy (using a needle to obtain a small tissue sample) investigations, along with new techniques in **thoracoscopy** (examination of the thoracic cavity with a lighted instrument called a thoracoscope) offer additional options in examining mediastinal masses. Mediastinoscopy may be required, however, when these other methods cannot be used or when the results they provide are inconclusive.

Precautions

Because mediastinoscopy is a surgical procedure, it should only be performed when the benefits of the exam’s findings outweigh the risks of surgery and anesthesia. Patients who previously had mediastinoscopy should not receive it again if there is scarring present from the first exam.

Several other medical conditions, such as impaired cerebral circulation, obstruction or distortion of the upper airway, or thoracic **aortic aneurysm** (abnormal dilation of the thoracic aorta) may also preclude mediastinoscopy. Anatomic structures that can be compressed by the mediastinoscope may complicate these pre-existing medical conditions.

Description

Mediastinoscopy is usually performed in a hospital under **general anesthesia**. An endotracheal tube is inserted first, after **local anesthesia** is applied to the throat. Once the patient is under general anesthesia, a small incision is made usually just below the neck or at the notch at the top of the breastbone. The surgeon

may clear a path and feel the patient's lymph nodes first to evaluate any abnormalities within the nodes. Next, the physician will insert the mediastinoscope through the incision. The scope is a narrow, hollow tube with an attached light that allows the surgeon to see inside the area. The surgeon can insert tools through the hollow tube to help perform biopsies. A sample of tissue from the lymph nodes or a mass can be extracted and sent for study under a microscope or on to a laboratory for further testing.

In some cases, analysis of the tissue sample which shows malignancy will suggest the need for immediate surgery while the patient is already prepared and under anesthesia. In other cases, the surgeon will complete the visual study and tissue extraction and stitch the small incision closed. The patient will remain in the surgery recovery area until it is determined that the effects of anesthesia have lessened and it is safe for the patient to leave the area. The entire procedure should take about an hour, not counting preparation and recovery time. Studies have shown that mediastinoscopy is a safe, thorough, and cost-effective diagnostic tool with less risk than some other procedures.

Preparation

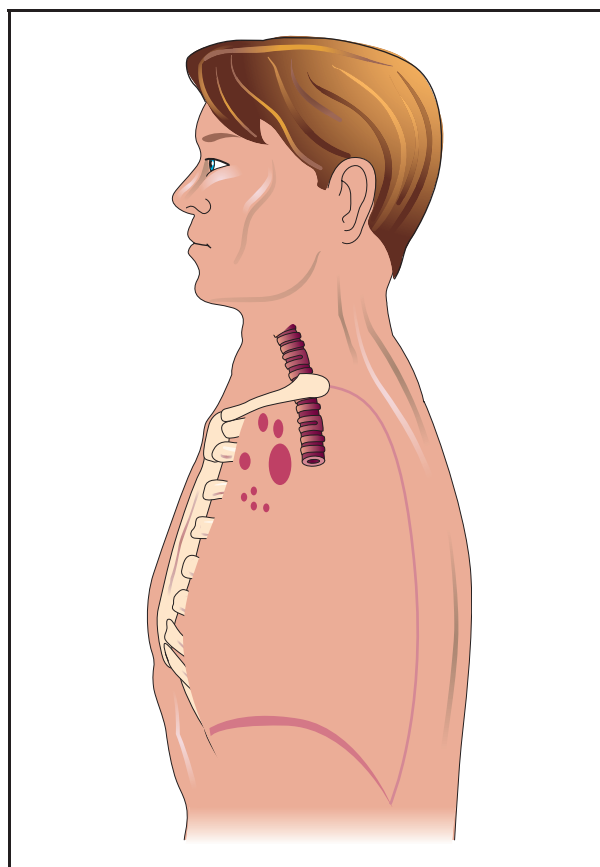
Patients are asked to sign a consent form after having reviewed the risks of mediastinoscopy and known risks or reactions to anesthesia. The physician will normally instruct the patient to fast from midnight before the test until after the procedure is completed. A physician may also prescribe a sedative the night before the exam and before the procedure. Often a local anesthetic will be applied to the throat to prevent discomfort during placement of the endotracheal tube.

Aftercare

Following mediastinoscopy, patients will be carefully monitored to watch for changes in vital signs or indications of complications of the procedure or the anesthesia. A patient may have a **sore throat** from the endotracheal tube, temporary chest **pain**, and soreness or tenderness at the site of incision.

Risks

Complications from the actual mediastinoscopy procedure are relatively rare—the overall complication rate in various studies has been 1.3–3.0%. However, the following complications, in decreasing order of frequency, have been reported:



Mediastinoscopy is a surgical procedure used to detect or stage lymphoma or lung cancer. In this procedure, the surgeon makes an incision below the neck and inserts a mediastinoscope (a narrow, hollow tube with an attached light) through it to reach the area behind the breastbone. The surgeon can then insert tools through the scope to collect tissue for laboratory analysis. (Illustration by Electronic Illustrators Group.)

- hemorrhage
- pneumothorax (air in the pleural space)
- recurrent laryngeal nerve injury, causing hoarseness
- infection
- tumor implantation in the wound
- phrenic nerve injury (injury to a thoracic nerve)
- esophageal injury
- chylothorax (chyle—a milky lymphatic fluid—in the pleural space)
- air **embolism** (air bubble)
- transient hemiparesis (**paralysis** on one side of the body)

The usual risks associated with general anesthesia also apply to this procedure.

KEY TERMS

Endotracheal—Placed within the trachea, also known as the windpipe.

Hodgkin's disease—A malignancy of lymphoid tissue found in the lymph nodes, spleen, liver, and bone marrow.

Lymph nodes—Small round structures located throughout the body; contain cells that fight infections.

Pleural space—Space between the layers of the pleura (membrane lining the lungs and thorax).

Sarcoidosis—A chronic disease characterized by nodules in the lungs, skin, lymph nodes and bones; however, any tissue or organ in the body may be affected.

Thymus—An unpaired organ in the mediastinal cavity that is important in the body's immune response.

Normal results

In the majority of procedures performed to diagnose cancer, a normal result involves evidence of small, smooth, normal-appearing lymph nodes and no abnormal tissue, growths, or signs of infection. In the case of lung cancer staging, results are related to the severity and progression of the cancer.

Abnormal results

Abnormal findings may indicate lung cancer, **tuberculosis**, the spread of disease from one body part to another, **sarcoidosis** (a disease that causes nodules, usually affecting the lungs), **lymphoma** (abnormalities in the lymph tissues), and **Hodgkin's disease**.

Resources

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Pagana, Kathleen Deska, and Timothy James Pagana. *Mosby's Manual of Diagnostic and Laboratory Tests*. St. Louis, MO: Mosby, 1998.

PERIODICALS

Deslauriers, Jean, and Jocelyn Gregoire. "Clinical and Surgical Staging of Non-Small Cell Lung Cancer." *Chest*, Supplement (April 2000): 96S–103S.

Tahara R. W., et al. "Is There a Role for Routine Mediastinoscopy in Patients With Peripheral T1 Lung Cancers?" *American Journal of Surgery* December 2000: 488–491.

ORGANIZATIONS

Alliance for Lung Cancer Advocacy, Support, and Education. P.O. Box 849, Vancouver, WA 98666. (800) 298-2436. <<http://www.alcase.org>>.

American Cancer Society. 1599 Clifton Rd. NE, Atlanta, GA 30329. 800-ACS-2345 <<http://www.cancer.org>>.

American Lung Association. 1740 Broadway, New York, NY 10019-4374. 800-LUNG-USA (800-586-4872). <<http://www.lungusa.org>>.

Teresa Odle

Meditation

Definition

Meditation is a practice of concentrated focus upon a sound, object, visualization, the breath, movement, or attention itself in order to increase awareness of the present moment, reduce **stress**, promote relaxation, and enhance personal and spiritual growth.

Purpose

Meditation benefits people with or without acute medical illness or stress. People who meditate regularly have been shown to feel less **anxiety** and depression. They also report that they experience more enjoyment and appreciation of life and that their relationships with others are improved. Meditation produces a state of deep relaxation and a sense of balance or equanimity. According to Michael J. Baime, "Meditation cultivates an emotional stability that allows the meditator to experience intense emotions fully while simultaneously maintaining perspective on them." Out of this experience of emotional stability, one may gain greater insight and understanding about one's thoughts, feelings, and actions. This insight in turn offers the possibility to feel more confident and in control of life. Meditation facilitates a greater sense of calmness, empathy, and acceptance of self and others.

Meditation can be used with other forms of medical treatment and is an important complementary therapy for both the treatment and prevention of many stress-related conditions. Regular meditation can reduce the number of symptoms experienced by patients with a wide range of illnesses and disorders.

MAHARISHI MAHESH YOGI (1911–)



(Archive. Reproduced by permission.)

Maharishi Mahesh Yogi is one of the most recognized spiritual leaders of the world. Almost single-handedly, the

Maharishi (meaning great sage) brought Eastern culture into Western consciousness. He emerged in the late 1950s in London and the United States as a missionary in the cause of Hinduism, the philosophy of which is called Vedanta—a belief that “holds that God is to be found in every creature and object, that the purpose of human life is to realize the godliness in oneself and that religious truths are universal.”

By 1967, the Maharishi became a leader among flower-children and an anti-drug advocate. The Maharishi’s sudden popularity was helped along by such early fans as the Beatles, Mia Farrow, and Shirley MacLaine. These people, and many others, practiced Transcendental Meditation (TM), a Hindu-influenced procedure that endures in America to this day.

When the 1960s drew to a close, the Maharishi began to fade from public view. The guru still had enough followers, though, to people the Maharishi International University, founded in 1971. One of the main draws of Maharishi International University was the study of TM-Sidha, an exotic form of Transcendental Meditation. Sidhas believe that group meditation can elicit the maharishi effect—a force strong enough to conjure world peace.

Based upon clinical evidence as well as theoretical understanding, meditation is considered to be one of the better therapies for **panic disorder**, **generalized anxiety disorder**, **substance dependence** and **abuse**, ulcers, colitis, chronic **pain**, **psoriasis**, and dysthymic disorder. It is considered to be a valuable adjunctive therapy for moderate **hypertension** (high blood pressure), prevention of cardiac arrest (**heart attack**), prevention of **atherosclerosis** (hardening of arteries), arthritis (including **fibromyalgia**), **cancer**, **insomnia**, migraine, and prevention of **stroke**. Meditation may also be a valuable complementary therapy for **allergies** and **asthma** because of the role stress plays in these conditions. Meditative practices have been reported to improve function or reduce symptoms in patients with some neurological disorders as well. These include people with Parkinson’s disease, people who experience **fatigue** with **multiple sclerosis**, and people with epilepsy who are resistant to standard treatment.

Overall, a 1995 report to the National Institutes of Health on alternative medicine concluded that, “More than 30 years of research, as well as the experience of a large and growing number of individuals and health

care providers, suggests that meditation and similar forms of relaxation can lead to better health, higher quality of life, and lowered health care costs. . .”

Description

Origins

Meditation techniques have been practiced for millennia. Originally, they were intended to develop spiritual understanding, awareness, and direct experience of ultimate reality. The many different religious traditions in the world have given rise to a rich variety of meditative practices. These include the contemplative practices of Christian religious orders, the Buddhist practice of sitting meditation, and the whirling movements of the Sufi dervishes. Although meditation is an important spiritual practice in many religious and spiritual traditions, it can be practiced by anyone regardless of their religious or cultural background to relieve stress and pain.

As Western medical practitioners begin to understand the mind’s role in health and disease, there has been more interest in the use of meditation in

medicine. Meditative practices are increasingly offered in medical clinics and hospitals as a tool for improving health and quality of life. Meditation has been used as the primary therapy for treating certain diseases; as an additional therapy in a comprehensive treatment plan; and as a means of improving the quality of life of people with debilitating, chronic, or terminal illnesses.

Sitting meditation is generally done in an upright seated position, either in a chair or cross-legged on a cushion on the floor. The spine is straight yet relaxed. Sometimes the eyes are closed. Other times the eyes are open and gazing softly into the distance or at an object. Depending on the type of meditation, the meditator may be concentrating on the sensation of the movement of the breath, counting the breath, silently repeating a sound, chanting, visualizing an image, focusing awareness on the center of the body, opening to all sensory experiences including thoughts, or performing stylized ritual movements with the hands.

Movement meditation can be spontaneous and free-form or involve highly structured, choreographed, repetitive patterns. Movement meditation is particularly helpful for those people who find it difficult to remain still.

Generally speaking, there are two main types of meditation. These types are concentration meditation and mindfulness meditation. Concentration meditation practices involve focusing attention on a single object. Objects of meditation can include the breath, an inner or external image, a movement pattern (as in **tai chi** or **yoga**), or a sound, word, or phrase that is repeated silently (mantra). The purpose of concentration practices is to learn to focus one's attention or develop concentration. When thoughts or emotions arise, the meditator gently directs the mind back to the original object of concentration.

Mindfulness meditation practices involve becoming aware of the entire field of attention. The meditator is instructed to be aware of all thoughts, feelings, perceptions or sensations as they arise in each moment. Mindfulness meditation practices are enhanced by the meditator's ability to focus and quiet the mind. Many meditation practices are a blend of these two forms.

The study and application of meditation to health care has focused on three specific approaches: 1. transcendental meditation (TM); 2. The "relaxation response," a general approach to meditation developed by Dr. Herbert Benson; and 3. mindfulness meditation, specifically the program of mindfulness-based **stress reduction** (MBSR) developed by Jon Kabat-Zinn.

Transcendental meditation

TM has its origins in the Vedic tradition of India and was introduced to the West by **Maharishi Mahesh Yogi**. TM has been taught to somewhere between two and four million people. It is one of the most widely practiced forms of meditation in the West. TM has been studied many times; these studies have produced much of the information about the physiology of meditation. In TM, the meditator sits with closed eyes and concentrates on a single syllable or word (mantra) for 20 minutes at a time, twice a day. When thoughts or feelings arise, the attention is brought back to the mantra. According to Charles Alexander, an important TM researcher, "During TM, ordinary waking mental activity is said to settle down, until even the subtlest thought is transcended and a completely unified wholeness of awareness... is experienced. In this silent, self-referential state of pure wakefulness, consciousness is fully awake to itself alone..." TM supporters believe that TM practices are more beneficial than other meditation practices.

The relaxation response

The relaxation response involves a similar form of mental focusing. Dr. Herbert Benson, one of the first Western doctors to conduct research on the effects of meditation, developed this approach after observing the profound health benefits of a state of bodily calm he calls "the relaxation response." In order to elicit this response in the body, he teaches patients to focus upon the repetition of a word, sound, prayer, phrase, or movement activity (including swimming, jogging, yoga, and even knitting) for 10–20 minutes at a time, twice a day. Patients are also taught not to pay attention to distracting thoughts and to return their focus to the original repetition. The choice of the focused repetition is up to the individual. Instead of Sanskrit terms, the meditator can choose what is personally meaningful, such as a phrase from a Christian or Jewish prayer.

Mindfulness meditation

Mindfulness meditation comes out of traditional Buddhist meditation practices. Psychologist Jon Kabat-Zinn has been instrumental in bringing this form of meditation into medical settings. In formal mindfulness practice, the meditator sits with eyes closed, focusing the attention on the sensations and movement of the breath for approximately 45–60 minutes at a time, at least once a day. Informal mindfulness practice involves bringing awareness to every



Girl in meditation. (Photograph by Robert J. Huffman. Field Mark Publications. Reproduced by permission.)

activity in daily life. Wandering thoughts or distracting feelings are simply noticed without resisting or reacting to them. The essence of mindfulness meditation is not what one focuses on but rather the quality of awareness the meditator brings to each moment. According to Kabat-Zinn, “It is this investigative, discerning observation of whatever comes up in the present moment that is the hallmark of mindfulness and differentiates it most from other forms of meditation. The goal of mindfulness is for you to be more aware, more in touch with life and whatever is happening in your own body and mind at the time it is happening—that is, the present moment.” The MBSR program consists of a series of classes involving meditation, movement, and group process. There are over 240 MBSR programs offered in health care settings around the world.

Meditation is not considered a medical procedure or intervention by most insurers. Many patients pay for meditation training themselves. Frequently, religious groups or meditation centers offer meditation instruction free of charge or for a nominal donation. Hospitals may offer MBSR classes at a reduced rate for their patients and a slightly higher rate for the general public.

KEY TERMS

Dervish—A member of the Sufi order. Their practice of meditation involves whirling ecstatic dance.

Mantra—A sacred word or formula repeated over and over to concentrate the mind.

Transcendental meditation (TM)—A meditation technique based on Hindu practices that involves the repetition of a mantra.

Precautions

Meditation appears to be safe for most people. There are, however, case reports and studies noting some adverse effects. Thirty-three to 50% of the people participating in long silent meditation retreats (two weeks to three months) reported increased tension, anxiety, confusion, and depression. On the other hand, most of these same people also reported very positive effects from their meditation practice. Kabat-Zinn notes that these studies fail to differentiate between serious psychiatric disturbances and normal emotional mood swings. These studies do suggest, however, that meditation may not be recommended for people with psychotic disorders, severe depression, and other severe **personality disorders** unless they are also receiving psychological or medical treatment.

Side effects

There are no reported side effects from meditation except for positive benefits.

Research and general acceptance

The scientific study of the physiological effects of meditation began in the early 1960s. These studies prove that meditation affects metabolism, the endocrine system, the central nervous system, and the autonomic nervous system. In one study, three advanced practitioners of Tibetan Buddhist meditation practices demonstrated the ability to increase “inner heat” as much as 61%. During a different meditative practice they were able to dramatically slow down the rate at which their bodies consumed oxygen. Preliminary research shows that mindfulness meditation is associated with increased levels of melatonin. These findings suggest a potential role for meditation in the treatment and prevention of breast and prostate cancer.

Despite the inherent difficulties in designing research studies, there is a large amount of evidence

of the medical benefits of meditation. Meditation is particularly effective as a treatment for chronic pain. Studies have shown meditation reduces symptoms of pain and pain-related drug use. In a four-year follow-up study, the majority of patients in a MBSR program reported “moderate to great improvement” in pain as a result of participation in the program.

Meditation has long been recommended as a treatment for high blood pressure; however, there is a debate over the amount of benefit that meditation offers. Although most studies show a reduction in blood pressure with meditation, medication is still more effective at lowering high blood pressure.

Meditation may also be an effective treatment for **coronary artery disease**. A study of 21 patients practicing TM for eight months showed increases in their amount of **exercise** tolerance, amount of workload, and a delay in the onset of ST-segment depression. Meditation is also an important part of Dean Ornish’s program, which has been proven to reverse coronary artery disease.

Research also suggests that meditation is effective in the treatment of chemical dependency. Gelderloos and others reviewed 24 studies and reported that all of them showed that TM is helpful in programs to stop **smoking** and also in programs for drug and alcohol abuse.

Studies also imply that meditation is helpful in reducing symptoms of anxiety and in treating anxiety-related disorders. Furthermore, a study in 1998 of 37 psoriasis patients showed that those practicing mindfulness meditation had more rapid clearing of their skin condition, with standard UV light treatment, than the control subjects. Another study found that meditation decreased the symptoms of fibromyalgia; over half of the patients reported significant improvement. Meditation was one of several stress management techniques used in a small study of HIV-positive men. The study showed improvements in the T-cell counts of the men, as well as in several psychological measures of well-being.

Resources

BOOKS

Astin, John A., et al. “Meditation” in *Clinician’s Complete Reference to Complementary and Alternative Medicine*, edited by Donald Novoy. St. Louis, MO: Mosby, 2000.

Baime, Michael J. “Meditation and Mindfulness” in *Essentials of Complementary and Alternative Medicine*, edited Wayne B. Jonas and Jeffrey S. Levin. New York: Lippencott, Williams and Wilkins, 1999.

ORGANIZATIONS

Insight Meditation Society. 1230 Pleasant, St. Barre, MA 01005. (978) 355-4378. FAX: (978) 355-6398.
< <http://www.dharma.org> >.

Mind-Body Medical Institute. Beth Israel Deaconess Medical Center. One Deaconess Road, Boston, MA 02215. (617) 632-9525.
< <http://www.mindbody.harvard.edu> >.

OTHER

Videos are available from the organizations listed above.

Linda Chrisman

Medullary sponge kidney

Definition

Medullary sponge kidney is a congenital defect of the kidneys where the kidneys fill with pools of urine.

Description

One of every 100 to 200 people have some form of this disease. The kidneys filter urine from the blood and direct it down tiny collecting tubes toward the ureters (ducts that carry urine from the kidney to the bladder). These tiny tubes gradually join together until they reach the renal pelvis, where the ureters begin. As the tubes join, they are supposed to get progressively bigger as they get fewer in number. In medullary sponge kidney, the tubes are irregular in diameter, forming pools of urine along the way. These pools encourage stone formation and infection.

Causes and symptoms

Although some cases of this disorder seem to be inherited, usually the cause is not known.

The symptoms associated with medullary sponge kidney are those related to infection and stone passage. Infection causes **fever**; back and flank **pain**; cloudy, frequent, and burning urine; and general discomfort. Stones cause pain in the flank or groin as they pass. They usually cause some bleeding. The bleeding may not be visible in the urine, but it is apparent under a microscope.

Diagnosis

Recurring kidney infections, bleeding, or stones will prompt x rays of the kidneys. The appearance

KEY TERMS

Congenital—Present at birth.

Intravenous pyelogram—X rays of the upper urinary system using a contrast agent that is excreted by the kidneys into the urine.

Thiazide diuretic—A particular class of medication that encourages urine production.

of medullary sponge kidney on an intravenous pyelogram (x rays of the upper urinary system) is characteristic.

Treatment

Many people never have trouble with this disorder. For those that do, infections and stones will need periodic treatment. Infections should be treated with **antibiotics** early in order to prevent kidney damage. Stones may need to be surgically removed. Often, removal can be accomplished without an incision but rather by reaching up with instruments through the lower urinary tract to grab the stones. There is also a new method of stone treatment called shock wave **lithotripsy**. A special machine delivers a focused blast of shock waves that breaks stones into sand so that they will pass out naturally. It is considered reasonably safe and usually effective.

Prognosis

Ignoring symptoms can result in progressive damage to the kidneys and ultimate kidney failure, but attentive early treatment will preserve kidney function.

Prevention

Diligent monitoring for infection at regular intervals and at the first symptom will give the best long-term results. By drinking extra liquids, most stones can be prevented. The most common kind of stones, calcium stones, can be deterred by regularly taking a medication that encourages urine production (thiazide diuretic).

Resources

ORGANIZATIONS

American Association of Kidney Patients. 100 S. Ashley Dr., #280, Tampa, FL 33602. (800) 749-2257. <<http://www.aakp.org>>.

American Kidney Foundation. 6110 Executive Boulevard, #1010, Rockville, MD 20852. 800-638-8299.
National Kidney Foundation. 30 East 33rd St., New York, NY 10016. (800) 622-9010. <<http://www.kidney.org>>.

J. Ricker Polsdorfer, MD

Medulloblastoma see **Brain tumor**

Mefloquine see **Antimalarial drugs**

Megalencephaly see **Congenital brain defects**

Melanoma see **Malignant melanoma**

Melioidosis

Definition

Melioidosis is an infectious disease of humans and animals caused by a gram-negative bacillus found in soil and water. It has both acute and chronic forms.

Description

Melioidosis, which is sometimes called *Pseudomonas pseudomallei* infection, is endemic (occurring naturally and consistently) in Southeast Asia, Australia, and parts of Africa. It was rare in the United States prior to recent immigration from Southeast Asia. Melioidosis is presently a public health concern because it is most common in **AIDS** patients and intravenous drug users.

Causes and symptoms

Melioidosis is caused by *Pseudomonas pseudomallei*, a bacillus that can cause disease in sheep, goats, pigs, horses, and other animals, as well as in humans. The organism enters the body through skin abrasions, **burns**, or **wounds** infected by contaminated soil; inhalation of dust; or by eating food contaminated with *P. pseudomallei*. Person-to-person transmission is unusual. Drug addicts acquire the disease from shared needles. The incubation period is two to three days.

Chronic melioidosis is characterized by **osteomyelitis** (inflammation of the bone) and pus-filled abscesses in the skin, lungs, or other organs. Acute melioidosis takes one of three forms: a localized skin infection that may spread to nearby lymph nodes; an infection of the lungs associated with high **fever** (102°F/38.9°C), **headache**, chest **pain**, and coughing;

KEY TERMS

Osteomyelitis—An inflammation of bone or bone marrow, often caused by bacterial infections. Chronic melioidosis may cause osteomyelitis.

Septicemia—Bacterial infection of the bloodstream. One form of melioidosis is an acute septicemic infection.

and septicemia (blood **poisoning**) characterized by disorientation, difficulty breathing, severe headache, and an eruption of pimples on the head or trunk. The third form is most common among drug addicts and may be rapidly fatal.

Diagnosis

Melioidosis is usually suspected based on the patient's history, especially travel, occupational exposure to infected animals, or a history of intravenous drug. Diagnosis must then be confirmed through laboratory tests. *P. pseudomallei* can be cultured from samples of the patient's sputum, blood, or tissue fluid from abscesses. Blood tests, including complement fixation (CF) tests and hemagglutination tests, also help to confirm the diagnosis. In acute infections, chest x rays and **liver function tests** are usually abnormal.

Treatment

Patients with mild or moderate infections are given a course of trimethoprim-sulfamethoxazole (TMP/SMX) and ceftazidime by mouth. Patients with acute melioidosis are given a lengthy course of ceftazidime followed by TMP/SMX. In patients with acute septicemia, a combination of **antibiotics** is administered intravenously, usually tetracycline, chloramphenicol, and TMP/SMX.

Prognosis

The mortality rate in acute cases of pulmonary melioidosis is about 10%; the mortality rate for the septicemic form is significantly higher (slightly above 50%). The prognosis for recovery from mild infections is excellent.

Prevention

There is no form of immunization for melioidosis. Prevention requires prompt cleansing of scrapes,

burns, or other open wounds in areas where the disease is common and avoidance of needle sharing among drug addicts.

Resources

BOOKS

Pollock, Matthew. "Infections Due to Pseudomonas Species and Related Organisms." In *Harrison's Principles of Internal Medicine*, edited by Anthony S. Fauci, et al. New York: McGraw-Hill, 1997.

Rebecca J. Frey, PhD

Membranous glomerulopathy see **Nephrotic syndrome**

Memory loss see **Amnesia**

Mènière's disease

Definition

Mènière's disease is a condition characterized by recurrent vertigo (**dizziness**), **hearing loss**, and **tinnitus** (a roaring, buzzing or ringing sound in the ears).

Description

Mènière's disease was named for the French physician Prosper Mènière, who first described the illness in 1861. It is an abnormality within the inner ear. A fluid called endolymph moves in the membranous labyrinth or semicircular canals within the bony labyrinth inside the inner ear. When the head or body moves, the endolymph moves, causing nerve receptors in the membranous labyrinth to send signals to the brain about the body's motion. A change in the volume of the endolymph fluid, or swelling or rupture of the membranous labyrinth, is thought to result in Mènière's disease symptoms.

Causes and symptoms

Causes

The cause of Mènière's disease is unknown as of 2002; however, scientists are studying several possible causes, including noise pollution, viral infections, or alterations in the patterns of blood flow in the structures of the inner ear. Since Mènière's disease sometimes runs in families, researchers are also looking into genetic factors as possible causes of the disorder.

One area of research that shows promise is the possible relationship between Ménère's disease and **migraine headache**. Dr. Ménère himself suggested the possibility of a link, but early studies yielded conflicting results. A rigorous German study published in late 2002 reported that the lifetime prevalence of migraine was 56% in patients diagnosed with Ménère's disease as compared to 25% for controls. The researchers noted that further work is necessary to determine the exact nature of the relationship between the two disorders.

A study published in late 2002 reported that there is a significant increase in the number of CD4 cells in the blood of patients having an acute attack of Ménère's disease. CD4 cells are a subtype of T cells, which are produced in the thymus gland and regulate the immune system's response to infected or malignant cells. Further research is needed to clarify the role of these cells in Ménère's disease.

Another possible factor in the development of Ménère's disease is the loss of myelin from the cells surrounding the vestibular nerve fibers. Myelin is a whitish fatty material in the cell membrane of the Schwann cells that form a sheath around certain nerve cells. It acts like an electrical insulator. A team of researchers at the University of Virginia reported in 2002 that the vestibular nerve cells in patients with unilateral Ménère's disease are demyelinated; that is, they have lost their protective "insulation." The researchers are investigating the possibility that a viral disease or disorder of the immune system is responsible for the demyelination of the vestibular nerve cells.

Symptoms

The symptoms of Ménère's disease are associated with a change in fluid volume within the labyrinth of the inner ear. Symptoms include severe dizziness or vertigo, tinnitus, hearing loss, and the sensation of **pain** or pressure in the affected ear. Symptoms appear suddenly, last up to several hours, and can occur as often as daily to as infrequently as once a year. A typical attack includes vertigo, tinnitus, and hearing loss; however, some individuals with Ménère's disease may experience a single symptom, like an occasional bout of slight dizziness or periodic, intense ringing in the ear. Attacks of severe vertigo can force the sufferer to have to sit or lie down, and may be accompanied by **headache, nausea, vomiting, or diarrhea**. Hearing tends to recover between attacks, but becomes progressively worse over time.

Ménère's disease usually starts between the ages of 20 and 50 years; however, it is not uncommon for

elderly people to develop the disease without a previous history of symptoms. Ménère's disease affects men and women in equal numbers. In most patients only one ear is affected but in about 15% both ears are involved.

Diagnosis

An estimated 3–5 million people in the United States have Ménère's disease, and almost 100,000 new cases are diagnosed each year. Diagnosis is based on medical history, **physical examination**, hearing and balance tests, and medical imaging with **magnetic resonance imaging (MRI)**.

Several types of tests may be used to diagnose the disease and to evaluate the extent of hearing loss. In patients with Ménère's disease, audiometric tests (hearing tests) usually indicate a sensory type of hearing loss in the affected ear. Speech discrimination or the ability to distinguish between words that sound alike is often diminished. In about 50% of patients, the balance function is reduced in the affected ear. An electronystagmograph (ENG) may be used to evaluate balance. Since the eyes and ears work together through the nervous system to coordinate balance, measurement of eye movements can be used to test the balance system. For this test, the patient is seated in a darkened room and recording electrodes, similar to those used with a heart monitor, are placed near the eyes. Warm and cool water or air are gently introduced into the each ear canal and eye movements are recorded.

Another test that may be used is an electrocochleograph (EcoG), which can measure increased inner ear fluid pressure.

Treatment

There is no cure for Ménère's disease, but medication, surgery, and dietary and behavioral changes, can help control or improve the symptoms.

Medications

Symptoms of Ménère's disease may be treated with a variety of oral medicine or through injections. **Antihistamines**, like diphenhydramine, meclizine, and cyclizine can be prescribed to sedate the vestibular system. A barbiturate medication like pentobarbital may be used to completely sedate the patient and relieve the vertigo. Anticholinergic drugs, like atropine or scopolamine, can help minimize **nausea and vomiting**. Diazepam has been found to be particularly effective for relief of vertigo and nausea in Ménère's

disease. There have been some reports of successful control of vertigo after **antibiotics** (gentamicin or streptomycin) or a steroid medication (dexamethasone) are injected directly into the inner ear. Some researchers have found that gentamicin is effective in relieving tinnitus as well as vertigo.

A newer medication that appears to be effective in treating the vertigo associated with Ménière's disease is flunarizine, which is sold under the trade name Sibelium. Flunarizine is a calcium channel blocker and anticonvulsant that is presently used to treat Parkinson's disease, migraine headache, and other circulatory disorders that affect the brain.

Surgical procedures

Surgical procedures may be recommended if the vertigo attacks are frequent, severe, or disabling and cannot be controlled by other treatments. The most common surgical treatment is insertion of a small tube or shunt to drain some of the fluid from the canal. This treatment usually preserves hearing and controls vertigo in about one-half to two-thirds of cases, but it is not a permanent cure in all patients.

The vestibular nerve leads from the inner ear to the brain and is responsible for conducting nerve impulses related to balance. A vestibular neurectomy is a procedure where this nerve is cut so the distorted impulses causing dizziness no longer reach the brain. This procedure permanently cures the majority of patients and hearing is preserved in most cases. There is a slight risk that hearing or facial muscle control will be affected.

A labyrinthectomy is a surgical procedure in which the balance and hearing mechanism in the inner ear are destroyed on one side. This procedure is considered when the patient has poor hearing in the affected ear. Labyrinthectomy results in the highest rates of control of vertigo attacks, however, it also causes complete deafness in the affected ear.

Alternative treatment

Changes in diet and behavior are sometimes recommended. Eliminating **caffeine**, alcohol, and salt may relieve the frequency and intensity of attacks in some people with Ménière's disease. Reducing **stress** levels and eliminating tobacco use may also help.

Acupuncture is an alternative treatment that has been shown to help patients with Ménière's disease. The World Health Organization (WHO) lists Ménière's disease as one of 104 conditions that can be treated effectively with acupuncture.

KEY TERMS

Myelin—A whitish fatty substance that acts like an electrical insulator around certain nerves in the peripheral nervous system. It is thought that the loss of the myelin surrounding the vestibular nerves may influence the development of Ménière's disease.

T cell—A type of white blood cell produced in the thymus gland that regulates the immune system's response to diseased or malignant cells. It is possible that a subcategory of T cells known as CD4 cells plays a role in Ménière's disease.

Tinnitus—A roaring, buzzing or ringing sound in the ears.

Transcutaneous electrical nerve stimulation (TENS)—A treatment in which a mild electrical current is passed through electrodes on the skin to stimulate nerves and block pain signals.

Vertigo—The medical term for dizziness or a spinning sensation.

Prognosis

Ménière's disease is a complex and unpredictable condition for which there is no cure. The vertigo associated with the disease can generally be managed or eliminated with medications and surgery. Hearing tends to become worse over time, and some of the surgical procedures recommended, in fact, cause deafness.

Prevention

Since the cause of Ménière's disease is unknown as of 2002, there are no current strategies for its prevention. Research continues on the environmental and biological factors that may cause Ménière's disease or induce an attack, as well as on the physiological components of the fluid and labyrinth system involved in hearing and balance. Preventive strategies and more effective treatment should become evident once these mechanisms are better understood.

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ORGANIZATIONS

- American Academy of Otolaryngology-Head and Neck Surgery, Inc. One Prince St., Alexandria VA 22314-3357. (703) 836-4444. <<http://www.entnet.org>>.
- Ménière's Network. 2000 Church St., P.O. Box 111, Nashville, TN 37236. (800) 545-4327. <<http://www.healthy.net/pan/cso/cioi/mn.htm>>.
- On-Balance, A Support Group for People with Ménière's Disease. <<http://www.midwestear.com/onbal.htm>>.
- Vestibular Disorders Association. P.O. Box 4467, Portland, OR 97208-4467. (800) 837-8428.

Altha Roberts Edgren
Rebecca J. Frey, PhD

Meningioma see **Brain tumor**

Meningitis

Definition

Meningitis is a serious inflammation of the meninges, the thin, membranous covering of the brain and the spinal cord. Meningitis is most commonly caused by infection (by bacteria, viruses, or fungi), although it can also be caused by bleeding into the meninges, **cancer**, diseases of the immune system, and an inflammatory response to certain types of **chemotherapy** or other chemical agents. The most serious and difficult-to-treat types of meningitis tend to be those caused by bacteria. In some cases, meningitis can be a potentially fatal condition.

Description

Meningitis is a particularly dangerous infection because of the very delicate nature of the brain. Brain cells are some of the only cells in the body that, once killed, will not regenerate themselves. Therefore, if enough brain tissue is damaged by an infection, serious, life-long handicaps will remain.

In order to learn about meningitis, it is important to have a basic understanding of the anatomy of the brain. The meninges are three separate membranes, layered together, which encase the brain and spinal cord:

- The dura is the toughest, outermost layer, and is closely attached to the inside of the skull.
- The middle layer, the arachnoid, is important because of its involvement in the normal flow of the cerebrospinal fluid (CSF), a lubricating and nutritive fluid that bathes both the brain and the spinal cord.
- The innermost layer, the pia, helps direct blood vessels into the brain.
- The space between the arachnoid and the pia contains CSF, which helps insulate the brain from trauma. Many blood vessels course through this space.

CSF, produced within specialized chambers deep inside the brain, flows over the surface of the brain and spinal cord. This fluid serves to cushion these relatively delicate structures, as well as supplying important nutrients for brain cells. CSF is reabsorbed by blood vessels located within the meninges. A careful balance between CSF production and reabsorption is important to avoid the accumulation of too much CSF.

Because the brain is enclosed in the hard, bony case of the skull, any disease that produces swelling

HATTIE ALEXANDER (1901–1968)



(Betmann/CORBIS. Reproduced by permission.)

Hattie Alexander, a dedicated pediatrician, medical educator, and researcher in microbiology, won international recognition for deriving a serum to combat influenzal meningitis, a common disease that previously had been nearly always fatal to infants and young children. Alexander subsequently investigated microbiological genetics and the processes whereby bacteria, through genetic mutation, acquire resistance to antibiotics. In 1964, as president of the American Pediatric Society, she

became one of the first women to head a national medical association.

As an intern at the Harriet Lane Home of Johns Hopkins Hospital from 1930 to 1931, Alexander became interested in influenzal meningitis. The source of the disease was *Hemophilus influenzae*, a bacteria that causes inflammation of the meninges, the membranes surrounding the brain and spinal cord. In 1931, Alexander began a second internship at the Babies Hospital of the Columbia-Presbyterian Medical Center in New York City. There, she witnessed first-hand the futility of medical efforts to save babies who had contracted influenzal meningitis.

Alexander's early research focused on deriving a serum (the liquid component of blood, in which antibodies are contained) that would be effective against influenzal meningitis. Serums derived from animals that have been exposed to a specific disease-producing bacterium often contain antibodies against the disease and can be developed for use in immunizing humans against it. Alexander knew that the Rockefeller Institute in New York City, however, had been able to prepare a rabbit serum for the treatment of pneumonia, another bacterial disease. Alexander therefore experimented with rabbit serums, and by 1939 was able to announce the development of a rabbit serum effective in curing infants of influenzal meningitis.

In the early 1940s, Alexander experimented with the use of drugs in combination with rabbit serum in the treatment of influenzal meningitis. Within the next two years, she saw infant deaths due to the disease drop by eighty percent.

will be damaging to the brain. The skull cannot expand at all, so when the swollen brain tissue pushes up against the skull's hard bone, the brain tissue becomes damaged and may ultimately die. Furthermore, swelling on the right side of the brain will not only cause pressure and damage to that side of the brain, but by taking up precious space within the tight confines of the skull, the left side of the brain will also be pushed up against the hard surface of the skull, causing damage to the left side of the brain as well.

Another way that infections injure the brain involves the way in which the chemical environment of the brain changes in response to the presence of an infection. The cells of the brain require a very well-regulated environment. Careful balance of oxygen, carbon dioxide, sugar (glucose), sodium, calcium, potassium, and other substances must be maintained in order

to avoid damage to brain tissue. An infection upsets this balance, and brain damage can occur when the cells of the brain are either deprived of important nutrients or exposed to toxic levels of particular substances.

The cells lining the brain's tiny blood vessels (capillaries) are specifically designed to prevent many substances from passing into brain tissue. This is commonly referred to as the blood-brain barrier. The blood-brain barrier prevents various substances that could be poisonous to brain tissue (toxins), as well as many agents of infection, from crossing from the blood stream into the brain tissue. While this barrier is obviously an important protective feature for the brain, it also serves to complicate treatment in the case of an infection by making it difficult for medications to pass out of the blood and into the brain tissue where the infection is located.

Causes and symptoms

The most common infectious causes of meningitis vary according to an individual's age, habits, living environment, and health status. While nonbacterial types of meningitis are more common, bacterial meningitis is the more potentially life-threatening. Three bacterial agents are responsible for about 80% of all bacterial meningitis cases. These bacteria are *Haemophilus influenzae* type b, *Neisseria meningitidis* (causing meningococcal meningitis), and *Streptococcus pneumoniae* (causing pneumococcal meningitis).

In newborns, the most common agents of meningitis are those that are contracted from the newborn's mother, including Group B streptococci (becoming an increasingly common infecting organism in the newborn period), *Escherichia coli*, and *Listeria monocytogenes*. The highest incidence of meningitis occurs in babies under a month old, with an increased risk of meningitis continuing through about two years of age.

Older children are more frequently infected by the bacteria *Haemophilus influenzae*, *Neisseria meningitidis*, and *Streptococci pneumoniae*.

Adults are most commonly infected by either *S. pneumoniae* or *N. meningitidis*, with pneumococcal meningitis the more common. Certain conditions predispose to this type of meningitis, including **alcoholism** and chronic upper respiratory tract infections (especially of the middle ear, sinuses, and mastoids).

N. meningitidis is the only organism that can cause epidemics of meningitis. In particular, these have occurred when a child in a crowded day-care situation or a military recruit in a crowded training camp has fallen ill with meningococcal meningitis.

Viral causes of meningitis include the herpes simplex virus, the **mumps** and **measles** viruses (against which most children are protected due to mass immunization programs), the virus that causes chicken pox, the **rabies** virus, and a number of viruses that are acquired through the bites of infected mosquitoes.

A number of medical conditions predispose individuals to meningitis caused by specific organisms. Patients with **AIDS** (acquired immune deficiency syndrome) are more prone to getting meningitis from fungi, as well as from the agent that causes **tuberculosis**. Patients who have had their spleens removed, or whose spleens are no longer functional (as in the case of patients with **sickle cell disease**) are more susceptible to other infections, including meningococcal and pneumococcal meningitis.

The majority of meningitis infections are acquired by blood-borne spread. A person may have another

type of infection (of the lungs, throat, or tissues of the heart) caused by an organism that can also cause meningitis. If this initial infection is not properly treated, the organism will continue to multiply, find its way into the blood stream, and be delivered in sufficient quantities to invade past the blood brain barrier. Direct spread occurs when an organism spreads to the meninges from infected tissue next to or very near the meninges. This can occur, for example, with a severe, poorly treated ear or sinus infection.

Patients who suffer from skull **fractures** possess abnormal openings to the sinuses, nasal passages, and middle ears. Organisms that usually live in the human respiratory system without causing disease can pass through openings caused by such fractures, reach the meninges, and cause infection. Similarly, patients who undergo surgical procedures or who have had foreign bodies surgically placed within their skulls (such as tubes to drain abnormal amounts of accumulated CSF) have an increased risk of meningitis.

Organisms can also reach the meninges via an uncommon but interesting method called intraneural spread. This involves an organism invading the body at a considerable distance away from the head, spreading along a nerve, and using that nerve as a kind of ladder into the skull, where the organism can multiply and cause meningitis. Herpes simplex virus is known to use this type of spread, as is the rabies virus.

The most classic symptoms of meningitis (particularly of bacterial meningitis) include **fever**, **headache**, **vomiting**, sensitivity to light (photophobia), irritability, severe **fatigue** (lethargy), stiff neck, and a reddish purple rash on the skin. Untreated, the disease progresses with seizures, confusion, and eventually **coma**.

A very young infant may not show the classic signs of meningitis. Early in infancy, a baby's immune system is not yet developed enough to mount a fever in response to infection, so fever may be absent. Some infants with meningitis have seizures as their only identifiable symptom. Similarly, debilitated elderly patients may not have fever or other identifiable symptoms of meningitis.

Damage due to meningitis occurs from a variety of phenomena. The action of infectious agents on the brain tissue is one direct cause of damage. Other types of damage may be due to the mechanical effects of swelling and compression of brain tissue against the bony surface of the skull. Swelling of the meninges may interfere with the normal absorption of CSF by blood vessels, causing accumulation of CSF and damage from the resulting pressure on the brain.

Interference with the brain's carefully regulated chemical environment may cause damaging amounts of normally present substances (carbon dioxide, potassium) to accumulate. Inflammation may cause the blood-brain barrier to become less effective at preventing the passage of toxic substances into brain tissue.

Diagnosis

A number of techniques are used when examining a patient suspected of having meningitis to verify the diagnosis. Certain manipulations of the head (lowering the head, chin towards chest, for example) are difficult to perform and painful for a patient with meningitis.

The most important test used to diagnose meningitis is the lumbar puncture (commonly called a spinal tap). Lumbar puncture (LP) involves the insertion of a thin needle into a space between the vertebrae in the lower back and the withdrawal of a small amount of CSF. The CSF is then examined under a microscope to look for bacteria or fungi. Normal CSF contains set percentages of glucose and protein. These percentages will vary with bacterial, viral, or other causes of meningitis. For example, bacterial meningitis causes a greatly lower than normal percentage of glucose to be present in CSF, as the bacteria are essentially "eating" the host's glucose, and using it for their own **nutrition** and energy production. Normal CSF should contain no infection-fighting cells (white blood cells), so the presence of white blood cells in CSF is another indication of meningitis. Some of the withdrawn CSF is also put into special lab dishes to allow growth of the infecting organism, which can then be identified more easily. Special immunologic and serologic tests may also be used to help identify the infectious agent.

In rare instances, CSF from a lumbar puncture cannot be examined because the amount of swelling within the skull is so great that the pressure within the skull (intracranial pressure) is extremely high. This pressure is always measured immediately upon insertion of the LP needle. If it is found to be very high, no fluid is withdrawn because doing so could cause herniation of the brain stem. Herniation of the brain stem occurs when the part of the brain connecting to the spinal cord is thrust through the opening at the base of the skull into the spinal canal. Such herniation will cause compression of those structures within the brain stem that control the most vital functions of the body (breathing, heart beat, consciousness). **Death** or permanent debilitation follows herniation of the brain stem.

KEY TERMS

Blood-brain barrier—An arrangement of cells within the blood vessels of the brain that prevents the passage of toxic substances, including infectious agents, from the blood and into the brain. It also makes it difficult for certain medications to pass into brain tissue.

Cerebrospinal fluid (CSF)—Fluid made in chambers within the brain which then flows over the surface of the brain and spinal cord. CSF provides nutrition to cells of the nervous system, as well as providing a cushion for the nervous system structures. It may accumulate abnormally in some disease processes, causing pressure on and damage to brain structures.

Lumbar puncture (LP)—A medical test in which a very narrow needle is inserted into a specific space between the vertebrae of the lower back in order to draw off a sample of CSF for further examination.

Meninges—The three-layer membranous covering of the brain and spinal cord, composed of the dura, arachnoid, and pia. It provides protection for the brain and spinal cord, as well as housing many blood vessels and participating in the appropriate flow of CSF.

Treatment

Antibiotic medications (forms of penicillin and **cephalosporins**, for example) are the most important element of treatment against bacterial agents of meningitis. Because of the effectiveness of the blood-brain barrier in preventing the passage of substances into the brain, medications must be delivered directly into the patient's veins (intravenously, or by IV), at very high doses. **Antiviral drugs** (acyclovir) may be helpful in shortening the course of viral meningitis, and antifungal medications are available as well.

Other treatments for meningitis involve decreasing inflammation (with steroid preparations) and paying careful attention to the balance of fluids, glucose, sodium, potassium, oxygen, and carbon dioxide in the patient's system. Patients who develop seizures will require medications to halt the seizures and prevent their return.

Prognosis

Viral meningitis is the least severe type of meningitis, and patients usually recover with no long-term effects from the infection. Bacterial infections,

however, are much more severe, and progress rapidly. Without very rapid treatment with the appropriate antibiotic, the infection can swiftly lead to coma and death in less than a day's time. While death rates from meningitis vary depending on the specific infecting organism, the overall death rate is just under 20%.

The most frequent long-term effects of meningitis include deafness and blindness, which may be caused by the compression of specific nerves and brain areas responsible for the senses of hearing and sight. Some patients develop permanent seizure disorders, requiring life-long treatment with anti-seizure medications. Scarring of the meninges may result in obstruction of the normal flow of CSF, causing abnormal accumulation of CSF. This may be a chronic problem for some patients, requiring the installation of shunt tubes to drain the accumulation regularly.

Prevention

Prevention of meningitis primarily involves the appropriate treatment of other infections an individual may acquire, particularly those that have a track record of seeding to the meninges (such as ear and sinus infections). Preventive treatment with **antibiotics** is sometimes recommended for the close contacts of an individual who is ill with meningococcal or *H. influenzae* type b meningitis. A meningococcal vaccine exists, and is sometimes recommended to individuals who are traveling to very high risk areas. A vaccine for *H. influenzae* type b is now given to babies as part of the standard array of childhood immunizations.

Resources

ORGANIZATIONS

American Academy of Neurology. 1080 Montreal Ave., St. Paul, MN 55116. (612) 695-1940. <<http://www.aan.com>>.

Meningitis Foundation of America. 7155 Shadeland Station, Suite 190, Indianapolis, IN 46256-3922. (800) 668-1129. <<http://www.musa.org/welcome.htm>>.

Rosalyn Carson-DeWitt, MD

Meningocele see **Spina bifida**

Meningococemia

Definition

Meningococemia is the presence of meningococcus in the bloodstream. Meningococcus, a bacteria

formally called *Neisseria meningitidis*, can be one of the most dramatic and rapidly fatal of all infectious diseases.

Causes and symptoms

Meningococemia, a relatively uncommon infection, occurs most commonly in children and young adults. In susceptible people, it may cause a very severe illness that can produce **death** within hours. The bacteria, which can spread from person to person, usually first causes a colonization in the upper airway, but without symptoms. From there, it can penetrate into the bloodstream to the central nervous system and cause **meningitis** or develop into a full-blown bloodstream infection (meningococemia). Fortunately in most colonized people, this does not happen and the result of this colonization is long-lasting immunity against the particular strain.

After colonization is established, symptoms can develop within one day to one to two weeks. After a short period of time (one hour up to one to two days) when the patient complains of **fever** and muscle aches, more severe symptoms can develop. Unfortunately during this early stage, a doctor cannot tell this illness from any other illness, such as a viral infection like **influenza**. Unless the case is occurring in a person known to have been exposed to or in the midst of an epidemic of meningococcal disease, there may be no specific symptoms or signs found that help the doctor diagnose the problem. Rarely, a low-grade bloodstream infection called chronic meningococemia can occur.

After this initial period, the patient will often complain of continued fever, shaking chills, overwhelming weakness, and even a feeling of impending doom. The organism is multiplying in the bloodstream, unchecked by the immune system. The severity of the illness and its dire complications are caused by the damage the organism does to the small blood vessel walls. This damage is called a **vasculitis**, an inflammation of a blood vessel. Damage to the small vessels causes them to become leaky. The first signs of the infection's severity are small bleeding spots seen on the skin (petechiae). A doctor should always suspect meningococemia when he/she finds an acutely ill patient with fever, chills, and petechiae.

Quickly (within hours), the blood vessel damage increases and large bleeding areas on the skin (purpura) are seen. The same changes are taking place in the affected person's internal organs. The blood pressure is often low and there may be signs of bleeding from other organs (like coughing up



A close-up image of a person's hand with meningococcemia. This disease is caused by the presence of meningococcus (*Neisseria meningitidis*) in the bloodstream. The organism can cause multiple illnesses and can damage small blood vessels. (Custom Medical Stock Photo. Reproduced by permission.)

blood, nose bleeds, blood in the urine). The organism not only damages the blood vessels by causing them to leak, but also causes clotting inside the vessels. If this clotting occurs in the larger arteries, it results in major tissue damage. Essentially, large areas of skin, muscle, and internal organs die from lack of blood and oxygen. Even if the disease is quickly diagnosed and treated, the patient has a high risk of dying.

Diagnosis

The diagnosis of meningococcemia can be made by the growth of the organism from blood cultures. Treatment should begin when the diagnosis is suspected and should not be delayed waiting for positive cultures. Obtaining fluid from a petechial spot and staining it in the laboratory can assist in quickly seeing the organism.

Treatment

Immediate treatment of a suspected case of meningococcemia begins with **antibiotics** that work against the organism. Possible choices include penicillin G, ceftriaxone (Rocephin), cefotaxime (Claforan), or trimethoprim/sulfamethoxazole (Bactrim, Septra). If the patient is diagnosed in a doctor's office, antibiotics should be given immediately if possible, even before transfer to the hospital and even if cultures cannot be obtained before treatment. It is most likely that the speed of initial treatment will affect the ultimate outcome.

KEY TERMS

Blood culture—A procedure where blood is collected from a vein and is placed in a small bottle that contains a special liquid; the liquid will make any organisms that are present in the blood sample grow. These organisms can then be grown and identified in the laboratory so that the proper antibiotic can be given to the patient.

Colonization—The presence of bacteria on a body surface (like on the skin, mouth, intestines or airway) without causing disease in the person.

Complement—One of several proteins in the blood that acts with other proteins to assist in killing bacteria.

Meningitis—Inflammation of the membranes of the brain or spinal cord.

Prognosis

As many as 15-20% of patients with meningococcemia will die as a result of the acute infection. A significant percentage of the survivors will have tissue damage that requires surgical treatment. This treatment may consist of skin grafts, or even partial or full amputations of an arm or leg. Certain people with immune system defects (particularly those with defects in the complement system) may have recurrent episodes of meningococcemia. These patients, however, seem to have a less serious outcome.

Prevention

Although a vaccine is available for meningococcus, it is still difficult at this time to produce a vaccine for the type B organism, the most common one in the United States. Because of this and the short time that the vaccine seems to offer protection, the product has not been routinely used in the United States. It can be used for travelers going to areas where meningococcal disease is more common or is epidemic. Recently, the vaccine has been suggested for use in incoming college freshman, particularly those living in dormitories. These students appear to have a somewhat higher risk of meningococcal infections.

It is, however, recommended that all people take certain antibiotics if they have had contact (like at home or in a daycare) with a person who has meningococcal infection. The most common antibiotics given are rifampin (Rifadin) or ciprofloxacin (Cipro). These medicines are usually taken by mouth twice a

day for two days. This treatment will decrease the risk of infection in these people who have been exposed. However, the overall risk to people who have been exposed, even without antibiotic use, is probably no more than 1-2%.

Resources

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Larry I. Lutwick, MD

Meningomyelocele see **Spina bifida**

Menkes' syndrome see **Mineral deficiency**

Menopause

Definition

Menopause represents the end of menstruation. While technically it refers to the final period, it is not an abrupt event, but a gradual process. Menopause is not a disease that needs to be cured, but a natural life-stage transition. However, women have to make important decisions about “treatment,” including the use of **hormone replacement therapy** (HRT).

Description

Many women have irregular periods and other problems of “pre-menopause” for years. It is not easy to predict when menopause begins, although doctors agree it is complete when a woman has not had a period for a year. Eight out of every 100 women stop menstruating before age 40. At the other end of the spectrum, five out of every 100 continue to have periods until they are almost 60. The average age of menopause is 51.

There is no mathematical formula to figure out when the ovaries will begin to scale back either, but a woman can get a general idea based on her family history, body type, and lifestyle. Women who began menstruating early will not necessarily stop having periods early as well. It is true that a woman will likely enter menopause at about the same age as her mother. Menopause may occur later than average among smokers.

Causes and symptoms

Once a woman enters **puberty**, each month her body releases one of the more than 400,000 eggs that are stored in her ovaries, and the lining of the womb (uterus) thickens in anticipation of receiving a fertilized egg. If the egg is not fertilized, progesterone levels drop and the uterine lining sheds and bleeds.

By the time a woman reaches her late 30s or 40s, her ovaries begin to shut down, producing less estrogen and progesterone and releasing eggs less often. The gradual decline of estrogen causes a wide variety of changes in tissues that respond to estrogen—including the vagina, vulva, uterus, bladder, urethra, breasts, bones, heart, blood vessels, brain, skin, hair, and mucous membranes. Over the long run, the lack of estrogen can make a woman more vulnerable to **osteoporosis** (which can begin in the 40s) and heart disease.

As the levels of hormones fluctuate, the menstrual cycle begins to change. Some women may have longer periods with heavy flow followed by shorter cycles and hardly any bleeding. Others will begin to miss periods completely. During this time, a woman also becomes less able to get pregnant.

The most common symptom of menopause is a change in the menstrual cycle, but there are a variety of other symptoms as well, including:

- hot flashes
- night sweats
- **insomnia**
- mood swings/irritability
- memory or concentration problems
- vaginal dryness
- heavy bleeding
- **fatigue**
- depression
- hair changes
- headaches
- heart **palpitations**
- sexual disinterest
- urinary changes
- weight gain

Diagnosis

The clearest indication of menopause is the absence of a period for one year. It is also possible to diagnose menopause by testing hormone levels. One

important test measures the levels of follicle-stimulating hormone (FSH), which steadily increases as a woman ages.

However, as a woman first enters menopause, her hormones often fluctuate wildly from day to day. For example, if a woman's estrogen levels are high and progesterone is low, she may have mood swings, irritability, and other symptoms similar to **premenstrual syndrome** (PMS). As hormone levels shift and estrogen level falls, hot flashes occur. Because of these fluctuations, a normal hormone level when the blood is tested may not necessarily mean the levels were normal the day before or will be the day after.

If it has been at least three months since a woman's last period, an FSH test might be more helpful in determining whether menopause has occurred. Most doctors believe that the FSH test alone cannot be used as proof that a woman has entered early menopause. A better measure of menopause is a test that checks the levels of estrogen, progesterone, testosterone and other hormones at mid-cycle, in addition to FSH.

Treatment

When a woman enters menopause, her levels of estrogen drop and symptoms (such as hot flashes and vaginal dryness) begin. Hormone replacement therapy can treat these symptoms by boosting the estrogen levels enough to suppress symptoms while also providing protection against heart disease and osteoporosis, which causes the bones to weaken. Experts disagree on whether HRT increases or decreases the risk of developing **breast cancer**. A Harvard study concluded that short-term use of hormones carries little risk, while HRT used for more than five years among women 55 and over seems to increase the risk of breast **cancer**.

There are two types of hormone treatments: hormone replacement therapy (HRT) and estrogen replacement therapy (ERT). HRT is the administration of estrogen and progesterone; ERT is the administration of estrogen alone. Only women who have had a **hysterectomy** (removal of the uterus) can take estrogen alone, since taking this "unopposed" estrogen can cause uterine cancer. The combination of progesterone and estrogen in HRT eliminates the risk of uterine cancer.

Most physicians do not recommend HRT until a woman's periods have stopped completely for one year. This is because women in early menopause who still have an occasional period are still producing estrogen; HRT would then provide far too much estrogen.

Most doctors believe that every woman (except those with certain cancers) should take hormones as they approach menopause because of the protection against heart disease, osteoporosis, and uterine cancer and the relatively low risk of breast cancer. Heart disease and osteoporosis are two of the leading causes of disability and **death** among post-menopausal women.

Critics say the benefit of taking hormonal drugs to ease symptoms is not worth the risk of breast cancer. Since menopause is not a disease, many argue that women should not take hormones to cure what is actually a natural process of **aging**. Advocates of HRT contend that the purpose of taking hormones is not to "treat" menopause but to prevent the development of other diseases.

There are risks with HRT and there are risks without it. In order to decide whether to take HRT, a woman should balance her risk of getting breast cancer against her risk of getting heart disease, and decide how bad her menopause symptoms are. Most doctors agree that short-term use of estrogen for those women with symptoms of hot flashes or night sweats is a sensible choice as long as they do not have a history of breast cancer.

For a woman who has no family history of cancer and a high risk of dying from heart disease, for example, the low risk of cancer might be worth the protective benefit of avoiding heart disease. Certainly, for Caucasian women aged 50 to 94, the risk of dying from heart disease is far greater than the risk of dying of breast cancer.

Women are poor candidates for hormone replacement therapy if they have:

- had breast or **endometrial cancer**
- a close relative (mother, sister, grandmother) who died of breast cancer or have two relatives who developed breast cancer before age 40
- had endometrial cancer
- had gallbladder or **liver disease**
- blood clots or phlebitis

Some women with liver or gallbladder disease, or who have clotting problems, may be able to go on HRT if they use a patch to administer the hormones through the skin, bypassing the liver.

Women would make a good candidate for HRT if they:

- need to prevent osteoporosis
- have had their ovaries removed

- need to prevent heart disease
- have significant symptoms.

Taking hormones can almost immediately eliminate hot flashes, vaginal dryness, **urinary incontinence** (depending on the cause), insomnia, moodiness, memory problems, heavy irregular periods, and concentration problems. Side effects of treatment include bloating, breakthrough bleeding, headaches, vaginal discharge, fluid retention, swollen breasts, or **nausea**. Up to 20% of women who try hormone replacement stop within nine months because of these side effects. However, some side effects can be lessened or prevented by changing the HRT regimen.

The decision should be made by a woman and her doctor after taking into consideration her medical history and situation. Women who choose to take hormones should have an annual mammogram, breast exam, and **pelvic exam** and should report any unusual vaginal bleeding or spotting (a sign of possible uterine cancer).

Anti-estrogens

A new type of hormone therapy offers some of the same protection against heart disease and bone loss as estrogen, but without the increased risk of breast cancer. This new class of drugs are known as anti-estrogens. The best known of these anti-estrogens is raloxifene, which mimics the effects of estrogen in the bones and blood, but blocks some of its negative effects elsewhere. It is called an anti-estrogen because for a long time these drugs had been used to counter the harmful effects of estrogen that caused breast cancer. Oddly enough, in other parts of the body these drugs mimic estrogen, protecting against heart disease and osteoporosis without putting a woman at risk for breast cancer.

Like estrogen, raloxifene works by attaching to an estrogen “receptor,” much like a key fits into a lock. When raloxifene clicks into the estrogen receptors in the breast and uterus, it blocks estrogen at these sites. This is the secret of its cancer-fighting property. Many tumors in the breast are fueled by estrogen; if the estrogen cannot get in the cell, then the cancer stops growing.

Women may prefer to take raloxifene instead of hormone replacement because the new drug does not boost the breast cancer risk and does not have side effects like uterine bleeding, bloating, or breast soreness. Unfortunately, the drug may worsen hot flashes. Raloxifene is basically a treatment to prevent osteoporosis. It does not help with common symptoms and

it is unclear if it has the same protective effect against heart disease as estrogen does.

Testosterone replacement

The ovaries also produce a small amount of male hormones, which decreases slightly as a woman enters menopause. The vast majority of women never need testosterone replacement, but it can be important if a woman has declining interest in sex. Testosterone can improve the libido, and decrease **anxiety** and depression; adding testosterone especially helps women who have had hysterectomies. Testosterone also eases breast tenderness and helps prevent bone loss. However, testosterone does have side effects. Some women experience mild **acne** and some facial hair growth, but because only small amounts of testosterone are prescribed, most women do not appear to have extreme masculine changes.

Birth control pills

Women who are still having periods but who have annoying menopausal symptoms may take low-dose birth control pills to ease the problems; this treatment has been approved by the FDA for perimenopausal symptoms in women under age 55. HRT is the preferred treatment for menopause, however, because it uses lower doses of estrogen.

Alternative treatment

Some women also report success in using natural remedies to treat the unpleasant symptoms of menopause. Not all women need estrogen and some women cannot take it. Many doctors don't want to give hormones to women who are still having their periods, however erratically. Indeed, only a third of menopausal women in the United States try HRT and of those who do, eventually half of them drop the therapy. Some are worried about breast cancer, some cannot tolerate the side effects, some do not want to medicate what they consider to be a natural occurrence.

Herbs

Herbs have been used to relieve menopausal symptoms for centuries. In general, most herbs are considered safe, and there is no substantial evidence that herbal products are a major source of toxic reactions. But because herbal products are not regulated in the United States, contamination or accidental overdose is possible. Herbs should be bought from a recognized company or through a qualified herbal practitioner.

Women who choose to take herbs for menopausal symptoms should learn as much as possible about herbs and work with a qualified practitioner (an herbalist, a traditional Chinese doctor, or a naturopathic physician). Pregnant women should avoid herbs because of unknown effects on a developing fetus.

The following list of herbs include those that herbalists most often prescribe to treat menstrual complaints:

- Black cohosh (*Cimicifuga racemosa*): hot flashes and other menstrual complaints
- Black currant: breast tenderness
- Chaste tree/chasteberry (*Vitex agnus-castus*): hot flashes, excessive menstrual bleeding, fibroids, and moodiness
- Evening primrose oil (*Oenothera biennis*): mood swings, irritability, and breast tenderness
- Fennel (*Foeniculum vulgare*): hot flashes, digestive gas, and bloating
- Flaxseed (linseed): excessive menstrual bleeding, breast tenderness, and other symptoms, including dry skin and vaginal dryness
- Gingko (*Ginkgo biloba*): memory problems
- Ginseng (*Panax ginseng*): hot flashes, fatigue and vaginal thinning.
- Hawthorn (*Crataegus laevigata*): memory problems, fuzzy thinking
- Lady's mantle: excessive menstrual bleeding
- Mexican wild yam (*Dioscorea villosa*) root: vaginal dryness, hot flashes and general menopause symptoms
- Motherwort (*Leonurus cardiaca*): night sweats, hot flashes
- Oat (*Avena sativa*) straw: mood swings, anxiety
- Red clover (*Trifolium pratense*): hot flashes
- Sage (*Salvia officinalis*): mood swings, headaches, night sweats
- Valerian (*Valeriana officinalis*): insomnia.

Natural estrogens (phytoestrogens)

Proponents of plant estrogens (including soy products) believe that plant estrogens are better than synthetic estrogen, but science has not yet proven this. The results of smaller preliminary trials suggest that the estrogen compounds in soy products can indeed relieve the severity of hot flashes and lower cholesterol. But no one yet has proven that soy can

provide all the benefits of synthetic estrogen without its negative effects.

It is true that people in other countries who eat foods high in plant estrogens (especially soy products) have lower rates of breast cancer and report fewer “symptoms” of menopause. While up to 80% of menopausal women in the United States complain of hot flashes, night sweats, and vaginal dryness, only 15% of Japanese women have similar complaints. When all other things are equal, a soy-based diet may make a difference (and soy is very high in plant estrogens).

The study of phytoestrogens is so new that there are not very many recommendations on how much a woman can consume. Herbal practitioners recommend a dose based on a woman's history, body size, lifestyle, diet, and reported symptoms. Research has indicated that some women were able to ease their symptoms by eating a large amount of fruits, vegetables, and whole grains, together with four ounces of tofu four times a week.

What concerns some critics of other alternative remedies is that many women think that “natural” or “plant-based” means “harmless.” In large doses, phytoestrogens can promote the abnormal growth of cells in the uterine lining. Unopposed estrogen of any type can lead to endometrial cancer, which is why women on conventional estrogen-replacement therapy usually take progesterone (progestin) along with their estrogen. However, a plant-based progesterone product can sometimes be effective alone, without estrogen, in assisting the menopausal woman in rebalancing her hormonal action throughout this transition time.

Yoga

Many women find that **yoga** (the ancient meditation/exercise developed in India 5,000 years ago) can ease menopausal symptoms. Yoga focuses on helping women unite the mind, body, and spirit to create balance. Because yoga has been shown to balance the endocrine system, some experts believe it may affect hormone-related problems. Studies have found that yoga can reduce **stress**, improve mood, boost a sluggish metabolism, and slow the heart rate. Specific yoga positions deal with particular problems, such as hot flashes, mood swings, vaginal and urinary problems, and other pains.

Exercise

Exercise helps ease hot flashes by lowering the amount of circulating FSH and LH and by raising endorphin levels that drop while having a hot flash. Even exercising 20 minutes three times a week can significantly reduce hot flashes.

KEY TERMS

Endometrium—The lining of the uterus that is shed with each menstrual period.

Estrogen—Female hormone produced by the ovaries and released by the follicles as they mature. Responsible for female sexual characteristics, estrogen stimulates and triggers a response from at least 300 tissues, and may help some types of breast cancer to grow. After menopause, the production of the hormone gradually stops.

Estrogen replacement therapy (ERT)—A treatment for menopause in which estrogen is given in pill, patch, or cream form.

Follicle-stimulating hormone (FSH)—The pituitary hormone that stimulates the ovary to mature egg capsules (follicles). It is linked with rising estrogen production throughout the cycle. An elevated FSH (above 40) indicates menopause.

Hormone—A chemical messenger secreted by a gland that is released into the blood, and that travels to distant cells where it exerts an effect.

Hormone replacement therapy (HRT)—The use of estrogen and progesterone to replace hormones that the ovary no longer supplies.

Hot flash—A wave of heat that is one of the most common perimenopausal symptoms, triggered by the hypothalamus' response to estrogen withdrawal.

Hysterectomy—Surgical removal of the uterus.

Ovary—One of the two almond-shaped glands in the female reproductive system responsible for producing eggs and the hormones estrogen and progesterone.

Ovulation—The monthly release of an egg from the ovary.

Pituitary gland—The “master gland” at the base of the brain that secretes a number of hormones responsible for growth, reproduction, and other activities. Pituitary hormones stimulate the ovaries to release estrogen and progesterone.

Progesterone—The hormone that is produced by the ovary after ovulation to prepare the uterine lining for a fertilized egg.

Testosterone—Male hormone produced by the testes and (in small amounts) in the ovaries. Testosterone is responsible for some masculine secondary sex characteristics such as growth of body hair and deepening voice.

Uterus—The female reproductive organ that contains and nourishes a fetus from implantation until birth. Also known as the womb.

Vagina—The tube-like passage from the vulva (a woman's external genital structures) to the cervix (the portion of the uterus that projects into the vagina).

Elimination

Regular, daily bowel movements to eliminate waste products from the body can be crucial in maintaining balance through menopause. The bowels are where circulating hormones are gathered and eliminated, keeping the body from recycling them and causing an imbalance.

Acupuncture

This ancient Asian art involves placing very thin needles into different parts of the body to stimulate the system and unblock energy. It is usually painless and has been used for many menopausal symptoms, including insomnia, hot flashes, and irregular periods. Practitioners believe that **acupuncture** can facilitate the opening of blocked energy channels, allowing the life force energy (chi) to flow freely. This allows the menopausal woman to keep her energy moving. Blocked energy usually increases the symptoms of menopause.

Acupressure and massage

Therapeutic massage involving **acupressure** can bring relief from a wide range of menopause symptoms by placing finger pressure at the same meridian points on the body that are used in acupuncture. There are more than 80 different types of massage, including foot **reflexology**, **Shiatsu** massage, or Swedish massage, but they are all based on the idea that boosting the circulation of blood and lymph benefits health.

Biofeedback

Some women have been able to control hot flashes through **biofeedback**, a painless technique that helps a person train her mind to control her body. A biofeedback machine provides information about body processes (such as heart rate) as the woman relaxes her body. Using this technique, it is possible to control the body's temperature, heart rate, and breathing.

Prognosis

Menopause is a natural condition of aging. Some women have no problems at all with menopause, while others notice significant unpleasant symptoms. A wide array of treatments, from natural to hormone replacement, mean that no woman needs to suffer through this time of her life.

Prevention

Menopause is a natural part of the aging process and not a disease that needs to be prevented. Most doctors recommend HRT for almost all post-menopausal women, usually for a few years. When HRT is then stopped, symptoms should be mild or non-existent. But HRT is not only useful in lessening the symptoms of menopause; it also protects against heart disease and osteoporosis.

Resources

BOOKS

Goldman, Lee, et al., editors. *Cecil Textbook of Medicine*. 21st ed. W. B. Saunders, 2000.

Goroll, Allan H., et al. *Primary Care Medicine*. 4th ed. Lippincott Williams & Wilkins, 2000.

OTHER

Menopause Online Page. <<http://www.menopause-online.com/links.htm>>.

Menopause Page. <<http://www.howdyneighbor.com/menopaus>>.

Meno Times Online. <<http://www.aimnet.com/~hyperion/meno/menotimes.index.html>>.

Laith Farid Gulli, M.D.

Menorrhagia see **Dysfunctional uterine bleeding**

Men's health

Definition

Men's health is concerned with identifying, preventing, and treating conditions that are most common or specific to men.

Purpose

Men live on average seven years less than women; life expectancy in the United States is 72 years for men and 79 years for women. The reasons for this

discrepancy are not completely understood. Men may have some genetic predisposition for lower life expectancy, as women tend to outlive men in most areas throughout the world. But men also have different lifestyle patterns that increase the wear and tear on their bodies. Studies have shown that men tend to drink and smoke more than women, men obtain medical care less frequently than women, and men generally have more stressful habits. It is clear to health professionals that men can benefit from increased knowledge of male medical issues and by understanding how lifestyle choices impact health.

According to the Centers for Disease Control (CDC), the 10 leading causes of **death** for men in the United States are:

- heart disease
- **cancer**
- **stroke**
- accidents
- lung disease (including **emphysema** and chronic bronchitis)
- pneumonia
- diabetes
- **suicide**
- liver disease
- homicides

Men also suffer regularly from conditions as diverse as **sexually transmitted diseases** (STDs), mental illness, arthritis, urinary tract infections, athletic injuries, hair and skin problems, and digestive disorders. The field of men's health strives to reduce the risks and incidence of men's conditions by researching preventive practices, designing testing procedures for early detection, and recommending specialized courses of treatment.

Description

Prevention

Preventive practices for men's health emphasize diet, **exercise** and **stress** management, as well as the elimination of risky behaviors such as **smoking** and excessive drinking. Four of the leading causes of death for American men are related to diet—heart disease, cancer, stroke, and diabetes. In addition men are more likely than women to suffer from diet-related conditions including **high cholesterol**, high blood pressure, and **obesity**, all of which increase the risk of certain diseases and premature death.

For American men, dietary problems are usually not the result of getting too little nourishment but of eating too much fat, sugar, and overall calories. The dietary change most likely to improve the health of males is reduced intake of fats, particularly cholesterol and saturated fats. Cholesterol and saturated fats are found mainly in meat and dairy products. Calories from fat should amount to no more than 30% of total daily calories. Eating adequate protein is generally not a problem for American men, so replacing some dairy and meat consumption with high fiber vegetable proteins such as beans and soy would be beneficial. Complex carbohydrates should provide the bulk of daily calories, such as those from whole grains and legumes, while sugar intake such as in soft drinks, desserts and processed foods, should be significantly reduced. Increasing dietary fiber is recommended by eating plenty of fresh fruits, vegetables, whole grains, and legumes. Other principles of a healthy diet are avoiding artificial and processed foods, eating food that is as fresh and natural as possible, drinking plenty of water, and avoiding hydrogenated or partially hydrogenated oils, which contain unhealthy substances called trans-fatty acids. Overeating should be avoided as should snacking between meals, and alcohol intake should be limited to one or two glasses per day.

Exercise

The health of men has been affected as work patterns have shifted. Physical labor has been replaced by machines and office work. Studies have estimated that more than 30% of Americans are now obese, which means that nearly one out of three people is significantly overweight. Obesity poses many risks including increased chance of heart disease, diabetes, and some cancers. Effective exercise programs help men control weight, reduce stress, increase energy levels, improve self-esteem, reduce **pain** and injuries, and improve sleep. Exercise programs should emphasize flexibility and stretching as well as plenty of aerobic activities, such as running and swimming. These activities exercise the heart and lungs and burn excess calories. Men may also choose anaerobic activities such as weight training to add muscles and increase strength. Routines should begin with warm-ups to reduce the chances of injuries and end with cool-down exercises to speed recovery.

Stress reduction

Stress is a silent killer; chronic (long-term) stress is a risk factor in many of the major diseases affecting men's mortality rates. Prolonged stress also may

cause ulcers, **sleep disorders**, addictions, depression, **anxiety**, and other conditions. Reduction of stress may require changes in both activities and attitudes. Exercise is recommended, as is reducing dependence on alcohol and nicotine. Men with extreme job-related stress may choose to spend more time with their families or in enjoyable activities. Men with stress levels that lead to destructive behaviors may need to pursue psychotherapy or significant lifestyle changes. **Nutrition**, social support, and healthy sleep patterns also reduce stress.

Alternative therapies may help with **stress reduction**. Their use has been adopted by many leading health centers. **Biofeedback** utilizes machines that monitor users' stress levels, helping people learn to control them. **Meditation** and other mind/body techniques are taught to enable the relaxation response, which has the opposite effects of stress in the body.

Testing

Routine physical examinations performed by physicians are recommended every three years for men in their twenties and thirties, every two years for men in their forties, and every year for men over 50. Physicians may order several screening tests as well, depending on the age and condition of the patient. Blood tests screen for diabetes, high cholesterol, cancer, infections, and HIV. The prostate-specific antigen (PSA) test is a blood screen for **prostate cancer**. The digital rectal exam is used to manually check the prostate gland for enlargement or irregularities. Urine tests check for infections, kidney problems, and diabetes. The **fecal occult blood test** examines the stool for indications of ulcers or cancer. A **sigmoidoscopy** checks the health of the rectum and lower colon. Electrocardiograms (ECGs) check the status of the heart. Older men may consult an ophthalmologist (eye specialist) every two years for vision and **glaucoma** testing.

Men may perform self-tests as preventative measures. During a skin cancer self-exam, the entire skin is checked closely for irregular or changing **moles**, lesions, or blemishes, usually red, white or blue in color. Abnormal findings should be reported to a physician. Like some forms of skin cancer, **testicular cancer** tends to spread rapidly and early detection is crucial. The testicular self-exam is best performed in the shower or bath, because warm water relaxes the scrotum. The testicles are gently rolled and massaged between the fingers and thumb to feel for bumps, swelling, tenderness, or irregularities. Some self-test kits are available in pharmacies, including kits for blood pressure, high cholesterol, colorectal cancer,

and blood glucose (diabetes). These do not take the place of proper medical care, and physicians should be consulted before their use.

Heart disease

Heart disease is the major cause of death among men. It claims nearly 500,000 lives each year in the United States and is more likely in men than women. Heart disease can take several forms but the most prevalent is coronary heart disease, in which the blood vessels that supply the heart with oxygen become blocked and the heart muscle becomes increasingly stressed. **Arteriosclerosis**, a major factor, is the hardening of arteries due to the accumulation of fatty materials. **Hypertension**, or high blood pressure, also poses major risks for both heart disease and stroke. **Angina pectoris** is the chest pain associated with the early stages of heart disease; more than three million American men suffer from it. When the blockage of blood supply to the heart becomes severe, a myocardial infarction (**heart attack**) may occur, which can be fatal.

The main symptom of angina pectoris is sharp pain on the left side of the chest that may radiate throughout the upper body. Other symptoms include **shortness of breath**, **dizziness**, **fatigue**, and swelling in the legs and ankles. Angina may be triggered by physical or emotional stress and lasts up to 30 minutes. Heart attacks have similar symptoms but with longer and more intense pain in the chest and upper body and may be accompanied by cold sweats and **vomiting**.

The American Heart Association lists the main risk factors for heart disease as being male, old age, having family history of the disease, smoking, high cholesterol, high blood pressure, diabetes, **alcoholism**, obesity, physical inactivity and stress. Clearly, lifestyle habits such as diet, exercise and stress control play major roles in the development and prevention of heart disease in men.

Cancer

The American Cancer Society (ACS) estimates that more than 1.2 million cases of cancer were reported in 2000. Men have a slightly higher risk for cancer than women. The World Health Organization (WHO) estimates that the number of cancer cases in most countries will double in the next 25 years, while men's prostate cancer is expected to go up 40% worldwide. The most common cancers in men are skin, prostate, lung, colorectal (colon and rectum), lymphoma (lymph glands), oral (mouth and throat), and

testicular cancer. The ACS lists seven warning signs of cancer:

- Unusual bleeding or discharge
- Changes in bowel or bladder patterns
- Persistent sores
- Lumps or irregularities on the body
- Difficulty swallowing or indigestion
- Changes in **warts** or moles
- Persistent **cough** or hoarseness in the throat

Although the causes of cancer are incompletely understood, there are several risk factors that increase its chances: family history of cancer, smoking, poor diet (high in fat, low in fiber), excessive alcohol consumption, skin damage from sunlight, and exposure to radiation, chemicals, and environmental pollutants.

The prostate gland is a walnut-sized organ in the male reproductive system, located near the rectum below the bladder. The ACS reported that nearly 232,000 new cases of prostate cancer would be diagnosed in 2005, causing more than 30,000 deaths, making prostate cancer the second most fatal cancer for men behind lung cancer. Worldwide studies have shown that about 12% of men in Western countries get prostate cancer, while 50% have enlarged prostates. Benign prostatic hyperplasia (BPH) is the enlargement of the prostate gland, called benign when it is non-cancerous although growth can be rapid.

With early detection, 98% of men with prostate cancer survive for five years. Symptoms of prostate cancer include difficulty in stopping or starting urination, frequent nighttime urination (nocturia), weak urine flow, and blood in the urine or semen.

Testicular cancer is most common in men between the ages of 15 and 34. The ACS estimated that there would be about 8,100 new cases of testicular cancer in 2005 in the United States. A 2004 study showed that cigarette smoking increased risk of testicular cancer and quitting smoking did not reduce the risk.

Stroke

Strokes occur when the blood supply to the brain is interrupted and brain function becomes impaired due to lack of oxygen. Ischemic strokes occur due to blood vessels becoming blocked while hemorrhagic strokes are the result of broken blood vessels in or near the brain. Ischemic strokes account for about 80% of all strokes. The American Heart Association estimates that more than 600,000 Americans suffer from strokes each year, with men having a 20% higher risk of stroke than women, although more

women die from strokes. Other risk factors are hypertension (high blood pressure), previous heart attacks, age, family history, high cholesterol, smoking, obesity, alcoholism, and physical inactivity. African Americans have 60% greater chances for strokes than whites.

Symptoms of strokes include sudden weakness or **numbness**, blurring or loss of vision, difficulty speaking or understanding, sudden severe **headache**, and dizziness or falling. Stroke victims should receive immediate emergency care.

Male urinary tract problems

The urinary system includes the kidneys and bladder, the ureters between the kidneys and bladder, and the urethra, the tube through which urine flows from the bladder. Symptoms of urinary tract problems include frequent urination, excessive urination at night, painful or burning urination, weak urination, blood in the urine, or incontinence (involuntary loss of urine). **Urethritis** is infection of the urethra, which is a major symptom of sexually transmitted diseases (STDs). **Kidney stones** (nephrolithiasis) are the most common urinary tract problems, accounting for nearly one out of every 100 hospital admissions in the United States. Eighty percent of kidney stone patients are men. About 12% of American men will develop kidney stones during their lifetimes. Kidney stones cause extreme pain when they move from the kidneys into the ureters. Ten percent of kidney stone cases require surgery. The best prevention for kidney stones is drinking plenty of fluids daily.

The male reproductive system

The male reproductive system includes the penis, testicles, scrotum, prostate and other organs. Problems include **orchitis**, or infection of the testicles, and hydrocele, the buildup of fluid on the testicles. **Epididymitis** is inflammation of the tube that transports sperm from the testicles, and can cause severe pain, swelling, and **fever**. A varicocele is a group of **varicose veins** in the scrotum that can cause swelling and damage sperm. **Peyronie's disease** is the abnormal curvature of the penis caused by accumulated scar tissue. **Testicular torsion** is considered a medical emergency, when a testicle becomes twisted and blood supply is cut off. This condition can lead to permanent damage if not treated quickly. It is most common in males between the ages of 12 and 18. **Prostatitis** is infection or inflammation of the prostate gland.

Sexually transmitted diseases include **genital warts**, chlamydia, **gonorrhea**, **syphilis**, **genital herpes**, hepatitis and HIV (human **immunodeficiency virus**). HIV is the leading cause of death for American men between the ages of 25 and 45. Symptoms of STDs include discharge of fluid from the penis; painful urination; sores, lesions, **itching**, or **rashes** in the genital area; and swelling of the lymph nodes in the groin. Prevention of STDs begins with safe sexual behavior: wearing condoms, limiting the number of sexual partners, not mixing sexual encounters with alcohol, and avoiding sexual contact with infected people, prostitutes and intravenous drug users. Men who engage in risky behaviors should have frequent HIV tests and medical examinations.

Male sexual health

Erectile dysfunction (ED), also called **impotence**, is a man's inability to maintain an erection for sexual intercourse. It affects nearly one in every 10 American men. Incidence of ED increases with age, but the problem can occur at any age. Up to 80% of ED is caused by physical problems, while 20% of cases are psychogenic, or psychological in origin. Causes of ED include hormonal problems, injuries, nerve damage, diseases, infections, diabetes, stress, depression, anxiety, drug **abuse**, and interactions with prescription drugs. ED also may be the first indication of circulation problems due to diabetes, high blood pressure, or **coronary artery disease**.

A self-test men can perform to determine whether ED is physical or psychological is the stamp test, or nocturnal penile tumescence test. Physically healthy men experience several prolonged erections during sleep. The stamp test is done by attaching a strip of stamps around the penis before bedtime; if the stamps are torn in the morning, it generally indicates that nocturnal (nightly) erections have occurred and thus ED is not physiological. Men with ED should see a urologist for further diagnosis and discussion of the several treatment options available including drugs, hormone injections, and surgical repair or implants. Several new prescription drugs have become available in recent years.

Infertility occurs when men lack an adequate supply of sperm to cause **pregnancy**. As many as 15% of American couples, or more than 5 million Americans, are affected by infertility in one or both partners. A World Health Organization (WHO) project found that in about 20% of infertile couples, the problem was due to the man, while in another 27% of couples both partners had infertility problems. Injuries, **birth defects**, infections, environmental pollutants, chronic

stress, drug abuse, and hormonal problems may account for male infertility, while one in four cases has no apparent cause and is termed idiopathic infertility. Declining sperm counts have been observed in industrialized countries, and possible explanations for this decrease are as diverse as increased environmental pollutants to the use of plastic diapers, which a German study claims damages infant testicles by keeping in excess heat. Male infertility can be diagnosed by sperm analysis, blood tests, radiographic scans of the testicles, and other tests.

Other types of **sexual dysfunction** include **premature ejaculation**, in which men cannot sustain intercourse long enough to bring their partners to climax, and retarded ejaculation (also called male orgasmic disorder) when male orgasm becomes difficult. Some men have periods of inadequate sexual desire (hypoactive sexual desire disorder), while sexual aversion disorder (SAD) is fear and repulsion of sexual activity. Dyspareunia is painful intercourse, and should be reported to physicians as it may indicate STDs or infections. In addition to medical care, sexual dysfunction may be treated by **sex therapy** or psychotherapy depending on its causes.

Vasectomies, a form of male birth control, are surgical operations that sever the tubes that transport sperm from the testicles. Vasectomies can be reversed but 10% of men become infertile due to the surgery. **Circumcision** is the surgical removal of the foreskin of the penis, for religious and medical reasons, performed on 60% of newborn males in the United States. Increasing controversy surrounds this procedure. Advocates of circumcision claim it prevents infections (called **balanitis**) on the head of the penis and reduces chances of **penile cancer**. Opponents of circumcision claim that the outdated procedure affords no medical benefits, that it causes unnecessary pain for infants, and that the lack of a foreskin may reduce sexual pleasure and performance.

Men's emotional health

Depression is a mood disorder marked by sadness, emotional pain, and the inability to feel pleasure. At least 10% of men will experience an episode of major depression at least once in their lives. Men with depression are five times more likely to commit suicide, a major cause of mortality in men. Men are half as likely to seek psychological help than women. Men may suffer depression and emotional problems between the ages of 50 and 65, called the midlife crisis as men face the major transition into retirement and older age.

KEY TERMS

Emphysema—Disease of severe lung deterioration and impairment.

Obesity—Condition defined as being overweight by 30% of normal limits.

Sigmoidoscopy—Test procedure using an optical instrument to view the internal rectum and colon.

Urologist—Physician specializing in male reproductive and urinary systems.

Panic attacks have symptoms of overwhelming fear, chest pain, shortness of breath, numbness, and increased heart rate. Men may mistake them as heart attacks. Men also are plagued by addictions to nicotine, alcohol, and other drugs, which are often the unhealthy escape routes from deeper emotional issues. Studies have estimated that as many as one-third of Americans have suffered from sleep disorders, which may be psychological in origin and related to anxiety, stress and lifestyle.

Mental illness can be particularly difficult for men because in our society men are taught to withhold rather than express emotions and feelings. Emotional problems can be strong signals for men to communicate and confront deeper issues. Help can be found from physicians, psychotherapists, and spiritual or religious counselors.

Osteoporosis

Osteoporosis often is thought of as disease more prevalent in women, but more than two million men have the disease characterized by decrease in bone mass and density. It develops about 10 to 15 years later in life in men than in women and risk of **fractures** from the disease can be greater in men. Men can decrease their risk by increasing calcium and vitamin D.

Resources

PERIODICALS

"Cigarette Smoking Influences Testicular Cancer Risk." *Medical Devices & Surgical Technology Week* March 28, 2004: 218.

"Men's Health: Erectile Dysfunction." *Medical Update* January 2004: 2.

"Osteoporosis Develops Later in Men, Hits Harder." *Internal Medicine News* March 15, 2004: 30.

ORGANIZATIONS

American Foundation for Urologic Disease. 1128 N. Charles St., Baltimore, MD 21201. (401) 468-1800. < <http://www.afud.org> >.

Center for Holistic Urology. 161 Fort Washington Ave., New York, NY 10032. (212) 305-0347. < <http://www.holisticurology.com> >.

OTHER

A Man's Life Online Magazine. < <http://www.manslife.com> >.

The Prostate Cancer Infolink. < <http://www.comed.com/prostate> >.

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Menstrual disorders

Definition

A menstrual disorder is a physical or emotional problem that interferes with the normal menstrual cycle, causing **pain**, unusually heavy or light bleeding, delayed menarche, or missed periods.

Description

Typically, a woman of childbearing age should menstruate every 28 days or so unless she is pregnant or moving into **menopause**. But numerous things can go wrong with the normal menstrual cycle, some the result of physical causes, others emotional. These include **amenorrhea**, or the cessation of menstruation, **menorrhagia**, or heavy bleeding, and **dysmenorrhea**, or severe menstrual cramps. Nearly every woman will experience one or more of these menstrual irregularities at some time in her life.

Amenorrhea

There are two types of amenorrhea: primary and secondary. Overall, they affect 2–5% of childbearing women, a number that is considerably higher among female athletes (possibly as high as 66%).

Primary amenorrhea occurs when a girl at least 16 years old is not menstruating. Young girls may not have regular periods for their first year or two, or their periods may be very light, a condition known as **oligomenorrhea**. A light flow is nothing to worry about. But if the period has not begun at all by age 16, there may be something wrong. Amenorrhea is most common in

girls who are severely underweight and/or **exercise** intensely, both of which affect the amount of body fat necessary to trigger the release of hormones that, in turn, begins **puberty**.

Secondary amenorrhea occurs in women of childbearing age after a period of normal menstruation and is diagnosed when menstruation has stopped for three months. It can occur in women of any age.

Dysmenorrhea

Characterized by menstrual cramps or painful periods, dysmenorrhea, which comes from the Greek words for “painful flow,” affects nearly every woman at some point in her life. It is the most common reproductive problem in women, resulting in numerous days absent from school, work, and other activities. There are two types: primary and secondary.

Primary, or normal cramps, affects up to 90% of all women, usually occurring in women about three years after they start menstruating and continuing through their mid-twenties or until they have a child. About 10% of women who have this type of dysmenorrhea cannot work, attend school, or participate in their normal activities. It may be accompanied by backache, **dizziness**, **headache**, **nausea**, **vomiting**, **diarrhea** and tenseness. The symptoms typically start a day or two before menstruation, usually ending when menstruation actually begins.

Secondary dysmenorrhea has an underlying physical cause and primarily affects older women, although it may also occur immediately after a woman begins menstruation.

Menorrhagia

Menorrhagia, or heavy bleeding, most commonly occurs in the years just before menopause or just after women start menstruating. It occurs in 15–20% of American women.

Premenstrual dysphoric disorder (PMDD)

The fourth edition of the *Diagnostic and Statistical Manual of Mental Disorders*, or DSM-IV, lists **premenstrual dysphoric disorder** (PMDD) in an appendix of criteria sets for further study. To meet full criteria for PMDD, a patient must have at least five out of 11 emotional or physical symptoms during the week preceding the menses for most menstrual cycles over the previous 12 months. Although the DSM-IV definition of PMDD as a mental disorder is controversial because of fear that it could be used to justify prejudice or job discrimination against women,

there is evidence that a significant proportion of premenopausal women suffer emotional distress or impairment in job functioning in the week before their menstrual period. One group of researchers estimates that 3–8% of women of childbearing age meet the strict DSM-IV criteria for PMDD, with another 13–18% having symptoms severe enough to interfere with their normal activities.

Causes and symptoms

Amenorrhea

The only symptom of primary amenorrhea is delayed menstruation. In addition to low body weight or excessive exercise, other causes of primary amenorrhea include Turner's syndrome, a birth defect related to the reproductive system, or ovarian problems. In 2003, a group of researchers reported on a new genetic mutation associated with primary amenorrhea. In secondary amenorrhea, the primary symptom is the ceasing of menstruation for at least three months. Causes include **pregnancy** or breastfeeding, sudden weight loss or gain, intense exercise, **stress**, endocrine disorders affecting the thyroid, pituitary or adrenal glands, including **Cushing's Syndrome** and **hyperthyroidism**, problems with or surgery on the ovaries, including removal of the ovaries, cysts or ovarian tumors.

Amenorrhea in athletes or dancers is frequently associated with two other disorders—osteopenia, or reduced bone mass, and eating disorders. This combination is sometimes called the female athlete triad. Osteopenia is of concern because it can lead to premature **osteoporosis**.

Dysmenorrhea

Primary dysmenorrhea is related to the production of prostaglandins, natural chemicals the body makes that cause an inflammatory reaction. They also cause the muscles of the uterus to contract, thus helping the uterus shed the lining built up during the first part of a woman's cycle. Women with severe menstrual pain have higher levels of prostaglandin in their menstrual blood than women who do not have such pain. In some women, prostaglandins can cause some of the smooth muscles in the gastrointestinal tract to contract, resulting in the nausea, vomiting and diarrhea some women experience. Prostaglandins also cause the arteries and veins to expand, so that blood collects in them rather than flowing freely through them, causing pain and heaviness. Yet another reason for severe cramps,

particularly in women who have not yet had a baby, is that the flow of the blood and clots through the tiny cervical opening is painful. After a woman has a baby, however, the cervix opening is larger.

Secondary dysmenorrhea is more serious and is related to some underlying cause. The pain may feel like regular menstrual cramps, but may last longer than normal and occur throughout the month. It may be stronger on one side of the body than the other. Possible causes include:

- A tipped uterus
- Endometriosis, a condition in which the same type of tissue found in the lining of the uterus occurs outside the uterus, usually elsewhere in the pelvic cavity
- Adenomyosis, a condition in which the endometrial lining grows into the muscle of the uterus
- Fibroids
- **Pelvic inflammatory disease (PID)**
- An **IUD**
- A uterine, ovarian, bowel or bladder tumor
- Uterine polyps
- Inflammatory bowel disease
- Scarring or **adhesions** from earlier surgery

Menorrhagia

Heavy bleeding during menstruation is usually related to a hormonal imbalance, although other causes include fibroids, cervical or endometrial polyps, the autoimmune disease lupus, pelvic inflammatory disease (PID), blood platelet disorder, a hereditary blood factor deficiency, or, possibly, some reproductive cancers. Thus, menorrhagia is actually a symptom of an underlying condition rather than a disease itself. It may also be related to the use of an IUD.

Women with menorrhagia experience not only significant inconvenience, but may feel very tired due to the loss of iron-rich blood. It is usually diagnosed when a woman soaks through a tampon or pad every hour for several hours or has a period lasting more than 7 days. Clots are not related to menorrhagia, although women with heavy cycles may pass clots. They are typically a normal part of menstruation, more common when a woman has been sitting or in a stationary position for a while

Diagnosis

Women should seek care from a gynecologist, family practitioner or internist for menstrual

irregularities. Depending on the problem, various tests and procedures will be performed, but the one common to any menstrual problem is a **pelvic exam**. This should be scheduled when women are not menstruating, simply for convenience.

Male doctors typically have a female nurse or assistant in the room. The examination begins by checking the external genitalia for any sores or irregularities. Then the doctor inserts a speculum (a metal duckbill-shaped device that holds open the vagina) into the vagina and peers throughout the opening to evaluate the health of the cervix (opening of the uterus), and inside the vagina, looking for growths or any other abnormalities.

The doctor will also manually examine the woman, inserting two fingers into the vagina while pressing on the abdomen, again feeling for any lumps or other abnormalities, checking the size and shape of the reproductive organs, and watching for any signs of infection, such as tenderness or pain. The exam is typically covered by insurance and takes about 10 minutes.

Other tests that will be done for menstrual irregularities include:

- **A pregnancy test.** The nurse takes some blood from a woman's arm and it is tested for the presence of certain hormones that indicate a pregnancy has occurred.
- **Ultrasound.** Typically performed by a trained ultrasound technologist, it involves using sound waves to get an image of the reproductive system. It is used to look for fibroids and other ovarian abnormalities that may cause heavy bleeding or cramps. Typically, the technologist will smear a jelly over the woman's stomach, then place a probe on her stomach and watch the images appear on a computer screen. It is painless. Women may be asked not to urinate for several hours prior to the test, as a full bladder makes it easier to see the other internal organs. The test takes about 20 minutes.
- **Endometrial biopsy.** Used to check the health of uterine tissue in women who have unusually heavy bleeding, this test should be performed by the physician. Women should take a pain reliever such as ibuprofen or naproxen prior to the procedure, as there may be some cramping. The woman lies back on the table with her feet in stirrups and the doctor inserts a speculum, then opens the cervix slightly with an instrument called a tenaculum. Then the doctor slides a small, hollow catheter into the uterus and sucks out a small piece of tissue from the uterine lining. The tissue is then examined for any

abnormalities in a laboratory. The test takes about 30 minutes and is typically covered by insurance. Some bleeding may result afterwards.

- **Blood, stool and urine tests** may also be conducted to check for levels of various hormones, blood cells, and other chemicals.
- **Dilation and curettage (D&C):** During this minor surgical procedure, the cervix is opened and the lining of the uterus scraped for a tissue sample.
- **Laparoscopy and hysteroscopy:** in some instances, these surgical procedures, in which a small camera is inserted into the woman to view the inside of the pelvis, abdomen or uterus.

Treatment

Amenorrhea

For primary amenorrhea with no underlying problem, no treatment is necessary, and a wait-and-see approach is often adopted. If women have genetic or hormonal abnormalities, amenorrhea is often treated with **oral contraceptives** that contain combinations of estrogen and progestin. Side effects include bloating, weight gain and **acne**, although some birth control pills actually improve acne. Progestins, or synthetic progesterone, are also used alone to "jump start" a woman's period. They include medroxyprogesterone (Provera, Amen, **Depo-Provera**), norethindrone acetate (Aygestin, Norlutate), and norgestrel (Ovrel). If the amenorrhea is due to a physical problem, such as a closed vagina, surgery may be required.

With secondary amenorrhea, treatment depends on the cause. Hormonal imbalances are treated with supplemental hormones. Tumors or cysts may require surgery. **Obesity** may require a diet and exercise regimen, while amenorrhea resulting from too much dieting or exercise necessitates lifestyle changes.

Dysmenorrhea

Primary dysmenorrhea is typically treated with nonsteroidal anti-inflammatory medications like ibuprofen and naproxen, which studies show help 64 to 100% of women. Birth control pills relieve pain and symptoms in about 90% of women by suppressing ovulation and reducing the amount of menstrual blood. It may take up to three cycles before a woman feels relief. Heat from a heating pad or hot bath, can also help relieve pain.

Treatment for secondary dysmenorrhea depends on the underlying cause of the condition.

Menorrhagia

If there are no other problems, and the bleeding is due to hormonal imbalances, birth control pills are often prescribed to bring the bleeding under control and regulate menstruation. Such medications as ibuprofen and naproxen can also help reduce the bleeding and any cramping associated with it. In severe cases, doctors may recommend removing the uterus during a **hysterectomy**, or performing some form of endometrial ablation, which removes the lining of the uterus. These procedures are typically only offered to women who have completed their families. A recent British study reported, however, that many women prefer endometrial ablation to hysterectomy because it is less invasive and safer. A new treatment that involves intrauterine hormonal therapy is gaining acceptance, but had not been approved by the FDA as of spring 2004.

Premenstrual dysphoric disorder (PMDD)

Medications that have been reported to be effective in treating PMDD include the **tricyclic antidepressants** and the **selective serotonin reuptake inhibitors** (SSRIs). Effective treatments other than medications include cognitive behavioral therapy (CBT), aerobic exercise, and dietary supplements containing calcium, magnesium, and vitamin B₆.

Alternative treatment

Amenorrhea

There are several herbal remedies that can bring on menstruation, including: black cohosh, cramp bark, chasteberry, celery, turmeric, and marshmallow. Numerous relaxation techniques, such as **meditation**, deep breathing, and **yoga** can help reduce stress and its affects on menstruation.

Dysmenorrhea

Numerous alternative treatments may help relieve the menstrual pain. These include:

- Transcutaneous **electrical nerve stimulation** (TENS), which several studies found, relieved pain in 42–60% of participants, working faster than naproxen in one study.
- **Acupuncture**: One study of 43 patients followed for a year found that 90% of those who had acupuncture once a week for three menstrual cycles had less pain, and 43% used less pain medication.
- **Omega-3 fatty acids**: Often sold as fish oil supplements, they are a known anti-inflammatory, working against the effects of prostaglandins. Studies found

that women with low amounts of omega-3 fatty acids in their **diets** were more likely to have menstrual cramps; those who took supplements had less pain.

- **Vitamin B-1**: One large study found that symptoms disappeared in 87% of women who took 100 mg a day for 90 days.
- **Magnesium supplements**: One study of 30 women who took 4.5 milligrams of oral magnesium three times daily for part of the month decreased their symptoms up to 84%.

Menorrhagia

Herbs used to treat menorrhagia include yarrow, nettles and shepherd's purse, as well as agrimony, particularly used in Chinese medicine, ladies mantle, vervain and red raspberry, which are thought to strengthen the uterus. Vitex is another herb recommended for a variety of menstrual disorders ranging from menorrhagia to PMS. Women may want to make sure they are taking an iron supplement to replace the iron lost during the heavy bleeding, although they should check with their doctor to make sure they do not suffer from a condition of having too much iron. Helpful **vitamins** include vitamin A, because women with heavy bleeding typically have lower levels of Vitamin A, K, which aids in clotting, and C and bioflavonoids which help strengthen veins and capillaries. Zinc may also help.

Prognosis

The prognosis for all menstrual irregularities is good once treatment is initiated.

Prevention

Amenorrhea

Simply following a healthy exercise and nutritional program can help prevent amenorrhea, as can reducing stress and learning relaxation techniques. Also, avoiding excessive alcohol intake and quitting **smoking** may prevent missed periods.

Dysmenorrhea

Prevention includes certain dietary supplements and vitamins described above. Exercise may also help.

Menorrhagia

There is little women can do to prevent this menstrual irregularity other than discovering the root

KEY TERMS

Adenomyosis—Uterine thickening caused when endometrial tissue, which normally lines the uterus, extends outward into the fibrous and muscular tissue of the uterus.

Cervical polyps—Growths originating from the surface of the cervix or endocervical canal. These small, fragile growths hang from a stalk and protrude through the cervical opening (the os).

Cushing's syndrome—A group of conditions caused by increased production of cortisol hormones or by the administration of glucocorticoid hormones (cortisone-like hormones).

Endometriosis—A condition in which the tissue that normally lines the uterus (endometrium) grows in other areas of the body, causing pain, irregular bleeding, and frequently, infertility.

Fibroids—Benign tumors of muscle and connective tissue that develop within or are attached to the uterine wall.

Hyperthyroidism—An imbalance in metabolism that occurs from overproduction of thyroid hormone.

Inflammatory bowel disease—A chronic inflammatory disease that can affect any part of the gastrointestinal tract but most commonly affects the ileum.

Lupus (systemic lupus erythematosus or SLE)—A chronic inflammatory autoimmune disorder that may affect many organ systems including the skin, joints, and internal organs.

Menarche—The first menstrual period or the establishment of the menstrual function.

Osteopenia—Reduction in bone mass, usually caused by a lowered rate of formation of new bone that is insufficient to keep up with the rate of bone destruction. Osteopenia often occurs together with amenorrhea and eating disorders in female athletes. It can lead to premature osteoporosis if left untreated.

Pelvic inflammatory disease (PID)—A general term referring to infection involving the lining of the uterus, the Fallopian tubes, or the ovaries.

Turner's syndrome—A disorder in women caused by an inherited chromosomal defect. This disorder inhibits sexual development and causes infertility. A symptom is absence of menstruation.

cause. One thing they can do, however, is stop using an IUD, which can often cause heavier bleeding.

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ORGANIZATIONS

- Advancement of Women's Health Research. 1828 L Street, N.W., Suite 625 Washington, DC 20036. 202-223-8224. <<http://www.womens-health.org>>.
- American College of Obstetricians and Gynecologists (ACOG). 409 12th Street, SW, P. O. Box 96920, Washington, DC 20090-6920. <<http://www.acog.org>>.
- American Psychiatric Association (APA). 1400 K Street, NW, Washington, DC 20005. (888) 357-7924. <<http://www.psych.org>>.

National Women's Health Resource Center. 120 Albany Street Suite 820, New Brunswick, NJ 08901. (877) 986-9472. < www.healthywomen.org >.

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Menstrual pain see **Dysmenorrhea**

Mental retardation

Definition

Mental retardation is a developmental disability that first appears in children under the age of 18. It is defined as an intellectual functioning level (as measured by standard tests for intelligence quotient) that is well below average and significant limitations in daily living skills (adaptive functioning).

Description

Mental retardation occurs in 2.5–3% of the general population. About 6–7.5 million mentally retarded individuals live in the United States alone. Mental retardation begins in childhood or adolescence before the age of 18. In most cases, it persists throughout adulthood. A diagnosis of mental retardation is made if an individual has an intellectual functioning level well below average and significant limitations in two or more adaptive skill areas. Intellectual functioning level is defined by standardized tests that measure the ability to reason in terms of mental age (intelligence quotient or IQ). Mental retardation is defined as IQ score below 70–75. Adaptive skills are the skills needed for daily life. Such skills include the ability to produce and understand language (communication); home-living skills; use of community resources; health, safety, leisure, self-care, and social skills; self-direction; functional academic skills (reading, writing, and arithmetic); and work skills.

In general, mentally retarded children reach developmental milestones such as walking and talking much later than the general population. Symptoms of mental retardation may appear at birth or later in childhood. Time of onset depends on the suspected cause of the disability. Some cases of mild mental retardation are not diagnosed before the child enters preschool. These children typically have difficulties

with social, communication, and functional academic skills. Children who have a neurological disorder or illness such as **encephalitis** or **meningitis** may suddenly show signs of cognitive impairment and adaptive difficulties.

Mental retardation varies in severity. *The Diagnostic and Statistical Manual of Mental Disorders*, Fourth Edition (*DSM-IV*) is the diagnostic standard for mental healthcare professionals in the United States. The *DSM-IV* classifies four different degrees of mental retardation: *mild*, *moderate*, *severe*, and *profound*. These categories are based on the functioning level of the individual.

Mild mental retardation

Approximately 85% of the mentally retarded population is in the mildly retarded category. Their IQ score ranges from 50–75, and they can often acquire academic skills up to the 6th grade level. They can become fairly self-sufficient and in some cases live independently, with community and social support.

Moderate mental retardation

About 10% of the mentally retarded population is considered moderately retarded. Moderately retarded individuals have IQ scores ranging from 35–55. They can carry out work and self-care tasks with moderate supervision. They typically acquire communication skills in childhood and are able to live and function successfully within the community in a supervised environment such as a group home.

Severe mental retardation

About 3–4% of the mentally retarded population is severely retarded. Severely retarded individuals have IQ scores of 20–40. They may master very basic self-care skills and some communication skills. Many severely retarded individuals are able to live in a group home.

Profound mental retardation

Only 1–2% of the mentally retarded population is classified as profoundly retarded. Profoundly retarded individuals have IQ scores under 20–25. They may be able to develop basic self-care and communication skills with appropriate support and training. Their retardation is often caused by an accompanying neurological disorder. The profoundly retarded need a high level of structure and supervision.

The American Association on Mental Retardation (AAMR) has developed another widely accepted diagnostic classification system for mental retardation. The AAMR classification system focuses on the capabilities of the retarded individual rather than on the limitations. The categories describe the level of support required. They are: *intermittent support*, *limited support*, *extensive support*, and *pervasive support*. To some extent, the AAMR classification mirrors the *DSM-IV* classification. Intermittent support, for example, is support needed only occasionally, perhaps during times of **stress** or crisis. It is the type of support typically required for most mildly retarded individuals. At the other end of the spectrum, pervasive support, or life-long, daily support for most adaptive areas, would be required for profoundly retarded individuals.

Causes and symptoms

Low IQ scores and limitations in adaptive skills are the hallmarks of mental retardation. Aggression, self-injury, and **mood disorders** are sometimes associated with the disability. The severity of the symptoms and the age at which they first appear depend on the cause. Children who are mentally retarded reach developmental milestones significantly later than expected, if at all. If retardation is caused by chromosomal or other genetic disorders, it is often apparent from infancy. If retardation is caused by childhood illnesses or injuries, learning and adaptive skills that were once easy may suddenly become difficult or impossible to master.

In about 35% of cases, the cause of mental retardation cannot be found. Biological and environmental factors that can cause mental retardation include:

Genetics

About 5% of mental retardation is caused by hereditary factors. Mental retardation may be caused by an inherited abnormality of the genes, such as **fragile X syndrome**. Fragile X, a defect in the chromosome that determines sex, is the most common inherited cause of mental retardation. Single gene defects such as **phenylketonuria** (PKU) and other inborn errors of metabolism may also cause mental retardation if they are not found and treated early. An accident or mutation in genetic development may also cause retardation. Examples of such accidents are development of an extra chromosome 18 (trisomy 18) and **Down syndrome**.

Down syndrome, also called mongolism or trisomy 21, is caused by an abnormality in the development of chromosome 21. It is the most common genetic cause of mental retardation.

Prenatal illnesses and issues

Fetal alcohol syndrome affects one in 600 children in the United States. It is caused by excessive alcohol intake in the first twelve weeks (trimester) of **pregnancy**. Some studies have shown that even moderate alcohol use during pregnancy may cause learning disabilities in children. Drug **abuse** and cigarette **smoking** during pregnancy have also been linked to mental retardation.

Maternal infections and illnesses such as glandular disorders, **rubella**, **toxoplasmosis**, and **cytomegalovirus infection** may cause mental retardation. When the mother has high blood pressure (**hypertension**) or blood **poisoning** (toxemia), the flow of oxygen to the fetus may be reduced, causing brain damage and mental retardation.

Birth defects that cause physical deformities of the head, brain, and central nervous system frequently cause mental retardation. Neural tube defect, for example, is a birth defect in which the neural tube that forms the spinal cord does not close completely. This defect may cause children to develop an accumulation of cerebrospinal fluid on the brain (**hydrocephalus**). Hydrocephalus can cause learning impairment by putting pressure on the brain.

Childhood illnesses and injuries

Hyperthyroidism, **whooping cough**, **chickenpox**, **measles**, and Hib disease (a bacterial infection) may cause mental retardation if they are not treated adequately. An infection of the membrane covering the brain (meningitis) or an inflammation of the brain itself (encephalitis) cause swelling that in turn may cause brain damage and mental retardation. Traumatic brain injury caused by a blow or a violent shake to the head may also cause brain damage and mental retardation in children.

Environmental factors

Ignored or neglected infants who are not provided the mental and physical stimulation required for normal development may suffer irreversible learning impairments. Children who live in poverty and suffer from **malnutrition**, unhealthy living conditions, and improper or inadequate medical care are at a higher risk. Exposure to lead can also cause mental retardation. Many children have developed **lead poisoning** by

eating the flaking lead-based paint often found in older buildings.

Diagnosis

If mental retardation is suspected, a comprehensive **physical examination** and medical history should be done immediately to discover any organic cause of symptoms. Conditions such as hyperthyroidism and PKU are treatable. If these conditions are discovered early, the progression of retardation can be stopped and, in some cases, partially reversed. If a neurological cause such as brain injury is suspected, the child may be referred to a neurologist or neuropsychologist for testing.

A complete medical, family, social, and educational history is compiled from existing medical and school records (if applicable) and from interviews with parents. Children are given intelligence tests to measure their learning abilities and intellectual functioning. Such tests include the Stanford-Binet Intelligence Scale, the Wechsler Intelligence Scales, the Wechsler Preschool and Primary Scale of Intelligence, and the Kaufmann Assessment Battery for Children. For infants, the Bayley Scales of Infant Development may be used to assess motor, language, and problem-solving skills. Interviews with parents or other caregivers are used to assess the child's daily living, muscle control, communication, and social skills. The Woodcock-Johnson Scales of Independent Behavior and the Vineland Adaptive Behavior Scale (VABS) are frequently used to test these skills.

Treatment

Federal legislation entitles mentally retarded children to free testing and appropriate, individualized education and skills training within the school system from ages 3–21. For children under the age of three, many states have established early intervention programs that assess, recommend, and begin treatment programs. Many day schools are available to help train retarded children in basic skills such as bathing and feeding themselves. Extracurricular activities and social programs are also important in helping retarded children and adolescents gain self-esteem.

Training in independent living and job skills is often begun in early adulthood. The level of training depends on the degree of retardation. Mildly retarded individuals can often acquire the skills needed to live independently and hold an outside job. Moderate to

profoundly retarded individuals usually require supervised community living.

Family therapy can help relatives of the mentally retarded develop coping skills. It can also help parents deal with feelings of guilt or anger. A supportive, warm home environment is essential to help the mentally retarded reach their full potential.

Prognosis

Individuals with mild to moderate mental retardation are frequently able to achieve some self-sufficiency and to lead happy and fulfilling lives. To reach these goals, they need appropriate and consistent educational, community, social, family, and vocational supports. The outlook is less promising for those with severe to profound retardation. Studies have shown that these individuals have a shortened life expectancy. The diseases that are usually associated with severe retardation may cause the shorter life span. People with Down syndrome will develop the brain changes that characterize **Alzheimer's disease** in later life and may develop the clinical symptoms of this disease as well.

Prevention

Immunization against diseases such as measles and Hib prevents many of the illnesses that can cause mental retardation. In addition, all children should undergo routine developmental screening as part of their pediatric care. Screening is particularly critical for those children who may be neglected or undernourished or may live in disease-producing conditions. Newborn screening and immediate treatment for PKU and hyperthyroidism can usually catch these disorders early enough to prevent retardation.

Good prenatal care can also help prevent retardation. Pregnant women should be educated about the risks of drinking and the need to maintain good **nutrition** during pregnancy. Tests such as **amniocentesis** and ultrasonography can determine whether a fetus is developing normally in the womb.

Resources

ORGANIZATIONS

American Association on Mental Retardation (AAMR).
444 North Capitol St., NW, Suite 846, Washington,
D.C. 20001-1512. (800) 424-3688. <<http://www.aamr.org>>.

The Arc. 900 Varum Street NE, Washington, D.C. 20017.
(202) 636-2950. <<http://thearc.org>>.

KEY TERMS

Amniocentesis—A test usually done between 16 and 20 weeks of pregnancy to detect any abnormalities in the development of the fetus. A small amount of the fluid surrounding the fetus (amniotic fluid) is drawn out through a needle inserted into the mother's womb. Laboratory analysis of this fluid can detect various genetic defects, such as Down syndrome, or neural tube defects.

Developmental delay—The failure to meet certain developmental milestones, such as sitting, walking, and talking, at the average age. Developmental delay may indicate a problem in development of the central nervous system.

Down syndrome—A disorder caused by an abnormality at the 21st chromosome. One symptom of Down syndrome is mental retardation.

Extensive support—Ongoing daily support required to assist an individual in a specific adaptive area, such as daily help with preparing meals.

Hib disease—An infection caused by *Haemophilus influenza* type b (Hib). This disease mainly affects children under the age of five. In that age group, it is the leading cause of bacterial meningitis, pneumonia, joint and bone infections, and throat inflammations.

Inborn error of metabolism—A rare enzyme deficiency; children with inborn errors of metabolism do

not have certain enzymes that the body requires to maintain organ functions. Inborn errors of metabolism can cause brain damage and mental retardation if left untreated. Phenylketonuria is an inborn error of metabolism.

Limited support—A predetermined period of assistance required to deal with a specific event, such as training for a new job.

Phenylketonuria (PKU)—An inborn error in metabolism that prevents the body from using phenylalanine, an amino acid necessary for normal growth and development.

Trisomy—An abnormality in chromosomal development. Chromosomes are the structures within a cell that carry its genetic information. They are organized in pairs. Humans have 23 pairs of chromosomes. In a trisomy syndrome, an extra chromosome is present so that the individual has three of a particular chromosome instead of the normal pair. An extra chromosome 18 (trisomy 18) causes mental retardation.

Ultrasonography—A process that uses the reflection of high-frequency sound waves to make an image of structures deep within the body. Ultrasonography is routinely used to detect fetal abnormalities.

OTHER

Americans With Disabilities Act (ADA) Page. <<http://www.usdoj.gov/crt/ada/adahom1.htm>>.

Paula Anne Ford-Martin

Mental status examination

Definition

A mental status examination (MSE) is an assessment of a patient's level of cognitive (knowledge-related) ability, appearance, emotional mood, and speech and thought patterns at the time of evaluation. It is one part of a full neurologic (nervous system) examination and includes the examiner's observations about the patient's attitude and cooperativeness as well as the patient's answers to specific questions.

The most commonly used test of cognitive functioning per se is the so-called Folstein Mini-Mental Status Examination (MMSE), developed in 1975.

Purpose

The purpose of a mental status examination is to assess the presence and extent of a person's mental impairment. The cognitive functions that are measured during the MSE include the person's sense of time, place, and personal identity; memory; speech; general intellectual level; mathematical ability; insight or judgment; and reasoning or problem-solving ability. Complete MSEs are most commonly given to elderly people and to other patients being evaluated for **dementia** (including AIDS-related dementia). Dementia is an overall decline in a person's intellectual function—including difficulties with language, simple calculations, planning or decision-making, and motor (muscular movement) skills as well as loss of memory. The MSE is an important part of the differential

diagnosis of dementia and other psychiatric symptoms or disorders. The MSE results may suggest specific areas for further testing or specific types of required tests. A mental status examination can also be given repeatedly to monitor or document changes in a patient's condition.

Precautions

The MSE cannot be given to a patient who cannot pay attention to the examiner, for example as a result of being in a **coma** or unconscious; or is completely unable to speak (aphasic); or is not fluent in the language of the examiner.

Description

The MMSE of Folstein evaluates five areas of mental status, namely, orientation, registration, attention and calculation, recall and language. A complete MSE is more comprehensive and evaluates the following ten areas of functioning:

- **Appearance.** The examiner notes the person's age, race, sex, civil status, and overall appearance. These features are significant because poor personal hygiene or grooming may reflect a loss of interest in self-care or physical inability to bathe or dress oneself.
- **Movement and behavior.** The examiner observes the person's gait (manner of walking), posture, coordination, eye contact, facial expressions, and similar behaviors. Problems with walking or coordination may reflect a disorder of the central nervous system.
- **Affect.** Affect refers to a person's outwardly observable emotional reactions. It may include either a lack of emotional response to an event or an overreaction.
- **Mood.** Mood refers to the underlying emotional "atmosphere" or tone of the person's answers.
- **Speech.** The examiner evaluates the volume of the person's voice, the rate or speed of speech, the length of answers to questions, the appropriateness and clarity of the answers, and similar characteristics.
- **Thought content.** The examiner assesses what the patient is saying for indications of **hallucinations**, **delusions**, obsessions, symptoms of dissociation, or thoughts of **suicide**. Dissociation refers to the splitting-off of certain memories or mental processes from conscious awareness. Dissociative symptoms include feelings of unreality, depersonalization, and confusion about one's identity.
- **Thought process.** Thought process refers to the logical connections between thoughts and their relevance to the main thread of conversation. Irrelevant detail, repeated words and phrases, interrupted thinking (thought blocking), and loose, illogical connections between thoughts, may be signs of a thought disorder.
- **Cognition.** Cognition refers to the act or condition of knowing. The evaluation assesses the person's orientation (ability to locate himself or herself) with regard to time, place, and personal identity; long- and short-term memory; ability to perform simple arithmetic (counting backward by threes or sevens); general intellectual level or fund of knowledge (identifying the last five Presidents, or similar questions); ability to think abstractly (explaining a proverb); ability to name specified objects and read or write complete sentences; ability to understand and perform a task (showing the examiner how to comb one's hair or throw a ball); ability to draw a simple map or copy a design or geometrical figure; ability to distinguish between right and left.
- **Judgment.** The examiner asks the person what he or she would do about a commonsense problem, such as running out of a prescription medication.
- **Insight.** Insight refers to a person's ability to recognize a problem and understand its nature and severity.

The length of time required for a mental status examination depends on the patient's condition. It may take as little as five minutes to examine a healthy person. Patients with speech problems or intellectual impairments, dementia, or other organic brain disorders may require fifteen or twenty minutes. The examiner may choose to spend more time on certain portions of the MSE and less time on others, depending on the patient's condition and answers.

Preparation

Preparation for a mental status examination includes a careful medical and psychiatric history of the patient. The history helps the examiner to interpret the patient's appearance and answers with greater accuracy, because some physical illnesses may produce psychiatric symptoms or require medications that influence the patient's mood or attentiveness. The psychiatric history should include a family history as well as the patient's personal history of development, behavior patterns, and previous treatment for mental disorders (if any). Symptoms of dissociation, for example, often point to a history of childhood

abuse, rape, or other severe emotional traumas in adult life. The examiner should also include information about the patient's occupation, level of education, marital status, and right- or left-handedness. Information about occupation and education helps in evaluating the patient's use of language, extent of memory loss, reasoning ability, and similar functions. Handedness is important in determining which half of the patient's brain is involved in writing, picking up a pencil, or other similar tasks that he or she may be asked to perform during the examination.

Aftercare

Depending on the examiner's specific observations, the patient may be given additional tests for follow-up. These tests might include blood or urine samples to test for drug or alcohol abuse, anemia, diabetes, disorders of the liver or kidneys, vitamin or thyroid deficiencies, medication side effects, or **syphilis** and **AIDS**. Brain imaging (CT, MRI, or **PET** scans) may be used to look for signs of seizures, strokes, head trauma, brain tumors, or other evidence of damage to specific parts of the brain. A spinal tap may be performed if the doctor thinks the patient may have an infection of the central nervous system.

Normal results

Normal results for a mental status examination depend to some extent on the patient's history, level of education, and recent life events. For example, a depressed mood is appropriate in the context of a recent **death** or other sad event in the patient's family but inappropriate in the context of a recent pay raise. Speech patterns are often influenced by racial or ethnic background as well as by occupation or schooling. In general, however, the absence of obvious delusions, hallucinations, or thought disorders together with the presence of insight, good judgment, and socially appropriate appearance and behavior are considered normal results. A normal numerical score for the MMSE is between 28 and 30.

Abnormal results

Abnormal results for a mental status examination include:

- Any evidence of organic brain damage.
- Evidence of thought disorders.
- A mood or affect that is clearly inappropriate to its context.
- Thoughts of suicide.

KEY TERMS

Aphasia—The loss of the ability to speak, or to understand written or spoken language. A person who cannot speak or understand language is said to be aphasic.

Cognition—The act or process of knowing or perceiving.

Coma—A state of prolonged unconsciousness in which a person cannot respond to spoken commands or mildly painful physical stimuli.

Delusion—A belief that is resistant to reason or contrary to actual fact. Common delusions include delusions of persecution, delusions about one's importance (sometimes called delusions of grandeur), or delusions of being controlled by others.

Dementia—A decline in a person's level of intellectual functioning. Dementia includes memory loss as well as difficulties with language, simple calculations, planning or decision-making, and motor (muscular movement) skills.

Dissociation—The splitting off of certain mental processes from conscious awareness. Specific symptoms of dissociation include feelings of unreality, depersonalization, and confusion about one's identity.

Hallucination—A sensory experience, usually involving either sight or hearing, of something that does not exist outside the mind.

Illusion—A false visual perception of an object that others perceive correctly. A common example is the number of sightings of "UFOs" that turn out to be airplanes or weather balloons.

Obsession—Domination of thoughts or feelings by a persistent idea, desire, or image.

Organic brain disorder—An organic brain disorder refers to impaired brain function due to damage or deterioration of brain tissue.

- Disturbed speech patterns.
- Dissociative symptoms.
- Delusions or hallucinations.

A score below 27 on the MMSE usually indicates an organic brain disorder.

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Rebecca J. Frey, PhD

Mesothelioma

Definition

Mesothelioma is an uncommon disease that causes malignant **cancer** cells to form within the lining of the chest, abdomen, or around the heart. Its primary cause is believed to be exposure to asbestos.

Description

Malignant mesothelioma is also known as asbestos cancer or simply "meso." Mesothelioma causes cancerous cells to develop in the body's mesothelium, where they can spread to and damage vital organs and tissue. These malignant cells can also metastasize to other regions of the body. Mesothelioma is very difficult to diagnose and responds poorly to most treatment modalities, accounting for a poor prognosis.

The disease derives its name from the mesothelium, a sac-like membrane that protects most of the body's internal organs. It is divided into two distinct protective layers of cells: the visceral (the layer directly surrounding the organ) and the parietal (a sac around the body cavity). By releasing a lubricating fluid, the mesothelium allows the organs to move more freely within the body cavity; for example, the contraction and expansion of the lungs. The mesothelium is also referred to according to where it is located in the body: pleura (chest), peritoneum (abdomen), and pericardium (heart).

Over two-thirds of all mesothelioma cases begin in the pleura region. Pleural mesothelioma spreads through the chest cavity, occasionally developing in the lungs as well. The disease most commonly causes **pleural effusion**, an excess build-up of fluid inside the chest cavity. This excess fluid increases pressure on the lungs and restricts breathing. In addition, malignant cells can cause the pleural lining to thicken and restrict the breathing space even further.

Peritoneal mesothelioma is the second most common form of the disease, accounting for less than 30% of all cases. Malignant cells form in the peritoneum, affecting the abdomen, bowel, liver, and spleen. Similar to pleural mesothelioma, the disease also causes a build up of excess fluid in the abdominal cavity. Normal bodily functions, such as digestion, can be hindered by the obstruction of organ movement.

Very rare forms of mesothelioma occur in the pericardium, as well as the mesothelium of the male and female reproductive organs. Cystic mesothelioma of the peritoneum, another rare form of the disease, occurs predominantly in women and is more benign in nature.

Malignant mesothelioma takes the form of one of three cell-types: epithelioid (50% to 70% of cases), sarcomatous (7% to 20% of cases), and biphasic/mixed (20% to 35% of cases). Of these cell-types, epithelioid mesothelioma carries the most favorable prognosis, followed by biphasic, and finally sarcomatous (very aggressive).

Mesothelioma remains relatively uncommon in the United States, with approximately 2,500 new cases reported annually. The incidence rates are much higher in Western Europe (over 5,000 cases reported annually). These numbers are expended to climb dramatically over the next 20 years. Older males (median age 60 at diagnosis) are three to five times more likely to develop mesothelioma than women. This is most likely due to male predominance in those professions with an increased risk of asbestos exposure.

Causes & symptoms

Approximately 80% of all mesothelioma patients have a history of asbestos exposure. The majority of these patients were employed in an industry that involved the use of asbestos in some fashion. In addition to occupational exposure, household exposure of family members is not uncommon. An exposed individual can carry the asbestos particles on their clothing, skin, and in their hair when they return home, resulting in paraoccupational exposure. Even brief exposure to asbestos, as little as one to two months, can result in long-term consequences. Although the dangers of asbestos have been known for decades, the long latency period of mesothelioma (30 to 40 years) means that majority of patients were already exposed as far back as the 1950s. It is estimated that up to eight million Americans have already been exposed. Several industries, in particular, show a higher incidence of asbestos exposure:

- Insulators (Asbestos workers)
- Boilermakers
- Ship-fitters
- Steel workers
- Maintenance workers
- Plumbers
- Brake mechanics

Symptoms

Mesothelioma is very aggressive once it truly takes hold. However, its initial symptoms are generally non-specific in nature and/or mimic other conditions, such as persistent **pneumonia** or gastronomic disorders. Some patients will exhibit no symptoms at all. As such, proper evaluation and diagnosis are commonly delayed and must be confirmed by a doctor.

Patients suffering from pleural mesothelioma most commonly exhibit signs of dyspnea, pleural effusions, and/or chest **pain**. The majority of pleural effusion symptoms will exhibit in the right lung (60% of the time). Patients may also exhibit persistent **cough**, weight loss, weakness, **fever**, and difficulty swallowing.

Patients suffering from peritoneal mesothelioma most commonly exhibit signs of pain and/or swelling in the abdomen from fluid retention or tumor growth. Weight loss, **nausea**, bowel obstruction, anemia, fever, and swelling in the legs and/or feet are also known symptoms.

Diagnosis

Only a physician can properly diagnose mesothelioma. A review of the patient's medical history, including any past exposure to asbestos, should be conducted for any patient displaying dyspnea, chest pain, fluid build-up, or pain and/or swelling in the abdomen. This review is followed up with a complete **physical examination**, which should involve the use of imaging techniques. X rays, computed tomography (CT) scans, and magnetic resonance (MRI) scans of the chest and/or abdomen, as well as lung function, can provide the doctor with critical diagnostic information. Although **positron emission tomography** scans are expensive and not covered under most insurance, this diagnostic tool has proven very useful in determining tumor sites and staging of the disease.

If indicated, the doctor may wish to internally examine the patient's chest and/or abdominal cavity. These diagnostic procedures, known as **thoracoscopy** (chest) and **peritoneoscopy** (abdomen), are usually

conducted in a hospital setting. Both procedures involve a fiber-optic imaging tool being inserted into the patient through an incision. These endoscopic tools will provide the doctor with a closer look at the body cavity, and any abnormal tissue or fluid build-up found therein. Excess fluid can be suctioned out through a needle or tube, in a process known as **thoracentesis** (for the chest) or **paracentesis** (for the abdomen). Additionally, the doctor may perform a biopsy of any abnormal tissue they discover during this time. Pathological examination of abnormal tissue, as well as fluid, remains the only effective method of confirming the diagnosis of mesothelioma. Biopsy will also assist the doctor in properly staging the disease's progression.

Once a confirmation of malignant mesothelioma has been established, the doctor will conduct further tests to determine the extent to which the primary disease has spread. This diagnostic process is known as "staging." Malignant pleural mesothelioma can be broken into four stages:

- **Localized Malignant Mesothelioma (Stage 1)**—Cancer is present in the right or left pleura. May involve the lung, the pericardium, or diaphragm on that side.
- **Advanced Malignant Mesothelioma (Stage 2)**—Cancer has spread beyond the right or left pleura to lymph nodes on that side. May involve the lung, the pericardium, or diaphragm on that side.
- **Advanced Malignant Mesothelioma (Stage 3)**—Cancer has spread into the chest wall, diaphragm, ribs, heart, esophagus, or through the abdominal lining. Nearby lymph nodes may or may not be involved.
- **Advanced Malignant Mesothelioma (Stage 4)**—Cancer shows evidence of metastasis or spread through the bloodstream to distant organs and/or tissues.

Recurrent malignant mesothelioma may also occur, where the cancer returns in its original location or elsewhere in the body even after treatment.

Treatment

There are three traditional treatment modalities for mesothelioma: surgery, **radiation therapy**, and **chemotherapy**. The location and the stage of the disease, as well as the patient's age and general health, will determine which treatment should be utilized. Additionally, these modalities can be combined if indicated. Indeed, the multimodality approach appears to provide the most positive results for treating mesothelioma.

Surgery, the most common treatment, involves the removal of the tumor. In the early stages of mesothelioma, this only involves removal of a section of the mesothelium and surrounding tissue, but may require removing part of the diaphragm as well. For more advanced stages of the disease, removing the entire lung may be the only option, which is known as pneumonectomy.

Radiation therapy, also known as radiotherapy, destroys and shrinks the cancer cells through various types of radiation. Both external (such as a machine) and internal (such as radioisotopes) radiation therapies can be utilized effectively to treat malignant mesothelioma.

Finally, chemotherapy, a systemic treatment modality, uses **anticancer drugs** to destroy the cancerous cells throughout the body. The majority of drugs used to treat mesothelioma are delivered intravenously. The effectiveness of intracavitary chemotherapy, the process of directly injecting the drugs into the chest or abdominal cavity, is being studied.

Pain and other symptoms caused by fluid build-up around the chest and/or abdomen can be treated by drain excess fluid through a needle or tube. These procedures are known as thoracentesis (chest) and paracentesis (abdomen). Drugs, radiotherapy, and surgery can also relieve or prevent further fluid accumulation.

Physicians are currently studying other treatment modalities, such as immunotherapy, **gene therapy**, and intraoperative photodynamic therapy.

Alternative treatment

Due to the poor prognosis associated with mesothelioma, regardless of proper treatment, in many cases palliative care is the preferred, and only, option available to patients. This is particularly true for the advanced stages of the disease. By treating the symptoms rather than the disease itself, the goal of this approach is to obtain “quality” of life instead of “quantity” of life. Palliative care aims to relieve the patient’s discomfort caused by dyspnea and pain. Chemotherapy, radiation, and/or surgical pleurodeisis, in combination with effective management of pain and respiratory function should form the basis of proper palliative care of mesothelioma. Techniques to reduce **stress**, such as **acupuncture**, **aromatherapy**, massage, and **reflexology**, can provide additional benefit to the patient’s sense of well-being.

Prognosis

The stage and location, what cell-type is involved, as well as the patient’s age and histology factor greatly

KEY TERMS

Asbestos—A naturally occurring mineral, utilized worldwide for its durability and heat resistant qualities. Extremely fibrous in nature, asbestos particles can easily enter the respiratory system and damage sensitive tissue. This damage can result in asbestosis, mesothelioma, and lung cancer.

Dyspnea—A difficulty in breathing or shortness of breath, typically associated with some form of heart or lung disease. Also known as air hunger.

Mesothelium—A membrane/sac that protects the body’s major internal organs and allows them freedom of movement (for example, lung contractions). The mesothelium is comprised of several regions, including the abdominal cavity (peritoneum), the chest cavity (pleura), and pericardium (heart).

Pleural effusion—An abnormal accumulation of fluid in the pleura, a fibrous membrane that lines the inside of the chest cavity and protects the lungs. This accumulation can cause shortness of breath, cough, and chest pain.

on life expectancy. Unfortunately, even with aggressive treatment, the prognosis for mesothelioma patients is poor. Pleural mesothelioma offers a median survival time of approximately 16 to 17 months after initial symptoms. Prognosis for peritoneal mesothelioma is poorer and has a median survival time of only ten months after initial symptoms. Unfortunately, the more advanced stages of mesothelioma may offer as little as four or five month’s survival time.

The survival time for patients with localized mesothelioma can be extended several months with aggressive therapy, with roughly 20% of patients surviving past the five-year mark. Therapy programs recently developed at leading cancer centers have extended this survival time even further. Dr. Sugarbaker, of the Brigham and Women’s Center in Boston, has achieved a median of 40% survival rate at five years with his treatment regimen for pleural mesothelioma, as reported in the *Journal of Thoracic and Cardiovascular Surgery*. Other programs are also exhibiting favorable results. However, despite such successes, no cure for mesothelioma currently exists.

Prevention

Avoiding or limiting exposure to asbestos is the best way to prevent mesothelioma. Unfortunately,

because of the significant delay between exposure and onset (30 to 40 years), it is probably too late to prevent the development of mesothelioma for most patients. Not **smoking** may slow the disease's progression and/or prevent other further complications associated with asbestos exposure.

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Mesothelioma Research Foundation of America. 5716 Corsa Ave., Suite 203, Westlake Village, CA 91362. (800) 281-9804. <<http://www.mesothelioma-rfa.org>>. The Mesothelioma Center. 1030 Fifth Ave., Pittsburgh, PA 15219. (412) 471-3980. <<http://mesotheliomacenter.org>>.

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Jason Fryer

Description

Metabolic acidosis, as a disruption of the body's acid/base balance, can be a mild symptom brought on by a lack of insulin, a **starvation** diet, or a gastrointestinal disorder like **vomiting** and **diarrhea**. Metabolic acidosis can indicate a more serious problem with a major organ like the liver, heart, or kidneys. It can also be one of the first signs of **drug overdose** or **poisoning**.

Causes and symptoms

Metabolic acidosis occurs when the body has more acid than base in it. Chemists use the term "pH" to describe how acidic or basic a substance is. Based on a scale of 14, a pH of 7.0 is neutral. A pH below 7.0 is an acid; the lower the number, the stronger the acid. A pH above 7.0 is a base; the higher the number, the stronger the base. Blood pH is slightly basic (alkaline), with a normal range of 7.36-7.44.

Acid is a natural by-product of the breakdown of fats and other processes in the body; however, in some conditions, the body does not have enough bicarbonate, an acid neutralizer, to balance the acids produced. This can occur when the body uses fats for energy instead of carbohydrates. Conditions where metabolic acidosis can occur include chronic **alcoholism**, **malnutrition**, and **diabetic ketoacidosis**. Consuming a diet low in carbohydrates and high in fats can also produce metabolic acidosis. The disorder may also be a symptom of another condition like kidney failure, liver failure, or severe diarrhea. The build up of lactic acid in the blood due to such conditions as **heart failure**, **shock**, or **cancer**, induces metabolic acidosis. Some poisonings and overdoses (**aspirin**, methanol, or ethylene glycol) also produce symptoms of metabolic acidosis.

In mild cases of metabolic acidosis, symptoms include **headache**, lack of energy, and sleepiness. Breathing may become fast and shallow. **Nausea**, vomiting, diarrhea, **dehydration**, and loss of appetite are also associated with metabolic acidosis. Diabetic patients with symptoms of metabolic acidosis may also have breath that smells fruity. The patient may lose consciousness or become disoriented. Severe cases can produce **coma** and **death**.

Diagnosis

Metabolic acidosis is suspected based on symptoms, but is usually confirmed by laboratory tests on blood and urine samples. Blood pH below 7.35 confirms the condition. Levels of other blood components, including potassium, glucose, ketones, or lactic

Metabolic acidosis

Definition

Metabolic acidosis is a pH imbalance in which the body has accumulated too much acid and does not have enough bicarbonate to effectively neutralize the effects of the acid.

KEY TERMS

Diabetic ketoacidosis—A condition caused by low insulin levels where the amount of sugar and ketones in the blood is high.

pH—A measurement of the acidity or alkalinity of a solution based on the amount of hydrogen ions available. Based on a scale of 14, a pH of 7.0 is neutral. A pH below 7.0 is an acid; the lower the number, the stronger the acid. A pH above 7.0 is a base; the higher the number, the stronger the base. Blood pH is slightly alkaline (basic) with a normal range of 7.36-7.44.

acid, may also be above normal ranges. The level of bicarbonate in the blood will be low, usually less than 22 mEq/L. Urine pH may fall below 4.5 in metabolic acidosis.

Treatment

Treatment focuses first on correcting the acid imbalance. Usually, sodium bicarbonate and fluids will be injected into the blood through a vein. An intravenous line may be started to administer fluids and allow for the quick injection of other drugs that may be needed. If the patient is diabetic, insulin may be administered. Drugs to regulate blood pressure or heart rate, to prevent seizures, or to control **nausea and vomiting** might be given. Vital signs like pulse, respiration, blood pressure, and body temperature will be monitored. The underlying cause of the metabolic acidosis must also be diagnosed and corrected.

Prognosis

If the metabolic acidosis is recognized and treated promptly, the patient may have no long-term complications, however, the underlying condition that caused the acidosis needs to be corrected or managed. Severe metabolic acidosis that is left untreated will lead to coma and death.

Prevention

Diabetic patients need to routinely test their urine for sugar and acetone, strictly follow their appropriate diet, and take any medications or insulin to prevent metabolic acidosis. Patients receiving **tube feedings** or intravenous feedings must be monitored to prevent

dehydration or the accumulation of ketones or lactic acid.

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Altha Roberts Edgren

Metabolic alkalosis

Definition

Metabolic alkalosis is a pH imbalance in which the body has accumulated too much of an alkaline substance, such as bicarbonate, and does not have enough acid to effectively neutralize the effects of the alkali.

Description

Metabolic alkalosis, as a disturbance of the body's acid/base balance, can be a mild condition, brought on by **vomiting**, the use of steroids or diuretic drugs, or the overuse of **antacids** or **laxatives**. Metabolic alkalosis can also indicate a more serious problem with a major organ such as the kidneys.

Causes and symptoms

Metabolic alkalosis occurs when the body has more base than acid in the system. Chemists use the term “pH” to describe how acidic or alkaline (also called basic) a substance is. Based on a scale of 14, a pH of 7.0 is neutral. A pH below 7.0 is an acid; the lower the number, the stronger the acid. A pH above 7.0 is alkaline; the higher the number, the stronger the alkali. Blood pH is slightly alkaline, with a normal range of 7.36-7.44. Conditions that lead to a reduced amount of fluid in the body, like vomiting or excessive urination due to use of diuretic drugs, change the balance of fluids and salts. The blood levels of potassium and sodium can decrease dramatically, causing symptoms of metabolic alkalosis.

In cases of metabolic alkalosis, slowed breathing may be an initial symptom. The patient may have episodes of apnea (not breathing) that may go on 15 seconds or longer. **Cyanosis**, a bluish or purplish discoloration of the skin, may also develop as a sign of

KEY TERMS

pH—A measurement of the acidity or alkalinity of a solution based on the amount of hydrogen ions available. Based on a scale of 14, a pH of 7.0 is neutral. A pH below 7.0 is an acid; the lower the number, the stronger the acid. A pH above 7.0 is a base; the higher the number, the stronger the base. Blood pH is slightly alkaline (basic) with a normal range of 7.36-7.44.

inadequate oxygen intake. **Nausea**, vomiting, and **diarrhea** may also occur. Other symptoms can include irritability, twitching, confusion, and picking at bedclothes. Rapid heart rate, irregular heart beats, and a drop in blood pressure are also symptoms. Severe cases can lead to convulsions and **coma**.

Diagnosis

Metabolic alkalosis may be suspected based on symptoms, but often may not be noticeable. The condition is usually confirmed by laboratory tests on blood and urine samples. Blood pH above 7.45 confirms the condition. Levels of other blood components, including salts like potassium, sodium, and chloride, fall below normal ranges. The level of bicarbonate in the blood will be high, usually greater than 29 mEq/L. Urine pH may rise to about 7.0 in metabolic alkalosis.

Treatment

Treatment focuses first on correcting the imbalance. An intravenous line may be started to administer fluids (generally normal saline, a salt water solution) and allow for the quick injection of other drugs that may be needed. Potassium chloride will be administered. Drugs to regulate blood pressure or heart rate, or to control **nausea and vomiting** might be given. Vital signs like pulse, respiration, blood pressure, and body temperature will be monitored. The underlying cause of the metabolic alkalosis must also be diagnosed and corrected.

Prognosis

If metabolic alkalosis is recognized and treated promptly, the patient may have no long-term complications; however, the underlying condition that caused the alkalosis needs to be corrected or managed.

Severe metabolic alkalosis that is left untreated will lead to convulsions, **heart failure**, and coma.

Prevention

Patients receiving **tube feedings** or intravenous feedings must be monitored to prevent an imbalance of fluids and salts, particularly potassium, sodium, and chloride. Overuse of some drugs, including **diuretics**, laxatives, and antacids, should be avoided.

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Metabolic encephalopathy see **Delirium**

Meth see **Muscle relaxants**

Methadone

Definition

Methadone is a powerful narcotic drug in the same class as heroin. This class is known as the opioids.

Purpose

Methadone, formerly known as dolophine, is a psycho-active drug, meaning that it affects the mind or behavior. It belongs to the class of opioids, drugs that share some of the analgesic properties, and mimic the action of some of the body's naturally occurring chemicals called peptides, such as endorphins and enkephalines.

Methadone is used to relieve chronic **pain in cancer** patients and as a maintenance drug to control withdrawal symptoms in people undergoing treatment for opiate **addiction**.

In opiate addiction treatment, methadone blocks the opioid receptors of the brain that bind opiates such as heroin. The blocking of these receptors leads to two major effects:

- because these chemical receptors remain blocked by methadone for up to 24 hours, even if a person addicted to heroin takes heroin after the administration of methadone, this person is not likely to feel the same effects of the heroin as he or she previously felt;
- because the action of methadone is associated with slower and less intense withdrawal symptoms than those of heroin, the patient can experience milder opiate effects while the addiction is being treated and avoid the unpleasant withdrawal symptoms associated with heroin.

Methadone has also been shown to reduce cravings for heroin while not altering a person's mood.

Precautions

Methadone magnifies the effects of alcohol and other **central nervous system depressants**, such as **anti-histamines**, cold medicines, sedatives, tranquilizers, other prescription and over-the-counter (OTC) pain medications, **barbiturates**, seizure medications, **muscle relaxants**, and certain anesthetics including some dental anesthetics. Alcohol and other central nervous system depressants should not be taken or consumed while methadone is being taken.

Methadone is a powerful narcotic. It can cause some people to feel drowsy, dizzy, or light-headed. People taking methadone should not drive a car or operate machinery.

Intentional or accidental overdose of methadone can lead to unconsciousness, **coma**, or **death**. The signs of methadone overdose include confusion, difficulty speaking, seizures, severe nervousness or restlessness, severe **dizziness**, severe drowsiness, and/or slow or troubled breathing. These symptoms are increased by alcohol or other central nervous system (CNS) depressants. Anyone who feels that he or she, or someone else, may have overdosed on methadone, or a combination of methadone and other central nervous system depressants, should seek emergency medical attention for that person at once.

Description

A typical adult dosage for methadone is 5–20 mg as an oral solution, 2.5–10 mg as an oral tablet or injection, every four to eight hours as necessary for pain. When used for **detoxification**, methadone is

initially given in a dose of 15–100 mg per day as an oral solution. This dose is then decreased until the patient no longer requires the medication. The injection form of methadone is only used for detoxification in patients who are unable to take the medication by mouth.

Preparation

No preparation is generally necessary prior to the intake of methadone as a pain reliever. In cases of maintenance treatments, it is important to be sure that the patient is not currently intoxicated by alcohol, heroin, other opioids, or taking other central nervous system depressants.

Aftercare

Patients receiving methadone should be monitored for adverse reactions to this drug, and/or possible accidental overdose.

Risks

Methadone can interfere with or exacerbate certain medical conditions. For these reasons, it is important that the prescribing physician be informed of any current case, or history of:

- alcohol abuse
- brain disease or head injury
- colitis
- drug dependency, particularly of **narcotics**
- emotional problems
- emphysema, **asthma**, or other chronic lung disease
- enlarged prostate
- gallstones or gallbladder disease
- heart disease
- kidney disease
- liver disease
- problems with urination
- seizures
- underactive thyroid

Side effects

The most common side effects of methadone include:

- constipation
- dizziness

- drowsiness
- itching
- nausea
- urine retention
- vomiting

Less common side effects of methadone include:

- abnormally fast or slow heartbeat
- blurred or double vision
- cold, clammy skin
- depression or other mood changes
- dry mouth
- fainting
- hallucinations
- hives
- loss of appetite
- nightmares or unusual dreams
- pinpoint pupils of the eyes
- redness or flushing of the face
- restlessness
- rigid muscles
- ringing or buzzing in the ears
- seizure
- severe drowsiness
- skin reaction at the site of injection
- stomach cramps or pain
- sweating
- trouble sleeping (insomnia)
- yellowing of the skin or whites of the eyes

Normal results

Normal results after the administration of methadone to treat chronic pain is the alleviation of that patient's pain, at least to the point where the pain is bearable.

Normal results of methadone treatment to control heroin addiction, is that the patient reduces heroin intake almost immediately upon starting methadone treatments, followed by complete abstinence, usually within two weeks after starting treatment.

KEY TERMS

Analgesic—Any agent that relieves pain.

Central nervous system (CNS) depressant—Any drug that tends to reduce the activity of the central nervous system. The major drug categories included in this classification are: alcohol, anesthetics, anti-anxiety medications, antihistamines, antipsychotics, hypnotics, narcotics, sedatives, and tranquilizers.

Endorphins—Any of several opiate peptides naturally produced in the brain that bind to certain neuron receptors and have the effect of relieving pain.

Enkephalines—Peptide produced by the body that have analgesic properties.

Morphine—Morphine is the naturally occurring opioid in the opium poppy, *Papaver somniferum*. It is a powerful narcotic analgesic, and its primary clinical use is in the management of moderately severe to severe pain. After heroin, morphine has the greatest potential for addiction of all narcotic analgesics.

Narcotic—Any drug that produces insensibility or stupor and/or generally causes effects similar to those caused by morphine.

Opiate—Any narcotic analgesic derived from a natural source, such as morphine from the opium poppy.

Opioid receptors—Receptors located in the brain and various organs that bind opiates or opioid substances.

Opioids—One of the major classes of semi or fully synthetic psycho-active drugs that includes methadone.

Psychoactive drugs—Any drug that affects the mind or behavior. There are five main classes of psychoactive drugs: opiates and opioids (e.g. heroin and methadone); stimulants (e.g. cocaine, nicotine), depressants (e.g. tranquilizers, antipsychotics, alcohol), hallucinogens (e.g. LSD), and marijuana and hashish.

Receptor—A molecular structure on the surface that selectively binds a specific substance resulting in a specific physiological effect.

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National Clearinghouse for Alcohol and Drug Information. 11426-28 Rockville Pike, Suite 200, Rockville, MD 20852. (800) 729-6686. < <http://www.health.org/> >.

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Methemoglobinemia

Definition

When excessive hemoglobin in the blood is converted to another chemical that cannot deliver oxygen to tissues, called methemoglobin.

Description

The molecule hemoglobin in the blood is responsible for binding oxygen to give to the body. When hemoglobin is oxidized to methemoglobin its structure changes and it is no longer able to bind oxygen. Hemoglobin is constantly under oxidizing stresses; however, normally less than 1% of a person's hemoglobin is in the methemoglobin state. This is due to the body's systems that reduce methemoglobin back to hemoglobin. Infants have a higher risk of acquiring methemoglobinemia because infant hemoglobin is more prone to be oxidized to methemoglobin.

Causes and symptoms

Methemoglobinemia can either be congenital or acquired.

There are two causes of the congenital form. One cause is a defect in the body's systems to reduce methemoglobin to hemoglobin. The other cause is a mutant form of hemoglobin called hemoglobin M that cannot bind to oxygen. Both of these forms are typically benign.

Acquired methemoglobinemia is caused by an external source, usually a drug or medication. Some of these medications include benzocaine, lidocaine and prilocaine. These medications can inhibit the body's systems of reducing methemoglobin to hemoglobin resulting in methemoglobinemia.

With a methemoglobin level of 3-15% skin can turn to a pale gray or blue (**cyanosis**). With levels above 25% the following symptoms may be present:

- Cyanosis unaffected by oxygen administration
- Blood that is dark or chocolate in color that will not change to red in the presence of oxygen
- Headache
- Weakness
- Confusion
- Chest pain

When methemoglobin levels are above 70% **death** may result if not treated immediately.

Diagnosis

Diagnosis is based on the symptoms and history. If these are indicative of methemoglobinemia blood tests are performed to confirm the presence and level of methemoglobin.

Treatment

For acquired methemoglobinemia the typical treatment is with methylene blue. This is administered with an IV over a five-minute period and results are typically seen within 20 minutes. Methylene blue reduces methemoglobin back to hemoglobin.

Though congenital methemoglobinemia is usually benign, the form due to a defective reducing system can be treated with ascorbic acid (vitamin C) taken daily. The other congenital form due to hemoglobin M has no treatment as of late.

Alternative treatment

There are not any known alternative treatments for methemoglobinemia. Methylene blue, or a similar treatment, is needed to reduce methemoglobin to hemoglobin.

Prognosis

If found early, acquired methemoglobinemia can be easily treated with no side effects. After treatment with methylene blue the patient can expect a full recovery.

Congenital methemoglobinemia is typically benign and should be observed. If methemoglobinemia symptoms occur the person should be taken to the hospital for treatment.

Prevention

If a person gets methemoglobinemia from a certain medication that medication should be avoided at

KEY TERMS

Cyanosis—When the body does not receive enough oxygen.

Oxidation—When a chemical element or compound loses an electron.

Reduction—When a chemical element or compound gains an electron.

all costs in the future. For people with congenital methemoglobinemia medications or other things that are known to oxidize hemoglobin should be avoided.

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Methylphenidate see **Central nervous system stimulants**

Metoprolol see **Beta blockers**

Metronidazole see **Antiprotozoal drugs**

Micronaz see **Antifungal drugs, topical**

Microphthalmia and anophthalmia

Definition

Anophthalmia is the complete absence of an eye. Microphthalmia is an eye that has an abnormal smallness.

Description

Anophthalmia is caused by a defect in embryonic development. The total absence of an eye is extremely rare and often a clinical sign associated with a broad range of genetic disorders or, more commonly, a sporadic mutation. Sporadic transmission occurs in the affected individual due to a genetic abnormality. It is not passed on from the parents, but usually due to a combination of environmental and genetic influences. More commonly anophthalmia clinically presents as a small cyst. The defect, which causes anophthalmia, is an absence of the optic vesicle, a structure important for eye development. The genetic abnormality usually occurs during weeks one to three after conception. It is estimated that the incidence of microphthalmia occurs 0.22 times per 1,000 live births. Anophthalmia can occur during adult life but not associated with a genetic cause.

Microphthalmia refers to an abnormally small eye. This clinical sign is often associated with autosomal dominant or recessively transmitted genetic disorders. Most disorders dominantly inherited with microphthalmia are associated with some visual capabilities in infancy and early childhood. Microphthalmia may be isolated (the only presenting sign) or associated with a range of ocular or systemic abnormalities. Isolated cases of microphthalmia may be sporadic or inherited. There is a variable degree of **visual impairment**. Microphthalmia occurs due to autosomal recessive transmission and is part of a syndrome associated with abnormalities in the retina or systemic lesions. Microphthalmia results from a developmental defect after formation of the optic vesicle. The developmental abnormality causes the optic vesicle to fold inwards, resulting in the formation of a cyst. The cyst will progressively swell from birth, and it may be situated along the optic nerve. The cyst may also be situated along other important eye structures.

Causes and symptoms

Microphthalmia and anophthalmia can be caused by sporadic or genetic mutations. Anophthalmia is characterized by a total absence of an eye. Anophthalmia in an adult is usually caused by trauma, infection, tumor, or advanced eye disease.

Diagnosis

Microscope examination confirms the diagnosis of true anophthalmia. The clinician examines a piece of tissue taken from the eye and notes eviscerated tissue. For microphthalmia the confirmation can be

KEY TERMS

Axial—A straight line passing through a spherical body between its two poles and about which the body may revolve.

Eviserated—Removal of eye contents.

Prostheses—A synthetic object that resembles a missing anatomical part.

Retina—A major portion of the eye responsible for reception of visual light rays.

established by eye measurements. Eyes, which have an axial length <21 mm in an adult, or <19 mm in a one-year-old child are described as having microphthalmia.

Treatment

Large cysts causing microphthalmia should be aspirated or removed surgically. There is no known cure for anophthalmia or microphthalmia. For anophthalmia a prosthetic eye can be fitted which may involve surgery. Treatment for microphthalmia depends on the complexity of eye involvement.

Prognosis

For anophthalmia, prosthetic eyes should be seen by a specialist two to three times per year to assess fit, mobility, and smoothness. They are usually well tolerated and have good appearance and mobility. The clinical course for microphthalmia depends on the extent of smallness, but usually patients progress favorably without major treatment. Since the smallness is distinctly noticeable, there may be individual cosmetic considerations.

Prevention

There is no known prevention for either, since these clinical signs are commonly associated with genetic inheritance.

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Middle ear infection see **Otitis media**

Mifeprex see **Mifepristone**

Mifepristone

Definition

Mifepristone is a pill that can be taken as an alternative to a surgical abortion.

Purpose

This medication most often is used for ending early pregnancies. In 2003, studies were surfacing reporting other possible uses for mifepristone, at least at low doses. These studies included its possible use in treating psychotic depression. Low-dose mifepristone also showed success as a treatment for **uterine fibroids**, or benign growths in the muscular tissue of a woman's uterus. However, many uses of the drug other than for abortions were still experimental, even if promising.

Mifepristone's primary use for medical abortions is preferred by many over surgical approaches to abortions. It has emerged as a form of **emergency contraception** when taken at low doses within a short time period following possible conception.

Precautions

Women who are more than seven weeks pregnant (or 49 days since their last menstrual period) should not take mifepristone. Other reasons to avoid mifepristone include: use of an intrauterine device (**IUD**), **ectopic pregnancy**, use of blood thinners, bleeding disorders, use of steroid medications, **allergies** to mifepristone or similar drugs and lack of access to medical help within two weeks after the treatment. When the drug is used at low doses as emergency **contraception**, it poses few side effects.

Description

Mifepristone, sold commercially under the name Mifeprex, also is known as RU-486, the abortion pill, the early option pill for medical abortion. While it has been used for many years in Europe, mifepristone has only been available for use in the United States since the U.S. Food and Drug Administration (FDA) approved it in 2000 for use in abortion. More than 37,000 abortions were performed using the pill in the first six months of 2001.

This drug causes **pregnancy** to end by blocking the female hormone progesterone. The lack of progesterone makes the uterus shed its lining, which causes bleeding similar to a menstrual period. Three days after taking mifepristone, women are given a second drug, misoprostol, to cause uterine contractions that expel the contents of the uterus. Most women are able remain in their own home while they pass the fetus.

Preparation

Before taking mifepristone, health care providers likely will give the woman a urine or blood test to be sure that she is, in fact, pregnant. They also may give her some counseling and support. Once she has made the decision to use mifepristone, they will ask her to sign a written statement that she has decided to end her pregnancy.

Aftercare

Using mifepristone and misoprostol causes heavy bleeding and cramping. Doctors can offer **pain** medicine, such as Motrin, to ease the cramps. For two weeks after treatment with mifepristone, health care providers likely will ask patients to abstain from sexual intercourse, heavy lifting and strenuous **exercise**. They also may advise against breast-feeding, since scientists are not sure if the drug is present in breast milk.

Physicians require patients to come in for a follow-up visit 14 days after their first dose of mifepristone to verify that they are no longer pregnant and that they are properly healing.

Risks

Other common side effects include: **fatigue**, headaches, **dizziness**, **nausea**, **vomiting**, **diarrhea** and low-back pain.

Since pregnancy hormones are in flux after a medical abortion, many women have emotional

KEY TERMS

Dilation and curettage (D and C)—During this surgical procedure, a physician dilates the cervix, then uses an instrument to scrape tissue away from the walls of the uterus.

Misoprostol—A drug used in combination with mifepristone to cause uterine contractions that expel the contents of the uterus.

side-effects, such as mood swings, depression or a mild case of the blues. These feelings usually subside when hormones stabilize a few weeks later. For those who feel stuck in their grief or anger about the situation, counseling or support groups may offer relief.

Normal results

Most women feel better after about two weeks. Bleeding and spotting usually occurs for 9-16 days, but may last for a month.

Abnormal results

In some cases, mifepristone does not completely end the pregnancy. If the fetus is still left inside the uterus, a doctor may recommend a surgical abortion, or a procedure called dilation and curettage (D and C). About five to eight of every 100 women who take mifepristone go on to have a surgical abortion, according to the FDA. During a D and C, which usually is done at a hospital or clinic under a local anesthetic, a physician dilates the cervix, then uses an instrument to scrape any residual tissue away from the walls of the uterus. This allows the heavy bleeding to eventually stop so a woman can return to her normal cycle sooner.

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Migraine headache

Definition

Migraine is a type of **headache** marked by severe head **pain** lasting several hours or more.

Description

Migraine is an intense and often debilitating type of headache. Migraines affect as many as 24 million people in the United States, and are responsible for billions of dollars in lost work, poor job performance, and direct medical costs. Approximately 18% of women and 6% of men experience at least one migraine attack per year. More than three million women and one million men have one or more severe headaches every month. Migraines often begin in adolescence, and are rare after age 60.

Two types of migraine are recognized. Eighty percent of migraine sufferers experience "migraine without aura" (common migraine). In "migraine with aura," or classic migraine, the pain is preceded or accompanied by visual or other sensory disturbances, including **hallucinations**, partial obstruction of the visual field, **numbness** or **tingling**, or a feeling of heaviness. Symptoms are often most prominent on one side of the head or body, and may begin as early as 72 hours before the onset of pain.

Causes and symptoms

Causes

The physiological basis of migraine has proved difficult to uncover. Genetics appear to play a part for many, but not all, people with migraine. There are a multitude of potential triggers for a migraine attack, and recognizing one's own set of triggers is the key to prevention.

PHYSIOLOGY. The most widely accepted hypothesis of migraine suggests that a migraine attack is precipitated when pain-sensing nerve cells in the brain (called nociceptors) release chemicals called neuropeptides. At least one of the neurotransmitters,

substance P, increases the pain sensitivity of other nearby nociceptors.

Other neuropeptides act on the smooth muscle surrounding cranial blood vessels. This smooth muscle regulates blood flow in the brain by relaxing or contracting, thus constricting the enclosed blood vessels and stimulating adjacent pain receptors. At the onset of a migraine headache, neuropeptides are thought to cause muscle relaxation, allowing vessel dilation and increased blood flow. Other neuropeptides increase the leakiness of cranial vessels, allowing fluid leak, and promote inflammation and tissue swelling. The pain of migraine is thought to result from this combination of increased pain sensitivity, tissue and vessel swelling, and inflammation. The aura seen during a migraine may be related to constriction in the blood vessels that dilate in the headache phase.

GENETICS. Susceptibility to some types of migraine is inherited. A child of a migraine sufferer has as much as a 50% chance of developing migraines. If both parents are affected, the chance rises to 70%. In 2002, a team of Australian researchers identified a region on human chromosome 1 that influences susceptibility to migraine. It is likely that more than one gene is involved in the inherited forms of the disorder. Many cases of migraine, however, have no obvious familial basis. It is likely that the genes that are involved set the stage for migraine, and that full development requires environmental influences, as well.

Two groups of Italian researchers have recently identified two loci on human chromosomes 1 and 14 respectively that are linked to migraine headaches. The locus on chromosome 1q23 has been linked to familial hemiplegic migraine type 2, while the locus on chromosome 14q21 is associated with migraine without aura.

TRIGGERS. A wide variety of foods, drugs, environmental cues, and personal events are known to trigger migraines. It is not known how most triggers set off the events of migraine, nor why individual migraine sufferers are affected by particular triggers but not others.

Common food triggers include:

- cheese
- alcohol
- **caffeine** products, and caffeine withdrawal
- chocolate
- intensely sweet foods

- dairy products
- fermented or pickled foods
- citrus fruits
- nuts
- processed foods, especially those containing nitrites, sulfites, or monosodium glutamate (msg)

Environmental and event-related triggers include:

- **stress** or time pressure
- menstrual periods, **menopause**
- sleep changes or disturbances, oversleeping
- prolonged overexertion or uncomfortable posture
- hunger or **fasting**
- odors, smoke, or perfume
- strong glare or flashing lights

Drugs which may trigger migraine include:

- oral contraceptives
- estrogen replacement therapy
- nitrates
- theophylline
- reserpine
- nifedipine
- indomethacin
- cimetidine
- decongestant overuse
- analgesic overuse
- benzodiazepine withdrawal

Symptoms

Migraine without aura may be preceded by elevations in mood or energy level for up to 24 hours before the attack. Other pre-migraine symptoms may include **fatigue**, depression, and excessive yawning.

Aura most often begins with shimmering, jagged arcs of white or colored light progressing over the visual field in the course of 10-20 minutes. This may be preceded or replaced by dark areas or other visual disturbances. **Numbness and tingling** is common, especially of the face and hands. These sensations may spread, and may be accompanied by a sensation of weakness or heaviness in the affected limb.

The pain of migraine is often present only on one side of the head, although it may involve both, or switch sides during attacks. The pain is usually throbbing, and may range from mild to incapacitating. It is often accompanied by **nausea** or **vomiting**, painful

sensitivity to light and sound, and intolerance of food or odors. Blurred vision is common.

Migraine pain tends to intensify over the first 30 minutes to several hours, and may last from several hours to a day or longer. Afterward, the affected person is usually weary, and sensitive to sudden head movements.

Diagnosis

Ideally, migraine is diagnosed by a careful medical history. Unfortunately, migraine is underdiagnosed because many doctors tend to minimize its symptoms as “just a headache.” According to a 2003 study, 64% of migraine patients in the United Kingdom and 77% of those in the United States never receive a correct medical diagnosis for their headaches.

So far, laboratory tests and such imaging studies as computed tomography (CT scan) or **magnetic resonance imaging** (MRI) scans have not been useful for identifying migraine. However, these tests may be necessary to rule out a **brain tumor** or other structural causes of migraine headache in some patients.

Treatment

Once a migraine begins, the person will usually seek out a dark, quiet room to lessen painful stimuli. Several drugs may be used to reduce the pain and severity of the attack.

Nonsteroidal anti-inflammatory drugs (NSAIDs) are helpful for early and mild headache. NSAIDs include **acetaminophen**, ibuprofen, naproxen, and others. A recent study concluded that a combination of acetaminophen, **aspirin**, and caffeine could effectively relieve symptoms for many migraine patients. One such over-the-counter preparation is available as Exedrin Migraine.

More severe or unresponsive attacks may be treated with drugs that act on serotonin receptors in the smooth muscle surrounding cranial blood vessels. Serotonin, also known as 5-hydroxytryptamine, constricts these vessels, relieving migraine pain. Drugs that mimic serotonin and bind to these receptors have the same effect. The oldest of them is ergotamine, a derivative of a common grain fungus. Ergotamine and dihydroergotamine are used for both acute and preventive treatment. Derivatives with fewer side effects have come onto the market in the past decade, including sumatriptan (Imitrex). Some of these drugs are available as nasal sprays, intramuscular injections, or rectal suppositories for patients in whom vomiting precludes oral

administration. Other drugs used for acute attacks include meperidine and metoclopramide.

Studies are showing that rizatriptan is a promising drug for the treatment of migraines. One study showed that 10mg of rizatriptan provided relief to 90% of the patients in the study group and kept 50% of them pain-free 2 hours after taking the medication. Sumatriptan has been on the market since 1993, while rizatriptan became available in 1998.

Sumatriptan and other triptan drugs (zolmitriptan, rizatriptan, naratriptan, almotriptan, and frovatriptan) should not be taken by people with any kind of vascular disease because they cause coronary artery narrowing. Otherwise these drugs have been shown to be very safe.

Continued use of some **antimigraine drugs** can lead to “rebound headache,” marked by frequent or chronic headaches, especially in the early morning hours. Rebound headache can be avoided by using antimigraine drugs under a doctor’s supervision, with the minimum dose necessary to treat symptoms. Tizanidine (Zanaflex) has been reported to be effective in treating rebound headaches when taken together with an NSAID.

Alternative treatments

Alternative treatments are aimed at prevention of migraine. Migraine headaches are often linked with **food allergies** or intolerances. Identification and elimination of the offending food or foods can decrease the frequency of migraines and/or alleviate these headaches altogether. Herbal therapy with feverfew (*Chrysanthemum parthenium*) may lessen the frequency of attacks. Learning to increase the flow of blood to the extremities through **biofeedback** training may allow a patient to prevent some of the vascular changes once a migraine begins. During a migraine, keep the lights low; put the feet in a tub of hot water and place a cold cloth on the occipital region (the back of the head). This treatment draws the blood to the feet and decreases the pressure in the head.

Prognosis

Most people with migraines can bring their attacks under control through recognizing and avoiding triggers, and by use of appropriate drugs when migraine occurs. Some people with severe migraines do not respond to preventive or drug therapy. Migraines usually wane in intensity by age 60 and beyond.

KEY TERMS

Aura—A group of visual or other sensations that precedes the onset of a migraine attack.

Coenzyme Q₁₀—A substance used by cells in the human body to produce energy for cell maintenance and growth. It is being studied as a possible preventive for migraine headaches.

Nociceptor—A specialized type of nerve cell that senses pain.

Prevention

The frequency of migraine may be lessened by avoiding triggers. It is useful to keep a headache journal, recording the particulars and noting possible triggers for each attack. Specific measures which may help include:

- Eating at regular times, and not skipping meals.
- Reducing the use of caffeine and pain-relievers.
- Restricting physical exertion, especially on hot days.
- Keeping regular sleep hours, but not oversleeping.
- Managing one’s time efficiently in order to avoid stress at work and home.

Some drugs can be used for migraine prevention, including specific members of these drug classes:

- beta blockers
- tricyclic antidepressants
- calcium channel blockers
- selective serotonin reuptake inhibitors (SSRIs)
- monoamine oxidase inhibitors (MAOIs)
- serotonin antagonists

One substance that is being studied as a possible migraine preventive is coenzyme Q₁₀, a compound used by cells to produce energy needed for cell growth and maintenance. Coenzyme Q₁₀ has been studied as a possible complementary treatment for **cancer**. Its use in preventing migraines is encouraging and merits further study.

A study published in early 2003 reported that three drugs currently used to treat disorders of muscle tone are being explored as possible preventive treatments for migraine. They are botulinum toxin type A (Botox), baclofen (Lioresal), and tizanidine (Zanaflex). Early results of open trials of these medications are positive.

Anti-epileptic drugs, which are also known as anticonvulsants, are also being studied as possible migraine preventives. As of 2003, sodium valproate (Epilim) is the only drug approved by the Food and Drug Administration (FDA) for prevention of migraine. Such newer anticonvulsants as gabapentin (Neurontin) and topiramate (Topamax) are presently being evaluated as migraine preventives.

A natural preparation made from butterbur root (*Petasites hybridus*) has been sold in Germany since the 1970s as a migraine preventive under the trade name Petadolex. Petadolex has been available in the United States since December 1998 and has passed several clinical safety and postmarketing surveillance trials.

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- American Council for Headache Education. 19 Mantua Road, Mt. Royal, NJ 08061. (609) 423-0043 or (800) 255-2243. <<http://www.achenet.org>>.
- National Headache Foundation. 428 West St. James Place, Chicago, IL 60614. (773) 388-6399 or (800) 843-2256. <<http://www.headaches.org>>.
- U. S. Food and Drug Administration (FDA). 5600 Fishers Lane, Rockville, MD 20857. (888) 463-6332. <<http://www.fda.gov>>.

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Miliaria see **Prickly heat**

Mineral deficiency

Definition

The term mineral deficiency means a condition where the concentration of any one of the **minerals** essential to human health is abnormally low in the body. In some cases, an abnormally low mineral concentration is defined as that which leads to an impairment in a function dependent on the mineral. In other cases, the convention may be to define an abnormally low mineral concentration as a level lower than that found in a specific healthy population.

The mineral nutrients are defined as all the inorganic elements or inorganic molecules that are required for life. As far as human **nutrition** is concerned, the inorganic nutrients include water, sodium, potassium, chloride, calcium, phosphate, sulfate, magnesium, iron, copper, zinc, manganese, iodine, selenium, and molybdenum. Some of the inorganic nutrients, such as water, do not occur as single atoms, but occur as molecules. Other inorganic nutrients that are molecules include phosphate, sulfate, and selenite. Phosphate contains an atom of phosphorus. Sulfate contains an atom of sulfur. We do not need to eat sulfate, since the body can acquire all the sulfate it needs from protein. Selenium occurs in foods as selenite and selenate.

There is some evidence that other inorganic nutrients, such as chromium and boron, play a part in human health, but their role is not well established. Fluoride has been proven to increase the strength of bones and teeth, but there is little or no reason to believe that is needed for human life.

The mineral content of the body may be measured by testing samples of blood plasma, red blood cells, or urine. In the case of calcium and phosphate deficiency, the diagnosis may also involve taking x rays of the skeleton. In the case of iodine deficiency, the diagnosis may include examining the patient's neck with the eyes and hands. In the case of iron deficiency, the diagnosis may include the performance of a stair-stepping test by the patient. Since all the minerals serve strikingly different functions in the body, the tests for the corresponding deficiency are markedly different from each other.

Description

Laboratory studies with animals have revealed that severe deficiencies in any one of the inorganic nutrients can result in very specific symptoms, and finally in **death**, due to the failure of functions associated with that nutrient. In humans, deficiency in one nutrient may occur less often than deficiency in several nutrients. A patient suffering from **malnutrition** is deficient in a variety of nutrients. In the United States, malnutrition is most often found among severe alcoholics. In part, this is because the alcohol consumption may supply half of the energy requirement, resulting in a mineral and vitamin intake of half the expected level. Deficiencies in one nutrient do occur, for example, in human populations living in iodine-poor regions of the world, and in iron deficient persons who lose excess iron by abnormal bleeding.

Inorganic nutrients have a great variety of functions in the body. Water, sodium, and potassium

deficiencies are most closely associated with abnormal nerve action and cardiac **arrhythmias**. Deficiencies in these nutrients tend to result not from a lack of content in the diet, but from excessive losses due to severe **diarrhea** and other causes. Iodine deficiency is a global public health problem. It occurs in parts of the world with iodine-deficient soils, and results in **goiter**, which involves a relatively harmless swelling of the neck, and cretinism, a severe birth defect. The only use of iodine in the body is for making thyroid hormone. However, since thyroid hormone has a variety of roles in development of the embryo, iodine deficiency during **pregnancy** results in a number of **birth defects**.

Calcium deficiency due to lack of dietary calcium occurs only rarely. However, calcium deficiency due to **vitamin D deficiency** can be found among certain populations. Vitamin D is required for the efficient absorption of calcium from the diet, and hence vitamin D deficiency in growing infants and children can result in calcium deficiency.

Dietary phosphate deficiency is rare because phosphate is plentiful in plant and animal foods, but also because phosphate is efficiently absorbed from the diet into the body. Iron deficiency causes anemia (lack of red blood cells), which results in tiredness and **shortness of breath**.

Dietary deficiencies in the remaining inorganic nutrients tend to be rare. Magnesium deficiency is uncommon, but when it occurs it tends to occur in chronic alcoholics, in persons taking diuretic drugs, and in those suffering from severe and prolonged diarrhea. Magnesium deficiency tends to occur with the same conditions that provoke deficiencies in sodium and potassium. Zinc deficiency is rare, but it has been found in impoverished populations in the Middle East, who rely on unleavened whole wheat bread as a major food source. Copper deficiency is also rare, but dramatic and health-threatening changes in copper metabolism occur in two genetic diseases, Wilson's disease and Menkes' disease.

Selenium deficiency may occur in regions of the world where the soils are poor in selenium. Low-selenium soils can produce foods that are also low in selenium. Premature infants may also be at risk for selenium deficiency. Manganese deficiency is very rare. Experimental studies with humans fed a manganese deficient diet have revealed that the deficiency produces a scaly, red rash on the skin of the upper torso. Molybdenum deficiency has probably never occurred, but indirect evidence suggests that if molybdenum deficiency could occur, it would result in **mental retardation** and death.

Causes and symptoms

Sodium deficiency (**hyponatremia**) and water deficiency are the most serious and widespread deficiencies in the world. These deficiencies tend to arise from excessive losses from the body, as during prolonged and severe diarrhea or **vomiting**. Diarrheal diseases are a major world health problem, and are responsible for about a quarter of the 10 million infant deaths that occur each year. Nearly all of these deaths occur in impoverished parts of Africa and Asia, where they result from contamination of the water supply by animal and human feces.

The main concern in treating diarrheal diseases is **dehydration**, that is, the losses of sodium and water which deplete the fluids of the circulatory system (the heart, veins, arteries, and capillaries). Severe losses of the fluids of the circulatory system result in **shock**. Shock nearly always occurs when dehydration is severe enough to produce a 10% reduction in body weight. Shock, which is defined as inadequate supply of blood to the various tissues of the body, results in a lack of oxygen to all the cells of the body. Although diarrheal fluids contain a number of electrolytes, the main concern in avoiding shock is the replacement of sodium and water.

Sodium deficiency and potassium deficiency also frequently result during treatment with drugs called **diuretics**. Diuretics work because they cause loss of sodium from the body. These drugs are used to treat high blood pressure (**hypertension**), where the resulting decline in blood pressure reduces the risk for cardiovascular disease. However, diuretics can lead to sodium deficiency, resulting in low plasma sodium levels. A side effect of some diuretics is excessive loss of potassium, and low plasma potassium (**hypokalemia**) may result.

Iodine deficiency tends to occur in regions of the world where the soil is poor in iodine. Where soil used in agriculture is poor in iodine, the foods grown in the soil will also be low in iodine. An iodine intake of 0.10–0.15 mg/day is considered to be nutritionally adequate, while iodine deficiency occurs at below 0.05 mg/day. Goiter, an enlargement of the thyroid gland (located in the neck), results from iodine deficiency. Goiter continues to be a problem in eastern Europe, parts of India and South America, and in Southeast Asia. Goiter has been eradicated in the United States because of the fortification of foods with iodine. Iodine deficiency during pregnancy results in cretinism in the newborn. Cretinism involves mental retardation, a large tongue, and sometimes deafness, muteness, and lameness.

Iron deficiency occurs due to periods of dietary deficiency, rapid growth, and excessive loss of the body's iron. Human milk and cow milk both contains low levels of iron. Infants are at risk for acquiring iron deficiency because their rapid rate of growth needs a corresponding increased supply of dietary iron, for use in making blood and muscles. Human milk is a better source of iron than cow milk, since about half of the iron in human breast milk is absorbed by the infant's digestive tract. In contrast, only 10% of the iron in cow milk is absorbed by the infant. Surveys of lower-income families in the United States have revealed that about 6% of the infants are anemic indicating a deficiency of iron in their **diets**. Blood loss that occurs with menstruation in women, as well as with a variety of causes of intestinal bleeding is a major cause of iron deficiency. The symptoms of iron deficiency are generally limited to anemia, and the resulting tiredness, weakness, and a reduced ability to perform physical work.

Calcium and phosphate are closely related nutrients. About 99% of the calcium and 85% of the phosphate in the body occur in the skeleton, where they exist as crystals of solid calcium phosphate. Both of these nutrients occur in a great variety of foods. Milk, eggs, and green, leafy vegetables are rich in calcium and phosphate. Whole cow milk, for example, contains about 1.2 g calcium and 0.95 g phosphorus per kg of food. Broccoli contains 1.0 g calcium and 0.67 g phosphorus per kg food. Eggs supply about one third of the calcium and phosphate of the overall population of the United States. Dietary deficiencies in calcium (**hypocalcemia**) or phosphate are extremely rare throughout the world. Vitamin D deficiency can be found among young infants, the elderly, and others who may be shielded from sunshine for prolonged periods of time. Vitamin D deficiency impairs the absorption of calcium from the diet, and in this way can provoke calcium deficiency even when the diet contains adequate calcium.

Zinc deficiency has been found among peasant populations in rural areas of the Middle East. Unleavened whole wheat bread can account for 75% of the energy intake in these areas. This diet, which does not contain meat, does contain zinc, but it also contains phytic acid at a level of about 3 g/day. The phytic acid, which naturally occurs in wheat, inhibits zinc absorption. The yeast used to leaven bread produces enzymes that inactivate the phytic acid. Unleavened bread does not contain yeast, and therefore, contains intact phytic acid. The symptoms of zinc deficiency include lack of sexual maturation, lack of pubic hair, and small stature. The amount of phytic

acid in a typical American diet cannot provoke zinc deficiency.

Zinc deficiency is relatively uncommon in the United States, but it may occur in adults with **alcoholism** or intestinal malabsorption problems. Low plasma zinc has been found in patients with alcoholic **cirrhosis**, **Crohn's disease**, and **celiac disease**. Experimental studies with humans have shown that the signs of zinc deficiency are detectable after two to five weeks of consumption of the zinc-free diet. The signs include a rash and diarrhea. The rash occurs on the face, groin, hands, and feet. These symptoms can easily be reversed by administering zinc. An emerging concern is that increased calcium intake can interfere with zinc absorption or retention. Hence, there is some interest in the question of whether persons taking calcium to prevent **osteoporosis** should also take zinc supplements.

Severe alterations in copper metabolism occur in two genetic diseases, Wilson's disease and Menkes' disease. Both of these diseases are rare and occur in about one in 100,000 births. Both diseases involve mutations in copper transport proteins, that is, in special channels that allow the passage of copper ions through cell membranes. Menkes' disease is a genetic disease involving mental retardation and death before the age of three years. The disease also results in steely or kinky hair. The hair is tangled, grayish, and easily broken. Menkes' disease involves a decrease in copper levels in the serum, liver, and brain, and increases in copper in the cells of the intestines and kidney.

Selenium deficiency may occur in premature infants, since this population naturally tends to have low levels of plasma selenium. Full term infants have plasma selenium levels of about 0.001-0.002 mM, while premature infants may have levels about one third this amount. Whether these lower levels result in adverse consequences is not clear. Selenium deficiency occurs in regions of the world containing low-selenium soils. These regions include Keshan Province in China, New Zealand, and Finland. In Keshan Province, a disease (Keshan disease) occurs which results in deterioration of regions of the heart and the development of fibers in these regions. Keshan disease, which may be fatal, is thought to result from a combination of selenium deficiency and a virus.

Diagnosis

The diagnosis of deficiencies in water, sodium, potassium, iron, calcium, and phosphate involve chemical testing of the blood plasma, urine, and red blood cells.

Iodine deficiency can be diagnosed by measuring the concentration of iodine in the urine. A urinary level greater than 0.05 mg iodine per gram creatinine means adequate iodine status. Levels under 0.025 mg iodine/g creatinine indicate a serious risk.

Normal blood serum magnesium levels are 1.2-2.0 mM. Magnesium deficiency results in hypomagnesemia, which is defined as serum magnesium levels below 0.8 mM. Magnesium levels below 0.5 mM provoke a decline in serum calcium levels. Hypomagnesemia can also result in low serum potassium. Some of the symptoms of hypomagnesemia, which include twitching and convulsions, actually result from the hypocalcemia. Other symptoms of hypomagnesemia, such as cardiac arrhythmias, result from the low potassium levels.

There is no reliable test for zinc deficiency. When humans eat diets containing normal levels of zinc (16 mg/day), the level of urinary zinc is about 0.45 mg/day, while humans consuming low-zinc diets (0.3 mg/day) may have urinary levels of about 0.150 mg/day. Plasma zinc levels tend to be maintained during a dietary deficiency in zinc. Plasma and urinary zinc levels can be influenced by a variety of factors, and for this reason cannot provide a clear picture of zinc status.

Selenium deficiency may be diagnosed by measuring the selenium in plasma (70 ng/mL) or red blood cells (90 ng/mL), where the normal values are indicated. There is also some interest in measuring the activity of an enzyme in blood platelets, in order to assess selenium status. This enzyme is glutathione peroxidase. Platelets are small cells of the bloodstream which are used mainly to allow the clotting of blood after an injury.

Treatment

The treatment of deficiencies in sodium, potassium, calcium, phosphate, and iron usually involves intravenous injections of the deficient mineral.

Iodine deficiency can be easily prevented and treated by fortifying foods with iodine. Table salt is fortified with 100 mg potassium iodide per kg sodium chloride. Goiter was once common in the United States in areas from Washington State to the Great Lakes region, but this problem has been eliminated by iodized salt. Public health programs in impoverished countries have involved injections of synthetic oils containing iodine. Goiter is reversible but, cretinism is not.

Magnesium deficiency can be treated with a magnesium rich diet. If magnesium deficiency is due to a

KEY TERMS

Recommended Dietary Allowance—The Recommended Dietary Allowances (RDAs) are quantities of nutrients that are required each day to maintain human health. RDAs are established by the Food and Nutrition Board of the National Academy of Sciences and may be revised every few years. A separate RDA value exists for each nutrient.

prolonged period of depletion, treatment may include injections of magnesium sulfate (2.0 mL of 50% MgSO₄). Where magnesium deficiency is severe enough to provoke convulsions, magnesium needs to be administered by injections or infusions. For infusion, 500 mL of a 1% solution (1 gram/100 mL) of magnesium sulfate is gradually introduced into a vein over the course of about five hours.

Zinc deficiency and copper deficiency are quite rare, but when they are detected or suspected, they can be treated by consuming zinc or copper, on a daily basis, at levels defined by the RDA.

Selenium deficiency in adults can be treated by eating 100 mg selenium per day for a week, where the selenium is supplied as selenomethionine. The incidence of Keshan disease in China has been reduced by supplementing children with 1.0 mg sodium selenite per week.

Prognosis

In iodine deficiency, the prognosis for treating goiter is excellent, however cretinism cannot be reversed. The effects of iron deficiency are not life-threatening and can be easily treated. The prognosis for treating magnesium deficiency is excellent. The symptoms may be relieved promptly or, at most, within two days of starting treatment. In cases of zinc deficiency in Iran and other parts of the Middle East, supplementation of affected young adults with zinc has been found to provoke the growth of pubic hair and enlargement of genitalia to a normal size within a few months.

Prevention

In the healthy population, all mineral deficiencies can be prevented by the consumption of inorganic nutrients at levels defined by the Recommended

Dietary Allowances (RDA). Where a balanced diet is not available, government programs for treating individuals, or for fortifying the food supply, may be used. Government sponsored programs for the prevention of iron deficiency and iodine deficiency are widespread throughout the world. Selenium treatment programs have been used in parts of the world where selenium deficiency exists. Attention to potassium status, and to the prevention of potassium deficiency, is an issue mainly in patients taking diuretic drugs. In many cases of mineral deficiency, the deficiency occurs because of disease, and individual medical attention, rather than preventative measures, is used. The prevention of calcium deficiency is generally not an issue or concern, however calcium supplements are widely used with the hope of preventing osteoporosis. The prevention of deficiencies in magnesium, zinc, copper, manganese, or molybdenum are not major health issues in the United States. Ensuring an adequate intake of these minerals, by eating a balanced diet or by taking mineral supplements, is the best way to prevent deficiencies.

Resources

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Brody, Tom. *Nutritional Biochemistry*. San Diego: Academic Press, 1998.

Tom Brody, PhD

Mineral excess see **Mineral toxicity**

Mineral toxicity

Definition

The term mineral toxicity means a condition where the concentration in the body of any one of the **minerals** is abnormally high, and where there is an adverse effect on health.

Description

In general, mineral toxicity results when there is an accidental consumption of too much of any mineral, as with drinking ocean water (sodium toxicity) or with overexposure to industrial pollutants, household chemicals, or certain drugs. Mineral toxicity may also apply to toxicity that can be the result of certain diseases or injuries. For example, **hemochromatosis** leads to iron toxicity; Wilson's disease results

in copper toxicity; severe trauma can lead to **hyperkalemia** (potassium toxicity).

The mineral nutrients are defined as all the inorganic elements or inorganic molecules that are required for life. As far as human **nutrition** is concerned, the inorganic nutrients include water, sodium, potassium, chloride, calcium, phosphate, sulfate, magnesium, iron, copper, zinc, manganese, iodine, selenium, and molybdenum.

The mineral content of the body may be measured by testing samples of blood plasma, red blood cells, and urine.

Causes and symptoms

An increase in the concentrations of sodium in the bloodstream can be toxic. The normal concentration of sodium in the blood plasma is 136-145 mM, while levels over 152 mM can result in seizures and **death**. Increased plasma sodium, which is called **hypernatremia**, causes various cells of the body, including those of the brain, to shrink. Shrinkage of the brain cells results in confusion, **coma**, **paralysis** of the lung muscles, and death. Death has occurred where table salt (sodium chloride) was accidentally used, instead of sugar, for feeding infants. Death due to sodium toxicity has also resulted when baking soda (sodium bicarbonate) was used during attempted therapy of excessive **diarrhea** or **vomiting**. Although a variety of processed foods contain high levels of sodium chloride, the levels used are not enough to result in sodium toxicity.

The normal level of potassium in the bloodstream is in the range of 3.5-5.0 mM, while levels of 6.3-8.0 mM (severe hyperkalemia) result in cardiac **arrhythmias** or even death due to cardiac arrest. Potassium is potentially quite toxic, however toxicity or death due to potassium **poisoning** is usually prevented because of the vomiting reflex. The consumption of food results in mild increases in the concentration of potassium in the bloodstream, but levels of potassium do not become toxic because of the uptake of potassium by various cells of the body, as well as by the action of the kidneys transferring the potassium ions from the blood to the urine. The body's regulatory mechanisms can easily be overwhelmed, however, when potassium chloride is injected intravenously, as high doses of injected potassium can easily result in death.

Iodine toxicity can result from an intake of 2.0 mg of iodide per day. The toxicity results in impairment of the creation of thyroid hormone, resulting in lower levels of thyroid hormone in the bloodstream. The

thyroid gland enlarges, as a consequence, and **goiter** is produced. This enlargement is also called **hyperthyroidism**. Goiter is usually caused by iodine deficiency. In addition to goiter, iodine toxicity produces ulcers on the skin. This condition has been called "kelp acne," because of its association with eating kelp, an ocean plant, which contains high levels of iodine. Iodine toxicity occurs in Japan, where large amounts of seaweed are consumed.

Iron toxicity is not uncommon, due to the wide distribution of iron pills. A lethal dose of iron is in the range of 200-250 mg iron/kg body weight. Hence, a child who accidentally eats 20 or more iron tablets may die as a result of iron toxicity. Within six hours of ingestion, iron toxicity can result in vomiting, diarrhea, abdominal **pain**, seizures, and possibly coma. A latent period, where the symptoms appear to improve, may occur but it is followed by **shock**, low blood glucose, liver damage, convulsions, and death, occurring 12-48 hours after toxic levels of iron are ingested.

Nitrite poisoning should be considered along with iron toxicity, since nitrite produces its toxic effect by reacting with the iron atom of hemoglobin. Hemoglobin is an iron-containing protein that resides within the red blood cells. This protein is responsible for the transport of nearly all of the oxygen, acquired from the lungs, to various tissues and organs of the body. Hemoglobin accounts for the red color of our red blood cells. A very small fraction of our hemoglobin spontaneously oxidizes per day, producing a protein of a slightly different structure, called methemoglobin. Normally, the amount of methemoglobin constitutes less than 1% of the total hemoglobin. Methemoglobin can accumulate in the blood as a result of nitrite poisoning. Infants are especially susceptible to poisoning by nitrite.

Nitrate, which is naturally present in green leafy vegetables and in the water supply is rapidly converted to nitrite by the naturally occurring bacteria residing on our tongue, as well as in the intestines, and then absorbed into the bloodstream. The amount of nitrate that is supplied by leafy vegetables and in drinking water is generally about 100-170 mg/day. The amount of nitrite supplied by a typical diet is much less, that is, than 0.1 mg nitrite/day. Poisoning by nitrite, or nitrate after its conversion to nitrite, results in the inability of hemoglobin to carry oxygen throughout the body. This condition can be seen by the blue color of the skin. Adverse symptoms occur when over 30% of the hemoglobin has been converted to methemoglobin, and these symptoms include cardiac arrhythmias, **headache**, **nausea and vomiting**, and in severe cases, seizures.

Calcium and phosphate are closely related nutrients. Calcium toxicity is rare, but overconsumption of calcium supplements may lead to deposits of calcium phosphate in the soft tissues of the body. Phosphate toxicity can occur with overuse of **laxatives** or **enemas** that contain phosphate. Severe phosphate toxicity can result in **hypocalcemia**, and in various symptoms resulting from low plasma calcium levels. Moderate phosphate toxicity, occurring over a period of months, can result in the deposit of calcium phosphate crystals in various tissues of the body.

Zinc toxicity is rare, but it can occur in metal workers who are exposed to fumes containing zinc. Excessive dietary supplements of zinc can result in **nausea**, vomiting, and diarrhea. The chronic intake of excessive zinc supplements can result in copper deficiency, as zinc inhibits the absorption of copper.

Severe alterations in copper metabolism occur in two genetic diseases, Wilson's disease and Menkes' disease. Both of these diseases are rare and occur in about one in 100,000 births. Both diseases involve mutations in the proteins that transport copper, that is, in special channels that allow the passage of copper ions through cell membranes. Wilson's disease tends to occur in teenagers and in young adults, and then remain for the lifetime. Copper accumulates in the liver, kidney, and brain, resulting in damage to the liver and nervous system. Wilson's disease can be successfully controlled by lifelong treatment with d-penicillamine. Treatment also involves avoiding foods that are high in copper, such as liver, nuts, chocolate, and mollusks. After an initial period of treatment with penicillamine, Wilson's disease may be treated with zinc (150 mg oral Zn/day). The zinc inhibits the absorption of dietary copper.

Selenium toxicity occurs in regions of the world, including some parts of China, where soils contain high levels of selenium. A daily intake of 0.75-5.0 mg selenium may occur in these regions, due to the presence of selenium in foods and water. Early signs of selenium toxicity include nausea, weakness, and diarrhea. With continued intake of selenium, changes in fingernails and hair loss results, and damage to the nervous system occurs. The breath may acquire a garlic odor, as a result of the increased production of dimethylselenide in the body, and its release via the lungs.

Manganese toxicity occurs in miners in manganese mines, where men breath air containing dust bearing manganese at a concentration of 5-250 mg/cubic meter. Manganese toxicity in miners has been documented in Chile, India, Japan, Mexico, and

elsewhere. Symptoms of manganese poisoning typically occur within several months or years of exposure. These symptoms include a mental disorder resembling **schizophrenia**, as well as hyperirritability, violent acts, **hallucinations**, and difficulty in walking.

Diagnosis

The initial diagnosis of mineral toxicity involves questioning the patient in order to determine any unusual aspects of the diet, unusual intake of drugs and chemicals, and possible occupational exposure. Diagnosis of mineral toxicities also involves measuring the metal concentration in the plasma or urine. Concentrations that are above the normal range can confirm the initial, suspected diagnosis.

Treatment

Iron toxicity is treated by efforts to remove remaining iron from the stomach, by use of a solution of 5% sodium bicarbonate. Where plasma iron levels are above 0.35 mg/dL, the patient is treated with deferoxamine. Treatment of manganese toxicity involves removal of the patient from the high manganese environment, as well as lifelong doses of the drug L-dopa. The treatment is only partially successful. Treatment of nitrite or nitrate toxicity involves inhalation of 100% oxygen for several hours. If oxygen treatment is not effective, then methylene blue may be injected, as a 1.0% solution, in a dose of 1.0 mg methylene blue/kg body weight.

Prognosis

The prognosis for treating toxicity due to sodium, potassium, calcium, and phosphate is usually excellent. Toxicity due to the deposit of calcium phosphate crystals is not usually reversible. The prognosis for treating iodine toxicity is excellent. For any mineral overdose that causes coma or seizures, the prognosis for recovery is often poor, and death results in a small fraction of patients. For any mineral toxicity that causes nerve damage, the prognosis is often fair to poor.

Prevention

When mineral toxicity results from the excessive consumption of mineral supplements, toxicity can be prevented by not using supplements. In the case of manganese, toxicity can be prevented by avoiding work in manganese mines. In the case of iodine, toxicity can be prevented by avoiding overconsumption of

seaweed or kelp. In the case of selenium toxicity that arises due to high-selenium soils, toxicity can be prevented by relying on food and water acquired from a low-selenium region.

Resources

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Brody, Tom. *Nutritional Biochemistry*. San Diego: Academic Press, 1998.

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Minerals

Definition

The minerals (inorganic nutrients) that are relevant to human **nutrition** include water, sodium, potassium, chloride, calcium, phosphate, sulfate, magnesium, iron, copper, zinc, manganese, iodine, selenium, and molybdenum. Cobalt is a required mineral for human health, but it is supplied by vitamin B₁₂. Cobalt appears to have no other function, aside from being part of this vitamin. There is some evidence that chromium, boron, and other inorganic elements play some part in human nutrition, but the evidence is indirect and not yet convincing. Fluoride seems not to be required for human life, but its presence in the diet contributes to long term dental health. Some of the minerals do not occur as single atoms, but occur as molecules. These include water, phosphate, sulfate, and selenite (a form of selenium). Sulfate contains an atom of sulfur. We do not need to eat sulfate, since the body can acquire all the sulfate it needs from protein.

The statement that various minerals, or inorganic nutrients, are required for life means that their continued supply in the diet is needed for growth, maintenance of body weight in adulthood, and for reproduction. The amount of each mineral that is needed to support growth during infancy and childhood, to maintain body weight and health, and to facilitate **pregnancy** and **lactation**, are listed in a table called the Recommended Dietary Allowances (RDA). This table was compiled by the Food and Nutrition Board, a committee that serves the United States government. All of the values listed in the RDA indicate the daily amounts that are expected to maintain health throughout most of the general population. The actual levels of each inorganic nutrient required by any given individual is likely to be less than that stated by the

RDA. The RDAs are all based on studies that provided the exact, minimal requirement of each mineral needed to maintain health. However, the RDA values are actually greater than the minimal requirement, as determined by studies on small groups of healthy human subjects, in order to accommodate the variability expected among the general population.

The RDAs for adult males are 800 mg of calcium, 800 mg of phosphorus, 350 mg of magnesium, 10 mg of iron, 15 mg of zinc, 0.15 mg of iodine, and 0.07 mg of selenium. The RDA for sodium is expressed as a range (0.5-2.4 g/day). The minimal requirement for chloride is about 0.75 g/day, and the minimal requirement for potassium is 1.6-2.0 g/day, though RDA values have not been set for these nutrients. The RDAs for several other minerals has not been determined, and here the estimated safe and adequate daily dietary intake has been listed by the Food and Nutrition Board. These values are listed for copper (1.5-3.0 mg), manganese (2-5 mg), fluoride (1.5-4.0 mg), molybdenum (0.075-0.25 mg), and chromium (0.05-0.2 mg). In noting the appearance of chromium in this list, one should note that the function of chromium is essentially unknown, and evidence for its necessity exists only for animals, and not for human beings. In considering the amount of any mineral used for treating **mineral deficiency**, one should compare the recommended level with the RDA for that mineral. Treatment at a level that is one tenth of the RDA might not be expected to be adequate, while treatment at levels ranging from 10-1,000 times the RDA might be expected to exert a toxic effect, depending on the mineral. In this way, one can judge whether any claim of action, for a specific mineral treatment, is likely to be adequate or appropriate.

Purpose

People are treated with minerals for several reasons. The primary reason is to relieve a mineral deficiency, when a deficiency has been detected. Chemical tests suitable for the detection of all mineral deficiencies are available. The diagnosis of the deficiency is often aided by tests that do not involve chemical reactions, such as the **hematocrit** test for the red blood cell content in blood for iron deficiency, the visual examination of the neck for iodine deficiency, or the examination of bones by densitometry for calcium deficiency. Mineral treatment is conducted after a test and diagnosis for iron-deficiency anemia, in the case of iron, and after a test and diagnosis for hypomagnesemia, in the case of magnesium, to give two examples.

A second general reason for mineral treatment is to prevent the development of a possible or expected deficiency. Here, minerals are administered when tests for possible mineral deficiency are not given. Examples include the practice of giving young infants iron supplements, and of the food industry's practice of supplementing infant formulas with iron. The purpose here is to reduce the risk for **iron deficiency anemia**. Another example is the practice of many women of taking calcium supplements, with the hope of reducing the risk of **osteoporosis**.

Most minerals are commercially available at supermarkets, drug stores, and specialty stores. There is reason to believe that the purchase and consumption of most of these minerals is beneficial to health for some, but not all, of the minerals. Potassium supplements are useful for reducing blood pressure, in cases of persons with high blood pressure. The effect of potassium varies from person to person. The consumption of calcium supplements is likely to have some effect on reducing the risk for osteoporosis. The consumption of selenium supplements is expected to be of value only for residents of Keshan Province, China, because of the established association of selenium deficiency in this region with "Keshan disease."

Precautions

During emergency treatment of sodium deficiency (**hyponatremia**), potassium deficiency (**hypokalemia**), and calcium deficiency (**hypocalcemia**) with intravenous injections, extreme caution must be taken to avoid producing toxic levels of each of these minerals (**hypernatremia**, **hyperkalemia**, and **hypercalcemia**), as **mineral toxicity** can be life-threatening in some instances. The latter three conditions can be life threatening. Selenium is distinguished among most of the nutrients in that dietary intakes at levels only ten times that of the RDA can be toxic. Hence, one must guard against any overdose of selenium. Calcium and zinc supplements, when taken orally, are distinguished among most of the other minerals in that their toxicity is relatively uncommon.

Description

Minerals are used in treatments by three methods, namely, by replacing a poor diet with a diet that supplies the RDA, by consuming oral supplements, or by injections or infusions. Injections are especially useful for infants, for mentally disabled persons, or where the physician wants to be totally sure of compliance. Infusions, as well as injections, are essential for medical emergencies, as during mineral deficiency

situations like hyponatremia, hypokalemia, hypocalcemia, and hypomagnesemia. Oral mineral supplements are especially useful for mentally alert persons who otherwise cannot or will not consume food that is a good mineral source, such as meat. For example, a vegetarian who will not consume meat may be encouraged to consume oral supplements of iron, as well as supplements of vitamin B₁₂.

Iron treatment is used for young infants, given as supplements of 7 mg of iron per day to prevent anemia. Iron is also supplied to infants via the food industry's practice of including iron at 12 mg/L in cow milk-based infant formulas, as well as adding powdered iron at levels of 50 mg iron per 100 g dry infant cereal.

Calcium supplements, along with estrogen and calcitonin therapy, are commonly used in the prevention and treatment of osteoporosis. Estrogen and calcitonin are naturally occurring hormones. Bone loss occurs with **diets** supplying under 400 mg Ca/day. Bone loss can be minimized with the consumption of the RDA for calcium. There is some thought that all postmenopausal women should consume 1,000–1,500 mg of calcium per day. These levels are higher than the RDA. There is some evidence that such supplementation can reduce bone losses in some bones, such as the elbow (ulna), but not in other bones. Calcium absorption by the intestines decreases with **aging**, especially after the age of 70. The regulatory mechanisms of the intestines that allow absorption of adequate calcium (500 mg Ca/day or less) may be impaired in the elderly. Because of these changes, there is much interest in increasing the RDA for calcium for older women.

Fluoride has been proven to reduce the rate of **tooth decay**. When fluoride occurs in the diet, it is incorporated into the structure of the teeth, and other bones. The optimal range of fluoride in drinking water is 0.7–1.2 mg/L. This level results in a reduction in the rate of tooth decay by about 50%. The American Dental Association recommends that persons living in areas lacking fluoridated water take fluoride supplements. The recommendation is 0.25 mg F/day from the ages of 0–2 years, 0.5 mg F/day for 2–3 years, and 1.0 mg F/day for ages 3–13 years.

Magnesium is often used to treat a dangerous condition, called **eclampsia**, that occasionally occurs during pregnancy. In this case, magnesium is used as a drug, and not to relieve a deficiency. High blood pressure is a fairly common disorder during pregnancy, affecting 1–5% of pregnant mothers. **Hypertension** during pregnancy can result in increased release of protein in the urine. In pregnancy, the combination

of hypertension with increased urinary protein is called **preeclampsia**. Preeclampsia is a concern during pregnancies as it may lead to eclampsia. Eclampsia involves convulsions and possibly **death** to the mother. Magnesium sulfate is the drug of choice for preventing the convulsions of eclampsia.

Treatment with cobalt, in the form of vitamin B₁₂, is used for relieving the symptoms of **pernicious anemia**. Pernicious anemia is a relatively common disease which tends to occur in persons older than 40 years. Free cobalt is never used for the treatment of any disease.

Preparation

Evaluation of a patient's mineral levels requires a blood sample, and the preparation of plasma or serum from the blood sample. An overnight fast is usually recommended as preparation prior to drawing the blood and chemical analysis. The reason for this is that any mineral present in the food consumed at breakfast may artificially boost the plasma mineral content beyond the normal **fasting** level, and thereby mask a mineral deficiency. In some cases, red blood cells are used for the mineral status assay.

Aftercare

The healthcare provider assesses the patient's response to mineral treatment. A positive response confirms that the diagnosis was correct. Lack of response indicates that the diagnosis was incorrect, that the patient had failed to take the mineral supplement, or that a higher dose of mineral was needed. The response to mineral treatment can be monitored by chemical tests, by an examination of red blood cells or white blood cells, or by physiological tests, depending on the exact mineral deficiency.

Risks

There are few risks associated with mineral treatment. In treating emergency cases of hyponatremia, hypokalemia, or hypocalcemia by intravenous injections, there exists a very real risk that giving too much sodium, potassium, or calcium, can result in hypernatremia, hyperkalemia, or hypercalcemia, respectively. Risk for toxicity is rare where treatment is by dietary means. This is because the intestines act as a barrier, and absorption of any mineral supplement is gradual. The gradual passage of any mineral through the intestines, especially when the mineral supplement is taken with food, allows the various organs of the body to acquire the mineral. Gradual passage of the mineral

into the bloodstream also allows the kidneys to excrete the mineral in the urine, should levels of the mineral rise to toxic levels in the blood.

Resources

BOOKS

Brody, Tom. *Nutritional Biochemistry*. San Diego: Academic Press, 1998.

Tom Brody, PhD

Minnesota multiphasic personality inventory (MMPI-2)

Definition

The Minnesota Multiphasic Personality Inventory (MMPI-2; MMPI-A) is a written psychological assessment, or test, used to diagnose mental disorders.

Purpose

The MMPI is used to screen for personality and **psychosocial disorders** in adults and adolescents. It is also frequently administered as part of a neuropsychological test battery to evaluate cognitive functioning.

Precautions

The MMPI should be administered, scored, and interpreted by a clinical professional trained in its use, preferably a psychologist or psychiatrist. The MMPI is only one element of psychological assessment, and should never be used alone as the sole basis for a diagnosis. A detailed history of the test subject and a review of psychological, medical, educational, or other relevant records are required to lay the groundwork for interpreting the results of any psychological measurement.

Cultural and language differences in the test subject may affect test performance and may result in inaccurate MMPI results. The test administrator should be informed before psychological testing begins if the test taker is not fluent in English and/or has a unique cultural background.

Description

The original MMPI was developed at the University of Minnesota and introduced in 1942. The

KEY TERMS

Neuropsychological testing—Tests used to evaluate patients who have experienced a traumatic brain injury, brain damage, or organic neurological problems (e.g., dementia). It may also be used to evaluate the progress of a patient who has undergone treatment or rehabilitation for a neurological injury or illness.

Norms—Normative or mean score for a particular age group.

Psychopathology—A mental disorder or illness, such as schizophrenia, personality disorder, or major depressive disorder.

Standardization—The process of determining established norms and procedures for a test to act as a standard reference point for future test results.

current standardized version for adults 18 and over, the MMPI-2, was released in 1989, with a subsequent revision of certain test elements in early 2001. The MMPI-2 has 567 items, or questions, and takes approximately 60 to 90 minutes to complete. There is a short form of the test that is comprised of the first 370 items on the long-form MMPI-2. There is also a version of the inventory for adolescents age 14 to 18, the MMPI-A.

The questions asked on the MMPI are designed to evaluate the thoughts, emotions, attitudes, and behavioral traits that comprise personality. The results of the test reflect an individual's personality strengths and weaknesses, and may identify certain disturbances of personality (psychopathologies) or mental deficits caused by neurological problems.

There are six validity scales and ten basic clinical or personality scales scored in the MMPI-2, and a number of supplementary scales and subscales that may be used with the test. The validity scales are used to determine whether the test results are actually valid (i.e., if the test-taker was truthful, answered cooperatively and not randomly) and to assess the test-taker's response style (i.e., cooperative, defensive). Each clinical scale uses a set or subset of MMPI-2 questions to evaluate a specific personality trait. The MMPI should always be administered in a controlled environment by a psychologist or other qualified mental health professional trained in its use.

Preparation

The administrator should provide the test subject with information on the nature of the test and its intended use, complete standardized instructions to taking the MMPI (including any time limits, and information on the confidentiality of the results).

Normal results

The MMPI should be scored and interpreted by a trained professional. When interpreting test results for test subjects, the test administrator will review what the test evaluates, its precision in evaluation and any margins of error involved in scoring, and what the individual scores mean in the context of overall norms for the test and the background of the test subject.

Resources

BOOKS

Graham, John R. *MMPI-2: Assessing Personality and Psychopathology*. 3rd ed. New York: Oxford University Press, 1999.

ORGANIZATIONS

ERIC Clearinghouse on Assessment and Evaluation. 1131 Shriver Laboratory Bldg 075, University of Maryland, College Park, MD 20742. (800) 464-3742. <<http://www.ericae.net>>.

Paula Anne Ford-Martin

Minor tranquilizers see **Antianxiety drugs**

Minority health

Definition

Minority health addresses the special medical and/or health needs associated with specific ethnic groups.

Description

The United States, as well as many other countries, experiences cultural diversity. This poses specific health issues that are specific to ethnic groups. Additionally, the propensity for certain diseases or illnesses is of concern in certain minority groups. These specific health issues include infant mortality rates, **cancer**, cardiovascular disease, diabetes, HIV infection, and immunizations.

DR. ANTONIA NOVELLO (1944–)



(Gamma Liaison. Reproduced by permission.)

Born Antonia Coello was born in Fajardo, Puerto Rico, on August 23, 1944, the oldest of three children. At eight

years old, she suffered two blows that she would carry all of her life. Her father, Antonio Coello, died, leaving her mother, Ana Delia Flores Coello, to raise her children alone until she later remarried Ramon Flores, an electrician. Novello was also diagnosed with a chronic condition called congenital megacolon, an illness in which her colon was overly large and not functioning properly, which required regular hospitalization. Although an operation would have helped Novello, it was not performed until she was 18 years old, and even after the surgery, complications followed her for years. Because of her childhood illness, Novello grew up wanting to be a doctor.

On October 17, 1989, President George Bush officially nominated Novello for Surgeon General. The fourteenth United States Surgeon General, Novello, sworn in on March 9, 1990, remarked that "the American dream is well and alive . . . today the West Side Story comes to the West Wing." Novello was the first woman and the first Hispanic to be appointed Surgeon General of the United States. Noted for her philosophy of "good science, good sense" and for her approachability, Novello was dedicated to the prevention of AIDS, substance abuse, and smoking, as well as to the education of the American public. Her special concerns were for women, children, and hispanics—populations often overlooked by public health services.

Infant mortality rates

Infant mortality rates (IMRs) in the United States and in all countries worldwide are an accurate indicator of health status. They provide information concerning programs about **pregnancy** education and counseling, technological advances, and procedures and aftercare. IMRs vary among racial groups. African Americans had an IMRs of 14.2 per 1,000 live births in 1996, approximately 2.5 times higher than Caucasians. The IMRs among American Native Indian groups varies greatly, with some communities possessing IMRs about two times more than national rates. Additionally Hispanic IMRs (7.6 per 1,000 live births) are also diverse for separate groups, since the IMRs for example among Puerto Ricans is higher (8.9 per 1,000 live births).

Cancer

Cancer is a serious national, worldwide, and minority health concern. It is the second cause of **death** in the United States, claiming over half a million lives each year. Approximately 50% of persons who develop cancer will die. There is great disparity

among the cancer rates in minority groups. Across genders, cancer death rates for African Americans are 35% higher when compared to statistics for Caucasians. The death rates for **prostate cancer** (two times more) and lung cancer (27 times more) are disproportionately higher when compared to Caucasians. There are also gender differences among ethnic groups and specific cancers. Lung cancers in African American and Hawaiian men are evaluated compared with Caucasian males. Vietnamese females who live in the United States have five times more new cases of **cervical cancer** when compared to Caucasian women. Hispanic females also have a greater incidence of cervical cancer than Caucasian females. Additionally, Alaskan native men and women have a greater propensity for cancers in the rectum and colon than do Caucasians.

Cardiovascular disease

Cardiovascular disease is responsible for the leading cause of disability and death rates about equal to death from all other diseases combined. Cardiovascular disease can affect the patient's lifestyle and function in addition to having an impact on family

members. The financial costs are very high. Among ethnic and racial groups cardiovascular disease is the leading cause of death. **Stroke** is the leading cause of cardiovascular related death, which occurs in higher numbers for Asian-American males when compared to Caucasian men. Mexican-American men and women and African-American males have a higher incidence of **hypertension**. African-American women have higher rates of being overweight, which is a major risk factor of cardiovascular disease.

Diabetes

Diabetes—a serious health problem in Americans and ethnic groups—is the seventh leading cause of death in the United States. The prevalence of diabetes in African Americans is about 70% higher than Caucasians.

HIV

HIV infection/AIDS is the most common cause of death for all persons age 25 to 44 years old. Ethnic groups account for 25% of the United States population and 54% of all **AIDS** cases. In addition to sexual transmission there is an increase in HIV among ethnic groups related to intravenous drug usage.

Immunizations

Immunization, the reduction of preventable disease by **vaccination**, was lower in 1996, but there has been a rapid increase in African Americans taking vaccinations. The coverage for immunization among African Americans and Hispanics for persons age 65 and over is currently below the general population. This may increase the death rates due to respiratory infections.

Causes and symptoms

IMRs are correlated with prenatal care. Women who receive adequate prenatal care tend to have better pregnancy outcomes when compared to little or no care. Women who receive inadequate prenatal care also have increased chances of delivering a very low birth weight (VLBW) infant, which is linked to risk of early death.

Cancer is related to several preventable lifestyle choices. Tobacco use, diet, and exposure to sun (skin cancer) can be prevented by lifestyle modifications. Additionally many cancers can occur due to lack of interest and/or lack of availability for screening and educational programs.

Cardiovascular diseases are higher among persons with high blood cholesterol and high blood pressure. Certain lifestyle choices may increase the chance for heart disease includes lack of **exercise**, overweight, and cigarette **smoking**. Cardiovascular disease is responsible for over 50% of the deaths in persons with diabetes.

HIV occurs at a higher frequency among homosexuals (the number of African-American males who have AIDS through sex with men has increased). Additionally unprotected sexual intercourse and sharing used needles for IV drug injection are strongly correlated with infection.

Vaccinations are an effective method of preventing certain disease such as **polio**, **tetanus**, pertussis, **diphtheria**, **influenza**, **hepatitis b**, and pneumococcal infections. Approximately 90% of influenza related mortality is associated with persons aged 65 and older. This is mostly due to neglect of vaccinations. About 45,000 adults each year die of diseases related to hepatitis B, pneumococcal, and influenza infections.

Diagnosis

The diagnosis of VLBW is by weight. Infants who weigh 1,500 g are at high risk for death. For cancer, the diagnosis can be made through screening procedures such as **mammography** (for **breast cancer**), PAP smear (for cervical cancer), and lifestyle modifications such as avoidance of sun, cigarette smoking, balanced **diets**, and adequate **nutrition**. Other specific screening tests (PSA, prostate surface antigen) are helpful for diagnosing prostate cancer. Cardiovascular diseases can be detected by medical check-up. Blood pressure and cholesterol levels can be measured. **Obesity** can be diagnosed by assessing a persons weight relative to height. Diabetes and its complications can be detected by blood tests, in-depth eye examinations and studies that assess the flow of blood through blood vessels in legs. HIV can be detected through a careful history/physical examination and analysis of blood using a special test called a western blot. Infections caused by lack of immunizations can either be detected by careful **physical examination** and culturing the specific microorganism in the laboratory.

Treatment

Treatment is directed at the primary causes(s) that minorities have increased chances of developing disease(s). Cancer may require treatment utilizing surgery, radiotherapy, or **chemotherapy**. Cardiovascular diseases may require surgical procedures for

KEY TERMS

Prevalence—Number of existing cases relative to time.

Propensity—A greater risk for developing a disease.

establishing a diagnosis and initiating treatment. Depending on the extent of disease cardiovascular management can become complicated requiring medications and daily lifestyle modifications. Treatment usually includes medications, dietary modifications, and—if complications arise—specific interventions tailored to alleviating the problem. HIV can be treated with specific medications and more often than not with symptomatic treatment as reported complications arise. Diseases caused by lack of immunizations are treated based on the primary disease. The best method of treatment is through prevention and generating public awareness through educational awareness.

Alternative treatment

Alternative therapies do exist, but more research is needed to substantiate present data. The diseases that relate to minority health are best treated with nationally accepted standards of care.

Prognosis

Generally the prognosis is related to the diagnosis, patients state of health, age, and if there is another disease or complication in addition to the presenting problem. The course for IMRs is related to educational programs and prenatal care, which includes medical and psychological treatments. The prognosis for chronic diseases such as cardiovascular problems, high blood pressure, cancer, and diabetes is variable. These diseases are not cured and control is achieved by standardized treatment options. Eventually complications, even with treatment can potentially occur. For HIV the clinical course at present is death even though this process may take years. Educational programs with an emphasis on disease prevention can potentially improve outcomes concerning pediatric and geriatric diseases.

Prevention

Prevention is accomplished best through educational programs specific to target populations. IMRs

can be prevented by increasing awareness, interest, and accessibility for prenatal care that address a comprehensive approach for the needs of each patient. Regular physicals and special screening tests can potentially prevent certain cancers in high-risk groups. Educational programs concerning lifestyle modifications, diet, exercise, and testing may prevent the development of cardiovascular disease and diabetes. Educational programs assemble to illicit IV drug abusers and persons who engage in unprotected sexual intercourse may decrease the incidence of HIV infection.

Resources

ORGANIZATIONS

Office of Minority Health. 2001. <<http://www.omhrc.gov/rah/3rdpgBlue/Cardio/3pgGoalsCardio.htm>>.

Laith Farid Gulli, M.D.
Nicole Mallory, M.S.

Minoxidil

Definition

Minoxidil is a drug available in two forms to treat different conditions. Oral minoxidil is used to treat high blood pressure and the topical solution form is used to treat hair loss and baldness.

Purpose

Minoxidil was the first drug approved by the FDA for the treatment of androgenetic **alopecia** (hair loss). Before that, minoxidil had been used as vasodilator drug prescribed as oral tablet to treat high blood pressure, with side effects that included hair growth and reversal of male baldness. In the 1980s, UpJohn Corporation came out with a topical solution of 2% minoxidil, called Rogaine, for the specific treatment of androgenetic alopecia. Since the 1990s, numerous generic forms of minoxidil have become available to treat hair loss while the oral form is still used to treat high blood pressure.

The popularity hair loss treatment is due to the general preference in the overall population for the cosmetic appearance of a full head of hair. Minoxidil is used to stimulate hair growth in areas of the scalp that have stopped growing hair. As of early 2001, the exact mechanism of action of minoxidil is not known.

Precautions

People who have had a prior unusual or allergic reaction to either minoxidil or propylene glycol, a non-active chemical in the Rogaine solution, should not use topical minoxidil. People who have had a previous allergic reaction to preservatives or dyes may also be at risk for having an allergic reaction to minoxidil.

People who are using cortisone, or cortisone-like drugs (**corticosteroids**), petroleum jelly (Vaseline), or tretinoin (Retin-A) on their scalps should consult their doctors prior to using minoxidil. The use of any of these products in conjunction with minoxidil may cause excessive minoxidil absorption into the body and increase the risk of side effects.

Also, people who have skin problems or irritations of the scalp, including **sunburn**, may absorb too much minoxidil and increase their risk of side effects.

As for oral minoxidil, the form prescribed for high blood pressure, patients should use minoxidil only under medical supervision to ensure that excessive amounts of the drug are not absorbed into their bodies. Large amounts of minoxidil may increase the severity of the symptoms and side effects of **hypertension**.

Minoxidil may pass from mother to child through breast milk. Therefore, women who are breastfeeding should not use minoxidil.

Description

For the treatment of hair loss, minoxidil is available as a topical solution that is generally either 2% or 5% minoxidil in propylene glycol. The propylene glycol ensures that the applied minoxidil is evenly spread across the affected area and easily absorbed through the skin. As of early 2001, the 5% solution is only approved by the FDA for use on men. Approximately 1 milliliter of minoxidil solution is applied to the scalp once a day using the fingertips or a pump spray. It should be applied from the center of the area being treated outward.

In the treatment of high blood pressure, oral minoxidil is usually prescribed when other medications have failed to treat the condition. Dosage is usually 2.5-100 mg per day as a single dose for adults and 200 micrograms to 1 mg per kg of body weight for children.

Preparation

Before using topical minoxidil, the hair and scalp should be clean and dry before the minoxidil solution is applied.

Aftercare

Hands, and any other areas of the body where hair growth is not desired that may have come into contact with topical minoxidil, should be washed immediately after applying the minoxidil solution on the scalp. Once applied, topical minoxidil should be allowed to air-dry for at least two to four hours before clothing are pulled on or off over the head, a hat is worn, or the patient goes to bed. Prior to this, the minoxidil solution may stain clothing, hats, or bed linens; or, it may be accidentally transferred from the patient's head to one of these objects, then back to other parts of the patient's body where hair growth is not desired. A blow dryer, or other drying methods, should not be used to speed the drying of the minoxidil as this may interfere with the absorption of the medicine. People using minoxidil should also not shampoo, wash, or rinse their hair for at least 4 hours after minoxidil is applied.

Risks

The most common side effects of topical minoxidil use are **itching** and skin irritation of the treated area of the scalp. Unwanted hair growth may also occur adjacent to treated areas or in areas where the medicine has been inadvertently transferred several times. This unwanted hair growth adjacent to the treatment area may be particularly distressing to women when the face is involved. The itching and irritation usually subside after the drug has been used for approximately two weeks. If symptoms persist after this time, minoxidil use should be halted until a physician has been consulted.

Extremely rare side effects that may occur if too much topically or orally administered minoxidil is being absorbed in the body include:

- changes in vision, most commonly blurred vision
- chest pain
- very low blood pressure
- decreased sexual desire
- fast or irregular heartbeat
- flushing of the skin
- headache
- lightheadedness
- numbness or **tingling** in the hands, feet, or face
- partial, or complete, impotence
- rapid weight gain
- swelling of the hands, feet, lower legs, or face

KEY TERMS

Androgenetic alopecia—Hair loss that develops into baldness and affects both men and women.

Hypertension—Persistently high arterial blood pressure.

Scalp—That part of the head that is usually covered with hair.

Topical drug—Drug or medication applied to a specific area of the skin and affecting only the area to which it is applied.

Vasodilation—The increase in the diameter of a blood vessel resulting from relaxation of smooth muscle within the wall of the vessel. Vasodilation activates the blood flow.

Vasodilators—Drugs or substances that cause vasodilation.

Normal results

Topical minoxidil is much more effective at treating baldness that occurs on the top, or crown, of the head than it is at causing hair growth on other parts of the head. Minoxidil does not work for everyone and there is no predictor, in early 2001, of whether or not it will be effective in any particular person. Clinical tests on the effectiveness of topical minoxidil in men with baldness on the top of the head showed that 48% of men who had used minoxidil for one year reported moderate to dense re-growth of hair within the treated area. Thirty-six percent reported minimal re-growth. While 16% reported no re-growth. Similar percentages have been reported in women.

In both men and women, hair re-growth generally does not begin until the medicine has been used for at least four months. The first signs that minoxidil may be effective in a particular person usually occur after approximately 90 days of treatment, when the patient notices that he or she is losing (shedding) much less hair than prior to beginning treatment.

When new growth begins, the first hairs may be soft and barely visible. For some patients, this is the extent to the effectiveness of this medication. For others, this down-like hair develops into hair of the same color and thickness as the other hairs on their heads.

Minoxidil is a treatment for hair loss, it is not a cure. Once a patient stops taking minoxidil, he or she will most likely lose all of the re-grown hair within

90 days of stopping the medication and no further hair growth will occur.

Resources

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ORGANIZATIONS

American Academy of Dermatology. 930 N. Meacham Road, PO Box 4014, Schaumburg, IL 60168-4014. (847) 330-0230. Fax: 847-330-0050. <<http://www.aad.org/>>.

American Hair Loss Council. 30 Grassy Plain Road, Bethel, CT 06801. (888) 873-9719. <<http://www.ahlc.org/>>.

Paul A. Johnson, Ed.M.

Miscarriage

Definition

Miscarriage means loss of an embryo or fetus before the 20th week of **pregnancy**. Most miscarriages occur during the first 14 weeks of pregnancy. The medical term for miscarriage is spontaneous abortion.

Description

Miscarriages are very common. Approximately 20% of pregnancies (one in five) end in miscarriage. The most common cause is a genetic abnormality of the fetus. Not all women realize that they are miscarrying and others may not seek medical care when it occurs.

A miscarriage is often a traumatic event for both partners, and can cause feelings similar to the loss of a child or other member of the family. Fortunately, 90% of women who have had one miscarriage subsequently have a normal pregnancy and healthy baby; 60% are able to have a healthy baby after two miscarriages. Even a woman who has had three miscarriages in a row still has more than a 50% chance of having a successful pregnancy the fourth time.

Causes and symptoms

There are many reasons why a woman's pregnancy ends in miscarriage. Often the cause is not clear. However, more than half the miscarriages that occur in the first eight weeks of pregnancy involve

serious chromosomal abnormalities or **birth defects** that would make it impossible for the baby to survive. These are different from inherited genetic diseases. They probably occur during development of the specific egg or sperm, and therefore are not likely to occur again.

In about 17% of cases, miscarriage is caused by an abnormal hormonal imbalance that interferes with the ability of the uterus to support the growing embryo. This is known as luteal phase defect. In another 10% of cases, there is a problem with the structure of the uterus or cervix. This can especially occur in women whose mothers used diethylstilbestrol (DES) when pregnant with them.

The risk of miscarriage is increased by:

- **Smoking** (up to a 50% increased risk)
- Infection
- Exposure to toxins (such as arsenic, lead, formaldehyde, benzene, and ethylene oxide)
- Multiple pregnancy
- Poorly-controlled diabetes.

The most common symptom of miscarriage is bleeding from the vagina, which may be light or heavy. However, bleeding during early pregnancy is common and is not always serious. Many women have slight vaginal bleeding after the egg implants in the uterus (about 7-10 days after conception), which can be mistaken for a threatened miscarriage. A few women bleed at the time of their monthly periods through the pregnancy. However, any bleeding in the first three months of pregnancy (first trimester) is considered a threat of miscarriage.

Women should not ignore vaginal bleeding during early pregnancy. In addition to signaling a threatened miscarriage, it could also indicate a potentially life-threatening condition known as **ectopic pregnancy**. In an ectopic pregnancy, the fetus implants at a site other than the inside of the uterus. Most often this occurs in the fallopian tube.

Cramping is another common sign of a possible miscarriage. The cramping occurs because the uterus attempts to push out the pregnancy tissue. If a pregnant woman experiences both bleeding and cramping the possibility of miscarriage is more likely than if only one of these symptoms is present.

If a woman experiences any sign of impending miscarriage, she should be examined by a practitioner. The doctor or nurse will perform a **pelvic exam** to check if the cervix is closed as it should be. If the cervix is open, miscarriage is inevitable and nothing can

preserve the pregnancy. Symptoms of an inevitable miscarriage may include dull relentless or sharp intermittent **pain** in the lower abdomen or back. Bleeding may be heavy. Clotted material and tissue (the placenta and embryo) may pass from the vagina.

A situation in which only some of the products in the uterus have been expelled is called an incomplete miscarriage. Pain and bleeding may continue and become severe. An incomplete miscarriage requires medical attention.

A “missed abortion” occurs when the fetus has died but neither the fetus nor placenta is expelled. There may not be any bleeding or pain, but the symptoms of pregnancy will disappear. The physician may suspect a missed abortion if the uterus does not continue to grow. The physician will diagnose a missed abortion with an ultrasound examination.

A woman should contact her doctor if she experiences any of the following:

- Any bleeding during pregnancy.
- Pain or cramps during pregnancy.
- Passing of tissue.
- Fever and chills during or after miscarriage.

Diagnosis

If a woman experiences any sign of impending miscarriage she should see a doctor or nurse for a pelvic examination to check if the cervix is closed, as it should be. If the cervix is open, miscarriage is inevitable.

An ultrasound examination can confirm a missed abortion if the uterus has shrunk and the patient has had continual spotting with no other symptoms.

Treatment

Threatened miscarriage

For women who experience bleeding and cramping, bed rest is often ordered until symptoms disappear. Women should not have sex until the outcome of the threatened miscarriage is determined. If bleeding and cramping are severe, women should drink fluids only.

Miscarriage

Although it may be psychologically difficult, if a woman has a miscarriage at home she should try to collect any material she passes in a clean container for analysis in a laboratory. This may help determine why the miscarriage occurred.

An incomplete miscarriage or missed abortion may require the removal of the fetus and placenta by a D&C (**dilatation and curettage**). In this procedure the contents of the uterus are scraped out. It is performed in the doctor's office or hospital.

After miscarriage, a doctor may prescribe rest or **antibiotics** for infection. There will be some bleeding from the vagina for several days to two weeks after miscarriage. To give the cervix time to close and avoid possible infection, women should not use tampons or have sex for at least two weeks. Couples should wait for one to three normal menstrual cycles before trying to get pregnant again.

Prognosis

A miscarriage that is properly treated is not life-threatening, and usually does not affect a woman's ability to deliver a healthy baby in the future.

Feelings of grief and loss after a miscarriage are common. In fact, some women who experience a miscarriage suffer from major depression during the six months after the loss. This is especially true for women who don't have any children or who have had depression in the past. The emotional crisis can be similar to that of a woman whose baby has died after birth.

Prevention

The majority of miscarriages cannot be prevented because they are caused by severe genetic problems determined at conception. Some doctors advise women who have a threatened miscarriage to rest in bed for a day and avoid sex for a few weeks after the bleeding stops. Other experts believe that a healthy woman (especially early in the pregnancy) should continue normal activities instead of protecting a pregnancy that may end in miscarriage later on, causing even more profound distress.

If miscarriage was caused by a hormonal imbalance (luteal phase defect), this can be treated with a hormone called progesterone to help prevent subsequent miscarriages. If structural problems have led to repeated miscarriage, there are some possible procedures to treat these problems. Other possible ways to prevent miscarriage are to treat genital infections, eat a well-balanced diet, and refrain from smoking and using recreational drugs.

Resources

ORGANIZATIONS

American College of Obstetricians and Gynecologists. 409 12th Street, S.W., P.O. Box 96920.

KEY TERMS

Diethylstilbestrol (DES)—This is a synthetic estrogen drug that is used to treat a number of hormonal conditions. However, it causes problems in developing fetuses and should not be taken during pregnancy. From about 1938 to 1971, DES was given to pregnant women because it was thought to prevent miscarriage. Children of women who took the drug during pregnancy are at risk for certain health problems.

Dilation and curettage (D&C)—A procedure in which the neck of the womb (cervix) is expanded and the lining of the uterus is scraped to remove pregnancy tissue or abnormal tissue.

Embryo—An unborn child in the first eight weeks after conception. After the eighth week until birth, the baby is called a fetus.

Hygeia Foundation, Inc. P.O. Box 3943 New Haven, CT 06525. (203) 387-3589. < <http://www.hygeia.org> >.

Carol A. Turkington

Mitral incompetence see **Mitral valve insufficiency**

Mitral regurgitation see **Mitral valve insufficiency**

Mitral stenosis see **Mitral valve stenosis**

Mitral valve insufficiency

Definition

Mitral valve insufficiency is a term used when the valve between the upper left chamber of the heart (atrium) and the lower left chamber (ventricle) does not close well enough to prevent back flow of blood when the ventricle contracts. Mitral valve insufficiency is also known as mitral valve regurgitation or mitral valve incompetence.

Description

Normally, blood enters the left atrium of the heart from the lungs and is pumped through the mitral valve into the left ventricle. The left ventricle contracts to

pump the blood forward into the aorta. The aorta is a large artery that sends oxygenated blood through the circulatory system to all of the tissues in the body. If the mitral valve is leaky due to mitral valve insufficiency, it allows some blood to get pushed back into the atrium. This extra blood creates an increase in pressure in the atrium, which then increases blood pressure in the vessels that bring the blood from the lungs to the heart. Increased pressure in these vessels can result in increased fluid buildup in the lungs.

Causes and symptoms

In the past, **rheumatic fever** was the most common cause of mitral valve insufficiency. However, the increased use of **antibiotics** for **strep throat** has made rheumatic fever rare in developed countries. In these countries, mitral valve insufficiency caused by rheumatic fever is seen mostly in the elderly. In countries with less developed health care, rheumatic fever is still common and is often a cause of mitral valve insufficiency.

Heart attacks that damage the structures that support the mitral valve are a common cause of mitral valve insufficiency. Myxomatous degeneration can cause a “floppy” mitral valve that leaks. In other cases, the valve simply deteriorates with age and becomes less efficient.

People with mitral valve insufficiency may not have any symptoms at all. It is often discovered during a doctor’s visit when the doctor listens to the heart sounds.

Both the left atrium and left ventricle tend to get a little bigger when the mitral valve does not work properly. The ventricle has to pump more blood so it gets bigger to increase the force of each beat. The atrium gets bigger to hold the extra blood. An enlarged ventricle can cause **palpitations**. An enlarged atrium can develop an erratic rhythm (atrial fibrillation), which reduces its efficiency and can lead to **blood clots** forming in the atrium.

Diagnosis

When the doctor listens to the heart sounds, mitral valve insufficiency is generally recognized by the sound the blood makes as it leaks backward. It sounds like a regurgitant murmur. The next step is generally a **chest x ray** and an electrocardiogram (ECG) to see if the heart is enlarged. The most definitive noninvasive test is **echocardiography**, a test that uses sound waves to make an image of the heart. This

KEY TERMS

Aorta—A large artery beginning at the base of the left ventricle.

Atrium—One of the two upper chambers of the heart.

Rheumatic fever—An illness that sometimes follows a streptococcal infection of the throat.

Ventricle—One of the two lower chambers of the heart.

test gives a picture of the valve in action and shows the severity of the problem.

Treatment

A severely impaired valve needs to be repaired or replaced. Either option will require surgery. Repairing the valve can fix the problem completely or reduce it enough to make it bearable and prevent damage to the heart. Valves can be replaced with either a mechanical valve or one that is partly mechanical and partly from a pig’s heart.

Mechanical valves are effective but can increase the incidence of blood clots. To prevent blood clots from forming, the patient will need to take drugs that prevent abnormal blood clotting (anticoagulants). The valves made partly from a pig’s heart do not have as great a risk of blood clots but don’t last as long as fully mechanical valves. If a valve wears out, it must be replaced again.

Damaged heart valves are easily infected. Anytime a procedure is contemplated that might allow infectious organisms to enter the blood, the person with mitral valve insufficiency should take antibiotics to prevent possible infection.

Prognosis

The diagnostic, medical and surgical procedures available to the person with mitral valve insufficiency are all likely to produce good results.

Prevention

The only possible way to prevent mitral valve insufficiency is to prevent rheumatic fever. This can be done by evaluating sore throats for the presence of the bacteria that causes strep throat. Strep throat is easily treated with antibiotics.

Resources

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300. <<http://www.americanheart.org>>.

OTHER

The Meck Page. <<http://www.merck.com>>.

Dorothy Elinor Stonely

Mitral valve prolapse

Definition

Mitral valve prolapse (MVP) is a ballooning of the support structures of the mitral heart valve into the left upper collection chamber of the heart.

Description

Other names for MVP include floppy valve and Barlow's syndrome. The mitral valve is located on the left side of the heart between the top chamber (left atrium) and the bottom chamber (left ventricle). The valve opens and closes according to the heartbeat and the pressure that is exerted upon it from the blood in both chambers.

The valve has supporting structures that attach to the heart muscle to help it open and close properly. When these structures weaken or lengthen abnormally, the valve may balloon into the left atrium. Sometimes this can cause the mitral valve to leak blood backward.

This condition may be inherited and occurs in approximately 10% of the population. It affects more women than men and often peaks after the age of 40.

Causes and symptoms

MVP may occur due to rheumatic heart disease but is usually found in healthy people. Changes that occur in the valve are caused by rapid multiplication of cells in the middle layer that presses on the outer layer. The outer layer weakens, causing a prolapse of the valve toward the left atrium.

Most persons do not have symptoms. Those that do may experience sharp, left-sided chest **pain**. Some complain of **fatigue**, or a pounding feeling in the chest. Others can have an irregular heart beat and even

pass out. Some persons may experience difficulty breathing, ankle swelling and fluid in the lungs. Other symptoms may include **anxiety**, headaches, morning tiredness and constantly cold hands and feet. **Death** from this condition is rare.

Diagnosis

The diagnosis of MVP is based on symptoms and physical exam. During the exam, the physician may hear a click and/or heart murmur with a stethoscope.

The best diagnostic test for MVP is the echocardiogram. The test reflects sound waves through the chest wall to give two-dimensional color flow pictures of the heart, its size, position, motion, chambers, and valves. Unfortunately, during the early 1980s, this diagnosis was often made excessively from faulty echocardiographic criteria prevalent at that time.

Any person with symptoms or family history of MVP should consider having an echocardiogram. The test takes 15-20 minutes and is done in doctor's offices and hospitals. It is performed by trained technicians and is read by cardiologists. Family physicians, internists, cardiologists, and nurse practitioners can treat MVP. Echocardiograms are recommended periodically depending on the extent of valve leakage.

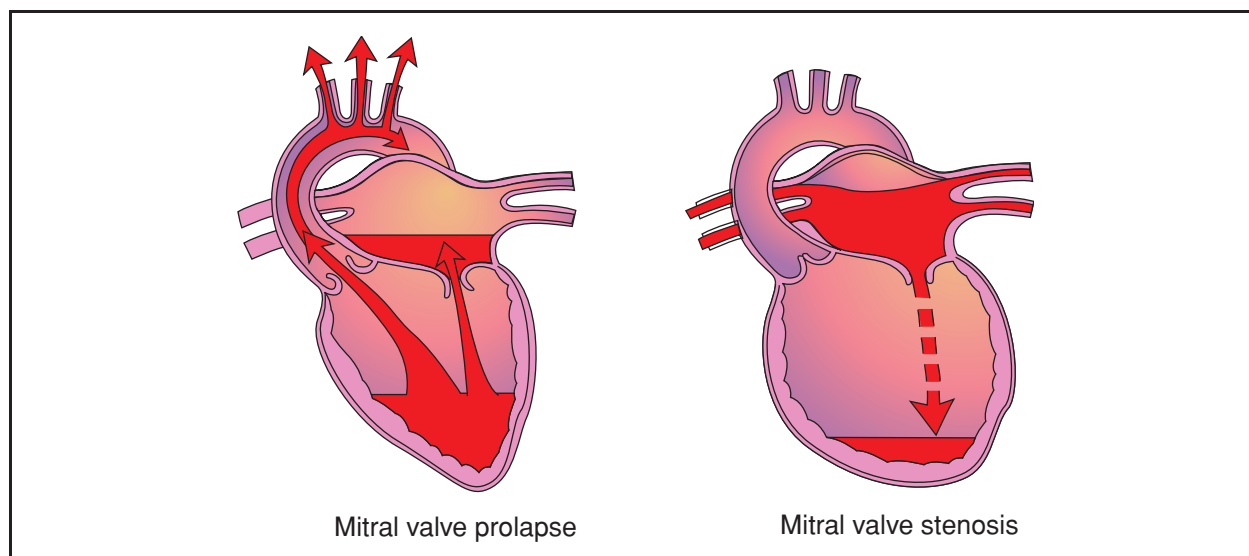
Treatment

Persons who experience certain types of an irregular heartbeat with MVP should be treated. Propranolol (Inderal) or other **beta blockers** or digoxin (Lanoxin) are often helpful. Persons who develop moderate to severe symptoms with a leaky mitral valve may require repair or replacement of the mitral valve with an artificial heart valve. Persons with MVP and a leaky valve need to protect themselves from heart or heart valve infections. **Antibiotics** should be taken before any surgical, dental or oral procedures according to the American Heart Association recommendations.

Other treatments include drinking lots of fluids during strenuous activity and hot weather. Water pills, **caffeine** and donating blood may aggravate the symptoms of MVP.

Prognosis

MVP is usually not a serious condition. However, dangerous, untreated irregular heartbeats may rarely cause sudden death. These persons should be carefully monitored.



Mitral valve prolapse occurs when the mitral valve does not open and close properly. When this happens, the valve may balloon into the left atrium of the heart, causing the mitral valve to leak blood backward. Mitral valve stenosis refers to the narrowing of the mitral valve, in which the flow of blood from the atrium to the ventricle becomes restricted. (Illustration by Electronic Illustrators Group.)

KEY TERMS

Heart murmur—Sound during the heartbeat caused by a heart valve that does not close properly.

Rheumatic heart disease—A condition caused by a streptococcus infection which can result in permanent heart damage.

Resources

PERIODICALS

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Lisa Papp, RN

Mitral valve stenosis

Definition

The term stenosis means an abnormal narrowing of an opening. Mitral valve stenosis refers to a condition in the heart in which one of the valve openings has

become narrow and restricts the flow of blood from the upper left chamber (left atrium) to the lower left chamber (left ventricle).

Description

In the heart, the valve that regulates the flow of blood between the left atrium and the left ventricle is called the mitral valve. If the mitral valve is abnormally narrow, due to disease or birth defect, blood flow from the atrium to the ventricle is restricted. This restricted flow leads to an increase in the pressure of blood in the left atrium. Over a period of time, this back pressure causes fluid to leak into the lungs. It can also lead to an abnormal heart rhythm (atrial fibrillation), which further decreases the efficiency of the pumping action of the heart.

Causes and symptoms

Mitral valve stenosis is almost always caused by **rheumatic fever**. As a result of rheumatic fever, the leaflets that form the opening of the valve are partially fused together. Mitral valve stenosis can also be present at birth. Babies born with this problem usually require surgery if they are to survive. Sometimes, growths or tumors can block the mitral valve, mimicking mitral valve stenosis.

If the restriction is severe, the increased blood pressure can lead to **heart failure**. The first symptoms

of heart failure, which are **fatigue** and **shortness of breath**, usually appear only during physical activity. As the condition gets worse, symptoms may also be felt even during rest. A person may also develop a deep red coloring in the cheeks.

Diagnosis

Mitral valve stenosis is usually detected by a physician listening to heart sounds. Normal heart valves open silently to permit the flow of blood. A stenotic valve makes a snapping sound followed by a “rumbling” murmur. The condition can be confirmed with a **chest x ray** and an electrocardiogram, both of which will show an enlarged atrium. **Echocardiography**, which produces images of the heart’s structure, is also helpful in making the diagnosis. If surgery is necessary, **cardiac catheterization** may be done to fully evaluate the heart before the operation.

Treatment

Drug therapy may help to slow the heart rate, strengthen the heart beat, and control abnormal heart rhythm. Drugs such as **beta blockers**, **calcium channel blockers**, and digoxin may be prescribed. A drug that prevents abnormal blood clotting (anticoagulant) called warfarin (Coumadin) may be recommended. If drug therapy does not produce satisfactory results, valve repair or replacement may be necessary.

Repair can be accomplished in two ways. In the first method, **balloon valvuloplasty**, the doctor will try to stretch the valve opening by threading a thin tube (catheter) with a balloon tip through a vein and into the heart. Once the catheter is positioned in the valve, the balloon is inflated, separating the fused areas. The second method involves opening the heart and surgically separating the fused areas.

If the valve is damaged beyond repair, it can be replaced with a mechanical valve or one that is partly mechanical and partly made from a pig’s heart.

Prognosis

Procedures available to treat mitral valve stenosis, whether medical or surgical, all produce effective results.

Prevention

The only possible way to prevent mitral valve stenosis is to prevent rheumatic fever. This can be done by evaluating sore throats for the presence of

KEY TERMS

Atrium—One of the two upper chambers of the heart.

Beta blocker—A drug that can be used to reduce blood pressure.

Rheumatic fever—An illness which sometimes follows a streptococcal infection of the throat.

Ventricle—One of the two lower chambers of the heart.

the bacteria that causes **strep throat**. Strep throat is easily treated with **antibiotics**.

Resources

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300. <<http://www.americanheart.org>>.

OTHER

The Meck Page. <<http://www.merck.com>>.

Dorothy Elinor Stonely

Molar pregnancy see **Hydatidiform mole**

Moles

Definition

A mole (nevus) is a pigmented (colored) spot on the outer layer of the skin (epidermis).

Description

Moles can be round, oval, flat, or raised. They can occur singly or in clusters on any part of the body. Most moles are brown, but colors can range from pinkish flesh tones to yellow, dark blue, or black.

Everyone has at least a few moles. They generally appear by the time a person is 20 and resemble freckles at first. A mole’s color and shape don’t usually change. Changes in hormone levels that occur during **puberty** and **pregnancy** can make moles larger and darker. New moles may also appear during this period.

A mole usually lasts about 50 years before beginning to fade. Some moles disappear completely, and

some never lighten at all. Some moles develop stalks that raise them above the skin's surface; these moles eventually drop off.

Types of moles

About 1–3% of all babies have one or more moles when they are born. Moles that are present at birth are called congenital nevi.

Other types of moles include:

- Junctional moles, which are usually brown and may be flat or slightly raised.
- Compound moles, which are slightly raised, range in color from tan to dark brown, and involve pigment-producing cells (melanocytes) in both the upper and lower layers of the skin (epidermis and dermis).
- Dermal moles, which range from flesh-color to brown, are elevated, most common on the upper body, and may contain hairs.
- Sebaceous moles, which are produced by over-active oil glands and are yellow and rough-textured.
- Blue moles, which are slightly raised, colored by pigment deep within the skin, and most common on the head, neck, and arms of women.

Most moles are benign, but atypical moles (dysplastic nevi) may develop into **malignant melanoma**, a potentially fatal form of skin **cancer**. Atypical moles are usually hereditary. Most are bigger than a pencil eraser, and the shape and pigmentation are irregular.

Congenital nevi are more apt to become cancerous than moles that develop after birth, especially if they are more than eight inches in diameter. Lentigo maligna (melanotic freckle of Hutchinson), most common on the face and after the age of 50, first appears as a flat spot containing two or more shades of tan. It gradually becomes larger and darker. One in three of these moles develop into a form of skin cancer known as lentigo maligna melanoma.

Causes and symptoms

The cause of moles is unknown, although atypical moles seem to run in families and result from exposure to sunlight.

In the past several years, researchers have identified two genes known as CDKN2A and CDK4 that govern susceptibility to melanoma in humans. Other susceptibility genes are being sought as of early 2003. Most experts, however, think that these susceptibility genes are not sufficient by themselves to account for moles becoming cancerous but are influenced by a

combination of other inherited traits and environmental factors.

Diagnosis

Only a small percentage of moles require medical attention. A mole that has the following symptoms should be evaluated by a dermatologist (a physician specializing in skin diseases).

- Appears after the age of 20
- Bleeds
- Itches
- Looks unusual or changes in any way.

A doctor who suspects skin cancer will remove all or part of the mole for microscopic examination. This procedure, which is usually performed in a doctor's office, is simple, relatively painless, and does not take more than a few minutes. It does leave a scar.

The doctor may also use a dermatoscope to examine the mole prior to removal. The dermatoscope, which can be used to distinguish between benign moles and melanomas, is an instrument that resembles an ophthalmoscope. An immersion oil is first applied to the mole to make the outer layers of skin transparent.

A combination of high-frequency ultrasound and color Doppler studies has also been shown to have a high degree of accuracy in distinguishing between melanomas and benign moles.

Treatment

If laboratory analysis confirms that a mole is cancerous, the dermatologist will remove the rest of the mole. Patients should realize that slicing off a section of a malignant mole will not cause the cancer to spread.

Removing a mole for cosmetic reasons involves numbing the area and using scissors or a scalpel to remove the elevated portion. The patient is left with a flat mole the same color as the original growth. Cutting out parts of the mole above and beneath the surface of the skin can leave a scar more noticeable than the mole.

Scissors or a razor can be used to temporarily remove hair from a mole. Permanent hair removal, however, requires electrolysis or surgical removal of the mole.

Prognosis

Moles are rarely cancerous and, once removed, unlikely to recur. A dermatologist should be consulted if a mole reappears after being removed.



Woman's birthmark being removed by laser. (Photograph by Alexander Tsiaras, Photo Researchers, Inc. Reproduced by permission.)

Prevention

Wearing a sunscreen and limiting sun exposure may prevent some moles. Anyone who has moles should examine them every month and see a dermatologist if changes in size, shape, color, or texture occur or if new moles appear.

A team of researchers at Duke University reported in 2003 that topical application of a combination of 15% vitamin C and 1% vitamin E over a four-day period offered significant protection against **sunburn**. The researchers suggest that this combination may protect skin against **aging** caused by sunlight as well.

Anyone with a family history of melanoma should see a dermatologist for an annual skin examination. Everyone should know the ABCDEs of melanoma:

- A: Asymmetry, which occurs when the two halves of the mole are not identical
- B: Borders that are irregular or indistinct
- C: Color that varies in a single mole

KEY TERMS

Dermatology—The branch of medicine that studies and treats disorders of the skin.

Malignant melanoma—A potentially fatal form of skin cancer that develops from melanocytes, which are skin cells containing melanin.

Melanin—A dark insoluble pigment found in humans in the skin, hair, choroid layer of the eye, and a part of the brain called the substantia nigra.

Nevus (plural, nevi)—The medical term for any anomaly of the skin that is present at birth, including moles and birthmarks.

- D: Diameter, which should be no larger than a pencil eraser (about 6 mm)
- E: Elevated above the surrounding tissue.

A mole with any of these characteristics should be evaluated by a dermatologist.

Advances in photographic technique have now made it easier to track the development of moles with the help of whole-body photographs. A growing number of hospitals are offering these photographs as part of outpatient mole-monitoring services.

Resources

BOOKS

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- Tucker, M. A., and A. M. Goldstein. "Melanoma Etiology: Where Are We?" *Oncogene* 22 (May 19, 2003): 3042–3052.

ORGANIZATIONS

- American Academy of Dermatology. 930 N. Meacham Road, P.O. Box 4014, Schaumburg, IL 60168-4014. (847) 330-0230. Fax: (847) 330-0050. <<http://www.aad.org>>.
- American Cancer Society. 1599 Clifton Road NE, Atlanta, GA 30329. (800) ACS-2345.
- National Cancer Institute (NCI). NCI Public Inquiries Office, Suite 3036A, 6116 Executive Boulevard, MSC8332, Bethesda, MD 20892-8322. (800) 4-CANCER or (800) 332-8615 (TTY). <<http://www.nci.nih.gov>>.

Nevus Outreach, Inc. 1601 Madison Blvd., Bartesville, OK 74006. (877) 426-3887. <<http://www.nevus.org>>.

Maureen Haggerty
Rebecca J. Frey, PhD

Molybdenum excess see **Mineral toxicity**

Mometasone see **Corticosteroids**

Monocytic ehrlichiosis see **Ehrlichiosis**

Mongolism see **Down syndrome**

Moniliasis see **Candidiasis**

Monkeypox

Definition

Monkeypox is an infectious disease caused by an orthopoxvirus. Orthopoxviruses are a genus of viruses that include the disease agents that cause human **smallpox**, cowpox, and camelpox as well as monkeypox. Monkeypox, which was first identified in humans in an outbreak in Africa in 1970, usually produces a less severe illness with fewer fatalities than smallpox. However, its symptoms are similar: **fever**, pus-filled blisters all over the body, and respiratory problems.

Monkeypox is classified as a **zoonosis**, which means that it is a disease of animals that can be transmitted to humans under natural conditions. The first cases of monkeypox reported in humans involved contact between humans and animals in the African rain forest. The outbreak that made headlines in the United States in June 2003, however, involved animals purchased as pets from pet stores. In nature, monkeypox has been found in monkeys, chimpanzees, rabbits, prairie dogs, Gambian rats, ground squirrels, and mice. It is not known as of late 2003 whether other wild or domestic animals can contract monkeypox.

Description

Prior to 2003, most monkeypox cases were diagnosed in remote areas of central and west Africa. Between February 1996 and October 1997, however, there were 511 suspected cases of monkeypox in the Democratic Republic of the Congo (DRC, formerly Zaire). This outbreak, the largest ever, raised fears that the virus had mutated and become more infectious.

In late 1997, the U.S. Centers for Disease Control and Prevention (CDC) and the World Health Organization (WHO) announced that this relatively large outbreak was likely due to human behavior, rather than virus mutation. During the outbreak, the DRC was embroiled in civil war. Food shortages increased reliance on hunting and raised chances that people would come into contact with infected animals.

The 2003 outbreak in the United States, which was the first confirmed instance of community-acquired monkeypox in North America, came to the attention of the CDC in early June, when a laboratory in Wisconsin identified the monkeypox virus in samples taken from the skin of an infected patient and lymph node tissue from the patient's pet prairie dog. By the end of June, cases of monkeypox in humans had been identified in six Midwestern states (Illinois, Indiana, Kansas, Missouri, Ohio, and Wisconsin). The patients acquired the virus from infected prairie dogs purchased as pets, which in turn were infected through contact with animals imported from Africa that were sold in the same pet stores.

Monkeypox is less severe than smallpox and can sometimes be confused with **chickenpox**. It seems partly preventable with smallpox **vaccination**, but vaccination programs were discontinued in the late 1970s. (Barring samples stored in laboratories, smallpox has been eradicated.) People under the age of 16—those born after smallpox vaccination ended—seem the most susceptible to monkeypox. During the 1996-97 outbreak, approximately 85% of the cases were in this age group.

Although the monkeypox virus is related to the smallpox virus, experts do not think (as of late 2003) that it is likely to be cultivated as an agent of bioterrorism. Monkeypox is much less easily transmitted person-to-person than smallpox and has a much lower fatality rate.

Causes and symptoms

The monkeypox virus is transmitted to humans through an infected animal's blood, body sores, or bite; or through handling an infected animal's bedding or cage. Initial symptoms of monkeypox in humans include fever, a bodywide rash (exanthem) of pus-filled blisters, and flu-like muscle aches and **fatigue**. These symptoms can be accompanied by **diarrhea**, swollen lymph nodes, a **sore throat**, and mouth sores. In some cases, a victim may experience trouble breathing. Symptoms are at their worst for 3–7 days, after which the fever lessens and blisters begin to form crusts.

The symptoms of monkeypox in pet rabbits, rats, or mice include inflammation of the eyes, a nasal

discharge, fever, loss of appetite, a skin rash, and tiredness. Pet monkeys typically develop a rash with pus-filled lesions on the palms of the hands, trunk, and tail. They may also have mouth ulcers.

Diagnosis

Since the symptoms of monkeypox resemble other diseases caused by orthopox viruses, definitive diagnosis may require laboratory testing to uncover the virus or evidence (from antibodies in the blood) that it is present. As of 2003, laboratory techniques that can be used to identify the monkeypox virus include electron microscopy, polymerase chain reaction (PCR), immunohistochemistry, and ELISA testing.

Treatment

Like most viruses, monkeypox cannot be resolved with medication. The only treatment option is symptomatic—that is, patients are made as comfortable as possible. In March 1998, the U.S. Army Medical Research Institute for Infectious Diseases reported that an antiviral drug called **cidofovir** may combat monkeypox infection. As of 2003, additional studies report that cidofovir appears to be safe and effective as a treatment for monkeypox in humans. The drug has worked successfully in primates, but further research is needed to determine its effectiveness in humans.

Prognosis

Children are more likely to contract the disease and have the highest **death** rate. Monkeypox is not as lethal as smallpox, but the death rate among young children may reach 2–10%. In some cases, hospitalization is required. Recovery is good among survivors, although some scarring may result from the blisters.

Prevention

Monkeypox is one of the diseases that physicians, veterinarians, and public health officials are required by law to report to the CDC.

Although smallpox vaccination offers some protection against monkeypox, experts do not generally recommend getting a smallpox vaccination simply to guard against monkeypox if one has not been exposed to it. However, the CDC recommends as of June 2003 that anyone who has had close contact with humans or animals infected with monkeypox, or has helped to care for them, should be vaccinated against smallpox. The vaccination can be administered as late as 14 days after exposure to the virus. In addition, veterinarians

KEY TERMS

Antiviral—Refers to a drug that can destroy viruses and help treat illnesses caused by them.

Bioterrorism—The intentional use of disease-causing microbes or other biologic agents to intimidate or terrorize a civilian population for political or military reasons.

Mutation—A change in an organism's genetic code that causes it to develop new characteristics.

Orthopoxvirus—The genus of viruses that includes monkeypox, smallpox, cowpox, and camelpox.

Symptomatic—Refers to treatment that addresses the symptoms of an illness, but not its underlying cause.

Zoonosis (plural, zoonoses)—Any disease of animals that can be transmitted to humans under natural conditions. Monkeypox is a zoonosis.

or public health personnel conducting field investigations should be vaccinated before any exposure to monkeypox.

As of late 2003, no cases of monkeypox were identified in cats or dogs belonging to people infected by the June outbreak. The American Veterinary Medicine Association (AVMA) recommends, however, that cats, dogs, or other mammals that have been in contact with an animal known to have monkeypox should be kept in quarantine for 30 days from the date of exposure.

People who have a pet with symptoms of monkeypox should *not* take it to an animal shelter or release it into the wild. They should isolate it from humans and other animals, and take it to a veterinarian in a closed, chew-proof container with air holes.

On June 11, 2003, the CDC and the Food and Drug Administration (FDA) issued a joint order prohibiting the importation of rats and other rodents from Africa. In addition, the agencies banned the sale and distribution of prairie dogs and six species of African rodents in the United States.

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ORGANIZATIONS

American Veterinary Medical Association (AVMA). 1931 North Meacham Road, Suite 100, Schaumburg, IL 60173-4360. <<http://www.avma.org>>.

Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

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Julia Barrett
Rebecca J. Frey, PhD

Monoamine oxidase inhibitors

Definition

Monoamine oxidase inhibitors (MAO inhibitors) are medicines that relieve certain types of mental depression.

Purpose

MAO inhibitors are a type of antidepressant and are used to treat mental depression. Like other **antidepressant drugs**, MAO inhibitors help reduce the extreme sadness, hopelessness, and lack of interest in life that are typical in people with depression. MAO inhibitors are especially useful in treating people whose depression is combined with other problems such as **anxiety**, panic attacks, **phobias**, or the desire to sleep too much.

Description

Discovered in the 1950s, MAO inhibitors work by correcting chemical imbalances in the brain. Normally, natural chemicals called neurotransmitters carry signals from one brain cell to another. Some neurotransmitters, such as serotonin and norepinephrine, play important roles in controlling mood. But other substances in the brain may interfere with mood control by breaking down these neurotransmitters. Researchers believe that MAO inhibitors work by blocking the chemicals that break down serotonin and norepinephrine. This gives the neurotransmitters more time to do their important work.

Because MAO inhibitors also affect other chemicals throughout the body, these drugs may produce many unwanted side effects. They can be especially dangerous when taken with certain foods, beverages and medicines. Anyone taking these drugs should ask his or her physician or pharmacist for a list of products to avoid.

MAO inhibitors are available only with a physician's prescription. They are sold in tablet form. Some commonly used MAO inhibitors are isocarboxazid (Marplan), phenelzine (Nardil), and tranylcypromine (Parnate).

Recommended dosage

The recommended dosage depends on the type of MAO inhibitor and the type of depression for which it is being taken. Dosages may be different for different patients. Check with the physician who prescribed the drug or the pharmacist who filled the prescription for the correct dosage.

Always take MAO inhibitors exactly as directed by your physician. Never take larger or more frequent doses, and do not take the drug for longer than directed. See the physician regularly while taking this medicine, especially in the first few months of treatment. The physician will check to make sure the medicine is

working as it should and will note unwanted side effects. The physician may also need to adjust the dosage during this period.

Several weeks may be needed for the effects of this medicine to be felt. Be sure to keep taking it as directed, even if it does not seem to be helping.

Do not stop taking this medicine suddenly. Tapering the dose may be necessary to reduce the chance of withdrawal symptoms. If it is necessary to stop taking the drug, check with the physician who prescribed it for instructions on how to stop.

MAO inhibitors may be taken with or without food, on a full or empty stomach. Check package directions or ask the physician or pharmacist for instructions on how to take the medicine. Remember that some foods and beverages must be avoided during treatment with MAO inhibitors.

Precautions

The effects of this medicine may continue for 2 weeks or more after patients stop taking it. All precautions should be observed during this period, as well as throughout treatment with MAO inhibitors.

MAO inhibitors may cause serious and possibly life-threatening reactions, such as sudden high blood pressure, when taken with certain foods, beverages, or medicines. The dangerous reactions may not begin until several hours after consuming these things. Aged cheeses, red wines, smoked or pickled meats, chocolate, caffeinated beverages, and foods containing monosodium glutamate (MSG) are among the foods and drinks to be avoided. Be sure to get a complete list from the physician who prescribed the medicine or the pharmacist who filled the prescription.

Do not drink any alcoholic beverages or reduced-alcohol or alcohol-free beer or wine while taking this medicine.

Anyone who is taking MAO inhibitors should not use any other medicine unless it has been approved or prescribed by a physician who knows that they are taking MAO inhibitors. This includes nonprescription (over-the-counter) medicines such as sleep aids; medicines for colds, **cough**, hay **fever**, or **asthma** (including nose drops or sprays); medicines to increase alertness or keep from falling asleep; and appetite control products.

Because MAO inhibitors work on the central nervous system, they may add to the effects of alcohol and other drugs that slow down the central nervous system, such as **antihistamines**, cold medicine, allergy

medicine, sleep aids, medicine for seizures, tranquilizers, some **pain** relievers, and **muscle relaxants**. Anyone taking MAO inhibitors should check with his or her physician before taking any of the above.

MAO inhibitors may interact with medicines used during surgery, dental procedures, or emergency treatment. These interactions could increase the chance of side effects. Anyone who is taking MAO inhibitors should be sure to tell the health care professional in charge before having any surgical or dental procedures or receiving emergency treatment.

Some people feel drowsy, dizzy, lightheaded, or less alert when using MAO inhibitors. The drugs may also cause blurred vision. For these reasons, anyone who takes these drugs should not drive, use machines or do anything else that might be dangerous until they have found out how the drugs affect them.

These medicines also make some people feel lightheaded, dizzy, or faint when they get up after sitting or lying down. To lessen the problem, get up gradually and hold onto something for support if possible.

Older people may be especially sensitive to the effects of MAO inhibitors. This may increase the chance of side effects, such as **dizziness** or lightheadedness.

Special conditions

People with certain medical conditions or who are taking certain other medicines can have problems if they take MAO inhibitors. Before taking these drugs, be sure to let the physician know about any of these conditions:

ALLERGIES. Anyone who has had unusual reactions to MAO inhibitors in the past should let his or her physician know before taking the drugs again. The physician should also be told about any **allergies** to foods, dyes, preservatives, or other substances.

PREGNANCY. Studies suggest that taking MAO inhibitors during **pregnancy** may increase the risk of **birth defects** or problems in the newborn after birth. Women who are pregnant or who may become pregnant should check with their physicians before using MAO inhibitors.

BREASTFEEDING. MAO inhibitors may pass into breast milk, but no problems have been reported in nursing babies whose mothers took the medicine. Women who are breastfeeding their babies should check with their physicians before using this medicine.

DIABETES. MAO inhibitors may affect blood sugar levels. Persons with diabetes who are taking this medicine and notice changes in their blood or urine tests should check with their physicians.

ANGINA. MAO inhibitors may make people feel unusually energetic and healthy. People with **angina** (chest pain) should be careful not to overexert themselves and should check with their physicians before increasing their levels of activity or **exercise**.

OTHER MEDICAL CONDITIONS. Before using MAO inhibitors, people with any of these medical problems should make sure their physicians are aware of their conditions:

- Alcohol **abuse**
- High blood pressure
- Recent **heart attack** or **stroke**
- Heart or blood vessel disease
- Liver disease
- Kidney disease
- Frequent or severe headaches
- Epilepsy
- Parkinson's disease
- Current or past mental illness
- Asthma or **bronchitis**
- Overactive thyroid
- Pheochromocytoma (a tumor of the adrenal gland).

USE OF CERTAIN MEDICINES. Taking MAO inhibitors with certain other drugs may affect the way the drugs work or may increase the chance of side effects.

Side effects

The most common side effects are dizziness, lightheadedness, drowsiness, tiredness, weakness, blurred vision, shakiness or trembling, restlessness, sleep problems or twitching during sleep, increased appetite (especially for sweets), weight gain, decreased sexual ability, decreased amount of urine, and mild **headache**. These problems usually go away as the body adjusts to the drug and do not require medical treatment unless they interfere with normal activities.

More serious side effects may occur. If any of the following side effects occur, stop taking the medicine and get emergency medical attention immediately:

- Severe chest pain
- Severe headache
- Stiff, sore neck
- Enlarged pupils
- Increased sensitivity of eyes to light
- Fast or slow heartbeat

KEY TERMS

Anxiety—Worry or tension in response to real or imagined stress, danger, or dreaded situations. Physical reactions, such as fast pulse, sweating, trembling, fatigue, and weakness may accompany anxiety.

Central nervous system—The brain and spinal cord.

Depression—A mental condition in which people feel extremely sad and lose interest in life. People with depression may also have sleep problems and loss of appetite and may have trouble concentrating and carrying out everyday activities.

Neurotransmitter—A chemical that carries messages from one nerve cell to another.

Phobia—An intense, abnormal, or illogical fear of something specific, such as heights or open spaces.

Withdrawal symptoms—A group of physical or mental symptoms that may occur when a person suddenly stops using a drug to which he or she has become dependent.

- Sweating, with or without fever or cold, clammy skin
- Nausea and vomiting.

Other side effects may occur. Anyone who has unusual or troublesome symptoms after taking MAO inhibitors should get in touch with his or her physician.

Interactions

MAO inhibitors may interact with many other medicines. When this happens, the effects of one or both of the drugs may change or the risk of side effects may be greater. *Anyone who takes MAO inhibitors must check with his or her physician before taking any other prescription or nonprescription (over-the-counter) medicine.* Among the drugs that may interact with MAO inhibitors are:

- Central nervous system (CNS) depressants such as medicine for allergies, colds, hay fever, and asthma; sedatives; tranquilizers; prescription pain medicine; muscle relaxants; medicine for seizures; sleep aids; **barbiturates**; and anesthetics
- Medicine for high blood pressure
- Other antidepressants, including **tricyclic antidepressants** (such as Tofranil and Norpramin), antidepressants that raise serotonin levels (such as Prozac and Zoloft), and bupropion (Wellbutrin)

- Diabetes medicines taken by mouth
- Insulin
- Water pills (diuretics).

The list above does not include every drug that may interact with MAO inhibitors. Check with a physician or pharmacist before combining MAO inhibitors with any other prescription or nonprescription (over-the-counter) medicine.

Nancy Ross-Flanigan

Mononucleosis see **Infectious mononucleosis**

Montezuma's revenge see **Traveler's diarrhea**

Mood disorders

Definition

Mood disorders are mental disorders characterized by periods of depression, sometimes alternating with periods of elevated mood.

Description

While many people go through sad or elated moods from time to time, people with mood disorders suffer from severe or prolonged mood states that disrupt their daily functioning. Among the general mood disorders classified in the fourth edition (1994) of the *Diagnostic and Statistical Manual of Mental Disorders (DSM-IV)* are major depressive disorder, **bipolar disorder**, and dysthymia.

In classifying and diagnosing mood disorders, doctors determine if the mood disorder is unipolar or bipolar. When only one extreme in mood (the depressed state) is experienced, this type of depression is called unipolar. Major depression refers to a single severe period of depression, marked by negative or hopeless thoughts and physical symptoms like **fatigue**. In major depressive disorder, some patients have isolated episodes of depression. In between these episodes, the patient does not feel depressed or have other symptoms associated with depression. Other patients have more frequent episodes.

Bipolar depression or bipolar disorder (sometimes called manic depression) refers to a condition in which people experience two extremes in mood. They

alternate between depression (the “low” mood) and **mania** or hypomania (the “high” mood). These patients go from depression to a frenzied, abnormal elevation in mood. Mania and hypomania are similar, but mania is usually more severe and debilitating to the patient.

Dysthymia is a recurrent or lengthy depression that may last a lifetime. It is similar to major depressive disorder, but dysthymia is chronic, long-lasting, persistent, and mild. Patients may have symptoms that are not as severe as major depression, but the symptoms last for many years. It seems that a mild form of the depression is always present. In some cases, people also may experience a major depressive episode on top of their dysthymia, a condition sometimes referred to as a “double depression.”

Causes and symptoms

Mood disorders tend to run in families. These disorders are associated with imbalances in certain chemicals that carry signals between brain cells (neurotransmitters). These chemicals include serotonin, norepinephrine, and dopamine. Women are more vulnerable to unipolar depression than are men. Major life stressors (like divorce, serious financial problems, **death** of a family member, etc.) will often provoke the symptoms of depression in susceptible people.

Major depression is more serious than just feeling “sad” or “blue.” The symptoms of major depression may include:

- Loss of appetite
- A change in sleep patterns, like not sleeping (**insomnia**) or sleeping too much
- Feelings of worthlessness, hopelessness, or inappropriate guilt
- Fatigue
- Difficulty in concentrating or making decisions
- Overwhelming and intense feelings of sadness or grief
- Disturbed thinking. The person may also have physical symptoms like stomachaches or headaches.

Bipolar disorder includes mania or hypomania. Mania is an abnormal elevation in mood. The person may be excessively cheerful, have grandiose ideas, and may sleep less. He or she may talk nonstop for hours, have unending enthusiasm, and demonstrate poor judgement. Sometimes the elevation in mood is marked by irritability and hostility rather than cheerfulness. While the person may at first seem normal with an

increase in energy, others who know the person well see a marked difference in behavior. The patient may seem to be in a frenzy and often will make poor, bizarre, or dangerous choices in his/her personal and professional lives. Hypomania is not as severe as mania and does not cause the level of impairment in work and social activities that mania can.

Diagnosis

Doctors diagnose mood disorders based on the patient’s description of the symptoms as well as the patient’s family history. The length of time the patient has had symptoms also is important. Generally patients are diagnosed with dysthymia if they feel depressed more days than not for at least two years. The depression is mild but long lasting. In major depressive disorder, the patient is depressed almost all day nearly every day of the week for at least two weeks. The depression is severe. Sometimes laboratory tests are performed to rule out other causes for the symptoms (like thyroid disease). The diagnosis may be confirmed when a patient responds well to medication.

Treatment

The most effective treatment for mood disorders is a combination of medication and psychotherapy. In fact, a 2003 report revealed that people on medication for bipolar disorder had better results if they also participated in family-focused therapy. The four different classes of drugs used in mood disorders are:

- Heterocyclic antidepressants (HCAs), like amitriptyline (Elavil)
- Selective serotonin reuptake inhibitors (**SSRI** inhibitors), like fluoxetine (Prozac), paroxetine (Paxil), and sertraline (Zoloft)
- Monoamine oxidase inhibitors (MAOI inhibitors), like phenelzine sulfate (Nardil) and tranylcypromine sulfate (Parnate)
- Mood stabilizers, like lithium carbonate (Eskalith) and valproate, often used in people with bipolar mood disorders.

A number of psychotherapy approaches are useful as well. Interpersonal psychotherapy helps the patient recognize the interaction between the mood disorder and interpersonal relationships. **Cognitive-behavioral therapy** explores how the patient’s view of the world may be affecting his or her mood and outlook.

When depression fails to respond to treatment or when there is a high risk of **suicide**, **electroconvulsive**

therapy (ECT) sometimes is used. ECT is believed to affect neurotransmitters like the medications do. Patients are anesthetized and given **muscle relaxants** to minimize discomfort. Then low-level electric current is passed through the brain to cause a brief convulsion. The most common side effect of ECT is mild, short-term memory loss.

Alternative treatment

There are many alternative therapies that may help in the treatment of mood disorders, including **acupuncture**, botanical medicine, homeopathy, **aromatherapy**, constitutional **hydrotherapy**, and light therapy. The therapy used is an individual choice. Short-term clinical studies have shown that the herb **St. John's wort** (*Hypericum perforatum*) can effectively treat some types of depression. Though it appears very safe, the herb may have some side effects and its long-term effectiveness has not been proven. It has not been tested in patients with bipolar disorder. Despite uncertainty concerning its effectiveness, a 2003 report said acceptance of the treatment continues to increase. A poll showed that about 41% of 15,000 science professionals in 62 countries said they would use St. John's wort for mild to moderate depression. Although St. John's wort appears to be a safe alternative to conventional antidepressants, care should be taken, as the herb can interfere with the actions of some pharmaceuticals. The usual dose is 300 mg three times daily. St. John's wort and **anti-depressant drugs** should not be taken simultaneously, so patients should tell their doctor if they are taking St. John's wort.

Prognosis

Most cases of mood disorders can be successfully managed if properly diagnosed and treated.

Prevention

People can take steps to improve mild depression and keep it from becoming worse. They can learn **stress** management (like relaxation training or breathing exercises), **exercise** regularly, and avoid drugs or alcohol.

Resources

PERIODICALS

- "Family-focused Therapy May Reduce Relapse Rate."
Health & Medicine Week September 29, 2003: 70.
- "St. John's Wort Healing Reputation Upheld."
Nutraceuticals International. September 2003.

KEY TERMS

Cognitive therapy—Psychotherapy technique designed to help people change their attitudes, perceptions, and patterns of thinking.

Electroconvulsive therapy (ECT)—Therapy for mood disorders that involves passing electrical current through the brain in order to create a brief convulsion.

Neurotransmitter—A chemical that aids or alters the transmission of impulses between the points that connect nerves.

Serotonin—A chemical messenger in the brain thought to play a role in mood regulation.

ORGANIZATIONS

American Psychiatric Association. 1400 K Street NW, Washington DC 20005. (888) 357-7924. <<http://www.psych.org>>.

National Depressive and Manic Depressive Association (NDMDA). 730 N. Franklin St., Ste. 501, Chicago, IL 60610. (800) 826-3632. <<http://www.ndmda.org>>.

National Institute of Mental Health. Mental Health Public Inquiries, 5600 Fishers Lane, Room 15C-05, Rockville, MD 20857. (888) 826-9438. <<http://www.nimh.nih.gov>>.

Robert Scott Dinsmoor
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Morning after pill see **Mifepristone**

Motion sickness

Definition

Motion sickness is the uncomfortable **dizziness**, **nausea**, and **vomiting** that people experience when their sense of balance and equilibrium is disturbed by constant motion. Riding in a car, aboard a ship or boat, or riding on a swing all cause stimulation of the vestibular system and visual stimulation that often leads to discomfort. While motion sickness can be bothersome, it is not a serious illness, and can be prevented.

PATRICIA SUZANNE COWINGS (1948–)

Patricia Suzanne Cowings was born on December 15, 1948, in New York City. She was one of four children born to Sadie and Albert Cowings, a grocery store owner. Cowings showed interest in science by the time she was eleven years old. She enrolled in the State University of New York at Stony Brook, earning her bachelor's degree with honors in 1970. She began her graduate work at the University of California at Davis where she was awarded both her master's and her doctoral degrees in 1973. Cowings also received an associateship from the National Research Council that same year, which allowed her to complete two years of research at NASA's Ames Research Center. She has held a position as a researcher with Ames since 1977.

Cowings's work at Ames' Psychophysiological Research Laboratory led to major breakthroughs for astronauts. Her pioneering experiments with bio-feedback as a method to control bodily functions has proven very effective for astronaut crews who experience "zero-gravity sickness syndrome." Her program was finally used during the 1992 *Endeavour* space flight. Presently, Cowings is researching exercises that will allow astronauts to maintain muscle strength while in zero gravity. She has published numerous papers with her colleague and husband, William B. Toscano. In addition, she has written articles including *The Relationship of Motion Sickness Susceptibility to Learned Autonomic Control for Symptom Suppression* (1982), *Autogenic-Feedback Training as a Preventive Method for Space Adaptation Syndrome* (1985), and *Autogenic-Feedback Training: A Preventive Method for Motion and Space Sickness* (1990).

Description

Motion sickness is a common problem, with nearly 80% of the general population suffering from it at one time in their lives. People with migraine headaches or Ménière's syndrome, however, are more likely than others to have recurrent episodes of motion sickness. Researchers at the Naval Medical Center in San Diego, California, reported in 2003 that 70% of research subjects with severe motion sickness had abnormalities of the vestibular system; these abnormalities are often found in patients diagnosed with migraines or **Ménière's disease**.

While motion sickness may occur at any age, it is more common in children over the age of two, with the majority outgrowing this susceptibility.

When looking at why motion sickness occurs, it is helpful to understand the role of the sensory organs. The sensory organs control a body's sense of balance by telling the brain what direction the body is pointing, the direction it is moving, and if it is standing still or turning. These messages are relayed by the inner ears (or labyrinth); the eyes; the skin pressure receptors, such as in those in the feet; the muscle and joint sensory receptors, which track what body parts are moving; and the central nervous system (the brain and spinal cord), which is responsible for processing all incoming sensory information.

Motion sickness and its symptoms surface when conflicting messages are sent to the central nervous system. An example of this is reading a book in the back seat of a moving car. The inner ears and skin receptors sense the motion, but the eyes register only the stationary pages of the book. This conflicting information may cause the usual motion sickness symptoms of dizziness, **nausea and vomiting**.

Causes and symptoms

While all five of the body's sensory organs contribute to motion sickness, excess stimulation to the vestibular system within the inner ear (the body's "balance center") has been shown to be one of the primary reasons for this condition. Balance problems, or vertigo, are caused by a conflict between what is seen and how the inner ear perceives it, leading to confusion in the brain. This confusion may result in higher heart rates, rapid breathing, nausea and sweating, along with dizziness and vomiting.

Pure optokinetic motion sickness is caused solely by visual stimuli, or what is seen. The optokinetic system is the reflex that allow the eyes to move when an object moves. Many people suffer when what they view is rotating or swaying, even if they are standing still.

Additional factors that may contribute to the occurrence of motion sickness include:

- Poor ventilation.
- Anxiety or fear. Both have been found to lower a person's threshold for experiencing motion sickness symptoms.
- Food. It is recommended that a heavy meal of spicy and greasy foods be avoided before and during a trip.
- Alcohol. A drink is often thought to help calm the nerves, but in this case it could upset the stomach further. A hangover for the next morning's trip may also lead to motion sickness.

- **Genetic factors.** Research suggests that some people inherit a predisposition to motion sickness. This predisposition is more marked in some ethnic groups than in others; one study published in 2002 found that persons of Chinese or Japanese ancestry are significantly more vulnerable to motion sickness than persons of British ancestry.
- **Pregnancy.** Susceptibility in women to vomiting during pregnancy appears to be related to motion sickness, although the precise connections are not well understood as of 2004.

Often viewed as a minor annoyance, some travelers are temporarily immobilized by motion sickness, and a few continue to feel its effects for hours and even days after a trip (the “mal d'embarquement” syndrome).

Diagnosis

Most cases of motion sickness are mild and self-treatable disorders. If symptoms such as dizziness become chronic, a doctor may be able to help alleviate the discomfort by looking further into a patient's general health. Questions regarding medications, head injuries, recent infections, and other questions about the ear and neurological system will be asked. An examination of the ears, nose, and throat, as well as tests of nerve and balance function, may also be completed.

Severe cases of motion sickness symptoms, and those that become progressively worse, may require additional, specific tests. Diagnosis in these situations deserves the attention and care of a doctor with specialized skills in diseases of the ear, nose, throat, equilibrium, and neurological system.

Treatment

There are a variety of medications to help ease the symptoms of motion sickness, and most of these are available without a prescription. Known as over-the-counter (OTC) medications, it is recommended that these be taken 30-60 minutes before traveling to prevent motion sickness symptoms, as well as during an extended trip.

Drugs

The following OTC drugs consist of ingredients that have been considered safe and effective for the treatment of motion sickness by the Food and Drug Administration:

- **Benadryl (and others).** Includes the active ingredient diphenhydramine and is not for use in children under age 6.
- **Dramamine (and others).** Includes the active ingredient dimenhydrinate and is not for use in children under age 2.
- **Bonine (and others).** Includes the active ingredient meclizine and is not for use in children under age 12.

Each of the active ingredients listed above are **antihistamines** whose main side effect is drowsiness. Caution should be used when driving a vehicle or operating machinery, and alcohol should be avoided when taking any drug for motion sickness. Large doses of OTC drugs for motion sickness may also cause **dry mouth** and occasional blurred vision.

The side effects of cinnarizine and the other antihistamine antiemetics indicate that they should not be used by members of flight crews responsible for the control of aircraft or for other tasks that require sustained attention and alertness.

The Food and Drug Administration (FDA) recommends that people with **emphysema**, chronic **bronchitis**, **glaucoma**, or difficulty urinating due to an **enlarged prostate** do not use OTC drugs for motion sickness unless directed by their doctor.

Longer trips may require a prescription medication called scopolamine (Transderm Scop). Formerly used in the transdermal skin patch (now discontinued), travelers must now ask their doctor to prescribe it in the form of a gel. In gel form, scopolamine is most effective when smeared on the arm or neck and covered with a bandage.

Another prescription drug that is sometimes given for motion sickness is ondansetron (Zofran), which was originally developed to treat nausea associated with **cancer chemotherapy**. Unlike cyclizine, ondansetron appears to be safe for use in children under the age of six.

Several newer antiemetic medications are under development as of early 2004. The most promising of these newer drugs is a class of compounds known as neurokinin-1 (substance P) antagonists. The neurokinins are being tested for the control of nausea following cancer chemotherapy as well as nausea related to motion sickness. In March 2003 the FDA approved the first of this new class of antiemetic drugs. Known as aprepitant, it is sold under the trade name Emend.

Alternative treatment

Alternative treatments for motion sickness have become widely accepted as a standard means of care. Ginger (*Zingiber officinale*) in its various forms is often used to calm the stomach, and it is now known that the oils it contains (gingerols and shogaols) appear to relax the intestinal tract in addition to mildly depressing the central nervous system. Some of the most effective forms of ginger include the powdered, encapsulated form; ginger tea prepared from sliced ginger root; or candied pieces. All forms of ginger should be taken on an empty stomach.

Placing manual pressure on the Neiguan or Pericardium-6 **acupuncture** point (located about three finger-widths above the wrist on the inner arm), either by acupuncture, **acupressure**, or a mild, electrical pulse, has shown to be effective against the symptoms of motion sickness. Elastic wristbands sold at most drugstores are also used as a source of relief due to the pressure it places in this area. Pressing the small intestine 17 (just below the earlobes in the indentations behind the jawbone) may also help in the functioning of the ear's balancing mechanism.

There are several homeopathic remedies that work specifically for motion sickness. They include *Cocculus*, *Petroleum*, and *Tabacum*.

Prognosis

While there is no cure for motion sickness, its symptoms can be controlled or even prevented. Most people respond successfully to the variety of treatments, or avoid the unpleasant symptoms through prevention methods.

Prevention

Because motion sickness is easier to prevent than treat once it has begun, the best treatment is prevention. The following steps may help deter the unpleasant symptoms of motion sickness before they occur:

- Avoid reading while traveling, and do not sit in a backward facing seat.
- Always ride where the eyes may see the same motion that the body and inner ears feel. Safe positions include the front seat of the car while looking at distant scenery; the deck of a ship where the horizon can be seen; and sitting by the window of an airplane. The least motion on an airplane is in a seat over the wings.
- Maintain a fairly straight-ahead view.

KEY TERMS

Acupressure—Often described as acupuncture without needles, acupressure is a traditional Chinese medical technique based on theory of *qi* (life energy) flowing in energy meridians or channels in the body. Applying pressure with the thumb and fingers to acupressure points can relieve specific conditions and promote overall balance and health.

Acupuncture—Based on the same traditional Chinese medical foundation as acupressure, acupuncture uses sterile needles inserted at specific points to treat certain conditions or relieve pain.

Neurological system—The tissue that initiates and transmits nerve impulses including the brain, spinal cord, and nerves.

Optokinetic—A reflex that causes a person's eyes to move when their field of vision moves.

Vertigo—The sensation of moving around in space, or objects moving around a person. It is a disturbance of equilibrium.

Vestibular system—The brain and parts of the inner ear that work together to detect movement and position.

- Eat a light meal before traveling, or if already nauseated, avoid food altogether.
- Avoid watching or talking to another traveler who is having motion sickness.
- Take motion sickness medicine at least 30-60 minutes before travel begins, or as recommended by a physician.
- Learn to live with the condition. Even those who frequently endure motion sickness can learn to travel by anticipating the conditions of their next trip. Research also suggests that increased exposure to the stimulation that causes motion sickness may help decrease its symptoms on future trips.

Resources

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PERIODICALS

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ORGANIZATIONS

- Civil Aerospace Medical Institute. P. O. Box 20582, Oklahoma City, OK 73125. (202) 366-4000. <<http://www.cami.jccbi.gov>>.
- National Aeronautics and Space Administration. Office of Biological and Physical Research. <<http://www.spaceresearch.nasa.gov>>.
- Vestibular Disorders Association. P.O. Box 4467, Portland, OR 97208-4467. (800) 837-8428. <<http://www.teleport.com/~veda>>.

Beth A. Kapes
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Mountain sickness see **Altitude sickness**

Mouth cancer see **Head and neck cancer**

Movement disorders

Definition

Movement disorders are a group of diseases and syndromes affecting the ability to produce and control movement.

Description

Though it seems simple and effortless, normal movement in fact requires an astonishingly complex system of control. Disruption of any portion of this system can cause a person to produce movements that are too weak, too forceful, too uncoordinated, or too poorly controlled for the task at hand. Unwanted movements may occur at rest. Intentional movement may become impossible. Such conditions are called movement disorders.

Abnormal movements themselves are symptoms of underlying disorders. In some cases, the abnormal movements are the only symptoms. Disorders causing abnormal movements include:

- Parkinson's disease
- Parkinsonism caused by drugs or poisons
- Parkinson-plus syndromes (**progressive supranuclear palsy**, multiple system atrophy, and cortical-basal ganglionic degeneration)
- Huntington's disease
- Wilson's disease
- Inherited ataxias (**Friedreich's ataxia**, Machado-Joseph disease, and spinocerebellar ataxias)
- **Tourette syndrome** and other tic disorders
- Essential tremor
- Restless leg syndrome
- Dystonia
- Stroke
- **Cerebral palsy**
- Encephalopathies
- Intoxication
- Poisoning by carbon monoxide, cyanide, methanol, or manganese.

Causes and symptoms

Causes

Movement is produced and coordinated by several interacting brain centers, including the motor

cortex, the cerebellum, and a group of structures in the inner portions of the brain called the basal ganglia. Sensory information provides critical input on the current position and velocity of body parts, and spinal nerve cells (neurons) help prevent opposing muscle groups from contracting at the same time.

To understand how movement disorders occur, it is helpful to consider a normal voluntary movement, such as reaching to touch a nearby object with the right index finger. To accomplish the desired movement, the arm must be lifted and extended. The hand must be held out to align with the forearm, and the forefinger must be extended while the other fingers remain flexed.

THE MOTOR CORTEX. Voluntary motor commands begin in the motor cortex located on the outer, wrinkled surface of the brain. Movement of the right arm is begun by the left motor cortex, which generates a large volley of signals to the involved muscles. These electrical signals pass along upper motor neurons through the midbrain to the spinal cord. Within the spinal cord, they connect to lower motor neurons, which convey the signals out of the spinal cord to the surface of the muscles involved. Electrical stimulation of the muscles causes contraction, and the force of contraction pulling on the skeleton causes movement of the arm, hand, and fingers.

Damage to or death of any of the neurons along this path causes weakness or **paralysis** of the affected muscles.

ANTAGONISTIC MUSCLE PAIRS. This picture of movement is too simple, however. One important refinement to it comes from considering the role of opposing, or antagonistic, muscle pairs. Contraction of the biceps muscle, located on the top of the upper arm, pulls on the forearm to flex the elbow and bend the arm. Contraction of the triceps, located on the opposite side, extends the elbow and straightens the arm. Within the spine, these muscles are normally wired so that willed (voluntary) contraction of one is automatically accompanied by blocking of the other. In other words, the command to contract the biceps provokes another command within the spine to prevent contraction of the triceps. In this way, these antagonist muscles are kept from resisting one another. Spinal cord or brain injury can damage this control system and cause involuntary simultaneous contraction and spasticity, an increase in resistance to movement during motion.

THE CEREBELLUM. Once the movement of the arm is initiated, sensory information is needed to guide the finger to its precise destination. In addition to sight,

the most important source of information comes from the “position sense” provided by the many sensory neurons located within the limbs (proprioception). Proprioception is what allows you to touch your nose with your finger even with your eyes closed. The balance organs in the ears provide important information about posture. Both postural and proprioceptive information are processed by a structure at the rear of the brain called the cerebellum. The cerebellum sends out electrical signals to modify movements as they progress, “sculpting” the barrage of voluntary commands into a tightly controlled, constantly evolving pattern. Cerebellar disorders cause inability to control the force, fine positioning, and speed of movements (ataxia). Disorders of the cerebellum may also impair the ability to judge distance so that a person under- or over-reaches the target (dysmetria). Tremor during voluntary movements can also result from cerebellar damage.

THE BASAL GANGLIA. Both the cerebellum and the motor cortex send information to a set of structures deep within the brain that help control involuntary components of movement (basal ganglia). The basal ganglia send output messages to the motor cortex, helping to initiate movements, regulate repetitive or patterned movements, and control muscle tone.

Circuits within the basal ganglia are complex. Within this structure, some groups of cells begin the action of other basal ganglia components and some groups of cells block the action. These complicated feedback circuits are not entirely understood. Disruptions of these circuits are known to cause several distinct movement disorders. A portion of the basal ganglia called the substantia nigra sends electrical signals that block output from another structure called the subthalamic nucleus. The subthalamic nucleus sends signals to the globus pallidus, which in turn blocks the thalamic nuclei. Finally, the thalamic nuclei send signals to the motor cortex. The substantia nigra, then, begins movement and the globus pallidus blocks it.

This complicated circuit can be disrupted at several points. For instance, loss of substantia nigra cells, as in Parkinson’s disease, increases blocking of the thalamic nuclei, preventing them from sending signals to the motor cortex. The result is a loss of movement (motor activity), a characteristic of Parkinson’s.

In contrast, cell loss in early Huntington’s disease decreases blocking of signals from the thalamic nuclei, causing more cortex stimulation and stronger but uncontrolled movements.

Disruptions in other portions of the basal ganglia are thought to cause tics, **tremors**, dystonia, and a

variety of other movement disorders, although the exact mechanisms are not well understood.

Some movement disorders, including Huntington's disease and inherited ataxias, are caused by inherited genetic defects. Some disease that cause sustained muscle contraction limited to a particular muscle group (focal dystonia) are inherited, but others are caused by trauma. The cause of most cases of Parkinson's disease is unknown, although genes have been found for some familial forms.

Symptoms

Abnormal movements are broadly classified as either hyperkinetic—too much movement—and hypokinetic—too little movement. Hyperkinetic movements include:

- **Dystonia.** Sustained muscle contractions, often causing twisting or repetitive movements and abnormal postures. Dystonia may be limited to one area (focal) or may affect the whole body (general). Focal dystonias may affect the neck (cervical dystonia or **torticollis**), the face (one-sided or hemifacial spasm, contraction of the eyelid or blepharospasm, contraction of the mouth and jaw or oromandibular dystonia, simultaneous spasm of the chin and eyelid or Meige syndrome), the vocal cords (laryngeal dystonia), or the arms and legs (writer's cramp, occupational cramps). Dystonia may be painful as well as incapacitating.
- **Tremor.** Uncontrollable (involuntary) shaking of a body part. Tremor may occur only when muscles are relaxed or it may occur only during an action or holding an active posture.
- **Tics.** Involuntary, rapid, nonrhythmic movement or sound. Tics can be controlled briefly.
- **Myoclonus.** A sudden, shock-like muscle contraction. Myoclonic jerks may occur singly or repetitively. Unlike tics, myoclonus cannot be controlled even briefly.
- **Chorea.** Rapid, nonrhythmic, usually jerky movements, most often in the arms and legs.
- **Ballism.** Like chorea, but the movements are much larger, more explosive and involve more of the arm or leg. This condition, also called ballismus, can occur on both sides of the body or on one side only (hemiballismus).
- **Akathisia.** Restlessness and a desire to move to relieve uncomfortable sensations. Sensations may include a feeling of crawling, **itching**, stretching, or creeping, usually in the legs.
- **Athetosis.** Slow, writhing, continuous, uncontrollable movement of the arms and legs.

Hypokinetic movements include:

- **Bradykinesia.** Slowness of movement.
- **Freezing.** Inability to begin a movement or involuntary stopping of a movement before it is completed.
- **Rigidity.** An increase in muscle tension when an arm or leg is moved by an outside force.
- **Postural instability.** Loss of ability to maintain upright posture caused by slow or absent righting reflexes.

Diagnosis

Diagnosis of movement disorders requires a careful medical history and a thorough physical and neurological examination. Brain imaging studies are usually performed. Imaging techniques include computed tomography scan (CT scan), **positron emission tomography (PET)**, or **magnetic resonance imaging (MRI)** scans. Routine blood and urine analyses are performed. A lumbar puncture (spinal tap) may be necessary. Video recording of the abnormal movement is often used to analyze movement patterns and to track progress of the disorder and its treatment. **Genetic testing** is available for some forms of movement disorders.

Treatment

Treatment of a movement disorder begins with determining its cause. Physical and occupational therapy may help make up for lost control and strength. Drug therapy can help compensate for some imbalances of the basal ganglionic circuit. For instance, levodopa (L-dopa) or related compounds can substitute for lost dopamine-producing cells in Parkinson's disease. Conversely, blocking normal dopamine action is a possible treatment in some hyperkinetic disorders, including tics. Oral medications can also help reduce overall muscle tone. Local injections of botulinum toxin can selectively weaken overactive muscles in dystonia and spasticity. Destruction of peripheral nerves through injection of phenol can reduce spasticity. All of these treatments may have some side effects.

Surgical destruction or inactivation of basal ganglionic circuits has proven effective for Parkinson's disease and is being tested for other movement disorders. Transplantation of fetal cells into the basal ganglia has produced mixed results in Parkinson's disease.

KEY TERMS

Botulinum toxin—Any of a group of potent bacterial toxins or poisons produced by different strains of the bacterium *Clostridium botulinum*. The toxins cause muscle paralysis, and thus force the relaxation of a muscle in spasm.

Cerebral palsy—A movement disorder caused by a permanent brain defect or injury present at birth or shortly after. It is frequently associated with premature birth. Cerebral palsy is not progressive.

Computed tomography (CT)—An imaging technique in which cross-sectional x rays of the body are compiled to create a three-dimensional image of the body's internal structures.

Encephalopathy—An abnormality in the structure or function of tissues of the brain.

Essential tremor—An uncontrollable (involuntary) shaking of the hands, head, and face. Also called familial tremor because it is sometimes inherited, it can begin in the teens or in middle age. The exact cause is not known.

Fetal tissue transplantation—A method of treating Parkinson's and other neurological diseases by grafting brain cells from human fetuses onto the basal ganglia. Human adults cannot grow new brain cells but developing fetuses can. Grafting fetal tissue stimulates the growth of new brain cells in affected adult brains.

Hereditary ataxia—One of a group of hereditary degenerative diseases of the spinal cord or cerebellum. These diseases cause tremor, spasm, and wasting of muscle.

Huntington's disease—A rare hereditary condition that causes progressive chorea (jerky muscle movements) and mental deterioration that ends in dementia. Huntington's symptoms usually appear in patients in their 40s. There is no effective treatment.

Levodopa (L-dopa)—A substance used in the treatment of Parkinson's disease. Levodopa can cross the blood-brain barrier that protects the brain. Once in the brain, it is converted to dopamine and thus can replace the dopamine lost in Parkinson's disease.

Magnetic resonance imaging (MRI)—An imaging technique that uses a large circular magnet and radio waves to generate signals from atoms in the body. These signals are used to construct images of internal structures.

Parkinson's disease—A slowly progressive disease that destroys nerve cells in the basal ganglia and thus causes loss of dopamine, a chemical that aids in transmission of nerve signals (neurotransmitter). Parkinson's is characterized by shaking in resting muscles, a stooping posture, slurred speech, muscular stiffness, and weakness.

Positron emission tomography (PET)—A diagnostic technique in which computer-assisted x rays are used to track a radioactive substance inside a patient's body. PET can be used to study the biochemical activity of the brain.

Progressive supranuclear palsy—A rare disease that gradually destroys nerve cells in the parts of the brain that control eye movements, breathing, and muscle coordination. The loss of nerve cells causes palsy, or paralysis, that slowly gets worse as the disease progresses. The palsy affects ability to move the eyes, relax the muscles, and control balance.

Restless legs syndrome—A condition that causes an annoying feeling of tiredness, uneasiness, and itching deep within the muscle of the leg. It is accompanied by twitching and sometimes pain. The only relief is in walking or moving the legs.

Tourette syndrome—An abnormal condition that causes uncontrollable facial grimaces and tics and arm and shoulder movements. Tourette syndrome is perhaps best known for uncontrollable vocal tics that include grunts, shouts, and use of obscene language (coprolalia).

Wilson's disease—An inborn defect of copper metabolism in which free copper may be deposited in a variety of areas of the body. Deposits in the brain can cause tremor and other symptoms of Parkinson's disease.

Alternative treatment

There are several alternative therapies that can be useful when treating movement disorders. The progress made will depend on the individual and

his/her condition. Among the therapies that may be helpful are **acupuncture**, homeopathy, touch therapies, postural alignment therapies, and **biofeedback**.

Prognosis

The prognosis for a patient with a movement disorder depends on the specific disorder.

Prevention

Prevention depends on the specific disorder.

Resources

ORGANIZATIONS

Worldwide Education and Awareness for Movement Disorders. One Gustave L. Levy Place, Box 1052, New York, NY 10029. (800) 437-6683. <<http://www.wemove.org>>.

Richard Robinson

Movement therapy

Definition

Movement therapy refers to a broad range of Eastern and Western movement approaches used to promote physical, mental, emotional, and spiritual well-being.

Purpose

The physical benefits of movement therapy include greater ease and range of movement, increased balance, strength and flexibility, improved muscle tone and coordination, joint resiliency, cardiovascular conditioning, enhanced athletic performance, stimulation of circulation, prevention of injuries, greater longevity, **pain** relief, and relief of rheumatic, neurological, spinal, **stress**, and respiratory disorders. Movement therapy can also be used as a **meditation** practice to quiet the mind, foster self-knowledge, and increase awareness. In addition, movement therapy is beneficial in alleviating emotional distress that is expressed through the body. These conditions include eating disorders, excessive clinging, and **anxiety** attacks. Since movements are related to thoughts and feelings, movement therapy can also bring about changes in attitude and emotions. People report an increase in self-esteem and self-image. Communication skills can be enhanced and tolerance of others increased. The physical openness facilitated by movement therapy leads to greater emotional openness and creativity.

Description

Origins

Movement is fundamental to human life. In fact movement is life. Contemporary physics tells us that the universe and everything in it is in constant motion. We can move our body and at the most basic level our body is movement. According to the somatic educator Thomas Hanna, "The living body is a moving body—indeed, it is a constantly moving body." The poet and philosopher Alan Watts eloquently states a similar view, "A living body is not a fixed thing but a flowing event, like a flame or a whirlpool." Centuries earlier, the great Western philosopher Socrates understood what modern physics has proven, "The universe is motion and nothing else."

Since the beginning of time, indigenous societies around the world have used movement and dance for individual and community healing. Movement and song were used for personal healing, to create community, to ensure successful crops, and to promote fertility. Movement is still an essential part of many healing traditions and practices throughout the world.

Western movement therapies generally developed out of the realm of dance. Many of these movement approaches were created by former dancers or choreographers who were searching for a way to prevent injury, attempting to recover from an injury, or who were curious about the effects of new ways of moving. Some movement therapies arose out of the fields of physical therapy, psychology, and bodywork. Other movement therapies were developed as way to treat an incurable disease or condition.

Eastern movement therapies, such as **yoga**, **qigong**, and t'ai chi began as a spiritual or self-defense practices and evolved into healing therapies. In China, for example, Taoist monks learned to use specific breathing and movement patterns in order to promote mental clarity, physical strength, and support their practice of meditation. These practices, later known as qigong and t'ai chi eventually became recognized as ways to increase health and prolong life.

There are countless approaches to movement therapy. Some approaches emphasize awareness and attention to inner sensations. Other approaches use movement as a form of psychotherapy, expressing and working through deep emotional issues. Some approaches emphasize alignment with gravity and specific movement sequences, while other approaches encourage spontaneous movement. Some approaches are primarily concerned with increasing the ease and efficiency of bodily movement. Other approaches

address the reality of the body “as movement” instead of the body as only something that runs or walks through space.

The term movement therapy is often associated with dance therapy. Some dance therapists work privately with people who are interested in personal growth. Others work in mental health settings with autistic, brain injured and learning disabled children, the elderly, and disabled adults.

Laban movement analysis (LMA), formerly known as Effort-Shape is a comprehensive system for discriminating, describing, analyzing, and categorizing movements. LMA can be applied to dance, athletic coaching, fitness, acting, psychotherapy, and a variety of other professions. Certified movement analysts can “observe recurring patterns, note movement preferences, assess physical blocks and dysfunctional movement patterns, and suggest new movement patterns.” As a student of Rudolf Laban, Irmgard Bartenieff developed his form of movement analysis into a system of body training or reeducation called Bartenieff fundamentals (BF). The basic premise of this work is that once the student experiences a physical foundation, emotional, and intellectual expression become richer. BF uses specific exercises that are practiced on the floor, sitting, or standing to engage the deeper muscles of the body and enable a greater range of movement.

Authentic movement (AM) is based upon Mary Starks Whitehouse’s understanding of dance, movement, and depth psychology. There is no movement instruction in AM, simply a mover and a witness. The mover waits and listens for an impulse to move and then follows or “moves with” the spontaneous movements that arise. These movements may or may not be visible to the witness. The movements may be in response to an emotion, a dream, a thought, pain, joy, or whatever is being experienced in the moment. The witness serves as a compassionate, non judgmental mirror and brings a “special quality of attention or presence.” At the end of the session the mover and witness speak about their experiences together. AM is a powerful approach for self development and awareness and provides access to preverbal memories, creative ideas, and unconscious movement patterns that limit growth.

Gabrielle Roth (5 Rhythms movement) and Anna Halprin have both developed dynamic movement practices that emphasize personal growth, awareness, expression, and community. Although fundamentally different forms, each of these movement/dance approaches recognize and encourage our inherent desire for movement.

Several forms of movement therapy grew out of specific bodywork modalities. **Rolfing** movement integration (RMI) and Rolfing rhythms are movement forms which reinforce and help to integrate the structural body changes brought about by the hands-on work of Rolfing (structural integration). RMI uses a combination of touch and verbal directions to help develop greater awareness of one’s vertical alignment and habitual movement patterns. RMI teacher Mary Bond says, “The premise of Rolfing Movement Integration... is that you can restore your structure to balance by changing the movement habits that perpetuate imbalance.” Rolfing rhythms is a series of lively exercises designed to encourage awareness of the Rolfing principles of ease, length, balance, and harmony with gravity.

The movement education component of **Aston-Patterning** bodywork is called neurokinetics. This movement therapy teaches ways of moving with greater ease throughout every day activities. These movement patterns can also be used to release tension in the body. Aston fitness is an **exercise** program which includes warm-up techniques, exercises to increase muscle tone and stability, stretching, and cardiovascular fitness.

Rosen method movement (an adjunct to Rosen method bodywork) consists of simple fun movement exercises done to music in a group setting. Through gentle swinging, bouncing, and stretching every joint in the body experiences a full range of movement. The movements help to increase balance and rhythm and create more space for effortless breathing.

The movement form of **Trager psychophysical integration** bodywork, Mentastics, consists of fun, easy swinging, shaking, and stretching movements. These movements, developed by Dr. **Milton Trager**, create an experience of lightness and freedom in the body, allowing for greater ease in movement. Trager also worked successfully with **polio** patients.

Awareness through movement, the movement therapy form of the **Feldenkrais method**, consists of specific structured movement experiences taught as a group lesson. These lessons reeducate the brain without tiring the muscles. Most lessons are done lying down on the floor or sitting. **Moshe Feldenkrais** designed the lessons to “improve ability ... turn the impossible into the possible, the difficult into the easy, and the easy into the pleasant.”

Ideokinesis is another movement approach emphasizing neuromuscular reeducation. Lulu Sweigart based her work on the pioneering approach of her teacher Mabel Elsworth Todd. Ideokinesis uses

imagery to train the nervous system to stimulate the right muscles for the intended movement. If one continues to give the nervous system a clear mental picture of the movement intended, it will automatically select the best way to perform the movement. For example, to enhance balance in standing, Sweigart taught people to visualize “lines of movement” traveling through their bodies. Sweigart did not train teachers in ideokinesis but some individuals use ideokinetic imagery in the process of teaching movement.

The Mensendieck system of functional movement techniques is both corrective and preventative. Bess Mensendieck, a medical doctor, developed a series of exercises to reshape, rebuild and revitalize the body. A student of this approach learns to use the conscious will to relax muscles and releases tension. There are more than 200 exercises that emphasize correct and graceful body movement through everyday activities. Unlike other movement therapy approaches this work is done undressed or in a bikini bottom, in front of mirrors. This allows the student to observe and feel where a movement originates. Success has been reported with many conditions including Parkinson’s disease, muscle and joint injuries, and repetitive strain injuries.

The **Alexander technique** is another functional approach to movement therapy. In this approach a teacher gently uses hands and verbal directions to subtly guide the student through movements such as sitting, standing up, bending and walking. The Alexander technique emphasizes balance in the neck-head relationship. A teacher lightly steers the students head into the proper balance on the tip of the spine while the student is moving in ordinary ways. The student learns to respond to movement demands with the whole body, in a light integrated way. This approach to movement is particularly popular with actors and other performers.

Pilates or physical mind method is also popular with actors, dancers, athletes, and a broad range of other people. Pilates consists of over 500 exercises done on the floor or primarily with customized exercise equipment. The exercises combine sensory awareness and physical training. Students learn to move from a stable, central core. The exercises promote strength, flexibility, and balance. Pilates training is increasingly available in sports medicine clinics, fitness centers, dance schools, spas, and physical therapy offices.

Many approaches to movement therapy emphasize awareness of internal sensations. Charlotte Selver, a student of somatic pioneer Elsa Gindler, calls her style of teaching sensory awareness (SA). This

approach has influenced the thinking of many innovators, including Fritz Perls, who developed **gestalt therapy**. Rather than suggesting a series of structured movements, visualizations, or body positions, in SA the teacher outlines experiments in which one can become aware of the sensations involved in any movement. A teacher might ask the student to feel the movement of her breathing while running, sitting, picking up a book, etc. This close attunement to inner sensory experience encourages an experience of body-mind unity in which breathing becomes less restricted and posture, coordination, flexibility, and balance are improved. There may also be the experience of increased energy and aliveness.

Gerda Alexander Eutony (GAE) is another movement therapy approach that is based upon internal awareness. Through GAE one becomes a master of self-sensing and knowing which includes becoming sensitive to the external environment, as well. For example, while lying on the floor sensing the breath, skin or form of the body, one also senses the connection with the ground. GAE is taught in group classes or private lessons which also include hands-on therapy. In 1987, after two years of observation in clinics throughout the world, GAE became the first mind-body discipline accepted by the World Health Organization (WHO) as an alternative health-care technique.

Kinetic awareness developed by dancer-choreographer Elaine Summers, emphasizes emotional and physical inquiry. Privately or in a group, a teacher sets up situations for the student to explore the possible causes of pain and movement restrictions within the body. Rubber balls of various sizes are used as props to focus attention inward, support the body in a stretched position and massage a specific area of the body. The work helps one to deal with chronic pain, move easily again after injuries and increase energy, flexibility, coordination, and comfort.

Body-mind centering (BMC) was developed by Bonnie Bainbridge Cohen and is a comprehensive educational and therapeutic approach to movement. BMC practitioners use movement, touch, **guided imagery**, developmental repatterning, dialogue, music, large balls, and other props in an individual session to meet the needs of each person. BMC encourages people to develop a sensate awareness and experience of the ligaments, nerves, muscles, skin, fluids, organs, glands, fat, and fascia that make up one’s body. It has been effective in preventing and rehabilitating from chronic injuries and in improving neuromuscular response in children with **cerebral palsy** and other neurological disorders.

Continuum movement has also been shown to be effective in treating neurological disorders including spinal cord injury. Developed by Emilie Conrad and Susan Harper, continuum movement is an inquiry into the creative flux of our body and all of life. Sound, breath, subtle and dynamic movements are explored that stimulate the brain and increase resonance with the fluid world of movement. The emphasis is upon unpredictable, spontaneous or spiral movements rather than a linear movement pattern. According to Conrad, "Awareness changes how we physically move. As we become more fluid and resilient so do the mental, emotional, and spiritual movements of our lives."

Eastern movement therapies such as yoga, t'ai chi, and qigong are also effective in healing and preventing a wide range of physical disorders, encouraging emotional stability, and enhancing spiritual awareness. There are a number of different approaches to yoga. Some emphasize the development of physical strength, flexibility, and alignment. Other forms of yoga emphasize inner awareness, opening, and meditation.

Precautions

People with acute injuries and chronic physical and mental conditions need to be careful when choosing a form of movement therapy. It is best to consult with a knowledgeable physician, physical therapist, or mental health therapist.

A special form of movement therapy known as constraint-induced movement therapy, or CIMT, is being used as of the early 2000s to rehabilitate the upper limbs of patients who have suffered a **stroke**, traumatic brain injury, or damage to the spinal cord. In CIMT, the arm that has been less affected by the injury is constrained by a sling for 90% of the patient's waking hours for a period of two weeks. The sling forces the patient to use the weaker arm more often; in addition, a physical therapist works with the patient to practice repetitive motions with the weaker arm. CIMT also appears to be useful in treating children with muscular weakness on one side of the body caused by cerebral palsy.

Research and general acceptance

Although research has documented the beneficial effects of dance therapy, qigong, t'ai chi, yoga, Alexander technique, awareness through movement (Feldenkrais), and Rolfing, other forms of movement therapy have not been as thoroughly researched.

CIMT has become widely accepted in **rehabilitation** medicine since its introduction in the mid-1990s, although some doctors still consider it experimental. Further research in CIMT is being carried out by the National Institute of Neurological Disorders and Stroke (NINDS), one of the National Institutes of Health.

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ORGANIZATIONS

National Institute of Neurological Disorders and Stroke (NINDS). NIH Neurological Institute, P. O. Box 5801, Bethesda, MD 20824. (800) 352-9424 or (301) 496-5751. < <http://www.ninds.nih.gov> >.

Linda Chrisman
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Mpell disease see **Ankylosing spondylitis**

MR see **Magnetic resonance imaging**

MRI see **Magnetic resonance imaging**

MS see **Multiple sclerosis**

M's disease see **Waldenström's macroglobulinemia**

Mucopolysaccharidoses

Definition

Mucopolysaccharidosis (MPS) is a general term for a number of inherited diseases that are caused by the accumulation of mucopolysaccharides, resulting in problems with an individual's development. With each condition, mucopolysaccharides accumulate in the cells and tissues of the body because of a deficiency of a specific enzyme. The specific enzyme that is deficient or absent is what distinguishes one type of MPS from another. However, before these enzymes were identified, the MPS disorders were diagnosed by the signs and symptoms that an individual expressed. The discovery of these enzymes resulted in a reclassification of some of the MPS disorders. These conditions are often referred to as MPS I, MPS II, MPS III, MPS IV, MPS VI, MPS VII, and MPS IX. However, these conditions are also referred to by their original names, which are Hurler, Hurler-Scheie, Scheie (all MPS I), Hunter (MPS II), Sanfilippo (MPS III), Morquio (MPS IV), Maroteaux-Lamy (MPS VI), Sly (MPS VII), and Hyaluronidase deficiency (MPS IX).

Description

Mucopolysaccharides are long chains of sugar molecules that are essential for building the bones, cartilage, skin, tendons, and other tissues in the body. Normally, the human body continuously breaks down and builds mucopolysaccharides. Another name for mucopolysaccharides is glycosaminoglycans (GAGs). There are many different types of GAGs and specific GAGs are unable to be broken down in each of the MPS conditions. There are several enzymes involved in breaking down each GAG and a deficiency or absence of any of the essential enzymes can cause the GAG to not be broken down completely and result in its accumulation in the tissues and organs in the body. In some MPS conditions, in addition to the GAG being stored in the body, some of the incompletely broken down GAGs can leave the body via the urine. When too much GAG is stored, organs and tissues can be damaged or not function properly.

Genetic profile

Except for MPS II, the MPS conditions are inherited in an autosomal recessive manner. MPS conditions occur when both of an individual's genes that produce the specific enzyme contain a mutation, causing them to not work properly. When both genes do not work properly, either none or a reduced amount of

the enzyme is produced. An individual with an autosomal recessive condition inherits one of those non-working genes from each parent. These parents are called "carriers" of the condition. When two people are known carriers for an autosomal recessive condition, they have a 25% chance with each **pregnancy** to have a child affected with the disease. Some individuals with MPS do have children of their own. Children of parents who have an autosomal recessive condition are all carriers of that condition. These children are not at risk to develop the condition unless the other parent is a carrier or affected with the same autosomal recessive condition.

Unlike the other MPS conditions, MPS II is inherited in an X-linked recessive manner. This means that the gene causing the condition is located on the X chromosome, one of the two sex chromosomes. Since a male has only one X chromosome, he will have the disease if the X chromosome inherited from his mother carries the defective gene. Females, because they have two X chromosomes, are called "carriers" of the condition if only one of their X chromosomes has the gene that causes the condition, while the other X chromosome does not.

Causes and symptoms

Each type of MPS is caused by a deficiency of one of the enzymes involved in breaking down GAGs. It is the accumulation of the GAGs in the tissues and organs in the body that cause the wide array of symptoms characteristic of the MPS conditions. The accumulating material is stored in cellular structures called lysosomes, and these disorders are also known as lysosomal storage diseases.

MPS I

MPS I is caused by a deficiency of the enzyme alpha-L-iduronidase. Three conditions, Hurler, Hurler-Scheie, and Scheie syndromes, all are caused by a deficiency of this enzyme. Initially, these three conditions were felt to be separate because each were associated with different physical symptoms and prognoses. However, once the underlying cause of these conditions was identified, it was realized that these three conditions were all variants of the same disorder. The gene involved with MPS I is located on chromosome 4p16.3.

MPS I H (HURLER SYNDROME). It has been estimated that approximately one baby in 100,000 will be born with Hurler syndrome. Individuals with Hurler syndrome tend to have the most severe form of MPS I. Symptoms of Hurler syndrome are often

evident within the first year or two after birth. Often these infants begin to develop as expected, but then reach a point where they begin to lose the skills that they have learned. Many of these infants may initially grow faster than expected, but their growth slows and typically stops by age three. Facial features also begin to appear “coarse.” They develop a short nose, flatter face, thicker skin, and a protruding tongue. Additionally, their heads become larger and they develop more hair on their bodies with the hair becoming coarser. Their bones are also affected, with these children usually developing joint **contractures** (stiff joints), **kyphosis** (a specific type of curve to the spine), and broad hands with short fingers. Many of these children experience breathing difficulties, and respiratory infections are common. Other common problems include heart valve dysfunction, thickening of the heart muscle (**cardiomyopathy**), enlarged spleen and liver, clouding of the cornea, **hearing loss**, and **carpal tunnel syndrome**. These children typically do not live past age 12.

MPS I H/S (HURLER-SCHIE SYNDROME). Hurler-Scheie syndrome is felt to be the intermediate form of MPS I, meaning that the symptoms are not as severe as those in individuals who have MPS I H but not as mild as those in MPS I S. Approximately one baby in 115,000 will be born with Hurler-Scheie syndrome. These individuals tend to be shorter than expected, and they can have normal intelligence, however, some individuals with MPS I H/S will experience learning difficulties. These individuals may develop some of the same physical features as those with Hurler syndrome, but usually they are not as severe. The prognosis for children with MPS I H/S is variable with some individuals dying during childhood, while others living to adulthood.

MPS I S (SCHEIE SYNDROME). Scheie syndrome is considered the mild form of MPS I. It is estimated that approximately one baby in 500,000 will be born with Scheie syndrome. Individuals with MPS I S usually have normal intelligence, but there have been some reports of individuals with MPS I S developing psychiatric problems. Common physical problems include corneal clouding, heart abnormalities, and orthopedic difficulties involving their hands and back. Individuals with MPS I S do not develop the facial features seen with MPS I H and usually these individuals have a normal life span.

MPS II (Hunter syndrome)

Hunter syndrome is caused by a deficiency of the enzyme iduronate-2-sulphatase. All individuals with Hunter syndrome are male, because the gene that

causes the condition is located on the X chromosome, specifically Xq28. Like many MPS conditions, Hunter syndrome is divided into two groups, mild and severe. It has been estimated that approximately 1 in 110,000 males are born with Hunter syndrome, with the severe form being three times more common than the mild form. The severe form is felt to be associated with progressive **mental retardation** and physical disability, with most individuals dying before age 15. In the milder form, most of these individuals live to adulthood and have normal intelligence or only mild mental impairments. Males with the mild form of Hunter syndrome develop physical differences similar to the males with the severe form, but not as quickly. Men with mild Hunter syndrome can have a normal life span and some have had children. Most males with Hunter syndrome develop joint stiffness, chronic **diarrhea**, enlarged liver and spleen, heart valve problems, hearing loss, kyphosis, and tend to be shorter than expected. These symptoms tend to progress at a different rate depending on if an individual has the mild or severe form of MPS II.

MPS III (Sanfilippo syndrome)

MPS III, like the other MPS conditions, was initially diagnosed by the individual having certain physical characteristics. It was later discovered that the physical symptoms associated with Sanfilippo syndrome could be caused by a deficiency in one of four enzymes. Each type of MPS III is now subdivided into four groups, labeled A-D, based on the specific enzyme that is deficient. All four of these enzymes are involved in breaking down the same GAG, heparan sulfate. Heparan sulfate is mainly found in the central nervous system and accumulates in the brain when it cannot be broken down because one of those four enzymes are deficient or missing.

MPS III is a variable condition with symptoms beginning to appear between ages two and six years of age. Because of the accumulation of heparan sulfate in the central nervous system, the central nervous system is severely affected. In MPS III, signs that the central nervous system is degenerating usually are evident in most individuals between ages six and 10. Many children with MPS III will develop seizures, sleeplessness, thicker skin, joint contractures, enlarged tongues, cardiomyopathy, behavior problems, and mental retardation. The life expectancy in MPS III is also variable. On average, individuals with MPS III live until they are teenagers, with some living longer and others not that long.

MPS IIIA (SANFILIPPO SYNDROME TYPE A). MPS IIIA is caused by a deficiency of the enzyme heparan

N-sulfatase. Type IIIA is felt to be the most severe of the four types, in which symptoms appear and **death** occurs at an earlier age. A study in British Columbia estimated that one in 324,617 live births are born with MPS IIIA. MPS IIIA is the most common of the four types in Northwestern Europe. The gene that causes MPS IIIA is located on the long arm of chromosome 17 (location 17q25).

MPS IIIB (SANFILIPPO SYNDROME TYPE B). MPS IIIB is due to a deficiency in N-acetyl-alpha-D-glucosaminidase (NAG). This type of MPS III is not felt to be as severe as Type IIIA and the characteristics vary. Type IIIB is the most common of the four in southeastern Europe. The gene associated with MPS IIIB is also located on the long arm of chromosome 17 (location 17q21).

MPS IIIC (SANFILIPPO SYNDROME TYPE C). A deficiency in the enzyme acetyl-CoA-alpha-glucosaminide acetyltransferase causes MPS IIIC. This is considered a rare form of MPS III. The gene involved in MPS IIIC is believed to be located on chromosome 14.

MPS IIID (SANFILIPPO SYNDROME TYPE D). MPS IIID is caused by a deficiency in the enzyme N-acetylglucosamine-6-sulfatase. This form of MPS III is also rare. The gene involved in MPS IIID is located on the long arm of chromosome 12 (location 12q14).

MPS IV (Morquio syndrome)

As with several of the MPS disorders, Morquio syndrome was diagnosed by the presence of particular signs and symptoms. However, it is now known that the deficiency of two different enzymes can cause the characteristics of MPS IV. These two types of MPS IV are called MPS IV A and MPS IV B. MPS IV is also variable in its severity. The intelligence of individuals with MPS IV is often completely normal. In individuals with a severe form, skeletal abnormalities can be extreme and include dwarfism, kyphosis (backward-curved spine), prominent breastbone, flat feet, and knock-knees. One of the earliest symptoms seen in this condition usually is a difference in the way the child walks. In individuals with a mild form of MPS IV, limb stiffness, and joint **pain** are the primary symptoms. MPS IV is one of the rarest MPS disorders, with approximately one baby in 300,000 born with this condition.

MPS IV A (MORQUIO SYNDROME TYPE A). MPS IV A is the “classic” or the severe form of the condition and is caused by a deficiency in the enzyme galactosamine-6-sulphatase. The gene involved with MPS IV A is located on the long arm of chromosome 16 (location 16q24.3).

MPS IV B (MORQUIO SYNDROME TYPE B). MPS IV B is considered the milder form of the condition. The enzyme, beta-galactosidase, is deficient in MPS IV B. The location of the gene that produces beta-galactosidase is located on the short arm of chromosome 3 (location 3p21).

MPS VI (Maroteaux-Lamy syndrome)

MPS VI, which is another rare form of MPS, is caused by a deficiency of the enzyme N-acetylglucosamine-4-sulphatase. This condition is also variable; individuals may have a mild or severe form of the condition. Typically, the nervous system or intelligence of an individual with MPS VI is not affected. Individuals with a more severe form of MPS VI can have airway obstruction, develop **hydrocephalus** (extra fluid accumulating in the brain) and have bone changes. Additionally, individuals with a severe form of MPS VI are more likely to die while in their teens. With a milder form of the condition, individuals tend to be shorter than expected for their age, develop corneal clouding, and live longer. The gene involved in MPS VI is believed to be located on the long arm of chromosome 5 (approximate location 5q11-13).

MPS VII (Sly syndrome)

MPS VII is an extremely rare form of MPS and is caused by a deficiency of the enzyme beta-glucuronidase. It is also highly variable, but symptoms are generally similar to those seen in individuals with Hurler syndrome. The gene that causes MPS VII is located on the long arm of chromosome 7 (location 7q21).

MPS IX (Hyaluronidase deficiency)

MPS IX is a condition that was first described in 1996 and has been grouped with the other MPS conditions by some researchers. MPS IX is caused by the deficiency of the enzyme hyaluronidase. In the few individuals described with this condition, the symptoms are variable, but some develop soft-tissue masses (growths under the skin). Also, these individuals are shorter than expected for their age. The gene involved in MPS IX is believed to be located on the short arm of chromosome 3 (possibly 3p21.3-21.2)

Many individuals with an MPS condition have problems with airway constriction. This constriction may be so serious as to create significant difficulties in administering **general anesthesia**. Therefore, it is recommended that surgical procedures be performed under **local anesthesia** whenever possible.

KEY TERMS

Cardiomyopathy—A thickening of the heart muscle.

Enzyme—A protein that catalyzes a biochemical reaction or change without changing its own structure or function.

Joint contractures—Stiffness of the joints that prevents full extension.

Kyphosis—An abnormal outward curvature of the spine, with a hump at the upper back.

Lysosome—Membrane-enclosed compartment in cells, containing many hydrolytic enzymes; where large molecules and cellular components are broken down.

Mucopolysaccharide—A complex molecule made of smaller sugar molecules strung together to form a chain. Found in mucous secretions and intercellular spaces.

Recessive gene—A type of gene that is not expressed as a trait unless inherited by both parents.

X-linked gene—A gene carried on the X chromosome, one of the two sex chromosomes.

Diagnosis

While a diagnosis for each type of MPS can be made on the basis of the physical signs described above, several of the conditions have similar features. Therefore, enzyme analysis is used to determine the specific MPS disorder. Enzyme analysis usually cannot accurately determine if an individual is a carrier for a MPS condition. This is because the enzyme levels in individuals who are not carriers overlaps the enzyme levels seen in those individuals who are carrier for a MPS. With many of the MPS conditions, several mutations have been found in each gene involved that can cause symptoms of each condition. If the specific mutation is known in a family, DNA analysis may be possible.

Once a couple has had a child with an MPS condition, prenatal diagnosis is available to them to help determine if a fetus is affected with the same MPS as their other child. This can be accomplished through testing samples using procedures such as an **amniocentesis** or **chorionic villus sampling** (CVS). Each of these procedures has its own risks, benefits, and limitations.

Treatment

There is no cure for mucopolysaccharidosis. There are several types of experimental therapies that are being investigated. Typically, treatment involves trying to relieve some of the symptoms. For MPS I and VI, **bone marrow transplantation** has been attempted as a treatment option. In those conditions, bone marrow transplantation has sometimes been found to help slow down the progression or reverse some of symptoms of the disorder in some children. The benefits of a bone marrow transplantation are more likely to be noticed when performed on children under two years of age. However it is not certain that a bone marrow transplant can prevent further damage to certain organs and tissues, including the brain. Furthermore, bone marrow transplantation is not felt to be helpful in some MPS disorders and there are risks, benefits, and limitations with this procedure. In 2000, ten individuals with MPS I received recombinant human alpha-L-iduronidase every week for one year. Those individuals showed an improvement with some of their symptoms. Additionally, there is ongoing research involving gene replacement therapy (the insertion of normal copies of a gene into the cells of patients whose gene copies are defective).

Prevention

No specific preventive measures are available for genetic diseases of this type. For some of the MPS diseases, biochemical tests are available that will identify healthy individuals who are carriers of the defective gene, allowing them to make informed reproductive decisions. There is also the availability of prenatal diagnosis for all MPS disease to detect affected fetuses.

Resources

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ORGANIZATIONS

- Canadian Society for Mucopolysaccharide and Related Diseases. PO Box 64714, Unionville, ONT L3R-OM9. Canada (905) 479-8701 or (800) 667-1846. <<http://www.mpssociety.ca>>.
- Children Living with Inherited Metabolic Diseases. The Quadrangle, Crewe Hall, Weston Rd., Crewe, Cheshire, CW1-6UR. UK 127 025 0221. Fax: 0870-7700-327. <<http://www.climb.org.uk>>.

Metabolic Information Network. PO Box 670847, Dallas, TX 75367-0847. (214) 696-2188 or (800) 945-2188.

National MPS Society. 102 Aspen Dr., Downingtown, PA 19335. (610) 942-0100. Fax: (610) 942-7188. info@mpsociety.org. < <http://www.mpsociety.org> >.

National Organization for Rare Disorders (NORD). PO Box 8923, New Fairfield, CT 06812-8923. (203) 746-6518 or (800) 999-6673. Fax: (203) 746-6481. < <http://www.rarediseases.org> >.

Society for Mucopolysaccharide Diseases. 46 Woodside Rd., Amersham, Buckinghamshire, HP6 6AJ. UK +44 (01494) 434156. < <http://www.mpsociety.co.uk> >.

Zain Hansen MPS Foundation. 23400 Henderson Rd., Covelo, CA 95420. (800) 767-3121.

OTHER

National Library of Medicine. National Institutes of Health. < <http://www.nlm.nih.gov> >.

“NINDS Mucopolysaccharidoses Information Page.” The National Institute of Neurological Disorders and Stroke. National Institutes of Health. < http://www.ninds.nih.gov/health_and_medical/disorders/mucopolysaccharidoses.htm >.

Online Mendelian Inheritance in Man (OMIM). National Center for Biotechnology Information. < <http://www.ncbi.nlm.nih.gov/Omim/> >.

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Mucormycosis

Definition

Mucormycosis is a rare but often fatal disease caused by certain fungi. It is sometimes called zygomycosis or phycomycosis. Mucormycosis is an opportunistic infection that typically develops in patients with weakened immune systems, diabetes, kidney failure, organ transplants, or **chemotherapy** for **cancer**. It may also develop in patients receiving an iron chelating drug called desferrioxamine (Desferal) as treatment for acute iron **poisoning**.

Description

In the United States, mucormycosis is most likely to develop in the patient's nasal area or in the lungs; however, it may also develop on the skin or in the digestive tract. Gastrointestinal disease usually develops only in severely malnourished patients. Cutaneous mucormycosis is most likely to develop under occlusive surgical dressings. Occlusive dressings are

intended to keep air out of incisions or other **wounds**, but they also trap body heat and moisture.

The incidence of the disease is difficult to evaluate because it is very rare; however, the rate seems to be increasing. One American cancer center reported in 2000 that mucormycosis was found in 0.7% of patients at **autopsy** and in 20 patients per 100,000 admissions to the center. The most recent mortality statistics from the Centers for Disease Control and Prevention (CDC) indicate that a total of 22 Americans died from mucormycosis in 2001—1 from pulmonary mucormycosis, 5 from rhinocerebral mucormycosis, 2 from disseminated mucormycosis, and 14 from unspecified forms of the disease.

As far as is known, mucormycosis affects members of either sex and all races equally, although the pulmonary form of the disease is somewhat more common in men than in women. Mucormycosis may develop in patients in any age group, including newborns.

Rhinocerebral mucormycosis

Rhinocerebral mucormycosis is an infection of the nose, eyes, and brain. The fungus destroys the tissue of the nasal passages, sinuses, or hard palate, producing a black or pus-filled discharge and visible patches of dying tissue. The patient will typically have **fever**, **pain**, and forward bulging of the eyes (proptosis). The fungus then invades the tissues around the eye socket and eventually the brain. At that point the patient may have convulsions or **paralysis** on one side of the body.

Pulmonary mucormycosis

Most patients with the pulmonary form of the disease are being treated for leukemia. The fungus enters the patient's lungs, where it eventually invades a major blood vessel, causing the patient to **cough** up blood or hemorrhage into the lungs.

Gastrointestinal mucormycosis

Gastrointestinal mucormycosis has been reported in premature or low-birth-weight infants as well as malnourished adults. It may lead to intestinal perforation and other complications requiring immediate surgery. A Spanish hospital reported in 2004 on an outbreak of gastrointestinal mucormycosis that affected five patients in an ICU over a 14-week period. Two of the patients died. The outbreak was eventually traced to a supply of wooden tongue depressors that had been contaminated by two species of *Rhizopus* fungi.

Causes and symptoms

Mucormycosis is caused by fungi of several different species, including *Mucor*, *Rhizopus*, *Absidia*, and *Rhizomucor*. When these organisms gain access to the mucous membranes of the patient's nose or lungs, they multiply rapidly and invade the nearby blood vessels. The fungi destroy soft tissue and bone, as well as the walls of blood vessels.

The early symptoms of rhinocerebral mucormycosis include fever, sinus pain, **headache**, and **cellulitis**. As the fungus reaches the eye tissues, the patient develops dilated pupils, drooping eyelids, a bulging eye, and eventually hemorrhage of the blood vessels in the brain—causing convulsions, partial paralysis, and **death**.

The symptoms of pulmonary mucormycosis include fever and difficulty breathing, with eventual bleeding from the lungs.

The symptoms of gastrointestinal mucormycosis are not unique to the disease, which may complicate diagnosis. Patients typically complain of pressure or pain in the abdomen, **nausea**, and **vomiting**.

Diagnosis

Diagnosis is usually based on a combination of the patient's medical history and a visual examination of the nose, throat, and eyes. The doctor will take a tissue sample for biopsy, or a PAS, potassium hydroxide (KOH), or Calcofluor stain in order to make a tentative diagnosis. Confirmation requires a laboratory culture.

Imaging studies are not needed to make the diagnosis. If the patient has mucormycosis, however, **magnetic resonance imaging** (MRI) and **computed tomography scans** (CT scans) will usually show the destruction of soft tissue or bone in patients with advanced disease. Chest x rays will sometimes show a cavity in the lung or an area filled with tissue fluid if the patient has pulmonary mucormycosis.

Treatment

Treatment is usually begun without waiting for laboratory reports because of the rapid spread and high mortality rate of the disease. Therapy includes intravenous amphotericin B (Fungizone); surgical removal of infected tissue; and careful monitoring of the disorder or condition that is responsible for the patient's vulnerability. Most patients who survive require a 4–6-week course of treatment.

Follow-up care includes educating patients about the signs of recurrent mucormycosis—particularly

KEY TERMS

Amphotericin B—An antibiotic used to treat mucormycosis and other severe fungal infections.

Opportunistic infection—An infection that develops only when a person's immune system is weakened.

Orbit—The bony cavity or socket surrounding the eye.

Zygomycosis—Another term for mucormycosis. The fungi that cause mucormycosis belong to a group called Zygomycetes.

facial swelling and a black discharge from the nose—and telling them to see a doctor at once if they notice these symptoms.

Patients who survive rhinocerebral mucormycosis are often left with severe facial disfigurement and usually require **plastic surgery** to restore their appearance.

Prognosis

The prognosis for recovery from mucormycosis is poor. The mortality rate is 30%–50% of patients with the rhinocerebral form, and even higher for patients with pulmonary mucormycosis. The disease is almost 100% fatal for patients with **AIDS**.

Prevention

Prevention depends on protecting high-risk patients from contact with sugary foods, decaying plants, moldy bread, manure, and other breeding grounds for fungi. In addition, health care professionals treating hospital inpatients should be careful to change occlusive dressings frequently and check the underlying skin for any signs of possible fungal infection.

Resources

BOOKS

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ORGANIZATIONS

Centers for Disease Control. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

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Mucoviscidosis see **Cystic fibrosis**

MUGA scan see **Multiple-gated acquisition (MUGA) scan**

Multiple-gated acquisition (MUGA) scan

Definition

The multiple-gated acquisition (MUGA) scan is a non-invasive nuclear test that uses a radioactive isotope called technetium to evaluate the functioning of the heart's ventricles.

Purpose

The MUGA scan is performed to determine if the heart's left and right ventricles are functioning properly and to diagnose abnormalities in the heart wall. It can be ordered in the following patients:

- With known or suspected **coronary artery disease**, to diagnose the disease and predict outcomes
- With lesions in their heart valves
- Who have recently had a **heart attack**, to assess damage to heart tissue and predict the likelihood of future cardiac events
- With congestive **heart failure**

- Who have undergone percutaneous transluminal coronary **angioplasty**, **coronary artery bypass graft surgery**, or medical therapy, to assess the efficacy of the treatment
- With low cardiac output after open-heart surgery
- Who are undergoing **chemotherapy**.

Precautions

Pregnant women and those who are breastfeeding should not be exposed to technetium.

Description

The MUGA scan measures the heart's function and the flow of blood through it. The strongest chamber in the heart is the left ventricle, which serves as the main pump of blood through the body. The left ventricular is assessed by measuring the amount of blood pumped with each heartbeat (the ejection fraction), ventricle filling, and the blood flow into the pumping chamber. A normal ejection fraction is 50% or more. The heart's ejection fraction is one of the most important measures of its performance. The right ventricle's ability to pump blood to the lungs is also assessed, and any abnormalities in the heart wall are identified. The MUGA scan is the most accurate, non-invasive test available to assess the heart's ventricles.

MUGA is a nuclear heart scan, which means that it involves the use of a radioactive isotope that targets the heart and a radionuclide detector that traces the absorption of the radioactive isotope. The isotope is injected into a vein and absorbed by healthy tissue at a known rate during a certain time period. The radionuclide detector, in this case a gamma scintillation camera, picks up the gamma rays emitted by the isotope.

During the MUGA scan, electrodes are placed on the patient's body so that an electrocardiogram (ECG) can be conducted. The imaging equipment and computer are synchronized with the ECG so that images of the heart can be recorded without motion or blur. Then a small amount of a mildly radioactive isotope called technetium Tc99m stannous pyrophosphate, usually called technetium, is injected, usually into an arm vein. While the patient lies motionless on the test table, a gamma scintillation camera follows the movement of the technetium through the blood circulating in the heart. The camera, which looks like an x-ray machine and is suspended above the table, moves back and forth over the patient. It displays multiple images of the heart in motion and records them on a computer for later analysis.

The MUGA scan is usually performed in a hospital's nuclear medicine department, but it can also be performed in an outpatient facility or at the patient's bedside if equipment is available. The scan is done immediately after injection of the technetium and usually takes about 30 minutes to one hour. It is also called multigated graft acquisition, multigated acquisition scan, cardiac blood-pool imaging, and equilibrium radionuclide **angiography**. Test results can be affected by patient movement during the test, electrocardiogram abnormalities, an irregular heartbeat, or long-acting nitrates.

The MUGA scan can be done with the patient at rest or exercising (called a **stress MUGA**). The stress MUGA is often performed in patients who have or are suspected of having coronary artery disease. The resting MUGA is compared to the stress MUGA and changes in the heart's pumping performance are analyzed. In some cases, the rest MUGA is compared to a nitroglycerin MUGA, in which a strong heart drug called nitroglycerin is administered to the patient before the scan. For the nitroglycerin MUGA, a cardiologist should be present.

The MUGA scan is not dangerous. The technetium is completely gone from the body within a few days of the test. The scan itself exposes the patient to about the same amount of radiation as a **chest x ray**. The patient can resume normal activities immediately after the test.

Normal results

If the patient's heart is normal, the technetium will appear to be evenly distributed in the scans. In a stress MUGA, patients with normal hearts will exhibit an increase in ejection fraction or no change.

Abnormal results

An uneven distribution of technetium in the heart indicates that the patient has coronary artery disease, a **cardiomyopathy**, or blood shunting within the heart. Abnormalities in a resting MUGA usually indicate a heart attack, while those that occur during **exercise** usually indicate **ischemia**. In a stress MUGA, patients with coronary artery disease may exhibit a decrease in ejection fraction.

Resources

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300. <<http://www.americanheart.org>>.

KEY TERMS

Ejection fraction—The fraction of all blood in the ventricle that is ejected at each heartbeat. One of the main advantages of the MUGA scan is its ability to measure ejection fraction, one of the most important measures of the heart's performance.

Electrocardiogram—A test in which electronic sensors called electrodes are placed on the body to record the heart's electrical activities.

Heart attack—A cardiac emergency that occurs when a clot blocks blood flow in one or more of the heart's arteries. Oxygen supply to the heart muscle is cut off, resulting in the death of heart tissue in the affected area.

Ischemia—A decreased supply of oxygenated blood to a body part or organ, often marked by pain and organ dysfunction, as in ischemic heart disease.

Non-invasive—A procedure that does not penetrate the body.

Radioactive isotope—One of two or more atoms with the same number of protons but a different number of neutrons with a nuclear composition. In nuclear scanning, radioactive isotopes are used as a diagnostic agent.

Technetium—A radioactive isotope frequently used in radionuclide scanning of the heart and other organs. It is produced during nuclear fission reactions.

Ventricles—The heart's lower chambers are called the left and right ventricles. They send blood to the lungs and throughout the body. The MUGA scan is performed to evaluate the ventricles.

Texas Heart Institute. Heart Information Service. P.O. Box 20345, Houston, TX 77225-0345. <<http://www.tmc.edu/thi>>.

Lori De Milto

Multiple chemical sensitivity

Definition

Multiple chemical sensitivity—also known as MCS syndrome, environmental illness, idiopathic environmental intolerance, chemical **AIDS**, total

allergy syndrome, or simply MCS—is a disorder in which a person develops symptoms from exposure to chemicals in the environment. With each incidence of exposure, lower levels of the chemical will trigger a reaction and the person becomes increasingly vulnerable to reactions triggered by other chemicals.

Medical experts disagree on the cause of the syndrome, and as to whether MCS is a clinically recognized illness. In a 1992 position statement that remained unchanged as of early 2000, the American Medical Association's Council on Scientific Affairs did not recognize MCS as a clinical condition due to a lack of accepted diagnostic criteria and controlled studies on the disorder. A more recent discussion of methodological problems in published studies of MCS, as well as recommendations for patient care, may be found in the 1999 position paper on MCS drafted by the American College of Occupational and Environmental Medicine (ACOEM). As of 2003, however, many researchers in Europe as well as the United States regard MCS as a contemporary version of neurasthenia, a concept first introduced by a physician named George Miller Beard in 1869.

Description

Multiple chemical sensitivity typically begins with one high-dose exposure to a chemical, but it may also develop with long-term exposure to a low level of a chemical. Chemicals most often connected with MCS include: formaldehyde; pesticides; solvents; petrochemical fuels such as diesel, gasoline, and kerosene; waxes, detergents, and cleaning products; latex; tobacco smoke; perfumes and fragrances; and artificial colors, flavors, and preservatives. People who develop MCS are commonly exposed in one of the following situations: on the job as an industrial worker; residing or working in a poorly ventilated building; or living in conditions of high air or water pollution. Others may be exposed in unique incidents.

Because MCS is difficult to diagnose, estimates vary as to what percentage of the population develops MCS. However, most MCS patients are female. The median age of MCS patients is 40 years old, and most experienced symptoms before they were 30 years old. There is also a large percentage of Persian Gulf War veterans who have reported symptoms of chemical sensitivity since their return from the Gulf in the early 1990s.

Causes and symptoms

Chemical exposure is often a result of indoor air pollution. Buildings that are tightly sealed for energy

conservation may cause a related illness called sick building syndrome, in which people develop symptoms from chronic exposure to airborne environmental chemicals such as formaldehyde from the furniture, carpet glues, and latex caulking. A person moving into a newly constructed building, which has not had time to degas, may experience the initial high-dose exposure that leads to MCS.

As of late 2002, the specific biochemical and physiological mechanisms in humans that lead to MCS are not well understood. A recent hypothesis, however, suggests that MCS is the end result of four different mechanisms of sensitization acting to reinforce one another. Further research is required to test this hypothesis.

The symptoms of MCS vary from person to person and are not chemical-specific. Symptoms are not limited to one physiological system, but primarily affect the respiratory and nervous systems. Symptoms commonly reported are **headache**, **fatigue**, weakness, difficulty concentrating, short-term memory loss, **dizziness**, irritability and depression, **itching**, **numbness**, burning sensation, congestion, **sore throat**, hoarseness, **shortness of breath**, **cough**, and stomach pains.

One commonly reported symptom of MCS is a heightened sensitivity to odors, including a stronger emotional reaction to them. A Japanese study published in late 2002 reported that patients diagnosed with MCS can identify common odors as accurately as most people, but regard a greater number of them as unpleasant.

One test that has been devised to evaluate patients with MCS is the capsaicin inhalation test. Capsaicin is an alkaloid found in hot peppers that is sometimes used in topical creams and rubs for the treatment of arthritis. When inhaled, capsaicin causes coughing in healthy persons as well as those with **allergies** that affect the airway; however, persons with MCS cough more deeply and frequently than control subjects when given a dose of capsaicin. Although the test is not diagnostic in the strict sense, it has been shown to be an effective way of identifying patients with MCS.

Diagnosis

Multiple chemical sensitivity is a twentieth-century disorder, becoming more prevalent as more human-made chemicals are introduced into the environment in greater quantities. It is especially difficult to diagnose because it presents no consistent or measurable set of symptoms and has no single diagnostic test or marker. For example, a 2002 study of **PET** scans of

MCS patients found no significant functional changes in the patients' brain tissues. Physicians are often either unaware of MCS as a condition, or refuse to accept that MCS exists. They may be unable to diagnose it, or may misdiagnose it as another degenerative disease, or may label it as a psychosomatic illness (a physical illness that is caused by emotional problems). Their lack of understanding generates frustration, **anxiety**, and distrust in patients already struggling with MCS. However, a new specialty of medicine is evolving to address MCS and related illnesses: occupational and environmental medicine. A physician looking for MCS will take a complete patient history and try to identify chemical exposures.

Some MCS patients may be helped by a psychological evaluation, particularly if they show signs of panic attacks or other **anxiety disorders**. It is known that many patients with MCS suffer from comorbid depression and anxiety. In addition, MCS patients appear to have high rates of **mood disorders** compared to **asthma** patients as well as normal test subjects.

Treatment

While doctors may recommend **antihistamines**, **analgesics**, and other medications to combat the symptoms, the most effective treatment is to avoid those chemicals which trigger the symptoms. This becomes increasingly difficult as the number of offending chemicals increases, and people with MCS often remain at home where they are able to control the chemicals in their environment. This isolation often limits their abilities to work and socialize, so supportive counseling may also be appropriate.

Alternative treatment

Some MCS patients find relief with **detoxification** programs of **exercise** and sweating, and chelation of heavy metals. Others support their health with nutritional regimens and immunotherapy vaccines. Some undergo food-allergy testing and testing for accumulated pesticides in the body to learn more about their condition and what chemicals to avoid. Homeopathy and **acupuncture** can give added support to any treatment program for MCS patients. Botanical medicine can help to support the liver and other involved organs.

Prognosis

Once MCS sets in, sensitivity continues to increase and a person's health continues to deteriorate. Strictly avoiding exposure to triggering chemicals for a year or more may improve health.

KEY TERMS

Capsaicin—An alkaloid found in hot peppers that is used in an inhalation test to identify patients with MCS.

Degas—To release and vent gases. New building materials often give off gases and odors and the air should be well circulated to remove them.

Neurasthenia—A term coined in the late nineteenth century to refer to a condition of chronic mental and physical weakness and fatigue. Some researchers regard MCS as a twentieth-century version of neurasthenia.

Sick building syndrome—An illness related to MCS in which a person develops symptoms in response to chronic exposure to airborne environmental chemicals found in a tightly sealed building.

Prevention

Multiple chemical sensitivity is difficult to prevent because even at high-dose exposures, different people react differently. Ensuring adequate ventilation in situations with potential for acute high-dose or chronic low-dose chemical exposure, as well as wearing the proper protective equipment in industrial situations, will minimize the risk.

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ORGANIZATIONS

American Academy of Environmental Medicine. P.O. Box CN 1001-8001, New Hope, PA 18938. (215) 862-4544.

American College of Occupational and Environmental Medicine (ACOEM). 1114 North Arlington Heights Road, Arlington Heights, IL 60004. (847) 818-1800. <www.acoem.org>.

OTHER

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Multiple endocrine adenomatosis see
Multiple endocrine neoplasia syndromes

Multiple endocrine neoplasia syndromes

Definition

The multiple endocrine neoplasia (MEN) syndromes are three related disorders affecting the thyroid and other hormonal (endocrine) glands of the

body. MEN has previously been known as familial endocrine adenomatosis.

Description

The three forms of MEN are MEN1 (Wermer’s syndrome), MEN2A (Sipple syndrome), and MEN2B (previously known as MEN3). Each is an autosomal dominant genetic condition which predisposes to hyperplasia (excessive growth of cells) and tumor formation in a number of endocrine glands.

Causes and symptoms

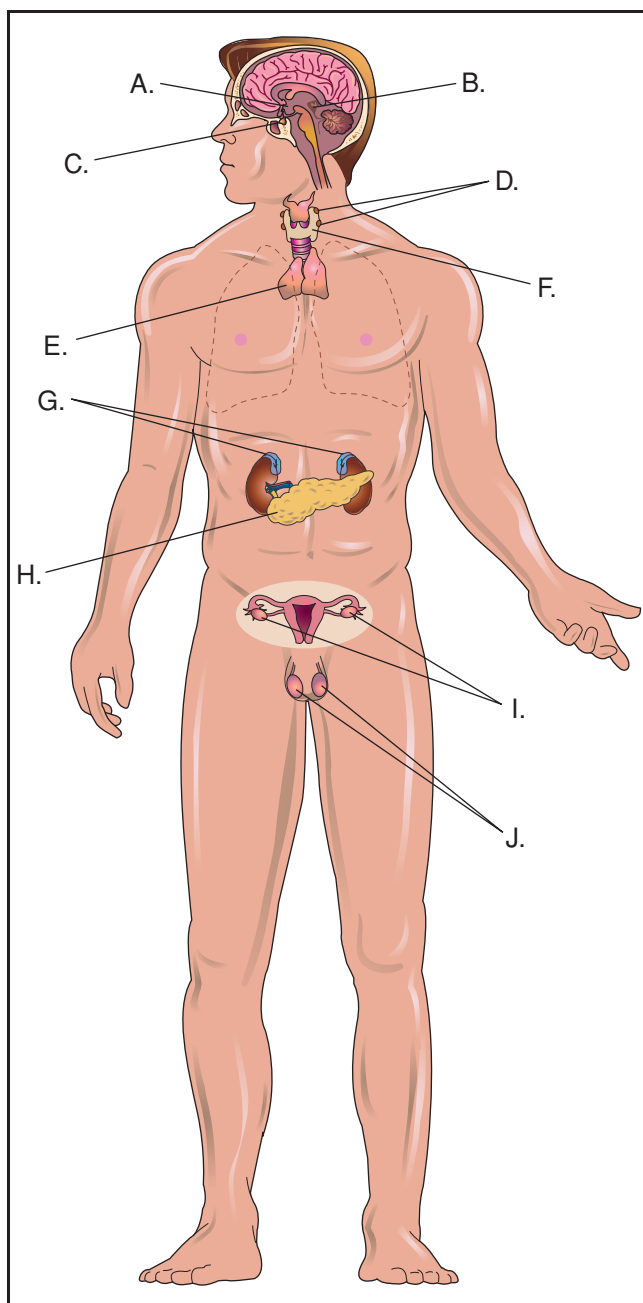
MEN1 patients experience hyperplasia or tumors of several endocrine glands, including the parathyroids, the pancreas, and the pituitary. The most frequent symptom of MEN1 is **hyperparathyroidism**. Overgrowth of the parathyroid glands leads to over secretion of parathyroid hormone, which leads to elevated blood calcium levels, **kidney stones**, weakened bones, and nervous system depression. Almost all MEN1 patients show parathyroid symptoms by age 40.

Tumors of the pancreas known as gastrinomas are also common in MEN1. Excessive secretion of gastrin (a hormone secreted into the stomach to aid in digestion) by these tumors can cause upper gastrointestinal ulcers. The anterior pituitary and the adrenal glands can also be affected. Unlike MEN2, the thyroid gland is rarely involved in MEN1 symptoms.

Patients with MEN2A and MEN2B experience two main symptoms, medullary **thyroid cancer** (MTC) and a tumor of the adrenal gland medulla known as **pheochromocytoma**. MTC is a slow-growing **cancer**, but one that can be cured in less than 50% of cases. Pheochromocytoma is usually a benign tumor that causes excessive secretion of adrenal hormones, which, in turn, can cause life-threatening **hypertension** and cardiac arrhythmia.

The two forms of MEN2 are distinguished by additional symptoms. MEN2A patients have a predisposition to increase in size (hypertrophy) and to develop tumors of the parathyroid gland. Although similar to MEN1, less than 20% of MEN2A patients will show parathyroid involvement.

MEN2B patients show a variety of additional conditions: a characteristic facial appearance with swollen lips; tumors of the mucous membranes of the eye, mouth, tongue, and nasal cavity; enlarged colon; and skeletal abnormalities. Symptoms develop early in life (often under five years of age) in cases of MEN2B and the tumors are more aggressive. MEN2B is about ten-fold less common than MEN2A.



The human endocrine system: A. Hypothalamus. B. Pineal. C. Pituitary. D. Parathyroid. E. Thymus. F. Thyroid. G. Adrenals. H. Pancreas. I. Ovaries (female). J. Testes (male). (Illustration by Electronic Illustrators Group.)

MEN1 is caused by mutation at the **PYGM** gene. **PYGM** is one of a group of genes known as tumor suppressor genes. A patient who inherits one defective copy of a tumor suppressor gene from either parent has a strong predisposition to the disease because of the high probability of incurring a second mutation in at least one dividing cell. That cell no longer possesses

even one normal copy of the gene. When both copies are defective, tumor suppression fails and tumors develop.

Both types of MEN2 are caused by mutations in another gene, known as **RET**. A mutation in only one copy of the **RET** gene is sufficient to cause disease. A number of different mutations can lead to MEN2A, but only one specific genetic alteration leads to MEN2B.

For all types of MEN, the children of an affected individual have a 50% chance of inheriting the defective gene.

Diagnosis

Classical diagnosis of MEN is based on clinical features and on testing for elevated hormone levels. For MEN1, the relevant hormone is parathyroid hormone. For both types of MEN2, the greatest concern is development of medullary thyroid cancer. MTC can be detected by measuring levels of the thyroid hormone, calcitonin. Numerous other hormone levels can be measured to assess the involvement of the various other endocrine glands.

Diagnosis of MEN2B can be made by **physical examination** alone. However, MEN2A shows no distinct physical features and must be identified by measuring hormone levels or by finding endocrine tumors.

Since 1994, genetic screening using DNA technology has been available for both MEN1 and MEN2. This new methodology allows diagnosis prior to the onset of symptoms.

In the past, there was no way of definitively identifying which children had inherited the defective gene. As a result, all children had to be considered at risk. In the case of MEN2A and MEN2B, children would undergo frequent calcitonin testing. Molecular techniques now allow a positive distinction to be made between children who are and are not actually at risk.

Children who are identified as carriers of the **RET** gene can be offered total **thyroidectomy** on a preventative (prophylactic) basis to prevent the development of MTC.

Treatment

No comprehensive treatment is available for genetic conditions such as MEN. However, some of the consequences of MEN can be symptomatically treated.

Pheochromocytoma in both types of MEN 2 can be cured by surgical removal of this slow growing tumor.

KEY TERMS

Endocrine—A term used to describe the glands that produce hormones in the body.

Hyperplasia—An overgrowth of normal cells within an organ or tissue.

Medullary thyroid cancer (MTC)—A slow-growing tumor associated with MEN.

Neoplasm—An abnormal formation of tissue; for example, a tumor.

Pheochromocytoma—A tumor of the medullary of the adrenal gland.

Treatment of MTC is by surgical removal of the thyroid, although doctors may disagree at what stage to remove the thyroid. After thyroidectomy, the patient will receive normal levels of thyroid hormone orally or by injection.

Even when surgery is performed early, metastatic spread of the cancer may have already occurred. Since this cancer is slow growing, metastasis may not be obvious. Metastasis is very serious in MTC because **chemotherapy** and **radiation therapy** are not effective in controlling its spread.

Prognosis

Diagnosed early, the prognosis for the MEN diseases is reasonably good, even for MEN2B, the most dangerous of the three forms. Even in the absence of treatment, a few individuals with MEN2A mutations will never show any symptoms at all. Analysis of at-risk family members using molecular genetic techniques will lead to earlier treatment and improved outcomes.

Prevention

One of the most serious consequences of MEN is MTC, which can be prevented by thyroidectomy. There is no preventive measure to block the occurrence of genetic mutations such as those that cause MEN.

Resources

ORGANIZATIONS

Canadian MEN Society. P.O. Box 100, Meola, Saskatchewan S0M 1X0. (306) 892-2080.

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Multiple myeloma

Definition

Multiple myeloma is a **cancer** in which antibody-producing plasma cells grow in an uncontrolled and invasive (malignant) manner.

Description

Multiple myeloma, also known as plasma cell myeloma, is the second-most common cancer of the blood. It is the most common type of plasma cell neoplasm. Multiple myeloma accounts for approximately 1% of all cancers and 2% of all deaths from cancer. Multiple myeloma is a disease in which malignant plasma cells spread through the bone marrow and hard outer portions of the large bones of the body. These myeloma cells may form tumors called plasmacytomas. Eventually, multiple soft spots or holes, called osteolytic lesions, form in the bones.

Bone marrow is the spongy tissue within the bones. The breastbone, spine, ribs, skull, pelvic bones, and the long bone of the thigh all are particularly rich in marrow. Bone marrow is a very active tissue that is responsible for producing the cells that circulate in the blood. These include the red blood cells that carry oxygen, the white blood cells that develop into immune system cells, and platelets, which cause blood to clot.

Plasma cells and immunoglobulins

Plasma cells develop from B-lymphocytes or B-cells, a type of white blood cell. B-cells, like all blood cells, develop from unspecialized stem cells in the bone marrow. Each B-cell carries a specific antibody that recognizes a specific foreign substance called an antigen. Antibodies are large proteins called immunoglobulins (Igs), which recognize and destroy foreign substances and organisms such as bacteria. When a B-cell encounters its antigen, it begins to divide rapidly to form mature plasma cells. These plasma cells are all identical (monoclonal). They produce large amounts of identical antibody that are specific for the antigen.

Malignant plasma cells

Multiple myeloma begins when the genetic material (DNA) is damaged during the development of a stem cell into a B-cell in the bone marrow. This causes the cell to develop into an abnormal or malignant plasmablast, a developmentally early form of plasma cell. Plasmablasts produce adhesive molecules that

allow them to bond to the inside of the bone marrow. A growth factor, called interleukin-6, promotes uncontrolled growth of these myeloma cells in the bone marrow and prevents their natural death. Whereas normal bone marrow contains less than 5% plasma cells, bone marrow of an individual with multiple myeloma contains over 10% plasma cells.

In most cases of multiple myeloma, the malignant plasma cells all make an identical Ig. Igs are made up of four protein chains that are bonded together. Two of the chains are light and two are heavy. There are five classes of heavy chains, corresponding to five types of Igs with different immune system functions. The Igs from myeloma cells are nonfunctional and are called paraproteins. All of the paraproteins from any one individual are monoclonal (identical) because the myeloma cells are identical clones of a single plasma cell. Thus, the paraprotein is a monoclonal protein or M-protein. The M-proteins crowd out the functional Igs and other components of the immune system. They also cause functional antibodies, which are produced by normal plasma cells, to rapidly break down. Thus, multiple myeloma depresses the immune system.

In about 75% of multiple myeloma cases, the malignant plasma cells also produce monoclonal light chains, or incomplete Igs. These are called Bence-Jones proteins and are secreted in the urine. Approximately 1% of multiple myelomas are called nonsecreters because they do not produce any abnormal Ig.

Osteolytic lesions

About 70% of individuals with multiple myeloma have soft spots or lesions in their bones. These lesions can vary from quite small to grapefruit-size. In part, these lesions occur because the malignant plasma cells rapidly outgrow the normal bone-forming cells. In addition, malignant myeloma cells produce factors that affect cells called osteoclasts. These are the cells that normally destroy old bone, so that new bone can be produced by cells called osteoblasts. The myeloma cell factors increase both the activation and the growth of osteoclasts. As the osteoclasts multiply and migrate, they destroy healthy bone and create lesions. **Osteoporosis**, or widespread bone weakness, may develop.

There are more than 40,000 multiple myeloma patients in the United States. The American Cancer Society predicts an additional 14,400 new cases in 2001. About 11,200 Americans will die of the disease in 2001. Multiple myeloma is one of the leading causes of cancer deaths among African Americans.

In Western industrialized countries, approximately four people in 100,000 develop multiple myeloma. The incidence of multiple myeloma among African Americans is 9.5 per 100,000, about twice that of Caucasians. Asians have a much lower incidence of the disease. In China, for example, the incidence of multiple myeloma is only one in 100,000. The offspring and siblings of individuals with multiple myeloma are at a slightly increased risk for the disease.

At diagnosis, the average age of a multiple myeloma patient is 68 to 70. Although the average age at onset is decreasing, most multiple myelomas still occur in people over 40. This cancer is somewhat more prevalent in men than in women.

Causes and symptoms

Associations

The cause of multiple myeloma has not been determined. However, a number of possible associations have been identified:

- decreased immune system function; the immune systems of older individuals may be less efficient at detecting and destroying cancer cells
- genetic (hereditary) factors, suggested by the increased incidence in some ethnic groups and among family members
- occupational factors, suggested by the increased incidence among agricultural, petroleum, wood, and leather workers, and cosmetologists
- long-term exposure to herbicides, pesticides, petroleum products, heavy metals, plastics, and dusts such as asbestos
- radiation exposure, as among Japanese atomic bomb survivors, nuclear weapons workers, and medical personnel such as radiologists
- Kaposi's sarcoma-associated herpes virus (also called human herpes virus-8 or HHV-8), found in the blood and bone marrow cells of many multiple myeloma patients

Early symptoms

The accumulation of malignant plasma cells can result in tiny cracks or **fractures** in bones. Malignant plasma cells in the bone marrow can suppress the formation of red and white blood cells and platelets. About 80% of individuals with multiple myeloma are anemic due to low red blood cell formation. Low white blood cell formation results in increased susceptibility to infection, since new, functional antibodies are not

produced. In addition, normal circulating antibodies are rapidly destroyed. Low platelet formation can result in poor blood clotting. It is rare, however, that insufficient white blood cell and platelet formations are presenting signs of multiple myeloma.

These factors cause the early symptoms of multiple myeloma:

- **pain** in the lower back or ribs
- fatigue and paleness due to anemia (low red blood cell count)
- frequent and recurring infections, including bacterial **pneumonia**, urinary-tract and kidney infections, and shingles
- bleeding

Bone destruction

Bone pain, particularly in the backbone, hips, and skull, is often the first symptom of multiple myeloma. As malignant plasma cells increase in the bone marrow, replacing normal marrow, they exert pressure on the bone. As overly-active osteoclasts (large cells responsible for the breakdown of bone) remove bone tissue, the bone becomes soft. Fracture and spinal cord compression may occur.

Plasmacytomas (malignant tumors of plasma cells) may weaken bones, causing fractures. Fractured bones or weak or collapsed spinal bones, in turn, may place unusual pressure on nearby nerves, resulting in nerve pain, burning, or **numbness** and muscle weakness. Proteins produced by myeloma cells also may damage nerves.

Calcium from the destroyed bone enters the blood and urine, causing **hypercalcemia**, a medical condition in which abnormally high concentrations of calcium compounds exist in the bloodstream. High calcium affects nerve cell and kidney function. The symptoms of hypercalcemia include:

- weakness and fatigue
- depression
- mental confusion
- constipation
- increased thirst
- increased urination
- nausea and vomiting
- kidney pain
- kidney failure

Hypercalcemia affects about one-third of multiple myeloma patients.

Serum proteins

The accumulation of M-proteins in the serum (the liquid portion of the blood) may cause additional complications, such as hyperviscosity syndrome, or thickening of the blood (though rare in multiple myeloma patients). Symptoms of hyperviscosity include:

- fatigue
- headaches
- shortness of breath
- mental confusion
- chest pain
- kidney damage and failure
- vision problems
- Raynaud's phenomenon

Poor blood circulation, or Raynaud's phenomenon, can affect any part of the body, but particularly the fingers, toes, nose, and ears.

Cryoglobulinemia occurs when the protein in the blood forms particles under cold conditions. These particles can block small blood vessels and cause pain and numbness in the toes, fingers, and other extremities during cold weather.

Amyloidosis is a rare complication of multiple myeloma. It usually occurs in individuals whose plasma cells produce only Ig light chains. These Bence-Jones proteins combine with other serum proteins to form amyloid protein. This starchy substance can invade tissues, organs, and blood vessels. In particular, amyloid proteins can accumulate in the kidneys, where they block the tiny tubules that are the kidney's filtering system. Indicators of amyloidosis include:

- carpal tunnel syndrome
- kidney failure
- liver failure
- heart failure

Diagnosis

Blood and urine tests

Often, the original diagnosis of multiple myeloma is made from routine blood tests that are performed for other reasons. Blood tests may indicate:

- anemia
- abnormal red blood cells
- high serum protein levels
- how levels of normal antibody

- high calcium levels
- high blood urea nitrogen (BUN) levels
- high creatinine levels

Urea and creatinine normally are excreted in the urine. High levels of urea and creatinine in the blood indicate that the kidneys are not functioning properly to eliminate these substances.

Protein electrophoresis is a laboratory technique that uses an electrical current to separate the different proteins in the blood and urine on the basis of size and charge. Since all of the multiple myeloma M-proteins in the blood and urine are identical, electrophoresis of blood and urine from a patient with multiple myeloma shows a large M-protein spike, corresponding to the high concentration of monoclonal Ig. Electrophoresis of the urine also can detect Bence-Jones proteins.

Bones

A **bone marrow aspiration** utilizes a very thin, long needle to remove a sample of marrow from the hip bone. Alternatively, a **bone marrow biopsy** with a larger needle removes solid marrow tissue. The marrow is examined under the microscope for plasma cells and tumors. If 10% to 30% of the cells are plasma cells, multiple myeloma is the usual diagnosis.

X rays are used to detect osteoporosis, osteolytic lesions, and fractures. Computer-assisted tomography (CAT or CT) scans can detect lesions in both bone and soft tissue. **Magnetic resonance imaging (MRI)** may give a more detailed image of a certain bone or a region of the body.

Treatment

Related disorders

Monoclonal gammopathy of undetermined significance (MGUS) is a common condition in which a monoclonal Ig is detectable. However, there are no tumors or other symptoms of multiple myeloma. MGUS occurs in about 1% of the general population and in about 3% of those over age 70. Over a period of years, about 16% to 20% of those with MGUS will develop multiple myeloma or a related cancer called malignant lymphoma.

Occasionally, only a single plasmacytoma develops, either in the bone marrow (isolated plasmacytoma of the bone) or other tissues or organs (extramedullary plasmacytoma). Some individuals with solitary plasmacytoma may develop multiple myeloma.



This x ray of the patient's left clavicle indicates an occurrence of myelomas in the bone. (Custom Medical Stock Photo. Reproduced by permission.)

Clinical stages

The Durie-Salmon system is used to stage multiple myeloma. Stage I multiple myeloma requires all of the following (1 gram = approx. 0.02 pints, 1 deciliter = approx. 0.33 ounces):

- hemoglobin (the oxygen-transporting molecule of red blood cells) above 10 grams/deciliter (g/dl)
- serum calcium below 12 mg/dl
- normal bone structure or only isolated plasmacytoma
- low M-protein, based on established guideline levels of Ig protein chains

Approximately 5% of multiple myeloma cases are not progressing at diagnosis, and may not progress for months or years. This is called smoldering myeloma. These patients have stage I blood chemistry but no symptoms.

Stage II multiple myeloma fits neither stage I nor stage III. Stage III multiple myeloma meets one or more the following criteria:

- hemoglobin below 8.5 g/dl
- serum calcium above 12 mg/dl
- advanced bone lesions
- high M-protein

Each stage is subclassified as A or B, based on serum creatinine indicators of normal or abnormal kidney function. Most patients have stage III multiple myeloma at diagnosis.

Prognostic indicators

Prognostic indicators for multiple myeloma may be used instead of, or in addition to, the staging system described above. Prognostic indicators are laboratory tests that help to define the stage of the disease at diagnosis, and its progression during treatment. These indicators are:

- plasmablastic multiple myeloma (presence of plasmablasts, the precursor malignant plasma cells)
- plasma cell labeling index (the percentage of plasma cells that are actively dividing)
- beta 2-microglobulin, a protein secreted by B-cells that correlates with the myeloma cell mass (also indicates kidney damage)

Since multiple myeloma often progresses slowly, and since the treatments can be toxic, the disease may not be treated until M-protein levels in the blood are quite high. In particular, MGUS and smoldering myeloma may be followed closely but not treated. Solitary plasmacytomas are treated with radiation and/or surgery and followed closely with examinations and laboratory tests.

Chemotherapy

Chemotherapy, or treatment with anti-cancer drugs, is used for multiple myeloma. **MP**, a combination of the drugs melphalan and prednisone, is the standard treatment. Usually, the drugs are taken by mouth every 3 to 4 weeks for 6 to 9 months or longer, until the M-protein levels in the blood stop decreasing. **MP** usually results in a 50% reduction in M-protein.

Dexamethasone, a corticosteroid, sometimes is used to treat the elderly or those in poor health. It can drop the M-protein levels by 40% in untreated individuals and by 20% to 40% in patients who have not responded to previous treatment. Other chemotherapy drugs, including cyclophosphamide, carmustine, doxorubicin, vincristine, and chlorambucil, may be used as well.

Multiple myeloma usually recurs within a year after the end of chemotherapy. Although the chemotherapy can be repeated after each recurrence, it is progressively less responsive to treatment.

Side effects of chemotherapy may include:

- anemia

- hair loss
- nausea
- vomiting
- diarrhea
- mood swings
- swelling
- acne

These side effects disappear after treatment is discontinued.

Other drug treatments

Bisphosphonates are drugs that inhibit the activity of osteoclasts. These drugs can slow the progression of bone disease, reduce pain, and help prevent bone fractures. Different types of bisphosphonates inhibit osteoclasts in different ways. They also reduce the production of interleukin-6 by bone marrow cells. Laboratory studies suggest that bisphosphonates may kill or inhibit the growth of multiple myeloma cells. Pamidronate is the most common bisphosphonate for treating multiple myeloma.

The drug thalidomide appears to have several anti-myeloma activities. Thalidomide affects the immune system in various ways and it appears to inhibit myeloma cells, both directly and indirectly. It also inhibits the growth of new blood vessels that are needed by tumors. However, if thalidomide is taken during **pregnancy**, it can cause severe **birth defects** or death of the fetus.

The drug allopurinol may be used to reduce high blood levels of uric acid that result from kidney dysfunction. **Diuretics** can improve kidney function. Infections require prompt treatment with **antibiotics**.

BONE AND PERIPHERAL BLOOD STEM CELL TRANSPLANTATION. Bone marrow or peripheral blood stem cell transplantations (PBSCT) are used to replace the stem cells of the bone marrow following high-dosage chemotherapy. Chemotherapy destroys the bone marrow stem cells that are necessary to produce new blood cells. In an autologous transplant, the patient's bone marrow stem cells or peripheral blood stem cells (immature bone marrow cells found in the blood) are collected, treated with drugs to kill any myeloma cells, and frozen prior to chemotherapy. Growth factors are used to increase the number of peripheral stem cells prior to collection. A procedure called apheresis is used to collect the peripheral stem cells. Following high-dosage chemotherapy, the stem cells are reinjected into the individual. In an allogeneic

KEY TERMS

Amyloidosis—A complication of multiple myeloma in which amyloid protein accumulates in the kidneys and other organs, tissues, and blood vessels.

Anemia—Any condition in which the red blood cell count is below normal.

Antibody—Immunoglobulin produced by immune system cells that recognizes and binds to a specific foreign substance (antigen).

Antigen—Foreign substance that is recognized by a specific antibody.

B-cell (B-lymphocyte)—Type of white blood cell that produces antibodies.

Bence-Jones protein—Light chain of an immunoglobulin that is overproduced in multiple myeloma and is excreted in the urine.

Beta 2-microglobulin—Protein produced by B-cells; high concentrations in the blood are indicative of multiple myeloma.

Cryoglobulinemia—Condition in which protein in the blood forms particles in the cold, blocking blood vessels, leading to pain and numbness of the extremities.

Electrophoresis—Use of an electrical field to separate proteins in a mixture (such as blood or urine), on the basis of the size and electrical charge of the proteins.

Hemoglobin—Protein in red blood cells that carries oxygen.

Hypercalcemia—Abnormally high levels of calcium in the blood.

Hyperviscosity—Thick, viscous blood, caused by the accumulation of large proteins, such as immunoglobulins, in the serum.

Immunoglobulin (Ig)—Antibody; large protein produced by B-cells that recognizes and binds to a specific antigen.

M-protein—Monoclonal or myeloma protein; paraprotein; abnormal antibody found in large amounts in the blood and urine of individuals with multiple myeloma.

Malignant—A characteristic of cancer cells that grow uncontrollably and invade other tissues.

Monoclonal—Identical cells or proteins; cells (clones) derived from a single, genetically-distinct cell, or proteins produced by these cells.

Monoclonal gammopathy of undetermined significance (MGUS)—Common condition in which M-protein is present, but there are no tumors or other symptoms of disease.

Neoplasm—Tumor made up of cancer cells.

Osteoblast—Bone-forming cell.

Osteoclast—Cell that absorbs bone.

Osteolytic lesion—Soft spot or hole in bone caused by cancer cells.

Osteoporosis—Condition in which the bones become weak and porous, due to loss of calcium and destruction of cells.

Paraprotein—M-protein; abnormal immunoglobulin produced in multiple myeloma.

Plasma cell—Type of white blood cell that produces antibodies; derived from an antigen-specific B-cell.

Platelet—Cell that is involved in blood clotting.

Stem cell—Undifferentiated cell that retains the ability to develop into any one of numerous cell types.

transplant, the donor stem cells come from a genetically-related individual such as a sibling.

Other treatments

Blood transfusions may be required to treat severe anemia.

Plasmapheresis, or plasma exchange **transfusion**, may be used to thin the blood to treat hyperviscosity syndrome. In this treatment, blood is removed and passed through a machine that separates the plasma, containing the M-protein, from the red and white blood cells and platelets. The blood cells are

transfused back into the patient, along with a plasma substitute or donated plasma.

Multiple myeloma may be treated with high-energy x rays directed at a specific region of the body. **Radiation therapy** is used for treating bone pain.

Alternative treatment

Interferon alpha, an immune-defense protein that is produced by some white blood cells and bone marrow cells, can slow the growth of myeloma cells. It usually is given to patients following chemotherapy, to prolong their remission. However, interferon may have toxic effects in older individuals with multiple myeloma.

Once multiple myeloma is in remission, calcium and vitamin D supplements can improve bone density. It is important not to take these supplements when the myeloma is active. Individuals with multiple myeloma must drink large amounts of fluid to counter the effects of hyperviscous blood.

Prognosis

The prognosis for individuals with MGUS or solitary plasmacytoma is very good. Most do not develop multiple myeloma. However, approximately 15% of all patients with multiple myeloma die within three months of diagnosis. About 60% respond to treatment and live for an average of two and a half to three years following diagnosis. Approximately 23% of patients die of other illnesses associated with advanced age.

The prognosis for a given individual may be based on the prognostic indicators described above. The median survival for those without plasmablasts, and with a low plasma cell labeling index (PCLI) and low beta 2-microglobulin, is 5.5 years. The median survival for patients with plasmablastic multiple myeloma, or with a high PCLI (1% or greater) and high beta 2-microglobulin (4 or higher), is 1.9 and 2.4 years, respectively. Many multiple myeloma patients are missing part or all of chromosome 13. The deletion of this chromosome, along with high beta 2-microglobulin, leads to a poor prognosis.

With treatment, multiple myeloma may go into complete remission. This is defined as:

- M-protein absent from the blood and urine
- myeloma cells not detectable in the bone marrow
- no clinical symptoms
- negative laboratory tests

However, with very sensitive testing, a few myeloma cells are usually detectable and eventually lead to a recurrence of the disease, in the bone or elsewhere in the body.

Prevention

There are no clearly-established risk factors for multiple myeloma and it is possible that a combination of factors interact to cause the disease. Thus, there is no method for preventing multiple myeloma.

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Margaret Alic, Ph.D.

Multiple personality disorder

Definition

Multiple personality disorder, or MPD, is a mental disturbance classified as one of the **dissociative disorders** in the fourth edition of the *Diagnostic and Statistical Manual of Mental Disorders (DSM-IV)*. It has been renamed dissociative identity disorder (DID). MPD or DID is defined as a condition in which "two or more distinct identities or personality states" alternate in controlling the patient's consciousness and behavior. Note: "Split personality" is not an accurate term for DID and should not be used as a synonym for **schizophrenia**.

Description

The precise nature of DID (MPD) as well as its relationship to other mental disorders is still a subject of debate. Some researchers think that DID may be a relatively recent development in western society. It may be a culture-specific syndrome found in western society, caused primarily by both childhood **abuse** and

unspecified long-term societal changes. Unlike depression or **anxiety disorders**, which have been recognized, in some form, for centuries, the earliest cases of persons reporting DID symptoms were not recorded until the 1790s. Most were considered medical oddities or curiosities until the late 1970s, when increasing numbers of cases were reported in the United States. Psychiatrists are still debating whether DID was previously misdiagnosed and underreported, or whether it is currently over-diagnosed. Because childhood trauma is a factor in the development of DID, some doctors think it may be a variation of **post-traumatic stress disorder (PTSD)**. DID and PTSD are conditions where dissociation is a prominent mechanism. The female to male ratio for DID is about 9:1, but the reasons for the gender imbalance are unclear. Some have attributed the imbalance in reported cases to higher rates of abuse of female children; and some to the possibility that males with DID are underreported because they might be in prison for violent crimes.

The most distinctive feature of DID is the formation and emergence of alternate personality states, or “alters.” Patients with DID experience their alters as distinctive individuals possessing different names, histories, and personality traits. It is not unusual for DID patients to have alters of different genders, sexual orientations, ages, or nationalities. Some patients have been reported with alters that are not even human; alters have been animals, or even aliens from outer space. The average DID patient has between two and 10 alters, but some have been reported with over one hundred.

Causes and symptoms

The severe dissociation that characterizes patients with DID is currently understood to result from a set of causes:

- An innate ability to dissociate easily
- Repeated episodes of severe physical or sexual abuse in childhood
- The lack of a supportive or comforting person to counteract abusive relative(s)
- The influence of other relatives with dissociative symptoms or disorders

The relationship of dissociative disorders to childhood abuse has led to intense controversy and lawsuits concerning the accuracy of childhood memories. The brain’s storage, retrieval, and interpretation of childhood memories are still not fully understood.

The major dissociative symptoms experienced by DID patients are **amnesia**, depersonalization, derealization, and identity disturbances.

Amnesia

Amnesia in DID is marked by gaps in the patient’s memory for long periods of their past, in some cases, their entire childhood. Most DID patients have amnesia, or “lose time,” for periods when another personality is “out.” They may report finding items in their house that they can’t remember having purchased, finding notes written in different handwriting, or other evidence of unexplained activity.

Depersonalization

Depersonalization is a dissociative symptom in which the patient feels that his or her body is unreal, is changing, or is dissolving. Some DID patients experience depersonalization as feeling to be outside of their body, or as watching a movie of themselves.

Derealization

Derealization is a dissociative symptom in which the patient perceives the external environment as unreal. Patients may see walls, buildings, or other objects as changing in shape, size, or color. DID patients may fail to recognize relatives or close friends.

Identity disturbances

Identity disturbances in DID result from the patient’s having split off entire personality traits or characteristics as well as memories. When a stressful or traumatic experience triggers the reemergence of these dissociated parts, the patient switches—usually within seconds—into an alternate personality. Some patients have histories of erratic performance in school or in their jobs caused by the emergence of alternate personalities during examinations or other stressful situations. Patients vary with regard to their alters’ awareness of one another.

Diagnosis

The diagnosis of DID is complex and some physicians believe it is often missed, while others feel it is over-diagnosed. Patients have been known to have been treated under a variety of other psychiatric diagnoses for a long time before being re-diagnosed with DID. The average DID patient is in the mental health care system for six to seven years before being diagnosed as a person with DID. Many DID patients are misdiagnosed as depressed because the primary or “core” personality

is subdued and withdrawn, particularly in female patients. However, some core personalities, or alters, may genuinely be depressed, and may benefit from antidepressant medications. One reason misdiagnoses are common is because DID patients may truly meet the criteria for **panic disorder** or somatization disorder.

Misdiagnoses include schizophrenia, borderline personality disorder, and, as noted, somatization disorder and panic disorder. DID patients are often frightened by their dissociative experiences, which can include losing awareness of hours or even days of time, meeting people who claim to know them by another name, or feeling “out of body.” Persons with the disorder may go to emergency rooms or clinics because they fear they are going insane.

When a doctor is evaluating a patient for DID, he or she will first rule out physical conditions that sometimes produce amnesia, depersonalization, or derealization. These conditions include head injuries; brain disease, especially seizure disorders; side effects from medications; **substance abuse** or intoxication; **AIDS dementia** complex; or recent periods of extreme physical **stress** and sleeplessness. In some cases, the doctor may give the patient an electroencephalograph (EEG) to exclude epilepsy or other seizure disorders. The physician also must consider whether the patient is **malinger**ing and/or offering fictitious complaints.

If the patient appears to be physically normal, the doctor will next rule out psychotic disturbances, including schizophrenia. Many patients with DID are misdiagnosed as schizophrenic because they may “hear” their alters “talking” inside their heads. If the doctor suspects DID, he or she can use a screening test called the Dissociative Experiences Scale (DES). If the patient has a high score on this test, he or she can be evaluated further with the Dissociative Disorders Interview Schedule (DDIS) or the Structured Clinical Interview for *DSM-IV* Dissociative Disorders (SCID-D). The doctor may also use the Hypnotic Induction Profile (HIP) or a similar test of the patient’s hypnotizability.

Treatment

Treatment of DID may last for five to seven years in adults and usually requires several different treatment methods.

Psychotherapy

Ideally, patients with DID should be treated by a therapist with specialized training in dissociation. This specialized training is important because the patient’s personality switches can be confusing or startling. In

addition, many patients with DID have hostile or suicidal alter personalities. Most therapists who treat DID patients have rules or contracts for treatment that include such issues as the patient’s responsibility for his or her safety. Psychotherapy for DID patients typically has several stages: an initial phase for uncovering and “mapping” the patient’s alters; a phase of treating the traumatic memories and “fusing” the alters; and a phase of consolidating the patient’s newly integrated personality.

Most therapists who treat multiples, or DID patients, recommend further treatment after personality integration, on the grounds that the patient has not learned the social skills that most people acquire in adolescence and early adult life. In addition, **family therapy** is often recommended to help the patient’s family understand DID and the changes that occur during personality reintegration.

Many DID patients are helped by group as well as individual treatment, provided that the group is limited to people with dissociative disorders. DID patients sometimes have setbacks in mixed therapy groups because other patients are bothered or frightened by their personality switches.

Medications

Some doctors will prescribe tranquilizers or antidepressants for DID patients because their alter personalities may have **anxiety** or **mood disorders**. However, other therapists who treat DID patients prefer to keep medications to a minimum because these patients can easily become psychologically dependent on drugs. In addition, many DID patients have at least one alter who abuses drugs or alcohol, substances which are dangerous in combination with most tranquilizers.

Hypnosis

While not always necessary, hypnosis is a standard method of treatment for DID patients. Hypnosis may help patients recover repressed ideas and memories. Further, hypnosis can also be used to control problematic behaviors that many DID patients exhibit, such as **self-mutilation**, or eating disorders like **bulimia nervosa**. In the later stages of treatment, the therapist may use hypnosis to “fuse” the alters as part of the patient’s personality integration process.

Alternative treatment

Alternative treatments that help to relax the body are often recommended for DID patients as an adjunct to psychotherapy and/or medication. These

KEY TERMS

Alter—An alternate or secondary personality in a patient with DID.

Amnesia—A general medical term for loss of memory that is not due to ordinary forgetfulness. Amnesia can be caused by head injuries, brain disease, or epilepsy as well as by dissociation.

Depersonalization—A dissociative symptom in which the patient feels that his or her body is unreal, is changing, or is dissolving.

Derealization—A dissociative symptom in which the external environment is perceived as unreal.

Dissociation—A psychological mechanism that allows the mind to split off traumatic memories or disturbing ideas from conscious awareness.

Dissociative identity disorder (DID)—Term that replaced Multiple Personality Disorder (MPD). A condition in which two or more distinctive identities or personality states alternate in controlling a person's consciousness and behavior.

Hypnosis—An induced trance state used to treat the amnesia and identity disturbances that occur in dissociative identity disorder (DID).

Multiple personality disorder (MPD)—The former, though often still used, term for dissociative identity disorder (DID).

Primary personality—The core personality of an DID patient. In women, the primary personality is often timid and passive, and may be diagnosed as depressed.

Trauma—A disastrous or life-threatening event that can cause severe emotional distress. DID is associated with trauma in a person's early life or adult experience.

treatments include **hydrotherapy**, botanical medicine (primarily herbs that help the nervous system), therapeutic massage, and **yoga**. Homeopathic treatment can also be effective for some people. **Art therapy** and the keeping of journals are often recommended as ways that patients can integrate their past into their present life. **Meditation** is usually discouraged until the patient's personality has been reintegrated.

Prognosis

Some therapists believe that the prognosis for recovery is excellent for children and good for most

adults. Although treatment takes several years, it is often ultimately effective. As a general rule, the earlier the patient is diagnosed and properly treated, the better the prognosis.

Prevention

Prevention of DID requires intervention in abusive families and treating children with dissociative symptoms as early as possible.

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Rebecca J. Frey, PhD

Multiple pregnancy

Definition

Multiple **pregnancy** is a pregnancy where more than one fetus develops simultaneously in the womb.

Description

Twins happen naturally about one in every 100 births. There are two types of twinning—identical and fraternal. Identical twins represent the splitting of a single fertilized zygote (union of two gametes or male/female sex cells that produce a developing fetus) into two separate individuals. They usually, but not always, have identical genes. When they do not separate completely, the result is Siamese (or conjoined) twins. Fraternal twins are three times as common as identical twins. They occur when two eggs are fertilized by separate sperm. Each has a different selection of its parents' genes. The natural incidence of multiple pregnancy has been upset by advances in fertility treatments, resulting in higher rates of multiple births in the United States. All these children are fraternal; they each arose from a separate egg and a separate sperm. Cloning produces identical twins.

The human female is designed to release one egg every menstrual cycle. A hormone called progesterone, released by the first egg to be produced, prevents any other egg from maturing during that cycle. When this control fails, fertilization of more than one egg is

LOUIS GERALD KEITH (1935–)

Louis Gerald Keith and his twin brother, Donald, were born on April 24, 1935, to Russian immigrants. Although the boys received much attention due to their status as twins, their parents encouraged them to pursue their own goals, and at age twelve, Louis decided he wanted to become a doctor. After completing his bachelor of science degree at the University of Illinois, Keith entered the Chicago Medical school, graduating with his degree in 1960. He joined the U.S. Public Health Service and was stationed in Puerto Rico after completing his residency and internship at Cook County Hospital. Keith returned to Chicago after receiving his certification in obstetrics and gynecology in 1967 and worked as a professor and a physician.

Keith began his research on twins when a friend's twin brother died of lymphoma. The fact that Keith was a twin made his research more personally fulfilling and worthwhile to him. Keith and Donald founded the Center for the Study of Multiple Birth in 1977 located in Chicago. The center, a non-profit facility, was the first multiple birth research organization in the United States. Keith has delivered many speeches and has published various study results. In addition to books and scientific articles written individually and cooperatively, he was co-author of *Multiple Pregnancy: Epidemiology, Gestation and Prenatal Outcome* (1996) with his brother Donald, and others. He is currently the chief obstetrician at the Prentice Women's Hospital and Maternity Center in Chicago.

possible. Fertility drugs inhibit these controls, allowing multiple pregnancy to occur. Multiple pregnancy is more difficult and poses more health risks than single pregnancy. Premature birth is greater with each additional fetus.

The problem with multiple births is that there is only so much room in even the most accommodating womb (uterus). Babies need to reach a certain size and gestational age before they can survive outside the uterus. **Prematurity** is the constant threat of multiple pregnancies. Twins have five times the **death** rate of single births. Triplets and higher die even more often. The principle threat of prematurity is that the lungs are not fully developed. A disease called hyaline membrane disease afflicts premature infants. Their lungs do not stay open after their first breath because they lack a chemical called **surfactant**. Survival of premature infants was greatly improved when surfactant was finally synthesized in a form that could be of benefit to premature babies. Tiny babies also have trouble regulating their body temperature.

KEY TERMS

Amniotic membranes—A thin membrane surrounding the fetus and containing serous fluid.

Genes—Hereditary determinants of identifying characteristics.

Gestational—Refers to pregnancy.

Ovulate—To release a mature egg for fertilization.

Zygote—The earliest stage of a fertilized egg.

Causes and symptoms

Fertility drugs prevent the normal process of single ovulation by permitting more than one egg at a time to mature and ovulate (move from the ovary to the uterus in anticipation of fertilization). This happens naturally to produce fraternal twins. The first drug to accomplish this was clomiphene. Subsequently, two natural hormones—follicle stimulating hormone and chorionic gonadotrophin—were developed and used.

Diagnosis

Multiple pregnancies cause the uterus to grow faster than usual. Obstetricians can detect this unusually rapid growth as the pregnancy progresses. Before birth, an ultrasound will also detect multiple babies in the uterus. After birth, physical appearance or a careful examination of the placenta and amniotic membranes will usually reveal whether the babies were in the same water bag or separate ones. One bag means identical twins.

A multiple pregnancy almost always means increased monitoring and surveillance for complications. This often means more frequent visits to the healthcare provider, serial ultrasounds to make sure that the babies are growing adequately, **amniocentesis** to check for lung development, and close monitoring for preterm labor.

Treatment

Mothers may be put on bedrest during a multiple pregnancy, in order to try to prevent pre-term labor and delivery. If preterm labor is impossible to control at home, the mother may be hospitalized, and medication may be used to attempt to control contractions and dilatation of the cervix. Multiple pregnancies more often end in cesarean deliveries than singleton pregnancies.

Babies from multiple pregnancies are often born early, and may require longer-than-normal hospitalization. The babies may need assistance with breathing, careful control of body temperature within an



An ultrasound image of identical twin male fetuses. The distortion is due to "twin B" being closer to the monitor. (Courtesy of Melissa Walsh Doig.)

incubator, and surveillance for other complications that frequently beset pre-term babies. While premature babies are fragile in many other ways, modern methods of intensive care have successfully stabilized babies as small as one pound.

Alternative treatment

There are no specific treatments to alleviate medical difficulties caused by multiple pregnancies, however there are supportive measures that may help both mother and children recover from the birthing process. There are treatments to encourage breast milk production and to combat postpartum difficulties. Various homeopathic remedies and massage can be helpful to both mother and children during the early adjustment period after birth.

Prognosis

With modern medical advances and excellent prenatal care, many multiple pregnancies reach fruition without difficulties. If the babies are born prematurely, immediate medical care increases the chance of survival without any complications.

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Rosalyn Carson-DeWitt, MD

Multiple sclerosis

Definition

Multiple sclerosis (MS) is a chronic autoimmune disorder affecting movement, sensation, and bodily functions. It is caused by destruction of the myelin insulation covering nerve fibers (neurons) in the central nervous system (brain and spinal cord).

Description

MS is a nerve disorder caused by destruction of the insulating layer surrounding neurons in the brain and spinal cord. This insulation, called myelin, helps electrical signals pass quickly and smoothly between the brain and the rest of the body. When the myelin is destroyed, nerve messages are sent more slowly and less efficiently. Patches of scar tissue, called plaques, form over the affected areas, further disrupting nerve communication. The symptoms of MS occur when the brain and spinal cord nerves no longer communicate properly with other parts of the body. MS causes a wide variety of symptoms and can affect vision, balance, strength, sensation, coordination, and bodily functions.

Multiple sclerosis affects more than a quarter of a million people in the United States. Most people have their first symptoms between the ages of 20 and 40; symptoms rarely begin before 15 or after 60. Women are almost twice as likely to get MS as men, especially in their early years. People of northern European heritage are more likely to be affected than people of other racial backgrounds, and MS rates are higher in the United States, Canada, and Northern Europe than in other parts of the world. MS is very rare among Asians, North and South American Indians, and Eskimos.

Causes and symptoms

Causes

Multiple sclerosis is an autoimmune disease, meaning its cause is an attack by the body's own immune system. For unknown reasons, immune cells attack and destroy the myelin sheath that insulates neurons in the brain and spinal cord. This myelin sheath, created by other brain cells called glia, speeds transmission and prevents electrical activity in one cell from short-circuiting to another cell. Disruption of communication between the brain and other parts of the body prevent normal passage of sensations and control messages, leading to the symptoms of MS.

The demyelinated areas appear as plaques, small round areas of gray neuron without the white myelin covering. The progression of symptoms in MS is correlated with development of new plaques in the portion of the brain or spinal cord controlling the affected areas. Because there appears to be no pattern in the appearance of new plaques, the progression of MS can be unpredictable.

Despite considerable research, the trigger for this autoimmune destruction is still unknown. At various times, evidence has pointed to genes, environmental factors, viruses, or a combination of these.

The risk of developing MS is higher if another family member is affected, suggesting the influence of genetic factors. In addition, the higher prevalence of MS among people of northern European background suggests some genetic susceptibility.

The role of an environmental factor is suggested by studies of the effect of migration on the risk of developing MS. Age plays an important role in determining this change in risk—young people in low-risk groups who move into countries with higher MS rates display the risk rates of their new surroundings, while older migrants retain the risk of their original home country. One interpretation of these studies is that an environmental factor, either protective or harmful, is acquired in early life; the risk of disease later in life reflects the effects of the early environment.

These same data can be used to support the involvement of a slow-acting virus, one that is acquired early on but begins its destructive effects much later. Slow viruses are known to cause other diseases, including **AIDS**. In addition, viruses have been implicated in other autoimmune diseases. Many claims have been made for the role of viruses, slow or otherwise, as the trigger for MS, but as of 2001 no strong candidate has emerged.

How a virus could trigger the autoimmune reaction is also unclear. There are two main models of virally induced autoimmunity. The first suggests the immune system is actually attacking a virus (one too well-hidden for detection in the laboratory), and the myelin damage is an unintentional consequence of fighting the infection. The second model suggests the immune system mistakes myelin for a viral protein, one it encountered during a prior infection. Primed for the attack, it destroys myelin because it resembles the previously-recognized viral invader.

Either of these models allows a role for genetic factors, since certain genes can increase the likelihood of autoimmunity. Environmental factors as well might change the sensitivity of the immune system or interact

with myelin to provide the trigger for the secondary immune response. Possible environmental triggers that have been invoked in MS include viral infection, trauma, electrical injury, and chemical exposure, although controlled studies do not support a causative role.

Symptoms

The symptoms of multiple sclerosis may occur in one of three patterns:

- The most common pattern is the “relapsing-remitting” pattern, in which there are clearly defined symptomatic attacks lasting 24 hours or more, followed by complete or almost complete improvement. The period between attacks may be a year or more at the beginning of the disease, but may shrink to several months later on. This pattern is especially common in younger people who develop MS.
- In the “primary progressive” pattern, the disease progresses without remission or with occasional plateaus or slight improvements. This pattern is more common in older people.
- In the “secondary progressive” pattern, the person with MS begins with relapses and remissions, followed by more steady progression of symptoms.

Between 10–20% of people have a benign type of MS, meaning their symptoms progress very little over the course of their lives.

Because plaques may form in any part of the central nervous system, the symptoms of MS vary widely from person-to-person and from stage-to-stage of the disease. Initial symptoms often include:

- Muscle weakness, causing difficulty walking
- Loss of coordination or balance
- **Numbness**, “pins and needles,” or other abnormal sensations
- Visual disturbances, including blurred or double vision.

Later symptoms may include:

- **Fatigue**
- Muscle spasticity and stiffness
- Tremors
- Paralysis
- **Pain**
- Vertigo
- Speech or swallowing difficulty
- Loss of bowel and bladder control
- Incontinence, **constipation**

- Sexual dysfunction
- Cognitive changes.

Weakness in one or both legs is common, and may be the first symptom noticed by a person with MS. Muscle spasticity, or excessive tightness, is also common and may be more disabling than weakness.

Double vision or eye tremor (**nystagmus**) may result from involvement of the nerve pathways controlling movement of the eye muscles. Visual disturbances result from involvement of the optic nerves (optic neuritis) and may include development of blind spots in one or both eyes, changes in color vision, or blindness. **Optic neuritis** usually involves only one eye at a time and is often associated with movement of the effected eye.

More than half of all people affected by MS have pain during the course of their disease, and many experience chronic pain, including pain from spasticity. Acute pain occurs in about 10% of cases. This pain may be a sharp, stabbing pain especially in the face, neck, or down the back. Facial numbness and weakness are also common.

Cognitive changes, including memory disturbances, depression, and personality changes, are found in people affected by MS, though it is not entirely clear whether these changes are due primarily to the disease or to the psychological reaction to it. Depression may be severe enough to require treatment in up to 25% of those with MS. A smaller number of people experience disease-related euphoria, or abnormally elevated mood, usually after a long disease duration and in combination with other psychological changes.

Symptoms of MS may be worsened by heat or increased body temperature, including **fever**, intense physical activity, or exposure to sun, hot baths, or showers.

Diagnosis

There is no single test that confirms the diagnosis of multiple sclerosis, and there are a number of other diseases with similar symptoms. While one person’s diagnosis may be immediately suggested by her symptoms and history, another’s may not be confirmed without multiple tests and prolonged observation. The distribution of symptoms is important: MS affects multiple areas of the body over time. The pattern of symptoms is also critical, especially evidence of the relapsing-remitting pattern, so a detailed medical history is one of the most important parts of the diagnostic process. A thorough search to exclude other causes of a patient’s symptoms is especially important if the following features are present: 1) family history of

neurologic disease, 2) symptoms and findings attributable to a single anatomic location, 3) persistent back pain, 4) age of onset over 60 or under 15 years of age, or 5) progressively worsening disease.

In addition to the medical history and a standard neurological exam, several lab tests are used to help confirm or rule out a diagnosis of MS:

- Magnetic resonance imaging (MRI) can reveal plaques on the brain and spinal cord. Gadolinium enhancement can distinguish between old and new plaques, allowing a correlation of new plaques with new symptoms. Plaques may be seen in several other diseases as well, including encephalomyelitis, neurosarcoidosis, and cerebral lupus. Plaques on MRI may be difficult to distinguish from small strokes, areas of decreased blood flow, or changes seen with trauma or normal **aging**.
- A lumbar puncture, or spinal tap, is done to measure levels of immune proteins, which are usually elevated in the cerebrospinal fluid of a person with MS. This test may not be necessary if other tests are diagnostic.
- Evoked potential tests, electrical tests of conduction speed in the nerves, can reveal reduced speeds consistent with the damage caused by plaques. These tests may be done with small electrical charges applied to the skin (somatosensory evoked potential), with light patterns flashed on the eyes (visual evoked potential), or with sounds presented to the ears (auditory evoked potential).

The clinician making the diagnosis, usually a neurologist, may classify the disease as “definite MS,” meaning the symptoms and test results all point toward MS as the cause. “Probable MS” and “possible MS” reflect less certainty and may require more time to pass to observe the progression of the disease and the distribution of symptoms.

Treatment

The three major drugs previously approved for the treatment of MS affect the course of the disease. None of these drugs is a cure, but they can slow disease progression in many patients.

Known as the ABC drugs, Avonex and Betaseron are forms of the immune system protein beta interferon, while Copaxone is glatiramer acetate (formerly called copolymer-1). All three have been shown to reduce the rate of relapses in the relapsing-remitting form of MS. Different measurements from tests of each have demonstrated other benefits as well: Avonex may slow the progress of physical impairment, Betaseron may reduce the severity of symptoms,

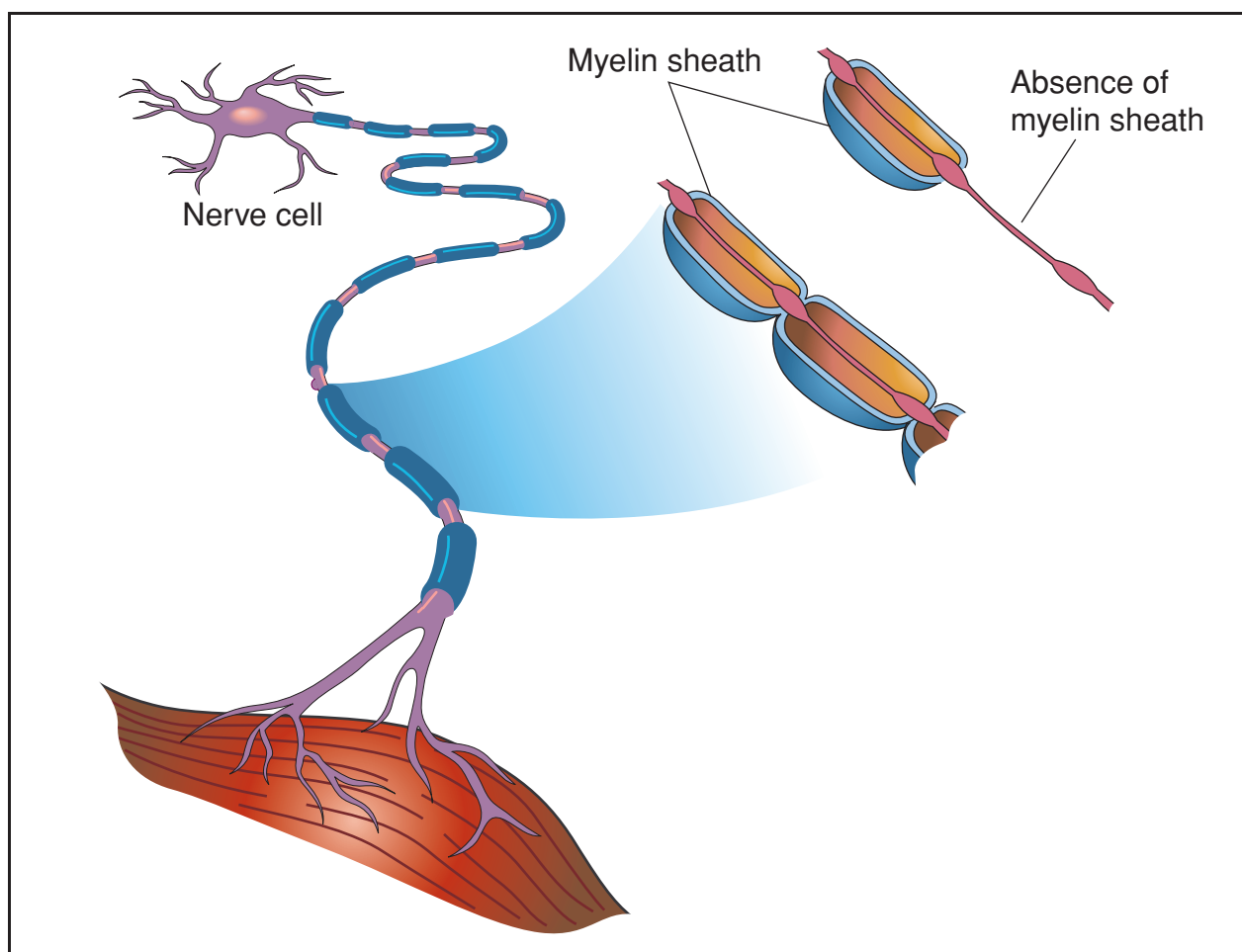
and Copaxone may decrease disability. All three drugs are administered by injection

Two major clinical studies were recently completed that focused on the question of whether disease-modifying therapy known to slow the disease, can postpone the development of clinically definitive MS in high risk patients. Data presented at the annual meeting of the American Academy of Neurology in May, 2000, highlighted the different effects of interferon therapy when it was initiated at the earliest recognizable stages of MS versus later. Previous studies with interferon beta-1b (Betaseron) and interferon beta-1a (Avonex, Rebif) clearly demonstrated benefits in patients with relapsing forms of MS. Moreover, previous treatment with High-dose **corticosteroids** also delays, but does not prevent the ultimate development of MS. The encouraging message from the CHAMPS study in the United states and the ETOMS study in Europe is that early intervention can reduce the probability of developing clinically definitive MS.

Although the ABC drugs stop relapses and may keep patients in relatively good health for the short-term, their long-term success has not been proven and they don't work well for patients who have reached a steadily progressive stage of MS. In the meantime, new approaches to using current therapies are being researched especially using combinations of different types of agents when one agent alone is not effective. Clinical trials are now evaluating the safety and efficacy of combining cyclophosphamide (Cytosan) and methylprednisolone (Medrol) in patients who do not respond to the ABC drugs, and of adding mitoxantrone (Novantrone), prednisone (Prelone), azathioprine (Imuran), or methotrexate (Rheumatrex) to beta-interferon for further benefit.

In addition, Mitoxantrone HCl (novantrone), a drug approved for **cancer** treatment, has been approved for treating patients with advanced or chronic multiple sclerosis. In clinical trials, mitoxantrone reduced the number of relapse episodes and slowed down the disease. Reserved for progressive forms of MS, it is given intravenously by a doctor to help maintain mobility and reduce the number of flare-ups. However, there are serious side effects with the drug including heart problems, **nausea**, and hair thinning.

As reported in the Spring, 2001, Volume 19, No 2 issue of InsideMS, the FDA recently approved the Copaxone Autoject and the Mixject vial adapters to help people using Copaxone self administer the drug. The autoject keeps the syringe steady and hides the needle. The same syringe may be used for both mixing



Multiple sclerosis (MS) is an autoimmune disease in which immune cells attack and destroy the myelin sheath which stimulates neurons in the brain and spinal cord. When the myelin is destroyed, nerve messages are sent more slowly and less efficiently. Scar tissue then forms over the affected areas, disrupting nerve communication. MS symptoms occur when the brain and spinal cord nerves cease to communicate properly with other parts of the body. (Illustration by Electronic Illustrators Group.)

and injecting with the Mixject vial adapters. A similar device is available for patients using Betaserson. Some patients are using the needlefree Biojector 2000 which uses a CO₂ cartridge to deliver doses of medication through the skin. The FDA has not approved its use and patients should discuss this with their physician for its use with either Copaxone or Betaseron. Avonex must be injected in the muscle.

Immunosuppressant drugs have been used for many years to treat acute exacerbations (relapses). Drugs used include corticosteroids such as prednisone and methylprednisone; the hormone adrenocorticotrophic hormone (ACTH); and azathioprine. Recent studies indicate that several days of intravenous methylprednisone may be more effective than other immunosuppressant treatments for acute symptoms. This treatment may require hospitalization.

MS causes a large variety of symptoms, and the treatments for these are equally diverse. Most symptoms can be treated and complications avoided with good care and attention from medical professionals. Good health and **nutrition** remain important preventive measures. **Vaccination** against **influenza** can prevent respiratory complications, and contrary to earlier concerns, is not associated with worsening of symptoms. Preventing complications such as **pneumonia**, bed sores, injuries from falls, or urinary infection requires attention to the primary problems which may cause them. Shortened life spans with MS are almost always due to complications rather than primary symptoms themselves.

Physical therapy helps the person with MS to strengthen and retrain affected muscles; to maintain

range of motion to prevent muscle stiffening; to learn to use assistive devices such as canes and walkers; and to learn safer and more energy-efficient ways of moving, sitting, and transferring. **Exercise** and stretching programs are usually designed by the physical therapist and taught to the patient and caregivers for use at home. Exercise is an important part of maintaining function for the person with MS. Swimming is often recommended, not only for its low-impact workout, but also because it allows strenuous activity without overheating.

Occupational therapy helps the person with MS adapt to her environment and adapt the environment to her. The occupational therapist suggests alternate strategies and assistive devices for activities of daily living, such as dressing, feeding, and washing, and evaluates the home and work environment for safety and efficiency improvements that may be made.

Training in bowel and bladder care may be needed to prevent or compensate for incontinence. If the urge to urinate becomes great before the bladder is full, some drugs may be helpful, including propantheline bromide (Probanthine), oxybutynin chloride (Ditropan), or imipramine (Tofranil). Baclofen (Lioresal) may relax the sphincter muscle, allowing full emptying. Intermittent catheterization is effective in controlling bladder dysfunction. In this technique, a catheter is used to periodically empty the bladder.

Spasticity can be treated with oral medications, including baclofen and diazepam (Valium), or by injection with botulinum toxin (Botox). Spasticity relief may also bring relief from chronic pain. Other more acute types of pain may respond to carbamazepine (Tegretol) or diphenylhydantoin (Dilantin). **Low back pain** is common from increased use of the back muscles to compensate for weakened legs. Physical therapy and over-the-counter pain relievers may help.

Fatigue may be partially avoidable with changes in the daily routine to allow more frequent rests. Amantadine (Symmetrel) and pemoline (Cylert) may improve alertness and lessen fatigue. Visual disturbances often respond to corticosteroids. Other symptoms that may be treated with drugs include seizures, vertigo, and tremor.

Myloral, an oral preparation of bovine myelin, has recently been tested in clinical trials for its effectiveness in reducing the frequency and severity of relapses. Preliminary data indicate no difference between it and placebo.

Alternative treatment

Bee venom has been suggested as a treatment for MS, but no studies or objective reports support this claim.

In British studies, **marijuana** has been shown to have variable effects on the symptoms of MS. Improvements have been documented for tremor, pain, and spasticity, and worsening for posture and balance. Side effects have included weakness, dizziness, relaxation, and incoordination, as well as euphoria. As a result, marijuana is not recommended as an alternative treatment.

Some studies support the value of high doses of **vitamins, minerals**, and other dietary supplements for controlling disease progression or improving symptoms. Alpha-linoleic and linoleic acids, as well as selenium and vitamin E, have shown effectiveness in the treatment of MS. The selenium and vitamin E act as antioxidants. In addition, the Swank diet (low in saturated fats), maintained over a long period of time, may retard the disease process.

Removal of mercury fillings has been touted as a possible cure, but is of no proven benefit.

Studies have also shown that t'ai chi can be an effective therapy for MS because it works to improve balance and increase strength.

There are conflicting views about **Echinacea** and its benefit to MS. Some medicine books recommend Echinacea for people with MS. However, Echinacea appears to stimulate different parts of the immune system, particularly immune cells known as macrophages. In MS these cells are very active already and further stimulation could worsen the disease.

Prognosis

It is difficult to predict how multiple sclerosis will progress in any one person. Most people with MS will be able to continue to walk and function at their work for many years after their diagnosis. The factors associated with the mildest course of MS are being female, having the relapsing-remitting form, having the first symptoms at a younger age, having longer periods of remission between relapses, and initial symptoms of decreased sensation or vision rather than of weakness or incoordination.

Less than 5% of people with MS have a severe progressive form, leading to **death** from complications within five years. At the other extreme, 10–20% have a benign form, with a very slow or no progression of their symptoms. The most recent studies show that

KEY TERMS

Clinical trial—All new drugs undergo clinical trials before approval. Clinical trials are carefully conducted tests in which effectiveness and side effects are studied, with the placebo effect eliminated.

Evoked potentials—Tests that measure the brain's electrical response to stimulation of sensory organs (eyes or ears) or peripheral nerves (skin). These tests may help confirm the diagnosis of multiple sclerosis.

Myelin—A layer of insulation that surrounds the nerve fibers in the brain and spinal cord.

Plaque—Patches of scar tissue that form where the layer of myelin covering the nerve fibers is destroyed by the multiple sclerosis disease process.

Primary progressive—A pattern of symptoms of multiple sclerosis in which the disease progresses without remission, or with occasional plateaus or slight improvements.

Relapsing-remitting—A pattern of symptoms of multiple sclerosis in which symptomatic attacks occur that last 24 hours or more, followed by complete or almost complete improvement.

Secondary progressive—A pattern of symptoms of multiple sclerosis in which there are relapses and remissions, followed by more steady progression of symptoms.

about seven out of 10 people with MS are still alive 25 years after their diagnosis, compared to about nine out of 10 people of similar age without disease. On average, MS shortens the lives of affected women by about six years, and men by 11 years. **Suicide** is a significant cause of death in MS, especially in younger patients.

The degree of disability a person experiences five years after onset is, on average, about three-quarters of the expected disability at 10–15 years. A benign course for the first five years usually indicates the disease will not cause marked disability.

Prevention

There is no known way to prevent multiple sclerosis. Until the cause of the disease is discovered, this is unlikely to change. Good nutrition; adequate rest; avoidance of **stress**, heat, and extreme physical exertion; and good bladder hygiene may improve quality of life and reduce symptoms.

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Ruthan Brodsky

Mumps

Definition

Mumps is a relatively mild short-term viral infection of the salivary glands that usually occurs during childhood. Typically, mumps is characterized by a painful swelling of both cheek areas, although the person could have swelling on one side or no perceivable swelling at all. The salivary glands are also called the parotid glands, therefore, mumps is sometimes referred to as an inflammation of the parotid glands (epidemic parotitis). The word mumps comes from an old English dialect, meaning lumps or bumps within the cheeks.

Description

Mumps is a very contagious infection that spreads easily in such highly populated areas as day care centers and schools. Although not as contagious as **measles** or **chickenpox**, mumps was once quite common. Prior to the release of a mumps vaccine in the United States in 1967, approximately 92% of all children had been exposed to mumps by the age of 15. In these pre-vaccine years, most children contracted mumps between the ages of four and seven. Mumps epidemics came in two to five year cycles. The greatest mumps epidemic was in 1941 when approximately 250 cases were reported for every 100,000 people. In 1968, the year after the live mumps vaccine was released, only 76 cases were reported for every 100,000 people. By 1985, less than 3,000 cases of mumps were reported throughout the entire United States, which works out to about 1 case per 100,000 people. The reason for the decline in mumps was the increased usage of the mumps vaccine. However, 1987 noted a five-fold increase in the incidence of the disease because of the reluctance of some states to adopt comprehensive school immunization laws. Since then, state-enforced school entry requirements have achieved student immunization rates of nearly 100% in kindergarten and first grade. In 1996, the Centers for Disease

Control and Prevention (CDC) reported only 751 cases of mumps nationwide, or, in other words, about one case for every five million people.

Causes and symptoms

The paramyxovirus that causes mumps is harbored in the saliva and is spread by sneezing, coughing, and other direct contact with another person's infected saliva. Once the person is exposed to the virus, symptoms generally occur in 14-24 days. Initial symptoms include chills, **headache**, loss of appetite, and a lack of energy. However, an infected person may not experience these initial symptoms. Swelling of the salivary glands in the face (parotitis) generally occurs within 12-24 hours of the above symptoms. Accompanying the swollen glands is **pain** on chewing or swallowing, especially with acidic beverages, such as lemonade. A **fever** as high as 104°F (40°C) is also common. Swelling of the glands reaches a maximum on about the second day and usually disappears by the seventh day. Once a person has contracted mumps, they become immune to the disease, despite how mild or severe their symptoms may have been.

While the majority of cases of mumps are uncomplicated and pass without incident, some complications can occur. Complications are, however, more noticeable in adults who get the infection. In 15% of cases, the covering of the brain and spinal cord becomes inflamed (**meningitis**). Symptoms of meningitis usually develop within four or five days after the first signs of mumps. These symptoms include a stiff neck, headache, **vomiting**, and a lack of energy. Mumps meningitis is usually resolved within seven days, and damage to the brain is exceedingly rare.

The mumps infection can spread into the brain causing inflammation of the brain (**encephalitis**). Symptoms of mumps encephalitis include the inability to feel pain, seizures, and high fever. Encephalitis can occur during the parotitis stage or one to two weeks later. Recovery from mumps encephalitis is usually complete, although complications, such as seizure disorders, have been noted. Only about 1 in 100 with mumps encephalitis dies from the complication.

About one-quarter of all post-pubertal males who contract mumps can develop a swelling of the scrotum (**orchitis**) about seven days after the parotitis stage. Symptoms include marked swelling of one or both testicles, severe pain, fever, **nausea**, and headache. Pain and swelling usually subside after five to seven days, although the testicles can remain tender for weeks.

Girls occasionally suffer an inflammation of the ovaries, or oophoritis, as a complication of mumps, but this condition is far less painful than orchitis in boys.

As of late 2002, some researchers in Europe are studying the possibility that mumps increases a person's risk of developing inflammatory bowel disease (IBD) in later life. This hypothesis will require further research, as present findings are inconclusive.

Diagnosis

When mumps reaches epidemic proportions, diagnosis is relatively easy on the basis of the physical symptoms. The doctor will take the child's temperature, gently palpate (touch) the skin over the parotid glands, and look inside the child's mouth. If the child has mumps, the openings to the ducts inside the mouth will be slightly inflamed and have a "pouty" appearance. With so many people vaccinated today, a case of mumps must be properly diagnosed in the event the salivary glands are swollen for reasons other than viral infection. For example, in persons with poor **oral hygiene**, the salivary glands can be infected with bacteria. In these cases, **antibiotics** are necessary. Also in rare cases, the salivary glands can become blocked, develop tumors, or swell due to the use of certain drugs, such as iodine. A test can be performed to determine whether the person with swelling of the salivary glands actually has the mumps virus.

As of late 2002, researchers in London have reported the development of a bioassay for measuring mumps-specific IgG. This test would allow a doctor to check whether an individual patient is immune to mumps, and allow researchers to measure the susceptibility of a local population to mumps in areas with low rates of **vaccination**.

Treatment

When mumps does occur, the illness is usually allowed to run its course. The symptoms, however, are treatable. Because of difficulty swallowing, the most important challenge is to keep the patient fed and hydrated. The individual should be provided a soft diet, consisting of cooked cereals, mashed potatoes, broth-based soups, prepared baby foods, or foods put through a home food processor. **Aspirin**, **acetaminophen**, or ibuprofen can relieve some of the pain due to swelling, headache, and fever. Avoid fruit juices and other acidic foods or beverages that can irritate the salivary glands. Avoid dairy products that can be hard to digest. In the event of complications, a physician



A young child with mumps. (Photo Researchers. Reproduced by permission.)

should be contacted at once. For example, if orchitis occurs, a physician should be called. Also, supporting the scrotum in a cotton bed on an adhesive-tape bridge between the thighs can minimize tension. Ice packs are also helpful.

Alternative treatment

Acupressure can be used effectively to relieve pain caused by swollen glands. The patient can, by using the middle fingers, gently press the area between the jawbone and the ear for two minutes while breathing deeply.

A number of homeopathic remedies can be used for the treatment of mumps. For example, belladonna may be useful for flushing, redness, and swelling. Bryonia (wild hops) may be useful for irritability, lack of energy, or thirst. Phytolacca (poke root) may be prescribed for extremely swollen glands. A homeopathic physician should always be consulted for appropriate doses for children, and remedies that do not work within one day should be stopped. A homeopathic preparation of the mumps virus can also be used prophylactically or as a treatment for the disease.

Several herbal remedies may be useful in helping the body recover from the infection or may help alleviate the discomfort associated with the disease. **Echinacea** (*Echinacea* spp.) can be used to boost the immune system and help the body fight the infection. Other herbs taken internally, such as cleavers (*Galium aparine*), calendula (*Calendula officinalis*), and phytolacca (poke root), target the lymphatic system and may help to enhance the activity of the body's internal filtration system. Since phytolacca can be toxic, it should only be used by patients under the care of a skilled practitioner. Topical applications are also

useful in relieving the discomfort of mumps. A cloth dipped in a heated mixture of vinegar and cayenne (*Capsicum frutescens*) can be wrapped around the neck several times a day. Cleavers or calendula can also be combined with vinegar, heated, and applied in a similar manner.

Prognosis

When mumps is uncomplicated, prognosis is excellent. However, in rare cases, a relapse occurs after about two weeks. Complications can also delay complete recovery.

Prevention

A vaccine exists to protect against mumps. The vaccine preparation (MMR) is usually given as part of a combination injection that helps protect against measles, mumps, and **rubella**. MMR is a live vaccine administered in one dose between the ages of 12-15 months, 4-6 years, or 11-12 years. Persons who are unsure of their mumps history and/or mumps vaccination history should be vaccinated. Susceptible health care workers, especially those who work in hospitals, should be vaccinated. Because mumps is still prevalent throughout the world, susceptible persons over age one who are traveling abroad would benefit from receiving the mumps vaccine.

The mumps vaccine is extremely effective, and virtually everyone should be vaccinated against this disease. There are, however, a few reasons why people should *not* be vaccinated against mumps:

- Pregnant women who contract mumps during **pregnancy** have an increased rate of **miscarriage**, but not **birth defects**. As a result, pregnant women should not receive the mumps vaccine because of the possibility of damage to the fetus. Women who have had the vaccine should postpone pregnancy for three months after vaccination.
- Unvaccinated persons who have been exposed to mumps should not get the vaccine, as it may not provide protection. The person should, however, be vaccinated if no symptoms result from the exposure to mumps.
- Persons with minor fever-producing illnesses, such as an upper respiratory infection, should not get the vaccine until the illness has subsided.
- Because mumps vaccine is produced using eggs, individuals who develop **hives**, swelling of the mouth or throat, **dizziness**, or breathing difficulties after eating eggs should not receive the mumps vaccine.

KEY TERMS

Asymptomatic—Persons who carry a disease and may be capable of transmitting the disease but who do not exhibit symptoms of the disease are said to be asymptomatic.

Autism—A severe developmental disorder that usually begins before three years of age and affects a child's social as well as intellectual development. Some researchers theorized that immunization with the MMR vaccine was a risk factor for autism.

Encephalitis—Inflammation of the brain.

Epidemic parotitis—The medical name for mumps.

Immunoglobulin G (IgG)—A group of antibodies against certain viral infections that circulate in the bloodstream. One type of IgG is specific against the mumps paramyxovirus.

Meningitis—Inflammation of the membranes covering the brain and spinal cord.

Orchitis—Inflammation or swelling of the scrotal sac containing the testicles.

Paramyxovirus—A genus of viruses that includes the causative agent of mumps.

Parotitis—Inflammation and swelling of the salivary glands.

- Persons with immune deficiency diseases and/or those whose immunity has been suppressed with anti-cancer drugs, **corticosteroids**, or radiation should not receive the vaccine. Family members of immunocompromised people, however, should get vaccinated to reduce the risk of mumps.
- The CDC recommends that all children infected with human **immunodeficiency** disease (HIV) who are asymptomatic should receive the MMR vaccine at 15 months of age.

The mumps vaccine has been controversial in recent years because of concern that its use was linked to a rise in the rate of childhood **autism**. The negative publicity given to the vaccine in the mass media led some parents to refuse to immunize their children with the MMR vaccine. One result has been an increase in the number of mumps outbreaks in several European countries, including Italy and the United Kingdom.

In the fall of 2002, the *New England Journal of Medicine* published a major Danish study disproving

the hypothesis of a connection between the MMR vaccine and autism. A second study in Finland showed that the vaccine is not associated with aseptic meningitis or encephalitis as well as autism. Since these studies were published, American primary care physicians have once again reminded parents of the importance of immunizing their children against mumps and other childhood diseases.

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ORGANIZATIONS

- American Academy of Pediatrics (AAP). 141 Northwest Point Boulevard, Elk Grove Village, IL 60007. (847) 434-4000. <www.aap.org>.
- Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

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Munchausen syndrome

Definition

Munchausen syndrome is a psychiatric disorder that causes an individual to self-inflict injury or illness or to fabricate symptoms of physical or mental illness, in order to receive medical care or hospitalization. In a variation of the disorder, Munchausen by proxy (MSBP), an individual, typically a mother, intentionally causes or fabricates illness in a child or other person under her care.

Description

Munchausen syndrome takes its name from Baron Karl Friederich von Munchausen, an 18th century German military man known for his tall tales. The disorder first appeared in psychiatric literature in the early 1950s when it was used to describe patients who sought hospitalization by inventing symptoms and complicated medical histories, and/or inducing illness and injury in themselves. Categorized as a factitious disorder (a disorder in which the physical or psychological symptoms are under voluntary control), Munchausen's syndrome seems to be motivated by a need to assume the role of a patient. Unlike **malinger-ing**, there does not seem to be any clear secondary gain (e.g., money) in Munchausen syndrome.

Individuals with Munchausen by proxy syndrome use their child (or another dependent person) to fulfill their need to step into the patient role. The disorder most commonly victimizes children from birth to 8 years old. Parents with MSBP may only exaggerate or fabricate their child's symptoms, or they may deliberately induce symptoms through various methods, including **poisoning**, suffocation, **starvation**, or infecting the child's bloodstream.

Causes and symptoms

The exact cause of Munchausen syndrome is unknown. It has been theorized that Munchausen patients are motivated by a desire to be cared for, a need for attention, dependency, an ambivalence toward doctors, or a need to suffer. Factors that may predispose an individual to Munchausen's include a serious illness in childhood or an existing personality disorder.

The Munchausen patient presents a wide array of physical or psychiatric symptoms, usually limited only by their medical knowledge. Many Munchausen patients are very familiar with medical terminology

KEY TERMS

Apnea—A cessation of breathing.

Factitious disorder—A disorder in which the physical or psychological symptoms are under voluntary control.

and symptoms. Some common complaints include fevers, **rashes**, abscesses, bleeding, and **vomiting**. Common Munchausen by proxy symptoms include apnea (cessation of breathing), **fever**, vomiting, and **diarrhea**. In both Munchausen and MSBP syndromes, the suspected illness does not respond to a normal course of treatment. Patients or parents may push for invasive diagnostic procedures and display an extraordinary depth of knowledge of medical procedures.

Diagnosis

Because Munchausen sufferers often go from doctor to doctor, gaining admission into many hospitals along the way, diagnosis can be difficult. They are typically detected rather than diagnosed. During a course of treatment, they may be discovered by a hospital employee who encountered them during a previous hospitalization. Their caregivers may also notice that symptoms such as high fever occur only when the patient is left unattended. Occasionally, unprescribed medication used to induce symptoms is found with the patient's belongings. When the patient is confronted, they often react with outrage and check out of the hospital to seek treatment at another facility with a new caregiver.

Treatment

There is no clearly effective treatment for Munchausen syndrome. Extensive psychotherapy may be helpful with some Munchausen patients. If Munchausen syndrome co-exists with other mental disorders, such as a personality disorder, the underlying disorder is typically treated first.

Prognosis

The infections and injuries Munchausen patients self-inflict can cause serious illness. Patients often undergo countless unnecessary surgeries throughout their lifetimes. In addition, because of their frequent hospitalizations, they have difficulty holding down a

job. Further, their chronic health complaints may damage interpersonal relationships with family and friends. Children victimized by sufferers of MSBP are at a real risk for serious injury and possible **death**. Those who survive physically unscathed may suffer developmental problems later in life.

Prevention

Because the cause of Munchausen syndrome is unknown, formulating a prevention strategy is difficult. Some medical facilities and healthcare practitioners have attempted to limit hospital admissions for Munchausen patients by sharing medical records. While these attempts may curb the number of hospital admissions, they do not treat the underlying disorder and may endanger Munchausen sufferers that have made themselves critically ill and require treatment. Children who are found to be victims of persons with Munchausen by proxy syndrome should be immediately removed from the care of the abusing parent or guardian.

Resources

ORGANIZATIONS

- American Psychiatric Association. 1400 K Street NW, Washington DC 20005. (888) 357-7924. <<http://www.psych.org>>.
- American Psychological Association (APA). 750 First St. NE, Washington, DC 20002-4242. (202) 336-5700. <<http://www.apa.org>>.
- National Alliance for the Mentally Ill (NAMI). Colonial Place Three, 2107 Wilson Blvd., Ste. 300, Arlington, VA 22201-3042. (800) 950-6264. <<http://www.nami.org>>.
- National Institute of Mental Health. Mental Health Public Inquiries, 5600 Fishers Lane, Room 15C-05, Rockville, MD 20857. (888) 826-9438. <<http://www.nimh.nih.gov>>.

Paula Anne Ford-Martin

Mupirocin see **Antibiotics, topical**

Murine (endemic) typhus see **Typhus**

Muscle cramps see **Muscle spasms and cramps**

Muscle relaxants

Definition

Skeletal muscle relaxants are drugs that relax striated muscles (those that control the skeleton).

They are a separate class of drugs from the muscle relaxant drugs used during intubations and surgery to reduce the need for anesthesia and facilitate intubation.

Purpose

Skeletal muscle relaxants may be used for relief of spasticity in neuromuscular diseases, such as **multiple sclerosis**, as well as for **spinal cord injury** and **stroke**. They may also be used for **pain** relief in minor strain injuries and control of the muscle symptoms of **tetanus**. Dantrolene (Dantrium) has been used to prevent or treat malignant hyperthermia in surgery.

Description

Although the muscle relaxants may be divided into only two groups, centrally acting and peripherally acting, the centrally acting group, which appears to act on the central nervous system, contains 10 drugs which are chemically different, while only dantrolene has a direct action at the level of the nerve-muscle connection.

Baclofen (Lioresal) may be administered orally or intrathecally for control of spasticity due to neuromuscular disease.

Carisoprodol (Soma), chlorphenesin (Maolate), chlorzoxazone (Paraflex), cyclobenzaprine (Flexeril), diazepam (Valium), metaxalone (Skelaxin), methocarbamol (Robaxin), and orphenadrine (Norflex) are used primarily as an adjunct for rest in management of acute **muscle spasms** associated with **sprains**. Muscle relaxation may also be an adjunct to physical therapy in **rehabilitation** following stroke, spinal cord injury, or other musculoskeletal conditions.

Diazepam and methocarbamol are also used by injection for relief of tetanus.

Recommended dosage

Dose varies with the drug, route of administration, and purpose. There may be individual variations in absorption that require doses higher than those usually recommended, particularly with methocarbamol. Consult specific references for further information.

Precautions

All drugs in this class may cause **sedation**. Baclofen, when administered intrathecally, may cause severe central nervous system (CNS) depression with cardiovascular collapse and **respiratory failure**.

Diazepam may be addictive. It is a controlled substance under federal law.

Dantrolene has a potential for hepatotoxicity. The incidence of symptomatic hepatitis is dose related, but may occur even with a short period of doses at or above. Even short periods of doses at or above 800 mg per day greatly increases the risk of serious liver injury. Overt hepatitis has been most frequently observed between the third and twelfth months of therapy. Risk of hepatic injury appears to be greater in women, in patients over 35 years of age and in patients taking other medications in addition to dantrolene.

Tizanidine may cause low blood pressure, but this may be controlled by starting with a low dose and increasing it gradually. The drug may rarely cause liver damage.

Methocarbamol and chlorzoxazone may cause harmless color changes in urine—orange or reddish-purple with chlorzoxazone and purple, brown, or green with methocarbamol. The urine will return to its normal color when the patient stops taking the medicine.

Most drugs in this class are well tolerated.

Not all drugs in this group have been evaluated for safety in **pregnancy** and breast feeding.

Baclofen is pregnancy category C. It has caused fetal abnormalities in rats at doses 13 times above the human dose. Baclofen passes into breast milk, and breast feeding while taking baclofen is not recommended.

Diazepam is category D. All **benzodiazepines** cross the placenta. Although the drugs appear to be safe for use during the first trimester of pregnancy, use later in pregnancy may be associated with **cleft lip and palate**. Diazepam should not be taken while breast feeding. Infants who were breast fed while their mothers took diazepam were excessively sleepy and lethargic.

Dantrolene is category C. In animal studies it has reduced the rate of survival of the newborn when given in doses seven times the normal human dose. Mothers should not breast feed while receiving dantrolene.

Interactions

Skeletal muscle relaxants have many potential **drug interactions**. Individual references should be consulted.

Because these drugs cause sedation, they should be used with caution with other drugs that may also cause drowsiness.

KEY TERMS

Central nervous system—The brain and spinal cord.

Intrathecal—Introduced into or occurring in the space under the arachnoid membrane which covers the brain and spinal cord.

Pregnancy category—A system of classifying drugs according to their established risks for use during pregnancy. Category A: Controlled human studies have demonstrated no fetal risk. Category B: Animal studies indicate no fetal risk, but no human studies, or adverse effects in animals, but not in well-controlled human studies. Category C: No adequate human or animal studies, or adverse fetal effects in animal studies, but no available human data. Category D: Evidence of fetal risk, but benefits outweigh risks. Category X: Evidence of fetal risk. Risks outweigh any benefits.

Sedative—Medicine used to treat nervousness or restlessness.

Spasm—Sudden, involuntary tensing of a muscle or a group of muscles.

Tranquilizer (minor)—A drug that has a calming effect and is used to treat anxiety and emotional tension.

The activity of diazepam may be increased by drugs that inhibit its metabolism in the liver. These include: Cimetidine, oral contraceptives, Disulfiram, Fluoxetine, Isoniazid, Ketoconazole, Metoprolol, Propoxyphene, Propranolol, and Valproic acid.

Dantrolene may have an interaction with estrogens. Although no interaction has been demonstrated, the rate of liver damage in women over the age of 35 who were taking estrogens is higher than in other groups.

Samuel D. Uretsky, PharmD

Muscle spasms and cramps

Definition

Muscle spasms and cramps are spontaneous, often painful muscle contractions.

Description

Most people are familiar with the sudden **pain** of a muscle cramp. The rapid, uncontrolled contraction, or spasm, happens unexpectedly, with either no stimulation or some trivially small one. The muscle contraction and pain last for several minutes, and then slowly ease. Cramps may affect any muscle, but are most common in the calves, feet, and hands. While painful, they are harmless, and in most cases, not related to any underlying disorder. Nonetheless, cramps and spasms can be manifestations of many neurological or muscular diseases.

The terms cramp and spasm can be somewhat vague, and they are sometimes used to include types of abnormal muscle activity other than sudden painful contraction. These include stiffness at rest, slow muscle relaxation, and spontaneous contractions of a muscle at rest (fasciculation). Fasciculation is a type of painless muscle spasm, marked by rapid, uncoordinated contraction of many small muscle fibers. A critical part of diagnosis is to distinguish these different meanings and to allow the patient to describe the problem as precisely as possible.

Causes and symptoms

Causes

Normal voluntary muscle contraction begins when electrical signals are sent from the brain through the spinal cord along nerve cells called motor neurons. These include both the upper motor neurons within the brain and the lower motor neurons within the spinal cord and leading out to the muscle. At the muscle, chemicals released by the motor neuron stimulate the internal release of calcium ions from stores within the muscle cell. These calcium ions then interact with muscle proteins within the cell, causing the proteins (actin and myosin) to slide past one another. This motion pulls their fixed ends closer, thereby shortening the cell and, ultimately, the muscle itself. Recapture of calcium and unlinking of actin and myosin allows the muscle fiber to relax.

Abnormal contraction may be caused by abnormal activity at any stage in this process. Certain mechanisms within the brain and the rest of the central nervous system help regulate contraction. Interruption of these mechanisms can cause spasm. Motor neurons that are overly sensitive may fire below their normal thresholds. The muscle membrane itself may be over sensitive, causing contraction without stimulation. Calcium ions may not be recaptured quickly enough, causing prolonged contraction.

Interruption of brain mechanisms and overly sensitive motor neurons may result from damage to the nerve pathways. Possible causes include **stroke**, **multiple sclerosis**, **cerebral palsy**, neurodegenerative diseases, trauma, **spinal cord injury**, and nervous system poisons such as strychnine, **tetanus**, and certain insecticides. Nerve damage may lead to a prolonged or permanent muscle shortening called contracture.

Changes in muscle responsiveness may be due to or associated with:

- **Prolonged exercise.** Curiously, relaxation of a muscle actually requires energy to be expended. The energy is used to recapture calcium and to unlink actin and myosin. Normally, sensations of pain and **fatigue** signal that it is time to rest. Ignoring or overriding those warning signals can lead to such severe energy depletion that the muscle cannot be relaxed, causing a cramp. The familiar advice about not swimming after a heavy meal, when blood flow is directed away from the muscles, is intended to avoid this type of cramp. Rigor mortis, the stiffness of a corpse within the first 24 hours after **death**, is also due to this phenomenon.
- **Dehydration** and salt depletion. This may be brought on by protracted **vomiting** or **diarrhea**, or by copious sweating during prolonged exercise, especially in high temperatures. Loss of fluids and salts—especially sodium, potassium, magnesium, and calcium—can disrupt ion balances in both muscle and nerves. This can prevent them from responding and recovering normally, and can lead to cramp.
- Metabolic disorders that affect the energy supply in muscle. These are inherited diseases in which particular muscle enzymes are deficient. They include deficiencies of myophosphorylase (McArdle's disease), phosphorylase b kinase, phosphofructokinase, phosphoglycerate kinase, and lactate dehydrogenase.
- Myotonia. This causes stiffness due to delayed relaxation of the muscle, but does not cause the spontaneous contraction usually associated with cramps. However, many patients with myotonia do experience cramping from exercise. Symptoms of myotonia are often worse in the cold. Myotonias include **myotonic dystrophy**, myotonia congenita, paramyotonia congenita, and neuromyotonia.

Fasciculations may be due to fatigue, cold, medications, metabolic disorders, nerve damage, or neurodegenerative disease, including **amyotrophic lateral sclerosis**. Most people experience brief, mild fasciculations from time to time, usually in the calves.

KEY TERMS

Motor neuron—Nerve cells within the central nervous system that carry nerve impulses controlling muscle movement.

Symptoms

The pain of a muscle cramp is intense, localized, and often debilitating. Coming on quickly, it may last for minutes and fade gradually. **Contractures** develop more slowly, over days or weeks, and may be permanent if untreated. Fasciculations may occur at rest or after muscle contraction, and may last several minutes.

Diagnosis

Abnormal contractions are diagnosed through a careful medical history, physical and neurological examination, and **electromyography** of the affected muscles. Electromyography records electrical activity in the muscle during rest and movement.

Treatment

Most cases of simple cramps require no treatment other than patience and stretching. Gently and gradually stretching and massaging the affected muscle may ease the pain and hasten recovery.

More prolonged or regular cramps may be treated with drugs such as carbamazepine, phenytoin, or quinine. Fluid and salt replacement, either orally or intravenously, is used to treat dehydration. Treatment of underlying metabolic or neurologic disease, where possible, may help relieve symptoms.

Alternative treatment

Cramps may be treated or prevented with Ginkgo (*Ginkgo biloba*) or Japanese quince (*Chaenomeles speciosa*). Supplements of vitamin E, niacin, calcium, and magnesium may also help. Taken at bedtime, they may help to reduce the likelihood of night cramps.

Prognosis

Occasional cramps are common, and have no special medical significance.

Prevention

The likelihood of developing cramps may be reduced by eating a healthy diet with appropriate levels of **minerals**, and getting regular exercise to build up energy reserves in muscle. Avoiding exercising in extreme heat helps prevent heat cramps. Heat cramps can also be avoided by taking salt tablets and water before prolonged exercise in extreme heat. Taking a warm bath before bedtime may increase circulation to the legs and reduce the incidence of nighttime leg cramps.

Resources

BOOKS

Bradley, Walter G., et al. *Neurology in Clinical Practice*. 2nd ed. Woburn, MA: Butterworth-Heinemann, 1995.

Richard Robinson

Muscular dystrophy

Definition

Muscular dystrophy is the name for a group of inherited disorders in which strength and muscle bulk gradually decline. Nine types of muscular dystrophies are generally recognized.

Description

The muscular dystrophies include:

- **Duchenne muscular dystrophy (DMD):** DMD affects young boys, causing progressive muscle weakness, usually beginning in the legs. It is the most severe form of muscular dystrophy. DMD occurs in about 1 in 3,500 male births, and affects approximately 8,000 boys and young men in the United States. A milder form occurs in very few female carriers.
- **Becker muscular dystrophy (BMD):** BMD affects older boys and young men, following a milder course than DMD. BMD occurs in about 1 in 30,000 male births.
- **Emery-Dreifuss muscular dystrophy (EDMD):** EDMD affects young boys, causing **contractures** and weakness in the calves, weakness in the shoulders and upper arms, and problems in the way electrical impulses travel through the heart to make it beat

(heart conduction defects). Fewer than 300 cases of EDMD have been identified.

- **Limb-girdle muscular dystrophy (LGMD):** LGMD begins in late childhood to early adulthood and affects both men and women, causing weakness in the muscles around the hips and shoulders. It is the most variable of the muscular dystrophies, and there are several different forms of the disease now recognized. Many people with suspected LGMD have probably been misdiagnosed in the past, and therefore the prevalence of the disease is difficult to estimate. The number of people affected in the United States may be in the low thousands.
- **Facioscapulohumeral muscular dystrophy (FSH):** FSH, also known as Landouzy-Dejerine disease, begins in late childhood to early adulthood and affects both men and women, causing weakness in the muscles of the face, shoulders, and upper arms. The hips and legs may also be affected. FSH occurs in about 1 out of every 20,000 people, and affects approximately 13,000 people in the United States.
- **Myotonic dystrophy:** also known as Steinert's disease, affects both men and women, causing generalized weakness first seen in the face, feet, and hands. It is accompanied by the inability to relax the affected muscles (myotonia). Symptoms may begin from birth through adulthood. It is the most common form of muscular dystrophy, affecting more than 30,000 people in the United States.
- **Oculopharyngeal muscular dystrophy (OPMD):** OPMD affects adults of both sexes, causing weakness in the eye muscles and throat. It is most common among French Canadian families in Quebec, and in Spanish-American families in the southwestern United States.
- **Distal muscular dystrophy (DD):** DD begins in middle age or later, causing weakness in the muscles of the feet and hands. It is most common in Sweden, and rare in other parts of the world.
- **Congenital muscular dystrophy (CMD):** CMD is present from birth, results in generalized weakness, and usually progresses slowly. A subtype, called Fukuyama CMD, also involves **mental retardation**. Both are rare; Fukuyama CMD is more common in Japan.

Causes and symptoms

Causes

Several of the muscular dystrophies, including DMD, BMD, CMD, and most forms of LGMD, are

due to defects in the genes for a complex of muscle proteins. This complex spans the muscle cell membrane to unite a fibrous network on the interior of the cell with a fibrous network on the outside. Current theory holds that by linking these two networks, the complex acts as a "shock absorber," redistributing and evening out the forces generated by contraction of the muscle, thereby preventing rupture of the muscle membrane. Defects in the proteins of the complex lead to deterioration of the muscle. Symptoms of these diseases set in as the muscle gradually exhausts its ability to repair itself. Both DMD and BMD are caused by flaws in the gene for the protein called dystrophin. The flaw leading to DMD prevents the formation of any dystrophin, while that of BMD allows some protein to be made, accounting for the differences in severity and onset between the two diseases. Differences among the other diseases in the muscles involved and the ages of onset are less easily explained.

The causes of the other muscular dystrophies are not as well understood:

- One form of LGMD is caused by defects in the gene for a muscle enzyme, calpain. The relationship between this defect and the symptoms of the disease is unclear.
- EDMD is due to a defect in the gene for a protein called emerin, which is found in the membrane of a cell's nucleus, but whose exact function is unknown.
- Myotonic dystrophy is linked to gene defects for a protein that may control the flow of charged particles within muscle cells. This gene defect is called a triple repeat, meaning it contains extra triplets of DNA code. It is possible that this mutation affects nearby genes as well, and that the widespread symptoms of myotonic dystrophy are due to a range of genetic disruptions.
- The gene for OPMD appears to also be mutated with a triple repeat. The function of the affected protein may involve translation of genetic messages in a cell's nucleus.
- The cause of FSH is unknown. Although the genetic region responsible for it has been localized on its chromosome, the identity and function of the gene or genes involved had not been determined as of 1997.
- The gene responsible for DD has not yet been found.

Genetics and patterns of inheritance

The muscular dystrophies are genetic diseases, meaning they are caused by defects in genes. Genes,

which are linked together on chromosomes, have two functions: They code for the production of proteins, and they are the material of inheritance. Parents pass along genes to their children, providing them with a complete set of instructions for making their own proteins.

Because both parents contribute genetic material to their offspring, each child carries two copies of almost every gene, one from each parent. For some diseases to occur, both copies must be flawed. Such diseases are called autosomal recessive diseases. Some forms of LGMD and DD exhibit this pattern of inheritance, as does CMD. A person with only one flawed copy, called a carrier, will not have the disease, but may pass the flawed gene on to his children. When two carriers have children, the chances of having a child with the disease is one in four for each **pregnancy**.

Other diseases occur when only one flawed gene copy is present. Such diseases are called autosomal dominant diseases. Other forms of LGMD exhibit this pattern of inheritance, as do DM, FSH, OPMD, and some forms of DD. When a person affected by the disease has a child with someone not affected, the chances of having an affected child is one in two.

Because of chromosomal differences between the sexes, some genes are not present in two copies. The chromosomes that determine whether a person is male or female are called the X and Y chromosomes. A person with two X chromosomes is female, while a person with one X and one Y is male. While the X chromosome carries many genes, the Y chromosome carries almost none. Therefore, a male has only one copy of each gene on the X chromosome, and if it is flawed, he will have the disease that defect causes. Such diseases are said to be X-linked. X-linked diseases include DMD, BMD, and EDMD. Women aren't usually affected by X-linked diseases, since they will likely have one unaffected copy between the two chromosomes. Some female carriers of DMD suffer a mild form of the disease, probably because their one unaffected gene copy is shut down in some of their cells.

Women carriers of X-linked diseases have a one in two chance of passing the flawed gene on to each child born. Daughters who inherit the disease gene will be carriers. A son born without the disease gene will be free of the disease and cannot pass it on to his children. A son born with the defect will have the disease. He will pass the flawed gene on to each of his daughters, who will then be carriers, but to none of his sons (because they inherit his Y chromosome).

Not all genetic flaws are inherited. As many as one third of the cases of DMD are due to new mutations

that arise during egg formation in the mother. New mutations are less common in other forms of muscular dystrophy.

Symptoms

All of the muscular dystrophies are marked by muscle weakness as the major symptom. The distribution of symptoms, age of onset, and progression differ significantly. **Pain** is sometimes a symptom of each, usually due to the effects of weakness on joint position.

DMD. A boy with Duchenne muscular dystrophy usually begins to show symptoms as a pre-schooler. The legs are affected first, making walking difficult and causing balance problems. Most patients walk three to six months later than expected and have difficulty running. Later on, the boy with DMD will push his hands against his knees to rise to a standing position, to compensate for leg weakness. About the same time, his calves will begin to swell, though with fibrous tissue rather than with muscle, and feel firm and rubbery; this condition gives DMD one of its alternate names, pseudohypertrophic muscular dystrophy. He will widen his stance to maintain balance, and walk with a waddling gait to advance his weakened legs. Contractures (permanent muscle tightening) usually begin by age five or six, most severely in the calf muscles. This pulls the foot down and back, forcing the boy to walk on tip-toes, called equinus, and further decreases balance. Frequent falls and broken bones are common beginning at this age. Climbing stairs and rising unaided may become impossible by age nine or ten, and most boys use a wheelchair for mobility by the age of 12. Weakening of the trunk muscles around this age often leads to **scoliosis** (a side-to-side spine curvature) and **kyphosis** (a front-to-back curvature).

The most serious weakness of DMD is weakness of the diaphragm, the sheet of muscles at the top of the abdomen that perform the main work of breathing and coughing. Diaphragm weakness leads to reduced energy and stamina, and increased lung infection because of the inability to **cough** effectively. Young men with DMD often live into their twenties and beyond, provided they have mechanical ventilation assistance and good respiratory hygiene.

About one third of boys with DMD experience specific learning disabilities, including trouble learning by ear rather than by sight and trouble paying attention to long lists of instructions. Individualized educational programs usually compensate well for these disabilities.

BMD. The symptoms of BMD usually appear in late childhood to early adulthood. Though the progression of symptoms may parallel that of DMD, the symptoms are usually milder and the course more variable. The same pattern of leg weakness, unsteadiness, and contractures occur later for the young man with BMD, often allowing independent walking into the twenties or early thirties. Scoliosis may occur, but is usually milder and progresses more slowly. Heart muscle disease (**cardiomyopathy**), occurs more commonly in BMD. Problems may include irregular heartbeats (**arrhythmias**) and congestive **heart failure**. Symptoms may include **fatigue**, **shortness of breath**, chest pain, and **dizziness**. Respiratory weakness also occurs, and may lead to the need for mechanical ventilation.

EDMD. This type of muscular dystrophy usually begins in early childhood, often with contractures preceding muscle weakness. Weakness affects the shoulder and upper arm originally, along with the calf muscles, leading to foot-drop. Most men with EDMD survive into middle age, although a defect in the heart's rhythm (**heart block**) may be fatal if not treated with a pacemaker.

LGMD. While there are at least a half-dozen genes that cause the various types of LGMD, two major clinical forms of LGMD are usually recognized. A severe childhood form is similar in appearance to DMD, but is inherited as an autosomal recessive trait. Symptoms of adult-onset LGMD usually appear in a person's teens or twenties, and are marked by progressive weakness and wasting of the muscles closest to the trunk. Contractures may occur, and the ability to walk is usually lost about 20 years after onset. Some people with LGMD develop respiratory weakness that requires use of a ventilator. Lifespan may be somewhat shortened. (Autosomal dominant forms usually occur later in life and progress relatively slowly.)

FSH. FSH varies in its severity and age of onset, even among members of the same family. Symptoms most commonly begin in the teens or early twenties, though infant or childhood onset is possible. Symptoms tend to be more severe in those with earlier onset. The disease is named for the regions of the body most severely affected by the disease: muscles of the face (facio-), shoulders (scapulo-), and upper arms (humeral). Hips and legs may be affected as well. Children with FSH often develop partial or complete deafness.

The first symptom noticed is often difficulty lifting objects above the shoulders. The weakness may be greater on one side than the other. Shoulder weakness

also causes the shoulder blades to jut backward, called scapular winging. Muscles in the upper arm often lose bulk sooner than those of the forearm, giving a "Popeye" appearance to the arms. Facial weakness may lead to loss of facial expression, difficulty closing the eyes completely, and inability to drink through a straw, blow up a balloon, or whistle. A person with FSH may not develop strong facial wrinkles. Contracture of the calf muscles may cause foot-drop, leading to frequent tripping over curbs or rough spots. People with earlier onset often require a wheelchair for mobility, while those with later onset rarely do.

MYOTONIC DYSTROPHY. Symptoms of Myotonic dystrophy include facial weakness and a slack jaw, drooping eyelids (**ptosis**), and muscle wasting in the forearms and calves. A person with this dystrophy has difficulty relaxing his grasp, especially if the object is cold. Myotonic dystrophy affects heart muscle, causing arrhythmias and heart block, and the muscles of the digestive system, leading to motility disorders and **constipation**. Other body systems are affected as well: Myotonic dystrophy may cause **cataracts**, retinal degeneration, low IQ, frontal balding, skin disorders, testicular atrophy, **sleep apnea**, and **insulin resistance**. An increased need or desire for sleep is common, as is diminished motivation. Severe disability affects most people with this type of dystrophy within 20 years of onset, although most do not require a wheelchair even late in life.

OPMD. OPMD usually begins in a person's thirties or forties, with weakness in the muscles controlling the eyes and throat. Symptoms include drooping eyelids, difficulty swallowing (dysphagia), and weakness progresses to other muscles of the face, neck, and occasionally the upper limbs. Swallowing difficulty may cause aspiration, or the introduction of food or saliva into the airways. **Pneumonia** may follow.

DD. DD usually begins in the twenties or thirties, with weakness in the hands, forearms, and lower legs. Difficulty with fine movements such as typing or fastening buttons may be the first symptoms. Symptoms progress slowly, and the disease usually does not affect life span.

CMD. CMD is marked by severe muscle weakness from birth, with infants displaying "floppiness" and very little voluntary movement. Nonetheless, a child with CMD may learn to walk, either with or without some assistive device, and live into young adulthood or beyond. In contrast, children with Fukuyama CMD are rarely able to walk, and have severe mental retardation. Most children with this type of CMD die in childhood.

Diagnosis

Diagnosis of muscular dystrophy involves a careful medical history and a thorough physical exam to determine the distribution of symptoms and to rule out other causes. Family history may give important clues, since all the muscular dystrophies are genetic conditions (though no family history will be evident in the event of new mutations).

Lab tests may include:

- Blood level of the muscle enzyme creatine kinase (CK). CK levels rise in the blood due to muscle damage, and may be seen in some conditions even before symptoms appear.
- Muscle biopsy, in which a small piece of muscle tissue is removed for microscopic examination. Changes in the structure of muscle cells and presence of fibrous tissue or other aberrant structures are characteristic of different forms of muscular dystrophy. The muscle tissue can also be stained to detect the presence or absence of particular proteins, including dystrophin.
- Electromyogram (EMG). This electrical test is used to examine the response of the muscles to stimulation. Decreased response is seen in muscular dystrophy. Other characteristic changes are seen in DM.
- Genetic tests. Several of the muscular dystrophies can be positively identified by testing for the presence of the mutated gene involved. Accurate genetic tests are available for DMD, BMD, DM, several forms of LGMD, and EDMD.
- Other specific tests as necessary. For EDMD and BMD, for example, an electrocardiogram may be needed to test heart function, and hearing tests are performed for children with FSH.

For most forms of muscular dystrophy, accurate diagnosis is not difficult when done by someone familiar with the range of diseases. There are exceptions, however. Even with a muscle biopsy, it may be difficult to distinguish between FSH and another muscle disease, **polymyositis**. Childhood-onset LGMD is often mistaken for the much more common DMD, especially when it occurs in boys. BMD with an early onset appears very similar to DMD, and a genetic test may be needed to accurately distinguish them. The muscular dystrophies may be confused with diseases involving the motor neurons, such as spinal muscular atrophy; diseases of the neuromuscular junction, such as **myasthenia gravis**; and other muscle diseases, as all involve generalized weakening of varying distribution.

Treatment

Drugs

There are no cures for any of the muscular dystrophies. Prednisone, a corticosteroid, has been shown to delay the progression of DMD somewhat, for reasons that are still unclear. Prednisone is also prescribed for BMD, though no controlled studies have tested its benefit. A related drug, deflazacort, appears to have similar benefits with fewer side effects. It is available and is prescribed in Canada and Mexico, but is unavailable in the United States. Albuterol, an adrenergic agonist, has shown some promise for FSH in small trials; larger trials are scheduled for 1998. No other drugs are currently known to have an effect on the course of any other muscular dystrophy.

Treatment of muscular dystrophy is mainly directed at preventing the complications of weakness, including decreased mobility and dexterity, contractures, scoliosis, heart defects, and respiratory insufficiency.

Physical therapy

Physical therapy, in particular regular stretching, is used to maintain the range of motion of affected muscles and to prevent or delay contractures. Braces are used as well, especially on the ankles and feet to prevent equinus. Full-leg braces may be used in DMD to prolong the period of independent walking. Strengthening other muscle groups to compensate for weakness may be possible if the affected muscles are few and isolated, as in the earlier stages of the milder muscular dystrophies. Regular, nonstrenuous **exercise** helps maintain general good health. Strenuous exercise is usually not recommended, since it may damage muscles further.

Surgery

When contractures become more pronounced, tenotomy surgery may be performed. In this operation, the tendon of the contracted muscle is cut, and the limb is braced in its normal resting position while the tendon regrows. In FSH, surgical fixation of the scapula can help compensate for shoulder weakness. For a person with OPMD, surgical lifting of the eyelids may help compensate for weakened muscular control. For a person with DM, sleep apnea may be treated surgically to maintain an open airway. Scoliosis surgery is often needed in DMD, but much less often in other muscular dystrophies. Surgery is recommended at a much lower degree of curvature for DMD than for scoliosis due to other conditions, since the decline in respiratory function in DMD makes surgery at a later time dangerous.

In this surgery, the vertebrae are fused together to maintain the spine in the upright position. Steel rods are inserted at the time of operation to keep the spine rigid while the bones grow together.

When any type of surgery is performed in patients with muscular dystrophy, anesthesia must be carefully selected. People with MD are susceptible to a severe reaction, known as malignant hyperthermia, when given halothane anesthetic.

Occupational therapy

The occupational therapist suggests techniques and tools to compensate for the loss of strength and dexterity. Strategies may include modifications in the home, adaptive utensils and dressing aids, compensatory movements and positioning, wheelchair accessories, or communication aids.

Nutrition

Good **nutrition** helps to promote general health in all the muscular dystrophies. No special diet or supplement has been shown to be of use in any of the conditions. The weakness in the throat muscles seen especially in OPMD and later DMD may necessitate the use of a **gastrostomy** tube, inserted in the stomach to provide nutrition directly.

Cardiac care

The arrhythmias of EDMD and BMD may be treatable with antiarrhythmia drugs such as mexiletine or nifedipine. A pacemaker may be implanted if these do not provide adequate control. Heart transplants are increasingly common for men with BMD.

Respiratory care

People who develop weakness of the diaphragm or other ventilatory muscles may require a mechanical ventilator to continue breathing deeply enough. Air may be administered through a nasal mask or mouthpiece, or through a tracheostomy tube, which is inserted through a surgical incision through the neck and into the windpipe. Most people with muscular dystrophy do not need a tracheostomy, although some may prefer it to continual use of a mask or mouthpiece. Supplemental oxygen is not needed. Good hygiene of the lungs is critical for health and longterm survival of a person with weakened ventilatory muscles. Assisted cough techniques provide the strength needed to clear the airways of secretions; an assisted cough machine is also available and provides excellent results.

Experimental treatments

Two experimental procedures aiming to cure DMD have attracted a great deal of attention in the past decade. In myoblast transfer, millions of immature muscle cells are injected into an affected muscle. The goal of the treatment is to promote the growth of the injected cells, replacing the defective host cells with healthy new ones. Despite continued claims to the contrary by a very few researchers, this procedure is widely judged a failure. Modifications in the technique may change that in the future.

Gene therapy introduces good copies of the dystrophin gene into muscle cells. The goal is to allow the existing muscle cells to use the new gene to produce the dystrophin it cannot make with its flawed gene. Problems have included immune rejection of the virus used to introduce the gene, loss of gene function after several weeks, and an inability to get the gene to enough cells to make a functional difference in the affected muscle. Nonetheless, after a number of years of refining the techniques in mice, researchers are beginning human trials in 1998.

Prognosis

The expected lifespan for a male with DMD has increased significantly in the past two decades. Most young men will live into their early or mid-twenties. Respiratory infections become an increasing problem as their breathing becomes weaker, and these infections are usually the cause of **death**.

The course of the other muscular dystrophies is more variable; expected life spans and degrees of disability are hard to predict, but may be related to age of onset and initial symptoms. Prediction is made more difficult because, as new genes are discovered, it is becoming clear that several of the dystrophies are not uniform disorders, but rather symptom groups caused by different genes.

People with dystrophies with significant heart involvement (BMD, EDMD, Myotonic dystrophy) may nonetheless have almost normal life spans, provided that cardiac complications are monitored and treated aggressively. The respiratory involvement of BMD and LGMD similarly require careful and prompt treatment.

Prevention

There is no way to prevent any of the muscular dystrophies in a person who has the genes responsible for these disorders. Accurate genetic tests, including

KEY TERMS

Autosomal dominant—Diseases that occur when a person inherits only one flawed copy of the gene.

Autosomal recessive—Diseases that occur when a person inherits two flawed copies of a gene—one from each parent.

Becker muscular dystrophy (BMD)—A type of muscular dystrophy that affects older boys and men, and usually follows a milder course than DMD.

Contractures—A permanent shortening (as of muscle, tendon, or scar tissue) producing deformity or distortion.

Distal muscular dystrophy (DD)—A form of muscular dystrophy that usually begins in middle age or later, causing weakness in the muscles of the feet and hands.

Duchenne muscular dystrophy (DMD)—The most severe form of muscular dystrophy, DMD usually affects young boys and causes progressive muscle weakness, usually beginning in the legs.

Dystrophin—A protein that helps muscle tissue repair itself. Both DMD and BMD are caused by flaws in the gene that instructs the body how to make this protein.

Facioscapulohumeral muscular dystrophy (FSH)—This form of muscular dystrophy, also known as Landouzy-Dejerine disease, begins in late childhood to early adulthood and affects both men and women, causing weakness in the muscles of the face, shoulders, and upper arms.

Limb-girdle muscular dystrophy (LGMD)—This form of muscular dystrophy begins in late childhood to early adulthood and affects both men and women, causing weakness in the muscles around the hips and shoulders.

Myotonic dystrophy—This type of muscular dystrophy, also known as Steinert's disease, affects both men and women, causing generalized weakness first seen in the face, feet, and hands. It is accompanied by the inability to relax the affected muscles (myotonia).

Oculopharyngeal muscular dystrophy (OPMD)—This type of muscular dystrophy affects adults of both sexes, causing weakness in the eye muscles and throat.

prenatal tests, are available for some of the muscular dystrophies. Results of these tests may be useful for purposes of family planning.

Resources

ORGANIZATIONS

Muscular Dystrophy Association. 3300 East Sunrise Drive, Tucson, AZ 85718. (800) 572-1717. <<http://www.mdaua.org>>.

Richard Robinson

Mushroom poisoning

Definition

Mushroom **poisoning** refers to the severe and often deadly effects of various toxins that are found in certain types of mushrooms. One type known as *Amanita phalloides*, appropriately called “death cap,” accounts for the majority of cases. The toxins initially cause severe abdominal cramping, **vomiting**, and watery **diarrhea**, and then lead to liver and kidney failure.

Description

The highest reported incidences of mushroom poisoning occur in western Europe, where a popular pastime is amateur mushroom hunting. Since the 1970s, the United States has seen a marked increase in mushroom poisoning due to an increase in the popularity of “natural” foods, the use of mushrooms as recreational hallucinogens, and the gourmet qualities of wild mushrooms. About 90% of the deaths due to mushroom poisoning in the United States and western Europe result from eating *Amanita phalloides*. This mushroom is recognized by its metallic green cap (the color may vary from light yellow to greenish brown), white gills (located under the cap), white stem, and bulb-shaped structure at the base of the stem. A pure white variety of this species also occurs. Poisoning results from ingestion of as few as one to three mushrooms. Higher **death** (mortality) rates of more than 50% occur in children less than 10 years of age.

Causes and symptoms

Poisonous mushrooms contain at least two different types of toxins, each of which can cause death if taken in large enough quantities. Some of the toxins

found in poisonous mushrooms are among the most potent ever discovered. One group of poisons, known as amatoxins, blocks the production of DNA, the basis of cell reproduction. This leads to the death of many cells, especially those that reproduce frequently such as in the liver, intestines, and kidney. Other mushroom poisons affect the proteins needed for muscle contraction, and therefore reduce the ability of certain muscle groups to perform.

Symptoms of *Amanita* poisoning occur in different stages or phases. These include:

- First phase. Abdominal cramping, **nausea**, vomiting, and severe watery diarrhea occur anywhere from 6-24 hours after eating the mushroom and last for about 24 hours. These intestinal symptoms can lead to **dehydration** and low blood pressure (hypotension).
- Second phase. A period of remission of symptoms that lasts 1-2 days. During this time, the patient feels better, but blood tests begin to show evidence of liver and kidney damage.
- Third phase. Liver and kidney failure develop at this point and either lead to death within about a week or recovery within 2-3 weeks.

Other symptoms are due to either a decrease in blood clotting factors that leads to internal bleeding or reduced muscle function, with the development of weakness and **paralysis**.

Diagnosis

In most cases, the fact that the patient has recently eaten wild mushrooms is the clue to the cause of symptoms. Moreover, the identification of any remaining mushrooms by a qualified mushroom specialist (mycologist) can be a key to diagnosis. When in doubt, the toxin known as alpha-amantin can be found in the blood, urine, or stomach contents of an individual who has ingested poisonous *Amanita* mushrooms.

Treatment

It is important to remember that there is no specific antidote for mushroom poisoning. However, several advances in therapy have decreased the death rate over the last several years. Early replacement of lost body fluids has been a major factor in improving survival rates.

Therapy is aimed at decreasing the amount of toxin in the body. Initially, attempts are made to remove toxins from the upper gastrointestinal tract by inducing vomiting or by gastric lavage (stomach



A poisonous mushroom, *Amanita muscaria* (Photo Researchers. Reproduced by permission.)

pumping). After that continuous aspiration of the upper portion of the small intestine through a nasogastric tube is done and oral charcoal (every four hours for 48 hours) is given to prevent absorption of toxin. These measures work best if started within six hours of ingestion.

In the United States, early removal of mushroom poison by way of an artificial kidney machine (dialysis) has become part of the treatment program. This is combined with the correction of any imbalances of salts (electrolytes) dissolved in the blood, such as sodium or potassium. An enzyme called thioctic acid and **corticosteroids** also appear to be beneficial, as well as high doses of penicillin. In Europe, a chemical taken from the milk thistle plant, *Silybum marianum*, is also part of treatment. When liver failure develops, **liver transplantation** may be the only treatment option.

Prognosis

The mortality rate has decreased with improved and rapid treatment. However, according to some medical reports death still occurs in 20-30% of cases,

with a higher mortality rate of 50% in children less than 10 years old.

Prevention

The most important factor in preventing mushroom poisoning is to avoid eating wild or noncultivated mushrooms. For anyone not expert in mushroom identification, there are generally no easily recognizable differences between nonpoisonous and poisonous mushrooms. It is also important to remember that most mushroom poisons are not destroyed or deactivated by cooking, canning, freezing, drying, or other means of food preparation.

Resources

OTHER

“Alerts from the CDC.” *Experience Lab Page*. <<http://www.medsitenavigator.com>>.

“Cyclopeptide-Containing Mushroom Toxicity.” *The Toxikon Multimedia Project Page*. <<http://www.uic.edu/com/er/toxikon/mushroo.htm>>.

“Mushroom Poisoning in Children.” *American Association of Family Physicians*. <<http://www.aafp.org/patient-info/mushroom.html>>.

Mushroom Toxins from the FDA. <<http://vm.cfsan.fda.gov/~mow/chap40.html>>.

David Kaminstein, MD

Music therapy

Definition

Music therapy is a technique of complementary medicine that uses music prescribed in a skilled manner by trained therapists. Programs are designed to help patients overcome physical, emotional, intellectual, and social challenges. Applications range from improving the well being of geriatric patients in nursing homes to lowering the **stress** level and **pain** of women in labor. Music therapy is used in many settings, including schools, **rehabilitation** centers, hospitals, hospice, nursing homes, community centers, and sometimes even in the home.

Purpose

Music can be beneficial for anyone. Although it can be used therapeutically for people who have physical, emotional, social, or cognitive deficits, even those who are healthy can use music to relax, reduce stress, improve mood, or to accompany **exercise**.

There are no potentially harmful or toxic effects. Music therapists help their patients achieve a number of goals through music, including improvement of communication, academic strengths, attention span, and motor skills. They may also assist with behavioral therapy and **pain management**.

Physical effects

Brain function physically changes in response to music. The rhythm can guide the body into breathing in slower, deeper patterns that have a calming effect. Heart rate and blood pressure are also responsive to the types of music that are listened to. The speed of the heartbeat tends to speed or slow depending on the volume and speed of the auditory stimulus. Louder and faster noises tend to raise both heart rate and blood pressure; slower, softer, and more regular tones produce the opposite result. Music can also relieve muscle tension and improve motor skills. It is often used to help rebuild physical patterning skills in rehabilitation clinics. Levels of endorphins, natural pain relievers, are increased while listening to music, and levels of stress hormones are decreased. This latter effect may partially explain the ability of music to improve immune function. A 1993 study at Michigan State University showed that even 15 minutes of exposure to music could increase interleukin-1 levels, a consequence which also heightens immunity.

Mental effects

Depending on the type and style of sound, music can either sharpen mental acuity or assist in relaxation. Memory and learning can be enhanced, and this used with good results in children with learning disabilities. This effect may also be partially due to increased concentration that many people have while listening to music. Better productivity is another outcome of an improved ability to concentrate. The term “Mozart effect” was coined after a study showed that college students performed better on math problems when listening to classical music.

Emotional effects

The ability of music to influence human emotion is well known, and is used extensively by movie-makers. A variety of musical moods may be used to create feelings of calmness, tension, excitement, or romance. Lullabies have long been popular for soothing babies to sleep. Music can also be used to express emotion nonverbally, which can be a very valuable therapeutic tool in some settings.

Description

Origins

Music has been used throughout human history to express and affect human emotion. In biblical accounts, King Saul was reportedly soothed by David's harp music, and the ancient Greeks expressed thoughts about music having healing effects as well. Many cultures are steeped in musical traditions. It can change mood, have stimulant or sedative effects, and alter physiologic processes such as heart rate and breathing. The apparent health benefits of music to patients in Veterans Administration hospitals following World War II lead to it being studied and formalized as a complementary healing practice. Musicians were hired to continue working in the hospitals. Degrees in music therapy became available in the late 1940s, and in 1950, the first professional association of music therapists was formed in the United States. The National Association of Music Therapy merged with the American Association of Music Therapy in 1998 to become the American Music Therapy Association.

Goals

Music is used to form a relationship with the patient. The music therapist sets goals on an individual basis, depending on the reasons for treatment, and selects specific activities and exercises to help the patient progress. Objectives may include development of communication, cognitive, motor, emotional, and social skills. Some of the techniques used to achieve this are singing, listening, instrumental music, composition, creative movement, **guided imagery**, and other methods as appropriate. Other disciplines may be integrated as well, such as dance, art, and psychology. Patients may develop musical abilities as a result of therapy, but this is not a major concern. The primary aim is to improve the patient's ability to function.

Techniques

Learning to play an instrument is an excellent musical activity to develop motor skills in individuals with developmental delays, brain injuries, or other motor impairment. It is also an exercise in impulse control and group cooperation. Creative movement is another activity that can help to improve coordination, as well as strength, balance, and gait. Improvisation facilitates the nonverbal expression of emotion. It encourages socialization and communication about feelings as well. Singing develops articulation, rhythm, and breath control. Remembering lyrics and melody is an exercise in sequencing for **stroke** victims and others who may be intellectually impaired.

Composition of words and music is one avenue available to assist the patient in working through fears and negative feelings. Listening is an excellent way to practice attending and remembering. It may also make the patient aware of memories and emotions that need to be acknowledged and perhaps talked about. Singing and discussion is a similar method, which is used with some patient populations to encourage dialogue. Guided Imagery and Music (GIM) is a very popular technique developed by music therapist Helen Bonny. Listening to music is used as a path to invoke emotions, picture, and symbols from the patient. This is a bridge to the exploration and expression of feelings.

Music and children

The sensory stimulation and playful nature of music can help to develop a child's ability to express emotion, communicate, and develop rhythmic movement. There is also some evidence to show that speech and language skills can be improved through the stimulation of both hemispheres of the brain. Just as with adults, appropriately selected music can decrease stress, **anxiety**, and pain. Music therapy in a hospital environment with those who are sick, preparing for surgery, or recovering postoperatively is appropriate and beneficial. Children can also experience improved self-esteem through musical activities that allow them to succeed.

Newborns may enjoy an even greater benefit of music. Those who are premature experience more rapid weight gain and hospital discharge than their peers who are not exposed to music. There is also anecdotal evidence of improved cognitive function.

Music and rehabilitation

Patients with brain damage from stroke, traumatic brain injury, or other neurologic conditions have been shown to exhibit significant improvement as a result of music therapy. This is theorized to be partially the result of entrainment, which is the synchronization of movement with the rhythm of the music. Consistent practice leads to gains in motor skill ability and efficiency. Cognitive processes and language skills often benefit from appropriate musical intervention.

Music and the elderly

The geriatric population can be particularly prone to anxiety and depression, particularly in nursing home residents. Chronic diseases causing pain are also not uncommon in this setting. Music is an excellent outlet to provide enjoyment, relaxation, relief from pain, and an opportunity to socialize and reminisce about music.

that has had special importance to the individual. It can have a striking effect on patients with **Alzheimer's disease**, even sometimes allowing them to focus and become responsive for a time. Music has also been observed to decrease the agitation that is so common with this disease. One study shows that elderly people who play a musical instrument are more physically and emotionally fit as they age than their nonmusical peers are.

Music and the mentally ill

Music can be an effective tool for the mentally or emotionally ill. **Autism** is one disorder that has been particularly researched. Music therapy has enabled some autistic children to relate to others and have improved learning skills. **Substance abuse**, **schizophrenia**, **paranoia**, and disorders of personality, anxiety, and affect are all conditions that may be benefited by music therapy. In these groups, participation and social interaction are promoted through music. Reality orientation is improved. Patients are helped to develop coping skills, reduce stress, and express their feelings.

Music and hospice

Pain, anxiety, and depression are major concerns with patients who are terminally ill, whether they are in hospice or not. Music can provide some relief from pain, through release of endorphins and promotion of relaxation. It can also provide an opportunity for the patient to reminisce and talk about the fears that are associated with **death** and dying. Music may help regulate the rapid breathing of a patient who is anxious, and soothe the mind. The Chalice of Repose project, headquartered at St. Patrick Hospital in Missoula, Montana, is one organization that attends and nurtures dying patients through the use of music, in a practice they called music-thanatology by developer Therese Schroeder-Sheker. Practitioners in this program work to relieve suffering through music prescribed for the individual patient.

Music and labor

Research has proven that mothers require less pharmaceutical pain relief during labor if they make use of music. Using music that is familiar and associated with positive imagery is the most helpful. During early labor, this will promote relaxation. Maternal movement is helpful to get the baby into a proper birthing position and dilate the cervix. Enjoying some "music to move by" can encourage the mother to stay active for as long as possible during labor. The rhythmic auditory stimulation may also prompt the body to release endorphins, which are a

KEY TERMS

Entrainment—The patterning of body processes and movements to the rhythm of music

Physiologic—Characteristic of normal, healthy functioning

natural form of pain relief. Many women select different styles of music for each stage of labor, with a more intense, or faster piece feeling like a natural accompaniment to the more difficult parts of labor. Instrumental music is often preferred.

Precautions

Patients making use of music therapy should not discontinue medications or therapies prescribed by other health providers without prior consultation.

Research and general acceptance

There is little disagreement among physicians that music can be of some benefit for patients, although the extent to which it can have physical effects is not as well acknowledged in the medical community. Research has shown that listening to music can decrease anxiety, pain, and recovery time. There is also good data for the specific subpopulations discussed. A therapist referral can be made through the AMTA.

Resources

ORGANIZATIONS

American Music Therapy Association, Inc. 8455 Colesville Road, Suite 1000 Silver Spring, MD 20910. (301) 589-3300. <<http://www.musictherapy.org>>.

Chalice of Repose Project at St. Patrick Hospital. 312 East Pine Street, Missoula, MT 59802. (406) 329-2810. Fax: (406) 329-5614. <<http://www.saintpatrick.org/chalice/>>.

Judith Turner

Mutism

Definition

Mutism is a rare childhood condition characterized by a consistent failure to speak in situations where talking is expected. The child has the ability to converse

normally, and does so, for example, in the home, but consistently fails to speak in specific situations such as at school or with strangers. It is estimated that one in every 1,000 school-age children are affected.

Description

Experts believe that this problem is associated with **anxiety** and fear in social situations such as in school or in the company of adults. It is therefore often considered a type of social phobia. This is not a communication disorder because the affected children can converse normally in some situations. It is not a developmental disorder because their ability to talk, when they choose to do so, is appropriate for their age level. This problem has been linked to anxiety, and one of the major ways in which both children and adults attempt to cope with anxiety is by avoiding whatever provokes the anxiety.

Affected children are typically shy, and are especially so in the presence of strangers and unfamiliar surroundings or situations. However, the behaviors of children with this condition go beyond **shyness**.

Causes and symptoms

Mutism is believed to arise from anxiety experienced in social situations where the child may be called upon to speak. Refusing to speak, or speaking in a whisper, spares the child from the possible humiliation or embarrassment of “saying the wrong thing.” When asked a direct question by teachers, for example, the affected child may act as if they are unable to answer. Some children may communicate via gestures, nodding, or very brief utterances. Additional features may include excessive shyness, oppositional behavior, and impaired learning at school.

Diagnosis

The diagnosis of mutism is fairly easy to make because the signs and symptoms are clear-cut and easily observable. However, other social disorders affecting social speech, such as **autism** or **schizophrenia**, must be considered in the diagnosis.

Treatment

There are two recommended treatments for mutism: behavior modification therapy and antidepressant medication. Treatment is most effective when individualized to each patient. It has been suggested that speech pathologists may also be able to help these children.

KEY TERMS

Behavior modification—A form of therapy that uses rewards to reinforce desired behavior. An example would be to give a child a piece of chocolate for grooming themselves appropriately.

Prognosis

The prognosis for mutism is good. Sometimes it disappears suddenly on its own. The negative impact on learning and school activities may, however, persist into adult life.

Prevention

Mutism cannot be prevented because the cause is not known. However, family conflict or problems at school contribute to the seriousness of the symptoms.

Resources

BOOKS

Diagnostic and Statistical Manual of Mental Disorders. 4th ed. Washington, DC: American Psychiatric Association, 1994.

Donald G. Barstow, RN

MVP see **Mitral valve prolapse**

Myasthenia gravis

Definition

Myasthenia gravis is an autoimmune disease that causes muscle weakness.

Description

Myasthenia gravis (MG) affects the neuromuscular junction, interrupting the communication between nerve and muscle, and thereby causing weakness. A person with MG may have difficulty moving their eyes, walking, speaking clearly, swallowing, and even breathing, depending on the severity and distribution of weakness. Increased weakness with exertion, and improvement with rest, is a characteristic feature of MG.

About 30,000 people in the United States are affected by MG. It can occur at any age, but is most common in women who are in their late teens and early twenties, and in men in their sixties and seventies.

Causes and symptoms

Myasthenia gravis is an autoimmune disease, meaning it is caused by the body's own immune system. In MG, the immune system attacks a receptor on the surface of muscle cells. This prevents the muscle from receiving the nerve impulses that normally make it respond. MG affects "voluntary" muscles, which are those muscles under conscious control responsible for movement. It does not affect heart muscle or the "smooth" muscle found in the digestive system and other internal organs.

A muscle is stimulated to contract when the nerve cell controlling it releases acetylcholine molecules onto its surface. The acetylcholine lands on a muscle protein called the acetylcholine receptor. This leads to rapid chemical changes in the muscle which cause it to contract. Acetylcholine is then broken down by acetylcholinesterase enzyme, to prevent further stimulation.

In MG, immune cells create antibodies against the acetylcholine receptor. Antibodies are proteins normally involved in fighting infection. When these antibodies attach to the receptor, they prevent it from receiving acetylcholine, decreasing the ability of the muscle to respond to stimulation.

Why the immune system creates these self-reactive "autoantibodies" is unknown, although there are several hypotheses:

- During fetal development, the immune system generates many B cells that can make autoantibodies, but B cells that could harm the body's own tissues are screened out and destroyed before birth. It is possible that the stage is set for MG when some of these cells escape detection.
- Genes controlling other parts of the immune system, called MHC genes, appear to influence how susceptible a person is to developing autoimmune disease.
- Infection may trigger some cases of MG. When activated, the immune system may mistake portions of the acetylcholine receptor for portions of an invading virus, though no candidate virus has yet been identified conclusively.
- About 10% of those with MG also have thymomas, or benign tumors of the thymus gland. The thymus is a principal organ of the immune system, and

researchers speculate that thymic irregularities are involved in the progression of MG.

Some or all of these factors (developmental, genetic, infectious, and thymic) may interact to create the autoimmune reaction.

The earliest symptoms of MG often result from weakness of the extraocular muscles, which control eye movements. Symptoms involving the eye (ocular symptoms) include double vision (diplopia), especially when not gazing straight ahead, and difficulty raising the eyelids (**ptosis**). A person with ptosis may need to tilt their head back to see. Eye-related symptoms remain the only symptoms for about 15% of MG patients. Another common early symptom is difficulty chewing and swallowing, due to weakness in the bulbar muscles, which are in the mouth and throat. **Choking** becomes more likely, especially with food that requires extensive chewing.

Weakness usually becomes more widespread within several months of the first symptoms, reaching their maximum within a year in two-thirds of patients. Weakness may involve muscles of the arms, legs, neck, trunk, and face, and affect the ability to lift objects, walk, hold the head up, and speak.

Symptoms of MG become worse upon exertion, and better with rest. Heat, including heat from the sun, hot showers, and hot drinks, may increase weakness. Infection and **stress** may worsen symptoms. Symptoms may vary from day to day and month to month, with intervals of no weakness interspersed with a progressive decline in strength.

"Myasthenic crisis" may occur, in which the breathing muscles become too weak to provide adequate respiration. Symptoms include weak and shallow breathing, **shortness of breath**, pale or bluish skin color, and a racing heart. Myasthenic crisis is an emergency condition requiring immediate treatment. In patients treated with anticholinesterase agents, myasthenic crisis must be differentiated from cholinergic crisis related to overmedication.

Pregnancy worsens MG in about one third of women, has no effect in one third, and improves symptoms in another third. About 12% of infants born to women with MG have "neonatal myasthenia," a temporary but potentially life-threatening condition. It is caused by the transfer of maternal antibodies into the fetal circulation just before birth. Symptoms include weakness, floppiness, feeble cry, and difficulty feeding. The infant may have difficulty breathing, requiring the use of a ventilator. Neonatal myasthenia usually clears up within a month.

Diagnosis

Myasthenia gravis is often diagnosed accurately by a careful medical history and a neuromuscular exam, but several tests are used to confirm the diagnosis. Other conditions causing worsening of bulbar and skeletal muscles must be considered, including drug-induced myasthenia, thyroid disease, Lambert-Eaton myasthenic syndrome, **botulism**, and inherited muscular dystrophies.

MG causes characteristic changes in the electrical responses of muscles that may be observed with an electromyogram, which measures muscular response to electrical stimulation. Repetitive nerve stimulation leads to reduction in the height of the measured muscle response, reflecting the muscle's tendency to become fatigued.

Blood tests may confirm the presence of the antibody to the acetylcholine receptor, though up to a quarter of MG patients will not have detectable levels. A **chest x ray** or chest computed tomography scan (CT scan) may be performed to look for **thymoma**.

Treatment

While there is no cure for myasthenia gravis, there are a number of treatments that effectively control symptoms in most people.

Edrophonium (Tensilon) blocks the action of acetylcholinesterase, prolonging the effect of acetylcholine and increasing strength. An injection of edrophonium rapidly leads to a marked improvement in most people with MG. An alternate drug, neostigmine, may also be used.

Pyridostigmine (Mestinon) is usually the first drug tried. Like edrophonium, pyridostigmine blocks acetylcholinesterase. It is longer-acting, taken by mouth, and well-tolerated. Loss of responsiveness and disease progression combine to eventually make pyridostigmine ineffective in tolerable doses in many patients.

Thymectomy, or removal of the thymus gland, has increasingly become standard treatment for MG. Up to 85% of people with MG improve after thymectomy, with complete remission eventually seen in about 30%. The improvement may take months or even several years to fully develop. Thymectomy is not usually recommended for children with MG, since the thymus continues to play an important immune role throughout childhood.

Immune-suppressing drugs are used to treat MG if response to pyridostigmine and thymectomy are not adequate. Drugs include **corticosteroids** such as

prednisone, and the non-steroids azathioprine (Imuran) and cyclosporine (Sandimmune).

Plasma exchange may be performed to treat myasthenic crisis or to improve very weak patients before thymectomy. In this procedure, blood plasma is removed and replaced with purified plasma free of autoantibodies. It can produce a temporary improvement in symptoms, but is too expensive for long-term treatment. Another blood treatment, intravenous immunoglobulin therapy, is also used for myasthenic crisis. In this procedure, large quantities of purified immune proteins (immunoglobulins) are injected. For unknown reasons, this leads to symptomatic improvement in up to 85% of patients. It is also too expensive for long-term treatment.

People with weakness of the bulbar muscles may need to eat softer foods that are easier to chew and swallow. In more severe cases, it may be necessary to obtain **nutrition** through a feeding tube placed into the stomach (**gastrostomy** tube).

Prognosis

Most people with MG can be treated successfully enough to prevent their condition from becoming debilitating. In some cases, however, symptoms may worsen even with vigorous treatment, leading to generalized weakness and disability. MG rarely causes early **death** except from myasthenic crisis.

Prevention

There is no known way to prevent myasthenia gravis. Thymectomy improves symptoms significantly in many patients, and relieves them entirely in some. Avoiding heat can help minimize symptoms.

Some drugs should be avoided by people with MG because they interfere with normal neuromuscular function.

Drugs to be avoided or used with caution include:

- Many types of **antibiotics**, including erythromycin, streptomycin, and ampicillin
- Some cardiovascular drugs, including Verapamil, betaxolol, and propranolol
- Some drugs used in psychiatric conditions, including chlorpromazine, clozapine, and lithium.

Many other drugs may worsen symptoms as well, so patients should check with the doctor who treats their MG before taking any new drugs.

A Medic-Alert card or bracelet provides an important source of information to emergency

KEY TERMS

Antibody—An immune protein normally used by the body for combating infection and which is made by B cells.

Autoantibody—An antibody that reacts against part of the self.

Autoimmune disease—A disease caused by a reaction of the body's immune system.

Bulbar muscles—Muscles that control chewing, swallowing, and speaking.

Neuromuscular junction—The site at which nerve impulses are transmitted to muscles.

Pyridostigmine bromide (Mestinon)—An anticholinesterase drug used in treating myasthenia gravis.

Tensilon test—A test for diagnosing myasthenia gravis. Tensilon is injected into a vein and, if the person has MG, their muscle strength will improve for about five minutes.

Thymus gland—A small gland located just above the heart, involved in immune system development.

providers about the special situation of a person with MG. They are available from health care providers.

Resources

ORGANIZATIONS

Muscular Dystrophy Association. 3300 East Sunrise Drive, Tucson, AZ 85718. (800) 572-1717. <<http://www.mdaua.org>>.

Myasthenia Gravis Foundation of America. 222 S. Riverside Plaza, Suite 1540, Chicago, IL 60606. (800) 541-5454. <<http://www.med.unc.edu>>.

Richard Robinson

Mycetoma

Definition

Mycetoma, or maduromycosis, is a slow-growing bacterial or fungal infection focused in one area of the body, usually the foot. For this reason—and because the first medical reports were from doctors in Madura, India—an alternate name for the disease is Madura

foot. The infection is characterized by an abnormal tissue mass beneath the skin, formation of cavities within the mass, and a fluid discharge. As the infection progresses, it affects the muscles and bones; at this advanced stage, disability may result.

Description

Although the bacteria and fungi that cause mycetoma are found in soil worldwide, the disease occurs mainly in tropical areas in India, Africa, South America, Central America, and southeast Asia. Mycetoma is an uncommon disease, affecting an unknown number of people annually.

There are more than 30 species of bacteria and fungi that can cause mycetoma. Bacteria or fungi can be introduced into the body through a relatively minor skin wound. The disease advances slowly over months or years, typically with minimal **pain**. When pain is experienced, it is usually due to secondary infections or bone involvement. Although it is rarely fatal, mycetoma causes deformities and potential disability at its advanced stage.

Causes and symptoms

Owing to a wound, bacteria or fungi gain entry into the skin. Approximately one month or more after the injury, a nodule forms under the skin surface. The nodule is painless, even as it increases in size over the following months. Eventually, the nodule forms a tumor, or mass of abnormal tissue. The tumor contains cavities—called sinuses—that discharge blood- or pus-tainted fluid. The fluid also contains tiny grains, less than two thousandths of an inch in size. The color of these grains depends on the type of bacteria or fungi causing the infection.

As the infection continues, surrounding tissue becomes involved, with an accumulation of scarring and loss of function. The infection can extend to the bone, causing inflammation, pain, and severe damage. Mycetoma may be complicated by secondary infections, in which new bacteria become established in the area and cause an additional set of problems.

Diagnosis

The primary symptoms of a tumor, sinuses, and grain-flecked discharge often provide enough information to diagnose mycetoma. In the early stages, prior to sinus formation, diagnosis may be more difficult and a biopsy, or microscopic examination of the tissue, may be necessary. If bone involvement is

KEY TERMS

Biopsy—A medical procedure in which a small piece of tissue is surgically removed for microscopic examination.

Grains—Flecks of hardened material such as bacteria or fungi spores.

Nodule—A hardened area or knot sometimes associated with infection.

Secondary infection—Illness caused by new bacteria, viruses, or fungi becoming established in the wake of an initial infection.

Sinuses—Cavities or hollow areas.

Tumor—A mass or clump of abnormal tissue, not necessarily caused by a cancer.

suspected, the area is x rayed to determine the extent of the damage. The species of bacteria or fungi at the root of the infection is identified by staining the discharge grains and inspecting them with a microscope.

Treatment

Combating mycetoma requires both surgery and drug therapy. Surgery usually consists of removing the tumor and a portion of the surrounding tissue. If the infection is extensive, **amputation** is sometimes necessary. Drug therapy is recommended in conjunction with surgery. The specific prescription depends on the type of bacteria or fungi causing the disease. Common medicines include antifungal drugs, such as ketoconazole and **antibiotics** (streptomycin sulfate, amikacin, sulfamethoxazole, penicillin, and rifampin).

Prognosis

Recovery from mycetoma may take months or years, and the infection recurs after surgery in at least 20% of cases. Drug therapy can reduce the chances of a re-established infection. The extent of deformity or disability depends on the severity of infection; the more deeply entrenched the infection, the greater the damage. By itself, mycetoma is rarely fatal, but secondary infections can be fatal.

Prevention

Mycetoma is a rare condition that is not contagious.

Resources

PERIODICALS

McGinnis, Michael R. "Mycetoma." *Dermatologic Clinics* 14, no. 1 (January 1996): 97.

Julia Barrett

Mycobacterial infections, atypical

Definition

Atypical mycobacterial infections are infections caused by several types of mycobacteria similar to the germ that causes **tuberculosis**. These atypical mycobacterial infections are a frequent complication in patients with human **immunodeficiency virus** (HIV) infection or **AIDS**.

Description

Mycobacteria are a group of rod-shaped bacteria that cause several diseases, among them **leprosy** and tuberculosis. For some time, scientists have known of bacteria that are similar to *Mycobacterium tuberculosis*, the cause of tuberculosis, but that grow and act differently. When tuberculosis was a much more widespread problem and microbiology was much less able to tell the difference between similar microbes, these atypical mycobacteria were ignored. Today, they have been classified more precisely as members of the same species and called atypical (or nontuberculosis) mycobacteria.

Although the medical profession has known about these atypical infections for a long time, they were not considered a serious problem until the early 1980s. It was then that many of these atypical infections were noticed among homosexuals and intravenous drug users in New York City. These bacteria rarely cause infection in humans other than those with HIV or AIDS.

Causes and symptoms

Although there are more than a dozen species of atypical mycobacteria, the two most common are *Mycobacterium kansasii* and *M. avium-intracellulare*. These microbes are found in many places in the environment: tap water, fresh and ocean water, milk, bird droppings, soil, and house dust. The manner in which

these bacteria are transmitted is not completely understood. There is no evidence that they are transmitted from person to person.

M. avium-intracellulare (MAC or MAI) is a rare cause of lung disease in otherwise healthy humans but a frequent cause of infection among those whose resistance has been lowered by another disorder (opportunistic infection). According to some experts, MAC infection is an almost inevitable complication of HIV. The infection is caused by one of two similar organisms, *M. avium* and *M. intracellulare*.

AIDS patients are almost always attacked by these mycobacteria. Once inside the body, the atypical mycobacterial organisms colonize and grow in the lungs like tuberculosis. Because AIDS patients have a poorly functioning immune system, the microbes multiply because they aren't stopped by the body's normal response to infection. Once they have colonized the lungs, the organisms enter the bloodstream and spread throughout the body, affecting almost every organ. These devastating infections can invade the lymph nodes, liver, spleen, bone marrow, gastrointestinal tract, skin, and brain.

Symptoms include **shortness of breath**, **fever**, night sweats, weight loss, appetite loss, **fatigue**, and progressively severe **diarrhea**, stomach **pain**, **nausea and vomiting**. If the infection spreads to the brain, the patient may experience weakness, headaches, vision problems, and loss of balance.

MAC and *M. kansasii* sometimes cause lung infections in middle-aged and elderly people with chronic lung conditions. MAC, *M. kansasii*, and *M. scrofulaceum* may cause inflammation of the lymph nodes in otherwise healthy young children. *M. fortuitum* and *M. chelonae* cause skin and wound infections and abscesses after trauma or surgical procedures. *M. marinum* causes a nodular inflammation, usually on the arms and legs. This infection is called "swimming pool granuloma" because it is associated with swimming pools, fish tanks, and other bodies of water. *M. ulcerans* infection causes chronic skin ulcerations, usually on an arm or leg. Atypical mycobacteria infections can also occur without causing any symptoms. In such cases, a **tuberculin skin test** may be positive.

Diagnosis

The diagnosis is made from the patient's symptoms and organisms grown in culture from the site of infection. In cases of lung infection, a diagnostic workup will include a **chest x ray** and tests on discharges from the respiratory passages (sputum).

KEY TERMS

Culture—A test in which a sample of body fluid, such as prostatic fluid, is placed on materials specially formulated to grow microorganisms. A culture is used to learn what type of bacterium is causing infection.

Human immunodeficiency virus (HIV)—The virus that causes AIDS.

Treatment

These nontypical mycobacteria are not easy to treat in any patient and the problem is complicated when the person has AIDS. **Antibiotics** are not particularly effective, although rifabutin (a cousin of the anti-tuberculosis drug rifampin) and clofazimine (an anti-leprosy drug) have helped some patients. It is also possible to contain the infection to some degree by combining different drugs, including ethionamide, cycloserine, ethambutol, and streptomycin.

Prognosis

Because drug therapy is not easily effective, the overwhelming infections caused by these mycobacteria in AIDS patients can be fatal.

Prevention

People with HIV infection can prevent or delay the onset of MAC by taking disease-preventing drugs such as rifabutin.

AIDS patients and persons with tissue damage, such as skin **wounds** or pulmonary disease, can make a number of lifestyle changes to help prevent MAC infection. Since these mycobacteria are found in most city water systems, in hospital water supplies, and in bottled water, at-risk persons should boil drinking water. Persons at risk should also avoid raw foods, especially salads, root vegetables, and unpasteurized milk or cheese. Fruits and vegetables should be peeled and rinsed thoroughly. Conventional cooking (baking, boiling or steaming) destroys mycobacteria, which are killed at 176°F (80°C).

Finally, at-risk patients should avoid contact with animals, especially birds and bird droppings. Pigeons in particular can transmit MAC.

Resources

ORGANIZATIONS

National AIDS Treatment Advocacy Project. 580 Broadway, Ste. 403, New York, NY 10012. (888) 266-2827. < <http://www.natap.org> >.

Carol A. Turkington

Mycobacterium leprae infection see **Leprosy**

Mycobacterium tuberculosis see **Tuberculosis**

Mycoplasma infections

Definition

Mycoplasma are the smallest of the free-living organisms. (Unlike viruses, mycoplasma can reproduce outside of living cells.) Many species within the genus *Mycoplasma* thrive as parasites in human, bird, and animal hosts. Some species can cause disease in humans.

Description

Mycoplasma are found most often on the surfaces of mucous membranes. They can cause chronic inflammatory diseases of the respiratory system, urogenital tract, and joints. The most common human illnesses caused by mycoplasma are due to infection with *M. pneumoniae*, which is responsible for 10-20% of all pneumonias. This type of **pneumonia** is also called atypical pneumonia, walking pneumonia, or community-acquired pneumonia. Infection moves easily among people in close contact because it is spread primarily when infected droplets circulate in the air (that is, become aerosolized), usually due to coughing, spitting, or sneezing.

Causes and symptoms

Atypical pneumonias can affect otherwise healthy people who have close contact with one another. Pneumonia caused by *M. pneumoniae* may start out with symptoms of an upper respiratory infection, probably a **sore throat** progressing to a dry **cough** within a few days. Gradually, **fever**, **fatigue**, muscle aches, and a cough that produces thin sputum (spit or phlegm) will emerge. Nonrespiratory symptoms may occur too: abdominal **pain**, **headache**, and **diarrhea**; about 20% of patients may have ear pain.

KEY TERMS

Community-acquired—Refers to an infectious disease that is passed among individuals who have close contact with one another.

Another mycoplasma species, *M. hominis*, is common in the mucous membranes of the genital area (including the cervix), and can cause infection in both males and females. Its presence does not always result in symptoms.

Diagnosis

Usually, mycoplasma pneumonia will be identified after other common diagnoses are set aside. For example, a type of antibiotic known as a beta-lactam might be prescribed for a respiratory infection producing fever and cough. If symptoms do not improve in 3-5 days, the organism causing the disease is not a typical one and not susceptible to these **antibiotics**. If a Gram's stain (a common test done on sputum) does not indicate a gram-positive pathogen, the doctor will suspect a gram-negative organism, such as mycoplasma. The actual underlying organism may not be identified (it is not in almost 50% of cases of atypical pneumonia). Although it is rare, a rash may appear along with pneumonia symptoms. This should trigger suspicion of mycoplasma pneumonia, even if laboratory tests are inconclusive.

Standard x rays may reveal a patchy material that has entered the tissue; this can be evident for months. Laboratory tests include cold agglutinins, complement fixation, culture, and enzyme immunoassay. The presence of infection with *M. pneumoniae* would be indicated by a fourfold rise in *M. pneumoniae*-specific antibody in serum, during the illness or convalescence. Highly sophisticated and specific polymerase chain reaction methods (PCR) have been developed for many respiratory pathogens, including *M. pneumoniae*. They are not readily available and are very expensive.

Treatment

A 2-3 week course of certain antibiotics (erythromycin, azithromycin, clarithromycin, dirithromycin, or doxycycline) is generally prescribed for atypical pneumonia. This disease is infectious for weeks, even after the patient starts antibiotics. A persistent cough may linger for 6 weeks.

Prognosis

Mycoplasma pneumonia may be involved in the onset of **asthma** in adults; other rare complications include meningoencephalitis, **Guillain-Barré syndrome**, mononeuritis multiplex, **myocarditis**, or **pericarditis**. This may increase the risk of acute **arrhythmias** leading to **sudden cardiac death**. However, with proper treatment and rest, recovery should be complete.

Prevention

At this time, there are no vaccines for mycoplasma infection. It is difficult to control its spread, especially in a group setting. The best measures are still the simplest ones. Avoid exposure to people with respiratory infections whenever possible. A person who has a respiratory infection should cover the face while coughing or sneezing.

Resources

BOOKS

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Jill S. Lasker

Mycoplasmal pneumonia see **Mycoplasma infections**

Myelocytic leukemia, acute see **Leukemias, acute**

Myelodysplastic syndrome

Definition

Myelodysplastic syndrome (MDS) is a disease that is associated with decreased production of blood cells. Blood cells are produced in the bone marrow, and the blood cells of people with MDS do not mature normally. There are three major types of blood cells — red blood cells, white blood cells and platelets. Patients with MDS can have decreased production of one, two, or all three types of blood cells.

Description

Blood cells are used in the body for many different and important functions, such as carrying oxygen (red

blood cells), fighting infection (white blood cells), and controlling bleeding (platelets). Blood cells are formed and stored in the bone marrow, which is the spongy tissue inside large bones. Stem cells, or immature blood cells, are stored in the bone marrow and have the ability to develop into all three types of mature blood cells. When the body needs a specific type of blood cell, the bone marrow uses its stockpile of stem cells to produce the kind of mature cells needed for that particular situation.

In patients who have MDS, blood cells fail to mature normally. In other words, the bone marrow is unable to develop a normal amount of mature blood cells, and is also not able to increase blood cell production when mature cells are needed. Sometimes, even the cells that are produced do not function normally. The marrow eventually becomes filled with the immature cells and there is not room for the normal cells to grow and develop. MDS therefore causes a shortage of functional blood cells.

Subtypes of MDS

MDS is divided into five different subtypes that are classified according to the number and appearance of blast cells in the bone marrow. It is important for doctors to know the type of MDS a patient has, because each subtype affects patients differently and requires specific treatment. The International Prognostic Scoring System (IPSS) can help the doctor to determine the best treatment for an individual patient. The subtypes are as follows:

- Refractory anemia (RA). Bone marrow with less than 5% blast cells and abnormal red blood cell blasts
- Refractory anemia with ring sideroblasts (RARS). Bone marrow with less than 5% blasts and characteristic abnormalities in red blood cells
- Refractory anemia with excess blasts (RAEB). Bone marrow with 5-20% blast cells, and higher risk of changing into acute leukemia over time
- Refractory anemia with excess blasts in transformation (RAEBT). Bone marrow with 21-30% blast cells. This form is most likely to change into acute leukemia.
- Chronic myelomonocytic leukemia (CMML). Marrow with 5-20% blasts and excess monocytes (a specific type of white blood cell).

Approximately 15,000 new cases are diagnosed annually in the United States. The average age at diagnosis is 70. The most common types are RA and

RARS. It is rare to have MDS before age 50. MDS is slightly more common in males than in females.

Causes and symptoms

Causes

There is no clear cause for the majority of MDS cases, which is referred to as primary or *de novo* myelodysplastic syndrome. In some cases, however, MDS results from earlier **cancer** treatments such as radiation and/or **chemotherapy**. This type of MDS is called secondary or treatment related MDS, is often seen 3 to 7 years after the exposure, and usually occurs in younger people.

Other possible causative agents for MDS include exposure to radiation, cigarette smoke or toxic chemicals such as benzene. Children with pre-existing chromosomal abnormalities such as **Down syndrome** have a higher risk of developing MDS. MDS does not appear to run in families, nor can it be spread to other individuals.

Symptoms

MDS symptoms are related to the type of blood cells that the body is lacking. The earliest symptoms are usually due to anemia, which results from a shortage of mature red blood cells. Anemia causes patients to feel tired and out of breath because there is a lack of cells transporting oxygen throughout the body. MDS may also lead to a shortage of white blood cells resulting in an increased likelihood of infections. Another symptom of MDS is increased bleeding (e.g., blood in stool, nose bleeds, increased **bruises** or bleeding gums) which is due to low level of platelets. These symptoms can occur in any combination, depending on a given patient's specific subtype of MDS.

Diagnosis

Blood tests

People who have MDS usually visit their primary care doctor first, with symptoms of **fatigue**, and are then referred to a hematologist (a physician who specializes in diseases of the blood). The diagnosis of MDS requires a complete analysis of the patient's blood and bone marrow, which is done by the hematologist. A complete **blood count** (CBC) is done to determine the number of each blood cell type within the sample. Low numbers of red blood cells, white blood cells, and or platelets are signs that a patient has MDS. Numerous other medical problems such as bleeding, nutritional deficiencies, or adverse reaction to a medication can also cause low blood counts. The

hematologist will investigate other causes for low blood counts before assigning a diagnosis of MDS. Blood cells in patients with MDS can also be abnormal when viewed under the microscope.

Bone marrow aspiration and biopsy

A **bone marrow biopsy** is required to confirm the diagnosis of MDS and determine the correct MDS subtype. This procedure involves a needle used to take a sample of marrow from inside the bone. The area of the skin where the needle is inserted is numbed and sometimes the patient is also sedated. Patients may experience some discomfort but the procedure is safe and is over fairly quickly. Marrow samples are usually taken from the back of the hip bone (iliac crest). A sample of the marrow, known as an aspirate, and a small piece of bone are both removed with the needle.

A hematologist or a pathologist (a specialist in diagnosing diseases through cell examination) will carefully examine the bone marrow sample through a microscope. Microscopic examination allows the doctor to determine the number and type of blast cells (immature cells) within the marrow in order to identify the MDS subtype. Cells from the bone marrow are also sent for cytogenetic testing, which analyzes the cells' chromosomes. Forty to seventy percent of MDS patients have abnormal bone marrow chromosomes as a result of the disease. The pattern of these abnormalities can be used to predict how a patient will respond to a particular treatment. Thus, the full set of information provided by a bone marrow biopsy and CBC will ultimately allow the doctor to recommend the most effective treatment plan.

International Prognostic Scoring System (IPSS) for MDS

Once a diagnosis of MDS is established, the doctor will calculate the IPSS score for each individual patient. The bone marrow blast percentage, chromosomal abnormalities and number of different blood types that are reduced determine the score. A score of 0 to 3.5 is assigned to each patient. Patients with lower score have a better prognosis and usually should not undertake treatment upon initial diagnosis. Patients with a higher score have more aggressive disease and should consider more aggressive treatment.

Treatments

Supportive care

Treatment for MDS is tailored to the patient's age, general health, specific MDS subtype, and IPSS

score. Treatment varies for each patient, but most treatment strategies are designed to control the symptoms of MDS. This approach is called supportive care and aims to improve the patient's quality of life.

Supportive care for the MDS patients commonly includes red blood cell transfusions to relieve symptoms related to anemia. Red cell transfusions are relatively safe and the physician will review risks and benefits with this approach. Transfusions of any type only last a certain amount of time and therefore need to be repeated at certain intervals. Platelet transfusions can also be a way to control excessive bleeding. The doctor will decide with each individual patient when it is appropriate to give a **transfusion**. **Antibiotics** are used when needed to combat infections that can occur more frequently in patients with low white blood cell counts.

Bone marrow transplantation

Bone marrow transplantation (BMT) is a type of treatment that attempts to provide MDS patients with a cure. This strategy requires the patients to be in fairly good health and are therefore more likely to be used in younger patients. Bone marrow transplantation (BMT) has been found to be a successful treatment for MDS patients under the age of 50 (and some over 50 in good health). Following BMT, many patients are able to achieve long-term, disease-free survival. Unfortunately, most MDS patients cannot receive a traditional bone marrow transplant because of older age or because they do not have a suitable donor. Bone marrow donors are usually siblings or are obtained from the national bone marrow registry. "Mini"-bone marrow transplants use less intense chemotherapy, and are currently being tested in older patients who would otherwise not be candidates for traditional bone marrow transplants.

Chemotherapy

Chemotherapy has been used to treat some MDS patients; however, the disease often recurs after a period of time. This type of therapy uses cell-killing drugs that may also damage healthy cells in the body. Most chemotherapy drugs are associated with some side effects. For these reasons, chemotherapy is generally not used until the MDS becomes more aggressive or the patient has a high IPSS score.

Growth factors

Growth factors are natural proteins that the body normally uses to control blood production. These substances stimulate the patient's bone marrow to

produce healthy blood cells. Growth factors that stimulate white cell production are G-CSF (also called neutropen or filgrastim) and GM-CSF (Leukine, sargramostim). In order to increase red cell production another growth factor, erythropoietin (Procrit) is used. These growth factors are safe with few side effects and are available only in the injectable form. The physician will decide if this treatment is appropriate for an individual patient.

Alternative treatment

There are no alternative therapies that have been proven to successfully treat MDS. Some of the available alternative drugs can have adverse side effects and therefore a physician should be informed if they are being used.

Prognosis

The prognosis for MDS patients depends on the subtype of their disease and the IPSS score. Patients with RA, RARS or low IPSS score rarely develop leukemia and may live with disease for some years. The higher-risk patients including those with RAEB, RAEBt, CMMoL or high IPSS scores progress more rapidly, and require intensive therapy to control the disease.

Managing MDS requires frequent doctor appointments to monitor disease progression and to evaluate the response to treatment. Fortunately for many patients, recent advances in therapy have significantly enhanced their ability to cope with MDS. Experimental drugs and a better understanding of the disease are likely to improve the overall prognosis in the future.

Prevention

MDS is usually impossible to prevent. Being careful about daily activities and avoiding the use of aspirin-like products that thin the blood may prevent secondary complications of MDS such as bruising and bleeding. Practicing good hygiene and avoiding crowds or people with infections can sometimes prevent infections. A well balanced diet is recommended to increase overall energy.

Resources

BOOKS

Aguayo, Alvaro, Jorge Cortes, and Hagop Kantarjian. "Myelodysplastic Syndromes." In *Cancer Management: A Multidisciplinary Approach*, edited by Richard Pazdur, et al., 4th ed. PRR, Inc, 2000.

ORGANIZATIONS

Aplastic Anemia Foundation of America. P.O. Box 613, Annapolis, MD 21404. (800) 747-2820. <<http://www.aplastic.org>>.

Leukemia Society of America. 600 Third Avenue, New York, NY 10016. (800) 955-4572. <<http://www.leukemia.org>>.

Myelodysplastic Syndromes Foundation. 464 Main Street, P.O. Box 477, Crosswicks, NJ 08515. (800) MDS-0839. <<http://www.mds-foundation.org>>.

Andrea Ruskin, M.D.

Myelofibrosis

Definition

Myelofibrosis is a rare disease of the bone marrow in which collagen builds up fibrous scar tissue inside the marrow cavity. This is caused by the uncontrolled growth of a blood cell precursor, which results in the accumulation of scar tissue in bone marrow. Myelofibrosis goes by many names including idiopathic myelofibrosis, agnogenic myeloid metaplasia, chronic myelosclerosis, aleukemic megakaryocytic myelosis, and leukoerythroblastosis.

Description

Myelofibrosis can be associated with many other conditions including **breast cancer**, **prostate cancer**, **Hodgkin's disease**, non-Hodgkin's lymphoma, acute myeloid leukemia, acute lymphocytic leukemia, **hairy cell leukemia**, **multiple myeloma**, myeloproliferative diseases, **tuberculosis**, Gaucher's disease, and **Paget's disease of bone**. Myelofibrosis typically becomes progressively worse and can cause **death**.

In myelofibrosis, abnormal cells (hematopoietic stem cells) grow out of control and begin to produce both immature blood cells and excess scar (fibrous) tissue. The fibrous tissue builds up (fibrosis) primarily in the bone marrow, the place where blood cells are produced. The fibrous tissue interferes with the production of normal blood cells. The outcome of this is that the blood made by the bone marrow is of poor quality. To compensate for this, blood cell production occurs in other parts of the body (extramedullary hematopoiesis), but most notably in the spleen and liver. This causes enlargement of the spleen (splenomegaly) and the liver (hepatomegaly). Extramedullary hematopoiesis is not effective and, combined with the

reduced production of blood cells by the bone marrow, a condition called anemia results.

The abnormal stem cells can spread throughout the body, settle in other organs, and form tumors that produce more abnormal blood cells and fibrous tissue. These tumors are most commonly found in the adrenals, kidneys, lymph nodes, breast, lungs, skin, bowel, thymus, thyroid, prostate, and urinary tract.

Most patients with myelofibrosis are over 50 years old; the average age at diagnosis is 65 years. However, myelofibrosis can occur at any age. Myelofibrosis occurs with equal frequency in women and men, but in children it affects girls twice as often as it does boys.

Causes and symptoms

Myelofibrosis is caused by an abnormality in a single stem cell, which causes it to grow out of control. Myelofibrosis tumors that have originated from a single cell are called monoclonal. The cause of the stem cell abnormality is unknown. Persons who were exposed to benzene or high doses of radiation have developed myelofibrosis. There may be an association between myelofibrosis and autoimmune diseases, such as **systemic lupus erythematosus** and **scleroderma**, in which the immune system treats certain molecules of the body as foreign invaders.

Symptoms usually appear slowly over a long period of time. About one quarter of all patients with myelofibrosis have no symptoms (asymptomatic). An enlarged spleen discovered at an annual medical examination may be the first clue. Symptoms of myelofibrosis include:

- fatigue
- weight loss
- paleness
- fever
- sweating
- weakness
- heart palpitations
- shortness of breath
- itchiness
- feeling full after eating a small amount of food
- stomach **pain** or discomfort
- pain in the left shoulder or upper left portion of the body
- unexpected bleeding
- bone pain, especially in the legs

Diagnosis

Because symptoms are similar to other diseases (mostly leukemias), myelofibrosis is not easy to diagnose. The doctor would use his or her hands to feel (palpate) for enlargement of the spleen and liver. Blood tests and urine tests would be performed. **Bone marrow aspiration and biopsy** can help make a diagnosis, but they often fail because of the fibrosis. X-ray imaging and **magnetic resonance imaging (MRI)** may be performed.

Treatment

Many asymptomatic patients, if stable, do not require treatment. There is no cure for myelofibrosis, although **bone marrow transplantation** is curative in some cases. Treatment is aimed at reducing symptoms and improving quality of life.

Medications

Male hormones (androgens) can be used to treat anemia but, in women, these drugs can cause the development of male characteristics (e.g., hair growth on the face and body). Glucocorticoid therapy is also an effective treatment of anemia and can improve myelofibrosis in children. Nutrients that stimulate blood formation (hematinics), such as iron, **folic acid**, and vitamin B₁₂, may reduce anemia. **Cancer chemotherapy** (usually hydroxyurea) can decrease splenomegaly and hepatomegaly, reduce symptoms of myelofibrosis, lessen anemia, and sometimes reduce bone marrow fibrosis. The bone marrow of myelofibrosis patients is often not strong enough to withstand the harsh chemotherapy drugs, so this treatment is not always an option. Interferon-alpha has been shown to reduce spleen size, reduce bone pain, and, in some cases, increase the number of blood platelets (structures involved in blood clotting).

Other treatments

In certain cases, the enlarged spleen may be removed (**splenectomy**). Conditions that warrant splenectomy include spleen pain, the need for frequent blood **transfusion**, very low levels of platelets (**thrombocytopenia**), and extreme pressure in the blood vessels of the liver (portal **hypertension**).

Radiation therapy is used to treat splenomegaly, spleen pain, bone pain, tumors in certain places such as next to the spinal cord, and fluid accumulation inside the abdomen (**ascites**). Patients who are not

KEY TERMS

Anemia—Low numbers of red blood cells in the blood.

Benzene—A colorless volatile flammable toxic liquid hydrocarbon used as a solvent and as a motor fuel.

Biopsy—Surgical removal of tissue for microscopic examination.

Fibrosis—Buildup of scar tissue.

Glucocorticoid therapy—Treatment using corticoids that are anti-inflammatory and immunosuppressive.

Leukemia—Cancer of white blood cells.

Portal hypertension—Extreme pressure on the blood vessels of the liver.

Stem cell—A cell that has the ability to become many different specialized cells.

strong enough to undergo splenectomy are often treated with radiation therapy.

Bone marrow transplantation may be used to treat some patients with myelofibrosis. This procedure may be performed on patients who are less than 50 years old, have a poor life expectancy, and have a brother or sister with blood-type similarities.

Patients with severe anemia may require blood transfusions.

Prognosis

Similar to leukemias, myelofibrosis is progressive and often requires therapy to control the disease. Myelofibrosis can progress to acute lymphocytic leukemia or lymphoma. Although a number of factors to predict the survival time have been proposed, advanced age or severe anemia are consistently associated with a poor prognosis. The average survival rate of patients diagnosed with myelofibrosis is five years. Death is usually caused by infection, bleeding, complications of splenectomy, **heart failure**, or progression to leukemia. Spontaneous remission is rare.

Prevention

Persons who have been exposed to radiation, benzene, or radioactive thorium dioxide (a chemical used during certain diagnostic radiological procedures) are at risk for myelofibrosis.

Resources

BOOKS

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Belinda Rowland, Ph.D.
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Myelogram see **Myelography**

Myelography

Definition

Myelography is an x-ray examination of the spinal canal. A contrast agent is injected through a needle into the space around the spinal cord to display the spinal cord, spinal canal, and nerve roots on an x ray.

Purpose

The purpose of a myelogram is to evaluate the spinal cord and/or nerve roots for suspected compression. Pressure on these delicate structures causes **pain** or other symptoms. A myelogram is performed when precise detail about the spinal cord is needed to make a definitive diagnosis. In most cases, myelography is used after other studies, such as **magnetic resonance imaging** (MRI) or a computed tomography scan (CT scan), have not yielded enough information to be sure of the disease process. Sometimes myelography followed by CT scan is an alternative for patients who cannot have an MRI scan, because they have a pacemaker or other implanted metallic device.

A herniated or ruptured intervertebral disc, popularly known as a slipped disc, is one of the most common causes for pressure on the spinal cord or nerve roots. Discs are pads of fiber and cartilage that contain rubbery tissue. They lie between the vertebrae, or individual bones, which make up the spine. Discs act as cushions, accommodating **strains**, shocks, and position changes. A disc may rupture suddenly, due to

injury, or a sudden straining with the spine in an unnatural position. In other cases, the problem may come on gradually as a result of progressive deterioration of the discs with **aging**. The lower back is the most common area for this problem, but it sometimes occurs in the neck, and rarely in the upper back. A myelogram can help accurately locate the disc or discs involved.

Myelography may be used when a tumor is suspected. Tumors can originate in the spinal cord, or in tissues surrounding the cord. Cancers that have started in other parts of the body may spread or metastasize in the spine. It is important to precisely locate the mass causing pressure, so effective treatment can be undertaken. Patients with known **cancer** who develop back pain may require a myelogram for evaluation.

Other conditions that may be diagnosed using myelography include arthritic bony growths, known as spurs, narrowing of the spinal canal, called **spinal stenosis**, or malformations of the spine.

Precautions

Patients who are unable to lie still or cooperate with positioning should not have this examination. Severe congenital spinal abnormalities may make the examination technically difficult to carry out. Patients with a history of severe allergic reaction to contrast material (x-ray dye) should report this to their physician. Pretreatment with medications to minimize the risk of severe reaction may be recommended.

Description

Myelograms can be performed in a hospital x-ray department or in an outpatient radiology facility. The patient lies on the x-ray table on his or her stomach. The radiologist first looks at the spine under fluoroscopy, where the images appear on a monitor screen. This is done to find the best location to position the needle. The skin is cleaned, then numbed with local anesthetic. The needle is inserted. Occasionally, a small amount of cerebrospinal fluid, the clear fluid which surrounds the spinal cord and brain, may be withdrawn through the needle and sent for laboratory studies. Then contrast material is injected. The contrast material (dye) is a liquid that shows up on x rays.

The x-ray table is tilted slowly. This allows the contrast material to reach different levels in the spinal canal. The flow is observed under fluoroscopy, then x rays are taken with the table tilted at various angles. A footrest and shoulder straps or supports will keep the patient from sliding.

In many instances, a CT scan of the spine will be performed immediately after a myelogram, while the contrast material is still in the spinal canal. This helps outline internal structures most clearly.

A myelogram takes approximately 30-60 minutes. A CT scan adds about another hour to the examination. If the procedure is done as an outpatient exam, some facilities prefer the patient to stay in a recovery area for up to four hours.

Preparation

Patients should be well hydrated at the time of a myelogram. Increasing fluids the day before the study is usually recommended. All food and fluid intake should be stopped approximately four hours before the myelogram.

Certain medications may need to be stopped for one to two days before myelography is performed. These include some antipsychotics, antidepressants, blood thinners, and diabetic medications. Patients should consult with their physician and/or the facility where the study is to be done.

Patients who smoke may be asked to stop the day before the test. This helps decrease the chance of **nausea** or headaches after the myelogram. Immediately before the examination, patients should empty their bowels and bladder.

Aftercare

After the examination is completed, the patient usually rests for several hours, with the head elevated. Extra fluids are encouraged, to help eliminate the contrast material and prevent headaches. A regular diet and routine medications may be resumed. Strenuous physical activity, especially any which involve bending over, may be discouraged for one or two days. The doctor should be notified if a **fever**, excessive **nausea and vomiting**, severe **headache**, or stiff neck develops.

Risks

Headache is a common complication of myelography. It may begin several hours to several days after the examination. The cause is thought to be changes in cerebrospinal fluid pressure, not a reaction to the dye. The headache may be mild and easily alleviated with rest and increased fluids. Sometimes, nonprescription medicine are recommended. In some instances, the headache may be more severe and require stronger medication or other measures for relief. Many factors influence whether the patient develops this problem.

KEY TERMS

Contrast agent—Also called a contrast medium, this is usually a barium or iodine dye that is injected into the area under investigation. The dye makes the interior body parts more visible on an x-ray film.

These include the type of needle used and the age and sex of the patient. Patients with a history of chronic or recurrent headache are more likely to develop a headache after a myelogram.

The chance of reaction to the contrast material is a very small, but potentially significant risk with myelography. It is estimated that only 5-10% of patients experience any effect from contrast exposure. The vast majority of reactions are mild, such as sneezing, nausea, or **anxiety**. These usually resolve by themselves. A moderate reaction, like **wheezing** or **hives**, may be treated with medication, but is not considered life threatening. Severe reactions, such as heart or **respiratory failure**, happen very infrequently. These require emergency medical treatment.

Rare complications of myelography include injury to the nerve roots from the needle, or from bleeding into the spaces around the roots. Inflammation of the delicate covering of the spinal cord, called arachnoiditis, or infections, can also occur. Seizures are another very uncommon complication reported after myelography.

Normal results

A normal myelogram would show a spinal canal of normal width, with no areas of constriction or obstruction.

Abnormal results

A myelogram may reveal a **herniated disk**, tumor, bone spurs, or narrowing of the spinal canal (spinal stenosis).

Resources

ORGANIZATIONS

Spine Center. 1911 Arch St., Philadelphia, PA 19103. (215) 665-8300. <<http://www.thespinecenter.com>>.

Ellen S. Weber, MSN

Myeloma see **Multiple myeloma**

Myers-Briggs type indicator

Definition

The Myers-Briggs Type Indicator (MBTI) is a widely-used personality inventory, or test, employed in vocational, educational, and psychotherapy settings to evaluate personality type in adolescents and adults age 14 and older.

Purpose

In an educational setting, the MBTI may be performed to assess student learning style. Career counselors use the test to help others determine what occupational field they might be best suited for, and it is also used in organizational settings to assess management skills and facilitate teamwork and problem-solving, including communication difficulties. Because the MBTI is also a tool for self-discovery, mental health professionals may administer the test in counseling sessions to provide their patients with insight into their behavior.

As of the early 2000s, the MBTI is also being used in the mental health field to assess vulnerability to **anxiety disorders** and depression. Preliminary results indicate that some of the 16 types are more susceptible to **mood disorders** than others. ISFPs, for example, are overrepresented among patients in treatment for unipolar depression, while the four ST types appear to be more vulnerable to **anxiety** states.

Precautions

The MBTI should be administered, scored, and interpreted only by a professional trained in its use. Cultural and language differences in the test subject may affect performance and may result in inaccurate test results. The test administrator should be informed before testing begins if the test taker is not fluent in English and/or he has a unique cultural background.

Description

In 2000, an estimated two million people took the MBTI, making it the most frequently used personality inventory available. The test was first introduced in 1942, the work of a mother and daughter, Katharine Cook Briggs and Isabel Briggs Myers. There are now several different versions of the test available. Form M, which contains 93 items, is the most commonly used.

The Myers-Briggs inventory is based on Carl Jung's theory of types, outlined in his 1921 work *Psychological Types*. Jung's theory holds that human

beings are either *introverts* or *extraverts*, and their behavior follows from these inborn psychological types. He also believed that people take in and process information different ways, based on their personality traits.

The Myers-Briggs evaluates personality type and preference based on the four Jungian psychological types:

- extraversion (E) or introversion (I)
- sensing (S) or intuition (N)
- thinking (T) or feeling (F)
- judging (J) or perceiving (P)

Preparation

Prior to the administration of the MBTI, the test subject should be fully informed about the nature of the test and its intended use. He or she should also receive standardized instructions for taking the test and any information on the confidentiality of the results.

Normal results

Myers-Briggs results are reported as a four-letter personality type (e.g., ESTP, ISFJ). Each letter corresponds to an individual's preference in each of the four pairs of personality indicators (i.e., E or I, S or N, T or F, and J or P). There are a total of sixteen possible combinations of personality types on the MBTI.

Letter One: E or I

Extraverts focus more on people and things in the outside world, introverts on internal thoughts and ideas.

Letter Two: S or N

Sensing dominant personalities prefer to perceive things through sight, sound, taste, touch, and smell, while intuition dominant types look to past experience and are more abstract in their thinking.

Letter Three: T or F

The third subtype is a measure of how people use judgment. Thinking types use logic to judge the world, while feeling types tend to view things on the basis of what emotions they elicit.

Letter Four: J or P

Everyone judges and perceives, but those who are judging dominant are said to be more methodical and

KEY TERMS

Multitasking—Performing multiple duties or taking on multiple responsibilities and roles simultaneously.

Vocational—Relating to an occupation, career, or job.

results-oriented, while perceiving dominant personalities are good at multitasking and are flexible.

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Paula Anne Ford-Martin
Rebecca J. Frey, PhD

Myocardial biopsy

Definition

Myocardial biopsy is a procedure wherein a small portion of tissue is removed from the heart muscle for testing. This test is also known as endomyocardial biopsy.

Purpose

The main reason for a biopsy is to secure tissue samples that will be useful in the diagnosis, treatment, and care of heart muscle disorders. The test is also used to detect rejection after a **heart transplantation** procedure.

Precautions

This procedure is not used when the patient is taking blood-thinning medication (anticoagulant therapy). It should not be done when the patient has leukemia and **aplastic anemia** or if there is a blood clot on the interior wall of the heart.

Description

A long, flexible tube, called a catheter, is inserted into a vein and threaded up into the heart. The doctor can guide the catheter by watching its movement on a TV monitor showing an x-ray image of the area. The tip of the catheter is fitted with tiny jaws that the doctor can open and close. Once the catheter is in place, the doctor will take several small snips of muscle for microscopic examination.

Preparation

Preparation for myocardial biopsy is quite extensive. The patient will be asked not to eat for several hours before the procedure. A technician will shave the hair from the area of the incision and will also insert an intravenous line in the arm. The patient will be given a sedative to relax but will not be fully anesthetized. The patient will be connected to an electrocardiograph (ECG) to monitor the heart, and a blood-pressure cuff will be placed. Finally, the patient will be covered with sterile drapes, so that the area of the biopsy is kept free of germs. The cardiologist will numb the area where the catheter will be inserted.

Aftercare

At the end of the biopsy, the catheter will be removed and pressure will be applied at the site where it entered the blood vessel in order to encourage healing. The patient will then be taken to the recovery room. It is advisable to remain flat and not to move about for 6-8 hours. After that time, most people begin walking around. Swelling and bruising at the puncture site are common and usually go away without need for further attention.

Risks

The risks involved with myocardial biopsy are small because the patient is monitored closely and attended by well-trained staff. Racing of the heart (**palpitations**) and quivering of the heart muscles (atrial fibrillation) are both possible during the procedure.

KEY TERMS

Anticoagulant—Medication that thins the blood and slows clot formation.

Aplastic anemia—A greatly decreased production of all of the formed elements of the blood caused by a failure of the cell-generating capacity of the bone marrow.

Electrocardiography—A test that uses electrodes attached to the chest with an adhesive gel to transmit the electrical impulses of the heart muscle to a recording device.

Leukemia—A disease characterized by an increasing number of abnormal cells in the blood.

Resources

ORGANIZATIONS

American Heart Association, 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300. <<http://www.americanheart.org>>.

Dorothy Elinor Stonely

Myocardial infarction see **Heart attack**

Myocardial resection

Definition

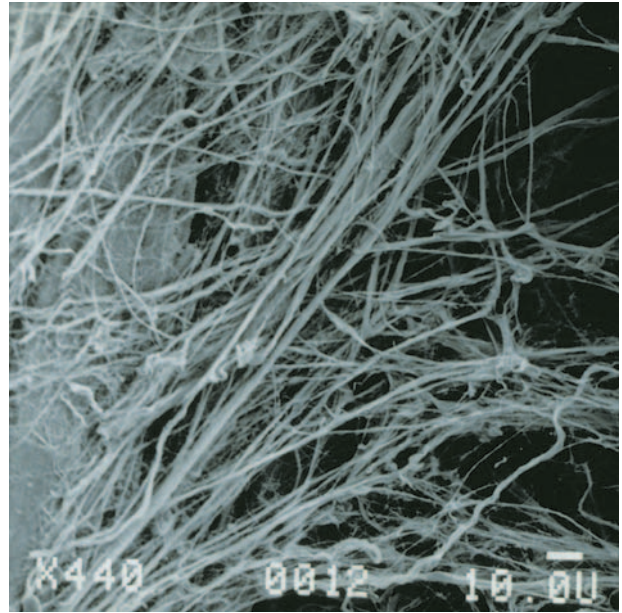
Myocardial resection is a surgical procedure in which a portion of the heart muscle is removed.

Purpose

Myocardial resection is done to improve the stability of the heart function or rhythm. Also known as endocardial resection, this open-heart surgery is done to destroy or remove damaged areas of the heart that cause life-threatening heart rhythms. This procedure is often performed in people who have had a **heart attack**, in order to prevent future rapid heart rates. It is also used in people who have **Wolff-Parkinson-White syndrome** (a condition resulting in abnormal heart rhythm).

Precautions

This is major surgery and should be the treatment of choice only after medications have failed and the



Once the catheter is threaded up into the heart, the surgeon will take several small samples of muscle for laboratory analysis. (Custom Medical Stock Photo. Reproduced by permission.)

use of an **implantable cardioverter-defibrillator** (a device that delivers electrical shock to control heart rhythm) has been ruled out.

Description

After receiving a general anesthetic, an incision will be made in the chest to expose the heart. When the exact source of the abnormal rhythm is identified, it is removed. If there are areas around the source that may contribute to the problem, they can be frozen with a special probe to further insure against dangerous heart rates. The amount of tissue removed is so small, usually only 2 or 3 millimeters, that there is no damage to the structure of the heart. On some occasions, aneurysms of the heart wall are removed as well.

Preparation

Prior to surgery, the physician will explain the procedure, routine blood tests will be completed, and consent forms will be signed.

Aftercare

Immediately after surgery, the patient will be moved to a recovery room until the affects of anesthesia have worn off. The patient will then be transferred to the intensive care unit for further recovery. In the

KEY TERMS

Implantable cardioverter-defibrillator—A device placed in the body to deliver an electrical shock to the heart in response to a serious abnormal rhythm.

Wolff-Parkinson-White syndrome—An abnormal, rapid heart rhythm, due to an extra pathway for the electrical impulses to travel from the atria to the ventricles.

intensive care unit, the heart will be monitored for any disturbances in rhythm and the patient will be watched for any signs of post-operative problems.

Risks

The risks of myocardial resection are based in large part on the person's underlying heart condition and, therefore, vary greatly. The procedure involves opening the heart, so the person is at risk for the complications associated with major heart surgery such as **stroke**, shock, infection, and hemorrhage.

Normal results

Anywhere from 5-25% of post-heart attack patients do not survive open-heart surgery. The survivors have a 90% arrhythmia-free one-year survival rate, (arrhythmia is an irregular heart beat).

Resources

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300. <<http://www.americanheart.org>>.

Dorothy Elinor Stonely

Myocarditis

Definition

Myocarditis is an inflammatory disease of the heart muscle (myocardium) that can result from a variety of causes. While most cases are produced by a viral infection, an inflammation of the heart muscle may also be instigated by toxins, drugs, and hypersensitive immune reactions. Myocarditis is a rare but

serious condition that affects both males and females of any age.

Description

Most cases of myocarditis in the United States originate from a virus, and the disease may remain undiagnosed by doctors due to its general lack of initial symptoms. The disease may also present itself as an acute, catastrophic illness that requires immediate treatment. Although the inflammation or degeneration of the heart muscle that myocarditis causes may be fatal, this disease often goes undetected. It may also disguise itself as ischemic, valvular, or hypertensive heart disease.

An inflammation of the heart muscle may occur as an isolated disorder or be the dominating feature of a systemic disease (one that affects the whole body, like **systemic lupus erythematosus**).

Causes and symptoms

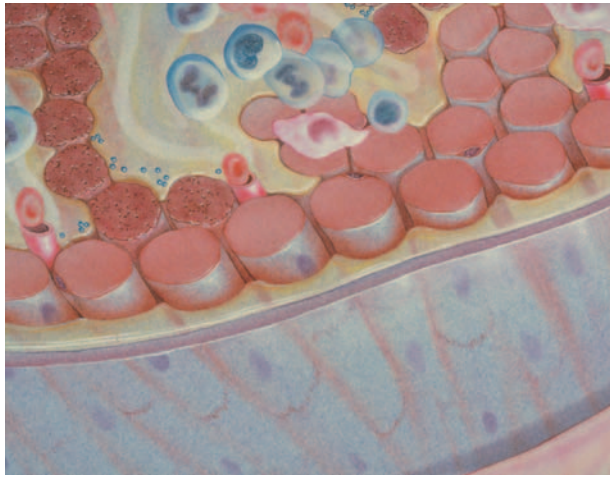
While there are several contributing factors that may lead to myocarditis, the primary cause is viral. Myocarditis usually results from the Coxsackie B virus, and may also result from **measles**, **influenza**, chicken pox, hepatitis virus, or the adenovirus in children. If an acute onset of severe myocarditis occurs, a patient may display the following symptoms:

- Rhythm disturbances of the heart
- Rapid heartbeat (**Ventricular tachycardia**)
- Left or right ventricular enlargement
- **Shortness of breath** (Dyspnea)
- Pulmonary **edema** (the accumulation of fluid in the lungs due to left-sided **heart failure**)
- Swollen legs.

Additional causes of myocarditis include:

- Bacterial infections, such as **tetanus**, **gonorrhea**, or **tuberculosis**
- Parasite infections, such as **Chagas' disease** (which is caused by an insect-borne protozoan most commonly seen in Central and South America)
- Rheumatic fever
- Surgery on the heart
- Radiation therapy for **cancer** that is localized in the chest, such as breast or lung cancer
- Certain medications.

As of 1996, research has shown that illegal drugs and toxic substances may also produce acute or



This illustration depicts the inflammation of the myocarditis, the middle muscular layer of the heart wall. (Custom Medical Stock Photo. Reproduced by permission.)

chronic injury to the myocardium. These studies also indicate an increase in the incidence of toxic results from the use of **cocaine**. This illegal drug causes coronary artery spasm, myocardial infarction (**heart attack**), and **arrhythmias**, as well as myocarditis.

Further studies conducted in 1996 indicate that **malnutrition** encourages the Coxsackie B virus to flourish, leading to the potential development of myocarditis. Human **immunodeficiency** virus (HIV) is also now recognized as a cause of myocarditis, though its prevalence is not known.

Symptoms of myocarditis may start as **fatigue**, shortness of breath, fever and aching of the joints, all characteristic of a flu-like illness. In contrast to this type of mild appearance, myocarditis may also appear suddenly in the form of heart failure, or **sudden cardiac death** without any prior symptoms. If an inflammation of the heart muscle leads to congestive heart failure, symptoms such as swollen feet and ankles, distended neck veins, a rapid heartbeat, and difficulty breathing while reclining may all appear.

Diagnosis

The best way to diagnose myocarditis may be through a person's observation of his or her own symptoms, followed by a thorough medical history and physical exam conducted by a doctor. Further tests usually include laboratory blood studies and **echocardiography**. An electrocardiogram (ECG) is also routinely used due to its ability to detect a mild case of the disease. **Cardiac catheterization** and **angiography** are additional diagnostic tests used to

determine the presence of myocarditis, or to rule out other possible heart diseases that may lead to heart failure.

Another measure used to diagnosis myocarditis is the endomyocardial biopsy procedure. This invasive catheterization procedure examines a biopsied, or "snipped," piece of the endocardium (the lining membrane of the inner surface of the heart). The tissue sample is examined to verify the presence of the disease, as well as to try to determine the infective cause. An approach used only with a patient's consent, this procedure may also confirm acute myocarditis, allowing close monitoring of potential congestive heart failure.

Treatment

While myocarditis is a serious condition, there is no medical treatment necessary if it results from a general viral infection. The only steps to recovery include rest and avoidance of physical exertion. Adequate rest becomes more important to recovery if the case is severe myocarditis with signs of dilated **cardiomyopathy** (disease of the heart muscles). In this case, medical treatment for congestive heart failure may include the following medications: angiotensin converting enzyme (ACE) inhibitors, **diuretics** to reduce fluid retention, digitalis to stimulate a stronger heartbeat, and low-dose beta-blockers.

If myocarditis is caused by a bacterial infection, the disease is treated with **antibiotics** to fight the infection. If severe rhythm disturbances are involved, cardiac assist devices, an "artificial heart," or **heart transplantation** may be the only option for complete recovery.

Prognosis

The outlook for a diagnosed case of myocarditis caused by a viral infection is excellent, with many cases healing themselves spontaneously. Severe or acute myocarditis may be controlled with medication to prevent heart failure. Because this disease may be mild or may be extreme and cause serious arrhythmias, the prognosis varies. Cases of myocarditis may vary from complete healing (with or without significant scarring), to severe congestive heart failure leading to **death** or requiring a heart transplant.

Inflammation of the myocardium may also cause acute **pericarditis** (inflammation of the outer lining of the heart). Due to the potential effects of the disease, including sudden death, it is imperative that proper medical attention is obtained.

KEY TERMS

Adenovirus—One type of virus that can cause upper respiratory tract infections.

Angiography—A procedure which uses x ray after injecting a radiopaque substance to examine the blood vessels and lymphatics.

Arrhythmia—An irregular heartbeat or action.

Cardiac catheterization—A diagnostic procedure that gives a comprehensive examination of how the heart and its blood vessels function; performed by inserting one or more catheters through a peripheral blood vessel in the arm or leg.

Coxsackie B virus—A mild virus belonging to a group of viruses (coxsackievirus) that may produce a variety of illnesses, including myocarditis.

Echocardiography—A noninvasive diagnostic procedure that uses ultrasound to examine internal cardiac structures.

Electrocardiogram—A record of the electrical activity of the heart, with each wave being labeled as P, Q, R, S, and T waves. Often used in the diagnosis of cases of abnormal cardiac rhythm and myocardial damage.

Hypertensive heart disease—High blood pressure resulting in a disease of the heart.

Ischemic heart disease—Insufficient blood supply to the heart muscle (myocardium).

Valvular heart disease—A disease of any one of the four valves that controls blood flow into, through, and out of the heart.

Ventricular tachycardia—An abnormally rapid heartbeat. It includes a series of at least three beats arising from a ventricular area at a rate of more than 100 beats per minute, usually ranging from 150-200 beats per minute.

Prevention

Although myocarditis is an unpredictable disease, the following measures may help prevent its onset. Individuals should:

- Take extra measures to avoid infections, and obtain appropriate treatment for infections.
- Limit alcohol consumption to no more than one or two drinks a day, if any.
- Maintain current immunizations against **diphtheria**, tetanus, measles, **rubella**, and **polio**.

- Avoid anything that may cause the abnormal heart to work too hard, including salt and vigorous exercise.

Resources

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300. <<http://www.americanheart.org>>.

National Heart, Lung and Blood Institute. P.O. Box 30105, Bethesda, MD 20824-0105. (301) 251-1222. <<http://www.nhlbi.nih.gov>>.

Beth A. Kapes

Myoglobin test

Definition

Myoglobin is a protein found in muscle. Myoglobin tests are done to evaluate a person who has symptoms of a **heart attack** (myocardial infarction) or other muscle damage.

Purpose

Myoglobin holds oxygen inside heart and skeletal muscle (muscles that attach to and move bones). It is continually released into the blood in small amounts due to normal turnover of muscle cells. Kidneys discard the myoglobin into urine.

When muscle is damaged, as in a heart attack, larger amounts of myoglobin are released and blood levels rise rapidly. Myoglobin is one of the first tests done to determine if a person with chest **pain** is having a heart attack, as it may be one of the first blood tests to become abnormal.

Damage or injury to skeletal muscle also causes myoglobin to be released into the blood.

Description

Heart attack must be diagnosed quickly. Medications to prevent heart damage are effective only within a limited number of hours. Yet, because of their risk for excessive bleeding, these medications are given only after a diagnosis of heart attack is made.

Myoglobin is one of several cardiac markers used to make the diagnosis. Cardiac markers are substances in blood whose levels rise in the hours following a

heart attack. Increased levels help diagnose a heart attack; persistent normal levels rule it out.

Each cardiac marker rises, peaks, and returns to a normal level according to its own timeline, or diagnostic window. Myoglobin is useful because it has the earliest diagnostic window. It is the first marker to rise after chest pain begins. Myoglobin levels rise within two to three hours, and sometimes as early as 30 minutes. They peak after six to nine hours. The levels return to normal within 24-36 hours.

Although a rise in myoglobin supports a diagnosis of heart attack, it is not conclusive. Simultaneous skeletal muscle damage could also cause the increase. Myoglobin rules out, rather than proves, a diagnosis in the following way. If myoglobin levels have not risen after more than five hours, a heart attack is unlikely. Normal levels in the first two to three hours do not rule out an infarction.

The myoglobin test is sometimes repeated every one to two hours to watch for the rise and peak. Results are available within 30 minutes.

Myoglobin in large amounts is toxic to the kidney. When a person has high amounts of myoglobin in the blood, kidney function must be monitored.

Preparation

This test requires 5 ml of blood. Collection of the sample takes only a few minutes. A urine myoglobin test requires 1 ml of urine collected into a urine collection cup.

Aftercare

Discomfort or bruising may occur at the puncture site or the person may feel dizzy or faint. Pressure to the puncture site until the bleeding stops reduces bruising. Warm packs to the puncture site relieve discomfort.

Normal results

Normal results vary based on the laboratory and method used.

Abnormal results

Myoglobin levels and levels of other cardiac markers are usually considered before finally confirming a diagnosis of heart attack. A level that has doubled after one to two hours, even if the level is still in the normal range, indicates a significant rise that may be due to heart attack.

KEY TERMS

Cardiac marker—A substance in the blood that rises following a heart attack.

Diagnostic window—A cardiac marker's timeline for rising, peaking, and returning to normal after a heart attack.

Myoglobin—A protein that holds oxygen in heart and skeletal muscle. It rises after damage to either of these muscle types.

Increased levels are also found with skeletal muscle damage or disease, such as an injury, **muscular dystrophy**, or **polymyositis**. Myoglobin levels also rise during renal failure because kidneys lose their ability to clear myoglobin from blood.

Resources

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Nancy J. Nordenson

Myomas see **Uterine fibroids**

Myomectomy

Definition

Myomectomy is the removal of fibroids (noncancerous tumors) from the wall of the uterus. Myomectomy is the preferred treatment for symptomatic fibroids in women who want to keep their uterus. Larger fibroids must be removed with an abdominal incision, but small fibroids can be taken out using **laparoscopy** or **hysteroscopy**.

Purpose

A myomectomy can remove **uterine fibroids** that are causing symptoms. It is an alternative to surgical removal of the whole uterus (**hysterectomy**). The procedure can relieve fibroid-induced menstrual symptoms that have not responded to medication. Myomectomy also may be an effective treatment for **infertility** caused by the presence of fibroids.

Precautions

There is a risk that removal of the fibroids may lead to such severe bleeding that the uterus itself will have to be removed. Because of the risk of blood loss during a myomectomy, patients may want to consider banking their own blood before surgery.

Description

Usually, fibroids are buried in the outer wall of the uterus and abdominal surgery is required. If they are on the inner wall of the uterus, uterine fibroids can be removed using hysteroscopy. If they are on a stalk (pedunculated) on the outer surface of the uterus, laparoscopy can be performed.

Removing fibroids through abdominal surgery is a more difficult and slightly more risky operation than a hysterectomy. This is because the uterus bleeds from the sites where the fibroids were, and it may be difficult or impossible to stop the bleeding. This surgery is usually performed under **general anesthesia**, although some patients may be given a spinal or epidural anesthesia.

The incision may be horizontal (the “bikini” incision) or a vertical incision from the navel downward. After separating the muscle layers underneath the skin, the surgeon makes an opening in the abdominal wall. Next, the surgeon makes an incision over each fibroid, grasping and pulling out each growth.

Every opening in the uterine wall is then stitched with sutures. The uterus must be meticulously repaired in order to eliminate potential sites of bleeding or infection. Then, the surgeon sutures the abdominal wall and muscle layers above it with absorbable stitches, and closes the skin with clips or nonabsorbable stitches.

When appropriate, a laparoscopic myomectomy may be performed. In this procedure, the surgeon removes fibroids with the help of a viewing tube (laparoscope) inserted into the pelvic cavity through an incision in the navel. The fibroids are removed through a tiny incision under the navel that is much smaller than the 4 or 5 inch opening required for a standard myomectomy.

If the fibroids are small and located on the inner surface of the uterus, they can be removed with a thin telescope-like device called a hysteroscope. The hysteroscope is inserted into the vagina through the cervix and into the uterus. This procedure does not require any abdominal incision, so hospitalization is shorter.

KEY TERMS

Epidural anesthesia—A method of pain relief for surgery in which local anesthetic is injected into the epidural space in the middle and lower back.

Preparation

Surgeons often recommend hormone treatment with a drug called leuprolide (Lupron) two to six months before surgery in order to shrink the fibroids. This makes the fibroids easier to remove. In addition, Lupron stops menstruation, so women who are anemic have an opportunity to build up their **blood count**. While the drug treatment may reduce the risk of excess blood loss during surgery, there is a small risk that temporarily-smaller fibroids might be missed during myomectomy, only to enlarge later after the surgery is completed.

Aftercare

Patients may need four to six weeks of recovery following a standard myomectomy before they can return to normal activities. Women who have had laparoscopic or hysteroscopic myomectomies, however, can leave the hospital the day after surgery and usually recovery completely within two to three days to one to three weeks.

Risks

The risks of a myomectomy performed by a skilled surgeon are about the same as hysterectomy (one of the most common and safest surgeries). Removing multiple fibroids is more difficult and slightly more risky.

Possible complications include:

- Infection.
- Blood loss.
- The wall of the uterus may be weakened if the removal of a large fibroid leaves a wound that extends the complete thickness of the wall. Special precautions may be needed in future pregnancies. For example, the delivery may need to be performed surgically (Caesarean section).
- Adverse reactions to anesthesia.
- Internal scarring (and possible infertility).

Since fibroids tend to appear and grow as a woman ages (until **menopause**), it is possible that new fibroids will appear after myomectomy.

Resources

OTHER

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Carol A. Turkington

Myopathies

Definition

Myopathies are diseases of skeletal muscle which are not caused by nerve disorders. These diseases cause the skeletal or voluntary muscles to become weak or wasted.

Description

There are many different types of myopathies, some of which are inherited, some inflammatory, and some caused by endocrine problems. Myopathies are rare and not usually fatal. Typically, effects are mild, largely causing muscle weakness and movement problems, and many are transitory. Only rarely will patients become dependent on a wheelchair. However, **muscular dystrophy** (which is technically a form of myopathy) is far more severe. Some types of this disease are fatal in early adulthood.

Causes and symptoms

Myopathies are usually degenerative, but they are sometimes caused by drug side effects, chemical **poisoning**, or a chronic disorder of the immune system.

Genetic myopathies

Among their many functions, genes are responsible for overseeing the production of proteins important in maintaining healthy cells. Muscle cells produce thousands of proteins. With each of the inherited myopathies, a genetic defect is linked to a lack of, or problem with, one of the proteins needed for normal muscle cell function.

There are several different kinds of myopathy caused by defective genes:

- Central core disease
- Centronuclear (myotubular) myopathy
- Myotonia congenita
- Nemaline myopathy
- Paramyotonia congenita
- Periodic **paralysis** (hypokalemic and hyperkalemic forms)
- Mitochondrial myopathies.

Most of these genetic myopathies are dominant, which means that a child needs to inherit only one copy of the defective gene from one parent in order to have the disease. The parent with the defective gene also has the disorder, and each of this parent's children has a 50% chance of also inheriting the disease. Male and female children are equally at risk.

However, one form of myotonia congenita and some forms of nemaline myopathy must be inherited from both parents, each of whom carry a recessive defective gene but who do not have symptoms of the disease. Each child of such parents has a 25% chance of inheriting both genes and showing signs of the disease, and a 50% chance of inheriting one defective gene from only one parent. If the child inherited just one defective gene, he or she would be a carrier but would not show signs of the disease.

A few forms of centronuclear myopathy develop primarily in males. Females who inherit the defective gene are usually carriers without symptoms, like their mothers, but they can pass on the disease to their sons. Mitochondrial myopathies are inherited through the mother, since sperm do not contain mitochondria. (Mitochondria play a key role in energy production in the body's cells.)

The major symptoms associated with the genetic myopathies include:

- Central core disease: mild weakness of voluntary muscles, especially in the hips and legs; hip displacement; delays in reaching developmental motor milestones; problems with running, jumping, and climbing stairs develop in childhood
- Centronuclear myopathy: weakness of voluntary muscles including those on the face, arms, legs, and trunk; drooping upper eyelids; facial weakness; foot drop; affected muscles almost always lack reflexes
- Myotonia congenita: voluntary muscles of the arms, legs, and face are stiff or slow to relax after contracting (myotonia); stiffness triggered by **fatigue**, **stress**, cold, or long rest periods, such as a night's sleep; stiffness can be relieved by repeated movement of the affected muscles
- Nemaline myopathy: moderate weakness of voluntary muscles in the arms, legs, and trunk; mild weakness of facial muscles; delays in reaching developmental

motor milestones; decreased or absent reflexes in affected muscles; long, narrow face; high-arched palate; jaw projects beyond upper part of the face

- Paramyotonia congenita: stiffness (myotonia) of voluntary muscles in the face, hands, and forearms; attacks spontaneous or triggered by cold temperatures; stiffness made worse by repeated movement; episodes of stiffness last longer than those seen in myotonia congenita
- Periodic paralysis: attacks of temporary muscle weakness (muscles work normally between attacks); in the hypokalemic (low calcium) form, attacks triggered by vigorous **exercise**, heavy meals (high in carbohydrates), insulin, stress, alcohol, infection, **pregnancy**; in the hyperkalemic (normal/high calcium) form, attacks triggered by vigorous exercise, stress, pregnancy, missing a meal, steroid drugs, high potassium intake
- Mitochondrial myopathies: symptoms vary quite widely with the form of the disease and may include progressive weakness of the eye muscles (ocular myopathy), weakness of the arms and legs, or multisystem problems primarily involving the brain and muscles.

Endocrine-related myopathies

In some cases, myopathies can be caused by a malfunctioning gland (or glands), which produces either too much or too little of the chemical messengers called hormones. Hormones are carried by the blood and one of their many functions is to regulate muscle activity. Problems in producing hormones can lead to muscle weakness.

Hyperthyroid myopathy and hypothyroid myopathy affect different muscles in different ways. Hyperthyroid myopathy occurs when the thyroid gland produces too much thyroxine, leading to muscle weakness, some muscle wasting in hips and shoulders, and, sometimes, problems with eye muscles. The hypothyroid type occurs when too little hormone is produced, leading to stiffness, cramps, and weakness of arm and leg muscles.

Inflammatory myopathies

Some myopathies are inflammatory, leading to inflamed, weakened muscles. Inflammation is a protective response of injured tissues characterized by redness, increased heat, swelling, and/or **pain** in the affected area. Examples of this type include **polymyositis**, **dermatomyositis**, and **myositis** ossificans.

Dermatomyositis is a disease of the connective tissue that also involves weak, tender, inflamed

muscles. In fact, muscle tissue loss may be so severe that the person may be unable to walk. Skin inflammation is also present. The cause is unknown, but viral infection and **antibiotics** are associated with the condition. In some cases, dermatomyositis is associated with rheumatologic disease or **cancer**. Polymyositis involves inflammation of many muscles usually accompanied by deformity, swelling, sleeplessness, pain, sweating, and tension. It, too, may be associated with cancer. Myositis ossificans is a rare inherited disease in which muscle tissue is replaced by bone, beginning in childhood.

Muscular dystrophy

While considered to be a separate group of diseases, the muscular dystrophies also technically involve muscle wasting and can be described as myopathies. These relatively rare diseases appear during childhood and adolescence, and are caused by muscle destruction or degeneration. They are a group of genetic disorders caused by problems in the production of key proteins.

The forms of muscular dystrophy (MD) differ according to the way they are inherited, the age of onset, the muscles they affect, and how fast they progress. The most common type is Duchenne MD, affecting one or two in every 10,000 boys. Other types of MD include Becker's, **myotonic dystrophy**, limb-girdle MD, and facioscapulohumeral MD.

Diagnosis

Early diagnosis of myopathy is important so that the best possible care can be provided as soon as possible. An experienced physician can diagnose a myopathy by evaluating a person's medical history and by performing a thorough physical exam. Diagnostic tests can help differentiate between the different types of myopathy, as well as between myopathy and other neuromuscular disorders. If the doctor suspects a genetic myopathy, a thorough family history will also be taken.

Diagnostic tests the doctor may order include:

- Measurements of potassium in the blood
- Muscle biopsy
- Electromyogram (EMG).

Treatment

Treatment depends on the specific type of myopathy the person has:

KEY TERMS

Electromyogram (EMG)—A diagnostic test that records the electrical activity of muscles. In the test, small electrodes are placed on or in the skin; the patterns of electrical activity are projected on a screen or over a loudspeaker. This procedure is used to test for muscle disorders, including muscular dystrophy.

Inflammation—A protective response of injured tissues characterized by redness, increased heat, swelling, and/or pain in the affected area.

Voluntary muscles—Muscles producing voluntary movement.

- Periodic paralysis: medication and dietary changes
- Hyperthyroid or hypothyroid myopathy: treatment of the underlying thyroid abnormality
- Myositis ossificans: medication may prevent abnormal bone formation, but there is no cure following onset
- Central core disease: no treatment
- Nemaline myopathy: no treatment
- Centronuclear (myotubular) myopathy: no treatment
- Paramyotonia congenita: treatment often unnecessary
- Myotonia congenita: drug treatment (if necessary), but drugs do not affect the underlying disease, and attacks may still occur.

Prognosis

The prognosis for patients with myopathy depends on the type and severity of the individual disease. In most cases, the myopathy can be successfully treated and the patient returned to normal life.

Muscular dystrophy, however, is generally a much more serious condition. Duchenne's MD is usually fatal by the late teens; Becker's MD is less serious and may not be fatal until the 50s.

Resources

ORGANIZATIONS

Muscular Dystrophy Association. 3300 East Sunrise Drive, Tucson, AZ 85718. (800) 572-1717. <<http://www.mdausa.org>>.

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Myopia

Definition

Myopia is the medical term for nearsightedness. People with myopia see objects more clearly when they are close to the eye, while distant objects appear blurred or fuzzy. Reading and close-up work may be clear, but distance vision is blurry.

Description

To understand myopia it is necessary to have a basic knowledge of the main parts of the eye's focusing system: the cornea, the lens, and the retina. The cornea is a tough, transparent, dome-shaped tissue that covers the front of the eye (not to be confused with the white, opaque sclera). The cornea lies in front of the iris (the colored part of the eye). The lens is a transparent, double-convex structure located behind the iris. The retina is a thin membrane that lines the rear of the eyeball. Light-sensitive retinal cells convert incoming light rays into electrical signals that are sent along the optic nerve to the brain, which then interprets the images.

In people with normal vision, parallel light rays enter the eye and are bent by the cornea and lens (a process called refraction) to focus precisely on the retina, providing a crisp, clear image. In the myopic eye, the focusing power of the cornea (the major refracting structure of the eye) and the lens is too great with respect to the length of the eyeball. Light rays are bent too much, and they converge in front of the retina. This inaccuracy is called a refractive error. In other words, an overfocused fuzzy image is sent to the brain.

There are many types of myopia. Some common types include:

- Physiologic
- Pathologic
- Acquired.

By far the most common form, physiologic myopia develops in children sometime between the ages of 5-10 years and gradually progresses until the eye is fully grown. Physiologic myopia may include refractive myopia (the cornea and lens-bending properties are too strong) and axial myopia (the eyeball is too long). Pathologic myopia is a far less common abnormality. This condition begins as physiologic myopia, but rather than stabilizing, the eye continues

to enlarge at an abnormal rate (progressive myopia). This more advanced type of myopia may lead to degenerative changes in the eye (degenerative myopia). Acquired myopia occurs after infancy. This condition may be seen in association with uncontrolled diabetes and certain types of **cataracts**. **Antihypertensive drugs** and other medications can also affect the refractive power of the lens.

Genetic profile

Eyecare professionals have debated the role of genetics in the development of myopia for many years. Some believe that a tendency toward myopia may be inherited, but the actual disorder results from a combination of environmental and genetic factors. Environmental factors include close work; work with computer monitors or other instruments that emit some light (electron microscopes, photographic equipment, lasers, etc.); emotional **stress**; and eye strain.

A variety of genetic patterns for inheriting myopia have been suggested. One explanation for lack of agreement is that the genetic profile of high myopia (defined as a refractive error greater than -6 diopters) may differ from that of low myopia. Some researchers think that high myopia is determined by genetic factors to a greater extent than low myopia.

Another explanation for disagreement regarding the role of heredity in myopia is the sensitivity of the human eye to very small changes in its anatomical structure. Since even small deviations from normal structure cause significant refractive errors, it may be difficult to single out any specific genetic or environmental factor as their cause.

Since 1992, genetic markers that may be associated with genes for myopia have been located on human chromosomes 1, 2, 12, and 18. There is some genetic information on the short arm of chromosome 2 in highly myopic people. Genetic information for low myopia appears to be located on the short arm of chromosome 1, but it is not known whether this information governs the structure of the eye itself or vulnerability to environmental factors.

In 1998 a team of American researchers presented evidence that a gene for familial high myopia with an autosomal dominant transmission pattern could be mapped to human chromosome 18 in eight North American families. The same group also found a second locus for this form of myopia on human chromosome 12 in a large German/Italian family. In 1999 a group of French researchers found no linkage between chromosome 18 and 32 French families with familial high myopia. These findings have been taken to

indicate that more than one gene is involved in the transmission of the disorder.

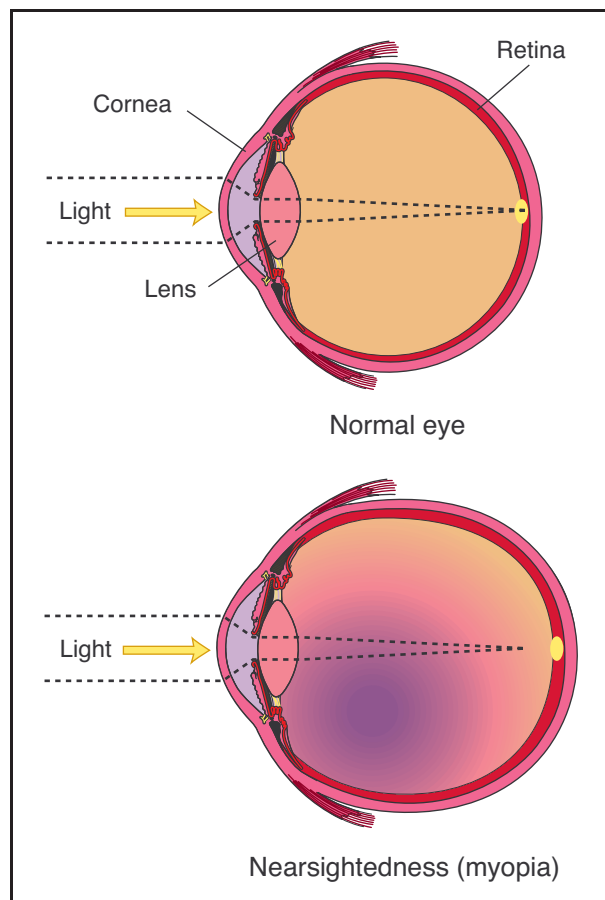
It has been known for some years that a family history of myopia is one of the most important risk factors for developing the condition. Only 6%-15% of children with myopia come from families in which neither parent is myopic. In families with one myopic parent, 23%-40% of the children develop myopia. If both parents are myopic, the rate rises to 33%-60% for their children. One American study found that children with two myopic parents are 6.42 times as likely to develop myopia themselves as children with only one or no myopic parents. The precise interplay of genetic and environmental factors in these family patterns, however, is not yet known.

One multigenerational study of Chinese subjects indicated that subjects in the third generation had a higher risk of developing myopia even if their parents were not myopic. The researchers concluded that, at least in China, the genetic factors in myopia have remained constant over the past three generations while the environmental factors have intensified. The increase in the percentage of people with myopia over the last 50 years in the United States has led American researchers to the same conclusion.

The debate continued with more recent reports. In the summer of 2004, one report said that scientists were close to identifying the myopia gene, located on chromosome 11. Another report reviewed several studies and claimed that lifestyle was to blame for myopia. For instance, a study found that 80% of 14- to 18-year old boys studying in schools in Israel that emphasize reading religious texts have myopia, while the rates for boys in state school was just 30%. It is likely that genes and environment play a role.

Myopia is the most common eye disorder in humans around the world. It affects between 25% and 35% of the adult population in the United States and the developed countries, but is thought to affect as much as 40% of the population in some parts of Asia. Some researchers have found slightly higher rates of myopia in women than in men.

The age distribution of myopia in the United States varies considerably. Five-year-olds have the lowest rate of myopia (less than 5%) of any age group. The prevalence of myopia rises among children and adolescents in school until it reaches the 25%-35% mark in the young adult population. It declines slightly in the over-45 age group; about 20% of 65-year-olds have myopia. The figure drops to 14% for Americans over 70.



Myopia, or nearsightedness, is a condition of the eye in which objects are seen more clearly when close to the eye while distant objects appear blurred or fuzzy. (Illustration by Electronic Illustrators Group.)

Other factors that affect the demographic distribution of myopia are income level and education. The prevalence of myopia is higher among people with above-average incomes and educational attainments. Myopia is also more prevalent among people whose work requires a great deal of close focusing, including work with computers.

Causes and symptoms

Myopia is said to be caused by an elongation of the eyeball. This means that the oblong (as opposed to normal spherical) shape of the myopic eye causes the cornea and lens to focus at a point in front of the retina. A more precise explanation is that there is an inadequate correlation between the focusing power of the cornea and lens and the length of the eye.

People are generally born with a small amount of **hyperopia** (farsightedness), but as the eye grows this

decreases and myopia does not become evident until later. This change is one reason why some researchers think that myopia is an acquired rather than an inherited trait.

The symptoms of myopia are blurred distance vision, eye discomfort, squinting, and eye strain.

Diagnosis

The diagnosis of myopia is typically made during the first several years of elementary school when a teacher notices a child having difficulty seeing the chalkboard, reading, or concentrating. The teacher or school nurse often recommends an **eye examination** by an ophthalmologist or optometrist. An ophthalmologist—M.D. or D.O. (Doctor of Osteopathy)—is a medical doctor trained in the diagnosis and treatment of eye problems. Ophthalmologists also perform eye surgery. An optometrist (O.D.) diagnoses and manages and/or treats eye and visual disorders. In many states, optometrists are licensed to use diagnostic and therapeutic drugs.

A patient's distance vision is tested by reading letters or numbers on a chart posted a set distance away (usually 20 ft). The doctor asks the patient to view images through a variety of lenses to obtain the best correction. The doctor also examines the inside of the eye and the retina. An instrument called a slit lamp is used to examine the cornea and lens. The eyeglass prescription is written in terms of diopters (D), which measure the degree of refractive error. Mild to moderate myopia usually falls between -1.00D and -6.00D. Normal vision is commonly referred to as 20/20 to describe the eye's focusing ability at a distance of 20 ft from an object. For example, 20/50 means that a myopic person must stand 20 ft away from an eye chart to see what a normal person can see at 50 ft. The larger the bottom number, the greater the myopia.

Treatment

People with myopia have three main options for treatment: eyeglasses, contact lenses, and for those who meet certain criteria, refractive eye surgery.

Eyeglasses

Eyeglasses are the most common method used to correct myopia. Concave glass or plastic lenses are placed in frames in front of the eyes. The lenses are ground to the thickness and curvature specified in the eyeglass prescription. The lenses cause the light rays to diverge so that they focus further back, directly on the retina, producing clear distance vision.

Contact lenses

Contact lenses are a second option for treatment. Contact lenses are extremely thin round discs of plastic that are worn on the eye in front of the cornea. Although there may be some initial discomfort, most people quickly grow accustomed to contact lenses. Hard contact lenses, made from a material called PMMA, are virtually obsolete. Rigid gas permeable lenses (RGP) are made of plastic that holds its shape but allows the passage of some oxygen into the eye. Some believe that RGP lenses may halt or slow the progression of myopia because they maintain a constant, gentle pressure that flattens the cornea. As of 2001, the National Eye Institute is conducting an ongoing study of RGP lenses called the Contact Lens and Myopia Progression (CLAMP) Study, with results to be published in 2003.

A procedure called orthokeratology acts on this principle of "corneal molding." However, when contact lenses are discontinued for a period of time, the cornea will generally go back to its original shape. Rigid gas permeable lenses offer crisp, clear, sight. Soft contact lenses are made of flexible plastic and can be up to 80% water. Soft lenses offer increased comfort and the advantage of extended wear; some can be worn continuously for up to one week. While oxygen passes freely through soft lenses, bacterial contamination and other problems can occur, requiring replacement of lenses on a regular basis. It is very important to follow the cleaning and disinfecting regimens prescribed because protein and lipid buildup can occur on the lenses, causing discomfort or increasing the risk of infection. Contact lenses offer several benefits over glasses, including: better vision, less distortion, clear peripheral vision, and cosmetic appeal. In addition, contacts do not steam up from perspiration or changes in temperature.

Refractive eye surgery

For people who find glasses and contact lenses inconvenient or uncomfortable, and who meet selection criteria regarding age, degree of myopia, general health, etc., refractive eye surgery is a third treatment alternative. There are three types of corrective surgeries available as of 2001: 1) **radial keratotomy (RK)**, 2) **photorefractive keratectomy (PRK)**, and 3) **laser-assisted in-situ keratomileusis (LASIK)**. Refractive eye surgery improves myopic vision by permanently changing the shape of the cornea so that light rays focus properly on the retina. These procedures are performed on an outpatient basis and generally take 10-30 minutes. LASIK is approved by the

FDA, though certain lasers are in various stages of approval at a given time. Patients should check with the FDA and ask for patient references when choosing a LASIK provider.

RADIAL KERATOTOMY. Radial keratotomy (RK), the first of these procedures made available, has a high associated risk. It was first developed in Japan and the Soviet Union, and introduced into the United States in 1978. The surgeon uses a delicate diamond-tipped blade, a microscope, and microscopic instruments to make several spoke-like "radial" incisions in the non-viewing (peripheral) portion of the cornea. As the incisions heal, the slits alter the curve of the cornea, making it more flat, which may improve the focus of images onto the retina.

PHOTOREFRACTIVE KERATECTOMY. Photorefractive keratectomy (PRK) involves the use of a computer to measure the shape of the cornea. Using these measurements, the surgeon applies a computer-controlled laser to make modifications to the cornea. The PRK procedure flattens the cornea by vaporizing small amounts of tissue from the cornea's surface. It is important to make sure the laser being used is FDA approved. Photorefractive keratectomy can be used to treat mild to moderate forms of myopia. The cost is approximately \$2,000 per eye.

LASER-ASSISTED IN-SITU KERATOMILEUSIS. Laser-assisted in-situ keratomileusis (LASIK) is the newest of these procedures. It is recommended for moderate to severe cases of myopia. A variation on the PRK method, LASIK uses lasers and a cutting tool called a microkeratome to cut a circular flap on the cornea. The flap is flipped back to expose the inner layers of the cornea. The cornea is treated with a laser to change the shape and focusing properties, then the flap is replaced.

Risks

All of these surgical procedures carry risks, the most serious being corneal scarring, corneal rupture, infection, cataracts, and loss of vision. In addition, a study published in March 2001 warned that mountain climbers who have had LASIK surgery should be aware of possible changes in their vision at high altitudes. The lack of oxygen at high altitudes causes temporary changes in the thickness of the cornea.

Since refractive eye surgery does not guarantee 20/20 vision, it is important to have realistic expectations before choosing this treatment. In a 10-year study conducted by the National Eye Institute between 1983 and 1993, more than 50% of people with radial keratotomy gained 20/20 vision, and 85% passed a

KEY TERMS

Accommodation—The ability of the lens to change its focus from distant to near objects. It is achieved through the action of the ciliary muscles that change the shape of the lens.

Cornea—The transparent structure of the eye over the lens that is continuous with the sclera in forming the outermost, protective, layer of the eye.

Diopter (D)—A unit of measure for describing refractive power.

Laser-assisted in-situ keratomileusis (LASIK)—A procedure that uses a cutting tool and a laser to modify the cornea and correct moderate to high levels of myopia.

Lens—The transparent, elastic, curved structure behind the iris (colored part of the eye) that helps focus light on the retina.

Ophthalmologist—A physician specializing in the medical and surgical treatment of eye disorders.

Optic nerve—A bundle of nerve fibers that carries visual messages from the retina in the form of electrical signals to the brain.

Optometrist—A medical professional who examines and tests the eyes for disease and treats visual disorders by prescribing corrective lenses and/or vision therapy. In many states, optometrists are licensed to use diagnostic and therapeutic drugs to treat certain ocular diseases.

Orthokeratology—A method of reshaping the cornea using a contact lens. It is not considered a permanent method to reduce myopia.

Peripheral vision—The ability to see objects that are not located directly in front of the eye. Peripheral vision allows people to see objects located on the side or edge of their field of vision.

Photorefractive keratectomy (PRK)—A procedure that uses an excimer laser to make modifications to the cornea and permanently correct myopia.

Radial keratotomy (RK)—A surgical procedure involving the use of a diamond-tipped blade to make several spoke-like slits in the peripheral (non-viewing) portion of the cornea to improve the focus of the eye and correct myopia by flattening the cornea.

Refraction—The bending of light rays as they pass from one medium through another. Used to describe the action of the cornea and lens on light rays as they enter the eye. Also used to describe the determination and measurement of the eye's focusing system by an optometrist or ophthalmologist.

Refractive eye surgery—A general term for surgical procedures that can improve or correct refractive errors by permanently changing the shape of the cornea.

Retina—The light-sensitive layer of tissue in the back of the eye that receives and transmits visual signals to the brain through the optic nerve.

Visual acuity—The ability to distinguish details and shapes of objects.

driving test (requiring 20/40 vision) after surgery, without glasses or contact lenses. Even if the patient gains near-perfect vision, however, there are potentially irritating side effects, such as postoperative **pain**, poor night vision, variation in visual acuity, light sensitivity and glare, and optical distortion. Refractive eye surgeries are considered elective procedures and are not always covered by insurance plans.

Myopia treatments under research include corneal implants and permanent surgically placed contact lenses.

Alternative treatments

Some eye care professionals recommend treatments to help improve circulation, reduce eye strain, and relax the eye muscles. It is possible that by

combining exercises with changes in behavior, the progression of myopia may be slowed or prevented. Alternative treatments include: visual therapy (also referred to as **vision training** or eye exercises); discontinuing close work; reducing eye strain (taking a rest break during periods of prolonged near vision tasks); and wearing bifocals to decrease the need to accommodate when doing close-up work.

Prognosis

Glasses and contact lenses can (but not always) correct the patient's vision to 20/20. Refractive surgery can make permanent improvements for the right candidates.

While the genetic factors that influence the transmission and severity of myopia cannot be changed,

some environmental factors can be modified. They include reducing close work; reading and working in good light; taking frequent breaks when working at a computer or microscope for long periods of time; maintaining good **nutrition**; and practicing visual therapy (when recommended).

Eye strain can be prevented by using sufficient light for reading and close work, and by wearing corrective lenses as prescribed. Everyone should have regular eye examinations to see if their prescription has changed or if any other problems have developed. This is particularly important for people with high (degenerative) myopia who are at a greater risk of developing **retinal detachment**, retinal degeneration, **glaucoma**, or other problems.

Resources

PERIODICALS

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- “Scientists close to identifying myopia gene.” *Chemistry and Industry* August 16, 2004: 7.
- Wu, MM, and MH Edwards. “The effect of having myopic parents: an analysis of myopia in three generations.” *Optometry and Visual Science* 76 (June 1999): 341-342.

ORGANIZATIONS

- American Academy of Ophthalmology. PO Box 7424, San Francisco, CA 94120-7424. (415) 561-8500.
< <http://www.eyenet.org> >.
- American Optometric Association. 243 North Lindbergh Blvd., St. Louis, MO 63141. (314) 991-4100.
< <http://www.aoanet.org> >.
- International Myopia Prevention Association. RD No. 5, Box 171, Ligonier, PA 15658. (412) 238-2101.
- Myopia International Research Foundation. 1265 Broadway, Room 608, New York, NY 10001. (212) 684-2777.
- National Eye Institute. Bldg. 31 Rm 6A32, 31 Center Dr., MSC 2510, Bethesda, MD 20892-2510. (301) 496-5248.
2020@nei.nih.gov. < <http://www.nei.nih.gov> >.

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Myositis

Definition

Myositis is a rare disease in which the muscle fibers and skin are inflamed and damaged, resulting in muscle weakness. There are several types of myositis that affect different parts of the body.

Description

The persistent inflammation that is associated with myositis develops slowly over weeks to months and often years, with progressive weakening of the muscles. Later in the course of the disease development muscle wasting or shortening (contracture) may develop. Myositis can range in severity from mild to debilitating.

The forms of myositis include:

- **Polymyositis (PM)** inflames and weakens muscles in many parts of the body, and especially those parts closest to the trunk. With polymyositis, dysphagia (difficulty, discomfort or **pain** in speaking or swallowing), **fatigue**, and pain in the muscles are common. PM rarely affects people under the age of 20, with the peak onset between the ages of 30 and 60.
- **Dermatomyositis (DM)** affects both the muscle fibers and skin by damaging the tiny blood vessels (capillaries) that supply blood to the muscle and skin, resulting in muscle weakness, pain, and fatigue. In addition the affected person develops a distinctive patchy, reddish rash on the eyelids, cheeks, bridge of the nose, back or upper chest, elbows, knees, and knuckles. There may also be hardened, tender bumps (possibly caused by inflammation of fat) under the skin. DM can occur at any age and is more common in females than males.
- **Inclusion Body Myositis (IBM)** typically begins after age 50, and is characterized by gradual weakening of muscles throughout the body, including the wrists or fingers, development of dysphagia, and atrophy of forearms and/or thigh muscles. Unlike the other types of myositis, IBM occurs more often in men than women, and also does not respond very well to drug therapy.
- **Juvenile myositis (JM)** involves muscle weakness, skin rash, and dysphagia in children. A common characteristic of JM is the formation of calcium deposits in the muscle (calcinosis). These deposits are hard and sometimes painful lumps of calcium under the skin that appear on the child's fingers, hands, elbows, and knees. Painful sores may appear if the lumps break through the skin. The child may

also suffer from **contractures**, which is muscle shortening that results in joints staying bent. About half of the children with JM will have pain in their muscles.

Myositis is rare, affecting about 10 in one million people each year. DM and PM affect mostly women in the forties and fifties but men and children can also be affected, some at a young age (between the ages of 5 and 15). About 40,000 people in the United States may have this disease, with about 3,000 to 5,000 children affected.

Causes and symptoms

Myositis is thought to be an autoimmune disease. The body normally fights infections and disease by producing antibodies and white blood cells called lymphocytes in a process called the immune response. In an autoimmune disease, the immune response is overactive, and the immune system attacks and destroys the body's own normal healthy tissues. There is no known cause to the autoimmune response that results in myositis. However, investigators are studying whether the disease is triggered by such environmental agents as the organism that causes **toxoplasmosis**, *Toxoplasma gondii*, the **Lyme disease** organism, *Borrelia burgdorferi*, the coxsackievirus, or by **food allergies**. Some cases of IBM are thought to be inherited.

The first symptoms of most types of myositis are weakness and pain in the muscles of the hips and shoulders. The affected person may have trouble getting up from a chair, lifting the arms above the head, or climbing stairs, and may be too tired to walk or stand. DM and PM mostly affects muscles that are close to and within the trunk of the body, while IBM involves a wider range of muscles. Myositis may make it difficult for the person to speak or swallow. When the disease affects the lungs or chest muscles, the person may have difficulty breathing. If the person has DM, they may develop characteristic **rashes**. Other symptoms may include **fever** and joint pain and swelling.

The first signs of JM is usually a red and patchy skin rash and/or a red or purplish rash on the eyelids or cheeks that look like **allergies**. Weak muscles may develop at the same time as the rash, or may develop days, weeks, or months after the appearance of the rash. Other symptoms of JM include falling, a weaker voice (dysphonia), or dysphagia. Calcinosis usually develops later during the course of the disease.

Diagnosis

Myositis can be a difficult disease to diagnose, because it is rare, because the symptoms develop

slowly, and because it can be mistaken for other diseases causing muscle weakness such as limb-girdle **muscular dystrophy**. Many cases of myositis go undiagnosed for years. The health care provider must rule out other conditions such as **hypothyroidism**, toxin exposure, drug reactions, and genetic disorders that can also affect muscles. The **physical examination** will include a complete medical history focusing on symptoms and when they occurred, and blood tests for autoantibodies and muscle enzymes (for example, creatine kinase (CK), which when present in the blood indicates muscle damage). Specialized tests may also be performed, including:

- an electromyogram, which measures the electrical pattern of the muscles
- a muscle biopsy, in which a small piece of muscle is removed, stained, and examined by microscopic techniques to determine if muscle fibers are damaged and whether the muscle fibers are being infiltrated by cells of the immune system
- magnetic resonance imaging (MRI) to identify areas of muscle inflammation.

Treatment

There is no cure for myositis. However, prompt and aggressive treatment may reduce muscle inflammation and prevent muscle weakness from progressing. Because of the many different kinds of symptoms and a wide range of reactions to different drugs, each person's treatment for myositis should be individualized.

Drugs that are used for treatment include **corticosteroids**, such as prednisone, to reduce inflammation and improve the body's reaction to infections. Corticosteroids are usually taken in the form of pills, but may also be injected. The amount of creatine kinase (CK) levels in the blood are monitored to determine how well the medicine is working. Corticosteroids may produce a number of side effects, such as weight gain, difficulties in fighting infections, psychiatric changes, sleeping troubles, water retention, bone thinning, facial swelling, diabetes, and **cataracts**. Corticosteroid therapy usually leads to improvement in myositis symptoms within two to three months, after which the dose can be lowered to avoid the side effects. If the dose of corticosteroids is going to be reduced, it is essential to lower the dose over a period of time.

Immunosuppressant drugs are used to slow down the immune system's attack on healthy tissue and improve skin rashes. Persons may be prescribed these

drugs to control myositis if they are unable to tolerate corticosteroids or if the corticosteroids are not accomplishing the desired degree of treatment. Immunosuppressant drugs may also be used in conjunction with corticosteroids so that lower doses of corticosteroids can be used. Immunosuppressant drugs include azathioprine, methotrexate, cyclosporine, tacrolimus, etanercept, and mycophenolate mofetil.

Intravenous immunoglobulin (IVIg) appears to aid in improving muscle strength in many persons with myositis, particularly those with DM. It may be less effective in PM, and its role in treating IBM requires more study, although it has been shown to help some patients with IBM if they are diagnosed early. Immunoglobulins are normal proteins in the blood that attack anything foreign in the body, such as viruses and bacteria. IVIg is made from donated blood plasma from people with normal immune systems. Side effects from the use of IVIg include **headache** and flu-like symptoms.

Topical cream or ointment forms of some of the medicines, such as prednisone and tacrolimus, can be used to heal and soothe the rash associated with DM. Non-steroidal anti-inflammatory drugs (NSAIDs) such as **aspirin** or ibuprofen can be used for pain relief. Calcinosis can be treated with prednisone, plaquenil (also called hydroxychloroquine), intravenous immunoglobulin (IVIg), cyclosporine, and methotrexate.

After drug treatment results in improvement, the affected person begins a program of regular stretching exercises to maintain range of motion in the weakened arms and legs. Physical therapy may be used to prevent permanent muscle shortening. Whirlpool baths, heat, and gentle massages may also provide relief. Adequate rest is necessary, and affected persons should take frequent breaks throughout the day and limit their activity.

Patients with throat problems should be evaluated by a speech therapist who can evaluate the swallowing-related problems and make recommendations regarding diet changes and safe swallowing techniques.

Before a woman with myositis becomes pregnant, she should discuss the medicines that she is taking with her health care provider and evaluate the possible risks that she and the baby face if she does become pregnant. Many of the drugs used in the treatment of myositis may be harmful to the fetus or to a breast-fed baby.

It is recommended that a doctor experienced in the treatment of myositis, assisted by a rheumatologist, dermatologist, or neurologist, be consulted. Oftentimes a patient may have to be treated at a

major medical center, where the disease has been seen and treated before.

Alternative treatment

Various supplements may be used in conjunction with traditional treatment to offset side effects of conventional drug treatment. The use of these supplements should be approved by the primary health care provider.

Immunosuppressant drugs such as methotrexate and cyclophosphamide increase the risk of infection, so a healthy well-balanced diet is required. Methotrexate impairs the body's ability to absorb **folic acid**, so foods high in folic acid, such as leafy green vegetables, fruits, and folate-fortified breads and cereals are recommended. The use of folate supplements may also be recommended by the health care provider. **Vitamins C** and **E** can be used to help with the pain and infections associated with calcinosis.

Corticosteroids may have multiple side effects when taken for long periods of time at high doses. Calcium and Vitamin D are recommended to lower the risk of **osteoporosis**, a common side effect of prednisone use. **Hypertension** and fluid retention may be controlled by a diet low in salt. Steroid-induced diabetes (hyperglycemia) can be aided by a diet low in sugar and other simple carbohydrates. Proteinuria, in which the body breaks down protein faster than normal, may mean that more protein should be included in the diet.

Weight gain associated with the use of corticosteroids can be managed by the use of the DASH (Dietary Approach to Stop Hypertension) diet, which is high in fruits, vegetables, dietary fiber, and low-fat dairy products. Information concerning this diet, developed by the National Institutes of Health, can be found at [www.nhlbi.nih.gov/health/public/heart/hbp/dash/]. Weight gain can overtax weakened muscles and should be avoided if possible.

Prognosis

The progression of PM and DM varies from person to person, but the lifespan of an affected person is not usually significantly affected. DM responds more favorably to therapy than PM. Overall, many patients do improve and have a functional recovery. About half of the patients recover and can discontinue treatment within 5 years of beginning treatment. In children the chances of a cure are better than in adults, although some children do suffer a relapse. Of the remaining 50%, about 20% will still have the active

KEY TERMS

Coxsackievirus—Enterovirus causing a disease resembling poliomyelitis but without paralysis

Dysphagia—Medical term for any difficulty, discomfort or pain when swallowing

Limb-girdle muscular dystrophy—An autosomal recessive form of muscular dystrophy that appears anywhere from late childhood to middle age and is characterized by progressive muscular weakness beginning either in the shoulder or pelvic girdle; the disease usually progresses slowly with cardiopulmonary complications in the later stages

disease and will require ongoing treatment, while up to 30% may have some remaining muscle weakness. However, IBM is disabling, and most patients will require the use of an assistive device such as a cane, walker, or wheel chair. The older the patient is when contracting IBM, the more rapidly the disease progresses.

Prevention

There is no known way to prevent myositis.

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- Icon Health Publications. *The Official Patient's Sourcebook on Inclusion Body Myositis: A Revised and Updated Directory for the Internet Age.* San Diego, CA: Icon Health Publications, 2002.

ORGANIZATIONS

- Muscular Dystrophy Association - USA, National Headquarters.* 3300 E. Sunrise Drive, Tucson, AZ 85718. Telephone: (800) 572-1717. Web site: www.mdausa.org/
- Myositis Support Group.* 205 Laurel Road, Athens, TX 75751. Telephone: (903) 675-6825. Fax: (903) 675-6823. Web site: www.myositissupportgroup.org/

The Myositis Association (TMA). 1233 20th St. NW, Suite 402 Washington, DC 20036. Telephone: (202) 887-0082; Fax: (202) 466-8940; Web site: www.myositis.org/

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Judith Sims

Myositis see **Myopathies**

Myotonia atrophica see **Myotonic dystrophy**

Myotonic dystrophy

Definition

Myotonic dystrophy is a progressive disease in which the muscles are weak and are slow to relax after contraction.

Description

Myotonic dystrophy (DM), also called dystrophia myotonica, myotonia atrophica, or Steinert's disease, is a common form of **muscular dystrophy**. DM is an inherited disease, affecting males and females approximately equally. About 30,000 people in the United States are affected. Symptoms may appear at any time from infancy to adulthood. DM causes general weakness, usually beginning in the muscles of the hands, feet, neck, or face. It slowly progresses to involve other muscle groups, including the heart. DM affects a wide variety of other organ systems as well.

A severe form of DM, congenital myotonic dystrophy or Thomsen's disease, may appear in newborns of mothers who have DM. Congenital means that the condition is present from birth. The incidence of congenital myotonic dystrophy is thought to be about 1:20,000.

DM occurs in about 1 per 7,000–8,000 people and has been described in people from all over the world.

Causes and symptoms

The most common type of DM is called DM1 and is caused by a mutation in a gene called myotonic

dystrophy protein kinase (DMPK). The DMPK gene is located on chromosome 19q. When there is a mutation in this gene, a person develops DM1. The specific mutation that causes DM1 is called a trinucleotide repeat expansion.

Some families with symptoms of DM do not have a mutation in the DMPK gene. As of early 2001, scientists have found that the DM in many of these families is caused by a mutation in a gene on chromosome 3. These families are said to have DM2.

Congenital myotonic dystrophy has been linked to a region on chromosome 7 that contains a muscle chloride channel gene.

Trinucleotide repeats

In the DMPK gene, there is a section of the genetic code called a CTG repeat. The letters stand for three nucleotides (complex organic molecules) known as cytosine, thymine, and guanine, and are repeated a certain number of times. In people who have DM1, this sequence of nucleotides is repeated too many times—more than the normal number of 37 times—and thus this section of the gene is too big. This enlarged section of the gene is called a trinucleotide repeat expansion.

People who have repeat numbers in the normal range will not develop DM1 and cannot pass it to their children. Having more than 50 repeats causes DM1. People who have 38–49 repeats have a premutation and will not develop DM1, but can pass DM1 onto their children. Having repeat numbers greater than 1000 causes congenital myotonic dystrophy.

In general, the more repeats in the affected range that someone has, the earlier the age of onset of symptoms and the more severe the symptoms. However, this is a general rule. It is not possible to look at a person's repeat number and predict at what age they will begin to have symptoms or how their condition will progress.

Exactly how the trinucleotide repeat expansion causes myotonia, the inability to relax muscles, is not yet understood. The disease somehow blocks the flow of electrical impulses across the muscle cell membrane. Without proper flow of charged particles, the muscle cannot return to its relaxed state after it has contracted.

Since 2001 it has been discovered that DM2 is caused by a CCTG (cytosine-cytosine-thymine-guanine) expansion on chromosome 3 at locus 3q21, but as of 2004 it is not known how this repeat affects muscle cell function.

Anticipation

Sometimes when a person who has repeat numbers in the affected or premutation range has children, the expansion grows larger. This is called anticipation. A larger expansion can result in an earlier age of onset in children than in their affected parent. Anticipation happens more often when a mother passes DM1 onto her children than when it is passed from the father. Occasionally repeat sizes stay the same or even get smaller when they are passed to a person's children.

Inheritance

DM is inherited through autosomal dominant inheritance. This means that equal numbers of males and females are affected. It also means that only one gene in the pair needs to have the mutation in order for a person to be affected. Since a person only passes one copy of each gene onto their children, there is a 50% or one in two chance that a person who has DM will pass it onto each of their children. This percentage is not changed by results of other pregnancies. A person with a premutation also has a 50%, or one in two, chance of passing the altered gene on to each of their children. However, whether or not their children will develop DM1 depends on whether the trinucleotide repeat becomes further expanded. A person who has repeat numbers in the normal range cannot pass DM1 onto their children.

There is a range in the severity of symptoms in DM and not everyone will have all of the symptoms listed here.

Myotonic dystrophy causes weakness and delayed muscle relaxation called myotonia. Symptoms of DM include facial weakness and a slack jaw, drooping eyelids called **ptosis**, and muscle wasting in the forearms and calves. A person with DM has difficulty relaxing his or her grasp, especially in the cold. DM affects the heart muscle, causing irregularities in the heartbeat. It also affects the muscles of the digestive system, causing **constipation** and other digestive problems. DM may cause **cataracts**, retinal degeneration, low IQ, frontal balding, skin disorders, atrophy of the testicles, and diabetes. It can also cause sleep apnea—a condition in which normal breathing is interrupted during sleep. DM increases the need for sleep and decreases motivation. Severe disabilities do not set in until about 20 years after symptoms begin. Most people with myotonic dystrophy maintain the ability to walk, even late in life.

A severe form of DM, congenital myotonic dystrophy, may appear in newborns of mothers who have DM1. Congenital myotonic dystrophy is marked by

severe weakness, poor sucking and swallowing responses, respiratory difficulty, delayed motor development, and **mental retardation**. **Death** in infancy is common in this type.

Some people who have a trinucleotide repeat expansion in their DMPK gene do not have symptoms or have very mild symptoms that go unnoticed. It is not unusual for a woman to be diagnosed with DM after she has an infant with congenital myotonic dystrophy.

Predictive testing

It is possible to test someone who is at risk for developing DM1 before they are showing symptoms to see whether they inherited an expanded trinucleotide repeat. This is called predictive testing. Predictive testing cannot determine the age of onset that someone will begin to have symptoms, or the course of the disease.

Diagnosis

Diagnosis of DM is not difficult once the disease is considered. However, the true problem may be masked because symptoms can begin at any age, can be mild or severe, and can occur with a wide variety of associated complaints. Diagnosis of DM begins with a careful medical history and a thorough physical exam to determine the distribution of symptoms and to rule out other causes. A family history of DM or unexplained weakness helps to establish the diagnosis.

A definitive diagnosis of DM1 is done by **genetic testing**, usually by taking a small amount of blood. The DNA in the blood cells is examined and the number of repeats in the DMPK gene is determined. Various other tests may be done to help establish the diagnosis, but only rarely would other testing be needed. An electromyogram (EMG) is a test is used to examine the response of the muscles to stimulation. Characteristic changes are seen in DM that helps distinguish it from other muscle diseases. Removing a small piece of muscle tissue for microscopic examination is called a muscle biopsy. DM is marked by characteristic changes in the structure of muscle cells that can be seen on a muscle biopsy. An electrocardiogram could be performed to detect characteristic abnormalities in heart rhythm associated with DM. These symptoms often appear later in the course of the disease.

Prenatal testing

Testing a **pregnancy** to determine whether an unborn child is affected is possible if genetic testing

in a family has identified a DMPK mutation. This can be done at 10–12 weeks gestation by a procedure called **chorionic villus sampling** (CVS) that involves removing a tiny piece of the placenta and analyzing DNA from its cells. It can also be done by **amniocentesis** after 14 weeks gestation by removing a small amount of the amniotic fluid surrounding the baby and analyzing the cells in the fluid. Each of these procedures has a small risk of **miscarriage** associated with it and those who are interested in learning more should check with their doctor or genetic counselor.

There is also another procedure called preimplantation diagnosis that allows a couple to have a child that is unaffected with the genetic condition in their family. This procedure is experimental and not widely available. Those interested in learning more about this procedure should check with their doctor or genetic counselor.

A group of researchers in Houston, Texas, reported in 2004 that they have successfully developed a technique for detecting the CCTG expansion that causes DM2 and estimating the size of the repeat expansion.

Treatment

Myotonic dystrophy cannot be cured, and no treatment can delay its progression. As of the early 2000s there is no standardized treatment for these disorders because the precise reasons for muscle weakness are not yet fully understood. However, many of the symptoms can be treated. Physical therapy can help preserve or increase strength and flexibility in muscles. Ankle and wrist braces can be used to support weakened limbs. Occupational therapy is used to develop tools and techniques to compensate for loss of strength and dexterity. A speech-language pathologist can provide retraining for weakness in the muscles controlling speech and swallowing.

Irregularities in the heartbeat may be treated with medication or a pacemaker. A yearly electrocardiogram is usually recommended to monitor the heartbeat. **Diabetes mellitus** in DM is treated in the same way that it is in the general population. A high-fiber diet can help prevent constipation. **Sleep apnea** may be treated with surgical procedures to open the airways or with nighttime ventilation. Treatment of sleep apnea may reduce drowsiness. Lens replacement surgery is available when cataracts develop. Pregnant woman should be followed by an obstetrician familiar with the particular problems of DM because complications can occur during pregnancy, labor and delivery.

KEY TERMS

Electrocardiogram (ECG, EKG)—A test that uses electrodes attached to the chest with an adhesive gel to transmit the electrical impulses of the heart muscle to a recording device.

Electromyography (EMG)—A test that uses electrodes to record the electrical activity of muscle. The information gathered is used to diagnose neuromuscular disorders.

Muscular dystrophy—A group of inherited diseases characterized by progressive wasting of the muscles.

Nucleotide—Any of a group of organic molecules that link together to form the building blocks of DNA or RNA.

Sleep apnea—Temporary cessation of breathing while sleeping.

Trinucleotide repeat expansion—A sequence of three nucleotides that is repeated too many times in a section of a gene.

Wearing a medical bracelet is advisable. Some emergency medications may have dangerous effects on the heart rhythm in a person with DM. Adverse reactions to **general anesthesia** may also occur.

Prognosis

The course of myotonic dystrophy varies. When symptoms appear earlier in life, disability tends to become more severe. Occasionally people with DM may require a wheelchair later in life. Children with congenital DM usually require special educational programs and physical and occupational therapy. For both types of DM, respiratory infections pose a danger when weakness becomes severe.

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ORGANIZATIONS

Muscular Dystrophy Association. 3300 East Sunrise Dr., Tucson, AZ 85718. (520) 529-2000 or (800) 572-1717. <<http://www.mdausa.org>>.

OTHER

Gene Clinics. <<http://www.geneclinics.org>>.

Myotonic Dystrophy Website. <http://www.umd.necker.fr/myotonic_dystrophy.html>.

NCBI Genes and Disease Web Page. <<http://www.ncbi.nlm.nih.gov/disease/Myotonic.html>>.

Karen M. Krajewski, M.S., C.G.C.
Rebecca J. Frey, Ph.D.

Myringotomy and ear tubes

Definition

Myringotomy is a surgical procedure in which a small incision is made in the eardrum (the tympanic membrane), usually in both ears. The word comes from *myringa*, modern Latin for drum membrane, and *tomē*, Greek for cutting. It is also called myringocentesis, tympanotomy, tympanostomy, or **paracentesis** of the tympanic membrane. Fluid in the middle ear can be sucked out through the incision.

Ear tubes, or tympanostomy tubes, are small tubes, open at both ends, that are inserted into the incisions in the eardrums during myringotomy. They come in various shapes and sizes and are made of plastic, metal, or both. They are left in place until they fall out by themselves or until they are removed by a doctor.

Purpose

Myringotomy with the insertion of ear tubes is an optional treatment for inflammation of the middle ear with fluid collection (effusion), also called glue ear, that

lasts more than three months (chronic **otitis media** with effusion) and does not respond to drug treatment. It is the recommended treatment if the condition lasts four to six months. Effusion is the collection of fluid that escapes from blood vessels or the lymphatic system. In this case, the fluid collects in the middle ear.

Initially, acute inflammation of the middle ear with effusion is treated with one or two courses of **antibiotics**. **Antihistamines** and **decongestants** have been used, but they have not been proven effective unless there is also hay **fever** or some other allergic inflammation that contributes to the problem. Myringotomy with or without the insertion of ear tubes is NOT recommended for initial treatment of otherwise healthy children with middle ear inflammation with effusion.

In about 10% of children, the effusion lasts for three months or longer, when the disease is considered chronic. In children with chronic disease, systemic steroids may help, but the evidence is not clear, and there are risks.

When medical treatment does not stop the effusion after three months in a child who is one to three years old, is otherwise healthy, and has **hearing loss** in both ears, myringotomy with insertion of ear tubes becomes an option. If the effusion lasts for four to six months, myringotomy with insertion of ear tubes is recommended.

The purpose of myringotomy is to relieve symptoms, to restore hearing, to take a sample of the fluid to examine in the laboratory in order to identify any microorganisms present, or to insert ear tubes.

Ear tubes can be inserted into the incision during myringotomy and left there. The eardrum heals around them, securing them in place. They usually fall out on their own in 6-12 months or are removed by a doctor.

While they are in place, they keep the incision from closing, keeping a channel open between the middle ear and the outer ear. This allows fresh air to reach the middle ear, allowing fluid to drain out, and preventing pressure from building up in the middle ear. The patient's hearing returns to normal immediately and the risk of recurrence diminishes.

Parents often report that children talk better, hear better, are less irritable, sleep better, and behave better after myringotomy with the insertion of ear tubes.

Description

The procedure is usually done in an ambulatory surgical unit under **general anesthesia**, although some physicians do it in the office with **sedation** and **local**

anesthesia, especially in older children. The ear is washed, a small incision made in the eardrum, the fluid sucked out, a tube inserted, and the ear packed with cotton to control bleeding.

There has been an effort to design ear tubes that are easier to insert or to remove, and to design tubes that stay in place longer. Therefore, ear tubes come in various shapes and sizes.

Preparation

The child may not have food or water for four to six hours before anesthesia. Antibiotics are usually not needed.

Aftercare

Use of antimicrobial drops is controversial. Water should be kept out of the ear canal until the eardrum is intact. A doctor should be notified if the tubes fall out.

Risks

The risks include:

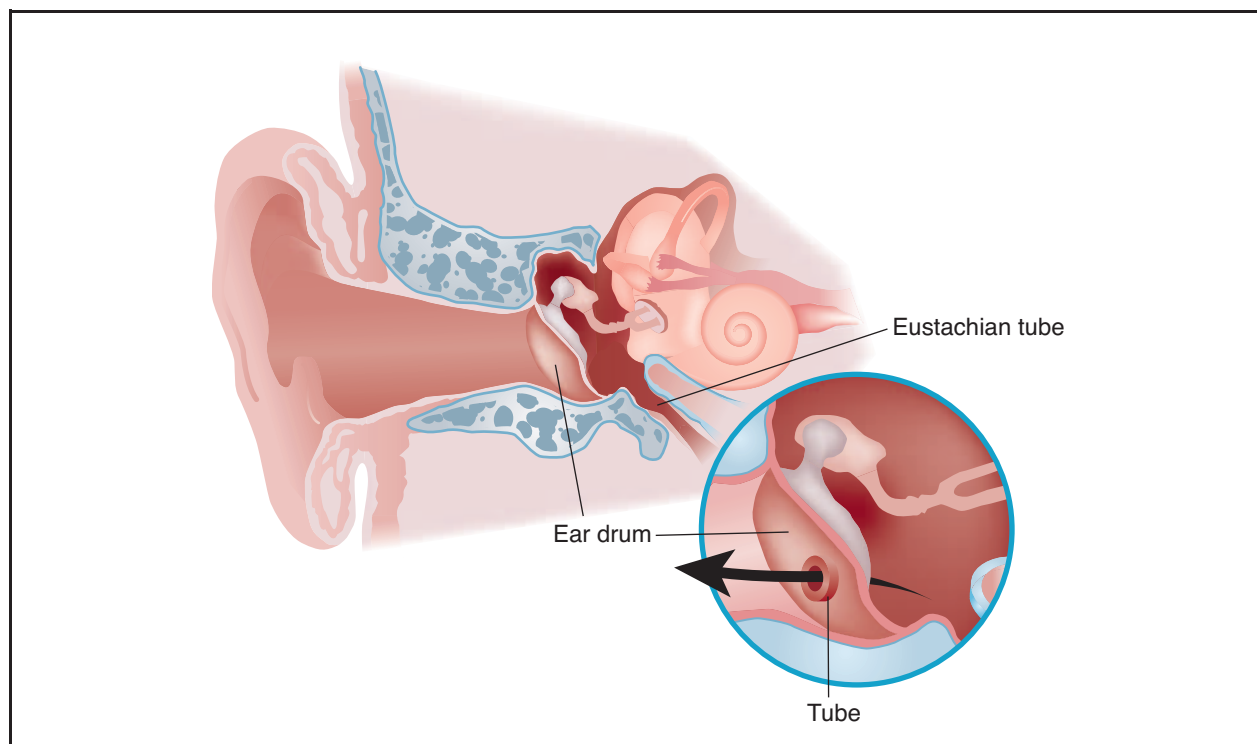
- Cutting the outer ear
- Formation at the myringotomy site of granular nodes due to inflammation
- Formation of a mass of skin cells and cholesterol in the middle ear that can grow and damage surrounding bone (cholesteatoma)
- Permanent perforation of the eardrum.

The risk of persistent discharge from the ear (otorrhea) is 13%.

If the procedure is repeated, structural changes in the eardrum can occur, such as loss of tone (flaccidity), shrinkage or retraction, or hardening of a spot on the eardrum (tympanosclerosis). The risk of hardening is 51%; its effects on hearing are not known, but they are probably insignificant.

It is possible that the incision will not heal properly, leaving a permanent hole in the eardrum, which can cause some hearing loss and increases the risk of infection.

It is also possible that the ear tube will move inward and get trapped in the middle ear, rather than move out into the external ear, where it either falls out on its own or can be retrieved by a doctor. The exact incidence of tubes moving inward is not known, but it could increase the risk of further episodes of middle-ear inflammation, inflammation of the eardrum or the part of the skull directly behind the ear, formation of a



The insertion of ear tubes in the eardrum helps to alleviate chronic middle ear infections. (Illustration by Argosy, Inc.)

mass in the middle ear, or infection due to the presence of a foreign body.

The surgery may not be a permanent cure. As many as 30% of children undergoing myringotomy with insertion of ear tubes need to undergo another procedure within five years.

The other risks include those associated with sedatives or general anesthesia.

An additional element of post-operative care is the recommendation by many doctors that the child use ear plugs to keep water out of the ear during bathing or swimming, to reduce the risk of infection and discharge.

Resources

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Mary Zoll, PhD

Myxedema see **Hypothyroidism**

KEY TERMS

Acute otitis media—Inflammation of the middle ear with signs of infection lasting less than three months.

Chronic otitis media—Inflammation of the middle ear with signs of infection lasting three months or longer.

Effusion—The escape of fluid from blood vessels or the lymphatic system and its collection in a cavity, in this case, the middle ear.

Middle ear—The cavity or space between the eardrum and the inner ear. It includes the eardrum, the three little bones (hammer, anvil, and stirrup) that transmit sound to the inner ear, and the eustachian tube, which connects the inner ear to the nasopharynx (the back of the nose).

Tympanic membrane—The eardrum. A thin disc of tissue that separates the outer ear from the middle ear.

Tympanostomy tube—Ear tube. A small tube made of metal or plastic that is inserted during myringotomy to ventilate the middle ear.

Myxoma

Definition

A myxoma is a rare, usually noncancerous, primary tumor (a new growth of tissue) of the heart. It is the most common of all benign heart tumors.

Description

Myxoma is an intracardiac tumor; it is found inside the heart. Seventy five percent of all myxomas are found in the left atrium, and almost all other myxomas are found in the right atrium. It is very rare for a myxoma to be found in either of the ventricles. The tumor takes one of two general shapes: a round, firm mass, or an irregular shaped, soft, gelatinous mass. They are attached to the endocardium, the inside lining of the heart. The cells that make up the tumor are spindle-shaped cells and are embedded in a matrix rich in mucopolysaccharides (a group of carbohydrates). Myxomas may contain calcium, which shows up on x rays. The tumor gets its blood supply from capillaries that bring blood from the heart to the tumor. Thrombi (**blood clots**) may be attached to the outside of the myxoma.

There are three major syndromes linked to myxomas: embolic events, obstruction of blood flow, and constitutional syndromes. Embolic events happen when fragments of the tumor, or the thrombi attached to the outside of the tumor, are released and enter the blood stream. Gelatinous myxomas are more likely to embolize than the more firm form of this tumor.

Myxomas may also obstruct blood flow in the heart, usually at a heart valve. The mitral valve is the heart valve most commonly affected. Blood flow restrictions can lead to pulmonary congestion and heart valve disease. Embolization can lead to severe consequences. In cases of left atrial myxoma, 40-50% of patients experience embolization. Emboli usually end up in the brain, kidneys, and extremities.

The third syndrome linked to myxomas are called constitutional syndromes, nonspecific symptoms caused by the myxoma.

Causes and symptoms

There is no known causative agent for myxoma. The main symptoms, if any, produced by myxoma are

KEY TERMS

Embolus—A piece of tissue, blood clot, etc. that travels through the blood system and can lodge in smaller blood vessels anywhere in the body.

Metastasis—The spread of a cancer or infectious agent from the site of origin to other areas of the body.

Raynaud's phenomenon—Intermittant ischemia (deficient blood flow) of the fingers or toes, sometimes also affecting the ears and nose.

generic and not specific. These include **fever**, weight loss, anemia, elevated white blood cell (WBC) count, decreased **platelet count** and Raynaud's phenomenon. Most patients with myxoma are between 30-60 years of age.

Diagnosis

Diagnosis is made following a suspicion that a myxoma might be present, and can usually be confirmed by echocardiogram

Treatment

Surgery is used to remove the tumor. Myxomas can regrow if they are not completely removed. The survival rate for this operation is excellent.

Prognosis

Successful removal of the tumor rids the patient of this disease. Emboli from a myxoma may survive in other areas of the body. However, there is no evidence that myxoma is truly metastatic (able to transfer disease from one area to another), causing tumors in other areas of the body.

Resources

OTHER

"Myxoma, Intracardiac." *OMIM Home Page, Online Mendelian Inheritance in Man*. <<http://www.ncbi.nlm.nih.gov/Omim>>.

John T. Lohr, PhD

N

Nail infections see **Onychomycosis**

Nail-patella syndrome

Definition

Nail-patella syndrome, is a genetic disease of the connective tissue that produces defects in the fingernails, knee caps, and kidneys.

Description

Nail-patella syndrome is also known as Fong Disease, Hereditary Onycho-Osteodysplasia (H.O.O.D.), Iliac Horn Disease, and Turner-Kieser syndrome. Patients who have nail-patella syndrome may show a variety of physical defects. The hallmark features of this syndrome are poorly developed fingernails, toenails, and patellae (kneecaps). Other common abnormalities include elbow deformities, abnormally shaped pelvis bone (hip bone), and kidney (renal) disease.

Less common medical findings include defects of the upper lip, the roof of the mouth, and unusual skeletal abnormalities. Skeletal abnormalities may include poorly developed scapulae (shoulder blades), sideways bent fingers (clinodactyly), clubfoot, scoliosis, and unusual neck bones. There are also other effects, such as thickening of the basement membrane in the skin and of the tiny clusters of capillaries (glomeruli) in the kidney. Scientists have recognized an association between nail-patella syndrome and **colon cancer**. Nail-patella syndrome is associated with open-angle **glaucoma**, which, if untreated, may lead to blindness. Patients may also have **cataracts**, drooping eyelids (**ptosis**), or corneal problems such as glaucoma.

People with nail-patella syndrome may display only a few or many of the recognized signs of this

disease. Symptoms vary widely from person to person. Signs even vary within a single family with multiple affected members.

The incidence of nail-patella syndrome is approximately one in 50,000 births. This disorder affects males and females equally. It is found throughout the world and occurs in all ethnic groups. The strongest risk factor for nail-patella syndrome is a family history of the disease.

Causes and symptoms

Nail-patella syndrome has been recognized as an inherited disorder for over 100 years. It is caused by mutations in a gene known as LIM Homeobox Transcription Factor 1-Beta (LMX1B), located on the long arm of chromosome 9.

The LMX1B gene codes for a protein that is important in organizing embryonic limb development. Mutations in this gene have been detected in many unrelated people with nail-patella syndrome. Scientists have also been able to interrupt this gene in mice to produce defects similar to those seen in human nail-patella syndrome.

Nail-patella syndrome is inherited in an autosomal dominant manner. This means that possession of only one copy of the defective gene is enough to cause disease. When a parent has nail-patella syndrome each of their children has a 50% chance to inherit the disease-causing mutation.

A new mutation causing nail-patella syndrome can also occur, causing disease in a person with no family history. This is called a sporadic occurrence and accounts for approximately 20% of cases of nail-patella syndrome. The children of a person with sporadic nail-patella syndrome are also at a 50% risk of developing signs of the disorder.

Medical signs of nail-patella syndrome vary widely between patients. Some patients with this

disorder do not display symptoms. These patients are discovered to have the nail-patella syndrome only when genetic studies trace their family history. Scientists are now working to learn what causes different people to display such different symptoms of nail-patella syndrome.

The most obvious sign associated with nail-patella syndrome is absent, poorly developed, or unusual fingernails. Fingernail abnormalities are found in over 80% of patients with this disorder. Abnormalities may be found in one or more fingernails. Only rarely are all fingernails affected. This disease most commonly affects the fingernails of the thumbs and index fingers. The pinky fingernail is least likely to be affected. Fingernails may be small and concave with pitting, ridges, splits, and/or discoloration. Toenails are less often affected. The lunulae, or light-colored crescent moons, at the base of the fingernail bed next to the cuticle are sometimes triangularly-shaped in people with nail-patella syndrome.

Kneecap abnormalities are the second most common sign associated with this disorder. Either or both kneecaps may be missing or poorly formed. If present, kneecaps are likely to be dislocated. The knees of people with nail-patella syndrome may have a square appearance. Besides the kneecap, other support structures including bones, ligaments, and tendons may also be malformed. These support structures stabilize the knee, therefore patients with some leg malformations may have difficulty in walking.

The hip bones of approximately 80% of patients with nail-patella syndrome have unusual bony projections called posterior iliac horns. These bony projections, or spurs, are internal and not obvious unless they are detected on x ray. This unusual pelvic anatomy is not associated with any other disease.

Kidney disease is present in at least 30% of people with nail-patella syndrome. Biopsy shows lesions that resemble those of inflammation of the clusters of capillaries in the kidneys (glomerulonephritis), but without any infection present. Kidney failure is the most dangerous consequence of nail-patella syndrome. It occurs in about 30% of patients who have kidney involvement. An early sign of kidney involvement is the presence of protein or blood in the urine (chronic, benign proteinuria and hematuria.) Kidney involvement is progressive, so early diagnosis and treatment of renal disease is important. Kidney disease has been reported in children with nail-patella syndrome, but renal involvement more commonly develops during adulthood.

Various skeletal symptoms may occur. Patients with nail-patella syndrome may not be able to fully

straighten their arms at the elbow. This may create a webbed appearance at the elbow joint. Patients may have sideways bent fingers, poorly developed shoulder blades, **clubfoot**, hip dislocation, unusual neck bones, or **scoliosis**.

Eye problems may be present and vary from person to person. Nail-patella syndrome is associated with open angle glaucoma. Open angle glaucoma is caused by fluid blocked into the front chamber of the eye. This blocked fluid builds increasing pressure into the eye. If untreated, this increased pressure may lead to permanent damage of the optic nerve and irreversible blindness. Some patients with nail-patella syndrome have ptosis, or drooping eyelids. Nail-patella syndrome has also been associated with abnormalities of the cornea, cataracts, and **astigmatism**. Additionally, the irises of the eye may be multi-colored, possibly displaying a clover-shaped pattern of color.

Diagnosis

As of early 2001, **genetic testing** for nail-patella syndrome is available only through research institutions that are working to further characterize this disorder. Genetic testing cannot predict which signs of the disease will develop. Nor can genetic testing predict the severity of disease symptoms. Improved genetic testing for nail-patella syndrome is anticipated in the future.

Diagnosis of this disease is most often made on visual medical clues such as the characteristic abnormalities of the fingernails and kneecaps. Diagnosis is confirmed by x-ray images of the affected bones and, when indicated, kidney biopsy. The bony pelvic spurs found in 80% of patients with nail-patella syndrome are not associated with any other disease.

Prenatal diagnosis for nail-patella syndrome by third-trimester ultrasound was documented in 1998. Prenatal diagnosis via genetic testing of cells obtained by chorionic villus sampling was reported the same year. As of 2001, prenatal genetic testing for nail-patella syndrome is not yet widely available. There is controversy surrounding the use of prenatal testing for such a variable disorder. Prenatal testing cannot predict the extent of an individual's disease.

Treatment

Treatment is usually not necessary. Treatment, when required, depends on each patient's specific symptoms. Severe kidney disease is treated with dialysis or a kidney transplant. Patients receiving kidney

KEY TERMS

Chorionic villus sampling (CVS)—A procedure used for prenatal diagnosis at 10–12 weeks gestation. Under ultrasound guidance a needle is inserted either through the mother's vagina or abdominal wall and a sample of cells is collected from around the early embryo. These cells are then tested for chromosome abnormalities or other genetic diseases.

Glomeruli—Tiny clusters of capillaries in the kidney.

Hematuria—The presence of blood in the urine.

Patella—The kneecap.

Proteinuria—Excess protein in the urine.

transplants do not develop nail-patella type renal complications in their new kidney.

A wheelchair may be required if walking becomes painful due to bone, tendon, ligament, or muscle defects. **Orthopedic surgery** may be necessary for congenital clubfoot deformity. Manipulation or surgery may be required to correct hip dislocation. Cataracts are also surgically treated. Medical treatment at early signs of glaucoma prevents progression of the disease to blindness.

Genetic counseling is offered to persons who have the disease. Parents with this disease have a 50% chance of passing it to each of their children. As of 2001, current genetic testing technology cannot predict the severity or scope of an individual's symptoms.

Because many possible manifestations of nail-patella syndrome exist, patients are advised to pursue extra medical care including regular **urinalysis** and special eye exams. Children with nail-patella syndrome should be screened for scoliosis.

Prognosis

Survival among patients with nail-patella syndrome is not decreased unless a they exhibit renal complications. It is estimated that 8% of individuals with nail-patella syndrome who come to medical attention eventually die of kidney disease.

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Nail removal

Definition

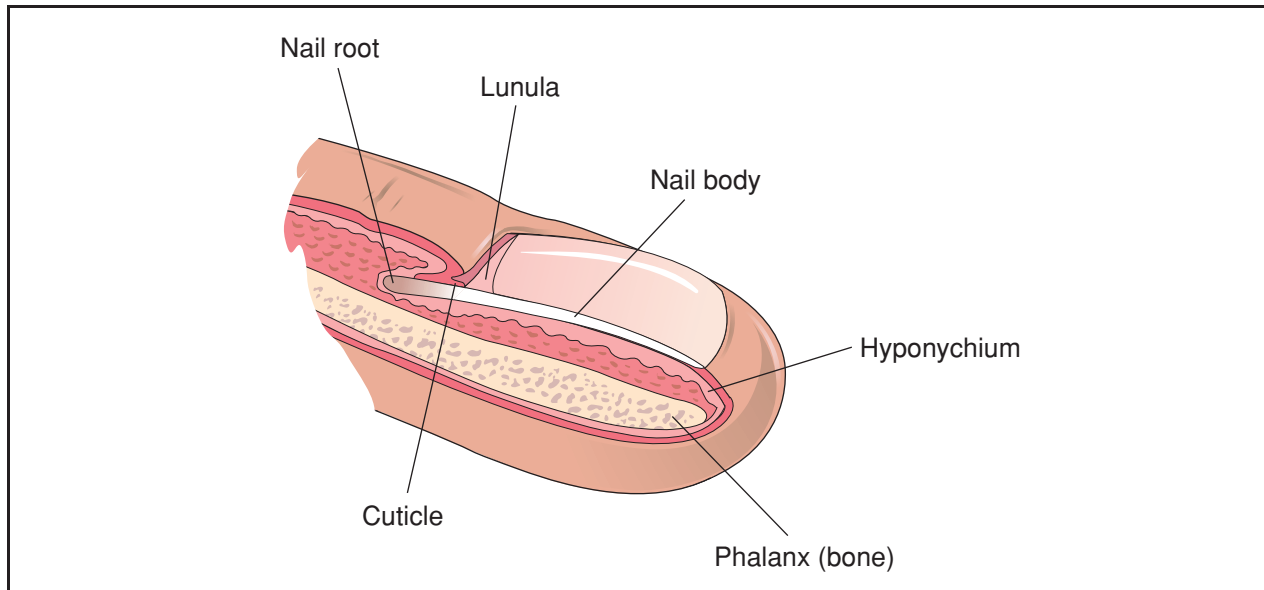
Nail removal is a form of treatment that is sometimes necessary following traumatic injuries or recurrent infections in the area of the nail. There are nonsurgical as well as surgical methods of nail removal.

Purpose

Nails are removed only when necessary to allow the skin beneath the nail (the nail bed) to heal or in some cases, to remove a nail that has been partially pulled out in an accident. In the case of toenails, it is occasionally necessary to remove the nail of the large toe due to a chronic condition caused by badly fitted shoes. In general, however, doctors prefer to try other forms of treatment before removing the nail. Depending on the cause, nail disorders are usually treated with oral medications; applying medicated gels or creams directly to the skin around the nail; avoiding substances that irritate the nail folds; surgical lancing of abscesses around the nail; or injecting **corticosteroids** under the nail fold.

The most common causes of nail disorders include:

- **Trauma.** The nails can be damaged by nail biting, using the fingernails as tools, and incorrect use of nail files and manicure scissors as well as by accidents and **sports injuries**.
- **Infections.** These include fungal infections under the nails, bacterial infections of cuts or breaks in the nail folds, or infections of the nails themselves caused by *Candida albicans*. Inflammation of the nail folds is called paronychia.
- **Exposure to harsh detergents, industrial chemicals, hot water, and other irritants.** People who work as dishwashers are especially vulnerable to separation of the nail itself from the nail bed (onycholysis).
- **Systemic diseases and disorders.** These include **psoriasis**, anemia, and certain congenital disorders.



The physiology of the human fingernail. The most common causes of nail disorders include trauma, infections, exposure to harsh detergents, hot water and other irritants, systemic diseases and disorders, and allergic reactions to nail polish, nail polish remover, and nail glue. (Illustration by Electronic Illustrators Group.)

- Allergic reactions to nail polish, polish remover, or the glue used to attach false nails.

Precautions

In the case of infections, it is necessary to distinguish between fungal, bacterial, and candidal infections before removing the nail. Cultures can usually be obtained from pus or tissue fluid from the affected nail.

Description

Surgical nail removal

If necessary, the surgeon can remove the nail at its base with an instrument called a needlepoint scalpel. In a few cases, the nail may need to be pulled out (avulsed) from its matrix.

Nonsurgical nail removal

Nails can be removed by applying a mixture of 40% urea, 20% anhydrous lanolin, 5% white wax, 25% white petroleum jelly, and silica gel type H.

Preparation

For nonsurgical nail removal, the nail fold is treated with tincture of benzoin and covered with adhesive tape. The nail itself is thickly coated with

KEY TERMS

Avulse—To pull or tear away forcibly. In some cases, a surgeon must remove a nail by avulsing it from its matrix.

Matrix—The tissue at the base of the nail, from which the nail grows.

Nail bed—The layer of tissue underneath the nail.

Onycholysis—The separation of a nail from its underlying bed. Onycholysis is a common symptom of candidal infections of the nail or of exposure to harsh chemicals and detergents.

Paronychia—Inflammation of the folds of skin that surround a nail.

the urea mixture, followed by a layer of plastic film and adhesive tape. The mixture is left on the nail for five to 10 days, after which the nail itself can be removed.

Aftercare

Aftercare of surgical removal is similar to the care of any minor surgical procedure. Aftercare of the urea paste method includes applying medication for the specific infection that is being treated.

Risks

Risks from either procedure are minimal.

Normal results

Normal results include the successful removal of the infected or damaged nail.

Resources

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Rebecca J. Frey, PhD

Nalidixic acid see **Urinary anti-infectives**

Narcissistic personality disorder see **Personality disorders**

Narcolepsy

Definition

Narcolepsy is a disorder marked by excessive daytime sleepiness, uncontrollable sleep attacks, and cataplexy (a sudden loss of muscle tone, usually lasting up to half an hour).

Description

Narcolepsy is the second-leading cause of excessive daytime sleepiness (after obstructive sleep apnea). Persistent sleepiness and sleep attacks are the hallmarks of this condition. The sleepiness has been compared to the feeling of trying to stay awake after not sleeping for two or three days.

People with narcolepsy fall asleep suddenly—anywhere, at any time, maybe even in the middle of a conversation. These sleep attacks can last from a few seconds to more than an hour. Depending on where they occur, they may be mildly inconvenient or even dangerous to the individual. Some people continue to function outwardly during the sleep episodes, such as talking or putting things away. But when they wake up, they have no memory of the event.

Narcolepsy is related to the deep, dreaming part of sleep known as rapid eye movement (REM) sleep. Normally when people fall asleep, they

experience 90 minutes of non-REM sleep, which is then followed by REM sleep. People with narcolepsy, however, enter REM sleep immediately. In addition, REM sleep occurs inappropriately throughout the day.

There has been debate over the incidence of narcolepsy. It is thought to affect between one in every 1,000 to 2,000 Americans. The known prevalence in other countries varies, from one in 600 in Japan to one in 500,000 in Israel. Reasons for these differences are not clear.

Causes and symptoms

In 1999 researchers identified the gene that causes narcolepsy. The gene allows cells in the hypothalamus (the part of the brain that regulates sleep behavior) to receive messages from other cells. When this gene is abnormal, cells cannot communicate properly, and abnormal sleeping patterns develop.

The disorder sometimes runs in families, but most people with narcolepsy have no relatives with the disorder. Researchers believe that the inheritance of narcolepsy is similar to that of heart disease. In heart disease, several genes play a role in being susceptible to the disorder, but it usually does not develop without an environmental trigger of some sort.

While the symptoms of narcolepsy usually appear during the teens or 20s, the disease may not be diagnosed for many years. Most often, the first symptom is an overwhelming feeling of fatigue. After several months or years, cataplexy and other symptoms appear.

Cataplexy is the most dramatic symptom of narcolepsy. It affects 75% of people with the disorder. During attacks, the knees buckle and the neck muscles go slack. In extreme cases, the person may become paralyzed and fall to the floor. This loss of muscle tone is temporary, lasting from a few seconds to half an hour, but frightening. The attacks can occur at any time but are often triggered by strong emotions, such as anger, joy, or surprise.

Other symptoms of narcolepsy include:

- sleep attacks: short, uncontrollable sleep episodes throughout the day
- sleep **paralysis**: a frightening inability to move shortly after awakening or dozing off
- auditory or visual **hallucinations**: intense, sometimes terrifying experiences at the beginning or end of a sleep period

- disturbed nighttime sleep: tossing and turning, nightmares, and frequent awakenings during the night

Diagnosis

If a person experiences both excessive daytime sleepiness and cataplexy, a diagnosis may be made on the patient history alone. Laboratory tests, however, can confirm a diagnosis. These may include an overnight polysomnogram—a test in which sleep is monitored with electrocardiography, video, and respiratory parameters. A Multiple Sleep Latency Test, which measures sleep latency (onset) and how quickly REM sleep occurs, may be used. People who have narcolepsy usually fall asleep in less than five minutes.

If a diagnosis is in question, a genetic blood test can reveal the existence of certain substances in people who have a tendency to develop narcolepsy. Positive test results suggest, but do not prove, the existence of narcolepsy.

Narcolepsy is a complex disorder, and it is often misdiagnosed. It takes 14 years, on average, for an individual to be correctly diagnosed.

Treatment

There is no cure for narcolepsy. It is not progressive, and it is not fatal, but it is chronic. The symptoms can be managed with medication or lifestyle adjustment. Amphetamine-like stimulant drugs are often prescribed to control drowsiness and sleep attacks. A drug called sodium oxybate (Xyrem) was tested for use in 2004 and was shown to reduce daytime sleepiness as well as cataplexy attacks when used along with stimulants. Patients who do not like taking high doses of stimulants may choose to take smaller doses and “manage” their lifestyles, such as by napping every few hours to relieve daytime sleepiness. Antidepressants are often effective in treating symptoms of abnormal REM sleep.

With the recent discovery of the gene that causes narcolepsy, researchers are hopeful that therapies can be designed to relieve the symptoms of the disorder.

Prognosis

Narcolepsy is not a degenerative disease, and patients do not develop other neurologic symptoms. However, narcolepsy can interfere with a person's ability to work, play, drive, and perform other daily activities. In severe cases, the disorder prevents people from living a normal life, leading to depression and a loss of independence.

KEY TERMS

Cataplexy—A symptom of narcolepsy in which there is a sudden episode of muscle weakness triggered by emotions. The muscle weakness may cause the person's knees to buckle, or the head to drop. In severe cases, the patient may become paralyzed for a few seconds to minutes.

Hypnagogic hallucinations—Dream-like auditory or visual hallucinations that occur while falling asleep.

Hypothalamus—A part of the forebrain that controls heartbeat, body temperature, thirst, hunger, body temperature and pressure, blood sugar levels, and other functions.

Sleep paralysis—An abnormal episode of sleep in which the patient cannot move for a few minutes, usually occurring on falling asleep or waking up. Often found in patients with narcolepsy.

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American Sleep Disorders Association. 1610 14th St. NW, Suite 300, Rochester, MN 55901. (507) 287-6006.

Narcolepsy Network. PO Box 42460, Cincinnati, OH 45242. (973) 276-0115.

National Center on Sleep Disorders Research. Two Rockledge Centre, 6701 Rockledge Dr., Bethesda, MD 20892. (301) 435-0199.

National Sleep Foundation. 1522 K St., NW, Suite 500, Washington, DC 20005. (202) 785-2300. <<http://www.sleepfoundation.org>>.

Stanford Center for Narcolepsy. 1201 Welch Rd-Rm P-112, Stanford, CA 94305. (415) 725-6517.

University of Illinois Center for Narcolepsy Research. 845 S. Damen Ave., Chicago, IL 60612. (312) 996-5176.

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Narcotics

Definition

Narcotics are natural opioid drugs derived from the Asian poppy *Palaver somniferous* or semi-synthetic or synthetic substitutes for these drugs.

Purpose

Narcotics are drugs that dull the sense of **pain** and cause drowsiness or sleep. They are the most effective tool a physician has to relieve severe pain. Narcotics are also given pre-operatively to relieve **anxiety** and induce anesthesia. Other common uses are to suppress **cough** and to control very severe **diarrhea**. In large doses, they can suppress the ability to breathe and cause **coma** and **death**. Narcotics are also taken illegally for recreational use.

Precautions

Narcotics should only be taken under the direction of a physician. These drugs depress the central nervous system and should not be taken with other drugs, such as alcohol, **barbiturates**, **antihistamines**, and **benzodiazepines** that also depress the central nervous system.

Opioids are broken down by the liver. Individuals with liver damage may not detoxify these substances as rapidly as healthy individuals, leading to potential accidental overdose. Street narcotics are of uncertain strength and may be contaminated with toxic chemicals or contain a mixture of drugs that can cause life-threatening reactions.

Description

Natural narcotics are derived directly from the sap of the unripe seed pods of the opium poppy. Morphine and codeine are the most familiar natural narcotics and are the narcotics most frequently used in medical settings. Often they are prescribed in combination with other non-narcotic drugs. Heroin is a semi-synthetic narcotic. It has no medical or legal uses. Other completely synthetic narcotics are made in the laboratory. These include drugs with medical uses such as fentanyl and oxycodone and illegal “designer drugs” synthesized for recreational use. Some man-made narcotics are hundreds of times more potent than natural narcotics.

Narcotics depress the central nervous system. They work by binding chemically with receptors in a

way that blocks the transmission of nerve impulses. These drugs do not cure the source of the pain; they simply block the individual’s perception of pain. When used to treat cough or diarrhea, they slow or block muscle contractions.

Morphine (Roxanol, Dura morphine, morphine sulfate, morphine hydrochloride) is the most commonly used medical narcotic for managing moderate to severe pain. It can be also be used to control extreme diarrhea caused by **cholera** or similar diseases. Morphine sulfate is a white powder that dissolves in water. It is usually given by injection into a muscle or intravenously by injection into a vein. When given intravenously, its effect occurs almost immediately. Individuals given morphine regularly have a high potential for developing dependence on the drug. Morphine can cause withdrawal symptoms if stopped abruptly. It is not a common street drug.

More codeine is prescribed medically than any other narcotic. Concentrations of codeine in the sap of the opium poppy are low, so most codeine is manufactured by chemical alteration of morphine. For pain control, codeine is combined with other non-narcotic painkillers such as **aspirin** (Empirin with Codeine,) **acetaminophen** (Tylenol with Codeine) or non-steroid anti-inflammatory drugs. These combination painkillers are manufactured as tablets (most common) or liquids and come in a variety of strengths based on the amount of codeine they contain. Codeine is also found in some cough syrups (Robitussin A-C, for example) and is used to control dry cough. Occasionally codeine is used to control severe diarrhea, although diphenoxylate (Lomotil) is used more often.

In Canada, certain low-dose codeine pain relievers are sold without prescription. In the United States pain medication with codeine requires a prescription. The likelihood of physical or psychological dependence on codeine is much lower than with morphine.

Hydromorphone (Dilaudid) is a narcotic synthetically produced from morphine. It is available in tablets or as an injectable solution and used for pain relief. It is one of the most common pain relievers prescribed for patients who are terminally ill, because it combines high effectiveness with low side effects.

Mederidine (Demerol) was originally developed to treat **muscle spasms** but is as of 2005 used mainly for pain relief. It is manufactured as tablets of varying strengths. Another synthetic pain relief narcotic whose use parallels mederidine is propoxyphene. When combined with aspirin this narcotic is known under the brand name Darvon.

Oxycodone (Oxycontin), a synthetic narcotic used for pain relief, is manufactured both alone and with aspirin (Percodan) or acetaminophen (Percoset) in tablets of various strengths. OxyContin is a controlled release formula of oxycodone that controls pain continuously for 12 hours at a time. Oxycodone has a high potential for prescription drug and street **abuse**. Hydrocodone with acetaminophen (Vicodin) is another synthetic narcotic whose use and potential abuse parallels oxycodone.

Fentanyl (Sublimaze, Actiq, Duragesic) is used as a surgical anesthetic. It is available as an injectable solution and as a skin patch.

Methadone is a synthetic narcotic used mainly as a substitute for heroin in heroin withdrawal treatment, although it does have pain-killing properties. Methadone, when taken by mouth (liquid, wafers, tablets) provides little of the euphoria of heroin, but it blocks heroin cravings and withdrawal symptoms.

The first international attempts to control narcotic drugs were made in 1909 with the formation of the Opium Commission Forum, which developed the first international drug control treaty in 1912. In the early 2000s narcotics are regulated internationally by the International Narcotics Control Board (INCB), established in 1961. The INCB regulates the cultivation of raw materials to make narcotics and natural and man-made drugs. **Cocaine** and **marijuana** also fall under the board's control, although they are not technically narcotics. Narcotic drugs are also regulated by federal and state governments. In law enforcement, the term narcotics is extended to include other, mainly illicit drugs such as cocaine that have little medical use.

Preparation

No special preparation is required before being treated with narcotics, although, as with all medications, individuals should tell their physician about all prescription and non-prescription drugs, supplements, and herbal remedies that they are taking, as certain medications may enhance the effects of narcotics.

Aftercare

When an individual is prescribed narcotics regularly for an extended period, tolerance may develop. With tolerance, the individual must take higher and higher doses to achieve the same level of pain control. In some cases, when narcotics are stopped abruptly, withdrawal symptoms may develop. These include:

KEY TERMS

Tapering—Gradually reducing the amount of a drug when stopping it abruptly would cause unpleasant withdrawal symptoms.

- anxiety
- irritability
- rapid breathing
- runny nose
- sweating
- vomiting and diarrhea
- confusion
- shaking
- lack of appetite

In order to prevent withdrawal symptoms, the dose of narcotics can be gradually diminished, a process known as tapering, until they can be discontinued completely without unpleasant effects. Individuals may also be treated with the drug clonidine (Catapres) to relieve some withdrawal symptoms.

Risks

All narcotics have the potential to become physically and psychologically addictive. When used regularly, tolerance can develop. Abuse and dependence on narcotic prescription drugs is an increasing problem among the elderly particularly and among members of the middle class generally.

Overdose and withdrawal symptoms and reactions caused by contamination with other drugs or toxic chemicals are common reasons for drug-related visits to the emergency room by individuals using street narcotics recreationally. Overdose is treated with the drug naloxone (ReVia). Naloxone blocks and reverses the effects of narcotics. When given intravenously it is effective within one to two minutes.

Normal results

When used as prescribed, narcotics are a generally safe and effective way to relieve pain and control cough and severe diarrhea. Individuals should not be afraid they will develop an **addiction** after a short-term course of narcotics following a dental or medical procedure, provided that they follow their physician's instructions for taking the drugs.

Resources

ORGANIZATION

National Institute on Drug Abuse/National Institutes of Health. 6001 Executive Boulevard, Room 5213, Bethesda, MD 20892-9561. (301) 443-1124.
< <http://www.nida.nih.gov> >.

United States Drug Enforcement Administration. Dr Mailstop: AXS, 2401 Jefferson Davis Highway, Alexandria, VA 22301. (202) 307-1000.
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Nasal culture see **Nasopharyngeal culture**

Nasal irrigation

Definition

Nasal irrigation is the practice of flushing the nasal cavity with a sterile solution. The solution may contain **antibiotics** or steroid medications.

Purpose

Nasal irrigation is used to clear infected sinuses or may be performed after surgery to the nose region. It may be performed by adding antibiotics to the solution to treat **nasal polyps**, nasal septal deviation, allergic nasal inflammation, chronic sinus infection, and swollen mucous membranes. One benefit of nasal irrigation in treating these conditions is that it usually lowers the amount of medication that the patient must take by mouth.

Irrigation is also used to treat long-term users of inhalants, such as illicit drugs (**cocaine**), or such occupational toxins as paint fumes, sawdust, pesticides, and coal dust.

Nasal irrigation may also be used in occupational medicine to monitor workers for exposure to airborne glass fibers, asbestos, and similar materials.

Precautions

Nasal irrigation should not be performed on people who have frequent nosebleeds; have recently had nasal surgery; or whose gag reflex is impaired, as fluid may enter the windpipe.

Description

Nasal irrigation can be performed by the patient at home or by a medical professional. A forced-flow instrument, such as a syringe, is filled with a warm saline solution. The solution can be commercially prepared (Ayr, NaSal) or can be prepared by the patient, using one half teaspoon salt with each eight ounces of warm water. Occasionally, antibiotics or steroids are added to the solution to kill bacteria and aid healing of irritated membrane. The syringe is then directed into the nostril. The irrigation solution loosens encrusted material in the nasal passage, and drainage takes place through the nose. The patient leans over a catch basin during irrigation, into which the debris flows. Irrigation continues until all debris is cleared from the passage. Nasal irrigation can be performed up to twice daily, unless the irrigation irritates the mucous membrane.

Preparation

Before nasal irrigation, the patient is instructed not to open his or her mouth or swallow during the procedure. Opening the mouth or swallowing may cause infectious material to move from the nasal passage into the sinuses or the ear.

Risks

Complications of nasal irrigation include irritation of the nasal passages due to extreme temperature of the irrigation solution. Rarely, irrigation fluid may enter the windpipe in people with a poor gag reflex.

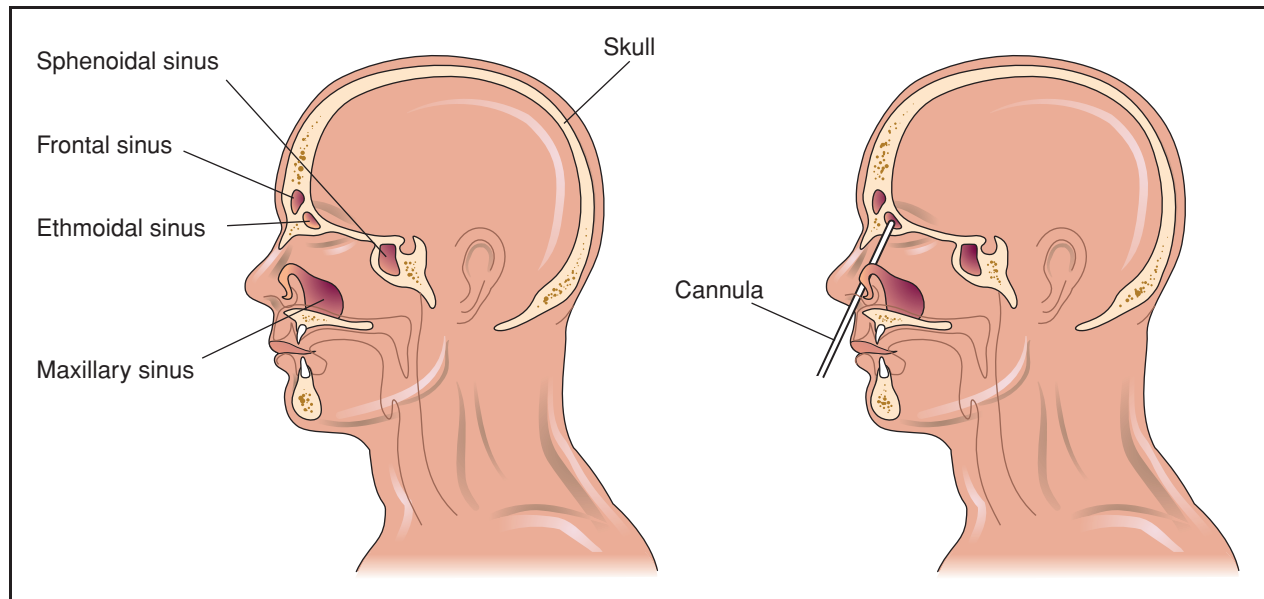
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Because surgery in the nasal area has a high incidence rate for contamination with pathogenic bacteria, nasal irrigation is performed to remove loose tissue and prevent infection. The illustration (right) shows a cannula in place while the sinus passages are being flushed. (Illustration by Electronic Illustrators Group.)

KEY TERMS

Irrigation—In medicine, the practice of washing out or flushing a wound or body opening with a stream of water or another liquid.

Saline—A solution made from salt and water.

American Academy of Otolaryngology, Head and Neck Surgery, Inc. One Prince St., Alexandria, VA 22314-3357. (703) 836-44 44. <<http://www.entnet.org>>.

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American Academy of Family Physicians (AAFP). 11400 Tomahawk Creek Parkway, Leawood, KS 66211-2672. (800) 274-2237 or (913) 906-6000. <<http://www.aafp.org>>.

Nasal packing

Definition

Nasal packing is the application of gauze or cotton packs to the nasal chambers.

Purpose

The most common purpose of nasal packing is to control bleeding following surgery to the septum or nasal reconstruction and to treat chronic nosebleeds. Packing is also used to provide support to the septum after surgery.

Description

Packing is the placement of gauze or cotton into the nasal area. Packing comes in three forms: gauze,

cotton balls, and preformed cotton wedges. Packing is usually coated with antibiotics and, sometimes, petrolatum. The end of the nose may be taped to keep the packings in place or to prevent the patient from pulling them out. In cases of surgery, packings are frequently removed within 24–48 hours following surgery. In the case of nosebleeds, packing is left in for extended periods of time to promote healing and to prevent the patient from removing scar tissue which might reopen the wound. If both sides of the nose are packed, the patient must breathe through his or her mouth while the packs are in place.

In patients who are chronic nose pickers, frequent bleeding is common and ulceration of nasal tissue is possible. To promote healing and to prevent nose picking, both sides of the nose are packed with cotton that contains **antibiotics**. The nose is taped shut with surgical tape to prevent the packing from being removed. The packing is left in the nose for seven to 10 days. If the wound is high up in the nasal cavity, gauze strips treated with petrolatum and antibiotics are used. The strips are placed into the nose one layer at a time, folding one layer on top of the other until the area is completely packed.

Local packing is a procedure used when only a small part of the nose must be packed. Typically, this occurs when one blood vessel is prone to bleeding, and there is no need to block breathing through the nose. Local packing is used when the pack can remain in place by itself. This situation can be found at the turbinates. Turbinates are folds of tissue on the insides of the nose. The folds are sufficiently firm to support packing. A small piece of gauze or cotton is wedged in between the turbinates where the blood vessel being treated is located. Local packing is left in place for up to 48 hours and then removed. The main advantage to this type of packing is that it enables the patient to breathe through his or her nose. Local packing is also more comfortable than complete packing, although the patient will still experience a sensation that something is in the nasal cavity. The patient must be instructed not to interfere with or probe the packing while it is in place.

A postnasal pack is used to treat bleeding in the postnasal area. This is a difficult area to pack. Packs used in this area are made from cotton balls or gauze that have been tied into a tubular shape with heavy gauge suture or umbilical tape. Long lengths of suture or tape are left free. The lengths of suture or tape are used to help position the pack during installation and to remove it. An alternative is to cut a vaginal tampon and reposition the strings. Balloons have been tried as a method to replace postnasal packing, but have not

KEY TERMS

Turbinate—Ridge-shaped cartilage or soft bony tissue inside the nose.

Ulcer—A sore on the skin or mucous tissue that produces pus and in which tissue is destroyed.

proved effective. After being tied, the pack is soaked with an antibiotic ointment. Generally, packs are formed larger than needed, so that they completely block the nasal passage. A catheter is passed through the nose and pulled out through the mouth. Strings from one end of the pack are tied to the catheter and the pack pulled into place by passing through the mouth and up the back of the nasal cavity. The pack is removed in a similar manner. Complications may occur if a pack compresses the Eustachian tube, causing ear problems. The ear should be examined to ensure that infection is not developing.

Packing of the anterior (front) part of the nose is also performed following surgery such as septoplasty and **rhinoplasty**. In these operations, the surgeon cuts through the skin flap covering cartilage and bone in the center, top, and bottom of the nose to correct the shape of the nose. At the conclusion of the surgery, the skin flap is sutured back into place. The purposes of packing are to absorb any drainage from the incision and mucus produced by nasal tissue, and to support the skin flap and cartilage. The packing used is either gauze or preformed adsorbent wedges of cotton. Both are usually treated with antibiotic to reduce the chance of infections at the incision site. Generally, there is little bleeding following septoplasty and rhinoplasty, and the incisions heal normally. These packs are left in place for 24 to 48 hours and then removed.

Aftercare

Ice chips or mouthwash can be used to moisten the mouth while packing is in place, as the mouth may be dry from breathing through it. Humidifiers may also help with breathing. After nasal packing, the nose should not be blown for two to three days.

Since one of the major reasons that packing is performed is to heal damage to nasal blood vessels from nose-picking, follow-up examination should be done to ensure that the patient is no longer practicing this habit. If the patient has restarted nose-picking, therapy to alter this behavior should be pursued. When the packing completely blocks the nasal cavity

and prevents breathing through the nose, the patient should adjust to breathing through the mouth. In elderly patients, adjustment may be more difficult. This leads to a drop in the blood oxygen content and an increase in blood carbon dioxide levels (CO₂). This, in turn, can cause respiratory and cardiac complications, including a racing pulse.

Risks

Nasal packing could cause a lack of oxygen in those who have difficulty breathing through their mouths. Rarely, sinus infection or middle ear infection may occur.

Resources

BOOKS

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Nasal papillomas

Definition

Nasal papillomas are **warts** located inside the nose.

Description

Two types of tumors can grow inside the nose: polyps and papillomas. By far the most common are polyps, which have smooth surfaces. On the contrary, papillomas have irregular surfaces and are, in fact, warts. Papillomas may be caused by the same viruses that cause warts elsewhere on the body. They are inside the nose, more often on the side near the cheek, and, because of their internal structure, they are much more likely to bleed than polyps.

There is a special type of nasal papilloma called an inverting papilloma because of its unique appearance. About 10 or 15% of these are or can become cancers.

Causes and symptoms

Like polyps, papillomas can plug up the nose and disable the sense of smell. Unlike polyps, papillomas often bleed.

KEY TERMS

Polyp—A tumor commonly found in the nasal cavity or intestine.

Diagnosis

A **physical examination** with special instruments will detect these tumors.

Treatment

Because of the possibility of **cancer**, all nasal papillomas must be removed surgically and sent to the laboratory for analysis. If a cancer is present, further surgery may be necessary to guarantee that all of the cancer has been removed. The initial surgery can be done in an office setting by a specialist in head and neck surgery, also known as otorhinolaryngology and popularly abbreviated ENT (ear, nose, and throat). Cancer surgery is more extensive and often requires hospitalization.

Prognosis

For benign (non-cancerous) lesions, removal is curative, although they tend to recur, just like warts elsewhere. The cancerous papillomas may occasionally escape complete surgical removal and spread to adjacent or distant sites. The prognosis is then much more complex.

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Nasal polyps

Definition

A polyp is the medical term for any overgrowth of tissue from the surface of a body organ. Polyps come in all shapes—round, droplet, and irregular being the most common. Nasal polyps are teardrop-shaped while growing and resemble peeled grapes when they

have reached their full size. The condition of nasal polyps is sometimes called nasal polyposis.

Description

Nasal polyps tend to occur in people with respiratory **allergies**. Hay fever (**allergic rhinitis**) is an irritation of the membranes of the nose by airborne particles or chemicals. These membranes secrete mucus. When irritated, they can also grow polyps. The nose is not only a passageway for air to reach the lungs; it also provides the connection between the sinuses and the outside world. Sinuses are lined with mucous membranes, just like the nose. Polyps can easily obstruct the drainage of mucus from the sinuses. When any fluid in the body is trapped so it cannot flow freely, it becomes infected. The result, **sinusitis**, is a common complication of allergic **rhinitis**.

Nasal polyps may also develop in children with **cystic fibrosis**.

Causes and symptoms

Some people who are allergic to **aspirin** develop both asthma and nasal polyps.

Nasal polyps often plug the nose, usually one side at a time. People with allergic rhinitis are so used to having a stopped-up nose they may not notice the difference when a polyp develops. Other polyps may be closer to a sinus opening, so airflow is not obstructed, but mucus becomes trapped in the sinus. In this case, there is a feeling of fullness in the head, no sense of smell, and perhaps a **headache**. The trapped mucus will eventually get infected, adding **pain**, fever, and perhaps bloody discharge from the nose.

Diagnosis

A **physical examination** will identify most polyps. Small polyps located higher up or further back may be hidden from view, but they will be detected with more sophisticated medical instruments. The otorhinolaryngologist is equipped to diagnose nasal polyps. In order to perform the examination, the doctor must apply medicine to reduce congestion in the swollen membranes. Cotton balls soaked with one of these agents and left in the nostrils for a few minutes provide adequate shrinkage.

Treatment

Most polyps can be removed by the head and neck surgeon as an office procedure called a nasal



A nasal polyp inside patient's right nostril. (Custom Medical Stock Photo. Reproduced by permission.)

KEY TERMS

Allergen—Any substance that irritates only those who are sensitive (allergic) to it.

Asthma—Wheezing (labored breathing) due to allergies or irritation of the lungs.

Decongestant—Medicines that shrink blood vessels and consequently mucus membranes. Pseudoephedrine, phenylephrine, and phenylpropanolamine are the most common.

Polyposis—The medical term for the development of multiple polyps on a body part.

Sinus—Air-filled cavities surrounding the eyes and nose are lined with mucus-producing membranes. They cleanse the nose, add resonance to the voice, and partially determine the structure of the face.

polypectomy. Bleeding, the only complication, is usually easy to control. Nose and sinus infections can be treated with **antibiotics** and **decongestants**, but if airflow is restricted, the infection will recur.

Prognosis

Polyps may reappear as long as the allergic irritation continues. In addition, one study of patients who had undergone nasal polypectomy reported that 60% had a recurrence of nasal polyposis, and 47% were advised to have revision surgery. The risk of recurrence is higher among patients with **asthma**.

Prevention

If aspirin is the cause of the polyps, all aspirin containing medications must be avoided.

Since most nasal polyps are the result of allergic rhinitis, they can be prevented by treating this condition. New treatments have greatly improved control of hay fever. There are now several spray medicines that are quite effective. Spray cortisone-like drugs, usually beclomethasone (Beconase, Vancenase) or flunisolide (Nasalide), are the most popular. Over-the-counter nasal decongestants have an irritating effect similar to the allergy they are supposed to be treating. Continued use can bring more trouble than relief and result in an addiction to nose sprays. The resulting disease, rhinitis medicamentosa, is more difficult to treat than allergic rhinitis.

Allergists and ENT surgeons both treat allergic rhinitis with a procedure called desensitization. After identifying suspect allergens using one of several methods, they will give the patient increasing doses of those allergens in order to produce blocking antibodies that will impede the allergic reaction. This approach is effective in a number of patients, but the treatment may take a period of months to years.

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American Academy of Otolaryngology–Head and Neck Surgery, Inc. One Prince Street, Alexandria, VA 22314-3357. (703) 836-4444. <<http://entnet.org>>.

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Nasal trauma

Definition

Nasal trauma is defined as any injury to the nose or related structure that may result in bleeding, a physical deformity, a decreased ability to breathe normally because of obstruction, or an impaired sense of smell. The injury may be either internal or external.

Description

The human nose is composed of bone, soft tissue, and cartilage. It serves as a passageway for air to flow from the outside environment into the lower respiratory tract and lungs. At the same time, the nasal passages warm and humidify the air that enters the body.

Internal injuries to the nose typically occur when a foreign object (including the fingers) is placed in the nose or when a person takes in drugs of **abuse** (inhalants or **cocaine**) through the nose. External injuries to the nose are usually blunt force injuries related to sports participation, criminal violence, parental abuse, or automobile or bicycle accidents. This type of injury may result in a nasal fracture. The nasal bones are the most frequently fractured facial bones due to their position on the face, and are the third most common type of bone fracture in general after **fractures** of the wrist and collarbone. A force of only 30g is required to break the nasal bones, compared to 70g for the bones in the jaw and 200 g for the bony ridge above the eyes. The pattern of the fracture depends on the direction of the blow to the nose, whether coming from the front, the side, or above the nose. Although not life-threatening by itself, a fractured nose may lead to difficulties in breathing as well as facial disfigurement.

Fractures resulting from trauma to the nose may involve the bones of the septum (the partition of bone and cartilage dividing the two nostrils) as well as the bones surrounding the eyes. These bones include the nasal, maxilla, lacrimal, and frontal bones. Direct trauma to the bridge of the nose may also result in damage to a part of the base of the skull known as the cribriform plate. This injury in turn may allow cerebrospinal fluid to leak out of the skull and leave the body through the nose. Fractures may also damage the membranes that line the nasal passages, leading to possible formation of scar tissue, obstruction of the airway, and damage to the child's sense of smell.

In addition to fractures, external injuries of the nose include soft-tissue injuries resulting from bites (human and animal), insect stings, cuts, or scrapes. Penetrating injuries to the nasal area caused by air gun or BB pellets



Fractured nose of an elderly patient. (Photograph by Dr. P Marazzi. Photo Researchers. Reproduced by permission.)

are also reported with increasing frequency in older children and adolescents. When fired at close range, these pellets can penetrate the skin and cheekbone and lodge in the nasal septum or the sinuses near the nose.

Lastly, nose **piercing** as a fashion trend is a type of intentional injury to the nose that has several possible complications, including infections of the cartilage and soft tissues in the nose; blockage of the airway due to a loosened stud or other nose ornament; and gastrointestinal emergencies caused by accidental swallowing of nose jewelry.

Causes and symptoms

Causes

External trauma to the nose may be accidental (transportation accidents, animal bites, air gun injuries, and **sports injuries**) or intentional (fights, criminal assault, domestic violence, nose piercing). Nasal injuries from athletic activities may result from contact with equipment (being hit in the face by a baseball, hockey ball, or other small ball hit at high speed, or by the bat or stick itself) or the bodies of other players (football, boxing, martial arts, rugby). Nasal injuries from piercing include bacterial infections of the skin and nasal cartilage, allergic reactions to the jewelry, tissue damage, and periodic bleeding.

In a few cases, external trauma to the nose may also be iatrogenic, or caused by medical care. Most of these injuries result from medical examination of the nose—particularly in emergency circumstances—or as complications of **plastic surgery**. In a few cases damage to the nose is caused by **radiation therapy** for **cancer**.

Internal injuries to the nose may be either mechanical (caused by **foreign objects** in the nose or

by picking or scratching the tissues lining the nose) or chemical (caused by environmental irritants or **substance abuse**).

Chemical injuries to the nose are caused by accidental or purposeful breathing or sniffing of irritating substances. These may include tobacco smoke; household cleaners (ammonia and chlorine bleach) and furniture polish; ozone and other air pollutants; cocaine; and glue, paint thinners, solvents, and similar household products that produce toxic vapors. An increasingly common form of chemical injury to the nasal membranes in toddlers is alkali **burns** caused by leakage from small batteries placed in the nose. While chemical damage to the nose is usually accidental in younger children, it is more often the result of substance abuse in adolescents. Taking cocaine through the nose (“snorting”) or inhalant abuse (“sniffing” or “huffing”) are the most common causes of chemical damage to the nose in older children or teenagers.

Symptoms

The symptoms of physical trauma to the nose may include:

- Flattening or other deformation of the shape of the nose
 - Infections of the cartilage or soft tissue
 - Epistaxis or bleeding from the nose
 - Crepitus. Crepitus is the crackling or crunching sound heard when the ends of a fractured bone are rubbed together
 - Pain and tissue swelling
 - Airway blockage from bleeding, fluid discharge, or tissue swelling
 - **Rhinitis**. Rhinitis is an inflammation of the mucous membranes lining the nose. In the case of a fracture, rhinitis may lead to increased tear production in the eyes and a runny nose
 - Septal hematoma. A septal hematoma is a mass of blood from torn tissue that may collect within the cartilage that divides the two nostrils. It may become infected and form an **abscess** that eventually destroys the cartilage
 - Bruising or discoloration (ecchymosis) of the tissues around the eye
 - Leakage of cerebrospinal fluid through the nostrils
- Chemical trauma to the nose may result in:
- Runny nose and watering of the eyes
 - Pain

- Loss of the sense of smell
- Nasal congestion and sneezing
- Reddening and swelling of the mucous membranes lining the nose
- Eventual destruction of the cartilage in the nasal septum and the tissues lining the nose

Some common irritants that may be encountered in the home and workplace include:

- cleaning solutions and powders
- ammonia
- environmental tobacco smoke
- bleach
- metalworking fluids
- ozone
- sulfur dioxide
- paint thinners
- arsenic
- chromic acid
- copper dust and mists

Sequelae following exposure to these chemicals are based not only on the concentration of the irritant but also on factors specific to the individual. Reactions vary among persons, even with similar exposures.

Diagnosis

Diagnosis of a fracture is normally based on a history of nasal trauma and clinical presentation. Epistaxis may or may not be present. An intranasal examination is performed in order to look for a septal hematoma that may result in serious consequences such as death of the septal cartilaginous tissue. The nose is also checked for tenderness, mobility, stability, and crepitation.

X rays are normally not indicated, however, in more severe fractures involving multiple bones, a computed tomography (CT) scan may be required. The physician should look for associated injuries such as periorbital (surrounding the eye) ecchymosis, watery eyes, or diplopia (double vision) that may indicate orbital injuries. In addition, dental fractures and a cerebrospinal fluid (CSF) leak should be looked for. CSF leaks indicate a more severe injury possibly involving an ethmoid bone fracture.

The physician may also ask for photographs taken prior to the injury in order to determine the extent of deformity. Photographs may also be taken to document the injury in regards to possible legal actions.

In order to diagnose trauma sustained by a chemical injury, a history of exposure to potentially toxic chemicals should be ascertained. In addition, the patient should also bring information related to the types of chemicals that he or she has been exposed to. If injury occurs in the workplace, Material Safety Data Sheets should be available in the employer's poison control center that list the chemical components of commercial materials. Measurements of air from the patient's work area may also be obtained. Symptomatic improvement on off-days followed by a subsequent return of symptoms when returning to work confirms that the illness is work related. The physician should perform an intranasal examination to determine the extent of the chemical injury. A **chest x ray** as well as a **pulmonary function test** may be ordered to determine if there is any subsequent lower respiratory tract involvement.

Treatment

Timing

Nasal injuries should be treated as promptly as possible to prevent complications. Batteries placed in the nose should be removed within 4 hours to prevent burns and other damage to the tissues from leaking chemicals. If a septal hematoma has developed, the doctor must remove it as quickly as possible to prevent infection or eventual death of the tissues in the nasal septum. Lastly, if the child has been bitten by an animal, the injury must be cleansed as soon as possible to lower the risk of **rabies**.

Treatment of nasal fractures is best performed during the first three hours after the injury. If this is impossible, management of a nasal fracture should be done within three to seven days. Timing is of utmost importance when treating nasal fractures because delays longer than seven to 10 days may allow the broken bones to set without proper alignment, or lead to such complications as scar tissue formation and airway obstruction. Poorly set nasal fractures usually require surgical correction.

Specific procedures

Foreign objects in the nose can be removed by nasal suction in most cases. Most nosebleeds are treated by 5–30 minutes of direct pressure on the nostrils, with the patient's head placed in an upright position. The doctor may also pack the nose with gauze coated with petroleum jelly. If the bleeding does not stop, or if it appears to originate in the upper nose, the doctor will consult a head and neck surgeon or an otolaryngologist for specialized evaluation of the bleeding.

Air gun or BB pellets that have penetrated the nose or nearby sinuses are generally removed with the help of an endoscope, which is a slender tubular instrument that allows the doctor to examine the inside of a body cavity.

Treatment of nasal fractures depends on the extent of the injury; the most difficult fractures to treat are those that involve the nasal septum. The doctor will usually reduce the fracture, which means that he or she will restore the damaged bones to their proper position and alignment. Although **local anesthesia** is usually sufficient for treating nasal fractures in adults and older teenagers, **general anesthesia** is usually given when treating these injuries in younger children.

Reductions of nasal fractures may be either open or closed. A closed reduction involves manipulation of the bones without cutting into the overlying skin. This type of reduction will be performed for fractures of the nasal bones that are limited in size and complexity. Open reductions are performed for more complex nasal fractures. In an open reduction, the nasal bones are moved back to their original location after the surgeon has made an incision in the overlying skin. This procedure is done for fractures involving dislocation of the septum as well as the nasal bones. In addition, an open reduction is necessary if the child has a septal hematoma or an open fracture in which the skin has been perforated. If a septal hematoma is present, the doctor will drain it and pack the nose to prevent subsequent accumulation of blood. The nasal bones are held in the proper position with external splints as well as the internal packing, and the splints are kept in place for 7–10 days. The patient will be given **antibiotics** to lower the risk of infection and may be referred to an otolaryngologist or plastic surgeon for further evaluation. Ice packs or cold compresses can be applied at home to lower swelling and ease discomfort.

In the case of animal bites, the patient may be given passive or active immunization against rabies if there is a chance that the dog or other animal is rabid. This precaution is particularly important for animal bites on the nose or other parts of the face, as the incubation period of the rabies virus is much shorter for bites on the head and neck than for bites elsewhere on the body.

Complications can arise following treatment and therefore follow-up is necessary. Problems that may occur resemble symptoms of nasal fractures. Others include infection, CSF leakage, scar tissue build-up, and a saddle nose deformity where the bridge of the nose is markedly depressed.

Treatment for trauma caused by irritant inhalation involves removing the patient from the

contaminated area or decreasing exposure time. Other measures include using a saline nasal spray or topical steroids. For acute injuries oxygen or supportive treatment for any subsequent lower respiratory tract involvement may be administered.

If the injury is occupation-related, changes should be made in order to eliminate future incidents. These changes may include having the patient wear a respiratory protection device while working. In addition, the employer should be made aware of the situation and employ measures to prevent future incidents.

Prognosis

Most types of nasal trauma have a good prognosis. Nosebleeds or tissue damage caused by scratching or picking at the nose usually clear completely once these habits are stopped. Infections or allergic reactions caused by foreign objects in the nose or piercing usually clear up promptly after the object or piece of jewelry is removed. Nasal fractures that do not involve the nasal septum or other facial bones and receive prompt treatment generally heal without deformities of the nose, cartilage destruction, or other complications. More extensive facial fractures, however, may require a second operation to correct the positioning of the bones and restore the appearance of the nose.

The prognosis for soft-tissue injuries to the nose depends on the cause and extent of the injuries. Such tearing or crushing injuries as those caused by bites take longer to heal than simple cuts, and may require plastic surgery at a later date to restore the appearance of the nose.

Damage to the tissues lining the nose caused by exposure to tobacco smoke or other irritants in the environment is usually reversible once the patient is removed from contact with the irritating substance. Erosion or destruction of the nasal cartilage as a result of inhalant or cocaine abuse, however, usually requires surgical treatment.

Prevention

Although most cases of nasal trauma happen inadvertently, some measures can be employed in order to prevent injury. Patients should be aware of the symptoms of nasal fracture and should seek medical attention as soon as possible to prevent more invasive reductions. Protective equipment should also be worn when playing sports. Employees should also be aware of irritating chemicals in their workplace and appropriate measures should be taken to avoid exposure.

KEY TERMS

Crepitus—A crackling or crunching sound heard when the ends of a fractured piece of bone rub against each other.

Diplopia—The medical term for seeing double.

Ecchymosis (plural, ecchymoses)—The medical term for a bruise. Ecchymoses may develop around the eyes following a nasal fracture.

Epistaxis—The medical term for a nosebleed.

Hematoma—A localized collection of blood that accumulates in an organ, tissue, or body space as the result of leakage from a broken blood vessel. Hematomas sometimes develop within the nasal cartilage when the nose is fractured.

Iatrogenic—Referring to injuries caused by a doctor. Nasal trauma may occasionally result from a doctor's examination of the nose or complications from plastic surgery.

Otolaryngologist—A doctor who specializes in diagnosing and treating disorders of the ears, nose, and throat.

Reduce—To restore a part of the body to its normal position or place, as in treating a fracture or dislocation. The repositioning of the bone or body part is called a reduction.

Rhinitis—An inflammation of the mucous membranes that line the nasal passages.

Rhinoplasty—Plastic surgery of the nose to repair or change the shape of the nose.

Septal hematoma—A mass of extravasated blood that is confined within the nasal septum.

Septum—The partition of bone and cartilage in the nose that separates the two nostrils.

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- American Academy of Family Physicians (AAFP). 11400 Tomahawk Creek Parkway, Leawood, KS 66211-2672. (800) 274-2237 or (913) 906-6000. <<http://www.aafp.org>>.
- American Academy of Otolaryngology—Head and Neck Surgery. One Prince Street, Alexandria, VA 22314-3357. (703) 836-4444. <<http://www.entnet.org>>.
- American College of Sports Medicine (ACSM). 401 West Michigan Street, Indianapolis, IN 46202-3233. (317) 637-9200. Fax: (317) 634-7817. <<http://www.acsm.org>>.

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Nasogastric suction

Definition

Nasogastric suction involves removing solids, liquids, or gasses from the stomach or small intestine by inserting a tube through the nose and suctioning the gastrointestinal material through the tube.

Purpose

Nasogastric suction may be done in the following situations:

- to decompress the stomach or small intestine when intestinal obstruction (**ileus**) is suspected
- prior to gastrointestinal operations
- to obtain a sample of the gastric contents for analysis
- to remove toxic substances
- to flush the stomach during gastrointestinal bleeding or poisonings

Nasogastric intubation, the insertion of a tube through the nose into the stomach or small intestine, is also done to temporarily feed certain patients. In this case, material is not suctioned out.

Precautions

Nasogastric tubes cannot be placed in patients who have blockages in their esophagus, enlarged esophageal veins or arteries that might bleed, or severe damage to the jaws and face. The tube cannot be inserted in a patient who is having convulsions, or who is losing or has lost consciousness unless a tube has been inserted into his or her airway (intubation).

Description

The patient sits upright while a lubricated tube is slipped through the nose and down the throat. The patient may be asked to sip water at a certain point in the procedure to facilitate the passage of the tube. If the tube is to be placed into the small intestine, the doctor may use an endoscope to help see where the tube is going. Once the tube is in place, material can be removed from the stomach or intestines with gentle suction.

There are several different types of nasogastric tubes, each with a different purpose. Tubes used for **stomach flushing** are called orogastric tubes and are the largest in diameter. Tubes that are threaded through the lower opening of the stomach (pylorus) and into the small intestine are stiffer and have a balloon tip. Other specialized tubes are used for long-term and short-term feeding.

Preparation

Little preparation is necessary for this procedure other than educating the patient as to what will happen. The patient should remove dental appliances before the nasogastric tube is inserted.

KEY TERMS

Endoscope—A piece of equipment with a camera and a light source in a thin tube that can be threaded through the nose into the gastrointestinal system so that the doctor can make a real-time visual examination.

Pylorus—The ring of muscle that controls the passage of material from the stomach into the small intestine.

Aftercare

After the tube is removed, no special care is needed. The patient's throat may feel irritated from the presence of the tube.

Risks

The most serious risk is that the patient will inhale some of the stomach contents into the lungs (aspiration). This may lead to bronchial infections and aspiration **pneumonia**. There is also the chance that the tube will be misplaced in the windpipe (trachea), causing violent coughing. Irritation to the throat and esophagus can cause bleeding.

Normal results

Nasogastric suctioning is normally well tolerated by patients and is a temporary treatment, performed in conjunction with other therapies.

Resources

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Tish Davidson, A.M.

Nasopharyngeal culture

Definition

A nasopharyngeal culture is used to identify pathogenic (disease-causing) organisms present in the nasal cavity that may cause upper respiratory tract symptoms.

Purpose

Some organisms that cause upper respiratory infections are carried primarily in the nasopharynx, or back of the nose. The person carrying these pathogenic bacteria may have no symptoms, but can still infect others with the pathogen and resulting illness. The most serious of these organisms is *Neisseria meningitidis*, which causes **meningitis** or blood stream infection in infants. By culturing a sample from the nasopharynx, the physician can identify this organism, and others, in the asymptomatic carrier. The procedure can also be used as a substitute for a **throat culture** in infants, the elderly patient, the debilitated patient, or in cases where a throat culture is difficult to obtain.

Precautions

The technician taking the specimen should wear gloves to prevent spreading infectious organisms. The patient should not be taking **antibiotics**, as these drugs may influence the test results.

Description

The patient should **cough** before collection of the specimen. Then, as the patient tilts his or her head backwards, the caregiver will inspect the back of the throat using a penlight and tongue depressor. A swab on a flexible wire is inserted into the nostril, back to the nasal cavity and upper part of the throat. The swab is rotated quickly and then removed. Next, the swab is placed into a sterile tube with culture fluid in it for transport to the microbiology laboratory. To prevent contamination, the swab should not touch the patient's tongue or side of the nostrils.

When the sample reaches the laboratory, the swab will be spread onto an agar plate and the agar plate incubated for 24–48 hours, to allow organisms present to grow. These organisms will be identified and any pathogenic organisms may also be tested for susceptibility to specific antibiotics. This allows the treating physician to determine which antibiotics will be effective.

Alternative procedures

In most cases of upper respiratory tract infections, a throat culture is more appropriate than a nasopharyngeal culture. However, the nasopharyngeal culture should be used in cases where throat cultures are difficult to obtain or to detect the carrier states of *Harmophilus influenzae* and meningococcal disease.

Some researchers regard the immunoblot method as preferable to a standard culture to detect certain

species of pneumococci and other organisms that cause **pneumonia**. The immunoblot method uses a membrane that changes color in response to a specific antigen-antibody reaction.

As of the early 2000s, polymerase chain reaction (PCR) analysis is considered more sensitive than standard culture in detecting *Bordetella pertussis*, the bacterium that causes **whooping cough**. PCR has the additional advantage of providing test results more rapidly than culture.

Preparation

The procedure of inserting the swab should be described to the patient, as there is a slight discomfort associated with taking the sample. Other than that, no special preparation is necessary.

Aftercare

None

Risks

There is little to no risk involved in a nasopharyngeal culture.

Normal results

Bacteria that normally grow in the nose cavity will be identified by a nasopharyngeal culture. These include nonhemolytic streptococci, alpha-hemolytic streptococci, some *Neisseria* species, and some types of staphylococci.

Abnormal results

Pathogenic organisms that might be identified by this culture include

- Group A beta-hemolytic streptococci
- *Bordetella pertussis*, the causative agent of whooping cough
- *Corynebacterium diphtheriae*, the causative agent of diphtheria
- *Staphylococcus aureus*, the causative agent of many staphylococcal infections.

Additional bacteria are abnormal if they are found in large amounts. These include

- *Haemophilus influenzae*, a causative agent for certain types of meningitis and chronic pulmonary disease.
- *Streptococcus pneumoniae*, a causative agent of pneumonia
- *Candida albicans*, the causative agent of thrush.

KEY TERMS

Antibiotic—A drug given to stop the growth of bacteria. Antibiotics are ineffective against viruses.

Nasopharynx—The back wall of the nasal cavity where it meets the throat.

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American Medical Association. 515 N. State St., Chicago, IL 60612. (312) 464-5000. <<http://www.ama-assn.org>>.

Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

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Native American health see **Minority health**

Naturopathic medicine

Definition

Naturopathic medicine is a branch of medicine in which a variety of natural medicines and treatments are used to heal illness. It uses a system of medical diagnosis and therapeutics based on the patterns of chaos and organization in nature. It is founded on the premise that people are naturally healthy, and that healing can occur through removing obstacles to a cure and by stimulating the body's natural healing abilities. The foundations of health in natural

medicine are diet, **nutrition**, homeopathy, physical manipulation, **stress** management, and **exercise**.

Naturopaths are general practitioners who treat a wide variety of illnesses. They believe in treating the "whole person"—the spirit as well as the physical body—and emphasize preventive care. They often recommend changes in diet and lifestyle to enhance the health of their patients.

Purpose

Naturopathic medicine is useful for treating chronic as well as acute diseases. It is sometimes used in conjunction with allopathic care to enhance wellness and relieve chronic symptoms, such as **fatigue** and **pain**. A naturopath treats a wide range of health problems, ranging from back pain to depression.

A naturopathic physician will spend extra time interviewing and examining the patient to find the underlying cause for a medical problem. Emotional and spiritual symptoms and patterns are included in the assessment. The naturopath often spends more time educating patients in preventive health, lifestyle, and nutrition than most M.D.s.

Description

Origins

People have always seen a connection to diet and disease, and many therapies are built around special **diets**. Naturopathy began in the eighteenth and nineteenth centuries, as the industrial revolution brought about unhealthy lifestyles, and the European custom of "taking the cure" at natural spas became popular. Benedict Lust, who believed deeply in natural medicine, organized naturopathy as a formal system of healthcare in the 1890s. By the early 1900s, it was flourishing.

The first naturopaths in the United States emphasized the healing properties of a nutritious diet, as did a number of their contemporaries. In the early twentieth century, for instance, John Kellogg, a physician and vegetarian, opened a sanitarium that used healing methods such as **hydrotherapy**, often prescribed by today's naturopaths. His brother Will produced health foods, such as corn flakes and shredded wheat. The Post brothers helped make naturopathic ideas popular and emphasized the value of whole grains over highly refined ones. Together with one of their employees, C.W. Post, they eventually went on to start the cereal companies that bear their names.

In the early 1900s, most states licensed naturopaths as physicians. There were 20 medical schools of

naturopathic medicine. From early on, naturopathic physicians were considered “eclectic,” since they drew on a variety of natural therapies and traditions for treating their patients.

In the 1930s, naturopathy dramatically declined for several reasons. Allopathic medicine finally stopped using therapies such as bloodletting and **heavy metal poisoning** as curatives. New therapies were more effective and less toxic. Allopathic medical schools became increasingly well-funded by foundations with links to the emerging drug industry. Also, allopathic physicians became much more organized and wielded political clout. Naturopathy has experienced a resurgence over the last 20 years, however. The lay public is aware of the connection between a healthy diet and lifestyle and avoiding chronic disease. In addition, conventional medicine is often unable to treat these chronic diseases. Patients are now health care consumers, and will seek their own resolution to health problems that cannot be resolved by conventional physicians. As a result, even medical groups which once considered naturopathy ineffective are now beginning to accept it.

Naturopathic medicine modalities include a variety of healing treatments, such as diet and clinical nutrition, homeopathy, botanical medicine, soft tissue and spinal manipulation, ultrasound, and therapeutic exercise. A naturopath provides complete diagnostic and treatment services in sciences such as obstetrics, pediatrics and obstetrics. Some are also licensed midwives.

Naturopaths consider health to be not just the absence of disease, but complete physical, mental and social well being. Naturopathic physicians often say that diseases must be healed not just by suppressing symptoms, but by rooting out the true cause. Symptoms are actually viewed as the body’s natural efforts to heal itself and restore balance.

A typical office visit to a naturopath takes an hour. During the first visit, the doctor will ask detailed questions about the patient’s symptoms, lifestyle, history of illness, and state of his or her emotions. The naturopath will take a complete medical history, and may order lab tests such as urine and blood tests. A naturopath may talk with the patient about the possible causes for an illness—poor diet, life stresses, occupational dangers, and mental, emotional, and spiritual problems. Naturopaths believe that even widely varying symptoms can sometimes be traced to one underlying cause. Often environmental or metabolic toxins or serious stress bring on an illness.

In some states, naturopaths prescribe pharmaceuticals. In these cases, naturopaths might prescribe natural medicines, such as natural hormones, glandular **thyroid hormones**, herbal extracts, **vitamins**, etc.

As with most doctors, treatment by a naturopath can range from one office visit to many. Some acute illnesses can be alleviated with one or two visits. Other chronic diseases need regular weekly or monthly attention. Clinical care provided by naturopathic physicians are covered by insurance in a number of states in the United States.

Preparations

There were about 1,500 naturopathic physicians in the United States practicing as of 2004; nearly 80% of these practitioners entered the profession following the revival of interest in naturopathy in the late 1970s. Consumers can find naturopaths by contacting the American Association of Naturopathic Physicians (AANP) or logging on to their web site. Naturopaths recommended by the AANP have met requirements for state licensure and have taken a national exam that qualifies them to practice. Qualified naturopaths can also be found through the local branch of the national or state association of naturopathic physicians. It is sometimes useful to request names from another health care provider who knows naturopathic practitioners in the community.

Some states license naturopathic physicians. As of late 2003, those states included Hawaii, Alaska, Washington, Oregon, Utah, Montana, Arizona, Connecticut, New Hampshire, Vermont, Maine, and Kansas, in addition to the territories of Puerto Rico and the Virgin Islands. Training via a correspondence school does not qualify a naturopath for licensure or to take the national qualifying examination.

Precautions

A good naturopath is always willing to work with the patient’s other physicians or health care providers. To avoid **drug interactions** and to coordinate care, it is important for a patient to inform his or her allopathic doctor about supplements prescribed by a naturopath.

Many naturopaths give childhood vaccinations, but some do not. If a parent is concerned about this, it is best to go to an allopathic doctor for vaccinations.

Naturopaths are not licensed to perform major surgery, or prescribe **narcotics** and **antidepressant drugs**. They must involve an oncologist when treating a **cancer** patient.

Side effects

Although naturopathic remedies are from natural sources and pose much less risk than traditional drugs do, there are some side effects with the use of some. One problem they can pose is the interaction with prescription medicines. It is important for a patient to inform his or her allopathic physician about any natural remedies or herbs prescribed by a naturopath.

It is also important to note that the U.S. Food and Drug Administration considers medicinal herbs as dietary supplements, not drugs, and so are not subject to the same regulations as drugs are. Because they come from natural sources, the active ingredients may not always be in the same concentration from bottle to bottle, since plants naturally vary. To guard against using too little or too much of a natural remedy, use herbs and supplements recommended by a naturopath or those produced by well-respected companies.

Research and general acceptance

Medical research in naturopathy has increased dramatically in the United States within the last 10 years. Naturopathic research often employs case histories, summaries of practitioners' clinical observations, and medical records. Some U.S. studies have also met today's scientific gold standard; they were double-blind and placebo-controlled. Much naturopathic research has also been done in Germany, France, England, India, and China.

Some mainstream medical practitioners remain distrustful of naturopathy, however. Such problems as health-food store employees without naturopathic credentials giving health-related advice to customers, or occasional rare cases of infections caused by naturopathic injections, continue to damage the reputation of this form of alternative medicine.

Research in naturopathy tends to focus on single treatments used by naturopaths, rather than naturopathy as a whole. In 1998, an extensive review of such single treatment studies found that naturopathic healing methods were effective for 15 different medical conditions, including **osteoarthritis**, **asthma**, and middle ear infections. A study of 8,341 men with damaged heart muscles in 1996 revealed that supplementation with niacin, a B vitamin, was associated with an 11% reduced risk of mortality over 15 years. In 1996, a study showed **St. John's wort** was as effective as prescription antidepressants in relieving depression, and had fewer side effects.

KEY TERMS

Clinical nutrition—The use of diet and nutritional supplements as a way to enhance health prevent disease.

Cryotherapy—The exposure of body tissue to extremely cold temperatures, often by applying a probe containing liquid nitrogen.

Herb—In naturopathy, a plant or plant derivative or extract prescribed for health or healing.

Homeopathy—The use of diluted remedies that have energetic rather than chemical properties. They are prescribed according to the axiom that "like cures like."

Hydrotherapy—The use of water as baths, poultices, and steams to heal.

Physical manipulation—The use of deep massage, spinal alignment, and joint manipulation to stimulate tissues.

Ultrasound—A therapy employing high frequency sound waves.

Studies have also demonstrated benefits in the arena of **women's health** issues. In one classic 1993 study, women with cervical dysplasia or abnormal Pap smears were treated by naturopaths with topical applications of herbs and dietary supplements. These medications included Bromelian, an enzyme from the pineapple; bloodroot; marigold; and zinc chloride; and suppositories made from herbal and nutritional ingredients, such as **echinacea**, vitamin A, and vitamin E. Thirty eight of the 43 women in the study had normal Pap smears and normal tissue biopsies after treatment. The study concluded that these protocols might benefit the health of patients undergoing more traditional treatments for cervical dysplasia, such as **cryotherapy**.

Other more recent research has documented the benefits of such nutritional foods as soy in relieving hot flashes and vaginal dryness. **Nutritional supplements** prescribed by naturopaths to enhance women's health during **menopause** have also proven effective; in general, naturopathy appears to be as useful as conventional medicine for treating menopausal symptoms. Research shows vitamin E supplements are helpful for 50% of postmenopausal women with thinning vaginal tissue. Studies also reveal that bioflavonoids with vitamin C and gamma-oryzanol, a

substance taken from rice bran oil, can relieve hot flashes.

Another area of women's health concerns that naturopathy has taken seriously is a growing preference for skin care and beauty products derived from natural sources rather than from chemical laboratories. Such products are often more beneficial to the skin and less likely to cause **rashes** or other allergic reactions.

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ORGANIZATIONS

The American Association of Naturopathic Physicians. 601 Valley Street, Suite 105, Seattle, WA 98109. (206) 298-0126. <<http://www.naturopathic.org>>.

Naturopathic Physicians Licensing Examination Board (NPLEX). P. O. Box 69657, Portland, OR 97201. (503) 250-9141. <<http://www.nabne.org/html/index2.html>>.

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Naturopathy see **Naturopathic medicine**

Nausea and vomiting

Definition

Nausea is the sensation of being about to vomit. Vomiting, or emesis, is the expelling of undigested food through the mouth.

Description

Nausea is a reaction to a number of causes that include overeating, infection, or irritation of the throat or stomach lining. Persistent or recurrent nausea and vomiting should be checked by a doctor.

A doctor should be called if nausea and vomiting occur:

- after eating rich or spoiled food or taking a new medication
- repeatedly or for 48 hours or longer
- following intense **dizziness**

It is important to see a doctor if nausea and vomiting are accompanied by:

- yellowing of the skin and whites of the eyes
- **pain** in the chest or lower abdomen
- trouble with swallowing or urination
- **dehydration** or extreme thirst
- drowsiness or confusion
- constant, severe abdominal pain
- a fruity breath odor

A doctor should be notified if vomiting is heavy and/or bloody, if the vomitus looks like feces, or if the patient has been unable to keep food down for 24 hours.

An ambulance or emergency response number should be called immediately if:

- Diabetic **shock** is suspected
- Nausea and vomiting continue after other symptoms of viral infection have subsided
- The patient has a severe **headache**
- The patient is sweating and having chest pain and trouble breathing
- The patient is known or suspected to have swallowed a **drug overdose** or poisonous substance
- The patient has a high body temperature, **muscle cramps**, and other signs of heat exhaustion or heat stroke

- Nausea, vomiting, and breathing problems occur after exposure to a known allergen

Causes and symptoms

Persistent, unexplained, or recurring nausea and vomiting can be symptoms of a variety of serious illnesses. It can be caused by simply overeating or drinking too much alcohol. It can be due to **stress**, certain medications, or illness. For example, people who are given morphine or other opioid medications for pain relief after surgery sometimes feel nauseated by the drug. Such poisonous substances as arsenic and other heavy metals cause nausea and vomiting. Morning sickness is a consequence of pregnancy-related hormone changes. **Motion sickness** can be induced by traveling in a vehicle, plane, or on a boat. Many patients experience nausea after eating spoiled food or foods to which they are allergic. Patients who suffer migraine headache often experience nausea. **Cancer** patients on **chemotherapy** are nauseated. **Gallstones**, **gastroenteritis** and stomach ulcer may cause nausea and vomiting. These symptoms should be evaluated by a physician.

Nausea and vomiting may also be psychological in origin. Some people vomit under such conditions of emotional stress as family arguments, academic tests, airplane travel, losing a job, and similar high-stress situations. In addition, some eating disorders are characterized by self-induced vomiting.

Diagnosis

Diagnosis is based on the severity, frequency, and duration of symptoms, and other factors that could indicate the presence of a serious illness.

Diagnosis is based on the taking of a careful patient history. In some cases, the doctor may order laboratory tests or imaging studies to determine the presence of drugs or poisonous substances in the patient's blood or urine, or evidence of head injuries or abnormalities in the digestive tract. If the nausea and vomiting appear to be related to **anxiety**, stress, or an eating disorder, the doctor may refer the patient to a psychiatrist for further evaluation.

Treatment

Getting a breath of fresh air or getting away from whatever is causing the nausea can solve the problem. Eating olives or crackers or sucking on a lemon can calm the stomach by absorbing acid and excess fluid. Coke syrup is another proven remedy.

Vomiting relieves nausea right away but can cause dehydration. Sipping clear juices, weak tea, and some sports drinks help replace lost fluid and minerals without irritating the stomach. Food should be reintroduced gradually, beginning with small amounts of dry, bland food like crackers and toast.

Medications that are given to relieve nausea and vomiting are called antiemetics. Meclizine (Bonine), a medication for motion sickness, also diminishes the feeling of queasiness in the stomach. Dimenhydrinate (Dramamine), another motion-sickness drug, is not effective on other types of nausea and may cause drowsiness.

Newer drugs that have been developed to treat postoperative or postchemotherapy nausea and vomiting include ondansetron (Zofran) and granisetron (Kytril). Another treatment that has been found to lower the risk of nausea after surgery is intravenous administration of supplemental fluid before the operation.

Alternative treatment

Advocates of alternative treatments suggest biofeedback, acupressure and the use of herbs to calm the stomach. Biofeedback uses **exercise** and deep relaxation to control nausea. Acupressure (applying pressure to specific areas of the body) can be applied by wearing a special wristband or by applying firm pressure to:

- the back of the jawbone
- the webbing between the thumb and index finger
- the top of the foot
- the inside of the wrist
- the base of the rib cage

Acupuncture is another alternative treatment found to be effective in relieving nausea. A few people, however, experience nausea as a side effect of acupuncture.

Chamomile (*Matricaria recutita*) or lemon balm (*Melissa officinalis*) tea may relieve symptoms. Ginger (*Zingiber officinale*), another natural remedy, can be drunk as tea or taken as candy or powered capsules.

Prevention

Massage, **meditation**, yoga, and other relaxation techniques can help prevent stress-induced nausea. Anti-nausea medication taken before traveling can prevent motion sickness. Sitting in the front seat,



These illustrations depict the mechanism and causes of vomiting in the human body. An impulse from the brain stimulates the vomiting center (top center) in the brain stem. Nerve impulses sent to the stomach, diaphragm, and abdominal wall (bottom center) result in stomach's contents being expelled. Other causes of vomiting include raised pressure in the skull due to injury or tumor (upper right), and hormonal changes during pregnancy. (Illustration by John Bavosi, Custom Medical Stock Photo. Reproduced by permission.)

KEY TERMS

Acupuncture—A treatment technique associated with traditional Chinese medicine, in which thin needles are inserted into specific points located along energy channels in the human body known as meridians.

Antiemetic—A preparation or medication that relieves nausea and vomiting. Coke syrup, ginger, and motion sickness medications are examples of antiemetics.

Dehydration—Loss of fluid and minerals following vomiting, prolonged diarrhea, or excessive sweating.

Diabetic coma—Reduced level of consciousness that requires immediate medical attention.

Emesis—The medical term for vomiting.

focusing on the horizon, and traveling after dark can also minimize symptoms.

Food should be fresh, properly prepared, and eaten slowly. Overeating, tight-fitting clothes, and strenuous activity immediately after a meal should be avoided.

Vomiting related to emotional upsets may be avoided by forms of psychotherapy that teach patients to manage stress in healthier ways.

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American Gastroenterological Association (AGA). 7910 Woodmont Ave., 7th Floor, Bethesda, MD 20814. (310) 654-2055. <<http://www.gastro.org/index.html>>. aga001@aol.com.

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Nberg disease see **Osteopetroses**

NCV see **Electromyography**

Near-drowning

Definition

Near-drowning is the term for survival after suffocation caused by submersion in water or another fluid. Some experts exclude from this definition cases of temporary survival that end in death within 24 hours, which they prefer to classify as drownings.

Description

An estimated 15,000–70,000 near-drownings occur in the United States each year (insufficient reporting prevents a better estimate). The typical victim is young and male. Nearly half of all drownings and near-drownings involve children less than four years old. Home swimming pools pose the greatest risk for children, being the site of 60–90% of drownings in the 0–4 age group. Teenage boys also face a heightened risk of drowning and near-drowning, largely because of their tendency to behave recklessly and use drugs and alcohol (drugs and alcohol are implicated in 40–50% of teenage drownings). Males, however, predominate even in the earliest age-groups, possibly because young boys are often granted more freedom from supervision than young girls enjoy, making it more likely that they will stumble into danger and less likely that they will attract an adult's attention in time for a quick rescue. Roughly four out of five drowning victims are males.

Causes and symptoms

The circumstances leading to near-drownings (and drownings also) cannot be reduced to a single scenario involving nonswimmers accidentally entering deep water. On many occasions, near-drownings are secondary to an event such as a **heart attack** that causes unconsciousness or a head or spinal injury that prevents a diver from resurfacing. Near-drownings, moreover, can occur in shallow as well as deep water. Small children have drowned or almost drowned in bathtubs, toilets, industrial-size cleaning buckets, and washing machines. Bathtubs are especially dangerous for infants six months to one year old, who can sit up straight in a bathtub but may lack the ability to pull themselves out of the water if they slip under the surface.

A reduced concentration of oxygen in the blood (hypoxemia) is common to all near-drownings. Human life, of course, depends on a constant supply of oxygen-laden air reaching the blood by way of the

lungs. When drowning begins, the larynx (an air passage) closes involuntarily, preventing both air and water from entering the lungs. In 10–15% of cases, hypoxemia results because the larynx stays closed. This is called “dry drowning.” Hypoxemia also occurs in “wet drowning,” the 85–90% of cases where the larynx relaxes and water enters the lungs. The physiological mechanisms that produce hypoxemia in wet drowning are different for freshwater and saltwater, but only a small amount of either kind of water is needed to damage the lungs and interfere with the body’s oxygen intake. All of this happens very quickly: within three minutes of submersion most people are unconscious, and within five minutes the brain begins to suffer from lack of oxygen. Abnormal heart rhythms (cardiac dysrhythmias) often occur in near-drowning cases, and the heart may stop pumping (cardiac arrest). An increase in blood acidity (acidosis) is another consequence of near-drowning, and under some circumstances near-drowning can cause a substantial increase or decrease in the volume of circulating blood. Many victims experience a severe drop in body temperature (**hypothermia**).

The signs and symptoms of near-drowning can differ widely from person to person. Some victims are alert but agitated, while others are comatose. Breathing may have stopped, or the victim may be gasping for breath. Bluish skin (**cyanosis**), coughing, and frothy pink sputum (material expelled from the respiratory tract by coughing) are often observed. Rapid breathing (tachypnea), a rapid heart rate (tachycardia), and a low-grade fever are common during the first few hours after rescue. Conscious victims may appear confused, lethargic, or irritable.

Diagnosis

Diagnosis relies on a **physical examination** of the victim and on a wide range of tests and other procedures. Blood is taken to measure oxygen levels and for many other purposes. Pulse oximetry, another way of assessing oxygen levels, involves attaching a device called a pulse oximeter to the patient’s finger. An electrocardiograph is used to monitor heart activity. X rays can detect head and neck injuries and excess tissue fluid (**edema**) in the lungs.

Treatment

Treatment begins with removing the victim from the water and performing **cardiopulmonary resuscitation (CPR)**. One purpose of CPR—which, of course, should be attempted only by people trained in its use—is to bring oxygen to the lungs, heart, brain, and other

organs by breathing into the victim’s mouth. When the victim’s heart has stopped, **CPR** also attempts to get the heart pumping again by pressing down on the victim’s chest. After CPR has been performed and emergency medical help has arrived on the scene, oxygen is administered to the victim. If the victim’s breathing has stopped or is otherwise impaired, a tube is inserted into the windpipe (trachea) to maintain the airway (this is called endotracheal intubation). The victim is also checked for head, neck, and other injuries, and fluids are given intravenously. Hypothermia cases require careful handling to protect the heart.

In the emergency department, victims continue receiving oxygen until blood tests show a return to normal. About one-third are intubated and initially need mechanical support to breathe. Rewarming is undertaken when hypothermia is present. Victims may arrive needing treatment for cardiac arrest or cardiac dysrhythmias. Comatose patients present a special problem: although various treatment approaches have been tried, none have proved beneficial. Patients can be discharged from the emergency department after four to six hours if their blood oxygen level is normal and no signs or symptoms of near-drowning are present. But because lung problems can arise 12 or more hours after submersion, the medical staff must first be satisfied that the patients are willing and able to seek further medical help if necessary. Admission to a hospital for at least 24 hours for further observation and treatment is a must for patients who do not appear to recover fully in the emergency department.

Prognosis

Neurological damage is the major long-term concern in the treatment of near-drowning victims. Patients who arrive at an emergency department awake and alert usually survive with brain function intact, as do about 90% of those who arrive mentally impaired (lethargic, confused, and so forth) but not comatose. **Death** or permanent neurological damage is very likely when patients arrive comatose. Early rescue of near-drowning victims (within five minutes of submersion) and prompt CPR (within less than 10 minutes of submersion) seem to be the best guarantees of a complete recovery. An analysis of 715 patients admitted to emergency departments in 1971–81 revealed that 69% recovered completely, 25% died, and 6% survived but suffered permanent neurological damage.

Prevention

Prevention depends on educating parents, other adults, and teenagers about water safety. Parents must

realize that young children who are left in or near water without adult supervision even for a short time can easily get into trouble, not just at the beach or next to a swimming pool, but in bathtubs and around toilets, buckets, washing machines, and other household articles where water can collect. Research on swimming pool drownings involving young children shows that the victims have usually been left unattended less than five minutes before the accident. Experts consider putting up a fence around a home swimming pool an essential precaution, and estimate that 50–90% of child drownings and near-drownings could be prevented if fences were widely adopted. The fence should be at least five feet high and unclimbable, have a self-closing and self-locking gate, and completely surround the pool.

Pool owners—and, indeed, all other adults—should consider learning CPR. Everyone, of course, should follow the rules for safe swimming and boating. Those who have a medical condition that can cause a seizure or otherwise threaten safety in the water are advised always to swim with a partner. And of course, people need to be aware that alcohol and drug use substantially increase the chances of an accident.

The danger of alcohol and drug use around water is a point that requires special emphasis where teenagers are concerned. Teenagers can also benefit from CPR training and safe swimming and boating classes.

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Howard Baker

Nearsightedness *see* **Myopia**

Necrotizing enterocolitis

Definition

Necrotizing enterocolitis is a serious bacterial infection in the intestine, primarily of sick or premature newborn infants. It can cause the death (necrosis) of intestinal tissue and progress to blood poisoning (septicemia).

Description

Necrotizing enterocolitis develops in approximately 10% of newborns weighing less than 800 g (under 2 lbs). It is a serious infection that can produce complications in the intestine itself—such as ulcers, perforations (holes) in the intestinal wall, and tissue necrosis—as well as progress to life-threatening septicemia. Necrotizing enterocolitis most commonly affects the lower portion of the small intestine (ileum). It is less common in the colon and upper small bowel.

Causes and symptoms

The cause of necrotizing enterocolitis is not clear. It is believed that the infection usually develops after the bowel wall has already been weakened or damaged by a lack of oxygen, predisposing it to bacterial invasion. Bacteria proliferate in the bowel and cause a deep infection that can kill bowel tissue and spread to the bloodstream.

Necrotizing enterocolitis almost always occurs in the first month of life. Infants who require **tube feedings** may have an increased risk for the disorder. A number of other conditions also make newborns susceptible, including **respiratory distress syndrome**, congenital heart problems, and episodes of apnea (cessation of breathing). The primary risk factor, however, is **prematurity**. Not only is the immature digestive tract less able to protect itself, but premature infants are subjected to many stresses on the body in their attempt to survive.

Early symptoms of necrotizing enterocolitis include an intolerance to formula, distended and tender abdomen, **vomiting**, and blood (visible or not) in the stool. One of the earliest signs may also be the need for mechanical support of the infant's breathing. If the infection spreads to the bloodstream, infants may develop lethargy, fluctuations in body temperature, and periodically stop breathing.

Diagnosis

The key to reducing the complications of this disease is early suspicion by the physician. A series of x rays of the bowel often reveals the progressive condition, and blood tests confirm infection.

Treatment

Over two-thirds of infants can be treated without surgery. Aggressive medical therapy is begun as soon as the condition is diagnosed or even suspected. Tube

feedings into the gastrointestinal tract (enteral **nutrition**) are discontinued, and tube feedings into the veins (parenteral nutrition) are used instead until the condition has resolved. Intravenous fluids are given for several weeks while the bowel heals.

Some infants are placed on a ventilator to help them breathe, and some receive transfusions of platelets, which help the blood clot when there is internal bleeding. **Antibiotics** are usually given intravenously for at least 10 days. These infants require frequent evaluations by the physician, who may order multiple abdominal x rays and blood tests to monitor their condition during the illness.

Sometimes, necrotizing enterocolitis must be treated with surgery. This is often the case when an infant's condition does not improve with medical therapy or there are signs of worsening infection.

The surgical treatment depends on the individual patient's condition. Patients with infection that has caused serious damage to the bowel may have portions of the bowel removed. It is sometimes necessary to create a substitute bowel by making an opening (**ostomy**) into the abdomen through the skin, from which waste products are discharged temporarily. But many physicians are avoiding this and operating to remove diseased bowel and repair the defect at the same time.

Postoperative complications are common, including wound infections and lack of healing, persistent **sepsis** and bowel necrosis, and a serious internal bleeding disorder known as disseminated intravascular coagulation.

Prognosis

Necrotizing enterocolitis is the most common cause of death in newborns undergoing surgery. The average mortality is 30–40%, even higher in severe cases.

Early identification and treatment are critical to improving the outcome for these infants. Aggressive nonsurgical support and careful timing of surgical intervention have improved overall survival; however, this condition can be fatal in about one-third of cases. With the resolution of the infection, the bowel may begin functioning within weeks or months. But infants need to be carefully monitored by a physician for years because of possible future complications.

About 10–35% of all survivors will eventually develop a stricture, or narrowing, of the intestine that occurs with healing. This can create an intestinal obstruction that will require surgery. Infants may also be more susceptible to future bacterial infections in the

KEY TERMS

Enteral nutrition—Liquid nutrition provided through tubes that enter the gastrointestinal tract, usually through the mouth or nose.

Necrosis—The death of cells, a portion of tissue, or a portion of an organ due to permanent damage of some sort, such as a lack of oxygen supply to the tissues.

Parenteral nutrition—Liquid nutrition provided through tubes that are placed in the veins.

Sepsis—The presence of pus-forming or other disease-causing organisms in the blood or tissues. Septicemia, commonly known as blood poisoning, is a common type of sepsis.

gastrointestinal tract and to a delay in growth. Infants with severe cases may also suffer neurological impairment.

The most serious long-term gastrointestinal complication associated with necrotizing enterocolitis is short-bowel, or short-gut, syndrome. This refers to a condition that can develop when a large amount of bowel must be removed, making the intestines less able to absorb certain nutrients and enzymes. These infants gradually evolve from tube feedings to oral feedings, and medications are used to control the malabsorption, **diarrhea**, and other consequences of this condition.

Prevention

In very small or sick premature infants, the risk for necrotizing enterocolitis may be diminished by beginning parenteral nutrition and delaying enteral feedings for several days to weeks.

Some have suggested that breast milk provides substances that may be protective, but there is no evidence that this reduces the risk of infection. A large multicenter trial showed that steroid drugs given to women in preterm labor may protect their offspring from necrotizing enterocolitis.

Sometimes necrotizing enterocolitis occurs in clusters, or outbreaks, in hospital newborn (neonatal) units. Because there is an infectious element to the disorder, infants with necrotizing enterocolitis may be isolated to avoid infecting other infants. Persons caring for these infants must also employ strict measures to prevent spreading the infection.

Resources

OTHER

Neonatology on the Web. < <http://www.neonatology.org> >.

Caroline A. Helwick

Necrotizing fasciitis see **Flesh-eating disease**

Neisseria gonorrhoeae infection see
Gonorrhea

Neisseria meningitidis bacteremia see
Meningococemia

Nelfinavir see **Protease inhibitors**

Neonatal jaundice

Definition

Neonatal **jaundice** (or hyperbilirubinemia) is a higher-than-normal level of bilirubin in the blood. Bilirubin is a by-product of the breakdown of red blood cells. This condition can cause a yellow discoloration of the skin and the whites of the eyes called jaundice.

Description

Bilirubin, a by-product of the breakdown of hemoglobin (the oxygen-carrying substance in red blood cells), is produced when the body breaks down old red blood cells. Normally, the liver processes the bilirubin and excretes it in the stool. Hyperbilirubinemia means there is a high level of bilirubin in the blood. This condition is particularly common in newborn infants. Before birth, an infant gets rid of bilirubin through the mother's blood and liver systems. After birth, the baby's liver has to take over processing bilirubin on its own. Almost all newborns have higher than normal levels of bilirubin. In most cases, the baby's systems continue to develop and can soon process bilirubin. However, some infants may need medical treatment to prevent serious complications which can occur due to the accumulation of bilirubin.

Causes and symptoms

In newborn infants, the liver and intestinal systems are immature and cannot excrete bilirubin as fast as the body produces it. This type of hyperbilirubinemia can cause jaundice to develop within a few days after birth. About one-half of all newborns develop



A newborn baby undergoes phototherapy with visible blue light to treat his jaundice. (Photograph by Ron Sutherland, Photo Researchers, Inc. Reproduced by permission.)

jaundice, while premature infants are much more likely to develop it. Hyperbilirubinemia is also more common in some populations, such as Native American and Asian. All infants with jaundice should be evaluated by a health care provider to rule out more serious problems.

Hyperbilirubinemia and jaundice can also be the result of other diseases or conditions. Hepatitis, **cirrhosis** of the liver, and mononucleosis are diseases that can affect the liver. **Gallstones**, a blocked bile duct, or the use of drugs or alcohol can also cause jaundice.

Extremely high levels of bilirubin in infants may cause kernicterus, a form of brain damage. Signs of severe hyperbilirubinemia include listlessness, high-pitched crying, apnea (periods of not breathing), arching of the back, and seizures. If severe hyperbilirubinemia is not treated, it can cause **mental retardation**, hearing loss, behavior disorders, **cerebral palsy**, or **death**.

KEY TERMS

Bilirubin—A yellowish-brown substance in the blood that forms as old red blood cells are broken down.

Hemoglobin—A protein, an oxygen-carrying pigment of the erythrocyte (red blood cell) formed in the bone marrow.

Jaundice—A yellow discoloration of the skin and whites of the eyes.

Kernicterus—A serious condition where high bilirubin levels cause brain damage in infants.

Diagnosis

The initial diagnosis of hyperbilirubinemia is based on the appearance of jaundice at physical examination. The child is often placed by an open window so he/she may be checked in natural light. Blood samples may be taken to determine the bilirubin level in the blood.

Treatment

Most cases of newborn jaundice resolve without medical treatment within two to three weeks, but should be checked by the health care provider. It is important that the infant is feeding regularly and having normal bowel movements. If bilirubin levels are extremely high, the infant may be treated with phototherapy—exposure of the baby's skin to fluorescent light. The bilirubin in the baby's skin absorbs the light and is changed to a substance that can be excreted in the urine. This treatment can be done in the hospital and is often done at home with special lights which parents can rent for the treatment. Treatment may be needed for several days before bilirubin levels in the blood return to normal. The baby's eyes are shielded to prevent the optic nerves from absorbing too much light. Another type of treatment uses a special fiberoptic blanket. There is no need to shield the baby's eyes with this treatment, and it can be done at home. In rare cases, where bilirubin levels are extremely high, the baby may need to receive a blood transfusion.

Prognosis

Most infants with hyperbilirubinemia and associated jaundice recover without medical treatment. **Phototherapy** is very effective in reducing bilirubin levels in the majority of infants who need it. There

are usually no long-term effects on the child from the hyperbilirubinemia or the phototherapy. It is very rare that a baby may need a blood transfusion for treatment of this condition.

Prevention

There is no way to predict which infants will be affected by hyperbilirubinemia. Newborns should be breastfed or given formula frequently, and feedings should begin as soon as possible after delivery to increase activity of the baby's digestive system.

Resources

OTHER

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"Neonatology on the Web." <<http://www.neonatology.org>>.

Altha Roberts Edgren

Nephrectomy

Definition

Nephrectomy is the surgical procedure of removing a kidney or section of a kidney.

Purpose

Nephrectomy, or kidney removal, is performed on patients with **cancer** of the kidney (renal cell carcinoma); a disease in which cysts (sac-like structures) displace healthy kidney tissue (**polycystic kidney disease**); and serious kidney infections. It is also used to remove a healthy kidney from a donor for the purposes of **kidney transplantation**.

Precautions

Because the kidney is responsible for filtering wastes and fluid from the bloodstream, kidney function is critical to life. Nephrectomy candidates suffering from serious

kidney disease, cancer, or infection usually have few treatment choices but to undergo the procedure. However, if kidney function is lost in the remaining kidney, the patient will require chronic dialysis treatments or transplantation of a healthy kidney to sustain life.

Description

Nephrectomy may involve removing a small portion of the kidney or the entire organ and surrounding tissues. In partial nephrectomy, only the diseased or infected portion of the kidney is removed. Radical nephrectomy involves removing the entire kidney, a section of the tube leading to the bladder (ureter), the gland that sits atop the kidney (adrenal gland), and the fatty tissue surrounding the kidney. A simple nephrectomy performed for transplant purposes requires removal of the kidney and a section of the attached ureter. A similar procedure is used to harvest cadaver kidneys, although both kidneys are typically removed at once (bilateral nephrectomy) and blood and cell samples for **tissue typing** are also taken.

The nephrectomy patient is administered **general anesthesia** and the surgeon makes an incision on the side or front of the abdomen. Muscle, fat, and tissue are cut away to reveal the kidney. The blood vessels connecting the kidney to the circulation are cut and clamped. Depending on the type of nephrectomy procedure being performed, the ureter, adrenal gland, and/or surrounding tissue may also be cut. The vessels and the ureter in the patient are then tied off and the incision is sewn up (sutured). The surgical procedure can take up to three hours, depending on the type of nephrectomy being performed.

Laparoscopic nephrectomy is a form of minimally-invasive surgery that utilizes instruments on long, narrow rods to view, cut, and remove the kidney. The surgeon views the kidney and surrounding tissue with a flexible videoscope. The videoscope and surgical instruments are maneuvered through four small incisions in the abdomen. Once the kidney is freed, it is secured in a bag and pulled through a fifth incision, approximately 3 in (7.6 cm) wide, in the front of the abdominal wall below the navel. Although this surgical technique takes slightly longer than a traditional nephrectomy, preliminary studies have shown that it promotes a faster recovery time, shorter hospital stays, and less post-operative **pain** for kidney donors.

Preparation

Prior to surgery, blood samples will be taken from the patient to type and crossmatch in case **transfusion**

KEY TERMS

Cadaver kidney—A kidney from a brain-dead organ donor used for purposes of kidney transplantation.

Polycystic kidney disease—A hereditary kidney disease that causes fluid- or blood-filled pouches of tissue called cysts to form on the tubules of the kidneys. These cysts impair normal kidney function.

Renal cell carcinoma—Cancer of the kidney.

is required during surgery. A catheter will also be inserted into the patient's bladder. The surgical procedure will be described to the patient, along with the possible risks.

Aftercare

Nephrectomy patients may experience considerable discomfort in the area of the incision. Patients may also experience **numbness**, caused by severed nerves, near or on the incision. Pain relievers are administered following the surgical procedure and during the recovery period on an as-needed basis. Although deep breathing and coughing may be painful due to the proximity of the incision to the diaphragm, breathing exercises are encouraged to prevent **pneumonia**. Patients should not drive an automobile for a minimum of two weeks.

Risks

Possible complications of a nephrectomy procedure include infection, bleeding (hemorrhage), and post-operative pneumonia. There is also the risk of kidney failure in a patient with impaired function or disease in the remaining kidney.

Normal results

Normal results of a nephrectomy are dependent on the purpose of the procedure and the type of nephrectomy performed. Immediately following the procedure, it is normal for patients to experience pain near the incision site, particularly when coughing or breathing deeply. Renal function of the patient is monitored carefully after nephrectomy surgery. If the remaining kidney is healthy, it will increase its functioning over time to compensate for the loss of the removed kidney.

Length of hospitalization depends on the type of nephrectomy procedure. Patients undergoing a laparoscopic radical nephrectomy may be released within two to four days after surgery. Traditional open nephrectomy patients are typically hospitalized for about a week. Recovery time will also vary, on average from three to six weeks.

Resources

ORGANIZATIONS

National Kidney Foundation. 30 East 33rd St., New York, NY 10016. (800) 622-9010. <<http://www.kidney.org>>.

Paula Anne Ford-Martin

Nephritic syndrome see **Glomerulonephritis**

Nephritis

Definition

Nephritis is inflammation of the kidney.

Description

The most prevalent form of acute nephritis is **glomerulonephritis**. This condition affects children and teenagers far more often than it affects adults. It is inflammation of the glomeruli, or small round filters located in the kidney. **Pyelonephritis** affects adults more than children, and is recognized as inflammation of the kidney and upper urinary tract. A third type of nephritis is hereditary nephritis, a rare inherited condition.

Causes and symptoms

Acute glomerulonephritis usually develops a few weeks after a strep infection of the throat or skin. Symptoms of glomerulonephritis include **fatigue**, high blood pressure, and swelling. Swelling is most notable in the hands, feet, ankles and face.

Pyelonephritis usually occurs suddenly, and the acute form of this disease is more common in adult women. The most common cause of this form of bacterial nephritis is the backward flow of infected urine from the bladder into the upper urinary tract. Its symptoms include **fever** and chills, fatigue, burning or frequent urination, cloudy or bloody urine, and aching **pain** on one or both sides of the lower back or abdomen.

Hereditary nephritis can be present at birth. The rare disease presents in many different forms and can

be responsible for up to 5% of end-stage renal disease in men.

Diagnosis

Diagnosis of nephritis is based on:

- the patient's symptoms and medical history
- physical examination
- laboratory tests
- kidney function tests
- imaging studies such as ultrasound or x rays to determine blockage and inflammation

Urinalysis can reveal the presence of:

- albumin and other proteins
- red and white blood cells
- pus, blood, or bacteria in the urine

Treatment

Treatment of glomerulonephritis normally includes drugs such as cortisone or cytotoxic drugs (those that are destructive to certain cells or antigens). **Diuretics** may be prescribed to increase urination. If high blood pressure is present, drugs may be prescribed to decrease the **hypertension**. Iron and vitamin supplements may be recommended if the patient becomes anemic.

Acute pyelonephritis may require hospitalization for severe illness. **Antibiotics** will be prescribed, with the length of treatment based on the severity of the infection. In the case of chronic pyelonephritis, a six-month course of antibiotics may be necessary to rid the infection. Surgery is sometimes necessary.

Treatment of hereditary nephritis depends of the variety of the disease and severity at the time of treatment.

Alternative treatment

Alternative treatment of nephritis should be used as a complement to medical care and under the supervision of a licensed practitioner. Some herbs thought to relieve symptoms of nephritis include cleavers (*Galium* spp.) and wild hydrangea.

Prognosis

Prognosis for most cases of glomerulonephritis is generally good. Ninety percent of children recover without complications. With proper medical

treatment, symptoms usually subside within a few weeks, or at the most, a few months.

Pyelonephritis in the acute form offers a good prognosis if diagnosed and treated early. Follow-up urinalysis studies will determine if the patient remains bacteria-free. If the infection is not cured or continues to recur, it can lead to serious complications such as **bacteremia** (bacterial invasion of the bloodstream), hypertension, chronic pyelonephritis and even permanent kidney damage.

If hereditary nephritis is not detected or treated, it can lead to complications such as eye problems, deafness or kidney failure.

Prevention

Streptococcal infections that may lead to glomerulonephritis can be prevented by avoiding exposure to strep infection and obtaining prompt medical treatment for **scarlet fever** or other infection.

Pyelonephritis can best be avoided if those with a history of urinary tract infections take care to drink plenty of fluids, urinate frequently, and practice good hygiene following urination.

Hereditary nephritis can not be prevented, but research to combat the disease continues.

Resources

ORGANIZATIONS

American Kidney Fund (AKF). Suite 1010, 6110 Executive Boulevard, Rockville, MD 20852. (800) 638-8299. <<http://216.248.130.102/Default.htm>>.

National Kidney Foundation. 30 East 33rd St., New York, NY 10016. (800) 622-9010. <<http://www.kidney.org>>.

OTHER

"Glomerulonephritis." *National Institute of Diabetes and Digestive and Kidney Disease*. <<http://www.niddk.nih.gov>>.

Maureen Haggerty

Nephroblastoma see **Wilms' tumor**

Nephrocarcinoma see **Kidney cancer**

Nephrotic syndrome

Definition

Nephrotic syndrome is a collection of symptoms which occur because the tiny blood vessels (the

glomeruli) in the kidney become leaky. This allows protein (normally never passed out in the urine) to leave the body in large amounts.

Description

The glomeruli (a single one is called a glomerulus) are tiny tufts of capillaries (the smallest type of blood vessels). Glomeruli are located in the kidneys, where they allow a certain amount of water and waste products to leave the blood, ultimately to be passed out of the body in the form of urine. Normally, proteins are unable to pass through the glomerular filter. Nephrotic syndrome, however, occurs when this filter becomes defective, allowing large quantities of protein to leave the blood circulation, and pass out of the body in the urine.

Patients with nephrotic syndrome are from all age groups, although in children there is an increased risk of the disorder between the ages of 18 months and four years. In children, boys are more frequently affected; in adults, the ratio of men to women is closer to equal.

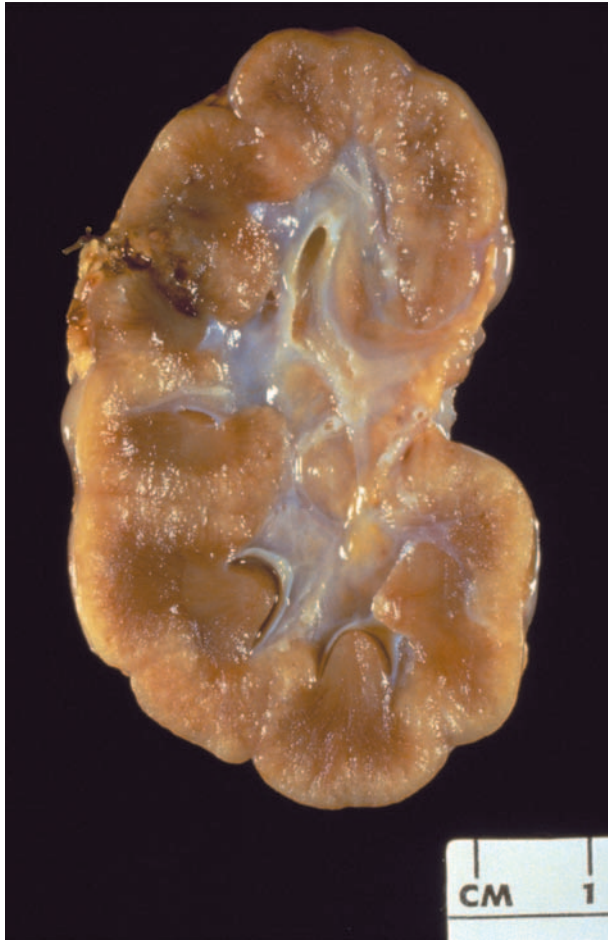
Causes and symptoms

Nephrotic syndrome can be caused by a number of different diseases. The common mechanism which seems to cause damage involves the immune system. For some reason, the immune system seems to become directed against the person's own kidney. The glomeruli become increasingly leaky as various substances from the immune system are deposited within the kidney.

A number of different kidney disorders are associated with nephrotic syndrome, including:

- minimal change disease or MCD (responsible for about 80% of nephrotic syndrome in children, and about 20% in adults) MCD is a disorder of the glomeruli
- focal glomerulosclerosis
- membranous glomerulopathy
- membranoproliferative glomerulonephropathy

Other types of diseases can also result in nephrotic syndrome. These include diabetes, sickle-cell anemia, **amyloidosis**, systemic lupus erythematosus, **sarcoidosis**, leukemia, lymphoma, cancer of the breast, colon, and stomach, reactions to drugs (including **nonsteroidal anti-inflammatory drugs**, lithium, and street heroine), allergic reactions (to insect stings, snake venom, and poison ivy), infections (**malaria**, various bacteria, **hepatitis B**, herpes zoster, and the virus which causes **AIDS**), and severe high blood pressure.



A specimen of a nephrotic human kidney. (Custom Medical Stock Photo. Reproduced by permission.)

The first symptom of nephrotic syndrome is often foamy urine. As the syndrome progresses, swelling (**edema**) is noticed in the eyelids, hands, feet, knees, scrotum, and abdomen. The patient feels increasingly weak and fatigued. Appetite is greatly decreased. Over time, the loss of protein causes the muscles to become weak and small (called muscle wasting). The patient may note abdominal **pain** and difficulty breathing. Because the kidneys are involved in blood pressure regulation, abnormally low or abnormally high blood pressure may develop.

Over time, the protein loss occurring in nephrotic syndrome will result in a generally malnourished state. Hair and nails become brittle, and growth is stunted. Bone becomes weak, and the body begins to lose other important nutrients (sugar, potassium, calcium). Infection is a serious and frequent complication, as are disorders of blood clotting. Acute kidney failure may develop.

KEY TERMS

Glomeruli—Tiny tufts of capillaries which carry blood within the kidneys. The blood is filtered by the glomeruli. The blood then continues through the circulatory system, but a certain amount of fluid and specific waste products are filtered out of the blood, to be removed from the body in the form of urine.

Immune system—The complex system within the body which serves to fight off harmful invaders, such as bacteria, viruses, fungi.

Kidney failure—The inability of the kidney to excrete toxic substances from the body.

Diagnosis

Diagnosis is based first on the laboratory examination of the urine and the blood. While the urine will reveal significant quantities of protein, the blood will reveal abnormally low amounts of circulating proteins. Blood tests will also reveal a high level of cholesterol. In order to diagnose one of the kidney disorders which cause nephrotic syndrome, a small sample of the kidney (biopsy) will need to be removed for examination. This biopsy can be done with a long, very thin needle which is inserted through the skin under the ribs.

Treatment

Treatment depends on the underlying disorder which has caused nephrotic syndrome. Medications which dampen down the immune system are a mainstay of treatment. The first choice is usually a steroid drug (such as prednisone). Some conditions may require even more potent medications, such as cyclophosphamide or cyclosporine. Treating the underlying conditions (lymphoma, cancers, heroine use, infections) which have led to nephrotic syndrome will often improve the symptoms of nephrotic syndrome as well. Some patients will require the use of specific medications to control high blood pressure. Occasionally, the quantity of fluid a patient is allowed to drink is restricted. Some patients benefit from the use of **diuretics** (which allow the kidney to produce more urine) to decrease swelling.

Prognosis

Prognosis depends on the underlying disorder. Minimal change disease has the best prognosis of all

the kidney disorders, with 90% of all patients responding to treatment. Other types of kidney diseases have less favorable outcomes, with high rates of progression to kidney failure. When nephrotic syndrome is caused by another, treatable disorder (infection, allergic or drug reaction), the prognosis is very good.

Resources

ORGANIZATIONS

American Kidney Fund (AKF). Suite 1010, 6110 Executive Boulevard, Rockville, MD 20852. (800) 638-8299. <<http://216.248.130.102/Default.htm>>.

National Kidney Foundation. 30 East 33rd St., New York, NY 10016. (800) 622-9010. <<http://www.kidney.org>>.

Rosalyn Carson-DeWitt, MD

Nephrotoxic injury

Definition

Nephrotoxic injury is damage to one or both of the kidneys that results from exposure to a toxic material, usually through ingestion.

Description

The kidneys are the primary organs of the urinary system, which purifies the blood by removing wastes from it and excreting them from the body in urine. Every day, the kidneys filter about 45 gal (180 l) of blood, about four times as much as the amount that passes through any other organ. Because of this high volume, the kidneys are more often exposed to toxic substances in the blood and are very vulnerable to injury from those sources.

Each kidney contains over one million structures called nephrons. Each nephron consists of two parts: the renal corpuscle and the renal tubule. The renal corpuscle is where the blood is filtered. It is made up of a network of capillaries (the glomerulus) and the structure that surrounds these capillaries (Bowman's capsule). Blood flows into the glomerulus, where the liquid part of the blood (plasma) passes through the walls of the capillaries and into Bowman's capsule (blood cells and some proteins are too big to pass through and therefore remain in the blood vessels). The plasma, now called filtrate, contains substances that the body needs, such as water, glucose, and other nutrients, as well as wastes, excess salts, and excess water. When the filtrate moves from Bowman's

capsule into the renal tubules, about 99% of it is taken back up as the action of the tubules allows beneficial substances to be reabsorbed into the blood stream. The remaining filtrate is then passed to the bladder as urine.

When the kidneys are exposed to a toxic agent, either accidentally or intentionally (as in a suicide attempt), damage can occur in a number of different ways, depending upon the agent. One toxin may directly affect the glomerulus or the renal tubules, causing the cells of these structures to die. Another toxin may create other substances or conditions that result in the same cell death. Nephrotoxic injury can lead to acute renal failure, in which the kidneys suddenly lose their ability to function, or chronic renal failure, in which kidney function slowly deteriorates. If unchecked, renal failure can result in death.

Causes and symptoms

Several different substances can be toxic to the kidneys. These include:

- **antibiotics**, primarily **aminoglycosides**, sulphonamides, amphotericin B, polymyxin, neomycin, bacitracin, rifampin, trimethoprim, cephaloridine, methicillin, aminosalicylic acid, oxy- and chlorotetracyclines
- **analgesics**, including **acetaminophen** (Tylenol), all nonsteroidal anti-inflammatory drugs (e.g. **aspirin**, ibuprofen), all prostaglandin synthetase inhibitors
- contrast agents used in some diagnostic tests, such as sodium iodide
- heavy metals, such as lead, mercury, arsenic, and uranium
- anti-cancer drugs, such as cyclosporin, cisplatin, and cyclophosphamide
- methemoglobin-producing agents
- solvents and fuels, such as carbon tetrachloride, methanol, amyl alcohol, and ethylene glycol
- herbicides and pesticides
- overproduction of uric acid

Nephrotoxic injury is most commonly caused by drugs, primarily antibiotics, analgesics, and contrast agents. In some cases, such as with aminoglycosides and amphotericin B, the drug itself will damage the kidneys. In others, such as with methicillin, sulphonamides, and some contrast agents, the drug provokes an allergic reaction that destroys the kidneys. Some chemicals found in certain drugs and industrial agents damage the kidneys by converting the hemoglobin of

red blood cells into methemoglobin, thereby interfering with the blood's transport of oxygen. In hospitals, the most common form of nephrotoxic injury is antibiotic nephropathy, which usually occurs when antibiotics are given to patients with already weakened kidneys. Analgesic nephropathy is another common form of nephrotoxic injury and occurs as a result of long-term abuse of analgesics, usually NSAIDs (e.g., ibuprofen). Analgesic nephropathy is most prevalent in women over 30. Lead nephropathy, arising from lead poisoning, and nephropathy, from ingestion of the solvent carbon tetrachloride, are also more common forms of nephrotoxic injury. Uric acid nephropathy is one form of nephropathy that is not caused by exposure to an external toxin; instead, it arises from the body's overproduction of uric acid, usually in persons with diseases of the lymph nodes or bone marrow.

Risk factors for nephrotoxic injury include:

- Age. The elderly are more likely to overdose on antibiotics or analgesics.
- Underlying **kidney disease**. Kidneys already weakened by conditions such as diabetes can be particularly susceptible to nephrotoxic injury.
- Severe **dehydration**.
- Prolonged exposure to heavy metals or solvents on the job or in the home.
- Presence of diseases that cause the overproduction of uric acid.

Symptoms of nephrotoxic injury are wide ranging and, in some cases, depend upon the type of toxin involved. In general, symptoms are similar to those of renal failure and include excess urea in the blood (azotemia), anemia, increased hydrogen ion concentration in the blood (acidosis), excess fluids in the body (**overhydration**), and high blood pressure (**hypertension**). Blood or pus may be present in the urine, as may uric acid crystals. A decrease in urinary output may also occur. If the toxin's effect on the kidneys remains unchecked, more serious symptoms of kidney failure may occur, including seizures and **coma**.

Diagnosis

Damage to the kidneys is assessed through a combination of physical examination, blood tests, urine tests, and imaging procedures. Diagnosis of nephrotoxic injury as the underlying cause results from a thorough investigation of the patient's history. Information regarding preexisting conditions, current prescriptions, and environmental exposures to toxins

KEY TERMS

Bowman's capsule—The structure surrounding the glomerulus.

Chelate—A chemical that binds to heavy metals in the blood, thereby helping the body to excrete them in urine.

Contrast agent—Substance ingested so as to highlight anatomical structures in x-ray imaging tests.

Diuretic—A drug that promotes the excretion of urine.

Glomerulus—A network of capillaries located in the nephron where wastes are filtered from the blood.

Methemoglobin—A compound formed from hemoglobin by oxidation.

Nephron—Basic functional unit of the kidney.

Nephrotoxin—Substance that is poisonous to the kidneys.

Renal failure—Disorder characterized by the kidney's inability to filter wastes from the blood. It may be acute (occurring suddenly and usually reversible) or chronic (developing slowly over time as a result of permanent damage).

aid the physician in determining what toxin, if any, has caused the kidneys to malfunction.

Treatment

Treatment of nephrotoxic injury takes place in the hospital and focuses on removing the toxin from the patient's system, while maintaining kidney function. Removal methods are targeted to specific toxins and may include the use of **diuretics** or chelates to enhance excretion of the toxin in urine, or, in extreme cases, the direct removal of toxins from the blood via hemodialysis or passing the blood over an absorbent substance such as charcoal. Support of kidney function depends on the extent of damage to the organs and ranges from monitoring fluid levels to dialysis.

Prognosis

The outcome of nephrotoxic injury is determined by the cause and severity of the damage. In cases where damage has not progressed beyond acute renal failure, kidney function can be fully restored once the toxin is removed from the system and equilibrium restored.

However, if permanent damage has resulted in chronic renal failure, lifelong dialysis or a kidney transplant may be required.

Prevention

Exposure to nephrotoxins can be minimized several different ways. When taking antibiotics or analgesics, recommended dosages should be strictly followed. Also, elderly patients on these medications (for example, those taking aspirin for heart problems or NSAIDs for arthritis) should be closely monitored to prevent accidental overdose. Health care workers should be aware of any underlying conditions, such as diabetes or **allergies** to antibiotics, that may heighten the effect of a potential nephrotoxin. When using solvents or handling heavy metals, procedures regarding their safe use should be employed.

Resources

ORGANIZATIONS

American Kidney Fund (AKF). Suite 1010, 6110 Executive Boulevard, Rockville, MD 20852. (800) 638-8299.
< <http://216.248.130.102/Default.htm> >.

National Kidney Foundation. 30 East 33rd St., New York, NY 10016. (800) 622-9010. < <http://www.kidney.org> >.

Bridget Travers

Nerve conduction velocity testing see

Electromyography

Neural hearing loss see **Hearing loss**

Neuralgia

Definition

Neuralgia is defined as an intense burning or stabbing pain caused by irritation of or damage to a nerve. The **pain** is usually brief but may be severe. It often feels as if it is shooting along the course of the affected nerve.

Description

Different types of neuralgia occur depending on the reason the nerve has been irritated. Neuralgia can be triggered by a variety of causes, including tooth decay, eye strain, or **shingles** (an infection caused by the herpes zoster virus). Pain is usually felt in the part of the body that is supplied by the irritated nerve.

Causes and symptoms

Neuralgia is caused by irritation or nerve damage from systemic disease, inflammation, infection, and compression or physical irritation of a nerve. The location of the pain depends on the underlying condition that is irritating the nerve or the location of the particular nerve that is being irritated.

Neuralgia can result from **tooth decay**, poor diet, eye strain, nose infections, or exposure to damp and cold. Postherpetic neuralgia is an intense debilitating pain felt at the site of a previous attack of shingles. **Trigeminal neuralgia** (also called tic douloureux, the most common type of neuralgia), causes a brief, searing pain along the trigeminal nerve, which supplies sensation to the face. The facial pain of migraine neuralgia lasts between 30 minutes and an hour and occurs at the same time on successive days. The cause is not known.

Glossopharyngeal neuralgia is an intense pain felt at the back of the tongue, in the throat, and in the ear—all areas served by the glossopharyngeal nerve. The pain may occur spontaneously, or it can be triggered by talking, eating, or swallowing (especially cold foods such as ice cream). Its cause is not known.

Occipital neuralgia is caused by a pinched occipital nerve. There are two occipital nerves, each located at the back of the neck, each supplying feeling to the skin over half of the back of the head. These nerves can be pinched due to factors ranging from arthritis to injury, but the result is the same: **numbness**, pain, or **tingling** over half the base of the skull.

Diagnosis

Neuralgia is a symptom of an underlying disorder; its diagnosis depends on finding the cause of the condition creating the pain.

To diagnose occipital neuralgia, a doctor can inject a small amount of anesthetic into the region of the occipital nerve. If the pain temporarily disappears, and there are no other physical reasons for the pain, the doctor may recommend surgery to deal with the pinched nerve.

Treatment

Glossopharyngeal, trigeminal, and postherpetic neuralgias sometimes respond to **anticonvulsant drugs**, such as carbamazepine or phenytoin, or to painkillers, such as **acetaminophen**. Trigeminal neuralgia may also be relieved by surgery in which the nerve is cut or decompressed. In some cases,

compression neuralgia (including occipital neuralgia) can be relieved by surgery.

People with shingles should see a doctor within three days of developing the rash, since aggressive treatment of the blisters that appear with the rash can ease the severity of the infection and minimize the risk of developing postherpetic neuralgia. However, it is not clear whether the treatment can prevent postherpetic neuralgia.

If postherpetic neuralgia develops, a variety of treatments can be tried, since their effectiveness varies from person-to-person.

- antidepressants such as amitriptyline (Elavil)
- anticonvulsants (phenytoin, valproate, or carbamazepine)
- capsaicin (Xostrix), the only medication approved by the FDA for treatment of postherpetic neuralgia
- topical painkillers
- desensitization
- TENS (transcutaneous **electrical nerve stimulation**)
- dorsal root zone (DREZ) surgery (a treatment of last resort)

Alternative treatment

B-complex **vitamins**, primarily given by intramuscular injection, can be an effective treatment. A whole foods diet with adequate protein, carbohydrates, and fats that also includes yeast, liver, wheat germ, and foods that are high in B vitamins may be helpful. **Acupuncture** is a very effective treatment, especially for postherpetic neuralgia. Homeopathic treatment can also be very effective when the correct remedy is used. Some botanical medicines may also be useful. For example, black cohosh (*Cimicifuga racemosa*) appears to have anti-inflammatory properties based on recent research.

Prognosis

The effectiveness of the treatment depends on the cause of the neuralgia, but many cases respond to pain relief.

Trigeminal neuralgia tends to come and go, but successive attacks may be disabling. Although neuralgia is not fatal, the patient's fear of being in pain can seriously interfere with daily life.

Some people with postherpetic neuralgia respond completely to treatment. Most people, however, experience some pain after treatment, and a few receive no

KEY TERMS

Desensitization—A technique of pain reduction in which the painful area is stimulated with whatever is causing the pain.

Dorsal root entry zone (DREZ)—A type of nerve surgery for postherpetic neuralgia that is occasionally used when the patient can get no other pain relief. The surgery destroys the area where damaged nerves join the central nervous system, thereby interfering with inappropriate pain messages from nerves to the brain.

Glossopharyngeal neuralgia—Sharp recurrent pain deep in the throat that extends to the area around the tonsils and possibly the ear. It is triggered by swallowing or chewing.

Migraine neuralgia—A variant of migraine pain, also called cluster headache, in which severe attacks of pain affect the eye and forehead on one side of the face.

Occipital neuralgia—Pain on one side of the back of the head caused by entrapment or pinching of an occipital nerve.

Postherpetic neuralgia—Persistent pain that occurs as a complication of a herpes zoster infection. Although the pain can be treated, the response is variable.

Shingles—A painful rash with blisters that appears along the course of a nerve. It is caused by infection with herpes zoster virus.

TENS—The abbreviation for transcutaneous electrical nerve stimulation, a technique used to control chronic pain. Electrodes placed over the painful area deliver a mild electrical impulse to nearby nerve pathways, thereby easing pain.

Trigeminal neuralgia—Brief episodes of severe shooting pain on one side of the face caused by inflammation of the root of the trigeminal nerve. Also referred to as tic douloureux.

relief at all. Some people live with this type of neuralgia for the rest of their lives, but for most, the condition gradually fades away within five years.

Resources

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American Chronic Pain Association. P.O. Box 850, Rocklin, CA 95677-0850. (916) 632-0922. <<http://members.tripod.com/~widdy/ACPA.html>>.
 National Chronic Pain Outreach Association. P.O. Box 274, Millboro, VA 24460. (540) 997-5004.
 Trigeminal Neuralgia/Tic Douloureux Association. P.O. Box 340, Barnegat Light, NJ 08006. (609) 361-1014.

Carol A. Turkington

Neuroblastoma

Definition

Neuroblastoma is a type of **cancer** that usually originates either in the tissues of the adrenal gland or in the ganglia of the abdomen or in the ganglia of the nervous system. (Ganglia are masses of nerve tissue or groups of nerve cells.) Tumors develop in the nerve tissue in the neck, chest, abdomen, or pelvis.

Description

Neuroblastoma is one of the few cancer types known to secrete hormones. It occurs most often in children, and it is the third most common cancer that occurs in children. Approximately 7.5% of the childhood cancers diagnosed in 2001 were neuroblastomas, affecting one in 80,000 to 100,000 children in the United States. Close to 50% of cases of neuroblastoma occur in children younger than two years old. The disease is sometimes present at birth, but is usually not noticed until later. By the time the disease is diagnosed, it has often spread to the lymph nodes, liver, lungs, bones, or bone marrow. Approximately one-third of neuroblastomas start in the adrenal glands.

Demographics

According to some reports, African-American children develop the disease at a slightly higher rate than Caucasian children (8.7 per million compared to 8.0 per million cases diagnosed).

Causes and symptoms

The causes of neuroblastoma are not precisely known. Current research holds that neuroblastomas develop when cells produced by the fetus (neuroblast cells) fail to mature into normal nerve or adrenal cells and keep growing and proliferating. The first

symptom of a neuroblastoma is usually an unusual growth or lump, found in most cases in the abdomen of the child, causing discomfort or a sensation of fullness and **pain**. Other symptoms such as **numbness** and **fatigue**, arise because of pressure caused by the tumor. Bone pain also occurs if the cancer has spread to the bone. If it has spread to the area behind the eye, the cancer may cause protruding eyes and dark circles around the eyes; in a few cases, blindness may be the presenting symptom. Or **paralysis** may result from compression of the spinal cord. **Fever** is also reported in one case out of four. High blood pressure, persistent **diarrhea**, rapid heartbeat, reddening of the skin and sweating occur occasionally. Some children may also have uncoordinated or jerky muscle movements, or uncontrollable eye movements, but these symptoms are rare. If the disease spreads to the skin, blue or purple patches are observed.

Diagnosis

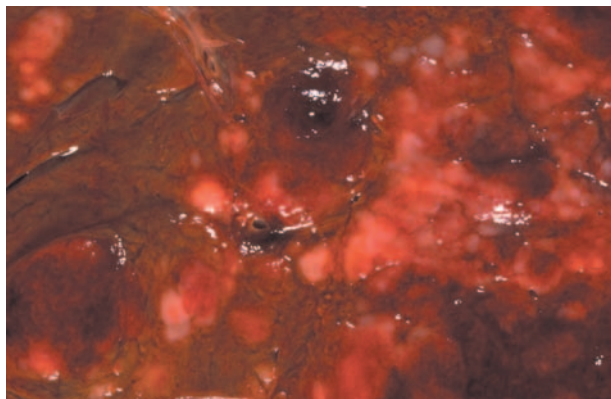
A diagnosis of neuroblastoma usually requires blood and urine tests to investigate the nature and quantity of chemicals (neurotransmitters) released by the nerve cells. These are broken down by the body and released in urine. Additionally, scanning techniques are used to confirm the diagnosis of neuroblastoma. These techniques produce images or pictures of the inside of the body and they include computed tomography scan (CT scan) and **magnetic resonance imaging** (MRI). To confirm the diagnosis, the physician will surgically remove some of the tissue from the tumor or bone marrow (biopsy), and examine the cells under the microscope.

Treatment

The treatment team usually consists of an oncologist specialized in the treatment of neuroblastoma, a surgeon to perform biopsies and possibly attempt surgical removal of the tumor, a **radiation therapy** team and, if indicated, a **bone marrow transplantation** team.

Staging

Once neuroblastoma has been diagnosed, the physician will perform more tests to determine if the cancer has spread to other tissues in the body. This process, called staging, is important for the physician to determine how to treat the cancer and check liver and kidney function. The staging system for neuroblastoma is based on how far the disease has spread from its original site to other tissues in the body.



A neuroblastoma appearing at the surface of the liver.
(Custom Medical Stock Photo. Reproduced by permission.)

Localized resectable (able to be cut out) neuroblastoma is confined to the site of origin, with no evidence that it has spread to other tissues, and the cancer can be surgically removed. Localized unresectable neuroblastoma is confined to the site of origin, but the cancer cannot be completely removed surgically. Regional neuroblastoma has extended beyond its original site, to regional lymph nodes, and/or surrounding organs or tissues, but has not spread to distant sites in the body. Disseminated neuroblastoma has spread to distant lymph nodes, bone, liver, skin, bone marrow, and/or other organs. Stage 4S (or IVS, or “special”) neuroblastoma has spread only to liver, skin, and/or, to a very limited extent, bone marrow. Recurrent neuroblastoma means that the cancer has come back, or continued to spread after it has been treated. It may come back in the original site or in another part of the body.

Treatments are available for children with all stages of neuroblastoma. More than one of these treatments may be used, depending on the stage of the disease. The four types of treatment used are:

- surgery (removing the tumor in an operation)
- radiation therapy (using high-energy x-rays to kill cancer cells)
- **chemotherapy** (using drugs to kill cancer cells)
- bone marrow transplantation (replacing the patient’s bone marrow cells with those from a healthy person)

Surgery is used whenever possible, to remove as much of the cancer as possible, and can generally cure the disease if the cancer has not spread to other areas of the body. Before surgery, chemotherapy may be used to shrink the tumor so that it can be more easily removed during surgery; this is called neoadjuvant chemotherapy. Radiation therapy is often used after

surgery; high-energy rays (radiation) are used to kill as many of the remaining cancer cells as possible. Chemotherapy (called adjuvant chemotherapy) may also be used after surgery to kill remaining cells. Bone marrow transplantation is used to replace bone marrow cells killed by radiation or chemotherapy. In some cases the patient’s own bone marrow is removed prior to treatment and saved for transplantation later. Other times the bone marrow comes from a “matched” donor, such as a sibling.

One novel approach to treatment of neuroblastomas is therapy with desferoxamine (DFO), which is ordinarily used to treat iron **poisoning**. DFO has been shown to have antitumor activity in neuroblastomas and other cancers of the central nervous system. It is thought that the drug works by lowering the increased iron levels in the body associated with cancer.

As of 2004, there are significant differences in treatment protocols for neuroblastoma between the major North American study group (Children’s Oncology Group) and its European counterpart, the Société Internationale d’Oncologie Pédiatrique (SIOP). These differences include biopsy techniques, the timing and extent of surgery, chemotherapy dosages, and the types of salvage therapy employed.

Alternative treatment

No alternative therapy has yet been reported to substitute for conventional neuroblastoma treatment. Complementary therapies—such as retinoic acid therapy—have been shown to be beneficial to patients when administered after a conventional course of chemotherapy or transplantation.

Prognosis

The chances of recovery from neuroblastoma depend on the stage of the cancer, the age of the child at diagnosis, the location of the tumor, and the state and nature of the tumor cells evaluated under the microscope. Infants have a higher rate of cure than do children over one year of age, even when the disease has spread. In general, the prognosis for a young child with neuroblastoma is good: the predicted five-year survival rate is approximately 85% for children who had the onset of the disease in infancy, and 35% for those whose disease developed later.

Prevention

Neuroblastoma may be a genetic disease passed down from the parents. In 2004, a group of German

KEY TERMS

Adjuvant chemotherapy—Treatment of the tumor with drugs after surgery to kill as many of the remaining cancer cells as possible.

Adrenal gland—Gland located above each kidney consisting of an outer wall (cortex) that produces steroid hormones and an inner section (medulla) that produces other important hormones, such as adrenaline and noradrenaline.

Alternative therapy—A therapy is generally called alternative when it is used instead of conventional cancer treatments.

Biopsy—A small sample of tissue removed from the site of the tumor to be examined under a microscope.

Conventional therapy—Treatments that are widely accepted and practiced by the mainstream medical community.

Complementary therapy—A therapy is called complementary when it is used in addition to conventional cancer treatments.

Disseminated—Spread to other tissues.

Hormone—A substance produced by specialized cells that affects the way the body carries out the biochemical and energy-producing processes required to maintain health (metabolism).

Localized—Confined to a small area.

Neoadjuvant chemotherapy—Treatment of the tumor with drugs before surgery to reduce the size of the tumor.

Neuroblast cells—Cells produced by the fetus which mature into nerve cells and adrenal medulla cells.

Monoclonal antibody—A protein substance which is produced in the laboratory by a single population of cells. They are being tested as a possible form of cancer treatment.

Resectable cancer—A tumor that can be surgically removed.

Salvage therapy—Treatment measures taken late in the course of a disease after other therapies have failed. It is also known as rescue therapy.

Staging system—A system based on how far the cancer has spread from its original site, developed to help the physician determine how best to treat the disease.

Unresectable cancer—A tumor that cannot be completely removed by surgery.

researchers reported that a series of neuroblastomas demonstrated a consistent pattern of deletions and overrepresentations on chromosomes 3, 10, 17q, and 20. There is currently no known method for its prevention.

Special concerns

After completion of a course of treatment for neuroblastoma, physicians sometimes recommend that the child undergo an investigative operation. This procedure allows the treatment team to evaluate how effective treatment has been, and may offer an opportunity to remove more of the tumor if it is still present.

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National Institutes of Health & National Cancer Institute. *Young People With Cancer: A Handbook for Parents*. <http://www.cancernet.nci.nih.gov/young_people/youngconts.html>.

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Neuroendocrine tumors

Definition

Neuroendocrine tumor refers to the type of cell that a tumor grows from rather than where that tumor is located. Neuroendocrine cells produce hormones or regulatory proteins, and so tumors of these cells usually have symptoms that are related to the specific hormones that they produce.

Description

Neuroendocrine cells have roles both in the endocrine system and the nervous system. They produce and secrete a variety of regulatory hormones, or neuropeptides, which include neurotransmitters and growth factors. When these cells become cancerous, they grow and overproduce their specific neuropeptide. Neuroendocrine tumors are generally rare. One type of neuroendocrine tumor is a carcinoid tumor. This type of tumor can occur in the intestinal tract, appendix, rectum, bronchial tubes, or ovary. Most carcinoid tumors secrete serotonin. When the blood concentration of this hormone is high enough, it causes carcinoid syndrome. This syndrome refers to a variety of symptoms that are caused by the excessive amount of hormone secreted rather than the tumor itself.

The total incidence of neuroendocrine tumors is thought to be between five and nine million people in the United States. It is possible that these tumors are underreported because they grow slowly and do not always produce dramatic symptoms.

Causes and symptoms

Many of the symptoms of carcinoid tumor are due to the hormones that the tumor secretes. These hormones can affect the whole body and cause what is referred to as carcinoid syndrome. The most common symptom of carcinoid syndrome is flushing, a sudden appearance of redness and warmth in the face and neck that can last from minutes to hours. Other symptoms of carcinoid syndrome are **diarrhea**, asthma-like symptoms and heart problems. Since most carcinoid tumors are found in the appendix, the symptoms are often similar to **appendicitis**, primarily **pain** in the abdomen. When these tumors are found in the small intestine, they can cause abdominal pain that is often initially diagnosed as bowel obstruction. Many patients have no symptoms and the carcinoids are found during routine endoscopy of the intestines.

Diagnosis

The diagnosis of carcinoid syndrome is made by the measurement of 5-hydroxy indole acetic acid (5-HIAA) in the urine. 5-HIAA is a breakdown (waste) product of serotonin. If the syndrome is diagnosed, the presence of carcinoid tumor is a given. When the syndrome is not present, diagnosis may be delayed, due to the vague symptoms present. Diagnosis can sometimes take up to two years. It is made by performing a number of tests, and the specific test used depends on the tumor's suspected location. The tests that may be performed include gastrointestinal endoscopy, **chest x ray**, computed tomography scan (CT scan), **magnetic resonance imaging**, or ultrasound. A biopsy of the tumor is performed for diagnosis. A variety of hormones can be measured in the blood as well to indicate the presence of a carcinoid.

Treatment

The only effective treatment for carcinoid tumor as of the early 2000s is surgical removal of the tumor. Although **chemotherapy** is sometimes used when metastasis has occurred, it is rarely effective. The treatment for carcinoid syndrome is typically meant to decrease the severity of symptoms. Patients should avoid **stress** as well as foods that bring on the syndrome. Some medications can be given for

KEY TERMS

Appendicitis—Inflammation of the appendix.

Growth factor—A local hormone produced by some cells that initiates growth.

Metastasis—The spread of disease from one part of the body to another, as when cancer cells appear in parts of the body remote from the site of the primary tumor.

Neurotransmitter—A chemical messenger used to transmit information in the nervous system.

symptomatic relief; for example, tumors of the gastrointestinal tract may be treated with octreotide (Sandostatin) or lanreotide (Somatuline) to relieve such symptoms as diarrhea and flushing. These drugs are known as somatostatin analogs.

Liver transplantation is a treatment option for patients with neuroendocrine tumors that have metastasized only to the liver. As of 2004, this approach is reported to offer patients long disease-free periods and relief of symptoms.

Prognosis

The prognosis of carcinoid tumors is related to the specific growth patterns of that tumor, as well as its location. For example, a group of researchers at the University of Wisconsin reported in 2004 that patients with gastrointestinal tumors in the hindgut had longer periods of disease-free survival than those with foregut or midgut cancers. For localized disease the five-year survival rate can be 94%, whereas for patients where metastasis has occurred, the average five-year survival rate is 18%. It is not unusual for patients with carcinoid tumors to live ten or fifteen years after the initial diagnosis.

Prevention

Neuroendocrine tumors such as carcinoid tumors are rare, and no information consequently is yet available on cause or prevention.

Resources

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ORGANIZATIONS

Carcinoid Cancer Foundation, Inc. 1751 York Ave., New York, NY 10128. (212) 722-3132. <<http://www.carcinoid.org>>.

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Neurofibromatosis

Definition

Neurofibromatosis (NF), or von Recklinghausen disease, is a genetic disease in which patients develop multiple soft tumors (neurofibromas). These tumors occur under the skin and throughout the nervous system.

Description

Neural crest cells are primitive cells which exist during fetal development. These cells eventually turn into:

- cells which form nerves throughout the brain, spinal cord, and body
- cells which serve as coverings around the nerves that course through the body
- pigment cells, which provide color to structures
- the meninges, the thin, membranous coverings of the brain and spinal cord
- cells which ultimately develop into the bony structures of the head and neck

In neurofibromatosis, a genetic defect causes these neural crest cells to develop abnormally. This results in numerous tumors and malformations of the nerves, bones, and skin.

Neurofibromatosis occurs in about one of every 4,000 births. Two types of NF exist, NF-1 (90% of all cases), and NF-2 (10% of all cases).

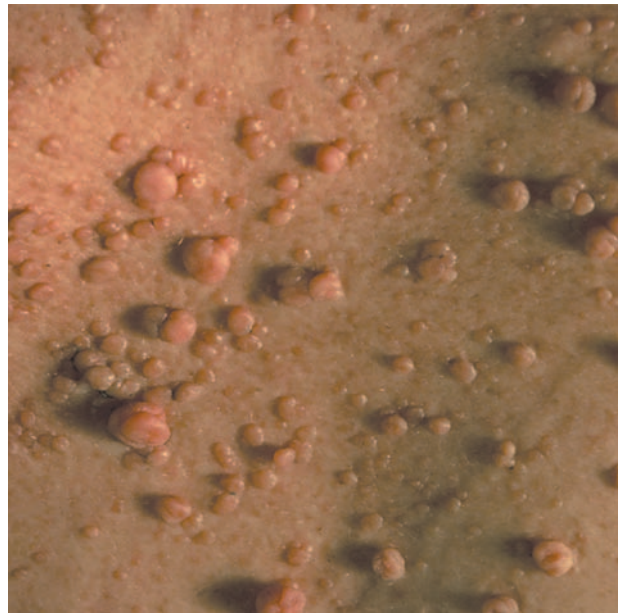
Causes and symptoms

Both forms of neurofibromatosis are caused by a defective gene. NF-1 is due to a defect on chromosome 17; NF-2 results from a defect on chromosome 22. Both of these disorders are inherited in a dominant fashion. This means that anybody who receives just one defective gene will have the disease. However, a family pattern of NF is only evident for about half of all cases of NF. The other cases of NF occur due to a spontaneous mutation (a permanent change in the structure of a specific gene). Once such a spontaneous mutation has been established in an individual, however, it is then possible to be passed on to any offspring. The chance of a person with NF passing on the NF gene to a child is 50%.

NF-1 has a number of possible signs and can be diagnosed if any two of the following are present:

- The presence of café-au-lait (French for coffee-with-milk) spots. These are patches of tan or light brown skin, usually about 5-15 mm in diameter. Nearly all patients with NF-1 will display these spots.
- Multiple freckles in the armpit or groin area.
- Ninety percent of patients with NF-1 have tiny tumors called Lisch nodules in the iris (colored area) of the eye.
- Neurofibromas. These soft tumors are the hallmark of NF-1. They occur under the skin, often located along nerves or within the gastrointestinal tract. Neurofibromas are small and rubbery, and the skin overlying them may be somewhat purple in color.
- Skeletal deformities, such as a twisted spine (scoliosis), curved spine (humpback), or bowed legs.
- Tumors along the optic nerve, which cause vision disturbance in about 20% of patients.
- The presence of NF-1 in a patient's parent, child, or sibling.

There are very high rates of speech impairment, learning disabilities, and attention deficit disorder in children with NF-1. Other complications include the development of a **seizure disorder**, or the abnormal accumulation of fluid within the brain (hydrocephalus).



This person's skin has multiple soft tumors, or neurofibromas. Such tumors develop underneath the skin. (Custom Medical Stock Photo. Reproduced by permission.)

A number of cancers are more common in patients with NF-1. These include a variety of types of malignant brain tumors, as well as leukemia, and cancerous tumors of certain muscles (rhabdomyosarcoma), the adrenal glands (**pheochromocytoma**), or the kidneys (Wilms' tumor).

Patients with NF-2 do not necessarily have the same characteristic skin symptoms (café-au-lait spots, freckling, and neurofibromas of the skin) that appear in NF-1. The characteristic symptoms of NF-2 are due to tumors along the acoustic nerve. Interfering with the function of this nerve results in the loss of hearing; and the tumor may spread to neighboring nervous system structures, causing weakness of the muscles of the face, **headache**, **dizziness**, poor balance, and uncoordinated walking. Cloudy areas on the lens of the eye (called **cataracts**) frequently develop at an unusually early age. As in NF-1, the chance of brain tumors developing is unusually high.

Diagnosis

Diagnosis is based on the symptoms outlined above. Diagnosis of NF-1 requires that at least two of the listed signs are present. Diagnosis of NF-2 requires the presence of either a mass on the acoustic nerve or another distinctive nervous system tumor. An important diagnostic clue for either NF-1 or NF-2 is the presence of the disorder in a patient's parent, child, or sibling.

KEY TERMS

Chromosome—A structure within the nucleus of every cell, which contains genetic information governing the organism's development.

Mutation—A permanent change to the genetic code of an organism. Once established, a mutation can be passed on to offspring.

Neurofibroma—A soft tumor usually located on a nerve.

Tumor—An abnormally multiplying mass of cells.

Monitoring the progression of neurofibromatosis involves careful testing of vision and hearing. X-ray studies of the bones are frequently done to watch for the development of deformities. CT scans and MRI scans are performed to track the development/progression of tumors in the brain and along the nerves. Auditory evoked potentials (the electric response evoked in the cerebral cortex by stimulation of the acoustic nerve) may be helpful to determine involvement of the acoustic nerve, and EEG (electroencephalogram, a record of electrical currents in the brain) may be needed for patients with suspected seizures.

Treatment

There are no available treatments for the disorders which underlie either type of neurofibromatosis. To some extent, the symptoms of NF-1 and NF-2 can be treated individually. Skin tumors can be surgically removed. Some brain tumors, and tumors along the nerves, can be surgically removed, or treated with drugs (**chemotherapy**) or x-ray treatments (**radiation therapy**). Twisting or curving of the spine and bowed legs may require surgical treatment, or the wearing of a special brace.

Prognosis

Prognosis varies depending on the types of tumors which an individual develops. As tumors grow, they begin to destroy surrounding nerves and structures. Ultimately, this destruction can result in blindness, deafness, increasingly poor balance, and increasing difficulty with the coordination necessary for walking. Deformities of the bones and spine can also interfere with walking and movement. When cancers develop, prognosis worsens according to the specific type of cancer.

Prevention

There is no known way to prevent the approximately 50% of all NF cases which occur due to a spontaneous change in the genes (mutation). New cases of inherited NF can be prevented with careful **genetic counseling**. A person with NF can be made to understand that each of his or her offspring has a 50% chance of also having NF. When a parent has NF, and the specific genetic defect causing the parent's disease has been identified, tests can be performed on the fetus (developing baby) during pregnancy. Amniocentesis or chorionic villus sampling are two techniques which allow small amounts of the baby's cells to be removed for examination. The tissue can then be examined for the presence of the parent's genetic defect. Some families choose to use this information in order to prepare for the arrival of a child with a serious medical problem. Other families may choose not to continue the **pregnancy**.

Resources

ORGANIZATIONS

March of Dimes Birth Defects Foundation. 1275

Mamaroneck Ave., White Plains, NY 10605. (914) 428-7100. resourcecenter@modimes.org. <<http://www.modimes.org>>.

National Neurofibromatosis Foundation, Inc. 95 Pine St., 16th Floor, New York, NY 10005. (800) 323-7938. <<http://nf.org>>.

Neurofibromatosis, Inc. 8855 Annapolis Rd., #110, Lanham, MD 20706-2924. (800) 942-6825.

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Neurogenic arthropathy see **Charcot's joints**

Neurogenic bladder

Definition

Neurogenic bladder is a dysfunction that results from interference with the normal nerve pathways associated with urination.

Description

Normal bladder function is dependent on the nerves that sense the fullness of the bladder (sensory nerves) and on those that trigger the muscle movements that either empty it or retain urine (motor nerves). The reflex to urinate is triggered when the bladder fills to 300-500 ml. The bladder is then

emptied when the contraction of the bladder wall muscles forces urine out through the urethra. The bladder, internal sphincters, and external sphincters may all be affected by nerve disorders that create abnormalities in bladder function.

There are two categories of neurogenic bladder dysfunction: overactive (spastic or hyper-reflexive) and underactive (flaccid or hypotonic). An overactive neurogenic bladder is characterized by uncontrolled, frequent expulsion of urine from the bladder. There is reduced bladder capacity and incomplete emptying of urine. An underactive neurogenic bladder has a capacity that is extremely large (up to 2000 ml). Due to a loss of the sensation of bladder filling, the bladder does not contract forcefully, and small amounts of urine dribble from the urethra as the bladder pressure reaches a breakthrough point.

Causes and symptoms

There are numerous causes for neurogenic bladder dysfunction and symptoms vary depending on the cause. An **overactive bladder** is caused by interruptions in the nerve pathways to the bladder occurring above the sacrum (five fused spinal vertebrae located just above the tailbone or coccyx). This nerve damage results in a loss of sensation and motor control and is often seen in **stroke**, Parkinson's disease, and most forms of spinal-cord injuries. An underactive bladder is the result of interrupted bladder stimulation at the level of the sacral nerves. This may result from certain types of surgery on the spinal cord, sacral spinal tumors, or congenital defects. It also may be a complication of various diseases, such as **syphilis**, **diabetes mellitus**, or **polio**.

Diagnosis

Neurogenic bladder is diagnosed by carefully recording fluid intake and urinary output and by measuring the quantity of urine remaining in the bladder after voiding (residual urine volume). This measurement is done by draining the bladder with a small rubber tube (catheter) after the person has urinated. Kidney function also is evaluated by regular laboratory testing of the blood and urine. **Cystometry** may be used to estimate the capacity of the bladder and the pressure changes within it. These measurements can help determine changes in bladder compliance in order to assess the effectiveness of treatment. Doctors may use a cystoscope to look inside the bladder and tubes that lead to it from the kidneys (ureters). **Cystoscopy** may be used to assess the loss of muscle fibers and elastic tissues and, in some cases, for removing small pieces of tissue for biopsy.

Treatment

Doctors begin treating neurogenic bladder by attempting to reduce bladder stretching (distension) through intermittent or continuous catheterization. In intermittent catheterization, a small rubber catheter is inserted at regular intervals (four to six times per day) to approximate normal bladder function. This avoids the complications that may occur when a catheter remains in the bladder's outside opening (urethra) continuously (an indwelling catheter). Intermittent catheterization should be performed using strict sterile technique (asepsis) by skilled personnel, and hourly fluid intake and output must be recorded. Patients who can use their arms may be taught to catheterize themselves.

Indwelling catheters avoid distension by emptying the bladder continuously into a bedside drainage collector. Individuals with indwelling catheters are encouraged to maintain a high fluid intake in order to prevent bacteria from accumulating and growing in the urine. Increased fluid intake also decreases the concentration of calcium in the urine, minimizing urine crystallization and the subsequent formation of stones. Moving around as much as possible and a low calcium diet also help to reduce stone formation.

Drugs may be used to control the symptoms produced by a neurogenic bladder. The unwanted contractions of an overactive bladder with only small volumes of urine may be suppressed by drugs that relax the bladder (anticholinergics) such as propantheline (Pro-Banthine) and oxybutynin (Ditropan). Contraction of an underactive bladder with normal bladder volumes may be stimulated with parasympathomimetics (drugs that mimic the action resulting from stimulation of the parasympathetic nerves) such as bethanechol (Urecholine).

Long-term management for the individual with an overactive bladder is aimed at establishing an effective spontaneous reflex voiding. The amount of fluid taken in is controlled in measured amounts during the waking hours, with sips only toward bedtime to avoid bladder distension. At regular intervals during the day (every four to six hours when fluid intake is two to three liters per 24 hours), the patient attempts to void using pressure over the bladder (Crede maneuver). The patient may also stimulate reflex voiding by abdominal tapping or stretching of the anal sphincter. The **Valsalva maneuver**, involving efforts similar to those used when straining to pass stool, produces an increase in intra-abdominal pressure that is sometimes adequate to completely empty the bladder. The amount of urine remaining in the bladder (residual

volume) is estimated by a comparison of fluid intake and output. The patient also may be catheterized immediately following the voiding attempt to determine residual urine. Catheterization intervals are lengthened as the residual urine volume decreases and catheterization may be discontinued when urine residuals are at an acceptable level to prevent urinary tract infection.

For an underactive bladder, the patient may be placed on a similar bladder routine with fluid intake and output adjusted to prevent bladder distension. If an adequate voiding reflex cannot be induced, the patient may be maintained on clean intermittent catheterization.

Some individuals who are unable to control urine output (**urinary incontinence**) due to deficient sphincter tone may benefit from perineal exercises. Although this is a somewhat dated technique, male patients with extensive sphincter damage may be helped by the use of a Cunningham clamp. The clamp is applied in a horizontal fashion behind the glans of the penis and must be removed approximately every four hours for bladder emptying to prevent bacteria from growing in the urine and causing an infection. Alternation of the Cunningham clamp with use of a **condom** collection device will reduce the skin irritation sometimes caused by the clamp.

Surgery is another treatment option for incontinence. Urinary diversion away from the bladder may involve creation of a urostomy or a continent diversion. The surgical implantation of an inflatable sphincter is another option for certain patients. An indwelling urinary catheter is sometimes used when all other methods of incontinence management have failed. The long-term use of an indwelling catheter almost inevitably leads to some urinary tract infections, and contributes to the formation of urinary stones (calculi). Doctors may prescribe **antibiotics** preventively to reduce recurrent urinary tract infection.

Alternative treatment

The cause of the bladder problem must be determined and treated appropriately. If nerve damage is not permanent, homeopathy and **acupuncture** may help restore function.

Prognosis

Individuals with an overactive bladder caused by spinal cord lesions at or above the seventh thoracic vertebra, are at risk for sympathetic dysreflexia, a life-threatening condition which can occur when the

KEY TERMS

Anticholinergic—An agent that blocks certain nerve impulses.

Catheterization—Insertion of a slender, flexible tube into the bladder to drain urine.

Compliance—A term used to describe how well a patient's behavior follows medical advice.

Cystometry—A test of bladder function in which pressure and volume of fluid in the bladder are measured during filling, storage, and voiding.

Cystoscopy—A direct method of bladder study and visualization using a cystoscope (self-contained optical lens system). The cystoscope can be manipulated to view the entire bladder, with a guide system to pass it up into the ureters (tubes leading from the kidneys to the bladder).

Glans penis—The bulbous tip of the penis.

Motor nerves—Nerves that cause movement when stimulated.

Parasympathomimetic—An agent whose effects mimic those resulting from stimulation of the parasympathetic nerves.

Perineal—The diamond-shaped region of the body between the pubic arch and the anus.

Reflex—An involuntary response to a particular stimulus.

Sensory nerves—Nerves that convey impulses from sense organs to the higher parts of the nervous system, including the brain.

Sphincter—A band of muscles that surrounds a natural opening in the body; these muscles can open or close the opening by relaxing or contracting.

Ureter—A tube leading from one of the kidneys to the bladder.

Urethra—The tube that leads from the bladder to the outside of the body.

Urostomy—A diversion of the urinary flow away from the bladder, resulting in output through the abdominal wall. The most common method involves use of a portion of intestine to conduct the urine out through the abdomen and into an external pouch worn for urine collection.

bladder (and/or rectum) becomes overly full. Initial symptoms include sweating (particularly on the forehead) and **headache**, with progression to slow heart

rate (bradycardia) and high blood pressure (**hypertension**). Patients should notify their physician promptly if symptoms do not subside after the bladder (or rectum) is emptied, or if the bladder (or rectum) is full and cannot be emptied.

Resources

ORGANIZATIONS

Bladder Health Council, American Foundation for Urologic Disease. 300 West Pratt St., Suite 401, Baltimore, MD 21201. (800) 242-2383 or (410) 727- 2908.

National Association for Continence. P.O. Box 8310, Spartanburg, SC 29305-8310. (800) 252-3337. <http://www.nafc.org>.

Simon Foundation for Continence. Box 835, Wilmette, IL 60091.

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Neuroleptics see **Antipsychotic drugs**

Neurolinguistic programming

Definition

Neurolinguistic programming (NLP) is aimed at enhancing the healing process by changing the conscious and subconscious beliefs of patients about themselves, their illnesses, and the world. These limiting beliefs are “reprogrammed” using a variety of techniques drawn from other disciplines including **hypnotherapy** and psychotherapy.

Purpose

Neurolinguistic programming has been used to change the limiting beliefs of patients about their prospects of recovery from a wide variety of medical conditions including Parkinson’s disease, **AIDS**, migraines, arthritis, and **cancer**. Practitioners claim to be able to cure most **phobias** in less than one hour, and to help in making lifestyle changes regarding **exercise**, diet, **smoking**, etc. NLP has also been used to treat **allergies**. In other fields, claimed benefits include improved relationships, communication, motivation, and business performance.

Description

Origins

NLP was originally developed during the early 1970s by linguistics professor John Grinder and

psychology and mathematics student Richard Bandler, both of the University of California at Santa Cruz.

Studying the well-known psychotherapist Virginia Satir, the hypnotherapist Milton Erickson, the anthropologist Gregory Bateson, and others whom they considered “charismatic superstars” in their fields, Grinder and Bandler identified psychological, linguistic and behavioral characteristics that they said contributed to the greatness of these individuals. On the other hand, they found that persons experiencing emotional difficulties could be similarly identified by posture, breathing pattern, choice of words, voice tone, eye movements, body language, and other characteristics.

Grinder and Bandler then focused on using these indicators to analyze and alter patterns of thought and behavior. After publishing their findings in two books in 1975, Grinder and Bandler parted company with themselves, with a number of other collaborators, and with the University of California, continuing their work on NLP outside the formal world of academia. As a result, NLP split into a number of competing schools.

Popularized by television “infomercial” personality Anthony Robbins and others, NLP was quickly adopted in management and self-improvement circles. During the 1990s, there was growing interest in NLP’s healing potential.

In a health-care context, practitioners of neurolinguistic programming first seek to identify the negative attitudes and beliefs with which a client has been “programmed” since birth. This is accomplished by asking questions and observing physical responses such as changes in skin color, muscle tension, etc. Then, a wide variety of techniques is employed to “reprogram” limiting beliefs. For example, clients with chronic illness such as **AIDS** or cancer might be asked to displace the despair and loss of identity caused by the disease by visualizing themselves in vigorous health. Treatment by NLP practitioners is often of shorter duration than that of other alternative practitioners, but NLP self-help seminars and courses can be quite expensive.

For those who wish to try self-treatment with NLP, a wide variety of books, audio tapes, and videos are available.

Precautions

NLP is particularly popular in the self-improvement and career-development fields, and some

trainers and practitioners have little experience in its use for healing. Practitioners should be specifically asked about this.

Because NLP is intended to enhance the healing process, it should not be used independently of other healing methods. In all cases of serious illness, a physician should be consulted.

Side effects

NLP is believed to be generally free of harmful side effects.

Research and general acceptance

Although some physicians and mental health practitioners employ principles of neurolinguistic programming, the field is generally considered outside of mainstream medical practice and academic thinking.

Resources

ORGANIZATIONS

Association for NLP. PO Box 78, Stourbridge, UK DY8 2YP.

International NLP Trainers Association, Ltd. Coombe House, Mill Road, Fareham, Hampshire, UK PO16 0TN. (044) 01489 571171.

Society of Neuro-Linguistic Programming. PO Box 424, Hopatcong, NJ 07843. (201) 770-3600.

David Helwig

Neurologic bladder dysfunction see
Neurogenic bladder

Neurologic exam

Definition

A neurological examination is an essential component of a comprehensive **physical examination**. It is a systematic examination that surveys the functioning of nerves delivering sensory information to the brain and carrying motor commands (Peripheral nervous system) and impulses back to the brain for processing and coordinating (Central nervous system).

Purpose

A careful neurological evaluation can help to determine the cause of impairment since a clinician can begin localizing the problem. Symptoms that

occur unexpectedly suggest a blood vessel or seizure problem. Those that are not so sudden suggest a possible tumor. Symptoms that have a waning course with recurrences and worsen over time suggest a disease that destroys nerve cells. Others that are chronic and progressive indicate a degenerative disorder. In cases of trauma, symptoms may be evident upon inspection and causes may be explained by third party witnesses. Some patients may require extensive neurological screening examination (NSE) and/or neurological examination (NE) to determine the cause. The NH will assist the clinician to diagnose illnesses such as seizure disorders, **narcolepsy**, migraine disorders, **dizziness**, and dementia.

Description

A neurological screening is an essential component of every comprehensive physical examination. In cases of neurological trauma, disease, or psychological disorders patients are usually given a very in-depth neurological examination. The examination is best performed in a systematic manner, which means that there is a recommended order for procedures.

Neurological screening examination

The NSE is basic procedure especially in patients who have a general neurological complaint or symptoms. The NSE consists of six areas of assessment:

- mental status: assessing normal orientation to time, place, space, and speech
- cranial nerves: checking the eyes with a special light source (ophthalmoscope), and also assessment of facial muscles strength and functioning
- motor: checking for tone, drift, heel, and toe and walking
- sensory: cold and vibration tests
- coordination: observing the patient walk and finger to nose testing
- reflexes: using a special instrument the clinician taps an area above a nerve to emit a reflex (usually movement of muscle groups)

Neurological examination

The NE should be performed on a patient suspected of having neurological trauma, neurological, or psychological diseases. The NE is performed in a systematic and comprehensive manner. The NE consists of several comprehensive and in-depth assessments of mental status, cranial nerves, motor examination,

reflexes, sensory examination, and posture and walking (gait) analysis.

MENTAL STATUS EXAMINATION (MSE). There are two types of MSE, informal and formal. The informal MSE is usually done when clinicians are obtaining historical information from a patient. The formal MSE is performed in a patient suspected of a neurological problem. The patient is commonly asked his/her name, the location, the day, and date. Retentive memory capability and immediate recall can be assessed by determining the number of digits that can be repeated in sequence. Recent memory is typically examined by testing recall potential of a series of objects after defined times, usually within five and 15 minutes. Remote memory can be assessed by asking the patient to review in a coherent and chronological fashion, his or her illness or personal life events that the patient feels comfortable talking about. Patient recall of common historical or current events can be utilized to assess general knowledge. Higher functioning (referring to brain processing capabilities) can be assessed by spontaneous speech, repetition, reading, naming, writing, and comprehension. The patient may be asked to perform further tasks such as identification of fingers, whistling, saluting, brushing teeth motions, combing hair, drawing, and tracing figures. These procedures will assess the intactness of what is called dominant (left-sided brain) functioning or higher cortical function referring to the portion of the brain that regulates these activities.

The MSE is particularly important in the specialty of psychotherapy. Psychotherapists recommend an in-depth MSE to all patients with possible organic (referring to the body) or psychotic disorders. This examination is also performed in a systematic and orderly manner. It is divided into several categories:

- **Appearance:** This assessment determines the patient's presentation, i.e., how the patient looks (clothes posture, grooming, and alertness).
- **Behavior:** This assesses the patient's motor (movements) activity such as walking, gestures, muscular twitching, and impulse control.
- **Speech:** the patient's speech can be examined concerning volume, rate of speech and coherence. Patients who exhibit latent or delayed speech can indicate depression, while a rapid or pressured speech may suggest possible **mania** or **anxiety**.
- **Mood and affect:** Normal mood is term euthymia. There is variation in mood presentations and patients may display a flat, labile, blunted, constructed or inappropriate mood. The patient can also be euphoric (elevated) or dysphoric (on the down side).
- **Thought processes and content:** This category is typically assessed by determining word usage (can indicate brain disease), thought stream (whether thoughts are slow, restricted, blocked, or overabundant), continuity of thought (referring to associations among ideas), and content of thought (delusional thoughts).
- **Perception:** This assessment examines the patient's ability to hear, see, touch, taste, and smell. Certain psychological states may cause hearing and visual **hallucinations**. Impairments of smell and touch are usually caused by medical (organic) causes or as side effects from certain medications.
- **Attention and concentration:** This clinician assesses the patient's ability to focus on a specific task or activity. Abnormalities in attention and concentration can indicate problems related to anxiety or hallucinations.
- **Orientation:** The patient is examined for orientation to time, place, and identification of self (asking the patient his/her name). Disturbances in orientation can be due to a medical condition (other than psychological), **substance abuse**, or as a side effect of certain medications such as those used to treat depression, anxiety or **psychosis** (since these medications usually have a sedative affect).
- **Memory:** Patients are examined for remote, recent, and immediate memory capabilities. Remote and recent memory can be assessed by the patient's ability to recall historical and current events. Immediate memory can be tested by naming three objects and asking the patient to repeat the named objects immediately, then after five and 15 minute intervals.
- **Judgment:** This category evaluates the patient's ability to **exercise** appropriate judgment. It also determines whether the patient has an understanding of consequences associated with their actions.
- **Intelligence and information:** The only precise measurement for this category can be obtained by administering specialized intelligence tests. However a preliminary assessment of intelligence can be made based on the patient's fund of information, general knowledge, awareness of current events, and the ability for abstract thinking (thinking of unique concepts).
- **Insight:** Insight in the MSE pertains to the patient's awareness of their problem that prompted them to seek professional examination. Insight concerning

the present illness can range from denial to fleeting admission of current illness.

CRANIAL NERVES (CN). Cranial nerves are specialized nerves that originate in the brain and connect to specialized structures such as the nose, eyes, muscles in the face, scalp, ear, and tongue.

- **CNI:** This nerve checks for visual capabilities. Patients are usually given the Snellen Chart (a chart with rows of large and small letters). Patients read letters with one eye at a time.
- **CN III, IV, and VI:** These nerves examine the pupillary (the circular center structure of the eye that light rays enter) reaction. The pupils get smaller, normally when exposed to the light. The eyelids are also examined for drooping or retraction. The eyeball is also checked for abnormalities in movement.
- **CNV:** The clinician can assess the muscles on both sides of the scalp muscles (the temporalis muscle). Additionally the jaw can be tested for motion resistance, opening, protrusion, and side-to-side mobility. The cornea located is a transparent tissue covering the eyeball and could be tested for intactness by lightly brushing a wisp of cotton directly on the outside of the eye.
- **CNVII:** Examination of CNVII assesses asymmetry of the face at rest and during spontaneous movements. The patient is asked to raise eyebrows, wrinkle forehead, close eyes, frown, smile, puff cheeks, purse lips, whistle, and contract chin muscles. Taste for the front and middle portions of the tongue can also be examined.
- **CNVIII:** Testing for this CN deals with hearing. The clinician usually uses a special instrument called a tuning fork and tests for air conduction and structural problems which can occur inside the ear.
- **CN IX and X:** These tests will evaluate certain structures in the mouth. The clinician will usually ask the patient to say “aah” and can detect abnormal positioning of certain structures such as the palatel-uvula. The examiner will also assess the sensation capabilities of the pharynx, by stimulating the area with a wooden tongue depressor, causing a gag reflex.
- **CNXI:** This nerve is usually examined by asking the patient to shrug shoulders (testing a muscle called the trapezius) and rotating the head to each side (testing a muscle called the sternocleidomastoid). These muscles are responsible for movement of the shoulders and neck. The test is usually done with resistance, meaning the examiner holds the area while the patient is asked to move. This is done to assess patient’s strength in these areas.
- **CNXII:** This nerve tests the bulk and power of the tongue. The examiner looks for tongue protrusion and/or abnormal movements.

MOTOR EXAMINATION. The motor examination assesses the patient’s muscle strength, tone, and shape. Muscles could be abnormally larger than expected (hypertrophy) or small due to tissues destruction (atrophy). It is important to assess if there is evidence of twitching or abnormal movements. Involuntary movements due to tics or myoclonus can be observed. Additionally, movements can be abnormal during maintained posture in neurological disorders such as Parkinson’s disease. Muscle tone is usually tested by applying resistance to passive motion of a relaxed limb. Power is assessed for movements at each joint. Decreases or increases in muscle tone can help the examiner localize the affected area.

REFLEXES. The patient’s reflexes are tested by using a special instrument that looks like a little hammer. The clinician will tap the rubber triangular shaped end in several different areas in the arms, knee, and Achilles heal area. The clinician will ask the patient to relax and gently tap the area. If there is a difference in response from the left to right knee, then there may be an underlying problem that merits further evaluation. A difference in reflexes between the arms and legs usually indicates of a lesion involving the spinal cord. Depressed reflexes in only one limb, while the other limb demonstrates a normal response usually indicates a peripheral nerve lesion.

SENSORY EXAMINATION. Although a very essential component of the NE, the sensory examination is the least informative and least exacting since it requires patient concentration and cooperation. Five primary sensory categories are assessed: vibration (using a tuning fork), joint position (examiner moves the limb side-to-side and in a downward position), light touch, pinprick, and temperature. Patients who have sensory abnormalities may have a lesion above the thalamus. Spinal cord lesions or disease can possibly be detected by pinprick and temperature assessment.

COORDINATION. The patient is asked to repetitively touch his nose using his index finger and then to touch the clinician’s outstretched finger. Coordination can also be assessed by asking the patient to alternate tapping the palm then the back of one hand on the thigh. For coordination in the lower extremities on legs, the patient lies on his or her back and is asked to slide the heel of each foot from the knee down the shin of the opposite leg and to raise the leg and touch the examiners index finger with the great toe.

KEY TERMS

Corticospinal tract—A tract of nerve cells that carries motor commands from the brain to the spinal cord.

Gait—Referring to walking motions.

Reflex—A response, usually a movement, elicited by tapping on the nerve with a special hammer-like instrument.

Thalamus—A part of the brain that filters incoming sensory information.

WALKING (GAIT). Normal walking is a complex process and requires usage of multiple systems such as power, coordination and sensation working together in a coordinated fashion. The examination of gait can detect a variety of disease states. Decreased arm swinging on one side is indicative of corticospinal tract disease. A stooped down posture and short-stepped gait may suggest Parkinson's syndrome. A high stepped, slapping gait may be the result of a peripheral nerve disease.

Preparation

The MSE is the first step in a continuous assessment to determine the diagnosis a psychotherapist should take a detailed medical history in the process of ruling out a general medical condition. If a general medical disease is suspected, referral is indicated to rule out this category. Once a medical condition has been fully excluded the therapist can then localize the components of an abnormal MSE to determine the underlying psychological disorder. Once this is determined treatment may include, but is not limited to therapy sessions and/or medication. For neurological diseases the clinician will use information gained from the NE for ordering further tests. These tests may include a complete blood analysis, liver function tests, **kidney function tests**, hormone tests, and a lumbar puncture to determine abnormalities in cerebrospinal fluid. In cases of trauma (car accident, sports injury) the NE is a quick and essential component of emergency assessment. Once a diagnosis is determined emergency measures may include further tests and/or surgery.

Aftercare

Care is usually specific once the final diagnosis has been determined. In psychological cases the treatment may include therapy and/or medication. In cases of

an acute insult such as stroke or trauma, the patient is usually admitted to the hospital for appropriate treatment. Some neurological diseases are chronic and require conservative (medical) treatment and frequent follow-up visits for monitoring and stability or progression of the disease state. The MSE and NE are good diagnostic tools. Further testing using advanced technological procedures is usually required for definitive diagnosis and initiation of disease-specific treatment.

The outcome depends ultimately on the final diagnosis. Neurological diseases typically follow a chronic course. Situations that present as trauma may require surgical intervention and intensive care with an outcome usually proportional to extent of injuries. Psychological disorders may require long term (chronic) treatment and/or medication(s). Most neurological conditions require follow-up and periodic monitoring.

Resources

BOOKS

- Behrman, Richard E., et al, editors. *Nelson Textbook of Pediatrics*. 16th ed. W. B. Saunders Company, 2000.
- Goldman, Lee, et al. *Cecil's Textbook of Medicine*. 21st ed. W. B. Saunders Company, 2000.

Laith Farid Gulli, M.D.
Bilal Nasser, M.Sc.

Neuromuscular junction disease see

Myasthenia gravis

Neuropathic bladder see **Neurogenic bladder**

Neutropenia

Definition

Neutropenia is an abnormally low level of neutrophils in the blood. Neutrophils are white blood cells (WBCs) produced in the bone marrow that ingest bacteria. Neutropenia is sometimes called agranulocytosis or granulocytopenia because neutrophils make up about 60% of WBCs and have granules inside their cell walls. Neutropenia is a serious disorder because it makes the body vulnerable to bacterial and fungal infections.

Description

The normal level of neutrophils in human blood varies slightly by age and race. Infants have lower counts than older children and adults, and African Americans have lower counts than Caucasians or Asians. The average adult level is 1500 cells/mm³ of blood. Neutrophil counts (in cells/mm³) are interpreted as follows:

- greater than 1000. Normal protection against infection
- 500–1000. Some increased risk of infection
- 200–500. Great risk of severe infection
- lower than 200. Risk of overwhelming infection; requires hospital treatment with antibiotics

Causes and symptoms

Causes

Neutropenia may result from three processes:

DECREASED WBC PRODUCTION. Lowered production of white blood cells is the most common cause of neutropenia. It can result from:

- medications that affect the bone marrow, including cancer drugs, chloramphenicol (Chloromycetin), anticonvulsant medications, and **antipsychotic drugs** (Thorazine, Prolixin, and other phenothiazines)
- hereditary and congenital disorders that affect the bone marrow, including familial neutropenia, cyclic neutropenia, and infantile agranulocytosis
- **cancer**, including certain types of leukemia
- radiation therapy
- exposure to pesticides
- vitamin B₁₂ and folate (folic acid) deficiency

DESTRUCTION OF WBCS. WBCs are used up at a faster rate by:

- acute bacterial infections in adults
- infections in newborns
- certain **autoimmune disorders**, including **systemic lupus erythematosus** (SLE)
- penicillin, phenytoin (Dilantin), and sulfonamide medications (Benemid, Bactrim, Gantanol)

SEQUESTRATION AND MARGINATION OF WBCS. Sequestration and margination are processes in which neutrophils are removed from the general blood circulation and redistributed within the body. These processes can occur because of:

- hemodialysis

- **Felty's syndrome** or **malaria**, the neutrophils accumulate in the spleen
- bacterial infections, the neutrophils remain in the infected tissues without returning to the bloodstream

Symptoms

Neutropenia has no specific symptoms except the severity of the patient's current infection. In severe neutropenia, the patient is likely to develop periodontal disease, oral and rectal ulcers, **fever**, and bacterial pneumonia. Fever recurring every 19–30 days suggests cyclical neutropenia.

Diagnosis

Diagnosis is made on the basis of a white blood cell count and differential. The cause of neutropenia is often difficult to establish and depends on a combination of the patient's history, genetic evaluation, **bone marrow biopsy**, and repeated measurements of the WBC.

Treatment

Treatment of neutropenia depends on the underlying cause.

Medications

Patients with fever and other signs of infection are treated for seven to 10 days with antibiotics. Nutritional deficiencies are corrected by green vegetables to supply **folic acid**, and by vitamin B supplements.

Medications known to cause neutropenia are stopped. Neutropenia related to pesticide exposure is treated by removing the patient from the contaminated environment.

Patients receiving **chemotherapy** for cancer may be given a blood growth factor called sargramostim (Leukine, Prokine) to stimulate WBC production.

Surgery

Patients with Felty's syndrome who have repeated infections may have their spleens removed.

Prognosis

The prognosis for mild or chronic neutropenia is excellent. Recovery from acute neutropenia depends on the severity of the patient's infection and the promptness of treatment.

KEY TERMS

Cyclical neutropenia—A rare genetic blood disorder in which the patient's neutrophil level drops below 500/mm³ for six to eight days every three weeks.

Differential—A blood cell count in which the percentages of cell types are calculated as well as the total number of cells.

Felty's syndrome—An autoimmune disorder in which neutropenia is associated with rheumatoid arthritis and an enlarged spleen.

Granulocyte—Any of several types of white blood cells that have granules in their cell substance. Neutrophils are the most common type of granulocyte.

Neutrophil—A granular white blood cell that ingests bacteria, dead tissue cells, and foreign matter.

Sargramostim—A medication made from yeast that stimulates WBC production. It is sold under the trade names Leukine and Prokine.

Sequestration and margination—The removal of neutrophils from circulating blood by cell changes that trap them in the lungs and spleen.

Resources

BOOKS

Linker, Charles A. "Blood." In *Current Medical Diagnosis and Treatment*, 1998, edited by Stephen McPhee, et al., 37th ed. Stamford: Appleton & Lange, 1997.

Rebecca J. Frey, PhD

Nevirapine see **Non-nucleoside reverse transcriptase inhibitors**

Nevus see **Moles**

Newborn life support see **Extracorporeal membrane oxygenation**

Niacin deficiency see **Pellagra**

Nicotine see **Smoking; Smoking-cessation drugs**

Nicotinic acid deficiency see **Pellagra**

Niemann-Pick disease see **Lipidoses**

Nifedipine see **Calcium channel blockers**

Night blindness see **Vitamin A deficiency**

Night terrors

Definition

Night terrors are a sleep disorder characterized by **anxiety** episodes with extreme panic, often accompanied by screaming, flailing, fast breathing, and sweating and that usually occur within a few hours after going to sleep.

Description

Night terrors occur most commonly in children between the ages of four and 12 but can also occur at all ages. Affected individuals usually suffer these episodes within a few hours after going to sleep. They appear to bolt up suddenly, and wake up screaming, sweating and panicked. The episode may last anywhere from five to 20 minutes. During this time, the individual is actually asleep, although the eyes may open. Quite often, nothing can be done to comfort the affected person. Very often, the person has no memory of the episode upon waking the next day.

Night terrors are differentiated from nightmares in that they have been shown to occur during Stage 4 of sleep, or in REM sleep, while nightmares can occur anytime throughout the sleep cycle.

Causes and symptoms

Suffering from night terrors seems to run in families. Extreme tension or **stress** can increase the incidence of the episodes. In adults, the use of alcohol also contributes to an increased incidence of night terrors. Episodes sometimes occur after an accident involving **head injury**. Other factors thought to contribute to episodic night terrors, but not actually cause them, include:

- medications
- excessive tiredness at bedtime
- eating a heavy meal prior to bedtime
- drug abuse

Diagnosis

Night terrors are primarily diagnosed by observing the person suffering from an episode. The

following symptoms are characteristic of a person suffering from a night terror:

- panic
- sweating
- gasping, moaning, crying or screaming during sleep
- little or no recollection of the episode upon awakening

Treatment

In most cases, the individual will still be asleep as the night terror episode happens and will prove difficult to awaken. The goal should be to help the affected person go back into a calm state of sleep. The lights should be turned on, and soothing comments should be directed at the person, avoiding brusque gestures such as shaking the person or shouting to startle them out of the episode. Any form of stress should be avoided.

Individuals affected by night terrors should be evaluated by a physician if they are really severe and occur frequently. A physician can recommend the best treatment for the particular circumstances of the night terrors. In some severe cases, the physician may prescribe a benzodiazepine tranquilizer, such as Diazepam, known to suppress Stage 4 of sleep. The physician may also refer the affected person for further evaluation by a sleep disorder specialist. It should be noted that episodic night terrors in children are normal and do not suggest the presence of psychological problems. In adults, night terrors are more likely to be related to a significant stress-related or emotional problem.

Prognosis

In children, night terror episodes in children usually end by the age of 12.

Prevention

If a child seems to have a regular pattern of night terror episodes, he should be gently awakened about 15 minutes before the episode usually happens. The child should be kept awake and out of the bed for a short period of time and then allowed to return to bed.

Since sleep deprivation is a strong trigger for night terror episodes, children should not be allowed to become overtired. Having children take a nap during the day may be useful.

Adults affected by night terror episodes should avoid stress, the consumption of alcohol and stimulants before going to sleep.

KEY TERMS

Benzodiazepines—A class of drugs that suppresses Stage 4 of sleep.

REM sleep—Rapid Eye Movement phase of sleep, a mentally active period during which dreaming occurs.

Sleep disorder—Any disorder that keep a person from falling asleep or staying asleep.

Resources

PERIODICALS

American Academy of Family Physicians. "Nightmares and Night Terrors in Children." October 2000.

Laberge, Luc, et al. "Development of Parasomnias from Childhood to Early Adolescence." *Pediatrics* July 2000: 67-74.

ORGANIZATIONS

American Sleep Disorders Association, 6301 Bandel Road Suite 101, Rochester, MN 55901. (507) 287-6008. <<http://www.asda.org>>.

National Foundation for Sleep and Related Disorders in Children. 4200 W. Peterson Suite 109, Chicago, IL 60646. (708) 971-1086.

Kim A. Sharp, M.Ln.

Nitrates see **Antiangina drugs**

Nitrofurantoin see **Urinary anti-infectives**

Nitrogen narcosis

Definition

Nitrogen narcosis is a condition that occurs in divers breathing compressed air. When divers go below depths of approximately 100 ft, increase in the partial pressure of nitrogen produces an altered mental state similar to alcohol intoxication.

Description

Nitrogen narcosis, commonly referred to as "rapture of the deep," typically becomes noticeable at 100 ft underwater and is incapacitating at 300 ft, causing stupor, blindness, unconsciousness, and even **death**. Nitrogen narcosis is also called "the martini effect" because divers experience an effect comparable to that

from one martini on an empty stomach for every 50 ft of depth beyond the initial 100 ft.

Causes and symptoms

Nitrogen narcosis is caused by gases in the body acting in a manner described by Dalton's Law of partial pressures: the total pressure of a gas mixture is equal to the sum of the partial pressures of gases in the mixture. As the total gas pressure increases with increasing dive depth, the partial pressure of nitrogen increases and more nitrogen becomes dissolved in the blood. This high nitrogen concentration impairs the conduction of nerve impulses and mimics the effects of alcohol or **narcotics**.

Symptoms of nitrogen narcosis include: wooziness; giddiness; euphoria; disorientation; loss of balance; loss of manual dexterity; slowing of reaction time; fixation of ideas; and impairment of complex reasoning. These effects are exacerbated by cold, **stress**, and a rapid rate of compression.

Diagnosis

A diagnosis must be made on circumstantial evidence of atypical behavior, taking into consideration the depth of the dive and the rate of compression. Nitrogen narcosis may be differentiated from toxicity of oxygen, carbon monoxide, or carbon dioxide by the absence of such symptoms as **headache**, seizure, and bluish color of the lips and nail beds.

Treatment

The effects of nitrogen narcosis are totally reversed as the gas pressure decreases. They are typically gone by the time the diver returns to a water depth of 60 ft. Nitrogen narcosis has no hangover or lasting effects requiring further treatment. However, a doctor should be consulted whenever a diver has lost consciousness.

Prognosis

When a diver returns to a safe depth, the effects of nitrogen narcosis disappear completely. Some evidence exists that certain divers may become partially acclimated to the effects of nitrogen narcosis with frequency—the more often they dive, the less the increased nitrogen seems to affect them.

Prevention

Helium may be used as a substitute for nitrogen to dilute oxygen for deep water diving. It is colorless,

KEY TERMS

Compressed air—Air that is held under pressure in a tank to be breathed by underwater divers. A tank of compressed air is part of a diver's scuba (self-contained underwater breathing apparatus) gear.

Compression—An increase in pressure from the surrounding water that occurs with increasing diving depth.

Partial pressure—The pressure exerted by one of the gases in a mixture of gases. The partial pressure of the gas is proportional to its concentration in the mixture. The total pressure of the gas mixture is the sum of the partial pressures of the gases in it (Dalton's Law) and as the total pressure increases, each partial pressure increases proportionally.

odorless, tasteless, and chemically inert. However, it is more expensive than nitrogen and drains body heat from a diver. In diving with rapid compression, the helium-oxygen mixture may produce **nausea**, **dizziness**, and trembling, but these adverse reactions are less severe than nitrogen narcosis.

Nitrogen narcosis can be avoided by limiting the depth of dives. The risk of nitrogen narcosis may also be minimized by following safe diving practices, including proper equipment maintenance, low work effort, proper buoyancy, maintenance of visual cues, and focused thinking. In addition, no alcohol should be consumed within 24 hours of diving.

Resources

ORGANIZATIONS

American College of Hyperbaric Medicine. P.O. Box 25914-130, Houston, Texas 77265. (713) 528-0657. <<http://www.hyperbaricmedicine.org>>.

Divers Alert Network. The Peter B. Bennett Center, 6 West Colony Place, Durham, NC 27705. (800) 446-2671. <<http://www.diversalertnetwork.org>>.

Undersea and Hyperbaric Medical Society. 10531 Metropolitan Ave., Kensington, MD 20895. (301) 942-2980. <<http://www.uhms.org>>.

Bethany Thivierge

Nitroglycerin see **Antiangina drugs**

Nlein purpura see **Allergic purpura**

NMR see **Magnetic resonance imaging**

Nocardia asteroides infection see
Nocardiosis

Nocardiosis

Definition

Nocardiosis is a serious infection caused by a fungus-like bacterium that begins in the lungs and can spread to the brain.

Description

Nocardiosis is found throughout the world among people of all ages, although it is most common in older people and males. While people with poor immunity are vulnerable to this infection, it sometimes strikes individuals with no history of other diseases. Nocardiosis is rare in AIDS patients. It is not transmitted by person-to-person contact.

Causes and symptoms

Nocardiosis is caused by a bacterium of the *Nocardia* species—usually *N. asteroides*, an organism that is normally found in the soil. The incubation period is not known, but is probably several weeks.

The bacteria can enter the human body when a person inhales contaminated dust. Less often, people can pick up the bacteria in contaminated puncture wounds or cuts.

Symptoms

The infection causes a **cough** similar to **pneumonia** or **tuberculosis**, producing thick, sometimes bloody, sputum. Other symptoms include chills, night sweats, chest **pain**, weakness, loss of appetite and weight loss. Nocardiosis does not, however, respond to short-term **antibiotics**.

Complications

In about one-third of patients, the infection spreads from the blood into the brain, causing brain abscesses. This complication can trigger a range of symptoms including severe **headache**, confusion, disorientation, dizziness, nausea and seizures, and problems in walking. If a brain abscess ruptures, it can lead to **meningitis**.

About a third of patients with nocardiosis also have abscesses in the skin or directly underneath the

KEY TERMS

Abscess—A localized area of infection in a body tissue. Abscesses in the brain or skin are possible complications of nocardiosis.

Meningitis—An infection of the outer covering of the brain (meninges) that can be caused by either bacteria or a virus.

skin. They may also have lesions in other organs, such as the kidneys, liver, or bones.

Diagnosis

Nocardia is not easily identified from cultures of sputum or discharge. A doctor can diagnose the condition using special staining techniques and taking a thorough medical history. Lung biopsies or x rays also may be required. Up to 40% of the time, however, a diagnosis can't be made until an **autopsy** is done.

Treatment

Treatment of nocardiosis includes bed rest and high doses of medication for a period of 12 to 18 months, including sulfonamide drugs or a combination of trimethoprim-sulfamethoxazole (Bactrim, Septra). If the patient doesn't respond to these drugs, antibiotics such as ampicillin (Amcill, Principen) or erythromycin (E-Mycin, Eryc) may be tried.

The abscesses may need to be drained and dead tissue cut away. Other symptoms are treated as necessary.

Prognosis

Nocardiosis is a serious disease with a high mortality rate. If it has been diagnosed early and caught before spreading to the brain, the prognosis is better. Even with appropriate treatment, however, the **death** rate is still 50%. Once the infection reaches the brain, the death rate is above 80%. This outcome is most commonly seen in patients with a weakened immune system.

Resources

BOOKS

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Carol A. Turkington

Nodule see **Skin lesions**

Non-Hodgkin's lymphomas see **Malignant lymphomas**

Non-melanoma skin cancer see **Skin cancer, non-melanoma**

Non-nucleoside reverse transcriptase inhibitors

Definition

This type of drug interferes with an enzyme that is key to the replication (reproduction) of the human **immunodeficiency virus (HIV)**. The drug is designed to help suppress the growth of HIV, but does not eliminate it.

Purpose

This medication is used to treat patients with the HIV virus and **AIDS** in combination with one or more other AIDS drugs. Combining NRTIs with older drugs improves their ability to lower the levels of HIV in the bloodstream, and strengthens the immune system.

HIV becomes rapidly resistant to this class of drugs when they are used alone. However, in combination with older drugs, they can interfere with the virus's ability to become resistant because they attack the virus on several fronts. As the virus tries to evade one drug, another attacks. This combination can lower the level of HIV in the blood to undetectable levels.

Precautions

Patients should not discontinue this drug even if symptoms improve without consultation with a physician.

Description

Nucleoside analogues, the first class of HIV drugs to be developed, worked by incorporating themselves into the virus's DNA, making the DNA incomplete and therefore unable to create a new virus. Non-nucleoside inhibitors work at the same stage as nucleoside analogues, but act in a completely different way, preventing the conversion of RNA to DNA.

This class of drugs includes nevirapine (Viramune) and delavirdine (Rescriptor). It may take several weeks or months before the full benefits are apparent.

KEY TERMS

Human immunodeficiency virus (HIV)—The virus that causes AIDS.

Depending on the drug prescribed, doses may start with a lower amount and be increased after a short period of time.

Risks

A mild skin rash is common; a severe skin rash can be a life threatening reaction. Other possible side effects include **fever**, blistering skin, mouth sores, aching joints, eye inflammation, **headache**, **nausea**, and tiredness.

Because the drug passes into breast milk, breast-feeding mothers should avoid the drug, or not nurse until the treatment is completed.

Resources

ORGANIZATIONS

National AIDS Treatment Advocacy Project. 580 Broadway, Ste. 403, New York, NY 10012. (888) 266-2827. <<http://www.natap.org>>.

Carol A. Turkington

Non-small cell lung cancer see **Lung cancer, non-small cell**

Non-A, non-B hepatitis see **Hepatitis C**

Non-tuberculous see **Mycobacterial infections, atypical**

Nonbacterial regional lymphaden see **Cat-scratch disease**

Noncholera vibrio infections see **Vibriosis**

Noneros see **Gastritis**

Nongonococcal urethritis

Definition

Any inflammation of the urethra not due to gonorrhea, almost always contracted through sexual intercourse and found far more often in men.

Description

Men between the ages of 15 and 30 who have multiple sex partners are most at risk for nongonococcal **urethritis** (NGU), which is believed to be the most common sexually transmitted disease in the United States.

Causes and symptoms

NGU is spread almost exclusively via sexual contact, and appears most often in men because a woman's urethra is less easily infected during sex. The infection is most often due to *Chlamydia trachomatis*, the organism that causes chlamydia. Those that aren't caused by *Chlamydia trachomatis* are usually due to another bacterium, *Ureaplasma urealyticum*. In 10% to 20% of NGU cases, the cause is unknown.

Symptoms appear within one to five weeks after infection, and include a slight clear discharge (the color of the discharge can vary from one patient to the next), and **itching** or burning during or after urination.

However, some men never develop symptoms, and women almost never show signs of infection. However, it's possible that symptoms of burning or itching in or around the vagina may be due to NGU.

The disease is communicable from the time of first infection until the patient is cured. Past infection doesn't make a person immune.

Diagnosis

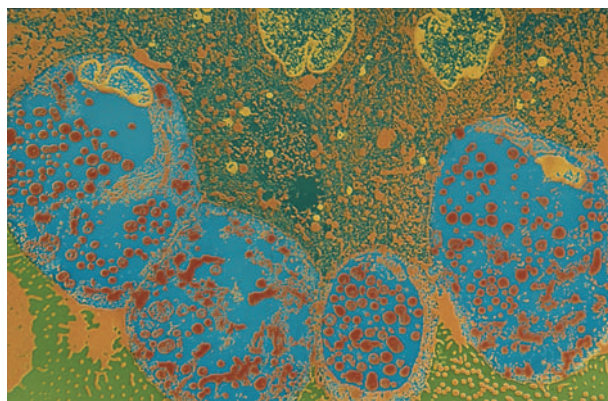
Nongonococcal urethritis is diagnosed by excluding other causes, since inflammation that is not caused by **gonorrhea** is classified as NGU. A microscopic and/or culture test of the discharge or urine can reveal the infection.

Since many people are infected with both NGU and syphilis at the same time, infected patients also should have a test for **syphilis** before treatment for NGU begins, and three months after treatment ends.

Treatment

Antibiotics such as tetracycline or azithromycin will cure NGU; both sexual partners should be treated at the same time.

Patients taking tetracycline should avoid milk or milk products and take the medication at least one hour before or two hours after meals. On the last day of treatment, a male should have a urine test to make



A microscopic image of non-specific urethritis. This sexually transmitted disease is usually caused by a bacterium of the genus *Chlamydia*. (Custom Medical Stock Photo. Reproduced by permission.)

KEY TERMS

Chlamydia—One of the most common sexually transmitted diseases in the United States. It causes discharge, inflammation and burning during urination. About half of the cases of nongonococcal urethritis are due to chlamydia.

Gonorrhea—A sexually transmitted disease that affects the genital mucous membranes of men and women.

Urethra—The tube that carries urine from the bladder through the outside of the body.

sure the infection has cleared. If it hasn't, he should take a second course of therapy. Men should use a **condom** during treatment and for several months after treatment is completed.

If urine tests indicate the infection is gone but symptoms persist, the doctor will check for signs of prostate inflammation.

Prognosis

NGU is completely curable with proper antibiotic treatment. Untreated, NGU can lead to sterility in both men and women, inflammation of the mouth of the uterus, and infections of the woman's internal sexual organs. An infection during **pregnancy** may lead to **pneumonia** or eye infections in the newborn child. Untreated men may develop swelling of the testicles and an infected prostate gland.

Prevention

People can prevent the spread of NGU by:

- using a condom
- limiting the number of sex partners
- washing the genital area after sex
- if infected, avoid sexual contact; take antibiotics, notify all partners

Resources

ORGANIZATIONS

American Social Health Association. P.O. Box 13827,
Research Triangle Park, NC 27709. (800) 227-8922.
<<http://www.ashstd.org>>.

OTHER

Sexually Transmitted Diseases Hotline. (800) 227-8922.

Carol A. Turkington

Nonsteroidal anti-inflammatory drugs

Definition

Nonsteroidal anti-inflammatory drugs are medicines that relieve **pain**, swelling, stiffness, and inflammation.

Purpose

Nonsteroidal anti-inflammatory drugs (NSAIDs) are prescribed for a variety of painful conditions, including arthritis, **bursitis**, **tendinitis**, **gout**, menstrual cramps, sprains, strains, and other injuries. They are also given to control the pain of **cancer** and the side effects of **radiation therapy**.

A group of researchers associated with the Women's Health Initiative reported in 2003 that regular use of **aspirin**, ibuprofen, and other NSAIDs may help to lower a woman's risk of developing **breast cancer**. Further clinical trials are needed, however, to confirm the group's findings.

Description

Nonsteroidal anti-inflammatory drugs relieve pain, stiffness, swelling, and inflammation, but they do not cure the diseases or injuries responsible for these problems. Two drugs in this category, ibuprofen and naproxen, also

reduce **fever**. Some nonsteroidal anti-inflammatory drugs can be bought over the counter; others are available only with a prescription from a physician or dentist.

Among the drugs in this group are diclofenac (Voltaren), etodolac (Lodine), flurbiprofen (Ansaid), ibuprofen (Motrin, Advil, Rufen), ketorolac (Toradol), nabumetone (Relafen), naproxen (Naprosyn); naproxen sodium (Aleve, Anaprox, Naprelan); and oxaprozin (Daypro). They are sold as tablets, capsules, caplets, liquids, and rectal suppositories and some are available in chewable, extended-release, or delayed-release forms.

A newer group of NSAIDs known as **COX-2 inhibitors** are being used successfully to treat patients with allergic reactions to the older NSAIDs. Their name comes from the fact that they block an enzyme known as cyclooxygenase-2, or COX-2, which is involved in the inflammation pathway. The COX-2 inhibitors are also less likely to affect the patient's digestive tract. They include such drugs as celecoxib (Celebrex), rofecoxib (Vioxx), etoricoxib (Arcoxia), and valdecoxib (Bextra). With regard to cancer treatment, some studies indicate that the use of COX-2 inhibitors may postpone the need to prescribe narcotic medications for severe pain.

Recommended dosage

Recommended doses vary, depending on the patient, the type of nonsteroidal anti-inflammatory drug prescribed, the condition for which the drug is prescribed, and the form in which it is used. Always take nonsteroidal anti-inflammatory drugs exactly as directed. If using non-prescription (over-the-counter) types, follow the directions on the package label. For prescription types, check with the physician who prescribed the medicine or the pharmacist who filled the prescription. Never take larger or more frequent doses, and do not take the drug for longer than directed. Patients who take nonsteroidal anti-inflammatory drugs for severe arthritis must take them regularly over a long time. Several weeks may be needed to feel the results, so it is important to keep taking the medicine, even if it does not seem to be working at first.

When taking nonsteroidal anti-inflammatory drugs in tablet, capsule, or caplet form, always take them with a full, 8-ounce glass of water or milk. Taking these drugs with food or an antacid will help prevent stomach irritation.

Precautions

Nonsteroidal anti-inflammatory drugs can cause a number of side effects, some of which may be very

serious (See Side effects). These side effects are more likely when the drugs are taken in large doses or for a long time or when two or more nonsteroidal anti-inflammatory drugs are taken together. Health care professionals can help patients weigh the risks of benefits of taking these medicines for long periods.

Do not take **acetaminophen**, aspirin, or other salicylates along with other nonsteroidal anti-inflammatory drugs for more than a few days unless directed to do so by a physician. Do not take ketorolac (Toradol) while taking other nonsteroidal anti-inflammatory drugs unless directed to do so by a physician.

Because older people are more sensitive than younger adults to nonsteroidal anti-inflammatory drugs, they may be more likely to have side effects. Some side effects, such as stomach problems, may also be more serious in older people.

Serious side effects are especially likely with one nonsteroidal anti-inflammatory drug, phenylbutazone. Patients age 40 and over are especially at risk of side effects from this drug, and the likelihood of serious side effects increases with age. Because of these potential problems, it is especially important to check with a physician before taking this medicine. Never take it for anything other than the condition for which it was prescribed, and never share it—or any other prescription drug—with another person.

Some nonsteroidal anti-inflammatory drugs can increase the chance of bleeding after surgery (including dental surgery), so anyone who is taking the drugs should alert the physician or dentist before surgery. Avoiding the medicine or switching to another type in the days prior to surgery may be necessary.

Some people feel drowsy, dizzy, confused, light-headed, or less alert when using these drugs. Blurred vision or other vision problems also are possible side effects. For these reasons, anyone who takes these drugs should not drive, use machines or do anything else that might be dangerous until they have found out how the drugs affect them.

Nonsteroidal anti-inflammatory drugs make some people more sensitive to sunlight. Even brief exposure to sunlight can cause severe **sunburn**, **rashes**, redness, **itching**, blisters, or discoloration. Vision changes also may occur. To reduce the chance of these problems, avoid direct sunlight, especially from mid-morning to mid-afternoon; wear protective clothing, a hat, and sunglasses; and use a sunscreen with a skin protection factor (SPF) rating of at least 15. Do not use sunlamps, tanning booths or tanning beds while taking these drugs.

Special conditions

People with certain medical conditions and people who are taking some other medicines can have problems if they take nonsteroidal anti-inflammatory drugs. Before taking these drugs, be sure to let the physician know about any of these conditions:

ALLERGIES. Let the physician know about any **allergies** to foods, dyes, preservatives, or other substances. Anyone who has had reactions to nonsteroidal anti-inflammatory drugs in the past should also check with a physician before taking them again.

PREGNANCY. Women who are pregnant or who plan to become pregnant should check with their physicians before taking these medicines. Whether nonsteroidal anti-inflammatory drugs cause birth defects in people is unknown, but some do cause birth defects in laboratory animals. If taken late in **pregnancy**, these drugs may prolong pregnancy, lengthen labor time, cause problems during delivery, or affect the heart or blood flow of the fetus.

BREASTFEEDING. Some nonsteroidal anti-inflammatory drugs pass into breast milk. Women who are breastfeeding should check with their physicians before taking these drugs.

OTHER MEDICAL CONDITIONS. A number of medical conditions may influence the effects of nonsteroidal anti-inflammatory drugs. Anyone who has any of the conditions listed below should tell his or her physician about the condition before taking nonsteroidal anti-inflammatory drugs.

- stomach or intestinal problems, such as colitis or Crohn's disease
- liver disease
- current or past **kidney disease**; current or past kidney stones
- heart disease
- high blood pressure
- blood disorders, such as anemia, low **platelet count**, low white blood cell count
- bleeding problems
- diabetes mellitus
- hemorrhoids, rectal bleeding, or rectal irritation
- asthma
- Parkinson's disease
- epilepsy
- systemic lupus erythematosus

- diseases of the blood vessels, such as polymyalgia rheumatica and **temporal arteritis**
- fluid retention
- alcohol **abuse**
- mental illness.

People who have sores or white spots in the mouth should tell the physician about them before starting to take nonsteroidal anti-inflammatory drugs. Sores or white spots that appear while taking the drug can be a sign of serious side effects.

SPECIAL DIETS. Some nonsteroidal anti-inflammatory drugs contain sugar or sodium, so anyone on a low-sugar or low-sodium diet should be sure to tell his or her physician.

SMOKING. People who smoke cigarettes may be more likely to have unwanted side effects from this medicine.

USE OF CERTAIN MEDICINES. Taking nonsteroidal anti-inflammatory drugs with certain other drugs may affect the way the drugs work or increase the risk of unwanted side effects. (See Interactions.)

Side effects

The most common side effects are stomach pain or cramps, **nausea**, **vomiting**, **indigestion**, **diarrhea**, **heartburn**, **headache**, **dizziness** or lightheadedness, and drowsiness. As the patient's body adjusts to the medicine, these symptoms usually disappear. If they do not, check with the physician who prescribed the medicine.

Serious side effects are rare, but do sometimes occur. If any of the following side effects occur, stop taking the medicine and get emergency medical care immediately:

- swelling or puffiness of the face
- swelling of the hands, feet, or lower legs
- rapid weight gain
- fainting
- breathing problems
- fast or irregular heartbeat
- tightness in the chest

Other side effects do not require emergency medical care, but should have medical attention. If any of the following side effects occur, stop taking the medicine and call the physician who prescribed the medicine as soon as possible:

- severe pain, cramps, or burning in the stomach or abdomen
- convulsions
- fever
- severe nausea, heartburn, or indigestion
- white spots or sores in the mouth or on the lips
- rashes or red spots on the skin
- any unusual bleeding, including nosebleeds, spitting up or vomiting blood or dark material
- black, tarry stool
- chest pain
- unusual bruising
- severe headaches

A number of less common, temporary side effects are also possible. They usually do not need medical attention and will disappear once the body adjusts to the medicine. If they continue or interfere with normal activity, check with the physician. Among these side effects are:

- gas, bloating, or **constipation**
- bitter taste or other taste changes
- sweating
- restlessness, irritability, **anxiety**
- trembling or twitching

Some patients who have had problems with side effects from NSAIDs may benefit from **acupuncture** as an adjunctive treatment in **pain management**. A recent study done in New York found that older patients with lower back pain related to cancer reported that their pain was relieved by acupuncture with fewer side effects than those caused by NSAIDs.

Interactions

Nonsteroidal anti-inflammatory drugs may interact with a variety of other medicines. When this happens, the effects of the drugs may change, and the risk of side effects may be greater. Anyone who takes these drugs should let the physician know all other medicines he or she is taking. Among the drugs that may interact with nonsteroidal anti-inflammatory drugs are:

- blood thinning drugs, such as warfarin (Coumadin)
- other nonsteroidal anti-inflammatory drugs
- heparin
- tetracyclines
- cyclosporine

KEY TERMS

Anemia—A lack of hemoglobin—the compound in blood that carries oxygen from the lungs throughout the body and brings waste carbon dioxide from the cells to the lungs, where it is released.

Bursitis—Inflammation of the tissue around a joint.

Colitis—Inflammation of the colon (large bowel).

COX-2 inhibitors—A class of newer NSAIDs that are less likely to cause side effects in the digestive tract. COX-2 inhibitors work by inhibiting the production of cyclooxygenase-2, an enzyme involved in inflammation.

Inflammation—Pain, redness, swelling, and heat that usually develop in response to injury or illness.

Salicylates—A group of drugs that includes aspirin and related compounds. Salicylates are used to relieve pain, reduce inflammation, and lower fever.

Tendinitis—Inflammation of a tendon, which is a tough band of tissue that connects muscle to bone.

- digitalis drugs
- lithium
- phenytoin (Dilantin)
- zidovudine (AZT, Retrovir).

NSAIDs may also interact with certain herbal preparations sold as dietary supplements. Among the herbs known to interact with NSAIDs are bearberry (*Arctostaphylos uva-ursi*), feverfew (*Tanacetum parthenium*), evening primrose (*Oenothera biennis*), and gossypol, a pigment obtained from cottonseed oil and used as a male contraceptive. In most cases, the herb increases the tendency of NSAIDs to irritate the digestive tract. It is just as important for patients to inform their doctor of herbal remedies that they take on a regular basis as it is to give the doctor a list of their other prescription medications.

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ORGANIZATIONS

- U. S. Food and Drug Administration (FDA). 5600 Fishers Lane, Rockville, MD 20857. (888) 463-6332. <<http://www.fda.gov>>.

Nancy Ross-Flanigan
Rebecca J. Frey, PhD

Nontropical sprue see **Celiac disease**

Nonvenereal syphilis see **Bejel**

Norfloxacin see **Fluoroquinolones**

Noroviruses

Definition

Noroviruses are a group of related, single-stranded RNA (ribonucleic acid) viruses that cause acute **gastroenteritis** in humans.

Description

Noroviral infection

Noroviruses are a major cause of viral gastroenteritis—an inflammation of the linings of the stomach and small and large intestines that causes **vomiting** and **diarrhea**. Viruses are responsible for 30–40% of all cases of infectious diarrhea and viral gastroenteritis is the second most common illness in the United States, exceeded only by the **common cold**.

Anyone can become infected with norovirus. During norovirus outbreaks there are high rates of infection among people of all ages. There are a large number of genetically-distinct strains of norovirus. Immunity appears to be specific for the norovirus strain and lasts for only a few months. Therefore norovirus infection can recur throughout a person's lifetime. Because of genetic (inherited) differences among humans, some people appear to be more susceptible to norovirus infection and may suffer more severe illness. People with type O blood are at the highest risk for severe infection.

Infected individuals are contagious from the first onset of symptoms until at least three days after full recovery. Some people may remain contagious for as long as two weeks after recovery.

Gastroenteritis

Gastroenteritis often is referred to as the stomach flu even though the flu is a respiratory illness caused by an **influenza** virus. Other common names for viral gastroenteritis include:

- food poisoning
- winter-vomiting disease
- non-bacterial gastroenteritis
- calicivirus infection.

The U.S. Centers for Disease Control and Prevention (CDC) estimate that noroviruses are responsible for some 23 million cases of acute gastroenteritis in the United States every year. Epidemiologists estimate that about 50,000 Americans are hospitalized annually and about 400 die as a result of norovirus infection. In developing countries noroviruses are a major cause of human illness.

Gastroenteritis caused by infection with a norovirus is rarely a serious illness. Typically an infected person suddenly feels very ill and may vomit many times in a single day. The symptoms, although quite unpleasant, usually last only 24–60 hours.

Transmission

Noroviruses are ubiquitous in the environment. They are highly contagious and are considered to be among the most infectious of viruses. The reasons for this include:

- Only a small number of viral particles—fewer than 100—are required for infection.
- Although noroviruses cannot reproduce outside of their human hosts, they can remain viable for weeks or even months on objects and surfaces.
- Human immunity to norovirus is short-lived and strain-specific.

Noroviruses are transmitted among people by a fecal-oral route, either by ingestion of food or water contaminated with feces or by contact with the vomit or feces of an infected person. Norovirus infection can occur by:

- consuming contaminated food or liquids
- hand contact with contaminated objects or surfaces, followed by hand contact with the mouth
- contact with an infected person, including caring for the sick person or sharing food or utensils
- aerosolized vomit that is swallowed or contaminates surfaces.

Environmental contamination or contact with infected clothing or linen also may be a source of transmission. Although there is no evidence that norovirus infection can occur via the respiratory system, the sudden and violent vomiting of noroviral gastroenteritis can lead to contamination of the surroundings and of public areas. Particles laden with virus can be suspended in the air and swallowed.

FOODBORNE TRANSMISSION. Noroviruses account for at least 50% of food-related outbreaks of gastroenteritis. A European study found that between 1995

and 2000 noroviruses were responsible for more than 85% of all foodborne non-bacterial gastroenteritis outbreaks. Restaurant or catered foods are common sources of norovirus transmission, with subsequent infection of household members. The majority of norovirus outbreaks occur via contamination by a food handler immediately before the food is consumed.

Foods that frequently are associated with norovirus outbreaks include:

- foods that are eaten without further cooking, including sandwiches, salads, and bakery products
- liquids such as salad dressing or cake icing in which the virus becomes evenly distributed
- food that is contaminated at its source, including oysters and clams from contaminated waters and raspberries irrigated with sewage-contaminated water
- food that becomes contaminated before distribution, including salads and frozen fruit.
- Shellfish, including oysters and clams, concentrate norovirus from contaminated water in their tissues. Steaming shellfish may not completely inactivate the virus.

WATERBORNE TRANSMISSION. There is widespread norovirus contamination of rivers and seas, often with more than one strain of the virus. Waterborne outbreaks of norovirus have been associated with:

- sewage-contaminated wells
- contaminated municipal water systems
- stream and lake water
- swimming pools and spas
- commercial ice.

Outbreaks

Norovirus infection can spread rapidly through daycare centers, schools, prisons, hospitals, nursing homes, camps, and other confined spaces. About 40% of group- or institutionally-related outbreaks of diarrhea are caused by norovirus. Outbreaks usually peak during the winter months.

Between July of 1997 and June of 2000, 232 norovirus outbreaks were reported to the CDC. It was determined that 57% of these outbreaks were due to foodborne transmission, 16% were spread by human contact, and 3% were due to waterborne transmission. The mode of transmission could not be determined in 23% of the outbreaks. Restaurants or catered food accounted for 36% of the outbreaks, 23% occurred

in nursing homes, 13% in schools, and 10% at resorts or on cruise ships. Outbreaks also have occurred at large family gatherings.

Cruise ships have become notorious for norovirus outbreaks among passengers and staff. Cruise ships and naval vessels are at increased risk for contamination when docking in regions that lack adequate sanitation and where contaminated food or water may be brought onboard. Outbreaks on cruise ships are exacerbated by close living quarters and the arrival of new, susceptible passengers every one to two weeks. Norovirus outbreaks have been reported to continue through more than 12 successive cruises on a single ship.

A study of 12 calicivirus outbreaks on cruise ships in 2002 found that 11 of the outbreaks were caused by noroviruses and seven of these were due to a previously unreported strain. In the same year, 10 out of 22 land-based outbreaks were attributed to this new strain.

Outbreaks of norovirus appear on the increase. In 2005 the CDC reported that norovirus outbreaks were increasing in hospitals, daycare centers, nursing homes, and schools across the country. The International Council of Cruise Lines reported that, although less than 1% of passengers become infected with norovirus each year, outbreaks on cruise ships also were on the increase. In the summer of 2004 an outbreak at Yellowstone National Park sickened 134 people. More than 1,100 people became ill in early 2004 after a norovirus outbreak at Las Vegas hotels. The following autumn more than 1,200 people became sick from a norovirus outbreak at a single Las Vegas hotel-casino.

Causes & symptoms

Norovirus strains

Noroviruses lack outer envelopes and their genetic material is carried as single-stranded RNA rather than DNA. Although noroviruses are not new, the extent of norovirus infection was not recognized until the 1990s. This has led to increased research on noroviruses and more monitoring of outbreaks.

Until 2004 noroviruses were commonly referred to as:

- Norwalk virus
- Norwalk-like viruses (NLVs)
- caliciviruses
- small, round-structured viruses (SRSVs).

Noroviruses are named after the original strain—the Norwalk virus—that caused an outbreak of gastroenteritis in a Norwalk, Ohio, school in 1968. The virus was identified in 1972. Since then many related viruses have been identified. In 2004 these viruses were grouped together in the genus *Norovirus* within the *Caliciviridae* family of viruses. Eight to ten distinct genogroups of norovirus have been found in various parts of the world. The most common genogroups are GI, GII, GIII, and GIV. Each of these groups can be further differentiated into at least 20 genetic clusters. Evidence suggests that noroviruses in different genetic clusters can recombine to form new, genetically-distinct noroviruses. GII strains, especially GII4, are the most prevalent. However the most common method of identifying noroviruses—the reverse transcription-polymerase chain reaction (RT-PCR)—may not always identify GII genetic clusters correctly.

The increased number of norovirus outbreaks in European countries in 2002—occurring in the spring and summer rather than in winter—were found to be associated with the emergence of a new variant of the GII4 strain. Increased international outbreaks in 2003 and 2004 also were caused by a GII4-related norovirus that was found to mutate rapidly. Mutations in the viral capsid—the virus's outer protective layer—were used to determine the predominant routes of norovirus transmission.

Symptoms

Symptoms of norovirus infection usually appear within 24–48 hours after exposure, with a median incubation period during outbreaks of 33–36 hours. However symptoms can occur as early as 12 hours or less after exposure.

Typical symptoms of norovirus infection are:

- nausea
- vomiting
- watery diarrhea without blood
- abdominal cramping.

Among children, vomiting is the predominant symptom, whereas diarrhea is more common in adults. Vomiting can be frequent and violent and may occur without warning.

Additional symptoms of norovirus infection may include:

- low-grade fever
- chills
- headache

- muscle aches
- fatigue.

Dehydration is the major risk from gastroenteritis caused by norovirus, particularly among infants, young children, the elderly, and those with underlying health conditions. Symptoms of dehydration include:

- dry mouth
- increased or excessive thirst
- low urine output
- nausea
- dizziness or faintness
- sunken eyes
- sunken fontanelle—the soft spot on an infant's head
- confusion.

As many as 30–50% of norovirus infections do not produce symptoms. It is not known whether individuals with asymptomatic infections can transmit the virus.

Diagnosis

Identifying noroviruses

Viral gastroenteritis usually is diagnosed on the basis of the symptoms. Many types of viruses cause gastroenteritis. Rotoviruses are a leading cause of gastroenteritis in children who then transmit the virus to adults. In addition to noroviruses, viral gastroenteritis in humans can be caused by another genus of viruses within the *Caliciviridae* family. Formerly known as the Sapporo-like virus, or classic or typical calicivirus, these now are grouped in the genus *Sapovirus*. Other genera in the *Caliciviridae* family are not pathogenic in humans. Some bacteria and parasites also cause illnesses that are similar to norovirus infection.

The cloning and sequencing of noroviruses in the early 1990s made it easier to identify norovirus outbreaks. RT-PCR is the most commonly used method for identifying norovirus. With this technique the virus's RNA is used as the template for transcribing the corresponding DNA using the enzyme reverse transcriptase. The DNA is amplified into many copies using the polymerase chain reaction. Many state public health laboratories use this method to detect norovirus in vomit and stools. The best identification usually comes from stool samples taken within 48–72 hours after the onset of symptoms; however norovirus can be detected in stool samples taken five days after

the onset of symptoms and sometimes even in samples taken up to two weeks after recovery.

Norovirus from fecal samples can be visualized using electron microscopy. With immune electron microscopy (IEM), antibodies against norovirus are collected from blood serum and used to trap and visualize the virus from fecal samples. However these methods require high concentrations of norovirus in the stool, as well as a fourfold increase in norovirus-specific antibodies in blood samples taken during the acute or recovery phases of gastroenteritis.

Enzyme-linked immunosorbent assays may be used to detect noroviruses in fecal samples. In these assays norovirus-specific antibodies bound to the virus are detected by the reaction of an enzyme that is attached to the antibody. Nucleic acid probes that hybridize with noroviral RNA also can be used for virus detection in feces.

As of 2005 a Japanese chemical company was producing a reagent kit that can be used to detect norovirus in two hours rather than the 12–24 hours needed for conventional detection. Other simpler methods for rapidly identifying norovirus are under development.

Investigating outbreaks

Epidemiological studies often involve sequencing the norovirus RNA. This can help to determine whether outbreaks in different geographical locations are connected to each other and can help trace the source of the norovirus to contaminated food or water. CaliciNet is a database that stores the RNA sequences of all norovirus strains that cause gastroenteritis in the United States.

Criteria that are sometimes used to determine whether an outbreak of gastroenteritis is caused by a norovirus include:

- a mean incubation period of 24–48 hours
- a mean duration time for illness of 12–60 hours
- vomiting in more than 50% of patients
- failure to find a bacterial cause for the illness.

During investigations of norovirus outbreaks, food handlers may be asked to provide a stool sample and possibly a blood sample. Food rarely is tested for norovirus since each type of food requires a specific assay. However tests are used to detect the virus in shellfish. When large amounts—1–26 gal. (5–100 L)—of water are processed through specially designed filters, the norovirus can be concentrated and assayed by RT-PCR.

Treatment

Gastroenteritis caused by noroviruses usually resolves itself without treatment within a very few days. As of 2005 there are no medications or vaccines that are effective against the norovirus. Viruses are not affected by **antibiotics** and antidiarrheal medications may prolong the infection.

Norovirus infections should be treated by:

- drinking plenty of fluids, such as water and juice, to prevent dehydration caused by vomiting and diarrhea
- intravenous fluids if severe nausea prevents drinking, particularly in small children
- drinking oral rehydration fluids (ORFs) to prevent dehydration and to replace electrolytes (salt and **minerals**) and glucose
- avoiding alcohol and **caffeine** which can increase urination.

Commercially available ORFs include Naturalyte, Pedialyte, Infalyte, and Rehydralyte.

Juice, soda, and water do not replace lost electrolytes; nor do sports drinks replace nutrients and minerals lost through vomiting and diarrhea. Those taking **diuretics** should ask their healthcare provider whether to stop taking the medication during acute diarrhea.

Since the risk of dehydration is higher for infants and young children, the number of wet diapers per day should be closely monitored. Severely dehydrated children may receive rapid **intravenous rehydration** in a hospital or emergency-room setting.

A health care provider should be consulted if:

- symptoms of dehydration appear
- diarrhea persists for longer than a few days
- there is blood in the stool.

Alternative treatment

An infusion of meadowsweet (*Filipendula ulmaria*) may reduce nausea. Once the symptoms are reduced, slippery elm (*Ulmus fulva*) may calm the digestive system. Castor oil packs placed on the abdomen can reduce inflammation and discomfort.

Homeopathic remedies for gastroenteritis include *Arsenicum album*, **ippecac**, and *Nux vomica*. Chinese patent herbal remedies include Po Chai and Pill Curing.

During recovery from viral gastroenteritis, live cultures of *Lactobacillus acidophilus*, found in

live-culture yogurt or as powder or capsules, may be useful for restoring the native flora of the digestive tract.

Prognosis

Norovirus infection is followed by complete recovery and there are no known long-term health effects. Infected persons do not become long-term carriers of the virus. Dehydration is the most serious possible consequence of noroviral infection and can be fatal, particularly among older people with debilitating medical conditions or impaired immune systems.

Prevention

Noroviruses are difficult to destroy. They can survive freezing as well as temperatures as high as 140° F (60° C). Noroviruses can survive chlorine levels as high as 10 parts per million (ppm), far higher than the levels present in most public water systems. A 2004 study from the Netherlands found that inactivation of norovirus with 70% ethanol was inefficient and that sodium hypochlorite solutions were effective only at concentrations above 300 ppm.

The best prevention against noroviral infection is frequent, thorough hand washing with soap and water. All soaped hand surfaces should be rubbed vigorously for at least 10 seconds. The hands should be thoroughly rinsed under a stream of water. In particular hands always should be washed before handling food and after using the toilet or changing diapers.

Other important measures for preventing norovirus infection include:

- proper handling of cold foods
- careful washing of fruits and vegetables
- steaming oysters before eating, although even this may be insufficient for destroying norovirus
- taking particular care when handling the diapers of children with diarrhea
- properly disposing of sewage and diapers
- excluding sick infants and children from food preparation areas.

To prevent further transmission of norovirus:

- All surfaces exposed to vomit or otherwise contaminated should be immediately cleaned and disinfected with a solution of 10% bleach, followed by rinsing.
- Contaminated clothing and linens should be removed immediately and washed with hot water and detergent on the maximum machine cycle and

with a minimum of handling, followed by machine drying.

- Vomit and feces should be discarded or flushed immediately and the toilet area should be kept clean.
- Exposed or contaminated food should be discarded.
- Masks may be worn while cleaning areas that have been badly contaminated with vomit or feces, such as in hospitals or nursing homes.

A 2004 study found that detergent-based cleaning with a cloth consistently failed to eliminate norovirus contamination. With fecal contamination, detergent-based cleaning, followed by cleaning with a combination hypochlorite/detergent formula containing 5000 ppm of available chlorine significantly reduced contamination. However norovirus still could be detected on as much as 28% of the surfaces. When this procedure failed to eliminate contamination, the virus was transmitted to the cleaner's hands. Contaminated fingers consistently transferred norovirus to up to seven different surfaces including doorknobs and telephones. However the contamination was diluted during secondary transmission and treatment with the combined bleach/detergent eliminated the virus without prior cleaning.

In situations where there is a periodic renewal of susceptible people, such as on cruise ships and at camps, the facility may have to be closed until cleaning is complete. Although many state and local health departments require that food handlers with gastroenteritis not return to work until 2–3 days following recovery, this may not be an adequate length of time to prevent noroviral transmission.

The prevention of norovirus outbreaks include reducing contamination of water supplies with human waste and using high-level chlorination—at least 10 ppm for more than 30 min. Surveillance of shorelines for potential sources of fecal contamination and for boats that are dumping human waste may help prevent shellfish-associated norovirus outbreaks.

In 2004 researchers at Washington University announced that they had succeeded in growing a mouse norovirus in the laboratory for the first time, with the goal of studying the virus and developing a vaccine against it. New surveillance systems also are being developed to detect norovirus outbreaks at an early stage.

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Antibody—A blood protein produced in response to foreign material such as a virus; the antibody attaches to the virus and destroys it.

Calicivirus—A member of the *Caliciviridae* family of viruses that includes noroviruses.

Capsid—The outer protein coat of a virus.

Gastroenteritis—An inflammation of the lining of the stomach and intestines, usually caused by a viral or bacterial infection.

Genetic cluster—A group of viral strains with very similar, yet distinct, nucleic acid sequences.

Genogroup—Related viruses within a genus; may be further subdivided into genetic clusters.

Reverse transcription-polymerase chain reaction; RT-PCR—A method of polymerase-chain-reaction amplification of nucleic acid sequences that uses RNA as the template for transcribing the corresponding DNA using reverse transcriptase.

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DC 20013-1133. healthfinder@nhic.org. <http://www.healthfinder.gov/>.

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Norplant see **Depo-Provera/Norplant**

Norwalk virus infection see **Gastroenteritis**

Nose injuries see **Nasal trauma**

Nose irrigation see **Nasal irrigation**

Nose job see **Rhinoplasty**

Nose packing see **Nasal packing**

Nose papillomas see **Nasal papillomas**

Nose polyps see **Nasal polyps**

Nosebleed

Definition

A nosebleed is bleeding from the nose; the medical term for it is epistaxis.

Description

Unexpected bleeding from anywhere is cause for alarm. Persistent bleeding should always be investigated because it may be the earliest sign of **cancer**. Fortunately, nosebleeds are rarely a sign of cancer. A much more common cause of nosebleeds is injury from picking or blowing or fisticuffs. People with

may **fever** have swollen membranes that are fragile and more likely to bleed.

Most nosebleeds (about 90%) come from the front of the septum, that plane of cartilage that separates the nostrils. These are called anterior nosebleeds. The lower front part of the septum has a mass of blood vessels on either side called Kiesselbach's plexus that is easy to injure. Nosebleeds from the more remote reaches of the nose are called posterior nosebleeds. They are less common, are less likely to have a benign cause, and are much harder to manage.

Nosebleeds are most likely to occur in children between the ages of two and 10 years, in part because younger children frequently insert small objects in the nose or pick at the tissues lining the nose. Nosebleeds in adolescents may indicate **cocaine abuse**. Nosebleeds in older adults may result from arteriosclerosis or high blood pressure.

Causes and symptoms

Nosebleeds may result from a number of different causes:

- local infections (colds, sinus infections)
- systemic infections (**scarlet fever**, **typhoid fever**, malaria)
- drying of the membranes lining the nose, often during heating season in colder climates
- medications, most commonly, overuse of nasal decongestant sprays
- trauma (from **foreign objects** in the nose; scratching or picking with the fingers; or blunt trauma to the face)
- tumors in the nasopharynx or paranasal sinuses
- cocaine abuse
- bleeding disorders (leukemia, **liver disease**, **hemophilia** and other hereditary clotting disorders)

Treatment

The first treatment is to pinch the patient's nostrils together, have them sit forward, and ask them to stay that way for 5–10 minutes. This method usually stops nosebleeds originating in Kiesselbach's plexus. The patient should not tilt his or her head backward, as this position may cause blood to drip backward into the throat or windpipe. It is best to hold the head upright.

In the case of small children, the doctor may examine the inside of the nose to check for foreign

bodies, evidence of scratching or picking, etc. Small foreign bodies (watch batteries, dried peas or beans, buttons, etc.) can be removed by suction if necessary. The doctor may also have to remove clotted blood by suction.

Bleeding that continues originates from the back of the nose in most cases and will flow down the throat. If that happens, emergency intervention is needed.

As an emergency procedure, the nose will be packed front and/or back with cotton gauze and a rubber balloon from a Foley catheter. This treatment is not comfortable. Having no place to flow, the blood should clot, giving the ear, nose and throat specialists (otorhinolaryngologists) a chance to find the source and permanently repair it. If the packing has to remain for any length of time, **antibiotics** and **pain** medication will be necessary—antibiotics because the sinuses will be plugged up and prone to infection. Nose packing may so interfere with breathing that the patient will need supplemental oxygen.

Newer options for controlling posterior nosebleeds include the use of Surgicel, Merocel, or other oxidized cellulose products that expand with moisture. These may control the bleeding without the need for bulky **nasal packing**.

Many bleeds are from small exposed blood vessels with no other disease. They can be destroyed by cautery, usually done by applying silver nitrate to the affected area. Larger vessels may not respond to cautery. The surgeon may have to tie them off, which is known as ligation. Another technique that is sometimes used with larger vessels is embolization, in which the doctor injects a chemical to block or close the blood vessel.

Alternative treatment

Estrogen cream, the same preparation used to revitalize vaginal tissue, can toughen fragile blood vessels in the anterior septum and forestall the need for cauterization. Botanical medicines known as styptics, which slow down and can stop bleeding, may be taken internally or applied topically. Some of the plants used are achillea (yarrow), trillium, geranium, and shepherd's purse (*capsella-bursa*). Homeopathic remedies can be one of the quickest and most effective treatments for epistaxis.

Prevention

Both before and after a nosebleed, the patient should blow the nose gently and avoid picking or

KEY TERMS

Cautery—The use of heat, electricity, or chemicals to destroy tissue.

Embolization—A technique for stopping bleeding by introducing a substance into larger blood vessels that blocks or closes them.

Epistaxis—The medical term for nosebleed.

Kiesselbach's plexus—An area on the anterior part of the nasal septum that has a rich supply of blood vessels and is a common site of nosebleeds. It is named for Wilhelm Kiesselbach, a nineteenth-century German otolaryngologist.

Septum—The partition that separates the two nostrils. It consists of membranes, cartilage, and bone.

Styptic—Any remedy with an astringent and hemostatic (stopping bleeding) quality.

scratching the tissues that line it. Children with recurrent nosebleeds during heating season may benefit from the use of a cool-mist vaporizer to humidify the bedroom at night, or from the application of a small quantity of petroleum jelly to the inside of each nostril. Treatment of hay fever helps reduce the fragility of the tissues.

Resources

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American Academy of Family Physicians (AAFP). 11400 Tomahawk Creek Parkway, Leawood, KS 66211-2672. (800) 274-2237 or (913) 906-6000. <<http://www.aafp.org>> .

American Academy of Otolaryngology—Head and Neck Surgery. One Prince Street, Alexandria, VA 22314-3357. (703) 836-4444. <<http://www.entnet.org>> .

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Nosocomial infections see **Hospital-acquired infections**

NS see **Nephrotic syndrome**

NSAIDs see **Nonsteroidal anti-inflammatory drugs**

Nther's disease see **Porphyrias**

Nuclear magnetic resonance see **Magnetic resonance imaging**

Nucleoside analogs see **Antiretroviral drugs**

Numbness and tingling

Definition

Numbness and tingling are decreased or abnormal sensations caused by altered sensory nerve function.

Description

The feeling of having a foot "fall asleep" is a familiar one. This same combination of numbness and tingling can occur in any region of the body and may be caused by a wide variety of disorders. Sensations such as these, which occur without any associated stimulus, are called paresthesias. Other types of paresthesias include feelings of cold, warmth, burning, **itching**, and skin crawling.

Causes and symptoms

Causes

Sensation is carried to the brain by neurons (nerve cells) running from the outer parts of the body to the spinal cord in bundles called nerves. In the spinal cord, these neurons make connections with other neurons that run up to the brain. Paresthesias are caused by disturbances in the function of neurons in the sensory pathway. This disturbance can occur in the central nervous system (the brain and spinal cord), the nerve roots that are attached to the spinal cord, or the

peripheral nervous system (nerves outside the brain and spinal cord).

Peripheral disturbances are the most common cause of paresthesias. “Falling asleep” occurs when the blood supply to a nerve is cut off—a condition called **ischemia**. Ischemia usually occurs when an artery is compressed as it passes through a tightly flexed joint. Sleeping with the arms above the head or sitting with the legs tightly crossed frequently cause numbness and tingling.

Direct compression of the nerve also causes paresthesias. Compression can be short-lived, as when a heavy backpack compresses the nerves passing across the shoulders. Compression may also be chronic. Chronic nerve compression occurs in entrapment syndromes. The most common example is **carpal tunnel syndrome**. Carpal tunnel syndrome occurs when the median nerve is compressed as it passes through a narrow channel in the wrist. Repetitive motion or prolonged vibration can cause the lining of the channel to swell and press on the nerve. Chronic nerve root compression, or radiculopathy, can occur in disk disease or spinal arthritis.

Other causes of paresthesias related to disorders of the peripheral nerves include:

- Metabolic or nutritional disturbances. These disturbances include diabetes, **hypothyroidism** (a condition caused by too little activity of the thyroid gland), alcoholism, **malnutrition**, and vitamin B₁₂ deficiency.
- Trauma. Trauma includes injuries that crush, sever, or pull on nerves.
- Inflammation.
- Connective tissue disease. These diseases include arthritis, systemic lupus erythematosus (a chronic inflammatory disease that affects many systems of the body, including the nervous system), polyarteritis nodosa (a vascular disease that causes widespread inflammation and ischemia of small and medium-size arteries), and **Sjögren’s syndrome** (a disorder marked by insufficient moisture in the tear ducts, salivary glands, and other glands).
- Toxins. Toxins include heavy metals (metallic elements such as arsenic, lead, and mercury which can, in large amounts, cause **poisoning**), certain **antibiotics** and **chemotherapy** agents, solvents, and overdose of pyridoxine (vitamin B₆).
- Malignancy.
- Infections. Infections include **Lyme disease**, human immunodeficiency virus (HIV), and leprosy.
- Hereditary disease. These diseases include Charcot-Marie-Tooth disease (a hereditary disorder that causes wasting of the leg muscles, resulting in malformation of the foot), porphyria (a group of inherited disorders in which there is abnormally increased production of substances called porphyrins), and Denny-Brown’s syndrome (a hereditary disorder of the nerve root).

Paresthesias can also be caused by central nervous system disturbances, including stroke, TIA (transient ischemic attack), tumor, trauma, **multiple sclerosis**, or infection.

Symptoms

Sensory nerves supply or innervate particular regions of the body. Determining the distribution of symptoms is an important way to identify the nerves involved. For instance, the median nerve innervates the thumb, the first two fingers, half of the ring finger, and the part of the hand to which they connect. The ulnar nerve innervates the other half of the ring finger, the little finger, and the remainder of the hand. Distribution of symptoms may also aid diagnosis of the underlying disease. Diabetes usually causes a symmetrical “glove and stocking” distribution in the hands and feet. Multiple sclerosis may cause symptoms in several, widely separated areas.

Other symptoms may accompany paresthesias, depending on the type and severity of the nerve disturbance. For instance, weakness may accompany damage to nerves that carry both sensory and motor neurons. (Motor neurons are those that carry messages outward from the brain.)

Diagnosis

A careful history of the patient is needed for a diagnosis of paresthesias. The medical history should focus on the onset, duration, and location of symptoms. The history may also reveal current related medical problems and recent or past exposure to drugs, toxins, infection, or trauma. The family medical history may suggest a familial disorder. A work history may reveal repetitive motion, chronic vibration, or industrial chemical exposure.

The physical and neurological examination tests for distribution of symptoms and alterations in reflexes, sensation, or strength. The distribution of symptoms may be mapped by successive stimulation over the affected area of the body.

Lab tests for paresthesia may include blood tests and **urinalysis** to detect metabolic or nutritional

abnormalities. Other tests are used to look for specific suspected causes. Nerve conduction velocity tests, **electromyography**, and imaging studies of the affected area may be employed. Nerve biopsy may be indicated in selected cases.

Treatment

Treatment of paresthesias depends on the underlying cause. For limbs that have “fallen asleep,” restoring circulation by stretching, exercising, or massaging the affected limb can quickly dissipate the numbness and tingling. If the paresthesia is caused by a chronic disease such as diabetes or occurs as a complication of treatments such as chemotherapy, most treatments are aimed at relieving symptoms. Anti-inflammatory drugs such as **aspirin** or ibuprofen are recommended if symptoms are mild. In more difficult cases, **antidepressant drugs** such as amitriptyline (Elavil) are sometimes prescribed. These drugs are given at a much lower dosage for this purpose than for relief of depression. They are thought to help because they alter the body’s perception of pain. In severe cases, opium derivatives such as codeine can be prescribed. Currently trials are being done to determine whether treatment with human nerve growth factor will be effective in regenerating the damaged nerves.

Alternative treatment

Several alternative treatments are available to help relieve symptoms of paresthesia. Nutritional therapy includes supplementation with **B complex vitamins**, especially vitamin B₁₂ (intramuscular injection of vitamin B₁₂ is most effective). Vitamin supplements should be used cautiously however. Overdose of Vitamin B₆ is one of the causes of paresthesias. People experiencing paresthesia should also avoid alcohol. **Acupuncture** and massage are said to relieve symptoms. Self-massage with aromatic oils is sometimes helpful. The application of topical ointments containing capsaicin, the substance that makes hot peppers hot, provides relief for some. It may also be helpful to wear loosely fitting shoes and clothing. None of these alternatives should be used in place of traditional therapy for the underlying condition.

Prognosis

Treating the underlying disorder may reduce the occurrence of paresthesias. Paresthesias resulting from damaged nerves may persist throughout or even beyond the recovery period. The overall prognosis depends on the cause.

KEY TERMS

Electromyography—A test that uses electrodes to record the electrical activity of muscle. The information gathered is used to diagnose neuromuscular disorders.

Motor nerve—Motor or efferent nerve cells carry impulses from the brain to muscle or organ tissue.

Nerve conduction velocity test—A test that measures the time it takes a nerve impulse to travel a specific distance over the nerve after electronic stimulation.

Nerve growth factor—A protein resembling insulin that affects growth and maintenance of nerve cells

Peripheral nervous system—The part of the nervous system that is outside the brain and spinal cord. Sensory, motor, and autonomic nerves are included.

Sensory nerves—Sensory or afferent nerves carry impulses of sensation from the periphery or outward parts of the body to the brain. Sensations include feelings, impressions, and awareness of the state of the body.

Prevention

Preventing the underlying disorder may reduce the incidence of paresthesias. For those with frequent paresthesias caused by ischemia, changes in posture may help.

Resources

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Richard Robinson

Nummular dermatitis see **Dermatitis**

Nutrition

Definition

Good nutrition can help prevent disease and promote health. There are six categories of nutrients that

the body needs to acquire from food: protein, carbohydrates, fat, fibers, **vitamins** and **minerals**, and water.

Proteins

Protein supplies amino acids to build and maintain healthy body tissue. There are 20 amino acids considered essential because the body must have all of them in the right amounts to function properly. Twelve of these are manufactured in the body but the other eight amino acids must be provided by the diet. Foods from animal sources such as milk or eggs often contain all these essential amino acids while a variety of plant products must be taken together to provide all these necessary protein components.

Fat

Fat supplies energy and transports nutrients. There are two families of fatty acids considered essential for the body: the omega-3 and omega-6 fatty acids. Essential fatty acids are required by the body to function normally. They can be obtained from canola oil, flaxseed oil, cold-water fish, or fish oil, all of which contain **omega-3 fatty acids**, and primrose or black currant seed oil, which contains omega-6 fatty acids. The American diet often contains an excess of omega-6 fatty acids and insufficient amounts of omega-3 fats. Increased consumption of omega-3 oils is recommended to help reduce risk of cardiovascular diseases and **cancer** and alleviate symptoms of rheumatoid arthritis, **premenstrual syndrome**, **dermatitis**, and inflammatory bowel disease.

Carbohydrates

Carbohydrates are the body's main source of energy and should be the major part of total daily intake. There are two types of carbohydrates: simple carbohydrates (such as sugar or honey) or complex carbohydrates (such as grains, beans, peas, or potatoes). Complex carbohydrates are preferred because these foods are more nutritious yet have fewer calories per gram compared to fat and cause fewer problems with overeating than fat or sugar. Complex carbohydrates also are preferred over simple carbohydrates by diabetics because they allow better blood glucose control.

Fiber

Fiber is the material that gives plants texture and support. Although it is primarily made up of carbohydrates, it does not have a lot of calories and is usually not broken down by the body for energy. Dietary fiber

is found in plant foods such as fruits, vegetables, legumes, nuts, and whole grains.

There are two types of fiber: soluble and insoluble. Insoluble fiber, as the name implies, does not dissolve in water because it contains high amount of cellulose. Insoluble fiber can be found in the bran of grains, the pulp of fruit and the skin of vegetables. Soluble fiber is the type of fiber that dissolves in water. It can be found in a variety of fruits and vegetables such as apples, oatmeal and oat bran, rye flour, and dried beans.

Although they share some common characteristics such as being partially digested in the stomach and intestines and have few calories, each type of fiber has its own specific health benefits. Insoluble fiber speeds up the transit of foods through the digestive system and adds bulk to the stools, therefore, it is the type of fiber that helps treat **constipation** or **diarrhea** and prevents colon cancer. On the other hand, only soluble fiber can lower blood cholesterol levels. This type of fiber works by attaching itself to the cholesterol so that it can be eliminated from the body. This prevents cholesterol from recirculating and being reabsorbed into the bloodstream. In 2003, the World Health Organization released a new report specifically outlining the link of a healthy diet rich in high-fiber plant foods to preventing cancer.

Vitamins and minerals

Vitamins are organic substances present in food and required by the body in a small amount for regulation of metabolism and maintenance of normal growth and functioning. The most commonly known vitamins are A, B₁ (thiamine), B₂ (riboflavin), B₃ (niacin), B₅ (pantothenic acid), B₆ (pyridoxine), B₇ (biotin), B₉ (folic acid), B₁₂ (cobalamin), C (ascorbic acid), D, E, and K. The B and C vitamins are water-soluble, excess amounts of which are excreted in the urine. The A, D, E, and K vitamins are fat-soluble and will be stored in the body fat.

Minerals are vital to our existence because they are the building blocks that make up muscles, tissues, and bones. They also are important components of many life-supporting systems, such as hormones, oxygen transport, and enzyme systems.

There are two kinds of minerals: the major (or macro) minerals and the trace minerals. Major minerals are the minerals that the body needs in large amounts. The following minerals are classified as major: calcium, phosphorus, magnesium, sodium, potassium, sulfur, and chloride. They are needed to build muscles, blood, nerve cells, teeth, and bones. They also are essential electrolytes that the body

requires to regulate blood volume and acid-base balance.

Unlike the major minerals, trace minerals are needed only in tiny amounts. Even though they can be found in the body in exceedingly small amounts, they are also very important to the human body. These minerals participate in most chemical reactions in the body. They also are needed to manufacture important hormones. The following are classified as trace minerals: iron, zinc, iodine, copper, manganese, fluoride, chromium, selenium, molybdenum, and boron.

Many vitamins (such as vitamins A, C, and E) and minerals (such as zinc, copper, selenium, or manganese) act as antioxidants. They protect the body against the damaging effects of free radicals. They scavenge or mop up these highly reactive radicals and change them into inactive, less harmful compounds. In so doing, these essential nutrients help prevent cancer and many other degenerative diseases, such as premature **aging**, heart disease, autoimmune diseases, arthritis, **cataracts**, **Alzheimer's disease**, and diabetes mellitus.

Water

Water helps to regulate body temperature, transports nutrients to cells, and rids the body of waste materials.

Origins

Unlike plants, human beings cannot manufacture most of the nutrients that they need to function. They must eat plants and/or other animals. Although nutritional therapy came to the forefront of the public's awareness in the late twentieth century, the notion that food affects health is not new. John Harvey Kellogg was an early health-food pioneer and an advocate of a high-fiber diet. An avowed vegetarian, he believed that meat products were particularly detrimental to the colon. In the 1870s, Kellogg founded the Battle Creek Sanitarium, where he developed a diet based on nut and vegetable products.

Purpose

Good nutrition helps individuals achieve general health and well-being. In addition, dietary modifications might be prescribed for a variety of complaints including **allergies**, anemia, arthritis, colds, depressions, **fatigue**, gastrointestinal disorders, high or low blood pressure, **insomnia**, headaches, **obesity**, **pregnancy**, premenstrual syndrome (PMS), respiratory conditions, and stress.

Nutritional therapy may also be involved as a complement to the allopathic treatments of cancer, diabetes, and Parkinson's disease. Other specific dietary measures include the elimination of food additives for attention deficit hyperactivity disorder (ADHD), gluten-free **diets** for schizophrenia, and dairy-free for chronic respiratory diseases.

A high-fiber diet helps prevent or treat the following health conditions:

- **High cholesterol levels.** Fiber effectively lowers blood cholesterol levels. It appears that soluble fiber binds to cholesterol and moves it down the digestive tract so that it can be excreted from the body. This prevents the cholesterol from being reabsorbed into the bloodstream.
- **Constipation.** A high-fiber diet is the preferred non-drug treatment for constipation. Fiber in the diet adds more bulk to the stools, making them softer and shortening the time foods stay in the digestive tract.
- **Hemorrhoids.** Fiber in the diet adds more bulk and softens the stool, thus, reducing painful hemorrhoidal symptoms.
- **Diabetes.** Soluble fiber in the diet slows down the rise of blood sugar levels following a meal and helps control diabetes.
- **Obesity.** Dietary fiber makes a person feel full faster.
- **Cancer.** Insoluble fiber in the diet speeds up the movement of the stools through the gastrointestinal tract. The faster food travels through the digestive tract, the less time there is for potential cancer-causing substances to work. Therefore, diets high in insoluble fiber help prevent the accumulation of toxic substances that cause cancer of the colon. Because fiber reduces fat absorption in the digestive tract, it also may prevent breast cancer.

A diet low in fat also promotes good health and prevents many diseases. Low-fat diets can help treat or control the following conditions:

- **Obesity.** High fat consumption often leads to excess caloric and fat intake, which increases body fat.
- **Coronary artery disease.** High consumption of saturated fats is associated with coronary artery disease.
- **Diabetes.** People who are overweight tend to develop or worsen existing diabetic conditions due to decreased insulin sensitivity.
- **Breast cancer.** A high dietary consumption of fat is associated with an increased risk of breast cancer.

Description

The four basic food groups, as outlined by the United States Department of Agriculture (USDA) are:

- dairy products (such as milk and cheese)
- meat and eggs (such as fish, poultry, pork, beef, and eggs)
- grains (such as bread cereals, rice, and pasta)
- fruits and vegetables

The USDA recommendation for adults is that consumption of meat, eggs, and dairy products should not exceed 20% of total daily caloric intake. The rest (80%) should be devoted to vegetables, fruits, and grains. For children age two or older, 55% of their caloric intake should be in the form of carbohydrates, 30% from fat, and 15% from proteins. In addition, saturated fat intake should not exceed 10% of total caloric intake. This low-fat, high-fiber diet is believed to promote health and help prevent many diseases, including heart disease, obesity, and cancer.

Allergenic and highly processed foods should be avoided. Highly processed foods do not contain significant amounts of essential trace minerals. Furthermore, they contain lots of fat and sugar as well as preservatives, artificial sweeteners and other additives. High consumption of these foods causes build up of unwanted chemicals in the body and should be avoided. Food allergies causes a variety of symptoms including food cravings, weight gain, bloating, and water retention. They also may worsen chronic inflammatory conditions such as arthritis.

Preparations

An enormous body of research exists in the field of nutrition. Mainstream Western medical practitioners point to studies that show that a balanced diet, based on the USDA Food Guide Pyramid, provides all of the necessary nutrients.

In 2004, the USDA was working on a revision of the Food Guide Pyramid to reflect changes in American lifestyle habits. The new eating guide was due for release in January 2005. The World Health Organization (WHO) also was weighing in on the obesity and nutrition issue, even struggling with objections from member nations that supply goods such as sugar, to endorse a global strategy in spring 2004 on diet, physical activity and health.

The Food Guide Pyramid recommends the following daily servings in six categories:

- grains: six or more servings
- vegetables: five servings
- fruits: two to four servings
- meat: two to three servings
- dairy: two to three servings
- fats and oils: use sparingly

Precautions

Individuals should not change their diets without the advice of nutritional experts or health care professionals. Certain individuals, especially children, pregnant and lactating women, and chronically ill patients, only should change their diets under professional supervision.

Side effects

It is best to obtain vitamins and minerals through food sources. Excessive intake of vitamins and mineral supplements can cause serious health problems. Likewise, eating too much of one type of food, as can happen with fad diets, can be harmful. The key to nutrition is moderation. If a person feels they are short on iron, for example, he or she should not go too far to the extreme in getting more iron through diet and supplements. A 2003 report said that too much stored iron in the body has possibly been linked with heart disease, cancer and diabetes.

The following is a list of possible side effects resulting from excessive doses of vitamins and minerals:

- vitamin A: **birth defects**, irreversible bone and liver damage
- vitamin B₁: deficiencies in B₂ and B₆
- vitamin B₆: damage to the nervous system
- vitamin C: affects the absorption of copper; diarrhea
- vitamin D: **hypercalcemia** (abnormally high concentration of calcium in the blood)
- phosphorus: affects the absorption of calcium
- zinc: affects absorption of copper and iron; suppresses the immune system

Research and general acceptance

Due to a large volume of scientific evidence demonstrating the benefits of the low-fat, high-fiber diet in disease prevention and treatment, these recommendations have been accepted and advocated by both complementary and allopathic practitioners.

Resources

BOOKS

U.S. Preventive Services Task Force Guidelines. "Counseling to Promote a Healthy Diet." *Guide to Clinical Preventive Services*. 2nd ed. <<http://cpmcnet.columbia.edu/texts/gcps/gcps0066.html>>.

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ORGANIZATIONS

American Association of Nutritional Consultants. 810 S. Buffalo Street, Warsaw, IN 46580. (888) 828-2262.

American Dietetic Association. 216 W. Jackson boulevard, Suite 800, Chicago, IL 60606-6995. (800) 366-1655. <<http://www.eatright.org>>.

Mai Tran
Teresa G. Odle

Nutrition through an intravenous line

Definition

Sterile solutions containing some or all of the nutrients necessary to support life, are injected into the body through a tube attached to a needle, which is inserted into a vein, either temporarily or for long-term treatment.

Purpose

Patients who cannot consume enough nutrients or who cannot eat at all due to an illness, surgery, or accident, can be fed through an intravenous (IV) line or tube. An IV can be used for as little as a few hours, to provide fluids to a patient during a short surgical procedure, or to rehydrate a patient after a viral illness.

Patients with more serious and long term illnesses and conditions may require months or even years of intravenous therapy to meet their nutritional needs. These patients may require a central **venous access** port. A specialized catheter (Silastic Broviac or Hickman) is inserted beneath the skin and positioned below the collarbone. Fluids can then be injected directly into the bloodstream for long periods of time. X rays are taken to ensure that the permanent catheter is properly positioned.

Precautions

Patients receiving IV therapy need to be monitored to ensure that the IV solutions are providing the correct amounts of fluids, **minerals**, and other nutrients needed.

Description

There are two types of IV, or parenteral, **nutrition**. Parenteral nutrition is that which is delivered through a system other than the digestive system. In this case, the nutrition is delivered through a vein. Partial parenteral nutrition (PPN) is given for short periods of time, to replace some of the nutrients required daily and only supplements a normal diet. **Total parenteral nutrition** (TPN) is given to someone who cannot eat anything and must receive all nutrients required daily through an intravenous line. Both of these types of nutrition can be performed in a medical facility or at the patient's home. Home parenteral nutrition (HPN) usually requires a central venous catheter, which must first be inserted in a fully equipped medical facility. After it is inserted, therapy can continue at home.

Basic IV solutions are sterile water with small amounts of sodium (salt) or dextrose (sugar) supplied in bottles or thick plastic bags that can hang on a stand mounted next to the patient's bed. Additional minerals, like potassium and calcium, **vitamins**, or drugs can be added to the IV solution by injecting them into the bottle or bag with a needle. These simple sugar and salt solutions can provide fluids, calories, and electrolytes necessary for short periods of time. If a patient requires intravenous feeding for more than a few days, additional nutrients like proteins and fats will be included. The amounts of each of the nutrients to be added will depend on the patient's age, medical condition, and particular nutritional requirements.

Preparation

A doctor orders the IV solution and any additional nutrients or drugs to be added to it. The doctor

KEY TERMS

Home parenteral nutrition (HPN)—Long-term parenteral nutrition, given through a central venous catheter and administered in the patient's home.

Intravenous—Into a vein; a needle is inserted into a vein in the back of the hand, inside the elbow, or some other location on the body. Fluids, nutrients, and drugs can be injected.

Parenteral—Not in or through the digestive system. Parenteral nutrition is given through the veins of the circulatory system, rather than through the digestive system.

Partial parenteral nutrition (PPN)—A solution, containing some essential nutrients, is injected into a vein to supplement other means of nutrition, usually a partially normal diet of food.

Total parenteral nutrition (TPN)—A solution containing all the required nutrients including protein, fat, calories, vitamins, and minerals, is injected over the course of several hours, into a vein. TPN provides a complete and balanced source of nutrients for patients who cannot consume a normal diet.

also specifies the rate at which the IV will be infused. The IV solutions are prepared under the supervision of a doctor, pharmacist, or nurse, using sanitary techniques that prevent bacterial contamination. Just like a prescription, the IV is clearly labeled to show its contents and the amounts of any additives. The skin around the area where the needle is inserted is cleaned and sanitized. Once the needle is in place, it will be taped to the skin to prevent it from dislodging.

In the case of HPN, the IV solution is delivered to the patient's home on a regular basis and should be kept refrigerated. Each bag will have an expiration date, by which time the bag should be used. The solution should be allowed to be warmed to room temperature before intravenous nutrition begins.

Aftercare

Patients who have been on IV therapy for more than a few days may need to have foods reintroduced gradually to give the digestive tract time to start working again. After the IV needle is removed, the site should be inspected for any signs of bleeding or infection.

When using HPN, the catheter should be kept clean at all times. The dressings around the site should be changed at least once a week and the catheter site should be monitored closely for signs of redness, swelling, and drainage. The patient's extremities should be watched for swelling, which is a sign of nutritional imbalance.

Risks

There is a risk of infection at the injection site, and for patients on long term IV therapy, the risk of an infection spreading to the entire body is fairly high. It is possible that the IV solution may not provide all of the nutrients needed, leading to a deficiency or an imbalance. If the needle becomes dislodged, it is possible that the solution may flow into tissues around the injection site rather than into the vein. The patient should be monitored regularly, particularly if receiving HPN, as intravenous nutrition can potentially cause infection at the site of the catheter, high blood sugar, and low blood potassium, which can all be life-threatening.

Resources

OTHER

"Clinical Management: Parenteral Nutrition" In *Revised Intravenous Nursing Standards of Practice*. <<http://www.insl.org>>.

Altha Roberts Edgren

Nutritional supplements

Definition

Nutritional supplements include **vitamins**, **minerals**, herbs, meal supplements, sports **nutrition** products, natural food supplements, and other related products used to boost the nutritional content of the diet.

Purpose

Nutritional supplements are used for many purposes. They can be added to the diet to boost overall health and energy; to provide immune system support and reduce the risks of illness and age-related conditions; to improve performance in athletic and mental activities; and to support the healing process during illness and disease. However, most of these products are treated as food and not regulated as drugs are.

Description

The Natural Nutritional Foods Association estimated that in 2003 nutritional supplements amounted to a \$19.8 billion market in the United States. By category, vitamins provided \$6.6 billion in sales, herbs \$4.2 billion, meal supplements \$2.5 billion, sports nutrition products \$2.0 billion, minerals \$1.8 billion, and specialty and other products totaling \$2.7 billion. The nutritional supplement industry provides a huge array of products for consumer needs.

Vitamins

Vitamins are micronutrients, or substances that the body uses in small amounts, as compared to macronutrients, which are the proteins, fats, and carbohydrates that make up all food. Vitamins are present in food, but adequate quantities of vitamins may be reduced when food is overcooked, processed, or improperly stored. For instance, processing whole wheat grain into white flour reduces the contents of vitamins B and E, fiber, and minerals, including zinc and iron. The body requires vitamins to support its basic biochemical functions, and deficiencies over time can lead to illness and disease.

Vitamins are either water-soluble or fat-soluble. Water-soluble vitamins dissolve in water and pass through the body quickly, meaning that the body needs them on a regular basis. Water-soluble vitamins include the B-complex vitamins and vitamin C. Fat-soluble vitamins are stored in the body's fatty tissue, meaning that they remain in the body longer. Fat-soluble vitamins include vitamins A, D, E, and K.

The amount of vitamins needed by the body has been the subject of much research. The U.S. government has published recommended dietary allowances (RDAs) for each vitamin for the general population. These figures can be used as guidelines, but individuals may have different needs depending on gender, age, and health conditions.

Vitamins can be natural or synthetic. Natural vitamins are extracted from food sources, while synthetic vitamins are formulated in laboratory processes. The only vitamin for which there is a noted difference between the natural and synthetic forms is vitamin E. The natural form is labeled d-alpha-tocopherol while the synthetic form is named dl-alpha-tocopherol, with the extra "l" signifying laboratory production. Natural vitamin E has been shown to be slightly more absorbable by the body than the synthetic version, although for other vitamins no significant differences in absorption have been noted.

Minerals

Minerals are micronutrients and are essential for the proper functioning of the body. Cells in the body require minerals as part of their basic make-up and chemical balance, and minerals are present in all foods. Minerals can either be bulk minerals, used by the body in larger quantities, or trace minerals, used by the body in minute or trace amounts. Bulk minerals include sodium, potassium, calcium, magnesium, and phosphorus. Trace minerals include iron, zinc, selenium, iodine, chromium, copper, manganese, and others. Some studies have shown that the amount of minerals, particularly trace minerals, may be decreasing in foods due to mineral depletion of the soil caused by unsustainable farming practices and soil erosion. Supplemental minerals are available in chelated form, in which they are bonded to proteins in order to improve their absorption by the body.

Herbs

Herbal supplements are added to the diet for both nutritional and medicinal purposes. Herbs have been used for centuries in many traditional medicine systems, and as sources of phytochemicals, or substances found in plants that have notable effects in the body. Chinese medicine and **Ayurvedic medicine** from India, two of the world's oldest healing systems, use hundreds of herbal medications. Naturopathy and homeopathy, two other systems of natural healing, also rely on herbal preparations as their main sources of medication. The medicinal effects of herbs are getting scientific validation; about one-fourth of all pharmaceuticals have been derived directly from plant sources, including **aspirin** (found in willow bark), codeine (from poppy seeds), paclitaxel (Taxol), a patented drug for ovarian and **breast cancer** (from the Pacific Yew tree), and many others.

Herbs can supplement the diet to aid in overall health or to stimulate healing for specific conditions. For instance, ginseng is used as a general tonic to increase overall health and vitality, while **echinacea** is a popular herb used to stimulate the body's resistance to colds and infections. Herbs come in many forms. They can be purchased as capsules and tablets, as well as in tinctures, teas, syrups, and ointments.

Meal supplements

Meal supplements are used to replace or fortify meals. They may be designed for people with special needs, or for people with illnesses that may affect digestion capabilities and nutritional requirements. Meal supplements may contain specific blends of

macronutrients, or proteins, carbohydrates, fats, and fiber. Some meal supplements consist of raw, unprocessed foods, or vegetarian or vegan options, or high protein and low fat composition. Meal supplements are available to support some popular diet programs. Meal supplements are often fortified with vitamins, minerals, herbs, and nutrient-dense foods.

Sports nutrition

Nutritional supplements may be designed to provide specialized support for athletes. Some of these consist of high-protein products, such as amino acid supplements, while other products contain nutrients that support metabolism, energy, and athletic performance and recovery. People engaging in intense athletic activity may have increased needs for water-soluble vitamins, antioxidants, and certain minerals, including chromium. Sports drinks contain blends of electrolytes (salts) that the body loses during exertion and sweating, as well as vitamins, minerals, and performance-supporting herbs.

Other nutritional supplements

Other nutritional supplements include nutrient-dense food products. Examples of these are brewer's yeast, spirulina (sea algae), bee pollen and royal jelly, fish oil and essential fatty acid supplements, colostrum (a specialty dairy product), psyllium seed husks (a source of fiber), wheat germ, wheatgrass, and medicinal mushrooms such as the shiitake and reishi varieties.

Specialty products may offer particular health benefits or are targeted for specific conditions. These products may consist of whole foods or may be isolated compounds from natural or synthetic sources. Examples include antioxidants, probiotics (supplements containing friendly bacteria for the digestive tract), digestive enzymes, shark cartilage, or other animal products, or chemical extracts such as the hormone DHEA (dehydroepiandrosterone) and coenzyme Q10, an antioxidant.

General guidelines

Considering average dietary needs and the prevalence of certain health conditions, some basic guidelines may provide the foundation for the effective use of nutritional supplements. First, a high quality, broad-spectrum multivitamin and mineral supplement, taken once per day, is recommended to provide a range of nutrients. This should contain the B-complex vitamins B6, B12, and **folic acid**, which may help prevent heart disease, and the minerals zinc and

copper, which aid immunity. In addition to a multivitamin, antioxidants can be added to a supplementation routine. These include vitamin A (or beta-carotene), vitamin C, and vitamin E, and the mineral selenium. Antioxidants may have several positive effects on the body, such as slowing the **aging** process, reducing the risks of **cancer** and heart disease, and reducing the risks of illness and infection by supporting the immune system. Coenzyme Q10 is another antioxidant in wide usage, as studies have shown it may improve the health of the heart and reduce the effects of heart disease. Essential fatty acids, particularly omega-3, are also recommended as they are involved in many important processes in the body, including brain function. Calcium supplementation is recommended for the elderly and for women, to strengthen bones and prevent bone loss. Calcium supplements that are balanced with magnesium have a less constipating effect and are better absorbed.

After basic nutritional requirements are supported, supplements may be used to target specific needs and health conditions. For instance, athletes, men, women, children, the elderly, and vegetarians have differing needs for nutrients, and an informed use of supplements would take these differences into account. People suffering from health conditions and diseases may use specific supplements to target their condition and to support the body's healing capacity by providing optimal amounts of nutrients.

Recommended dosage

Dosages of nutritional supplements vary widely, depending on the product and individual needs. For vitamins and minerals, U.S. RDA's are essential guidelines. For other products, manufacturers' guidelines, consumer information sources such as nutritional books and magazines, and practitioners including nutritionists and naturopathic physicians may be consulted.

Precautions

Overall diet is an important first consideration for those considering nutritional supplementation. Healthy dietary habits can help optimize nutrition and the absorption of supplements, and nutritional supplements cannot substitute for a diet that is not nutritionally balanced in the first place. Supplements are best used moderately to supply any extra nutritional requirements. Sound **diets** contain a variety of wholesome foods. At least five servings per day of fruits and vegetables are recommended, as well as the inclusion of whole grains in the diet. Variety in the diet

is important to provide a full range of vitamins and minerals. Overeating inhibits digestion and absorption of nutrients, while regular **exercise** contributes to sound nutrition, by improving metabolism and digestion. Drinking plenty of clean water prevents **dehydration**, improves digestion, and helps the body flush out impurities.

Generally, nutrients from food sources are more efficiently utilized by the body than isolated substances. For instance, fresh fruit and vegetable juice could be used to provide concentrated amounts of particular nutrients, such as vitamins A and C, to the diet. As another example, eating plenty of leafy green vegetables is a healthy option for those wishing to add calcium to the diet.

Vitamins and minerals are most easily digested with food. Fat-soluble vitamins should be taken with food that contains fat. Vitamins tend to work synergistically, meaning that they work together in order to be effective. For instance, vitamin E requires some of the B-complex vitamins and the minerals selenium and zinc for most effective absorption. Some minerals may not be absorbed or may inhibit each other when taken in improper ratios. Generally, a high quality, broad-spectrum vitamin and mineral supplement will be formulated to prevent unfavorable interactions.

Vitamin A can become toxic when taken in large amounts (over 100,000 International Units) on a daily basis over time, as can vitamin D. Substituting beta-carotene for vitamin A can alleviate this risk. Very large doses of minerals taken over long periods may have toxic effects in the body. Dosages far exceeding RDA's of vitamins are not recommended, nor are large doses of other supplements.

Consumers can make wise choices for nutritional supplementation by consulting professional nutritionists and naturopathic physicians. Nutritional supplements are best added into the diet slowly, starting with small dosages and working up to the manufacturers' recommended amounts over time. Also, some supplements, such as herbal medications that may stimulate processes in the body, are best taken intermittently, allowing the body occasional rest periods without the supplement. To avoid unfavorable interactions, nutritional supplements are best used moderately and individually, rather than taking handfuls of capsules and tablets for various needs and conditions at the same time. Finally, consumers should be wary of excessive or grandiose health claims made by manufacturers of nutritional supplements and rely on scientific information to validate these claims.

KEY TERMS

Antioxidants—A class of biochemicals that have been found to protect cells from free-radical damage.

Enzymes—Chemical catalysts that help initiate biochemical processes.

Essential fatty acids—Sources of fat in the diet, including omega-3 and omega-6 fatty acids.

Naturopathic physicians—Physicians specializing in the treatment of disease using a variety of natural methods and plant-based medicines.

Side effects

Some nutritional supplements can cause upset stomach and allergic reactions, including **rashes**, flushing, **nausea**, sweating, and headaches.

Interactions

Herbal preparations and nutritional supplements may interact unfavorably with pharmaceutical drugs. For instance, some nutritional supplements recommended for nervous system function may not be recommended for those taking pharmaceutical antidepressants, such as taking 5-HTP, a nutritional supplement for the brain, or the herb **St. John's Wort**, with prescription antidepressants. Vitamin C should not be taken with aspirin, as it can irritate the stomach and limit absorption. Minerals should be taken in proper proportions to prevent unfavorable interactions; large amounts of zinc may deplete the body of the mineral copper, while too much calcium adversely affects the magnesium levels in the body. Balanced mineral supplements are recommended to alleviate these interactions.

Resources

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ORGANIZATIONS

Center for Science in the Public Interest. 1875 Connecticut Avenue NW, Suite 300, Washington, D.C. 20009. (202)332-9110. <<http://www.cspinet.org>>. Publishes *Nutrition Action Newsletter*.

National Nutritional Foods Association. 1220 19th Street NW, Washington, D.C. 20036. (202)223-0101. <<http://www.nnfa.org>>.

OTHER

USDA Food and Nutrition Information Center. <<http://www.nal.usda.gov/fnic>>.

Douglas Dupler

Nystagmus

Definition

Rhythmic, oscillating motions of the eyes are called nystagmus. The to-and-fro motion is generally involuntary. Vertical nystagmus occurs much less frequently than horizontal nystagmus and is often, but not necessarily, a sign of serious brain damage. Nystagmus can be a normal physiological response or a result of a pathologic problem.

Description

The eyes play a critical role in maintaining balance. They are directly connected to other organs of equilibrium, most important of which is the inner ear. Paired structures called the semicircular canals deep in the skull behind the ears sense motion and relay that information to balance control centers in the brain. The eyes send visual information to the same centers. A third set of sensors consists of nerve endings all over the body, particularly in joints, that detect position. All this information is integrated to allow the body to navigate in space and gravity.

It is possible to fool this system or to overload it with information so that it malfunctions. A spinning ride at the amusement park is a good way to overload it with information. The system has adapted to the spinning, expects it to go on forever, and carries that momentum for some time after it is over. Nystagmus is the lingering adjustment of the eyes to tracking the world as it revolves around them.

Nystagmus can be classified depending upon the type of motion of the eyes. In pendular nystagmus the speed of motion of the eyes is the same in both directions. In jerk nystagmus there is a slow and fast phase. The eyes move slowly in one direction and then seem to jerk back in the other direction.

Nystagmus can be present at birth (congenital) or acquired later on in life. A certain type of acquired nystagmus, called spasmus nutans, includes a head tilt and head bobbing and generally occurs between four to 12 months of age. It may last a few months to a few years, but generally goes away by itself.

Railway nystagmus is a physiological type of nystagmus. It happens when someone is on a moving train (thus the term railway) and is watching a stationary object which appears to be going by. The eyes slowly follow the object and then quickly jerk back to start over. Railway nystagmus (also called optokinetic nystagmus) is a type of jerk nystagmus. This phenomenon can be used to check vision in infants. Nystagmus can also be induced by fooling the semicircular canals. Caloric stimulation refers to a medical method of testing their connections to the brain, and therefore to the eyes. Cold or warm water flushed into the ear canal will generate motion signals from the inner ear. The eyes will respond to this signal with nystagmus if the pathways are intact.

Causes and symptoms

There are many causes of nystagmus. Nystagmus may be present at birth. It may be a result of the lack of development of normal binocular fixation early on in life. This can occur if there is a cataract at birth or a problem is some other part of the visual system. Some other conditions that nystagmus may be associated with include:

- **Albinism**. This condition is caused by a decrease in pigmentation and may affect the eyes.
- **Disorders of the eyes**. This may include optic atrophy, color blindness, very high nearsightedness (**myopia**) or severe **astigmatism**, or opacities in the structures of the eyes.
- **Acute labyrinthitis**. This is an inflammation in the inner ear. The patient may have **dizziness** (vertigo), nausea and **vomiting**, and nystagmus.
- **Brain lesions**. Disease in many parts of the brain can result in nystagmus.
- **Alcohol and drugs**. Alcohol and some medications (e.g., anti-epilepsy medications) can induce or exaggerate nystagmus.

- Multiple sclerosis. A disease of the central nervous system.

Diagnosis

Nystagmus is a sign, not a disease. If abnormal, it indicates a problem in one of the systems controlling it. An ophthalmologist and/or neuro-ophthalmologist should be consulted.

Treatment

There is one kind of nystagmus that seems to occur harmlessly by itself. The condition, benign positional vertigo, produces vertigo and nystagmus when the head is moved in certain directions. It can arise spontaneously or after a concussion. Motion sickness medicines sometimes help. But the reaction will dissipate if continuously evoked. Each morning a patient is asked to produce the symptom by moving

his or her head around until it no longer happens. This prevents it from returning for several hours or the entire day.

Prisms, contact lenses, eyeglasses, or eye muscle surgery are some possible treatments. These therapies may reduce the nystagmus but may not alleviate it. Again, because nystagmus may be a symptom, it is important to determine the cause.

Resources

ORGANIZATIONS

American Academy of Ophthalmology. 655 Beach Street, P.O. Box 7424, San Francisco, CA 94120-7424. <<http://www.eyenet.org>>.

American Optometric Association. 243 North Lindbergh Blvd., St. Louis, MO 63141. (314) 991-4100. <<http://www.aoanet.org>>.

J. Ricker Polsdorfer, MD



Obesity

Definition

Obesity is an abnormal accumulation of body fat, usually 20% or more over an individual's ideal body weight. Obesity is associated with increased risk of illness, disability, and **death**.

The branch of medicine that deals with the study and treatment of obesity is known as bariatrics. As obesity has become a major health problem in the United States, bariatrics has become a separate medical and surgical specialty.

Description

Obesity traditionally has been defined as a weight at least 20% above the weight corresponding to the lowest death rate for individuals of a specific height, gender, and age (ideal weight). Twenty to forty percent over ideal weight is considered mildly obese; 40–100% over ideal weight is considered moderately obese; and 100% over ideal weight is considered severely, or morbidly, obese. More recent guidelines for obesity use a measurement called BMI (body mass index) which is the individual's weight multiplied by 703 and then divided by twice the height in inches. BMI of 25.9–29 is considered overweight; BMI over 30 is considered obese. Measurements and comparisons of waist and hip circumference can also provide some information regarding risk factors associated with weight. The higher the ratio, the greater the chance for weight-associated complications. Calipers can be used to measure skin-fold thickness to determine whether tissue is muscle (lean) or adipose tissue (fat).

Much concern has been generated about the increasing incidence of obesity among Americans. Some studies have noted an increase from 12% to 18% occurring between 1991 and 1998. Other studies have actually estimated that a full 50% of all

Americans are overweight. The World Health Organization terms obesity a worldwide epidemic, and the diseases which can occur due to obesity are becoming increasingly prevalent.

Excessive weight can result in many serious, potentially life-threatening health problems, including **hypertension**, Type II **diabetes mellitus** (non-insulin dependent diabetes), increased risk for coronary disease, increased unexplained **heart attack**, hyperlipidemia, **infertility**, and a higher prevalence of colon, prostate, endometrial, and, possibly, **breast cancer**. Approximately 300,000 deaths a year are attributed to obesity, prompting leaders in public health, such as former Surgeon General C. Everett Koop, M.D., to label obesity “the second leading cause of preventable deaths in the United States.”

Causes and symptoms

The mechanism for excessive weight gain is clear—more calories are consumed than the body burns, and the excess calories are stored as fat (adipose) tissue. However, the exact cause is not as clear and likely arises from a complex combination of factors. Genetic factors significantly influence how the body regulates the appetite and the rate at which it turns food into energy (metabolic rate). Studies of adoptees confirm this relationship—the majority of adoptees followed a pattern of weight gain that more closely resembled that of their birth parents than their adoptive parents. A genetic predisposition to weight gain, however, does not automatically mean that a person will be obese. Eating habits and patterns of physical activity also play a significant role in the amount of weight a person gains. Recent studies have indicated that the amount of fat in a person's diet may have a greater impact on weight than the number of calories it contains. Carbohydrates like cereals, breads, fruits, and vegetables and protein (fish, lean meat, turkey breast, skim milk) are converted to fuel almost as soon as they are consumed.

Height And Weight Goals

Men

Height	Small Frame	Medium Frame	Large Frame
5'2" 5'3" 5'4"	128–134 lbs. 130–136 132–138	131–141 lbs. 133–143 135–145	138–150 lbs. 140–153 142–153
5'5" 5'6" 5'7"	134–140 136–142 138–145	137–148 139–151 142–154	144–160 146–164 149–168
5'8" 5'9" 5'10"	140–148 142–151 144–154	145–157 148–160 151–163	152–172 155–176 158–180
5'11" 6'0" 6'1"	146–157 159–160 152–164	154–166 157–170 160–174	161–184 164–188 168–192
6'2" 6'3" 6'4"	155–168 158–172 162–176	164–178 167–182 171–187	172–197 176–202 181–207

Women

Height	Small Frame	Medium Frame	Large Frame
4'10" 4'11" 5'0"	102–111 lbs. 103–113 104–115	109–121 lbs. 111–123 113–126	118–131 lbs. 120–134 112–137
5'1" 5'2" 5'3"	106–118 108–121 111–124	115–129 118–132 121–135	125–140 128–143 131–147
5'4" 5'5" 5'6"	114–127 117–130 120–133	124–141 127–141 130–144	137–151 137–155 140–159
5'7" 5'8" 5'9"	123–136 126–139 129–142	133–147 136–150 139–153	143–163 146–167 149–170
5'10" 5'11" 6'0"	132–145 135–148 138–151	142–156 145–159 148–162	152–176 155–176 158–179

Most fat calories are immediately stored in fat cells, which add to the body's weight and girth as they expand and multiply. A sedentary lifestyle, particularly prevalent in affluent societies, such as in the United States, can contribute to weight gain. Psychological factors, such as depression and low self-esteem may, in some cases, also play a role in weight gain.

At what stage of life a person becomes obese can affect his or her ability to lose weight. In childhood, excess calories are converted into new fat cells (hyperplastic obesity), while excess calories consumed in adulthood only serve to expand existing fat cells (hypertrophic obesity). Since dieting and **exercise** can only reduce the size of fat cells, not eliminate them, persons who were obese as children can have great difficulty losing weight, since they may have up to five times as many fat cells as someone who became overweight as an adult.

Obesity can also be a side effect of certain disorders and conditions, including:

- Cushing's syndrome, a disorder involving the excessive release of the hormone cortisol
- hypothyroidism, a condition caused by an underactive thyroid gland
- neurologic disturbances, such as damage to the hypothalamus, a structure located deep within the brain that helps regulate appetite
- consumption of such drugs as steroids, antipsychotic medications, or antidepressants

The major symptoms of obesity are excessive weight gain and the presence of large amounts of fatty tissue. Obesity can also give rise to several secondary conditions, including:

- arthritis and other orthopedic problems, such as lower back pain
- hernias
- heartburn
- adult-onset asthma
- gum disease
- high cholesterol levels
- gallstones
- high blood pressure
- menstrual irregularities or cessation of menstruation (amenorrhea)
- decreased fertility, and **pregnancy** complications
- shortness of breath that can be incapacitating
- sleep apnea and sleeping disorders
- skin disorders arising from the bacterial breakdown of sweat and cellular material in thick folds of skin or from increased friction between folds
- emotional and social problems

Diagnosis

Diagnosis of obesity is made by observation and by comparing the patient's weight to ideal weight charts. Many doctors and obesity researchers refer to the body mass index (BMI), which uses a height-weight relationship to calculate an individual's ideal weight and personal risk of developing obesity-related health problems. Physicians may also obtain direct measurements of an individual's body fat content by using calipers to measure skin-fold thickness at the back of the upper arm and other sites. The most

accurate means of measuring body fat content involves immersing a person in water and measuring relative displacement; however, this method is very impractical and is usually only used in scientific studies requiring very specific assessments. Women whose body fat exceeds 30% and men whose body fat exceeds 25% are generally considered obese.

Doctors may also note how a person carries excess weight on his or her body. Studies have shown that this factor may indicate whether or not an individual has a predisposition to develop certain diseases or conditions that may accompany obesity. “Apple-shaped” individuals who store most of their weight around the waist and abdomen are at greater risk for **cancer**, heart disease, **stroke**, and diabetes than “pear-shaped” people whose extra pounds settle primarily in their hips and thighs.

Treatment

Treatment of obesity depends primarily on how overweight a person is and his or her overall health. However, to be successful, any treatment must affect life-long behavioral changes rather than short-term weight loss. “Yo-yo” dieting, in which weight is repeatedly lost and regained, has been shown to increase a person’s likelihood of developing fatal health problems than if the weight had been lost gradually or not lost at all. Behavior-focused treatment should concentrate on:

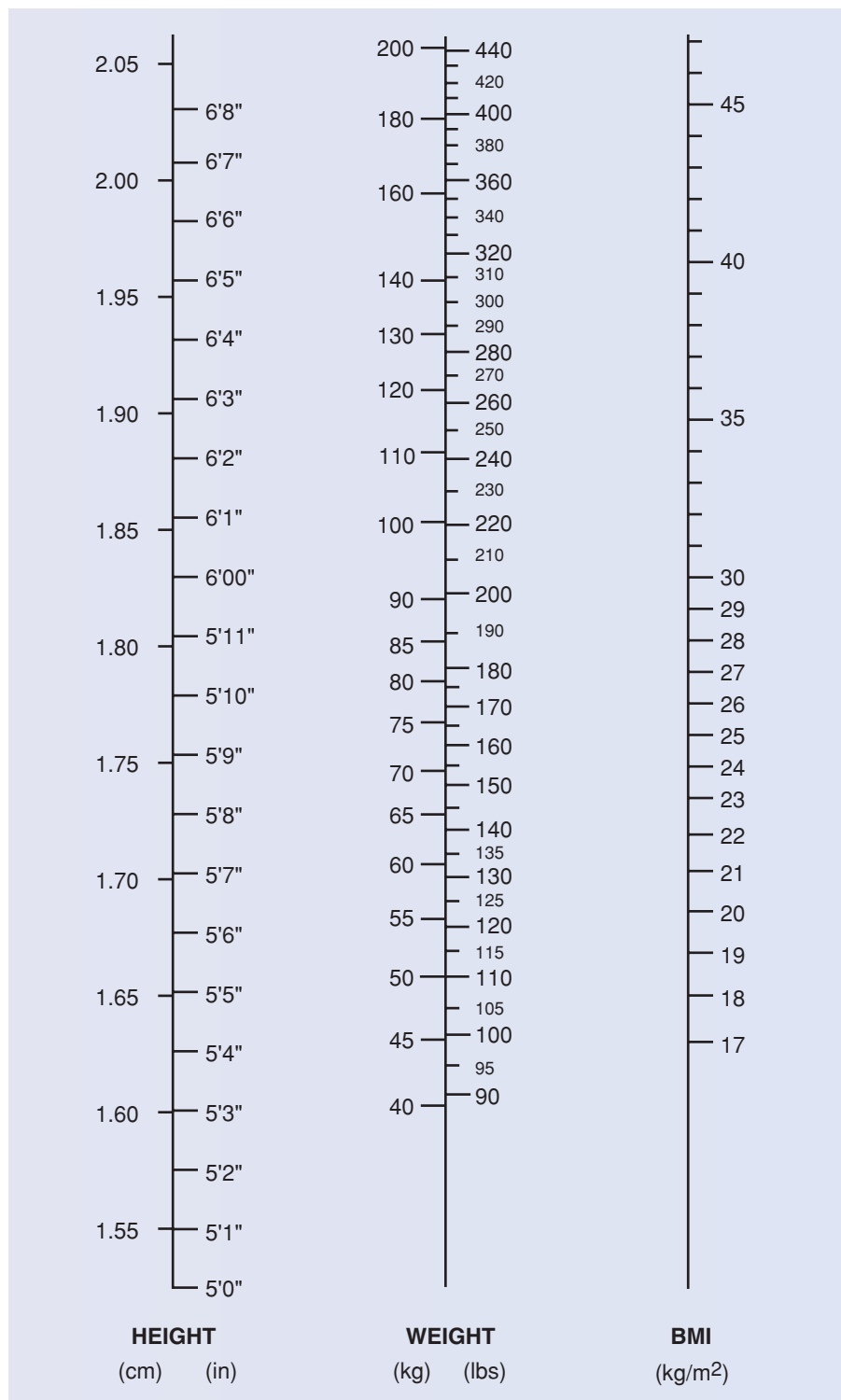
- What and how much a person eats. This aspect may involve keeping a food diary and developing a better understanding of the nutritional value and fat content of foods. It may also involve changing grocery-shopping habits (e.g., buying only what is on a prepared list and only going on a certain day), timing of meals (to prevent feelings of hunger, a person may plan frequent, small meals), and actually slowing down the rate at which a person eats.
- How a person responds to food. This may involve understanding what psychological issues underlie a person’s eating habits. For example, one person may binge eat when under **stress**, while another may always use food as a reward. In recognizing these psychological triggers, an individual can develop alternate coping mechanisms that do not focus on food.
- How they spend their time. Making activity and exercise an integrated part of everyday life is a key to achieving and maintaining weight loss. Starting slowly and building endurance keeps individuals from becoming discouraged. Varying routines and trying new activities also keeps interest high.

For most individuals who are mildly obese, these behavior modifications entail life-style changes they can make independently while being supervised by a family physician. Other mildly obese persons may seek the help of a commercial weight-loss program (e.g., Weight Watchers). The effectiveness of these programs is difficult to assess, since programs vary widely, drop-out rates are high, and few employ members of the medical community. However, programs that emphasize realistic goals, gradual progress, sensible eating, and exercise can be very helpful and are recommended by many doctors. Programs that promise instant weight loss or feature severely restricted **diets** are not effective and, in some cases, can be dangerous.

For individuals who are moderately obese, medically supervised behavior modification and weight loss are required. While doctors will put most moderately obese patients on a balanced, low-calorie diet (1200–1500 calories a day), they may recommend that certain individuals follow a very-low-calorie liquid protein diet (400–700 calories) for as long as three months. This therapy, however, should not be confused with commercial liquid protein diets or commercial weight-loss shakes and drinks. Doctors tailor these diets to specific patients, monitor patients carefully, and use them for only a short period of time. In addition to reducing the amount and type of calories consumed by the patient, doctors will recommend professional therapists or psychiatrists who can help the individual effectively change his or her behavior in regard to eating.

For individuals who are severely obese, dietary changes and behavior modification may be accompanied by surgery to reduce or bypass portions of the stomach or small intestine. Although **obesity surgery** is less risky as of 2003 because of recent innovations in equipment and surgical technique, it is still performed only on patients for whom other strategies have failed and whose obesity seriously threatens their health. Other surgical procedures are not recommended, including **liposuction**, a purely cosmetic procedure in which a suction device is used to remove fat from beneath the skin, and **jaw wiring**, which can damage gums and teeth and cause painful **muscle spasms**.

Appetite-suppressant drugs are sometimes prescribed to aid in weight loss. These drugs work by increasing levels of serotonin or catecholamine, which are brain chemicals that control feelings of fullness. Appetite suppressants, though, are not considered truly effective, since most of the weight lost while taking them is usually regained after stopping them. Also, suppressants containing amphetamines can be



Body/mass index can be calculated by locating your height and weight on the chart and drawing a diagonal line between the two. Where the line crosses over the third bar is the approximate BMI. (Illustration by Argosy Inc.)

potentially abused by patients. While most of the immediate side-effects of these drugs are harmless, the long-term effects of these drugs, in many cases, are unknown. Two drugs, dexfenfluramine hydrochloride (Redux) and fenfluramine (Pondimin) as well as a combination fenfluramine-phentermine (Fen/Phen) drug, were taken off the market when they were shown to cause potentially fatal heart defects. In November 1997, the United States Food and Drug Administration (FDA) approved a new weight-loss drug, sibutramine (Meridia). Available only with a doctor's prescription, Meridia can significantly elevate blood pressure and cause **dry mouth**, **headache**, **constipation**, and **insomnia**. This medication should not be used by patients with a history of congestive **heart failure**, heart disease, stroke, or uncontrolled high blood pressure.

Other weight-loss medications available with a doctor's prescription include:

- diethylpropion (Tenuate, Tenuate dospan)
- mazindol (Mazanor, Sanorex)
- phendimetrazine (Bontril, Plegine, Prelu-2, X-Trozone)
- phentermine (Adipex-P, Fastin, Ionamin, Oby-trim)

Phenylpropanolamine (Acutrim, Dextarim) is the only nonprescription weight-loss drug approved by the FDA. These over-the-counter diet aids can boost weight loss by 5%. Combined with diet and exercise and used only with a doctor's approval, prescription anti-obesity medications enable some patients to lose 10% more weight than they otherwise would. Most patients regain lost weight after discontinuing use of either prescription medications or nonprescription weight-loss products.

Prescription medications or over-the-counter weight-loss products can cause:

- constipation
- dry mouth
- headache
- irritability
- **nausea**
- nervousness
- sweating

None of them should be used by patients taking monoamine oxidase inhibitors (MAO inhibitors).

Doctors sometimes prescribe fluoxetine (Prozac), an antidepressant that can increase weight loss by about 10%. Weight loss may be temporary and side effects of this medication include **diarrhea**, **fatigue**, **insomnia**, **nausea**, and **thirst**. Weight-loss drugs

currently being developed or tested include ones that can prevent fat absorption or digestion; reduce the desire for food and prompt the body to burn calories more quickly; and regulate the activity of substances that control eating habits and stimulate overeating.

Alternative treatment

The Chinese herb ephedra (*Ephedra sinica*), combined with **caffeine**, exercise, and a low-fat diet in physician-supervised weight-loss programs, can cause at least a temporary increase in weight loss. However, the large doses of ephedra required to achieve the desired result can also cause:

- anxiety
- heart **arrhythmias**
- heart attack
- high blood pressure
- insomnia
- irritability
- nervousness
- seizures
- strokes
- death

Ephedra should not be used by anyone with a history of diabetes, heart disease, or thyroid problems. In fact, an article that appeared in the *Journal of the American Medical Association* in early 2003 advised against the use of ephedra.

Diuretic herbs, which increase urine production, can cause short-term weight loss but cannot help patients achieve lasting weight control. The body responds to heightened urine output by increasing thirst to replace lost fluids, and patients who use **diuretics** for an extended period of time eventually start retaining water again anyway. In moderate doses, psyllium, a mucilaginous herb available in bulk-forming **laxatives** like Metamucil, absorbs fluid and makes patients feel as if they have eaten enough. Red peppers and mustard help patients lose weight more quickly by accelerating the metabolic rate. They also make people more thirsty, so they crave water instead of food. Walnuts contain serotonin, the brain chemical that tells the body it has eaten enough. Dandelion (*Taraxacum officinale*) can raise metabolism and counter a desire for sugary foods.

Acupressure and **acupuncture** can also suppress food cravings. Visualization and **meditation** can create and reinforce a positive self-image that enhances the

patient's determination to lose weight. By improving physical strength, mental concentration, and emotional serenity, **yoga** can provide the same benefits. Also, patients who play soft, slow music during meals often find that they eat less food but enjoy it more.

Getting the correct ratios of protein, carbohydrates, and good-quality fats can help in weight loss via enhancement of the metabolism. Support groups that are informed about healthy, nutritious, and balanced diets can offer an individual the support he or she needs to maintain this type of eating regimen.

Prognosis

As many as 85% of dieters who do not exercise on a regular basis regain their lost weight within two years. In five years, the figure rises to 90%. Repeatedly losing and regaining weight (yo yo dieting) encourages the body to store fat and may increase a patient's risk of developing heart disease. The primary factor in achieving and maintaining weight loss is a life-long commitment to regular exercise and sensible eating habits.

Prevention

Obesity experts suggest that a key to preventing excess weight gain is monitoring fat consumption rather than counting calories, and the National Cholesterol Education Program maintains that only 30% of calories should be derived from fat. Only one-third of those calories should be contained in saturated fats (the kind of fat found in high concentrations in meat, poultry, and dairy products). Because most people eat more than they think they do, keeping a detailed food diary is a useful way to assess eating habits. Eating three balanced, moderate-portion meals a day—with the main meal at mid-day—is a more effective way to prevent obesity than **fasting** or crash diets. Exercise increases the metabolic rate by creating muscle, which burns more calories than fat. When regular exercise is combined with regular, healthful meals, calories continue to burn at an accelerated rate for several hours. Finally, encouraging healthful habits in children is a key to preventing childhood obesity and the health problems that follow in adulthood.

New directions in obesity treatment

The rapid rise in the incidence of obesity in the United States since 1990 has prompted researchers to look for new treatments. One approach involves the application of antidiabetes drugs to the treatment of

KEY TERMS

Adipose tissue—Fat tissue.

Appetite suppressant—Drug that decreases feelings of hunger. Most work by increasing levels of serotonin or catecholamine, chemicals in the brain that control appetite.

Bariatrics—The branch of medicine that deals with the prevention and treatment of obesity and related disorders.

Ghrelin—A recently discovered peptide hormone secreted by cells in the lining of the stomach. Ghrelin is important in appetite regulation and maintaining the body's energy balance.

Hyperlipidemia—Abnormally high levels of lipids in blood plasma.

Hyperplastic obesity—Excessive weight gain in childhood, characterized by the creation of new fat cells.

Hypertension—High blood pressure.

Hypertrophic obesity—Excessive weight gain in adulthood, characterized by expansion of already existing fat cells.

Ideal weight—Weight corresponding to the lowest death rate for individuals of a specific height, gender, and age.

Leptin—A protein hormone that affects feeding behavior and hunger in humans. At present it is thought that obesity in humans may result in part from insensitivity to leptin.

obesity. Metformin (Glucophage), a drug that was approved by the Food and Drug Administration (FDA) in 1994 for the treatment of type 2 diabetes, shows promise in treating obesity associated with **insulin resistance**.

Another field of obesity research is the study of hormones, particularly leptin, which is produced by fat cells in the body, and ghrelin, which is secreted by cells in the lining of the stomach. Both hormones are known to affect appetite and the body's energy balance. Leptin is also related to reproductive function, while ghrelin stimulates the pituitary gland to release growth hormone. Further studies of these two hormones may lead to the development of new medications to control appetite and food intake.

A third approach to obesity treatment involves research into the social factors that encourage or

reinforce weight gain in humans. Researchers are looking at such issues as the advertising and marketing of food products; media stereotypes of obesity; the development of eating disorders in adolescents and adults; and similar questions.

Resources

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ORGANIZATIONS

- American Dietetic Association. (800) 877-1600. <www.eatright.org>.
- American Obesity Association (AOA). 1250 24th Street NW, Suite 300, Washington, DC 20037. (202) 776-7711 or (800) 98-OBESE. <www.obesity.org>.
- American Society for Bariatric Surgery. 7328 West University Avenue, Suite F, Gainesville, FL 32607. (352) 331-4900. <www.asbs.org>.
- American Society of Bariatric Physicians. 5453 East Evans Place, Denver, CO 80222-5234. (303) 770-2526. <www.asbp.org>.
- HCF Nutrition Research Foundation, Inc. P.O. Box 22124, Lexington, KY 40522. (606) 276-3119.
- National Institute of Diabetes and Digestive and Kidney Diseases. 31 Center Drive, USC2560, Building 31, Room 9A-04, Bethesda, MD 20892-2560. (301) 496-3583. <www.niddk.nih.gov>.
- National Obesity Research Foundation. Temple University, Weiss Hall 867, Philadelphia, PA 19122.
- Weight-Control Information Network. 1 Win Way, Bethesda, MD 20896-3665. (301) 951-1120. <www.navigator.tufts.edu/special/win.html>.

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Obesity surgery

Definition

Obesity surgery is an operation that reduces or bypasses the stomach or small intestine so that severely overweight people can achieve significant and permanent weight loss.

Purpose

Obesity surgery, also called **bariatric surgery**, is performed only on severely overweight people who are more than twice their ideal weight. This level of obesity often is referred to as morbid obesity since it can result in many serious, and potentially deadly, health problems, including **hypertension**, **Type II diabetes mellitus** (non-insulin dependent diabetes), increased risk for coronary disease, increased unexplained **heart attack**, hyperlipidemia, and a higher prevalence of colon, prostate, endometrial, and, possibly, **breast cancer**. In 2003, researchers concluded that obesity surgery could cure Type II diabetes in many people who were not yet morbidly obese. Therefore, this surgery is performed on people whose risk of complications of surgery is outweighed by the

need to lose weight to prevent health complications, and for whom supervised weight loss and **exercise** programs have repeatedly failed. Obesity surgery, however, does not make people thin. Most people lose about 60% of their excess weight through this treatment. Changes in diet and exercise still are required to maintain a normal weight.

The theory behind obesity surgery is that if the volume the stomach holds is reduced and the entrance into the intestine is made smaller to slow stomach emptying, or part of the small intestine is bypassed or shortened, people will not be able to consume and/or absorb as many calories. With obesity surgery the volume of food the stomach can hold is reduced from about four cups to about 1/2 cup.

Insurers may consider obesity surgery elective surgery and not cover it under their policies. Documentation of the necessity for surgery and approval from the insurer should be sought before this operation is performed.

Precautions

Obesity surgery should not be performed on people who are less than twice their ideal weight. It also is not appropriate for people who have substance addictions or who have psychological disorders. Other considerations in choosing candidates for obesity surgery include the general health of the person and his or her willingness to comply with follow-up treatment.

Description

Obesity surgery is usually performed in a hospital by a surgeon who has experience with obesity surgery or at a center that specializes in the procedure. **General anesthesia** is used, and the operation takes 2–3 hours. The hospital stay lasts about a week.

Three procedures are currently used for obesity surgery:

- **Gastric bypass surgery.** Probably the most common type of obesity surgery, gastric bypass surgery has been performed in the United States for about 25 years. In this procedure, the volume of the stomach is reduced by four rows of stainless steel staples that separate the main body of the stomach from a small, newly created pouch. The pouch is attached at one end to the esophagus. At the other end is a very small opening into the small intestine. Food flows through this pouch, bypassing the main portion of the stomach and emptying slowly into the small intestine where it is absorbed.

- **Vertical banding gastroplasty.** In this procedure, an artificial pouch is created using staples in a different section of the stomach. Plastic mesh is sutured into part of the pouch to prevent it from dilating. In both surgeries the food enters the small intestine farther along than it would enter if exiting the stomach normally. This reduces the time available for absorption of nutrients. The procedure is normally done laparoscopically, meaning that the surgeon makes one or more small incisions in the abdomen and inserts the necessary tools and instruments through the tiny holes. He or she can view the patient's organs via an inserted camera that displays pictures on a monitor. This method makes for a faster and easier recovery than a large incision.
- **Jejunioleal bypass.** Now a rarely performed procedure, jejunioleal bypass involves shortening the small intestine. Because of the high occurrence of serious complications involving chronic **diarrhea** and **liver disease**, it has largely been abandoned for the other, safer procedures

Preparation

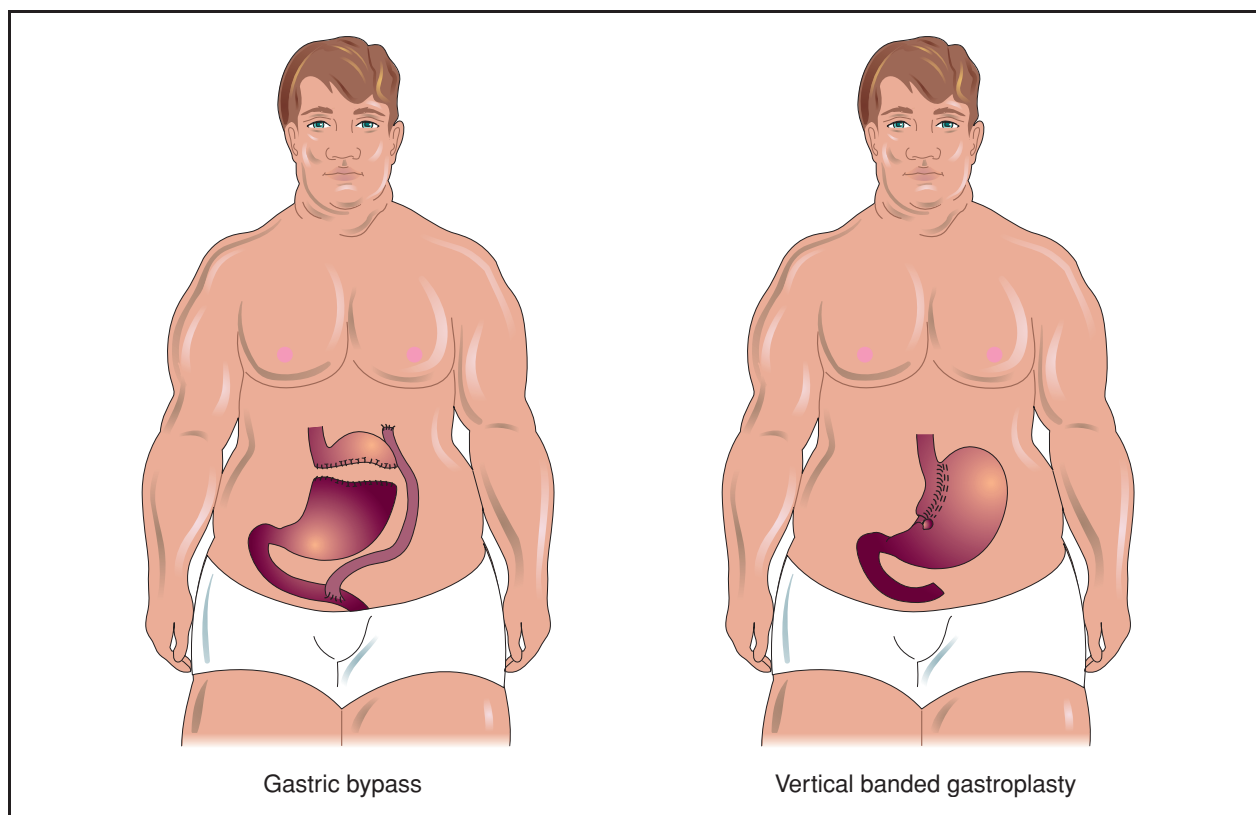
After patients are carefully selected as appropriate for obesity surgery, they receive standard preoperative blood and urine tests and meet with an anesthesiologist to discuss how their health may affect the administration of anesthesia. Pre-surgery counseling is done to help patients anticipate what to expect after the operation.

Aftercare

Immediately after the operation, most patients are restricted to a liquid diet for 2–3 weeks; however, some may remain on it for up to 12 weeks. Patients then move on to a diet of pureed food for about a month, and, after about two months, most can tolerate solid food. High fat food is restricted because it is hard to digest and causes diarrhea. Patients are expected to work on changing their eating and exercise habits to assist in weight loss. Most people eat 3–4 small meals a day once they return to solid food. Eating too quickly or too much after obesity surgery can cause **nausea and vomiting** as well as intestinal “dumping,” a condition in which undigested food is shunted too quickly into the small intestine, causing **pain**, diarrhea, weakness, and **dizziness**.

Risks

As in any abdominal surgery, there is always a risk of excessive bleeding, infection, and allergic reaction to anesthesia. Specific risks associated with obesity



The purpose of obesity surgery is to reduce the size of the stomach and slow the stomach emptying process by narrowing the entrance into the intestine. With this surgery, the volume of food the stomach can hold is reduced from approximately 4 cups to approximately one-half cup. There are two types of procedures used for obesity surgery: gastric bypass surgery and vertical banded gastroplasty, as shown in the illustration above. (Illustration by Electronic Illustrators Group.)

surgery include leaking or stretching of the pouch and loosening of the gastric staples. Although the average **death** rate associated with this procedure is less than one percent, the rate varies from center to center, ranging from 0–4%. Long-term failure rates can reach 50%, sometimes making additional surgery necessary. Other complications of obesity surgery include an intolerance to foods high in fats, **lactose intolerance**, bouts of **vomiting**, diarrhea, and intestinal discomfort

Studies on the risks of these surgeries continue. A 2003 report showed that gastric bypass surgery risk increases with age, weight and male gender. Patients age 55 and older experienced more complications than did younger patients and male patients had more life-threatening complications than female patients, particularly those who were more severely obese.

Normal results

Many people lose about 60% of the weight they need to reach their ideal weight through obesity surgery. However, surgery is not a magic weight-loss

operation, and success also depends on the patient's willingness to exercise and eat low-calorie foods. A 2003 report showed that super obese patients had a lower success rate with laparoscopic vertical banding gastroplasty than those considered morbidly obese. However, the overall success rate was nearly 77% of patients carrying less than 50% excess weight four years after the procedure.

Resources

PERIODICALS

- “Gastric Bypass Surgery Risk Increases with Age, Weight, and Male Gender.” *Medical Devices and Surgical Technology Week* January 19, 2003: 29.
- “Laparoscopic Vertical Banding Gastroplasty Safe and Effective for Morbid Obesity.” *Medical Devices and Surgical Technology Week* January 19, 2003: 2.
- Sadovsky, Richard. “Obesity Surgery May Cure Diabetes in Nonobese Patients.” *American Family Physician* 56 (February 15, 2003): 866.

Tish Davidson, A.M.
Teresa G. Odle

Obsessive-compulsive disorder

Definition

Obsessive-compulsive disorder (OCD) is a type of anxiety disorder. **Anxiety** disorder is the experience of prolonged, excessive worry about circumstances in one's life. OCD is characterized by distressing repetitive thoughts, impulses or images that are intense, frightening, absurd, or unusual. These thoughts are followed by ritualized actions that are usually bizarre and irrational. These ritual actions, known as compulsions, help reduce anxiety caused by the individual's obsessive thoughts. Often described as the "disease of doubt," the sufferer usually knows the obsessive thoughts and compulsions are irrational but, on another level, fears they may be true.

Description

Almost one out of every 40 people will suffer from obsessive-compulsive disorder at some time in their lives. The condition is two to three times more common than either **schizophrenia** or manic depression, and strikes men and women of every ethnic group, age and social level. Because the symptoms are so distressing, sufferers often hide their fears and rituals but cannot avoid acting on them. OCD sufferers are often unable to decide if their fears are realistic and need to be acted upon.

Most people with obsessive-compulsive disorder have both obsessions and compulsions, but occasionally a person will have just one or the other. The degree to which this condition can interfere with daily living also varies. Some people are barely bothered, while others find the obsessions and compulsions to be profoundly traumatic and spend much time each day in compulsive actions.

Obsessions are intrusive, irrational thoughts that keep popping up in a person's mind, such as "my hands are dirty, I must wash them again." Typical obsessions include fears of dirt, germs, contamination, and violent or aggressive impulses. Other obsessions include feeling responsible for others' safety, or an irrational fear of hitting a pedestrian with a car. Additional obsessions can involve excessive religious feelings or intrusive sexual thoughts. The patient may need to confess frequently to a religious counselor or may fear acting out the strong sexual thoughts in a hostile way. People with obsessive-compulsive disorder may have an intense preoccupation with order and symmetry, or be unable to throw anything out.

Compulsions usually involve repetitive rituals such as excessive washing (especially handwashing or bathing), cleaning, checking and touching, counting, arranging or hoarding. As the person performs these acts, he may feel temporarily better, but there is no long-lasting sense of satisfaction or completion after the act is performed. Often, a person with obsessive-compulsive disorder believes that if the ritual is not performed, something dreadful will happen. While these compulsions may temporarily ease **stress**, short-term comfort is purchased at a heavy price—time spent repeating compulsive actions and a long-term interference with life.

The difference between OCD and other compulsive behavior is that while people who have problems with gambling, overeating or with **substance abuse** may appear to be compulsive, these activities also provide pleasure to some degree. The compulsions of OCD, on the other hand, are never pleasurable.

OCD may be related to some other conditions, such as the continual urge to pull out body hair (trichotillomania) fear of having a serious disease (**hypochondriasis**) or preoccupation with imagined defects in personal appearance disorder (body dysmorphia). Some people with OCD also have **Tourette syndrome**, a condition featuring tics and unwanted vocalizations (such as swearing). OCD is often linked with depression and other anxiety disorders.

Causes and symptoms

While no one knows for sure, research suggests that the tendency to develop obsessive-compulsive disorder is inherited. There are several theories behind the cause of OCD. Some experts believe that OCD is related to a chemical imbalance within the brain that causes a communication problem between the front part of the brain (frontal lobe) and deeper parts of the brain responsible for the repetitive behavior. Research has shown that the orbital cortex located on the underside of the brain's frontal lobe is overactive in OCD patients. This may be one reason for the feeling of alarm that pushes the patient into compulsive, repetitive actions. It is possible that people with OCD experience overactivity deep within the brain that causes the cells to get "stuck," much like a jammed transmission in a car damages the gears. This could lead to the development of rigid thinking and repetitive movements common to the disorder. The fact that drugs which boost the levels of serotonin, a brain messenger substance linked to emotion and many different anxiety disorders, in the brain can reduce OCD symptoms may indicate that to some degree OCD is related to levels of serotonin in the brain.

Recently, scientists have identified an intriguing link between childhood episodes of strep throat and the development of OCD. It appears that in some vulnerable children, strep antibodies attack a certain part of the brain. Antibodies are cells that the body produces to fight specific diseases. That attack results in the development of excessive washing or germ **phobias**. A phobia is a strong but irrational fear. In this instance the phobia is fear of disease germs present on commonly handled objects. These symptoms would normally disappear over time, but some children who have repeated infections may develop full-blown OCD. Treatment with **antibiotics** has resulted in lessening of the OCD symptoms in some of these children.

If one person in a family has obsessive-compulsive disorder, there is a 25% chance that another immediate family member has the condition. It also appears that stress and psychological factors may worsen symptoms, which usually begin during adolescence or early adulthood.

Diagnosis

People with obsessive-compulsive disorder feel ashamed of their problem and often try to hide their symptoms. They avoid seeking treatment. Because they can be very good at keeping their problem from friends and family, many sufferers do not get the help they need until the behaviors are deeply ingrained habits and hard to change. As a result, the condition is often misdiagnosed or underdiagnosed. All too often, it can take more than a decade between the onset of symptoms and proper diagnosis and treatment.

While scientists seem to agree that OCD is related to a disruption in serotonin levels, there is no blood test for the condition. Instead, doctors diagnose OCD after evaluating a person's symptoms and history.

Treatment

Obsessive-compulsive disorder can be effectively treated by a combination of **cognitive-behavioral therapy** and medication that regulates the brain's serotonin levels. Drugs that are approved to treat obsessive-compulsive disorder include fluoxetine (Prozac), fluvoxamine (Luvox), paroxetine (Paxil), and sertraline (Zoloft), all selective serotonin reuptake inhibitors (SSRIs) that affect the level of serotonin in the brain. Older drugs include the antidepressant clomipramine (Anafranil), a widely-studied drug in the treatment of OCD, but one that carries a greater risk of side effects. Drugs should be taken for at least 12 weeks before deciding whether or not they are effective.

Cognitive-behavioral therapy (CBT) teaches patients how to confront their fears and obsessive thoughts by making the effort to endure or wait out the activities that usually cause anxiety without compulsively performing the calming rituals. Eventually their anxiety decreases. People who are able to alter their thought patterns in this way can lessen their preoccupation with the compulsive rituals. At the same time, the patient is encouraged to refocus attention elsewhere, such as on a hobby.

In a few severe cases where patients have not responded to medication or behavioral therapy, brain surgery may be tried as a way of relieving the unwanted symptoms. Surgery can help up to a third of patients with the most severe form of OCD. The most common operation involves removing a section of the brain called the cingulate cortex. The serious side effects of this surgery for some patients include seizures, personality changes and less ability to plan.

Alternative treatment

Because OCD sometimes responds to **SSRI** antidepressants, a botanical medicine called St. John's wort (*Hypericum perforatum*) might have some beneficial effect as well, according to herbalists. Known popularly as "Nature's Prozac," **St. John's wort** is prescribed by herbalists for the treatment of anxiety and depression. They believe that this herb affects brain levels of serotonin in the same way that SSRI antidepressants do. Herbalists recommend a dose of 300 mg., three times per day. In about one out of 400 people, St. John's wort (like Prozac) may initially increase the level of anxiety. Homeopathic constitutional therapy can help rebalance the patient's mental, emotional, and physical well-being, allowing the behaviors of OCD to abate over time.

Prognosis

Obsessive-compulsive disorder is a chronic disease that, if untreated, can last for decades, fluctuating from mild to severe and worsening with age. When treated by a combination of drugs and behavioral therapy, some patients go into complete remission. Unfortunately, not all patients have such a good response. About 20% of people cannot find relief with either drugs or behavioral therapy. Hospitalization may be required in some cases.

Despite the crippling nature of the symptoms, many successful doctors, lawyers, business people, performers and entertainers function well in society despite their condition. Nevertheless, the emotional and financial cost of obsessive-compulsive disorder can be quite high.

KEY TERMS

Anxiety disorder—This is the experience of prolonged, excessive worry about circumstances in one's life. It disrupts daily life.

Cognitive-behavior therapy—A form of psychotherapy that seeks to modify behavior by manipulating the environment to change the patient's response.

Compulsion—A rigid behavior that is repeated over and over each day.

Obsession—A recurring, distressing idea, thought or impulse that feels “foreign” or alien to the individual.

Selective serotonin reuptake inhibitors (SSRIs)—A class of antidepressants that work by blocking the reabsorption of serotonin in brain cells, raising the level of the chemical in the brain. SSRIs include Prozac, Zoloft, Luvox, and Paxil.

Serotonin—One of three major neurotransmitters found in the brain that is related to emotion, and is linked to the development of depression and obsessive-compulsive disorder.

Resources

ORGANIZATIONS

Anxiety Disorders Association of America. 11900 Park Lawn Drive, Ste. 100, Rockville, MD 20852. (800) 545-7367. <<http://www.adaa.org>>.

National Alliance for the Mentally Ill (NAMI). Colonial Place Three, 2107 Wilson Blvd., Ste. 300, Arlington, VA 22201-3042. (800) 950-6264. <<http://www.nami.org>>.

National Anxiety Foundation. 3135 Custer Dr., Lexington, KY 40517. (606) 272-7166. <<http://www.lexington-on-line.com/naf.html>>.

Obsessive-Compulsive Anonymous. P.O. Box 215, New Hyde Park, NY 11040. (516) 741-4901. <west24th@aol.com> <<http://members.aol.com/west24th/index.html>>.

Obsessive-Compulsive Foundation. P.O. Box 70, Milford, CT 06460. (203) 874-3843. <JPHS28A@Prodigy.com> <<http://pages.prodigy.com/alwillen/ocf.html>>.

Carol A. Turkington

Obsessive compulsive personality disorder
see **Personality disorders**

Obstetric sonogram see **Pelvic ultrasound**

Obstetrical emergencies

Definition

Obstetrical emergencies are life-threatening medical conditions that occur in **pregnancy** or during or after labor and delivery.

Description

There are a number of illnesses and disorders of pregnancy that can threaten the well-being of both mother and child. Obstetrical emergencies may also occur during active labor, and after delivery (postpartum).

Obstetrical emergencies of pregnancy

ECTOPIC PREGNANCY. An ectopic, or tubal, pregnancy occurs when the fertilized egg implants itself in the fallopian tube rather than the uterine wall. If the pregnancy is not terminated at an early stage, the fallopian tube will rupture, causing internal hemorrhaging and potentially resulting in permanent infertility.

PLACENTAL ABRUPTION. Also called *abruptio placenta*, placental abruption occurs when the placenta separates from the uterus prematurely, causing bleeding and contractions. If over 50% of the placenta separates, both the fetus and mother are at risk.

PLACENTA PREVIA. When the placenta attaches to the mouth of the uterus and partially or completely blocks the cervix, the position is termed *placenta previa* (or low-lying placenta). **Placenta previa** can result in premature bleeding and possible postpartum hemorrhage.

PREECLAMPSIA/ECLAMPSIA. **Preeclampsia** (toxemia), or pregnancy-induced high blood pressure, causes severe edema (swelling due to water retention) and can impair kidney and liver function. The condition occurs in approximately 5% of all United States pregnancies. If it progresses to **eclampsia**, toxemia is potentially fatal for mother and child.

PREMATURE RUPTURE OF MEMBRANES (PROM). **Premature rupture of membranes** is the breaking of the bag of waters (amniotic fluid) before contractions or labor begins. The situation is only considered an emergency if the break occurs before thirty-seven weeks and results in significant leakage of amniotic fluid and/or infection of the amniotic sac.

Obstetrical emergencies during labor and delivery

AMNIOTIC FLUID EMBOLISM. A rare but frequently fatal complication of labor, this condition occurs when amniotic fluid embolizes from the amniotic sac and through the veins of the uterus and into the circulatory system of the mother. The fetal cells present in the fluid then block or clog the pulmonary artery, resulting in **heart attack**. This complication can also happen during pregnancy, but usually occurs in the presence of strong contractions.

INVERSION OR RUPTURE OF UTERUS. During labor, a weak spot in the uterus (such as a scar or a uterine wall that is thinned by a multiple pregnancy) may tear, resulting in a uterine rupture. In certain circumstances, a portion of the placenta may stay attached to the wall and will pull the uterus out with it during delivery. This is called uterine inversion.

PLACENTA ACCRETA. *Placenta accreta* occurs when the placenta is implanted too deeply into the uterine wall, and will not detach during the late stages of **childbirth**, resulting in uncontrolled bleeding.

PROLAPSED UMBILICAL CORD. A prolapse of the umbilical cord occurs when the cord is pushed down into the cervix or vagina. If the cord becomes compressed, the oxygen supply to the fetus could be diminished, resulting in brain damage or possible **death**.

SHOULDER DYSTOCIA. Shoulder dystocia occurs when the baby's shoulder(s) becomes wedged in the birth canal after the head has been delivered.

Obstetrical emergencies postpartum

POSTPARTUM HEMORRHAGE OR INFECTION. Severe bleeding or uterine infection occurring after delivery is a serious, potentially fatal situation.

Causes and symptoms

Obstetrical emergencies can be caused by a number of factors, including **stress**, trauma, genetics, and other variables. In some cases, past medical history, including previous pregnancies and deliveries, may help an obstetrician anticipate the possibility of complications.

Signs and symptoms of an obstetrical emergency include, but are not limited to:

- Diminished fetal activity. In the late third trimester, fewer than ten movements in a two hour period may indicate that the fetus is in distress.
- Abnormal bleeding. During pregnancy, brown or white to pink vaginal discharge is normal, bright

red blood or blood containing large clots is not. After delivery, continual blood loss of over 500 ml indicates hemorrhage.

- Leaking amniotic fluid. Amniotic fluid is straw-colored and may easily be confused with urine leakage, but can be differentiated by its slightly sweet odor.
- Severe abdominal **pain**. Stomach or lower back pain can indicate preeclampsia or an undiagnosed ectopic pregnancy. Postpartum stomach pain can be a sign of infection or hemorrhage.
- Contractions. Regular contractions before 37 weeks of gestation can signal the onset of preterm labor due to obstetrical complications.
- Abrupt and rapid increase in blood pressure. Hypertension is one of the first signs of toxemia.
- Edema. Sudden and significant swelling of hands and feet caused by fluid retention from toxemia.
- Unpleasant smelling vaginal discharge. A thick, malodorous discharge from the vagina can indicate a postpartum infection.
- **Fever**. Fever may indicate an active infection.
- Loss of consciousness. **Shock** due to blood loss (hemorrhage) or amniotic **embolism** can precipitate a loss of consciousness in the mother.
- Blurred vision and headaches. Vision problems and **headache** are possible symptoms of preeclampsia.

Diagnosis

Diagnosis of an obstetrical emergency typically takes place in a hospital or other urgent care facility. A specialist will take the patient's medical history and perform a pelvic and general physical examination. The mother's vital signs are taken, and if preeclampsia is suspected, blood pressure may be monitored over a period of time. The fetal heartbeat is assessed with a doppler stethoscope, and diagnostic blood and urine tests of the mother may also be performed, including laboratory analysis for protein and/or bacterial infection. An **abdominal ultrasound** may aid in the diagnosis of any condition that involves a malpositioned placenta, such as placenta previa or placenta abruption.

In cases where an obstetrical complication is suspected, a fetal heart monitor is positioned externally on the mother's abdomen. If the fetal heart rate is erratic or weak, or if it does not respond to movement, the fetus may be in distress. A biophysical profile (BPP) may also be performed to evaluate the health of the fetus. The BPP uses data from an

ultrasound examination to analyze the fetus size, movement, heart rate, and surrounding amniotic fluid.

If the mother's membranes have ruptured and her cervix is partially dilated, an internal fetal scalp electrode can be inserted through the vagina to assess heart rate. A fetal oximetry monitor that measures the oxygen saturation levels of the fetus may also be attached to the scalp.

Treatment

Obstetrical emergencies of pregnancy

ECTOPIC PREGNANCY. Treatment of an **ectopic pregnancy** is laparoscopic surgical removal of the fertilized ovum. If the fallopian tube has burst or been damaged, further surgery will be necessary.

PLACENTAL ABRUPTION. In mild cases of **placental abruption**, bed rest may prevent further separation of the placenta and stem bleeding. If a significant abruption (over 50%) occurs, the fetus may have to be delivered immediately and a blood **transfusion** may be required.

PLACENTA PREVIA. Hospitalization or highly restricted at-home bed rest is usually recommended if placenta previa is diagnosed after the twentieth week of pregnancy. If the fetus is at least 36 weeks old and the lungs are mature, a **cesarean section** is performed to deliver the baby.

PREECLAMPSIA/ECLAMPSIA. Treatment of preeclampsia depends upon the age of the fetus and the acuteness of the condition. A woman near full term who has only mild toxemia may have labor induced to deliver the child as soon as possible. Severe preeclampsia in a woman near term also calls for immediate delivery of the child, as this is the only known cure for the condition. However, if the fetus is under 28 weeks, the mother may be hospitalized and steroids may be administered to try to hasten lung development in the fetus. If the life of the mother or fetus appears to be in danger, the baby is delivered immediately, usually by cesarean section.

PREMATURE RUPTURE OF MEMBRANES (PROM). If PROM occurs before 37 weeks and/or results in significant leakage of amniotic fluid, a course of intravenous **antibiotics** is started. A culture of the cervix may be taken to analyze for the presence of bacterial infection. If the fetus is close to term, labor is typically induced if contractions do not start within 24 hours of rupture.

Obstetrical emergencies during labor and delivery

AMNIOTIC FLUID EMBOLISM. The stress of contractions can cause this complication, which has a high mortality rate. Administering steroids to the mother and delivering the fetus as soon as possible is the standard treatment.

INVERSION OR RUPTURE OF UTERUS. An inverted uterus is either manually or surgically replaced to the proper position. A ruptured uterus is repaired if possible, although if the damage is extreme, a **hysterectomy** (removal of the uterus) may be performed. A blood transfusion may be required in either case if hemorrhaging occurs.

PLACENTA ACCRETA. Women who experience placenta accreta will typically need to have their placenta surgically removed after delivery. Hysterectomy is necessary in some cases.

PROLAPSED UMBILICAL CORD. Saline may be infused into the vagina to relieve the compression. If the cord has prolapsed out the vaginal opening, it may be replaced, but immediate delivery by cesarean section is usually indicated.

Obstetrical emergencies postpartum

POSTPARTUM HEMORRHAGE OR INFECTION. The source of the hemorrhage is determined, and blood transfusion and IV fluids are given as necessary. Oxytocic drugs may be administered to encourage contraction of the uterus. Retained placenta is a frequent cause of persistent bleeding, and surgical removal of the remaining fragments (curettage) may be required. Surgical repair of lacerations to the birth canal or uterus may be required. Drugs that encourage coagulation (clotting) of the blood may be administered to stem the bleeding. Infrequently, hysterectomy is required.

In cases of infection, a course of intravenous antibiotics is prescribed. Most postpartum infections occur in the endometrium, or lining of the uterus, and may be also caused by a piece of retained placenta. If this is the case, it will also require surgical removal.

SHOULDER DYSTOCIA. The mother is usually positioned with her knees to her chest, known as the McRoberts maneuver, in an effort to free the child's shoulder. An **episiotomy** is also performed to widen the vaginal opening. If the shoulder cannot be dislodged from the pelvis, the baby's clavicle (collarbone) may have to be broken to complete the delivery before a lack of oxygen causes brain damage to the infant.

KEY TERMS

Amniotic fluid—The liquid in the placental sac that cushions the fetus and regulates temperature in the placental environment. Amniotic fluid also contains fetal cells.

Cesarean section—The surgical delivery of a fetus through an incision in the uterus.

Embolism—Blood vessel obstruction by a blood clot or other substance (i.e., air, cell matter).

Episiotomy—Incision of the perineum, the area between the vulva and the anus, to assist delivery and avoid severe tearing of the perineum.

Laparoscopic—A minimally-invasive surgical or diagnostic procedure that uses a flexible endoscope (laparoscope) to view and operate on structures in the abdomen.

Postpartum—After childbirth.

Prognosis

If a fetus is close to full-term (37 weeks) and the complication is detected early enough, the prognosis is usually good for mother and child. With advances in neonatal care, approximately 85% of infants weighing less than 3 lbs 5 oz survive, and these infants are being delivered at 28 weeks and younger. However, preterm infants have a greater chance of serious medical problems, and developmental disabilities occur in 25–50%. They also have a higher incidence of learning disorders, and are four to six times more likely to be diagnosed with attention-deficit hyperactivity disorder (ADHD).

Prevention

Proper prenatal care is the best prevention for obstetrical emergencies. When complications of pregnancy do arise, pregnant women who see their OB/GYN on a regular basis are more likely to get an early diagnosis, and with it, the best chance for fast and effective treatment. In addition, eating right and taking prenatal **vitamins** and supplements as recommended by a physician will also contribute to the health of both mother and child.

Resources

PERIODICALS

Chamberlain, Geoffrey, and Phillip Steer. "Obstetric Emergencies." *British Medical Journal*. 318, no.7194 (May 1999): 1342.

ORGANIZATIONS

National Institute of Child Health and Human Development (NICHD) Clearinghouse. Bldg 31, Room 2A32, MSC 2425, 31 Center Drive, Bethesda, MD 20892-2425. (800) 370-2943. <<http://www.nichd.nih.gov/publications/health.htm>>.

Paula Anne Ford-Martin

Occupational asthma

Definition

Occupational **asthma** is a form of lung disease in which the breathing passages shrink, swell, or become inflamed or congested as a result of exposure to irritants in the workplace.

Description

As many as 15% of all cases of asthma may be related to on-the-job exposure to:

- animal hair
- dander
- dust composed of bacteria, protein, or organic matter like cereal, grains, cotton, and flax
- fumes created by metal soldering
- insulation and packaging materials
- mites and other insects
- paints

Hundreds of different types of jobs involve exposure to substances that could trigger occupational asthma, but only a small fraction of people who do such work develop this disorder. Occupational asthma is most apt to affect workers who have personal or family histories of allergies or asthma, or who are often required to handle or breathe dust or fumes created by especially irritating material.

Causes and symptoms

Although occupational asthma is not new, today, more than 240 causes of occupational asthma have been identified. It was probably first recorded in 1713 when one of the fathers of occupational health, Bernadina Ramazzini said bakers and textile workers had problems with coughing **shortness of breath**, hoarseness and asthma. Even short-term exposure to low levels of one or more irritating substances can cause

ALICE HAMILTON (1869–1970)



(AP/Wide World Photos. Reproduced by permission.)

Alice Hamilton was born on February 27, 1869, in New York City, the second of five children born to Montgomery Hamilton, a wholesale grocer, and Gertrude (Pond) Hamilton. She earned a medical degree from the University of Michigan in 1893, without having completed

an undergraduate degree and taking surprisingly few science courses. Realizing that she wanted to pursue research rather than medical practice, Hamilton went on to do further studies both in the United States and abroad: from 1895–1896 at Leipzig and Munich; 1896–1897 at Johns Hopkins; and 1902 in Paris at the Pasteur Institute. In 1897 she accepted a post as professor of pathology at the Women's Medical College at Northwestern University in Chicago.

In Chicago Hamilton became a resident of Hull House, the pioneering settlement designed to give care and advice to the poor of Chicago. Here, under the influence of Jane Addams, the founder of Hull House, Hamilton saw the effects of poverty up close, leading her to a lifelong career focused on industrial medicine.

Alice Hamilton was a pioneer in correcting the medical problems caused by industrialization, awakening the country in the early twentieth century to the dangers of industrial poisons and hazardous working conditions. Through her untiring efforts, toxic substances in the lead, mining, painting, pottery, and rayon industries were exposed and legislation passed to protect workers. She was also a champion of worker's compensation laws, and was instrumental in bringing about this type of legislation in the state of Illinois. A medical doctor and researcher, she was the first woman of faculty status at Harvard University, and was a consultant on governmental commissions, both domestic and foreign.

Occupations Associated With Asthma

Animal Handling
Bakeries
Health Care
Jewelry Making
Laboratory Work
Manufacturing Detergents
Nickel Plating
Soldering
Snow Crab and Egg Processing
Tanneries

a very sensitive person to develop symptoms of occupational asthma. A person who has occupational asthma has one or more symptoms, including coughing, shortness of breath, tightness in the chest, and

wheezing. Symptoms may appear less than 24 hours after the person is first exposed to the irritant or develop two or three years later.

At first, symptoms appear while the person is at work or several hours after the end of the workday. Symptoms disappear or diminish when the person spends time away from the workplace and return or intensify when exposure is renewed.

As the condition becomes more advanced, symptoms sometimes occur even when the person is not in the workplace. Symptoms may also develop in response to minor sources of lung irritation.

Diagnosis

An allergist, occupational medicine specialist, or a doctor who treats lung disease performs a thorough **physical examination** and takes a medical history that explores:

- the kind of work the patient has done
- the types of exposures the patient may have experienced
- what symptoms the patient has had
- when, how often, and how severely symptoms have occurred

Performed before and after work, pulmonary function tests can show how job-related exposures affect the airway. Laboratory analysis of blood and sputum may confirm a diagnosis of workplace asthma. To pinpoint the cause more precisely, the doctor may ask the patient to inhale specific substances and monitor the body's response to them. This is called a challenge test.

Treatment

The most effective treatment for occupational asthma is to reduce or eliminate exposure to symptom-producing substances.

Medication may be prescribed for workers who can not prevent occasional exposure. Leukotriene modifiers (montelukast and zafirlukast) are new drugs that help manage asthma. They work by counteracting leukotrienes, which are substances released by white blood cells in the lung that cause the air passages to constrict and promote mucus secretion. Leukotriene modifiers also fight off some forms of **rhinitis**, an added bonus for people with asthma. Medication, physical therapy, and breathing aids may be needed to relieve symptoms of advanced occupational asthma involving airway damage.

A patient who has occupational asthma should learn what causes symptoms and how to control them, and what to do when an asthma attack occurs.

Because asthma symptoms and the substances that provoke them can change, a patient who has occupational asthma should be closely monitored by a family physician, allergist, or doctor who specializes in occupational medicine or lung disease.

Prognosis

Occupational can be reversible. However, continued exposure to the symptom-producing substance can cause permanent lung damage. Follow-up studies of people with occupational asthma show that some cannot be protected from the exposure or are forced to change jobs, lose their jobs, or have worse prospects for future jobs based on their **allergies** and asthma.

In time, occupational asthma can cause asthma-like symptoms to occur when the patient is exposed to tobacco smoke, household dust, and other ordinary irritants.

Smoking aggravates symptoms of occupational asthma. Patients who eliminate workplace exposure and stop smoking are more apt to recover fully than those who change jobs but continue to smoke.

Prevention

Industries and environments where employees have a heightened exposure to substances known to cause occupational asthma can take measures to diminish or eliminate the amount of pollution in the atmosphere or decrease the number of exposed workers.

Regular medical screening of workers in these environments may enable doctors to diagnose occupational asthma before permanent lung damage takes place.

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American College of Allergy, Asthma and Immunology. 85 West Algonquin Road, Suite 550, Arlington Heights, IL 60005. (847) 427-1200.

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Occupational therapy see **Rehabilitation**

Ocular myopathy see **Ophthalmoplegia**

Ocular rosacea see **Rosacea**

Ofloxacin see **Fluoroquinolones**

Ohio Valley disease see **Histoplasmosis**

Oligomenorrhea

Definition

Medical dictionaries define oligomenorrhea as infrequent or very light menstruation. But physicians typically apply a narrower definition, restricting the diagnosis of oligomenorrhea to women whose periods were regularly established before they developed problems with infrequent flow. With oligomenorrhea, menstrual periods occur at intervals of greater than 35 days, with only four to nine periods in a year.

Description

True oligomenorrhea can not occur until menstrual periods have been established. In the United States, 97.5% of women have begun normal menstrual cycles by age 16. The complete absence of menstruation, whether menstrual periods never start or whether they stop after having been established, is called **amenorrhea**. Oligomenorrhea can become amenorrhea if menstruation stops for six months or more.

It is quite common for women at the beginning and end of their reproductive lives to miss or have irregular periods. This is normal and is usually the result of imperfect coordination between the hypothalamus, the pituitary gland, and the ovaries. For no apparent reason, a few women menstruate (with ovulation occurring) on a regular schedule as infrequently as once every two months. For them that schedule is normal and not a cause for concern.

Women with **polycystic ovary syndrome** (PCOS) are also likely to suffer from oligomenorrhea. PCOS is a condition in which the ovaries become filled with small cysts. Women with PCOS show menstrual irregularities that range from oligomenorrhea and amenorrhea on the one hand to very heavy, irregular periods on the other. The condition affects about 6% of premenopausal women and is related to excess androgen production.

Other physical and emotional factors also cause a woman to miss periods. These include:

- emotional **stress**
- chronic illness
- poor **nutrition**
- eating disorders such as **anorexia nervosa**
- excessive **exercise**
- estrogen-secreting tumors
- illicit use of anabolic steroid drugs to enhance athletic performance

Professional ballet dancers, gymnasts, and ice skaters are especially at risk for oligomenorrhea because they combine strenuous physical activity with a diet intended to keep their weight down. Menstrual irregularities are now known to be one of the three disorders comprising the so-called “female athlete triad,” the other disorders being disordered eating and **osteoporosis**. The triad was first formally named at the annual meeting of the American College of Sports Medicine in 1993, but doctors were aware of the combination of bone mineral loss, stress **fractures**, eating disorders, and participation in women’s sports for several decades before the triad was named. Women’s coaches have become increasingly aware of the problem since the early 1990s, and are encouraging female athletes to seek medical advice.

Causes and symptoms

Symptoms of oligomenorrhea include:

- menstrual periods at intervals of more than 35 days
- irregular menstrual periods with unpredictable flow
- some women with oligomenorrhea may have difficulty conceiving.

Oligomenorrhea that occurs in adolescents is often caused by immaturity or lack of synchronization between the hypothalamus, pituitary gland, and ovaries. The hypothalamus is part of the brain that controls body temperature, cellular metabolism, and basic functions such as eating, sleeping, and reproduction. It secretes hormones that regulate the pituitary gland.

The pituitary gland is then stimulated to produce hormones that affect growth and reproduction. At the beginning and end of a woman’s reproductive life, some of these hormone messages may not be synchronized, causing menstrual irregularities.

In PCOS, oligomenorrhea is probably caused by inappropriate levels of both female and male hormones. Male hormones are produced in small quantities by all women, but in women with PCOS, levels of male hormone (androgens) are slightly higher than in other women. More recently, however, some researchers are hypothesizing that the ovaries of women with PCOS are abnormal in other respects. In 2003, a group of researchers in London reported that there are fundamental differences between the development of egg follicles in normal ovaries and follicle development in the ovaries of women with PCOS.

In athletes, models, actresses, dancers, and women with anorexia nervosa, oligomenorrhea occurs because the ratio of body fat to weight drops too low.

Diagnosis

History and physical examination

Diagnosis of oligomenorrhea begins with the patient informing the doctor about infrequent periods. The doctor will ask for a detailed description of the problem and take a history of how long it has existed and any patterns the patient has observed. A woman can assist the doctor in diagnosing the cause of oligomenorrhea by keeping a record of the time, frequency, length, and quantity of bleeding. She should also tell the doctor about any recent illnesses, including long-standing conditions like **diabetes mellitus**. The doctor may also inquire about the patient's diet, exercise patterns, sexual activity, contraceptive use, current medications, or past surgical procedures.

The doctor will then perform a **physical examination** to evaluate the patient's weight in proportion to her height, to check for signs of normal sexual development, to make sure the heart rhythm and other vital signs are normal, and to palpate (feel) the thyroid gland for evidence of swelling.

In the case of female athletes, the doctor may need to establish a relationship of trust with the patient before asking about such matters as diet, practice and workout schedules, and the use of such drugs as steroids or ephedrine. The presence of stress fractures in young women should be investigated. In some cases, the doctor may give the patients the Eating Disorder Inventory (EDI) or a similar screening questionnaire to help determine whether the patient is at risk for developing anorexia or bulimia.

Laboratory tests

After taking the woman's history, the gynecologist or family practitioner does a pelvic examination and **Pap test**. To rule out specific causes of oligomenorrhea, the doctor may also do a **pregnancy** test and blood tests to check the level of thyroid hormone. Based on the initial test results, the doctor may want to do tests to determine the level of other hormones that play a role in reproduction.

As of 2003, more sensitive monoclonal assays have been developed for measuring hormone levels in the blood serum of women with PCOS, thus allowing earlier and more accurate diagnosis.

Imaging studies

In some cases the doctor may order an ultrasound study of the pelvic region to check for anatomical abnormalities, or x rays or a bone scan to check for bone fractures. In a few cases the doctor may order an

MRI to rule out tumors affecting the hypothalamus or pituitary gland.

Treatment

Treatment of oligomenorrhea depends on the cause. In adolescents and women near **menopause**, oligomenorrhea usually needs no treatment. For some athletes, changes in training routines and eating habits may be enough to return the woman to a regular menstrual cycle.

Most patients suffering from oligomenorrhea are treated with birth control pills. Other women, including those with PCOS, are treated with hormones. Prescribed hormones depend on which particular hormones are deficient or out of balance. When oligomenorrhea is associated with an eating disorder or the female athlete triad, the underlying condition must be treated. Consultation with a psychiatrist and nutritionist is usually necessary to manage an eating disorder. Female athletes may require physical therapy or **rehabilitation** as well.

Alternative treatment

As with conventional medical treatments, alternative treatments are based on the cause of the condition. If a hormonal imbalance is revealed by laboratory testing, hormone replacements that are more "natural" for the body (including tri-estrogen and natural progesterone) are recommended. Glandular therapy can assist in bringing about a balance in the glands involved in the reproductive cycle, including the hypothalamus, pituitary, thyroid, ovarian, and adrenal glands. Since homeopathy and **acupuncture** work on deep, energetic levels to rebalance the body, these two modalities may be helpful in treating oligomenorrhea. Western and Chinese herbal medicines also can be very effective. Herbs used to treat oligomenorrhea include dong quai (*Angelica sinensis*), black cohosh (*Cimicifuga racemosa*), and chaste tree (*Vitex agnus-castus*). Herbal preparations used to bring on the menstrual period are known as emmenagogues. For some women, **meditation**, **guided imagery**, and visualization can play a key role in the treatment of oligomenorrhea by relieving emotional stress.

Diet and adequate nutrition, including adequate protein, essential fatty acids, whole grains, and fresh fruits and vegetables, are important for every woman, especially if deficiencies are present or if she regularly exercises very strenuously. Female athletes at the high school or college level should consult a nutritionist to make sure that they are eating a well-balanced diet

KEY TERMS

Anorexia nervosa—A disorder of the mind and body in which people starve themselves in a desire to be thin, despite being of normal or below normal body weight for their size and age.

Cyst—An abnormal sac containing fluid or semi-solid material.

Emmenagogue—A medication or herbal preparation given to bring on a woman's menstrual period.

Female athlete triad—A combination of disorders frequently found in female athletes that includes disordered eating, osteoporosis, and oligo- or amenorrhea. The triad was first officially named in 1993.

Osteoporosis—The excessive loss of calcium from the bones, causing the bones to become fragile and break easily. Women who are not menstruating are especially vulnerable to this condition because estrogen, a hormone that protects bones against calcium loss, decreases drastically after menopause.

that is adequate to maintain a healthy weight for their height. Girls participating in dance or in sports that emphasize weight control or a slender body type (gymnastics, track and field, swimming, and cheerleading) are at higher risk of developing eating disorders than those that are involved in such sports as softball, weight lifting, or basketball. In some cases the athlete may be given calcium or vitamin D supplements to lower the risk of osteoporosis.

Many women, including those with PCOS, are successfully treated with hormones for oligomenorrhea. They have more frequent periods and begin ovulating during their menstrual cycle, restoring their fertility.

For women who do not respond to hormones or who continue to have an underlying condition that causes oligomenorrhea, the outlook is less positive. Women who have oligomenorrhea may have difficulty conceiving children and may receive fertility drugs. The absence of adequate estrogen increases risk for bone loss (osteoporosis) and cardiovascular disease. Women who do not have regular periods also are more likely to develop uterine **cancer**. Oligomenorrhea can become amenorrhea at any time, increasing the chance of having these complications.

Prevention

Oligomenorrhea is preventable only in women whose low body fat to weight ratio is keeping them

from maintaining a regular menstrual cycle. Adequate nutrition and a less vigorous training schedules will normally prevent oligomenorrhea. When oligomenorrhea is caused by hormonal factors, it is not preventable, but it is often treatable.

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American Academy of Child and Adolescent Psychiatry. 3615 Wisconsin Avenue, NW, Washington, DC 20016-3007. (202) 966-7300. Fax: (202) 966-2891. <<http://www.aacap.org>> .

American College of Sports Medicine (ACSM). 401 West Michigan Street, Indianapolis, IN 46202-3233. (317) 637-9200. Fax: (317) 634-7817. <<http://www.acsm.org>>.

Polycystic Ovarian Syndrome Association. P.O. Box 80517, Portland, OR 97280. (877) 775-7267. <<http://www.pcosupport.org>>.

OTHER

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Omega-3 Fatty Acids

Definition

Essential to human health, omega-3 fatty acids are a form of polyunsaturated fats that are not made by the body and must be obtained from a person's food.

Purpose

Eating foods rich in omega-3 fatty acids is part of a healthy diet and helps people maintain their health.

Description

In recent years, a great deal of attention has been placed on the value of eating a low fat diet. In some cases, people have taken this advice to the extreme by adopting a diet that is far too low in fat or, worse yet, a diet that has no fat at all. But the truth is that not all fat is bad. Although it is true that trans and saturated fats, which are found in high amounts in red meat, butter, whole milk, and some prepackaged foods, have been shown to raise a person's total cholesterol, polyunsaturated fats can actually play a part in keeping cholesterol low. Two especially good fats are the omega-3 fatty acids and the omega-6 fatty acids, which are polyunsaturated.

Two types of omega-3 fatty acids are eicosapentaenoic acid (EPA) and docosahexanoic acid (DHA), which are found mainly in oily cold-water fish, such as tuna, salmon, trout, herring, sardines, bass, swordfish, and mackerel. With the exception of seaweed, most plants do not contain EPA or DHA. However, alpha-linolenic acid (ALA), which is another kind of omega-3 fatty acid, is found in dark green leafy vegetables, flaxseed oil, fish oil, and canola oil, as well as nuts and

beans, such as walnuts and soybeans. Enzymes in a person's body can convert ALA to EPA and DHA, which are the two kinds of omega-3 fatty acids easily utilized by the body.

Many experts agree that it is important to maintain a healthy balance between omega-3 fatty acids and omega-6 fatty acids. As Dr. Penny Kris-Etherton and her colleagues reported in their article published in the *American Journal of Nutrition* an over consumption of omega-6 fatty acids has resulted in an unhealthy dietary shift in the American diet. The authors point out that what used to be a 1:1 ratio between omega-3 and omega-6 fatty acids is now estimated to be a 10:1 ratio. This poses a problem, researchers say, because consuming some of the beneficial effects gained from omega-3 fatty acids are negated by an over consumption of omega-6 fatty acids. For example, omega-3 fatty acids have anti-inflammatory properties, whereas omega-6 fatty acids tend to promote inflammation. Cereals, whole grain bread, margarine, and vegetable oils, such as corn, peanut, and sunflower oil, are examples of omega-6 fatty acids. In addition, people consume a lot of omega-6 fatty acid simply by eating the meat of animals that were fed grain rich in omega-6. Some experts suggest that eating one to four times more omega-6 fatty acids than omega-3 fatty acids is a reasonable ratio. In other words, as dietitians often say, the key to a healthy diet is moderation and balance.

The health benefits of omega-3 fatty acids

There is strong evidence that omega-3 fatty acids protect a person against **atherosclerosis** and therefore against heart disease and **stroke**, as well as abnormal heart rhythms that cause **sudden cardiac death**, and possibly **autoimmune disorders**, such as lupus and **rheumatoid arthritis**. In fact, Drs. Dean Ornish and Mehmet Oz, renowned heart physicians, said in a 2002 article published in *O Magazine* that the benefits derived from consuming the proper daily dose of omega-3 fatty acids may help to reduce sudden cardiac **death** by as much as 50%. In fact, in an article published by *American Family Physician*, Dr. Maggie Covington, a clinical assistant professor at the University of Maryland, also emphasized the value of omega-3 fatty acids with regard to cardiovascular health and referred to one of the largest clinical trials to date, the GISSI-Prevenzione Trial, to illustrate her point. In the study, 11,324 patients with coronary heart disease were divided into four groups: one group received 300 mg of vitamin E, one group received 850 mg of omega-3 fatty acids, one group received the vitamin E and fatty acids, and one group served as the control group. After a little more than three years,

“the group given omega-3 fatty acids only had a 45% reduction in sudden death and a 20% reduction in all-cause mortality,” as stated by Dr. Covington.

According to the American Heart Association (AHA), the ways in which omega-3 fatty acids may reduce cardiovascular disease are still being studied. However, the AHA indicates that research as shown that omega-3 fatty acids:

- decrease the risk of arrhythmias, which can lead to sudden cardiac death
- decrease triglyceride levels
- decrease the growth rate of atherosclerotic plaque
- lower blood pressure slightly

In fact, numerous studies show that a diet rich in omega-3 fatty acids not only lowers bad cholesterol, known as LDL, but also lowers triglycerides, the fatty material that circulates in the blood. Interestingly, researchers have found that the cholesterol levels of Inuit Eskimos tend to be quite good, despite the fact that they have a high fat diet. The reason for this, research has found, is that their diet is high in fatty fish, which is loaded with omega-3 fatty acids. The same has often been said about the typical Mediterranean-style diet.

Said to reduce joint inflammation, omega-3 fatty acid supplements have been the focus of many studies attempting to validate its effectiveness in treating rheumatoid arthritis. According to a large body of research in the area, omega-3 fatty acid supplements are clearly effective in reducing the symptoms associated with rheumatoid arthritis, such as joint tenderness and stiffness. In some cases, a reduction in the amount of medication needed by rheumatoid arthritis patients has been noted.

More research needs to be done to substantiate the effectiveness of omega-3 fatty acids in treating eating disorders, attention deficit disorder, and depression. Some studies have indicated, for example, that children with behavioral problems and attention deficit disorder have lower than normal amounts of omega-3 fatty acids in their bodies. However, until there is more data in these very important areas of research, a conservative approach should be taken, especially when making changes to a child's diet. Parents should talk to their child's pediatrician to ascertain if adding more omega-3 fatty acids to their child's diet is appropriate. In addition, parents should take special care to avoid feeding their children fish high in mercury. A food list containing items rich in omega-3 fatty acids can be obtained from a licensed dietitian.

Mercury levels and concerns about safety

A great deal of media attention has been focused on the high mercury levels found in some types of fish. People concerned about fish consumption and mercury levels can review public releases on the subject issued by the U. S. Food and Drug Administration and the Environmental Protection Agency. Special precautions exist for children and pregnant or breast-feeding women. They are advised to avoid shark, mackerel, swordfish, and tilefish. However, both the U.S. Food and Drug Administration and the Environmental Protection Agency emphasize the importance of dietary fish. Fish, they caution, should not be eliminated from the diet. In fact, Robert Oh, M.D., stated in his 2005 article, which was published in *The Journal of the American Board of Family Practice* “with the potential health benefits of fish, women of childbearing age should be encouraged to eat 1 to 2 low-mercury fish meals per week.”

Contaminants and concerns about safety

Other concerns regarding fish safety have also been reported. In 2004, Hites and colleagues assessed organic contaminants in salmon in an article published in *Science*. Their conclusion that farmed salmon had higher concentrations of polychlorinated biphenyls than wild salmon prompted public concerns and a response from the American Cancer Society. Farmed fish in Europe was found to have higher levels of mercury than farmed salmon in North and South America; however, the American Cancer Society reminded the public that the “levels of toxins Hites and his colleagues found in the farmed salmon were still below what the U.S. Food and Drug Administration, which regulates food, considers hazardous.” The American Cancer Society still continues to promote a healthy, varied diet, which includes fish as a food source.

Recommended dosage

The AHA recommends that people eat two servings of fish, such as tuna or salmon, at least twice a week. A person with coronary heart disease, according to the AHA, should consume 1 gram of omega-3 fatty acids daily through food intake, most preferably through the consumption of fatty fish. The AHA also states that “people with elevated triglycerides may need 2 to 4 grams of EPA and DHA per day provided as a supplement,” which is available in liquid or capsule form. Ground or cracked flaxseed can easily be incorporated into a person's diet by sprinkling it over salads, soup, and cereal.

Sources differ, but here are some general examples:

- 3 ounces of pickled herring = 1.2 grams of omega-3 fatty acids
- 3 ounces of salmon = 1.3 grams of omega-3 fatty acids
- 3 ounces of halibut = 1.0 grams of omega-3 fatty acids
- 3 ounces of mackerel = 1.6 grams of omega-3 fatty acids
- 1 1/2 teaspoons of flaxseeds = 3 grams of omega-3 fatty acids

Precautions

In early 2004, the U.S. Food and Drug Administration along with the the Environmental Protection Agency issued a statement that women who are or may be pregnant, as well as breastfeeding mothers and children, should avoid eating some types of fish thought to contain high levels of mercury. Fish that typically contain high levels of mercury are shark, swordfish, and mackerel, whereas shrimp, canned light tuna, salmon, and catfish are generally thought to have low levels of mercury. Because many people engage in fishing as a hobby, women should be sure before they eat any fish caught by friends and family that the local stream or lake is considered low in mercury.

Conflicting information exists whether it is safe for patients with **macular degeneration** to take omega-3 fatty acids in supplement form. Until more data becomes available, it is better for people with macular degeneration to receive their omega-3 fatty acids from the food they eat.

Side effects

Fish oil supplements can cause **diarrhea** and gas. Also, the fish oil capsules tend to have a fishy aftertaste.

Interactions

Although there are no significant **drug interactions** associated with eating foods containing omega-3 fatty acids, patients who are being treated with blood-thinning medications should not take omega-3 fatty acid supplements without seeking the advice of their physicians. Excessive bleeding could result. For the same reason, some patients who plan to take more than 3 grams of omega-3 fatty acids in supplement form should first seek the approval of their physicians.

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Lee Ann Paradise

Omeprazole *see* **Antiulcer drugs**
 Omphalocele *see* **Abdominal wall defects**
 Onchocerciasis *see* **Filariasis**

Onychomycosis

Definition

Onychomycosis is a fungal infection of the fingernails or toenails. The actual infection is of the bed of the nail and of the plate under the surface of the nail.

Description

Onychomycosis is the most common of all diseases of the nails in adults. In North America, the incidence falls roughly between 2–13%. The incidence of onychomycosis is also greater in older adults, and up to 90% of the elderly may be affected. Men are more commonly infected than women.

Individuals who are especially susceptible include those with chronic diseases such as diabetes and circulatory problems and those with diseases that suppress the immune system. Other risk factors include a family history, previous trauma to the nails, warm climate, and occlusive or tight footwear.

Causes and symptoms

Onychomycosis is caused by three types of fungi, called dermatophytes, yeasts, and nondermatophyte molds. Fungi are simple parasitic plant organisms that do not need sunlight to grow. Toenails are especially susceptible because fungi prefer dark damp places. Swimming pools, locker rooms, and showers typically harbor fungi. Chronic diseases such as diabetes, problems with the circulatory system, or immune deficiency disease are risk factors. A history of athlete's foot and excess perspiration are also risk factors.

Onychomycosis can be present for years without causing **pain** or disturbing symptoms. Typically, the nail becomes thicker and changes to a yellowish-brown. Foul smelling debris may collect under the nail. The infection can spread to the surrounding nails and even the skin.

Diagnosis

To make a diagnosis of onychomycosis, the clinician must collect a specimen of the nail in which

infection is suspected. A clipping is taken from the nail plate, and a sample of the debris from underneath the nail bed is also taken, usually with a sharp curette. Debris from the nail surface may also be taken. These will be sent for microscopic analysis to a laboratory, as well as cultured to determine what types of fungus are growing there.

Treatment

Onychomycosis is very difficult and sometimes impossible to treat, and therapy is often long-term. Therapy consists of topical treatments that are applied directly to the nails, as well as two systemic drugs, griseofulvin and ketoconazole. Topical therapy is reserved for only the mildest cases. The use of griseofulvin and ketoconazole is problematic, and there are typically high relapse rates of 50–85%. In addition, treatment must be continued for a long duration (10–18 months for toenails), with monthly laboratory monitoring for several side effects, including liver toxicity. Individuals taking these medications must also abstain from alcohol consumption.

In the past few years, newer oral antifungal agents have been developed, and include itraconazole (Sporanox), terbinafine (Lamisil), and fluconazole (Diflucan). These agents, when taken orally for as little as 12 weeks, bring about better cure rates and fewer side effects than either griseofulvin or ketoconazole. The most common side effect is stomach upset. Patients taking oral antifungal therapy must have a complete **blood count** and liver enzyme workup every four to six weeks. Terbinafine in particular has markedly less toxicity to the liver, one of the more severe side effects of the older agents, griseofulvin and ketoconazole.

Treatment should be continued until microscopic exam or culture shows no more fungal infection. Nails may, however, continue to look damaged even after a clinical cure is achieved. Nails may take up to a full year to return to normal. If the nail growth slows or stops, additional doses of antifungal therapy should be taken.

Nail **debridement** is another treatment option, but it is considered by many to be primitive compared with topical or systemic treatment. Clinicians perform nail debridement in their offices. The nail is cut and then thinned using surgical tools or chemicals, and then the loose debris under the nail is removed. The procedure is painless, and often improves the appearance of the nails immediately. In addition, it helps whatever medication being used to penetrate the newly thinned nail. Patients with very thickened nails will sometimes

KEY TERMS

Curette—Spoon-shaped instrument for removing debris, growths, or infected nail matter.

Dermatophytes, yeasts, and nondermatophyte molds—Three types of fungi responsible for fungal infections of the nails.

undergo chemical removal of a nail. A combination of oral, topical, and surgical removal can increase the chances of curing the infection.

Alternative treatment

For controlling onychomycosis, as opposed to curing it, some experts advocate using Lotrimin cream, available over the counter. The cream should be thoroughly rubbed into the nail daily in order to control the infection.

In general, **nutrition** may also play a role in promoting good nail health and thus preventing nail disease. Adequate protein and **minerals**, in the form of nuts, seeds, whole grains, legumes, fresh vegetables, and fish, should be consumed. Sugars, alcohol, and **caffeine** should be avoided. Certain supplements may also be beneficial, including vitamin A (10,000 IU per day), zinc (15–30 mg per day), iron (ferrous glycinate 100 mg per day, vitamin B₁₂ (1,000 mcg per day), and essential fatty acids in the form of flax, borage, or evening primrose oil (1,000–1,500 mg twice daily).

Herbal remedies may also relieve some of the symptoms of onychomycosis. A combination of cone-flower, oregano, spilanthes, usnea, Oregon grape root, and myrrh can be used as a tincture (20 drops four times daily).

Undiluted grapefruit seed extract and tea tree oil are also said to be beneficial when applied topically to the infected nails.

Prognosis

Onychomycosis is typically quite difficult to cure completely. Even if a clinical cure is achieved after long therapy with either topical or oral drugs, normal regrowth takes four to six months in the fingernails, and eight to 12 months in the toenails, which grow more slowly. Relapse is common, and often, the nail or nail bed is permanently damaged. For toenails infected with onychomycosis, terbinafine seems to offer the highest cure rate (35–50%). Itraconazole cure rates typically

range from 25–40%, and those with fluconazole, which was recently approved in the United States, have not been documented by long-term trials

Prevention

Keeping the feet clean and dry, and washing with soap and water and drying thoroughly are important preventive steps to take to prevent onychomycosis. Other preventive measures include keeping the nails cut short and wearing shower shoes whenever walking or showering in public places. Daily changes of shoes, socks, or hosiery are also helpful. Excessively tight hose or shoes promote moisture, which in turn, provides a wonderful environment for onychomycotic infections. To prevent this, individuals should wear only socks made of synthetic fibers, which can absorb moisture more quickly than those made of cotton or wools. Manicure and pedicure tools should be disinfected after each use. Finally, nail polish should not be applied to nails that are infected, as this causes the water or moisture that collects under the surface of the nail to not evaporate and be trapped.

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American Academy of Dermatology. 930 N. Meacham Road, PO Box 4014, Schaumburg, IL 60168-4014. (847) 330-0230. Fax: (847) 330-0050. <<http://www.aad.org>>.

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Liz Meszaros

Oophorectomy

Definition

Oophorectomy is the surgical removal of one or both ovaries. It is also called ovariectomy or ovarian ablation. If one ovary is removed, a woman may continue to menstruate and have children. If both ovaries

are removed, menstruation stops and a woman loses the ability to have children.

Purpose

Oophorectomy is performed to:

- remove cancerous ovaries
- remove the source of estrogen that stimulates some cancers
- remove large **ovarian cysts** in women with polycystic ovarian syndrome (PCOS)
- excise an abscess
- treat **endometriosis**
- lower the risk of an ectopic **pregnancy**
- lower the risk of **cancer** in a woman with a family history of ovarian or **breast cancer**

In an oophorectomy, one or a portion of one ovary may be removed or both ovaries may be removed. When oophorectomy is done to treat **ovarian cancer** or other spreading cancers, both ovaries are always removed. This is called a bilateral oophorectomy. Oophorectomies are sometimes performed on premenopausal women who have estrogen-sensitive breast cancer in an effort to remove the main source of estrogen from their bodies. This procedure has become less common than it was in the 1990s. Today, **chemotherapy** drugs are available that alter the production of estrogen and tamoxifen blocks any of the effects any remaining estrogen may have on cancer cells.

In younger women with low-grade or early-stage ovarian tumors who have not yet completed their families, the surgeon may perform a unilateral oophorectomy. This approach is called fertility-saving or fertility-sparing surgery. Women who are appropriate candidates for this type of oophorectomy do not have higher rates of cancer recurrence than women who have both ovaries removed.

Until the 1980s, women over age 40 having hysterectomies (surgical removal of the uterus) routinely had healthy ovaries and fallopian tubes removed at the same time. This operation is called a bilateral **salpingo-oophorectomy**. Many physicians reasoned that a woman over 40 was approaching **menopause** and soon her ovaries would stop secreting estrogen and releasing eggs. Removing the ovaries would eliminate the risk of ovarian cancer and only accelerate menopause by a few years.

In the 1990s, the thinking about routine oophorectomy began to change. The risk of ovarian cancer in women who have no family history of the disease

is less than 1%. Meanwhile, removing the ovaries increases the risk of cardiovascular disease and accelerates **osteoporosis** unless a woman takes prescribed hormone replacements. In addition, other studies indicate that a bilateral oophorectomy increases a woman's risk of developing **thyroid cancer**. Women with mild endometriosis can often be successfully treated with birth control pills or other hormone medications without having to undergo surgery.

Under certain circumstances, oophorectomy may still be the treatment of choice to prevent breast and ovarian cancer in certain high-risk women. A study done at the University of Pennsylvania and released in 2000 showed that healthy women who carried the BRCA1 or BRCA2 genetic mutations that predisposed them to breast cancer had their risk of breast cancer drop from 80% to 19% when their ovaries were removed before age 40. Women between the ages of 40 and 50 showed less risk reduction, and there was no significant reduction of breast cancer risk in women over age 50.

Overall, ovarian cancer still ranks low on a woman's list of health concerns: It accounts for only 4% of all cancers in women. But the lifetime risk for developing ovarian cancer in women who have mutations in BRCA1 is significantly increased over the general population and may cause an ovarian cancer risk of 30% by age 60. For women at increased risk, oophorectomy may be considered after the age of 35 if childbearing is complete.

The value of ovary removal in preventing both breast and ovarian cancer has been documented. However, there are disagreements within the medical community about when and at what age this treatment should be offered. Preventative oophorectomy, called preventative bilateral oophorectomy (PBO), is not always covered by insurance. One study conducted in 2000 at the University of California at San Francisco found that only 20% of insurers paid for PBO. Another 25% had a policy against paying for the operation, and the remaining 55% said that they would decide about payment on an individual basis.

Precautions

There are situations in which oophorectomy is a medically wise choice for women who have a family history of breast or ovarian cancer. However, women with healthy ovaries who are undergoing **hysterectomy** for reasons other than cancer should discuss with their doctors the benefits and disadvantages of having their ovaries removed at the time of the hysterectomy. It is important for women to ask questions about the long-

term risks of a bilateral oophorectomy; one study published in 2003 reported that many women awaiting surgery felt that they did not have adequate information about their treatment options and were unaware of the possible long-term consequences to health.

Description

Oophorectomy is done under **general anesthesia**. It is performed through the same type of incision, either vertical or horizontal, as an abdominal hysterectomy. Horizontal incisions leave a less noticeable scar, but vertical incisions give the surgeon a better view of the abdominal cavity.

After the incision is made, the abdominal muscles are pulled apart, not cut, so that the surgeon can see the ovaries. Then the ovaries, and often the fallopian tubes, are removed.

Oophorectomy can sometimes be done with a laparoscopic procedure. With this surgery, a tube containing a tiny lens and light source is inserted through a small incision in the navel. A camera can be attached that allows the surgeon to see the abdominal cavity on a video monitor. When the ovaries are detached, they are removed through a small incision at the top of the vagina. The ovaries can also be cut into smaller sections and removed.

The advantages of abdominal incision are that the ovaries can be removed even if a woman has many **adhesions** from previous surgery. The surgeon gets a good view of the abdominal cavity and can check the surrounding tissue for disease. A vertical abdominal incision is mandatory if cancer is suspected. The disadvantages are that bleeding is more likely to be a complication of this type of operation. The operation is more painful than a laparoscopic operation and the recovery period is longer. A woman can expect to be in the hospital two to five days and will need three to six weeks to return to normal activities.

Preparation

Before surgery, the doctor will order blood and urine tests, and any additional tests such as ultrasound or x rays to help the surgeon visualize the woman's condition. The woman may also meet with the anesthesiologist to evaluate any special conditions that might affect the administration of anesthesia. A colon preparation may be done, if extensive surgery is anticipated.

On the evening before the operation, the woman should eat a light dinner, then take nothing by mouth, including water or other liquids, after midnight.

Aftercare

After surgery a woman will feel some discomfort. The degree of discomfort varies and is generally greatest with abdominal incisions, because the abdominal muscles must be stretched out of the way so that the surgeon can reach the ovaries.

When both ovaries are removed, women who do not have cancer are started on **hormone replacement therapy** to ease the symptoms of menopause that occur because estrogen produced by the ovaries is no longer present. If even part of one ovary remains, it will produce enough estrogen that a woman will continue to menstruate, unless her uterus was removed in a hysterectomy. **Antibiotics** are given to reduce the risk of post-surgery infection.

Return to normal activities takes anywhere from two to six weeks, depending on the type of surgery. When women have cancer, chemotherapy or radiation are often given in addition to surgery. Some women have emotional trauma following an oophorectomy, and can benefit from counseling and support groups.

Risks

Oophorectomy is a relatively safe operation, although, like all major surgery, it does carry some risks. These include unanticipated reaction to anesthesia, internal bleeding, **blood clots**, accidental damage to other organs, and post-surgery infection.

Complications after an oophorectomy include changes in sex drive, hot flashes, and other symptoms of menopause if both ovaries are removed. Women who have both ovaries removed and who do not take estrogen replacement therapy run an increased risk for cardiovascular disease and osteoporosis. Women with a history of psychological and emotional problems before an oophorectomy are more likely to experience psychological difficulties after the operation.

Normal results

If the surgery is successful, the ovaries will be removed without complication, and the underlying problem resolved. In the case of cancer, all the cancer will be removed.

Abnormal results

Complications may arise if the surgeon finds that cancer has spread to other places in the abdomen. If the cancer cannot be removed by surgery, it must be treated with chemotherapy and radiation.

KEY TERMS

Cyst—An abnormal sac containing fluid or semi-solid material.

Ectopic pregnancy—A pregnancy that develops when a fertilized egg implants outside the uterus, usually in the Fallopian tubes, but sometimes in the ovary itself.

Endometriosis—A benign condition that occurs when cells from the lining of the uterus begin growing outside the uterus.

Fallopian tubes—Slender tubes that carry ova from the ovaries to the uterus.

Hysterectomy—Surgical removal of the uterus.

Osteoporosis—The excessive loss of calcium from the bones, causing the bones to become fragile and break easily.

Polycystic ovarian syndrome (PCOS)—A condition in which the eggs are not released from the ovaries and instead form multiple cysts.

Ovarian remnant syndrome is a complication that results in about 18% of women who have had an oophorectomy for severe endometriosis. The syndrome is characterized by chronic pelvic **pain** and/or a pelvic mass; it is treated by further surgery to remove the remaining ovarian tissue.

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- American College of Obstetricians and Gynecologists (ACOG). 409 12th Street, SW, P. O. Box 96920, Washington, DC 20090-6920. <<http://www.acog.org>>.
- National Cancer Institute (NCI). Public Inquiries Office, Suite 3036A, 6116 Executive Blvd., MSC 8322, Bethesda, MD 20892-8322. (800) 4-CANCER. <<http://www.nci.nih.gov>>.

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- Ovarian Cancer Home Page, National Cancer Institute. <http://www.nci.nih.gov/cancer_information/cancer_type/ovarian/>.

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Open fracture reduction see **Fracture repair**

Ophthalmic antibiotics see **Antibiotics, ophthalmic**

Ophthalmoplegia

Definition

Ophthalmoplegia is a **paralysis** or weakness of one or more of the muscles that control eye movement. The condition can be caused by any of several neurologic disorders. It may be myopathic, meaning that the muscles controlling eye movement are directly involved, or neurogenic, meaning that the nerve

pathways controlling eye muscles are affected. Diseases associated with ophthalmoplegia are ocular myopathy, which affects muscles, and internuclear ophthalmoplegia, a disorder caused by **multiple sclerosis**, a disease which affects nerves.

Description

Because the eyes do not move together in ophthalmoplegia, patients may complain of double vision. Double vision is especially troublesome if the ophthalmoplegia comes on suddenly or affects each eye differently. Because ophthalmoplegia is caused by another, underlying disease, it is often associated with other neurologic symptoms, including limb weakness, lack of coordination, and **numbness**.

Causes and symptoms

Ocular myopathy is also known as mitochondrial encephalomyelopathy with ophthalmoplegia or progressive external ophthalmoplegia. Because it is so often associated with diseases affecting many levels of the neurologic system, it is often referred to as “ophthalmoplegia plus.” The main feature is progressive limitation of eye movements, usually with drooping of the eyelids (**ptosis**). Ptosis may occur years before other symptoms of ophthalmoplegia. Because both eyes are equally involved and because ability to move the eyes lessens gradually over the course of years, double vision is rare. On examination, the eyelids may appear thin. This disease usually begins in childhood or adolescence but may start later.

When ophthalmoplegia is caused by muscle degeneration (myopathic), muscle biopsy, in which a small piece of muscle is surgically removed and examined microscopically, will find characteristic abnormal muscle fibers called ragged red fibers. In this form of ophthalmoplegia, the patient may experience weakness of the face, the muscles involved in swallowing, the neck, or the limbs.

Progressive external ophthalmoplegia is sometimes associated with specific neurologic syndromes. These syndromes include familial forms of spastic paraplegia, spinocerebellar disorders, or sensorimotor **peripheral neuropathy**. Kearns-Sayre syndrome causes ophthalmoplegia along with loss of pigment in the retina, the light-sensitive membrane lining the eye. In addition, the disease may cause **heart block** that must be corrected with a pacemaker, increased protein in the cerebrospinal fluid, and a progressively disabling lack of muscular coordination (cerebellar syndrome). Symptoms of the disease appear before age 15.

Some of the progressive external ophthalmoplegia syndromes are unusual in that inheritance is controlled by DNA in the mitochondria. The mitochondria are rod-shaped structures within a cell that convert food to usable energy. Most inherited diseases are passed on by DNA in the cell nucleus, the core that contains the hereditary material. Mitochondrial inheritance tends to be passed on by the mother. Other forms of progressive external ophthalmoplegia are not inherited but occur sporadically with no clear family history. It is not known why some forms are neurogenic and others are myopathic. In the forms inherited through mitochondrial DNA, it is not known which gene product is affected.

Internuclear ophthalmoplegia in multiple sclerosis is caused by damage to a bundle of fibers in the brainstem called the medial longitudinal fasciculus. In this syndrome, the eye on the same side as the damaged medial longitudinal fasciculus is unable to look outward (that is, the left eye cannot look left). The other eye exhibits jerking movements (**nystagmus**) when the patient tries to look left. Internuclear ophthalmoplegia may be seen rarely without multiple sclerosis in patients with certain types of **cancer** or with Chiari type II malformation.

Eye **movement disorders** and ophthalmoplegia can also be seen with **progressive supranuclear palsy**, thyroid disease, **diabetes mellitus**, brainstem tumors, migraine, basilar artery **stroke**, pituitary stroke, **myasthenia gravis**, **muscular dystrophy**, and the Fisher variant of **Guillain-Barré syndrome**. A tumor or aneurysm in the cavernous sinus, located behind the eyes, can cause painful ophthalmoplegia. Painful ophthalmoplegia can also be caused by an inflammatory process in the same area, called Tolosa-Hunt syndrome.

Diagnosis

The patient's medical and family history and the examination findings will usually help differentiate the various syndromes associated with ophthalmoplegia. In addition, each syndrome is associated with characteristic features, such as nystagmus or ptosis. All patients with progressive external ophthalmoplegia should have a muscle biopsy to look for ragged red fibers or changes suggesting muscular dystrophy. A sample should be sent for analysis of mitochondrial DNA. Electromyogram (EMG), measurement of electrical activity in the muscle, helps diagnose myopathy.

Computed tomography scan (CT scan) or **magnetic resonance imaging** (MRI) scans of the brain may be needed to rule out **brain tumor**, stroke, aneurysm, or

KEY TERMS

Cerebellar—Involving the cerebellum, which controls walking, balance, and coordination.

Cerebrospinal fluid—Fluid bathing the brain and spinal cord.

Heart block—A problem with electrical conduction in the heart muscle that may lead to irregular heart beat and require a pacemaker for treatment.

Mitochondria—Spherical or rod shaped parts of the cell. Mitochondria contain genetic material (DNA and RNA) and are responsible for converting food to energy.

multiple sclerosis. When multiple sclerosis is suspected, evoked potential testing of nerve response may also be helpful. Analysis of cerebrospinal fluid may show changes characteristic of multiple sclerosis or Kearns-Sayre syndrome. Other tests that may be helpful in Kearns-Sayre include electrocardiogram (measuring electrical activity of the heart muscles), retinal examination, and a hearing test (audiogram). For possible myasthenia gravis, the Tensilon (edrophonium) test should be done. Tests should also be done to measure activity of the cell-surface receptors for acetylcholine, a chemical that helps pass electrical impulses along nerve cells in the muscles. Thyroid disease and diabetes mellitus should be excluded by appropriate blood work.

Treatment

There are no specific cures for ocular myopathy or progressive external ophthalmoplegia. Vitamin E therapy has been used to treat Kearns-Sayre syndrome. Coenzyme Q (ubiquinone), a naturally occurring substance similar to vitamin K, is widely used to treat other forms of progressive external ophthalmoplegia, but the degree of success varies. Specific treatments are available for multiple sclerosis, myasthenia gravis, diabetes mellitus, and thyroid disease. Symptoms of ophthalmoplegia can be relieved by mechanical treatment. Surgical procedures can lift drooping eyelids or a patch over one eye can be used to relieve double vision. Because there is no blink response, a surgically lifted eyelid exposes the cornea of the eye so that it may become dry or be scratched. These complications must be avoided by using artificial tears and wearing eyepatches at night. In Kearns-Sayre syndrome, a pacemaker may be needed.

Prognosis

The prognosis of progressive external ophthalmoplegia depends on the associated neurological problems; in particular, whether there is severe limb weakness or cerebellar symptoms that may be mild or disabling. As with most chronic neurologic diseases, mortality increases with disability. Progressive external ophthalmoplegia itself is not a life-threatening condition. Kearns-Sayre syndrome is disabling, probably shortens the life span, and few if any patients have children. Overall life expectancy for multiple sclerosis patients is seven years less than normal; **death** rates are higher for women than for men.

Prevention

There is no way to prevent ophthalmoplegia.

Resources

ORGANIZATIONS

American Academy of Neurology, 1080 Montreal Ave., St. Paul, MN 55116. (612) 695-1940. <<http://www.aan.com>>.

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Ophthalmoscopic examination see **Eye examination**

Opiate withdrawal see **Withdrawal syndromes**

Opioid analgesics see **Analgesics, opioid**

Oppositional defiant disorder

Definition

Oppositional defiant disorder (ODD) is defined by the *Diagnostic and Statistical Manual of Mental Disorders*, fourth edition (DSM-IV), as a recurring pattern of negative, hostile, disobedient, and defiant behavior in a child or adolescent, lasting for at least six months without serious violation of the basic rights of others. The incidence of ODD in the American population varies somewhat according to the sample studied; DSM-IV gives the rate as between 2–16% while the American Academy of Child and Adolescent Psychiatry (AACAP) gives a figure of 5%–15%, and a researcher at a children's hospital gives a rate of 6–10%.

Description

In order to meet DSM-IV criteria for ODD, the behavior disturbances must cause clinically significant problems in social, school, or work functioning. The course of oppositional defiant disorder varies among patients. In males, the disorder is more common among those who had problem temperaments or high motor activity in the preschool years. During the school years, patients may have low self-esteem, changing moods, and a low frustration tolerance. Patients may swear and use alcohol, tobacco, or illicit drugs at an early age. There are frequent conflicts with parents, teachers, and peers.

Children with this disorder show their negative and defiant behaviors by being persistently stubborn and resisting directions. They may be unwilling to compromise, give in, or negotiate with adults. Patients may deliberately or persistently test limits, ignore orders, argue, and fail to accept blame for misdeeds. Hostility is directed at adults or peers and is shown by verbal aggression or deliberately annoying others.

Causes and symptoms

Oppositional defiant disorder is more common in boys than girls before **puberty**; the disorder typically begins by age eight. After puberty the male:female ratio is about 1:1. Although the specific causes of the disorder are unknown, parents who are overly concerned with power and control may cause an eruption to occur. Symptoms often appear at home, but over time may appear in other settings as well. Usually the disorder occurs gradually over months or years. Several theories about the causes of oppositional defiant disorder are being investigated. Oppositional defiant disorder may be related to:

- the child's temperament and the family's response to that temperament
- an inherited predisposition to the disorder in some families
- marital discord or violence between husband and wife
- frequent or multiple geographical moves
- a neurological cause, like a head injury
- a chemical imbalance in the brain (especially with the brain chemical serotonin)

Oppositional defiant disorder appears to be more common in families where at least one parent has a history of a mood disorder, **conduct disorder**, **attention deficit/hyperactivity disorder**, antisocial

personality disorder, or a substance-related disorder. Additionally, some studies suggest that mothers with a depressive disorder are more likely to have children with oppositional behavior. However, it is unclear to what extent the mother's depression results from or causes oppositional behavior in children.

Symptoms include a pattern of negative, hostile, and defiant behavior lasting at least six months. During this time four or more specific behaviors must be present. These behaviors include the child who:

- often loses his/her temper
- often argues with adults
- often actively defies or refuses to comply with adults' requests or rules
- often deliberately annoys people
- often blames others for his/her mistakes or misbehavior
- is often touchy or easily annoyed by others
- is often angry and resentful
- is often spiteful or vindictive
- misbehaves
- swears or uses obscene language
- has a low opinion of him/herself

The diagnosis of oppositional defiant disorder is not made if the symptoms occur exclusively in psychotic or **mood disorders**. Criteria are not met for conduct disorder, and, if the child is 18 years old or older, criteria are not met for antisocial personality disorder. In other words, a child with oppositional defiant disorder does not show serious aggressive behaviors or exhibit the physical cruelty that is common in other disorders.

Additional problems may be present, including:

- learning problems
- a depressed mood
- hyperactivity (although attention deficit/hyperactivity disorder must be ruled out)
- substance **abuse** or dependence
- dramatic and erratic behavior

The patient with oppositional defiant disorder is moody, easily frustrated, and may abuse drugs.

Diagnosis

While psychological testing may be needed, the doctor must examine and talk with the child, talk

with the parents, and review the medical history. Diagnosis is complicated because oppositional defiant disorder rarely travels alone. Children with attention/deficit hyperactive disorder will also have oppositional defiant disorder 50% of the time. Children with depression/anxiety will have oppositional defiant disorder 10–29% of the time. Because all of the features of this disorder are usually present in conduct disorder, oppositional defiant disorder is not diagnosed if the criteria are met for conduct disorder.

A diagnosis of oppositional defiant disorder should be considered only if the behaviors occur more frequently and have more serious consequences than is typically observed in other children of a similar developmental stage. Further, the behavior must lead to significant impairment in social, school, or work functioning.

As of 2004 a new evaluation scale known as the Oppositional Defiant Behavior Inventory (ODBI) has been developed as an aid to diagnosis. The ODBI appears to meet accepted standards of reliability and validity.

Treatment

Treatment of oppositional defiant disorder usually consists of group, individual and/or family therapy, and education. Of these, individual therapy is the most common. Therapy can provide a consistent daily schedule, support, consistent rules, discipline, and limits. It can also help train patients to get along with others and modify behaviors. Therapy can occur in residential, day treatment, or medical settings. Additionally, having a healthy role model as an example is important for the patient.

Parent management training focuses on teaching the parents specific and more effective techniques for handling the child's opposition and defiance. Research has shown that parent management training is more effective than **family therapy**. One variation of parent management training known as Parent-Child Interaction Therapy (PCIT) appears to be helpful over the long term; a group of Australian researchers reported in 2004 that families who were given a course of PCIT maintained their gains two years after the program ended.

As of the early 2000s, elementary school teachers are being trained to deal more effectively with classroom disruptions caused by children with ODD. The long-term effectiveness of these interventions, however, will require further study.

Whether involved in therapy or working on this disorder at home, the patient must work with his or

KEY TERMS

Attention deficit/hyperactivity disorder—A persistent pattern of inattention, hyperactivity and/or impulsiveness; the pattern is more frequent and severe than is typically observed in people at a similar level of development.

Conduct disorder—A repetitive and persistent pattern of behavior in which the basic rights of others are violated or major age-appropriate rules of society are broken.

her parents' guidance to make the fullest possible recovery. According to the New York Hospital/Cornell Medical Center, the patients must:

- use self timeouts
- identify what increases anxiety
- talk about feelings instead of acting on them
- find and use ways to calm themselves
- frequently remind themselves of their goals
- get involved in tasks and physical activities that provide a healthy outlet for energy
- learn how to talk with others
- develop a predictable, consistent, daily schedule of activity
- develop ways to obtain pleasure and feel good
- learn how to get along with other people
- find ways to limit stimulation
- learn to admit mistakes in a matter-of-fact way

Stimulant medication is used only when oppositional defiant disorder coexists with attention deficit/hyperactivity disorder. Currently, no research is currently available on the use of other psychiatric medications in the treatment of oppositional defiant disorder.

Prognosis

The outcome varies. In some children the disorder evolves into a conduct disorder or a mood disorder. Later in life, oppositional defiant disorder can develop into passive aggressive personality disorder or antisocial personality disorder. Some children respond well to treatment and some do not. Generally, with treatment, reasonable adjustment in social settings and in the workplace can be made in adulthood.

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American Academy of Child and Adolescent Psychiatry (AACAP). 3615 Wisconsin Avenue, NW, Washington, DC 20016-3007. (202) 966-7300. Fax: (202) 966-2891. <<http://www.aacap.org>>.

American Psychiatric Association. 1400 K Street, NW, Washington, DC 20005. <<http://www.psych.org>>.

Families Anonymous. Westchester County, Westchester, NY. (212) 354-8525.

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Optic atrophy

Definition

Optic atrophy can be defined as damage to the optic nerve resulting in a degeneration or destruction of the optic nerve. Optic atrophy may also be referred to as optic nerve head pallor because of the pale appearance of the optic nerve head as seen at the back of the eye. Possible causes of optic atrophy include:

optic neuritis, Leber's hereditary optic atrophy, toxic or nutritional optic neuropathy, **glaucoma**, vascular disorders, trauma, and other systemic disorders.

Description

The process of vision involves light entering the eye and triggering chemical changes in the retina, a pigmented layer lining the back of the eye. Nerve impulses created by this process travel to the brain via the optic nerve. Using a hand-held instrument called an ophthalmoscope, the doctor can see the optic nerve head (optic disc) which is the part of the optic nerve that enters at the back of the eyeball. In optic atrophy, the disc is pale and has fewer blood vessels than normal.

Causes and symptoms

Symptoms of optic atrophy are a change in the optic disc and a decrease in visual function. This change in visual function can be a decrease in sharpness and clarity of vision (visual acuity) or decreases in side (peripheral) vision. Color vision and contrast sensitivity can also be affected.

There are many possible causes of optic atrophy. The causes can range from trauma to systemic disorders. Some possible causes of optic atrophy include:

- **Optic neuritis.** Optic neuritis is an inflammation of the optic nerve. It may be associated with eye **pain** worsened by eye movement. It is more common in young to middle-aged women. Some patients with optic neuritis may develop **multiple sclerosis** later on in life.
- **Leber's hereditary optic neuropathy.** This is a disease of young men (late teens, early 20s), characterized by an onset over a few weeks of painless, severe, central visual loss in one eye, followed weeks or months later by the same process in the other eye. At first the optic disc may be slightly swollen, but eventually there is optic atrophy. The visual loss is generally permanent. This condition is hereditary. If a patient knows that Leber's runs in the family, **genetic counseling** should be considered.
- **Toxic optic neuropathy.** Nutritional deficiencies and poisons can be associated with gradual vision loss and optic atrophy, or with sudden vision loss and optic disc swelling. Toxic and nutritional optic neuropathies are uncommon in the United States, but took on epidemic proportions in Cuba in 1992–1993. The most common toxic optic neuropathy is known as tobacco-alcohol **amblyopia**, thought to be caused by exposure to cyanide from tobacco **smoking**, and by low levels of vitamin B₁₂ because of poor **nutrition** and poor absorption associated with drinking alcohol.

Other possible toxins included ethambutol, methyl alcohol (moonshine), ethylene glycol (antifreeze), cyanide, lead, and carbon monoxide. Certain medications have also been implicated. Nutritional optic neuropathy may be caused by deficiencies of protein, or of the **B vitamins** and folate, associated with **starvation**, malabsorption, or **alcoholism**.

- **Glaucoma.** Glaucoma may be caused by an increase of pressure inside the eye. This increased pressure may eventually affect the optic nerve if left untreated.
- **Compressive optic neuropathy.** This is the result of a tumor or other lesion putting pressure on the optic nerve. Another possible cause is enlargement of muscles involved in eye movement seen in **hyperthyroidism** (Graves' disease).
- **Retinitis pigmentosa.** This is a hereditary ocular disorder.
- **Syphilis.** Left untreated, this disease may result in optic atrophy.

Diagnosis

Diagnosis involves recognizing the characteristic changes in the optic disc with an ophthalmoscope, and measuring visual acuity, usually with an eye chart. Visual field testing can test peripheral vision. Color vision and contrast sensitivity can also be tested. Family history is important in the diagnosis of inherited conditions. Exposure to poisons, drugs, and even medications should be determined. Suspected **poisoning** can be confirmed through blood and urine analysis, as can vitamin deficiency.

Brain **magnetic resonance imaging** (MRI) may show a tumor or other structure putting pressure on the optic nerve, or may show plaques characteristic of multiple sclerosis, which is frequently associated with optic neuritis. However, similar MRI lesions may appear in Leber's hereditary optic neuropathy. Mitochondrial DNA testing can be done on a blood sample, and can identify the mutation responsible for Leber's.

Visual evoked potentials (VEP), which measure speed of conduction over the nerve pathways involved in sight, may detect abnormalities in the clinically unaffected eye in early cases of Leber's. Fluorescein **angiography** gives more detail about blood vessels in the retina.

Treatment

Treatment of optic neuritis with steroids is controversial. As of mid-1998, there is no known treatment for Leber's hereditary optic neuropathy.

KEY TERMS

Atrophy—A destruction or dying of cells, tissues, or organs.

Cerebellar—Involving the part of the brain (cerebellum), which controls walking, balance, and coordination.

Mitochondria—A structure in the cell responsible for producing energy. A defect in the DNA in the mitochondria is involved in Leber's optic neuropathy.

Neuritis—An inflammation of the nerves.

Neuropathy—A disturbance of the nerves, not caused by an inflammation. For example, the cause may be toxins, or unknown.

Treatment of other causes of optic atrophy varies depending upon the underlying disease.

Prognosis

Many patients with optic neuritis eventually develop multiple sclerosis. Most patients have a gradual recovery of vision after a single episode of optic neuritis, even without treatment. Prognosis for visual improvement in Leber's hereditary optic neuropathy is poor, with the specific rate highly dependent on which mitochondrial DNA mutation is present. If the cause of toxic or nutritional deficiency optic neuropathy can be found and treated early, such as stopping smoking and taking vitamins in tobacco-alcohol amblyopia, vision generally returns to near normal over several months' time. However, visual loss is often permanent in cases of long-standing toxic or nutritional deficiency optic neuropathy.

Prevention

People noticing a decrease in vision (central and/or side vision) should ask their eye care practitioner for a check up. Patients should also go for regular vision exams. Patients should ask their doctor how often that should be, as certain conditions may warrant more frequent exams. Early detection of inflammations or other problems lessens the chance of developing optic atrophy.

As of mid 1998, there are no preventive measures that can definitely abort Leber's hereditary optic neuropathy in those genetically at risk, or in those at risk based on earlier involvement of one eye. However, some doctors recommend that their patients take

vitamin C, vitamin E, coenzyme Q₁₀, or other antioxidants, and that they avoid the use of tobacco or alcohol. Patients should ask their doctors about the use of vitamins. Avoiding toxin exposure and nutritional deficiency should prevent toxic or nutritional deficiency optic neuropathy.

Resources

ORGANIZATIONS

American Academy of Neurology. 1080 Montreal Ave., St. Paul, MN 55116. (612) 695-1940. <<http://www.aan.com>>.

Prevent Blindness America. 500 East Remington Road, Schaumburg, IL 60173. (800) 331-2020. <<http://www.preventblindness.org>>.

Laurie Barclay, MD

Optic neuritis

Definition

Optic neuritis is a vision disorder characterized by inflammation of the optic nerve.

Description

Optic neuritis occurs when the optic nerve, the pathway that transmits visual information to the brain, becomes inflamed and the myelin sheath that surrounds the nerve is destroyed (a process known as demyelination). It typically occurs in one eye at a time (70%), and the resulting vision loss is rapid and progressive, but only temporary. Thirty percent of patients experience occurrence in both eyes. Optic neuritis tends to afflict young adults with an average age in their 30s. Seventy-five percent of patients with optic neuritis are women.

Nerve damage that occurs in the section of the optic nerve located behind the eyeball, is called *retrobulbar neuritis*, and is most often associated with multiple sclerosis. Optic nerve inflammation and **edema** (swelling) caused by intracranial pressure at the place where the nerve enters the eyeball is termed *papillitis*.

Causes and symptoms

Symptoms of optic neuritis include one or more of the following:

- blurred or dimmed vision
- blind spots, particularly with central vision
- pain with eye movement
- headache
- sudden color blindness
- impaired night vision
- impaired contrast sensitivity

Optic neuritis is most commonly associated with **multiple sclerosis** (MS). Other causes include viral or fungal infections, encephalomyelitis, autoimmune diseases, or pressure on the nerve from tumors or vascular diseases (i.e., temporal arteritis). Some toxins, such as methanol and lead, can also damage the optic nerve, as can long-term abuse of alcohol and tobacco. Patients with non-MS related optic neuritis are usually immunocompromised in some way.

Diagnosis

An ophthalmologist, a physician trained in diseases of the eye, will typically make a diagnosis of optic neuritis. A complete visual exam, including a visual acuity test, color vision test, and examination of the retina and optic disc with an ophthalmoscope, will be performed. Clinical signs such as impaired pupil response may be apparent during an eye exam, but in some cases the eye may appear normal. A medical history will also be performed to determine if exposure to toxins such as lead may have caused the optic neuritis.

Further diagnostic testing such as **magnetic resonance imaging** (MRI) may be necessary to confirm a diagnosis of optic neuritis. An MRI can also reveal signs of multiple sclerosis.

Treatment

Treatment of optic neuritis depends on the underlying cause of the condition. Vision loss resulting from a viral condition usually resolves itself once the virus is treated, and optic neuritis resulting from toxin damage may improve once the source of the toxin is removed.

A course of intravenous **corticosteroids** (steroids) followed by oral steroids has been found to be helpful in restoring vision quickly to patients with MS-related episodes of optic neuritis, but its efficacy in preventing relapse is debatable. The Optic Neuritis Treatment Trial (ONTT) has shown that IV steroids may be effective in reducing the onset of MS for up to two years, but further studies are necessary. Oral prednisone has been found to increase the likelihood of

KEY TERMS

Atrophy—Cell wasting or death.

Multiple sclerosis—An autoimmune disease of the central nervous system characterized by damage to the myelin sheath that covers nerves.

Temporal arteritis—Also known as giant cell arteritis. Inflammation of the large arteries located in the temples which is marked by the presence of giant cells and symptoms of headache and facial pain.

Visual acuity test—An eye examination that determines sharpness of vision, typically performed by identifying objects and/or letters on an eye chart.

recurrent episodes of optic neuritis, and is not recommended for treating the disorder.

Prognosis

The vision loss associated with optic neuritis is usually temporary. Spontaneous remission occurs in two to eight weeks. Sixty-five to eighty percent of patients can expect 20/30 or better vision after recovery. Long-term prognosis depends on the underlying cause of the condition. If a viral infection has triggered the episode, it frequently resolves itself with no after effects. If optic neuritis is associated with multiple sclerosis, future episodes are not uncommon. Thirty-three percent of optic neuritis cases recur within five years. Each recurrence results in less recovery and worsening vision. There is a strong association between optic neuritis and MS. In those without multiple sclerosis, half who experience an episode of vision loss related to optic neuritis will develop the disease within 15 years.

Prevention

Regular annual eye exams are critical to maintaining healthy vision. Early treatment of vision problems can prevent permanent optic nerve damage (atrophy).

Resources

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ORGANIZATIONS

Prevent Blindness America. 500 East Remington Road, Schaumburg, IL 60173. (800) 331-2020. <<http://www.prevent-blindness.org>>.

Paula Anne Ford-Martin

Oral cancer see **Head and neck cancer**

Oral cholecystography see **Gallbladder x rays**

Oral contraceptives

Definition

Oral contraceptives are medicines taken by mouth to help prevent **pregnancy**. They are also known as the Pill, OCs, or birth control pills.

Purpose

Oral contraceptives, also known as birth control pills, contain artificially made forms of two hormones produced naturally in the body. These hormones, estrogen and progestin, regulate a woman's menstrual cycle. When taken in the proper amounts, following a specific schedule, oral contraceptives are very effective in preventing pregnancy. Studies show that less than one of every one hundred women who use oral contraceptives correctly becomes pregnant during the first year of use.

These pills have several effects that help prevent pregnancy. For pregnancy to occur, an egg must become mature inside a woman's ovary, be released, and travel to the fallopian tube. A male sperm must also reach the fallopian tube, where it fertilizes the egg. Then the fertilized egg must travel to the woman's uterus (womb), where it lodges in the uterus lining and develops into a fetus. The main way that oral contraceptives prevent pregnancy is by keeping an egg from ripening fully. Eggs that do not ripen fully cannot be fertilized. In addition, birth control pills thicken mucus in the woman's body through which the sperm has to swim. This makes it more difficult for the sperm to reach the egg. Oral contraceptives also change the uterine lining so that a fertilized egg cannot lodge there to develop.

Birth control pills may cause good or bad side effects. For example, a woman's menstrual periods are regular and usually lighter when she is taking oral contraceptives, and the pills may reduce the risk of **ovarian cysts**, breast lumps, **pelvic inflammatory disease**, and other medical problems. However, taking

birth control pills increases the risk of **heart attack**, **stroke**, and **blood clots** in certain women. Serious side effects such as these are more likely in women over 35 years of age who smoke cigarettes and in those with specific health problems such as high blood pressure, diabetes, or a history of breast or uterine **cancer**. A woman who wants to use oral contraceptives should ask her physician for the latest information on the risks and benefits of all types of birth control and should consider her age, health, and medical history when deciding what to use.

Precautions

No form of birth control (except not having sex) is 100% effective. However, oral contraceptives can be highly effective when used properly. Discuss the options with a health care professional.

Oral contraceptives do not protect against **AIDS** or other sexually transmitted diseases. For protection against such diseases, use a latex **condom**.

Oral contraceptives are not effective immediately after a woman begins taking them. Physicians recommend using other forms of birth control for the first 1–3 weeks. Follow the instructions of the physician who prescribed the medicine.

Smoking cigarettes while taking oral contraceptives greatly increases the risk of serious side effects. *Women who take oral contraceptives should not smoke cigarettes.*

Seeing a physician regularly while taking this medicine is very important. The physician will note unwanted side effects. Follow his or her advice on how often you should be seen.

Anyone taking oral contraceptives should be sure to tell the health care professional in charge before having any surgical or dental procedures, laboratory tests, or emergency treatment.

This medicine may increase sensitivity to sunlight. Women using oral contraceptives should avoid too much sun exposure and should not use tanning beds, tanning booths, or sunlamps until they know how the medicine affects them. Some women taking oral contraceptives may get brown splotches on exposed areas of their skin. These usually go away over time after the women stop taking birth control pills.

Oral contraceptives may cause the gums to become tender and swollen or to bleed. Careful brushing and flossing, gum massage, and regular cleaning may help prevent this problem. Check with a physician or dentist if gum problems develop.

Women who have certain medical conditions or who are taking certain other medicines may have problems if they take oral contraceptives. Before taking these drugs, be sure to let the physician know about any of these conditions:

ALLERGIES. Anyone who has had unusual reactions to estrogens or progestins in the past should let her physician know before taking oral contraceptives. The physician should also be told about any allergies to foods, dyes, preservatives, or other substances.

PREGNANCY. Women who become pregnant or think they may have become pregnant while taking birth control pills should stop taking them immediately and check with their physicians. Women who want to start taking oral contraceptives again after pregnancy should not refill their old prescriptions without checking with their physicians. The physician may need to change the prescription.

BREASTFEEDING. Women who are breastfeeding should check with their physicians before using oral contraceptives. The hormones in the pills may reduce the amount of breast milk and may cause other problems in breastfeeding. They may also cause **jaundice** and enlarged breasts in nursing babies whose mothers take the medicine.

OTHER MEDICAL CONDITIONS. Oral contraceptives may improve or worsen some medical conditions. The possibility that they may make a condition worse does not necessarily mean they cannot be used. In some cases, women may need only to be tested or followed more closely for medical problems while using oral contraceptives. Before using oral contraceptives, women with any of these medical problems should make sure their physicians are aware of their conditions:

- Female conditions such as menstrual problems, **endometriosis**, or fibroid tumors of the uterus. Birth control pills usually make these problems better, but may sometimes make them worse or more difficult to diagnose.
- Heart or circulation problems; recent or past blood clots or stroke. Women who already have these problems may be at greater risk of developing blood clots or circulation problems if they use oral contraceptives. However, healthy women who do not smoke may lower their risk of circulation problems and heart disease by taking the pills.
- Breast cysts, lumps, or other noncancerous breast problems. Oral contraceptives generally protect against these conditions, but physicians may recommend more frequent breast exams for women taking the pills.

- **Breast cancer** or other cancer (now or in the past, or family history). Oral contraceptives may make some existing cancers worse. Women with a family history of breast cancer may need more frequent screening for the disease if they decide to take birth control pills.
- Migraine headaches. This condition may improve or may get worse with the use of birth control pills.
- Diabetes. Blood sugar levels may increase slightly when oral contraceptives are used. Usually this increase is not enough to affect the amount of diabetes medicine needed. However, blood sugar will need to be monitored closely while taking oral contraceptives.
- Depression. This condition may worsen in women who already have it or may (rarely) occur again in women who were depressed in the past.
- Gallbladder disease, **gallstones**, high blood cholesterol, or chorea gravidarum (a nervous disorder). Oral contraceptives may make these conditions worse.
- Epilepsy, high blood pressure, heart or circulation problems. By increasing fluid build-up, oral contraceptives may make these conditions worse.

Description

Oral contraceptives (birth control pills) come in a wide range of estrogen-progestin combinations. The pills in use today contain much lower doses of estrogen than those available in the past, and this change has reduced the likelihood of serious side effects. Some pills contain only progestin. These are prescribed mainly for women who need to avoid estrogens and may not be as effective in preventing pregnancy as the estrogen-progestin combinations.

These medicines come in tablet form, in containers designed to help women keep track of which tablet to take each day. The tablets are different colors, indicating amounts of hormones they contain. Some may contain no hormones at all. These are included simply to help women stay in the habit of taking a pill every day, as the hormone combination needs to be taken only on certain days of the menstrual cycle. Keeping the tablets in their original container and taking them exactly on schedule is very important. They will not be as effective if they are taken in the wrong order or if doses are missed.

Oral contraceptives are available only with a physician's prescription. Some commonly used brands are Demulen, Desogen, Loestrin, Lo/Ovral, Nordette, Ortho-Novum, Ortho-Tri-Cyclen, Estrostep, Orthocept, Alesse, Levlite and Ovcon.

The dose schedule depends on the type of oral contraceptive. The two basic schedules are a 21-day schedule and a 28-day schedule. On the 21-day schedule, take one tablet a day for 21 days, then skip 7 days and repeat the cycle. On the 28-day schedule, take one tablet a day for 28 days; then repeat the cycle. Be sure to carefully follow the instructions provided with the medicine. For additional information or explanations, check with the physician who prescribed the medicine or the pharmacist who filled the prescription.

Taking doses more than 24 hours apart may increase the chance of side effects or pregnancy. Try to take the medicine at the same time every day. Take care not to run out of pills. If possible, keep an extra month's supply on hand and replace it every month with the most recently filled prescription.

Try not to miss a dose, as this increases the risk of pregnancy. If a dose is missed, follow the package directions or check with the physician who prescribed the medicine for instructions. It may be necessary to use another form of birth control for some time after missing a dose.

Taking this medicine with food or at bedtime will help prevent **nausea**, a side effect that sometimes occurs during the first few weeks. This side effect usually goes away as the body adjusts to the medicine.

Taking oral contraceptives may have several benefits outside of their ability to prevent pregnancy. Research indicates that with 10 to 12 years of oral contraceptive use, a woman's risk of ovarian cancer is reduced by up to 80 %. There may also be an approximate 50% decrease in the rate of endometrial cancers in women. One other well-known, noncontraceptive benefit of oral contraceptives is an improvement in **acne**. The combination oral contraceptive ethinyl estradiol/norgestimate has been approved by the Food and Drug Administration for the treatment of acne. Another positive effect of oral contraceptive use is improvement in abnormal uterine bleeding. Older women may also benefit from using oral contraceptives, because the pills can increase bone mass as women enter their menopausal years, when **osteoporosis** is a growing concern.

Oral contraceptives may also be used on an emergency basis as a means of preventing pregnancy in women who have had unprotected intercourse. Two products specifically designed for this purpose are Preven and Plan B. In 2001, the American College of Obstetricians and Gynecologists (ACOG) recommended that emergency oral contraceptives be available as an over-the-counter medicine. The Food and

Drug Administration, however, has not yet approved any measures that would allow this to happen.

Risks

Taking oral contraceptives with certain other drugs may affect the way the drugs work or may increase the chance of side effects.

Side Effects

Serious side effects are rare in healthy women who do not smoke cigarettes. In women with certain health problems, however, oral contraceptives may cause problems such as **liver cancer**, noncancerous liver tumors, blood clots, or stroke. Health care professionals can help women weigh the benefits of being protected against unwanted pregnancy against the risks of possible health problems.

The most common minor side effects are nausea; **vomiting**; abdominal cramping or bloating; breast **pain**, tenderness or swelling; swollen ankles or feet; tiredness; and acne. These problems usually go away as the body adjusts to the drug and do not need medical attention unless they continue or they interfere with normal activities.

Other side effects should be brought to the attention of the physician who prescribed the medicine. Check with the physician as soon as possible if any of the following side effects occur:

- menstrual changes, such as lighter periods or missed periods, longer periods, or bleeding or spotting between periods
- headaches
- vaginal infection, **itching**, or irritation
- increased blood pressure

Women who have any of the following symptoms should get emergency help right away. These symptoms may be signs of blood clots:

- sudden changes in vision, speech, breathing, or coordination
- severe or sudden **headache**
- coughing up blood
- sudden, severe, or continuing pain in the abdomen or stomach
- pain in the chest, groin, or leg (especially in the calf)
- weakness, **numbness**, or pain in an arm or leg

Oral contraceptives may continue to affect the menstrual cycle for some time after a woman stops taking them. Women who miss periods for several

months after stopping this medicine should check with their physicians.

Other rare side effects may occur. Anyone who has unusual symptoms while taking oral contraceptives should get in touch with her physician.

Interactions

Oral contraceptives may interact with a number of other medicines. When this happens, the effects of one or both of the drugs may change or the risk of side effects may be greater. Anyone who takes oral contraceptives should let the physician know all other medicines she is taking and should ask whether the possible interactions can interfere with drug therapy.

These drugs may make oral contraceptives less effective in preventing pregnancy. Anyone who takes these drugs should use an additional birth control method for the entire cycle in which the medicine is used:

- ampicillin
- penicillin V
- rifampin (Rifadin)
- tetracyclines
- griseofulvin (Gris-PEG, Fulvicin)
- corticosteroids
- barbiturates
- carbamazepine (Tegretol)
- phenytoin (Dilantin)
- primidone (Mysoline)
- ritonavir (Norvir)

In addition, taking these medicines with oral contraceptives may increase the risk of side effects or interfere with the medicine's effects:

- theophylline—effects of this medicine may increase, along with the chance of unwanted side effects
- cyclosporine—effects of this medicine may increase, along with the chance of unwanted side effects
- troleandomycin (TAO)—chance of liver problems may increase. Effectiveness of oral contraceptive may also decrease, raising the risk of pregnancy

The list above does not include every drug that may interact with oral contraceptives. Be sure to check with a physician or pharmacist before combining oral contraceptives with any other prescription or nonprescription (over-the-counter) medicine.

As with any medication, the benefits and risks should be discussed with a physician.

KEY TERMS

Cyst—An abnormal sac or enclosed cavity in the body, filled with liquid or partially solid material.

Endometriosis—A condition in which tissue like that normally found in the lining of the uterus is present outside the uterus. The condition often causes pain and bleeding.

Fallopian tube—One of a pair of slender tubes that extend from each ovary to the uterus. Eggs pass through the fallopian tubes to reach the uterus.

Fetus—A developing baby inside the womb.

Fibroid tumor—A noncancerous tumor formed of fibrous tissue.

Hormone—A substance that is produced in one part of the body, then travels through the bloodstream to another part of the body where it has its effect.

Jaundice—Yellowing of the eyes and skin due to the build up of a bile pigment (bilirubin) in the blood.

Migraine—A throbbing headache that usually affects only one side of the head. Nausea, vomiting, increased sensitivity to light, and other symptoms often accompany migraine.

Mucus—Thick fluid produced by the moist membranes that line many body cavities and structures.

Ovary—A reproductive organ in females that produces eggs and hormones.

Pelvic inflammatory disease—Inflammation of the female reproductive tract, caused by any of several microorganisms. Symptoms include severe abdominal pain, high fever, and vaginal discharge. Severe cases can result in sterility. Also called PID.

Uterus—A hollow organ in a female in which a fetus develops until birth.

Resources

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Deanna M. Swartout-Corbeil, R.N.

Oral herpes see **Cold sore**

Oral hygiene

Definition

Oral hygiene is the practice of keeping the mouth clean and healthy by brushing and flossing to prevent **tooth decay** and gum disease.

Purpose

The purpose of oral hygiene is to prevent the build-up of plaque, the sticky film of bacteria and food that forms on the teeth. Plaque adheres to the crevices and fissures of the teeth and generates acids that, when not removed on a regular basis, slowly eat away, or decay, the protective enamel surface of the teeth, causing holes (cavities) to form. Plaque also irritates gums and can lead to gum disease (**periodontal disease**) and tooth loss. Toothbrushing and flossing remove plaque from teeth, and antiseptic mouthwashes kill some of the bacteria that help form plaque. Fluoride—in toothpaste, drinking water, or dental treatments—also helps to protect teeth by binding with enamel to make it stronger. In addition to such daily oral care, regular visits to the dentist promote oral health. Preventative services that he or she can perform include fluoride treatments, sealant application, and scaling (scraping off the hardened plaque, called tartar). The dentist can also perform such diagnostic services as x-ray imaging and oral **cancer** screening as well as such treatment services as fillings, crowns, and bridges.

Precautions

Maintaining oral hygiene should be a lifelong habit. An infant's gums and, later, teeth should be kept clean by wiping them with a moist cloth or a soft toothbrush. However, only a very small amount (the size of a pea) of toothpaste containing fluoride should be used since too much fluoride may be toxic to infants.

An adult who has partial or full dentures should also maintain good oral hygiene. Bridges and dentures

must be kept clean to prevent gum disease. Dentures should be relined and adjusted by a dentist as necessary to maintain proper fit so the gums do not become red, swollen, and tender.

Brushing and flossing should be performed thoroughly but not too vigorously. Rough mechanical action may irritate or damage sensitive oral tissues. Sore or bleeding gums may be experienced for the first few days after flossing is begun. However, bleeding continuing beyond one week should be brought to the attention of a dentist. As a general rule, any sore or abnormal condition that does not disappear after 10 days should be examined by a dentist.

Description

Brushing

Brushing should be performed with a toothbrush and a fluoride toothpaste at least twice a day and preferably after every meal and snack. Effective brushing must clean each outer tooth surface, inner tooth surface, and the flat chewing surfaces of the back teeth. To clean the outer and inner surfaces, the toothbrush should be held at a 45-degree angle against the gums and moved back and forth in short strokes (no more than one toothwidth distance). To clean the inside surfaces of the front teeth, the toothbrush should be held vertically and the bristles at the tip (called the toe of the brush) moved gently up and down against each tooth. To clean the chewing surfaces of the large back teeth, the brush should be held flat and moved back and forth. Finally, the tongue should also be brushed using a back-to-front sweeping motion to remove food particles and bacteria that may sour the breath.

Toothbrushes wear out and should be replaced every three months. Consumers should look for toothbrushes with soft, nylon, rounded bristles in a size and shape that allows them to reach all tooth surfaces easily.

Holding a toothbrush may be difficult for people with limited use of their hands. The toothbrush handle may be modified by inserting it into a rubber ball for easier gripping.

Flossing

Flossing once a day helps prevent gum disease by removing food particles and plaque at and below the gumline as well as between teeth. To begin, most of an 18-in (45-cm) strand of floss is wrapped around the third finger of one hand. A 1-in (2.5-cm) section is then grasped firmly between the thumb and forefinger of each hand. The floss is eased between two teeth and worked gently up and down several times with a

rubbing motion. At the gumline, the floss is curved first around one tooth and then the other with gentle sliding into the space between the tooth and gum. After each tooth contact is cleaned, a fresh section of floss is unwrapped from one hand as the used section of floss is wrapped around the third finger of the opposite hand. Flossing proceeds between all teeth and behind the last teeth. Flossing should also be performed around the abutment (support) teeth of a bridge and under any artificial teeth using a device called a floss threader.

Dental floss comes in many varieties (waxed, unwaxed, flavored, tape) and may be chosen on personal preference. For people who have difficulty handling floss, floss holders and other types of interdental (between the teeth) cleaning aids, such as brushes and picks, are available.

Risks

Negative consequences arise from improper or infrequent brushing and flossing. The five major oral health problems are plaque, tartar, gingivitis, periodontitis, and tooth decay.

Plaque is a soft, sticky, colorless bacterial film that grows on the hard, rough surfaces of teeth. These bacteria use the sugar and starch from food particles in the mouth to produce acid. Left to accumulate, this acid destroys the outer enamel of the tooth, irritates the gums to the point of bleeding, and produces foul breath. Plaque starts forming again on teeth four to 12 hours after brushing, so brushing a minimum of twice a day is necessary for adequate oral hygiene.

When plaque is not regularly removed by brushing and flossing, it hardens into a yellow or brown mineral deposit called tartar or calculus. This formation is crusty and provides additional rough surfaces for the growth of plaque. When tartar forms below the gumline, it can lead to periodontal (gum) disease.

Gingivitis is an early form of periodontal disease, characterized by inflammation of the gums with painless bleeding during brushing and flossing. This common condition is reversible with proper dental care but if left untreated, it will progress into a more serious periodontal disease, periodontitis.

Periodontitis is a gum disease that destroys the structures supporting the teeth, including bone. Without support, the teeth will loosen and may fall out or have to be removed. To diagnose periodontitis, a dentist looks for gums that are red, swollen, bleeding, and shrinking away from the teeth, leaving widening spaces between teeth and exposed root surfaces vulnerable to decay.

KEY TERMS

Calculus—A hardened yellow or brown mineral deposit from unremoved plaque; also called tartar.

Cavity—A hole or weak spot in the tooth surface caused by decay.

Gingivitis—Inflammation of the gums, seen as painless bleeding during brushing and flossing.

Interdental—Between the teeth.

Periodontal—Pertaining to the gums.

Periodontitis—A gum disease that destroys the structures supporting the teeth, including bone.

Plaque—A thin, sticky, colorless film of bacteria that forms on teeth.

Tartar—A hardened yellow or brown mineral deposit from unremoved plaque; also called calculus.

Tooth decay, also called dental caries or cavities, is a common dental problem that results when the acid produced by plaque bacteria destroys the outer surface of a tooth. A dentist will remove the decay and fill the cavity with an appropriate dental material to restore and protect the tooth; left untreated, the decay will expand, destroying the entire tooth and causing significant pain.

Normal results

With proper brushing and flossing, oral hygiene may be maintained and oral health problems may be avoided. Older adults may no longer assume that they will lose all of their teeth in their lifetime. Regular oral care preserves speech and eating functions, thus prolonging the quality of life.

Resources

ORGANIZATIONS

American Dental Association. 211 E. Chicago Ave., Chicago, IL 60611. (312) 440-2500. <<http://www.ada.org>>.

American Dental Hygienists' Association. 444 North Michigan Ave., Chicago, IL 60611. (800) 847-6718.

OTHER

Healthtouch Online Page. <<http://www.healthtouch.com>>.

Bethany Thivierge

Oral hypoglycemics see **Antidiabetic drugs**

Orbital and periorbital cellulitis

Definition

Periorbital **cellulitis** is an inflammation and infection of the eyelid and the skin surrounding the eye. Orbital cellulitis affects the eye socket (orbit) as well as the skin closest to it.

Description

Inside the eyelid is a septum. The septum divides the eyelid into outer and inner areas. This orbital septum helps prevent the spread of infection to the eye socket. Periorbital and orbital cellulitis are more common in children than in adults. Periorbital cellulitis, which accounts for 85–90% of all ocular cellulitis, usually occurs in children under the age of five. Responsible for the remaining 10–15% of these infections, orbital cellulitis is most common in children over the age of five.

These conditions usually begin with swelling or inflammation of one eye. Infection spreads rapidly and can cause serious problems that affect the eye or the whole body.

Causes and symptoms

Orbital and periorbital cellulitis are usually caused by infection of the sinuses near the nose. Insect bites or injuries that break the skin cause about one-third of these cellulitis infections. Orbital and periorbital cellulitis may also occur in people with a history of dental infections.

The blood of about 33 of every 100 patients with orbital or periorbital cellulitis contains bacteria known to cause:

- acute ear infections
- inflammation of the epiglottis (the cartilage flap that covers the opening of the windpipe during swallowing)
- **meningitis** (inflammation of the membranes that enclose and protect the brain)
- pneumonia
- sinus infection.

People with periorbital cellulitis will have swollen, painful lids and redness, but probably no **fever**. About one child in five has a runny nose, and 20% have conjunctivitis. **Conjunctivitis**, also called pinkeye, is an inflammation of the mucous membrane that lines

the eyelid and covers the front white part of the eye. It can be caused by allergy, irritation, or bacterial or viral infection.

As well as a swollen lid, other symptoms of orbital cellulitis include:

- bulging or displacement of the eyeball (proptosis)
- Chemosis (swelling of the mucous membrane of the eyeball and eyelid as a result of infection, injury, or systemic disorders like anemia or **kidney disease**)
- diminished ability to see clearly
- eye **pain**
- fever
- paralysis of nerves that control eye movements (**ophthalmoplegia**)

Diagnosis

An eye doctor may use special instruments to open a swollen lid in order to:

- examine the position of the eyeball
- evaluate eye movement
- test the patient's vision.

If the source of infection is not apparent, the position of the eyeball may suggest its location. **Computed tomography scans** (CT scans) can indicate which sinuses and bones are involved or whether abscesses have developed.

Treatment

A child who has orbital or periorbital cellulitis should be hospitalized without delay. Antibiotics are used to stop the spread of infection and prevent damage to the optic nerve, which transmits visual images to the brain.

Symptoms of optic-nerve damage or infection that has spread to sinus cavities close to the brain include:

- very limited ability to move the eye
- impaired response of the pupil to light and other stimulus
- loss of visual acuity
- papilledema (swelling of the optic disk—where the optic nerve enters the eye)

One or both eyes may be affected, and eye sockets or sinus cavities may have to be drained. These surgical procedures should be performed by an ophthalmologist or otolaryngologist.

Prognosis

If diagnosed promptly and treated with **antibiotics**, most orbital and periorbital cellulitis can be cured. These conditions are serious and need prompt treatment.

Infections that spread beyond the eye socket can cause:

- abscesses in the brain or in the protective membranes that enclose it
- bacterial meningitis
- blood clots
- vision loss

Resources

ORGANIZATIONS

American Academy of Ophthalmology. 655 Beach Street, P.O. Box 7424, San Francisco, CA 94120-7424.
< <http://www.eyenet.org> >.

American Optometric Association. 243 North Lindbergh Blvd., St. Louis, MO 63141. (314) 991-4100.
< <http://www.aoanet.org> >.

Maureen Haggerty

Orchiectomy see **Testicular surgery**

Orchiopexy see **Testicular surgery**

Orchitis

Definition

Orchitis is an inflammation of the testis, accompanied by swelling, **pain**, **fever**, and a sensation of heaviness in the affected area.

Description

Viral **mumps** is the most common cause of orchitis. Bacterial infections associated with the disorder are **tuberculosis**, **syphilis**, **gonorrhea**, and chlamydia. A mechanical injury to the groin area may also cause orchitis. Fifteen to twenty-five percent of males past the age of **puberty** with mumps develop orchitis. Epididymo-orchitis (inflammation of both testes and part of the spermatic duct) is the most common bacterial type of Orchitis. This form of the condition occurs most often in sexually active males fifteen years and older, and in men over 45 with enlarged prostates.

Causes and symptoms

The people most susceptible to orchitis are those with inadequate mumps inoculation and, in the case of sexually transmitted orchitis, those who practice unsafe sex or have a history of sexually transmitted disease. Inadequate protection of the groin area during contact sports or other potentially harmful physical activities may result in injury leading to orchitis. Symptoms of orchitis include swelling of one or both testicles, tenderness in the groin area, fever, **headache**, and **nausea**. Symptoms may also include bloody discharge from the penis, and pain during urination, intercourse, or ejaculation.

Diagnosis

In most cases, Orchitis can be diagnosed by an urologist, general practitioner, or emergency room physician. Diagnosis is usually based on the results of a **physical examination** and patient history. Other testing may include a **urinalysis** and **urine culture**, screening for chlamydia and gonorrhea, ultrasound imaging, or blood tests.

Treatment

Elevation and support of the scrotum, and the application of cold packs to the groin area give some relief from the pain of orchitis. Medication for pain such as codeine and meperidine may be given. Only the symptoms of viral mumps orchitis are treated. **Antibiotics** are used to alleviate orchitis that is bacterial in origin. Sexually transmitted orchitis (especially when resultant from chlamydia or gonorrhea) is often treated with the antibiotic Ceftriaxone in conjunction with azithromycin or doxycycline.

Alternative treatment

For relief from swelling, the drinking of dandelion tea is recommended in **traditional Chinese medicine** (TCM). Another traditional Chinese treatment for swelling is the application of a poultice of ground dandelion and aloe to the affected area. Homeopathic remedies to reduce swelling include apis mel, belladonna, and pulsatilla. Consult a homeopathic physician before taking or administering these remedies to ensure safe and correct dosage.

Prognosis

Orchitis is usually unilateral and lasts between one and two weeks. Atrophy of the scrotum occurs in 60% of orchitis cases. However, hormonal function is not affected and resulting sterility is rare from mumps.

KEY TERMS

Atrophy—A wasting away or withering.

Epididymo-orchitis—Inflammation of both the testes and a part of the spermatic duct system.

Unilateral—Affecting only one side.

Prevention

Keeping mumps inoculations current and diligently practicing safe sex are the best ways to prevent orchitis from occurring. For males involved in contact sports or other potentially harmful physical activities, the wearing of a protective cup over the genitals will help guard against mechanical injuries that could lead to orchitis.

Resources

PERIODICALS

Rodriguez, Rod. "Acute Scrotum Due to Epididymo-Orchitis in Male Children." *Impotence & Male Health Weekly Plus* February 1999.

OTHER

Mycyck, Mark, MD. "Orchitis from Emergency Medicine/Genitourinary." *Emedicine, Instant Access to the Minds of Medicine*. February 2001. <<http://www.emedicine.com/emerg/topic344.htm>>.

Mary Jane Tenerelli, MS

Organophosphates see **Insecticide poisoning**

Oriental sore see **Leishmaniasis**

Ornithosis see **Parrot fever**

Oroya fever see **Bartonellosis**

Orthopedic surgery

Definition

Orthopedic (sometimes spelled orthopaedic) surgery is surgery performed by a medical specialist, such as an orthopedist or orthopedic surgeon, trained to deal with problems that develop in the bones, joints, and ligaments of the human body.

Purpose

Orthopedic surgery corrects problems that arise in the skeleton and its attachments, the ligaments and tendons. It may also deal with some problems of the nervous system, such as those that arise from injury of the spine. These problems can occur at birth, through injury, or as the result of **aging**. They may be acute, as in injury, or chronic, as in many aging-related problems.

Orthopedics comes from two Greek words, *ortho*, meaning straight and *pais*, meaning child. Originally orthopedic surgeons dealt with bone deformities in children, using braces to straighten the child's bones. With the development of anesthesia and an understanding of the importance of aseptic technique in surgery, orthopedic surgeons extended their role to include surgery involving the bones and related nerves and connective tissue.

The terms orthopedic surgeon and orthopedist are used interchangeably today to indicate a medical doctor with special certification in orthopedics.

Many orthopedic surgeons maintain a general practice, while some specialize in one particular aspect of orthopedics, such as hand surgery, joint replacements, or disorders of the spine. Orthopedics treats both acute and chronic disorders. Some orthopedists specialize in trauma medicine and can be found in emergency rooms and trauma centers treating injuries. Others find their work overlapping with plastic surgeons, geriatric specialists, pediatricians, or podiatrists (**foot care** specialists). A rapidly growing area of orthopedics is sports medicine, and many sports medicine doctors are board certified orthopedists.

Precautions

Choosing an orthopedist is an important step in obtaining appropriate treatment. Patients looking for a qualified orthopedist should inquire if they are "board certified" by their accrediting organization.

Description

The range of treatments done by orthopedists is enormous. It can cover anything from **traction** to **amputation**, hand reconstruction to spinal fusion or joint replacements. They also treat broken bones, **strains** and **sprains**, and **dislocations**. Some specific procedures done by orthopedic surgeons are listed as separate entries in this book, including **arthroplasty**, **arthroscopic surgery**, **bone grafting**, **fasciotomy**, **fracture repair**, **kneecap removal**, and traction.

In general orthopedists are attached to a hospital, medical center, trauma center, or free-standing surgical center where they work closely with a surgical team including an anesthesiologist and surgical nurse. Orthopedic surgery can be performed under general, regional, or **local anesthesia**.

Much of the work of the surgeon involves adding foreign material to the body in the form of screws, wires, pins, tongs, and prosthetics to hold damaged bones in their proper alignment or to replace damaged bone or connective tissue. Great improvements have been made in the development of artificial limbs and joints, and in the materials available to repair damage to bones and connective tissue. As developments occur in the fields of metallurgy and plastics, changes will take place in orthopedic surgery that will allow the surgeon to more nearly duplicate the natural functions of the bones, joints, and ligaments, and to more accurately restore damaged parts to their original range of motion.

Preparation

Patients are usually referred to an orthopedic surgeon by a general physical or family doctor. Prior to any surgery, the patient undergoes extensive testing to determine the proper corrective procedure. Tests may include x rays, **computed tomography scans** (CT scans), **magnetic resonance imaging** (MRI), myelograms, diagnostic arthroplasty, and blood tests. The orthopedist will determine the history of the disorder and any treatments that were tried previously. A period of rest to the injured part may be recommended before surgery is prescribed.

Patients undergo standard blood and urine tests before surgery and, for major surgery, may be given an electrocardiogram or other diagnostic tests prior to the operation. Patients may choose to give some of their own blood to be held in reserve for their use in major surgery, such as knee replacement, where heavy bleeding is common.

Aftercare

Rehabilitation from orthopedic injuries can be a long, arduous task. The doctor will work closely with physical therapists to assure that the patient is receiving treatment that will enhance the range of motion and return function to the affected part.

Risks

As with any surgery, there is always the risk of excessive bleeding, infection, and allergic reaction to

KEY TERMS

Arthroplasty—The surgical reconstruction or replacement of a joint.

Prosthesis—A synthetic replacement for a missing part of the body, such as a knee or a hip.

Range of motion—The normal extent of movement (flexion and extension) of a joint.

anesthesia. Risks specifically associated with orthopedic surgery include inflammation at the site where foreign material (pins, prosthesis) is introduced into the body, infection as the result of surgery, and damage to nerves or to the spinal cord.

Normal results

Thousands of people have successful orthopedic surgery each year to recover from injuries or restore lost function. The degree of success in individual recoveries depends on the age and general health of the patient, the medical problem being treated, and the patient's willingness to comply with rehabilitative therapy after the surgery.

Resources

ORGANIZATIONS

American Academy of Orthopaedic Surgeons. 6300 North River Road, Rosemont, IL 60018-4262. (800) 823-8125. <<http://www.AAOS.org>>.

American Osteopathic Board of Orthopedic Surgery. <<http://www.netincom.com/aobos/about.html>>.

OTHER

Orthogate. <<http://owl.orthogate.org/>>.

Tish Davidson, A.M.

Orthopedic x rays see **Bone x rays**

Orthostatic hypotension

Definition

Orthostatic **hypotension** is an abnormal decrease in blood pressure when a person stands up. This may lead to **fainting**.

Description

When a person stands upright, a certain amount of blood normally pools in the veins of the ankles and legs. This pooling means that there is slightly less blood for the heart to pump and causes a drop in blood pressure. Usually, the body responds to this drop so quickly, a person is unaware of the change. The brain tells the blood vessels to constrict so they have less capacity to carry blood, and at the same time tells the heart to beat faster and harder. These responses last for a very brief time. If the body's response to a change in vertical position is slow or absent, the result is orthostatic hypotension. It is not a true disease, but the inability to regulate blood pressure quickly.

Causes and symptoms

Orthostatic hypotension has many possible causes. The most common cause is medications used to treat other conditions. **Diuretics** reduce the amount of fluid in the body which reduces the volume of blood. Medicines used to expand the blood vessels increase the vessel's ability to carry blood and so lower blood pressure.

If there is a severe loss of body fluid from **vomiting**, **diarrhea**, untreated diabetes, or even excessive sweating, blood volume will be reduced enough to lower blood pressure. Severe bleeding can also result in orthostatic hypotension.

Any disease or **spinal cord injury** that damages the nerves which control blood vessel diameter can cause orthostatic hypotension.

Symptoms of orthostatic hypotension include faintness, **dizziness**, confusion, or blurry vision, when standing up quickly. An excessive loss of blood pressure can cause a person to pass out.

Diagnosis

When a person experiences any of the symptoms above, a physician can confirm orthostatic hypotension if the person's blood pressure falls significantly on standing up and returns to normal when lying down. The physician will then look for the cause of the condition.

Treatment

When the cause of orthostatic hypotension is related to medication, it is often possible to treat it by reducing dosage or changing the prescription.

If it is caused by low blood volume, an increase in fluid intake and retention will solve the problem.

Medications designed to keep blood pressure from falling can be used when they will not interfere with other medical problems.

When orthostatic hypotension cannot be treated, the symptoms can be significantly reduced by remembering to stand up slowly or by wearing elastic stockings.

Prognosis

The prognosis for people who have orthostatic hypotension depends on the underlying cause of the problem.

Prevention

There is no way to prevent orthostatic hypotension, since it is usually the result of another medical condition.

Resources

ORGANIZATIONS

National Heart, Lung and Blood Institute. P.O. Box 30105, Bethesda, MD 20824-0105. (301) 251-1222. <<http://www.nhlbi.nih.gov>>.

National Organization for Rare Disorders. P.O. Box 8923, New Fairfield, CT 06812-8923. (800) 999-6673. <<http://www.rarediseases.org>>.

Dorothy Elinor Stonely

Orthotopic transplantation see **Liver transplantation**

Osgood-Schlatter disease see **Osteochondroses**

Osteitis deformans see **Paget's disease of bone**

Osteoarthritis

Definition

Osteoarthritis (OA), which is also known as osteoarthrosis or degenerative joint disease (DJD), is a progressive disorder of the joints caused by gradual loss of cartilage and resulting in the development of bony spurs and cysts at the margins of the joints. The

name osteoarthritis comes from three Greek words meaning bone, joint, and inflammation.

Description

OA is one of the most common causes of disability due to limitations of joint movement, particularly in people over 50. It is estimated that 2% of the United States population under the age of 45 suffers from osteoarthritis; this figure rises to 30% of persons between 45 and 64, and 63–85% in those over 65. About 90% of the American population will have some features of OA in their weight-bearing joints by age 40. Men tend to develop OA at earlier ages than women.

OA occurs most commonly after 40 years of age and typically develops gradually over a period of years. Patients with OA may have joint **pain** on only one side of the body and it primarily affects the knees, hands, hips, feet, and spine.

Causes and symptoms

Osteoarthritis results from deterioration or loss of the cartilage that acts as a protective cushion between bones, particularly in weight-bearing joints such as the knees and hips. As the cartilage is worn away, the bone forms spurs, areas of abnormal hardening, and fluid-filled pockets in the marrow known as subchondral cysts. As the disorder progresses, pain results from deformation of the bones and fluid accumulation in the joints. The pain is relieved by rest and made worse by moving the joint or placing weight on it. In early OA, the pain is minor and may take the form of mild stiffness in the morning. In the later stages of OA, inflammation develops; the patient may experience pain even when the joint is not being used; and he or she may suffer permanent loss of the normal range of motion in that joint.

Until the late 1980s, OA was regarded as an inevitable part of **aging**, caused by simple “wear and tear” on the joints. This view has been replaced by recent research into cartilage formation. OA is now considered to be the end result of several different factors contributing to cartilage damage, and is classified as either primary or secondary.

Primary osteoarthritis

Primary OA results from abnormal stresses on weight-bearing joints or normal stresses operating on weakened joints. Primary OA most frequently affects the finger joints, the hips and knees, the cervical and lumbar spine, and the big toe. The enlargements of the

finger joints that occur in OA are referred to as Heberden's and Bouchard's nodes. Some gene mutations appear to be associated with OA. **Obesity** also increases the pressure on the weight-bearing joints of the body. Finally, as the body ages, there is a reduction in the ability of cartilage to repair itself. In addition to these factors, some researchers have theorized that primary OA may be triggered by enzyme disturbances, bone disease, or liver dysfunction.

Secondary osteoarthritis

Secondary OA results from chronic or sudden injury to a joint. It can occur in any joint. Secondary OA is associated with the following factors:

- trauma, including sports injuries
- repetitive **stress** injuries associated with certain occupations (like the performing arts, construction or assembly line work, computer keyboard operation, etc.)
- repeated episodes of **gout** or septic arthritis
- poor posture or bone alignment caused by developmental abnormalities
- metabolic disorders

Diagnosis

History and physical examination

The two most important diagnostic clues in the patient's history are the pattern of joint involvement and the presence or absence of **fever**, rash, or other symptoms outside the joints. As part of the physical examination, the doctor will touch and move the patient's joint to evaluate swelling, limitations on the range of motion, pain on movement, and crepitus (a cracking or grinding sound heard during joint movement).

Diagnostic imaging

There is no laboratory test that is specific for osteoarthritis. Treatment is usually based on the results of diagnostic imaging. In patients with OA, x-rays may indicate narrowed joint spaces, abnormal density of the bone, and the presence of subchondral cysts or bone spurs. The patient's symptoms, however, do not always correlate with x-ray findings. Magnetic resonance imaging (MRI) and computed tomography scans (CT scans) can be used to determine more precisely the location and extent of cartilage damage.

Treatment

Treatment of OA patients is tailored to the needs of each individual. Patients vary widely in the location

of the joints involved, the rate of progression, the severity of symptoms, the degree of disability, and responses to specific forms of treatment. Most treatment programs include several forms of therapy.

Patient education and psychotherapy

Patient education is an important part of OA treatment because of the highly individual nature of the disorder and its potential impacts on the patient's life. Patients who are depressed because of changes in employment or recreation usually benefit from counseling. The patient's family should be involved in discussions of coping, household reorganization, and other aspects of the patient's disease and treatment regimen.

Medications

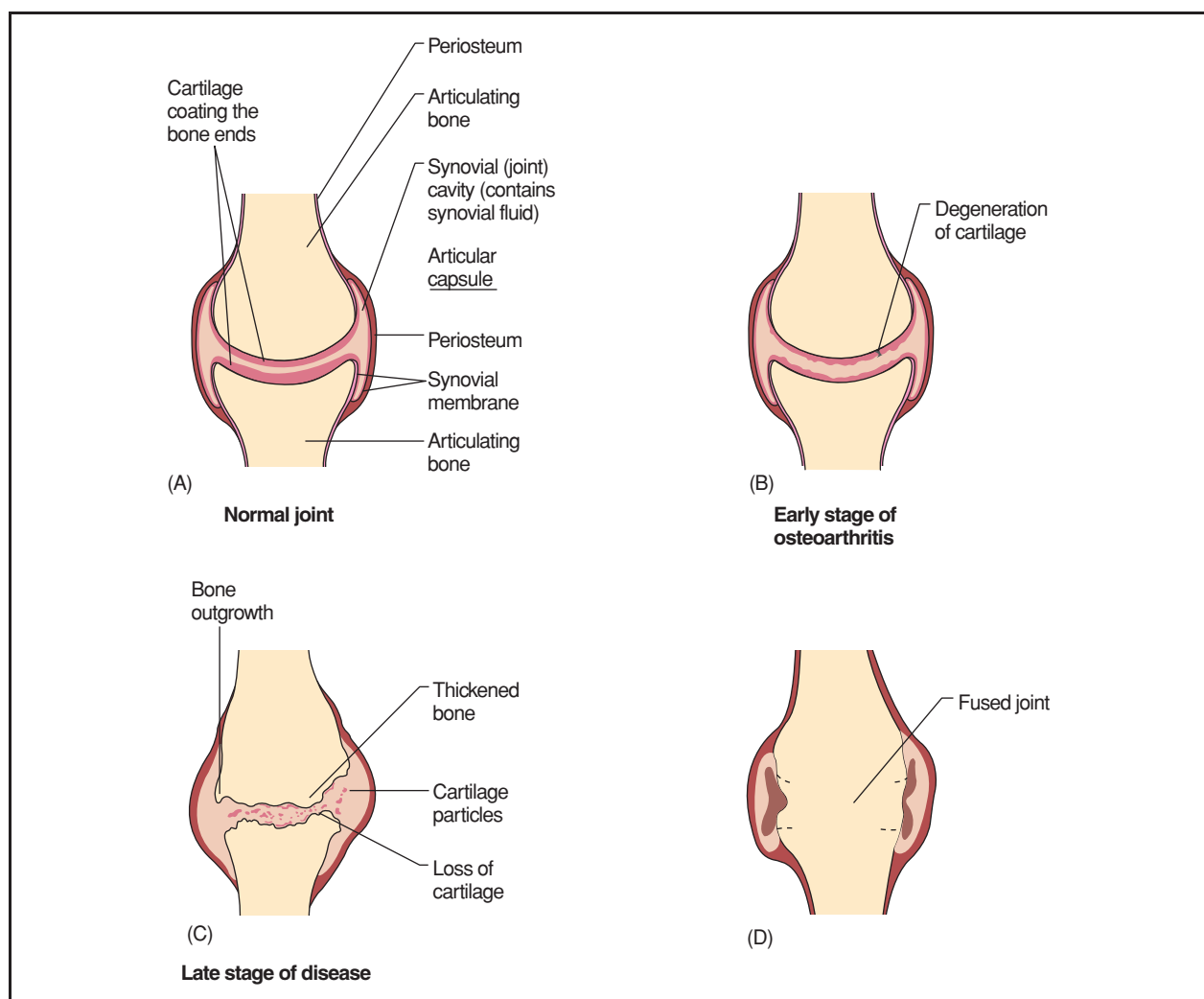
Patients with mild OA may be treated only with pain relievers such as **acetaminophen** (Tylenol). Most patients with OA, however, are given **nonsteroidal anti-inflammatory drugs**, or NSAIDs. These include compounds such as ibuprofen (Motrin, Advil), ketoprofen (Orudis), and flurbiprofen (Ansaid). The NSAIDs have the advantage of relieving inflammation as well as pain. They also have potentially dangerous side effects, including stomach ulcers, sensitivity to sun exposure, kidney disturbances, and nervousness or depression.

Some OA patients are treated with **corticosteroids** injected directly into the joints to reduce inflammation and slow the development of Heberden's nodes. Injections should not be regarded as a first-choice treatment and should be given only two or three times a year.

Most recently, a new class of NSAIDs, known as the cyclo-oxygenase-2 (COX-2) inhibitors have been studied and approved for the treatment of OA. These **COX-2 inhibitors** work to block the enzyme COX-2, which stimulates inflammatory responses in the body. They work to decrease both the inflammation and joint pain of OA, but without the high risk of gastrointestinal ulcers and bleeding seen with the traditional NSAIDs. This is due to the fact that they do not block COX-1, which is another enzyme that has protective effects on the stomach lining. The COX-2 inhibitors included celecoxib (Celebrex) and rofecoxib (Vioxx). Celecoxib is taken once or twice daily, and rofecoxib once daily.

Physical therapy

Patients with OA are encouraged to **exercise** as a way of keeping joint cartilage lubricated. Exercises that increase balance, flexibility, and range of motion



The progression of osteoarthritis. (Illustration by Hans & Cassady.)

are recommended for OA patients. These may include walking, swimming and other water exercises, **yoga** and other stretching exercises, or isometric exercises.

Physical therapy may also include massage, moist hot packs, or soaking in a hot tub.

Surgery

Surgical treatment of osteoarthritis may include the replacement of a damaged joint with an artificial part or appliance; surgical fusion of spinal bones; scraping or removal of damaged bone from the joint; or the removal of a piece of bone in order to realign the bone.

Protective measures

Depending on the location of the affected joint, patients with OA may be advised to use neck braces or

collars, crutches, canes, hip braces, knee supports, bed boards, or elevated chair and toilet seats. They are also advised to avoid unnecessary knee bending, stair climbing, or lifting of heavy objects.

New treatments

Since 1997, several new methods of treatment for OA have been investigated. Although they are still being developed and tested, they appear to hold promise. They include:

- **Disease-modifying drugs.** These compounds may be useful in assisting the body to form new cartilage or improve its repair of existing cartilage.
- **Hyaluronic acid.** Injections of this substance may help to lubricate and protect cartilage, thereby promoting flexibility and reduced pain. These agents

include hyaluronan (Hyalgan) and hylan G-F20 (Synvisc).

- Cartilage transplantation. This technique is presently used in Sweden.

Alternative treatment

Diet

Food intolerance can be a contributing factor in OA, although this is more significant in rheumatoid arthritis. Dietary suggestions that may be helpful for people with OA include emphasizing high-fiber, complex-carbohydrate foods, while minimizing fats. Plants in the Solanaceae family, such as tomatoes, eggplant, and potatoes, should be avoided, as should refined and processed foods. Foods that are high in bioflavonoids (berries as well as red, orange, and purple fruits and vegetables) should be eaten often.

Nutritional supplements

In the past several years, a combination of glucosamine and chondroitin sulfate has been proposed as a dietary supplement that helps the body maintain and repair cartilage. Studies conducted in Europe have shown the effectiveness of this treatment in many cases. These substances are nontoxic and do not require prescriptions. Other supplements that may be helpful in the treatment of OA include the antioxidant **vitamins** and **minerals** (vitamins A, C, E, selenium, and zinc) and the B vitamins, especially vitamins B₆ and B₅.

Naturopathy

Naturopathic treatment for OA includes hydrotherapy, diathermy (deep-heat therapy), **nutritional supplements**, and botanical preparations, including yucca, devil's claw (*Harpagophytum procumbens*), and hawthorn (*Crataegus laevigata*) berries.

Traditional Chinese medicine (TCM)

Practitioners of Chinese medicine treat arthritis with suction cups, massage, moxibustion (warming an area of skin by burning a herbal wick a slight distance above the skin), the application of herbal poultices, and internal doses of Chinese herbal formulas.

Other alternatives

Recently, several alternative treatments for OA have received considerable attention and study. These include:

- transcutaneous **electrical nerve stimulation** (TENS)
- magnet therapy

KEY TERMS

Bouchard's nodes—Swelling of the middle joint of the finger.

Cartilage—Elastic connective tissue that covers and protects the ends of bones.

Heberden's nodes—Swelling or deformation of the finger joints closest to the fingertips.

Primary osteoarthritis—OA that results from hereditary factors or stresses on weight-bearing joints.

Secondary osteoarthritis—OA that develops following joint surgery, trauma, or repetitive joint injury.

Subchondral cysts—Fluid-filled sacs that form inside the marrow at the ends of bones as part of the development of OA.

- therapeutic touch
- acupuncture
- yoga

Prognosis

OA is a progressive disorder without a permanent cure. In some patients, the rate of progression can be slowed by weight loss, appropriate exercise, surgical treatment, and the use of alternative therapies.

Resources

PERIODICALS

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Liz Meszaros

Osteoarthrosis see **Osteoarthritis**

Osteochondroses

Definition

Osteochondroses is a group of diseases of children and adolescents in which localized tissue death (necrosis) occurs, usually followed by full regeneration of healthy bone tissue. The singular term is osteochondrosis.

Description

During the years of rapid bone growth, blood supply to the growing ends of bones (epiphyses) may become insufficient resulting in necrotic bone, usually near joints. The term avascular necrosis is used to describe osteochondrosis. Since bone is normally undergoing a continuous rebuilding process, the necrotic areas are most often self-repaired over a period of weeks or months.

Osteochondrosis can affect different areas of the body and is often categorized by one of three locations: articular, non-articular, and physeal.

Physeal osteochondrosis is known as Scheuermann's disease. It occurs in the spine at the intervertebral joints (physes), especially in the chest (thoracic) region.

Articular disease occurs at the joints (articulations). One of the more common forms is Legg-Calvé-Perthes disease, occurring at the hip. Other forms include Köhler's disease (foot), Freiberg's disease (second toe), and Panner's disease (elbow). Freiberg's disease is the one type of osteochondrosis that is more common in females than in males. All others affect the sexes equally.

Non-articular osteochondrosis occurs at any other skeletal location. For instance, Osgood-Schlatter disease of the tibia (the large inner bone of the leg between the knee and ankle) is relatively common.

Osteochondritis dissecans is a form of osteochondrosis in which loose bone fragments may form in a joint.

Causes and symptoms

Many theories have been advanced to account for osteochondrosis, but none has proven fully satisfactory. **Stress** and **ischemia** (reduced blood supply) are two of the most commonly mentioned factors. Athletic young children are often affected when they overstress their developing limbs with a particular repetitive motion. Many cases are idiopathic, meaning that no specific cause is known.

The most common symptom for most types of osteochondrosis is simply **pain** at the affected joint,

especially when pressure is applied. Locking of a joint or limited range of motion at a joint can also occur.

Scheuermann's disease can lead to serious **kyphosis** (hunchback condition) due to erosion of the vertebral bodies. Usually, however, the kyphosis is mild, causing no further symptoms and requiring no special treatment.

Diagnosis

Diagnosis can be confirmed by x-ray findings.

Treatment

Conservative treatment is usually attempted first. In many cases, simply resting the affected body part for a period of days or weeks will bring relief. A cast may be applied if needed to prevent movement of a joint.

Surgical intervention may be needed in some cases of osteochondritis dissecans to remove abnormal bone fragments in a joint.

Prognosis

Accurate prediction of the outcome for individual patients is difficult with osteochondrosis. Some patients will heal spontaneously. Others will heal with little treatment other than keeping weight or stress off the affected limb. The earlier the age of onset, the better the prospects for full recovery. Surgical intervention is often successful in osteochondritis dissecans.

Prevention

No preventive measures are known.

Resources

BOOKS

Eilert, Robert E., and Gaia Geogopoulos. "Orthopedics." In *Current Pediatric Diagnosis and Treatment*, edited by W. W. Hay, Jr., et al. Stamford: Appleton & Lange, 1997.

Victor Leipzig, PhD

Osteogenesis imperfecta

Definition

Osteogenesis imperfecta (OI) is a group of genetic diseases of collagen in which the bones are formed improperly, making them fragile and prone to breaking.

Description

Collagen is a fibrous protein material. It serves as the structural foundation of skin, bone, cartilage, and ligaments. In osteogenesis imperfecta, the collagen produced is abnormal and disorganized. This results in a number of abnormalities throughout the body, the most notable being fragile, easily broken bones.

There are four forms of OI, Types I through IV. Of these, Type II is the most severe, and is usually fatal within a short time after birth. Types I, III, and IV have some overlapping and some distinctive symptoms, particularly weak bones.

Evidence suggests that OI results from abnormalities in the collagen gene COL1A1 or COL1A2, and possibly abnormalities in other genes. In OI Type I, II, and III, the gene map locus is 17q21.31-q22, 7q22.1, and in OI Type IV, the gene map locus is 17q21.31-q22.

In OI, the genetic abnormality causes one of two things to occur. It may direct cells to make an altered collagen protein and the presence of this altered collagen causes OI Type II, III, or IV. Alternately, the dominant altered gene may fail to direct cells to make any collagen protein. Although some collagen is produced by instructions from the normal gene, an overall decrease in the total amount of collagen produced results in OI Type I.

A child with only one parent who is a carrier of a single altered copy of the gene has no chance of actually having the disease, but a 50% chance of being a carrier.

If both parents have OI caused by an autosomal dominant gene change, there is a 75% chance that the child will inherit one or both OI genes. In other words, there is a 25% chance the child will inherit only the mother's OI gene (and the father's unaffected gene), a 25% chance the child will inherit only the father's OI gene (and the mother's unaffected gene), and a 25% chance the child will inherit both parents' OI genes. Because this situation has been uncommon, the outcome of a child inheriting two OI genes is hard to predict. It is likely that the child would have a severe, possibly lethal, form of the disorder.

About 25% of children with OI are born into a family with no history of the disorder. This occurs when the gene spontaneously mutates in either the sperm or the egg before the child's conception. No triggers for this type of mutation are known. This is called a new dominant mutation. The child has a 50% chance of passing the disorder on to his or her children. In most cases, when a family with no history of

OI has a child with OI, they are not at greater risk than the general population for having a second child with OI, and unaffected siblings of a person with OI are at no greater risk of having children with OI than the general population.

In studies of families into which infants with OI Type II were born, most of the babies had a new dominant mutation in a collagen gene. In some of these families, however, more than one infant was born with OI. Previously, researchers had seen this recurrence as evidence of recessive inheritance of this form of OI. More recently, however, researchers have concluded that the rare recurrence of OI to a couple with a child with autosomal dominant OI is more likely due to gonadal mosaicism. Instead of mutation occurring in an individual sperm or egg, it occurs in a percentage of the cells that give rise to a parent's multiple sperm or eggs. This mutation, present in a percentage of his or her reproductive cells, can result in more than one affected child without affecting the parent with the disorder. An estimated 2%–4% of families into which an infant with OI Type II is born are at risk of having another affected child because of gonadal mosaicism.

Demographics

OI affects equal numbers of males and females. It occurs in about one of every 20,000 births.

Causes and symptoms

OI is usually inherited as an autosomal dominant condition. In autosomal dominant inheritance, a single abnormal gene on one of the autosomal chromosomes (one of the first 22 "non-sex" chromosomes) from either parent can cause the disease. One of the parents will have the disease (since it is dominant) and is the carrier. Only one parent needs to be a carrier in order for the child to inherit the disease. A child who has one parent with the disease has a 50% chance of also having the disease.

Type I

This is the most common and mildest type. Among the common features of Type I are the following:

- bones are predisposed to fracture, with most fractures occurring before **puberty**; people with OI type I typically have about 20–40 **fractures** before puberty.
- stature is normal or near-normal
- joints are loose and muscle tone is low

- usually sclera (whites of the eyes) have blue, purple, or gray tint
- face shape is triangular
- tendency toward **scoliosis** (a curvature of the spine)
- bone deformity is absent or minimal
- dentinogenesis imperfecta may occur, causing brittle teeth
- hearing loss is a possible symptom, often beginning in early 20s or 30s
- structure of collagen is normal, but the amount is less than normal

Type II

Sometimes called the lethal form, Type II is the most severe form of OI. Among the common features of Type II are the following:

- frequently, OI Type II is lethal at or shortly after birth, often as a result of respiratory problems
- fractures are numerous and bone deformity is severe
- stature is small with underdeveloped lungs
- collagen is formed improperly

Type III

Among the common features of Type III are the following:

- bones fracture easily (fractures are often present at birth, and x rays may reveal healed fractures that occurred before birth; people with OI Type III may have more than 100 fractures before puberty)
- stature is significantly shorter than normal
- sclera (whites of the eyes) have blue, purple, or gray tint
- joints are loose and muscle development is poor in arms and legs
- rib cage is barrel-shaped
- face shape is triangular
- scoliosis (a curvature of the spine) is present
- respiratory problems are possible
- bones are deformed and deformity is often severe
- dentinogenesis imperfecta may occur, causing brittle teeth
- hearing loss is possible
- collagen is formed improperly

Type IV

OI Type IV falls between Type I and Type III in severity. Among the common features of Type IV are the following:

- bones fracture easily, with most fractures occurring before puberty
- stature is shorter than average
- sclera (whites of the eyes) are normal in color, appearing white or near-white
- bone deformity is mild to moderate
- scoliosis (curvature of the spine) is likely
- rib cage is barrel-shaped
- face is triangular in shape
- dentinogenesis imperfecta may occur, causing brittle teeth
- hearing loss is possible
- collagen is formed improperly

Diagnosis

It is often possible to diagnose OI solely on clinical features and x-ray findings. Collagen or DNA tests may help confirm a diagnosis of OI. These tests generally require several weeks before results are known. Approximately 10–15% of individuals with mild OI who have collagen testing, and approximately 5% of those who have genetic testing, test negative for OI despite having the disorder.

Diagnosis is usually suspected when a baby has bone fractures after having suffered no apparent injury. Another indication is small, irregular, isolated bones in the sutures between the bones of the skull (wormian bones). Sometimes the bluish sclera serves as a diagnostic clue. Unfortunately, because of the unusual nature of the fractures occurring in a baby who cannot yet move, some parents have been accused of **child abuse** before the actual diagnosis of osteogenesis imperfecta was reached.

Prenatal diagnosis

Testing is available to assist in prenatal diagnosis. Women with OI who become pregnant, or women who conceive a child with a man who has OI, may wish to explore prenatal diagnosis. Because of the relatively small risk (2–4%) of recurrence of OI Type II in a family, families may opt for ultrasound studies to determine if a developing fetus has the disorder.

Ultrasound is the least invasive procedure for prenatal diagnosis, and carries the least risk. Using

ultrasound, a doctor can examine the fetus's skeleton for bowing of the leg or arm bones, fractures, shortening, or other bone abnormalities that may indicate OI. Different forms of OI may be detected by ultrasound in the second trimester. The reality is that when it occurs as a new dominant mutation, it is found inadvertently on ultrasound, and it may be difficult to know the diagnosis until after delivery since other genetic conditions can cause bowing and/or fractures prenatally.

Chorionic villus sampling is a procedure to obtain chorionic villi tissue for testing. Examination of fetal collagen proteins in the tissue can reveal information about the quantitative or qualitative collagen defects that leads to OI. When a parent has OI, it is necessary for the affected parent to have the results of his or her own collagen test available. Chorionic villus sampling can be performed at 10–12 weeks of **pregnancy**.

Amniocentesis is a procedure that involves inserting a thin needle into the uterus, into the amniotic sac, and withdrawing a small amount of amniotic fluid. DNA can be extracted from the fetal cells contained in the amniotic fluid and tested for the specific mutation known to cause OI in that family. This technique is useful only when the mutation causing OI in a particular family has been identified through previous **genetic testing** of affected family members, including previous pregnancies involving a baby with OI. Amniocentesis is performed at 16–18 weeks of pregnancy.

Treatment

There are no treatments available to cure OI, nor to prevent most of its complications. Most treatments are aimed at treating the fractures and bone deformities caused by OI. Splints, casts, braces, and rods are all used. Rodding refers to a surgical procedure in which a metal rod is implanted within a bone (usually the long bones of the thigh and leg). This is done when bowing or repeated fractures of these bones has interfered with a child's ability to begin to walk.

Other treatments include **hearing aids** and early capping of teeth. Patients may require the use of a walker or wheelchair. Pain may be treated with a variety of medications. **Exercise** is encouraged as a means to promote muscle and bone strength. Swimming is a form of exercise that puts a minimal amount of strain on muscles, joints, and bones. Walking is encouraged for those who are able.

Smoking, excessive alcohol and **caffeine** consumption, and steroid medications may deplete bone and exacerbate bone fragility.

KEY TERMS

Collagen—The main supportive protein of cartilage, connective tissue, tendon, skin, and bone.

Ligament—A type of connective tissue that connects bones or cartilage and provides support and strength to joints.

Mutation—A permanent change in the genetic material that may alter a trait or characteristic of an individual, or manifest as disease, and can be transmitted to offspring.

Sclera—The tough white membrane that forms the outer layer of the eyeball.

Scoliosis—An abnormal, side-to-side curvature of the spine.

Alternative treatment such as **acupuncture**, naturopathic therapies, hypnosis, relaxation training, visual imagery, and **biofeedback** have all been used to try to decrease the constant **pain** of fractures.

Prognosis

Lifespan for people with OI Type I, III, and IV is not generally shortened. The prognosis for people with these types of OI is quite variable, depending on the severity of the disorder and the number and severity of the fractures and bony deformities.

Fifty percent of all babies with OI Type II are stillborn. The rest of these babies usually die within a very short time after birth. In recent years, some people with Type II have lived into young adulthood.

Resources

PERIODICALS

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ORGANIZATIONS

Children's Brittle Bone Foundation. 7701 95th St., Pleasant Prairie, WI 53158. (847) 433-498. <<http://www.cbbf.org>>.

OTHER

“Osteogenesis Imperfecta.” *National Institutes of Health Osteoporosis and Related Bone Diseases–National Resource Center*. < <http://www.osteoporosis.org/oi.html> >.

Jennifer F. Wilson, MS

Osteogenic sarcoma see **Sarcomas**

Osteomalacia see **Vitamin D deficiency**

Osteomyelitis

Definition

Osteomyelitis refers to a bone infection, almost always caused by a bacteria. Over time, the result can be destruction of the bone itself.

Description

Bone infections may occur at any age. Certain conditions increase the risk of developing such an infection, including sickle cell anemia, injury, the presence of a foreign body (such as a bullet or a screw placed to hold together a broken bone), intravenous drug use (such as heroin), diabetes, **kidney dialysis**, surgical procedures to bony areas, untreated infections of tissue near a bone (for example, extreme cases of untreated sinus infections have led to osteomyelitis of the bones of the skull).

Causes and symptoms

Staphylococcus aureus, a bacterium, is the most common organism involved in osteomyelitis. Other types of organisms include the mycobacterium which causes **tuberculosis**, a type of Salmonella bacteria in patients with sickle cell anemia, *Pseudomonas aeruginosa* in drug addicts, and organisms which usually reside in the gastrointestinal tract in the elderly. Extremely rarely, the viruses which cause **chickenpox** and **smallpox** have been found to cause a viral osteomyelitis.

There are two main ways that infecting bacteria find their way to bone, resulting in the development of osteomyelitis. These include:

- Spread via the bloodstream; 95% of these types of infections are due to *Staphylococcus aureus*. In this situation, the bacteria travels through the bloodstream to reach the bone. In children, the most likely site of infection is within one of the long bones,

particularly the thigh bone (femur), one of the bones of the lower leg (tibia), or the bone of the upper arm (humerus). This is because in children these bones have particularly extensive blood circulation, making them more susceptible to invasion by bacteria. Different patterns of blood circulation in adults make the long bones less well-served by the circulatory system. These bones are therefore unlikely to develop osteomyelitis in adult patients. Instead, the bones of the spine (vertebrae) receive a lot of blood flow. Therefore, osteomyelitis in adults is most likely to affect a vertebra. Drug addicts may have osteomyelitis in the pubic bone or clavicle.

- Spread from adjacent infected soft tissue; about 50% of all such cases are infected by *Staphylococcus aureus*. This often occurs in cases where recent surgery or injury has result in a soft tissue infection. The bacteria can then spread to nearby bone, resulting in osteomyelitis. Patients with diabetes are particularly susceptible to this source of osteomyelitis. The diabetes interferes with both nerve sensation and good blood flow to the feet. Diabetic patients are therefore prone to developing poorly healing **wounds** to their feet, which can then spread to bone, causing osteomyelitis.

Acute osteomyelitis refers to an infection which develops and peaks over a relatively short period of time. In children, acute osteomyelitis usually presents itself as **pain** in the affected bone, tenderness to pressure over the infected area, **fever** and chills. Patients who develop osteomyelitis, due to spread from a nearby area of soft tissue infection, may only note poor healing of the original wound or infection.

Adult patients with osteomyelitis of the spine usually have a longer period of dull, aching pain in the back, and no fever. Some patients note pain in the chest, abdomen, arm, or leg. This occurs when the inflammation in the spine causes pressure on a nerve root serving one of these other areas. The lower back is the most common location for osteomyelitis. When caused by tuberculosis, osteomyelitis usually affects the thoracic spine (that section of the spine running approximately from the base of the neck down to where the ribs stop).

When osteomyelitis is not properly treated, a chronic (long-term) type of infection may occur. In this case, the infection may wax and wane indefinitely, despite treatment during its active phases. An abnormal opening in the skin overlaying the area of bone infection (called a sinus tract) may occasionally drain pus. This type of smoldering infection may also result in areas of dead bone, called sequestra. These areas occur when the infection interferes with blood flow to

a particular part of the bone. Such sequestra lack cells called osteocytes, which in normal bone are continuously involved in the process of producing bony material.

Diagnosis

Diagnosis of osteomyelitis involves several procedures. Blood is usually drawn and tested to demonstrate an increased number of the infection-fighting white blood cells (particularly elevated in children with acute osteomyelitis). Blood is also cultured in a laboratory, a process which allows any bacteria present to multiply. A specimen from the culture is then specially treated, and examined under a microscope to try to identify the causative bacteria.

Injection of certain radioactive elements into the bloodstream, followed by a series of x-ray pictures, called a scan (radionuclide scanning), will reveal areas of bone inflammation. Another type of scan used to diagnose osteomyelitis is called **magnetic resonance imaging**, or MRI

When pockets of pus are available, or overlaying soft tissue infection exists, these can serve as sources for samples which can be cultured to allow identification of bacteria present. A long, sharp needle can be used to obtain a specimen of bone (biopsy), which can then be tested to attempt to identify any bacteria present.

Treatment

Antibiotics are medications used to kill bacteria. These medications are usually given through a needle in a vein (intravenously) for at least part of the time. In children, these antibiotics can be given by mouth after initial treatment by vein. In adults, four to six weeks of intravenous antibiotic treatment is usually recommended, along with bed-rest for part or all of that time. Occasionally, a patient will have such extensive osteomyelitis that surgery will be required to drain any pockets of pus, and to clean the infected area.

Alternative treatment

General recommendations for the treatment of infections include increasing vitamin supplements, such as **vitamins** A and C. Liquid garlic extract is sometimes suggested. **Guided imagery** can help induce relaxation and improve pain, both of which are considered to improve healing. Herbs such as **echinacea** (*Echinacea* spp.), goldenseal (*Hydrastis canadensis*), Siberian ginseng (*Eleutherococcus senticosus*), and

KEY TERMS

Abscess—A pus-filled pocket of infection.

Femur—The thighbone.

Humerus—The bone of the upper arm.

Thoracic—Pertaining to the area bounded by the rib cage.

Tibia—One of the two bones of the lower leg.

myrrh (*Commiphora molmol*) are all suggested for infections. Juice therapists recommend drinking combinations of carrot, celery, beet, and cantaloupe juices. A variety of homeopathic remedies may be helpful, especially those used to counter inflammation.

Prognosis

Prognosis varies depending on how quickly an infection is identified, and what other underlying conditions exist to complicate the infection. With quick, appropriate treatment, only about 5% of all cases of acute osteomyelitis will eventually become chronic osteomyelitis. Patients with chronic osteomyelitis may require antibiotics periodically for the rest of their lives.

Prevention

About the only way to have any impact on the development of osteomyelitis involves excellent care of any wounds or injuries.

Resources

PERIODICALS

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Rosalyn Carson-DeWitt, MD

Osteopathic medicine see **Osteopathy**

Osteopathy

Definition

Osteopathy is a system and philosophy of health care that separated from traditional (allopathic) medical practice about a century ago. It places emphasis on

the musculoskeletal system, hence the name—osteo refers to bone and path refers to disease. Osteopaths also believe strongly in the healing power of the body and do their best to facilitate that strength. During this century, the disciplines of osteopathy and allopathic medicine have been converging.

Purpose

Osteopathy shares many of the same goals as traditional medicine, but places greater emphasis on the relationship between the organs and the musculoskeletal system as well as on treating the whole individual rather than just the disease.

Precautions

Pain is the chief reason patients seek musculoskeletal treatment. Pain is a symptom, not a disease by itself. Of critical importance is first to determine the cause of the pain. Cancers, brain or spinal cord disease, and many other causes may be lying beneath this symptom. Once it is clear that the pain is originating in the musculoskeletal system, treatment that includes manipulation is appropriate.

Description

History

Osteopathy was founded in the 1890s by Dr. Andrew Taylor, who believed that the musculoskeletal system was central to health. The primacy of the musculoskeletal system is also fundamental to **chiropractic**, a related health discipline. The original theory behind both approaches presumed that energy flowing through the nervous system is influenced by the supporting structure that encase and protect it—the skull and vertebral column. A defect in the musculoskeletal system was believed to alter the flow of this energy and cause disease. Correcting the defect cured the disease. Defects were thought to be misalignments—parts out of place by tiny distances. Treating misalignments became a matter of restoring the parts to their natural arrangement by adjusting them.

As medical science advanced, defining causes of disease and discovering cures, schools of osteopathy adopted modern science, incorporated it into their curriculum, and redefined their original theory of disease in light of these discoveries. Near the middle of the 20th century the equivalence of medical education between osteopathy and allopathic medicine was recognized, and the D.O. degree (Doctor of Osteopathy) was granted official parity with the

M.D. (Doctor of Medicine) degree. Physicians could adopt either set of initials.

However, osteopaths have continued their emphasis on the musculoskeletal system and their traditional focus on “whole person” medicine. As of 1998, osteopaths constitute 5.5% of American physicians, approximately 45,000. They provide 100 million patient visits a year. From its origins in the United States, osteopathy has spread to countries all over the world.

Practice

Osteopaths, chiropractors, and physical therapists are the experts in manipulations (adjustments). The place of manipulation in medical care is far from settled, but millions of patients find relief from it. Particularly backs, but also necks, command most of the attention of the musculoskeletal community. This community includes orthopedic surgeons, osteopaths, general and family physicians, orthopedic physicians, chiropractors, physical therapists, massage therapists, specialists in orthotics and prosthetics, and even some dentists and podiatrists. Many types of headaches also originate in the musculoskeletal system. Studies comparing different methods of treating musculoskeletal back, head, and neck pain have not reached a consensus, in spite of the huge numbers of people that suffer from it.

The theory behind manipulation focuses on joints, mostly those of the vertebrae and ribs. Some believe there is a very slight offset of the joint members—a subluxation. Others believe there is a vacuum lock of the joint surfaces, similar to two suction cups stuck together. Such a condition would squeeze joint lubricant out and produce abrasion of the joint surfaces with movement. Another theory focuses on weakness of the ligaments that support the joint, allowing it freedom to get into trouble. Everyone agrees that the result produces pain, that pain produces **muscle spasms and cramps**, which further aggravates the pain.

Some, but not all, practitioners in this field believe that the skull bones can also be manipulated. The skull is, in fact, several bones that are all moveable in infants. Whether they can be moved in adults is controversial. Other practitioners manipulate peripheral joints to relieve arthritis and similar afflictions.

Manipulation returns the joint to its normal configuration. There are several approaches. Techniques vary among practitioners more than between disciplines. Muscle relaxation of some degree is often required for the manipulation to be successful. This



Osteopathic physician demonstrating the articulation of a foot. (Photo Researchers, Inc. Reproduced by permission.)

can be done with heat or medication. Muscles can also be induced to relax by gentle but persistent stretching. The manipulation is most often done by a short, fast motion called a thrust, precisely in the right direction. A satisfying “pop” is evidence of success. Others prefer steady force until relaxation permits movement.

Return of the joint to its normal status may be only the first step in treating these disorders. There is a reason for the initial event. It may be a fall, a stumble, or a mild impact, in which case the manipulation is a cure. On the other hand, there may be a postural misalignment (such as a short leg), a limp, or a stretched ligament that permits the joint to slip back into dysfunction. Tension, as well as pain, for emotional reasons causes muscles to tighten. If the pain has been present for any length of time, there will also be muscle deterioration. The osteopathic approach to the whole person takes all these factors into account in returning the patient to a state of health.

Other repairs may be needed. A short leg is thought by some to be a subluxation in the pelvis that may be manipulated back into position. Other short legs may require a lift in one shoe. Long-standing pain requires additional methods of physical therapy

KEY TERMS

Orthotics—Mechanical devices that assist function.

Prosthetics—Mechanical devices that replace missing body parts.

to rehabilitate muscles, correct posture, and extinguish habits that arose to compensate for the pain. Medications that relieve muscle spasm and pain are usually part of the treatment. Psychological problems may need attention and medication.

Risks

Manipulation has rarely caused problems. Once in a while too forceful a thrust has damaged structures in the neck and caused serious problems. The most common adverse event, though, is misdiagnosis. Cancers have been missed; surgical back disease has been ignored until spinal nerves have been permanently damaged.

Normal results

Many patients find that one or a series of manipulations cures long-standing pain. Other patients need repeated treatments. Some do not respond at all. It is always a good idea to reassess any treatment that is not producing the expected results.

Resources

ORGANIZATIONS

American Association of Colleges of Osteopathic Medicine.
5550 Friendship Blvd., Suite 310, Chevy Chase, MD
20815-7231. (301) 968-4100. <<http://www.aacom.org>>.

American Osteopathic Association. <osteomed@wwa.com>
<<http://www.am-osteo-assn.org>>.

J. Ricker Polsdorfer, MD

Osteopetroses

Definition

Osteopetrosis (plural osteopetroses) is a rare hereditary disorder that makes bones increase in both density and fragility. A potentially fatal condition

that can deform bone structure and distort the appearance, osteopetrosis is also called chalk bones, ivory bones, or marble bones.

Description

Osteopetrosis occurs when bones are spongy or porous, or new bone is repeatedly added to calcified cartilage (hardened connective tissue).

Bone density begins to increase at birth or earlier, but symptoms may not become evident until adulthood. In mild cases, bone density increases at gradual, irregular intervals until full adult height is attained. Some bones are not affected.

More severe osteopetrosis progresses at a rapid pace and destroys bone structure. This condition involves bones throughout the body, but the lower jaw is never affected.

Types of osteopetroses

In early-onset osteopetrosis ends of the long bones of the arms and legs appear clubbed (widened and thickened) at birth, and bone density continues to increase sporadically or without pause. Early-onset osteopetroses can be a fatal condition, resulting in **death** before the age of two.

Malignant infantile osteopetrosis is most often discovered by the time a baby is a few months old. One-third of all malignant infantile osteopetroses cases result in death before the age of 10.

Intermediate osteopetrosis generally appears in children under 10. This condition, usually less severe than early-onset or malignant infantile osteopetrosis, is not life-threatening.

Symptoms of adult or delayed-onset osteopetrosis may not become evident until the child becomes a teenager or adult.

Relatively common in many parts of the world, Albers-Schönberg disease is a mild form of this condition. People who have this disease are born with normal bone structure. Bone density increases as they age but does not affect appearance, health, intelligence, or life span.

Causes and symptoms

Osteopetrosis is the result of a genetic defect that causes the body to add new bone more rapidly than existing bone disintegrates.

When fibrous or bony tissue invades bone marrow and displaces red blood cells, the patient may develop

anemia. Infection results when excess bone impairs the immune system, and hemorrhage can occur when platelet production is disrupted. When the skeleton grows so thick that nerves are unable to pass between bones, the patient may have a **stroke** or become blind or deaf.

Other symptoms associated with osteopetrosis include:

- bones that break easily and do not heal properly
- bruising
- convulsions
- enlargement of the liver, lymph glands, or spleen
- failure to thrive (delayed growth, weight gain, and development)
- hydrocephalus (fluid on the brain)
- macrocephaly (abnormal enlargement of the head)
- paralysis or loss of control of muscles in the face or eyes

Diagnosis

Osteopetrosis is usually diagnosed when x rays reveal abnormalities or increases in bone density. **Bone biopsy** can confirm the presence of osteopetrosis, but additional tests may be needed to distinguish one type of the disorder from another.

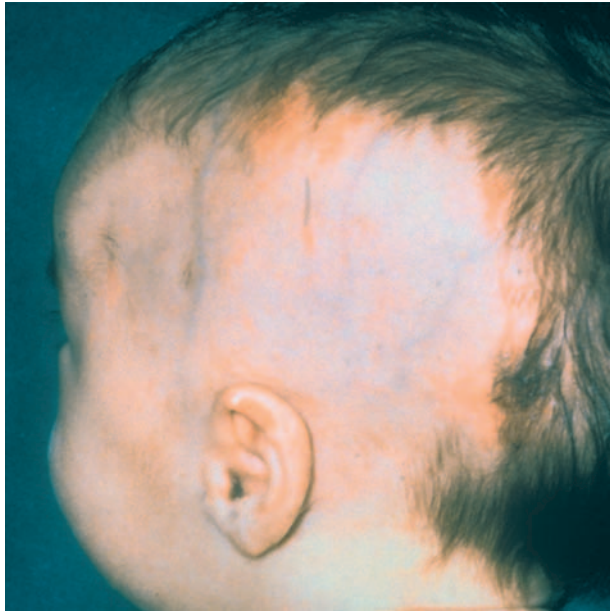
Treatment

High doses of vitamin D can stimulate cells responsible for disintegration of old bone and significantly alleviate symptoms of severe disease. Experimental interferon gamma 1-b therapy has been shown to reduce the risk of infection experienced by patients who are severely ill.

When bone overgrowth deforms the shape of the skull, surgery may be required to relieve pressure on the brain. Orthodontic treatment is sometimes necessary to correct **malocclusion** (a condition that shifts the position of the teeth and makes closing the mouth impossible).

Professional counseling can help patients cope with the emotional aspects of deformed features.

Bone marrow transplants (BMT) have cured some cases of early-onset and malignant infantile osteopetrosis. Because 30–60% of children who undergo BMT do not survive, this procedure is rarely performed.



This infant has osteopetrosis, a condition which thickens and hardens the bone. (Custom Medical Stock Photo. Reproduced by permission.)

Prognosis

The severity of anemia seems to determine the course of an individual's osteopetrosis. When pronounced symptoms are present at the time of birth, the child's condition deteriorates rapidly. Death usually occurs within two years. When mild or moderate disease develops in older children or adults and symptoms can be controlled, the patient is likely to survive.

Resources

ORGANIZATIONS

Osteoporosis and Related Bone Diseases-National Resource Center. 1150 17th S. NW, Ste. 500, Washington, DC 20036. (800) 624-2663.

Maureen Haggerty

Osteoporosis

Definition

The word osteoporosis literally means "porous bones." It occurs when bones lose an excessive amount of their protein and mineral content, particularly calcium. Over time, bone mass, and therefore bone

strength, is decreased. As a result, bones become fragile and break easily. Even a sneeze or a sudden movement may be enough to break a bone in someone with severe osteoporosis.

Description

Osteoporosis is a serious public health problem. Some 44 million people in the United States are at risk for this potentially debilitating disease, which is responsible for 1.5 million **fractures** (broken bones) annually. These fractures, which are often the first sign of the disease, can affect any bone, but the most common locations are the hip, spine, and wrist. Breaks in the hip and spine are of special concern because they almost always require hospitalization and major surgery, and may lead to other serious consequences, including permanent disability and even **death**.

To understand osteoporosis, it is helpful to understand the basics of bone formation. Bone is living tissue that is constantly being renewed in a two-stage process (resorption and formation) that occurs throughout life. In the resorption stage, old bone is broken down and removed by cells called osteoclasts. In the formation stage, cells called osteoblasts build new bone to replace the old. During childhood and early adulthood, more bone is produced than removed, reaching its maximum mass and strength by the mid-30s. After that, bone is lost at a faster pace than it is formed, so the amount of bone in the skeleton begins to slowly decline. Most cases of osteoporosis occur as an acceleration of this normal **aging** process, which is referred to as primary osteoporosis. The condition also can be caused by other disease processes or prolonged use of certain medications that result in bone loss. If so, this is called secondary osteoporosis.

Osteoporosis occurs most often in older people and in women after **menopause**. It affects nearly half of men and women over the age of 75. Women are about five times more likely than men to develop the disease. They have smaller, thinner bones than men to begin with, and they lose bone mass more rapidly after menopause (usually around age 50), when they stop producing a bone-protecting hormone called estrogen. In the five to seven years following menopause, women can lose about 20% of their bone mass. By age 65 or 70, though, men and women lose bone mass at the same rate. As an increasing number of men reach an older age, there is more awareness that osteoporosis is an important health issue for them as well. In fact, a 2003 report noted that one in every eight men over age 50 will suffer a hip fracture as a result of osteoporosis.

Causes and symptoms

A number of factors increase the risk of developing osteoporosis. They include:

- **Age.** Osteoporosis is more likely as people grow older and their bones lose tissue.
- **Gender.** Women are smaller and start out with less bone. They also lose bone tissue more rapidly as they age. While women commonly lose 30–50% of their bone mass over their lifetimes, men lose only 20–33%.
- **Race.** Caucasian and Asian women are most at risk for the disease, but African American and Hispanic women can get it too.
- **Figure type.** Women with small bones and those who are thin are more liable to have osteoporosis.
- **Early menopause.** Women who stop menstruating early because of heredity, surgery or lots of physical **exercise** may lose large amounts of bone tissue early in life. Conditions such as anorexia and bulimia also may lead to early menopause and osteoporosis.
- **Lifestyle.** People who smoke or drink too much, or do not get enough exercise have an increased chance of osteoporosis.
- **Diet.** Those who do not get enough calcium or protein may be more likely to have osteoporosis. That is why people who constantly diet are more prone to the disease.
- **Genetics.** Research in Europe reported in 2003 that variations of a gene on chromosome 20 might make some postmenopausal women more likely to have osteoporosis. Studies were continuing on how to identify the gene and use information from the research to prevent osteoporosis in carriers.

Osteoporosis is often called the “silent” disease, because bone loss occurs without symptoms. People often do not know they have the disease until a bone breaks, frequently in a minor fall that would not normally cause a fracture. A common occurrence is compression fractures of the spine. These can happen even after a seemingly normal activity, such as bending or twisting to pick up a light object. The fractures can cause severe back **pain**, but sometimes go unnoticed—either way, the vertebrae collapse down on themselves, and the person actually loses height. The hunchback appearance of many elderly women, sometimes called “dowager’s” hump or “widow’s” hump, is due to this effect of osteoporosis on the vertebrae.

Diagnosis

Certain types of doctors may have more training and experience than others in diagnosing and treating people with osteoporosis. These include a geriatrician, who specializes in treating the aged; an endocrinologist, who specializes in treating diseases of the body’s endocrine system (glands and hormones); and an orthopedic surgeon, who treats fractures such as those caused by osteoporosis.

Before making a diagnosis of osteoporosis, the doctor usually takes a complete medical history, conducts a physical exam, and orders x rays, as well as blood and urine tests, to rule out other diseases that cause loss of bone mass. The doctor also may recommend a **bone density test**. This is the only way to know for certain if osteoporosis is present. It also can show how far the disease has progressed.

Several diagnostic tools are available to measure bone density. The ordinary x ray is one, though it is the least accurate for early detection of osteoporosis, because it does not reveal bone loss until the disease is advanced and most of the damage has already been done. Two other tools that are more likely to catch osteoporosis at an early stage are **computed tomography scans** (CT scans) and machines called densitometers, which are designed specifically to measure bone density.

The CT scan, which takes a large number of x rays of the same spot from different angles, is an accurate test, but uses higher levels of radiation than other methods. The most accurate and advanced of the densitometers uses a technique called DEXA (dual energy x-ray absorptiometry). With the DEXA scan, a double x-ray beam takes pictures of the spine, hip, or entire body. It takes about 20 minutes to do, is painless, and exposes the patient to only a small amount of radiation—about one-fiftieth that of a **chest x ray**.

Doctors do not routinely recommend the test, partly because access to densitometers is still not widely available. People should talk to their doctors about their risk factors for osteoporosis and if, and when, they should get the test. Ideally, women should have bone density measured at menopause, and periodically afterward, depending on the condition of their bones. Men should be tested around age 65. Men and women with additional risk factors, such as those who take certain medications, may need to be tested earlier.

Treatment

There are a number of good treatments for primary osteoporosis, most of them medications.

Two medications, alendronate and calcitonin (in nose spray form), have been approved by the Food and Drug Administration (FDA). They provide people who have osteoporosis with a variety of choices for treatment. For people with secondary osteoporosis, treatment may focus on curing the underlying disease.

Drugs

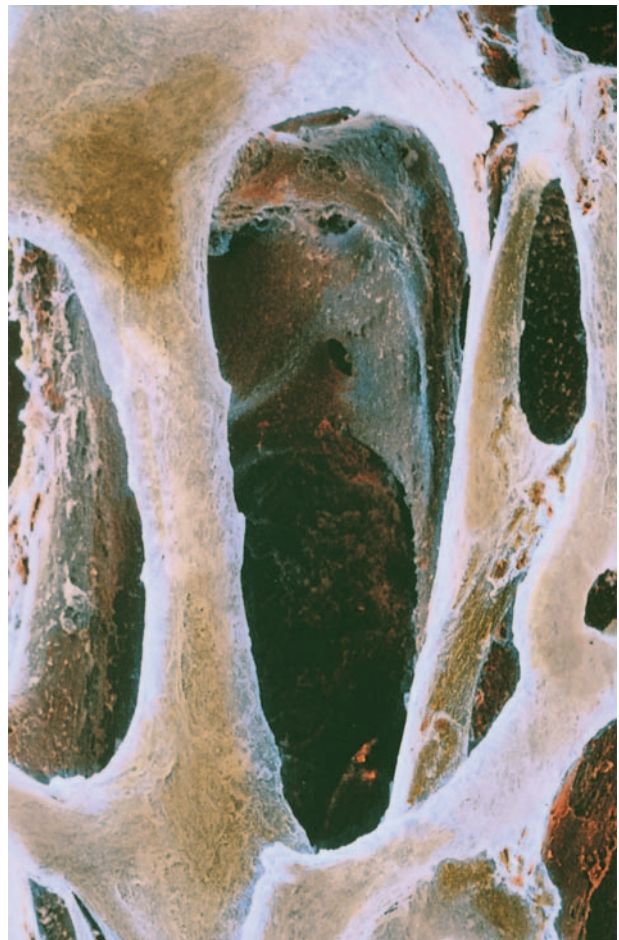
For many women who have gone through menopause, the treatment of choice for osteoporosis has been **hormone replacement therapy** (HRT), also called estrogen replacement therapy. Many women choose HRT when they undergo menopause to alleviate symptoms such as hot flashes, but hormones increase a woman's supply of estrogen, which helps build new bone, while preventing further bone loss. A 2002 report from a large clinical trial called the Women's Health Initiative helped verify HRT's positive effects in preventing osteoporosis in postmenopausal women.

However, the WHI also revealed several risks with taking combined HRT (estrogen and progesterone). In fact, the trial was stopped early because the incidence of invasive **breast cancer** in women on HRT passed a threshold that was considered too risky for the benefits they were receiving. The study also found that the women on combined hormone therapy were at increased risk for coronary heart disease and **stroke**. Whether or not a woman takes hormones and for how long is a decision she should make carefully with her doctor. Women should talk to their doctors about personal risks for osteoporosis, as well as their risks for heart disease and breast **cancer**.

Since estrogen may no longer be recommended for prevention of osteoporosis, selective use of alendronate and calcitonin are possible alternatives. Alendronate and calcitonin both stop bone loss, help build bone, and decrease fracture risk by as much as 50%. Alendronate (sold under the name Fosamax) is the first nonhormonal medication for osteoporosis ever approved by the FDA. It attaches itself to bone that has been targeted by bone-eating osteoclasts, protecting the bone from these cells. Osteoclasts help the body break down old bone tissue.

Calcitonin is a hormone that has been used as an injection for many years. A new version is on the market as a nasal spray. It too slows down bone-eating osteoclasts.

Side effects of these drugs are minimal, but calcitonin builds bone by only 1.5% a year, which may not be enough for some women to recover the bone they lose. Fosamax has proven safe in large, multi-year studies, but not much is known about the effects of



A scanning electron microscopy (SEM) image of cancellous (spongy) bone from an osteoporosis patient. Osteoporosis is characterized by increased brittleness of the bones and a greater risk of fractures. This is reflected here in the thin appearance of the bony network of the cancellous bone that forms the core of the body's long bones. (Photograph by Professor P. Motta, Photo Researchers, Inc. Reproduced by permission.)

its long-term use. Several medications under study include other bisphosphonates that slow bone breakdown (like alendronate), sodium fluoride, vitamin D metabolites, and selective estrogen receptor modulators. Some of these treatments are already being used in other countries, but have not yet been approved by the FDA for use in the United States.

In early 2003, a report announced that the FDA had recently approved the first drug that could form bone in osteoporosis patients. The drug is a form of the human parathyroid hormone called teriparatide. It shows promise for those patients at highest risk for fracture from the disease. There are some patients who cannot use the drug, so all considering the new

treatment must check with their physician and may need to undergo bone densitometry scans or other testing.

Surgery

Unfortunately, much of the treatment for osteoporosis is for fractures that result from advanced stages of the disease. For complicated fractures, such as broken hips, hospitalization and a surgical procedure are required. In hip replacement surgery, the broken hip is removed and replaced with a new hip made of plastic, or metal and plastic. Though the surgery itself is usually successful, complications of the hip fracture can be serious. Those individuals have a 5–20% greater risk of dying within the first year following the injury than do others in their age group. A large percentage of those who survive are unable to return to their previous level of activity, and many move self-care to a supervised living situation or nursing home. That is why getting early treatment and taking steps to reduce bone loss are vital.

Alternative treatment

Alternative treatments for osteoporosis focus on maintaining or building strong bones. A healthy diet low in fats and animal products and containing whole grains, fresh fruits and vegetables, and calcium-rich foods (such as dairy products, dark-green leafy vegetables, sardines, salmon, and almonds), along with **nutritional supplements** (such as calcium, magnesium, and vitamin D), and weight-bearing exercises are important components of both conventional prevention and treatment strategies and alternative approaches to the disease. In addition, alternative practitioners recommend a variety of botanical medicines or herbal supplements. Herbal supplements designed to help slow bone loss emphasize the use of calcium-containing plants, such as horsetail (*Equisetum arvense*), oat straw (*Avena sativa*), alfalfa (*Medicago sativa*), licorice (*Glycyrrhiza galbra*), marsh mallow (*Althaea officinalis*), and yellow dock (*Rumex crispus*). Homeopathic remedies focus on treatments believed to help the body absorb calcium. These remedies are likely to include such substances as *Calcarea carbonica* (calcium carbonate) or silica. In **traditional Chinese medicine**, practitioners recommend herbs thought to slow or prevent bone loss, including dong quai (*Angelica sinensis*) and Asian ginseng (*Panax ginseng*). Natural hormone therapy, using plant estrogens (from soybeans) or progesterone (from wild yams), may be recommended for women who cannot or choose not to take synthetic hormones.

Prognosis

There is no cure for osteoporosis, but it can be controlled. Most people who have osteoporosis fare well once they receive treatment. The medicines available now build bone, protect against bone loss, and halt the progress of this disease.

Prevention

Building strong bones, especially before the age of 35, and maintaining a healthy lifestyle are the best ways to prevent osteoporosis. To build as much bone mass as early as possible in life, and to help slow the rate of bone loss later in life, doctors advise:

Getting calcium from foods

Experts recommend 1,500 milligrams (mg) of calcium per day for adolescents, pregnant or breastfeeding women, older adults (over 65), and postmenopausal women not using hormone replacement therapy. All others should get 1,000 mg per day. Foods are the best source for this important mineral. Milk, cheese, and yogurt have the highest amounts. Other foods that are high in calcium are green leafy vegetables, tofu, shellfish, Brazil nuts, sardines, and almonds.

Taking calcium supplements

Many people, especially those who do not like or can not eat dairy foods, do not get enough calcium in their **diets** and may need calcium supplements. Supplements vary in the amount of calcium they contain. Those with calcium carbonate have the most amount of useful calcium. Supplements should be taken with meals and accompanied by six to eight glasses of water a day.

Getting vitamin D

Vitamin D helps the body absorb calcium. People can get vitamin D from sunshine with a quick (15–20 minute) walk each day or from foods such as liver, fish oil, and vitamin-D fortified milk. During the winter months it may be necessary to take supplements. Four hundred mg daily is usually the recommended amount.

Avoiding smoking and alcohol

Smoking reduces bone mass, as does heavy drinking. Avoiding smoking and limiting alcoholic drinks to no more than two per day reduces risks. An alcoholic drink is one-and-a-half ounces of hard liquor, 12 ounces of beer, or five ounces of wine.

KEY TERMS

Alendronate—A nonhormonal drug used to treat osteoporosis in postmenopausal women.

Anticonvulsants—Drugs used to control seizures, such as in epilepsy.

Biphosphonates—Compounds (like alendronate) that slow bone loss and increase bone density.

Calcitonin—A hormonal drug used to treat postmenopausal osteoporosis

Estrogen—A female hormone that also keeps bones strong. After menopause, a woman may take hormonal drugs with estrogen to prevent bone loss.

Glucocorticoids—Any of a group of hormones (like cortisone) that influence many body functions and are widely used in medicine, such as for treatment of rheumatoid arthritis inflammation.

Hormone replacement therapy (HRT)—Also called estrogen replacement therapy, this controversial treatment is used to relieve the discomforts of menopause. Estrogen and another female hormone, progesterone, are usually taken together to replace the estrogen no longer made by the body.

Menopause—The ending of a woman's menstrual cycle, when production of bone-protecting estrogen decreases.

Osteoblasts—Cells in the body that build new bone tissue.

Osteoclasts—Cells that break down and remove old bone tissue.

Selective estrogen receptor modulator—A hormonal preparation that offers the beneficial effects of hormone replacement therapy without the increased risk of breast and uterine cancer associated with HRT.

Exercise

Exercising regularly builds and strengthens bones. Weight-bearing exercises—where bones and muscles work against gravity—are best. These include aerobics, dancing, jogging, stair climbing, tennis, walking, and lifting weights. People who have osteoporosis may want to attempt gentle exercise, such as walking, rather than jogging or fast-paced aerobics, which increase the chance of falling. Exercising three to four times per week for 20–30 minutes each time helps.

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Barbara Boughton
Teresa G. Odle

Osteosarcoma see **Sarcomas**

Ostomy

Definition

A surgical procedure creating an opening in the body for the discharge of body wastes.

Purpose

Certain diseases of the bowel or urinary tract involve removing all or part of the intestine or bladder. This creates a need for an alternate way for feces or urine to leave the body. An opening is surgically created in the abdomen for body wastes to pass through. The surgical procedure is called an ostomy. The opening that is created at the end of the bowel or ureter is called a stoma, which is pulled through the abdominal wall.

Description

Different types of ostomy are performed depending on how much and what part of the intestines or bladder is removed.

The three most common types of ostomies are:

- colostomy
- ileostomy
- urostomy

Colostomy

A colostomy is when a small portion of the colon (large intestine) is brought to the surface of the abdominal wall to allow stool to be eliminated. A colostomy may be temporary or permanent. A permanent colostomy usually involves the loss of the rectum.

A colostomy might be performed due to cancer, diverticulitis, imperforate anus, **Hirschsprung's disease**, or trauma to the affected area.

Ileostomy

An ileostomy is an opening created in the small intestine to bypass the colon for stool elimination. The end of the ileum, which is the lowest part of the small intestine, is brought through the abdominal wall to form a stoma.

Ileoanal reservoir surgery is an alternative to a permanent ileostomy. It requires two surgical procedures. The first removes the colon and rectum and a temporary ileostomy is created. The second procedure creates an internal pouch from a portion of the small intestine to hold stool. This is then attached to the anus. Since the muscle of the rectum is left in place, there is control over bowel movements.

An ileostomy might be performed due to ulcerative colitis, Crohn's disease, or familial polyposis.

Urostomy

A urostomy is a surgical procedure that diverts urine away from a diseased or defective bladder. Among several methods to create the urostomy, the most common method is called an ileal or cecal conduit. Either a section at the end of the small intestine (ileum) or at the beginning of the large intestine (cecum) is relocated surgically to form a stoma for urine to pass out of the body. Other common names for this procedure are ileal loop or colon conduit.

A urostomy may be performed due to **bladder cancer**, spinal cord injuries, malfunction of the bladder, and **birth defects** such as spina bifida.

Since colostomy, ileostomy, and urostomy bypass the sphincter muscle there is no voluntary control over bowel movements and an external pouch must be worn to catch the discharge.

Preparation

Aftercare

The skin around the stoma, called the peristomal skin, must be protected from direct contact with discharge. The discharge can be irritating to the stoma since it is very high in digestive enzymes. The peristomal skin should be cleansed with plain soap and rinsed with water at each change of the pouch.

The stoma can change in size due to weight gain/loss or several other situations. To ensure proper fit of discharge pouch the stoma should be measured each time supplies are purchased.

Risks

People with ostomies can be prone to certain types of skin infections. Skin irritations or rashes around the stoma may be caused by leakage from around the pouch due to an improperly fitted pouch. Correctly fitting the pouch and carefully cleaning the skin around the stoma after each change are the best ways of preventing skin irritation.

Urinary tract infections are common among people who have urostomies. Preventative measures include drinking plenty of fluids, emptying the pouch regularly and using a pouch with an anti-reflux valve to prohibit the discharge from moving back into the stoma.

Normal results

Most ostomy pouches are inconspicuous and can be worn under almost any kind of clothing. There are typically no restrictions of activity, sport, or travel with an ostomy. Certain contact sports would warrant special protection for the stoma.

After recovery from surgery, most people with ostomies can resume a balanced diet.

Ostomy surgery does not generally interfere with a person's sexual or reproductive capacities.

Abnormal results

After an ileostomy, water and electrolyte loss may occur. It may be necessary to drink a significant amount of fluid or fruit juice each day to prevent **dehydration**.

After any type of ostomy surgery digestion and absorption of medications may also be affected.

High-fiber foods can cause blockages in the ileum, especially after surgery. Chewing food well helps

KEY TERMS

Crohn's disease—A chronic inflammatory disease, primarily involving the small and large intestine, but which can affect other parts of the digestive system as well.

Diverticulitis—Inflammation of the diverticula (small outpouchings) along the wall of the colon, the large intestine.

Familial polyposis—An inherited condition in which several hundred polyps develop in the colon and rectum.

Hirschsprung disease—Hirschsprung disease is a congenital abnormality (birth defect) of the bowel in which there is absence of the ganglia (nerves) in the wall of the bowel. Nerves are missing starting at the anus and extending a variable distance up the bowel. This results in megacolon (massive enlargement of the bowel) above the point where the nerves are missing. (The nerves are needed to assist in the natural movement of the muscles in the lining of the bowels that move bowel contents through.)

Ileum—The lowest part of the small intestine, located beyond the duodenum and jejunum, just before the large intestine (the colon).

Imperforate anus—A congenital malformation (a birth defect) in which the rectum is a blind alley (a cul-de-sac) and there is no anus.

Spina bifida—A birth defect (a congenital malformation) in which there is a bony defect in the vertebral column so that part of the spinal cord, which is normally protected within the vertebral column, is exposed. People with spina bifida can suffer from bladder and bowel incontinence, cognitive (learning) problems and limited mobility.

break fiber into smaller pieces and makes it less likely to accumulate at a narrow point in the bowel. Drinking plenty of fluids can also help.

Resources

ORGANIZATIONS

Crohn's & Colitis Foundation of America, Inc. 386 Park Avenue South 17th Floor, New York, NY 10016-8804. (800) 932-2423 or (212) 685-3440.

International Foundation for Functional Gastrointestinal Disorders. P.O. Box 17864, Milwaukee, WI 53217. (414) 964-1799.

National Digestive Diseases Clearinghouse. 2 Information Way, Bethesda, MD 20892-3570. < <http://www.niddk.nih.gov/> >.

United Ostomy Association. 19772 MacArthur Boulevard, Suite 200 Irvine, CA 92612-2405. (800) 826-0826 or (949) 660-8624.

Gary Gilles

Otitis externa

Definition

Otitis externa refers to an infection of the ear canal, the tube leading from the outside opening of the ear in towards the ear drum.

Description

The external ear canal is a tube approximately 1 in (2.5 cm) in length. It runs from the outside opening of the ear to the start of the middle ear, designated by the ear drum or tympanic membrane. The canal is partly cartilage and partly bone. In early childhood, the first two-thirds of the canal is made of cartilage, and the last one-third is made of bone. By late childhood, and lasting throughout all of adulthood, this proportion is reversed, so that the first one-third is cartilage, and the last two-thirds is bone. The lining of the ear canal is skin, which is attached directly to the covering of the bone. Glands within the skin of the canal produce a waxy substance called cerumen (popularly called earwax). Cerumen is designed to protect the ear canal, repel water, and keep the ear canal too acidic to allow bacteria to grow.

Causes and symptoms

Bacteria, fungi, and viruses have all been implicated in causing ear infections called otitis externa. The most common cause of otitis externa is bacterial infection. The usual offenders include *Pseudomonas aeruginosa*, *Enterobacter aerogenes*, *Proteus mirabilis*, *Klebsiella pneumoniae*, *Staphylococcus epidermidis*, and bacteria of the family called Streptococci. Occasionally, fungi may cause otitis externa. These include *Candida* and *Aspergillus*. Two types of viruses, called herpesvirus hominis and varicella-zoster virus, have also been identified as causing otitis externa.

Otitis externa occurs most often in the summer months, when people are frequenting swimming pools

and lakes. Continually exposing the ear canal to moisture may cause significant loss of cerumen. The delicate skin of the ear canal, unprotected by cerumen, retains moisture and becomes irritated. Without cerumen, the ear canal stops being appropriately acidic, which allows bacteria the opportunity to multiply. Thus, the warm, moist, dark environment of the ear canal becomes a breeding ground for bacteria.

Other conditions predisposing to otitis externa include the use of cotton swabs to clean the ear canals. This pushes cerumen and normal skin debris back into the ear canal, instead of allowing the ear canal's normal cleaning mechanism to work, which would ordinarily move accumulations of cerumen and debris out of the ear. Also, putting other items into the ear can scratch the canal, making it more susceptible to infection.

The first symptom of otitis externa is often **itching** of the ear canal. Eventually, the ear begins to feel extremely painful. Any touch, movement, or pressure on the outside structure of the ear (auricle) may cause quite severe **pain**. This is because of the way in which the skin lining the ear canal is directly attached to the covering of the underlying bone. If the canal is sufficiently swollen, hearing may become muffled. The canal may appear swollen and red, and there may be evidence of greenish-yellow pus.

In severe cases, otitis externa may have an accompanying **fever**. Often, this indicates that the outside ear structure (auricle) has become infected as well. It will become red and swollen, and there may be enlarged and tender lymph nodes in front of, or behind, the auricle.

A serious and life-threatening otitis externa is called malignant otitis externa. This is an infection which most commonly affects patients who have diabetes, especially the elderly. It can also occur in other patients who have weakened immune systems. In malignant otitis externa, a patient has usually had minor symptoms of otitis externa for some months, with pain and drainage. The causative bacteria is usually *Pseudomonas aeruginosa*. In malignant otitis externa, this bacteria spreads from the external canal into all of the nearby tissues, including the bones of the skull. Swelling and destruction of these tissues may lead to damage of certain nerves, resulting in spasms of the jaw muscles or paralysis of the facial muscles. Other, more severe, complications of this very destructive infection include meningitis (swelling and infection of the coverings of the spinal cord and brain), brain infection, or **brain abscess** (the development of a pocket of infection with pus).



A close-up image of the ear of an elderly man suffering from non-infectious otitis externa. The skin in the ear canal and outer ear is scaly. (Photograph by Dr. P. Marazzi, Custom Medical Stock Photo. Reproduced by permission.)

Diagnosis

Diagnosis of uncomplicated otitis externa is usually quite simple. The symptoms alone, of ear pain worsened by any touch to the auricle, are characteristic of otitis externa. Attempts to examine the ear canal will usually reveal redness and swelling. It may be impossible (due to pain and swelling) to see much of the ear canal, but this inability itself is diagnostic.

If there is any confusion about the types of organisms causing otitis externa, the canal can be gently swabbed to obtain a specimen. The organisms present in the specimen can then be cultured (allowed to multiply) in a laboratory, and then viewed under a microscope to allow identification of the causative organisms.

If the rare disease malignant otitis externa is suspected, **computed tomography scan** (CT scan) or

magnetic resonance imaging (MRI) scans will be performed to determine how widely the infection has spread within bone and tissue. A swab of the external canal will not necessarily reveal the actual causative organism, so some other tissue sample (biopsy) will need to be obtained. The CT or MRI will help the practitioner decide where the most severe focus of infection is located, in order to guide the choice of a biopsy site.

Treatment

Antibiotics which can be applied directly to the skin of the ear canal (**topical antibiotics**) are usually excellent for treatment of otitis externa. These are often combined in a preparation which includes a steroid medication. The steroid helps cut down on the inflammation and swelling within the ear canal. Some practitioners prefer to insert a cotton wick into the ear canal, leaving it there for about 48 hours. The medications are applied directly to the wick, enough times per day to allow the wick to remain continuously saturated. After the wick is removed, the medications are then put directly into the ear canal three to four times each day.

In malignant otitis externa, antibiotics will almost always need to be given through a needle in the vein (intravenously or IV). If the CT or MRI scan reveals that the infection has spread extensively, these IV antibiotics will need to be continued for six to eight weeks. If the infection is in an earlier stage, two weeks of IV antibiotics can be followed by six weeks of antibiotics by mouth.

Prognosis

The prognosis is excellent for otitis externa. It is usually easily treated, although it may tend to recur in certain susceptible individuals. Left untreated, malignant otitis externa may spread sufficiently to cause death.

Prevention

Keeping the ear dry is an important aspect of prevention of otitis externa. Several drops of a mixture of alcohol and acetic acid can be put into the ear canal after swimming to insure that it dries adequately.

The most serious complications of malignant otitis externa can be avoided by careful attention to early symptoms of ear pain and drainage from the ear canal. Patients with conditions that put them at higher risk for this infection (diabetes, conditions

KEY TERMS

Auricle—The external structure of the ear.

Biopsy—The removal and examination, usually under a microscope, of tissue from the living body. Used for diagnosis.

Cerumen—Earwax.

which weakened the immune system) should always report new symptoms immediately.

Resources

ORGANIZATIONS

American Academy of Otolaryngology-Head and Neck Surgery, Inc. One Prince St., Alexandria VA 22314-3357. (703) 836-4444. <<http://www.entnet.org>>.

Rosalyn Carson-DeWitt, MD

Otitis media

Definition

Otitis media is an infection of the middle ear space, behind the eardrum (tympanic membrane). It is characterized by **pain**, **dizziness**, and partial loss of hearing.

Description

A little knowledge of the basic anatomy of the middle ear will be helpful for understanding the development of otitis media. The external ear canal is that tube which leads from the outside opening of the ear to the structure called the tympanic membrane. Behind the tympanic membrane is the space called the middle ear. Within the middle ear are three tiny bones, called ossicles. Sound (in the form of vibration) causes movement in the eardrum, and then the ossicles. The ossicles transmit the sound to a structure within the inner ear, which sends it to the brain for processing.

The nasopharynx is that passageway behind the nose which takes inhaled air into the breathing tubes leading to the lungs. The eustachian tube is a canal which runs between the middle ear and the nasopharynx. One of the functions of the eustachian tube is to keep the air pressure in the middle ear equal to that

outside. This allows the eardrum and ossicles to vibrate appropriately, so that hearing is normal.

By age three, almost 85% of all children will have had otitis media at least once. Babies and children between the ages of six months and six years are most likely to develop otitis media. Children at higher risk factors for otitis media include boys, children from poor families, Native Americans, Native Alaskans, children born with **cleft palate** or other defects of the structures of the head and face, and children with **Down syndrome**. Exposure to cigarette smoke significantly increases the risk of otitis media as well as other problems affecting the respiratory system. Also, children who enter daycare at an early age have more upper respiratory infections (URIs or colds), and thus more cases of otitis media. The most usual times of year for otitis media to strike are in winter and early spring (the same times URIs are most common).

Otitis media is an important problem, because it often results in fluid accumulation within the middle ear (effusion). The effusion can last for weeks to months. Effusion within the middle ear can cause significant hearing impairment. When such hearing impairment occurs in a young child, it may interfere with the development of normal speech.

In adults, acute otitis media can lead to such complications as **paralysis** of the facial nerves. Recovery from these complications may take from two weeks to as long as three months.

Causes and symptoms

The first precondition for the development of acute otitis media is exposure to an organism capable of causing the infection. Otitis media can be caused by either viruses or bacteria. Virus infections account for about 15% of cases. The three most common bacterial pathogens are *Streptococcus pneumoniae*, *Haemophilus influenzae*, or *Moraxella catarrhalis*. As of 2003, about 75% of ear infections caused by *S. pneumoniae* are reported to be penicillin-resistant.

Otitis media may also be caused by other disease organisms, including *Bordetella pertussis*, the causative agent of **whooping cough**, and *Pneumocystis carinii*, which often causes opportunistic infections in patients with **AIDS**.

There are other factors which make the development of an ear infection more likely. Because the eustachian tube has a more horizontal orientation and is considerably shorter in early childhood, material from the nasopharynx (including infection-causing

organisms) is better able to reach the middle ear. Children also have a lot of lymph tissue (commonly called the adenoids) in the area of the eustachian tube. These adenoids may enlarge with repeated respiratory tract infections (colds), ultimately blocking the eustachian tubes. When the eustachian tube is blocked, the middle ear is more likely to fill with fluid. This fluid, then, increases the risk of infection, and the risk of **hearing loss** and delayed speech development.

Most cases of acute otitis media occur during the course of a URI. Symptoms include fever, ear pain, and problems with hearing. Babies may have difficulty feeding. When significant fluid is present within the middle ear, pain may increase depending on position. Lying down may cause an increase in painful pressure within the middle ear, so that babies may fuss if not held upright. If the fluid build-up behind the eardrum is sufficient, the eardrum may develop a hole (perforate), causing bloody fluid or greenish-yellow pus to drip from the ear. Although pain may be significant leading up to such a perforation, the pain is usually relieved by the reduction of pressure brought on by a perforation.

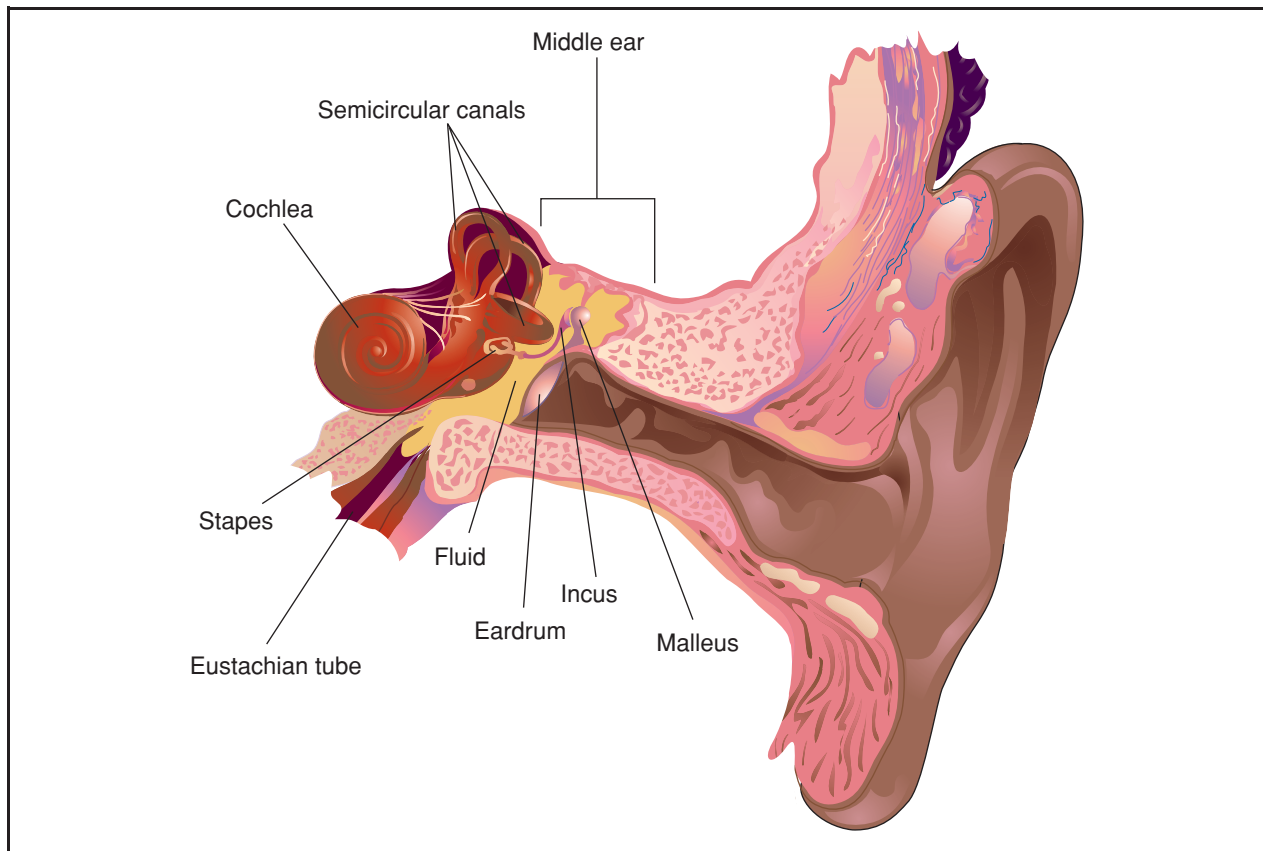
Recent advances in gene mapping have led to the discovery of genetic factors that increase a child's susceptibility to otitis media. Researchers are hoping to develop molecular diagnostic assays that will help to identify children at risk for severe ear infections.

Diagnosis

Diagnosis is usually made simply by looking at the eardrum through a special lighted instrument called an otoscope. The eardrum will appear red and swollen, and may appear either abnormally drawn inward, or bulging outward. Under normal conditions, the ossicles create a particular pattern on the eardrum, referred to as "landmarks." These landmarks may be obscured. Normally, the light from the otoscope reflects off of the eardrum in a characteristic fashion. This is called the "cone of light." In an infection, this cone of light may be shifted or absent.

A special attachment to the otoscope allows a puff of air to be blown lightly into the ear. Normally, this should cause movement of the eardrum. In an infection, or when there is fluid behind the eardrum, this movement may be decreased or absent.

If fluid or pus is draining from the ear, it can be collected. This sample can then be processed in a laboratory to allow any organisms present to multiply sufficiently (cultured) to permit the organisms to be viewed under a microscope and identified.



Otitis media is an ear infection in which fluid accumulates within the middle ear. A common condition occurring in childhood, it is estimated that 85% of all American children will develop otitis media at least once. (Illustration by Electronic Illustrators Group.)

Treatment

Medications

Antibiotics are the treatment of choice for acute otitis media (AOM). Different antibiotics are used depending on the type of bacteria most likely to be causing the infection. This decision involves knowledge of the types of antibiotics that have worked on other ear infections occurring within a particular community at a particular time. Options include sulfa-based antibiotics, as well as a variety of **penicillins**, **cephalosporins**, and others. The patient's sensitivity to certain medications, as well as previously demonstrated resistant strains, also contributes to the choice of antibiotic. As of 2003, an 0.3% topical solution of ofloxacin has been recommended as a more effective medication than other oral or **topical antibiotics**.

Some controversy exists regarding whether overuse of antibiotics is actually contributing to the development of bacteria, which may evolve and become

able to avoid being killed by antibiotics. Research is being done to try to help determine whether there may be some ear infections that will clear up without antibiotic treatment. In the meantime, the classic treatment of an ear infection continues to involve a seven to 10-day course of antibiotic medication.

Some medical practitioners prescribe the use of special nosedrops, **decongestants**, or **antihistamines** to improve the functioning of the eustachian tube.

Whether or not antibiotics are used, such pain relievers as Tylenol or Motrin can be very helpful in reducing the pain and inflammation associated with otitis media.

Surgery

In a few rare cases, a surgical perforation to drain the middle ear of pus may be performed. This procedure is called a **myringotomy**. The hole created by the myringotomy generally heals itself in about a week.

In 2002 a new minimally invasive procedure was introduced that uses a laser to perform the myringotomy. It can be performed in the doctor's office and heals more rapidly than the standard myringotomy.

Although some doctors have recommended removing the adenoids to prevent recurrent otitis media in young children, recent studies indicate that surgical removal of the adenoids does not appear to offer any advantages over a myringotomy as a preventive measure.

Alternative treatment

Some practitioners believe that **food allergies** may increase the risk of ear infections, and they suggest eliminating suspected food allergens from the diet. The top food allergens are wheat, dairy products, corn, peanuts, citrus fruits, and eggs. Elimination of sugar and sugar products can allow the immune system to work more effectively. A number of herbal treatments have been recommended, including ear drops made with goldenseal (*Hydrastis canadensis*), mullein (*Verbascum thapsus*), **St. John's wort** (*Hypericum perforatum*), and echinacea (*Echinacea* spp.). Among the herbs often recommended for oral treatment of otitis media are **echinacea** and cleavers (*Galium aparine*), or black cohosh (*Cimicifuga racemosa*) and ginkgo (*Ginkgo biloba*). Homeopathic remedies that may be prescribed include aconite (*Aconitum napellus*), *Ferrum phosphoricum*, belladonna, chamomile, *Lycopodium*, pulsatilla (*Pulsatilla nigricans*), or silica. **Craniosacral therapy** uses gentle manipulation of the bones of the skull to relieve pressure and improve eustachian tube function.

Prognosis

With treatment, the prognosis for acute otitis media is very good. However, long-lasting accumulations of fluid within the middle ear are a risk both for difficulties with hearing and speech, and for the repeated development of ear infections. Furthermore, without treatment, otitis media can lead to an infection within the nearby mastoid bone, called **mastoiditis**.

Prevention

Although otitis media seems somewhat inevitable in childhood, some measures can be taken to decrease the chance of repeated infections and fluid accumulation. Breastfeeding provides some protection against URIs, which in turn protects against the development of otitis media. If a child is bottle-

fed, parents should be advised to feed him or her upright, rather than allowing the baby to lie down with the bottle. General good hygiene practices (especially handwashing) help to decrease the number of upper respiratory infections in a household or day-care center.

The use of pacifiers should be avoided or limited. They may act as fomites, particularly in a daycare setting. In children who are more susceptible to otitis media, pacifier use can increase by as much as 50% the number of ear infections experienced.

Two vaccines can prevent otitis media associated with certain strains of bacteria. One is designed to prevent **meningitis** and other diseases, including otitis media, that result from infection with *Haemophilus influenzae* type B. Another is a vaccine against *Streptococcus pneumoniae*, a very common cause of otitis media. Children who are at high risk or have had severe or chronic infections may be good candidates for these vaccines; in fact, a recent consensus report among pediatricians recommended routine administration of the pneumococcal conjugate vaccine to children younger than two years, as well as those at high risk for AOM. Parents should consult a health care provider concerning the advisability of this treatment.

Another vaccine that appears to lower the risk of AOM in children is the intranasal vaccine that was recently introduced for preventing **influenza**. Although the flu vaccine was not developed to prevent AOM directly, one team of researchers found that children who were given the vaccine before the start of flu season were 43% less likely to develop AOM than children who were not vaccinated.

As of early 2003, there is no vaccine effective against *M. catarrhalis*. Researchers are working on developing such a vaccine, as well as a tribacterial vaccine that would be effective against all three pathogens that commonly cause otitis media.

A nutrition-based approach to preventive treatment is undergoing clinical trials as of late 2002. This treatment involves giving children a dietary supplement of lemon-flavored cod liver oil plus a multivitamin formula containing selenium. The pilot study found that children receiving the supplement had fewer cases of otitis media, and that those who did develop it recovered with a shorter course of antibiotic treatment than children who were not receiving the supplement.

After a child has completed treatment for otitis media, a return visit to the practitioner should be scheduled. This visit should occur after the

KEY TERMS

Adenoid—A collection of lymph tissue located in the nasopharynx.

Effusion—A collection of fluid which has leaked out into some body cavity or tissue.

Eustachian tube—A small tube which runs between the middle ear space and the nasopharynx.

Fomite—An inanimate object that can transmit infectious organisms.

Myringotomy—A surgical procedure performed to drain an infected middle ear. A newer type of myringotomy uses a laser instead of a scalpel.

Nasopharynx—The part of the airway into which the nose leads.

Ossicles—Tiny bones located within the middle ear which are responsible for conveying the vibrations of sound through to the inner ear.

Perforation—A hole.

Topical—Referring to a medication applied to the skin or outward surface of the body. Ear drops are one type of topical medication.

antibiotic has been completed, and allows the practitioner to evaluate the patient for the persistent presence of fluid within the middle ear. In children who have a problem with recurrent otitis media, a small daily dose of an antibiotic may prevent repeated full attacks of otitis media. In children who have persistent fluid, a procedure to place tiny tubes within the eardrum may help equalize pressure between the middle ear and the outside, thus preventing further fluid accumulation.

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ORGANIZATIONS

American Academy of Otolaryngology, Head and Neck Surgery, Inc. One Prince Street, Alexandria, VA 22314-3357. (703) 836-4444.

American Academy of Pediatrics (AAP). 141 Northwest Point Boulevard, Elk Grove Village, IL 60007. (847) 434-4000. <www.aap.org>.

American Osteopathic Association (AOA). 142 East Ontario Street, Chicago, IL 60611. (800) 621-1773. <www.aoa-net.org>.

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Otosclerosis

Definition

Otosclerosis is an excessive growth in the bones of the middle ear which interferes with the transmission of sound.

Description

The middle ear consists of the eardrum and a chamber which contains three bones called the hammer, the anvil, and the stirrup (or stapes). Sound waves passing through the ear cause the ear drum to vibrate. This vibration is transmitted to the inner ear by the three bones. In the inner ear, the vibrations are changed into impulses which are carried by the nerves, to the brain. If excessive bone growth interferes with the stapes ability to vibrate and transmit sound waves, hearing loss will result.

Otosclerosis is classified as a conductive disorder because it involves the bones of the ear, which conduct the sound to the nerve. If a person has **hearing loss** classified as neural, the nerve conducting the impulses to the brain is involved.

Otosclerosis is a common hereditary condition. About 10% of the caucasion population has some form of otosclerosis, however, it is rare among other ethnic backgrounds. Women are more likely than men to suffer from otosclerosis. It is the most common cause of conductive hearing loss between the ages of 15–50, but if the bony growth affects only the hammer or anvil, there are no symptoms and the condition goes undetected. Disease affecting the stapes is also associated with progressive hearing loss.

Causes and symptoms

Otosclerosis is hereditary. Acquired illness and accidents have no relationship to its development.

The primary symptom of otosclerosis is loss of hearing. In addition, many people experience **tinnitus** (noise originating inside the ear). The amount of tinnitus is not necessarily related to the kind or severity of hearing loss.

Diagnosis

Hearing loss due to otosclerosis is usually first noticed in the late teens or early twenties. Hearing loss usually occurs in the low frequencies first, followed by high frequencies, then middle frequencies. Extensive hearing tests will confirm the diagnosis.

KEY TERMS

Tinnitus—Tinnitus is noise originating in the ear, not in the environment. The noise can range from faint ringing to roaring.

Treatment

People with otosclerosis often benefit from a properly fitted hearing aid.

The surgical replacement of the stapes has become a common procedure to improve conductive hearing problems. During this operation, called a **stapedectomy**, the stapes is removed and replaced with an artificial device. The operation is performed under **local anesthesia** and is usually an outpatient procedure. Surgery is done on only one ear at a time, with a one year wait between procedures. The degree of hearing improvement reaches its maximum about four months after the surgery. Over 80% of these procedures successfully improve or restore hearing.

Prognosis

People with otosclerosis almost never become totally deaf, and will usually be able to hear with a hearing aid or with surgery plus a hearing aid. In older people, the tendency for additional hearing loss is diminished due to the hardening of the bones.

Prevention

Otosclerosis cannot be prevented.

Resources

ORGANIZATIONS

- American Tinnitus Association. P.O. Box 5, Portland, OR 97207. (503) 248-9985. <tinnitus@ata.org>.
- National Association of the Deaf. 814 Thayer Ave., Silver Spring, MD, 20910. (301) 587-1788. <<http://nad.policy.net>>.
- NIDCD Hereditary Hearing Impairment Resource Registry. c/o Boys Town National Research Hospital. 555 N. 30th St., Omaha NE 68131. (800) 320-1171.
- Self Help for Hard of Hearing People, Inc. 7800 Wisconsin Ave., Bethesda, MD 20814. (301) 657-2248. <<http://www.shhh.org>>.

Dorothy Elinor Stonely

Otoscopy examination see **Ear exam with an otoscope**

Ototoxicity

Definition

Ototoxicity is damage to the hearing or balance functions of the ear by drugs or chemicals.

Description

Ototoxicity is drug or chemical damage to the inner ear. This section of the ear contains both the hearing mechanism and the vestibulocochlear nerve, the nerve that sends hearing and balance information to the brain. Because of this, ototoxic drugs may cause lack of hearing, and loss of sense of balance.

The extent of ototoxicity varies with the drug, the dose, and other conditions. In some cases, there is full recovery after the drug has been discontinued. In other cases, the extent of damage is limited, and may even be too small to be noticed. This may occur in high-frequency **hearing loss**, where the damage to the ear makes it difficult to hear high pitched musical notes, but does not affect the ability to hear the spoken word, or carry on a conversation. In extreme cases, there may be permanent and complete deafness.

Although ototoxicity is undesirable, the ear damage can actually be used to help people with **Ménière's disease**. This is a disease of no known cause that is marked by sudden episodes of **dizziness** and vertigo. Other symptoms include a feeling of "fullness" in the ears, roaring in the ears, and ringing in the ears. While most people with this condition can be controlled with medication, about 10% require surgery. However, use of some ototoxic drugs can actually improve this condition, while causing less damage to the hearing mechanism than traditional treatments.

Causes and symptoms

Many drugs can cause ototoxicity.

Antibiotics

- amikacin (Amikin)
- streptomycin
- neomycin
- gentamicin (Garamycin)
- erythromycin (E-Mycin, Eryc)
- kanamycin (Kantrex)
- tobramycin (Nebcin)

- netilmycin (Netromycin)
- vancomycin (Vancocin)

Anti-cancer drugs

- cisplatin (Platinol AQ)
- bleomycin (Blenoxane)
- vincristine (Oncovin)

Diuretics

- acetazolamide (Diamox)
- furosemide (Lasix)
- bumetanide (Bumex)
- ethacrynic acid (Edecrine)

A number of other drugs and chemicals may also cause ototoxicity. **Aspirin** overdose causes ringing in the ears. The **antimalarial drugs** quinine and chloroquine may also cause ear damage. Among the environmental chemicals that can cause ear damage are tin, lead, mercury, carbon monoxide, and carbon disulfide. This list is not complete, and many other drugs and chemicals, such as industrial solvents, may cause ear problems.

Diagnosis

Ototoxicity often goes undiagnosed. This occurs when the hearing loss is slight, or when it is restricted to the higher frequencies. Patients may notice a change in their hearing, but it may not be significant enough to report.

In other cases, the loss of hearing may be very significant, or the ototoxicity may take the form of ringing in the ears, or other sensations.

When physicians are administering medications that are known to cause hearing loss, it is often recommended that the patient receive regular hearing tests. By monitoring hearing on a regular basis, it may be possible to discontinue the medication, or reduce the dose so that no further damage is done.

Ototoxicity that causes loss of balance may be even more difficult to diagnose. These changes may take place gradually, over time, and may be confused with the effects of the condition the drugs are meant to treat. If ototoxicity is suspected, balance tests are available, including a platform balance test, and a rotary chair. These, and other tests, determine how a patient responds to motion and changes in body position.

Treatment

There are no current treatments to reverse the effects of ototoxicity.

People who suffer permanent hearing loss may elect to use hearing aids, or, when appropriate, receive a cochlear implant. For those who have balance problems, physical therapy may often be helpful. Physical therapists can help people with balance problems learn to rely more on vision and the sensations from muscles to achieve balance.

Prognosis

The prognosis depends on the drugs that caused the ototoxicity, and their dose.

The aminoglycoside **antibiotics**, gentamicin, kanamycin, netilmycin and tobramycin all cause hearing loss to varying degrees. These drugs may be used to treat life-threatening infections that are resistant to other classes of drugs, and so there may be no choice but to use them. Careful dosing can minimize, but not eliminate the risk. It is estimated that the chances of recovery are 10-15%. The hearing loss usually begins at the higher frequencies, and is usually not recognized immediately.

Erythromycin may cause hearing loss that affects all frequencies. This hearing loss usually reverses itself over time.

Aspirin and the **non-steroidal anti-inflammatory drugs** (NSAIDS) may cause ringing in the ears (**tinnitus**). This stops when the drug is discontinued.

The **diuretics** may cause a hearing loss with a rapid onset. This will usually, but not always, reverse itself when the drugs are stopped.

In some cases, the prognosis is not really clear. Vancomycin appears to cause hearing loss, but this may only occur when vancomycin is used at the same time as other ototoxic drugs, such as gentamicin or erythromycin.

Prevention

Since most ototoxicity occurs when the harmful drugs are used in high doses, careful dose calculations are the best method of prevention. Sometimes it is possible to replace the ototoxic drugs with drugs that have less severe adverse effects.

Resources

BOOKS

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New York: New York Academy of Sciences, June 1999.

KEY TERMS

Antibiotic—Drugs that kill or inhibit the growth of bacteria.

Cochlea—A division of the inner ear.

Diuretic—A drug that increases water loss through increased urination.

Ménière's disease—A disorder of the membranous labyrinth of the inner ear that is marked by recurrent attacks of dizziness, tinnitus, and deafness—also called Ménière's syndrome. It is named after Prosper Ménière (1799–1862), a French physician who was among the first people to study diseases of the ear, nose, and throat.

Tinnitus—Ringing sounds in the ears.

ORGANIZATIONS

Deafness Research Foundation. 1225 I St. NW, No. 500, Washington, DC 20005.

Ear Research Foundation. 1901 Floyd St., Sarasota, FL 34239-2909.

National Institute on Deafness and Other Communication Disorders. NIH Bldg. 10, Rm. 5C-306 9000, Rockville Park, Bethesda, MD 20892.

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Ova & parasites collection see **Stool O & P test**

Ovarian cancer

Definition

Ovarian **cancer** is cancer of the ovaries, the egg-releasing and hormone-producing organs of the female reproductive tract. Cancerous, or malignant, cells divide and multiply in an abnormal fashion.

Description

The ovaries are small, almond-shaped organs, located in the pelvic region, one on either side of the uterus. When a woman is in her childbearing years, the ovaries alternate to produce and release an egg each month during the menstrual cycle. The released egg is picked up by the adjacent fallopian tube, and continues down toward the uterus. The ovaries also

produce and secrete the female hormones estrogen and progesterone, which regulate the menstrual cycle and pregnancy, as well as support the development of the secondary female sexual characteristics (breasts, body shape, and body hair). During **pregnancy** and when women take certain medications, such as **oral contraceptives**, the ovaries are given a rest from their usual monthly duties.

Types of ovarian cancers

Ninety percent of all ovarian cancers develop in the cells lining the surface, or epithelium, of the ovaries and so are called epithelial cell tumors. About 15% of epithelial cancers are considered low malignant potential or LMP tumors. These tumors occur more often in younger women, and are more likely to be caught early, so prognosis is good.

Germ cell tumors develop in the egg-producing cells of the ovary, and comprise about 5% of ovarian tumors. These tumors are usually found in teenage girls or young women. The prognosis is good if found early, but as with other ovarian cancers, early detection is difficult.

Primary peritoneal carcinoma (PPC) is a cancer of the peritoneum, the lining of the abdominal cavity where the internal organs are located. Although it is a distinct disease, it is linked with ovarian cancer. This is because the ovarian and peritoneal cells have the same embryonic origin. This means that the very early cells of the embryo that will ultimately develop into the ovaries and the peritoneum share a common origin. The term “primary” means that the cancer started first in the peritoneum, as opposed to the cancer starting in the ovary and then moving, or metastasizing, into the peritoneum.

Ovarian cancer can develop at any age, but is most likely to occur in women who are 50 years or older. More than half the cases are among women who are aged 65 years and older. Industrialized countries have the highest incidence of ovarian cancer. Native American and Caucasian women, especially Caucasians of Ashkenazi Jewish descent, are at somewhat higher risk; Hispanic, African-American, and Asian women are at a slightly lower risk. The risk of developing the disease increases with age. Ovarian cancer is the fifth most common cancer among women in the United States, and the second most common gynecologic cancer. It accounts for 4% of all cancers in women. However, because of poor early detection, the **death** rate for ovarian cancer is higher than for that of any other cancer among women. About 1 in 70 women in the United States will

eventually develop ovarian cancer, and 1 in 100 will die from it. The American Cancer Society estimates that 26,000 new cases of ovarian cancer will be diagnosed in the United States in 2004, and that 16,000 women will die from the disease.

Only 50% of the women who are diagnosed with ovarian cancer will survive five years after initial diagnosis. This is due to the cancer being at an advanced stage at the time of diagnosis. With early detection, however, survival at five years post diagnosis may be 95%.

Causes and symptoms

Causes

The actual cause of ovarian cancer remains unknown, but several factors are known to increase one's chances of developing the disease. These are called risk factors. Women at a higher risk than average of developing ovarian cancer include women who:

- have never been pregnant or had children,
- are Caucasian, especially of Northern European or Ashkenazi Jewish descent,
- are over 50 (half of all diagnosed cases are in women over 65),
- have a family history of breast, ovarian, endometrial (uterine), prostate, or colon cancer,
- have had breast cancer,
- have a first-degree relative (mother, daughter, sister) who has had ovarian cancer. (The risk is greater if two or more first-degree relatives had the disease. Having a grandmother, aunt or cousin with ovarian cancer also puts a woman at higher-than-average risk.)
- have the genetic mutation BRCA1 or BRCA2. (Not all women with these genetic breast cancer mutations will develop ovarian cancer. By age 70, a woman who has the BRCA1 mutation carries about a 40–60% risk of developing ovarian cancer. Women with the genetic mutation BRCA2 have a 15% increased risk of developing ovarian cancer. However, heredity only plays a role in about 5–10% of cases of ovarian cancer.)

Women who have a strong familial history may benefit from genetic counseling to better understand their risk factors.

In addition to the above risk factors, the following factors appear to play a role in affecting a women's chances of developing ovarian cancer.

Reproduction and hormones. Early menstruation (before age 12) and late **menopause** seem to put women at a higher risk for ovarian cancer. This appears to be

because the longer, or more often, a woman ovulates, the higher her risk for ovarian cancer. As mentioned above, women who were never pregnant have a higher risk of developing the disease than women with one or more pregnancies. It is not yet clear from research studies whether a pregnancy that ends in **miscarriage** or **stillbirth** lowers the risk factor to the same degree as the number of term pregnancies. The use of postmenopausal estrogen supplementation for 10 years or more may double a woman's risk of ovarian cancer. Short-term use does not seem to alter one's risk factor.

Infertility drug-stimulated ovulation. Research studies have reported mixed findings on this issue. It appears that women who take medications to stimulate ovulation, yet do not become pregnant, are at higher risk of developing ovarian cancer. Women who do become pregnant after taking fertility drugs do not appear to be at higher risk. One study reported that the use of the fertility drug clomiphene citrate for more than a year increased the risk of developing LMP tumors. LMP tumors respond better to treatment than other ovarian tumors.

Talc. The use of talcum powder in the genital area has been implicated in ovarian cancer in many studies. It may be because talc contains particles of asbestos, a known carcinogen. Female workers exposed to asbestos had a higher-than-normal risk of developing ovarian cancer. Genital deodorant sprays may also present an increased risk. Not all studies have brought consistent results.

Fat. A high-fat diet has been reported in some studies to increase the risk of developing ovarian cancer. In one study the risk level increased with every 10 grams of saturated fat added to the diet. This may be because of its effect on estrogen production.

Symptoms

Most of the literature on ovarian cancer states that there are usually no early warning symptoms for the disease. Ovarian cancer is often referred to as a silent killer, because women either are unaware of having it, or have symptoms that are not accurately diagnosed until the disease is in an advanced state. However, a November 2000 study reported in the medical journal *Cancer* analyzed more than 1,700 questionnaires completed by women with stage III and stage IV ovarian cancer. The researchers found that 95% of the women reported having had early symptoms that they brought to their doctors. Most symptoms were somewhat vague and either abdominal or gastrointestinal in nature, and consequently were either not properly diagnosed or were recognized

as being ovarian in nature only after a significant length of time had passed.

The following symptoms are warning signs of ovarian cancer, but could also be due to other causes. Symptoms that persist for two to three weeks, or symptoms that are unusual for the particular woman should be evaluated by a doctor right away.

- digestive symptoms, such as gas, **indigestion**, **constipation**, or a feeling of fullness after a light meal
- bloating, distention or cramping
- abdominal or low-back discomfort
- pelvic pressure or frequent urination
- unexplained changes in bowel habits
- **nausea** or **vomiting**
- **pain** or swelling in the abdomen
- loss of appetite
- **fatigue**
- unexplained weight gain or loss
- pain during intercourse
- vaginal bleeding in post-menopausal women

Diagnosis

In the best-case scenario a woman is diagnosed with ovarian cancer while it is still contained in just one ovary. Early detection can bring five-year survival to near 95%. Unfortunately, about 75% of women (3 out of 4) have advanced ovarian cancer at the time of diagnosis. (Advanced cancer is at stage III or stage IV when it has already spread to other organs.) Five-year survival for women with stage IV ovarian cancer may be less than 5%.

Diagnostic tests and techniques

If ovarian cancer is suspected, several of the following tests and examinations will be necessary to make a diagnosis:

- a complete medical history to assess all the risk factors
- a thorough bi-manual pelvic examination
- CA-125 assay
- one or more various imaging procedures
- a lower GI series, or **barium enema**
- diagnostic laparoscopy

BI-MANUAL PELVIC EXAMINATION. The exam should include feeling the following organs for any abnormalities in shape or size: the ovaries, fallopian

tubes, uterus, vagina, bladder, and rectum. Because the ovaries are located deep within the pelvic area, it is unlikely that a manual exam will pick up an abnormality while the cancer is still localized. However, a full examination provides the practitioner with a more complete picture. An enlarged ovary does not confirm cancer, as the ovary may be large because of a cyst or **endometriosis**. While women should have an annual **Pap test**, this test screens for cervical cancer. Cancerous ovarian cells, however, might be detected on the slide. Effectiveness of using Pap smears for ovarian cancer detection is about 10-30%.

CA-125 ASSAY. This is a blood test to determine the level of CA-125, a tumor marker. A tumor marker is a measurable protein-based substance given off by the tumor. A series of CA-125 tests may be done to see if the amount of the marker in the blood is staying stable, increasing or decreasing. A rising CA-125 level usually indicates cancer, while a stable or declining value is more characteristic of a cyst. The CA-125 level should never be used alone to diagnose ovarian cancer. It is elevated in about 80% of women with ovarian cancer, but in 20% of cases is not. In addition, it could be elevated because of a non-ovarian cancer, or it can be elevated with non-malignant gynecologic conditions, such as endometriosis or **ectopic pregnancy**. During menstruation the CA-125 level may be elevated, so the test is best done when the woman is not in her menses.

IMAGING. There are several different imaging techniques used in ovarian cancer evaluation. A fluid-filled structure such as a cyst creates a different image than does a solid structure, such as a tumor. An ultrasound uses high-frequency sound waves that create a visual pattern of echoes of the structures at which they are aimed. It is painless, and is the same technique used to check the developing fetus in the womb. Ultrasound may be done externally through the abdomen and lower pelvic area, or with a transvaginal probe.

Other painless imaging techniques are computed tomography (CT) and **magnetic resonance imaging** (MRI). Color Doppler analysis provides additional contrast and accuracy in distinguishing masses. It remains unclear whether Doppler is effective in reducing the high number of false-positives with transvaginal ultrasonography. These imaging techniques allow better visualization of the internal organs and can detect abnormalities without having to perform surgery.

LOWER GI SERIES. A lower GI series, or barium enema, uses a series of x rays to highlight the colon

and rectum. To provide contrast, the patient drinks a chalky liquid containing barium. This test might be done to see if the cancer had spread to these areas.

DIAGNOSTIC LAPAROSCOPY. This technique uses a thin hollow lighted instrument inserted through a small incision in the skin near the belly button to visualize the organs inside of the abdominal cavity. If the ovary is believed to be malignant, the entire ovary is removed (**oophorectomy**) and its tissue sent for evaluation to the pathologist, even though only a small piece of the tissue is needed for evaluation. If cancer is present, great care must be taken not to cause the rupture of the malignant tumor, as this would cause spreading of the cancer to adjacent organs. If the cancer is completely contained in the ovary, its removal functions also as the treatment. If the cancer has spread or is suspected to have spread, then a saline solution may be instilled into the cavity and then drawn out again. This technique is called peritoneal lavage. The aspirated fluid will be evaluated for the presence of cancer cells. If peritoneal fluid is present, called **ascites**, a sample of this will also be drawn and examined for malignant cells. If cancer cells are present in the peritoneum, then treatment will be directed at the abdominal cavity as well.

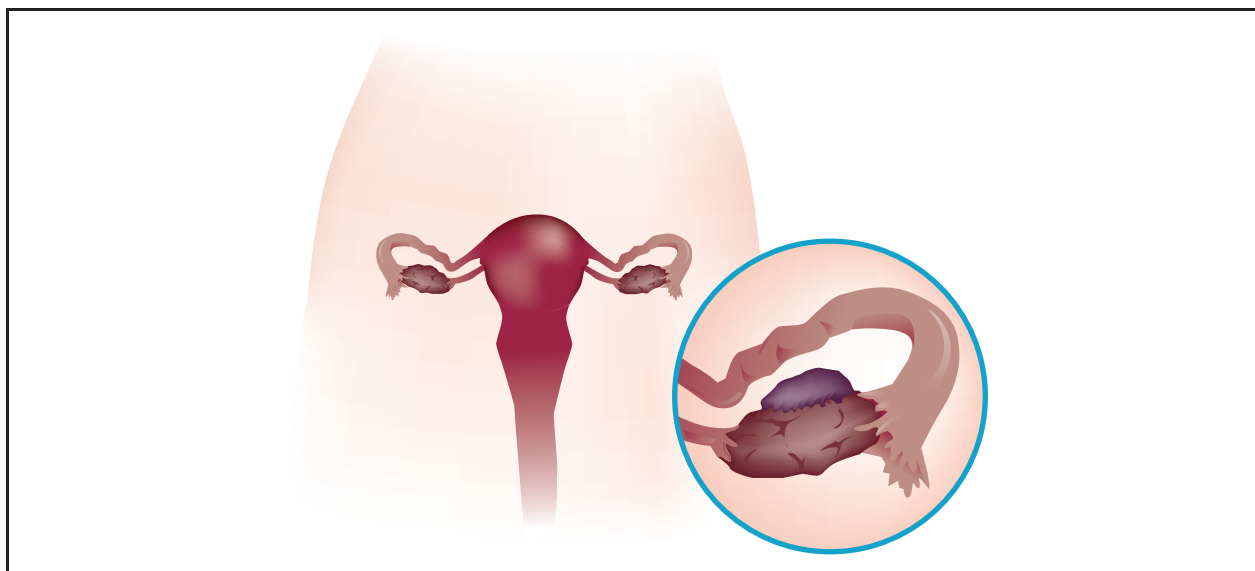
RESEARCH AND NEW DIAGNOSTIC TESTS. Many cancer researchers recognize the urgency of developing a new diagnostic test for ovarian cancer that is both sensitive and reliable. Some experts in the field look to proteomics, which is the large-scale identification and analysis of all the proteins in an organism or organ, to lead eventually to the development of a useful new test for ovarian cancer.

A group of researchers in Canada reported in 2003 that human kallikrein gene 14 (KLK14) might serve as a new biomarker for ovarian cancer. Kallikreins are a group of compounds that help to split up complex protein molecules into smaller units; prostate-specific antigen, or PSA, is a kallikrein. Early results of tests for KLK14 indicate that about 65% of women known to have ovarian cancer have elevated levels of this kallikrein.

Treatment

Clinical staging

Staging is the term used to determine if the cancer is localized or has spread, and if so, how far and to where. Staging helps define the cancer, and will determine the course of suggested treatment. Staging involves examining any tissue samples that have been taken from the ovary, nearby lymph nodes, as well as



A stage I tumor of the ovary. (Illustration by Argosy Inc.)

from any nearby organs or structures where metastasis was suspected. This may include the diaphragm, lungs, stomach, intestines and omentum (the tissue covering internal organs), and any fluid as described above.

The National Cancer Institute Stages for ovarian cancer are:

- Stage I: Cancer is confined to one or both ovaries.
- Stage II: Cancer is found in one or both ovaries and/or has spread to the uterus, fallopian tubes, and/or other body parts within the pelvic cavity.
- Stage III: Cancer is found in one or both ovaries and has spread to lymph nodes or other body parts within the abdominal cavity, such as the surfaces of the liver or intestines.
- Stage IV: Cancer is found in one or both ovaries and has spread to other organs such as the liver or lung.

The individual stages are also further broken down in detail, such as Ia, Ib, etc. Accurate staging is important for several reasons. Treatment plans are based on staging, in part because of trying to duplicate the best results achieved in prior research trials. When staging is inconsistent, it becomes more difficult to know how different research studies compare, so the results themselves cannot be relied upon.

Treatment offered will primarily depend on the stage of the cancer and the woman's age. It is always appropriate to consider getting a second opinion, especially when treatment involves surgery, **chemotherapy**, and possible radiation. Before the patient

makes her decision as to which course of treatment to take, she should feel that she has the information necessary with which to make an informed decision. The diagnostic tools mentioned above are used to determine the course of treatment. However, the treatment plan may need to be revised if the surgeon sees that the tumor has spread beyond the scope of what was seen during diagnostic tests.

Surgery

Surgery is done to remove as much of the tumor as possible (called tissue debulking), utilizing chemotherapy and/or radiation to target cancer cells that have remained in the body, without jeopardizing the woman's health. This can be hard to balance once the cancer has spread. Removal of the ovary is called oophorectomy, and removal of both ovaries is called bilateral oophorectomy. Unless it is very clear that the cancer has not spread, the fallopian tubes are usually removed as well (**salpingo-oophorectomy**). Removal of the uterus is called **hysterectomy**.

If the woman is very young, all attempts will be made to spare the uterus. It is crucial that a woman discuss with her surgeon her childbearing plans prior to surgery. Unfortunately, ovarian cancer spreads easily and often swiftly throughout the reproductive tract. It may be necessary to remove all reproductive organs as well as part of the lining of the peritoneum to provide the woman with the best possible chance of long-term survival. Fertility-sparing surgery can be successful if the ovarian cancer is caught very early.

Side effects of the surgery will depend on the extent of the surgery, but may include pain and temporary difficulty with bladder and bowel function, as well as reaction to the loss of hormones produced by the organs removed. A hormone replacement patch may be applied to the woman's skin in the recovery room to help with the transition. An emotional side effect may be the feeling of loss stemming from the removal of reproductive organs.

Chemotherapy

Chemotherapy is used to target cells that have traveled to other organs, and throughout the body via the lymphatic system or the blood stream. Chemotherapy drugs are designed to kill cancer cells, but may also be harmful to healthy cells as well. Chemotherapy may be administered through a vein in the arm (intravenous, IV), may be taken in tablet form, and/or may be given through a thin tube called a catheter directly into the abdominal cavity (intraperitoneal). IV and oral chemotherapy drugs travel throughout the body; intraperitoneal chemotherapy is localized in the abdominal cavity.

Side effects of chemotherapy can vary greatly depending on the drugs used. Currently, chemotherapy drugs are often used in combinations to treat advanced ovarian cancer, and usually the combination includes a platinum-based drug (such as cisplatin) with a taxol agent, such as paclitaxel. Some of the combinations used or being studied include: carboplatin/paclitaxel, cisplatin/paclitaxel, cisplatin/topotecan, and cisplatin/carboplatin. As new drugs are evaluated and developed, the goal is always for maximum effectiveness with minimum side effects. Side effects include **nausea and vomiting**, **diarrhea**, decreased appetite and resulting weight loss, fatigue, headaches, loss of hair, and **numbness and tingling** in the hands or feet. Managing these side effects is an important part of cancer treatment.

After the full course of chemotherapy has been given, the surgeon may perform a "second look" surgery to examine the abdominal cavity again to evaluate the success of treatment.

Radiation

Radiation uses high-energy, highly focused x rays to target very specific areas of cancer. This is done using a machine that generates an external beam. Very careful measurements are taken so that the targeted area can be as focused and small as possible. Another form of radiation uses a radioactive liquid that is administered into the abdominal cavity in the

same fashion as intraperitoneal chemotherapy. Radiation is usually given on a daily Monday through Friday schedule and for several weeks continuously. Radiation is not painful, but side effects can include skin damage at the area exposed to the external beam, and extreme fatigue. The fatigue may hit suddenly in the third week or so of treatment, and may take a while to recover even after treatments have terminated. Other side effects may include nausea, vomiting, diarrhea, loss of appetite, weight loss and urinary difficulties. For patients with incurable ovarian cancer, radiation may be used to shrink tumor masses to provide pain relief and improve quality of life.

Once the full course of treatment has been undertaken, it is important to have regular follow-up care to monitor for any long-term side effects as well as for future relapse or metastases.

Alternative treatment

The term alternative therapy refers to therapy utilized instead of conventional treatment. By definition, these treatments have not been scientifically proven or investigated as thoroughly and by the same standards as conventional treatments. The terms complementary or integrative therapies denote practices used in conjunction with conventional treatment. Regardless of the therapies chosen, it is key for patients to inform their doctors of any alternative or complementary therapies being used or considered. (Some alternative and complementary therapies adversely affect the effectiveness of conventional treatments.) Some common complementary and alternative medicine techniques and therapies include:

- prayer and faith healing
- **meditation**
- mind/body techniques such as support groups, visualization, guided imagery and hypnosis
- energy work such as **therapeutic touch** and Reiki
- **Acupuncture** and Chinese herbal medicine
- body work such as **yoga**, massage and **t'ai chi**
- **vitamins** and herbal supplements
- diets such as **vegetarianism** and macrobiotic

Mind/body techniques along with meditation, prayer, yoga, T'ai Chi and acupuncture have been shown to reduce **stress** levels, and the relaxation provided may help boost the body's immune system. The effectiveness of other complementary and alternative treatments is being studied by the National Institutes of Health's National Center for Complementary and

Alternative Medicine (NCCAM). For a current list of the research studies occurring, results of recent studies, or publications available, patients can visit the NCCAM web site or call at (888) 644-6226.

Some programs for treatment of ovarian cancer integrate alternative or complementary treatments with conventional surgery or chemotherapy. As of early 2003, the University of Kansas Medical Center is conducting a study of the effectiveness of adding four well-known antioxidants (vitamins A, C, E, and beta-carotene) to standard chemotherapy regimens for ovarian cancer.

Prognosis

Prognosis for ovarian cancer depends greatly on the stage at which it is first diagnosed. While stage I cancer may have a 95% success rate, stages III and IV may have a survival rate of 17-30% at five years post-diagnosis. Early detection remains an elusive, yet hopeful, goal of research. Also, clinical trials are addressing new drug and treatment combinations to prolong survival in women with more advanced disease. Learning one's family history may assist in early detection, and genetic studies may clarify who is at greater risk for the disease.

Research studies are usually designed to compare a new treatment method against the standard method, or the effectiveness of a drug against a placebo (an inert substance that would be expected to have no effect on the outcome). Since the research is experimental in nature, there are no guarantees about the outcome. New drugs being used may have harmful, unknown side effects. Some people participate to help further knowledge about their disease. For others, the study may provide a possible treatment that is not yet available otherwise. If one participates in a study and is in the group receiving the standard care or the placebo, and the treatment group gets clear benefit, it may be possible to receive the experimental treatment once one's original participation role is over. Participants will have to meet certain criteria before being admitted into the study. It is important to fully understand one's role in the study, and weigh the potential risks versus benefits when deciding whether or not to participate.

Prevention

Since the cause of ovarian cancer is not known, it is not possible to fully prevent the disease. However, there are ways to reduce one's risks of developing the disease.

KEY TERMS

Biomarker—A biochemical substance that can be detected in blood samples and indicates the presence of a cancerous tumor.

Gynecologic oncologist—A physician specializing in the treatment of cancers of the female reproductive tract.

Kallikrein—Any of a group of compounds in the body known as serine endopeptidases that help to break down proteins into smaller units. Prostate-specific antigen belongs to this group of chemicals. A recently discovered kallikrein may be useful as a biomarker for ovarian cancer.

Lymphatic system—A connected network of nodes, or glands, that carry lymph throughout the body. Lymph is a fluid that contains the infection-fighting white blood cells that form part of the body's immune system. Because the network goes throughout the body, cancer cells that enter the lymphatic system can travel to and be deposited at any point into the tissues and organs and form new tumors there.

Paclitaxel—A drug derived from the common yew tree (*Taxus baccata*) that is the mainstay of chemotherapy for ovarian cancer.

Pathologist—The pathologist is a doctor specializing in determining the presence and type of disease by looking at cells and tissue samples.

Decrease ovulation. Pregnancy gives a break from ovulation, and multiple pregnancies appear to further reduce the risk of ovarian cancer. The research is not clear as to whether the pregnancy must result in a term delivery to have full benefit. Women who breastfeed their children also have a lower risk of developing the disease. Since oral contraceptives suppress ovulation, women who take birth control pills (BCPs), even for as little as 3 to 6 months have a lower incidence of the disease. It appears that the longer a woman takes BCPs, the lower her risk for ovarian cancer. Also, this benefit may last for up to 15 years after a woman has stopped taking them. However, since BCPs alter a woman's hormonal status, her risk for other hormonally related cancers may change. For this reason it is very important to discuss all the risks and benefits with one's health care provider.

Genetic testing. Tests are available which can help to determine whether a woman who has a family

history of breast, endometrial, or ovarian cancer has inherited the mutated BRCA gene that predisposes her to these cancers. If the woman tests positive for the mutation, then she may be able to choose to have her ovaries removed. Even without testing for the mutated gene, some women with strong family histories of ovarian cancer may consider having their ovaries removed as a preventative measure (prophylactic oophorectomy). This procedure diminishes but does not completely remove the risk of cancer, as some women may still develop primary peritoneal carcinoma after oophorectomy.

Surgery. Procedures such as **tubal ligation** (in which the fallopian tubes are blocked or cut off) and hysterectomy (in which the uterus is removed) appear to reduce the risk of ovarian cancer. However, any removal of the reproductive tract organs has surgical as well as hormonal side effects.

Screening. There are no definitive tests or screening procedures to detect ovarian cancer in its early stages, although a blood test for early detection of asymptomatic ovarian cancer is under development as of early 2003. Women at high risk should consult with their physicians about regular screenings, which may include **transvaginal ultrasound** and the blood test for the CA-125 protein.

The American Cancer Society recommends annual pelvic examinations for all women after age 40, in order to increase the chances of early detection of ovarian cancer.

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ORGANIZATIONS

- American Cancer Society. (800) ACS-2345. <<http://www.cancer.org>>.
- Cancer Research Institute. 681 Fifth Avenue, New York, NY 10022. (800) 992-2623. <<http://www.cancerresearch.org>>.
- National Cancer Institute. Building 31, Room 10A31, 31 Center Drive, MSC 2580, Bethesda, MD 20892-2580. (301) 435-3848. <<http://www.nci.nih.gov>>.
- National Center for Complementary and Alternative Medicine. NCCAM Clearinghouse, P.O. Box 8218, Silver Spring, MD 20907-8218. (888) 644-6226. <<http://nccam.nih.gov>>.
- Oncolink at the University of Pennsylvania. <<http://www.oncolink.upenn.edu>>.
- Women's Cancer Network. c/o Gynecologic Cancer Foundation, 401 N. Michigan Avenue, Chicago, IL 60611. (312) 644-6610. <<http://www.wcn.org>>.

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Ovarian cysts

Definition

Ovarian cysts are sacs containing fluid or semi-solid material that develop in or on the surface of an ovary.

Description

Ovarian cysts are common and the vast majority are harmless. Because they cause symptoms that may be the same as ovarian tumors that may be cancerous, ovarian cysts should always be checked out. The most common types of ovarian cysts are follicular and corpus luteum, which are related to the menstrual cycle. Follicular cysts occur when the cyst-like follicle on the ovary in which the egg develops does not burst and release the egg. They are usually small and harmless, disappearing within two to three menstrual cycles. Corpus luteum cysts occur when the corpus luteum—a small, yellow body that secretes hormones—does not dissolve after the egg is released. They usually disappear in a few weeks but can grow to more than 4 in (10 cm) in diameter and may twist the ovary.

Ovarian cysts can develop at any time in a female’s life from infancy to **puberty** to menopause, including during **pregnancy**. Follicular cysts occur frequently during the years when a woman is menstruating, and are nonexistent in postmenopausal women or any woman who is not ovulating. Corpus luteum cysts occur occasionally during the menstrual years and during early pregnancy. (Dermoid cysts, which may contain hair, teeth, or skin derived from the outer layer of cells of an embryo, are also occasionally found in the ovary.)

Causes and symptoms

Causes

Follicular cysts are caused by the formation of too much fluid around a developing egg. Corpus luteum cysts are caused by excessive accumulation of blood

during the menstrual cycle, hormone therapy, or other types of ovarian tumors.

There is also a condition known as **polycystic ovary syndrome** (PCOS) in which the eggs and follicles are not released from the ovaries and instead form multiple cysts. **Obesity** is linked to this condition, as 50% of women with PCOS are also obese. Hormonal imbalances play a major role in this condition, including high levels of the hormone androgen and low levels of progesterone, the female hormone necessary for egg release. High levels of insulin, the hormone that regulates blood sugar, are often found in women with PCOS. PCOS is also characterized by irregular menstrual periods, **infertility**, and **hirsutism** (excessive hair growth on the body and face). Although PCOS was formerly thought to be an adult-onset condition, more recent research indicates that it begins in childhood, possibly even during fetal development.

PCOS is also known to run in families, which suggests that genetic factors contribute to its development. As of 2002, the specific gene or genes responsible for PCOS have not yet been identified; however, several groups of researchers in different countries have been investigating genetic variations associated with increased risk of type 2 diabetes in order to determine whether the same genetic variations may be involved in PCOS.

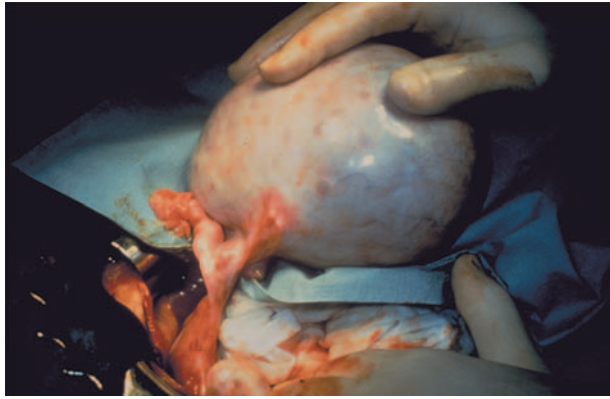
In adolescent girls, ovarian cysts may be associated with a genetic disorder known as McCune-Albright syndrome, which is characterized by abnormal bone growth, discoloration of the skin, and early onset of puberty. The ovarian cysts are responsible for the early sexual maturation.

As of early 2003, McCune-Albright syndrome is known to be associated with mutations in the *GNAS1* gene. The mutation is sporadic, which means that it occurs during the child’s development in the womb and that the syndrome is not inherited.

Symptoms

Many ovarian cysts have no symptoms. When the growth is large or there are multiple cysts, the patient may experience any of the following symptoms:

- Fullness or heaviness in the abdomen.
- Pressure on the rectum or bladder.
- Pelvic **pain** that is a constant dull ache and may spread to the lower back and thighs, occurs shortly before the beginning or end of menstruation, or occurs during intercourse.



An ovarian cyst is being surgically removed from a 25-year-old female patient. (Photograph by Art Siegel, Custom Medical Stock Photo. Reproduced by permission.)

Diagnosis

Non-symptomatic ovarian cysts are often felt by a doctor examining the ovaries during a routine **pelvic exam**. Symptomatic ovarian cysts are diagnosed through a pelvic exam and ultrasound. Ultrasonography is a painless test that uses a hand-held wand to send and receive sound waves to create images of the ovaries on a computer screen. The images are photographed for later analysis. It takes about 15 minutes and is usually done in a hospital or a physician's office.

Ovarian cysts can be diagnosed in female fetuses by transabdominal ultrasound during the mother's pregnancy.

Treatment

Watchful waiting

Many follicular and corpus luteum cysts require no treatment and disappear on their own. Often the physician will wait and re-examine the patient in four to six weeks before taking any action. Follicular cysts do not require treatment, but birth control pills may be taken if the cysts interfere with the patient's daily activities.

Most uncomplicated ovarian cysts in female infants resolve on their own shortly after delivery. Complicated cysts are treated by **laparoscopy** or laparotomy after the baby is born.

Medications

McCune-Albright syndrome is treated with testosterone (Teslac), an anti-estrogen drug that corrects the hormonal imbalance caused by the ovarian cysts.

Long-term management of PCOS has been complicated in the past by lack of a clear understanding of the causes of the disorder. Most commonly, hormonal therapy has been recommended, including estrogen and progesterone and such other hormone-regulating drugs as ganirelix (Antagon). Birth control pills have also been prescribed by doctors to regulate the menstrual cycle and to shrink functional cysts.

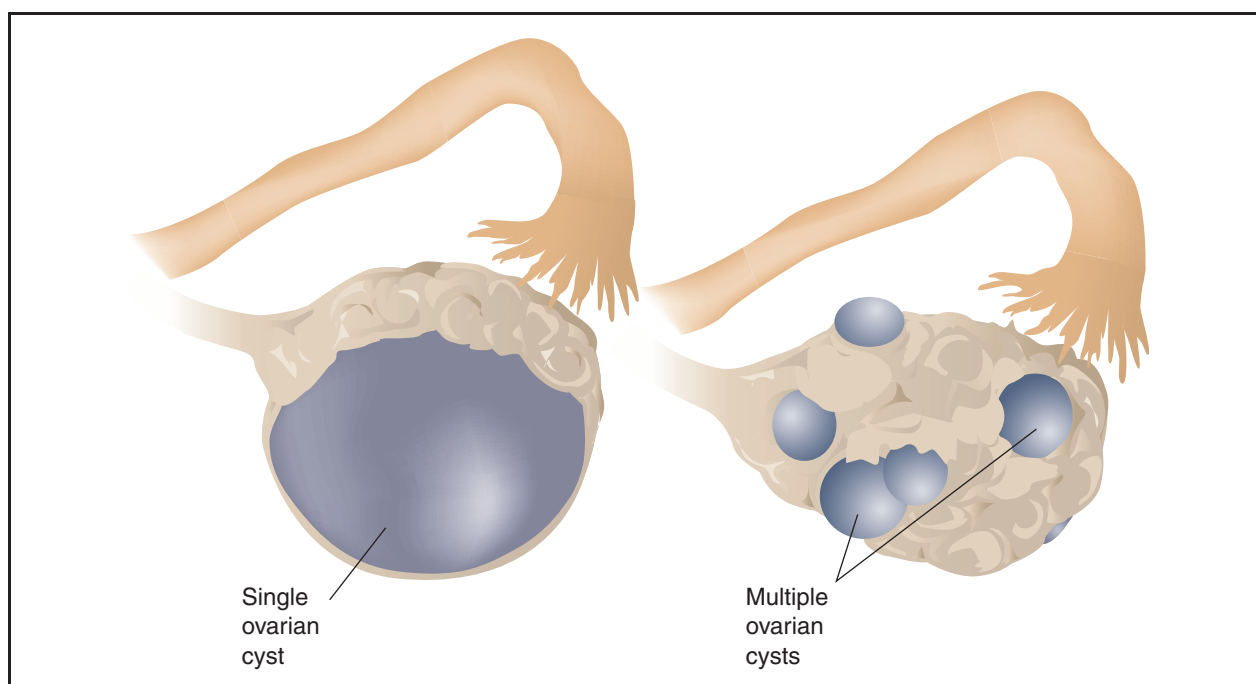
More recent studies have shown that increasing sensitivity to insulin in women with PCOS leads to improvement in both the hormonal and metabolic symptoms of the disorder. As of 2002, this sensitivity is increased by either weight loss and **exercise** programs or by medications. Metformin (Glucophage), a drug originally developed to treat type 2 diabetes, has been shown to be effective in reducing the symptoms of hyperandrogenism as well as **insulin resistance** in women with PCOS.

Another strategy that is being tried with PCOS is administration of flutamide (Eulexin), a drug normally used to treat **prostate cancer** in men. Preliminary results indicate that the antiandrogenic effects of flutamide benefit patients with PCOS by increasing blood flow to the uterus and ovaries.

Surgery

Surgery is usually indicated for patients who have not reached puberty and have an ovarian mass and in postmenopausal patients. Surgery is also indicated if the growth is larger than 4 in (10 cm), complex, growing, persistent, solid and irregularly shaped, on both ovaries, or causes pain or other symptoms. Ovarian cysts are curable with surgery but often recur without it.

Surgical options include removal of the cyst or removal of one or both ovaries. More than 90% of benign ovarian cysts can be removed using laparoscopy, a minimally invasive outpatient procedure. In laparoscopic **cystectomy**, the patient receives a general or local anesthetic, then a small incision is made in the abdomen. The laparoscope is inserted into the incision and the cyst or the entire ovary is removed. Laparoscopic cystectomy enables the patient to return to normal activities within two weeks. Surgical cystectomy to remove cysts and/or ovaries is performed under general anesthesia in a hospital and requires a stay of five to seven days. After an incision is made in the abdomen, the muscles are separated and the membrane surrounding the abdominal cavity (peritoneum) is opened. Blood vessels to the ovaries are clamped and tied. The cyst is located and removed. The peritoneum is closed, and the abdominal muscles and skin



(Illustration by Argosy Inc.)

are closed with sutures or clips. Recovery takes four weeks.

A surgical procedure known as ovarian wedge resection appears to improve fertility in women with PCOS who have not responded to drug treatments. In an ovarian wedge resection, the surgeon removes a portion of the polycystic ovary in order to induce ovulation.

Alternative treatment

Alternative treatments for ovarian problems—herbal therapies, **nutrition** and diet, and homeopathy—should be used to supplement, not replace, conventional treatment. General herbal tonics for female reproductive organs that can be taken in tea or tincture (an alcohol-based herbal extract) form include blue cohosh (*Caulophyllum thalictroides*) and false unicorn root (*Chamaelirium luteum*). Recommendations to help prevent and treat ovarian cysts include a vegan diet (no dairy or animal products) that includes beets, carrots, dark-green leafy vegetables, and lemons; antioxidant supplements including zinc and **vitamins** A, E, and C; as well as black currant oil, borage oil, and evening primrose oil (*Oenothera biennis*) supplements. Homeopathic treatments—tablets, powders, and liquids prepared from plant, mineral, and animal extracts—may also be

effective in treating ovarian cysts. Castor oil packs can help reduce inflammation. Hydrotherapy applied to the abdomen can help prevent rupture of the cyst and assist its reabsorption.

Prognosis

The prognosis for non-cancerous ovarian cysts is excellent.

Prevention

Ovarian cysts cannot be prevented.

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KEY TERMS

Corpus luteum—A small, yellow structure that forms in the ovary after an egg has been released.

Cystectomy—Surgical removal of a cyst.

Dermoid—A skin-like benign growth that may appear on the ovary and resemble a cyst.

Endocrine—Internal secretions, usually in the systemic circulation.

Follicular—Relating to one of the round cells in the ovary that contain an ovum.

Hirsutism—A condition marked by excessive hair growth on the face and body.

Functional cyst—A benign cyst that forms on the ovary and resolves on its own without treatment.

McCune-Albright syndrome (MCAS)—A genetic syndrome characterized in girls by the development of ovarian cysts and puberty before the age of 8, together with abnormalities of bone structure and skin pigmentation.

Ovulation—The phase of the female monthly cycle when a developed egg is released from the ovary into the fallopian tube for possible fertilization.

Polycystic ovarian syndrome (PCOS)—A condition in which the eggs are not released from the ovaries and instead form multiple cysts.

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ORGANIZATIONS

American College of Obstetricians and Gynecologists (ACOG). 409 12th Street, SW, P. O. Box 96920, Washington, DC 20090-6920. <<http://www.acog.org>>.

The Health Resource. 209 Katherine Drive. Conway, AR 72032. (501) 329-5272.

Polycystic Ovarian Syndrome Association. P. O. Box 80517, Portland, OR 97280. (877) 775-PCOS. <www.pcosupport.org>.

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Ovarian torsion

Definition

Ovarian torsion is the twisting of the ovary due to the influence of another condition or disease. This results in extreme lower abdominal **pain**.

Description

Ovarian torsion occurs infrequently only in females. It can occur in women of all ages, but most women that experience this are younger. Approximately 70-75% of cases occur in women under 30 years old. About 20% of all reported cases are in pregnant women. It is the fifth most common gynecological emergency which can include surgical intervention.

Ovarian torsion usually arises in only one ovary at a time. They can occur in either normal or enlarged ovaries and fallopian tubes, and occasionally they develop in both.

Causes and symptoms

There are a variety of conditions that can cause torsion of the ovary ranging from changes in normal ovaries to congenital and developmental abnormalities or even a disease that affects the tube or ovary. Normal ovaries that experience spasms or changes in the blood vessels in the mesosalpinx can become twisted. For example, if the veins in the mesosalpinx become congested, the ovaries will undergo torsion.

Developmental abnormalities of the fallopian tube such as extremely longer-than-normal tubes or a missing mesosalpinx will cause ovarian torsion. Diseases such as **ovarian cysts** or fibromas, tumor of the ovary or tubes, and trauma to either the ovaries or the tubes will also cause ovarian torsion.

The characteristic symptom of ovarian torsion is the sudden onset of extreme lower abdominal pain that radiates to the back, side and thigh. **Nausea**, **vomiting**, **diarrhea**, and **constipation** can accompany the pain. The patient may also experience tenderness in the lower abdominal area, a mild **fever** and tachycardia.

Diagnosis

The diagnosis of ovarian torsions usually occurs in an emergency room due to the suddenness of extreme pain. Emergency room physicians may consult with another physician specializing in obstetrics and gynecology. Since 20% of ovarian torsions occur in pregnant women, physicians will order a **pregnancy test**. Visualization with an ultrasound and CT scan (computed tomography) will help pinpoint the ovarian structures and allow physicians to diagnose. Diagnosis is often confirmed through **laparoscopy**.

Treatment

Ovarian torsions need to be repaired. This is done through surgery, and for less severe cases laparoscopic

KEY TERMS

Congenital—condition present at birth

Laparoscopy—endoscope used to observe structures in the abdomen

Mesosalpinx—a ligament connected to the fallopian tube

Ovary—female reproductive gland that contains the ova (eggs)

Tachycardia—rapidly beating heart

Torsion—the action of twisting

surgery is used. Medications such as NSAIDs are given to control pain.

Prognosis

If ovarian torsions are diagnosed and treated early, then the prognosis is favorable. However, if diagnosis is delayed, the torsions can worsen and cut off arterial blood flow into and venous blood flow out of the ovary. This results in necrosis (death) of the ovarian tissue. Delayed diagnosis can also result in problems when trying to conceive due to infertility.

Prevention

Currently, there are no known methods for prevention of ovarian torsion.

Sally C. McFarlane-Parrott

Ovary and fallopian tube removal see **Salpingo-oophorectomy**

Ovary removal see **Oophorectomy**

Overactive bladder

Definition

Overactive bladder is the leakage of large amounts of urine at unexpected times, including during sleep.

Description

People who lose urine for no apparent reason while suddenly feeling the need or urge to urinate

may have overactive bladder. The condition affects 17 million Americans. The most common cause of overactive bladder is inappropriate bladder contractions. Medical professionals describe such a bladder as “unstable,” “spastic,” or “overactive.” A doctor might call the condition “reflex incontinence” if it results from overactive nerves controlling the bladder. Having an overactive bladder can mean that the bladder empties during sleep, after drinking a small amount of water, or when touching water or hearing it running (as when someone else is taking a shower or washing dishes). Involuntary actions of bladder muscles can occur because of damage to the nerves of the bladder, to the nervous system (spinal cord and brain), or to muscles themselves. **Multiple sclerosis**, Parkinson’s disease, **Alzheimer’s disease**, **stroke**, brain tumors, and injury—including injury that occurs during surgery—all can harm bladder nerves or muscles.

Causes and symptoms

People with overactive bladder lose urine as soon as they feel a strong need to go to the bathroom. People with overactive bladder may leak urine:

- When they can not get to the bathroom quickly enough
- When they drink even a small amount of liquid, or when they hear or touch running water

People with overactive bladder may also go to the bathroom very often; for example, every two hours during the day and night. They may even wet the bed.

Diagnosis

To diagnose the problem, a doctor will first ask about symptoms and medical history. Other obvious factors that can help define the problem include straining and discomfort, use of drugs, recent surgery, and illness. If the patient’s medical history does not define the problem, it will at least suggest which tests are needed. The doctor will physically examine the patient for signs of medical conditions causing the overactive bladder, such as tumors that block the urinary tract, stool impaction, and poor reflexes or sensations, which may be evidence of a nerve-related cause. Overactive bladder is often treated by general or family practitioners but the patient may be referred to a urologist, who specializes in the urinary tract, or a urogynecologist, who focuses on urological problems in women.

Common tests used to diagnose overactive bladder include:

- Blood tests to examine blood for levels of various chemicals

- Cystoscopy to look for abnormalities in the bladder and lower urinary tract. It works by inserting a small tube into the bladder that has a telescope for the doctor to look through
- Post-void residual (PVR) measurement to see how much urine is left in the bladder after urinating by placing a small soft tube into the bladder or by using ultrasound (sound waves)
- Urinalysis to examine urine for signs of infection, blood, or other abnormalities
- Urodynamic testing to examine bladder and urethral sphincter function (may involve inserting a small tube into the bladder; x rays also can be used to see the bladder)

Treatment

Medications can reduce many types of leakage. Some drugs inhibit contractions of an overactive bladder. Others, such as *solifenacin succinate* (Vesicare), relax muscles, leading to more complete bladder emptying during urination. Some drugs tighten muscles at the bladder neck and urethra, preventing leakage. Among the drugs used are *oxybutynin* (Ditropan XL), 5-30 mg daily; *solifenacin* (Vesicare), 5-10 mg a day; *darifenacin* (Enablex), 3.75-15 mg daily; and *tolterodine* (Detrol), 2-4 mg daily. A one-month supply of these drugs costs \$90-125. Some medications, especially hormones such as estrogen, are believed to cause muscles involved in urination to function normally. Some of these medications can produce harmful side effects if used for long periods. In particular, estrogen therapy has been associated with an increased risk for cancers of the breast and the lining of the uterus. Patients should talk to their doctor about the risks and benefits of long-term use of medications.

Alternative treatment

Adjusting dietary habits and avoiding acidic and spicy foods, alcohol, **caffeine**, and other bladder irritants can help to prevent urinary leaking. Eat recommended amounts of whole grains, fruits, and vegetables to avoid **constipation**. **Bladder training**, used to treat urge incontinence, can also be a useful treatment tool. The technique involves placing a patient on a toileting schedule. The time interval between urination is then gradually increased until an acceptable time period between bathroom breaks is consistently achieved.

Biofeedback techniques can teach overactive bladder patients to control the urge to urinate.

KEY TERMS

Alzheimer's disease—A degenerative disorder that affects the brain and causes dementia, especially late in life.

Biofeedback—The use of monitoring devices that display information about the operation of a bodily function, for example, heart rate or blood pressure, that is not normally consciously controlled.

Cystoscopy—The use of a narrow tubular instrument that is passed through the urethra to examine the interior of the urethra and the urinary bladder.

Estrogen—Any of several steroid hormones, produced mainly in the ovaries, that stimulate estrus and the development of female secondary sexual characteristics.

Multiple sclerosis—A serious progressive disease of the central nervous system.

Parkinson's disease—An incurable nervous disorder marked by the symptoms of trembling hands, lifeless face, monotone voice, and a slow, shuffling walk.

Sphincter—A circular band of muscle that surrounds an opening or passage in the body and narrows or closes the opening by contracting.

Urethral—Referring to the tube in humans that carries urine from the bladder out of the body.

Urogynecologist—A physician that deals with women's health, especially with the health of women's reproductive organs and urinary tract.

Urologist—A physician who deals with the study and treatment of disorders of the urinary tract in women and the urogenital system in men.

Biofeedback uses sensors to monitor temperature and muscle contractions in the vagina to help overactive bladder patients learn to increase their control over the pelvic muscles.

An infusion, or tea, of horsetail (*Equisetum arvense*), agrimony (*Agrimonia eupatoria*), and sweet sumach (*Rhus aromatica*) may be prescribed by an herbalist or naturopath to an overactive bladder. These herbs are natural astringents, and encourage toning of the digestive and urinary tracts. Other herbs, such as urtica, or stinging nettle (*Urtica urens*), plantain (*Plantago major*), or maize (*Zea mays*) may be helpful. Homeopathic remedies may include pulsatilla and causticum. Chinese herbalists might recommend golden lock tea, a mixture of several herbs that helps the body retain fluids.

Prognosis

With proper treatment, the prognosis for controlling the disorder is very good. There is no cure for overactive bladder.

Prevention

There are no known preventative measures for overactive bladder.

Resources

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ORGANIZATIONS

National Bladder Foundation. P.O. Box 1095, Ridgefield, CT 06877. (877) 252-3337. debsla@aol.com. <http://www.bladder.org>.

Ken R. Wells

Overhydration

Definition

Overhydration, also called water excess or water intoxication, is a condition in which the body contains too much water.

Description

Overhydration occurs when the body takes in more water than it excretes and its normal sodium level is diluted. This can result in digestive problems, behavioral changes, brain damage, seizures, or **coma**. An adult whose heart, kidneys, and pituitary gland are functioning properly would have to drink more than two gallons of water a day to develop water intoxication. This condition is most common in patients whose kidney function is impaired and may occur when doctors, nurses, or other healthcare professionals administer greater amounts of water-producing fluids and medications than the patient's body can excrete. Overhydration is the most common electrolyte imbalance in hospitals, occurring in about 2% of all patients.

Infants seem to be at greater risk for developing overhydration. The Centers for Disease Control and Prevention has declared that babies are especially susceptible to oral overhydration during the first month of life, when the kidneys' filtering mechanism is too immature to excrete fluid as rapidly as older infants do. Breast milk or formula provide all the fluids a healthy baby needs. Water should be given slowly, sparingly, and only during extremely hot weather. Overhydration, which has been cited as a hazard of infant swimming lessons, occurs whenever a baby drinks too much water, excretes too little fluid, or consumes and retains too much water.

Causes and symptoms

Drinking too much water rarely causes overhydration when the body's systems are working normally. People with heart, kidney, or **liver disease** are more likely to develop overhydration because their kidneys are unable to excrete water normally. It may be necessary for people with these disorders to restrict the amount of water they drink and/or adjust the amount of salt in their diets.

Since the brain is the organ most susceptible to overhydration, a change in behavior is usually the first symptom of water intoxication. The patient may become confused, drowsy, or inattentive. Shouting and **delirium** are common. Other symptoms of overhydration may include blurred vision, **muscle cramps** and twitching, **paralysis** on one side of the body, poor coordination, nausea and **vomiting**, rapid breathing, sudden weight gain, and weakness. The patient's complexion is normal or flushed. Blood pressure is sometimes higher than normal, but elevations may not be noticed even when the degree of water intoxication is serious.

Overhydration can cause acidosis (a condition in which blood and body tissues have an abnormally high acid content), anemia, **cyanosis** (a condition that occurs when oxygen levels in the blood drop sharply), hemorrhage, and **shock**. The brain is the organ most vulnerable to the effects of overhydration. If excess fluid levels accumulate gradually, the brain may be able to adapt to them and the patient will have only a few symptoms. If the condition develops rapidly, confusion, seizures, and coma are likely to occur.

Risk factors

Chronic illness, **malnutrition**, a tendency to retain water, and kidney diseases and disorders increase the likelihood of becoming overhydrated. Infants and the elderly seem to be at increased risk for overhydration, as are people with certain mental disorders or **alcoholism**.

Diagnosis

Before treatment can begin, a doctor must determine whether a patient's symptoms are due to overhydration, in which excess water is found within and outside cells, or excess blood volume, in which high sodium levels prevent the body from storing excess water inside the cells. Overhydration is characterized by excess water both within and around the body's cells, while excess blood volume occurs when the body has too much sodium and can not move water to reservoirs within the cells. In cases of overhydration, symptoms of fluid accumulation do not usually occur. On the other hand, in cases of excess blood volume, fluid tends to accumulate around cells in the lower legs, abdomen, and chest. Overhydration can occur alone or in conjunction with excess blood volume, and differentiating between these two conditions may be difficult.

Treatment

Mild overhydration can generally be corrected by following a doctor's instructions to limit fluid intake. In more serious cases, **diuretics** may be prescribed to increase urination, although these drugs tend to be most effective in the treatment of excess blood volume. Identifying and treating any underlying condition (such as impaired heart or kidney function) is a priority, and fluid restrictions are a critical component of every treatment plan.

In patients with severe neurologic symptoms, fluid imbalances must be corrected without delay. A powerful diuretic and fluids to restore normal sodium

concentrations are administered rapidly at first. When the patient has absorbed 50% of the therapeutic substances, blood levels are measured. Therapy is continued at a more moderate pace in order to prevent brain damage as a result of sudden changes in blood chemistry.

Prognosis

Mild water intoxication is usually corrected by drinking less than a quart of water a day for several days. Untreated water intoxication can be fatal, but this outcome is quite rare.

Resources

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Maureen Haggerty

Oxycodo see **Analgesics, opioid**

Oxygen inhalation therapy see **Oxygen/ozone therapy**

Oxygen/ozone therapy

Definition

Oxygen/ozone therapy is a term that describes a number of different practices in which oxygen, ozone, or hydrogen peroxide are administered via gas or water to kill disease microorganisms, improve cellular function, and promote the healing of damaged tissues. The rationale behind bio-oxidative therapies, as they are sometimes known, is the notion that as long as the body's needs for antioxidants are met, the use of certain oxidative substances will stimulate the movement of oxygen atoms from the bloodstream to the cells. With higher levels of oxygen in the tissues, bacteria and viruses are killed along with defective tissue cells. The healthy cells survive and multiply more rapidly. The result is a stronger immune system.

Ozone itself is a form of oxygen, O₃, produced when ultraviolet light or an electric spark passes through air or oxygen. It is a toxic gas that creates free radicals, the opposite of what antioxidant **vitamins** do. Oxidation, however, is good when it occurs in harmful foreign organisms that have invaded the

body. Ozone inactivates many disease bacteria and viruses.

Purpose

Oxygen and ozone therapies are thought to benefit patients in the following ways:

- stimulating white blood cell production
- killing viruses (ozone and hydrogen peroxide)
- improving the delivery of oxygen from the blood stream to the tissues of the body
- speeding up the breakdown of petrochemicals
- increasing the production of interferon and tumor necrosis factor, thus helping the body to fight infections and cancers
- increasing the efficiency of antioxidant enzymes
- increasing the flexibility and efficiency of the membranes of red blood cells
- speeding up the citric acid cycle, which in turn stimulates the body's basic metabolism

Description

Origins

The various forms of oxygen and ozone therapy have been in use since the late nineteenth century. The earliest recorded use of oxygen to treat a patient was by Dr. J. A. Fontaine in 1879. In the 1950s, hyperbaric oxygen treatment was used by cancer researchers. The term "hyperbaric" means that the oxygen is given under pressure higher than normal air pressure. Recently, oxygen therapy has also been touted as a quick purification treatment for mass-market consumers. Oxygen bars can be found in airports and large cities, and provide pure oxygen in 20-minute sessions for approximately \$16. While proponents claim that breathing oxygen will purify the body, most medical doctors do not agree. What is more, oxygen can be harmful to people with severe lung diseases, and these people should never self-treat with oxygen.

Ozone has been used since 1856 to disinfect operating rooms in European hospitals, and since 1860 to purify the water supplies of several large German cities. Ozone was not, however, used to treat patients until 1915, when a German doctor named Albert Wolff began to use it to treat skin diseases. During World War I, the German Army used ozone to treat **wounds** and **anaerobic infections**. In the 1950s, several German physicians used ozone to treat **cancer** alongside mainstream therapeutic methods. It is estimated

that as of the late 1990s, about 8,000 practitioners in Germany were using ozone in their practices. This figure includes medical doctors as well as naturopaths and homeopaths.

Hydrogen peroxide is familiar to most people as an over-the-counter preparation that is easily available at supermarkets as well as pharmacies, and is used as an antiseptic for cleansing minor cuts and scrapes. It was first used as an intravenous infusion in 1920 by a British physician in India, T. H. Oliver, to treat a group of 25 Indian patients who were critically ill with **pneumonia**. Oliver's patients had a mortality rate of 48%, compared to the standard mortality rate of 80% for the disease. In the 1920s, an American physician named William Koch experimented with hydrogen peroxide as a treatment for cancer. He left the United States after a legal battle with the FDA. In the early 1960s, researchers at Baylor University studied the effects of hydrogen peroxide in removing plaque from the arteries as well as its usefulness in treating cancer, but their findings were largely ignored.

Oxygen, ozone, and hydrogen peroxide are used therapeutically in a variety of different ways.

Hyperbaric oxygen therapy (HBO)

Hyperbaric oxygen therapy (HBO) involves putting the patient in a pressurized chamber in which he or she breathes pure oxygen for a period of 90 minutes to two hours. HBO may also be administered by using a tight-fitting mask, similar to the masks used for anesthesia. A nasal catheter may be used for small children.

Ozone therapy

Ozone therapy may be administered in a variety of ways.

- **Intramuscular injection:** A mixture of oxygen and ozone is injected into the muscles of the buttocks.
- **Rectal insufflation:** A mixture of oxygen and ozone is introduced into the rectum and absorbed through the intestines.
- **Autohemotherapy:** Between 10–15 mL of the patient's blood is removed, treated with a mixture of oxygen and ozone and reinjected into the patient.
- **Intra-articular injection:** Ozone-treated water is injected into the patient's joints to treat arthritis, rheumatism and other joint diseases.
- **Ozonated water:** Ozone is bubbled through water that is used to cleanse wounds, **burns**, and skin infections, or to treat the mouth after dental surgery.

- **Ozonated oil:** Ozone is bubbled through olive or safflower oil, forming a cream that is used to treat fungal infections, insect bites, **acne**, and skin problems.
- **Ozone bagging:** Ozone and oxygen are pumped into an airtight bag that surrounds the area to be treated, allowing the body tissues to absorb the mixture.

Hydrogen peroxide

Hydrogen peroxide may be administered intravenously in a 0.03% solution. It is infused slowly into the patient's vein over a period of one to three hours. Treatments are given about once a week for chronic illness but may be given daily for such acute illnesses as pneumonia or influenza. A course of intravenous hydrogen peroxide therapy may range from one to 20 treatments, depending on the patient's condition and the type of illness being treated. Injections of 0.03% hydrogen peroxide have also been used to treat rheumatoid and osteoarthritis. The solution is injected directly into the inflamed joint.

Hydrogen peroxide is also used externally to treat stiff joints, **psoriasis**, and fungal infections. The patient soaks for a minimum of 20 minutes in a tub of warm water to which 1 pint of 35% food-grade hydrogen peroxide (a preparation used by the food industry as a disinfectant) has been added.

Preparations

Oxygen is usually delivered to the patient as a gas; ozone as a gas mixed with oxygen or bubbled through oil or water; and hydrogen peroxide as an 0.03% solution for intravenous injection or a 35% solution for external **hydrotherapy**.

Precautions

Patients interested in oxygen/ozone therapies must consult with a physician before receiving treatment. Hyperbaric oxygen treatment should not be given to patients with untreated pneumothorax, a condition in which air or gas is present in the cavity surrounding the lungs. Patients with a history of **pneumothorax**, chest surgery, **emphysema**, middle **ear surgery**, uncontrolled high fevers, upper respiratory infections, seizures, or disorders of the red blood cells are not suitable candidates for oxygen/ozone therapy. In addition, patients should be aware that oxygen is highly flammable. If treatments are administered incorrectly or by an unskilled person, there is a risk of fire.

KEY TERMS

Autohemotherapy—A form of ozone therapy in which a small quantity of the patient's blood is withdrawn, treated with a mixture of ozone and oxygen, and reinfused into the patient.

Hydrogen peroxide—A colorless, unstable compound of hydrogen and oxygen (H_2O_2). An aqueous solution of hydrogen peroxide is used as an antiseptic and bleaching agent.

Hyperbaric oxygen therapy (HBO)—A form of oxygen therapy in which the patient breathes oxygen in a pressurized chamber.

Ozone—A form of oxygen with three atoms in its molecule (O_3), produced by an electric spark or ultraviolet light passing through air or oxygen. Ozone is used therapeutically as a disinfectant and oxidative agent.

Side effects

Typical side effects of oxygen or ozone therapy can include elevated blood pressure and ear pressure similar to that experienced while flying. Side effects may also include **headache**, **numbness** in the fingers, temporary changes in the lens of the eye, and seizures.

Research and general acceptance

Oxygen/ozone therapies are far more widely accepted in Europe than in the United States. The most intensive research in these therapies is presently being conducted in the former Soviet Union and in Cuba. In the United States, the work of the Baylor researchers was not followed up. In 2000, the Office of Alternative Medicine of the National Institutes of Health (presently the National Center for Complementary and Alternative Medicine, or NCCAM) indicated interest in conducting clinical trials of oxygen/ozone therapies; as of 2003, however, these studies have not been carried out.

Recent European research in ozone therapy includes studies in the oxygenation of resting muscles, the treatment of vascular disorders, and the relief of **pain** from herniated lumbar disks. No corresponding studies are being done in the United States as of late 2003.

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ORGANIZATIONS

- International Bio-Oxidative Medicine Foundation (IBOMF). P.O. Box 891954. Oklahoma City, OK 73109. (405) 634-7855. Fax (405) 634-7320.
- International Ozone Association, Ind. Pan American Group. 31 Strawberry Hill Ave. Stamford, CT 06902. (203) 348-3542. Fax (203) 967-4845.
- NIH National Center for Complementary and Alternative Medicine (NCCAM). NCCAM Clearinghouse. P. O. Box 8218. Silver Spring, MD 20907-8218. TTY/TDY: (888) 644-6226. <<http://nccam.nih.gov>>.

OTHER

- Oxygen Healing Therapies. <<http://www.oxygenhealingtherapies.com>>.

Amy Cooper
Rebecca J. Frey, PhD

Oxymetazoline see **Decongestants**

Oxytocin see **Drugs used in labor**

Ozone therapy see **Oxygen/ozone therapy**

P

PAC see **Atrial ectopic beats**

Pacemaker implantation see **Pacemakers**

Pacemakers

Definition

A pacemaker is a surgically-implanted electronic device that regulates a slow or erratic heartbeat.

Purpose

Pacemakers are implanted to regulate irregular contractions of the heart (arrhythmia). They are most frequently prescribed to speed the heartbeat of patients who have a heart rate well under 60 beats per minute (severe symptomatic bradycardia). They are also used in some cases to slow a fast heart rate (tachycardia).

Precautions

The symptoms of **fatigue** and lightheadedness that are characteristic of bradycardia can also be caused by a number of other medical conditions, including anemia. Certain prescription medications can also slow the heart rate. A doctor should take a complete medical history and perform a full physical work-up to rule out all non-cardiac causes of bradycardia.

Patients with cardiac pacemakers should not undergo a magnetic resonance imaging (MRI) procedure. Devices that emit electromagnetic waves (including magnets) may alter pacemaker programming or functioning. A 1997 study found that cellular phones often interfere with pacemaker programming and cause irregular heart rhythm. However, advances in pacemaker design and materials have greatly reduced the risk of pacemaker interference from electromagnetic fields.

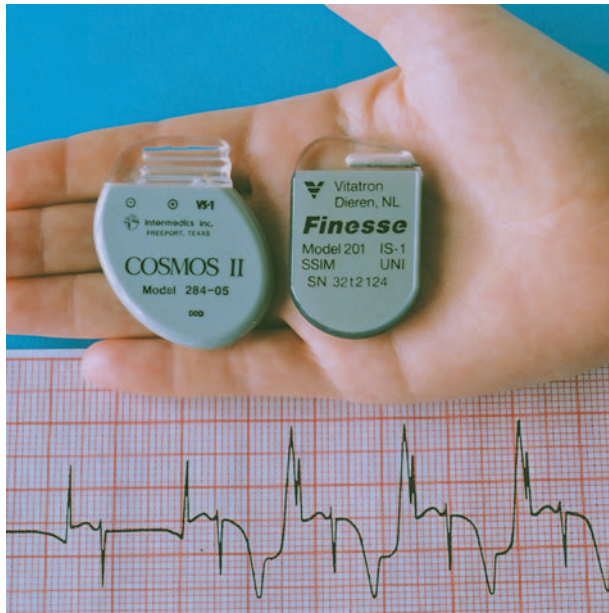
Description

Approximately 500,000 Americans have an implantable permanent pacemaker device. A pacemaker implantation is performed under **local anesthesia** in a hospital by a surgeon assisted by a cardiologist. An insulated wire called a lead is inserted into an incision above the collarbone and guided through a large vein into the chambers of the heart. Depending on the configuration of the pacemaker and the clinical needs of the patient, as many as three leads may be used in a pacing system. Current pacemakers have a double, or bipolar, electrode attached to the end of each lead. The electrodes deliver an electrical charge to the heart to regulate heartbeat. They are positioned on the areas of the heart that require stimulation. The leads are then attached to the pacemaker device, which is implanted under the skin of the patient's chest.

Patients undergoing surgical pacemaker implantation usually stay in the hospital overnight. Once the procedure is complete, the patient's vital signs are monitored and a **chest x ray** is taken to ensure that the pacemaker and leads are properly positioned.

Modern pacemakers have sophisticated programming capabilities and are extremely compact. The smallest weigh less than 13 grams (under half an ounce) and are the size of two stacked silver dollars. The actual pacing device contains a pulse generator, circuitry programmed to monitor heart rate and deliver stimulation, and a lithiumiodide battery. Battery life typically ranges from seven to 15 years, depending on the number of leads the pacemaker is configured with and how much energy the pacemaker uses. When a new battery is required, the unit can be exchanged in a simple outpatient procedure.

A temporary pacing system is sometimes recommended for patients who are experiencing irregular heartbeats as a result of a recent **heart attack** or other acute medical condition. The implantation procedure for the pacemaker leads is similar to that for a



Pacemakers like these are usually implanted under the skin below the collarbone. The pacemaker is connected to the heart by a wire inserted into a major vein in the neck and guided down into the heart. (Photograph by Eamonn McNulty, Photo Researchers, Inc. Reproduced by permission.)

permanent pacing system, but the actual pacemaker unit housing the pulse generator remains outside the patient's body. Temporary pacing systems may be replaced with a permanent device at a later date.

Preparation

Patients being considered for pacemaker implantation will undergo a full battery of cardiac tests, including an electrocardiogram (ECG) or an electrophysiological study or both to fully evaluate the bradycardia or tachycardia.

Patients are advised to abstain from eating 6-8 hours before the surgical procedure. The patient is usually given a sedative to help him or her relax for the procedure. An intravenous (IV) line will also be inserted into a vein in the patient's arm before the procedure begins in case medication or blood products are required during the insertion.

Aftercare

Pacemaker patients should schedule a follow-up visit with their cardiologist approximately six weeks after the surgery. During this visit, the doctor will make any necessary adjustments to the settings of the pacemaker. Pacemakers are programmed externally with a handheld electromagnetic device. Pacemaker

KEY TERMS

Electrocardiogram (ECG)—A recording of the electrical activity of the heart. An ECG uses externally attached electrodes to detect the electrical signals of the heart.

Electrophysiological study—A test that monitors the electrical activity of the heart in order to diagnose arrhythmia. An electrophysiological study measures electrical signals through a cardiac catheter that is inserted into an artery in the leg and guided up into the atrium and ventricle of the heart.

Embolism—A blood clot, air bubble, or clot of foreign material that blocks the flow of blood in an artery. When an embolism blocks the blood supply to a tissue or organ, the tissue the artery feeds dies (infarction). Without immediate and appropriate treatment, an embolism can be fatal.

Magnetic resonance imaging (MRI)—An imaging technique that uses a large circular magnet and radio waves to generate signals from atoms in the body. These signals are used to construct images of internal structures.

batteries must be checked regularly. Some pacing systems allow patients to monitor battery life through a special telephone monitoring service that can read pacemaker signals.

Risks

Because pacemaker implantation is an invasive surgical procedure, internal bleeding, infection, hemorrhage, and **embolism** are all possible complications. Infection is more common in patients with temporary pacing systems. Antibiotic therapy given as a precautionary measure can reduce the risk of pacemaker infection. If infection does occur, the entire pacing system may have to be removed.

The placing of the leads and electrodes during the implantation procedure also presents certain risks for the patient. The lead or electrode could perforate the heart or cause scarring or other damage. The electrodes can also cause involuntary stimulation of nearby skeletal muscles.

A complication known as *pacemaker syndrome* develops in approximately 7% of pacemaker patients with single-chamber pacing systems. The syndrome is characterized by the low blood pressure and **dizziness** that are symptomatic of bradycardia. It can usually be

corrected by the implantation of a dual-chamber pacing system.

Normal results

Pacemakers that are properly implanted and programmed can correct a patient's arrhythmia and resolve related symptoms.

Resources

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300. <<http://www.americanheart.org>>.

Paula Anne Ford-Martin

Packed cell volume see **Hematocrit**

Packed red blood cell volume see **Hematocrit**

Paget's disease of bone

Definition

Paget's disease of bone (*osteitis deformans*) is the abnormal formation of bone tissue that results in weakened and deformed bones.

Description

Named for Sir James Paget (1814–1899), this disease affects 1–3% of people over 50 years of age, but affects over 10% of people over 80 years of age. Paget's disease can affect one or more bones in the body. Most often, the pelvis, bones in the skull, the long bones (the large bones that make up the arms and legs), and the collarbones are affected by Paget's disease. In addition, the joints between bones (the knees or elbows, for example) can develop arthritis because of this condition.

Paget's disease is characterized by changes in the normal mechanism of bone formation. Bone is a living material made by the body through the continual processes of formation and breakdown (resorption). The combination of these two actions is called remodeling and is used by the body to build bone tissue that is strong and healthy. Strong bones are formed when bone tissue is made up of plate-shaped crystals of **minerals** called hydroxyapatite. Normal wear and tear on the skeletal system is repaired throughout life

by the ongoing process of remodeling. In fact, the entire human skeleton is remodeled every five years.

Healthy bone tissue has an ordered structure that gives the bone its strength. Bones affected by Paget's disease, however, have a structure that is disorganized. This disorganized structure weakens the diseased bone and makes people suffering from this disease more likely to have fractures. These **fractures** are slow to heal.

Paget's disease of bone is most commonly found in Europe, England, Australia, New Zealand, and North America. In these areas, up to 3% of all people over 55 years of age are affected with the disease. It is interesting to note that Paget's disease is rare in Asia, possibly showing that this disease may affect some ethnic groups and geographic areas more than others.

Causes and symptoms

The cause of Paget's disease is not known. Various viruses have been suggested to be involved in this disease, but the relationship between viral infections and Paget's disease remains uncertain. There also may be a genetic component to this disease since it may appear in more than one person within the same family.

Paget's disease usually begins without any symptoms. And, in its early stages, the symptoms that do occur are often confused with symptoms of arthritis. However, as the disease progresses, bone and joint **pain** develop. A unique feature of Paget's disease is the enlargement of areas of affected bone. This type of enlargement is clearly identifiable on an x ray.

If the bones of the skull are affected by Paget's disease, enlargement of the skull can occur and may result in a loss of hearing. When the long bones in the legs are affected, they can become bent under the body's weight because of their weakness. Little or no injury to a bone can cause fractures in the weakened bones. Fractures that occur when no traumatic injury is present are known as spontaneous fractures.

Although rare, bone **cancer** occurs in less than 1% of patients with Paget's disease. Such cancer is often accompanied by an abrupt increase in the intensity of pain at the diseased site. Unfortunately, this type of cancer has a poor prognosis; the average survival time from the onset of symptoms is generally one to three years.

Diagnosis

Paget's disease is often found when an individual is having x rays taken for medical reasons unrelated to this bone disease. A diagnosis of Paget's disease can also be made when higher than normal levels of a



This woman's legs are bowed due to Paget's disease. (Custom Medical Stock Photo. Reproduced by permission.)

chemical called alkaline phosphatase are found in the blood. Alkaline phosphatase is a substance involved in the bone formation process, so if its levels are abnormally high this indicates that the balance between bone formation and resorption is upset.

Treatment

Treatment, given only when symptoms are present, consists of the following types:

Drugs

Paget's disease is most often treated with drug therapy, with bone pain lessening within weeks of starting the treatment. While non-steroidal anti-inflammatory drugs can reduce bone pain, two additional categories of drugs are used to treat this disease.

HORMONE TREATMENT. Calcitonin, a hormone which is made naturally by the thyroid gland, is used

to treat Paget's disease. This chemical rapidly decreases the amount of bone breakdown or loss (resorption). After approximately two to three weeks of treatment with extra calcitonin, bone pain lessens and new bone tissue forms. Calcitonin is commonly given as daily injections for one month, followed by three injections each week for several additional months. The total dose of calcitonin given to an individual depends upon the amount of disease present and how well the individual's condition responds to the treatment.

Although calcitonin is effective in slowing the progression of Paget's disease, the favorable effects of the drug do not continue for very long once administration of the drug is stopped. In addition, some temporary side effects can occur with this drug. **Nausea** and flushing are the most common side effects and have been found in 20-30% of individuals taking calcitonin. Vomiting, diarrhea, and abdominal pain can also occur, but these effects are also temporary. A form of calcitonin taken nasally causes fewer side effects, but requires higher doses because less of the drug reaches the diseased bone.

BISPHOSPHONATES. The bisphosphonate group of drugs are drugs that bind directly to bone minerals because of their specific chemical structure. Once bound to the bone, these drugs inhibit bone loss by reducing the action of bone cells that normally degrade bone during the remodeling process. Unlike treatment with calcitonin, the positive effects of increased bone formation and reduced pain can continue for many months or even years after bisphosphonate treatment is stopped. Bisphosphonates are considered the treatment of choice for Paget's disease and are usually given for 3-6 months at a time.

Bisphosphonate drugs suitable for the treatment of Paget's disease are alendronate, clodronate, etidronate, pamidronate, risedronate, and tiludronate. The main side effects of these drugs include a flu-like reaction (pamidronate), gastrointestinal disturbances (alendronate, clodronate), and abnormal bone formation (etidronate, when taken in high doses). Risedronate is the newest of these drugs. It is about 1,000 times more potent than etidronate and 3 to 5 times more potent than alendronate. Because of the greater potency of this drug, lower doses and a shorter duration of treatment are required. This leads to fewer side effects with similar, or better, clinical results in the patient.

Surgery

Treatment of Paget's disease usually begins with drug therapy. However, various surgical treatments can also be used to treat skeletal conditions that occur in patients with Paget's disease.

KEY TERMS

Bisphosphonate—A class of drugs used to treat Paget's disease. These drugs bind to the minerals in bone tissue and lessen the amount of bone loss associated with Paget's disease.

Calcitonin—A naturally occurring hormone made by the thyroid gland that can be used as a drug to treat Paget's disease.

Remodeling—The ongoing process of bone formation and breakdown that results in healthy bone development.

In patients with severe arthritis of the hip or knee, a joint replacement operation can be beneficial. However, in addition to the malformation of bone tissue caused by this condition, there are greater numbers of blood vessels that form in the diseased bone relative to a healthy bone. This makes surgery on bones affected with Paget's disease more difficult.

Prognosis

There is no cure for Paget's disease. However, the development of potent bisphosphonate drugs like risedronate has resulted in the ability to slow the progress of the disease.

Paul A. Johnson, Ed.M.

Paget's disease of the breast

Definition

Paget's disease of the breast is a rare form of breast cancer which makes up approximately 1-4 % of all breast tumors. While sharing its name with **Paget's disease of bone**, these are two medically unrelated conditions. They are simply named after the same doctor who first described them.

Description

Paget's disease of the breast is generally associated with an underlying **breast cancer**. It is generally seen in people between the ages of 40 and 80 years. Cases in men have been identified, but they are extremely rare.

Paget's disease of the breast may also be called mammary Paget's disease (MPD). There is a much rarer form of this disease called extramammary Paget's disease (EMPD). MPD affects the breast nipple and is also called Paget's disease of the nipple. EMPD can affect the skin of the external genital tissues in both women and men, as well as the skin of the eyelids and external ear canal. MPD is believed to develop from a tumor growth within the milk ducts of the breast. EMPD may represent a spreading (metastasis) of MPD to other parts of the body.

Causes and symptoms

The cause of Paget's disease of the breast is unknown, but it is usually associated with an underlying **cancer** of the breast.

The symptoms of Paget's disease of the breast include:

- red scaly patches of skin on the nipple and sometimes also on the dark area of skin around the nipple (areola)
- crusting, bleeding, or ulceration of the skin of the affected area
- a discharge of fluid from the nipple
- a turning inward (inversion) of the nipple

In approximately 30-40 % of cases of Paget's disease of the breast, there is also a detectable lump in the breast.

Diagnosis

Paget's disease of the breast is often confused with other skin conditions, such as eczema, dermatitis, or **psoriasis**. These misdiagnoses often lead to delays in appropriate treatment. Misdiagnosis is more common when both breast are affected and no lump in the breast is detected. When only one breast is affected, or when the presence of a lump in the breast is also detected, a correct initial diagnosis is more likely to occur.

Once Paget's disease of the breast is suspected, it can be definitively confirmed by biopsy of the affected tissue. In this procedure, a small piece of the affected skin and the underlying tissue is removed and sent to a laboratory for examination under a microscope. The shape and other characteristics of the cells in the biopsied sample will allow the laboratory personnel to determine if the sample is affected with Paget's disease of the breast, or some other condition.

KEY TERMS

Metastasis—The spread of a cancer from one part of the body (where the cancer originated) to another part of the body.

Ulceration—The formation of an ulcer, a patch of tissue that is discontinuous with the surrounding tissue because the tissue within the ulcer has decayed or died and been swept away.

Topical steroid creams are usually used to treat eczema, **dermatitis**, and psoriasis. These creams will have no effect on the skin conditions caused by Paget's disease of the breast.

Treatment

Surgery is the main treatment for Paget's disease of the breast. Removal of the breast (mastectomy) may be recommended if the cancer is seen in a wide area away from the nipple or appears to be deep into the breast tissue. Breast conservation surgery, aimed at keeping as much of the breast as possible, may be recommended in cases where the disease is diagnosed early enough and the cancer has not spread far from the surface of the nipple.

Some people will require further treatment after surgery. This treatment may include radiation therapy, **chemotherapy**, or a combination of both. **Radiation therapy** involves using high-energy x rays to destroy any cancer cells that may remain after surgical removal of the primary tumor. Radiation therapy is most common after breast conservation surgery. Chemotherapy involves the use of medicinal drugs to destroy the growth of any cancer cells that may remain after removal of the primary cancer. Chemotherapy treatments are most common after **mastectomy**.

Alternative treatment

Alternative treatments for Paget's disease of the breast include: the use of cartilage from cows or sharks; a diet known as Gerson therapy; administration of the chemicals hydrazine sulfate or laetrile; and, the injection of solutions derived from the mistletoe plant.

Prognosis

The prognosis for Paget's disease of the breast depends on the underlying cancer that is causing this condition and whether or not this cancer has spread (metastasized) to other parts of the body.

Prevention

Because the cause of Paget's disease of the breast is not known, prevention of this disease is not possible. In instances where this condition arises from other underlying cancers of the breast, it may be possible to prevent Paget's disease of the breast from occurring if the underlying cause is diagnosed and successfully treated prior to the development of Paget's disease of the breast.

Resources

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ORGANIZATIONS

American Cancer Society. 1599 Clifton Road NE, Atlanta, GA 30329. (800)ACS-2345. <<http://www.cancer.org>>.

National Alliance of Breast Cancer Organizations. 9 East 37th Street, 10th Floor, New York, NY 10016. (888) 806-2226. Fax: 212-689-1213. <<http://www.nabco.org/>>.

National Breast Cancer Coalition. 1707 L Street Northwest, Suite 1060, Washington, DC 20036. (800) 622-2838. Fax: 202-265-6854. <<http://www.natlbcc.org/>>.

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Paul A. Johnson, Ed.M.

Pain

Definition

Pain is an unpleasant feeling that is conveyed to the brain by sensory neurons. The discomfort signals actual or potential injury to the body. However, pain is more than a sensation, or the physical awareness of pain; it also includes perception, the subjective interpretation of the discomfort. Perception gives information on the pain's location, intensity, and

something about its nature. The various conscious and unconscious responses to both sensation and perception, including the emotional response, add further definition to the overall concept of pain.

Description

Pain arises from any number of situations. Injury is a major cause, but pain may also arise from an illness. It may accompany a psychological condition, such as depression, or may even occur in the absence of a recognizable trigger.

Acute pain

Acute pain often results from tissue damage, such as a skin burn or broken bone. Acute pain can also be associated with headaches or **muscle cramps**. This type of pain usually goes away as the injury heals or the cause of the pain (stimulus) is removed.

To understand acute pain, it is necessary to understand the nerves that support it. Nerve cells, or neurons, perform many functions in the body. Although their general purpose, providing an interface between the brain and the body, remains constant, their capabilities vary widely. Certain types of neurons are capable of transmitting a pain signal to the brain.

As a group, these pain-sensing neurons are called nociceptors, and virtually every surface and organ of the body is wired with them. The central part of these cells is located in the spine, and they send threadlike projections to every part of the body. Nociceptors are classified according to the stimulus that prompts them to transmit a pain signal. Thermoreceptive nociceptors are stimulated by temperatures that are potentially tissue damaging. Mechanoreceptive nociceptors respond to a pressure stimulus that may cause injury. Polymodal nociceptors are the most sensitive and can respond to temperature and pressure. Polymodal nociceptors also respond to chemicals released by the cells in the area from which the pain originates.

Nerve cell endings, or receptors, are at the front end of pain sensation. A stimulus at this part of the nociceptor unleashes a cascade of neurotransmitters (chemicals that transmit information within the nervous system) in the spine. Each neurotransmitter has a purpose. For example, substance P relays the pain message to nerves leading to the spinal cord and brain. These neurotransmitters may also stimulate nerves leading back to the site of the injury. This response prompts cells in the injured area to release chemicals that not only trigger an immune response, but also influence the intensity and duration of the pain.

Chronic and abnormal pain

Chronic pain refers to pain that persists after an injury heals, **cancer** pain, pain related to a persistent or degenerative disease, and long-term pain from an unidentifiable cause. It is estimated that one in three people in the United States will experience chronic pain at some point in their lives. Of these people, approximately 50 million are either partially or completely disabled.

Chronic pain may be caused by the body's response to acute pain. In the presence of continued stimulation of nociceptors, changes occur within the nervous system. Changes at the molecular level are dramatic and may include alterations in genetic transcription of neurotransmitters and receptors. These changes may also occur in the absence of an identifiable cause; one of the frustrating aspects of chronic pain is that the stimulus may be unknown. For example, the stimulus cannot be identified in as many as 85% of individuals suffering lower back pain.

Scientists have long recognized a relationship between depression and chronic pain. In 2004, a survey of California adults diagnosed with major depressive disorder revealed that more than one-half of them also suffered from chronic pain.

Other types of abnormal pain include allodynia, hyperalgesia, and phantom limb pain. These types of pain often arise from some damage to the nervous system (neuropathic). Allodynia refers to a feeling of pain in response to a normally harmless stimulus. For example, some individuals who have suffered nerve damage as a result of viral infection experience unbearable pain from just the light weight of their clothing. Hyperalgesia is somewhat related to allodynia in that the response to a painful stimulus is extreme. In this case, a mild pain stimulus, such as a pin prick, causes a maximum pain response. Phantom limb pain occurs after a limb is amputated; although an individual may be missing the limb, the nervous system continues to perceive pain originating from the area.

Causes and symptoms

Pain is the most common symptom of injury and disease, and descriptions can range in intensity from a mere ache to unbearable agony. Nociceptors have the ability to convey information to the brain that indicates the location, nature, and intensity of the pain. For example, stepping on a nail sends an information-packed message to the brain: the foot has experienced a puncture wound that hurts a lot.

Pain perception also varies depending on the location of the pain. The kinds of stimuli that cause a pain response on the skin include pricking, cutting, crushing, burning, and freezing. These same stimuli would not generate much of a response in the intestine. Intestinal pain arises from stimuli such as swelling, inflammation, and distension.

Diagnosis

Pain is considered in view of other symptoms and individual experiences. An observable injury, such as a broken bone, may be a clear indicator of the type of pain a person is suffering. Determining the specific cause of internal pain is more difficult. Other symptoms, such as fever or **nausea**, help narrow down the possibilities. In some cases, such as lower back pain, a specific cause may not be identifiable. Diagnosis of the disease causing a specific pain is further complicated by the fact that pain can be referred to (felt at) a skin site that does not seem to be connected to the site of the pain's origin. For example, pain arising from fluid accumulating at the base of the lung may be referred to the shoulder.

Since pain is a subjective experience, it may be very difficult to communicate its exact quality and intensity to other people. There are no diagnostic tests that can determine the quality or intensity of an individual's pain. Therefore, a medical examination will include a lot of questions about where the pain is located, its intensity, and its nature. Questions are also directed at what kinds of things increase or relieve the pain, how long it has lasted, and whether there are any variations in it. An individual may be asked to use a pain scale to describe the pain. One such scale assigns a number to the pain intensity; for example, 0 may indicate no pain, and 10 may indicate the worst pain the person has ever experienced. Scales are modified for infants and children to accommodate their level of comprehension.

Treatment

There are many drugs aimed at preventing or treating pain. Nonopioid **analgesics**, narcotic analgesics, anticonvulsant drugs, and **tricyclic antidepressants** work by blocking the production, release, or uptake of neurotransmitters. Drugs from different classes may be combined to handle certain types of pain.

Nonopioid analgesics include common over-the-counter medications such as **aspirin**, **acetaminophen** (Tylenol), and **ibuprofen** (Advil). These are most often used for minor pain, but there are some prescription-strength medications in this class.

Narcotic analgesics are only available with a doctor's prescription and are used for more severe pain, such as cancer pain. These drugs include codeine, morphine, and **methadone**. **Addiction** to these painkillers is not as common as once thought. Many people who genuinely need these drugs for pain control typically do not become addicted. However, narcotic use should be limited to patients thought to have a short life span (such as people with terminal cancer) or patients whose pain is only expected to last for a short time (such as people recovering from surgery). In August 2004, the Drug Enforcement Administration (DEA) issued new guidelines to help physicians prescribe **narcotics** appropriately without fear of being arrested for prescribing the drugs beyond the scope of their medical practice. DEA is trying to work with physicians to ensure that those who need to drugs receive them but to ensure opioids are not abused.

Anticonvulsants, as well as **antidepressant drugs**, were initially developed to treat seizures and depression, respectively. However, it was discovered that these drugs also have pain-killing applications. Furthermore, since in cases of chronic or extreme pain, it is not unusual for an individual to suffer some degree of depression; antidepressants may serve a dual role. Commonly prescribed anticonvulsants for pain include phenytoin, carbamazepine, and clonazepam. Tricyclic antidepressants include doxepin, amitriptyline, and imipramine.

Intractable (unrelenting) pain may be treated by injections directly into or near the nerve that is transmitting the pain signal. These root blocks may also be useful in determining the site of pain generation. As the underlying mechanisms of abnormal pain are uncovered, other pain medications are being developed.

Drugs are not always effective in controlling pain. Surgical methods are used as a last resort if drugs and local anesthetics fail. The least destructive surgical procedure involves implanting a device that emits electrical signals. These signals disrupt the nerve and prevent it from transmitting the pain message. However, this method may not completely control pain and is not used frequently. Other surgical techniques involve destroying or severing the nerve, but the use of this technique is limited by side effects, including unpleasant **numbness**.

Alternative treatment

Both physical and psychological aspects of pain can be dealt with through alternative treatment. Some of the most popular treatment options include **acupressure** and **acupuncture**, massage, chiropractic,

KEY TERMS

Acute pain—Pain in response to injury or another stimulus that resolves when the injury heals or the stimulus is removed.

Chronic pain—Pain that lasts beyond the term of an injury or painful stimulus. Can also refer to cancer pain, pain from a chronic or degenerative disease, and pain from an unidentified cause.

Neuron—A nerve cell.

Neurotransmitters—Chemicals within the nervous system that transmit information from or between nerve cells.

Nociceptor—A neuron that is capable of sensing pain.

Referred pain—Pain felt at a site different from the location of the injured or diseased part of the body. Referred pain is due to the fact that nerve signals from several areas of the body may “feed” the same nerve pathway leading to the spinal cord and brain.

Stimulus—A factor capable of eliciting a response in a nerve.

and relaxation techniques such as **yoga**, hypnosis, and **meditation**. Herbal therapies are gaining increased recognition as viable options; for example, capsaicin, the component that makes cayenne peppers spicy, is used in ointments to relieve the joint pain associated with arthritis. Contrast **hydrotherapy** can also be very beneficial for pain relief.

Lifestyles can be changed to incorporate a healthier diet and regular **exercise**. Regular exercise, aside from relieving **stress**, has been shown to increase endorphins, painkillers naturally produced in the body.

Prognosis

Successful pain treatment is highly dependent on successful resolution of the pain’s cause. Acute pain will stop when an injury heals or when an underlying problem is treated successfully. Chronic pain and abnormal pain are more difficult to treat, and it may take longer to find a successful resolution. Some pain is intractable and will require extreme measures for relief.

Prevention

Pain is generally preventable only to the degree that the cause of the pain is preventable. For example, improved surgical procedures, such as those done

through a thin tube called a laparoscope, minimize post-operative pain. Anesthesia techniques for surgeries also continuously improve. Some disease and injuries are often unavoidable. However, pain from some surgeries and other medical procedures and continuing pain are preventable through drug treatments and alternative therapies.

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ORGANIZATIONS

American Chronic Pain Association. P.O. Box 850, Rocklin, CA 95677-0850. (916) 632-0922. <<http://members.tripod.com/~widdy/ACPA.html>>.

American Pain Society. 4700 W. Lake Ave., Glenview, IL 60025. (847) 375-4715. <<http://www.ampainsoc.org>>.

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Pain management

Definition

Pain management encompasses pharmacological, nonpharmacological, and other approaches to prevent, reduce, or stop pain sensations.

Purpose

Pain serves as an alert to potential or actual damage to the body. The definition for damage is quite broad; pain can arise from injury as well as disease. After the message is received and interpreted, further pain can be counter-productive. Pain can have a negative impact on a person’s quality of life and impede recovery from illness or injury. Unrelieved pain can become a syndrome in its own right and cause a downward spiral in a person’s health and outlook. Managing pain properly facilitates recovery, prevents additional health complications, and improves an individual’s quality of life.

Description

What is pain?

Before considering pain management, a review of pain definitions and mechanisms may be useful. Pain is the means by which the peripheral nervous system (PNS) warns the central nervous system (CNS) of injury or potential injury to the body. The CNS comprises the brain and spinal cord, and the PNS is composed of the nerves that stem from and lead into the CNS. The PNS includes all nerves throughout the body except the brain and spinal cord.

A pain message is transmitted to the CNS by special PNS nerve cells called nociceptors. Nociceptors are distributed throughout the body and respond to different stimuli depending on their location. For example, nociceptors that extend from the skin are stimulated by sensations such as pressure, temperature, and chemical changes.

When a nociceptor is stimulated, neurotransmitters are released within the cell. Neurotransmitters are chemicals found within the nervous system that facilitate nerve cell communication. The nociceptor transmits its signal to nerve cells within the spinal cord, which conveys the pain message to the thalamus, a specific region in the brain.

Once the brain has received and processed the pain message and coordinated an appropriate response, pain has served its purpose. The body uses natural pain killers, called endorphins, that are meant to derail further pain messages from the same source. However, these natural pain killers may not adequately dampen a continuing pain message. Also, depending on how the brain has processed the pain information, certain hormones, such as prostaglandins, may be released. These hormones enhance the pain message and play a role in immune system responses to injury, such as inflammation. Certain neurotransmitters, especially substance P and calcitonin gene-related peptide, actively enhance the pain message at the injury site and within the spinal cord.

Pain is generally divided into two categories: acute and chronic. Nociceptive pain, or the pain that is transmitted by nociceptors, is typically called acute pain. This kind of pain is associated with injury, headaches, disease, and many other conditions. It usually resolves once the condition that caused it is resolved.

Following some disorders, pain does not resolve. Even after healing or a cure has been achieved, the brain continues to perceive pain. In this situation, the pain may be considered chronic. The time limit used to define chronic pain typically ranges from three to six

months, although some healthcare professionals prefer a more flexible definition, and consider chronic pain as pain that endures beyond a normal healing time. The pain associated with **cancer**, persistent and degenerative conditions, and neuropathy, or nerve damage, is included in the chronic category. Also, unremitting pain that lacks an identifiable physical cause, such as the majority of cases of **low back pain**, may be considered chronic. The underlying biochemistry of chronic pain appears to be different from regular nociceptive pain.

Some researchers have said that uninterrupted and unrelenting pain can induce changes in the spinal cord. In the past, intractable pain has been treated by severing a nerve's connection to the CNS. However, the lack of any sensory information being relayed by that nerve can cause pain transmission in the spinal cord to go into overdrive, as evidenced by the phantom limb pain experienced by amputees. Evidence is accumulating that unrelenting pain or the complete lack of nerve signals increases the number of pain receptors in the spinal cord. Nerve cells in the spinal cord may also begin secreting pain-amplifying neurotransmitters independent of actual pain signals from the body. Immune chemicals, primarily cytokines, may play a prominent role in such changes.

Scientists have long recognized a relationship between depression and chronic pain. In 2004, a survey of California adults diagnosed with major depressive disorder revealed that more than one-half of them also suffered from chronic pain.

Managing pain

Considering the different causes and types of pain, as well as its nature and intensity, management can require an interdisciplinary approach. The elements of this approach include treating the underlying cause of pain, pharmacological and nonpharmacological therapies, and some invasive (surgical) procedures.

Treating the cause of pain underpins the idea of managing it. Injuries are repaired, diseases are diagnosed, and certain encounters with pain can be anticipated and treated prophylactically (by prevention). However, there are no guarantees of immediate relief from pain. Recovery can be impeded by pain and quality of life can be damaged. Therefore, pharmacological and other therapies have developed over time to address these aspects of disease and injury.

PHARMACOLOGICAL OPTIONS. Pain-relieving drugs, otherwise called **analgesics**, include nonsteroidal anti-inflammatory drugs (NSAIDs), **acetaminophen**, **narcotics**, antidepressants, anticonvulsants, and

others. NSAIDs and acetaminophen are available as over-the-counter and prescription medications, and are frequently the initial pharmacological treatment for pain. These drugs can also be used as adjuncts to the other drug therapies, which might require a doctor's prescription.

NSAIDs include **aspirin**, ibuprofen (Motrin, Advil, Nuprin), naproxen sodium (Aleve), and ketoprofen (Orudis KT). These drugs are used to treat pain from inflammation and work by blocking production of pain-enhancing neurotransmitters, such as prostaglandins. Acetaminophen is also effective against pain, but its ability to reduce inflammation is limited.

NSAIDs and acetaminophen are effective for most forms of acute (sharp, but of a short course) pain, but moderate and severe pain may require stronger medication. Narcotics handle intense pain effectively, and are used for cancer pain and acute pain that does not respond to NSAIDs and acetaminophen. Narcotics are classified as either opiates or opioids, and are available only with a doctor's prescription. Opiates include morphine and codeine, which are derived from opium, a substance naturally found in some poppy species. Opioids are synthetic drugs based on the structure of opium. This drug class includes drugs such as oxycodone, **methadone**, and meperidine (Demerol). In August 2004, the Drug Enforcement Administration (DEA) issued new guidelines to help physicians prescribe narcotics appropriately without fear of being arrested for prescribing the drugs beyond the scope of their medical practice. DEA is trying to work with physicians to ensure that those who need to drugs receive them but to ensure opioids are not abused.

Narcotics may be ineffective against some forms of chronic pain, especially since changes in the spinal cord may alter the usual pain signaling pathways. In such situations, pain can be managed with the help of with antidepressants and anticonvulsants, which are also only available with a doctor's prescription.

Although **antidepressant drugs** were developed to treat depression, it has been discovered that they are also effective in combating some chronic headaches, cancer pain, and pain associated with nerve damage. Antidepressants that have been shown to have analgesic (pain reducing) properties include amitriptyline (Elavil), trazodone (Desyrel), and imipramine (Tofranil). **Anticonvulsant drugs** share a similar background with antidepressants. Developed to treat epilepsy, certain anticonvulsants were found to relieve pain as well. Drugs such as phenytoin (Dilantin) and carbamazepine (Tegretol) are prescribed to treat the pain associated with nerve damage.

Other prescription drugs are used to treat specific types of pain or specific pain syndromes. For example, **corticosteroids** are effective against pain caused by inflammation and swelling, and sumatriptan (Imitrex) was developed to treat migraine headaches.

Drug administration depends on the drug type and the required dose. Some drugs are not absorbed very well from the stomach and must be injected or administered intravenously. Injections and intravenous administration may also be used when high doses are needed or if an individual is nauseous. Following surgery and other medical procedures, patients may have the option of controlling the pain medication themselves. By pressing a button, they can release a set dose of medication into an intravenous solution. This procedure has also been employed in other situations requiring pain management. Another mode of administration involves implanted catheters that deliver pain medication directly to the spinal cord. Delivering drugs in this way can reduce side effects and increase the effectiveness of the drug.

NONPHARMACOLOGICAL OPTIONS. Pain treatment options that do not use drugs are often used as adjuncts to, rather than replacements for, drug therapy. One of the benefits of non-drug therapies is that an individual can take a more active stance against pain. Relaxation techniques, such as **yoga** and **meditation**, are used to decrease muscle tension and reduce **stress**. Tension and stress can also be reduced through **biofeedback**, in which an individual consciously attempts to modify skin temperature, muscle tension, blood pressure, and heart rate.

Participating in normal activities and exercising can also help control pain levels. Through physical therapy, an individual learns beneficial exercises for reducing stress, strengthening muscles, and staying fit. Regular **exercise** has been linked to production of endorphins, the body's natural pain killers.

Acupuncture involves the insertion of small needles into the skin at key points. **Acupressure** uses these same key points, but involves applying pressure rather than inserting needles. Both of these methods may work by prompting the body to release endorphins. Applying heat or being massaged are very relaxing and help reduce stress. Transcutaneous electrical nerve stimulation (TENS) applies a small electric current to certain parts of nerves, potentially interrupting pain signals and inducing release of endorphins. To be effective, use of TENS should be medically supervised.

INVASIVE PROCEDURES. There are three types of invasive procedures that may be used to manage or treat pain: anatomic, augmentative, and ablative.

These procedures involve surgery, and certain guidelines should be followed before carrying out a procedure with permanent effects. First, the cause of the pain must be clearly identified. Next, surgery should be done only if noninvasive procedures are ineffective. Third, any psychological issues should be addressed. Finally, there should be a reasonable expectation of success.

Anatomic procedures involve correcting the injury or removing the cause of pain. Relatively common anatomic procedures are decompression surgeries, such as repairing a herniated disk in the lower back or relieving the nerve compression related to **carpal tunnel syndrome**. Another anatomic procedure is neurolysis, also called a nerve block, which involves destroying a portion of a peripheral nerve.

Augmentative procedures include electrical stimulation or direct application of drugs to the nerves that are transmitting the pain signals. Electrical stimulation works on the same principle as TENS. In this procedure, instead of applying the current across the skin, electrodes are implanted to stimulate peripheral nerves or nerves in the spinal cord. Augmentative procedures also include implanted drug-delivery systems. In these systems, catheters are implanted in the spine to allow direct delivery of drugs to the CNS.

Ablative procedures are characterized by severing a nerve and disconnecting it from the CNS. However, this method may not address potential alterations within the spinal cord. These changes perpetuate pain messages and do not cease even when the connection between the sensory nerve and the CNS is severed. With growing understanding of neuropathic pain and development of less invasive procedures, ablative procedures are used less frequently. However, they do have applications in select cases of **peripheral neuropathy**, cancer pain, and other disorders.

Preparation

Prior to beginning management, pain is thoroughly evaluated. Pain scales or questionnaires are used to attach an objective measure to a subjective experience. Objective measurements allow health care workers a better understanding of the pain being experienced by the patient. Evaluation also includes physical examinations and diagnostic tests to determine underlying causes. Some evaluations require assessments from several viewpoints, including neurology, psychiatry and psychology, and physical therapy. If pain is due to a medical procedure, management consists of anticipating the type and intensity of associated pain and managing it preemptively.

KEY TERMS

Acute—Referring to pain in response to injury or other stimulus that resolves when the injury heals or the stimulus is removed.

Chronic—Referring to pain that endures beyond the term of an injury or painful stimulus. Can also refer to cancer pain, pain from a chronic or degenerative disease, and pain from an unidentified cause.

CNS or central nervous system—The part of the nervous system that includes the brain and the spinal cord.

Iatrogenic—Resulting from the activity of the physician.

Neuropathy—Nerve damage.

Neurotransmitter—Chemicals within the nervous system that transmit information from or between nerve cells.

Nociceptor—A nerve cell that is capable of sensing pain and transmitting a pain signal.

Nonpharmacological—Referring to therapy that does not involve drugs.

Pharmacological—Referring to therapy that relies on drugs.

PNS or peripheral nervous system—Nerves that are outside of the brain and spinal cord.

Stimulus—A factor capable of eliciting a response in a nerve.

Risks

Owing to toxicity over the long term, some drugs can only be used for acute pain or as adjuncts in chronic pain management. NSAIDs have the well-known side effect of causing gastrointestinal bleeding, and long-term use of acetaminophen has been linked to kidney and liver damage. Other drugs, especially narcotics, have serious side effects, such as **constipation**, drowsiness, and **nausea**. Serious side effects can also accompany pharmacological therapies; mood swings, confusion, bone thinning, cataract formation, increased blood pressure, and other problems may discourage or prevent use of some analgesics.

Nonpharmacological therapies carry little or no risk. However, it is advised that individuals recovering from serious illness or injury consult with their health care providers or physical therapists before making use of adjunct therapies. Invasive procedures carry

risks similar to other surgical procedures, such as infection, reaction to anesthesia, iatrogenic (injury as a result of treatment) injury, and failure.

A traditional concern about narcotics use has been the risk of promoting **addiction**. As narcotic use continues over time, the body becomes accustomed to the drug and adjusts normal functions to accommodate to its presence. Therefore, to elicit the same level of action, it is necessary to increase dosage over time. As dosage increases, an individual may become physically dependent on narcotic drugs.

However, physical dependence is different from psychological addiction. Physical dependence is characterized by discomfort if drug administration suddenly stops, while psychological addiction is characterized by an overpowering craving for the drug for reasons other than pain relief. Psychological addiction is a very real and necessary concern in some instances, but it should not interfere with a genuine need for narcotic pain relief. However, caution must be taken with people with a history of addictive behavior.

Normal results

Effective application of pain management techniques reduces or eliminates acute or chronic pain. This treatment can improve an individual's quality of life and aid in recovery from injury and disease.

Resources

PERIODICALS

Finn, Robert. "More than Half of Patients With Major Depression Have Chronic Pain." *Family Practice News* October 15, 2004: 38.

"New Guidelines Set for Better Pain Treatment." *Medical Letter on the CDC & FDA* September 5, 2004: 95.

ORGANIZATIONS

American Chronic Pain Association. P.O. Box 850, Rocklin, CA 95677-0850. (916) 632-0922. <<http://members.tripod.com/~widdy/ACPA.html>>.

American Pain Society. 4700 West Lake Ave., Glenview, IL 60025. (847) 375-4715. <<http://www.ampainsoc.org>>.

National Chronic Pain Outreach Association, Inc. P.O. Box 274, Millboro, VA 24460-9606. (540) 997-5004.

Julia Barrett
Teresa G. Odle

Pain relievers see **Analgesics**

Painful menstruation see **Dysmenorrhea**

Palliative cancer therapy see **Cancer therapy, palliative**

Palpitations

Definition

A sensation in which a person is aware of an irregular, hard, or rapid heartbeat.

Description

Palpitations mean that the heart is not behaving normally. It can appear to skip beats, beat rapidly, beat irregularly, or thump in the chest. Although palpitations are very common and often harmless, they can be frightening to the person, who is usually unaware of his or her heartbeat.

Palpitations can also be a sign of serious heart trouble. Palpitations that are caused by certain types of abnormal heart rhythms (**arrhythmias**) can be serious, and even fatal if left untreated. Recognizable arrhythmias are present in a small number of patients who have palpitations. Immediate medical attention should be sought for palpitations that feel like a very fast series of heartbeats, last more than two or three minutes, and are unrelated to strenuous physical activity, obvious fright, or anger. Medical attention should also be sought if palpitations are accompanied by chest **pain**, **dizziness**, **shortness of breath**, or an overall feeling of weakness.

Most people have experienced a skipped or missed heartbeat, which is really an early beat and not a skipped beat at all. After a premature heartbeat, the heart rests for an instant then beats with extra force, making the person feel as if the heart has skipped a beat. This type of palpitation is nothing to worry about unless it occurs frequently. Severe palpitations feel like a thudding or fluttering sensation in the chest. After chest pain, palpitations are the most common reason that people are referred for cardiology evaluation.

Causes and symptoms

Palpitations can be caused by **anxiety**, arrhythmias, caffeine, certain medications, **cocaine** and other amphetamines, emotional **stress**, overeating, panic, somatization, and vigorous **exercise**. There may be no other symptoms. But, anxiety, dizziness, shortness of breath, and chest pain may be signs of more severe arrhythmias.

Diagnosis

Palpitations are diagnosed through a medical history, a physical examination, an electrocardiogram

(ECG), and screening for psychiatric disorders. It is often difficult to distinguish palpitations from **panic disorder**, a common problem in which the person experiences frequent and unexplained “fight-or-flight” responses, which is the body’s natural physical reaction to extreme danger or physical exertion, but without the obvious external stimulus.

To accurately diagnose palpitations, one of the irregular heartbeats must be “captured” on an EKG, which shows the heart’s activity. Electrodes covered with a type of gel that conducts electrical impulses are placed on the patient’s chest, arms, and legs. These electrodes send impulses of the heart’s activity to a recorder, which traces them on paper. This **electrocardiography** test takes about 10 minutes and is performed in a physician’s office or hospital. Because the palpitations are unlikely to occur during a standard EKG, **Holter monitoring** is often performed. In this procedure, the patient wears a small, portable tape recorder that is attached to a belt or shoulder strap and connected to electrode disks on his or her chest. The Holter monitor records the heart’s rhythm during normal activities. Some medical centers are now using event recorders that the patient can carry for weeks or months. When the palpitations occur, the patient presses a button on the device, which captures the information about the palpitations for physician evaluation. Later the recording can be transmitted over the telephone line for analysis.

Treatment

Most palpitations require no treatment. Persistent palpitations can be treated with small doses of a beta blocker. **Beta blockers** are drugs that tend to lower blood pressure. They slow the heart rate and decrease the force with which the heart pumps. If the cause of the palpitations is determined to be an arrhythmia, medical or surgical treatment may be prescribed, although surgery is rarely needed.

Alternative treatment

Alternative treatments for palpitations should be used only as a complement to traditional medicine. Alternative treatments include: **aromatherapy**, Chinese herbs, herbal therapies, **homeopathic medicine**, exercise, mind/body medicine, and diet and nutrition. In aromatherapy, adding citrus oils to bath water may help with minor palpitations. Some Chinese herbs can also help, but others can worsen arrhythmias, so a qualified herbalist should be consulted. Herbal therapies such as hawthorn

KEY TERMS

Arrhythmia—Any variation from the normal heart-beat. Some arrhythmias are harmless, while others, such as ventricular tachycardia, ventricular fibrillation, and ventricular standstill, can be fatal.

Somatization—Anxiety converted into physical symptoms. Somatization is a sign of panic disorder.

(*Crataegus laevigata*) and motherwort (*Leonurus cardiaca*) can help with palpitations. Homeopathic remedies such as *Lachesis*, *Digitalis*, and *Aconite* (*Aconitum napellus*) may be used to control palpitations but should be taken only when prescribed by a homeopathic physician. Mind/body medicine such as **meditation** and **yoga** can help the person relax, eliminating or reducing palpitations caused by anxiety or stress. Reducing or eliminating tea, cola, coffee, and chocolate, and consuming adequate amounts of the **minerals** calcium, magnesium, and potassium can help reduce or eliminate palpitations.

Prognosis

Most palpitations are harmless, but some can be a sign of heart trouble, which could be fatal if left untreated.

Prevention

Palpitations not caused by arrhythmias can be prevented by reducing or eliminating anxiety and emotional stress, and reducing or eliminating consumption of tea, cola, coffee, and chocolate. Exercise can also help, but a treadmill **stress test** performed by a physician should be considered first to make sure the exercise is safe.

Resources

PERIODICALS

Mayou, Richard. “Chest Pain, Palpitations, and Panic.” *Journal of Psychosomatic Research* 44 (1998): 53-70.

Lori De Milto

Panax quinquefolius see **Ginseng**

Pancreas removal see **Pancreatectomy**

Pancreas transplantation

Definition

Pancreas transplantation is a surgical procedure in which a diseased pancreas is replaced with a healthy pancreas that has been obtained immediately after **death** from an immunologically compatible donor.

Purpose

The pancreas secretes insulin to regulate glucose (sugar) metabolism. Failure to regulate glucose levels leads to diabetes. Over one million patients in the United States have insulin dependent (type I) **diabetes mellitus**. Successful pancreas transplantation allows the body to make and secrete its own insulin, and establishes insulin independence for these patients.

Pancreas transplantation is major surgery that requires suppression of the immune system to prevent the body from rejecting the transplanted pancreas. Immunosuppressive drugs have serious side effects. Because of these side effects, in 1996, 85% of pancreas transplants were performed simultaneously with kidney transplants, 10% after a kidney transplant, and only 5% were performed as a pancreas transplant alone.

The rationale for this is that patients will already be receiving immunosuppressive treatments for the kidney transplant, so they might as well receive the benefit of a pancreas transplant as well. Patients considering pancreas transplantation alone must decide with their doctors whether life-long treatment with immunosuppressive drugs is preferable to life-long insulin dependence.

The best candidates for pancreas transplantation are:

- between the ages of 20–40
- those who have extreme difficulty regulating their glucose levels
- those who have few secondary complications of diabetes
- those who are in good cardiovascular health.

Precautions

Many people with diabetes are not good candidates for a pancreas transplant. Others do not have tissue compatibility with the donor organ. People who are successfully controlling their diabetes with insulin

National Transplant Waiting List By Organ Type (June 2000)

Organ Needed	Number Waiting
Kidney	48,349
Liver	15,987
Heart	4,139
Lung	3,695
Kidney-Pancreas	2,437
Pancreas	942
Heart-Lung	212
Intestine	137

injections are usually not considered for pancreas transplants.

Description

Once a donor pancreas is located, the patient is prepared for surgery. Since only about 1,000 pancreas transplants are performed each year in the United States, the operation usually occurs at a hospital where surgeons have special expertise in the procedure.

The surgeon makes an incision under the ribs and locates the pancreas and duodenum. The pancreas and duodenum (part of the small intestine) are removed. The new pancreas and duodenum are then connected to the patient's blood vessels.

Replacing the duodenum allows the pancreas to drain into the gastrointestinal system. The transplant can also be done creating a bladder drainage. Bladder drainage makes it easier to monitor organ rejection. Once the new pancreas is in place, the abdomen and skin are closed. This surgery is often done at the same time as kidney transplant surgery.

Preparation

After the patient and doctor have decided on a pancreas transplant, a complete immunological study is done to match the patient to a donor. All body functions are evaluated. The timing of surgery depends on the availability of a donated organ.

Aftercare

Patients receiving a pancreas transplantation are monitored closely for organ rejection, and all vital body functions are monitored also. The average hospital stay is three weeks. It takes about six months to recover from surgery. Patients will take immunosuppressive drugs for the rest of their lives.



A surgeon harvests the islets of Langerhans from a donor pancreas. (Photograph by Daniel Portnoy. AP/Wide World Photo. Reproduced by permission.)

KEY TERMS

Duodenum—The section of the small intestine immediately after the stomach.

Risks

Diabetes and poor kidney function greatly increase the risk of complications from anesthesia during surgery. Organ rejection, excessive bleeding, and infection are other major risks associated with this surgery.

Normal results

During a nine year period from 1987 to 1996, the patient survival rate for all types of pancreas

transplants (with or without associated kidney transplant) was 92% after one year and 86% after three years. In a successful transplant, the pancreas begins producing insulin, bringing the regulation of glucose back under normal body control. Natural availability of insulin prevents the development of additional damage to the kidneys and blindness associated with diabetes. Many patients report an improved quality of life.

Resources

ORGANIZATIONS

American Diabetes Association. 1701 North Beauregard Street, Alexandria, VA 22311. (800) 342-2383.
< <http://www.diabetes.org> >.

Tish Davidson, A.M.

Pancreatectomy

Definition

Pancreatectomy is the surgical removal of the pancreas. Pancreatectomy may be total, in which case the whole organ is removed, or partial, referring to the removal of part of the pancreas.

Purpose

Pancreatectomy is the most effective treatment for **cancer** of the pancreas, an abdominal organ that secretes digestive enzymes, insulin, and other hormones. The thickest part of the pancreas near the duodenum (small intestine) is called the head, the middle part is called the body, and the thinnest part adjacent to the spleen is called the tail.

While surgical removal of tumors in the pancreas is preferred, it is only possible in the 10-15% of patients who are diagnosed early enough for a potential cure. Patients who are considered suitable for surgery usually have small tumors in the head of the pancreas (close to the duodenum, or first part of the small intestine), have **jaundice** as their initial symptom, and have no evidence of metastatic disease (spread of cancer to other sites).

Pancreatectomy is sometimes necessary when the pancreas has been severely injured by trauma, especially injury to the body and tail of the pancreas. While such surgery removes normal pancreatic tissue as well, the long-term consequences of this surgery are minimal, with virtually no effects on the production of insulin, digestive enzymes, and other hormones.

Chronic **pancreatitis** is another condition for which pancreatectomy is occasionally performed. Chronic pancreatitis—or continuing inflammation of the pancreas that results in permanent damage to this organ—can develop from long-standing, recurring episodes of acute (periodic) pancreatitis. This painful condition usually results from alcohol abuse or the presence of **gallstones**. In most patients with alcohol-induced disease, the pancreas is widely involved, therefore, surgical correction is almost impossible.

Precautions

Pancreatectomy is only performed when surgery provides a clear benefit. Patients who have tumors that are obviously not operable should be carefully excluded from consideration.

Description

Pancreatectomy sometimes entails removal of the entire pancreas, called a total pancreatectomy, but more often involves removal of part of the pancreas, which is called a subtotal pancreatectomy, or distal pancreatectomy, when the body and tail of the pancreas are removed. When the duodenum is removed along with all or part of the pancreas, the procedure is called a pancreaticoduodenectomy, which surgeons sometimes refer to as “Whipple’s procedure.” Pancreaticoduodenectomy is being used increasingly for treatment of a variety of malignant and benign diseases of the pancreas.

Regional lymph nodes are usually removed during pancreaticoduodenectomy. In distal pancreatectomy, the spleen may also be removed.

Preparation

Patients with symptoms of a pancreatic disorder usually undergo a number of tests before surgery is even considered. These can include ultrasonography, x-ray examinations, **computed tomography scans** (CT scan), and endoscopic retrograde cholangiopancreatography (ERCP), an x-ray imaging technique. Tests may also include **angiography**, an x-ray technique for visualizing the arteries feeding the pancreas, and needle aspiration cytology, in which cells are drawn from areas suspected to contain cancer. Such tests aid in the diagnosis of the pancreatic disorder and in the planning of the operation.

Since many patients with pancreatic cancer are undernourished, appropriate nutritional support, sometimes by **tube feedings**, may be required prior to surgery.

Some patients with pancreatic cancer deemed suitable for pancreatectomy will undergo chemotherapy and/or radiation therapy. This treatment is aimed at shrinking the tumor, which will improve the chances for successful surgical removal. Sometimes, patients who are not initially considered surgical candidates may respond so well to chemoradiation that surgical treatment becomes possible. **Radiation therapy** may also be applied during the surgery (intraoperatively) to improve the patient’s chances of survival, but this treatment is not yet in routine use. Some studies have shown that intraoperative radiation therapy extends survival by several months.

Patients undergoing distal pancreatectomy that involves removal of the spleen may receive preoperative medication to decrease the risk of infection.

Aftercare

Pancreatectomy is major surgery. Therefore, extended hospitalization is usually required. Some studies report an average hospital stay of about two weeks.

Some cancer patients may also receive combined **chemotherapy** and radiation therapy after surgery. This additional treatment has been clearly shown to enhance survival from pancreatic cancer.

Removal of all or part of the pancreas can lead to a condition called pancreatic insufficiency, in which food cannot be normally processed by the body, and insulin secretion may be inadequate. These conditions can be treated with pancreatic enzyme replacement therapy, to supply digestive enzymes, and insulin injections, to supply insulin.

Risks

The mortality rate for pancreatectomy has improved in recent years to 5–10%, depending on the extent of the surgery and the experience of the surgeon. A study of 650 patients at Johns Hopkins Medical Institution, Baltimore, found that only nine patients, or 1.4%, died from complications related to surgery.

There is still, however, a fairly high risk of complications following any form of pancreatectomy. The Johns Hopkins study documented complications in 41% of cases. The most devastating complication is postoperative bleeding, which increases the mortality risk to 20–50%. In cases of postoperative bleeding, the patient may be returned to surgery to find the source of hemorrhage, or may undergo other procedures to stop the bleeding.

One of the most common complications from a pancreaticoduodenectomy is delayed gastric emptying, a condition in which food and liquids are slow to leave the stomach. This complication occurred in 19% of patients in the Johns Hopkins study. To manage this problem, many surgeons insert feeding tubes at the original operation site, through which nutrients can be fed directly into the patient's intestines. This procedure, called enteral nutrition, maintains the patient's **nutrition** if the stomach is slow to recover normal function. Certain medications, called promotility agents, can help move the nutritional contents through the gastrointestinal tract.

The other most common complication is pancreatic anastomotic leak. This is a leak in the connection that the surgeon makes between the remainder of the pancreas and the other structures in the abdomen.

KEY TERMS

Chemotherapy—A treatment of the cancer with synthetic drugs that destroy the tumor either by inhibiting the growth of the cancerous cells or by killing the cancer cells.

Computed tomography (CT) scan—A medical procedure where a series of x rays are taken and put together by a computer in order to form detailed pictures of areas inside the body.

Magnetic resonance imaging (MRI)—A medical procedure used for diagnostic purposes where pictures of areas inside the body can be created using a magnet linked to a computer.

Pancreas—A large gland located on the back wall of the abdomen, extending from the duodenum (first part of the small intestine) to the spleen. The pancreas produces enzymes essential for digestion, and the hormones insulin and glucagon, which play a role in diabetes.

Pancreaticoduodenectomy—Removal of all or part of the pancreas along with the duodenum. Also known as “Whipple’s procedure” or “Whipple’s operation.”

Pancreatitis—Inflammation of the pancreas, either acute (sudden and episodic) or chronic, usually caused by excessive alcohol intake or gallbladder disease.

Radiation therapy—A treatment using high energy radiation from x-ray machines, cobalt, radium, or other sources.

Ultrasonogram—A procedure where high-frequency sound waves that cannot be heard by human ears are bounced off internal organs and tissues. These sound waves produce a pattern of echoes which are then used by the computer to create sonograms or pictures of areas inside the body.

Most surgeons handle the potential for this problem by assuring that there will be adequate drainage from the surgical site.

Normal results

Unfortunately, pancreatic cancer is the most lethal form of gastrointestinal malignancy. However, for a highly selective group of patients, pancreatectomy offers a chance for cure, especially when performed by experienced surgeons. The overall five-year survival rate for patients who undergo

pancreatectomy for pancreatic cancer is about 10%; patients who undergo pancreaticoduodenectomy have a 4–5% survival at five years. The risk for tumor recurrence is thought to be unaffected by whether the patient undergoes a total pancreatectomy or a pancreaticoduodenectomy, but is increased when the tumor is larger than 3 cm and the cancer has spread to the lymph nodes or surrounding tissue.

After total pancreatectomy, the body loses the ability to secrete insulin, enzymes, and other substances, therefore, certain medications will be required to compensate for this. In some cases of pancreatic disease, the pancreas ceases to function normally, then total pancreatectomy may be preferable to other less radical forms of the operation.

When pancreatectomy is performed for chronic pancreatitis, the majority of patients obtain some relief from **pain**. Some studies report that one half to three quarters of patients become free of pain.

Resources

BOOKS

Bastidas, J. Augusto, and John E. Niederhuber. "The Pancreas." In *Fundamentals of Surgery*, edited by John E. Niederhuber. Stamford: Appleton & Lange, 1998.

Caroline A. Helwick

Pancreatic cancer, endocrine

Definition

Endocrine pancreatic **cancer** is a disease in which cancerous cells originate within the tissues of the pancreas that produce hormones.

Description

The pancreas is a 6–8 in (15–20 cm) long, slipper-shaped gland located in the abdomen. It lies behind the stomach, within a loop formed by the small intestine. Other nearby organs include the gallbladder, spleen, and liver. The pancreas has a wide end (head), a narrow end (tail), and a middle section (body). A healthy pancreas is important for normal food digestion and plays a critical role in the body's metabolic processes. The pancreas has two main functions, each performed by distinct types of tissue. The exocrine tissue secretes fluids into the other organs of the digestive system, while the endocrine tissue

secretes substances that are circulated in the bloodstream. The exocrine pancreas makes up the vast majority of the gland; it produces pancreatic juices containing enzymes that help break down proteins and fatty food. The endocrine tissue of the pancreas makes up only 2% of the gland's total mass. It consists of small patches of cells that produce hormones (like insulin) that control how the body stores and uses nutrients. These patches are called islets (islands) of Langerhans or islet cells and are interspersed evenly throughout the pancreas. Each islet contains approximately 1,000 endocrine cells and a dense network of capillaries (tiny blood vessels), which allows immediate entry of hormones into the circulatory system.

Pancreatic tumors are classified as either exocrine or endocrine tumors depending on which type of tissue they arise from within the gland. Endocrine tumors of the pancreas are very rare, accounting for only 5% of all pancreatic cancers. The majority of endocrine pancreatic tumors are functional adenocarcinomas that overproduce a specific hormone. There are several types of islet cells and each produces its own hormone or peptide (small protein molecule). Functional endocrine tumors are named after the hormone they secrete. Insulinoma is the most common tumor of the endocrine pancreas. Patients with this disease usually develop hypoglycemia due to increased insulin production that leads to abnormally low blood sugar levels. **Gastrinoma**, a disease in which gastrin (hormone that stimulates stomach acid production) is overproduced, causes multiple ulcers in the upper gastrointestinal (GI) tract. Gastrinoma was first described in patients with a rare form of severe peptic ulcer disease known as Zollinger-Ellison syndrome (ZES). The less common glucagonoma causes mild diabetes due to excess glucagon (hormone that stimulates glucose production) secretion. Other rare islet cell tumors include vipoma (vasoactive intestinal peptide) and somatostatinoma. Nonfunctional pancreatic endocrine tumors are not associated with an excess production of any hormone and can be difficult to distinguish from exocrine pancreatic cancer. Cancers of the endocrine pancreas are relatively slow-growing compared to the more common ductal adenocarcinomas of the exocrine pancreas.

Between one and four cases of insulinoma occur per million people per year, and 90% of these tumors are benign. They occur mostly between the ages of 50 and 60 and affect men and women equally. Less than three cases of gastrinoma per million people are diagnosed each year, but it is the most common functional islet cell tumor in patients with multiple endocrine tumors, a condition known as multiple endocrine

neoplasia (MEN) syndrome. Vipoma and glucagonoma are even rarer and they occur more frequently in women. Somatostatinoma is exceedingly uncommon, and less than 100 cases have been reported worldwide. Nonfunctional islet cell cancers account for approximately one-third of all cancers of the endocrine pancreas, and the majority of these are malignant.

Causes and symptoms

There are no known causes of islet cell cancer, but a small percentage of cases occur due to hereditary syndromes such as MEN. This is a condition that frequently causes more than one tumor in several endocrine glands, such as the parathyroid and pituitary, in addition to the islet cells of the pancreas. Twenty-five percent of gastrinomas and less than 10% of insulinomas occur in MEN patients. Von Hippel-Lindau (VHL) syndrome is another genetic disorder that causes multiple tumors, and 10–15% of VHL patients will develop islet cell cancer.

Symptoms vary among the different islet cell cancer types. Insulinoma causes repeated episodes of **hypoglycemia**, sweating, and **tremors**, while patients with gastrinoma have inflammation of the esophagus, epigastric **pain**, multiple ulcers, and possibly **diarrhea**. Symptoms of glucagonoma include a distinctive skin rash, inflammation of the stomach, glucose intolerance, weight loss, weakness, and anemia (less common). Patients with vipoma have episodes of profuse, watery diarrhea, even after **fasting**. Somatostatinoma causes mild diabetes, diarrhea/steatorrhea (fatty stools), weight loss, and gallbladder disease. Nonfunctional endocrine tumors frequently produce the same symptoms as cancer of the exocrine pancreas such as abdominal pain, jaundice, and weight loss.

Diagnosis

A thorough physical exam is usually performed when a patient presents with the above symptoms, however, functional endocrine tumors of the pancreas tend to be small and are not detected by palpating the abdomen. Once other illnesses such as infection are ruled out, the doctor will order a series of blood and urine tests. The functional endocrine tumors can be identified through increased levels of hormone in the bloodstream.

Functional endocrine tumors can occur in multiple sites in the pancreas and are often small (less than 1 cm), making them difficult to diagnose. Nonfunctional tumors tend to be larger, which makes

them difficult to distinguish from tumors of the exocrine pancreas. Methods such as computed tomography (CT) scan and **magnetic resonance imaging** (MRI) are used to take pictures of the internal organs and allow the doctor to determine whether a tumor is present. Somatostatin receptor scintigraphy (trade name OctreoScan) is an imaging system used to localize endocrine tumors, especially gastrinomas and somatostatinomas. Endoscopic ultrasound (EUS) is a more sensitive technique that may be used if a CT scan fails to detect a tumor. Endocrine tumors usually have many blood vessels, so **angiography** may be useful in the doctor's assessment and staging of the tumor. Surgical exploration is sometimes necessary in order to locate very small tumors that occur in multiple sites. These techniques also help the doctor evaluate how far the tumor has spread. A biopsy can be taken to confirm diagnosis, but more often, doctors look at the size and local invasion of the tumor in order to plan a treatment strategy.

Treatment

Staging

The staging system for islet cell cancer is still evolving, but the tumors typically fall into three categories: cancers that arise in one location within the pancreas, cancers that arise in several locations within the pancreas, and cancers that have spread to nearby lymph nodes or to other organs in the body.

Surgery is the only curative method for islet cell cancers, and studies have shown that an aggressive surgical approach can improve survival and alleviate symptoms of the disease. As with most forms of cancer, the earlier it is diagnosed, the greater the chance for survival. With the exception of insulinoma, the majority of islet cell tumors are malignant at the time of diagnosis, and more than half are metastatic. However, surgery and **chemotherapy** have been shown to improve the outcome of patients even if they have metastatic disease. Surgery may include partial or total removal of the pancreas, and in patients with gastrinoma, the stomach may be removed as well. Streptozotocin, doxorubicin, and 5-fluorouracil (5-FU) are chemotherapeutic agents commonly used in the treatment of islet cell cancer. Patients may experience **nausea** and vomiting as well as kidney toxicity from streptozotocin, and bone marrow suppression from doxorubicin. Hormone therapy is used to relieve the symptoms of functional tumors by inhibiting excess hormone production. Other techniques may be used to block blood flow to the liver in an attempt to kill the cancer cells that have

KEY TERMS

Adenocarcinoma—A malignant tumor that arises within the tissues of a gland and retains its glandular structure.

Angiography—Diagnostic technique used to study blood vessels in a tumor.

Biopsy—Removal and microscopic examination of cells to determine whether they are cancerous.

Chemotherapy—Drug treatment administered to kill cancerous cells.

Endocrine—Refers to glands that secrete hormones circulated in the bloodstream.

Endoscopic ultrasonography (EUS)—Diagnostic imaging technique where an ultrasound probe is inserted down a patient's throat to determine if a tumor is present.

Gastrinoma—Tumor that arises from the gastrin-producing cells in the pancreas.

Insulinoma—Tumor that arises from the insulin-producing cells in the pancreas.

Islets of Langerhans—Clusters of cells in the pancreas that make up the endocrine tissue.

spread there. Abdominal pain, nausea, **vomiting** and **fever** may result from this type of treatment. Radiation has little if any role in the treatment of islet cell cancer.

Prognosis

Islet cell cancers overall have a more favorable prognosis than cancers of the exocrine pancreas, and the median survival from diagnosis is three and a half years. This is mainly due to their slow-growing nature. Insulinomas have a five-year survival rate of 80% and gastrinomas have 65%. When malignant, islet cell cancers do not generally respond well to chemotherapy, and the treatment is mainly palliative. Most patients with metastasis do not survive five years. Islet cell cancer tends to spread to the surrounding lymph nodes, stomach, small intestine, and liver.

Prevention

There are no known risk factors associated with sporadic islet cell cancer. Therefore, it is not clear how to prevent its occurrence. Individuals with MEN syndrome or VHL, however, have a genetic predisposition to developing islet cell cancer and should be screened regularly in an effort to catch the disease early.

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Pancreatic cancer, exocrine

Definition

Exocrine pancreatic **cancer** is a disease in which cancerous cells originate within the tissues of the pancreas that produce digestive juices.

Description

The pancreas is a 6-8 in (15-20 cm) long, slipper-shaped gland located in the abdomen. It lies behind the stomach, within a loop formed by the small intestine. Other nearby organs include the gallbladder, spleen, and liver. The pancreas has a wide end (head), a narrow end (tail), and a middle section (body). A healthy pancreas is important for normal food digestion and also plays a critical role in the body's metabolic processes. The pancreas has two main functions, and each are performed by distinct types of tissue. The exocrine tissue makes up the vast majority of the gland and secretes fluids into the other

organs of the digestive system. The endocrine tissue secretes hormones (like insulin) that are circulated in the bloodstream, and these substances control how the body stores and uses nutrients. The exocrine tissue of the pancreas, comprised mostly of acinar cells and ductal cells, produces pancreatic (digestive) juices. These juices contain several enzymes that help break down proteins and fatty foods. The exocrine pancreas forms an intricate system of channels or ducts, which are tubular structures that carry pancreatic juices to the small intestine where they are used for digestion.

Pancreatic tumors are classified as either exocrine or endocrine tumors depending on which type of tissue they arise from within the gland. Ninety-five percent of pancreatic cancers occur in the tissues of the exocrine pancreas. Ductal adenocarcinomas arise in the cells that line the ducts of the exocrine pancreas and account for 80% to 90% of all tumors of the pancreas. Unless specified, nearly all reports on pancreatic cancer refer to ductal adenocarcinomas. Less common types of pancreatic exocrine tumors include acinar cell carcinoma, cystic tumors that are typically benign but may become cancerous, and papillary tumors that grow within the pancreatic ducts. Pancreatoblastoma is a very rare disease that primarily affects young children. Two-thirds of pancreatic tumors occur in the head of the pancreas, and tumor growth in this area can lead to the obstruction of the nearby common bile duct that empties bile fluid into the small intestine. When bile cannot be passed into the intestine, patients may develop yellowing of the skin and eyes (jaundice) due to the buildup of bilirubin (a component of bile) in the bloodstream. Tumor blockage of bile or pancreatic ducts may also cause digestive problems since these fluids contain critical enzymes in the digestive process. Depending on their size, pancreatic tumors may cause abdominal **pain** by pressing on the surrounding nerves. Because of its location deep within the abdomen, pancreatic cancer often remains undetected until it has spread to other organs such as the liver or lung. Pancreatic cancer tends to rapidly spread to other organs, even when the primary (original) tumor is relatively small.

Though pancreatic cancer accounts for only 3% of all cancers, it is the fifth most frequent cause of cancer deaths. In 2001, an estimated 29,200 new cases of pancreatic cancer will be diagnosed in the United States. Pancreatic cancer is primarily a disease associated with advanced age, with 80% of cases occurring between the ages of 60 and 80. Men are almost twice as likely to develop this disease than women. Countries with the highest frequencies of pancreatic cancer include the United States, New Zealand,

Western European nations, and Scandinavia. The lowest occurrences of the disease are reported in India, Kuwait and Singapore. African Americans have the highest incidence of pancreatic cancer of any ethnic group worldwide. Whether this difference is due to diet or environmental factors remains unclear.

Causes and symptoms

Although the exact cause for pancreatic cancer is not known, several risk factors have been shown to increase susceptibility to this particular cancer, the greatest of which is cigarette smoking. Approximately one-third of pancreatic cancer cases occur among smokers. People who have diabetes develop pancreatic cancer twice as often as non-diabetics. Numerous studies suggest that a family history of pancreatic cancer is another strong risk factor for developing the disease, particularly if two or more relatives in the immediate family have the disease. Other risk factors include chronic (long-term) inflammation of the pancreas (**pancreatitis**), **diets** high in fat, and occupational exposure to certain chemicals such as petroleum.

Pancreatic cancer often does not produce symptoms until it reaches an advanced stage. Patients may then present with the following signs and symptoms:

- upper abdominal and/or back pain
- **jaundice**
- weight loss
- loss of appetite
- diarrhea
- weakness
- **nausea**

These symptoms may also be caused by other illnesses; therefore, it is important to consult a doctor for an accurate diagnosis.

Diagnosis

Pancreatic cancer is difficult to diagnose, especially in the absence of symptoms, and there is no current screening method for early detection. The most sophisticated techniques available often do not detect very small tumors that are localized (have not begun to spread). At advanced stages where patients show symptoms, a number of tests may be performed to confirm diagnosis and to assess the stage of the disease. Approximately half of all pancreatic cancers

are metastatic (have spread to other sites) at the time of diagnosis.

The first step in diagnosing pancreatic cancer is a thorough medical history and complete physical examination. The abdomen will be palpated to check for fluid accumulation, lumps, or masses. If there are signs of jaundice, blood tests will be performed to rule out the possibility of liver diseases such as hepatitis. Urine and stool tests may be performed as well.

Non-invasive imaging tools such as computed tomography (CT) scans and magnetic resonance imaging (MRI) can be used to produce detailed pictures of the internal organs. CT is the tool most often used to diagnose pancreatic cancer, as it allows the doctor to determine if the tumor can be removed by surgery or not. It is also useful in staging a tumor by showing the extent to which the tumor has spread. During a CT scan, patients receive an intravenous injection of a contrast dye so the organs can be visualized more clearly. MRI may be performed instead of CT if a patient has an allergy to the CT contrast dye. In some cases where the tumor is impinging on blood vessels or nearby ducts, MRI may be used to generate an image of the pancreatic ducts.

If the doctor suspects pancreatic cancer and no visible masses are seen with a CT scan, a patient may undergo a combination of invasive tests to confirm the presence of a pancreatic tumor. Endoscopic ultrasound (EUS) involves the use of an ultrasound probe at the end of a long, flexible tube that is passed down the patient's throat and into the stomach. This instrument can detect a tumor mass through high frequency sound waves and echoes. EUS can be accompanied by fine needle aspiration (FNA), where a long needle, guided by the ultrasound, is inserted into the tumor mass in order to take a biopsy sample. **Endoscopic retrograde cholangiopancreatography (ERCP)** is a technique often used in patients with severe jaundice because it enables the doctor to relieve blockage of the pancreatic ducts. The doctor, guided by endoscopy and x rays, inserts a small metal or plastic stent into the duct to keep it open. During ERCP, a biopsy can be done by collecting cells from the pancreas with a small brush. The cells are then examined under the microscope by a pathologist, who determines the presence of any cancerous cells.

In some cases, a biopsy may be performed during a type of surgery called laparoscopy, which is done under **general anesthesia**. Doctors insert a small camera and instruments into the abdomen after a minor incision is made. Tissue samples are removed for

examination under the microscope. This procedure allows a doctor to determine the extent to which the disease has spread and decide if the tumor can be removed by further surgery.

An **angiography** is a type of test that studies the blood vessels in and around the pancreas. This test may be done before surgery so that the doctor can determine the extent to which the tumor invades and interacts with the blood vessels within the pancreas. The test requires local anesthesia and a catheter is inserted into the patient's upper thigh. A dye is then injected into blood vessels that lead into the pancreas, and x rays are taken.

As of April 2001, doctors at major cancer research institutions such as Memorial Sloan-Kettering Cancer Center in New York are investigating CT angiography, an imaging technique that is less invasive than angiography alone. CT angiography is similar to a standard CT scan, but allows doctors to take a series of pictures of the blood vessels that support tumor growth. A dye is injected as in a CT scan (but at rapid intervals) and no catheter or **sedation** is required. A computer generates 3D images from the pictures that are taken, and the information is gathered by the surgical team who will develop an appropriate strategy if the patient's disease can be operated on.

Treatment

Staging

After cancer of the pancreas has been diagnosed, doctors typically use a TNM staging system to classify the tumor based on its size and the degree to which it has spread to other areas in the body. T indicates the size and local advancement of the primary tumor. Since cancers often invade the lymphatic system before spreading to other organs, regional lymph node involvement (N) is an important factor in staging. M indicates whether the tumor has metastasized (spread) to distant organs. In stage I, the tumor is localized to the pancreas and has not spread to surrounding lymph nodes or other organs. Stage II pancreatic cancer has spread to nearby organs such as the small intestine or bile duct, but not the surrounding lymph nodes. Stage III indicates lymph node involvement, whether the cancer has spread to nearby organs or not. Stage IVA pancreatic cancer has spread to organs near the pancreas such as the stomach, spleen, or colon. Stage IVB is a cancer that has spread to distant sites (liver, lung). If pancreatic cancer has been treated with success and then appears again in the pancreas or in other organs, it is referred to as recurrent disease.

Treatment of pancreatic cancer will depend on several factors, including the stage of the disease and the patient's age and overall health status. A combination of therapies is often employed in the treatment of this disease to improve the patient's chances for survival. Surgery is used whenever possible and is the only means by which cancer of the pancreas can be cured. However, less than 15% of pancreatic tumors can be removed by surgery. By the time the disease is diagnosed (usually at stage III), therapies such as radiation and **chemotherapy** or both are used in addition to surgery to relieve a patient's symptoms and enhance quality of life. For patients with metastatic disease, chemotherapy and radiation are used mainly as palliative (pain alleviating) treatments.

Surgery

Three types of surgery are used in the treatment of pancreatic cancer, depending on what section of the pancreas the tumor is located in. A Whipple procedure removes the head of the pancreas, part of the small intestine and some of the surrounding tissues. This procedure is most common since the majority of pancreatic cancers occur in the head of the organ. A total pancreatectomy removes the entire pancreas and the organs around it. Distal pancreatectomy removes only the body and tail of the pancreas. Chemotherapy and radiation may precede surgery (neoadjuvant therapy) or follow surgery (adjuvant therapy). Surgery is also used to relieve symptoms of pancreatic cancer by draining fluids or bypassing obstructions. Side effects from surgery can include pain, weakness, **fatigue**, and digestive problems. Some patients may develop diabetes or malabsorption as a result of partial or total removal of the pancreas.

Radiation therapy

Radiation therapy is sometimes used to shrink a tumor before surgery or to remove remaining cancer cells after surgery. Radiation may also be used to relieve pain or digestive problems caused by the tumor if it cannot be removed by surgery. External radiation therapy refers to radiation applied externally to the abdomen using a beam of high-energy x rays. High-dose intraoperative radiation therapy is sometimes used during surgery on tumors that have spread to nearby organs. Internal radiation therapy refers to the use of small radioactive seeds implanted in the tumor tissue. The seeds emit radiation over a period of time to kill tumor cells. Radiation treatment may cause side effects such as fatigue, tender or itchy skin, nausea, **vomiting**, and digestive problems.

Chemotherapy

Chemotherapeutic agents are powerful drugs that are used to kill cancer cells. They are classified according to the mechanism by which they induce cancer cell death. Multiple agents are often used to increase the chances of tumor cell death. Gemcitabine is the standard drug used to treat pancreatic cancers and can be used alone or in combination with other drugs, such as 5-fluorouracil (5-FU). Other drugs are being tested in combination with gemcitabine in several ongoing clinical trials, specifically irinotecan (CPT-11) and oxaliplatin. Chemotherapy may be administered orally or intravenously in a series of doses over several weeks. During treatment, patients may experience fatigue, nausea, vomiting, hair loss, and mouth sores, depending on which drugs are used.

Biological treatments

Numerous vaccine treatments are being developed in an effort to stimulate the body's immune system into attacking cancer cells. This is also referred to as immunotherapy. Another type of biological treatment involves using a targeted monoclonal antibody to inhibit the growth of cancer cells. The antibody is thought to bind to and neutralize a protein that contributes to the growth of the cancer cells. Investigational treatments such as these may be considered by patients with metastatic disease who would like to participate in a clinical trial. Biological treatments typically cause flu-like symptoms (chills, **fever**, loss of appetite) during the treatment period.

Alternative treatment

Acupuncture or **hypnotherapy** may be used in addition to standard therapies to help relieve the pain associated with pancreatic cancer. Because of the poor prognosis associated with pancreatic cancer, some patients may try special diets with vitamin supplements, certain exercise programs, or unconventional treatments not yet approved by the FDA. Patients should always inform their doctors of any alternative treatments they are using as they could interfere with standard therapies. As of the year 2000, the National Cancer Institute (NCI) is funding phase III clinical trials of a controversial treatment for pancreatic cancer that involves the use of supplemental pancreatic enzymes (to digest cancerous cells) and coffee **enemas** (to stimulate the liver to detoxify the cancer). These theories remain unproven and the study is widely criticized in the medical community. It remains to be seen whether this method of treatment has any advantage over the standard

KEY TERMS

Acinar cell carcinoma—A malignant tumor arising from the acinar cells of the pancreas.

Angiography—Diagnostic technique used to study blood vessels in a tumor.

Biopsy—Removal and microscopic examination of cells to determine whether they are cancerous.

Cancer vaccines—A treatment that uses the patient's immune system to attack cancer cells.

Chemotherapy—Drug treatment administered to kill cancerous cells.

Ductal adenocarcinoma—A malignant tumor arising from the duct cells within a gland.

Endoscopic retrograde cholangiopancreatography (ERCP)—Diagnostic technique used to obtain a biopsy. Also a surgical method of relieving biliary obstruction caused by a tumor.

Endoscopic ultrasonography (EUS)—Diagnostic imaging technique in which an ultrasound probe is inserted down a patient's throat to determine if a tumor is present.

Exocrine—Refers to glands which secrete their products through a duct.

Laparoscopic surgery—Minimally invasive surgery in which a camera and surgical instruments are inserted through a small incision.

Pancreatectomy—Partial or total surgical removal of the pancreas.

Radiation therapy—Use of radioisotopes to kill tumor cells. Applied externally through a beam of x rays, intraoperatively (during surgery), or deposited internally by implanting radioactive seeds in tumor tissue.

Whipple procedure—Surgical removal of the head of the pancreas, part of the small intestine, and some surrounding tissue.

chemotherapeutic regimen in prolonging patient survival or improving quality of life.

Prognosis

Unfortunately, cancer of the pancreas is often fatal, and median survival from diagnosis is less than six months, while the five-year survival rate is 4%. This is mainly due to the lack of screening methods available for early detection of the disease. Yet, even when localized tumors can be removed by surgery,

patient survival after five years is only 10% to 15%. These statistics demonstrate the aggressive nature of most pancreatic cancers and their tendency to recur. Pancreatic cancers tend to be resistant to radiation and chemotherapy and these modes of treatment are mainly used to relieve pain and tumor burden.

Prevention

Although the exact cause of pancreatic cancer is not known, there are certain risk factors that may increase a person's chances of developing the disease. Quitting **smoking** will certainly reduce the risk for pancreatic cancer and many other cancers. The American Cancer Society recommends a diet rich in fruits, vegetables, and dietary fiber in order to reduce the risk of pancreatic cancer. According to the NCI, workers who are exposed to petroleum and other chemicals may be at greater risk for developing the disease and should follow their employer's safety precautions. People with a family history of pancreatic cancer are at greater risk than the general population, as a small percentage of pancreatic cancers are considered hereditary.

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Hirshberg Foundation for Pancreatic Cancer Research. 375 Homewood Rd., Los Angeles, CA 90049. (310) 472-6310. <<http://www.pancreatic.org>>.

National Pancreas Foundation. PO Box 935, Wexford, PA 15090-0935. <<http://www.pancreasfoundation.org>>.

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Pancreatitis

Definition

Pancreatitis is an inflammation of the pancreas, an organ that is important in digestion. Pancreatitis can be acute (beginning suddenly, usually with the patient recovering fully) or chronic (progressing slowly with continued, permanent injury to the pancreas).

Description

The pancreas is located in the midline of the back of the abdomen, closely associated with the liver, stomach, and duodenum (the first part of the small intestine). The pancreas is considered a gland. A gland is an organ whose primary function is to produce chemicals that pass either into the main blood circulation (called an endocrine function), or pass into another organ (called an exocrine function). The pancreas is unusual because it has both endocrine and exocrine functions. Its endocrine function produces three hormones. Two of these hormones, insulin and glucagon, are central to the processing of sugars in the diet (carbohydrate metabolism or breakdown). The third hormone produced by the endocrine cells of the pancreas affects gastrointestinal functioning. This hormone is called vasoactive intestinal polypeptide (VIP). The pancreas's exocrine function produces a variety of digestive enzymes (trypsin, chymotrypsin, lipase, and amylase, among others). These enzymes are passed into the duodenum through a channel called the pancreatic duct. In the duodenum, the enzymes

begin the process of breaking down a variety of food components, including, proteins, fats, and starches.

Acute pancreatitis occurs when the pancreas suddenly becomes inflamed but improves. Patients recover fully from the disease, and in almost 90% of cases the symptoms disappear within about a week after treatment. The pancreas returns to its normal architecture and functioning after healing from the illness. After an attack of acute pancreatitis, tissue and cells of the pancreas return to normal. With chronic pancreatitis, damage to the pancreas occurs slowly over time. Symptoms may be persistent or sporadic, but the condition does not disappear and the pancreas is permanently impaired. Pancreatic tissue is damaged, and the tissue and cells function poorly.

Causes and symptoms

There are a number of causes of acute pancreatitis. The most common, however, are gallbladder disease and **alcoholism**. These two diseases are responsible for more than 80% of all hospitalizations for acute pancreatitis. Other factors in the development of pancreatitis include:

- certain drugs
- infections
- structural problems of the pancreatic duct and bile ducts (channels leading from the gallbladder to the duodenum)
- injury to the abdomen resulting in injury to the pancreas (including injuries occurring during surgery)
- abnormally high levels of circulating fats in the bloodstream
- malfunction of the parathyroid gland, with high blood levels of calcium
- complications from kidney transplants
- a hereditary tendency toward pancreatitis.

Pancreatitis caused by drugs accounts for about 5% of all cases. Some drugs that are definitely related to pancreatitis include:

- Azathioprine, 6-mercaptopurine (Imuran)
- Dideoxyinosine (Videx)
- Estrogens (birth control pills)
- Furosemide (Lasix)
- Pentamidine (NebuPent)
- Sulfonamides (Urobak, Azulfidine)
- Tetracycline

- Thiazide **diuretics** (Diuril, Enduron)
- Valproic acid (Depakote).

Some drugs that are probably related to pancreatitis include:

- Acetaminophen (Tylenol)
- Angiotensin-converting enzyme (ACE) inhibitors (Capoten, Vasotec)
- Erythromycin
- Methyldopa (Aldomet)
- Metronidazole (Flagyl, Protostat)
- Nitrofurantoin (Furadantin, Furan)
- **Nonsteroidal anti-inflammatory drugs** (NSAIDs) (Aleve, Naprosyn, Motrin)
- Salicylates (aspirin).

All of these causes of pancreatitis seem to have a similar mechanism in common. Under normal circumstances, many of the extremely potent enzymes produced by the pancreas are not active until they are passed into the duodenum, where contact with certain other chemicals allow them to function. In pancreatitis, something allows these enzymes to become prematurely activated, so that they actually begin their digestive functions within the pancreas. The pancreas, in essence, begins digesting itself. A cycle of inflammation begins, including swelling and loss of function. Digestion of the blood vessels in the pancreas results in bleeding. Other active pancreatic chemicals cause blood vessels to become leaky, and fluid begins leaking out of the normal circulation into the abdominal cavity. The activated enzymes also gain access to the bloodstream through leaky, eroded blood vessels, and begin circulating throughout the body.

Pain is a major symptom in pancreatitis. The pain is usually quite intense and steady, located in the upper right hand corner of the abdomen, and often described as “boring.” This pain is also often felt all the way through to the patient’s back. The patient’s breathing may become quite shallow because deeper breathing tends to cause more pain. Relief of pain by sitting up and bending forward is characteristic of pancreatic pain. **Nausea and vomiting**, and abdominal swelling are all common as well. A patient will often have a slight **fever**, with an increased heart rate and low blood pressure.

Classic signs of **shock** may appear in more severely ill patients. Shock is a very serious syndrome that occurs when the volume (quantity) of fluid in the blood is very low. In shock, a patient’s arms and legs become extremely cold, the blood pressure drops dangerously low,

the heart rate is quite fast, and the patient may begin to experience changes in mental status.

In very severe cases of pancreatitis (called necrotizing pancreatitis), the pancreatic tissue begins to die, and bleeding increases. Due to the bleeding into the abdomen, two distinctive signs may be noted in patients with necrotizing pancreatitis. Turner’s sign is a reddish-purple or greenish-brown color to the flank area (the area between the ribs and the hip bone). Cullen’s sign is a bluish color around the navel.

Some of the complications of pancreatitis are due to shock. When shock occurs, all of the body’s major organs are deprived of blood (and, therefore, oxygen), resulting in damage. Kidney, respiratory, and **heart failure** are serious risks of shock. The pancreatic enzymes that have begun circulating throughout the body (as well as various poisons created by the abnormal digestion of the pancreas by those enzymes) have severe effects on the major body systems. Any number of complications can occur, including damage to the heart, lungs, kidneys, lining of the gastrointestinal tract, liver, eyes, bones, and skin. As the pancreatic enzymes work on blood vessels surrounding the pancreas, and even blood vessels located at a distance, the risk of **blood clots** increases. These blood clots complicate the situation by blocking blood flow in the vessels. When blood flow is blocked, the supply of oxygen is decreased to various organs and the organ can be damaged.

The pancreas may develop additional problems, even after the pancreatitis decreases. When the entire organ becomes swollen and suffers extensive cell death (pancreatic necrosis), the pancreas becomes extremely susceptible to serious infection. A local collection of pus (called a pancreatic **abscess**) may develop several weeks after the illness subsides, and may result in increased fever and a return of pain. Another late complication of pancreatitis, occurring several weeks after the illness begins, is called a pancreatic pseudocyst. This occurs when dead pancreatic tissue, blood, white blood cells, enzymes, and fluid leaked from the circulatory system accumulate. In an attempt to enclose and organize this abnormal accumulation, a kind of wall forms from the dead tissue and the growing scar tissue in the area. Pseudocysts cause additional abdominal pain by putting pressure on and displacing pancreatic tissue (resulting in more pancreatic damage). Pseudocysts also press on other nearby structures in the gastrointestinal tract, causing more disruption of function. Pseudocysts are life-threatening when they become infected (abscess) and when they rupture. Simple rupture of a pseudocyst causes death 14% of the time. Rupture complicated by bleeding causes death 60% of the time.

As the pancreatic tissue is increasingly destroyed in chronic pancreatitis, many digestive functions become disturbed. The quantity of hormones and enzymes normally produced by the pancreas begins to seriously decrease. Decreases in the production of enzymes result in the inability to appropriately digest food. Fat digestion, in particular, is impaired. A patient's stools become greasy as fats are passed out of the body. The inability to digest and use proteins results in smaller muscles (wasting) and weakness. The inability to digest and use the nutrients in food leads to **malnutrition**, and a generally weakened condition. As the disease progresses, permanent injury to the pancreas can lead to diabetes.

Diagnosis

Diagnosis of pancreatitis can be made very early in the disease by noting high levels of pancreatic enzymes circulating in the blood (amylase and lipase). Later in the disease, and in chronic pancreatitis, these enzyme levels will no longer be elevated. Because of this fact, and because increased amylase and lipase can also occur in other diseases, the discovery of such elevations are helpful but not mandatory in the diagnosis of pancreatitis. Other abnormalities in the blood may also point to pancreatitis, including increased white blood cells (occurring with inflammation and/or infection), changes due to **dehydration** from fluid loss, and abnormalities in the blood concentration of calcium, magnesium, sodium, potassium, bicarbonate, and sugars.

X rays or ultrasound examination of the abdomen may reveal **gallstones**, perhaps responsible for blocking the pancreatic duct. The gastrointestinal tract will show signs of inactivity (**ileus**) due to the presence of pancreatitis. Chest x rays may reveal abnormalities due to air trapping from shallow breathing, or due to lung complications from the circulating pancreatic enzyme irritants. **Computed tomography scans** (CT scans) of the abdomen may reveal the inflammation and fluid accumulation of pancreatitis, and may also be useful when complications like an abscess or a pseudocyst are suspected.

In the case of chronic pancreatitis, a number of blood tests will reveal the loss of pancreatic function that occurs over time. Blood sugar (glucose) levels will rise, eventually reaching the levels present in diabetes. The levels of various pancreatic enzymes will fall, as the organ is increasingly destroyed and replaced by non-functioning scar tissue. Calcification of the pancreas can also be seen on x rays. Endoscopic retrograde cholangiopancreatography (ERCP) may be

used to diagnose chronic pancreatitis in severe cases. In this procedure, the doctor uses a medical instrument fitted with a fiber-optic camera to inspect the pancreas. A magnified image of the area is shown on a television screen viewed by the doctor. Many endoscopes also allow the doctor to retrieve a small sample (biopsy) of pancreatic tissue to examine under a microscope. A contrast product may also be used for radiographic examination of the area.

Treatment

Treatment of pancreatitis involves quickly and sufficiently replacing lost fluids by giving the patient new fluids through a needle inserted in a vein (intravenous or IV fluids). These IV solutions need to contain appropriate amounts of salts, sugars, and sometimes even proteins, in order to correct the patient's disturbances in blood chemistry. Pain is treated with a variety of medications. In order to decrease pancreatic function (and decrease the discharge of more potentially harmful enzymes into the bloodstream), the patient is not allowed to eat. A thin, flexible tube (nasogastric tube) may be inserted through the patient's nose and down into his or her stomach. The nasogastric tube can empty the stomach of fluid and air, which may accumulate due to the inactivity of the gastrointestinal tract. Oxygen may need to be administered by nasal prongs or by a mask.

The patient will need careful monitoring in order to identify complications that may develop. Infections (often occurring in cases of necrotizing pancreatitis, abscesses, and pseudocysts) will require **antibiotics** through the IV. Severe necrotizing pancreatitis may require surgery to remove part of the dying pancreas. A pancreatic abscess can be drained by a needle inserted through the abdomen and into the collection of pus (percutaneous needle aspiration). If this is not sufficient, an abscess may also require surgical removal. Pancreatic pseudocysts may shrink on their own (in 25–40% of cases) or may continue to expand, requiring needle aspiration or surgery. When diagnostic exams reveal the presence of gallstones, surgery may be necessary for their removal. When a patient is extremely ill from pancreatitis, however, such surgery may need to be delayed until any infection is treated, and the patient's condition stabilizes.

Because chronic pancreatitis often includes repeated flares of acute pancreatitis, the same kinds of basic treatment are necessary. Patients cannot take solids or fluids by mouth. They receive IV replacement fluids, receive pain medication, and are monitored for complications. Treatment of chronic pancreatitis

caused by alcohol consumption requires that the patient stop drinking alcohol entirely. As chronic pancreatitis continues and insulin levels drop, a patient may require insulin injections in order to be able to process sugars in his or her diet. Pancreatic enzymes can be replaced with oral medicines, and patients sometimes have to take as many as eight pills with each meal. As the pancreas is progressively destroyed, some patients stop feeling the abdominal pain that was initially so severe. Others continue to have constant abdominal pain, and may even require a surgical procedure for relief. Drugs can be used to reduce the pain, but when narcotics are used for pain relief there is danger of the patient becoming addicted.

Prognosis

A number of systems have been developed to help determine the prognosis of an individual with pancreatitis. A very basic evaluation of a patient will allow some prediction to be made based on the presence of dying pancreatic tissue (necrosis) and bleeding. When necrosis and bleeding are present, as many as 50% of patients may die.

More elaborate systems have been created to help determine the prognosis of patients with pancreatitis. The most commonly used system identifies 11 different signs (Ranson's signs) that can be used to determine the severity of the disease. The first five categories are evaluated when the patient is admitted to the hospital:

- age over 55 years
- blood sugar level over 200 mg/Dl
- serum lactic dehydrogenase over 350 IU/L (increased with increased breakdown of blood, as would occur with internal bleeding, and with heart or liver damage)
- AST over 250 mu (a measure of liver function, as well as a gauge of damage to the heart, muscle, brain, and kidney)
- white **blood count** over 16,000 u L

The next six of Ranson's signs are reviewed 48 hours after admission to the hospital. These are:

- greater than 10% decrease in **hematocrit** (a measure of red blood cell volume)
- increase in BUN greater than 5 mg/dL (blood urea nitrogen, an indicator of kidney function)
- blood calcium less than 8 mg/dL
- PaO₂ less than 60 mm Hg (a measure of oxygen in the blood)
- base deficit greater than 4 mEq/L (a measure of change in the normal acidity of the blood)

KEY TERMS

Abscess—A pocket of infection; pus.

Acute—Of short and sharp course. Illnesses that are acute appear quickly and can be serious or life-threatening. The illness ends and the patient usually recovers fully.

Chronic—Of long duration and slow progression. Illnesses that are chronic develop slowly over time, and do not end. Symptoms may be continual or intermittent, but the patient usually has the condition for life.

Diabetes—A disease characterized by an inability to process sugars in the diet, due to a decrease in or total absence of insulin production. May require injections of insulin before meals to aid in the metabolism of sugars.

Duodenum—The first section of the small intestine that receives partly digested material from the stomach.

Endocrine—A system of organs that produces chemicals that go into the bloodstream to reach other organs whose functioning they affect.

Enzyme—A chemical that speeds up or makes a particular chemical reaction more efficient. In the digestive system, enzymes are involved in breaking down large food molecules into smaller molecules that can be processed and utilized by the body.

Exocrine—A system of organs that produces chemicals that go through a duct (or tube) to reach other organs whose functioning they affect.

Gland—Collections of tissue that produce chemicals needed for chemical reactions elsewhere in the body.

Hormone—A chemical produced in one part of the body that travels to another part of the body in order to exert an effect.

- fluid sequestration greater than 6 L (an estimation of the quantity of fluid that has leaked out of the blood circulation and into other body spaces)

Once a doctor determines how many of Ranson's signs are present and gives the patient a score, the doctor can better predict the risk of death. The more signs present, the greater the chance of fatal complications. A patient with less than three positive Ranson's signs has a 95% survival rate. A patient with three to four positive Ranson's signs has a 80-85% survival rate.

The results of a CT scan can also be used to predict the severity of pancreatitis. Slight swelling of the pancreas indicates mild illness. Significant swelling, especially with evidence of destruction of the pancreas and/or fluid build-up in the abdominal cavity, indicates more severe illness. With severe illness, there is a worse prognosis.

Prevention

Alcoholism is essentially the only preventable cause of pancreatitis. Patients with chronic pancreatitis must stop drinking alcohol entirely. The drugs that cause or may cause pancreatitis should also be avoided.

Resources

ORGANIZATIONS

National Digestive Diseases Information Clearinghouse. 2 Information Way, Bethesda, MD 20892-3570. (800) 891-5389. <<http://www.niddk.nih.gov/health/digest/nddic.htm>>.

Rosalyn Carson-DeWitt, MD

Panic attack see **Panic disorder**

Panic disorder

Definition

A panic attack is a sudden, intense experience of fear coupled with an overwhelming feeling of danger, accompanied by physical symptoms of anxiety, such as a pounding heart, sweating, and rapid breathing. A person with panic disorder may have repeated panic attacks (at least several a month) and feel severe **anxiety** about having another attack.

Description

Each year, panic disorder affects one out of 63 Americans. While many people experience moments of anxiety, panic attacks are sudden and unprovoked, having little to do with real danger.

Panic disorder is a chronic, debilitating condition that can have a devastating impact on a person's family, work, and social life. Typically, the first attack strikes without warning. A person might be walking down the street, driving a car, or riding an escalator when suddenly panic strikes. Pounding heart, sweating palms, and an overwhelming feeling of impending doom are common features. While the attack may last

only seconds or minutes, the experience can be profoundly disturbing. A person who has had one panic attack typically worries that another one may occur at any time.

As the fear of future panic attacks deepens, the person begins to avoid situations in which panic occurred in the past. In severe cases of panic disorder, the victim refuses to leave the house for fear of having a panic attack. This fear of being in exposed places is often called **agoraphobia**.

People with untreated panic disorder may have problems getting to work or staying on the job. As the person's world narrows, untreated panic disorder can lead to depression, substance abuse, and in rare instances, **suicide**.

Causes and symptoms

Scientists are not sure what causes panic disorder, but they suspect the tendency to develop the condition can be inherited. Some experts think that people with panic disorder may have a hypersensitive nervous system that unnecessarily responds to nonexistent threats. Research suggests that people with panic disorder may not be able to make proper use of their body's normal stress-reducing chemicals.

People with panic disorder usually have their first panic attack in their 20s. Four or more of the following symptoms during panic attacks would indicate panic disorder if no medical, drug-related, neurologic, or other psychiatric disorder is found:

- pounding, skipping or palpitating heartbeat
- shortness of breath or the sensation of smothering
- dizziness or lightheadedness
- nausea or stomach problems
- chest pains or pressure
- choking sensation or a "lump in the throat"
- chills or hot flashes
- sweating
- fear of dying
- feelings of unreality or being detached
- tingling or **numbness**
- shaking and trembling
- fear of losing control or going crazy

A panic attack is often accompanied by the urge to escape, together with a feeling of certainty that **death** is imminent. Others are convinced they are about to have a **heart attack**, suffocate, lose control,

or “go crazy.” Once people experience a panic attack, they tend to worry so much about having another attack that they avoid the place or situation associated with the original episode.

Diagnosis

Because its physical symptoms are easily confused with other conditions, panic disorder often goes undiagnosed. A thorough **physical examination** is needed to rule out a medical condition. Because the physical symptoms are so pronounced and frightening, panic attacks can be mistaken for a heart problem. Some people experiencing a panic attack go to an emergency room and endure batteries of tests until a diagnosis is made.

Once a medical condition is ruled out, a mental health professional is the best person to diagnose panic attack and panic disorder, taking into account not just the actual episodes, but how the patient feels about the attacks, and how they affect everyday life.

Most health insurance policies include some limited amount of mental health coverage, although few completely cover outpatient mental health care.

Treatment

Most patients with panic disorder respond best to a combination of **cognitive-behavioral therapy** and medication. Cognitive-behavioral therapy usually runs from 12–15 sessions. It teaches patients:

- how to identify and alter thought patterns so as not to misconstrue bodily sensations, events, or situations as catastrophic
- how to prepare for the situations and physical symptoms that trigger a panic attack
- how to identify and change unrealistic self-talk (such as “I’m going to die!”) that can worsen a panic attack
- how to calm down and learn breathing exercises to counteract the physical symptoms of panic
- how to gradually confront the frightening situation step by step until it becomes less terrifying
- how to “desensitize” themselves to their own physical sensations, such as rapid heart rate

At the same time, many people find that medications can help reduce or prevent panic attacks by changing the way certain chemicals interact in the brain. People with panic disorder usually notice whether or not the drug is effective within two months,

but most people take medication for at least six months to a year.

Several kinds of drugs can reduce or prevent panic attacks, including:

- selective serotonin reuptake inhibitor (**SSRI**) antidepressants like paroxetine (Paxil) or fluoxetine (Prozac), are approved specifically for the treatment of panic
- tricyclic antidepressants such as clomipramine (Anafranil)
- **benzodiazepines** such as alprazolam (Xanax) and clonazepam (Klonopin)

Finally, patients can make certain lifestyle changes to help keep panic at bay, such as eliminating **caffeine** and alcohol, **cocaine**, amphetamines, and **marijuana**.

Alternative treatment

One approach used in several medical centers focuses on teaching patients how to accept their fear instead of dreading it. In this method, the therapist repeatedly stimulates a person’s body sensations (such as a pounding heartbeat) that can trigger fear. Eventually, the patient gets used to these sensations and learns not to be afraid of them. Patients who respond report almost complete absence of panic attacks.

A variety of other alternative therapies may be helpful in treating panic attacks. Neurolinguistic programming and **hypnotherapy** can be beneficial, since these techniques can help bring an awareness of the root cause of the attacks to the conscious mind. Herbal remedies, including lemon balm (*Melissa officinalis*), oat straw (*Avena sativa*), passionflower (*Passiflora incarnata*), and skullcap (*Scutellaria lateriflora*), may help significantly by strengthening the nervous system. Homeopathic medicine, nutritional supplementation (especially with B vitamins, magnesium, and antioxidant **vitamins**), creative visualization, guided imagery, and relaxation techniques may help some people experiencing panic attacks. Hydrotherapies, especially hot epsom salt baths or baths with essential oil of lavender (*Lavandula officinalis*), can help patients relax.

Prognosis

While there may be occasional periods of improvement, the episodes of panic rarely disappear on their own. Fortunately, panic disorder responds very well to treatment; panic attacks decrease in up

KEY TERMS

Agoraphobia—Fear of open spaces.

Benzodiazepines—A class of drugs that have a hypnotic and sedative action, used mainly as tranquilizers to control symptoms of anxiety or panic.

Cognitive-behavioral therapy—A type of psychotherapy used to treat anxiety disorders (including panic disorder) that emphasizes behavioral change together with alteration of negative thought patterns.

Selective serotonin reuptake inhibitors (SSRIs)—A class of antidepressants used to treat panic that affects mood by boosting the levels of the brain chemical serotonin.

Tricyclic antidepressants—A class of antidepressants named for their three-ring structure that increase the levels of serotonin and other brain chemicals. They are used to treat depression and anxiety disorders, but have more side effects than the newer class of antidepressants called SSRIs.

to 90% of people after 6-8 weeks of a combination of cognitive-behavioral therapy and medication.

Unfortunately, many people with panic disorder never get the help they need. If untreated, panic disorder can last for years and may become so severe that a normal life is impossible. Many people who struggle with untreated panic disorder and try to hide their symptoms end up losing their friends, family, and jobs.

Prevention

There is no way to prevent the initial onset of panic attacks. **Antidepressant drugs** or benzodiazepines can prevent future panic attacks, especially when combined with cognitive-behavioral therapy. There is some suggestion that avoiding stimulants (including caffeine, alcohol, or over-the-counter cold medicines) may help prevent attacks as well.

Resources

ORGANIZATIONS

American Psychiatric Association. 1400 K Street NW, Washington, DC 20005. (888) 357-7924. <<http://www.psych.org>>.

Anxiety Disorders Association of America. 11900 Park Lawn Drive, Ste. 100, Rockville, MD 20852. (800) 545-7367. <<http://www.adaa.org>>.

Freedom From Fear. 308 Seaview Ave., Staten Island, NY 10305. (718) 351-1717.

National Alliance for the Mentally Ill (NAMI). Colonial Place Three, 2107 Wilson Blvd., Ste. 300, Arlington, VA 22201-3042. (800) 950-6264. <<http://www.nami.org>>.

National Anxiety Foundation. 3135 Custer Dr., Lexington, KY 40517. (606) 272-7166. <<http://www.lexington-online.com/naf.html>>.

National Institute of Mental Health, Panic Campaign. Rm 15C-05, 5600 Fishers Lane, Rockville, MD 20857. (800) 647-2642. <<http://www.nimh.nih.gov>>.

National Mental Health Association. 1021 Prince St., Alexandria, VA 22314. (703) 684-7722. <<http://www.nmha.org>>.

OTHER

The Anxiety and Panic Internet Resource. <<http://www.algy.com/anxiety>>.

Anxiety Network Page. <<http://www.anxietynetwork.com>>.

National Institute of Mental Health Page. <<http://www.nimh.nih.gov>>.

“Panic Disorder.” Internet Mental Health Page. <<http://www.mentalhealth.com>>.

Carol A. Turkington

Pap test

Definition

The Pap test is a procedure in which a physician scrapes cells from the cervix or vagina to check for **cervical cancer**, vaginal **cancer**, or abnormal changes that could lead to cancer. It often is called a “Pap smear.”

Purpose

The Pap test is used to detect abnormal growth of cervical cells at an early stage so that treatment can be started when the condition is easiest to treat. This microscopic analysis of cells can detect cervical cancer, precancerous changes, inflammation (vaginitis), infections, and some **sexually transmitted diseases** (STDs). The Pap test can occasionally detect endometrial (uterine) cancer or **ovarian cancer**, although it was not designed for this purpose.

Women should begin to have Pap tests at the age of 21 or within three years of becoming sexually active, whichever comes first. Young people are more likely to have multiple sex partners, which increases their risk of certain diseases that can cause cancer, such as human papillomavirus (HPV).

The American Cancer Society (ACS) updated its guidelines concerning Pap test frequency in late 2002. In brief, women should continue screening every year with regular Pap tests until age 30, every two years if using the liquid-based Pap test. Once a woman age 30 and older has had three normal results in a row, she may get screened every two to three years. A doctor may suggest more frequent screening if a woman has certain risk factors for cervical cancer. Women who have had total hysterectomies including the removal of the cervix do not need Pap tests unless the **hysterectomy** resulted from cervical cancer. Those over age 70 who have had three normal results generally do not need to continue having Pap tests under the new guidelines.

Women with certain risk factors may have yearly tests. Those at highest risk for cervical cancer are women who started having sex before age 18, those with many sex partners (especially if they did not use condoms, which protect against STDs), those who have had STDs such as **genital herpes** or **genital warts**, and those who smoke. Women older than 40 may have the test yearly, if experiencing bleeding after **menopause**. Women who have had a positive test result in the past may need screening every six months. Women who have had cervical cancer or precancer should have regular Pap smears.

Other women also benefit from the Pap test. Women over age 65 account for 25% of all cases of cervical cancer and 41% of deaths from this disease. Women over age 65 who have never had a Pap smear benefit the most from the test. Some women have the cervix left in place after hysterectomy and will continue to receive regular Pap tests. Finally, a pregnant woman should have a Pap test as part of her first prenatal examination.

The Pap smear is a screening test. It identifies women who are at increased risk of cervical dysplasia (abnormal cells) or cervical cancer. Only an examination of the cervix with a special lighted instrument (**colposcopy**) and samples of cervical tissue (biopsies) can actually diagnose these problems.

Precautions

The Pap test is usually not done during the menstrual period because of the presence of blood cells. The best time is in the middle of the menstrual cycle.

Description

The Pap test is an extremely cost-effective and beneficial exam. Cervical cancer used to be a leading

cause of cancer deaths in American women, but widespread use of this diagnostic procedure reduced the **death** rate from this disease by 74% between 1955 and 1992. A 2003 study reported that the test reduces rates of invasive cervical cancer by as much as 94%. In 2003, the FDA approved a new screening test that combines DNA testing for the HPV type that causes the most cases of cervical cancer with the standard Pap test, increasing its screening value.

The Pap test, sometimes called a cervical smear, is the microscopic examination of cells scraped from both the outer cervix and the cervical canal. (The cervix is the opening between the vagina and the uterus, or womb.) It is called the “Pap” test after its developer, Dr. George N. Papanicolaou. This simple procedure is performed during a gynecologic examination and is usually covered by insurance. For those with coverage, Medicare will pay for one screening Pap smear every three years.

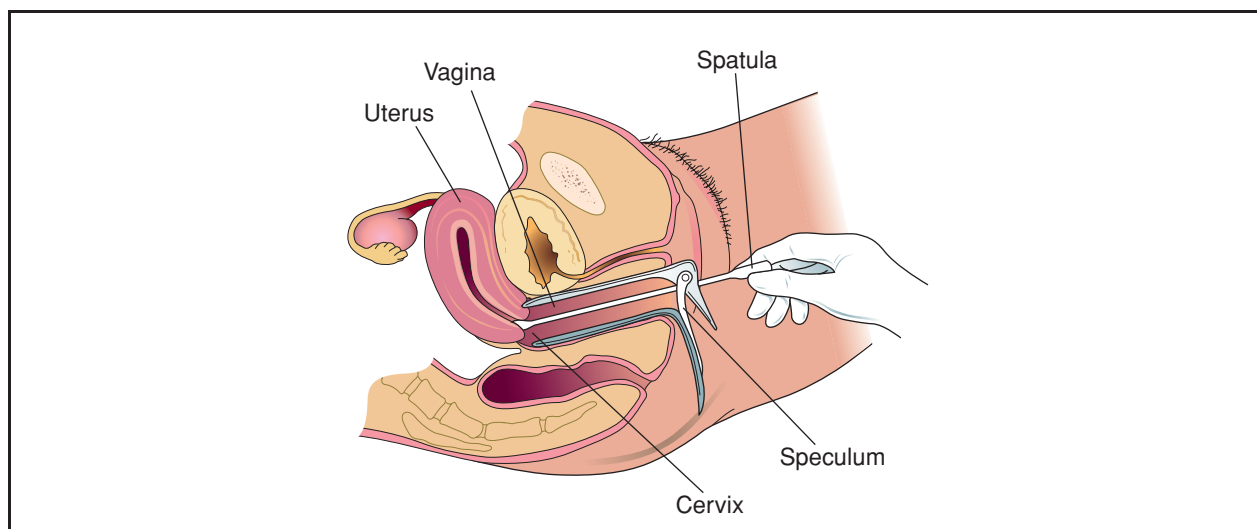
During the pelvic examination, an instrument called a speculum is inserted into the vagina to open it. The doctor then uses a tiny brush, or a cotton-tipped swab and a small spatula to wipe loose cells off the cervix and to scrape them from the inside of the cervix. The cells are transferred or “smeared” onto glass slides, the slides are treated to stabilize the cells, and the slides are sent to a laboratory for microscopic examination. The entire procedure is usually painless and takes five to 10 minutes at most.

The newer method called liquid-based cytology, or the liquid-based Pap test, involves spreading the cells more evenly on a slide after removing them from the sample. The liquid-based method prevents cells from drying out and becoming distorted. Studies show that liquid-based testing slightly improves cancer detection and greatly improves detection of precancers, but it costs more than the traditional Pap test. Trade names in 2003 for liquid-based Pap smears were ThinPrep and AutoCyte.

Preparation

The Pap test may show abnormal results when a woman is healthy or normal results in women with cervical abnormalities as much as 25% of the time. It may even miss up to 5% of cervical cancers. Some simple preparations may help to ensure that the results are reliable. Among the measures that may help increase test reliability are:

- avoiding sexual intercourse for two days before the test
- not using douches for two or three days before the test



The Pap test is a procedure used to detect abnormal growth of cervical cells which may be a precursor to cancer of the cervix. It is administered by a physician who inserts a speculum into the vagina to open and separate the vaginal walls. A spatula is then inserted to scrape cells from the cervix. These cells are transferred onto glass slides for laboratory analysis. The Pap test may also identify vaginitis, some sexually transmitted diseases, and cancers of the uterus and ovaries. (Illustration by Electronic Illustrators Group.)

- avoiding tampons, vaginal creams, or birth control foams or jellies for two to three days before the test
- scheduling the Pap smear when not menstruating.

However, most women are not routinely advised to make any special preparations for a Pap test.

If possible, women may want to ensure that their test is performed by an experienced gynecologist, physician, or provider and sent to a reputable laboratory. The physician should be confident in the accuracy of the chosen lab.

Before the exam, the physician will take a complete sexual history to determine a woman's risk status for cervical cancer. Questions may include date and results of the last Pap test, any history of abnormal Pap tests, date of last menstrual period and any irregularity, use of hormones and birth control, family history of gynecologic disorders, and any vaginal symptoms. These topics are relevant to the interpretation of the Pap test, especially if any abnormalities are detected. Immediately before the Pap test, the woman should empty her bladder to avoid discomfort during the procedure.

Aftercare

Harmless cervical bleeding is possible immediately after the test; a woman may need to use a sanitary napkin. She should also be sure to comply with her doctor's orders for follow-up visits.

Risks

No appreciable health risks are associated with the Pap test. However, abnormal results (whether valid or due to technical error) can cause significant **anxiety**. Women may wish to have their sample double-checked, either by the same laboratory or by the new technique of computer-assisted rescreeing. The Food and Drug Administration (FDA) has approved the use of AutoPap and PAPNET to doublecheck samples that have been examined by technologists. AutoPap may also be used to perform initial screening of slides, which are then checked by a technologist. Any abnormal Pap test should be followed by colposcopy, not by double checking the Pap test.

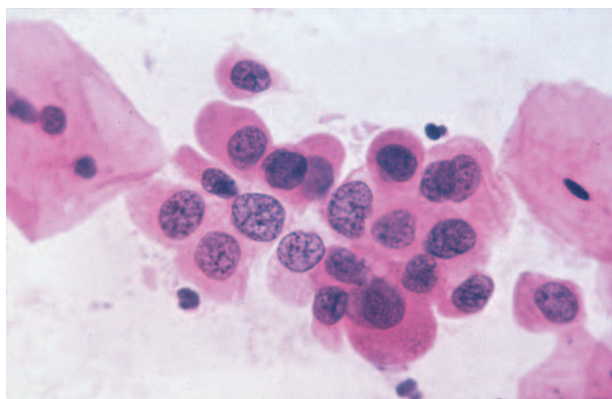
Normal results

Normal (negative) results from the laboratory exam mean that no atypical, dysplastic, or cancer cells were detected, and the cervix is normal.

Abnormal results

Terminology

Abnormal cells found on the Pap test may be described using two different grading systems. Although this can be confusing, the systems are quite similar. The Bethesda system is based on the term "squamous intraepithelial lesion" (SIL). Precancerous cells are classified as atypical squamous cells of



These malignant cells were taken from a woman's cervix during a Pap test. (Photograph by Parviz M. Pour, Photo Researchers, Inc. Reproduced by permission.)

undetermined significance, low-grade SIL, or high-grade SIL. Low-grade SIL includes mild dysplasia (abnormal cell growth) and abnormalities caused by HPV; high-grade SIL includes moderate or severe dysplasia and carcinoma in situ (cancer that has not spread beyond the cervix).

Another term that may be used is “cervical intraepithelial neoplasia” (CIN). In this classification system, mild dysplasia is called CIN I, moderate is CIN II, and severe dysplasia or carcinoma in situ is CIN III.

Regardless of terminology, it is important to remember that an abnormal (positive) result does not necessarily indicate cancer. Results may be falsely abnormal after infection or irritation of the cervix. Up to 40% of mild dysplasia reverts to normal tissue without treatment, and only 1% of mild abnormalities ever develop into cancer.

Changes of unknown cause

ASCUS or LSIL cells are found in 5%–10% of all Pap tests. The most common abnormality is atypical squamous cells of undetermined significance, which are found in 4% of all Pap tests. Sometimes these results are described further as either reactive or precancerous. Reactive changes suggest that the cervical cells are responding to inflammation, such as from a yeast infection. These women may be treated for infection and then undergo repeat Pap testing in three to six months. If those results are negative, no further treatment is necessary. This category may also include atypical “glandular” cells, which could imply a more severe type of cancer and requires repeat testing and further evaluation.

Dysplasia

The next most common finding (in about 25 of every 1,000 tests) is low-grade SIL, which includes mild dysplasia or CIN I and changes caused by HPV. Unlike cancer cells, these cells do not invade normal tissues. Women are most susceptible to cervical dysplasia between the ages of 25 and 35. Typically, dysplasia causes no symptoms, although women may experience abnormal vaginal bleeding. Because dysplasia is precancerous, it should be treated if it is moderate or severe.

Treatment of dysplasia depends on the degree of abnormality. In women with no other risk factors for cervical cancer, mild precancerous changes may be simply observed over time with repeat testing, perhaps every four to six months. This strategy works only if women are diligent about keeping later appointments. Premalignant cells may remain that way without causing cancer for five to ten years, and may never become malignant.

In women with positive results or risk factors, the gynecologist must perform colposcopy and biopsy. A colposcope is an instrument that looks like binoculars, with a light and a magnifier, used to view the cervix. Biopsy, or removal of a small piece of abnormal cervical or vaginal tissue for analysis, is usually done at the same time.

High-grade SIL (found in three of every 50 Pap tests) includes moderate to severe dysplasia or carcinoma in situ (CIN II or III). After confirmation by colposcopy and biopsy, it must be removed or destroyed to prevent further growth. Several outpatient techniques are available: conization (removal of a cone-shaped piece of tissue), **laser surgery**, **cryotherapy** (freezing), or the “loop electrosurgical excision procedure.” Cure rates are nearly 100% after prompt and appropriate treatment of carcinoma in situ. Of course, frequent checkups are then necessary.

Cancer

HPV, the most common STD in the United States, may be responsible for many cervical cancers. Cancer may be manifested by unusual vaginal bleeding or discharge, bowel and bladder problems, and **pain**. Women are at greatest risk of developing cervical cancer between the ages of 30 and 40 and between the ages of 50 and 60. Most new cancers are diagnosed in women between 50 and 55. Although the likelihood of developing this disease begins to level off for Caucasian women at the age of 45, it increases steadily for African Americans for another

KEY TERMS

Carcinoma in situ—Malignant cells that are present only in the outer layer of the cervix.

Cervical intraepithelial neoplasia (CIN)—A term used to categorize degrees of dysplasia arising in the epithelium, or outer layer, of the cervix.

Dysplasia—Abnormal changes in cells.

Human papillomavirus (HPV)—The most common STD in the United States. Various types of HPV are known to cause cancer.

Neoplasia—Abnormal growth of cells, which may lead to a neoplasm, or tumor.

Squamous intraepithelial lesion (SIL)—A term used to categorize the severity of abnormal changes arising in the squamous, or outermost, layer of the cervix.

40 years. Biopsy is indicated when any abnormal growth is found on the cervix, even if the Pap test is negative.

Doctors have traditionally used **radiation therapy** and surgery to treat cervical cancer that has spread within the cervix or throughout the pelvis. In severe cases, postoperative radiation is administered to kill any remaining cancer cells, and **chemotherapy** may be used if cancer has spread to other organs. Recent studies have shown that giving chemotherapy and radiation at the same time improves a patient's chance of survival. The National Cancer Institute has urged physicians to strongly consider using both chemotherapy and radiation to treat patients with invasive cervical cancer. The survival rate at five years after treatment of early invasive cancer is 91%; rates are below 70% for more severe invasive cancer. That is why prevention, risk reduction, and frequent Pap tests are the best defense for a woman's gynecologic health.

The Pap test is a procedure used to detect abnormal growth of cervical cells which may be a precursor to cancer of the cervix. It is administered by a physician who inserts a speculum into the vagina to open and separate the vaginal walls. A spatula is then inserted to scrape cells from the cervix. These cells are transferred onto glass slides for laboratory analysis. The Pap test may also identify vaginitis, some sexually transmitted diseases, and cancers of the uterus and ovaries.

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Teresa G. Odle

Papanicolaou test see **PAP test**

Papilledema

Definition

Papilledema is a swelling of the optic nerve, at the point where this nerve joins the eye, that is caused by an increase in fluid pressure within the skull (intracranial pressure). Swelling of the optic nerve due to other causes such as infection or inflammatory disease is not called papilledema.

Description

The optic nerve is the nerve that transmits signals from the eye to the brain. Papilledema is a swelling of this nerve where it meets the eye (the optic disc) caused by an increase in intracranial pressure. Almost all cases of papilledema are bilateral (affect both eyes). Papilledema can be observed in people of any age, but is relatively uncommon in infants because the bones of the skull are not fully fused together at this age.

Causes and symptoms

Papilledema is caused by an increase in the pressure of the fluid (cerebrospinal fluid) that is present between the brain and the skull, inside the head. This increase in intracranial pressure may be caused by any of a variety of conditions within the skull, brain, or spinal cord. The most common causes of papilledema are:

- tumor of the brain, spinal cord, skull, spinal column, or optic nerve
- abscess (the accumulation of pus within a confined space)
- craniosynostosis (an abnormal closure of the bones of the skull)
- hemorrhage (bleeding)
- hydrocephalus (an accumulation of cerebrospinal fluid within the skull)
- intracranial infection (any infection within the skull such as **meningitis** and encephalitis)
- head injury

The symptoms of papilledema include:

- headaches, which are usually worse upon awakening and exacerbated by coughing, holding the breath, or other maneuvers that tend to increase intracranial pressure.
- nausea and vomiting.
- changes in vision, such as temporary and transient blurring, graying, flickering, or double vision

Diagnosis

A diagnosis of papilledema is achieved by visual examination of the eye with an ophthalmoscope. This instrument shines light through the pupil of the eye and illuminates the retina while the clinician looks through it. Eye drops to dilate the pupils are used to insure a thorough examination.

Treatment

Treatment of papilledema is generally aimed at the treatment of the underlying disorder that is causing papilledema.

Diuretic drugs combined with a weight reduction program may be useful in cases of papilledema that are caused by an abnormally high production of cerebrospinal fluid.

Corticosteroids have been shown to be effective in relieving the symptoms in some patients with papilledema caused by inflammatory disorders.

Alternative treatment

Alternative treatments for conditions that cause the occurrence of papilledema include acupuncture, aromatherapy, hydrotherapy, massage, and herbal remedies.

KEY TERMS

Craniosynostosis—A premature closure of one or more of the joints (fissures) between the bones of the skull, which causes an abnormally shaped skull.

Hydrocephalus—The accumulation of cerebrospinal fluid within the skull.

Ophthalmoscope—A medical instrument which shines a light through the pupil of the patient's eye and illuminates the retina (back) of the eye, allowing a visual examination of the interior of the eye.

Prognosis

With prompt medical care to treat the underlying cause of papilledema, a person affected with papilledema will not have permanent damage to his or her eye-sight. However, prolonged papilledema can result in permanent damage to the optic nerve which could lead to blindness.

Prevention

Preventing papilledema is only possible if the underlying condition causing the papilledema can be found. Treatment of this underlying condition may prevent recurrences of papilledema.

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National Eye Institute. 2020 Vision Place, Bethesda, MD 20892-3655. (301) 496-5248. <<http://www.nei.nih.gov/>>.

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Paul A. Johnson, Ed.M.

Papillomavirus infection see **Genital warts**

Papule see **Skin lesions**

Paracentesis

Definition

Paracentesis is a procedure during which fluid from the abdomen is removed through a needle.

Purpose

There are two reasons to take fluid out of the abdomen. One is to analyze it. The other is to relieve pressure.

Liquid that accumulates in the abdomen is called ascites. Ascites seeps out of organs for several reasons related either to disease in the organ or fluid pressures that are changing.

Liver disease

All the blood flowing through the intestines passes through the liver on its way back to the heart. When progressive disease such as alcohol damage or hepatitis destroys enough liver tissue, the scarring that results shrinks the liver and constricts the blood flow. Such scarring of the liver is called **cirrhosis**. Pressure builds up in the intestinal circulation, slowing flow and pushing fluid into the tissues. Slowly the fluid accumulates in areas with the lowest pressure and greatest capacity. The free space around abdominal organs receives most of it. This space is called the peritoneal space because it is enclosed by a thin membrane called the peritoneum. The peritoneum wraps around nearly every organ in the abdomen, providing many folds and spaces for the fluid to gather.

Infections

Peritonitis is an infection of the peritoneum. Infection changes the dynamics of body fluids, causing them to seep into tissues and spaces. Peritonitis can develop in several ways. Many abdominal organs contain germs that do not belong elsewhere in the body. If they spill their contents into the peritoneum, infection is the result. The gall bladder, the stomach, any part of the intestine, and most especially the appendix—all cause peritonitis when they leak or rupture. **Tuberculosis** can infect many organs in the body; it is not confined to the lungs. Tuberculous peritonitis causes **ascites**.

Other inflammations

Peritoneal fluid is not just produced by infections. The pancreas can cause a massive sterile peritonitis when it leaks its digestive enzymes into the abdomen.

Cancer

Any **cancer** that begins in or spreads to the abdomen can leak fluid. One particular tumor of the ovary that leaks fluid, the resulting presentation of the disease, is Meigs' syndrome.

Kidney disease

Since the kidneys are intimately involved with the body's fluid balance, diseases of the kidney often cause excessive fluid to accumulate. Nephrosis and nephrotic syndrome are the general terms for diseases that cause the kidneys to retain water and provoke its movement into body tissues and spaces.

Heart failure

The ultimate source of fluid pressure in the body is the heart, which generates blood pressure. All other pressures in the body are related to blood pressure. As the heart starts to fail, blood backs up, waiting to be pumped. This increases back pressure upstream, particularly below the heart where gravity is also pulling blood away from the heart. The extra fluid from **heart failure** is first noticed in the feet and ankles, where gravitational effects are most potent. In the abdomen, the liver swells first, then it and other abdominal organs start to leak.

Pleural fluid

The other major body cavity is the chest. The tissue in the chest corresponding to the peritoneum is called the pleura, and the space contained within the pleura, between the ribs and the lungs, is called the pleural space. Fluid is often found in both cavities, and fluid from one cavity can find its way into the other.

Fluid that accumulates in the abdomen creates abnormal pressures on organs in the abdomen. Digestion is hindered; blood flow is slowed. Pressure upward on the chest compromises breathing. The kidneys function poorly in the presence of such external pressures and may even fail with tense, massive ascites.

Description

During paracentesis, special needles puncture the abdominal wall, being careful not to hit internal organs. If fluid is needed only for analysis, just a bit is removed. If pressure relief is an additional goal, many quarts may be removed. Rapid removal of large amounts of fluid can cause blood pressure to drop suddenly. For this reason, the physician will often leave a tube in place so that fluid can be removed slowly, giving the circulation time to adapt.

KEY TERMS

Ectopic pregnancy—A pregnancy occurring outside the womb that often ruptures and requires surgical removal.

A related procedure called culdocentesis removes ascitic fluid from the very bottom of the abdominal cavity through the back of the vagina. This is used mostly to diagnose female genital disorders like **ectopic pregnancy** that bleed or exude fluid into the peritoneal space.

Fluid is sent to the laboratory for testing, where cancer and blood cells can be detected, infections identified, and chemical analysis can direct further investigations.

Aftercare

An adhesive bandage and perhaps a single stitch close the hole. Nothing more is required.

Risks

Risks are negligible. It is remotely possible that an organ could be punctured and bleed or that an infection could be introduced.

Normal results

A diagnosis of the cause and/or relief from accumulated fluid pressure are the expected results.

Abnormal results

Fluid will continue to accumulate until the cause is corrected. Repeat procedures may be needed.

Resources

BOOKS

Glickman, Robert M. "Abdominal Swelling and Ascites." In *Harrison's Principles of Internal Medicine*, edited by Anthony S. Fauci, et al. New York: McGraw-Hill, 1997.

J. Ricker Polsdorfer, MD

Paracoccidioidomycosis see **South American blastomycosis**

Paragonamiasis see **Fluke infections**

Paralysis

Definition

Paralysis is defined as complete loss of strength in an affected limb or muscle group.

Description

The chain of nerve cells that runs from the brain through the spinal cord out to the muscle is called the motor pathway. Normal muscle function requires intact connections all along this motor pathway. Damage at any point reduces the brain's ability to control the muscle's movements. This reduced efficiency causes weakness, also called paresis. Complete loss of communication prevents any willed movement at all. This lack of control is called paralysis. Certain inherited abnormalities in muscle cause **periodic paralysis**, in which the weakness comes and goes.

The line between weakness and paralysis is not absolute. A condition causing weakness may progress to paralysis. On the other hand, strength may be restored to a paralyzed limb. Nerve regeneration or regrowth is one way in which strength can return to a paralyzed muscle. Paralysis almost always causes a change in muscle tone. Paralyzed muscle may be flaccid, flabby, and without appreciable tone, or it may be spastic, tight, and with abnormally high tone that increases when the muscle is moved.

Paralysis may affect an individual muscle, but it usually affects an entire body region. The distribution of weakness is an important clue to the location of the nerve damage that is causing the paralysis. Words describing the distribution of paralysis use the suffix "-plegia," from the Greek word for "stroke." The types of paralysis are classified by region:

- monoplegia, affecting only one limb
- diplegia, affecting the same body region on both sides of the body (both arms, for example, or both sides of the face)
- hemiplegia, affecting one side of the body
- paraplegia, affecting both legs and the trunk
- quadriplegia, affecting all four limbs and the trunk

Causes and symptoms

Causes

The nerve damage that causes paralysis may be in the brain or spinal cord (the central nervous system) or it may be in the nerves outside the spinal cord (the

peripheral nervous system). The most common causes of damage to the brain are:

- **stroke**
- tumor
- trauma (caused by a fall or a blow)
- Multiple sclerosis (a disease that destroys the protective sheath covering nerve cells)
- **cerebral palsy** (a condition caused by a defect or injury to the brain that occurs at or shortly after birth)
- metabolic disorder (a disorder that interferes with the body's ability to maintain itself)

Damage to the spinal cord is most often caused by trauma, such as a fall or a car crash. Other conditions that may damage nerves within or immediately adjacent to the spine include:

- tumor
- herniated disk (also called a ruptured or slipped disk)
- spondylosis (a disease that causes stiffness in the joints of the spine)
- rheumatoid arthritis of the spine
- neurodegenerative disease (a disease that damages nerve cells)
- multiple sclerosis

Damage to peripheral nerves may be caused by:

- trauma
- compression or entrapment (such as carpal tunnel syndrome)
- **Guillain-Barré syndrome** (a disease of the nerves that sometimes follows **fever** caused by a viral infection or immunization)
- chronic inflammatory demyelinating polyradiculoneuropathy (CIDP) (a condition that causes **pain** and swelling in the protective sheath covering nerve cells)
- radiation
- inherited demyelinating disease (a condition that destroys the protective sheath around the nerve cell)
- toxins or poisons

Symptoms

The distribution of paralysis offers important clues to the site of nerve damage. Hemiplegia is almost always caused by brain damage on the side opposite the paralysis, often from a stroke. Paraplegia occurs after injury to the lower spinal cord, and quadriplegia

occurs after damage to the upper spinal cord at the level of the shoulders or higher (the nerves controlling the arms leave the spine at that level). Diplegia usually indicates brain damage, most often from cerebral palsy. Monoplegia may be caused by isolated damage to either the central or the peripheral nervous system. Weakness or paralysis that occurs only in the arms and legs may indicate demyelinating disease. Fluctuating symptoms in different parts of the body may be caused by multiple sclerosis.

Sudden paralysis is most often caused by injury or stroke. Spreading paralysis may indicate degenerative disease, inflammatory disease such as Guillain-Barré syndrome or CIDP, metabolic disorders, or inherited demyelinating disease.

Other symptoms often accompany paralysis from any cause. These symptoms may include numbness and **tingling**, pain, changes in vision, difficulties with speech, or problems with balance. **Spinal cord injury** often causes loss of function in the bladder, bowel, and sexual organs. High spinal cord injuries may cause difficulties in breathing.

Diagnosis

Careful attention should be paid to any events in the patient's history that might reveal the cause of the paralysis. The examiner should look for incidents such as falls or other traumas, exposure to toxins, recent infections or surgery, unexplained **headache**, preexisting metabolic disease, and family history of weakness or other neurologic conditions. A neurologic examination tests strength, reflexes, and sensation in the affected area and normal areas.

Imaging studies, including **computed tomography scans** (CT scans), **magnetic resonance imaging** (MRI) scans, or **myelography** may reveal the site of the injury. **Electromyography** and nerve conduction velocity tests are performed to test the function of the muscles and peripheral nerves.

Treatment

The only treatment for paralysis is to treat its underlying cause. The loss of function caused by long-term paralysis can be treated through a comprehensive **rehabilitation** program. Rehabilitation includes:

- Physical therapy. The physical therapist focuses on mobility. Physical therapy helps develop strategies to compensate for paralysis by using those muscles that still have normal function, helps maintain and build any strength and control that remain in the affected muscles, and helps maintain range of motion in the

KEY TERMS

Computed tomography (CT)—An imaging technique in which cross-sectional x rays of the body are compiled to create a three-dimensional image of the body's internal structures.

Electromyography—A test that uses electrodes to record the electrical activity of muscle. The information gathered is used to diagnose neuromuscular disorders.

Magnetic resonance imaging (MRI)—An imaging technique that uses a large circular magnet and radio waves to generate signals from atoms in the body. These signals are used to construct images of internal structures.

Myelin—The insulation covering nerve cells. Demyelinating disease causes a breakdown of myelin.

Myelography—An x-ray process that uses a dye or contrast medium injected into the space around the spine.

Nerve conduction velocity test—A test that measures the time it takes a nerve impulse to travel a specific distance over the nerve after electronic stimulation.

affected limbs to prevent muscles from shortening (contracture) and becoming deformed. If nerve regrowth is expected, physical therapy is used to retrain affected limbs during recovery. A physical therapist also suggests adaptive equipment such as braces, canes, or wheelchairs.

- Occupational therapy. The occupational therapist focuses on daily activities such as eating and bathing. Occupational therapy develops special tools and techniques that permit self-care and suggests ways to modify the home and workplace so that a patient with an impairment may live a normal life.
- Other specialties. The nature of the impairment may mean that the patient needs the services of a respiratory therapist, vocational rehabilitation counselor, social worker, speech-language pathologist, nutritionist, special education teacher, recreation therapist, or clinical psychologist.

Prognosis

The likelihood of recovery from paralysis depends on what is causing it and how much damage has been done to the nervous system.

Prevention

Prevention of paralysis depends on prevention of the underlying causes. Risk of stroke can be reduced by controlling high blood pressure and cholesterol levels. Seatbelts, air bags, and helmets reduce the risk of injury from motor vehicle accidents and falls. Good prenatal care can help prevent premature birth, which is a common cause of cerebral palsy.

Resources

BOOKS

Bradley, Walter G., et al., editors. *Neurology in Clinical Practice*. 2nd ed. Boston: Butterworth-Heinemann, 1996.

Richard Robinson

Paralysis agitans see **Parkinson's disease**

Paralytic shellfish poisoning see **Fish and shellfish poisoning**

Paranoia

Definition

Paranoia is an unfounded or exaggerated distrust of others, sometimes reaching delusional proportions. Paranoid individuals constantly suspect the motives of those around them, and believe that certain individuals, or people in general, are “out to get them.”

Description

Paranoid perceptions and behavior may appear as features of a number of mental illnesses, including depression and **dementia**, but are most prominent in three types of psychological disorders: paranoid schizophrenia, delusional disorder (persecutory type), and paranoid personality disorder (PPD).

Individuals with paranoid **schizophrenia** and persecutory delusional disorder experience what is known as persecutory **delusions**: an irrational, yet unshakable, belief that someone is plotting against them. Persecutory delusions in paranoid schizophrenia are bizarre, sometimes grandiose, and often accompanied by auditory **hallucinations**. Delusions experienced by individuals with delusional disorder are more plausible than those experienced by paranoid schizophrenics; not bizarre, though still unjustified. Individuals with delusional disorder may seem offbeat

or quirky rather than mentally ill, and, as such, may never seek treatment.

Persons with paranoid personality disorder tend to be self-centered, self-important, defensive, and emotionally distant. Their paranoia manifests itself in constant suspicions rather than full-blown delusions. The disorder often impedes social and personal relationships and career advancement. Some individuals with PPD are described as “litigious,” as they are constantly initiating frivolous law suits. PPD is more common in men than in women, and typically begins in early adulthood.

Causes and symptoms

The exact cause of paranoia is unknown. Potential causal factors may be genetics, neurological abnormalities, changes in brain chemistry, and stress. Paranoia is also a possible side effect of drug use and **abuse** (for example, alcohol, marijuana, amphetamines, **cocaine**, PCP). Acute, or short term, paranoia may occur in some individuals overwhelmed by **stress**.

The *Diagnostic and Statistical Manual of Mental Disorders*, fourth edition (*DSM-IV*), the diagnostic standard for mental health professionals in the United States, lists the following symptoms for paranoid personality disorder:

- suspicious; unfounded suspicions; believes others are plotting against him/her
- preoccupied with unsupported doubts about friends or associates
- reluctant to confide in others due to a fear that information may be used against him/her
- reads negative meanings into innocuous remarks
- bears grudges
- perceives attacks on his/her reputation that are not clear to others, and is quick to counterattack
- maintains unfounded suspicions regarding the fidelity of a spouse or significant other

Diagnosis

Patients with paranoid symptoms should undergo a thorough physical examination and patient history to rule out possible organic causes (such as dementia) or environmental causes (such as extreme stress). If a psychological cause is suspected, a psychologist will conduct an interview with the patient and may administer one of several clinical inventories, or tests, to evaluate mental status.

KEY TERMS

Persecutory delusion—A fixed, false, and inflexible belief that others are engaging in a plot or plan to harm an individual.

Treatment

Paranoia that is symptomatic of paranoid schizophrenia, delusional disorder, or paranoid personality disorder should be treated by a psychologist and/or psychiatrist. Antipsychotic medication such as thioridazine (Mellaril), haloperidol (Haldol), chlorpromazine (Thorazine), clozapine (Clozaril), or risperidone (Risperdal) may be prescribed, and cognitive therapy or psychotherapy may be employed to help the patient cope with their paranoia and/or persecutory delusions. Antipsychotic medication, however, is of uncertain benefit to individuals with paranoid personality disorder and may pose long-term risks.

If an underlying condition, such as depression or drug abuse, is found to be triggering the paranoia, an appropriate course of medication and/or psychosocial therapy is employed to treat the primary disorder.

Prognosis

Because of the inherent mistrust felt by paranoid individuals, they often must be coerced into entering treatment. As unwilling participants, their recovery may be hampered by efforts to sabotage treatment (for example, not taking medication or not being forthcoming with a therapist), a lack of insight into their condition, or the belief that the therapist is plotting against them. Albeit with restricted lifestyles, some patients with PPD or persecutory delusional disorder continue to function in society without treatment.

Resources

ORGANIZATIONS

American Psychiatric Association. 1400 K Street NW, Washington, DC 20005. (888) 357-7924.
< <http://www.psych.org> >.

American Psychological Association (APA). 750 First St. NE, Washington, DC 20002-4242. (202) 336-5700.
< <http://www.apa.org> >.

National Alliance for the Mentally Ill (NAMI). Colonial Place Three, 2107 Wilson Blvd., Ste. 300, Arlington, VA 22201-3042. (800) 950-6264. < <http://www.nami.org> >.

National Institute of Mental Health. Mental Health Public Inquiries, 5600 Fishers Lane, Room 15C-05, Rockville, MD 20857. (888) 826-9438. <http://www.nimh.nih.gov>.

Paula Anne Ford-Martin

Parapharyngeal abscess see **Abscess**

Paraphilias see **Sexual perversions**

Paraplegia see **Paralysis**

Parasomnia see **Sleep disorders**

Parathyroid gland removal see
Parathyroidectomy

Parathyroid hormone test

Definition

The parathyroid hormone (PTH) test is a blood test performed to determine the serum levels of a hormone secreted by the parathyroid gland in response to low blood calcium levels. PTH works together with vitamin D to maintain healthy bones. The parathyroid glands are small paired glands located near the thyroid gland at the base of the neck.

Purpose

The PTH level is measured to evaluate the level of blood calcium. It is routinely monitored in patients with a kidney disorder called chronic renal failure (CRF). Because PTH is one of the major factors affecting calcium metabolism, the PTH test helps to distinguish nonparathyroid from parathyroid causes of too much calcium in the blood (**hypercalcemia**).

Differential diagnosis of hyperparathyroidism

PTH is also useful in the differential diagnosis of overactive parathyroid glands (hyperparathyroidism). Primary **hyperparathyroidism** is most often caused by a benign tumor in one or more of the parathyroid glands. It is rarely caused by parathyroid **cancer**. Patients with this condition have high PTH and calcium levels.

Secondary hyperparathyroidism is often seen in patients with chronic renal failure (CRF). The kidneys fail to excrete sufficient phosphate, and the parathyroid gland secretes PTH in an effort to lower calcium levels to balance the calcium-phosphate ratio. Because

of the constant stimulation of the parathyroid, CRF patients have high PTH and normal or slightly low calcium levels.

Tertiary hyperparathyroidism occurs when CRF causes a severe imbalance in the calcium-phosphate ratio, leading to very high PTH production that results in hypercalcemia. Patients with this condition have high PTH and high calcium levels.

Specific PTH assays

PTH is broken down in the body into three different molecular forms: the intact PTH molecule and several smaller fragments which include an amino acid or N-terminal, a midregion or midmolecule, and a carboxyl or C-terminal. Two tests are currently used to measure intact PTH and its terminal fragments. While both tests are used to diagnose hyper- or **hypo-parathyroidism**, each test also has specific applications as well. The C-terminal PTH assay is used to diagnose the ongoing disturbances in PTH metabolism that occur with secondary and tertiary hyperparathyroidism. The assay for intact PTH and the N-terminal fragment, which are both measured at the same time, is more accurate in detecting sudden changes in the PTH level. For this reason, the N-terminal PTH assay is used to monitor a patient's response to therapy.

Precautions

Drug interactions

Some prescription drugs affect the results of PTH tests. Drugs that *increase* PTH levels include phosphates, anticonvulsants, steroids, isoniazid, lithium, and rifampin. Drugs that *decrease* PTH include cimetidine and propranolol.

Timing

PTH levels are subject to daily variation, ranging from a peak around 2:00 A.M. to a low point around 2:00 P.M. Specimens are usually drawn at 8:00 A.M. The laboratory should be notified if the patient works a night shift so that this difference in biological rhythm can be taken into account.

Other serum level tests

Due to the relationship between PTH and calcium, calcium levels should be tested at the same time as PTH. Most laboratories have established reference values to indicate what PTH level is normal for a particular calcium level. In addition, the effects of PTH on kidney function and bone strength indicate

KEY TERMS

Assay—An analysis of the chemical composition or strength of a substance.

Hypercalcemia—Abnormally high levels of blood calcium.

Hyperparathyroidism—Overactivity of the parathyroid glands. Symptoms include generalized aches and pains, depression, and abdominal pain.

Hypoparathyroidism—Insufficient production of parathyroid hormone, which results in low levels of blood calcium.

that serum calcium, phosphorus, and creatinine levels should be measured together with PTH. The **creatinine test** measures kidney function and aids in the diagnosis of parathyroid dysfunction.

Description

The PTH test is performed on a sample of the patient's blood, withdrawn from a vein into a vacuum tube. The procedure, which is called a venipuncture, takes about five minutes.

Preparation

The patient should have nothing to eat or drink from midnight of the day of the test.

Risks

Risks for this test are minimal, but may include slight bleeding from the puncture site, a small bruise or swelling in the area, or **fainting** or feeling lightheaded.

Normal results

Reference ranges for PTH tests vary somewhat depending on the laboratory, and must be interpreted in association with calcium results. The following ranges are typical:

- Intact PTH: 10–65 pg/mL
- PTH N-terminal (includes intact PTH): 8–24 pg/mL
- PTH C-terminal (includes C-terminal, intact PTH, and midmolecule): 50–330 pg/mL.

Abnormal results

When measured with serum calcium levels, abnormally *high* PTH values may indicate primary,

secondary, or tertiary hyperparathyroidism, chronic renal failure, **malabsorption syndrome**, and **vitamin D deficiency**. Abnormally *low* PTH levels may indicate hypoparathyroidism, hypercalcemia, and certain malignancies.

Resources

BOOKS

Pagana, Kathleen Deska. *Mosby's Manual of Diagnostic and Laboratory Tests*. St. Louis: Mosby, Inc., 1998.

Janis O. Flores

Parathyroid scan

Definition

A parathyroid scan is sometimes called a parathyroid localization scan or parathyroid scintigraphy. This scan uses radioactive pharmaceuticals that are readily taken up by cells in the parathyroid glands to obtain an image of the glands and any abnormally active areas within them.

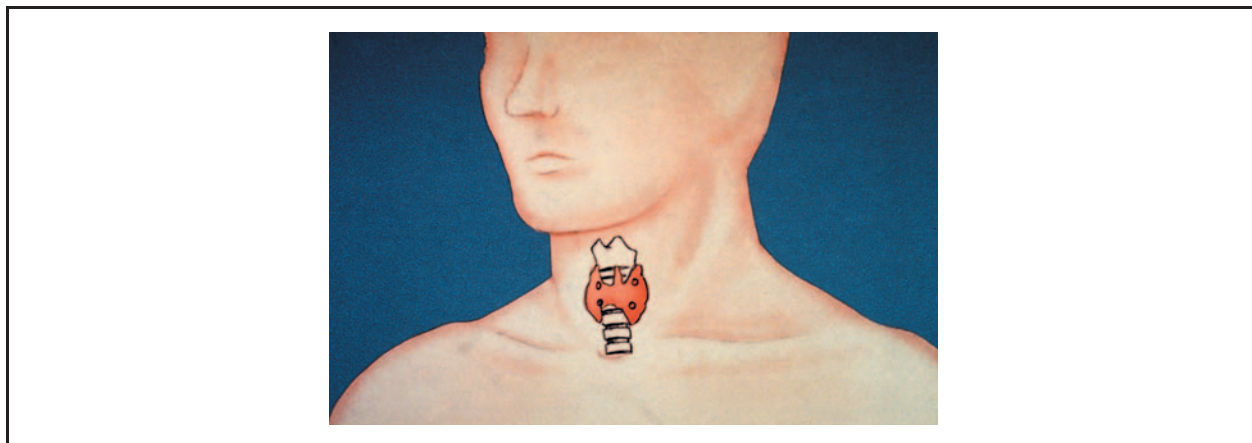
Purpose

The parathyroid glands, embedded in the thyroid gland in the neck, but separate from the thyroid in function, control calcium metabolism in the body. The parathyroid glands produce parathyroid hormone (PTH). PTH regulates the level of calcium in the blood.

Calcium is critical to cellular metabolism, as well as being the main component of bones. If too much PTH is secreted, the bones release calcium into the bloodstream. Over time, the bones become brittle and more likely to break. A person with levels of calcium in the blood that are too high feels tired, run down, irritable, and has difficulty sleeping. Additional signs of too much calcium in the blood are **nausea and vomiting**, frequent urination, kidney stones and bone **pain**. A parathyroid scan is administered when the parathyroid appears to be overactive and a tumor is suspected.

Precautions

Parathyroid scans are not recommended for pregnant women because of the potential harm to the developing fetus. People who have had another recent nuclear medicine procedure or an intravenous contrast test may need to wait until the earlier radioactive



The parathyroid glands, embedded in the thyroid gland in the neck but separate from the thyroid gland in function, control calcium metabolism in the body by producing parathyroid hormone, or PTH. (Custom Medical Stock Photo. Reproduced by permission.)

markers have been eliminated from their system in order to obtain accurate results from the parathyroid scan.

Description

A parathyroid scan is a non-invasive procedure that uses two radiopharmaceuticals (drugs with a radioactive marker) to obtain an image of highly active areas of the parathyroid glands. The test can be done in two ways.

Immediate scan

If the test is to be performed immediately, the patient lies down on an imaging table with his head and neck extended and immobilized. The patient is injected with the first radiopharmaceutical. After waiting 20 minutes, the patient is positioned under the camera for imaging. Each image takes five minutes. It is essential that the patient remain still during imaging.

After the first image, the patient is injected with a second radiopharmaceutical, and imaging continues for another 25 minutes. Total time for the test is about one hour: injection 10 minutes, waiting period 20 minutes, and imaging 30 minutes.

Another way to do this test is as follows. After the first images are acquired, the patient returns two hours later for additional images. Time for this procedure totals about three hours: injection 10 minutes, waiting period two hours and 20 minutes, and imaging 30 minutes.

Delayed scan

In a delayed parathyroid scan, the patient is asked to swallow capsules containing the first

KEY TERMS

Cyst—An abnormal sac containing fluid or semi-solid material.

Goiter—Chronic enlargement of the thyroid gland.

Neoplasm—An uncontrolled growth of new tissue.

radiopharmaceutical. The patient returns after a four hour waiting period, and the initial image is made. Then the patient is injected with the second radiopharmaceutical. Imaging continues for another 25 minutes. The total time is about four hours and 40 minutes: waiting period four hours, injection 10 minutes, and imaging 30 minutes.

Preparation

No special preparations are necessary for this test. It is not necessary to fast or maintain a special diet. The patient should wear comfortable clothing and no metal jewelry around the neck.

Aftercare

The patient should not feel any adverse effects of the test and can resume normal activities immediately.

Risks

The only risk associated with this test is to the fetus of a pregnant woman.

Normal results

Normal results will show no unusual activity in the parathyroid glands.

Abnormal results

A concentration of radioactive materials in the parathyroid gland beyond background levels suggests excessive activity and the presence of a tumor. False positive results sometimes result from the presence of multinodular **goiter**, neoplasm, or cysts. False positive tests are tests that interpret the results as abnormal when this is not true.

Resources

OTHER

"Parathyroid Scan." *HealthGate Page*. June 13, 1998. <<http://www.healthgate.com>>.

Tish Davidson, A.M.

Parathyroidectomy

Definition

Parathyroidectomy is the removal of one or more of the parathyroid glands. The parathyroid glands are usually four in number, although the exact number may vary from three to seven. They are located in the neck in front of the Adam's apple and are closely linked to the thyroid gland. The parathyroid glands regulate the balance of calcium in the body.

Purpose

Parathyroidectomy is usually done to treat **hyperparathyroidism** (abnormal over-functioning of the parathyroid glands).

Precautions

Parathyroidectomy should only be done when other non-operative methods have failed to control the patient's hyperparathyroidism.

Description

Parathyroidectomy is an operation done most commonly by a general surgeon, or occasionally by an otolaryngologist, in the operating room of a

hospital. The operation begins when the anesthesiologist anesthetizes or puts the patient to sleep. The surgeon makes an incision in the front of the neck where a tight-fitting necklace would rest. All of the parathyroid glands are identified. The surgeon then identifies the gland or glands with the disease and confirms the diagnosis by sending a piece of the gland(s) to the pathology department for immediate microscopic examination. The glands are then removed and the incision is closed and a dressing is placed over the incision.

Patients generally stay overnight in the hospital after completion of the operation and may remain for one or two additional days. These procedures are reimbursed by insurance companies. Surgeon's fees typically range from \$1,000–\$2,000. Anesthesiologists charge for their services based on the medical status of the patient and the length of the operative procedure. Hospitals charge for use of the operating suite, equipment, lab and diagnostic tests, and medications.

Preparation

Prior to the operation, the diagnosis of hyperparathyroidism should be confirmed using lab tests. Occasionally, physicians order **computed tomography scans** (CT scans), ultrasound exams, and/or **magnetic resonance imaging** (MRI) tests to determine the total number of parathyroid glands and their location prior to the procedure.

Aftercare

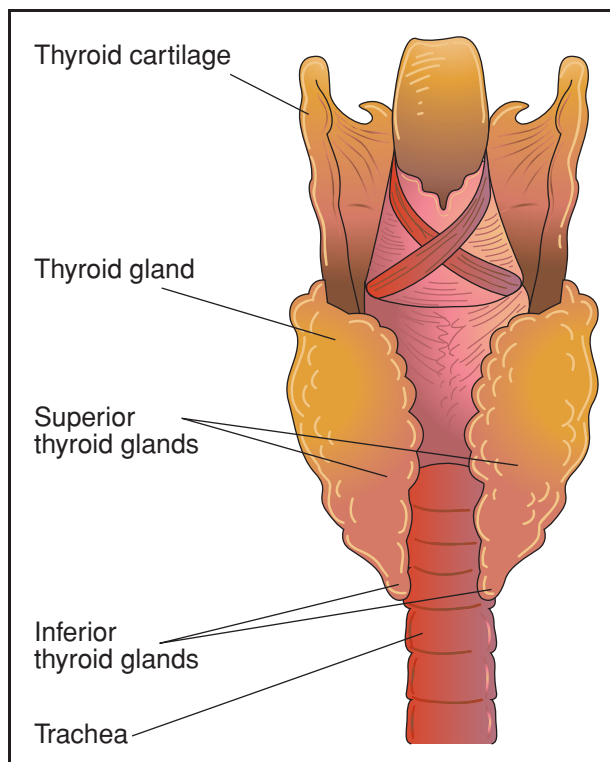
The incision should be watched for signs of infection. In general, no specific wound care is required.

The level of calcium in the body should be monitored during the first 48 hours after the operation by obtaining frequent blood samples for laboratory analysis.

Risks

The major risk of parathyroidectomy is injury to the recurrent laryngeal nerve (a nerve that lies very near the parathyroid glands and serves the larynx or voice box). If this nerve is injured, the voice may become hoarse or weak.

Occasionally, too much parathyroid tissue is removed, and the patient may develop hypoparathyroidism (under-functioning of the parathyroid glands). If this occurs, the patient will require daily calcium supplements.



Parathyroidectomy refers to the surgical removal of one or more of the parathyroid glands due to hyperparathyroidism (an abnormal over-functioning of the parathyroid glands). It is usually done after other non-operative methods have failed to control or correct this condition. (Illustration by Electronic Illustrators Group.)

Sometimes not all of the parathyroid glands are found in the initial operation. A fifth or sixth gland may be located in an aberrant location such as the chest (ectopic parathyroid). If this occurs, the patient's hyperparathyroidism may not be corrected, and a second procedure may be required to find the other gland(s).

Hematoma formation (collection of blood under the incision) is a possible complication of any operative procedure. However, in procedures that involve the neck it is of particular concern, because a rapidly enlarging hematoma can obstruct the airway.

Infection of the surgical incision may occur, as with any operative procedure, but this is not common.

Normal results

Most patients require only two or three days of hospitalization to recover from the operation. They usually can resume most of their normal activities within one to two weeks.

KEY TERMS

Anesthesiologist—A physician who specializes in anesthetizing patients for operations.

Ectopic parathyroid tissue—A condition where the thyroid tissue is located in an abnormal place.

Hyperparathyroidism—Abnormal over-functioning of the parathyroid glands.

Hypoparathyroidism—Abnormal under-functioning of the parathyroid glands.

Otolaryngologist—A surgeon who treats people with abnormalities in the head and neck regions of the body.

Resources

OTHER

"Parathyroidectomy." *ThriveOnline*.
< <http://thriveonline.oxygen.com> >.

Mary Jeanne Krob, MD, FACS

Paratyphoid fever

Definition

Paratyphoid **fever**, which is sometimes called *Salmonella paratyphi* infection, is a serious contagious disease caused by a gram-negative bacterium. It is also grouped together with **typhoid fever** under the name enteric fever.

Description

Enteric fever is increasingly rare in the United States. Of the 500 cases reported in an average year, about 60% are infections acquired during travel in Mexico, India, or South America.

Paratyphoid fever has three stages: an early stage marked by high fever; a toxic stage with abdominal **pain** and intestinal symptoms, and a long period of recovery from fever (defervescence). In adults, these three phases may cover a period of four to six weeks; in children, they are shorter and may cover 10 days to two weeks. During the toxic stage there is a 1–10% chance of intestinal perforation or hemorrhage.

Causes and symptoms

Paratyphoid fever is caused by any of three strains of *Salmonella paratyphi*: *S. paratyphi A*; *S. schottmuel-leri* (also called *S. paratyphi C*); or *S. hirschfeldii* (also called *S. paratyphi B*). It can be transmitted from animals or animal products to humans or from person to person. The incubation period is one to two weeks but is often shorter in children. Symptom onset may be gradual in adults but is often sudden in children.

Paratyphoid fever is marked by high fever, **head-ache**, loss of appetite, **vomiting**, and **constipation** or **diarrhea**. The patient typically develops an enlarged spleen. About 30% of patients have rose spots on the front of the chest during the first week of illness. The rose spots develop into small hemorrhages that may be hard to see in African or Native Americans.

Patients with intestinal complications have symptoms resembling those of **appendicitis**: intense cramp- ing pain with soreness in the right lower quadrant of the abdomen.

Diagnosis

The diagnosis is usually made on the basis of a history of recent travel and culturing the paratyphoid organism. Because the disease is unusual in the United States, the doctor may not consider paratyphoid in the diagnosis unless the patient has the classic symptoms of an enlarged spleen and rose spots. The doctor will need to rule out other diseases with high fevers, including typhus, **brucellosis**, **tular-emia** (rabbit fever), psittacosis (**parrot fever**), mono- nucleosis, and Kawasaki syndrome. *S. paratyphi* is easily cultured from samples of blood, stool, urine, or bone marrow.

Treatment

Medications

Paratyphoid fever is treated with **antibiotics** over a two- to three-week period with trimethoprim- sulfamethoxazole (Bactrim, Septra); amoxicillin (Amoxil, Novamoxin); and ampicillin (Amcill). Third- generation **cephalosporins** (ceftriaxone [Rocephin], cefotaxime [Claforan], or cefixime [Suprax]) or chlor- amphenicol (Chloromycetin) may be given if the specific strain is resistant to other antibiotics.

Surgery

Patients with intestinal perforation or hemor- rhage may need surgery if the infection cannot be controlled by antibiotics.

KEY TERMS

Defervescence—Return to normal body tempera- ture after high fever.

Enteric fever—A term that is sometimes used for either typhoid or paratyphoid fever.

Rose spots—Small slightly raised reddish pimples that are a distinguishing feature of typhoid or para- typhoid infection.

Supportive care

Patients with paratyphoid fever need careful mon- itoring for signs of complications as well as bed rest and nutritional support. Patients with severe infec- tions may require fluid replacement or blood transfusions.

Prognosis

Most patients with paratyphoid fever recover completely, although intestinal complications can result in **death**. With early treatment, the mortality rate is less than 1%.

Prevention

Immunization

Vaccination against paratyphoid fever is not necessary within the United States but is recom- mended for travel to countries with high rates of enteric fever.

Hygienic measures

Travelers in countries with high rates of para- typhoid fever should be careful to wash hands before eating and to avoid meat, egg, or poultry dishes unless they have been thoroughly cooked.

Resources

BOOKS

Fauci, Anthony S., et al., editors. *Harrison's Principles of Internal Medicine*. New York: McGraw-Hill, 1997.

Rebecca J. Frey, PhD

Paresthesias see **Numbness and tingling**

Parkinson disease

Definition

Parkinson disease (PD) is a progressive movement disorder marked by **tremors**, rigidity, slow movements (bradykinesia), and posture instability. It occurs when cells in one of the movement-control centers of the brain begin to die for unknown reasons. PD was first noted by British physician James Parkinson in the early 1800s.

Description

Usually beginning in a person's late fifties or early sixties, Parkinson disease causes a progressive decline in movement control, affecting the ability to control initiation, speed, and smoothness of motion. Symptoms of PD are seen in up to 15% of those ages 65–74, and almost 30% of those ages 75–84.

Most cases of PD are sporadic. This means that there is a spontaneous and permanent change in nucleotide sequences (the building blocks of genes). Sporadic mutations also involve unknown environmental factors in combination with genetic defects. The abnormal gene (mutated gene) will form an altered end-product or protein. This will cause abnormalities in specific areas in the body where the protein is used. Some evidence suggests that the disease is transmitted by autosomal dominant inheritance. This implies that an affected parent has a 50% chance of transmitting the disease to any child. This type of inheritance is not commonly observed. The most recent evidence is linking PD with a gene that codes for a protein called alpha-synuclein. Further research is attempting to fully understand the relationship with this protein and nerve cell degeneration.

PD affects approximately 500,000 people in the United States, both men and women, with as many as 50,000 new cases each year.

Causes and symptoms

The immediate cause of PD is degeneration of brain cells in the area known as the substantia nigra, one of the movement control centers of the brain. Damage to this area leads to the cluster of symptoms known as “parkinsonism.” In PD, degenerating brain cells contain Lewy bodies, which help identify the disease. The cell death leading to parkinsonism may be caused by a number of conditions, including infection, trauma, and **poisoning**. Some drugs given for **psychosis**, such as haloperidol (Haldol) or

chlorpromazine (thorazine), may cause parkinsonism. When no cause for nigral cell degeneration can be found, the disorder is called idiopathic parkinsonism, or Parkinson disease. Parkinsonism may be seen in other degenerative conditions, known as the “parkinsonism plus” syndromes, such as **progressive supranuclear palsy**.

The substantia nigra, or “black substance,” is one of the principal movement control centers in the brain. By releasing the neurotransmitter known as dopamine, it helps to refine movement patterns throughout the body. The dopamine released by nerve cells of substantia nigra stimulates another brain region, the corpus striatum. Without enough dopamine, the corpus striatum cannot control its targets, and so on down the line. Ultimately, the movement patterns of walking, writing, reaching for objects, and other basic programs cannot operate properly, and the symptoms of parkinsonism are the result.

There are some known toxins that can cause parkinsonism, most notoriously a chemical called MPTP, found as an impurity in some illegal drugs. Parkinsonian symptoms appear within hours of ingestion, and are permanent. MPTP may exert its effects through generation of toxic molecular fragments called free radicals, and reducing free radicals has been a target of several experimental treatments for PD using antioxidants.

It is possible that early exposure to some as-yet-unidentified environmental toxin or virus leads to undetected nigral cell death, and PD then manifests as normal age-related decline brings the number of functioning nigral cells below the threshold needed for normal movement. It is also possible that, for genetic reasons, some people are simply born with fewer cells in their substantia nigra than others, and they develop PD as a consequence of normal decline.

Symptoms

The identifying symptoms of PD include:

- Tremors, usually beginning in the hands, often occurring on one side before the other. The classic tremor of PD is called a “pill-rolling tremor,” because the movement resembles rolling a pill between the thumb and forefinger. This tremor occurs at a frequency of about three per second.
- Slow movements (bradykinesia) occur, which may involve slowing down or stopping in the middle of familiar tasks such as walking, eating, or shaving. This may include freezing in place during movements (akinesia).

- Muscle rigidity or stiffness, occurring with jerky movements replacing smooth motion.
- Postural instability or balance difficulty occurs. This may lead to a rapid, shuffling gait (festination) to prevent falling.
- In most cases, there is a “masked face,” with little facial expression and decreased eye-blinking.

In addition, a wide range of other symptoms may often be seen, some beginning earlier than others:

- depression
- speech changes, including rapid speech without inflection changes
- problems with sleep, including restlessness and nightmares
- emotional changes, including fear, irritability, and insecurity
- incontinence
- **constipation**
- handwriting changes, with letters becoming smaller across the page (micrographia)
- progressive problems with intellectual function (**dementia**)

Diagnosis

The diagnosis of Parkinson disease involves a careful medical history and a neurological exam to look for characteristic symptoms. There are no definitive tests for PD, although a variety of lab tests may be done to rule out other causes of symptoms, especially if only some of the identifying symptoms are present. Tests for other causes of parkinsonism may include brain scans, blood tests, lumbar puncture, and x rays.

Treatment

There is no cure for Parkinson disease. Most drugs treat the symptoms of the disease only, although one drug, selegiline (Eldepryl), may slow degeneration of the substantia nigra.

Exercise, nutrition, and physical therapy

Regular, moderate **exercise** has been shown to improve motor function without an increase in medication for a person with PD. Exercise helps maintain range of motion in stiff muscles, improve circulation, and stimulate appetite. An exercise program designed by a physical therapist has the best chance of meeting the specific needs of the person with PD. A physical therapist may also suggest strategies for balance

compensation and techniques to stimulate movement during slowdowns or freezes.

Good **nutrition** is important to maintenance of general health. A person with PD may lose some interest in food, especially if depressed, and may have **nausea** from the disease or from medications, especially those known as dopamine agonists. Slow movements may make it difficult to eat quickly, and delayed gastric emptying may lead to a feeling of fullness without having eaten much. Increasing fiber in the diet can improve constipation, soft foods can reduce the amount of needed chewing, and a prokinetic drug such as cisapride (Propulsid) can increase the movement of food through the digestive system.

People with PD may need to limit the amount of protein in their **diets**. The main drug used to treat PD, L-dopa, is an amino acid, and is absorbed by the digestive system by the same transporters that pick up other amino acids broken down from proteins in the diet. Limiting protein, under the direction of the physician or a nutritionist, can improve the absorption of L-dopa.

No evidence indicates that vitamin or mineral supplements can have any effect on the disease other than in the improvement of the patient's general health. No antioxidants used to date have shown promise as a treatment except for selegiline, an MAO-B inhibitor which is discussed in the Drugs section. A large, carefully controlled study of vitamin E demonstrated that it could not halt disease progression.

Drugs

The pharmacological treatment of Parkinson disease is complex. While there are a large number of drugs that can be effective, their effectiveness varies with the patient, disease progression, and the length of time the drug has been used. Dose-related side effects may preclude using the most effective dose, or require the introduction of a new drug to counteract them. There are five classes of drugs currently used to treat PD.

DRUGS THAT REPLACE DOPAMINE. One drug that helps replace dopamine, levodopa (L-dopa), is the single most effective treatment for the symptoms of PD. L-dopa is a derivative of dopamine, and is converted into dopamine by the brain. It may be started when symptoms begin, or when they become serious enough to interfere with work or daily living.

L-dopa therapy usually remains effective for five years or longer. Following this, many patients develop motor fluctuations, including peak-dose “dyskinesias” (abnormal movements such as tics, twisting, or

restlessness), rapid loss of response after dosing (known as the “on-off” phenomenon), and unpredictable drug response. Higher doses are usually tried, but may lead to an increase in dyskinesias. In addition, side effects of L-dopa include **nausea and vomiting**, and low blood pressure upon standing (**orthostatic hypotension**), which can cause **dizziness**. These effects usually lessen after several weeks of therapy.

ENZYME INHIBITORS. Dopamine is broken down by several enzyme systems in the brain and elsewhere in the body, and blocking these enzymes is a key strategy to prolonging the effect of dopamine. The two most commonly prescribed forms of L-dopa contain a drug to inhibit the amino acid decarboxylase (an AADC inhibitor), one type of enzyme that breaks down dopamine. These combination drugs are Sinemet (L-dopa plus carbidopa) and Madopar (L-dopa plus benzaseride). Controlled-release formulations also aid in prolonging the effective interval of an L-dopa dose.

The enzyme monoamine oxidase B (MAO-B) inhibitor selegiline may be given as add-on therapy for L-dopa. Research indicates selegiline may have a neuroprotective effect, sparing nigral cells from damage by free radicals. Because of this, and the fact that it has few side effects, it is also frequently prescribed early in the disease before L-dopa is begun. Entacapone and tolcapone, two inhibitors of another enzyme system called catechol-o-methyl transferase (COMT), may soon reach the market as early studies suggest that they effectively treat PD symptoms with fewer motor fluctuations and decreased daily L-dopa requirements.

DOPAMINE AGONISTS. Dopamine works by stimulating receptors on the surface of corpus striatum cells. Drugs that also stimulate these cells are called dopamine agonists, or DAs. DAs may be used before L-dopa therapy, or added on to avoid requirements for higher L-dopa doses late in the disease. DAs available in the United States as of early 1998, include bromocriptine (Permax, Parlodel), pergolide (Permax), and pramipexole (Mirapex). Two more, cabergoline (Dostinex) and ropinirole (Requip), are expected to be approved soon. Other dopamine agonists in use elsewhere include lisuride (Dopergine) and apomorphine. Side effects of all the DAs are similar to those of dopamine, plus confusion and **hallucinations** at higher doses.

ANTICHOLINERGIC DRUGS. Anticholinergics maintain dopamine balance as levels decrease. However, the side effects of anticholinergics (**dry mouth**, constipation, confusion, and blurred vision) are usually

too severe in older patients or in patients with dementia. In addition, anticholinergics rarely work for very long. They are often prescribed for younger patients who have predominant shaking. Trihexyphenidyl (Artane) is the drug most commonly prescribed.

DRUGS WHOSE MODE OF ACTION IS UNCERTAIN. Amantadine (Symmetrel) is sometimes used as an early therapy before L-dopa is begun, and as an add-on later in the disease. Its anti-parkinsonian effects are mild, and are not seen in many patients. Clozapine (Clozaril) is effective especially against psychiatric symptoms of late PD, including psychosis and hallucinations.

Surgery

Two surgical procedures are used for treatment of PD that cannot be controlled adequately with drug therapy. In PD, a brain structure called the globus pallidus (GPi) receives excess stimulation from the corpus striatum. In a pallidotomy, the GPi is destroyed by heat, delivered by long thin needles inserted under anesthesia. Electrical stimulation of the GPi is another way to reduce its action. In this procedure, fine electrodes are inserted to deliver the stimulation, which may be adjusted or turned off as the response dictates. Other regions of the brain may also be stimulated by electrodes inserted elsewhere. In most patients, these procedures lead to significant improvement for some motor symptoms, including peak-dose dyskinesias. This allows the patient to receive more L-dopa, since these dyskinesias are usually what causes an upper limit on the L-dopa dose.

A third procedure, transplant of fetal nigral cells, is still highly experimental. Its benefits to date have been modest, although improvements in technique and patient selection are likely to change that.

Alternative treatment

Currently, the best treatments for PD involve the use of conventional drugs such as levodopa. Alternative therapies, including **acupuncture**, massage, and **yoga**, can help relieve some symptoms of the disease and loosen tight muscles. Alternative practitioners have also applied herbal and dietary therapies, including amino acid supplementation, antioxidant (**vitamins A, C, E, selenium, and zinc**) therapy, B vitamin supplementation, and calcium and magnesium supplementation, to the treatment of PD. Anyone using these therapies in conjunction with conventional drugs should check with their doctor to

KEY TERMS

AADC inhibitors—Drugs that block the amino acid decarboxylase; one type of enzyme that breaks down dopamine. Also called DC inhibitors, they include carbidopa and benserazide.

Akinesia—A loss of the ability to move; freezing in place.

Bradykinesia—Extremely slow movement.

COMT inhibitors—Drugs that block catechol-o-methyl transferase, an enzyme that breaks down dopamine. COMT inhibitors include entacapone and tolcapone.

Dopamine—A neurochemical made in the brain that is involved in many brain activities, including movement and emotion.

Dyskinesia—Impaired ability to make voluntary movements.

MAO-B inhibitors—Inhibitors of the enzyme monoamine oxidase B. MAO-B helps break down dopamine; inhibiting it prolongs the action of dopamine in the brain. Selegiline is an MAO-B inhibitor.

Orthostatic hypotension—A sudden decrease in blood pressure upon sitting up or standing. May be a side effect of several types of drugs.

Substantia nigra—One of the movement control centers of the brain.

avoid the possibility of adverse interactions. For example, vitamin B₆ (either as a supplement or from foods such as whole grains, bananas, beef, fish, liver, and potatoes) can interfere with the action of L-dopa when the drug is taken without carbidopa.

Prognosis

Despite medical treatment, the symptoms of Parkinson disease worsen over time, and become less responsive to drug therapy. Late-stage psychiatric symptoms are often the most troubling, including difficulty sleeping, nightmares, intellectual impairment (dementia), hallucinations, and loss of contact with reality (psychosis).

Prevention

There is no known way to prevent Parkinson disease.

Resources

ORGANIZATIONS

National Parkinson Foundation. 1501 NW Ninth Ave., Bob Hope Road, Miami, FL 33136.

< <http://www.parkinson.org> >.

Parkinson Disease Foundation. 710 West 168th St., New York, NY 10032. (800) 457-6676.

< <http://www.apdaparkinson.com> >.

Worldwide Education and Awareness for Movement Disorders. One Gustave Levy Place, New York, NY 10029. (800) 437-MOV2. < <http://www.wemove.org> >.

OTHER

AWAKENINGS. < <http://www.parkinsonsdisease.com> >.

Laith Farid Gulli, M.D.

Parkinsonism see **Parkinson disease**

Parotid gland removal see **Parotidectomy**

Parotid gland scan see **Salivary gland scan**

Parotidectomy

Definition

Parotidectomy is the removal of the parotid gland, a salivary gland near the ear.

Purpose

The main purpose of parotidectomy is to remove cancerous tumors in the parotid gland. A number of tumors can develop in the parotid gland. Many of these are tumors that have spread from other areas of the body, entering the parotid gland by way of the lymphatic system. Among the tumors seen in the parotid gland are lymphoma, melanoma, and squamous cell carcinoma.

Description

The parotid gland is the largest of the salivary glands. There are two parotid glands, one on each side of the face. They lie just in front of the ears and a duct runs from each to the inside of the cheek. Each parotid gland has several lobes. Surgery is recommended as part of the treatment for all cancers in the parotid gland. Superficial or localized parotidectomy is recommended by some authorities, unless a lipoma or Warthin's tumor is present. One of the advantages to this approach is that nerves to facial muscles are left intact. Many facial nerves run through the same area

KEY TERMS

Fistula—An abnormal opening or duct through tissue that results from injury, disease, or other trauma.

Salivary gland—Three pairs of glands that secrete into the mouth and aid digestion.

as the parotid gland and can be damaged during more complete parotidectomies. Most authorities recommend total parotidectomy, especially if **cancer** is found in both the superficial and deep lobes of the parotid gland. If the tumor has spread to involve the facial nerve, the operation is expanded to include parts of bone behind the ear (mastoid) to remove as much tumor as possible. Some authorities recommend post-surgery radiation as follow-up treatment for cancer.

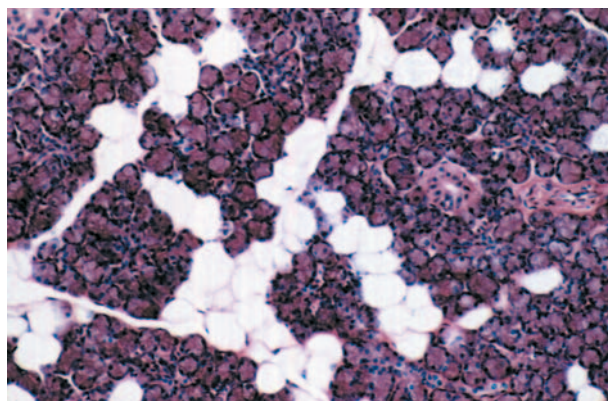
Aftercare

After surgery, the patient will remain in the hospital for one to three days. The site of incision will be watched closely for signs of infection and heavy bleeding (hemorrhage). The incision site should be kept clean and dry until it is completely healed. The patient should not wash their hair until the stitches have been removed. If the patient has difficulty smiling, winking, or drinking fluids, the physician should be contacted immediately. These are signs of facial nerve damage.

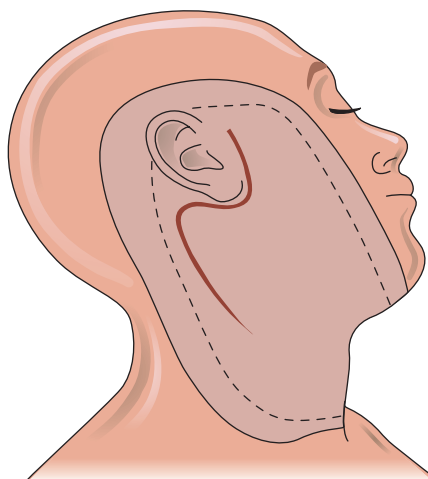
Risks

There are a number of complications that follow parotidectomy. Facial nerve **paralysis** after minor surgery should be minimal. During surgery, it is possible to repair cut nerves. After major surgery, a graft is attempted to restore nerve function to facial muscles. Salivary fistulas can occur when saliva collects in the incision site or drains through the incision. Reoccurrence of cancer is the single most important consideration for patients who have undergone parotidectomy. Long term survival rates are largely dependent on the tumor types and the stage of tumor development at the time of the operation.

Other risks include **blood clots** (hematoma) and infection. The most common long-term complication of parotidectomy is redness and sweating in the cheek, known as Frey's syndrome. Rarely, paralysis may extend throughout all the branches of the facial nervous system.



A micrograph of a normal human parotid gland. One of the salivary glands, the parotid consists of acini arranged in lobes. This image shows a junction between several lobes; the clear spaces represent the interlobular connective tissue. The masses of secretory cells produce a watery secretion which is passed to the intralobular. (Photograph by Astrid and Hanns-Frieder Michler, Custom Medical Stock Photo. Reproduced by permission.)



Parotidectomy is a surgical procedure performed to remove cancerous tumors in the parotid gland, a salivary gland near the ear. Among the tumors seen in the parotid gland are lymphoma, melanoma, and squamous cell carcinoma. The illustration above shows the facial incision sites for this procedure. (Illustration by Electronic Illustrators Group.)

Resources

BOOKS

Bentz, M. L. *Pediatric Plastic Surgery*. Stamford: Appleton & Lange, 1998.

Mary K. Fyke

Parotitis, epidemic see **Mumps**

Paroxetine see **Selective serotonin reuptake inhibitors**

Paroxysmal atrial tachycardia

Definition

A period of very rapid and regular heart beats that begins and ends abruptly. The heart rate is usually between 160 and 200 beats per minute. This condition is also known as paroxysmal supraventricular tachycardia.

Description

The term paroxysmal means that the event begins suddenly, without warning and ends abruptly. Atrial tachycardia means that the upper chambers of the heart are beating abnormally fast. Paroxysmal atrial tachycardia can occur without any heart disease being present. It is usually more annoying than dangerous.

Causes and symptoms

Paroxysmal atrial tachycardia may be caused by several different things. The fast rate may be triggered by a premature atrial beat that sends an impulse along an abnormal electrical path to the ventricles. Other causes stem from **anxiety**, stimulants, overactive thyroid, and in some women, the onset of menstruation.

Though seldom life-threatening, paroxysmal atrial tachycardia produces annoying symptoms which can include lightheadedness, chest pain, palpitations, anxiety, sweating, and shortness of breath.

Diagnosis

Diagnosis is not always easy, because the event is usually over by the time the patient sees a doctor. A careful description of the episode will aid the doctor in his diagnosis. If the rapid heart rate is still occurring, an electrocardiograph (ECG) will show the condition. If the event is over, physicians often recommend a period of ambulatory electrocardiographic monitoring (called Holter monitoring) to confirm the diagnosis.

Treatment

The doctor may suggest that during an episode of paroxysmal atrial tachycardia the following practice

KEY TERMS

Premature atrial beat—A beat that occurs before it would normally be expected.

Supraventricular—A term for an event that occurs in the upper chambers (atria) of the heart.

may help. Briefly hold the nose and mouth closed and breathe out, or by bearing down, as though straining at a bowel movement. The doctor may try to stop the episode by gently massaging an area in the neck called the carotid sinus.

If these conservative measures do not work, an injection of the drug verapamil or adenosine should stop the episode quickly.

In rare cases, the drugs do not work and electrical shock (**cardioversion**) may be necessary, particularly if serious symptoms are also present with the tachycardia.

Prognosis

Paroxysmal atrial tachycardia is not a disease, and is seldom life-threatening. The episodes are usually more unpleasant than they are dangerous, and the prognosis is generally good.

Prevention

Frequent episodes are usually cause for medication. In rare cases, the doctor may recommend a procedure called **catheter ablation**, which will remove (or ablate) the precise area of the heart responsible for triggering the fast heart rate.

In a catheter ablation procedure, the doctor will place a special catheter against the area of the heart responsible for the problem. Radio-frequency energy is then passed to the tip of the catheter, so that it heats up and destroys the target area. Catheter ablation is considered a non-surgical technique.

Resources

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave., Dallas, TX 75231. (214) 373-6300.
< <http://www.americanheart.org> >.

Dorothy Elinor Stonely

Paroxysmal supraventricular tachycardia see **Paroxysmal atrial tachycardia**

Parrot fever

Definition

Parrot **fever** is a rare infectious disease that causes **pneumonia** in humans. It is transmitted from pet birds or poultry. The illness is caused by a chlamydia, which is a type of intracellular parasite closely related to bacteria. Parrot fever is also called chlamydiosis, psittacosis, or ornithosis.

Description

Parrot fever, which is referred to as avian psittacosis when it infects birds, is caused by *Chlamydia psittaci*. Pet birds in the parrot family, including parrots, parakeets, macaws, and cockatiels, are the most common carriers of the infection. Other birds that may also spread *C. psittaci* include pigeons, doves, mynah birds, and turkeys. Birds that are carrying the organism may appear healthy, but can shed it in their feces. The symptoms of avian psittacosis include inactivity, loss of appetite and ruffled feathers, **diarrhea**, runny eyes and nasal discharge, and green or yellow-green urine. Sick birds can be treated with **antibiotics** by a veterinarian.

C. psittaci is usually spread from birds to humans through exposure to infected bird feces during cage cleaning or by handling infected birds. In humans, parrot fever can range in severity from minor flu-like symptoms to severe and life-threatening pneumonia.

Causes and symptoms

Parrot fever is usually transmitted by inhaling dust from dried bird droppings or by handling infected birds. Humans can also spread the disease by person-to-person contact, but that is very rare. The symptoms usually develop within five to 14 days of exposure and include fever, **headache**, chills, loss of appetite, **cough**, and tiredness. In the most severe cases of parrot fever, the patient develops pneumonia. People who work in pet shops or who keep pet birds are the most likely to become infected.

Diagnosis

Only 100–200 cases of parrot fever are reported each year in the United States. It is possible, however, that the illness is more common since it is easily confused with other types of influenza or pneumonia. Doctors are most likely to consider a diagnosis of parrot fever if the patient has a recent history of exposure to birds. The diagnosis can be confirmed by blood

KEY TERMS

Avian chlamydiosis—An illness in pet birds and poultry caused by *Chlamydia psittaci*. It is also known as parrot fever in birds.

Chlamydia psittaci—An organism related to bacteria that infects some types of birds and can be transmitted to humans to cause parrot fever.

Chlamydiosis, psittacosis, or ornithosis—Other names for parrot fever in humans.

tests for antibodies, usually complement fixation or immunofluorescence tests. The organism is difficult to culture. A **chest x ray** may also be used to diagnose the pneumonia caused by *C. psittaci*.

Treatment

Psittacosis is treated with an antibiotic, usually tetracycline (Achromycin, Sumycin); doxycycline (Doxy, Vibramycin); or erythromycin (Eryc, Ilotycin). Oral medication is typically prescribed for at least 10–14 days. Severely ill patients may be given intravenous antibiotics for the first few days of therapy.

Prognosis

The prognosis for recovery is excellent; with antibiotic treatment, more than 99% of patients with parrot fever will recover. Severe infections, however, may be fatal to the elderly, untreated persons, and persons with weak immune systems.

Prevention

As of 1998, there is no vaccine that is effective against parrot fever. Birds that are imported into the country as pets should be quarantined to ensure that they are not infected before they can be sold. Health authorities recommend that breeders and importers feed imported birds a special blend of feed mixed with antibiotics for 45 days to ensure that any *C. psittaci* organisms are destroyed. In addition, bird cages and food and water bowls should be cleaned daily.

Resources

ORGANIZATIONS

Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

OTHER

“Psittacosis (Parrot Fever; Ornithosis).” *ThriveOnline*.
< <http://thriveonline.oxygen.com> >.

Altha Roberts Edgren

Partial birth abortion see **Abortion, partial birth**

Partial thromboplastin time

Definition

The partial thromboplastin time (PTT) test is a blood test that is done to investigate bleeding disorders and to monitor patients taking an anticoagulating drug (heparin).

Purpose

Diagnosis

Blood clotting (coagulation) depends on the action of substances in the blood called clotting factors. Measuring the partial thromboplastin time helps to assess which specific clotting factors may be missing or defective.

Monitoring

Certain surgical procedures and diseases cause **blood clots** to form within blood vessels. Heparin is used to treat these clots. The PTT test can be used to monitor the effect of heparin on a patient's coagulation system.

Precautions

Certain medications besides heparin can affect the results of the PPT test. These include **antihistamines**, vitamin C (ascorbic acid), **aspirin**, and chlorpromazine (Thorazine).

Description

When a body tissue is injured and begins to bleed, it starts a sequence of clotting factor activities called the coagulation cascade, which leads to the formation of a blood clot. The cascade has three pathways: extrinsic, intrinsic, and common. Many of the thirteen known clotting factors in human blood are shared by both pathways; several are found in only one. The PTT test evaluates the factors found in the intrinsic

and common pathways. It is usually done in combination with other tests, such as the prothrombin test, which evaluate the factors of the extrinsic pathway. The combination of tests narrows the list of possible missing or defective factors.

Heparin prevents clotting by blocking certain factors in the intrinsic pathway. The PTT test allows a doctor to check that there is enough heparin in the blood to prevent clotting, but not so much as to cause bleeding. The test is done before the first dose of heparin or whenever the dosage level is changed; and again when the heparin has reached a constant level in the blood. The PTT test is repeated at scheduled intervals.

The PTT test uses blood to which a chemical has been added to prevent clotting before the test begins. About 5 mL of blood are drawn from a vein in the patient's inner elbow region. Collection of the sample takes only a few minutes. The blood is spun in a centrifuge, which separates the pale yellow liquid part of blood (plasma) from the cells. Calcium and activating substances are added to the plasma to start the intrinsic pathway of the coagulation cascade. The partial thromboplastin time is the time it takes for a clot to form, measured in seconds.

The test can be done without activators, but they are usually added to shorten the clotting time, making the test more useful for monitoring heparin levels. When activators are used, the test is called activated partial thromboplastin time or APTT.

Test results can be obtained in less than one hour. The test is usually covered by insurance.

Preparation

The doctor should check to see if the patient is taking any of the medications that may influence the test results. If the patient is on heparin therapy, the blood sample is drawn one hour before the next dose of heparin.

Aftercare

Aftercare includes routine care of the puncture site. In addition, patients on heparin therapy must be watched for signs of spontaneous bleeding. The patient should not be left alone until the doctor or nurse is sure that bleeding has stopped. Patients should also be advised to watch for bleeding gums, bruising easily, and other signs of clotting problems; to avoid activities that might cause minor cuts or **bruises**; and to avoid using aspirin.

KEY TERMS

Activated partial thromboplastin time—Partial thromboplastin time test that uses activators to shorten the clotting time, making it more useful for heparin monitoring.

Clotting factors—Substances in the blood that act in sequence to stop bleeding by forming a clot.

Coagulation—The process of blood clotting.

Coagulation cascade—The sequence of biochemical activities, involving clotting factors, that stop bleeding by forming a clot.

Common pathway—The pathway that results from the merging of the extrinsic and intrinsic pathways. The common pathway includes the final steps before a clot is formed.

Extrinsic pathway—One of three pathways in the coagulation cascade.

Heparin—A medication that prevents blood clots.

Intrinsic pathway—One of three pathways in the coagulation cascade.

Partial thromboplastin time—A test that checks the clotting factors of the intrinsic pathway.

Plasma—The fluid part of blood, as distinguished from blood cells.

Risks

The patient may develop a bruise or swelling around the puncture site, which can be treated with moist warm compresses. People with coagulation problems may bleed for a longer period than normal.

Normal results

Normal results vary based on the method and activators used. Normal APTT results are usually between 25–40 seconds; PTT results are between 60–70 seconds. APTT results for a patient on heparin should be 1.5–2.5 times normal values. An APTT longer than 100 seconds indicates spontaneous bleeding.

Abnormal results

Increased levels in a person with a bleeding disorder indicate a clotting factor may be missing or defective. Further tests are done to identify the factor involved. **Liver disease** decreases production of factors, increasing the PTT.

Low levels in a patient on heparin indicate too little heparin is in the blood to prevent clots. High levels indicate too much heparin is present, placing the person at risk of excessive bleeding.

Resources

PERIODICALS

Berry, Brian R., and Stephen Nantel. "Heparin Therapy: Current Regimens and Principles of Monitoring." *Postgraduate Medicine* 99 (June 1996): 64-76.

Nancy J. Nordenson

Paruresis

Definition

The inability to urinate in the presence of others.

Description

Paruresis, also known as shy or bashful bladder, is the inability or difficulty to urinate in the presence of other people, when under time pressure, or on vehicles such as trains or airplanes. Urination is normal when those constraints or factors are absent, typically when in the bathroom at home. Research suggests up to 17 million Americans, 3.25 million Canadians, and 51 million Europeans suffer from the social **anxiety** disorder. Paruresis ranges in intensity from mild, in which the person can urinate in public facilities under certain circumstances, to severe, in which the person can only urinate when alone at home. The condition almost exclusively affects males although it can occur in females.

Paruresis can be socially disabling and can often completely take over a person's life. Examples include avoiding travel, social functions, and sports arenas. Just as serious are the psychological consequences, such as depression, and anxiety. Job choices and career decisions are often adversely affected. People with the condition often avoid jobs where there is mandatory drug testing done by the supervised collection of a urine sample.

Causes and symptoms

Paruretics (people who suffer from paruresis) commonly refer to three triggers that influence them when in public restrooms. For the typical paruretic, these triggers must be removed, or the person must try another toilet, for urination to occur on a particular occasion. First, the condition occurs much more

frequently when strangers are present in the restroom as opposed to friends or relatives. Second, proximity plays a role in the problem. Proximity for the paruretic is both physical, involving the relative closeness of others in or near the restroom, and psychological, involving the need for privacy. The most frequent complaint about physical stimuli in public facilities is the absence of suitable partitions and doors on urinals or stalls. Third, temporary psychological states, especially anxiety, anger, and fear can interfere with urination.

Diagnosis

The condition is diagnosed on the basis of the sufferer's account of their symptoms. In severe cases, sufferers can waste considerable time waiting for everyone else to leave the toilet before they can urinate, and might totally avoid urinating in public toilets. The condition is usually self-diagnosed when any or all of the three main triggers of paruresis are present and the condition is chronic.

Treatment

The most well documented current treatment is based upon **cognitive-behavioral therapy**, of which the aim is to reorganize the "abnormal" emotional schemes arising from the anxiety generating elements that trigger this problem. This can be done individually in a self-help situation, in a support group, or through psychotherapy with a psychologist or psychiatrist.

Therapy includes three separate but linked components:

- **Cognitive**—An attempt to modify the abnormal thoughts and ideas around the object of anxiety, such as the thought, "When I use a toilet, everybody looks at me and wonders what I'm doing."
- **Behavioral**—Step by step desensitization by very gradual exposure to the feared situation, the aim being to achieve a series of small successes, and thus reassure the subconscious mind that it is "safe" to urinate in a situation that previously led to panic and failure. This can be thought of as relearning urination in a social situation.
- **Relaxation**—Learning techniques that facilitate relaxation, both mental and physical, such as sphincter relaxation exercises.

Drug treatments, usually with medications used to treat benign prostate hyperplasia (BPH), an enlargement of the prostate gland, such as *terazosin* (Hytrin), *tamsulosin* (Flomax), and *alfuzosin* (Uroxatral) are the subject of much debate and usually produce poor results.

KEY TERMS

Benign prostate hyperplasia (BPH)—Enlargement of the prostate gland.

Psychotherapy—The treatment of mental disorders by psychological methods, usually by a psychiatrist or psychologist.

Sphincter—A circular band of muscle that surrounds an opening or passage in the body and narrows or closes the opening by contracting.

Urethra—The tube in humans that carries urine from the bladder out of the body.

Alternative treatment

One possible alternative medicine treatment is **saw palmetto**, used to treat urinary problems in men with BPH, an enlargement of the prostate gland. BPH results in a swelling of the prostate gland that obstructs the urethra. This causes painful urination, reduced urine flow, difficulty starting or stopping the flow, dribbling after urination, and more frequent nighttime urination. A typical dose is 320 mg per day of standardized extract. It may take up to four weeks of use before beneficial effects are seen.

Prognosis

Most people who suffer from the condition never seek help or treatment. Many never even discuss the problem with anyone. But anecdotal evidence suggests that those who do seek help have a good success rate at overcoming their fear or anxiety over time, sometimes a year or longer.

Prevention

There is no known way to prevent a person from developing paruresis. Anecdotal evidence suggests it often does not occur until around the age of **puberty**. One suggestion for prevention is to condition children from an early age to urinate in public restrooms.

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Parvovirus B19 infection see **Fifth disease**

Pasteurellosis see **Animal bite infections**

Patau syndrome

Definition

Patau syndrome, also called trisomy 13, is a congenital (present at birth) disorder associated with the presence of an extra copy of chromosome 13. The extra chromosome 13 causes numerous physical and mental abnormalities, especially heart defects. Patau syndrome is named for Dr. Klaus Patau, who reported the syndrome and its association with trisomy in 1960. It is sometimes called Bartholin-Patau syndrome, named in part for Thomas Bartholin, a French physician who described an infant with the syndrome in 1656.

Description

Children normally inherit 23 chromosomes from each parent, for a total of 46 chromosomes. A typical human being has 46 chromosomes: 22 pairs of non-sex linked chromosomes and one pair of sex-linked chromosomes, that determine that child's sex. Sometimes a child may end up with more than 46 chromosomes because of problems with the father's sperm or the mother's egg; or, because of mutations that occurred after the sperm and the egg fused to form the embryo (conception).

Normally, there are two copies of each of the 23 chromosomes: one from each parent. A condition called trisomy occurs when three, instead of two, copies of a chromosome are present in a developing human embryo. An extra copy of a particular chromosome can come either from the egg or sperm, or because of mutations that occur after conception.

The best-known trisomy-related disorder is **Down syndrome** (trisomy 21), in which the developing embryo has an extra copy of chromosome 21. Patau syndrome is trisomy 13, in which the developing embryo has three copies of chromosome 13.

An extra copy of chromosome 13 is not the only cause of Patau syndrome. Other changes in chromosome 13, such as mispositioning (translocation), can also result in the characteristics classified as Patau syndrome. In these cases, an error occurs that causes a portion of chromosome 13 to be exchanged for a portion of another chromosome. There is no production of extra chromosomes, but a portion of each affected chromosome is "misplaced" (translocated) to another chromosome.

Patau syndrome causes serious physical and mental abnormalities including: heart defects; incomplete brain development; such unusual facial features as a sloping forehead, a smaller than average head (microcephaly), small or missing eyes, low-set ears, and **cleft palate** or hare lip; extra fingers and toes (**polydactyly**); abnormal genitalia; spinal defects; seizures; gastrointestinal hernias, particularly at the navel (omphalocele); and mental retardation. Due to the severity of these conditions, fewer than 20% of those affected with Patau syndrome survive beyond infancy. Most infants with the syndrome die within the first three months of life; the average life expectancy of the survivors is about 10 years.

Genetic profile

When an extra copy (trisomy) of a chromosome is made, it may either be a total trisomy (in which an extra copy of the entire chromosome is made), or partial trisomy (in which only one part of the chromosome is made an extra time).

In most cases of trisomy, errors in chromosome duplication occur at conception because of problems with the egg or the sperm that are coming together to produce an offspring. In these cases, every cell in the body of the offspring has an extra copy of the affected chromosome. However, errors in chromosome duplication may also occur during the rapid cell division that takes place immediately after conception. In these cases, only some cells of the body have the extra chromosome error. The condition in which only some of the cells in the body have the extra chromosome is called mosaicism.

Seventy-five to 80 percent of the cases of Patau syndrome are caused by a trisomy of chromosome 13. Some of these cases are the result of a total trisomy, while others are the result of a partial trisomy. Partial

trisomy generally causes less severe physical symptoms than full trisomy. Ten percent of these cases are of the mosaic type, in which only some of the body's cells have the extra chromosome. The physical symptoms of the mosaic form of Patau syndrome depends on the number and type of cells that carry the trisomy.

Most cases of trisomy are not passed on from one generation to the next. Usually they result from a malfunction in the cell division (mitosis) that occurs after conception. At least 75% of the cases of Patau syndrome are caused by errors in chromosome replication that occur after conception. The remaining 25% are caused by the inheritance of translocations of chromosome 13 with other chromosomes within the parental chromosomes. In these cases, a portion of another chromosome switches places with a portion of chromosome 13. This leads to errors in the genes on both chromosome 13 and the chromosome from which the translocated portion originated.

Patau syndrome occurs in approximately one in 8,000–12,000 live births in the United States. In many cases, spontaneous abortion (**miscarriage**) occurs and the fetus does not survive to term. In other cases, the affected individual is stillborn. As appears to be the case in all trisomies, the risks of Patau syndrome seem to increase with the mother's age, particularly if she is over 30 when pregnant. Male and female children are equally affected, and the syndrome occurs in all races and ethnic groups. Females with Patau syndrome, however, have a better chance of surviving past infancy than males.

Causes and symptoms

The severity and symptoms of Patau syndrome vary with the type of chromosomal anomaly, from extremely serious conditions to nearly normal appearance and functioning.

Full trisomy 13, which is present in the majority of the cases, results in the most severe and numerous internal and external abnormalities. Commonly, the forebrain fails to divide into lobes or hemispheres (holoprosencephaly) and the entire head is unusually small (microcephaly). The spinal cord may protrude through a defect in the vertebrae of the spinal column (myelomeningocele). Children who survive infancy have profound **mental retardation** and may experience seizures. In a few rare cases Patau syndrome may coexist with Klinefelter's syndrome or other chromosomal abnormalities.

Incomplete development of the optic (sight) and olfactory (smell) nerves often accompany the brain

defects described above. The eyes may be unusually small (**microphthalmia**) or one eye may be absent (**anophthalmia**). The eyes are sometimes set close together (hypotelorism) or even fused into a single structure. Incomplete development of any structures in the eye (coloboma) or failure of the retina to develop properly (retinal dysplasia) will also produce vision problems. Patau syndrome affected individuals may be born either partially or totally deaf and many are subject to recurring ear infections.

The facial features of many Patau syndrome-affected individuals appear flattened. The ears are generally malformed and lowset. Frequently, a child with trisomy 13 has a **cleft lip**, a cleft palate, or both. Other physical characteristics include loose folds of skin at the back of the neck, extra fingers or toes (polydactyly), permanently flexed (closed) fingers (camptodactyly), noticeably prominent heels, "rocker-bottom foot," and missing ribs. Genital malformations are common in individuals affected with Patau syndrome and include undescended testicles (cryptorchidism), an abnormally developed scrotum, and ambiguous genitalia in males, or an abnormally formed uterus (bicornuate uterus) in females.

In nearly all cases, Patau syndrome affected infants have respiratory difficulties and heart defects, including atrial and ventricular septal defects (holes between chambers of the heart); malformed ducts that cause abnormal direction of blood flow (patent ductus arteriosus); holes in the valves of the lungs and the heart (pulmonary and aortic valves); and misplacement of the heart in the right, rather than the left, side of the chest (dextrocardia). The kidneys and gastrointestinal system may also be affected with cysts similar to those seen in **polycystic kidney disease**. These defects are frequently severe and life-threatening.

Partial trisomy of the distal segment of chromosome 13 results in generally less severe, but still serious, symptoms and a distinctive facial appearance including a short upturned nose, a longer than usual area between the nose and upper lip (philtrum), bushy eyebrows, and tumors made up of blood capillaries on the forehead (frontal capillary hemangiomata). Partial trisomy of the proximal segment of chromosome 13 is much less likely to be fatal and has been associated with a variety of facial features including a large nose, a short upper lip, and a receding jaw. Both forms of partial trisomy also result in severe mental retardation.

Beyond one month of age, other symptoms that are seen in individuals with Patau syndrome are:

feeding difficulties and **constipation**, reflux disease, slow growth rates, curvature of the spine (**scoliosis**), irritability, sensitivity to sunlight, low muscle tone, high blood pressure, sinus infections, urinary tract infections, and ear and eye infections.

Diagnosis

Patau syndrome is detectable during **pregnancy** through the use of ultrasound imaging, **amniocentesis**, and **chorionic villus sampling** (CVS). At birth, the newborn's numerous malformations indicate a possible chromosomal abnormality. Trisomy 13 is confirmed by examining the infant's chromosomal pattern through karyotyping or another procedure. Karyotyping involves the separation and isolation of the chromosomes present in cells taken from an individual. These cells are generally extracted from cells found in a blood sample. The 22 non-sex linked chromosomes are identified by size, from largest to smallest, as chromosomes 1 through 22. The sex-determining chromosomes are also identified. The diagnosis of Patau syndrome is confirmed by the presence of three, rather than the normal two, copies of the thirteenth largest chromosome.

A newer method of diagnosing trisomies that has the advantages of speed and lower cost is the quantitative fluorescent PCR (QF-PCR) assay. QF-PCR testing allows a doctor to determine the presence of a chromosomal abnormality within 24 hours with a very high degree of accuracy.

Treatment

Some infants born with Patau syndrome have severe and incurable birth defects. However, children with better prognoses require medical treatment to correct structural abnormalities and associated complications. For feeding problems, special formulas, positions, and techniques may be used. Tube feeding or the placement of a gastric tube (**gastrostomy**) may be required. Structural abnormalities such as cleft lip and cleft palate can be corrected through surgery. Special diets, hearing aids, and vision aids can be used to mitigate the symptoms of Patau syndrome. Physical therapy, speech therapy, and other types of developmental therapy will help the child reach his or her potential.

Since the translocation form of Patau syndrome is genetically transmitted, **genetic counseling** for the parents should be part of the management of the disease.

KEY TERMS

Amniocentesis—A procedure performed at 16–18 weeks of pregnancy in which a needle is inserted through a woman's abdomen into her uterus to draw out a small sample of the amniotic fluid from around the baby. Either the fluid itself or cells from the fluid can be used for a variety of tests to obtain information about genetic disorders and other medical conditions in the fetus.

Chorionic villus sampling (CVS)—A procedure used for prenatal diagnosis at 10–12 weeks gestation. Under ultrasound guidance a needle is inserted either through the mother's vagina or abdominal wall and a sample of cells is collected from around the fetus. These cells are then tested for chromosome abnormalities or other genetic diseases.

Chromosome—A microscopic thread-like structure found within each cell of the body consisting of a complex of proteins and DNA. Humans have 46 chromosomes arranged into 23 pairs. Changes in either the total number of chromosomes or their shape and size (structure) may lead to physical or mental abnormalities.

Karyotyping—A laboratory procedure in which chromosomes are separated from cells, stained, and arranged so that their structure can be studied under the microscope.

Mosaicism—A genetic condition resulting from a mutation, crossing over, or nondisjunction of chromosomes during cell division, causing a variation in the number of chromosomes in the cells.

Translocation—The transfer of one part of a chromosome to another chromosome during cell division. A balanced translocation occurs when pieces from two different chromosomes exchange places without loss or gain of any chromosome material. An unbalanced translocation involves the unequal loss or gain of genetic information between two chromosomes.

Trisomy—The condition of having three identical chromosomes instead of the normal two in a cell.

Ultrasound—An imaging technique that uses sound waves to help visualize internal structures in the body.

Prognosis

Approximately 45% of trisomy 13 babies die within their first month of life; up to 70% in the first

six months; and over 70% by one year of age. Survival to adulthood is very rare. Only one adult is known to have survived to age 33.

Most survivors have profound mental and physical disabilities; however, the capacity for learning in children with Patau syndrome varies from case to case. Older children may be able to walk with or without a walker. They may also be able to understand words and phrases, follow simple commands, use a few words or signs, and recognize and interact with others.

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Patent ductus arteriosus

Definition

Patent ductus arteriosus (PDA) is a heart defect that occurs when the ductus arteriosus (the temporary fetal blood vessel that connects the aorta and the pulmonary artery) does not close at birth.

Description

The ductus arteriosus is a temporary fetal blood vessel that connects the aorta and the pulmonary artery before birth. The ductus arteriosus should be present and open before birth while the fetus is developing in the uterus. Since oxygen and nutrients are received from the placenta and the umbilical cord instead of the lungs, the ductus arteriosus acts as a "short cut" that allows blood to bypass the deflated lungs and go straight out to the body. After birth, when the lungs are needed to add oxygen to the blood, the ductus arteriosus normally closes. The closure of the ductus arteriosus ensures that blood goes to the lungs to pick up oxygen before going out to the body. Closure of the ductus arteriosus usually occurs at birth as levels of certain chemicals, called prostaglandins, change and the lungs fill with air. If the ductus arteriosus closes correctly, the blood pumped from the heart goes to the lungs, back into the heart, and then out to the body through the aorta. The blood returning from the lungs and moving out of the aorta carries oxygen to the cells of the body.

In some infants, the ductus arteriosus remains open (or patent) and the resulting heart defect is known as patent ductus arteriosus. In most cases, a small PDA does not result in physical symptoms. If the PDA is larger, health complications may occur.

In an average individual's body, the power of blood being pumped by the heart and other forces leads to a certain level of pressure between the heart and lungs.

The pressure between the heart and lungs of an individual affected by PDA causes some of the oxygenated blood that should go out to the body (through the aorta) to return back through the PDA into the pulmonary artery. The pulmonary artery takes the blood immediately back to the lungs. The recycling of the already oxygenated blood forces the heart to work harder as it tries to supply enough oxygenated blood to the body. In this case, usually the left side of the heart grows larger as it works harder and must contain all of the extra blood moving back into the heart. This is known as a left-to-right or aortic-pulmonary shunt.

As noted, the size of the PDA determines how much harder the heart has to work and how much bigger the heart becomes. If the PDA is large, the bottom left side of the heart is forced to pump twice as much blood because it must supply enough blood to recycle back to the lungs and move out to the body. As the heart responds to the increased demands for more oxygenated blood by pumping harder, the pulmonary artery has to change in size and shape in order to adapt to the increased amount and force of the blood. In some cases, the increase in size and shape changes the pressure in the pulmonary artery and lungs. If the pressure in the lungs is higher than that of the heart and body, blood returning to the heart will take the short cut back into the aorta from the pulmonary artery through the PDA instead of going to the lungs. This backward flowing of blood does not carry much oxygen. If blood without much oxygen is being delivered to the body, the legs and toes will turn blue or cyanotic. This is called a shunt reversal.

When a PDA results in a large amount of blood being cycled in the wrong order, either through a left-to-right shunt or shunt reversal, the overworked, enlarged heart may stop working (congestive **heart failure**) and the lungs can become filled with too much fluid (**pulmonary edema**). At this time, there is also an increased risk for a bacterial infection that can inflame the lining of the heart (**endocarditis**). These three complications are very serious.

PDA is a very common heart defect. Though an exact incidence of PDA is difficult to determine, one review in 1990 found that approximately 8% of live births were found to be affected by PDA. PDA can occur in full-term infants, but it is seen most frequently in preterm infants, infants born at a high altitude, and babies whose mothers were affected by the German measles (**rubella**) during pregnancy. PDA is two to three times more common in females than males. PDA occurs in individuals of every ethnic origin and does not occur more frequently in any one country or ethnic population.

Causes and symptoms

PDA can be a result of an environmental exposure before birth, inheriting a specific changed or mutated gene or genes, a symptom of a genetic syndrome, or be caused by a combination of genetic and environmental factors (multifactorial).

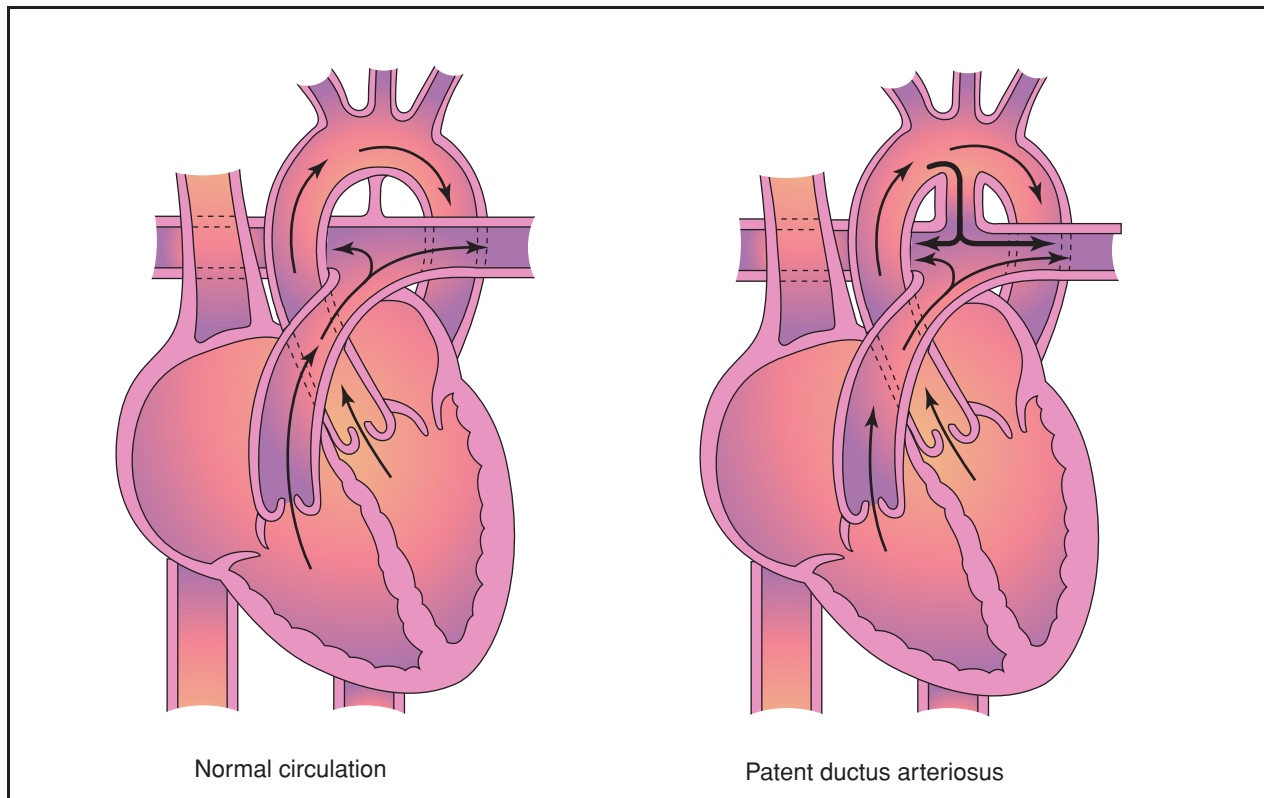
Environmental exposures that can increase the chance for a baby to be affected by PDA include fetal exposure to rubella before birth, preterm delivery, and birth at a high altitude location.

PDA can be an inherited condition running in families as isolated PDA or part of a genetic syndrome. In either case, there are specific gene changes or mutations which lead to a defect in the elastic tissue forming the walls of the ductus arteriosus. The genes causing isolated PDA have not been identified, but it is known that PDA can be inherited through a family in an autosomal dominant pattern or an autosomal recessive pattern.

Every person has approximately 30,000 genes, which tell our bodies how to grow and develop correctly. Each gene is present in pairs since one is inherited from our mother, and one is inherited from our father. In an autosomal dominant condition, only one specific changed or mutated copy of the gene for PDA is necessary for a person to have PDA. If a parent has an autosomal dominant form of PDA, there is a 50% chance for each child to have the same or similar condition.

PDA can also be inherited in an autosomal recessive manner. A recessive condition occurs when a child receives two changed or mutated copies of the gene for a particular condition, such as PDA (one copy from each parent). Individuals with a single changed or mutated copy of a gene for a recessive condition, are known as "carriers," and have no health problems related to the condition. In fact, each of us carries between five and 10 genes for harmful, recessive conditions. However, when two people who each carry a changed or mutated copy of the same gene for a recessive condition meet, there is a chance, with each **pregnancy**, for the child to inherit the two changed or mutated copies from each parent. In this case, the child would have PDA. For two known carriers, there is a 25% risk with each child to have a child with PDA, a 50% chance to have a child who is a carrier, and a 25% chance to have a child who is neither affected nor a carrier.

Most cases of PDA occur as the result of multifactorial inheritance which is caused by the combination of genetic factors and environmental factors. The



Patent ductus arteriosus (PDA) is the failure of the ductus arteriosus to close after birth, allowing blood to inappropriately flow from the aorta into the pulmonary artery. (Illustration by Electronic Illustrators Group.)

combined factors lead to isolated defects in the elastic tissue forming the walls of the ductus arteriosus. Family studies can provide different recurrence risks depending on the family member affected by multifactorial PDA. If an individual is affected by isolated, multifactorial PDA, they have a 2–4% chance of having a child affected by PDA. If a couple has one child with isolated, multifactorial PDA, there is a 3% chance that another of their children could be affected by PDA. If a couple has two children affected by isolated, multifactorial PDA, there is a 10–25% chance that they could have another child affected by PDA.

Unless a specific pattern of inheritance, preterm delivery, or known exposure is found through the examination of a detailed pregnancy and family history, the multifactorial family studies are used to estimate the possible risk of recurrence of PDA in a family.

The main sign of PDA is a constant heart murmur that sounds like the hum of a refrigerator or other machinery. This murmur is usually heard by the doctor using a stethoscope. Otherwise, there are no

specific symptoms of PDA, unless the ductus arteriosus size is large. Children and adults with a large ductus arteriosus can show difficulty in breathing during moderate physical exercise, an enlarged heart, and failure to gain weight. In some cases, heart failure and pulmonary congestion can indicate a PDA.

Diagnosis

Diagnosis is most often made by detecting the characteristic “machinery” heart murmur heard by a doctor through a stethoscope. Tests such as a chest x ray, echocardiograph, and ECG are used to support the initial diagnosis. Other indications of PDA include failure to gain weight, frequent chest infections, heavy breathing during mild physical exertion, congestive heart failure, and pulmonary edema. Prenatal ultrasounds are unable to detect PDA because the heart defect does not occur until the time of birth.

Treatment

The treatment and management of PDA depends upon the size of the PDA and symptoms being

KEY TERMS

Aorta—The main artery located above the heart which pumps oxygenated blood out into the body. Many congenital heart defects affect the aorta.

Catheterization—The process of inserting a hollow tube into a body cavity or blood vessel.

Ductus arteriosus—The temporary channel or blood vessel between the aorta and pulmonary artery in the fetus.

Echocardiograph—A record of the internal structures of the heart obtained from beams of ultrasonic waves directed through the wall of the chest.

Electrocardiogram (ECG, EKG)—A test used to measure electrical impulses coming from the heart in order to gain information about its structure or function.

Endocarditis—A dangerous infection of the heart valves caused by certain bacteria.

Oxygenated blood—Blood carrying oxygen through the body.

Pulmonary artery—An artery that carries blood from the heart to the lungs.

Pulmonary edema—A problem caused when fluid backs up into the veins of the lungs. Increased pressure in these veins forces fluid out of the vein and into the air spaces (alveoli). This interferes with the exchange of oxygen and carbon dioxide in the alveoli.

experienced by the affected individual. In some cases, a PDA can correct itself in the first months of life. In preterm infants experiencing symptoms, the first step in correcting a PDA is treatment through medications such as indomethacin. In preterm infants whose PDA is not closed through medication, full term infants affected by PDA, and adults, surgery is an option for closing the ductus arteriosus. In 2000 and 2001, medicine has developed and reviewed alternatives to surgical closure such as interventional **cardiac catheterization** and video-assisted thorascopic surgical repair. A cardiologist can help individuals determine the best method for treatment based on their physical symptoms and medical history.

Prognosis

Adults and children can survive with a small opening remaining in the ductus arteriosus. Treatment, including surgery, of a larger PDA is usually successful

and frequently occurs without complications. Proper treatment allows children and adults to lead normal lives.

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PCV see **Hematocrit**

Pediculosis see **Lice infestation**

Pedophilia see **Sexual perversions**

Pellagra

Definition

Pellagra is a disorder brought on by a deficiency of the nutrient called niacin or nicotinic acid, one of the B-complex **vitamins**.

Description

Nicotinic acid plays a crucial role in the cellular process called respiration. Respiration is the process by which nutrients (specifically sugar, or glucose) and oxygen are taken in, chemical reactions take place, energy is produced and stored, and carbon dioxide and wastes are given off. This process is absolutely central to basic cell functioning, and thus the functioning of the body as a whole.

Niacin is a B vitamin found in such foods as yeast, liver, meat, fish, whole-grain cereals and breads, and

legumes. Niacin can also be produced within the body from the essential amino acid called tryptophan. Dietary requirements for niacin depend on the age, gender, size, and activity level of the individual. Niacin requirements range from 5 mg in infants up to 20 mg in certain adults.

Causes and symptoms

Pellagra can be either primary or secondary. Primary pellagra results when the diet is extremely deficient in niacin-rich foods. A classic example occurs in geographic locations where Indian corn (maize) is the dietary staple. Maize does contain niacin, but in a form which cannot be absorbed from the intestine (except when it has been treated with alkali, as happens in the preparation of tortillas). People who rely on maize as their major food source often develop pellagra. Pellagra can also occur when a hospitalized patient, unable to eat for a very prolonged period of time, is given fluids devoid of vitamins through a needle in the vein (intravenous or IV fluids).

Secondary pellagra occurs when adequate quantities of niacin are present in the diet, but other diseases or conditions interfere with its absorption and/or processing. This is seen in various diseases that cause prolonged **diarrhea**, with **cirrhosis** of the liver and **alcoholism**, with long-term use of the anti-tuberculosis drug called isoniazid, in patients with malignant carcinoid tumor, and in patients suffering from **Hartnup disease** (an inherited disorder which results in disordered absorption of amino acids from the intestine and kidney).

Pellagra causes a variety of symptoms affecting the skin; mucous membranes (moist linings of the mouth, organs, etc.); central nervous system (including the brain and nerves); and the gastrointestinal system. The classic collection of symptoms includes redness and swelling of the mouth and tongue, diarrhea, skin rash, and abnormal mental functioning, including memory loss. While early patients may simply have a light skin rash, over time the skin becomes increasingly thickened, pigmented, and may slough off in places. Areas of the skin may become prone to bacterial infection. The mouth and tongue, and sometimes the vagina, become increasingly thick, swollen, and red. Abdominal **pain** and bloating occur, with **nausea and vomiting**, and bloody diarrhea to follow. Initial mental changes appear as inability to sleep (**insomnia**), **fatigue**, and a sense of disconnectedness (apathy). These mental changes progress to memory loss, confusion, depression, and **hallucinations** (in which the individual sees sights or hears sounds that

KEY TERMS

Niacinamide—A form of niacin, which is usually used as a dietary supplement for people with insufficient niacin.

Respiration—Respiration is the process by which nutrients (specifically sugar, or glucose) and oxygen are taken in to a cell; chemical reactions take place; energy is produced and stored; and carbon dioxide and wastes are given off.

do not really exist). The most severe states include stiffness of the arms and legs, with resistance to attempts to move the limbs; variations in level of consciousness; and the development of involuntary sucking and grasping motions. This collection of symptoms is called “encephalopathic syndrome.”

Diagnosis

Diagnosis is purely based on the patient’s collection of symptoms, together with information regarding the patient’s diet. When this information points to niacin deficiency, replacement is started, and the diagnosis is then partly made by evaluating the patient’s response to increased amounts of niacin. There are no chemical tests available to definitively diagnose pellagra.

Treatment

Treatment of pellagra usually involves supplementing the individual’s diet with a form of niacin called niacinamide (niacin itself in pure supplementation form causes a number of unpleasant side effects, including sensations of **itching**, burning, and flushing). The niacinamide can be given by mouth (orally) or by injection (when diarrhea would interfere with its absorption). The usual oral dosage is 300–500 mg each day; the usual dosage of an injection is 100–250 mg, administered two to three times each day. When pellagra has progressed to the point of the encephalopathic syndrome, a patient will require 1,000 mg of niacinamide orally, and 100–250 mg of niacinamide by injection. Once the symptoms of pellagra have subsided, a maintenance dose of niacin can be calculated, along with attempting (where possible) to make appropriate changes in the diet. Because many B-complex vitamin deficiencies occur simultaneously, patients will usually require the administration of other B-complex vitamins as well.

Prognosis

Untreated pellagra will continue progressing over the course of several years, and is ultimately fatal. Often, **death** is due to complications from infections, massive **malnutrition** brought on by continuous diarrhea, blood loss due to bleeding from the gastrointestinal tract, or severe encephalopathic syndrome.

Prevention

Prevention of pellagra is completely possible; what is required is either a diet adequate in niacin-rich foods, or appropriate supplementation. However, in many geographic locations in the world such foods are unavailable to the general population, and pellagra becomes an unavoidable complication of poverty.

Resources

ORGANIZATIONS

American Dietetic Association, 216 W. Jackson Blvd.,
Chicago, IL 60606-6995. (800) 745-0775.
< <http://www.eatright.org/cdr.html> >.

Rosalyn Carson-DeWitt, MD

Pelvic endoscopy see **Laparoscopy**

Pelvic exam

Definition

A pelvic examination is a routine procedure used to assess the well being of the female patients' lower genito-urinary tract. This is done as part of a usual health screening and prevention tool, and is an element of the total health care for the female patient.

Purpose

Pelvic exams are useful as a screening tool for sexually transmitted diseases such as **gonorrhea**, **chlamydia**, **genital warts**, **herpes**, and **syphilis**. In addition, exams detect some forms of **cancer** that may affect the genitalia. By analyzing the cervical region with a Papanicolaou or Pap smear, clinicians are able to look for signs of **cervical cancer**. The American Cancer Society and the American College of Obstetricians and Gynecologists recommend pelvic exams with **Pap tests** for women starting at age 18. It is also recommended that exams start earlier if the teenager requests oral **contraception**. Pap smears

should continue once yearly for three years and at the physicians discretion following this time. Various groups differ in opinions on when to discontinue screening for cervical cancer, however, the United States Preventative Services Task Force recommends screening continue until age 65 if the patient has not had previous abnormal results. Women who have undergone a total **hysterectomy** for reasons other than cervical cancer do not need to be screened.

Precautions

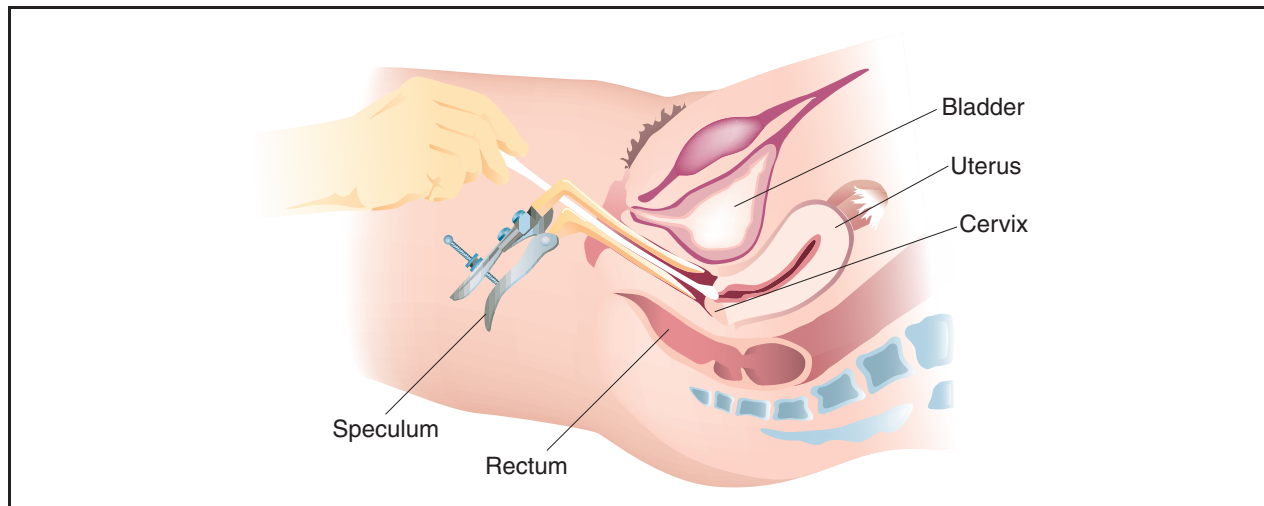
Pelvic examinations are safe procedures, thus no precautions are necessary.

Description

The first part of the examination involves visual inspection and palpation of the external genitalia. The examiner will note the characteristics of the labia majora, labia minora, clitoris, urethral orifice, and the Skene's and Bartholin's glands. In addition, the perineum and anus will be checked. The clinician will be examining these areas for any indication of swelling, inflammation, abnormal discharge, polyps, abnormal odor, or other lesions.

The next part involves examining the internal genitalia. The examiner will first insert a gloved finger into the vagina in order to palpate the cervix. Next an instrument called a speculum is inserted. This device is made of plastic or metal and used to open the vaginal cavity in order for the examiner to be able to view the vaginal walls and cervix. Any lesions, bleeding, or abnormal discharge can be visualized with the speculum in place. If indicated, a Pap smear will then be performed. With the speculum still in place, the examiner gently scrapes the patient's cervix with a wooden or plastic spatula as well as a cylindrical-type brush. The spatula collects cells from the outer surface of the cervix, while the brush is used to collect cells from the inner-cervix. The collected cells are then spread on a glass slide, sprayed with a fixative, and sent to a laboratory for analysis. The examiner may then insert a cotton or Dacron swab into the cervix. This will be held in place for 10–30 seconds and when withdrawn spread on a plastic plate or into a tube containing a reagent for the specimen. This procedure may be repeated again with the anus. Such swab tests are used to check for gonorrhea and chlamydia, or bacterial vaginitis, which is a bacterial infection resulting in inflammation of the vagina.

Following the Pap smear is the bimanual examination during which the examiner will place an index



During a pelvic exam, cells from the cervix are scraped on a spatula and are tested for abnormalities. (Illustration by Argosy Inc.)

and middle finger into the vagina to first examine the vaginal walls for any irregularities or tenderness. The cervix will then be palpated in order to note its shape, consistency, mobility, and any tenderness. The examiner will then place his or her other hand on the abdomen and gently push down while pushing the cervix up. This is done to assess the size and shape of the uterus, and also to note any tenderness or abnormal lesions. During this time, the ovaries are also checked for any masses, or tenderness.

The last part of the pelvic exam is the rectovaginal examination. This allows the clinician to better examine the pelvic organs and structures. The examiner will place their index finger into the vagina and a lubricated, gloved middle finger against the anus. During this part, the patient may feel an urgency to have a bowel movement. However, this is a natural feeling and a bowel movement will not occur. The patient will then be asked to strain down in order for the anal sphincter to relax. As relaxation occurs the examiner will insert the middle finger into the rectum, enabling the position and shape of the uterus to be better assessed. In addition, any masses or tenderness can be evaluated at this point. The anal canal and rectum can also be examined for any polyps, or other lesions at this time. After the rectovaginal exam, the patient will be allowed clean off any excess lubricant and get dressed. The examiner will then discuss the procedure and any findings with the patient.

Preparation

Pelvic exams require the patient to void prior to starting, as a full bladder can add to discomfort and

make palpation difficult for the examiner. Even though some tests cannot be done on a menstruating patient, an examination can still be performed. Any tampons should be removed prior to the exam. Douching is not recommended before an exam due to the hazard of washing away cells that are needed for examination. If a Pap smear is to be done, the patients should also refrain from sexual intercourse or using vaginal suppositories for 24 to 48 hours prior to the exam. The patient will be asked to undress and put on a gown. The examiner will instruct the patient to lie on the examination table on her back and may assist her in putting her feet in stirrups. The buttocks are then slid to the edge of the table in order for a full view of the area to be examined.

Aftercare

Even with the invasiveness of this procedure, the patient should be able to immediately resume normal daily activities.

Risks

Other than minor discomfort, there are no risks associated with a routine pelvic examination.

Normal results

No significant findings by the examiner indicate a normal pelvic examination. The external and internal genitalia will be free of any lesions or abnormal discharge. The Pap smear will not reveal cervical dysplasia or abnormal tissue development, and there will not

KEY TERMS

Bacterial vaginitis—This is the term for inflammation of the vagina due to a bacterial infection.

Bartholin's glands—These glands are embedded in the vestibule of the vagina and function to maintain moisture.

Cervical dysplasia—Dysplasia is the abnormal growth of the epithelial cells. This is what a Pap smear will detect in the cervix.

Colposcopy—This procedure is done when a Pap smear reveals abnormal results. With an endoscope placed through the vagina and into the cervix, a physician can determine exactly where lesions of the cervix are.

Hematoma—Hematomas are masses of blood (or clotted blood) that accumulate in tissues and may result from trauma.

Myoma—These are benign (non-malignant) tumors of the uterus.

Papanicolaou or Pap smear—This is a screening test for cancer of the cervix. Cells are scraped from the cervix, smeared on a glass plate, and sent to a laboratory to examine for any abnormal cells or dysplasia. This test may also detect other cells seen in certain vaginal infections.

Skene's glands—These are the glands of the female urethra.

Speculum—A speculum is an instrument that is used during the internal genitalia examination. It can be made of plastic or metal and is used to open up the vaginal cavity in order for the examiner to view the walls of the vagina and the cervix.

Urethral meatus—This is the external opening of the urethra.

be any abnormal masses or tenderness upon palpation.

Abnormal results

The examiner may discover abnormal lesions during the course of the exam that may require additional tests. Ulcerations, bumps, sores, blisters, or vesicles on the external genitalia may be signs of a sexually transmitted disease. Some of the **sexually transmitted diseases** that may cause lesions to the external genitalia include venereal warts, syphilis, and **genital herpes**. Gonorrhea or chlamydia may

also cause inflammation to the urethral meatus or the external opening of the urethra. These, in addition to bacterial infections, can also cause inflammation of the Skene's glands, Bartholin's glands, and vulva. Infections may result in an irritating discharge. Discharge may also be noted in yeast infections. Other abnormal findings of the external genitalia include carcinomas, vulvar tumors, or hematomas. Hematomas are masses of accumulated blood that appears as a bluish swelling of the labium that may occur following trauma to this area. Examination of the internal genitalia may reveal similar findings in regards to sexually transmitted diseases and carcinomas. Cervical abnormalities can also be found and may include lacerations, infections, ulcers, cysts, and polyps. All of these will require further evaluation in order to determine the underlying cause.

Since Pap smears screen for cervical cancer, abnormal results require special attention. Due to the incidence of false-positives or false-negatives, the test may be repeated or the physician may choose to have the patient undergo a **colposcopy**. This procedure uses an endoscope and will examine the vagina and cervix in more depth. This will identify 100% of lesions present. A biopsy may then be taken of the lesion in order to determine the exact type of abnormality. Several new techniques are now available that improve the accuracy of the Pap smear including automated analysis machines. Bimanual and rectovaginal exams may reveal abnormalities of the uterus or other pelvic structures. One commonly encountered finding is a myoma, which is a benign uterine tumor. In addition, the uterus may be positioned abnormally by being angled too far forward or backward. Ovarian cysts and tumors, as well as some disorders of the fallopian tubes, can be findings of these two exams.

Resources

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Laith Farid Gulli, M.D.
Robert Ramirez, B.S.

Pelvic fracture

Definition

A pelvic fracture is a break in one or more bones of the pelvis.

Description

The pelvis is a butterfly-shaped group of bones located at the base of the spine. The pelvis consists of the pubis, ilium, and ischium bones (among others) held together by tough ligaments. With a cavity in its center, the pelvis forms one major ring and two smaller rings of bone that support and protect internal organs such as the bladder, intestines, and rectum. In women, the pelvis also surrounds the uterus and vagina. The pelvis is wider and has a larger cavity in females than in males because it must accommodate **childbirth**.

Fractures of the pelvis are uncommon, accounting for only 0.3–6% of all fractures. Pelvic rings often break in more than one place. Pelvic fractures range widely in severity. Disruption of the major ring is usually a severe injury while disruption of a minor ring is often not serious. A mild fracture (for example, one that occurs due to the impact of jogging) may heal in several weeks without surgery. However, a serious pelvic fracture can be a life-threatening event requiring emergency medical care and lengthy **rehabilitation**. The latter type of injury may involve damage to nearby internal organs.

Pelvic fractures are classified as stable or unstable, and as open or closed. A stable fracture is one in which the pelvis remains stable and involves one break-point in the pelvic ring with minimal hemorrhage. An unstable fracture is one in which the pelvis is unstable with two or more break-points in the pelvic ring with moderate to severe hemorrhage. All types of pelvic

fractures are further divided into "open" or "closed," depending on whether open skin **wounds** are present or not in the lower abdomen.

Causes and symptoms

Most pelvic fractures occur during high-speed accidents (such as car or motorcycle crashes) or falls from great heights. The greater the force, the greater the opportunity for a severe fracture. Pelvic fractures can also occur spontaneously or after minor falls in people with bone-weakening diseases such as **osteoporosis**. Less commonly, pelvic fractures may occur during athletic activities such as football, hockey, skiing, and long-distance running.

The primary symptom of a pelvic fracture is **pain** in the groin, hip, or lower back, which may worsen when walking or moving the legs. Other symptoms may include abdominal pain; numbness/tingling in the groin or legs; bleeding from the vagina, urethra (urine tube), or rectum; difficulty urinating; and difficulty walking or standing. A stress fracture that occurs while jogging may cause pain in the thigh or buttock.

Diagnosis

A pelvic fracture is typically diagnosed by an emergency physician looking for bone tenderness, limitations of movement, difficulty walking, and any loss of nerve function in the lower part of the body. In addition, the physician looks for signs of injury to nearby organs of the intestinal or genitourinary systems. This search may include checking the rectum, vagina, and urethra for signs of bleeding. The physician will order a plain x ray of the pelvis; this will usually detect the presence of a fracture. Blood and urine tests may also be done. A computed tomography (CT) scan will be performed in complicated cases. Depending on the severity of the fracture, other imaging procedures may be required as well, such as contrasting studies involving the injection of a radioactive dye; the pictures can be used to evaluate organs and structures in the pelvic area, such as the urethra, bladder, and blood vessels.

Treatment

In the case of a potentially serious pelvic fracture (such as that occurring after an accident or high fall), emergency assistance should be summoned. The person with the injury should be covered with a blanket or jacket (to maintain body heat), and should not be moved by non-trained personnel, especially if there is severe pain or signs of possible nerve injury.

KEY TERMS

Fracture—A break in a bone.

Computed tomography (CT) scan—An imaging procedure that produces a series of thin x-ray slices of internal body organs or structures.

Orthopedist—A doctor who specializes in disorders of the musculoskeletal system.

Osteoporosis—A decrease in the amount of bone mass, leading to fractures.

Shock—A condition of profound physiological disturbance characterized by failure of the circulatory system to maintain adequate blood supply to vital organs.

Stress fracture—A crack in a bone (usually the result of overuse).

Treatment depends on the severity of the injury. In the case of a minor fracture, treatment may consist of bed rest and over-the-counter (OTC) or prescription pain killers. Physical therapy, the use of crutches, and surgery may also be recommended. Healing can take anywhere from a few weeks to several months.

Severe injuries to the pelvis (such as those involving more than one break) can be life threatening, resulting in **shock**, extensive internal bleeding, and damage to internal organs. In these situations, the immediate goal is to control the bleeding and stabilize the injured person's condition. Resuscitation procedures may be required as well as large amounts of intravenous fluids and blood transfusions if internal bleeding is present. These injuries often require extensive surgery as well as lengthy rehabilitation.

Alternative treatment

To speed up the healing process, some practitioners of alternative medicine recommend magnetic field therapy, hydrogen peroxide therapy, calcium, vitamin D, vitamin B complex, and zinc.

Prognosis

The prognosis for minor pelvic fractures is excellent, with most people gaining full mobility in a matter of weeks or months. Severe pelvic fractures can be fatal due to internal bleeding or damage to nearby organs, or result in chronic pain and physical disabilities.

Prevention

People with bone-weakening conditions such as osteoporosis or **cancer**, or tendencies to fall are more vulnerable to bone fractures. They should follow their treatment regimens and make use of canes and other walking aids as well as safety devices in the home (bars, non skidding mats) and avoid climbing up, even on a small stool.

Resources

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Korovessis, P., et al. "Medium- and long-term results of open reduction and internal fixation for unstable pelvic ring fractures." *Orthopedics* November 2000: 1165-71.

Malavaud, B., et al. "Evaluation of male sexual function after pelvic trauma by the International Index of Erectile Function." *Urology* June 2000: 842-6.

ORGANIZATIONS

American Academy of Orthopaedic Surgeons. 6300 North River Road, Rosemont, IL 60018-4262. (800) 346-AAOS. <<http://www.aaos.org>>.

Greg Annussek

Pelvic gynecologic sonogram see **Pelvic ultrasound**

Pelvic inflammatory disease

Definition

Pelvic inflammatory disease (PID) is a term used to describe any infection in the lower female reproductive tract that spreads to the upper female reproductive tract. The lower female genital tract consists of the vagina and the cervix. The upper female genital tract consists of the body of the uterus, the fallopian or uterine tubes, and the ovaries.

Description

PID is the most common and the most serious consequence of infection with **sexually transmitted diseases** (STD) in women. Over one million cases of PID are diagnosed annually in the United States, and it is the most common cause for hospitalization of reproductive-age women. Sexually active women aged 15–25 are at highest risk for developing PID. The disease can also occur, although less frequently, in women having monogamous sexual relationships.

The most serious consequences of PID are increased risk of infertility and ectopic pregnancy.

To understand PID, it is helpful to understand the basics of inflammation. Inflammation is the body's response to disease-causing (pathogenic) microorganisms. The affected body part may swell due to accumulation of fluid in the tissue or may become reddened due to an excessive accumulation of blood. A discharge (pus) may be produced that consists of white blood cells and dead tissue. Following inflammation, scar tissue may form by the proliferation of scar-forming cells and is called fibrosis. **Adhesions** of fibrous tissue form and cause organs or parts of organs to stick together.

PID may be used synonymously with the following terms:

- salpingitis (Inflammation of the fallopian tubes)
- endometritis (Inflammation of the inside lining of the body of the uterus)
- tubo-ovarian abscesses (Abscesses in the tubes and ovaries)
- pelvic **peritonitis** (Inflammation inside of the abdominal cavity surrounding the female reproductive organs)

Causes and symptoms

A number of factors affect the risk of developing PID. They include:

- Age. The incidence of PID is very high in younger women and decreases as a woman ages.
- Race. The incidence of PID is 8–10 times higher in nonwhites than in whites.
- Socioeconomic status. The higher incidence of PID in women of lower socioeconomic status is due in part to a woman's lack of education and awareness of health and disease and her accessibility to medical care.
- Contraception. Induced abortion, use of an **IUD**, non-use of barrier contraceptives such as condoms, and frequent douching are all associated with a higher risk of developing PID.
- Lifestyle. High risk behaviors, such as drug and alcohol **abuse**, early age of first intercourse, number of sexual partners, and **smoking** all are associated with a higher risk of developing PID.
- Types of sexual practices. Intercourse during menses and frequent intercourse may offer more opportunities for the admission of pathogenic organisms to the inside of the uterus.

- Disease. Sixty to 75% of cases of PID are associated with STDs. A prior episode of PID increases the chances of developing subsequent infections.

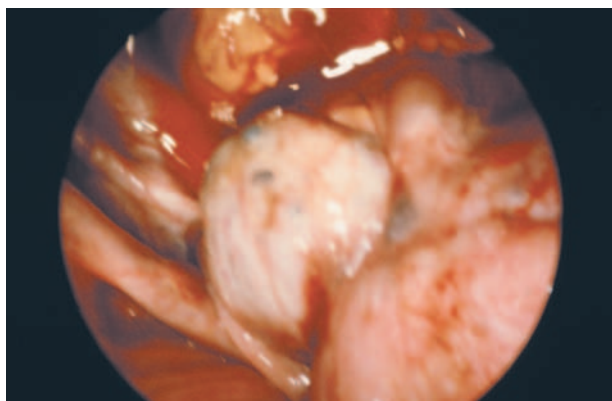
The two major causes of STDs are the organisms *Neisseria gonorrhoeae* and *Chlamydia trachomatis*. The main symptom of *N. gonorrhoeae* infection (**gonorrhea**) is a vaginal discharge of mucus and pus. Sometimes bacteria from the colon normally in the vaginal cavity may travel upward to infect the upper female genital organs, facilitated by the infection with gonorrhea. Infections with *C. trachomatis* and other nongonoccal organisms are more likely to have mild or no symptoms.

Normally, the cervix produces mucus which acts as a barrier to prevent disease-causing microorganisms, called pathogens, from entering the uterus and moving upward to the tubes and ovaries. This barrier may be breached in two ways. A sexually transmitted pathogen, usually a single organism, invades the lining cells, alters them, and gains entry. Another way for organisms to gain entry happens when trauma or alteration to the cervix occurs. **Childbirth**, spontaneous or induced abortion, or use of an intrauterine contraceptive device (IUD) are all conditions that may alter or weaken the normal lining cells, making them susceptible to infection, usually by several organisms. During menstruation, the cervix widens and may allow pathogens entry into the uterine cavity.

Recent evidence suggests that **bacterial vaginosis** (BV), a bacterial infection of the vagina, may be associated with PID. BV results from the alteration of the balance of normal organisms in the vagina, by douching, for example. While the balance is altered, conditions are formed that favor the overgrowth of anaerobic bacteria, which thrive in the absence of free oxygen. A copious discharge is usually present. Should some trauma occur in the presence of anaerobic bacteria, such as menses, abortion, intercourse, or childbirth, these organisms may gain entrance to the upper genital organs.

The most common symptom of PID is pelvic pain. However, many women with PID have symptoms so mild that they may be unaware that an infection exists.

In acute salpingitis, a common form of PID, swelling of the fallopian tubes may cause tenderness on **physical examination**. **Fever** may be present. Abscesses may develop in the tubes, ovaries, or in the surrounding pelvic cavity. Infectious discharge may leak into the peritoneal cavity and cause peritonitis, or abscesses may rupture causing a life-threatening surgical emergency.



Laparoscopic view of pelvic inflammatory disease. (Custom Medical Stock Photo. Reproduced by permission.)

Chronic salpingitis may follow an acute attack. Subsequent to inflammation, scarring and resulting adhesions may result in chronic **pain** and irregular menses. Due to blockage of the tubes by scar tissue, women with chronic salpingitis are at high risk of having an **ectopic pregnancy**. The fertilized ovum is unable to travel down the fallopian tube to the uterus and implants itself in the tube, on the ovary, or in the peritoneal cavity. This condition can also be a life-threatening surgical emergency.

IUD

IUD usage has been strongly associated with the development of PID. Bacteria may be introduced to the uterine cavity while the IUD is being inserted or may travel up the tail of the IUD from the cervix into the uterus. Uterine tissue in association with the IUD shows areas of inflammation that may increase its susceptibility to pathogens.

Susceptibility to STDs

Susceptibility to STDs involves many factors, some of which are not known. The ability of the organism to produce disease and the circumstances that place the organism in the right place at a time when a trauma or alteration to the lining cells has occurred are factors. The individual's own immune response also helps to determine whether infection occurs.

Diagnosis

If PID is suspected, the physician will take a complete medical history and perform an internal pelvic examination. Other diseases that may cause pelvic pain, such as **appendicitis** and **endometriosis**, must be

ruled out. If pelvic examination reveals tenderness or pain in that region, or tenderness on movement of the cervix, these are good physical signs that PID is present.

Specific diagnosis of PID is difficult to make because the upper pelvic organs are hard to reach for samplings. The physician may take samples directly from the cervix to identify the organisms that may be responsible for infection. Two blood tests may help to establish the existence of an inflammatory process. A positive **C-reactive protein** (CRP) and an elevated erythrocyte sedimentation rate (ESR) indicate the presence of inflammation. The physician may take fluid from the cavity surrounding the ovaries called the *cul de sac*; this fluid may be examined directly for bacteria or may be used for culture. Diagnosis of PID may also be done using a laparoscope, but **laparoscopy** is expensive, and it is an invasive procedure that carries some risk for the patient.

Treatment

The goals of treatment are to reduce or eliminate the clinical symptoms and abnormal physical findings, to get rid of the microorganisms, and to prevent long term consequences such as infertility and the possibility of ectopic **pregnancy**. If acute salpingitis is suspected, treatment with antibiotics should begin immediately. Early intervention is crucial to keep the fallopian tubes undamaged. The patient is usually treated with at least two broad spectrum **antibiotics** that can kill both *N. gonorrhoeae* and *C. trachomatis* plus other types of bacteria that may have the potential to cause infection. Hospitalization may be required to ensure compliance. Treatment for chronic PID may involve **hysterectomy**, which may be helpful in some cases.

If a woman is diagnosed with PID, she should see that her sexual partner is also treated to prevent the possibility of reinfection.

Alternative treatment

Alternative therapy should be complementary to antibiotic therapy. For pain relief, an experienced practitioner may apply castor oil packs, or use **acupressure** or **acupuncture**. Some herbs, such as *Echinacea* (*Echinacea* spp.) and calendula (*Calendula officinalis*) are believed to have antimicrobial activity and may be taken to augment the action of prescribed antibiotics. General tonic herbs, as well as good **nutrition** and rest, are important in recovery and strengthening after an episode of PID. Blue cohosh

KEY TERMS

Adhesion—The joining or sticking together of parts of an organ that are not normally joined together.

C-reactive protein (CRP)—A protein present in blood serum in various abnormal states, like inflammation.

Ectopic—Located away from normal position; ectopic pregnancy results in the attachment and growth of the fertilized egg outside of the uterus, a life-threatening condition.

Endometriosis—The presence and growth of functioning endometrial tissue in places other than the uterus; often results in severe pain and infertility.

Erythrocyte sedimentation rate (ESR)—The rate at which red blood cells settle out in a tube of unclotted blood, expressed in millimeters per hour; elevated sedimentation rates indicate the presence of inflammation.

Fibrosis—The formation of fibrous, or scar, tissue which may follow inflammation and destruction of normal tissue.

Hysterectomy—Surgical removal of the uterus.

Laparoscope—A thin flexible tube with a light on the end that is used to examine the inside of the abdomen; the tube is inserted into the abdomen by way of a small incision just below the navel.

(*Caulophyllum thalictroides*) and false unicorn root (*Chamaelirium luteum*) are recommended as tonics for the general well-being of the female genital tract.

Prognosis

PID can be cured if the initial infection is treated immediately. If infection is not recognized, as frequently happens, the process of tissue destruction and scarring that results from inflammation of the tubes results in irreversible changes in the tube structure that cannot be restored to normal. Subsequent bouts of PID increase a woman's risks manyfold. Thirty to forty percent of cases of female **infertility** are due to acute salpingitis.

With modern antibiotic therapy, **death** from PID is almost nonexistent. In rare instances, death may occur from the rupture of tubo-ovarian abscesses and the resulting infection in the abdominal cavity. One recent study has linked infertility, a consequence of PID, with a higher risk of **ovarian cancer**.

Prevention

The prevention of PID is a direct result of the prevention and prompt recognition and treatment of STDs or of any suspected infection involving the female genital tract. The main symptom of infection is an abnormal discharge. To distinguish an abnormal discharge from the mild fluctuations of normal discharge associated with the menstrual cycle takes vigilance and self-awareness. Sexually active women must be able to detect symptoms of lower genital tract disease. Ideally these women will be able to have a frank dialogue regarding their sexual history, risks for PID, and treatment options with their physicians. Also, these women should have open discussions with their sexual partners regarding disclosure of significant symptoms of possible infection.

Lifestyle changes should be geared to preventing the transfer of organisms when the body's delicate lining cells are unprotected or compromised. Barrier contraceptives, such as condoms, diaphragms, and cervical caps should be used. Women in monogamous relationships should use barrier contraceptives during menses and take their physician's advice regarding intercourse following abortion, childbirth, or biopsy procedures.

Resources

BOOKS

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Karen J. Wells

Pelvic relaxation

Definition

Pelvic relaxation is a weakening of the supportive muscles and ligaments of the pelvic floor. This condition, which affects women and is usually caused by **childbirth**, **aging**, and problems with support, causes the pelvic floor to sag and press into the wall of the vagina.

Description

The pelvic floor normally holds the uterus and the bladder in position above the vagina. When the pelvic floor becomes stretched and damaged,

these organs can sag into the vagina, sometimes bulging out through the vaginal opening. A sagging uterus is referred to as a uterine prolapse, pelvic floor **hernia**, or pudendal hernia. A sagging bladder is referred to as a bladder prolapse, or cystocele. Other organs, such as the rectum and intestine, can also sag into the vagina as a result of a weakened pelvic floor.

Causes and symptoms

Childbirth increases the risk of pelvic relaxation. Other causes include **constipation**, a chronic **cough**, **obesity**, and heavy lifting. Some women develop the condition after **menopause**, when the body loses the estrogen that helps maintain muscle tone. Mild pelvic relaxation may cause no symptoms. More severe pelvic relaxation can cause the following symptoms:

- an aching sensation in the vagina, lower abdomen, groin or lower back
- heaviness or pressure in the vaginal area, as if something is about to “fall out” of the vagina
- bladder control problems that worsen with heavy lifting, coughing, or sneezing
- frequent urinary tract infections
- difficulty having a bowel movement

Diagnosis

A thorough **pelvic exam** can help diagnose pelvic relaxation, as can tests of bladder function.

Treatment

Exercises called Kegel exercises can strengthen pelvic floor muscles and lessen the symptoms of pelvic relaxation. These exercises involve squeezing the muscles that stop the flow of urine. The pelvic floor can also be strengthened by estrogen supplements. Physicians sometimes prescribe the insertion of a supportive ring-shaped device called a pessary into the vagina, to prevent the uterus and bladder from pressing into the vagina. Sometimes surgery is recommended to repair a sagging bladder or uterus, and sometimes surgical removal of the uterus (**hysterectomy**) is recommended. Patients are often advised to adhere to a high-fiber diet to reduce the strain of bowel movements, maintain a moderate weight, and avoid activities that strain the pelvic floor. They are sometimes prescribed medications to help control urination and prevent leakage.

KEY TERMS

Cystocele—Bulging of the bladder into the vagina.

Cystourethrocele—Bulging of the bladder neck into the vagina.

Enterocoele—Bulging of the intestine into the upper part of the vagina.

Kegel exercises—Pelvic muscle exercises that strengthen bladder and bowel control.

Pessary—A device inserted into the vagina to support sagging organs.

Rectocele—Bulging of the rectum into the vaginal wall.

Uterine prolapse—Bulging of the uterus into the vagina.

Vaginal prolapse—Bulging of the top of the vagina into the lower vagina or outside the opening of the vagina.

Prognosis

Mild cases of pelvic relaxation can sometimes be reversed through Kegel exercises, while severe cases usually do not respond to **exercise** or estrogen therapy, but usually require pessary support or surgery.

Prevention

To limit **stress** on the pelvic support system, women are advised to maintain a normal body weight, limit heavy lifting, and avoid unnecessary straining to have bowel movements.

Resources

ORGANIZATIONS

- American College of Obstetricians and Gynecologists. 409 12th St., S.W., P.O. Box 96920, Washington, DC 20090-6920. <<http://www.acog.org>>.
- American Foundation for Urologic Disease. 1128 North Charles Street, Baltimore, MD 21201. (800) 242-2383. <<http://www.afud.org>>.
- National Association For Continence. P.O. Box 8310, Spartanburg, SC 29305-8310. (800) BLADDER. <<http://www.nafc.org>>.
- National Kidney and Urologic Diseases Information Clearinghouse. 3 Information Way, Bethesda, MD, 20892. NIH Publication No. 97-4195. (800) 891-5390. <<http://www.niddk.nih.gov/health/kidney/nkudic.htm>>.

Ann Quigley

Pelvic ultrasound

Definition

Pelvic ultrasound is a procedure where harmless, high-frequency sound waves are projected into the abdomen. These waves reflect off of the internal structures and create shadowy black and white pictures on a display screen.

Purpose

Ultrasound is performed routinely during **pregnancy**. Early in the pregnancy (at about seven weeks), it might be used to determine the size of the uterus or the fetus, to detect multiple or **ectopic pregnancy**, to confirm that the fetus is alive (or viable), or to confirm the due date. Toward the middle of the pregnancy (at about 16–20 weeks), ultrasound may be used to confirm fetal growth, to reveal defects in the anatomy of the fetus, and to check the placenta. Toward the end of pregnancy, it may be used to evaluate fetal size, position, growth, or to check the placenta. Doctors may use ultrasound during diagnostic procedures like **amniocentesis** and **chorionic villus sampling**. Both of these tests use long needles inserted through the mother's abdomen into the uterus or placenta to gather cells. Ultrasound can also be used in men or women to examine other internal organs, such as the liver, gallbladder, kidney, and heart. The procedure can be useful in detecting cysts, tumors, and cancer of the uterus, ovaries, and breasts.

Precautions

There are no special precautions recommended before an ultrasound examination. Unlike x rays, ultrasound does not produce any harmful radiation and does not pose a risk to the mother or the fetus. While many women have an ultrasound as part of their prenatal care, there may be no medical need to perform the procedure.

Description

Ultrasound examinations can be done in a doctor's office, clinic, or hospital setting. Typically, the pregnant woman will lie on an examination table with her abdomen exposed. Gel or oil is applied to the area. The doctor or technician will move a hand-held scanner (called a transducer) over the abdomen. The transducer emits high-frequency sound waves (usually in the range of 3.5–7.0 megahertz) into

the abdomen. The waves are reflected back to the transducer and the wave patterns are shown as an image on a display screen. An ultrasound scan reveals the shapes, densities, and even movements of organs and tissues. Although the pictures transmitted by an ultrasound scan appear gray and grainy, a trained technician can identify the fetus within the uterus, monitor its heartbeat, and sometimes determine its sex. Using computerized tools, the technician can measure various structures shown on the screen. For example, the length of the upper thigh bone (femur) or the distance between the two sides of the skull can indicate the age of the fetus.

Ultrasound technology has been used safely in medical settings for over 30 years, and several significant improvements have been made to the procedure. A specially designed transducer probe can be placed in the vagina to provide better ultrasound images. This transvaginal or endovaginal scan is particularly useful in early pregnancy or in cases where ectopic pregnancy is suspected. Doppler ultrasound uses enhanced sound waves to monitor subtle events, like the flow of fetal blood through the heart and arteries. Color imaging is a recent addition to ultrasound technology. With this process, color can be assigned to the various shades of gray for better visualization of subtle tissue details. A new technology under development is three-dimensional ultrasound, which has the potential for detecting even very subtle fetal defects.

Preparation

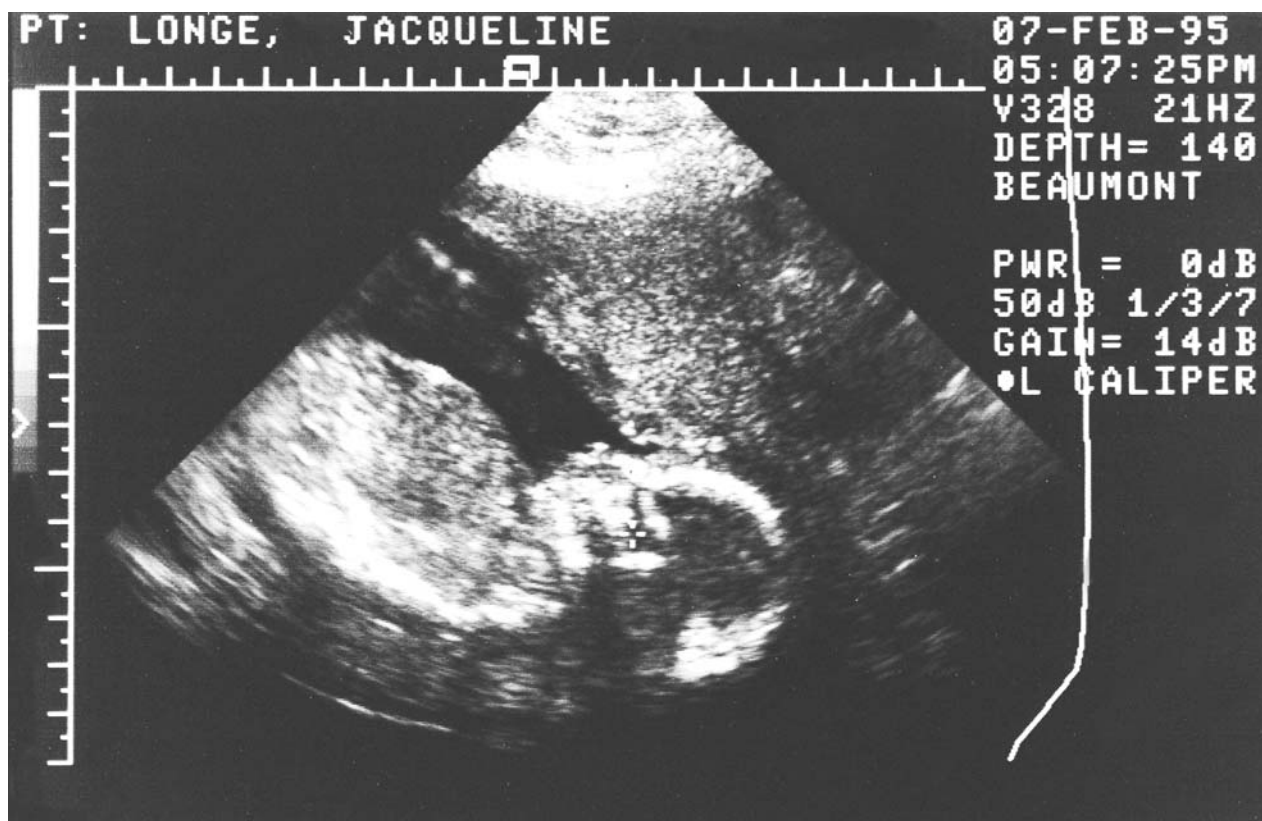
Before undergoing a pelvic ultrasound, a woman may be asked to drink several glasses of water and to avoid urinating for about one hour before the examination. When the bladder is full, the uterus and fetus are easier to see. A lubricating gel or mineral oil may be applied to the area to make moving the transducer easier.

Aftercare

The lubricating jelly or oil applied to the abdomen is wiped off at the end of the procedure. After an ultrasound examination, a patient can immediately resume normal activities.

Risks

There are no known risks, to either the mother or the fetus, associated with the use of ultrasound.



An ultrasound image of Anabelle Ashlyn Longe at 20 weeks. (Courtesy of Jacqueline Longe.)

KEY TERMS

Amniocentesis—A procedure where a needle is inserted through the pregnant mother's abdomen and into the uterus to draw off some of the amniotic fluid surrounding the fetus.

Chorionic villus sampling—A procedure where a needle is inserted into the placenta to draw off some of the placenta's inner wall cells surrounding the fetus.

Ectopic pregnancy—A pregnancy where the fertilized egg becomes implanted somewhere other than in the uterus. A tubal pregnancy is when the fertilized egg implants in the fallopian tube.

Fetus—A term for an unborn baby, usually from the end of week eight to the moment of birth.

Placenta—The organ that allows interchange between the fetus and the mother. Blood from the fetus and the mother do not directly mix, but the thin placental membrane allows the fetus to absorb nutrients and oxygen from the mother. Waste products from the fetus can exit through the placenta.

Ultrasonography—Another term for ultrasound.

Normal results

The reliability of ultrasound readings depends on the skill of the technician or doctor performing the scan. Patients should be aware that fetal abnormalities cannot be detected with 100% accuracy using ultrasound. A normal ultrasound result does not necessarily guarantee that the fetus will be normal.

Abnormal results

Ultrasound examinations in obstetrics may detect abnormalities or defects in the fetus. This information may reveal that the fetus cannot survive on its own after birth or that it will require extensive treatment or care. Some surgical procedures can be performed to correct defects while the fetus is still in the uterus. Parents faced with information regarding possible **birth defects** may require counseling to consider their choice to either continue or end the pregnancy.

The diagnostic use of ultrasound may reveal the presence of cysts, tumors, or **cancer** in internal organs.

Resources

ORGANIZATIONS

American Institute of Ultrasound in Medicine. 14750 Sweitzer Lane, Suite 100, Laurel, MD 20707-5906. (800) 638-5352. <<http://www.aium.org>>.

Altha Roberts Edgren

Penicillin V see **Penicillins**

Penicillins

Definition

Penicillins are medicines that kill bacteria or prevent their growth.

Purpose

Penicillins are **antibiotics** (medicines used to treat infections caused by microorganisms). There are several types of penicillins, each used to treat different kinds of infections, such as skin infections, dental infections, ear infections, respiratory tract infections, urinary tract infections, **gonorrhea**, and other infections caused by bacteria. These drugs will *not* work for colds, flu, and other infections caused by viruses.

Description

Examples of penicillins are penicillin V (Beepen-VK, Pen-Vee K, V-cillin K, Veetids) and amoxicillin (Amoxil, Polymox, Trimox, Wymox). Penicillins are sometimes combined with other ingredients called beta-lactamase inhibitors, which protect the penicillin from bacterial enzymes that may destroy it before it can do its work. The drug Augmentin, for example, contains a combination of amoxicillin and a beta-lactamase inhibitor, clavulanic acid.

Penicillins are available only with a physician's prescription. They are sold in capsule, tablet (regular and chewable), liquid, and injectable forms.

Recommended dosage

The recommended dosage depends on the type of penicillin, the strength of the medicine, and the medical problem for which it is being taken. Check with the physician who prescribed the drug or the pharmacist who filled the prescription for the correct dosage.

Always take penicillins exactly as directed. Never take larger, smaller, more frequent, or less frequent doses. To make sure the infection clears up completely, take the medicine for as long as it has been prescribed. Do not stop taking the drug just because symptoms begin to improve. This is important with all types of infections, but it is especially important with "strep" infections, which can lead to serious heart problems if they are not cleared up completely.

Take this medicine only for the infection for which it was prescribed. Different kinds of penicillins cannot be substituted for one another. Do not save some of the medicine to use on future infections. It may not be the right treatment for other kinds of infections, even if the symptoms are the same.

Penicillins work best when they are at constant levels in the blood. To help keep levels constant, take the medicine in doses spaced evenly through the day and night. Do not miss any doses.

Some penicillins, notably penicillin V, should be taken on an empty stomach, but others may be taken with food. Check package directions or ask the physician or pharmacist for instructions on how to take the medicine.

Precautions

Symptoms should begin to improve within a few days of beginning to take this medicine. If they do not, or if they get worse, check with the physician who prescribed the medicine.

Penicillins may cause **diarrhea**. Certain diarrhea medicines may make the problem worse. Check with a physician before using any diarrhea medicine to treat diarrhea caused by taking penicillin. If diarrhea is severe, check with a physician as soon as possible. This could be a sign of a serious side effect.

Penicillins may change the results of some medical tests. Before having medical tests, patients who are taking penicillin should be sure to let the physician in charge know that they are taking this medicine.

Special conditions

People with certain medical conditions or who are taking certain other medicines can have problems if they take penicillins. Before taking these drugs, be sure to let the physician know about any of these conditions:

ALLERGIES. People who have had **fever**, **asthma**, **eczema**, or other general **allergies** (or who have had such allergies in the past) may be more likely to have severe reactions to penicillins. They should be sure their health care provider knows about their allergies.

Anyone who has had unusual reactions to penicillins or **cephalosporins** in the past should let his or her physician know before taking the drugs again. The physician should also be told about any allergies to foods, dyes, preservatives, or other substances.

LOW-SODIUM DIET. Some penicillin medicines contain large enough amounts of sodium to cause problems for people on low-sodium **diets**. Anyone on such a diet should make sure that the physician treating the infection knows about the special diet.

DIABETES. Penicillins may cause false positive results on urine sugar tests for diabetes. People with diabetes should check with their physicians to see if they need to change their diet or the doses of their diabetes medicine.

PHENYLKETONURIA. Some formulations of Augmentin contain phenylalanine. People with **phenylketonuria** (PKU) should consult a physician before taking this medicine.

OTHER MEDICAL CONDITIONS. Before using penicillins, people with any of these medical problems should make sure their physicians are aware of their conditions:

- bleeding problems
- congestive **heart failure**
- **cystic fibrosis**
- kidney disease
- mononucleosis (“mono”)
- stomach or intestinal problems, especially ulcerative colitis

USE OF CERTAIN MEDICINES. Taking penicillins with certain other drugs may affect the way the drugs work or may increase the chance of side effects.

Side effects

The most common side effects are mild diarrhea, headache, vaginal **itching** and discharge, sore mouth or tongue, or white patches in the mouth or on the tongue. These problems usually go away as the body adjusts to the drug and do not require medical treatment unless they continue or they are bothersome.

More serious side effects are not common, but may occur. If any of the following side effects occur, get emergency medical help immediately:

- breathing problems, such as **shortness of breath** or fast or irregular breathing
- fever
- sudden lightheadedness or faintness

KEY TERMS

Enzyme—A type of protein that brings about or speeds up chemical reactions.

Microorganism—An organism that is too small to be seen with the naked eye.

Mononucleosis—An infectious disease with symptoms that include severe fatigue, fever, sore throat, and swollen lymph nodes in the neck and armpits. Also called “mono.”

- joint **pain**
- skin rash, **hives**, itching, or red, scaly skin
- swelling or puffiness in the face

Other rare side effects may occur. Anyone who has unusual symptoms after taking penicillin should get in touch with his or her physician.

Interactions

Birth control pills may not work properly when taken at the same time as penicillin. To prevent **pregnancy**, use additional methods of birth control while taking penicillin, such as latex condoms or spermicide.

Penicillins may interact with many other medicines. When this happens, the effects of one or both of the drugs may change or the risk of side effects may be greater. Anyone who takes penicillin should let the physician know all other medicines he or she is taking. Among the drugs that may interact with penicillins are:

- Acetaminophen (Tylenol) and other medicines that relieve pain and inflammation
- medicine for overactive thyroid
- male hormones (androgens)
- female hormones (estrogens)
- other antibiotics
- blood thinners
- Disulfiram (Antabuse), used to treat alcohol **abuse**
- antiseizure medicines such as Depakote and Depakene
- blood pressure drugs such as Capoten, Monopril, and Lotensin

The list above does not include every drug that may interact with penicillins. Be sure to check with a

physician or pharmacist before combining penicillins with any other prescription or nonprescription (over-the-counter) medicine.

Nancy Ross-Flanigan

Penile cancer

Definition

Penile **cancer** is the growth of malignant cells on the external skin and in the tissues of the penis.

Description

Penile cancer is a disease in which cancerous cells appear on the penis. If left untreated, this cancer can grow and spread from the penis to the lymph nodes in the groin and eventually to other parts of the body.

Demographics

Penile cancer is a rare form of cancer that develops in about one out of 100,000 men per year in the United States. Penile cancer is more common in other parts of the world, particularly Africa and Asia. In Uganda, penile cancer is the most common form of cancer for men.

Causes and symptoms

The cause of penile cancer is unknown. The most common symptoms of penile cancer are:

- a tender spot, an open sore, or a wart-like lump on the penis
- unusual liquid discharges from the penis
- pain or bleeding in the genital area

Diagnosis

In order to diagnose penile cancer, the doctor examines the patient's penis for lumps or other abnormalities. A tissue sample, or biopsy, may be ordered to distinguish cancerous cells from **syphilis** and penile **warts**. If the results confirm a diagnosis of cancer, additional tests are done to determine whether the disease has spread to other parts of the body.

Treatment

In Stage I penile cancer, malignant cells are found only on the surface of the head (glans) and on the foreskin of the penis. If the cancer is limited to the foreskin, treatment may involve wide local excision and **circumcision**. Wide local excision is a form of surgery that removes only cancer cells and a small amount of normal tissue adjacent to them. Circumcision is removal of the foreskin.

If the Stage I cancer is only on the glans, treatment may involve the use of a fluorouracil cream (Adrucil, Efudex), and/or microsurgery. Microsurgery removes cancerous tissue and the smallest possible amount of normal tissue. During microsurgery, the doctor uses a special instrument that provides a comprehensive view of the area where cancer cells are located and makes it possible to determine that all malignant cells have been removed.

In Stage II, the penile cancer has spread to the surface of the glans, tissues beneath the surface, and the shaft of the penis. The treatment recommended may be **amputation** of all or part of the penis (total or partial penectomy). If the disease is diagnosed early enough, surgeons are often able to preserve enough of the organ for urination and sexual activity. Treatment may also include microsurgery and external **radiation therapy**, in which a machine provides radiation to the affected area. **Laser surgery** is an experimental treatment for Stage II cancers. Laser surgery uses an intense precisely focused beam of light to dissolve or burn away cancer cells.

In Stage III, malignant cells have spread to lymph nodes in the groin, where they cause swelling. The recommended treatment may include amputation of the penis and removal of the lymph nodes on both sides. Radiation therapy may also be suggested. More advanced disease requires systemic treatments using drugs (**chemotherapy**). In chemotherapy, medicines are administered intravenously or taken by mouth. These drugs enter the bloodstream and kill cancer cells that have spread to any part of the body.

In Stage IV, the disease has spread throughout the penis and lymph nodes in the groin, or has traveled to other parts of the body. Treatments are similar to that for Stage III cancer.

Recurrent penile cancer is disease that recurs in the penis or develops in another part of the body after treatment has eradicated the original cancer cells.

KEY TERMS

Circumcision—Surgical removal of the foreskin of the penis. It is usually performed shortly after birth.

Fluorouracil—A cell-killing (cytotoxic) medication that can be applied in cream form to treat cancer of the penis.

Cure rates are high for cancers diagnosed in Stage I or II, but much lower for Stages III and IV, by which time cancer cells have spread to the lymph nodes.

Alternative treatment

In addition to the treatments previously described, biological therapy is another treatment that is currently being studied. Biological therapy is a type of treatment that is sometimes called biological response modifier (BRM) therapy. It uses natural or artificial substances to boost, focus, or reinforce the body's disease-fighting resources.

Prevention

Conditions which increase a person's chance of getting penile cancer include:

- infection with **genital warts** (human papillomavirus, or HPV)
- a skin disease called psoriasis
- a condition called **phimosis**, in which the foreskin becomes difficult to retract
- other conditions that result in repeated irritation of the penis.
- a history of smoking.

There appears to be a connection between development of the disease and lack of personal hygiene. Failure to regularly and thoroughly cleanse the part of the penis covered by the foreskin increases the risk of developing the disease. Penile cancer is also more common in uncircumcised men.

Resources

ORGANIZATIONS

American Cancer Society. 1599 Clifton Road NE, Atlanta, GA 30329. (800) ACS-2345.
< <http://www.cancer.org> >.

Cancer Group Institute. 17620 9th Ave. NE,
North Miami Beach, FL 33162.
(305) 493-1980. < <http://www.cancergroup.com> >.

OTHER

CancerNet: Penile Cancer. March 25, 2001. [cited June 28, 2001]. < http://www.cancernet.nci.nih.gov/cancer_types/penile_cancer.shtml >.

Maureen Haggerty
Paul A. Johnson, Ed.M.

Penile implant surgery see **Penile prostheses**

Penile prostheses

Definition

Penile prostheses are semirigid or inflatable devices that are implanted into penises to alleviate **impotence**.

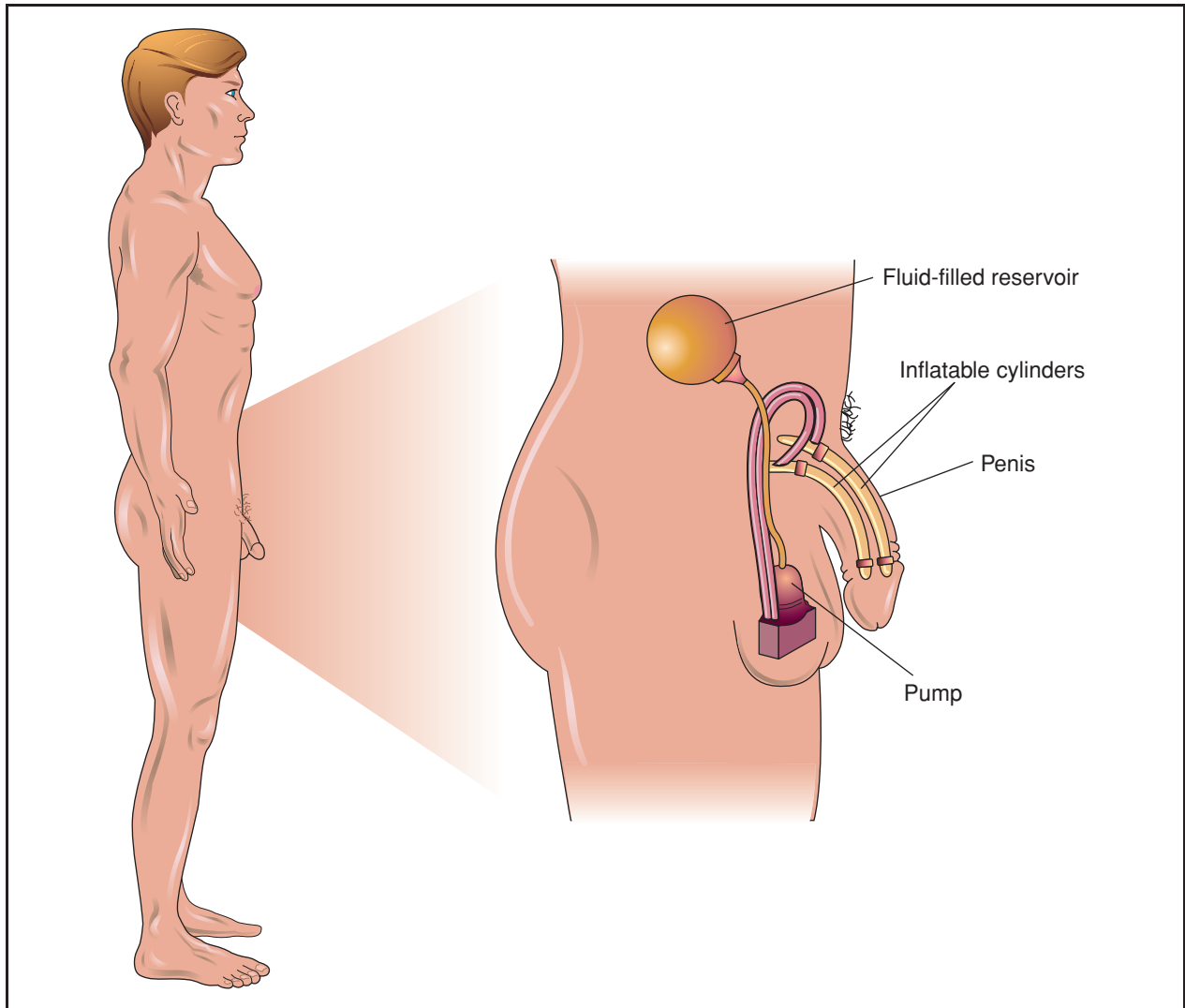
Purpose

The penis is composed of one channel for urine and semen and three compartments with tough, fibrous walls containing "erectile tissue." With appropriate stimulation, the blood vessels that lead out of these compartments constrict, trapping blood. Blood pressure fills and hardens the compartments producing an erection of sufficient firmness to perform sexual intercourse. Additional stimulation leads to ejaculation, where semen is pumped out of the urethra. When this system fails, impotence (failure to create and maintain an erection) occurs.

Impotence can be caused by a number of conditions, including diabetes, **spinal cord injury**, prolonged drug **abuse**, and removal of a prostate gland. If the medical condition is irreversible, a penile prosthesis may be considered. Patients whose impotence is caused by psychological problems are not recommended for implant surgery.

Description

Penile implant surgery is conducted on patients who have exhausted all other areas of treatment. The semirigid device consists of two rods that are easier and less expensive to implant than the inflatable cylinders. Once implanted, the semirigid device needs no follow-up adjustments, however it produces a penis which constantly remains semi-erect. The inflatable



The inflatable implant is a common penile prosthesis. This device connects through a tube to a flexible fluid reservoir and a pump. The pump is shaped like a testicle and inserted in the scrotum. When the pump is squeezed, the fluid is forced into the inflatable cylinders implanted inside the penis, producing an erection. (Illustration by Electronic Illustrators Group.)

cylinders produce a more natural effect. The patient is able to simulate an erection by using a pump located in the scrotum.

With the patient asleep under **general anesthesia**, the device is inserted into the erectile tissue of the penis through an incision in the fibrous wall. In order to implant the pump for the inflatable implant, incisions are made in the abdomen and the perineum (area between the anus and the genitals). A fluid reservoir is inserted into the groin and the pump is placed in the scrotum. The cylinders, reservoir, and pump are connected by tubes and tested before the incisions are closed.

Preparation

Surgery always requires an adequately informed patient, both as to risks and benefits. In this case, the sexual partner should also be involved in the discussion. Prior to surgery, antibacterial cleansing occurs and the surrounding areas are shaved.

Aftercare

To minimize swelling, ice packs are applied to the penis for the first 24 hours following surgery. The incision sites are cleansed daily to prevent infection. **Pain** relievers may be taken.

KEY TERMS

General anesthesia—Deep sleep induced by a combination of medicines that allows surgery to be performed.

Genital—Sexual organ.

Perineum—Area between the anus and genitals.

Scrotum—The external pouch containing the male reproductive glands (testes) and part of the spermatic cord.

Risks

With any implant, there is a slightly greater risk of infection. The implant may irritate the penis and cause continuous pain. The inflatable prosthesis may need follow-up surgery to repair leaks in the reservoir or to reconnect the tubing.

Resources

BOOKS

Jordan, Gerald H., et al. "Surgery of the Penis and Urethra." In *Campbell's Urology*, edited by Patrick C. Walsh, et al. Philadelphia: W. B. Saunders Co., 1998.

J. Ricker Polsdorfer, MD

Pentoxifylline see **Blood-viscosity reducing drugs**

Peptic ulcer disease see **Helicobacteriosis**

Percutaneous renal biopsy see **Kidney biopsy**

Percutaneous transhepatic cholangiography

Definition

Percutaneous transhepatic cholangiography (PTHC) is an x-ray test used to identify obstructions either in the liver or bile ducts that slow or stop the flow of bile from the liver to the digestive system.

Purpose

Because the liver and bile ducts are not normally seen on x rays, the doctor injects the liver with a special dye that will show up on the resulting picture. This dye distributes evenly to fill the whole liver drainage

system. If the dye does not distribute evenly, this is indicative of a blockage, which may be caused by a gallstone or a tumor in the liver, bile ducts, or pancreas.

Precautions

Patients should report allergic reactions to:

- anesthetics
- dyes used in medical tests
- iodine
- shellfish

PTHC should not be performed on anyone who has **cholangitis** (inflammation of the bile duct), massive **ascites**, a severe allergy to iodine, or a serious uncorrectable or uncontrollable bleeding disorder. Patients who have diabetes should inform their doctor.

Description

PTHC is performed in a hospital, doctor's office, or outpatient surgical or x-ray facility. The patient lies on a movable x-ray table and is given a local anesthetic. The patient will be told to hold his or her breath, and a doctor, nurse, or laboratory technician will inject a special dye into the liver as the patient exhales.

The patient may feel a twinge when the needle penetrates the liver, a pressure or fullness, or brief discomfort in the upper right side of the back. Hands and feet may become numb during the 30–60 minute procedure.

The x-ray table will be rotated several times during the test, and the patient helped to assume a variety of positions. A special x-ray machine called a fluoroscope will track the dye's movement through the bile ducts and show whether the fluid is moving freely or if its passage is obstructed.

PTHC costs about \$1,600. The test may have to be repeated if the patient moves while x rays are being taken.

Preparation

An intravenous antibiotic may be given every 4–6 hours during the 24 hours before the test. The patient will be told to fast overnight. Having an empty stomach is a safety measure in case of complications, such as bleeding, that might require emergency repair surgery. Medications such as **aspirin**, or **non-steroidal**

KEY TERMS

Ascites—Abnormal accumulation of fluid in the abdomen.

Bile ducts—Tubes that carry bile, a thick yellowish-green fluid that is made by the liver, stored in the gallbladder, and helps the body digest fats.

Cholangitis—Inflammation of the bile duct.

Fluoroscope—An x-ray machine that projects images of organs.

Granulomatous disease—Characterized by growth of tiny blood vessels and connective tissue.

Jaundice—Disease that causes bile to accumulate in the blood, causing the skin and whites of the eyes to turn yellow. Obstructive jaundice is caused by blockage of bile ducts, while non-obstructive jaundice is caused by disease or infection of the liver.

anti-inflammatory drugs that thin the blood, should be stopped three–seven days prior to taking the PRHC test. Patients may also be given a sedative a few minutes before the test begins.

Aftercare

A nurse will monitor the patient's vital signs and watch for:

- itching
- flushing
- nausea and vomiting
- sweating
- excessive flow of saliva
- possible serious allergic reactions to contrast dye

The patient should stay in bed for at least six hours after the test, lying on the right side to prevent bleeding from the injection site. The patient may resume normal eating habits and gradually resume normal activities. The doctor should be informed right away if **pain** develops in the right abdomen or shoulder or in case of **fever**, **dizziness**, or a change in stool color to black or red.

Risks

Septicemia (blood **poisoning**) and bile **peritonitis** (a potentially fatal infection or inflammation of the membrane covering the walls of the abdomen) are rare but serious complications of this procedure. Dye

occasionally leaks from the liver into the abdomen, and there is a slight risk of bleeding or infection.

Normal results

Normal x rays show dye evenly distributed throughout the bile ducts. **Obesity**, gas, and failure to fast can affect test results.

Abnormal results

Enlargement of bile ducts may indicate:

- obstructive or non-obstructive **jaundice**
- cholelithiasis (gallstones)
- hepatitis (inflammation of the liver)
- cirrhosis (chronic liver disease)
- granulomatous disease
- pancreatic cancer
- bile duct or gallbladder cancers

Resources

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Percutaneous Transhepatic Cholangiography. < <http://207.25.144.143/health/Library/medtests/> >.

Percutaneous Transhepatic Cholangiography (PTHC). < <http://www.uhs.org/frames/health/test/test3554.htm> >.

Test Universe Site: Percutaneous Transhepatic Cholangiography. < <http://www.testuniverse.com/mdx/MDX-3055.html> >.

Maureen Haggerty

Perforated eardrum

Definition

A perforated eardrum exists when there is a hole or rupture in the eardrum, the thin membrane that separates the outer ear canal from the middle ear. A perforated eardrum may cause temporary **hearing loss** and occasional discharge.

Description

The eardrum (tympanic membrane) is a thin wall that separates the outer ear from the middle ear, vibrating when sound waves strike the membrane. The middle ear is connected to the nose by the Eustachian tube.

In addition to conducting sound, the eardrum also protects the middle ear from bacteria. When it is perforated, bacteria can more easily get into this part of the ear, causing ear infections.

In general, the larger the hole in the eardrum, the greater the temporary loss of hearing. The location of the perforation also affects the degree of hearing loss. Severe hearing loss may follow a skull fracture that disrupts the bones in the middle ear. Eardrum perforation caused by a loud noise may result in ringing in the ear (**tinnitus**), in addition to a temporary hearing loss. Over time, this hearing loss improves and the ringing usually fades in a few days.

Causes and symptoms

The eardrum can become damaged by a direct injury. It is possible to perforate the eardrum:

- with a cotton-tipped swab or another foreign object
- by hitting the ear with an open hand
- after a skull fracture
- after a loud explosion or other loud noise

In addition, an ear infection can rupture the eardrum as pressure within the middle ear rises when fluid builds up. If the eardrum is punctured by pressure from an ear infection, there may be infected or bloody drainage from the ear.

Rarely, a small hole may remain in the eardrum after a pressure-equalizing tube falls out or is removed by a doctor.

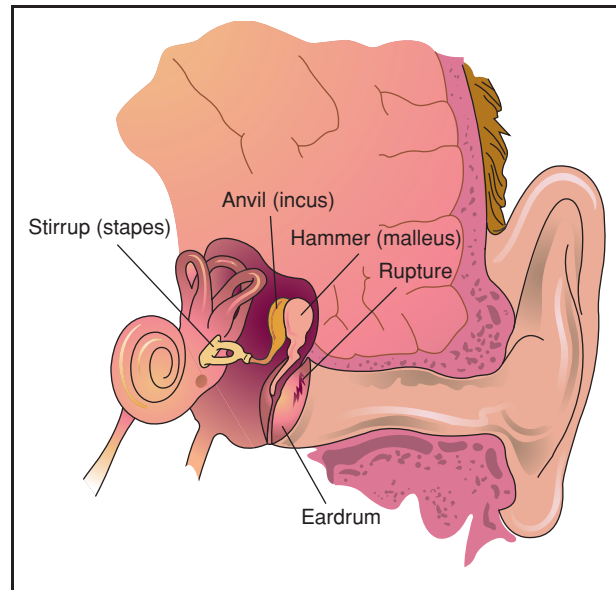
Symptoms include an earache or **pain** in the ear, which may be severe, or a sudden decrease in ear pain, followed by ear drainage of clear, bloody, or pus-filled fluid, hearing loss, or ear noise/buzzing.

Diagnosis

A doctor can diagnose a perforated eardrum by direct inspection with an otoscope. Hearing tests may reveal a hearing loss.

Treatment

A perforated eardrum usually heals by itself within two months. **Antibiotics** may be given to



A perforated eardrum is caused by a hole or rupture in the eardrum, the thin membrane that separates the outer ear canal from the middle ear. It may result in temporary hearing loss and occasional discharge. (Illustration by Electronic Illustrators Group.)

KEY TERMS

Eustachian tube—The air duct that connects the area behind the nose to the middle ear.

Otoscope—An instrument used to examine the ear, to inspect the outer ear canal and the eardrum, and to detect diseases in the middle ear.

prevent infection or to treat an existing ear infection. Painkillers can relieve any ear pain.

Sometimes, a paper patch is placed over the eardrum until the membrane heals. Three or four patches may be needed before the perforation closes completely. If the eardrum does not heal on its own, surgical repair (tympanoplasty) may be necessary.

The ear should be kept clean and dry while the eardrum heals; patients should insert cotton balls into the ear when showering or shampooing to block any water from getting into the ear. Pain in the ear may be eased by applying warm compresses.

Prognosis

While a perforated eardrum may be uncomfortable, it usually heals on its own. Any hearing loss that accompanies the perforation is usually temporary.

Prevention

A perforated eardrum can be prevented by avoiding insertion of any object into the ear to clean it. If a foreign object becomes lodged in the ear, only a doctor should try to remove it.

Promptly treating all ear infections is another way to guard against a ruptured eardrum.

Resources

ORGANIZATIONS

American Academy of Otolaryngology-Head and Neck Surgery, Inc. One Prince St., Alexandria VA 22314-3357. (703) 836-4444. <<http://www.entnet.org>>.

Better Hearing Institute. 515 King Street, Suite 420, Alexandria, VA 22314. (703) 684-3391.

Carol A. Turkington

Perforated septum

Definition

A perforated septum is a hole in the nasal septum, the vertical plane of tissue that separates the nostrils.

Description

The nasal septum is a thin structure in the middle of the nose. In front, it is cartilage, further back it is bone. On either side, it is covered with mucus membranes. The cartilage depends upon the blood vessels in the mucus membranes on either side for its **nutrition**. If that blood supply is shut off, the cartilage dies, producing a hole or perforation.

Causes and symptoms

There are several causes of a perforated septum.

- Wearing ornaments in the nose. To hang an ornament from the middle of the nose requires that the tissue directly in front of the septal cartilage be pierced or perforated.
- Sniffing **cocaine**. Cocaine is a potent vasoconstrictor, which means that it causes small blood vessels to

KEY TERMS

Systemic lupus erythematosus—A collagen-vascular disease in the autoimmune category that causes damage to many different parts of the body.

close. It is used in nose surgery because it shrinks mucus membranes, permitting better visualization and access into the nose. Used continuously, tissues are deprived of blood and die. The nasal septum is the most vulnerable to this effect of sniffing cocaine.

- Getting the septum cauterized. Nosebleeds usually come from the front part of the nasal septum, which is rich in blood vessels. Uncontrolled repeated bleeding from these vessels may require cautery—burning the vessels with electricity or chemicals to close them off. Injudicious cautery of both sides of the septum has in the past led to death of tissue and consequent perforation.
- More and more people are having **cosmetic surgery** done on their nose. The procedure, called **rhinoplasty**, occasionally damages the septum's blood supply.
- Contracting certain diseases. Several diseases—typhoid, **syphilis**, **systemic lupus erythematosus**, and tuberculosis—can infect this tissue and destroy it.
- Being exposed to harmful vapors. Toxic air pollutant-like acid fumes, phosphorus, and copper vapor—and sometimes even cortisone sprays—can destroy nasal tissue.

Perforation is not serious. It causes irritation, mostly complaints of dryness and crusting. Sometimes air blowing past it whistles. Picking at the crusts can cause bleeding.

Treatment

Surgical repair is not difficult. The surgeon may devise a plastic button that fits exactly into the defect and stays in place like a collar button.

Alternative treatment

Saline nasal sprays may be sufficient to control symptoms and prevent the need for surgery.

Prevention

Nosebleeds from the septum can usually be controlled with pinching. Vaginal estrogen cream has also been used successfully to toughen the blood vessels.

Resources

BOOKS

Ballenger, John Jacob. *Disorders of the Nose, Throat, Ear, Head, and Neck*. Philadelphia: Lea & Febiger, 1991.

J. Ricker Polsdorfer, MD

Pericardiocentesis

Definition

Pericardiocentesis is the removal by needle of pericardial fluid from the sac surrounding the heart for diagnostic or therapeutic purposes.

Purpose

The pericardium, the sac (or membrane) that surrounds the heart muscle, normally contains a small amount of fluid that cushions and lubricates the heart as the heart expands and contracts. When too much fluid gathers in the pericardial cavity, the space between the pericardium and the outer layers of the heart, a condition known as pericardial effusion occurs. Abnormal amounts of fluid may result from:

- **pericarditis** (caused by infection, inflammation)
- trauma (producing blood in the pericardial sac)
- surgery or other invasive procedures performed on the heart
- **cancer** (producing malignant effusions)
- myocardial infarction, congestive heart failure
- renal failure

Possible causes of pericarditis include chest trauma, systemic infection (bacterial, viral, or fungal), myocardial infarction (**heart attack**), or **tuberculosis**. When pericarditis is suspected, pericardiocentesis may be advisable in order to obtain a fluid sample for laboratory analysis to identify the underlying cause of the condition.

Pericardiocentesis is also used in emergency situations to remove excessive accumulations of blood or fluid from the pericardial sac, such as with **cardiac tamponade**. When fluid builds up too rapidly or excessively in the pericardial cavity, the resulting compression on the heart impairs the pumping action of the vascular system. Cardiac tamponade is a life-threatening condition that requires immediate treatment.

Precautions

Whenever possible, an echocardiogram (ultrasound test) should be performed to confirm the presence of the pericardial effusion and to guide the pericardiocentesis needle during the procedure. Because of the risk of accidental puncture to major arteries or organs in pericardiocentesis, surgical drainage may be a preferred treatment option for pericardial effusion in non-emergency situations.

Description

The patient's vital signs are monitored throughout the procedure, and an ECG tracing is continuously run. If time allows, **sedation** is administered, the puncture site is cleaned with an antiseptic iodine solution, and a local anesthetic is injected into the skin to numb the area. The patient is instructed to remain still. The physician performing pericardiocentesis will insert a syringe with an attached cardiac needle slowly into the chest wall until the needle tip reaches the pericardial sac. The patient may experience a sensation of pressure as the needle enters the membrane. When the needle is in the correct position, the physician will aspirate, or withdraw, fluid from the pericardial sac.

When the procedure is performed for diagnostic purposes, the fluid will be collected into specimen tubes for laboratory analysis. If the pericardiocentesis is performed to treat a cardiac tamponade or other significant fluid build-up, a pericardial catheter may be attached to the needle to allow for continuous drainage.

After the cardiac needle is removed, pressure is applied to the puncture site for approximately five minutes, and the site is then bandaged.

Preparation

Prior to pericardiocentesis, the test procedure is explained to the patient, along with the risks and possible complications involved, and the patient is asked to sign an informed consent form. If the patient is incapacitated, the same steps are followed with a family member.

No special diet or **fasting** is required for the test. After the patient changes into a hospital gown, an intravenous line is inserted into a vein in the arm. The IV will be used to administer sedation, and any required medications or blood products. Leads for an electrocardiogram (ECG) tracing are attached to the patient's right and left arms and legs, and the fifth lead is attached to the cardiac needle used for the procedure. The patient is instructed to lie flat on the table, with the upper body elevated to a 60 degree angle.

KEY TERMS

Cardiac tamponade—Compression and restriction of the heart that occurs when the pericardium fills with blood or fluid. This increase in pressure outside the heart interferes with heart function and can result in shock and/or death.

Catheter—A long, thin, flexible tube used to drain or administer fluids.

Echocardiogram—An imaging test using high-frequency sound waves to obtain pictures of the heart and surrounding tissues.

Electrocardiogram—A cardiac test that measures the electrical activity of the heart.

Myocardium—The middle layer of the heart wall.

Pericardium—A double membranous sac that envelops and protects the heart.

Aftercare

The site of the puncture and any drainage catheter should be checked regularly for signs of infection such as redness and swelling. Blood pressure and pulse are also monitored following the procedure. Patients who experience continued bleeding or abnormal swelling of the puncture site, sudden **dizziness**, difficulty breathing, or chest pains in the days following a pericardiocentesis procedure should seek immediate medical attention.

Risks

Pericardiocentesis is an invasive procedure, and infection of the puncture site or pericardium is always a risk. Possible complications include perforation of a major artery, lung, or liver. The myocardium, the outer muscle layer of the heart, could also be damaged if the cardiac needle is inserted too deeply.

Normal results

Normal pericardial fluid is clear to straw-colored in appearance with no bacteria, blood, cancer cells or pathogens. There is typically a minimal amount of the fluid (10–50 ml) in the pericardial cavity.

Abnormal results

A large volume of pericardial fluid (over 50 ml) indicates the presence of pericardial effusion. Laboratory analysis of the fluid can aid in the diagnosis of the cause of pericarditis. The presence of an

infectious organism such as *staphylococcus aureus* is a sign of bacterial pericarditis. Excessive protein is present in cases of **systemic lupus erythematosus** or myocardial infarction (heart attack). An elevated white **blood count** may point to a fungal infection. If the patient has a hemorrhage, a cardiac rupture, or cancer, there may be blood in the pericardial fluid.

Resources

BOOKS

Weinstock, Doris et al. eds. “Body system Tests: Cardiovascular System.” In *Illustrated Guide to Diagnostic Tests*, 2nd edition. Springhouse, PA: Springhouse Corporation, 1998.

ORGANIZATION

The American Heart Association. National Center. 7272 Greenville Avenue, Dallas, Texas 75231. (800) AHA-USA 1. <<http://www.americanheart.org>>.

Paula Anne Ford-Martin

Pericarditis

Definition

Pericarditis is an inflammation of the two layers of the thin, sac-like membrane that surrounds the heart. This membrane is called the pericardium, so the term pericarditis means inflammation of the pericardium.

Description

Pericarditis is fairly common. It affects approximately one in 1,000 people. The most common form is caused by infection with a virus. People in their 20s and 30s who have had a recent upper respiratory infection are most likely to be affected, along with men aged 20–50. One out of every four people who have had pericarditis will get it again, but after two years these relapses are less likely.

Causes and symptoms

The viruses that cause pericarditis include those that cause **influenza**, **polio**, and **rubella** (German measles). In children, the most common viruses that cause pericarditis are the adenovirus and the cock-sackievirus (which is most likely to affect children during warmer weather).

Although pericarditis is usually caused by a virus, it also can be caused by an injury to the heart or it can

follow a **heart attack**. It may also be caused by certain inflammatory diseases such as **rheumatoid arthritis** or systemic lupus erythematosus. Bacteria, fungi, parasites, **tuberculosis**, **cancer** or kidney failure may also affect the pericardium. Sometimes the cause is unknown.

There are several forms of pericarditis, depending on the cause.

Acute pericarditis

This is caused by infection with a virus, bacteria, or fungus—usually in the lungs and upper respiratory tract. This form of the disease causes a sharp, severe **pain** that starts in the region of the breastbone. If the pericarditis is caused by a bacteria, it is called bacterial or purulent pericarditis.

Cardiac tamponade

Sometimes fluid collects between the heart and the pericardium. This is called pericardial effusion, and may lead to a condition called cardiac tamponade. When the fluid accumulates, it can squeeze the heart and prevent it from filling with blood. This keeps the rest of the body from getting the necessary supply of oxygen and can cause dangerously low blood pressure. A **cardiac tamponade** can happen when the chest is injured during surgery, radiation therapy, or an accident. Cardiac tamponade is a serious medical emergency and must be treated immediately.

Constrictive pericarditis

When the pericardium is scarred or thickened, the heart has difficulty contracting. This is because the pericardium has shrunk or tightened around the heart, constricting the muscle's heart movement. This usually occurs as a result of tuberculosis, which now is rarely found in the United States, except in immigrant, **AIDS**, and prison populations.

Symptoms of pericarditis

Symptoms likely to be associated with pericarditis include:

- rapid breathing
- breathlessness
- dry **cough**
- fever and chills
- weakness
- broken blood vessels (hemorrhages) in the mucus membrane of the eyes, the back, the chest, fingers, and toes
- feelings of anxiety

- A sharp or dull pain that starts in the front of the chest under the breastbone and radiates to the left side of the neck, upper abdomen, and left shoulder the pain is less intense when the patient sits up or leans forward and worsens when lying down; it may worsen with a deep breath, like pleurisy, which may accompany pericarditis

In cardiac tamponade, neck veins may be swollen and blood pressure may be very low.

Diagnosis

The heart of a person with pericarditis is likely to produce a grating sound (friction rub) when heard through a stethoscope. This sound occurs because the roughened pericardium surfaces are rubbing against each other.

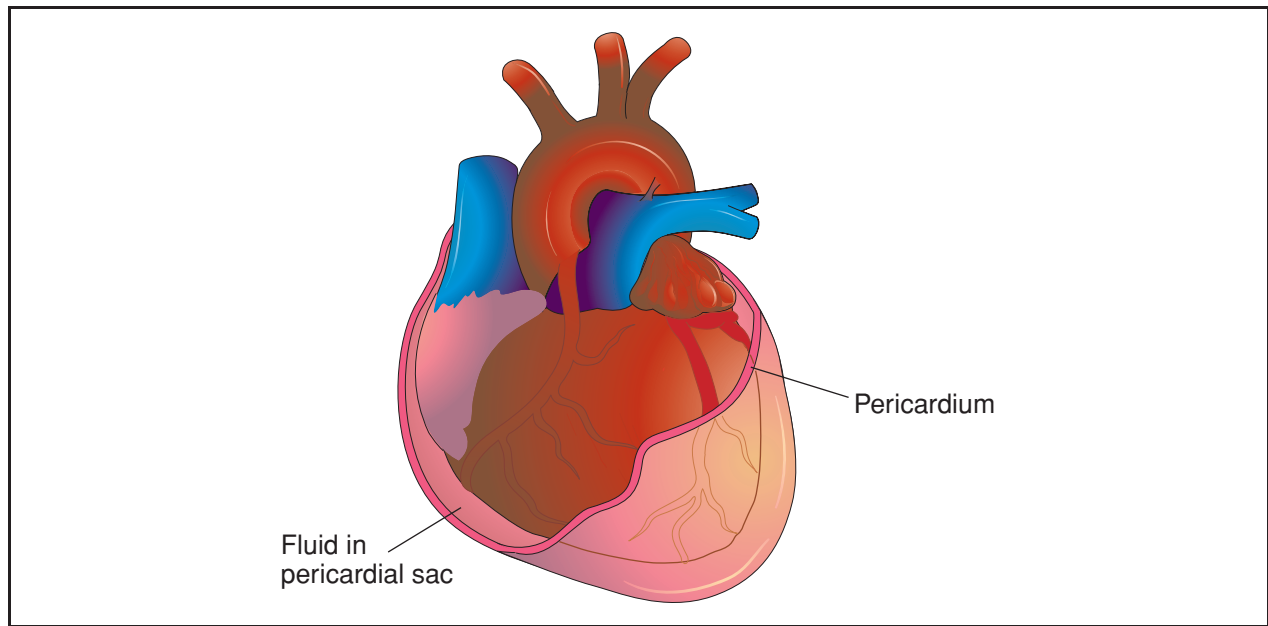
The following tests will also help diagnose pericarditis and what is causing it:

- electrocardiograph (ECG) and echocardiogram to distinguish between pericarditis and a heart attack.
- x ray to show the traditional “water bottle” shadow around the heart that is often seen in pericarditis where there is a sufficient fluid build up.
- computed tomography scan (CT scan) of the chest.
- heart catheterization to view the heart's chambers and valves.
- **pericardiocentesis** to test for viruses, bacteria, fungus, cancer, and tuberculosis.
- blood tests such as LDH and CPK to measure cardiac enzymes and distinguish between a heart attack and pericarditis, as well as a complete blood count (CBC) to look for infection.

Treatment

Since most pericarditis is caused by a virus and will heal naturally, there is no specific, curative treatment. Ordinary **antibiotics** do not work against viruses. Pericarditis that comes from a virus usually clears up in two weeks to three months. Medications may be used to reduce inflammation, however. They include **non-steroidal anti-inflammatory drugs** (NSAIDs), such as ibuprofen and aspirin. **Corticosteroids** are helpful if the pericarditis was caused by a heart attack or **systemic lupus erythematosus**. **Analgesics** (painkillers such as **aspirin** or **acetaminophen**) also may be given.

If the pericarditis recurs, removal of all or part of the pericardium (pericardiectomy) may be necessary. In the case of constrictive pericarditis, the pericardiectomy may be necessary to remove the stiffened



Cardiac tamponade occurs when fluid collects in the pericardial sac between the heart and the surrounding pericardium. A medical emergency, cardiac tamponade deprives the body of oxygen and requires immediate treatment. (Illustration by Electronic Illustrators Group.)

KEY TERMS

Computed tomography (CT) scan—A CT scan uses x rays to scan the body from many angles. A computer compiles the x rays into a picture of the area being studied. The images are viewed on a monitor and printed-out.

Echocardiogram—An echocardiogram bounces sound waves off the heart to create a picture of its chambers and valves.

Electrocardiogram (ECG)—An ECG is a test to measure electrical activity in the heart.

Heart catheterization—A heart catheterization is used to view the heart's chamber and valves. A tube (catheter) is inserted into an artery, usually in the groin. A dye is then put into the artery through the tube. The dye makes its way to the heart to create an image of the heart on x-ray film. The image is photographed and stored for further examination.

Pericardiocentesis—Pericardiocentesis is a procedure used to test for viruses, bacteria, and fungus. The physician puts a small tube through the skin, directly into the pericardial sac, and withdraws fluid. The fluid then is tested for viruses, bacteria, and fungus.

Pericardium—The pericardium is the thin, sac-like membrane that surrounds the heart. It has two layers: the serous pericardium and the fibrous pericardium.

parts of the pericardium that are preventing the heart from beating correctly.

If a cardiac tamponade is present, it may be necessary to drain excess fluid from the pericardium. Pericardiocentesis, the same procedure used for testing, will be used to withdraw the fluid.

For most people, home care with rest and medications to relieve pain are sufficient. A warm heating pad or compress also may help relieve pain. Sitting in an upright position and bending forward helps relieve discomfort. A person with pericarditis may also be kept in bed, with the head of the bed elevated to reduce the heart's need to work hard as it pumps blood. Along with painkillers and antibiotics, diuretic drugs ("water pills") to reduce fluids may also be used judiciously.

Prognosis

Prognosis is good. Most people recover within three weeks to several months and do not need any additional treatment.

Prevention

There is no way to prevent pericarditis, but a healthy lifestyle with proper **nutrition** and **exercise** will help keep the body's immune system strong and more likely to fight off invading microorganisms.

Resources

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300 or (800) 242-8721. inquire@heart.org. <<http://www.americanheart.org>>.

National Heart, Lung and Blood Institute. P.O. Box 30105, Bethesda, MD 20824-0105. (301) 251-1222. <<http://www.nhlbi.nih.gov>>.

Christine Kuehn Kelly

Perinatal infection

Definition

An infection caused by a bacteria or virus that can be passed from a mother to her baby during **pregnancy** or delivery is called a perinatal infection.

Description

Perinatal infections include bacterial or viral illnesses that can be passed from a mother to her baby either while the baby is still in the uterus, during the delivery process, or shortly after birth. Maternal infection can, in some cases, cause complications at birth. The mother may or may not experience active symptoms of the infection during the pregnancy. The most serious and most common perinatal infections, and the impact of these diseases on the mother and infant, are discussed below in alphabetical order. It is important to note that men can become infected and can transmit many of these infections to other women. The sexual partners of women who have these infections also should seek medical treatment.

Causes and symptoms

Chlamydia

Chlamydia trachomatis is the most common bacterial sexually transmitted disease in the United States, causing more than 4 million infections each year. The majority of women with chlamydial infection experience no obvious symptoms. The infection affects the reproductive tract and causes pelvic inflammatory disease, **infertility**, and ectopic pregnancy (the fertilized egg implants somewhere other than in the uterus). This infection can cause premature rupture of the membranes and early labor. It can be passed to the infant during delivery and can cause ophthalmia neonatorum (an eye infection) within the first month of

life and **pneumonia** within one to three months of age. Symptoms of **chlamydial pneumonia** are a repetitive **cough** and rapid breathing. **Wheezing** is rare and the infant does not develop a **fever**.

Cytomegalovirus

Cytomegalovirus (CMV) is a common virus in the herpes virus family. It is found in saliva, urine, and other body fluids and can be spread through sexual contact or other more casual forms of physical contact like kissing. In adults, CMV may cause mild symptoms of swollen lymph glands, fever, and **fatigue**. Many people who carry the virus experience no symptoms at all. Infants can become infected with CMV while still in the uterus if the mother becomes infected or develops a recurrence of the infection during pregnancy. Most infants exposed to CMV before birth develop normally and do not show any symptoms. As many as 6,000 infants who were exposed to CMV before birth are born with serious complications each year. CMV interferes with normal fetal development and can cause mental retardation, blindness, deafness, or epilepsy in these infants.

Genital herpes

Genital herpes, which is usually caused by herpes simplex virus type 2 (HSV-2), is a sexually transmitted disease that causes painful sores on the genitals. Women who have their first outbreak of genital herpes during pregnancy are at high risk of miscarriage or delivering a low birth weight baby. The infection can be passed to the infant at the time of delivery if the mother has an active sore. The most serious risk to the infant is the possibility of developing HSV-2 **encephalitis**, an inflammation of the brain, with symptoms of irritability and poor feeding.

Hepatitis B

Hepatitis B is a contagious virus that causes liver damage and is a leading cause of chronic **liver disease** and cirrhosis. Approximately 20,000 infants are born each year to mothers who test positive for the hepatitis B virus. These infants are at high risk for developing hepatitis B infection through exposure to their mothers blood during delivery.

Human immunodeficiency virus (HIV)

Human **immunodeficiency virus (HIV)** is a serious, contagious virus that causes acquired immunodeficiency syndrome (**AIDS**). About one-fourth of pregnant women with HIV pass the infection on to their

newborn infants. An infant with HIV usually develops AIDS and dies before the age of two.

Human papillomavirus

Human papillomavirus (HPV) is a sexually transmitted disease that causes **genital warts** and can increase the risk of developing some cancers. HPV appears to be transferred from the mother to the infant during the birth process.

Rubella (German measles)

Rubella is a virus that causes **German measles**, an illness that includes rash, fever, and symptoms of an upper respiratory tract infection. Most people are exposed to rubella during childhood and develop antibodies to the virus so they will never get it again. Rubella infection during early pregnancy can pass through the placenta to the developing infant and cause serious **birth defects** including heart abnormalities, mental retardation, blindness, and deafness.

Streptococcus

Group B streptococcus (GBS) infection is the most common bacterial cause of infection and **death** in newborn infants. Although rates have declined in the United States since the introduction of **antibiotics** to at-risk women during labor in the 1980s, about 1,600 cases and 80 newborn deaths still occur each year. In women, GBS can cause vaginitis and urinary tract infections. Both infections can cause premature birth and the bacteria can be transferred to the infant in the uterus or during delivery. GBS causes pneumonia, **meningitis**, and other serious infections in infants.

Syphilis

Syphilis is a sexually transmitted bacterial infection that can be transferred from a mother to an infant through the placenta before birth. Up to 50% of infants born to mothers with syphilis will be premature, stillborn, or will die shortly after birth. Infected infants may have severe birth defects. Those infants who survive infancy may develop symptoms of syphilis up to two years later.

Diagnosis

Chlamydia

Chlamydial bacteria can be diagnosed by taking a cotton swab sample of the cervix and vagina during

the third trimester of the pregnancy. Chlamydial cell cultures take three to seven days to grow but many laboratories are not equipped to run the tests necessary to confirm the diagnosis.

Cytomegalovirus

Past or recent infection with CMV can be identified by antibody tests and CMV can be grown from body fluids.

Genital herpes

The appearance of a genital sore is enough to suspect an outbreak of genital herpes. The sore can be cultured and tested to confirm that HSV-2 is present.

Hepatitis B

A blood test can be used to screen pregnant women for the hepatitis B surface antigen (HBsAg) in prenatal health programs.

Human immunodeficiency virus (HIV)

HIV can be detected using a blood test and is part of most prenatal screening programs.

Human papillomavirus

HPV causes the growth of **warts** in the genital area. The wart tissue can be removed with a scalpel and tested to determine what type of HPV virus caused the infection.

Rubella (German measles)

Pregnant women are usually tested for antibodies to rubella, which would indicate that they have been previously exposed to the virus and therefore would not develop infection during pregnancy if exposed.

Streptococcus

GBS can be detected by a vaginal or rectal swab culture, and sometimes from a **urine culture**. Blood tests can be used to confirm GBS infection in infants who exhibit symptoms.

Syphilis

Pregnant women are usually tested for syphilis as part of the prenatal screening.

Treatment

Chlamydia

Pregnant women can be treated during the third trimester with oral erythromycin, for seven–14 days depending on the dose used. Newborn infants can be treated with erythromycin liquid for 10–14 days at a dosage determined by their body weight.

Cytomegalovirus

No drugs or vaccines are currently available for prevention or treatment of CMV.

Genital herpes

The **antiviral drugs** acyclovir or famciclovir can be administered to the mother during pregnancy. Little is known about the risks of these drugs to the fetus, however, the risk of birth defects does not seem to be any higher than for women who do not take these medications. Infants with suspected HSV-2 can be treated with acyclovir. Delivery of the infant by **cesarean section** is recommended if the mother has an active case of genital herpes.

Hepatitis B

Infants born to mothers who test positive to the HBsAg test should be treated with hepatitis B immune globulin at birth to give them immediate protection against developing hepatitis B. These infants, as well as all infants, should also receive a series of three hepatitis B vaccine injections as part of their routine immunizations.

Human immunodeficiency virus (HIV)

Recent studies have shown that prenatal care and HIV testing before delivery are major opportunities to prevent perinatal HIV infection. Pregnant women with HIV should be treated as early in the pregnancy as possible with zidovudine (AZT). Other newer drugs designed to treat HIV/AIDS also may be used during pregnancy with the knowledge that these drugs may have unknown effects on the infant. The risks and benefits of such treatments need to be discussed. Infants born with HIV should receive aggressive drug treatment to prevent development of AIDS.

Human papillomavirus

Genital warts are very difficult to treat and frequently recur even after treatment. They can be removed by **cryotherapy** (freezing), laser or

electrocauterization (burning), or surgical excision (cutting) of the warts. Some medications (imiquimod 5% cream, podophyllin, trichloroacetic acid or topical 5-fluorouracil) can be applied to help dissolve genital warts. Cesarean delivery rather than vaginal delivery seems to reduce the risk of transmission of HPV from mothers to infants.

Rubella (German measles)

No treatment is available. Some health care providers may recommend giving the mother an injection of immune globulin (to boost the immune system to fight off the virus) if she is exposed to rubella early in the pregnancy. However, no evidence to support the use of these injections exists. Exposure to rubella early in pregnancy poses a high risk that the infant will have serious birth defects. Termination of the pregnancy may be considered. Women who have not been previously exposed to rubella will usually be vaccinated immediately after the first pregnancy to protect infants of future pregnancies.

Streptococcus

Pregnant women diagnosed with GBS late in the pregnancy should be treated with antibiotics injected intravenously to prevent **premature labor**. In 2003, the Centers for Disease Control and Prevention (CDC) issued revised guidelines for preventing perinatal GBS disease. They began recommending that women not only be tested as soon as they learn of their pregnancy, but again at 35 to 37 weeks gestation. The CDC also recommended updated **prophylaxis** regimens for women with penicillin **allergies**, as well as new guidelines for patients with threatened preterm deliveries and other new recommendations. If transmission of GBS to the newborn infant already is suspected or if the baby develops symptoms of infection, infants often are treated with antibiotics.

Syphilis

Antibiotic therapy, usually penicillin, given early in the pregnancy can be used to treat the infection and may prevent transmission to the infant.

Prognosis

Chlamydia

Without treatment, the most serious consequences of chlamydial infection are related to complications of premature delivery. Treatment of the mother with antibiotics during the third trimester can prevent premature delivery and the transfer of

KEY TERMS

Cesarean section—A surgical procedure in which an incision is made in a woman's abdomen to deliver the infant from the uterus.

Ectopic pregnancy—A condition that ends in miscarriage, in which the fertilized ovum attaches somewhere other than in the uterus (for example in the fallopian tube or abdomen).

Encephalitis—Inflammation or swelling of the brain.

Perinatal—The period of time around the time of pregnancy and delivery.

Pneumonia—An infection and inflammation of the lungs that usually causes shortness of breath, cough, fever, and chest pain.

the infection to the baby. Infants treated with antibiotics for eye infection or pneumonia generally recover.

Cytomegalovirus

The chance for recovery after exposure to CMV is very good for both the mother and the infant. Exposure to CMV can be serious and even life threatening for mothers and infants whose immune systems are compromised, for example those receiving **chemotherapy** or who have HIV/AIDS. Those infants who develop birth defects after CMV exposure may have serious, lifelong complications.

Genital herpes

Once a woman or infant is infected, outbreaks of genital herpes sores can recur at any point during their lifetimes.

Hepatitis B

Infants treated at birth with immune globulin and the series of vaccinations will be protected from development of hepatitis B infection. Infants infected with hepatitis B develop a chronic, mild form of hepatitis and are at increased risk for developing liver disease.

Human immunodeficiency virus (HIV)

Treatment with AZT during pregnancy significantly reduces the chance that the infant will be infected with HIV from the mother.

Human papillomavirus

Once infected with HPV, there is a lifelong risk of developing warts and an increased risk of some cancers.

Rubella (German measles)

Infants exposed to rubella virus in the uterus are at high risk for severe birth defects including heart defects, blindness, and deafness.

Streptococcus

Infection of the urinary tract or genital tract of pregnant women can cause premature birth. Infants infected with GBS can develop serious, life-threatening infections.

Syphilis

Premature birth, birth defects, or the development of serious syphilis symptoms is likely to occur in untreated pregnant women.

Prevention

Use of a barrier method of contraceptive (**condom**) can prevent transmission of some of the infections. Intravenous drug use and sexual intercourse with infected partners increases the risks of exposure to most of these infections. Pregnant women can be tested for many of the bacterial or viral infections described; however, effective treatment may not be available to protect the infant. New studies show that a woman's nutritional status may contribute to her ability to fight off infections, particularly in cases of **malnutrition**. Proper prenatal care may improve outcomes and prevent some infections.

Resources

PERIODICALS

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- Morantz, Carrie A. "CDC Updates Guidelines for Prevention of Perinatal Group B Streptococcal Disease." *American Family Physician* March 1, 2003: 1121.
- Peters, Vicki, et al. "Missed Opportunities for Perinatal HIV Prevention Among HIV-exposed Infants Born 1996–2000, Pediatric Spectrum of HIV Disease Cohort." *Pediatrics* May 2003: S1186.

Altha Roberts Edgren
Teresa G. Odle

Periodic paralysis

Definition

Periodic **paralysis** (PP) is the name for several rare, inherited muscle disorders marked by temporary weakness, especially following rest, sleep, or exercise.

Description

Periodic paralysis disorders are genetic disorders that affect muscle strength. There are two major forms, hypokalemic and hyperkalemic, each caused by defects in different genes.

In hypokalemic PP, the level of potassium in the blood falls in the early stages of a paralytic attack, while in hyperkalemic PP, it rises slightly or is normal. (The root of both words, “kali,” refers to potassium.) Hyperkalemic PP is also called potassium-sensitive PP.

Causes and symptoms

Causes

Both forms of PP are caused by inheritance of defective genes. Both genes are dominant, meaning that only one copy of the defective gene is needed for a person to develop the disease. A parent with the gene has a 50% chance of passing it along to each offspring, and the likelihood of passing it on is unaffected by the results of previous pregnancies.

The gene for hypokalemic PP is present equally in both sexes, but leads to noticeable symptoms more often in men than in women. The normal gene is responsible for a muscle protein controlling the flow of calcium during muscle contraction.

The gene for hyperkalemic PP affects virtually all who inherit it, with no difference in male-vs.-female expression. The normal gene is responsible for a muscle protein controlling the flow of sodium during muscle contraction.

Symptoms

The attacks of weakness in hypokalemic PP usually begin in late childhood or early adolescence and often become less frequent during middle age. The majority of patients develop symptoms before age 16. Since they begin in the school years, the symptoms of hypokalemic PP are often first seen during physical education classes or after-school sports, and may be mistaken for laziness, or lack of interest on the part of the child.

Attacks are most commonly brought on by:

- strenuous **exercise** followed by a short period of rest
- large meals, especially ones rich in carbohydrates or salt
- emotional **stress**
- alcohol use
- infection
- **pregnancy**

The weakness from a particular attack may last from several hours to as long as several days, and may be localized to a particular limb, or might involve the entire body.

The attacks of weakness of hyperkalemic PP usually begin in infancy or early childhood, and may become less severe later in life. As in the hypokalemic form, attacks are brought on by stress, pregnancy, and exercise followed by rest. In contrast, though, hyperkalemic attacks are not associated with a heavy meal but rather with missing a meal, with high potassium intake, or use of glucocorticoid drugs such as prednisone. (Glucocorticoids are a group of steroids that regulate metabolism and affect muscle tone.)

Weakness usually lasts less than three hours, and often persists for only several minutes. The attacks are usually less severe, but more frequent, than those of the hypokalemic form. Weakness usually progresses from the lower limbs to the upper, and may involve the facial muscles as well.

Diagnosis

Diagnosis of either form of PP begins with a careful medical history and a complete physical and neurological exam. A family medical history may reveal other affected relatives. Blood and urine tests done at the onset of an attack show whether there are elevated or depressed levels of potassium. Electrical tests of muscle and a muscle biopsy show characteristic changes.

Challenge tests, to aid in diagnosis, differ for the two forms. In hypokalemic PP, an attack of weakness can be brought on by administration of glucose and insulin, with exercise if necessary. An attack of hyperkalemic PP can be induced with administration of potassium after exercise during **fasting**. These tests are potentially hazardous and require careful monitoring.

Genetic tests are available at some research centers and are usually recommended for patients with a known family history. However, the number of

KEY TERMS

Gene—A biologic unit of heredity transmitted from parents to offspring.

different possible mutations leading to each form is too great to allow a single comprehensive test for either form, thus limiting the usefulness of **genetic testing**.

Treatment

Severe respiratory weakness from hypokalemic PP may require intensive care to ensure adequate ventilation. Potassium chloride may be given by mouth or intravenously to normalize blood levels.

Attacks requiring treatment are much less common in hyperkalemic PP. Glucose and insulin may be prescribed. Eating carbohydrates may also relieve attacks.

Prognosis

Most patients learn to prevent their attacks well enough that no significant deterioration in the quality of life occurs. Strenuous exercise must be avoided, however. Attacks often lessen in severity and frequency during middle age. Frequent or severe attacks increase the likelihood of permanent residual weakness, a risk in both forms of periodic paralysis.

Prevention

There is no way to prevent the occurrence of either disease in a person with the gene for the disease. The likelihood of an attack of either form of PP may be lessened by avoiding the triggers (the events or combinations of circumstances which cause an attack) for each.

Hypokalemic PP attacks may be prevented with use of acetazolamide (or another carbonic anhydrase inhibitor drug) or a diuretic to help retain potassium in the bloodstream. These attacks may also be prevented by avoiding such triggers as salty food, large meals, a high-carbohydrate diet, and strenuous exercise.

Attacks of hyperkalemic PP may be prevented with frequent small meals high in carbohydrates, and the avoidance of foods high in potassium such as orange juice or bananas. Acetazolamide or thiazide (a diuretic) may be prescribed.

Resources

ORGANIZATIONS

Muscular Dystrophy Association. 3300 East Sunrise Drive, Tucson, AZ 85718. (800) 572-1717. <<http://www.mdausa.org>>.

Periodic Paralysis Association. 5225 Canyon Crest Drive #71-351, Riverside, CA 92507. (909) 781-4401. <<http://www.periodicparalysis.org>>.

Richard Robinson

Periodontal disease

Definition

Periodontal diseases are a group of diseases that affect the tissues that support and anchor the teeth. Left untreated, periodontal disease results in the destruction of the gums, alveolar bone (the part of the jaws where the teeth arise), and the outer layer of the tooth root.

Description

Periodontal disease is usually seen as a chronic inflammatory disease. An acute infection of the periodontal tissue may occur, but is not usually reported to the dentist. The tissues that are involved in periodontal diseases are the gums, which include the gingiva, periodontal ligament, cementum, and alveolar bone. The gingiva is a pink-colored mucus membrane that covers parts of the teeth and the alveolar bone. The periodontal ligament is the main part of the gums. The cementum is a calcified structure that covers the lower parts of the teeth. The alveolar bone is a set of ridges from the jaw bones (maxillary and mandible) in which the teeth are embedded. The main area involved in periodontal disease is the gingival sulcus, a pocket between the teeth and the gums. Several distinct forms of periodontal disease are known. These are gingivitis, acute necrotizing ulcerative gingivitis, adult periodontitis, and localized juvenile periodontitis. Although periodontal disease is thought to be widespread, serious cases of periodontitis are not common. Gingivitis is also one of the early signs of leukemia in some children.

Gingivitis

Gingivitis is an inflammation of the outermost soft tissue of the gums. The gingivae become red and

inflamed, lose their normal shape, and bleed easily. Gingivitis may remain a chronic disease for years without affecting other periodontal tissues. Chronic gingivitis may lead to a deepening of the gingival sulcus. Acute necrotizing ulcerative gingivitis is mainly seen in young adults. This form of gingivitis is characterized by painful, bleeding gums, and death (necrosis) and erosion of gingival tissue between the teeth. It is thought that stress, malnutrition, **fatigue**, and poor oral hygiene are among the causes for acute necrotizing ulcerative gingivitis.

Adult periodontitis

Adult periodontitis is the most serious form of the periodontal diseases. It involves the gingiva, periodontal ligament, and alveolar bone. A deep periodontal pocket forms between the teeth, the cementum, and the gums. Plaque, calculus, and debris from food and other sources collect in the pocket. Without treatment, the periodontal ligament can be destroyed and resorption of the alveolar bone occurs. This allows the teeth to move more freely and eventually results in the loss of teeth. Most cases of adult periodontitis are chronic, but some cases occur in episodes or periods of tissue destruction.

Localized juvenile periodontitis

Localized juvenile periodontitis is a less common form of periodontal disease and is seen mainly in young people. Primarily, localized juvenile periodontitis affects the molars and incisors. Among the distinctions that separate this form of periodontitis are the low incidence of bacteria in the periodontal pocket, minimal plaque formation, and mild inflammation.

Herpetic gingivostomatitis

Herpes infection of the gums and other parts of the mouth is called herpetic gingivostomatitis and is frequently grouped with periodontal diseases. The infected areas of the gums turn red in color and have whitish herpetic lesions. There are two principal differences between this form of periodontal diseases and most other forms. Herpetic gingivostomatitis is caused by a virus, Herpes simplex, not by bacteria, and the viral infection tends to heal by itself in approximately two weeks. Also, herpetic gingivostomatitis is infectious to other people who come in contact with the herpes lesions or saliva that contains virus from the lesion.

Pericoronitis

Pericoronitis is a condition found in children who are in the process of producing molar teeth. The



An extreme case of juvenile periodontitis. (Custom Medical Stock Photo. Reproduced by permission.)

disease is seen more frequently in the lower molar teeth. As the molar emerges, a flap of gum still covers the tooth. The flap of gum traps bacteria and food, leading to a mild irritation. If the upper molar fully emerges before the lower one, it may bite down on the flap during chewing. This can increase the irritation of the flap and lead to an infection. In bad cases, the infection can spread to the neck and cheeks.

Desquamative gingivitis

Desquamative gingivitis occurs mainly in postmenopausal women. The cause of the disease is not understood. The outer layers of the gums slough off, leaving raw tissue and exposed nerves.

Trench mouth

Trench mouth is an acute, necrotizing (causing tissue death), ulcerating (causing open sores) form of gingivitis. It causes **pain** in the affected gums. **Fever** and **fatigue** are usually present also. Trench mouth, also known as Vincent's disease, is a complication of mild cases of gingivitis. Frequently, poor **oral hygiene** is the main cause. **Stress**, an unbalanced diet, or lack of sleep are frequent cofactors in the development of trench mouth. This form of periodontal disease is more common in people who smoke. The term "trench mouth" was created in World War I, when the disease was common in soldiers who lived in the trenches. Symptoms of trench mouth appear suddenly. The initial symptoms include painful gums and foul breath. Gum tissue between teeth becomes infected and dies, and starts to disappear. Often, what appears to be remaining gum is dead tissue. Usually, the gums bleed easily, especially when chewing. The pain can increase to the point where eating and swallowing become difficult. Inflammation or infection from



Gingivitis, an inflammation of the gums, is a common periodontal disease. (Photograph by Edward H. Gill, Custom Medical Stock Photo. Reproduced by permission.)

trench mouth can spread to nearby tissues of the face and neck.

Periodontitis

Periodontitis is a condition in which gingivitis has extended down around the tooth and into the supporting bone structure. Periodontitis is also called pyorrhea. Plaque and tarter buildup sometimes lead to the formation of large pockets between the gums and teeth. When this happens, anaerobic bacteria grow in the pockets. The pockets eventually extend down around the roots of the teeth where the bacteria cause damage to the bone structure supporting the teeth. The teeth become loose and tooth loss can result. Some medical conditions are associated with an increased likelihood of developing periodontitis. These diseases include diabetes, **Down syndrome**, Cohn's disease, **AIDS**, and any disease that reduces the number of white blood cells in the body for extended periods of time.

Causes and symptoms

Several factors play a role in the development of periodontal disease. The most important are age and oral hygiene. The number and type of bacteria present on the gingival tissues also play a role in the development of periodontal diseases. The presence of certain species of bacteria in large enough numbers in the gingival pocket and related areas correlates with the development of periodontal disease. Also, removal of the bacteria correlates with reduction or elimination of disease. In most cases of periodontal disease, the bacteria remain in the periodontal pocket and do not invade surrounding tissue.

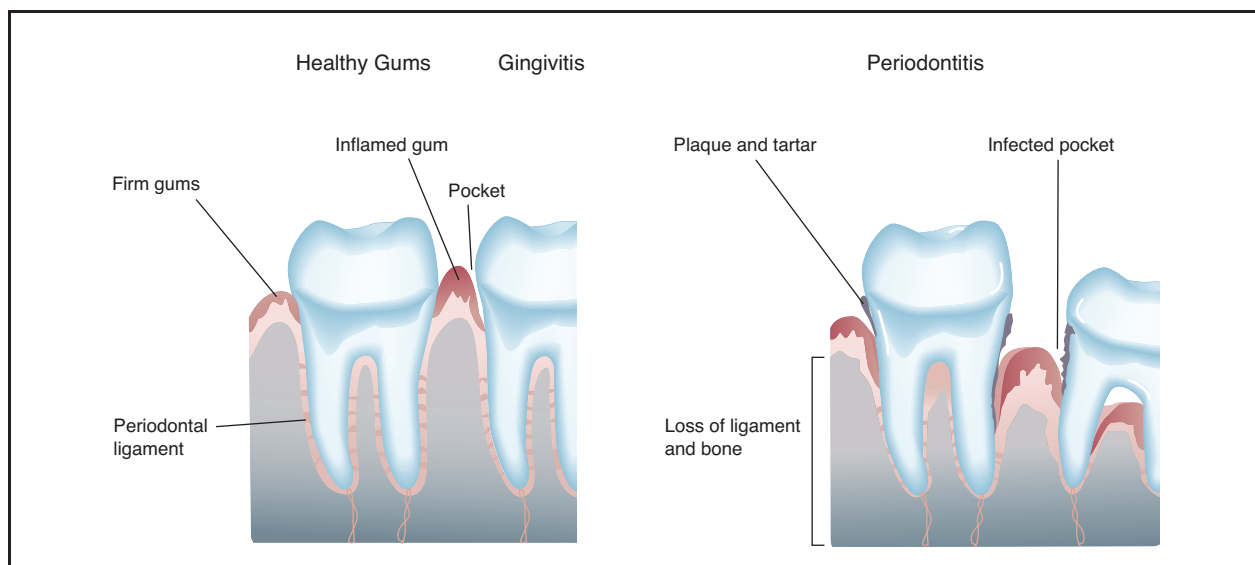
The mechanisms by which bacteria in the periodontal pocket cause tissue destruction in the surrounding region are not fully understood. Several bacterial products that diffuse through tissue are thought to play a role in disease formation. Bacterial endotoxin is a toxin produced by some bacteria that can kill cells. Studies show that the amount of endotoxin present correlates with the severity of periodontal disease. Other bacterial products include proteolytic enzymes, molecules that digest protein found in cells, thereby causing cell destruction. The immune response has also been implicated in tissue destruction. As part of the normal immune response, white blood cells enter regions of inflammation to destroy bacteria. In the process of destroying bacteria, periodontal tissue is also destroyed.

Gingivitis usually results from inadequate oral hygiene. Proper brushing of the teeth and flossing decreases plaque buildup. The bacteria responsible for causing gingivitis reside in the plaque. Plaque is a sticky film that is largely made from bacteria. Tartar is plaque that has hardened. Plaque can turn into tartar in as little as three days if not brushed off. Tartar is difficult to remove by brushing. Gingivitis can be aggravated by hormones, and sometimes becomes temporarily worse during **pregnancy**, **puberty**, and when the patient is taking birth control pills. Interestingly, some drugs used to treat other conditions can cause an overgrowth of the gingival tissue that can result in gingivitis because plaque builds up more easily. Drugs associated with this condition are phenytoin, used to treat seizures; cyclosporin, given to organ transplant patients to reduce the likelihood of organ rejection; and calcium blockers, used to treat several different heart conditions. **Scurvy**, a vitamin C deficiency, and **pellagra**, a niacin deficiency, can also lead to bleeding gums and gingivitis.

The initial symptoms of periodontitis are bleeding and inflamed gums, and **bad breath**. Periodontitis follows cases of gingivitis, which may not be severe enough to cause a patient to seek dental help. Although the symptoms of periodontitis are also seen in other forms of periodontal diseases, the key characteristic in periodontitis is a large pocket that forms between the teeth and gums. Another characteristic of periodontitis is that pain usually does not develop until late in the disease, when a tooth loosens or an **abscess** forms.

Diagnosis

Diagnosis is made by observation of infected gums. Usually, a dentist is the person to diagnose



Healthy gums support the teeth. When gingivitis goes untreated, the gums become weak and pockets form around the teeth. Plaque and tartar build up in the pockets, the gum recedes, and periodontitis occurs. (Illustration by Argosy Inc.)

KEY TERMS

Anaerobic bacteria—Microorganisms that grow in the absence of oxygen.

Inflammation—A painful redness and swelling of an area of tissue in response to infection or injury.

and characterize the various types of periodontal disease. In cases such as acute herpetic gingivostomatitis, there are characteristic herpetic lesions. Many of the periodontal diseases are distinguished based on the severity of the infection and the number and type of tissues involved.

Diagnosis of periodontitis includes measuring the size of the pockets formed between the gums and teeth. Normal gingival pockets are shallow. If periodontal disease is severe, jaw bone loss will be detected in x-ray images of the teeth. If too much bone is lost, the teeth become loose and can change position. This will also be seen in x-ray images.

Treatment

Tartar can only be removed by professional dental treatment. Following treatment, periodontal tissues usually heal quickly. Gingivitis caused by vitamin deficiencies is treated by administering the needed vitamin. There are no useful drugs to treat herpetic

gingivostomatitis. Because of the pain associated with the herpes lesions, patients may not brush their teeth while the lesions are present. Herpes lesions heal by themselves without treatment. After the herpetic lesions have disappeared, the gums usually return to normal if good oral hygiene is resumed. Pericoronitis is treated by removing debris under the flap of gum covering the molar. This operation is usually performed by a dentist. Surgery is used to remove molars that are not likely to form properly.

Treatment for trench mouth starts with a complete cleaning of the teeth, removal of all plaque, tartar, and dead tissue on the gums. For the first few days after cleaning, the patient uses hydrogen peroxide mouth washes instead of brushing. After cleaning, the gum tissue will be very raw and rinsing minimizes damage to the gums that might be caused by the toothbrush. For the first few days, the patient should visit the dentist daily for checkups and then every second or third day for the next two weeks. Occasionally, antibiotic treatment is used to supplement dental cleaning of the teeth and gums. Surgery may be needed if the damage to the gums is extensive and they do not heal properly.

Treatment of periodontitis requires professional dental care. The pockets around the teeth must be cleaned, and all tartar and plaque removed. In periodontitis, tartar and plaque can extend far down the tooth root. Normal dental hygiene, brushing and flossing, cannot reach deep enough to be effective in treating periodontitis. In cases where pockets are very deep (more than 0.25 in [0.64 cm] deep), surgery is

required to clean the pocket. This is performed in a dental office. Sections of gum that are not likely to reattach to the teeth may be removed to promote healing by healthy sections of gum. Abscesses are treated with a combination of **antibiotics** and surgery. The antibiotics may be delivered directly to the infected gum and bone tissues to ensure that high concentrations of the antibiotic reach the infected area. Abscess infections, especially of bone, are difficult to treat and require long term antibiotic treatments to prevent a reoccurrence of infection.

Prognosis

Periodontal diseases can be easily treated. The gums usually heal and resume their normal shape and function. In cases where they do not, prostheses or surgery can restore most of the support for proper functioning of the teeth.

Prevention

Most forms of periodontal disease can be prevented with good dental hygiene. Daily use of a toothbrush and flossing is sufficient to prevent most cases of periodontal disease. Tartar control toothpastes help prevent tartar formation, but do not remove tartar once it has formed.

Resources

BOOKS

Gorbach, S. L., J. G. Bartlett, and N. R. Blacklow. *Infectious Diseases*. 2nd ed. Philadelphia: W. B. Saunders Co., 1998.

John T. Lohr, PhD

Periodontitis see **Periodontal disease**

Periorbital cellulitis see **Orbital and periorbital cellulitis**

Peripheral arterial disease see **Peripheral vascular disease**

Peripheral neuritis see **Peripheral neuropathy**

Peripheral neuropathy

Definition

The term peripheral neuropathy encompasses a wide range of disorders in which the nerves outside

of the brain and spinal cord—peripheral nerves—have been damaged. Peripheral neuropathy may also be referred to as peripheral neuritis, or if many nerves are involved, the terms polyneuropathy or polyneuritis may be used.

Description

Peripheral neuropathy is a widespread disorder, and there are many underlying causes. Some of these causes are common, such as diabetes, and others are extremely rare, such as acrylamide **poisoning** and certain inherited disorders. The most common worldwide cause of peripheral neuropathy is leprosy. Leprosy is caused by the bacterium *Mycobacterium leprae*, which attacks the peripheral nerves of affected people. According to statistics gathered by the World Health Organization, an estimated 1.15 million people have **leprosy** worldwide.

Leprosy is extremely rare in the United States, where diabetes is the most commonly known cause of peripheral neuropathy. It has been estimated that more than 17 million people in the United States and Europe have diabetes-related polyneuropathy. Many neuropathies are idiopathic, meaning that no known cause can be found. The most common of the inherited peripheral neuropathies in the United States is **Charcot-Marie-Tooth disease**, which affects approximately 125,000 persons.

Another of the better known peripheral neuropathies is **Guillain-Barré syndrome**, which arises from complications associated with viral illnesses, such as cytomegalovirus, Epstein-Barr virus, and human **immunodeficiency** virus (HIV), or bacterial infection, including *Campylobacter jejuni* and **Lyme disease**. The worldwide incidence rate is approximately 1.7 cases per 100,000 people annually. Other well-known causes of peripheral neuropathies include chronic **alcoholism**, infection of the varicella-zoster virus, **botulism**, and poliomyelitis. Peripheral neuropathy may develop as a primary symptom, or it may be due to another disease. For example, peripheral neuropathy is only one symptom of diseases such as amyloid neuropathy, certain cancers, or inherited neurologic disorders. Such diseases may affect the peripheral nervous system (PNS) and the central nervous system (CNS), as well as other body tissues.

To understand peripheral neuropathy and its underlying causes, it may be helpful to review the structures and arrangement of the PNS.

Nerve cells and nerves

Nerve cells are the basic building block of the nervous system. In the PNS, nerve cells can be

threadlike—their width is microscopic, but their length can be measured in feet. The long, spidery extensions of nerve cells are called axons. When a nerve cell is stimulated, by touch or pain, for example, the message is carried along the axon, and neurotransmitters are released within the cell. Neurotransmitters are chemicals within the nervous system that direct nerve cell communication.

Certain nerve cell axons, such as the ones in the PNS, are covered with a substance called myelin. The myelin sheath may be compared to the plastic coating on electrical wires—it is there both to protect the cells and to prevent interference with the signals being transmitted. Protection is also given by Schwann cells, special cells within the nervous system that wrap around both myelinated and unmyelinated axons. The effect is similar to beads threaded on a necklace.

Nerve cell axons leading to the same areas of the body may be bundled together into nerves. Continuing the comparison to electrical wires, nerves may be compared to an electrical cord—the individual components are coated in their own sheaths and then encased together inside a larger protective covering.

Peripheral nervous system

The nervous system is classified into two parts: the CNS and the PNS. The CNS is made up of the brain and the spinal cord, and the PNS is composed of the nerves that lead to or branch off from the CNS.

The peripheral nerves handle a diverse array of functions in the body. This diversity is reflected in the major divisions of the PNS—the afferent and the efferent divisions. The afferent division is in charge of sending sensory information from the body to the CNS. When afferent nerve cell endings, called receptors, are stimulated, they release neurotransmitters. These neurotransmitters relay a signal to the brain, which interprets it and reacts by releasing other neurotransmitters.

Some of the neurotransmitters released by the brain are directed at the efferent division of the PNS. The efferent nerves control voluntary movements, such as moving the arms and legs, and involuntary movements, such as making the heart pump blood. The nerves controlling voluntary movements are called motor nerves, and the nerves controlling involuntary actions are referred to as autonomic nerves. The afferent and efferent divisions continually interact with each other. For example, if a person were to touch a hot stove, the receptors in the skin would transmit a message of heat and **pain** through the sensory nerves to the

brain. The message would be processed in the brain and a reaction, such as pulling back the hand, would be transmitted via a motor nerve.

Neuropathy

NERVE DAMAGE. When an individual has a peripheral neuropathy, nerves of the PNS have been damaged. Nerve damage can arise from a number of causes, such as disease, physical injury, poisoning, or malnutrition. These agents may affect either afferent or efferent nerves. Depending on the cause of damage, the nerve cell axon, its protective myelin sheath, or both may be injured or destroyed.

CLASSIFICATION. There are hundreds of peripheral neuropathies. Reflecting the scope of PNS activity, symptoms may involve sensory, motor, or autonomic functions. To aid in diagnosis and treatment, the symptoms are classified into principal neuropathic syndromes based on the type of affected nerves and how long symptoms have been developing. Acute development refers to symptoms that have appeared within days, and subacute refers to those that have evolved over a number of weeks. Early chronic symptoms are those that take months to a few years to develop, and late chronic symptoms have been present for several years.

The classification system is composed of six principal neuropathic syndromes, which are subdivided into more specific categories. By narrowing down the possible diagnoses in this way, specific medical tests can be used more efficiently and effectively. The six syndromes and a few associated causes are listed below:

- Acute motor **paralysis**, accompanied by variable problems with sensory and autonomic functions. Neuropathies associated with this syndrome are mainly accompanied by motor nerve problems, but the sensory and autonomic nerves may also be involved. Associated disorders include Guillain-Barré syndrome, diphtheritic polyneuropathy, and porphyritic neuropathy.
- Subacute sensorimotor paralysis. The term sensorimotor refers to neuropathies that are mainly characterized by sensory symptoms, but also have a minor component of motor nerve problems. Poisoning with heavy metals (e.g., lead, mercury, and arsenic), chemicals, or drugs are linked to this syndrome. Diabetes, Lyme disease, and **malnutrition** are also possible causes.
- Chronic sensorimotor paralysis. Physical symptoms may resemble those in the above syndrome, but the time scale of symptom development is

extended. This syndrome encompasses neuropathies arising from cancers, diabetes, leprosy, inherited neurologic and metabolic disorders, and **hypothyroidism**.

- Neuropathy associated with mitochondrial diseases. Mitochondria are organelles—structures within cells—responsible for handling a cell's energy requirements. If the mitochondria are damaged or destroyed, the cell's energy requirements are not met and it can die.
- Recurrent or relapsing polyneuropathy. This syndrome covers neuropathies that affect several nerves and may come and go, such as Guillain-Barré syndrome, porphyria, and chronic inflammatory demyelinating polyneuropathy.
- Mononeuropathy or plexopathy. Nerve damage associated with this syndrome is limited to a single nerve or a few closely associated nerves. Neuropathies related to physical injury to the nerve, such as **carpal tunnel syndrome** and **sciatica**, are included in this syndrome.

Causes and symptoms

Typical symptoms of neuropathy are related to the type of affected nerve. If a sensory nerve is damaged, common symptoms include **numbness**, **tingling** in the area, a prickling sensation, or pain. Pain associated with neuropathy can be quite intense and may be described as cutting, stabbing, crushing, or burning. In some cases, a nonpainful stimulus may be perceived as excruciating or pain may be felt even in the absence of a stimulus. Damage to a motor nerve is usually indicated by weakness in the affected area. If the problem with the motor nerve has continued over a length of time, muscle shrinkage (atrophy) or lack of muscle tone may be noticeable. Autonomic nerve damage is most noticeable when an individual stands upright and experiences problems such as lightheadedness or changes in blood pressure. Other indicators of autonomic nerve damage are lack of sweat, tears, and saliva; **constipation**; urinary retention; and impotence. In some cases, heart beat irregularities and respiratory problems can develop.

Symptoms may appear over days, weeks, months, or years. Their duration and the ultimate outcome of the neuropathy are linked to the cause of the nerve damage. Potential causes include diseases, physical injuries, poisoning, and malnutrition or alcohol **abuse**. In some cases, neuropathy is not the primary disorder, but a symptom of an underlying disease.

Disease

Diseases that cause peripheral neuropathies may either be acquired or inherited; in some cases, it is difficult to make that distinction. The diabetes-peripheral neuropathy link has been well established. A typical pattern of diabetes-associated neuropathic symptoms includes sensory effects that first begin in the feet. The associated pain or pins-and-needles, burning, crawling, or prickling sensations form a typical “stocking” distribution in the feet and lower legs. Other diabetic neuropathies affect the autonomic nerves and have potentially fatal cardiovascular complications.

Several other metabolic diseases have a strong association with peripheral neuropathy. Uremia, or **chronic kidney failure**, carries a 10–90% risk of eventually developing neuropathy, and there may be an association between liver failure and peripheral neuropathy. Accumulation of lipids inside blood vessels (**atherosclerosis**) can choke-off blood supply to certain peripheral nerves. Without oxygen and nutrients, the nerves slowly die. Mild polyneuropathy may develop in persons with low thyroid hormone levels. Individuals with abnormally enlarged skeletal extremities (acromegaly), caused by an overabundance of growth hormone, may also develop mild polyneuropathy.

Neuropathy can also result from severe vasculitides, a group of disorders in which blood vessels are inflamed. When the blood vessels are inflamed or damaged, blood supply to the nerve can be affected, injuring the nerve.

Both viral and bacterial infections have been implicated in peripheral neuropathy. Leprosy is caused by the bacteria *M. leprae*, which directly attack sensory nerves. Other bacterial illness may set the stage for an immune-mediated attack on the nerves. For example, one theory about Guillain-Barré syndrome involves complications following infection with *Campylobacter jejuni*, a bacterium commonly associated with **food poisoning**. This bacterium carries a protein that closely resembles components of myelin. The immune system launches an attack against the bacteria; but, according to the theory, the immune system confuses the myelin with the bacteria in some cases and attacks the myelin sheath as well. The underlying cause of neuropathy associated with Lyme disease is unknown; the bacteria may either promote an immune-mediated attack on the nerve or inflict damage directly.

Infection with certain viruses is associated with extremely painful sensory neuropathies. A primary

example of such a neuropathy is caused by **shingles**. After a case of **chickenpox**, the causative virus, varicella-zoster virus, becomes inactive in sensory nerves. Years later, the virus may be reactivated. Once reactivated, it attacks and destroys axons. Infection with HIV is also associated with peripheral neuropathy, but the type of neuropathy that develops can vary. Some HIV-linked neuropathies are noted for myelin destruction rather than axonal degradation. Also, HIV infection is frequently accompanied by other infections, both bacterial and viral, that are associated with neuropathy.

Several types of peripheral neuropathies are associated with inherited disorders. These inherited disorders may primarily involve the nervous system, or the effects on the nervous system may be secondary to an inherited metabolic disorder. Inherited neuropathies can fall into several of the principal syndromes, because symptoms may be sensory, motor, or autonomic. The inheritance patterns also vary, depending on the specific disorder. The development of inherited disorders is typically drawn out over several years and may herald a degenerative condition—that is, a condition that becomes progressively worse over time. Even among specific disorders, there may be a degree of variability in inheritance patterns and symptoms. For example, Charcot-Marie-Tooth disease is usually inherited as an autosomal dominant disorder, but it can be autosomal recessive or, in rare cases, linked to the X chromosome. Its estimated frequency is approximately one in 2,500 people. Age of onset and sensory nerve involvement can vary between cases. The main symptom is a degeneration of the motor nerves in legs and arms, and resultant muscle atrophy. Other inherited neuropathies have a distinctly metabolic component. For example, in familial amyloid polyneuropathies, protein components that make up the myelin are constructed and deposited incorrectly.

Physical injury

Accidental falls and mishaps during sports and recreational activities are common causes of physical injuries that can result in peripheral neuropathy. The common types of injuries in these situations occur from placing too much pressure on the nerve, exceeding the nerve's capacity to stretch, blocking adequate blood supply of oxygen and nutrients to the nerve, and tearing the nerve. Pain may not always be immediately noticeable, and obvious signs of damage may take a while to develop.

These injuries usually affect one nerve or a group of closely associated nerves. For example, a common injury encountered in contact sports such as football is

the “burner,” or “stinger,” syndrome. Typically, a stinger is caused by overstretching the main nerves that span from the neck into the arm. Immediate symptoms are numbness, tingling, and pain that travels down the arm, lasting only a minute or two. A single incident of a stinger is not dangerous, but recurrences can eventually cause permanent motor and sensory loss.

Poisoning

The poisons, or toxins, that cause peripheral neuropathy include drugs, industrial chemicals, and environmental toxins. Neuropathy that is caused by drugs usually involves sensory nerves on both sides of the body, particularly in the hands and feet, and pain is a common symptom. Neuropathy is an unusual side effect of medications; therefore, most people can use these drugs safely. A few of the drugs that have been linked with peripheral neuropathy include metronidazole, an antibiotic; phenytoin, an anticonvulsant; and simvastatin, a cholesterol-lowering medication.

Certain industrial chemicals have been shown to be poisonous to nerves (neurotoxic) following work-related exposures. Chemicals such as acrylamide, allyl chloride, and carbon disulfide have all been strongly linked to development of peripheral neuropathy. Organic compounds, such as N-hexane and toluene, are also encountered in work-related settings, as well as in glue-sniffing and solvent abuse. Either route of exposure can produce severe sensorimotor neuropathy that develops rapidly.

Heavy metals are the third group of toxins that cause peripheral neuropathy. Lead, arsenic, thallium, and mercury usually are not toxic in their elemental form, but rather as components in organic or inorganic compounds. The types of metal-induced neuropathies vary widely. Arsenic poisoning may mimic Guillain-Barré syndrome; lead affects motor nerves more than sensory nerves; thallium produces painful sensorimotor neuropathy; and the effects of mercury are seen in both the CNS and PNS.

Malnutrition and alcohol abuse

Burning, stabbing pains and numbness in the feet, and sometimes in the hands, are distinguishing features of alcoholic neuropathy. The level of alcohol consumption associated with this variety of peripheral neuropathy has been estimated as approximately 3 L of beer or 300 mL of liquor daily for three years. However, it is unclear whether alcohol alone is responsible for the neuropathic symptoms, because chronic alcoholism is strongly associated with malnutrition.

Malnutrition refers to an extreme lack of nutrients in the diet. It is unknown precisely which nutrient deficiencies cause peripheral neuropathies in alcoholics and famine and **starvation** patients, but it is suspected that the **B vitamins** have a significant role. For example, thiamine (vitamin B₁) deficiency is the cause of **beriberi**, a neuropathic disease characterized by **heart failure** and painful polyneuropathy of sensory nerves. **Vitamin E deficiency** seems to have a role in both CNS and PNS neuropathy.

Diagnosis

Clinical symptoms can indicate peripheral neuropathy, but an exact diagnosis requires a combination of medical history, medical tests, and possibly a process of exclusion. Certain symptoms can suggest a diagnosis, but more information is commonly needed. For example, painful, burning feet may be a symptom of alcohol abuse, diabetes, HIV infection, or an underlying malignant tumor, among other causes. Without further details, effective treatment would be difficult.

During a **physical examination**, an individual is asked to describe the symptoms very carefully. Detailed information about the location, nature, and duration of symptoms can help exclude some causes or even pinpoint the actual problem. The person's medical history may also provide clues as to the cause, because certain diseases and medications are linked to specific peripheral neuropathies. A medical history should also include information about diseases that run in the family, because some peripheral neuropathies are genetically linked. Information about hobbies, recreational activities, alcohol consumption, and work place activities can uncover possible injuries or exposures to poisonous substances.

The physical examination also includes blood tests, such as those that check levels of glucose and creatinine to detect diabetes and kidney problems, respectively. A **blood count** is also done to determine levels of different blood cell types. Iron, vitamin B₁₂, and other factors may be measured as well, to rule out malnutrition. More specific tests, such as an assay for heavy metals or poisonous substances, or tests to detect **vasculitis**, are not typically done unless there is reason to suspect a particular cause.

An individual with neuropathy may be sent to a doctor that specializes in nervous system disorders (neurologist). By considering the results of the physical examination and observations of the referring doctor, the neurologist may be able to narrow down the possible diagnoses. Additional tests, such as nerve conduction studies and **electromyography**, which tests

muscle reactions, can confirm that nerve damage has occurred and may also be able to indicate the nature of the damage. For example, some neuropathies are characterized by destruction of the myelin. This type of damage is shown by slowed nerve conduction. If the axon itself has suffered damage, the nerve conduction may be slowed, but it will also be diminished in strength. Electromyography adds further information by measuring nerve conduction and muscle response, which determines whether the symptoms are due to a neuropathy or to a muscle disorder.

In approximately 10% of peripheral neuropathy cases, a nerve biopsy may be helpful. In this test, a small part of the nerve is surgically removed and examined under a microscope. This procedure is usually the most helpful in confirming a suspected diagnosis, rather than as a diagnostic procedure by itself.

Treatment

Treat the cause

Attacking the underlying cause of the neuropathy can prevent further nerve damage and may allow for a better recovery. For example, in cases of bacterial infection such as leprosy or Lyme disease, **antibiotics** may be given to destroy the infectious bacteria. Viral infections are more difficult to treat, because antibiotics are not effective against them. Neuropathies associated with drugs, chemicals, and toxins are treated in part by stopping exposure to the damaging agent. Chemicals such as ethylenediaminetetraacetic acid (EDTA) are used to help the body concentrate and excrete some toxins. Diabetic neuropathies may be treated by gaining better control of blood sugar levels, but chronic kidney failure may require dialysis or even kidney transplant to prevent or reduce nerve damage. In some cases, such as compression injury or tumors, surgery may be considered to relieve pressure on a nerve.

In a crisis situation, as in the onset of Guillain-Barré syndrome, plasma exchange, intravenous immunoglobulin, and steroids may be given. Intubation, in which a tube is inserted into the trachea to maintain an open airway, and ventilation may be required to support the respiratory system. Treatment may focus more on symptom management than on combating the underlying cause, at least until a definitive diagnosis has been made.

Supportive care and long-term therapy

Some peripheral neuropathies cannot be resolved or require time for resolution. In these cases,

KEY TERMS

Afferent—Refers to peripheral nerves that transmit signals to the spinal cord and the brain. These nerves carry out sensory function.

Autonomic—Refers to peripheral nerves that carry signals from the brain and that control involuntary actions in the body, such as the beating of the heart.

Autosomal dominant or autosomal recessive—Refers to the inheritance pattern of a gene on a chromosome other than X or Y. Genes are inherited in pairs—one gene from each parent. However, the inheritance may not be equal, and one gene may overshadow the other in determining the final form of the encoded characteristic. The gene that overshadows the other is called the dominant gene; the overshadowed gene is the recessive one.

Axon—A long, threadlike projection that is part of a nerve cell.

Central nervous system (CNS)—The part of the nervous system that includes the brain and the spinal cord.

Efferent—Refers to peripheral nerves that carry signals away from the brain and spinal cord. These nerves carry out motor and autonomic functions.

Electromyography—A medical test that assesses nerve signals and muscle reactions. It can determine

if there is a disorder with the nerve or if the muscle is not capable of responding.

Inheritance pattern—Refers to dominant or recessive inheritance.

Motor—Refers to peripheral nerves that control voluntary movements, such as moving the arms and legs.

Myelin—The protective coating on axons.

Nerve biopsy—A medical test in which a small portion of a damaged nerve is surgically removed and examined under a microscope.

Nerve conduction—The speed and strength of a signal being transmitted by nerve cells. Testing these factors can reveal the nature of nerve injury, such as damage to nerve cells or to the protective myelin sheath.

Neurotransmitter—Chemicals within the nervous system that transmit information from or between nerve cells.

Peripheral nervous system (PNS)—Nerves that are outside of the brain and spinal cord.

Sensory—Refers to peripheral nerves that transmit information from the senses to the brain.

long-term monitoring and supportive care is necessary. Medical tests may be repeated to chart the progress of the neuropathy. If autonomic nerve involvement is a concern, regular monitoring of the cardiovascular system may be carried out.

Because pain is associated with many of the neuropathies, a pain management plan may need to be mapped out, especially if the pain becomes chronic. As in any chronic disease, **narcotics** are best avoided. Agents that may be helpful in neuropathic pain include amitriptyline, carbamazepine, and capsaicin cream. Physical therapy and physician-directed exercises can help maintain or improve function. In cases in which motor nerves are affected, braces and other supportive equipment can aid an individual's ability to move about.

Prognosis

The outcome for peripheral neuropathy depends heavily on the cause. Peripheral neuropathy ranges from a reversible problem to a potentially fatal complication. In the best cases, a damaged nerve

regenerates. Nerve cells cannot be replaced if they are killed, but they are capable of recovering from damage. The extent of recovery is tied to the extent of the damage and a person's age and general health status. Recovery can take weeks to years, because neurons grow very slowly. Full recovery may not be possible and it may also not be possible to determine the prognosis at the outset.

If the neuropathy is a degenerative condition, such as Charcot-Marie-Tooth disease, an individual's condition will become worse. There may be periods of time when the disease seems to reach a plateau, but cures have not yet been discovered for many of these degenerative diseases. Therefore, continued symptoms, potentially worsening to disabilities are to be expected.

A few peripheral neuropathies are eventually fatal. Fatalities have been associated with some cases of **diphtheria**, botulism, and others. Some diseases associated with neuropathy may also be fatal, but the ultimate cause of **death** is not necessarily related to the neuropathy, such as with **cancer**.

Prevention

Peripheral neuropathies are preventable only to the extent that the underlying causes are preventable. Steps that a person can take to prevent potential problems include vaccines against diseases that cause neuropathy, such as **polio** and diphtheria. Treatment for physical injuries in a timely manner can help prevent permanent or worsening damage to nerves. Precautions when using certain chemicals and drugs are well advised in order to prevent exposure to neurotoxic agents. Control of chronic diseases such as diabetes may also reduce the chances of developing peripheral neuropathy.

Although not a preventive measure, genetic screening can serve as an early warning for potential problems. Genetic screening is available for some inherited conditions, but not all. In some cases, presence of a particular gene may not mean that a person will necessarily develop the disease, because there may be environmental and other components involved.

Resources

ORGANIZATIONS

American Diabetes Association. 1701 North Beauregard Street, Alexandria, VA 22311. (800) 342-2383. <<http://www.diabetes.org>>.

Myelin Project Headquarters. Suite 225, 2001 Pennsylvania Ave., N.W., Washington, D.C. 20006-1850. (202) 452-8994. <<http://www.myelin.org>>.

Neuropathy Association. 60 E. 42nd St., Suite 942, New York, NY 10165. (800) 247-6968. <<http://www.neuropathy.org/association.html>>.

Julia Barrett

Peripheral vascular disease

Definition

Peripheral vascular disease is a narrowing of blood vessels that restricts blood flow. It mostly occurs in the legs, but is sometimes seen in the arms.

Description

Peripheral vascular disease includes a group of diseases in which blood vessels become restricted or blocked. Typically, the patient has peripheral vascular disease from **atherosclerosis**. Atherosclerosis is a disease in which fatty plaques form in the inside walls of blood vessels. Other processes, such as **blood clots**,

further restrict blood flow in the blood vessels. Both veins and arteries may be affected, but the disease is usually arterial. All the symptoms and consequences of peripheral vascular disease are related to restricted blood flow. Peripheral vascular disease is a progressive disease that can lead to **gangrene** of the affected area. Peripheral vascular disease may also occur suddenly if an **embolism** occurs or when a blood clot rapidly develops in a blood vessel already restricted by an atherosclerotic plaque, and the blood flow is quickly cut off.

Causes and symptoms

There are many causes of peripheral vascular disease. One major risk factor is **smoking** cigarettes. Other diseases predispose patients to develop peripheral vascular disease. These include diabetes, Buerger's disease, hypertension, and Raynaud's disease. The main symptom is **pain** in the affected area. Early symptoms include an aching, tired sensation in the affected muscles. Since this disease is seen mainly in the legs, these sensations usually occur when walking. The symptoms may disappear when resting. As the disease becomes worse, symptoms occur even during light exertion and, eventually, occur all the time, even at rest. In the severe stages of the disease the leg and foot may be cold to the touch and will feel numb. The skin may become dry and scaly. If the leg is even slightly injured, ulcers may form because, without a good blood supply, proper healing can not take place. At the most severe stage of the disease, when the blood flow is greatly restricted, gangrene can develop in those areas lacking blood supply. In some cases, peripheral vascular disease occurs suddenly. This happens when an embolism rapidly blocks blood flow to a blood vessel. The patient will experience a sharp pain, followed by a loss of sensation in the affected area. The limb will become cold and numb, and lose color or turn bluish.

Diagnosis

Peripheral vascular disease can be diagnosed by comparing blood pressures taken above and below the point of pain. The area below the pain (downstream from the obstruction) will have a much lower or undetectable blood pressure reading. Doppler ultrasonography and **angiography** can also be used to diagnose and define this disease.

Treatment

If the person is a smoker, they should stop smoking immediately. **Exercise** is essential to treating this

KEY TERMS

Embolism—The blockage of a blood vessel by air, blood clot, or other foreign body.

Plaque—A deposit, usually of fatty material, on the inside wall of a blood vessel.

disease. The patient should walk until pain appears, rest until the pain disappears, and then resume walking. The amount of walking a patient can do should increase gradually as the symptoms improve. Ideally, the patient should walk 30–60 minutes per day. Infections in the affected area should be treated promptly. Surgery may be required to attempt to treat clogged blood vessels. Limbs with gangrene must be amputated to prevent the **death** of the patient.

Prognosis

The prognosis depends on the underlying disease and the stage at which peripheral vascular disease is discovered. Removal of risk factors, such as smoking, should be done immediately. In many cases, peripheral vascular disease can be treated successfully but coexisting cardiovascular problems may ultimately prove to be fatal.

Resources

BOOKS

Alexander, R. W., R. C. Schlant, and V. Fuster, editors. *The Heart*. 9th ed. New York: McGraw-Hill, 1998.

John T. Lohr, PhD

Peritoneal dialysis see **Dialysis, kidney**

Peritoneal endoscopy see **Laparoscopy**

Peritoneal fluid analysis see **Paracentesis**

Peritonitis

Definition

Peritonitis is an inflammation of the membrane which lines the inside of the abdomen and all of the internal organs. This membrane is called the peritoneum.

Description

Peritonitis may be primary (meaning that it occurs spontaneously, and not as the result of some other medical problem) or secondary (meaning that it results from some other condition). It is most often due to infection by bacteria, but may also be due to some kind of a chemical irritant (such as spillage of acid from the stomach, bile from the gall bladder and biliary tract, or enzymes from the pancreas during the illness called **pancreatitis**). Peritonitis has even been seen in patients who develop a reaction to the cornstarch used to powder gloves worn during surgery. Peritonitis with no evidence of bacteria, chemical irritant, or foreign body has occurred in such diseases as systemic lupus erythematosus, porphyria, and **familial Mediterranean fever**. When the peritoneum is contaminated by blood, the blood can both irritate the peritoneum and serve as a source of bacteria to cause an infection. Blood may leak into the abdomen due to a burst tubal **pregnancy**, an injury, or bleeding after surgery.

Causes and symptoms

Primary peritonitis usually occurs in people who have an accumulation of fluid in their abdomens (**ascites**). Ascites is a common complication of severe cirrhosis of the liver (a disease in which the liver grows increasingly scarred and dysfunctional). The fluid that accumulates creates a good environment for the growth of bacteria.

Secondary peritonitis most commonly occurs when some other medical condition causes bacteria to spill into the abdominal cavity. Bacteria are normal residents of a healthy intestine, but they should have no way to escape and enter the abdomen, where they could cause an infection. Bacteria can infect the peritoneum due to conditions in which a hole (perforation) develops in the stomach (due to an ulcer eating its way through the stomach wall) or intestine (due to a large number of causes, including a ruptured appendix or a ruptured diverticulum). Bacteria can infect the peritoneum due to a severe case of pelvic inflammatory disease (a massive infection of the female organs, including the uterus and fallopian tubes). Bacteria can also escape into the abdominal cavity due to an injury that causes the intestine to burst, or an injury to an internal organ which bleeds into the abdominal cavity.

Symptoms of peritonitis include **fever** and abdominal pain. An acutely ill patient usually tries to lie very still, because any amount of movement causes excruciating **pain**. Often, the patient lies with the knees bent, to decrease strain on the tender peritoneum. There is often **nausea** and vomiting. The usual sounds made by

KEY TERMS

Ascites—An accumulation of fluid within the abdominal cavity.

Cirrhosis—A progressive liver disease in which the liver grows increasingly more scarred. The presence of scar tissue then interferes with liver function.

Diverticulum—An outpouching of the intestine.

Laparotomy—An open operation on the abdomen.

Pancreatitis—An inflammation of the pancreas.

Perforation—A hole.

Peritoneum—The membrane that lines the inside of the abdominal cavity, and all of the internal organs.

the active intestine and heard during examination with a stethoscope will be absent, because the intestine usually stops functioning. The abdomen may be rigid and boardlike. Accumulations of fluid will be notable in primary peritonitis due to ascites. Other signs and symptoms of the underlying cause of secondary peritonitis may be present.

Diagnosis

A diagnosis of peritonitis is usually based on symptoms. Discovering the underlying reason for the peritonitis, however, may require some work. A blood sample will be drawn in order to determine the white blood cell count. Because white blood cells are produced by the body in an effort to combat foreign invaders, the white blood cell count will be elevated in the case of an infection. A long, thin needle can be used to take a sample of fluid from the abdomen in an effort to diagnose primary peritonitis. The types of immune cells present are usually characteristic in this form of peritonitis. X-ray films may be taken if there is some suspicion that a perforation exists. In the case of a perforation, air will have escaped into the abdomen and will be visible on the picture. When a cause for peritonitis cannot be found, an open exploratory operation on the abdomen (laparotomy) is considered to be a crucial diagnostic procedure, and at the same time provides the opportunity to begin treatment.

Treatment

Treatment depends on the source of the peritonitis, but an emergency laparotomy is usually

performed. Any perforated or damaged organ is usually repaired at this time. If a clear diagnosis of **pelvic inflammatory disease** or pancreatitis can be made, however, surgery is not usually performed. Peritonitis from any cause is treated with antibiotics given through a needle in the vein, along with fluids to prevent **dehydration**.

Prognosis

Prognosis for untreated peritonitis is poor, usually resulting in **death**. With treatment, the prognosis is variable, dependent on the underlying cause.

Prevention

There is no way to prevent peritonitis, since the diseases it accompanies are usually not under the voluntary control of an individual. However, prompt treatment can prevent complications.

Resources

BOOKS

Isselbacher, Kurt J., and Alan Epstein. "Diverticular, Vascular, and Other Disorders of the Intestine and Peritoneum." In *Harrison's Principles of Internal Medicine*, edited by Anthony S. Fauci, et al. New York: McGraw-Hill, 1997.

Rosalyn Carson-DeWitt, MD

Permanent pacemakers see **Pacemakers**

Pernicious anemia

Definition

Pernicious anemia is a disease in which the red blood cells are abnormally formed, due to an inability to absorb vitamin B₁₂. True pernicious anemia refers specifically to a disorder of atrophied parietal cells leading to absent intrinsic factor, resulting in an inability to absorb B₁₂.

Description

Vitamin B₁₂, or cobalamin, plays an important role in the development of red blood cells. It is found in significant quantities in liver, meats, milk and milk products, and legumes. During the course of the digestion of foods containing B₁₂, the B₁₂ becomes attached

to a substance called intrinsic factor. Intrinsic factor is produced by parietal cells that line the stomach. The B₁₂-intrinsic factor complex then enters the intestine, where the vitamin is absorbed into the bloodstream. In fact, B₁₂ can only be absorbed when it is attached to intrinsic factor.

In pernicious anemia, the parietal cells stop producing intrinsic factor. The intestine is then completely unable to absorb B₁₂. So, the vitamin passes out of the body as waste. Although the body has significant amounts of stored B₁₂, this will eventually be used up. At this point, the symptoms of pernicious anemia will develop.

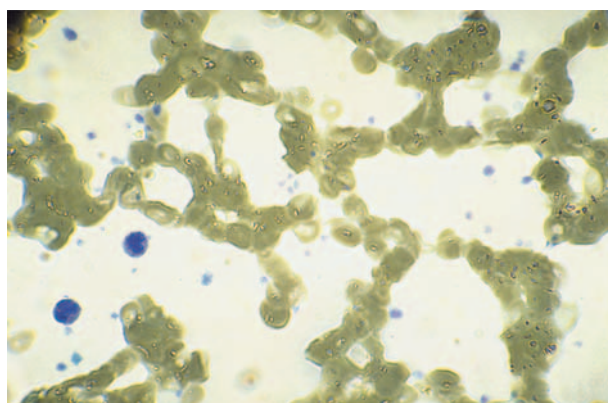
Pernicious anemia is most common among people from northern Europe and among African Americans. It is far less frequently seen among people from southern Europe and Asia. Pernicious anemia occurs in equal numbers in both men and women. Most patients with pernicious anemia are older, usually over 60. Occasionally, a child will have an inherited condition that results in defective intrinsic factor. Pernicious anemia seems to run in families, so that anyone with a relative diagnosed with the disease has a greater likelihood of developing it as well.

Causes and symptoms

Intrinsic factor is produced by specialized cells within the stomach called parietal cells. When these parietal cells shrink in size (atrophy), they produce less and less intrinsic factor. Eventually, the parietal cells stop functioning altogether. Other important products of parietal cells are also lessened, including stomach acid, and an enzyme involved in the digestion of proteins.

People with pernicious anemia seem to have a greater chance of having certain other conditions. These conditions include **autoimmune disorders**, particularly those affecting the thyroid, parathyroid, and adrenals. It is thought that the immune system, already out of control in these diseases, incorrectly becomes directed against the parietal cells. Ultimately, the parietal cells seem to be destroyed by the actions of the immune system.

As noted, true pernicious anemia refers specifically to a disorder of atrophied parietal cells leading to absent intrinsic factor, resulting in an inability to absorb B₁₂. However, there are other related conditions that result in decreased absorption of B₁₂. These conditions cause the same types of symptoms as true pernicious anemia. Other conditions that interfere with either the production of intrinsic factor, or the



A smear of red blood cells indicating folic acid (vitamin B₁₂) deficiency. (Custom Medical Stock Photo. Reproduced by permission.)

body's use of B₁₂, include conditions that require surgical removal of the stomach, or poisonings with corrosive substances which destroy the lining of the stomach. Certain structural defects of the intestinal system can result in an overgrowth of normal bacteria. These bacteria then absorb B₁₂ themselves, for use in their own growth. Intestinal worms (especially one called fish tapeworm) may also use B₁₂, resulting in anemia. Various conditions that affect the first part of the intestine (the ileum), from which B₁₂ is absorbed, can also cause anemia due to B₁₂ deficiency. These ileum-related disorders include tropical sprue, Whipple's disease, **Crohn's disease**, **tuberculosis**, and the Zollinger-Ellison syndrome.

Symptoms of pernicious anemia and decreased B₁₂ affect three systems of the body: the system that is involved in the formation of blood cells (hematopoietic system); the gastrointestinal system; and the nervous system.

The hematopoietic system is harmed because B₁₂ is required for the proper formation of red blood cells. Without B₁₂, red blood cell production is greatly reduced. Those red blood cells that are produced are abnormally large and abnormal in shape. Because red blood cells are responsible for carrying oxygen around the body, decreased numbers (termed anemia) result in a number of symptoms, including **fatigue**, **dizziness**, ringing in the ears, pale or yellowish skin, fast heart rate, enlarged heart with an abnormal heart sound (murmur) evident on examination, and chest **pain**.

Symptoms that affect the gastrointestinal system include a sore and brightly red tongue, loss of appetite, weight loss, **diarrhea**, and abdominal cramping.

KEY TERMS

Anemia—A condition in which those elements of the blood responsible for oxygen delivery throughout the body (red blood cells, hemoglobin) are decreased in quantity or defective in some way.

Atrophy—Refers to the shrinking in size of an organ or cell.

Autoimmune disorder—A disorder in which the immune system, (responsible for fighting off such foreign invaders as bacteria and viruses), begins to attack and damage a part of the body as if it were foreign.

Hematopoietic system—The system in the body which is responsible for the production of blood cells.

Intrinsic factor—A substance produced by the parietal cells of the stomach. In order to be absorbed by the intestine, vitamin B₁₂ must form a complex with intrinsic factor.

Parietal cells—Specific cells which line the inside of the stomach. These cells are responsible for producing intrinsic factor and hydrochloric acid.

Reticulocyte—An early, immature form of a red blood cell. Over time, the reticulocyte develops to become a mature, oxygen-carrying red blood cell.

The nervous system is severely affected when pernicious anemia goes untreated. Symptoms include **numbness, tingling**, or burning in the arms, legs, hands, and feet; muscle weakness; difficulty and loss of balance while walking; changes in reflexes; irritability, confusion, and depression.

Diagnosis

Diagnosis of pernicious anemia is suggested when a blood test reveals abnormally large red blood cells. Many of these will also be abnormally shaped. The earliest, least mature forms of red blood cells (reticulocytes) will also be low in number. White blood cells and platelets may also be decreased in number. Measurements of the quantity of B₁₂ circulating in the bloodstream will be low.

Once these determinations are made, it will be important to diagnose the cause of the anemia. True pernicious anemia means that the parietal cells of the stomach are atrophied, resulting in decreased production of intrinsic factor. This diagnosis is made by the Schilling test. In this test, a patient is given radioactive B₁₂ under two different sets of conditions: once alone,

and once attached to intrinsic factor. Normally, large amounts of B₁₂ are absorbed through the intestine, then circulate through the blood, and enter the kidneys, where a certain amount of B₁₂ is then passed out in the urine. When a patient has pernicious anemia, the dose of B₁₂ given by itself will not be absorbed by the intestine, so it will not pass into the urine. Therefore, levels of B₁₂ in the urine will be low. When the B₁₂ is given along with intrinsic factor, the intestine is able to absorb the vitamin. Urine levels of B₁₂ will therefore be higher.

Treatment

Treatment of pernicious anemia requires the administration of lifelong injections of B₁₂. Vitamin B₁₂ given by injection enters the bloodstream directly, and does not require intrinsic factor. At first, injections may need to be given several times a week, in order to build up adequate stores of the vitamin. After this, the injections can be given on a monthly basis. Other substances required for blood cell production may also need to be given, iron and vitamin C.

Prognosis

Prognosis is generally good for patients with pernicious anemia. Many of the symptoms improve within just a few days of beginning treatment, although some of the nervous system symptoms may take up to 18 months to improve. Occasionally, when diagnosis and treatment have been delayed for a long time, some of the nervous system symptoms may be permanent.

Because an increased risk of **stomach cancer** has been noted in patients with pernicious anemia, careful monitoring is necessary, even when all the symptoms of the original disorder have improved.

Resources

BOOKS

Babior, Bernard M., and H. Franklin Bunn. "Megaloblastic Anemias." In *Harrison's Principles of Internal Medicine*, edited by Anthony S. Fauci, et al. New York: McGraw-Hill, 1997.

PERIODICALS

Toh, Ban-Hock, et al. "Pernicious Anemia." *The New England Journal of Medicine* 337, no. 20 (November 13, 1997): 52+.

Rosalyn Carson-DeWitt, MD

Peroneal muscular atrophy see **Charcot-Marie-Tooth disease**

Peroxisomal disorders

Definition

Peroxisomal disorders are a group of congenital (existing from birth) diseases characterized by the absence of normal peroxisomes in the cells of the body. Peroxisomes are special parts (organelles) within a cell that contain enzymes responsible for critical cellular processes, including oxidation of fatty acids, biosynthesis of membrane phospholipids (plasmalogens), cholesterol, and bile acids, conversion of amino acids into glucose, reduction of hydrogen peroxide by catalase, and prevention of excess synthesis of oxalate (which can form crystals with calcium, resulting in **kidney stones**). Peroxisomal disorders are subdivided into two major categories. The first are disorders resulting from a failure to form intact, normal peroxisomes, resulting in multiple metabolic abnormalities, which are referred to as peroxisome biogenesis disorders (PBD) or as generalized peroxisomal disorders. The second category includes those disorders resulting from the deficiency of a single peroxisomal enzyme. There are about 25 peroxisomal disorders known, although the number of diseases that are considered to be separate, distinct peroxisomal disorders varies among researchers and health care practitioners.

Description

A cell can contain several hundred peroxisomes, which are round or oval bodies with diameters of about 0.5 micron, that contain proteins that function as enzymes in metabolic processes. By definition, a peroxisome must contain catalase, which is an enzyme that breaks down hydrogen peroxide.

Approximately 50 different biochemical reactions occur entirely or partially within a peroxisome. Some of the processes are anabolic, or constructive, resulting in the synthesis of essential biochemical compounds, including bile acids, cholesterol, plasmalogens, and docosahexanoic acid (DHA), which is a long chain fatty acid that is a component of complex lipids, including the membranes of the central nervous system. Other reactions are catabolic, or destructive, and lead to the destruction of some fatty acids, including very long chain fatty acids (VLCFAs, fatty acids with more than 22 carbon atoms in their chains), phytanic acid, pipecolic acid, and the prostoglandins. The peroxisome is involved in breaking down VLCFAs to lengths that the body can use or get rid of.

When VLCFAs accumulate due to abnormal functioning of the peroxisomes, they are disruptive

to the structure and stability of certain cells, especially those associated with the central nervous system and the myelin sheath, which is the fatty covering of nerve fibers. The peroxisomal disorders that include effects on the growth of the myelin sheath are considered to be part of a group of genetic disorders referred to as leukodystrophies. While metachromatic leukodystrophy (MLD) usually has its onset in infants or juveniles, there have been reports of its onset in young adults.

There are many other metabolic deficiencies that can occur in those who have peroxisomal disorders, which result in other types of detrimental effects, and together result in the abnormalities associated with the peroxisomal disorders. Unfortunately, it is not known how these abnormalities, and combinations of abnormalities, cause the disabilities seen in those afflicted with the disease.

Peroxisomal disorders form a heterogeneous disease group, with different degrees of severity. Included in the group referred to as PBD are:

- Zellweger syndrome (ZS), which is usually fatal within the first year of life,
- neonatal **adrenoleukodystrophy** (NALD), which is usually fatal within the first 10 years,
- infantile Refsum disease (IRD), which is not as devastating as ZS and NALD, as the children with this disorder with time and patience can develop some degree of motor, cognitive, and communication skills, although **death** generally occurs during the second decade of life.
- rhizomelic chondrodysplasia punctata (RCDP), which in its most severe form is fatal within the first year or two of life. However, survival into the teens has been known to occur. It is characterized by shortening of the proximal limbs (i.e., the legs from knee to foot, and the arms from elbow to hand).
- Zellweger-like syndrome, which is fatal in infancy, and is known to be a defect of three particular enzymes.

The differences among these disorders are continuous, with overlap between abnormalities. The range of disease abnormalities may be a result of a corresponding range of peroxisome failure; that is, in severe cases of ZS, the failure is nearly complete, while in IRD, there is some degree of peroxisome activity.

In peroxisomal single-enzyme disorders, the peroxisome is intact and functioning, but there is a defect in only one enzymatic process, with only one corresponding biochemical abnormality. However, these disorders can be as severe as those in which peroxisomal activity is nearly or completely absent.

X-linked adrenoleukodystrophy (X-ALD) is the most common of the peroxisomal disorders, affecting about one in 20,000 males. It is estimated that there are about 1,400 people in the United States with the disorder. In X-ALD there is a deficiency in the enzyme that breaks down VLCFAs, which then accumulate in the myelin and adrenal glands. Onset of X-ALD-related neurological symptoms occurs at about five–12 years of age, with death occurring within one to 10 years after onset of symptoms. In addition to physical abnormalities seen in other types of peroxisomal disorders, common symptoms of X-ALD also include behavioral changes such as abnormal withdrawal or aggression, poor memory, **dementia**, and poor academic performance. Other symptoms are muscle weakness and difficulties with hearing, speech, and vision. As the disease progresses, muscle tone deteriorates, swallowing becomes difficult and the patient becomes comatose. Unless treated with a diet that includes Lorenzo's oil, the disease will result in **paralysis**, **hearing loss**, blindness, **vegetative state**, and death. There are also milder forms of X-ALD: an adult onset ALD that typically begins between the ages of 21 and 35, and a form that is occasionally seen in women who are carriers of the disorder. In addition to X-ALD, there are at least 10 other single-enzyme peroxisomal disorders, each with its own specific abnormalities.

Causes and symptoms

Most peroxisomal disorders are inherited autosomal recessive diseases, with X-ALD as an exception. They occur in all countries, among all races and ethnic groups. They are extremely rare, with frequencies reported at one in 30,000 to one in 150,000, although these numbers are only estimates.

In general, developmental delay, **mental retardation**, and vision and hearing impairment are common in those who have these disorders. Acquisition of speech appears to be especially difficult, and because of the reduced communication abilities, **autism** is common in those who live longer. Peroxisomal disorder patients have decreased muscle tone (hypotonia), which in the most severe cases is generalized, while in less severe cases, is usually restricted to the neck and trunk muscles. Sometimes this lack of control is only noticeable by a curved back in the sitting position. Head control and independent sitting is delayed, with most patients unable to walk independently.

Failure to thrive is a common characteristic of patients with peroxisomal disorder, along with an enlarged liver, abnormalities in liver enzyme function, and loss of fats in stools (steatorrhea).

Peroxisomal disorders are also associated with facial abnormalities, including high forehead, frontal bossing (swelling), small face, low set ears, and slanted eyes. These characteristics may not be prominent in some children, and are especially difficult to identify in an infant.

Diagnosis

Since hearing and vision deficiencies may be difficult to identify in infants, peroxisomal disorders are usually detected by observations of failure to thrive, hypotonia, mental retardation, widely open fontanel, abnormalities in liver enzymes, and an enlarged liver. If peroxisomal disorders are suspected, blood plasma assays for VLCFAs, phytanic acid, and pipecolic acid are conducted. Additional tests include plasmalogen biosynthesis potential.

Treatment

For many of the peroxisomal disorders, there is no standard course of treatment, with supportive treatment strategies focusing on alleviation of complications and symptoms. In general, most treatments that are attempted are dietary, whereby attempts are made to artificially correct biochemical abnormalities associated with the disorders. Therapies include supplementation of the diet with antioxidant **vitamins**, or limitation of intake of fatty acids, especially VLCFAs.

Another area of dietary therapy that is being investigated is the supplementation of the diet with pure DHA, given as early in life as possible, in conjunction with a normal well-balanced diet. Some results have indicated that if given soon enough during development, DHA therapy may prevent some of the devastating consequences of peroxisomal disorders, including brain damage and the loss of vision.

Other treatment strategies include addition of important missing chemicals. For example, in disorders where there is faulty adrenal function, replacement adrenal hormone therapy is used.

Any dietary changes should be monitored biochemically to determine if the supplements are having their desired effects and are not causing additional adverse effects.

Bone marrow transplants may be used to treat X-ALD, and can be effective if done early in the course of the childhood form of the disease.

Physical and psychological therapies are important for all types of peroxisomal disorders.

KEY TERMS

Autosomal recessive inheritance—Two copies of an altered gene located on one of the autosomes must be present for an individual to be affected with the trait or condition determined by that gene:

- an affected individual (homozygote) has two parents who are unaffected but each parent carries the altered gene (heterozygote).
- the risk of two heterozygotes, or carriers, having an affected child is 25%, one in four, for each child that they have; similarly, there is a three in four chance that each child will not be affected.
- males and females are at equal risk for being affected.
- two affected individuals usually produce children, all of whom are affected as well.

Autosome—A chromosome not involved in sex determination.

Fontanel—One of the membranous intervals between the uncompleted angles of the parietal and neighboring bones of a fetal or young skull; so called because it exhibits a rhythmical pulsation.

Metabolic—Relating to the chemical changes in living cells.

Organelle—Specialized structure within a cell, which is separated from the rest of the cell by a membrane composed of lipids and proteins, where chemical and metabolic functions take place.

Alternative treatment

Patients with peroxisomal disorders, and particularly X-ALD, have been treated with a mixture of glycerol trioleate-glycerol trierucate (4:1 by volume), prepared from olive and rapeseed oils, and referred to as Lorenzo's oil (developed by parents of a son, Lorenzo, who had X-ALD, whose story was documented in the 1992 movie, *Lorenzo's Oil*), to decrease the levels of VLCFA. Other **diets** that have been tried include dietary supplementation with plasmalogen precursors to increase plasmalogen levels and with cholic acid to normalize bile acids. However, there has been only limited success demonstrated with the use of these treatments. More research is needed to determine the long-term safety and effectiveness of these treatment strategies.

Prognosis

Peroxisomal disorders range from life-threatening to cases in which people may function with some degree of mental and motor retardation. As of 2001, there was not yet a cure. Enzyme replacement therapies, including enzyme infusion, transplantation, and **gene therapy**, may hold promise for future advances in the treatment of these disorders. Research is being conducted to increase scientific understanding of these disorders and to find ways to prevent, treat, and cure them.

Prevention

Unfortunately not enough is yet known about these diseases to develop comprehensive strategies for prevention. **Genetic counseling** is recommended for known or suspected carriers. As genes are identified that result in the disorders, **genetic testing** is being developed to identify carriers, who then can manage their reproduction to avoid the possibility of children being born with these deficiencies. As the genetic bases for the disorders are defined, prenatal diagnosis and identification of carriers will be facilitated. For example, for X-ALD, diagnosis can be made from cultured skin fibroblasts or amniotic fluid cells. This allows prenatal diagnosis and carrier identification in 90% of those affected. More recently it has been shown that biochemical diagnosis can be performed through chorionic villi biopsy, a procedure performed very early in the first trimester of **pregnancy**.

Animal models of ZS and X-ADL have been developed and are providing researchers with methods to define pathogenic mechanisms and to evaluate new therapies.

Resources

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National Institute of Neurological Disorders and Stroke, NIH Neurological Institute. P.O. Box 5801 Bethesda, MD 20824. (800) 352-9424. <<http://www.ninds.nih.gov/index.htm>>.

National Organization for Rare Disorders. P.O. Box 8923, New Fairfield, CT 06812-8923.

OTHER

The Peroxisome Website. <<http://www.peroxisome.org>>.

Judith Sims
Teresa G. Odle

Persantine-thallium heart scan see **Thallium heart scan**

Personality disorders

Definition

Personality disorders are a group of mental disturbances defined by the fourth edition, text revision (2000) of the *Diagnostic and Statistical Manual of Mental Disorders (DSM-IV)* as “enduring pattern[s] of inner experience and behavior” that are sufficiently rigid and deep-seated to bring a person into repeated conflicts with his or her social and occupational environment. *DSM-IV* specifies that these dysfunctional patterns must be regarded as nonconforming or deviant by the person’s culture, and cause significant emotional **pain** and/or difficulties in relationships and occupational performance. In addition, the patient usually sees the disorder as being consistent with his or her self-image (ego-syntonic) and may blame others for his or her social, educational, or work-related problems.

Description

To meet the diagnosis of personality disorder, which is sometimes called character disorder, the patient’s problematic behaviors must appear in two or more of the following areas:

- perception and interpretation of the self and other people
- intensity and duration of feelings and their appropriateness to situations
- relationships with others
- ability to control impulses

Personality disorders have their onset in late adolescence or early adulthood. Doctors rarely give a diagnosis of personality disorder to children on the grounds that children’s personalities are still in the process of formation and may change considerably by the time they are in their late teens. In retrospect, however, many individuals with personality disorders could be judged to have shown evidence of the problems in childhood.

It is difficult to give close estimates of the percentage of the population that has personality disorders. Patients with certain personality disorders, including antisocial and borderline disorders, are more likely to get into trouble with the law or otherwise attract attention than are patients whose disorders chiefly affect their capacity for intimacy. On the other hand, some patients, such as those with narcissistic or obsessive-compulsive personality disorders, may be outwardly successful because their symptoms are useful within their particular occupations. It has, however, been estimated that about 15% of the general population of the United States has a personality disorder, with higher rates in poor or troubled neighborhoods. The rate of personality disorders among patients in psychiatric treatment is between 30% and 50%. It is possible for patients to have a so-called dual diagnosis; for example, they may have more than one personality disorder, or a personality disorder together with a substance-abuse problem.

By contrast, *DSM-IV* classifies personality disorders into three clusters based on symptom similarities:

- Cluster A (paranoid, schizoid, schizotypal): Patients appear odd or eccentric to others.
- Cluster B (antisocial, borderline, histrionic, narcissistic): Patients appear overly emotional, unstable, or self-dramatizing to others.
- Cluster C (avoidant, dependent, obsessive-compulsive): Patients appear tense and anxiety-ridden to others.

The *DSM-IV* clustering system does not mean that all patients can be fitted neatly into one of the three clusters. It is possible for patients to have symptoms of more than one personality disorder or to have symptoms from different clusters.

Some psychiatrists maintain that the *DSM-IV* classification is inadequate and should be expanded to include three additional categories: passive-aggressive personality disorder, characterized by a need to control or punish others through frustrating them or sabotaging plans; cyclothymic personality disorder, characterized by intense mood swings alternating

between high spirits and moroseness or gloom; and depressive personality disorder, characterized by a negative and pessimistic approach to life.

Since the criteria for personality disorders include friction or conflict between the patient and his or her social environment, these syndromes are open to redefinition as societies change. Successive editions of *DSM* have tried to be sensitive to cultural differences, including changes over time, when defining personality disorders. One category that had been proposed for *DSM-III-R*, self-defeating personality disorder, was excluded from *DSM-IV* on the grounds that its definition reflected prejudice against women. *DSM-IV* recommends that doctors take a patient's background, especially recent immigration, into account before deciding that he or she has a personality disorder. One criticism that has been made of the general category of personality disorder is that it is based on Western notions of individual uniqueness. Its applicability to people from cultures with different definitions of human personhood is thus open to question. Furthermore, even within a culture, it can be difficult to define the limits of "normalcy."

The personality disorders defined by *DSM-IV* are as follows:

Paranoid

Patients with paranoid personality disorder are characterized by suspiciousness and a belief that others are out to harm or cheat them. They have problems with intimacy and may join cults or groups with paranoid belief systems. Some are litigious, bringing lawsuits against those they believe have wronged them. Although not ordinarily delusional, these patients may develop psychotic symptoms under severe **stress**. It is estimated that 0.5–2.5% of the general population meet the criteria for paranoid personality disorder.

Schizoid

Schizoid patients are perceived by others as "loners" without close family relationships or social contacts. Indeed, they are aloof and really do prefer to be alone. They may appear cold to others because they rarely display strong emotions. They may, however, be successful in occupations that do not require personal interaction. About 2% of the general population has this disorder. It is slightly more common in men than in women.

Schizotypal

Patients diagnosed as schizotypal are often considered odd or eccentric because they pay little

attention to their clothing and sometimes have peculiar speech mannerisms. They are socially isolated and uncomfortable in parties or other social gatherings. In addition, people with schizotypal personality disorder often have oddities of thought, including "magical" beliefs or peculiar ideas (for example, a belief in telepathy or UFOs) that are outside of their cultural norms. It is thought that 3% of the general population has schizotypal personality disorder. It is slightly more common in males. Schizotypal disorder should not be confused with **schizophrenia**, although there is some evidence that the disorders are genetically related.

Antisocial

Patients with antisocial personality disorder are sometimes referred to as sociopaths or psychopaths. They are characterized by lying, manipulateness, and a selfish disregard for the rights of others; some may act impulsively. People with antisocial personality disorder are frequently chemically dependent and sexually promiscuous. It is estimated that 3% of males in the general population and 1% of females have antisocial personality disorder.

Borderline

Patients with borderline personality disorder (BPD) are highly unstable, with wide mood swings, a history of intense but stormy relationships, impulsive behavior, and confusion about career goals, personal values, or sexual orientation. These often highly conflictual ideas may correspond to an even deeper confusion about their sense of self (identity). People with BPD frequently cut or burn themselves, or threaten or attempt **suicide**. Many of these patients have histories of severe childhood **abuse** or neglect. About 2% of the general population have BPD; 75% of these patients are female.

Histrionic

Patients diagnosed with this disorder impress others as overly emotional, overly dramatic, and hungry for attention. They may be flirtatious or seductive as a way of drawing attention to themselves, yet they are emotionally shallow. Histrionic patients often live in a romantic fantasy world and are easily bored with routine. About 2–3% of the population is thought to have this disorder. Although historically the disorder has been more associated with women in clinical settings, there may be bias toward diagnosing women with the histrionic personality disorder.

Narcissistic

Narcissistic patients are characterized by self-importance, a craving for admiration, and exploitative attitudes toward others. They have unrealistically inflated views of their talents and accomplishments, and may become extremely angry if they are criticized or outshone by others. Narcissists may be professionally successful but rarely have long-lasting intimate relationships. Fewer than 1% of the population has this disorder; about 75% of those diagnosed with it are male.

Avoidant

Patients with avoidant personality disorder are fearful of rejection and shy away from situations or occupations that might expose their supposed inadequacy. They may reject opportunities to develop close relationships because of their fears of criticism or humiliation. Patients with this personality disorder are often diagnosed with dependent personality disorder as well. Many also fit the criteria for social phobia. Between 0.5–1.0% of the population have avoidant personality disorder.

Dependent

Dependent patients are afraid of being on their own and typically develop submissive or compliant behaviors in order to avoid displeasing people. They are afraid to question authority and often ask others for guidance or direction. Dependent personality disorder is diagnosed more often in women, but it has been suggested that this finding reflects social pressures on women to conform to gender stereotyping or bias on the part of clinicians.

Obsessive-compulsive

Patients diagnosed with this disorder are preoccupied with keeping order, attaining perfection, and maintaining mental and interpersonal control. They may spend a great deal of time adhering to plans, schedules, or rules from which they will not deviate, even at the expense of openness, flexibility, and efficiency. These patients are often unable to relax and may become “workaholics.” They may have problems in employment as well as in intimate relationships because they are very stiff and formal, and insist on doing everything their way. About 1% of the population has obsessive-compulsive personality disorder; the male/female ratio is about 2:1.

Causes and symptoms

Personality disorders are thought to result from a bad interface, so to speak, between a child’s

temperament and character on one hand and his or her family environment on the other. Temperament can be defined as a person’s innate or biologically shaped basic disposition. Human infants vary in their sensitivity to light or noise, their level of physical activity, their adaptability to schedules, and similar traits. Even such traits as **shyness** or novelty-seeking may be at least in part determined by the biology of the brain and the genes one inherits.

Character is defined as the set of attitudes and behavior patterns that the individual acquires or learns over time. It includes such personal qualities as work and study habits, moral convictions, neatness or cleanliness, and consideration of others. Since children must learn to adapt to their specific families, they may develop personality disorders in the course of struggling to survive psychologically in disturbed or stressful families. For example, nervous or high-strung parents might be unhappy with a baby who is very active and try to restrain him or her at every opportunity. The child might then develop an avoidant personality disorder as the outcome of coping with constant frustration and parental disapproval. As another example, **child abuse** is believed to play a role in shaping borderline personality disorder. One reason that some therapists use the term developmental damage instead of personality disorder is that it takes the presumed source of the person’s problems into account.

Some patients with personality disorders come from families that appear to be stable and healthy. It has been suggested that these patients are biologically hypersensitive to normal family stress levels. Levels of the brain chemical (neurotransmitter) dopamine may influence a person’s level of novelty-seeking, and serotonin levels may influence aggression.

Other factors that have been cited as affecting children’s personality development are the mass media and social or group **hysteria**, particularly after the events of September 11, 2001. Cases of so-called mass sociogenic illness have been identified, in which a group of children began to vomit or have other physical symptoms brought on in response to an imaginary threat. In two such cases, the children were reacting to the suggestion that toxic fumes were spreading through their school. Some authors believe that overly frequent or age-inappropriate discussions of terrorist attacks or bioterrorism may make children more susceptible to sociogenic illness as well as other distortions of personality.

Diagnosis

Diagnosis of personality disorders is complicated by the fact that affected persons rarely seek help until

they are in serious trouble or until their families (or the law) pressure them to get treatment. The reason for this slowness is that the problematic traits are so deeply entrenched that they seem normal (ego-syntonic) to the patient. Diagnosis of a personality disorder depends in part on the patient's age. Although personality disorders originate during the childhood years, they are considered adult disorders. Some patients, in fact, are not diagnosed until late in life because their symptoms had been modified by the demands of their job or by marriage. After retirement or the spouse's **death**, however, these patients' personality disorders become fully apparent. In general, however, if the onset of the patient's problem is in mid- or late-life, the doctor will rule out **substance abuse** or personality change caused by medical or neurological problems before considering the diagnosis of a personality disorder. It is unusual for people to develop personality disorders "out of the blue" in mid-life.

There are no tests that can provide a definitive diagnosis of personality disorder. Most doctors will evaluate a patient on the basis of several sources of information collected over a period of time in order to determine how long the patient has been having difficulties, how many areas of life are affected, and how severe the dysfunction is. These sources of information may include:

Interviews

The doctor may schedule two or three interviews with the patient, spaced over several weeks or months, in order to rule out an adjustment disorder caused by job loss, **bereavement**, or a similar problem. An office interview allows the doctor to form an impression of the patient's overall personality as well as obtain information about his or her occupation and family. During the interview, the doctor will note the patient's appearance, tone of voice, body language, eye contact, and other important non-verbal signals, as well as the content of the conversation. In some cases, the doctor may contact other people (family members, employers, close friends) who know the patient well in order to assess the accuracy of the patient's perception of his or her difficulties. It is quite common for people with personality disorders to have distorted views of their situations or to be unaware of the impact of their behavior on others.

Psychologic testing

Doctors use psychologic testing to help in the diagnosis of a personality disorder. Most of these tests require interpretation by a professional with

specialized training. Doctors usually refer patients to a clinical psychologist for this type of test.

PERSONALITY INVENTORIES. Personality inventories are tests with true/false or yes/no answers that can be used to compare the patient's scores with those of people with known personality distortions. The single most commonly used test of this type is the Minnesota Multiphasic Personality Inventory, or MMPI. Another test that is often used is the Millon Clinical Multiaxial Inventory, or MCMI.

PROJECTIVE TESTS. Projective tests are unstructured. Unstructured means that instead of giving one-word answers to questions, the patient is asked to talk at some length about a picture that the psychologist has shown him or her, or to supply an ending for the beginning of a story. Projective tests allow the clinician to assess the patient's patterns of thinking, fantasies, worries or anxieties, moral concerns, values, and habits. Common projective tests include the Rorschach, in which the patient responds to a set of ten inkblots; and the Thematic Apperception Test (TAT), in which the patient is shown drawings of people in different situations and then tells a story about the picture.

Treatment

At one time psychiatrists thought that personality disorders did not respond very well to treatment. This opinion was derived from the notion that human personality is fixed for life once it has been molded in childhood, and from the belief among people with personality disorders that their own views and behaviors are correct, and that others are the ones at fault. More recently, however, doctors have recognized that humans can continue to grow and change throughout life. Most patients with personality disorders are now considered to be treatable, although the degree of improvement may vary. The type of treatment recommended depends on the personality characteristics associated with the specific disorder.

Hospitalization

Inpatient treatment is rarely required for patients with personality disorders, with two major exceptions: borderline patients who are threatening suicide or suffering from drug or alcohol withdrawal; and patients with paranoid personality disorder who are having psychotic symptoms.

Psychotherapy

Psychoanalytic psychotherapy is suggested for patients who can benefit from insight-oriented

treatment. These patients typically include those with dependent, obsessive-compulsive, and avoidant personality disorders. Doctors usually recommend individual psychotherapy for narcissistic and borderline patients, but often refer these patients to therapists with specialized training in these disorders. Psychotherapeutic treatment for personality disorders may take as long as three to five years.

Insight-oriented approaches are not recommended for patients with paranoid or antisocial personality disorders. These patients are likely to resent the therapist and see him or her as trying to control or dominate them.

Supportive therapy is regarded as the most helpful form of psychotherapy for patients with schizoid personality disorder.

Cognitive-behavioral therapy

Cognitive-behavioral approaches are often recommended for patients with avoidant or dependent personality disorders. Patients in these groups typically have mistaken beliefs about their competence or likableness. These assumptions can be successfully challenged by cognitive-behavioral methods. More recently, Aaron Beck and his coworkers have successfully extended their approach to cognitive therapy to all ten personality disorders as defined by DSM-IV.

Group therapy

Group therapy is frequently useful for patients with schizoid or avoidant personality disorders because it helps them to break out of their social isolation. It has also been recommended for patients with histrionic and antisocial personality disorders. These patients tend to act out, and pressure from peers in group treatment can motivate them to change. Because patients with antisocial personality disorder can destabilize groups that include people with other disorders, it is usually best if these people meet exclusively with others who have APD (in homogeneous groups).

Family therapy

Family therapy may be suggested for patients whose personality disorders cause serious problems for members of their families. It is also sometimes recommended for borderline patients from overinvolved or possessive families.

Medications

Medications may be prescribed for patients with specific personality disorders. The type of medication

depends on the disorder. In general, however, patients with personality disorders are helped only moderately by medications.

ANTIPSYCHOTIC DRUGS. **Antipsychotic drugs**, such as haloperidol (Haldol), may be given to patients with paranoid personality disorder if they are having brief psychotic episodes. Patients with borderline or schizotypal personality disorder are sometimes given antipsychotic drugs in low doses; however, the efficacy of these drugs in treating personality disorder is less clear than in schizophrenia.

MOOD STABILIZERS. Carbamazepine (Tegretol) is a drug that is commonly used to treat seizures, but is also helpful for borderline patients with rage outbursts and similar behavioral problems. Lithium and valproate may also be used as mood stabilizers, especially among people with borderline personality disorder.

ANTIDEPRESSANTS AND ANTI-ANXIETY MEDICATIONS. Medications in these categories are sometimes prescribed for patients with schizoid personality disorder to help them manage **anxiety** symptoms while they are in psychotherapy. Antidepressants are also commonly used to treat people with borderline personality disorder.

Treatment with medications is not recommended for patients with avoidant, histrionic, dependent, or narcissistic personality disorders. The use of potentially addictive medications should be avoided in people with borderline or antisocial personality disorders. However, some avoidant patients who also have social phobia may benefit from monoamine oxidase inhibitors (MAO inhibitors), a particular class of antidepressant.

Prognosis

The prognosis for recovery depends in part on the specific disorder. Although some patients improve as they grow older and have positive experiences in life, personality disorders are generally life-long disturbances with periods of worsening (exacerbations) and periods of improvement (remissions). Others, particularly schizoid patients, have better prognoses if they are given appropriate treatment. Beck and his coworkers estimate that effective cognitive therapy with patients with personality disorders takes two to three years on average. Patients with paranoid personality disorder are at some risk for developing delusional disorders or schizophrenia.

The personality disorders with the poorest prognoses are the antisocial and the borderline. Borderline patients are at high risk for developing substance abuse disorders or bulimia. About 80% of

KEY TERMS

Character—An individual's set of emotional, cognitive, and behavioral patterns learned and accumulated over time.

Character disorder—Another name for personality disorder.

Cognitive therapy—A form of psychotherapy that focuses on changing people's patterns of emotional reaction by correcting distorted patterns of thinking and perception.

Developmental damage—A term that some therapists prefer to personality disorder, on the grounds that it is more respectful of the patient's capacity for growth and change.

Ego-syntonic—Consistent with one's sense of self, as opposed to ego-alien or dystonic (foreign to one's sense of self). Ego-syntonic traits typify patients with personality disorders.

Neuroleptic—Another name for older antipsychotic medications, such as haloperidol. The term does not apply to such newer atypical agents as clozapine (Clozaril).

Personality—The organized pattern of behaviors and attitudes that makes a human being distinctive. Personality is formed by the ongoing interaction of temperament, character, and environment.

Projective tests—Psychological tests that probe into personality by obtaining open-ended responses to such materials as pictures or stories. Projective tests are often used to evaluate patients with personality disorders.

Rorschach test—A well-known projective test that requires the patient to describe what he or she sees in each of 10 inkblots. It is named for the Swiss psychiatrist who invented it.

Temperament—A person's natural or genetically determined disposition.

hospitalized borderline patients attempt suicide at some point during treatment, and about 5% succeed in committing suicide. Borderline patients are also the most likely to sue their mental health professional for malpractice.

Prevention

The most effective preventive strategy for personality disorders is early identification and treatment of

children at risk. High-risk groups include abused children, children from troubled families, children with close relatives diagnosed with personality disorders, children of substance abusers, and children who grow up in cults or extremist political groups.

Resources

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ORGANIZATIONS

- American Academy of Child and Adolescent Psychiatry (AACAP). 3615 Wisconsin Avenue, NW, Washington, DC 20016-3007. (202) 966-7300. Fax: (202) 966-2891. < <http://www.aacap.org> >.
- American Psychiatric Association (APA). 1400 K Street, NW, Washington, DC 20005. < <http://www.psych.org> >.
- National Institute of Mental Health (NIMH). 6001 Executive Boulevard, Room 8184, MSC 9663, Bethesda, MD 20892-9663. (301) 443-4513. < <http://www.nimh.nih.gov> >.

Rebecca J. Frey, PhD

Perthes disease see **Osteochondroses**

Pertussis see **Whooping cough**

Pervasive developmental disorders

Definition

Pervasive developmental disorders include five different conditions: Asperger's syndrome, autistic disorder, childhood disintegrative disorder (CDD), pervasive developmental disorder not otherwise specified (PDDNOS), and Rett's syndrome. They are grouped together because of the similarities between them. The three most common shared problems involve communication skills, motor skills, and social skills. Since there are no clear diagnostic boundaries separating these conditions it is sometimes difficult to distinguish one from the other for diagnostic purposes.

Asperger's syndrome, autistic disorder, and childhood disintegrative disorder are four to five times more common in boys, and Rett's syndrome has been diagnosed primarily in girls. All of these disorders are rare.

Description

Asperger's syndrome

Children afflicted with Asperger's syndrome exhibit difficulties in social relationships and communication. They are reluctant to make eye contact, do not respond to social or emotional contacts, do not initiate play activities with peers, and do not give or receive attention or affection. To receive this diagnosis the individual must demonstrate normal development of language, thinking and coping skills. Due to an impaired coordination of muscle movements, they appear to be clumsy. They usually become deeply involved in very few interests, which tend to occupy most of their time and attention.

Autistic disorder

Autistic disorder is frequently evident within the first year of life, and must be diagnosed before age three. It is associated with moderate **mental retardation** in three out of four cases. These children do not want to be held, rocked, cuddled or played with. They are unresponsive to affection, show no interest in peers or adults and have few interests. Other traits include avoidance of eye contact, an expressionless face and the use of gestures to express needs. Their actions are repetitive, routine and restricted. Rocking, hand and arm flapping, unusual hand and finger movements, and attachment to objects rather than pets and people

are common. Speech, play, and other behaviors are repetitive and without imagination. They tend to be overactive, aggressive, and self-injurious. They are often highly sensitive to touch, noise, and smells and do not like changes in routine. **Autism** and several disorders classified with it have increased significantly in recent years so that they now are diagnosed more often in children than **spina bifida**, **cancer**, or **Down syndrome**. This may be due partly to improved recognition and diagnosis.

Childhood disintegrative disorder

Childhood disintegrative disorder is also called Heller's disease and most often develops between two and ten years of age. Children with CDD develop normally until two to three years of age and then begin to disintegrate rapidly. Signs and symptoms include deterioration of the ability to use and understand language to the point where they are unable to carry on a conversation. This is accompanied by loss of control of the bladder and bowels. Any interest or ability to play and engage in social activities is lost. The behaviors are nearly identical with those that are characteristic of autistic disorder. However, childhood disintegrative disorder becomes evident later in life and results in developmental regression, or loss of previously attained skills, whereas autistic disorder can be detected as early as the first month of life and results in a failure to progress.

Pervasive developmental disorder not otherwise specified

The term pervasive developmental disorder not otherwise specified (PDDNOS) is also referred to as atypical personality development, atypical PDD, or atypical autism. Individuals with this disorder share some of the same signs and symptoms of autism or other conditions under the category of pervasive developmental disorders, but do not meet all of the criteria for diagnosis for any of the four syndromes included in this group of diseases. Because the children diagnosed with PDDNOS do not all exhibit the same combination of characteristics, it is difficult to do research on this disorder, but the limited evidence available suggests that patients are seen by medical professionals later in life than is the case for autistic children, and they are less likely to have intellectual deficits.

Rett's syndrome

Rett's syndrome was first described in 1966 and is found almost exclusively in girls. It is a disease in

which cells in the brain experience difficulty in communicating with each other. At the same time the growth of the head falls behind the growth of the body so that these children are usually mentally retarded. These conditions are accompanied by deficits in movement (motor) skills and a loss of interest in social activities.

The course of the illness has been divided into four stages. In stage one the child develops normally for six to 18 months. In stage two, development slows down and stops. Stage three is characterized by a loss of the speech and motor skills already acquired. Typically this happens between nine months and three years of age. Stage four begins with a return to learning that will continue across the lifespan, but at a very slow rate. Problems with coordination and walking are likely to continue and even worsen. Other conditions that can occur with Rett's syndrome are convulsions, **constipation**, breathing problems, impaired circulation in the feet and legs, and difficulty chewing or swallowing.

Causes and symptoms

The causes of these disorders are unknown although brain structure abnormalities, genetic mutation, and alterations in brain function are believed to play a role. Still, no single brain abnormality or location has been connected to a cause. In 2004, scientists reported finding the gene mutation (on gene MECP2) that is present in 80% of people affected with Rett's syndrome. In 2004, comprehensive review of research on twins revealed that interactions between multiple genes may play a role in the cause of autism. A number of neurological conditions, such as convulsions, are commonly found to accompany these disorders.

Diagnosis

The diagnosis of pervasive developmental disorder is made by medical specialists based on a thorough examination of the patient, including observing behavior and gathering information from parents and caregivers. Because many symptoms are common to more than one condition, distinctions between conditions must be carefully made. The following summary describes the distinction between three common disorders.

PDDNOS:

- impairment of two-way social interaction
- Repetitive and predictable behavior patterns and activities

Autism:

- all listed for PDDNOS
- severe impairment in communication
- abnormal social interaction and use of language for social communication or imaginative play before age of three
- not better accounted for by another psychiatric order

Asperger's disorder:

- all listed for PDDNOS
- clinically significant impairment in social, occupational, or other areas of functioning
- no general delay in language
- no delay in cognitive development, self-help skills, or adaptive behavior
- not better accounted for by another pervasive developmental disorder or schizophrenia

Rett's syndrome:

- a period of normal development between 6–18 months
- normal head circumference at birth, followed by a slowing of head growth
- retardation
- repetitive hand movements

CDD:

- normal development for at least two years
- loss of skills in at least two of the following areas: language, social skills, bowel or bladder control, play, movement skills
- abnormal functioning in at least two of the following areas: social interaction, communication, behavior patterns
- not better accounted for by another PDD or mental illness

Treatment

Treatment for children with pervasive developmental disorders is limited. Those who can be enrolled in educational programs will need a highly structured learning environment, a teacher-student ratio of not more than 1:2, and a high level of parental involvement that provides consistent care at home. Psychotherapy and social skills training can prove helpful to some. There is no specific medication available for treating the core symptoms of any of these disorders, though research is promising. Some psychiatric medications may be helpful in controlling

KEY TERMS

Hydrotherapy—This term literally means “water treatment” and involves the use of water in physical therapy as well as treatment of physical and emotional illness.

Mutation—A mutation is a change in a gene. Since genes determine how a body is structured and functions, any change in a gene will produce some change in these areas.

Neurological conditions—A condition that has its origin in some part of the patient’s nervous system.

particular behavior difficulties, such as agitation, mood instability, and self-injury. Music, massage, and **hydrotherapy** may exert a calming effect on behavior. Treatment may also include physical and occupational therapy.

Prognosis

In general, the prognosis in all of these conditions is tied to the severity of the illness.

The prognosis for Asperger’s syndrome is more hopeful than that for other diseases in this cluster. These children are likely to grow up to be functional independent adults, but will always have problems with social relationships. They are also at greater risk for developing serious mental illness than the general population.

The prognosis for autistic disorder is not as good, although great strides have been made in recent years in its treatment. The higher the patient’s IQ (intelligence quotient) and ability to communicate, the better the prognosis. However, many patients will always need some level of custodial care. In the past, most of these individuals were confined to institutions, but many are now able to live in group homes or supervised apartments. The prognosis for childhood disintegrative disorder is even less favorable. These children will require intensive and long-term care. Children diagnosed with PDDNOS have a better prognosis because their initial symptoms are usually milder, IQ scores are higher, and language development is stronger.

Prevention

The causes of pervasive developmental disorders are not understood, although research efforts are

getting closer to understanding the problem. Until the causes are discovered, it will remain impossible to prevent these conditions.

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PET scan see **Positron emission tomography (PET)**

Pet therapy

Definition

Animal-assisted therapy (AAT), also known as pet therapy, utilizes trained animals and handlers to



This autistic child is encouraged to interact with the guinea pig in an effort to improve his social interaction. (Helen B. Senisi. Photo Researchers, Inc. Reproduced by permission.)

achieve specific physical, social, cognitive, and emotional goals with patients.

Purpose

Studies have shown that physical contact with a pet can lower high blood pressure, and improve survival rates for **heart attack** victims. There is also evidence that petting an animal can cause endorphins to be released. Endorphins are chemicals in the body that suppress the **pain** response. These are benefits that can be enjoyed from pet ownership, as well as from visiting therapeutic animals.

Many skills can be learned or improved with the assistance of a therapy animal. Patient rehabilitation can be encouraged by such activities as walking or running with a dog, or throwing objects for the animal to retrieve. Fine motor skills may be developed by petting, grooming, or feeding the animal. Patient communication is encouraged by the response of the animal to either verbal or physical commands. Activities such as writing or talking about the therapy animals or past pets also develop cognitive skills and

communication. Creative inclusion of an animal in the life or therapy of a patient can make a major difference in the patient's comfort, progress, and recovery.

Description

Origins

The enjoyment of animals as companions dates back many centuries, perhaps even to prehistoric times. The first known therapeutic use of animals started in Gheel, Belgium in the ninth century. In this town, learning to care for farm animals has long been an important part of an assisted living program designed for people with disabilities.

Some of the earliest uses of animal-assisted healing in the United States were for psychiatric patients. The presence of the therapy animals produced a beneficial effect on both children and adults with mental health issues. It is only in the last few decades that AAT has been more formally applied in a variety of therapeutic settings, including schools and prisons, as

KEY TERMS

Endorphins—A group of chemicals resembling opiates that are released in the body in response to trauma or stress. Endorphins react with opiate receptors in the brain to reduce pain sensations.

well as hospitals, hospices, nursing homes, and outpatient care programs.

The way in which AAT is undertaken depends on the needs and abilities of the individual patient. Dogs are the most common visiting therapy animals, but cats, horses, birds, rabbits, and other domestic pets can be used as long as they are appropriately screened and trained.

For patients who are confined, small animals can be brought to the bed if the patient is willing and is not allergic to the animal. A therapeutic plan may include a simple interaction aimed at improving communication and small motor skills, or a demonstration with educational content to engage the patient cognitively.

If the patient is able to walk or move around, more options are available. Patients can walk small animals outside, or learn how to care for farm animals. Both of these activities develop confidence and motor abilities. Horseback riding has recently gained great therapeutic popularity. It offers an opportunity to work on balance, trunk control, and other skills. Many patients who walk with difficulty, or not at all, get great emotional benefit from interacting with and controlling a large animal.

One advantage of having volunteers provide this service is that cost and insurance are not at issue.

Precautions

AAT does not involve just any pet interacting with a patient. Standards for the training of the volunteers and their animals are crucial in order to promote a safe, positive experience for the patient. Trained volunteers will understand how to work with other medical professionals to set goals for the patient and keep records of progress. Animals that have been appropriately trained are well socialized to people, other animals, and medical equipment. They are not distracted by the food and odors that may be present in the therapy environment and will not chew inappropriate objects or mark territory.

Animals participating in AAT should be covered by some form of liability insurance.

Research and general acceptance

While the research evidence supporting the efficacy of AAT is slim, the anecdotal support is vast. Although it may not be given much credence by medical personnel as a therapy with the potential to assist the progress of the patients, some institutions do at least allow it as something that will uplift the patients or distract them from their discomforts.

Resources

ORGANIZATIONS

Delta Society. <<http://www.deltasociety.org>>.

Judith Turner

Peyronie's disease

Definition

Peyronie's disease is a condition characterized by a bent penis.

Description

The cause of Peyronie's disease is unknown and the disease is often difficult to treat. For some reason, a thick scar develops in the penis and bends it. Almost a third of patients with Peyronie's disease also have similar contracting **scars** on their hands, a disease called Dupuytren's contractures. Some cases are associated with diabetes, and others appear after prostate surgery. Because prostate surgery always requires a catheter in the bladder, there is some suspicion that catheters can cause the scarring. However, many cases of Peyronie's disease arise without any use of a catheter. There is also a congenital form of penile deviation, again with no known cause. Most of the scars are located in the mid-line, therefore most of the angulations are either up or down.

Causes and symptoms

Peyronie's disease occurs in about 1% of men, most of them between 45–60 years old. Although there is no good research data to back it up, the suspicion exists that Peyronie's disease is the result of injury. If not a catheter, then sudden, forceful bending

KEY TERMS

Catheter—A flexible tube placed into a body vessel or cavity.

Congenital—Present at birth.

Plastic surgery—The restoring and reshaping of the skin and its appendages to improve their function and appearance.

Prostate—A gland that surrounds the outlet to the male bladder.

Prosthesis—Artificial substitute for a body part.

during sexual intercourse could easily tear the supporting tissues and lead to scarring.

The symptom is bending of an erect penis, sometimes with **pain**. It often interferes with sexual intercourse. Erectile failure associated with the angulation often precedes it.

Treatment

Attempts have been made to reduce the angulation with injections of cortisone-like drugs directly into the scar, but they are rarely successful. Surgery seems to be the better answer. After the scar is removed, plastic repair of the penis is attempted, often with a graft of tissue from somewhere else on the body. The Nesbit procedure is one of the more successful methods of doing this. The other surgical approach is to implant a penile prosthesis that overcomes the angulation mechanically. Results with these procedures are reported to be 60–80% satisfactory, including the return of orgasm.

Prognosis

Sometimes the condition disappears spontaneously. A careful look for other causes of impotence should be done before surgery.

Resources

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J. Ricker Polsdorfer, MD

Pharmacogenetics

Definition

Pharmacogenetics is the study of how the actions of and reactions to drugs vary with the patient's genes.

Description

Genes are the portions of chromosomes that determine many of the traits in every living thing. In humans, genes influence race, hair and eye color, gender, height, weight, aspects of behavior, and even the likelihood of developing certain diseases. Although some traits are a combination of genetics and environment, researchers are still discovering new ways in which people are affected by their genes.

Pharmacogenetics is the study of how people respond to drug therapy. Although this science is still new, there have been many useful discoveries. It has long been known that genes influence the risk of developing certain diseases, or that genes could determine traits such as hair and eye color. Genes can also alter the risk of developing different diseases. It has long been known that people of African descent were more likely to have sickle cell anemia than people of other races. People of Armenian, Arab, and Turkish heritage are more prone to familiar Mediterranean **fever** than people of other nationalities. More recently, discoveries have shown that genes can determine other aspects of each individual, down to the level of the enzymes produced in the liver. Since these enzymes determine how quickly a drug is removed from the body, they can make major differences in the way people respond to drugs. Some of the most basic work concerns the way race and gender influence drug reactions—and race and gender are genetically determined.

Women often respond differently than men to drugs at the same dose levels. For example, women are more likely to have a good response to the **anti-depressant drugs** that act as serotonin specific reuptake inhibitors (SSRIs, the group that includes Prozac and Paxil) than they are to the older group of **tricyclic antidepressants** (the group that includes Elavil and Tofranil). Women have a greater response to some narcotic **pain** relieving drugs than do men, but get less relief from some non-narcotic pain medications. Women may show a greater response to some steroid hormones than men do, but have a lower level of response to some anti-anxiety medications than men.

Race may also affect the way people respond to some medications. In this case, race implies specific genetic factors that are generally, but not always, found among members of specific ethnic groups. For example, the angiotensin II inhibitor enalapril (Vasotec), which is used to lower blood pressure, works better in Caucasians than in Blacks. Carvedilol (Coreg), a beta-adrenergic blocking agent that is also used to lower blood pressure, is more effective than other drugs in the same class when used to treat Black patients. Black patients with **heart failure** appear to respond better to a combination of hydralazine and isosorbide than do Caucasian patients using the same medication.

More specific research has identified individual genes than may influence drugs response, without relying on group information such as gender and race. Specific genes have been identified that may determine how patients will respond to specific drugs. For example, some genes may determine whether people will get pain relief from codeine, or how well they will respond to drugs used to treat **cancer**.

Causes and symptoms

Genes alter responses to drugs because the genes influence many parts of the body itself. One of the simplest examples is the gene that influences body weight. Since many drugs are soluble in body fat, people with large amounts of fat will have these drug deposited into their fat stores. This means that there are lower levels of the drug that can reach the actual organs on which they work.

In the case of gender responses to antidepressants, women show greater response to serotonin specific antidepressants because women naturally have lower levels of serotonin than men do. This makes women more likely to develop a type of depression marked by low serotonin levels, but it also means that women will respond better to replacement of serotonin.

Because people of the same race carry similar genes, studies based on race were the earliest types of pharmacogenetic studies. One study evaluated the levels of alcohol dehydrogenase in people of different nationalities. This is an enzyme involved in the metabolism of alcohol. When people with high levels of this enzyme, or people in whom the enzyme acts more rapidly than in other people, drink alcohol, they are subject to facial flushing and slowing of the heartbeat. The activity of this enzyme is determined by genetics, and different levels can be seen in different races because these people belong to the same gene pools.

Among Asiatic people, 85% have high levels of this enzyme, compared to 20% of Swiss people, and only 5–10% of British people.

Another trait that is influenced by genes is a liver enzyme, CYP2D6. This enzyme metabolizes some drugs, convert them to a form that can be removed from the body. Genes determine the level of this enzyme in the liver. People with low levels of CYP2D6 will metabolize drugs slowly. Slow metabolism means the drugs will act for a longer period of time. Slow metabolizers respond to smaller doses of medications that are eliminated by this enzyme, while fast metabolizers, people who have a lot of the enzyme, will need larger drug doses to get the same effects. At the same time, low levels of CYP2D6 means that people taking the drugs that are metabolized by this enzyme will have higher drug levels, and are more likely to have unwanted side effects.

Another enzyme that can be important in drug dosing is called 2C9, and this enzyme is responsible for metabolizing the anticoagulant drug warfarin (Coumadin). Most people take warfarin in a dose of about 5 milligrams a day, but people who have low levels of 2C9 normally require a dose of only 1–5 milligrams a week.

Yet another mechanism of drug activity is the presence or absence of a specific drug receptor site. Drugs act by binding to specific chemicals, receptor sites, within body cells. Genes may help determine how many of these cells there are. The action of the widely used antipsychotic drug haloperidol (Haldol) depends on its ability to bind to the dopamine (D2) receptor site. The number of these sites are determined by genetics. In one study, 63% of patients whose genes caused a large number of these receptor sites had a response to treatment with haloperidol, while only about 29% of patients with a smaller number of dopamine (D2) receptor sites did well on the drug.

Other genetic studies indicate that genes may affect how people respond to foods as well as to drugs. An Australian study of **osteoporosis** (softening of the bones that often occurs in elderly people), reported that separate genes may affect response to vitamin D, calcium, and estrogens.

Implications

Although the study is still new, pharmacogenetics promises to offer great benefits in drug effectiveness and safety.

At the present time, most drug treatment is done by trial and error. Physicians prescribe medication, and the patient tries the drug. The drug may work,

KEY TERMS

Enzyme—Proteins produced by living cells that help produce specific biochemical reactions in the body.

Metabolism—The process by which foods and drugs are broken down for use and removal from the body.

Sickle cell anemia—A severe, inheritable disease, most common among people of African descent, marked by deformation and destruction of red blood cells, and by adherence of blood cells to the walls of blood vessels.

or it may not. It may cause adverse effects, or it may be safe. If the drug does not work, the dose is increased. If it causes harmful or unpleasant effects, a new drug is tried until, finally, the right drug is found. In some cases this procedure may take weeks or even months.

In other cases, drugs are carefully tested, and appear to be safe and effective. Only after they are approved for general use are reports of serious adverse effects that did not appear in the initial studies documented. This can occur if there is a rare gene that affects the way in which the drug acts, or the way in which the drug is metabolized.

With increasing understanding of how genes determine the way people respond to drugs, it will be possible to select drugs and doses based on a greater understanding of each individual patient. This promises more effective drug therapy, with greater safety and fewer treatment failures.

Physicians may be able to compare the person's genetic make-up with the properties of specific drugs, and make informed decisions about which drug in a group will work most effectively or most safely.

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University of California, Los Angeles Harbor-UCLA
Medical Center Research and Education Institute 1124
W Carson St., B-4 South Torrance, CA 90502.

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Pharyngeal pouch see **Esophageal pouches**

Pharyngitis see **Sore throat**

Phenelzine see **Monoamine oxidase inhibitors**

Phenobarbital see **Barbiturates**

Phenol see **Antiseptics**

Phenolphthalein see **Laxatives**

Phenylalaninemia see **Phenylketonuria**

Phenylketonuria

Definition

Phenylketonuria (PKU) can be defined as a rare metabolic disorder caused by a deficiency in the production of the hepatic (liver) enzyme phenylalanine hydroxylase (PAH). PKU is the most serious form of a class of diseases referred to as "hyperphenylalaninemia," all of which involve above normal (elevated) levels of phenylalanine in the blood. The primary symptom of untreated PKU, **mental retardation**, is the result of consuming foods that contain the amino acid phenylalanine, which is toxic to brain tissue.

PKU is an inherited, autosomal recessive disorder. It is the most common genetic disease involving "amino acid metabolism." PKU is incurable, but early, effective treatment can prevent the development of serious mental incapacity.

Description

PKU is a disease caused by the liver's inability to produce a particular type of PAH enzyme. This enzyme converts (metabolizes) the amino acid called phenylalanine into another amino acid, tyrosine. This is the only role of PAH in the body. A lack of PAH results in the buildup of abnormally high phenylalanine concentrations (or levels) in the blood and brain. Above normal levels of phenylalanine are toxic to the cells that make up the nervous system and causes irreversible abnormalities in brain structure and function in PKU patients. Phenylalanine is a type of

teratogen. Teratogens are any substance or organism that can cause **birth defects** in a developing fetus.

The liver is the body's chief protein processing center. Proteins are one of the major food nutrients. They are generally very large molecules composed of strings of smaller building blocks or molecules called amino acids. About twenty amino acids exist in nature. The body breaks down proteins from food into individual amino acids and then reassembles them into "human" proteins. Proteins are needed for growth and repair of cells and tissues, and are the key components of enzymes, antibodies, and other essential substances.

PKU affects on the human nervous system

The extensive network of nerves in the brain and the rest of the nervous system are made up of nerve cells. Nerve cells have specialized extensions called dendrites and axons. Stimulating a nerve cell triggers nerve impulses, or signals, that speed down the axon. These nerve impulses then stimulate the end of an axon to release chemicals called "neurotransmitters" that spread out and communicate with the dendrites of neighboring nerve cells.

Many nerve cells have long, wire-like axons that are covered by an insulating layer called the myelin sheath. This covering helps speed nerve impulses along the axon. In untreated PKU patients, abnormally high phenylalanine levels in the blood and brain can produce nerve cells with "deformed" axons and dendrites, and cause imperfections in the myelin sheath referred to as hypomyelination and demyelination. This loss of myelin can "short circuit" nerve impulses (messages) and interrupt cell communication. A number of brain scan studies also indicate a degeneration of the "white matter" in the brains of older patients who have not maintained adequate dietary control.

PKU can also affect the production of one of the major neurotransmitters in the brain, called dopamine. The brain makes dopamine from the amino acid tyrosine. PKU patients who do not consume enough tyrosine in their diet cannot produce sufficient amounts of dopamine. Low dopamine levels in the brain disrupt normal communication between nerve cells, which results in impaired cognitive (mental) function.

Some preliminary research suggests that nerve cells of PKU patients also have difficulty absorbing tyrosine. This abnormality may explain why many PKU patients who receive sufficient dietary tyrosine still experience some form of learning disability.

Behavior and academic performance

IQ (intelligence quotient) tests provide a measure of cognitive function. The IQ of PKU patients is generally lower than the IQ of their healthy peers. Students with PKU often find academic tasks difficult and must struggle harder to succeed than their non-PKU peers. They may require special tutoring and need to repeat some of their courses. Even patients undergoing treatment programs may experience problems with typical academic tasks as math, reading, and spelling. Visual perception, visual-motor skills, and critical thinking skills can also be affected. Ten years of age seems to be an important milestone for PKU patients. After age 10, variations in a patient's diet seems to have less influence on their IQ development.

People with PKU tend to avoid contact with others, appear anxious and show signs of depression. However, some patients may be much more expressive and tend to have hyperactive, talkative, and impulsive personalities. It is also interesting to note that people with PKU are less likely to display such "antisocial" habits as lying, teasing, and active disobedience. It should be emphasized that current research findings are still quite preliminary and more extensive research is needed to clearly show how abnormal phenylalanine levels in the blood and brain might affect behavior and academic performance.

One in fifty individuals in the United States have inherited a gene for PKU. About five million Americans are PKU carriers. About one in 15,000 babies test positive for PKU in the United States. Studies indicate that the incidence of this disease in Caucasian and Native American populations is higher than in African-American, Hispanic, and Asian populations.

Causes and symptoms

PKU symptoms are caused by alterations or "mutations" in the genetic code for the PAH enzyme. Mutations in the PAH gene prevent the liver from producing adequate levels of the PAH enzyme needed to break down phenylalanine. The PAH gene and its PKU mutations are found on chromosome 12 in the human genome. In more detail, PKU mutations can involve many different types of changes, such as deletions and insertions, in the DNA of the gene that codes for the PAH enzyme.

PKU is described as an inherited, autosomal recessive disorder. The term autosomal means that the gene for PKU is not located on either the X or Y

sex chromosome. The normal PAH gene is dominant to recessive PKU mutations. A recessive genetic trait, such as PKU, is one that is expressed—or shows up—only when two copies are inherited (one from each parent).

A person with one normal and one PKU gene is called a carrier. A carrier does not display any symptoms of the disease because their liver produces normal quantities of the PAH enzyme. However, PKU carriers can pass the PKU genetic mutation onto their children. Two carrier parents have a 25% chance of producing a baby with PKU symptoms, and a 50% chance having a baby that is a carrier for the disease. Although PKU conforms to these basic genetic patterns of inheritance, the actual expression, or phenotype, of the disease is not strictly an “either/or” situation. This is because there are at least 400 different types of PKU mutations. Although some PKU mutations cause rather mild forms of the disease, others can initiate much more severe symptoms in untreated individuals. The more severe the PKU mutation, the greater the effect on cognitive development and performance (mental ability).

Untreated PKU patients develop a broad range of symptoms related to severely impaired cognitive function, sometimes referred to as mental retardation. Other symptoms can include extreme patterns of behavior, delayed speech development, seizures, a characteristic body odor, and light body pigmentation. The light pigmentation is due to a lack of melanin, which normally colors the hair, skin and eyes. Melanin is made from the amino acid tyrosine, which is lacking in untreated cases of PKU. Physiologically, PKU patients show high levels of phenylalanine and low levels of tyrosine in the blood. Babies do not show any visible symptoms of the disease for the first few months of life. However, typical PKU symptoms usually do show up by a baby's first birthday.

Diagnosis

The primary diagnostic test for PKU is the measurement of phenylalanine levels in a drop of blood taken from the heel of a newborn baby's foot. This screening procedure is referred to as the Guthrie test (Guthrie bacterial inhibition assay). In this test, PKU is confirmed by the appearance of bacteria growing around high concentrations of phenylalanine in the blood spot. PKU testing was introduced in the early 1960s and is the largest genetic screening program in the United States. It is required by law in all 50 states. Early diagnosis is critical. It ensures early the treatment PKU babies need to develop normally and avoid the ravages of PKU.

The American Academy of Pediatrics recommends that this test should be performed on infants between 24 hours and seven days after birth. The preferred time for testing is after the baby's first feeding. If the initial PKU test produces a positive result, then follow-up tests are performed to confirm the diagnosis and to determine if the elevated phenylalanine levels may be caused by some medical condition other than PKU. Treatment for PKU is recommended for babies that show a blood phenylalanine level of 7–10 mg/dL or higher for more than a few consecutive days. Another, more accurate test procedure for PKU measures the ratio (comparison) of the amount of phenylalanine to the amount of tyrosine in the blood.

Newer diagnostic procedures (called mutation analysis and genotype determination) can actually identify the specific types of PAH gene mutations inherited by PKU infants. Large-scale studies have helped to clarify how various mutations affect the ability of patients to process phenylalanine. This information can help doctors develop more effective customized treatment plans for each of their PKU patients.

Treatment

The severity of the PKU symptoms experienced by people with this disease is determined by both lifestyle as well as genetic factors. In the early 1950s, researchers first demonstrated that phenylalanine-restricted **diets** could eliminate most of the typical PKU symptoms—except for mental retardation. Today, dietary therapy (also called **nutrition** therapy) is the most common form of treatment for PKU patients. PKU patients who receive early and consistent dietary therapy can develop fairly normal mental capacity to within about five IQ points of their healthy peers. By comparison, untreated PKU patients generally have IQ scores below 50.

Infants with PKU should be put on a specialized diet as soon as they are diagnosed to avoid progressive brain damage and other problems caused by an accumulation of phenylalanine in the body. A PKU diet helps patients maintain very low blood levels of phenylalanine by restricting the intake of natural foods that contain this amino acid. Even breast milk is a problem for PKU babies. Special PKU dietary mixtures or formulas are usually obtained from medical clinics or pharmacies.

Phenylalanine is actually an essential amino acid. This means that it has to be obtained from food because the body cannot produce this substance on its own. Typical diets prescribed for PKU patients provide very small amounts of phenylalanine and

higher quantities of other amino acids, including tyrosine. The amount of allowable phenylalanine can be increased slightly as a child becomes older.

In addition, PKU diets include all the nutrients normally required for good health and normal growth, such as carbohydrates, fats, **vitamins**, and **minerals**. High protein foods like meat, fish, chicken, eggs, nuts, beans, milk, and other dairy products are banned from PKU diets. Small amounts of moderate protein foods (such as grains and potatoes) and low protein foods (some fruits and vegetables, low protein breads and pastas) are allowed. Sugar-free foods, such as diet soda, which contain the artificial sweetener aspartame, are also prohibited foods for PKU patients. That is because aspartame contains the amino acid phenylalanine.

Ideally, school-age children with PKU should be taught to assume responsibility for managing their diet, recording food intake, and for performing simple blood tests to monitor their phenylalanine levels. Blood tests should be done in the early morning when phenylalanine levels are highest. Infants and young children require more frequent blood tests than older children and adults. The amount of natural foods allowed in a diet could be adjusted to ensure that the level of phenylalanine in the blood is kept within a safe range—two to 6 mg/dL before 12 years of age and 2–15 mg/dL for PKU patients over 12 years old.

A specialized PKU diet can cause abnormal fluctuations in tyrosine levels throughout the day. Thus, some health professionals recommend adding time released tyrosine that can provide a more constant supply of this amino acid to the body. It should be noted that some PKU patients show signs of learning disabilities even with a special diet containing extra tyrosine. Research studies suggests that these PKU patients may not be able to process tyrosine normally.

For PKU caregivers, providing a diet that is appealing as well as healthy and nutritious is a constant challenge. Many PKU patients, especially teenagers, find it difficult to stick to the relatively bland PKU diet for extended periods of time. Some older patients decide to go off their diet plan simply because they feel healthy. However, many patients who abandon careful nutritional management develop cognitive problems, such as difficulties remembering, maintaining focus, and paying attention. Many PKU health professionals contend that all PKU patients should adhere to a strictly controlled diet for life.

One promising line of PKU research involves the synthesis (manufacturing) of a new type of enzyme that can break down phenylalanine in food consumed

by the patient. This medication would be taken orally and could prevent the absorption of digested phenylalanine into the patient's bloodstream.

In general, medical researchers express concern about the great variation in treatment programs currently available to PKU patients around the world. They have highlighted the urgent need for new, consistent international standards for proper management of PKU patients, which should emphasize comprehensive psychological as well as physiological monitoring and assessment.

PKU and Pregnancy

Women with PKU must be especially careful with their diets if they want to have children. They should ensure that phenylalanine blood levels are under control before conception and throughout her **pregnancy**. Mothers with elevated (higher than normal) phenylalanine levels are high risk for having babies with significant birth defects, such as microencephaly (smaller than normal head size), and **congenital heart disease** (abnormal heart structure and function), stunted growth, mental retardation, and psychomotor (coordination) difficulties. This condition is referred to as maternal PKU and can even affect babies who do not have the PKU disease.

Prognosis

Early newborn screening, careful monitoring, and a life-long strict dietary management can help PKU patients to live normal, healthy, and long lives.

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KEY TERMS

Amino acid—Organic compounds that form the building blocks of protein. There are 20 types of amino acids (eight are “essential amino acids” which the body cannot make and must therefore be obtained from food).

Axon—Skinny, wire-like extension of nerve cells.

Enzyme—A protein that catalyzes a biochemical reaction or change without changing its own structure or function.

Gene—A building block of inheritance, which contains the instructions for the production of a particular protein, and is made up of a molecular sequence found on a section of DNA. Each gene is found on a precise location on a chromosome.

Genetic disease—A disease that is (partly or completely) the result of the abnormal function or expression of a gene; a disease caused by the inheritance and expression of a genetic mutation.

IQ—Abbreviation for Intelligence Quotient. Compares an individual's mental age to his/her true or chronological age and multiplies that ratio by 100.

Metabolism—The total combination of all of the chemical processes that occur within cells and tissues of a living body.

Mutation—A permanent change in the genetic material that may alter a trait or characteristic of an individual, or manifest as disease, and can be transmitted to offspring.

Myelin—A fatty sheath surrounding nerves in the peripheral nervous system, which help them conduct impulses more quickly.

Nervous system—The complete network of nerves, sense organs, and brain in the body.

Phenylalanine—An essential amino acid that must be obtained from food since the human body cannot manufacture it.

Protein—Important building blocks of the body, composed of amino acids, involved in the formation of body structures and controlling the basic functions of the human body.

Recessive—Genetic trait expressed only when present on both members of a pair of chromosomes, one inherited from each parent.

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Centers for Disease Control. GDP Office, 4770 Buford Highway NE, Atlanta, GA 30341-3724. (770) 488-3235. <<http://www.cdc.gov/genetics>>.

Children's PKU Network. 1520 State St., Suite 111, San Diego, CA 92101-2930. (619) 233-3202. Fax: (619) 233 0838. pkunetwork@aol.com.

March of Dimes Birth Defects Foundation. 1275 Mamaroneck Ave., White Plains, NY 10605. (888) 663-4637. resourcecenter@modimes.org. <<http://www.modimes.org>>.

National PKU News. Virginia Schuett, editor/dietician. 6869 Woodlawn Avenue NE #116, Seattle, WA 98115-5469. (206) 525-8140. Fax: (206) 525-5023. <<http://www.pkunews.org>>.

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Marshall G. Letcher, MA

Phenylpropanolamine see **Decongestants**

Phenytoin see **Anticonvulsant drugs**

Pheochromocytoma

Definition

Pheochromocytoma is a tumor of special cells (called chromaffin cells), most often found in the middle of the adrenal gland.

Description

Because pheochromocytomas arise from chromaffin cells, they are occasionally called chromaffin tumors. Most (90%) are benign tumors so they do not spread to other parts of the body. However, these tumors can cause many problems and if they are not treated and can result in death.

Pheochromocytomas can be found anywhere chromaffin cells are found. They may be found in the heart and in the area around the bladder, but most (90%) are found in the adrenal glands. Every individual has two adrenal glands that are located above the kidneys in the back of the abdomen. Each adrenal gland is made up of two parts: the outer part (called the adrenal cortex) and the inner part (called the adrenal medulla). Pheochromocytomas are found in the adrenal medulla. The adrenal medulla normally secretes two substances, or hormones, called norepinephrine and epinephrine. These two substances, when considered together, are known as adrenaline. Adrenaline is released from the adrenal gland, enters the bloodstream and helps to regulate many things in the body including blood pressure and heart rate. Pheochromocytomas cause the adrenal medulla to secrete too much adrenaline, which in turn causes high blood pressure. The high blood pressure usually causes the other symptoms of the disease.

Pheochromocytomas are rare tumors. They have been reported in babies as young as five days old as well as adults 92 years old. Although they can be found

at any time during life, they usually occur in adults between 30 and 40 years of age. Pheochromocytomas are somewhat more common in women than in men.

Causes and symptoms

The cause of most pheochromocytomas is not known. A small minority (about 10-20%) of pheochromocytomas arise because a person has an inherited susceptibility to them. Inherited pheochromocytomas are associated with four separate syndromes: Multiple Endocrine Neoplasia, type 2A (MEN2A), Multiple Endocrine Neoplasia, type 2B (MEN2B), von Hippel-Lindau disease (VHL) and **Neurofibromatosis** type 1 (NF1).

Individuals with pheochromocytomas as part of any of these four syndromes usually have other medical conditions as well. People with MEN2A often have **cancer** (usually **thyroid cancer**) and other hormonal problems. Individuals with MEN2B can also have cancer and hormonal problems, but also have other abnormal physical features. Both MEN2A and MEN2B are due to genetic alterations or mutations in a gene called RET, found at chromosome 10q11.2. Individuals with VHL often have other benign tumors of the central nervous system and pancreas, and can sometimes have renal cell cancer. This syndrome is caused by a mutation in the VHL gene, found at chromosome 3p25-26. Individuals with NF1 often have neurofibromas (benign tumors of the peripheral nervous system). NF1 is caused by mutations in the NF1 gene, found at chromosome 17q11.

All of these disorders are inherited in an autosomal dominant inheritance pattern. With autosomal dominant inheritance, men and women are equally likely to inherit the syndrome. In addition, children of individuals with the disease are at 50% risk of inheriting it. **Genetic testing** is available for these four syndromes (MEN2A, MEN2B, VHL and NF1) but, due to the complexity, **genetic counseling** should be considered before testing.

Most people (90%) with pheochromocytoma have **hypertension**, or high blood pressure. The other symptoms of the disease are extremely variable. These symptoms usually occur in episodes (or attacks) called paroxysms and include:

- headaches
- excess sweating
- racing heart
- rapid breathing
- anxiety/nervousness

- nervous shaking
- pain in the lower chest or upper abdomen
- nausea
- heat intolerance

The episodes can occur as often as 25 times a day or, as infrequently as once every few months. They can last a few minutes, several hours, or days. Usually, the attacks occur several times a week and last for about 15 minutes. After the episode is over, the person feels exhausted and fatigued.

Between the attacks, people with pheochromocytoma can experience the following:

- increased sweating
- cold hands and feet
- weight loss
- constipation

Diagnosis

If a pheochromocytoma is suspected, urine and/or a blood test are usually recommended. A test called “24-hour urinary catecholamines and metanephrines” will be done. This test is designed to look for adrenaline and the break-down products of adrenaline. Since the body gets rid of these hormones in the urine, those testing will need to collect their urine for 24 hours. The laboratory will determine whether or not the levels of hormones are too high. This test is very good at making the diagnosis of pheochromocytoma. Another test called “serum catecholamines” measures the level of adrenaline compounds in the blood. It is not as sensitive as the 24-hour urine test, but can still provide some key information if it shows that the level of adrenaline compounds is too high.

One of the difficulties with these tests is that a person needs to have an attack of symptoms either during the 24-hour urine collection time period or shortly before the blood is drawn for a serum test to ensure the test’s accuracy. If a person did not have an episode during that time, the test can be a “false negative”. If a doctor suspects the patient has had a “false negative” test, additional tests called “pharmacologic tests” can be ordered. During these tests, a specific drug is given to the patient (usually through an IV) and the levels of hormones are monitored from the patient’s blood. These types of tests are only done rarely.

Once a person has been diagnosed with a pheochromocytoma, he or she will undergo tests to identify exactly where in the body the tumor is located. The

imaging techniques used are usually computed tomography scan (CT scan) and magnetic resonance imaging (MRI). A CT scan creates pictures of the interior of the body from computer-analyzed differences in x rays passing through the body. CT scans are performed at a hospital or clinic and take only a few minutes. An MRI is a computerized scanning method that creates pictures of the interior of the body using radio waves and a magnet. An MRI is usually performed at a hospital and takes about 30 minutes.

Treatment

Once a pheochromocytoma is found, more tests will be done to see if the tumor is benign (not cancer) or malignant (cancer). If the tumor is malignant, tests will be done to see how far the cancer has spread. There is no accepted staging system for pheochromocytoma; but an observation of the tumor could provide one of these four indications:

- Localized benign pheochromocytoma means that the tumor is found only in one area, is not cancer, and cannot spread to other tissues of the body.
- Regional pheochromocytoma means that the tumor is malignant and has spread to the lymph nodes around the original cancer. Lymph nodes are small structures found all over the body that make and store infection-fighting cells.
- Metastatic pheochromocytoma means that the tumor is malignant and has spread to other, more distant parts of the body.
- Recurrent pheochromocytoma means that a malignant tumor that was removed has come back.

Treatment in all cases begins with surgical removal of the tumor. Before surgery, medications such as alpha-adrenergic blockers are given to block the effect of the hormones and normalize blood pressure. These medications are usually started seven to 10 days prior to surgery. The surgery of choice is laparoscopic laparotomy, which is a minimally invasive outpatient procedure performed under general or **local anesthesia**. A small incision is made in the abdomen, the laparoscope is inserted, and the tumor is removed. The patient can usually return to normal activities within two weeks. If a laparoscopic laparotomy cannot be done, a traditional laparotomy will be performed. This is a more invasive surgery done under spinal or **general anesthesia** and requires five to seven days in the hospital. Usually patients are able to return to normal activities after four weeks. After surgery, blood and urine tests will be done to make sure hormone levels return to normal. If the hormone levels are

KEY TERMS

Adrenal medulla—The central core of the adrenal gland.

Laparoscope—An instrument used to examine body cavities during certain types of surgery; for example, surgeries to remove fibroid tumors, or gall bladders, are often removed through the navel rather than cutting into the body.

Paroxysm—A sudden attack of symptoms.

still above normal, it may mean that some tumor tissue was not removed. If not all tumor can be removed (as in malignant pheochromocytoma, for example) drugs will be given to control high blood pressure.

If a pheochromocytoma is malignant, radiation therapy and/or **chemotherapy** may be used. **Radiation therapy** uses high-energy x rays to kill cancer cells and shrink tumors. Because there is no evidence that radiation therapy is effective in the treatment of malignant pheochromocytoma, it is not often used for treatment. However, it is useful in the treatment of painful bone metastases if the tumor has spread to the bones. Chemotherapy uses drugs to kill cancer cells. Like radiation therapy, it has not been shown to be effective in the treatment of malignant pheochromocytoma. Chemotherapy, therefore, is only used in rare instances.

Untreated pheochromocytoma can be fatal due to complications of the high blood pressure. In the vast majority of cases, when the tumor is surgically removed, pheochromocytoma is cured. In the minority of cases (10%) where pheochromocytoma is malignant, prognosis depends on how far the cancer has spread, and the patient's age and general health. The overall median five-year survival from the initial time of surgery and diagnosis is approximately 43%.

Prevention

Unfortunately, little is known about environmental and other causes of pheochromocytoma. Some of the tumors are due to inherited predisposition. Because of these factors, pheochromocytoma can not be prevented.

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Phimosis

Definition

A tightening of the foreskin of the penis that may close the opening of the penis.

Description

The foreskin of a newborn boy is always closely contracted around the penis head (glans). Only a small passage allows the urine to pass through. In the first months the foreskin is stuck to the glans and cannot be pulled back and one should not attempt to do so. During the first couple of years, the foreskin will become gradually looser and in many boys it can in time be pulled back without trouble. Half of all three-year-olds can pull back their foreskin. It is not advisable to try pulling the foreskin back using force, since this may cause small cuts in the foreskin with **scars** which could finally cause a regular foreskin contraction.

Foreskin contraction, called phimosis, can last throughout life and not cause any trouble at all. It is a voluntary decision whether to have a **circumcision** operation or not. If any problems do arise, they happen after **puberty**. The contraction may occur for the first time as an adult and usually requires circumcision, the surgical removal of the foreskin.

Causes & symptoms

Phimosis is caused by the inability of the foreskin to retract from around the opening of the penis. In adults, phimosis can lead to chronic inflammation and cancer.

Diagnosis

A physician usually diagnoses phimosis when there are persistent problems urinating, when there are recurrent infections under the foreskin, or when the opening to the penis is completely blocked by the foreskin. Phimosis is a tight ring of foreskin often made of scar tissue preventing retraction of the foreskin. It may be primary, or secondary to recurrent infection. It may produce urinary obstruction with ballooning of the foreskin. Phimosis is different than having a non-retractable foreskin, which is normal in many boys.

Treatment

If the foreskin cannot be pulled back into place treatment should be sought. If the blood flow to the penis is restricted then emergency treatment is required and if the foreskin cannot be pulled back a surgical cut to the trapped foreskin may be required. Failure to seek treatment can result in permanent damage to the penis. Once phimosis is diagnosed, the available treatments include topical **corticosteroids**, manual stretching, foreskin surgical repair or **plastic surgery**, and circumcision. Conservative treatments should be tried in the first instance and surgery used as the treatment of last resort.

A number of studies show that phimosis can be safely and effectively treated by the application of topical steroids in 80–90% of cases. Betamethasone cream 0.05% should be applied to the exterior and interior of the tip of the foreskin two or three times a day. The treatment should be discontinued as ineffective after three months if the foreskin has not become retractile during this time.

A number of corrections are available for the adult or adolescent non-retractable foreskin. These include surgery to repair the foreskin, in which an incision is made through the constrictive band of the foreskin. The underlying tissue is spread with forceps to expose the Buck's fascia (the deep, connective tissue of the penis) and the incision is closed with absorbable sutures. This procedure has less risk of disease and infection than circumcision, and allows the foreskin to be retained.

Circumcision is very traumatic to a child. It is essentially irreversible and should be the treatment of

KEY TERMS

Balanitis xerotica obliterans (BXO)—A chronic, progressive, hardening skin inflammation of the penis.

Buck's fascia—The deep connective tissue of the penis.

Circumcision—The removal of all or part of the foreskin from the penis.

Corticosteroids—A synthetic drug similar or identical to a natural corticosteroid, used to reduce inflammation.

Glans—The head of the penis.

Paraphimosis—The entrapment of a retracted foreskin behind the coronal sulcus, a groove that separates the shaft and head of the penis.

last resort. Phimosis due to *balanitis xerotica obliterans* (BXO), a chronic, progressive, hardening skin inflammation of the penis, has been considered the one common absolute indication for circumcision.

Alternative treatment

There are no alternative medicine treatments for phimosis.

Prognosis

In most men, phimosis is not a serious problem and will not require treatment. However, it is not expected to improve on its own. With treatment, phimosis in most males can be managed or corrected.

Prevention

Proper hygiene is the most important preventative measure. The American Academy of Pediatrics recommends that the immature foreskin of boys not be forced back for cleaning. The only person who should clean and retract the foreskin is the boy himself. Bubble bath products and other chemical irritants can cause the foreskin to tighten and it is recommended they should be avoided by males with foreskins.

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Phlebitis see **Thrombophlebitis**

Phlebotomy

Definition

Phlebotomy is the act of drawing or removing blood from the circulatory system through a cut (incision) or puncture in order to obtain a sample for analysis and diagnosis. Phlebotomy is also done as part of the patient's treatment for certain blood disorders.

Purpose

Treatment

Phlebotomy that is part of treatment (therapeutic phlebotomy) is performed to treat polycythemia vera, a condition that causes an elevated red blood cell volume (**hematocrit**). Phlebotomy is also prescribed for patients with disorders that increase the amount of iron in their blood to dangerous levels, such as hemochromatosis, **hepatitis B**, and hepatitis C. Patients with pulmonary edema may undergo phlebotomy procedures to decrease their total blood volume.

Diagnosis

Phlebotomy is also used to remove blood from the body during **blood donation** and for analysis of the substances contained within it.

Precautions

Patients who are anemic or have a history of cardiovascular disease may not be good candidates for phlebotomy.

Description

Phlebotomy, which is also known as venesection, is performed by a nurse or a technician known as a phlebotomist. Blood is usually taken from a vein on the back of the hand or inside of the elbow. Some blood tests, however, may require blood from an artery. The skin over the area is wiped with an antiseptic, and an elastic band is tied around the arm. The band acts as a tourniquet, slowing the blood flow in the arm and making the veins more visible. The patient is asked to make a fist, and the technician feels the veins in order to select an appropriate one. When a vein is selected, the technician inserts a needle into the vein and releases the elastic band. The appropriate amount of blood is drawn and the needle is withdrawn from the vein. The patient's pulse and blood pressure may be monitored during the procedure.

For some tests requiring very small amounts of blood for analysis, the technician uses a finger stick. A lance, or small needle, makes a small cut in the surface of the fingertip, and a small amount of blood is collected in a narrow glass tube. The fingertip may be squeezed to get additional blood to surface.

The amount of blood drawn depends on the purpose of the phlebotomy. Blood donors usually contribute a unit of blood (500 mL) in a session. The volume of blood needed for laboratory analysis varies widely with the type of test being conducted. Therapeutic phlebotomy removes a larger amount of blood than donation and blood analysis require. Phlebotomy for treatment of **hemochromatosis** typically involves removing a unit of blood—or 250 mg of iron—once a week. Phlebotomy sessions are required until iron levels return to a consistently normal level, which may take several months to several years. Phlebotomy for polycythemia vera removes enough blood to keep the patient's hematocrit below 45%. The frequency and duration of sessions depends on the patient's individual needs.

Preparation

Patients having their blood drawn for analysis may be asked to discontinue medications or to avoid food (to fast) for a period of time before the blood test. Patients donating blood will be asked for a brief

KEY TERMS

Finger stick—A technique for collecting a very small amount of blood from the fingertip area.

Hemochromatosis—A genetic disorder known as iron overload disease. Untreated hemochromatosis may cause osteoporosis, arthritis, cirrhosis, heart disease, or diabetes.

Thrombocytosis—A vascular condition characterized by high blood platelet counts.

Tourniquet—Any device that is used to compress a blood vessel to stop bleeding or as part of collecting a blood sample. Phlebotomists usually use an elastic band as a tourniquet.

Venesection—Another name for phlebotomy.

medical history, have their blood pressure taken, and have their hematocrit checked with a finger stick test prior to donation.

Aftercare

After blood is drawn and the needle is removed, pressure is placed on the puncture site with a cotton ball to stop bleeding, and a bandage is applied. It is not uncommon for a patient to feel dizzy or nauseated during or after phlebotomy. The patient may be encouraged to rest for a short period once the procedure is completed. Patients are also instructed to drink plenty of fluids and eat regularly over the next 24 hours to replace lost blood volume. Patients who experience swelling of the puncture site or continued bleeding after phlebotomy should get medical help at once.

Risks

Most patients will have a small bruise or mild soreness at the puncture site for several days. Therapeutic phlebotomy may cause **thrombocytosis** and chronic iron deficiency (anemia) in some patients. As with any invasive procedure, infection is also a risk. This risk can be minimized by the use of prepackaged sterilized equipment and careful attention to proper technique.

Normal results

Normal results include obtaining the needed amount of blood with the minimum of discomfort to the patient.

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Paula Anne Ford-Martin

Phobias

Definition

A phobia is an intense but unrealistic fear that can interfere with the ability to socialize, work, or go about everyday life, brought on by an object, event or situation.

Description

Just about everyone is afraid of something—an upcoming job interview or being alone outside after dark. But about 18% of all Americans are tormented by irrational fears that interfere with their daily lives. They are not “crazy”—they know full well their fears are unreasonable—but they can not control the fear. These people have phobias.

Phobias belong to a large group of mental problems known as **anxiety disorders** that include **obsessive-compulsive disorder (OCD)**, **panic disorder**, and post-traumatic stress disorder. Phobias themselves can be divided into three specific types:

- specific phobias (formerly called “simple phobias”)
- social phobia
- agoraphobia

Specific phobias

As its name suggests, a specific phobia is the fear of a particular situation or object, including anything from airplane travel to dentists. Found in one out of every 10 Americans, specific phobias seem to run in families and are roughly twice as likely to appear in women. If the person rarely encounters the feared object, the phobia does not cause much harm. However, if the feared object or situation is common, it can seriously disrupt everyday life. Common examples of specific phobias, which can begin at any age, include fear of snakes, flying, dogs, escalators, elevators, high places, or open spaces.

Social phobia

People with social phobia have deep fears of being watched or judged by others and being embarrassed in public. This may extend to a general fear of social situations—or be more specific or circumscribed, such as a fear of giving speeches or of performing (stage fright). More rarely, people with social phobia may have trouble using a public restroom, eating in a restaurant, or signing their name in front of others.

Social phobia is not the same as **shyness**. Shy people may feel uncomfortable with others, but they don't experience severe **anxiety**, they don't worry excessively about social situations beforehand, and they don't avoid events that make them feel self-conscious. On the other hand, people with social phobia may not be shy—they may feel perfectly comfortable with people except in specific situations. Social phobias may be only mildly irritating, or they may significantly interfere with daily life. It is not unusual for people with social phobia to turn down job offers or avoid relationships because of their fears.

Agoraphobia

Agoraphobia is the intense fear of feeling trapped and having a panic attack in a public place. It usually begins between ages 15 and 35, and affects three times as many women as men—about 3% of the population.

An episode of spontaneous panic is usually the initial trigger for the development of agoraphobia. After an initial panic attack, the person becomes afraid of experiencing a second one. Patients literally “fear the fear,” and worry incessantly about when and where the next attack may occur. As they begin to avoid the places or situations in which the panic attack occurred, their fear generalizes. Eventually the person completely avoids public places. In severe cases, people with agoraphobia can no longer leave their homes for fear of experiencing a panic attack.

Causes and symptoms

Experts don't really know why phobias develop, although research suggests the tendency to develop phobias may be a complex interaction between heredity and environment. Some hypersensitive people have unique chemical reactions in the brain that cause them to respond much more strongly to **stress**. These people also may be especially sensitive to **caffeine**, which triggers certain brain chemical responses.

Advances in neuroimaging have also led researchers to identify certain parts of the brain and specific neural pathways that are associated with phobias. One

part of the brain that is currently being studied is the amygdala, an almond-shaped body of nerve cells involved in normal fear conditioning. Another area of the brain that appears to be linked to phobias is the posterior cerebellum.

While experts believe the tendency to develop phobias runs in families and may be hereditary, a specific stressful event usually triggers the development of a specific phobia or agoraphobia. For example, someone predisposed to develop phobias who experiences severe turbulence during a flight might go on to develop a phobia about flying. What scientists don't understand is why some people who experience a frightening or stressful event develop a phobia and others do not.

Social phobia typically appears in childhood or adolescence, sometimes following an upsetting or humiliating experience. Certain vulnerable children who have had unpleasant social experiences (such as being rejected) or who have poor social skills may develop social phobias. The condition also may be related to low self-esteem, unassertive personality, and feelings of inferiority.

A person with agoraphobia may have a panic attack at any time, for no apparent reason. While the attack may last only a minute or so, the person remembers the feelings of panic so strongly that the possibility of another attack becomes terrifying. For this reason, people with agoraphobia avoid places where they might not be able to escape if a panic attack occurs. As the fear of an attack escalates, the person's world narrows.

While the specific trigger may differ, the symptoms of different phobias are remarkably similar: e.g., feelings of terror and impending doom, rapid heartbeat and breathing, sweaty palms, and other features of a panic attack. Patients may experience severe anxiety symptoms in anticipating a phobic trigger. For example, someone who is afraid to fly may begin having episodes of pounding heart and sweating palms at the mere thought of getting on a plane in two weeks.

Diagnosis

A mental health professional can diagnose phobias after a detailed interview and discussion of both mental and physical symptoms. Social phobia is often associated with other anxiety disorders, depression, or **substance abuse**.

Treatment

People who have a specific phobia that is easy to avoid (such as snakes) and that doesn't interfere with their lives may not need to get help. When phobias do

interfere with a person's daily life, a combination of psychotherapy and medication can be quite effective. While most health insurance covers some form of mental health care, most do not cover outpatient care completely, and most have a yearly or lifetime maximum.

Medication can block the feelings of panic, and when combined with **cognitive-behavioral therapy**, can be quite effective in reducing specific phobias and agoraphobia.

Cognitive-behavioral therapy adds a cognitive approach to more traditional behavioral therapy. It teaches patients how to change their thoughts, behavior, and attitudes, while providing techniques to lessen anxiety, such as deep breathing, muscle relaxation, and refocusing.

One cognitive-behavioral therapy is desensitization (also known as exposure therapy), in which people are gradually exposed to the frightening object or event until they become used to it and their physical symptoms decrease. For example, someone who is afraid of snakes might first be shown a photo of a snake. Once the person can look at a photo without anxiety, he might then be shown a video of a snake. Each step is repeated until the symptoms of fear (such as pounding heart and sweating palms) disappear. Eventually, the person might reach the point where he can actually touch a live snake. Three fourths of patients are significantly improved with this type of treatment.

Another more dramatic cognitive-behavioral approach is called flooding. It exposes the person immediately to the feared object or situation. The person remains in the situation until the anxiety lessens.

Several drugs are used to treat specific phobias by controlling symptoms and helping to prevent panic attacks. These include anti-anxiety drugs (**benzodiazepines**) such as alprazolam (Xanax) or diazepam (Valium). Blood pressure medications called **beta blockers**, such as propranolol (Inderal) and atenolol (Tenormin), appear to work well in the treatment of circumscribed social phobia, when anxiety gets in the way of performance, such as public speaking. These drugs reduce overstimulation, thereby controlling the physical symptoms of anxiety.

In addition, some antidepressants may be effective when used together with cognitive-behavioral therapy. These include the monoamine oxidase inhibitors (MAOI inhibitors) phenelzine (Nardil) and tranylcypromine (Parnate), as well as **selective serotonin reuptake inhibitors** (SSRIs) like fluoxetine (Prozac), paroxetine (Paxil), sertraline (Zoloft) and fluvoxamine (Luvox).

A medication that shows promise as a treatment for social phobia is valproic acid (Depakene or Depakote), which is usually prescribed to treat seizures or to prevent migraine headaches. Researchers conducting a twelve-week trial with 17 patients found that about half the patients experienced a significant improvement in their social anxiety symptoms while taking the medication. Further studies are underway.

In all types of phobias, symptoms may be eased by lifestyle changes, such as:

- eliminating caffeine
- cutting down on alcohol
- eating a good diet
- getting plenty of **exercise**
- reducing stress

Treating agoraphobia is more difficult than other phobias because there are often so many fears involved, such as open spaces, traffic, elevators, and escalators. Treatment includes cognitive-behavioral therapy with antidepressants or anti-anxiety drugs. Paxil and Zoloft are used to treat panic disorders with or without agoraphobia.

Prognosis

Phobias are among the most treatable mental health problems; depending on the severity of the condition and the type of phobia, most properly treated patients can go on to lead normal lives. Research suggests that once a person overcomes the phobia, the problem may not return for many years—if at all.

Untreated phobias are another matter. Only about 20% of specific phobias will go away without treatment, and agoraphobia will get worse with time if untreated. Social phobias tend to be chronic, and will not likely go away without treatment. Moreover, untreated phobias can lead to other problems, including depression, **alcoholism**, and feelings of shame and low self-esteem.

A group of researchers in Boston reported in 2003 that phobic anxiety appears to be a risk factor for Parkinson's disease (PD) in males, although it is not yet known whether phobias cause PD or simply share an underlying biological cause.

While most specific phobias appear in childhood and subsequently fade away, those that remain in adulthood often need to be treated. Unfortunately, most people never get the help they need; only about 25% of people with phobias ever seek help to deal with their condition.

KEY TERMS

Agoraphobia—An intense fear of being trapped in a crowded, open, or public space where it may be hard to escape, combined with the dread of having a panic attack.

Benzodiazepine—A class of drugs that have a hypnotic and sedative action, used mainly as tranquilizers to control symptoms of anxiety.

Beta blockers—A group of drugs that are usually prescribed to treat heart conditions, but that also are used to reduce the physical symptoms of anxiety and phobias, such as sweating and palpitations.

Monoamine oxidase inhibitors (MAO inhibitors)—A class of antidepressants used to treat social phobia.

Neuroimaging—The use of x ray studies and magnetic resonance imaging (MRIs) to detect abnormalities or trace pathways of nerve activity in the central nervous system.

Selective serotonin reuptake inhibitors (SSRIs)—A class of antidepressants that work by blocking the reabsorption of serotonin in the brain, raising the levels of serotonin. SSRIs include Prozac, Zoloft, and Paxil.

Serotonin—One of three major types of neurotransmitters found in the brain that is linked to emotions.

Social phobia—Fear of being judged or ridiculed by others; fear of being embarrassed in public.

Prevention

There is no known way to prevent the development of phobias. Medication and cognitive-behavioral therapy may help prevent the recurrence of symptoms once they have been diagnosed.

Resources

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ORGANIZATIONS

Agoraphobics Building Independent Lives. 1418 Lorraine Ave., Richmond, VA 23227.

Agoraphobics In Motion. 605 W. 11 Mile Rd., Royal Oak, MI 48067.

American Psychiatric Association (APA). 1400 K Street, NW, Washington, DC 20005. (888) 357-7924. <<http://www.psych.org>>.

Anxiety Disorders Association of America. 11900 Parklawn Dr., Ste. 100, Rockville, MD 20852. (301) 231-9350.

National Anxiety Foundation. 3135 Custer Dr., Lexington, KY 40517. (606) 272-7166. <<http://www.lexington-online.com/naf.html>>.

National Institute of Mental Health (NIMH) Office of Communications. 6001 Executive Boulevard, Room 8184, MSC 9663, Bethesda, MD 20892-9663. (866) 615-NIMH or (301) 443-4513. <<http://www.nimh.nih.gov>>.

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Carol A. Turkington
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Phospholipidosis see **Pulmonary alveolar proteinosis**

Phosphorus imbalance

Definition

Phosphorus imbalance refers to conditions in which the element phosphorus is present in the body at too high a level (hyperphosphatemia) or too low a level (hypophosphatemia).

Description

Almost all of the phosphorus in the body occurs as phosphate (phosphorus combined with four oxygen atoms), and most of the body's phosphate (85%) is located in the skeletal system, where it combines with calcium to give bones their hardness. The remaining amount (15%) exists in the cells of the body, where it plays an important role in the formation of key nucleic acids, such as DNA, and in the process by which the body turns food into energy (metabolism). The body regulates phosphate levels in the blood through the controlled release of parathyroid hormone (PTH) from the parathyroid gland and calcitonin from the thyroid gland. PTH keeps phosphate levels from becoming too high by stimulating the excretion of phosphate in urine and causing the release of calcium from bones (phosphate blood levels are inversely proportional to calcium blood levels). Calcitonin keeps phosphate blood levels in check by moving phosphates out of the blood and into the bone matrix to form a mineral salt with calcium.

Most phosphorus imbalances develop gradually and are the result of other conditions or disorders, such as **malnutrition**, poor kidney function, or a malfunctioning gland.

Causes and symptoms

Hypophosphatemia

Hypophosphatemia (low blood phosphate) has various causes. **Hyperparathyroidism**, a condition in which the parathyroid gland produces too much PTH, is one primary cause. Poor kidney function, in which the renal tubules do not adequately reabsorb phosphorus, can result in hypophosphatemia, as can overuse of **diuretics**, such as theophylline, and antacids containing aluminum hydroxide. Problems involving the intestinal absorption of phosphate, such as chronic **diarrhea** or a deficiency of vitamin D (needed by the intestines to properly absorb phosphates) can cause the condition. Malnutrition due to chronic alcoholism can result in an inadequate intake of phosphorus. Recovery from conditions such as **diabetic ketoacidosis** or severe burns can provoke hypophosphatemia, since the body must use larger-than-normal amounts of phosphate. Respiratory alkalosis, brought on by hyperventilation, can also result in temporary hypophosphatemia.

Symptoms generally occur only when phosphate levels have decreased profoundly. They include muscle weakness, **tingling** sensations, **tremors**, and bone weakness. Hypophosphatemia may also result in confusion and memory loss, seizures, and coma.

Hyperphosphatemia

Hyperphosphatemia (high blood phosphate) also has various causes. It is most often caused by a decline in the normal excretion of phosphate in urine as a result of kidney failure or impaired function. **Hypoparathyroidism**, a condition in which the parathyroid gland does not produce enough PTH, or **pseudoparathyroidism**, a condition in which the kidneys lose their ability to respond to PTH, can also contribute to decreased phosphate excretion. Hyperphosphatemia can also result from the overuse of **laxatives** or enemas that contain phosphate. **Hypocalcemia** (abnormally low blood calcium) can cause phosphate blood levels to increase abnormally. A side-effect of hyperphosphatemia is the formation of calcium-phosphate crystals in the blood and soft tissue.

Hyperphosphatemia is generally asymptomatic; however, it can occur in conjunction with hypocalcemia, the symptoms of which are **numbness and tingling** in the extremities, **muscle cramps** and spasms, depression, memory loss, and convulsions. When calcium-phosphate crystals build up in the blood vessels, they can cause arteriosclerosis, which can lead to heart attacks or strokes. When the crystals build up in the skin, they can cause severe itching.

Diagnosis

Disorders of phosphate metabolism are assessed by measuring serum or plasma levels of phosphate and calcium. Hypophosphatemia is diagnosed if the blood phosphate level is less than 2.5 milligrams per deciliter of blood. Hyperphosphatemia is diagnosed if the blood phosphate level is above 4.5 milligrams per deciliter of blood. Appropriate tests are also used to determine if the underlying cause of the imbalance, including assessments of kidney function, dietary intake, and appropriate hormone levels.

Treatment

Treatment of phosphorus imbalances focuses on correcting the underlying cause of the imbalance and restoring equilibrium. Treating the underlying condition may involve surgical removal of the parathyroid gland in the case of hypophosphatemia caused by hyperparathyroidism; initiating hormone therapy in cases of hyperphosphatemia caused by hypoparathyroidism; ceasing intake of drugs or medications that contribute to phosphorus imbalance; or instigating measures to restore proper kidney function.

Restoring phosphorus equilibrium in cases of mild hypophosphatemia may include drinking a prescribed

solution that is rich in phosphorus; however, since this solution can cause diarrhea, many doctors recommend that patients drink 1 qt (0.9 L) of skim milk per day instead, since milk and other dairy products are significant sources of phosphate. Other phosphate-rich foods include green, leafy vegetables; peas and beans; nuts; chocolate; beef liver; turkey; and some cola drinks. Severe hypophosphatemia may be treated with the administration of an intravenous solution containing phosphate.

Restoring phosphorus equilibrium in cases of mild hyperphosphatemia involves restricting intake of phosphorus-rich foods and taking a calcium-based antacid that binds to the phosphate and blocks its absorption in the intestines. In cases of severe hyperphosphatemia, an intravenous infusion of calcium gluconate may be administered. Dialysis may also be required in severe cases to help remove excess phosphate from the blood.

Prognosis

The prognosis for treating hyperphosphatemia and hypophosphatemia are excellent, though in cases where these problems are due to genetic disease, life-long hormone treatment may be necessary.

Prevention

Phosphorus imbalances caused by hormonal disorders or other genetically determined conditions cannot be prevented. Hypophosphatemia resulting from poor dietary intake can be prevented by eating foods rich in phosphates, and hypophosphatemia caused by overuse of diuretics or **antacids** can be prevented by strictly following instructions concerning proper dosages, as can hyperphosphatemia due to excessive use of **enemas** or laxative. Finally, patients on dialysis or who are being fed intravenously should be monitored closely to prevent phosphorus imbalances.

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Tom Brody, PhD

Photoallergy see **Photosensitivity**

Photokeratitis see **Keratitis**

Photorefractive keratectomy and laser-assisted in-situ keratomileusis

Definition

Photorefractive keratectomy (PRK) and laser-assisted in-situ keratomileusis (LASIK) are two similar surgical techniques that use an excimer laser to correct nearsightedness (**myopia**) by reshaping the cornea. The cornea is the clear outer structure of the eye that lies in front of the colored part of the eye (iris). PRK and LASIK are two forms of vision-correcting (refractive) surgery. The two techniques differ in how the surface layer of the cornea is treated. As of mid 1998, two eximer lasers (Summit and Visx) had been approved for laser vision correction (refractive surgery using a laser) in the PRK procedure. Since then, Visx, Summit, and other lasers have received approval by the Food and Drug Administration (FDA) for use in LASIK procedures.

Purpose

The purpose of both LASIK and PRK is to correct nearsightedness in persons who don't want to, or can't, wear eye glasses or contact lenses. Most patients are able to see well enough to pass a driver's license exam without glasses or contact lenses after the operation. After approximately age 40, the lens in the eye stiffens making it harder to focus up close. Because laser vision correction only affects the cornea, the procedures do not eliminate the need for reading glasses. Patients should be wary of any ads that "guarantee" 20/20 vision. Patients should also make sure that the laser being used is approved by the FDA.

Precautions

Patients should be over 18 years of age, have healthy corneas, and have vision that has been stable for the past year. People who may not be good candidates for these procedures are pregnant women or women who are breastfeeding (vision may not be stable); people with scarred corneas or macular disease; people with autoimmune diseases (i.e., systemic lupus erythematosus or **rheumatoid arthritis**); or people with diabetes. Patients with **glaucoma** should not have LASIK because the intraocular pressure (IOP) of the eye is raised during the procedure. A patient with persistent lid infections (i.e., blepharitis) may not be a good candidate because of an increased risk of infection. An ophthalmologist who specializes in laser vision correction can determine who would be likely to

benefit from the operation and suggest which of the two operations might be more appropriate for any given patient.

If a patient is thinking of having **cataract surgery**, they should discuss it with the doctor. During cataract surgery an intracocular lens (IOL) will be inserted and that alone may correct distance vision.

Description

PRK and LASIK are both performed with an excimer laser, which uses a cold beam of ultraviolet light to sculpt or reshape the cornea so that light will focus properly on the retina. The cornea is the major focusing structure of the eye. The retina sends the image focused on it to the brain. In myopia, the cornea is either too steep or the eye is too long for a clear image to be focused on the retina. PRK and LASIK flatten out the cornea so that the image will focus more precisely on the retina.

In PRK, the surface of the cornea is removed by the laser. In LASIK, the outer layer of the cornea is sliced, lifted, moved aside while the cornea is reshaped with the laser, then replaced to speed healing. Both procedures cause the cornea to become flatter, which corrects the nearsighted vision.

At least one laser has been approved to treat mild **astigmatism** as of 2000. Correcting farsightedness (**hyperopia**) may be possible in the future.

These laser vision-correcting procedures are rapidly replacing radial keratotomy (RK), an earlier form of refractive surgery that involved cutting the cornea with a scalpel in a pattern of radiating spokes. RK has declined in popularity since the approval of the excimer laser in 1995, falling from a high of 250,000 procedures performed per year in 1994 to 50,000 in 1997.

For both LASIK and PRK, the patient's eye is numbed with anesthetic drops. No injections are necessary. The patient is awake and relaxed during the procedure.

LASIK is sometimes referred to as a "flap and zap" procedure because a thin flap of tissue is temporarily removed from the surface of the cornea and the underlying cornea is then "zapped" with a laser. Prior to the surgery, the surface of the cornea is marked with a dye marker so that the flap of cornea can be precisely aligned when it is replaced. The doctor places a suction ring on the eye to hold it steady. During this part of the operation, which lasts only a few seconds, the patient is not able to see. A surgical instrument called a microkeratome is passed over the

cornea to create a very thin flap of tissue. The IOP is increased at this time which is why it is contraindicated in patients with glaucoma. This thin tissue layer is folded back. The cornea is reshaped with the laser beam and the cell layer is replaced. Because the cell layer is not permanently removed, patients have a faster recovery time and experience far less discomfort than with PRK. An antibiotic drop is put in and the eye is patched until the following day's checkup.

In PRK, a small area of the surface layer of the cornea is vaporized. It takes about three days for the surface cells to grow back and vision will be blurred. Some patients describe it as "looking through Vaseline." PRK is generally recommended for patient's with mild to moderate myopia (usually under -5.00 diopters).

With both PRK and LASIK, there is a loud tapping sound from the laser and a burning smell as the cornea is reshaped. The surgery itself is painless and takes only a minute or two. Patients are usually able to return home immediately after surgery. Most patients wait (up to six months) before they have the second one done. This allows the first eye to heal and to see if there were complications from the surgery.

The cost of these procedures can vary with geographic area and the doctor. In general, the procedure costs \$1,350–\$2,500 per eye for PRK and about \$500 more per eye for LASIK. PRK and LASIK are generally not covered by insurance. However, insurance may cover these procedures for people in certain occupations, such as police officers and firefighters.

Preparation

If a patient wears contact lenses, they should not be worn for a few weeks prior to surgery. It also is important to discontinue contact lens wear prior to the visual exams to make sure vision is stable. The doctor should be advised of contact lens wear.

Upon arrival at the doctor's office on the day of surgery, patients are given some eye drops and a sedative, such as Valium, to relax them. Their vision is tested. They rest while waiting for the sedative to take effect. Immediately before the surgery, patients are given local anesthetic eye drops.

Aftercare

After surgery, antibiotic drops are placed in the eye and the eye may be patched. The patient returns for a follow-up visit the next day. The patient is usually given a prescription for eyedrops (usually antibiotic and anti-inflammatory). Patients who have had PRK usually feel mild discomfort for one to three days after

the procedure. They may need a bandage contact lens. Patients who have had LASIK generally have less, or even no discomfort after the surgery. After LASIK, antibiotic and anti-inflammatory drops are generally necessary for one week. After PRK, steroidal eye drops may be necessary for months. Because steroids may increase the possibility of glaucoma or **cataracts**, it is a big drawback to the procedure. The patient should speak with the doctor to see how long follow-up medications will be necessary.

Most patients return to work within one to three days after the procedure, although visual recovery from PRK may take as long as four weeks. An eye shield may be used for about one week at night and patients may be sensitive to bright light for a few days. Patients may be asked by their doctor to keep water out of their eye for a week and to avoid mascara or eyeliner during this period.

Risks

There is a risk of under- or over-correction with either of these procedures. If vision is under-corrected, a second procedure can be performed to achieve results that may be closer to 20/20 vision. About 5–10% of PRK patients return for an adjustment, as do 10–25% of LASIK patients. People with higher degrees of myopia have vision that is harder to correct and usually have LASIK surgery rather than PRK. This may account for the higher incidence of adjustments for LASIK patients. Patients with very high myopia (over -15.00 diopters) may experience improvement after LASIK, but they are not likely to achieve 20/40 vision without glasses. However, their glasses will not need to be as thick or heavy after the surgery. However, most patients, especially those with less extreme myopia, do not need glasses after the surgery.

Haze is another possible side effect. Although hazy vision is unlikely, it is more likely to occur after PRK than after LASIK. This haze usually clears up. Corneal scarring, halos, or glare at night, or an irritating bump on the cornea are other possible side effects. As with any eye surgery, infection is possible, but rare. Loss of vision is possible with these procedures, but this complication is extremely rare.

Most complications from LASIK are related to the creation and realignment of the flap. The microkeratome must be in good-working order and sharp. LASIK requires a great deal of skill on the part of the surgeon and the complication rate is related to the experience level of the surgeon. In one study, the rate of LASIK complications declined from 3% for

KEY TERMS

Blepharitis—An inflammation of the eyelid.

Cataract—A condition in which the lens of the eye turns cloudy and interferes with vision.

Cornea—The clear, curved tissue layer in front of the eye. It lies in front of the colored part of the eye (iris) and the black hole in the center of the iris (pupil).

Diopter (D)—A unit of measure of the power or strength of a lens.

Excimer laser—An instrument that is used to vaporize tissue with a cold, coherent beam of light with a single wavelength in the ultraviolet range.

Intraocular lens (IOL) implant—A small, plastic device (IOL) that is usually implanted in the lens capsule of the eye to correct vision after the lens of the eye is removed. This is the implant is used in cataract surgery.

Macular degeneration—A condition usually associated with age in which the area of the retina called the macula is impaired due to hardening of the arteries (arteriosclerosis). This condition interferes with vision.

Microkeratome—A precision surgical instrument that can slice an extremely thin layer of tissue from the surface of the cornea.

Myopia—A vision problem in which distant objects appear blurry. Myopia results when the cornea is too steep or the eye is too long and the light doesn't focus properly on the retina. People who are myopic or nearsighted can usually see near objects clearly, but not far objects.

Refractive surgery—A surgical procedure that corrects visual defects.

Retina—The sensory tissue in the back of the eye that is responsible for collecting visual images and sending them to the brain.

surgeons during their first three months using this technique, to 1% after a year's experience in the technique, to 0% after 18 months experience.

Normal results

Most patients experience improvement in their vision immediately after the operation and about half of LASIK patients are able to see 20/30 within one day of the surgery. Vision tends to become sharper

over the next few days and then stabilizes; however, it is possible to have shifts in myopia for the next few months. Vision clears and stabilizes faster after LASIK than after PRK. Final vision is achieved within three to six months with LASIK and six to eight months with PRK. The vast majority of patients (95% for people with low to moderate myopia and 75% for people with high levels of myopia) are able to see 20/40 after either of these procedures and are able to pass a driver's license test without glasses or contact lenses.

LASIK is more complicated than PRK because of the addition of the microkeratome procedure. However, LASIK generally has faster recovery time, less **pain**, and less chance of halos and scarring than PRK. LASIK can treat higher degrees of myopia (-5.00–25.00 diopters). LASIK also requires less use of steroids. Patients need to speak with qualified, experienced eye surgeons to help in choosing the procedure that is right for them.

Resources

ORGANIZATIONS

American Academy of Ophthalmology. 655 Beach Street, P.O. Box 7424, San Francisco, CA 94120-7424. < <http://www.eyenet.org> >.

American Society of Cataract and Refractive Surgery. 4000 Legato Road, Suite 850, Fairfax, VA 22033-4055. (703) 591-2220. < <http://www.ascrs.org> >.

Louann W. Murray, PhD

Photosensitivity

Definition

Photosensitivity refers to any increase in the reactivity of the skin to sunlight.

Description

The skin is a carefully designed interface between our bodies and the outside world. It is infection-proof when intact, nearly waterproof, and filled with protective mechanisms. Sunlight threatens the health of the skin. Normal skin is highly variable in its ability to resist sun damage. Natural skin pigmentation is its main protection. The term photosensitivity refers to any increase beyond what is considered normal variation.



A skin rash on the front of a woman's neck caused by a photosensitive reaction to sunlight. (Photograph by Dr. P. Marazzi, Photo Researchers, Inc. Reproduced by permission.)

Causes and symptoms

There are over three dozen diseases, two dozen drugs, a variety of herbal preparations, and several perfume and cosmetic components that can cause photosensitivity. There are also several different types of reaction to sunlight—phototoxicity, photoallergy, and polymorphous light eruption. In addition, prolonged exposure to sunlight, even in normal skin, leads to skin **aging** and **cancer**. These effects are accelerated in patients who have photosensitivity.

- Phototoxicity is a severely exaggerated reaction to sunlight caused by a new chemical in the skin. The primary symptom is **sunburn**, which is rapid and can be severe enough to blister (a second degree burn). The chemicals associated with phototoxicity are usually drugs. The list includes several common antibiotics—quinolones, **sulfonamides**, and **tetracyclines**; **diuretics** (water pills); major tranquilizers; oral diabetes medication; and cancer medicines. There are also some dermatologic drugs, both topical and oral, that can sensitize skin.
- Photoallergy produces an intense **itching** rash on exposure to sunlight. Patients develop chronic skin changes (lichen simplex) as a result of scratching. Some of the agents that cause phototoxicity can also cause photoallergy. Some cosmetic and perfume ingredients, including a compound that was formerly used in sunscreens—para-amino benzoic acid (PABA)—can do this. Most sunscreen preparations in the early 2000s, however, no longer include PABA.
- Polymorphous light eruption (PLE) resembles photoallergy in its production of intensely itching **rashes** in sunlight. However, this condition lessens with continued light exposure, and so is seen mostly in the spring. Also, there does not seem to be an identifiable chemical involved. PLE is most likely to develop in

fair-skinned individuals. It is estimated to affect about 10% of the United States population compared to 21% of the Swedish population. The female: male ratio is 2.5: 1, but it is thought that the imbalance may be due to the fact that women are more likely than men to seek treatment for PLE.

- There is a form of inherited PLE that affects Native Americans. The inheritance pattern is autosomal dominant.

Diseases of several kinds increase skin sensitivity:

- A hereditary disease called xeroderma pigmentosum includes a defect in repair mechanisms that greatly accelerates skin damage from sunlight.
- A family of metabolic diseases called **porphyrias** produce chemicals (porphyrins) that absorb sunlight in the skin and thereby cause damage.
- Albinos lack skin pigment through a genetic defect and are thus very sensitive to light.
- Malnutrition, specifically a deficiency of niacin known as **pellagra**, sensitizes the skin.
- Several diseases like **acne**, **systemic lupus erythematosus**, **rosacea**, and herpes simplex (**fever blisters**) decrease the resistance of the skin to sun damage. Rosacea is sometimes described as a photoaggravated skin disorder because its symptoms increase in severity when patients are exposed to sunlight.
- Photosensitivity is increasingly recognized as a common development in HIV-positive patients. Risk factors for photosensitivity in this group include African American ethnicity and treatment with highly active antiretroviral therapy (HAART).

Diagnosis

The pattern of appearance on the skin, a history of drug or chemical exposure, and the timing of the symptoms often suggests a diagnosis. A **skin biopsy** may be needed for further clarification.

Treatment

Removal of the offending drug or chemical is primary. Direct sunlight exposure should be limited. Some people must avoid sunlight altogether, while others can tolerate some direct sunlight with the aid of **sunscreens**.

Prevention

A sunscreen with an SPF of 15 or greater protects most skin from damage. Such protective garments as hats and long-sleeved shirts are highly recommended in addition.



This person had a phototoxic reaction after taking an antibiotic drug. (Photo Researchers, Inc. Reproduced by permission.)

KEY TERMS

Albino—A person or animal lacking normal coloring in the eyes, hair, and skin due to a hereditary inability to produce the skin pigment melanin. The condition itself is called albinism.

Biopsy—Surgical removal of tissue for examination.

Rosacea—A chronic skin disease characterized by persistent redness of the skin and periodic outbreaks of pustules, usually affecting the middle third of the face.

Resources

BOOKS

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- United States Food and Drug Administration (FDA). 5600 Fishers Lane, Rockville, MD 20857-0001. (888) INFO-FDA. <<http://www.fda.gov>>.

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Phototherapy

Definition

Phototherapy, or light therapy, is the administration of doses of bright light in order to normalize the body's internal clock and/or relieve depression.

Purpose

Phototherapy is prescribed primarily to treat seasonal affective disorder (SAD), a mood disorder characterized by depression in the winter months, and is occasionally employed to treat **insomnia** and **jet lag**. The exact mechanisms by which the treatment works are not known, but the bright light employed in phototherapy may act to readjust the body's circadian (daily) rhythms, or internal clock. Other popular theories are that light triggers the production of serotonin, a neurotransmitter believed to be related to **depressive disorders**, or that it influences the body's production of melatonin, a hormone derived from serotonin that may be related to circadian rhythms.

Precautions

Patients with eye problems should see an ophthalmologist regularly, both before and during phototherapy. Because some ultraviolet rays are

emitted by the light boxes used in phototherapy, patients taking photosensitizing medications (medications making the skin more sensitive to light) and those who have sun-sensitive skin should consult with their physician before beginning treatment. Patients with medical conditions that make them sensitive to ultraviolet rays should also be seen by a physician before starting phototherapy. Patients who have a history of mood swings or **mania** should be monitored closely, since phototherapy may cause excessive mood elevation in some individuals.

Description

Phototherapy is generally administered at home. The most commonly used phototherapy equipment is a portable lighting device known as a light box. The box may be mounted upright to a wall, or slanted downwards towards a table. The patient sits in front of the box for a prescribed period of time (anywhere from 15 minutes to several hours). Some patients with SAD undergo phototherapy sessions two or three times a day, others only once. The time of day and number of times treatment is administered depend on the physical needs and lifestyle of the individual patient. If phototherapy has been prescribed for the treatment of SAD, it typically begins in the fall months as the days begin to shorten, and continues throughout the winter and possibly the early spring.

The light from a slanted light box is designed to focus on the table it sits upon, so patients may look down to read or do other sedentary activities during therapy. Patients using an upright light box must face the light source (although they need not look directly into the light). The light sources in these light boxes typically range from 2,500–10,000 lux. (In contrast, average indoor lighting is 300–500 lux; a sunny summer day is about 100,000 lux).

Phototherapy prescribed for the treatment of SAD may be covered by insurance. Individuals requiring phototherapy should check with their insurance company to see if the cost of renting or purchasing a light box is covered.

Aftercare

Patients beginning light therapy for SAD may need to adjust the length, frequency, and timing of their phototherapy sessions to achieve the maximum benefit. These patients should keep their doctor informed of their progress and the status of their depressive symptoms. Occasionally, antidepressants and/or psychotherapy may be recommended as an adjunct to phototherapy.

KEY TERMS

Circadian rhythm—The rhythmic repetition of certain phenomena in living organisms at about the same time each day.

Lux—A standard unit of measure for illumination.

Neurotransmitter—A chemical in the brain that transmits messages between neurons, or nerve cells.

Photosensitivity—An abnormally heightened reaction to light.

Seasonal affective disorder (SAD)—A mood disorder characterized by depression during the winter months. An estimated 11 million Americans experience SAD.

Risks

An abnormally elevated or expansive mood (hypomania) may occur, but it is usually temporary. Some patients undergoing phototherapy treatment report side effects of eyestrain, headaches, insomnia, **fatigue**, **sunburn**, and dry eyes or nose. Most of these effects can be managed by adjusting the timing and duration of the phototherapy sessions. A strong sun block and eye and nose drops can alleviate the other problems. Long-term studies have shown no negative effects to the eye function of individuals undergoing phototherapy treatments.

Normal results

Patients with SAD typically report an alleviation of depressive symptoms within two to 14 days after beginning phototherapy.

Resources

ORGANIZATIONS

National Institute of Mental Health. Mental Health Public Inquiries, 5600 Fishers Lane, Room 15C-05, Rockville, MD 20857. (888) 826-9438. <<http://www.nimh.nih.gov>>.

Society for Light Treatment and Biological Rhythms. P.O. Box 591687, 174 Cook St., San Francisco, CA 94159-1687. <<http://www.websciences.org/sltbr>>.

Paula Anne Ford-Martin

Phototoxic reaction see **Photosensitivity**

Phycomycosis see **Mucormycosis**

Physical allergy

Definition

Physical **allergies** are allergic reactions to cold, sunlight, heat, or minor injury.

Description

The immune system is designed to protect the body from harmful invaders such as germs. Occasionally, it goes awry and attacks harmless or mildly noxious agents, doing more harm than good. This event is termed allergy if the target is from the outside—like pollen or bee venom—and autoimmunity if it is caused by one of the body's own components.

The immune system usually responds only to certain kinds of chemicals, namely proteins. However, non-proteins can trigger the same sort of response, probably by altering a protein to make it look like a target. Physical allergy refers to reactions in which a protein is not the initial inciting agent.

Sometimes it takes a combination of elements to produce an allergic reaction. A classic example is drugs that are capable of sensitizing the skin to sunlight. The result is phototoxicity, which appears as an increased sensitivity to sunlight or as localized skin **rashes** on sun-exposed areas.

Causes and symptoms

- Minor injury, such as scratching, causes itchy welts to develop in about 5% of people. The presence of itchy welts (urticaria) is a condition called dermatographism.
- Cold can change certain proteins in the blood so that they induce an immune reaction. This may indicate that there are abnormal proteins in the blood from a disease of the bone marrow. The reaction may also involve the lungs and circulation, producing **wheezing** and **fainting**.
- Heat allergies can be caused by **exercise** or even strong emotions in sensitive people.
- Sunlight, even without drugs, causes immediate urticaria in some people. This may be a symptom of porphyria—a genetic metabolic defect.
- Elements like nickel and chromium, although not proteins, commonly cause skin rashes, and iodine allergy causes skin rashes and sores in the mouth in allergic individuals.
- Pressure or vibration can also cause urticaria.
- Water contact can cause aquagenic urticaria, presumably due to chlorine or some other trace chemical

KEY TERMS

Antihistamine—Drugs that block histamine, a major cause of itching.

Hemolysis—Destruction of red blood cells.

Inflammation—Heat, redness, swelling, and pain caused by an immune response.

in the water, although distilled water has been known to cause this reaction.

When the inflammatory reaction involves deeper layers of the skin, urticaria becomes angioedema. The skin, especially the lips and eyelids, swells. The tongue, throat, and parts of the digestive tract may also be involved. Angioedema may be due to physical agents. Often the cause remains unknown.

Diagnosis

Visual examination of the symptoms usually diagnoses the reaction. Further skin tests and review of the patient's **photosensitivity** may reveal a cause.

Treatment

Removing the offending agent is the first step to treatment. If sun is involved, shade and sunscreens are necessary. The reaction can usually be controlled with epinephrine, **antihistamines**, or cortisone-like drugs. Urticaria may be treated with antihistamines such as diphenhydramine (Benadryl) or desloratadine (Clarinet). Clarinet is non-sedating, meaning it will not make patients drowsy. Itching can be controlled with cold packs or commercial topical agents that contain menthol, camphor, eucalyptus oil, aloe, antihistamines, or cortisone preparations.

Prognosis

If the causative agent has been diagnosed, avoidance of or protection against the allergen cures the allergy. Usually, allergies can be managed through treatment.

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Physical examination

Definition

A physical examination is an evaluation of the body and its functions using inspection, palpation (feeling with the hands), percussion (tapping with the fingers), and auscultation (listening). A complete health assessment also includes gathering information about a person's medical history and lifestyle, doing laboratory tests, and screening for disease.

Purpose

The annual physical examination has been replaced by the periodic health examination. How often this is done depends on the patient's age, sex, and risk factors for disease. The United States Preventative Services Task Force (USPSTF) has developed guidelines for preventative health examinations that health care professionals widely follow. Organizations that promote detection and prevention of specific diseases, like the American Cancer Society, generally recommend more intensive or frequent examinations.

A comprehensive physical examination provides an opportunity for the healthcare professional to obtain baseline information about the patient for future use, and to establish a relationship before problems happen. It provides an opportunity to answer questions and teach good health practices. Detecting a problem in its early stages can have good long-term results.

Precautions

The patient should be comfortable and treated with respect throughout the examination. As the examination proceeds, the examiner should explain what he or she is doing and share any relevant findings.

Description

A complete physical examination usually starts at the head and proceeds all the way to the toes. However, the exact procedure will vary according to the needs of the patient and the preferences of the examiner. An average examination takes about 30 minutes. The cost of the examination will depend on the charge for the professional's time and any tests that are done. Most health plans cover routine physical examinations including some tests.

The examination

First, the examiner will observe the patient's appearance, general health, and behavior, along with measuring height and weight. The vital signs—including pulse, breathing rate, body temperature, and blood pressure—are recorded.

With the patient sitting up, the following systems are reviewed:

- **Skin.** The exposed areas of the skin are observed; the size and shape of any lesions are noted.
- **Head.** The hair, scalp, skull, and face are examined.
- **Eyes.** The external structures are observed. The internal structures can be observed using an ophthalmoscope (a lighted instrument) in a darkened room.
- **Ears.** The external structures are inspected. A lighted instrument called an otoscope may be used to inspect internal structures.
- **Nose and sinuses.** The external nose is examined. The nasal mucosa and internal structures can be observed with the use of a penlight and a nasal speculum.
- **Mouth and pharynx.** The lips, gums, teeth, roof of the mouth, tongue, and pharynx are inspected.
- **Neck.** The lymph nodes on both sides of the neck and the thyroid gland are palpated (examined by feeling with the fingers).
- **Back.** The spine and muscles of the back are palpated and checked for tenderness. The upper back, where the lungs are located, is palpated on the right and left sides and a stethoscope is used to listen for breath sounds.
- **Breasts and armpits.** A woman's breasts are inspected with the arms relaxed and then raised. In both men and women, the lymph nodes in the armpits are felt with the examiner's hands. While the patient is still sitting, movement of the joints in the hands, arms, shoulders, neck, and jaw can be checked.

Then while the patient is lying down on the examining table, the examination includes:

- **Breasts.** The breasts are palpated and inspected for lumps.
- **Front of chest and lungs.** The area is inspected with the fingers, using palpation and percussion. A stethoscope is used to listen to the internal breath sounds.

The head should be slightly raised for:

- **Heart.** A stethoscope is used to listen to the heart's rate and rhythm. The blood vessels in the neck are observed and palpated.

The patient should lie flat for:

- **Abdomen.** Light and deep palpation is used on the abdomen to feel the outlines of internal organs including the liver, spleen, kidneys, and aorta, a large blood vessel.
- **Rectum and anus.** With the patient lying on the left side, the outside areas are observed. An internal digital examination (using a finger), is usually done if the patient is over 40 years old. In men, the prostate gland is also palpated.
- **Reproductive organs.** The external sex organs are inspected and the area is examined for hernias. In men, the scrotum is palpated. In women, a pelvic examination is done using a speculum and a Papamnicolaou test (**Pap test**) may be taken.
- **Legs.** With the patient lying flat, the legs are inspected for swelling, and pulses in the knee, thigh, and foot area are found. The groin area is palpated for the presence of lymph nodes. The joints and muscles are observed.
- **Musculoskeletal system.** With the patient standing, the straightness of the spine and the alignment of the legs and feet is noted.
- **Blood vessels.** The presence of any abnormally enlarged veins (varicose), usually in the legs, is noted.

In addition to evaluating the patient's alertness and mental ability during the initial conversation, additional inspection of the nervous system may be indicated:

- **Neurologic screen.** The patient's ability to take a few steps, hop, and do deep knee bends is observed. The strength of the hand grip is felt. With the patient sitting down, the reflexes in the knees and feet can be tested with a small hammer. The sense of touch in the hands and feet can be evaluated by testing reaction to **pain** and vibration.
- Sometimes additional time is spent examining the 12 nerves in the head (cranial) that are connected directly to the brain. They control the sense of smell, strength of muscles in the head, reflexes in the eye, facial movements, gag reflex, and muscles in the jaw. General muscle tone and coordination, and the reaction of the abdominal area to stimulants like pain, temperature, and touch would also be evaluated.

Preparation

Before visiting the health care professional, the patient should write down important facts and dates about his or her own medical history, as well as those

KEY TERMS

Auscultation—The process of listening to sounds that are produced in the body. Direct auscultation uses the ear alone, such as when listening to the grating of a moving joint. Indirect auscultation involves the use of a stethoscope to amplify the sounds from within the body, like a heartbeat.

Hernia—The bulging of an organ, or part of an organ, through the tissues normally containing it; also called a rupture.

Inspection—The visual examination of the body using the eyes and a lighted instrument if needed. The sense of smell may also be used.

Ophthalmoscope—Lighted device for studying the interior of the eyeball.

Otoscope—An instrument with a light for examining the internal ear.

Palpation—The examination of the body using the sense of touch. There are two types: light and deep.

Percussion—An assessment method in which the surface of the body is struck with the fingertips to obtain sounds that can be heard or vibrations that can be felt. It can determine the position, size, and consistency of an internal organ. It is done over the chest to determine the presence of normal air content in the lungs, and over the abdomen to evaluate air in the loops of the intestine.

Reflex—An automatic response to a stimulus.

Speculum—An instrument for enlarging the opening of any canal or cavity in order to facilitate inspection of its interior.

Stethoscope—A Y-shaped instrument that amplifies body sounds such as heartbeat, breathing, and air in the intestine. Used in auscultation.

Varicose veins—The permanent enlargement and twisting of veins, usually in the legs. They are most often seen in people with occupations requiring long periods of standing, and in pregnant women.

of family members. He or she should have a list of all medications with their doses or bring the actual bottles of medicine along. If there are specific concerns about anything, writing them down is a good idea.

Before the physical examination begins, the bladder should be emptied and a urine specimen can be collected in a small container. For some blood tests, the patient may be told ahead of time not to eat or drink after midnight.

The patient usually removes all clothing and puts on a loose-fitting hospital gown. An additional sheet is provided to keep the patient covered and comfortable during the examination.

Aftercare

Once the physical examination has been completed, the patient and the examiner should review what laboratory tests have been ordered and how the results will be shared with the patient. The medical professional should discuss any recommendations for treatment and follow-up visits. Special instructions should be put in writing. This is also an opportunity for the patient to ask any remaining questions about his or her own health concerns.

Normal results

Normal results of a physical examination correspond to the healthy appearance and normal functioning of the body. For example, appropriate reflexes will be present, no suspicious lumps or lesions will be found, and vital signs will be normal.

Abnormal results

Abnormal results of a physical examination include any findings that indicated the presence of a disorder, disease, or underlying condition. For example, the presence of lumps or lesions, fever, muscle weakness or lack of tone, poor reflex response, heart arrhythmia, or swelling of lymph nodes will point to a possible health problem.

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Physical therapy see **Rehabilitation**

Pica

Definition

Pica is the persistent craving and compulsive eating of nonfood substances. The Diagnostic and Statistical Manual of Mental Disorders, fourth edition, classifies it as a feeding and eating disorder of childhood.

Description

The puzzling phenomenon of pica has been recognized and described since ancient times. Pica has been observed in ethnic groups worldwide, in both primitive and modernized cultures, in both sexes, and in all age groups. The word pica comes from the Latin name for magpie, a bird known for its unusual and indiscriminate eating habits. In addition to humans, pica has been observed in other animals, including the chimpanzee.

Pica in humans has many different subgroups, defined by the substance that is ingested. Some of the most commonly described types of pica are eating earth, soil or clay (geophagia), ice (pagophagia) and starch (amylophagia). However, pica involving dozens of other substances, including cigarette butts and ashes, hair, paint chips, and paper have also been reported. In one unusual case, the patient ingested transdermal patches of fentanyl, an opioid medication given for severe **pain**. Eating the skin patch increased the patient's dose of the drug by a factor of 10.

Although pica can occur in individuals of any background, a higher incidence of pica is associated with:

- **pregnancy**
- developmental delay and **mental retardation**
- psychiatric disease and **autism**
- early childhood
- poor **nutrition** or low blood levels of iron and other minerals
- certain cultural or religious traditions

Causes and symptoms

Evidence suggests that there may be several causes of pica. One widely held theory points to iron deficiency as a major cause of pica. Several reports have described pica in individuals with documented iron deficiency, although there has been uncertainty as to whether the iron deficiency was a cause of pica or a result of it. Because some substances, such as clay, are believed to block the absorption of iron into the bloodstream, it was thought that low blood levels of iron could be the direct result of pica. However, some studies have shown that pica cravings in individuals with iron deficiency stop once iron supplements are given to correct the deficiency. Another study looked specifically at the rate of iron absorption during pica conditions and normal dietary behavior, and showed that the iron absorption was not decreased by pica. In addition, low blood levels of iron commonly occur in

pregnant women and those with poor nutrition, two populations at higher risk for pica. Such findings offer strong support of iron deficiency as a cause, rather than result, of pica.

Other reports suggest that pica may have a psychological basis and may even fall into the spectrum of **obsessive-compulsive disorder**. Pica has a higher incidence in populations with an underlying diagnosis involving mental functioning. These diagnoses include psychiatric conditions like schizophrenia, developmental disorders including autism, and conditions with mental retardation. These conditions are not characterized by iron deficiency, which supports a psychological component in the cause of pica.

Cultural and religious traditions may also play a role in pica behavior. In some cultures, nonfood substances are believed to have positive health or spiritual effects. Among some African Americans in the south, ingesting a particular kind of white clay is believed to promote health and reduce morning sickness during pregnancy. Other cultures practice pica out of belief that eating a particular substance may promote fertility or bring good luck.

The hallmark feature of pica, consistently consuming nonfood substances, often does not present publicly. People may be embarrassed to admit to these unusual eating habits, and may hide it from their family and physician. In other cases, an individual may not report the pica to a physician simply because of a lack of knowledge of pica's potential medical significance.

Because the eating behaviors of pica are not usually detected or reported, it is the complications of the behavior that bring it to attention. Complications vary, depending on the type of pica. Geophagia has potential side effects that most commonly affect the intestine and bowel. Complications can include **constipation**, cramping, pain, obstruction caused by formation of an indigestible mass, perforation from sharp objects like rocks or gravel, and contamination and infection from soil-dwelling parasites.

Amylophagia usually involves the consumption of cornstarch and, less frequently, laundry starch. The high caloric content of starch can cause excessive weight gain, while at the same time leading to **malnutrition**, as starch contributes "empty" calories lacking **vitamins** and **minerals**. Amylophagia during pregnancy can mimic **gestational diabetes** in its presentation and even in its potential harmful effects on the fetus.

Pica involving the ingestion of substances such as lead-based paint or paper containing mercury can cause symptoms of toxic **poisoning**. Compulsive

consumption of even a seemingly harmless substance like ice (pagophagia) can have negative side effects, including decreased absorption of nutrients by the gut.

Diagnosis

In order for the diagnosis of pica to be made, there must be a history of persistent consumption of a nonfood substance continuing for a minimum period of one month. Infants and toddlers are typically excluded from this diagnosis since mouthing objects is a normal developmental behavior at that age. Individuals with mental retardation who function at or below an approximate cognitive level of 18 months may also be exempt from this diagnosis.

Pica is most often diagnosed when a report of such behaviors can be provided by the patient or documented by another individual. In other cases, pica is diagnosed after studies have been performed to assess the presenting symptoms. For example, imaging studies ordered to assess severe gastrointestinal complaints may reveal intestinal blockage with an opaque substance; such a finding is suggestive of pica. Biopsy of intestinal contents can also reveal findings, such as parasitic infection, consistent with pica. Pica may also be suspected if abnormal levels of certain minerals or chemicals are detected in the blood.

Pica in pregnant women is sometimes diagnosed after **childbirth** because of a health problem in the newborn caused by the substance(s) ingested by the mother. In one instance reported in Chicago, a newborn girl was treated for **lead poisoning** caused by her mother's eating fragments of lead-glazed pottery during pregnancy.

Treatment

Treatment of pica will often depend on the cause and type of pica. Conventional medical treatment may be appropriate in certain situations. For example, supplementation with iron-containing vitamins has been shown to cause the unusual cravings to subside in some iron-deficient patients.

Medical complications and health threats, including high lead levels, bowel perforation or intestinal obstruction, will require additional medical management, beyond addressing the underlying issue of pica.

Alternative treatment

Because most cases of pica do not have an obvious medical cause, treatment with counseling, education, and nutritional management is often more successful

KEY TERMS

Amylophagia—The compulsive eating of purified starch, typically cornstarch or laundry starch.

Geophagia—The compulsive eating of earthy substances, including sand, soil, and clay.

Pagophagia—The compulsive eating of ice.

and more appropriate than treatment with medication. Some therapists specializing in eating disorders may have expertise in treating pica.

Prognosis

The prognosis for individuals with pica varies greatly, according to the type and amount of substance ingested, the extent of presenting side effects, and the success of treatment. Many of the side effects and complications of pica can be reversed once the behavior is stopped, while other complications, including infection and bowel perforation, pose significant health threats and if not successfully treated may result in **death**.

When seen in children, pica behavior tends to lessen with age. However, individuals with a history of pica are more likely to experience it again. Counseling and nutritional education can reduce the risk of recurrence.

Prevention

There are no known methods of preventing pica. However, once pica is known or suspected, measures can be taken to reduce further ingestion of nonfood substances. Removing the particular substance from readily accessible areas can be helpful. Close observation of the individual with pica may limit inappropriate eating behaviors.

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3615 Wisconsin Avenue, NW, Washington, DC
20016-3007. (202) 966-7300. Fax: (202) 966-2891.
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Pickwickian syndrome

Definition

A group of symptoms that generally accompany massive **obesity**.

Description

Pickwickian syndrome is a complex of symptoms that primarily affect patients with extreme obesity. The syndrome is named after a character in a Charles Dickens novel, *The Pickwick Papers*, who seemed to show some of the traits of this disease.

KEY TERMS

Latency—The period of inactivity between the time a stimulus is provided and the time a response occurs.

Obesity—Exceeding one's normal weight by 20%. A person suffering from extreme obesity would exceed their normal weight by a much higher percentage.

Pulmonary system—Lungs and respiratory system of the body.

The major health problem that occurs in patients with this disease is **sleep apnea**. This is caused in part by the excess amounts of fatty tissue surrounding the chest muscles. This excess fat places a strain on the heart, lungs, and diaphragm of the patient, making it difficult to breathe.

Causes and symptoms

The major cause of Pickwickian syndrome is extreme obesity. This obesity places an excessive load on the pulmonary system. The role of genetics is also being studied. Symptoms of Pickwickian syndrome include excessive daytime sleepiness, shortness of breath due to elevated blood carbon dioxide pressure, disturbed sleep at night, and flushed face. The skin can also have a bluish tint, and the patient may have high blood pressure, an enlarged liver, and an abnormally high red blood cell count.

Diagnosis

Some tests that can be used to diagnose this condition include **echocardiography** to determine heart enlargement or **pulmonary hypertension**. Giving the patient multiple sleep latency tests can help give an objective measurement of daytime sleepiness. Magnetic resonance imaging (MRI), computed tomography (CT) scans, or fiberoptic evaluation of the upper airway may also be used.

Treatment

The primary treatment for Pickwickian syndrome is focused on weight loss and increased physical activity. Also, medroxyprogesterone may help improve the condition.

Prognosis

Pickwickian syndrome is entirely reversible if it is diagnosed and treated properly. If the problem goes undiagnosed, the outcome can be fatal.

Prevention

Prevention of Pickwickian syndrome can be achieved by maintaining a healthy body weight and getting the proper amount of **exercise**. For prevention of the sleep apnea that generally accompanies Pickwickian syndrome, there are several possible treatments. If the sleep apnea is only present when the patient is flat on their back, a tennis ball can be sewn into the sleep clothes to remind the patient not to sleep on their back. For more severe cases of sleep apnea, a **tonsillectomy** or the use of dental appliances may be recommended.

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Kim A. Sharp, M.Ln.

PID see **Pelvic inflammatory disease**

Piercing and tattoos

Definition

Body piercing and tattoos are a popular form of body art that have been practiced throughout history by various cultures.

Description

Various cultures have embraced adorning their bodies with piercings and tattoos throughout history. In 1992, the 4,000-year-old body of a tattooed man was discovered in a glacier on the Austrian border,

and historical research has shown that Egyptians identified tattooing with fertility and nobility in the period from 4000–2000 BC. Similar to tattooing, body piercing also has a rich history, which includes being used as a symbol of royalty and courage. In some hunting and gathering societies, body piercing and tattoos have long been used in initiation rites and as socialization/enculturation symbols.

In today's industrialized cultures, tattoos and piercing are a popular art form shared by people of all ages. They also are indicative of a psychology of **self-mutilation**, defiance, independence, and belonging, as for example in prison or gang cultures.

Popular piercing sites include the ear, nasal septum, eyebrow, tongue, cheek, navel, labia, and penis. Tattoos permanently mark various areas on the body.

Piercing is performed quickly and without anesthesia by either a spring-loaded ear-piercing gun or piercing needles, with the needle diameter varying from six to 18 gauge. The skin is cleaned, then the needle and jewelry are inserted through the tissue in one swift motion. Piercing is typically completed in tattoo or beauty parlors.

Originating from the Tahitian word *tattau*, meaning “to mark,” tattoos are relatively permanent marks or designs on the skin. An electric needle injects colored pigment into small deep holes made in the skin to form the tattoo. Prison tattoo techniques are usually very crude, in marked contrast to the highly skilled art practiced in Japan and also performed in America and Europe. In recent years, the ancient art of Mehndi, or temporary tattooing of the skin with a paste made of henna, has become popular in America and around the world. Henna is a stain normally made for hair, and therefore exempt from U.S. Food and Drug Administration (FDA) regulation. Although seemingly safe because it does not pierce the skin, henna tattoos using black henna, a paste that contains para-phenylenediamine, can actually be dangerous when absorbed into the skin of some people.

Causes and symptoms

While piercing and tattooing are popular, both present definite health risks. Tattoos can lead to the transmission of infectious diseases, such as hepatitis B and C, and theoretically HIV, when proper sterilization and safety procedures are not followed. Black henna tattoos can cause significant **allergies** and **rashes**, leading to renal (kidney) failure and even **death** in those who are sensitive to their ingredients. These types of tattoos have appeared particularly

dangerous to young children. Body piercing also presents the risk of chronic infection, scarring, **hepatitis B** and C, **tetanus**, and skin allergies to the jewelry that is used. A recent Mayo Clinic study reported that 17% of college students with piercings suffered a medical complication such as infection or tearing. Use of piercing guns and preferences for upper ear piercing have led to increased infections in recent years. The force of the gun's delivery further complicates matters around the delicate cartilage of the upper ear and some people require surgical intervention.

Body piercing and tattooing are unregulated in most United States, but illegal in some. The American Dental Association (ADA) opposes oral (tongue, lip or cheek) piercing, and the American Academy of Dermatology is against all forms of body piercing except ear lobe piercing.

Diagnosis

Some of the signs of an infection from either piercing or tattoos are obvious, such as inflammation of the pierced or tattooed area, while the symptoms of hepatitis C, the most common blood-borne infection in the United States, may not be so obvious. Allergic responses to tattoos may occur due to the pigment compounds used, such as oxides of iron, mercury, chromium, cadmium, and cobalt and synthetic organic dyes. Symptoms of an allergic reaction include swelling, redness and severe **itching**. Symptoms of henna tattoo reactions are an eczema-like rash around the tattoo site. The patch should be tested for reaction severity before it proceeds to anaphylactic **shock**, or severe allergic reaction.

Most infections from piercing are due to the use of non-sterile techniques. The skin pathogens streptococcus and staphylococcus are most frequently involved in skin infections from piercing. The fleshy tissue around the pierced area may weaken and tear, leading for example, to a badly disfigured earlobe. Other common complications include contact dermatitis and **scars**. Piercing can result in **endocarditis**, urethral rupture, and a serious infection of the penis foreskin leading to severe disability or even death.

Treatment

Treatment of a local infection from piercing includes warm compresses and antibacterial ointment for local infections, to a five-day course of oral antibiotic therapy. If hepatitis B or C is confirmed, a series of diet and lifestyle changes, such as the elimination of alcohol, is recommended to control the disease.



Tattoo artist Michael Wilson displaying his own tattoos and piercings. (Photograph by Susan Mc Cartney, Photo Researchers. Reproduced by permission.

There are four methods to remove tattoos, including: surgical removal that involves cutting the tattoo away; sanding the skin with a wire brush to remove the epidermis and dermis layers in a process called dermabrasion; using a salt solution to soak the tattooed skin (salabrasion); and scarification, removing the tattoo with an acid solution to form a scar in its place. Topical steroids can often treat reactions to henna tattoos, but improvement may take several weeks.

Prognosis

Depending on the type of infection resulting from either piercing or tattoos, the treatment and prognosis vary. Minor infections respond well to antibiotic therapy, while blood borne diseases such as hepatitis B and C cause life-altering results. Disfigurement may or may not be fully correctable by later **plastic surgery**.

KEY TERMS

Endocarditis—Infection of a valve inside the heart.

Hepatitis—Inflammation of the liver.

Socialization—Process by which new members of a social group are integrated in the group.

Patients particularly sensitive or allergic to the ingredients in black henna may suffer serious consequences, even death, if their reaction progresses. Others may be left with scarring or altered pigmentation along the tattoo design.

Prevention

The best way to prevent infection from piercing or tattoos is not to get one in the first place. Procedures should be performed in a sterile environment by an experienced professional. The person performing the procedure should remove a new needle from the plastic in front of you and should put on a new pair of sterile gloves. Anyone considering a henna tattoo should require proof from the artists that he or she is using pure, safe brown henna, not the unsafe black henna.

Piercing should be completed with smoothly polished jewelry made of 14 or 18 carat gold, titanium, surgical steel or niobium. An allergic reaction can result with the use of jewelry made of brass plate or containing a nickel alloy. Healing time from a piercing range from six months to two years. A piercing should be completed in a sterile environment that uses every precaution to reduce the risk of infection. Excessive force, such as exerting a strong pull, should never be applied to jewelry inserted into pierced body parts to avoid tearing and injuring the tissues.

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Pilates

Definition

Pilates or Physical Mind method, is a series of non-impact exercises designed by Joseph Pilates to develop strength, flexibility, balance, and inner awareness.

Purpose

Pilates is a form of strength and flexibility training that can be done by someone at any level of fitness. The exercises can also be adapted for people who have limited movement or who use wheel chairs. It is an engaging **exercise** program that people want to do. Pilates promotes a feeling of physical and mental well-being and also develops inner physical awareness. Since this method strengthens and lengthens the muscles without creating bulk, it is particularly beneficial for dancers and actors. Pilates is also helpful in preventing and rehabilitating from injuries, improving posture, and increasing flexibility, circulation, and balance. Pregnant women who do these exercises can develop body alignment, improve concentration, and develop body shape and tone after **pregnancy**. According to Joseph Pilates, "You will feel better in 10 sessions, look better in 20 sessions and have a completely new body in 30 sessions."

Although Pilates is often associated with dancers, athletes, and younger people in general who are interested in improving their physical strength and flexibility, a simplified version of some Pilates exercises is also being used as of 2003 to lower the risk of hospital-related deconditioning in older adults. A Canadian study of hospitalized patients over the age of 70 found that those who were given a set of Pilates exercises that could be performed in bed recovered more rapidly than a control group given a set of passive range-of-motion exercises.

Description

Origins

Joseph Pilates (pronounced pie-LAH-tes), the founder of the Pilates method (also simply referred to as “the method”) was born in Germany in 1880. As a frail child with **rickets**, **asthma**, and **rheumatic fever**, he was determined to become stronger. He dedicated himself to building both his body and his mind through practices which included **yoga**, **zen**, and ancient Roman and Greek exercises. His conditioning regime worked and he became an accomplished gymnast, skier, boxer, and diver.

While interned in England during World War I for being a German citizen, Pilates became a nurse. During this time, he designed a unique system of hooking springs and straps to a hospital bed in order to help his disabled and immobilized patients regain strength and movement. It was through these experiments that he recognized the importance of training the core abdominal and back muscles to stabilize the torso and allow the entire body to move freely. This experimentation provided the foundation for his style of conditioning and the specialized exercise equipment associated with the Pilates method.

Pilates emigrated to the United States in 1926 after the German government invited him to use his conditioning methods to train the army. That same year he opened the first Pilates studio in New York City. Over the years, dancers, actors, and athletes flocked to his studio to heal, condition, and align their bodies.

Joseph Pilates died at age 87 in a fire at his studio. Although his strength enabled him to escape the flames by hanging from the rafters for over an hour, he died from **smoke inhalation**. He believed that ideal fitness is “the attainment and maintenance of a uniformly developed body with a sound mind fully capable of naturally, easily, and satisfactorily performing our many and varied daily tasks with spontaneous zest and pleasure.”

During the initial meeting, an instructor will analyze the client’s posture and movement and design a specific training program. Once the program has been created, the sessions usually follow a basic pattern. A session generally begins with mat work and passive and active stretching. In passive stretching, the instructor moves and presses the client’s body to stretch and elongate the muscles. During the active stretching period, the client preforms the stretches while the instructor watches their form and breathing. These exercises warm up the muscles in preparation for the machine work. The machines help the client to maintain the correct positioning required for each exercise.

KEY TERMS

Yoga—A system of physical, mental, and breathing exercises developed in India.

Zen—A form of meditation that emphasizes direct experience.

There are over 500 exercises that were developed by Joseph Pilates. “Classical” exercises, according to the Pilates Studio in New York involve several principles. These include concentration, centering, flowing movement, and breath. Some instructors teach only the classical exercises originally taught by Joseph Pilates. Others design new exercises that are variations upon these classical forms in order to make the exercises more accessible for a specific person.

There are two primary exercise machines used for Pilates, the Universal Reformer and the Cadillac, and several smaller pieces of equipment. The Reformer resembles a single bed frame and is equipped with a carriage that slides back and forth and adjustable springs that are used to regulate tension and resistance. Cables, bars, straps, and pulleys allow the exercises to be done from a variety of positions. Instructors usually work with their clients on the machines for 20–45 minutes. During this time, they are observing and giving feedback about alignment, breathing, and precision of movement. The exercises are done slowly and carefully so that the movements are smooth and flowing. This requires focused concentration and muscle control. The session ends with light stretching and a cool-down period.

Once the basics are learned from an instructor, from either one-on-one lessons or in a class, it is possible to train at home using videos. Exercise equipment for use at home is also available and many exercises can be preformed on a mat.

A private session costs between \$45–75 dollars, depending on the part of the country one is in. This method is not specifically covered by insurance although it may be covered when the instructor is a licensed physical therapist.

Precautions

The Pilates method is not a substitute for good physical therapy, although it has been increasingly used and recommended by physical therapists since the mid-1980s. People with chronic injuries are advised to see a physician.

Research and general acceptance

As of early 2004, several physical therapists and gerontologists have done research studies on the Pilates method, although much more work needs to be done in this area. One recent finding is that the method should not be used by patients with lower back **pain**, as it appears to be ineffective in treating this condition.

The appeal of the Pilates method to a wide population, coupled with a new interest in it on the part of **rehabilitation** therapists, suggests that further studies may soon be underway. Dancers and actors originally embraced the Pilates method as a form of strength training that did not create muscle bulk. Professional and amateur athletes also use these exercises to prevent reinjury. Sedentary people find Pilates to be a gentle, non-impact approach to conditioning. Pilates equipment and classes can be found in hospitals, health clubs, spas, and gyms.

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ORGANIZATIONS

Physical Mind Institute. 1807 Second Street, Suite 15/16, Santa Fe, New Mexico 87505. (505) 988-1990 or (800) 505-1990. Fax: (505) 988-2837. themethod@trail.com.

Pilates Studio. 2121 Broadway, Suite 201, New York, New York, 10023-1786. (800)474-5283 or (888) 474-5283 or

(212) 875-0189. Fax: (212) 769-2368. <http://www.pilates-studio.com>.

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Piles see **Hemorrhoids**

Pinguecula and pterygium

Definition

Pinguecula and pterygium are both non-malignant, slow-growing proliferations of conjunctival connective tissue in the eye. Pterygia, but not pingueculae, extend over the cornea.

Description

The outer layer of the eyeball consists of the tough white sclera and the transparent cornea. The cornea lies in front of the colored part of the eye (iris). Overlying the sclera is a transparent mucous membrane called the conjunctiva. The conjunctiva lines the inside of the lids (palpebral conjunctiva) and covers the sclera (bulbar conjunctiva).

Pingueculae and pterygia are common in adults, and their incidence increases with age. Pterygia are less common than pingueculae.

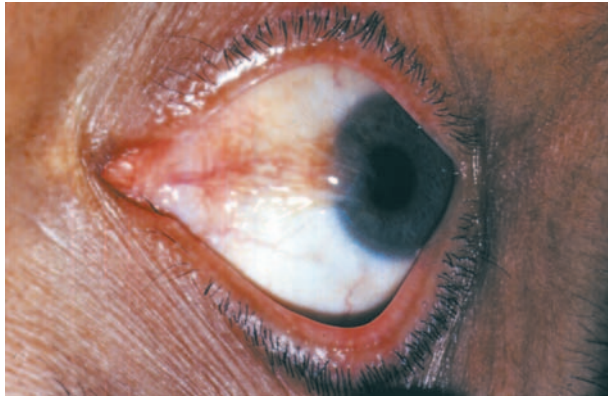
Pingueculae are seen as small, raised, thickenings of the conjunctiva. They may be yellow, gray, white, or colorless. They are almost always to one side of the iris—not above or below—and usually on the side closest to the nose. A pinguecula may develop into a pterygium.

Pterygia are conjunctival thickenings that may have blood vessels associated with them. They often have a triangular-shaped appearance. The pterygia may also grow over the cornea and may therefore affect vision.

Causes and symptoms

Causes

The cause or causes of these disorders are unknown, but they are more frequent in people who live in sunny and windy climates and people whose jobs expose them to ultraviolet (UV) light (for example, farmers and arc welders). Pingueculae and pterygia also occur in older people. It is thought these growths are the result of UV or infrared light and



Pterygium, an overgrowth of the cornea, is usually on the inner side of the eye by thickened and degenerative conjunctiva. (Photo Researchers, Inc. Reproduced by permission.)

irritation. It is also believed that prolonged exposure to these risk factors (that is, UV light) increases the chances of occurrence.

Symptoms

Although some people with pinguecula constantly feel like they have a foreign body in their eye, most are asymptomatic. Because the lids can no longer spread the tears over a smooth area, dry areas may result. Some people with a pterygium are also asymptomatic; some feel like they have a foreign body in their eye. Because a pterygium can stretch and distort the cornea, some people acquire **astigmatism** from a pterygium.

Diagnosis

An eye doctor (ophthalmologist or optometrist) can usually diagnose pingueculae and pterygia by external observation, generally using an instrument called a slit lamp. A slit lamp is a microscope with a light source and magnifies the structures of the eye for the examiner. However, because pingueculae and pterygia can sometimes look similar to more serious eye growths, it is important for people to have them checked by an eye care professional.

Treatment

Usually, no treatment is needed. Artificial tears can be used to relieve the sensation of a foreign body in the eye and to protect against dryness. Surgery to remove the pinguecula or pterygium is advisable when the effect on the cornea causes visual defects or when the thickening is causing excessive and recurrent discomfort or inflammation. Sometimes surgical

KEY TERMS

Astigmatism—Asymetric vision problems due to irregularities in the cornea.

Beta radiation—Streams of electrons emitted by beta emitters like carbon-14 and radium.

Conjunctiva—The mucous membrane that covers the white part of the eyes and lines the eyelids.

Cornea—The clear outer covering of the front of the eye. It is in front of the colored part of the eye (iris) and the iris's central black hole (pupil).

removal is also performed for cosmetic reasons. However, healing from this type of surgery, although usually painless, takes many weeks, and there is a high rate of recurrence (as high as 50–60% in some regions). Accordingly, surgery is avoided unless problems due to the pinguecula or pterygium are significant.

Several methods have been used to attempt to reduce the recurrence of the pinguecula or pterygium after surgery. One method that should be abandoned is beta radiation. Although it is effective at slowing the regrowth of pingueculae and pterygia, it can cause **cataracts**. A preferable method is the topical application of the anticancer drug, mitomycin-C.

Prognosis

Most pingueculae and pterygia grow slowly and almost never cause significant damage, so the prognosis is excellent. Again, a diagnosis must be made to rule out other more serious disorders.

Prevention

There is nothing that has been clearly shown to prevent these disorders, or to prevent a pinguecula from progressing to a pterygium. However, the presence of pingueculae and pterygia have been linked to exposure to UV radiation. For that reason, UV exposure should be reduced. The American Optometric Association (AOA) suggests that sunglasses should block 99–100% of UV-A and UV-B rays. Patients should speak to their eye care professionals about protective coatings on sunglasses or regular spectacles. Protecting the eyes from sunlight, dust, and other environmental irritants is a good idea.

Resources

ORGANIZATIONS

New York University (NYU), Department of Ophthalmology.
550 First Ave., New York, NY 10016. (212) 263-6433.
< <http://ophth-www.med.nyu.edu/Ophth> >.

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Pinkeye see **Conjunctivitis**

Pinta

Definition

A bacterial infection of the skin which causes red to bluish-black colored spots.

Description

Pinta is a skin infection caused by the bacterium *Treponema carateum*, a relative of the bacterium which causes **syphilis**. The word “pinta” comes from the Spanish and means “painted.” Pinta is also known as “azula” (blue), and “mal de pinto” (pinto sickness). It is one of several infections caused by different *Treponema* bacteria, which are called “endemic” or “non-venereal” treponematoses.

Pinta is primarily found in rural, poverty-stricken areas of northern South America, Mexico, and the Caribbean. The disease is usually acquired during childhood and is spread from one person to another by direct skin-to-skin contact. The bacteria enter the skin through a small cut, scratch, or other skin damage. Once inside the skin, the warmth and moisture allow the bacteria to multiply. The bacterial infection causes red, scaly lesions on the skin.

Causes and symptoms

Pinta is caused by an infection with the bacterium *Treponema carateum*. Persons at risk for pinta are those who live in rural, poverty-stricken, overcrowded regions of South America, Mexico, and the Caribbean. Symptoms of pinta occur within two to four weeks after exposure to the bacteria. The first sign of infection is a red, scaly, slowly enlarging bump on the skin. This is called the “primary lesion.” The primary lesion usually appears at the site where the bacteria entered the skin. This is often on the arms, legs, or face. The smaller lesions which form around the primary lesion are called “satellite lesions.” Lymph nodes

KEY TERMS

Lesion—An abnormal change in skin due to disease.

located near the infected area will become enlarged, but are painless.

The second stage of pinta occurs between one and 12 months after the primary lesion stage. Many flat, red, scaly, itchy lesions called “pintids” occur either near the primary lesion, or scattered around the body. Pintid lesions progress through a range of color changes, from red to bluish-black. The skin of older lesions will become depigmented (loss of normal color).

Diagnosis

Pinta can be diagnosed by dermatologists (doctors who specialize in skin diseases) and infectious disease specialists. The appearance of the lesions helps in the diagnosis. A blood sample will be taken from the patient’s arm to test for antibodies to *Treponema carateum*. A scraping of a lesion will be examined under the microscope to look for *Treponema* bacteria. The results of these tests should be available within one to two days.

Treatment

Pinta is treated with benzathine penicillin G (Bicillin), given as a single injection.

Prognosis

Treatment will result in a complete cure but will not undo any skin damage caused by the late stages of disease. Spread of pinta to the eyes can cause eyelid deformities.

Prevention

Good personal hygiene and general health may help prevent infections. In general, avoid physical contact with persons who have **skin lesions**.

Resources

OTHER

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Belinda Rowland, PhD

Pinworm infection see **Enterobiasis**

Pituitary adenoma see **Pituitary tumors**

Pituitary dwarfism

Definition

Dwarfism is a condition in which the growth of the individual is very slow or delayed. There are many forms of dwarfism. The word pituitary is in reference to the pituitary gland in the body. This gland regulates certain chemicals (hormones) in the body. Therefore, pituitary dwarfism is decreased bodily growth due to hormonal problems. The end result is a proportionate little person, because the height as well as the growth of all other structures of the individual are decreased.

Description

Pituitary dwarfism is caused by problems arising in the pituitary gland. The pituitary gland is also called the hypophysis. The pituitary gland is divided into two halves: the anterior (front) and posterior (back) halves. The anterior half produces six hormones: growth hormone, adrenocorticotropin (corticotropin), thyroid stimulating hormone (thyrotropin), prolactin, follicle stimulating hormone, and lutenizing hormone. The posterior pituitary gland only produces two hormones. It produces antidiuretic hormone (vasopressin) and oxytocin.

Most forms of dwarfism are a result of decreased production of hormones from the anterior half of the pituitary gland. The most common form is due to decreases of growth hormone which will be discussed here. These decreases during childhood cause the individual's arms, legs, and other structures to develop normal proportions for their bodies, but at a decreased rate.

When all of the hormones of the anterior pituitary gland are not produced, this is called panhypopituitarism. Another type of dwarfism occurs when only the growth hormone is decreased. Dwarfism can also result from a lack of somatomedin C (also called insulin like growth factor, IGF-1) production. Somatomedin C is a hormone produced in the liver that increases bone growth when growth hormone is present. The African pygmy and the Levi-Lorain dwarfs lack the ability to produce somatomedin C in response to growth hormone. All causes of dwarfism lead to a proportionate little person.

Growth is the body's response to different hormones. The forebrain contains a small organ called the hypothalamus, which is responsible for releasing hormones in response to the body's needs for purposes of regulation. Growth hormone is produced in the anterior pituitary gland when growth hormone-releasing hormone (GHRH), is released by the hypothalamus. Growth hormone is then released and stimulates the liver to produce IGF-1. In return, IGF-1 stimulates the long bones to grow in length. Thus, growth can be slowed down or stopped if there is a problem making any of these hormones or if there is a problem with the cells receiving these hormones.

Some estimates show that there are between 10,000 and 15,000 children in the United States who have growth problems due to a deficiency of growth hormone.

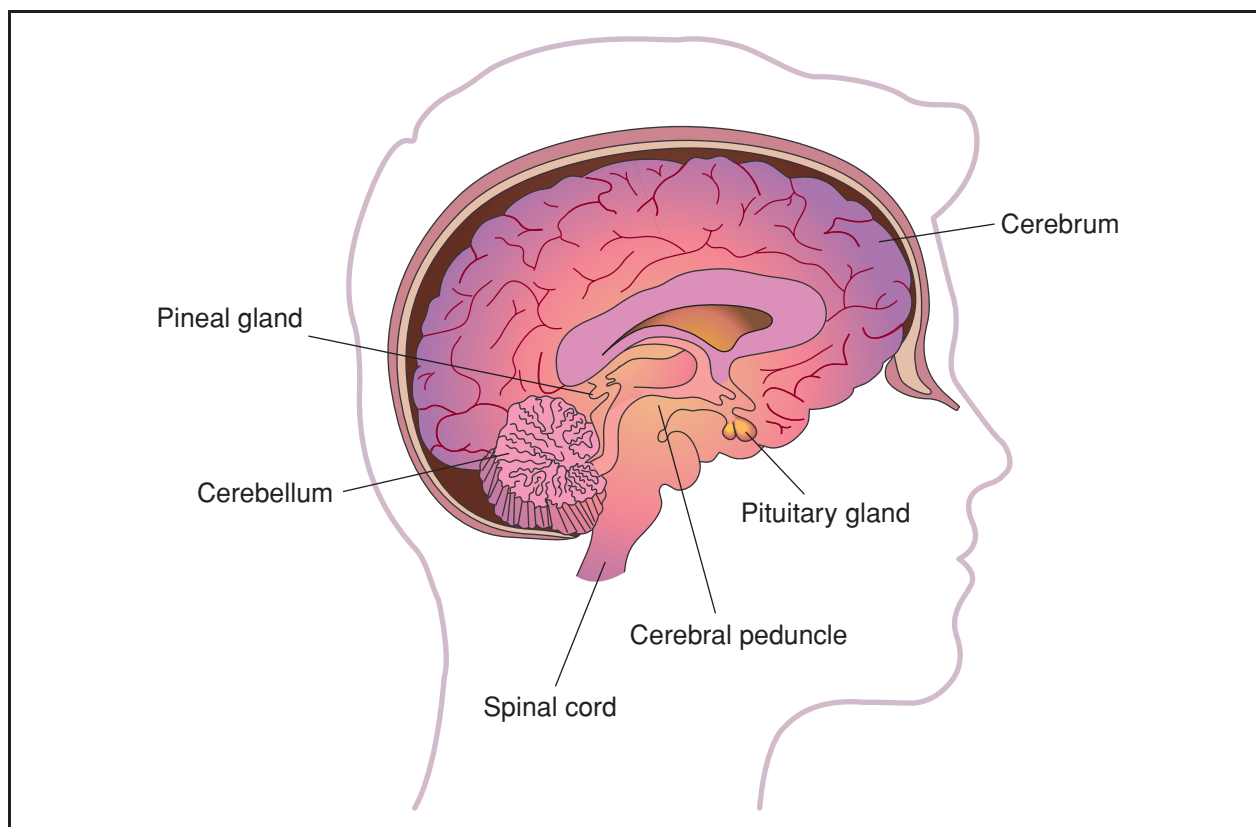
Causes and symptoms

Pituitary dwarfism has been shown to run in families. New investigations are underway to determine the specific cause and location of the gene responsible for dwarfism. The human cell contains 46 chromosomes arranged in 23 pairs. Most of the genes in the two chromosomes of each pair are identical or almost identical with each other. However, with dwarfism, there appears to be disruption on different areas of chromosome 3 and 7. Some studies have isolated defects for the production of pituitary hormones to the short arm (the "p" end) of chromosome 3 at a specific location of 3p11. Other studies have found changes on the short arm of chromosome 7.

A child with a growth hormone deficiency is often small with an immature face and chubby body build. The child's growth will slow down and not follow the normal growth curve patterns. In cases of tumor, most commonly craniopharyngioma (a tumor near the pituitary gland), children and adolescents may present with neurological symptoms such as headaches, **vomiting**, and problems with vision. The patient may also have symptoms of double vision. Symptoms such as truly bizarre and excessive drinking behaviors (polydipsia) and sleep disturbances may be common.

Diagnosis

The primary symptom of pituitary dwarfism is lack of height. Therefore, a change in the individual's growth habits will help lead to a diagnosis. Another diagnostic technique uses an x ray of the child's hand to determine the child's bone age by comparing this to the child's actual chronological age. The bone age in



Pituitary dwarfism is a condition of growth retardation characterized by patients who are very short but have normal body proportions. It is caused by a dysfunction of the pituitary gland, the pea-sized mass of tissue located at the base of the brain. (Illustration by Electronic Illustrators Group.)

affected children is usually two years or more behind the chronological age. This means that if a child is ten years old, his or her bones will look like they are those of an eight-year-old child. The levels of growth hormone and somatomedin C must also be measured with blood tests.

Hypopituitarism may be gained or acquired following birth for several reasons. It could be due to trauma to the pituitary gland such as a fall or following surgery to the brain for removal of a tumor. It may also be due to the child's environment (deprivational dwarfism).

On examination by the doctor there may be optic nerve atrophy, if the dwarfism is due to a type of tumor. X rays of the area where the pituitary gland is located (sella turcica) or more advanced imaging such as **magnetic resonance imaging (MRI)** or computed tomography CT may show changes of the pituitary gland itself. Computed tomography, is an advanced form of x ray that will help determine the integrity of the bone and how much calcification the tumor is producing. Magnetic resonance imaging, will also

help in the diagnosis. MRI is a type of imaging device that can visualize soft tissues such as muscle and fat.

If the dwarfism is due to environmental and emotional problems, the individual may be hospitalized to monitor hormone levels. Following a few days of hospitalized, hormone levels may become normal due to avoidance of the original environment.

Treatment

The main course of therapy is growth **hormone replacement therapy** when there is lack of growth hormone in the body. A pediatric endocrinologist, a doctor specializing in the hormones of children, usually administers this type of therapy before a child's growth plates have fused or joined together. Once the growth plates have fused, GH replacement therapy is rarely effective.

Growth hormone used to be collected from recently deceased humans. However, frequent disease complications resulting from human growth hormone collected from deceased bodies, lead to the banning of

KEY TERMS

Adrenocorticotropin (corticotrophin)—A hormone that acts on cells of the adrenal cortex, causing them to produce male sex hormones and hormones that control water and mineral balance in the body.

Antidiuretic hormone (vasopressin)—A hormone that acts on the kidneys to regulate water balance.

Craniopharyngioma—A tumor near the pituitary gland in the craniopharyngeal canal that often results in intracranial pressure.

Deprivational dwarfism—A condition where emotional disturbances are associated with growth failure and abnormalities of pituitary function.

Follicle-stimulating hormone (FSH)—A hormone that in females stimulates estrogen and in males stimulates sperm production.

Growth hormone—A hormone that eventually stimulates growth. Also called somatotropin.

Hormone—A chemical messenger produced by the body that is involved in regulating specific bodily functions such as growth, development, and reproduction.

Luteinizing hormone—A hormone secreted by the pituitary gland that regulates the menstrual cycle and triggers ovulation in females. In males it stimulates the testes to produce testosterone.

Oxytocin—A hormone that stimulates the uterus to contract during child birth and the breasts to release milk.

Panhypopituitarism—Generalized decrease of all of the anterior pituitary hormones.

Prolactin—A hormone that helps the breast prepare for milk production during pregnancy.

Puberty—Point in development when the gonads begin to function and secondary sexual characteristics begin to appear.

Thyroid stimulating hormone (thyrotropin)—A hormone that stimulates the thyroid gland to produce hormones that regulate metabolism.

this method. In the mid-1980s, techniques were discovered that could produce growth hormones in the lab. Now, the only growth hormone used for treatment is that made in a laboratory.

A careful balancing of all of the hormones produced by the pituitary gland is necessary for patients

with panhypopituitarism. This form of dwarfism is very difficult to manage.

Prognosis

The prognosis for each type of dwarfism varies. A panhypopituitarism dwarf does not pass through the initial onset of adult sexual development (**puberty**) and never produces enough gonadotropic hormones to develop adult sexual function. These individuals also have several other medical conditions. Dwarfism due to only growth hormone deficiency has a different prognosis. These individuals do pass through puberty and mature sexually, however, they remain proportionately small in stature.

If the individual is lacking only growth hormone then growth hormone replacement therapy can be administered. The success of treatment with growth hormone varies however. An increase in height of 4–6 in (10–15 cm) can occur in the first year of treatment. Following this first year, the response to the hormone is not as successful. Therefore the amount of growth hormone administered must be tripled to maintain this rate. Long-term use is considered successful if the individual grows at least 0.75 in (2 cm) per year more than they would without the hormone. However, if the growth hormone treatment is not administered before the long bones—such as the legs and arms—fuse, then the individual will never grow. This fusion is completed by adult age.

Improvement for individuals with dwarfism due to other causes such as a tumor, varies greatly. If the dwarfism is due to deprevational causes, then removing a child from that environment should help to alleviate the problem.

Resources

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ORGANIZATIONS

Human Growth Foundation. 997 Glen Cove Ave., Glen Head, NY 11545. (800) 451-6434. Fax: (516) 671-4055. <<http://www.hgfound.org>>.

Little People of America, Inc. National Headquarters, PO Box 745, Lubbock, TX 79408. (806) 737-8186 or (888) LPA-2001. lpadatabase@juno.com. <<http://www.lpaonline.org>>.

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Jason S. Schliesser, DC

Pituitary gland removal see
Hypophysectomy

Pituitary tumors

Definition

Pituitary tumors are abnormal growths on the pituitary gland. Some tumors secrete hormones normally made by the pituitary gland.

Description

Located in the center of the brain, the pituitary gland manufactures and secretes hormones that regulate growth, sexual development and functioning, and the fluid balance of the body. About 10% of all cancers in the skull are pituitary tumors. Pituitary adenomas (adenomas are tumors that grow from gland tissues) and pituitary tumors in children and adolescents (craniopharyngiomas) are the most common types of pituitary tumors. They are usually benign and grow slowly. Even malignant pituitary tumors rarely spread to other parts of the body.

Pituitary adenomas do not secrete hormones but are likely to be larger and more invasive than tumors that do. Craniopharyngiomas are benign tumors that are extremely difficult to remove. Radiation does not stop them from spreading throughout the pituitary gland. Craniopharyngiomas account for less than 5% of all brain tumors. Pituitary tumors usually develop between the ages of 30 and 40, but half of all craniopharyngiomas occur in children, with symptoms most often appearing between the ages of five and ten.

Causes and symptoms

The cause of pituitary tumors is not known, but may be genetic. Symptoms related to tumor location, size, and pressure on neighboring structures include:

- persistent **headache** on one or both sides, or in the center of the forehead
- blurred or double vision; loss of peripheral vision
- drooping eyelid caused by pressure on nerves leading to the eye
- seizures

Symptoms related to hormonal imbalance include:

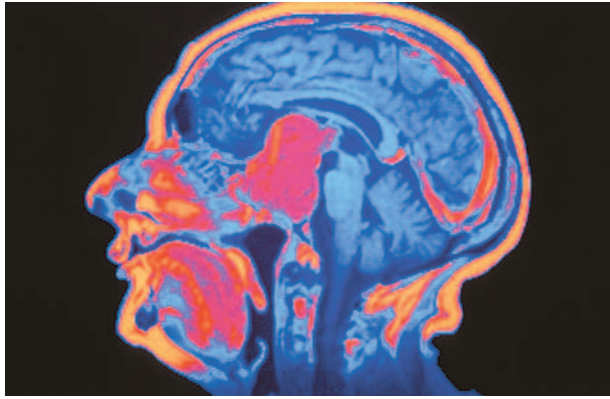
- excessive sweating
- loss of appetite
- loss of interest in sex
- inability to tolerate cold temperatures
- nausea
- high levels of sodium in the blood
- menstrual problems
- excessive thirst
- frequent urination
- dry skin
- constipation
- premature or delayed **puberty**
- delayed growth in children
- galactorrhea (milk secretion in the absence of **pregnancy** or breast feeding)
- low blood pressure
- low blood sugar

Diagnosis

As many as 40% of all pituitary tumors do not release excessive quantities of hormones into the blood. Known as clinically nonfunctioning, these tumors are difficult to distinguish from tumors that produce similar symptoms. They may grow to be quite large before they are diagnosed.

Endocrinologists and neuroendocrinologists base the diagnosis of pituitary tumors on:

- the patient's own observations and medical history
- physical examination
- laboratory studies of the patient's blood and cerebrospinal fluid



Colorized MRI showing large pituitary tumor at center in pink.
(Mehau Kulyk, Photo Researchers. Reproduced by permission.)

- x rays of the skull and other studies that provide images of the inside of the brain (CT, MRI)
- vision tests
- urinalysis

Treatment

Some pituitary tumors stabilize without treatment, but a neurosurgeon will operate at once to remove the tumor (adenectomy) or pituitary gland (**hypophysectomy**) of a patient whose vision is deteriorating rapidly. Patients who have pituitary apoplexy may experience very severe headaches, have symptoms of stiff neck, and sensitivity to light. This condition is considered an emergency. Magnetic resonance imaging (MRI) is the best imaging technique for patients with these symptoms. If the tumor is small, surgery may be done through the nose. If the tumor is large, it may require opening the skull for **tumor removal**. Selected patients do well with proton beam radiosurgery (the use of high energy particles in the form of a high energy beam to destroy an overactive gland).

Treatment is determined by the type of tumor and by whether it has invaded tissues adjacent to the pituitary gland. Hormone-secreting tumors can be successfully treated with surgery, radiation, bromocriptine (Parlodel), Sandostatin (Octreotide), or other somatostatin analogues (drugs similar to somatostatin). Surgery is usually used to remove all or part of a tumor within the gland or the area surrounding it, and may be combined with radiation therapy to treat tumors that extend beyond the pituitary gland. Removal of the pituitary gland requires life-long **hormone replacement therapy**.

KEY TERMS

Agonist—A drug that increases the effectiveness of another drug.

Analogue—A drug that is similar to the drug from which it is derived.

Radiation therapy can provide long-term control of the disease if it recurs after surgery, and radioactive pellets can be implanted in the brain to treat cranio-pharyngiomas. CV205-502, a new dopamine agonist (a drug that increases the effect of another, in this instance dopamine) can control symptoms of patients who do not respond to bromocriptine.

Prognosis

Pituitary tumors are usually curable. Following surgery, adults may gradually resume their normal activities, and children may return to school when the effects of the operation have diminished, and appetite and sense of well-being have returned. Patients should wear medical identification tags identifying their condition and the hormonal replacement medicines they take.

Resources

ORGANIZATIONS

American Brain Tumor Association. 2770 River Road, Des Plaines, IL 60018. (800) 886-2289. <<http://www.abta.org>>.

Brain Tumor Information Services. Box 405, Room J341, University of Chicago Hospitals, 5841 S. Maryland Ave., Chicago, IL 60637. (312) 684-1400.

Maureen Haggerty

Pityriasis rosea

Definition

Pityriasis rosea is a mild, noncontagious skin disorder common among children and young adults, and characterized by a single round spot on the body, followed later by a rash of colored spots on the body and upper arms.



The torso of a man covered with pityriasis rosea. The cause of this disorder is thought to be due to a viral infection. It often appears on the torso and upper parts of the limbs of young people and may be contagious. (Photograph by Dr. P. Marazzi, Photo Researchers, Inc. Reproduced by permission.)

Description

Pityriasis rosea is most common in young adults, and appears up to 50% more often in women. Its cause is unknown; however, some scientists believe that the rash is an immune response to some type of infection in the body.

Causes and symptoms

Doctors do not think that pityriasis rosea is contagious, but the cause is unknown. Some experts suspect the rash, which is most common in spring and fall, may be triggered by a virus, but no infectious agent has ever been found.

It is not sexually transmitted, and does not appear to be contagious from one person to the next.

KEY TERMS

Antihistamines—A group of drugs that block the effects of histamine, a chemical released during an allergic reaction.

Steroids—A group of drugs that includes the corticosteroids, similar to hormones produced by the adrenal glands, and used to relieve inflammation and itching.

Sometimes, before the symptoms appear, people experience preliminary sensations including **fever**, malaise, sore throat, or **headache**. Symptoms begin with a single, large round spot called a “herald patch” on the body, followed days or weeks later by slightly raised, scaly-edged round or oval pink-copper colored spots on the trunk and upper arms. The spots, which have a wrinkled center and a sharp border, sometimes resemble a Christmas tree. They may be mild to severely itchy, and they can spread to other parts of the body.

Diagnosis

A physician can diagnose the condition with blood tests, skin scrapings, or a biopsy of the lesion.

Treatment

The rash usually clears up on its own, although a physician should rule out other conditions that may cause a similar rash (such as **syphilis**).

Treatment includes external and internal medications for **itching** and inflammation. Mild inflammation and itching can be relieved with antihistamine drugs or calamine lotion, zinc oxide, or other mild lubricants or anti-itching creams. Gentle, soothing strokes should be used to apply the ointments, since vigorous rubbing may cause the lesions to spread. More severe itching and inflammation is treated with topical steroids. Moderate exposure to sun or ultraviolet light may help heal the lesions, but patients should avoid being sunburned.

Soap makes the rash more uncomfortable; patients should bathe or shower with plain lukewarm water, and apply a thin coating of bath oil to freshly-dried skin afterwards.

Prognosis

These spots, which may be itchy, last for 3–12 weeks. Symptoms rarely recur.

Resources

ORGANIZATIONS

American Academy of Dermatology. 930 N. Meacham Road, P.O. Box 4014, Schaumburg, IL 60168-4014. (847) 330-0230. Fax: (847) 330-0050. <<http://www.aad.org>>.

Carol A. Turkington

PKU see **Phenylketonuria**

Placenta previa

Definition

Placenta previa is a condition that occurs during **pregnancy** when the placenta is abnormally placed, and partially or totally covers the cervix.

Description

The uterus is the muscular organ that contains the developing baby during pregnancy. The lowest segment of the uterus is a narrowed portion called the cervix. This cervix has an opening (the os) that leads into the vagina, or birth canal. The placenta is the organ that attaches to the wall of the uterus during pregnancy. The placenta allows nutrients and oxygen from the mother's blood circulation to pass into the developing baby (the fetus) via the umbilical cord.

During labor, the muscles of the uterus contract repeatedly. This allows the cervix to begin to grow thinner (called effacement) and more open (dilatation). Eventually, the cervix will become completely effaced and dilated, and the baby can leave the uterus and enter the birth canal. Under normal circumstances, the baby will emerge through the mother's vagina during birth.

In placenta previa, the placenta develops in an abnormal location. Normally, the placenta should develop relatively high up in the uterus, on the front or back wall. In about one in 200 births, the placenta will be located low in the uterus, partially or totally covering the os. This causes particular problems in late pregnancy, when the lower part of the uterus begins to take on a new formation in preparation for delivery. As the cervix begins to efface and dilate, the attachments of the placenta to the uterus are damaged, resulting in bleeding.

Causes and symptoms

While the actual cause of placenta previa is unknown, certain factors increase the risk of a woman developing the condition. These factors include:

- having abnormalities of the uterus
- being older in age
- having had other babies
- having a prior delivery by **cesarean section**
- smoking cigarettes

When a pregnancy involves more than one baby (twins, triplets, etc.), the placenta will be considerably larger than for a single pregnancy. This also increases the chance of placenta previa.

Placenta previa may cause a number of problems. It is thought to be responsible for about 5% of all miscarriages. It frequently causes very light bleeding (spotting) early in pregnancy. Sometime after 28 weeks of pregnancy (most pregnancies last about 40 weeks), placenta previa can cause episodes of significant bleeding. Usually, the bleeding occurs suddenly and is bright red. The woman rarely experiences any accompanying **pain**, although about 10% of the time the placenta may begin separating from the uterine wall (called abruptio placentae), resulting in pain. The bleeding usually stops on its own. About 25% of such patients will go into labor within the next several days. Sometimes, placenta previa does not cause bleeding until labor has already begun.

Placenta previa puts both the mother and the fetus at high risk. The mother is at risk of severe and uncontrollable bleeding (hemorrhage), with dangerous blood loss. If the mother's bleeding is quite severe, this puts the fetus at risk of becoming oxygen deprived. The fetus' only source of oxygen is the mother's blood. The mother's blood loss, coupled with certain changes that take place in response to that blood loss, decreases the amount of blood going to the placenta, and ultimately to the fetus. Furthermore, placenta previa increases the risk of preterm labor, and the possibility that the baby will be delivered prematurely.

Diagnosis

Diagnosis of placenta previa is suspected whenever bright red, painless vaginal bleeding occurs during the course of a pregnancy. The diagnosis can be confirmed by performing an ultrasound examination. This will allow the location of the placenta to be evaluated.

KEY TERMS

Cesarean section—Delivery of a baby through an incision in the mother's abdomen instead of through the vagina.

Labor—The process during which the uterus contracts, and the cervix opens to allow the passage of a baby into the vagina.

Placenta—The organ that provides oxygen and nutrition from the mother to the baby during pregnancy. The placenta is attached to the wall of the uterus and leads to the baby via the umbilical cord.

Umbilical cord—The blood vessels that allow the developing baby to receive nutrition and oxygen from its mother; the blood vessels also eliminate the baby's waste products. One end of the umbilical cord is attached to the placenta and the other end is attached to the baby's belly button (umbilicus).

Vagina—The birth canal; the passage from the cervix of the uterus to the opening leading outside of a woman's body.

While many conditions during pregnancy require a pelvic examination, in which the health care provider's fingers are inserted into the patient's vagina, such an examination should never be performed if there is any suspicion of placenta previa. Such an examination can disturb the already susceptible placenta, resulting in hemorrhage.

Sometimes placenta previa is found early in a pregnancy, during an ultrasound examination performed for another reason. In these cases, it is wise to have a repeat ultrasound performed later in pregnancy (during the last third of the pregnancy, called the third trimester). A large percentage of these women will have a low-lying placenta, but not a true placenta previa where some or all of the os is covered.

Treatment

Treatment depends on how far along in the pregnancy the bleeding occurs. When the pregnancy is less than 36 weeks along, the fetus is not sufficiently developed to allow delivery without a high risk of complications. Therefore, a woman with placenta previa is treated with bed rest, blood transfusions as necessary, and medications to prevent labor. After 36 weeks, the baby can be delivered via cesarean section. This is

almost always the preferred method of delivery in order to avoid further bleeding from the low-lying placenta.

Prognosis

In cases of placenta previa, the prognosis for the mother is very good. However, there is a 15–20% chance the infant will not survive. This is 10 times the **death** rate associated with normal pregnancies. About 60% of these deaths occur because the baby delivered was too premature to survive.

Prevention

There are no known ways to insure the appropriate placement of the placenta in the uterus. However, careful treatment of the problem can result in the best chance for a good outcome for both mother and baby.

Resources

ORGANIZATIONS

American College of Obstetricians and Gynecologists. 409 12th Street, S.W., P.O. Box 96920, Washington, DC 20090-6920. .

Rosalyn Carson-DeWitt, MD

Placental abruption

Definition

Placental abruption occurs when the placenta separates from the wall of the uterus prior to the birth of the baby. This can result in severe, uncontrollable bleeding (hemorrhage).

Description

The uterus is the muscular organ that contains the developing baby during **pregnancy**. The lowest segment of the uterus is a narrowed portion called the cervix. This cervix has an opening (the os) that leads into the vagina, or birth canal. The placenta is the organ that attaches to the wall of the uterus during pregnancy. The placenta allows nutrients and oxygen from the mother's blood circulation to pass into the developing baby (the fetus) via the umbilical cord.

During labor, the muscles of the uterus contract repeatedly. This allows the cervix to begin to grow thinner (called effacement) and more open

(dilatation). Eventually, the cervix will become completely effaced and dilated, and the baby can leave the uterus and enter the birth canal. Under normal circumstances, the baby will go through the mother's vagina during birth.

During a normal labor and delivery, the baby is born first. Several minutes to 30 minutes later, the placenta separates from the wall of the uterus and is delivered. This sequence is necessary because the baby relies on the placenta to provide oxygen until he or she begins to breathe independently.

Placental abruption occurs when the placenta separates from the uterus before the birth of the baby. Placental abruption occurs in about one out of every 200 deliveries. African-American and Latin-American women have a greater risk of this complication than do Caucasian women. It was once believed that the risk of placental abruption increased in women who gave birth to many children, but this association is still being researched.

Causes and symptoms

The cause of placental abruption is unknown. However, a number of risk factors have been identified. These factors include:

- older age of the mother
- history of placental abruption during a previous pregnancy
- high blood pressure
- certain disease states (diabetes, collagen vascular diseases)
- the presence of a type of uterine tumor called a leiomyoma
- twins, triplets, or other multiple pregnancies
- cigarette **smoking**
- heavy alcohol use
- **cocaine** use
- malformations of the uterus
- malformations of the placenta
- injury to the abdomen (as might occur in a car accident)

Symptoms of placental abruption include bleeding from the vagina, severe **pain** in the abdomen or back, and tenderness of the uterus. Depending on the severity of the bleeding, the mother may experience a drop in blood pressure, followed by symptoms of organ failure as her organs are deprived of oxygen. Sometimes, there is no visible vaginal bleeding.

Instead, the bleeding is said to be "concealed." In this case, the bleeding is trapped behind the placenta, or there may be bleeding into the muscle of the uterus. Many patients will have abnormal contractions of the uterus, particularly extremely hard, prolonged contractions. Placental abruption can be total (in which case the fetus will almost always die in the uterus), or partial.

Placental abruption can also cause a very serious complication called consumptive coagulopathy. A series of reactions begin that involve the elements of the blood responsible for clotting. These clotting elements are bound together and used up by these reactions. This increases the risk of uncontrollable bleeding and may contribute to severe bleeding from the uterus, as well as causing bleeding from other locations (nose, urinary tract, etc.).

Placental abruption is risky for both the mother and the fetus. It is dangerous for the mother because of blood loss, loss of clotting ability, and oxygen deprivation to her organs (especially the kidneys and heart). This condition is dangerous for the fetus because of oxygen deprivation, too, since the mother's blood is the fetus' only source of oxygen. Because the abrupting placenta is attached to the umbilical cord, and the umbilical cord is an extension of the fetus' circulatory system, the fetus is also at risk of hemorrhaging. The fetus may die from these stresses, or may be born with damage due to oxygen deprivation. If the abruption occurs well before the baby was due to be delivered, early delivery may cause the baby to suffer complications of premature birth.

Diagnosis

Diagnosis of placental abruption relies heavily on the patient's report of her symptoms and a **physical examination** performed by a health care provider. Ultrasound can sometimes be used to diagnose an abruption, but there is a high rate of missed or incorrect diagnoses associated with this tool when used for this purpose. Blood will be taken from the mother and tested to evaluate the possibility of life-threatening problems with the mother's clotting system.

Treatment

The first line of treatment for placental abruption involves replacing the mother's lost blood with blood transfusions and fluids given through a needle in a vein. Oxygen will be administered, usually by a mask or through tubes leading to the nose. When the placental separation is severe, treatment may require

KEY TERMS

Cesarean section—Delivery of a baby through an incision in the mother's abdomen, instead of through the vagina.

Labor—The process during which the uterus contracts, and the cervix opens to allow the passage of a baby into the vagina.

Placenta—The organ that provides oxygen and nutrition from the mother to the baby during pregnancy. The placenta is attached to the wall of the uterus and leads to the baby via the umbilical cord.

Umbilical cord—The blood vessels that allow the developing baby to receive nutrition and oxygen from its mother; the blood vessels also eliminate the baby's waste products. One end of the umbilical cord is attached to the placenta and the other end is attached to the baby's belly button (umbilicus).

Uterus—The muscular organ that contains the developing baby during pregnancy.

Vagina—The birth canal; the passage from the cervix of the uterus to the opening leading outside of a woman's body.

prompt delivery of the baby. However, delivery may be delayed when the placental separation is not as severe, and when the fetus is too immature to insure a healthy baby if delivered. The baby is delivered vaginally when possible. However, a **cesarean section** may be performed to deliver the baby more quickly if the abruption is quite severe or if the baby is in distress.

Prognosis

The prognosis for cases of placental abruption varies, depending on the severity of the abruption. The risk of **death** for the mother ranges up to 5%, usually due to severe blood loss, **heart failure**, and kidney failure. In cases of severe abruption, 50–80% of all fetuses die. Among those who survive, nearly half will have lifelong problems due to oxygen deprivation in the uterus and premature birth.

Prevention

Some of the causes of placental abruption are preventable. These include cigarette smoking, alcohol **abuse**, and cocaine use. Other causes of abruption may

not be avoidable, like diabetes or high blood pressure. These diseases should be carefully treated. Patients with conditions known to increase the risk of placental abruption should be carefully monitored for signs and symptoms of this complication.

Resources

ORGANIZATIONS

American College of Obstetricians and Gynecologists. 409 12th Street, S.W., P.O. Box 96920, Washington, DC 20090-6920.

Rosalyn Carson-DeWitt, MD

Plague

Definition

Plague is a serious, potentially life-threatening infectious disease that is usually transmitted to humans by the bites of rodent fleas. It was one of the scourges of early human history. There are three major forms of the disease: bubonic, septicemic, and pneumonic.

Description

Plague has been responsible for three great world pandemics, which caused millions of deaths and significantly altered the course of history. A pandemic is a disease occurring in epidemic form throughout the entire population of a country, a people, or the world. Although the cause of the plague was not identified until the third pandemic in 1894, scientists are virtually certain that the first two pandemics were plague because a number of the survivors wrote about their experiences and described the symptoms.

The first great pandemic appeared in AD 542 and lasted for 60 years. It killed millions of citizens, particularly people living along the Mediterranean Sea. This sea was the busiest, coastal trade route at that time and connected what is now southern Europe, northern Africa, and parts of coastal Asia. This pandemic is sometimes referred to as the Plague of Justinian, named for the great emperor of Byzantium who was ruling at the beginning of the outbreak. According to the historian Procopius, this outbreak of plague killed 10,000 people per day at its height just within the city of Constantinople.

The second pandemic occurred during the fourteenth century, and was called the Black Death because its main symptom was the appearance of black patches (caused by bleeding) on the skin. It was also a subject found in many European paintings, drawings, plays, and writings of that time. The connections between large active trading ports, rats coming off the ships, and the severe outbreaks of the plague were understood by people at the time. This was the most severe of the three, beginning in the mid-1300s with an origin in central Asia and lasting for 400 years. Between a fourth and a third of the entire European population died within a few years after plague was first introduced. Some smaller villages and towns were completely wiped out.

The final pandemic began in northern China, reaching Canton and Hong Kong by 1894. From there, it spread to all continents, killing millions.

The great pandemics of the past occurred when wild rodents spread the disease to rats in cities, and then to humans when the rats died. Another route for infection came from rats coming off ships that had traveled from heavily infected areas. Generally, these were busy coastal or inland trade routes. Plague was introduced into the United States during this pandemic and it spread from the West towards the Midwest and became endemic in the Southwest of the United States.

About 10–15 Americans living in the southwestern United States contract plague each year during the spring and summer. The last rat-borne epidemic in the United States occurred in Los Angeles in 1924–25. Since then, all plague cases in this country have been sporadic, acquired from wild rodents or their fleas. Plague can also be acquired from ground squirrels and prairie dogs in parts of Arizona, New Mexico, California, Colorado, and Nevada. Around the world, there are between 1,000 and 2,000 cases of plague each year. Recent outbreaks in humans occurred in Africa, South America, and Southeast Asia.

Some people and/or animals with bubonic plague go on to develop **pneumonia** (pneumonic plague). This can spread to others via infected droplets during coughing or sneezing.

Plague is one of three diseases still subject to international health regulations. These rules require that all confirmed cases be reported to the World Health Organization (WHO) within 24 hours of diagnosis. According to the regulations, passengers on an international voyage who have been to an area where there is an epidemic of pneumonic plague must be placed in isolation for six days before being allowed to leave.

While plague is found in several countries, there is little risk to United States travelers within endemic areas (limited locales where a disease is known to be present) if they restrict their travel to urban areas with modern hotel accommodations.

Over the past few years, this infection primarily of antiquity has become a modern issue. This change has occurred because of the concerns about the use of plague as a weapon of biological warfare or terrorism (bioterrorism). Along with **anthrax** and **smallpox**, plague is considered to be a significant risk. In this scenario, the primary manifestation is likely to be pneumonic plague transmitted by clandestine aerosols. It has been reported that during World War II the Japanese dropped “bombs” containing plague-infected fleas in China as a form of biowarfare.

Causes and symptoms

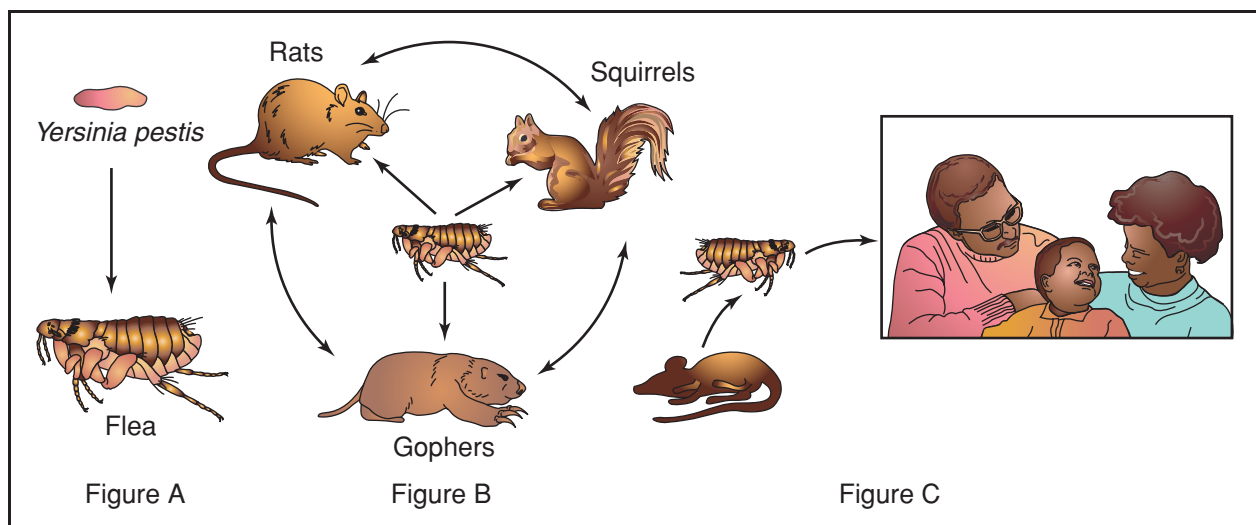
Fleas carry the bacterium *Yersinia pestis*, formerly known as *Pasteurella pestis*. The plague bacillus can be stained with Giemsa stain and typically looks like a safety pin under the microscope. When a flea bites an infected rodent, it swallows the plague bacteria. The bacteria are passed on when the fleas, in turn, bite a human. Interestingly, the plague bacterium grows in the gullet of the flea, obstructing it and not allowing the flea to eat. Transmission occurs during abortive feeding with regurgitation of bacteria into the feeding site. Humans also may become infected if they have a break or cut in the skin and come in direct contact with body fluids or tissues of infected animals.

More than 100 species of fleas have been reported to be naturally infected with plague; in the western United States, the most common source of plague is the golden-manteled ground squirrel flea. Chipmunks and prairie dogs have also been identified as hosts of infected fleas.

Since 1924, there have been no documented cases in the United States of human-to-human spread of plague from droplets. All but one of the few pneumonic cases have been associated with handling infected cats. While dogs and cats can become infected, dogs rarely show signs of illness and are not believed to spread disease to humans. However, plague has been spread from infected coyotes (wild dogs) to humans. In parts of central Asia, gerbils have been identified as the source of cases of bubonic plague in humans.

Bubonic plague

Two to five days after infection, patients experience a sudden **fever**, chills, seizures, and severe



Plague is a serious infectious disease transmitted by the bites of rat fleas. There are three major forms of plague: bubonic, pneumonic, and septicemic. As illustrated above, fleas carry the bacterium *Yersinia pestis*. When a flea bites an infected rodent, it becomes a vector and then passes the plague bacteria when it bites a human. (Illustration by Electronic Illustrators Group.)

headaches, followed by the appearance of swellings or “buboes” in armpits, groin, and neck. The most commonly affected sites are the lymph glands near the site of the first infection. As the bacteria multiply in the glands, the lymph node becomes swollen. As the nodes collect fluid, they become extremely tender. Occasionally, the bacteria will cause an ulcer at the point of the first infection.

Septicemic plague

Bacteria that invade the bloodstream directly (without involving the lymph nodes) cause septicemic plague. (Bubonic plague also can progress to septicemic plague if not treated appropriately.) Septicemic plague that does not involve the lymph glands is particularly dangerous because it can be hard to diagnose the disease. The bacteria usually spread to other sites, including the liver, kidneys, spleen, lungs, and sometimes the eyes, or the lining of the brain. Symptoms include fever, chills, prostration, abdominal **pain**, **shock**, and bleeding into the skin and organs.

Pneumonic plague

Pneumonic plague may occur as a direct infection (primary) or as a result of untreated bubonic or septicemic plague (secondary). Primary pneumonic plague is caused by inhaling infective drops from another person or animal with pneumonic plague. Symptoms, which appear within one to three days after infection, include a severe, overwhelming pneumonia, with shortness of breath, high fever, and blood

in the phlegm. If untreated, half the patients will die; if blood **poisoning** occurs as an early complication, patients may die even before the buboes appear.

Life-threatening complications of plague include shock, high fever, problems with blood clotting, and convulsions.

Diagnosis

Plague should be suspected if there are painful buboes, fever, exhaustion, and a history of possible exposure to rodents, rabbits, or fleas in the West or Southwest. The patient should be isolated. Chest x rays are taken, as well as blood cultures, antigen testing, and examination of lymph node specimens. Blood cultures should be taken 30 minutes apart, before treatment.

A group of German researchers reported in 2004 on a standardized enzyme-linked immunosorbent assay (ELISA) kit for the rapid diagnosis of plague. The test kit was developed by the German military and has a high degree of accuracy as well as speed in identifying the plague bacillus. The kit could be useful in the event of a bioterrorist attack as well as in countries without advanced microbiology laboratories.

Treatment

As soon as plague is suspected, the patient should be isolated, and local and state departments notified. Drug treatment reduces the risk of **death** to less

KEY TERMS

Bioterrorism—The use of disease agents to terrorize or intimidate a civilian population.

Buboes—Smooth, oval, reddened, and very painful swellings in the armpits, groin, or neck that occur as a result of infection with the plague.

Endemic—A disease that occurs naturally in a geographic area or population group.

Epidemic—A disease that occurs throughout part of the population of a country.

Pandemic—A disease that occurs throughout a regional group, the population of a country, or the world.

Septicemia—The medical term for blood poisoning, in which bacteria have invaded the bloodstream and circulates throughout the body.

than 5%. The preferred treatment is streptomycin administered as soon as possible. Alternatives include gentamicin, chloramphenicol, tetracycline, or trimethoprim/sulfamethoxazole.

Prognosis

Plague can be treated successfully if it is caught early; the mortality rate for treated disease is 1–15% but 40–60% in untreated cases. Untreated pneumonic plague is almost always fatal, however, and the chances of survival are very low unless specific antibiotic treatment is started within 15–18 hours after symptoms appear. The presence of plague bacteria in a blood smear is a grave sign and indicates septicemic plague. Septicemic plague has a mortality rate of 40% in treated cases and 100% in untreated cases.

Prevention

Anyone who has come in contact with a plague pneumonia victim should be given antibiotics, since untreated pneumonic plague patients can pass on their illness to close contacts throughout the course of the illness. All plague patients should be isolated for 48 hours after antibiotic treatment begins. Pneumonic plague patients should be completely isolated until sputum cultures show no sign of infection.

Residents of areas where plague is found should keep rodents out of their homes. Anyone working in a rodent-infested area should wear insect repellent on skin and clothing. Pets can be treated with insecticidal

dust and kept indoors. Handling sick or dead animals (especially rodents and cats) should be avoided.

Plague vaccines have been used with varying effectiveness since the late nineteenth century. Experts believe that **vaccination** lowers the chance of infection and the severity of the disease. However, the effectiveness of the vaccine against pneumonic plague is not clearly known.

Vaccinations against plague are not required to enter any country. Because immunization requires multiple doses over a 6–10 month period, plague vaccine is not recommended for quick protection during outbreaks. Moreover, its unpleasant side effects make it a poor choice unless there is a substantial long-term risk of infection. The safety of the vaccine for those under age 18 has not been established. Pregnant women should not be vaccinated unless the need for protection is greater than the risk to the unborn child. Even those who receive the vaccine may not be completely protected. The inadequacy of the vaccines available as of the early 2000s explains why it is important to protect against rodents, fleas, and people with plague. A team of researchers in the United Kingdom reported in the summer of 2004 that an injected subunit vaccine is likely to offer the best protection against both bubonic and pneumonic forms of plague.

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National Institute of Allergies and Infectious Diseases, Division of Microbiology and Infectious Diseases. Bldg. 31, Rm. 7A-50, 31 Center Drive MSC 2520, Bethesda, MD 20892.

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OTHER

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World Health Organization. <<http://www.who.ch/>>.

Arnold Cua, MD
Rebecca J. Frey, PhD

Plaque see **Skin lesions**

Plasma cell myeloma see **Multiple myeloma**

Plasma renin activity

Definition

Renin is an enzyme released by the kidney to help control the body's sodium-potassium balance, fluid volume, and blood pressure.

Purpose

Plasma renin activity (PRA), also called plasma renin assay, may be used to screen for high blood pressure (**hypertension**) of kidney origin, and may help plan treatment of essential hypertension, a genetic disease often aggravated by excess sodium intake. PRA is also used to further evaluate a diagnosis of excess aldosterone, a hormone secreted by the adrenal cortex, in a condition called Conn's syndrome.

Precautions

Patients taking **diuretics**, antihypertensives, **vasodilators**, **oral contraceptives**, and licorice should discontinue use of these substances for two to four

weeks before the test. It should be noted that renin is increased in **pregnancy** and in **diets** with reduced salt intake. Also, since renin is affected by body position, as well as by diurnal (daily) variation, blood samples should be drawn in the morning, and the position of the patient (sitting or lying down) should be noted.

Description

When the kidneys release the enzyme renin in response to certain conditions (high blood potassium, low blood sodium, decreased blood volume), it is the first step in what is called the renin-angiotensin-aldosterone cycle. This cycle includes the conversion of angiotensinogen to angiotensin I, which in turn is converted to angiotensin II, in the lung. Angiotensin II is a powerful blood vessel constrictor, and its action stimulates the release of aldosterone from an area of the adrenal glands called the adrenal cortex. Together, angiotensin and aldosterone increase the blood volume, the blood pressure, and the blood sodium to re-establish the body's sodium-potassium and fluid volume balance. Primary aldosteronism, the symptoms of which include hypertension and low blood potassium (**hypokalemia**), is considered "low-renin aldosteronism."

Renin itself is not actually measured in the PRA test, because renin can be measured only with great difficulty even in research laboratories. In the most commonly used renin assay, the test actually determines, by a procedure called radioimmunoassay, the rate of angiotensin I generation per unit time, while the PRC (plasma renin concentration) measures the maximum renin effect.

Both the PRA and the PRC are extremely difficult to perform. Not only is renin itself unstable, but the patient's body position and the time of day affect the results. Also, the sample must be collected properly: drawn into a chilled syringe and collection tube, placed on ice, and sent to the performing laboratory immediately. Even if all these procedures are followed, results can vary significantly.

A determination of the PRA and a measurement of the plasma aldosterone level are used in the differential diagnosis of primary and secondary **hyperaldosteronism**. Patients with primary hyperaldosteronism (caused by an adrenal tumor that overproduces aldosterone) will have an increased aldosterone level with decreased renin activity. Conversely, patients with secondary hyperaldosteronism (caused by certain types of **kidney disease**) will have increased levels of renin.

KEY TERMS

Aldosteronism—A disorder caused by excessive production of the hormone aldosterone, which is produced by a part of the adrenal glands called the adrenal cortex. Causes include a tumor of the adrenal gland (Conn's syndrome), or a disorder reducing the blood flow through the kidney. This leads to overproduction of renin and angiotensin, and in turn causes excessive aldosterone production. Symptoms include hypertension, impaired kidney function, thirst and muscle weakness.

Conn's syndrome—A disorder caused by excessive aldosterone secretion by a benign tumor of one of the adrenal glands. This results in malfunction of the body's salt and water balance and subsequently causes hypertension. Symptoms include thirst, muscle weakness, and excessive urination.

Renin stimulation test

The renin stimulation test is performed to help diagnose and distinguish the two forms of hyperaldosteronism. With the patient having been on a low-salt diet and lying down for the test, a blood sample for PRA is obtained. The PRA is repeated with the patient still on the low salt diet but now standing upright. In cases of primary hyperaldosteronism, the blood volume is greatly expanded, and a change in position or reduced salt intake does not result in decreased kidney blood flow or decreased blood sodium. As a result, renin levels do not increase. However, in secondary hyperaldosteronism, blood sodium levels decrease with a lowered salt intake, and when the patient is standing upright, the kidney blood flow decreases as well. Consequently, renin levels do increase.

Captopril test

The captopril test is a screening test for hypertension of kidney origin (**renovascular hypertension**). For this test, a baseline PRA test is done first, then the patient receives an oral dose of captopril, which is an angiotensin-converting enzyme (ACE) inhibitor. Blood pressure measurements are taken at this time and again at 60 minutes when another PRA test is done. Patients with kidney-based hypertension demonstrate greater falls in blood pressure and increases in PRA after captopril administration

than do those with essential hypertension. Consequently, the captopril test is an excellent screening procedure to determine the need for a more invasive radiographic evaluation such as renal arteriography.

Preparation

This test requires a blood sample. For the PRA, the patient should maintain a normal diet with a restricted amount of sodium (approximately 3 g per day) for three days before the test. It is recommended that the patient be **fasting** (nothing to eat or drink) from midnight the day of the test.

Risks

Risks for this test are minimal, but may include slight bleeding from the puncture site, fainting or feeling lightheaded after venipuncture, or hematoma (blood accumulating under the puncture site).

Normal results

Reference values for the PRA test are laboratory-specific and depend upon the kind of diet (sodium restricted or normal), the age of the patient, and the patient's posture at the time of the test. Values are also affected if renin has been stimulated or if the patient has received an ACE inhibitor, like captopril.

Abnormal results

Increased PRA levels are seen in essential hypertension (uncommon), malignant hypertension, and kidney-based (renovascular) hypertension. Renin-producing renal tumors, while rare, can also cause elevated levels, as can **cirrhosis**, low blood volume due to hemorrhage, and diminished adrenal function (Addison's disease). Decreased renin levels may indicate increased blood volume due to a high-sodium diet, salt-retaining steroids, primary aldosteronism, licorice ingestion syndrome, or essential hypertension with low renin levels.

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Janis O. Flores

Plasmapheresis

Definition

Plasmapheresis is a blood purification procedure used to treat several autoimmune diseases. It is also known as therapeutic plasma exchange.

Purpose

In an autoimmune disease, the immune system attacks the body's own tissues. In many autoimmune diseases, the chief weapons of attack are antibodies, proteins that circulate in the bloodstream until they meet and bind with the target tissue. Once bound, they impair the functions of the target, and signal other immune components to respond as well.

Plasmapheresis is used to remove antibodies from the bloodstream, thereby preventing them from attacking their targets. It does not directly affect the immune system's ability to make more antibodies, and therefore may only offer temporary benefit. This procedure is most useful in acute, self-limited disorders such as **Guillain-Barré syndrome**, or when chronic disorders, such as **myasthenia gravis**, become more severe in symptoms. In these instances, a rapid improvement could save the patient's life. Neurologic diseases comprise 90% of the diseases that could profit from plasmapheresis.

Precautions

Patients with clotting disorders may not be suitable candidates for plasmapheresis.

Description

The basic procedure consists of removal of blood, separation of blood cells from plasma, and return of these blood cells to the body's circulation, diluted with fresh plasma or a substitute. Because of concerns over viral infection and allergic reaction, fresh plasma is not routinely used. Instead, the most common substitute is saline solution with sterilized human albumin protein. During the course of a single session, two to three liters of plasma is removed and replaced.

Plasmapheresis requires insertion of a venous catheter, either in a limb or central vein. Central veins allow higher flow rates and are more convenient for repeat procedures, but are more often the site of complications, especially bacterial infection.

When blood is outside the body, it must be treated to prevent it from clotting. While most of the

anticoagulating agent is removed from the blood during treatment, some is returned to the patient.

Three procedures are available:

- “Discontinuous flow centrifugation.” Only one venous catheter line is required. Approximately 300 ml of blood is removed at a time and centrifuged to separate plasma from blood cells.
- “Continuous flow centrifugation.” Two venous lines are used. This method requires slightly less blood volume to be out of the body at any one time.
- “Plasma filtration.” Two venous lines are used. The plasma is filtered using standard hemodialysis equipment. It requires less than 100 ml of blood to be outside the body at one time.

A single plasmapheresis session may be effective, although it is more common to have several sessions per week over the course of two weeks or more.

Preparation

Good **nutrition** and plenty of rest make the procedure less stressful. The treating physician determines which of the patient's medications should be discontinued before the plasmapheresis session.

Aftercare

The patient may experience **dizziness**, **nausea**, numbness, tingling, or lightheadedness during or after the procedure. These effects usually pass quickly, allowing the patient to return to normal activities the same day.

Risks

Reinfusion (replacement) with human plasma may cause **anaphylaxis**, a life threatening allergic reaction. All procedures may cause a mild allergic reaction, leading to **fever**, chills, and rash. Bacterial infection is a risk, especially when a central venous catheter is used. Reaction to the citrate anticoagulant used may cause cramps and **numbness**, though these usually resolve on their own. Patients with impaired kidney function may require drug treatment for the effects of citrate metabolism.

Plasma contains clotting agents, chemicals that allow the blood to coagulate into a solid clot. Plasma exchange removes these. Bleeding complications are rare following plasmapheresis, but may require replacement of clotting factors.

KEY TERMS

Anaphylaxis—Also called anaphylactic shock, it is a severe allergic reaction to a foreign substance that the patient has had contact with. Penicillin is an example of a substance that causes severe allergic reactions for some people.

Antibody—Chemicals produced by the body to defend it against bacteria, viruses, or other cells foreign to the body (antigens). Each specific antibody reacts against a specific foreign body. Antibodies are also termed immunoglobulins.

Autoimmune—Autoimmune refers to the body's development of intolerance of the antigens on its own cells.

Hemodialysis—A method to take out unwanted parts of the blood. The patient's blood is run through a catheter and tubing into a machine called a dialyzer, which filters out the unwanted blood component.

Plasma—Plasma makes up 50% of human blood. It is a watery fluid that carries red cells, white cells, and platelets throughout the body.

Normal results

Plasmapheresis is an effective temporary treatment for:

- Guillain-Barré syndrome (an acute neurological disorder following a viral infection that produces progressive muscle weakness and **paralysis**)
- Myasthenia gravis (an autoimmune disease that causes muscle weakness)
- chronic inflammatory demyelinating polyneuropathy (a chronic neurological disorder caused by destruction of the myelin sheath of peripheral nerves, which produces symptoms similar to Guillain-Barré syndrome)
- thrombotic thrombocytopenic purpura (a rare blood disorder)
- Paraproteinemic peripheral neuropathies (a neurological disorder affecting the peripheral nerves)
- blood that is too thick (hyperviscosity)

Other conditions may respond to plasmapheresis as well. Beneficial effects are usually seen within several days. Effects commonly last up to several months, although longer-lasting changes are possible, presumably by inducing shifts in immune response.

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Richard Robinson

Plasmodium infection see **Malaria**

Plastic, cosmetic, and reconstructive surgery

Definition

Plastic, cosmetic, and reconstructive surgery refers to a variety of operations performed in order to repair or restore body parts to look normal, or to change a body part to look better. These types of surgery are highly specialized. They are characterized by careful preparation of the patient's skin and tissues, by precise cutting and suturing techniques, and by care taken to minimize scarring. Recent advances in the development of miniaturized instruments, new materials for artificial limbs and body parts, and improved surgical techniques have expanded the range of plastic surgery operations that can be performed.

Purpose

Although these three types of surgery share some common techniques and approaches, they have somewhat different emphases. Plastic surgery is usually performed to treat **birth defects** and to remove skin blemishes such as **warts**, **acne scars**, or **birthmarks**. Cosmetic surgery procedures are performed to make the patient look younger or enhance his or her appearance in other ways. Reconstructive surgery is used to reattach body parts severed in combat or accidents, to perform skin grafts after severe **burns**, or to reconstruct parts of the patient's body that were missing at birth or removed by surgery. Reconstructive surgery is the oldest form of plastic surgery, having developed out of the need to treat wounded soldiers in wartime.

Precautions

Medical

Some patients should not have plastic surgery because of certain medical risks. These groups include:



A patient undergoing abdominoplasty. (Photo Researchers, Inc. Reproduced by permission)

- patients recovering from a **heart attack**, severe infection (for example, **pneumonia**), or other serious illness
- patients with infectious hepatitis or HIV infection
- **cancer** patients whose cancer might spread (metastasize)
- patients who are extremely overweight. Patients who are more than 30% overweight should not have **liposuction**
- patients with blood clotting disorders

Psychological

Plastic, cosmetic, and reconstructive surgeries have an important psychological dimension because of the high value placed on outward appearance in Western society. Many people who are born with visible deformities or disfigured by accidents later in life develop emotional problems related to social rejection. Other people work in fields such as acting, modeling, media journalism, and even politics, where their employment depends on how they look. Some people have unrealistic expectations of cosmetic surgery and

think that it will solve all their life problems. It is important for anyone considering nonemergency plastic or cosmetic surgery to be realistic about its results. One type of psychiatric disorder, called **body dysmorphic disorder**, is characterized by an excessive preoccupation with imaginary or minor flaws in appearance. Patients with this disorder frequently seek unnecessary plastic surgery.

Description

Plastic surgery

Plastic surgery includes a number of different procedures that usually involve skin. Operations to remove excess fat from the abdomen (“tummy tucks”), dermabrasion to remove acne scars or **tattoos**, and reshaping the cartilage in children’s ears (otoplasty) are common applications of plastic surgery.

Cosmetic surgery

Most cosmetic surgery is done on the face. It is intended either to correct disfigurement or to enhance the patient’s features. The most common cosmetic

The Top 10 Elective Cosmetic Surgeries In The U.S. (1999)

Procedure	Female Patients	Male Patients	Total
Liposuction	201,083	29,782	230,865
Breast augmentation	167,318	0	167,318
Eyelid surgery	120,160	21,859	142,033
Face lift	66,096	6,697	72,793
Tummy tuck	52,888	2,089	54,977
Collagen injections	48,989	4,208	53,197
Chemical peel	47,359	4,215	51,589
Laser skin resurfacing	46,162	4,343	50,505
Rhinoplasty	34,761	11,831	46,596
Forehead lift	36,995	3,962	40,969

procedure for children is correction of a **cleft lip** or palate. In adults, the most common procedures are remodeling of the nose (rhinoplasty), removal of baggy skin around the eyelids (**blepharoplasty**), face-lifts (rhytidectomy), or changing the size of the breasts (mammoplasty). Although many people still think of cosmetic surgery as only for women, growing numbers of men are choosing to have facelifts and eyelid surgery, as well as hair transplants and “tummy tucks.”

Reconstructive surgery

Reconstructive surgery is often performed on cancer patients as well as on burn and accident victims. It may involve the rebuilding of severely fractured bones, as well as skin grafting. Reconstructive surgery includes such procedures as the reattachment of an amputated finger or toe, or implanting a prosthesis. Prostheses are artificial structures and materials that are used to replace missing limbs or teeth, or arthritic hip and knee joints.

In cancer patients, reconstructive surgery is done to restore the function as well as the appearance of the face and other parts of the body. The most commonly performed reconstructive surgeries of cancer patients are **breast reconstruction**, **laceration repair**, scar revision, and **tumor removal**.

The most challenging area of reconstructive surgery involves the structures of the face, neck, and jaw because trauma or cancer treatments often affect the patient's ability to see, eat, taste, swallow, speak, and hear as well as his or her external appearance. The surgeon must try to retain as much sensation as possible when performing skin or bone grafts in the head and neck as well as recreate a reasonably normal appearance.

Preparation

Preparation for nonemergency plastic or reconstructive surgery includes patient education, as well as medical considerations. Some operations, such as nose reshaping or the removal of warts, small birthmarks, and tattoos can be done as outpatient procedures under local anesthesia. Most plastic and reconstructive surgery, however, involves a stay in the hospital and general anesthesia. Patients are typically asked to stop taking certain prescription medications that affect bleeding for about two weeks before the procedure and to stop **smoking** for at least a week before the operation.

Preparation for reconstructive surgery following cancer treatment may require much more extensive counseling and discussion of possible alternatives to the surgery.

Medical preparation

Preparation for plastic surgery includes the surgeon's detailed assessment of the parts of the patient's body that will be involved. Skin grafts require evaluating suitable areas of the patient's skin for the right color and texture to match the skin at the graft site. Facelifts and cosmetic surgery in the eye area require very close attention to the texture of the skin and the placement of surgical cuts (incisions).

Patients scheduled for plastic surgery under **general anesthesia** will be given a **physical examination**, blood and urine tests, and other tests to make sure that they do not have any previously undetected health problems or blood clotting disorders. The doctor will check the list of prescription medications that the patient may be taking to make sure that none of them will interfere with normal blood clotting or interact with the anesthetic.

Patients are asked to avoid using **aspirin** or medications containing aspirin for a week to two weeks before surgery, because these drugs lengthen the time of blood clotting. Smokers are asked to stop smoking two weeks before surgery because smoking interferes with the healing process. For some types of plastic surgery, the patient may be asked to donate several units of his or her own blood before the procedure, in case a **transfusion** is needed during the operation. The patient will be asked to sign a consent form before the operation.

Patient education

The doctor will meet with the patient before the operation is scheduled, in order to explain the

procedure and to be sure that the patient is realistic about the expected results. This consideration is particularly important if the patient is having cosmetic surgery.

Aftercare

Medical

Medical aftercare following plastic surgery under general anesthesia includes bringing the patient to a recovery room, monitoring his or her vital signs, and giving medications to relieve pain as necessary. Patients who have had fat removed from the abdomen may be kept in bed for as long as two weeks. Patients who have had mammoplasties, breast reconstruction, and some types of facial surgery typically remain in the hospital for a week after the operation. Patients who have had liposuction or eyelid surgery are usually sent home in a day or two.

Patients who have had outpatient procedures are usually given **antibiotics** to prevent infection and are sent home as soon as their vital signs are normal.

Psychological

Some patients may need follow-up psychotherapy or counseling after plastic or reconstructive surgery. These patients typically include children whose schooling and social relationships have been affected by birth defects, as well as patients of any age whose deformities or disfigurements were caused by trauma from accidents, war injuries, or violent crime.

Risks

The risks associated with plastic, cosmetic, and reconstructive surgery include the postoperative complications that can occur with any surgical operation under anesthesia. These complications include wound infection, internal bleeding, pneumonia, and reactions to the anesthesia.

In addition to these general risks, plastic, cosmetic, and reconstructive surgery carry specific risks:

- formation of undesirable scar tissue
- persistent **pain**, redness, or swelling in the area of the surgery
- infection inside the body related to inserting a prosthesis. These infections can result from contamination at the time of surgery or from bacteria migrating into the area around the prosthesis at a later time
- anemia or fat embolisms from liposuction
- rejection of skin grafts or tissue transplants

KEY TERMS

Blepharoplasty—Surgical reshaping of the eyelid.

Dermabrasion—A technique for removing the upper layers of skin with planing wheels powered by compressed air.

Facelift—Plastic surgery performed to remove sagging skin and wrinkles from the patient's face.

Liposuction—A surgical technique for removing fat from under the skin by vacuum suctioning.

Mammoplasty—Surgery performed to change the size of breasts.

Rhinoplasty—Surgery performed to change the shape of the nose.

- loss of normal feeling or function in the area of the operation (for example, it is not unusual for women who have had mammoplasties to lose sensation in their nipples)
- complications resulting from unforeseen technological problems (the best-known example of this problem was the discovery in the mid-1990s that breast implants made with silicone gel could leak into the patient's body).

Normal results

Normal results include the patient's recovery from the surgery with satisfactory results and without complications.

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ORGANIZATIONS

- American Academy of Facial Plastic and Reconstructive Surgery (AAFPRS). 310 South Henry Street, Alexandria, VA 22314. (703) 299-9291. <<http://www.facemd.org>>.
- American Medical Association. 515 N. State St., Chicago, IL 60612. (312) 464-5000. <<http://www.ama-assn.org>>.
- American Society of Plastic Surgeons (ASPS). 444 East Algonquin Road, Arlington Heights, IL 60005. (847) 228-9900. <www.plasticsurgery.org>.

Rebecca J. Frey, PhD

Platelet aggregation test

Definition

Platelets are disk-shaped blood cells that are also called thrombocytes. They play a major role in the blood-clotting process. The platelet aggregation test is a measure of platelet function.

Purpose

The platelet aggregation test aids in the evaluation of bleeding disorders by measuring the rate and degree to which platelets form a clump (aggregate) after the addition of a chemical that stimulates clumping (aggregation).

Precautions

There are many medications that can affect the results of the platelet aggregation test. The patient should discontinue as many as possible beforehand. Some of the drugs that can decrease platelet aggregation include **aspirin**, some **antibiotics**, **beta blockers**, dextran (Macrodex), alcohol, heparin (Lipo-Hepin), nonsteroidal anti-inflammatory drugs (NSAIDs), **tricyclic antidepressants**, and warfarin (Coumadin).

Description

There are many factors involved in blood clotting (coagulation). One of the first steps in the process

involves small cells in the bloodstream called platelets, which are produced in the bone marrow. Platelets gather at the site of an injury and clump together to form a plug, or aggregate, that helps to limit the loss of blood and promote healing.

Inherited bleeding disorders (e.g., **hemophilia** or von Willebrand's disease) and acquired bleeding problems that occur because of another disorder or a medication can affect the number of platelets and their level of function. When these problems are present, the result is a drop in platelet aggregation and a lengthened bleeding time.

The platelet aggregation test uses a machine called an aggregometer to measure the cloudiness (turbidity) of blood plasma. Several different substances called agonists are used in the test. These agonists include adenosine diphosphate, epinephrine, thrombin, collagen, and ristocetin. The addition of an agonist to a plasma sample causes the platelets to clump together, making the fluid more transparent. The aggregometer then measures the increased light transmission through the specimen.

Preparation

The test requires a blood sample. The patient should either avoid food and drink altogether for eight hours before the test, or eat only nonfat foods. High levels of fatty substances in the blood can affect test results.

Because the use of aspirin and/or aspirin compounds can directly affect test results, the patient should avoid these medications for two weeks before the test. If the patient must take aspirin and the test cannot be postponed, the laboratory should be notified and asked to verify the presence of aspirin in the blood plasma. If the results are abnormal, aspirin use must be discontinued and the test repeated in two weeks.

Aftercare

Because the platelet aggregation test is ordered when some type of bleeding problem is suspected, the patient should be cautioned to watch the puncture site for signs of additional bleeding.

Risks

Risks for this test are minimal in normal individuals. Patients with bleeding disorders, however, may have prolonged bleeding from the puncture wound or the formation of a bruise (hematoma) under the skin where the blood was withdrawn.

KEY TERMS

Aggregation—The blood cell clumping process that is measured in the platelet aggregation test.

Agonist—A chemical that is added to the blood sample in the platelet aggregation test to stimulate the clumping process.

Hemophilia—An inherited bleeding disorder caused by a deficiency of factor VIII, one of a series of blood proteins essential for blood clotting.

Platelets—Small, round, disk-shaped blood cells that are involved in clot formation. The platelet aggregation test measures the clumping ability of platelets.

Turbidity—The cloudiness or lack of transparency of a solution.

von Willebrand's disease—An inherited lifelong bleeding disorder caused by an abnormal gene, similar to hemophilia. The gene defect results in a decreased blood concentration of a substance called von Willebrand's factor (vWF).

Normal results

The normal time for platelet aggregation varies somewhat depending on the laboratory, the temperature, the shape of the vial in which the test is performed, and the patient's response to different agonists. For example, the difference between the response to ristocetin and other products should be noted because ristocetin triggers aggregation through a different mechanism than other agonists.

Abnormal results

Prolonged platelet aggregation time can be found in such congenital disorders as hemophilia and von Willebrand's disease, as well as in some connective tissue disorders. Prolonged aggregation times can also occur in leukemia or myeloma; after recent heart/lung bypass or **kidney dialysis**; and after taking certain drugs.

Resources

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Janis O. Flores

Platelet count

Definition

A platelet count is a diagnostic test that determines the number of platelets in the patient's blood. Platelets, which are also called thrombocytes, are small disk-shaped blood cells produced in the bone marrow and involved in the process of blood clotting. There are normally between 150,000-450,000 platelets in each microliter of blood. Low platelet counts or abnormally shaped platelets are associated with bleeding disorders. High platelet counts sometimes indicate disorders of the bone marrow.

Purpose

The primary functions of a platelet count are to assist in the diagnosis of bleeding disorders and to monitor patients who are being treated for any disease involving bone marrow failure. Patients who have leukemia, **polycythemia vera**, or **aplastic anemia** are given periodic platelet count tests to monitor their health.

Description

Blood collection and storage

Platelet counts use a freshly-collected blood specimen to which a chemical called EDTA has been added to prevent clotting before the test begins. About 5 mL of blood are drawn from a vein in the patient's inner elbow region. Blood drawn from a vein helps to produce a more accurate count than blood drawn from a fingertip. Collection of the sample takes only a few minutes.

After collection, the mean platelet volume of EDTA-blood will increase over time. This increase is caused by a change in the shape of the platelets after removal from the body. The changing volume is relatively stable for a period of one to three hours after collection. This period is the best time to count the sample when using electronic instruments, because the platelets will be within a standard size range.

Counting methods

Platelets can be observed in a direct blood smear for approximate quantity and shape. A direct smear is made by placing a drop of blood onto a microscope slide and spreading it into a thin layer. After staining to make the various blood cells easier to see and distinguish, a laboratory technician views the smear

through a light microscope. Accurate assessment of the number of platelets requires other methods of counting. There are three methods used to count platelets; hemacytometer, voltage-pulse counting, and electro-optical counting.

HEMACYTOMETER COUNTING. The microscopic method uses a phase contrast microscope to view blood on a hemacytometer slide. A sample of the diluted blood mixture is placed in a hemacytometer, which is an instrument with a grid etched into its surface to guide the counting. For a proper count, the platelets should be evenly distributed in the hemacytometer. Counts made from samples with platelet clumping are considered unreliable. Clumping can be caused by several factors, such as clotting before addition of the anticoagulant and allowing the blood to remain in contact with a capillary blood vessel during collection. Errors in platelet counting are more common when blood is collected from capillaries than from veins.

ELECTRONIC COUNTING. Electronic counting of platelets is the most common method. There are two types of electronic counting, voltage-pulse and electro-optical counting systems. In both systems, the collected blood is diluted and counted by passing the blood through an electronic counter. The instruments are set to count only particles within the proper size range for platelets. The upper and lower levels of the size range are called size exclusion limits. Any cells or material larger or smaller than the size exclusion limits will not be counted. Any object in the proper size range is counted, however, even if it isn't a platelet. For these instruments to work properly, the sample must not contain other material that might mistakenly be counted as platelets. Electronic counting instruments sometimes produce artificially low platelet counts. If a platelet and another blood cell pass through the counter at the same time, the instrument will not count the larger cell because of the size exclusion limits, which will cause the instrument to accidentally miss the platelet. Clumps of platelets will not be counted because clumps exceed the upper size exclusion limit for platelets. In addition, if the patient has a high white blood cell count, electronic counting may yield an unusually low platelet count because white blood cells may filter out some of the platelets before the sample is counted. On the other hand, if the red blood cells in the sample have burst, their fragments will be falsely counted as platelets.

Aftercare

Because platelet counts are sometimes ordered to diagnose or monitor bleeding disorders, patients with

KEY TERMS

Capillaries—The smallest of the blood vessels that bring oxygenated blood to tissues.

EDTA—A colorless compound used to keep blood samples from clotting before tests are run. Its chemical name is ethylene-diamine-tetra-acetic acid.

Hemacytometer—An instrument used to count platelets or other blood cells.

Phase contrast microscope—A light microscope in which light is focused on the sample at an angle to produce a clearer image.

Thrombocyte—Another name for platelet.

Thrombocytopenia—An abnormally low platelet count.

Thrombocytosis—An abnormally high platelet count. It occurs in polycythemia vera and other disorders in which the bone marrow produces too many platelets.

these disorders should be cautioned to watch the puncture site for signs of additional bleeding.

Risks

Risks for a platelet count test are minimal in normal individuals. Patients with bleeding disorders, however, may have prolonged bleeding from the puncture wound or the formation of a bruise (hematoma) under the skin where the blood was withdrawn.

Normal results

The normal range for a platelet count is 150,000-450,000 platelets per microliter of blood.

Abnormal results

An abnormally low platelet level (**thrombocytopenia**) is a condition that may result from increased destruction of platelets, decreased production, or increased usage of platelets. In **idiopathic thrombocytopenic purpura** (ITP), platelets are destroyed at abnormally high rates. **Hypersplenism** is characterized by the collection (sequestration) of platelets in the spleen. Disseminated intravascular coagulation (DIC) is a condition in which **blood clots** occur within blood vessels in a number of tissues. All of these diseases produce reduced platelet counts.

Abnormally high platelet levels (**thrombocytosis**) may indicate either a benign reaction to an infection, surgery, or certain medications; or a disease like polycythemia vera, in which the bone marrow produces too many platelets too quickly.

Resources

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Berketow, Robert, et al., editors. *Merck Manual of Medical Information*. Whitehouse Station, NJ: Merck Research Laboratories, 2004.

John T. Lohr, PhD

Platelet function disorders

Definition

Platelets are elements within the bloodstream that recognize and cling to damaged areas inside blood vessels. When they do this, the platelets trigger a series of chemical changes that result in the formation of a blood clot. There are certain hereditary disorders that affect platelet function and impair their ability to start the process of blood clot formation. One result is the possibility of excessive bleeding from minor injuries or menstrual flow.

Description

Platelets are formed in the bone marrow—a spongy tissue located inside the long bones of the body—as fragments of a large precursor cell (a megakaryocyte). These fragments circulate in the bloodstream and form the first line of defense against blood escaping from injured blood vessels.

Damaged blood vessels release a chemical signal that increases the stickiness of platelets in the area of the injury. The sticky platelets adhere to the damaged area and gradually form a platelet plug. At the same time, the platelets release a series of chemical signals that prompt other factors in the blood to reinforce the platelet plug. Between the platelet and its reinforcements, a sturdy clot is created that acts as a patch while the damaged area heals.

There are several hereditary disorders characterized by some impairment of the platelet's action. Examples include von Willebrand's disease, Glanzmann's thrombasthenia, and **Wiskott-Aldrich syndrome**. Vulnerable aspects of platelet function

include errors in the production of the platelets themselves or errors in the formation, storage, or release of their chemical signals. These defects can prevent platelets from responding to injuries or from prompting the action of other factors involved in clot formation.

Causes and symptoms

Platelet function disorders can be inherited, but they may also occur as a symptom of acquired diseases or as a side effect of certain drugs, including **aspirin**. Common symptoms of platelet function disorders include bleeding from the nose, mouth, vagina, or anus; pinpoint **bruises** and purplish patches on the skin; and abnormally heavy menstrual bleeding.

Diagnosis

In diagnosing platelet function disorders, specific tests are needed to determine whether the problem is caused by low numbers of platelets or impaired platelet function. A blood **platelet count** and **bleeding time** are common screening tests. If these tests confirm that the symptoms are due to impaired platelet function, further tests are done—such as platelet aggregation or an analysis of the platelet proteins—that pinpoint the exact nature of the defect.

Treatment

Treatment is intended to prevent bleeding and stop it quickly when it occurs. For example, patients are advised to be careful when they brush their teeth to reduce damage to the gums. They are also warned against taking medications that interfere with platelet function. Some patients may require iron and folate supplements to counteract potential anemia. Platelet transfusions may be necessary to prevent life-threatening hemorrhaging in some cases. Bone marrow transplantation can cure certain disorders but also carries some serious risks. Hormone therapy is useful in treating heavy menstrual bleeding. Von Willebrand's disease can be treated with desmopressin (DDAVP, Stimate).

Prognosis

The outcome depends on the specific disorder and the severity of its symptoms. Platelet function disorders range from life-threatening conditions to easily treated or little-noticed problems.

Prevention

Inherited platelet function disorders cannot be prevented except by **genetic counseling**; however,

KEY TERMS

Anemia—A condition in which inadequate quantities of hemoglobin and red blood cells are produced.

Bone marrow—A spongy tissue located within the body's flat bones—including the hip and breast bones and the skull. Marrow contains stem cells, the precursors to platelets and red and white blood cells.

Hemoglobin—The substance inside red blood cells that enables them to carry oxygen.

Megakaryocyte—A large bone marrow cell with a lobed nucleus that is the precursor cell of blood platelets.

Platelets—Fragments of a large precursor cell (a megakaryocyte) found in the bone marrow. These fragments adhere to areas of blood vessel damage and release chemical signals that direct the formation of a blood clot.

some acquired function disorders may be guarded against by avoiding substances that trigger the disorder.

Resources

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Julia Barrett

Pleural biopsy

Definition

The pleura is the membrane that lines the lungs and chest cavity. A pleural biopsy is the removal of pleural tissue for examination.

Purpose

Pleural biopsy is done to differentiate between benign and malignant disease, to diagnose viral, fungal, or parasitic diseases, and to identify a condition called collagen vascular disease of the pleura. It is also

ordered when a **chest x ray** indicates a pleural-based tumor, reaction, or thickening of the lining.

Precautions

Because pleural biopsy is an invasive procedure, it is not recommended for patients with severe bleeding disorders.

Description

Pleural biopsy is usually ordered when pleural fluid obtained by another procedure called thoracentesis (aspiration of pleural fluid) suggests infection, signs of **cancer**, or **tuberculosis**. Pleural biopsies are 85–90% accurate in diagnosing these diseases.

The procedure most often performed for pleural biopsy is called a percutaneous (passage through the skin by needle puncture) needle biopsy. The procedure takes 30–45 minutes, although the biopsy needle itself remains in the pleura for less than one minute. This type of biopsy is usually performed by a physician at bedside, if the patient is hospitalized, or in the doctor's office under local anesthetic.

The actual procedure begins with the patient in a sitting position, shoulders and arms elevated and supported. The skin overlying the biopsy site is anesthetized and a small incision is made to allow insertion of the biopsy needle. This needle is inserted with a cannula (a plastic or metal tube) until fluid is removed. Then the inner needle is removed and a trocar (an instrument for withdrawing fluid from a cavity) is inserted to obtain the actual biopsy specimen. As many as three separate specimens are taken from different sites during the procedure. These specimens are then placed into a fixative solution and sent to the laboratory for tissue (histologic) examination.

Preparation

Preparations for this procedure vary, depending on the type of procedure requested. Pleural biopsy can be performed in several ways: percutaneous needle biopsy (described above), by thoracoscopy (insertion of a visual device called a laparoscope into the pleural space for inspection), or by open pleural biopsy, which requires general anesthesia.

Aftercare

Potential complications of this procedure include bleeding or injury to the lung, or a condition called **pneumothorax**, in which air enters the pleural cavity (the space between the two layers of pleura lining the

lungs and the chest wall). Because of these possibilities, the patient is to report any shortness of breath, and to note any signs of bleeding, decreased blood pressure, or increased pulse rate.

Risks

Risks for this procedure include respiratory distress on the side of the biopsy, as well as bleeding, possible shoulder **pain**, pneumothorax (immediate) or pneumonia (delayed).

Normal results

Normal findings indicate no evidence of any pathologic or disease conditions.

Abnormal results

Abnormal findings include tumors called neoplasms (any new or abnormal growth) that can be either benign or malignant. Pleural tumors are divided into two classifications: primary (mesothelioma), or metastatic (arising from cancer sites elsewhere in the body). These tumors are often associated with an accumulation of fluid between the pleural layers called a **pleural effusion**, which itself may be caused by **pneumonia**, **heart failure**, cancer, or blood clot in the lungs (pulmonary embolism).

Other causes of abnormal findings include viral, fungal, or parasitic infections, and tuberculosis.

Resources

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Pagana, Kathleen Deska. *Mosby's Manual of Diagnostic and Laboratory Tests*. St. Louis: Mosby, Inc., 1998.

Janis O. Flores

Pleural effusion

Definition

Pleural effusion occurs when too much fluid collects in the pleural space (the space between the two layers of the pleura). It is commonly known as “water on the lungs.” It is characterized by **shortness of breath**, chest **pain**, gastric discomfort (**dyspepsia**), and cough.

Description

There are two thin membranes in the chest, one (the visceral pleura) lining the lungs, and the other (the parietal pleura) covering the inside of the chest wall. Normally, small blood vessels in the pleural linings produce a small amount of fluid that lubricates the opposed pleural membranes so that they can glide smoothly against one another during breathing movements. Any extra fluid is taken up by blood and lymph vessels, maintaining a balance. When either too much fluid forms or something prevents its removal, the result is an excess of pleural fluid—an effusion. The most common causes are disease of the heart or lungs, and inflammation or infection of the pleura.

Pleural effusion itself is not a disease as much as a result of many different diseases. For this reason, there is no “typical” patient in terms of age, sex, or other characteristics. Instead, anyone who develops one of the many conditions that can produce an effusion may be affected.

There are two types of pleural effusion: the transudate and the exudate. This is a very important point because the two types of fluid are very different, and which type is present points to what sort of disease is likely to have produced the effusion. It also can suggest the best approach to treatment.

Transudates

A transudate is a clear fluid, similar to blood serum, that forms not because the pleural surfaces themselves are diseased, but because the forces that normally produce and remove pleural fluid at the same rate are out of balance. When the heart fails, pressure in the small blood vessels that remove pleural fluid is increased and fluid “backs up” in the pleural space, forming an effusion. Or, if too little protein is present in the blood, the vessels are less able to hold the fluid part of blood within them and it leaks out into the pleural space. This can result from disease of the liver or kidneys, or from **malnutrition**.

Exudates

An exudate—which often is a cloudy fluid, containing cells and much protein—results from disease of the pleura itself. The causes are many and varied. Among the most common are infections such as bacterial **pneumonia** and **tuberculosis**; **blood clots** in the lungs; and connective tissue diseases, such as rheumatoid arthritis. **Cancer** and disease in organs such as the pancreas also may give rise to an exudative pleural effusion.

Special types of pleural effusion

Some of the pleural disorders that produce an exudate also cause bleeding into the pleural space. If the effusion contains half or more of the number of red blood cells present in the blood itself, it is called hemothorax. When a pleural effusion has a milky appearance and contains a large amount of fat, it is called chylothorax. Lymph fluid that drains from tissues throughout the body into small lymph vessels finally collects in a large duct (the thoracic duct) running through the chest to empty into a major vein. When this fluid, or chyle, leaks out of the duct into the pleural space, chylothorax is the result. Cancer in the chest is a common cause.

Causes and symptoms

Causes of transudative pleural effusion

Among the most important specific causes of a transudative pleural effusion are:

- Congestive **heart failure**. This causes pleural effusions in about 40% of patients and is often present on both sides of the chest. Heart failure is the most common cause of bilateral (two-sided) effusion. When only one side is affected it usually is the right (because patients usually lie on their right side).
- Pericarditis. This is an inflammation of the pericardium, the membrane covering the heart.
- Too much fluid in the body tissues, which spills over into the pleural space. This is seen in some forms of **kidney disease**; when patients have bowel disease and absorb too little of what they eat; and when an excessive amount of fluid is given intravenously.
- Liver disease. About 5% of patients with a chronic scarring disease of the liver called cirrhosis develop pleural effusion.

Causes of exudative pleural effusions

A wide range of conditions may be the cause of an exudative pleural effusion:

- Pleural tumors account for up to 40% of one-sided pleural effusions. They may arise in the pleura itself (**mesothelioma**), or from other sites, notably the lung.
- Tuberculosis in the lungs may produce a long-lasting exudative pleural effusion.
- Pneumonia affects about three million persons each year, and four of every ten patients will develop pleural effusion. If effective treatment is not provided, an extensive effusion can form that is very difficult to treat.

- Patients with any of a wide range of infections by a virus, fungus, or parasite that involve the lungs may have pleural effusion.
- Up to half of all patients who develop blood clots in their lungs (**pulmonary embolism**) will have pleural effusion, and this sometimes is the only sign of **embolism**.
- Connective tissue diseases, including **rheumatoid arthritis**, lupus, and **Sjögren's syndrome** may be complicated by pleural effusion.
- Patients with disease of the liver or pancreas may have an exudative effusion, and the same is true for any patient who undergoes extensive abdominal surgery. About 30% of patients who undergo heart surgery will develop an effusion.
- Injury to the chest may produce pleural effusion in the form of either hemothorax or chylothorax.

Symptoms

The key symptom of a pleural effusion is shortness of breath. Fluid filling the pleural space makes it hard for the lungs to fully expand, causing the patient to take many breaths so as to get enough oxygen. When the parietal pleura is irritated, the patient may have mild pain that quickly passes or, sometimes, a sharp, stabbing pleuritic type of pain. Some patients will have a dry cough. Occasionally a patient will have no symptoms at all. This is more likely when the effusion results from recent abdominal surgery, cancer, or tuberculosis. Tapping on the chest will show that the usual crisp sounds have become dull, and on listening with a stethoscope the normal breath sounds are muted. If the pleura is inflamed, there may be a scratchy sound called a "pleural friction rub."

Diagnosis

When pleural effusion is suspected, the best way to confirm it is to take chest x rays, both straight-on and from the side. The fluid itself can be seen at the bottom of the lung or lungs, hiding the normal lung structure. If heart failure is present, the x-ray shadow of the heart will be enlarged. An ultrasound scan may disclose a small effusion that caused no abnormal findings during chest examination. A computed tomography scan is very helpful if the lungs themselves are diseased.

In order to learn what has caused the effusion, a needle or catheter is often used to obtain a fluid sample, which is examined for cells and its chemical make-up. This procedure, called a thoracentesis, is the way to determine whether an effusion is a transudate or

exudate, giving a clue as to the underlying cause. In some cases—for instance when cancer or bacterial infection is present—the specific cause can be determined and the correct treatment planned. Culturing a fluid sample can identify the bacteria that cause tuberculosis or other forms of pleural infection. The next diagnostic step is to take a tissue sample, or **pleural biopsy**, and examine it under a microscope. If the effusion is caused by lung disease, placing a viewing tube (bronchoscope) through the large air passages will allow the examiner to see the abnormal appearance of the lungs.

Treatment

The best way to clear up a pleural effusion is to direct treatment at what is causing it, rather than treating the effusion itself. If heart failure is reversed or a lung infection is cured by antibiotics, the effusion will usually resolve. However, if the cause is not known, even after extensive tests, or no effective treatment is at hand, the fluid can be drained away by placing a large-bore needle or catheter into the pleural space, just as in diagnostic **thoracentesis**. If necessary, this can be repeated as often as is needed to control the amount of fluid in the pleural space. If large effusions continue to recur, a drug or material that irritates the pleural membranes can be injected to deliberately inflame them and cause them to adhere close together—a process called sclerosis. This will prevent further effusion by eliminating the pleural space. In the most severe cases, open surgery with removal of a rib may be necessary to drain all the fluid and close the pleural space.

Prognosis

When the cause of pleural effusion can be determined and effectively treated, the effusion itself will reliably clear up and should not recur. In many other cases, sclerosis will prevent sizable effusions from recurring. Whenever a large effusion causes a patient to be short of breath, thoracentesis will make breathing easier, and it may be repeated if necessary. To a great extent, the outlook for patients with pleural effusion depends on the primary cause of effusion and whether it can be eliminated. Some forms of pleural effusion, such as that seen after abdominal surgery, are only temporary and will clear without specific treatment. If heart failure can be controlled, the patient will remain free of pleural effusion. If, on the other hand, effusion is caused by cancer that cannot be controlled, other effects of the disease probably will become more important.

KEY TERMS

Culture—A test that exposes a sample of body fluid or tissue to special material to see whether bacteria or another type of microorganism is present.

Dyspepsia—A vague feeling of being too full and having heartburn, bloating, and nausea. Usually felt after eating.

Exudate—The type of pleural effusion that results from inflammation or other disease of the pleura itself. It features cloudy fluid containing cells and proteins.

Pleura or pleurae—A delicate membrane that encloses the lungs. The pleura is divided into two areas separated by fluid—the visceral pleura, which covers the lungs, and the parietal pleura, which lines the chest wall and covers the diaphragm.

Pleural cavity—The area of the thorax that contains the lungs.

Pleural space—The potential area between the visceral and parietal layers of the pleurae.

Pneumonia—An acute inflammation of the lungs, usually caused by bacterial infection.

Sclerosis—The process by which an irritating material is placed in the pleural space in order to inflame the pleural membranes and cause them to stick together, eliminating the pleural space and recurrent effusions.

Thoracentesis—Placing a needle, tube, or catheter in the pleural space to remove the fluid of pleural effusion. Used for both diagnosis and treatment.

Transudate—The type of pleural effusion seen with heart failure or other disorders of the circulation. It features clear fluid containing few cells and little protein.

Prevention

Because pleural effusion is a secondary effect of many different conditions, the key to preventing it is to promptly diagnose the primary disease and provide effective treatment. Timely treatment of infections such as tuberculosis and pneumonia will prevent many effusions. When effusion occurs as a drug side-effect, withdrawing the drug or using a different one may solve the problem. On rare occasions, an effusion occurs because fluid meant for a vein is mistakenly injected into the pleural space. This can be prevented by making sure that proper technique is used.

Resources

ORGANIZATIONS

American Lung Association. 1740 Broadway, New York, NY 10019. (800) 586-4872. <<http://www.lungusa.org>>.

National Heart, Lung and Blood Institute. P.O. Box 30105, Bethesda, MD 20824-0105. (301) 251-1222. <<http://www.nhlbi.nih.gov>>.

David A. Cramer, MD

Pleural fluid analysis see **Thoracentesis**

Pleurisy

Definition

Pleurisy is an inflammation of the membrane that surrounds and protects the lungs (the pleura). Inflammation occurs when an infection or damaging agent irritates the pleural surface. As a consequence, sharp chest pains are the primary symptom of pleurisy.

Description

Pleurisy, also called pleuritis, is a condition that generally stems from an existing respiratory infection, disease, or injury. In people who have otherwise good health, respiratory infections or **pneumonia** are the main causes of pleurisy. This condition used to be more common, but with the advent of **antibiotics** and modern disease therapies, pleurisy has become less prevalent.

The pleura is a double-layered structure made up of an inner membrane, which surrounds the lungs, and an outer membrane, which lines the chest cavity. The pleural membranes are very thin, close together, and have a fluid coating in the narrow space between them. This liquid acts as a lubricant, so that when the lungs inflate and deflate during breathing, the pleural surfaces can easily glide over one another.

Pleurisy occurs when the pleural surfaces rub against one another, due to irritation and inflammation. Infection within the pleural space is the most common irritant, although the abnormal presence of air, blood, or cells can also initiate pleurisy. These disturbances all act to displace the normal pleural fluid, which forces the membranes to rub, rather than glide, against one another. This rubbing irritates nerve endings in the outer membrane and causes **pain**. Pleurisy

also causes a chest noise that ranges from a faint squeak to a loud creak. This characteristic sound is called a “friction rub.”

Pleurisy cases are classified either as having pleural effusion or as being “dry.” **Pleural effusion** is more common and refers to an accumulation of fluid within the pleural space; dry pleurisy is inflammation without fluid build-up. Less pain occurs with pleural effusion because the fluid forces the membrane surfaces apart. However, pleural effusion causes additional complications because it places pressure on the lungs. This leads to respiratory distress and possible lung collapse.

Causes and symptoms

A variety of conditions can give rise to pleurisy. The following list represents the most common sources of pleural inflammation.

- infections, including pneumonia, **tuberculosis**, and other bacterial or viral respiratory infections
- immune disorders, including systemic lupus erythematosus, rheumatoid arthritis, and **sarcoidosis**
- diseases, including **cancer**, **pancreatitis**, liver **cirrhosis**, and heart or kidney failure
- injury, from a rib fracture, collapsed lung, esophagus rupture, blood clot, or material such as asbestos
- drug reactions, from certain drugs used to treat tuberculosis (isoniazid), cancer (methotrexate, procarbazine), or the immune disorders mentioned above (hydralazine, procainamide, phenytoin, quinidine)

Symptomatic pain

The hallmark symptom of pleurisy is sudden, intense chest pain that is usually located over the area of inflammation. Although the pain can be constant, it is usually most severe when the lungs move during breathing, coughing, sneezing, or even talking. The pain is usually described as shooting or stabbing, but in minor cases it resembles a mild cramp. When pleurisy occurs in certain locations, such as near the diaphragm, the pain may be felt in other areas such as the neck, shoulder, or abdomen (referred pain). Another indication of pleurisy is that holding one's breath or exerting pressure against the chest causes pain relief.

Breathing difficulties

Pleurisy is also characterized by certain respiratory symptoms. In response to the pain, pleurisy

patients commonly have a rapid, shallow breathing pattern. Pleural effusion can also cause **shortness of breath**, as excess fluid makes expanding the lungs difficult. If severe breathing difficulties persist, patients may experience a blue colored complexion (**cyanosis**).

Additional symptoms of pleurisy are specific to the illness that triggers the condition. Thus, if infection is the cause, then chills, **fever**, and **fatigue** will be likely pleurisy symptoms.

Diagnosis

The distinctive pain of pleurisy is normally the first clue physicians use for diagnosis. Doctors usually feel the chest to find the most painful area, which is the likely site of inflammation. A stethoscope is also used to listen for abnormal chest sounds as the patient breathes. If the doctor hears the characteristic friction rub, the diagnosis of pleurisy can be confirmed. Sometimes, a friction rub is masked by the presence of pleural effusion and further examination is needed for an accurate diagnosis.

Identifying the actual illness that causes pleurisy is more difficult. To make this diagnosis, doctors must evaluate the patient's history, additional symptoms, and laboratory test results. A chest x ray may also be taken to look for signs of accumulated fluid and other abnormalities. Possible causes, such as pneumonia, fractured ribs, esophagus rupture, and lung tumors may be detected on an x ray. Computed tomography scan (CT scan) and ultrasound scans are more powerful diagnostic tools used to visualize the chest cavity. Images from these techniques more clearly pinpoint the location of excess fluid or other suspected problems.

The most helpful information in diagnosing the cause of pleurisy is a fluid analysis. Once the doctor knows the precise location of fluid accumulation, a sample is removed using a procedure called **thoracentesis**. In this technique, a fine needle is inserted into the chest to reach the pleural space and extract fluid. The fluid's appearance and composition is thoroughly examined to help doctors understand how the fluid was produced. Several laboratory tests are performed to analyze the chemical components of the fluid. These tests also determine whether infection-causing bacteria or viruses are present. In addition, cells within the fluid are identified and counted. Cancerous cells can also be detected to learn whether the pleurisy is caused by a malignancy.

In certain instances, such as dry pleurisy, or when a fluid analysis is not informative, a biopsy of the

pleura may be needed for microscopic analysis. A sample of pleural tissue can be obtained several ways: with a biopsy needle, by making a small incision in the chest wall, or by using a thoracoscope (a video-assisted instrument for viewing the pleural space and collecting samples).

Treatment

Pain management

The pain of pleurisy is usually treated with analgesic and anti-inflammatory drugs, such as acetaminophen, ibuprofen, and indomethacin. People with pleurisy may also receive relief from lying on the painful side. Sometimes, a painful **cough** will be controlled with codeine-based cough syrups. However, as the pain eases, a person with pleurisy should try to breathe deeply and cough to clear any congestion, otherwise pneumonia may occur. Rest is also important to aid in the recovery process.

Treating the source

The treatment used to cure pleurisy is ultimately defined by the underlying cause. Thus, pleurisy from a bacterial infection can be successfully treated with antibiotics, while no treatment is given for viral infections that must run their course. Specific therapies designed for more chronic illnesses can often cause pleurisy to subside. For example, tuberculosis pleurisy is treated with standard anti-tuberculosis drugs. With some illnesses, excess fluid continues to accumulate and causes severe respiratory distress. In these individuals, the fluid may be removed by thoracentesis, or the doctor may insert a chest tube to drain large amounts. If left untreated, a more serious infection may develop within the fluid, called **empyema**.

Alternative treatment

Alternative treatments can be used in conjunction with conventional treatment to help heal pleurisy. **Acupuncture** and botanical medicines are alternative approaches for alleviating pleural pain and breathing problems. An herbal remedy commonly recommended is pleurisy root (*Asclepias tuberosa*), so named because of its use by early American settlers who learned of this medicinal plant from Native Americans. Pleurisy root helps to ease pain, inflammation, and breathing difficulties brought on by pleurisy. This herb is often used in conjunction with mullein (*Verbascum thapsus*) or elecampane (*Inula helenium*), which serve as **expectorants** to clear excess mucus from the lungs. In addition, there are many

KEY TERMS

Effusion—The accumulation of fluid within a cavity, such as the pleural space.

Empyema—An infection that causes pus to accumulate in the pleural space. The pus may cause a tear in the pleural membrane, which allows the infection to spread to other areas in the body. Intravenous antibiotics are often given to control the infection.

Inflammation—An accumulation of fluid and cells within tissue that is often caused by infection and the immune response that occurs as a result.

Pneumonia—A condition caused by bacterial or viral infection that is characterized by inflammation of the lungs and fluid within the air passages. Pneumonia is often an underlying cause of pleurisy.

Referred pain—The presence of pain in an area other than where it originates. In some pleurisy cases, referred pain occurs in the neck, shoulder, or abdomen.

other respiratory herbs that are used as expectorants or for other actions on the respiratory system. Herbs thought to combat infection, such as **echinacea** (*Echinacea* spp.) are also included in herbal pleurisy remedies. Antiviral herbs, such as *Lomatium dissectum* and *Ligusticum porteri*, can be used if the pleurisy is of viral origin. **Traditional Chinese medicine** uses the herb ephedra (*Ephedra sinica*), which acts to open air passages and alleviate respiratory difficulties in pleurisy patients. Dietary recommendations include eating fresh fruits and vegetables, adequate protein, and good quality fats (omega-3 fatty acids are anti-inflammatory and are found in fish and flax oil). Taking certain nutritional supplements, especially large doses of vitamin C, may also provide health benefits to people with pleurisy. Contrast **hydrotherapy** applied to the chest and back, along with compresses (cloths soaked in an herbal solution) or poultices (crushed herbs applied directly to the skin) of respiratory herbs, can assist in the healing process. Homeopathic treatment, guided by a trained practitioner, can be effective in resolving pleurisy.

Prognosis

Prompt diagnosis, followed by appropriate treatment, ensures a good recovery for most pleurisy

patients. Generally speaking, the prognosis for pleurisy is linked to the seriousness of its cause. Therefore, the outcome of pleurisy caused by a disease such as cancer will vary depending on the type and location of the tumor.

Prevention

Preventing pleurisy is often a matter of providing early medical attention to conditions that can cause pleural inflammation. Along this line, appropriate antibiotic treatment of bacterial respiratory infections may successfully prevent some cases of pleurisy. Maintaining a healthy lifestyle and avoiding exposure to harmful substances (for example, asbestos) are more general preventative measures.

Resources

ORGANIZATIONS

American Lung Association. 1740 Broadway, New York, NY 10019. (800) 586-4872. <<http://www.lungusa.org>>.

National Heart, Lung and Blood Institute. P.O. Box 30105, Bethesda, MD 20824-0105. (301) 251-1222. <<http://www.nhlbi.nih.gov>>.

Julie A. Gelderloos

Pleuritis see **Pleurisy**

Plumbism see **Lead poisoning**

PMS see **Premenstrual syndrome**

Pneumococcal pneumonia

Definition

Pneumococcal **pneumonia** is a common but serious infection and inflammation of the lungs. It is caused by the bacterium *Streptococcus pneumoniae*.

Description

The gram-positive, spherical bacteria, *Streptococcus pneumoniae*, is the cause of many human diseases, including pneumonia. Although the bacteria can normally be found in the nose and throat of healthy individuals, it can grow and cause infection when the immune system is weakened. Infection usually begins with the upper respiratory tract and then travels into the lungs. Pneumonia occurs when the bacteria find their way deep into the lungs, to the

area called the alveoli, or air sacs. This is the functional part of the lungs where oxygen is absorbed into the blood. Once in the alveoli, *Streptococcus pneumoniae* begin to grow and multiply. White blood cells and immune proteins from the blood also accumulate at the site of infection in the alveoli. As the alveoli fill with these substances and fluid, they can no longer function in the exchange of oxygen. This fluid filling of the lungs is how pneumonia is defined.

Those people most at risk of developing pneumococcal pneumonia have a weakened immune system. This includes the elderly, infants, **cancer** patients, AIDS patients, post-operative patients, alcoholics, and those with diabetes. Pneumococcal pneumonia is a disease that has a high rate of hospital transmission, putting hospital patients at greater risk. Prior lung infections also makes someone more likely to develop pneumococcal pneumonia. The disease can be most severe in patients who have had their spleen removed. It is the spleen that is responsible for removing the bacteria from the blood. Cases of pneumonia, which is spread by close contact, seem to occur most often between November through April. If not treated, the disease can spread, causing continually decreasing lung function, heart problems, and arthritis.

Causes and symptoms

Symptoms of bacterial pneumonia include a **cough**, sputum (mucus) production that may be pus-like or bloody, shaking and chills, **fever**, and chest **pain**. Symptoms often have an abrupt beginning and occur after an upper respiratory infection such as a cold. Symptoms may differ somewhat in the elderly, with minimal cough, no sputum and no fever, but rather tiredness and confusion leading to **hypothermia** and **shock**.

Diagnosis

The presence of symptoms and a physical exam that reveals abnormal lung sounds usually suggest the presence of pneumonia. Diagnosis is typically made from an x ray of the lungs, which indicates the accumulation of fluid. Additional tests that may be done include a complete blood count, a sputum sample for microscopic examination and culture for *Streptococcus pneumoniae*, and possibly blood cultures.

Treatment

Depending on the severity of the disease, **antibiotics** are given either at home or in the hospital. Historically, the treatment for pneumococcal

pneumonia has been penicillin. An increasing number of cases of pneumococcal pneumonia have become partially or completely resistant to penicillin, making it less effective in treating this disease. Other effective antibiotics include amoxicillin and erythromycin. If these antibiotics are not effective, vancomycin or cephalosporin may alternatively be used.

Symptoms associated with pneumococcal pneumonia can also be treated. For instance, fever can be treated with **aspirin** or **acetaminophen**. Supplemental oxygen and intravenous fluids may help. Patients are advised to get plenty of rest and take increased amounts of fluids. Coughing should be promoted because it helps to clear the lungs of fluid.

Alternative treatment

Being a serious, sometimes fatal disease, pneumococcal pneumonia is best treated as soon as possible with antibiotics. However, there are alternative treatments that both support this conventional treatment and prevent recurrences. Maintaining a healthy immune system is important. One way to do this is by taking the herb, **echinacea** (*Echinacea* spp.). Getting plenty of rest and reducing **stress** can help the body heal. Some practitioners feel that mucus-producing foods (including dairy products, eggs, gluten-rich grains such as wheat, oats, rye, as well as sugar) can contribute to the lung congestion that accompanies pneumonia. Decreasing these foods and increasing the amount of fresh fruits and vegetables may help to decrease lung congestion. Adequate protein in the diet is also essential for the body to produce antibodies. Contrast and constitutional **hydrotherapy** can be very helpful in treating cases of pneumonia. Other alternative therapies, including acupuncture, Chinese herbal medicine, and homeopathy, can be very useful during the recovery phase, helping the body to rebuild after the illness and contributing to the prevention of recurrences.

Prognosis

Simple, uncomplicated cases of pneumococcal pneumonia will begin to respond to antibiotics in 48 to 72 hours. Full recovery from pneumonia, however, is greatly dependent on the age and overall health of the individual. Normally, healthy and younger patients can recover in only a few days, while the elderly or otherwise weakened individuals may not recover for several weeks. Complications may develop which give a poorer prognosis. Even when promptly and properly diagnosed, such weakened patients may die of their pneumonia.

KEY TERMS

Acetaminophen—A drug used for pain relief as well as to decrease fever. A common trade name for the drug is Tylenol.

Aspirin—A commonly used drug for pain relief and to decrease fever.

Bronchi—Two main branches of the trachea that go into the lungs. This then further divides into the bronchioles and alveoli.

Sputum—A substance that comes up from the throat when coughing or clearing the throat. It is important since it contains materials from the lungs.

Prevention

Vaccination

Recently, a **vaccination** has become available for the prevention of pneumococcal pneumonia. This vaccination is generally recommended for people with a high likelihood of developing pneumococcal infection or for those in whom a serious complication of infection is likely to develop. This would include persons over the age of 65, as well as those with:

- chronic pulmonary disease
- advanced cardiovascular disease
- diabetes mellitus
- alcoholism
- cirrhosis
- chronic kidney disease
- spleen dysfunction, or removal of spleen
- immunosuppression (cancer, organ transplant or AIDS)
- sickle cell anemia

Unfortunately, those people for whom the vaccination is most recommended are also those who are least likely to respond favorably to a vaccination. Therefore, the overall effectiveness of this vaccine remains questionable.

Antibiotics

The use of oral penicillin to prevent infection may be recommended for some patients at high risk, such as children with **sickle cell disease** and those with a spleen removed. This treatment, however, must be

weighed with the increased likelihood of developing penicillin-resistant infections.

Resources

ORGANIZATIONS

American Lung Association. 1740 Broadway, New York, NY 10019. (800) 586-4872. <<http://www.lungusa.org>>.

Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

Cindy L. A. Jones, PhD

Pneumocystis pneumonia

Definition

Pneumocystis **pneumonia** is a lung infection that occurs primarily in people with weakened immune systems—especially people who are HIV-positive. The disease agent is an organism whose biological classification is still uncertain. *Pneumocystis carinii* was originally thought to be a one-celled organism (a protozoan), but more recent research suggests that it is a fungus. Although its life cycle is known to have three stages, its method of reproduction is not yet completely understood. The complete name of the disease is *Pneumocystis carinii* pneumonia, often shortened to PCP. PCP is also sometimes called pneumocystosis.

Description

Pneumonia as a general term refers to a severe lung inflammation. In pneumocystis pneumonia, this inflammation is caused by the growth of *Pneumocystis carinii*, a fungus-like organism that is widespread in the environment. PCP is ordinarily a rare disease, affecting only people with weakened immune systems. Many of these people are patients receiving drugs for organ transplants or **cancer** treatment. With the rising incidence of **AIDS**, however, PCP has become primarily associated with AIDS patients. In fact, as many as 75% of AIDS patients have developed PCP. It has also been the leading cause of **death** in AIDS patients.

Transmission

The organism that causes PCP is widely distributed in nature and is transmitted through the air. When the organism is inhaled, it enters the upper respiratory tract and infects the tiny air sacs at the ends of the

smaller air tubes (bronchioles) in the lungs. These tiny air sacs are called alveoli. Under a microscope, alveoli look like groups of hollow spheres resembling grape clusters. The exchange of oxygen with the blood takes place in the alveoli. It appears that *P. carinii* lives in the fluid in the lining of the alveoli.

Person-to-person infection does not appear to be very common; however, clusters of PCP outbreaks in hospitals and groups of immunocompromised people indicate that patients with active PCP should not be exposed to others with weakened immune systems. It is thought that many people actually acquire mild *Pneumocystis carinii* infections from time to time, but are protected by their immune systems from developing a full-blown case of the disease.

Causes and symptoms

Causes

P. carinii is an opportunistic organism. This means that it causes disease only under certain conditions, as when a person is immunocompromised. Under these circumstances, *P. carinii* can multiply and cause pneumonia. The mechanisms of the organism's growth within the alveoli are not fully understood. As the pneumocystis organism continues to replicate, it gradually fills the alveoli. As the pneumonia becomes more severe, fluid accumulates and tissue scarring occurs. These changes result in decreased respiratory function and lower levels of oxygen in the blood.

High-risk groups

Some patients are at greater risk of developing PCP. These high-risk groups include:

- premature infants
- patients with **immunodeficiency** diseases, including **severe combined immunodeficiency disease (SCID)** and acquired immunodeficiency syndrome (AIDS),
- patients receiving immunosuppressive drugs, especially cortisone-like drugs (**corticosteroids**)
- Patients with protein malnutrition.

AIDS is currently the most common risk factor for PCP in the United States. PCP is, however, also found in countries with widespread hunger and poor hygiene.

Symptoms

The incubation period of PCP is not definitely known, but is thought to be between four and eight

weeks. The major symptoms include **shortness of breath**, **fever**, and a nonproductive **cough**. Less common symptoms include production of sputum, blood in the sputum, difficulty breathing, and chest **pain**. Most patients will have symptoms for one to two weeks before seeing a physician. Occasionally, the disease will spread outside of the lung to other organs, including the lymph nodes, spleen, liver, or bone marrow.

Diagnosis

The diagnosis of PCP begins with a thorough **physical examination** and blood tests. Although imaging studies are helpful in identifying abnormal areas in the lungs, the diagnosis of PCP must be confirmed by microscopic identification of the organism in the lung. Samples may be taken from the patient's sputum, or may be obtained via **bronchoscopy** or **lung biopsy**. Because of the severity of the disease, many physicians will proceed to treat patients with symptoms of pneumocystis pneumonia if they belong to a high-risk group, without the formality of an actual diagnosis. The severity of PCP can be measured by x-ray studies and by determining the amount of oxygen and carbon dioxide present in the patient's blood.

Treatment

Treatment for PCP involves the use of **antibiotics**. These include trimethoprim-sulfamethoxazole (TMP-SMX, Bactrim, Septra) and pentamidine isethionate (Nebupent, Pentam 300). Both of these anti-microbial drugs are equally effective. AIDS patients are typically treated for 21 days, whereas non-AIDS patients are treated for 14 days. TMP-SMX may be highly toxic in AIDS patients, causing severe side effects that include fever, rash, decreased numbers of white blood cells and platelets, and hepatitis. Pentamidine also causes side effects in immunocompromised patients. These side effects include decreased blood pressure, irregular heart beats, the accumulation of nitrogenous waste products in the blood (azotemia), and electrolyte imbalances. Pentamidine can be given in aerosol form to minimize side effects. Alternative drugs can be used for patients experiencing these side effects.

P. carinii appears to be developing resistance to TMP-SMX. In addition, some patients are allergic to the standard antibiotics given for PCP. As a result, other antibiotics for the treatment of PCP are continually under investigation. Some drugs proven to be effective against *P. carinii* include

KEY TERMS

Alveoli—Small, hollow air sacs found in the lungs at the end of the smaller airways (bronchioles). Air exchange occurs in the alveoli.

Azotemia—The presence of excess nitrogenous wastes in the blood.

Biopsy—A procedure in which a piece of tissue is obtained for microscopic study.

Bronchoscopy—A procedure that uses a fiber-optic scope to view the airways in the lung.

Fungus—A member of a group of simple organisms related to yeasts and molds.

Pentamidine isoethionate—An antibiotic used to treat and prevent PCP.

Pneumocystosis—Another name for active PCP infection.

Protozoan—A microorganism belonging to the Protista, which includes the simplest one-celled organisms.

Sputum—A substance obtained from the lungs and bronchial tubes by clearing the throat or coughing. Sputum can be tested for evidence of PCP infection.

Trimethoprim-sulfamethoxazole (TMP-SMX)—An antibiotic used to treat and prevent PCP.

dapsone (DDS) with trimethoprim (Trimplex), clindamycin (Cleocin) with primaquine, as well as atovaquone (Mepron). Paradoxically, corticosteroids have been found to improve the ability of TMP-SMX or pentamidine to treat PCP. As a treatment of last resort, trimetrexate with leucovorin (Wellcovorin) can also be used.

Prognosis

If left untreated, PCP will cause breathing difficulties that will eventually cause death. The prognosis for this disease depends on the amount of damage to the patient's lungs prior to treatment. Prognosis is usually better at a facility that specializes in caring for AIDS patients. Antibiotic treatment of PCP is about 80% effective.

Prevention

Medications

For patients at serious risk for PCP infection, low doses of TMP-SMX, given daily or three times a week,

are effective in preventing PCP. The drug is, however, highly toxic. Researchers are currently evaluating the effectiveness and toxicity of aerosol pentamidine and dapsone in preventing PCP.

Lifestyle modifications

Patients who have previously had PCP often experience a recurrence. Healthy lifestyle choices, including exercising, eating well, and giving up **smoking** may keep the disease at bay.

Resources

ORGANIZATIONS

American Lung Association. 1740 Broadway, New York, NY 10019. (800) 586-4872. <<http://www.lungusa.org>>.

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Cindy L. A. Jones, PhD

Pneumonectomy see **Lung surgery**

Pneumonia

Definition

Pneumonia is an infection of the lung that can be caused by nearly any class of organism known to cause human infections. These include bacteria, amoebae, viruses, fungi, and parasites. In the United States, pneumonia is the sixth most common disease leading to **death**; 2 million Americans develop pneumonia each year, and 40,000–70,000 die from it. Pneumonia is also the most common fatal infection acquired by already hospitalized patients. In developing countries, pneumonia ties with **diarrhea** as the most common cause of death. Even in nonfatal cases, pneumonia is a significant economic burden on the health care system. One study estimates that people in the American workforce who develop pneumonia cost employers five times as much in health care as the average worker.

According to the Centers for Disease Control and Prevention (CDC), however, the number of deaths from pneumonia in the United States has declined slightly since 2001.

Description

Anatomy of the lung

To better understand pneumonia, it is important to understand the basic anatomic features of the respiratory system. The human respiratory system begins at the nose and mouth, where air is breathed in (inspired) and out (expired). The air tube extending from the nose is called the nasopharynx. The tube carrying air breathed in through the mouth is called the oropharynx. The nasopharynx and the oropharynx merge into the larynx. The oropharynx also carries swallowed substances, including food, water, and salivary secretion, which must pass into the esophagus and then the stomach. The larynx is protected by a trap door called the epiglottis. The epiglottis prevents substances that have been swallowed, as well as substances that have been regurgitated (thrown up), from heading down into the larynx and toward the lungs.

A useful method of picturing the respiratory system is to imagine an upside-down tree. The larynx flows into the trachea, which is the tree trunk, and thus the broadest part of the respiratory tree. The trachea divides into two tree limbs, the right and left bronchi. Each one of these branches off into multiple smaller bronchi, which course through the tissue of the lung. Each bronchus divides into tubes of smaller and smaller diameter, finally ending in the terminal bronchioles. The air sacs of the lung, in which oxygen-carbon dioxide exchange actually takes place, are clustered at the ends of the bronchioles like the leaves of a tree. They are called alveoli.

The tissue of the lung which serves only a supportive role for the bronchi, bronchioles, and alveoli is called the lung stroma.

Function of the respiratory system

The main function of the respiratory system is to provide oxygen, the most important energy source for the body's cells. Inspired air (the air you breath in) contains the oxygen, and travels down the respiratory tree to the alveoli. The oxygen moves out of the alveoli and is sent into circulation throughout the body as part of the red blood cells. The oxygen in the inspired air is exchanged within the alveoli for the waste product of human metabolism, carbon dioxide. The air you breathe out contains the gas called carbon dioxide. This gas leaves the alveoli during expiration. To restate this exchange of gases simply, you breathe in oxygen, you breathe out carbon dioxide

Respiratory system defenses

The healthy human lung is sterile. There are no normally resident bacteria or viruses (unlike the upper respiratory system and parts of the gastrointestinal system, where bacteria dwell even in a healthy state). There are multiple safeguards along the path of the respiratory system. These are designed to keep invading organisms from leading to infection.

The first line of defense includes the hair in the nostrils, which serves as a filter for large particles. The epiglottis is a trap door of sorts, designed to prevent food and other swallowed substances from entering the larynx and then trachea. Sneezing and coughing, both provoked by the presence of irritants within the respiratory system, help to clear such irritants from the respiratory tract.

Mucus, produced through the respiratory system, also serves to trap dust and infectious organisms. Tiny hair like projections (cilia) from cells lining the respiratory tract beat constantly. They move debris trapped by mucus upwards and out of the respiratory tract. This mechanism of protection is referred to as the mucociliary escalator.

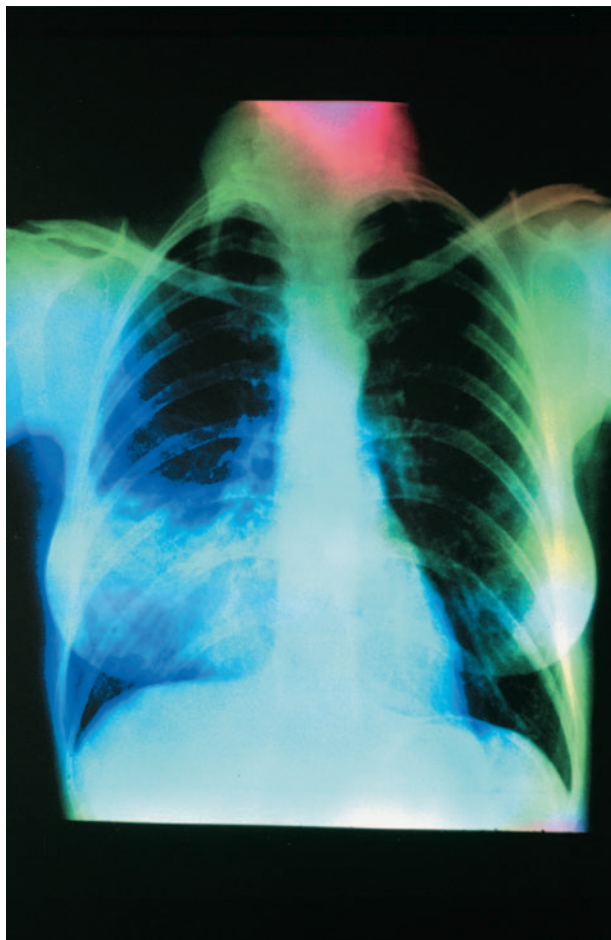
Cells lining the respiratory tract produce several types of immune substances which protect against various organisms. Other cells (called macrophages) along the respiratory tract actually ingest and kill invading organisms.

The organisms that cause pneumonia, then, are usually carefully kept from entering the lungs by virtue of these host defenses. However, when an individual encounters a large number of organisms at once, the usual defenses may be overwhelmed, and infection may occur. This can happen either by inhaling contaminated air droplets, or by aspiration of organisms inhabiting the upper airways.

Conditions predisposing to pneumonia

In addition to exposure to sufficient quantities of causative organisms, certain conditions may make an individual more likely to become ill with pneumonia. Certainly, the lack of normal anatomical structure could result in an increased risk of pneumonia. For example, there are certain inherited defects of cilia which result in less effective protection. Cigarette smoke, inhaled directly by a smoker or second-hand by a innocent bystander, interferes significantly with ciliary function, as well as inhibiting macrophage function.

Stroke, seizures, alcohol, and various drugs interfere with the function of the epiglottis. This leads to a



A chest x ray showing lobar pneumonia in the lower lobe of a patient's right lung. The alveoli (air sacs) of the lung become blocked with pus, which forces air out and causes the lung to become solidified. (Photo Researchers, Inc. Reproduced by permission.)

leaky seal on the trap door, with possible contamination by swallowed substances and/or regurgitated stomach contents. Alcohol and drugs also interfere with the normal **cough** reflex. This further decreases the chance of clearing unwanted debris from the respiratory tract.

Viruses may interfere with ciliary function, allowing themselves or other microorganism invaders (such as bacteria) access to the lower respiratory tract. One of the most important viruses is **HIV** (Human **Immunodeficiency** virus), the causative virus in **AIDS** (acquired immunodeficiency syndrome). In recent years this virus has resulted in a huge increase in the incidence of pneumonia. Because **AIDS** results in a general decreased effectiveness of many aspects of the host's immune system, a patient with **AIDS** is susceptible to all kinds of pneumonia. This includes some

previously rare parasitic types which would be unable to cause illness in an individual possessing a normal immune system.

The elderly have a less effective mucociliary escalator, as well as changes in their immune system. This causes this age group to be more at risk for the development of pneumonia.

Various chronic conditions predispose a person to infection with pneumonia. These include **asthma**, **cystic fibrosis**, and neuromuscular diseases which may interfere with the seal of the epiglottis. **Esophageal disorders** may result in stomach contents passing upwards into the esophagus. This increases the risk of aspiration into the lungs of those stomach contents with their resident bacteria. Diabetes, sickle cell anemia, lymphoma, leukemia, and **emphysema** also predispose a person to pneumonia.

Genetic factors also appear to be involved in susceptibility to pneumonia. Certain changes in DNA appear to affect some patients' risk of developing such complications of pneumonia as **septic shock**.

Pneumonia is also one of the most frequent infectious complications of all types of surgery. Many drugs used during and after surgery may increase the risk of aspiration, impair the cough reflex, and cause a patient to underfill their lungs with air. **Pain** after surgery also discourages a patient from breathing deeply enough, and from coughing effectively.

Radiation treatment for **breast cancer** increases the risk of pneumonia in some patients by weakening lung tissue.

Causes and symptoms

The list of organisms which can cause pneumonia is very large, and includes nearly every class of infecting organism: viruses, bacteria, bacteria-like organisms, fungi, and parasites (including certain worms). Different organisms are more frequently encountered by different age groups. Further, other characteristics of an individual may place him or her at greater risk for infection by particular types of organisms:

- Viruses cause the majority of pneumonias in young children (especially respiratory syncytial virus, parainfluenza and **influenza** viruses, and adenovirus).
- Adults are more frequently infected with bacteria (such as *Streptococcus pneumoniae*, *Haemophilus influenzae*, and *Staphylococcus aureus*).
- Pneumonia in older children and young adults is often caused by the bacteria-like *Mycoplasma*

pneumoniae (the cause of what is often referred to as “walking” pneumonia).

- *Pneumocystis carinii* is an extremely important cause of pneumonia in patients with immune problems (such as patients being treated for **cancer** with **chemotherapy**, or patients with AIDS). Classically considered a parasite, it appears to be more related to fungi.
- People who have reason to come into contact with bird droppings, such as poultry workers, are at risk for pneumonia caused by the organism *Chlamydia psittaci*.
- A very large, serious outbreak of pneumonia occurred in 1976, when many people attending an American Legion convention were infected by a previously unknown organism. Subsequently named *Legionella pneumophila*, it causes what is now called “Legionnaire’s disease.” The organism was traced to air conditioning units in the convention’s hotel.

Pneumonia is suspected in any patient who has **fever**, cough, chest pain, **shortness of breath**, and increased respirations (number of breaths per minute). Fever with a shaking chill is even more suspicious. Many patients cough up clumps of sputum, commonly known as spit. These secretions are produced in the alveoli during an infection or other inflammatory condition. They may appear streaked with pus or blood. Severe pneumonia results in the signs of oxygen deprivation. This includes blue appearance of the nail beds or lips (**cyanosis**).

The invading organism causes symptoms, in part, by provoking an overly-strong immune response in the lungs. In other words, the immune system, which should help fight off infections, kicks into such high gear, that it damages the lung tissue and makes it more susceptible to infection. The small blood vessels in the lungs (capillaries) become leaky, and protein-rich fluid seeps into the alveoli. This results in less functional area for oxygen-carbon dioxide exchange. The patient becomes relatively oxygen deprived, while retaining potentially damaging carbon dioxide. The patient breathes faster and faster, in an effort to bring in more oxygen and blow off more carbon dioxide.

Mucus production is increased, and the leaky capillaries may tinge the mucus with blood. Mucus plugs actually further decrease the efficiency of gas exchange in the lung. The alveoli fill further with fluid and debris from the large number of white blood cells being produced to fight the infection.

Consolidation, a feature of bacterial pneumonias, occurs when the alveoli, which are normally hollow

air spaces within the lung, instead become solid, due to quantities of fluid and debris.

Viral pneumonias and mycoplasma pneumonias do not result in consolidation. These types of pneumonia primarily infect the walls of the alveoli and the stroma of the lung.

Severe acute respiratory syndrome (SARS)

Severe acute respiratory syndrome, or **SARS**, is a contagious and potentially fatal disease that first appeared in the form of a multi-country outbreak in early February 2003. Later that month, the CDC began to work with the World Health Organization (WHO) to investigate the cause(s) of SARS and to develop guidelines for **infection control**. SARS has been described as an “atypical pneumonia of unknown etiology;” by the end of March 2003, the disease agent was identified as a previously unknown coronavirus.

The early symptoms of SARS include a high fever with chills, **headache**, **muscle cramps**, and weakness. This early phase is followed by respiratory symptoms, usually a dry cough and painful or difficult breathing. Some patients require mechanical ventilation. The mortality rate of SARS is thought to be about 3%.

As of the end of March 2003, the CDC did not have clearly defined recommendations for treating SARS. Treatments that have been used include **antibiotics** known to be effective against bacterial pneumonia; ribavirin and other **antiviral drugs**; and steroids.

Diagnosis

For the most part, diagnosis is based on the patient’s report of symptoms, combined with examination of the chest. Listening with a stethoscope will reveal abnormal sounds, and tapping on the patient’s back (which should yield a resonant sound due to air filling the alveoli) may instead yield a dull thump if the alveoli are filled with fluid and debris.

Laboratory diagnosis can be made of some bacterial pneumonias by staining sputum with special chemicals and looking at it under a microscope. Identification of the specific type of bacteria may require culturing the sputum (using the sputum sample to grow greater numbers of the bacteria in a lab dish.).

X-ray examination of the chest may reveal certain abnormal changes associated with pneumonia. Localized shadows obscuring areas of the lung may indicate a bacterial pneumonia, while streaky or

patchy appearing changes in the x-ray picture may indicate viral or mycoplasma pneumonia. These changes on x ray, however, are known to lag in time behind the patient's actual symptoms.

Treatment

Prior to the discovery of penicillin antibiotics, bacterial pneumonia was almost always fatal. Today, antibiotics, especially given early in the course of the disease, are very effective against bacterial causes of pneumonia. Erythromycin and tetracycline improve recovery time for symptoms of mycoplasma pneumonia. They, do not, however, eradicate the organisms. Amantadine and acyclovir may be helpful against certain viral pneumonias.

A newer antibiotic named linezolid (Zyvox) is being used to treat penicillin-resistant organisms that cause pneumonia. Linezolid is the first of a new line of antibiotics known as oxazolidinones. Another new drug known as ertapenem (Invanz) is reported to be effective in treating bacterial pneumonia.

Prognosis

Prognosis varies according to the type of organism causing the infection. Recovery following pneumonia with *Mycoplasma pneumoniae* is nearly 100%. *Staphylococcus pneumoniae* has a death rate of 30–40%. Similarly, infections with a number of gram negative bacteria (such as those in the gastrointestinal tract which can cause infection following aspiration) have a death rate of 25–50%. *Streptococcus pneumoniae*, the most common organism causing pneumonia, produces a death rate of about 5%. More complications occur in the very young or very old individuals who have multiple areas of the lung infected simultaneously. Individuals with other chronic illnesses (including **cirrhosis** of the liver, congestive **heart failure**, individuals without a functioning spleen, and individuals who have other diseases that result in a weakened immune system, experience complications. Patients with immune disorders, various types of cancer, transplant patients, and AIDS patients also experience complications.

Prevention

Because many bacterial pneumonias occur in patients who are first infected with the influenza virus (the flu), yearly **vaccination** against influenza can decrease the risk of pneumonia for certain patients. This is particularly true of the elderly and people with chronic diseases (such as asthma, cystic

KEY TERMS

Alveoli—The little air sacs clustered at the ends of the bronchioles, in which oxygen-carbon dioxide exchange takes place.

Aspiration—A situation in which solids or liquids which should be swallowed into the stomach are instead breathed into the respiratory system.

Cilia—Hair-like projections from certain types of cells.

Consolidation—A condition in which lung tissue becomes firm and solid rather than elastic and air-filled because it has accumulated fluids and tissue debris.

Coronavirus—One of a family of RNA-containing viruses known to cause severe respiratory illnesses. In March 2003, a previously unknown coronavirus was identified as the causative agent of severe acute respiratory syndrome, or SARS.

Cyanosis—A bluish tinge to the skin that can occur when the blood oxygen level drops too low.

Sputum—Material produced within the alveoli in response to an infectious or inflammatory process.

Stroma—A term used to describe the supportive tissue surrounding a particular structure. An example is that tissue which surrounds and supports the actually functional lung tissue.

fibrosis, other lung or heart diseases, **sickle cell disease**, diabetes, **kidney disease**, and forms of cancer).

A specific vaccine against *Streptococcus pneumoniae* is very protective, and should also be administered to patients with chronic illnesses.

Patients who have decreased immune resistance are at higher risk for infection with *Pneumocystis carinii*. They are frequently put on a regular drug regimen of trimethoprim sulfa and/or inhaled pentamidine to avoid **pneumocystis pneumonia**.

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ORGANIZATIONS

- American Lung Association. 1740 Broadway, New York, NY 10019. (800) 586-4872. <<http://www.lungusa.org>>.
- Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

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Pneumonitis see **Pneumonia**

Pneumothorax

Definition

Pneumothorax is a collection of air or gas in the chest or pleural space that causes part or all of a lung to collapse.

Description

Normally, the pressure in the lungs is greater than the pressure in the pleural space surrounding the lungs. However, if air enters the pleural space, the

pressure in the pleura then becomes greater than the pressure in the lungs, causing the lung to collapse partially or completely. Pneumothorax can be either spontaneous or due to trauma.

If a pneumothorax occurs suddenly or for no known reason, it is called a spontaneous pneumothorax. This condition most often strikes tall, thin men between the ages of 20 to 40. In addition, people with lung disorders, such as emphysema, cystic fibrosis, and **tuberculosis**, are at higher risk for spontaneous pneumothorax. Traumatic pneumothorax is the result of accident or injury due to medical procedures performed to the chest cavity, such as **thoracentesis** or mechanical ventilation. Tension pneumothorax is a serious and potentially life-threatening condition that may be caused by traumatic injury, chronic lung disease, or as a complication of a medical procedure. In this type of pneumothorax, air enters the chest cavity, but cannot escape. This greatly increased pressure in the pleural space causes the lung to collapse completely, compresses the heart, and pushes the heart and associated blood vessels toward the unaffected side.

Causes and symptoms

The symptoms of pneumothorax depend on how much air enters the chest, how much the lung collapses, and the extent of lung disease. Symptoms include the following, according to the cause of the pneumothorax:

- **Spontaneous pneumothorax.** Simple spontaneous pneumothorax is caused by a rupture of a small air sac or fluid-filled sac in the lung. It may be related to activity in otherwise healthy people or may occur during scuba diving or flying at high altitudes. Complicated spontaneous pneumothorax, also generally caused by rupture of a small sac in the lung, occurs in people with lung diseases. The symptoms of complicated spontaneous pneumothorax tend to be worse than those of simple pneumothorax, due to the underlying lung disease. Spontaneous pneumothorax is characterized by dull, sharp, or stabbing chest **pain** that begins suddenly and becomes worse with deep breathing or coughing. Other symptoms are **shortness of breath**, rapid breathing, abnormal breathing movement (that is, little chest wall movement when breathing), and cough.
- **Tension pneumothorax.** Following trauma, air may enter the chest cavity. A penetrating chest wound allows outside air to enter the chest, causing the lung to collapse. Certain medical procedures performed in the chest cavity, such as thoracentesis,



An x ray of a patient undergoing pneumothorax treatment. ECG electrodes attached to chest monitor heartbeat while endotracheal tube is inserted in windpipe. (Photo Researchers. Reproduced by permission.)

also may cause a lung to collapse. Tension pneumothorax may be the immediate result of an injury; the delayed complication of a hidden injury, such as a fractured rib, that punctures the lung; or the result of lung damage from asthma, chronic **bronchitis**, or emphysema. Symptoms of tension pneumothorax tend to be severe with sudden onset. There is marked **anxiety**, distended neck veins, weak pulse, decreased breath sounds on the affected side, and a shift of the mediastinum to the opposite side.

Diagnosis

To diagnose pneumothorax, it is necessary for the health care provider to listen to the chest (auscultation) during a **physical examination**. By using a stethoscope, the physician may note that one part of the chest does not transmit the normal sounds of breathing. A **chest x ray** will show the air pocket and the collapsed lung. An electrocardiogram (ECG) will be performed to record the electrical impulses that control the heart's activity. Blood samples may be taken to check for the level of arterial blood gases.

KEY TERMS

Electrocardiogram—A test that provides a typical record of normal heart action.

Mediastinum—The space between the right and left lung.

Pleural—Pleural refers to the pleura or membrane that enfolds the lungs.

Thoracentesis—Also called a pleural fluid tap, this procedure involves aspiration of fluid from the pleural space using a long, thin needle inserted between the ribs.

Treatment

A small pneumothorax may resolve on its own, but most require medical treatment. The object of treatment is to remove air from the chest and allow the lung to re-expand. This is done by inserting a needle and syringe (if the pneumothorax is small) or chest tube through the chest wall. This allows the air to escape without allowing any air back in. The lung will then re-expand itself within a few days. Surgery may be needed for repeat occurrences.

Prognosis

Most people recover fully from spontaneous pneumothorax. Up to half of patients with spontaneous pneumothorax experience recurrence. Recovery from a collapsed lung generally takes one to two weeks. Tension pneumothorax can cause **death** rapidly due to inadequate heart output or insufficient blood oxygen (hypoxemia), and must be treated as a medical emergency.

Prevention

Preventive measures for a non-injury related pneumothorax include stopping **smoking** and seeking medical attention for respiratory problems. If the pneumothorax occurs in both lungs or more than once in the same lung, surgery may be needed to prevent it from occurring again.

Resources

ORGANIZATIONS

American Association for Respiratory Care. 11030 Ables Lane, Dallas, Texas 75229. (972) 243-2272. <<http://www.aarc.org>>.

American Lung Association. 1740 Broadway, New York, NY 10019. (800) 586-4872. <<http://www.lungusa.org>>.

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Lorraine Steefel, RN

Podiatry see **Foot care**

Poison ivy and poison oak

Definition

Poison ivy and poison oak are plants that cause an allergic skin reaction in most people who are exposed to them.

Description

Poison ivy, which is generally thought of as a climbing vine, can also grow as a shrub or bush. It has leaves that are elliptical in shape and grow in groups of three on a stem. Poison ivy is common in the United States, except in the southwest, Alaska, and Hawaii. Poison oak, which grows as a shrub, has leaves that are shaped like oak leaves and also grow in groups of three to a stem. Poison oak is common in the United States, especially on the west coast from Mexico to Canada.

Not everyone is sensitive to poison ivy and poison oak; however, nine out of ten people who come in contact with either of the plants will have an allergic reaction to some degree. All parts of the plants are poisonous and the amount of time it takes for an allergic reaction to develop varies from person to person. The extent and severity of the reaction depends on the length of exposure, type of contact, and how sensitive the person is to the plants. If a person is going to have an allergic reaction, it will usually occur within one or two days of exposure. However, some people have a reaction within an hour, whereas others don't experience a reaction until five days after the exposure.

Causes and symptoms

The substance that causes the allergic reaction is the same for both plants. It is an oily resin called urushiol. It only takes a small amount of the resin to cause a reaction. The resin can be transferred to the skin by directly touching the plant or indirectly by coming in contact with something that has touched the plant, such as tools, animals, or clothing. Although

animals are rarely affected, they can carry the resin on their fur and transfer it to humans. According to the experts at the University of Maryland Medical Center, the “chemical [resin] can remain active for more than a year.”

The symptoms for poison ivy and poison oak are the same. Usually the first symptoms to appear are itchiness and swelling in the areas of contact. The itchy rash that follows is made up of small pimple-like bumps (sometimes referred to as papules), as well as blisters that later break open, ooze, and crust over.

Diagnosis

A diagnosis is made based on the symptoms and a **physical examination** of the patient.

In some cases, people have jobs that make it difficult for them to avoid poison ivy and poison oak, such as people who work in wooded areas or on construction sites, for example. Employees without health insurance may be covered by workers compensation.

Treatment

Anyone who comes in contact with either plant should wash the exposed area with soap and water immediately. Taking a bath immediately after contact is not recommended, because that could spread the resin to other areas of the body. All clothing, including shoes and shoelaces, should be removed carefully and either washed separately or discarded.

For minor cases, hydrocortisone cream and Calamine lotion can provide relief until the symptoms disappear. Over-the-counter Benadryl capsules help with the **itching**. Some people find oatmeal or baking soda baths to be soothing as well. Oral steroids, such as prednisone, are available for more serious cases, especially those affecting the face, eyes, mouth, or genitals. If signs of infection develop, such as pus and a **fever**, patients should contact their doctors.

Patients should consult their physicians before they use any ointments that contain benzocaine or zirconium, because they can cause an allergic reaction that worsens the condition. Antihistamine ointments are not recommended for the same reason. The experts at the Alabama Cooperative Extension System caution that “some people have severe allergic reactions to these plants and can have swelling in the throat, breathing problems, weakness, **dizziness**, and bluish lips.” Emergency medical care should be sought if any serious reactions occur.

Prognosis

In most cases, the condition goes away in two weeks.

Prevention

The best prevention is know what the plants look like and to avoid them. A common saying should be kept in mind: Leaves of three, let them be.

People who plan to be in an area where poison ivy and poison oak might be found should wear protective clothing, such as long-sleeved shirts and long pants.

Eradication of the plants should be handled with care. As stated by the experts at the Alabama Cooperative Extension System, "burning can be dangerous and is not recommended for disposal or as a control measure, because the toxic oil from the plant can be carried in the smoke." Instead they recommend spraying the plants with glyphosate, which is commonly known as the brands Roundup or Kleenup.

Resources

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Lee Ann Paradise

Poisoning

Definition

Poisoning occurs when any substance interferes with normal body functions after it is swallowed, inhaled, injected, or absorbed. The branch of medicine that deals with the detection and treatment of poisons is known as toxicology.

Description

Poisonings are a common occurrence. About 10 million cases of poisoning occur in the United States each year. In 80% of the cases, the victim is a child under the age of five. About 50 children die each year from poisonings. Curiosity, inability to read warning

labels, a desire to imitate adults, and inadequate supervision lead to childhood poisonings.

The elderly are the second most likely group to be poisoned. Mental confusion, poor eyesight, and the use of multiple drugs are the leading reasons why this group has a high rate of accidental poisoning. A substantial number of poisonings also occur as **suicide** attempts or drug overdoses.

Poisons are common in the home and workplace, yet there are basically two major types. One group consists of products that were never meant to be ingested or inhaled, such as shampoo, paint thinner, pesticides, houseplant leaves, and carbon monoxide. The other group contains products that can be ingested in small quantities, but which are harmful if taken in large amounts, such as pharmaceuticals, medicinal herbs, or alcohol. Other types of poisons include the bacterial toxins that cause **food poisoning**, such as *Escherichia coli*; heavy metals, such as the lead found in the paint on older houses; and the venom found in the bites and stings of some animals and insects. The staff at a poison control center and emergency room doctors have the most experience diagnosing and treating poisoning cases.

Causes and symptoms

The effects of poisons are as varied as the poisons themselves; however, the exact mechanisms of only a few are understood. Some poisons interfere with the metabolism. Others destroy the liver or kidneys, such as heavy metals and some pain relief medications, including **acetaminophen** (Tylenol) and nonsteroidal anti-inflammatory drugs (Advil, Ibuprofen). A poison may severely depress the central nervous system, leading to coma and eventual respiratory and circulatory failure. Potential poisons in this category include anesthetics (e.g. ether and chloroform), opiates (e.g., morphine and codeine), and **barbiturates**. Some poisons directly affect the respiratory and circulatory system. Carbon monoxide causes **death** by binding with hemoglobin that would normally transport oxygen throughout the body. Certain corrosive vapors trigger the body to flood the lungs with fluids, effectively drowning the person. Cyanide interferes with respiration at the cellular level. Another group of poisons interferes with the electrochemical impulses that travel between neurons in the nervous system. Yet another group, including cocaine, ergot, strychnine, and some snake venoms, causes potentially fatal seizures.

Severity of symptoms can range from **headache** and nausea to convulsions and death. The type of poison, the amount and time of exposure, and the

Common Household, Industrial, And Agricultural Products Containing Toxic Substances

Alcohol (rubbing)	Fuel/Antifreeze	Floor/furniture polish
Arsenic	Gasoline	Art and craft supplies
Glues/adhesives	Automotive fluids	Hemlock
Batteries, automotive	Kerosene	Batteries, household
Mercury	Building products	Metal primers
Cleaning products	Metalworking materials	Cosmetics/personal care items
Mothballs	Cyanide	Oven cleaners
Daffodil bulbs	Paint strippers/thinners	Dieffenbachia
Paints, oil-based or alkyds	Disinfectants/air fresheners	Paints, water-based or latex
Drain openers	Pesticides	Flea collars/insect repellent
English nightshade	Stains/finishes	Ethanol
Strychnine	Foxglove	Wood preservatives

age, size, and health of the victim are all factors which determine the severity of symptoms and the chances for recovery.

Plant poisoning

There are more than 700 species of poisonous plants in the United States. Plants are second only to medicines in causing serious poisoning in children under age five. There is no way to tell by looking at a plant if it is poisonous. Some plants, such as the yew shrub, are almost entirely toxic: needles, bark, seeds, and berries. In other plants, only certain parts are poisonous. The bulb of the hyacinth and daffodil are toxic, but the flowers are not; while the flowers of the jasmine plant are the poisonous part. Moreover, some plants are confusing because portions of them are eaten as food while other parts are poisonous. For example, the fleshy stem (tuber) of the potato plant is nutritious; however, its roots, sprouts, and vines are poisonous. The leaves of tomatoes are poisonous, while the fruit is not. Rhubarb stalks are good to eat, but the leaves are poisonous. Apricots, cherries, peaches, and apples all produce healthful fruit, but their seeds contain a form of cyanide that can kill a child if chewed in sufficient quantities. One hundred milligrams (mg) of moist, crushed apricot seeds can produce 217 mg of cyanide.

Common houseplants that contain some poisonous parts include:

- Aloe
- Amaryllis
- Cyclamen
- Dumb cane (also called Dieffenbachia)
- Philodendron

Common outdoor plants that contain some poisonous part include:

- Bird of paradise flower
- Buttercup
- Castor bean
- Chinaberry tree
- Daffodil
- English ivy
- Eucalyptus
- Foxglove
- Holly
- Horse chestnut
- Iris
- Jack-in-the-pulpit
- Jimsonweed (also called thornapple)
- Larkspur
- Lily-of-the-valley
- Morning glory
- Nightshade (several varieties)
- Oleander
- Potato
- Rhododendron
- Rhubarb
- Sweet pea
- Tomato
- Wisteria
- Yew

Symptoms of plant poisoning range from irritation of the skin or mucous membranes of the mouth and throat to **nausea, vomiting**, convulsions, irregular heartbeat, and even death. It is often difficult to tell if a person has eaten a poisonous plant because there are

no tell-tale empty containers and no unusual lesions or odors around the mouth.

Many cases of plant poisoning involve plants that contain hallucinogens, such as peyote cactus buttons, certain types of mushrooms, and **marijuana**. A recent case of plant poisoning in France concerned *Datura*, or moonflower, a plant that has become popular with young people trying to imitate Native American **puberty** rites.

Other cases of plant poisoning result from the use of herbal dietary supplements that have been contaminated by toxic substances. The Food and Drug Administration (FDA) has the authority to monitor herbal products on the market and issue warnings about accidental poisoning or other adverse affects associated with these products. For example, in 2002 a manufacturer of nettle capsules found to contain lead recalled the product following a warning from the FDA. Other dietary supplements have been found to contain small quantities of prescription medications or even toxic plants.

Household chemicals

Many products used daily in the home are poisonous if swallowed. These products often contain strong acids or strong bases (alkalis). Toxic household cleaning products include:

- ammonia
- bleach
- dishwashing liquids
- drain openers
- floor waxes and furniture polishes
- laundry detergents, spot cleaners, and fabric softeners
- mildew removers
- oven cleaners
- toilet bowl cleaners

Personal care products found in the home can also be poisonous. These include:

- deodorant
- hairspray
- hair straighteners
- nail polish and polish remover
- perfume
- shampoo

Signs that a person has swallowed one of these substances include evidence of an empty container

nearby, nausea or vomiting, and **burns** on the lips and skin around the mouth if the substance was a strong acid or alkali. The chemicals in some of these products may leave a distinctive odor on the breath.

Pharmaceuticals

Both over-the-counter and prescription medicines can help the body heal if taken as directed. However, when taken in large quantities, or with other drugs where there may be an adverse interaction, they can act as poisons. Drug overdoses, both accidental and intentional, are the leading cause of poisoning in adults. Medicinal herbs should be treated like pharmaceuticals and taken only in designated quantities under the supervision of a knowledgeable person. Herbs that have healing qualities when taken in small doses can be toxic in larger doses, or may interact with prescription medications in unpredictable ways.

Drug overdoses cause a range of symptoms, including excitability, sleepiness, confusion, unconsciousness, rapid heartbeat, convulsions, nausea, and changes in blood pressure. The best initial evidence of a **drug overdose** is the presence of an empty container near the victim.

Other causes of poisonings

People can be poisoned by fumes they inhale. Carbon monoxide is the most common form of inhaled poison. Other toxic substances that can be inhaled include:

- farm and garden insecticides and herbicides
- gasoline fumes
- insect repellent
- paint thinner fumes

Diagnosis

Initially, poisoning is suspected if the victim shows changes in behavior and signs or symptoms previously described. **Hallucinations** or other psychiatric symptoms may indicate poisoning by a hallucinogenic plant. Evidence of an empty container or information from the victim are helpful in determining exactly what substance has caused the poisoning. Some acids and alkalis leave burns on the mouth. Petroleum products, such as lighter fluid or kerosene, leave a distinctive odor on the breath. The vomit may be tested to determine the exact composition of the poison. Once hospitalized, the patient may be given blood and urine tests to determine his or her metabolic condition.

Treatment

Treatment for poisoning depends on the poison swallowed or inhaled. Contacting the poison control center or hospital emergency room is the first step in getting proper treatment. The poison control center's telephone number is often listed with emergency numbers on the inside cover of the telephone book, or it can be reached by dialing the operator. The poison control center will ask for specific information about the victim and the poison, then give appropriate first aid instructions. If the patient is to be taken to a hospital, a sample of vomit and the poison container should be taken along, if they are available.

Most cases of plant poisoning are treated by inducing vomiting, if the patient is fully conscious. Vomiting can be induced by taking syrup of **ipecac**, an over-the-counter emetic available at any pharmacy.

For acid, alkali, or petroleum product poisonings, the patient should not vomit. Acids and alkalis can burn the esophagus if they are vomited, and petroleum products can be inhaled into the lungs during vomiting, resulting in **pneumonia**.

Once under medical care, doctors have the option of treating the patient with a specific remedy to counteract the poison (antidote) or with **activated charcoal** to absorb the substance inside the patient's digestive system. In some instances, pumping the stomach may be required. This technique, which is known as gastric lavage, involves introducing 20–30 mL of tap water or 9% saline solution into the patient's digestive tract and removing the stomach contents with a siphon or syringe. The process is repeated until the washings are free of poison. Medical personnel will also provide supportive care as needed, such as intravenous fluids or mechanical ventilation.

If the doctor suspects that the poisoning was not accidental, he or she is required to notify law enforcement authorities. Most cases of malicious poisoning concern family members or acquaintances of the victim, but the number of intentional random poisonings of the general public has increased in recent years. A case reported in 2003 involved the use of nicotine to poison 1700 pounds of ground beef in a Michigan supermarket. Over a hundred persons fell ill after eating the poisoned beef.

Prognosis

The outcome of poisoning varies from complete recovery to death, and depends on the type and amount of the poison, the health of the victim, and the speed with which medical care is obtained.

KEY TERMS

Antidote—A medication or remedy for counteracting the effects of a poison.

Emetic—A medication or substance given to induce vomiting.

Gastric lavage—A technique for washing poison out of the stomach by instilling water or saline solution through a tube, removing the stomach contents by suction, and repeating the process until the washings are free of poison. It is also called stomach pumping.

Toxicology—The branch of medicine that deals with the effects, detection, and treatment of poisons.

Prevention

Most accidental poisonings are preventable. The number of deaths of children from poisoning has declined from about 450 per year in the 1960s to about 50 each year in the 1990s. This decline has occurred mainly because of better packaging of toxic materials and better public education.

Actions to prevent poisonings include:

- removing plants that are poisonous
- keeping medicines and household chemicals locked and in a place inaccessible to children
- keeping medications in child-resistant containers
- never referring to medicine as “candy”
- keeping cleaners and other poisons in their original containers
- disposing of outdated prescription medicines
- not purchasing over-the-counter medications with damaged protective seals or packaging
- avoiding the use of herbal preparations not made by a reputable manufacturer

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American Association of Poison Control Centers (AAPCC). 3201 New Mexico Avenue, Suite 330, Washington, DC 20016. (202) 362-7217. POISONING EMERGENCIES: (800) 222-1222. <<http://www.aapcc.org>>.

National Toxicology Program (NTP) of the National Institute of Environmental Health Sciences (NIEHS). P. O. Box 12233, Research Triangle Park, NC 27709. (919) 541-3419. <<http://www.ntp-server.niehs.nih.gov>>.

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Polarity therapy

Definition

Polarity therapy is a holistic, energy-based system that includes bodywork, diet, **exercise**, and lifestyle counseling for the purpose of restoring and maintaining proper energy flows throughout the body. The underlying concept of polarity therapy is that all energy within the human body is based in electromagnetic force and that disease results from improperly dissipated energy.

Purpose

Polarity therapy unblocks and recharges the flow of life energy and realigns unbalanced energy as a means of eliminating disease. Patients learn to release tension by addressing the source of the **stress** and by maintaining a healthy demeanor accordingly.

This treatment may be effective to promote health and healing to anyone willing to embrace the appropriate lifestyle. Polarity therapy is reportedly effective for anyone who has been exposed to toxic poisons. Likewise, HIV-positive individuals may find comfort in polarity therapy. Additionally this is an appropriate therapy for relieving general stress, back **pain**, stomach cramps, and other recurring maladies and conditions.

Description

Origins

Austrian-American chiropractor, osteopath, and naturopath Randolph Stone (1888–1981) developed polarity therapy as an integration of Eastern and Western principles and techniques of healing. Stone discovered the ancient principles of the Ayurvedic philosophy in the course of his travels during a sojourn in India. On a life-long quest to learn the fundamentals of human vitality, he also studied **reflexology** and traditional Chinese medicine.

Stone became committed to the principles of Ayurvedic medicine, which he interpreted in conjunction with his scientific and medical knowledge to define polarity therapy. According to the philosophy of Ayurved, which is based in a set of principles called the tridosha—the energy of the human body is centered in five organs or regions (the brain; the cardiopulmonary [heart and lungs] region, the diaphragm, the smaller intestine, and the larger intestine). One of five airs or energy forms controls each respective region: prana in the brain, vyana in the heart and lungs, udana in the diaphragm, samana in the smaller intestine, and apana in the larger intestine. The five airs control all directional motion in the body, with each air in command of a different type of movement. Stone established further that the prana, centered in the brain, ultimately controlled the combined forces of the body. Any impediment or restriction to the flow of prana in turn affects the health of the entire body. The prana force is nurtured through the flow of food and air into the body as well as through our interactions with other living beings and through the intake of the five sensory organs.

Stone devoted much of his life to defining an elaborately detailed cause and effect relationship between the human anatomy and illness, based on

the energy flow of the prana. He further attributed electromagnetic energy as the basis of the energy forces. He used the medical symbol of the Caduceus to define the patterns of the flow and described the energy movement in detail in charts of the human body. Polarity therapy is based in charted energy flows. The primary energy pattern is defined in a spiral motion that radiates from the umbilicus and defines the original energy flow of the fetus in the womb.

After determining the exact source of a patient's energy imbalance, the therapist begins the first of a series of bodywork sessions designed to rechannel and release the patient's misdirected prana. This therapy, akin to massage, is based in energetic pressure and involves circulating motions. In performing the regimen, the therapist pays strict attention to the pressure exerted at each location—even to which finger is used to apply pressure at any given point of the patient's anatomy. This technique, which comprises the central regimen or focal point of polarity therapy is very gentle and is unique to polarity therapy. It typically involves subtle rocking movements and cranial holds to stimulate body energy. Although firm, deep pushing touches are employed in conjunction with the massage technique, the polarity therapist never exerts a particularly forceful contact.

To support the bodywork, the therapist often prescribes a diet for the patient, to encourage cleansing and eliminate waste. The precepts of polarity therapy take into consideration specific interactions between different foods and the human energy fields.

Likewise, a series of exercises is frequently prescribed. These exercises, called polarity yoga include squats, stretches, rhythmic movements, deep breathing, and expression of sounds. They can be both energizing and relaxing. Counseling may be included whenever appropriate as a part of a patient's highly individual therapy regimen to promote balance.

Preparations

Therapists take a comprehensive case history from every patient prior to beginning treatment. This preliminary verbal examination often monopolizes the first therapy session. Depending upon circumstances, a therapist might have a need to assess the patient's physical structural balance through observation and physical examination.

Precautions

Polarity therapy is safe for virtually anyone, even the elderly and the most frail patients, because of the intrinsic gentleness of the **massage therapy**.

KEY TERMS

Apana—Life sustaining energy centered in the larger intestine; the fifth of the five airs of Ayurvedic philosophy; the life force governing expulsion activity.

Ayurveda—(Sanskrit, *Ayur*, life, and *veda*, knowledge) Translated as “knowledge of life” or “science of longevity.” It became established as the traditional Hindu system of medicine.

Caduceus—The ancient and universal symbol of medicine consisting of the winged staff of Mercury and two intertwining serpents.

Primary energy pattern—A spiral motion that radiates from the umbilicus; the energy pattern associated with a child in the womb.

Prana—Life sustaining energy centered in the human brain; the first of the five airs of Ayurvedic philosophy; the life force governing inspiration and the conscious intellect.

QV—Quantum vacuum, a theory coined by physicists, which defines the interactions of energy that combine to form reality.

Reflexology—Belief that reflex areas in the feet correspond to every part of the body, including organs and glands, and that stimulating the correct reflex area can affect the body part.

Samana—Life sustaining energy of the smaller intestine; the fourth of the five airs of Ayurvedic philosophy; the life force governing side-to-side motion.

Tridosha—The combination of three basic principles of energy, or biological humor, that comprise life, according to Ayurvedic philosophy.

Udana—Life sustaining energy of the diaphragm, the third of the five airs of Ayurvedic philosophy, the life force governing upward motion.

Vyana—Life sustaining energy of the heart and lungs; the second of the five airs of Ayurvedic philosophy; the life force governing circular motion.

Side effects

Highly emotional releases of energy (laughter, tears, or a combination of both) are associated with this therapy.

Research and general acceptance

This is a complementary therapy of holistic, spiritually based treatment, which may be used in

conjunction with a medical approach. Polarity therapy is practiced worldwide, but the majority of practitioners are based in the United States. Modern physicists employ concepts similar to Stone's basic theories of polarity in defining the quantum vacuum (QV) as a foundation of all reality. Still, by 2000, this holistic regimen had not achieved the widespread acceptance anticipated by Stone before his death in 1981.

When St. Paul Fire and Marine insurers offered a liability insurance package to therapy providers, the company recognized polarity therapy as an alternative medical treatment along with acupuncture, biofeedback, homeopathy, reflexology, and others.

Resources

PERIODICALS

Modern Medicine August 1, 1999: 15.

ORGANIZATIONS

American Polarity Therapy Association. P.O. Box 19858, Boulder Colorado 80308. (303) 545-2080. Fax: (303) 545-2161.

Trans-Hyperboreau Institute of Science. P.O. Box 2344 Sausalito, California 94966. (415) 331-0230. (800) 485-8095. Fax: (415) 331-0231.

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Gloria Cooksey

Polio

Definition

Poliomyelitis, also called polio or infantile **paralysis**, is a highly infectious viral disease that may attack the central nervous system and is characterized by symptoms that range from a mild nonparalytic infection to total paralysis in a matter of hours.

Description

There are three known types of polioviruses (called 1, 2, and 3), each causing a different strain of the disease and all are members of the viral family of enteroviruses (viruses that infect the gastrointestinal tract). Type 1 is the cause of epidemics and many cases of paralysis, which is the most severe manifestation of the infection. The virus is usually a harmless parasite of human beings.

Some statistics quote one in 200 infections as leading to paralysis while others state that one in 1,000 cases reach the central nervous system (CNS). When it does reach the CNS, inflammation and destruction of the spinal cord motor cells (anterior horn cells) occurs, which prevents them from sending out impulses to muscles. This causes the muscles to become limp or soft and they cannot contract. This is referred to as flaccid paralysis and is the type found in polio. The extent of the paralysis depends on where the virus strikes and the number of cells that it destroys. Usually, some of the limb muscles are paralyzed; the abdominal muscles or muscles of the back may be paralyzed, affecting posture. The neck muscles may become too weak for the head to be lifted. Paralysis of the face muscles may cause the mouth to twist or the eyelids to droop. Life may be threatened if paralysis of the throat or of the breathing muscles occurs.

Man is the only natural host for polioviruses and it most commonly infects younger children, although older children and adults can be infected. Crowded living conditions and poor hygiene encourage the spread of poliovirus. Risk factors for this paralytic illness include older age, **pregnancy**, abnormalities of the immune system, recent tonsillectomy, and a recent episode of excessively strenuous **exercise** concurrent with the onset of the CNS phase.

Causes and symptoms

Poliovirus can be spread by direct exposure to an infected individual, and more rarely, by eating foods contaminated with waste products from the intestines (feces) and/or droplets of moisture (saliva) from an infected person. Thus, the major route of transmission is fecal-oral, which occurs primarily with poor sanitary conditions. The virus is believed to enter the body through the mouth with primary multiplication occurring in the lymphoid tissues in the throat, where it can persist for about one week. During this time, it is absorbed into the blood and lymphatics from the gastrointestinal tract where it can reside and multiply, sometimes for as long as 17 weeks. Once absorbed, it is widely distributed throughout the body until it ultimately reaches the CNS (the brain and spinal cord). The infection is passed on to others when poor hand-washing allows the virus to remain on the hands after eating or using the bathroom. Transmission remains possible while the virus is being excreted and it can be transmitted for as long as the virus remains in the throat or feces. The incubation period ranges from three to 21 days, but cases are most infectious from seven to 10 days before and after the onset of symptoms.

DR. JONAS E. SALK (1914–1995)



(Library of Congress.)

Jonas Salk was born in New York, New York, on October 28, 1914. He received his medical degree from New York University in 1939. In 1942, Salk began working for a former teacher, Thomas Francis, Jr., to produce

influenza vaccines, a project that continued until 1949. That year, as a research professor, Salk began a three-year project sponsored by the National Foundation for Infantile Paralysis, also known as the March of Dimes. Caused by the poliomyelitis virus, polio was also known as infantile paralysis. Periodic outbreaks of the disease, which attacks the nervous system, caused death or a lifetime of paralysis, especially in children. It was a difficult disease to study because sufficient viruses could not be obtained. Unlike bacteria, which can be grown in cultures, viruses need living tissue on which to grow. Once a method for preparing viruses was discovered and improved, sufficient viruses became available for research.

Salk first set out to confirm that there were three virus types responsible for polio and then began to experiment with ways to kill the virus and yet retain its ability to produce an immune response. By 1952, he had produced a dead virus vaccine that worked against the three virus types. He began testing. First the vaccine was tested on monkeys, then on children who had recovered from the disease, and finally on Salk's own family and children, none of whom had ever had the disease. Following large-scale trials in 1954, the vaccine was finally released for public use in 1955. The Salk vaccine was not the first vaccine against polio, but it was the first to be found safe and effective. By 1961, there was a 96 percent reduction in polio cases in the United States.

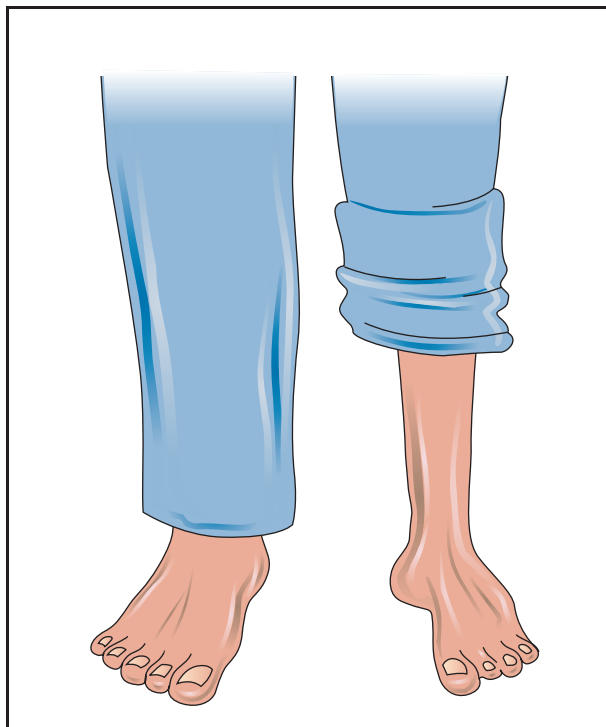
There are two basic patterns to the virus: the minor illness (abortive type) and the major illness (which may be paralytic or nonparalytic). The minor illness accounts for 80–90% of clinical infections and is found mostly in young children. It is mild and does not involve the CNS. Symptoms include a slight **fever**, **fatigue**, **headache**, sore throat, and **vomiting**, which generally develop three to five days after exposure. Recovery from the minor illness occurs within 24–72 hours. Symptoms of the major illness usually appear without a previous minor illness and generally affect older children and adults.

About 10% of people infected with poliovirus develop severe headache and **pain** and stiffness of the neck and back. This is due to an inflammation of the meninges (tissues which cover the spinal cord and brain). This syndrome is called “aseptic meningitis.” The term “aseptic” is used to differentiate this type of **meningitis** from those caused by bacteria. The patient usually recovers completely from this illness within several days.

About 1% of people infected with poliovirus develop the most severe form. Some of these patients

may have two to three symptom-free days between the minor illness and the major illness but the symptoms often appear without any previous minor illness. Symptoms again include headache and back and neck pain. The major symptoms, however, are due to invasion of the motor nerves, which are responsible for movement of the muscles. This viral invasion causes inflammation, and then destruction of these nerves. The muscles, therefore, no longer receive any messages from the brain or spinal cord. The muscles become weak, floppy, and then totally paralyzed. All muscle tone is lost in the affected limb and the muscle becomes soft (flaccid). Within a few days, the muscle will begin to decrease in size (atrophy). The affected muscles may be on both sides of the body (symmetric paralysis), but are often on unbalanced parts of the body (asymmetric paralysis). Sensation or the ability to feel is not affected in these paralyzed limbs.

When poliovirus invades the brainstem (the stalk of brain which connects the two cerebral hemispheres with the spinal cord, called bulbar polio), a person may begin to have trouble breathing and swallowing.



In its most severe form, polio causes paralysis of the muscles of the legs, arms, and respiratory system. All muscle tone is lost in the affected limb, and the muscle becomes flaccid and begins to atrophy, as shown in the illustration above. (Illustration by Electronic Illustrators Group.)

If the brainstem is severely affected, the brain's control of such vital functions as heart rate and blood pressure may be disturbed. This can lead to **death**.

The maximum state of paralysis is usually reached within just a few days. The remaining, unaffected nerves then begin the process of attempting to grow branches which can compensate for the destroyed nerves. Fortunately, the nerve cells are not always completely destroyed. By the end of a month, the nerve impulses start to return to the apparently paralyzed muscle and by the end of six months, recovery is almost complete. If the nerve cells are completely destroyed, however, paralysis is permanent.

Diagnosis

Fever and asymmetric flaccid paralysis without sensory loss in a child or young adult almost always indicate poliomyelitis. Using a long, thin needle inserted into the lower back to withdraw spinal fluid (lumbar puncture) will reveal increased white blood cells and no bacteria (aseptic meningitis). Nonparalytic poliomyelitis cannot be distinguished

clinically from aseptic meningitis due to other agents. Virus isolated from a throat swab and/or feces or blood tests demonstrating the rise in a specific antibody is required to confirm the diagnosis.

Treatment

There is no specific treatment for polio except symptomatic. Therapy is designed to make the patient more comfortable (pain medications and hot packs to soothe the muscles), and intervention if the muscles responsible for breathing fail (for instance, a ventilator to take over the work of breathing). During active infection, rest on a firm bed is indicated. Physical therapy is the most important part of management of paralytic polio during recovery.

Prognosis

When poliovirus causes only the minor illness or simple aseptic meningitis, the patient can be expected to recover completely. When the patient is diagnosed with the major illness, about 50% will recover completely. About 25% of such patients will have slight disability, and about 25% will have permanent and serious disability. Approximately 1% of all patients with major illness die. The greatest return of muscle function occurs in the first six months, but improvements may continue for two years.

A recently described phenomenon called post-polio syndrome may begin many years after the initial illness. This syndrome is characterized by a very slow, gradual decrease in muscle strength.

Prevention

There are two types of polio immunizations available in the United States. Both of these vaccines take advantage of the fact that infection with polio leads to an immune reaction, which will give the person permanent, lifelong immunity from re-infection with the form of poliovirus for which the person was vaccinated.

The Sabin vaccine (also called the oral polio vaccine or OPV) is given to infants by mouth at the same intervals as the DPT (three doses). It contains the live, but weakened, poliovirus, which make the recipient immune to future infections with poliovirus. Because OPV uses live virus, it has the potential to cause infection in individuals with weak immune defenses (both in the person who receives the vaccine and in close contacts). This is a rare complication, however, occurring in only one in 6.8 million doses administered and

KEY TERMS

Aseptic—Sterile; containing no microorganisms, especially no bacteria.

Asymmetric—Not occurring evenly on both sides of the body.

Atrophy—Shrinking; growing smaller in size.

Brainstem—The stalk of the brain which connects the two cerebral hemispheres with the spinal cord.

DPT—Diphtheria, Pertussis and Tetanus injections.

Epidemic—Refers to a situation in which a particular disease rapidly spreads among many people in the same geographical region over a small time period.

Flaccid—Weak, soft, floppy.

Gastrointestinal—Pertaining to the stomach and intestines.

Lymph/lymphatic—One of the three body fluids that is transparent and a slightly yellow liquid that is collected from the capillary walls into the tissues and circulates back to the blood supply.

Paralysis—The inability to voluntarily move.

Symmetric—Occurring on both sides of the body, in a mirror-image fashion.

one in every 6.4 million doses from having close contact with someone who received the vaccine.

The Salk vaccine (also called the killed polio vaccine or inactivated polio vaccine) consists of a series of three shots that are given just under the skin. This immunization contains no live virus, just the components of the virus that provoke the recipient's immune system to react as if the recipient were actually infected with the poliovirus. The recipient thus becomes immune to infection with the poliovirus in the future.

In the 13 years following the launching of the Global Polio Eradication Initiative, the number of cases has fallen 99% from an estimated 350,000 cases to less than 3,500 cases worldwide in 2000. At the end of 2000, the number of polio-infected countries was approximately 20, down from 125. The goal of the World Health Organization (WHO) is to have polio eliminated from the planet by the year 2005. The virus has still been identified in Africa and parts of Asia, so travelers to those areas may want to check with their physicians concerning booster vaccinations.

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Linda K. Bennington, CNS

Poliomyelitis see **Polio**

Polyangiitis overlap syndrome see **Vasculitis**

Polyarteritis nodosa see **Vasculitis**

Polycystic kidney disease

Definition

Polycystic **kidney disease** (PKD) is one of the most common of all life-threatening human genetic disorders. It is an incurable genetic disorder characterized by the formation of fluid-filled cysts in the kidneys of affected individuals. These cysts multiply over time. It was originally believed that the cysts eventually caused kidney failure by crowding out the healthy kidney tissue. It is now thought that the kidney damage seen in PKD is actually the result of the body's immune system. The immune system, in its attempts to rid the kidney of the cysts, instead progressively destroys the formerly healthy kidney tissue.

Description

A healthy kidney is about the same size as a human fist. PKD cysts, which can be as small as the head of a pin or as large as a grapefruit, can expand the kidneys until each one is bigger than a football and weighs as much as 38lb (17 kg).

There are two types of PKD: infantile PKD, which generally shows symptoms prior to birth; and adult onset PKD. Individuals affected with infantile PKD are often stillborn. Among the liveborn individuals affected with infantile PKD, very few of these children survive to the age of two. The adult onset form of PKD is much more common. The time and degree of symptom onset in the adult form of PKD can vary widely, even within a single family with two or more affected individuals. Symptoms of this form of PKD usually start to appear between the ages of 20 and 50. Organ deterioration progresses more slowly in adult onset PKD than it does in the infantile form; but, if left untreated, adult onset PKD also eventually leads to kidney failure.

One of the most common of all life-threatening genetic diseases, PKD affects more than 60,000 Americans. Over 12.5 million people worldwide are affected with PKD. Approximately one in every 400 to 1,000 people is affected with ADPKD. Another one in 10,000 affected with ARPKD. PKD is observed in equal numbers in both males and females. PKD is also observed with equal frequency among ethnic groups.

Causes and symptoms

Polycystic kidney disease is expressed as both a recessive and a dominant trait. A recessive genetic trait will not cause disease in a child unless it is inherited from both parents. A dominant genetic trait can be inherited from just one parent. Those people affected with autosomal dominant PKD (ADPKD) have the much more common adult onset form. Those with autosomal recessive PKD (ARPKD) have the infantile form.

There are mutations on at least three genes that cause adult onset PKD. Approximately 85% of these cases are known to arise from mutations in the PKD1 gene that has been mapped to a region on the short arm of chromosome 16 (16p13.3-p13.12). Another 10–15% of cases of adult onset PKD are thought to be caused by mutations in the PKD2 gene that has been mapped to a region on the long arm of chromosome 4 (4q21-q23). As of early 2001, it is thought that the remainder of the cases of PKD are caused by mutations in the PKD3 gene, which has not yet been mapped. This unidentified “PKD3 gene” may, in fact, be more than one gene.

Adult onset PKD is transmitted from parents to their offspring as a non-sex linked (autosomal) dominant trait. This means that if either parent carries this genetic mutation, there is a 50% chance that their child will inherit this disease. In the case of two

affected parents, there is a 75% probability that their children will be affected with adult onset PKD.

Infantile PKD is caused by a non-sex linked (autosomal) recessive genetic mutation that has been mapped to a region on the short arm of chromosome 6 (6p21). Both parents must be carriers of this mutation for their children to be affected with infantile PKD. In the case of two carrier parents, the probability is 25% that their child will be affected by infantile PKD.

A baby born with infantile PKD has floppy, low-set ears, a pointed nose, a small chin, and folds of skin surrounding the eyes (epicanthal folds). Large, rigid masses can be felt on the back of both thighs (flanks), and the baby usually has trouble breathing.

In the early stages of adult onset PKD, many people show no symptoms. Generally, the first symptoms to develop are: high blood pressure (**hypertension**); general **fatigue**; **pain** in the lower back or the backs of the thighs; headaches; and/or urinary tract infections accompanied by frequent urination.

As PKD becomes more advanced, the kidneys' inability to function properly becomes more pronounced. The cysts on the kidney may begin to rupture and the kidneys tend to be much larger than normal. Individuals affected with PKD have a much higher rate of **kidney stones** than the rest of the population at this, and later stages, of the disease. Approximately 60% of those individuals affected with PKD develop cysts in the liver, while 10% develop cysts in the pancreas.

Because the kidneys are primarily responsible for cleaning the blood, individuals affected with PKD often have problems involving the circulatory system. These include: an underproduction of red blood cells which results in an insufficient supply of oxygen to the tissues and organs (anemia); an enlarged heart (cardiac hypertrophy) probably caused by long term hypertension; and, a leakage of the valve between the left chambers (auricle and ventricle) of the heart (**mitral valve prolapse**). Less common (affecting approximately 5% of PKD patients) are brain aneurysms. An aneurysm is an abnormal and localized bulging of the wall of a blood vessel. If an aneurysm within the brain leaks or bursts, it may cause a **stroke** or even **death**.

Other health problems associated with adult onset PKD include: chronic leg or back pain; frequent infections; and herniations of the groin and abdomen, including herniation of the colon (diverticular disease). A herniation, or **hernia**, is caused when a tissue, designed to hold the shape of an underlying tissue,



A pair of human kidneys. The left a polycystic kidney, the right a normal kidney. (Photograph by A. Glauber, Photo Researchers, Inc. Reproduced by permission.)

becomes weakened at a particular spot. The underlying tissue pushes against this weakened area until the area is no longer able to hold back the underlying tissue and the area forms an abnormal bulge through which the underlying tissue projects. Diverticular disease is caused by a weakening of the muscles that hold the shape of the organs of the digestive tract. These muscles weaken allowing these organs, particularly one section of the colon, to form sac-like projections that can trap feces and become infected, or rupture.

In the final stages of PKD, the major symptom is kidney (renal) failure. Renal failure is indicated by an increase of nitrogen (in the form of urea) in the blood (uremia, or uremic **poisoning**). Uremia is a rapidly fatal condition without treatment.

Diagnosis

Many patients who have PKD do not have any symptoms. Their condition may not be discovered unless tests that detect it are performed for other reasons.

When symptoms of PKD are present, the diagnostic procedure begins with a family medical history and **physical examination** of the patient. If several family members have PKD, there is a strong likelihood that the patient has it too. If the disease is advanced, the doctor will be able to feel the patient's enlarged kidneys. Heart murmur, high blood pressure, and other signs of cardiac impairment can also be detected.

Urinalysis and a blood test called creatine clearance can indicate how effectively the kidneys are functioning. Scanning procedures using intravenous dye reveal kidney enlargement or deformity and scarring

caused by cysts. Ultrasound and **computed tomography scans** (CT scans) can reveal kidney enlargement and the cysts that caused it. CT scans can highlight cyst-damaged areas of the kidneys. A sampling of the kidney cells (biopsy) may be performed to verify the diagnosis.

Treatment

There is no way to prevent cysts from forming or becoming enlarged, or to prevent PKD from progressing to kidney failure. Treatment goals include preserving healthy kidney tissue; controlling symptoms and, preventing infection and other complications.

If adult PKD is diagnosed before symptoms become evident, urinalysis and other diagnostic tests are performed at six-week intervals to monitor the patient's health status. If results indicate the presence of infection or another PKD-related health problem, aggressive antibiotic therapy is initiated to prevent inflammation that can accelerate disease progression; iron supplements or infusion of red blood cells are used to treat anemia; and surgery may be needed to drain cysts that bleed, cause pain, have become infected, or interfere with normal kidney function.

Lowering high blood pressure can slow loss of kidney function. Blood-pressure control, which is the cornerstone of PKD treatment, is difficult to achieve. Therapy may include antihypertensive medications, diuretic medications, and/or a low-salt diet. As kidney function declines, some patients need dialysis and/or a kidney transplant.

There is no known way to prevent PKD, but certain lifestyle modifications can help control symptoms. People who have PKD should not drink heavily or smoke. They should not use **aspirin**, non-steroidal anti-inflammatory drugs (NSAIDs), or other prescription or over-the-counter medications that can impair kidney function. Individuals affected with PKD should eat a balanced diet, **exercise** regularly, and maintain a weight appropriate for their height, age, and body type. Regular medical monitoring is also recommended.

Prognosis

There is no known cure for PKD. Those affected with infantile PKD generally die before the age of two. In adults, untreated disease can be rapidly fatal or continue to progress slowly, even after symptoms of kidney failure appear. About half of all adults with PKD also develop kidney failure. Unless the patient undergoes dialysis or has a kidney transplant, this

KEY TERMS

Biopsy—The surgical removal and microscopic examination of living tissue for diagnostic purposes.

Cancer—A disease caused by uncontrolled growth of the body's cells.

Computed tomography (CT) scan—An imaging procedure that produces a three-dimensional picture of organs or structures inside the body, such as the brain.

Cyst—An abnormal sac or closed cavity filled with liquid or semisolid matter.

Diuretics—Medications that increase the excretion of urine.

Kidney—Either of two organs in the lumbar region that filter the blood, excreting the end products of the body's metabolism in the form of urine and regulating the concentrations of hydrogen, sodium, potassium, phosphate and other ions in the body.

Magnetic resonance imaging (MRI)—A technique that employs magnetic fields and radio waves to create detailed images of internal body structures and organs, including the brain.

Ultrasonogram—A procedure where high-frequency sound waves that cannot be heard by human ears are bounced off internal organs and tissues. These sound waves produce a pattern of echoes, which are then used by the computer to create sonograms or pictures of areas inside the body.

Uremic poisoning—Accumulation of waste products in the body.

condition usually leads to death within four years of diagnosis.

Although medical treatment can temporarily alleviate symptoms of PKD, the expanding cysts continue to increase pressure on the kidneys. Kidney failure and uremic poisoning (accumulation of waste products the body is unable to eliminate) generally cause death about 10 years after symptoms first appear.

Medications used to fight **cancer** and reduce elevated cholesterol levels have slowed the advance of PKD in laboratory animals. They may soon be used to treat adults and children who have the disease. Researchers are also evaluating the potential benefits of anti-inflammatory drugs, which may prevent the scarring that destroys kidney function.

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Polycystic ovary syndrome

Definition

Polycystic ovary syndrome (PCOS) is a condition characterized by the accumulation of numerous cysts (fluid-filled sacs) on the ovaries associated with high male hormone levels, chronic anovulation (absent ovulation), and other metabolic disturbances. Classic symptoms include excess facial and body hair, **acne**, **obesity**, irregular menstrual cycles, and **infertility**.

Description

PCOS, also called Stein-Leventhal syndrome, is a group of symptoms caused by underlying hormonal and metabolic disturbances that affect about 6% of premenopausal women. PCOS symptoms appear as early as adolescence in the form of **amenorrhea** (missed periods), obesity, and **hirsutism**, the abnormal growth of body hair.

A disturbance in normal hormonal signals prevents ovulation in women with PCOS. Throughout the cycle, estrogen levels remain steady, luteinizing hormone (LH) levels are high, and follicle stimulating hormone (FSH) and progesterone levels are low. Since eggs are rarely or never released from their follicles, multiple **ovarian cysts** develop over time.

One of the most important characteristics of PCOS is hyperandrogenism, the excessive production of male hormones (androgens), particularly testosterone, by the ovaries. This accounts for the male hair-growth patterns and acne in women with PCOS. Hyperandrogenism has been linked with **insulin resistance** (the inability of the body to respond to insulin) and hyperinsulinemia (high blood insulin levels), both of which are common in PCOS.

Causes and symptoms

While the exact cause of PCOS is unknown, it runs in families, so the tendency to develop the syndrome may be inherited. The interaction of hyperinsulinemia and hyperandrogenism is believed to play a role in chronic anovulation in susceptible women.

The numbers and types of PCOS symptoms that appear vary among women. These include:

- Hirsutism. Related to hyperandrogenism, this occurs in 70% of women.
- Obesity. Approximately 40–70% of persons with PCOS are overweight.
- Anovulation and menstrual disturbances. Anovulation appears as amenorrhea in 50% of women, and as heavy uterine bleeding in 30% of women. However, 20% of women with PCOS have normal menstruation.
- Male-pattern hair loss. Some women with PCOS develop bald spots.
- Infertility. Achieving **pregnancy** is difficult for many women with PCOS.
- Polycystic ovaries. Most, but not all, women with PCOS have multiple cysts on their ovaries.
- Skin discoloration. Some women with PCOS have dark patches on their skin.
- Abnormal blood chemistry. Women with PCOS have high levels of low-density lipoprotein (LDL or “bad”) cholesterol and triglycerides, and low levels of high-density lipoprotein (HDL or “good”) cholesterol.
- Hyperinsulinemia. Some women with PCOS have high blood insulin levels, particularly if they are overweight.

Diagnosis

PCOS is diagnosed when a woman visits her doctor for treatment of symptoms such as hirsutism, obesity, menstrual irregularities, or infertility. Women with PCOS are treated by a gynecologist, a doctor who treats diseases of the female reproductive organs, or a reproductive endocrinologist, a specialist who treats diseases of the body’s endocrine (hormones and glands) system and infertility.

PCOS can be difficult to diagnose because its symptoms are similar to those of many other diseases or conditions, and because all of its symptoms may not occur. A doctor takes a complete medical history, including questions about menstruation and reproduction, and weight gain. **Physical examination** includes a pelvic examination to determine the size of the ovaries, and visual inspection of the skin for hirsutism, acne, or other changes. Blood tests are performed to measure levels of luteinizing hormone, follicle stimulating hormone, estrogens, androgens, glucose, and insulin. A glucose-tolerance test may be administered. An ultrasound examination of the ovaries is performed to evaluate their size and shape. Most insurance plans cover the costs of diagnosing and treating PCOS and its related problems.

Treatment

PCOS treatment is aimed at correcting anovulation, restoring normal menstrual periods, improving fertility, eliminating hirsutism and acne, and preventing future complications related to high insulin and blood lipid (fat) levels. Treatment consists of weight loss, drugs or surgery, and hair removal, depending upon which symptoms are most bothersome, and whether a woman desires pregnancy.

Weight loss

In overweight women, weight loss (as little as 5%) through diet and **exercise** may correct hyperandrogenism, and restore normal ovulation and fertility. This is often tried first.

Drugs

HORMONAL DRUGS. Women who do not want to become pregnant and require **contraception** (spontaneous ovulation occurs occasionally among women with PCOS) are treated with low-dose oral contraceptive pills (OCPs). OCPs bring on regular menstrual periods and correct heavy uterine bleeding, as well as hirsutism, although improvement may not be seen for up to a year.

If an infertile woman desires to become pregnant, the first drug usually given to help induce ovulation is clomiphene citrate (Clomid), which results in pregnancy in about 70% of women but can cause multiple births. In the 20–25% of women who do not respond to clomiphene, other drugs that stimulate follicle development and induce ovulation, such as human menstrual gonadotropin (Pergonal) and human chorionic gonadotropin (HCG), are given. However, these drugs have a lower pregnancy rate (less than 30%), a higher rate of **multiple pregnancy** (from 5–30%, depending on the dose of the drug), and a higher risk of medical problems. Women with PCOS have a high rate of **miscarriage** (30%), and may be treated with the gonadotropin-releasing hormone agonist leuprolide (Lupron) to reduce this risk.

Since women with PCOS do not have regular endometrial shedding due to high estrogen levels, they are at increased risk for overgrowth of this tissue and **endometrial cancer**. The drug medroxyprogesterone acetate, when taken for the first 10 days of each month, causes regular shedding of the endometrium, and reduces the risk of **cancer**. However, in most cases, oral contraceptive pills are used instead to bring about regular menstruation.

OTHER DRUGS. Another drug that helps to trigger ovulation is the steroid hormone dexamethasone. This drug acts by reducing the production of androgens by the adrenal glands.

The antiandrogen spironolactone (Aldactazide), which is usually given with an oral contraceptive, improves hirsutism and male-pattern baldness by reducing androgen production, but has no effect on fertility. The drug causes abnormal uterine bleeding and is linked with **birth defects** if taken during pregnancy. Another antiandrogen used to treat hirsutism, flutamide (Eulexin), can cause liver abnormalities, **fatigue**, mood swings, and loss of sexual desire. A drug used to reduce insulin levels, metformin (Glucophage), has shown promising results in women with PCOS hirsutism, but its effects on infertility and other PCOS symptoms are unknown. Drug treatment of hirsutism is long-term, and improvement may not be seen for up to a year or longer.

Acne is treated with **antibiotics**, antiandrogens, and other drugs such as retinoic acids (vitamin A compounds).

Surgical treatment

Surgical treatment of PCOS may be performed if drug treatment fails, but it is not common. A wedge resection, the surgical removal of part of the ovary and

cysts through a laparoscope (an instrument inserted into the pelvis through a small incision), or an abdominal incision, reduces androgen production and restores ovulation. Although laparoscopic surgery is less likely to cause scar tissue formation than abdominal surgery, both are associated with the potential for scarring that may require additional surgery. Laparoscopic ovarian drilling is another type of laparoscopic surgery used to treat PCOS. The ovarian cysts are penetrated with a laser beam and some of the fluid is drained off. Between 50–65% of women may become pregnant after either type of surgery.

Some cases of severe hirsutism are treated by removal of the uterus (**hysterectomy**) and the ovaries (oophorectomy), followed by estrogen replacement therapy.

Other treatment

Hirsutism may be treated by hair removal techniques such as shaving, depilatories (chemicals that break down the structure of the hair), tweezing, waxing, electrolysis (destruction of the hair root by an electrical current), or the destruction of hair follicles by laser therapy. However, the treatments may have to be repeated.

Alternative treatment

PCOS can be addressed using many types of alternative treatment. The rebalancing of hormones is a primary focus of all these therapies. **Acupuncture** works on the body's energy flow according to the meridian system. Chinese herbs, such as *gui zhi fu ling wan*, can be effective. In **naturopathic medicine**, treatment focuses on helping the liver function more optimally in the hormonal balancing process. Dietary changes, including reducing animal products and fats, while increasing foods that nourish the liver such as carrots, dark green vegetables, lemons, and beets, can be beneficial. Essential fatty acids, including flax oil, evening primrose oil (*Oenothera biennis*), and black currant oil, act as anti-inflammatories and hormonal regulators. Western herbal medicine uses phytoestrogen and phytoprogesteric herbs, such as blue cohosh (*Caulophyllum thalictroides*) and false unicorn root (*Chamaelirium luteum*), as well as liver herbs, like dandelion (*Taraxacum mongolicum*), to work toward hormonal balance. Supplementation with antioxidants, including zinc, and **vitamins** A, E, and C, is also recommended. Constitutional homeopathy can bring about a deep level of healing with the correct remedies.

KEY TERMS

Androgens—Male sex hormones produced by the adrenal glands and testes, the male sex glands.

Anovulation—The absence of ovulation.

Antiandrogens—Drugs that inhibit androgen production.

Estrogens—Hormones produced by the ovaries, the female sex glands.

Follicle stimulating hormone—A hormone that stimulates the growth and maturation of mature eggs in the ovary.

Gynecologist—A physician with specialized training in diseases and conditions of the female reproductive system.

Hirsutism—An abnormal growth of hair on the face and other parts of the body caused by an excess of androgens.

Hyperandrogenism—The excessive secretion of androgens.

Hyperinsulinemia—High blood insulin levels.

Insulin resistance—An inability to respond to insulin, a hormone produced by the pancreas that helps the body to use glucose.

Laparoscope—An instrument inserted into the pelvis through a small incision.

Luteinizing hormone—A hormone that stimulates the secretion of sex hormones by the ovary.

Ovarian follicles—Structures found within the ovary that produce eggs.

Prognosis

With proper diagnosis and treatment, most PCOS symptoms can be adequately controlled or eliminated. Infertility can be corrected and pregnancy achieved in most women although, in some, hormonal disturbances and anovulation may recur. Women should be monitored for endometrial cancer. Because of the high rate of hyperinsulinemia seen in PCOS, women with the disorder should have their glucose levels checked regularly to watch for the development of diabetes. Blood pressure and cholesterol screening are also needed because these women also tend to have high levels of LDL cholesterol and triglycerides, which put them at risk for developing heart disease.

Prevention

There is no known way to prevent PCOS, but if diagnosed and treated early, risks for complications such as heart disease and diabetes may be minimized. Weight control through diet and exercise stabilizes hormones and lowers insulin levels.

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ORGANIZATIONS

- American Academy of Family Physicians. 11400 Tomahawk Creek Parkway, Leawood, KS 66211-2672. (913) 906-6000. <<http://www.aafp.org/>>. fp@aafp.org.
- American Medical Association. 515 N. State Street, Chicago, IL 60610. (312) 464-5000. <<http://www.ama-assn.org/>>.
- Polycystic Ovarian Syndrome Association. PO Box 80517, Portland, OR 97280. (877) 775-7267. <<http://www.pcossupport.org/>>. info@pcossupport.org.

OTHER

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- Women's Health-UK. <<http://www.womens-health.co.uk/pcos.htm>>.

L. Fleming Fallon, Jr., MD, DrPH

Polycythemia see **Secondary polycythemia**

Polycythemia rubra vera see **Polycythemia vera**

Polycythemia vera

Definition

Polycythemia vera (PV) is a chronic blood disorder marked by an abnormal increase in three types of blood cells produced by bone marrow: red blood cells

(RBCs), white blood cells (WBCs), and platelets. PV is called a myeloproliferative disorder, which means that the bone marrow is producing too many cells too quickly. Most of the symptoms of PV are related to the increased volume of the patient's blood and its greater thickness (high viscosity). PV sometimes evolves into a different myeloproliferative disorder or into acute leukemia.

Description

Polycythemia vera is a relatively common progressive disorder that develops over a course of 10–20 years. In the United States, PV affects about one person in every 200,000. PV has several other names, including splenomegalic polycythemia, Vaquez-Osler syndrome, erythremia, and primary polycythemia. Primary polycythemia means that the disorder is not caused or triggered by other illnesses. PV most commonly affects middle-aged adults. It is rarely seen in children or young adults and does not appear to run in families. The male/female ratio is 2:1.

Risk factors for polycythemia vera include:

- Caucasian race
- male sex
- age between 40 and 60

Causes and symptoms

The cause of PV remains uncertain. In general, the increased mass of red blood cells in the patient's blood causes both hemorrhage and abnormal formation of **blood clots** in the circulatory system (thrombosis). The reasons for these changes in clotting patterns are not yet fully understood.

Early symptoms

The symptoms of early PV may be minimal—it is not unusual for the disorder to be discovered during a routine blood test. More often, however, patients have symptoms that include headaches, ringing in the ears, tiring easily, memory problems, difficulty breathing, giddiness or lightheadedness, **hypertension**, visual problems, or **tingling** or burning sensations in their hands or feet. Another common symptom is **itching** (pruritus). Pruritus related to PV is often worse after the patient takes a warm bath or shower.

Some patients' early symptoms include unusually heavy bleeding from minor cuts, nosebleeds, stomach ulcers, or bone **pain**. In a few cases, the first symptom is the development of blood clots in an unusual part of the circulatory system (e.g., the liver).

Later symptoms and complications

As the disease progresses, patients with PV may have episodes of hemorrhage or thrombosis. Thrombosis is the most frequent cause of **death** from PV. Other complications include a high level of uric acid in the blood and an increased risk of peptic ulcer disease. About 10% of PV patients eventually develop **gout**; another 10% develop peptic ulcers.

Spent phase

The spent phase is a development in late PV that affects about 30% of patients. The bone marrow eventually fails and the patient becomes severely anemic, requiring repeated blood transfusions. The spleen and liver become greatly enlarged—in the later stages of PV, the patient's spleen may fill the entire left side of the abdomen.

Diagnosis

Physical examination

PV is often a diagnosis of exclusion, which means that the doctor will first rule out other possible causes of the patient's symptoms. The doctor can detect some signs of the disorder during a **physical examination**. Patients with PV will have an enlarged spleen (splenomegaly) in 75% of cases. About 50% will have a slightly enlarged liver. The doctor can feel these changes when he or she presses on (palpates) the patient's abdomen while the patient is lying flat. An **eye examination** will usually reveal swollen veins at the back of the eye. Patients with PV often have unusually red complexions; mottled red patches on their legs, feet, or hands; or swelling at the ends of the fingers.

Diagnostic criteria for PV

Accurate diagnosis of PV is critical because its treatment may require the use of drugs with the potential to cause leukemia. The results of the patient's blood tests are evaluated according to criteria worked out around 1970 by the Polycythemia Vera Study Group. The patient is considered to have PV if all three major criteria are met; or if the first two major criteria and any two minor criteria are met.

Major criteria:

- red blood cell mass greater than 36 mL/kg in males, greater than 32 mL/kg in females
- arterial oxygen level greater than 92%
- splenomegaly

Minor criteria:

- platelet count greater than 400,000/mm³
- WBC greater than 12,000/mm³ without **fever** or infection
- leukocyte alkaline phosphatase (LAP) score greater than 100 with increased blood serum levels of vitamin B₁₂

Laboratory testing

BLOOD TESTS. The diagnosis of PV depends on a set of findings from blood tests. The most important single measurement is the patient's red blood cell mass as a proportion of the total blood volume. This measurement is made by tagging RBCs with radioactive chromium (⁵¹Cr) in order to determine the patient's RBC volume. While a few patients with PV may have a red cell mass level within the normal range if they have had recent heavy bleeding, a high score may eliminate the need for some other tests. A score higher than 36 mL/kg for males and 32 mL/kg for females on the ⁵¹Cr test suggests PV. Measurements of the oxygen level in the patient's arterial blood, of the concentration of vitamin B₁₂ in the blood serum, and of leukocyte alkaline phosphatase (LAP) staining can be used to distinguish PV from certain types of leukemia or from other types of polycythemia. LAP staining measures the intensity of enzyme activity in a type of white blood cell called a neutrophil. In PV, the LAP score is higher than normal whereas in leukemia it is below normal.

BONE MARROW TESTS. Bone marrow testing can be used as part of the diagnostic process. A sample of marrow can be cultured to see if red blood cell colonies develop without the addition of a hormone that stimulates RBC production. The growth of a cell colony without added hormone indicates PV. Bone marrow testing is also important in monitoring the progress of the disease, particularly during the spent phase.

GENETIC TESTING. Genetic testing can be used to rule out the possibility of chronic myeloid leukemia. Patients with this disease have a characteristic chromosomal abnormality called the Philadelphia chromosome. The Philadelphia chromosome does *not* occur in patients with PV.

Imaging studies

Imaging studies are not necessary to make the diagnosis of PV. In some cases, however, imaging studies can detect enlargement of the spleen that the doctor may not be able to feel during the physical examination.

Treatment

Treatment of PV is tailored to the individual patient according to his or her age, the severity of the symptoms and complications, and the stage of the disease.

Phlebotomy

Phlebotomy is the withdrawal of blood from a vein. It is the first line of treatment for patients with PV. Phlebotomy is used to bring down the ratio of red blood cells to fluid volume (the **hematocrit**) in the patient's blood to a level below 45%. In most cases the doctor will withdraw about 500 mL of blood (about 15 fluid ounces) once or twice a week until the hematocrit is low enough. Phlebotomy is considered the best course of treatment for patients younger than 60 and for women of childbearing age. Its drawback is that patients remain at some risk for either thrombosis or hemorrhage.

Myelosuppression

Myelosuppressive therapies are used to slow down the body's production of blood cells. They are given to patients who are older than 60 and at high risk for thrombosis. These therapies, however, increase the patient's risk of developing leukemia. The substances most frequently used as of 1998 include hydroxyurea (Hydrea), interferon alfa (Intron), or radioactive phosphorus (^{32}P). ^{32}P is used only in elderly patients with life expectancies of less than five years because it causes leukemia in about 10% of patients. Interferon alfa is expensive and causes side effects resembling the symptoms of influenza but is an option for some younger PV patients.

Investigational treatment

The Food and Drug Administration (FDA) has approved the use of anagrelide, an orphan drug, for investigational use in the treatment of PV. Anagrelide has moderate side effects and controls the platelet level in over 90% of patients.

Treatment of complications

The itching caused by PV is often difficult to control. Patients with pruritus are given diphenhydramine (Benadryl) or another antihistamine. Patients with high levels of uric acid are usually given allopurinol (Lopurin, Zyloprim) by mouth. Supportive care includes advice about diet—splenomegaly often makes patients feel full after eating only a little food. This

KEY TERMS

Anagrelide—An orphan drug that is approved for treating PV patients on an investigational basis. Anagrelide works by controlling the level of platelets in the blood.

Leukocyte alkaline phosphatase (LAP) test—A blood test that measures the level of enzyme activity in a type of white blood cell called neutrophils.

Myeloproliferative disorder—A disorder in which the bone marrow produces too many cells too rapidly.

Myelosuppressive therapy—Any form of treatment that is aimed at slowing down the rate of blood cell production.

Orphan drug—A drug that is known to be useful in treatment but lacks sufficient funding for further research and development.

Philadelphia chromosome—An abnormal chromosome that is found in patients with a chronic form of leukemia but not in PV patients.

Phlebotomy—Drawing blood from a patient's vein as part of diagnosis or therapy. Phlebotomy is sometimes called venesection. It is an important part of the treatment of PV.

Pruritus—An itching sensation or feeling. In PV the itching is not confined to a specific part of the body and is usually worse after a warm bath or shower.

Spent phase—A late development in PV leading to failure of the bone marrow and severe anemia.

Splenomegaly—Abnormal enlargement of the spleen. Splenomegaly is a major diagnostic criterion of PV.

problem can be minimized by advising patients to eat small meals followed by rest periods.

Because of the clotting problems related to PV, patients should not undergo surgery until their blood counts are close to normal levels. Female patients of childbearing age should be warned about the dangers of **pregnancy** related to their clotting abnormalities.

Prognosis

The prognosis for untreated polycythemia vera is poor; 50% of patients die within 18 months after diagnosis. Death usually results from heart failure, leukemia, or hemorrhage. Patients being treated for

PV can expect to live between 11 and 15 years on average after diagnosis.

Resources

ORGANIZATIONS

National Heart, Lung and Blood Institute. P.O. Box 30105, Bethesda, MD 20824-0105. (301) 251-1222. <<http://www.nhlbi.nih.gov>>.

National Organization for Rare Disorders. P.O. Box 8923, New Fairfield, CT 06812-8923. (800) 999-6673. <<http://www.rarediseases.org>>.

Rebecca J. Frey, PhD

Polydactyly and syndactyly

Definition

Polydactyly and syndactyly are congenital irregularities of the hands and feet. Polydactyly is the occurrence of extra fingers or toes, and syndactyly is the webbing or fusing together of two or more fingers or toes.

Description

Polydactyly can vary from an unnoticeable rudimentary finger or toe to fully developed extra digits.

Syndactyly also exhibits a large degree of variation. Digits can be partially fused or fused along their entire length. The fusion can be simple with the digits connected only by skin, or it can be complicated with shared bones, nerves, vessels, or nails.

Polydactyly and syndactyly can occur simultaneously when extra digits are fused. This condition is known as polysyndactyly.

Causes and symptoms

Polydactyly and syndactyly are due to errors in the process of fetal development. For example, syndactyly results from the failure of the programmed cell death that normally occurs between digits. Most often these errors are due to genetic defects.

Polydactyly and syndactyly can both occur by themselves as isolated conditions or in conjunction with other symptoms as one aspect of a multi-symptom disease. There are several forms of isolated syndactyly and several forms of isolated polydactyly; each of these, where the genetics is understood, is caused by an autosomal dominant gene. This means that since



Polydactyly is the occurrence of extra or partial fingers or toes. (Custom Medical Stock Photo. Reproduced by permission.)

the gene is autosomal (not sex-linked), males and females are equally likely to inherit the trait. This also means that since the gene is dominant, children who have only one parent with the trait have a 50% chance of inheriting it. However, people in the same family carrying the same gene can have different degrees of polydactyly or syndactyly.

Polydactyly and syndactyly are also possible outcomes of a large number of rare inherited and developmental disorders. One or both of them can be present in over 100 different disorders where they are minor features compared to other characteristics of these diseases.

For example, polydactyly is a characteristic of Meckel syndrome and Laurence-Moon-Biedl syndrome. Polydactyly may also be present in Patau's syndrome, asphyxiating thoracic dystrophy, hereditary spherocytic hemolytic anemia, Moebius syndrome, VACTERL association, and Klippel-Trenaunay syndrome.

Syndactyly is a characteristic of Apert syndrome, Poland syndrome, Jarcho-Levin syndrome, oral-facial-digital syndrome, Pfeiffer syndrome, and Edwards syndrome. Syndactyly may also occur with Gordon syndrome, Fraser syndrome, Greig cephalopolysyndactyly, phenylketonuria, Saethre-Chotzen syndrome, Russell-Silver syndrome, and triploidy.

In some isolated cases of polydactyly or syndactyly, it is not possible to determine the cause. Some of these cases might nevertheless be due to genetic defects; sometimes there is too little information to demonstrate



Syndactyly is the webbing or fusing together of two or more fingers or toes. (Photograph by Keith, Custom Medical Stock Photo. Reproduced by permission.)

a genetic cause. Some cases might be due to external factors like exposure to toxins or womb anomalies.

Diagnosis

Polydactyly and syndactyly can be diagnosed by external observation, x ray, and fetal sonogram.

Treatment

Polydactyly can be corrected by surgical removal of the extra digit or partial digit. Syndactyly can also be corrected surgically, usually with the addition of a skin graft from the groin.

Prognosis

The prognosis for isolated polydactyly and syndactyly is excellent. When polydactyly or syndactyly are part of a larger condition, the prognosis depends on the condition. Many of these conditions are quite serious, and early death may be the probable outcome.

Prevention

There is no known prevention for these conditions.

Resources

ORGANIZATIONS

March of Dimes Birth Defects Foundation. 1275 Mamaroneck Ave., White Plains, NY 10605. (914) 428-7100. resourcecenter@modimes.org. < <http://www.modimes.org> >.

National Institute of Child Health and Human Development. Bldg 31, Room 2A32, MSC 2425, 31 Center Drive, Bethesda, MD 20892-2425. (800) 505-2742. < <http://www.nichd.nih.gov/sids/sids.htm> >.

KEY TERMS

Autosomal chromosome—One of the non-X or non-Y chromosomes.

Congenital—A condition present at birth.

Digit—A finger or a toe.

Dominant trait—A genetic trait that will always express itself when present as one of a pair of genes (as opposed to a recessive trait where two copies of the gene are necessary to give the individual the trait).

Gene—A portion of a DNA molecule that either codes for a protein or RNA molecule or has a regulatory function.

Triploidy—The condition where an individual has three entire sets of chromosomes instead of the usual two.

Trisomy—An abnormal condition where three copies of one chromosome are present in the cells of an individual's body instead of two, the normal number.

OTHER

OMIM Home Page, *Online Mendelian Inheritance in Man*. < <http://www.ncbi.nlm.nih.gov/Omim> >.

Lorraine Lica, PhD

Polyendocrine deficiency syndromes see
Polyglandular deficiency syndromes

Polyglandular deficiency syndromes

Definition

Polyglandular deficiency syndromes are disorders characterized by the failure of more than one endocrine gland to make hormones in sufficient quantities for the body to function normally.

Description

the endocrine system is a diverse group of glands located all over the body that work together to regulate the body's metabolic activities. It includes:

- the pituitary gland, located deep in the brain, is considered the “master gland” that regulates many of the others
- the thyroid gland is located in the neck and sets the metabolic speed of many processes,
- the parathyroid glands, attached to the back of the thyroid, regulate calcium balance,
- the adrenal glands are located on top of the kidneys and make four separate kinds of hormones,
- the gonads (sex organs) produce sex hormones,
- the pancreas is responsible for the production of digestive juices, insulin, and glucagon.

There are over a dozen different syndromes that involve failure of more than one endocrine gland.

Causes and symptoms

The cause of polyglandular deficiency syndromes is usually an autoimmune response—a condition in which the body generates antibodies to its own tissues. The immune system may attack one or more glands; however, because of their inter dependence, the destruction of one gland can often lead to the impairment of another. Other causes may include infectious disease; insufficient blood flow to the glands due to an obstruction such as a blood clot; or the presence of a tumor.

Doctors usually group polyglandular deficiency syndromes into three types:

- Type I occurs during childhood and is characterized by failure of the adrenals, parathyroids, thyroid, and gonads combined with hepatitis, hair loss, skin pigment changes, and inability of the bowel to absorb adequate **nutrition**. These children also get a persistent skin fungus infection called candidiasis.
- Type II occurs during adulthood and is characterized by failure of the adrenals, thyroid (Schmidt’s syndrome), and gonads combined with similar nutritional failures and hair and skin changes. These patients also have **myasthenia gravis**. This type of polyglandular deficiency syndrome often produces insulin-dependent diabetes mellitus (IDDM).
- Type III disease may produce diabetes or adrenal failure combined with thyroid problems. It may also include baldness (**alopecia**), anemia, and **vitiligo** (condition characterized by white patches on normally pigmented skin).

Not all symptoms of any syndrome appear at once or in the same patient.

KEY TERMS

Antibody—A weapon in the body’s immune defense arsenal that attacks a specific antigen.

Congenital—Present at birth.

Myasthenia gravis—A disease that causes muscle weakness.

Rubella—German measles.

Syndrome—A collection of abnormalities that occur often enough to suggest they have a common cause.

Diagnosis

Because these diseases evolve over time, the final diagnosis may not appear for years. A family history is very helpful in knowing what to expect. Any single endocrine abnormality should heighten suspicion that there are others, since they so often occur together, both as underproduction and overproduction of hormones. Most hormone levels can be monitored through blood tests. Many of the antibodies that characterize these conditions can also be found by blood testing.

Treatment

Fortunately there are replacements available for all the missing hormones. Careful balancing of them all can provide a reasonably comfortable quality of life for these patients.

Resources

BOOKS

Sherman, Steven I., and Robert F. Gagel. “Disorders Affecting Multiple Endocrine Systems.” In *Harrison’s Principles of Internal Medicine*, edited by Anthony S. Fauci, et al. New York: McGraw-Hill, 1997.

J. Ricker Polsdorfer, MD

Polyhydramnios and oligohydramnios

Definition

Polyhydramnios is a high level and oligohydramnios is low level of amniotic fluid.

Description

Amniotic fluid is the liquid that surrounds the developing fetus during **pregnancy**. It is contained within the amniotic membrane that forms the amniotic sac (bag of waters). During the first three months after conception (first trimester), amniotic fluid is mainly derived from the blood plasma that diffuses through the thin tissues of the fetus into the surrounding space. After the fetal kidneys form and become functional at about 10–11 weeks, fetal urine becomes the main source of amniotic fluid and remains so for the rest of the pregnancy. In addition, the lungs also produce liquid that becomes part of the amniotic fluid. Other contributions come from fetal oral and nasal secretions and from the fetal surface of the placenta. Amniotic fluid removal is largely due to fetal swallowing and absorption into the fetal blood. Uptake also occurs across the placental surface. The volume of amniotic fluid normally increases throughout pregnancy, reaching a peak at about 32–33 weeks and remaining fairly constant or decreasing slightly thereafter. There is a wide range of normal fluid volumes with an average of 700–800 ml at 32–33 weeks. Through the processes of swallowing and urination, a fetus can recycle the entire volume in less than 24 hours. Because the normal values for amniotic fluid volume increase during pregnancy, the actual volume that constitutes polyhydramnios is dependent on the gestational age of the fetus. During the last two months of pregnancy, polyhydramnios usually refers to amniotic fluid volumes greater than 1,700–1,900 ml. Severe cases are associated with much greater fluid volume excesses. The range of fluid values diagnostic of oligohydramnios is not as wide as that for polyhydramnios. Less than 300 ml, or lower than the 5% percentile for gestational age, is usually considered the upper threshold.

Causes and symptoms

Polyhydramnios, also referred to as hydramnios, can have any one of a number of causes related either to an underlying maternal or fetal condition. Maternal diabetes, which is associated with a macrosomic (enlarged) fetus, is a common cause. The medication lithium, used to treat depression, can also increase amniotic fluid levels. Twin gestations are prone to polyhydramnios. Infections passed from mother to fetus such as **rubella**, cytomegalovirus, and **toxoplasmosis**, can also result in damage to the fetus and elevated amniotic fluid levels. Fetal abnormalities, including many that are life-threatening or lead to a significant impairment in the quality of life, are found in up to a quarter of

all patients. For this reason, the initial finding of excess amniotic fluid should be followed by thorough diagnostic studies to determine the cause and the prognosis.

Because fetal swallowing is a major factor in amniotic fluid removal, fetal abnormalities that prevent fluid uptake should be investigated. These include gastrointestinal obstructions such as **esophageal atresia** and duodenal atresia, as well as neurological conditions that affect swallowing including anencephaly. Certain cardiac abnormalities, kidney disorders, and genetic conditions such as **myotonic dystrophy** and alpha-thalassemia can also cause polyhydramnios. Fetal chromosome abnormalities are frequently associated with elevated amniotic fluid levels. The more severe the polyhydramnios the more likely it is that fetal abnormalities will be present. In addition, there are other, infrequent causes, and in a number of cases, no cause can be found. Polyhydramnios can lead to maternal abdominal discomfort and respiratory difficulties as well as preterm labor. When polyhydramnios is associated with fetal abnormalities, perinatal mortality is significantly increased.

Oligohydramnios is most commonly associated with abnormalities of the fetal kidneys. Since fetal urine is the main source of amniotic fluid in the latter two-thirds of pregnancy, any condition that interferes with fetal urine production can lead to oligohydramnios. Renal agenesis, cystic kidneys, and bladder outlet obstructions are common. Meckel-Gruber syndrome, a lethal autosomal recessive genetic disorder featuring brain and kidney abnormalities and extra digits is one specific cause. Placental insufficiency and fetal growth retardation can also result in oligohydramnios. **Premature rupture of membranes**, especially between 16 and 24 weeks is another cause and, because amniotic fluid is important in lung growth, it can lead to underdevelopment of the lungs (pulmonary hypoplasia). In general, regardless of the cause, oligohydramnios that arises early in a pregnancy, can cause hypoplastic lungs. It can also result in space limitations within the amniotic sac that cause fetal compression and orthopedic abnormalities such as clubbed feet in the newborn. In general, oligohydramnios that begins near the time of delivery is associated with a better outcome than cases that have an onset earlier in pregnancy.

Diagnosis

In current obstetrical practice, polyhydramnios and oligohydramnios are usually detected during a

routine prenatal ultrasound. If the ultrasonographer suspects that excess or reduced fluid is present, it is customary to take measurements of pockets of fluid visualized around the fetus, calculate the amniotic fluid index (AFI), and compare it to AFI values found in standard tables. Subsequent ultrasound measurements can then be used to track the increase or decrease in fluid.

It is extremely important that the cause of an abnormal AFI be sought. Because of the high risk of fetal abnormalities, detailed ultrasound exams (targeted exams) should then be performed. The mother should be counseled about the possible complications and offered additional testing as necessary. For example, an **amniocentesis** for prenatal chromosome analysis may be important because of the high risk of fetal chromosome abnormalities. This test is usually indicated if fetal abnormalities are suspected on the basis of the ultrasound exam. An amniocentesis can also be used to check for fetal infections and some rare single gene defects.

Treatment

Effective treatments for polyhydramnios and oligohydramnios are limited. To relieve maternal discomfort, an excess fluid level can be reduced by inserting a needle into the amniotic sac and using a syringe to withdraw excess fluid. This can be done repeatedly, if necessary. In oligohydramnios, the opposite approach of adding fluid either by increasing oral intake in the mother or by directly infusing saline into the amniotic sac has been tried in select cases. If the cause of oligohydramnios is a fetal bladder obstruction, it may be possible to place a small tube in the bladder to shunt the fluid into the amniotic sac.

Alternative treatment

In select cases where polyhydramnios is thought to be due to an increased output of fetal urine, the drug indomethacin has been used with some success, but there is concern about side effects, particularly on the fetus. Another similar drug, sulindac, is currently being investigated. If oligohydramnios is due to premature rupture of the membranes, a protocol to manage complications should be instituted.

Prognosis

The prognosis for both polyhydramnios and oligohydramnios depends on the cause. If excess or reduced amniotic fluid is the result of an underlying

KEY TERMS

Alpha-thalassemia—An inherited disorder that interferes with the normal production of hemoglobin.

Anencephaly—Congenital absence of the brain. Occurs during the first month of embryonic development.

Autosomal recessive—A pattern of inheritance in which both copies of an autosomal gene must be abnormal for a genetic condition or disease to occur. An autosomal gene is one that is located on one of the autosomes or non-sex chromosomes. When both parents have one abnormal copy of the same gene, they have a 25% chance with each pregnancy that their offspring will have the disorder.

Congenital—Present at birth.

Duodenal atresia—Closure or blockage of the duodenum, the upper section of the small intestine.

Esophageal atresia—Blockage or closure of the esophagus, the tube leading from the mouth to the stomach.

Gestational age—The estimated age of a fetus expressed in weeks, calculated from the first day of the last normal menstrual period.

Myotonic dystrophy—A genetic defect resulting in abnormal muscle function.

Placenta—The flat, spongy structure that forms within the uterus during pregnancy and provides nourishment to the developing fetus.

Renal agenesis—Failure of the fetal kidneys to form. Oligohydramnios usually associated with absence of both kidneys.

fetal abnormality, the nature of that abnormality will determine the prognosis. This is one reason why it is important to perform the necessary follow-up studies. A woman who has been diagnosed with polyhydramnios or oligohydramnios needs to be made fully aware of the types of testing available and carefully counseled about the diagnosis and its impact on the chance for a successful pregnancy outcome and a healthy infant.

Prevention

In order to prevent polyhydramnios or oligohydramnios, it would be necessary to prevent the underlying cause. Good control of maternal diabetes and the prevention of infections transmittable from

mother to fetus are two approaches for a subset of cases, but, in general, prevention is not possible.

Resources

BOOKS

Cunningham, F. Gary. *Williams Obstetrics*. New York; McGraw-Hill, 2001.

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PERIODICALS

Kilpatrick, Sarah J. "Therapeutic Interventions for Oligohydramnios: Amnioinfusion and Maternal Hydration" *Clinical Obstetric and GYN* 40, no. 2 (June 1996): 266-279.

Sallie Boineau Freeman, PhD

Polymyalgia rheumatica

Definition

Polymyalgia rheumatica is a syndrome that causes **pain** and stiffness in the hips and shoulders of people over the age of 50.

Description

Although the major characteristics of this condition are just pain and stiffness, there are reasons to believe it is more than just old-fashioned rheumatism. Patients are commonly so afflicted that their muscles atrophy from disuse. A similar complaint of such weakness is also seen in serious muscle diseases. Moreover, some patients develop arthritis or a disease called giant cell arteritis or **temporal arteritis**.

Causes and symptoms

This condition may arise as often as once in every 2,000 people. Rarely does it affect people under 50 years old. The average age is 70; women are afflicted twice as often as men. Along with the pain and stiffness of larger muscles, **headache** may add to the discomfort. The scalp is often tender. Pain is usually worse at night. There may be fever and weight loss before the full disease appears. Patients complain that stiffness is worse in the morning and returns if they have been inactive for any period of time, a condition called gelling. Sometimes the stiffness is severe enough that it causes frozen shoulder.

KEY TERMS

Anemia—A condition in which the blood lacks enough red blood cells (hemoglobin).

Atrophy—Wasting away of a body part.

Frozen shoulder—A shoulder that becomes scarred and cannot move.

Giant cell arteritis—Also called temporal arteritis. A condition which causes the inflammation of temporal arteries. It can cause blindness when the inflammation effects the ophthalmic artery.

NSAIDs—Non-steroidal anti-inflammatory drugs like aspirin, ibuprofen, and naproxen.

Syndrome—A collection of abnormalities that occur often enough to suggest they have a common cause.

Diagnosis

Symptoms are usually present for over a month by the time patients seek medical attention. A mild anemia is often present. One blood test, called an erythrocyte sedimentation rate, is very high, much more so than in most other diseases. The most important issue in evaluating polymyalgia rheumatica is to check for giant cell arteritis. Giant cell arteritis can lead to blindness if left untreated.

Treatment

Polymyalgia rheumatica responds dramatically to cortisone-like drugs in modest doses. In fact, one part of confirming the diagnosis is to observe the response to this treatment. It may also respond to **nonsteroidal anti-inflammatory drugs** (NSAIDs). Temporal arteritis is also treated with cortisone, but in higher doses.

Prognosis

The disease often remits after a while, with no further treatment required.

Resources

BOOKS

Griggs, Robert C. "Episodic Muscle Spasms, Cramps, and Weakness." In *Harrison's Principles of Internal Medicine*, edited by Anthony S. Fauci, et al. New York: McGraw-Hill, 1997.

J. Ricker Polsdorfer, MD

Polymyositis

Definition

Polymyositis is an inflammatory muscle disease causing weakness and **pain**. **Dermatomyositis** is identical to polymyositis with the addition of a characteristic skin rash.

Description

Polymyositis (PM) is an inflammatory disorder in which muscle tissue becomes inflamed and deteriorates, causing weakness and pain. It is one of several types of inflammatory muscle disease, or myopathy. Others include dermatomyositis (DM) and inclusion body **myositis**. All three types are progressive conditions, usually beginning in adulthood. A fourth type, juvenile dermatomyositis, occurs in children. Although PM and DM can occur at any age, 60% of cases appear between the ages of 30 and 60. Females are affected twice as often as males.

Causes and symptoms

Causes

The cause of PM and DM is not known, but it is suspected that a variety of factors may play a role in the development of these diseases. PM and DM may be autoimmune diseases, caused by the immune system's attack on the body's own tissue. The reason for this attack is unknown, although some researchers believe that a combination of immune system susceptibility and an environmental trigger may explain at least some cases. Known environmental agents associated with PM and DM include infectious agents such as *Toxoplasma*, *Borrelia* (**Lyme disease** bacterium), and coxsackievirus. Most cases, however, have no obvious triggers (direct causative agents). There may also be a genetic component in the development of PM and DM.

Symptoms

The early symptoms of PM and DM are slowly progressing muscle weakness, usually symmetrical between the two sides of the body. PM and DM affect primarily the muscles of the trunk and those closest to the trunk, while the hands, feet, and face usually are not involved. Weakness may cause difficulty walking, standing, and lifting objects. Rarely, the muscles of breathing may be affected. Weakness of the muscles used for swallowing can cause difficulty with swallowing (dysphagia). Joint pain and/or swelling also may

be present. Later in the course of these diseases, muscle wasting or shortening (contracture) may develop in the arms or legs. Heart abnormalities, including electrocardiogram (ECG) changes and **arrhythmias**, develop at some time during the course of these diseases in about 30% of patients.

Dermatomyositis is marked by a skin rash. The rash is dusky, reddish, or lilac in color, and is most often seen on the eyelids, cheeks, bridge of the nose, and knuckles, as well as on the back, upper chest, knees, and elbows. The rash often appears before the muscle weakness.

Diagnosis

PM and DM are often difficult diseases to diagnose, because they are rare, because symptoms come on slowly, and because they can be mistaken for other diseases that cause muscle weakness, especially limb girdle **muscular dystrophy**.

Accurate diagnosis involves:

- A neurological exam.
- Blood tests to determine the level of the muscle enzyme creatine kinase, whose presence in the circulation indicates muscle damage.
- Electromyography, an electrical test of muscle function.
- Muscle biopsy, in which a small sample of affected muscle is surgically removed for microscopic analysis. A biopsy revealing muscle cells surrounded by immune system cells is a strong indicator of myositis.

Treatment

PM and DM respond to high doses of immunosuppressant drugs in most cases. The most common medication used is the corticosteroid prednisone. Prednisone therapy usually leads to improvement within two or three months, at which point the dose can be tapered to a lower level to avoid the significant side effects associated with high doses of prednisone. Unresponsive patients are often given a replacement or supplementary immunosuppressant, such as azathioprine, cyclosporine, or methotrexate. Intravenous immunoglobulin treatments may help some people who are unresponsive to other immunosuppressants.

Pain can usually be controlled with an over-the-counter analgesic, such as **aspirin**, ibuprofen, or naproxen. A speech-language therapist can help suggest exercises and tips to improve difficulty in

KEY TERMS

Autoimmune disease—A diseases in which the body's immune system, responsible for fighting off foreign invaders such as bacteria and viruses, begins to attack and damage a part of the body as if it were foreign.

Immunosuppressant—A drug that reduces the body's natural immunity by suppressing the natural functioning of the immune system.

swallowing. Avoiding weight gain helps prevent overtaxing weakened muscles.

Alternative treatment

As with all autoimmune conditions, food allergies/intolerances and environmental triggers may be contributing factors. For **food allergies** and intolerances, an elimination challenge diet can be used under the supervision of a trained practitioner, naturopath, or nutritionist, to identify trigger foods. These foods can then be eliminated from the person's diet. For environmental triggers, it is helpful to identify the source so that it can be avoided or eliminated. A thorough **detoxification** program can help alleviate symptoms and change the course of the disease. Dietary changes from processed foods to whole foods that do not include allergen triggers can have significant results. Nutrient supplements, especially the antioxidants zinc, selenium, and **vitamins A, C, and E**, can be beneficial. Constitutional homeopathic treatment can work at a deep level to rebalance the whole person. **Acupuncture** and Chinese herbs can be effective in symptom alleviation and deep healing. Visualization, guided imagery, and hypnosis for **pain management** are also useful.

Prognosis

The progression of PM and DM varies considerably from person to person. Immunosuppressants can improve strength, although not all patients respond, and relapses may occur. PM and DM can lead to increasing weakness and disability, although the life span usually is not significantly affected. About half of the patients recover and can discontinue treatment within five years of the onset of their symptoms. About 20% still have active disease requiring ongoing treatment after five years, and about 30% have inactive disease but some remaining muscle weakness.

Prevention

There is no known way to prevent myositis, except to avoid exposure to those environmental agents that may be associated with some cases.

Resources

ORGANIZATIONS

Dermatomyositis and Polymyositis Support Group. 146 Newtown Road, Southampton, SO2 9HR, U.K.

Muscular Dystrophy Association. 3300 East Sunrise Drive, Tucson, AZ 85718. (800) 572-1717. < <http://www.mdasa.org> >.

Myositis Association of America. 600-D University Boulevard, Harrisonburg, VA 22801. (540) 433-7686. < <http://www.myositis.org> >.

National Institutes of Health. National Institute of Arthritis and Musculoskeletal and Skin Diseases. 900 Rockville Pike, Bethesda, MD 20892. (301) 496-8188. < <http://www.hih.gov.niams> >.

Richard Robinson

Polyneuritis see **Peripheral neuropathy**

Polysomnography

Definition

The word polysomnography, derived from the Greek roots “poly,” meaning many, “somno,” meaning sleep, and “graphy” meaning to write, refers to multiple tests performed on patients while they sleep. Polysomnography is an overnight test to evaluate **sleep disorders**. Polysomnography generally includes monitoring of the patient's airflow through the nose and mouth, blood pressure, electrocardiographic activity, blood oxygen level, brain wave pattern, eye movement, and the movement of respiratory muscle and limbs.

Purpose

Polysomnography is used to help diagnose and evaluate a number of sleep disorders. For instance, it can help diagnose **sleep apnea**, a common disorder in middle-aged and elderly obese men, in which the muscles of the soft palate in the back of the throat relax and close off the airway during sleep. This may cause the person to snore loudly and gasp for air at night, and to be excessively sleepy and doze off during the day. Another syndrome often evaluated by polysomnography is **narcolepsy**. In narcolepsy, people have

sudden attacks of sleep and/or cataplexy (temporary loss of muscle tone caused by moments of emotion, such as fear, anger, or surprise, which causes people to slump or fall over), sleep **paralysis** or **hallucinations** at the onset of sleep. Polysomnography is often used to evaluate parasomnias (abnormal behaviors or movements during sleep), such as sleep walking, talking in one's sleep, nightmares, and bedwetting. It can also be used to detect or evaluate seizures that occur in the middle of the night, when the patient and his or her family are unlikely to be aware of them.

Precautions

Polysomnography is extremely safe and no special precautions need to be taken.

Description

Polysomnography requires an overnight stay in a sleep laboratory. During this stay, while the patient sleeps, he or she is monitored in a number of ways that can provide very useful information.

One form of monitoring is **electroencephalography** (EEG), in which electrodes are attached to the patient's scalp in order to record his or her brain wave activity. The electroencephalograph records brain wave activity from different parts of the brain and charts them on a graph. The EEG not only helps doctors establish what stage of sleep the patient is in, but may also detect seizures.

Another form of monitoring is continuous electro-oculography (EOG), which records eye movement and is used to determine when the patient is going through a stage of sleep called rapid-eye-movement (REM) sleep. Both EEG and EOG can be helpful in determining sleep latency (the time that transpires between lights out and the onset of sleep), total sleep time, the time spent in each sleep stage, and the number of arousals from sleep.

The air flow through the patient's nose and mouth are measured by heat-sensitive devices called thermistors. This can help detect episodes of apnea (stopped breathing), or hypnopea (inadequate breathing). Another test called pulse oximetry measures the amount of oxygen in the blood, and can be used to assess the degree of oxygen **starvation** during episodes of hypnopea or apnea.

The electrical activity of the patient's heart is also measured on an electrocardiogram, or ECG. Electrodes are affixed to the patient's chest and they pick up electrical activity from various areas of the heart. They help detect cardiac arrhythmias (abnormal

heart rhythms), which may occur during periods of sleep apnea. Blood pressure is also measured: sometimes episodes of sleep apnea can dangerously elevate blood pressure.

In some cases, sleep laboratories monitor the movement of limbs during sleep. This can be helpful in detecting such sleep disorders as periodic limb movements.

Preparation

The patient may be asked to discontinue taking any medications used to help him/her sleep. Before the patient goes to sleep, the technician hooks him or her up to all of the monitors being used.

Aftercare

Once the test is over, the monitors are detached from the patient. No special measures need to be taken after polysomnography.

Normal results

A normal result in polysomnography shows normal results for all parameters (EEG, ECG, blood pressure, eye movement, air flow, pulse oximetry, etc.) monitored throughout all stages of sleep.

Abnormal results

Polysomnography may yield a number of abnormal results, indicating a number of potential disorders. For instance, abnormal transitions in and out of various stages of sleep, as documented by the EEG and the EOG, may be a sign of narcolepsy. Reduced air flow through the nose and mouth, along with a fall in oxygenation of the blood, may indicate apnea or hypopnea. If apnea is accompanied by abnormalities in ECG or elevations in blood pressure, this can indicate that sleep apnea may be particularly harmful. Frequent movement of limbs may indicate a sleep disorder called periodic limb movement.

Resources

ORGANIZATIONS

National Heart, Lung and Blood Institute. P.O. Box 30105, Bethesda, MD 20824-0105. (301) 251-1222.
< <http://www.nhlbi.nih.gov> >.

Robert Scott Dinsmoor

Pompe's disease see **Glycogen storage diseases**

Porphyrias

Definition

The porphyrias are disorders in which the body produces too much porphyrin and insufficient heme (an iron-containing nonprotein portion of the hemoglobin molecule). Porphyrin is a foundation structure for heme and certain enzymes. Excess porphyrins are excreted as waste in the urine and stool. Overproduction and overexcretion of porphyrins causes low, unhealthy levels of heme and certain important enzymes, creating various physical symptoms.

Description

Biosynthesis of heme is a multistep process that begins with simple molecules and ends with a large, complex heme molecule. Each step of the chemical pathway is directed by its own task-specific protein, called an enzyme. As a heme precursor molecule moves through each step, an enzyme modifies the precursor in some way. If a precursor molecule is not modified, it cannot proceed to the next step, causing a buildup of that specific precursor.

This situation is the main characteristic of the porphyrias. Owing to a defect in one of the enzymes of the heme biosynthesis pathway, protoporphyrins or porphyrins (heme precursors) are prevented from proceeding further along the pathway. These precursors accumulate at the stage of the enzyme defect causing an array of physical symptoms in an affected person. Specific symptoms depend on the point at which heme biosynthesis is blocked and which precursors accumulate. In general, the porphyrias primarily affect the skin and the nervous system. Symptoms can be debilitating or life threatening in some cases. Porphyria is most commonly an inherited condition. It can also, however, be acquired after exposure to poisonous substances.

Heme

Heme is produced in several tissues in the body, but its primary biosynthesis sites are the liver and the bone marrow. Heme synthesis for immature red blood cells, namely the erythroblasts and the reticulocytes, occurs in the bone marrow.

Although production is concentrated in the liver and bone marrow, heme is utilized in various capacities in virtually every tissue in the body. In most cells, heme is a key building block in the construction of factors that oversee metabolism and transport of

oxygen and energy. In the liver, heme is a component of several vital enzymes, particularly cytochrome P450. Cytochrome P450 is involved in the metabolism of chemicals, **vitamins**, fatty acids, and hormones; it is very important in transforming toxic substances into easily excretable materials. In immature red blood cells, heme is the featured component of hemoglobin. Hemoglobin is the red pigment that gives red blood cells their characteristic color and their essential ability to transport oxygen.

Heme biosynthesis

The heme molecule is composed of porphyrin and an iron atom. Much of the heme biosynthesis pathway is dedicated to constructing the porphyrin molecule. Porphyrin is a large molecule shaped like a four-leaf clover. An iron atom is placed at its center point in the last step of heme biosynthesis.

The production of heme may be compared to a factory assembly line. At the start of the line, raw materials are fed into the process. At specific points along the line, an addition or adjustment is made to further development. Once additions and adjustments are complete, the final product rolls off the end of the line.

The heme “assembly line” is an eight-step process, requiring eight different and properly functioning enzymes:

1. delta-aminolevulinic acid synthase
2. delta-aminolevulinic acid dehydratase
3. porphobilogen deaminase
4. uroporphyrinogen III cosynthase
5. uroporphyrinogen decarboxylase
6. coproporphyrinogen oxidase
7. protoporphyrinogen oxidase
8. ferrochelatase

The control of heme biosynthesis is complex. Various chemical signals can trigger increased or decreased production. These signals can affect the enzymes themselves or the production of these enzymes, starting at the genetic level. For example, one point at which heme biosynthesis may be controlled is at the first step. When heme levels are low, greater quantities of delta-aminolevulinic acid (ALA) synthase are produced. As a result, larger quantities of heme precursors are fed into the biosynthesis pathway to step up heme production.

Porphyrias

Under normal circumstances, when heme concentrations are at an appropriate level, precursor

production decreases. However, a glitch in the biosynthesis pathway—represented by a defective enzyme—means that heme biosynthesis does not reach completion. Because heme levels remain low, the synthesis pathway continues to churn out precursor molecules in an attempt to correct the heme deficit.

The net effect of this continued production is an abnormal accumulation of precursor molecules and development of some type of porphyria. Each type of porphyria corresponds with a specific enzyme defect and an accumulation of the associated precursor. Although there are eight steps in heme biosynthesis, there are only seven types of porphyrias; a defect in ALA synthase activity does not have a corresponding porphyria.

Enzymes involved in heme biosynthesis display subtle, tissue-specific variations; therefore, heme biosynthesis may be impeded in the liver, but normal in the immature red blood cells, or vice versa. Incidence of porphyria varies widely between types and occasionally by geographic location. Although certain porphyrias are more common than others, their greater frequency is only relative to other types. All porphyrias are considered to be rare disorders.

In the past, the porphyrias were divided into two general categories based on the location of the porphyrin production. Porphyrins affecting heme biosynthesis in the liver were referred to as hepatic porphyrias. Porphyrins that affect heme biosynthesis in immature red blood cells were referred to as erythropoietic porphyrias (erythropoiesis is the process through which red blood cells are produced). As of 2001, porphyrias were usually grouped into acute and non-acute types. Acute porphyrias produce severe attacks of **pain** and neurological effects. Non-acute porphyrias present as chronic diseases.

The acute porphyrias, and the heme biosynthesis steps at which enzyme defects occur, are:

- ALA dehydratase deficiency porphyria (step 2). This porphyria type is very rare. The inheritance pattern appears to be autosomal recessive. In autosomal recessively inherited disorders, a person must inherit two defective genes, one from each parent. A parent with only one gene for an autosomal recessive disorder does not display symptoms of the disease.
- Acute intermittent porphyria (step 3). Acute intermittent porphyria (AIP) is also known as Swedish porphyria, pyroloporphyria, and intermittent acute porphyria. AIP is inherited as an autosomal dominant trait, which means that only one copy of the defective gene needs to be present for the disorder to

occur. Simply inheriting this gene, however, does not necessarily mean that a person will develop the disease. Approximately five to 10 per 100,000 people in the United States carry a gene for AIP, but only 10% of these people ever develop symptoms of the disease.

- Hereditary coproporphyria (step 6). Hereditary coproporphyria (HCP) is inherited in an autosomal dominant manner. As with all porphyrias, it is an uncommon ailment. By 1977, only 111 cases of HCP were recorded; in Denmark, the estimated incidence is two in one million people.
- Variegate porphyria (step 7). Variegate porphyria (VP) is also known as porphyria variegata, protocoproporphyria, South African genetic porphyria, and Royal malady (supposedly King George III of England and Mary, Queen of Scots, suffered from VP). VP is inherited in an autosomal dominant manner and is especially prominent in South Africans of Dutch descent. Among that population, the incidence is approximately three in 1,000 persons. It is estimated that there are 10,000 cases of VP in South Africa. Interestingly, it appears that the affected South Africans are descendants of two Dutch settlers who came to South Africa in 1680. Among other populations, the incidence of VP is estimated to be one to two cases per 100,000 persons.

The non-acute porphyrias, and the steps of heme biosynthesis at which they occur, are:

- Congenital erythropoietic porphyria (step 4). Congenital erythropoietic porphyria (CEP) is also called Gunther's disease, erythropoietic porphyria, congenital porphyria, congenital hematoporphyria, and erythropoietic uroporphyria. CEP is inherited in an autosomal recessive manner. It is a rare disease, estimated to affect fewer than one in one million people. Onset of dramatic symptoms usually occurs in infancy, but may hold off until adulthood.
- Porphyria cutanea tarda (step 5). Porphyria cutanea tarda (PCT) is also called symptomatic porphyria, porphyria cutanea symptomatica, and idiosyncratic porphyria. PCT may be acquired, typically as a result of disease (especially **hepatitis C**), drug or alcohol use, or exposure to certain poisons. PCT may also be inherited as an autosomal dominant disorder, however most people remain latent—that is, symptoms never develop. PCT is the most common of the porphyrias, but the incidence of PCT is not well defined.
- Hepatoerythropoietic porphyria (step 5). Hepatoerythropoietic porphyria (HEP) affects heme biosynthesis in both the liver and the bone marrow. HEP results from a defect in uroporphyrinogen

decarboxylase activity (step 5), and is caused by defects in the same gene as PCT. Disease symptoms, however, strongly resemble congenital erythropoietic porphyria. HEP seems to be inherited in an autosomal recessive manner.

- Erythropoietic protoporphyria (step 8). Also known as protoporphyria and erythrohepatic protoporphyria, erythropoietic protoporphyria (EPP) is more common than CEP; more than 300 cases have been reported. In these cases, onset of symptoms typically occurred in childhood.

Causes and symptoms

General characteristics

The underlying cause of all porphyrias is a defective enzyme important to the heme biosynthesis pathway. Porphyrias are inheritable conditions. In virtually all cases of porphyria an inherited factor causes the enzyme's defect. An environmental trigger—such as diet, drugs, or sun exposure—may be necessary before any symptoms develop. In many cases, symptoms do not develop. These asymptomatic individuals may be completely unaware that they have a gene for porphyria.

All of the hepatic porphyrias—except porphyria cutanea tarda—follow a pattern of acute attacks separated by periods during which no symptoms are present. For this reason, this group is often referred to as the acute porphyrias. The erythropoietic porphyrias and porphyria cutanea tarda do not follow this pattern and are considered to be chronic conditions.

The specific symptoms of each porphyria vary based on which enzyme is affected and whether that enzyme occurs in the liver or in the bone marrow. The severity of symptoms can vary widely, even within the same type of porphyria. If the porphyria becomes symptomatic, the common factor between all types is an abnormal accumulation of protoporphyrins or porphyrin.

ALA dehydratase porphyria (ADP)

ADP is characterized by a deficiency of ALA dehydratase. ADP is caused by mutations in the delta-aminolevulinate dehydratase gene (ALAD) at 9q34. Of the few cases on record, the prominent symptoms are **vomiting**, pain in the abdomen, arms, and legs, and neuropathy. (Neuropathy refers to nerve damage that can cause pain, **numbness**, or paralysis.) The nerve damage associated with ADP could cause breathing impairment or lead to weakness or **paralysis** of the arms and legs.

Acute intermittent porphyria (AIP)

AIP is caused by a deficiency of porphobilogen deaminase, which occurs due to mutations in the hydroxymethylbilane synthase gene (HMBS) located at 11q23.3. Symptoms of AIP usually do not occur unless a person with the deficiency encounters a trigger substance. Trigger substances can include hormones (for example oral contraceptives, menstruation, **pregnancy**), drugs, and dietary factors. Most people with this deficiency never develop symptoms.

Attacks occur after **puberty** and commonly feature severe abdominal pain, **nausea**, vomiting, and **constipation**. Muscle weakness and pain in the back, arms, and legs are also typical symptoms. During an attack, the urine is a deep reddish color. The central nervous system may also be involved. Possible psychological symptoms include **hallucinations**, confusion, seizures, and mood changes.

Congenital erythropoietic porphyria (CEP)

CEP is caused by a deficiency of uroporphyrinogen III cosynthase due to mutations in the uroporphyrinogen III cosynthase gene (UROS) located at 10q25.2-q26.3. Symptoms are often apparent in infancy and include reddish urine and possibly an enlarged spleen. The skin is unusually sensitive to light and blisters easily if exposed to sunlight. (Sunlight induces protoporphyrin changes in the plasma and skin. These altered protoporphyrin molecules can cause skin damage.) Increased hair growth is common. Damage from recurrent blistering and associated skin infections can be severe. In some cases facial features and fingers may be lost to recurrent damage and infection. Deposits of protoporphyrins can sometimes lead to red staining of the teeth and bones.

Porphyria cutanea tarda (PCT)

PCT is caused by deficient uroporphyrinogen decarboxylase. PCT is caused by mutations in the uroporphyrinogen decarboxylase gene (UROD) located at 1p34. PCT may occur as an acquired or an inherited condition. The acquired form usually does not appear until adulthood. The inherited form may appear in childhood, but often demonstrates no symptoms. Early symptoms include blistering on the hands, face, and arms following minor injuries or exposure to sunlight. Lightening or darkening of the skin may occur along with increased hair growth or loss of hair. Liver function is abnormal but the signs are mild.

Hepatoerythropoietic porphyria (HEP)

HEP is linked to a deficiency of uroporphyrinogen decarboxylase in both the liver and the bone marrow. HEP is an autosomal recessive disease caused by mutations in the gene responsible for PCT, the uroporphyrinogen decarboxylase gene (UROD), located at 1p34. The gene is shared, but the mutations, inheritance, and specific symptoms of these two diseases are different. The symptoms of HEP resemble those of CEP.

Hereditary coproporphyria (HCP)

HCP is similar to AIP, but the symptoms are typically milder. HCP is caused by a deficiency of coproporphyrinogen oxidase due to mutations in a gene by the same name at 3q12. The greatest difference between HCP and AIP is that people with HCP may have some skin sensitivity to sunlight. However, extensive damage to the skin is rarely seen.

Variegate porphyria (VP)

VP is caused by a deficiency of protoporphyrinogen oxidase. There is scientific evidence that VP is caused by mutation in the gene for protoporphyrinogen oxidase located at 1q22. Like AIP, symptoms of VP occur only during attacks. Major symptoms of this type of porphyria include neurological problems and sensitivity to light. Areas of the skin that are exposed to sunlight are susceptible to burning, blistering, and scarring.

Erythropoietic protoporphyria (EPP)

Owing to deficient ferrochelatase, the last step in the heme biosynthesis pathway—the insertion of an iron atom into a porphyrin molecule—cannot be completed. This enzyme deficiency is caused by mutations in the ferrochelatase gene (FECH) located at 18q21.3. The major symptoms of this disorder are related to sensitivity to light—including both artificial and natural light sources. Following exposure to light, a person with EPP experiences burning, **itching**, swelling, and reddening of the skin. Blistering and scarring may occur but are neither common nor severe. EPP is associated with increased risks for **gallstones** and liver complications. Symptoms can appear in childhood and tend to be more severe during the summer when exposure to sunlight is more likely.

Diagnosis

Depending on the array of symptoms an individual may exhibit, the possibility of porphyria may not immediately come to a physician's mind. In the absence of a family history of porphyria, non-specific

symptoms, such as abdominal pain and vomiting, may be attributed to other disorders. Neurological symptoms, including confusion and hallucinations, can lead to an initial suspicion of psychiatric disease. Diagnosis is more easily accomplished in cases in which non-specific symptoms appear in combination with symptoms more specific to porphyria, like neuropathy, sensitivity to sunlight, or certain other manifestations. Certain symptoms, such as urine the color of port wine, are hallmark signs very specific to porphyria. DNA analysis is not yet of routine diagnostic value.

A common initial test measures protoporphyrins in the urine. However, if skin sensitivity to light is a symptom, a blood plasma test is indicated. If these tests reveal abnormal levels of protoporphyrins, further tests are done to measure heme precursor levels in red blood cells and the stool. The presence and estimated quantity of porphyrin and protoporphyrins in biological samples are easily detected using spectrofluorometric testing. Spectrofluorometric testing uses a spectrofluorometer that directs light of a specific strength at a fluid sample. The porphyrins and protoporphyrins in the sample absorb the light energy and fluoresce, or glow. The spectrofluorometer detects and measures fluorescence, which indicates the amount of porphyrins and protoporphyrins in the sample.

Whether heme precursors occur in the blood, urine, or stool gives some indication of the type of porphyria, but more detailed biochemical testing is required to determine their exact identity. Making this determination yields a strong indicator of which enzyme in the heme biosynthesis pathway is defective; which, in turn, allows a diagnosis of the particular type of porphyria.

Biochemical tests rely on the color, chemical properties, and other unique features of each heme precursor. For example, a screening test for acute intermittent porphyria (AIP) is the Watson-Schwartz test. In this test, a special dye is added to a urine sample. If one of two heme precursors—porphobilinogen or urobilinogen—is present, the sample turns pink or red. Further testing is necessary to determine whether the precursor present is porphobilinogen or urobilinogen—only porphobilinogen is indicative of AIP.

Other biochemical tests rely on the fact that heme precursors become less soluble in water (able to be dissolved in water) as they progress further through the heme biosynthesis pathway. For example, to determine whether the Watson-Schwartz urine test is positive for porphobilinogen or urobilinogen, chloroform is added to the test tube. Chloroform is a water-insoluble substance. Even after vigorous mixing, the water and chloroform separate into two distinct layers.

Urobilinogen is slightly insoluble in water, while porphobilinogen tends to be water soluble. The porphobilinogen mixes more readily in water than chloroform, so if the water layer is pink (from the dye added to the urine sample), that indicates the presence of porphobilinogen, and a diagnosis of AIP is probable.

As a final test, measuring specific enzymes and their activities may be done for some types of porphyrias; however, such tests are not done as a screening method. Certain enzymes, such as porphobilinogen deaminase (the defective enzyme in AIP), can be easily extracted from red blood cells; other enzymes, however, are less readily collected or tested. Basically, an enzyme test involves adding a certain amount of the enzyme to a test tube that contains the precursor it is supposed to modify. Both the production of modified precursor and the rate at which it appears can be measured using laboratory equipment. If a modified precursor is produced, the test indicates that the enzyme is doing its job. The rate at which the modified precursor is produced can be compared to a standard to measure the efficiency of the enzyme.

Treatment

Treatment for porphyria revolves around avoiding acute attacks, limiting potential effects, and treating symptoms. Treatment options vary depending on the specific type of porphyria diagnosed. **Gene therapy** has been successful for both CEP and EPP. In the future, scientists expect development of gene therapy for the remaining porphyrias. Given the rarity of ALA dehydratase porphyria, definitive treatment guidelines for this rare type have not been developed.

Acute intermittent porphyria, hereditary coproporphyria, and variegate porphyria

Treatment for acute intermittent porphyria, hereditary coproporphyria, and variegate porphyria follows the same basic regime. A person who has been diagnosed with one of these porphyrias can prevent most attacks by avoiding precipitating factors, such as certain drugs that have been identified as triggers for acute porphyria attacks. Individuals must maintain adequate nutrition, particularly with respect to carbohydrates. In some cases, an attack can be stopped by increasing carbohydrate consumption or by receiving carbohydrates intravenously. In 2004, a report from Turkey revealed successful treatment of an acute intermittent porphyria attack with a drug called fluoxetine.

When attacks occur prompt medical attention is necessary. Pain is usually severe, and narcotic

analgesics are the best option for relief. Phenothiazines can be used to counter nausea, vomiting, and **anxiety**, and chloral hydrate or diazepam is useful for **sedation** or to induce sleep. Hematin, a drug administered intravenously, may be used to halt an attack. Hematin seems to work by signaling the pathway of heme biosynthesis to slow production of precursors. Women, who tend to develop symptoms more frequently than men owing to hormonal fluctuations, may find ovulation-inhibiting hormone therapy to be helpful.

Gene therapy is a possible future treatment for these porphyrias. An experimental animal model of AIP has been developed and research is in progress.

Congenital erythropoietic porphyria

The key points of congenital erythropoietic porphyria treatment are avoiding exposure to sunlight and prevention of skin trauma or skin infection. Liberal use of **sunscreens** and consumption of beta-carotene supplements can provide some protection from sun-induced damage. Medical treatments such as removing the spleen or administering transfusions of red blood cells can create short-term benefits, but these treatments do not offer a cure. Remission can sometimes be achieved after treatment with oral doses of activated charcoal. Severely affected patients may be offered bone marrow transplantation which appears to confer long-term benefit.

Porphyria cutanea tarda

As with other porphyrias, the first line of defense is avoidance of factors, especially alcohol, that could bring about symptoms. Regular blood withdrawal is a proven therapy for pushing symptoms into remission. If an individual is anemic or cannot have blood drawn for other reasons, chloroquine therapy may be used.

Erythropoietic protoporphyria

Avoiding sunlight, using sunscreens, and taking beta-carotene supplements are typical treatment options for erythropoietic protoporphyria. The drug cholestyramine may reduce the skin's sensitivity to sunlight as well as the accumulated heme precursors in the liver. **Liver transplantation** has been used in cases of liver failure. In 2004, a report in a medical journal told of one case of successful treatment of a 19-year-old patient with acute intermittent porphyria with liver transplantation. While she had only been studied for 1.5 years, the authors said her quality of life was good and they hoped that the procedure would offer cure for select patients with severe forms of the disease.

Alternative treatment

Acute porphyria attacks can be life-threatening events, so attempts at self-treatment can be dangerous. Alternative treatments can be useful adjuncts to conventional therapy. For example, some people may find relief for the pain associated with acute intermittent porphyria, hereditary coproporphyria, or variegate porphyria through **acupuncture** or hypnosis. Relaxation techniques, such as **yoga** or **meditation**, may also prove helpful in pain management.

Prognosis

Even when porphyria is inherited, symptom development depends on a variety of factors. In the majority of cases, a person remains asymptomatic throughout life. About one percent of acute attacks can be fatal. Other symptoms may be associated with temporarily debilitating or permanently disfiguring consequences. Measures to avoid these consequences are not always successful, regardless of how diligently they are pursued. Although pregnancy has been known to trigger porphyria attacks, dangers associated with pregnancy as not as great as was once thought.

Prevention

For the most part, the porphyrias are attributable to inherited genes; such inheritance cannot be prevented. However, symptoms can be limited or prevented by avoiding factors that trigger symptom development.

People with a family history of an acute porphyria should be screened for the disease. Even if symptoms are absent, it is useful to know about the presence of the gene to assess the risks of developing the associated porphyria. This knowledge also reveals whether a person's offspring may be at risk. Prenatal testing for certain porphyrias is possible. Prenatal diagnosis of congenital erythropoietic porphyria has been successfully accomplished. Any prenatal tests, however, would not indicate whether a child would develop porphyria symptoms; only that the potential is there.

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KEY TERMS

Autosomal dominant—A pattern of genetic inheritance in which only one abnormal gene is needed to display the trait or disease.

Autosomal recessive—A pattern of genetic inheritance in which two abnormal genes are needed to display the trait or disease.

Biosynthesis—The manufacture of materials in a biological system.

Bone marrow—A spongy tissue located in the hollow centers of certain bones, such as the skull and hip bones. Bone marrow is the site of blood cell generation.

Enzyme—A protein that catalyzes a biochemical reaction or change without changing its own structure or function.

Erythropoiesis—The process through which new red blood cells are created; it begins in the bone marrow.

Erythropoietic—Referring to the creation of new red blood cells.

Gene—A building block of inheritance, which contains the instructions for the production of a particular protein, and is made up of a molecular sequence found at a section of DNA. Each gene is found on a precise location on a chromosome.

Hematin—A drug administered intravenously to halt an acute porphyria attack. It causes heme biosynthesis to decrease, preventing the further accumulation of heme precursors.

Heme—The iron-containing molecule in hemoglobin that serves as the site for oxygen binding.

Hemoglobin—Protein-iron compound in the blood that carries oxygen to the cells and carries carbon dioxide away from the cells.

Hepatic—Referring to the liver.

Neuropathy—A condition caused by nerve damage. Major symptoms include weakness, numbness, paralysis, or pain in the affected area.

Porphyrin—A large molecule shaped like a four-leaf clover. Combined with an iron atom, it forms a heme molecule.

Protoporphyrin—A precursor molecule to the porphyrin molecule.

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OTHER

Gene Clinics. <<http://www.geneclinics.org>>.

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Online Mendelian Inheritance in Man (OMIM). <<http://www3.ncbi.nlm.nih.gov/Omim>>.

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Port-wine stain see **Birthmarks**

Portacaval shunting see **Portal vein bypass**

Portal-systemic encephalopathy see **Liver encephalopathy**

Portal vein bypass

Definition

Portal vein bypass surgery diverts blood from the portal vein into another vein. It is performed when pressure in the portal vein is so high that it causes internal bleeding from blood vessels in the esophagus.

Purpose

The portal vein carries blood from the stomach and abdominal organs to the liver. It is a major vein that splits into many branches. High pressure in the portal vein causes swelling and bleeding from blood vessels in the esophagus. This situation occurs when the liver is damaged from **cirrhosis** of the liver, a condition usually caused by prolonged, excessive alcohol consumption.

Massive internal bleeding caused by high pressure in the portal vein occurs in about 40% of patients with cirrhosis. It is initially fatal in at least half of these patients. Patients who survive are likely to experience bleeding recurrence. Portal vein bypass, also called portacaval shunting, is performed on these surviving patients to control bleeding.

KEY TERMS

Cirrhosis—A chronic degenerative liver disease common among alcoholics.

Inferior vena cava—A large vein that returns blood from the legs, pelvis, and abdomen to the heart.

Portal vein—Formed by a fusion of small veins that end in a network of capillaries, the portal vein delivers blood to the liver.

Precautions

Most patients who need portal vein bypass surgery not only have **liver disease** and poor liver function, but also suffer from an enlarged spleen, jaundice, and damage to the vascular system brought on by years of **alcoholism**. They are likely to experience serious complications during surgery. Some patients are aggressively uncooperative with medical personnel. Under these conditions, half the patients may not survive the operation.

Description

A choice of portal vein bypasses is available. Portal vein bypass is usually performed as an emergency operation in a hospital under **general anesthesia**. The surgeon makes an abdominal incision and finds the portal vein. In portacaval shunting, blood from the portal vein is diverted into the inferior vena cava. This is the most common bypass. In splenorenal shunting, the splenic vein (a part of the portal vein), is connected to the renal vein. A mesocaval shunt connects the superior mesenteric vein (another part of the portal vein) to the inferior vena cava.

Portal pressure can also be reduced in a procedure called transvenous intrahepatic portosystemic shunt (TIPS). A catheter is threaded into the portal vein, and an expandable balloon or wire mesh is inserted to divert blood from the portal vein to the hepatic vein. The rate of serious complications in TIPS is only 1–2%. The operation cannot be performed at all hospitals, but is becoming the preferred treatment for reducing portal pressure.

Preparation

Standard preoperative blood and urine tests are performed, and liver function is evaluated. The heart and arterial blood pressure are monitored both during and after the operation.

Aftercare

The patient will be connected to a heart monitor and fed through a nasogastric tube. Vital functions are monitored through blood and urine tests. Patients receive **pain** medication and **antibiotics**. Once released from the hospital, patients are expected to abstain from alcohol and follow a diet and medication schedule designed to reduce the risks of re-bleeding.

Risks

Portal vein bypass surgery is high risk because it is performed on patients who are generally in poor health. Only half the patients survive, although the chances of survival are greater with TIPS surgery. Those patients who survive the operation still face the risk of **heart failure**, brain disease due to a decrease in the liver's conversion of waste products (**liver encephalopathy**), hemorrhage, lung complications, infection, **coma**, and **death**.

Normal results

The survival rate is directly related to the amount of liver damage patients have. The less damage, the more likely the patient is to recover. Cooperation with restrictions on alcohol and diet affect long-term survival.

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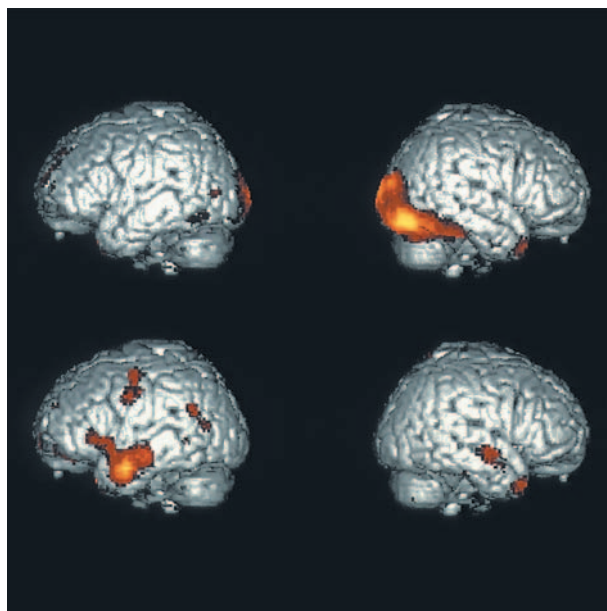
Positron emission tomography (PET)

Definition

Positron emission tomography (PET) is a scanning technique used in conjunction with small amounts of radiolabeled compounds to visualize brain anatomy and function.

Purpose

PET was the first scanning method to provide information on brain function as well as anatomy. This information includes data on blood flow, oxygen



A PET scan showing brain activity while patient recognizes faces—left sides at left/right sides at right. Activity is prevalent in temporal lobe (bottom scans). (Photo Researchers. Reproduced by permission.)

consumption, glucose metabolism, and concentrations of various molecules in brain tissue.

PET has been used to study brain activity in various neurological diseases and disorders, including **stroke**; epilepsy; Alzheimer's disease, Parkinson's disease, and Huntington's disease; and in some psychiatric disorders, such as schizophrenia, depression, **obsessive-compulsive disorder**, **attention-deficit/hyperactivity disorder**, and Tourette syndrome. PET studies have helped to identify the brain mechanisms that operate in drug **addiction**, and to shed light on the mechanisms by which individual drugs work. PET is also proving to be more accurate than other methods in the diagnosis of many types of **cancer**. In the treatment of cancer, PET can be used to determine more quickly than conventional tests whether a given therapy is working. PET scans also give accurate and detailed information on heart disease, particularly in women, in whom breast tissue can interfere with other types of tests.

Description

A very small amount of a radiolabeled compound is inhaled by or injected into the patient. The injected or inhaled compound accumulates in the tissue to be studied. As the radioactive atoms in the compound decay, they release smaller particles called positrons, which are positively charged. When

KEY TERMS

Electron—One of the small particles that make up an atom. An electron has the same mass and amount of charge as a positron, but the electron has a negative charge.

Gamma ray—A high-energy photon, emitted by radioactive substances.

Half-life—The time required for half of the atoms in a radioactive substance to disintegrate.

Photon—A light particle.

Positron—One of the small particles that make up an atom. A positron has the same mass and amount of charge as an electron, but the positron has a positive charge.

a positron collides with an electron (negatively charged), they are both annihilated, and two photons (light particles) are emitted. The photons move in opposite directions and are picked up by the detector ring of the PET scanner. A computer uses this information to generate three-dimensional, cross-sectional images that represent the biological activity where the radiolabeled compound has accumulated.

A related technique is called single photon emission computed tomography (CT) scan (SPECT). SPECT is similar to PET, but the compounds used contain heavier, longer-lived radioactive atoms that emit high-energy photons, called gamma rays, instead of positrons. SPECT is used for many of the same applications as PET, and is less expensive than PET, but the resulting picture is usually less sharp than a PET image and reveals less information about the brain.

Risks

Some of radioactive compounds used for PET or SPECT scanning can persist for a long time in the body. Even though only a small amount is injected each time, the long half-lives of these compounds can limit the number of times a patient can be scanned.

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Lisa Christenson, PhD

Post-concussion syndrome

Definition

Post-concussion syndrome (PCS) is a common but controversial disorder that presents with variety of symptoms including—but not limited to—headache, **dizziness**, **fatigue**, and personality changes.

Description

PCS occurs in approximately 23–93% of persons with mild to severe head injuries. It is estimated that a neurologist (a physician who specializes in nerve and brain disorders) sees five patients with PCS per month. There is no accurate correlation between the severity of injury and the development of PCS symptoms, since signs of the disorder can occur in someone who was just dazed by an injury. Some studies suggest that PCS symptoms occur at a higher rate in patients who were unconscious after trauma.

Causes and symptoms

PCS is most commonly caused by minor **head injury** called a **concussion**. The majority of patients with minor head injury characteristically develop PCS with distinct symptoms. Patients may report problems with concentration, recent memory, and abstract thinking. Additionally, patients may develop dizziness, irritability, fatigue, and personality changes. Elderly patients are particularly affected by disequilibrium and chronic dizziness even after minor trauma.

Diagnosis

There are no specific or reliable tests to diagnose PCS. A neuropsychologist can perform an in-depth neuropsychologic assessment that can determine presence or absence and extent of impairment. These tests may be performed for medical purposes.

Treatment

Treatment for PCS can be extensive. Medications for **headache** and **pain** may be indicated (analgesics and muscle relaxants). Antidepressants may be given to improve **insomnia**, irritability, or anxiety. Pain control could be achieved with **acupuncture**, nerve blocks, or transcutaneous electrical nerve stimulation (TENS, electrical stimulation of muscle groups). It is important for clinicians to educate caretakers and to provide

KEY TERMS

Disequilibrium—Difficulty with equilibrium that can mean a deficiency in balance and/or orientation.

Neuropsychologist—A clinical psychologist who specializes in assessing psychological status caused by a brain disorder.

referrals for family therapy and cognitive **rehabilitation** for the affected person.

Prognosis

The overall outcome is difficult to assess. Limited interpretation in literature is primarily due to the subjective nature of symptoms. Patient recovery is directed and evaluated by cognitive function changes, subjective symptoms, and return to work. Most cases of PCS can be a financial strain and threaten family stability. There may be compensation and litigation claims, which is often stressful and aggravates symptoms.

Resources

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Laith Farid Gulli, M.D.

Post-herpetic neuralgia see **Neuralgia**

Post-traumatic stress disorder

Definition

Post-traumatic **stress** disorder (PTSD) is a debilitating psychological condition triggered by a major traumatic event, such as **rape**, war, a terrorist act, **death** of a loved one, a natural disaster, or a catastrophic accident. It is marked by upsetting memories or thoughts of the ordeal, “blunting” of emotions, increased arousal, and sometimes severe personality changes.

Description

Officially termed post-traumatic stress disorder since 1980, PTSD was once known as shell shock or battle **fatigue** because of its more common manifestation in war veterans. However in the past 20 years, PTSD has been diagnosed in rape victims and victims of violent crime; survivors of natural disasters; the families of loved ones lost in the downing of Flight 103 over Lockerbie, Scotland; and survivors of the 1993 World Trade Center bombing, the 1995 Oklahoma City bombing, the random school and workplace shootings, and the release of poisonous gas in a Japanese subway; and, most recently, in the September 11, 2001, World Trade Center and Pentagon terrorist attacks. PTSD can affect adults of all ages. Statistics gathered from past events indicate that the risk of PTSD increases in order of the following factors.

- female gender
- middle-aged (40 to 60 years old)
- little or no experience coping with traumatic events
- ethnic minority
- lower socioeconomic status (SES)
- children in the home
- women with spouses exhibiting PTSD symptoms
- pre-existing psychiatric conditions
- primary exposure to the event including injury, life-threatening situation, and loss
- living in traumatized community

For example, over a third of the Oklahoma City bombing survivors developed PTSD and over half showed signs of **anxiety**, depression, and alcohol **abuse**. Over one year later, Oklahomans in general had a increased use of alcohol and tobacco products, as well as PTSD symptoms.

Children are also susceptible to PTSD and their risk is increased exponentially as their exposure to the event increases. Children experiencing abuse, the death of a parent, or those located in a community suffering a traumatic event can develop PTSD. Two years after the Oklahoma City bombing, 16% of children in a 100 mile radius of Oklahoma City with no direct exposure to the bombing had increased symptoms of PTSD. Weak parental response to the event, having a parent suffering from PTSD symptoms, and increased exposure to the event via the media all increase the possibility of the child developing PTSD symptoms.

Causes and symptoms

Specific causes for the onset of PTSD following a trauma aren't clearly defined, although experts suspect it may be influenced both by the severity of the event, by the person's personality and genetic make-up, and by whether or not the trauma was expected. First response emergency personnel and individuals directly involved in the event or those children and families who have lost loved ones are more likely to experience PTSD. Natural disasters account for about a 5% rate of PTSD, while there is a 50% rate of PTSD among rape and Holocaust survivors.

Media coverage plays a new role in both adult and pediatric onset of PTSD symptoms. The heightened level of news footage of actual traumatic events, such as the Oklahoma City bombing and the terrorist attack on the World Trade Center and the Pentagon, increases the exposure to the violence, injury, and death associated with the event and may reinforce PTSD symptoms in individuals, especially young children who cannot distinguish between the actual event and the repeated viewing of the event in the media.

PTSD symptoms are distinct and prolonged stress reactions that naturally occur during a highly stressful event. Common symptoms are:

- hyperalertness
- fear and anxiety
- nightmares and flashbacks
- sight, sound, and smell recollection
- avoidance of recall situations
- anger and irritability
- guilt
- depression
- increased substance abuse
- negative world view
- decreased sexual activity

Symptoms usually begin within three months of the trauma, although sometimes PTSD doesn't develop until years after the initial trauma occurred. Once the symptoms begin, they may fade away again within six months. Others suffer with the symptoms for far longer and in some cases, the problem may become chronic.

Among the most troubling symptoms of PTSD are flashbacks, which can be triggered by sounds, smells, feelings, or images. During a flashback, the person relives the traumatic event and may completely lose touch with reality, suffering through the trauma

for minutes or hours at a time, believing that the traumatizing event is actually happening all over again.

For a diagnosis of PTSD, symptoms must include at least one of the following so-called "intrusive" symptoms:

- flashbacks
- sleep disorders: nightmares or night terrors
- intense distress when exposed to events that are associated with the trauma

In addition, the person must have at least three of the following "avoidance" symptoms that affect interactions with others:

- trying to avoid thinking or feeling about the trauma
- inability to remember the event
- inability to experience emotion, as well as a loss of interest in former pleasures (psychic numbing or blunting)
- a sense of a shortened future

Finally, there must be evidence of increased arousal, including at least two of the following:

- problems falling asleep
- startle reactions: hyperalertness and strong reactions to unexpected noises
- memory problems
- concentration problems
- moodiness
- violence

In addition to the above symptoms, children with PTSD may experience learning disabilities and memory or attention problems. They may become more dependent, anxious, or even self-abusing.

Recovery may be slowed by injuries, damage to property, loss of employment, or other major problems in the community due to disaster.

Diagnosis

Not every person who experiences a traumatic event will experience PTSD. A mental health professional will diagnose the condition if the symptoms of stress last for more than a month after a traumatic event. While a formal diagnosis of PTSD is made only in the wake of a severe trauma, it is possible to have a mild PTSD-like reaction following less severe stress.

Treatment

Several factors have shown to be important in the treatment of post-traumatic stress. These include proximity of the treatment to the site of the event, immediate intervention of therapy as soon as possible, and the expectation that the individual will eventually return to more normal functions. The most helpful treatment of prolonged PTSD appears to be a combination of medication along with supportive and cognitive-behavioral therapies.

Emergency care

Immediate intervention is important for individuals directly affected by the traumatic event. Emergency care workers focus on achieving the following during the hours and days following the trauma.

- protect survivors from further danger
- treat immediate injuries
- provide food, shelter, fluids, and clothing
- provide safe zone
- locate separated loved ones
- reconnect loved ones
- provide normal social contact
- help reestablish routines
- help resolve transportation, housing, or other issues caused by disaster
- provide grief counseling, **stress reduction**, and other consultation to enable survivors and families to return to normal life

As well as providing care to others, emergency personnel often need the same support as the survivors. Operational debriefing is used to organize the emergency response and to disseminate information and sense of purpose to the first responders. Critical Incident Stress Debriefing (CISD) is a formal group invention designed to include various crisis intervention, such as information disbursement, one-on-one counseling, consultation, family crisis intervention, and referrals. CISD is not useful for survivors and is an interim support for first responders until they are able to receive therapy.

Medications

Medications used to reduce the symptoms of PTSD include anxiety-reducing medications and antidepressants, especially the **selective serotonin reuptake inhibitors** (SSRIs) such as fluoxetine (Prozac) and sertraline HCl (Zoloft). In 2001, the U.S. Food and Drug Administration (FDA) approved Zoloft as a long-term

treatment for PTSD. In a controlled study, Zoloft was effective in safely improving symptoms of PTSD over a period of 28 weeks and reducing the risk of relapse.

Sleep problems can be lessened with brief treatment with an anti-anxiety drug, such as a benzodiazepine like alprazolam (Xanax), but long-term usage can lead to disturbing side effects, such as increased anger, drug tolerance, dependency, and abuse.

Therapy

Several types of therapy may be useful and they are often combined in a multi-faceted approach to understand and treat this condition.

- Cognitive-behavioral therapy focuses on changing specific actions and thoughts through repetitive review of traumatic events, identification of negative behaviors and thoughts, and stress management.
- Group therapy has been useful in decreasing psychological distress, depression, and anxiety in some PTSD sufferers such as sexually abused women and war veterans.
- Psychological debriefing has been widely used to treat victims of natural disasters and other traumatic events such as bombings and workplace shootings, however, recent research shows that psychological debriefing may increase the stress response. Since this type of debriefing focuses on the emotional response of the survivor, it is not recommended for individuals experiencing an extreme level of grief.

Alternative treatment

Several means of alternative treatment may be helpful in combination with conventional therapy for reduction of the symptoms of post-traumatic stress disorder. These include relaxation training, breathing techniques, spiritual treatment, and drama therapy where the event is re-enacted.

Prognosis

The severity of the illness depends in part on whether the trauma was unexpected, the severity of the trauma, how chronic the trauma was (such as for victims of sexual abuse), and the person's readiness to embrace the recovery process. With appropriate medication, emotional support, counseling, and follow-up care, most people show significant improvement. However, prolonged exposure to severe trauma, such as experienced by victims of prolonged physical or sexual abuse and survivors of the Holocaust, may cause permanent psychological **scars**.

KEY TERMS

Benzodiazepine—A class of drugs that have a hypnotic and sedative action, used mainly as tranquilizers to control symptoms of anxiety.

Cognitive-behavioral therapy—A type of psychotherapy used to treat anxiety disorders (including PTSD) that emphasizes behavioral change, together with alteration of negative thought patterns.

Selective serotonin reuptake inhibitor (SSRI)—A class of antidepressants that work by blocking the reabsorption of serotonin in the brain, raising the levels of serotonin. SSRIs include Prozac, Zoloft, and Paxil.

Prevention

More studies are needed to determine if PTSD can actually be prevented. Some measures that have been explored include controlling exposure to traumatic events through safety and security measures, psychological preparation for individuals who will be exposed to traumatic events (i.e. policemen, paramedics, soldiers), and stress inoculation training (rehearsal of the event with small doses of the stressful situation).

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Freedom From Fear. 308 Seaview Ave., Staten Island, NY 10305. (718) 351-1717.

National Anxiety Foundation. 3135 Custer Dr., Lexington, KY 40517. (606) 272-7166.

National Center for Post-Traumatic Stress Disorder. < <http://www.dartmouth.edu/dms/ptsd> >.

National Institute of Mental Health. Rm 15C-05, 5600 Fishers Lane, Rockville, MD 20857.

Society for Traumatic Stress Studies, 60 Revere Dr., Ste. 500, Northbrook, IL 60062. (708) 480-9080.

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Postmenopausal bleeding

Definition

Postmenopausal bleeding is bleeding from the reproductive system that occurs six months or more after menstrual periods have stopped due to **menopause**.

Description

Menopause, the end of ovulation and menstrual periods, naturally occurs for most women age 40–55 years. The process of ending ovulation and menstruation is gradual, spanning one to two years.

Postmenopausal bleeding is bleeding that occurs after menopause has been established for at least six months. It is different from infrequent, irregular periods (**oligomenorrhea**) that occur around the time of menopause.

Many women experience some postmenopausal bleeding. However, postmenopausal bleeding is not normal. Because it can be a symptom of a serious medical condition, any episodes of postmenopausal bleeding should be brought to the attention of a woman's doctor.

Women taking estrogen (called hormone replacement therapy or HRT) are more likely to experience postmenopausal bleeding. So are obese women, because fat cells transform male hormones (androgens) secreted by the adrenal gland into estrogen.

Causes and symptoms

Postmenopausal bleeding can originate in different parts of the reproductive system. Bleeding from

the vagina may occur because when estrogen secretion stops, the vagina dries out and can diminish (atrophy). This is the most common cause of bleeding from the lower reproductive tract.

Lesions and cracks on the vulva may also bleed. Sometimes bleeding occurs after intercourse. Bleeding can occur with or without an associated infection.

Bleeding from the upper reproductive system can be caused by:

- hormone replacements
- **endometrial cancer**
- endometrial polyps
- cervical **cancer**
- cervical lesions
- uterine tumors
- ovarian cancer
- estrogen-secreting tumors in other parts of the body

The most common cause of postmenopausal bleeding is HRT. The estrogen in the replacement therapy eases the symptoms of menopause (like hot flashes), and decreases the risk of osteoporosis. Sometimes this supplemental estrogen stimulates the uterine lining to grow. When the lining is shed, postmenopausal bleeding occurs. Most women on HRT usually take the hormone progesterone with the estrogen, and may have monthly withdrawal bleeding. This is a normal side effect.

About 5–10% of postmenopausal bleeding is due to endometrial cancer or its precursors. Uterine hyperplasia, the abnormal growth of uterine cells, can be a precursor to cancer.

Diagnosis

Diagnosis of postmenopausal bleeding begins with the patient. The doctor will ask for a detailed history of how long postmenopausal bleeding has occurred. A woman can assist the doctor by keeping a record of the time, frequency, length, and quantity of bleeding. She should also tell the doctor about any medications she is taking, especially any estrogens or steroids.

After taking the woman's history, the doctor does a pelvic examination and **Pap test**. The doctor will examine the vulva and vagina for signs of atrophy, and will feel for any sign of uterine polyps. Depending on the results of this examination, the doctor may want to do more extensive testing.

Invasive diagnostic procedures

Endometrial biopsy allows the doctor to sample small areas of the uterine lining, while cervical biopsy allows the cervix to be sampled. Tissues are then examined for any abnormalities. This is a simple office procedure.

Dilatation and curettage (D & C) is often necessary for definitive diagnosis. This is done under either general or **local anesthesia**. After examining the tissues collected by an endometrial biopsy or D & C, the doctor may order additional tests to determine if an estrogen-secreting tumor is present on the ovaries or in another part of the body.

Non-invasive diagnostic procedures

With concerns about the rising cost of health care, vaginal probe ultrasound is increasingly being used more than endometrial biopsy to evaluate women with postmenopausal bleeding. Vaginal ultrasound measures the thickness of the endometrium. When the endometrial stripe is less than 0.2 in (5 mm) thick, the chance of cancer is less than 1%. The disadvantage of vaginal ultrasound is that it often does not show polyps and fibroids in the uterus.

A refinement of vaginal probe ultrasound is saline infusion sonography (SIS). A salt water (saline) solution is injected into the uterus with a small tube (catheter) before the vaginal probe is inserted. The presence of liquid in the uterus helps make any structural abnormalities more distinct. These two non-invasive procedures cause less discomfort than endometrial biopsies and D & Cs, but D & C still remains the definitive test for diagnosing uterine cancer.

Treatment

It is common for women just beginning HRT to experience some bleeding. Most women who are on HRT also take progesterone with the estrogen and may have monthly withdrawal bleeding. Again, this is a normal side effect that usually does not require treatment.

Postmenopausal bleeding due to bleeding of the vagina or vulva can be treated with local application of estrogen or HRT.

When diagnosis indicates cancer, some form of surgery is required. The uterus, cervix, ovaries, and fallopian tubes may all be removed depending on the type and location of the cancer. If the problem is estrogen- or androgen-producing tumors elsewhere in the body, these must also be surgically removed.

KEY TERMS

Dilation and curettage (D & C)—A procedure performed under anesthesia during which the cervix is opened more (or dilated) and tissue lining the uterus is scraped out with a metal, spoon-shaped instrument or a suction tube. The procedure can be used to diagnose a problem or to remove growths (polyps).

Endometrial biopsy—The removal of uterine tissue samples either by suction or scraping; the cervix is not dilated. The procedure has a lower rate of diagnostic accuracy than D & C, but can be done as an office procedure under local anesthesia.

Endometrium—The tissue lining the inside of the uterus.

Fibroid tumors—Non-cancerous (benign) growths in the uterus. These growths occur in 30–40% of women over age 40, and do not need to be removed unless they are causing symptoms that interfere with a woman's normal activities.

Osteoporosis—The excessive loss of calcium from the bones, causing the bones to become fragile and break easily. Postmenopausal women are especially vulnerable to this condition because estrogen, a hormone that protects bones against calcium loss, decreases drastically after menopause.

Postmenopausal bleeding that is not due to cancer and cannot be controlled by any other treatment usually requires a **hysterectomy**.

Prognosis

Response to treatment for postmenopausal bleeding is highly individual and is not easy to predict. The outcome depends largely on the reason for the bleeding. Many women are successfully treated with hormones. As a last resort, hysterectomy removes the source of the problem by removing the uterus. However, this operation is not without risk and the possibility of complications. The prognosis for women who have various kinds of reproductive cancer varies with the type of cancer and the stage at which the cancer is diagnosed.

Prevention

Postmenopausal bleeding is not a preventable disorder. However, maintaining a healthy weight will decrease the chances of it occurring.

Resources

ORGANIZATIONS

American Cancer Society. 1599 Clifton Rd., NE, Atlanta, GA 30329-4251. (800) 227-2345.

< <http://www.cancer.org> >.

National Cancer Institute. Building 31, Room 10A31, 31 Center Drive, MSC 2580, Bethesda, MD 20892-2580. (800) 422-6237. < <http://www.nci.nih.gov> >.

Tish Davidson, A.M.

Postpartum blues see **Postpartum depression**

Postpartum depression

Definition

Postpartum depression is a mood disorder that begins after **childbirth** and usually lasts beyond six weeks.

Description

The onset of postpartum depression tends to be gradual and may persist for many months, or develop into a second bout following a subsequent pregnancy. Postpartum depression affects approximately 15% of all childbearing women. Mild to moderate cases are sometimes unrecognized by women themselves. Many women feel ashamed if they are not coping and so may conceal their difficulties. This is a serious problem that disrupts women's lives and can have effects on the baby, other children, her partner, and other relationships. Levels of depression for fathers also increase significantly.

Postpartum depression is often divided into two types: early onset and late onset. An early onset most often seems like the "blues," a mild brief experience during the first days or weeks after birth. During the first week after the birth up to 80% of mothers will experience the "baby blues." This is usually a time of extra sensitivity and symptoms include tearfulness, irritability, **anxiety**, and mood changes, which tend to peak between three to five days after childbirth. The symptoms normally disappear within two weeks without requiring specific treatment apart from understanding, support, skill, and practice. In short, some depression, tiredness, and anxiety may fall within the "normal" range of reactions to giving birth.

Late onset appears several weeks after the birth. This involves a slowly growing feeling of sadness,

depression, lack of energy, chronic tiredness, inability to sleep, change in appetite, significant weight loss or gain, and difficulty caring for the baby.

Causes and symptoms

As of 2006, experts cannot say what causes postpartum depression. Most likely, it is caused by many factors that vary from individual to individual. Mothers commonly experience some degree of depression during the first weeks after birth. **Pregnancy** and birth are accompanied by sudden hormonal changes that affect emotions. Additionally, the 24-hour responsibility for a newborn infant represents a major psychological and lifestyle adjustment for most mothers, even after the first child. These physical and emotional stresses are usually accompanied by inadequate rest until the baby's routine stabilizes, so **fatigue** and depression are not unusual.

Experiences vary considerably but usually include several symptoms.

Feelings:

- persistent low mood
- inadequacy, failure, hopelessness, helplessness
- exhaustion, emptiness, sadness, tearfulness
- guilt, shame, worthlessness
- confusion, anxiety, and panic
- fear for the baby and of the baby
- fear of being alone or going out

Behaviors:

- lack of interest or pleasure in usual activities
- insomnia or excessive sleep, nightmares
- not eating or overeating
- decreased energy and motivation
- withdrawal from social contact
- poor self-care
- inability to cope with routine tasks

Thoughts:

- inability to think clearly and make decisions
- lack of concentration and poor memory
- running away from everything
- fear of being rejected by partner
- worry about harm or **death** to partner or baby
- ideas about suicide

Some symptoms may not indicate a severe problem. However, persistent low mood or loss of interest or pleasure in activities, along with four other symptoms occurring together for a period of at least two weeks, indicate clinical depression, and require adequate treatment.

There are several important risk factors for postpartum depression, including:

- **stress**
- lack of sleep
- poor **nutrition**
- lack of support from one's partner, family or friends
- family history of depression
- labor/delivery complications for mother or baby
- premature or postmature delivery
- problems with the baby's health
- separation of mother and baby
- A difficult baby (temperament, feeding, sleeping, settling problems)
- preexisting neurosis or **psychosis**

Diagnosis

There is no diagnostic test for postpartum depression. However, it is important to understand that it is, nonetheless, a real illness, and like a physical ailment, it has specific symptoms.

Treatment

Several treatment options exist, including medication, psychotherapy, counseling, and group treatment and support strategies, depending on the woman's needs. One effective treatment combines antidepressant medication and psychotherapy. These types of medication are often effective when used for 3 to 4 weeks. Any medication use must be carefully considered if the woman are breast-feeding, but with some medications, continuing breast-feeding is safe. Nevertheless, medication alone is never sufficient and should always be accompanied by counseling or other support services.

Alternative treatment

Postpartum depression can be effectively alleviated through counseling and support groups, so that the mother doesn't feel she is alone in her feelings. Constitutional homeopathy can be the most effective treatment of the alternative therapies because it acts

on the emotional level where postpartum depression is felt. **Acupuncture**, Chinese herbs, and Western herbs can all help the mother suffering from postpartum depression come back to a state of balance. Seeking help from a practitioner allows the new mother to feel supported and cared for and allows for more effective treatment.

A new mother also should remember that this time of stress does not last forever. In addition, there are useful things she can do for herself, including:

- valuing her role as a mother and trusting her own judgment
- making each day as simple as possible
- avoiding extra pressures or unnecessary tasks
- trying to involve her partner more in the care of the baby from the beginning
- discussing with her partner how both can share the household chores and responsibilities
- scheduling frequent outings, such as walks and short visits with friends
- having the baby sleep in a separate room so she sleeps more restfully
- sharing her feelings with her partner or a friend who is a good listener
- talking with other mothers to help keep problems in perspective
- trying to sleep or rest when the baby is sleeping
- taking care of her health and well-being.
- not losing her sense of humor

Prognosis

With support from friends and family, mild postpartum depression usually disappears quickly. If depression becomes severe, a mother cannot care for herself and the baby, and in rare cases, hospitalization may be necessary. Yet, medication, counseling, and support from others usually cures even severe depression in 3–6 months.

Prevention

Exercise can help enhance a new mother's emotional well-being. New mothers should also try to cultivate good sleeping habits and learn to rest when they feel physically or emotionally tired. It's important for a woman to learn to recognize her own warning signs of fatigue, respond to them by taking a break.

Resources

ORGANIZATIONS

Depression After Delivery (D.A.D.). P.O. Box 1282, Morrisville, PA 19067. (800) 944-4773.

Postpartum Support International. 927 North Kellogg Ave., Santa Barbara, CA 93111. (805) 967-7636.

David James Doermann

Postpartum psychosis see **Postpartum depression**

Postpolio syndrome

Definition

Postpolio syndrome (PPS) is a condition that strikes survivors of the disease **polio**. PPS occurs about 20–30 years after the original bout with polio, and causes slow but progressive weakening of muscles.

Description

Polio is a disease caused by the poliovirus. It most commonly infects younger children, although it can also infect older children and adults. About 90% of people infected by poliovirus develop only a mild case or no illness at all. However, infected people can continue to spread the virus to others. In its most severe form polio causes **paralysis** of the muscles of the legs, arms, and respiratory system.

About 1% of all people infected with poliovirus develop the actual disease known as polio. In these cases, the virus (which enters the person's body through the mouth) multiplies rapidly within the intestine. The viruses then invade the nearby lymphatic system. Eventually, poliovirus enters the bloodstream, which allows it to gain access to the central nervous system or CNS (the brain and spinal cord). The virus may actually infect a nerve elsewhere in the body, and then spread along that nerve to enter the brain.

The major illness associated with poliovirus often follows a mild illness, which has symptoms of **fever**, **nausea**, and **vomiting**. However, after a symptom-free interval of several days, the patient who is on the way to a major illness develops new symptoms such as **headache** and back and neck **pain**. These symptoms are due to invasion of the nervous system. The motor nerves (those nerves responsible for movement of the muscles) become inflamed, injured, and destroyed. The muscles, therefore, no longer receive any messages

from the brain or spinal cord. The muscles become weak, floppy, and then totally paralyzed (unable to move). All muscle tone is lost in the affected limb, and the muscle begins to decrease in size (atrophy). The affected muscles are often only on one side (asymmetric paralysis) of the body. Sensation (the person's ability to feel) is not affected in these paralyzed limbs.

The maximum state of paralysis is usually reached within just a few days. The remaining, unaffected nerves then begin the process of attempting to grow branches to compensate (make up for) the destroyed nerves. This process continues for about six months. Whatever function has not been regained in this amount of time will usually be permanently lost.

Causes and symptoms

PPS occurs in about 25% of patients, several decades after their original infection with polio. However, long-term follow-up indicates that two thirds of polio survivors may experience new weakness. Several theories exist as to the cause of this syndrome.

One such theory has looked at the way function is regained by polio survivors. Three mechanisms seem to be at work:

- injured nerves recuperate and begin functioning again
- muscles that still have working nerve connections grow in size and strength, in order to take over for other paralyzed muscles
- working nerves begin to send small branches out to muscles whose original nerves were destroyed by polio

As a person ages, injured nerves that were able to regain function may fail again, as may muscles that have been over-worked for years in order to compensate for other paralyzed muscles. Even the uninjured nerves that provided new nerve twigs to the muscles may begin to falter after years of relative over-activity. This theory, then, suggests that the body's ability to compensate for destroyed nerves may eventually begin to fail. The compensating nerves and muscles grow older, and because they've been working so much harder over the years, they wear out relatively sooner than would be expected of normal nerves and muscles. Some researchers look at this situation as a form of premature **aging**, brought on by overuse.

Other researchers note that normal aging includes the loss of a fair number of motor nerves. When a patient has already lost motor nerves through polio, normal loss of motor nerves through aging may cause

the number of remaining working nerves to drop low enough to cause symptoms of weakness.

Other theories of PPS include the possibility that particles of the original polioviruses remain in the body. These particles may exert a negative effect, decades later, or they may cause the body's immune system to produce substances originally intended to fight the invading virus, but which may accidentally set off a variety of reactions within the body that actually serve to interfere with the normal functioning of the nerves and muscles.

Still other researchers are looking at the possibility that polio patients have important spinal cord changes which, over time, affect the nerves responsible for movement.

The symptoms of PPS include generalized **fatigue**, low energy, progressively increasing muscle weakness, shrinking muscle size (atrophy), involuntary twitching of the muscle fibers (fasciculations), painful muscles and joints, difficulties with breathing and swallowing, and sleep problems.

Survivors of polio may also develop arthritis of the spine, shoulders, or arms, related to the long-term use of crutches or overcompensation for weak leg muscles.

Diagnosis

Diagnosis is primarily through history. When a patient who has recovered from polio some decades previously begins to experience muscle weakness, PPS must be strongly suspected.

Treatment

Just as there are no treatments available to reverse the original damage of polio, there are also no treatments available to reverse the damaging effects of post polio syndrome. Attempts can be made to relieve some of the symptoms, however.

Pain and inflammation of the muscles and joints can be treated with anti-inflammatory medications, application of hot packs, stretching exercises, and physical therapy. Exercises to maintain/increase flexibility are particularly important. However, an **exercise** regimen must be carefully designed, so as not to strain already fatigued muscles and nerves.

Some patients will require new types of braces to provide support for weakening muscles. Others will need to use wheelchairs or motorized scooters to maintain mobility.

Sleep problems and respiratory difficulties may be related to each other. If breathing is labored during

KEY TERMS

Asymmetric—Not occurring equally on both sides of the body.

Atrophy—Shrinking, growing smaller in size.

Flaccid—Weak, soft, floppy.

Paralysis—The inability to voluntarily move.

sleep, the blood's oxygen content may drop low enough to interfere with the quality of sleep. This may require oxygen supplementation, or even the use of a machine to aid in breathing.

Prognosis

Prognosis for patients with post polio syndrome is relatively good. It is a very slow, gradually progressing syndrome. Only about 20% of all patients with PPS will need to rely on new aids for mobility or breathing. It appears that the PPS symptoms reach their most severe about 30–34 years after original diagnosis of polio.

Prevention

There is no way to prevent PPS. However, paying attention to what types of exertion worsen symptoms may slow the progression of the syndrome.

Resources

ORGANIZATIONS

International Polio Network. 4207 Lindell Blvd., Suite 110, St. Louis, MO 63108-2915. (314) 534-0475.

March of Dimes Birth Defects Foundation. 1275 Mamaroneck Ave., White Plains, NY 10605. (914) 428-7100. resourcecenter@modimes.org. <<http://www.modimes.org>>.

Polio Survivors Association. 12720 Lareina Ave., Downey, CA 90242. (310) 862-4508.

Rosalyn Carson-DeWitt, MD

Postpoliomyelitis muscular atrophy see **Postpolio syndrome**

Postpoliomyelitis syndrome see **Postpolio syndrome**

Poststreptococcal glomerulonephritis see **Acute poststreptococcal glomerulonephritis**

Postural drainage see **Chest physical therapy**

Postural hypotension see **Orthostatic hypotension**

Postviral thrombocytopenia see **Idiopathic thrombocytopenic purpura**

Potassium hydroxide test see **KOH test**

Potassium imbalance see **Hyperkalemia; Hypokalemia**

PPD skin test see **Tuberculin skin test**

Prader-Willi syndrome

Definition

Prader-Willi syndrome (PWS) is a genetic condition caused by the absence of chromosomal material from chromosome 15. The genetic basis of PWS is complex. Characteristics of the syndrome include developmental delay, poor muscle tone, short stature, small hands and feet, incomplete sexual development, and unique facial features. Insatiable appetite is a classic feature of PWS. This uncontrollable appetite can lead to health problems and behavior disturbances.

Description

The first patients with features of PWS were described by Dr. Prader, Dr. Willi, and Dr. Lambert in 1956. Since that time, the complex genetic basis of PWS has begun to be understood. Initially, scientists found that individuals with PWS have a portion of genetic material deleted (erased) from chromosome 15. In order to have PWS, the genetic material must be deleted from the chromosome 15 received from one's father. If the deletion is on the chromosome 15 inherited from one's mother a different syndrome develops. This was an important discovery. It demonstrated for the first time that the genes inherited from one's mother can be expressed differently than the genes inherited from one's father.

Over time, scientists realized that some individuals with PWS do not have genetic material deleted from chromosome 15. Further studies found that these patients inherit both copies of chromosome 15 from their mother. This is not typical. Normally, an individual receives one chromosome 15 from their father and one chromosome 15 from their mother.

When a person receives both chromosomes from the same parent it is called “uniparental disomy”. “When a person receives both chromosomes from his or her mother it is called “maternal uniparental disomy.”

Scientists are still discovering other causes of PWS. A small number of patients with PWS have a change (mutation) in the genetic material on the chromosome 15 inherited from their father. This mutation prevents certain genes on chromosome 15 from working properly. PWS develops when these genes do not work normally.

Newborns with PWS generally have poor muscle tone, (hypotonia) and do not feed well. This can lead to poor weight gain and **failure to thrive**. Genitalia can be smaller than normal. Hands and feet are also typically smaller than normal. Some patients with PWS have unique facial characteristics. These unique facial features are typically subtle and detectable only by physicians.

As children with PWS age, development is typically slower than normal. Developmental milestones, such as crawling, walking and talking occur later than usual. Developmental delay continues into adulthood for approximately 50% of individuals with PWS. At about one to two years of age, children with PWS develop an uncontrollable, insatiable appetite. Left to their own devices, individuals with PWS will eat until they suffer from life-threatening **obesity**. The desire to eat can lead to significant behavior problems.

The symptoms and features of PWS require life long support and care. If food intake is strictly monitored and various therapies provided, individuals with PWS have a normal life expectancy.

PWS affects approximately 1 in 10,000 to 25,000 live births. It is the most common genetic cause of life-threatening obesity. It affects both males and females. PWS can be seen in all races and ethnic groups.

Causes and symptoms

In order to comprehend the various causes of PWS, the nature of chromosomes and genes must be well understood. Human beings have 46 chromosomes in the cells of their body. Chromosomes contain genes, which regulate the function and development of the body. An individual's chromosomes are inherited from his/her parents. Each parent normally gives a child 23 chromosomes. A child receives 23 chromosomes from the egg and 23 chromosomes from the sperm.

The 46 chromosomes in the human body are divided into pairs based on their physical

characteristics. Each pair is assigned a number or a letter. When viewed under a microscope, chromosomes within the same pair appear identical because they contain the same genes.

Most chromosomes have a constriction near the center called the centromere. The centromere separates the chromosome into long and short arms. The short arm of a chromosome is called the “p arm.” the long arm and is called the “q arm,”

Chromosomes in the same pair contain the same genes. However, some genes work differently depending on if they were inherited from the egg or the sperm. Sometimes, genes are silenced when inherited from the mother. Other times, genes are silenced when inherited from the father. When genes in a certain region on a chromosome are silenced, they are said to be “imprinted.” Imprinting is a normal process that does not typically cause disease. If normal imprinting is disrupted a genetic disease can develop.

Individuals have two complete copies of chromosome 15. One chromosome 15 is inherited from the mother, or “maternal” in origin. The other chromosome 15 is inherited from the father, or is “paternal” in origin.

Chromosome 15 contains many different genes. There are several genes found on the q arm of chromosome 15 that are imprinted. A gene called “SNPRN” is an example of one of these genes. It is normally imprinted, or silenced, if inherited from the mother. The imprinting of this group of maternal genes does not typically cause disease. The genes in this region should not be imprinted if paternal in origin. Normal development depends on these paternal genes being present and active. If these genes are deleted, not inherited, or incorrectly imprinted PWS develops.

Seventy percent of the cases of PWS are caused when a piece of material is deleted, or erased, from the paternal chromosome 15. This deletion occurs in a specific region on the q arm of chromosome 15. The piece of chromosomal material that is deleted contains genes that must be present for normal development. These paternal genes must be working normally, because the same genes on the chromosome 15 inherited from the mother are imprinted. When these paternal genes are missing, the brain and other parts of the body do not develop as expected. This is what causes the symptoms associated with PWS.

In 99% of the cases of PWS the deletion is sporadic. This means that it happens randomly and there is not an apparent cause. It does not run in the family. If a child has PWS due to a sporadic deletion in the paternal chromosome 15, the chance the parents

could have another child with PWS is less than 1%. In fewer than 1% of the cases of PWS there is a chromosomal rearrangement in the family which causes the deletion. This chromosomal rearrangement is called a “translocation.” If a parent has a translocation the risk of having a child with PWS is higher than 1%.

PWS can also develop if a child receives both chromosome 15s from his/her mother. This is seen in approximately 25% of the cases of PWS. “Maternal uniparental disomy”. Maternal uniparental disomy for chromosome 15 leads to PWS because the genes on chromosome 15 that should have been inherited from the father are missing, and the genes on both the chromosome 15s inherited from the mother are imprinted.

PWS caused by maternal uniparental is sporadic. This means that it occurs randomly and there is not an apparent cause. If a child has PWS due to maternal uniparental disomy the chance the parents could have another child with PWS is less than 1%.

Approximately 3–4% of patients with PWS have a change (mutation) in a gene located on the q arm of chromosome 15. This mutation leads to incorrect imprinting. This mutation causes genes inherited from the father to be imprinted or silenced, which should not normally be imprinted. If a child has PWS due to a mutation that changes imprinting, the chance the parents could have another child with PWS is approximately 5%.

Infants with PWS have weak muscle tone (hypotonia). This hypotonia causes problems with sucking and eating. Infants with PWS may have problems gaining weight. Some infants with PWS are diagnosed with “failure to thrive” due to slow growth and development. During infancy, babies with PWS may also sleep more than normal and have problems controlling their temperature.

Some of the unique physical features associated with PWS can be seen during infancy. Genitalia that is smaller than normal is common. This may be more evident in males with PWS. Hands and feet may also be smaller than average. The unique facial features seen in some patients with PWS may be difficult to detect in infancy. These facial features are very mild and do not cause physical problems.

As early as six months, but more commonly at one to two years a compulsive desire to eat develops. This uncontrollable appetite is a classic feature of PWS. Individuals with PWS lack the ability to feel full or satiated. This uncontrollable desire to eat is thought to be related to a difference in the brain, which controls hunger. Over-eating (hyperphagia), a lack of a desire to exercise, and a slow metabolism places individuals with

PWS at high risk for severe obesity. Some individuals with PWS may also have a reduced ability to vomit.

Behavior problems are a common feature of PWS. Some behavior problems develop from the desire to eat. Other reported problems include obsessive/compulsive behaviors, depression, and temper tantrums. Individuals with PWS may also pick their own skin (skin picking). This unusual behavior may be due to a reduced **pain** threshold.

Developmental delay, learning disabilities, and mental retardation are associated with PWS. Approximately 50% of individuals with PWS have developmental delay. The remaining 50% are described as having mild mental retardation. The **mental retardation** can occasionally be more severe. Infants and children with PWS are often delayed in development.

Puberty may occur early or late, but it is usually incomplete. In addition to the effects on sexual development and fertility, individuals do not undergo the normal adolescent growth spurt and may be short as adults. Muscles often remain underdeveloped and body fat is increased.

Diagnosis

During infancy the diagnosis of PWS may be suspected if poor muscle tone, feeding problems, small genitalia, or the unique facial features are present. If an infant has these features, testing for PWS should be performed. This testing should also be offered to children and adults who display features commonly seen in PWS (developmental delay, uncontrollable appetite, small genitalia, etc.). There are several different genetic tests that can detect PWS. All of these tests can be performed from a blood sample.

Methylation testing detects 99% of the cases of PWS. Methylation testing can detect the absence of the paternal genes that should be normally active on chromosome 15. Although methylation testing can accurately diagnose PWS, it can not determine if the PWS is caused by a deletion, maternal uniparental disomy, or a mutation that disrupts imprinting. This information is important for **genetic counseling**. Therefore, additional testing should be performed.

Chromosome analysis can determine if the PWS is the result of a deletion in the q arm of chromosome 15. Chromosome analysis, also called “karyotyping,” involves staining the chromosomes and examining them under a microscope. In some cases the deletion of material from chromosome 15 can be easily seen. In

other cases, further testing must be performed. FISH (fluorescence in-situ hybridization) is a special technique that detects small deletions that cause PWS.

More specialized DNA testing is required to detect maternal uniparental disomy or a mutation that disrupts imprinting. This DNA testing identifies unique DNA patterns in the mother and father. The unique DNA patterns are then compared with the DNA from the child with PWS.

PWS can be detected before birth if the mother undergoes **amniocentesis** testing or chorionic villus sampling (CVS). This testing is only recommended if the mother or father is known to have a chromosome rearrangement, or if they already have a child with PWS syndrome.

Treatment

There is currently not a cure for PWS. Treatment during infancy includes therapies to improve muscle tone. Some infants with PWS also require special nipples and feeding techniques to improve weight gain.

Treatment and management during childhood, adolescence, and adulthood is typically focused on weight control. Strict control of food intake is vital to prevent severe obesity. In many cases food must be made inaccessible. This may involve unconventional measures such as locking the refrigerator or kitchen cabinets. A lifelong restricted-calorie diet and regular **exercise** program are also suggested. Unfortunately, diet medications have not been shown to significantly prevent obesity in PWS. However, growth hormone therapy has been shown to improve the poor muscle tone and reduced height typically associated with PWS.

Special education may be helpful in treating developmental delays and behavior problems. Individuals with PWS typically excel in highly structured environments.

Prognosis

Life expectancy is normal and the prognosis good, if weight gain is well controlled.

Resources

BOOKS

Couch, Cheryl. *My Rag Doll*. Couch Publishing, October 2000.

PERIODICALS

Butler, Merlin G. and Travis Thompson. "Prader-Willi Syndrome: Clinical and Genetic Findings." *The Endocrinologist* 10 (2000): 3S-16S.

KEY TERMS

Amniocentesis—A procedure in which a needle is inserted through a pregnant woman's abdomen and into her uterus. Amniotic fluid is then removed from around the fetus and may be used for genetic testing.

Centromere—Major constriction in a chromosome.

Deletion—Removal of a piece of genetic material.

DNA—Deoxyribonucleic acid. Genes are made of sections of DNA.

FISH—(fluorescence in-situ hybridization) Technique used to detect small deletions or rearrangements in chromosomes.

Gene—Segment of DNA that controls the development and function of the body. Genes are contained within chromosomes.

Hyperphagia—Over-eating.

Hypotonia—Low muscle tone.

Imprinting—Process that silences a gene or group of genes. The genes are silenced depending on if they are inherited through the egg or the sperm.

Maternal—From one's mother.

Maternal uniparental disomy—Chromosome abnormality in which both chromosomes in a pair are inherited from one's mother.

Methylation testing—DNA testing that detects if a gene is active or imprinted.

Mutation—A change in a gene.

Paternal—From one's father.

Translocation—Chromosome abnormality in which chromosomes are rearranged and placed together.

Uniparental disomy—Chromosome abnormality in which both chromosomes in a pair are inherited from the same parent.

State, Matthew W., and Elisabeth Dykens. "Genetics of Childhood Disorders: XV. Prader-Willi Syndrome: Genes, Brain and Behavior." *J. Am. Acad. Child. Adolesc. Psychiatry* 39, no. 6 (June 2000): 797-800.

ORGANIZATIONS

Alliance of Genetic Support Groups. 4301 Connecticut Ave. NW, Suite 404, Washington DC 20008. (202) 966-5557. Fax: (202) 966-8553. <<http://www.geneticalliance.org>>.

International Prader-Willi Syndrome Organization. <<http://www.ipwsp.org>>.

National Organization for Rare Disorders, Inc. P.O. Box 8923, New Fairfield, CT 06812. (800) 999-6673.
< <http://www.rarediseases.org> >.

Prader-Willi Foundation. 223 Main Street, Port Washington, NY 11050. (800)253- 7993.
< <http://www.prader-willi.org> >.

Prader-Willi Syndrome Association (USA). 5700 Midnight Pass Rd., Sarasota, FL 34242. (800)926-4797.
< <http://www.pwsusa.org> >.

OTHER

Gene Clinics. < <http://www.geneclinics.org/profiles/pws/details.html> >.

OMIM. < <http://www.ncbi.nlm.nih.gov/htbin-port/Omim/dispim?176270> >.

Holly Ann Ishmael, M.S.

Praziquantel see **Antihelminthic drugs**

Precocious puberty

Definition

Sexual development before the age of eight in girls, and age 10 in boys.

Description

Not every child reaches **puberty** at the same time, but in most cases it's safe to predict that sexual development will begin at about age 11 in girls and 12 or 13 in boys. However, occasionally a child begins to develop sexually much earlier. Between four to eight times more common in girls than boys, precocious puberty occurs in one out of every 5,000 to 10,000 U.S. children.

Precocious puberty often begins before age 8 in girls, triggering the development of breasts and hair under the arms and in the genital region. The onset of ovulation and menstruation also may occur. In boys, the condition triggers the development of a large penis and testicles, with spontaneous erections and the production of sperm. Hair grows on the face, under arms and in the pubic area, and **acne** may become a problem.

While the early onset of puberty may seem fairly benign, in fact it can cause problems when hormones trigger changes in the growth pattern, essentially halting growth before the child has reached normal adult height. Girls may never grow above 5 ft (152 cm) and boys often stop growing by about 5 ft 2 in (157 cm).

The abnormal growth patterns are not the only problem, however. Children with this condition look

noticeably different than their peers, and may feel rejected by their friends and socially isolated. Adults may expect these children to act more maturely simply because they look so much older. As a result, many of these children—especially boys—are much more aggressive than others their own age, leading to behavior problems both at home and at school.

Causes and symptoms

Puberty begins when the brain secretes a hormone that triggers the pituitary gland to release gonadotropins, which in turn stimulate the ovaries or testes to produce sex hormones. These sex hormones (especially estrogen in girls and testosterone in boys) are what causes the onset of sexual maturity.

The hormonal changes of precious puberty are normal—it's just that the whole process begins a few years too soon. Especially in girls, there is not usually any underlying problem that causes the process to begin too soon. However, some boys do inherit the condition; the responsible gene may be passed directly from father to son, or inherited indirectly from the maternal grandfather through the mother, who does not begin early puberty herself. This genetic condition in girls can be traced in only about 1% of cases.

In about 15% of cases, there is an underlying cause for the precocious puberty, and it is important to search for these causes. The condition may result from a benign tumor in the part of the brain that releases hormones. Less commonly, it may be caused by other types of brain tumors, central nervous system disorders, or adrenal gland problems.

Diagnosis

Physical exams can reveal the development of sexual characteristics in a young child. Bone x rays can reveal bone age, and **pelvic ultrasound** may show an enlarged uterus and rule out ovarian or adrenal tumors. Blood tests can highlight higher-than-normal levels of hormones. MRI or CAT scans should be considered to rule out intracranial tumors.

Treatment

Treatment aims to halt or reverse sexual development so as to stop the accompanying rapid growth that will limit a child's height. There are two possible approaches: either treat the underlying condition (such as an ovarian or intracranial tumor) or change the hormonal balance to stop sexual development. It may not be possible to treat the underlying condition;

for this reason, treatment is usually aimed at adjusting hormone levels.

There are several drugs that have been developed to do this:

- histrelin (Supprelin)
- nafarelin (Synarel)
- synthetic gonadotropin-releasing hormone agonist
- deslorelin
- ethylamide
- triptorelin
- leuprolide

Prognosis

Drug treatments can slow growth to 2–3 in (5–7.5 cm) a year, allowing these children to reach normal adult height, although the long-term effects aren't known.

Resources

ORGANIZATIONS

National Institute of Child Health and Human Development. Bldg 31, Room 2A32, MSC 2425, 31 Center Drive, Bethesda, MD 20892-2425. (800) 505-2742. <<http://www.nichd.nih.gov/sids/sids.htm>> .

Carol A. Turkington

Prednis see **Corticosteroids**

Preeclampsia and eclampsia

Definition

Preeclampsia and eclampsia are complications of **pregnancy**. In preeclampsia, the woman has dangerously high blood pressure, swelling, and protein in the urine. If allowed to progress, this syndrome will lead to eclampsia.

Description

Blood pressure is a measurement of the pressure of blood on the walls of blood vessels called arteries. The arteries deliver blood from the heart to all of the tissues in the body. Blood pressure is reported as two numbers. For example, a normal blood pressure is reported as 110/70 mm Hg (read as 110 over 70 millimeters of mercury; or just 110 over 70). These two

numbers represent two measurements, the systolic pressure and the diastolic pressure. The systolic pressure (the first number in the example; 110/70 mm Hg) measures the peak pressure of the blood against the artery walls. This higher pressure occurs as blood is being pumped out of the heart and into the circulatory system. The pumping chambers of the heart (ventricles) squeeze to force the blood out of the heart. The diastolic pressure (the second number in the example 110/70 mm Hg) measures the pressure, during the filling of the ventricles. At this point, the atria contract to fill the ventricles. Because the ventricles are relatively relaxed, and are not pumping blood into the arteries, pressure in the arteries is lower as well.

High blood pressure in pregnancy (**hypertension**) is a very serious complication. It puts both the mother and the fetus (developing baby) at risk for a number of problems. Hypertension can exist in several different forms:

- The preeclampsia-eclampsia continuum (also called pregnancy-induced hypertension or PIH). In this type of hypertension, high blood pressure is first noted sometime after week 20 of pregnancy and is accompanied by protein in the urine and swelling.
- Chronic hypertension. This type of hypertension usually exists before pregnancy or may develop before week 20 of pregnancy.
- Chronic hypertension with superimposed preeclampsia. This syndrome occurs when a woman with pre-existing chronic hypertension begins to have protein in the urine after week 20 of pregnancy.
- Late hypertension. This is a form of high blood pressure occurring after week 20 of pregnancy and is unaccompanied by protein in the urine and does not progress the way preeclampsia-eclampsia does.

Preeclampsia is most common among women who have never given birth to a baby (called nulliparas). About 7% of all nulliparas develop preeclampsia. The disease is most common in mothers under the age of 20, or over the age of 35. African-American women have higher rates of preeclampsia than do Caucasian women. Other risk factors include poverty, multiple pregnancies (twins, triplets, etc.), pre-existing chronic hypertension or **kidney disease**, diabetes, excess amniotic fluid, and a condition of the fetus called nonimmune hydrops. The tendency to develop preeclampsia appears to run in families. The daughters and sisters of women who have had preeclampsia are more likely to develop the condition.

Causes and symptoms

Experts are still trying to understand the exact causes of preeclampsia and eclampsia. It is generally accepted that preeclampsia and eclampsia are problematic because these conditions cause blood vessels to leak. The effects are seen throughout the body.

- **General body tissues.** When blood vessels leak, they allow fluid to flow out into the tissues of the body. The result is swelling in the hands, feet, legs, arms, and face. While many pregnant women experience swelling in their feet, and sometimes in their hands, swelling of the upper limbs and face is a sign of a more serious problem. As fluid is retained in these tissues, the woman may experience significant weight gain (two or more pounds per week).
- **Brain.** Leaky vessels can cause damage within the brain, resulting in seizures or **coma**.
- **Eyes.** The woman may experience problems seeing, and may have blurry vision or may see spots. The retina may become detached.
- **Lungs.** Fluid may leak into the tissues of the lungs, resulting in shortness of breath.
- **Liver.** Leaky vessels within the liver may cause it to swell. The liver may be involved in a serious complication of preeclampsia, called the HELLP syndrome. In this syndrome, red blood cells are abnormally destroyed, chemicals called liver enzymes are abnormally high, and cells involved in the clotting of blood (platelets) are low.
- **Kidneys.** The small capillaries within the kidneys can leak. Normally, the filtration system within the kidney is too fine to allow protein (which is relatively large) to leave the bloodstream and enter the urine. In preeclampsia, however, the leaky capillaries allow protein to be dumped into the urine. The development of protein in the urine is very serious, and often results in a low birth weight baby. These babies have a higher risk of complications, including **death**.
- **Blood pressure.** In preeclampsia, the volume of circulating blood is lower than normal because fluid is leaking into other parts of the body. The heart tries to make up for this by pumping a larger quantity of blood with each contraction. Blood vessels usually expand in diameter (dilate) in this situation to decrease the work load on the heart. In preeclampsia, however, the blood vessels are abnormally constricted, causing the heart to work even harder to pump against the small diameters of the vessels. This causes an increase in blood pressure.

The most serious consequences of preeclampsia and eclampsia include brain damage in the mother due to brain swelling and oxygen deprivation during seizures. Mothers can also experience blindness, kidney failure, liver rupture, and placental abruption. Babies born to preeclamptic mothers are often smaller than normal, which makes them more susceptible to complications during labor, delivery, and in early infancy. Babies of preeclamptic mothers are also at risk of being born prematurely, and can suffer the complications associated with **prematurity**.

Diagnosis

Diagnosing preeclampsia may be accomplished by noting painless swelling of the arms, legs, and/or face, in addition to abnormal weight gain. The patient's blood pressure is taken during every doctor's visit during pregnancy. An increase of 30 mm Hg in the systolic pressure, or 15 mm Hg in the diastolic pressure, or a blood pressure reading greater than 140/90 mm Hg is considered indicative of preeclampsia. A simple laboratory test in the doctor's office can indicate the presence of protein in a urine sample (a dipstick test). A more exact measurement of the amount of protein in the urine can be obtained by collecting urine for 24 hours, and then testing it in a laboratory to determine the actual quantity of protein present. A 24-hour urine specimen containing more than 500 mg of protein is considered indicative of preeclampsia.

Treatment

With mild preeclampsia, treatment may be limited to bed rest, with careful daily monitoring of weight, blood pressure, and urine protein via dipstick. This careful monitoring will be required throughout pregnancy, labor, delivery, and even for 2–4 days after the baby has been born. About 25% of all cases of eclampsia develop in the first few days after the baby's birth. If the diastolic pressure does not rise over 100 mm Hg prior to delivery, and no other symptoms develop, the woman can continue pregnancy until the fetus is mature enough to be delivered safely. Ultrasound tests can be performed to monitor the health and development of the fetus.

If the diastolic blood pressure continues to rise over 100 mm Hg, or if other symptoms like headache, vision problems, abdominal **pain**, or blood abnormalities develop, then the patient may require medications to prevent seizures. Magnesium sulfate is commonly given through a needle in a vein (intravenous, or IV). Medications that lower blood pressure

KEY TERMS

Capillary—The tiniest blood vessels with the smallest diameter. These vessels receive blood from the arterioles and deliver blood to the venules.

Diastolic—The phase of blood circulation in which the heart's pumping chambers (ventricles) are being filled with blood. During this phase, the ventricles are at their most relaxed, and the pressure against the walls of the arteries is at its lowest.

Placenta—The organ that provides oxygen and nutrition from the mother to the fetus during pregnancy. The placenta is attached to the wall of the uterus and leads to the fetus via the umbilical cord.

Placental abruption—An abnormal separation of the placenta from the uterus before the birth of the baby, with subsequent heavy uterine bleeding. Normally, the baby is born first and then the placenta is delivered within a half hour.

Systolic—The phase of blood circulation in which the heart's pumping chambers (ventricles) are actively pumping blood. The ventricles are squeezing (contracting) forcefully, and the pressure against the walls of the arteries is at its highest.

Urine dipstick test—A test using a small, chemically treated strip that is dipped into a urine sample; when testing for protein, an area on the strip changes color depending on the amount of protein (if any) in the urine.

Uterus—The muscular organ that contains the developing baby during pregnancy.

Ventricles—The two chambers of the heart that are involved in pumping blood. The right ventricle pumps blood into the lungs to receive oxygen. The left ventricle pumps blood into the circulation of the body to deliver oxygen to all of the body's organs and tissues.

(antihypertensive drugs) are reserved for patients with very high diastolic pressures (over 110 mm Hg), because lowering the blood pressure will decrease the amount of blood reaching the fetus. This places the fetus at risk for oxygen deprivation. If preeclampsia appears to be progressing toward true eclampsia, then medications may be given in order to start labor. Babies can usually be delivered vaginally. After the baby is delivered, the woman's blood pressure and other vital signs will usually begin to return to normal quickly.

Prognosis

The prognosis in preeclampsia and eclampsia depends on how carefully a patient is monitored. Very careful, consistent monitoring allows quick decisions to be made, and improves the woman's prognosis. Still, the most common causes of death in pregnant women are related to high blood pressure.

About 33% of all patients with preeclampsia will have the condition again with later pregnancies. Eclampsia occurs in about 1 out of every 200 women with preeclampsia. If not treated, eclampsia is almost always fatal.

Prevention

More information on how preeclampsia and eclampsia develop is needed before recommendations can be made on how to prevent these conditions. Research is being done with patients in high risk groups to see if calcium supplementation, **aspirin**, or fish oil supplementation may help prevent preeclampsia. Most importantly, it is clear that careful monitoring during pregnancy is necessary to diagnose preeclampsia early. Although even carefully monitored patients may develop preeclampsia and eclampsia, close monitoring by practitioners will help decrease the complications of these conditions.

Resources

ORGANIZATIONS

American College of Obstetricians and Gynecologists. 409 12th Street, S.W., P.O. Box 96920.

Rosalyn Carson-DeWitt, MD

Pregnancy-induced high blood pressure see
Preeclampsia and eclampsia

Pregnancy

Definition

The period from conception to birth. After the egg is fertilized by a sperm and then implanted in the lining of the uterus, it develops into the placenta and embryo, and later into a fetus. Pregnancy usually lasts 40 weeks, beginning from the first day of the woman's last menstrual period, and is divided into three trimesters, each lasting three months.

Description

Pregnancy is a state in which a woman carries a fertilized egg inside her body. Due to technological advances, pregnancy is increasingly occurring among older women in the United States.

First month

At the end of the first month, the embryo is about a third of an inch long, and its head and trunk—plus the beginnings of arms and legs—have started to develop. The embryo receives nutrients and eliminates waste through the umbilical cord and placenta. By the end of the first month, the liver and digestive system begin to develop, and the heart starts to beat.

Second month

In this month, the heart starts to pump and the nervous system (including the brain and spinal cord) begins to develop. The 1 in (2.5 cm) long fetus has a complete cartilage skeleton, which is replaced by bone cells by month's end. Arms, legs and all of the major organs begin to appear. Facial features begin to form.

Third month

By now, the fetus has grown to 4 in (10 cm) and weighs a little more than an ounce (28 g). Now the major blood vessels and the roof of the mouth are almost completed, as the face starts to take on a more recognizably human appearance. Fingers and toes appear. All the major organs are now beginning to form; the kidneys are now functional and the four chambers of the heart are complete.

Fourth month

The fetus begins to kick and swallow, although most women still can't feel the baby move at this point. Now 4 oz (112 g), the fetus can hear and urinate, and has established sleep-wake cycles. All organs are now fully formed, although they will continue to grow for the next five months. The fetus has skin, eyebrows, and hair.

Fifth month

Now weighing up to a 1 lb (454 g) and measuring 8–12 in (20–30 cm), the fetus experiences rapid growth as its internal organs continue to grow. At this point, the mother may feel her baby move, and she can hear the heartbeat with a stethoscope.

Sixth month

Even though its lungs are not fully developed, a fetus born during this month can survive with intensive care. Weighing 1–1.5 lbs (454–681 g), the fetus is red, wrinkly, and covered with fine hair all over its body. The fetus will grow very fast during this month as its organs continue to develop.

Seventh month

There is a better chance that a fetus born during this month will survive. The fetus continues to grow rapidly, and may weigh as much as 3 lb (1.3 kg) by now. Now the fetus can suck its thumb and look around its watery womb with open eyes.

Eighth month

Growth continues but slows down as the baby begins to take up most of the room inside the uterus. Now weighing 4–5 lbs (1.8–2.3 kg) and measuring 16–18 in (40–45 cm) long, the fetus may at this time prepare for delivery next month by moving into the head-down position.

Ninth month

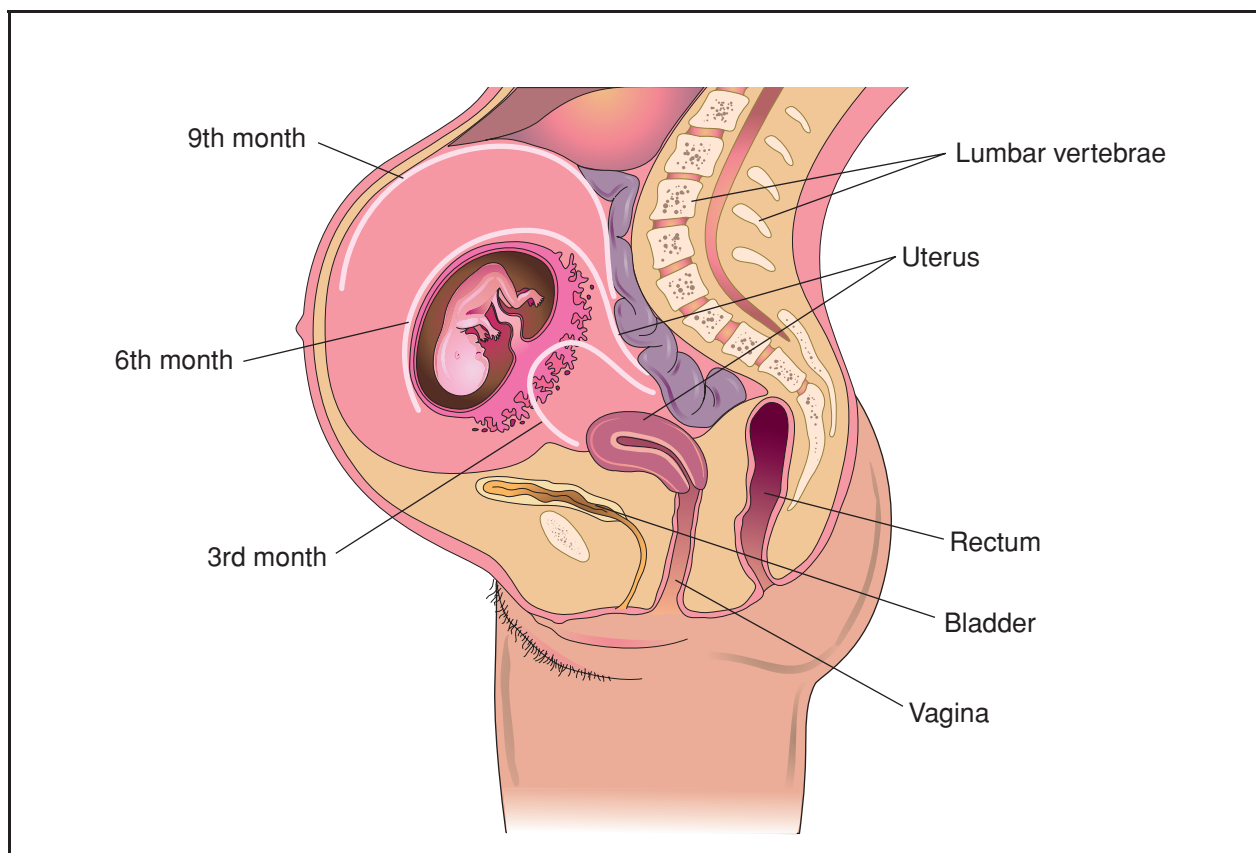
Adding 0.5 lb (227 g) a week as the due date approaches, the fetus drops lower into the mother's abdomen and prepares for the onset of labor, which may begin any time between the 37th and 42nd week of gestation. Most healthy babies will weigh 6–9 lb (2.7–4 kg) at birth, and will be about 20 in. long.

Causes and symptoms

The first sign of pregnancy is usually a missed menstrual period, although some women bleed in the beginning. A woman's breasts swell and may become tender as the mammary glands prepare for eventual breastfeeding. Nipples begin to enlarge and the veins over the surface of the breasts become more noticeable.

Nausea and vomiting are very common symptoms and are usually worse in the morning and during the first trimester of pregnancy. They are usually caused by hormonal changes, in particular, increased levels of progesterone. Women may feel worse when their stomach is empty, so it is a good idea to eat several small meals throughout the day, and to keep things like crackers on hand to eat even before getting out of bed in the morning.

Many women also feel extremely tired during the early weeks. Frequent urination is common, and there



Pregnancy usually lasts 40 weeks in humans, beginning from the first day of the woman's last menstrual period, and is divided into three trimesters. The illustration above depicts the position of the developing fetus during each trimester. (Illustration by Electronic Illustrators Group.)

may be a creamy white discharge from the vagina. Some women crave certain foods, and an extreme sensitivity to smell may worsen the **nausea**. Weight begins to increase.

In the second trimester (13–28 weeks) a woman begins to look noticeably pregnant and the enlarged uterus is easy to feel. The nipples get bigger and darker, skin may darken, and some women may feel flushed and warm. Appetite may increase. By the 22nd week, most women have felt the baby move. During the second trimester, nausea and **vomiting** often fade away, and the pregnant woman often feels much better and more energetic. Heart rate increases as does the volume of blood in the body.

By the third trimester (29–40 weeks), many women begin to experience a range of common symptoms. Stretch marks may develop on abdomen, breasts, and thighs, and a dark line may appear from the navel to pubic hair. A thin fluid may be expressed from the nipples. Many women feel hot, sweat easily and often find it hard to get comfortable. Kicks from

an active baby may cause sharp pains, and lower backaches are common. More rest is needed as the woman copes with the added **stress** of extra weight. Braxton Hicks contractions may get stronger.

At about the 36th week in a first pregnancy (later in repeat pregnancies), the baby's head drops down low into the pelvis. This may relieve pressure on the upper abdomen and the lungs, allowing a woman to breathe more easily. However, the new position places more pressure on the bladder.

A healthy gain for most women is between 25 and 35 pounds. Women who are overweight should gain less; and women who are underweight should gain more. On average, pregnant women need an additional 300 calories a day. Generally, women will gain three to five pounds in the first three months, adding one to two pounds a week until the baby is born. An average, healthy full-term baby at birth weighs 7.5 lb (3.4 kg), and the placenta and fluid together weigh another 3.5 lb. The remaining weight that a woman gains during pregnancy is mostly due to water

retention and fat stores. Her breasts, for instance, gain about 2 lb. in weight, and she gains another 4 lb due to the increased blood volume of pregnancy.

In addition to the typical, common symptoms of pregnancy, some women experience other problems that may be annoying, but which usually disappear after delivery. **Constipation** may develop as a result of food passing more slowly through the intestine. **Hemorrhoids** and **heartburn** are fairly common during late pregnancy. Gums may become more sensitive and bleed more easily; eyes may dry out, making contact lenses feel painful. **Pica** (a craving to eat substances other than food) may occur. Swollen ankles and **varicose veins** may be a problem in the second half of pregnancy, and chloasma may appear on the face.

Chloasma, also known as the “mask of pregnancy” or melasma, is caused by hormonal changes that result in blotches of pale brown skin appearing on the forehead, cheeks, and nose. These blotches may merge into one dark mask. It usually fades gradually after pregnancy, but it may become permanent or recur with subsequent pregnancies. Some women also find that the line running from the top to the bottom of their abdomen darkens. This is called the *linea nigra*.

While the above symptoms are all considered to be normal, there are some symptoms that could be a sign of a more dangerous underlying problem. A pregnant woman with any of the following signs should contact her doctor immediately:

- abdominal pain
- rupture of the amniotic sac or leaking of fluid from the vagina
- bleeding from the vagina
- no fetal movement for 24 hours (after the fifth month)
- continuous headaches
- marked, sudden swelling of eyelids, hands, or face during the last three months
- dim or blurry vision during last three months
- persistent vomiting

Diagnosis

Many women first discover they are pregnant after a positive home pregnancy test. Pregnancy urine tests check for the presence of human chorionic gonadotropin (hCG), which is produced by a placenta. The newest home tests can detect pregnancy on the day of the missed menstrual period.

Home pregnancy tests are more than 97% accurate if the result is positive, and about 80% accurate if the result is negative. If the result is negative and there is no menstrual period within another week, the pregnancy test should be repeated. While home pregnancy tests are very accurate, they are less accurate than a pregnancy test conducted at a lab. For this reason, women may want to consider having a second pregnancy test conducted at their doctor's office to be sure of the accuracy of the result.

Blood tests to determine pregnancy are usually used only when a very early diagnosis of pregnancy is needed. This more expensive test, which also looks for hCG, can produce a result within nine to 12 days after conception.

Once pregnancy has been confirmed, there are a range of screening tests that can be done to screen for **birth defects**, which affect about 3% of unborn children. Two tests are recommended for all pregnant women: alpha-fetoprotein (AFP) and the triple marker test.

Other tests are recommended for women at higher risk for having a child with a birth defect. This would include women over age 35, who had another child or a close relative with a birth defect, or who have been exposed to certain drugs or high levels of radiation. Women with any of these risk factors may want to consider **amniocentesis**, **chorionic villus sampling** (CVS) or ultrasound.

Other prenatal tests

There are a range of other prenatal tests that are routinely performed, including:

- PAP test
- gestational diabetes screening test at 24–28 weeks
- tests for sexually transmitted diseases
- urinalysis
- blood tests for anemia or blood type
- screening for immunity to various diseases, such as German measles

Treatment

Prenatal care is vitally important for the health of the unborn baby. A pregnant woman should be sure to eat a balanced, nutritious diet of frequent, small meals. Women should begin taking 400 mcg of **folic acid** several months before becoming pregnant, as folic acid has been shown to reduce the risk of spinal cord defects, such as **spina bifida**.

KEY TERMS

Alpha-fetoprotein—A substance produced by a fetus' liver that can be found in the amniotic fluid and in the mother's blood. Abnormally high levels of this substance suggests there may be defects in the fetal neural tube, a structure that will include the brain and spinal cord when completely developed. Abnormally low levels suggest the possibility of Down' syndrome.

Braxton Hicks' contractions—Short, fairly painless uterine contractions during pregnancy that may be mistaken for labor pains. They allow the uterus to grow and help circulate blood through the uterine blood vessels.

Chloasma—A skin discoloration common during pregnancy, also known as the "mask of pregnancy" or melasma, in which blotches of pale brown skin appear on the face. It is usually caused by hormonal changes. The blotches may appear in the forehead, cheeks, and nose, and may merge into one dark mask. It usually fades gradually after pregnancy, but it may become permanent or recur with subsequent pregnancies. Some women may also find that the line running from the top to the bottom of their abdomen darkens. This is called the linea nigra.

Embryo—An unborn child during the first eight weeks of development following conception (fertilization with sperm). For the rest of pregnancy, the embryo is known as a fetus.

Fetus—An unborn child from the end of the eighth week after fertilization until birth.

Human chorionic gonadotropin (hCG)—A hormone produced by the placenta during pregnancy.

Placenta—The organ that develops in the uterus during pregnancy that links the blood supplies of the mother and baby.

Rhythm method—The oldest method of contraception with a very high failure rate, in which partners periodically refrain from having sex during ovulation. Ovulation is predicted on the basis of a woman's previous menstrual cycle.

Spina bifida—A congenital defect in which part of the vertebrae fail to develop completely, leaving a portion of the spinal cord exposed.

No medication (not even a nonprescription drug) should be taken except under medical supervision, since it could pass from the mother through the placenta to the developing baby. Some drugs, called

teratogens, have been proven harmful to a fetus, but no drug should be considered completely safe (especially during early pregnancy). Drugs taken during the first three months of a pregnancy may interfere with the normal formation of the baby's organs, leading to birth defects. Drugs taken later on in pregnancy may slow the baby's growth rate, or they may damage specific fetal tissue (such as the developing teeth), or cause preterm birth.

To have the best chance of having a healthy baby, a pregnant woman should avoid:

- smoking
- alcohol
- street drugs
- large amounts of **caffeine**
- artificial sweeteners

Nutrition

Women should begin following a healthy diet even before they become pregnant. This means cutting back on high-calorie, high-fat, high-sugar snacks, and increasing the amount of fruits, vegetables and whole grains in her diet. Once she becomes pregnant, she should make sure to get at least six to 11 servings of breads and other whole grains, three to five servings of vegetables, two to four servings of fruits, four to six servings of milk and milk products, three to four servings of meat and protein foods, and six to eight glasses of water. She should limit caffeine to no more than one soft drink or cup of coffee per day.

Prognosis

Pregnancy is a natural condition that usually causes little discomfort provided the woman takes care of herself and gets adequate prenatal care. **Childbirth** education classes for the woman and her partner help prepare the couple for labor and delivery.

Prevention

There are many ways to avoid pregnancy. A woman has a choice of many methods of **contraception** which will prevent pregnancy, including (in order of least to most effective):

- spermicide alone
- natural (rhythm) method
- diaphragm or cap alone
- condom alone

- diaphragm with spermicide
- condom with spermicide
- intrauterine device (IUD)
- contraceptive pill
- sterilization (either a man or woman)
- avoiding intercourse

Resources

ORGANIZATIONS

Healthy Mothers, Healthy Babies National Coalition. 409 12th St., Washington, DC 20024. (202) 638-5577.

National Institute of Child Health and Human Development. 9000 Rockville Pike, Bldg. 31, Rm. 2A32, Bethesda, MD 20892. (301) 496-5133.

Positive Pregnancy and Parenting Fitness. 51 Saltrock Rd., Baltic, CT 06330. (203) 822-8573.

OTHER

Doulas of North America <<http://www.dona.com>>.

Planned Parenthood. <<http://www.plannedparenthood.org>>.

Pregnancy Information. <<http://www.childbirth.org>>.

Debra Gordon

Pregnancy test see **Human chorionic gonadotropin pregnancy test**

Preleukemia see **Myelodysplastic syndrome**

Premature atrial contractions see **Atrial ectopic beats**

Premature birth see **Prematurity**

Premature ejaculation

Definition

Premature ejaculation occurs when male sexual climax (orgasm) occurs before a man wishes it or too quickly during intercourse to satisfy his partner.

Description

Premature ejaculation is the most commonly reported sexual complaint of men and couples. The highest number of complaints is among teenage, young adult, and sexually inexperienced males. Increased risk is associated with sexual inexperience and lack of knowledge of normal male sexual responses.

Causes and symptoms

There are several reasons why a man may ejaculate prematurely. For some men, the cause is due to an innate reflex or psychological predisposition of the nervous system. Sometimes it can be caused by certain drugs, such as non-prescription cold medications. Psychological factors, such as **stress**, fear, or guilt can also play a role. Examples of psychological factors include guilt that the sexual activity is wrong or sinful, fear of getting caught, or stress from problems at work or home.

In general, symptoms are when a male reaches climax in less than two minutes or when it occurs before the male or couple want it to occur.

Diagnosis

There are no tests used to diagnose premature ejaculation. It is usually determined by the male involved based on his belief that he reached orgasm too quickly. General guidelines for premature ejaculation is if it occurs in two minutes or less, or prior to about 15 thrusts during sexual intercourse.

Treatment

In 1966, William H. Masters and Virginia E. Johnson published *Human Sexual Response*, in which they broke the first ground in approaching this topic from a new perspective. Their method was devised by Dr. James Seman and has been modified subsequently by Dr. Helen Singer Kaplan and others.

A competent and orthodox sex therapist will spend much more time focusing on the personal than the sexual relationship between the two people who come for treatment. Without emotional intimacy, sexual relations are superficial and sexual problems such as premature ejaculation are not always overcome.

With that foremost in mind, a careful plan is outlined that requires dedication, patience, and commitment by both partners. It necessarily begins by prohibiting intercourse for an extended period of time—at least a week, often a month. This is very important to the man because “performance anxiety” is the greatest enemy of performance. If he knows he cannot have intercourse he is able to relax and focus on the exercises. The first stage is called “sensate focus” and involves his concentration on the process of sexual arousal and climax. He should learn to recognize each step in the process, most particularly the moment just before the “point of no return.” Ideally, this stage of treatment requires the man’s

partner to be devoted to his sensations. In order to regain equality, he should in turn spend separate time stimulating and pleasing his mate, without intercourse.

At this point the techniques diverge. The original “squeeze technique” requires that the partner become expert at squeezing the head of the penis at intervals to prevent orgasm. The modified procedure, described by Dr. Ruth Westheimer, calls upon the man to instruct the partner when to stop stimulating him to give him a chance to draw back. A series of stages follows, each offering greater stimulation as the couple gains greater control over his arousal. This whole process has been called “outercourse.” After a period of weeks, they will have together retrained his response and gained satisfactory control over it. In addition, they will each have learned much about the other’s unique sexuality and ways to increase each other’s pleasure.

With either technique, the emphasis is on the mutual goal of satisfactory sexual relations for both partners.

However, the 1990s ushered in a new era in the treatment of premature ejaculation when physicians discovered that certain antidepressant drugs had a side effect of delaying ejaculation. Clinical studies have shown that a class of antidepressants called selective serotonin reuptake inhibitors (SSRIs) can be very effective in prolonging the time to ejaculation. The individual drugs and the average amount of time they delay ejaculation are fluoxetine (Prozac), one to two minutes with doses of 20–40 milligrams per day (mg/day) and eight minutes with 60 mg/day; paroxetine (Paxil), three to 10 minutes with doses of 20–40 mg/day; and sertraline (Zoloft), two to five minutes with doses of 50–200 mg/day.

Alternative treatment

There are several alternative products, usually found in health food and **nutrition** stores, designed to be sprayed or rubbed on the penis to delay ejaculation. Although the products promise results, there are no valid clinical studies to support the claims. A device called a testicular restraint, sold through erotic mail-order magazines, sometimes helps men delay ejaculation. The Velcro-like device restrains the testicles from their natural tendency to move during sex. Testicular movement can cause premature ejaculation.

Prognosis

The “squeeze technique” has elicited a 95% success rate, whereby the patient is able to control ejaculation.

Treatment with SSRIs is effective in 85–90% of cases. However, the effectiveness begins to decrease after five weeks of daily administration. Although more studies are needed, this suggests the SSRIs are more effective when used on an as-needed basis.

Prevention

The best prevention is obtaining adequate information on normal sexual responses of males before having sex. It is also helpful to have sex in a comfortable, relaxed, private setting, free of guilt, stress, and fear.

Resources

PERIODICALS

“Lengthen Your Fuse.” *Men’s Health* November 1999: 56. Rowland, David L., and Arthur L. Burnett.

“Pharmacotherapy in the Treatment of Male Sexual Dysfunction.” *The Journal of Sex Research* August 2000: 226+.

ORGANIZATIONS

American Association for Marriage and Family Therapy. 1133 15th St. NW, Suite 300, Washington, DC 20005-2710. (202) 452-0109. <<http://www.aamft.org>>.

American Association of Sex Educators, Counselors, and Therapists. P.O. Box 5488, Richmond, VA 23220. <<http://www.aasect.org>>.

Sexuality Information and Education Council of the U.S. 130 W. 42nd St., Ste. 350, New York, NY 10036. (212) 819-9770. <<http://www.siecus.org>>.

Ken R. Wells

Premature labor

Definition

Premature labor is the term to describe contractions of the uterus that begin at weeks 20–36 of a **pregnancy**.

Description

The usual length of a human pregnancy is 38–42 weeks after the first day of the last menstrual period. Labor is a natural series of events that indicate that the birth process is starting. Premature labor is defined as contractions that occur after 20 weeks and before 37 weeks during the term of pregnancy. The baby is more likely to survive and be healthy if it remains in the uterus for the full term of the

pregnancy. It is estimated that around 10% of births in the United States occur during the premature period. Premature birth is the greatest cause of newborn illness and death. In the United States, **prematurity** has a greater impact on African-Americans.

Causes and symptoms

The causes of premature labor cannot always be determined. Some research suggests that infection of the urinary or reproductive tract may stimulate premature labor and premature births. Multiple pregnancies (twins, triplets, etc.) are more likely to result in to premature labor. Smoking, alcohol use, drug **abuse**, and poor **nutrition** can increase the risk of premature labor and birth. Adolescent mothers are also at higher risk for premature delivery. Women whose mothers took diethylstilbestrol (DES) when they carried them are more likely to deliver prematurely, as are women who have had previous surgery on the cervix.

The symptoms of premature labor can include contractions of the uterus or tightening of the abdomen, which occurs every ten minutes or more frequently. These contractions usually increase in frequency, duration, and intensity, and may or may not be painful. Other symptoms associated with premature labor can include menstrual-like cramps, abdominal cramping with or without diarrhea, pressure or **pain** in the pelvic region, low backache, or a change in the color or amount of vaginal discharge. As labor progresses, the cervix or opening of the uterus will open (dilate) and the tissue around it will become thinner (efface). **Premature rupture of membranes** (when the water breaks) may also occur.

An occasional contraction can occur anytime during the pregnancy and does not necessarily indicate that labor is starting. Premature contractions are sometimes confused with Braxton Hicks contractions, which can occur throughout the pregnancy. Braxton Hicks contractions do not cause the cervix to open or efface, and are considered “false labor.”

Diagnosis

The health care provider will conduct a physical examination and ask about the timing and intensity of the contractions. A vaginal examination is the only way to determine if the cervix has started to dilate or efface. Urine and blood samples may be collected to screen for infection. A vaginal culture (a cotton-tipped

KEY TERMS

Braxton Hicks contractions—Tightening of the uterus or abdomen that can occur throughout pregnancy. These contractions do not cause changes to the cervix and are sometimes called false labor or practice contractions.

Cervix —The opening at the bottom of the uterus, which dilates or opens in order for the fetus to pass into the vagina or birth canal during the delivery process.

Contraction—A tightening of the uterus during pregnancy. Contractions may or may not be painful and may or may not indicate labor.

swab is used to collect some fluid and cells from the vagina) may be done to look for a vaginal infection. A fetal heart monitor may be placed on the mother's abdomen to record the heartbeat of the fetus and to time the contractions. A fetal ultrasound may be performed to determine the age and weight of the fetus, the condition of the placenta, and to see if there is more than one fetus present. **Amniocentesis** will sometimes be performed. This is a procedure where a needle-like tube is inserted through the mother's abdomen to draw out some of the fluid surrounding the fetus. Analysis of the amniotic fluid can determine if the baby's lungs are mature. A baby with mature lungs is much more likely to survive outside the uterus.

Treatment

The goal of treatment is to stop the premature labor and prevent the fetus from being delivered before it is full term. A first recommendation may be for the woman with premature contractions to lie down with feet elevated and to drink juice or other fluids. If contractions continue or increase, medical attention should be sought. In addition to bed rest, medical care may include intravenous fluids. Sometimes, this extra fluid is enough to stop contractions. In some cases, oral or injectable drugs like terbutaline sulfate, ritodrine, magnesium sulfate, or nifedipine must be given to stop the contractions. These are generally very effective; however, as with any drug therapy, there are risks of side effects. Some women may need to continue on medication for the duration of the pregnancy. **Antibiotics** may be prescribed if a vaginal or urinary tract infection is

detected. If the membranes have already ruptured, it may be difficult or impossible to stop premature labor. If infection of the membranes that cover the fetus (chorioamnionitis) develops, the baby must be delivered.

Prognosis

If premature labor is managed successfully, the pregnancy may continue normally for the delivery of a healthy infant. Once symptoms of preterm labor occur during the pregnancy, the mother and fetus need to be monitored regularly since it is likely that premature labor will occur again. If the preterm labor cannot be stopped or controlled, the infant will be delivered prematurely. These infants that are born prematurely have an increased risk of health problems including **birth defects**, lung problems, mental retardation, blindness, deafness, and developmental disabilities. If the infant is born too early, its body systems may not be mature enough for it to survive. Evaluating the infant's lung maturity is one of the keys to determining its chance of survival. Fetuses delivered further into pregnancy and those with more mature lungs are more likely to survive.

Prevention

Smoking, poor nutrition, and drug or alcohol abuse can increase the risk of premature labor and early delivery. Smoking and drug or alcohol use should be stopped. A healthy diet and prenatal vitamin supplements (prescribed by the health care provider) are important for the growth of the fetus and the health of the mother. Pregnant women are advised to see a health care provider early in the pregnancy and receive regular prenatal examinations throughout the pregnancy. The health care provider should be informed of any medications that the mother is receiving and any health conditions that exist before and during the pregnancy.

Resources

ORGANIZATIONS

March of Dimes Birth Defects Foundation. 1275 Mamaroneck Ave., White Plains, NY 10605. (914) 428-7100. resourcecenter@modimes.org. <<http://www.modimes.org>>.

OTHER

"Am I in Labor?" *The Virtual Hospital Page*. University of Iowa. <<http://www.vh.org>>.

Altha Roberts Edgren

Premature menopause

Definition

The average age at which American women go through **menopause** is 51 years. If menopause (hormonal changes at the end of the female reproductive years) occurs before age 40, it is said to be premature menopause. Possible causes include autoimmune problems and common cancer treatments.

Description

About half of all women will go through menopause before age 51 and the rest will go through it after. Most women will finish menopause between the ages of 42 and 58. A small number of women will find that their periods stop prematurely, before age 40.

Causes and symptoms

There are many possible causes of premature menopause. Women who have premature menopause often have **autoimmune disorders** like thyroid disease or **diabetes mellitus**. In these diseases, the body produces antibodies to one or more of its own organs. These antibodies interfere with the normal function of the organ. Just as antibodies might attack the thyroid or the pancreas (causing thyroid disease or diabetes), antibodies may attack the ovaries and stop the production of female hormones.

Cancer treatments like **chemotherapy** or radiation can cause premature menopause. The risk depends on the type and length of treatment and the age of the woman when she first begins radiation or chemotherapy.

If the ovaries are surgically removed (during a **hysterectomy**, for example) menopause will occur within a few days, no matter how old the woman is.

The symptoms of premature menopause are similar to those of menopause at any time. Menstrual periods stop and women may notice hot flashes, vaginal dryness, mood swings, and sleep problems. Sometimes the first symptom of premature menopause is **infertility**. A woman may find that she cannot become pregnant because she is not ovulating (producing eggs) anymore.

When menopause occurs after the ovaries are surgically removed, the symptoms begin within several days after surgery and tend to be more severe. This happens because the drop in the level of estrogen is dramatic, unlike the gradual drop that usually occurs.

KEY TERMS

Autoimmune diseases—Diseases in which the body creates antibodies that attack one of its own organs.

Follicle stimulating hormone (FSH)—A female hormone that regulates ovulation and menstruation.

Hormone replacement therapy (HRT)—Replacement of estrogen and progesterone lost by women who have gone through menopause. Hormone replacement therapy has been shown to lower the risk of osteoporosis and heart disease in elderly women.

Luteinizing hormone (LH)—A female hormone that regulates ovulation and menstruation.

Menopause—The end of a woman's reproductive years. The hormonal changes that accompany menopause include the hot flashes, vaginal dryness, mood swings, sleep problems, and the end of menstrual periods. Commonly known as "the change" or "the change of life."

Diagnosis

Premature menopause can be confirmed by blood tests to measure the levels of follicle stimulating hormone (FSH) and luteinizing hormone (LH). The levels of these hormones will be higher if menopause has occurred.

Because premature menopause is often associated with other hormonal problems, women who have premature menopause should be screened for diabetes, thyroid disease, and similar diseases.

Treatment

There is no treatment to reverse premature menopause. Hormone replacement therapy (HRT) can prevent the common symptoms of menopause and lower the long-term risk of **osteoporosis**. Women who have premature menopause should take HRT. Estrogen relieves the unpleasant symptoms of menopause, including the hot flashes and the vaginal dryness. Estrogen is especially important for women who go through premature menopause. The long-term health risks of menopause (osteoporosis and increased risk of heart disease) are even more likely to occur after premature menopause. However, women who have certain medical conditions (like **liver disease**, uterine cancer, or breast cancer) may not be candidates for estrogen.

If a woman still has her uterus after premature menopause, she will also need to take progesterone along with the estrogen. If her uterus has been removed, estrogen alone will be enough.

Women who wish to become pregnant after premature menopause now have the option of fertility treatments using donor eggs. This is similar to in vitro fertilization, but the eggs come from a donor instead of the woman who is trying to become pregnant.

Prevention

Premature menopause cannot be prevented.

Resources

BOOKS

Hall, Janet E. "Amenorrhea." In *Primary Care of Women*, edited by Karen J. Carlson and Stephanie A. Eisenstat. St. Louis: Mosby-Year Book, Inc., 1995.

Amy B. Tuteur, MD

Premature rupture of membranes

Definition

Premature rupture of membranes (PROM) is an event that occurs during **pregnancy** when the sac containing the developing baby (fetus) and the amniotic fluid bursts or develops a hole prior to the start of labor.

Description

During pregnancy, the unborn baby (fetus) is surrounded and cushioned by a liquid called amniotic fluid. This fluid, along with the fetus and the placenta, is enclosed within a sac called the amniotic membrane. The amniotic fluid is important for several reasons. It cushions and protects the fetus, allowing the fetus to move freely. The amniotic fluid also allows the umbilical cord to float, preventing it from being compressed and cutting off the fetus's supply of oxygen and nutrients. The amniotic membrane contains the amniotic fluid and protects the fetal environment from the outside world. This barrier protects the fetus from organisms (like bacteria or viruses) that could travel up the vagina and potentially cause infection.

Although the fetus is almost always mature at between 36–40 weeks and can be born without complication, a normal pregnancy lasts an average of 40

weeks. At the end of 40 weeks, the pregnancy is referred to as being “term.” At term, labor usually begins. During labor, the muscles of the uterus contract repeatedly. This allows the cervix to begin to grow thinner (called effacement) and more open (dilatation). Eventually, the cervix will become completely effaced and dilated. In the most common sequence of events (about 90% of all deliveries), the amniotic membrane breaks (ruptures) around this time. The baby then leaves the uterus and enters the birth canal. Ultimately, the baby will be delivered out of the mother’s vagina. In the 30 minutes after the birth of the baby, the placenta should separate from the wall of the uterus and be delivered out of the vagina.

Sometimes the membranes burst before the start of labor, and this is called premature rupture of membranes (PROM). There are two types of PROM. One occurs at a point in pregnancy before normal labor and delivery should take place. This is called preterm PROM. The other type of PROM occurs at 36–40 weeks of pregnancy.

PROM occurs in about 10% of all pregnancies. Only about 20% of these cases are preterm PROM. Preterm PROM is responsible for about 34% of all premature births.

Causes and symptoms

The causes of PROM have not been clearly identified. Some risk factors include **smoking**, multiple pregnancies (twins, triplets, etc.), and excess amniotic fluid (**polyhydramnios**). Certain procedures carry an increased risk of PROM, including **amniocentesis** (a diagnostic test involving extraction and examination of amniotic fluid) and cervical cerclage (a procedure in which the uterus is sewn shut to avoid **premature labor**). A condition called **placental abruption** is also associated with PROM, although it is not known which condition occurs first. In some cases of preterm PROM, it is believed that bacterial infection of the amniotic membrane causes it to weaken and then break. However, most cases of PROM and infection occur in the opposite order, with PROM occurring first followed by an infection.

The main symptom of PROM is fluid leaking from the vagina. It may be a sudden, large gush of fluid, or it may be a slow, constant trickle of fluid. The complications that may follow PROM include premature labor and delivery of the fetus, infections of the mother and/or the fetus, and compression of the umbilical cord (leading to oxygen deprivation in the fetus).

Labor almost always follows PROM, although the delay between PROM and the onset of labor varies. When PROM occurs at term, labor almost always begins within 24 hours. Earlier in pregnancy, labor can be delayed up to a week or more after PROM. The chance of infection increases as the time between PROM and labor increases. While this may cause doctors to encourage labor in the patient who has reached term, the risk of complications in a premature infant may cause doctors to try delaying labor and delivery in the case of preterm PROM.

The types of infections that can complicate PROM include amnionitis and endometritis. Amnionitis is an infection of the amniotic membrane. Endometritis is an infection of the innermost lining of the uterus. Amnionitis occurs in 0.5–1% of all pregnancies. In the case of PROM at term, amnionitis complicates about 3–15% of pregnancies. About 15–23% of all cases of preterm PROM will be complicated by amnionitis. The presence of amnionitis puts the fetus at great risk of developing an overwhelming infection (**sepsis**) circulating throughout its bloodstream. Preterm babies are the most susceptible to this life-threatening infection. One type of bacteria responsible for overwhelming infections in newborn babies is called group B streptococci.

Diagnosis

Depending on the amount of amniotic fluid leaking from the vagina, diagnosing PROM may be easy. Some doctors note that amniotic fluid has a very characteristic musty smell. A **pelvic exam** using a sterile medical instrument (speculum) may reveal a trickle of amniotic fluid leaving the cervix, or a pool of amniotic fluid collected behind the cervix. One of two easy tests can be performed to confirm that the liquid is amniotic fluid. A drop of the fluid can be placed on nitrazine paper. Nitrazine paper is made so that it turns from yellowish green to dark blue when it comes in contact with amniotic fluid. Another test involves smearing a little of the fluid on a slide, allowing it to dry, and then viewing it under a microscope. When viewed under the microscope, dried amniotic fluid will be easy to identify because it will look “feathery” like a fern.

Once PROM has been diagnosed, efforts are made to accurately determine the age of the fetus and the maturity of its lungs. Premature babies are at great risk if they have immature lungs. These evaluations can be made using amniocentesis and ultrasound measurements of the fetus’ size. Amniocentesis also allows the practitioner to check for infection. Other indications of infection include a **fever** in the mother,

increased heart rate of the mother and/or the fetus, high white blood cell count in the mother, foul smelling or pus-filled discharge from the vagina, and a tender uterus.

Treatment

Treatment of PROM depends on the stage of the patient's pregnancy. In PROM occurring at term, the mother and baby will be watched closely for the first 24 hours to see if labor will begin naturally. If no labor begins after 24 hours, most doctors will use medications to start labor. This is called inducing labor. Labor is induced to avoid a prolonged gap between PROM and delivery because of the increased risk of infection.

Preterm PROM presents more difficult treatment decisions. The younger the fetus, the more likely it may die or suffer serious permanent damage if delivered prematurely. Yet the risk of infection to the mother and/or the fetus increases as the length of time from PROM to delivery increases. Depending on the age of the fetus and signs of infection, the doctor must decide either to try to prevent labor and delivery until the fetus is more mature, or to induce labor and prepare to treat the complications of **prematurity**. However, the baby will need to be delivered to avoid serious risks to both it and the mother if infection is present, regardless of the risks of prematurity.

A variety of medications may be used in PROM:

- Medication to induce labor (oxytocin) may be used, either in the case of PROM occurring at term or in the case of preterm PROM and infection.
- Tocolytics may be given to halt or prevent the start of labor. These may be used in the case of preterm PROM, when there are no signs of infection. Delaying the start of labor may give the fetus time to develop more mature lungs.
- Steroids may be used to help the fetus' lungs mature early. Steroids may be given in preterm PROM if the fetus must be delivered early because of infection or labor that cannot be stopped.
- **Antibiotics** can be given to fight infections. Research is being done to determine whether antibiotics should be given prior to any symptoms of infection to avoid the development of infection.

Prognosis

The prognosis in PROM varies. It depends in large part on the maturity of the fetus and the development of infection.

KEY TERMS

Amniocentesis—A medical procedure during which a long, thin needle is inserted through the abdominal and uterine walls, and into the amniotic sac. A sample of amniotic fluid is withdrawn through the needle for examination.

Amniotic fluid—The fluid within the amniotic sac; the fluid surrounds, cushions, and protects the fetus.

Amniotic membrane—The thin tissue that creates the walls of the amniotic sac.

Cervical cerclage—A procedure in which the cervix is sewn closed; used in cases when the cervix starts to dilate too early in a pregnancy to allow the birth of a healthy baby.

Placenta—The organ that provides oxygen and nutrition from the mother to the fetus during pregnancy. The placenta is attached to the wall of the uterus, and leads to the fetus via the umbilical cord.

Prevention

The only controllable factor associated with PROM is smoking. Cigarette smoking should always be discontinued during a pregnancy.

Resources

ORGANIZATIONS

American College of Obstetricians and Gynecologists. 409
12th Street, S.W., P.O. Box 96920

Rosalyn Carson-DeWitt, MD

Premature ventricular contractions see

Ventricular ectopic beats

Prematurity

Definition

The length of a normal **pregnancy** or gestation is considered to be 40 weeks (280 days) from the date of conception. Infants born before 37 weeks gestation are considered premature and may be at risk for complications.

Description

More than one out of every ten infants born in the United States is born prematurely. Advances in medical technology have made it possible for infants born as young as 23 weeks gestational age (17 weeks premature) to survive. These premature infants, however, are at higher risk for **death** or serious complications, which include heart defects, respiratory problems, blindness, and brain damage.

Causes and symptoms

The birth of a premature baby can be brought on by several different factors, including premature labor; placental abruption, in which the placenta detaches from the uterus; placenta previa, in which the placenta grows too low in the uterus; **premature rupture of membranes**, in which the amniotic sac is torn, causing the amniotic fluid to leak out; **incompetent cervix**, in which the opening to the uterus opens too soon; and maternal toxemia, or blood **poisoning**. While one of these conditions are often the immediate reason for a premature birth, its underlying cause is usually unknown. Prematurity is much more common in multiple pregnancy and for mothers who have a history of miscarriages or who have given birth to a premature infant in the past. One of the few, and most important, identifiable causes of prematurity is drug **abuse**, particularly **cocaine**, by the mother.

Infants born prematurely may experience major complications due to their low birth weight and the immaturity of their body systems. Some of the common problems among premature infants are **jaundice** (yellow discoloration of the skin and whites of the eyes), apnea (a long pause in breathing), and inability to breast or bottle feed. Body temperature, blood pressure, and heart rate may be difficult to regulate in premature infants. The lungs, digestive system, and nervous system (including the brain) are underdeveloped in premature babies, and are particularly vulnerable to complications. Some of the more common risks and complications of prematurity are described below.

Respiratory distress syndrome (RDS) is the most common problem seen in premature infants. Babies born too soon have immature lungs that have not developed **surfactant**, a protective film that helps air sacs in the lungs to stay open. With RDS, breathing is rapid and the center of the chest and rib cage pull inward with each breath. Extra oxygen can be supplied to the infant through tubes that fit into the nostrils of the nose, or by placing the baby under an oxygen hood. In more serious cases, the baby may have to

have a breathing tube inserted and receive air from a respirator or ventilator. A surfactant drug can be given in some cases to coat the lung tissue. Extra oxygen may be needed for a few days or weeks, depending on how small and premature the baby was at birth. Bronchopulmonary dysplasia is the development of scar tissue in the lungs, and can occur in severe cases of RDS.

Necrotizing enterocolitis (NEC) is a further complication of prematurity. In this condition, part of the baby's intestines are destroyed as a result of bacterial infection. In cases where only the innermost lining of the bowel dies, the infant's body can regenerate it over time; however, if the full thickness of a portion dies, it must be removed surgically and an opening (ostomy) must be made for the passage of wastes until the infant is healthy enough for the remaining ends to be sewn together. Because NEC is potentially fatal, doctors are quick to respond to its symptoms, which include lethargy, **vomiting**, a swollen and/or red abdomen, **fever**, and blood in the stool. Measures include taking the infant off mouth feedings and feeding him or her intravenously; administering **antibiotics**; and removing air and fluids from the digestive tract via a nasal tube. Approximately 70% of NEC cases can be successfully treated without surgery.

Intraventricular hemorrhage (IVH) is another serious complication of prematurity. It is a condition in which immature and fragile blood vessels within the brain burst and bleed into the hollow chambers (ventricles) normally reserved for cerebrospinal fluid and into the tissue surrounding them. Physicians grade the severity of IVH according to a scale of I–IV, with I being bleeding confined to a small area around the burst vessels and IV being an extensive collection of blood not only in the ventricles, but in the brain tissue itself. Grades I and II are not uncommon, and the baby's body usually reabsorbs the blood with no ill effects. However, more severe IVH can result in **hydrocephalus**, a potentially fatal condition in which too much fluid collects in the ventricles, exerting increased pressure on the brain and causing the baby's head to expand abnormally. To drain fluid and relieve pressure on the brain, doctors will either perform lumbar punctures, a procedure in which a needle is inserted into the spinal canal to drain fluids; install a reservoir, a tube that drains fluid from a ventricle and into an artificial chamber under or on top of the scalp; or install a ventricular shunt, a tube that drains fluid from the ventricles and into the abdomen, where it is reabsorbed by the body. Infants who are at high risk for IVH usually have an ultrasound taken of their brain in the first week after birth, followed by others

if bleeding is detected. IVH cannot be prevented; however, close monitoring can ensure that procedures to reduce fluid in the brain are implemented quickly to minimize possible damage.

Apnea of prematurity is a condition in which the infant stops breathing for periods lasting up to 20 seconds. It is often associated with a slowing of the heart rate. The baby may become pale, or the skin color may change to a blue or purplish hue. Apnea occurs most commonly when the infant is asleep. Infants with serious apnea may need medications to stimulate breathing or oxygen through a tube inserted in the nose. Some infants may be placed on a ventilator or respirator with a breathing tube inserted into the airway. As the baby gets older, and the lungs and brain tissues mature, the breathing usually becomes more regular. A group of researchers in Cleveland reported in 2003, however, that children who were born prematurely are 3–5 times more likely to develop sleep-disordered breathing by age 10 than children who were full-term babies.

As the fetus develops, it receives the oxygen it needs from the mother's blood system. Most of the blood in the infant's system bypasses the lungs. Once the baby is born, its own blood must start pumping through the lungs to get oxygen. Normally, this bypass duct closes within the first few hours or days after birth. If it does not close, the baby may have trouble getting enough oxygen on its own. **Patent ductus arteriosus** is a condition in which the duct that channels blood between two main arteries does not close after the baby is born. In some cases, a drug, indomethacin, can be given to close the duct. Surgery may be required if the duct does not close on its own as the baby develops.

Retinopathy of prematurity is a condition in which the blood vessels in the baby's eyes do not develop normally, and can, in some cases, result in blindness. Premature infants are also more susceptible to infections. They are born with fewer antibodies, which are necessary to fight off infections.

Diagnosis

Many of the problems associated with prematurity depend on how early the baby is born and how much it weighs at birth. The most accurate way of determining the gestational age of an infant in utero is calculating from a known date of conception or using ultrasound imaging to observe development. When a baby is born, doctors can use the Dubowitz exam to estimate gestational age. This standardized test scores responses to 33 specific neurological stimuli

to estimate the infant's neural development. Once the baby's gestational age and weight are determined, further tests and **electronic fetal monitoring** may need to be used to diagnose problems or to track the baby's condition. A blood pressure monitor may be wrapped around the arm or leg. Several types of monitors can be taped to the skin. A heart monitor or cardiorespiratory monitor may be attached to the baby's chest, abdomen, arms, or legs with adhesive patches to monitor breathing and heart rate. A thermometer probe may be taped on the skin to monitor body temperature. Blood samples may be taken from a vein or artery. X rays or ultrasound imaging may be used to examine the heart, lungs, and other internal organs.

Treatment

Treatment depends on the types of complications that are present. It is not unusual for a premature infant to be placed in a heat-controlled unit (an incubator) to maintain its body temperature. Infants that are having trouble breathing on their own may need oxygen either pumped into the incubator, administered through small tubes placed in their nostrils, or through a respirator or ventilator, which pumps air into a breathing tube inserted into the airway. The infant may require fluids and nutrients to be administered through an intravenous line in which a small needle is inserted into a vein in the hand, foot, arm, leg, or scalp. If the baby needs drugs or medications, they may also be administered through the intravenous line. Another type of line may be inserted into the baby's umbilical cord. This can be used to draw blood samples or to administer medications or nutrients. If heart rate is irregular, the baby may have heart monitor leads taped to the chest. Many premature infants require time and support with breathing and feeding until they mature enough to breathe and eat unassisted. Depending on the complications, the baby may require drugs or surgery.

A form of treatment that is being recommended by many mainstream practitioners as of 2003 is **massage therapy**. Research has shown that the risks of massaging preterm infants are minimal, and that the infants benefit from improved developmental scores, more rapid weight gain, and earlier discharge from the hospital. An additional benefit of massage therapy is closer bonding between the parents and their newborn child.

Prognosis

Advances in medical care have made it possible for many premature infants to survive and develop normally. However, whether or not a premature infant

will survive is still intimately tied to his or her gestational age:

- 21 weeks or less: 0% survival rate
- 22 weeks: 0–10% survival rate
- 23 weeks: 10–35% survival rate
- 24 weeks: 40–70% survival rate
- 25 weeks: 50–80% survival rate
- 26 weeks: 80–90% survival rate
- 27 weeks: greater than 90% survival rate

Physicians cannot predict long-term complications of prematurity; some consequences may not become evident until the child is school-aged. Minor disabilities like learning problems, poor coordination, or short attention span may be the result of premature birth, but can be overcome with early intervention. The risks of serious long-term complications depend on many factors, including how premature the infant was at birth, weight at birth, and the presence or absence of breathing problems. Gender is a definite factor: a Swedish study published in 2003 found that boys are at greater risk of death or serious long-term consequences of prematurity than girls; for example, 60% of boys born at 24 weeks' gestation die, compared to 38% mortality for girls. The development of infection or the presence of a birth defect can also affect long-term prognosis. Infections in premature and very low birth weight infants are a risk factor for later disorders of the nervous system; a study done at Johns Hopkins reported that 77 out of a group of 213 premature infants developed neurologic disorders. Severe disabilities like brain damage, blindness, and chronic lung problems are possible and may require ongoing care.

Prevention

Some of the risks and complications of premature delivery can be reduced if the mother receives good prenatal care, follows a healthy diet, avoids alcohol or drug consumption, and refrains from cigarette **smoking**. In some cases of **premature labor**, the mother may be placed on bed rest or given drugs that can stop labor contractions for days or weeks, giving the developing infant more time to develop before delivery. The physician may prescribe a steroid medication to be given to the mother before the delivery to help speed up the baby's lung development. The availability of neonatal intensive care unit, a special hospital unit equipped and trained to deal with premature infants, can also increase the chances of survival.

A new medication may help to prevent spontaneous premature births. Researchers at Wake Forest

KEY TERMS

Apnea—A long pause in breathing.

Dubowitz exam—A standardized test that scores responses to 33 specific neurological stimuli to estimate an infant's neural development and, hence, gestational age.

Intraventricular hemorrhage (IVH)—A condition in which blood vessels within the brain burst and bleed into the hollow chambers (ventricles) normally reserved for cerebrospinal fluid and into the tissue surrounding them.

Jaundice—Yellow discoloration of skin and whites of the eyes that results from excess bilirubin in the body's system.

Necrotizing enterocolitis (NEC)—A condition in which part of the intestines are destroyed as a result of bacterial infection.

Respiratory distress syndrome (RDS)—Condition in which a premature infant with immature lungs does not develop surfactant, a protective film that helps air sacs in the lungs to stay open. RDS is the most common problem seen in premature infants.

Retinopathy of prematurity—A condition in which the blood vessels in a premature infant's eyes do not develop normally, and can, in some cases, result in blindness.

Surfactant—A protective film that helps air sacs in the lungs to stay open. Premature infants may not have developed this protective layer before birth and are more susceptible to respiratory problems without it. Some surfactant drugs are available. These can be given through a respirator and will coat the lungs when the baby breathes the drug in.

University reported in June 2003 that a drug known as 17 alpha-hydroxyprogesterone caproate not only reduced the number of premature births in a group of women who received weekly injections of the drug compared to a placebo group, but also lowered the rates of necrotizing enterocolitis, intraventricular hemorrhage, and need for supplemental oxygen in their infants.

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Premenstrual dysphoric disorder

Definition

Premenstrual dysphoric disorder (PMDD) is a collection of physical and emotional symptoms that occurs 5 to 11 days before a woman's period begins, and goes

away once menstruation starts. The most severe form of **premenstrual syndrome (PMS)** is PMDD.

Description

PMS is estimated to affect 70–90% of childbearing age. The more severe form of the disorder, PMDD, affects 3–5% of women of childbearing age. Up to 40% of women have PMDD symptoms that are so severe they interfere with their daily activities. It is more common in women in their late 20s and early 40s, who have at least one child and a history of depression, anxiety/tension, affective lability, or irritability/anger.

Causes and symptoms

Although the actual cause of PMDD is not known, it is believed to be related to hormonal changes that occur before menstruation. There are more than 150 signs and symptoms attributed to PMDD, and every woman experiences different ones at different times. There seem to be socioeconomic and genetic factors that precipitate PMDD. Twin studies have demonstrated a positive correlation with heritability and PMDD symptoms. Anti-anxiety medications have been shown to help improve symptoms associated with PMDD. The most common symptoms include **headache**, swelling of ankles, feet, and hands, backache, abdominal cramps, heaviness or **pain**, bloating and/or gas, **muscle spasms**, breast tenderness, weight gain, recurrent cold sores, **acne**, **nausea**, **constipation** or **diarrhea**, food cravings, **anxiety** or panic, confusion, difficulty concentrating and forgetfulness, poor judgment, and depression.

Diagnosis

PMDD is diagnosed when symptoms occur during the second half of the menstrual cycle (14 days or more after the first day of a woman's period), are absent for about seven days after the period ends, increase in severity as the cycle progresses, go away when the menstrual flow begins or shortly thereafter, and occur for at least three consecutive menstrual cycles. There are no tests to diagnose it. The diagnosis of PMDD emphasizes and requires psychologically important mood symptoms.

Treatment

Recently, the Food and Drug Administration approved the first prescription drug for the treatment of PMDD, Serafem (fluoxetine). Additionally, **non-steroidal anti-inflammatory drugs**, such as ibuprofen

KEY TERMS

Antidepressant—A medication used to relieve the symptoms of clinical depression.

Beta blockers—Class of drug, including Corgard (nadolol), and Lanoxin (digoxin), that primarily work by blunting the action of adrenaline, the body's natural fight-or-flight chemical.

Nonsteroidal anti-inflammatory drugs—This class of drugs includes aspirin and ibuprofen, and primarily works by interfering with the formation of prostaglandins, enzymes implicated in pain and inflammation.

and **aspirin**, may help with bloating and pain; beta-blockers may help with migraines; anti-anxiety medications, such as buspirone or alpraxolam, may help with anxiety; and certain other antidepressants in addition to Serafem may help with depression.

Alternative treatment

Non-pharmaceutical treatments include a variety of lifestyle changes, such as following a healthy diet, **exercise**, **stress** relief therapies, and even such alternative therapies as **aromatherapy**. Certain **vitamins** and supplements may also help, such as vitamin B6, calcium, magnesium, and vitamin E. Certain herbs may also help with symptom relief, including vitex, black cohosh, valerian, kava kava, and **St. John's wort**.

Prognosis

The prognosis varies for each woman, and is largely dependent on how much work she is willing to do in terms of lifestyle changes. Additionally, planning for PMDD symptoms, joining a support group, and communicating with her spouse and family can help minimize the negative effects of PMDD and its impact on a woman's home and work environments.

Prevention

Some women may find their PMDD disappears periodically. Diet and **nutritional supplements** can have the greatest impact in preventing PMDD.

Resources

ORGANIZATIONS

Advancement of Women's Health Research. 1828 L Street, N.W., Suite 625 Washington, DC 20036. 202-223-8224. < <http://www.womens-health.org> >.

National Association for Premenstrual Syndrome. 7 Swift's Court, High Street, Seal, Kent TN15 0EG UK +44 (0) 1732 760011 < www.PMDD.org.uk >.

Premenstrual syndrome

Definition

Premenstrual syndrome (PMS) refers to symptoms that occur between ovulation and the onset of menstruation. The symptoms include both physical symptoms, such as breast tenderness, back **pain**, abdominal cramps, **headache**, and changes in appetite, as well as psychological symptoms of **anxiety**, depression, and unrest. Severe forms of this syndrome are referred to as **premenstrual dysphoric disorder** (PMDD). These symptoms may be related to hormones and emotional disorders.

Description

Approximately 75% of all menstruating women experience some symptoms that occur before or during menstruation. PMS encompasses symptoms severe enough to interfere with daily life. About 3–seven% of women experience the more severe PMDD. These symptoms can last 4–10 days and can have a substantial impact on a woman's life.

The reason some women get severe PMS while others have none is not understood. PMS symptoms usually begin at about age 20–30 years. The disease may run in families and is also more prone to occur in women with a history of psychological problems. Overall however, it is difficult to predict who is most at risk for PMS.

Causes and symptoms

Because PMS is restricted to the second half of a woman's menstrual cycle, after ovulation, it is thought that hormones play a role. During a woman's monthly menstrual cycle, which lasts 24–35 days, hormone levels change. The hormone estrogen gradually rises during the first half of a woman's cycle, the preovulatory phase, and falls dramatically at ovulation. After ovulation, the postovulatory phase, progesterone levels gradually increase until menstruation occurs. Both estrogen and progesterone are secreted by the ovaries, which are responsible for producing the eggs.

The main role of these hormones is to cause thickening of the lining of the uterus (endometrium). However, estrogen and progesterone also affect other parts of the body, including the brain. In the brain and nervous system, estrogen can affect the levels of neurotransmitters, such as serotonin. Serotonin has long been known to have an effect on emotions, as well as eating behavior. It is thought that when estrogen levels go down during the postovulatory phase of the menstrual cycle, decreases in serotonin levels follow. Whether these changes in estrogen, progesterone, and serotonin are responsible for the emotional aspects of PMS is not known with certainty. However, most researchers agree that the chemical transmission of signals in the brain and nervous system is in some way related to PMS. This is supported by the fact that the times following **childbirth** and **menopause** are also associated with both depression and low estrogen levels.

Symptoms for PMS are varied and many, including both physical and emotional aspects that range from mild to severe. The physical symptoms include: bloating, headaches, food cravings, abdominal cramps, headaches, tension, and breast tenderness. Emotional aspects include mood swings, irritability, and depression.

Diagnosis

The best way to diagnose PMS is to review a detailed diary of a woman's symptoms for several months. PMS is diagnosed by the presence of physical, psychological, and behavioral symptoms that are cyclic and occur in association with the premenstrual period of time. PMDD, which is far less common, was officially recognized as a disease in 1987. Its diagnosis depends on the presence of at least five symptoms related to mood that disappear within a few days of menstruation. These symptoms must interfere with normal functions and activities of the individual. The diagnosis of PMDD has caused controversy in fear that it may be used against women, labeling them as being impaired by their menstrual cycles.

Treatment

There are many treatments for PMS and PMDD depending on the symptoms and their severity. For mild cases, treatment includes **vitamins**, **diuretics**, and pain relievers. Vitamins E and B₆ may decrease breast tenderness and help with **fatigue** and mood swings in some women. Diuretics that remove excess fluid from the body seem to work for some women. For more severe cases and for PMDD, treatments available include **antidepressant drugs**, hormone treatment, or

(only in extreme cases) surgery to remove the ovaries. Hormone treatment usually involves **oral contraceptives**. This treatment, as well as removal of the ovaries, is used to prevent ovulation and the changes in hormones that accompany ovulation. Recent studies, however, indicate that hormone treatment has little effect over placebo.

Antidepressants

The most progress in the treatment of PMS and PMDD has been through the use of antidepressant drugs. The most effective of these include sertraline (Zoloft), fluoxetine (Prozac), and paroxetine (Paxil). They are termed **selective serotonin reuptake inhibitors** (SSRIs) and act by indirectly increasing the brain serotonin levels, thus stabilizing emotions. Some doctors prescribe antidepressant treatment for PMS throughout the cycle, while others direct patients to take the drug only during the latter half of the cycle. Antidepressants should be avoided by women wanting to become pregnant. A recent clinical study found that women who took sertraline had a significant improvement in productivity, social activities, and relationships compared with a placebo group. Side effects of sertraline were found to include **nausea**, **diarrhea**, and decreased libido.

Alternative treatment

There are alternative treatments that can both affect serotonin and hormone responses, as well as affect some of the physical symptoms of PMS.

Vitamins and minerals

Some women find relief with the use of vitamin and mineral supplements. Magnesium can reduce the fluid retention that causes bloating, while calcium may decrease both irritability and bloating. Magnesium and calcium also help relax smooth muscles and this may reduce cramping. Vitamin E may reduce breast tenderness, nervous tension, fatigue, and **insomnia**. Vitamin B₆ may decrease fluid retention, fatigue, irritability, and mood swings. Vitamin B₅ supports the adrenal glands and may help reduce fatigue.

Phytoestrogens and Natural Progesterone

The Mexican wild yam (*Dioscorea villosa*) contains a substance that may be converted to progesterone in the body. Because this substance is readily absorbed through the skin, it can be found as an ingredient in many skin creams. (Some products also have natural progesterone added to them.) Some

KEY TERMS

Antidepressant—A drug used to control depression.

Estrogen—A female hormone important in the menstrual cycle.

Neurotransmitter—A chemical messenger used to transmit an impulse from one nerve to the next.

Phytoestrogens—Compounds found in plants that can mimic the effects of estrogen in the body.

Progesterone—A female hormone important in the menstrual cycle.

Serotonin—A neurotransmitter important in regulating mood.

herbalists believe that these products can have a progesterone-like effect on the body and decrease some of the symptoms of PMS.

The most important way to alter hormone levels may be by eating more phytoestrogens. These plant-derived compounds have an effect similar to estrogen in the body. One of the richest sources of phytoestrogens is soy products, such as tofu. Additionally, many supplements can be found that contain black cohosh (*Cimicifugacemosa*) or dong quai (*Angelica sinensis*), which are herbs high in phytoestrogens. Red clover (*Trifolium pratense*), alfalfa (*Medicago sativa*), licorice (*Glycyrrhiza glabra*), hops (*Humulus lupulus*), and legumes are also high in phytoestrogens. Increasing the consumption of phytoestrogens is also associated with decreased risks of **osteoporosis**, **cancer**, and heart disease.

Antidepressant Alternatives

Many antidepressants act by increasing serotonin levels. An alternative means of achieving this is to eat more carbohydrates. For instance, two cups of cereal or a cup of pasta have enough carbohydrates to effectively increase serotonin levels. An herb known as **St. John's wort** (*Hypericum perforatum*) has stood up to scientific trials as an effective antidepressant. As with the standard antidepressants, however, it must be taken continuously and does not show an effect until used for 46 weeks. There are also herbs, such as skullcap (*Scutellaria lateriflora*) and kava (*Piper methysticum*), that can relieve the anxiety and irritability that often accompany depression. An advantage of these herbs is that they can be taken when symptoms occur rather than continually. Chaste tree (*Vitex agnus-castus*) in addition to helping rebalance estrogen and

progesterone in the body, also may relieve the anxiety and depression associated with PMS.

Prognosis

The prognosis for women with both PMS and PMDD is good. Most women who are treated for these disorders do well.

Prevention

Maintaining a good diet, one low in sugars and fats and high in phytoestrogens and complex carbohydrates, may prevent some of the symptoms of PMS. Women should try to **exercise** three times a week, keep in generally good health, and maintain a positive self image. Because PMS is often associated with **stress**, avoidance of stress or developing better means to deal with stress can be important.

Resources

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Cindy L. A. Jones, PhD

Prenatal surgery

Definition

Prenatal surgery is a surgical procedure performed on a fetus prior to birth.

Purpose

In most cases prenatal surgery is performed only when the fetus is not expected to survive delivery or live long after birth without prenatal intervention. The most common prenatal surgeries are for conditions in which the newborn will not be able to breathe on its own.

Most prenatal surgeries are performed for:

- urinary tract obstructions in males, usually caused by a narrowing of the urinary tract, in which urine backs up and injures the kidneys. About 10% of fetal urinary tract obstructions require prenatal surgery to

prevent multiple abnormalities and depleted amniotic fluid.

- congenital diaphragmatic **hernia** (CDH), a condition in which the diaphragm—the muscle that separates the chest and the abdomen—does not form completely. Without surgery about 50% of fetuses with CDH do not survive after birth because of underdeveloped lungs.
- congenital cystic adenomatoid malformation (CCAM), a condition in which one or more lobes of the lungs become fluid-filled sacs called cysts. Large CCAMs may prevent lung development, cause **heart failure**, or prevent the fetus from ingesting amniotic fluid.
- sacrococcygeal teratoma (SCT), tumors at the base of the tailbone. The most common tumor in newborns, occurring in one out of every 35,000–40,000 births, some prenatal SCTs are very large, hard, and full of blood vessels, and can **stress** the heart.
- twin-twin **transfusion** syndrome (TTTS), a condition in which, because of abnormal blood-vessel connections in the placenta, one twin pumps the circulating blood for both twins. Affecting up to 15% of twins sharing a placenta (monochorionic), TTTS can lead to a variety of problems including heart failure.
- twin:twin reverse arterial perfusion (TRAP) sequence, a condition in which one twin lacks a heart. Occurring in about 1% of monochorionic twins, the healthy twin pumps all of the blood and, if untreated, 50–75% of these normal twins die.

Other conditions that may be treated by prenatal surgery include:

- various congenital defects that block air passages and will prevent the newborn from breathing on its own
- various lung malformations
- omphalocele, a birth defect in which portions of the stomach, liver, and intestines protrude through an opening in the abdominal wall
- fetal gastroschisis—a birth defect in which the stomach and intestines protrude through improperly formed abdominal wall muscles and float in the amniotic fluid
- bowel obstructions, usually caused by a narrowing in the small intestine
- hypoplastic left heart syndrome, in which the blood flow through the left side of the heart is obstructed
- X-linked **severe combined immunodeficiency** syndrome

- **spina bifida** (myelomeningocele)—the second most common birth defect in the United States, affecting one out of every 2,000 newborns. It is a lesion or hole where the nerves of the spinal cord are not completely enclosed and is not considered to be life-threatening.

Precautions

Prenatal surgery involves:

- serious risks for the mother and fetus
- travel to a hospital that performs the procedure
- possibly having to stay near the hospital until delivery
- extended postoperative bed rest, sometimes until delivery
- a significant financial commitment.

Description

Prenatal surgery may be referred to as fetal surgery, antenatal surgery, or maternal-fetal surgery. There are only about 600 candidates for prenatal surgery in the United States each year. Of these, only about 10% actually undergo the procedure. Most prenatal surgeries are performed between 18 and 26 weeks of gestation. Some surgeries may not be covered by insurance.

Prenatal surgery usually requires a general anesthetic, although sometimes an epidural anesthetic to numb the abdominal region may be used. The fetus receives the anesthetic via the mother's blood. An anesthesiologist and a perinatologist monitor the heart rates of the mother and fetus during the procedure.

Prenatal surgeries include:

- inserting a device into the fetal bladder to drain urine into the amniotic sac for treating urinary tract obstruction
- draining or removing CCAMs
- destroying blood vessels leading to a large SCT
- amnioreduction for TTTS, in which a syringe through the mother's abdomen is used to remove fluid from the overfilled amniotic sac and replace it in the depleted sac of the twin pumping the blood. The procedure that may be repeated during the course of the pregnancy.
- destroying abnormal blood vessel connections in the placenta of TTTS twins
- severing the connections between TRAP sequence twins

- experimental hematopoietic-stem-cell transplants for X-linked severe combined **immunodeficiency** syndrome
- closing the lesion in spina bifida

Open surgeries

In open prenatal surgeries incisions are made through the mother's abdominal wall and the fetus is partially removed from the uterus or the entire uterus is removed through the mother's abdomen. Using ultrasound as a guide, the surgeon feels for the affected fetal part. The surgeon may knead and push on the uterus to move or flip the fetus away from the placenta, the disk-shaped organ within the uterus that supplies the fetal blood. A narrow tube is placed through a tiny hole in the uterine wall to drain and collect the amniotic fluid. Opening the uterus is the riskiest part of prenatal surgery. The first incision is made at a point away from the placenta to prevent damaging it. Following the procedure the fetus is replaced in the uterus and the incision is stitched. Prior to the final stitch the amniotic fluid is re-injected into the uterus. The uterus is repositioned in the mother's body cavity and her abdominal wall is closed.

The first successful open fetal surgery was performed in 1981 for a urinary tract obstruction. The first successful open fetal surgery for CDH was performed in 1989.

Prenatal open surgery for CCAM requires opening the fetus's chest. If a large cyst does not have a hard component, procedures called thoracoamniotic shunting or catheter decompression may be used to drain it. Otherwise the surgeon must remove part or all of the cyst. The first successful resection (removal) of a CCAM from a fetal lung was performed in 1990. The first resectioning of a fetal SCT was performed in 1992.

In prenatal surgery for spina bifida, An incision the size of a small fist is made in the uterus. The surgeon loosens and lifts the tissues of the spinal-canal lesion and stitches them closed. Between 1997 and 2004, more than 200 open surgeries were performed for spina bifida. As of 2005 the surgery was available only as part of a prospective randomized clinical trial.

Less invasive procedures

For urinary tract obstructions a needle may be used to insert a catheter through the mother's abdomen and uterus and into the fetal bladder where it drains the urine into the amniotic fluid. The catheter

may have a wire mesh that expands in the bladder to prevent it from plugging up or dislodging.

The first successful fetoscopic temporary tracheal occlusion for CDH was performed in 1996. Small openings are made in the uterus and a tiny fiber-optic fetoscope is inserted to guide the operation. A needle-like instrument is used to place a balloon in the fetus's trachea to prevent lung fluid from escaping through the mouth, enabling the lungs to expand, grow, and push the abdominal organs out of the chest. The balloon is removed at birth.

Hypoplastic left heart syndrome is treated by passing a needle, guided by ultrasound, through the mother's abdominal wall, into the uterus, and the fetal heart. A catheter is passed through the needle across the fetus's aortic valve. A balloon is inflated, opening the valve and allowing blood to flow through the left side of the heart.

RADIOFREQUENCY ABLATION. Radiofrequency ablation (RFA) sometimes is used for SCT. Guided by ultrasound a needle is inserted through the mother's abdomen and uterus and into the tumor. Radiofrequency waves sent through the needle destroy the blood supply to the tumor with heat. This slows the tumor's growth and may enable the fetus to survive until delivery. The first RFA of a SCT was performed in 1998.

TRAP sequence also may be treated by RFA. A 3-mm needle targets the exact point where the blood enters the twin without a heart. Using an echocardiographic device, RFA is applied until the blood vessels and surrounding tissue are destroyed and the blood flow is halted. This procedure has eliminated the need for open surgery to treat TRAP sequence.

LASER TREATMENT. If TTTS does not respond to amnioreduction, laser treatment to halt the abnormal blood circulation may be attempted. A thin fetoscope is inserted through the mother's abdominal and uterine walls and into the amniotic cavity of the recipient twin to examine the surface placental blood vessels. The abnormal blood vessel connections are located and eliminated with a laser beam. The first successful fetoscopic laser treatment for TTTS was performed in 1999.

EXIT. Ex utero intrapartum treatment (EXIT) is a surgery performed for a congenital defect that blocks a fetal airway. The fetus is removed from the womb by **cesarean section** but the umbilical cord is left intact so that the mother's placenta continues to sustain the fetus. After the air passage is cleared, the umbilical cord is cut and the newborn can breathe on its own. The EXIT procedure is used for various types of airway obstruction including CCAM.

Preparation

The decision to perform prenatal surgery is made on the basis of detailed ultrasound imaging of the fetus—including echocardiograms that use ultrasound to obtain images of the heart—as well as other diagnostic tools. Consultations include a perinatologist, a neonatologist, a pediatric surgeon, a clinical nurse specialist, and a social worker. Since additional congenital defects preclude prenatal surgery, **amniocentesis** or chorionic villi sampling (CVS) are used to check for chromosomal abnormalities in the fetus.

Prior to surgery the mother must:

- arrange for postoperative bed rest to prevent preterm labor
- prepare for the possibility of remaining near the hospital until delivery
- receive betamethasone, a steroid, in two intramuscular injections 12–24 hours apart to accelerate fetal lung maturation
- wear a fetal/uterine monitor

The mother usually receives medications called tocolytics to prevent contractions and labor during and after surgery:

- terbutalin
- indocin suppositories before surgery and up to 48 hours after surgery
- magnesium sulfate for one to two days after surgery with careful monitoring
- nifedipine every four to six hours as the indocin is decreased, continuing until 37 weeks of gestation or delivery

Aftercare

In addition to usual post-surgical care, the mother:

- usually remains in the hospital for four to seven days
- lies on her side to help prevent contractions and ensure the best possible fetal circulation
- has a transparent dressing over the abdominal incision for fetal monitoring
- has continuous electronic fetal/uterine monitoring to check the fetal heart, the uterine response to tocolytics, and to watch for signs of preterm labor.

After discharge from the hospital the mother is on modified bed rest, lying on her side, until 37 weeks of gestation. This increases blood flow to the fetus and reduces pressure on the cervix to help prevent uterine

contractions. She sees a perinatologist once a week and has at least one ultrasound per week.

Risks

Most prenatal surgeries are high risk and may be considered experimental. The greatest risk is that the placenta will be nicked during surgery, causing blood hemorrhaging, uterine contractions, and birth of a premature infant who may not survive. Preterm labor is the most common complication of prenatal surgery. Fetoscopic surgeries are less dangerous and traumatic than open fetal surgery and reduce the risk of **premature labor**. Subsequent children of a mother who has undergone prenatal surgery usually are delivered by cesarean section because of uterine scarring.

Maternal risks

Risks to the mother include:

- extensive blood loss
- complications from general anesthesia
- side effects—potentially fatal—from medications to control premature labor
- rupture of the uterine incision
- infection of the wound or uterus
- psychological stress
- inability to have additional children
- **death**.

Fetal risks

All fetuses that undergo surgery are born prematurely. Those born even six weeks early are at risk for walking and talking delays and learning disabilities. Infants born at 30 weeks of gestation or less are at risk for blindness, **cerebral palsy**, and brain hemorrhages.

About 25% of women undergoing prenatal surgery lose some amniotic fluid, often because of leakage at the uterine incision. Amniotic fluid is essential for lung development and protects the fetus from injury and infection. If all of the amniotic fluid is lost, the fetal lungs may not develop properly. Without the fluid cushion in which the fetus floats, the umbilical cord may be compressed, causing death.

Other risks to the fetus include:

- birth during surgery
- separation of the tissues surrounding the amniotic fluid sac and the uterus, causing early delivery or interference with blood flow to some fetal body part such as an arm or leg

KEY TERMS

Amniocentesis—Withdrawal of amniotic fluid through the mother's abdominal wall, using a needle and syringe, to test for fetal disorders.

Amniotic fluid—The watery fluid within the amniotic sac that surrounds the fetus.

Cesarean section—C-section; incision through the abdominal and uterine walls to deliver a baby.

Chorion—The outermost membrane of the sac enclosing the fetus.

Chorionic villus sampling (CVS)—The removal of fetal cells from the chorion for the diagnosis of genetic disorders.

Congenital cystic adenomatoid malformation (CCAM)—A condition in which one or more lobes of the fetal lungs develop into fluid-filled sacs called cysts.

Congenital diaphragmatic hernia (CDH)—A condition in which the fetal diaphragm—the muscle dividing the chest and abdominal cavity—does not close completely.

Echocardiography—Ultrasonic examination of the heart.

Ex utero intrapartum treatment (EXIT)—A cesarean section in which the infant is removed from the uterus but the umbilical cord is not cut until after surgery for a congenital defect that blocks an air passage.

Fetoscope—A fiber-optic instrument for viewing the fetus inside the uterus.

Monochorionic twins—Twins that share a single placenta.

Omphalocele—A congenital hernia in which a small portion of the fetal abdominal contents, covered by a membrane sac, protrudes into the base of the umbilical cord.

Placenta—The organ within the uterus that provides nourishment to the fetus.

Radiofrequency ablation (RFA)—A procedure in which radiofrequency waves are used to destroy blood vessels and tissues.

Sacroccygeal teratoma (SCT)—A tumor occurring at the base of the fetus's tailbone.

Spina bifida—Myelomeningocele; a congenital defect in which the fetal backbone and spinal canal do not close completely, allowing the spinal cord and its surrounding membranes to protrude.

Tocolytic—A medication that inhibits uterine contractions.

Twin:twin reverse arterial perfusion (TRAP) sequence—A condition in which one fetus lacks a heart and the other fetus pumps the blood for both.

Twin-twin transfusion syndrome (TTTS)—A condition in monochorionic twins in which there is a connection between the two circulatory systems so that the donor twin pumps the blood to the recipient twin without a return of blood to the donor.

Ultrasound—A procedure that uses high-frequency sound waves to image a fetus.

- intrauterine infection requiring immediate birth of the fetus
- further damage to the spinal cord and nerves during surgery to treat spina bifida
- brain damage
- physical deformities
- death.

Normal results

Although fetal surgeries heal without scarring, it is difficult to predict their outcome because relatively few have been performed:

- Fetal surgery for CDH lessens the severity of the condition so that the fetus usually survives delivery and lives long enough to undergo corrective surgery.

- Thoracoamniotic shunting for CCAM usually results in infant survival.
- The infant survival rate following prenatal removal of solid CCAMs is about 50%.
- RFA to slow the growth of a tumor usually enables the fetus to survive delivery, after which the tumor can be removed.
- The infant survival rate following prenatal treatment for TTTS is about 70%. Since TTTS is a progressive disorder, early intervention may prevent later complications.

Spina bifida arises during the first month of fetal development. Fluid leaking from the spinal cord and exposure of the cord to amniotic fluid causes damage throughout gestation. Lesions higher up in the spinal cord can cause severe deformities, **paralysis**, and

mental retardation. Prenatal surgery may reduce the abnormalities, although it does not cure the condition. Babies who survive prenatal surgery appear to be 33–50% less likely to have **hydrocephalus**, a condition that requires surgically implanted tubes or shunts to remove fluid from the ventricles (cavities) of the brain. The surgery also appears to reverse hindbrain herniation, in which the back of the brain slips down into the spinal canal, causing breathing and swallowing problems and death in 15% of affected children. Children who had prenatal surgery to treat spina bifida appear to have better brain function than those who did not. However prenatal surgery does not prevent two of the most serious conditions associated with spina bifida: leg movement and bladder and bowel control. As of 2005 the long-term prognosis for these children was not known.

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- Management of Myelomeningocele Study (MOMS). Catherine Shaer, M.D., The George Washington University Biostatistics Center. 6110 Executive Blvd., Suite 750, Rockville, MD 20852. 866-ASK-MOMS. <<http://www.spinabifidamoms.com>>.

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Margaret Alic, Ph.D.

Prepregnancy counseling

Definition

Prepregnancy **pregnancy** counseling is advice supplied by an obstetrician, nurse, certified nurse-midwife, or **childbirth** educator about those steps a mother-to-be and father-to-be can take in preparation for pregnancy. Basically, it is a checklist for people to see if they are living lives that are most accommodating to having a healthy pregnancy. Prepregnancy counseling gives time for one to make changes before pregnancy.

Purpose

The purpose and goal of prepregnancy counseling is to help patients have full-term, healthy pregnancies and babies. The counseling and education are important because lifestyle habits such as **smoking** or alcohol usage can be hazardous to a developing fetus.

Precautions

Women who have diabetes should take special precautions before pregnancy. This counseling, usually provided by a team of professionals including a registered dietitian, diabetes educators, an obstetrician, and others, helps to prevent early pregnancy loss and congenital malformations in infants of diabetic mothers.

Women who have a history of genetic disease can opt to have **genetic testing**. Prepregnancy counseling can include referrals to those specialists.

Women who are over 40 have higher **cesarean section** rates. They are also more likely than younger

women to have conditions such as high blood pressure, and are more likely to have babies with genetic problems, such as **Down syndrome**.

Women who are considering pregnancy should avoid exposure to hazards such as chemicals, illicit drugs, alcohol, and smoking. They should reduce their **caffeine** intake and be careful not to let their body temperatures rise to dangerous levels.

Description

Prepregnancy counseling involves communicating important aspects about **nutrition**, medication use, and lifestyle months in advance of getting pregnant. Issues include diet, nutrition, **exercise**, smoking, alcohol, drugs, emotional health, and referral to **genetic counseling** if a patient knows of a history of inherited disease.

Preparation

The mother-to-be should stop using birth control pills to allow for at least two regular menstrual cycles to occur before conception. This requires that she stop taking birth control pills several months before getting pregnant.

Other steps to prepare for pregnancy include:

Being at optimal weight. Women should not go on prepregnancy weight loss **diets** unless they are under the care of a physician because abrupt weight loss can affect the mother's menstrual cycle and reduce fertility.

Eating a balanced diet. This is achieved by taking a prenatal vitamin provided by a health care provider and focusing on nutrients that are important for a developing fetus. These include folate, or **folic acid**, which is important for the development of the baby's brain and spinal cord. Folate can be found in fortified cereals, citrus fruits, and green leafy vegetables. Calcium is important for baby and mother. It helps the baby's bones to develop normally and keeps the mother from suffering a calcium deficiency during pregnancy. Iron keeps the mother from developing anemia during pregnancy. Good sources of iron are green leafy vegetables, red meat, beans, and fortified cereal. Fiber helps mothers avoid **constipation**, a common occurrence during pregnancy. Good sources of fiber include beans, fruits, and vegetables.

Exercising on a regular basis. One should exercise for general overall health.

Undergoing routine physical and dental exams. These include having a physical and breast

KEY TERMS

Preeclampsia—Also called toxemia, preeclampsia is a condition during pregnancy that results in high blood pressure, swelling that doesn't go away and large amounts of protein in the urine. Without treatment, it can progress to a dangerous condition called eclampsia, in which a woman goes into convulsions.

examination and **Pap test**. Other tests might be recommended according to a woman's health and genetic history. They should also report any prescription drugs, over-the-counter medications, or natural **vitamins** and herbs they are taking. This is the time for a woman to make sure she is up to date on her immunizations. A dental exam, with x rays, can eliminate the need to have x rays while pregnant.

Getting psychological support. Mental support is also important in the prepregnancy stage. This can help a woman to relax and better prepare mentally and physically for what lies ahead.

Risks

About 10–15% of couples in the United States experience **infertility**. When couples should seek medical evaluation and an infertility work-up depends on their ages. Generally, it takes longer for older couples to conceive. Prepregnancy counseling might include a referral to such a specialist. While infertility is often treatable, treatment can be expensive, emotionally difficult, and time-consuming. About 10 percent of the time, doctors cannot detect a reason for the infertility.

There always is the risk that a pregnancy goes awry or a baby is born with a medical condition, regardless of whether or not a person has had prepregnancy counseling.

Normal results

The counseling can provide guidelines for people so that they can maximize their chances to have emotionally and physically healthy pregnancies and healthy babies.

Abnormal results

Many abnormal results, such as genetic conditions, **miscarriage**, **preeclampsia** (also known as

toxemia), and preterm births, cannot be avoided even with prepregnancy counseling. Still, some abnormal results, such as miscarriages and preterm births, may occur when mothers and fathers lead unhealthy lifestyles despite their counseling.

Resources

ORGANIZATIONS

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"General pre pregnancy guidelines." <<http://parentsplace.com>>.

"Preconception care of women with diabetes." American Diabetes Association. <<http://journal.diabetes.org/FullText/Supplements/DiabetesCare/Supplement100/S65.htm>>.

"Take a health inventory before pregnancy." UC Davis Health System. UC Davis Medical Center. 2315 Stockton Blvd. Sacramento, CA 95817. (916) 734-2011. <<http://www.ucdmc.ucdavis.edu>>.

Trish Booth, M.A., LCCE, FACCE, Childbirth Educator, Educational Process Consultant. 7507 Northfield Lane. Manlius, N.Y. 13104. (315) 682-2922.

Lisette Hilton

Presbyopia

Definition

The term presbyopia means "old eye" and is a vision condition involving the loss of the eye's ability to focus on close objects.

Description

Presbyopia is a condition that occurs as a part of normal **aging** and is not considered to be an eye disease. The process occurs gradually over a number of years. Symptoms are usually noticeable by age 40–45 and continue to develop until the process stabilizes some 10–20 years later. Presbyopia occurs without regard to other eye conditions.

Causes and symptoms

In the eye, the crystalline lens is located just behind the iris and the pupil. Tiny ciliary muscles pull and push the lens, adjusting its curvature, and thereby adjusting the eye's focal power to bring objects into focus. As individuals age, the lens becomes less flexible and elastic, and the muscles become less powerful. Because these

changes result in inadequate adjustment of the lens of the eye for various distances, objects that are close will appear blurry. The major cause of presbyopia is loss of elasticity of the lens of the eye. Loss of ciliary muscle power, however, is also believed to contribute to the problem.

Symptoms of presbyopia result in the inability to focus on objects close at hand. As the lens hardens, it is unable to focus the rays of light that come from nearby objects. Individuals typically have difficulty reading small print, such as that in telephone directories and newspaper advertisements, and may need to hold reading materials at arm's length. Symptoms include headache and eyestrain when doing close work, blurry vision, and eye **fatigue**. Symptoms may be worse early in the morning or when individuals are fatigued. Dim lighting may also aggravate the problem.

Diagnosis

Presbyopia is officially diagnosed during an eye examination conducted by eye specialists, such as optometrists or ophthalmologists. After completing optometric college, doctors of optometry screen patients for eye problems and prescribe glasses and contact lenses. In contrast, ophthalmologists are medical doctors who specialize in eye diseases. They perform eye surgery, treat eye diseases, and also prescribe glasses and contact lenses.

A comprehensive **eye examination** requires at least 30 minutes. Part of the examination will assess vision while reading by using various strength lenses. If the pupils are dilated with drugs to permit a thorough examination of the retina, an additional hour is required. The cost of eye examinations can range from \$40 to \$250 depending on the complexity and site of the examination and the qualifications and reputation of the examiner. Some insurers cover the cost of routine eye examinations, while others do not. A thorough eye examination is recommended at regular intervals during the adult and aging years to monitor and diagnose eye conditions. However, individuals frequently self-diagnose presbyopia by trying on inexpensive mass-produced reading glasses until they find a pair that permits reading without strain.

Treatment

Presbyopia cannot be cured, but individuals can compensate for it by wearing reading, bifocal, or trifocal eyeglasses. A convex lens is used to make up for the lost automatic focusing power of the eye. Half-glasses can be worn, which leave the top open and uncorrected for distance vision. Bifocals achieve the same goal by

allowing correction of other refractive errors (improper focusing of images on the retina of the eye).

In addition to glasses, contact lenses have also been found to be useful in the treatment of presbyopia. The two common types of contact lenses prescribed for this condition are bifocal and monovision contact lenses. Bifocal contact lenses are similar to bifocal glasses. The top portion of the lens serves as the distance lens while the lower serves as the near vision lens. To prevent rotation while in the eye, bifocal contacts use a specially manufactured type of lens. Good candidates for bifocal lenses are those patients who have a good tear film (moist eyes), good binocular vision (ability to focus both eyes together) and visual acuity in each eye, and no disease or abnormalities in the eyelids. The bifocal contact lens wearer must be motivated to invest the time it requires to maintain contact lenses and be involved in occupations that do not impose high visual demands. Further, bifocal contact lenses may limit binocular vision. Bifocal contact lenses are relatively expensive, in part due to the time it takes the patient to be accurately fitted.

An alternative to wearing eyeglasses or bifocal contact lenses is monovision contact lenses. Monovision fitting provides one contact lens that corrects for near vision and a second contact lens for the alternate eye that corrects for distance vision. If distance vision is normal, the individual wears only a single contact lens for near vision. Monovision works by having one eye focus for distant objects while the other eye becomes the reading eye. The brain learns to adapt to this and will automatically use the correct eye depending on the location of material in view. Advantages of monovision are patient acceptability, convenience, and lower cost.

Several problems exist with the use of contact lenses in the treatment of presbyopia. Some individuals experience **headache** and fatigue during the adjustment period or find the slight decrease in visual acuity unacceptable. Monovision contact lenses usually result in a small reduction in high-contrast visual acuity when compared with bifocal contact lenses.

Prognosis

The changes in vision due to aging usually start in a person's early 40s and continue for several decades. At some point, there is no further development of presbyopia, as the ability to accommodate is virtually gone.

Prevention

There is no known way to prevent presbyopia.

KEY TERMS

Accommodation—The ability of the eye to change its focus from near to distant objects.

Binocular vision—Using both eyes at the same time to see an image.

Ciliary muscles—The small muscles that permit the lens to change its shape in order to focus on near or distant objects.

Lens (or crystalline lens)—The eye structure behind the iris and pupil that helps focus light on the retina.

Visual acuity—Sharpness or clearness of vision.

Resources

ORGANIZATIONS

American Academy of Ophthalmology. 655 Beach Street, P.O. Box 7424, San Francisco, CA 94120-7424.
< <http://www.eyenet.org> >.

American Optometric Association. 243 North Lindbergh Blvd., St. Louis, MO 63141. (314) 991-4100.
< <http://www.aoanet.org> >.

Lighthouse National Center for Vision and Aging. 111 E. 59th St., New York, NY 10022. (800) 334-5497.
< <http://www.lighthouse.org> >.

National Eye Institute. 2020 Vision Place, Bethesda, MD 20892-3655. (301) 496-5248.
< <http://www.nei.nih.gov> >.

Elaine Souder, PhD

Presenile dementia see **Alzheimer's disease**

Pressure sores see **Bedsore**

Preterm labor see **Premature labor**

Priapism

Definition

Priapism is a rare condition that causes a persistent, and often painful, penile erection.

Description

Priapism is drug induced, injury related, or caused by disease, not sexual desire. As in a normal erection, the penis fills with blood and becomes erect. However, unlike a normal erection that dissipates after sexual

KEY TERMS

Antineoplastic—A drug used to inhibit the growth and spread of cancerous cells.

Doppler ultrasound—An imaging technique using ultrasound that can detect moving liquids.

Infarction—Death of tissue due to inadequate blood supply.

Nuclear scanning—Use of injected radioactive elements to analyze blood flow.

Sickle cell anemia—A hereditary abnormality of blood cells in which some are deformed and may plug up small blood vessels.

activity ends, the persistent erection caused by priapism is maintained because the blood in the penile shaft does not drain. The shaft remains hard, while the tip of the penis is soft. If it is not relieved promptly, priapism can lead to permanent scarring of the penis and inability to have a normal erection.

Causes and symptoms

Priapism is caused by leukemia, **sickle cell disease**, or **spinal cord injury**. It has also been associated as a rare side effect to trazodone (Desyrel), a drug prescribed to treat depression. An overdose of self-injected chemicals to counteract **impotence** has also been responsible for priapism. The chemicals are directly injected into the penis, and at least a quarter of all men who have used this method of treatment for over three months develop priapism.

Diagnosis

A **physical examination** is needed to diagnose priapism. Further testing, including nuclear scanning or Doppler ultrasound, will diagnose the underlying cause of the condition.

Treatment

There are three methods of treatment. The most effective is the injection of medicines into the penis that allow the blood to escape. Cold packs may also be applied to alleviate the condition, but this method becomes ineffective after about eight hours. For the most serious cases and those that do not respond to the first two treatments, a needle can be used to remove the blood. The tissues may need to be flushed with saline or diluted medications by the same needle

method. That failing, there are more extensive surgical procedures available. One of them shuts off much of the blood supply to the penis so that it can relax. If the problem is due to a sickle cell crisis, treatment of the crisis with oxygen or **transfusion** may suffice.

Prognosis

If priapism is relieved within the first 12–24 hours, there is usually no residual damage. After that, permanent impotence may result, since the high pressure in the penis compromises blood flow and leads to tissue death (infarction).

Prevention

An antineoplastic drug (hydroxyurea) may prevent future episodes of priapism for patients with sickle cell disease.

Resources

BOOKS

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PERIODICALS

Werthman, P., and J. Rajfer. "MUSE Therapy: Preliminary Clinical Observations." *Urology* 50 (November 1997): 809-811.

J. Ricker Polsdorfer, MD

Prickly heat

Definition

Also known as sweat retention syndrome or miliaria rubra, prickly heat is a common disorder of the sweat glands.

Description

The skin contains two types of glands: one produces oil and the other produces sweat. Sweat glands are coil-shaped and extend deep into the skin. They are capable of plugging up at several different depths, producing four distinct skin **rashes**.

- Miliaria crystallina is the most superficial of the occlusions. At this level, only the thin upper layer of skin is affected. Little blisters of sweat that cannot escape to the surface form. A bad **sunburn** as it just starts to blister can look exactly like this.

- Deeper plugging causes miliaria rubra as the sweat seeps into the living layers of skin, where it irritates and itches.
- Miliaria pustulosais (a complication of miliaria rubra) occurs when the sweat is infected with pyogenic bacteria and turns to pus.
- Deeper still is miliaria profunda. The skin is dry, and goose bumps may or may not appear.

There are two requirements for each of these phases of sweat retention: hot enough weather to induce sweating, and failure of the sweat to reach the surface.

Causes and symptoms

Best evidence as of 2001, suggests that bacteria form the plugs in the sweat glands. These bacteria are probably normal inhabitants of the skin, and why they suddenly interfere with sweat flow is still not known.

Infants are more likely to get miliaria rubra than adults. All the sweat retention rashes are also more likely to occur in hot, humid weather.

Besides **itching**, these conditions prevent sweat from cooling the body, which it is supposed to do by evaporating from the skin surface. Sweating is the most important cooling mechanism available in hot environments. If it does not work effectively, the body can rapidly become too hot, with severe and even lethal consequences. Before entering this phase of heat **stroke**, there will be a period of heat exhaustion symptoms—dizziness, thirst, weakness—when the body is still effectively maintaining its temperature. Then the temperature rises, often rapidly, to 104–5°F (40°C) and beyond. This is an emergency of the first order, necessitating immediate and rapid cooling. The best method is immersion in ice water.

Diagnosis

Rash and dry skin in hot weather are usually sufficient to diagnose these conditions.

Treatment

The rash itself may be treated with topical antipruritics (itch relievers). Preparations containing aloe, menthol, camphor, eucalyptus oil, and similar ingredients are available commercially. Even more effective, particularly for widespread itching in hot

KEY TERMS

Ambient—Surrounding.

Pyogenic—Capable of generating pus. *Streptococcus*, *Staphylococcus* and bowel bacteria are the primary pyogenic organisms.

Syndrome—A collection of abnormalities that occur together often enough to suggest they have a common cause.

weather, are cool baths with corn starch and/or oatmeal (about 0.5 lb [224 g] of each per bathtub-full).

Dermatologists can peel off the upper layers of skin using a special ultraviolet light. This will remove the plugs and restore sweating, but is not necessary in most cases.

Much more important, however, is to realize that the body cannot cool itself adequately without sweating. Careful monitoring for symptoms of heat disease is important. If they appear, some decrease in the ambient temperature must be achieved by moving to the shade, taking a cool bath or shower, or turning up the air conditioner.

Prognosis

The rash disappears in a day with cooler temperatures, but the skin may not recover its ability to sweat for two weeks—the time needed to replace the top layers of skin with new growth from below.

Prevention

Experimental application of topical **antiseptics** like hexachlorophene almost completely prevented these rashes.

Resources

BOOKS

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J. Ricker Polsdorfer, MD

Primaquine see **Antimalarial drugs**

Primary biliary cirrhosis

Definition

Primary biliary **cirrhosis** is the gradual destruction of the biliary system for unknown reasons.

Description

Although the cause of this serious condition is not known, it has many features to suggest that it is an autoimmune disease. Autoimmunity describes the process whereby the body's defense mechanisms are turned against itself. The immune system is supposed to recognize and attack only dangerous foreign invaders like germs, but many times it attacks, for no apparent reason, the cells of the body itself. Autoimmune reactions occur in many different tissues of the body, creating a great variety of diseases.

Primary biliary cirrhosis progressively destroys the system that drains bile from the liver into the intestines. Bile is a collection of waste products excreted by the liver. As the disease progresses it also **scars** the liver, leading to cirrhosis. In some patients, the disease destroys the liver in as little as five years. In others, it may lie dormant for a decade or more.

Causes and symptoms

Ninety percent of patients with this disease are women between the ages of 35 and 60. The first sign of it may be an abnormal blood test on routine examination. **Itching** is a common early symptom, caused by a buildup of bile in the skin. Fatigue is also common in the early stages of the disease. Later symptoms include **jaundice** from the accumulation of bile and specific nutritional deficiencies—bruising from vitamin K deficiency, bone **pain** from vitamin D deficiency, night blindness from vitamin A deficiency, and skin **rashes**, possibly from vitamin E or essential fatty acid deficiency. All these vitamin problems are related to the absence of bile to assist in the absorption of nutrients from the intestines.

Diagnosis

Blood tests strongly suggest the correct diagnosis, but a liver biopsy is needed for confirmation. It is also usually necessary to x ray the biliary system to look for other causes of obstruction.

Treatment

Of the many medicines tried to relieve the symptoms and slow the progress of this disease, only one



A close-up image indicating biliary cirrhosis of the liver.
(Custom Medical Stock Photo. Reproduced by permission.)

KEY TERMS

Biopsy—Surgical removal of tissue for examination.

Cirrhosis—Scarring, usually referring to the liver.

Immunosuppression—Techniques to prevent transplant graft rejection by the body's immune system.

has had consistently positive results. Ursodeoxycholic acid, a chemical that dissolves gall stones, provides substantial symptomatic relief. It is still unclear if it slows liver damage.

Primary biliary cirrhosis is a major reason for liver transplantation. Patients do so well that this is becoming the treatment of choice. As experience, technique, and immunosuppression progressively improve, patients with this disease will come to transplant surgery earlier and earlier in their disease course.

Prognosis

So far, this disease has not returned in a transplanted liver.

Resources

ORGANIZATIONS

American Liver Foundation. 1425 Pompton Ave., Cedar Grove, NJ 07009. (800) 223-0179. <<http://www.liverfoundation.org>>.

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Primary degenerative dementia *see*

Alzheimer's disease

Primary polycythemia *see* **Polycythemia vera**

Primary pulmonary hypertension *see*

Pulmonary hypertension

PRK *see* **Photorefractive keratectomy and laser-assisted in-**

Pro time *see* **Prothrombin time**

Probenecid *see* **Gout drugs**

Procainamide *see* **Antiarrhythmic drugs**

Prochlorperazine *see* **Antinausea drugs**

Proctitis

Definition

Proctitis is an inflammation of the rectum.

Description

Proctitis affects mainly adolescents and adults. It is most common in men around age 30. Proctitis is caused by several different sexually transmitted diseases. Male homosexuals and people who practice anal intercourse are more likely to suffer from proctitis. Patients who have **AIDS** or who are immunocompromised are also more at risk.

Causes and symptoms

Proctitis is caused most often by **sexually transmitted diseases**, including **gonorrhea**, **syphilis**, herpes simplex (genital herpes), **candidiasis**, and chlamydia. It can also be caused by inflammatory bowel diseases, such as **Crohn's disease**, or **ulcerative colitis** (a chronic recurrent ulceration in the colon)—with which it is a very common component. Occasionally it is caused by an amoeba that causes **dysentery**.

Discharge of blood and mucus and intense **pain** in the area of the rectum and anus are all signs of proctitis. Patients feel the urge to have frequent bowel movements even when there is nothing present to eliminate. They may also have **constipation**, **diarrhea**, **fever**, and open sores around the anus. Other symptoms include cramping, lower back pain, difficulty urinating, and **impotence**.

Diagnosis

Proctitis is diagnosed by a patient history and physical examination. It is confirmed by a proctoscopy (examination of the rectum with an endoscope inserted through the anus). Proctoscopy usually shows a red, sore, inflamed lining of the rectum. Biopsies, smears, and lab cultures of rectal material are used to determine the exact cause of the inflammation so that the underlying cause can be treated appropriately.

Since the two problems often occur together, in the presence of proctitis, the large bowel should be examined for ulcerative colitis.

Treatment

Once the underlying cause of the inflammation is diagnosed, appropriate treatment begins. Antibiotics are given for bacterial infections. There is no cure for genital herpes, but the antiviral drug, acyclovir, is often prescribed to reduce symptoms. Corticosteroid suppositories or ointments such as hydrocortisone are used to lessen discomfort, and the patient is encouraged to take warm baths to ease painful symptoms. Ulcerative proctitis often responds well to corticosteroid **enemas** or foam, or to sulfasalazine and related drugs.

Alternative treatment

Depending on the cause of proctitis, alternative medicine has several types of treatments available. If proctitis is related to gonorrhea, syphilis, or chlamydia, appropriate antibiotic treatment is recommended. Supplementation with *Lactobacillus acidophilus* is also recommended during and following antibiotic therapy to help rebuild normal gut flora that is destroyed by **antibiotics**. If proctitis is herpes-related, antiviral herbs taken internally, as well as applied topically, can be helpful. Sitz baths and compresses of herbal infusions (herbs steeped in hot water) and decoctions (herbal extracts prepared by boiling the herb in water) can be very effective. Among the herbs recommended are calendula (*Calendula officinalis*), comfrey (*Symphytum officinale*), and plantain (*Plantago major*). Proctitis related to candidiasis requires dietary alterations, especially elimination of sugar from the diet. Any immunocompromised person needs close medical attention. If proctitis is related to inflammatory bowel diseases, the resolution of the underlying condition should contribute to resolution of the proctitis. **Acupuncture** and homeopathic treatment can be very useful in resolving inflammatory bowel diseases.

KEY TERMS

Candidiasis—A common fungal infection caused by yeast that thrives in moist, warm areas of the body.

Chlamydia—A gonorrhea-like bacterial infection.

Proctoscopy—A procedure in which a thin tube containing a camera and a light is inserted into the rectum so that the doctor can visually inspect it.

Rectum—The final section of the large intestine.

Ulcerative colitis—Chronic ulceration of the colon and rectum.

Prognosis

Proctitis caused by bacteria is curable with antibiotics. **Genital herpes** is not curable. Although symptoms can be suppressed, proctitis may reoccur. Patients with AIDS are especially susceptible to candidiasis infections, which may be hard to control. Recovering from proctitis caused by inflammatory bowel diseases is variable and depends on successful management of those diseases. Severe proctitis can result in permanent narrowing of the anus.

Prevention

Proctitis is best prevented by using condoms and practicing safer sex to prevent acquiring sexually transmitted diseases. Avoiding anal intercourse also helps prevent damage to the rectum.

Resources

OTHER

ThriveOnline. "Proctitis." <<http://thriveonline.oxygen.com>>.

Tish Davidson, A.M.

Proctosigmoidoscopy see **Sigmoidoscopy**

Progesterone assay see **Sex hormones tests**

Progressive multifocal leukoencephalopathy

Definition

Progressive multifocal leukoencephalopathy (PML) is a rapidly progressive neuromuscular disease

caused by opportunistic infection of brain cells (oligodendrocytes and astrocytes) by the JC virus (JCV).

Description

PML is an opportunistic infection associated with **AIDS** and certain cancers. It occurs in people with inadequate immune response and carries a poor prognosis. The incidence of PML, once quite rare, is rising as the numbers of people living with persistently compromised immune systems rises. An estimated 2–7% of people with HIV disease will develop PML. The infection also occurs among people undergoing long-term chemotherapy for **cancer**. PML is not considered a contagious disease. According to the Centers for Disease Control definition of AIDS, PML in the presence of HIV infection is sufficient to form a diagnosis of AIDS.

Causes and symptoms

Although at least 80% of the adults in the United States have been exposed to JC virus (as evidenced by the presence of antibodies to this virus), very few will develop PML. Little is certain about what causes JCV to produce active disease, but the virus persists in the kidneys of otherwise healthy people without making them ill. Recent evidence suggests that after prolonged compromise of the immune system, the virus changes into a form that can reach brain tissue and cause disease. In PML, the JCV infects and kills the cells (oligodendrocytes) that produce myelin, which is needed to form the sheath that surrounds and protects nerves.

About 45% of people with PML experience vision problems, most often a blindness affecting half of the visual field of each eye. Mental impairment affects about 38% of people with PML. Eventually, about 75% experience extreme weakness. Other symptoms include lack of coordination, **paralysis** on one side of the body (hemiparesis), and problems in speaking or using language.

Diagnosis

Diagnosis is difficult, but usually relies on a neurologist and radiologist assessing the white matter of the brain on a computed tomography scan or a magnetic resonance imaging (MRI). Tests of the cerebrospinal fluid can help distinguish between PML and other diseases, such as **multiple sclerosis** and acute hemorrhagic leukoencephalopathy. The rapid clinical progression in immunocompromised patients is another distinguishing factor.

KEY TERMS

Multifocal—Having many focal points. In progressive multifocal leukoencephalopathy, it means that damage caused by the disease occurs at multiple sites.

Opportunistic infection—A illness caused by infecting organisms that would not be able to produce disease in a person with a healthy immune system, but are able to take advantage of an impaired immune response.

Treatment

Currently, there is no known cure for PML, although it sometimes responds to treatment in patients with AIDS who are taking anti-HIV drugs (such as AZT, alpha-interferon, and peptide T). Although several agents have shown some potential in the last few years, such as the highly toxic cancer drug cytarabine, none are safe enough or sufficiently effective to be approved for PML.

Prognosis

PML is usually a very aggressive disease. The time between the onset of symptoms and death can be as little as one to six months. However, some patients infected with HIV have improved without receiving treatment specifically for PML.

Resources

PERIODICALS

Royal III, Walter. "Update on Progressive Multifocal Leukoencephalopathy." *The Hopkins HIV Report* 9 (March 1997).

Jill S. Lasker

Progressive supranuclear ophthalmoplegia
see **Progressive supranuclear palsy**

Progressive supranuclear palsy

Definition

Progressive supranuclear palsy (PSP; also known as Steele-Richardson-Olszewski syndrome) is a rare

disease that gradually destroys nerve cells in the parts of the brain that control eye movements, breathing, and muscle coordination. The loss of nerve cells causes palsy, or paralysis, that slowly gets worse as the disease progresses. The palsy affects ability to move the eyes, relax the muscles, and control balance.

Description

Progressive supranuclear palsy is a disease of middle age. Symptoms usually begin in the 60s, rarely before age 45 or after age 75. Men develop PSP more often than women do. It affects three to four people per million each year.

Causes and symptoms

PSP affects the brainstem, the basal ganglia, and the cerebellum. The brainstem is located at the top of the spinal cord. It controls the most basic functions needed for survival—the involuntary (unwilled) movements such as breathing, blood pressure, and heart rate. The brainstem has three parts: the medulla oblongata, the pons, and the midbrain. The parts affected by PSP are the pons, which controls facial nerves and the muscles that turn the eye outward, and the midbrain, the visual center. The basal ganglia are islands of nerve cells located deep within the brain. They are involved in the initiation of voluntary (willed) movement and control of emotion. Damage to the basal ganglia causes muscle stiffness (spasticity) and tremors. The cerebellum is located at the base of the skull. It controls balance and muscle coordination.

Vision is controlled by groups of cells called *nuclei* in the brainstem. In PSP, the nuclei continue to function, but the mechanisms that control the nuclei are destroyed. The term *supranuclear* means that the damage is done above (*supra*) the nuclei. Patients with PSP have difficulty with voluntary (willed) eye movement. At first, the difficulty only occurs in trying to look down. As the disease progresses, ability to move the eyes right and left is also affected. However, reflex or unwilled eye movements remain normal. Thus, when the patient's head is tilted upwards, the eyes move to look down. These reflex movements remain normal until late in the course of the disease. The upper eyelids may be pulled back, the eyebrows raised, and the brow wrinkled, causing a typical wide-eyed stare. Rate of blinking may decrease from the normal 20–30 per minute to three to five per minute. It becomes difficult to walk downstairs, to maintain eye contact during conversation, or to move the eyes up and down to read.

The earliest symptoms of PSP may be frequent falls or stiff, slow movements of the arms and legs. These symptoms may appear as much as five years before the characteristic vision problems. Walking becomes increasingly awkward, and some patients tend to lean and fall backward. Facial muscles may be weak, causing slurred speech and difficulty swallowing. Sleep may be disturbed and thought processes slowed. Although memory remains intact, the slowed speech and thought patterns and the rigid facial expression may be mistaken for senile **dementia** or **Alzheimer's disease**. Emotional responses may become exaggerated and inappropriate, and the patient may experience **anxiety**, depression, and agitation.

The cause of PSP is not known. Most people who develop PSP come from families with no history of the disease, so it does not seem to be inherited, except in certain rare instances. People who have PSP seem to lack the neurotransmitters dopamine and homovanillic acid in the basal ganglia. Neurotransmitters are chemicals that help carry electrical impulses along the nervous system. Transmitting structures in brain cells called neurofibrils become disorganized (neurofibrillary tangles). Neurofibrillary tangles are also found in Alzheimer's disease, but the pattern is somewhat different.

Diagnosis

PSP is sometimes mistaken for Parkinson's disease, which is also associated with stiffness, frequent falls, slurred speech, difficulty swallowing, and decreased spontaneous movement. The facial expression in Parkinson's, however, is blank or mask-like, whereas in PSP it is a grimace and wide-eyed stare. PSP does not cause the uncontrolled shaking (tremor) in muscles at rest that is associated with Parkinson's disease. Posture is stooped in Parkinson's disease, but erect in PSP. Speech is of low volume in both diseases, but is more slurred and irregular in rhythm in PSP.

Multiple strokes or abnormal accumulations of fluid within the skull (**hydrocephalus**) can also cause balance problems similar to PSP. **Magnetic resonance imaging (MRI)** scans of the brain may be needed to rule out these conditions. In advanced cases, MRI shows characteristic abnormalities in the brainstem described as "mouse ears."

Treatment

PSP cannot be cured. Drugs are sometimes given to relieve symptoms, but drug treatment is usually disappointing. Dopaminergic medications used in

KEY TERMS

Basal ganglia—Brain structure at the base of the cerebral hemispheres, involved in controlling movement.

Brainstem—Brain structure closest to the spinal cord, involved in controlling vital functions, movement, sensation, and nerves supplying the head and neck.

Cerebellum—The part of the brain involved in coordination of movement, walking, and balance.

Magnetic resonance imaging (MRI)—An imaging technique that uses a large circular magnet and radio waves to generate signals from atoms in the body. These signals are used to construct images of internal structures.

Parkinson's disease—A slowly progressive disease that destroys nerve cells. Parkinson's is characterized by shaking in resting muscles, a stooping posture, slurred speech, muscular stiffness, and weakness.

Parkinson's disease, such as levodopa (Sinemet), sometimes decrease stiffness and ease spontaneous movement. Anticholinergic medications, such as trihexyphenidyl (Artane), which restore function to neurotransmitters, or tricyclic drugs, such as amitriptyline (Elavil) may improve speech, walking, and inappropriate emotional responses.

Speech therapy may help manage the swallowing and speech difficulty in PSP. As the disease progresses, the difficulty in swallowing may cause the patient to choke and get small amounts of food in the lungs. This condition can cause aspiration **pneumonia**. The patient may also lose too much weight. In these cases, a feeding tube may be needed. The home environment should be modified to decrease potential injury from falls. Walkers can be weighted in front, to prevent backward falls and handrails can be installed in the bathroom. Because the patient cannot look down, low objects like throw rugs and coffee tables should be removed. Dry eyes from infrequent blinking can be treated with drops or ointments.

Prognosis

The patient's condition gradually deteriorates. After about seven years, balance problems and stiffness make it nearly impossible for the patient to walk. Persons with PSP become more and more immobile

and unable to care for themselves. **Death** is not caused by the PSP itself. It is usually caused by pneumonia related to **choking** on secretions or by **starvation** related to swallowing difficulty. It usually occurs within 10 years, but if good general health and nutrition are maintained, the patient may survive longer.

Prevention

PSP cannot be prevented.

Resources

ORGANIZATIONS

American Academy of Neurology. 1080 Montreal Ave., St. Paul, MN 55116. (612) 695-1940. <<http://www.aan.com>>.

Society for Progressive Supranuclear Palsy, Inc. Suite #5065 Johns Hopkins Outpatient Center, 601 N. Caroline St., Baltimore, MD 21287. (800) 457-4777. <<http://www.psp.org>>.

Laurie Barclay, MD

Progressive systemic sclerosis see

Scleroderma

Prolactin test

Definition

Prolactin is a hormone secreted by the anterior portion of the pituitary gland (sometimes called the “master gland”). Its role in the male has not been demonstrated, but in females, prolactin promotes **lactation**, or milk production, after **childbirth**.

Purpose

The prolactin test is used to diagnose pituitary dysfunction that might be caused by a tumor called an adenoma. In some circumstances, the test is also used to evaluate absence of menstrual periods (**amenorrhea**), or spontaneous production of milk (**galactorrhea**) by a woman who is not pregnant or lactating.

Precautions

Stress from trauma, illness, surgery, or even nervousness about a blood test can elevate prolactin levels. Drugs that may increase prolactin include phenothiazines, **oral contraceptives**, opiates, histamine antagonists, monoamine oxidase inhibitors

(MAO inhibitors), estrogen, and antihistamines. Drugs that can decrease values include levodopa and dopamine.

Description

Prolactin is also known as the lactogenic hormone or lactogen. It is essential for the development of the mammary glands for lactation during pregnancy, and for stimulating and maintaining lactation after childbirth. Like the human growth hormone, prolactin acts directly on tissues, and its levels rise in response to sleep and to physical or emotional stress. During sleep, prolactin levels can increase to the circulating levels found in pregnant women (as high as ten to twenty times the normal level).

Prolactin secretion is controlled by prolactin-releasing and prolactin-inhibiting chemicals (factors) secreted by an area of the brain called the hypothalamus. Another hormone, thyroid-releasing hormone, or TRH, can also stimulate prolactin.

Tumors of the pituitary, called adenomas, are the most common cause of excessive levels of prolactin. Depending on the type of cell involved, these tumors are also called prolactin-secreting pituitary acidophilic or chromophobic adenomas. Moderately high prolactin levels are found to a lesser extent in women with secondary amenorrhea, **galactorrhea**, low thyroid, anorexia, and a disorder known as polycystic ovary syndrome, a disease whose cause is not well-known.

Because high prolactin levels are more likely due to pituitary adenoma than other causes, the prolactin level is used to diagnose and monitor this type of tumor. Several stimulation and suppression tests, with TRH or levodopa, respectively, have been designed to differentiate pituitary adenoma from other causes of prolactin overproduction.

Preparation

This test requires a blood sample that should be drawn in the morning at least two hours after the patient wakes (samples drawn earlier may show sleep-induced peak levels). The patient need not restrict food or fluids nor limit physical activity, but should relax for approximately 30 minutes before the test.

Risks

Risks posed by this test are minimal, but may include slight bleeding from the blood-drawing site, **fainting** or lightheadedness after venipuncture,

KEY TERMS

Adenoma—A benign tumor

Amenorrhea—The absence or abnormal stoppage of menstrual periods.

Factor—Any of several substances necessary to produce a result or activity in the body. The term is used when the chemical nature of the substance is unknown. In endocrinology, when the chemical nature is known, factors are renamed hormones.

Galactorrhea—Excessive or spontaneous flow of milk.

Pituitary gland—A gland located at the base of the brain, and controlled by the hypothalamus. It controls most endocrine functions and is responsible for things such as kidney function, lactation, and growth and development.

or hematoma (blood accumulating under the puncture site).

Normal results

Reference ranges vary from laboratory to laboratory but are generally within the following values:

- adult male: 0–20 ng/ml
- adult female: 0–20 ng/ml
- pregnant female: 20–400 ng/ml

Abnormal results

Increased prolactin levels are found in galactorrhea, amenorrhea, prolactin-secreting pituitary tumor, infiltrative diseases of the hypothalamus, and metastatic **cancer** of the pituitary gland. Higher levels than normal are seen in stress which may be produced by **anorexia nervosa**, surgery, strenuous exercise, trauma, and in renal (kidney) failure.

Decreased prolactin levels are seen in Sheehan's syndrome, a condition of severe hemorrhage after obstetric delivery that causes decreased blood supply to the pituitary.

Resources

BOOKS

Pagana, Kathleen Deska. *Mosby's Manual of Diagnostic and Laboratory Tests*. St. Louis: Mosby, Inc., 1998.

Janis O. Flores

Prolactinoma see **Galactorrhea**

Prolapsed disk see **Herniated disk**

Prolonged QT syndrome

Definition

Prolonged QT syndrome, also known as long QT syndrome (LQTS), refers to a group of disorders that increase the risk for sudden **death** due to an abnormal heartbeat.

Description

Abnormal heartbeats (cardiac **arrhythmias**) are a primary cause of sudden death, especially in the young population. In the United States, an estimated 1 in 300,000 individuals per year die suddenly due to irregular heart rhythms. One of the better understood causes of these arrhythmias is LQTS.

The QT of LQTS refers to an interval between two points (Q and T) on the common electrocardiogram (ECG, EKG) used to record the electrical activity of the heart. This electrical activity, in turn, is the result of small molecules (ions such as sodium and potassium) passing in and out of channels in the membranes surrounding heart cells. A prolonged QT interval indicates an abnormality in electrical activity that leads to irregularities in heart muscle contraction. One of these irregularities is a specific pattern of very rapid contractions (tachycardia) of the lower chambers of the heart called torsade de pointes, a type of **ventricular tachycardia**. The rapid contractions, which are not effective in pumping blood to the body, result in a decreased flow of oxygen-rich blood to the brain. This can result in a sudden loss of consciousness (syncope) and death.

Causes and symptoms

Both inherited and acquired forms of LQTS have been identified. Most acquired forms are thought to be due to certain drugs including adrenaline (epinephrine), several **antihistamines** and **antibiotics**, specific heart medications, **diuretics**, and others. It has been proposed, but not yet documented, that individuals who experience LQTS after using one of these medications, may actually have a genetic defect that increases their tendency to cardiac arrhythmias. Severe weight loss such as is associated with **anorexia nervosa** can

also disrupt ion balances in the heart and result in prolongation of the QT interval.

Three inherited forms of LQTS have been described to date. Jervell and Lange-Neilsen syndrome, named for the physicians who described the condition in 1957, is associated with congenital deafness and is inherited as an autosomal recessive trait. Romano-Ward syndrome, the most common inherited form of LQTS, was first described in the 1960's. It is inherited in an autosomal dominant pattern and is not associated with other physical impairments such as deafness. In 1995, a third type of LQTS was reported in to occur in association with bilateral **syndactyly**. Little is known about the inheritance of this form, except that reported cases have been sporadic with no associated family history of LQTS.

As of early 2001, six different genes had been associated with the inherited forms of LQTS, and mutations in at least four of these genes had been reported in a number of affected individuals and families. The genes involved in LQTS play important roles in the formation of ion channels in the cell membrane, and, thus, mutations in these genes disrupt normal cardiac rhythms.

LQTS usually presents with symptoms that constitute a life-threatening emergency. Sudden loss of consciousness or cardiac arrest can be brought on by emotional or physical **stress** in young, otherwise healthy individuals, both female and male. Fright, anger, surprise, sudden awakening as a result of loud sounds (alarm clock, telephone), as well as physical activities, especially swimming, have all been reported to precipitate an episode of cardiac arrhythmia in susceptible individuals. Sudden death often occurs. Although the information is preliminary, recent research has also suggested that a small number of SIDS (**sudden infant death syndrome**) cases may be due to mutations in one or more of the genes associated with LQTS.

Diagnosis

Problems exist in diagnosing LQTS. Although the method of diagnosis is the electrocardiogram, most young, healthy people do not routinely undergo this test, and, thus, their first, and possibly fatal, episode of LQTS comes without warning. In some cases, a non-fatal episode is mistakenly treated as a seizure, and, therefore, a follow-up assessment does not include an electrocardiogram. In addition, some cases of LQTS cannot be diagnosed by a routine electrocardiogram. That is, the QT interval is not found to be prolonged in routine testing. If LQTS is suspected either because of

a previous episode of syncope or because of a family member with LQTS, an **exercise** electrocardiogram should be performed. In all instances where an individual is diagnosed with LQTS, family members should be thoroughly evaluated, and a detailed family history should be taken noting any individuals with episodes of sudden loss of consciousness and any cases of unexplained sudden death. Because many of the genes involved in LQTS have been identified, **genetic testing** can offer a more reliable means of diagnosis of other family members at risk. The first step in determining if this type of testing is appropriate in any particular situation is to consult a genetic counselor or medical geneticist.

Treatment

A conventional treatment is the oral administration of beta-blockers, medications that decrease the input from the sympathetic nervous system to the heart. Although beta-blockers do not correct the abnormalities in the ion channels of the heart cells, they do appear to decrease the occurrence of cardiac arrhythmias. However, these medications are not helpful in all cases, and are actually contraindicated in some individuals. Potassium supplementation is also being explored as a treatment in certain cases. As the genetics of LQTS becomes better understood, it should be possible to tailor treatments that will be effective for each of the various gene mutations.

Alternative treatment

In some patients, severing of the sympathetic nerve to the heart has decreased the occurrence of arrhythmias. **Pacemakers** and defibrillators appear to hold promise as new forms of treatment. As devices of this type are developed that are smaller in size, they may come into more widespread use, either alone or in conjunction with specific medications.

Prognosis

LQTS is a life-long condition. Individuals who are not diagnosed and treated are at an increased risk of syncope and sudden death. Adequate treatment can decrease this risk. There is no cure. Individuals with one of the inherited forms of LQTS are at risk of passing the mutation and the disease to their offspring.

Prevention

The risk of cardiac arrhythmias due to acquired forms of LQTS can be decreased by avoiding the

KEY TERMS

Anorexia nervosa—A loss of appetite for food not explainable by local disease. It is thought to have a psychological basis.

Autosomal dominant—A pattern of inheritance in which only one of the two copies of an autosomal gene must be abnormal for a genetic condition or disease to occur. An autosomal gene is a gene that is located on one of the autosomes or non-sex chromosomes. A person with an autosomal dominant disorder has a 50% chance of passing it to each of their offspring.

Autosomal recessive—A pattern of inheritance in which both copies of an autosomal gene must be abnormal for a genetic condition or disease to occur. An autosomal gene is a gene that is located on one of the autosomes or non-sex chromosomes. When both parents have one abnormal copy of the same gene, they have a 25% chance with each pregnancy that their offspring will have the disorder.

Diuretic—An agent that increases the production of urine.

Electrocardiogram—A record of the electrical activity of the heart showing certain waves called P, Q, R, S, and T waves. The Q, R, S, T waves are associated with contraction of the ventricles, the lower two chambers of the heart.

Sympathetic nervous system—A division of the autonomic nervous system, the portion of the nervous system that controls involuntary bodily functions such as heart rate.

Syndactyly—A fusion of two or more toes or fingers.

medications and situations that trigger episodes. At present there is no genetic therapy to correct the gene mutations present in the inherited forms of LQTS, but individuals who are known to have an inherited form may also be able to lessen the risk of a life-threatening episode by avoiding such environmental triggers and by taking the appropriate medications.

Resources

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Vizgirda, Vida M. "The Genetic Basis for Cardiac Dysrhythmias and the Long QT Syndrome" *J. Cardiovasc Nursing* 13, no.4 (1999): 34–45.

ORGANIZATIONS

Sudden Arrhythmia Death Syndromes Foundation. 540 Arapsee Drive, Suite 207, Salt Lake City Utah, 84108, (800) STOP SAD, <<http://www.sads.org>> or <<http://www.ihc.com/research/longqt.html>>.

OTHER

American Heart Association. <http://www.Americanheart.org/Heart_and_Stroke_A_Z_Guide/longqt.html>.

NORD (National Organization for Rare Disorders, Inc.). <<http://www.rarediseases.org>>.

Sallie Boineau Freeman, PhD

PROM see **Premature rupture of membranes**

Promethaz see **Antihistamines**

Prophylaxis

Definition

A prophylaxis is a measure taken to maintain health and prevent the spread of disease. Antibiotic prophylaxis is the focus of this article and refers to the use of **antibiotics** to prevent infections.

Purpose

Antibiotics are well known for their ability to treat infections. But some antibiotics also are prescribed to *prevent* infections. This usually is done only in certain situations or for people with particular medical problems. For example, people with abnormal heart valves have a high risk of developing heart valve infections after even minor surgery. This happens because bacteria from other parts of the body get into the bloodstream during surgery and travel to the heart valves. To prevent these infections, people with heart valve problems often take antibiotics before having any kind of surgery, including dental surgery.

Antibiotics also may be prescribed to prevent infections in people with weakened immune systems, such as people with **AIDS** or people who are having

chemotherapy treatments for cancer. But even healthy people with strong immune systems may occasionally be given preventive antibiotics—if they are having certain kinds of surgery that carry a high risk of infection, or if they are traveling to parts of the world where they are likely to get an infection that causes **diarrhea**, for example.

In all of these situations, a physician should be the one to decide whether antibiotics are necessary. Unless a physician says to do so, it is not a good idea to take antibiotics to prevent ordinary infections.

Because the overuse of antibiotics can lead to resistance, drugs taken to prevent infection should be used only for a short time.

Description

Among the drugs used for antibiotic prophylaxis are amoxicillin (a type of penicillin) and fluoroquinolones such as ciprofloxacin (Cipro) and trovafloxacin (Trovan). These drugs are available only with a physician's prescription and come in tablet, capsule, liquid, and injectable forms.

Recommended dosage

The recommended dosage depends on the type of antibiotic prescribed and the reason it is being used. For the correct dosage, check with the physician or dentist who prescribed the medicine or the pharmacist who filled the prescription. Be sure to take the medicine exactly as prescribed. Do not take more or less than directed, and take the medicine only for as long as the physician or dentist says to take it.

Precautions

If the medicine causes **nausea**, **vomiting**, or diarrhea, check with the physician or dentist who prescribed it as soon as possible. Patients who are taking antibiotics before surgery should not wait until the day of the surgery to report problems with the medicine. The physician or dentist needs to know right away if problems occur.

For other specific precautions, see the entry on the type of drug prescribed such as penicillins or fluoroquinolones.

Side effects

Antibiotics may cause a number of side effects. For details, see entries on specific types of antibiotics. Anyone who has unusual or disturbing symptoms

KEY TERMS

AIDS—Acquired immunodeficiency syndrome. A disease caused by infection with the human immunodeficiency virus (HIV). In people with this disease, the immune system breaks down, opening the door to other infections and some types of cancer.

Antibiotic—A medicine used to treat infections.

Chemotherapy—Treatment of an illness with chemical agents. The term is usually used to describe the treatment of cancer with drugs.

Immune system—The body's natural defenses against disease and infection.

after taking antibiotics should get in touch with his or her physician.

Interactions

Whether used to treat or to prevent infection, antibiotics may interact with other medicines. When this happens, the effects of one or both of the drugs may change or the risk of side effects may be greater. Anyone who takes antibiotics for any reason should inform the physician about all the other medicines he or she is taking and should ask whether any possible interactions may interfere with drugs' effects. For details of **drug interactions**, see entries on specific types of antibiotics.

Nancy Ross-Flanigan

Proportionate dwarfism see **Pituitary dwarfism**

Proptosis see **Exophthalmos**

Prostaglandins see **Drugs used in labor**

Prostate biopsy

Definition

Prostate biopsy is a surgical procedure that involves removing a small piece of prostate tissue for microscopic examination.

Purpose

This test is usually done to determine whether the patient has prostate cancer. Occasionally, it may also be used to diagnose a condition called benign prostatic hyperplasia that causes enlargement of the prostate. In the United States, prostate cancer is the most common **cancer** among men over 50, and is the second leading cause of cancer deaths. According to statistics released in 2003, African American men in the United States are at much greater risk of developing **prostate cancer** after age 50 than Caucasian or Asian American men.

Prostate biopsy is recommended when a digital rectal examination (a routine screening test for prostate diseases) reveals a lump or some other abnormality in the prostate. In addition, if blood tests reveal that the levels of certain markers, such as PSA, are above normal, the doctor may order a biopsy.

Description

The prostate gland is one of the three male sex glands and lies just below the urinary bladder, in the area behind the penis and in front of the rectum. It secretes semen, the liquid portion of the ejaculate. The urethra carries the urine from the urinary bladder and the semen from the sex glands to the outside of the body.

Prostate biopsies can be performed in three different ways. They can be performed by inserting a needle through the perineum (the area between the base of the penis and the rectum), by inserting a needle through the wall of the rectum, or by cytoscopy. Before the procedure is performed, the patient may be given a sedative to help him relax. Patients undergoing cytoscopy may be given either **general anesthesia** or **local anesthesia**. The doctor will ask the patient to have an enema before carrying out the biopsy. The patient is also given **antibiotics** to prevent any possible infection.

Needle biopsy via the perineum

The patient lies either on one side or on his back with his knees up. The skin of the perineum is thoroughly cleansed with an iodine solution. A local anesthetic is injected at the site where the biopsy is performed. Once the area is numb, the doctor makes a small (1 in) incision in the perineum. The doctor places one finger in the rectum to guide the placement of the needle. The needle is then inserted into the prostate, a small amount of tissue is collected, and the needle is withdrawn. The needle is then re-inserted into another part of the prostate. Tissue may be taken from several areas. Pressure is then applied at the

biopsy site to stop the bleeding. The procedure generally takes 15–30 minutes and is usually done in a physician's office or in a hospital operating room. Although it sounds painful, it typically causes only slight discomfort.

Needle biopsy via the rectum

This procedure is also done in the physician's office or in the hospital operating room, and is usually done without any anesthetic, although some doctors prefer to inject a local anesthetic, usually lidocaine. The patient is asked to lie on his side or on his back with his legs in stirrups. The doctor attaches a curved needle guide to his finger and then inserts the finger into the rectum. After firmly placing the needle guide in the rectum, the biopsy needle is pushed along the guide, through the wall of the rectum and into the prostate. The needle is rotated gently, prostate tissue samples are collected and the needle withdrawn. When an ultrasound probe is used to guide the needle, the procedure is called a transrectal ultrasound-guided biopsy, or TRUS.

Cytoscopy

For this procedure, the patient is given either a general or a local anesthetic. An instrument called a cystoscope (a thin-lighted tube with telescopic lenses) is passed through the urethra. By looking through the cystoscope, the doctor can see if there is any blockage in the urethra and remove it. Tissue samples from the urinary bladder or the prostate can be collected for microscopic examination.

This test is generally performed in an operating room or in a physician's office. An hour before the procedure, the patient is given a sedative to help him relax. An intravenous (IV) line will be placed in a vein in the arm to give medications and fluids if necessary. The patient is asked to lie on a special table with his knees apart and stirrups are used to support his feet and thighs. The genital area is cleansed with an anti-septic solution. If general anesthesia is being used, the patient is given the medication through the IV tube or inhaled gases or both. If a local anesthetic is being used, the anesthetic solution is gently instilled into the urethra.

After the area is numb, a cystoscope is inserted into the urethra and slowly pushed into the prostate. Tiny forceps or scissors are inserted through the cystoscope to collect small pieces of tissue that are used for biopsy. The cystoscope is then withdrawn. The entire procedure may take 30–45 minutes. Sometimes a catheter (tube) is left in the urinary bladder to help

the urine drain out, until the swelling in the urethra has subsided.

Alternate procedures

Many different tests can be performed to diagnose prostate diseases and cancer. A routine screening test called digital **rectal examination** (DRE) can identify any lumps or abnormality with the prostate. Blood tests that measure the levels of certain protein markers, such as PSA, can indicate the presence of prostate cancer cells. X rays and other imaging techniques (such as computed tomography scans, magnetic resonance imaging, and ultrasonograms), where detailed pictures of areas inside the body are put together by a computer, can also be used to determine the extent and spread of the disease. However, a prostate biopsy and examination of the cells under a microscope remains the most definitive test for diagnosing and grading prostate cancer.

Preparation

Before scheduling the biopsy, the doctor should be made aware of all the medications that the patient is taking, if the patient is allergic to any medication, and if he has any bleeding problems. The patient may be given an antibiotic shortly before the test to reduce the risk of any infection afterwards. If the biopsy is done through the perineum, there are no special preparations. If it is being done through the rectum, the patient is asked to take an enema and is instructed on how to do it.

If a cystoscopy is being performed, the patient is asked to sign a consent form. The patient is also asked to take antibiotics before and for several days after the test to prevent infection due to insertion of the instruments. If a general anesthetic is going to be used, food and liquids will be restricted for at least eight hours before the test.

Aftercare

Following a needle biopsy, the patient may experience some **pain** and discomfort. He should avoid strenuous activities for the rest of the day. He may also notice some blood in his urine for two to three days after the test and some amount of rectal bleeding. If there is persistent bleeding, pain, or **fever**, and if the patient is unable to urinate for 24 hours, the doctor should be notified immediately.

When a cystoscopy is performed under a local anesthetic, the patient is asked to lie down for 30 minutes after the test and is then allowed to go. If

general anesthesia is used, the patient is taken to the recovery room and kept there until he wakes up and is able to walk. He is allowed food and liquids after he wakes up. After general anesthesia, the patient may experience some tiredness and aching of the muscles throughout the body. If local anesthesia was administered, there is a brief burning sensation and a strong urge to urinate when the cystoscope is removed.

After the procedure, it is common to experience frequent urination with a burning sensation for a few days. Drinking a lot of fluids will help reduce the burning sensation and the chances of an infection. There may also be some blood in the urine. However, if **blood clots** are seen, or if the patient is unable to pass urine eight hours after the cystoscopy, the doctor should be notified. In addition, if the patient develops a high fever, and complains of chills or abdominal pain after the procedure, he should see the doctor right away. Although serious infections are rare, a few patients develop such severe illnesses as **meningitis** following a prostate biopsy.

Risks

Prostate biopsy performed with a needle is a low-risk procedure. The possible complications include some bleeding into the urethra, bleeding from the rectum, an infection, a temporarily lowered sperm count, or an inability to urinate. These complications are treatable and the doctor should be notified of them.

Cystoscopy is generally a very safe procedure. The most common complication is an inability to urinate due to a swelling of the urethra. A catheter (tube) may have to be inserted to help drain out the urine. If there is an infection after the procedure, antibiotics are given to treat it. In very rare instances, the urethra or the urinary bladder may be perforated because of the insertion of the instrument. If this complication occurs, surgery may be needed to repair the damage.

Normal results

If the prostate tissue samples show no sign of inflammation, and if no cancerous cells are detected, the results are normal.

Abnormal results

Analysis of the prostate tissue under the microscope reveals any abnormalities. In addition, the presence of cancerous cells can be detected. If a tumor is

KEY TERMS

Benign prostatic hyperplasia (BPH)—A noncancerous condition of the prostate that causes overgrowth of the prostate tissue, thus enlarging the prostate and obstructing urination.

Biopsy—The surgical removal and microscopic examination of living tissue for diagnostic purposes.

Computed tomography (CT) scan—A medical procedure in which a series of x rays are taken and put together by a computer in order to form detailed pictures of areas inside the body.

Digital rectal examination—A routine screening test that is used to detect any lumps in the prostate gland or any hardening or other abnormality of the prostate tissue. The doctor inserts a gloved and lubricated finger (digit) into the patient's rectum, which lies just behind the prostate. Typically, since a majority of tumors develop in the posterior region of the prostate, they can be detected through the rectum.

Magnetic resonance imaging (MRI)—A medical procedure used for diagnostic purposes where pictures of areas inside the body are created using a magnet linked to a computer.

Pathologist—A doctor who specializes in the diagnosis of disease by studying cells and tissues under a microscope.

Ultrasonogram—A procedure in which high-frequency sound waves that cannot be heard by human ears are bounced off internal organs and tissues. These sound waves produce a pattern of echoes that are then used by the computer to create sonograms or pictures of areas inside the body.

Urethra—The tube that carries the urine from the urinary bladder and (in males) the semen from the sex glands to the outside of the body.

present, the pathologist “grades” the tumor, in order to estimate how aggressive the tumor is. The most commonly used grading system is called the “Gleason system.”

Normal prostate tissue has certain characteristic features that the cancerous tissue lacks. In the Gleason system, prostate cancers are graded by how closely they resemble normal prostate tissue. The system assigns a grade ranging from 1 to 5. The grades

assigned to two areas of cancer are added up for a combined score that is between 2 and 10. A score between 2 and 4 is called low and implies that the cancer is a slow-growing one. A Gleason score of 8 to 10 is high and indicates that the cancer is aggressive. The higher the Gleason score, the more likely it is that the cancer is fast-growing and may have already grown out of the prostate and spread to other areas (metastasized).

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- American Urologic Association. 1120 N. Charles St., Baltimore, MD 21201. (410) 223-4310.
- National Prostate Cancer Coalition. 1300 19th Street NW, Suite 400, Washington, DC 20036. (202) 842-3600 ext. 214.
- Prostate Cancer InfoLink. <<http://www.comed.com/Prostate/index.html>>.

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Prostate cancer

Definition

Prostate **cancer** is a disease in which cells in the prostate gland become abnormal and start to grow uncontrollably, forming tumors.

Description

Prostate cancer is a malignancy of one of the major male sex glands. Along with the testicles and the seminal vesicles, the prostate secretes the fluid that makes up semen. The prostate is about the size of a walnut and lies just behind the urinary bladder. A tumor in the prostate interferes with proper control of the bladder and normal sexual functioning. Often the first symptom of prostate cancer is difficulty in urinating. However, because a very common, non-cancerous condition of the prostate, benign prostatic hyperplasia (BPH), also causes the same problem, difficulty in urination is not necessarily due to cancer.

Cancerous cells within the prostate itself are generally not deadly on their own. However, as the tumor grows, some of the cells break off and spread to other parts of the body through the lymph or the blood, a process known as metastasis. The most common sites for prostate cancer to metastasize are the seminal vesicles, the lymph nodes, the lungs, and various bones around the hips and the pelvic region. The effects of these new tumors are what can cause **death**.

As of the early 2000s, prostate cancer is the most commonly diagnosed malignancy among adult males in Western countries. Although prostate cancer is often very slow growing, it can be aggressive, especially in younger men. Given its slow growing nature, many men with the disease die of other causes rather than from the cancer itself.

Prostate cancer affects African-American men twice as often as white men; the mortality rate among African-Americans is also two times higher. African-Americans have the highest rate of prostate cancer of any world population group.

Causes and symptoms

The precise cause of prostate cancer is not known as of the early 2000s. However, there are several known risk factors for disease including age over 55, African-American heritage, a family history of the disease, occupational exposure to cadmium or rubber, and a high fat diet. Men with high plasma testosterone

levels may also have an increased risk for developing prostate cancer.

Frequently, prostate cancer has no symptoms and the disease is diagnosed when the patient goes for a routine screening examination. However, when the tumor is big or the cancer has spread to the nearby tissues, the following symptoms may be seen:

- weak or interrupted flow of the urine
- frequent urination (especially at night)
- difficulty starting urination
- inability to urinate
- **pain** or burning sensation when urinating
- blood in the urine
- persistent pain in lower back, hips, or thighs (bone pain)
- painful ejaculation

Diagnosis

Prostate cancer is curable when detected early. Yet the early stages of prostate cancer are often asymptomatic, so the disease often goes undetected until the patient has a routine **physical examination**. Diagnosis of prostate cancer can be made using some or all of the following tests.

Digital rectal examination (DRE)

In order to perform this test, the doctor puts a gloved, lubricated finger (digit) into the rectum to feel for any lumps in the prostate. The rectum lies just behind the prostate gland, and a majority of prostate tumors begin in the posterior region of the prostate. If the doctor does detect an abnormality, he or she may order more tests in order to confirm these findings.

Blood tests

Blood tests are used to measure the amounts of certain protein markers, such as prostate-specific antigen (PSA), found circulating in the blood. The cells lining the prostate generally make this protein and a small amount can be detected normally in the bloodstream. In contrast, prostate cancers produce a lot of this protein, significantly raising the circulating levels. A finding of a PSA level higher than normal for the patient's age group therefore suggests that cancer is present.

Transrectal ultrasound

A small probe is placed in the rectum and sound waves are released from the probe. These sound waves

bounce off the prostate tissue and an image is created. Since normal prostate tissue and prostate tumors reflect the sound waves differently, the test is an efficient and accurate way to detect tumors. Though the insertion of the probe into the rectum may be slightly uncomfortable, the procedure is generally painless and takes only 20 minutes.

Prostate biopsy

If cancer is suspected from the results of any of the above tests, the doctor will remove a small piece of prostate tissue with a hollow needle. This sample is then checked under the microscope for the presence of cancerous cells. **Prostate biopsy** is the most definitive diagnostic tool for prostate cancer.

Prostate cancer can also be diagnosed based on the examination of the tissue removed during a transurethral resection of the prostate (TURP). This procedure is performed to help alleviate the symptoms of BPH, a benign enlargement of the prostate. Like a biopsy, this is a definitive diagnostic method for prostate cancer.

X rays and imaging techniques

A **chest x ray** may be ordered to determine whether the cancer has spread to the lungs. Imaging techniques (such as **computed tomography scans** (CT) and **magnetic resonance imaging** (MRI)), where a computer is used to generate a detailed picture of the prostate and areas nearby, may be done to get a clearer view of the internal organs. A bone scan may be used to check whether the cancer has spread to the bone.

Treatment

Once cancer is detected during the microscopic examination of the prostate tissue during a biopsy or TURP, doctors will determine two different numerical scores that will help define the patient's treatment and prognosis.

Tumor grading

Initially, the pathologist will grade the tumor based on his or her examination of the biopsy tissue. The pathologist scores the appearance of the biopsy sample using the Gleason system. This system uses a scale of one to five based on the sample's similarity or dissimilarity to normal prostate tissue. If the tissue is very similar to normal tissue, it is still well differentiated and given a low grading number, such as one or two. As the tissue becomes more and more abnormal (less and less differentiated), the grading number

increases, up to five. Less differentiated tissue is considered more aggressive and more likely to be the source of metastases.

The Gleason grading system is best predictive of the prognosis of a patient if the pathologist gives two scores to a particular sample—a primary and a secondary pattern. The two numbers are then added together and that is the Gleason score reported to the patient. Thus, the lowest Gleason score available is two (a primary and secondary pattern score of one each). A typical Gleason score is five (which can be a primary score of two and a secondary score of three or visa-versa). The highest score available is 10, with a pure pattern of very undifferentiated tissue, that is, of grade five. The higher the score, the more abnormal behavior of the tissue, the greater the chance for metastases, and the more serious the prognosis after surgical treatment. A study found that the ten-year cancer survival rate without evidence of disease for grade two, three, and four cancers is 94% of patients. The rate is 91% for grade five cancers, 78% for grade six, 46% for grade seven, and 23% for grade eight, nine, and ten cancers.

Cancer staging

The second numeric score determined by the doctor will be the stage of the cancer, which takes into account the grade of the tumor determined by the pathologist. Based on the recommendations of the American Joint Committee on Cancer (AJCC), two kinds of data are used for staging prostate cancer. Clinical data are based on the external symptoms of the cancer, while histopathological data is based on surgical removal of the prostate and examination of its tissues. Clinical data are most useful to make treatment decisions, while pathological data is the best predictor of prognosis. For this reason, the staging of prostate cancer takes into account both clinical and histopathologic information. Specifically, doctors look at tumor size (T), lymph node involvement (N), the presence of visceral (internal organ) involvement (metastasis = M), and the grade of the tumor (G).

The classification of tumor as T1 means the cancer that is confined to the prostate gland and the tumor that is too small to be felt during a DRE. T1 tumors are often found after examination of tissue removed during a TURP. The T1 definition is subdivided into those cancers that show less than 5% cancerous cells in the tissue sample (T1a) or more than 5% cancerous cells in the tissue sample (T1b). T1c means that the biopsy was performed based on an elevated PSA result. The second tumor classification is T2, where the tumor is large enough to be felt during the DRE.

T2a indicates that only the left or the right side of the gland is involved, while T2b means both sides of the prostate gland has tumor.

With a T3 tumor the cancer has spread to the connective tissue near the prostate (T3a) or to the seminal vesicles as well (T3b). T4 indicates that cancer has spread within the pelvis to tissue next to the prostate such as the bladder's sphincter, the rectum, or the wall of the pelvis. Prostate cancer tends to spread next into the regional lymph nodes of the pelvis, indicated as N1. Prostate cancer is said to be at the M1 stage when it has metastasized outside the pelvis in distant lymph nodes (M1a), bone (M1b) or organs such as the liver or the brain (M1c). Pain, weight loss, and **fatigue** often accompany the M1 stage.

The grade of the tumor (G) can be assessed during a biopsy, TURP surgery, or after removal of the prostate. There are three grades recognized: G1, G2, and G3, indicating the tumor is well, moderately, or poorly differentiated, respectively. The G, LN, M descriptions are combined with the T definition to determine the stage of the prostate cancer.

Stage I prostate cancer comprises patients that are T1a, N0, M0, G1. Stage II includes a variety of condition combinations including T1a, N0, M0, G2, 3 or 4; T1b, N0, M0, Any G; T1c, N0, M0, Any G; T1, N0, M0, Any G or T2, N0, M0, Any G. The prognosis for cancers at these two stages is very good. For men treated with stage I or stage II disease, over 95% are alive after five years.

Stage III prostate cancer occurs when conditions are T3, N0, M0, any G. Stage IV is T4, N0, M0, any G; any T, N1, M0, any G; or any T, any N, M1, Any G. Although the cancers of Stage III are more advanced, the five year prognosis is still good, with 70% of men diagnosed at these stage still living. The spread of the cancer into the pelvis (T4), lymph (N1), or distant locations (M1) are very significant events, as the five year survival rate drops to 30% for Stage IV.

Treatment options

The doctor and the patient will decide on the treatment mode after considering many factors. For example, the patient's age, the stage of the disease, his general health, and the presence of any co-existing illnesses have to be considered. In addition, the patient's personal preferences and the risks and benefits of each treatment protocol are also taken into account before any decision is made.

SURGERY. For stage I and stage II prostate cancer, surgery is the most common method of treatment because it theoretically offers the chance of completely

removing the cancer from the body. Radical **prostatectomy** involves complete removal of the prostate. The surgery can be done using a perineal approach, where the incision is made between the scrotum and the anus, or using a retropubic approach, where the incision is made in the lower abdomen. Perineal approach is also known as nerve-sparing prostatectomy, as it is thought to reduce the effect on the nerves and thus reduce the side effects of **impotence** and incontinence. However, the retropubic approach allows for the simultaneous removal of the pelvic lymph nodes, which can give important pathological information about the tumor spread.

The drawback to surgical treatment for early prostate cancer is the significant risk of side effects that impact the quality of life of the patient. Even using nerve-sparing techniques, studies run by the National Cancer Institute (NCI) found that 60–80% of men treated with radical prostatectomy reported themselves as impotent (unable to achieve an erection sufficient for sexual intercourse) two years after surgery. This side effect can be sometimes countered by prescribing **sildenafil citrate** (Viagra). Furthermore, 8% to 10% of patients were incontinent in that time span. Despite the side effects, the majority of men were reported as satisfied with their treatment choice. Additionally, there is some evidence that the skill and the experience of the surgeon are central factors in the ultimate side effects seen.

A second method of surgical treatment of prostate cancer is cryosurgery. Guided by ultrasound, surgeons insert up to eight cryoprobes through the skin and into close proximity with the tumor. Liquid nitrogen (temperature of -320.8 °F, or -196° C) is circulated through the probe, freezing the tumor tissue. In prostate surgery, a warming tube is also used to keep the urethra from freezing. Patients currently spend a day or two in the hospital following the surgery, but it could be an outpatient procedure in the near future. Recovery time is about one week. Side effects have been reduced in recent years, although impotence still affects almost all who have had cryosurgery for prostate cancer. Cryosurgery is considered a good alternative for those too old or sick to have traditional surgery or radiation treatments or when these more traditional treatments are unsuccessful. There is a limited amount of information about the long-term efficacy of this treatment for prostate cancer.

Radiation therapy

Radiation therapy involves the use of high-energy x rays to kill cancer cells or to shrink tumors. It can be used instead of surgery for stage I and II cancer. The

radiation can either be administered from a machine outside the body (external beam radiation), or small radioactive pellets can be implanted in the prostate gland in the area surrounding the tumor, called brachytherapy or interstitial implantation. Pellets containing radioactive iodine (I-125), palladium (Pd 103), or iridium (Ir 192) can be implanted on an outpatient basis, where they remain permanently. The radioactive effect of the seeds last only about a year.

The side effects of radiation can include inflammation of the bladder, rectum, and small intestine as well as disorders of blood clotting (coagulopathies). Impotence and incontinence are often delayed side effects of the treatment. A study indicated that bowel control problems were more likely after radiation therapy when compared to surgery, but impotent and incontinence were more likely after surgical treatment. Long-term results with radiation therapy are dependent on stage. A review of almost 1000 patients treated with megavoltage irradiation showed 10 year survival rates to be significantly different by T-stage: T1 (79%), T2 (66%), T3 (55%), and T4 (22%). There does not appear to be a large difference in survival between external beam or interstitial treatments.

HORMONE THERAPY. Hormone therapy is commonly used when the cancer is in an advanced stage and has spread to other parts of the body, such as stage III or stage IV. Prostate cells need the male hormone testosterone to grow. Decreasing the levels of this hormone or inhibiting its activity will cause the cancer to shrink. Hormone levels can be decreased in several ways. Orchiectomy is a surgical procedure that involves complete removal of the testicles, leading to a decrease in the levels of testosterone. Another method tricks the body by administering the female hormone estrogen. When estrogen is given, the body senses the presence of a sex hormone and stops making the male hormone testosterone. However, there are some unpleasant side effects to hormone therapy. Men may have “hot flashes,” enlargement and tenderness of the breasts, or impotence and loss of sexual desire, as well as **blood clots**, heart attacks, and strokes, depending on the dose of estrogen. Another side effect is **osteoporosis**, or loss of bone mass leading to brittle and easily fractured bones.

WATCHFUL WAITING. Watchful waiting means no immediate treatment is recommended, but doctors keep the patient under careful observation. This is often done using periodic PSA tests. This option is generally used in older patients when the tumor is not very aggressive and the patients have other, more life-threatening, illnesses. Prostate cancer in older men tends to be slow-growing. Therefore, the risk of the

patient dying from prostate cancer, rather than from other causes, is relatively small.

Treatments for prostate cancer that are under investigation in the early 2000s include evaluation of combination therapies, such as postoperative radiation delivery, use of cytotoxic agents, and hormonal treatment using luteinizing hormone-releasing hormone (LHRH) agonists and/or antiandrogens to shut down the growth of the hormone-dependent tumors. Other drugs that are being tested as of 2003 are chemoprotective agents like amifostine (Ethyol), which are given to prostate cancer patients to counteract the harmful side effects of radiation treatment.

Alternative treatment

Alternative treatments that have been found helpful in coping with the emotional **stress** associated with prostate cancer include **meditation**, **guided imagery**, and relaxation techniques. **Acupuncture** is effective in relieving pain in some patients.

A variety of herbal products have been used to treat prostate cancer, including various compounds used in **traditional Chinese medicine** as well as single agents like Reishi mushrooms (*Ganoderma lucidum*). One herbal compound that was under investigation by the National Center for Complementary and Alternative Medicine (NCCAM) as a possible treatment for prostate cancer was PC-SPES, a mixture of eight herbs adapted from traditional Chinese medicine. In the summer of 2002, however, NCCAM put its studies of PC-SPES on hold when the Food and Drug Administration (FDA) determined that samples of the product were contaminated with undeclared prescription drug ingredients. PC-SPES was withdrawn from the American market in late 2002.

Prevention

Because the cause of the cancer is not known, there is no definite way to prevent prostate cancer. Given its common occurrence and the low cost of screening, the American Cancer Society (ACS) and the National Comprehensive Cancer Network (NCCN) recommends that all men over age 40 have an annual **rectal examination** and that men have an annual PSA test beginning at age 50. African-American men and men with a family history of prostate cancer, who have a higher than average risk, should begin annual PSA testing even earlier, starting at age 45.

However, mandatory screening for prostate cancer is controversial. Because the cancer is so slow

KEY TERMS

Antiandrogen—A substance that blocks the action of androgens, the hormones responsible for male characteristics. Used to treat prostate cancers that require male hormones for growth.

Benign prostatic hyperplasia (BPH)—A non-cancerous swelling of the prostate.

Brachytherapy—A method of treating cancers, such as prostate cancer, involving the implantation near the tumor of radioactive seeds.

Gleason grading system—A method of predicting the tendency of a tumor in the prostate to metastasize based on how similar the tumor is to normal prostate tissue.

Granulocyte/macrophage colony stimulating factor (GM-CSF)—A substance produced by cells of the immune system that stimulates the attack upon foreign cells. Used to treat prostate cancers as a genetically engineered component of a vaccine that stimulates the body to attack prostate tissue.

Histopathology—The study of diseased tissues at a minute (microscopic) level.

Luteinizing hormone releasing hormone (LHRH) agonist—A substance that blocks the action of LHRH, a hormone that stimulates the production of testosterone (a male hormone) in men. Used to treat prostate cancers that require testosterone for growth.

Orchiectomy—Surgical removal of the testes that eliminates the production of testosterone to treat prostate cancer.

Radical prostatectomy—Surgical removal of the entire prostate, a common method of treating prostate cancer.

Prostate-specific antigen—A protein made by the cells of the prostate that is increased by both BPH and prostate cancer.

Transurethral resection of the prostate (TURP)—Surgical removal of a portion of the prostate through the urethra, a method of treating the symptoms of an enlarged prostate, whether from BPH or cancer.

growing, and the side effects of the treatment can have significant impact on patient quality of life, some medical organizations question the wisdom of yearly exams. Some organizations have even noted that the effect of screening is discovering the cancer at an early

stage when it may never grow to have any outward effect on the patient during their lifetime. Nevertheless, the NCI reports that the current aggressive screening methods have achieved a reduction in the death rate of prostate cancer of about 2.3% for African-Americans and about 4.6% for Caucasians since the mid-1990s, with a 20% increase in overall survival rate during that period.

A low-fat diet may slow the progression of prostate cancer. To reduce the risk or progression of prostate cancer, the American Cancer Society recommends a diet rich in fruits, vegetables and dietary fiber, and low in red meat and saturated fats.

Resources

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ORGANIZATIONS

Association for the Cure of Cancer of the Prostate (CaPCure). 1250 Fourth St., Suite 360, Santa Monica, CA 90401. (800) 757-CURE. <<http://www.capcure.org>>.

National Cancer Institute. Building 31, Room 10A31 31 Center Drive, MSC 2580, Bethesda, MD 20892-2580. (800) 4-CANCER. <<http://cancernet.nci.nih.gov>>.

National Center for Complementary and Alternative Medicine (NCCAM) Clearinghouse. P. O. Box 7923, Gaithersburg, MD 20898. (888) 644-6226. <<http://nccam.nih.gov>>.

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FDA MedWatch Safety Alert for PC-SPES, SPES, updated September 20, 2002. <<http://www.fda.gov/medwatch/SAFETY/2002/safety02.htm#spes>>.

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Prostate gland removal see **Prostatectomy**

Prostate sonogram see **Prostate ultrasound**

Prostate-specific antigen test

Definition

Prostate-specific antigen, or PSA, is a protein produced by the prostate gland that may be found in elevated levels in the blood when a person develops certain diseases of the prostate, notably **prostate cancer**. PSA is *specific*, because it is present only in prostate tissue. It is not specific for prostate *cancer*, however, as it may also be elevated in men with benign enlargement of this organ. The PSA test has been called the “male PAP test.”

Purpose

The blood test for PSA is used to screen older men to detect prostate **cancer** at an early stage, and also to

monitor its response to treatment. After lung cancer, prostate cancer is the most common form of cancer in men in the United States. Any routine physical exam of a man aged 50 and older should include a digital rectal examination (DRE), in which the doctor’s finger probes the surface of the prostate gland to detect any suspicious area of hardness or a tumor mass. If the examination suggests that a tumor may in fact be present or if the examiner is uncertain the logical next step is a PSA test. If the PSA test is positive, a sample of prostate tissue (biopsy) may be taken to confirm that cancer is present. If negative, the test may be repeated immediately to confirm the diagnosis, or repeated the next year. Many physicians today routinely do both a DRE and a PSA test each year on their older male patients, so that, if cancer does develop, it will be found at an early stage will be easier to treat. The combination of a DRE and a PSA test can detect approximately 80% of all prostate cancers.

At present, the PSA test is widely accepted as a way of telling whether a patient with definite cancer is responding to treatment. Because only the prostate produces PSA, its presence in the blood following complete removal of the prostate (radical **prostatectomy**) indicates that some cancer has been left behind.

Precautions

There is no physical reason not to do a PSA test. Although, the level of PSA usually is elevated in men with prostate cancer, it also may be abnormally high (though usually not *as* high) in men with non-cancerous enlargement of the prostate (benign prostatic hyperplasia or BPH). If thousands of men have the PSA test routinely each year, many of them will have unnecessary tests (such as biopsy or an ultrasound study) to confirm cancer. If a “false-positive” result is obtained, where the PSA level seems high but really is not, some men may even be treated for prostate cancer when no cancer is present. Both the American Cancer Society and the American Urological Association urge annual PSA testing to detect early cancers, but the National Cancer Institute does not.

Description

The PSA test is a radioimmunoassay. Any antigen causes the body to produce antibodies in an attempt to neutralize or eliminate the antigen, often a substance that harms body tissues. In the laboratory, a sample of the patient’s blood is exposed to the antibody against PSA, so that the amount of antigen (PSA) can be measured. The results generally are available the next day.

KEY TERMS

Antibody—A substance formed in the body in reaction to some foreign material invading the body, or sometimes to diseased body tissue such as prostate cancer. An antibody also may be prepared in the laboratory and used to measure the amount of antigen in the blood.

Antigen—Either a foreign substance such as a virus or bacterium, or a protein produced by diseased or injured body tissue.

Biopsy—A procedure using a hollow needle to obtain a small sample of tissue, such as from the prostate. Often done to determine whether cancer is present.

BPH—Benign prostatic hyperplasia, a noncancerous disorder that causes the prostate to enlarge.

Preparation

No special measures are needed when doing a PSA test other than taking the usual precautions to prevent infection at the needle puncture site.

Normal results

Each laboratory has its own normal range for PSA. In fact, they may redefine the normal range whenever starting to use a new batch of test chemicals.

Abnormal results

Some experts believe that more than 90% of men with prostate cancer will have an elevated PSA level. Others claim that as many as one-third of cancers will be missed. The amount of PSA in the blood drops when cancer is successfully treated, but rises again if the tumor recurs, especially if it spreads to other parts of the body. A new variation of the PSA test shows how much of the material is bound to other protein in the blood and how much is “free.” This procedure may be more accurate and could well indicate whether either prostate cancer or BPH is present.

Resources

ORGANIZATIONS

Prostate Health Council, American Foundation for Urologic Disease. 1128 N. Charles St., Baltimore, MD 21201.

OTHER

Prostate Health Council *Important Information About Prostate-Specific Antigen (PSA)*. American Foundation for Urologic Disease, 1128 N. Charles St., Baltimore, MD 21201.

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Prostate ultrasound

Definition

A prostate ultrasound is a diagnostic test used to detect potential problems with a man's prostate. An ultrasound test uses very high frequency sound waves that are passed through the body. The pattern of reflected sound waves, or “echoes,” shows the outline of the prostate. This test can show whether the prostate is enlarged, and whether an abnormal growth that might be cancer is present.

Purpose

The prostate is a chestnut-shaped organ surrounding the beginning of the urethra in men. It produces a milky fluid that is part of the seminal fluid discharged during ejaculation. The prostate can become enlarged, particularly in men over age 50. Also, **cancer** of the prostate can develop, which tends to affect older men.

During a **physical examination**, a doctor may perform a digital **rectal examination**. In this examination, the doctor uses a gloved and lubricated finger inserted in the rectum to feel for any abnormalities. If this examination shows that the prostate is enlarged or a hard lump is present, an ultrasound may be done. Another reason a doctor might perform an ultrasound is if a blood test shows abnormal levels of a substance called prostate-specific antigen (PSA). Abnormal levels of PSA may indicate the presence of cancer.

If there is a suspicious lump, the doctor will want to take a sample of some of the tissue (prostate biopsy) to test it to see whether it is in fact cancer. Doing an ultrasound first will show the doctor what part of the prostate should be taken as a sample. Ultrasound can also show whether cancerous tissue is still only within the prostate or whether it has begun to spread to other locations. If **prostate cancer** is present and the doctor decides to treat it with a

surgical freezing procedure, ultrasound is used as an aid in the procedure.

An ultrasound can reveal other types of prostate disease as well. For example, it can show if there is inflammation of the prostate (**prostatitis**). Sometimes it is used to learn why a man is unable to father children (**infertility**).

Precautions

A prostate ultrasound study is generally not performed on men who have recently had surgery on their lower bowel. This is because the test requires placing an ultrasound probe about the size of a finger into the rectum.

Description

Prostate ultrasound is generally done using a technique called the transrectal method. This procedure can be done in an outpatient clinic. The cylinder-shaped ultrasound probe is gently placed in the rectum as the patient lies on his left side with the knees bent. The probe is rocked back and forth to obtain images of the entire prostate. The procedure takes about 15–25 minutes to perform. After the test, the patient's doctor can be notified right away, and usually he or she will have a written report within 36 hours.

Preparation

To prepare for a prostate ultrasound, an enema is taken two to four hours before the exam. The patient should not urinate for one hour before the test. If biopsies may be done, the doctor will prescribe an antibiotic that usually is taken in four doses starting the night before the biopsy, the morning of the test, that evening, and the following morning.

Aftercare

There is some discomfort, but less than most patients expect. In fact, worrying ahead of time is usually the hardest part. Generally, the patient is allowed to leave after a radiologist or urologist has reviewed the results. There may be some mucus or a small amount of bleeding from the rectum after the ultrasound. Some patients notice a small amount of blood in the urine for up to two days after the test. Blood may also be present in the semen. As long as the amount of blood is small, there is no cause for concern.

KEY TERMS

Benign prostatic hypertrophy (BPH)—Benign prostatic hypertrophy is an enlargement of the prostate that is not cancerous. However, it may cause problems with urinating or other symptoms.

Prostate-specific antigen (PSA)—A substance that often is produced by cancers of the prostate. It can be detected in a blood test.

Urethra—The tube through which urine passes from the bladder and is excreted to outside the body.

Risks

There are no serious risks from a prostate ultrasound study. Infection is rare and probably is a result of biopsy rather than the sonogram itself. If the ultrasound probe is moved too vigorously, some bleeding may continue for a few days.

Normal results

Modern ultrasound techniques can display both the smooth-surfaced outer shell of the prostate and the core tissues surrounding the urethra. The entire volume of the prostate should be less than 20 milliliters, and its outline should appear as a smooth echo-reflecting (echogenic) rim. Some irregularities within the substance of the gland and calcium deposits are normal findings.

Abnormal results

An **enlarged prostate** with dimmed echoes may indicate either prostatitis or benign enlargement of the gland, called benign prostatic hypertrophy (BPH). A distinct lump of tissue more likely means cancer. Cancer also often appears as an irregular area within the gland that distorts the normal pattern of echoes. In either case, a biopsy should clarify the diagnosis.

Resources

ORGANIZATIONS

Prostate Health Council, American Foundation for Urologic Disease, 1128 N. Charles St., Baltimore, MD 21201. 800-242-AFUD.

David A. Cramer, MD

Prostatectomy

Definition

Prostatectomy refers to the surgical removal of part of the prostate gland (transurethral resection, a procedure performed to relieve urinary symptoms caused by benign enlargement), or all of the prostate (radical prostatectomy, the curative surgery most often used to treat **prostate cancer**).

Purpose

Benign disease

When men reach their mid-40s, the prostate gland begins to enlarge. This condition, benign prostatic hyperplasia (BPH) is present in more than half of men in their 60s and as many as 90% of those over 90. Because the prostate surrounds the urethra, the tube leading urine from the bladder out of the body, the enlarging prostate narrows this passage and makes urination difficult. The bladder does not empty completely each time a man urinates, and, as a result, he must urinate with greater frequency, night and day. In time, the bladder can overfill, and urine escapes from the urethra, resulting in incontinence. An operation called transurethral resection of the prostate (TURP) relieves symptoms of BPH by removing the prostate tissue that is blocking the urethra. No incision is needed. Instead a tube (retroscope) is passed through the penis to the level of the prostate, and tissue is either removed or destroyed, so that urine can freely pass from the body.

Malignant disease

Prostate **cancer** is the single most common form of non-skin cancer in the United States and the most common cancer in men over 50. Half of men over 70 and almost all men over the age of 90 have prostate cancer, and the American Cancer Society estimates that 198,000 new cases will be diagnosed in 2001. This condition does not always require surgery. In fact, many elderly men adopt a policy of “watchful waiting,” especially if their cancer is growing slowly. Younger men often elect to have their prostate gland totally removed along with the cancer it contains—an operation called radical prostatectomy. The two main types of this surgery, radical retropubic prostatectomy and radical perineal prostatectomy, are performed only on patients whose cancer is limited to the prostate. If cancer has broken out of the capsule surrounding the prostate gland and spread in the area or to distant sites, removing the prostate will not prevent

the remaining cancer from growing and spreading throughout the body.

Precautions

Potential complications of TURP include bleeding, infection, and reactions to general or **local anesthesia**. About one man in five will need to have the operation again within 10 years.

Open (incisional) prostatectomy for cancer should not be done if the cancer has spread beyond the prostate, as serious side effects may occur without the benefit of removing all the cancer. If the bladder is retaining urine, it is necessary to insert a catheter before starting surgery. Patients should be in the best possible general condition before radical prostatectomy. Before surgery, the bladder is inspected using an instrument called a cystoscope to help determine the best surgical technique to use, and to rule out other local problems.

Description

TURP

This procedure does not require an abdominal incision. With the patient under either general or spinal anesthesia, a cutting instrument or heated wire loop is inserted to remove as much prostate tissue as possible and seal blood vessels. The excised tissue is washed into the bladder, then flushed out at the end of the operation. A catheter is left in the bladder for one to five days to drain urine and blood. Advanced laser technology enables surgeons to safely and effectively burn off excess prostate tissue blocking the bladder opening with fewer of the early and late complications associated with other forms of prostate surgery. This procedure can be performed on an outpatient basis, but urinary symptoms do not improve until swelling subsides several weeks after surgery.

Radical prostatectomy

RADICAL RETROPUBIC PROSTATECTOMY. This is a useful approach if the prostate is very large, or cancer is suspected. With the patient under general or spinal anesthesia or an epidural, a horizontal incision is made in the center of the lower abdomen. Some surgeons begin the operation by removing pelvic lymph nodes to determine whether cancer has invaded them, but recent findings suggest there is no need to sample them in patients whose likelihood of lymph node metastases is less than 18%. A doctor who removes the lymph nodes for examination will not continue the operation if they contain cancer cells, because the surgery will

not cure the patient. Other surgeons remove the prostate gland before examining the lymph nodes. A tube (catheter) inserted into the penis to drain fluid from the body is left in place for 14–21 days.

Originally, this operation also removed a thin rim of bladder tissue in the area of the urethral sphincter—a muscular structure that keeps urine from escaping from the bladder. In addition, the nerves supplying the penis often were damaged, and many men found themselves impotent (unable to achieve erections) after prostatectomy. A newer surgical method called potency-sparing radical prostatectomy preserves sexual potency in 75% of patients and fewer than 5% become incontinent following this procedure.

RADICAL PERINEAL PROSTATECTOMY. This procedure is just as curative as radical retropubic prostatectomy but is performed less often because it does not allow the surgeon to spare the nerves associated with erection or, because the incision is made above the rectum and below the scrotum, to remove lymph nodes. Radical perineal prostatectomy is sometimes used when the cancer is limited to the prostate and there is no need to spare nerves or when the patient's health might be compromised by the longer procedure. The perineal operation is less invasive than retropubic prostatectomy. Some parts of the prostate can be seen better, and blood loss is limited. The absence of an abdominal incision allows patients to recover more rapidly. Many urologic surgeons have not been trained to perform this procedure. Radical prostatectomy procedures last one to four hours, with radical perineal prostatectomy taking less time than radical retropubic prostatectomy. The patient remains in the hospital three to five days following surgery and can return to work in three to five weeks. Ongoing research indicates that laparoscopic radical prostatectomy may be as effective as open surgery in treatment of early-stage disease.

Cryosurgery

Also called **cryotherapy** or cryoablation, this minimally invasive procedure uses very low temperatures to freeze and destroy cancer cells in and around the prostate gland. A catheter circulates warm fluid through the urethra to protect it from the cold. When used in connection with ultrasound imaging, cryosurgery permits very precise tissue destruction. Traditionally used only in patients whose cancer had not responded to radiation, but now approved by Medicare as a primary treatment for prostate cancer, cryosurgery can safely be performed on older men, on patients who are not in good enough general health to undergo radical prostatectomy, or to treat recurrent

disease. Recent studies have shown that total cryosurgery, which destroys the prostate, is at least as effective as radical prostatectomy without the trauma of major surgery.

Preparation

As with any type of major surgery done under **general anesthesia**, the patient should be in optimal condition. Most patients having prostatectomy are in the age range when cardiovascular problems are frequent, making it especially important to be sure that the heart is beating strongly, and that the patient is not retaining too much fluid. Because long-standing prostate disease may cause kidney problems from urine “backing up,” it also is necessary to be sure that the kidneys are working properly. If not, a period of catheter drainage may be necessary before doing the surgery.

Aftercare

Following TURP, a catheter is placed in the bladder to drain urine and remains in place for two to three days. A solution is used to irrigate the bladder and urethra until the urine is clear of blood, usually within 48 hours after surgery. Whether **antibiotics** should be routinely given remains an open question. Catheter drainage also is used after open prostatectomy. The bladder is irrigated only if **blood clots** block the flow of urine through the catheter. Patients are given intravenous fluids for the first 24 hours, to ensure good urine flow. Patients resting in bed for long periods are prone to blood clots in their legs (which can pass to the lungs and cause serious breathing problems). This can be prevented by elastic stockings and by periodically exercising the patient's legs. The patient remains in the hospital one to two days following surgery and can return to work in one to two weeks.

Risks

The complications and side effects that may occur during and after prostatectomy include:

- Excessive bleeding, which in rare cases may require blood transfusion.
- Incontinence when, during retropubic prostatectomy, the muscular valve (sphincter) that keeps urine in the bladder is damaged. Less common today, when care is taken not to injure the sphincter.
- Impotence, occurring when nerves to the penis are injured during the retropubic operation. Today's

KEY TERMS

BPH—Benign prostatic hypertrophy, a very common noncancerous cause of prostatic enlargement in older men.

Catheter—A tube that is placed through the urethra into the bladder in order to provide free drainage of urine and blood following either TURP or open prostatectomy.

Cryosurgery—In prostatectomy, the use of a very low-temperature probe to freeze and thereby destroy prostatic tissue.

Impotence—The inability to achieve and sustain penile erections.

Incontinence—The inability to retain urine in the bladder until a person is ready to urinate voluntarily.

Prostate gland—The gland surrounding the male urethra just below the base of the bladder. It secretes a fluid that constitutes a major portion of the semen.

Urethra—The tube running from the bladder to the tip of the penis that provides a passage for eliminating urine from the body.

“nerve-sparing” technique has drastically cut down on this problem.

- Some patients who receive a large volume of irrigating fluid after TURP develop high blood pressure, **vomiting**, trouble with their vision, and mental confusion. This condition is caused by a low salt level in the blood, and is reversed by giving salt solution.
- A permanent narrowing of the urethra called a stricture occasionally develops when the urethra is damaged during TURP.
- There is about a 34% chance that the cancer will recur within 10 years of the procedure. In addition, about 25% of patients experience what is known as biochemical recurrence, which means that the level of prostate-specific antigen (PSA) in the patient’s blood serum begins to rise rapidly. Recurrence of the tumor or biochemical recurrence can be treated with **radiation therapy** or androgen deprivation therapy.

Normal results

In patients with BPH who have the TURP operation, urination should become much easier and less frequent, and dribbling or incontinence should cease.

In patients having radical prostatectomy for cancer, a successful operation will remove the tumor and prevent its spread to other areas of the body (metastasis). If examination of lymph nodes shows that cancer already had spread beyond the prostate at the time of surgery, other measures are available to control the tumor.

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ORGANIZATIONS

- Cancer Research Institute. 681 Fifth Ave., New York, NY 10022. (800) 99CANCER. <<http://www.cancerresearch.org>>.
- National Prostate Cancer Coalition. 1156 15th St., NW, Washington, DC 20005. (202) 463-9455. <<http://www.4npcc.org>>.
- Prostate Health Council. American Foundation for Urologic Disease. 1128 N. Charles St., Baltimore, MD 21201-5559. (800) 828-7866. <<http://www.afud.org>>.

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Prostatic acid phosphatase test see **Acid phosphatase test**

Prostatitis

Definition

Prostatitis is an inflammation of the prostate gland, a common condition in adult males. Often caused by infection, prostatitis may develop rapidly (*acute*) or slowly (*chronic*).

Description

Prostatitis may be the symptom-producing disease of the genitourinary tract for which men most often seek medical help. About 40% of visits to a specialist in genitourinary problems (urologist) are for prostatitis. Forms of prostate inflammation include acute and chronic bacterial prostatitis and inflammation not caused by bacterial infection. A painful condition called *prostatodynia*, which may be caused by abnormal nerves or muscles in the region, is also thought to be a form of prostatitis. The chronic bacterial form is sometimes experienced by men whose sex partners have a bacterial infection of the vagina, making this a sexually transmitted disease. Other cases occur when small stones form within the prostate and become infected. Sometimes infection is caused by poor hygiene, surgical procedures, or even swimming in polluted water.

The sexually transmitted disease **gonorrhea** may sometimes cause prostatitis, and **tuberculosis** may spread to the prostate. Parasites and fungi may infect the prostate gland. Some men whose prostatitis is not caused by any microorganism have microscopic collections of cells called *granulomas* in their prostate tissue. Whether viruses also may cause prostatitis is debatable.

Causes and symptoms

However the inflammation may begin, it causes blockages in the tiny glands within the prostate so that secretions build up, and the prostate swells. In acute cases, this swelling can occur very suddenly and cause considerable **pain**. When prostatitis develops gradually, trouble with the flow of urine may be the first symptom. Small stones may form, because the body attempts to neutralize bacteria by coating them with calcium. These stones may become infected themselves and make the condition worse.

Symptoms and signs that are typically experienced by men with prostatitis include:

- Difficulties in urinating. Most urinary problems are caused when the swollen prostate blocks the tube

that carries urine from the bladder to the outside of the body (urethra). Patients feel the need to urinate more often than usual, often urgently. Urination is sometimes painful. It is hard to start the flow of urine and difficult to totally empty the bladder. Patients wake up at night to urinate. The stream may be weak or split. Dribbling after attempts to urinate may leave embarrassing wet spots on clothing. In severe prostatitis blood or sand-like particles (small calcium collections) may be passed in the urine.

- Pain. Besides pain when urinating, caused by prostate swelling, stimulation of nerves in the prostate gland may cause pain in the penis, one or both testicles, the lower stomach, the low back, and the area between the scrotum and the anus (perineum). Some patients experience pain during or after ejaculation, whenever they sit down or walk, or during bowel movements.
- Sex and fertility. The pain of prostatitis can make it impossible to enjoy sex. Men with prostatitis may be troubled by early release of sperm (premature ejaculation). Occasionally there is blood in the semen. Some of the drugs prescribed to ease the flow of the urine can dampen the desire to have sex. Because the normal prostate secretions make up part of the semen, prostatitis may lower fertility by severely lowering the number of sperm and making them less mobile.
- Psychological problems. A man with prostatitis who feels that nothing can be done and he “just has to live with it” may experience serious depression. Low sexual desire certainly contributes to depression.

A person with *acute prostatitis* may suddenly develop fever and chills, along with rapidly developing urinary symptoms and pain in the perineum or low back. This state is a medical emergency that demands immediate medical help.

Diagnosis

Most often the symptoms and physical findings are enough to form a diagnosis of prostatitis. When the examiner inserts a finger in the rectum, the swollen prostate can be felt; it may be extremely tender when probed. Squeezing the gland slightly will produce a few drops of fluid that may be *cultured* to learn whether bacteria are present. The fluid typically contains a large number of white blood cells, especially the cells used to fight off infection (*macrophages*). Note: too much pressure on the prostate can force bacteria into the blood and cause a serious general infection. Many patients with chronic bacterial prostatitis also have recurring urinary tract infections (diagnosed by examining and culturing urine samples). These infections can be an

important clue to the diagnosis. If doubt remains, the urologist may insert a special instrument called a *cystoscope* through the penis to directly view the prostate from inside and see whether it looks inflamed.

Treatment

Acute prostatitis is first treated with **antibiotics**. Even though it may be difficult for drugs to actually get into the inflamed prostate, most patients do quickly get better. If intravenous antibiotics are needed or the bladder is retaining urine, a hospital stay may be necessary. Broad-spectrum antibiotics that work against most bacteria are used first. At the same time tests are done with samples of prostatic fluid to determine which bacterium is causing the infection, so that drugs can be prescribed to fight the specific germ. In chronic cases, the best results are obtained with a combination of the antibiotics trimethoprim and sulfamethoxazole. Oral antibiotics should be given for one to three months; longer, if necessary. If a fungus or some other organism is causing infection, special drugs are available. If chronic prostatitis continues despite all medical efforts and is seriously affecting the patient's life, the prostate may be removed surgically.

Nonbacterial prostatitis requires other measures to relieve urinary symptoms. These measures include drugs that fight inflammation (steroids or nonsteroids) and a type of drug called an alpha-blocker that reduces muscle tension. Reduced muscle tension eases urine flow, allowing the bladder to empty. A narrowed urethra may be widened by placing a collapsed balloon at the site of obstruction and expanding it. This procedure is called *balloon dilation*. The effects of such dilation are usually temporary. Some physicians believe that **stress** is an important factor in prostatitis, and therefore prescribe diazepam (Valium) or another tranquilizer. The type of prostatitis known as prostatodynia is usually treated with a combination of muscle relaxing drugs, heat, special exercises, and sometimes a tranquilizer.

There are a number of "tips" for relieving symptoms of prostatitis. They are especially helpful early on, before antibiotics have a chance to cure infection, or for patients with chronic or non-bacterial prostatitis:

- **Hot sitz baths.** Exposing the perineum to very hot water for 20 minutes or longer often relieves pain.
- **Ice.** When heat does not help, ice packs, or simply placing a small ice cube in the rectum, may relieve pain for hours.
- **Water.** A patient who has to urinate very often may want to cut back on his fluid intake but this will cause

dehydration and increase the risk of bladder infection. Instead, it is best to drink plenty of water.

- **Diet.** Most doctors recommend cutting out—or cutting down on—caffeine (as in coffee or tea), alcohol, and spicy or acid foods. **Constipation** should be avoided because large, hard bowel movements may press on the swollen prostate and cause great pain. Bran cereals and whole-grain breads are helpful.
- **Exercise.** It is especially important for patients with chronic prostatitis to keep up their activity level. Simply walking often will help (unless walking happens to make the pain worse).
- **Frequent ejaculation.** Ejaculating two or three times a week often is recommended, especially when taking antibiotics.

Alternative treatment

A treatment popularized in the Philippines is called "prostate drainage." At regular intervals, a finger is inserted into the rectum, to exert pressure on the prostate at the same time that an antibiotic treatment is given. **Acupuncture** and Chinese herbal medicine also can be effective in treating prostatitis. **Nutritional supplements** that support the prostate, including zinc, **omega-3 fatty acids**, several amino acids, and anti-inflammatory nutrients and herbs, can help reduce pain and promote healing. Western herbal medicine recommends saw palmetto (*Serenoa repens*) to support the prostate gland. Hot and cold contrast sitz baths can help reduce inflammation.

Prognosis

Most patients with acute bacterial prostatitis are cured if they receive proper antibiotic treatment. Every effort should be made to get a cure at the acute stage because chronic prostatitis can be much more difficult to eliminate. If the acute illness is *not* controlled, complications such as a localized infection (prostatic **abscess**), kidney infection, or infection of the blood (septicemia) may develop. When chronic prostatitis cannot be cured, it still is possible to keep urinary symptoms under control and keep the patient active by using low doses of antibiotics and other measures. If a man with any form of prostatitis develops serious psychological problems, he should be referred to a psychiatric specialist.

Prevention

Potential sources of infection should be avoided. Good perineal hygiene should be maintained and sex

KEY TERMS

Culture—A test in which a sample of body fluid, such as prostatic fluid, is placed on materials specially formulated to grow microorganisms. A culture is used to learn what type of bacterium is causing infection.

Cystoscope—A viewing instrument that is passed up the urethra into the region of the prostate to get a good look at the organ “from the inside.”

Ejaculation—The process by which semen (made up in part of prostatic fluid) is ejected by the erect penis.

Granuloma—A cluster of cells that form in tissue which has been inflamed for some time.

Perineum—An area close to the prostate, between the scrotum and anus.

should be avoided when one's partner has an active bacterial vaginal infection. If the kidneys, bladder, or other genitourinary organs are infected, prompt treatment may prevent the development of prostatitis. By far the best way of preventing chronic prostatitis is to treat an initial *acute* episode promptly and effectively.

Resources

ORGANIZATIONS

Prostate Health Council, American Foundation for Urologic Disease. 1128 N. Charles St., Baltimore, MD 21201. (800) 242-AFUD.

Prostatitis Foundation, Information Distribution Center. 2029 Ireland Grove Park, Bloomington, IL 61704. (309) 664-6222. < <http://www.prostate.org> >.

David A. Cramer, MD

Prosthetic joint infection see **Infectious arthritis**

Protease inhibitors

Definition

A protease inhibitor is a type of drug that cripples the enzyme protease. An enzyme is a substance that triggers chemical reactions in the body. The human **immunodeficiency** virus (HIV) uses protease in the final stages of its reproduction (replication) process.

Purpose

The drug is used to treat selected patients with HIV infection. Blocking protease interferes with HIV reproduction, causing it to make copies of itself that cannot infect new cells. The drug may improve symptoms and suppress the infection but does not cure it.

Precautions

Patients should not discontinue this drug even if symptoms improve without consulting a doctor.

These drugs do not necessarily reduce the risk of transmitting HIV to others through sexual contact, so patients should avoid sexual activities or use condoms.

Description

Protease inhibitors are considered one of the most potent medications for HIV developed so far.

This class of drugs includes indinavir (Crixivan), ritonavir (Norvir), nelfinavir (Viracept), amprenavir (Agenerase), lopinavir plus ritonavir (Kaletra), saquinavir (Fortovase), and a new drug called atazanavir (Reyataz). Reyataz received approval from the U.S. Food and Drug Administration (FDA) in mid-2003 and was the first protease inhibitor approved for once-daily dosing. Several weeks or months of drug therapy may be required before the full benefits are apparent.

The drug should be taken at the same time each day. Some types should be taken with a meal to help the body absorb them. Each of the types of protease inhibitor may have to be taken in a different way. In most cases, protease inhibitors are part of a combination therapy, used in conjunction with other classes of HIV drugs.

Risks

Common side effects include **diarrhea**, stomach discomfort, nausea, and mouth sores. Less often, patients may experience rash, muscle **pain**, **headache**, or weakness. Rarely, there may be confusion, severe skin reaction, or seizures. Some of these drugs can have interactions with other medication, and indinavir can be associated with kidney stones. Diabetes or high blood pressure may become worse when these drugs are taken. Reyataz has been shown to have fewer side effects than some protease inhibitors, though it can interact with other medications, including certain heart medications and antidepressants.

Experts do not know whether the drugs pass into breast milk, so breastfeeding mothers should avoid them or should stop nursing until the treatment is completed.

KEY TERMS

Human immunodeficiency virus (HIV)—The virus that causes AIDS.

Resources

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ORGANIZATIONS

National AIDS Treatment Advocacy Project. 580 Broadway, Ste. 403, New York, NY 10012. (888) 266-2827. <<http://www.natap.org>>.

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Protein-calorie malnutrition see **Protein-energy malnutrition**

Protein components test

Definition

Protein components tests measure the amounts and types of protein in the blood. Proteins are constituents of muscle, enzymes, hormones, transport proteins, hemoglobin, and other functional and structural elements of the body. Albumin and globulin make up most of the protein within the body and are measured in the total protein of the blood and other body fluids. Thus, the serum (blood) protein components test measures the total protein, as well as its albumin and globulin components in the blood.

Purpose

The protein components test is used to diagnose diseases that either affect proteins as a whole, or that involve a single type of protein. The test is also used to monitor the course of disease in certain cancers, intestinal and kidney protein-wasting states, immune disorders, liver dysfunction, and impaired **nutrition**.

Precautions

Drugs that may cause increased protein levels include the anabolic steroids, androgens (male hormones), growth hormone, insulin, and progesterone. Drugs that may decrease protein levels include estrogen, drugs poisonous to the liver, and oral contraceptives.

Description

Proteins are large molecules (complex organic compounds) that consist of amino acids, sugars, and lipids. There are two main types of proteins: those that are made of fiber and form the structural basis of body tissues, such as hair, skin, muscle, tendons, and cartilage; and globular proteins (generally water soluble), which interact with many hormones, various other proteins in the blood, including hemoglobin and antibodies, and all the enzymes (substances that promote biochemical reactions in the body).

Proteins are needed in the diet to supply the body with amino acids. Ingested proteins are broken down in the digestive system to amino acids, which are then absorbed and rebuilt into new body proteins. One of the most important functions of proteins in the body is to contribute to the osmotic pressure (the movement of water between the bloodstream and tissues). An example of this is seen in diseases that result in damage to the filtering units of the kidneys (**nephrotic syndrome**). A severe loss of protein from the bloodstream into the urine (proteinuria) results, lowering the protein content of the blood and resulting in fluid retention, or **edema**.

Albumin and globulin are two key components of protein. Albumin is made in the liver and constitutes approximately 60% of the total protein. The main function of albumin is to maintain osmotic pressure and to help transport certain blood constituents around the body via the bloodstream. Because albumin is made in the liver, it is one element that is used to monitor liver function.

Globulin is the basis for antibodies, glycoproteins (protein-carbohydrate compounds), lipoproteins (proteins involved in fat transport), and clotting factors. Globulins are divided into three main groups, the alpha-, beta-, and gammaglobulins. Alphaglobulins include enzymes produced by the lungs and liver, and haptoglobin, which binds hemoglobin together. The betaglobulins consist mostly of low-density lipoproteins (LDLs), substances involved in fat transport. All of the gammaglobulins are antibodies, proteins produced by the immune system in response to infection, during allergic reaction, and after organ transplants.

Both serum albumin and globulin are measures of nutrition. Malnourished patients, especially after surgery, demonstrate greatly decreased protein levels, while burn patients and those who have protein-losing syndromes show low levels despite normal synthesis. **Pregnancy** in the third trimester is also associated with reduced protein levels.

The relationship of albumin to globulin is determined by ratio, so when certain diseases cause the albumin levels to drop, the globulin level will be increased by the body in an effort to maintain a normal total protein level. For example, when the liver is unable to synthesize sufficient albumin in chronic **liver disease**, the albumin level will be low, but the globulin levels will be normal or higher than normal. In such cases, the protein components test is an especially valuable diagnostic aid because it determines the ratio of albumin to globulin, as well as the total protein level. It should be noted, however, that when globulin is provided as a calculation (total protein – albumin = globulin), the result is much less definitive than other methods of determining globulin.

Consequently, when the albumin/globulin ratio (A/G ratio) is less than 1.0, more precise tests should be ordered. These tests include protein electrophoresis, a method of separating the different blood proteins into groups. If the protein electrophoresis indicates a rise, or “spike” at the globulin level, an even more specific test for globulins, called **immuno-electrophoresis**, should be ordered to separate out the various globulins according to type. Some diseases characterized by dysproteinemia (derangement of the protein content of the blood), have typical electrophoretic globulin peaks.

Preparation

Unless this is requested by the physician, there is no need that the patient restrict food or fluids before the test.

Risks

Risks posed by this test are minimal, but may include slight bleeding from the blood-drawing site, **fainting** or lightheadedness after venipuncture, or hematoma (blood accumulating under the puncture site).

Normal results

Reference values vary from laboratory to laboratory, but can generally be found within the following ranges: Total protein: 6.4–8.3 g/dL; albumin: 3.5–5.0 g/dL; globulin: 2.3–3.4 g/dL.

KEY TERMS

Nephrotic syndromes—A collection of symptoms that result from damage to the filtering units of the kidney (glomeruli) causing severe loss of protein from the blood into the urine.

Abnormal results

Increased total protein levels are seen in **dehydration**, in some cases of chronic liver disease (like **auto-immune hepatitis** and **cirrhosis**), and in certain tropical diseases (for example, **leprosy**). Very low total protein levels (less than 4.0 g/dl) and low albumin cause the edema (water retention) usually seen in nephrotic syndromes. Decreased protein levels may be seen in pregnancy, chronic **alcoholism**, prolonged immobilization, heart failure, starvation, and malabsorption or **malnutrition**.

Increased albumin levels are found in dehydration. Decreased albumin levels are indicative of liver disease, protein-losing syndromes, malnutrition, inflammatory disease, and familial idiopathic (of unknown cause) dysproteinemia, a genetic disease in which the albumin is significantly reduced and globulins increased.

Increased globulin levels are found in multiple myeloma and Waldenström’s macroglobulinemia, two cancers characterized by overproduction of gammaglobulin from proliferating plasma cells. Increased globulin levels are also found in chronic inflammatory diseases such as **rheumatoid arthritis**, acute and chronic infection, and cirrhosis. Decreased globulin levels are seen in genetic immune disorders and secondary immune deficiency.

Resources

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Janis O. Flores

Protein electrophoresis

Definition

Electrophoresis is a technique used to separate different elements (fractions) of a blood sample into

individual components. Serum protein electrophoresis (SPEP) is a screening test that measures the major blood proteins by separating them into five distinct fractions: albumin, alpha₁, alpha₂, beta, and gamma proteins. Protein electrophoresis can also be performed on urine.

Purpose

Protein electrophoresis is used to evaluate, diagnose, and monitor a variety of diseases and conditions. It can be used for these purposes because the levels of different blood proteins rise or fall in response to such disorders as **cancer**, intestinal or kidney protein-wasting syndromes, disorders of the immune system, liver dysfunction, impaired nutrition, and chronic fluid-retaining conditions.

Precautions

Certain other diagnostic tests or prescription medications can affect the results of SPEP tests. The administration of a contrast dye used in some other tests may falsely elevate protein levels. Drugs that can alter results include **aspirin**, bicarbonates, chlorpromazine (Thorazine), **corticosteroids**, isoniazid (INH), and neomycin (Mycifradin).

Description

Proteins are major components of muscle, enzymes, hormones, hemoglobin, and other body tissues. Proteins are composed of elements that can be separated from one another by several different techniques: chemical methods, ultracentrifuge, or electrophoresis. There are two major types of electrophoresis: protein electrophoresis and **immunoelectrophoresis**. Immunoelectrophoresis is used to assess the blood levels of specific types of proteins called immunoglobulins. An immunoelectrophoresis test is usually ordered if a SPEP test has a “spike,” or rise, at the immunoglobulin level. Protein electrophoresis is used to determine the total amount of protein in the blood, and to establish the levels of other types of proteins called albumin, alpha₁ globulin, alpha₂ globulin, and beta-globulin.

Blood proteins

ALBUMIN. Albumin is a protein that is made in the liver. It helps to retain elements like calcium, some hormones, and certain drugs in the circulation by binding to them to prevent their being filtered out by the kidneys. Albumin also acts to regulate the movement of water between the tissues and the bloodstream

by attracting water to areas with higher concentrations of salts or proteins.

GLOBULINS. Globulins are another type of protein, larger in size than albumin. They are divided into three main groups: alpha, beta, and gamma.

- **Alphaglobulins.** These proteins include alpha₁ and alpha₂ globulins. Alpha₁ globulin is predominantly alpha₁ antitrypsin, an enzyme produced by the lungs and liver. Alpha₂ globulin, which includes serum haptoglobin, is a protein that binds hemoglobin to prevent its excretion by the kidneys. Various other alphaglobulins are produced as a result of inflammation, tissue damage, autoimmune disorders, or certain cancers.
- **Betaglobulins.** These include low-density substances involved in fat transport (lipoproteins), iron transport (transferrin), and blood clotting (plasminogen and complement).
- **Gammaglobulins.** All of the gammaglobulins are antibodies—proteins produced by the immune system in response to infection, allergic reactions, and organ transplants. If serum protein electrophoresis has demonstrated a significant rise at the **gammaglobulin** level, immunoelectrophoresis is done to identify the specific globulin that is involved.

Electrophoretic measurement of proteins

All proteins have an electrical charge. The SPEP test is designed to make use of this characteristic. There is some difference in method, but basically the sample is placed in or on a special medium (e.g., a gel), and an electric current is applied to the gel. The protein particles move through the gel according to the strength of their electrical charges, forming bands or zones. An instrument called a densitometer measures these bands, which can be identified and associated with specific diseases. For example, a decrease in albumin with a rise in the alpha₂ globulin usually indicates an acute reaction of the type that occurs in infections, **burns**, **stress**, or heart attack. On the other hand, a slight decrease in albumin, with a slight increase in gammaglobulin, and a normal alpha₂ globulin is more indicative of a chronic inflammatory condition, as might be seen in **cirrhosis** of the liver.

Protein electrophoresis is performed on urine samples to classify kidney disorders that cause protein loss. Here also certain band patterns are specific for disease. For example, the identification of a specific protein called the Bence Jones protein (by performing the **Bence Jones protein test**) during the procedure suggests **multiple myeloma**.

Preparation

The serum protein electrophoresis test requires a blood sample. It is not necessary for the patient to restrict food or fluids before the test. The urine protein electrophoresis test requires either an early morning urine sample or a 24-hour urine sample according to the physician's request. The doctor should check to see if the patient is taking any medications that may affect test results.

Risks

Risks posed by the blood test are minimal but may include slight bleeding from the puncture site, **fainting** or lightheadedness after the blood is drawn, or the development of a small bruise at the puncture site.

Normal results

The following values are representative, although there is some variation among laboratories and specific methods. These values are based on the agarose system.

- total protein: 5.9–8.0 g/dL
- albumin: 4.0–5.5 g/dL
- alpha₁ globulin: 0.15–0.25 g/dL
- alpha₂ globulin: 0.43–0.75 g/dL
- beta-globulin: 0.5–1.0 g/dL
- gammaglobulin: 0.6–1.3 g/dL

Abnormal results

Albumin levels are increased in **dehydration**. They are decreased in **malnutrition**, **pregnancy**, **liver disease**, inflammatory diseases, and such protein-losing syndromes as **malabsorption syndrome** and certain kidney disorders.

Alpha₁ globulins are increased in inflammatory diseases. They are decreased or absent in juvenile pulmonary **emphysema**, which is a genetic disease.

Alpha₂ globulins are increased in a kidney disorder called nephrotic syndrome. They are decreased in patients with an overactive thyroid gland (**hyperthyroidism**) or severe liver dysfunction.

Betaglobulin levels are increased in conditions of high cholesterol levels (**hypercholesterolemia**) and iron deficiency anemia. They are decreased in malnutrition.

Gammaglobulin levels are increased in chronic inflammatory disease (for example, rheumatoid

KEY TERMS

Albumin—A blood protein that is made in the liver and helps to regulate water movement in the body.

Electrophoresis—A technique for separating various blood fractions by running an electric current through a gel containing a blood sample.

Globulins—A group of proteins in blood plasma whose levels can be measured by electrophoresis in order to diagnose or monitor a variety of serious illnesses.

Haptoglobin—A protein in blood plasma that binds hemoglobin.

Immunoglobulins—Any of several types of globulin proteins that function as antibodies.

arthritis, systemic lupus erythematosus); cirrhosis; acute and chronic infection; and a cancerous disease characterized by uncontrolled multiplication of plasma cells in the bone marrow (multiple myeloma). Gammaglobulins are decreased in a variety of genetic immune disorders, and in secondary immune deficiency related to steroid use, leukemia, or severe infection.

Resources

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Pagana, Kathleen Deska. *Mosby's Manual of Diagnostic and Laboratory Tests*. St. Louis: Mosby, Inc., 1998.

Janis O. Flores

Protein-energy malnutrition

Definition

Protein-energy **malnutrition** (PEM) is a potentially fatal body-depletion disorder. It is the leading cause of **death** in children in developing countries.

Description

PEM is also referred to as protein-calorie malnutrition. It develops in children and adults whose consumption of protein and energy (measured by calories) is insufficient to satisfy the body's nutritional needs. While pure protein deficiency can occur when a person's diet provides enough energy but lacks the protein

minimum, in most cases the deficiency will be dual. PEM may also occur in persons who are unable to absorb vital nutrients or convert them to energy essential for healthy tissue formation and organ function.

Although PEM is not prevalent among the general population of the United States, it is often seen in elderly people who live in nursing homes and in children whose parents are poor. PEM occurs in one of every two surgical patients and in 48% of all other hospital patients.

Types of PEM

Primary PEM results from a diet that lacks sufficient sources of protein and/or energy. Secondary PEM is more common in the United States, where it usually occurs as a complication of AIDS, **cancer**, **chronic kidney failure**, inflammatory bowel disease, and other illnesses that impair the body's ability to absorb or use nutrients or to compensate for nutrient losses. PEM can develop gradually in a patient who has a chronic illness or experiences chronic semi-starvation. It may appear suddenly in a patient who has an acute illness.

Kwashiorkor

Kwashiorkor, also called wet protein-energy malnutrition, is a form of PEM characterized primarily by protein deficiency. This condition usually appears at the age of about 12 months when breastfeeding is discontinued, but it can develop at any time during a child's formative years. It causes fluid retention (**edema**); dry, peeling skin; and hair discoloration.

Marasmus

Primarily caused by energy deficiency, marasmus is characterized by stunted growth and wasting of muscle and tissue. Marasmus usually develops between the ages of six months and one year in children who have been weaned from breast milk or who suffer from weakening conditions like chronic **diarrhea**.

Causes and symptoms

Secondary PEM symptoms range from mild to severe, and can alter the form or function of almost every organ in the body. The type and intensity of symptoms depend on the patient's prior nutritional status and on the nature of the underlying disease and the speed at which it is progressing.

Mild, moderate, and severe classifications have not been precisely defined, but patients who lose 10–20% of their body weight without trying are usually said to have moderate PEM. This condition is also

characterized by a weakened grip and inability to perform high-energy tasks.

Losing 20% of body weight or more is generally classified as severe PEM. People with this condition can't eat normal-sized meals. They have slow heart rates and low blood pressure and body temperatures. Other symptoms of severe secondary PEM include baggy, wrinkled skin; constipation; dry, thin, brittle hair; lethargy; pressure sores and other **skin lesions**.

Kwashiorkor

People who have kwashiorkor often have extremely thin arms and legs, but liver enlargement and **ascites** (abnormal accumulation of fluid) can distend the abdomen and disguise weight loss. Hair may turn red or yellow. Anemia, diarrhea, and fluid and **electrolyte disorders** are common. The body's immune system is often weakened, behavioral development is slow, and mental retardation may occur. Children may grow to normal height but are abnormally thin.

Kwashiorkor-like secondary PEM usually develops in patients who have been severely burned, suffered trauma, or had **sepsis** (tissue-destroying infection) or another life-threatening illness. The condition's onset is so sudden that body fat and muscle mass of normal-weight people may not change. Some obese patients even gain weight.

Marasmus

Profound weakness accompanies severe marasmus. Since the body breaks down its own tissue to use as calories, people with this condition lose all their body fat and muscle strength, and acquire a skeletal appearance most noticeable in the hands and in the temporal muscle in front of and above each ear. Children with marasmus are small for their age. Since their immune systems are weakened, they suffer from frequent infections. Other symptoms include loss of appetite, diarrhea, skin that is dry and baggy, sparse hair that is dull brown or reddish yellow, mental retardation, behavioral retardation, low body temperature (**hypothermia**), and slow pulse and breathing rates.

The absence of edema distinguishes marasmus-like secondary PEM, a gradual wasting process that begins with weight loss and progresses to mild, moderate, or severe malnutrition (cachexia). It is usually associated with cancer, chronic obstructive pulmonary disease (COPD), or another chronic disease that is inactive or progressing very slowly.

Some individuals have both kwashiorkor and marasmus at the same time. This most often occurs

when a person who has a chronic, inactive condition develops symptoms of an acute illness.

Hospitalized patients

Difficulty chewing, swallowing, and digesting food, pain, nausea, and lack of appetite are among the most common reasons that many hospital patients don't consume enough nutrients. Nutrient loss can be accelerated by bleeding, diarrhea, abnormally high sugar levels (glycosuria), **kidney disease**, malabsorption disorders, and other factors. **Fever**, infection, surgery, and benign or malignant tumors increase the amount of nutrients hospitalized patients need. So do trauma, **burns**, and some medications.

Diagnosis

A thorough **physical examination** and a health history that probes eating habits and weight changes, focuses on body-fat composition and muscle strength, and assesses gastrointestinal symptoms, underlying illness, and nutritional status is often as accurate as blood tests and urinalyses used to detect and document abnormalities.

Some doctors further quantify a patient's nutritional status by:

- comparing height and weight to standardized norms
- calculating body mass index (BMI)
- measuring skinfold thickness or the circumference of the upper arm

Treatment

Treatment is designed to provide adequate **nutrition**, restore normal body composition, and cure the condition that caused the deficiency. Tube feeding or intravenous feeding is used to supply nutrients to patients who can't or won't eat protein-rich foods.

In patients with severe PEM, the first stage of treatment consists of correcting fluid and electrolyte imbalances, treating infection with **antibiotics** that don't affect protein synthesis, and addressing related medical problems. The second phase involves replenishing essential nutrients slowly to prevent taxing the patient's weakened system with more food than it can handle. Physical therapy may be beneficial to patients whose muscles have deteriorated significantly.

Prognosis

Most people can lose up to 10% of their body weight without side effects, but losing more than

40% is almost always fatal. Death usually results from heart failure, an electrolyte imbalance, or low body temperature. Patients with certain symptoms, including semiconsciousness, persistent diarrhea, **jaundice**, and low blood sodium levels, have a poorer prognosis than other patients. Recovery from marasmus usually takes longer than recovery from kwashiorkor. The long-term effects of childhood malnutrition are uncertain. Some children recover completely, while others may have a variety of lifelong impairments, including an inability to properly absorb nutrients in the intestines and **mental retardation**. The outcome appears to be related to the length and severity of the malnutrition, as well as to the age of the child when the malnutrition occurred.

Prevention

Breastfeeding a baby for at least six months is considered the best way to prevent early-childhood malnutrition. Preventing malnutrition in developing countries is a complicated and challenging problem. Providing food directly during famine can help in the short-term, but more long-term solutions are needed, including agricultural development, public health programs (especially programs that monitor growth and development, as well as programs that provide nutritional information and supplements), and improved food distribution systems. Programs that distribute infant formula and discourage breastfeeding should be discontinued, except in areas where many mothers are infected with HIV.

Every patient being admitted to a hospital should be screened for the presence of illnesses and conditions that could lead to PEM. The nutritional status of patients at higher-than-average risk should be more thoroughly assessed and periodically reevaluated during extended hospital stays or nursing home residence.

Resources

ORGANIZATIONS

American College of Nutrition. 722 Robert E. Lee Drive, Wilmington, NC 20412-0927. (919) 152-1222.

American Institute of Nutrition. 9650 Rockville Pike, Bethesda, MD 20814-3990. (301) 530-7050.

Food and Nutrition Information Center. 10301 Baltimore Boulevard, Room 304, Beltsville, MD 20705-2351.
< <http://www.nalusda.gov/fnic> >.

Maureen Haggerty

Protein-modified diet see **Diets**

Prothrombin time

Definition

The prothrombin time test belongs to a group of blood tests that assess the clotting ability of blood. The test is also known as the pro time or PT test.

Purpose

The PT test is used to monitor patients taking certain medications as well as to help diagnose clotting disorders.

Diagnosis

Patients who have problems with delayed blood clotting are given a number of tests to determine the cause of the problem. The prothrombin test specifically evaluates the presence of factors VIIa, V, and X, prothrombin, and fibrinogen. Prothrombin is a protein in the liquid part of blood (plasma) that is converted to thrombin as part of the clotting process. Fibrinogen is a type of blood protein called a globulin; it is converted to fibrin during the clotting process. A drop in the concentration of any of these factors will cause the blood to take longer to clot. The PT test is used in combination with the **partial thromboplastin time** (PTT) test to screen for **hemophilia** and other hereditary clotting disorders.

Monitoring

The PT test is also used to monitor the condition of patients who are taking warfarin (Coumadin). Warfarin is a drug that is given to prevent clots in the deep veins of the legs and to treat **pulmonary embolism**. It interferes with blood clotting by lowering the liver's production of certain clotting factors.

Description

A sample of the patient's blood is obtained by venipuncture. The blood is collected in a tube that contains sodium citrate to prevent the clotting process from starting before the test. The blood cells are separated from the liquid part of blood (plasma). The PT test is performed by adding the patient's plasma to a protein in the blood (thromboplastin) that converts prothrombin to thrombin. The mixture is then kept in a warm water bath at 37°C for one to two minutes. Calcium chloride is added to the mixture in order to counteract the

sodium citrate and allow clotting to proceed. The test is timed from the addition of the calcium chloride until the plasma clots. This time is called the prothrombin time.

Preparation

The doctor should check to see if the patient is taking any medications that may affect test results. This precaution is particularly important if the patient is taking warfarin, because there are a number of medications that can interact with warfarin to increase or decrease the PT time.

Aftercare

Aftercare consists of routine care of the area around the puncture mark. Pressure is applied for a few seconds and the wound is covered with a bandage.

Risks

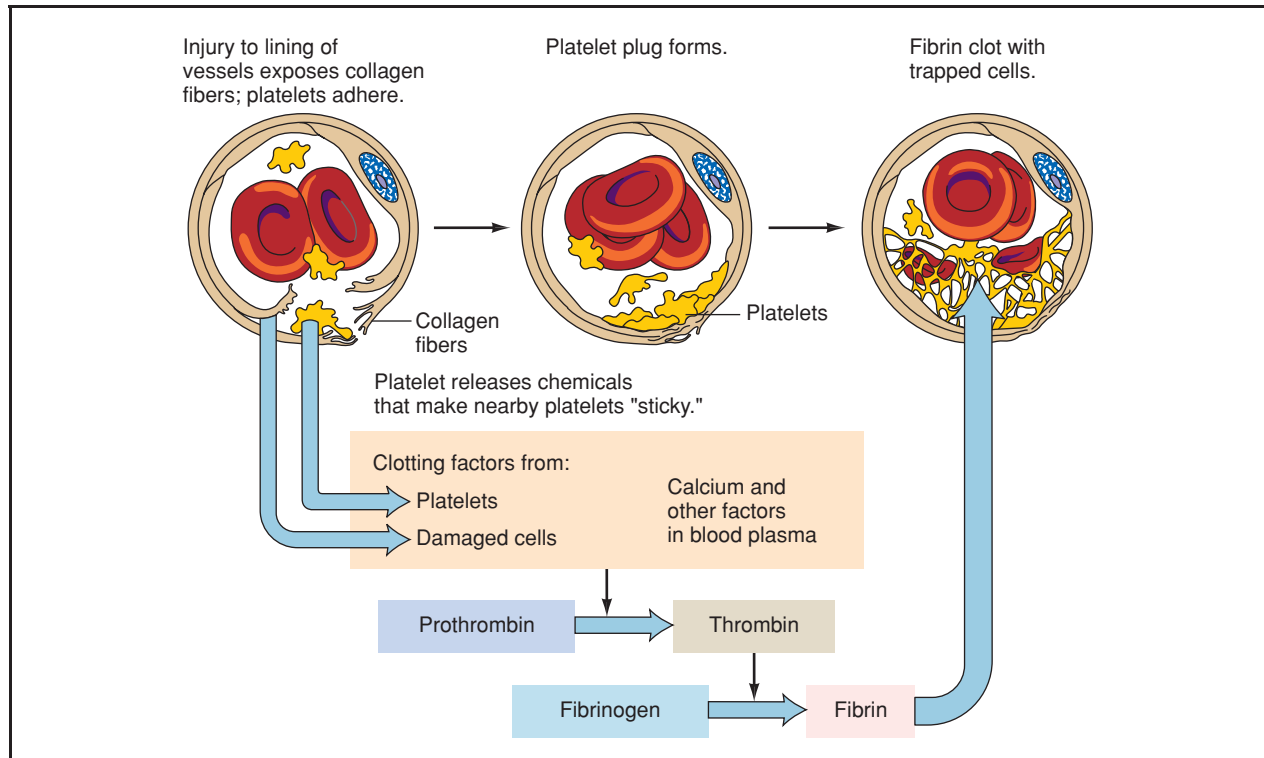
The primary risk is mild **dizziness** and the possibility of a bruise or swelling in the area where the blood was drawn. The patient can apply moist warm compresses.

Normal results

The normal prothrombin time is 11–15 seconds, although there is some variation depending on the source of the thromboplastin used in the test. (For this reason, laboratories report a normal control value along with patient results.) A prothrombin time within this range indicates that the patient has normal amounts of clotting factors VII and X.

Abnormal results

A prolonged PT time is considered abnormal. The prothrombin time will be prolonged if the concentration of any of the tested factors is 10% or more below normal plasma values. A prolonged prothrombin time indicates a deficiency in any of factors VII, X, V, prothrombin, or fibrinogen. It may mean that the patient has a vitamin K deficiency, a **liver disease**, or disseminated intravascular coagulation (DIC). The prothrombin time of patients receiving warfarin therapy will also be prolonged—usually in the range of one and one half to two times the normal PT time. A PT time that exceeds approximately two and a half times the control value (usually 30 seconds or longer) is grounds for concern, as abnormal bleeding may occur.



The blood clotting process. (Illustration by Hans & Cassady.)

KEY TERMS

Disseminated intravascular coagulation (DIC)—A condition in which spontaneous bleeding and clot formation occur throughout the circulatory system. DIC can be caused by transfusion reactions and a number of serious illnesses.

Fibrin—The protein formed as the end product of the blood clotting process when fibrinogen interacts with thrombin.

Fibrinogen—A type of blood protein called a globulin that interacts with thrombin to form fibrin.

Plasma—The liquid part of blood, as distinct from blood cells.

Prothrombin—A protein in blood plasma that is converted to thrombin during the clotting process.

Thrombin—An enzyme in blood plasma that helps to convert fibrinogen to fibrin during the last stage of the clotting process.

Thromboplastin—A protein in blood that converts prothrombin to thrombin.

Warfarin—A drug given to control the formation of blood clots. The PT test can be used to monitor patients being treated with warfarin.

Resources

BOOKS

Berkow, Robert, et al., editors. *Merck Manual of Medical Information*. Whitehouse Station, NJ: Merck Research Laboratories, 2004.

John T. Lohr, PhD

Proton pump inhibitors

Definition

The proton pump inhibitors are a group of drugs that reduce the secretion of gastric (stomach) acid. They act by binding with the enzyme H^+ , $K^+(+)-ATPase$, *hydrogen/potassium adenosine triphosphatase*, which is sometimes referred to as the proton pump. This enzyme causes parietal cells of the stomach lining to produce acid.

Although they perform much the same functions as the histamine H-2 receptor blockers, the proton pump inhibitors reduce stomach acid more and over a longer period.

Purpose

Proton pump inhibitors are used to treat ulcers; gastroesophageal reflux disease (GERD), a condition in which backward flow of acid from the stomach causes **heartburn** and injury of the food pipe (esophagus); and conditions in which the stomach produces too much acid, such as Zollinger-Ellison syndrome. Omeprazole is used in combination with other medications to treat recurrent ulcers caused by helicobacter pylori infections.

Two of the proton pump inhibitors, lansoprazole and omeprazole, have been used to improve pancreatic enzyme absorption in **cystic fibrosis** patients with intestinal malabsorption.

Proton pump inhibitors may be used to protect against the ulcerogenic effects of non-steroidal anti-inflammatory drugs and to help heal ulcers caused by these drugs.

Description

There are five drugs in this class: esomeprazole (Nexium), lansoprazole (Prevacid), omeprazole (Prilosec), pantoprazole (Protonix), and rabeprazole (Aciphex). They act in a similar manner, and their cautions and adverse effects are similar.

The products are generally formulated as enteric-coated granules. Absorption does not start until the granules have left the stomach and reached the intestine, so the onset of action is delayed about an hour, subject to gastric emptying time. Since they act slowly, proton pump inhibitors are not a suitable alternative to **antacids** which have a rapid effect.

Although these drugs are eliminated from the body relatively quickly, usually within 90 minutes of absorption, they all work for over 24 hours after a dose. This is because the factor that determines duration of action is how long it takes the body to replace the H^+ , $K(+)$ -ATPase. There is some build up over time. For example, a single dose of lansoprazole reduces stomach acid by 71%, but after a week of regular dosing, the acid reduction rises to 80%.

For treatment of recurrent ulcers, the proton pump inhibitors are part of combination therapy that uses an antibiotic (occasionally two **antibiotics**) and proton pump inhibitor. There are a number of regimens, and while they may vary in the selection of specific drugs, or even types of drugs used, usually they include a proton pump inhibitor. The cure rates are all within similar ranges for these regimens.

Recommended dosage

Dose varies with the indication. The following are commonly prescribed doses:

- Esomeprazole: 20 to 40 mg once a day.
- Lansoprazole: 15 to 30 mg once a day.
- Omeprazole: 20 to 40 mg once a day.
- Pantoprazole: 40 mg once or twice a day.
- Rabeprazole: 20 mg once a day. In hypersecretory conditions, doses as high as 60 mg twice daily have been reported.

In the above examples, the lower dose is usually adequate for GERD, while the higher dose may be required for ulcer therapy or hypersecretory conditions.

Precautions

Proton pump inhibitors should not be given to any patient who has shown a reaction to any of the components of the drug or a related drug. Proton pump inhibitors should also not be given to patients with severe **liver disease**.

Omeprazole is **pregnancy** category C, while esomeprazole, lansoprazole, rabeprazole, and pantoprazole are category B. As of 2005, there are no adequate and well-controlled studies concerning the effects of these drugs on pregnant women. These drugs should be used during pregnancy only if the potential benefit justifies the risk to the fetus. Because the proton pump inhibitors are excreted into breast milk, they should not be used by women who are breastfeeding their babies.

The proton pump inhibitors may mask the symptoms of **stomach cancer**.

Side effects

The proton pump inhibitors are relatively safe drugs. The most commonly observed adverse effects are **constipation**, **diarrhea**, **dizziness**, **headache**, skin itch, and skin rash. Less often, the following adverse effects have been reported: abdominal **pain** with cramps, appetite changes, and **nausea**.

The following adverse effects are extremely rare but have been reported with this class of drugs:

- acute pancreatitis
- anxiety
- cough
- depression

KEY TERMS

Antacid—A substance that counteracts or neutralizes acidity, usually of the stomach. Antacids have a rapid onset of action compared to histamine H-2 receptor blockers and proton pump inhibitors, but they have a short duration of action and require frequent dosing.

Cystic fibrosis—A hereditary disease that appears in early childhood, involves functional disorder of digestive glands, and is marked especially by faulty digestion due to a deficiency of pancreatic enzymes, by difficulty in breathing due to mucus accumulation in airways, and by excessive loss of salt in the sweat.

Enteric coat—A coating put on some tablets or capsules to prevent their disintegration in the stomach. The contents of coated tablets or capsules will be released only when the dose reaches the intestine. This may be done to protect the drug from stomach acid, to protect the stomach from drug irritation, or to delay the onset of action of the drug.

GERD—A chronic condition in which the lower esophageal sphincter allows gastric acids to reflux into the esophagus, causing heartburn, acid indigestion, and possible injury to the esophageal lining.

Malabsorption—Defective or inadequate absorption of nutrients from the intestinal tract.

Parietal cells—Cells of the gastric glands that secrete hydrochloric acid.

Recurrent ulcer—Stomach ulcers that return after apparently complete healing. These ulcers appear to be caused by helicobacter pylori infections and can generally be successfully treated with a combination of antibiotics and gastric acid reducing compounds, particularly the proton pump inhibitors.

- drug toxin-related hepatitis
- erythema multiforme
- flu-like symptoms
- myalgia
- Stevens-Johnson syndrome
- thrombocytopenia
- toxic epidermal necrolysis
- ulcerative colitis

- upper respiratory hypersensitivity reaction
- upper respiratory infection
- vomiting

Interactions

Proton pump inhibitors should not be used in conjunction with the anti-retroviral (anti-AIDS) drug atazanavir (Rayataz). The conjunction may reduce the effectiveness of the atazanavir. Proton pump inhibitors should not be used in combination with the anti-fungal drugs itraconazole or ketoconazole. This combination may reduce the effectiveness of the anti-fungal drugs.

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ORGANIZATION

International Foundation for Functional Gastrointestinal Disorders. PO Box 170864, Milwaukee, WI 53217-8076. (414)964-1799. <www.iffgd.org>.

Samuel D. Uretsky, Pharm.D.

Pruritis see **Itching**

PSA test see **Prostate-specific antigen test**

PSDD see **Premenstrual dysphoric disorder**

Pseudoephedrin see **Decongestants**

Pseudogout

Definition

Pseudogout is a form of arthritis that causes **pain**, redness, and inflammation in one or more joints.

Description

Pseudogout is also known by another name: calcium pyrophosphate dihydrate deposition disease (CPPD), the basis of which is derived from the calcium deposits that collect in the joint. The deposits or crystals, as they are sometimes called, cause pain and inflammation in the joint. According to the Arthritis Foundation, this can eventually weaken the cartilage, which serves as padding between the bones, “allowing bone to rub against bone.” Pseudogout typically affects the large joints, such as the knees, wrists, and ankles. In general, it occurs with equal frequency in men and women.

Most often seen in older adults, pseudogout can also affect younger patients, especially those with diseases that put them at a greater risk of developing it, such as **hemochromatosis**, **hypercalcemia**, **hypothyroidism**, **ochronosis**, or Wilson’s disease. Some people, according to an article for the American College of Rheumatology, experience attacks of pseudogout “following joint surgery or other surgery. Because many older people have calcium crystal deposits in their joints, any kind of insult to the joint can trigger the release of the calcium crystals, which then induce a painful inflammatory response.” Pseudogout affects about 3% of elderly people. Not all will experience severe attacks. By their nineties, 50% of people will have joint deposits. Although researchers have noticed that some people with pseudogout also have a family history of the disease, it is not clear what role genetics might play in its development.

Causes and symptoms

As the Arthritis Foundation points out, it is unclear what causes the crystals to form, but some speculation exists that “an abnormality in the cartilage cells or connective tissue could be responsible” for their development. Acute pain and fluid accumulation that leads to joint swelling are typical symptoms of pseudogout. When the crystals move into a joint, the Arthritis Foundation categorizes the pain as “sudden and severe.” Many patients report that joint motion is limited. Statistically

speaking, in 50% of the cases, the patient will run a **fever**. Half of all the acute pseudogout attacks will involve a knee. The experts at MedlinePlus identify “chronic (long term) arthritis” as a symptom that can be present at the time of an acute pseudogout attack. The word “acute” implies short term; therefore, acute attacks of pseudogout will come and go, but chronic arthritis may remain. In addition, progressive degenerative arthritis is sometimes seen in numerous joints.

Diagnosis

Pseudogout and **gout** have similar symptoms, which can be confusing. However, uric acid is associated with gout, whereas calcium pyrophosphate crystals are associated with pseudogout. After a patient’s detailed medical history is obtained, a diagnosis can be made based on the symptoms and medical tests.

Using a needle, the physician can take a sample of the synovial fluid from the swollen or painful joint to ascertain the presence of calcium pyrophosphate crystals. The fluid will also contain white blood cells, which can be counted to assist in the diagnosis. Synovial fluid is the lubricating fluid that’s secreted by the membranes that line the joints.

X rays may also be taken to confirm the presence of crystals. The x-rays may also show joint damage or that crystals have led to a condition called chondrocalcinosis, which is calcification of the cartilage. Other possible causes such as gout, **rheumatoid arthritis**, or infection must be ruled out too. Blood tests can also help to confirm the diagnosis. Indeed, as the experts at MedlinePlus indicated, “careful workup, with analysis of crystals found in the joints, should ultimately lead to the correct diagnosis.”

Treatment

There are a variety of treatment options. If the patients have an adequate support system, such as family and friends willing to help, it makes it easier for patients to recover faster. Patients are often advised to avoid putting pressure on the affected joint. In some cases, it is appropriate for the patient to engage in special isometric exercises designed to help their specific condition heal faster. Once the inflammation and pain subsides, exercises are sometimes suggested to regain range of motion.

Medications can be prescribed to ease the pain, which typically fall into the nonsteroidal anti-inflammatory

KEY TERMS

hemochromatosis—This disease refers to when a person's body absorbs too much iron.

hypercalcemia—This disease refers to when a person's bones absorb too much calcium.

hypothyroidism—This disease refers to when a person's thyroid gland is underactive.

ochronosis—People with this rare hereditary condition tend to develop arthritis in adulthood.

Wilson's disease—Wilson's disease causes the body to retain copper, which ultimately can lead to liver damage.

category. For example, ibuprofen (Motrin) and naproxen (Aleve) are **nonsteroidal anti-inflammatory drugs** (NSAIDs) that are used quite often, because they are generally well tolerated and highly effective. However, in the article Dr. Schumacher wrote for the American College of Rheumatology he cautions patients by stating that, "people with poor kidney function, a history of stomach ulcers, and those who are on blood thinners often cannot take NSAIDs." Indomethacin (Indocin) is also a first-line drug that is commonly used to treat pseudogout that falls into this category.

When no infection is present, steroids, such as prednisone, may be prescribed. Much of the literature discussing treatment options also suggests a medication called colchicine, which is only available as a generic. It is generally prescribed in low doses and should not be used by anyone with significant bone marrow dysfunction or renal insufficiency. Patients should talk with their physicians regarding any other reasons why colchicine may not be suitable for them.

In order to relieve some of the pressure, the excess fluid around the joint can be removed (or aspirated, as it is sometimes called) with a needle.

Treatment options to reduce inflammation are valuable, "because they may slow the progression of joint degeneration that often accompanies pseudogout," according to Dr. Schumacher in his American College of Rheumatology article. If joint degeneration does occur, surgery is available to replace or repair damaged joints; however, it is better for patients to engage in preventative measures that

will help them avoid the need for surgery. As the old saying goes, "an ounce of prevention is worth a pound of cure."

Prognosis

With regard to an acute attack of pseudogout, the prognosis is usually very good. The symptoms usually go away within two weeks. However, over time, joint degeneration can occur.

Prevention

There are not specific techniques applicable to every patient to prevent the formulation of the crystals; however, some patients with certain diseases are at greater risk of developing them. Diagnoses and treatment of underlying disorders is one of the most important aspects of managing crystal-induced arthropathies. Once a causative crystal is identified and a diagnosis has been established, long-term management and prevention can be devised.

Resources

PERIODICALS

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Lee Ann Paradise

Pseudohermaphroditism see **Intersex states**

Pseudomembraneous enterocolitis see **Antibiotic-associated colitis**

Pseudomonas aeruginosa infection see **Pseudomonas infections**

Pseudomonas infections

Definition

A pseudomonas infection is caused by a bacterium, *Pseudomonas aeruginosa*, and may affect any part of the body. In most cases, however, pseudomonas infections strike only persons who are very ill, usually hospitalized.

Description

P. aeruginosa is a rod-shaped organism that can be found in soil, water, plants, and animals. Because it rarely causes disease in healthy persons, but infects those who are already sick or who have weakened immune systems, it is called an opportunistic pathogen. Opportunistic pathogens are organisms that do not ordinarily cause disease, but multiply freely in persons whose immune systems are weakened by illness or medication. Such persons are said to be immunocompromised. Patients with **AIDS** have an increased risk of developing serious pseudomonas infections. Hospitalized patients are another high-risk group, because *P. aeruginosa* is often found in hospitals. Infections that can be acquired in the hospital are sometimes called nosocomial diseases.

Of the two million nosocomial infections each year, 10% are caused by *P. aeruginosa*. The bacterium is the second most common cause of nosocomial pneumonia and the most common cause of intensive care unit (ICU) pneumonia. Pseudomonas infections can be spread within hospitals by health care workers, medical equipment, sinks, disinfectant solutions, and food. These infections are a very serious problem in hospitals for two reasons. First, patients who are critically ill can die from a pseudomonas infection. Second, many *Pseudomonas* bacteria are resistant to certain **antibiotics**, which makes them difficult to treat.

P. aeruginosa is able to infect many different parts of the body. Several factors make it a strong opponent. These factors include:

- the ability to stick to cells
- minimal food requirements
- resistance to many antibiotics
- production of proteins that damage tissue
- a protective outer coat

Infections that can occur in specific body sites include:

- Heart and blood. *P. aeruginosa* is the fourth most common cause of bacterial infections of the blood (**bacteremia**). Bacteremia is common in patients with blood **cancer** and patients who have pseudomonas infections elsewhere in the body. *P. aeruginosa* infects the heart valves of intravenous drug abusers and persons with artificial heart valves.
- Bones and joints. Pseudomonas infections in these parts of the body can result from injury, the spread of infection from other body tissues, or bacteremia. Persons at risk for pseudomonas infections of the bones and joints include diabetics, intravenous drug abusers, and bone surgery patients.
- Central nervous system. *P. aeruginosa* can cause inflammation of the tissues covering the brain and spinal cord (**meningitis**) and brain abscesses. These infections may result from brain injury or surgery, the spread of infection from other parts of the body, or bacteremia.
- Eye and ear. *P. aeruginosa* can cause infections in the external ear canal—so-called “swimmer’s ear”—that usually disappear without treatment. The bacterium can cause a more serious ear infection in elderly patients, possibly leading to hearing problems, facial **paralysis**, or even **death**. Pseudomonas infections of the eye usually follow an injury. They can cause ulcers of the cornea that may cause rapid tissue destruction and eventual blindness. The risk factors for pseudomonas eye infections include: wearing soft extended-wear contact lenses; using topical corticosteroid eye medications; being in a **coma**; having extensive **burns**; undergoing treatment in an ICU; and having a tracheostomy or endotracheal tube.
- Urinary tract. Urinary tract infections can be caused by catheterization, medical instruments, and surgery.
- Lung. Risk factors for *P. aeruginosa* **pneumonia** include: cystic fibrosis; chronic lung disease; immunocompromised condition; being on antibiotic therapy or a respirator; and congestive heart failure. Patients with **cystic fibrosis** often develop pseudomonas infections as children and suffer recurrent attacks of pneumonia.
- Skin and soft tissue. Even healthy persons can develop a pseudomonas skin rash following exposure to the bacterium in contaminated hot tubs, water parks, whirlpools, or spas. This skin disorder is called pseudomonas or “hot tub” **folliculitis**, and is often confused with **chickenpox**. Severe skin infection may occur in patients with *P. aeruginosa* bacteremia. The bacterium is the second most common cause of burn wound infections in hospitalized patients.

Causes and symptoms

P. aeruginosa can be sudden and severe, or slow in onset and cause little **pain**. Risk factors for acquiring a pseudomonas infection include: having a serious illness; being hospitalized; undergoing an invasive procedure such as surgery; having a weakened immune system; and being treated with antibiotics that kill many different kinds of bacteria (broad-spectrum antibiotics).

Each of the infections listed above has its own set of symptoms. *Pseudomonas* bacteremia resembles other bacteremias, producing **fever**, tiredness, muscle pains, joint pains, and chills. Bone infections are marked by swelling, redness, and pain at the infected site and possibly fever. *Pseudomonas* meningitis causes fever, **headache**, irritability, and clouded consciousness. Ear infection is associated with pain, ear drainage, facial paralysis, and reduced hearing. Pseudomonas infections of the eye cause ulcers that may spread to cover the entire eye, pain, reduced vision, swelling of the eyelids, and pus accumulation within the eye.

P. aeruginosa pneumonia is marked by chills, fever, productive **cough**, difficult breathing, and blue-tinted skin. Patients with cystic fibrosis with pseudomonas lung infections experience coughing, decreased appetite, weight loss, tiredness, **wheezing**, rapid breathing, fever, blue-tinted skin, and abdominal enlargement. Skin infections can cause a range of symptoms from a mild rash to large bleeding ulcers. Symptoms of pseudomonas folliculitis include a red itchy rash, headache, dizziness, earache, sore eyes, nose, and throat, breast tenderness, and stomach pain. Pseudomonas wound infections may secrete a blue-green colored fluid and have a fruity smell. Burn wound infections usually occur one to two weeks after the burn and cause discoloration of the burn scab, destruction of the tissue below the scab, early scab loss, bleeding, swelling, and a blue-green drainage.

Diagnosis

Diagnosis and treatment of pseudomonas infections can be performed by specialists in infectious disease. Because *P. aeruginosa* is commonly found in hospitals, many patients carry the bacterium without having a full-blown infection. Consequently, the mere presence of *P. aeruginosa* in patients does not constitute a diagnostic finding. Cultures, however, can be easily done for test purposes. The organism grows readily in laboratory media; results are usually available in two to three days. Depending on the location of

the infection, body fluids that can be tested for *P. aeruginosa* include blood, urine, cerebrospinal fluid, sputum, pus, and drainage from an infected ear or eye. X rays and other imaging techniques can be used to assess infections in deep organ tissues.

Treatment

Medications

Because *P. aeruginosa* is commonly resistant to antibiotics, infections are usually treated with two antibiotics at once. Pseudomonas infections may be treated with combinations of ceftazidime (Ceftaz, Fortraz, Tazicef), ciprofloxacin (Cipro), imipenem (Primaxin), gentamicin (Garamycin), tobramycin (Nebcin), ticarcillin-clavulanate (Timentin), or piperacillin-tazobactam (Zosyn). Most antibiotics are administered intravenously or orally for two to six weeks. Treatment of an eye infection requires local application of antibiotic drops.

Surgery

Surgical treatment of pseudomonas infections is sometimes necessary to remove infected and damaged tissue. Surgery may be required for brain abscesses, eye infections, bone and joint infections, ear infections, heart infections, and wound infections. Infected **wounds** and burns may cause permanent damage requiring arm or leg amputation.

Prognosis

Most pseudomonas infections can be successfully treated with antibiotics and surgery. In immunocompromised persons, however, *P. aeruginosa* infections have a high mortality rate, particularly following bacteremia or infections of the lower lung. Mortality rates range from 15 to 20% of patients with severe ear infections to 89% of patients with infections of the left side of the heart.

Prevention

Most hospitals have programs for the prevention of nosocomial infections. Patients with cystic fibrosis may be given periodic doses of antibiotics to prevent episodes of pseudomonas pneumonia.

Minor skin infections can be prevented by avoiding hot tubs with cloudy water; avoiding public swimming pools at the end of the day; removing wet swimsuits as soon as possible; bathing after sharing a hot tub or using a public pool; cleaning hot tub filters

KEY TERMS

Bacteremia—Bacterial infection of the blood.

“Hot tub” folliculitis—A skin infection caused by *P. aeruginosa* that often follows bathing in a hot tub or public swimming pool.

Immunocompromised—Having a weak immune system due to disease or the use of certain medications.

Nosocomial infection—An infection that is acquired in the hospital.

Opportunistic—Causing disease only under certain conditions, as when a person is already sick or has a weak immune system.

Pathogen—Any microorganism that produces disease.

every six weeks; and using appropriate amounts of chlorine in the water.

Resources

OTHER

Centers for Disease Control. <<http://www.cdc.gov/nccdphp/ddt/ddthome.htm>>.

Belinda Rowland, PhD

Pseudomonas pseudomallei infection see
Melioidosis

Pseudost see **Strabismus**

Pseudotuberculosis see **Sarcoidosis**

Pseudoxanthoma elasticum

Definition

Pseudoxanthoma elasticum (PXE) is an inherited connective tissue disorder in which the elastic fibers present in the skin, eyes, and cardiovascular system gradually become calcified and inelastic.

Description

PXE was first reported in 1881 by Rigal, but the defect in elastic fibers was described in 1986 by Darier, who gave the condition its name. PXE is also known as

Grönblad-Strandberg-Touraine syndrome and systemic elastorrhexis.

The course of PXE varies greatly between individuals. Typically it is first noticed during adolescence as yellow-orange bumps on the side of the neck. Similar bumps may appear at other places where the skin bends a lot, like the backs of the knees and the insides of the elbows. The skin in these areas tends to get thick, leathery, inelastic, and acquire extra folds. These skin problems have no serious consequences, and for some people, the disease progresses no further.

Bruch's membrane, a layer of elastic fibers in front of the retina, becomes calcified in some people with PXE. Calcification causes cracks in Bruch's membrane, which can be seen through an ophthalmoscope as red, brown, or gray streaks called angioid streaks. The cracks can eventually (e.g., in 10–20 years) cause bleeding, and the usual resultant scarring leads to central vision deterioration. However, peripheral vision is unaffected.

Arterial walls and heart valves contain elastic fibers that can become calcified. This leads to a greater susceptibility to the conditions that are associated with hardening of the arteries in the normal **aging** population—high blood pressure, **heart attack**, **stroke**, and arterial obstruction—and, similarly, **mitral valve prolapse**. Heart disease and **hypertension** associated with PXE have been reported in children as young as four to 13 years of age. Although often appearing at a younger age, the overall incidence of these conditions is only slightly higher for people with PXE than it is in the general population.

Arterial inelasticity can lead to bleeding from the gastrointestinal tract and, rarely, acute vomiting of blood.

PXE is rare and occurs in about 1 in every 160,000 people in the general population. It is likely, though, that PXE is underdiagnosed, because of the presence of mild symptoms in some affected persons and the lack of awareness of the condition among primary care physicians.

Causes and symptoms

PXE is caused by changes in the genetic material, called mutations, that are inherited in either a dominant or recessive mode. A person with the recessive form of the disease (which is most common) must possess two copies of the PXE gene to be affected, and, therefore, must have received one from each parent. In the dominant form, one copy of the defective gene is sufficient to cause the disease. In some

cases, a person with the dominant form inherits the abnormal gene from a parent with PXE. More commonly, the mutation arises as a spontaneous change in the genetic material of the affected person. These cases are called “sporadic” and do not affect parents or siblings, although each child of a person with sporadic PXE has a 50% risk to inherit the condition.

Both males and females develop PXE, although the skin findings seem to be somewhat more common in females.

The actual genetic causes of this condition were not discovered until 2000. The recessive, dominant, and sporadic forms of PXE all appear to be caused by different mutations or deletions in a single gene called *ABCC6* (also known as *MRP6*), located on chromosome 16. Although the responsible gene has been identified, how it causes PXE is still unknown.

Genetic researchers have since identified mutations in a number of persons with PXE, most of whom have been found to have the recessive type. Affected individuals in these families had mutations in both copies of the gene and parents, who are obligate carriers, had a mutation in only one copy. Contrary to the usual lack of symptoms in carriers of recessive genes, some carriers of recessive PXE have been found to have cardiovascular symptoms typical of PXE.

Although the recessive type is the most common, there are also familial and sporadic cases that have been found to be caused by dominant mutations in the *ABCC6* gene.

A wide range in the type and severity of symptoms exists between people with PXE. The age of onset also varies, although most people notice initial symptoms during adolescence or early adulthood. Often, the first symptoms to appear are thickened skin with yellow bumps in localized areas such as the folds of the groin, arms, knees, and armpits. These changes can also occur in the mucous membranes, most often in the inner portion of the lower lip. The appearance of the skin in PXE has been likened to a plucked chicken or Moroccan leather.

Angioid streaks in front of the retina are present in most people with PXE and an ophthalmologic examination can be used as an initial screen for the condition. Persons with PXE often complain of sensitivity to light. Because of the progressive breakdown of Bruch's membrane, affected persons are at increased risk for bleeding and scarring of the retina, which can lead to decreased central vision but does not usually cause complete blindness.

Calcium deposits in the artery walls contribute to early-onset **atherosclerosis**, and another condition called claudication, inadequate blood flow that results in **pain** in the legs after exertion. Abnormal bleeding, caused by calcification of the inner layer of the arteries, can occur in the brain, retina, uterus, bladder, and joints but is most common in the gastrointestinal tract.

Diagnosis

The presence of calcium in elastic fibers, as revealed by microscopic examination of biopsied skin, unequivocally establishes the diagnosis of PXE.

Treatment

PXE cannot be cured, but **plastic surgery** can treat PXE skin lesions, and **laser surgery** is used to prevent or slow the progression of vision loss. Excessive blood loss due to bleeding into the gastrointestinal tract or other organ systems may be treated by **transfusion**. Mitral valve prolapse (protrusion of one or both cusps of the mitral heart valve back into the atrium during heart beating) can be corrected by surgery, if necessary.

Measures should be taken to prevent or lessen cardiovascular complications. People with PXE should control their cholesterol and blood pressure, and maintain normal weight. They should exercise for cardiovascular health and to prevent or reduce claudication later in life. They should also avoid the use of tobacco, thiazide **antihypertensive drugs**, blood thinners like coumadin, and **nonsteroidal anti-inflammatory drugs** like **aspirin** and ibuprofen. In addition, they should avoid strain, heavy lifting, and contact sports, since these activities could trigger retinal and gastrointestinal bleeding.

People with PXE should have regular eye examinations by an ophthalmologist and report any eye problems immediately. Regular check-ups with a physician are also recommended, including periodic blood pressure readings.

Some people have advocated a calcium-restricted diet, but it is not yet known whether this aids the problems brought about by PXE. It is known, however, that calcium-restriction can lead to bone disorders.

Prognosis

The prognosis is for a normal life span with an increased chance of cardiovascular and circulatory

KEY TERMS

Angioid streaks—Gray, orange, or red wavy branching lines in Bruch’s membrane.

Bruch’s membrane—A membrane in the eye between the choroid membrane and the retina.

Carrier—A person who possesses a gene for an abnormal trait without showing signs of the disorder. The person may pass the abnormal gene on to offspring.

Claudication—Pain in the lower legs after exercise caused by insufficient blood supply.

Connective tissue—A group of tissues responsible for support throughout the body; includes cartilage, bone, fat, tissue underlying skin, and tissues that support organs, blood vessels, and nerves throughout the body.

Deletion—The absence of genetic material that is normally found in a chromosome. Often, the genetic material is missing due to an error in replication of an egg or sperm cell.

Dominant trait—A genetic trait in which one copy of the gene is sufficient to yield an outward display of the trait; dominant genes mask the presence of recessive genes; dominant traits can be inherited from a single parent.

Elastic fiber—Fibrous, stretchable connective tissue made primarily from proteins, elastin, collagen, and fibrillin.

Gene—A building block of inheritance, which contains the instructions for the production of a particular protein, and is made up of a molecular sequence found on a section of DNA. Each gene is found on a precise location on a chromosome.

Mitral valve—The heart valve that prevents blood from flowing backwards from the left ventricle into the left atrium. Also known as bicuspid valve.

Mutation—A permanent change in the genetic material that may alter a trait or characteristic of an individual, or manifest as disease, and can be transmitted to offspring.

Recessive trait—An inherited trait or characteristic that is outwardly obvious only when two copies of the gene for that trait are present.

problems, hypertension, gastrointestinal bleeding, and impaired vision. However, now that the gene for PXE has been identified, the groundwork for research to provide effective treatment has been laid. Studying the

role of the ABCC6 protein in elastic fibers may lead to drugs that will ameliorate or arrest the problems caused by PXE.

Genetic tests are now available that can provide knowledge needed to both diagnose PXE in symptomatic persons and predict it prior to the onset of symptoms in persons at risk. Prenatal diagnosis of PXE, by testing fetal cells for mutations in the ABCC6 gene, can be done in early pregnancy by procedures such as **amniocentesis** or chorionic villus sampling. For most people, PXE is compatible with a reasonably normal life, and prenatal diagnosis is not likely to be highly desired.

Genetic testing to predict whether an at-risk child will develop PXE may be helpful for medical management. A child who is found to carry a mutation can be monitored more closely for eye problems and bleeding, and can begin the appropriate lifestyle changes to prevent cardiovascular problems.

Resources

PERIODICALS

Ringpfeil, F., et al. “Pseudoxanthoma Elasticum: Mutations in the MRP6 Gene Encoding a Transmembrane ABC-binding Cassette (ABC) Transporter.” *Proceedings of the National Academy of Sciences* 97 (May 2000): 6001–6.

Sherer, D.W., et al. “Pseudoxanthoma Elasticum: An Update.” *Dermatology* 199 (1999): 3–7.

ORGANIZATIONS

National Association for Pseudoxanthoma Elasticum. 3500 East 12th Avenue, Denver, CO 80206. (303) 355-3866. Fax: (303) 355-3859. Pxnape@estreet.com. <http://www.napxe.org>.

PXE International, Inc. 23 Mountain Street, Sharon, MA 02067. (781) 784-3817. Fax: (781) 784-6672. PXEInter@aol.com. <http://www.pxe.org/>.

Barbara J. Pettersen

Psittacosis see **Parrot fever**

Psoas abscess see **Abscess**

Psoriasis

Definition

Named for the Greek word *psōra* meaning “itch,” psoriasis is a chronic, non-contagious disease characterized by inflamed lesions covered with silvery-white scabs of dead skin.

Description

Psoriasis, which affects at least four million Americans, is slightly more common in women than in men. Although the disease can develop at any time, 10–15% of all cases are diagnosed in children under 10, and the average age at the onset of symptoms is 28. Psoriasis is most common in fair-skinned people and extremely rare in dark-skinned individuals.

Normal skin cells mature and replace dead skin every 28–30 days. Psoriasis causes skin cells to mature in less than a week. Because the body can't shed old skin as rapidly as new cells are rising to the surface, raised patches of dead skin develop on the arms, back, chest, elbows, legs, nails, folds between the buttocks, and scalp.

Psoriasis is considered mild if it affects less than 5% of the surface of the body; moderate, if 5–30% of the skin is involved, and severe, if the disease affects more than 30% of the body surface.

Types of psoriasis

Dermatologists distinguish different forms of psoriasis according to what part of the body is affected, how severe symptoms are, how long they last, and the pattern formed by the scales.

PLAQUE PSORIASIS. Plaque psoriasis (psoriasis vulgaris), the most common form of the disease, is characterized by small, red bumps that enlarge, become inflamed, and form scales. The top scales flake off easily and often, but those beneath the surface of the skin clump together. Removing these scales exposes tender skin, which bleeds and causes the plaques (inflamed patches) to grow.

Plaque psoriasis can develop on any part of the body, but most often occurs on the elbows, knees, scalp, and trunk.

SCALP PSORIASIS. At least 50 of every 100 people who have any form of psoriasis have scalp psoriasis. This form of the disease is characterized by scale-capped plaques on the surface of the skull.

NAIL PSORIASIS. The first sign of nail psoriasis is usually pitting of the fingernails or toenails. Size, shape, and depth of the marks vary, and affected nails may thicken, yellow, or crumble. The skin around an affected nail is sometimes inflamed, and the nail may peel away from the nail bed.

GUTTATE PSORIASIS. Named for the Latin word *gutta*, which means “a drop,” guttate psoriasis is characterized by small, red, drop-like dots that enlarge rapidly and may be somewhat scaly. Often found on

the arms, legs, and trunk and sometimes in the scalp, guttate psoriasis can clear up without treatment or disappear and resurface in the form of plaque psoriasis.

PUSTULAR PSORIASIS. Pustular psoriasis usually occurs in adults. It is characterized by blister-like lesions filled with non-infectious pus and surrounded by reddened skin. Pustular psoriasis, which can be limited to one part of the body (localized) or can be widespread, may be the first symptom of psoriasis or develop in a patient with chronic plaque psoriasis.

Generalized pustular psoriasis is also known as Von Zumbusch pustular psoriasis. Widespread, acutely painful patches of inflamed skin develop suddenly. Pustules appear within a few hours, then dry and peel within two days.

Generalized pustular psoriasis can make life-threatening demands on the heart and kidneys.

Palmar-plantar pustulosis (PPP) generally appears between the ages of 20 and 60. PPP causes large pustules to form at the base of the thumb or on the sides of the heel. In time, the pustules turn brown and peel. The disease usually becomes much less active for a while after peeling.

Acrodermatitis continua of Hallopeau is a form of PPP characterized by painful, often disabling, lesions on the fingertips or the tips of the toes. The nails may become deformed, and the disease can damage bone in the affected area.

INVERSE PSORIASIS. Inverse psoriasis occurs in the armpits and groin, under the breasts, and in other areas where skin flexes or folds. This disease is characterized by smooth, inflamed lesions and can be debilitating.

ERYTHRODERMIC PSORIASIS. Characterized by severe scaling, **itching**, and **pain** that affects most of the body, erythrodermic psoriasis disrupts the body's chemical balance and can cause severe illness. This particularly inflammatory form of psoriasis can be the first sign of the disease, but often develops in patients with a history of plaque psoriasis.

PSORIATIC ARTHRITIS. About 10% of patients with psoriasis develop a complication called **psoriatic arthritis**. This type of arthritis can be slow to develop and mild, or it can develop rapidly. Symptoms of psoriatic arthritis include:

- joint discomfort, swelling, stiffness, or throbbing
- swelling in the toes and ankles
- pain in the digits, lower back, wrists, knees, and ankles
- eye inflammation or pink eye (conjunctivitis)

Causes and symptoms

The cause of psoriasis is unknown, but research suggests that an immune-system malfunction triggers the disease. Factors that increase the risk of developing psoriasis include:

- family history
- stress
- exposure to cold temperatures
- injury, illness, or infection
- steroids and other medications
- race

Trauma and certain bacteria may trigger psoriatic arthritis in patients with psoriasis.

Diagnosis

A complete medical history and examination of the skin, nails, and scalp are the basis for a diagnosis of psoriasis. In some cases, a microscopic examination of skin cells is also performed.

Blood tests can distinguish psoriatic arthritis from other types of arthritis. **Rheumatoid arthritis**, in particular, is diagnosed by the presence of a particular antibody present in the blood. That antibody is not present in the blood of patients with psoriatic arthritis.

Treatment

Age, general health, lifestyle, and the severity and location of symptoms influence the type of treatment used to reduce inflammation and decrease the rate at which new skin cells are produced. Because the course of this disease varies with each individual, doctors must experiment with or combine different treatments to find the most effective therapy for a particular patient.

Mild-moderate psoriasis

Steroid creams and ointments are commonly used to treat mild or moderate psoriasis, and steroids are sometimes injected into the skin of patients with a limited number of lesions. In mid-1997, the United States Food and Drug Administration (FDA) approved the use of tazarotene (Tazorac) to treat mild-to-moderate plaque psoriasis. This water-based gel has chemical properties similar to vitamin A.

Brief daily doses of natural sunlight can significantly relieve symptoms. **Sunburn** has the opposite effect.



Psoriasis, a chronic skin disorder, may appear on any area of the body, including the elbow, as shown above. (Photo Researchers, Inc. Reproduced by permission.)

Moisturizers and bath oils can loosen scales, soften skin, and may eliminate the itch. So can adding a cup of oatmeal to a tub of bath water. Salicylic acid (an ingredient in **aspirin**) can be used to remove dead skin or increase the effectiveness of other therapies.

Moderate psoriasis

Administered under medical supervision, ultraviolet light B (UVB) is used to control psoriasis that covers many areas of the body or that has not responded to topical preparations. Doctors combine UVB treatments with topical medications to treat some patients and sometimes prescribe home **phototherapy**, in which the patient administers his or her own UVB treatments.

Photochemotherapy (PUVA) is a medically supervised procedure that combines medication with exposure to ultraviolet light (UVA) to treat localized or widespread psoriasis. An individual with widespread psoriasis that has not responded to treatment may enroll in one of the day treatment programs conducted at special facilities throughout the United States. Psoriasis patients who participate in these intensive sessions are exposed to UVB and given other treatments for six to eight hours a day for two to four weeks.

Severe psoriasis

Methotrexate (MTX) can be given as a pill or as an injection to alleviate symptoms of severe psoriasis or psoriatic arthritis. Patients who take MTX must be carefully monitored to prevent liver damage.

Psoriatic arthritis can also be treated with non steroidal anti-inflammatory drugs (NSAID), like

acetaminophen (Tylenol) or aspirin. Hot compresses and warm water soaks may also provide some relief for painful joints.

Other medications used to treat severe psoriasis include etretinate (Tegison) and isotretinoin (Accutane), whose chemical properties are similar to those of vitamin A. Most effective in treating pustular or erythrodermic psoriasis, Tegison also relieves some symptoms of plaque psoriasis. Tegison can enhance the effectiveness of UVB or PUVA treatments and reduce the amount of exposure necessary.

Accutane is a less effective psoriasis treatment than Tegison, but can cause many of the same side effects, including nosebleeds, inflammation of the eyes and lips, bone spurs, hair loss, and **birth defects**. Tegison is stored in the body for an unknown length of time, and should not be taken by a woman who is pregnant or planning to become pregnant. A woman should use reliable birth control while taking Accutane and for at least one month before and after her course of treatment.

Cyclosporin emulsion (Neoral) is used to treat stubborn cases of severe psoriasis. Cyclosporin is also used to prevent rejection of transplanted organs, and Neoral, approved by the FDA in 1997, should be particularly beneficial to psoriasis patients who are young children or African-Americans, or those who have diabetes.

Other conventional treatments for psoriasis include:

- Capsaicin (*Capsicum frutescens*), an ointment that can stop production of the chemical that causes the skin to become inflamed and halts the runaway production of new skin cells. Capsaicin is available without a prescription, but should be used under a doctor's supervision to prevent **burns** and skin damage.
- Hydrocortisone creams, topical ointments containing a form of vitamin D called calcitriol, and coal-tar shampoos and ointments can relieve symptoms. Hydrocortisone creams have been associated with such side effects as **folliculitis** (inflammation of the hair follicles), while coal-tar preparations have been associated with a heightened risk of skin cancer.

Alternative treatment

Non-traditional psoriasis treatments include:

- Soaking in warm water and German chamomile (*Matricaria recutita*) or bathing in warm salt water.
- Drinking as many as three cups a day of hot tea made with one or a combination of the following herbs: burdock (*Arctium lappa*) root, dandelion

KEY TERMS

Arthritis—An inflammation of joints.

(*Taraxacum mongolicum*) root, Oregon grape (*Mahonia aquifolium*), sarsaparilla (*Smilax officinalis*), and balsam pear (*Momordica charantia*).

- Taking two 500-mg capsules of evening primrose oil (*Oenothera biennis*) a day. Pregnant women should not use evening primrose oil, and patients with **liver disease** or **high cholesterol** should use it only under a doctor's supervision.
- Eating a diet that includes plenty of fish, turkey, celery (for cleansing the kidneys), parsley, lettuce, lemons (for cleansing the liver), limes, fiber, and fruit and vegetable juices.
- Eating a diet that eliminates animal products high in saturated fats, since they promote inflammation.
- Drinking plenty of water (at least eight glasses) each day.
- Taking **nutritional supplements** including **folic acid**, lecithin, vitamin A (specific for the skin), vitamin E, selenium, and zinc.
- Regularly imagining clear, healthy skin.

Other helpful alternative approaches include identifying and eliminating food allergens from the diet, enhancing the function of the liver, augmenting the hydrochloric acid in the stomach, and completing a **detoxification** program. Constitutional homeopathic treatment, if properly prescribed, can also help resolve psoriasis.

Prognosis

Most cases of psoriasis can be controlled, and most people who have psoriasis can live normal lives.

Some people who have psoriasis are so self-conscious and embarrassed about their appearance that they become depressed and withdrawn. The Social Security Administration grants disability benefits to about 400 psoriasis patients each year, and a comparable number die from complications of the disease.

Prevention

A doctor should be notified if:

- psoriasis symptoms appear or reappear after treatment

- pustules erupt on the skin and the patient experiences **fatigue**, muscle aches, and **fever**
- unfamiliar, unexplained symptoms appear.

Resources

ORGANIZATIONS

American Academy of Dermatology. 930 N. Meacham Road, P.O. Box 4014, Schaumburg, IL 60168-4014. (847) 330-0230. Fax: (847) 330-0050. <<http://www.aad.org>>.

American Skin Association, Inc. 150 E. 58th St., 3rd floor, New York, NY 10155-0002. (212) 688-6547.

National Psoriasis Foundation. 6600 S.W. 92nd Ave., Suite 300, Portland, OR 97223. (800) 723-9166. <<http://www.psoriasis.org>>.

Maureen Haggerty

Psoriatic arthritis

Definition

Psoriatic arthritis is a form of arthritic joint disease associated with the chronic skin scaling and fingernail changes seen in **psoriasis**.

Description

Physicians recognize a number of different forms of psoriatic arthritis. In some patients, the arthritic symptoms will affect the small joints at the ends of the fingers and toes. In others, symptoms will affect joints on one side of the body but not on the other. In addition, there are patients whose larger joints on both sides of the body simultaneously become affected, as in rheumatoid arthritis. Some people with psoriatic arthritis experience arthritis symptoms in the back and spine; in rare cases, called psoriatic arthritis mutilans, the disease destroys the joints and bones, leaving patients with gnarled and club-like hands and feet. In many patients, symptoms of psoriasis precede the arthritis symptoms; a clue to possible joint disease is pitting and other changes in the fingernails.

Most people develop psoriatic arthritis at ages 35–45, but it has been observed earlier in adults and children. Both the skin and joint symptoms will come and go; there is no clear relationship between the severity of the psoriasis symptoms and arthritis **pain** at any given time. It is unclear how common psoriatic arthritis is. Recent surveys suggest that between 1 in 5

people and 1 in 2 people with psoriasis may also have some arthritis symptoms.

Causes and symptoms

The cause of psoriatic arthritis is unknown. As in psoriasis, genetic factors appear to be involved. People with psoriatic arthritis are more likely than others to have close relatives with the disease, but they are just as likely to have relatives with psoriasis but no joint disease. Researchers believe genes increasing the susceptibility to developing psoriasis may be located on chromosome 6p and chromosome 17, but the specific genetic abnormality has not been identified. Like psoriasis and other forms of arthritis, psoriatic arthritis also appears to be an autoimmune disorder, triggered by an attack of the body's own immune system on itself.

Symptoms of psoriatic arthritis include dry, scaly, silver patches of skin combined with joint pain and destructive changes in the feet, hands, knees, and spine. Tendon pain and nail deformities are other hallmarks of psoriatic arthritis.

Diagnosis

Skin and nail changes characteristic of psoriasis with accompanying arthritic symptoms are the hallmarks of psoriatic arthritis. A blood test for rheumatoid factor, antibodies that suggest the presence of **rheumatoid arthritis**, is negative in nearly all patients with psoriatic arthritis. X rays may show characteristic damage to the larger joints on either side of the body as well as fusion of the joints at the ends of the fingers and toes.

Treatment

Treatment for psoriatic arthritis is meant to control the skin lesions of psoriasis and the joint inflammation of arthritis. **Nonsteroidal anti-inflammatory drugs**, gold salts, and sulfasalazine are standard arthritis treatments, but have no effect on psoriasis. Antimalaria drugs and systemic corticosteroids should be avoided because they can cause dermatitis or exacerbate psoriasis when they are discontinued.

Several treatments are useful for both the **skin lesions** and the joint inflammation of psoriatic arthritis. Etretinate, a vitamin A derivative; methotrexate, a potent suppressor of the immune system; and ultraviolet light therapy have all been successfully used to treat psoriatic arthritis.

KEY TERMS

Psoriasis—A common recurring skin disease that is marked by dry, scaly, and silvery patches of skin that appear in a variety of sizes and locations on the body.

Psoriatic arthritis mutilans—A severe form of psoriatic arthritis that destroys the joints of the fingers and toes and causes the bones to fuse, leaving patients with gnarled and club-like hands and feet.

Rheumatoid arthritis—A systemic disease that primarily affects the joints, causing inflammation, changes in structure, and loss of function.

Rheumatoid factor—A series of antibodies that signal the presence of rheumatoid arthritis. May also be present in Sjögren's syndrome and systemic lupus erythematosus, among others.

Alternative treatment

Food allergies/intolerances are believed to play a role in most **autoimmune disorders**, including psoriatic arthritis. Identification and elimination of food allergens from the diet can be helpful. Constitutional homeopathy can work deeply and effectively with this condition, if the proper prescription is given. **Acupuncture**, Chinese herbal medicine, and western herbal medicine can all be useful in managing the symptoms of psoriatic arthritis. Nutritional supplements can contribute added support to the healing process. Alternative treatments recommended for psoriasis and rheumatoid arthritis may also be helpful in treating psoriatic arthritis.

Prognosis

The prognosis for most patients with psoriatic arthritis is good. For many the joint and other arthritis symptoms are much milder than those experienced in rheumatoid arthritis. One in five people with psoriatic arthritis, however, face potentially crippling joint disease. In some cases, the course of the arthritis can be far more mutilating than in rheumatoid arthritis.

Prevention

There are no preventive measures for psoriatic arthritis.

Resources

ORGANIZATIONS

American Academy of Dermatology. 930 N. Meacham Road, P.O. Box 4014, Schaumburg, IL 60168-4014. (847) 330-0230. Fax: (847) 330-0050. <<http://www.aad.org>>.

The American College of Rheumatology. 1800 Century Place, Suite 250, Atlanta, GA 30345. (404) 633-3777. <<http://www.rheumatology.org>>.

Richard H. Camer

PSP see **Progressive supranuclear palsy**

Psychiatric confinement

Definition

Psychiatric confinement is the use of restraints to detain a person in need of care and further evaluation.

Purpose

The purpose of restraint and confinement are crucial since they have a medico legal implication. The primary purpose for such intervention is typically an urgent or emergent condition that could cause danger to the affected person or others, or cause severe disability to the extent whereby the affected person is unable to care for his/herself.

Precautions

Clinicians utilizing this form of patient and public safety should perform a comprehensive mental status examination and document the findings. This approach can potentially provide clear records establishing the specific presenting problems and symptoms that can lead to ambiguities and potential legal action.

Description

Confinement with restraints can be categorized as urgent or emergent. Emergent causes can include patients exhibiting abnormal vital signs (breathing, pulse rate, temperature, blood pressure), threatening or violent behavior, and those who present with signs and symptoms of alcohol or illicit drug intoxication. Urgent use of confinement is indicated in patients showing suicidal thoughts, extreme **anxiety**, homicidal tendencies, violence, or a danger to self or the public at large.

Preparation

Those categorized as emergent should be prepared for further testing that can include blood chemistry and psychological assessment and evaluation. Initially the patient is restrained with four point leather restraints (both arms and both legs) and placed in a quiet room with a sitter. For those with urgent needs, restraint is initiated and initial management is directed to assess for an underlying medical cause and address psychological needs.

Aftercare

Further assessment, testing, and evaluation is necessary for a definitive diagnosis and devising an appropriate treatment plan.

Risks

A deficiency in record-keeping can lead to legal problems. The criteria and specifications for confinement should be clearly indicated. Meticulous clinical examination and documentation is essential for a definitive diagnosis. Persons who are confined due to **substance abuse** problems may have legal issues. As of 2001 there is proposed legislation concerning the misuse of restraints for psychiatric inpatients, which in the past has been responsible for numerous wrongful deaths. There are currently no federal laws that regulate the use of inpatient restraints nor any requirements for reporting injuries or **death**.

Resources

PERIODICALS

Reeves, R., et al. "Medicolegal Errors in the ED Related to the Involuntary Confinement of Psychiatric Patients." *American Journal of Emergency Medicine* November 1998.

Laith Farid Gulli, M.D.
Catherine Seeley
Keith Tatarelli, J.D.

Psychoanalysis

Definition

Psychoanalysis is a form of psychotherapy used by qualified psychotherapists to treat patients who have a range of mild to moderate chronic life problems. It is related to a specific body of theories about the

relationships between conscious and unconscious mental processes, and should not be used as a synonym for psychotherapy in general. Psychoanalysis is done one-on-one with the patient and the analyst; it is not appropriate for group work.

Purpose

Psychoanalysis is the most intensive form of an approach to treatment called psychodynamic therapy. Psychodynamic refers to a view of human personality that results from interactions between conscious and unconscious factors. The purpose of all forms of psychodynamic treatment is to bring unconscious mental material and processes into full consciousness so that the patient can gain more control over his or her life.

Classical psychoanalysis has become the least commonly practiced form of psychodynamic therapy because of its demands on the patient's time, as well as on his or her emotional and financial resources. It is, however, the oldest form of psychodynamic treatment. The theories that underlie psychoanalysis were worked out by Sigmund Freud (1856–1939), a Viennese physician, during the early years of the twentieth century. Freud's discoveries were made in the context of his research into hypnosis. The goal of psychoanalysis is the uncovering and resolution of the patient's internal conflicts. The treatment focuses on the formation of an intense relationship between the therapist and patient, which is analyzed and discussed in order to deepen the patient's insight into his or her problems.

Psychoanalytic psychotherapy is a modified form of psychoanalysis that is much more widely practiced. It is based on the same theoretical principles as psychoanalysis, but is less intense and less concerned with major changes in the patient's character structure. The focus in treatment is usually the patient's current life situation and the way problems relate to early conflicts and feelings, rather than an exploration of the unconscious aspects of the relationship that has been formed with the therapist.

Not all patients benefit from psychoanalytic treatment. Potential patients should meet the following prerequisites:

- The capacity to relate well enough to form an effective working relationship with the analyst. This relationship is called a therapeutic alliance.
- At least average intelligence and a basic understanding of psychological theory.

CARL GUSTAV JUNG (1875–1961)



(Library of Congress.)

Carl Gustav Jung was born in Kesswil, Switzerland, on July 26, 1875, to a Protestant clergyman who moved his family to Basel when Jung was four. While growing up,

Jung exhibited an interest in many diverse areas of study but finally decided to pursue medicine at the University of Basel and the University of Zurich, earning his degree in 1902. He also studied psychology in Paris. In 1903, Jung married Emma Rauschenbach, his companion and collaborator. The couple had five children.

Jung's professional career began in 1900 at the University of Zurich where he worked as an assistant to Eugene Bleuler in the psychiatric clinic. During his internship, he and some co-workers used an experiment that revealed groups of ideas in the unconscious psyche which he named *complexes*. Jung sent his publication *Studies in Word Association* (1904) to Sigmund Freud after finding his own beliefs confirmed by Freud's work. Jung and Freud became friends and collaborators until 1913 when Jung's ideas began to conflict with Freud's. During the time following this split, Jung published *Two Essays on Analytical Psychology* (1916, 1917) and *Psychological Types* (1921). Jung's later work developed from the concepts in his *Two Essays* publication and he became known as a founder of modern depth psychology.

In 1944, Jung gave up his psychological practice and his explorations after he suffered a severe heart attack. Jung received honorary doctorates from numerous universities and in 1948 he founded the C. G. Jung Institute in Zurich. Jung died on June 6, 1961.

- The ability to tolerate frustration, sadness, and other painful emotions.
- The capacity to distinguish between reality and fantasy.

People considered best suited to psychoanalytic treatment include those with depression, character disorders, neurotic conflicts, and chronic relationship problems. When the patient's conflicts are long-standing and deeply entrenched in his or her personality, psychoanalysis may be preferable to psychoanalytic psychotherapy, because of its greater depth.

Precautions

Psychoanalysis is not usually considered suitable for patients suffering from severe depression or such psychotic disorders as **schizophrenia**, although some analysts have successfully treated patients with psychoses. It is also not appropriate for people with addictions or substance dependency, disorders of aggression or impulse control, or acute crises; some of these people may benefit from psychoanalysis after the crisis has been resolved.

Description

In both psychoanalysis and psychoanalytic psychotherapy, the therapist does not tell the patient how to solve problems or offer moral judgments. The focus of treatment is exploration of the patient's mind and habitual thought patterns. Such therapy is termed "non-directed." It is also "insight-oriented," meaning that the goal of treatment is increased understanding of the sources of one's inner conflicts and emotional problems. The basic techniques of psychoanalytical treatment include:

Therapist neutrality

Neutrality means that the analyst does not take sides in the patient's conflicts, express feelings about the patient, or talk about his or her own life. Therapist neutrality is intended to help the patient stay focused on issues rather than be concerned with the therapist's reactions. In psychoanalysis, the patient lies on a couch facing away from the therapist. In psychodynamic psychotherapy, however, the patient and therapist usually sit in comfortable chairs facing each other.

Free association

Free association means that the patient talks about whatever comes into mind without censoring or editing the flow of ideas or memories. Free association allows the patient to return to earlier or more childlike emotional states (“regress”). Regression is sometimes necessary in the formation of the therapeutic alliance. It also helps the analyst to understand the recurrent patterns of conflict in the patient’s life.

Therapeutic alliance and transference

Transference is the name that psychoanalysts use for the patient’s repetition of childlike ways of relating that were learned in early life. If the therapeutic alliance has been well established, the patient will begin to transfer thoughts and feelings connected with siblings, parents, or other influential figures to the therapist. Discussing the transference helps the patient gain insight into the ways in which he or she misreads or misperceives other people in present life.

Interpretation

In psychoanalytic treatment, the analyst is silent as much as possible, in order to encourage the patient’s free association. However, the analyst offers judiciously timed interpretations, in the form of verbal comments about the material that emerges in the sessions. The therapist uses interpretations in order to uncover the patient’s resistance to treatment, to discuss the patient’s transference feelings, or to confront the patient with inconsistencies. Interpretations may be either focused on present issues (“dynamic”) or intended to draw connections between the patient’s past and the present (“genetic”). The patient is also often encouraged to describe dreams and fantasies as sources of material for interpretation.

Working through

“Working through” occupies most of the work in psychoanalytic treatment after the transference has been formed and the patient has begun to acquire insights into his or her problems. Working through is a process in which the new awareness is repeatedly tested and “tried on for size” in other areas of the patient’s life. It allows the patient to understand the influence of the past on his or her present situation, to accept it emotionally as well as intellectually, and to use the new understanding to make changes in present life.

Working through thus helps the patient to gain some measure of control over inner conflicts and to resolve them or minimize their power.

Although psychoanalytic treatment is primarily verbal, medications are sometimes used to stabilize patients with severe **anxiety**, depression, or other mood disorders during the analysis.

The cost of either psychoanalysis or psychoanalytic psychotherapy is prohibitive for most patients without insurance coverage. A full course of psychoanalysis usually requires three to five weekly sessions with a psychoanalyst over a period of three to five years. A course of psychoanalytic psychotherapy involves one to three meetings per week with the therapist for two to five years. Each session or meeting typically costs between \$80 and \$200, depending on the locale and the experience of the therapist. The increasing reluctance of most HMOs and other managed care organizations to pay for long-term psychotherapy is one reason that these forms of treatment are losing ground to short-term methods of treatment and the use of medications to control the patient’s emotional symptoms. It is also not clear as of 2003 that long-term psychoanalytically oriented approaches are more beneficial than briefer therapy methods for many patients. On the other hand, patients who can benefit from a psychoanalytic approach but cannot afford private fees may wish to contact the American Psychoanalytic Association, which maintains a list of analysts in training who offer treatment for reduced fees.

Preparation

Some patients may need evaluation for possible medical problems before entering psychoanalysis because numerous diseases—including virus infections and certain vitamin deficiencies—have emotional side effects or symptoms. The therapist will also want to know whether the patient is taking any prescription medications that may affect the patient’s feelings or ability to concentrate. In addition, it is important to make sure that the patient is not abusing drugs or alcohol.

Risks

The primary risk to the patient is related to the emotional **pain** resulting from new insights and changes in long-standing behavior patterns. In some patients, psychoanalysis produces so much anxiety that they cannot continue with this treatment method. In other cases, the therapist’s lack of skill or

KEY TERMS

Free association—A technique used in psychoanalysis in which the patient allows thoughts and feelings to emerge without trying to organize or censor them.

Interpretation—A verbal comment made by the analyst in response to the patient's free association. It is intended to help the patient gain new insights.

Neurosis—A mental and emotional disorder that affects only part of the personality and is accompanied by a significantly less distorted perception of reality than in psychosis.

Psychodynamic—An approach to psychotherapy based on the interplay of conscious and unconscious factors in the patient's mind. Psychoanalysis is one type of psychodynamic therapy.

Regression—The process in which the patient reverts to earlier or less mature feelings and behaviors.

Therapeutic alliance—The working relationship between a therapist and a patient that is necessary to the success of therapy.

Transference—The process that develops during psychoanalytic work during which the patient redirects feelings about early life figures toward the analyst.

Working through—The repeated testing of insights, which takes up most of the work in psychoanalysis after the therapeutic alliance has been formed.

differences in cultural background may prevent the formation of a solid therapeutic alliance.

Normal results

Psychoanalysis and psychoanalytic psychotherapy both have the goal of basic changes in the patient's personality structure and level of functioning, although psychoanalysis typically aims at more extensive and more profound change. In general, this approach to treatment is considered successful if the patient has shown:

- reduction in intensity or number of symptoms
- some resolution of basic emotional conflicts
- increased independence and self-esteem
- improved functioning and adaptation to life

Attempts to compare the effectiveness of psychoanalytical treatment to other modes of therapy are difficult to evaluate. Some aspects of Freudian theory have been questioned since the 1970s on the grounds of their limited applicability to women and to people from non-Western cultures. In particular, some psychiatrists with cross-cultural experience maintain that psychoanalysis presupposes a highly individualistic Western concept of human personhood that is alien to traditional Asian and African societies. There is, however, general agreement that psychoanalytic approaches work well for certain types of patients, specifically those with neurotic conflicts.

Resources

PERIODICALS

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International Journal of Psychoanalysis 84 (August 2003): 929–943.

Gabbard, G. O., and D. Westen. "Rethinking Therapeutic Action." *International Journal of Psychoanalysis* 84 (August 2003): 823–841.

Lombardi, R. "Mental Models and Language Registers in the Psychoanalysis of Psychosis: An Overview of a Thirteen-Year Analysis." *International Journal of Psychoanalysis* 84 (August 2003): 843–863.

Roland, A. "Psychoanalysis Across Civilizations: A Personal Journey." *Journal of the American Academy of Psychoanalysis and Dynamic Psychiatry* 31 (Summer 2003): 275–295.

ORGANIZATIONS

American Psychoanalytic Association. 309 East 49th Street, New York, NY 10017. (212) 752-0450. <<http://www.apsa-co.org>>.

Rebecca J. Frey, PhD

Psychological tests

Definition

Psychological tests are written, visual, or verbal evaluations administered to assess the cognitive and emotional functioning of children and adults.

Purpose

Psychological tests are used to assess a variety of mental abilities and attributes, including achievement and ability, personality, and neurological functioning.

Achievement and ability tests

For children, academic achievement, ability, and intelligence tests may be used as a tool in school placement, in determining the presence of a learning disability or a developmental delay, in identifying giftedness, or in tracking intellectual development. Intelligence testing may be used with adults to determine vocational ability (e.g., in career counseling) or to assess adult intellectual ability in the classroom.

Personality tests

Personality tests are administered for a wide variety of reasons, from diagnosing psychopathology (e.g., personality disorder, depressive disorder) to screening job candidates. They may be used in an educational or vocational setting to determine personality strengths and weaknesses, or in the legal system to evaluate parolees.

Neuropsychological tests

Patients who have experienced a traumatic brain injury, brain damage, or organic neurological problems (for example, **dementia**) are administered neuropsychological tests to assess their level of functioning and identify areas of mental impairment. They may also be used to evaluate the progress of a patient who has undergone treatment or **rehabilitation** for a neurological injury or illness. In addition, certain neuropsychological measures may be used to screen children for developmental delays and/or learning disabilities.

Precautions

Psychological testing requires a clinically trained examiner. All psychological tests should be administered, scored, and interpreted by a trained professional, preferably a psychologist or psychiatrist with expertise in the appropriate area.

Psychological tests are only one element of a psychological assessment. They should never be used alone as the sole basis for a diagnosis. A detailed history of the test subject and a review of psychological, medical, educational, or other relevant records are required to lay the groundwork for interpreting the results of any psychological measurement.

Cultural and language differences in the test subject may affect test performance and may result in inaccurate test results. The test administrator should be informed before psychological testing begins if the test taker is not fluent in English and/or belongs to a

minority culture. In addition, the subject's motivation and motives may also affect test results.

Description

Psychological tests are formalized measures of mental functioning. Most are objective and quantifiable; however, certain projective tests may involve some level of subjective interpretation. Also known as inventories, measurements, questionnaires, and scales, psychological tests are administered in a variety of settings, including preschools, primary and secondary schools, colleges and universities, hospitals, outpatient healthcare settings, social agencies, prisons, and employment or human resource offices. They come in a variety of formats, including written, verbal, and computer administered.

Achievement and ability tests

Achievement and ability tests are designed to measure the level of an individual's intellectual functioning and cognitive ability. Most achievement and ability tests are standardized, meaning that norms were established during the design phase of the test by administering the test to a large representative sample of the test population. Achievement and ability tests follow a uniform testing protocol, or procedure (i.e., test instructions, test conditions, and scoring procedures) and their scores can be interpreted in relation to established norms. Common achievement and ability tests include the **Wechsler intelligence test** (WISC-III and WAIS) and the **Stanford-Binet intelligence scales**.

Personality tests

Personality tests and inventories evaluate the thoughts, emotions, attitudes, and behavioral traits that comprise personality. The results of these tests determine an individual's personality strengths and weaknesses, and may identify certain disturbances in personality, or psychopathology. Tests such as the *Minnesota multiphasic personality inventory* (MMPI-2) and the *Millon clinical multiaxial Inventory III* (MMPI-III), are used to screen individuals for specific psychopathologies or emotional problems.

Another type of personality test is the projective personality assessment. A projective test asks a subject to interpret some ambiguous stimuli, such as a series of inkblots. The subject's responses provide insight into his or her thought processes and personality traits. For example, the Rorschach inkblot test and the **Holtzman ink blot test** (HIT) use a series of inkblots that the test

KEY TERMS

Norms—A fixed or ideal standard; normative or mean score for a particular age group.

Psychopathology—A mental disorder or illness, such as schizophrenia, personality disorder, or major depressive disorder.

Quantifiable—Can be expressed as a number. The results of quantifiable psychological tests can be translated into numerical values, or scores.

Representative sample—A random sample of people that adequately represent the test taking population in age, gender, race, and socioeconomic standing.

Standardization—The process of determining established norms and procedures for a test to act as a standard reference point for future test results.

subject is asked to identify. Another projective assessment, the **Thematic apperception test** (TAT), asks the subject to tell a story about a series of pictures. Some consider projective tests to be less reliable than objective personality tests. If the examiner is not well-trained in psychometric evaluation, subjective interpretations may affect the evaluation of these tests.

Neuropsychological tests

Many insurance plans cover all or a portion of diagnostic neuropsychological or psychological testing. As of 1997, Medicare reimbursed for psychological and neuropsychological testing. Billing time typically includes test administration, scoring and interpretation, and reporting.

Preparation

Prior to the administration of any psychological test, the administrator should provide the test subject with information on the nature of the test and its intended use, complete standardized instructions for taking the test (including any time limits and penalties for incorrect responses), and information on the confidentiality of the results. After these disclosures are made, informed consent should be obtained from the test subject before testing begins (except in cases of legally mandated testing, where consent is not required of the subject).

Normal results

All psychological and neuropsychological assessments should be administered, scored, and interpreted by a trained professional. When interpreting test results for test subjects, the test administrator will review with subjects: what the test evaluates, its precision in evaluation, any margins of error involved in scoring, and what the individual scores mean in the context of overall test norms and the background of the test subject.

Resources

ORGANIZATIONS

American Psychological Association (APA). 750 First St. NE, Washington, DC 20002-4242. (202) 336-5700. < <http://www.apa.org> >.

Paula Anne Ford-Martin

Psychosis

Definition

Psychosis is a symptom or feature of mental illness typically characterized by radical changes in personality, impaired functioning, and a distorted or non-existent sense of objective reality.

Description

Patients suffering from psychosis have impaired reality testing; that is, they are unable to distinguish personal subjective experience from the reality of the external world. They experience **hallucinations** and/or **delusions** that they believe are real, and may behave and communicate in an inappropriate and incoherent fashion. Psychosis may appear as a symptom of a number of mental disorders, including mood and **personality disorders**. It is also the defining feature of **schizophrenia**, schizophreniform disorder, **schizoaffective disorder**, delusional disorder, and the psychotic disorders (i.e., brief psychotic disorder, shared psychotic disorder, psychotic disorder due to a general medical condition, and substance-induced psychotic disorder).

Causes and symptoms

Psychosis may be caused by the interaction of biological and psychosocial factors depending on the

disorder in which it presents; psychosis can also be caused by purely social factors, with no biological component.

Biological factors that are regarded as contributing to the development of psychosis include genetic abnormalities and substance use. With regard to chromosomal abnormalities, studies indicate that 30% of patients diagnosed with a psychotic disorder have a microdeletion at chromosome 22q11. Another group of researchers has identified the gene G72/G30 at chromosome 13q33.2 as a susceptibility gene for childhood-onset schizophrenia and psychosis not otherwise specified.

With regard to **substance abuse**, several different research groups reported in 2004 that cannabis (**marijuana**) use is a risk factor for the onset of psychosis.

Migration is a social factor that influences people's susceptibility to psychotic disorders. Psychiatrists in Europe have noted the increasing rate of schizophrenia and other psychotic disorders among immigrants to almost all Western European countries. Black immigrants from Africa or the Caribbean appear to be especially vulnerable. The stresses involved in migration include family breakup, the need to adjust to living in large urban areas, and social inequalities in the new country.

Schizophrenia, schizophreniform disorder, and schizoaffective disorder

Psychosis in schizophrenia and perhaps schizophreniform disorder appears to be related to abnormalities in the structure and chemistry of the brain, and appears to have strong genetic links; but its course and severity can be altered by social factors such as **stress** or a lack of support within the family. The cause of schizoaffective disorder is less clear cut, but biological factors are also suspected.

Delusional disorder

The exact cause of delusional disorder has not been conclusively determined, but potential causes include heredity, neurological abnormalities, and changes in brain chemistry. Some studies have indicated that delusions are generated by abnormalities in the limbic system, the portion of the brain on the inner edge of the cerebral cortex that is believed to regulate emotions. Delusional disorder is also more likely to develop in persons who are isolated from others in their society by language difficulties and/or cultural differences.

Brief psychotic disorder

Trauma and stress can cause a short-term psychosis (less than a month's duration) known as brief psychotic disorder. Major life-changing events such as the **death** of a family member or a natural disaster have been known to stimulate brief psychotic disorder in patients with no prior history of mental illness.

Psychotic disorder due to a general medical condition

Psychosis may also be triggered by an organic cause, termed a psychotic disorder due to a general medical condition. Organic sources of psychosis include neurological conditions (for example, epilepsy and cerebrovascular disease), metabolic conditions (for example, porphyria), endocrine conditions (for example, hyper- or **hypothyroidism**), renal failure, electrolyte imbalance, or **autoimmune disorders**.

Substance-induced psychotic disorder

Psychosis is also a known side effect of the use, **abuse**, and withdrawal from certain drugs. So-called recreational drugs, such as hallucinogenics, PCP, amphetamines, **cocaine**, marijuana, and alcohol, may cause a psychotic reaction during use or withdrawal. Certain prescription medications such as steroids, anticonvulsants, chemotherapeutic agents, and antiparkinsonian medications may also induce psychotic symptoms. Toxic substances such as carbon monoxide have also been reported to cause substance-induced psychotic disorder.

Shared psychotic disorder

Shared psychotic disorder, also known as *folie à deux* or psychosis by association, is a relatively rare delusional disorder involving two (or more) people with close emotional ties. In the West, shared psychosis most commonly develops between two sisters or between husband and wife, while in Japan the most common form involves a parent and a son or daughter. Shared psychosis occasionally involves an entire nuclear family.

Psychosis is characterized by the following symptoms:

- **Delusions.** Those delusions that occur in schizophrenia and its related forms are typically bizarre (i.e., they could not occur in real life). Delusions occurring in delusional disorder are more plausible, but still patently untrue. In some cases, delusions may be accompanied by feelings of **paranoia**.

- Hallucinations. Psychotic patients see, hear, smell, taste, or feel things that aren't there. Schizophrenic hallucinations are typically auditory or, less commonly, visual; but psychotic hallucinations can involve any of the five senses.
- Disorganized speech. Psychotic patients, especially those with schizophrenia, often ramble on in incoherent, nonsensical speech patterns.
- Disorganized or catatonic behavior. The catatonic patient reacts inappropriately to his/her environment by either remaining rigid and immobile or by engaging in excessive motor activity. Disorganized behavior is behavior or activity that is inappropriate for the situation, or unpredictable.

Diagnosis

Patients with psychotic symptoms should undergo a thorough **physical examination** and history to rule out such possible organic causes as seizures, **delirium**, or alcohol withdrawal, and such other psychiatric conditions as dissociation or panic attacks. If a psychiatric cause such as schizophrenia is suspected, a mental health professional will typically conduct an interview with the patient and administer one of several clinical inventories, or tests, to evaluate mental status. This assessment takes place in either an outpatient or hospital setting.

Psychotic symptoms and behaviors are considered psychiatric emergencies, and persons showing signs of psychosis are frequently taken by family, friends, or the police to a hospital emergency room. A person diagnosed as psychotic can be legally hospitalized against his or her will, particularly if he or she is violent, threatening to commit **suicide**, or threatening to harm another person. A psychotic person may also be hospitalized if he or she has become malnourished or ill as a result of failure to feed, dress appropriately for the climate, or otherwise take care of him- or herself.

Treatment

Psychosis that is symptomatic of schizophrenia or another psychiatric disorder should be treated by a psychologist and/or psychiatrist. An appropriate course of medication and/or psychosocial therapy is employed to treat the underlying primary disorder. If the patient is considered to be at risk for harming himself or others, inpatient treatment is usually recommended.

Treatment of shared psychotic disorder involves separating the affected persons from one another as

well as using antipsychotic medications and psychotherapy.

Antipsychotic medication such as thioridazine (Mellaril), haloperidol (Haldol), chlorpromazine (Thorazine), clozapine (Clozaril), sertindole (Serlect), olanzapine (Zyprexa), or risperidone (Risperdal) is usually prescribed to bring psychotic symptoms under control and into remission. Possible side effects of antipsychotics include **dry mouth**, drowsiness, muscle stiffness, and **tardive dyskinesia** (involuntary movements of the body). Agranulocytosis, a potentially serious but reversible health condition in which the white blood cells that fight infection in the body are destroyed, is a possible side effect of clozapine. Patients treated with this drug should undergo weekly blood tests to monitor white blood cell counts for the first six months, then every two weeks.

After an acute psychotic episode has subsided, antipsychotic drug maintenance treatment is typically employed and psychosocial therapy and living and vocational skills training may be attempted.

Prognosis

Prognosis for brief psychotic disorder is quite good; for schizophrenia, less so. Generally, the longer and more severe a psychotic episode, the poorer the prognosis is for the patient. Early diagnosis and treatment are critical to improving outcomes for the patient across all psychotic disorders.

Approximately 10% of America's permanently disabled population is comprised of schizophrenic individuals. The mortality rate of schizophrenic individuals is also high—approximately 10% of schizophrenics commit suicide, and 20% attempt it. However, early diagnosis and long-term follow up care can improve the outlook for these patients considerably. Roughly 60% of patients with schizophrenia will show substantial improvement with appropriate treatment.

Resources

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- Beers, Mark H., MD, and Robert Berkow, MD., editors. "Psychiatric Emergencies." In *The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck Research Laboratories, 2004.
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KEY TERMS

Brief psychotic disorder—An acute, short-term episode of psychosis lasting no longer than one month. This disorder may occur in response to a stressful event.

Delirium—An acute but temporary disturbance of consciousness marked by confusion, difficulty paying attention, delusions, hallucinations, or restlessness. Delirium may be caused by drug intoxication, high fever related to infection, head trauma, brain tumors, kidney or liver failure, or various metabolic disturbances.

Delusional disorder—Individuals with delusional disorder suffer from long-term, complex delusions that fall into one of six categories: persecutory, grandiose, jealousy, erotomanic, somatic, or mixed.

Delusions—An unshakable belief in something untrue which cannot be explained by religious or cultural factors. These irrational beliefs defy normal reasoning and remain firm even when overwhelming proof is presented to refute them.

Hallucinations—False or distorted sensory experiences that appear to be real perceptions to the person experiencing them.

Paranoia—An unfounded or exaggerated distrust of others, sometimes reaching delusional proportions.

Porphyria—A disease of the metabolism characterized by skin lesions, urine problems, neurologic disorders, and/or abdominal pain.

Schizoaffective disorder—Schizophrenic symptoms occurring concurrently with a major depressive and/or manic episode.

Schizophrenia—A debilitating mental illness characterized by delusions, hallucinations, disorganized speech and behavior, and inappropriate or flattened affect (a lack of emotions) that seriously hampers the afflicted individual's social and occupational functioning. Approximately 2 million Americans suffer from schizophrenia.

Schizophreniform disorder—A short-term variation of schizophrenia that has a total duration of one to six months.

Shared psychotic disorder—Also known as *folie à deux*, shared psychotic disorder is an uncommon disorder in which the same delusion is shared by two or more individuals.

Tardive dyskinesia—Involuntary movements of the face and/or body which are a side effect of the long-term use of some older antipsychotic (neuroleptic) drugs. Tardive dyskinesia affects 15-20% of patients on long-term neuroleptic treatment.

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ORGANIZATIONS

American Psychiatric Association. 1400 K Street NW, Washington DC 20005. (888) 357-7924. <<http://www.psych.org>>.

American Psychological Association (APA). 750 First St. NE, Washington, DC 20002-4242. (202) 336-5700. <<http://www.apa.org>>.

National Alliance for the Mentally Ill (NAMI). Colonial Place Three, 2107 Wilson Blvd., Ste. 300, Arlington, VA 22201-3042. (800) 950-6264. <<http://www.nami.org>>.

National Institute of Mental Health (NIMH). 6001 Executive Boulevard, Room 8184, MSC 9663, Bethesda, MD 20892-9663. (301) 443-4513. <<http://www.nimh.nih.gov>>.

OTHER

The Schizophrenia Page. < <http://www.schizophrenia.com> >.

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Psychosocial disorders

Definition

A psychosocial disorder is a mental illness caused or influenced by life experiences, as well as maladjusted cognitive and behavioral processes.

Description

The term psychosocial refers to the psychological and social factors that influence mental health. Social influences such as peer pressure, parental support, cultural and religious background, socioeconomic status, and interpersonal relationships all help to shape personality and influence psychological makeup. Individuals with psychosocial disorders frequently have difficulty functioning in social situations and may have problems effectively communicating with others.

In the American Psychiatric Association it distinguishes 16 different subtypes (or categories) of mental illness. Although psychosocial variables arguably have some degree of influence on all subtypes of mental illness, the major categories of mental disorders thought to involve significant psychosocial factors include:

- **Substance-related disorders.** Disorders related to alcohol and drug use, **abuse**, dependence, and withdrawal.
- **Schizophrenia** and other psychotic disorders. These include the schizoid disorders (schizophrenia, schizopreniform, and **schizoaffective disorder**), delusional disorder, and psychotic disorders.
- **Mood disorders.** Affective disorders such as depression (major, dysthymic) and bipolar disorders.
- **Anxiety disorders.** Disorders in which a certain situation or place triggers excessive fear and/or anxiety symptoms (i.e., **dizziness**, racing heart), such as **panic disorder**, **agoraphobia**, social phobia, **obsessive-compulsive disorder**, **post-traumatic stress disorder**, and generalized anxiety disorders.

- **Somatoform disorders.** Somatoform disorders involve clinically significant physical symptoms that cannot be explained by a medical condition (e.g., somatization disorder, conversion disorder, **pain disorder**, **hypochondriasis**, and body dysmorphic disorder).
- **Factitious disorders.** Disorders in which an individual creates and complains of symptoms of a non-existent illness in order to assume the role of a patient (or sick role).
- **Sexual and gender identity disorders.** Disorders of sexual desire, arousal, and performance. It should be noted that the categorization of **gender identity disorder** as a mental illness has been a point of some contention among mental health professionals.
- **Eating disorders.** Anorexia and bulimia nervosa.
- **Adjustment disorders.** Adjustment disorders involve an excessive emotional or behavioral reaction to a stressful event.
- **Personality disorders.** Maladjustments of personality, including paranoid, schizoid, schizotypal, antisocial, borderline, histrionic, narcissistic, avoidant, dependent, and obsessive-compulsive personality disorder (not to be confused with the anxiety disorder OCD).
- Disorders usually first diagnosed in infancy childhood, or adolescence. Some learning and developmental disorders (i.e., **ADHD**) may be partially psychosocial in nature.

Causes and symptoms

It is important to note that the causes of mental illness are diverse and not completely understood. The majority of psychological disorders are thought to be caused by a complex combination of biological, genetic (hereditary), familial, and social factors or biopsychosocial influences. In addition, the role that each of these play can differ from person to person, so that a disorder such as depression that is caused by genetic factors in one person may be caused by a traumatic life event in another.

The symptoms of psychosocial disorders vary depending on the diagnosis in question. In addition to disorder-specific symptoms, individuals with psychosocial dysfunction usually have difficulty functioning normally in social situations and may have trouble forming and maintaining close interpersonal relationships.

Diagnosis

Patients with symptoms of psychosocial disorders or other mental illness should undergo a thorough **physical examination** and patient history to rule out an organic cause for the illness (such as a neurological disorder). If no organic cause is suspected, a psychologist or other mental healthcare professional will meet with the patient to conduct an interview and take a detailed social and medical history. If the patient is a minor, interviews with a parent or guardian may also be part of the diagnostic process. The physician may also administer one or more **psychological tests** (also called clinical inventories, scales, or assessments).

Treatment

Counseling is typically a front-line treatment for psychosocial disorders. A number of counseling or talk therapy approaches exist, including psychotherapy, cognitive therapy, behavioral therapy, and **group therapy**. Therapy or counseling may be administered by social workers, nurses, licensed counselors and therapists, psychologists, or psychiatrists.

Psychoactive medication may also be prescribed for symptom relief in patients with mental disorders considered psychosocial in nature. For disorders such as major depression or **bipolar disorder**, which may have psychosocial aspects but also have known organic causes, drug therapy is a primary treatment approach. In cases such as personality disorder that are thought to not have biological roots, psychoactive medications are usually considered a secondary, or companion treatment to psychotherapy.

Many individuals are successful in treating psychosocial disorders through regular attendance in self-help groups or 12-step programs such as Alcoholics Anonymous. This approach, which allows individuals to seek advice and counsel from others in similar circumstances, can be extremely effective.

In some cases, treating mental illness requires hospitalization of the patient. This hospitalization, also known as inpatient treatment, is usually employed in situations where a controlled therapeutic environment is critical for the patient's recovery (e.g., **rehabilitation** treatment for **alcoholism** or other drug addictions), or when there is a risk that the patient may harm himself (**suicide**) or others. It may also be necessary when the patient's physical health has deteriorated to a point where life-sustaining treatment is necessary, such as with severe **malnutrition** associated with **anorexia nervosa**.

KEY TERMS

Affective disorder—An emotional disorder involving abnormal highs and/or lows in mood.

Bipolar disorder—An affective mental illness that causes radical emotional changes and mood swings, from manic highs to depressive lows. The majority of bipolar individuals experience alternating episodes of mania and depression.

Bulimia—An eating disorder characterized by binge eating and inappropriate compensatory behavior such as vomiting, misusing laxatives, or excessive exercise.

Cognitive processes—Thought processes (i.e., reasoning, perception, judgment, memory).

Learning disorders—Academic difficulties experienced by children and adults of average to above-average intelligence that involve reading, writing, and/or mathematics, and which significantly interfere with academic achievement or daily living.

Schizophrenia—A debilitating mental illness characterized by delusions, hallucinations, disorganized speech and behavior, and flattened affect (i.e., a lack of emotions) that seriously hampers normal functioning.

Alternative treatment

Therapeutic approaches such as **art therapy** that encourage self-discovery and empowerment may be useful in treating psychosocial disorders. Art therapy, the use of the creative process to express and understand emotion, encompasses a broad range of humanistic disciplines, including visual arts, dance, drama, music, film, writing, literature, and other artistic genres. This use of the creative process is believed to provide the patient/artist with a means to gain insight to emotions and thoughts they might otherwise have difficulty expressing. After the artwork is created, the patient/artist continues the therapeutic journey by interpreting its meaning under the guidance of a trained therapist.

Prognosis

According to the National Institute of Mental Health, more than 90% of Americans who commit suicide have a diagnosable mental disorder, so swift and appropriate treatment is important. Because of the diversity of types of mental disorders influenced

by psychosocial factors, and the complexity of diagnosis and treatment, the prognosis for psychosocial disorders is highly variable. In some cases, they can be effectively managed with therapy and/or medication. In others, mental illness can cause long-term disability.

Prevention

Patient education (i.e., therapy or self-help groups) can encourage patients to take an active part in their treatment program and to recognize symptoms of a relapse of their condition. In addition, educating friends and family members on the nature of the psychosocial disorder can assist them in knowing how and when to provide support to the patient.

Resources

PERIODICALS

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ORGANIZATIONS

National Institute of Mental Health. 6001 Executive Boulevard, Rm. 8184, MSC 9663, Bethesda, MD 20892-9663. (301) 443-4513.

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Psychosurgery

Definition

Psychosurgery involves severing or otherwise disabling areas of the brain to treat a personality disorder, behavior disorder, or other mental illness. Modern psychosurgical techniques target the pathways between the limbic system (the portion of the brain on the inner edge of the cerebral cortex) that is believed to regulate emotions, and the frontal cortex, where thought processes are seated.

Purpose

Lobotomy is a psychosurgical procedure involving selective destruction of connective nerve fibers or tissue. It is performed on the frontal lobe of the

brain and its purpose is to alleviate mental illness and chronic **pain** symptoms. The bilateral cingulotomy, a modern psychosurgical technique which has replaced the lobotomy, is performed to alleviate mental disorders such as major depression, bipolar disorder, or **obsessive-compulsive disorder** (OCD), which have not responded to psychotherapy, behavioral therapy, electroshock, or pharmacologic treatment. Bilateral cingulotomies are also performed to treat chronic pain in **cancer** patients.

Precautions

Psychosurgery should be considered only after all other non-surgical psychiatric therapies have been fully explored. Much is still unknown about the biology of the brain and how psychosurgery affects brain function.

Description

Psychosurgery, and lobotomy in particular, reached the height of use just after World War II. Between 1946 and 1949, the use of the lobotomy grew from 500 to 5,000 annual procedures in the United States. At that time, the procedure was viewed as a possible solution to the overcrowded and understaffed conditions in state-run mental hospitals and asylums. Known as prefrontal or transorbital lobotomy, depending on the surgical technique used and area of the brain targeted, these early operations were performed with surgical knives, electrodes, suction, or ice picks, to cut or sweep out portions of the frontal lobe.

Today's psychosurgical techniques are much more refined. Instead of going in "blind" to remove large sections on the frontal lobe, as in these early operations, neurosurgeons use a computer-based process called stereotactic magnetic resonance imaging to guide a small electrode to the limbic system (brain structures involved in autonomic or automatic body functions and some emotion and behavior). There an electrical current **burns** in a small lesion [usually 0.5 in (1.3 cm) in size]. In a bilateral cingulotomy, the cingulate gyrus, a small section of brain that connects the limbic region of the brain with the frontal lobes, is targeted. Another surgical technique uses a non-invasive tool known as a gamma knife to focus beams of radiation at the brain. A lesion forms at the spot where the beams converge in the brain.

Preparation

Candidates for cingulotomies or other forms of psychosurgery undergo a rigorous screening process

KEY TERMS

Gamma knife—A surgical tool that focuses beams of radiation at the head, which converge in the brain to form a lesion.

Lesion—Any discontinuity of tissue. Often a cut or wound.

Limbic system—A portion of the brain on the inner edge of the cerebral cortex that is thought to regulate emotions.

Psychosurgery—Brain surgery performed to alleviate chronic psychological conditions such as obsessive-compulsive disorder (OCD), depression, and bipolar disorder.

Stereotactic technique—A technique used by neurosurgeons to pinpoint locations within the brain. It employs computer imaging to create an external frame of reference.

to ensure that all possible non-surgical psychiatric treatment options have been explored. Psychosurgery is only performed with the patient's informed consent.

Aftercare

Ongoing behavioral and medication therapy is often required in OCD patients who undergo cingulotomy. All psychosurgery patients should remain under a psychiatrist's care for follow-up evaluations and treatment.

Risks

As with any type of brain surgery, psychosurgery carries the risk of permanent brain damage, though the advent of non-invasive neurosurgical techniques, such as the gamma knife, has reduced the risk of brain damage significantly.

Normal results

In a 1996 study at Massachusetts General Hospital, over one-third of patients undergoing cingulotomy demonstrated significant improvements after the surgery. And, in contrast to the bizarre behavior and personality changes reported with lobotomy patients in the 1940s and 1950s, modern psychosurgery patients have demonstrated little post-surgical losses of memory or other high level thought processes.

Resources

ORGANIZATIONS

Massachusetts General Hospital. Functional and Stereotactic Neurosurgery Cingulotomy Unit. Fruit St., Boston, MA 02114. (617) 726-2000. <<http://neurosurgery.mgh.harvard.edu/cingulot.htm>>.

National Alliance for the Mentally Ill (NAMI). Colonial Place Three, 2107 Wilson Blvd., Ste. 300, Arlington, VA 22201-3042. (800) 950-6264. <<http://www.nami.org>>.

National OCD Headquarters. P.O. Box 70, Milford, CT 06460. (203) 878-5669.

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Psyllium preparations see **Laxatives**

PT see **Prothrombin time**

Pterygium see **Pinguecula and pterygium**

Ptomaine poisoning see **Food poisoning**

Ptosis

Definition

Ptosis is the term used for a drooping upper eyelid. Ptosis, also called blepharoptosis, can affect one or both eyes.

Description

The eyelids serve to protect and lubricate the outer eye. The upper eyelid is lifted by a muscle called the levator muscle. Inside the back part of the lid is a tarsal plate which adds rigidity to the lid. The levator muscle is attached to the tarsal plate by a flat tendon called the levator aponeurosis. When the muscle cannot lift the eyelid or lifts it only partially, the person is said to have a ptosis.

There are two types of ptosis, acquired and congenital. Acquired ptosis is more common. Congenital ptosis is present at birth. Both congenital and acquired ptosis can be, but are not necessarily, hereditary.

Causes and symptoms

Ptosis may occur because the levator muscle's attachment to the lid is weakening with age. Acquired ptosis can also be caused by a number of different things, such as disease that impairs the nerves, diabetes, injury, tumors, inflammation, or



A close-up view of a drooping upper eyelid (ptosis) on an elderly woman's face. Ptosis is normally due to a weakness of the levator muscle of the upper eyelid or to interference with the nerve supply to the muscle. (Photograph by Dr. P. Marazzi, Photo Researchers, Inc. Reproduced by permission.)

aneurysms. Congenital ptosis may be caused by a problem with nerve innervation or a weak muscle. Drooping eyelids may also be the result of diseases such as **myotonic dystrophy** or **myasthenia gravis**.

The primary symptom of ptosis is a drooping eyelid. Adults will notice a loss of visual field because the upper portion of the eye is covered. Children who are born with a ptosis usually tilt their head back in an effort to see under the obstruction. Some people raise their eyebrows in order to lift the lid slightly and therefore may appear to be frowning.

Diagnosis

Diagnosis of ptosis is usually made by observing the drooping eyelid. Finding the cause of the condition will require testing for any of the illnesses or injuries known to have this effect. Some possible tests include x rays and blood tests.

Treatment

Ptosis is usually treated surgically. Surgery can generally be done on an outpatient basis under local anesthetic. For minor drooping, a small amount of the eyelid tissue can be removed. For more pronounced ptosis the approach is to surgically shorten the levator muscle or connect the lid to the muscles of the eyebrow. Or, the aponeurosis can be reattached to the tarsal plate if it has separated. Correcting the ptosis is usually done only after determining the cause of the condition. For example, myasthenia gravis must be ruled out before performing any surgery. As with

KEY TERMS

Congenital—A condition existing at birth.

Hereditary—A condition passed from parent to child. In other words, a genetic condition.

any surgery, there are risks, and they should be discussed with the surgeon.

Children with ptosis need not have surgery immediately, however their vision must be checked periodically to prevent lazy eye (**amblyopia**).

“Ptosis crutches” are also available. These can be attached to the frame of eyeglasses to hold up the eyelid. These devices are uncomfortable and usually not well tolerated.

Prognosis

After diagnosing the cause of a drooping eyelid, then correcting the condition, most people have no further problems related to the ptosis. The correction, however, may still not make the eyes symmetrical. Patients should have reasonable expectations and discuss the outcome with their doctor prior to surgery.

Prevention

Ptosis cannot be prevented.

Resources

ORGANIZATIONS

American Academy of Ophthalmology. 655 Beach Street, P.O. Box 7424, San Francisco, CA 94120-7424. < <http://www.eyenet.org> >.

American Medical Association. 515 N. State St., Chicago, IL 60612. (312) 464-5000. < <http://www.ama-assn.org> >.

American Optometric Association. 243 North Lindbergh Blvd., St. Louis, MO 63141. (314) 991-4100. < <http://www.aoanet.org> >.

U.S. Department of Health and Human Services. 200 Independence Ave., SW, Washington, DC 20201, (202) 619-0257

Dorothy Elinor Stonely

PTSD see **Post-traumatic stress disorder**

PTT see **Partial thromboplastin time**

Puberty

Definition

Puberty is the period of human development during which physical growth and sexual maturation occurs.

Description

Beginning as early as age eight in girls—and two years later, on average, in boys—the hypothalamus (part of the brain) signals hormonal change that stimulates the pituitary. In turn, the pituitary releases its own hormones called gonadotrophins that stimulate the gonads and adrenals. From these glands come a flood of sex hormones—androgens and testosterone in the male, estrogens and progestins in the female—that regulate the growth and function of the sex organs. It is interesting to note that the gonadotrophins are the same for males and females, but the sex hormones they induce are different.

In the United States, the first sign of puberty occurs on average at age 11 in girls, with menstruation and fertility following about two years later. Boys lag behind by about two years. Puberty may not begin until age 16 in boys and continue in a desultory fashion on past age 20. In contrast to puberty, adolescence is more of a social/cultural term referring to the interval between childhood and adulthood.

Diagnosis

Puberty has been divided into five Sexual Maturity Rating (SMR) stages by two doctors, W. Marshall and J. M. Tanner. These ratings are often referred to as Tanner Stages 1–5. Staging is based on pubic hair growth, on male genital development, and female breast development. Staging helps determine whether development is normal for a given age. Both sexes also grow axillary (arm pit) hair and pimples. Males develop muscle mass, a deeper voice, and facial hair. Females redistribute body fat. Along with the maturing of the sex organs, there is a pronounced growth spurt averaging 3–4 in (8–10 cm) and culminating in full adult stature. Puberty can be precocious (early) or delayed. It all depends upon the sex hormones.

Puberty falling outside the age limits considered normal for any given population should prompt a search for the cause. As health and **nutrition** have improved over the past few generations, there has been a gradual decrease in the average age for the normal onset of puberty.

- Excess hormone stimulation is the cause for precocious puberty. It can come from the brain in the form of gonadotrophins or from the gonads and adrenals. Overproduction may be caused by functioning tumors or simple overactivity. Brain overproduction can also be the result of brain infections or injury.
- Likewise, delayed puberty is due to insufficient hormone. If the pituitary output is inadequate, so will be the output from the gonads and adrenals. On the other hand, a normal pituitary will overproduce if it senses there are not enough hormones in the circulation.
- There are several congenital disorders (polyglandular deficiency syndromes) that include failure of hormone output. These children do not experience normal puberty, but it may be induced by giving them the proper hormones at the proper time.
- Finally, there are in females abnormalities in hormone production that produce male characteristics—so called virilizing syndromes. Should one of these appear during adolescence, it will disturb the normal progress of puberty. Notice that virilizing requires abnormal hormones in the female, while feminizing results from absent hormones in the male. Each embryo starts out life as female. Male hormones transform it if they are present.

Delayed or **precocious puberty** requires measurement of the several hormones involved to determine which are lacking or which are in excess. There are blood tests for each one. If a tumor is suspected, imaging of the suspect organ needs to be done with x rays, **computed tomography scans** (CT scans), or magnetic resonance imaging (MRI).

Treatment

Puberty is a period of great **stress**, both physically and emotionally. The psychological changes and challenges of puberty are made infinitely greater if its timing is off.

In precocious puberty, the offending gland or tumor may require surgical attention, although there are several drugs now that counteract hormone effects. If delayed, puberty can be stimulated with the correct hormones. Treatment should not be delayed because necessary bone growth is also affected.

Prognosis

Properly administered hormones can restore the normal growth pattern.

KEY TERMS

Adrenals—Glands on top of the kidneys that produce four different types of hormones.

Computed tomography scan (CT)—A method of creating images of internal organs using x rays.

Embryo—The life in the womb during the first two months.

Hormone—A chemical produced in one place that has an effect somewhere else in the body.

Hypothalamus—Part of the brain located deep in the center of the skull and just above the pituitary.

Gonads—Glands that make sex hormones and reproductive cells—testes in the male, ovaries in the female.

Magnetic resonance imaging (MRI)—A method of creating images of internal organs. Magnetic resonance imaging (MRI) uses magnet fields and radio-frequency signals.

Pituitary—The “master gland” of the body, controlling many of the others by releasing stimulating hormones.

Syndrome—A collection of abnormalities that occur often enough to suggest they have a common cause.

Resources

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Pubic lice see **Lice infestation**

Puerperal infection

Definition

The term puerperal infection refers to a bacterial infection following **childbirth**. The infection may also be referred to as puerperal or postpartum fever. The genital tract, particularly the uterus, is the most commonly infected site. In some cases infection can spread to other points in the body. Widespread infection, or **sepsis**, is a rare, but potentially fatal complication.

Description

Puerperal infection affects an estimated 1–8% of new mothers in the United States. Given modern medical treatment and **antibiotics**, it very rarely advances to the point of threatening a woman's life. An estimated 2–4% of new mothers who deliver vaginally suffer some form of puerperal infection, but for cesarean sections, the figure is five–10 times that high.

Deaths related to puerperal infection are very rare in the industrialized world. It is estimated three in 100,000 births result in maternal **death** due to infection. However, the death rate in developing nations may be 100 times higher.

Postpartum **fever** may arise from several causes, not necessarily infection. If the fever is related to infection, it often results from endometritis, an inflammation of the uterus. Urinary tract, breast, and wound infections are also possible, as well as septic **thrombophlebitis**, a blood clot-associated inflammation of veins. A woman's susceptibility to developing an infection is related to such factors as cesarean section, extended labor, **obesity**, anemia, and poor prenatal **nutrition**.

Causes and symptoms

The primary symptom of puerperal infection is a fever at any point between birth and 10 days postpartum. A temperature of 100.4°F (38°C) on any two days during this period, or a fever of 101.6°F (38.6 °C) in the first 24 hours postpartum, is cause for suspicion. An assortment of bacterial species may cause puerperal infection. Many of these bacteria are normally found in the mother's genital tract, but other bacteria may be introduced from the woman's intestine and skin or from a healthcare provider.

The associated symptoms depend on the site and nature of the infection. The most typical site of infection is the genital tract. Endometritis, which affects the uterus, is the most prominent of these infections. Endometritis is much more common if a small part of the placenta has been retained in the uterus. Typically, several species of bacteria are involved and may act synergistically—that is, the bacteria's negative effects are multiplied rather than simply added together. Synergistic action by the bacteria can result in a stubborn infection such as an **abscess**. The major symptoms of a genital tract infection include fever, malaise, abdominal **pain**, uterine tenderness, and abnormal vaginal discharge. If these symptoms do not respond to antibiotic therapy, an abscess or blood clot may be suspected.

Other causes of postpartum fever include urinary tract infections, wound infections, septic thrombophlebitis, and **mastitis**. Mastitis, or breast infection, is indicated by fever, malaise, achy muscles, and reddened skin on the affected breast. It is usually caused by a clogged milk duct that becomes infected. Infections of the urinary tract are indicated by fever, frequent and painful urination, and back pain. An **episiotomy** and a **cesarean section** carry the risk of a wound infection. Such infections are suggested by a fever and pus-like discharge, inflammation, and swelling at wound sites.

Diagnosis

Fever is not an automatic indicator of puerperal infection. A new mother may have a fever owing to prior illness or an illness unconnected to childbirth. However, any fever within 10 days postpartum is aggressively investigated. Physical symptoms such as pain, malaise, loss of appetite, and others point to infection.

Many doctors initiate antibiotic therapy early in the fever period to stop an infection before it advances. A pelvic examination is done and samples are taken from the genital tract to identify the bacteria involved in the infection. The pelvic examination can reveal the extent of infection and possibly the cause. Blood samples may also be taken for blood counts and to test for the presence of infectious bacteria. A **urinalysis** may also be ordered, especially if the symptoms are indicative of a urinary tract infection.

If the fever and other symptoms resist antibiotic therapy, an ultrasound examination or computed tomography scan (CT scan) is done to locate potential abscesses or **blood clots** in the pelvic region. **Magnetic resonance imaging (MRI)** may be useful as well, in addition to a heparin challenge test if blood clots are suspected. If a lung infection is suspected, a **chest x ray** may also be ordered.

Treatment

Antibiotic therapy is the backbone of puerperal infection treatment. Initial antibiotic therapy may consist of clindamycin and gentamicin, which fight a broad array of bacteria types. If the fever and other symptoms do not respond to these antibiotics, a third, such as ampicillin, is added. Other antibiotics may be used depending on the identity of the infective bacteria and the possibility of an allergic reaction to certain antibiotics.

Antibiotics taken together are effective against a wide range of bacteria, but may not be capable of clearing up the infection alone, especially if an abscess

KEY TERMS

Abscess—A pus-filled area with definite borders.

Blood clot—A dense mat formed by certain components of the blood stream to prevent blood loss.

Cesarean section—Incision through the abdomen and uterus to facilitate delivery.

Computed tomography scan (CT scan)—Cross-sectional x rays of the body are compiled to create a three-dimensional image of the body's internal structures.

Episiotomy—Incision of the vulva (external female genitalia) during vaginal delivery to prevent tissue tearing.

Heparin—A blood component that controls the amount of clotting. It can be used as a drug to reduce blood clot formation.

Heparin challenge test—A medical test to evaluate how readily the blood clots.

Magnetic resonance imaging (MRI)—An imaging technique that uses a large circular magnet and radio waves to generate signals from atoms in the body. These signals are used to construct images of internal structures.

Postpartum—Referring to the time period following childbirth.

Prophylactic—Measures taken to prevent disease.

Sepsis—The presence of viable bacteria in the blood or body tissues.

Septic—Referring to the presence of infection.

Thrombophlebitis—An inflammation of veins accompanied by the formation of blood clots.

Ultrasound examination—A medical test in which high frequency sound waves are directed at a particular internal area of the body. As the sound waves are reflected by internal structures, a computer uses the data to construct an image of the structures.

Warfarin—A drug that reduces the ability of the blood to clot.

or blood clot is present. Heparin is combined with the antibiotic therapy in order to break apart blood clots. Heparin is used for five-seven days, and may be followed by warfarin for the following month. If the infection is complicated, it may be necessary to surgically drain the infected site. Infected episiotomies can be opened and allowed to drain, but abscesses and blood clots may require surgery.

Prognosis

Antibiotic therapy and other treatment measures are virtually always successful in curing puerperal infections.

Prevention

Careful attention to antiseptic procedures during childbirth is the basic underpinning of preventing infection. With some procedures, such as cesarean section, a doctor may administer prophylactic antibiotics as a preemptive strike against infectious bacteria.

Resources

PERIODICALS

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Julia Barrett

Pulmonary alveolar proteinosis

Definition

Pulmonary alveolar proteinosis (PAP) is a rare disease of the lungs.

Description

In this disease, also called alveolar proteinosis or phospholipidosis, gas exchange in the lungs is progressively impaired by the accumulation of phospholipids, compounds widely found in other living cells of the body. The alveoli are filled with this substance that renders them less effective in protecting the lung. This may explain why infections are often associated with the disease.

Pulmonary alveolar proteinosis most commonly affects people ages 20–50, although it has been reported in children and the elderly. The incidence is 5 out of every 1 million people. The disease is more common among males.

Causes and symptoms

The cause of this disease is unknown. In some people, however, it appears to result from infection,

KEY TERMS

Alveoli—The small cavities, or air sacs, in the lungs.

Bronchoscopy—A bronchoscopy is the examination of the bronchi, the primary divisions of the trachea that penetrate the lung, through a tube called a bronchoscope.

Clubbing—Clubbing is the rounding of the ends and swelling of fingers found in people with lung disease.

Remission—Lessening of severity, or abatement of symptoms.

Transtacheal biopsy—A transtracheal biopsy is the removal of a small piece of tissue from across the trachea or windpipe for examination under a microscope.

immune deficiency, or from exposure to silica, aluminum oxide, and a variety of dusts and fumes.

Symptoms include mild **shortness of breath** associated with a nonproductive or minimally productive **cough**, weight loss, and **fatigue**. Acute symptoms such as **fever** or progressive shortness of breath suggest a complicating infection.

Diagnosis

Physical examination may reveal clubbing of the fingers or a bluish coloration of the skin as a result of decreased oxygen.

A **chest x ray** may show alveolar disease. An arterial blood gas reveals low oxygen levels in the blood. **Bronchoscopy** with transtracheal biopsy shows alveolar proteinosis. Specific diagnosis requires a **lung biopsy**.

Treatment

Treatment consists of periodic whole-lung lavage, a washing out of the phospholipids from the lung with a special tube placed in the trachea. This is performed under **general anesthesia**.

Prognosis

In some, spontaneous remission occurs, while in others progressive **respiratory failure** develops. Disability from respiratory insufficiency is common,

but **death** rarely occurs. Repeated lavage may be necessary. Lung transplant is a last resort option.

Prevention

There is no known prevention for this very rare disorder.

Resources

ORGANIZATIONS

American Association for Respiratory Care. 11030 Ables Lane, Dallas, Texas 75229. (972) 243-2272. <<http://www.aarc.org>>.

American Lung Association. 1740 Broadway, New York, NY 10019. (800) 586-4872. <<http://www.lungusa.org>>.

Lorraine Steefel, RN

Pulmonary artery catheterization

Definition

Pulmonary artery catheterization is a diagnostic procedure in which a small catheter is inserted through a neck, arm, chest, or thigh vein and maneuvered into the right side of the heart, in order to measure pressures at different spots in the heart.

Purpose

Pulmonary artery catheterization is performed to:

- evaluate **heart failure**
- monitor therapy after a **heart attack**
- check the fluid balance of a patient with serious burns, kidney disease, or after heart surgery
- check the effect of medications on the heart

Precautions

Pulmonary artery catheterization is a potentially complicated and invasive procedure. The doctor must decide if the value of the information obtained will outweigh the risk of catheterization.

Description

Pulmonary artery catheterization, sometimes called Swan-Ganz catheterization, is usually

performed at the bedside of a patient in the intensive care unit. A catheter is threaded through a vein in the arm, thigh, chest, or neck until it passes through the right side of the heart. This procedure takes about 30 minutes. **Local anesthesia** is given to reduce discomfort.

Once the catheter is in place, the doctor briefly inflates a tiny balloon at its end. This temporarily blocks the blood flow and allows the doctor to make a pressure measurement in the pulmonary artery system. Pressure measurements are usually recorded for the next 48–72 hours in different parts of the heart. During this time, the patient must stay in bed so the catheter stays in place. Once the pressure measurements are no longer needed, the catheter is removed.

Preparation

Before and during the test, the patient will be connected to an electrocardiograph, which makes a recording of the electrical stimuli that cause the heart to contract. The insertion site is sterilized and prepared. The catheter is often sutured to the skin to prevent dislodgment.

Aftercare

The patient is observed for any sign of infection or complications from the procedure.

Risks

Pulmonary artery catheterization is not without risks. Possible complications from the procedure include:

- infection at the site where the catheter was inserted
- pulmonary artery perforation
- blood clots in the lungs
- irregular heartbeat

Normal results

Normal pressures reflect a normally functioning heart with no fluid accumulation. These normal pressure readings are:

- right atrium: 1–6 mm of mercury (mm Hg)
- right ventricle during contraction (systolic): 20–30 mm Hg
- right ventricle at the end of relaxation (end diastolic): less than 5 mm Hg

KEY TERMS

Cardiac shunt—A defect in the wall of the heart that allows blood from different chambers to mix.

- pulmonary artery during contraction (systolic): 20–30 mm Hg
- pulmonary artery during relaxation (diastolic): about 10 mm Hg
- mean pulmonary artery: less than 20 mm Hg
- pulmonary artery wedge pressure: 6–12 mm Hg
- left atrium: about 10 mm Hg

Abnormal results

Abnormally high right atrium pressure can indicate:

- pulmonary disease
- right side heart failure
- fluid accumulation
- compression of the heart after hemorrhage (cardiac tamponade)
- right heart valve abnormalities
- pulmonary **hypertension** (high blood pressure)

Abnormally high right ventricle pressure may indicate:

- pulmonary hypertension (high blood pressure)
- pulmonary valve abnormalities
- right ventricle failure
- defects in the wall between the right and left ventricle
- congestive heart failure
- serious heart inflammation

Abnormally high pulmonary artery pressure may indicate:

- diversion of blood from a left-to-right cardiac shunt
- pulmonary artery hypertension
- chronic obstructive pulmonary disease or emphysema
- blood clots in the lungs
- fluid accumulation in the lungs
- left ventricle failure

Abnormally high pulmonary artery wedge pressure may indicate:

- left ventricle failure
- mitral valve abnormalities
- cardiac insufficiency
- compression of the heart after hemorrhage

Resources

BOOKS

“Pulmonary Artery Catheterization.” In *The Patient’s Guide to Medical Tests*, ed. Barry L. Zaret, et al., Boston: Houghton Mifflin, 1997.

Tish Davidson, A.M.

Pulmonary edema

Definition

Pulmonary **edema** is a condition in which fluid accumulates in the lungs, usually because the heart’s left ventricle does not pump adequately.

Description

The build-up of fluid in the spaces outside the blood vessels of the lungs is called pulmonary edema. Pulmonary edema is a common complication of heart disorders, and most cases of the condition are associated with **heart failure**. Pulmonary edema can be a chronic condition, or it can develop suddenly and quickly become life threatening. The life-threatening type of pulmonary edema occurs when a large amount of fluid suddenly shifts from the pulmonary blood vessels into the lung, due to lung problems, **heart attack**, trauma, or toxic chemicals. It can also be the first sign of coronary heart disease.

In heart-related pulmonary edema, the heart’s main chamber, the left ventricle, is weakened and does not function properly. The ventricle does not completely eject its contents, causing blood to back up and cardiac output to drop. The body responds by increasing blood pressure and fluid volume to compensate for the reduced cardiac output. This, in turn, increases the force against which the ventricle must expel blood. Blood backs up, forming a pool in the pulmonary blood vessels. Fluid leaks into the spaces between the tissues of the lungs and begins to accumulate. This process makes it more difficult for the lungs to expand. It also impedes the exchange of air and gases between the lungs and blood moving through lung blood vessels.

Causes and symptoms

Most cases of pulmonary edema are caused by failure of the heart's main chamber, the left ventricle. It can be brought on by an acute heart attack, severe **ischemia**, volume overload of the heart's left ventricle, and mitral stenosis. Non-heart-related pulmonary edema is caused by lung problems like **pneumonia**, an excess of intravenous fluids, some types of **kidney disease**, bad **burns**, **liver disease**, nutritional problems, and **Hodgkin's disease**. Non-heart-related pulmonary edema can also be caused by other conditions where the lungs do not drain properly, and conditions where the respiratory veins are blocked.

Early symptoms of pulmonary edema include:

- shortness of breath upon exertion
- sudden respiratory distress after sleep
- difficulty breathing, except when sitting upright
- coughing

In cases of severe pulmonary edema, these symptoms will worsen to:

- labored and rapid breathing
- frothy, bloody fluid containing pus coughed from the lungs (sputum)
- a fast pulse and possibly serious disturbances in the heart's rhythm (atrial fibrillation, for example)
- cold, clammy, sweaty, and bluish skin
- a drop in blood pressure resulting in a thready pulse

Diagnosis

A doctor can usually diagnose pulmonary edema based on the patient's symptoms and a physical exam. Patients with pulmonary edema will have a rapid pulse, rapid breathing, abnormal breath and heart sounds, and enlarged neck veins. A **chest x ray** is often used to confirm the diagnosis. Arterial blood gas testing may be done. Sometimes **pulmonary artery catheterization** is performed to confirm that the patient has pulmonary edema and not a disease with similar symptoms (called **adult respiratory distress syndrome** or "noncardiogenic pulmonary edema").

Treatment

Pulmonary edema requires immediate emergency treatment. Treatment includes: placing the patient in a sitting position, oxygen, assisted or mechanical ventilation (in some cases), and drug therapy. The goal of treatment is to reduce the amount of fluid in the lungs,

KEY TERMS

Edema—Swelling caused by accumulation of fluid in body tissues.

Ischemia—A condition in which the heart muscle receives an insufficient supply of blood and slowly starves.

Left ventricle—The large chamber on the lower left side of the heart. The left ventricle sends blood to the aorta and the rest of the body.

Mitral stenosis—Narrowing or constricting of the mitral valve, which separates the left atrium from the left ventricle.

Pulmonary—Referring to the lungs and respiratory system.

improve gas exchange and heart function, and, where possible, to correct the underlying disease.

To help the patient breathe better, he/she is placed in a sitting position. High concentrations of oxygen are administered. In cases where respiratory distress is severe, a mechanical ventilator and a tube down the throat (tracheal intubation) will be used to improve the delivery of oxygen. Non-invasive pressure support ventilation is a new treatment for pulmonary edema in which the patient breathes against a continuous flow of positive airway pressure, delivered through a face or nasal mask. Non-invasive pressure support ventilation decreases the effort required to breathe, enhances oxygen and carbon dioxide exchange, and increases cardiac output.

Drug therapy could include morphine, nitroglycerin, **diuretics**, angiotensin-converting enzyme (ACE) inhibitors, and **vasodilators**. Vasopressors are used for cardiogenic **shock**. Morphine is very effective in reducing the patient's **anxiety**, easing breathing, and improving blood flow. Nitroglycerin reduces pulmonary blood flow and decreases the volume of fluid entering the overloaded blood vessels. Diuretics, like furosemide (Lasix), promote the elimination of fluids through urination, helping to reduce pressure and fluids in the blood vessels. ACE inhibitors reduce the pressure against which the left ventricle must expel blood. In patients who have severe **hypertension**, a vasodilator such as nitroprusside sodium (Nipride) may be used. For cardiogenic shock, an adrenergic agent (like dopamine hydrochloride [Intropin], dobutamine hydrochloride [Dobutrex], or epinephrine) or a bipyridine (like amrinone lactate [Inocor] or milrinone lactate [Primacor]) are given.

Prognosis

Most patients with pulmonary edema who seek immediate treatment can be treated quickly and effectively.

Prevention

Cardiogenic pulmonary edema can sometimes be prevented by treating the underlying heart disease. These treatments, can including maintaining a healthy diet, taking appropriate medications correctly, and avoiding excess alcohol and salt.

Resources

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Lori De Milto

Pulmonary embolism

Definition

Pulmonary **embolism** is an obstruction of a blood vessel in the lungs, usually due to a blood clot, which blocks a coronary artery.

Description

Pulmonary embolism is a fairly common condition that can be fatal. According to the American Heart Association, an estimated 600,000 Americans develop pulmonary embolism annually; 60,000 die from it. As many as 25,000 Americans are hospitalized each year for pulmonary embolism, which is a relatively common complication in hospitalized patients. Even without warning symptoms, pulmonary embolism can cause sudden **death**. Treatment is not always successful.

Pulmonary embolism is difficult to diagnose. Less than 10% of patients who die from pulmonary embolism were diagnosed with the condition. It occurs when emboli block a pulmonary artery, usually due to a blood clot that breaks off from a large vein and travels to the lungs. More than 90% of cases of pulmonary embolism are complications of deep vein thrombosis, **blood clots** from the leg or pelvic veins. Emboli can

also be comprised of fat, air, or tumor tissue. When emboli block the main pulmonary artery, pulmonary embolism can quickly become fatal.

Causes and symptoms

Pulmonary embolism is caused by emboli that travel through the blood stream to the lungs and block a pulmonary artery. When this occurs, circulation and oxygenation of blood is compromised. The emboli are usually formed from blood clots but are occasionally comprised of air, fat, or tumor tissue. Risk factors include: prolonged bed rest, surgery, **childbirth**, **heart attack**, **stroke**, congestive **heart failure**, **cancer**, **obesity**, a broken hip or leg, **oral contraceptives**, sickle cell anemia, congenital **coagulation disorders**, chest trauma, certain congenital heart defects, and old age.

Common symptoms of pulmonary embolism include:

- labored breathing, sometimes accompanied by chest pain.
- a rapid pulse.
- a **cough** that produces bloody sputum.
- a low **fever**.
- fluid build-up in the lungs.

Less common symptoms include:

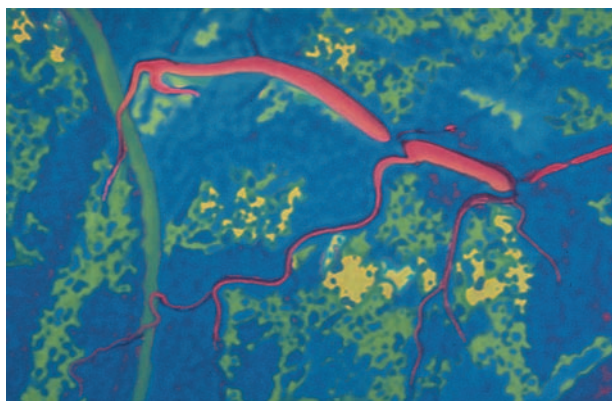
- coughing up a lot of blood
- pain caused by movement
- leg swelling
- bluish skin
- fainting
- swollen neck veins

In some cases there are no symptoms.

Diagnosis

Pulmonary embolism can be diagnosed through the patient's history, a physical exam, and diagnostic tests including **chest x ray**, lung scan, pulmonary angiography, electrocardiography, arterial blood gas measurements, and leg vein ultrasonography or **venography**.

A chest x ray can be normal or show fluid or other signs and rule out other diseases. The lung scan shows poor flow of blood in areas beyond blocked arteries. The patient inhales a small amount of radiopharmaceutical and pictures of airflow into the lungs are taken with a gamma camera. Then a different



An angiography of a pulmonary embolism. (Custom Medical Stock Photo. Reproduced by permission.)

radiopharmaceutical is injected into an arm vein and lung blood flow is scanned. A normal result essentially rules out pulmonary embolism. A lung scan can be performed in a hospital or an outpatient facility and takes about 45 minutes.

Pulmonary **angiography** is the most reliable test for diagnosing pulmonary embolism but it is not used often, because it carries some risk and is expensive, invasive, and not readily available in many hospitals. Pulmonary angiography is a radiographic test which involves injection of a pharmaceutical “contrast agent” to show up the pulmonary arteries. A cinematic camera records the blood flow through the lungs of the patient, who lies on a table. Pulmonary angiography is usually performed in a hospital’s radiology department and takes 30 minutes to one hour.

An electrocardiograph shows the heart’s electrical activity and helps distinguish pulmonary embolism from a heart attack. Electrodes covered with conducting jelly are placed on the patient’s chest, arms, and legs. Impulses of the heart’s activity are traced on paper. The test takes about 10 minutes and can be performed in a physician’s office or hospital lab.

Arterial blood gas measurements can be helpful, but they are rarely diagnostic for pulmonary embolism. Blood is taken from an artery instead of a vein, usually in the wrist and it is analyzed for oxygen, carbon dioxide and acid levels.

Venography is used to look for the most likely source of pulmonary embolism, deep vein thrombosis. It is very accurate, but it is not used often, because it is painful, expensive, exposes the patient to a fairly high dose of radiation, and can cause complications. Venography identifies the location, extent, and degree of attachment of the blood clots and enables the

KEY TERMS

Deep vein thrombosis—A blood clot in the calf’s deep vein. This frequently leads to pulmonary embolism if untreated.

Emboli—Clots or other substances that travel through the blood stream and get stuck in an artery, blocking circulation.

Thrombosis—The development of a blood clot inside a blood vessel.

condition of the deep leg veins to be assessed. A contrast solution is injected into a foot vein through a catheter. The physician observes the movement of the solution through the vein with a fluoroscope while a series of x rays are taken. Venography takes between 30–45 minutes and can be done in a physician’s office, a laboratory, or a hospital. Radionuclide venography, in which a radioactive isotope is injected, is occasionally used, especially if a patient has had reactions to contrast solutions. Most commonly performed are ultrasound and Doppler studies of leg veins.

Treatment

Patients with pulmonary embolism are hospitalized and generally treated with clot-dissolving and clot-preventing drugs. **Oxygen therapy** is often needed to maintain normal oxygen concentrations. For people who can’t take anticoagulants and in some other cases, surgery may be needed to insert a device that filters blood returning to the heart and lungs. The goal of treatment is to maintain the patient’s cardiovascular and respiratory functions while the blockage resolves, which takes 10–14 days, and to prevent the formation of other emboli.

Thrombolytic therapy to dissolve blood clots is the aggressive treatment for very severe pulmonary embolism. Streptokinase, urokinase, and recombinant tissue plasminogen activator (TPA) are thrombolytic agents. Heparin is the injectable anticoagulant (clot-preventing) drug of choice for preventing formation of blood clots. Warfarin, an oral anticoagulant, is usually continued when the patient leaves the hospital and doesn’t need heparin any longer.

Prognosis

About 10% of patients with pulmonary embolism die suddenly within the first hour of onset of the

condition. The outcome for all other patients is generally good; only 3% of patients who are properly diagnosed and treated die. In cases of undiagnosed pulmonary embolism, about 30% of patients die.

Prevention

Pulmonary embolism risk can be reduced in certain patients through judicious use of antithrombotic drugs such as heparin, venous interruption, gradient elastic stockings and/or intermittent pneumatic compression of the legs.

Resources

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300 or (800) 242-8721. inquire@heart.org <<http://www.americanheart.org>>.

Lori De Milto

Pulmonary fibrosis

Definition

Pulmonary fibrosis is scarring in the lungs.

Description

Pulmonary fibrosis develops when the alveoli, tiny air sacs that transfer oxygen to the blood, become damaged and inflamed. The body tries to heal the damage with **scars**, but these scars collapse the alveoli and make the lungs less elastic. If the cycle of inflammation and scarring continues, the lungs become increasingly unable to deliver oxygen to the blood. Changes in the lungs can also increase the blood pressure in the pulmonary artery. This condition, called pulmonary **hypertension**, makes the heart work harder and it may fail.

Pulmonary fibrosis can result from many different lung diseases including **sarcoidosis**, drug reactions, autoimmune diseases, environmental **allergies** such as Farmer's lung, and exposure to toxic dusts and gases.

Pulmonary fibrosis that develops without a known cause is called idiopathic pulmonary fibrosis. This disease is equally common in men and women. It is usually diagnosed between the ages of 40 and 60.

Causes and symptoms

The causes and risk factors vary with the underlying disease. They may include genetics, environmental factors, and infections.

The first symptom of pulmonary fibrosis is usually shortness of breath—at first, during **exercise**, but later also while resting. Patients may also have a dry **cough**, a rapid heartbeat, or enlargement of the fingertips and ends of the toes. Some people feel tired or have a **fever**, weight loss, muscle or joint pains. In late stages of the disease, the lack of oxygen in the blood can give the skin and mucus membranes a blue tinge known as **cyanosis**.

Diagnosis

Pulmonary fibrosis is often referred to a lung specialist. Several tests are usually needed to diagnose this disease and determine its cause. They include a physical examination, detailed history of the symptoms, chest x rays, lung function tests, and blood tests, including a measurement of the amount of oxygen in the blood. Computed tomography (CT scan) may give a more detailed picture of the lungs. **Bronchoscopy** may be done to examine the air passages and analyze the cells found deep in the lungs.

Lung biopsies are necessary to diagnose some diseases. Lung biopsies can be done through a needle inserted into the chest through the skin, during bronchoscopy, or as a surgical procedure under **general anesthesia**.

Treatment

The treatment of pulmonary fibrosis depends on the underlying cause. Many diseases are treated by suppressing inflammation with **corticosteroids**. Stronger immune suppressants such as cyclophosphamide (Cytoxan) or azathioprine (Imuran) may also be tried. Some patients need supplemental oxygen. A lung transplant may be an option for incurable diseases. Approximately 60–80% of patients live for at least two years after the transplant.

There is no good treatment for idiopathic pulmonary fibrosis. Only 10–20% of patients with this disease respond to corticosteroids.

Alternative treatment

Anxiety and fear can make breathing difficulties worse. Some patients find that activities such as **yoga**, prayer or **meditation**, **music therapy**, or **biofeedback** help to relax them.

KEY TERMS

Alveoli—Tiny air sacs in the lungs where oxygen and carbon dioxide are exchanged with the blood.

Autoimmune disease—A disease that develops when the immune system attacks normal cells or organs.

Bronchoscopy Scan—The examination of the air passages through a flexible or rigid tube inserted into the nostril (or mouth). Sometimes cells are collected by washing the lungs with a small amount of fluid.

Computed tomography(CT)—A special x-ray technique that produces a cross sectional image of the organs inside the body.

Corticosteroids—A class of drugs, related to hormones naturally found in the body, that suppress the immune system. One example is prednisone, sold under many brand names including Deltasone.

End-stage lung disease—The final stages of lung disease, when the lung can no longer keep the blood supplied with oxygen. End-stage lungs in pulmonary fibrosis have large air spaces separated by bands of inflammation and scarring.

Farmer's lung—An allergic reaction to moldy hay, most often seen in farmers, that results in lung disease.

Immune suppressant drug—Any drug that dampens immune responses and decreases inflammation.

Inflammation—The body's reaction to an irritant, characterized by the accumulation of immune cells, redness, and swelling.

Lung function tests—Tests of how much air the lungs can move in and out, and how quickly and efficiently this can be done. Lung function tests are usually done by breathing into a device that measures air flow.

Mucous membranes—The moist coverings that line the mouth, nose, intestines, and other internal organs.

Pulmonary artery—The blood vessel that delivers blood from the heart to the lungs.

Sarcoidosis—A disease of unknown origin that results in clumps of immune cells and inflammation in organs throughout the body.

Prognosis

The prognosis depends on the specific disease. Some cases may stop progressing or improve, particularly if the cause can be identified and treated. Others may develop quickly or slowly into end-stage lung disease. The course of idiopathic pulmonary fibrosis is very difficult to predict; however, average survival is approximately five to seven years.

Prevention

There is no known prevention for idiopathic pulmonary fibrosis.

Some ways to prevent other causes of pulmonary fibrosis are:

- avoid exposure to particle dust such as asbestos, coal dust, and silica
- avoid exposure to chemical fumes
- do not smoke

Resources

BOOKS

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ORGANIZATIONS

The American Lung Association. 1740 Broadway, New York, NY 10019. (212) 315-8700. <<http://www.lungusa.org>>.

Pulmonary Fibrosis Association. P.O. Box 75004, Seattle, WA 98125-0004. (206) 417-0949. <<http://pulmonaryfibrosisassn.com>>.

Pulmonary Fibrosis Foundation. 1075 Santa Fe Drive, Denver, Colorado 80204. (720) 932-7850. <<http://pulmonaryfibrosis.org>>.

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Anna Rovid Spickler, D.V.M., Ph.D.

Pulmonary fibrosis *see* **Idiopathic infiltrative lung diseases**

Pulmonary function test

Definition

Pulmonary function tests are a group of procedures that measure the function of the lungs, revealing problems in the way a patient breathes. The tests can determine the cause of **shortness of breath** and may help confirm lung diseases, such as **asthma**, **bronchitis** or **emphysema**. The tests also are performed before any major **lung surgery** to make sure the person won't be disabled by having a reduced lung capacity.

Purpose

Pulmonary function tests can help a doctor diagnose a range of respiratory diseases which might not otherwise be obvious to the doctor or the patient. The tests are important since many kinds of lung problems can be successfully treated if detected early.

The tests are also used to measure how a lung disease is progressing, and how serious the lung disease has become. Pulmonary function tests also can be used to assess how a patient is responding to different treatments.

One of the most common of the pulmonary function tests is spirometry (from the Greco-Latin term meaning "to measure breathing"). This test, which can be given in a hospital or doctor's office, measures how much and how fast the air is moving in and out of the lungs. Specific measurements taken during the test include the volume of air from start to finish, the

fastest flow that is achieved, and the volume of air exhaled in the first second of the test.

A peak flow meter can determine how much a patient's airways have narrowed. A test of blood gases is a measurement of the concentration of oxygen and carbon dioxide in the blood, which shows how efficient the gas exchange is in the lungs.

Another lung function test reveals how efficient the lungs are in absorbing gas from the blood. This is measured by testing the volume of carbon monoxide a person breathes out after a known volume of the gas has been inhaled.

Precautions

Pulmonary function tests shouldn't be given to patients who have had a recent **heart attack**, or who have certain other types of heart disease. It is crucial that the patient cooperate with the health care team if accurate results are to be obtained.

Description

The patient places a clip over the nose and breathes through the mouth into a tube connected to a machine known as a spirometer. First the patient breathes in deeply, and then exhales as quickly and forcefully as possible into the tube. The exhale must last at least six seconds for the machine to work properly. Usually the patient repeats this test three times, and the best of the three results is considered to be the measure of the lung function. The results will help a doctor figure out which type of treatment to pursue.

Preparation

The patient should not eat a heavy meal before the test, nor smoke for four to six hours beforehand. The patient's doctor will issue specific instructions about whether or not to use specific medications, including **bronchodilators** or inhalers, before the test. Sometimes, medication may be administered as part of the test.

Risks

The risk is minimal for most people, although the test carries a slight risk of a collapsed lung in some patients with lung disease.

Normal results

Normal results are based on a person's age, height, and gender. Normal results are expressed as a percentage



This young girl is undergoing a pulmonary function test in order to measure the functionality of her lungs. (Custom Medical Stock Photo. Reproduced by permission.)

KEY TERMS

Emphysema—A disease in which the small air sacs in the lungs become damaged, causing shortness of breath. In severe cases it can lead to respiratory or heart failure.

of the predicted lung capacity. The prediction takes into account the patient's age, height, and sex.

Abnormal results

Abnormal results mean that the person's lung capacity is less than 80% of the predicted value. Such findings usually mean that there is some degree of chest or lung disease.

Resources

BOOKS

Ruppel, Gregg L. *Manual of Pulmonary Function Testing*. St. Louis: Mosby, 1997.

Carol A. Turkington

Pulmonary heart disease see **Cor pulmonale**

Pulmonary hypertension

Definition

Pulmonary **hypertension** is a rare lung disorder characterized by increased pressure in the pulmonary artery. The pulmonary artery carries oxygen-poor blood from the lower chamber on the right side of the heart (right ventricle) to the lungs where it picks up oxygen.

Description

Pulmonary hypertension is present when the blood pressure in the circulation of the lungs is measured at greater than 25 mm of mercury (Hg) at rest or 30 mm Hg during **exercise**. Pulmonary hypertension can be either primary or secondary:

- Primary pulmonary hypertension. The cause of pulmonary hypertension is unknown. It is rare, affecting one people per-million. The illness most often occurs in young adults, especially women.
- Secondary pulmonary hypertension. Secondary pulmonary hypertension is increased pressure of the blood vessels of the lungs as a result of other medical conditions.

Regardless of whether pulmonary hypertension is primary or secondary, the disorder results in thickening of the pulmonary arteries and narrowing of these blood vessels. In response, the right side of the heart works harder to move the blood through these arteries and it becomes enlarged. Eventually overworking the right side of the heart may lead to right-sided **heart failure**, resulting in **death**.

Causes and symptoms

While the cause of primary pulmonary hypertension is uncertain, researchers think that in most people who develop the disease, the blood vessels are sensitive to certain factors that cause them to narrow. Diet suppressants, **cocaine**, and **pregnancy** are some of the

KEY TERMS

Hypertension—The medical term for abnormally high blood pressure.

Perfusion lung scan—A scan that shows the pattern of blood flow in the lungs.

Pulmonary—Having to do with the lungs.

Pulmonary function test—A test that measures how much air the lungs hold and the air flow in and out of the lungs.

Right-heart cardiac catheterization—A medical procedure during which a physician threads a catheter into the right side of the heart to measure the blood pressure in the right side of the heart and the pulmonary artery. The right heart's pumping ability can also be evaluated.

factors that are thought to trigger constriction or narrowing of the pulmonary artery. In about 6–10% of cases, primary pulmonary hypertension is inherited.

Secondary pulmonary hypertension can be associated with breathing disorders such as emphysema and **bronchitis**, or diseases such as **scleroderma**, systemic lupus erythematosus (SLE) or congenital heart disease involving heart valves, and pulmonary thromboembolism.

Symptoms of pulmonary hypertension include shortness of breath with minimal exertion, general fatigue, dizziness, and **fainting**. Swelling of the ankles, bluish lips and skin, and chest **pain** are among other symptoms of the disease.

Diagnosis

Pulmonary hypertension is rarely detected during routine physical examinations and, therefore, often progresses to later stages before being diagnosed. In addition to listening to heart sounds with a stethoscope, physicians also use electrocardiogram, pulmonary function tests, perfusion lung scan, and/or right-heart **cardiac catheterization** to diagnose pulmonary hypertension.

Treatment

The aim of treatment for pulmonary hypertension is to treat the underlying cause, if it is known. For example, thromboendarterectomy is a surgical procedure performed to remove a blood clot on the lung

that is causing the pulmonary hypertension. Lung transplants are another surgical treatment.

Some patients are helped by taking medicines that make the work of the heart easier. Anticoagulants, drugs that thin the blood, decrease the tendency of the blood to clot and allow blood to flow more freely. **Diuretics** decrease the amount of fluid in the body and reduce the amount of work the heart has to do. Calcium channel blockers relax the smooth muscle in the walls of the heart and blood vessels and improve the ability of the heart to pump blood.

One effective medical treatment that dilates blood vessels and seems to help prevent blood clots from forming is epoprostenol (prostacyclin). Prostacyclin is given intravenously to improve survival, exercise duration, and well-being. It is sometimes used as a bridge to help people who are waiting for a lung transplant. In other cases it is used for long-term treatment.

Some people require supplemental oxygen through nasal prongs or a mask if breathing becomes difficult.

Prognosis

Pulmonary hypertension is chronic and incurable with an unpredictable survival rate. Length of survival has been improving, with some patients able to live 15–20 years or longer with the disorder.

Prevention

Since the cause of primary pulmonary hypertension is still unknown, there is no way to prevent or cure this disease. A change in lifestyle may assist patients with daily activities. For example, relaxation exercises help to reduce **stress**. Good health habits such as a healthy diet, not **smoking**, and getting plenty of rest should be maintained.

Resources

ORGANIZATIONS

American Association for Respiratory Care. 11030 Ables Lane, Dallas, TX 75229. (972) 243-2272. <<http://www.aarc.org>>.

Pulmonary Hypertension Association. P.O. Box 24733, Speedway, IN 46224-0733. (800) 748-7274. <<http://www.phassociation.org>>.

OTHER

"Primary Pulmonary Hypertension." *National Heart, Lung, and Blood Institute*. <<http://www.nhlbi.nih.gov/nhlbi/lung/other/gp/pph.txt>>.

Lorraine Steefel, RN

Pulmonary incompetence see **Pulmonary valve insufficiency**

Pulmonary regurgitation see **Pulmonary valve insufficiency**

Pulmonary stenosis see **Pulmonary valve stenosis**

Pulmonary valve insufficiency

Definition

Pulmonary valve insufficiency is a disorder involving a defect of the valve located in the pulmonary artery.

Description

This disorder is also known as pulmonary valve regurgitation or pulmonary incompetence. The pulmonary valve is the structure in the pulmonary artery consisting of three flaps, which open and close during each heartbeat. The flaps keep blood from flowing back into the heart from the pulmonary artery—the artery that supplies blood to the lungs. With pulmonary valve insufficiency, the flaps may allow the blood to flow backward, resulting in a distinct murmur. The disorder may be congenital, but also often occurs in patients with severe **pulmonary hypertension**.

Causes and symptoms

There are generally few to no symptoms with pulmonary valve insufficiency. It may be initially noticed as a murmur in a routine exam of the heart and chest with a stethoscope. The most common causes of the disorder are severe pulmonary **hypertension**, or the presence of high pressure in the arteries and veins of the lungs. Pulmonary hypertension is usually caused by chronic lung disease, lung **blood clots**, and sometimes other diseases, such as **endocarditis**, an inflammation of the lining of the heart and valves. Previous surgery for **congenital heart disease** may also cause pulmonary valve insufficiency.

Diagnosis

The pitch and location of the murmur will help a physician determine if the cause is pulmonary valve insufficiency. An electrocardiogram (EKG) can detect flow changes. **Echocardiography** with

KEY TERMS

Congenital—Used to describe a condition or defect present at birth.

Endocarditis—Inflammation of the lining of the heart and valves.

Prophylaxis—Preventive. Antibiotic prophylaxis is the use of antibiotics to prevent a possible infection.

Pulmonary—Refers to the lungs and the breathing system and function.

Pulmonary hypertension—High blood pressure in the veins and arteries of the lungs.

color Doppler can usually detect regurgitation of blood in the area. This exam is done with ultrasound imaging. A **chest x ray** may show prominence of the pulmonary artery. In some cases, angiocardiology, or x ray of the arteries and vessels with injection of a dye, may be ordered.

Treatment

On its own, pulmonary valve insufficiency is seldom severe enough to require treatment. **Antibiotics** are usually recommended before dental work to reduce the possibility of bacterial endocarditis. Management of the primary condition, such as medications to manage pulmonary hypertension, may help control pulmonary valve insufficiency.

Alternative treatment

Since there are few or no symptoms and the disorder is a structural defect, alternative treatment may have only limited usefulness. Proper diet, **exercise**, and **stress reduction** may help control hypertension. Coenzyme Q10 and hawthorn (*Crataegus laevigata*) are two important nutrients to nourish the heart. Antioxidant supplements (including **vitamins** A, C, and E, selenium, and zinc) can help keep the tissues of the whole body, including the heart, in optimal condition.

Prognosis

Patients with this disorder may never experience limitations from pulmonary valve insufficiency. The disorder may only show up if complicated by pulmonary hypertension. There is an increased incidence of

bacterial endocarditis in patients with pulmonary valve insufficiency. Endocarditis can progress rapidly and be fatal.

Prevention

Pulmonary valve insufficiency resulting from chronic lung diseases can be prevented by behaviors and interventions to prevent those primary diseases. Bacterial endocarditis resulting from pulmonary valve insufficiency can usually be prevented with the use of antibiotic **prophylaxis** in preparation for dental procedures or other procedures which may introduce bacteria into the bloodstream.

Resources

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300 or (800) 242-8721. inquire@heart.org <<http://www.americanheart.org>>.

National Heart, Lung and Blood Institute. P.O. Box 30105, Bethesda, MD 20824-0105. (301) 251-1222. <<http://www.nhlbi.nih.gov>>.

Teresa Odle

Pulmonary valve stenosis

Definition

Pulmonary valve stenosis is a congenital heart defect in which blood flow from the heart to the pulmonary artery is blocked.

Description

Pulmonary valve stenosis is an obstruction in the pulmonary valve, located between the right ventricle and the pulmonary artery. Normally, the pulmonary valve opens to let blood flow from the right ventricle to the lungs. When the pulmonary valve is malformed, it forces the right ventricle to pump harder to overcome the obstruction. In its most severe form, pulmonary valve stenosis can be life-threatening.

Patients with pulmonary valve stenosis are at increased risk for getting valve infections and must take antibiotics to help prevent this before certain dental and surgical procedures. Pulmonary valve stenosis is also called pulmonary stenosis.

Causes and symptoms

Pulmonary valve stenosis is caused by a congenital malformation in which the pulmonary valve does not open properly. In most cases, scientists don't know why it occurs. In cases of mild or moderate stenosis, there are often no symptoms. With more severe obstruction, symptoms include a bluish skin tint and signs of **heart failure**.

Diagnosis

Diagnosis of pulmonary valve stenosis begins with the patient's medical history and a physical exam. Tests to confirm the diagnosis include **chest x ray**, echocardiogram, electrocardiogram, and catheterization. An electrocardiograph shows the heart's activity. Electrodes covered with conducting jelly are placed on the patient. The electrodes send impulses that are traced on a recorder. **Echocardiography** uses sound waves to create an image of the heart's chambers and valves. The technician applies gel to a wand (transducer) and presses it against the patient's chest. The returning sound waves are converted into an image displayed on a monitor. Catheterization is an invasive procedure used to diagnose, and in some cases treat, heart problems. A thin tube, called a catheter, is inserted into a blood vessel and threaded up into the heart, enabling physicians to see and sometimes correct the problems.

Treatment

Patients with mild to moderate pulmonary valve stenosis, and few or no symptoms, do not require treatment. In more severe cases, the blocked valve will be opened surgically, either through **balloon valvuloplasty** or surgical valvulotomy. For initial treatment, balloon valvuloplasty is the procedure of choice. This is a catheterization procedure in which a special catheter containing a deflated balloon is inserted in a blood vessel and threaded up into the heart. The catheter is positioned in the narrowed heart valve and the balloon is inflated to stretch the valve open.

In some cases, surgical valvulotomy may be necessary. This is open heart surgery performed with a heart-lung machine. The valve is opened with an incision and in some cases, hypertrophied muscle in the right ventricle is removed. Rarely does the pulmonary valve need to be replaced.

Alternative treatment

Pulmonary valve stenosis can be life threatening and always requires a physician's care. In mild to

KEY TERMS

Congenital—Present at birth.

Pulmonary—Relating to the opening leading from the right large chamber of the heart into the lung artery.

Stenosis—A narrowing or constriction, in this case of various heart valves. Stenosis reduces or cuts off the flow of blood.

Valve—Tissue between the heart's upper and lower chambers that controls blood flow.

moderate cases of pulmonary valve stenosis, general lifestyle changes, including dietary modifications, **exercise**, and **stress reduction**, can contribute to maintaining optimal wellness.

Prognosis

Patients with the most severe form of pulmonary valve stenosis may die in infancy. The prognosis for children with more severe stenosis who undergo balloon valvuloplasty or surgical valvulotomy is favorable. Patients with mild to moderate pulmonary stenosis can lead a normal life, but they require regular medical care.

Prevention

Pulmonary valve stenosis cannot be prevented.

Resources

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300. <<http://www.americanheart.org>>.

Children's Health Information Network. 1561 Clark Drive, Yardley, PA 19067. (215) 493-3068. <<http://www.tchin.org>>.

Congenital Heart Anomalies Support, Education & Resources, Inc. 2112 North Wilkins Road, Swanton, OH 43558. (419) 825-5575. <<http://www.csun.edu/~hfmth006/chaser>>.

Texas Heart Institute. Heart Information Service. P.O. Box 20345, Houston, TX 77225-0345. <<http://www.tmc.edu/thi>>.

Lori De Milto

Punctures see **Wounds**

Purple coneflower see **Echinacea**

Purpura hemorrhagica see **Idiopathic thrombocytopenic purpura**

Pustule see **Skin lesions**

Pyelography see **Intravenous urography**

Pyelonephritis

Definition

Pyelonephritis is an inflammation of the kidney and upper urinary tract that usually results from non-contagious bacterial infection of the bladder (**cystitis**).

Description

Acute pyelonephritis is most common in adult females but can affect people of either sex and any age. Its onset is usually sudden, with symptoms that often are mistaken as the results of straining the lower back. Pyelonephritis often is complicated by systemic infection. Left untreated or unresolved, it can progress to a chronic condition that lasts for months or years, leading to scarring and possible loss of kidney function.

Causes and symptoms

The most common cause of pyelonephritis is the backward flow (reflux) of infected urine from the bladder to the upper urinary tract. Bacterial infections also may be carried to one or both kidneys through the bloodstream or lymph glands from infection that began in the bladder. Kidney infection sometimes results from urine that becomes stagnant due to obstruction of free urinary flow. A blockage or abnormality of the urinary system, such as those caused by stones, tumors, congenital deformities, or loss of bladder function from nerve disease, increases a person's risk of pyelonephritis. Other risk factors include diabetes mellitus, pregnancy, chronic bladder infections, a history of analgesic abuse, paralysis from **spinal cord injury**, or tumors. Catheters, tubes, or surgical procedures may also trigger a kidney infection.

The bacteria most likely to cause pyelonephritis are those that normally occur in the feces. *Escherichia coli* causes about 85% of acute bladder and kidney infections in patients with no obstruction or history of surgical procedures. *Klebsiella*, *Enterobacter*, *Proteus*, or *Pseudomonas* are other common causes of infection. Once these organisms enter the urinary tract,

they cling to the tissues that line the tract and multiply in them.

Symptoms of acute pyelonephritis typically include fever and chills, burning or frequent urination, aching **pain** on one or both sides of the lower back or abdomen, cloudy or bloody urine, and **fatigue**. The patient also may have **nausea**, **vomiting**, and **diarrhea**. The flank pain may be extreme. The symptoms of chronic pyelonephritis include weakness, loss of appetite, hypertension, anemia, and protein and blood in the urine.

Diagnosis

The diagnosis of pyelonephritis is based on the patient's history, a **physical examination**, and the results of laboratory and imaging tests. During the physical examination, the doctor will touch (palpate) the patient's abdomen carefully in order to rule out **appendicitis** or other causes of severe abdominal pain.

Laboratory tests

In addition to collecting urine samples for **urinalysis** and urine culture and sensitivity tests, the doctor will take a sample of the patient's blood for a blood cell count. If the patient has pyelonephritis, the urine tests will show the presence of white blood cells, and bacteria in the urine. Bacterial counts of 100,000 organisms or higher per milliliter of urine point to a urinary tract infection. The presence of antibody-coated bacteria (ACB) in the urine sample distinguishes kidney infection from bladder infection, because bacteria in the kidney trigger an antibody response that coats the bacteria. The blood cell count usually indicates a sharp increase in the number of white blood cells.

Imaging studies

The doctor may order ultrasound imaging of the kidney area if he or she suspects that there is an obstruction blocking the flow of urine. X rays may demonstrate scarring of the kidneys and ureters resulting from long-standing infection.

Treatment

Treatment of acute pyelonephritis may require hospitalization if the patient is severely ill or has complications. Therapy most often involves a two- to three-week course of **antibiotics**, with the first few days of treatment given intravenously. The choice of antibiotic is based on laboratory sensitivity studies. The antibiotics used most often include ciprofloxacin

KEY TERMS

Bacteremia—The presence of bacteria in the bloodstream.

Cystitis—Inflammation of the bladder, usually caused by bacterial infection.

Reflux—The backward flow of a fluid in the body. Pyelonephritis is often associated with the reflux of urine from the bladder to the upper urinary tract.

(Cipro), ampicillin (Omnipen), or trimethoprim-sulfamethoxazole (Bactrim, Septra). Several advances in antibiotic therapy have been made in recent years. In 2003, the U.S. Food and Drug Administration (FDA) approved Cipro extended release tablets (Cipro XR) that could be taken once daily for acute uncomplicated pyelonephritis. A study in Europe also showed that a shorter course than that normally used in the United States could eradicate the bacteria that cause the disease. The primary objective of antimicrobial therapy is the permanent eradication of bacteria from the urinary tract. The early symptoms of pyelonephritis usually disappear within 48 to 72 hours of the start of antibacterial treatment. Repeat urine cultures are done in order to evaluate the effectiveness of the medication.

Chronic pyelonephritis may require high doses of antibiotics for as long as six months to clear the infection. Other medications may be given to control **fever**, **nausea**, and **pain**. Patients are encouraged to drink extra fluid to prevent **dehydration** and increase urine output. Surgery sometimes is necessary if the patient has complications caused by **kidney stones** or other obstructions, or to eradicate infection. Urine cultures are repeated as part of the follow-up of patients with chronic pyelonephritis. These repeat tests are necessary to evaluate the possibility that the patient's urinary tract is infected with a second organism as well as to assess the patient's response to the antibiotic. Some persons are highly susceptible to reinfection, and a second antibiotic may be necessary to treat the organism.

Prognosis

The prognosis for most patients with acute pyelonephritis is quite good if the infection is caught early and treated promptly. The patient is considered cured if the urine remains sterile for a year. Untreated or recurrent kidney infection can lead to bacterial

invasion of the bloodstream (bacteremia), **hypertension**, chronic pyelonephritis with scarring of the kidneys, and permanent kidney damage. In 2003, a report on long-term follow-up of adults with acute pyelonephritis looked at kidney scarring and resulting complications. Kidney damage that causes complications is rare after 10 to 20 years, even though many women showed renal scarring.

Prevention

Persons with a history of urinary tract infections should urinate frequently, and drink plenty of fluids at the first sign of infection. Women should void after intercourse which may help flush bacteria from the bladder. Girls should be taught to wipe their genital area from front to back after urinating to avoid getting fecal matter into the opening of the urinary tract.

Resources

PERIODICALS

Jancin, Bruce. "Short-course Cipro for Pyelonephritis: Unapproved Regimen Shows Promise." *OBGYN News* November 1, 2003: 5.

Mangan, Doreen. "The FDA has Approved Ciprofloxacin Extended Release Tablets (Cipro XR), a Once-daily Formulation, for the Treatment of Complicated Urinary Tract Infections (cUTIs) and Acute Pyelonephritis (AUP), or Kidney Infection." *RN* November 2003: 97.

Raz, Paul, et al. "Long-term Follow-up of Women Hospitalized for Acute Pyelonephritis." *Clinical Infectious Diseases* (October 15, 2003):1014–1017.

ORGANIZATIONS

American Foundation for Urologic Disease. 1128 N. Charles St., Baltimore, MD 21201. (401) 468-1800. < <http://www.afud.org> >.

Kathleen D. Wright, RN
Teresa G. Odle

Pyloric stenosis

Definition

Pyloric stenosis refers to a narrowing of the passage between the stomach and the small intestine. The condition, which affects infants during the first several weeks of life, can be corrected effectively with surgery.

Description

Frequent **vomiting** may be an indication of pyloric stenosis. The pylorus is the passage between the stomach and the small intestine. During the digestive process food passes through the pylorus, which is located near the bottom of the stomach, on its way to the intestines. In pyloric stenosis, the muscular wall of the passage becomes abnormally thickened. This causes the pylorus to become too narrow, which prevents food from emptying out of the stomach in a normal fashion. The partially digested contents of the stomach are forced upwards into the mouth. As a result, a baby with pyloric stenosis often vomits after feedings.

The condition affects one in 4,000 infants. Most are diagnosed between three and five weeks old, though some babies may show symptoms during the first or second week of life. Infants with a family history of pyloric stenosis are more at risk for the condition, which tends to occur less often in females, blacks, and Asians. Pyloric stenosis is also referred to as hypertrophic pyloric stenosis.

Causes and symptoms

The cause of pyloric stenosis is not known. The main symptom is vomiting after feedings. These episodes of vomiting usually get worse over time, happening more often and becoming more forceful (forceful vomiting is often called "projectile" vomiting). Other symptoms include increased appetite, weight loss, infrequent bowel movements, belching, and **diarrhea**. Due to **dehydration**, the infant may also have fewer wet diapers.

Diagnosis

The clinician will examine the baby and talk with the parents about their infant's symptoms. If a child has the condition, the doctor should be able to feel a hard mass (about 2 cm wide and olive shaped) in the area above the bellybutton. If the doctor cannot detect the mass, ultrasonography will be done to confirm the diagnosis. A blood test may also be performed to see if the infant is dehydrated, in which case intravenous fluids can be used to correct the problem.

Treatment

Pyloric stenosis can be cured with a surgical procedure called a pyloromyotomy. In this operation, the surgeon makes an incision in the baby's

KEY TERMS

Laparoscope—A thin, camera-fitted tube that can be inserted into the abdomen in order to view internal organs.

Stenosis—The narrowing of a passage (such as the pylorus).

Ultrasonography—A non-invasive imaging procedure that uses high-frequency sound waves.

abdomen. Then a small cut is made in the thickened muscle of the pylorus and it is spread apart. In this manner, the passage can be widened without removing any tissue. (The procedure may be performed with the aid of a laparoscope.) After surgery, the pylorus will heal itself. The thickening gradually goes away and the passage resumes a normal shape. The whole procedure (including anesthesia) takes about an hour.

Most babies go home one or two days after surgery. Any mild discomfort can be controlled with Tylenol. The infant may still vomit occasionally after surgery, but this is not usually a cause for alarm. However, if vomiting occurs three or more times a day, or for several consecutive days, the baby's pediatrician should be notified.

Alternative treatment

None known.

Prognosis

Surgery is often a complete cure. Most infants do not experience complications or long-term effects.

Prevention

It is not known how to prevent pyloric stenosis.

Resources

BOOKS

Behrman, Richard E., et al., *Nelson Textbook of Pediatrics*. Philadelphia: WB Saunders, 2000.

PERIODICALS

Yoshizawa J, et al. Ultrasonographic Features of Normalization of the Pylorus after Pyloromyotomy for Hypertrophic Pyloric Stenosis. "Journal of Pediatric Surgery" 36 (April 2001): 582-6.

ORGANIZATIONS

American Academy of Family Physicians. 11400 Tomahawk Creek Parkway, Leawood, KS 66211-2672. (913) 906-6000. <<http://www.aafp.org/>> . fp@aafp.org.

American Academy of Pediatrics. 141 Northwest Point Boulevard, Elk Grove Village, IL 60007-1098. (847) 434-4000. <<http://www.aap.org>>.

Greg Annussek

Pyloroplasty

Definition

Pyloroplasty is an elective surgical procedure in which the lower portion of the stomach, the pylorus, is cut and resutured, to relax the muscle and widen the opening into the intestine. Pyloroplasty is a treatment for high-risk patients for gastric or peptic ulcer disease. A peptic ulcer is a well-defined sore on the stomach where the lining of the stomach or duodenum has been eaten away by stomach acid and digestive juices.

Purpose

The end of the pylorus is surrounded by a strong band of muscle (pyloric sphincter), through which stomach contents are emptied into the duodenum (the first part of the small intestine). Pyloroplasty widens this opening into the duodenum.

A pyloroplasty is performed to treat complications of gastric ulcer disease, or when conservative treatment is unsatisfactory. The longitudinal cut made in the pylorus is closed transversely, permitting the muscle to relax. By establishing an enlarged outlet from the stomach into the intestine, the stomach empties more quickly. A pyloroplasty is often done in conjunction with a **vagotomy**, a procedure in which the nerves that stimulate stomach acid production and gastric motility (movement) are cut. As these nerves are cut, gastric emptying may be delayed, and the pyloroplasty compensates for that effect.

Preparation

As with any surgical procedure, the patient will be required to sign a consent form after the procedure is explained thoroughly. Blood and urine studies, along with various x rays may be ordered as the doctor deems necessary. Food and fluids will be prohibited

KEY TERMS

Gastric (or peptic) ulcer—An ulcer (sore) of the stomach, duodenum or other part of the gastrointestinal system. Though the causes are not fully understood, they include excessive secretion of gastric acid, stress, heredity, and the use of certain drugs, especially acetylsalicylic acid and nonsteroidal antiinflammatory drugs.

Pylorus—The valve which releases food from the stomach into the intestines.

Vagotomy—Cutting of the vagus nerve. If the vagus nerves are cut as they enter the stomach (truncal vagotomy), gastric secretions are decreased, as is intestinal motility (movement) and stomach emptying. In a selective vagotomy, only those branches of the vagus nerve are cut that stimulate the secretory cells.

after midnight before the procedure. Cleansing **enemas** may be ordered to empty the intestine. If **nausea** or **vomiting** are present, a suction tube to empty the stomach may be used.

Aftercare

Post-operative care for the patient who has had a pyloroplasty, as for those who have had any major surgery, involves monitoring of blood pressure, pulse, respiration, and temperature. Breathing tends to be shallow because of the effect of anesthesia and the patient's reluctance to breathe deeply and experience **pain** that is caused by the abdominal incision. The patient is shown how to support the operative site while breathing deeply and coughing, and given pain medication as necessary. Fluid intake and output is measured, and the operative site is observed for color and wound drainage. Fluids are given intravenously for 24–48 hours, until the patient's diet is gradually advanced as bowel activity resumes. The patient is generally allowed to walk approximately eight hours after surgery and the average hospital stay, dependent upon overall recovery status, ranges from six to eight days.

Risks

Potential complications of this abdominal surgery include:

- excessive bleeding
- surgical wound infection

- incisional **hernia**
- recurrence of gastric ulcer
- chronic **diarrhea**
- malnutrition

Normal results

Complete healing is expected without complications. Four to six weeks should be allowed for recovery from the surgery.

Abnormal results

The doctor should be made aware of any of the following problems after surgery:

- increased pain, swelling, redness, drainage, or bleeding in the surgical area
- headache, muscle aches, **dizziness**, or **fever**
- increased abdominal pain or swelling, **constipation**, nausea or vomiting, rectal bleeding, or black, tarry stools

Resources

OTHER

“Peptic ulcer surgery.” *ThriveOnline*. April 20, 1988. < <http://thriveonline.oxygen.com> >.

Kathleen D. Wright, RN

Pylorus repair see **Pyloroplasty**

Pyorrhea see **Periodontal disease**

Pyrazinamide see **Antituberculosis drugs**

Pyridoxine deficiency see **Vitamin B₆ deficiency**

Pyrimethamine see **Antimalarial drugs**

Pyruvate kinase deficiency

Definition

Pyruvate kinase deficiency (PKD) is part of a group of disorders called hereditary nonspherocytic hemolytic **anemias**. Hereditary nonspherocytic anemias are rare genetic conditions that affect the red blood cells. PKD is caused by a deficiency in the enzyme, pyruvate kinase. Although PKD is the second most common of the hereditary nonspherocytic

anemias, it is still rare, with the incidence estimated to be 51 cases per million in the Caucasian population.

Description

In PKD, there is an functional abnormality with the enzyme, pyruvate kinase. Usually, pyruvate kinase acts as a catalyst in the glycolysis pathway, and is considered an essential component in this pathway. Glycolysis is the method by which cells produce their own energy. A problem with any of the key components in glycolysis can alter the amount of energy produced. In the red blood cells, glycolysis is the only method available to produce energy. Without the proper amount of energy, the red blood cells do not function normally. Since pyruvate kinase is one of the key components in glycolysis, when there is a problem with this enzyme in the red blood cells, there is a problem with the production of energy, causing the red blood cells not to function properly.

There are four different forms of the pyruvate kinase enzyme in the human body. These forms, called isozymes, all perform the same function but each isozyme of pyruvate kinase is structurally different and works in different tissues and organs. The four isozymes of pyruvate kinase are labeled M1, M2, L, and R. The isozyme M1 is found in the skeletal muscle and brain, isozyme M2 can be found in most fetal and adult tissues, isozyme L works in the liver, and isozyme R works in the red blood cells. In PKD, only the pyruvate kinase isozyme found in red blood cells, called PKR, is abnormal. Therefore, PKD only affects the red blood cells and does not directly affect the energy production in the other organs and tissues of the body.

In general, PKD not does appear to affect one gender more than another or be more common in certain regions. However, there are studies of an Amish group in Pennsylvania where a severe form of PKD is more common. As previously mentioned, the three mutations found in the PKLR gene have been linked to individuals of specific decents. Caucasians of northern and central European decent are more likely to have the 1529A mutations, individuals of southern European descent usually have the 1456T mutation, and individuals of Asian descent are more likely to have the 1468T mutation.

Causes and symptoms

There are two PK genes and each gene produces two of the four isozymes of pyruvate kinase. The M1 and M2 isozymes are produced by the pyruvate kinase

gene called PKM2 and pyruvate kinase isozymes, L and R, are products of the pyruvate kinase gene, PKLR. The PKLR gene is located on chromosome 1, on the q arm (the top half of the chromosome), in region 21 (written as 1q21). As of 2001, there have been over 125 different mutations described in the PKLR gene that have been detected in individuals with PKD.

PKD is mainly inherited in an autosomal recessive manner. There have been a few families where it appeared that PKD was inherited in either an autosomal dominant manner or where the carriers of PKD exhibited mild problems with their red blood cells. As with all autosomal recessive conditions, affected individuals have a mutation in both pair of genes. Most individuals with PKD are compound heterozygotes, meaning that each PKLR gene in a pair contains a different mutation. There are individuals who have the same mutation on each PKLR gene, but these individuals tend to be children of parents who are related to each other.

There are three mutations in the PKLR gene called, 1529A, 1456T, and 1468T, that are seen more frequently in individuals with PKD than the other mutations. The mutation 1529A is most frequently seen in Caucasians of northern and central European descent and is the most common mutation seen in PKD. The mutation 1456T is more common in individuals of southern European descent and the mutation 1468T is more common in individuals of Asian descent.

In general, the more severe the PKD, the earlier in life symptoms tend to be detected. Individuals with the more severe form of PKD often show symptoms soon after birth, but most individuals with PKD begin to exhibit symptoms during infancy or childhood. In individuals with the more mild form of PKD, the condition is sometimes not diagnosed until late adulthood, after an acute illness, or during a **pregnancy** evaluation.

For most of the mutations seen in the PKLR gene, no correlation between the specific mutation and the severity of the disorder has been observed. However, for two of the mutations, there has been speculation on their effect on the severity of PKD. When the mutation 1456T has been seen in the homozygous state (when both PKLR genes contain the same mutation), those rare individuals experienced very mild symptoms of PKD. Also, there have been individuals who were homozygous for the 1529A mutation. These individuals had a very severe form of PKD. Therefore, it is thought that the 1456T mutation is associated

KEY TERMS

Anemia—When the amount of red blood cells is less than normal.

Catalyst—A substance that causes the next step in a pathway or reaction to occur, but is not physically changed by the process.

Compound heterozygotes—Individuals who have one gene in a pair with one mutation and the other gene in the pair has a different mutation.

Enzyme—A protein produced by cells that acts as a catalyst to allow a change or function to occur.

Glycolysis—The pathway in which a cell breaks down glucose into energy.

Hemolytic anemia—Anemia that results from red blood cells being destroyed sooner than normal.

Heterozygote/Heterozygotes—Are individuals who have one gene in a pair that has a mutation while the other gene in the pair is unaffected.

Homozygote/Homozygotes—Are individuals who have both genes in a pair with the same mutation.

Homozygous—When both genes in a pair have the same mutation.

Isozyme/Isoenzyme—Are a group of enzymes that perform the same function, but are different from one another in their structure or how they move.

Mutation—A change in the gene that causes it to alter its function

Nonspherocytic—Literally means not sphere-shaped. Often in inherited hemolytic anemias, the red blood cells are sphere-shaped. In Nonspherocytic Hemolytic Anemias, the red blood cells are not sphere-shaped.

with a milder form of the disease and the 1529A mutation is associated with a more severe form of the disease. It is not known how these mutations affect the severity of PKD when paired with different mutations.

Symptoms of PKD are similar to those symptoms seen in individuals who have long-term **hemolytic anemia**. The more common symptoms include variable degrees of **jaundice** (a yellowish pigment of the skin), slightly to moderately enlarged spleen (splenomegaly), and increased incidence of **gallstones**. Other physical effects of PKD can include smaller head size and the forehead appearing prominent and rounded (called frontal bossing). If a child with PKD has their spleen removed, their growth tends to improve. Even within

the same family, individuals can have different symptoms and severity of PKD.

In individuals with PKD, the red blood cells are taken out of their circulation earlier than normal (shorten life-span). Because of this, individuals with PKD will have hemolytic anemia. Additionally, the anemia or other symptoms of PKD may worsen during a sudden illness or pregnancy.

Diagnosis

A diagnosis of PKD can be made by measuring the amount of pyruvate kinase in red blood cells. Individuals with PKD tend to have 5–25% of the normal amount of pyruvate kinase. Carriers of PKD also can have less pyruvate kinase in their red blood cells, approximately 40–60% of the normal value. However, there is an overlap between the normal range of pyruvate kinase and the ranges seen with carriers of PKD. Therefore, measuring the amount of pyruvate kinase in the red blood cells is not a good method of detecting carriers of PKD. If the mutations causing PKD in a family are known, it may be possible to perform mutation analysis to determine carrier status of an individual and to help diagnose individuals with PKD.

Treatment

In the severest cases, individuals with PKD will require multiple blood transfusions. In some of those cases, the spleen may be removed (**splenectomy**). Red blood cells are normally removed from circulation by the spleen. By removing an individual's spleen (usually a child), the red blood cells are allowed to stay in circulation longer than normal; thereby, reducing the severity of the anemia. After a splenectomy, or once an individual with PKD is older, the number of transfusions tends to decrease.

Prognosis

The prognosis of PKD is extremely variable. Early intervention and treatment of symptoms frequently improves the individual's health. Without treatment, individuals may experience severe complications that may become lethal. Individuals with a mild form of PKD may appear to have no symptoms at all.

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Psychogenic disorder see **Somatoform disorders**

Q

Q fever

Definition

Q fever is an illness caused by a type of bacteria, *Coxiella burnetii*, resulting in a **fever** and rash.

Description

C. burnetii lives in many different kinds of animals, including cattle, sheep, goats, ticks, cats, rabbits, birds, and dogs. In sheep and cattle, for example, the bacteria tends to accumulate in large numbers in the female's uterus (the organ where lambs and calves develop) and udder. Other animals have similar patterns of bacterial accumulation within the females. As a result, *C. burnetii* can cause infection through contaminated milk, or when humans come into contact with the fluids or tissues produced when a cow or sheep gives birth. Also, the bacteria can survive in dry dust for months; therefore, if the female's fluids contaminate the ground, humans may become infected when they come in contact with the contaminated dust.

Persons most at risk for Q fever include anybody who works with cattle or sheep, or products produced from them. These include farm workers, slaughterhouse workers, workers in meat-packing plants, veterinarians, and wool workers. Since September 2001, however, Q fever has become an additional concern because of its potential as an agent of bioterrorism.

Q fever has been found all over the world, except in some areas of Scandinavia, Antarctica, and New Zealand.

Causes and symptoms

C. burnetii causes infection when a human breathes in tiny droplets, or drinks milk, containing the bacteria. After three to 30 days, symptoms of the illness appear.

The usual symptoms of Q fever include fever, chills, heavy sweating, **headache**, **nausea and vomiting**, **diarrhea**, **fatigue**, and **cough**. Also, a number of other problems may present themselves, including inflammation of the liver (hepatitis); inflammation of the sac containing the heart (**pericarditis**); inflammation of the heart muscle itself (**myocarditis**); inflammation of the coverings of the brain and spinal cord, or of the brain itself (meningoencephalitis); and **pneumonia**.

Chronic Q fever occurs most frequently in patients with other medical problems, including diseased heart valves, weakened immune systems, or **kidney disease**. Such patients usually have about a year's worth of vague symptoms, including a low fever, enlargement of the spleen and/or liver, and fatigue. Testing almost always reveals that these patients have inflammation of the lining of the heart (**endocarditis**).

Diagnosis

Q fever is diagnosed by demonstrating that the patient's immune system is making increasing numbers of antibodies (special immune cells) against markers (antigens) that are found on *C. burnetii*.

Treatment

Doxycycline and quinolone **antibiotics** are effective for treatment of Q fever. Treatment usually lasts for two weeks. Rifampin and doxycycline together are given for chronic Q fever. Chronic Q fever requires treatment for at least three years.

Minocycline has been found to be useful in treating post-Q fever fatigue. The dosage is 100 mg per day for three months.

Prognosis

Death is rare from Q fever. Most people recover completely, although some patients with endocarditis

KEY TERMS

Antibodies—Specialized cells of the immune system that can recognize organisms that invade the body (such as bacteria, viruses, and fungi). The antibodies are then able to set off a complex chain of events designed to kill these foreign invaders.

Antigens—Markers on the outside of bacteria or viruses which can be recognized by antibodies.

Bioterrorism—The use of disease microorganisms to intimidate or terrorize a civilian population.

Immune system—The system of specialized organs, lymph nodes, and blood cells throughout the body which work together to prevent foreign invaders (bacteria, viruses, fungi, etc.) from taking hold and growing.

Inflammation—The body's response to tissue damage. Includes increased heat, swelling, redness, and pain in the affected part.

will require surgery to replace their damaged heart valves.

Prevention

Q fever can be prevented by the appropriate handling of potentially infective substances. For example, milk should always be pasteurized, and people who work with animals giving birth should carefully dispose of the tissues and fluids associated with birth. Industries which process animal materials (meat, wool) should take care to prevent the contamination of dust within the plant.

Vaccines are available for workers at risk for Q fever.

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Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

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Qigong

Definition

Qigong (pronounced "chee-gung," also spelled *chi kung*) is translated from the Chinese to mean "energy cultivation" or "working with the life energy." Qigong is an ancient Chinese system of postures, exercises, breathing techniques, and meditations. Its techniques are designed to improve and enhance the body's *qi*. According to traditional Chinese philosophy, *qi* is the fundamental life energy responsible for health and vitality.

Purpose

Qigong may be used as a daily routine to increase overall health and well-being, as well as for disease prevention and longevity. It can be used to increase energy and reduce **stress**. In China, qigong is used in conjunction with other medical therapies for many chronic conditions, including **asthma**, **allergies**, **AIDS**, **cancer**, headaches, **hypertension**, depression, mental illness, strokes, heart disease, and **obesity**.

Qigong is presently being used in Hong Kong to relieve depression and improve the overall psychological and social well-being of elderly people with chronic physical illnesses.

Description

Origins

Qigong originated before recorded history. Scholars estimate qigong to be as old as 5,000–7,000 years old. Tracing the exact historical development of qigong is difficult, because it was passed down in secrecy among monks and teachers for many generations. Qigong survived through many years before paper was invented, and it also survived the Cultural Revolutions in China of the 1960s and 1970s, which banned many traditional practices.

Qigong has influenced and been influenced by many of the major strands of Chinese philosophy. The Taoist philosophy states that the universe operates within laws of balance and harmony, and that people must live within the rhythms of nature—ideas that pervade qigong. When Buddhism was brought from India to China around the seventh century A.D., **yoga** techniques and concepts of mental and spiritual awareness were introduced to qigong masters. The Confucian school was concerned with how people should live their daily lives, a concern of qigong as well. The martial arts were highly influenced by qigong, and many of them, such as **t'ai chi** and kung fu, developed directly from it. **Traditional Chinese medicine** also shares many of the central concepts of qigong, such as the patterns of energy flow in the body. **Acupuncture** and **acupressure** use the same points on the body that qigong seeks to stimulate. In China, qigong masters have been renowned physicians and healers. Qigong is often prescribed by Chinese physicians as part of the treatment.

Due to the political isolation of China, many Chinese concepts have been shrouded from the Western world. Acupuncture was “discovered” by American doctors only in the 1970s, although it had been in use for thousands of years. With an increased exchange of information, more Americans have gained access to the once-secret teachings of qigong. In 1988, the First World Conference for Academic Exchange of Medical Qigong was held in Beijing, China, where many studies were presented to attendees from around the world. In 1990, Berkeley, California, hosted the First International Congress of Qigong. In the past decade, more Americans have begun to discover the beneficial effects of qigong, which motivate an estimated 60 million Chinese to practice it every day.

Basic concepts

In Chinese thought, qi, or chi, is the fundamental life energy of the universe. It is invisible but present in

air, water, food and sunlight. In the body, qi is the unseen vital force that sustains life. We are all born with inherited amounts of qi, and we also get acquired qi from the food we eat and the air we breathe. In qigong, the breath is believed to account for the largest quantity of acquired qi, because the body uses air more than any other substance. The balance of our physical, mental, and emotional levels also affect qi levels in the body.

Qi travels through the body along channels called meridians. There are 12 main meridians, corresponding to the 12 principal organs as defined by the traditional Chinese system: the lung, large intestines, stomach, spleen, heart, small intestine, urinary bladder, kidney, liver, gallbladder, pericardium, and the “triple warmer,” which represents the entire torso region. Each organ has qi associated with it, and each organ interacts with particular emotions on the mental level. Qigong techniques are designed to improve the balance and flow of energy throughout the meridians, and to increase the overall quantity and volume of qi. In qigong philosophy, mind and body are not separated as they often are in Western medicine. In qigong, the mind is present in all parts of the body, and the mind can be used to move qi throughout the body.

Yin and yang are also important concepts in qigong. The universe and the body can be described by these two separate but complementary principles, which are always interacting, opposing, and influencing each other. One goal of qigong is to balance yin and yang within the body. Strong movements or techniques are balanced by soft ones, leftward movements by rightward, internal techniques by external ones, and so on.

Practicing qigong

There are thousands of qigong exercises. The specific ones used may vary depending on the teacher, school, and objective of the practitioner. Qigong is used for physical fitness, as a martial art, and most frequently for health and healing. Internal qigong is performed by those wishing to increase their own energy and health. Some qigong masters are renowned for being able to perform external qigong, by which the energy from one person is passed on to another for healing. This transfer may sound suspect to Western logic, but in the world of qigong there are some amazing accounts of healing and extraordinary capabilities demonstrated by qigong masters. Qigong masters generally have deep knowledge of the concepts of Chinese medicine and healing. In China, there are hospitals that use medical qigong to heal patients, along with

herbs, acupuncture, and other techniques. In these hospitals, qigong healers use external qigong and also design specific internal qigong exercises for patients' problems.

There are basic components of internal qigong sessions. All sessions require warm-up and concluding exercises. Qigong consists of postures, movements, breathing techniques, and mental exercises. Postures may involve standing, sitting, or lying down. Movements include stretches, slow motions, quick thrusts, jumping, and bending. Postures and movements are designed to strengthen, stretch, and tone the body to improve the flow of energy. One sequence of postures and movements is known as the "Eight Figures for Every Day." This sequence is designed to quickly and effectively work the entire body, and is commonly performed daily by millions in China.

Breathing techniques include deep abdominal breathing, chest breathing, relaxed breathing, and holding breaths. One breathing technique is called the "Six Healing Sounds." This technique uses particular breathing sounds for each of six major organs. These sounds are believed to stimulate and heal the organs.

Meditations and mind exercises are used to enhance the mind and move qi throughout the body. These exercises are often visualizations that focus on different body parts, words, ideas, objects, or energy flowing along the meridians. One mental **exercise** is called the "Inner Smile," during which the practitioner visualizes joyful, healing energy being sent sequentially to each organ in the body. Another mental exercise is called the "Microscopic Orbit Meditation," in which the practitioner intently meditates on increasing and connecting the flow of qi throughout major channels.

Discipline is an important dimension of qigong. Exercises are meant to be performed every morning and evening. Sessions can take from 15 minutes to hours. Beginners are recommended to practice between 15–30 minutes twice a day. Beginners may take classes once or twice per week, with practice outside of class. Classes generally cost between \$10–\$20 per session.

Preparations

Qigong should be practiced in a clean, pleasant environment, preferably outdoors in fresh air. Loose and comfortable clothing is recommended. Jewelry should be removed. Practitioners can prepare for success at qigong by practicing at regular hours each day to promote discipline. Qigong teachers also recommend that students prepare by adopting lifestyles that promote balance, moderation, proper rest,

and healthy **diets**, all of which are facets of qigong practice.

Precautions

Beginners should learn from an experienced teacher, as performing qigong exercises in the wrong manner may cause harm. Practitioners should not perform qigong on either full or completely empty stomachs. Qigong should not be performed during extreme weather, which may have negative effects on the body's energy systems. Menstruating and pregnant women should perform only certain exercises.

Side effects

Side effects may occur during or after qigong exercises for beginners, or for those performing exercises incorrectly. Side effects may include **dizziness**, **dry mouth**, **fatigue**, headaches, **insomnia**, rapid heartbeat, **shortness of breath**, heaviness or **numbness** in areas of the body, emotional instability, **anxiety**, or decreased concentration. Side effects generally clear up with rest and instruction from a knowledgeable teacher.

Research and general acceptance

Western medicine generally does not endorse any of the traditional Chinese healing systems that utilize the concept of energy flow in the body, largely because this energy has yet to be isolated and measured scientifically. New research is being conducted using sophisticated equipment that may verify the existence of energy channels as defined by the Chinese system. Despite the lack of scientific validation, the results of energy techniques including qigong and acupuncture have gained widespread interest and respect. One California group of qigong practitioners now conducts twice-yearly retreats to improve their skills and energy level. Furthermore, qigong masters have demonstrated to Western observers astounding control over many physical functions, and some have even shown the ability to increase electrical voltage measured on their skin's surface. Most of the research and documentation of qigong's effectiveness for medical conditions has been conducted in China, and is slowly becoming more available to English readers. Papers from the World Conferences for Academic Exchange of Medical Qigong are available in English, and address many medical studies and uses of qigong. A video is now available that presents the basic concepts of medical qigong as well as specific exercise prescriptions for the treatment of **breast cancer**. The exercise

KEY TERMS

Martial arts—Group of diverse activities originating from the ancient fighting techniques of the Orient.

Meridians—Channels or conduits through which Qi travels in the body.

Qi—Basic life energy, according to traditional Chinese medicine.

Yin/Yang—Universal characteristics used to describe aspects of the natural world.

prescriptions consist of movements, postures, visualizations, and positive affirmations.

In terms of mainstream research in the United States, the first ongoing long-term study of qigong began in 1999 at the Center for Alternative and Complementary Medicine Research in Heart Disease at the University of Michigan; it focuses on the speed of healing of graft **wounds** in patients undergoing coronary bypass surgery. The National Center for Complementary and Alternative Medicine (NCCAM) has been funding studies of qigong since 2000. The first such study was conducted by a researcher in Arizona with patients using heart devices (**pacemakers**, etc.).

The breathing techniques of qigong are being studied intensively by Western physicians as of 2003 as a form of therapy for anxiety-related problems and for disorders involving the vocal cords. Qigong is also being used in the **rehabilitation** of patients with severe asthma or chronic obstructive pulmonary disease (COPD).

Training and certification

In China, qigong has been subject to much government regulation, from banning to increased requirements for teachers. In the United States at this time, qigong has not been regulated. Different schools may provide teacher training, but there are no generally accepted training standards. Qigong teachings may vary, depending on the founder of the school, who is often an acknowledged Chinese master. The organizations listed below can provide further information to consumers.

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ORGANIZATIONS

Chinese National Chi Kung Institute. PO Box 31578. San Francisco, CA 94131. (800) 824-2433.

International Chi Kung/Qi Gong Directory. 2730 29th Street. Boulder, CO 80301. (303) 442-3131.

National Center for Complementary and Alternative Medicine (NCCAM) Clearinghouse. P.O. Box 7923, Gaithersburg, MD 20898-7923. (888) 644-6226. <<http://nccam.nih.gov>>.

Qigong Human Life Research Foundation. PO Box 5327. Cleveland, OH 44101. (216) 475-4712.

OTHER

Qigong Magazine. PO Box 31578. San Francisco, CA 94131. (800) 824-2433.

Qi: The Journal of Traditional Eastern Health and Fitness. PO Box 221343. Chantilly, VA 22022. (202) 378 3859.

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Quadriplegia see **Paralysis**

Quarantine see **Isolation**

Quinidine see **Antiarrhythmic drugs**

Quinine see **Antimalarial drugs**

R

Rabbit fever see **Tularemia**

Rabies

Definition

Rabies is an acute viral disease of the central nervous system that affects humans and other mammals but is most common in carnivores (flesh-eaters). It is sometimes referred to as a **zoonosis**, or disease of animals that can be communicated to humans. Rabies is almost exclusively transmitted through saliva from the bite of an infected animal. Another name for the disease is *hydrophobia*, which literally means “fear of water,” a symptom shared by half of all people infected with rabies. Other symptoms include **fever**, depression, confusion, painful **muscle spasms**, sensitivity to touch, loud noise, and light, extreme thirst, painful swallowing, excessive salivation, and loss of muscle tone. If rabies is not prevented by immunization, it is almost always fatal.

Description

Cases of rabies in humans are very infrequent in the United States and Canada, averaging one or two a year (down from over 100 cases annually in 1900), but the worldwide incidence is estimated to be between 30,000 and 50,000 cases each year. These figures are based on data collected by the World Health Organization (WHO) in 1997 and updated in 2002. Rabies is most common in developing countries in Africa, Latin America, and Asia, particularly India. Dog bites are the major origin of infection for humans in developing countries, but other important host animals may include the wolf, mongoose, raccoon, jackal, and bat. A group of researchers in India found that monkeys as well as dogs were frequent vectors of rabies. The team also reported that the male:female ratio of rabies patients in India is 4:1.

Most deaths from rabies in the United States and Canada result from bat bites; the most recent fatality was a 66-year-old man in California who died in September 2003. The **death** of a nine-year-old girl in Quebec in the fall of 2000 was the first case of human rabies in Canada since 1985. Public health officials eventually determined that the girl had been bitten while she was sleeping by a silver-haired bat that had gotten into the family’s home.

On October 18, 2004, a Wisconsin teenager was diagnosed with full-blown rabies after suffering from a minor bat bite on September 12, 2004. Miraculously, she was cured of rabies after doctors induced **coma** and administered four **antiviral drugs** to her. Since the therapy was only given and successful for one case, its curative properties needs to be corroborated by other cases before it will be considered a viable treatment option. The case and the physicians’ findings will be published in a medical journal.

People whose work frequently brings them in contact with animals are considered to be at higher risk than the general population. This would include those in the fields of veterinary medicine, animal control, wildlife work, and laboratory work involving live rabies virus. People in these occupations and residents of or travelers to areas where rabies is a widespread problem should consider being immunized.

In late 2002, rabies re-emerged as an important public health issue. Dr. Charles E. Rupprecht, director of the World Health Organization (WHO) Collaborating Center for Rabies Reference and Research, has listed several factors responsible for the increase in the number of rabies cases worldwide:

- Rapid evolution of the rabies virus. Bats in the United States have developed a particularly infectious form of the virus.
- Increased diversity of animal hosts for the disease.

- Changes in the environment that are bringing people and domestic pets into closer contact with infected wildlife.
- Increased movement of people and animals across international borders. In one recent case, a man who had contracted rabies in the Philippines was not diagnosed until he began to feel ill in the United Kingdom.
- Lack of advocacy about rabies.

Causes and symptoms

Rabies is caused by a rod- or bullet-shaped virus that belongs to the family *Rhabdoviridae*. The rabies virus is a member of a genus of viruses called lyssaviruses, which include several related viruses that infect insects as well as mammals. The rabies virus is usually transmitted via an animal bite, however, cases have also been reported in which the virus penetrated the body through infected saliva, moist tissues such as the eyes or lips, a scratch on the skin, or the transplantation of infected tissues. Inhalation of the virus in the air, as might occur in a highly populated bat cave, is also thought to occur.

From the bite or other area of penetration, the virus multiplies as it spreads along nerves that travel away from the spinal cord and brain (efferent nerves) and into the salivary glands. The rabies virus may lie dormant in the body for several weeks or months, but rarely much longer, before symptoms appear. Initially, the area around the bite may burn and be painful. Early symptoms may also include a **sore throat**, low-grade fever, **headache**, loss of appetite, **nausea and vomiting**, and **diarrhea**. Painful spasms develop in the muscles that control breathing and swallowing. The individual may begin to drool thick saliva and may have dilated or irregular pupils, increased tears and perspiration, and low blood pressure.

Later, as the disease progresses, the patient becomes agitated and combative and may exhibit increased mental confusion. The affected person usually becomes sensitive to touch, loud noises, and bright lights. The victim also becomes extremely thirsty, but is unable to drink because swallowing is painful. Some patients begin to dread water because of the painful spasms that occur. Other severe symptoms during the later stage of the disease include excessive salivation, **dehydration**, and loss of muscle tone. Death usually occurs three to 20 days after symptoms have developed. Unfortunately, recovery is very rare.

Diagnosis

After the onset of symptoms, blood tests and **cerebrospinal fluid (CSF) analysis** tests will be conducted. CSF will be collected during a procedure called a lumbar puncture in which a needle is used to withdraw a sample of CSF from the area around the spinal cord. The CSF tests do not confirm diagnosis but are useful in ruling out other potential causes for the patient's altered mental state.

The two most common diagnostic tests are the fluorescent antibody test and isolation of the rabies virus from an individual's saliva or **throat culture**. The fluorescent antibody test involves taking a small sample of skin (biopsy) from the back of the neck of the patient. If specific proteins, called antibodies, that are produced only in response to the rabies virus are present, they will bind with the fluorescent dye and become visible. Another diagnostic procedure involves taking a corneal impression in which a swab or slide is pressed lightly against the cornea of the eye to determine whether viral material is present.

Treatment

Until the most recent successful cure of a late-term rabies case can be validated with further success and validation from the medical community, the historic treatment options for rabies prevention immediately following a bite remains the most viable treatment. Because of the extremely serious nature of a rabies infection, the need for rabies immunizations will be carefully considered for anyone who has been bitten by an animal, based on a personal history and results of diagnostic tests.

If necessary, treatment includes the following:

- The wound is washed thoroughly with medicinal soap and water. Deep puncture **wounds** should be flushed with a catheter and soapy water. Unless absolutely necessary, a wound should not be sutured.
- Tetanus toxoid and **antibiotics** will usually be administered.
- Rabies **vaccination** may or not be given, based on the available information. If the individual was bitten by a domestic animal and the animal was captured, the animal will be placed under observation in quarantine for ten days. If the animal does not develop rabies within four to seven days, then no immunizations are required. If the animal is suspected of being rabid, it is killed, and the brain is examined for evidence of rabies infection. In cases involving bites from domestic animals where the animal is not available for examination, the decision for vaccination is

made based on the prevalence of rabies within the region where the bite occurred. If the bite was from a wild animal and the animal was captured, it is generally killed because the incubation period of rabies is unknown in most wild animals.

- If necessary, the patient is vaccinated immediately, generally through the administration of human rabies immune globulin (HRIG) for passive immunization, followed by human diploid cell vaccine (HDCV) or rabies vaccine adsorbed (RVA) for active immunization. Passive immunization is designed to provide the individual with antibodies from an already immunized individual, while active immunization involves stimulating the individual's own immune system to produce antibodies against the rabies virus. Both rabies vaccines are equally effective and carry a lower risk of side effects than some earlier treatments. Unfortunately, however, in underdeveloped countries, these newer vaccines are usually not available. Antibodies are administered to the patient in a process called passive immunization. To do this, the HRIG vaccine is administered once, at the beginning of treatment. Half of the dose is given around the bite area, and the rest is given in the muscle. Inactivated viral material (antigenic) is then given to stimulate the patient's own immune system to produce antibodies against rabies. For active immunization, either the HDCV or RVA vaccine is given in a series of five injections. Immunizations are typically given on days one, three, seven, 14, and 28.

In those rare instances in which rabies has progressed beyond the point where immunization would be effective, the groundbreaking treatment involving a drug-induced coma and the administration of four different antiviral drugs will most likely be a radical treatment option. The traditional approach prior to October 2004 was to provide as much relief from **pain** and suffering as possible through medical intervention while waiting to see if survival was possible. The patient would be given medication to prevent seizures, relieve some of the **anxiety**, and relieve painful muscle spasms. Pain relievers would also be given. In the later stages, aggressive supportive care would be provided to maintain breathing and heart function. Survival via the traditional treatment is rare but can occur.

Prognosis

If preventative treatment is sought promptly, rabies need not be fatal. Immunization is almost always effective if started within two days of the bite. Chance of effectiveness declines, however, the longer vaccination is put off. It is, however, important to start

KEY TERMS

Active immunization—Treatment that provides immunity by challenging an individual's own immune system to produce antibody against a particular organism, in this case the rabies virus.

Antibody—A specific protein produced by the immune system in response to a specific foreign protein or particle called an antigen.

Biopsy—The removal of a small sample of tissue for diagnostic purposes.

Efferent nerves—Nerves that convey impulses away from the central nervous system to the periphery.

Fluorescent antibody test (FA test)—A test in which a fluorescent dye is linked to an antibody for diagnostic purposes.

Lumbar puncture—A procedure that involves withdrawing a small sample of cerebrospinal fluid from the back around the spinal cord.

Lyssavirus—A genus of viruses that includes the rabies virus and related viruses that infect insects as well as mammals.

Passive immunization—Treatment that provides immunity through the transfer of antibodies obtained from an immune individual.

Rhabdovirus—A type of virus named for its rod- or bullet-like shape. The rabies virus belongs to a family of viruses called Rhabdoviridae.

Vector—An animal or insect that carries a disease-producing organism.

Zoonosis—Any disease of animals that can be transmitted to humans. Rabies is an example of a zoonosis.

immunizations, even if it has been weeks or months following a suspected rabid animal bite, because the vaccine can be effective even in these cases. If immunizations do not prove effective or are not received, rabies is nearly always fatal with a few days of the onset of symptoms.

As of October 2004, the medical community awaits the publication of findings by the doctors that administered a life-saving treatment that cured a Wisconsin teenager of full-blown rabies. Further test cases will prove whether or not this treatment option will be a historic development in the search for a cure for rabies.

Prevention

One promising preventive strategy that has been used since the early 2000s is the distribution of wildlife baits containing an oral vaccine against rabies. This strategy has been used in Germany to vaccinate wild foxes, which are frequent carriers of the disease in Europe. In the United States, veterinary researchers at Kansas State University have developed an oral vaccine for fruit bats; early trials of the vaccine have given promising results.

The following precautions should be observed in environments where humans and animals may likely come into contact.

- Domesticated animals, including household pets, should be vaccinated against rabies. If a pet is bitten by an animal suspected to have rabies, its owner should contact a veterinarian immediately and notify the local animal control authorities. Domestic pets with current vaccinations should be revaccinated immediately; unvaccinated dogs, cats, or ferrets are usually euthanized (put to sleep). Further information about domestic pets and rabies is available on the American Veterinary Medical Association (AVMA) web site.
- Wild animals should not be touched or petted, no matter how friendly they may appear. It is also important not to touch an animal that appears ill or passive, or whose behavior seems odd, such as failing to show the normal fear of humans. These are all possible signs of rabies. Many animals, such as raccoons and skunks, are nocturnal and their activity during the day should be regarded as suspicious.
- People should not interfere in fights between animals.
- Because rabies is transmitted through saliva, a person should wear rubber gloves when handling a pet that has had an encounter with a wild animal.
- Garbage or pet food should not be left outside the house or camp site because it may attract wild or stray animals.
- Windows and doors should be screened. Some victims of rabies have been attacked by infected animals, particularly bats, that entered through unprotected openings.
- State or county health departments should be consulted for information about the prevalence of rabies in an area. Some areas, such as New York City, have been rabies-free, only to have the disease reintroduced at a later time.
- Preventative vaccination against rabies should be considered if one's occupation involves frequent contact with wild animals or non-immunized domestic animals.
- Bites from mice, rats, or squirrels rarely require rabies prevention because these rodents are typically killed by any encounter with a larger, rabid animal, and would, therefore, not be carriers.
- Travelers should ask about the prevalence of the disease in countries they plan to visit.

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Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

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Radial keratotomy

Definition

Radial keratotomy (RK) is a type of eye surgery used to correct **myopia** (nearsightedness). It works by changing the shape of the cornea—the transparent part of the eye that covers the iris and the pupil.

Purpose

About 25–30% of all people in the world are nearsighted and need eyeglasses or contact lenses for distance vision to be clear. For a number of reasons, some people don't like wearing corrective lenses. Some feel unattractive in eyeglasses. Others worry about not being able to see without their glasses in an emergency, such as a house fire or a burglary. Both glasses and contact lenses can be scratched, broken, or lost. In addition, contact lenses require special care and can irritate the eyes.

Radial keratotomy was introduced in North America in 1978. Since then doctors have improved the technique, and its results have become more predictable. Radial keratotomy is one of several surgical techniques to correct nearsightedness, reducing or eliminating the need for corrective lenses. It is most successful in patients with a low to moderate amount of nearsightedness—people whose eyes require up to -5.00 diopters of correction. A diopter (D) is a unit of measure of focusing power. Minus lenses correct nearsightedness.

Precautions

Not every nearsighted person is a good candidate for radial keratotomy. This type of surgery cannot help people whose nearsightedness is caused by keratoconus, a rare condition in which the cornea is cone shaped. The procedure usually is not done on patients under 18, because their eyes are still growing and changing shape. It is important that visual status is stable. Women who are pregnant, have just given birth, or are breast-feeding should not have the surgery because hormonal changes may cause temporary changes in the cornea. In addition, anyone with **glaucoma** or with any disease that interferes with healing (e.g., **rheumatoid arthritis**, lupus erythematosus, or uncontrolled diabetes) should not have RK.

Radial keratotomy weakens the cornea, making it vulnerable to injuries even long after the surgery. Getting hit in the head after having RK can cause the cornea to tear and can lead to blindness. For this reason, the procedure is not recommended for people who engage in sports that could result in a blow to the head (i.e., karate or racquetball).

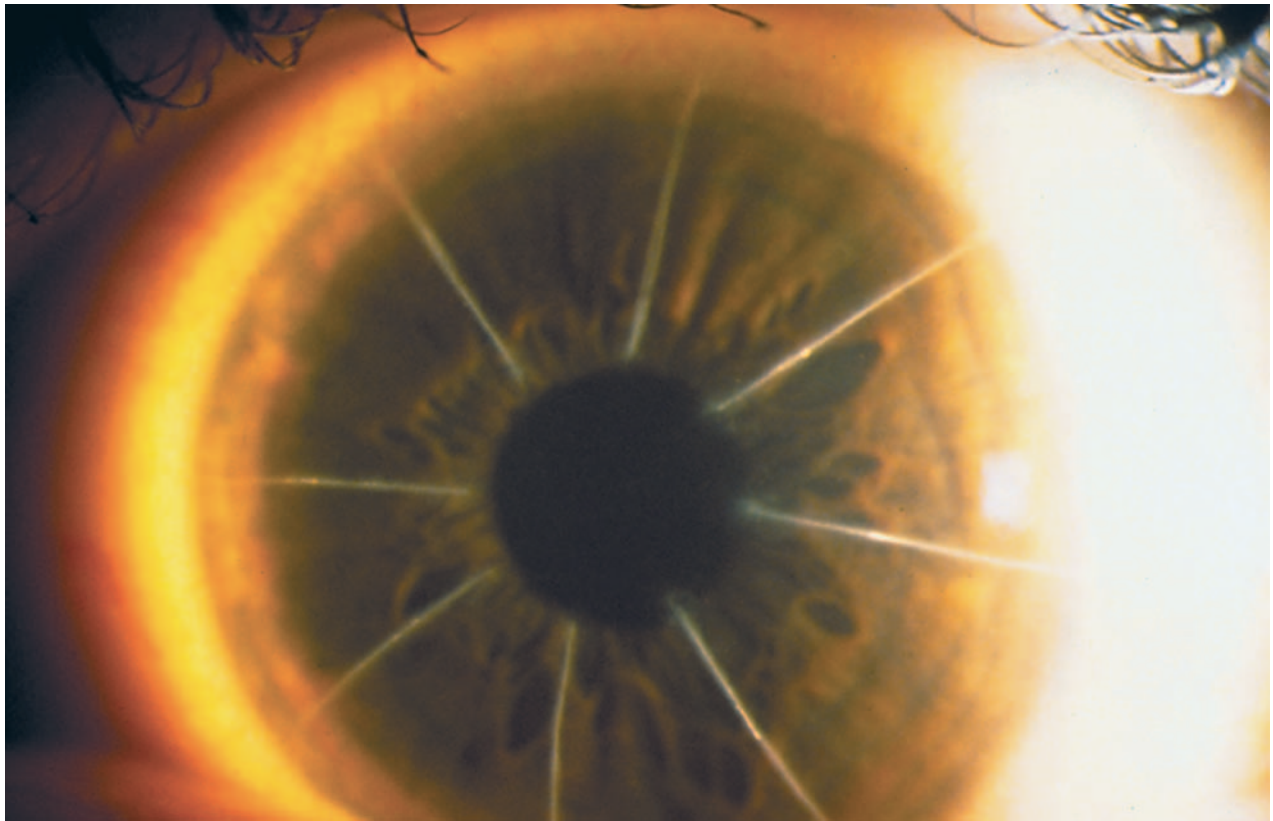
It is important to keep in mind that RK is a permanent procedure and that success cannot be guaranteed. An experienced eye surgeon can estimate how likely it is that the surgery will help a particular patient, but that is just an estimate. There is no way to know for sure whether the surgery will improve eyesight enough to eliminate the need for corrective lenses. Vision usually improves after RK, but it is not always perfect. Anyone who decides to have RK should be prepared to accept less-than-perfect vision after surgery, which may necessitate the continued use of glasses or contact lenses. This surgery does not eliminate the need for reading glasses. Actually, someone who didn't need reading glasses before surgery because their myopia allowed near vision to be clear may find themselves needing reading glasses. Patients must ask about this prior to surgery.

Anyone considering RK should also be aware that certain professions, including branches of the military, are not open to people who have had the procedure.

A reputable ophthalmologist will discuss the risks of the procedure and should tell anyone considering it that perfect vision can't be guaranteed. Patients should be wary of any doctor who tries too hard to "sell" them on RK.

Description

In a person with clear vision, light passes through the cornea and the lens of the eye and focuses on a



Radial keratotomy scars on the cornea of an eye. (Photograph by Bob Masini, Phototake NYC. Reproduced by permission.)

membrane lining the back of the eye called the retina. In a person with myopia, the eyeball is usually too long, so light focuses in front of the retina. Radial keratotomy reduces myopia by flattening the cornea. This reduces the focusing power of the cornea allowing light to focus further back onto the retina (or at least closer to it), forming a clearer image.

A surgeon performing RK uses a very small diamond-blade knife to make four to eight radial incisions around the edge of the cornea. These slits are made in a pattern that resembles the spokes of a wheel. As the cornea heals, its center flattens out.

Radial keratotomy is usually performed in an ophthalmologist's office. Before the surgery begins, the patient may be given medicine to help him or her relax. A local anesthetic—usually in the form of eye drops—is used to numb the eye, but the patient remains conscious during the procedure. The surgeon looks through a surgical microscope while making the slits. The treatment usually takes no more than 30 minutes.

Some ophthalmologists will perform RK on both eyes at once but others prefer to do one eye at a time. It once was thought that surgeons could use the results of

the first eye to predict how well the procedure would work on the second eye. However, a study published in 1997 found that this was not the case. The authors of the study cautioned that there might be other reasons not to operate on both eyes at once, such as increased risk of infection and other complications.

The cost for RK depends on the surgeon, but usually ranges from \$1,000–\$1,500 per eye. Medical insurance usually does not cover RK, because it is considered an elective procedure—one that people choose to have done.

Preparation

Before beginning the procedure, the surgeon marks an area in the center of the cornea called the optical zone. This is the part of the cornea that one sees through (it is the area over the pupil). No cuts are made in this region. The surgeon also measures the cornea's thickness, to decide how deep the slits should be.

Aftercare

After the surgery is over, the anesthetic wears off. Some patients feel slight **pain** and are given eye drops

and medications to relieve their discomfort. For several days after the surgery, the eye that was treated may feel scratchy and look red. This is normal. The eye may also water, burn slightly, and be sensitive to light.

As with any type of surgery, it is important to guard against infection. Patients are given eye drops to protect against infection and may be told to use them for several weeks after the surgery. Because RK weakens the cornea it is important to protect the head and eyes.

The cornea heals slowly, and full recovery can take several months (another reason not to have the surgery done on both eyes at the same time). While the cornea is healing, patients may experience these problems:

- Variations in vision. Eyesight may be better in the morning than in the evening or vice versa.
- Temporary pain.
- Increased glare.
- Starburst or halo effects. Rays or rings of light around lights at night.
- Hyperopic shift. As the cornea flattens, vision may become more farsighted (hyperopic). For this reason, the surgeon may initially undercorrect the patient. This gradual shift may occur over several years.

If RK does not completely correct a person's nearsightedness, glasses or contact lenses may be needed. In general, people who were able to wear contact lenses before the procedure can still wear them afterward. Even patients whose nearsightedness was corrected may still need glasses for reading. This is especially true for middle-aged and older patients. The lens of the eye stiffens with age, making reading glasses necessary (**presbyopia**). Radial keratotomy does not correct this problem.

The surgeon who performs the RK procedure will tell the patient how often to return for follow-up visits. Often, two to four visits are needed, including one the day after surgery. It is also important to know what side effects should be reported immediately to the surgeon (e.g., pain or **nausea**).

Risks

Complications from RK are rare, but they can occur. These include:

- cataract a clouding of the lens of the eye, resulting in partial or total loss of vision
- serious infection
- lasting pain

KEY TERMS

Cornea—The transparent part of the eye that covers the iris and the pupil.

Diopter (D)—Unit describing the amount of focusing power of a lens.

Iris—The colored part of the eye.

Laser-assisted in situ keratomileusis (LASIK)—A type of refractive eye surgery using a laser and another instrument to change the shape of the cornea.

Local anesthetic—Used to numb an area where surgery or another procedure is to be done, without causing the patient to lose consciousness.

Myopia—Nearsightedness. People with myopia cannot see distant objects clearly.

Ophthalmologist—A physician who specializes in treating eyes.

Photorefractive keratectomy (PRK)—A type of refractive eye surgery using a laser to change the shape of the cornea.

Pupil—The part of the eye that looks like a black circle in the center of the iris. It is actually an opening through which light passes.

Retina—A membrane lining the back of the eye onto which light is focused to form images.

- rips along an incision, especially after being hit in the head or eye
- loss of vision
- chance of overcorrection (hyperopic shift)

The chances of complications are reduced when the surgery is done by an ophthalmologist with a lot of experience in RK. Younger patients also tend to heal faster.

Normal results

The desired result of radial keratotomy is a reduction in myopia. A major study by the National Eye Institute, reported in 1994, tracked the success of RK in 374 patients who had had the procedure done 10 years earlier. The study found that:

- 85% had at least 20/40 vision (the acuity considered good enough to drive without glasses)
- 70% did not need glasses or contact lenses for distance vision

- 53% had 20/20 vision without glasses
- 30% still needed glasses or contact lenses to see clearly
- 1–3% had worse vision than before they had RK
- 40% had a hyperopic shift.

As with all surgeries, RK has risks. These risks include having worse vision than before the surgery; halos; glare; and although rare, blindness. Some aftereffects, such as halos or glare may last for years. Other refractive surgeries, such as photorefractive keratectomy (PRK) and laser-assisted in situ keratomileusis (LASIK) use lasers to change the shape of the cornea and they may produce fewer side effects. It is important to speak with an experienced eye surgeon who has done many refractive surgeries to fully understand the options and risks involved before making a decision.

Resources

ORGANIZATIONS

American Academy of Ophthalmology. 655 Beach Street, P.O. Box 7424, San Francisco, CA 94120-7424. <<http://www.eyenet.org>>.

American Optometric Association. 243 North Lindbergh Blvd., St. Louis, MO 63141. (314) 991-4100. <<http://www.aoanet.org>>.

American Society of Cataract & Refractive Surgery. 4000 Legato Road, Suite 850, Fairfax, VA 22033. (703) 591-2220. <<http://www.ascrs.org>>.

Nancy Ross-Flanigan

Radiation injuries

Definition

Radiation injuries are caused by ionizing radiation emitted by sources such as the sun, x-ray and other diagnostic machines, tanning beds, and radioactive elements released in nuclear power plant accidents and detonation of nuclear weapons during war and as terrorist acts.

Description

Ionizing radiation is made up of unstable atoms that contain an excess amount of energy. In an attempt to stabilize, the atoms emit the excess energy into the atmosphere, creating radiation. Radiation can either be electromagnetic or particulate.

The energy of electromagnetic radiation is a direct function of its frequency. The high-energy, high-frequency waves that can penetrate solids to various depths cause damage by separating molecules into electrically charged pieces, a process known as ionization. X rays are a type of electromagnetic radiation. Atomic particles come from radioactive isotopes as they decay to stable elements. Electrons are called beta particles when they radiate. Alpha particles are the nuclei of helium atoms—two protons and two neutrons—without the surrounding electrons. Alpha particles are too large to penetrate a piece of paper unless they are greatly accelerated in electric and magnetic fields. Both beta and alpha particles are types of particulate radiation. When over-exposure to ionizing radiation occurs, there is chromosomal damage in deoxyribonucleic acid (DNA). DNA is very good at repairing itself; both strands of the double helix must be broken to produce genetic damage.

Because radiation is energy, it can be measured. There are a number of units used to quantify radiation energy. Some refer to effects on air, others to effects on living tissue. The roentgen, named after Wilhelm Conrad Roentgen, who discovered x rays in 1895, measures ionizing energy in air. A rad expresses the energy transferred to tissue. The rem measures tissue response. A roentgen generates about a rad of effect and produces about a rem of response. The gray and the sievert are international units equivalent to 100 rads and rems, respectively. A curie, named after French physicists who experimented with radiation, is a measure of actual radioactivity given off by a radioactive element, not a measure of its effect. The average annual human exposure to natural background radiation is roughly 3 milliSieverts (mSv).

Any amount of ionizing radiation will produce some damage, however, there is radiation everywhere, from the sun (cosmic rays) and from traces of radioactive elements in the air (radon) and the ground (uranium, radium, carbon-14, potassium-40 and many others). Earth's atmosphere protects us from most of the sun's radiation. Living at 5,000 feet altitude in Denver, Colorado, doubles exposure to radiation, and flight in a commercial airliner increases it 150-fold by lifting us above 80% of that atmosphere. Because no amount of radiation is perfectly safe and because radiation is ever present, arbitrary limits have been established to provide some measure of safety for those exposed to unusual amounts. Less than 1% of them reach the current annual permissible maximum of 20 mSv.

A 2001 ruling by the Federal Court of Australia indicated that two soldiers died from **cancer** caused by

minimal exposure to radiation while occupying Hiroshima in 1945. The soldiers were exposed to less than 5 mSv of radiation. The international recommendation for workers is safety level of up to 20 mSv. The ruling and its support by many international agencies suggests that even extremely low doses of radiation can be potentially harmful.

Ultraviolet (UV) radiation exposure from the sun and tanning beds

UV radiation from the sun and tanning beds and lamps can cause skin damage, premature **aging**, and skin cancers. **Malignant melanoma** is the most dangerous of skin cancers and there is a definite link between type UVA exposure used in tanning beds and its occurrence. UVB type UV radiation is associated with **sunburn**, and while not as penetrating as UVA, it still damages the skin with over exposure. Skin damage accumulates over time, and effects do not often manifest until individuals reach middle age. Light-skinned people who most often burn rather than tan are at a greater risk of skin damage than darker-skinned individuals that almost never burn. The U.S. Food and Drug Administration (FDA) and the Centers for Disease Control (CDC) discourage the use of tanning beds and sun lamps and encourage the use of sunscreen with at least an SPF of 15 or greater.

Over exposure during medical procedures

Ionizing radiation has many uses in medicine, both in diagnosis and in treatment. X rays, CT scanners, and fluoroscopes use it to form images of the body's insides. Nuclear medicine uses radioactive isotopes to diagnose and to treat medical conditions. In the body, radioactive elements localize to specific tissues and give off tiny amounts of radiation. Detecting that radiation provides information on both anatomy and function. During the past 10 years, skin injuries caused by too much exposure during a medical procedure have been documented. In 1995, the FDA issued a recommendation to physicians and medical institutions to record and monitor the dosage of radiation used during medical procedures on patients in order to minimize the amount of skin injuries. The FDA suggested doses of radiation not exceed 1 Grey (Gy). (A Grey is roughly equivalent to a sievert.) As of 2001, the FDA was preparing further guidelines for fluoroscopy, the procedure most often associated with medical-related radiation skin injuries such as **rashes** and more serious **burns** and tissue death. Injuries occurred most often during **angioplasty** procedures using fluoroscopy.

CT scans of children have also been problematic. Oftentimes the dosage of radiation used for an adult isn't decreased for a child, leading to radiation over exposure. Children are more sensitive to radiation and a February 2001 study indicates 1,500 out of 1.6 million children under 15 years of age receiving CT scans annually will develop cancer. Studies show that decreasing the radiation by half for CT scans of children will effectively decrease the possibility of over exposure while still providing an effective diagnostic image. The benefits to receiving the medical treatment utilizing radiation is still greater than the risks involved, however, more stringent control over the amount of radiation used during the procedures will go far to minimize the risk of radiation injury to the patient.

Radiation exposure from nuclear accidents, weaponry, and terrorist acts

Between 1945 and 1987, there were 285 nuclear reactor accidents, injuring over 1,550 people and killing 64. The most striking example was the meltdown of the graphite core nuclear reactor at Chernobyl in 1986, which spread a cloud of radioactive particles across the entire continent of Europe. Information about radiation effects is still being gathered from that disaster, however 31 people were killed in the immediate accident and 1,800 children have thus far been diagnosed with **thyroid cancer**. In a study published in May 2001 by the British Royal Society, children born to individuals involved in the cleanup of Chernobyl and born after the accident are 600% more likely to have genetic mutations than children born before the accident. These findings indicate that exposure to low doses of radiation can cause inheritable effects.

Since the terrorist attack on the World Trade Center and the Pentagon on September 11, 2001, the possibility of terrorist-caused nuclear accidents has been a growing concern. All 103 active nuclear power plants in the United States are on full alert, but they are still vulnerable to sabotage such as bombing or attack from the air. A no-fly zone of 12 miles below 18,000 feet has been established around nuclear power plants by the Federal Aviation Administration (FAA). There is also growing concern over the security of spent nuclear fuel—more than 40,000 tons of spent fuel is housed in buildings at closed plants around the country. Unlike the active nuclear reactors that are enclosed in concrete-reinforced buildings, the spent fuel is stored in non-reinforced buildings. Housed in cooling pools, the spent fuel could emit dangerous levels of radioactive material if exploded or used in makeshift weaponry. Radioactive medical and industrial waste could also be used to make “dirty



This person's nose is inflamed and scaly due to radiation exposure. (Custom Medical Stock Photo. Reproduced by permission.)

bombs.” Since 1993, the Nuclear Regulatory Commission (NRC) has reported 376 cases of stolen radioactive materials.

Causes and symptoms

Radiation can damage every tissue in the body. The particular manifestation will depend upon the amount of radiation, the time over which it is absorbed, and the susceptibility of the tissue. The fastest growing tissues are the most vulnerable, because radiation as much as triples its effects during the growth phase. Bone marrow cells that make blood are the fastest growing cells in the body. A fetus in the womb is equally sensitive. The germinal cells in the testes and ovaries are only slightly less sensitive. Both can be rendered useless with very small doses of radiation. More resistant are the lining cells of the body—skin and intestines. Most resistant are the brain cells, because they grow the slowest.

The length of exposure makes a big difference in what happens. Over time the accumulating damage, if not enough to kill cells outright, distorts their growth and causes scarring and/or cancers. In addition to leukemias, cancers of the thyroid, brain, bone, breast, skin, stomach, and lung all arise after radiation. Damage depends, too, on the ability of the tissue to repair itself. Some tissues and some types of damage produce much greater consequences than others.

There are three types of radiation injuries.

- **External irradiation:** as with x-ray exposure, all or part of the body is exposed to radiation that either is absorbed or passes through the body.
- **Contamination:** as with a nuclear accident, the environment and its inhabitants are exposed to radiation.

People are affected internally, externally, or with both internal and external exposure.

- **Incorporation:** dependent on contamination, the bodies of individuals affected incorporate the radiation chemicals within cells, organs, and tissues and the radiation is dispersed throughout the body.

Immediately after sudden irradiation, the fate of those affected depends mostly on the total dose absorbed. This information comes mostly from survivors of the atomic bomb blasts over Japan in 1945.

- Massive doses incinerate immediately and are not distinguishable from the heat of the source.
- A sudden whole body dose over 50 Sv produces such profound neurological, heart, and circulatory damage that patients die within the first two days.
- Doses in the 10–20 Sv range affect the intestines, stripping their lining and leading to death within three months from **vomiting, diarrhea, starvation, and infection.**
- Victims receiving 6–10 Sv all at once usually escape an intestinal death, facing instead bone marrow failure and death within two months from loss of blood coagulation factors and the protection against infection provided by white blood cells.
- Between 2–6 Sv gives a fighting chance for survival if victims are supported with blood transfusions and **antibiotics.**
- One or two Sv produces a brief, non-lethal sickness with vomiting, loss of appetite, and generalized discomfort.

Treatment

It is clearly important to have some idea of the dose received as early as possible, so that attention can be directed to those victims in the 2–10 Sv range that might survive with treatment. Blood transfusions, protection from infection in damaged organs, and possibly the use of newer stimulants to blood formation can save many victims in this category.

Local radiation exposures usually damage the skin and require careful wound care, removal of dead tissue, and **skin grafting** if the area is large. Again **infection control** is imperative.

One of the best known, and perhaps even mainstream, treatments of radiation injury is the use of *Aloe vera* preparations on damaged areas of skin. It has demonstrated remarkable healing properties even for chronic ulcerations resulting from radiation exposure.

KEY TERMS

DNA—Deoxyribonucleic acid. The chemical of chromosomes and hence the vehicle of heredity.

Isotope—An unstable form of an element that gives off radiation to become stable. Elements are characterized by the number of electrons around each atom. One electron's negative charge balances the positive charge of each proton in the nucleus. To keep all those positive charges in the nucleus from repelling each other (like the same poles of magnets), neutrons are added. Only certain numbers of neutrons work. Other numbers cannot hold the nucleus together, so it splits apart, giving off ionizing radiation. Sometimes one of the split products is not stable either, so another split takes place. The process is called radioactivity.

Alternative treatment

There is considerable interest these days in benevolent chemicals called “free radical scavengers.” How well they work is yet to be determined, but population studies strongly suggest that certain **diets** are better than others, and that those diets are full of free radical scavengers, otherwise known as antioxidants. The recommended ingredients are beta-carotene, **vitamins** E and C, and selenium, all available as commercial preparations. Beta-carotene is yellow-orange and is present in yellow and orange fruits and vegetables. Vitamin C can be found naturally in citrus fruits. **Traditional Chinese medicine** (TCM) and **acupuncture**, botanical medicine, and homeopathy all have contributions to make to recovery from the damage of radiation injuries. The level of recovery will depend on the exposure. Consulting practitioners trained in these modalities will result in the greatest benefit.

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Jacqueline L. Longe

Radiation sickness see **Radiation injuries**

Radiation therapy

Definition

Radiation therapy, sometimes called radiotherapy, x-ray therapy radiation treatment, cobalt therapy, electron beam therapy, or irradiation uses high energy, penetrating waves or particles such as x rays, gamma rays, proton rays, or neutron rays to destroy **cancer** cells or keep them from reproducing.

Purpose

The purpose of radiation therapy is to kill or damage cancer cells. Radiation therapy is a common form of cancer therapy. It is used in more than half of all cancer cases. Radiation therapy can be used:

- alone to kill cancer
- before surgery to shrink a tumor and make it easier to remove
- during surgery to kill cancer cells that may remain in surrounding tissue after the surgery (called intraoperative radiation)
- after surgery to kill cancer cells remaining in the body
- to shrink an inoperable tumor in order to and reduce **pain** and improve quality of life.
- in combination with **chemotherapy**

For some kinds of cancers such as early-stage **Hodgkin's disease**, non-Hodgkin's lymphoma, and certain types of prostate, or brain cancer, radiation therapy alone may cure the disease. In other cases, radiation therapy used in conjunction with surgery, chemotherapy, or both, increases survival rates over any of these therapies used alone.

Precautions

Radiation therapy does not make the person having the treatments radioactive. In almost all cases, the benefits of this therapy outweigh the risks. However radiation therapy can have serious consequences, so anyone contemplating it should be sure understand why the treatment team believes it is the best possible treatment option for their cancer. Radiation therapy is often not appropriate for pregnant women, because the radiation can damage the cells of the developing baby. Women who think they might be pregnant should discuss this with their doctor.

Description

Radiation therapy is a local treatment. It is painless. The radiation acts only on the part of the body that is exposed to the radiation. This is very different from chemotherapy in which drugs circulate throughout the whole body. There are two main types of radiation therapy. In external radiation therapy a beam of radiation is directed from outside the body at the cancer. In internal radiation therapy, called brachytherapy or implant therapy, where a source of radioactivity is surgically placed inside the body near the cancer.

How radiation therapy works

The protein that carries the code controlling most activities in the cell is called deoxyribonucleic acid or DNA. When a cell divides, its DNA must also double and divide. High-energy radiation kills cells by damaging their DNA, thus blocking their ability to grow and increase in number.

One of the characteristics of cancer cells is that they grow and divide faster than normal cells. This makes them particularly vulnerable to radiation. Radiation also damages normal cells, but because normal cells are growing more slowly, they are better able to repair radiation damage than are cancer cells. In order to give normal cells time to heal and reduce side effects, radiation treatments are often given in small doses over a six or seven week period.

External radiation therapy

External radiation therapy is the most common kind of radiation therapy. It is usually done during outpatient visits to a hospital clinic and is usually covered by insurance.

Once a doctor, called a radiation oncologist, determines the proper dose of radiation for a particular cancer, the dose is divided into smaller doses called fractions. One fraction is usually given each day, five days a week for six to seven weeks. However, each radiation plan is individualized depending on the type and location of the cancer and what other treatments are also being used. The actual administration of the therapy usually takes about half an hour daily, although radiation is administered for only from one to five minutes at each session. It is important to attend every scheduled treatment to get the most benefit from radiation therapy.

Recently, trials have begun to determine if there are ways to deliver radiation fractions so that they kill more cancer cells or have fewer side effects. Some trials use smaller doses given more often. Up-to-date information

on voluntary participation in clinical trials and where they are being held is available by entering the search term "radiation therapy" at the following web sites:

- National Cancer Institute. <<http://cancertrials.nci.nih.gov>> or (800) 4-CANCER.
- National Institutes of Health Clinical Trials. <<http://clinicaltrials.gov>>
- Center Watch: A Clinical Trials Listing. <<http://www.centerwatch.com>>.

The type of machines used to administer external radiation therapy and the material that provides the radiation vary depending on the type and location of the cancer. Generally, the patient puts on a hospital gown and lies down or sits in a special chair. Parts of the body not receiving radiation are covered with special shields that block the rays. A technician then directs a beam of radiation to a pre-determined spot on the body where the cancer is located. The patient must stay still during the administration of the radiation so that no other parts of the body are affected. As an extra precaution in some treatments, special molds are made to make sure the body is in the same position for each treatment. However, the treatment itself is painless, like having a bone x-rayed.

Internal radiation therapy

Internal radiation therapy is called brachytherapy, implant therapy, interstitial radiation, or intracavitary radiation. With internal radiation therapy, a bit of radioactive material is sealed in an implant (sometimes called a seed or capsule). The implant is then placed very close to the cancer. The advantage of internal radiation therapy is that it concentrates the radiation near the cancer and lessens the chance of damage to normal cells. Many different types of radioactive materials can be used in the implant, including cesium, iridium, iodine, phosphorus, and palladium.

How the implant is put near the cancer depends on the size and location of the cancer. Internal radiation therapy is used for some cancers of the head, neck, thyroid, breast, female reproductive system, and prostate. Most people will have the radioactive capsule implanted by a surgeon while under either general or **local anesthesia** at a hospital or surgical clinic.

Patients receiving internal radiation therapy do become temporarily radioactive. They must remain in the hospital during the time that the implant stays in place. The length of time is determined by the type of cancer and the dose of radioactivity to be delivered. During the time the implant is in place, the patient will have to stay in bed and remain reasonably still.

While the implant is in place, the patient's contact with other people will be limited. Healthcare workers will make their visits as brief as possible to avoid exposure to radiation, and visitors, especially children and pregnant women, will be limited.

The implant usually can be removed in a simple procedure without an anesthetic. As soon as the implant is out of the body, the patient is no longer radioactive, and restrictions on being with other people are lifted. Generally people can return to a level of activity that feels comfortable to them as soon as the implant is removed. Occasionally the site of the implant is sore for some time afterwards. This discomfort may limit specific activities.

In some cases, an implant is left permanently inside the body. People who have permanent implants need to stay in the hospital and away from other people for the first few days. Gradually the radioactivity of the implant decreases, and it is safe to be around other people.

Radioimmunotherapy

Radioimmunotherapy is a promising way to treat cancer that has spread (metastasized) to multiple locations throughout the body. Antibodies are immune system proteins that specifically recognize and bind to only one type of cell. They can be designed to bind only with a certain type of cancer cell. To carry out radioimmunotherapy, antibodies with the ability to bind specifically to a patient's cancer cells are attached to radioactive material and injected into the patient's bloodstream. When these man-made antibodies find a cancer cell, they bind to it. Then the radiation kills the cancer cell. This process is still experimental, but because it can be used to selectively attack only cancer cells, it holds promise for eliminating cancers that have spread beyond the primary tumor.

Radiation used to treat cancer

PHOTON RADIATION. Early radiation therapy used x rays like those used to take pictures of bones, or gamma rays. X rays and gamma rays are high energy rays composed of massless particles of energy (like light) called photons. The distinction between the two is that gamma rays originate from the decay of radioactive substances (like radium and cobalt-60), while x rays are generated by devices that excite electrons (such as cathode ray tubes and linear accelerators). These high energy rays act on cells by disrupting the electrons of atoms within the molecules inside cells, disrupting cell functions, and most importantly stop their ability to divide and make new cells.

PARTICLE RADIATION. Particle radiation is radiation delivered by particles that have mass. Proton therapy has been used since the early 1990s. Proton rays consist of protons, a type of positively charged atomic particle, rather than photons, which have neither mass nor charge. Like x rays and gamma rays, proton rays disrupt cellular activity. The advantage of using proton rays is that they can be shaped to conform to the irregular shape of the tumor more precisely than x rays and gamma rays. They allow delivery of higher radiation doses to tumors without increasing damage to the surrounding tissue.

Neutron therapy is another type of particle radiation. Neutron rays are very high-energy rays. They are composed of neutrons, which are particles with mass but no charge. The type of damage they cause to cells is much less likely to be repaired than that caused by x rays, gamma rays, or proton rays.

Neutron therapy can treat larger tumors than conventional radiation therapy. Conventional radiation therapy depends on the presence of oxygen to work. The center of large tumors lack sufficient oxygen to be susceptible to damage from conventional radiation. Neutron radiation works in the absence of oxygen, making it especially effective for the treatment of inoperable **salivary gland tumors**, bone cancers, and some kinds of advanced cancers of the pancreas, bladder, lung, prostate, and uterus.

Recent advances in radiation therapy

A newer mode of treating brain cancers with radiation therapy is known as stereotactic radiosurgery. As of the early 2000s, this approach is limited to treating cancers of the head and neck because only these parts of the body can be held completely still throughout the procedure. Stereotactic radiosurgery allows the doctor to deliver a single high-level dose of precisely directed radiation to the tumor without damaging nearby healthy brain tissue. The treatment is planned with the help of three-dimensional computer-aided analysis of CT and MRI scans. The patient's head and neck are held steady in a skeletal fixation device during the actual treatment. Stereotactic radiosurgery can be used in addition to standard surgery to treat a recurrent **brain tumor**, or in place of surgery if the tumor cannot be reached by standard surgical techniques.

Two major forms of stereotactic radiosurgery are in use as of 2003. The gamma knife is a stationary machine that is most useful for small tumors, blood vessels, or similar targets. Because it does not move, it can deliver a small, highly localized and precise beam of radiation. Gamma knife treatment is done all at once in

a single hospital stay. The second type of radiosurgery uses a movable linear accelerator-based machine that is preferred for larger tumors. This treatment is delivered in several small doses given over several weeks. Radiosurgery that is performed with divided doses is known as fractionated radiosurgery. The total dose of radiation is higher with a linear accelerator-based machine than with gamma knife treatment.

Another advance in intraoperative radiotherapy (IORT) is the introduction of mobile devices that allow the surgeon to use radiotherapy in early-stage disease and to operate in locations where it would be difficult to transport the patient during surgery for radiation treatment. Mobile IORT units have been used successfully as of 2003 in treating early-stage **breast cancer** and **rectal cancer**.

Radiation sensitizers are another recent innovation in radiation therapy. Sensitizers are medications that are given to make cancer cells easier to kill by radiation than normal cells. Gemcitabine (Gemzar) is one of the drugs most commonly used for this purpose.

Preparation

Before radiation therapy, the size and location of the patient's tumor are determined very precisely using **magnetic resonance imaging (MRI)** and/or **computed tomography scans (CT scans)**. The correct radiation dose, the number of sessions, the interval between sessions, and the method of application are calculated by a radiation oncologist based on the tumor type, its size, and the sensitivity of the nearby tissues.

The patient's skin is be marked with a semi-permanent ink to help the radiation technologist achieve correct positioning for each treatment. Molds may be built to hold tissues in exactly the right place each time.

Aftercare

Many patients experience skin burn, **fatigue**, **nausea**, and **vomiting** after radiation therapy regardless of the where radiation is applied. After treatment, the skin around the site of the treatment may also become sore. Affected skin should be kept clean and can be treated like **sunburn**, with skin lotion or vitamin A and D ointment. Patients should avoid perfume and scented skin products and protect affected areas from the sun.

Nausea and vomiting are most likely to occur when the radiation dose is high or if the abdomen or another part of the digestive tract is irradiated. Sometimes nausea and vomiting occur after radiation to other regions, but in these cases the symptoms usually disappear within a

KEY TERMS

Anemia—Insufficient red blood cells in the body.

Antibody—Protein molecule that recognizes and binds specifically to a foreign substance in the body in order to eliminate it.

Chemotherapy—Injecting drugs into the body where they circulate and kill cancer cells.

Computed tomography (CT or CAT) scan—Using x rays taken from many angles and computer modeling, CT scans help locate and size tumors and provide information on whether they can be surgically removed.

Fractionation—A procedure for dividing a dose of radiation into smaller treatment doses.

Gamma rays—Short wavelength, high energy electromagnetic radiation emitted by radioactive substances.

Hodgkin's disease—Cancer of the lymphatic system, characterized by lymph node enlargement and the presence of a large polyploid cells called Reed-Sternberg cells.

Magnetic resonance imaging (MRI)—MRI uses magnets and radio waves to create detailed cross-sectional pictures of the interior of the body.

Stereotactic—Characterized by precise positioning in space. When applied to radiosurgery, stereotactic refers to a system of three-dimensional coordinates for locating the target site.

few hours after treatment. Nausea and vomiting can be treated with **antacids**, Compazine, Tigan, or Zofran.

Fatigue frequently starts after the second week of therapy and may continue until about two weeks after the therapy is finished. Patients may need to limit their activities, take naps, and get extra sleep at night.

Patients should see their oncologist (cancer doctor) at least once within the first few weeks after their final radiation treatment. They should also see an oncologist every six to twelve months for the rest of their lives so they can be checked to see if the tumor has reappeared or spread.

Risks

Radiation therapy can cause anemia, nausea, vomiting, **diarrhea**, hair loss, skin burn, sterility, and rarely **death**. However, the benefits of radiation therapy almost always exceed the risks. Patients

should discuss the risks with their doctor and get a second opinion about their treatment plan.

Normal results

The outcome of radiation treatment varies depending on the type, location, and stage of the cancer. For some cancers such as Hodgkin's disease, about 75% of the patients are cured. **Prostate cancer** also responds well to radiation therapy. Radiation to painful bony metastases is usually a dramatically effective form of pain control. Other cancers may be less sensitive to the benefits of radiation.

Resources

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- Lawrence, T. S. "Radiation Sensitizers and Targeted Therapies." *Oncology (Huntington)* 17 (December 2003): 23–28.
- Merrick, H. W. IIIrd, L. L. Gunderson, and F. A. Calvo. "Future Directions in Intraoperative Radiation Therapy." *Surgical Oncology Clinics of North America* 12 (October 2003): 1099–1105.
- Nag, S., and K. S. Hu. "Intraoperative High-Dose-Rate Brachytherapy." *Surgical Oncology Clinics of North America* 12 (October 2003): 1079–1097.
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ORGANIZATIONS

- American Cancer Society. 1599 Clifton Rd. NE, Atlanta GA 30329-4251. (800) ACS-2345. <<http://www.cancer.org>>.
- International Radiosurgery Support Association (IRSA). 3005 Hoffman Street, Harrisburg, PA 17110. (717) 260-9808. <<http://www.irsa.org>>.
- National Association for Proton Therapy. 7910 Woodmont Ave., Suite 1303, Bethesda, MD 20814. (301) 913-9360. <<http://www.proton-therapy.org/Default.htm>>.

OTHER

- Radiation Therapy and You. A Guide to Self-Help During Treatment.* National Cancer Institute CancerNet Information Service. <<http://cancernet.nci.nih.gov>>.

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Radiation treatments see **Radiation therapy**

Radical neck dissection

Definition

Radical neck dissection is an operation used to remove cancerous tissue in the head and neck.

Purpose

The purpose of radical neck dissection is to remove lymph nodes and other structures in the head and neck that are likely or proven to be malignant. Variations on neck dissections exist depending on the extent of the **cancer**. A radical neck dissection removes the most tissue. It is done when the cancer has spread widely in the neck. A modified neck dissection removes less tissue, and a selective neck dissection even less.

Precautions

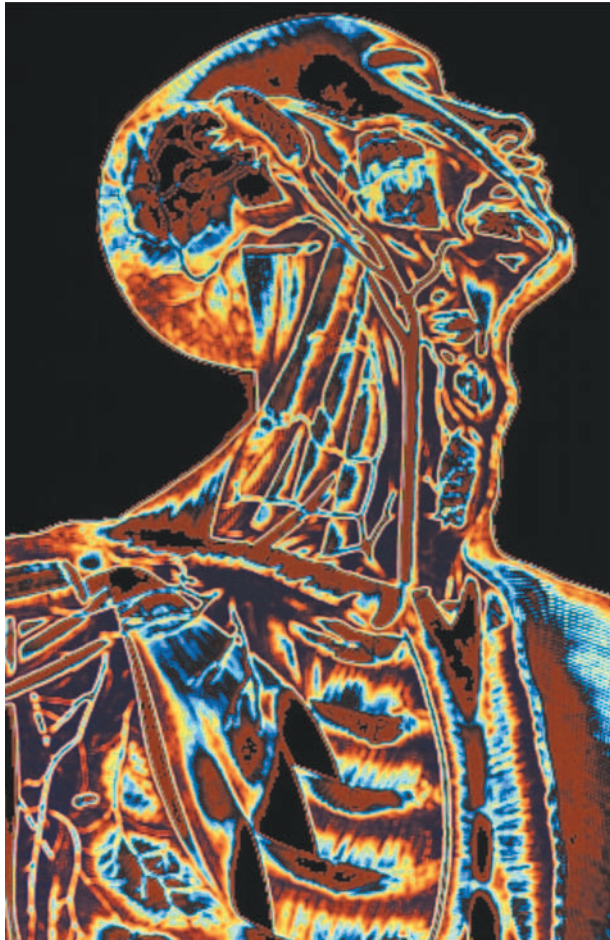
This operation should not be done if cancer has metastasized (spread) beyond the head and neck, or if the cancer has invaded the bones of the cervical vertebrae (the first seven vertebrae of the spinal column) or the skull. In these cases, the surgery will not effectively contain the cancer.

Description

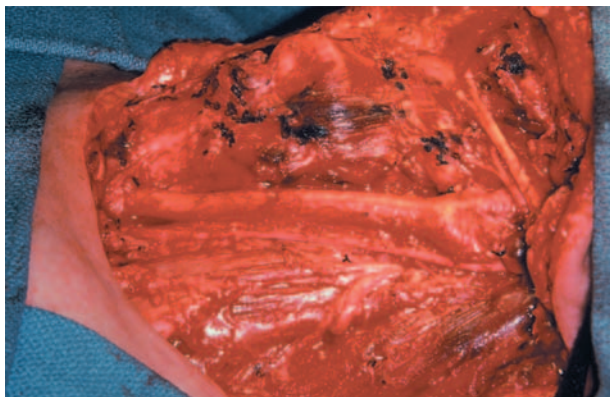
Cancers of the head and neck (sometimes inaccurately called throat cancer) often spread to nearby tissues and into the lymph nodes. Removing these structures is one way of controlling the cancer.

Of the 600 hundred lymph nodes in the body, about 200 are in the neck. Only a small number of these are removed during a neck dissection. In addition, other structures such as muscles, veins, and nerves may be removed during a radical neck dissection. These include the sternocleidomastoid muscle (one of the muscles that functions to flex the head), internal jugular (neck) vein, submandibular gland (one of the salivary glands), and the spinal accessory nerve (a nerve that helps control speech, swallowing and certain movements of the head and neck). The goal is always to remove all the cancer but to save as many components surrounding the nodes as possible.

Radical neck dissections are done in a hospital under **general anesthesia** by a head and neck surgeon. An incision is made in the neck, and the skin is pulled back to reveal the muscles and lymph nodes. The surgeon is guided in what to remove by tests done



A digitized illustration of the human head and chest showing nasal passages, sinuses, trachea, vascular nerves, as well as ribs and parts of the lungs. (Custom Medical Stock Photo. Reproduced by permission.)



A specimen taken from radical neck surgery. (Custom Medical Stock Photo. Reproduced by permission.)

prior to surgery and by examination of the size and texture of the lymph nodes.

Preparation

Radical neck dissection is a major operation. Extensive tests are done before the operation to try to determine where and how far the cancer has spread. These may include lymph node biopsies, CT (computed tomography) scans, MRI scans, and barium swallows. In addition, standard pre-operative blood and **liver function tests** are performed, and the patient will meet with an anesthesiologist before the operation. The patient should tell the anesthesiologist about all drug **allergies** and all medication (prescription, non-prescription, or herbal) that he or she is taking.

Aftercare

A person who has had a radical neck dissection will stay in the hospital several days after the operation, and sometimes longer if surgery to remove the primary tumor was done at the same time. Drains are inserted under the skin to remove the fluid that accumulates in the neck area. Once the drains are removed and the incision appears to be healing well, patients are usually discharged from the hospital, but will require follow-up doctor visits. Depending on how many structures are removed, a person who has had a radical neck dissection may require physical therapy to regain use of the arm and shoulder.

Risks

The greatest risk in a radical neck dissection is damage to the nerves, muscles, and veins in the neck. Nerve damage can result in **numbness** (either temporary or permanent) to different regions on the neck and loss of function (temporary or permanent) to parts of the neck, throat, and shoulder. The more extensive the neck dissection, the more function the patient is likely to lose. As a result, it is common following radical neck dissection for a person to have stooped shoulders, limited ability to lift the arm, and limited head and neck rotation and flexion due to the removal of nerves and muscles. Other risks are the same as for all major surgery: potential bleeding, infection, and allergic reaction to anesthesia.

Normal results

Normal lymph nodes are small and show no cancerous cells under the microscope.

KEY TERMS

Barium swallow—Barium is used to coat the throat in order to take x-ray pictures of the tissues lining the throat.

Computed tomography (CT or CAT) scan—Using x rays taken from many angles and computer modeling, CT scans help size and locate tumors and provide information on whether they can be surgically removed.

Lymphatic system—Primary defense against infection in the body. The tissues, organs, and channels (similar to veins) that produce, store, and transport lymph and white blood cells to fight infection.

Lymph nodes—Small, bean-shaped collections of tissue found in lymph vessels. They produce cells and proteins that fight infection and filter lymph. Nodes are sometimes called lymph glands.

Malignant—Cancerous. Cells tend to reproduce without normal controls on growth and form tumors or invade other tissues.

Metastasize—Spread of cells from the original site of the cancer to other parts of the body where secondary tumors are formed.

Magnetic resonance imaging (MRI)—MRI uses magnets and radio waves to create detailed cross-sectional pictures of the interior of the body.

Abnormal results

Abnormal lymph nodes may be enlarged and show malignant cells when examined under the microscope.

Resources

ORGANIZATIONS

American Cancer Society. 1599 Clifton Road NE, Atlanta, GA 30329. (800) ACS-2345. <<http://www.cancer.org>>.

Cancer Information Service. National Cancer Institute, Building 31, Room 10A19, 9000 Rockville Pike, Bethesda, MD 20892. (800) 4-CANCER. <<http://www.nci.nih.gov/cancerinfo/index.html>>.

OTHER

The Voice Center at Eastern Virginia Medical School. February 17, 2001. [cited June 7, 2001]. <<http://www.voice-center.com>>.

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Radioactive implants

Definition

Radioactive implants are devices that are placed directly within cancerous tissue or tumors, in order to deliver **radiation therapy** intended to kill cancerous cells. The practice of internal radiation therapy also is referred to as brachytherapy.

Purpose

With the use of radioactive implants, the tumor is subjected to radioactive activity over a longer period of time, as compared to external beam therapy.

Precautions

The patient is required to remain in his or her bed or room during the treatment. During the period of greatest radioactivity (24–72 hours), health care providers will limit the amount of time spent with the patient to that required for essential care. Some radiation exposure to family members has been noted with certain radiation implants. However, a study in 2003 found that exposure to family members of **prostate cancer** patients, an increasingly more commonly treated group of patients to receive implants, is low.

Description

Interstitial radiation therapy places the sources of radiation directly into the tumor and surrounding tissue. Most commonly used in tumors of the head, neck, prostate, and breast, it also may be used in combination with external radiation therapy. The implant may be permanent or removable. A permanent implant of radioactive seeds, such as gold or iodine, is placed directly into the organ. Over several weeks or months, the seeds slowly deliver radiation to the tumor. More commonly used is the removable implant that requires an operation under **general anesthesia** to place narrow, hollow stainless steel needles through the tumor. Teflon tubes are inserted through the needles, and the needles are then removed. After the patient returns to his or her room, radioactive seeds are inserted into the tubes in a procedure called afterloading. Once the desired dosage is reached, the tubes and seeds are removed.

The planning and procedures used for treatment with radioactive implants is becoming increasingly accurate and sophisticated as technology develops. Special imaging tools and computer software help physicians and radiation therapists visualize implant

KEY TERMS

Ageusia—The loss of taste perception.

Alopecia—The loss of hair, or baldness.

Dysgeusia—Unpleasant alteration of taste sensation, often with a metallic taste.

Hypogeusia—Diminished taste perception.

placements. Further, improved radiologic imaging techniques help physicians track the progress of this and other **cancer** therapies.

Intracavity radiation often is used for gynecologic cancers. Under general or spinal anesthesia, hollow applicators are placed directly inside the affected organ. Correct positioning is confirmed by x rays, and once the patient has returned to his or her room, a small plastic tube containing the radioactive isotope is inserted into the hollow applicator. The treatment is delivered over 48–72 hours, after which time the applicator and radioactive sources are removed. Very high doses of radiation can be delivered to the tumor, while the rapid removal of the radioactive dose limits damage to the surrounding structures.

Abnormal results

Normal cells are subjected to the effects of radiation; any tissue near the radiation site may be damaged or destroyed. Some side effects are acute and temporary, while others develop over time and may be permanent. Skin reactions, such as redness, **itching**, flaking, or stripping of the top layer, usually are temporary; long-term effects can include scarring, and changes in texture. Radiation recall is a delayed skin side effect in which the area that had been exposed to radiation becomes irritated or blistered after the patient receives certain **chemotherapy**.

Following treatment for tumors of the head and neck region, the lining of the mouth and throat can become inflamed or irritated, resulting in a condition known as mucositis or **stomatitis**. Injury to the salivary glands can decrease saliva production, resulting in a condition known as xerostomia, or **dry mouth**. There also may be alteration in the patient's taste buds, resulting in decrease or loss of taste sensation (hypogeusia or ageusia), or the presence of unpleasant taste, sometimes described as metallic (dysgeusia). Patients may experience **nausea and vomiting** as a result of the effect of radiation on the brain. Hair loss (**alopecia**) may result from radiation's effect on hair follicles.

Radiation's effect on the rapidly growing cells of the gastrointestinal tract may result in **diarrhea** or abdominal cramping. Pelvic radiation can affect the bowel, bladder, or sexual function. Radiation also can affect production of blood cell components in the bone marrow.

Resources

PERIODICALS

"Brachytherapy Software Receives FDA Approval." *Urology Times* August 2003: 48.

"Radiation Exposure Among Family Members of Prostate Brachytherapy Patients Appears to be Low." *Urology Times* September 2003: 12.

ORGANIZATIONS

American Cancer Society. 1599 Clifton Rd., NE, Atlanta, GA 30329-4251. (800) 227-2345.
< <http://www.cancer.org> >.

National Cancer Institute. Building 31, Room 10A31, 31 Center Drive, MSC 2580, Bethesda, MD 20892-2580. (800) 422-6237. < <http://www.nci.nih.gov> >.

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Radioactive iodine uptake test see **Thyroid nuclear medicine scan**

Radioallergosorbent test (RAST) see **Allergy tests**

Radiotherapy see **Radiation therapy**

Raloxifene see **Bone disorder drugs**

Range-of-motion exercises see **Exercise**

Rape and sexual assault

Definition

The various definitions of rape range from the broad (coercing a person to engage in any sexual act) to the specific (forcing a woman to submit to sexual intercourse). The United States Code includes the crime of rape under the more comprehensive term "sexual abuse." Two types of sexual assault are defined in the code: sexual **abuse** and aggravated sexual abuse. Sexual abuse includes acts in which an individual is forced to engage in sexual activity by use of threats or other fear tactics, or instances in which an individual is physically unable to decline. Aggravated sexual abuse occurs when an individual is forced to submit to sexual acts by use of physical

force; threats of **death**, injury, or kidnapping; or substances that render that individual unconscious or impaired.

Description

Many misconceptions exist about rape and sexual assault. It is often assumed that rape victims are all women who have been attacked by a total stranger and forced into having sexual intercourse. In reality, sexual assault can take many forms—it may be violent or nonviolent; the victim may be male or female, child or adult; the offender may be a stranger, relative, friend, authority figure, or spouse.

The number of sexual assaults reported depends on how those abuses are defined. The United States Code uses two terms to distinguish between different sexual activities:

- Sexual act: contact between penis and vagina or penis and anus that involves penetration; contact between the mouth and genitals or anus; penetration of the vagina or anus with an object; or direct touching (not through clothing) of the genitals of an individual under the age of 16.
- Sexual contact: intentional touching of the genitals, breasts, buttocks, anus, inner thigh, or groin with no sexual penetration.

National statistics

According to the Federal Bureau of Investigation's *Uniform Crime Reports*, there were 95,136 forcible rapes reported to United States law enforcement agencies in 2002. Sixty-five out of every 100,000 women were reported to be victims of rape that year, up 4.7% from 2001 but down 3.9% from 1998. The actual number of rapes and sexual assaults, however, is in reality much larger; estimates of unreported rape range between 2 and 10 times the number reported to law enforcement. The National Violence Against Women Survey, jointly sponsored by the Centers for Disease Control and Prevention (CDC) and the National Institute of Justice (NIJ) and conducted in the mid-1990s, found that one in six women (18%) and one in 33 men (3%) has experienced an attempted or completed rape. The survey estimated that approximately 17,722,672 women and 2,782,440 men in the United States have been raped or have had rape attempted as a child or adult, and that 302,091 women and 92,748 men were raped in the 12 months prior to the study.

There are numerous reasons why the majority of sexual assaults are never reported. Often the victim

What To Do If You Are Raped

Don't bathe
Don't blame yourself
Retain all evidence
Get examined
Consider the "morning-after" pill

fears retaliation from the offender. He or she may be afraid of family, friends, the community, or the media learning about the offense. There may be a concern about being judged or blamed by others. The victim may think that no one will believe the assault occurred.

THE VICTIMS. The 2000 "Victim, Incident, and Offender Characteristics," published by the National Center for Juvenile Justice (NCJJ), analyzed sexual assault data collected by law enforcement agencies over a five-year span. The following characteristics were found to be significant among victims of sexual assault:

- Age: Over two-thirds of reported victims of sexual assault were juveniles under the age of 18. Twelve to 18 year olds represented the largest group of victims at 33%; 20% were between the ages of six and 11; children less than five years old and adults between 18 and 24 years of age each constituted 14% of victims; 12% were between the ages of 25 and 34; and 7% were over the age of 34. Persons over the age of 54 represented 1% of all victims. One out of every seven victims surveyed in the study were under the age of six.
- Gender: Females were more than six times more likely to be a victim of sexual assault than males; more than 86% of victims were females. The great majority (99%) of the victims of forcible rapes were women, while men constituted the majority (54%) of the victims of forcible sodomy (oral or anal intercourse). Females are most likely to be the victim of sexual assault at age 14, while males are at most risk at age four.
- Location: The residence of the victim was the most commonly noted location of sexual assault (70%). Other common locations included schools, hotels/motels, fields, woods, parking lots, roadways, and commercial/office buildings.
- Weapons: A personal weapon (hands, feet, or fists) was used in 77% of cases. No weapon was noted in 14% of assaults; other weapons (knives, clubs, etc.) were used in 6% of cases. Firearms were involved in only 2% of assaults.

THE OFFENDERS. Similar statistics were gathered by the NCJJ regarding the perpetrators of rape and sexual assault. These characteristics included:

- **Age:** Over 23% of offenders were under the age of 18; juveniles were more likely to be perpetrators of forcible sodomy and fondling. The remaining 77% of offenders were adults and were responsible for 67% of juvenile victims. For younger juvenile victims (under the age of 12), juvenile offenders were responsible for approximately 40% of assaults.
- **Gender:** The great majority of all reported offenders were male (96%). The number of female offenders rose for victims under the age of six (12%), in contrast to 6% for victims aged six through 12, 3% for victims aged 12 through 17, and 1% for adult victims.
- **Relationship with offender:** Approximately 59% of offenders were acquaintances of their victims, compared to family members (27%) or strangers (14%). Family members were more likely to be perpetrators against juveniles (34%) than against adults (12%). In contrast, strangers accounted for 27% of adult victims and 7% of juveniles.
- **Past offenses:** In 19% of juvenile cases, the victim was not the only individual to be assaulted by the offender, compared to only 4% of adult cases.

Consequences

Victims of sexual assault may sustain a range of injuries; male victims are more likely than females to suffer severe physical trauma. The National Women's Study, funded by the National Institute of Drug Abuse, found that more than 70% of rape victims report no physical injuries as a result of their assault; only 4% sustain serious injuries that require hospitalization. At least 49% of victims, however, state that they feared severe injuries or death during their assault. Fatalities occur in approximately 0.1% of rape cases.

Sexually transmitted diseases (STDs) are a source of concern for many victims of sexual assault. The most commonly transmitted diseases are **gonorrhea** (caused by *Neisseria gonorrhoeae*), **chlamydia** (caused by *Chlamydia trachomatis*), **trichomoniasis** (caused by *Trichomonas vaginalis*), and **genital warts** (caused by human papillomavirus). **Syphilis** (caused by *Treponema pallidum*) and human **immunodeficiency virus (HIV)** are also noted among some sexual assault victims. The transmission rate of STDs is estimated to be between 3.6% and 30% of rapes.

According to the National Women's Study, approximately 5% of adult female rape victims become pregnant as a result of their assault, leading

Common Misconceptions Of Males Perpetrating Date Rape

Since I took her out and paid for the date, she should have sex with me.
When she says no, she really means yes.
If she's aroused, she wants to have sex.
She wouldn't go parking with me if she didn't want to have sex.
If she didn't want to have sex, why did she let me go as far as she did?
If she gets me erect, then it's her responsibility to do something about it.
She's slept with other people, so she should sleep with me.
We've had sex before, and she didn't say no then.

to 32,100 pregnancies a year among women 18 years of age or older. Approximately 50% of pregnant rape victims had an abortion, 6% put the child up for adoption, and 33% kept the child (the remaining pregnancies resulted in **miscarriage**).

MENTAL HEALTH PROBLEMS. Also known as rape trauma syndrome, **post-traumatic stress disorder (PTSD)** is a mental health disorder that describes a range of symptoms often experienced by someone who has undergone a severely traumatic event. Approximately 31% of rape victims develop PTSD as a result of their assault; victims are more than six times more likely to develop PTSD than women who have not been victimized.

The symptoms of PTSD include:

- recurrent memories or flashbacks of the incident
- nightmares
- insomnia
- mood swings
- difficulty concentrating
- panic attacks
- emotional numbness
- depression
- **anxiety**

Persons who have been sexually assaulted have also been noted to have increased risk for developing other mental health problems. Over those who have not been victimized, rape victims are:

- three times more likely to have a major depressive episode
- four times more likely to have contemplated suicide
- thirteen times more likely to develop alcohol dependency problems
- twenty-six times more likely to develop drug abuse problems

Treatment

Once a victim of sexual assault reports the crime to local authorities, calls a rape crisis hotline, or arrives at the emergency room to be treated for injuries, a multidisciplinary team is often formed to address his or her physical, psychological, and judicial needs. This team usually includes law enforcement officers, physicians, nurses, mental health professionals, victim advocates, and/or prosecutors.

The victim of sexual assault may continue to feel fear and anxiety for some time after the incident, and in some instances this may significantly impact his or her personal or professional life. Follow-up counseling should therefore be provided for the victim, particularly if symptoms of PTSD become evident.

Forensic medical examination

Because rape is a crime, there are certain requirements for medical evaluation of the patient and for record keeping. The forensic medical examination is an invaluable tool for collecting evidence against a perpetrator that may be admissible in court. Since the great majority of victims know their assailant, the purpose of the medical examination is often not to establish identity but to establish nonconsensual sexual contact. The Sexual Assault Nurse Examiner program is an effective model that is used in many United States hospitals and clinics to collect and document evidence, evaluate and treat for STDs and **pregnancy**, and refer victims to follow-up medical care and counseling. The "Sexual Assault Nurse Examiner Development and Operation Guide," prepared by the Sexual Assault Resource Service, describes the ideal protocol for collecting evidence from a sexual assault victim. This includes:

- performing the medical examination within 72 hours of the assault
- taking a history of the assault
- documenting the general health of the victim, including menstrual cycle, potential **allergies**, and pregnancy status
- assessment for trauma and taking photographic evidence of injuries
- taking fingernail clippings or scrapings
- taking samples for sperm or seminal fluid
- combing head/pubes hair for foreign hairs, fibers, and other substances
- collection of bloody, torn, or stained clothing
- taking samples for **blood typing** and DNA screening

Prevention

STD Prevention

While the concern of sexual assault victims of contracting an STD is often high, the actual risk of transmission is relatively low; the CDC estimates that the risk of contracting gonorrhea from an offender is between 6% and 12%, chlamydia between 4% and 17%, syphilis between 0.5% and 3%, and HIV less than 1%. Nonetheless, post-exposure **prophylaxis** (preventative treatment) against certain STDs is often provided for the victim. Treatment with zidovudine, for example, is recommended for individuals who are at a high risk of exposure to HIV. The CDC recommends the following prophylactic regimen be provided for victims of sexual assaults in which vaginal, oral, or anal penetration took place:

- a single dose of ceftriaxone, an antibiotic effective against *Neisseria gonorrhoeae*
- a single dose of metronidazole, an antibiotic effective against *Trichomonas vaginalis*
- a single dose of azithromycin or doxycycline, **antibiotics** effective against *Chlamydia trachomatis*
- inoculation with the post-exposure **hepatitis B** vaccine

In some instances, cultures may be taken during the medical examination and at time points afterward to test for gonorrhea or chlamydia. It is important that the victim receive information regarding the symptoms of STDs and be counseled to return for further examination if any of these symptoms occur.

Pregnancy prevention

Female victims at risk for becoming pregnant after an assault should be counseled on the availability of **emergency contraception**. According to the Food and Drug Administration (FDA), emergency **contraception** is not effective if there is no pregnancy but works to prevent pregnancy from occurring by delaying or preventing ovulation, by affecting the transport of sperm, and/or by thinning the inner layer of the uterus (endometrium) so that implantation is prevented. It is therefore not a form of abortion.

A number of options are available for women if they choose to use emergency contraceptives to prevent pregnancy following a sexual assault. The Yuzpe regimen uses two oral contraceptive pills that contain both of the hormones estrogen and progestin. The risk of pregnancy is reduced by 75% after use of the Yuzpe regimen, reducing the average number of pregnancies after unprotected sex from eight in 100 to two in 100.

KEY TERMS

Aggravated sexual abuse—When an individual is forced to submit to sexual acts by use of physical force; threats of death, injury, or kidnapping; or substances that render that individual unconscious or impaired.

Forcible sodomy—Forced oral or anal intercourse.

Forensic—Pertaining to or used during legal proceedings.

Post-traumatic stress disorder (PTSD)—Also known as rape trauma syndrome; a mental health disorder that describes a range of symptoms often experienced by someone who has undergone a severely traumatic event.

Sexual abuse—When an individual is forced to engage in sexual activity by use of threats or other fear tactics, or instances in which an individual is physically unable to refuse.

Sexual assault nurse examiner (SANE)—A registered nurse who is trained to collect and document evidence from a sexual assault victim, evaluate and treat for STDs and pregnancy, and refer victims to follow-up medical care and counseling.

Yupze regimen—A form of emergency contraception in which two oral contraceptive pills that contain both of the hormones estrogen and progesterin are taken to prevent pregnancy.

Progesterin-only **oral contraceptives** are also available and reduce the risk of pregnancy by 89% to 95%.

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American Psychiatric Association (APA). 1400 K Street, NW, Washington, DC 20005. (888) 357-7924. <<http://www.psych.org>>.

Federal Bureau of Investigation. J. Edgar Hoover Building, 935 Pennsylvania Avenue, NW, Washington, DC 20535-0001. (202) 324-3000. <<http://www.fbi.gov>>.

Rape, Abuse, and Incest National Network. 635-B Pennsylvania Ave. SE, Washington, DC 20003. (800) 656-HOPE.

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Stéphanie Dionne
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Rashes

Definition

The popular term for a group of spots or red, inflamed skin that is usually a symptom of an underlying condition or disorder. Often temporary, a rash is only rarely a sign of a serious problem.

Description

A rash may occur on only one area of the skin, or it could cover almost all of the body. Also, a rash may or may not be itchy. Depending on how it looks, a rash may be described as:

- blistering (raised oval or round collections of fluid within or beneath the outer layer of skin)
- macular (flat spots)
- nodular (small, firm, knotty rounded mass)
- papular (small solid slightly raised areas)
- pustular (pus-containing skin blister).

Causes and symptoms

There are many theories as to the development of skin rashes, but experts are not completely clear what causes some of them. Generally a skin rash is an intermittent symptom, fading and reappearing. Rashes may accompany a range of disorders and conditions, such as:

- Infectious illness. A rash is symptom of many different kinds of childhood infectious illnesses, including **chickenpox** and **scarlet fever**. It may be triggered by other infections, such as **Rocky Mountain spotted fever** or **ringworm**.
- Allergic reactions. One of the most common symptoms of an allergic reaction is an itchy rash. **Contact dermatitis** is a rash that appears after the skin is exposed to an allergen, such as metal, rubber, some cosmetics or lotions, or some types of plants (e.g. poison ivy). Drug reactions are another common allergic cause of rash; in this case, a rash is only one of a variety of possible symptoms, including **fever**, seizures, **nausea and vomiting**, **diarrhea**, heartbeat irregularities, and breathing problems. This rash usually appears soon after the first dose of the course of medicine is taken.
- **Autoimmune disorders**. Conditions in which the immune system turns on the body itself, such as **systemic lupus erythematosus** or **purpura**, often have a characteristic rash.

- nutritional disorders. For example, **scurvy**, a disease caused by a lack of Vitamin C, has a rash as one of its symptoms.
- **cancer**. A few types of cancer, such as chronic lymphocytic leukemia, can be the underlying cause of a rash.

Rashes in infancy

Rashes are extremely common in infancy, and are usually not serious at all and can be treated at home.

Diaper rash is caused by prolonged skin contact with bacteria and the baby's waste products in a damp diaper. This rash has red, spotty sores and there may be an ammonia smell. In most cases the rash will respond within three days to drying efforts. A diaper rash that does not improve in this time may be a yeast infection requiring prescription medication. A doctor should be consulted if the rash is solid, bright red, causes fever, or the skin develops blisters, **boils**, or pus.

Infants also can get a rash on cheeks and chin caused by contact with food and stomach contents. This rash will come and go, but usually responds to a good cleaning after meals. About a third of all infants develop "acne" usually after the third week of life in response to their mothers' hormones before birth. This rash will disappear between weeks and a few months. Heat rash is a mass of tiny pink bumps on the back of the neck and upper back caused by blocked sweat glands. The rash usually appears during hot, humid weather, although a baby with a fever can also develop the rash.

A baby should see a doctor immediately if the rash:

- appears suddenly and looks purple or blood-colored
- looks like a burn
- appears while the infant seems to be sick

Diagnosis

A physician can make a diagnosis based on the medical history and the appearance of the rash, where it appears, and any other accompanying symptoms.

Treatment

Treatment of rashes focuses on resolving the underlying disorder and providing relief of the **itching** that often accompanies them. Soothing lotions or oral **anti-histamines** can provide some relief, and **topical antibiotics** may be administered if the patient, particularly a child, has caused a secondary infection by scratching.



An unidentified rash on young boy's back. (Custom Medical Stock Photo. Reproduced by permission.)

KEY TERMS

Purpura—A group of disorders characterized by purple or red brown areas of discoloration visible through the skin.

Scurvy—A nutritional disorder that causes skin bruising and hemorrhages.

The rash triggered by **allergies** should disappear as soon as the allergen is removed; drug rashes will fade when the patient stops taking the drug causing the allergy. For the treatment of diaper rash, the infant's skin should be exposed to the air as much as possible; ointments are not needed unless the skin is dry and cracked. Experts also recommend switching to cloth diapers and cleaning affected skin with plain water.

Prognosis

Most rashes that have an acute cause, such as an infection or an allergic reaction, will disappear as soon as the infection or irritant is removed from the body's system. Rashes that are caused by chronic conditions, such as autoimmune disorders, may remain indefinitely or fade and return periodically.

Prevention

Some rashes can be prevented, depending on the triggering factor. A person known to be allergic to certain drugs or substances should avoid those things in order to prevent a rash. Diaper rash can be prevented by using cloth diapers and keeping the diaper area very clean, breast feeding, and changing diapers often.

Resources

ORGANIZATIONS

American Academy of Dermatology. 930 N. Meacham Road, P.O. Box 4014, Schaumburg, IL 60168-4014. (847) 330-0230. Fax: (847) 330-0050. <<http://www.aad.org>>.

Carol A. Turkington

Rat-bite fever

Definition

Rat-bite **fever** refers to an infection which develops after having been bitten or scratched by an infected animal.

Description

Rat-bite fever occurs most often among laboratory workers who handle lab rats in their jobs, and among people who live in poor conditions, with rodent infestation. Children are particularly likely to be bitten by rodents infesting their home, and are therefore most likely to contract rat-bite fever. Other animals that can carry the types of bacteria responsible for this illness include mice, squirrels, weasels, dogs, and cats. One of the causative bacteria can cause the same illness if it is ingested, for example in unpasteurized milk.

Causes and symptoms

There are two variations of rat-bite fever, caused by two different organisms. In the United States, the

bacteria *Streptobacillus moniliformis* is the most common cause (causing streptobacillary rat-bite fever). In other countries, especially Africa, *Spirillum minus* causes a different form of the infection (called spirillary rat-bite fever).

Streptobacillary rat-bite fever occurs up to 22 days after the initial bite or scratch. The patient becomes ill with fever, chills, **nausea and vomiting**, **headache**, and **pain** in the back and joints. A rash made up of tiny pink bumps develops, covering the palms of the hands and the soles of the feet. Without treatment, the patient is at risk of developing serious infections of the lining of the heart (**endocarditis**), the sac containing the heart (**pericarditis**), the coverings of the brain and spinal cord (**meningitis**), or lungs (**pneumonia**). Any tissue or organ throughout the body may develop a pocket of infection and pus, called an **abscess**.

Spirillary rat-bite fever occurs some time after the initial injury has already healed, up to about 28 days after the bite or scratch. Although the wound had appeared completely healed, it suddenly grows red and swollen again. The patient develops a fever. Lymph nodes in the area become swollen and tender, and the patient suffers from fever, chills, and headache. The skin in the area of the original wound sloughs off. Although rash is less common than with streptobacillary rat-bite fever, there may be a lightly rosy, itchy rash all over the body. Joint and muscle pain rarely occur. If left untreated, the fever usually subsides, only to return again in repeated two- to four-day cycles. This can go on for up to a year, although, even without treatment, the illness usually resolves within four to eight weeks.

Diagnosis

In streptobacillary rat-bite fever, found in the United States, diagnosis can be made by taking a sample of blood or fluid from a painful joint. In a laboratory, the sample can be cultured, to allow the growth of organisms. Examination under a microscope will then allow identification of the bacteria *Streptobacillus moniliformis*.

In spirillary rat-bite fever, diagnosis can be made by examining blood or a sample of tissue from the wound for evidence of *Spirillum minus*.

Treatment

Shots of procaine penicillin G or penicillin V by mouth are effective against both streptobacillary and spirillary rat-bite fever. When a patient is allergic to

KEY TERMS

Abscess—A pocket of infection; a collection of pus.

Endocarditis—An inflammation of the lining of the heart.

Meningitis—An inflammation of the tissues covering the brain and spinal cord.

Pasteurization—A process during which milk is heated up and maintained at a particular temperature long enough to kill bacteria.

Pericarditis—An inflammation of the sac containing the heart.

the **penicillins**, erythromycin may be given by mouth for streptobacillary infection, or tetracycline by mouth for spirillary infection.

Prognosis

With treatment, prognosis is excellent for both types of rat-bite fever. Without treatment, the spirillary form usually resolves on its own, although it may take up to a year to do so.

The streptobacillary form, found in the United States, however, can progress to cause extremely serious, potentially fatal complications. In fact, before **antibiotics** were available to treat the infection, streptobacillary rat-bite fever frequently resulted in **death**.

Prevention

Prevention involves avoiding contact with those animals capable of passing on the causative organisms. This can be an unfortunately difficult task for people whose economic situations do not allow them to move out of rat-infested buildings. Because streptobacillary rat-bite fever can occur after drinking contaminated milk or water, only pasteurized milk, and water from safe sources, should be ingested.

Resources

ORGANIZATIONS

Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

Rosalyn Carson-DeWitt, MD

Rational-emotive therapy see **Cognitive-behavioral therapy**

Raynaud's disease

Definition

Raynaud's disease refers to a disorder in which the fingers or toes (digits) suddenly experience decreased blood circulation. It is characterized by repeated episodes of color changes of the skin of digits on cold exposure or emotional **stress**.

Description

Raynaud's disease can be classified as one of two types: primary (or idiopathic) and secondary (also called Raynaud's phenomenon). Primary Raynaud's disease has no predisposing factor, is more mild, and causes fewer complications. About half of all cases of Raynaud's disease are of this type. Women are five times more likely than men to develop primary Raynaud's disease. The average age of diagnosis is between 20 and 40 years. Approximately three out of ten people with primary Raynaud's disease eventually progress to secondary Raynaud's disease after diagnosis. About 15% of individuals improve.

Secondary Raynaud's disease is the same as primary Raynaud's disease, but occurs in individuals with a predisposing factor, usually a form of collagen vascular disease. What is typically identified as primary Raynaud's is later identified as secondary once a predisposing disease is diagnosed. This occurs in approximately 30% of patients. As a result, the secondary type is often more complicated and severe, and is more likely to worsen.

Several related conditions that predispose persons to secondary Raynaud's disease include **scleroderma**, **systemic lupus erythematosus**, **rheumatoid arthritis** and **polymyositis**. **Pulmonary hypertension** and some nervous system disorders such as herniated discs and tumors within the spinal column, strokes, and **polio** can progress to Raynaud's disease. Finally, injuries due to mechanical trauma caused by vibration (such as that associated with chain saws and jackhammers), repetitive motion (**carpal tunnel syndrome**), electrical shock, and exposure to extreme cold can lead to the development of Raynaud's disease. Some drugs used to control high blood pressure or migraine headaches have been known to cause Raynaud's disease.

The prevalence of Reynaud's Phenomena in the general population varies 4–15%. Females are seven times more likely to develop Raynaud's diseases than are men. The problem has not been correlated with coffee consumption, dietary habits, occupational

history (excepting exposure to vibration) and exposure to most drugs. An association between Raynaud's disease and migraine headaches has been reported. Secondary Raynaud's disease is common among individuals with systemic lupus erythematosus in tropical countries.

Causes and symptoms

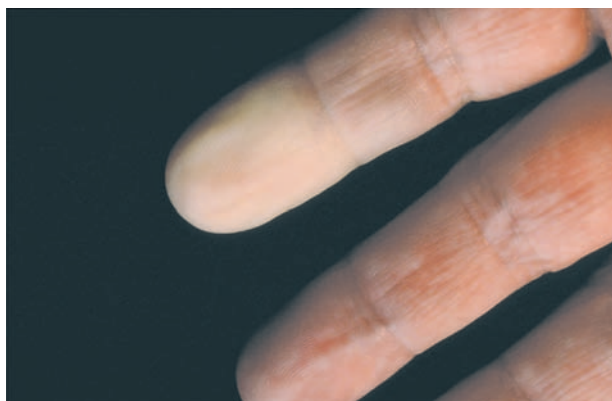
There is significant familial aggregation of primary Raynaud's disease. However, no causative gene has been identified.

Risk factors for Raynaud's disease differ between males and females. Age and **smoking** seem to be associated with Raynaud's disease only in men, while the associations of marital status and alcohol use with Raynaud's disease are usually only observed in women. These findings suggest that different mechanisms influence the expression of Raynaud's disease in men and women.

Both primary and secondary Raynaud's disease signs and symptoms are thought to be due to arterioles over-reacting to stimuli. Cold normally causes the tiny muscles in the walls of arteries to contract, thus reducing the amount of blood that can flow through them. In people with Raynaud's disease, the extent of constriction is extreme, thus severely restricting blood flow. Attacks or their effects may be brought on or worsened by **anxiety** or emotional distress.

There are three distinct phases to an episode of Raynaud's disease. When first exposed to cold, small arteries respond with intense contractions (vasoconstriction). The affected fingers or toes (in rare instances, the tip of the nose or tongue) become pale and white because they are deprived of blood and, thus, oxygen. In response, capillaries and veins expand (dilate). Because these vessels are carrying deoxygenated blood, the affected area then becomes blue in color. The area often feels cold and tingly or numb. After the area begins to warm up, the arteries dilate. Blood flow is significantly increased. This changes the color of the area to a bright red. During this phase, persons often describe the affected area as feeling warm and throbbing painfully.

Raynaud's disease may initially affect only the tips of fingers or toes. As the disease progresses, it may eventually involve all of one or two digits. Ultimately, all the fingers or toes may be affected. About one person in ten, will experience a complication called sclerodactyly. In sclerodactyly, the skin over the involved digits becomes tight, white, thick, smooth and shiny. In approximately 1% of cases of



A close-up view of a patient's fingers afflicted with Raynaud's disease. While this disorder may initially only affect the tips of the fingers and toes, eventually blood circulation of the entire finger or toe is affected. (Custom Medical Stock Photo. Reproduced by permission.)

Raynaud's disease, deep sores (ulcers) may develop in the skin. In rare cases of frequent, repetitive bouts of severe **ischemia** (decreased supply of oxygenated blood to tissues or organs), tissue loss, or **gangrene** may result and **amputation** may be required.

Diagnosis

Primary Raynaud's disease is diagnosed following the Allen Brown criteria. There are four components. The certainty of the diagnosis and severity of the disease increase as more criteria are met. The first is that at least two of the three color changes must occur during attacks provoked by cold and or stress. The second is that episodes must periodically occur for at least two years. The third is that attacks must occur in both the hands and the feet in the absence of vascular occlusive disease. The last is that there is no other identifiable cause for the Raynaud's episodes.

A cold stimulation test may also be performed to help to confirm a diagnosis of Raynaud's disease. The temperature of affected fingers or toes is taken. The hand or foot is then placed completely into a container of ice water for 20 seconds. After removal from the water, the temperature of the affected digits is immediately recorded. The temperature is retaken every five minutes until it returns to the pre-immersion level. Most individuals recover normal temperature within 15 minutes. People with Raynaud's disease may require 20 minutes or more to reach their pre-immersion temperature.

Laboratory testing is performed frequently. However, these results are often inconclusive for several reasons. Provocative testing such as the ice

emergence just described, is difficult to interpret because there is considerable overlap between normal and abnormal results. The **antinuclear antibody test** of blood is usually negative in Raynaud's disease. Capillary beds under finger nails usually appear normal. Erythrocyte sedimentation rates are often abnormal in people with connective tissue diseases. Unfortunately, this finding is not consistent in people with Raynaud's disease.

Treatment

There is no known way to prevent the development of Raynaud's disease. Further, there is no known cure for this condition. Therefore, avoidance of the trigger is the best supportive management available. Most cases of primary Raynaud's disease can be controlled with proper medical care and avoidance.

Many people are able to find relief by simply adjusting their lifestyles. Affected individuals need to stay warm, and keep their hands and feet well covered in cold weather. Layered clothing, scarves, heavy coats, heavy socks, and mittens under gloves are suggested because gloves alone allow heat to escape. It is also recommended that patients cover or close the space between their sleeves and mittens. Indoors, they should wear socks and comfortable shoes. Smokers should quit as nicotine will worsen the problem. Avoid the use of vibrating tools as well.

People with severe cases of Raynaud's disease may need to be treated with medications to help keep the arterioles relaxed and dilated. Medications such as calcium-channel blockers, reserpine or nitroglycerin may be prescribed to relax artery walls and improve blood flow.

Alternative Treatment

Because episodes of Raynaud's disease have also been associated with stress and emotional upset, the condition may be improved by learning to manage stress. Regular **exercise** is known to decrease stress and lower anxiety. Hypnosis, relaxation techniques, and visualization are also useful methods to help control emotions.

Biofeedback training is a technique during which a patient is given continuous information on the temperature of his or her digits, and then taught to voluntarily control this temperature. Some alternative practitioners believe that certain dietary supplements and herbs may be helpful in decreasing the vessel spasm of Raynaud's disease. Suggested supplements include vitamin E (found in fruits, vegetables, seeds,

KEY TERMS

Arteriole—The smallest type of artery.

Artery—A blood vessel that carries blood away from the heart to peripheral tissues.

Gangrene—Death of a tissue, usually caused by insufficient blood supply and followed by bacterial infection of the tissue.

Idiopathic—Of unknown origin.

Polymyositis—An inflammation of many muscles.

Pulmonary hypertension—A severe form of high blood pressure caused by diseased arteries in the lung.

Rheumatoid arthritis—Chronic, autoimmune disease marked by inflammation of the membranes surrounding joints.

Scleroderma—A relatively rare autoimmune disease affecting blood vessels and connective tissue that makes skin appear thickened.

Systemic lupus erythematosus—A chronic inflammatory disease that affects many tissues and parts of the body including the skin.

and nuts), magnesium (found in seeds, nuts, fish, beans, and dark green vegetables), and fish oils. The circulatory herbs cayenne, ginger and prickly ash may help enhance circulation to affected areas.

Prognosis

The prognosis for most people with Raynaud's disease is very good. In general, primary Raynaud's disease has the best prognosis, with a relatively small chance (1%) of serious complications. Approximately half of all affected individuals do well by taking simple precautions, and never require medication. The prognosis for people with secondary Raynaud's disease (or phenomenon) is less predictable. This prognosis depends greatly on the severity of other associated conditions such as scleroderma, lupus, or Sjögren syndrome.

Prevention

There is no way to prevent the development of Raynaud's disease. Once an individual realizes that he or she suffers from this disorder, however, steps can be taken to reduce the frequency and severity of episodes.

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Voulgari, P. V., et al. "Prevalence of Raynaud's Phenomenon in a Healthy Greek Population." *Annals of Rheumatic Disease* 59, no. 3 (March 2000): 206-210.

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave., Dallas, TX 75231-4596. (214) 373-6300 or (800) 242-8721. inquire@heart.org. <http://www.americanheart.org>.

Irish Raynaud's and Scleroderma Society. PO Box 2958 Foxrock, Dublin 18, Ireland (01) 235 0900. irss@indigo.ie.

National Heart, Lung, and Blood Institute. PO Box 30105, Bethesda, MD 20824-0105. (301) 592-8573. nhlbiinfo@rover.nhlbi.nih.gov. <http://www.nhlbi.nih.gov>.

National Organization for Rare Disorders (NORD). PO Box 8923, New Fairfield, CT 06812-8923. (203) 746-6518 or (800) 999-6673. Fax: (203) 746-6481. <http://www.rarediseases.org>.

Raynaud's & Scleroderma Association (UK). 112 Crewe Road, Alsager, Cheshire, ST7 2JA. UK (44) (0) 1270 872776. webmaster@raynauds.demon.co.uk. <http://www.raynauds.demon.co.uk>.

OTHER

Arthritis Foundation. <http://www.arthritis-foundation.com/>.

British Sjögren's Syndrome Association. <http://ourworld.copmpuserve.com/homepages/BSSAssociation>.

Raynaud's & Scleroderma Association. <http://www.Raunaud's.demon.co.uk/>.

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L. Fleming Fallon, Jr., MD, PhD, DrPH

RDS see **Respiratory distress syndrome**

Reactive airway disease see **Asthma**

Reactive polycythemia see **Secondary polycythemia**

Reading disorder see **Learning disorders**

Recompression treatment

Definition

Recompression treatment is the use of elevated pressure to treat conditions within the body after it has been subjected to a rapid decrease in pressure. It also includes hyperbaric **oxygen therapy**.

Purpose

Recompression treatment is used to overcome the adverse effects of **gas embolism** and **decompression sickness** (sometimes called the bends) in underwater divers who breathe compressed air. It is also approved for treatment of severe **smoke inhalation**, **carbon monoxide poisoning**, gas **gangrene**, radiation tissue damage, thermal **burns**, extreme blood loss, crush injuries, and **wounds** that won't heal.

Precautions

Hyperbaric oxygen therapy delivers greater amounts of oxygen more quickly to the body than breathing room air (which is only 21% oxygen) at regular pressure. Unmonitored, increased oxygen can produce toxic effects. Treatments must follow safe time-dose limits and may only be administered by a doctor.

Description

Recompression treatment is performed in a **hyperbaric chamber**, a sealed compartment in which the patient breathes normal air or "enhanced" air with up to 100% oxygen while exposed to controlled pressures up to three times normal atmospheric pressure. The patient may receive the oxygen through a face mask, a hood or tent around the head, or an endotracheal tube down the windpipe if the patient is already on a ventilator. When used to treat decompression sickness or gas **embolism**, the increased pressure reduces the size of gas bubbles in the patient's body. The increased oxygen concentration speeds the diffusion of the nitrogen within the bubbles out of the patient's body. As gas bubbles deflate, the trauma of gas embolism and decompression sickness begins to resolve. Treatment for diving emergencies typically involves one session, lasting four to six hours, at three atmospheres of pressure.

When used to treat other conditions, the increased pressure allows oxygen and other gases to dissolve more rapidly into the blood and thus be carried to

oxygen-starved tissues to enhance healing. Elevated oxygen levels can also purge toxins such as carbon monoxide from the body. In addition, when body tissues are super-saturated with oxygen, the destruction of some bacteria is enhanced and the spread of certain toxins is halted. This makes hyperbaric oxygen therapy useful in treating gas gangrene and infections that cause tissue necrosis (death). Hyperbaric oxygen therapy also promotes the growth of new blood vessels.

Preparation

Oxygen is often administered to a patient as first aid while he or she is being transported to a hyperbaric chamber. The treatment begins with chamber compression; as the pressure of the chamber atmosphere increases, the temperature also rises and the patient's ears may fill as they would during an airplane landing. Swallowing and yawning are ways to relieve the inner ear pressure. Once the desired pressure is achieved, the patient is given pure oxygen to breathe. Because treatment is lengthy, patients are encouraged to sleep or listen to music. In larger chambers, patients may also read or watch videos.

Aftercare

Depending on the reason for treatment and the treatment outcome, the patient may be taken to a hospital for further care, or examined and released.

Risks

There is minimal risk when recompression treatment is administered by a competent physician. However, some common side effects are sinus **pain**, temporary changes in vision, and **fatigue**.

Normal results

With prompt and appropriate recompression treatment, most patients show marked improvement in their blood oxygen levels and tissue circulation, as well as other signs of healing. Divers treated for gas embolism or decompression sickness may recover with no lasting effects.

Abnormal results

When recompression treatment is not begun promptly or not conducted at adequate time-dose levels, patients with decompression sickness may develop bone necrosis. This significant destruction of

KEY TERMS

Atmosphere—A measurement of pressure. One atmosphere equals the pressure of air at sea level (14.7 pounds per square inch [psi]).

Compressed air—Air that is held under pressure in a tank to be breathed underwater by divers. A tank of compressed air is part of a diver's scuba (self-contained underwater breathing apparatus) gear.

Decompression—A decrease in pressure from the surrounding water that occurs with decreasing diving depth.

Decompression sickness—A condition found in divers in which gas bubbles of nitrogen form in tissues and blood vessels as a result of decreasing surrounding pressure, such as in ascent from a dive. It may be a painful condition, especially as nitrogen bubbles invade the joints; persons stricken may walk stooped over in pain, in a bent stance that led to it being called "the bends."

Gas embolism—The presence of gas bubbles in the bloodstream that obstruct circulation. Also called air embolism.

Hyperbaric chamber—A sealed compartment in which patients are exposed to controlled pressures up to three times normal atmospheric pressure. Hyperbaric treatment may be used to regulate blood gases, reduce gas bubbles, and provide higher levels of oxygen more quickly.

Recompression—Restoring the elevated pressure of the diving environment to treat decompression sickness and gas embolism by decreasing bubble size.

bone, most commonly found in the hip and shoulder, produces chronic pain and severe disability. Another result of delayed or inadequate treatment may be permanent neurological damage. When decompression sickness involves the spinal cord, partial **paralysis** may occur.

Resources

ORGANIZATIONS

American College of Hyperbaric Medicine. P.O. Box 25914-130, Houston, Texas 77265. (713) 528-0657. <<http://www.hyperbaricmedicine.org>>.

Divers Alert Network. The Peter B. Bennett Center, 6 West Colony Place, Durham, NC 27705. (800) 446-2671. <<http://www.diversalertnetwork.org>>.

Undersea and Hyperbaric Medical Society. 10531 Metropolitan Ave., Kensington, MD 20895. (301) 942-2980. <<http://www.uhms.org>>.

Bethany Thivierge

Reconstructive surgery see **Plastic, cosmetic, and reconstructive surgery**

Rectal cancer

Definition

The rectum is the portion of the large bowel that lies in the pelvis, terminating at the anus. **Cancer** of the rectum is the disease characterized by the development of malignant cells in the lining or epithelium of the rectum. Malignant cells have changed such that they lose normal control mechanisms governing growth. These cells may invade surrounding local tissue or they may spread throughout the body and invade other organ systems.

Description

The rectum is the continuation of the colon (part of the large bowel) after it leaves the abdomen and descends into the pelvis. It is divided into equal thirds: the upper, mid, and lower rectum.

The pelvis and other organs in the pelvis form boundaries to the rectum. Behind, or posterior to the rectum is the sacrum (the lowest portion of the spine, closest to the pelvis). Laterally, on the sides, the rectum is bounded by soft tissue and bone. In front, the rectum is bounded by different organs in the male and female. In the male, the bladder and prostate are present. In the female, the vagina, uterus, and ovaries are present.

The upper rectum receives its blood supply from branches of the inferior mesenteric artery from the abdomen. The lower rectum has blood vessels entering from the sides of the pelvis. Lymph, a protein-rich fluid that bathes the cells of the body, is transported in small channels known as lymphatics. These channels run with the blood supply of the rectum. Lymph nodes are small filters through which the lymph flows on its way back to the blood stream. Cancer spreads elsewhere in the body by invading the lymph and vascular systems.

When a cell or cells lining the rectum become malignant, they first grow locally and may invade

partially or totally through the wall of the rectum. The tumor here may invade surrounding tissue or the organs that bound it, a process known as local invasion. In this process, the tumor penetrates and may invade the lymphatics or the capillaries locally and gain access to the circulation in this way. As the malignant cells work their way to other areas of the body, they again become locally invasive in the new area to which they have spread. These tumor deposits, originating in the primary tumor in the rectum, are then known as metastasis. If metastases are found in the regional lymph nodes, they are known as regional metastases. If they are distant from the primary tumor, they are known as distant metastases. The patient with distant metastases may have widespread disease, also referred to as systemic disease. Thus the cancer originating in the rectum begins locally and, given time, may become systemic.

By the time the primary tumor is originally detected, it is usually larger than 1 cm (about 0.39 in) in size and has over one million cells. This amount of growth is estimated to take about three to seven years. Each time the cells double in number, the size of the tumor quadruples. Thus like most cancers, the part that is identified clinically is later in the progression than would be desired. Screening becomes a very important endeavor to aid in earlier detection of this disease.

Passage of red blood with the stool, (noticeable bleeding with defecation), is much more common in rectal cancer than that originating in the colon because the tumor is much closer to the anus. Other symptoms (**constipation** and/ or **diarrhea**) are caused by obstruction and, less often, by local invasion of the tumor into pelvic organs or the sacrum. When the tumor has spread to distant sites, these metastases may cause dysfunction of the organ they have spread to. Distant metastasis usually occurs in the liver, less often to the lung(s), and rarely to the brain.

There are about 36,500 cases of rectal cancer diagnosed per year in the United States. Together, colon and rectal cancers account for 10% of cancers in men and 11% of cancers in women. It is the second most common site-specific cancer affecting both men and women. Nearly 57,000 people died from colon and rectal cancer in the United States in 2003. In recent years the incidence of this disease is decreasing very slightly, as has the mortality rate. It is difficult to tell if the decrease in mortality reflects earlier diagnosis, less **death** related to the actual treatment of the disease, or a combination of both factors.

Cancer of the rectum is felt to arise sporadically in about 80% of those who develop the disease. About

20% of cases probably arise from genetic predisposition; some people have a family history of rectal cancer occurring in a first-degree relative. Development of rectal cancer at an early age suggests a genetically transmitted form of the disease as opposed to the sporadic form.

Causes and symptoms

Causes of rectal cancer are probably environmental in sporadic cases (80%), and genetic in the hereditary-predisposed (20%) cases. Since malignant cells have a changed genetic makeup, this means that in 80% of cases, the environment spontaneously induces change. Those born with a genetic predisposition are either destined to get the cancer, or it will take less environmental exposure to induce the cancer. Exposure to agents in the environment that may induce mutation is the process of carcinogenesis and is caused by agents known as carcinogens. Specific carcinogens have been difficult to identify; dietary factors, however, seem to be involved.

Rectal cancer is more common in industrialized nations. Dietary factors may be the reason. **Diets** high in fat, red meat, total calories, and alcohol seem to add to increased risk. Diets high in fiber are associated with a decreased risk. High-fiber diets may be related to less exposure of the rectal epithelium to carcinogens from the environment as the transit time through the bowel is faster with a high-fiber diet than with a low-fiber diet.

Age plays a definite role in rectal cancer risk. Rectal cancer is rare before age 40. This incidence increases substantially after age 50 and doubles with each succeeding decade.

There also is a slight increase of risk for rectal cancer in the individual who smokes.

Patients who suffer from an inflammatory disease of the colon known as **ulcerative colitis** are also at increased risk.

On chromosome 5 is the APC gene associated with familial adenomatous polyposis (FAP) syndrome. There are multiple mutations that occur at this site, yet they all cause a defect in tumor suppression that results in early and frequent development of **colon cancer**. This is transmitted to 50% of offspring and each of those affected will develop colon or rectal cancer, usually at an early age. Another syndrome, hereditary non-polypoid colon cancer (HNPCC), is related to mutations in any of four genes responsible for DNA mismatch repair. In patients with colon or rectal cancer, the p53 gene is mutated 70% of the time.

When the p53 gene is mutated and ineffective, cells with damaged DNA escape repair or destruction, allowing the damaged cell to multiply. Continued replication of the damaged DNA may lead to tumor development. Though these syndromes (FAP and HNPCC) have a very high incidence of colon or rectal cancer, family history without the syndromes is also a substantial risk factor. When considering first-degree relatives, history of one with colon or rectal cancer raises the baseline risk from 2% to 6%; the presence of a second raises the risk to 17%.

The development of polyps of the colon or rectum commonly precedes the development of rectal cancer. Polyps are growths of the rectal lining. They can be unrelated to cancer, pre-cancerous, or malignant. Polyps, when identified, are removed for diagnosis. If the polyp, or polyps, are benign, the patient should undergo careful surveillance for the development of more polyps or the development of colon or rectal cancer.

Symptoms of rectal cancer most often result from the local presence of the tumor and its capacity to invade surrounding pelvic structure:

- bright red blood present with stool
- abdominal distention (stretching from internal pressure), bloating, inability to have a bowel movement
- narrowing of the stool, so-called ribbon stools
- pelvic pain
- unexplained weight loss
- persistent chronic **fatigue**
- rarely, urinary infection or passage of air in urine in males (late symptom)
- rarely, passage of feces through vagina in female (late symptom)

If the tumor is large and obstructing the rectum, the patient will not be evacuating stool normally and will get bloated and have abdominal discomfort. The tumor itself may bleed and, since it is near the anus, the patient may see bright red blood on the surface of the stool. Blood alone (without stool) may also be passed. Thus, **hemorrhoids** are often incorrectly blamed for bleeding, delaying the diagnosis. If anemia develops, which is rare, the patient will experience chronic fatigue. If the tumor invades the bladder in the male or the vagina in the female, stool will get where it does not belong and cause infection or discharge. (This condition is also rare.) Patients with widespread disease lose weight secondary to the chronic illness.

Diagnosis

Screening evaluation of the colon and rectum are accomplished together. Screening involves physical exam, simple laboratory tests, and the visualization of the lining of the rectum and colon. X rays (indirect visualization) and endoscopy (direct visualization) are used to visualize the organs' lining.

The **physical examination** involves the performance of a digital rectal exam (DRE). At the time of this exam, the physician checks the stool on the examining glove with a chemical to see if any occult (invisible), blood is present. At home, after having a bowel movement, the patient is asked to swipe a sample of stool obtained with a small stick on a card. After three such specimens are on the card, the card is then easily chemically tested for occult blood. These exams are accomplished as an easy part of a routine yearly physical exam.

Proteins are sometimes produced by cancers and these may be elevated in the patients blood. When this occurs the protein produced is known as a tumor marker. There is a tumor marker for cancer of the colon and rectum; it is known as carcinoembryonic antigen, (CEA). Unfortunately, this may be made by other adenocarcinomas as well, or it may not be produced by a particular colon or rectal cancer. Therefore, screening by chemical analysis for CEA has not been helpful. CEA has been helpful in patients treated for colon or rectal cancer if their tumor makes the protein. It is used in a follow-up role, not a screening role.

Direct visualization of the lining of the rectum is accomplished using a scope or endoscope. The physician introduces the instrument into the rectum and is able to see the epithelium of the rectum directly. A simple rigid tubular scope may be used to see the rectal epithelium; however, screening of the colon is done at the same time. The lower colon may be visualized using a fiberoptic flexible scope in a procedure known as flexible **sigmoidoscopy**. When the entire colon is visualized, the procedure is known as total **colonoscopy**. Each type of endoscopy requires pre-procedure preparation (evacuation) of the rectum and colon.

The American Cancer Society has recommended the following screening protocol for colon and rectal cancers those over age 50:

- yearly fecal occult blood test
- flexible sigmoidoscopy at age 50
- flexible sigmoidoscopy repeated every 5 years

- double contrast **barium enema** every five years
- colonoscopy every 10 years

If there are predisposing factors such as positive family history, history of polyps, or a familial syndrome, screening evaluations should start sooner.

Evaluation of patients with symptoms

When patients visit their physician because they are experiencing symptoms that could possibly be related to colon or rectal cancer, the entire colon and rectum must be visualized. Even if a rectal lesion is identified, the entire colon must be screened to rule out a syndromous polyp or cancer of the colon. The combination of a flexible sigmoidoscopy and double contrast barium enema may be performed, but the much preferred evaluation of the entire colon and rectum is that of complete colonoscopy. Colonoscopy allows direct visualization, photography, as well as the opportunity to obtain a biopsy, (a sample of tissue), of any abnormality visualized. If, for technical reasons the entire colon is not visualized endoscopically, a double contrast barium enema should complement the colonoscopy. A patient who is identified to have a problem in one area of the colon or rectum is at greater risk to have a similar problem in area of the colon or rectum. Therefore the entire colon and rectum need to be visualized during the evaluation.

The diagnosis of rectal cancer is actually made by the performance of a biopsy of any abnormal lesion in the rectum. Many rectal cancers are within reach of the examiner's finger. Identifying how close to the anus the cancer has developed is very important in planning the treatment. Another characteristic ascertained by exam is whether the tumor is mobile or fixed to surrounding structure. Again, this will have implications related to primary treatment. As a general rule, it is easier to identify and adequately obtain tissue for evaluation in the rectum as opposed to the colon. This is because the lesion is closer to the anus.

If the patient has advanced disease, areas where the tumor has spread, such as the liver, may require biopsy. Such biopsies are usually obtained using a special needle under **local anesthesia**.

Once a diagnosis of rectal cancer has been established by biopsy, in addition to the physical exam, an **endorectal ultrasound** will be performed to assess the extent of the disease. For rectal cancer, endorectal ultrasound is the most preferred method for staging both depth of tumor penetration and local lymph node status. Endorectal ultrasound:

- differentiates areas of invasion within large rectal adenomas that seem benign
- determines the depth of tumor penetration into the rectal wall
- determines the extent of regional lymph node invasion
- can be combined with other tests (chest x rays and **computed tomography scans**, or CT scans) to determine the extent of cancer spread to distant organs, such as the lungs or liver

The resulting rectal cancer staging allows physicians to determine the need for—and order of—radiation, surgery, and **chemotherapy**. In 2003, it was reported that **magnetic resonance imaging** (MRI) also may be useful in staging rectal cancer. MRI may help physicians determine if a tumor can be resected and risk of cancer recurrence.

Treatment

Once the diagnosis has been confirmed by biopsy and the endorectal ultrasound has been performed, the clinical stage of the cancer is assigned. The treating physicians use staging to plan the specific treatment protocol for the patient. In addition, the stage of the cancer at the time of presentation gives a statistical likelihood of the treatment outcome (prognosis).

Clinical staging

Rectal cancer first invades locally and then progresses to spread to regional lymph nodes or to other organs. Stage is derived using the characteristics of the primary tumor, its depth of penetration through the rectum, local invasion into pelvic structure, and the presence or absence of regional or distant metastases. A CT scan of the pelvis is helpful in staging because tumor invasion into the sacrum or pelvic sidewalls may mean surgical therapy is not initially possible. On this basis, clinical staging is used to begin treatment. The pathologic stage is defined when the results of analyzing the surgical specimen are available. (typically stage I and II).

Rectal cancer is assigned stages I through IV, based on the following general criteria:

- Stage I: the tumor is confined to the epithelium or has not penetrated through the first layer of muscle in the rectal wall.
- Stage II: the tumor has penetrated through to the outer wall of the rectum or has gone through it, possibly invading other local tissue or organs.

- Stage III: Any depth or size of tumor associated with regional lymph node involvement.
- Stage IV: any of previous criteria associated with distant metastasis.

Surgery

The first, or primary, treatment modality utilized in the treatment of rectal cancer is surgery. Stage I, II, and even suspected stage III disease are treated by surgical removal of the involved section of the rectum along with the complete vascular and lymphatic supply. Most Stage II and Stage III rectal cancers (based on endorectal ultrasound, CT scan, and **chest x ray**) are treated with radiation and possibly chemotherapy prior to surgery.

When determining primary treatment for rectal cancer, the surgeon's ability to reconnect the ends of the rectum. The pelvis is a confining space that makes the performance of the hook-up more difficult to do safely when the tumor is in the lower rectum. The upper rectum does not usually present a substantial problem to the surgeon restoring bowel continuity after the cancer has been removed. Mid-rectal tumors, (especially in males where the pelvis is usually smaller than a woman's), may present technical difficulties in hooking the proximal bowel to the remaining rectum. Technical advances in stapling instrumentation have largely overcome these difficulties. If the anastomosis (hook-up) leaks postoperatively, infection can occur. In the past, this was a major cause of complications in resection of rectal cancers. Today, utilizing the stapling instrumentation, a hook-up at the time of original surgery is much safer. If the surgeon feels that the hook-up is compromised or may leak, a **colostomy** may be performed. A colostomy is performed by bringing the colon through the abdominal wall and sewing it to the skin. In these cases the stool is diverted away from the hook-up, allowing it to heal and preventing the infectious complications associated with leak. Later, when the hook-up has completely healed, the colostomy can be taken down and bowel continuity restored.

Stapling devices have allowed the surgeon to get closer to the anus and still allow the technical performance of a hook-up, but there are limits. It is generally felt that there should be at least three centimeters of normal rectum below the tumor or the risk of recurrence locally will be excessive. In addition, if there is no residual native rectum, the patient will not have normal sensation or control and will have problems with uncontrollable soilage, (incontinence). For these reasons, patients presenting with low rectal tumors may

undergo total removal of the rectum and anus. This procedure is known as an abdominal-perineal resection. A permanent colostomy is performed in the lower left abdomen.

Radiation

As mentioned, for many late stage II or stage III tumors, **radiation therapy** can shrink the tumor prior to surgery. The other roles for radiation therapy are as an aid to surgical therapy in locally advanced disease that has been removed, and in the treatment of certain distant metastases. Especially when utilized in combination with chemotherapy, radiation used postoperatively has been shown to reduce the risk of local recurrence in the pelvis by 46% and death rates by 29%. Such combined therapy is recommended in patients with locally advanced primary tumors that have been removed surgically. Radiation has been helpful in treating effects of distant metastases, particularly in the brain. In very few cases, radiation alone may be the curative treatment for rectal cancer.

Chemotherapy

Adjuvant chemotherapy, (treating the patient who has no evidence of residual disease but who is at high risk for recurrence), is considered in patients whose tumors deeply penetrate or locally invade (late stage II and stage III). If the tumor was not locally advanced, this form of chemotherapeutic adjuvant therapy may be recommended without radiation. This therapy is identical to that of colon cancer and leads to similar results. Standard therapy is treatment with 5-fluorouracil, (5-FU) combined with leucovorin for a period of six to 12 months. 5-FU is an antimetabolite and leucovorin improves the response rate. Another agent, levamisole, (which seems to stimulate the immune system), may be substituted for leucovorin. These protocols reduce rate of recurrence by about 15% and reduce mortality by about 10%. The regimens have some toxicity but usually are tolerated fairly well.

Similar chemotherapy is administered for stage IV disease or if a cancer progresses and metastasis develops. Results show response rates of about 20%. A response is a temporary regression of the cancer in response to the chemotherapy. Unfortunately, these patients eventually succumb to the disease. Clinical trials have now shown that the results can be improved with the addition of another agent to this regimen. Irinotecan does not seem to increase toxicity but has improved response rates to 39%, added

two to three months to disease free survival, and prolonged overall survival by a little more than two months.

Alternative treatment

Most alternative therapies have not been studied in clinical trials. Large doses of **vitamins**, fiber, and green tea are among therapies tried. A 2003 report on a large Harvard University study showed that people who took multivitamins for at least 15 years had a 34% reduction in risk of rectal cancer. Before initiating any alternative therapies, the patient should consult his or her physician to be sure that these therapies do not complicate or interfere with the recommended therapy.

Prognosis

Prognosis is the long-term outlook or survival after therapy. Overall, about 50% of patients treated for colon and rectal cancer survive the disease. As expected, the survival rates are dependent upon the stage of the cancer at the time of diagnosis, making early detection crucial.

About 15% of patients present with stage I disease, or are diagnosed with Stage I disease when they initially visit a doctor, and 85-90% survive. Stage II represents 20-30% of cases and 65-75% survive; 30-40% comprise the stage III presentation, of which 55% survive. The remaining 20-25% present with stage IV disease and are rarely cured.

Prevention

There is not an absolute method for preventing colon or rectal cancer. An individual can lessen risk or identify the precursors of colon and rectal cancer. The patient with a familial history can enter screening and surveillance programs earlier than the general population. High-fiber diets and vitamins, avoiding **obesity**, and staying active lessen the risk. In fact, a 2003 report said that vigorous **exercise** (to the point of sweating or feeling out of breath) lowered risk of rectal cancer by nearly 40% compared to those who exercised less. Avoiding cigarettes and alcohol may be helpful. By controlling these environmental factors, an individual can lessen risk and to this degree prevent the disease.

By undergoing appropriate screening when uncontrollable genetic risk factors have been identified, an individual may be rewarded by the identification of benign polyps that can be treated as opposed

KEY TERMS

Adenocarcinoma—Type of cancer beginning in glandular epithelium.

Adjuvant therapy—Treatment involving radiation, chemotherapy (drug treatment), or hormone therapy, or a combination of all three given after the primary treatment for the possibility of residual microscopic disease.

Anastomosis—Surgical re-connection of the ends of the bowel after removal of a portion of the bowel.

Anemia—The condition caused by too few circulating red blood cells, often manifest in part by fatigue.

Carcinogens—Substances in the environment that cause cancer, presumably by inducing mutations, with prolonged exposure.

Defecation—The act of having a bowel movement.

Epithelium—Cells composing the lining of an organ.

Lymphatics—Channels that are conduits for lymph.

Lymph nodes—Cellular filters through which lymphatics flow.

Malignant—Cells that have been altered such that they have lost normal control mechanisms and are capable of local invasion and spread to other areas of the body.

Metastasis—Site of invasive tumor growth that originated from a malignancy elsewhere in the body.

Mutation—A change in the genetic makeup of a cell that may occur spontaneously or be environmentally induced.

Occult blood—Presence of blood that cannot be appreciated visually.

Polyps—Localized growths of the epithelium that can be benign, pre-cancerous, or harbor malignancy.

Resect—To remove surgically.

Sacrum—Posterior bony wall of the pelvis.

Systemic—Referring to throughout the body.

to having these growths degenerate into a malignancy.

Resources

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ORGANIZATIONS

American Cancer Society. 1599 Clifton Road NE, Atlanta, GA 30329. (800)ACS-2345. Lt;<http://www.cancer.org> >.

Cancer Information Service of the NCI. 9000 Rockville Pike, Building 31, Suite 10A18, Bethesda, MD 20892. (800) 4-CANCER. <<http://wwwwic.nci.nih.gov>>.

OTHER

Colon Cancer Alliance. <<http://www.ccalliance.org>>.

National Cancer Institute Clinical Trials. <<http://www.cancertrials.nci.nih.gov>>.

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Purpose

DRE is used as a screening tool to locate **rectal cancer** and **prostate cancer**. It is also used as a diagnostic test to find non-cancerous abnormalities within the rectum like **hemorrhoids**, anal fissures, or congenital deformities that can cause chronic **constipation**.

Precautions

There are no precautions when performing DRE, aside from routine sanitary procedures.

Description

DRE is performed in most instances as an annual routine procedure in colorectal **cancer** screening. Digital palpitation of the rectum can often find abnormal growths which may require further testing or commonplace hemorrhoids. It is a critical initial clinical test and is important in the assessment of the size and location of tumors.

This procedure is often not performed routinely on patients over 70, even though this population is at high risk for colorectal cancer. It also is not done as often in elderly women as in elderly men.

DRE has also been used as a screening tool for prostate cancer. It seems to be very effective for larger masses found in the prostate and correlated well with higher prostate-specific antigens.

Of less predictive value was DRE in routine rectovaginal examinations of women under the age of 50. These instances of DRE did not locate colorectal cancer or any other abnormality.

More gastroenterologists are recommending that pediatricians and family physicians perform DRE on pediatric patients exhibiting chronic constipation before those patients are referred to intestinal specialists. The pediatrician or family physician could identify fecal compaction and treat it themselves, and then only refer patients who have a specific abnormality to gastroenterologists.

Preparation

The physician must conduct DRE using a gloved hand. Some sort of lubricant should be used so that penetration of the rectum is easier and does not create the damage that the procedure is seeking.

Aftercare

There is no aftercare after a DRE is performed.

Rectal examination

Definition

Rectal examination or digital rectal examination (DRE) is performed by means of inserting a gloved, lubricated finger into the rectum and palpating (feeling) for lumps.

KEY TERMS

DRE—Digital rectal examination.

Gastroenterologist—A physician who specializes in diseases of the digestive system.

Rectum—The last eight to ten inches of the colon, of which the anus is a part and the opening through which wastes are removed from the body.

Risks

There are no risks to DRE and it is virtually painless.

Normal results

The physician finds a normal rectal canal with no abnormalities.

Abnormal results

Growths, tears, anal fissures, or congenital structural defects can be found inside the rectum with DRE.

Resources

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Janie F. Franz

Rectal polyps

Definition

Rectal polyps are tissue growths that arise from the wall of the rectum and protrude into it. They may be either benign or malignant (cancerous).

Description

The rectum is the last segment of the large intestine, ending in the anus, the opening to the exterior of the body. Rectal polyps are quite common. They occur in 7–50% of all people, and in two thirds of people over age 60.

Rectal polyps can be either benign or malignant, large or small. There are several different types of polyps. The type is determined by taking a sample of the polyp and examining it microscopically. Most polyps are benign. They are of concern, however, because 90% of colon and rectal cancers arise from polyps that are initially benign. For this reason, rectal polyps are usually removed when they are discovered.

Causes and symptoms

The cause of most rectal polyps is unknown, however a diet high in animal fat and red meat, and low in fiber, is thought to encourage polyp formation. Some types of polyps are hereditary. In an inherited disease called **familial polyposis**, hundreds of small, malignant and pre-malignant polyps are produced before the age of 40. Also, inflammatory bowel disease may cause growth of polyps and pseudo-polyps. Juvenile polyps (polyps in children) are usually benign and often outgrow their blood supply and disappear at **puberty**.

Most rectal polyps produce no symptoms and are discovered on routine digital or endoscopic examination of the rectum. Rectal bleeding is the most common complaint when symptoms do occur. Abdominal cramps, **pain**, or obstruction of the intestine occur with some large polyps. Certain types of polyps cause mucous-filled or watery **diarrhea**.

Diagnosis

Rectal polyps are commonly found by **sigmoidoscopy** (visual inspection with an instrument consisting of a tube and a light) or **colonoscopy**. If polyps are found in the rectum, a complete examination of the large intestine is done, as multiple polyps are common. Polyps do not show up on regular x rays, but they do appear on **barium enema** x rays.

Treatment

Normally polyps are removed when they are found. Polypectomy is the name for the surgery that removes these growths. Polypectomy is performed at a hospital, outpatient surgical facility or in a doctor's

KEY TERMS

Colon—The part of the large intestine that extends from the cecum to the rectum. The sigmoid colon is the area of the intestine just above the rectum; linking the descending colon with the rectum. It is shaped like the letter S.

Rectum—The final part of the large intestine, ending in the anus.

Sigmoidoscopy—A procedure where a thin tube containing a camera and a light is inserted into the lower section of the large intestine so that the doctor can visually inspect the lower (sigmoid) colon and rectum. Colonoscopy examines the entire large intestine using the same techniques.

office, depending on the number and type of polyps to be removed, and the age and health of the patient. The procedure can be done by a surgeon, gastroenterologist, or family practitioner.

Before the operation, a colonoscopy (examination of the intestine with an endoscope) is performed, and standard pre-operative blood and urine studies are done. The patient is also given medicated **enemas** to cleanse the bowel.

The patient is given a sedative and a narcotic pain killer. A colonoscope is inserted into the rectum. The polyps are located and removed with a wire snare, ultrasound, or laser beam. After they are removed, the polyps are examined to determine if they are malignant or benign. When polyps are malignant, it may be necessary to remove a portion of the rectum or colon to completely remove cancerous tissue.

Alternative treatment

In addition to a diet low in animal fat and high in fiber, nutritionists recommend antioxidant supplements (including **vitamins** A, C, and E, selenium, and zinc) to reduce rectal polyps.

Prognosis

For most people, the removal of polyps is an uncomplicated procedure. Benign polyps that are left in place can give rise to **rectal cancer**. People who have had rectal polyps once are more likely to have them again and should have regular screening examinations.

Prevention

Eating a diet low in red meat and animal fat, and high in fiber, is thought to help prevent rectal polyps.

Resources

ORGANIZATIONS

American Cancer Society. 1599 Clifton Rd., NE, Atlanta, GA 30329-4251. (800) 227-2345. <<http://www.cancer.org>>.

National Cancer Institute. Building 31, Room 10A31, 31 Center Drive, MSC 2580, Bethesda, MD 20892-2580. (800) 422-6237. <<http://www.nci.nih.gov>>.

Tish Davidson, A.M.

Rectal prolapse

Definition

Rectal prolapse is protrusion of rectal tissue through the anus to the exterior of the body. The rectum is the final section of the large intestine.

Description

Rectal prolapse can be either partial or complete. In partial prolapse, only the mucosa layer (mucous membrane) of the rectum extends outside the body. The projection is generally 0.75–1.5 in (2–4 cm) long. In complete prolapse, called procidentia, the full thickness of the rectum protrudes for up to 4.5 in (12 cm).

Rectal prolapse is most common in people over age 60, and occurs much more frequently in women than in men. It is also more common in psychiatric patients. Prolapse can occur in normal infants, where it is usually transient. In children it is often an early sign of **cystic fibrosis** or is due to neurological or anatomical abnormalities.

Although rectal prolapse in adults may initially reduce spontaneously after bowel movements, it eventually becomes permanent. Adults who have had prior rectal or vaginal surgery, who have chronic **constipation**, regularly depend on **laxatives**, have **multiple sclerosis** or other neurologic diseases, **stroke**, or **paralysis** are more likely to experience rectal prolapse.

Causes and symptoms

Rectal prolapse in adults is caused by a weakening of the sphincter muscle or ligaments that hold the

rectum in place. Weakening can occur because of **aging**, disease, or in rare cases, surgical trauma. Prolapse is brought on by straining to have bowel movements, chronic laxative use, or severe **diarrhea**.

Symptoms of rectal prolapse include discharge of mucus or blood, **pain** during bowel movements, and inability to control bowel movements (**fecal incontinence**). Patients may also feel the mass of tissue protruding from the anus. With large prolapses, the patient may lose the normal urge to have a bowel movement.

Diagnosis

Prolapse is initially diagnosed by taking a patient history and giving a **rectal examination** while the patient is in a squatting position. It is confirmed by **sigmoidoscopy** (inspection of the colon with a viewing instrument called an endoscope) **Barium enema** x rays and other tests are done to rule out neurologic (nerve) disorders or disease as the primary cause of prolapse.

Treatment

In infants, conservative treatment, consisting of strapping the buttocks together between bowel movements and eliminating any causes of bowel straining, usually produces a spontaneous resolution of prolapse. For partial prolapse in adults, excess tissue is surgically tied off with special bands causing the tissue to wither in a few days.

Complete prolapse requires surgery. Different surgical techniques are used, but all involve anchoring the rectum to other parts of the body, and using plastic mesh to reinforce and support the rectum. In patients too old, or ill, to tolerate surgery, a wire or plastic loop can be inserted to hold the sphincter closed and prevent prolapse. Treatment should be undertaken as soon as prolapse is diagnosed, since the longer the condition exists, the more difficult it is to reverse.

Alternative treatment

Alternative therapies can act as support for conventional treatment, especially if surgery is required. **Acupuncture**, homeopathy, and botanical medicine can all be used to assist in resolution of the prolapse or in recovery from surgery.

Prognosis

Successful resolution of rectal prolapse involves prompt treatment and the elimination of any underlying causes of prolapse. Infants and children usually

KEY TERMS

Rectum—The part of the large intestine that ends at the anal canal.

recover completely without complications. Recovery in adults depends on age, general health, and the extent of the prolapse.

Prevention

Reducing constipation by eating a diet high in fiber, drinking plenty of fluids, and avoiding straining during bowel movements help prevent the onset of prolapse. Exercises that strengthen the anal sphincter may also be helpful.

Resources

OTHER

“Rectal Prolapse.” *ThriveOnline*. <<http://thriveonline.oxygen.com>>.

Tish Davidson, A.M.

Recurrent fever see **Relapsing fever**

Recurrent miscarriage

Definition

Recurrent **miscarriage** is defined as three or more miscarriages of a fetus before 20 weeks of gestation (i.e., before the fetus can live outside the womb).

Description

Also referred to as spontaneous abortion, miscarriage occurs in 15–20% of all conceptions. The majority of miscarriages occur during the first trimester. The number of previous miscarriages does not affect subsequent full-term pregnancies.

Causes and symptoms

Recurrent miscarriage can be caused by several factors, including fetal, placental, or maternal abnormalities.

- In over half of all miscarriages, the fetus is abnormal. The abnormality can either be genetic or

developmental. The fetus is very sensitive to ionizing radiation. Tobacco and even moderate alcohol consumption are known to cause fetal damage that may lead to miscarriage. There is some evidence that over four cups of coffee a day, because of the **caffeine**, adversely affect **pregnancy**, as well.

- Placental abnormalities, including abnormal implantation in the placental wall and premature separation of the placenta, can cause miscarriage.
- Maternal abnormalities include insufficient hormones (usually progesterone) to support the pregnancy, an **incompetent cervix** (mouth of the womb does not stay closed), or a deformed uterus (womb). A deformed uterus can be caused by diethylstilbestrol (DES) given to the mother's mother during her pregnancy. Some immunologic abnormalities may cause the mother to reject the fetus as if it were an infection or a transplant. Maternal blood clotting abnormalities may cut-off blood supply to the fetus, causing miscarriage.
- Maternal **diabetes mellitus** causes miscarriage if the diabetes is poorly controlled. Maternal infections may occasionally lead to miscarriage. There is some evidence that conceptions that take place between old eggs (several days after ovulation) or old sperm (that start out several days before ovulation) may be more likely to miscarry.

Symptoms of miscarriage include pink or brown colored discharge for several weeks, which develops into painful cramping and increased vaginal bleeding; dilation of the cervix; and expulsion of the fetus.

Diagnosis

A pelvic examination can detect a deformed uterus, and frequent examinations during pregnancy can detect an incompetent cervix. Blood tests can detect the presence of immunologic or blood-clotting problems in the mother. **Genetic testing** can also determine if chromosomal abnormalities may be causing the miscarriages.

Treatment

If a uterus is deformed, it may be surgically repaired. If a cervix is incompetent, it can be surgically fortified, until the fetus matures, by a procedure known as circlage (tying the cervix closed). Supplemental progesterone may also help sustain a pregnancy. Experimental treatment of maternal immunologic abnormalities with white cell immunization (injecting the mother with white cells from the father) has been successful in some cases of recurrent

KEY TERMS

Fetus—A developing embryo in the womb after the first eight weeks of gestation.

Ionizing radiation—Radiation produced by x rays and radioactivity.

Ovulation—Release of an egg for fertilization from the ovary that happens about fourteen days before each menstrual period.

miscarriage. Clotting abnormalities can be treated with **anticoagulant drugs**, such as heparin and **aspirin**, to keep blood flowing to the fetus.

Prognosis

If there is no underlying disease or abnormality present, the rate of successful pregnancy after several miscarriages approaches normal. Seventy to eighty-five percent of women with three or more miscarriages will go on to complete a healthy pregnancy.

Resources

BOOKS

Cunningham, F. Gary, et al., editor. *Williams Obstetrics*. Stamford: Appleton & Lange, 1997.

J. Ricker Polsdorfer, MD

Red blood cell indices

Definition

Red blood cell indices are measurements that describe the size and oxygen-carrying protein (hemoglobin) content of red blood cells. The indices are used to help in the differential diagnosis of anemia. They are also called red cell absolute values or erythrocyte indices.

Purpose

Anemia includes a variety of conditions with the same outcome: a person's blood cannot carry as much oxygen as it should. A healthy person has an adequate number of correctly sized red blood cells that contain enough hemoglobin to carry sufficient oxygen to all

the body's tissues. An anemic person has red blood cells that are either too small or too few in number. As a result, the heart and lungs must work harder to make up for the lack of oxygen delivered to the tissues by the blood.

Anemia is caused by many different diseases or disorders. The first step in finding the cause is to determine what type of anemia the person has. Red blood cell indices help to classify the **anemias**.

Precautions

Certain prescription medications may affect the test results. These drugs include zidovudine (Retrovir), phenytoin (Dilantin), and azathioprine (Imuran).

Description

Overview

Anemia has several general causes: blood loss; a drop in production of red blood cells; or a rise in the number of red blood cells destroyed. Blood loss can result from severe hemorrhage or a chronic slow bleed, such as the result of an accident or an ulcer. Lack of iron, vitamin B₁₂, or **folic acid** in the diet, as well as certain chronic diseases, lower the number of red blood cells produced by the bone marrow. Inherited disorders affecting hemoglobin, severe reactions to blood transfusions, prescription medications, or poisons can cause red blood cells to burst (hemolyze) well before the end of their usual 120-day lifespan.

Anemia of any type affects the results of one or more of the common blood tests. These tests are the **hematocrit**, hemoglobin, and red blood cell count. The hematocrit is a measure of red blood cell mass, or how much space in the blood is occupied by red blood cells. The **hemoglobin test** is a measure of how much hemoglobin protein is in the blood. The red blood cell count (RBC) measures the number of red blood cells present in the blood. Red blood cell indices are additional measurements of red blood cells based on the relationship of these three test results.

The relationships between the hematocrit, the hemoglobin level, and the RBC are converted to red blood cell indices through mathematical formulas. These formulas were worked out and first applied to the classification of anemias by Maxwell Wintrobe in 1934.

The indices include these measurements: mean corpuscular volume (MCV); mean corpuscular hemoglobin (MCH); mean corpuscular hemoglobin

concentration (MCHC); and red cell distribution width (RDW). They are usually calculated by an automated instrument as part of a complete **blood count** (CBC). Indices are covered by insurance when medically necessary. Results are available the same day that the blood is drawn or the following day.

Mean corpuscular volume (MCV)

MCV is the index most often used. It measures the average volume of a red blood cell by dividing the hematocrit by the RBC. The MCV categorizes red blood cells by size. Cells of normal size are called normocytic, smaller cells are microcytic, and larger cells are macrocytic. These size categories are used to classify anemias. Normocytic anemias have normal-sized cells and a normal MCV; microcytic anemias have small cells and a decreased MCV; and macrocytic anemias have large cells and an increased MCV. Under a microscope, stained red blood cells with a high MCV appear larger than cells with a normal or low MCV.

Mean corpuscular hemoglobin concentration (MCHC)

The MCHC measures the average concentration of hemoglobin in a red blood cell. This index is calculated by dividing the hemoglobin by the hematocrit. The MCHC categorizes red blood cells according to their concentration of hemoglobin. Cells with a normal concentration of hemoglobin are called normochromic; cells with a lower than normal concentration are called hypochromic. Because there is a physical limit to the amount of hemoglobin that can fit in a cell, there is no hyperchromic category.

Just as MCV relates to the size of the cells, MCHC relates to the color of the cells. Hemoglobin contains iron, which gives blood its characteristic red color. When examined under a microscope, normal red blood cells that contain a normal amount of hemoglobin stain pinkish red with a paler area in the center. These normochromic cells have a normal MCHC. Cells with too little hemoglobin are lighter in color with a larger pale area in the center. These hypochromic cells have a low MCHC. Anemias are categorized as hypochromic or normochromic according to the MCHC index.

Mean corpuscular hemoglobin (MCH)

The average weight of hemoglobin in a red blood cell is measured by the MCH. The formula for this index is the sum of the hemoglobin multiplied by 10

and divided by the RBC. MCH values usually rise or fall as the MCV is increased or decreased.

Red cell distribution width (RDW)

The RDW measures the variation in size of the red blood cells. Usually red blood cells are a standard size. Certain disorders, however, cause a significant variation in cell size.

Obtaining the blood sample

The RBC indices test requires 0.17–24 oz (5–7 ml) of blood. A healthcare worker ties a tourniquet on the person's upper arm, locates a vein in the inner elbow region, and inserts a needle into that vein. Vacuum action draws the blood through the needle into an attached tube. Collection of the sample takes only a few minutes.

Preparation

The doctor should check to see if the patient is taking any medications that may affect test results. The patient does not need to fast before the test.

Aftercare

Aftercare consists of routine care of the area around the puncture mark. Pressure is applied for a few seconds and the wound is covered with a bandage.

Risks

The primary risk is mild **dizziness** and the possibility of a bruise or swelling in the area where the blood was drawn. The patient can apply moist warm compresses.

Normal results

Normal results for red blood cell indices are as follows:

- MCV 82–98 fl (femtoliters)
- MCHC 31–37 g/dl
- MCH 26–34 pg (picograms)
- RDW 11.5–14.5%.

Abnormal results

The category into which a person's anemia is placed based on the indices provides a significant clue as to the cause of the anemia, but further testing is needed to confirm a specific diagnosis.

KEY TERMS

Anemia—A variety of conditions in which a person's blood can't carry as much oxygen as it should due to a decreased number or size of red blood cells.

Hypochromic—A descriptive term applied to a red blood cell with a decreased concentration of hemoglobin.

Macrocytic—A descriptive term applied to a larger than normal red blood cell.

Mean corpuscular hemoglobin (MCH)—A measurement of the average weight of hemoglobin in a red blood cell.

Mean corpuscular hemoglobin concentration (MCHC)—The measurement of the average concentration of hemoglobin in a red blood cell.

Mean corpuscular volume (MCV)—A measure of the average volume of a red blood cell.

Microcytic—A descriptive term applied to a smaller than normal red blood cell.

Normochromic—A descriptive term applied to a red blood cell with a normal concentration of hemoglobin.

Normocytic—A descriptive term applied to a red blood cell of normal size.

Red blood cell indices—Measurements that describe the size and hemoglobin content of red blood cells.

Red cell distribution width (RDW)—A measure of the variation in size of red blood cells.

The most common causes of macrocytic anemia (high MCV) are vitamin B₁₂ and folic acid deficiencies. Lack of iron in the diet, **thalassemia** (a type of hereditary anemia), and chronic illness are the most common causes of microcytic anemia (low MCV). Normocytic anemia (normal MCV) can be caused by kidney and **liver disease**, bone marrow disorders, or excessive bleeding or hemolysis of the red blood cells.

Lack of iron in the diet and thalassemia are the most common causes of hypochromic anemia (low MCHC). Normocytic anemias are usually also normochromic and share the same causes (normal MCHC).

The RDW is increased in anemias caused by deficiencies of iron, vitamin B₁₂, or folic acid. Abnormal hemoglobins, such as in sickle cell anemia, can change

the shape of red blood cells as well as cause them to hemolyze. The abnormal shape and the cell fragments resulting from hemolysis increase the RDW. Conditions that cause more immature cells to be released into the bloodstream, such as severe blood loss, will increase the RDW. The larger size of immature cells creates a distinct size variation.

Resources

BOOKS

Pagana, Kathleen Deska, and Timothy James Pagana, editors. *Mosby's Manual of Diagnostic and Laboratory Tests*. St. Louis: Mosby, Inc., 1998.

Nancy J. Nordenson

Red blood cell test see **Hemoglobin test**

Reflex sympathetic dystrophy

Definition

Reflex sympathetic dystrophy is the feeling of **pain** associated with evidence of minor nerve injury.

Description

Historically, reflex sympathetic dystrophy (RSD) was noticed during the civil war in patients who suffered pain following gunshot **wounds** that affected the median nerve (a major nerve in the arm). In 1867 the condition was called causalgia from the Greek term meaning “burning pain.” Causalgia refers to pain associated with major nerve injury. The exact causes of RSD are still unclear. Patients usually develop a triad of phases. In the first phase, pain and sympathetic activity is increased. Patients will typically present with swelling (**edema**), stiffness, pain, increased vascularity (increasing warmth), hyperhidrosis, and x-ray changes demonstrating loss of **minerals** in bone (demineralization). The second phase develops three to nine months later. It is characterized by increased stiffness and changes in the extremity that include a decrease in warmth and atrophy of the skin and muscles. The late phase commencing several months to years later presents with a pale, cold, painful, and atrophic extremity. Patients at this stage will also have **osteoporosis**.

It has been thought that each phase relates to a specific nerve defect that involves nerve tracts from the periphery spinal cord to the brain. Both sexes are

KEY TERMS

Atrophy—Abnormal changes in a cell that lead to loss of cell structure and function.

Osteoporosis—Reduction in the quantity of bone.

affected, but the number of new cases is higher in women, adolescents, and young adults. RDS has been associated with other terms such as Sudeck’s atrophy, post-traumatic osteoporosis, causalgia, shoulder-hand syndrome, and reflex neuromuscular dystrophy.

Causes and symptoms

The exact causes of RSD at present is not clearly understood. There are several theories such as sympathetic overflow (over activity), abnormal circuitry in nerve impulses through the sympathetic system, and as a post-operative complication for both elective and traumatic surgical procedures. Patients typically develop pain, swelling, temperature, color changes, and skin and muscle wasting.

Diagnosis

The diagnosis is simple and confirmed by a local anesthetic block along sympathetic nerve paths in the hand or foot, depending on whether an arm or leg is affected. A test called the **erythrocyte sedimentation rate** (ESR) can be performed to rule out diseases with similar presentation and arising from other causes.

Treatment

The preferred method to treat RSD includes sympathetic block and physical therapy. Pain is improved as motion of the affected limb improves. Patients may also require tranquilizers and mild **analgesics**. Patients who received repeated blocks should consider surgical sympathectomy (removal of the nerves causing pain).

Prognosis

The prognosis for treatment during phase one is favorable. As the disease progresses undetected into phase two or three the prognosis for recovery is poor.

Prevention

There is no known prevention since the cause is not clearly understood.

Resources

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Goetz, Christopher G., et al, editors. *Textbook of Clinical Neurology*. 1st ed. W. B. Saunders Company, 1999.

Ruddy, Shaun, et al, editors. *Kelly's Textbook of Rheumatology*. 6th ed. W. B. Saunders Company, 2001.

OTHER

Reflex Sympathetic Dystrophy Syndrome Association of America. <<http://www.rsds.org/fact.html>>.

Laith Farid Gulli, M.D.
Robert Ramirez, B.S.

Reflex tests

Definition

Reflex tests are simple physical tests of nervous system function.

Purpose

A reflex is a simple nerve circuit. A stimulus, such as a light tap with a rubber hammer, causes sensory neurons (nerve cells) to send signals to the spinal cord. Here, the signals are conveyed both to the brain and to nerves that control muscles affected by the stimulus. Without any brain intervention, these muscles may respond to an appropriate stimulus by contracting.

Reflex tests measure the presence and strength of a number of reflexes. In so doing, they help to assess the integrity of the nerve circuits involved. Reflex tests are performed as part of a neurological exam, either a “mini-exam” done to quickly confirm integrity of the spinal cord, or a more complete exam performed to diagnose the presence and location of **spinal cord injury** or neuromuscular disease.

Deep tendon reflexes are responses to muscle stretch. The familiar “knee-jerk” reflex is an example; this reflex tests the integrity of the spinal cord in the lower back region. The usual set of deep tendon reflexes tested, involving increasingly higher regions of the spinal cord, are:

- ankle
- knee
- abdomen
- forearm

- biceps
- triceps

Another type of reflex test is called the Babinski test, which involves gently stroking the sole of the foot to assess proper development of the spine and cerebral cortex.

Precautions

Reflex tests are entirely safe, and no special precautions are needed.

Description

The examiner positions the patient in a comfortable position, usually seated on the examination table with legs hanging free. The examiner uses a rubber mallet to strike different points on the patient’s body, and observes the response. The examiner may position, or hold, one of the limbs during testing, and may require exposure of the ankles, knees, abdomen, and arms. Reflexes can be difficult to elicit if the patient is paying too much attention to the stimulus. To compensate for this, the patient may be asked to perform some muscle contraction, such as clenching teeth or grasping and pulling the two hands apart. When performing the Babinski reflex test, the doctor will gently **stroke** the outer soles of the patient’s feet with the mallet while checking to see whether or not the big toe extends out as a result.

Normal results

The strength of the response depends partly on the strength of the stimulus. For this reason, the examiner will attempt to elicit the response with the smallest stimulus possible. Learning the range of normal responses requires some clinical training. Responses should be the same for both sides of the body. A normal response to the Babinski reflex test depends upon the age of the person being examined. In children under the age of one and a half years, the big toe will extend out with or without the other toes. This is due to the fact that the fibers in the spinal cord and cerebral cortex have not been completely covered in myelin, the protein and lipid sheath that aids in processing neural signals. In adults and children over the age of one and a half years, the myelin sheath should be completely formed, and, as a result, all the toes will curl under (planter flexion reflex).

Abnormal results

Weak or absent response may indicate damage to the nerves outside the spinal cord (**peripheral**

neuropathy), damage to the motor neurons just before or just after they leave the spinal cord (motor neuron disease), or muscle disease. Excessive response may indicate spinal cord damage above the level controlling the hyperactive response. Different responses on the two sides of the body may indicate early onset of progressive disease, or localized nerve damage, as from trauma. An adult or older child who responds to the Babinski with an extended big toe may have a lesion in the spinal cord or cerebral cortex.

Resources

OTHER

Rathe, Richard. "The Neurological Exam." *A Healthy Me Page*. July 2, 1997. <<http://www.ahealthyme.com>>.

Richard Robinson

Reflexology

Definition

Reflexology is a therapeutic method of relieving **pain** by stimulating predefined pressure points on the feet and hands. This controlled pressure alleviates the source of the discomfort. In the absence of any particular malady or abnormality, reflexology may be as effective for promoting good health and for preventing illness as it may be for relieving symptoms of **stress**, injury, and illness.

Reflexologists work from maps of predefined pressure points that are located on the hands and feet. These pressure points are reputed to connect directly through the nervous system and affect the bodily organs and glands. The reflexologist manipulates the pressure points according to specific techniques of reflexology therapy. By means of this touching therapy, any part of the body that is the source of pain, illness, or potential debility can be strengthened through the application of pressure at the respective foot or hand location.

Purpose

Reflexology promotes healing by stimulating the nerves in the body and encouraging the flow of blood. In the process, reflexology not only quells the sensation of pain, but relieves the source of the pain as well.

Anecdotally, reflexologists claim success in the treatment of a variety of conditions and injuries. One

EUNICE INGHAM(1889–1974)

Eunice D. Ingham was born on February 24, 1889. A physical therapist by occupation, she was a colleague of Dr. Shelby Riley, who along with Dr. W. H. Fitzgerald actively developed zone therapy, a similar but distinct therapy from reflexology. Unlike reflexology, zone therapy does not connect the zones with the body as a whole. In the 1930s, Ingham discovered an unmistakable pattern of reflexes on the human foot; she subsequently devoted the rest of her life to publicizing the message of reflexology until shortly before her death on December 10, 1974.

Ingham traveled and lectured widely about reflexology, initially to audiences of extremely desperate or aging patients who had lost hope in finding relief. Because of their sometimes astonishing improvement, reflexology became better known and respected among the medical community and gained credibility for its therapeutic value. Ingham described her theories of reflexology in her 1938 book, entitled *Stories the Feet Can Tell*, which included a map of the reflex points on the feet and the organs that they parallel. The book was translated into seven languages, although it was erroneously published as *Zone Therapy* in some countries, an error which led to misunderstanding about the true nature of reflexology and inaccurately linked it to zone therapy.

condition is **fibromyalgia**. People with this disease are encouraged to undergo reflexology therapy to alleviate any of a number of chronic bowel syndromes associated with the condition. Frequent brief sessions of reflexology therapy are also recommended as an alternative to drug therapy for controlling the muscle pain associated with fibromyalgia and for relieving difficult breathing caused by tightness in the muscles of the patient's neck and throat.

Reflexology applied properly can alleviate allergy symptoms, as well as stress, back pain, and chronic **fatigue**. The techniques of reflexology can be performed conveniently on the hand in situations where a session on the feet is not practical, although the effectiveness of limited hand therapy is less pronounced than with the foot pressure therapy.

Description

Origins

Reflexology is a healing art of ancient origin. Although its origins are not well documented, there are reliefs on the walls of a Sixth Dynasty Egyptian tomb (c. 2450 B.C.) that depict two seated men

receiving massage on their hands and feet. From Egypt, the practice may have entered the Western world during the conquests of the Roman Empire. The concepts of reflexology have also been traced to pre-dynastic China (possibly as early as 3000 B.C.) and to ancient Indian medicine. The Inca civilization may have subscribed to the theories of reflexology and passed on the practice of this treatment to the Native Americans in the territories that eventually entered the United States.

In recent times, Sir Henry Head first investigated the concepts underlying reflexology in England in the 1890s. Therapists in Germany and Russia were researching similar notions at approximately the same time, although with a different focus. Less than two decades later, a physician named William H. Fitzgerald presented a similar concept that he called zone analgesia or zone therapy. Fitzgerald's zone analgesia was a method of relieving pain through the application of pressure to specific locations throughout the entire body. Fitzgerald divided the body into 10 vertical zones, five on each side, that extended from the head to the fingertips and toes, and from front to back. Every aspect of the human body appears in one of these 10 zones, and each zone has a reflex area on the hands and feet. Fitzgerald and his colleague, Dr. Edwin Bowers, demonstrated that by applying pressure on one area of the body, they could anesthetize or reduce pain in a corresponding part. In 1917, Fitzgerald and Bowers published *Relieving Pain at Home*, an explanation of zone therapy.

Later, in the 1930s, a physical therapist, Eunice D. Ingham, explored the direction of the therapy and made the startling discovery that pressure points on the human foot were situated in a mirror image of the corresponding organs of the body with which the respective pressure points were associated. Ingham documented her findings, which formed the basis of reflexology, in *Stories the Feet Can Tell*, published in 1938. Although Ingham's work in reflexology was inaccurately described as zone therapy by some, there are differences between the two therapies of pressure analgesia. Among the more marked differences, reflexology defines a precise correlation between pressure points and afflicted areas of the body. Furthermore, Ingham divided each foot and hand into 12 respective pressure zones, in contrast to the 10 vertical divisions that encompass the entire body in Fitzgerald's zone therapy.

In 1968 two siblings, Dwight Byers and Eusebia Messenger, established the National Institute of Reflexology. By the early 1970s the institute had grown and was renamed the International Institute of Reflexology.

In a typical reflexology treatment, the therapist and patient have a preliminary discussion prior to therapy, to enable the therapist to focus more accurately on the patient's specific complaints and to determine the appropriate pressure points for treatment.

A reflexology session involves pressure treatment that is most commonly administered in foot therapy sessions of approximately 40–45 minutes in duration. The foot therapy may be followed by a brief 15-minute hand therapy session. No artificial devices or special equipment are associated with this therapy. The human hand is the primary tool used in reflexology. The therapist applies controlled pressure with the thumb and forefinger, generally working toward the heel of the foot or the outer palm of the hand. Most reflexologists apply pressure with their thumbs bent; however, some also use simple implements, such as the eraser end of a pencil. Reflexology therapy is not massage, and it is not a substitute for medical treatment.

Reflexology is a complex system that identifies and addresses the mass of 7,000 nerve endings that are contained in the foot. Additional reflexology addresses the nerves that are located in the hand. This is a completely natural therapy that affords relief without the use of drugs. The Reflexology Association of America (RAA) formally discourages the use of oils or other preparations in performing this hands-on therapy.

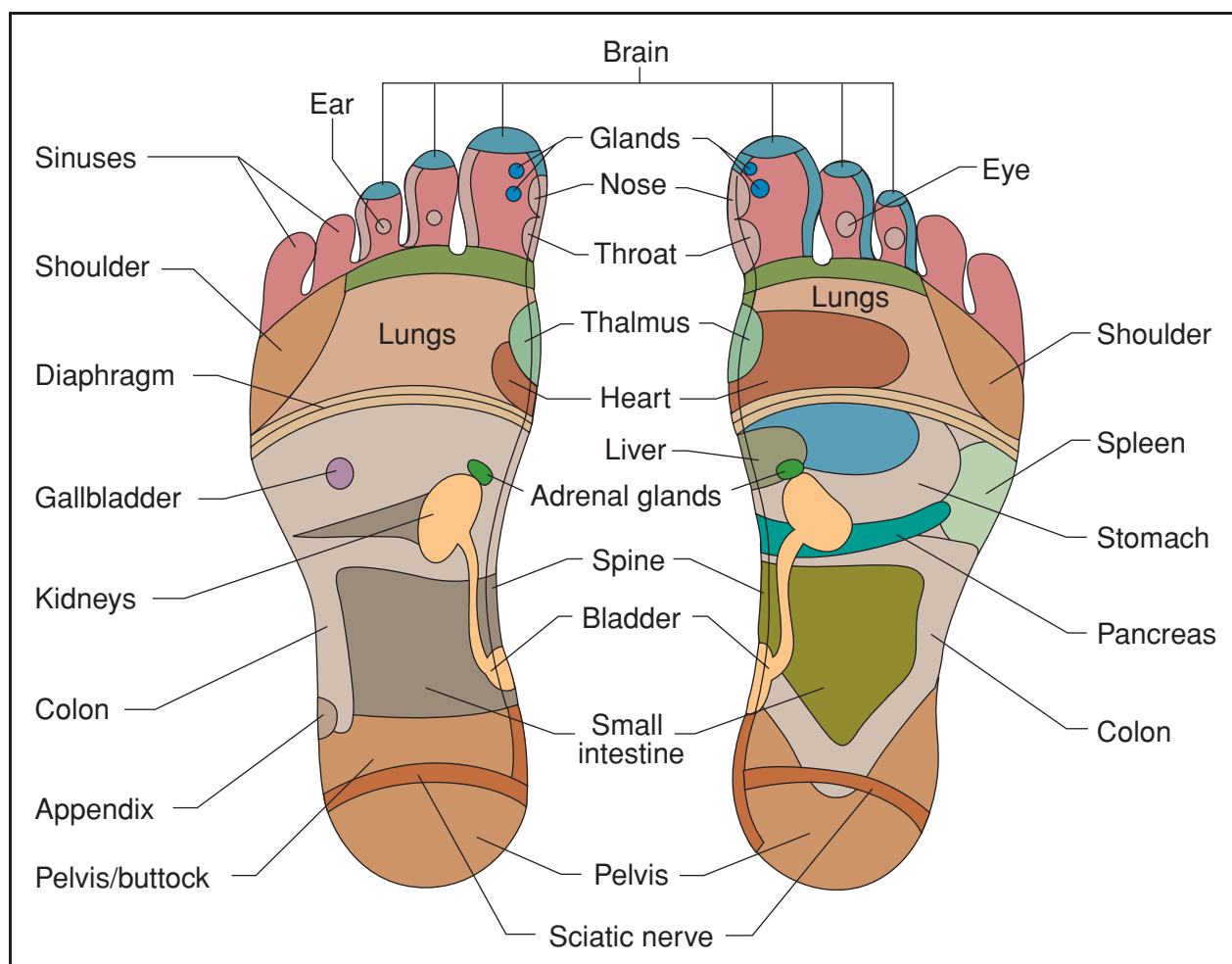
Preparations

In order to realize maximum benefit from a reflexology session, the therapist as well as the patient should be situated so as to afford optimal comfort for both. Patients in general receive treatment in a reclining position, with the therapist positioned as necessary—to work on the bare feet, or alternately on the bare hands.

A reflexology patient removes both shoes and socks in order to receive treatment. No other preparation is involved. No prescription drugs, creams, oils, or lotions are used on the skin.

Precautions

Reflexology is extremely safe. It may even be self-administered in a limited form whenever desired. The qualified reflexologist offers a clear and open disclaimer that reflexology does not constitute medical treatment in any form, nor is reflexology given as a substitute for medical advice or treatment. The ultimate purpose of the therapy is to promote wellness; fundamentally it is a form of preventive therapy.



Reflexology employs the principle that the reflex points on the feet, when hand pressure is applied, will reflexively stimulate energy to a related muscle or organ in the body and promote healing. (Illustration by Electronic Illustrators Group.)

People with serious and long-term medical problems are urged to seek the advice of a physician. Diabetes patients in particular are urged to approach this therapy cautiously. Likewise pregnant women are cautioned emphatically to avoid reflexology during the early phases of **pregnancy** altogether, as accidentally induced labor and subsequent premature delivery can result from reflexology treatment.

A consultation with a reflexologist is recommended in order to determine the safety and appropriateness of reflexology therapy for a specific health problem or condition.

Side effects

Because reflexology is intended to normalize the body functions, the therapy does not cause a condition to worsen. Most patients find that pain diminishes

over the course of the therapy. It has been noted, however, that some patients experience greater discomfort in the second session than in the first session, because a significant easing of pain and tension is generally associated with the initial therapy session. As a result, when pressure is reapplied to the tender points of the foot during the second session, the sensitivity has been heightened. This increase in sensitivity may cause minor additional discomfort for the patient.

Research and general acceptance

Although only one controlled trial of reflexology therapy, done in 1993, has been documented in medical journals, this therapy is practiced worldwide at different levels of medical care. In Russia, for example, only licensed physicians may legally perform

reflexology treatment. In contrast, the practice is a commonplace homestyle remedy in the Netherlands. The Internet “Home of Reflexology” lists at least 66 professional organizations worldwide, including New Zealand and Malaysia. Associations include the following:

- Academy of Reflexology Austria
- Association of Finnish Reflexologists
- Chinese Society of Reflexologists
- Hellenic Association of Reflexologists
- Indian Society for Promotion of Reflexology
- International Council of Reflexologists (HQ: San Diego, USA)
- Israeli Reflexology Association
- New Zealand Reflexology Association
- Polish Instytut of Reflexology (Polish Language)
- Reflexology Association of America
- Reflexology Association of Australia
- Rwo-Shr Health Institute International (Malaysia)
- The South African Reflexology Society

Regulatory status

Ongoing legislative debate ensued during the 1990s regarding the legal status of the reflexology trade. The reflexology community, along with legislators and other bodywork practitioners, engaged in reassessment of the reflexology business and its relationship to **massage therapy** and massage parlors. Organizations and individuals brought judicial appeals of certain court cases that threatened the legitimate licensing of reflexologists as practitioners of alternative medicine. Such professional reflexology interests as the RAA documented in detail the disparities between reflexology and massage, citing the purpose of reflexology, which is to stimulate internal body functions (glands and organs) as opposed to the topical muscular and joint relief associated with massage. In a status update in 1998 the Association reported that 19 states had laws requiring the licensing of massage/reflexology therapists. Licensing laws established educational requirements and required candidates to pass written, oral, and/or practical examinations.

Also at issue was a trend among municipalities to license massage parlors (and reflexologists) under the business codes affecting the adult entertainment business. B. and K. Kunz reported that judicial decisions in two states—Tennessee and New Mexico—had excluded the practice of reflexology practice from the laws pertaining to massage parlors. Those courts held

KEY TERMS

Pressure points—Specific locations on the feet and hands that correspond to nerve endings that connect to the organs and glands of the human body via the spinal cord.

Zone therapy—Also called zone analgesia, a method of relieving pain by applying pressure to specific points on the body. It was developed in the early twentieth century by Dr. William Fitzgerald.

that reflexology is a business separate and distinct from massage parlors, and deserving of its own respective licensing standards. In Sacramento, California, reflexologists petitioned successfully to become licensed as practitioners of somatic therapy rather than as providers of adult entertainment. Likewise, in the Canadian province of Ontario, a nonprofit organization to register reflexology practitioners was established in order to define a distinct classification for therapists separate from erotic body rubbers, which was the original classification given to reflexologists. Other states where court proceedings or legislative attempts to legitimize reflexology have stalled include Pennsylvania, Florida, New Jersey, and New York

Training and certification

Training programs

Reflexology is taught by means of a series of seminars, classes, and training films. Certification is earned after a six month program that includes 200 hours of training. The certification training breaks down as follows: 28 hours of preliminary seminar training; 14 hours of advanced seminar training; 58 hours of self-directed study; and 100 hours of practical experience, including administering reflexology to a minimum of 15 people.

Specific aspects of the training include instruction in the assessment of the pressure points on the feet and hands through a study of human anatomy. Students also learn to give reflexology sessions to patients along with specific techniques for working with the hands.

Certification and advanced certification

As part of its function, the independently organized American Reflexology Certification Board (ARCB) certifies the competency of reflexology

practitioners on an individual basis. The ARCB does not evaluate schools and teachers. Prerequisites for individual certification include completion of educational requirements and passing a standard qualifying examination. Successful candidates receive the title of Board Certified Reflexologist.

Minimum qualifications to take the certification examination include attendance at an advanced seminar within two years prior to taking the examination. In addition, the applicant must have attended preliminary seminars for two full days—in addition to the required day of advanced seminar training—and the applicant is required to have a minimum of six months of practical experience in administering the therapy. Applicants are examined by means of both written tests and practical demonstrations.

Continuing education certification is available. Advanced training focuses on mastering the ability to perform hand reflexology. The therapist also receives instruction in new and advanced techniques of basic reflexology. Some reflexology training classes may be applied toward degree programs in other disciplines, depending on the specific course of study and the certification of the respective training institutions involved.

The RAA provides published standards of practice for reflexologists.

Resources

ORGANIZATIONS

International Institute of Reflexology. P.O. Box 12642. St. Petersburg, FL 33733-2642. (727) 343-4811. Fax: (727) 381-2807. E-mail: ftflex@concentric.net.
 Reflexology Association of America. 4012 Rainbow St. KPMB#585. Las Vegas, NV 89103-2059.

Gloria Cooksey

Refsum's syndrome see **Lipidoses**

Regional anesthetic see **Anesthesia, local**

Regional enteritis see **Crohn's disease**

Rehabilitation

Definition

Rehabilitation is a treatment or treatments designed to facilitate the process of recovery from injury, illness, or disease to as normal a condition as possible.

Purpose

The purpose of rehabilitation is to restore some or all of the patient's physical, sensory, and mental capabilities that were lost due to injury, illness, or disease. Rehabilitation includes assisting the patient to compensate for deficits that cannot be reversed medically. It is prescribed after many types of injury, illness, or disease, including amputations, arthritis, **cancer**, cardiac disease, neurological problems, orthopedic injuries, spinal cord injuries, **stroke**, and traumatic brain injuries. The Institute of Medicine has estimated that as many as 14% of all Americans may be disabled at any given time.

Precautions

Rehabilitation should be carried out only by qualified therapists. Exercises and other physical interventions must take into account the patient's deficit. An example of a deficit is the loss of a limb.

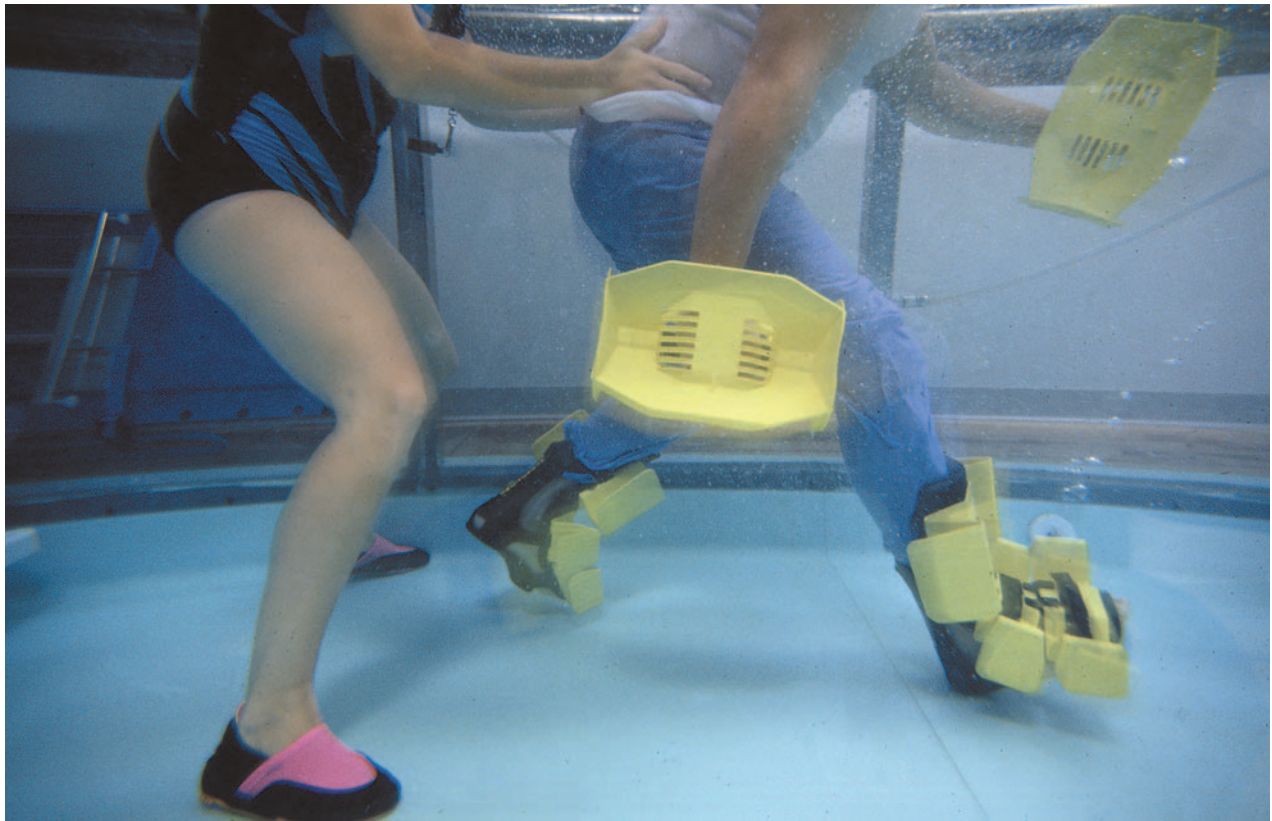
Description

A proper and adequate rehabilitation program can reverse many disabling conditions or can help patients cope with deficits that cannot be reversed by medical care. Rehabilitation addresses the patient's physical, psychological, and environmental needs. It is achieved by restoring the patient's physical functions and/or modifying the patient's physical and social environment. The main types of rehabilitation are physical, occupational, and speech therapy.

Each rehabilitation program is tailored to the individual patient's needs and can include one or more types of therapy. The patient's physician usually coordinates the efforts of the rehabilitation team, which can include physical, occupational, speech, or other therapists; nurses; engineers; physiatrists (physical medicine); psychologists; orthotists (makes devices such as braces to straighten out curved or poorly shaped bones); prosthetists (a therapist who makes artificial limbs or prostheses); and vocational counselors. Family members are often actively involved in the patient's rehabilitation program.

Physical therapy

Physical therapy helps the patient restore the use of muscles, bones, and the nervous system through the use of heat, cold, massage, whirlpool baths, ultrasound, **exercise**, and other techniques. It seeks to relieve **pain**, improve strength and mobility, and train the patient to perform important everyday tasks. Physical therapy may be prescribed to rehabilitate a



A patient (holding paddles) is undergoing a hydrotherapy treatment. (Photograph by Will & Deni McIntyre, Photo Researchers, Inc. Reproduced by permission.)

patient after amputations, arthritis, **burns**, cancer, cardiac disease, cervical and lumbar dysfunction, neurological problems, orthopedic injuries, pulmonary disease, spinal cord injuries, stroke, traumatic brain injuries, and other injuries/illnesses. The duration of the physical therapy program varies depending on the injury/illness being treated and the patient's response to therapy.

Exercise is the most widely used and best known type of physical therapy. Depending on the patient's condition, exercises may be performed by the patient alone or with the therapist's help, or with the therapist moving the patient's limbs. Exercise equipment for physical therapy could include an exercise table or mat, a stationary bicycle, walking aids, a wheelchair, practice stairs, parallel bars, and pulleys and weights.

Heat treatment, applied with hot-water compresses, infrared lamps, short-wave radiation, high frequency electrical current, ultrasound, paraffin wax, or warm baths, is used to stimulate the patient's circulation, relax muscles, and relieve pain. Cold treatment is applied with ice packs or cold-water soaking. Soaking in a whirlpool can ease muscle spasm pain

and help strengthen movements. Massage aids circulation, helps the patient relax, relieves pain and **muscle spasms**, and reduces swelling. Very low strength electrical currents applied through the skin stimulate muscles and make them contract, helping paralyzed or weakened muscles respond again.

Occupational therapy

Occupational therapy helps the patient regain the ability to do normal everyday tasks. This may be achieved by restoring old skills or teaching the patient new skills to adjust to disabilities through adaptive equipment, orthotics, and modification of the patient's home environment. Occupational therapy may be prescribed to rehabilitate a patient after **amputation**, arthritis, cancer, cardiac disease, head injuries, neurological injuries, orthopedic injuries, pulmonary disease, spinal cord disease, stroke, and other injuries/illnesses. The duration of the occupational therapy program varies depending on the injury/illness being treated and the patient's response to therapy.

Occupational therapy includes learning how to use devices to assist in walking (artificial limbs,

KEY TERMS

Orthotist—A health care professional who is skilled in making and fitting orthopedic appliances.

Physiatrist—A physician who specializes in physical medicine.

Prosthetist—A health care professional who is skilled in making and fitting artificial parts (prosthetics) for the human body.

canes, crutches, walkers), getting around without walking (wheelchairs or motorized scooters), or moving from one spot to another (boards, lifts, and bars). The therapist will visit the patient's home and analyze what the patient can and cannot do. Suggestions on modifications to the home, such as rearranging furniture or adding a wheelchair ramp, will be made. Health aids to bathing and grooming could also be recommended.

Speech therapy

Speech therapy helps the patient correct **speech disorders** or restore speech. Speech therapy may be prescribed to rehabilitate a patient after a brain injury, cancer, neuromuscular diseases, stroke, and other injuries/illnesses. The duration of the speech therapy program varies depending on the injury/illness being treated and the patient's response to therapy.

Performed by a speech pathologist, speech therapy involves regular meetings with the therapist in an individual or group setting and home exercises. To strengthen muscles, the patient might be asked to say words, smile, close his mouth, or stick out his tongue. Picture cards may be used to help the patient remember everyday objects and increase his vocabulary. The patient might use picture boards of everyday activities or objects to communicate with others. Workbooks might be used to help the patient recall the names of objects and practice reading, writing, and listening. Computer programs are available to help sharpen speech, reading, recall, and listening skills.

Other types of therapists

Inhalation therapists, audiologists, and registered dietitians are other types of therapists. Inhalation therapists help the patient learn to use respirators and other breathing aids to restore or support breathing. Audiologists help diagnose the patient's **hearing loss** and recommend solutions. Dietitians provide

dietary advice to help the patient recover from or avoid specific problems or diseases.

Rehabilitation centers

Rehabilitation services are provided in a variety of settings including clinical and office practices, hospitals, skilled-care nursing homes, sports medicine clinics, and some health maintenance organizations. Some therapists make home visits. Advice on choosing the appropriate type of therapy and therapist is provided by the patient's medical team.

Resources

ORGANIZATIONS

National Rehabilitation Association. 633 S. Washington St., Alexandria, VA 22314. (703) 836-0850.

National Rehabilitation Information Center. 8455 Colesville Road, Suite 935, Silver Spring, MD 20910. (800) 34-NARIC.

Rehabilitation International. 25 East 21st St., New York, NY 10010. (212) 420-1500.

Lori De Milto

Rehydration see **Intravenous rehydration**

Reiki

Definition

Reiki is a form of therapy that uses simple hands-on, no-touch, and visualization techniques, with the goal of improving the flow of life energy in a person. Reiki (pronounced *ray-key*) means “universal life energy” in Japanese, and Reiki practitioners are trained to detect and alleviate problems of energy flow on the physical, emotional, and spiritual level. Reiki touch therapy is used in much the same way to achieve similar effects that traditional **massage therapy** is used—to relieve **stress** and **pain**, and to improve the symptoms of various health conditions.

Purpose

Reiki claims to provide many of the same benefits as traditional massage therapy, such as reducing stress, stimulating the immune system, increasing energy, and relieving the pain and symptoms of health conditions. Practitioners have reported success in helping patients with acute and chronic illnesses, from **asthma** and arthritis to trauma and recovery from surgery. Reiki is a gentle and safe technique,

MIKAO USUI (1865–1926)

Mikao Usui, born in the Gifu Prefecture (Japan), was an ethereal child who sought to unravel the mysteries of the universe. As an adult he developed an interest in the metaphysical healing talent of Buddha. Usui became determined to regenerate the healing secrets of Buddha in order to improve the lot of humanity. He traveled to many temples and spoke with holy people, but all said that the secret of Buddha's powers were lost to the world due to lack of use.

Eventually the abbot of a Zen monastery encouraged Usui to study the ancient writings containing the secrets on healing. Usui learned two new languages, Chinese and Sanskrit, in order to understand the writings better, and from his reading he obtained the formula for healing. The Sutras in particular provided the enlightenment that he sought.

Usui next set out to obtain the power to heal. It is widely believed that he developed that ability after spending 21 days in retreat and in fasting on the holy Mountain of Kori-yama, where he had a vision of light and received the knowledge of the symbols of reiki and their use in healing. He officially formulated Usui Reiki therapy in 1922 and touted as many as one million followers during his lifetime.

Prior to the transition (death) of Usui, he imparted the secrets of healing to 16 teachers in order that the secrets would not be lost again.

and has been used successfully in some hospitals. It has been found to be very calming and reassuring for those suffering from severe or fatal conditions. Reiki can be used by doctors, nurses, psychologists and other health professionals to bring touch and deeper caring into their healing practices.

Description

Origins

Reiki was developed in the mid-1800s by Dr. **Mikao Usui**, a Japanese scholar of religion. According to the story that has been passed down among reiki teachers, Usui was a Christian who was intrigued by the idea that Christ could heal sick people by touching them with his hands. Searching for clues that would explain the secrets of healing with hands, Usui made a long pilgrimage around the world, visiting many ancient religious sects and studying ancient books. Some reiki teachers claim that Usui found clues

leading back nearly 10,000 years to healing arts that originated in ancient Tibet. During his intense studies, Usui claimed he had a spiritual experience, which enabled him to heal with his own hands by becoming aware of and tapping into the universal life force. After that, he dedicated his life to helping the sick and poor. His reputation grew as he healed sick people for many years in Kyoto, Japan. Before his death, Usui passed on his healing insights using universal life energy to Dr. Chujiru Hayashi, a close acquaintance. Hayashi, in turn, passed on the healing techniques in 1938 to Hawayo Takata, a Japanese woman from Hawaii, whom he had cured of life-threatening illness using reiki methods. Takata became a firm believer and proponent of reiki, and during the 1970s formed an initiation program for training reiki masters to preserve Usui's teachings. Before she died, she prepared her granddaughter, Phyllis Lei Furumoto, to continue the lineage. Takata had personally trained 21 practitioners before she died at the age of 80 in 1980. Along with other reiki masters authorized by Takata, Furumoto formed the reiki Alliance. A faction led by Barbara Ray, formed the American Reiki Association, which was known as Radiance Technique Association International. Today, there are over 1,000 reiki masters practicing around the world, whose methods can all be traced back directly to Dr. Usui.

The basic philosophy of reiki

The basic concept underlying reiki is that the body has an energy field that is central to its health and proper functioning, and this energy travels in certain pathways that can become blocked or weakened. This idea of energy flow in the body is also a central concept in **Ayurvedic medicine** and **traditional Chinese medicine**, including **acupuncture**.

Reiki practitioners believe that everyone has the potential to access the universal life energy, but that over time most people's systems become blocked and the energy becomes weakened in them. A reiki practitioner is trained to be able to detect these blockages, and practitioners will use their hands, thoughts, and own energy fields to improve the energy flow in a patient. Reiki is one of the more esoteric alternative medical practices, because no one is sure exactly how it works on the physiological level. Practitioners claim that it works on very subtle energy levels, or possibly works on the *chakra* system. The chakras are the system of seven energy centers along the middle of the body believed to be connected with the nervous and endocrine systems, as defined by **yoga** and Ayurvedic medicine. Reiki masters claim that healing

energy can even be sent to a person from far away, noting that reiki works on the same principles that enables praying to work for some patients, although a practitioner needs advanced training to be able to send energy from afar.

According to the original principles of Usui, patients must also have a proper attitude for reiki to work most effectively. Patients must take responsibility for their own health, and must want to be healed. Furthermore, when energy is received from a reiki healer, patients must be willing to give back energy to others, and to compensate the healer in some way, as well. Finally, Usui claimed that a healing attitude was free from worry and fear, was filled with gratitude for life and for others, and placed emphasis on each person finding honest and meaningful work in their lives—all this, in order to complete the picture of overall health.

A reiki session

Reiki sessions can take various forms, but most commonly resemble typical bodywork appointments, where the receiver lies clothed on his or her back on a flat surface or massage table. A session generally lasts from an hour to an hour and a half. Reiki is a simple procedure, consisting of calm and concentrated touching, with the practitioner focusing on healing and giving energy to specific areas on the receiver's body. Practitioners place their hands over positions on the body where the organs and endocrine glands reside, and the areas that correspond to the chakra centers. Practitioners also use mental visualization to send healing energy to areas of the receiver's body that need it. In special cases or with injuries, a no-touch technique is used, where the practitioner's hands are sometimes held just above the body without touching it. Advanced practitioners rely on intuition and experience to determine which areas of a body need the most energy healing.

The practitioner's hands are held flat against the receiver's body, with the fingertips touching. There can be over 20 positions on both sides of the body where the hands are placed. The positions begin at the crown of the head and move towards the feet. The receiver usually turns over once during the session. The practitioner's hands are held in each position for a usually five minutes, to allow the transfer of energy and the healing process to take place. In each position, the hands are kept stationary, unlike typical massage where the hands move, and both the giver and receiver attempt to maintain an attitude of awareness, openness, and caring.

Reiki practitioners recommend that those receiving reiki for the first time go through a series of three to four initial treatments over the course of about a week, to allow for cleansing and the initial readjustment of energy. Reiki sessions can cost from \$30–100 per session. Insurance coverage is rare, and consumers should consult their individual policies as to whether or not such therapies are included.

Self-treatment with reiki

Although reiki practitioners believe that formal training is necessary to learn the proper methods of energy channeling and healing, individuals can still use some of the basic positions of reiki to relieve stress and to stimulate healing on themselves or another. The positions can be performed anywhere and for however long they are needed. Positions generally move from the top of the body down, but positions can be used wherever there is pain or stress. Mental attitude is important during reiki; the mind should be cleared of all stressful thoughts and concentrated on compassion, love, and peace as forms of energy that are surrounding, entering, and healing the body.

The following positions are illustrated in *Reiki: Energy Medicine*:

- Position one: Hands are placed on the top of the head, with the wrists near the ears and the fingertips touching on the crown of the head. Eyes should be closed. Hold for five minutes or more, until the mind feels clear and calm.
- Position two: Cup the hands slightly and place the palms over the closed eyes, with the fingers resting on the forehead.
- Position three: Place the hands on the sides of the head, with the thumbs behind the ear and the palms over the lower jaws, with the fingers covering the temples.
- Position four: Place one hand on the back of the neck, at the base of the skull, and put the other hand on the head just above it, parallel to it.
- Position five: Wrap the hands around the front of the throat, and rest them there gently with the heels of the hands touching in front.
- Position six: Place each hand on top of a shoulder, close to the side of neck, on top of the trapezius muscle.
- Position seven: Form a T-shape with the hands over the chest, with the left hand covering the heart and the right hand above it, covering the upper part of the chest.

KEY TERMS

Attunement—Life energy teaching given by Reiki master to a student.

Chakra—One of seven major energy centers in the body, as defined by Hindu and yoga philosophy.

Relaxation response—The human body's response to relaxation techniques, during which metabolism and stress levels decrease and immune response increases.

- **Position eight:** The hands are placed flat against the front of the body with fingertips touching. Hold for five minutes or so, and repeat four or five times, moving down a hand-width each time until the pelvic region is reached, which is covered with a v-shape of the hands. Then, for the final position, repeat this technique on the back, beginning as close to the shoulders as the hands can reach, and ending by forming a T-shape with the hands at the base of the spine.

Side effects

Reiki generally has no side effects, as it is a very low impact and gentle procedure. Some receivers report **tingling** or sensations of heat or cold during treatment. Others have reported sadness or **anxiety** during treatment, which practitioners claim are buried or repressed emotions being released by the new energy flow.

Research and general acceptance

Reiki has been used in major clinics and hospitals as part of alternative healing practice, and doctors, dentists, nurses and other health professionals have been trained to use its gentle touch techniques as part of their practice. To date, the little scientific research that has been conducted with reiki implies that its techniques bring about the *relaxation response*, in which stress levels decrease, and immune response increases. Reiki practitioners claim that the most important measurement of their technique is whether the individual feels better after treatment. They also claim that science cannot measure the subtle energy changes that they are attempting to bring about.

As of the early 2000s, there are differences of opinion within the mainstream medical community regarding the acceptability of reiki. On the one hand,

medical professionals in Canada have proposed strategies to limit the popularity of reiki as well as several other alternative therapies by resisting the integration of these therapies with mainstream treatments and by opposing government research in complementary and alternative medicine. On the other hand, the U. S. National Center for Complementary and Alternative Medicine (NCCAM) is conducting a series of clinical trials to evaluate the efficacy of reiki. As of the summer of 2004, there are four NCCAM trials for reiki, measuring its effectiveness in treating such disorders as **fibromyalgia**, neuropathy, **prostate cancer**, and advanced **AIDS**.

Training and certification

Reiki practitioners undergo a series of *attunements*, which are sessions with reiki masters that teach the basic methods of energy healing. Several organizations provide resources for reiki training. Reiki practitioners believe these attunements are necessary for correct technique. The masters teach each person how to activate the universal life energy in themselves before they can pass it on to others. These initiations often are held during weekend workshops. Trainees can achieve up to four levels of attunements, until they reach the level of master themselves. The certification process is not a formal one; masters approve students when they feel satisfied with their progress.

Resources

PERIODICALS

Hallett, A. "Narratives of Therapeutic Touch." *Nursing Standard* 19 (September 15, 2004): 33–37.

Kelner, M., B. Wellman, H. Boon, and S. Welch. "Responses of Established Healthcare to the Professionalization of Complementary and Alternative Medicine in Ontario." *Social Science and Medicine* 59 (September 2004): 915–930.

ORGANIZATIONS

International Association of Reiki Professionals. P.O. Box 481, Winchester, MA 01890. <<http://www.iarp.org>>.

International Center for Reiki Training. 21421 Hilltop Street, Unit #28, Southfield, MI 48034. (800) 332-8112 or (248) 948-8112. Fax: (248) 948-9534. <<http://www.reiki.org>>.

National Center for Complementary and Alternative Medicine (NCCAM) Clearinghouse. P. O. Box 7923. Gaithersburg, MD 20898. (888) 644-6226. Fax: (866) 464-3616. <<http://nccam.nih.gov>>.

OTHER

American Reiki Masters Association (ARMA). PO Box 130, Lake City, FL 32056-0130. (904) 755-9638.

Global Reiki Healing Network. <<http://www.reiki.org>>.
 NCCAM Reiki Clinical Trials. <<http://nccam.nih.gov/clinicaltrials/reiki.htm>>.
 Reiki Alliance. P.O. Box 41, Cataldo, ID 83810-1041, phone
 (208) 682-3535.

Douglas Dupler, MA
 Rebecca J. Frey, PhD

Reiter's syndrome

Definition

Reiter's syndrome (RS), which is also known as arthritis urethritica, venereal arthritis, reactive arthritis, and polyarteritis enterica, is a form of arthritis that affects the eyes, urethra, and skin, as well as the joints. It was first described by Hans Reiter, a German physician, during World War I.

Description

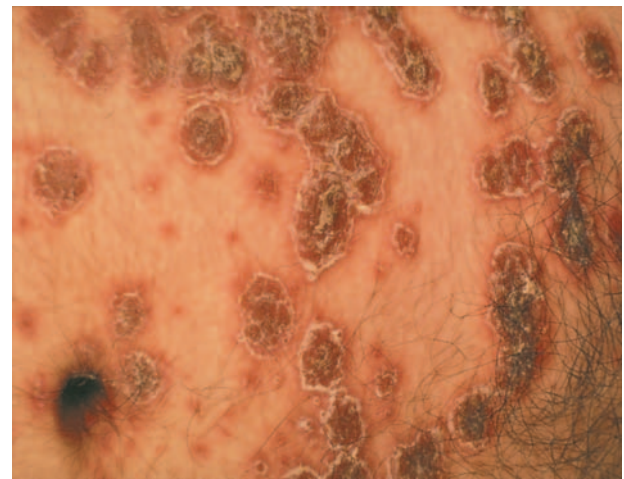
Reiter's syndrome is marked by a cluster of symptoms in different organ systems of the body that may or may not appear simultaneously. The disease may be acute or chronic, with spontaneous remissions or recurrences.

RS primarily affects sexually active males between ages 20–40, particularly males who are HIV positive. Most women and children who develop RS acquire the disease in its intestinal form.

Causes and symptoms

The cause of Reiter's syndrome was unknown as of early 1998, but scientists think the disease results from a combination of genetic vulnerability and various disease agents. Over 80% of Caucasian patients and 50–60% of African Americans test positive for HLA-B27, which suggests that the disease has a genetic component. In sexually active males, most cases of RS follow infection with *Chlamydia trachomatis* or *Ureaplasma urealyticum*. Other patients develop the symptoms following gastrointestinal infection with *Shigella*, *Salmonella*, *Yersinia*, or *Campylobacter* bacteria.

The initial symptoms of RS are inflammation either of the urethra or the intestines, followed by acute arthritis four to 28 days later. The arthritis usually affects the fingers, toes, and weight-bearing joints in the legs. Other symptoms include:



Keratoderma, a skin condition characterized by horny patches, is one symptom of Reiter's syndrome. (Photograph by Milton Reisch, M.D., Corbis Images. Reproduced by permission.)

- inflammation of the urethra, with painful urination and a discharge from the penis
- mouth ulcers
- inflammation of the eye
- keratoderma blennorrhagica, these are patches of scaly skin on the palms, soles, trunk, or scalp of RS patients

Diagnosis

Patient history

Diagnosis of Reiter's syndrome can be complicated by the fact that different symptoms often occur several weeks apart. The patient does not usually draw a connection between the arthritis and previous sexual activity. The doctor is likely to consider Reiter's syndrome when the patient's arthritis occurs together with or shortly following inflammation of the eye and the genitourinary tract lasting a month or longer.

Laboratory tests

There is no specific test for diagnosing RS, but the physician may have the urethral discharge cultured to rule out **gonorrhea**. Blood tests of RS patients are typically positive for the HLA-B27 genetic marker, with an elevated white blood cell (WBC) count and an increased sedimentation rate of red blood cells. The patient may also be mildly anemic.

KEY TERMS

Acute—Having a sudden onset and lasting a short time.

Chronic—Of long duration.

Keratoderma blennorrhagica—The medical name for the patches of scaly skin that occur on the arms, legs, and trunk of RS patients.

Reactive arthritis—Another name for Reiter's syndrome.

Diagnostic imaging

X rays do not usually reveal any abnormalities unless the patient has had recurrent episodes of the disease. Joints that have been repeatedly inflamed may show eroded areas, signs of **osteoporosis**, or bony spurs when x rayed.

Treatment

There is no specific treatment for RS. Joint inflammation is usually treated with **nonsteroidal anti-inflammatory drugs** (NSAIDs.) Skin eruptions and eye inflammation can be treated with **corticosteroids**. Gold treatments may be given for eroded bone.

Patients with chronic arthritis are also given physical therapy and advised to **exercise** regularly.

Prognosis

The prognosis varies. Most patients recover in three to four months, but about 50% have recurrences for several years. Some patients develop complications that include inflammation of the heart muscle, stiffening inflammation of the vertebrae, **glaucoma**, eventual blindness, deformities of the feet, or accumulation of fluid in the lungs.

Prevention

In males, Reiter's syndrome can be prevented by sexual abstinence or the use of condoms.

Resources

BOOKS

Berktoiw, Robert, editor. "Musculoskeletal and Connective Tissue Disorders: Reiter's Syndrome (RS)." In *The Merck Manual of Diagnosis and Therapy*. 16th ed. Rahway, NJ: Merck Research Laboratories, 1992.

Hellman, David B. "Arthritis & Musculoskeletal Disorders." In *Current Medical Diagnosis and Treatment*, 1998, edited by Stephen McPhee, et al., 37th ed. Stamford: Appleton & Lange, 1997.

Rebecca J. Frey, PhD

Relapsing fever

Definition

Relapsing **fever** refers to two similar illnesses, both of which cause high fevers. The fevers resolve, only to recur again within about a week.

Description

Relapsing fever is caused by spiral-shaped bacteria of the genus *Borrelia*. This bacterium lives in rodents and in insects, specifically ticks and body lice. The form of relapsing fever acquired from ticks is slightly different from that acquired from body lice.

In tick-borne relapsing fever (TBRF), rodents (rats, mice, chipmunks, and squirrels) which carry *Borrelia* are fed upon by ticks. The ticks then acquire the bacteria, and are able to pass it on to humans. TBRF is most common in sub-Saharan Africa, parts of the Mediterranean, areas in the Middle East, India, China, and the south of Russia. Also, *Borrelia* causing TBRF exist in the western regions of the United States, particularly in mountainous areas. The disease is said to be endemic to these areas, meaning that the causative agents occur naturally and consistently within these locations.

In louse-borne relapsing fever (LBRF), lice acquire *Borrelia* from humans who are already infected. These lice can then go on to infect other humans. LBRF is said to be epidemic, as opposed to endemic, meaning that it can occur suddenly in large numbers in specific communities of people. LBRF occurs in places where poverty and overcrowding predispose to human infestation with lice. LBRF has flared during wars, when conditions are crowded and good hygiene is next to impossible. At this time, LBRF is found in areas of east and central Africa, China, and in the Andes Mountains of Peru.

Causes and symptoms

In TBRF, humans contract *Borrelia* when they are fed upon by ticks. Ticks often feed on humans at night,

so many people who have been bitten are unaware that they have been. The bacteria is passed on to humans through the infected body fluids of the tick.

In LBRF, a louse must be crushed or smashed in order for *Borrelia* to be released. The bacteria then enter the human body through areas where the person may have scratched him or herself.

Both types of relapsing fever occur some days after having acquired the bacteria. About a week after becoming infected, symptoms begin. The patient spikes a very high fever, with chills, sweating, terrible **headache**, **nausea**, **vomiting**, severe **pain** in the muscles and joints, and extreme weakness. The patient may become dizzy and confused. The eyes may be blood-shot and very sensitive to light. A **cough** may develop. The heart rate is greatly increased, and the liver and spleen may be swollen. Because the substances responsible for blood clotting may be disturbed during the illness, tiny purple marks may appear on the skin, which are evidence of minor bleeding occurring under the skin. The patient may suffer from a **nose-bleed**, or may cough up bloody sputum. All of these symptoms last for about three days in TBRF, and about five days in LBRF.

With or without treatment, a crisis may occur as the bacteria are cleared from the blood. This crisis, called a Jarisch-Herxheimer reaction, results in a new spike in fever, chills, and an initial rise in blood pressure. The blood pressure then falls drastically, which may deprive tissues and organs of appropriate blood flow (**shock**). This reaction usually lasts for about a day.

Recurrent episodes of fever with less severe symptoms occur after about a week. In untreated infections, fevers recur about three times in TBRF, and only once or twice in LBRF.

Diagnosis

Diagnosis of relapsing fever is relatively easy, because the causative bacteria can be found by examining a sample of blood under the microscope. The characteristically spiral-shaped bacteria are easily identifiable. The blood is best drawn during the period of high fever, because the bacteria are present in the blood in great numbers at that time.

Treatment

Either tetracycline or erythromycin is effective against both forms of relapsing fever. The medications are given for about a week for cases of TBRF; LBRF requires only a single dose. Children and pregnant women should receive either erythromycin or penicillin.

KEY TERMS

Endemic—Refers to a particular organism which consistently exists in a particular location under normal conditions.

Epidemic—Refers to a condition suddenly acquired by a large number of people within a specific community, and which spreads rapidly throughout that community.

Shock—A state in which the blood pressure is so low that organs and tissues are not receiving an appropriate flow of blood.

Because of the risk of the Jarish-Herxheimer reaction, patients must be very carefully monitored during the initial administration of antibiotic medications. Solutions containing salts must be given through a needle in the vein (intravenously) to keep the blood pressure from dropping too drastically. Patients with extreme reactions may need medications to improve blood circulation until the reaction resolves.

Prognosis

In epidemics of LBRF, **death** rates among untreated victims have run as high as 30%. With treatment, and careful monitoring for the development of the Jarish-Herxheimer reaction, prognosis is good for both LBRF and TBRF.

Prevention

Prevention of TBRF requires rodent control, especially in and near homes. Careful use of insecticides on skin and clothing is important for people who may be enjoying outdoor recreation in areas known to harbor the disease-carrying ticks.

Prevention of LBRF is possible, but probably more difficult. Good hygiene and decent living conditions would prevent the spread of LBRF, but these may be difficult for those people most at risk for the disease.

Resources

ORGANIZATIONS

Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

Rosalyn Carson-DeWitt, MD

Relapsing polychondritis

Definition

Relapsing polychondritis is a disease characterized by autoimmune-like episodic or progressive inflammation of cartilage and other connective tissue, such as the nose, ears, throat, joints, kidneys, and heart.

Description

Cartilage is a tough, flexible tissue that turns into bone in many places in the body. Bones all start out as cartilage in the fetus. Consequently, children have more cartilage than adults. Cartilage persists in adults in the linings of joints, the ears, the nose, the airway and the ribs near the breast bone. All these sites are attacked by relapsing polychondritis, which usually occurs equally in middle-aged males and females. It is frequently diagnosed along with **rheumatoid arthritis**, **systemic lupus erythematosus**, and other connective tissue diseases.

Causes and symptoms

The most common first symptom of relapsing polychondritis is **pain** and swelling of the external ear. Usually, both ears turn red or purple and are tender to the touch. The swelling can extend into the ear canal and beyond, causing ear infections, **hearing loss**, balance disturbances with vertigo and **vomiting**, and eventually a droopy ear. The nose is often afflicted as well and can deteriorate into a flattened nose bridge called saddle nose. Inflammation of the eye occurs less frequently, but can lead to blindness.

As relapsing polychondritis advances, it causes more dangerous symptoms such as deterioration of the cartilage that holds the windpipe open. Progressive disease can destroy the integrity of the airway and compromise breathing. Destruction of the rib cartilage can collapse the chest, again hindering breathing. Joints everywhere are involved in episodes of arthritis, with pain and swelling. Other tissues besides cartilage are also involved, leading to a variety of problems with the skin and other tissues. Occasionally, the aorta or heart valves are damaged.

The disease may occur in episodes with complete remission between, or it may smolder along for years, causing progressive destruction.

KEY TERMS

Aorta—The biggest artery in the body, receiving blood directly from the heart.

Connective tissue—Several types of tissue that hold the body's parts together—tendons, ligaments, fascia, and cartilage.

Inflammation—The body's immune reaction to presumed foreign substances like germs. Inflammation is characterized by increased blood supply and activation of defense mechanisms. It produces redness, swelling, heat, and pain.

Diagnosis

A characteristic array of symptoms and physical findings will yield a diagnosis of relapsing polychondritis. Laboratory tests are sometime helpful. Biopsies of the affected cartilage may confirm the diagnosis. Further diagnostic tests are done to confirm other associated conditions such as rheumatoid arthritis. It is important to evaluate the airway, although only 10% of patients will die from airway complications.

Treatment

Mild inflammations can be treated with **aspirin** or **nonsteroidal anti-inflammatory drugs** (NSAIDs) such as ibuprofen. **Corticosteroids** (most often prednisone) are usually prescribed for more advanced conditions and do improve the disease. They may have to be continued over long periods of time, in which case their usage must be closely watched to avoid complications. Immune suppression with cyclophosphamide, azathioprine, cyclosporine, or dapsone is reserved for more aggressive cases. A collapsed chest or airway may require surgical support, and a heart valve or aorta may need repair or replacing.

Prognosis

There is no known cure for relapsing polychondritis. It can only be combated with each onset of inflammation and deterioration of cartilaginous tissue. As the disease progresses over a period of years, the mortality rate increases. At five years duration, relapsing polychondritis has a 30% mortality rate.

Resources

BOOKS

Gilliland, Bruce C. "Relapsing Polychondritis and Other Arthritides." In *Harrison's Principles of Internal Medicine*, edited by Anthony S. Fauci, et al. New York: McGraw-Hill, 1997.

J. Ricker Polsdorfer, MD

Renal artery occlusion

Definition

Renal artery occlusion is a blockage of the major arteries that supply blood to the kidneys caused by thrombosis or **embolism**.

Description

Renal artery occlusion occurs when the flow of blood from the arteries leading to the kidneys becomes blocked by a blood clot or cholesterol emboli. The lack of oxygenation can lead to necrosis (tissue death) and ultimately, **chronic kidney failure**.

Causes and symptoms

Renal arterial occlusion occurs when a thrombus or embolism (blood clot or cholesterol plaque) breaks free and blocks the arteries leading to one or both kidneys.

Symptoms of an acute renal arterial occlusion may include:

- **hypertension**
- fever
- sudden **pain** in the lower back or flank
- nausea and vomiting
- protein and/or blood in the urine

An individual with renal arterial occlusion may have no overt symptoms, particularly if only one kidney is affected or if the blockage is only partial. Health problems from secondary complications such as chronic kidney failure may be the first indication that something is wrong.

Diagnosis

The high blood pressure that is sometimes associated with a renal artery blockage may be the first sign

KEY TERMS

Angioplasty—A non-surgical procedure that uses a balloon-tipped catheter to open a blocked artery.

Artherosclerotic plaque—A deposit of fatty and calcium substances that accumulate in the lining of the artery wall, restricting blood flow.

Atrophy—Cell or tissue wasting or death.

Chronic kidney failure—End-stage renal disease (ESRD); chronic kidney failure is diagnosed as ESRD when kidney function falls to 5–10% of capacity.

Embolism—Blood vessel obstruction by a blood clot or other substance (i.e., air).

Thrombus—Formation of a blood clot within the vascular system. A thrombus becomes an embolism if it breaks away and blocks a blood vessel.

that it is present, particularly if the hypertension is not responding to standard treatment. Urine and blood tests may or may not be useful in diagnosing this condition. Blood tests may show an elevated plasma creatinine level. If kidney tissue infarction (cell death caused by a lack of blood supply) has occurred, lactic dehydrogenase (LDH) may also be present in the urine and blood.

An arteriogram, an x-ray study of the arteries that uses a radiopaque substance, or dye, to make the arteries visible under x ray, may also be performed. This test is used with caution in patients with impaired kidney function, as the contrast medium can cause further kidney damage. In patients with whom this is not an issue, a spiral computed tomography (CT) scan with contrast medium may also be used.

Treatment

Occlusions may be treated with anticoagulant (blood thinning) or thrombolytic (clot destroying) drugs. If the blockage is significant, surgical intervention or **angioplasty** may be required. Between 1996 and 2000, the number of these procedures performed on Medicare patients more than doubled, said a 2004 report.

Alternative treatment

Renal arterial occlusion is a serious and potentially life-threatening condition, and should always be treated by a healthcare professional familiar with the disorder.

Prognosis

The outcome of renal arterial occlusion depends on the speed with which it is treated. Once the blood supply is minimized or cut off to the kidney, tissue death soon results, ultimately leading to chronic kidney failure (end-stage renal disease).

Prevention

Atherosclerosis may encourage the formation of cholesterol emboli, a potential cause of renal artery occlusion. Strategies for avoiding vascular disease include eating right, maintaining a desirable weight, quitting **smoking**, managing **stress**, and exercising regularly. People prone to emboli from **blood clots** can take blood thinning drugs to prevent potential emboli from lodging in the renal artery.

Resources

PERIODICALS

Bloch, M. J., and T. Pickering. "Renal Vascular Disease: Medical Management, Angioplasty, and Stenting." *Seminars in Nephrology* 20, no. 5 (September 2000): 474-88.

"Explosive Growth Seen in Renal Artery Interventional Procedures." *Heart Disease Weekly* September 26, 2004: 20.

Truelove, Christiane. "First for Pulmonary Embolism." *Med Ad News* August 2004: 82.

ORGANIZATIONS

American Kidney Fund (AKF). Suite 1010, 6110 Executive Boulevard, Rockville, MD 20852. (800) 638-8299. <<http://www.arbon.com/kidney/>>.

National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK). Natcher Building, 6AS-13K, 45 Center Drive, Bethesda, MD 20892-6600. <<http://www.niddk.nih.gov>>.

National Kidney Foundation. 30 East 33rd Street, New York, NY 10016. (800)622-9020. <<http://www.kidney.org>>.

Paula Anne Ford-Martin
Teresa G. Odle

Renal artery stenosis

Definition

Renal artery stenosis is a blockage or narrowing of the major arteries that supply blood to the kidneys.

Description

Renal artery stenosis occurs when the flow of blood from the arteries leading to the kidneys is constricted by tissue or atherosclerotic plaque. This narrowing of the arteries diminishes the blood supply to the kidneys, which can cause them to atrophy and may ultimately lead to kidney failure. It may also cause **renovascular hypertension**, or high blood pressure related to renal artery blockage.

Causes and symptoms

The two main causes of renal artery stenosis are **atherosclerosis** and fibromuscular disease. Fibromuscular diseases such as fibromuscular dysplasia cause growth of fibrous tissues on the arterial wall. Stenosis may also occur when scar tissue forms in the renal artery after trauma to the kidney.

Renal arterial stenosis has no overt symptoms. Eventually, untreated renal arterial stenosis causes secondary complications such as **chronic kidney failure**, which may be characterized by frequent urination, anemia, **edema**, headaches, **hypertension**, lower back **pain**, and other signs and symptoms.

Diagnosis

The high blood pressure that is sometimes associated with renal artery stenosis may be the first sign that it is present, particularly if the hypertension is not responding to standard treatment. Presence of a *bruit*, a swooshing sound from the artery that indicates an obstruction, may be heard through a stethoscope.

An arteriogram, an x-ray study of the arteries that uses a radiopaque substance, or dye, to make the arteries visible under x ray, may also be performed. This test is used with caution in patients with impaired kidney function, as the contrast medium may cause further kidney damage.

Treatment

Treatment for renal artery stenosis is either surgical, pharmaceutical, or with **angioplasty** or stenting. Angioplasty involves guiding a balloon catheter down into the renal artery and inflating the balloon to clear the blockage. A stent may be inserted into the artery to widen the opening. Some patients may be candidates for surgical revascularization, which involves restoring blood flow with an arterial bypass. Drugs known as angiotension-converting enzyme (ACE) inhibitors may be prescribed for some patients. The chosen treatment approach depends on the cause of the stenosis.

KEY TERMS

Artherosclerotic plaque—A deposit of fatty and calcium substances that accumulate in the lining of the artery wall, restricting blood flow.

Atrophy—Cell or tissue wasting or death.

Chronic kidney failure—End-stage renal disease (ESRD); chronic kidney failure is diagnosed as ESRD when kidney function falls to 5–10% of capacity.

Edema—Swelling which occurs when body tissues retain fluid.

Stent—An expandable “scaffold-like” device, usually constructed of a stainless steel material, that is inserted into an artery to expand the inside passage and improve blood flow.

and factors such as the patient’s kidney function and blood pressure control.

Alternative treatment

Renal artery stenosis is a serious and potentially life-threatening condition, and should always be treated by a healthcare professional familiar with the disorder.

Prognosis

Untreated renal artery stenosis can cause hypertension (high blood pressure) and may ultimately lead to chronic kidney failure (end-stage renal disease).

Prevention

Maintaining a heart healthy lifestyle can help to prevent cases of renal arterial stenosis attributable to atherosclerosis. Strategies for avoiding vascular disease include eating right, maintaining a desirable weight, quitting **smoking**, managing **stress**, and exercising regularly.

Resources

PERIODICALS

- Bloch, M. J., and T. Pickering. “Renal Vascular Disease: Medical Management, Angioplasty, and Stenting.” *Seminars in Nephrology* 20, no. 5 (September 2000): 474-88.
- Fenves, A. Z., and C. V. Ram. “Fibromuscular Dysplasia of the Renal Arteries.” *Current Hypertension Reports* 1, no. 6 (December 1999): 546-9.

ORGANIZATIONS

- American Kidney Fund (AKF). Suite 1010, 6110 Executive Boulevard, Rockville, MD 20852. (800)638-8299. <<http://www.arbon.com/kidney/>>.
- National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK). Natcher Building, 6AS-13K, 45 Center Drive, Bethesda, MD 20892-6600. <<http://www.niddk.nih.gov>>.
- National Kidney Foundation. 30 East 33rd Street, New York, NY 10016. (800)622-9020. <<http://www.kidney.org>>.

Paula Anne Ford-Martin

Renal calculi see **Kidney stones**

Renal failure see **Acute kidney failure;**
Chronic kidney failure

Renal nuclear medicine scan see **Kidney nuclear medicine scan**

Renal tubular acidosis

Definition

Renal tubular acidosis (RTA) is a condition characterized by too much acid in the body due to a defect in kidney function.

Description

Chemical balance is critical to the body’s functioning. Therefore, the body controls its chemicals very strictly. The acid-base balance must be between a pH of 7.35 and 7.45 or trouble will start. Every other chemical in the body is affected by the acid-base balance. The most important chemicals in this system are sodium, chloride, potassium, calcium, ammonium, carbon dioxide, oxygen, and phosphates.

The lungs rapidly adjust acid-base balance by the speed of breathing, because carbon dioxide dissolved in water is an acid-carbonic acid. Faster breathing eliminates more carbon dioxide, decreases the carbonic acid in the blood and increases the pH. Holding your breath does the opposite. Blood acidity from carbon dioxide controls the rate of breathing, not oxygen.

The kidneys also regulate acid-base balance somewhat more slowly than the lungs. They handle all the chemicals, often trading one for another that is more or less acidic. The trading takes place between the blood and the urine, so that extra chemicals end up passing out of the body. If the kidneys do not

effectively eliminate acid, it builds up in the blood, leading to a condition called **metabolic acidosis**. These conditions are called renal tubular acidosis.

Causes and symptoms

There are three types of renal tubular acidosis. They include:

- Distal renal tubular acidosis (type 1) may be a hereditary condition or may be triggered by an autoimmune disease, lithium therapy, **kidney transplantation**, or chronic obstruction.
- Proximal renal tubular acidosis (type 2) is caused by hereditary diseases, such as **Fanconi's syndrome**, fructose intolerance, and Lowe's syndrome. It can also develop with **vitamin D deficiency**, kidney transplantation, **heavy metal poisoning**, and treatment with certain drugs.
- Type 4 renal tubular acidosis is not hereditary, but is associated with **diabetes mellitus**, sickle cell anemia, an autoimmune disease, or an obstructed urinary tract.

Symptoms vary with the underlying mechanism of the defect and the readjustment of chemicals required to compensate for the defect.

- Distal RTA results in high blood acidity and low blood potassium levels. Symptoms include mild **dehydration**; muscle weakness or **paralysis** (due to potassium deficiency); **kidney stones** (due to excess calcium in the urine); and bone fragility and pain.
- Proximal RTA also results in high blood acidity and low blood potassium levels. Symptoms include mild dehydration.
- Type 4 RTA is characterized by high blood acidity and high blood potassium levels; it rarely causes symptoms unless potassium levels rise so high as to cause heart **arrhythmias** or muscle paralysis.

Diagnosis

RTA is suspected when a person has certain symptoms indicative of the disease or when routine tests show high blood acid levels and low blood potassium levels. From there, more testing of blood and urine chemicals will help determine the type of RTA present.

Treatment

The foundation of treatment for RTA types 1 and 2 is replacement of alkali (base) by drinking a bicarbonate solution daily. Potassium may also have to be replaced, and other chemicals added to maintain balance. In type 4 RTA acidity will normalize if

KEY TERMS

Autoimmune disease—Type of diseases characterized by antibodies that attack the body's own tissues.

Fanconi's syndrome—A disorder of the kidneys characterized by glucose in the urine.

Lowe's syndrome—A rare inherited disorder that is distinguished by congenital cataracts, glaucoma, and severe mental retardation.

Rickets—A deficiency disease that affects the bone development of growing bodies, usually causing soft bones.

potassium is reduced. This is done by changing the diet and by using diuretic medicines that promote potassium excretion in the urine.

Prognosis

Careful balancing of body chemicals will usually produce good results. If there is an underlying disease responsible for the kidney malfunction, it may be the determining factor in the prognosis.

Prevention

Relatives of patients with the possibly hereditary forms of renal tubular acidosis should be tested.

Resources

BOOKS

Chesney, Russell W. "Specific Renal Tubular Disorders." In *Cecil Textbook of Medicine*, edited by J. Claude Bennett and Fred Plum. Philadelphia: W. B. Saunders Co., 1996.

J. Ricker Polsdorfer, MD

Renal ultrasound see **Abdominal ultrasound**

Renal vein thrombosis

Definition

Renal vein thrombosis develops when a blood clot forms in the renal vein, which carries blood from the kidneys back to the heart. The disorder is not common.

Description

Renal vein thrombosis occurs in both infants and adults. Onset of the disorder can be rapid (acute) or gradual. The number of people who suffer from renal vein thrombosis is difficult to determine, as many people do not show symptoms, and the disorder is diagnosed only by specific tests. Ninety percent of childhood cases occur in children under one year old, and 75% occur in infants under one month of age. In adult women, oral contraceptive use increases the risk of renal vein thrombosis.

Causes and symptoms

In children, renal vein thrombosis almost always occurs rapidly after an episode of severe **dehydration**. Severe dehydration decreases blood volume and causes the blood to clot more readily.

In adults, renal vein thrombosis can be caused by injury to the abdomen or back, as a result of malignant kidney tumors growing into the renal vein, or as a result of kidney diseases that cause degenerative changes in the cells of the renal tubules (**nephrotic syndrome**).

Acute onset of renal vein thrombosis at any age causes **pain** in the lower back and side, **fever**, bloody urine, decreased urine output, and sometimes kidney failure. In adults, when the onset of the disorder is gradual, there is a slow decrease in kidney function, and protein appears in the urine. Many adults with renal vein thrombosis show few symptoms.

Diagnosis

Renal **venography**, where a contrast material (dye) is injected into the renal vein before x rays are taken, is one of the best ways to detect renal vein thrombosis. Other useful tests to detect a clot include **computed tomography scans** (CT scans), **magnetic resonance imaging** (MRI), and ultrasound.

Treatment

One of the major goals of treatment is to prevent the blood clot in the renal vein from detaching and moving into the lungs, where it can cause serious complications as a **pulmonary embolism**. The enzyme streptokinase may be given to help dissolve the renal clot. Anticoagulant medications are usually prescribed to prevent clots from recurring. Rarely, when there is a complete blockage of the renal vein in infants, the kidney must be surgically removed.

Prognosis

Most cases of renal vein thrombosis resolve without any permanent damage. **Death** from renal vein thrombosis is rare, and is often caused by the blood clot detaching and lodging in the heart or lungs.

Prevention

There is no specific prevention for renal vein thrombosis. Preventing dehydration reduces the risk that it will occur.

Resources

ORGANIZATIONS

National Kidney Foundation. 30 East 33rd St., New York, NY 10016. (800) 622-9010. <<http://www.kidney.org>>.

OTHER

"Renal Vein Thrombosis." *HealthAnswers.com*. <<http://www.healthanswers.com/database/ami/converted/ooo513.html>>.

Tish Davidson, A.M.

Rendu-Osler-Weber disease see **Hereditary hemorrhagic telangiectasia**

Renin assay see **Plasma renin activity**

Renovascular hypertension

Definition

Renovascular **hypertension** is a secondary form of high blood pressure caused by a narrowing of the renal artery.

Description

Primary hypertension, or high blood pressure, affects millions of Americans. It accounts for over 90% of all cases of hypertension and develops without apparent causes. It is helpful for the clinician to know if a secondary disease is present and may be contributing to the high pressure. If clinical tests indicate this is so, the term used for the rise in blood pressure is secondary hypertension.

Renal hypertension is the most common form of secondary hypertension and affects no more than one percent of all adults with primary hypertension. There are two forms of renovascular hypertension.

In atherosclerotic renovascular hypertension disease, plaque is deposited in the renal artery. The deposits narrow the artery, disrupting blood flow. Atherosclerotic renovascular hypertension is most often seen in men over age 45 and accounts for two-thirds of the cases of renovascular hypertension. In most patients, it affects the renal arteries to both kidneys.

Renovascular hypertension caused by fibromuscular dysplasia occurs mainly in women under age 45. It is also the cause of hypertension in 10% of children with the disorder. In fibromuscular dysplasia, cells from the artery wall overgrow and cause a narrowing of the artery channel.

The risk of having hypertension is related to age, lifestyle, environment, and genetics. **Smoking, stress, obesity**, a diet high in salt, exposure to heavy metals, and an inherited predisposition toward hypertension all increase the chances that a person will develop both primary and renovascular hypertension.

Causes and symptoms

Narrowing of the renal artery reduces the flow of blood to the kidney. In response, the kidney produces the protein renin. Renin is released into the blood stream. Through a series of steps, renin is converted into an enzyme that causes sodium (salt) retention and constriction of the arterioles. In addition to atherosclerotic and fibromuscular dysplasia, narrowing of the renal artery can be caused by compression from an injury or tumor, or by **blood clots**.

Renovascular hypertension is suspected when hypertension develops suddenly in patients under 30 or over 55 years of age or abruptly worsens in any patient. Symptoms are often absent or subtle.

Diagnosis

No single test for renovascular hypertension is definitive. About half of patients with renovascular hypertension have a specific cardiovascular sound that is heard when a doctor listens to the upper abdomen with a stethoscope. Other diagnostic tests give occasional false positive and false negative results. Most tests are expensive, and some involve serious risks.

Imaging studies are used to diagnose renovascular hypertension. In **intravenous urography**, a dye is injected into the kidney, pictures are made, and the kidneys compared. In renal arteriography, contrast material is inserted into the renal artery and cinematic x rays (showing motion within the kidney) are taken. Studies of kidney function are performed. Tests are done to measure renin

production. The results of these tests taken together are used to diagnose renovascular hypertension.

Treatment

Renovascular hypertension may not respond well to anti-hypertensive drugs. Percutaneous transluminal **angioplasty** (PTA), where a balloon catheter is used to dilate the renal artery and remove the blockage, is effective in improving the condition of about 90% of patients with fibromuscular dysplasia. One year later, 60% remain cured. It is less successful in patients with **atherosclerosis**, where renovascular hypertension recurs in half the patients. Where kidney damage occurs, surgery to repair or bypass the renal artery blockage is often effective. In some cases, the damaged kidney must be removed.

Alternative treatment

Alternative treatment stresses eliminating the root causes of hypertension. With renovascular hypertension, as with primary hypertension, the root causes generally cannot be totally reversed by any method. Lifestyle changes are recommended. These include stopping smoking, eating a diet low in animal fats and salt, avoiding exposure to heavy metals, stress control through **meditation**, and anger management. Herbal medicine practitioners recommend garlic (*Allium sativum*) to help lower blood pressure. Constitutional homeopathy and **acupuncture** also can be helpful in lowering blood pressure.

Prognosis

PTA is effective in many younger patients with fibromuscular dysplasia. Older patients are less responsive to this treatment. Surgery is also more risky and less successful in older patients.

Prevention

Renovascular hypertension is possibly preventable through lifestyles that prevent atherosclerosis and primary hypertension. It is unknown how to prevent fibromuscular hyperplasia.

Resources

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300. <<http://www.americanheart.org>>.

Tish Davidson, A.M.

Respiratory acidosis

Definition

Respiratory acidosis is a condition in which a build-up of carbon dioxide in the blood produces a shift in the body's pH balance and causes the body's system to become more acidic. This condition is brought about by a problem either involving the lungs and respiratory system or signals from the brain that control breathing.

Description

Respiratory acidosis is an acid imbalance in the body caused by a problem related to breathing. In the lungs, oxygen from inhaled air is exchanged for carbon dioxide from the blood. This process takes place between the alveoli (tiny air pockets in the lungs) and the blood vessels that connect to them. When this exchange of oxygen for carbon dioxide is impaired, the excess carbon dioxide forms an acid in the blood. The condition can be acute with a sudden onset, or it can develop gradually as lung function deteriorates.

Causes and symptoms

Respiratory acidosis can be caused by diseases or conditions that affect the lungs themselves, such as **emphysema**, chronic **bronchitis**, **asthma**, or severe **pneumonia**. Blockage of the airway due to swelling, a foreign object, or vomit can induce respiratory acidosis. Drugs like anesthetics, sedatives, and **narcotics** can interfere with breathing by depressing the respiratory center in the brain. Head injuries or brain tumors can also interfere with signals sent by the brain to the lungs. Such neuromuscular diseases as **Guillain-Barré syndrome** or **myasthenia gravis** can impair the muscles around the lungs making it more difficult to breathe. Conditions that cause chronic **metabolic alkalosis** can also trigger respiratory acidosis.

The most notable symptom will be slowed or difficult breathing. **Headache**, drowsiness, restlessness, tremor, and confusion may also occur. A rapid heart rate, changes in blood pressure, and swelling of blood vessels in the eyes may be noted upon examination. This condition can trigger the body to respond with symptoms of metabolic alkalosis, which may include **cyanosis**, a bluish or purplish discoloration of the skin due to inadequate oxygen intake. Severe cases of respiratory acidosis can lead to **coma** and **death**.

KEY TERMS

pH—A measurement of acid or alkali (base) of a solution based on the amount of hydrogen ions available. Based on a scale of 14, a pH of 7.0 is neutral. A pH below 7.0 is an acid; the lower the number, the stronger the acid. A pH above 7.0 is a base; the higher the number, the stronger the base. Blood pH is slightly alkali with a normal range of 7.36–7.44.

Diagnosis

Respiratory acidosis may be suspected based on symptoms. A blood sample to test for pH and arterial blood gases can be used to confirm the diagnosis. In this type of acidosis, the pH will be below 7.35. The pressure of carbon dioxide in the blood will be high, usually over 45 mmHg.

Treatment

Treatment focuses on correcting the underlying condition that caused the acidosis. In patients with chronic lung diseases, this may include use of a bronchodilator or steroid drugs. Supplemental oxygen supplied through a mask or small tubes inserted into the nostrils may be used in some conditions, however, an oversupply of oxygen in patients with lung disease can make the acidosis worse. **Antibiotics** may be used to treat infections. If the acidosis is related to an overdose of narcotics, or a **drug overdose** is suspected, the patient may be given a dose of naloxone, a drug that will block the respiratory-depressing effects of narcotics. Use of mechanical ventilation like a respirator may be necessary. If the respiratory acidosis has triggered the body to compensate by developing metabolic alkalosis, symptoms of that condition may need to be treated as well.

Prognosis

If the underlying condition that caused the respiratory acidosis is treated and corrected, there may be no long term effects. Respiratory acidosis may occur chronically along with the development of lung disease or **respiratory failure**. In these severe conditions, the patient may require the assistance of a respirator or ventilator. In extreme cases, the patient may experience coma and death.

Prevention

Patients with chronic lung diseases and those who receive sedatives and narcotics need to be monitored closely for development of respiratory acidosis.

Resources

BOOKS

“Fluid, Electrolyte, and Acid-Base Disorders.” In *Family Medicine Principles and Practices*. 5th ed. New York: Springer-Verlag, 1998.

Altha Roberts Edgren

Respiratory alkalosis

Definition

Respiratory alkalosis is a condition where the amount of carbon dioxide found in the blood drops to a level below normal range. This condition produces a shift in the body's pH balance and causes the body's system to become more alkaline (basic). This condition is brought on by rapid, deep breathing called hyperventilation.

Description

Respiratory alkalosis is an alkali imbalance in the body caused by a lower-than-normal level of carbon dioxide in the blood. In the lungs, oxygen from inhaled air is exchanged for carbon dioxide from the blood. This process takes place between the alveoli (tiny air pockets in the lungs) and the blood vessels that connect to them. When a person hyperventilates, this exchange of oxygen for carbon dioxide is speeded up, and the person exhales too much carbon dioxide. This lowered level of carbon dioxide causes the pH of the blood to increase, leading to alkalosis.

Causes and symptoms

The primary cause of respiratory alkalosis is hyperventilation. This rapid, deep breathing can be caused by conditions related to the lungs like **pneumonia**, lung disease, or **asthma**. More commonly, hyperventilation is associated with **anxiety**, **fever**, **drug overdose**, **carbon monoxide poisoning**, or serious infections. Tumors or swelling in the brain or nervous system can also cause this type of respiration. Other stresses to the body, including **pregnancy**, liver failure,

KEY TERMS

Hyperventilation—Rapid, deep breathing, possibly exceeding 40 breaths/minute. The most common cause is anxiety, although fever, aspirin overdose, serious infections, stroke, or other diseases of the brain or nervous system.

pH—A measurement of acid or alkali (base) of a solution based on the amount of hydrogen ions available. Based on a scale of 14, a pH of 7.0 is neutral. A pH below 7.0 is an acid; the lower the number, the stronger the acid. A pH above 7.0 is a base; the higher the number, the stronger the base. Blood pH is slightly alkali with a normal range of 7.36-7.44.

high elevations, or **metabolic acidosis** can also trigger hyperventilation leading to respiratory alkalosis.

Hyperventilation, the primary cause of respiratory alkalosis, is also the primary symptom. This symptom is accompanied by **dizziness**, light headedness, agitation, and **tingling** or numbing around the mouth and in the fingers and hands. Muscle twitching, spasms, and weakness may be noted. Seizures, irregular heart beats, and tetany (**muscle spasms** so severe that the muscle locks in a rigid position) can result from severe respiratory alkalosis.

Diagnosis

Respiratory alkalosis may be suspected based on symptoms. A blood sample to test for pH and arterial blood gases can be used to confirm the diagnosis. In this type of alkalosis, the pH will be elevated above 7.44. The pressure of carbon dioxide in the blood will be low, usually under 35 mmHg.

Treatment

Treatment focuses on correcting the underlying condition that caused the alkalosis. Hyperventilation due to anxiety may be relieved by having the patient breath into a paper bag. By rebreathing the air that was exhaled, the patient will inhale a higher amount of carbon dioxide than he or she would normally. **Antibiotics** may be used to treat pneumonia or other infections. Other medications may be required to treat fever, seizures, or irregular heart beats. If the alkalosis is related to a drug overdose, the patient may require treatment for **poisoning**. Use of mechanical ventilation

like a respirator may be necessary. If the respiratory alkalosis has triggered the body to compensate by developing metabolic acidosis, symptoms of that condition may need to be treated, as well.

Prognosis

If the underlying condition that caused the respiratory alkalosis is treated and corrected, there may be no long-term effects. In severe cases of respiratory alkalosis, the patient may experience seizures or heart beat irregularities that may be serious and life threatening.

Resources

BOOKS

“Fluid, Electrolyte, and Acid-Base Disorders.” In *Family Medicine Principles and Practices*. 5th ed. New York: Springer-Verlag, 1998.

Altha Roberts Edgren

Respiratory distress syndrome

Definition

Respiratory distress syndrome (RDS) of the newborn, also known as infant RDS, is an acute lung disease present at birth, which usually affects premature babies. Layers of tissue called hyaline membranes keep the oxygen that is breathed in from passing into the blood. The lungs are said to be “airless.” Without treatment, the infant will die within a few days after birth, but if oxygen can be provided, and the infant receives modern treatment in a neonatal intensive care unit, complete recovery with no after-effects can be expected.

Description

If a newborn infant is to breathe properly, the small air sacs (alveoli) at the ends of the breathing tubes must remain open so that oxygen in the air can get into the tiny blood vessels that surround the alveoli. Normally, in the last months of **pregnancy**, cells in the alveoli produce a substance called **surfactant**, which keep the surface tension inside the alveoli low so that the sacs can expand at the moment of birth, and the infant can breathe normally. Surfactant is produced starting at about 34 weeks of pregnancy

and, by the time the fetal lungs mature at 37 weeks, a normal amount is present.

If an infant is born prematurely, enough surfactant might not have formed in the alveoli causing the lungs to collapse and making it very difficult for the baby to get enough air (and the oxygen it contains). Sometimes a layer of fibrous tissue called a hyaline membrane forms in the air sacs, making it even harder for oxygen to get through to the blood vessels. RDS in newborn infants used to be called hyaline membrane disease.

Causes and symptoms

RDS nearly always occurs in premature infants, and the more premature the birth, the greater is the chance that RDS will develop. RDS also is seen in some infants whose mothers are diabetic. Paradoxically, RDS is less likely in the presence of certain states or conditions which themselves are harmful: abnormally slow growth of the fetus; high blood pressure, a condition called **preeclampsia** in the mother; and early rupture of the birth membranes.

Labored breathing (the “respiratory distress” of RDS) may begin as soon as the infant is born, or within a few hours. Breathing becomes very rapid, the nostrils flare, and the infant grunts with each breath. The ribs, which are very flexible in young infants, move inwards each time a breath is taken. Before long the muscles that move the ribs and diaphragm, so that air is drawn into the lungs, become fatigued. When the oxygen level in the blood drops severely the infant’s skin turns bluish in color. Tiny, very premature infants may not even have signs of trouble breathing. Their lungs may be so stiff that they cannot even start breathing when born.

There are two major complications of RDS. One is called **pneumothorax**, which means “air in the chest.” When the infant itself or a breathing machine applies pressure on the lungs in an attempt to expand them, a lung may rupture, causing air to leak into the chest cavity. This air causes the lung to collapse further, making breathing even harder and interfering with blood flow in the lung arteries. The blood pressure can drop suddenly, cutting the blood supply to the brain. The other complication is called intraventricular hemorrhage; this is bleeding into the cavities (ventricles) of the brain, which may be fatal.

Diagnosis

When a premature infant has obvious trouble breathing when born or within a few hours of birth,

RDS is an obvious possibility. If premature birth is expected, or there is some condition that calls for delivery as soon as possible, the amount of surfactant in the amniotic fluid will indicate how well the lungs have matured. If little surfactant is found in an amniotic fluid sample taken by placing a needle in the uterus (**amniocentesis**), there is a definite risk of RDS. Often this test is done at regular intervals so that the infant can be delivered as soon as the lungs are mature. If the membranes have ruptured, surfactant can easily be measured in a sample of vaginal fluid.

The other major diagnostic test is a **chest x ray**. Collapsed lung tissue has a typical appearance, and the more lung tissue is collapsed, the more severe the RDS. An x ray also can demonstrate pneumothorax (air or gas in the area around the lung), if this complication has occurred. The level of oxygen in the blood can be measured by taking a blood sample from an artery, or, more easily, using a device called an oximeter, which is clipped to an earlobe. Pneumothorax may have occurred if the infant suddenly becomes worse while on ventilation; x rays can help make the diagnosis.

Treatment

If only a mild degree of RDS is present at birth, placing the infant in an oxygen hood may be enough. It is important to guard against too much oxygen, as this may damage the retina and cause loss of vision. Using an oximeter to keep track of the blood oxygen level, repeated artery punctures or heel sticks can be avoided. In more severe cases a drug very like natural surfactant (Exosurf Neonatal or Surfactant), can be dripped into the lungs through a fine tube (endotracheal tube) placed in the infant's windpipe (trachea). Typically the infant will be able to breathe more easily within a few days at the most, and complications such as lung rupture are less likely to occur. The drug is continued until the infant starts producing its own surfactant. There is a risk of bleeding into the lungs from surfactant treatment; about 10% of the smallest infants are affected.

Infants with severe RDS may require treatment with a ventilator, a machine that takes over the work of the lungs and delivers air under pressure. In tiny infants who do not breathe when born, ventilation through a tracheal tube is an emergency procedure. Assisted ventilation must be closely supervised, as too much pressure can cause further lung damage. A gentler way of assisting breathing, continuous positive airway pressure or CPAP, delivers an oxygen mixture through nasal prongs or a tube placed through the nose rather than an endotracheal tube. CPAP may be tried before resorting to a ventilator, or after an infant placed on a

ventilator begins to improve. Drugs that stimulate breathing may speed the recovery process.

Pneumothorax is an emergency that must be treated right away. Air may be removed from the chest using a needle and syringe. A tube then is inserted into the lung cavity, and suction applied.

Prognosis

If an infant born with RDS is not promptly treated, lack of an adequate oxygen supply will damage the body's organs and eventually cause them to stop functioning altogether. **Death** is the result. The central nervous system in particular—made up of the brain and spinal cord—is very dependent on a steady oxygen supply and is one of the first organ systems to feel the effects of RDS. On the other hand, if the infant's breathing is supported until the lungs mature and make their own surfactant, complete recovery within three to five days is the rule.

If an air leak causes pneumothorax, immediate removal of air from the chest will allow the lungs to re-expand. Bleeding into the brain is a very serious condition that worsens the outlook for an infant with RDS.

Prevention

The best way of preventing RDS is to delay delivery until the fetal lungs have matured and are producing enough surfactant—generally at about 37 weeks of pregnancy. If delivery cannot be delayed, the mother may be given a steroid hormone, similar to a natural substance produced in the body, which crosses the barrier of the placenta and helps the fetal lungs to produce surfactant. The steroid should be given at least 24 hours before the expected time of delivery. If the infant does develop RDS, the risk of bleeding into the brain will be much less if the mother has been given a dose of steroid.

If a very premature infant is born without symptoms of RDS, it may be wise to deliver surfactant to its lungs. This may prevent RDS, or make it less severe if it does develop. An alternative is to wait until the first symptoms of RDS appear and then immediately give surfactant. Pneumothorax may be prevented by frequently checking the blood oxygen content, and limiting oxygen treatment under pressure to the minimum needed.

Resources

ORGANIZATIONS

American Lung Association. 1740 Broadway, New York, NY 10019. (800) 586-4872. <<http://www.lungusa.org>>.

KEY TERMS

Alveoli—The small air sacs located at the ends of the breathing tubes of the lung, where oxygen normally passes from inhaled air to blood vessels.

Amniotic fluid—The fluid bathing the fetus, which may be sampled using a needle to determine whether the fetus is making enough surfactant.

Endotracheal tube—A metal or plastic tube inserted in the windpipe which may be attached to a ventilator. It also may be used to deliver medications such as surfactant.

Hyaline membranes—A fibrous layer that settles in the alveoli in RDS, and prevents oxygen from escaping from inhaled air to the bloodstream.

Pneumothorax—Air in the chest, often a result of the lung rupturing when oxygen is delivered under too high a pressure.

Preeclampsia—A disease of pregnancy in which the mother's blood pressure is elevated; associated with both maternal and fetal complications, and sometimes with fetal death.

Steroid—A natural body substance that often is given to women before delivering a very premature infant to stimulate the fetal lungs to produce surfactant, hopefully preventing RDS (or making it less severe).

Surfactant—A material normally produced in the fetal lungs in the last months of pregnancy, which helps the air sacs to open up at the time of birth so that the newborn infant can breathe freely.

Ventilator—A machine that can breathe for an infant having RDS until its lungs are producing enough surfactant and are able to function normally.

National Respiratory Distress Syndrome Foundation. P.O. Box 723, Montgomeryville, PA 18936.

David A. Cramer, MD

Respiratory failure

Definition

Respiratory failure is nearly any condition that affects breathing function or the lungs themselves

and can result in failure of the lungs to function properly. The main tasks of the lungs and chest are to get oxygen from the air that is inhaled into the bloodstream, and, at the same time, to eliminate carbon dioxide (CO₂) from the blood through air that is breathed out. In respiratory failure, the level of oxygen in the blood becomes dangerously low, and/or the level of CO₂ becomes dangerously high. There are two ways in which this can happen. Either the process by which oxygen and CO₂ are exchanged between the blood and the air spaces of the lungs (a process called “gas exchange”) breaks down, or the movement of air in and out of the lungs (ventilation) does not take place properly.

Description

Respiratory failure often is divided into two main types. One of them, called hypoxemic respiratory failure, occurs when something interferes with normal gas exchange. Too little oxygen gets into the blood (hypoxemia), and all organs and tissues in the body suffer as a result. One common type of hypoxemic failure, occurring in both adults and prematurely born infants, is **respiratory distress syndrome**, a condition in which fluid or tissue changes prevent oxygen from passing out of the air sacs of the lungs into the circulating blood. Hypoxemia also may result from spending time at high altitudes (where there is less oxygen in the air); various forms of lung disease that separate oxygen from blood in the lungs; severe anemia (“low blood”); and blood vessel disorders that shunt blood away from the lungs, thus precluding the lungs from picking up oxygen.

The other main type of respiratory failure is ventilatory failure, occurring when, for any reason, breathing is not strong enough to rid the body of CO₂. Then CO₂ builds up in the blood (hypercapnia). Ventilatory failure can result when the respiratory center in the brainstem fails to drive breathing; when muscle disease keeps the chest wall from expanding when breathing in; or when a patient has **chronic obstructive lung disease** that makes it very difficult to exhale air with its CO₂. Many of the specific diseases and conditions that cause respiratory failure cause both too little oxygen in the blood (hypoxemia) and abnormal ventilation.

Causes and symptoms

Several different abnormalities of breathing function can cause respiratory failure. The major categories, with specific examples of each, are:

- Obstruction of the airways. Examples are chronic **bronchitis** with heavy secretions; **emphysema**; **cystic fibrosis**; **asthma** (a condition in which it is very hard to get air in and out through narrowed breathing tubes).
- Weak breathing. This can be caused by drugs or alcohol, which depress the respiratory center; extreme **obesity**; or **sleep apnea**, where patients stop breathing for long periods while sleeping.
- Muscle weakness. This can be caused by a muscle disease called myasthenia; **muscular dystrophy**; **polio**; a **stroke** that paralyzes the respiratory muscles; injury of the spinal cord; or Lou Gehrig's disease.
- Lung diseases, including severe **pneumonia**. **Pulmonary edema**, or fluid in the lungs, can be the source of respiratory failure. Also, it can often be a result of heart disease; respiratory distress syndrome; **pulmonary fibrosis** and other scarring diseases of the lung; radiation exposure; burn injury when smoke is inhaled; and widespread lung **cancer**.
- An abnormal chest wall (a condition that can be caused by **scoliosis** or severe injury of the chest wall).

A majority of patients with respiratory failure are short of breath. Both low oxygen and high carbon dioxide can impair mental functions. Patients may become confused and disoriented and find it impossible to carry out their normal activities or do their work. Marked CO₂ excess can cause headaches and, in time, a semi-conscious state, or even **coma**. Low blood oxygen causes the skin to take on a bluish tinge. It also can cause an abnormal heart rhythm (arrhythmia). **Physical examination** may show a patient who is breathing rapidly, is restless, and has a rapid pulse. Lung disease may cause abnormal sounds heard when listening to the chest with a stethoscope: **wheezing** in asthma, "crackles" in obstructive lung disease. A patient with ventilatory failure is prone to gasp for breath, and may use the neck muscles to help expand the chest.

Diagnosis

The symptoms and signs of respiratory failure are not specific. Rather, they depend on what is causing the failure and on the patient's condition before it developed. Good general health and some degree of "reserve" lung function will help see a patient through an episode of respiratory failure. The key diagnostic determination is to measure the amount of oxygen, carbon dioxide, and acid in the blood at regular intervals. A sudden low oxygen level in the lung tissue may cause the arteries of the lungs to narrow. This, in turn,

causes the resistance in these vessels to increase, which can be measured using a special catheter. A high blood level of CO₂ may cause increased pressure in the fluid surrounding the brain and spinal cord; this, too, can be measured.

Treatment

Nearly all patients are given oxygen as the first treatment. Then the underlying cause of respiratory failure must be treated. For example, **antibiotics** are used to fight a lung infection, or, for an asthmatic patient, a drug to open up the airways is commonly prescribed.

A patient whose breathing remains very poor will require a ventilator to aid breathing. A plastic tube is placed through the nose or mouth into the windpipe and is attached to a machine that forces air into the lungs. This can be a lifesaving treatment and should be continued until the patient's own lungs can take over the work of breathing. It is very important to use no more pressure than is necessary to provide sufficient oxygen; otherwise ventilation may cause further lung damage. Drugs are given to keep the patient calm, and the amount of fluid in the body is carefully adjusted so that the heart and lungs can function as normally as possible. Steroids, which combat inflammation, may sometimes be helpful but they can cause complications, including weakening the breathing muscles.

The respiratory therapist has a number of methods available to help patients overcome respiratory failure. They include:

- Suctioning the lungs through a small plastic tube passed through the nose, in order to remove secretions from the airways that the patient cannot **cough** up.
- Postural drainage, in which the patient is propped up at an angle or tilted to help secretions drain out of the lungs. The therapist may clap the patient on the chest or back to loosen the secretions, or a vibrator may be used for the same purpose.
- Breathing exercises often are prescribed after the patient recovers. They make the patient feel better and help to strengthen the muscles that aid breathing. One useful method is for the patient to suck on a tube attached to a clear plastic hosing containing a ball so as to keep the ball lifted. Regular deep breathing exercises are simpler and often just as helpful. Another technique is to have the patient breathe out against pursed lips to increase pressure in the airways and keep them from collapsing.

KEY TERMS

Chronic obstructive lung disease—A common form of lung disease in which breathing, and therefore gas exchange, is labored and increasingly difficult.

Gas exchange—The process by which oxygen is extracted from inhaled air into the bloodstream, and, at the same time, carbon dioxide is eliminated from the blood and exhaled.

Hypoxemia—An abnormally low amount of oxygen in the blood, the major consequence of respiratory failure, when the lungs no longer are able to perform their chief function of gas exchange.

Pulmonary fibrosis—An end result of many forms of lung disease (especially chronic inflammatory conditions). Normal lung tissue is converted to scarred, “fibrotic” tissue that cannot carry out gas exchange.

Prognosis

The outlook for patients with respiratory failure depends chiefly on its cause. If the underlying disease can be effectively treated, with the patient’s breathing supported in the meantime, the outlook is usually good.

Care is needed not to expose the patient to polluting substances in the atmosphere while recovering from respiratory failure; this could tip the balance against recovery. When respiratory failure develops slowly, pressure may build up in the lung’s blood vessels, a condition called **pulmonary hypertension**. This condition may damage the vessels, worsen hypoxemia, and cause the heart to fail. If it is not possible to provide enough oxygen to the body, complications involving either the brain or the heart may prove fatal.

If the kidneys fail or the diseased lungs become infected, the prognosis is worse. In some cases, the primary disease causing the lungs to fail is irreversible. The patient, family, and physician together then must decide whether to prolong life by ventilator support. Occasionally, **lung transplantation** is a possibility, but it is a highly complex procedure and is not widely available.

Prevention

Because respiratory failure is not a disease itself, but the end result of many lung disorders, the best

prevention is to treat any lung disease promptly and effectively. It is also important to make sure that any patient who has had lung disease is promptly treated for any respiratory infection (even of the upper respiratory tract). Patients with lung problems should also avoid exposure to pollutants, as much as is possible. Once respiratory failure is present, it is best for a patient to receive treatment in an intensive care unit, where specialized personnel and all the needed equipment are available. Close supervision of treatment, especially mechanical ventilation, will help minimize complications that would compound the problem.

Resources

ORGANIZATIONS

National Heart, Lung and Blood Institute. P.O. Box 30105, Bethesda, MD 20824-0105. (301) 251-1222. <<http://www.nhlbi.nih.gov>>.

National Respiratory Distress Syndrome Foundation. P.O. Box 723, Montgomeryville, PA 18936.

David A. Cramer, MD

Respiratory syncytial virus infection

Definition

Respiratory syncytial virus (RSV) is a virus that can cause severe lower respiratory infections in children under the age of two, and milder upper respiratory infections in older children and adults. RSV infection is also called **bronchiolitis**, because it is marked in young children by inflammation of the bronchioles. Bronchioles are the narrow airways that lead from the bronchi to the tiny air sacs (alveoli) in the lungs. The result is **wheezing**, difficulty breathing, and sometimes fatal **respiratory failure**.

Description

RSV infection is caused by a group of viruses found worldwide. There are two different subtypes of the virus with numerous different strains. Taken together, these viruses account for a significant number of deaths in infants.

RSV infection is primarily a disease of winter or early spring, with waves of illness sweeping through a community. The rate of RSV infection is estimated to be 11.4 cases in every 100 children during their first

year of life. In the United States, RSV infection occurs most frequently in infants between the ages of two months and six months.

RSV infection shows distinctly different symptoms, depending on the age of the infected person. In children under two, the virus causes a serious lower respiratory infection in the lungs. In older children and healthy adults, it causes a mild upper respiratory infection often mistaken for the **common cold**.

Although anyone can get this disease, infants suffer the most serious symptoms and complications. Breast feeding seems to provide partial protection from the virus. Conditions in infants that increase their risk of infection include:

- premature birth
- lower socio-economic environment
- congenital heart disease
- chronic lung diseases, such as cystic fibrosis
- immune system deficiencies, including HIV infection
- immunosuppressive therapy given to organ transplant patients

Many older children and adults get RSV infection, but the symptoms are so similar to the common cold that the true cause is undiagnosed. People of any age with weakened immune systems, either from such diseases as **AIDS** or leukemia, or as the result of **chemotherapy** or corticosteroid medications, are more at risk for serious RSV infections. So are people with chronic lung disease.

Causes and symptoms

Respiratory syncytial virus is spread through close contact with an infected person. It has been shown that if a person with RSV infection sneezes, the virus can be carried to others within a radius of 6 f (1.8 m). This group of viruses is hardy. They can live on the hands for up to half an hour and on toys or other inanimate objects for several hours.

Scientists have yet to understand why RSV viruses attack the lower respiratory system in infants and the upper respiratory system in adults. In infants, RSV begins with such cold symptoms as a low **fever**, runny nose, and **sore throat**. Soon, other symptoms appear that suggest an infection which involves the lower airways. Some of these symptoms resemble those of **asthma**. RSV infection is suggested by:

- wheezing and high-pitched, whistling breathing
- rapid breathing (more than 40 breaths per minute)

- shortness of breath
- labored breathing out (exhalations)
- bluish tinge to the skin (cyanosis)
- croupy, seal-like, barking **cough**
- high fever

Breathing problems occur in RSV infections because the bronchioles swell, making it difficult for air to get in and out of the lungs. If the child is having trouble breathing, immediate medical care is needed. Breathing problems are most common in infants under one year of age; they can develop rapidly.

Diagnosis

Physical examination and imaging studies

RSV infection is usually diagnosed during a **physical examination** by the pediatrician or primary care doctor. The doctor listens with a stethoscope for wheezing and other abnormal lung sounds in the patient's chest. The doctor will also take into consideration whether there is a known outbreak of RSV infection in the area. Chest x rays give some indication of whether the lungs are hyperinflated from an effort to move air in and out. X rays may also show the presence of a secondary bacterial infection, such as **pneumonia**.

Laboratory tests

A blood test can also detect RSV infection. This test measures the level of antibodies the body has formed against the virus. The blood test is less reliable in infants than in older children because antibodies in the infant's blood may have come from the mother during **pregnancy**. If infants are hospitalized, other tests such as an arterial **blood gas analysis** are done to determine if the child is receiving enough oxygen.

Treatment

Home care

Home treatment for RSV infection is primarily supportive. It involves taking steps to ease the child's breathing. **Dehydration** can be a problem, so children should be encouraged to drink plenty of fluids. **Antibiotics** have no effect on viral illnesses. In time, the body will make antibodies to fight the infection and return itself to health.

Home care for keeping a child with RSV comfortable and breathing more easily includes:

- Use a cool mist room humidifier to ease congestion and sore throat.
- Raise the baby's head by putting books under the head end of the crib.
- Give **acetaminophen** (Tylenol, Pandol, Temptra) for fever. **Aspirin** should not be given to children because of its association with **Reye's syndrome**, a serious disease.
- For babies too young to blow their noses, suction away any mucus with an infant nasal aspirator.

Hospital treatment

In the United States, RSV infections are responsible for 90,000 hospitalizations and 4,500 deaths each year. Children who are hospitalized receive oxygen and humidity through a mist tent or vaporizer. They also are given intravenous fluids to prevent dehydration. Mechanical ventilation may be necessary. Blood gases are monitored to assure that the child is receiving enough oxygen.

Medications

Bronchodilators, such as albuterol (Proventil, Ventolin), may be used to keep the airways open. Ribavirin (Virazole) is used for desperately ill children to stop the growth of the virus. Ribavirin is both expensive and has toxic side effects, so its use is restricted to the most severe cases.

Alternative treatment

Alternative medicine has little to say specifically about bronchiolitis, especially in very young children. Practitioners emphasize that people get viral illnesses because their immune systems are weak. Prevention focuses on strengthening the immune system by eating a healthy diet low in sugars and high in fresh fruits and vegetables, reducing **stress**, and getting regular, moderate **exercise**. Like traditional practitioners, alternative practitioners recommend breastfeeding infants so that the child may benefit from the positive state of health of the mother. Inhaling a steaming mixture of lemon oil, thyme oil, eucalyptus, and tea tree oil (**aromatherapy**) may make breathing easier.

Prognosis

RSV infection usually runs its course in seven to 14 days. The cough may linger weeks longer. There are no medications that can speed the body's production of antibodies against the virus. Opportunistic bacterial infections that take advantage of a weakened

KEY TERMS

Alveoli—Small air sacs or cavities in the lung that give the tissue a honeycomb appearance and expand its surface area for the exchange of oxygen and carbon dioxide.

Antibody—A protein produced by specialized white blood cells in response to the presence of a foreign protein such as a virus. Antibodies help the body fight infection.

Reye's syndrome—A rare disorder in children that follows a viral infection and is associated with a reaction to aspirin. Its symptoms include vomiting, damaged liver function, and swelling of the brain.

respiratory system may cause ear, sinus, and throat infections or pneumonia.

Hospitalization and **death** are much more likely to occur in children whose immune systems are weakened or who have underlying diseases of the lungs and heart. People do not gain permanent immunity to respiratory syncytial virus and can be infected many times. Children who suffer repeated infections seem to be more likely to develop asthma in later life.

Prevention

As of 1998 there are no vaccines against RSV. Respiratory syncytial virus infection is so common that prevention is impossible. However, steps can be taken to reduce a child's contact with the disease. People with RSV symptoms should stay at least six feet away from young children. Frequent handwashing, especially after contact with respiratory secretions, and the correct disposal of used tissues help keep the disease from spreading. Parents should try to keep their children under 18 months old away from crowded environments—for example, shopping malls during holiday seasons—where they are likely to come in contact with older people who have only mild symptoms of the disease. Child care centers should regularly disinfect surfaces that children touch.

Resources

PERIODICALS

Hemming, Val, et al. "Bracing for the Cold and Flu Season." *Patient Care* 31 (September 1997): 47-54.

Tish Davidson, A.M.

Restless legs syndrome

Definition

Restless legs syndrome (RLS) is characterized by unpleasant sensations in the limbs, usually the legs, that occur at rest or before sleep and are relieved by activity such as walking. These sensations are felt deep within the legs and are described as creeping, crawling, aching, or fidgety.

Description

Restless legs syndrome, also known as Ekbom syndrome, Wittmaack-Ekbom syndrome, *anxietas tibiarum*, or *anxietas tibialis*, affects up to 10–15% of the population. Some studies show that RLS is more common among elderly people. Almost half of patients over age 60 who complain of **insomnia** are diagnosed with RLS. In some cases, the patient has another medical condition with which RLS is associated. In idiopathic RLS, no cause can be found. In familial cases, RLS may be inherited from a close relative, most likely a parent.

Causes and symptoms

Most people experience mild symptoms. They may lie down to rest at the end of the day and, just before sleep, will experience discomfort in their legs that prompts them to stand up, massage the leg, or walk briefly. Eighty-five percent of RLS patients either have difficulty falling asleep or wake several times during the night, and almost half experience daytime **fatigue** or sleepiness. It is common for the symptoms to be intermittent. They may disappear for several months and then return for no apparent reason. Two-thirds of patients report that their symptoms become worse with time. Some older patients claim to have had symptoms since they were in their early 20s, but were not diagnosed until their 50s. Suspected under-diagnosis of RLS may be attributed to the difficulty experienced by patients in describing their symptoms.

More than 80% of patients with RLS experience periodic limb movements in sleep (PLMS). These random movements of arms or legs may result in further sleep disturbance and daytime fatigue. Most patients have restless feelings in both legs, but only one leg may be affected. Arms may be affected in nearly half of patients.

There is no known cause for the disorder, but recent research has focused on several key areas. These include:

- Central nervous system (CNS) abnormalities. Several types of drugs have been found to reduce the symptoms of RLS. Based on an understanding of how these drugs work, theories have been developed to explain the cause of the disorder. Levodopa and other drugs that correct problems with signal transmission within the central nervous system (CNS) can reduce the symptoms of RLS. It is therefore suspected that the source of RLS is a problem related to signal transmission systems in the CNS.
- Iron deficiency. The body stores iron in the form of ferritin. There is a relationship between low levels of iron (as ferritin) stored in the body and the occurrence of RLS. Studies have shown that older people with RLS often have low levels of ferritin. Supplements of iron sulfate have been shown to significantly reduce RLS symptoms for these patients.

Diagnosis

A careful history enables the physician to distinguish RLS from similar types of disorders that cause night time discomfort in the limbs, such as **muscle cramps**, burning feet syndrome, and damage to nerves that detect sensations or cause movement (polyneuropathy).

The most important tool the doctor has in diagnosis is the history obtained from the patient. There are several common medical conditions that are known to either cause or to be closely associated with RLS. The doctor may link the patient's symptoms to one of these conditions, which include anemia, diabetes, disease of the spinal nerve roots (lumbosacral radiculopathy), Parkinson's disease, late-stage **pregnancy**, kidney failure (uremia), and complications of stomach surgery. In order to identify or eliminate such a primary cause, blood tests may be performed to determine the presence of serum iron, ferritin, folate, vitamin B₁₂, creatinine, and thyroid-stimulating hormones. The physician may also ask if symptoms are present in any close family members, since it is common for RLS to run in families and this type is sometimes more difficult to treat.

In some cases, sleep studies such as **polysomnography** are undertaken to identify the presence of PLMS that are reported to affect 70–80% of people who suffer from RLS. The patient is often unaware of these movements, since they may not cause him to wake. However, the presence of PLMS with RLS can leave the person more tired, because it interferes with deep sleep. A patient who also displays evidence of some neurologic disease may undergo **electromyography** (EMG). During EMG, a very small, thin needle is

inserted into the muscle and electrical activity of the muscle is recorded. A doctor or technician usually performs this test at a hospital outpatient department.

Treatment

The first step in treatment is to treat existing conditions that are known to be associated with RLS and that will be identified by blood tests. If the patient is anemic, iron (iron sulfate) or vitamin supplements (folate or vitamin B₁₂) will be prescribed. If **kidney disease** is identified as a cause, treatment of the kidney problem will take priority.

Prescription drugs

In some people whose symptoms cannot be linked to a treatable associated condition, drug therapy may be necessary to provide relief and restore a normal sleep pattern. Prescription drugs that are normally used for RLS include:

- **Benzodiazepines** and low-potency opioids. These drugs are prescribed for use only on an “as needed” basis, for patients with mild RLS. Benzodiazepines appear to reduce nighttime awakenings due to PLMS. The benzodiazepine most commonly used to treat RLS is clonazepam (Klonopin, Rivotril). The main disadvantage of this drug type is that it causes daytime drowsiness. It also causes unsteadiness that may lead to accidents, especially for an elderly patient. Opioids are narcotic **pain** relievers. Those commonly used for mild RLS are low potency opioids, such as codeine (Tylenol #3) and propoxyphene (Darvocet). Studies have shown that these can be successfully used in the treatment of RLS on a long-term basis without risk of **addiction**. However, **narcotics** can cause **constipation** and difficulty urinating.
- Levodopa (L-dopa) and carbidopa (Sinemet). Levodopa is the drug most commonly used to treat moderate or severe RLS. It acts by supplying a chemical called dopamine to the brain. It is often taken in conjunction with carbidopa to prevent or decrease side effects. Although it is effective against RLS, levodopa may also cause a worsening of symptoms during the afternoon or early evening in 50–80% of patients. This phenomenon is known as “restless legs augmentation,” and if it occurs, the physician will probably discontinue Levodopa for a brief period while an alternate drug is used. Levodopa can often be reintroduced after a short break.
- Pergolide (Permax). Pergolide acts on the same part of the brain as Levodopa. It is less likely than

Levodopa to cause daytime worsening of symptoms (occurs in about 25% of patients). However, it is not recommended as the first choice in drug therapy since it causes a high rate of minor side effects. Pergolide is often used only if Levodopa has been discontinued.

- High potency opioids. If the symptoms of RLS are difficult to treat with the above medication, higher dose opioids will be used. These include **methadone** (Dolophine), oxycodone, and clonidine (Catapres, Combipres, Dixarit). A significant disadvantage of these drugs is risk of addiction.
- Anticonvulsants. Some cases of RLS may be improved by **anticonvulsant drugs**, such as carbamazepine (Tegretol).
- Combination therapy. Some patients respond well to combinations of drugs such as a benzodiazepine and Levodopa.

Many drugs have been investigated for treatment of RLS, but it seems as though the perfect therapy has not yet been found. However, careful monitoring of side effects and good communication between patient and doctor can result in a flexible program of therapy that minimizes side effects and maximizes effectiveness.

Alternative treatment

It is likely that the best alternative therapy will combine both conventional and alternative approaches. Levodopa may be combined with a therapy that relieves pain, relaxes muscles, or focuses in general on the nervous system and the brain. Any such combined therapy that allows a reduction in dosage of levodopa is advantageous, since this will reduce the likelihood of unacceptable levels of drug side effects. Of course, the physician who prescribes the medication should monitor any combined therapy. Alternative methods may include:

- **Acupuncture**. Patients who also suffer from **rheumatoid arthritis** may especially benefit from acupuncture to relieve RLS symptoms. Acupuncture is believed to be effective in arthritis treatment and may also stimulate those parts of the brain that are involved in RLS.
- **Homeopathy**. Homeopaths believe that disorders of the nervous system are especially important because the brain controls so many other bodily functions. The remedy is tailored to the individual patient and is based on individual symptoms as well as the general symptoms of RLS.

KEY TERMS

Anemia—A condition that affects the size and number of red blood cells. It often results from lack of iron or certain B vitamins and may be treated with iron or vitamin supplements.

Insomnia—Trouble sleeping. People who suffer from RLS often lose sleep either because they spend time walking to relieve discomfort or because they have PLMS, which causes them to wake often during the night.

Periodic limb movements in sleep (PLMS)—Random movements of the arms or legs that occur at regular intervals of time during sleep.

- **Reflexology.** Reflexologists claim that the brain, head, and spine all respond to indirect massage of specific parts of the feet.
- Nutritional supplements. Supplementation of the diet with vitamin E, calcium, magnesium, and **folic acid** may be helpful for people with RLS.

Some alternative methods may treat the associated condition that is suspected to cause restless legs. These include:

- Anemia or low ferritin levels. Chinese medicine will emphasize stimulation of the spleen as a means of improving blood circulation and vitamin absorption. Other treatments may include acupuncture and herbal therapies, such as ginseng (*Panax ginseng*) for anemia-related fatigue.
- Late-stage pregnancy. There are few conventional therapies available to pregnant women, since most of the drugs prescribed are not recommended for use during pregnancy. Pregnant women may benefit from alternative techniques that focus on body work, including **yoga**, reflexology, and acupuncture.

Prognosis

RLS usually does not indicate the onset of other neurological disease. It may remain static, although two-thirds of patients get worse with time. The symptoms usually progress gradually. Treatment with Levodopa is effective in moderate to severe cases that may include significant PLMS. However, this drug produces significant side effects, and continued successful treatment may depend on carefully monitored use of combination drug therapy. The prognosis is usually best if RLS symptoms are recent and can be

traced to another treatable condition that is associated with RLS. Some associated conditions are not treatable. In these cases, such as for rheumatoid arthritis, alternative therapies such as acupuncture may be helpful.

Prevention

Diet is key in preventing RLS. A preventive diet will include an adequate intake of iron and the **B vitamins**, especially B₁₂ and folic acid. Strict vegetarians should take vitamin supplements to obtain sufficient vitamin B₁₂. Ferrous gluconate may be easier on the digestive system than ferrous sulfate, if iron supplements are prescribed. Some medications may cause symptoms of RLS. Patients should check with their doctor about these possible side effects, especially if symptoms first occur after starting a new medication. **Caffeine**, alcohol, and nicotine use should be minimized or eliminated. Even a hot bath before bed has been shown to prevent symptoms for some sufferers.

Resources

PERIODICALS

Silber, Michael H. "Concise Review for Primary-Care Physicians. Restless Legs Syndrome." *Mayo Clinical Proceedings* 72 (March 1997): 261-264.

ORGANIZATIONS

Restless Legs Syndrome Foundation. 1904 Banbury Road, Raleigh, NC 27608-4428. (919) 781-4428. <<http://www.rls.org>>.

Ann M. Haren

Restrictive cardiomyopathy

Definition

Cardiomyopathy is an ongoing disease process that damages the muscle wall of the lower chambers of the heart. Restrictive cardiomyopathy is a form of cardiomyopathy in which the walls of the heart become rigid.

Description

Restrictive cardiomyopathy is the least common type of cardiomyopathy in the United States. The stiffened heart walls cannot stretch properly to allow enough blood to fill the ventricles between heartbeats. As the stiffening worsens, **heart failure** occurs. The

blood backs up into the blood vessels, causing fluid buildup in tissues (congestion and **edema**).

Causes and symptoms

Restrictive cardiomyopathy can be caused by a number of diseases. Often, the cause is unknown. The rigidity of the heart walls may be caused by fibrosis, the replacement of muscle cells with tough, fibrous tissue. In some disorders, proteins and other substances are deposited in the heart wall. **Amyloidosis** is the accumulation of a protein material, called amyloid, in the tissue of the heart wall and other organs. In **hemochromatosis**, there is too much iron in the body and some of the excess iron can build up in the heart. **Sarcoidosis** causes the formation of many small lesions, called granulomas, in the heart wall and other tissues of the body. These granulomas contain inflammatory white blood cells and other cells that decrease the flexibility of the heart.

People with restrictive cardiomyopathy usually feel tired and weak, and have **shortness of breath**, especially during **exercise**. If blood is backing up in the circulation they may also experience edema (large amounts of fluid in tissues) of the legs and feet.

Diagnosis

The diagnosis is usually based on a **physical examination**, **echocardiography**, and other tests as needed. The physician listens to the heart with a stethoscope to detect abnormal heart rhythms and heart sounds.

Echocardiography uses sound waves to make images of the heart. These images provide information about the structures of the heart and its heart valves. Echocardiography can also be used to find out how much blood the heart is pumping. It determines the amount of blood in the ventricle, called the ventricular volume, and the amount of blood the ventricle pumps each time it beats, called the ejection fraction. A healthy heart pumps at least one half the amount of blood in the left ventricle with each heartbeat.

Computed tomography scan (CT scan) and **magnetic resonance imaging** (MRI) are imaging tests that can also provide information about the structure of the heart. However, these tests are rarely needed for diagnosis.

Cardiac catheterization may be needed to confirm a diagnosis or cause. In cardiac catheterization, a small tube called a catheter is inserted into an artery and passed into the heart. It is used to measure pressure in the heart and the amount of blood

KEY TERMS

Amyloidosis—Build up of amyloid, a protein substance, in tissues of the body, including the heart.

Cardiac catheterization—A diagnostic test for evaluating heart disease; a catheter is inserted into an artery and passed into the heart.

Edema—Swelling caused by fluid buildup in tissues.

Fibrosis—Replacement of normal tissue with tough, fibrous tissue.

Hemochromatosis—A disease in which there is too much iron in the body; iron deposits can build up in the heart muscle and other tissues.

Sarcoidosis—A chronic disease causing the formation of many small lesions called granulomas in the heart wall and other tissues of the body.

pumped by the heart. A small tissue sample (biopsy) of the heart muscle can be removed through the catheter for microscopic examination. Fibrous tissue or deposits in the heart muscle can be identified in this biopsy.

Treatment

There is no effective treatment for restrictive cardiomyopathy. Treatment of a causative disease may reduce or stop the damage to the heart, but existing damage cannot be reversed. Medications may be used to lessen the workload on the heart and to control the heart rhythm. Drugs normally used to treat other types of cardiomyopathy and heart failure may cause problems for patients with restrictive cardiomyopathy. For example, medicines that reduce the heart's workload may lower blood pressure too much.

A heart transplant may be necessary for patients who develop severe heart failure.

Prognosis

The prognosis for patients with restrictive cardiomyopathy is poor. If the disease process causing the problem can be treated, the damage to the heart muscle may be stopped. Also, medicines may relieve symptoms. However, for most patients, restrictive cardiomyopathy eventually causes heart failure. A heart transplant may be necessary when heart failure becomes too severe to treat with medicines.

Prevention

Obtaining early treatment for diseases that might cause restrictive cardiomyopathy might prevent or slow the development of heart wall stiffness. Anyone experiencing symptoms of shortness of breath, tiredness, and weakness should see a physician.

Resources

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300. <<http://www.americanheart.org>>.

National Heart, Lung and Blood Institute. P.O. Box 30105, Bethesda, MD 20824-0105. (301) 251-1222. <<http://www.nhlbi.nih.gov>>.

Texas Heart Institute. Heart Information Service. P.O. Box 20345, Houston, TX 77225-0345. <<http://www.tmc.edu/thi>>.

Toni Rizzo

Reticulocyte count

Definition

A reticulocyte count is a blood test performed to assess the body's production of immature red blood cells (reticulocytes). A reticulocyte count is usually performed when patients are evaluated for anemia and response to its treatment. It is sometimes called a retic count.

Purpose

Diagnosis

A reticulocyte count provides information about the rate at which the bone marrow is producing red cells. A normal count means that the production is adequate; a decreased count means it is not. This information helps determine whether a lack of red cells in an anemic person is caused by a bone marrow problem, by excessive bleeding, or by red cell destruction.

Monitoring

The test is also used to monitor the response of bone marrow response to treatment for anemia. The reticulocyte count rises within days if the treatment is successful. It is also used following bone marrow transplant to evaluate the new marrow's cell production.

Description

Reticulocytes were first described as transitional forms of red blood cells by Wilhelm H. Erb in 1865. A red cell begins in the bone marrow as a large bluish cell filled with ribonucleic acid (RNA). As the cell matures, it shrinks. Its color gradually changes from blue to pink as its load of oxygen-carrying protein (hemoglobin) increases and the RNA decreases. The center of the cell (nucleus) becomes clumped. It is expelled three days before the cell leaves the bone marrow. The cell is now a reticulocyte. On its fourth and final day of maturation, the reticulocyte enters the bloodstream. One day later, it is a mature red blood cell.

The first step in a retic count is drawing the patient's blood sample. About 17 oz (5 ml) of blood is withdrawn from a vein into a vacuum tube. The procedure, which is called a venipuncture, takes about five minutes.

After the sample is collected, the blood is mixed with a dye (methylene blue) in a test tube. The RNA remaining in the reticulocytes picks up a deep blue stain. Drops of the mixture are smeared on slides and examined under a microscope. Reticulocytes appear as cells containing dark blue granules or a blue network. The laboratory technologist counts 1,000 red cells, keeping track of the number of reticulocytes. The number of reticulocytes is reported as a percentage of the total red cells. When the red cell count is low, the percentage of reticulocytes is inaccurately high, suggesting that more reticulocytes are present than there are in reality. The percentage is mathematically corrected for greater accuracy. This figure is called the corrected reticulocyte count or reticulocyte index.

Reticulocyte counts can also be done on automated instruments, such as flow cytometers, using fluorescent stains. These instruments are able to detect small changes in the reticulocyte count because they count a larger number of cells (10,000–50,000).

Preparation

The doctor should make a note of any prescription medications that the patient is taking. Some drugs lower the red blood cell count.

Aftercare

Aftercare consists of routine care of the area around the puncture mark. Pressure is applied for a few seconds and the wound is covered with a bandage.

KEY TERMS

Anemia—A condition marked by a decrease in the number or size of red blood cells

Methylene blue—A dye that is used to stain the blood cells for the reticulocyte count.

Reticulocyte—An immature red blood cell.

Risks

The primary risk is mild **dizziness** and the possibility of a bruise or swelling in the area where the blood was drawn. The patient can apply moist warm compresses.

Normal results

Adults have reticulocyte counts of 0.5–2.5%. Women and children usually have higher reticulocyte counts than men.

Abnormal results

A low reticulocyte count indicates that the bone marrow is not producing a normal number of red blood cells. Low production may be caused by a lack of vitamin B₁₂, **follic acid**, or iron in the diet; or by an illness affecting the bone marrow (for example, **cancer**). Further tests are needed to diagnose the specific cause.

The reticulocyte count rises when the bone marrow makes more red cells in response to blood loss or treatment of anemia.

Resources

PERIODICALS

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Nancy J. Nordenson

Retinal artery occlusion

Definition

Retinal artery occlusion refers to the closure of the central retinal artery and usually results in complete loss

of vision in one eye. Occlusion of its branches causes loss of vision in only a portion of the field of vision.

Description

Retinal artery occlusion (RAO) occurs when the central retinal artery, the main source of blood supply to the retina, or one of its branches becomes blocked.

Causes and symptoms

The main causes of RAO are the following:

- embolism (the sudden obstruction of a blood vessel by a blood clot)
- atherosclerotic disease that results in the progressive narrowing of the arteries over time
- endarteritis (the chronic inflammation of the inner layer of arteries)
- angiospasm (a spasmodic contraction of a blood vessel with increase in blood pressure)

The most common symptom of RAO is an acute, painless loss of vision in one eye. The degree of loss depends on the location of the occlusion. If the occlusion occurs in the central artery of the retina, damage usually results in complete loss of vision in the affected eye. If occlusion occurs in a branch artery, vision loss will be partial and may even go unnoticed if only a section of the peripheral vision is affected.

People affected by RAO typically have high blood pressure, heart disease, or diabetes as an underlying condition. Other conditions that may increase the risk of RAO include **high cholesterol** and **glaucoma**. Incidence is slightly more common in men and in people age 60 or older.

Diagnosis

RAO is diagnosed by examination of the retina with an ophthalmoscope.

Treatment

Central retinal artery occlusion (CRAO) is an emergency. If treatment begins within an hour, the patient has the highest possibility of regaining vision in the affected eye, although complete restoration is unlikely.

A common treatment is inhalation of carbon dioxide so as to dilate the retinal vessels and move the occlusion from the central retinal artery to a branch artery. This movement reduces the area of the retina affected and may restore a certain amount of vision. Eyeball massage may also be performed, also in

KEY TERMS

Angiospasm—Spasmodic contraction of a blood vessel with increase in blood pressure.

Arterioles—Small blood vessels that carry arterial (oxygenated) blood.

Atherosclerotic disease—The progressive narrowing and hardening of the arteries over time.

Central retinal artery—A branch of the ophthalmic artery that supplies blood to the retina and branches to form the arterioles of the retina.

Embolism—The sudden obstruction of a blood vessel by a blood clot.

Endarteritis—Chronic inflammation of the inner layer of arteries.

Hyperbaric oxygenation—Administration of oxygen in a compression chamber at an ambient pressure greater than 1 atmosphere, in order to increase the amount of oxygen in organs and tissues.

Occlusion—Momentary complete closure of some area or channel of the body.

Ophthalmic artery—The artery supplying the eye and adjacent structures with blood.

Ophthalmoscope—An instrument used for viewing the inside of the eye that consists of a concave mirror with a hole in the middle through which the physician examines the eye, and a light source that is reflected into the eye by the mirror.

Retina—Light sensitive layer of the eye, that consists of four major layers: the outer neural layer, containing nerve cells and blood vessels, the photoreceptor layer, a single layer that contains the light sensing rods and cones, the pigmented retinal epithelium (PRE) and the choroid, consisting of connective tissue and capillaries.

an effort to remove the occlusion. The physician may also consider puncturing the eyeball.

Drug therapy includes the use of carbonic anhydrase inhibitors to reduce the internal eye pressure and enhance movement of the occlusion. Both of the treatments would be used within the first 24 hours of noticeable vision loss.

Alternative treatment

Hyperbaric **oxygen therapy** may be beneficial if started within 90 minutes of the onset of symptoms.

Some studies indicate a 40% improvement of visual acuity using this method.

Prognosis

The prognosis for central retinal visual acuity is poor with only about one-third of patients recovering useful vision. The longest delay in getting treatment that has been associated with significant visual recovery was approximately 72 hours.

Branch retinal artery occlusions (BRAO) have a recovery rate of 80% where vision is restored to 20/40 or better.

Prevention

Individuals affected by underlying conditions such as high blood pressure, heart disease, diabetes, glaucoma, and elevated cholesterol should treat their conditions appropriately to minimize the possibility of a retinal artery occlusion.

Resources

ORGANIZATIONS

American Academy of Ophthalmology. P.O. Box 7424, San Francisco, CA 94120-7424. (415) 561-8500. Fax: (415) 561-8533. <<http://www.eyenet.org/>>.

American Diabetes Association. 1701 North Beauregard Street, Alexandria, VA 22311. (800) DIABETES. <<http://www.diabetes.org/>>.

American Heart Association National Center. 7272 Greenville Avenue, Dallas, Texas 75231. (877) 242-4277. <<http://www.americanheart.org/>>.

CliniWeb International. <<http://www.ohsu.edu/clinweb/C11/C11.768.html>>.

Gary Gilles

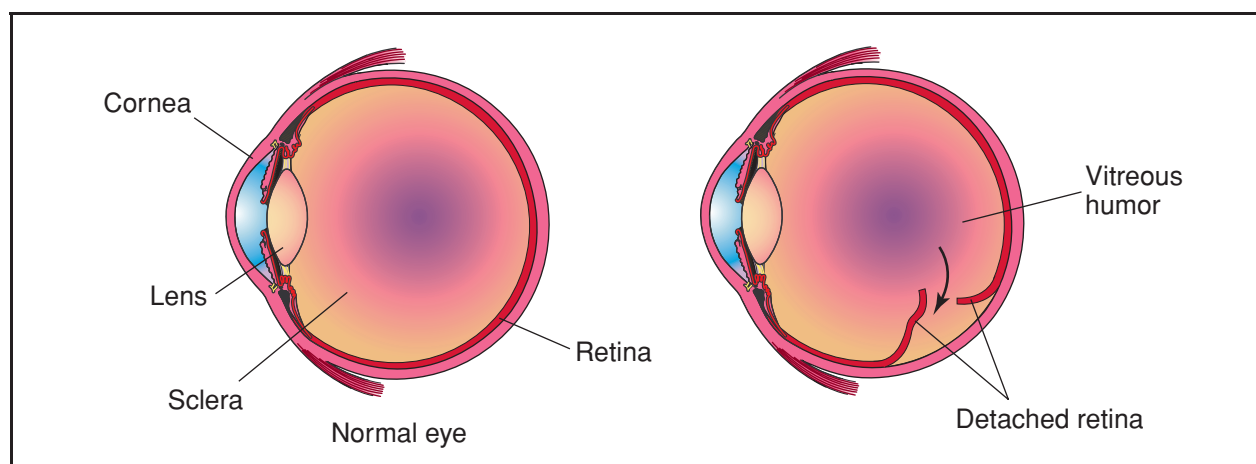
Retinal detachment

Definition

Retinal detachment is movement of the transparent sensory part of the retina away from the outer pigmented layer of the retina. In other words, the moving away of the retina from the outer wall of the eyeball.

Description

There are three layers of the eyeball. The outer, tough, white sclera. Lining the sclera is the choroid, a



Retinal detachment refers to the movement of the retina away from the inner wall of the eyeball, resulting in a sudden defect in vision. Persons suffering from diabetes have a high incidence of developing retinal disease. (Illustration by Electronic Illustrators Group.)

thin membrane that supplies nutrients to part of the retina. The innermost layer is the retina.

The retina is the light-sensitive membrane that receives images and transmits them to the brain. It is made up of several layers. One layer contains the photoreceptors. The photoreceptors, the rods and cones, send the visual message to the brain. Between the photoreceptor layer (also called the sensory layer) and the choroid is the pigmented epithelium.

The vitreous is a clear gel-like substance that fills up most of the inner space of the eyeball. It lies behind the lens and is in contact with the retina.

A retinal detachment occurs between the two outermost layers of the retina—the photoreceptor layer and the outermost pigmented epithelium. Because the choroid supplies the photoreceptors with nutrients, a detachment can basically starve the photoreceptors. If a detachment is not repaired within 24–72 hours, permanent damage may occur.

Causes and symptoms

Several conditions may cause retinal detachment:

- Scarring or shrinkage of the vitreous can pull the retina inward.
- Small tears in the retina allow liquid to seep behind the retina and push it forward.
- Injury to the eye can simply knock the retina loose.
- Bleeding behind the retina, most often due to diabetic retinopathy or injury, can push it forward.

- Retinal detachment may be spontaneous. This occurs more often in the elderly or in very near-sighted (myopic) eyes.
- Cataract surgery causes retinal detachment 2% of the time.
- Tumors can cause the retina to detach.

Retinal detachment will cause a sudden defect in vision. It may look as if a curtain or shadow has just descended before the eye. If most of the retina is detached, there may be only a small hole of vision remaining. If just a part of the retina is involved, there will be a blind spot that may not even be noticed. It is often associated with *floaters*—little dark spots that float across the eye and can be mistaken for flies in the room. There may also be *flashes* of light. Anyone experiencing a sudden onset of flashes and/or floaters should contact their eye doctor immediately, as this may signal a detachment.

Diagnosis

If the eye is clear—that is, if there is no clouding of the liquids inside the eye—the detachment can be seen by looking into the eye with a hand-held instrument called an ophthalmoscope. To evaluate the blood vessels in the retina, a fluorescent dye (fluorescein) may be injected into a vein and photographed with ultraviolet light as it passes through the retina. Further studies may include computed tomography scan (CT scan), **magnetic resonance imaging** (MRI), or ultrasound study. Other lenses may be used to examine the back of the eyes. One example is binocular indirect ophthalmoscopy. The doctor dilates the patient's eyes with

KEY TERMS

Cauterize—To damage with heat or cold so that tissues shrink. It is an effective way to stop bleeding.

Diabetic retinopathy—Disease that damages the blood vessels in the back of the eye caused by diabetes.

Saline—A salt solution equivalent to that in the body—0.9% salt in water.

eyedrops and then examines the back of the eyes with a handheld lens.

Treatment

Reattaching the retina to the inner surface of the eye requires making a scar that will hold it in place and then bringing the retina close to the scarred area. The scar can be made from the outside, through the sclera, using either a laser or a freezing cold probe (cryopexy). Bringing the retina close to the scar can be done in two ways. A tiny belt tightened around the eyeball will bring the sclera in until it reaches the retina. This procedure is called scleral buckling and may be done under **general anesthesia**. Using this procedure permits the repair of retinal detachments without entering the eyeball. Sometimes, the eye must be entered to pump in air or gas, forcing the retina outward against the sclera and its scar. This is called pneumatic retinopexy and can generally be done under **local anesthesia**.

If all else fails, and especially if there is disease in the vitreous, the vitreous may have to be removed in a procedure called **vitrectomy**. This can be done through tiny holes in the eye, through which equally tiny instruments are placed to suck out the vitreous and replace it with saline, a salt solution. The procedure must maintain pressure inside the eye so that the eye does not collapse.

Prognosis

Retinal reattachment has an 80–90% success rate.

Prevention

In diseases such as diabetes, with a high incidence of retinal disease, routine eye examinations can detect early changes. Early treatment can prevent both progressing to detachment and blindness from other events like hemorrhage. The most common problem

is weakness of blood vessels that causes them to break down and bleed. When enough vessels have been damaged, new vessels grow to replace them. These new vessels may grow into the vitreous, producing blind spots and scarring. The scarring can in turn pull the retina loose. Other diseases can cause the tiny holes and tears in the retina through which fluid can leak. Preventive treatment uses a laser to cauterize the blood vessels, so that they do not bleed and the holes, so they do not leak.

Good control of diabetes can help prevent diabetic eye disease. Blood pressure control can prevent **hypertension** from damaging the retinal blood vessels. Eye protection can prevent direct injury to the eyes. Regular eye exams can also detect changes that the patient may not be aware of. This is important for patients with high **myopia** who may be more prone to detachment.

Resources

ORGANIZATIONS

American Academy of Ophthalmology. 655 Beach Street, P.O. Box 7424, San Francisco, CA 94120-7424. <<http://www.eyenet.org>>.

American Optometric Association. 243 North Lindbergh Blvd., St. Louis, MO 63141. (314) 991-4100. <<http://www.aoanet.org>>.

J. Ricker Polsdorfer, MD

Retinal hemorrhage

Definition

Retinal hemorrhage is the abnormal bleeding of the blood vessels in the retina, the membrane in the back of the eye.

Description

The retina is the part of the eye that converts light into nerve signals that are processed by the brain into visual images. The retina is the inside surface of the back of the eye, consisting of millions of densely arranged, light-sensitive cells called rods and cones. Blood flow to the retina is maintained by the retinal vein and artery, and a dense network of small blood vessels (capillaries) supplies the area with circulation. These blood vessels can become damaged by injury and disease and may bleed (hemorrhage) and cause temporary or permanent loss of visual accuracy.

Because the cells of the retina are so dense and sensitive, even small injuries to the blood vessels can translate into vision problems. Diseases that affect the health of the circulatory system, such as diabetes and high blood pressure, also affect the blood vessels of the eye. Damage to the blood vessels in the retina, including hemorrhage, is termed retinopathy.

Causes and symptoms

Retinal hemorrhages can be caused by injuries, usually forceful blows to the head during accidents and falls, as well as by adverse health conditions. In infants, retinal hemorrhage is frequently associated with **child abuse** and has been termed **shaken baby syndrome**. A condition called retinopathy of **prematurity** occurs in prematurely born infants or infants with low birth weights. When children are born prematurely, the blood vessels in the eye may not have had time to fully develop and may become damaged easily, leaking or hemorrhaging. The condition must be determined by an ophthalmologist, as the symptoms are not readily observable.

Diabetic retinopathy is a common eye problem associated with diabetes. Diabetes, by stressing the circulatory system, can cause damage, including hemorrhaging, to the small blood vessels of the retina. Non-proliferative retinopathy occurs when the damaged or leaking blood vessels do not spread. Symptoms of this disorder include vision spots, floaters (floating areas of blurred vision), decreased or loss of vision, or loss of fine vision for detailed activities such as reading. Proliferative retinopathy occurs when new blood vessels begin to form in damaged areas of the retina, and may lead to spots, floaters, decreased vision, or sudden loss of vision. Sudden vision loss may occur if one of the newly formed blood vessels ruptures. Due to increased pressure in the area, the retina may detach from the back of the eye, a serious condition and a cause of blindness.

People with high blood pressure (**hypertension**) may develop hypertensive retinopathy, in which blood vessels in the retina become damaged from increased blood pressure. Symptoms are typically not pronounced, but blurred or decreased vision may be caused by the disorder.

Central serous retinopathy is a condition in which the vessels behind the retina leak and cause fluid to collect in small blisters behind the retina. Symptoms include sudden blurry areas in the vision, blind spots, distorted vision areas, and loss of vision. This condition is most common in males between 20 and 50 years of age.

KEY TERMS

Diabetes—Disease in which the body does not properly produce or use insulin, resulting in fluctuating blood sugar levels.

Hypertension—Condition caused by high blood pressure.

Ophthalmologist—Physician specializing in the diagnosis and treatment of disorders of the eye.

Diagnosis

Diagnosis of retinopathy is performed by an ophthalmologist, particularly one who specializes in disorders of the retina (retinal specialist). The ophthalmologist may perform an ophthalmoscopy, using an instrument called an ophthalmoscope to examine the inside of the eye. For a detailed view of the blood vessels of the retina, a fluorescein **angiography** test might be performed, in which a fluorescent dye is injected into the patient's bloodstream and photographs record the status of the blood vessels in the retina. Vision tests, patient history, and blood tests might also be ordered by the diagnosing physician.

Treatment

Laser surgery by an ophthalmologist is a common treatment for retinal hemorrhages, in which a laser beam is used to remove or seal off damaged or bleeding blood vessels in the retina. Some vision loss occurs with this technique. For retinal hemorrhages associated with diabetes and high blood pressure, treating the overall condition is required.

Alternative treatment

Alternative treatment of retinal hemorrhages focuses on providing nutrients to strengthen and heal the injured blood vessels. **Nutritional supplements** include antioxidant **vitamins A, C, and E**; vitamin B-complex including B₆ and B₁₂; the mineral zinc; and essential fatty acids including omega-3 from fish oil and flaxseed oil. Herbal supplements include bilberry, grape seed extract, pine bark extract (pycnogenol), and lutein.

Prognosis

For retinal hemorrhages associated with retinopathy of prematurity, nearly 85 percent of cases heal

without treatment. Diabetic retinopathy is the leading cause of blindness for those between 20 and 65 years old in the U.S. Diabetic retinopathy typically takes years to develop in people with diabetes, but occurs in nearly 80 percent of those with diabetes for over 20 years and who are treated with insulin. Regular monitoring and treatments can slow the degeneration of the eye, while advanced cases of the disorder lead to blindness. **Retinopathies** requiring laser treatment have a partial loss of vision due to the surgery. For hypertensive retinopathy, most vision problems go away when high blood pressure is treated and lowered. The majority of cases of central serous retinopathy disappear after three to four months, and full vision generally returns within six months, although recurrence of the disorder is common.

Prevention

The first step in sound prevention is for people with vision problems, including visual spots, flashes or floaters in the vision, and loss or distortion of visual accuracy, to see an ophthalmologist as soon as possible. To prevent complications of retinal hemorrhages in infants, the prevention includes regular prenatal care and monitoring of infants with high risks of the disorder (born prematurely or with weight less than four pounds and six ounces). For diabetic retinopathy, control of blood sugar and blood pressure fluctuations is necessary, as well as frequently scheduled eye exams by an ophthalmologist. For retinal hemorrhages associated with hypertension, controlling high blood pressure through diet, **exercise**, and **stress reduction** is recommended. Central serous retinopathy has been associated with high **stress** levels, so preventative care for this disorder includes stress management practices.

Resources

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- Grossman, Marc. *Natural Eye Care: An Encyclopedia*. McGraw Hill, 1999.

ORGANIZATIONS

- American Academy of Ophthalmology. P.O. Box 7424, San Francisco, CA 94120. (415) 561-8500. <<http://www.aao.org>>.

National Eye Institute. 2020 Vision Place, Bethesda, MD 20892-3655. (301) 496-5248. <<http://www.nei.nih.gov>>.

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Retinal vein occlusion

Definition

Retinal vein occlusion refers to the closure of the central retinal vein that drains the retina or to that of one of its branches.

Description

Retinal vein occlusion (RVO) occurs when the central retinal vein, the blood vessel that drains the retina, or one of its branches becomes blocked. RVO may be categorized by the anatomy of the occluded vein and the degree of **ischemia** produced. The two major RVO types are central retinal vein occlusion (CRVO) and branch retinal vein occlusion (BRVO). CRVO has been diagnosed in patients as young as nine months to patients of 90 years. The age of affected individuals is usually low to mid 60s. Approximately 90% of patients are over 50 at the time of diagnosis, with 57% of them being male and 43% being female. BRVO accounts for some 30% of all vein occlusions.

Causes and symptoms

CRVO is a painless loss of vision that can be caused by a swollen optic disk, the small area in the retina where the optic nerve enters the eye, by dilated retinal veins, and by retinal hemorrhages. CRVO is also called venous stasis retinopathy, or hemorrhagic retinopathy.

In BRVO, the superotemporal branch vein is the most often affected vessel. Retinal hemorrhages follow, often occurring at the crossing of two vessels near the optic disk. Initially the hemorrhage may be extensive and underlie the fovea.

The exact cause of RVO is not yet identified, but the following mechanisms been proposed:

- external compression between the central connective strand and the cribriform plate
- venous disease
- blood clot formation

Conditions associated with RVO risk include:

- hypertension
- hyperlipidemia
- diabetes mellitus
- hyperviscosity
- hypercoagulability
- **glaucoma**
- trauma

Diagnosis

A complete physical evaluation is recommended for CRVO and BRVO, including complete blood tests, and glucose tolerance test (for non-diabetics). In the case of a **head injury** when bleeding around the optic nerve is a possibility, an MRI may be performed.

Treatment

Following a patient with RVO is vital. Patients should be seen at least monthly for the first three months to monitor for signs of other complications, such as the abnormal formation of blood vessels (neovascularization) in the iris of the eye or glaucoma.

The treatment for retinal vein occlusion varies for each case and should be given based on the doctor's best recommendation. Although treatments for occlusion itself are limited, surgical treatment of the occlusion provides an option.

Treatments may include anticoagulants with heparin, bishydroxycoumarin, and streptokinase. When the blood is highly viscous, dilution of the blood may be useful. Ideally, an alternate pathway is needed to allow venous drainage. Recent reports published in 1999 suggest that use of a laser to create a retinal choroidal hole may be useful to treat CRVO. Laser therapy depends on the type of occlusion. The management of laser therapy should be controlled by an ophthalmologist.

Alternative treatment

There are no documented alternative treatment methods.

Prognosis

The outlook for people with RVO is fairly good whether it is treated early or not. With no treatment at

KEY TERMS

Anticoagulants—Drugs that act by lowering the capacity of the blood to coagulate, thus facilitating removal of blood clots.

Central retinal vein—Central blood vessel and its branches that drains the retina.

Cribiform plate—The horizontal bone plate perforated with several holes for the passage of olfactory nerve filaments from the nasal cavity.

Fovea—A small area of the retina responsible for acute vision.

Glaucoma—A group of eye diseases characterized by an increase in eyeball pressure.

Hyperlipidemia—A general term for elevated concentrations of any or all of the lipids in the plasma.

Iris—The contractile diaphragm located in the fluid in front of the lens of the eye and is perforated by the eye pupil.

Ischemia—A state of low oxygen in a tissue usually due to organ dysfunction.

Neovascularization—Abnormal or excessive formation of blood vessels as in some retinal disorders.

Occlusion—Momentary complete closure of some area or channel of the body.

Optic disk—The small area in the retina where the optic nerve enters the eye that is not sensitive to light. Also called the blind spot.

Retina—Light sensitive layer of the eye, that consists of four major layers: the outer neural layer, containing nerve cells and blood vessels, the photoreceptor layer, a single layer that contains the light sensing rods and cones, the pigmented retinal epithelium (PRE) and the choroid, consisting of connective tissue and capillaries.

all, approximately 60% of all patients recover 20/40 vision or better within a year.

Prevention

Retinal vein occlusion is difficult to prevent because the exact cause is still uncertain. Ethnic factors may play a role since in the UK the disease is rare in Asians and West Indians.

Resources

BOOKS

Spaide, Richard F., MD. *Diseases of the Retina and Vitreous*. New York: W.B. Saunders Co., 1999.

Michael Sherwin Walston
Ronald Watson, PhD

Retinitis pigmentosa

Definition

Retinitis pigmentosa (RP) refers to a group of inherited disorders that slowly lead to blindness due to abnormalities of the photoreceptors (primarily the rods) in the retina.

Description

The retina lines the interior surface of the back of the eye. The retina is made up of several layers. One layer contains two types of photoreceptor cells referred to as the rods and cones. The cones are responsible for sharp, central vision and color vision and are primarily located in a small area of the retina called the fovea. The area surrounding the fovea contains the rods, which are necessary for peripheral vision and night vision (scotopic vision). The number of rods increases in the periphery. The rod and cone photoreceptors convert light into electrical impulses and send the message to the brain via the optic nerve. Another layer of the retina is called the retinal pigmented epithelium (RPE).

In RP, the photoreceptors (primarily the rods) begin to deteriorate and lose their ability to function. Because the rods are primarily affected, it becomes harder to see in dim light, thus causing a loss of night vision. As the condition worsens, peripheral vision disappears, which results in tunnel vision. The ability to see color is eventually lost. In the late stages of the disease, there is only a small area of central vision remaining. Ultimately, this too is lost.

There are many forms of retinitis pigmentosa. Sometimes the disorder is classified by the age of onset or the inheritance pattern. RP can also accompany other conditions. This entry discusses “non-syndromic” RP, the type that is not associated with other organ or tissue dysfunction.

Approximately 100,000 Americans have RP. It is estimated to affect about one in every 4,000 Americans

and Europeans. For other parts of the world, there are no published data. Nor is there any known ethnic difference in the occurrence of RP.

Causes and symptoms

Retinitis pigmentosa is an inherited disease that has many different modes of inheritance. It is known to be caused by more than 100 different genetic mutations. RP, with any inheritance pattern, may be either familial (multiple family members affected) or isolated (only one affected person). In the non-sex-linked, or autosomal, form, it can either be a dominant or recessive trait. In the sex-linked form, called x-linked recessive, it is a recessive trait. This x-linked form is more severe than the autosomal forms. Two rare forms of RP are the digenic and mitochondrial forms.

Isolated RP cases represent 10–40% of all cases. Some of these cases may be the result of new gene mutations (changes in the genes). Other isolated cases are those in which the person has a relative with a mutation in the gene, but the relative is not affected by the condition.

Autosomal dominant RP (AdRP) occurs in about 15–25% of affected individuals. At least 12 different genes have been identified as causing AdRP. People with AdRP will usually have an affected parent. The risk for affected siblings or children is 50%.

Autosomal recessive RP (ArRP) occurs in about 5–20% of affected individuals. More than 16 genes have been identified that cause this type of RP. In ArRP, each parent of the affected person is a carrier of an abnormal gene that causes RP. Neither of these carrier parents is affected. There is a two-thirds chance that an unaffected sibling is a carrier of RP. All of the children of an affected person would be a carrier of the ArRP gene.

Five to 15% of individuals with RP have x-linked recessive RP (XLRP). Six different genes have been identified as the cause of this type of RP. Usually in this type of inheritance, males are affected carriers, while females are unaffected carriers or have a milder form of the disease. The mother may be a carrier of the mutation on the X-chromosome. It is also possible that a new mutation can occur for the first time in an affected person. For families with one affected male, there is a mathematical formula called the Bayesian analysis that can be applied to the family history. It takes into account the number of unaffected males to determine whether a female is likely to be a carrier or not. If a mother is a carrier, her children have a 50% chance of inheriting the RP gene. For affected males,



A fundus camera image showing the degeneration of the retina due to retinitis pigmentosa. The pattern of dark spots across the retina corresponds to the extent of loss of vision. (Custom Medical Stock Photo. Reproduced by permission.)

all of their daughters will be carriers but none of their sons will be affected.

The digenic form of RP occurs when the affected person has inherited one copy of an altered ROM1 gene from one parent and one copy of an altered peripherin/RDS gene from the other parent. The parents are asymptomatic. Mitochondrial inheritance occurs when the gene mutation is in a mitochondrial gene. People with this type of RP have progressive **hearing loss** and mild myopathy. Both of these types of RP are very rare.

The first symptoms, a loss of night vision followed by a loss of peripheral vision, usually begin in early adolescence or young adulthood. Occasionally, the loss of the ability to see color occurs before the loss of peripheral vision. Another possible symptom is seeing twinkling lights or small flashes of lights.

Diagnosis

When a person complains of a loss of night vision, a doctor will examine the interior of the eye with an ophthalmoscope to determine if there are changes in the retina. For people with advanced RP, the condition is characterized by the presence of clumps of black pigment in the inner retina (intraretinal). However, the appearance of the retina is not enough for an RP diagnosis since there are other disorders that may give the retina a similar appearance. There are also other reasons someone may have night blindness. Consequently, certain electrodiagnostic tests must be performed. An electroretinogram (ERG) determines the functional status of the photoreceptors by exposing the retina to light. The ERG uses a contact lens in

KEY TERMS

Ophthalmoscope—An instrument, with special lighting, designed to view structures in the back of the eye.

the eye, and the output is measured on a special instrument called an oscilloscope. The functional assessments of visual fields, visual acuity, or color vision may also be performed.

The diagnosis of RP can be established when the following criteria are met:

- rod dysfunction measured by dark adaptation test or ERG,
- progressive loss in photoreceptor function,
- loss of peripheral (side) vision,
- both eyes affected (bilaterality).

Molecular **genetic testing** is available on a research basis. Prenatal diagnosis for this condition has not yet been achieved.

Treatment

There are no medications or surgery to treat RP. However, researchers continue to seek possible treatments. In 2004, scientists injected stem cells to the back of mouse eyes and stopped retinal degeneration. Scientists are also exploring the possibility of retinal transplantation. Some doctors believe **vitamins A** and **E** will slightly slow the progression of the disease in some people. However, large doses of certain vitamins may be toxic and affected individuals should speak to their doctors before taking supplements.

If a person with RP must be exposed to bright sunlight, some doctors recommend wearing dark sunglasses to reduce the effect on the retina. Affected people should talk to their eye doctors about the correct lenses to wear outdoors.

Because there is no cure for RP, the affected person should be monitored for visual function and counseled about low-vision aids (for example, field-expansion devices). **Genetic counseling** is also appropriate. A three-generation family history with attention to other relatives with possible RP can help to clarify the inheritance pattern. For some people however, the inheritance pattern cannot be discerned.

Prognosis

There is no known cure for RP, which will eventually lead to blindness. The more severe forms will lead to blindness sooner than milder forms.

Resources

PERIODICALS

“Grant Boosts RP Research Into Transplantation.”

Ophthalmology Times August 1, 2004: 6.

“Stem Cells Delivered Into Back of Eye Hold Promise for People With Retinitis Pigmentosa, Other Retinal Degenerations; Potential Treatment for Untreatable Blindness Shows Promise in Mice.” *Ascribe Health News Service* September 15, 2004.

ORGANIZATIONS

American Academy of Ophthalmology. PO Box 7424, San Francisco, CA 94120-7424. (415) 561-8500. <<http://www.eyenet.org>>.

American Association of the Deaf-Blind. 814 Thayer Ave., Suite 302, Silver Spring, MD 20910. (301) 588-6545.

American Optometric Association. 243 North Lindbergh Blvd., St. Louis, MO 63141. (314) 991-4100. <<http://www.aoanet.org>>.

Foundation Fighting Blindness. Executive Plaza 1, Suite 800, 11350 McCormick Rd., Hunt Valley, MD 21031. (888) 394-3937. <<http://www.blindness.org>>.

National Retinitis Pigmentosa Foundation. 11350 McCormick Rd., Executive Plaza 1, Suite 800, Hunt Valley, MD 21031-1014. (800) 683-5555. <<http://www.blindness.org>>.

Prevent Blindness America. 500 East Remington Rd., Schaumburg, IL 60173. (800) 331-2020. <<http://www.prevent-blindness.org>>.

OTHER

Genetic Alliance. <<http://www.geneticalliance.org>>.

National Federation of the Blind. <<http://www.nfb.org>>.

“OMIM—Online Mendelian Inheritance in Man.” National Center for Biotechnology Information. <<http://www.ncbi.nlm.nih.gov/Omim/searchomim.html>>.

Retinitis Pigmentosa International. <<http://www.rpinternational.org>>.

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Description

The eye has three layers, the sclera, the choroid, and the retina. The sclera is the outer protective white coating of the eye. The choroid is the middle layer and contains blood vessels that nourish the eye. The front portion of the choroid is colored and is called the iris. The opening in the iris is called the pupil. The pupil is responsible for allowing light into the eye and usually appears black. When the pupil is exposed to bright light it contracts (closes), and when it is exposed to low light conditions it dilates (opens) so that the appropriate amount of light enters the eye. Light that enters through the pupil hits the lens of the eye. The lens then focuses the light onto the retina, the innermost of the three layers. The job of the retina is to transform the light into information that can be transmitted to the optic nerve, which will transmit this information to the brain. It is through this process that people are able to see the world around them.

Occasionally a tumor, called a retinoblastoma, will develop in the retina of the eye. Usually this tumor forms in young children but it can occasionally occur in adults. Most people with retinoblastoma develop only one tumor (unifocal) in only one eye (unilateral). Some, however, develop multiple tumors (multifocal) in one or both eyes. When retinoblastoma occurs independently in both eyes, it is then called bilateral retinoblastoma.

Occasionally, children with retinoblastoma develop trilateral retinoblastoma. Trilateral retinoblastoma results from the development of an independent **brain tumor** that often forms in a part of the brain called the pineal gland. In order for retinoblastoma to be classified as trilateral retinoblastoma, the tumor must have developed independently and not as the result of the spread of the retinal **cancer**. The prognosis for trilateral retinoblastoma is quite poor.

The retinal tumor which characterizes retinoblastoma is malignant, meaning that it can metastasize (spread) to other parts of the eye and eventually other parts of the body. In most cases, however, retinoblastoma is diagnosed before it spreads past the eye to other parts of the body (intraocular) and the prognosis is quite good. The prognosis is poorer if the cancer has spread beyond the eye (extraocular).

Retinoblastoma can be inherited or can arise spontaneously. Approximately 40% of people with retinoblastoma have an inherited form of the condition and approximately 60% have a sporadic (not inherited) form. Individuals with multiple independent tumors, bilateral retinoblastoma, or trilateral

Retinoblastoma

Definition

Retinoblastoma is a malignant tumor of the retina that occurs predominantly in young children.

retinoblastoma are more likely to be affected with the inherited form of retinoblastoma.

Approximately 1 in 15,000 to 1 in 30,000 infants in Western countries are born with retinoblastoma, making it the most common childhood **eye cancer**. It is, however, a relatively rare childhood cancer and accounts for approximately 3% of childhood cancers. The American Academy of Ophthalmology estimates that 300–350 cases of retinoblastoma occur in the United States each year.

Retinoblastoma is found mainly in children under the age of five but can occasionally be seen in older children and adults. Retinoblastoma is found in individuals of all ethnic backgrounds and is found equally frequently in males and females. The incidence of bilateral retinoblastoma in the United States is thought to be slightly higher among black children than among either Caucasian or Asian American children.

Causes and symptoms

Causes

Retinoblastoma is caused by changes in or absence of a gene called RB1. RB1 is located on the long arm of chromosome 13. Cells of the body, with the exception of the egg and sperm cells, contain 23 pairs of chromosomes. All of the cells of the body excluding the egg and the sperm cells are called the somatic cells. The somatic cells contain two of each chromosome 13 and therefore two copies of the RB1 gene. Each egg and sperm cell contains only one copy of chromosome and therefore only one copy of the RB1 gene.

RB1 produces a tumor suppressor protein that normally helps to regulate the cell cycle of cells such as those of the retina. A normal cell of the retina goes through a growth cycle during which it produces new cells. Genes such as tumor suppressor genes tightly regulate this growth cycle.

Cells that lose control of their cell cycle and replicate out of control are called cancer cells. These undergo many cell divisions, often at a quicker rate than normal cells, and do not have a limited lifespan. A group of adjacent cancer cells can form a mass called a tumor. Malignant (cancerous) tumors can spread to other parts of the body. A malignant tumor of the retina (retinoblastoma) can result when just one retinal cell loses control of its cell cycle and replicates out of control.

Normally the tumor suppressor protein produced by RB1 prevents a retinal cell from becoming

cancerous. Each RB1 gene produces tumor suppressor protein. Only one functioning RB1 gene in a retinal cell is necessary to prevent the cell from becoming cancerous. If both RB1 genes in a retinal cell become non-functional, then a retinal cell can become cancerous and retinoblastoma can result. An RB1 gene is non-functional when it is changed or missing (deleted) and no longer produces normal tumor suppressor protein.

Approximately 40% of people with retinoblastoma have inherited a non-functional or deleted RB1 gene from either their mother or father. Therefore, they have a changed/deleted RB1 gene in every somatic cell. A person with an inherited missing or non-functional RB1 gene will develop a retinal tumor if the remaining RB1 gene becomes changed or deleted in a retinal cell. The remaining RB1 gene can become non-functional when exposed to environmental triggers such as chemicals and radiation. In most cases, however, the triggers are unknown. Approximately 90% of people who inherit a changed or missing RB1 gene will develop retinoblastoma.

People with an inherited form of retinoblastoma are more likely to have a tumor in both eyes (bilateral) and are more likely to have more than one independent tumor (multifocal) in one or both eyes. The average age of onset for the inherited form of retinoblastoma is one year, which is earlier than the sporadic form of retinoblastoma. Although most people with the inherited form of retinoblastoma develop bilateral tumors, approximately 15% of people with a tumor in only one eye (unilateral) are affected with an inherited form of retinoblastoma.

A person with an inherited missing or non-functional RB1 gene has a 50% chance of passing on this abnormal gene to his or her offspring. The chance that their children will inherit the changed/deleted gene and actually develop retinoblastoma is approximately 45%.

Some people with retinoblastoma have inherited a non-functioning or missing RB1 gene from either their mother or father even though their parents have never developed retinoblastoma. It is possible that one parent has a changed or missing RB1 gene in every somatic cell but has not developed retinoblastoma because their remaining RB1 gene has remained functional. It is also possible that the parent had developed a retinal tumor that was destroyed by the body. In other cases, one parent has two normal RB1 genes in every somatic cell, but some of their egg or sperm cells contain a changed or missing RB1 gene. This is called gonadal mosaicism.

Retinoblastoma can also result when both RB1 genes become spontaneously changed or deleted in a retinal cell but the RB1 genes are normal in all the other cells of the body. Approximately 60% of people with retinoblastoma have this type of disease, called sporadic retinoblastoma. A person with sporadic retinoblastoma does not have a higher chance of having children with the disease. Their relatives do not have a higher risk of developing retinoblastoma themselves or having children who develop retinoblastoma. Sporadic retinoblastoma is usually unifocal and has an average age of onset of approximately two years.

Symptoms

The most common symptom of retinoblastoma is leukocoria. Leukocoria results when the pupil reflects a white color rather than the normal black or red color that is seen on a flash photograph. It is often most obvious in flash photographs; since the pupil is exposed to a lot of light and the duration of the exposure is so short, the pupil does not have time to constrict. Children with retinoblastoma can also have problems seeing and this can cause them to appear cross-eyed (**strabismus**). People with retinoblastoma may also experience red, painful, and irritated eyes, inflamed tissue around the eye, enlarged pupils, and possibly different-colored eyes.

Diagnosis

Children who have symptoms of retinoblastoma are usually first evaluated by their pediatrician. The pediatrician will often perform a red reflex test to diagnose or confirm leukocoria. Prior to this test the doctor inserts medicated eye drops into the child's eyes so that the pupils will remain dilated and not contract when exposed to bright light. The doctor then examines the eyes with an ophthalmoscope, which shines a bright light into the eyes and allows the doctor to check for leukocoria. Leukocoria can also be diagnosed by taking a flash Polaroid photograph of a patient who has been in a dark room for three to five minutes.

If the pediatrician suspects retinoblastoma on the basis of these evaluations, he or she will most likely refer the patient to an ophthalmologist (eye doctor) who has experience with retinoblastoma. The ophthalmologist will examine the eye using an indirect ophthalmoscope. The ophthalmoscope shines a bright light into the eye, which helps the doctor to visualize the retina. This evaluation is usually done under general anesthetic, although some very young or older patients may not require it. Prior to the

examination, medicated drops are put into the eyes to dilate the pupils, and anesthetic drops may also be used. A metal clip is used to keep the eyes open during the evaluation. During the examination, a cotton swab or a metal instrument with a flattened tip is used to press on the outer lens of the eye so that a better view of the front areas of the retina can be obtained. Sketches or photographs of the tumor as seen through the ophthalmoscope are taken during the procedure.

An ultrasound evaluation is used to confirm the presence of the tumor and to evaluate its size. Computed axial tomography (CT scan) is used to determine whether the tumor has spread outside of the eye and to the brain. Sometimes **magnetic resonance imaging** (MRI) is also used to look at the eyes, eye sockets, and the brain to see if the cancer has spread.

In most cases the cancer has not spread beyond the eye, and other evaluations are unnecessary. If the cancer appears to have spread beyond the eye, then other assessments such as a blood test, spinal tap (lumbar puncture), and/or **bone marrow biopsy** may be recommended. During a spinal tap, a needle is inserted between the vertebrae of the spinal column and a small sample of the fluid surrounding the spinal cord is obtained. In a bone marrow biopsy, a small amount of tissue (bone marrow) is taken from inside the hip or breast bone for examination.

Genetic testing

Establishing whether someone is affected with an inherited or non-inherited form of retinoblastoma can help to ascertain whether other family members such as siblings, cousins, and offspring are at increased risk for developing retinoblastoma. It can also sometimes help guide treatment choices, since patients with an inherited form of retinoblastoma may be at increased risk for developing recurrent tumors or other types of cancers, particularly when treated with radiation. It is helpful for the families of a child diagnosed with retinoblastoma to meet with a genetic specialist such as a genetic counselor and/or geneticist. These specialists can help to ascertain the chances that the retinoblastoma is inherited and facilitate **genetic testing** if desired.

If a patient with unilateral or bilateral retinoblastoma has a relative or relatives with retinoblastoma, it can be assumed that they have an inherited form of retinoblastoma. However, it cannot be assumed that a patient without a family history of the disease has a sporadic form.

Even when there is no family history, most cases of bilateral and trilateral retinoblastoma are inherited, as are most cases of unilateral, multifocal retinoblastoma. However, only 15% of unilateral, unifocal retinoblastoma cases are inherited.

The only way to establish whether someone has an inherited form of retinoblastoma is to see if the retinoblastoma gene is changed or deleted in the blood cells obtained from a blood sample. Approximately 5–8% of individuals with retinoblastoma possess a chromosomal abnormality involving the RB1 gene that can be detected by looking at their chromosomes under the microscope. The chromosomes can be seen by obtaining a blood sample. If this type of chromosomal abnormality is detected in a child, then analysis of the parents' chromosomes should be performed. If one of the parents possesses a chromosomal abnormality, then they are at higher risk for having other offspring with retinoblastoma. Chromosome testing would be recommended for the blood relatives of the parent with the abnormality.

Usually, however, a chromosomal abnormality is not detected in a child with retinoblastoma. In this case, specialized DNA tests that look for small RB1 gene changes need to be performed on the blood cells. DNA testing can be difficult, time consuming, and expensive, since there are many possible RB1 gene changes that can cause the gene to become nonfunctional.

If a sample of tumor is available, then it is recommended that DNA testing be performed on the tumor cells prior to DNA testing of the blood cells. This testing can usually identify the gene changes/deletions in the RB1 genes that caused the tumor to develop. In some cases, RB1 gene changes/deletions are not found in the tumor cells (as of 2001, approximately 20% of RB1 gene changes or deletions are not detectable). In these cases, DNA testing of the blood cells will not be able to ascertain whether someone is affected with an inherited or non-inherited form of retinoblastoma.

If the changes in both RB1 genes are detected in the tumor cell, then these same changes can be looked for in the blood cells. If an RB1 gene is deleted or changed in all of the blood cells tested, the patient can be assumed to have been born with a changed/deleted RB1 gene in all of their cells. This person has a 50% chance of passing the RB1 gene change/deletion on to his or her children. Most of the time, this change/deletion has been inherited from a parent. Occasionally the gene change/deletion occurred spontaneously in the original cell that was formed when the egg and sperm came together at conception (*de novo*).



This child's right eye is completely covered with a tumor associated with retinoblastoma. (Custom Medical Stock Photo. Reproduced by permission.)

If an RB1 gene change/deletion is found in all of the blood cells tested, both parents should undergo blood testing to check for the same RB1 gene change/deletion. If the RB1 gene change/deletion is identified in one of the parents, it can be assumed that the retinoblastoma was inherited and that siblings have a 50% chance of inheriting the altered gene. More distant blood relatives of the parent with the identified RB1 gene change/deletion may also be at risk for developing retinoblastoma. Siblings and other relatives could undergo DNA testing to see if they have inherited the RB1 gene change/deletion.

If the RB1 gene change/deletion is not identified in either parent, then the results can be more difficult to interpret. In this case, there is a 90–94% chance that the retinoblastoma was not inherited.

In some cases, a person with retinoblastoma will have an RB1 gene change/deletion detected in some of their blood cells and not others. It can be assumed that this person did not inherit the retinoblastoma from either parent. Siblings and other relatives would therefore not be at increased risk for developing retinoblastoma. Offspring would be at increased risk since some of the egg or sperm cells could have the changed/deleted RB1 gene. The risks to offspring would probably be less than 50%.

In families where there are multiple family members affected with retinoblastoma, blood samples from

multiple family members are often analyzed and compared through DNA testing. Ninety-five percent of the time, this type of analysis is able to detect patterns in the DNA that are associated with a changed RB1 gene in that particular family. When a pattern is detected, at-risk relatives can be tested to establish whether they have inherited an RB1 gene change/deletion.

PRENATAL TESTING. If chromosome or DNA testing identifies an RB1 gene/deletion in someone's blood cells, then prenatal testing can be performed on this person's offspring. An **amniocentesis** or **chorionic villus sampling** can be used to obtain fetal cells which can be analyzed for the RB1 gene change/deletion or chromosomal abnormality.

Treatment

A number of different classification (staging) systems are used to establish the severity of retinoblastoma and aid in choosing an appropriate treatment plan. The most widely used staging system is the Reese-Ellsworth system. This system is used to classify intraocular tumors and predict which tumors are favorable enough that sight can be maintained. The Reese-Ellsworth classification system is divided into:

- Group I (very favorable for maintenance of sight): small solitary or multiple tumors, less than 6.4 mm in size (1 inch = 25.4 mm), located at or below the equator of the eye
- Group II (favorable for maintenance of sight): solitary or multiple tumors, 6.4–16 mm in size, located at or behind the equator of the eye
- Group III (possible for maintenance of sight): any tumor located in front of the equator of the eye, or a solitary tumor larger than 16 mm in size and located behind the equator of the eye
- Group IV (unfavorable for maintenance of sight): multiple tumors, some larger than 16 mm in size, or any tumor extending in front of the outer rim of the retina (ora serrata)
- Group V (very unfavorable for maintenance of sight): large tumors involving more than half of the retina, or vitreous seeding, in which small pieces of tumor are broken off and floating around the inside of the eye

When choosing a treatment plan, the first important criteria to ascertain is whether the cancer is localized within the eye (intraocular) or has spread to other parts of the body (extraocular). An intraocular retinoblastoma may only involve the retina or could involve other parts of the eye. An extraocular

retinoblastoma could involve only the tissues around the eye or could result from the spread of cancer to the brain or other parts of the body.

It is also important to establish whether the cancer is unilateral (one eye) or bilateral (both eyes), multifocal or unifocal. In order for the tumors to be considered multifocal, they must have arisen independently and not as the result of the spread of cancer cells. It is also important to check for trilateral retinoblastoma.

Treatments

The treatment chosen depends on the size and number of tumors, whether the cancer is unilateral or bilateral, and whether the cancer has spread to other parts of the body. The goal of treatment is to cure the cancer and prevent as much loss of vision as possible. Since the late 1990s, doctors treating patients with retinoblastoma have tended to avoid enucleation and external beam **radiation therapy** whenever possible, in favor of **chemotherapy** to reduce the tumor in addition to focal therapies. Improved methods of chemoreduction have led to increasing success in saving patients' eyes, often with some visual function.

TREATMENT OF INTRAOCULAR TUMORS. Surgical removal of the affected eye (enucleation) is used when the tumor(s) are so large and extensive that preservation of sight is not possible. This surgery is performed under general anesthetic and usually takes less than an hour. Most children who have undergone this surgery can leave the hospital on the same day. A temporary ball is placed in the eye socket after the surgery. Approximately three weeks after the operation, a plastic artificial eye (prosthesis) that looks like the normal eye is inserted into the eye socket.

Radiation therapy is often used for treatment of large tumors when preservation of sight is possible. External beam radiation therapy involves focusing a beam of radiation on the eye. If the tumor has not spread extensively, the radiation beam can be focused on the cancerous retinal cells. If the cancer is extensive, radiation treatment of the entire eye may be necessary. External beam radiation is performed on an outpatient basis and usually occurs over a period of three to four weeks. Some children may need sedatives prior to the treatment. This type of therapy can result in a temporary loss of a patch of hair on the back of the head and a small area of "sun-burned" skin. Long-term side effects of radiation treatment can include **cataracts**, vision problems, bleeding from the retina, and decreased growth of the bones on the side of the head. People with an inherited form of retinoblastoma

have an increased risk of developing other cancers as a result of this therapy. Some consideration should therefore be given to alternative treatment therapies for those with an inherited form of retinoblastoma.

Photocoagulation therapy is often used in conjunction with radiation therapy but may be used alone to treat small tumors that are located on the back of the eye. Photocoagulation involves using a laser to destroy the cancer cells. This type of treatment is done under local or **general anesthesia** and is usually not associated with post-procedural **pain**.

Thermotherapy is also often used in conjunction with radiation therapy or drug therapy (chemotherapy). Thermotherapy involves the use of heat to help shrink tumor cells. The heat is either used on the whole eye or localized to the tumor area. It is done under local or general anesthesia and is usually not painful.

Cryotherapy is a treatment often used in conjunction with radiation therapy but can also be used alone on small tumors located on the front part of the retina. Cryotherapy involves the use of intense cold to destroy cancer cells and can result in harmless, temporary swelling of the external eye and eyelids that can last for up to five days. Eye drops or ointment are sometimes provided to reduce the swelling.

Brachytherapy involves the application of radioactive material to the outer surface of the eye at the base of the tumor. It is generally used for tumors of medium size. A patient undergoing this type of procedure is usually hospitalized for three to seven days. During that time, he or she undergoes one surgery to attach the radioactive material and one surgery to remove it. Eye drops are often administered for three to four weeks following the operation to prevent inflammation and infection. The long-term side effects of this treatment can include cataracts and damage to the retina, which can lead to impaired vision.

Intravenous treatment with one or more drugs (chemotherapy) is often used for treatment of both large and small tumors. Chemotherapy is sometimes used to shrink tumors prior to other treatments such as radiation therapy or brachytherapy. Occasionally, it is also used alone to treat very small tumors.

TREATMENT OF INTRAOCULAR AND UNILATERAL RETINOBLASTOMA. Often, by the time that unilateral retinoblastoma is diagnosed, the tumor is so large that useful vision cannot be preserved. In these cases removal of the eye (enucleation) is the treatment of choice. Other therapies are unnecessary if enucleation is used to treat intraocular unilateral retinoblastoma. If the tumor is small enough, other therapies such as external beam radiation therapy, photocoagulation,

cryotherapy, thermotherapy, chemotherapy, and brachytherapy may be considered.

TREATMENT OF INTRAOCULAR AND BILATERAL RETINOBLASTOMA. If vision can be preserved in both eyes, radiation therapy of both eyes may be recommended. Smaller, more localized tumors can sometimes be treated by local therapies such as cryotherapy, photocoagulation therapy, thermotherapy or brachytherapy. Some centers may use chemotherapy in place of radiation therapy when the tumors are too large to be treated by local therapies or are found over the optic nerve of the eye. Many centers are moving away from radiation treatment and toward chemotherapy because it is less likely to induce future tumors. Enucleation is performed on the more severely affected eye if sight cannot be preserved in both.

EXTRAOCULAR RETINOBLASTOMA. There is no proven effective therapy for the treatment of extraocular retinoblastomas. Commonly, radiation treatment of the eyes and chemotherapy is provided.

Alternative treatment

There are no alternative or complementary therapies specific to the treatment of retinoblastoma. Since most people diagnosed with retinoblastoma are small children, most drug-based alternative therapies designed to treat general cancer would not be recommended. Many specialists would, however, **stress** the importance of establishing a well-balanced diet, including certain fruits, vegetables, and vitamin supplements, to ensure that the body is strengthened in its fight against cancer. Some advocate the use of visualization strategies, in which patients would visualize the immune cells of their body attacking and destroying the cancer cells.

Prognosis

Individuals with intraocular retinoblastoma who do not have trilateral retinoblastoma usually have a good survival rate with a 90% chance of disease-free survival for five years. Those with extraocular retinoblastoma have less than a 10% chance of disease-free survival for the same amount of time. Trilateral retinoblastoma generally has a very poor prognosis. Patients with trilateral retinoblastoma who receive treatment have an average survival rate of approximately eight months, while those who remain untreated have an average survival rate of approximately one month. Patients with trilateral retinoblastoma who are asymptomatic at the time of diagnosis

may have a better prognosis than those who experience symptoms.

Patients with an inherited form of unilateral retinoblastoma have a 70% chance of developing retinoblastoma in the other eye. Retinoblastoma reoccurs in the other eye in approximately 5% of people with a non-inherited form of retinoblastoma, so it is advisable for even these patients to be closely monitored. People with an inherited form of retinoblastoma who have not undergone radiation treatment have approximately a 26% chance of developing cancer in another part of the body within 50 years of the initial diagnosis. Those with an inherited form who have undergone radiation treatment have a 58% chance of developing a secondary cancer by 50 years after the initial diagnosis. Most of the secondary cancers are skin cancers, bone tumors (osteosarcomas), and soft-tissue **sarcomas**. Soft-tissue sarcomas are malignant tumors of the muscle, nerves, joints, blood vessels, deep skin tissues, or fat. The prognosis for retinoblastoma patients who develop secondary cancers, however, is very poor as of the early 2000s.

Survivors of retinoblastoma are likely to have visual field defects after their cancer treatment is completed, most commonly scotomas, which are areas of lost or depressed vision within an area of normal vision. The size and type of these visual defects are determined by the size and type of the original tumor and the form of therapy used to treat it.

Prevention

Although retinoblastoma cannot be prevented, appropriate screening and surveillance should be applied to all at-risk individuals to ensure that the tumor(s) are diagnosed at an early stage. The earlier the diagnosis, the more likely that an eye can be salvaged and vision maintained.

Screening of people diagnosed with retinoblastoma

Children who have been diagnosed with retinoblastoma should receive periodic dilated retinal examinations until the age of five. Young children will need to undergo these evaluations under anesthetic. After five years of age, periodic eye examinations are recommended. It may be advisable for patients with bilateral retinoblastoma or an inherited form of retinoblastoma to undergo periodic screening for the brain tumors found in trilateral retinoblastoma. There are no specific screening protocols designed to detect non-ocular tumors. All lumps and complaints of bone pain, however, should be thoroughly evaluated.

Screening of relatives

When a child is diagnosed with retinoblastoma, it is recommended that parents and siblings receive a dilated retinal examination by an ophthalmologist who is experienced in the diagnosis and treatment of the disease. It is also recommended that siblings continue to undergo periodic retinal examinations under anesthetic until they are three years of age. From three to seven years of age, periodic eye examinations are recommended. The retinal examinations can be avoided if DNA testing indicates that the patient has a non-inherited form of retinoblastoma or if the sibling has not inherited the RB1 gene change/deletion. Any relatives who are found through DNA testing to have inherited an RB1 gene change/deletion should undergo the same surveillance procedures as siblings.

The children of someone diagnosed with retinoblastoma should also undergo periodic retinal examinations under anesthetic. Retinal surveillance should be performed unless DNA testing proves that their child does not possess the RB1 gene change/deletion. If desired, prenatal detection of tumors using ultrasound may also be performed. During the ultrasound procedure, a hand-held instrument is placed on the maternal abdomen or inserted vaginally. The ultrasound produces sound waves that are reflected back from the body structures of the fetus, producing a picture that can be seen on a video screen. If a tumor is detected through this evaluation, the affected baby may be delivered a couple of weeks earlier. This can allow for earlier intervention and treatment.

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KEY TERMS

Amniocentesis—Prenatal testing performed at 16 to 20 weeks of pregnancy that involves inserting a needle through the abdomen of a pregnant mother and obtaining a small sample of fluid from the amniotic sack, which contains the fetus. Often is used to obtain a sample of the fetus' cells for biochemical or DNA testing.

Benign tumor—An abnormal proliferation of cells that does not spread to other parts of the body.

Bilateral—Affecting both eyes.

Brachytherapy—Cancer treatment that involves the application of radioactive material to the site of the tumor.

Cryotherapy—Cancer treatment in which the tumor is destroyed by exposure to intense cold.

Chromosome—A microscopic structure found within each cell of the body, made of a complex of proteins and DNA.

Chorionic villus sampling (CVS)—Prenatal testing performed at 10 to 12 weeks of pregnancy, which involves inserting a catheter through the vagina of a pregnant mother or inserting a needle through the abdomen of the mother and obtaining a sample of placenta. Often is used to obtain a sample of the fetus' cells for biochemical or DNA testing.

DNA (deoxyribonucleic acid)—The hereditary material that makes up genes; influences the development and functioning of the body.

DNA testing—Testing for a change or changes in a gene or genes.

Enucleation—Surgical removal of the eye.

Equator—Imaginary line encircling the eyeball and dividing the eye into a front and back half.

Extraocular retinoblastoma—Cancer that has spread from the eye to other parts of the body.

Gene—A building block of inheritance, made up of a compound called DNA (deoxyribonucleic acid) and containing the instructions for the production of a particular protein. Each gene is found in a specific location on a chromosome.

Intraocular retinoblastoma—Cancer that is limited to the eye and has not spread to other parts of the body.

Malignant tumor—An abnormal proliferation of cells that can spread to other sites.

Multifocal—More than one tumor present.

Ophthalmologist—Physician specializing in the diseases of the eye.

Optic nerve—The part of the eye which contains nerve fibers that transmit signals from the eye to the brain.

Oncologist—A physician specializing in the diagnosis and treatment of cancer

Photocoagulation—Cancer treatment in which the tumor is destroyed by an intense beam of laser light.

Prenatal testing—Testing for a disease such as a genetic condition in an unborn baby.

Protein—A substance produced by a gene that is involved in creating the traits of the human body, such as hair and eye color, or is involved in controlling the basic functions of the human body, such as control of the cell cycle.

Retina—The light-sensitive layer of the eye that receives images and sends them to the brain.

Scotoma—An area of lost or depressed vision within the visual field surrounded by an area of normal vision. Survivors of retinoblastoma frequently develop scotomas.

Somatic cells—All the cells of the body with the exception of the egg and sperm cells.

Tumor—A growth of tissue resulting from the uncontrolled proliferation of cells.

Tumor-suppressor gene—Gene involved in controlling normal cell growth and preventing cancer.

Unifocal—Only one tumor present in one eye.

Unilateral—Affecting only one eye.

Vitreous—The transparent gel that fills the back part of the eye.

Vitreous seeding—When small pieces of tumor have broken off and are floating around the vitreous.

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- American Academy of Ophthalmology (AAO). P. O. Box 7424, San Francisco, CA 94120-7424. (415) 561-8500. Fax: (415) 561-8533. <<http://www.aao.org>>.
- Institute for Families with Blind Children. PO Box 54700, Mail Stop 111, Los Angeles, CA 90054-0700. (213) 669-4649.
- National Retinoblastoma Parents Group. PO Box 317, Watertown, MA 02471. (800) 562-6265. Fax: (617) 972-7444. napvi@perkins.pvt.k12.ma.us.
- Retinoblastoma International. 4650 Sunset Blvd., Mail Stop #88, Los Angeles, CA 90027. (323) 669-2299. info@retinoblastoma.net. <http://www.retinoblastoma.net/rbi/index_rbi.htm>.
- Retinoblastoma Society. Saint Bartholomew's Hospital, London, UK EC1A 7BE. Phone: 020 7600 3309 Fax: 020 7600 8579. <<http://ds.dial.pipex.com/rbinfo>>.

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Retinol deficiency **see** **Vitamin A deficiency**

Retinopathies

Definition

Retinopathy is a noninflammatory disease of the retina. There are many causes and types of retinopathy.

Description

The retina is the thin membrane that lines the back of the eye and contains light-sensitive cells

(photoreceptors). Light enters the eye and is focused onto the retina. The photoreceptors send a message to the brain via the optic nerve. The brain then "interprets" the electrical message sent to it, resulting in vision. The macula is a specific area of the retina responsible for central vision. The fovea is about 1.5 mm in size and located in the macula. The fovea is responsible for sharp vision. When looking at something, the fovea should be directed at the object.

Retinopathy, or damage to the retina, has various causes. A hardening or thickening of the retinal arteries is called arteriosclerotic retinopathy. High blood pressure in the arteries of the body can damage the retinal arteries and is called hypertensive retinopathy. The spreading of a **syphilis** infection to the retinal blood vessels causes syphilitic retinopathy, and diabetes damages the retinal vessels resulting in a condition called diabetic retinopathy. Sick cell anemia also affects the blood vessels in the eyes. Exposure to the sun (or looking at the sun during an eclipse) can cause damage (solar retinopathy), as well as certain drugs (for example, chloroquine, thioridazine, and large doses of tamoxifen). The arteries and veins can become blocked, thus resulting in a retinal artery or vein occlusion. These are just some of the causes of the various retinopathies.

Retinopathies are divided into two broad categories, *simple* or *nonproliferative* retinopathies and *proliferative* retinopathies. The simple retinopathies include the defects identified by bulging of the vessel walls, by bleeding into the eye, by small clumps of dead retinal cells called cotton wool exudates, and by closed vessels. This form of retinopathy is considered mild. The proliferative, or severe, forms of retinopathies include the defects identified by newly grown blood vessels, by scar tissue formed within the eye, by closed-off blood vessels that are badly damaged, and by the retina breaking away from the mesh of blood vessels that nourish it (**retinal detachment**).

While each disease has its own specific effect on the retina, a general scenario for many of the retinopathies is as follows (note: not all retinopathies necessarily affect the blood vessels). Blood flow to the retina is disrupted, either by blockage or breakdown of the various vessels. This can lead to bleeding (hemorrhage) and fluids, cells, and proteins leaking into the area (exudates). There can be a lack of oxygen to surrounding tissues (hypoxia) or decreased blood flow (**ischemia**). Chemicals produced by the body then can cause new blood vessels to grow (neovascularization), however, these new vessels generally leak and cause more problems. Neovascularization even can grow on the colored part of the eye (iris). The retina can swell and vision will be affected.

Diabetic retinopathy is the leading cause of blindness in people ages 20 to 74. Diabetic retinopathy will occur in 90% of persons with type 1 diabetes (insulin-dependent, or insulin requiring) and 65% of persons with type II diabetes (non-insulin-dependent, or not requiring insulin) by about 10 years after the beginning of diabetes. In the United States, new cases of blindness most often are caused by diabetic retinopathy. Among these new cases of blindness, 12% are people between the ages of 20 to 44 years, and 19% are people between the ages of 45 to 64 years.

Causes and symptoms

There are many causes of retinopathy. Some of the more common ones are listed below.

Diabetic retinopathy

Diabetes is a complex disorder characterized by an inability of the body to properly regulate the levels of sugar and insulin (a hormone made by the pancreas) in the blood. As diabetes progresses, the blood vessels that feed the retina become damaged in different ways. The damaged vessels can have bulges in their walls (aneurysms); they can leak blood into the surrounding jelly-like material (vitreous) that fills the inside of the eyeball; they can become completely closed; or new vessels can begin to grow where there would not normally be blood vessels. However, although these new blood vessels are growing in the eye, they cannot nourish the retina and they bleed easily, releasing blood into the inner region of the eyeball, which can cause dark spots and cloudy vision.

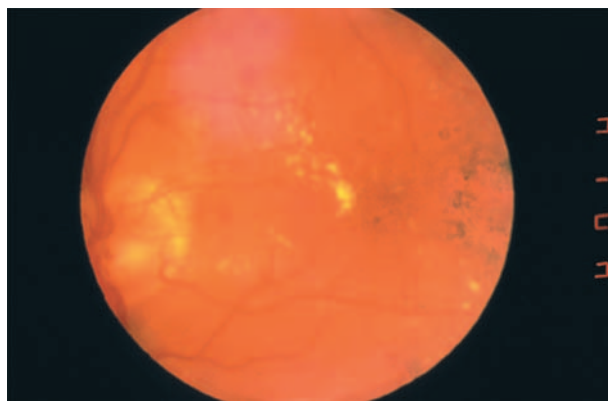
Diabetic retinopathy begins prior to any outward signs of disease being noticed. Once symptoms are noticed, they include poorer than normal vision, fluctuating or distorted vision, cloudy vision, dark spots, episodes of temporary blindness, or permanent blindness.

Hypertensive retinopathy

High blood pressure can affect the vessels in the eyes. Some blood vessels can narrow. The blood vessels can thicken and harden (arteriosclerosis). There will be flame-shaped hemorrhages and macular swelling (**edema**). This edema may cause distorted or decreased vision.

Sickle cell retinopathy

Sickle cell anemia occurs mostly in blacks and is a hereditary disease that affects the red blood cells. The



A slit lamp view of a human eye with diabetic retinopathy.
(Custom Medical Stock Photo. Reproduced by permission.)

sickle-shaped blood cell reduces blood flow. People will not have visual symptoms early in the disease. However, patients need to be followed closely in case neovascularization occurs.

Retinal vein and artery occlusion

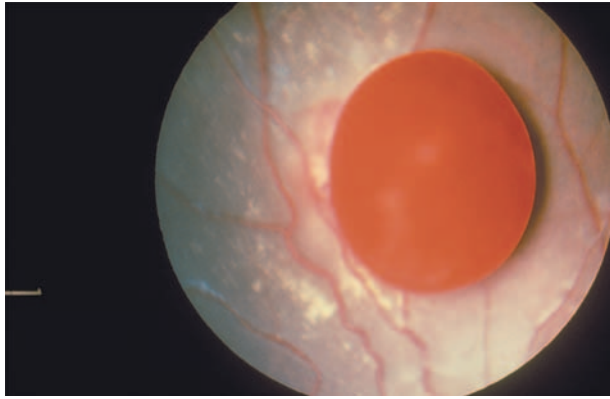
Retinal vein occlusion generally occurs in the elderly. There is usually a history of other systemic disease, such as diabetes or high blood pressure. The central retinal vein (CRV), or the retinal veins branching off of the CRV, can become compressed, thus stopping the drainage of blood from the retina. This may occur if the central retinal artery hardens.

Symptoms of retinal vein occlusion include a sudden, painless loss of vision or field of vision in one eye. There may be a sudden onset of floating spots (floaters) or flashing lights. Vision may be unchanged or decrease dramatically.

Retinal artery occlusion generally is the result of an **embolism** that dislodges from somewhere else in the body and travels to the eye. Transient loss of vision may precede an occlusion. Symptoms of a central retinal artery or branch occlusion include a sudden, painless loss of vision or decrease in visual field. Ten percent of the cases of a retinal artery occlusion occur because of giant cell arteritis (a chronic vascular disease).

Solar retinopathy

Looking directly at the sun or watching an eclipse can cause damage. There may be a loss of the central visual field or decreased vision. The symptoms can occur hours to days after the incident.



A close-up view of a human eye following retinal hemorrhage.
(Custom Medical Stock Photo. Reproduced by permission.)

Drug-related retinopathies

Certain medications can affect different areas of the retina. Doses of 20–40 mg a day of tamoxifen usually do not cause a problem, but much higher doses may cause irreversible damage.

Patients taking chloroquine for lupus, **rheumatoid arthritis**, or other disorders may notice a decrease in vision. If so, discontinuing medication will stop, but not reverse, any damage. However, patients should never discontinue medication without the advice of their physician.

Patients taking thioridazine may notice a decrease in vision or color vision.

These drug-related retinopathies generally only affect patients taking large doses. However, patients need to be aware if any medication they are taking will affect the eyes. Patients should inform their doctors of any visual effects.

Diagnosis

The damaged retinal blood vessels and other retinal changes are visible to an eye doctor when an examination of the retina (fundus exam) is done. This can be done using a hand-held instrument called an ophthalmoscope or another instrument called a binocular indirect ophthalmoscope. This allows the doctor to see the back of the eye. Certain retinopathies have classic signs (for example, vascular “sea fans” in sickle cell, dot and blot hemorrhages in diabetes, flame-shaped hemorrhages in high blood pressure). Patients then may be referred for other tests to confirm the underlying cause of the retinopathy. These tests include blood tests and measurement of blood pressure.

Fluorescein **angiography**, where a dye is injected into the patient and the back of the eyes are viewed and photographed, helps to locate leaky vessels. Sometimes patients may become nauseated from the dye.

A newer diagnostic method called digital retinal photography can be used to screen those at high risk for retinopathies, in particular, diabetics. Some researchers say the technique could lead to more cost-effective screening for people with diabetic retinopathy.

Treatment

Retinal specialists are ophthalmologists who specialize in retinal disorders. Retinopathy is a disorder of the retina that can result from different underlying systemic causes, so general physicians should be consulted as well. For drug-related retinopathies, the treatment generally is discontinuation of the drug (only under the care of a physician).

Surgery with lasers can help to prevent blindness or lessen any losses in vision. The high-energy light from a laser is aimed at the weakened blood vessels in the eye, destroying them. **Scars** will remain where the laser treatment was performed. For that reason, laser treatment cannot be performed everywhere. For example, laser photocoagulation at the fovea would destroy the area for sharp vision. Panretinal photocoagulation may be performed. This is a larger area of treatment in the periphery of the retina; hopefully it will decrease neovascularization. Prompt treatment of proliferative retinopathy may reduce the risk of severe vision loss by 50%.

Patients with retinal artery occlusion should be referred to a cardiologist. Patients with retinal vein occlusion need to be referred to a physician, as they may have an underlying systemic disorder, such as high blood pressure.

Prognosis

Nonproliferative retinopathy has a better prognosis than proliferative retinopathy. Prognosis depends on the extent of the retinopathy, the cause, and promptness of treatment.

Prevention

Complete eye examinations done regularly can help detect early signs of retinopathy. Patients on certain medications should have more frequent eye exams. They also should have a baseline eye exam

KEY TERMS

Exudate—Cells, protein, fluid, or other material that passes through blood vessel walls to accumulate in the surrounding tissue.

Neovascularization—New blood vessel formation—usually leaky vessels.

Nonproliferative retinopathy—Retinopathy without the growth of new blood vessels.

Proliferative retinopathy—Retinopathy with the growth of new blood vessels (neovascularization).

when starting the drug. People with diabetes must take extra care to have thorough, periodic eye exams, especially if early signs of **visual impairment** are noticed. A 2003 report recommended re-screening eye exams every two years for diabetics whose blood sugar had remained in control, and more frequent exams if visual symptoms appear. Anyone experiencing a sudden loss of vision, decrease in vision or visual field, flashes of light, or floating spots should contact their eye doctor right away.

Proper medical treatment for any of the systemic diseases known to cause retinal damage will help prevent retinopathy. For diabetics, maintaining proper blood sugar and blood pressure levels is important as well; however, some form of retinopathy usually will occur in diabetics, given enough time. A proper diet, particularly for those persons with diabetes, and stopping **smoking** also will help delay retinopathy.

Frequent, thorough eye exams and control of systemic disorders are the best prevention.

Resources

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ORGANIZATIONS

American Academy of Ophthalmology. 655 Beach Street, P.O. Box 7424, San Francisco, CA 94120-7424. <<http://www.eyenet.org>>.

American Diabetes Association. 1701 North Beauregard Street, Alexandria, VA 22311. (800) 342-2383. <<http://www.diabetes.org>>.

American Optometric Association. 243 North Lindbergh Blvd., St. Louis, MO 63141. (314) 991-4100. <<http://www.aoanet.org>>.

Foundation Fighting Blindness. Executive Plaza I, Suite 800, 11350 McCormick Road, Hunt Valley, MD 21031-1014. (888) 394-3937. <<http://www.blindness.org>>.

Prevent Blindness America. 500 East Remington Road, Schaumburg, IL 60173. (800) 331-2020. <<http://www.preventblindness.org>>.

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Retrocaval ureter see **Congenital ureter anomalies**

Retrograde cystography

Definition

A retrograde cystogram provides x-ray visualization of the bladder with injection of sterile dye.

Purpose

A retrograde cystogram is performed to evaluate the structure of the bladder and identify bladder disorders, such as tumors, or recurrent urinary tract infections. The presence of urine reflux (backward flow) into the ureters may also be visualized with this x-ray study.

Precautions

The doctor should be made aware of any previous history of reactions to shellfish, iodine, or any iodine-containing foods or dyes. Allergic reactions during previous dye studies is not necessarily a contraindication, as dye is not infused into the bloodstream for this study. Other conditions to be considered by the physician prior to proceeding with the test include active urinary tract infection, **pregnancy**, recent bladder surgery, or presence of obstruction that interferes with passage of a urinary catheter.

Description

After administration of anesthesia, the doctor will insert a thin, tubelike instrument called a catheter through the patient's urethra and into the bladder. The contrast medium is then injected through the catheter into the bladder. X-ray pictures are taken at various stages of filling, from various angles, to visualize the bladder. Additional films are taken after

KEY TERMS

Bladder—A balloon-like organ located in the lower abdomen that stores urine.

Catheter—A thin tube used to inject or withdraw fluids from the body.

Stones—Also known as calculi, stones result from an excessive build-up of mineral crystals in the kidney. Symptoms of stones include intense pain in the lower back or abdomen, urinary tract infection, fever, burning sensation on urination, and/or blood in the urine.

Ureter—Tube that carries urine from the kidney to the bladder.

Urethra—Tube that empties urine from the bladder to outside the body.

drainage of the dye. The procedure takes approximately one to one and one-half hours and the patient may be asked to wait while films are developed.

Alternately, instead of a contrast dye and x-ray pictures, the test can be done with a radioactive tracer and a different camera. This is known as a “radio-nuclide” retrograde cystogram.

Preparation

The patient will be required to sign a consent form after the risks and benefits of the procedure have been explained. **Laxatives** or **enemas** may be necessary before the procedure, as the bowel must be relatively empty of stool and gas to provide visualization of the urinary tract. Immediately before the procedure, the patient should remove all clothing and jewelry, and put on a surgical gown.

Aftercare

Sometimes, pulse, blood pressure, breathing status, and temperature are checked at regular intervals after the procedure, until they are stable. The patient may have some burning on urination for a few hours after the test, due to the irritation of the urethra from the catheter. The discomfort can be reduced by liberal fluid intake, in order to dilute the urine. The appearance and amount of urine output should be noted, and the doctor should be notified if blood appears in the urine after three urinations. Also, patients should report any signs of urinary infection, including chills, **fever**, rapid pulse, and rapid breathing rate.

Normal results

A normal result would reveal no anatomical or functional abnormalities.

Abnormal results

Abnormal results may indicate:

- stones
- blood clots
- tumors
- reflex (urine passing backward from the bladder into the ureters)

Resources

ORGANIZATIONS

American Kidney Fund (AKF). Suite 1010, 6110 Executive Boulevard, Rockville, MD 20852. (800) 638-8299.

< <http://216.248.130.102/Default.htm> >.

National Kidney Foundation. 30 East 33rd St., New York, NY 10016. (800) 622-9010. < <http://www.kidney.org> >.

Kathleen D. Wright, RN

Retrograde ureteropyelography

Definition

A retrograde ureteropyelogram provides x-ray visualization of the bladder, ureters, and the kidney (renal) pelvis by injection of sterile dye into the renal collecting system.

Purpose

A retrograde ureteropyelogram is performed to determine the exact location of a ureteral obstruction when it cannot be visualized on an intravenous pyelogram (a dye is injected and an x ray taken of the kidneys and the tubes that carry urine to the bladder). This may occur due to poor renal function and inadequate excretion of the contrast medium (dye).

Precautions

The doctor should be made aware of any previous history of reactions to shellfish, iodine, or any iodine-containing foods or dyes. Allergic reactions during previous dye studies is not necessarily a

contraindication, as dye is not infused into the bloodstream for this study. Other conditions to be considered by the physician prior to proceeding with the test include **pregnancy** and active urinary tract infection.

Description

After administration of anesthesia, the doctor will insert a thin, tubelike instrument (catheter) through the patient's urethra and into the bladder. A catheter is then placed into the affected ureter to instill the contrast medium. X-ray pictures are taken to visualize the ureter. If complete obstruction is found, a ureteral catheter may be left in place and secured to an indwelling urethral catheter to facilitate drainage of urine. The procedure takes approximately one hour.

Preparation

Laxatives or **enemas** may be necessary before the procedure, as the bowel must be relatively empty to provide visualization of the urinary tract. When **general anesthesia** is used for insertion of the ureteral catheter, there should be no eating and drinking after midnight prior to the procedure.

Aftercare

Even if no catheters are left in place after the procedure, the patient may have some burning on urination for a few hours after the procedure due to the irritation of the urethra. The discomfort can be reduced by liberal fluid intake, in order to dilute the urine. The appearance and amount of urine output should be noted for 24 hours after the procedure. If a stone was found, all urine should be strained to allow chemical analysis of any stones passed spontaneously. This will allow the doctor to provide advice on measures to prevent recurrent stone formation. **Antibiotics** are usually given after the procedure to prevent urinary tract infection.

Normal results

A normal result would reveal no anatomical or functional abnormalities.

Abnormal results

Abnormal results may indicate:

- congenital abnormalities
- fistulas or false passages
- renal stones

- strictures
- tumors

Resources

ORGANIZATIONS

American Kidney Fund (AKF). Suite 1010, 6110 Executive Boulevard, Rockville, MD 20852. (800) 638-8299. < <http://216.248.130.102/Default.htm> >.

National Kidney Foundation. 30 East 33rd St., New York, NY 10016. (800) 622-9010. < <http://www.kidney.org> >.

Kathleen D. Wright, RN

Retrograde urethrography

Definition

Retrograde urethrography involves the use of x-ray pictures to provide visualization of structural problems or injuries to the urethra.

Purpose

Retrograde urethrography is used, in combination with a doctor's observation and other tests, to establish a diagnosis for individuals, almost exclusively men, who may have structural problems of the urethra.

Precautions

The doctor should be made aware of any previous history of reactions to shellfish, iodine, or any iodine-containing foods or dyes. An earlier allergic reaction during a dye study is not necessarily something that makes the test inadvisable (a contraindication) as no dye is injected into the bloodstream for this study. Other conditions that should be considered by the physician before the test is done include **pregnancy**, recent urethral surgery, or severe inflammation of the urethra, bladder, or prostate.

Description

The urethra is first visually examined by the doctor, and the opening is cleansed with an antiseptic solution. A flexible rubber or plastic catheter is then inserted into the urethra, and dye is injected into the catheter. A clamp is applied to hold the dye in place while x-ray pictures are taken of the urethral structure. The clamp and catheter are then removed. The

KEY TERMS

Bladder—The balloonlike organ in the lower abdomen that holds urine.

Catheter—Tube used to inject into or withdraw fluids from the bladder.

Renal—Relating to the kidneys, from the Latin word for kidneys, *renes*.

Urethra—Tube that carries the urine from the bladder out of the body.

Visualization—The process of making an internal organ visible. A radiopaque substance is introduced into the body, then an x-ray picture of the desired organ is taken.

procedure takes approximately 15 minutes. However, the patient may be asked to wait while films are developed, which also permits the patient to be observed for any immediate side effects from the dye. The test may be performed in a hospital, doctor's office, outpatient center, or freestanding surgical facility. The time involved for reporting of test results to the doctor may vary from a few minutes to a few days.

Preparation

The patient will be asked to sign a consent form after the risks and benefits of the procedure have been explained. No diet or activity changes are necessary in preparation for the procedure. The patient will be asked to remove all clothing and put on a surgical gown before the test begins.

Normal results

The presence of no anatomical or functional abnormalities is considered a normal result.

Abnormal results

Abnormal findings may indicate:

- congenital abnormalities
- fistulas or false passages
- lacerations
- strictures
- valves, known as “posterior urethral valves”
- tumors

Resources

ORGANIZATIONS

American Kidney Fund (AKF). Suite 1010, 6110 Executive Boulevard, Rockville, MD 20852. (800) 638-8299. < <http://216.248.130.102/Default.htm> >.

National Kidney Foundation. 30 East 33rd St., New York, NY 10016. (800) 622-9010. < <http://www.kidney.org> >.

Kathleen D. Wright, RN

Retrograde urography see **Retrograde urethrography**

Retropharyngeal abscess see **Abscess**

Reye's syndrome

Definition

Reye's syndrome is a disorder principally affecting the liver and brain, marked by rapid development of life-threatening neurological symptoms.

Description

Reye's syndrome is an emergency illness chiefly affecting children and teenagers. It almost always follows a viral illness such as a cold, the flu, or chicken pox. Reye's syndrome may affect all the organs of the body, but most seriously affects the brain and liver. Rapid development of severe neurological symptoms, including lethargy, confusion, seizures, and **coma**, make Reye's syndrome a life-threatening emergency.

Reye's syndrome is a rare illness, even rarer now than when first described in the early 1970s. The incidence of the disorder peaked in 1980, with 555 cases reported. The number of cases declined rapidly thereafter due to decreased use of **aspirin** compounds for childhood **fever**, an important risk factor for Reye's syndrome development. Because of its rarity, it is often misdiagnosed as **encephalitis**, **meningitis**, diabetes, or **poisoning**, and the true incidence may be higher than the number of reported cases indicates.

Causes and symptoms

Reye's syndrome causes fatty accumulation in the organs of the body, especially the liver. In the brain, it causes fluid accumulation (**edema**), which leads to a rise in intracranial pressure. This pressure squeezes blood vessels, preventing blood from entering the

brain. Untreated, this pressure increase leads to brain damage and **death**.

Although the cause remains unknown, Reye's syndrome appears to be linked to an abnormality in the energy-converting structures (mitochondria) within the body's cells.

Reye's syndrome usually occurs after a viral, fever-causing illness, most often an upper respiratory tract infection. Its cause is unknown. It is most often associated with use of aspirin during the fever, and for this reason aspirin and aspirin-containing products are not recommended for people under the age of 19 during fever. Reye's syndrome may occur without aspirin use, and in adults, although very rarely.

After the beginning of recovery from the viral illness, the affected person suddenly becomes worse, with the development of persistent **vomiting**. This may be followed rapidly by quietness, lethargy, agitation or combativeness, seizures, and coma. In infants, **diarrhea** may be more common than vomiting. Fever is usually absent at this point.

Diagnosis

Reye's syndrome may be suspected in a child who begins vomiting three to six days after a viral illness, followed by an alteration in consciousness. Diagnosis involves blood tests to determine the levels of certain liver enzymes, which are highly elevated in Reye's syndrome. Other blood changes may occur as well, including an increase in the level of ammonia and amino acids, a drop in blood sugar, and an increase in clotting time. A **liver biopsy** may also be done after clotting abnormalities are corrected with vitamin K or blood products. A lumbar puncture (spinal tap) may be needed to rule out other possible causes, including meningitis or encephalitis.

Treatment

Reye's syndrome is a life-threatening emergency that requires intensive management. The likelihood of recovery is greatest if it is recognized early and treated promptly. Children with Reye's syndrome should be managed in an intensive-care unit.

Treatment in the early stages includes intravenous sugar to return levels to normal and plasma **transfusion** to restore normal clotting time. Intracranial pressure is monitored, and if elevated, is treated with intravenous mannitol and hyperventilation to constrict the blood vessels in the brain.

KEY TERMS

Acetylsalicylic acid—Aspirin; an analgesic, antipyretic, and antirheumatic drug prescribed to reduce fever and for relief of pain and inflammation.

Edema—The abnormal accumulation of fluid in interstitial spaces of tissue.

Mitochondria—Small rodlike, threadlike, or granular organelle within the cytoplasm that function in metabolism and respiration.

If the pressure remains high, **barbiturates** may be used.

Prognosis

The mortality rate for Reye's syndrome is between 30–50%. The likelihood of recovery is increased to 90% by early diagnosis and treatment. Almost all children who survive Reye's syndrome recover fully, although recovery may be slow. In some patients, permanent neurologic damage may remain, requiring physical or educational special services and equipment.

Prevention

Because Reye's syndrome is so highly correlated with use of aspirin for fever in young people, avoidance of aspirin use by children is strongly recommended. Aspirin is in many over-the-counter and prescription drugs, including drugs for **headache**, fever, menstrual cramps, muscle **pain**, **nausea**, upset stomach, and arthritis. It may be used in drugs taken orally or by suppository.

Any of the following ingredients indicates that aspirin is present:

- aspirin
- acetylsalicylate
- acetylsalicylic acid
- salicylic acid
- salicylate

Teenagers who take their own medications without parental consultation should be warned not to take aspirin-containing drugs.

Resources

ORGANIZATIONS

National Reye's Syndrome Foundation. P.O. Box 829, Bryan, OH 43506-0829. (800) 233-7393. <<http://www.bright.net/~reyessyn>>.

Richard Robinson

Rh disease see **Erythroblastosis fetalis**

Rh incompatibility see **Erythroblastosis fetalis**

Rh typing see **Blood typing and crossmatching**

Rheumatic fever

Definition

Rheumatic **fever** (RF) is an illness which arises as a complication of untreated or inadequately treated **strep throat** infection. Rheumatic fever can seriously damage the valves of the heart.

Description

Throat infection with a member of the Group A streptococcus (strep) bacteria is a common problem among school-aged children. It is easily treated with a ten-day course of **antibiotics** by mouth. However, when such a throat infection occurs without symptoms, or when a course of medication is not taken for the full ten days, there is a 3% chance of that person developing rheumatic fever. Other types of strep infections (such as of the skin) do not put the patient at risk for RF.

Children between the ages of five and fifteen are most susceptible to strep throat, and therefore most susceptible to rheumatic fever. Other risk factors include poverty, overcrowding (as in military camps), and lack of access to good medical care. Just as strep throat occurs most frequently in fall, winter, and early spring, so does rheumatic fever.

Causes and symptoms

Two different theories exist as to how a bacterial throat infection can develop into the disease called rheumatic fever. One theory, less supported by research evidence, suggests that the bacteria produce

some kind of poisonous chemical (toxin). This toxin is sent into circulation throughout the bloodstream, thus affecting other systems of the body.

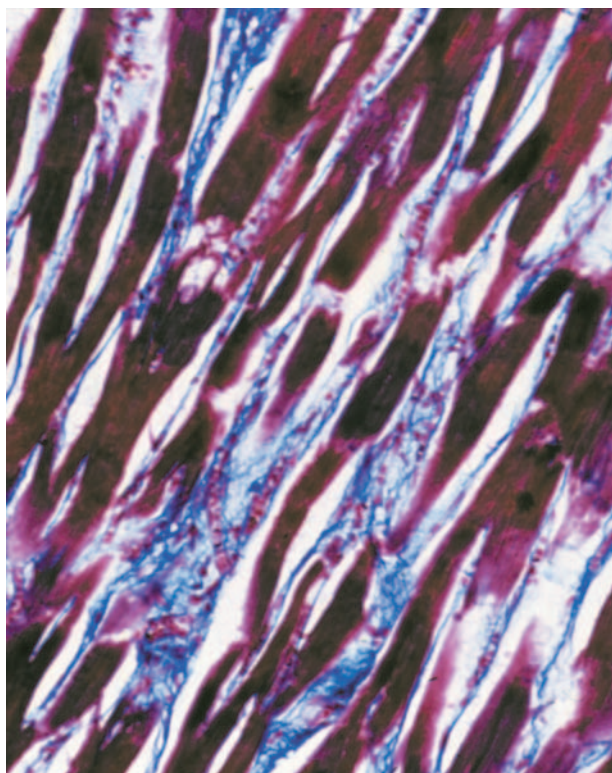
Research seems to point to a different theory, however. This theory suggests that the disease is caused by the body's immune system acting inappropriately. The body produces immune cells (called antibodies), which are specifically designed to recognize and destroy invading agents; in this case, streptococcal bacteria. The antibodies are able to recognize the bacteria because the bacteria contain special markers called antigens. Due to a resemblance between Group A streptococcus bacteria's antigens and antigens present on the body's own cells, the antibodies mistakenly attack the body itself.

It is interesting to note that members of certain families seem to have a greater tendency to develop rheumatic fever than do others. This could be related to the above theory, in that these families may have cell antigens which more closely resemble streptococcal antigens than do members of other families.

In addition to fever, in about 75% of all cases of RF one of the first symptoms is arthritis. The joints (especially those of the ankles, knees, elbows, and wrists) become red, hot, swollen, shiny, and extraordinarily painful. Unlike many other forms of arthritis, the arthritis may not occur symmetrically (affecting a particular joint on both the right and left sides, simultaneously). The arthritis of RF rarely strikes the fingers, toes, or spine. The joints become so tender that even the touch of bedsheets or clothing is terribly painful.

A peculiar type of involuntary movement, coupled with emotional instability, occurs in about 10% of all RF patients (the figure used to be about 50%). The patient begins experiencing a change in coordination, often first noted by changes in handwriting. The arms or legs may flail or jerk uncontrollably. The patient seems to develop a low threshold for anger and sadness. This feature of RF is called **Sydenham's chorea** or St. Vitus' Dance.

A number of skin changes are common to RF. A rash called erythema marginatum develops (especially in those patients who will develop heart problems from their illness), composed of pink splotches, which may eventually spread into each other. It does not itch. Bumps the size of peas may occur under the skin. These are called subcutaneous nodules; they are hard to the touch, but not painful. These nodules most commonly occur over the knee and elbow joint, as well as over the spine.



A magnified image of cardiac muscle damaged by chronic myocarditis caused by rheumatic fever. (*Science Photo Library. Custom Medical Stock Photo. Reproduced by permission.*)

The most serious problem occurring in RF is called pancarditis (“pan” means total; “carditis” refers to inflammation of the heart). Pancarditis is an inflammation that affects all aspects of the heart, including the lining of the heart (endocardium), the sac containing the heart (pericardium), and the heart muscle itself (myocardium). About 40-80% of all RF patients develop pancarditis. This RF complication has the most serious, long-term effects. The valves within the heart (structures which allow the blood to flow only in the correct direction, and only at the correct time in the heart’s pumping cycle) are frequently damaged during the course of pancarditis. This may result in blood which either leaks back in the wrong direction, or has a difficult time passing a stiff, poorly moving valve. Either way, damage to a valve can result in the heart having to work very hard in order to move the blood properly. The heart may not be able to “work around” the damaged valve, which may result in a consistently inadequate amount of blood entering the circulation.

Diagnosis

Diagnosis of RF is done by carefully examining the patient. A list of diagnostic criteria has been

created. These “Jones Criteria” are divided into major and minor criteria. A patient can be diagnosed with RF if he or she has either two major criteria (conditions), or one major and two minor criteria. In either case, it must also be proved that the individual has had a previous infection with streptococcus.

The major criteria include:

- carditis
- arthritis
- chorea
- subcutaneous nodules
- erythema marginatum

The minor criteria include:

- fever
- joint **pain** (without actual arthritis)
- evidence of electrical changes in the heart (determined by measuring electrical characteristics of the heart’s functioning during a test called an electrocardiogram, or EKG)
- evidence (through a blood test) of the presence in the blood of certain proteins, which are produced early in an inflammatory/infectious disease.

Tests are also performed to provide evidence of recent infection with group A streptococcal bacteria. A swab of the throat can be taken, and smeared on a substance in a petri dish, to see if bacteria will multiply and grow over 24-72 hours. These bacteria can then be specially processed, and examined under a microscope, to identify streptococcal bacteria. Other tests can be performed to see if the patient is producing specific antibodies; that are only made in response to a recent strep infection.

Treatment

A 10-day course of penicillin by mouth, or a single injection of penicillin G-is the first line of treatment for RF. Patients will need to remain on some regular dose of penicillin to prevent recurrence of RF. This can mean a small daily dose of penicillin by mouth, or an injection every three weeks. Some practitioners keep patients on this regimen for five years, or until they reach 18 years of age (whichever comes first). Other practitioners prefer to continue treating those patients who will be regularly exposed to streptococcal bacteria (teachers, medical workers), as well as those patients with known RF heart disease.

Arthritis quickly improves when the patient is given a preparation containing **aspirin**, or some other anti-

KEY TERMS

Antibodies—Specialized cells of the immune system which can recognize organisms that invade the body (such as bacteria, viruses, and fungi). The antibodies are then able to set off a complex chain of events designed to kill these foreign invaders.

Antigen—A special, identifying marker on the outside of cells.

Arthritis—Inflammation of the joints.

Autoimmune disorder—A disorder in which the body's antibodies mistake the body's own tissues for foreign invaders. The immune system therefore attacks and causes damage to these tissues.

Chorea—Involuntary movements in which the arms or legs may jerk or flail uncontrollably.

Immune system—The system of specialized organs, lymph nodes, and blood cells throughout the body, which work together to prevent foreign invaders (bacteria, viruses, fungi, etc.) from taking hold and growing.

Inflammation—The body's response to tissue damage. Includes hotness, swelling, redness, and pain in the affected part.

Pancarditis—Inflammation of the lining of the heart, the sac around the heart, and the muscle of the heart.

inflammatory agent (ibuprofen). Mild carditis will also improve with such anti-inflammatory agents, although more severe cases of carditis will require steroid medications. A number of medications are available to treat the involuntary movements of chorea, including diazepam for mild cases, and haloperidol for more severe cases.

Prognosis

The long-term prognosis of an RF patient depends primarily on whether he or she develops carditis. This is the only manifestation of RF which can have permanent effects. Those patients with no or mild carditis have an excellent prognosis. Those with more severe carditis have a risk of **heart failure**, as well as a risk of future heart problems, which may lead to the need for valve replacement surgery.

Prevention

Prevention of the development of RF involves proper diagnosis of initial strep throat infections,

and adequate treatment within 10 days with an appropriate antibiotic. Prevention of RF recurrence requires continued antibiotic treatment, perhaps for life. Prevention of complications of already-existing RF heart disease requires that the patient always take a special course of antibiotics when he or she undergoes any kind of procedure (even dental cleanings) that might allow bacteria to gain access to the bloodstream.

Resources

ORGANIZATIONS

Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

Rosalyn Carson-DeWitt, MD

Rheumatoid arthritis

Definition

Rheumatoid arthritis (RA) is a chronic autoimmune disease that causes inflammation and deformity of the joints. Other problems throughout the body (systemic problems) may also develop, including inflammation of blood vessels (**vasculitis**), the development of bumps (called rheumatoid nodules) in various parts of the body, lung disease, blood disorders, and weakening of the bones (**osteoporosis**).

Description

The skeletal system of the body is made up of different types of strong, fibrous tissue called connective tissue. Bone, cartilage, ligaments, and tendons are all forms of connective tissue that have different compositions and different characteristics.

The joints are structures that hold two or more bones together. Some joints (synovial joints) allow for movement between the bones being joined (articulating bones). The simplest synovial joint involves two bones, separated by a slight gap called the joint cavity. The ends of each articular bone are covered by a layer of cartilage. Both articular bones and the joint cavity are surrounded by a tough tissue called the articular capsule. The articular capsule has two components, the fibrous membrane on the outside and the synovial membrane (or synovium) on the inside. The fibrous membrane may include tough bands of tissue called ligaments, which are responsible for providing

support to the joints. The synovial membrane has special cells and many tiny blood vessels (capillaries). This membrane produces a supply of synovial fluid that fills the joint cavity, lubricates it, and helps the articular bones move smoothly about the joint.

In rheumatoid arthritis (RA), the synovial membrane becomes severely inflamed. Usually thin and delicate, the synovium becomes thick and stiff, with numerous infoldings on its surface. The membrane is invaded by white blood cells, which produce a variety of destructive chemicals. The cartilage along the articular surfaces of the bones may be attacked and destroyed, and the bone, articular capsule, and ligaments may begin to wear away (erode). These processes severely interfere with movement in the joint.

RA exists all over the world and affects men and women of all races. In the United States alone, about two million people suffer from the disease. Women are three times more likely than men to have RA. About 80% of people with RA are diagnosed between the ages of 35-50. RA appears to run in families, although certain factors in the environment may also influence the development of the disease.

Causes and symptoms

The underlying event that promotes RA in a person is unknown. Given the known genetic factors involved in RA, some researchers have suggested that an outside event occurs that triggers the disease cycle in a person with a particular genetic makeup.

Many researchers are examining the possibility that exposure to an organism (like a bacteria or virus) may be the first event in the development of RA. The body's normal response to such an organism is to produce cells that can attack and kill the organism, protecting the body from the foreign invader. In an autoimmune disease like RA, this immune cycle spins out of control. The body produces misdirected immune cells, which accidentally identify parts of the person's body as foreign. These immune cells then produce a variety of chemicals that injure and destroy parts of the body.

RA can begin very gradually, or it can strike quickly. The first symptoms are **pain**, swelling, and stiffness in the joints. The most commonly involved joints include hands, feet, wrists, elbows, and ankles, although other joints may also be involved. The joints are affected in a symmetrical fashion. This means that if the right wrist is involved, the left wrist is also involved. Patients frequently experience painful joint stiffness when they first get up in the morning, lasting

for perhaps an hour. Over time, the joints become deformed. The joints may be difficult to straighten, and affected fingers and toes may be permanently bent (flexed). The hands and feet may curve outward in an abnormal way.

Many patients also notice increased **fatigue**, loss of appetite, weight loss, and sometimes **fever**. Rheumatoid nodules are bumps that appear under the skin around the joints and on the top of the arms and legs. These nodules can also occur in the tissue covering the outside of the lungs and lining the chest cavity (pleura), and in the tissue covering the brain and spinal cord (meninges). Lung involvement may cause **shortness of breath** and is seen more in men. Vasculitis (inflammation of the blood vessels) may interfere with blood circulation. This can result in irritated pits (ulcers) in the skin, tissue death (**gangrene**), and interference with nerve functioning that causes **numbness and tingling**.

Juvenile RA is a chronic inflammatory disease that affects the joints of children less than 16 years old. It is estimated to affect as many as 250,000 children in the United States alone. Most children with juvenile RA have arthritis when the illness starts, which affects multiple joints in 50% of these children, and only one joint in 30%. In all, 20% of the children affected by juvenile RA have the acute systemic form of the disease, which is characterized by fever, joint inflammation, rash, **liver disease**, and gastrointestinal disease.

Two periods of childhood are associated with an increased incidence of onset of juvenile RA. The first is from one to three years of age, and the second, from eight to 12 years. When more than four joints are affected, the disease is described as being polyarticular. If less than four joints are affected, the disease is known as pauciarticular. Juvenile RA and this particular manifestation falls into two categories. The first occurs in girls aged one to four years old, and the onset of joint involvement is in the knees, ankles, or elbows. The second form occurs in boys aged eight years and older, and involves the larger joints, such as those of the hips and legs.

Diagnosis

There are no tests available that can absolutely diagnose RA. Instead, a number of tests exist that can suggest the diagnosis of RA. Blood tests include a special test of red blood cells (called **erythrocyte sedimentation rate**), which is positive in nearly 100% of patients with RA. However, this test is also positive in a variety of other diseases. Tests for anemia are usually

positive in patients with RA, but can also be positive in many other unrelated diseases. Rheumatoid factor is another diagnostic test that measures the presence and amounts of rheumatoid factor in the blood. Rheumatoid factor is an autoantibody found in about 80% of patients with RA. It is often not very specific however, because it is found in about 5% of all healthy people and in 10-20% of healthy people over the age of 65. In addition, rheumatoid factor is also positive in a large number of other autoimmune diseases and other infectious diseases, including **systemic lupus erythematosus**, bacterial **endocarditis**, **malaria**, and **syphilis**. In addition, young people who have a process called juvenile rheumatoid arthritis often have no rheumatoid factor present in their blood.

Finally, the clinician may examine the synovial fluid, by inserting a thin needle into a synovial joint. In RA, this fluid has certain characteristics that indicate active inflammation. The fluid is cloudy, with increased protein and decreased or normal glucose. It also contains a higher than normal number of white blood cells. While these findings suggest inflammatory arthritis, they are not specific to RA.

Treatment

There is no cure available for RA. However, treatment is available to combat the inflammation in order to prevent destruction of the joints, and to prevent other complications of the disease. Efforts are also made to maintain flexibility and mobility of the joints.

The “first line” agents for the treatment of RA include nonsteroidal anti-inflammatory agents (NSAIDs) and **aspirin**, which are used to decrease inflammation and to treat pain. The NSAIDs include naproxen (Naprosyn), ibuprofen (Advil, Medipren, Motrin), and etodolac (Lodine). While these medications can be helpful, they do not interrupt the progress of the disease. Low-dose steroid medications can be helpful at both managing symptoms and slowing the progress of RA. Disease-modifying **antirheumatic drugs**, including gold compounds, D-penicillamine, certain antimalarial-like drugs, and sulfasalazine (Azulfadine) are also often the first agents clinicians use to treat RA, but in patients with the aggressive destructive type of RA, more slow-acting medications are needed. Methotrexate, azathioprine, and cyclophosphamide are all drugs that suppress the immune system and can decrease inflammation. All of the drugs listed have significant toxic side effects, which require healthcare professionals to carefully compare the risks associated with these medications versus the benefits.

Recently, several categories of drugs have been explored and developed for the treatment of RA. The first is a category of agents known as biological response modifiers. These work to reduce joint inflammation by blocking a substance called tumor necrosis factor (TNF). TNF is a protein that triggers inflammation during the body’s normal immune responses. When TNF production is not regulated, the excess TNF can cause inflammation. Three agents in this class have become “second line” drugs for the treatment of RA. These are etanercept (Enbrel), leflunomide (Arava), and infliximab (Remicade), and they are recommended for patients in whom other medications have not been effective. Etanercept is approved by the FDA but is not recommended for patients with active infection. It is given twice weekly via subcutaneous injections by either the patient or a health care professional. Because this agent is so new, long-term side effects have not been fully studied. Infliximab is given intravenously once every eight weeks, and is approved for combined use with methotrexate to combat RA.

The cyclo-oxygenase-2 (COX-2) inhibitors are another category of drugs used to treat RA. Like the traditional NSAIDs, the **COX-2 inhibitors** work to block COX-2, which is an enzyme that stimulates inflammatory responses in the body. Unlike the NSAIDs, the COX-2 inhibitors do not carry a high risk of gastrointestinal ulcers and bleeding, because they do not inhibit COX-1, which is the enzyme that protects the stomach lining. These new agents include celecoxib (Celebrex) and rofecoxib (Vioxx). Celecoxib has been approved by the FDA for the treatment of RA and **osteoarthritis**, and is taken once or twice daily by mouth. Rofecoxib is approved for RA and osteoarthritis, and for acute pain caused by primary **dysmenorrhea** and surgery.

Total bed rest is sometimes prescribed during the very active, painful phases of RA. Splints may be used to support and rest painful joints. Later, after inflammation has somewhat subsided, physical therapists may provide a careful **exercise** regimen in an attempt to maintain the maximum degree of flexibility and mobility. **Joint replacement** surgery, particularly for the knee and the hip joints, is sometimes recommended when these joints have been severely damaged.

Alternative treatment

A variety of alternative therapies has been recommended for patients with RA. **Meditation**, hypnosis, **guided imagery**, and relaxation techniques have been

used effectively to control pain. **Acupressure** and **acupuncture** have also been used for pain. Bodywork can be soothing, decreasing **stress** and tension, and is thought to improve/restore chemical balance within the body.

A multitude of **nutritional supplements** can be useful for RA. Fish oils, the enzymes bromelain and pancreatin, and the antioxidants (**vitamins** A, C, and E, selenium, and zinc) are the primary supplements to consider.

Many herbs also are useful in the treatment of RA. Anti-inflammatory herbs may be very helpful, including tumeric (*Curcuma longa*), ginger (*Zingiber officinale*), feverfew (*Chrysanthemum parthenium*), devil's claw (*Harpagophytum procumbens*), Chinese thoroughwax (*Bupleuri falcatum*), and licorice (*Glycyrrhiza glabra*). Lobelia (*Lobelia inflata*) and cramp bark (*Viburnum opulus*) can be applied topically to the affected joints.

Homeopathic practitioners recommended *Rhus toxicodendron* and *Bryonia* (*Bryonia alba*) for acute prescriptions, but constitutional treatment, generally used for chronic problems like RA, is more often recommended. **Yoga** has been used for RA patients to promote relaxation, relieve stress, and improve flexibility. Nutritionists suggest that a vegetarian diet low in animal products and sugar may help to decrease both inflammation and pain from RA. Beneficial foods for patients with RA include cold water fish (mackerel, herring, salmon, and sardines) and flavonoid-rich berries (cherries, blueberries, hawthorn berries, blackberries, etc.).

RA, considered an autoimmune disorder, is often connected with food allergies/intolerances. An elimination/challenge diet can help to decrease symptoms of RA as well as identify the foods that should be eliminated to prevent flare-ups and recurrences. **Hydrotherapy** can help to greatly reduce pain and inflammation. Moist heat is more effective than dry heat, and cold packs are useful during acute flare-ups.

Prognosis

About 15% of all RA patients will have symptoms for a short period of time and will ultimately get better, leaving them with no long-term problems. A number of factors are considered to suggest the likelihood of a worse prognosis. These include:

- race and gender (female and Caucasian).
- more than 20 joints involved.
- extremely high erythrocyte sedimentation rate.

KEY TERMS

Articular bones—Two or more bones connected to each other via a joint.

Joint—Structures holding two or more bones together.

Pauciarticular juvenile RA—Rheumatoid arthritis found in children that affects less than four joints.

Polyarticular juvenile RA—Rheumatoid arthritis found in children that affects more than four joints.

Synovial joint—A type of joint that allows articular bones to move.

Synovial membrane—The membrane that lines the inside of the articular capsule of a joint and produces a lubricating fluid called synovial fluid.

- extremely high levels of rheumatoid factor.
- consistent, lasting inflammation.
- evidence of erosion of bone, joint, or cartilage on x rays.
- poverty.
- older age at diagnosis.
- rheumatoid nodules.
- other coexisting diseases.
- certain genetic characteristics, diagnosable through testing.

Patients with RA have a shorter life span, averaging a decrease of three to seven years of life. Patients sometimes die when very severe disease, infection, and gastrointestinal bleeding occur. Complications due to the side effects of some of the more potent drugs used to treat RA are also factors in these deaths.

Prevention

There is no known way to prevent the development of RA. The most that can be hoped for is to prevent or slow its progress.

Resources

BOOKS

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PERIODICALS

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Liz Meszaros

Rheumatoid spondylit see **Ankylosing spondylitis**

Rhinitis

Definition

Rhinitis is inflammation of the mucous lining of the nose.

Description

Rhinitis is a nonspecific term that covers infections, **allergies**, and other disorders whose common feature is the location of their symptoms. In rhinitis, the mucous membranes become infected or irritated, producing a discharge, congestion, and swelling of the tissues. The most widespread form of infectious rhinitis is the **common cold**.

The common cold is the most frequent viral infection in the general population, causing more absenteeism from school or work than any other illness. Colds are self-limited, lasting about 3-10 days, although they are sometimes followed by a bacterial infection. Children are more susceptible than adults; teenage boys more susceptible than teenage girls; and adult women more susceptible than adult men. In the United States, colds are most frequent during the late fall and winter.

Causes and symptoms

Colds can be caused by as many as 200 different viruses. The viruses are transmitted by sneezing and coughing, by contact with soiled tissues or handkerchiefs, or by close contact with an infected person. Colds are easily spread in schools, offices, or any place where people live or work in groups. The incubation period ranges between 24 and 72 hours.

The onset of a cold is usually sudden. The virus causes the lining of the nose to become inflamed and produce large quantities of thin, watery mucus. Children sometimes run a **fever** with a cold. The inflammation spreads from the nasal passages to the throat and upper airway, producing a dry **cough**, **headache**, and watery eyes. Some people develop muscle or joint aches and feel generally tired or weak. After several days, the nose becomes less inflamed and the watery discharge is replaced by a thick, sticky mucus. This change in the appearance of the nasal discharge helps to distinguish rhinitis caused by a viral infection from rhinitis caused by an allergy.

Diagnosis

There is no specific test for viral rhinitis. The diagnosis is based on the symptoms. In children, the doctor will examine the child's throat and glands to rule out **measles** and other childhood illnesses that have similar early symptoms. Adults whose symptoms last longer than a week may require further testing to rule out a secondary bacterial infection, or an allergy. Bacterial infections can usually be identified from a laboratory culture of the patient's nasal discharge. Allergies can be evaluated by blood tests, skin testing for specific substances, or nasal smears.

Treatment

There is no cure for the common cold; treatment is given for symptom relief. Medications include **aspirin** or **nonsteroidal anti-inflammatory drugs** (NSAIDs) for headache and muscle **pain**, and **decongestants** to relieve stuffiness or runny nose. Patients should be warned against overusing decongestants, because they can cause a rebound effect. Over-the-counter (OTC) **antihistamines** are also available; however, most antihistamines carry warnings of drowsiness and the inability to do some tasks while medicated. Claritin is a prescription-strength OTC non-drowsy antihistamine that helps relieve symptoms of rhinitis. **Antibiotics** are not given for colds because they do not kill viruses.

Supportive care includes bed rest and drinking plenty of fluid.

Treatments under investigation include the use of ultraviolet light and injections of interferon.

Many prescription and over-the-counter drugs are available to help control the symptoms of **allergic rhinitis**. The most common class is antihistamines.

KEY TERMS

Interferon—A protein produced by cells infected by a virus that stimulates the body's resistance to the virus.

Alternative treatment

Homeopaths might prescribe any of 10 different remedies, depending on the appearance of the nasal discharge, the patient's emotional state, and the stage of infection. Naturopaths would recommend vitamin A and zinc supplements, together with botanical preparations made from goldenseal (*Hydrastis canadensis*), licorice (*Glycyrrhiza glabra*), or astragalus (*Astragalus membranaceus*) root.

At one time, the herb (*Echinacea* spp.) was touted as a remedy to relieve cold and rhinitis symptoms. However, a study published in 2004 reported that the herb failed to relieve cold symptoms in 400 children taking it and caused skin **rashes** in some children.

Prognosis

Most colds resolve completely in about a week. Complications are unusual but may include **sinusitis** (inflammation of the nasal sinuses), bacterial infections, or infections of the middle ear.

Prevention

There is no vaccine effective against colds, and infection does not confer immunity. Prevention depends on:

- washing hands often, especially before touching the face
- minimizing contact with people already infected
- not sharing hand towels, eating utensils, or water glasses.

Resources

PERIODICALS

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February 16, 2004: 25.

Rebecca J. Frey, PhD
Teresa G. Odle

Rhinoplasty

Definition

The term rhinoplasty means “nose molding” or “nose forming.” It refers to a procedure in **plastic surgery** in which the structure of the nose is changed. The change can be made by adding or removing bone or cartilage, grafting tissue from another part of the body, or implanting synthetic material to alter the shape of the nose.

Rhinoplasty is the most frequently performed cosmetic surgical procedure in the United States as of the early 2000s. According to the American Society of Plastic Surgeons (ASPS), 356,554 rhinoplasties were performed in the United States in 2003, compared to 254,140 breast augmentations and 128,667 facelifts.

Purpose

Rhinoplasty is most often performed for cosmetic reasons. A nose that is too large, crooked, misshapen, malformed at birth, or deformed by an injury or **cancer** surgery can be given a more pleasing appearance. If breathing is impaired due to the form of the nose or to an injury, it can often be improved with rhinoplasty.

Precautions

The best candidates for rhinoplasty are those with relatively minor deformities. Nasal anatomy and proportions are quite varied and the final look of any rhinoplasty operation is the result of the patient's anatomy, as well as of the surgeon's skill.

The quality of the skin plays a major role in the outcome of rhinoplasty. Patients with extremely thick skin may not see a definite change in the underlying bone structure after surgery. On the other hand, thin skin provides almost no cushion to hide the most minor of bone irregularities or imperfections.

A cosmetic change in the shape of the nose will change a person's appearance, but it will not change self-image. A person who expects a different lifestyle after rhinoplasty is likely to be disappointed.

Rhinoplasty should not be performed until the pubertal growth spurt is complete, between ages 14-15 for girls and older for boys.

The cost of rhinoplasty depends on the difficulty of the work required and on the specialist chosen. Prices run from about \$3,000 to over \$6,000. If the problem was caused by an injury, insurance will usually cover the cost. A rhinoplasty done only to

change a person's appearance is not usually covered by insurance.

Description

The external nose is composed of a series of inter-related parts which include the skin, the bony pyramid, cartilage, and the tip of the nose, which is both cartilage and skin. The strip of skin separating the nostrils is called the columella.

Surgical approaches to nasal reconstruction are varied. Internal rhinoplasty involves making all incisions inside the nasal cavity. The external or "open" technique involves a skin incision across the base of the nasal columella. An external incision allows the surgeon to expose the bone and cartilage more fully and is most often used for complicated procedures. During surgery, the surgeon will separate the skin from the bone and cartilage support. The framework of the nose is then reshaped in the desired form. Shape can be altered by removing bone, cartilage, or skin. The remaining skin is then replaced over the new framework. If the procedure requires adding to the structure of the nose, the donated bone, cartilage, or skin can come from the patient or from a synthetic source.

When the operation is over, the surgeon will apply a splint to help the bones maintain their new shape. The nose may also be packed, or stuffed with a dressing, to help stabilize the septum.

When a local anesthetic is used, light **sedation** is usually given first, after which the operative area is numbed. It will remain insensitive to **pain** for the length of the surgery. A general anesthetic is used for lengthy or complex procedures or if the doctor and patient agree that it is the best option.

Simple rhinoplasty is usually performed in an outpatient surgery center or in the surgeon's office. Most procedures take only an hour or two, and patients may return home right away. Complex procedures may be done in the hospital and require a short stay.

Preparation

During the initial consultation, the patient and surgeon will determine what changes can be made in the shape of the nose. Most doctors take photographs at the same time. The surgeon will also explain the techniques and anesthesia options available to the patient.

For legal reasons, many plastic surgeons now screen patients for psychological stability as well as

general physical fitness for surgery. When a person consults a plastic surgeon about a rhinoplasty, the doctor will spend some time talking with the patient about his or her motives for facial surgery. The following are considered psychological warning signs:

- The patient is considering surgery to please someone else, most often a spouse or partner.
- The patient expects facial surgery to guarantee career advancement.
- The patient has a history of multiple cosmetic procedures and/or complaints about previous surgeons.
- The patient thinks that the surgery will solve all his or her life problems.
- The patient has an unrealistic notion of what he or she will look like after surgery.
- The patient seems otherwise emotionally unstable.

The patient and surgeon should also discuss guidelines for eating, drinking, **smoking**, taking or avoiding certain medications, and washing of the face.

Aftercare

Patients usually feel fine immediately after surgery; however, most surgery centers do not allow patients to drive themselves home after an operation.

The first day after surgery there will be some swelling of the face. Patients should stay in bed with their heads elevated for at least a day. The nose may hurt and a **headache** is not uncommon. The surgeon will prescribe medication to relieve these conditions. Swelling and bruising around the eyes will increase for a few days, but will begin to diminish after about the third day. Slight bleeding and stuffiness are normal, and vary according to the extensiveness of the surgery performed. Most people are up in two days, and back to school or work in a week. No strenuous activities are allowed for two to three weeks.

Patients are given a list of postoperative instructions, which include requirements for hygiene, **exercise**, eating, and follow-up visits to the doctor. Patients should not blow their noses for the first week to avoid disruption of healing. It is extremely important to keep the surgical dressing dry. Dressings, splints, and stitches are removed in one to two weeks. Patients should avoid **sunburn**.

Patients should remember that it may take as long as a year for the nose to assume its final shape; the tip of the nose in particular may be mildly swollen for several months.

KEY TERMS

Cartilage—Firm supporting tissue that does not contain blood vessels.

Columella—The strip of skin running from the tip of the nose to the upper lip, which separates the nostrils.

Septum—The dividing wall in the nose.

Risks

Any type of surgery carries a degree of risk. There is always the possibility of unexpected events, such as an infection or a reaction to the anesthesia. Some patients may have a so-called foreign body reaction to a nasal implant made from synthetic materials. In these cases the surgeon can replace the implant with a piece of cartilage from the patient's own body.

Some risks of rhinoplasty are social or psychological. The ASPS patient brochure about rhinoplasty mentions the possibility of criticism or rejection by friends or family if they feel threatened by the patient's new look. This type of reaction sometimes occurs with rhinoplasty if the friends or relatives consider the shape of the nose an important family or ethnic trait.

When the nose is reshaped or repaired from inside, the **scars** are not visible, but if the surgeon needs to make the incision on the outside of the nose, there will be some slight scarring. In addition, tiny blood vessels may burst, leaving small red spots on the skin. These spots are barely visible but may be permanent.

About 10% of patients require a second procedure; however, the corrections required are usually minor.

Resources

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Alloplastic Implant Rejection." *Archives of Facial and Plastic Surgery* 6 (May-June 2004): 192–196.

ORGANIZATIONS

- American Academy of Facial Plastic and Reconstructive Surgery (AAFPRS). 310 South Henry Street, Alexandria, VA 22314. (703) 299-9291. <<http://www.facemd.org>>.
- American Society of Plastic Surgeons (ASPS). 444 East Algonquin Road, Arlington Heights, IL 60005. (847) 228-9900. <<http://www.plasticsurgery.org>>.

OTHER

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Rhinovirus infection see **Common cold**

Rhytidoplasty see **Face lift**

Riboflavin deficiency

Definition

Riboflavin deficiency occurs when the chronic failure to eat sufficient amounts of foods that contain riboflavin produces lesions of the skin, lesions of smooth surfaces in the digestive tract, or nervous disorders.

Description

Riboflavin, also called vitamin B₂, is a water-soluble vitamin. The recommended dietary allowance (RDA) for riboflavin is 1.7 mg/day for an adult man and 1.3 mg/day for an adult woman. The best sources of this vitamin are meat, dairy products, and dark green vegetables, especially broccoli. Grains and legumes (beans and peas) also contribute riboflavin to the diet. Riboflavin is required for the processing of dietary fats, carbohydrates, and proteins to convert these nutrients to energy. Riboflavin is also used for the continual process of renewal and regeneration of all cells and tissues in the body.

Riboflavin is sensitive to light. For this reason, commercially available milk is sometimes supplied in cartons, rather than in clear bottles. Riboflavin is not rapidly destroyed by cooking. Milk contains about 1.7 mg riboflavin/kg. Cheese contains about 4.3 mg/kg, while beef has 2.4 mg/kg and broccoli has about 2.0

mg/kg. Apples, a food that is low in all nutrients, except water, contains only 0.1 mg riboflavin per kg.

Causes and symptoms

A deficiency only in riboflavin has never occurred in the natural environment. In contrast, diseases where people are deficient in one vitamin, such as thiamin, vitamin C, and vitamin D, for example, have been clearly documented. Poorer populations in the United States may be deficient in riboflavin, but when this happens, they are also deficient in a number of other nutrients as well. When riboflavin deficiency is actually detected, it is often associated with low consumption of milk, chronic **alcoholism**, or chronic **diarrhea**.

The symptoms of riboflavin deficiency include:

- swelling and fissuring of the lips (cheilosis)
- ulceration and cracking of the angles of the mouth (angular stomatitis)
- oily, scaly skin **rashes** on the scrotum, vulva, or area between the nose and lips
- inflammation of the tongue
- red, itchy eyes that are sensitive to light

The nervous symptoms of riboflavin deficiency include:

- numbness of the hands
- decreased sensitivity to touch, temperature, and vibration

Diagnosis

Riboflavin status is diagnosed using a test conducted on red blood cells that measures the activity of an enzyme called glutathione reductase. An extract of the red blood cells is placed in two test tubes. One test tube contains no added riboflavin, while the second test tube contains a derivative of riboflavin, called flavin adenine dinucleotide. The added riboflavin derivative results in little or no stimulation of enzyme activity in patients with normal riboflavin levels. A stimulation of 20% or less is considered normal. A stimulation of over 20% means that the patient is deficient in riboflavin.

Treatment

Riboflavin deficiency can be treated with supplemental riboflavin (0.5 mg/kg body weight per day) until the symptoms disappear.

KEY TERMS

Recommended dietary allowance—The recommended daily allowances (RDAs) are quantities of nutrients of the diet that are required to maintain human health. RDAs are established by the Food and Nutrition Board of the National Academy of Sciences and may be revised every few years. A separate RDA value exists for each nutrient. The RDA values refer to the amount of nutrient needed to maintain health in a population of people. The actual amounts of each nutrient required to maintain health in any specific individual differs from person to person.

Water-soluble vitamin—Water-soluble vitamins can be dissolved in water or juice. Fat-soluble vitamins can be dissolved in oil or in melted fat.

Prognosis

The prognosis for correcting riboflavin deficiency is excellent.

Prevention

Riboflavin deficiency can be prevented by including milk, cheese, yogurt, meat, and/or certain vegetables in the daily diet. Of the vegetables, broccoli, asparagus, and spinach are highest in riboflavin. These vegetables have a riboflavin content that is similar to that of milk, yogurt, or meat.

Resources

BOOKS

Brody, Tom. *Nutritional Biochemistry*. San Diego: Academic Press, 1998.

Tom Brody, PhD

Rickets

Definition

Rickets is a childhood condition caused by serious **vitamin D deficiency**. This lacking in vitamin D results in weak, soft bones, along with slowed growth and skeletal development. Rickets is, by definition, a

disorder which begins in childhood. If this problem occurs only later in life it is known as osteomalacia.

Description

Rickets occurs when the body has a severe lack of vitamin D during the developmental years. Vitamin D is essential to the development of strong, healthy bones. A child with rickets can experience stunted growth and will most likely be short in stature as an adult. This is because, without proper vitamin D levels, decreased mineralization of the bones at the growth plate level affects the strength, size and shape of the bones. A related condition called osteomalacia can occur in adults with the same sort of vitamin D deficiency, but osteomalacia occurs only in adulthood after the growth plates of the bones have closed.

Most vitamin D is produced by the body, although some can be directly supplied by diet. In order to accomplish production of vitamin D, the body requires both cholesterol and ultraviolet light. Most often, the cholesterol comes from digesting animal tissue, oils, fats, and egg yolks. The ultraviolet light is usually supplied by direct sunlight. Only when this light is available can the skin alter the cholesterol molecule to make vitamin D. Children who do not receive enough sunlight are at greater risk of developing rickets, as are children with darker skin, which can block the ultraviolet rays. Vitamin D is found naturally in the foods listed above, but more often children receive vitamin D supplements through foods which have had the vitamin added, as in milk or infant formula.

Vitamin D is necessary in the body, because it can be converted into a hormone which stimulates calcium intake by the intestines. This conversion begins in the liver, where vitamin D becomes a hormone called 25-OH-D, and is completed when the kidneys convert 25-OH-D into a hormone called 1,25-diOH-D. This is the hormone that causes the intestines to absorb calcium from the person's diet. Without proper levels of vitamin D, there is not enough 1,25-diOH-D produced, which results in lower levels of calcium in the body. Adequate calcium is needed by the bones for both development and maintenance.

Causes and symptoms

Rickets is directly caused by insufficient calcium for bone mineralization during growth and development. This is caused by vitamin D deficiency which can be a result of too little cholesterol, ultraviolet light, or vitamin D supplement. During the Industrial

Revolution, rickets was quite common in cities because pollution in the air blocked much of the sunlight needed for vitamin D production in the body. There is also a hereditary type of rickets, called X-linked hypophosphatemia, that causes the kidneys to be unable to retain phosphate.

The most commonly recognized symptoms of rickets occur in the arms and legs, where **stress** on the underdeveloped bones can cause bowing. Children with rickets may feel **pain** or tenderness in the bones of their arms, legs, spine, pelvis, and ribs. The skull may develop an odd or asymmetrical shape. Calcium levels in the blood will be low and overall growth is often impaired.

Diagnosis

The initial approach to diagnosing rickets involves a musculoskeletal examination followed by an x ray is often. Affected children may have obviously widened spaces between their joints or bowing of the bones in their arms and legs. Some children may not experience normal dental development as well. A doctor may also assess levels of serum calcium, alkaline phosphatase and other indicator chemicals by using a blood test. While calcium levels can be normal or slightly low, alkaline phosphatase levels in a child with rickets can be high even compared to a normal adult. While x rays can prove misleading, diagnosis by chemical analysis is highly accurate.

Treatment

The treatment for rickets primarily involves corrections of the conditions which led to the disorder. This can be as simple as a change in diet to include foods high in vitamin D such as milk, fish, or liver. Treatment might also mean a gradual increase in the amount sunlight received by the child. In more severe cases, bracing or surgery may be necessary to aid in the correction and repair of bones. Treatment is usually mild and bone deformities usually reduce over time.

Alternative Treatment

There is currently little known about any alternative method for treating rickets. Treatments which involve raising vitamin D levels and ultraviolet light exposure are usually simple and effective.

Prognosis

Children with rickets are likely to suffer from stunted growth, bone abnormalities and bone pain,

KEY TERMS

25-hydroxy-vitamin D—This is the form of vitamin D that is measured in order to assess vitamin D deficiency.

Cholesterol—A fat-soluble steroid alcohol (sterol) found in animal fats and oils, and in egg yolks. The human body needs cholesterol to produce vitamin D.

Growth plate—The place in long bones where growth occurs during childhood.

International unit (IU)—A measurement of biological activity in which one IU is equal to one mg (milligram).

Mineralization—The process by which the body uses minerals to build bone structure.

X-linked hypophosphatemia—A type of rickets caused by genetic factors which prevent the kidneys from retaining phosphate.

however these symptoms often disappear with treatment. In women, deformation of the pelvic bone structure can prevent vaginal **childbirth** later in life. Most deformities correct with growth when proper levels of vitamin D are restored and normal bone calcification is maintained.

Prevention

Rickets caused by vitamin D deficiency is simple to prevent. Commercially available infant formula is usually fortified with more than enough vitamin D for infants. For parents who breastfeed their children, it is recommended by the U.S. Department of Health and Human Services that children also receive 400 international units (10 micrograms) of vitamin D supplement. This is because human breast milk contains little vitamin D. It is also important that children are allowed decent amounts of sunlight. As little as twenty minutes each day can be sufficient. For children living in cities, where pollution is likely to block ultraviolet light, and children with dark skin, which can block ultraviolet light, vitamin D supplement is especially important.

Resources

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Tish Davidson, A.M.

Rickets see **Vitamin D deficiency**

Rickettsia rickettsii infection see **Rocky Mountain spotted fever**

Rickettsialpox

Definition

Rickettsialpox is a relatively mild disease caused by a member of the bacterial family called *Rickettsia*. Rickettsialpox causes rash, **fever**, chills, heavy sweating, **headache**, eye **pain** (especially when exposed to light), weakness, and achy muscles.

Description

Like other members of the family of *Rickettsia*, the bacteria causing rickettsialpox live in mice. Tiny mites feed on these infected mice, thus acquiring the organism. When these mites feed on humans, the bacteria can be transmitted.

Rickettsialpox occurs mostly within cities. In the United States, the disease has cropped up in such places as New York City, Boston, Philadelphia, Pittsburgh, and Cleveland. It has also been identified in Russia, Korea, and Africa.

Causes and symptoms

The specific bacteria responsible for rickettsialpox is called *Rickettsia akari*. A person contracts this bacteria through the bite of an infected mite. After a person has been bitten by an infected mite, there is a delay of about 10 days to three weeks prior to the onset of symptoms.

The first symptom is a bump which appears at the site of the original bite. The bump (papule) develops a tiny, fluid-filled head (vesicle). The vesicle sloughs

KEY TERMS

Eschar—A crusty, blackish scab.

Papule—A bump on the skin.

Vesicle—A fluid-filled head on a papule.

away, leaving a crusty black scab in its place (eschar). In about a week, the patient develops a fever, chills, heavy sweating, headache, eye pain (especially when exposed to light), weakness, and achy muscles. The fever rises and falls over the course of about a week. A bumpy rash spreads across the body. Each individual papule follows the same progression: papule, then vesicle, then eschar. The rash does not affect the palms of the hands or the soles of the feet.

Diagnosis

Most practitioners are able to diagnose rickettsialpox simply on the basis of its rising and falling fever, and its characteristic rash. Occasionally, blood will be drawn and tests performed to demonstrate the presence of antibodies (immune cells directed against specific bacterial agents) which would confirm a diagnosis of rickettsialpox.

Treatment

Because rickettsialpox is such a mild illness, some practitioners choose to simply treat the symptoms (giving **acetaminophen** for fever and achiness, pushing fluids to avoid **dehydration**). Others will give their patients a course of the antibiotic tetracycline, which will shorten the course of the illness to about one to two days.

Prognosis

Prognosis for full recovery from rickettsialpox is excellent. No deaths have ever been reported from this illness, and even the skin rash heals without scarring.

Prevention

As with all mite- or tick-borne illnesses, prevention includes avoidance of areas known to harbor the insects, and/or careful application of insect repellents. Furthermore, because mice pass the bacteria on to the mites, it is important to keep mice from nesting in or around residences.

Resources

ORGANIZATIONS

Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

Rosalyn Carson-DeWitt, MD

Rifampin see **Antituberculosis drugs**

Ringing ears see **Tinnitus**

Ringworm

Definition

Ringworm is a common fungal infection of the skin. The name is a misnomer since the disease is not caused by a worm.

Description

More common in males than in females, ringworm is characterized by patches of rough, reddened skin. Raised eruptions usually form the circular pattern that gives the condition its name. Ringworm may also be referred to as dermatophyte infection.

As lesions grow, the centers start to heal. The inflamed borders expand and spread the infection.

Types of ringworm

Ringworm is a term that is commonly used to encompass several types of fungal infection. Sometimes, however, only body ringworm is classified as true ringworm.

Body ringworm (*tinea corporis*) can affect any part of the body except the scalp, feet, and facial area where a man's beard grows. The well-defined, flaky sores can be dry and scaly or moist and crusty.

Scalp ringworm (*tinea capitis*) is most common in children. It causes scaly, swollen blisters or a rash that looks like black dots. Sometimes inflamed and filled with pus, scalp ringworm lesions can cause crusting, flaking, and round bald patches. Most common in black children, scalp ringworm can cause scarring and permanent hair loss.

Ringworm of the groin (*tinea cruris* or **jock itch**) produces raised red sores with well-marked edges. It can spread to the buttocks, inner thighs, and external genitals.



Ringworm on a man's chin. These infections are most common on the feet, scalp, or in toenails, but they can infect any part of the skin. (Custom Medical Stock Photo. Reproduced by permission.)

Ringworm of the nails (tinea unguium) generally starts at the tip of one or more toenails, which gradually thicken and discolor. The nail may deteriorate or pull away from the nail bed. Fingernail infection is far less common.

Causes and symptoms

Ringworm can be transmitted by infected people or pets or by towels, hairbrushes, or other objects contaminated by them. Symptoms include inflammation, scaling, and sometimes, **itching**.

Diabetes mellitus increases susceptibility to ringworm. So do dampness, humidity, and dirty, crowded living areas. Braiding hair tightly and using hair gel also raise the risk.

Diagnosis

Diagnosis is based on microscopic examination of scrapings taken from lesions. A dermatologist may also study the scalp of a patient with suspected tinea capitis under ultraviolet light.

Treatment

Some infections disappear without treatment. Others respond to such topical antifungal medications as naftifine (Caldesene Medicated Powder) or tinactin (Desenex) or to griseofulvin (Fulvicin), which is taken by mouth. Medications should be continued for two weeks after lesions disappear.

A person with body ringworm should wear loose clothing and check daily for raw, open sores. Wet dressings applied to moist sores two or three times a day can lessen inflammation and loosen scales. The doctor may suggest placing special pads between folds of infected skin, and anything the patient has touched or worn should be sterilized in boiling water.

Infected nails should be cut short and straight and carefully cleared of dead cells with an emery board.

Patients with jock itch should:

- wear cotton underwear and change it more than once a day
- keep the infected area dry
- apply antifungal ointment over a thin film of antifungal powder

Shampoo containing selenium sulfide can help prevent spread of scalp ringworm, but prescription shampoo or oral medication is usually needed to cure the infection.

Alternative treatment

The fungal infection ringworm can be treated with homeopathic remedies. Among the homeopathic remedies recommended are:

- *sepia* for brown, scaly patches
- *tellurium* for prominent, well-defined, reddish sores
- *graphites* for thick scales or heavy discharge
- *sulphur* for excessive itching.

Topical applications of antifungal herbs and essential oils also can help resolve ringworm. Tea tree oil (*Melaleuca* spp.), thuja (*Thuja occidentalis*), and lavender (*Lavandula officinalis*) are the most common. Two drops of essential oil in 1/4 oz of carrier oil is the dose recommended for topical application. Essential oils should not be applied to the skin undiluted. Botanical medicine can be taken internally to enhance the body's immune response. A person must be susceptible to exhibit this overgrowth of fungus on the skin. **Echinacea** (*Echinacea* spp.) and astragalus (*Astragalus membranaceus*) are the two most common immune-enhancing herbs. A well-balanced diet, including protein, complex carbohydrates, fresh fruits and vegetables, and good quality fats, is also important in maintaining optimal immune function.

Prognosis

Ringworm can usually be cured, but recurrence is common. Chronic infection develops in one patient in five.

It can take six to 12 months for new hair to cover bald patches, and three to 12 months to cure infected fingernails. Toenail infections do not always respond to treatment.

Prevention

Likelihood of infection can be lessened by avoiding contact with infected people or pets or contaminated objects and staying away from hot, damp places.

Resources

OTHER

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Maureen Haggerty

Rinne test see **Hearing tests with a tuning fork**

Ritonavir see **Protease inhibitors**

River blindness see **Filariasis**

RMSF see **Rocky Mountain spotted fever**

Rocky Mountain spotted fever

Definition

Rocky Mountain spotted **fever** (RMSF) is a tick-borne illness caused by a bacteria, resulting in a high fever and a characteristic rash.

Description

The bacteria causing RMSF is passed to humans through the bite of an infected tick. The illness begins within about two weeks of such a bite. RMSF is the most widespread tick-borne illness in the United States, occurring in every state except Alaska and Hawaii. The states in the mid-Atlantic region, the Carolinas, and the Virginias have a great deal of tick activity during the spring and summer months, and the largest number of RMSF cases come from those states. About 5% of all ticks carry the causative bacteria. Children under the age of 15 years have the majority of RMSF infections.

Causes and symptoms

The bacterial culprit in RMSF is called *Rickettsia rickettsii*. It causes no illness in the tick carrying it, and can be passed on to the tick's offspring. When a tick attaches to a human, the bacteria is passed. The tick must be attached to the human for about six hours for this passage to occur. Although prompt tick removal will cut down on the chance of contracting RMSF, removal requires great care. If the tick's head and body are squashed during the course of removal, the bacteria can be inadvertently rubbed into the tiny bite wound.

Symptoms of RMSF begin within two weeks of the bite of the infected tick. Symptoms usually begin suddenly, with high fever, chills, **headache**, severe weakness, and muscle **pain**. Pain in the large muscle of the calf is very common, and may be particularly severe. The patient may be somewhat confused and delirious. Without treatment, these symptoms may last two weeks or more.

The rash of RMSF is quite characteristic. It usually begins on the fourth day of the illness, and occurs in at least 90% of all patients with RMSF. It starts around the wrists and ankles, as flat pink marks (called macules). The rash spreads up the arms and legs, toward the chest, abdomen, and back. Unlike **rashes** which accompany various viral infections, the rash of RMSF does spread to the palms of the hands and the soles of the feet. Over a couple of days, the macules turn a reddish-purple color. They are now called petechiae, which are tiny areas of bleeding under the skin (pinpoint hemorrhages). This signifies a new phase of the illness. Over the next several days, the individual petechiae may spread into each other, resulting in larger patches of hemorrhage.

The most severe effects of RMSF occur due to damage to the blood vessels, which become leaky. This accounts for the production of petechiae. As blood and fluid leak out of the injured blood vessels, other tissues and organs may swell and become damaged, and:

- breathing difficulties may arise as the lungs are affected.
- heart rhythms may become abnormal
- kidney failure occurs in very ill patients
- liver function drops
- the patient may experience **nausea, vomiting**, abdominal pain, and diarrhea
- the brain may swell (**encephalitis**) in about 25% of all RMSF patients (brain injury can result in seizures, changes in consciousness, actual **coma**, loss of coordination, imbalance on walking, **muscle spasms**, loss of bladder control, and various degrees of paralysis)
- the clotting system becomes impaired, and blood may be evident in the stools or vomit

Diagnosis

Diagnosis of RMSF is almost always made on the basis of the characteristic symptoms, coupled with either a known tick bite (noted by about 60–70% of patients) or exposure to an area known to harbor ticks. Complex tests exist to nail down a diagnosis of RMSF, but these are performed in only a few laboratories. Because the results of these tests take so long to obtain, they are seldom used. This is because delaying treatment is the main cause of **death** in patients with RMSF.

Treatment

It is essential to begin treatment absolutely as soon as RMSF is seriously suspected. Delaying treatment can result in death.

Antibiotics are used to treat RMSF. The first choice is a form of tetracycline; the second choice (used in young children and pregnant women) is chloramphenicol. If the patient is well enough, treatment by oral intake of medicine is perfectly effective. Sicker patients will need to be given the medication through a needle in the vein (intravenously). Penicillin and sulfa drugs are not suitable for treatment of RMSF, and their use may increase the death rate by delaying the use of truly effective medications.

Very ill patients will need to be hospitalized in an intensive care unit. Depending on the types of complications a particular patient experiences, a variety of treatments may be necessary, including intravenous fluids, blood transfusions, anti-seizure medications, **kidney dialysis**, and mechanical ventilation (a breathing machine).

Alternative treatment

Although alternative treatments should never be used in place of conventional treatment with antibiotics, they can be useful adjuncts to antibiotic therapy. The use of *Lactobacillus acidophilus* and *L. bifidus* supplementaion during and after antibiotic treatment can help rebalance the intestinal flora. **Acupuncture**, homeopathy, and botanical medicine can all be beneficial supportive therapies during recovery from this disease.

Prognosis

Prior to the regular use of antibiotics to treat RMSF, the death rate was about 25%. Although the death rate from RMSF has improved greatly with an understanding of the importance of early use of antibiotics, there is still a 5% death rate. This rate is believed to be due to delays in the administration of appropriate medications.

Certain risk factors suggest a worse outcome in RMSF. Death rates are higher in males and increase as people age. It is considered a bad prognostic sign to develop symptoms of RMSF within only two to five days of a tick bite.

Prevention

The mainstay of prevention involves avoiding areas known to harbor ticks. However, because

KEY TERMS

Encephalitis—Inflammation of the tissues of the brain.

Macule—A flat, discolored area on the skin.

Petechia—A small, round, reddish purple spot on the skin, representing a tiny area of bleeding under the skin.

many people enjoy recreational activities in just such areas, the following steps can be taken:

- Wear light colored clothing (so that attached ticks are more easily noticed).
- Wear long sleeved shirts and long pants; tuck the pants legs into socks.
- Spray clothing with appropriate tick repellents.
- Examine. Anybody who has been outside for any amount of time in an area known to have a population of ticks should examine his or her body carefully for ticks. Parents should examine their children at the end of the day.
- Remove any ticks using tweezers, so that infection doesn't occur due to handling the tick. Grasp the tick's head with the tweezers, and pull gently but firmly so that the head and body are entirely removed.
- Keep areas around homes clear of brush, which may serve to harbor ticks.

Resources

ORGANIZATIONS

Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

Rosalyn Carson-DeWitt, MD

Rogaine see **Minoxidil**

Rolfing

Definition

Rolfing, also called Rolf therapy or structural integration, is a holistic system of bodywork that uses deep manipulation of the body's soft tissue to

IDA P. ROLF, PH.D. (1896–1979)

Born in New York City and raised in the Bronx, Ida P. Rolf attended school in the New York area, graduating from Barnard College in 1916. In 1920, she graduated from the Columbia University College of Physicians and Surgeons with a doctorate in biological chemistry. For the next 12 years, she worked in the departments of chemotherapy and organic chemistry at the Rockefeller Institute. During an extended leave of absence, she studied atomic physics and mathematics at the Swiss Technical University in Zurich and homeopathic medicine in Geneva. During the 1930s, she studied osteopathy, chiropractic medicine, tantric yoga, the Alexander Technique of tension reduction through body movement, and the philosophy of altered states of consciousness of Alfred H.S. Korzybski.

Her interest in body structure, movement, and manipulation began after being kicked by a horse shortly after graduating from Barnard. The accident left her with acute pneumonia. Dissatisfied with conventional medical treatment, she began her quest for more natural and effective ways of treating the body.

By 1940, Dr. Rolf had developed a technique of body movement she called structural integration, also known today as Rolfing. The therapy reshapes the body's muscular structure by applying pressure and energy, freeing the body from physical and emotional traumas. In 1977, she authored *Rolfing: The Integration of Human Structures*. She continued to teach and refine her therapy until her death in 1979. Dr. Rolf's desire to teach her work to others led to her establishing the Guild for Structural Engineering, now known as the Rolf Institute of Structural Integration, 205 Canyon Blvd., Boulder, CO 80302.

realign and balance the body's myofascial structure. Rolfing improves posture, relieves chronic **pain**, and reduces **stress**.

Purpose

Rolfing helps to improve posture and bring the body's natural structure into proper balance and alignment. This can bring relief from general aches and pains, improve breathing, increase energy, improve self-confidence, and relieve physical and mental stress. Rolfing has also been used to treat such specific physical problems as chronic back, neck, shoulder, and joint pain, and repetitive stress injuries, including **carpal tunnel syndrome**. Many amateur and professional athletes, including Olympic skaters and skiers, use Rolfing to keep in top condition, to prevent injuries, and to more quickly recover from injuries.

Description

Origins

Ida Pauline Rolf (1896–1979) was a biochemist from New York who developed structural integration over the course of many years after an accident as a young woman. She was kicked by a horse's hoof on a trip out West and developed symptoms resembling those of acute **pneumonia**. She made her way to a hospital in Montana, where she was treated by a physician who called in an osteopath to assist in her treatment. After the osteopath treated her, she was able to breathe normally. After her return to New York, her mother took her to a blind osteopath for further treatment. He taught her about the body's structure and function, after which Rolf became dissatisfied with conventional medical treatment. Following completion of a doctorate in biochemistry from Columbia University in 1920, Rolf studied atomic physics, mathematics, and **homeopathic medicine** in Europe. After 1928, when her father died and left her an inheritance that allowed her to pursue her own studies, she explored various forms of alternative treatment, including **osteopathy**, **chiropractic** medicine, **tantric yoga**, the **Alexander technique** of tension reduction through body movement, and Alfred Korzybski's philosophy of altered states of consciousness.

By 1940, Rolf had synthesized what she had learned from these various disciplines into her own technique of body movement that she called structural integration, which later became known as Rolfing. During the Second World War, Rolf continued to study with an osteopath in California named Amy Cochran. In the mid-1960s, Gestalt therapist Fritz Perls invited Rolf to Esalen, where she began to develop a following among people involved in the human potential movement. In 1977, she published *Rolfing: The Integration of Human Structures*, the definitive book on structural integration bodywork. She continued to refine the therapy until her death in 1979. Rolf's work is carried on through her Guild for Structural Integration, now known as the Rolf Institute of Structural Integration, which she founded in 1971 in Boulder, Colo.

Rolfing is more than just a massage of the body's surface. It is a system that reshapes the body's myofascial structure by applying pressure and energy, thereby freeing the body from the effects of physical and emotional traumas. Although Rolfing is used extensively to treat **sports injuries** and back pain, it is not designed as a therapy for any particular condition. Rather, it is a systematic approach to overall wellness. It works by counteracting the effects of gravity, which

over time pulls the body out of alignment. This pull causes the body's connective tissue to become harder and stiffer, and the muscles to atrophy. Signs of this stiffening and contraction include slouching or an overly erect posture.

Rolfing identifies the vertical line as the ideal that the body should approximate. The mission statement of the Guild for Structural Integration describes Rolfing as "a method and a philosophy of personal growth and integrity.... The vertical line is our fundamental concept. The physical and psychological embodiment of the vertical line is a way of Being in the physical world [that] forms a basis for personal growth and integrity."

The basic ten

Basic Rolfing treatment consists of 10 sessions, each lasting 60–90 minutes and costing about \$100 each. The sessions are spaced a week or longer apart. After a period of integration, specialized or advanced treatment sessions are available. A "tuneup" session is recommended every six months. In each session, the Rolfer uses his or her fingers, hands, knuckles, and elbows to rework the connective tissue over the entire body. The tissues are worked until they become pliable, allowing the muscles to lengthen and return to their normal alignment. The deep tissue manipulation improves posture and agility, and increases the body's range of movement. Rolfers also believe that the blocked energy accumulated in the tissue from emotional tension is released through Rolfing treatment, causing the patient to feel more energetic and have a more positive frame of mind.

Clients are asked to wait for a period of six to 12 months before scheduling advanced work, known as the PostTen/Advanced Series. This period allows the body to integrate the work done in the "Basic Ten."

Rolfing movement integration

Rolfing movement integration, or RMI, is intended to help clients develop better awareness of their vertical alignment and customary movement patterns. They learn to release tension and discover better ways to use body movement effectively.

Rolfing rhythms

Rolfing rhythms are a series of exercises intended to remind participants of the basic principles of Rolfing: ease, length, balance, and harmony with gravity. In addition, Rolfing rhythms improve the client's flexibility as well as muscle tone and coordination.

KEY TERMS

Atrophy—A progressive wasting and loss of function of any part of the body.

Carpal tunnel syndrome—A condition caused by compression of the median nerve in the carpal tunnel of the hand, characterized by pain.

Fascia—The sheet of connective tissue that covers the body under the skin and envelops every muscle, bone, nerve, gland, organ, and blood vessel. Fascia helps the body to retain its basic shape.

Osteopathy—A system of medical practice that believes that the human body can make its own remedies to heal infection. It originally used manipulative techniques but also added surgical, hygienic, and medicinal methods when needed.

Parasympathetic nervous system—A part of the autonomic nervous system that is concerned with conserving and restoring energy. It is the part of the nervous system that predominates in a state of relaxation.

Structural integration—The term used to describe the method and philosophy of life associated with Rolfing. Its fundamental concept is the vertical line.

Preparations

No pre-procedure preparations are needed to begin Rolfing treatment. The treatment is usually done on a massage table with the patient wearing only undergarments. Prior to the first session, however, the client is asked to complete a health questionnaire, and photographs are taken to assist with evaluation of his or her progress.

Precautions

Since Rolfing involves vigorous deep tissue manipulation, it is often described as uncomfortable and sometimes painful, especially during the first several sessions. In the past decade, however, Rolfers have developed newer techniques that cause less discomfort to participants. Since Rolfing is a bodywork treatment that requires the use of hands, it may be a problem for people who do not like or are afraid of being touched. It is not recommended as a treatment for any disease or a chronic inflammatory condition such as arthritis, and can worsen such a condition. Anyone with a serious medical condition, including heart disease, diabetes, or respiratory problems,

should consult with a medical practitioner before undergoing Rolfing.

Side effects

There are no reported serious side effects associated with Rolfing when delivered by a certified practitioner to adults and juveniles.

Research and general acceptance

There is a growing amount of mainstream scientific research documenting the effectiveness of Rolf therapy. A 1988 study published in the *Journal of the American Physical Therapy Association* indicated that Rolfing stimulates the parasympathetic nervous system, which can help speed the recovery of damaged tissue. Other studies done in the 1980s concerned the effectiveness of Rolfing in treating figure skaters and children with **cerebral palsy**. In 1992 a presentation was made to the National Center of Medical **Rehabilitation** Research regarding Rolfing in the treatment of degenerative joint disease. A 1997 article in *The Journal of Orthopedic and Sports Physical Therapy* reported that Rolfing can provide effective and sustained pain relief from lower back problems.

Resources

BOOKS

Levine, Andrew S., and Valerie J. Levine. *The Bodywork and Massage Sourcebook*. Lincolnwood, IL: Lowell House. 1999.

ORGANIZATIONS

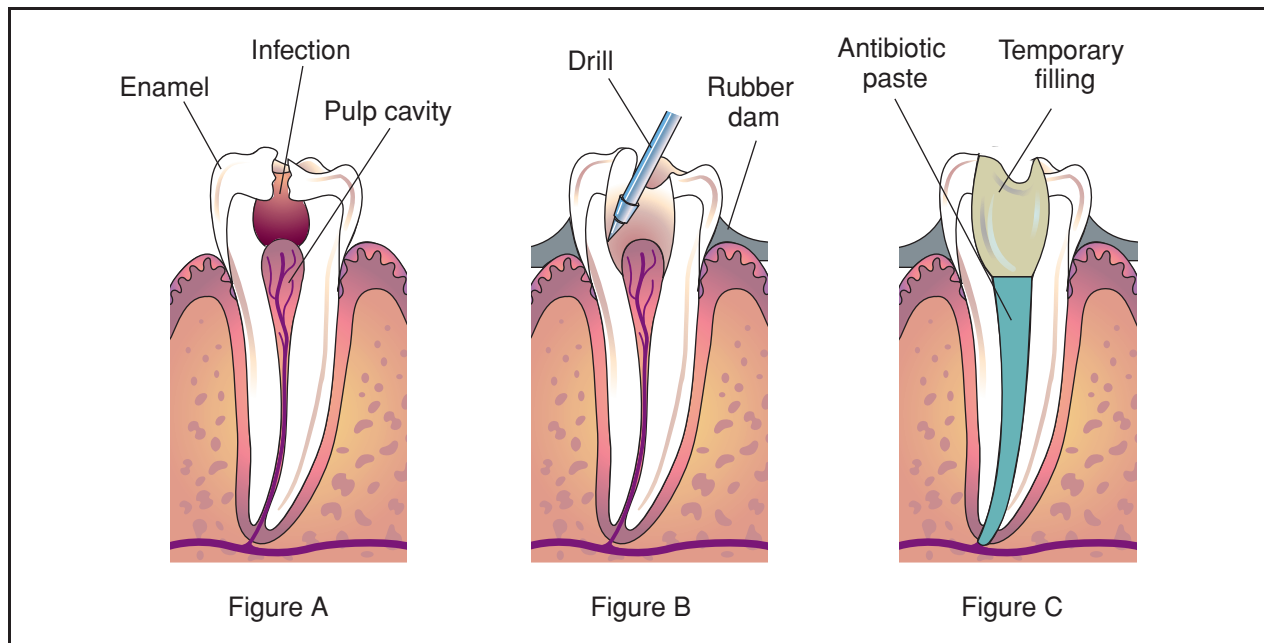
Rolf Institute of Structural Integration. 209 Canyon Blvd. P.O. Box 1868. Boulder, CO 80306-1868. (303) 449-5903. (800) 530-8875. <<http://www.rolf.org>>.

Ken R. Wells

Root canal treatment

Definition

Root canal treatment, also known as endodontic treatment, is a dental procedure in which the diseased or damaged pulp (core) of a tooth is removed and the inside areas (the pulp chamber and root canals) are filled and sealed.



Root canal treatment is a dental procedure in which the diseased pulp of a tooth is removed and the inside areas are filled and sealed. In figure A, the infection can be seen above the pulp cavity. The dentist drills into the enamel and the pulp cavity is extracted (figure B). Finally, the dentist fills the pulp cavity with antibiotic paste and a temporary filling (figure C). (Illustration by Electronic Illustrators Group.)

Purpose

Inflamed or infected pulp (pulpitis) most often causes a **toothache**. To relieve the **pain** and prevent further complications, the tooth may be extracted (surgically removed) or saved by root canal treatment. Root canal treatment has become a common dental procedure; more than 14 million are performed every year, with a 95% success rate, according to the American Association of Endodontists.

Precautions

Once root canal treatment is performed, the patient must have a crown placed over the tooth to protect it. The cost of the treatment and the crown may be expensive. However, replacing an extracted tooth with a fixed bridge, a removable partial denture, or an implant to maintain the space and restore the chewing function is typically even more expensive.

Description

Root canal treatment may be performed by a general dentist or by an endodontist, a dentist who specializes in endodontic (literally “inside of the tooth”) procedures. Inside the tooth, the pulp’s soft tissue contains the blood supply, by which the

tooth gets its nutrients, and the nerve, by which the tooth senses hot and cold. This tissue is vulnerable to damage from deep dental decay, accidental injury, tooth fracture, or trauma from repeated dental procedures (such as multiple fillings over time). If a tooth becomes diseased or injured, bacteria build up inside the pulp, spreading infection from the natural crown of the tooth to the root tips in the jawbone. Pus accumulates at the ends of the roots, forming a painful **abscess** which can damage the bone supporting the teeth. Such an infection may produce pain that is severe, constant, or throbbing, as well as prolonged sensitivity to heat or cold, swelling and tenderness in the surrounding gums, facial swelling, and discoloration of the tooth. However, in some cases, the pulp may die so gradually that there is little noticeable pain.

Root canal treatment is performed under **local anesthesia**. A thin sheet of rubber, called a rubber dam, is placed in the mouth to isolate the tooth. The dentist removes any **tooth decay** and makes an opening through the natural crown of the tooth into the pulp chamber. Creating this access also relieves the pressure inside the tooth and can dramatically ease pain.

The dentist determines the length of the root canals, usually with a series of x rays. Small wire-like

files are then used to clean the entire canal space of diseased pulp tissue and bacteria. The debris is flushed out with large amounts of water (irrigation). The canals are also slightly enlarged and shaped to receive an inert (non-reactive) filling material called gutta percha. However, the tooth is not filled and permanently sealed until it is completely free of active infection. The dentist may place a temporary seal, or leave the tooth open to drain, and prescribe an antibiotic to counter any spread of infection from the tooth. This is why root canal treatment may require several visits to the dentist.

Once the canals are completely clean, they are filled with gutta percha and a sealer cement to prevent bacteria from entering the tooth in the future. A metal post may be placed in the pulp chamber for added structural support and better retention of the crown restoration. The tooth is protected by a temporary filling or crown until a permanent restoration may be made. This restoration is usually a gold or porcelain crown, although it may be a gold inlay, or an amalgam or composite filling (paste fillings that harden).

Preparation

There is no typical preparation for root canal treatment. Once the tooth is opened to drain, the dentist may prescribe an antibiotic, then the patient should take the full prescribed course. With the infection under control, local anesthetic is more effective, so that the root canal procedure may be performed without discomfort.

Aftercare

The tooth may be sore for several days after filling. Pain relievers, such as ibuprofen (Advil, Motrin) may be taken to ease the soreness. The tissues around the tooth may also be irritated. Rinsing the mouth with hot salt water several times a day will help. Chewing on that side of the mouth should be avoided for the first few days following treatment. A follow-up appointment should be scheduled with the dentist for six months after treatment to make sure the tooth and surrounding structures are healthy.

Risks

There is a possibility that the root canal treatment will not be successful the first time. If infection and inflammation recur and an x ray indicates retreatment is feasible, the old filling material is removed and the canals are thoroughly cleaned out. The dentist will try

KEY TERMS

Abscess—A hole in the tooth or gum tissue filled with pus as the result of infection. Its swelling exerts pressure on the surrounding tissues, causing pain.

Apicoectomy—Also called root resectioning. The root tip of a tooth is accessed in the bone and a small amount is shaved away. The diseased tissue is removed and a filling is placed to reseal the canal.

Crown—The natural crown of a tooth is that part of the tooth covered by enamel. Also, a restorative crown is a protective shell that fits over a tooth.

Endodontic—Pertaining to the inside structures of the tooth, including the dental pulp and tooth root, and the periapical tissue surrounding the root.

Endodontist—A dentist who specializes in the diagnosis and treatment of disorders affecting the inside structures of the tooth.

Extraction—The surgical removal of a tooth from its socket in the bone.

Gutta percha—An inert latex-like substance used for filling root canals.

Pulp—The soft innermost layer of a tooth, containing blood vessels and nerves.

Pulp chamber—The area within the natural crown of the tooth occupied by dental pulp.

Pulpitis—Inflammation of the pulp of a tooth involving the blood vessels and nerves.

Root canal—The space within a tooth that runs from the pulp chamber to the tip of the root.

Root canal treatment—The process of removing diseased or damaged pulp from a tooth, then filling and sealing the pulp chamber and root canals.

to identify and correct problems with the first root canal treatment before filling and sealing the tooth a second time.

In cases where an x ray indicates that retreatment cannot correct the problem, endodontic surgery may be performed. In a procedure called an apicoectomy, or root resectioning, the root end of the tooth is accessed in the bone, and a small amount is shaved away. The area is cleaned of diseased tissue and a filling is placed to reseal the canal.

In some cases, despite root canal treatment and endodontic surgery, the tooth dies anyway and must be extracted.

Normal results

With successful root canal treatment, the tooth will no longer cause pain. However, because it does not contain an internal nerve, it no longer has sensitivity to hot, cold, or sweets. These are signs of dental decay, so the patient must receive regular dental check-ups with periodic x rays to avoid further disease in the tooth. The restored tooth could last a lifetime; however, with routine wear, the filling or crown may eventually need to be replaced.

Resources

ORGANIZATIONS

American Association of Endodontists. 211 East Chicago Ave., Ste. 1100, Chicago, IL 60611-2691. (800) 872-3636. <<http://www.aae.org>>.

American Dental Association. 211 E. Chicago Ave., Chicago, IL 60611. (312) 440-2500. <<http://www.ada.org>>.

OTHER

"Endodontic (Root Canal) Therapy." *Tooth Talk and Your Health with Dr. Frank Gober*. <<http://www.toothtalk.com>>.

"Root Canal Treatment." *Annapolis Endodontics*. <<http://users.erols.com/canals/index.html>>.

"Root Canal Treatment." *Value Added Benefits*. <<http://www.vab.com>>.

Thivierge, Bethany. "What is the Value (and Cost) of a Root Canal?" *A Healthy Me Page*. <<http://www.ahealthyme.com/topic/rootcanal>>.

Bethany Thivierge

Rosacea

Definition

Rosacea is a skin disease typically appearing in people during their 30s and 40s. It is marked by redness (erythema) of the face, flushing of the skin, and the presence of hard pimples (papules) or pus-filled pimples (pustules), and small visible spider-like veins called telangiectasias. In later stages of the disease, the face may swell and the nose may take on a bulb-like appearance called rhinophyma.

Description

Rosacea produces redness and flushing of the skin, as well as pustules and papules. Areas of the face, including the nose, cheeks, forehead, and chin,

are the primary sites, but some people experience symptoms on their necks, backs, scalp, arms, and legs.

The similarity in appearance of rosacea to **acne** led people in the past to erroneously call the disease acne rosacea or adult acne. Like acne, the skin can have pimples and papules. Unlike acne, however, people with rosacea do not have blackheads.

In early stages of rosacea, people typically experience repeated episodes of flushing. Later, areas of the face are persistently red, telangiectasia appear on the nose and cheeks, as well as inflamed papules and pustules. Over time, the skin may take on a roughened, orange peel texture. Very late in the disorder, a small group of patients with rosacea will develop rhinophyma, which can give the nose a bulb-like look.

Up to one half of patients with rosacea may experience symptoms related to their eyes. Ocular rosacea, as it is called, frequently precedes the other manifestations on the skin. Most of these eye symptoms do not threaten sight, however. Telangiectasia may appear around the borders of the eyelid, the eyelids may be chronically inflamed, and small lumps called chalazions may develop. The cornea of the eye, the transparent covering over the lens, can also be affected, and in some cases vision will be affected.

Causes and symptoms

There is no known specific cause of rosacea. A history of redness and flushing precedes the disease in most patients. The consensus among many experts is that multiple factors may lead to an overreaction of the facial blood vessels, which triggers flushing. Over time, persistent episodes of redness and flushing leave the face continually inflamed. Pimples and blood-vessel changes follow.

Certain genetic factors may also come into play, although these have not been fully described. The disease is more common in women and light-skinned, fair-haired people. It may be more common in people of Celtic background, although this is an area of disagreement among experts.

Certain **antibiotics** are useful in the treatment of rosacea, leading some researchers to suspect a bacterium or other infectious agent may be the cause. One of the newest suspects is a bacterium called *Helicobacter pylori*, which has been implicated in causing many cases of stomach ulcers but the evidence here is mixed.

Other investigators have observed that a particular parasite, the mite *Demodex folliculorum*, can be found in areas of the skin affected by rosacea. The mite can also be detected, however, in the skin of

people who do not have the disease. It is likely that the mite does not cause rosacea, but merely aggravates it.

Diagnosis

Diagnosis of rosacea is made by the presence of clinical symptoms. There is no specific test for the disease. Episodes of persistent flushing, redness (erythema) of the nose, cheeks, chin, and forehead, accompanied by pustules and papules are hallmarks of the disease. A dermatologist will attempt to rule out a number of other diseases that have similar symptoms. Acne vulgaris is perhaps the disorder most commonly mistaken for rosacea, but redness and spider-like veins are not observed in patients with acne. Blackheads and cysts, however, are seen in acne patients, but not in those with rosacea.

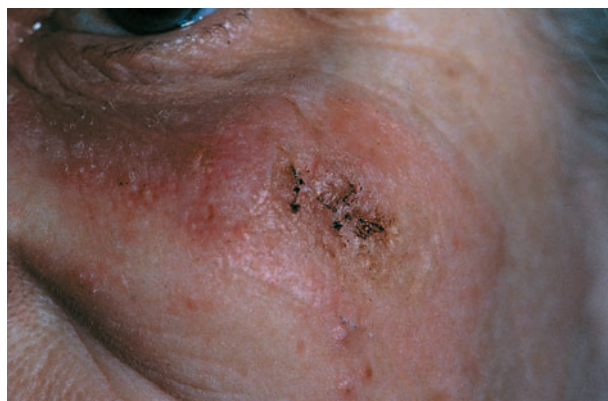
Other diseases that produce some of the same symptoms as rosacea include perioral **dermatitis** and **systemic lupus erythematosus**.

Treatment

The mainstay of treatment for rosacea is oral antibiotics. These appear to work by reducing inflammation in the small blood vessels and structure of the skin, not by destroying bacteria that are present. Among the more widely used oral antibiotics is tetracycline. In many patients, antibiotics are effective against the papules and pustules that can appear on the face, but they appear less effective against the background redness, and they have no effect on telangiectasia. Patients frequently take a relatively high dose of antibiotics until their symptoms are controlled, and then they slowly reduce their daily dose to a level that just keeps their symptoms in check. Other oral antibiotics used include erythromycin and minocycline.

Some patients are concerned about long-term use of oral antibiotics. For them, a topical agent applied directly to the face may be tried in addition to an oral antibiotic, or in its place. **Topical antibiotics** are also useful for controlling the papules and pustules of rosacea, but do not control the redness, flushing, and telangiectasias. The newest of these topical agents is metronidazole gel, which can be applied twice daily. Like the oral antibiotics, topical preparations appear to work by reducing inflammation, not by killing bacteria.

Vitamin A derivatives, called retinoids, also appear useful in the treatment of rosacea. An oral retinoid, called isotretinoin, which is used in severe cases of acne also reduces the pustules and papules in severe cases of rosacea that do not respond to antibiotics. Isotretinoin must be taken with care, however,



Rosacea on a woman's cheek. (Custom Medical Stock Photo. Reproduced by permission.)

particularly in women of childbearing age. They must agree to a reliable form of **contraception**, because the drug is known to cause **birth defects**.

Topical vitamin A derivatives that are used in the treatment of acne also may have a role in the treatment of rosacea. Accumulating evidence suggests that topical isotretinoin and topical azelaic acid can reduce the redness and pimples. Some patients who use these medications experience skin irritation that tends to resolve with time.

For later stages of the disorder, a surgical procedure may be needed to improve the appearance of the skin. To remove the telangiectasias, a dermatologist may use an electrocautery device to apply a current to the blood vessel in order to destroy it. Special lasers, called tunable dye lasers, can also be adjusted to selectively destroy these tiny blood vessels.

A variety of surgical techniques can be used to improve the shape and appearance of a bulbous nose in the later stages of the disease. Surgeons may use a scalpel or laser to remove excess tissue from the nose and restore a more natural appearance.

Alternative treatment

Alternative treatments have not been extensively studied in rosacea. Some reports advocate gentle circular massage for several minutes daily to the nose, cheeks, and forehead. Scientifically controlled studies are lacking, however.

Many people are able to avoid outbreaks by reducing things that trigger flushing. Alcoholic beverages, hot beverages, and spicy foods are among the more common factors in the diet that can provoke flushing. Reducing or eliminating these items in the diet can help limit rosacea outbreaks in many people. Exposure to

KEY TERMS

Blackhead—A plug of fatty cells capped with a blackened mass.

Erythema—A diffuse red and inflamed area of the skin.

Papule—A small hard elevation of the skin.

Pustule—A small pus-filled elevation of the skin.

Retinoid—A synthetic vitamin A derivative used in the treatment of a variety of skin disorders.

Rhinophyma—Long-term swelling and overgrowth in skin tissue of the nose that leaves it with a knobby bulb-like look.

Telangiectasia—Small blood veins visible at the surface of the skin of the nose and cheeks.

heat, cold, and sunlight are also known triggers of flushing. The specific things that provoke flushing vary considerably from person to person, however. It usually takes some trial and error to figure these out.

A deficiency in hydrochloric acid (HCl) in the stomach may be a cause of rosacea, and supplementation with HCl capsules may bring relief in some cases.

Prognosis

The prognosis for controlling symptoms of rosacea and improving the appearance of the face is good. Many people require life-long treatment and achieve good results. There is no known cure for the disorder.

Prevention

Rosacea cannot be prevented, but once correctly diagnosed, outbreaks can be treated and repeated episodes can be limited.

Use mild soaps

Avoiding anything that irritates the skin is a good preventive measure for people with rosacea. Mild soaps and cleansers are recommended. Astringents and alcohol should be avoided.

Learn what triggers flushing

Reducing factors in the diet and environment that cause flushing of the face is another good preventive strategy. Alcoholic and hot beverages, and spicy foods are among the more common triggers.

Use sunscreen

Limiting exposure of the face to excesses of heat and cold can also help. A sunscreen with a skin protection factor (SPF) of 15 or greater used daily can limit the damage to the skin and small blood vessels caused by the sun, and reduce outbreaks.

Resources

ORGANIZATIONS

American Academy of Dermatology. 930 N. Meacham Road, P.O. Box 4014, Schaumburg, IL 60168-4014. (847) 330-0230. Fax: (847) 330-0050.
< <http://www.aad.org> >.

National Rosacea Society. 800 S. Northwest Highway, Suite 200, Barrington, IL 60010. (888) 662-5874.
< <http://www.rosacea.org> >.

Richard H. Camer

Rosary bead esophagus see **Diffuse esophageal spasm**

Roseola

Definition

Roseola is a common disease of babies or young children, in which several days of very high **fever** are followed by a rash.

Description

Roseola is an extraordinarily common infection, caused by a virus. About 90% of all children have been exposed to the virus, with about 33% actually demonstrating the syndrome of fever followed by rash.

The most common age for a child to contract roseola is between six and twelve months. Roseola infection strikes boys and girls equally. The infection may occur at any time of year, although late spring and early summer seem to be peak times for it.

Causes and symptoms

About 85% of the time, roseola is caused by a virus called Human Herpesvirus 6, or HHV-6. Although the virus is related to those herpesviruses known to cause sores on the lips or genitalia, HHV-6 causes a very different type of infection. HHV-6 is believed to be passed between people via infected saliva. A few other viruses (called enteroviruses) can



Roseola rash on infant's back and shoulders. (Custom Medical Stock Photo. Reproduced by permission.)

produce a similar fever-then-rash illness, which is usually also called roseola.

Researchers believe that it takes about 5-15 days to develop illness after having been infected by HHV-6. Roseola strikes suddenly, when a previously-well child spikes an impressively high fever. The temperature may reach 106°F. As is always the case with sudden fever spikes, the extreme change in temperature may cause certain children to have seizures. About 5-35% of all children with roseola will have these “febrile seizures.”

The most notable thing about this early phase of roseola is the absence of symptoms, other than the high fever. Although some children have a slightly reddened throat, or a slightly runny nose, most children have no symptoms whatsoever, other than the sudden development of high fever. This fever lasts for between three and five days.

Somewhere around the fifth day, a rash begins on the body. The rash is usually composed of flat pink patches or spots, although there may be some raised patches as well. The rash usually starts on the chest, back, and abdomen, and then spreads out to the arms and neck. It may or may not reach the legs and face. The rash lasts for about three days, then fades.

Very rarely, roseola will cause more serious disease. Patients so afflicted will experience significant swelling of the lymph nodes, the liver, and the spleen. The liver may become sufficiently inflamed to interfere with its functioning, resulting in a yellowish color to the whites of the eyes and the skin (**jaundice**). This syndrome (called a mononucleosis-like syndrome, after the disease called mononucleosis that causes many of the same symptoms) has occurred in both infants and adults.

KEY TERMS

Jaundice—The development of a yellowish tone to the skin and the whites of the eyes, caused by poor liver function.

Mononucleosis—An infection which causes swelling of lymph nodes, spleen, and liver, usually accompanied by extremely sore throat, fever, headache, and intense long-lasting fatigue.

Diagnosis

The diagnosis of roseola is often made by carefully examining the feverish child to make sure that other illnesses are not causing the temperature spike. Once it is clear that no **pneumonia**, ear infection, **strep throat**, or other common childhood illness is present, the practitioner usually feels comfortable waiting to see if the characteristic rash of roseola begins.

Treatment

There are no treatments available to stop the course of roseola. **Acetaminophen** or ibuprofen is usually given to try to lower the fever. Children who are susceptible to seizures may be given a sedative medication when the fever first spikes, in an attempt to prevent such a seizure.

Prognosis

Children recover quickly and completely from roseola. The only complications are those associated with seizures, or the rare mononucleosis-like syndrome.

Prevention

Other than the usual good hygiene practices always recommended to decrease the spread of viral illness, no methods are available to specifically prevent roseola.

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Rosalyn Carson-DeWitt, MD

Roseola infantum see **Roseola**

Ross River Virus

Definition

Ross River Virus (RRV) is Australia's most common and widespread mosquito-borne pathogen. Also known as RRV disease, it can cause debilitating polyarthritis, rash, **fever**, and constitutional symptoms.

Description

Originally known as epidemic polyarthritis, RRV is a member of the *Togaviridae* family of arboviruses. RRV is transmitted in an animal host-vector-human cycle, where the vector is the mosquito. Serological investigations have indicated that native macropods are the main vertebrate hosts of RRV, although other animals can become infected as well. The RRV lives in the blood stream of an infected animal. When a mosquito feeds on the infected animal, the virus is transmitted to the insect where it rapidly multiplies. The virus is then passed onto the next animal or person the mosquito bites. It has been proposed that human-mosquito-human transmission can occur during RRV epidemics. One-third of all humans bitten by an infected mosquito will develop the RRV disease.

The RRV disease occurs throughout continental Australia. However, the majority of RRV infections occur in the northern states and along coastal areas; in particular, the state of Queensland. Of the 4,800 cases reported annually in Australia, approximately 2,700 of these occur in Queensland. In addition to these cases, many more go unreported. Infection can occur year round, but outbreaks typically coincide with the increased mosquito activity of the wet season (between late November and the end of April). Also, areas with intensive irrigation and those near salt marches have higher mosquito populations, and, thus, tend to exhibit a higher number of RRV cases.

In addition to continental Australia, RRV is endemic to the Solomon Islands, East Timor, Papua New Guinea, and the adjacent islands of Indonesia. Epidemics have also been reported in the Cook Islands, Fiji, French Polynesia, New Caledonia, and Western Samoa.

Causes and symptoms

Many people that are infected with RRV will never develop symptoms. However, 25% to 45% of cases will develop symptoms within three days to three weeks (averaging nine days) of the infection. Symptoms will vary between patients, but typically

include arthralgia, arthritis, myalgia, skin rash, fever, **fatigue**, **headache**, and swollen lymph nodes. **Tingling** and **pain** in the palms of the hands and soles of the feet can accompany these symptoms. Other, less frequent, symptoms can include general malaise, **nausea**, sore eyes, and **sore throat**.

Most patients with RRV disease (83% to 98%) experience symptoms of polyarthritis involving the wrists, knees, ankles, and small joints of the extremities. Less frequently affected joints include the elbows, toes, tarsal joints, vertebral joints, shoulders, and hips. Symptoms can range from restricted joint movement to prominent swell and severe pain. Although severe joint pain can last for only 2 to 6 weeks, over half of patients will continue to experience joint pain for 6 to 12 months after RRV infection. Symptoms typically diminish over time, but relapses are common and have been known to persist for several years. This persistent polyarthritis can lead to fatigue and myalgia, contributing to RRV diseases high morbidity.

Diagnosis

Diagnosis usually consists of serological tests to determine the presence and increase of RRV antibodies. Samples should be taken during the acute or convalescent stages of the illness. Testing will help clinicians differentiate between RRV disease and Barmah Forest virus disease, a very similar arbovirus. Virological tests can help distinguish between RRV disease and other causes of arthritis.

Treatment

No cure for RRV disease currently exists, so only its symptoms can be treated. In one of the few studies on RRV disease treatment, Cordon and Rouse (1995) found that roughly one-third of patients (36.4%) reported that **nonsteroidal anti-inflammatory drugs** (NSAIDS) provided them with the best symptomatic relief. In addition to NSAIDS, others patients found that rest (24.1%), **aspirin** and paracetamol (16.4%), or physical therapies (10.3%), such as **hydrotherapy**, massage, and physiotherapy, were their only source of symptom relief. Unfortunately, 20% of patients found none of these interventions effective. Health providers typically use one or a combination of these treatments for their patients. In particular, paracetamol has been found to be effective for treating the pain and fever associated with RRV disease.

Although some clinicians have found the use of oral corticosteroid useful and effective, this practice is considered unwise and unnecessary. The adverse

KEY TERMS

Arthralgia—Sharp, severe pain, extending along a nerve or group of nerves, experienced in a joint and/or joints.

Arbovirus viruses—Also known as arthropod-borne viruses, these viruses are maintained in nature through biological transmission between vertebrate hosts and blood-feeding arthropods. Infection occurs when an infected arthropod, such as a mosquito, feeds off a vertebrate, such as a human.

Macropods—Derived from the Greek, macropod literally means “large footed.” Macropods are marsupials belonging to the family Macropodidae, which includes kangaroos, wallabies, tree kangaroos, pademelons, and several others.

Myalgia—Muscular pain or tenderness, typically of a diffuse and/or nonspecific nature.

Polyarthritis—A nonspecific term for arthritis involving two or more joints, typically associated with auto-immune forms of arthritis. Symptoms usually include pain, inflammation, and/or swelling in multiple joints.

effects associated with **corticosteroids** may outweigh their benefits, and may even worsen the RRV disease. More study is required on this and other treatment interventions of RRV disease.

Prognosis

Patients infected with RRV disease will fully recover within four to seven months. Although milder cases can recover within a few weeks, many cases have persisted for several years. Only the symptoms can be treated during this time, not the disease. Fortunately, RRV infection usually provides the patient with life-long immunity to future infection.

Prevention

Prevention techniques of RRV typically coincide with measures used to avoid mosquito bites; the primary source of the virus. These include the use of insect repellent (with 5% to 20% DEET) on exposed body parts, wearing loose-fitting clothes over the limbs and torso while outdoors, using mosquito coils and/or citronella candles outdoors, and limiting outdoor activities during peak biting periods and/or in areas with high mosquito density. While camping outdoors, knock-down spray or bed netting with pyrethrum is suggested.

Additional steps for reducing risk of being bitten include using screens in homes and removing mosquito-breeding areas near the home, such as uncovered water containers and old tires. Mosquito eradication programs can assist in reducing insect populations. An RRV vaccine is currently being developed.

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Jason Fryer

Rotator cuff injury

Definition

A rotator cuff injury is a tear or inflammation of the rotator cuff tendons in the shoulder.

Description

Rotator cuff injury is known by several names, including pitcher's shoulder, swimmer's shoulder, and tennis shoulder. As these names imply, the injury occurs most frequently in athletes practicing sports that require the arm to be moved over the head repeatedly, such as pitching, swimming, tennis, and weight

lifting. Rotator cuff tendonitis is an inflammation of the shoulder tendons while a rotator cuff tear is a ripping of one or more of the tendons.

The tendons of four muscles make up the rotator cuff. The muscles are the supraspinatus, infraspinatus, teres minor, and subscapularis. The tendons attach the muscles to four shoulder bones: the shoulder blade (scapula), the upper arm bone (humerus), and the collarbone (clavicle.) The rotator cuff tendons can also degenerate due to age, usually starting around age 40. Rotator cuff injury may also be caused by falling on the outstretched arm or joint of the elbow. Either of these may produce enough force to drive the humerus into the shoulder socket.

Causes and symptoms

Some areas of the rotator cuff tendons have poor blood supply. Thus, the tissue is very slow to heal and maintain itself during normal use. Tearing and inflammation in athletes is usually due to hard and repetitive use, especially in baseball pitchers. In non-athletes over age 40, the injuries usually occur as a result of lifting heavy objects. The two primary symptoms are **pain** and weakness in the shoulder or arm, especially with arm movement or at night. A partial tear may cause pain but still allow normal arm movement. A complete tear usually leaves the injured person unable to raise the arm away from the side.

Diagnosis

Diagnosis is usually made after a **physical examination**, often by a sports medicine physician. X rays are also sometimes used in diagnosis as well as an arthrogram. However, the arthrogram is an invasive procedure and may be painful afterwards. For this reason, **magnetic resonance imaging (MRI)** is preferred to determine tendon tears as it also show greater detail than the arthrogram.

Treatment

The primary treatment is resting the shoulder and, for minor tears and inflammation, applying ice packs. Anti-inflammatory medications may also be prescribed. As soon as pain decreases, physical therapy is usually started to help regain normal motion. If pain persists after several weeks, the physician may inject cortisone into the affected area.

Serious tears to the rotator cuff tendons usually require surgery to repair. An instrument called an arthroscope is used to view the shoulder joint and

KEY TERMS

Arthrogram—A test done by injecting dye into the shoulder joint and then taking x-rays. Areas where the dye leaks out indicate a tear in the tendons.

Arthroscope—An instrument for the visual examination of the interior of a joint.

Arthroscopy—Examination of a joint with an arthroscope or joint surgery using an arthroscope.

Cortisone—A hormone produced naturally by the adrenal glands or made synthetically.

Magnetic resonance imaging (MRI) scan—A special radiological test that uses magnetic waves to create pictures of an area, including bones, muscles, and tendons.

Spur—Any projection from a bone.

confirm the presence of a tear. The arthroscope can also be used to remove any bone spurs that may be present in the shoulder area. Current arthroscopic procedures usually involve a 2 in (5.1 cm) incision in the outer shoulder. Through this incision the torn rotator edge may be reattached to the humerus with stitches.

Alternative treatment

There are no effective alternative medicine treatments for rotator cuff injuries.

Prognosis

The prognosis for recovery from minor rotator cuff injuries is excellent. For serious injuries, the prognosis is usually good, some six weeks of physical therapy being required following surgery. Full recovery may take several more months. In some cases, the injury is so severe that it requires tendon grafts and muscle transfers. In rare cases, a severe injury is not repairable, usually because the tendon has been torn for too long a time.

Prevention

The best prevention is to avoid repetitive overhead arm movements and to develop shoulder strength.

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< <http://www.aaos.org>. >.

American Orthopaedic Society for Sports Medicine. 6300 N. River Road, Ste. 200, Rosemont, IL 60018. (847)

292-4900. < <http://www.sportsmed.org>. >.

OTHER

WebMD. < <http://www.my.webmd.com> >.

Ken R. Wells

Rotavirus infections

Definition

Rotavirus is the major cause of **diarrhea** and **vomiting** in young children worldwide. The infection is highly contagious and may lead to severe **dehydration** (loss of body fluids) and even **death**. In the United States, more than 50,000 children are hospitalized and up to 125 die each year as a result of rotavirus infection.

Description

Gastroenteritis, or inflammation of the stomach and the intestine, is the second most common illness in the United States, after the **common cold**. More than one-third of such cases are caused by viruses. Many different viruses can cause gastroenteritis, but the most common ones are the rotavirus and the Norwalk virus.

The name rotavirus comes from the Latin word "rota" for wheel and is given because the viruses have a distinct wheel-like shape. Rotavirus infection is also known as infantile diarrhea, or winter diarrhea, because it mainly targets infants and young children. The outbreaks are usually in the cooler months of winter.

The virus is classified into different groups (Group A through group G), depending on the type of protein marker (antigen) that is present on its surface. The diarrheal infection of children is caused by the Group A rotaviruses. Group B rotaviruses have caused major epidemics of adult diarrhea in China.

Group C rotavirus has been associated with rare cases of diarrheal outbreaks in Japan and England. Groups D through G have not been detected in humans.

Causes and symptoms

The main symptoms of the rotavirus infection are **fever**, stomach cramps, vomiting, and diarrhea (this could lead to severe dehydration). The symptoms last anywhere from four to six days. If a child has dry lips and tongue, dry skin, sunken eyes, and wets fewer than six diapers a day, it is a sign of dehydration and a physician needs to be notified. Because of the excellence of healthcare in this country, rotavirus is rarely fatal to American children. However, it causes deaths of up to a million children in the third world countries, every year.

The virus is usually spread by the "fecal-oral route." In other words, a child can catch a rotavirus infection if she puts her finger in her mouth after touching toys or things that have been contaminated by the stool of another infected child. This usually happens when children do not wash their hands after using the toilet, or before eating food.

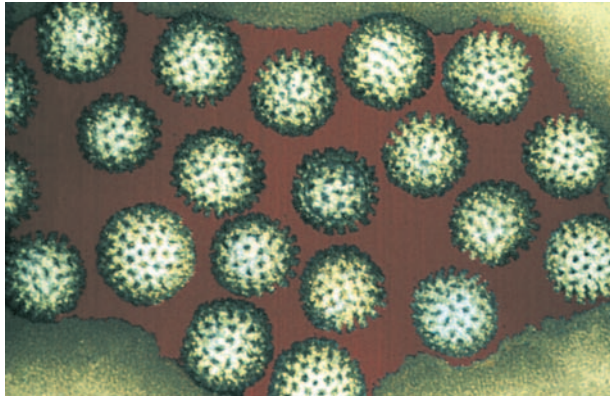
The viruses can also spread by way of contaminated food and drinking water. Infected food handlers who prepare salads, sandwiches, and other foods that require no cooking can spread the disease. Generally, symptoms appear within 4–48 hours after exposure to the contaminated food or water.

Children between the ages of six months and two years, especially in a daycare setting, are the most susceptible to this infection. Breastfed babies may be less likely to become infected, because breast milk contains antibodies (proteins produced by the white blood cells of the immune system) that fight the illness. Nearly every child by the age of four has been infected by this virus, and has rotavirus antibodies in their body. The disease also targets the elderly and people who have weak immune systems.

Children who have been infected once can be infected again. However, second infections are less severe than the first infections. By the time a child has had two infections, the chances of subsequent severe infection is remote.

Diagnosis

The rotavirus infection is diagnosed by identifying the virus in the patient's stool. This is done using electron microscopy. Immunological tests such as ELISA (Enzyme-linked immunosorbent assay) are



Rotaviruses are probably the most common viruses to infect humans and animals. These viruses are associated with gastroenteritis and diarrhea in humans and other animals. (Photograph by Dr. Linda Stannard, Photo Researchers, Inc. Reproduced by permission.)

also widely used for diagnosis, and several commercial kits are available.

Treatment

“Oral rehydration therapy,” or drinking enough fluids to replace those lost through bowel movements and vomiting, is the primary aim of the treatment. Electrolyte and fluid replacement solutions are available over the counter in food and drug stores. Dehydration is one of the greatest dangers for infants and young children. If the diarrhea becomes severe, it may be necessary to hospitalize the patient so that fluids can be administered intravenously.

Anti-diarrheal medication should not be given to children unless directed to do so by the physician. Antibiotic therapy is not useful in viral illness. Specific drugs for the virus are not available.

Prognosis

Most of the infections resolve spontaneously. Dehydration due to severe diarrhea is one of the major complications.

Prevention

The best way to prevent the disease is by proper food handling and thorough hand washing, after using the toilet and whenever hands are soiled. In child care centers and hospital settings, the staff should be educated about personal and environmental hygiene. All dirty diapers should be regarded as infectious and disposed of in a sanitary manner.

Vaccines that prevent rotavirus in young children have been tested in nationwide trials. Researchers report that the vaccines appear to prevent the infection in 80% of the tested children. The vaccine is intended to be given orally (by mouth) at two, four, and six months of age. The only side effect of the vaccine is a low-grade fever in a small percentage of the children, three to four days after the **vaccination**. Within the next few years, a rotavirus vaccine may become part of every child’s immunization schedule.

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Lata Cherath, PhD

Roundworm infections

Definition

Roundworm infections are diseases of the digestive tract and other organ systems caused by nematodes. Nematodes are parasitic worms with long, cylindrical bodies.

Description

Roundworm infections are widespread throughout the world, with some regional differences. Ascariasis and trichuriasis are more common in warm, moist climates where people use human or animal feces for fertilizer. Anisakiasis is most common in countries where raw or pickled fish or squid is a popular food item.

Causes and symptoms

The causes and symptoms of roundworm infection vary according to the species. Humans acquire most types of roundworm infection from contaminated food or by touching the mouth with unwashed hands.

Anisakiasis

Anisakiasis is caused by anisakid roundworms. Humans are not the primary host for these parasites. Anisakid roundworms infest whales, seals, and dolphins; crabs then ingest roundworm eggs from the

feces of these animals. In the crabs, the eggs hatch into larvae that can infect fish. The larvae enter the muscles of marine animals further up the food chain, including squid, mackerel, herring, cod, salmon, tuna, and halibut. Humans become accidental hosts when they eat raw or undercooked fish containing anisakid larvae. The larvae attach themselves to the tissues lining the stomach and intestine, and eventually die inside the inflamed tissue.

In humans, anisakiasis can produce a severe syndrome that affects the stomach and intestines, or a mild chronic disease that may last for weeks or years. In acute anisakiasis, symptoms begin within one to seven hours after the patient eats infected seafood. Patients are often violently sick, with **nausea**, **vomiting**, **diarrhea**, and severe abdominal **pain** that may resemble **appendicitis**. In chronic anisakiasis, the patient has milder forms of stomach or intestinal irritation that resemble stomach ulcers or **irritable bowel syndrome**. In some cases, the acute form of the disease is followed by chronic infestation.

Ascariasis

Ascariasis, which is caused by *Ascaris lumbricoides*, is one of the most widespread parasitic infections in humans, affecting over 1.3 billion people worldwide. Ascarid roundworms cause a larger burden on the human host than any other parasite; adult worms can grow as long as 12 or 14 inches, and release 200,000 eggs per day. The eggs infect people who eat unwashed vegetables from contaminated soil or touch their mouths with unwashed hands. Once inside the digestive tract, the eggs release larvae that penetrate the intestinal wall and migrate to the lungs through the liver and the bloodstream. After about 10 days in the lungs, the larvae migrate further into the patient's upper lung passages and airway, where they are swallowed. When they return to the intestine, they mature into adults and reproduce. The time period from the beginning of the infection to egg production is 60–75 days.

The first symptoms of infection may occur when the larvae reach the lungs. The patient may develop chest pain, coughing, difficulty breathing, and inflammation of the lungs. In some cases, the patient's sputum is streaked with blood. This phase of the disease is sometimes called Loeffler's syndrome. It is marked by an accumulation of parasites in the lung tissue and by eosinophilia (an abnormal increase in the number of a specific type of white blood cell). The intestinal phase of ascariasis is marked by stomach pain, cramping, nausea, and intestinal blockage in severe cases.

Toxocariasis

Toxocariasis is sometimes called visceral larva migrans (VLM) because the larval form of the organism hatches inside the intestines and migrates throughout the body to other organs (viscera). The disease is caused by *Toxocara canis* and *T. cati*, which live within the intestines of dogs and cats. Most human patients are children between the ages of two and four years, who become infected after playing in sandboxes or soil contaminated by pet feces, although adults are also susceptible. The eggs can survive in soil for as long as seven years.

The organism's eggs hatch inside the human intestine and release larvae that are carried in the bloodstream to all parts of the body, including the eyes, liver, lungs, heart, and brain. The patient usually has a **fever**, with coughing or **wheezing** and a swollen liver. Some patients develop skin **rashes** and inflammation of the lungs. The larvae may survive inside the body for months, producing allergic reactions and small granulomas, which are tissue swellings or growths produced in response to inflammation. Infection of the eye can produce ocular larva migrans (OLM), which is the first symptom of toxocariasis in some patients.

Trichuriasis

Trichuriasis, caused by *Trichuris trichiura*, is sometimes called whipworm because the organism has a long, slender, whiplike front end. The adult worm is slightly less than an inch long. Trichuriasis is most common in warm, humid climates, including the southeastern United States. The number of people with trichuriasis may be as high as 800 million worldwide.

Whipworm larvae hatch from swallowed eggs in the small intestine and move on to the upper part of the large intestine, where they attach themselves to the lining. The adult worms produce eggs that are passed in the feces and mature in the soil. Patients with mild infections may have few or no symptoms. In cases of heavy infestation, the patient may have abdominal cramps and other symptoms resembling amebic **dysentery**. In children, severe trichuriasis may cause anemia and developmental retardation.

Diagnosis

Since the first symptoms of roundworm infection are common to a number of illnesses, a doctor is most likely to consider the possibility of a parasitic disease on the basis of the patient's history—especially in children.

The definite diagnosis is based on the results of stool or tissue tests. In trichuriasis, adult worms may also be visible in the lining of the patient's rectum. In ascariasis, adult worms may appear in the patient's feces or vomit; they can also be detected by x ray and ultrasound. In toxocariasis, larvae are sometimes found in tissue samples taken from a granuloma. If a patient with toxocariasis develops OLM, it is important to obtain a granuloma sample in order to distinguish between OLM and **retinoblastoma** (a type of eye tumor).

Anisakiasis is one of two roundworm infections that cannot be diagnosed from stool specimens. Instead, the diagnosis is made by x rays of the patient's stomach and small intestine. The larvae may appear as small threads when double contrast x rays are used. In acute cases, the doctor may use an endoscope (an instrument for examining the interior of a body cavity) to look for or remove larvae.

Blood tests cannot be used to differentiate among different types of roundworm infections, but the presence of eosinophilia can help to confirm the diagnosis.

Patients with trichuriasis or ascariasis should be examined for signs of infection by other roundworm species; many patients are infected by several parasites at the same time.

Treatment

Trichuriasis, ascariasis, and toxocariasis are treated with anthelmintic medications. These are drugs that destroy roundworms either by paralyzing them or by blocking them from feeding. Anthelmintic drugs include pyrantel pamoate, piperazine, albendazole, and mebendazole. Mebendazole cannot be given to pregnant women because it may harm the fetus. Treatment with anthelmintic drugs does not prevent reinfection.

There is no drug treatment for anisakiasis; however, symptoms usually resolve in one to two weeks when the larvae die. In some cases, the larvae are removed with an endoscope or by surgery.

Patients with an intestinal obstruction caused by ascariasis may be given **nasogastric suction**, followed by anthelmintic drugs, in order to avoid surgery. If suction fails, the worms must be removed surgically to prevent intestinal rupture or blockage.

Prognosis

The prognosis for recovery from roundworm infections is good for most patients. The severity of

KEY TERMS

Eosinophilia—An abnormal increase in the number of a specific type of white blood cell. Eosinophilia is a characteristic of all types of roundworm infections.

Granuloma—A tissue swelling produced in response to inflammation. Granulomas are important in diagnosing toxocariasis.

Loeffler's syndrome—The respiratory phase of ascariasis, marked by inflammation of the lungs and eosinophilia.

Nematode—A parasitic roundworm with a long, cylindrical body.

Ocular larva migrans (OLM)—A syndrome associated with toxocariasis, in which the eye is invaded by migrating larvae.

Visceral larva migrans (VLM)—Another name for toxocariasis. The name is derived from the life cycle of the organism.

Whipworm—Another name for trichuriasis. The name comes from the organism's long whiplike front end.

infection, however, varies considerably from person to person. Children are more likely to have heavy infestations and are also more likely to suffer from malabsorption and **malnutrition** than adults.

Ascariasis is the only roundworm infection with a significant mortality rate. *A. lumbricoides* grows large enough to perforate the bile or pancreatic ducts; in addition, a mass of worms in the digestive tract can cause rupture or blockage of the intestines. It is estimated that 20,000 children die every year from intestinal ascariasis.

Prevention

There are no effective vaccines against any of the soil-transmitted roundworms, nor does infection confer immunity. Prevention of infection or reinfection requires adequate hygiene and sanitation measures, including regular and careful handwashing before eating or touching the mouth with the hands.

With respect to specific infections, anisakiasis can be prevented by avoiding raw or improperly prepared fish or squid. Trichuriasis, ascariasis, and toxocariasis can be prevented by keeping children from playing in soil contaminated by human or animal feces; by

teaching children to wash their hands before eating; and by having pets dewormed regularly by a veterinarian.

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Rebecca J. Frey, PhD

Routine urinalysis see **Urinalysis**

RSV see **Respiratory syncytial virus infection**

RTA see **Renal tubular acidosis**

RU-486 see **Mifepristone**

Rubella

Definition

Rubella is a highly contagious viral disease, spread through contact with discharges from the nose and throat of an infected person. Although rubella causes only mild symptoms of low **fever**, swollen glands, joint **pain**, and a fine red rash in most children and adults, it can have severe complications for women in their first trimester of **pregnancy**. These complications include severe **birth defects** or **death** of the fetus.

Description

Rubella is also called German **measles** or three-day measles. This disease was once a common childhood illness, but its occurrence has been drastically reduced since vaccine against rubella became available in 1969. In the 20 years following the introduction of the vaccine, reported rubella cases dropped 99.6%. Only 229 cases of rubella were reported in the United States in 1996.

Rubella is spread through contact with fluid droplets expelled from the nose or throat of an infected person. A person infected with the rubella virus is contagious for about seven days before any symptoms appear and continues to be able to spread the disease for about four days after the appearance of symptoms. Rubella has an incubation period of 12–23 days.

Although rubella is generally considered a childhood illness, people of any age who have not been vaccinated or previously caught the disease can become infected. Having rubella once or being immunized against rubella normally gives lifetime immunity. This is why **vaccination** is so effective in reducing the number of rubella cases.

Women of childbearing age who do not have immunity against rubella should be the most concerned about getting the disease. Rubella infection during the first three months of pregnancy can cause a woman to miscarry or cause her baby to be born with birth defects. Although it has been practically eradicated in the United States, rubella is still common in less developed countries because of poor immunization penetration, creating a risk to susceptible travelers. Some countries have chosen to target rubella vaccination to females only and outbreaks in foreign-born males have occurred on cruise ships and at U.S. summer camps.

Causes and symptoms

Rubella is caused by the rubella virus (*Rubivirus*). Symptoms are generally mild, and complications are rare in anyone who is not pregnant.

The first visible sign of rubella is a fine red rash that begins on the face and rapidly moves downward to cover the whole body within 24 hours. The rash lasts about three days, which is why rubella is sometimes called the three-day measles. A low fever and swollen glands, especially in the head (around the ears) and neck, often accompany the rash. Joint pain and sometimes joint swelling can occur, more often in women. It is quite common to get rubella and not show any symptoms (subclinical infection).

Symptoms disappear within three to four days, except for joint pain, which may linger for a week or two. Most people recover fully with no complications. However, severe complications may arise in the unborn children of women who get rubella during the first three months of their pregnancy. These babies may be miscarried or stillborn. A high percentage are born with birth defects. Birth defects are reported to occur in 50% of women who contract the disease during the first month of pregnancy, 20% of those who contract it in the second month, and 10% of those who contract it in the third month.

The most common birth defects resulting from congenital rubella infection are eye defects such as **cataracts**, **glaucoma**, and blindness; deafness; congenital heart defects; and **mental retardation**. Taken together, these conditions are called congenital rubella



A red rash is one characteristic of rubella, or German measles, as seen on this man's arm. (Custom Medical Stock Photo. Reproduced by permission.)

syndrome (CRS). The risk of birth defects drops after the first trimester, and by the 20th week, there are rarely any complications.

Diagnosis

The rash caused by the rubella virus and the accompanying symptoms are so similar to other viral infections that it is impossible for a physician to make a confirmed diagnosis on visual examination alone. The only sure way to confirm a case of rubella is by isolating the virus with a blood test or in a laboratory culture.

A blood test is done to check for rubella antibodies. When the body is infected with the rubella virus, it produces both immunoglobulin G (IgG) and immunoglobulin M (IgM) antibodies to fight the infection. Once IgG exists, it persists for a lifetime, but the special IgM antibody usually wanes over six months. A blood test can be used either to confirm a recent infection (IgG

and IgM) or determine whether a person has immunity to rubella (IgG only). The lack of antibodies indicates that a person is susceptible to rubella.

All pregnant women should be tested for rubella early in pregnancy, whether or not they have a history of vaccination. If the woman lacks immunity, she is counseled to avoid anyone with the disease and to be vaccinated after giving birth.

Treatment

There is no drug treatment for rubella. Bed rest, fluids, and **acetaminophen** for pain and temperatures over 102°F (38.9°C) are usually all that is necessary.

Babies born with suspected CRS are isolated and cared for only by people who are sure they are immune to rubella. Congenital heart defects are treated with surgery.

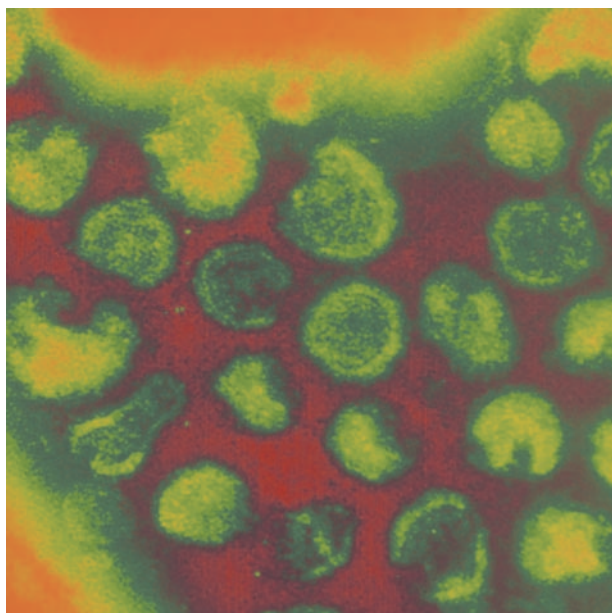
Alternative treatment

Rather than vaccinating a healthy child against rubella, many alternative practitioners recommend allowing the child to contract the disease naturally at the age of five or six years, since the immunity conferred by contracting the disease naturally lasts a lifetime. It is, however, difficult for a child to contract rubella naturally when everyone around him or her has been vaccinated.

Ayurvedic practitioners recommend making the patient comfortable and giving the patient ginger or clove tea to hasten the progress of the disease. **Traditional Chinese medicine** uses a similar approach. Believing that inducing the skin rash associated with rubella hastens the progress of the disease, traditional Chinese practitioners prescribe herbs such as peppermint (*Mentha piperita*) and *chai-hu* (*Bupleurum chinense*). Cicada is often prescribed as well. Western herbal remedies may be used to alleviate rubella symptoms. Distilled witch hazel (*Hamamelis virginiana*) helps calm the **itching** associated with the skin rash and an eyewash made from a filtered diffusion of eyebright (*Euphrasia officinalis*) can relieve eye discomfort. Antiviral western herbal or Chinese remedies can be used to assist the immune system in establishing equilibrium during the healing process. Depending on the patient's symptoms, among the remedies a homeopath may prescribe are *Belladonna*, *Pulsatilla*, or *Phytolacca*.

Prognosis

Complications from rubella infection are rare in children, pregnant women past the 20th week of



A digitized image of rubella virus particles. (Custom Medical Stock Photo. Reproduced by permission.)

KEY TERMS

Incubation period—The time it takes for a person to become sick after being exposed to a disease.

Trimester—The first third or 13 weeks of pregnancy.

pregnancy, and other adults. For women in the first trimester of pregnancy, there is a high likelihood of the child being born with one or more birth defect. Unborn children exposed to rubella early in pregnancy are also more likely to be miscarried, stillborn, or have a low birthweight. Although the symptoms of rubella pass quickly for the mother, the consequences to the unborn child can last a lifetime.

Prevention

Vaccination is the best way to prevent rubella and is normally required by law for children entering school. Rubella vaccine is usually given in conjunction with measles and **mumps** vaccines in a shot referred to as MMR (mumps, measles, and rubella). Children receive one dose of MMR vaccine at 12-15 months and another dose at four to six years.

Pregnant women should not be vaccinated, and women who are not pregnant should avoid conceiving for at least three months following vaccination. To

date, however, accidental rubella vaccinations during pregnancy have not clearly been associated with the same risk as the natural infection itself. Women may be vaccinated while they are breastfeeding. People whose immune systems are compromised, either by the use of drugs such as steroids or by disease, should discuss possible complications with their doctor before being vaccinated.

Resources

ORGANIZATIONS

March of Dimes Birth Defects Foundation. 1275 Mamaroneck Ave., White Plains, NY 10605. (914) 428-7100. resourcecenter@modimes.org. < <http://www.modimes.org> >.

National Organization for Rare Disorders. P.O. Box 8923, New Fairfield, CT 06812-8923. (800) 999-6673. < <http://www.rarediseases.org> >.

Tish Davidson, A.M.

Rubella test

Definition

The **rubella** test is a routine blood test performed as part of prenatal care of pregnant women. It is sometimes also used to screen women of childbearing age before the first **pregnancy**.

Purpose

The test is given to evaluate whether a woman is immune to rubella (German **measles**) as a result of childhood exposure or immunization, or whether she may be presently infected with the disease. The question of a current infection is particularly urgent for pregnant women. Although the disease itself is not serious in adults, it can cause **miscarriage**, **stillbirth**, or damage to the fetus during the first trimester (three months) of pregnancy. The rubella test is regarded as a more reliable indication of the patient's immune status than her history, because reinfection with rubella is possible even after immunization. The results of the test may influence decisions to terminate a pregnancy.

Description

The rubella test belongs to a category of blood tests called hemagglutination inhibition (HI) tests. Hemagglutination refers to the clumping or clustering of red blood cells caused by a disease antibody, virus,

or certain other substances. Inhibition refers to interference with the clumping process. The presence of rubella antibodies inhibits the cell clumping caused by the rubella virus. Thus, the addition of the virus to a sample of the patient's blood allows a doctor to determine the presence and concentration of rubella antibodies and the patient's immunity to the disease.

When a person is infected with the rubella virus, the body produces both immunoglobulin G (IgG) and immunoglobulin M (IgM) antibodies to fight the infection. Once IgG exists, it persists for a lifetime, but the special IgM antibody usually wanes over six months. The rubella test can either confirm that a recent infection has occurred (both IgG and IgM are present) or that a patient has immunity to rubella (IgG only is present).

When the test is performed to confirm the diagnosis of rubella in a woman already pregnant, two blood samples are drawn. One is drawn during the acute phase of the illness about three days after the rash breaks out, and the second is drawn during the convalescent phase about three weeks later. The specimens are then tested simultaneously by a single laboratory. Alternatively, a pregnant woman with a rash suspected to be rubella can be tested for IgM antibody. If the test shows that IgM antibody is present, then a recent rubella infection has occurred.

Because there have been cases of children born with rubella syndrome even though the mother's blood test indicated that she was sufficiently immune to rubella, some researchers are presently recommending a second test, known as a synthetic peptide enzyme-linked immunosorbent assay (ELISA). This test screens for the presence of rubella virus neutralizing (RVN) antibodies in the mother's blood.

Normal results

If the patient has been successfully immunized against rubella or has had the disease, the HI antibody titer (concentration) will be greater than 1:10–1:20. The red blood cells will fail to clump when the rubella virus is added to the blood serum.

In the case of paired testing for pregnant women, a fourfold rise in antibody titer between the first and second blood samples indicates the suspicious rash was caused by rubella. The alternative test for IgM antibody confirms recent rubella infection if IgM is found in the patient's blood.

Abnormal results

If the patient has little or no immunity to rubella, her HI antibody titer will be 1:8 or less. Women

KEY TERMS

Antibody—A protein molecule produced by the immune system that is specific to a virus, such as the rubella virus. The antibody combines with the virus and disables it.

Hemagglutination—The clumping or clustering of red blood cells caused by certain viruses, antibodies, or other substances.

Inhibition—Restraint of or interference with a biological process, such as the clumping of blood cells.

Titer—The concentration of a substance in a given sample of blood or other tissue fluid.

without immunity should receive immunization against rubella provided that they avoid pregnancy for a period of three months following immunization. Women with disease of the immune system or who are taking corticosteroid medications should receive immune serum globulin rather than rubella vaccine to prevent infection.

Resources

BOOKS

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PERIODICALS

Andrews, J. I. "Diagnosis of Fetal Infections" *Current Opinion in Obstetrics and Gynecology* 16 (April 2004): 163–166.

Giessauf, A., T. Letschka, G. Walder, et al. "A Synthetic Peptide ELISA for the Screening of Rubella Virus Neutralizing Antibodies in Order to Ascertain Immunity." *Journal of Immunological Methods* 287 (April 2004): 1–11.

ORGANIZATIONS

American Academy of Family Physicians (AAFP). 11400 Tomahawk Creek Parkway, Leawood, KS 66211-2672. (800) 274-2237 or (913) 906-6000. <<http://www.aafp.org>>.

Rebecca J. Frey, PhD

Rubeola see **Measles**

Ruptured disk see **Herniated disk**

RVT see **Renal vein thrombosis**

S

Sacroiliac disease

Definition

Sacroiliac disease is high-impact trauma to the sacroiliac joint that can cause **death**, or bone, and nerve damage.

Description

The sacroiliac joint is a strong, weight bearing synovial joint between the ilium and sacrum bones of the pelvis. The bones are held in place and allowed limited movements by a system of sacroiliac ligaments. Relaxation of this and other joints and ligaments is important during **pregnancy** and is accomplished by a special hormone called relaxin. Usually the sacroiliac is damaged by high-impact injuries. These injuries may be life threatening and mortality is approximately 20% if neighboring structures are also damaged. Injuries to this area often includes neurological deficits. Dislocation and nerve damage are frequently missed in the diagnosis.

Causes and symptoms

The primary cause of **dislocations**, **fractures**, and accompanying damage is usually a traumatic accident. Patients receiving such injuries require emergency medical attention. There may be severe blood loss due to breakage of large bones and resuscitative measures may be required for stabilization.

Diagnosis

The diagnosis can be difficult since nerve damage can mimic other conditions with similar symptoms (i.e., **low back pain** in persons with **sciatica**). Additionally imaging studies and **physical examination** maneuvers will miss the diagnosis. The definitive method for diagnosing sacroiliac

pathology would be injection of local anesthetic in the correct area of the affected sacroiliac joint. This procedure is usually performed using advance guidance systems (CT or fluoroscopic assisted guidance). If the **pain** is relieved by anesthetic injection, then the diagnosis is confirmed. There are three typical patterns of pain: pain directly over the joint, pain in the groin extending down the affected leg that can mimic the signs associated with a herniated lumbar disc, and pain widely dispersed in the affected leg.

Treatment

Treatment initially can include emergency interventions, but usually is conservative. Treatment includes physical therapy, manipulation, and medications for pain control. In some cases a sacroiliac belt can help with symptoms. In sacroiliac joint disease that has already progressed and is chronic and severe, corrective joint fusion may be indicated.

Prognosis

Outcome is variable and takes into account the extent of injuries, early diagnosis, and responsiveness to conservative treatment.

Prevention

There is no known prevention since the disease is secondary to an accident.

Resources

BOOKS

Goetz, Christopher G., et al, editors. *Textbook of Clinical Neurology*. 1st ed. W. B. Saunders Company, 1999.

Ruddy, Shaun, et al, editors. *Kelly's Textbook of Rheumatology*. 6th ed. W. B. Saunders Company, 2001.

KEY TERMS

Herniated disk—A protrusion in a disk located in the spinal column.

Ligament—Fibrous tissue which connect bones.

Synovial joint—A joint that allows for bone movement.

OTHER

American Academy of Orthopaedic Surgeons.
< <http://www.aaos.org> >.

Laith Farid Gulli, M.D.

SAD see **Seasonal affective disorder**

St. Anthony's fire see **Erysipelas**

St. John's wort

Definition

Hypericum perforatum is the most medicinally important species of the *Hypericum* genus, commonly known as St. John's wort. There are as many as 400 species in the genus, which is part of the Guttiferae family. Native to Europe, St. John's wort is found throughout the world. It thrives in sunny fields, open woods, and gravelly roadsides. Early colonists brought this valuable medicinal to North America, and the plant has become naturalized in the eastern United States and California, as well as in Australia, New Zealand, eastern Asia, and South America.

The entire plant, particularly the round, black seed, exudes a slight, turpentine-like odor. The woody, branched root spreads from the base with runners that produce numerous stalks. The simple, dark green leaves are veined and grow in opposite, oblong-obovate pairs on round, branching stalks that reach 3 ft (91.4 cm) high. Tiny holes, visible when the leaf is held to the light, are actually transparent oil glands containing the chemical photo sensitizer known as hypericin. These characteristic holes inspired the species name, *perforatum*, Latin for perforated. The bright yellow, star-shaped flowers, often clustered in a trio, have five petals. Each blossom has many showy stamens. Black dots along the margins of the blossom contain more of the red-pigmented chemical hypericin. The herb is also useful as a dye. The flowers bloom in branching, flat-

topped clusters atop the stalks in mid-summer, around the time of the summer solstice. St. John's wort, sometimes called devil's flight or grace of God, was believed to have magical properties to ward off evil spirits. Its generic name *hypericum* is derived from a Greek word meaning "over an apparition." The herb was traditionally gathered on mid-summer's eve, June 23. This date was later christianized as the eve of the feast day of St. John the Baptist. This folk custom gave the plant its popular name. The Anglo-Saxon word *wort* means medicinal herb.

Purpose

St. John's wort has been known for its numerous medicinal properties as far back as Roman times. It was a valued remedy on the Roman battlefields where it was used to promote healing from trauma and inflammation. The herb is vulnerary and can speed the healing of **wounds**, **bruises**, ulcers, and **burns**. It is popularly used as a nervine for its calming effect, easing tension and **anxiety**, relieving mild depression, and soothing emotions during **menopause**. The bitter-sweet herb is licensed in Germany for use in cases of mild depression, anxiety, and sleeplessness. It is useful in circumstances of nerve injury and trauma, and has been used to speed healing after brain surgery. Its antispasmodic properties can ease uterine cramping and menstrual difficulties. St. John's wort acts medicinally as an astringent, and may also be used as an expectorant. The hypericin in St. John's wort possesses anti-viral properties that may be active in combating certain cancers, including many brain cancers. An infusion of the plant, taken as a tea, has been helpful in treating night-time incontinence in children. The oil, taken internally, has been used to treat **colic**, intestinal worms, and abdominal **pain**. The medicinal parts of St. John's wort are the fresh leaves and flowers. This valuable remedy has been extensively tested in West Germany, and is dispensed throughout Germany as a popular medicine called, *Johnniskraut*. Commercially prepared extracts are commonly standardized to 0.3% hypericin.

Clinical studies

A 1988 study at New York University found the antiviral properties in hypericin, a chemical component of *Hypericum*, to be useful in combating the virus that causes **AIDS**. Additional studies are under way through the Federal Drug Administration (FDA) to determine the effectiveness of the herb as a treatment for AIDS. Hypericin extract has also been reported to inhibit a form of leukemia that sometimes occurs after **radiation therapy**. Numerous clinical studies have

found hypericum preparations to have an antidepressive effect when used in standardized extracts for treatment of mild depression. Clinical trials continue with this important herbal anti-depressant, particularly in view of its relative lack of undesirable side effects in humans.

Preparations

An oil extract can be purchased commercially or prepared by combining fresh flowers and leaves of St. John's wort in a glass jar and sunflower or olive oil. Seal the container with an airtight lid and leave on a sunny windowsill for four to six weeks, shaking daily. The oil will absorb the red pigment. Strain through muslin or cheesecloth, and store in a dark container. The medicinal oil will maintain its potency for two years or more. The oil of St. John's wort has been known in folk culture as "Oil of Jesus." This oil makes a good rub for painful joints, **varicose veins**, muscle strain, arthritis, and rheumatism. Used in a compress it can help to heal wounds and inflammation, and relieve the pain of deep bruising.

An infusion is made by pouring one pint of boiling water over 1 oz (28 g) of dried herb, or 2 oz (57 g) of fresh, minced flower and leaf. Steep in a glass or enamel pot for five to 10 minutes. Strain and cover. Drink the tea warm. A general dose is one cupful, up to three times daily.

Capsule: Dry the leaves and flowers and grind with mortar and pestle into a fine powder. Place in gelatin capsules. The potency of the herb varies with the soil, climate and harvesting conditions of the plant. A standardized extract of 0.3% hypericin extract, commercially prepared from a reputable source, is more likely to yield reliable results. Standard dosage is up to three 300 mg capsules of 0.3% standardized extract daily.

A tincture is prepared by combining one part fresh herb to three parts alcohol (50% alcohol/water solution) in glass container. Set aside in dark place, shaking the mixture daily for two weeks. Strain through muslin or cheesecloth, and store in dark bottle. The tincture should maintain potency for two years. Standard dosage, unless otherwise prescribed, is 0.24–1 tsp added to 8 oz (237 ml) of water, up to three times daily.

A salve is made by warming 2 oz (59 ml) of prepared oil extract in double boiler. Once warmed, 1 oz (28 g) of grated beeswax is added and mixed until melted. Pour into a glass jar and cool. The salve can be stored for up to one year. The remedy keeps best if refrigerated after preparation. The salve is useful in



St. John's wort flowers. (Photo Researchers, Inc. Reproduced by permission.)

treating burns, wounds, and soothing painful muscles. It is also a good skin softener. St. John's wort salve may be prepared in combination with calendula extract (*Calendula officinalis*) for application on bruises.

Precautions

Consult a physician prior to use. Pregnant or lactating women should not use the herb. Individuals taking prescribed psychotropic medications classified as **selective serotonin reuptake inhibitors**, or **SSRI**, such as Prozac, should not simultaneously use St. John's wort. Many herbalists also discourage use of St. John's wort by individuals taking any other anti-depressant medication.

Cattlemen dislike the shrub because there have been some reports of toxicity to livestock that overgraze in fields abundant with the wild herb. Toxic

KEY TERMS

Antispasmodic—Relieves mild cramping or muscle spasm.

Expectorant—Promotes the discharge of mucus from respiratory system.

Nervine—Soothes and calms the nervous system.

Vulnerary—Heals wounds, bruises, sprains, and ulcers.

effects in livestock include reports of **edema** of the ears, eyelids, and the face due to photosensitization after ingestion of the herb. Exposure to sunlight activates the hypericin in the plant. Adverse effects have been reported in horses, sheep, and swine and include staggering, and blistering and peeling of the skin. Toxicity is greater in smaller mammals, such as rabbits.

Side effects

When used either internally or externally, the herb may cause photo-dermatitis in humans with fair or sensitive skin when exposed to sun light or other ultraviolet light source. There have been some reports of changes in **lactation** in some nursing women taking the hypericum extract. Changes in the nutritional quality and flavor of the milk, and reduction or cessation of lactation have also been reported. It can also cause headaches, stiff neck, **nausea and vomiting**, and high blood pressure.

Interactions

St. John's wort can interact with amphetamines, **asthma** inhalants, **decongestants**, diet pills, **narcotics**, and amino acid tryptophan and tyrosine, as well as certain foods. Reactions range from **nausea** to increased high blood pressure. Consult a practitioner prior to using St. John's wort.

Resources

ORGANIZATIONS

American Botanical Council. PO Box 201660, Austin, TX 78720-1660.

OTHER

Herb Research Foundation. <<http://www.herbs.org>>.

Clare Hanrahan

St. Vitus' dance see **Sydenham's chorea**

Salivary gland scan

Definition

A salivary gland scan is a nuclear medicine test that examines the uptake and secretion in the salivary glands of a radioactively labeled marker substance. The pattern of uptake and secretion shows if these glands are functioning normally.

Purpose

A salivary gland scan is done to help diagnose the cause of **dry mouth**. It is a test that is done when Sjogren's syndrome, salivary duct obstruction, asymmetric hypertrophy, or growths such as Warthin's tumors are suspected.

Precautions

Salivary gland scans are a safe and effective way to diagnose problems associated with dry mouth. The level of radioactivity in the marker substance is low and poses no threat to health. The only people who should not undergo this test are pregnant women.

Other recent nuclear medicine tests may affect the results of this scan. It may be necessary to wait until earlier radiopharmaceuticals have been cleared from the body before undergoing this scan.

Description

A salivary gland scan, also called a parotid gland scan, is a noninvasive test. The patient is positioned under a gamma scintillation camera that detects radiation. The patient then is injected with a low-level radioactive marker, usually technetium-99m or technetium pertechnetate.

Immediately after the injection, imaging begins. For accurate results, the patient must stay still during imaging. After several images, the patient is given lemon drop candies to suck on, which stimulate the salivary glands. Another set of images is made for comparison purposes. The entire process takes about ten minutes for the injection and 30–45 minutes for the scan.

Preparation

No special preparations are needed for this test. It is not necessary to fast or to restrict medications before testing. Any blood that needs to be drawn for other tests should be taken before the radiopharmaceutical is injected.

KEY TERMS

Hypertrophy—Overgrowth of tissue not due to a tumor.

Parotid gland—The salivary gland that lies below and in front of each ear.

Radiopharmaceutical—A radioactive pharmaceutical or chemical (usually radioactive iodine or cobalt) used for diagnostic or therapeutic purposes.

Sjögren's syndrome—A disease often associated with rheumatoid arthritis, that causes dry mouth, lesions on the skin, and enlargement of the parotid glands. It is often seen in menopausal women.

Technetium—A synthetic element used in nuclear medicine; it is obtained from the fission of uranium.

Warthin's tumor—A benign tumor of the parotid gland.

Aftercare

Patients can return to normal activities immediately.

Risks

A salivary gland scan is a safe test. The only risk is to the fetus of a pregnant woman. Women who are pregnant should discuss the risks and benefits of this procedure with their doctor.

Normal results

Normally functioning salivary glands take up the radiopharmaceutical then secrete it when stimulated by the lemon drops.

Abnormal results

Abnormally functioning salivary glands fail to exhibit a normal uptake and secretion pattern. This test does not differentiate between benign and malignant lesions.

Resources

BOOKS

Beers, Mark H., MD, and Robert Berkow, MD., editors. "Disorders of the Oral Region: Neoplasms." Section 9, Chapter 105 In *The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck Research Laboratories, 2004.

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Chae, S. W., J. H. Sohn, H. S. Shin, et al. "Unilateral, Multicentric Warthin's Tumor Mimicking a Tumor Metastatic to a Lymph Node. A Case Report." *Acta Cytologica* 48 (March-April 2004): 229–233.

ORGANIZATIONS

Society of Nuclear Medicine (SNM). 1850 Samuel Morse Drive, Reston, VA 20190-5316. (703) 708-9000. Fax: (703) 708-9015. <<http://www.snm.org>>.

OTHER

"Salivary Gland Scan." *HealthGate Page*. <<http://www3.healthgate.com>>.

Tish Davidson, AM
Rebecca J. Frey, PhD

Salivary gland tumors

Definition

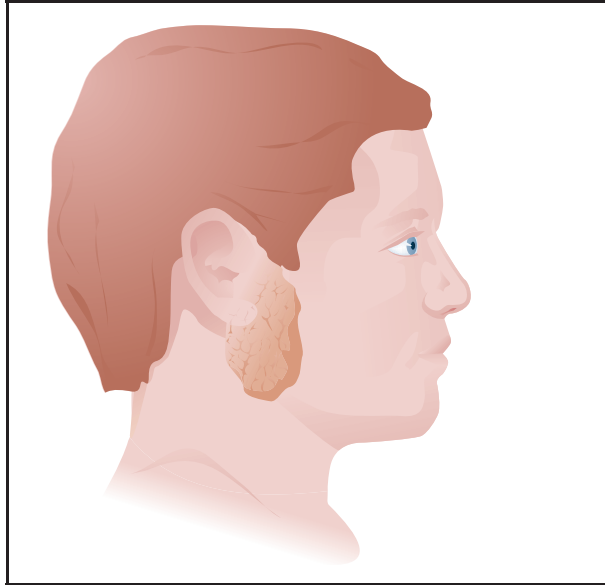
A salivary gland tumor is an uncontrolled growth of cells that originates in one of the many saliva-producing glands in the mouth.

Description

The tongue, cheeks, and palate (the hard and soft areas at the roof of the mouth) contain many glands that produce saliva. In saliva there are enzymes, or catalysts, that begin the breakdown (digestion) of food while it is still in the mouth. The glands are called salivary glands because of their function.

There are three big pairs of salivary glands in addition to many smaller ones. The parotid glands, submandibular glands and sublingual glands are the large, paired salivary glands. The parotids are located inside the cheeks, one below each ear. The submandibular glands are located on the floor of the mouth, with one on the inner side of each part of the lower jaw, or mandible. The sublingual glands are also in the floor of the mouth, but they are under the tongue.

The parotids are the salivary glands most often affected by tumors. Yet most of the tumors that grow in the parotid glands are benign, or not cancerous. Approximately 8 out of 10 salivary tumors diagnosed are in a parotid gland. One in 10 diagnosed is in a



A salivary gland with a tumor. (Illustration by Argosy Inc.)

submandibular gland. The remaining 10% are diagnosed in other salivary glands.

In general, glands more likely to show tumor growth are also glands least likely to show malignant tumor growth. Thus, although tumors of the sublingual glands are rare, almost all of them are malignant. In contrast, about one in four tumors of the parotid glands is malignant.

Cancers of the salivary glands begin to grow in epithelial cells, or the flat cells that cover body surfaces. Thus, they are called carcinomas, cancers that by definition begin in epithelial cells.

Demographics

Cancers in the mouth account for fewer than 2% of all cases of **cancer** and about 1.5% of cancer deaths. About 7% of all cancers diagnosed in the head and neck region are diagnosed in a salivary gland. Men and women are at equal risk.

Mortality from salivary gland tumors in the United States is higher among male African Americans below the age of 50 than among older workers of any race or either sex. The reasons for these findings are not clear as of early 2004.

Causes and symptoms

When survivors of the 1945 atomic bombings of Nagasaki and Hiroshima began to develop salivary gland tumors at a high rate, radiation was suspected as

a cause. Ionizing radiation, particularly gamma radiation, is a factor that contributes to tumor development. So is **radiation therapy**. Adults who received radiation therapy for enlarged adenoids or tonsils when they were children are at greater risk for salivary gland tumors.

Another reported risk factor is an association between wood dust inhalation and adenocarcinoma of the minor salivary glands of the nose and paranasal sinuses. There is also evidence that people infected with herpes viruses may be at greater risk for salivary gland tumors. And individuals infected with human **immunodeficiency** virus (HIV) have more salivary gland disease in general, and may be at greater risk for salivary gland tumors.

Although there has been speculation that the electromagnetic fields generated by cell phones increase the risk of salivary gland tumors, a recent study done in Denmark has concluded that the use of cell phones, pagers, and similar devices is not a risk factor.

There seems to be some link between **breast cancer** and salivary gland tumors. Women with breast cancer are more likely to be diagnosed with salivary gland tumors. Also linked to salivary gland tumors is alcohol use, exposure to sunlight (ultraviolet radiation) and hair dye use. There is evidence that people infected with herpes viruses may be at greater risk for salivary gland tumors. Individuals infected with human immunodeficiency virus (HIV) have more salivary gland disease in general, and may be at greater risk for salivary gland tumors.

Symptoms are often absent until the tumor is large or has metastasized (spread to other sites). In many cases, the tumor is first discovered by the patient's dentist. During regular dental examinations, the dentist looks for masses on the palate or under the tongue or in the cheeks, and such checkups are a good way to detect tumors early. Some symptoms are:

- a lump or mass in the mouth
- swelling in the face
- pain in the jaw or the side of the face
- difficulty swallowing
- difficulty breathing
- difficulty speaking

Diagnosis

A tissue sample will be taken for study via a biopsy. Usually an incision is necessary to take the tissue sample. Sometimes it is possible to take a tissue sample with a needle.

Magnetic resonance imaging (MRI) and computed tomography (CT) scans are also used to evaluate the tumor. They help determine whether the cancer has spread to sites adjacent to the salivary gland where it is found. MRI offers a good way to examine the tonsils and the back of the tongue, which are soft tissues. CT is tapped as a way of studying the jaw, which is bone.

Treatment

To assess the stage of growth of a salivary gland tumor, many features are examined, including how big it is and the type of abnormal cell growth. Analysis of the types of abnormal cell growth in tissue is so specific that many salivary gland tumors are given unique names.

In stage I cancer the tumor is less than one inch in size and it has not spread. Stage II salivary gland cancers are larger than one inch and smaller than two and one-half inches, but they have not spread. Stage III cancers are smaller than one inch, but they have spread to a lymph node. Stage IV cancers have spread to adjacent sites in the head, which may include the base of the skull and nearby nerves, or they are larger than two and one-half inches and have invaded a lymph node.

Surgical removal (excision) of the tumor is the most common treatment. **Chemotherapy** and radiation therapy may be part of the treatment, particularly if the cancer has metastasized, or spread to other sites; chemotherapy of salivary gland cancers, however, does not appear to extend survival or improve the patient's quality of life. Because there are many nerves and blood vessels near the three major pairs of salivary glands, particularly the parotids, the surgery can be quite complicated. A complex surgery is especially true if the tumor has spread.

A promising form of treatment for patients at high risk of tumor recurrence in the salivary glands near the base of the skull is gamma knife surgery. Used as a booster treatment following standard neutron radiotherapy, gamma knife surgery appears to be well tolerated by the patients and to have minimal side effects.

Alternative treatment

Any technique, such as **yoga**, **meditation** or **bio-feedback**, that helps a patient cope with **anxiety** over the condition and discomfort from treatment is useful and should be explored as an option.

KEY TERMS

Adenoids—Common name for the pharyngeal tonsils, which are lymph masses in the wall of the air passageway (pharynx) just behind the nose.

Biopsy—Tissue sample is taken from the body for examination.

Computed tomography (CT)—X rays are aimed at slices of the body (by rotating equipment) and results are assembled with a computer to give a three-dimensional picture of a structure.

Lymph—Tissue that is part of the lymphatic system, the system that collects and returns fluid to the blood vessels and produces substances that fight infection.

Magnetic resonance imaging (MRI)—Magnetic fields and radio frequency waves are used to take pictures of the inside of the body.

Tonsils—Common name for the palatine tonsils, which are lymph masses in the back of the mouth, on either side of the tongue.

Prognosis

Tumors in small salivary glands that are localized can usually be removed without much difficulty. The outlook for survival once the tumor is removed is very good if it has not metastasized.

For parotid cancers, the five-year survival rate is more than 85% whether or not a lymph node is involved at diagnosis. Ten-year survival rate is just under 50%.

Most early stage salivary gland tumors are removed, and they do not return. Those that do return, or recur, are the most troublesome and reduce the chance an individual will remain cancer-free.

Prevention

Minimizing intake of beverages containing alcohol may be important. Avoiding unnecessary exposure of the head to radiation and to sunlight may also be considered preventative. Anything that reduces the risk of contracting a sexually transmitted disease, such as the use of condoms, also may lower the risk of salivary gland cancer.

A salivary gland with a tumor.

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SPOHNC, Support for People with Oral and Head and Neck Cancer. P.O. Box 53, Locust Valley, NY 11560-0053. 800-377-0928. <<http://www.spohnc.org>>.

OTHER

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Diane M. Calabrese
Rebecca J. Frey, PhD

Salmonella food poisoning

Definition

Salmonella food poisoning is a bacterial food poisoning caused by the *Salmonella* bacterium. It results in the swelling of the lining of the stomach and intestines (gastroenteritis). While domestic and wild animals, including poultry, pigs, cattle, and pets such as turtles, iguanas, chicks, dogs, and cats can

transmit this illness, most people become infected by ingesting foods contaminated with significant amounts of *Salmonella*.

Description

Salmonella food poisoning occurs worldwide, however it is most frequently reported in North America and Europe. Only a small proportion of infected people are tested and diagnosed, and as few as 1% of cases are actually reported. While the infection rate may seem relatively low, even an attack rate of less than 0.5% in such a large number of exposures results in many infected individuals. The poisoning typically occurs in small, localized outbreaks in the general population or in large outbreaks in hospitals, restaurants, or institutions for children or the elderly. In the United States, *Salmonella* is responsible for about 15% of all cases of food poisoning.

Improperly handled or undercooked poultry and eggs are the foods which most frequently cause Salmonella food poisoning. Chickens are a major carrier of *Salmonella* bacteria, which accounts for its prominence in poultry products. However, identifying foods which may be contaminated with *Salmonella* is particularly difficult because infected chickens typically show no signs or symptoms. Since infected chickens have no identifying characteristics, these chickens go on to lay eggs or to be used as meat.

At one time, it was thought that *Salmonella* bacteria were only found in eggs which had cracked, thus allowing the bacteria to enter. Ultimately, it was learned that, because the egg shell has tiny pores, even uncracked eggs which sat for a time on a surface (nest) contaminated with *Salmonella* could themselves become contaminated. It is known also that the bacteria can be passed from the infected female chicken directly into the substance of the egg before the shell has formed around it.

Anyone may contract Salmonella food poisoning, but the disease is most serious in infants, the elderly, and individuals with weakened immune systems. In these individuals, the infection may spread from the intestines to the blood stream, and then to other body sites, causing **death** unless the person is treated promptly with **antibiotics**. In addition, people who have had part or all of their stomach or their spleens removed, or who have sickle cell anemia, **cirrhosis** of the liver, leukemia, lymphoma, **malaria**, louse-borne **relapsing fever**, or Acquired Immunodeficiency Syndrome (**AIDS**) are particularly susceptible to Salmonella food poisoning.

Causes and symptoms

Salmonella food poisoning can occur when someone drinks unpasteurized milk or eats undercooked chicken or eggs, or salad dressings or desserts which contain raw eggs. Even if *Salmonella*-containing foods such as chicken are thoroughly cooked, any food can become contaminated during preparation if conditions and equipment for food preparation are unsanitary.

Other foods can then be accidentally contaminated if they come into contact with infected surfaces. In addition, children have become ill after playing with turtles or iguanas, and then eating without washing their hands. Because the bacteria are shed in the feces for weeks after infection with *Salmonella*, poor hygiene can allow such a carrier to spread the infection to others.

Symptoms appear about one-two days after infection, and include **fever** (in 50% of patients), **nausea** and vomiting, **diarrhea**, and abdominal cramps and pain. The diarrhea is usually very liquid, and rarely contains mucus or blood. Diarrhea usually lasts for about four days. The illness usually ends in about five-seven days.

Serious complications are rare, occurring most often in individuals with other medical illnesses. Complications occur when the *Salmonella* bacteria make their way into the bloodstream (**bacteremia**). Once in the bloodstream, the bacteria can enter any organ system throughout the body, causing disease. Other infections which can be caused by *Salmonella* include:

- bone infections (**osteomyelitis**)
- joint infections (arthritis)
- infection of the sac containing the heart (pericarditis)
- infection of the tissues which cover the brain and spinal cord (**meningitis**)
- infection of the liver (hepatitis)
- lung infections (**pneumonia**)
- infection of aneurysms (aneurysms are abnormal outpouchings which occur in weak areas of the walls of blood vessels)
- infections in the center of already-existing tumors or cysts.

Diagnosis

Under appropriate laboratory conditions, *Salmonella* can be grown and then viewed under a microscope for identification. Early in the infection, the blood is far more likely to positively show a



***Salmonella enteritidis*.** Exposure to this bacterium usually occurs by contact with contaminated food. (Photograph by Oliver Meckes, Photo Researchers, Inc. Reproduced by permission.)

presence of the *Salmonella* bacterium when a sample is grown on a nutrient substance (culture) for identification purposes. Eventually, however, positive cultures can be obtained from the stool and in some cases from a urine culture.

Treatment

Even though Salmonella food poisoning is a bacterial infection, most practitioners do not treat simple cases with antibiotics. Studies have shown that using antibiotics does not usually reduce the length of time that the patient is ill. Paradoxically, it appears that antibiotics do, however, cause the patient to shed bacteria in their feces for a *longer* period of time. In order to decrease the length of time that a particular individual is a carrier who can spread the disease, antibiotics are generally not given.

In situations where an individual has a more severe type of infection with *Salmonella* bacteria, a number of antibiotics may be used. Chloramphenicol was the first antibiotic successfully used to treat Salmonella food poisoning. It is still a drug of choice in developing countries because it is so inexpensive, although some resistance has developed to it. Ampicillin and trimethoprim-sulfonamide have been used successfully in the treatment of infections caused by chloramphenicol-resistant strains. Newer types of antibiotics, such as cephalosporin or quinolone, are also effective. These drugs can be given by mouth or through a needle in the vein (intravenously) for very ill patients. With effective antibiotic therapy, patients feel better in 24–48 hours, the temperature returns to normal in three-five days, and the patient is generally recovered by 10–14 days.

KEY TERMS

Carrier—Someone who has an organism (bacteria, virus, fungi) in his or her body, without signs of illness. The individual may therefore pass the organism on to others.

Gastroenteritis—Inflammation of the stomach and intestines. Usually causes nausea, vomiting, diarrhea, abdominal pain and cramps.

Alternative treatment

A number of alternative treatments have been recommended for food poisoning. One very effective treatment that is strongly recommended is supplementation with *Lactobacillus acidophilus*, *L. bulgaricus*, and/or *Bifidobacterium* to restore essential bacteria in the digestive tract. These preparations are available as powders, tablets, or capsules from health food stores; yogurt with live *L. acidophilus* cultures can also be eaten. Fasting or a liquid-only diet is often used for food poisoning. Homeopathic treatment can work very effectively in the treatment of *Salmonella* food poisoning. The appropriate remedy for the individual and his/her symptoms must be used to get the desired results. Some examples of remedies commonly used are *Chamomilla*, *Nux vomica*, *Ipecac*, and *Colchicum*. Juice therapy, including carrot, beet, and garlic juices, is sometimes recommended, although it can cause discomfort for some people. Charcoal tablets can help absorb toxins and remove them from the digestive tract through bowel elimination. A variety of herbs with antibiotic action, including citrus seed extract, goldenseal (*Hydrastis canadensis*), and Oregon grape (*Mahonia aquifolium*), may also be effective in helping to resolve cases of food poisoning.

Prognosis

The prognosis for uncomplicated cases of *Salmonella* food poisoning is excellent. Most people recover completely within a week's time. In cases where other medical problems complicate the illness, prognosis depends on the severity of the other medical conditions, as well as the specific organ system infected with *Salmonella*.

Prevention

Prevention of *Salmonella* food poisoning involves the proper handling and cooking of foods likely to carry the bacteria. This means that recipes utilizing

uncooked eggs (Caesar salad dressing, meringue toppings, mousses) need to be modified to eliminate the raw eggs. Not only should chicken be cooked thoroughly, until no pink juices flow, but all surfaces and utensils used on raw chicken must be carefully cleaned to prevent *Salmonella* from contaminating other foods. Careful handwashing is a must before, during, and after all food preparation involving eggs and poultry. Handwashing is also important after handling and playing with pets such as turtles, iguanas, chicks, dogs and cats.

Resources

ORGANIZATIONS

Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

Rosalyn Carson-DeWitt, MD

Salmonella paratyphi infection see

Paratyphoid fever

Salmonella typhi infection see **Typhoid fever**

Salpingectomy

Definition

Salpingectomy is the removal of one or both of a woman's fallopian tubes, the tubes through which an egg travels from the ovary to the uterus.

Purpose

A salpingectomy may be performed for several different reasons. Removal of one tube (unilateral salpingectomy) is usually performed if the tube has become infected (a condition known as salpingitis).

Salpingectomy is also used to treat an **ectopic pregnancy**, a condition in which a fertilized egg has implanted in the tube instead of inside the uterus. In most cases, the tube is removed only after drug treatments designed to save the structure have failed. (Women with one remaining fallopian tube are still able to get pregnant and carry a **pregnancy** to term.) The other alternative to salpingectomy is surgery to remove the fetus from the fallopian tube, followed by surgery to repair the tube.

A bilateral salpingectomy (removal of both the tubes) is usually done if the ovaries and uterus are also going to be removed. If the fallopian tubes and the

KEY TERMS

Ectopic pregnancy—The development of a fetus at a site other than the inside of the uterus; most commonly, the egg implants itself in the fallopian tube.

Laparoscope—A surgical instrument with a light attached that is inserted through the abdominal wall to allow the surgeon to see the organs in the abdomen.

ovaries are both removed at the same time, this is called a **salpingo-oophorectomy**. A salpingo-oophorectomy is necessary when treating ovarian and endometrial cancer because the fallopian tubes and ovaries are the most common sites to which cancer may spread.

Description

Regional or **general anesthesia** may be used. Often a laparoscope (a hollow tube with a light on one end) is used in this type of operation, which means that the incision can be much smaller and the recovery time much shorter.

In this procedure, the surgeon makes a small incision just beneath the navel. The surgeon inserts a short hollow tube into the abdomen and, if necessary, pumps in carbon dioxide gas in order to move intestines out of the way and better view the organs. After a wider double tube is inserted on one side for the laparoscope, another small incision is made on the other side through which other instruments can be inserted. After the operation is completed, the tubes and instruments are withdrawn. The tiny incisions are sutured and there is very little scarring.

In the case of a pelvic infection, the surgeon makes a horizontal (bikini) incision 4-6 in (10-15 cm) long in the abdomen right above the pubic hairline. This allows the doctor to remove the scar tissue. (Alternatively, a surgeon may use a vertical incision from the pubic bone toward the navel, although this is less common.)

Preparation

The patient is given an injection an hour before surgery to encourage drowsiness.

Aftercare

Aftercare varies depending on whether the tube was removed by **laparoscopy** or through an abdominal

incision. Even when major surgery is performed, most women are out of bed and walking around within three days. Within a month or two, a woman can slowly return to normal activities such as driving, exercising, and working.

Risks

All surgery, especially under general anesthesia, carries certain risks, such as the risk of scarring, hemorrhaging, infection, and reactions to the anesthesia. Pelvic surgery can also cause internal scarring which can lead to discomfort years afterward.

Resources

ORGANIZATIONS

National Women's Health Resource Center. 120 Albany St., Suite 820, New Brunswick, NJ 08901. (877) 986-9472. <<http://www.healthywomen.org>>.

Carol A. Turkington

Salpingitis see **Pelvic inflammatory disease**

Salpingo-oophorectomy

Definition

The surgical removal of a fallopian tube and an ovary.

Purpose

This surgery is performed to treat ovarian or other gynecological cancers, or infections as a result of **pelvic inflammatory disease**. Occasionally, removal of one or both ovaries may be done to treat **endometriosis**. If only one tube and ovary are removed, the woman may still be able to conceive and carry a **pregnancy** to term.

Description

If the procedure is performed through a laparoscope, the surgeon can avoid a large abdominal incision and can shorten recovery. With this technique, the surgeon makes a small cut through the abdominal wall just below the navel. When the laparoscope is used, the patient can be given either regional or **general anesthesia**; if there are no complications, the patient can leave the hospital in a day or two.

If a laparoscope is not used, the surgery involves an incision 4–6 in (10–long into the abdomen either extending vertically up from the pubic bone toward the navel, or horizontally (the “bikini incision”) across the pubic hairline. The scar from a bikini incision is less noticeable, but some surgeons prefer the vertical incision because it provides greater visibility while operating.

Preparation

A spinal block or general anesthesia may be given before surgery.

Aftercare

If performed through an abdominal incision, salpingo-oophorectomy is major surgery that requires three to six weeks for full recovery. However, if performed laparoscopically, the recovery time can be much shorter. There may be some discomfort around the incision for the first few days after surgery, but most women are walking around by the third day. Within a month or so, patients can gradually resume normal activities such as driving, exercising, and working.

Immediately following the operation, the patient should avoid sharply flexing the thighs or the knees. Persistent back **pain** or bloody or scanty urine indicates that a ureter may have been injured during surgery.

If both ovaries are removed in a premenopausal woman as part of the operation, the sudden loss of estrogen will trigger an abrupt **premature menopause** that may involve severe symptoms of hot flashes, vaginal dryness, painful intercourse, and loss of sex drive. (This is also called “surgical menopause.”) In addition to these symptoms, women who lose both ovaries also lose the protection these hormones provide against heart disease and **osteoporosis** many years earlier than if they had experienced natural menopause. Women who have had their ovaries removed are seven times more likely to develop coronary heart disease and much more likely to develop bone problems at an early age than are premenopausal women whose ovaries are intact.

For these reasons, some form of estrogen replacement therapy (ERT) may be prescribed to relieve the symptoms of surgical **menopause** and to help prevent heart and bone disease.

In addition, to help offset the higher risks of heart and bone disease after loss of the ovaries, women

KEY TERMS

Androgens—Hormones (specifically testosterone) responsible for male sex characteristics.

Endometriosis—A painful disease in which cells from the lining of the uterus (endometrium) aren’t shed during menstruation, but instead attach themselves to other organs in the pelvic cavity. The condition is hard to diagnose and often causes severe pain as well as infertility.

Fallopian tubes—Tubes that extend from either end of the uterus that convey the egg from the ovary to the uterus during each monthly cycle.

Ureter—The tube that carries urine from the bladder to the kidneys.

should get plenty of **exercise**, maintain a low-fat diet, and ensure intake of calcium is adequate.

Reaction to the removal of fallopian tubes and ovaries depends on a wide variety of factors, including the woman’s age, the condition that required the surgery, her reproductive history, how much social support she has, and any previous history of depression. Women who have had many gynecological surgeries or chronic pelvic pain seem to have a higher tendency to develop psychological problems after the surgery.

Risks

Major surgery always involves some risk, including infection, reactions to the anesthesia, hemorrhage, and **scars** at the incision site. Almost all pelvic surgery causes some internal scars, which, in some cases, can cause discomfort years after surgery.

Resources

ORGANIZATIONS

Midlife Women’s Network, 5129 Logan Ave. S.,
Minneapolis, MN 55419.(800) 886-4354.

Carol A. Turkington

San Joaquin fever see **Coccidioidomycosis**

Sanfilippo’s syndrome see

Mucopolysaccharidoses

Saquinavir see **Protease inhibitors**

Sarcoidosis

Definition

Sarcoidosis is a disease which can affect many organs within the body. It causes the development of granulomas. Granulomas are masses resembling little tumors. They are made up of clumps of cells from the immune system.

Description

Sarcoidosis is a very puzzling disorder. In addition to having no clear-cut understanding of the cause of sarcoidosis, researchers are also puzzled by its distribution in the world population. In the United States, for example, 10-17 times as many African-Americans are affected as white Americans. In Europe, whites are primarily affected.

Prevalence is a way of measuring the number of people affected per 100,000 people in a given population. The prevalence figures for sarcoidosis are very unusual. In the United States, prevalence figures range from five (5/100,000 in the United States) for whites to 40 for blacks. In Europe, prevalence ranges from three in Poland, to 10 in France, to 64 in Sweden, to 200 for Irish women living in London. Furthermore, a person from a group with very low prevalence who leaves his or her native land for a second location with a higher prevalence will then have the same risk as anyone living in that second location.

Sarcoidosis affects both men and women, although women are more likely to have the disorder. The average age for diagnosis is around 20-40 years.

Causes and symptoms

The cause of sarcoidosis is not known. Because the granulomas are primarily made up of cells from the immune system (macrophages and lymphocytes), an immune connection is strongly suspected. One of the theories which has been put forth suggests that exposure to some toxic or infectious material starts up an immune response. For some reason, the body is unable to stop the response, and it spreads from the original organ to other organs.

Because sarcoidosis has been noted to occur in family groups, a genetic cause has also been suggested. Research shows that identical twins are more likely to both have sarcoidosis than are nonidentical twins or other siblings.

Some cases of sarcoidosis occur without the patient even noting any symptoms. These cases are

often discovered by chance during routine chest x rays. Most cases of sarcoidosis, however, begin with very nonspecific symptoms, such as decreased energy, weakness, and a dry **cough**. Occasionally, the cough is accompanied by some mild **pain** in the breast-bone (sternum). Some patients note that they are having unusual **shortness of breath** while exercising. Some patients develop **fever**, decreased appetite, and weight loss.

Virtually every system of the body has the potential to suffer the effects of sarcoidosis:

- tender reddish bumps (nodules) or patches often appear on the skin
- the eyes may become red and teary, and the vision blurry
- the joints may become swollen and painful (arthritis)
- lymph nodes in the neck, armpits, and groin become enlarged and tender, lymph nodes within the chest, around the lungs, also become enlarged
- fluid may accumulate around the lungs (pleural effusion), making breathing increasingly difficult
- nasal stuffiness is common, as well as a hoarse sound to the voice
- cysts in the bone may cause pain in the hands and feet, or in other bony areas
- the bone marrow may decrease the production of all blood cells; decreased number of red blood cells causes anemia, fewer white blood cells increases the chance of infections, fewer platelets can increase the chance of bleeding
- the body's ability to process calcium often becomes abnormal, so that excess calcium passes through the kidneys and into the urine; this may cause kidney stones to form
- the liver may become enlarged
- the heart may suffer a variety of complications, including abnormal or missed beats (arrhythmias), inflammation of the covering of the heart (**pericarditis**), and an increasing tendency toward weak, ineffective pumping of the blood (heart failure)
- the nervous system may display the effects of sarcoidosis by hearing loss, chronic inflammation of the coverings of the brain and spinal cord (**meningitis**), abnormalities of the nerve that is involved in vision (optic nerve dysfunction), seizures, and the development of psychiatric disorders

Any, all, or even none of the above symptoms may be present in sarcoidosis.

Diagnosis

Diagnosis depends on information from a number of sources, including the patient's symptoms, the **physical examination**, x-ray pictures of the chest, and a number of other laboratory examinations of blood or other tissue. None of these categories of information are sufficient to make the diagnosis of sarcoidosis. There is no one test or sign or symptom which clearly points to sarcoidosis, excluding all other types of diseases. This is because nearly all of the symptoms and laboratory results in sarcoidosis also occur in other diseases. Diagnosis, then, requires careful consideration of many facts.

The physical examination in sarcoidosis may reveal the characteristic **skin lesions**. Wheezes may be heard throughout the lungs. The liver may be enlarged. Examination of the eyes using a special light called a slit-lamp may reveal changes indicative of sarcoidosis.

The **chest x ray** will show some pattern of abnormalities, which may include enlargement of the lymph nodes which drain the lung, scarring and abnormalities to the tissue of the lungs, and fluid accumulation around the lungs.

Lung function tests measure such things as the amount of air an individual can breathe in and breathe out, the speed at which the air flows in and out, and the amount of air left in the lung after blowing out as much as possible in one second. A variety of lung function tests may show abnormal results in sarcoidosis.

Other types of tests may be abnormal in sarcoidosis. The abnormal test results may also indicate other diseases. They include an elevation of a substance called angiotensin-converting enzyme in the blood, and an increased amount of calcium present in 24 hours worth of urine.

Bronchoscopy is a very helpful diagnostic test. This involves passing a tiny tube (bronchoscope) through the nose or mouth, down the trachea, and into the airways (bronchial tubes). The bronchial tubes can be inspected through the bronchoscope. The bronchoscope is also designed in such a way as to allow biopsies to be obtained. Bronchoalveolar lavage involves washing the surfaces with a sterile saltwater (saline) solution. The saline is then retrieved and examined in a laboratory. Cells and debris from within the bronchial tubes and the tiny sacs of the lung (the alveoli) will be obtained in this way, and can be studied for the presence of an abnormally large number of white blood cells. A tiny piece of the lung tissue can also be obtained through the bronchoscope. This can be studied under a microscope to look for the characteristic granulomas and inflammation of sarcoidosis.

KEY TERMS

Granuloma—Masses made up of a variety of immune cells, as well as fibroblasts (cells which make up connective tissue).

Immune system—The system of specialized organs, lymph nodes, and blood cells throughout the body which work together to prevent foreign invaders (bacteria, viruses, fungi, etc.) from taking hold and growing.

A gallium 67 scan involves the injection of a radioactive material called gallium 67. In sarcoidosis, areas of the body which are inflamed will retain the gallium 67. These areas will then show up on the scan.

Treatment

Many cases of sarcoidosis resolve without treatment. If treatment is needed, the most effective one for sarcoidosis is the administration of steroid medications. These medications work to decrease inflammation throughout the body. The long-term use of steroid medications has serious potential side-effects. Patients are only treated with steroids when the problems caused by sarcoidosis are particularly serious. Many cases of sarcoidosis resolve without treatment.

Prognosis

The prognosis for sarcoidosis is quite good. About 60-70% of the time, sarcoidosis cures itself within a year or two. In about 20-30% of patients, permanent damage occurs to the lungs. About 15-20% of all patients go on to develop a chronic, relapsing form of sarcoidosis. Death can be blamed on sarcoidosis in about 10% of all sarcoidosis cases.

Prevention

Until researchers are able to pinpoint the cause of sarcoidosis, there will be no available recommendations for how to prevent it.

Resources

PERIODICALS

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Rosalyn Carson-DeWitt, MD

Sarcomas

Definition

A sarcoma is a bone tumor that contains **cancer** (malignant) cells. A benign bone tumor is an abnormal growth of noncancerous cells.

Description

A primary bone tumor originates in or near a bone. Most primary bone tumors are benign, and the cells that compose them do not spread (metastasize) to nearby tissue or to other parts of the body.

Malignant primary bone tumors account for fewer than 1% of all cancers diagnosed in the United States. They can infiltrate nearby tissues, enter the bloodstream, and metastasize to bones, tissues, and organs far from the original malignancy. Malignant primary bone tumors are characterized as either:

- bone cancers which originate in the hard material of the bone.
- soft-tissue sarcomas which begin in blood vessels, nerves, or tissues containing muscles, fat, or fiber.

Types of bone tumors

Osteogenic sarcoma, or osteosarcoma, is the most common form of bone cancer, accounts for 6% of all instances of the disease, and for about 5% of all cancers that occur in children. Nine hundred new cases of osteosarcoma are diagnosed in the United States every year. The disease usually affects teenagers, and is almost twice as common in boys as in girls.

Osteosarcomas, which grow very rapidly, can develop in any bone but most often occur along the edge or on the end of one of the fast-growing long bones that support the arms and legs. About 80% of all osteosarcomas develop in the parts of the upper and lower leg nearest the knee (the distal femur or in the proximal tibia). The next likely location for an osteosarcoma is the bone of the upper arm closest to the shoulder (the proximal humerus).

Ewing's sarcoma is the second most common form of childhood bone cancer. Accounting for fewer than 5% of bone tumors in children, Ewing's sarcoma usually begins in the soft tissue (the marrow) inside bones of the leg, hips, ribs, and arms. It rapidly infiltrates the lungs, and may metastasize to bones in other parts of the body.

More than 80% of patients who have Ewing's sarcoma are white, and the disease most frequently

affects children between ages 5-9, and young adults between ages 20-30. About 27% of all cases of Ewing's sarcoma occur in children under age 10, and 64% occur in adolescents between ages 10-20.

Chondrosarcomas are cancerous bone tumors that most often appear in middle age. Usually originating in strong connective tissue (cartilage) in ribs or leg or hip bones, chondrosarcomas grow slowly. They rarely spread to the lungs. It takes years for a chondrosarcoma to metastasize to other parts of the body, and some of these tumors never spread.

Parosteal osteogenic sarcomas, fibrosarcomas, and chordomas are rare. Parosteal osteosarcomas generally involve both the bone and the membrane that covers it. Fibrosarcomas originate in the ends of the bones in the arm or leg, and then spread to soft tissue. Chordomas develop on the skull or spinal cord.

Osteochondromas, which usually develop between age 10-20, are the most common noncancerous primary bone tumors. Giant cell tumors generally develop in a section of the thigh bone near the knee. Giant cell tumors are originally benign but sometimes become malignant.

Causes and symptoms

The cause of bone cancer is unknown, but the tendency to develop it may be inherited. Children who have bone tumors are often tall for their age, and the disease seems to be associated with growth spurts that occur during childhood and adolescence. Injuries can make the presence of tumors more apparent but do not cause them.

A bone that has been broken or exposed to high doses of radiation used to treat other cancers is more likely than other bones to develop osteosarcoma. A history of noncancerous bone disease also increases bone-cancer risk.

The amount of radiation in diagnostic x rays poses little or no danger of bone-cancer development, but children who have a family history of the most common childhood cancer of the eye (**retinoblastoma**), or who have inherited rare cancer syndromes have a greater-than-average risk of developing bone cancer. Exposure to chemicals found in some paints and dyes can slightly raise the risk.

Both benign and malignant bone tumors can distort and weaken bone and cause **pain**, but benign tumors are generally painless and asymptomatic.

It is sometimes possible to feel a lump or mass, but pain in the affected area is the most common early

symptom of bone cancer. Pain is not constant in the initial stages of the disease, but it is aggravated by activity and may be worse at night. If the tumor is located on a leg bone, the patient may limp. Swelling and weakness of the limb may not be noticed until weeks after the pain began.

Other symptoms of bone cancer include:

- a bone that breaks for no apparent reason
- difficulty moving the affected part of the body
- fatigue
- fever
- a lump on the trunk, an arm or leg, or another bone
- persistent, unexplained back pain
- weight loss

Diagnosis

Physical examination and routine x rays may yield enough evidence to diagnose benign bone tumors, but removal of tumor tissue for microscopic analysis (biopsy) is the only sure way to rule out malignancy.

A needle biopsy involves using a fine, thin needle to remove small bits of tumor, or a thick needle to extract tissue samples from the innermost part (the core) of the growth. An excisional biopsy is the surgical removal of a small, accessible tumor. An incisional biopsy is performed on tumors too large or inaccessible to be completely removed. The surgeon performing an incisional biopsy cuts into the patient's skin and removes a portion of the exposed tumor. Performed under local or general anesthetic, biopsy reveals whether a tumor is benign or malignant and identifies the type of cancer cells the malignant tumor contains.

Bone cancer is usually diagnosed about three months after symptoms first appear, and 20% of malignant tumors have metastasized to the lungs or other parts of the body by that time.

Imaging techniques

The following procedures are used, in conjunction with biopsy, to diagnose bone cancer:

- Bone x rays. These x rays usually provide a clear image of osteosarcomas.
- Computerized axial tomography (CAT scan) is a specialized x ray that uses a rotating beam to obtain detailed information about an abnormality and its physical relationship to other parts of the body. A CAT scan can differentiate between osteosarcomas and other types of bone tumors, illustrate how tumor

cells have infiltrated other tissues, and help surgeons decide which portion of a growth would be best to biopsy. Because more than four of every five malignant bone tumors metastasize to the lungs, a CAT scan of the chest is performed to see if these organs have been affected. Chest and abdominal CAT scans are used to determine whether Ewing's sarcoma has spread to the lungs, liver, or lymph nodes.

- Magnetic resonance imaging (MRI) is a specialized scan that relies on radio waves and powerful magnets to reflect energy patterns created by tissue abnormalities and specific diseases. An MRI provides more detailed information than does a CAT scan about tumors and marrow cavities of the bone, and can sometimes detect clusters of cancerous cells that have separated from the original tumor. This valuable information helps surgeons select the most appropriate approach for treatment.
- Radionuclide bone scans. These scans involve injecting a small amount of radioactive material into a vein. Primary tumors or cells that have metastasized absorb the radioactive material and show up as dark spots on the scan.

Cytogenic and molecular genetic studies, which assess the structure and composition of chromosomes and genes, may also be used to diagnose osteosarcoma. These tests can sometimes indicate what form of treatment is most appropriate.

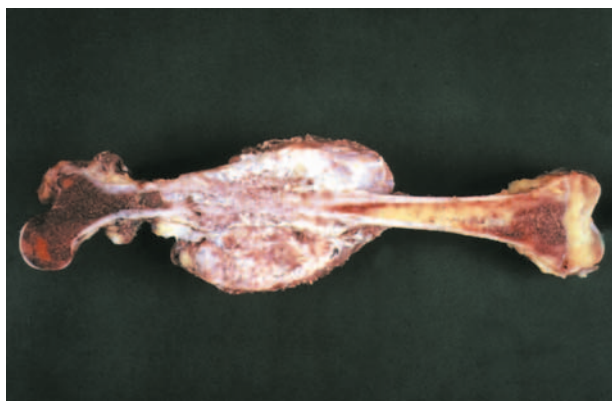
Laboratory studies

A complete **blood count** (CBC) reveals abnormalities in the blood, and may indicate whether bone marrow has been affected. A blood test that measures levels of the enzyme lactate dehydrogenase (LDH) can predict the likelihood of a specific patient's survival.

Immunohistochemistry involves adding special antibodies and chemicals, or stains, to tumor samples. This technique is effective in identifying cells that are found in Ewing's sarcoma but are not present in other malignant tumors.

Reverse transcription polymerase chain reaction (RT-PCR) relies on chemical analysis of the substance in the body that transmits genetic information (RNA) to:

- evaluate the effectiveness of cancer therapies
- identify mutations consistent with the presence of Ewing's sarcoma
- reveal cancer that recurs after treatment has been completed



A specimen of a femur bone indicating the cancerous growth around the knee. Osteosarcoma is the most common primary cancer of the bone. (Photo Researchers, Inc. Reproduced by permission.)

Staging

Once bone cancer has been diagnosed, the tumor is staged. This process indicates how far the tumor has spread from its original location. The stage of a tumor suggests which form of treatment is most appropriate, and predicts how the condition will probably respond to therapy.

An osteosarcoma may be localized or metastatic. A localized osteosarcoma has not spread beyond the bone where it arose or beyond nearby muscles, tendons, and other tissues. A metastatic osteosarcoma has spread to the lungs, to bones not directly connected to the bone in which the tumor originated, or to other tissues or organs.

Treatment

Since the 1960s, when **amputation** was the only treatment for bone cancer, new **chemotherapy** drugs and innovative surgical techniques have improved survival with intact limbs. Because osteosarcoma is so rare, patients should consider undergoing treatment at a major cancer center staffed by specialists familiar with the disease.

A treatment plan for bone cancer, developed after the tumor has been diagnosed and staged, may include:

- **Amputation.** Amputation may be the only therapeutic option for large tumors involving nerves or blood vessels that have not responded to chemotherapy. MRI scans indicate how much of the diseased limb must be removed, and surgery is planned to create a cuff, formed of muscles and skin, around the amputated bone. Following surgery, an artificial

(prosthetic) leg is fitted over the cuff. A patient who actively participates in the **rehabilitation** process may be walking independently as soon as three months after the amputation.

- **Chemotherapy.** Chemotherapy is usually administered in addition to surgery, to kill cancer cells that have separated from the original tumor and spread to other parts of the body. Although chemotherapy can increase the likelihood of later development of another form of cancer, the American Cancer Society maintains that the need for chemotherapeutic bone-cancer treatment is much greater than the potential risk.
- **Surgery.** Surgery, coordinated with diagnostic biopsy, enhances the probability that limb-salvage surgery can be used to remove the cancer while preserving nearby blood vessels and bones. A metal rod or bone graft is used to replace the area of bone removed, and subsequent surgery may be needed to repair or replace rods that have loosened or broken. Patients who have undergone limb-salvage surgery need intensive rehabilitation. It may take as long as a year for a patient to regain full use of a leg following limb-salvage surgery, and patients who have this operation may eventually have to undergo amputation.
- **Radiation therapy.** Radiation therapy is used often to treat Ewing's sarcoma.
- **Rotationoplasty.** Rotationoplasty, sometimes performed after a leg amputation, involves attaching the lower leg and foot to the thigh bone, so that the ankle replaces the knee. A prosthetic is later added to make the leg as long as it should be. Prosthetic devices are not used to lengthen limbs that remain functional after amputation to remove osteosarcomas located on the upper arm. When an osteosarcoma develops in the jaw bone, the entire lower jaw is removed. Bones from other parts of the body are later grafted on remaining bone to create a new jaw.

Follow-up treatments

After a patient completes the final course of chemotherapy, CAT scans, bone scans, x rays, and other diagnostic tests may be repeated to determine if any traces of tumor remain. If none are found, treatment is discontinued, but patients are advised to see their oncologist and orthopedic surgeon every two or three months for the next year. X rays of the chest and affected bone are taken every four months. An annual echocardiogram is recommended to evaluate any adverse effect chemotherapy may have had on the heart, and CT scans are performed every six months.

Patients who have received treatment for Ewing's sarcoma are examined often - at gradually lengthening intervals - after completing therapy. Accurate growth measurements are taken during each visit and blood is drawn to be tested for side effects of treatment. X rays, CT scans, bone scans, and other imaging studies are generally performed every three months during the first year. If no evidence of tumor growth or recurrence is indicated, these tests are performed less frequently in the following years.

Some benign bone tumors shrink or disappear without treatment. However, regular examinations are recommended to determine whether these tumors have changed in any way.

Alternative treatment

Alternative treatments should never be substituted for conventional bone-cancer treatments or used without the approval of a physician. However, some alternative treatments can be used as adjunctive and supportive therapies during and following conventional treatments.

Dietary adjustments can be very helpful for patients with cancer. Whole foods, including grains, beans, fresh fruits and vegetables, and high quality fats, should be emphasized in the diet, while processed foods should be avoided. Increased consumption of fish, especially cold water fish like salmon, mackerel, halibut, and tuna, provides a good source of **omega-3 fatty acids**. **Nutritional supplements** can build strength and help maintain it during and following chemotherapy, radiation, or surgery. These supplements should be individually prescribed by an alternative practitioner who has experience working with cancer patients.

Many cancer patients claim that **acupuncture** alleviates pain, **nausea**, and **vomiting**. It can also be effective in helping to maintain energy and relative wellness during surgery, chemotherapy, and radiation. Massage, **reflexology**, and relaxation techniques are said to relieve pain, tension, **anxiety**, and depression. **Exercise** can be an effective means of reducing mental and emotional **stress**, while increasing physical strength. **Guided imagery**, **biofeedback**, hypnosis, body work, and progressive relaxation can also enhance quality of life.

Claims of effectiveness in fighting cancer have been made for a variety of herbal medicines. These botanical remedies work on an individual basis and should only be used when prescribed by a practitioner familiar with cancer treatment.

Treating cancer is a complex and individual task. It should be undertaken by a team of support

practitioners with varying specialties who can work together for healing the person with cancer.

Prognosis

Benign brain tumors rarely recur, but sarcomas can reappear after treatment was believed to have eliminated every cell.

Likelihood of long-term survival depends on:

- the type and location of the tumor
- how much the tumor has metastasized, and on what organs, bones, or tissues have been affected

More than 85% of patients survive for more than five years after complete surgical removal of low-grade osteosarcomas (tumors that arise in mature tissue and contain a small number of cancerous cells). About 25-30% of patients diagnosed with high-grade osteosarcomas (tumors that develop in immature tissue and contain a large number of cancer cells) will die of the disease.

Two-thirds of all children diagnosed with Ewing's sarcoma will live for more than five years after the disease is detected. The outlook is most favorable for children under age 10, and least favorable in patients whose cancer is not diagnosed until after it has metastasized: fewer than three of every 10 of these patients remain alive five years later. More than 80% of patients whose Ewing's sarcoma is confined to a small area and surgically removed live, for at least five years. Postsurgical radiation and chemotherapy add years to their lives. More than 70% of patients live five years or more with a small Ewing's sarcoma that cannot be removed, but only three out of five patients with large, unremovable tumors survive that long.

Prevention

There is no known way to prevent bone cancer.

Resources

ORGANIZATIONS

American Cancer Society. 1599 Clifton Rd., NE, Atlanta, GA 30329-4251. (800) 227-2345.
< <http://www.cancer.org> >.

CancerCare, Inc. 1180 Avenue of the Americas, New York, NY 10036. (800) 813-4673.
< <http://www.cancercare.org> >.

National Cancer Institute. Building 31, Room 10A31, 31 Center Drive, MSC 2580, Bethesda, MD 20892-2580. (800) 422-6237. < <http://www.nci.nih.gov> >.

Maureen Haggerty

Saw palmetto

Definition

Saw palmetto is an extract derived from the deep purple berries of the saw palmetto fan palm (*Serenoa repens*), a plant indigenous to the coastal regions of the southern United States and southern California. There is an estimated one million acres of wild saw palmetto palms in Florida, where the bulk of commercial saw palmetto is grown.

Purpose

Saw palmetto is used by natural health practitioners to treat a variety of ailments in men and women, such as testicular inflammation, urinary tract inflammation, coughs, and respiratory congestion. It is also used to strengthen the thyroid gland, balance the metabolism, stimulate appetite, and aid digestion. According to the American Dietetic Association, saw palmetto is one of the most commonly used dietary supplements among Americans between the ages of 50 and 76.

Most of the evidence supporting these uses is anecdotal and has not been proven by controlled clinical trials. However, there is much scientific documentation outlining the effectiveness of the herb in treating irritable bladder and urinary problems in men with benign prostate hyperplasia (BPH), an enlargement of the prostate gland. BPH results in a swelling of the prostate gland that obstructs the urethra. This causes painful urination, reduced urine flow, difficulty starting or stopping the flow, dribbling after urination, and more frequent nighttime urination. Saw palmetto does not reduce prostate enlargement. Instead, it is thought to work in a variety of ways. First, it inhibits the conversion of testosterone into dihydrotestosterone (DHT). BPH is thought to be caused by an increase in testosterone to DHT. Secondly, saw palmetto is believed to interfere with the production of estrogen and progesterone, hormones associated with DHT production.

In addition to causing **pain** and embarrassment, BPH can lead to serious kidney problems if undiagnosed and left untreated. It is a common problem in men over the age of 40. Estimates are that 50–60% of all men will develop BPH in their lifetimes. It is estimated that there are six million men between the ages of 50–79 who have BPH serious enough to require some type of therapy. Yet only half of them seek treatment from physicians. Health practitioners in both the allopathic and natural medicine communities recommend annual prostate examinations for men

over the age of 50, and an annual blood test that measures prostate specific antigen, a marker for **prostate cancer**.

Recently, a number of clinical trials have confirmed the effectiveness of saw palmetto in treating BPH. Many of these trials have shown saw palmetto works better than the most commonly used prescription drug, Proscar. Saw palmetto is effective in nearly 90% of patients after six weeks of use, while Proscar is effective in less than 50% of patients. In addition, Proscar may take up to six months to achieve its full effect. Since Proscar blocks the production of testosterone, it can cause **impotence** and breast enlargement. Also, saw palmetto is significantly less expensive than Proscar. A one-month supply of saw palmetto costs \$12–25, while a one month supply of Proscar costs \$65–75. Other prescription drugs used to treat BPH are Cardura (doxazosin), Hytrin (terazosin), and Flomax (tamsulosin hydrochloride). Originally prescribed to treat **hypertension**, Cardura and Hytrin can drop blood pressure, causing lightheadedness and **fainting**. Presently, saw palmetto is being evaluated by the U.S. Food and Drug Administration (FDA) for treatment of BPH. If approved, it would become the first herbal product to be licensed by the agency as a treatment for a specific condition. Saw palmetto is also used as a treatment for prostate complaints and irritable bladder.

Since the 1960s, extensive clinical studies of saw palmetto have been done in Europe. A 1998 review of 24 European trials involved nearly 3,000 men, some taking saw palmetto, others taking Proscar, and a third group taking a placebo. The men taking saw palmetto had a 28% improvement in urinary tract symptoms, a 24% improvement in peak urine flow, and 43% improvement in overall urine flow. The results were nearly comparable to the group taking Proscar and superior to the men taking a placebo.

On the other hand, saw palmetto does not appear to be useful in treating **prostatitis** or chronic pelvic pain syndrome (CPPS) in men. A group of researchers at Columbia University reported in early 2004 that men given saw palmetto for CP/CPPS showed no appreciable improvement at the end of a year-long trial.

Uses in women

There is very little documentation or scientific research into saw palmetto use in women. However, several studies in the 1990s show that the BPH drug Proscar can be effective in stopping unwanted facial and body hair growth, and in treating thinning hair in



Saw palmetto leaves. (Photo Researchers, Inc. Reproduced by permission.)

women. It works by blocking the action of an enzyme called 5-alpha reductase. Anecdotal reports suggest that saw palmetto may be as effective as Proscar in treating unwanted hair growth and thinning hair, and in preventing some types of **acne**. It has also been used to treat urinary tract inflammation and help relieve the symptoms of menstruation. There are claims it can be used to enlarge breasts, but these claims have not been scientifically tested.

History

Saw palmetto berries have been used in American folk medicine for several hundred years as an aphrodisiac and for treating prostate problems. Native Americans in the southeast United States have used saw palmetto since the 1700s to treat male urinary problems. In the 1800s, medical botanist John Lloyd noted that animals that ate saw palmetto appeared healthier and fatter than other livestock. Early American settlers noticed the same effects and used the juice from saw palmetto berries to gain weight, to improve general disposition, as a sedative, and to promote reproductive health.

In the United States, the medicinal uses of saw palmetto were first documented in 1879 by Dr. J. B. Read, a physician in Savannah, Georgia, who published a paper on the medicinal benefits of the herb in the April 1879 issue of *American Journal of Pharmacy*. He found the herb useful in treating a wide range of conditions. "By its peculiar soothing power on the mucous membrane it induces sleep, relieves the most troublesome coughs, promotes expectoration, improves digestion, and increases fat, flesh and strength. Its sedative and diuretic properties are remarkable," Read wrote. "Considering the great and diversified power of the saw palmetto as a therapeutic agent, it seems strange that it should have so long escaped the notice of the medical profession."

A pungent tea made from saw palmetto berries was commonly used in the early 1900s to treat prostate enlargement and urinary tract infections. It was also used in men to increase sperm production and sex drive, although these uses are discounted today. One of the first published medical recommendations that saw palmetto was effective in treating prostate problems appeared in the 1926 edition of *United States Dispensatory*. In the late 1920s, the use of medicinal

plants, including saw palmetto, began to decline in the United States, while at the same time, it was on the rise in Europe.

Preparations

The National Institute on Aging recommends that people taking saw palmetto should obtain it only from reputable sources. In addition, people should use only standardized extracts that contain 85–95% fatty acids and sterols. Dosages vary depending on the type of saw palmetto used. A typical dose is 320 mg per day of standardized extract (1–2 g) per day of ground dried whole berries. It may take up to four weeks of use before beneficial effects are seen. In late 1999, the web-based independent consumer organization ConsumerLab.com tested 27 leading brands of saw palmetto for fatty acid and sterol content. Ten of the brands contained less than the minimum recommended level of 85% fatty acids and sterols.

Precautions

There are no special precautions associated with taking saw palmetto, even in high doses. However, BPH can become a serious problem if left untreated. Men who are experiencing symptoms should be examined by a physician, since the symptoms of BPH are similar to those of prostate **cancer**. Men over the age of 50 should have a yearly prostate exam. Saw palmetto should only be used under a doctor's supervision by people with prostate cancer, **breast cancer**, or any sex hormone related diseases. Although the effects of saw palmetto on a fetus is unknown, pregnant women are advised not to take saw palmetto. Saw palmetto can alter hormonal activity that could have an adverse effect on the fetus. Women taking birth control pills or estrogen replacement products should consult a physician before taking saw palmetto. Persons taking testosterone or other anabolic steroids should not take saw palmetto without first consulting their doctor.

In rare cases, allergic reactions to saw palmetto have been reported. Symptoms include difficulty breathing, constricting of the throat, **hives**, and swelling of the lips, tongue, or face. Persons experiencing any of these symptoms should stop taking saw palmetto and seek immediate medical attention.

Side effects

The only reported minor side effects are rare and include cramps, **nausea**, **diarrhea**, and **headache**.

KEY TERMS

Anabolic steroids—A group of mostly synthetic hormones sometimes taken by athletes to temporarily increase muscle size.

Aphrodisiac—Any substance that excites sexual desire.

Estrogen—A hormone that stimulates development of female secondary sex characteristics.

Placebo—An inert or innocuous substance used in controlled experiments testing the efficacy of another substance.

Progesterone—A steroid hormone that is a biological precursor to corticoid (another steroid hormone) and androgen (a male sex hormone).

Testosterone—A male hormone produced in the testes or made synthetically that is responsible for male secondary sex characteristics.

Urethra—The canal that carries urine from the bladder.

Interactions

Saw palmetto may interfere with such hormone-related drugs as testosterone and estrogen replacements, including Premarin, Cenestin, Vivelle, Fempatch, and Climara. It may also interact with such birth control pills as Triphasil, Ovral, Lo-Ovral, Nordette, Alesse, Demulen, and Ortho-Novum. Anyone on these types of medications should consult their doctor before taking saw palmetto. There are no known restrictions on food, beverages, or physical activity while taking saw palmetto.

Several herbs and **minerals** have been used in conjunction with saw palmetto in treating BPH. A 1996 European study showed positive results in treating patients with a daily dose of 320 mg of saw palmetto extract and 240 mg of nettle root extract. Many alternative health practitioners also recommend saw palmetto be used in combination with the herb pygeum africanum, pumpkin seeds, zinc, flaxseed oil, certain amino acids, antioxidants, and **diets** high in protein and soy products. Some factors that can impair the effectiveness of saw palmetto include beer, cigarette smoke, and some chemical pesticides used on fruit and vegetables. Some physicians recommend using saw palmetto in addition to a prescription medicine, such as Proscar, Hytrin, or Cardura.

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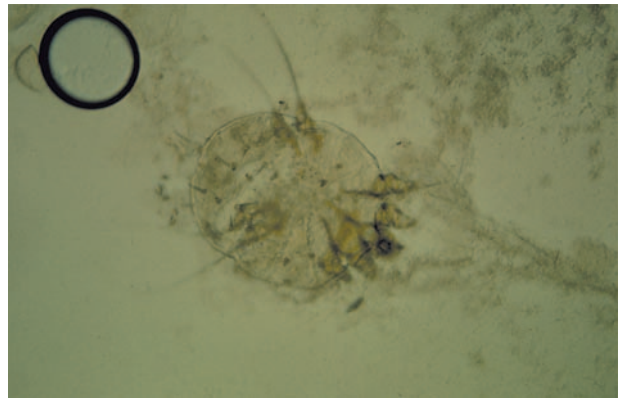
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ORGANIZATIONS

National Institute on Aging (NIA) Information Center. P. O. Box 8057, Gaithersburg, MD 20892-8057. (800) 222-2225. < <http://www.nih.gov/nia> >.

Ken R. Wells
Rebecca J. Frey, PhD



An enhanced image of a scab mite. (Custom Medical Stock Photo. Reproduced by permission.)

The intense **itching** almost always caused by scabies is due to a reaction within the skin to the feces of the mite. The first time someone is infected with scabies, he or she may not notice any itching for a number of weeks (four to six weeks). With subsequent infections, the itchiness will begin within hours of picking up the first mite.

Causes and symptoms

Scabies is most common among people who live in overcrowded conditions, and whose ability to practice good hygiene is limited. Scabies can be passed between people by close skin contact. Although the mites can only live away from human skin for about three days, sharing clothing or bedclothes can pass scabies among family members or close contacts. In May 2002, the Centers for Disease Control (CDC) included scabies in its updated guidelines for the treatment of **sexually transmitted diseases**.

The itching, or pruritus, from scabies is worse after a hot shower and at night. Burrows are seen as winding, slightly raised gray lines along the skin. The female mite may be seen at one end of the burrow, as a tiny pearl-like bump underneath the skin. Because of the intense itching, burrows may be obscured by scratch marks left by the patient. The most common locations for burrows include the sides of the fingers, between the fingers, the top of the wrists, around the elbows and armpits, around the nipples of the breasts in women, in the genitalia of men, around the waist (beltline), and on the lower part of the buttocks. Babies may have burrows on the soles of their feet, palms of their hands, and faces.

Scratching seems to serve some purpose in scabies, as the mites are apparently often inadvertently

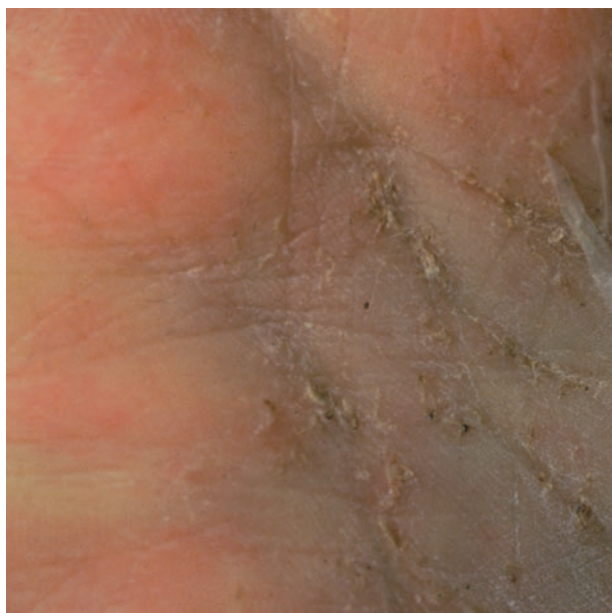
Scabies

Definition

Scabies is a relatively contagious infection caused by a tiny mite (*Sarcoptes scabiei*).

Description

Scabies is caused by a tiny insect about 0.3 mm long called a mite. When a human comes in contact with the female mite, the mite burrows under the skin, laying eggs along the line of its burrow. These eggs hatch, and the resulting offspring rise to the surface of the skin, mate, and repeat the cycle either within the skin of the original host, or within the skin of its next victim.



Scab mites have penetrated under the skin of this person's hand. (Custom Medical Stock Photo. Reproduced by permission.)

removed. Most infestations with scabies are caused by no more than 15 mites altogether.

Infestation with huge numbers of mites (on the order of thousands to millions) occurs when an individual does not scratch, or when an individual has a weakened immune system. These patients include those who live in institutions; are mentally retarded, or physically infirm; have other diseases which affect the amount of sensation they have in their skin (**leprosy** or syringomyelia); have leukemia or diabetes; are taking medications which lower their immune response (**cancer chemotherapy**, drugs given after organ transplantation); or have other diseases which lower their immune response (such as acquired **immunodeficiency** syndrome or **AIDS**). This form of scabies, with its major infestation, is referred to as crusted scabies or Norwegian scabies. Infected patients have thickened, crusty areas all over their bodies, including over the scalp. Their skin is scaly. Their fingernails may be thickened and horny.

Diagnosis

Diagnosis can be made simply by observing the characteristic burrows of the mites causing scabies. A sterilized needle can be used to explore the pearly bump at the end of a burrow, remove its contents, and place it on a slide to be examined. The mite itself may then be identified under a microscope.

Occasionally, a type of mite carried on dogs (*Sarcoptes scabiei* var. *canis*) may infect humans. These mites cannot survive for very long on humans, and so the infection is very light.

Treatment

Several types of lotions (usually containing 5% permethrin) can be applied to the body, and left on for 12–24 hours. One topical application is usually sufficient, although the scabicide may be reapplied after a week if mites remain. Preparations containing lindane are no longer recommended for treating scabies as of 2003 because of the potential for damage to the nervous system. Itching can be lessened by the use of calamine lotion or antihistamine medications.

In addition to topical medications, the doctor may prescribe oral ivermectin. Ivermectin is a drug that was originally developed for veterinary practice as a broad-spectrum antiparasite agent. Studies done in humans, however, have found that ivermectin is as safe and effective as topical medications for treating scabies. A study published in 2003 reported that ivermectin is safe for people in high-risk categories, including those with compromised immune systems.

Prognosis

The prognosis for complete recovery from scabies infestation is excellent. In patients with weak immune systems, the biggest danger is that the areas of skin involved with scabies will become secondarily infected with bacteria.

Prevention

Good hygiene is essential in the prevention of scabies. When a member of a household is diagnosed with scabies, all that person's recently-worn clothing and bedding should be washed in very hot water.

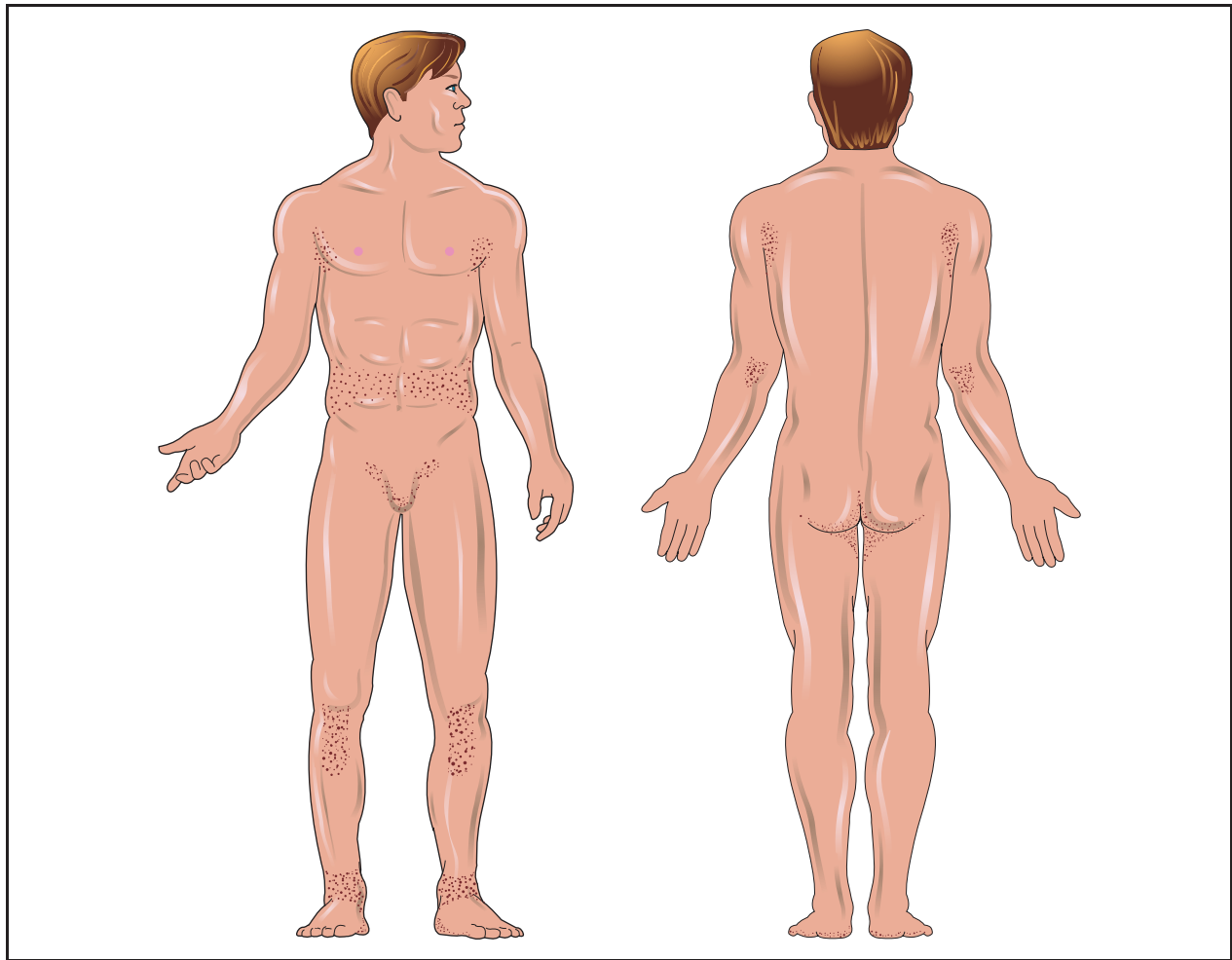
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PERIODICALS

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Scabies is a contagious skin infection common among people who live in overcrowded, less than ideal hygienic environments. It is caused by the infestation of female scab mites that, upon contact, burrows under the victim's skin and lays eggs along the lines of passage. Once the eggs hatch, the new mites rise to the skin's surface, mate, and repeats the infestation. Scabies can occur anywhere on the body, including the armpit, groin, buttocks, genital area, and ankles, as shown in the illustration above. (Illustration by Electronic Illustrators Group.)

KEY TERMS

Mite—An insect parasite belonging to the order Acarina. The organism that causes scabies is a mite.

Pruritus—An unpleasant itching sensation. Scabies is characterized by intense pruritus.

Topical—A type of medication applied to the skin or body surface.

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ORGANIZATIONS

American Academy of Dermatology (AAD). 930 East Woodfield Road, Schaumburg, IL 60173. (847) 330-0230. <<http://www.aad.org>>.

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Scarlatina see **Scarlet fever**

Scarlet fever

Definition

Scarlet **fever** is an infection that is caused by a bacteria called streptococcus. The disease is characterized by a sore throat, fever, and a sandpaper-like rash on reddened skin. It is primarily a childhood disease. If scarlet fever is untreated, serious complications such as **rheumatic fever** (a heart disease) or kidney inflammation (**glomerulonephritis**) can develop.

Description

Scarlet fever, also known as scarlatina, gets its name from the fact that the patient's skin, especially on the cheeks, is flushed. A **sore throat** and raised rash over much of the body are accompanied by fever and sluggishness (lethargy). The fever usually subsides within a few days and recovery is complete by two weeks. After the fever is gone, the skin on the face and body flakes; the skin on the palms of the hands and soles of the feet peels more dramatically.

This disease primarily affects children ages two to ten. It is highly contagious and is spread by sneezing, coughing, or direct contact. The incubation period is three to five days, with symptoms usually beginning on the second day of the disease, and lasting from four to ten days.

Early in the twentieth century, severe scarlet fever epidemics were common. Today, the disease is rare. Although this decline is due in part to the availability of **antibiotics**, that is not the entire reason since the decline began before the widespread use of antibiotics. One theory is that the strain of bacteria that causes scarlet fever has become weaker with time.

Causes and symptoms

Scarlet fever is caused by Group A streptococcal bacteria (*S. pyogenes*). Group A streptococci can be highly toxic microbes that can cause strep throat, wound or skin infections, **pneumonia**, and serious kidney infections, as well as scarlet fever. The Group A streptococci are; hemolytic bacteria, which means that the bacteria have the ability to lyse or break red blood cells. The strain of streptococcus that causes scarlet fever is slightly different from the strain that causes most strep throats. The scarlet fever strain of bacteria produces a toxin, called an erythrogenic toxin. This toxin is what causes the skin to flush.

GLADYS DICK (1881–1963)

Before 1922, not much was known about the then-endemic disease of scarlet fever, which primarily affected children in Europe and North America, killing about 25% of the children who contracted it. Additionally, scarlet fever had many complications, some of which were severe and could be crippling. Gladys Dick, with her husband, George Dick, successfully isolated the bacteria which caused scarlet fever, developed a test for human vulnerability to the disease, and devised preventive methods. The couple patented their findings, specifically the way their scarlet fever toxin and antitoxin were prepared, although this decision was controversial at the time.

In 1923, the Dicks published papers in which they proved that scarlet fever was caused by hemolytic streptococcus. Within a few years, the Dicks also published papers on how to prevent, test, diagnose, and treat scarlet fever. Their groundbreaking work ensured that the disease was finally understood and brought under control.

Dick and her husband announced the development of what came to be known as the Dick test in 1924. This skin test showed whether the patient was susceptible or immune to scarlet fever. The test involved injecting a toxin-containing substance in the arm and determining if the skin around the area became inflamed. If it did, the patient was vulnerable to scarlet fever. This test was also useful in predicting if pregnant women would develop puerperal infection during childbirth.

The main symptoms and signs of scarlet fever are fever, lethargy, sore throat, and a bumpy rash that blanches under pressure. The rash appears first on the upper chest and spreads to the neck, abdomen, legs, arms, and in folds of skin such as under the arm or groin. In scarlet fever, the skin around the mouth tends to be pale, while the cheeks are flushed. The patient usually has a “strawberry tongue,” in which inflamed bumps on the tongue rise above a bright red coating. Finally, dark red lines (called Pastia's lines) may appear in the creases of skin folds.

Diagnosis

Cases of scarlet fever are usually diagnosed and treated by pediatricians or family medicine practitioners. The chief diagnostic signs of scarlet fever are the characteristic rash, which spares the palms and soles of the feet, and the presence of a strawberry tongue in children. Strawberry tongue is rarely seen in adults.

The doctor will take note of the signs and symptoms to eliminate the possibility of other diseases. Scarlet fever can be distinguished from **measles**, a viral infection that is also associated with a fever and rash, by the quality of the rash, the presence of a sore throat in scarlet fever, and the absence of the severe eye inflammation and severe runny nose that usually accompany measles.

The doctor will also distinguish between a **strep throat**, a viral infection of the throat, and scarlet fever. With a strep infection, the throat is sore and appears beefy and red. White spots appear on the tonsils. Lymph nodes under the jawline may swell and become tender. However, none of these symptoms are specific for strep throat and may also occur with a viral infection. Other signs are more characteristic of bacterial infections. For example, inflammation of the lymph nodes in the neck is typical in strep infections, but not viral infections. On the other hand, cough, **laryngitis**, and stuffy nose tend to be associated with viral infections rather than strep infections. The main feature that distinguishes scarlet fever from a mere strep throat is the presence of the sandpaper-red rash.

Laboratory tests are needed to make a definitive diagnosis of a strep infection and to distinguish a strep throat from a viral sore throat. One test that can be performed is a blood cell count. Bacterial infections are associated with an elevated white blood cell count. In viral infections, the white blood cell count is generally below normal.

A **throat culture** can distinguish between a strep infection and a viral infection. A throat swab from the infected person is brushed over a nutrient gel (a sheep blood agar plate) and incubated overnight to detect the presence of hemolytic bacteria. In a positive culture, a clear zone will appear in the gel surrounding the bacterium, indicating that a strep infection is present.

Treatment

Although scarlet fever will often clear up spontaneously within a few days, antibiotic treatment with either oral or injectable penicillin is usually recommended to reduce the severity of symptoms, prevent complications, and prevent spread to others. Antibiotic treatment will shorten the course of the illness in small children but may not do so in adolescents or adults. Nevertheless, treatment with antibiotics is important to prevent complications.

Since penicillin injections are painful, oral penicillin may be preferable. If the patient is unable to tolerate penicillin, alternative antibiotics such as

KEY TERMS

Clindamycin—An antibiotic that can be used instead of penicillin.

Erythrogenic toxin—A toxin or agent produced by the scarlet fever-causing bacteria that causes the skin to turn red.

Erythromycin—An antibiotic that can be used instead of penicillin.

Glomerulonephritis—A serious inflammation of the kidneys that can be caused by streptococcal bacteria; a potential complication of untreated scarlet fever.

Hemolytic bacteria—Bacteria that are able to burst red blood cells.

Lethargy—The state of being sluggish.

Pastia's lines—Red lines in the folds of the skin, especially in the armpit and groin, that are characteristic of scarlet fever.

Penicillin—An antibiotic that is used to treat bacterial infections.

Procaine penicillin—An injectable form of penicillin that contains an anesthetic to reduce the pain of the injection.

Rheumatic fever—A heart disease that is a complication of a strep infection.

Sheep blood agar plate—A petri dish filled with a nutrient gel containing red blood cells that is used to detect the presence of streptococcal bacteria in a throat culture. Streptococcal bacteria will lyse or break the red blood cells, leaving a clear spot around the bacterial colony.

Strawberry tongue—A sign of scarlet fever in which the tongue appears to have a red coating with large raised bumps.

erythromycin or clindamycin may be used. However, the entire course of antibiotics, usually 10 days, will need to be followed for the therapy to be effective. Because symptoms subside quickly, there is a temptation to stop therapy prematurely. It is important to take all of the pills in order to kill the bacteria. Not completing the course of therapy increases the risk of developing rheumatic fever and kidney inflammation.

If the patient is considered too unreliable to take all of the pills or is unable to take oral medication, daily injections of procaine penicillin can be given in the hip or thigh muscle. Procaine is an anesthetic that makes the injections less painful.

Bed rest is not necessary, nor is **isolation** of the patient. Aspirin or Tylenol (acetaminophen) may be given for fever or relief of **pain**.

Prognosis

If treated promptly with antibiotics, full recovery is expected. Once a patient has had scarlet fever, they develop immunity and cannot develop it again.

Prevention

Avoiding exposure to children who have the disease will help prevent the spread of scarlet fever.

Resources

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Sally J. Jacobs, EdD

Scars

Definition

Scars are marks created during the healing of damage to the skin or tissues.

Description

A scar is a manifestation of the skin's healing process. After skin or tissue is wounded, the body releases collagen to mend the damage. Collagen, a protein, reattaches the damaged skin. As the wound heals, a temporary crust forms and covers it. The crust is a scab that protects the damaged area.

Causes of scars include cuts, sores, surgery, and **burns**. Severe **acne** and chicken pox may also scar skin. The degree that skin scars depends on more than the size and depth of the wound. Age also affects the process. The healing process is stronger in younger skin. That results in scars that are thicker than those of older people. Other factors affecting the type of scar are ethnicity, heredity, and the location of the injury.

Children are active and susceptible to cuts and injuries. They and people with fair complexions tend to get hypertrophic scars. While Asians and blacks are likely to have keloid scars, people from other ethnic groups also experience this form of scarring.

Keloid and hypertrophic scars have similar appearances. However, the keloid scar expands beyond the original wound.

The location of the wound also has an effect on its size. If the scar is located on places like the knee or shoulder, it will eventually widen because these areas are in motion.

Treatment could minimize a scar but will not erase the mark.

Causes and symptoms

Scarring is the natural process of repairing an open wound, injury, surgical incision, or other conditions like acne. Initially, a scar is red because blood vessels are created while the body forms scar tissue. The damaged area is covered by a protective scab that eventually falls off. The scar may become brown or pink. It generally fades over time and becomes less visible.

The healing process takes from one year to 18 months. Some scars heal naturally. Other scars require additional treatment.

Hypertrophic scars and keloids

Hypertrophic scars and **keloids** are caused by an over-active healing process. This produces an excessive amount of collagen at the wound site. Both types of scars are red, thick, and raised above the wound.

Hypertrophic scars do not extend beyond the wound site. The scar may itch and usually heals without professional treatment in about a year.

Keloids are large scars that could form after surgery, an injury, burn, or body piercing. This scarring often occurs on the ear lobe or chest. Sometimes keloids develop spontaneously.

The keloid is raised, rigid, and grows beyond the wound. The keloid can continue to grow. Scars are generally harmless, but may itch or feel tender. In addition, a person may feel self-conscious about the scar's appearance.

Contracture scars

Contracture scars are caused by the loss of a large section of skin due to burns or other injury. The scar contracts or tightens around the wound. This contraction could impact a person's mobility. If the scar deepens, it could affect muscles and nerves.

Acne scars

Acne scars may appear after the severe stage of acne, a skin condition usually caused by hormonal changes. The inflammatory condition is seen in adolescence, but acne can occur later in life.

Severe acne is triggered by clogged pores that cause bacteria to multiply. It occurs more frequently in adolescent boys than girls. If the acne is not treated, there could be scarring. The types of scars include pit-like pockmarks.

Diagnosis

Since visible scars could make people self-conscious, they will probably seek treatment rather than a diagnosis. Medical professionals who treat scars include dermatologists and plastic surgeons. Dermatologists are physicians who care for the skin. Their expertise includes three or more years of medical and surgical training.

Scar treatment is usually not covered by insurance. Cosmetic procedures, those done to improve a person's appearance, are considered elective surgery and are paid for by the patient. However, if scars cause a physical impairment, coverage may be issued. Examples of impairments include burn scars and keloids that restrict motion. For coverage to be approved, it is helpful for the primary care doctor to document the patient's case in writing.

Treatment

A scar is permanent and cannot be completely removed. However, treatment can alter a scar's appearance. These procedures range from the application of over-the-counter ointment to surgery. Scar treatment should start after an injury because wound care affects scarring. The wound should be cleaned and covered. Picking at the scab breaks the collagen and allows germs to enter the wound. Time also helps with healing. Scars become smaller, and the color fades.

However, additional treatment is required for some scars. While some procedures are more effective for keloids and hypertrophic scars, the procedure for acne scars is based on the type of scarring. Treatment for burn scars may include skin grafts surgery.

Surgery

The surgical procedure for scars is referred to as scar revision because the procedure modifies the scar's appearance. The cost of scar revision averaged \$1,129

in 2003, according to a membership survey of the American Academy of Facial Plastic and **Reconstructive Surgery** (AAFPRS).

This procedure works well on scars that are wide or long. Other treatments may be recommended for keloids because a surgical incision could cause a new scar and create another keloid. To reduce the risk of another scar, surgery may be followed by the injection of cortisone steroids.

Steroid Injections

Steroid injection is a singular form of treatment for scars, particularly keloid and hypertrophic scars. **Corticosteroids** are an anti-inflammatory drug that helps to lessen the scar's red color and thickness. The treatment flattens the scar and helps with **itching**. Injection costs vary and could cost \$150 per scar, according to a member of the American Academy of Dermatologists (AAD).

Cryosurgery

Cryosurgery involves the freezing of freezable tissue with a probe containing nitrous oxide. It is used to modify scars, especially keloid and hypertrophic scars. Treatment vary could cost \$175 per lesion, according to the AAD member.

Dermabrasion

Dermabrasion is the removal of a layer of the skin's surface. Scars including those caused by acne are smoothed or sanded by an instrument. The procedure costs approximately \$150 per treatment.

Silicone gel sheets

Silicone gel sheets can be purchased over-the-counter. The sheets are worn over the scar area to seal moisture. The treatment helps with itching and to reduce scar thickness and color. Cost of sheets for small **wounds** ranges from \$30 to \$50.

Alternative treatment

Alternate methods of treating scars range from applying Vitamin E to massaging the skin. People should consult with a doctor or other health care professional before starting treatment involving contact with the scarred area.

These procedures include applying obtained Vitamin E, aloe vera, or cocoa butter to the scar. Vitamin E is sold as an oil or obtained by opening a vitamin capsule. Aloe is an African plant and is sold I

capsule form and as a skin care product. Cocoa butter is a fat made from cacao seeds.

Those items are thought to help with healing so that a scar is less visible. However, time also helps to lessen the scar's appearance. Those substances should be applied only after a scar is well-healed.

Massaging mild scars is done to relax rigid scar tissue. The scar is massaged for about two minutes. Afterwards, Vitamin E oil is applied to the skin. The process should be discontinued if the area becomes sore or red.

Prognosis

The prognosis for scar treatment depends on factors including the type and severity of the scar. Keloids may return, and all scars are permanent. If treatment does not completely minimize a scar to the patient's satisfaction, the person can apply make-up to the scarred area.

Prevention

The primary way to prevent scarring is to avoid injuries. People should wear protective gear when participating in sports. Furthermore, acne should be treated before the condition reaches the severe stage.

If injured, a person should immediately treat the wound because this reduces the risk of scarring. The wound should be cleaned and covered. If stitches aren't needed, a butterfly bandage is effective at keeping the wound closed. Moreover, a balanced diet also helps with the healing process.

Picking at the scab should be avoided because this interferes with the healing process and raises the risk of scarring.

Resources

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PERIODICALS

ORGANIZATIONS

American Academy of Dermatology. 1350 I Street NW, Suite 870, Washington, DC 20005-4355. 202-842-3555. < <http://www.aad.org> >.

American Academy of Facial Plastic and Reconstructive Surgery. 310 South Henry Street, Alexandria, VA 22314. 703) 299-9291. < <http://www.aafprs.org> >.

American Society of Plastic Surgeons, Plastic Surgery Educational Foundation. 444 East Algonquin Road, Arlington Heights, IL 60005. 847-228-9900. < <http://www.plasticsurgery.org> >.

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Liz Swain

Schatzki's ring see **Lower esophageal ring**

Schistosomiasis

Definition

Schistosomiasis, also known as bilharziasis or snail **fever**, is a primarily tropical parasitic disease caused by the larvae of one or more of five types of flatworms or blood flukes known as schistosomes. The name bilharziasis comes from Theodor Bilharz, a German pathologist, who identified the worms in 1851.

Description

Infections associated with worms present some of the most universal health problems in the world. In fact, only **malaria** accounts for more diseases than schistosomiasis. The World Health Organization (WHO) estimates that 200 million people are infected and 120 million display symptoms. Another 600 million people are at risk of infection. Schistosomes are prevalent in rural and outlying city areas of 74 countries in Africa, Asia, and Latin America. In Central China and Egypt, the disease poses a major health risk.

There are five species of schistosomes that are prevalent in different areas of the world and produce somewhat different symptoms:

- *Schistosoma mansoni* is widespread in Africa, the Eastern-Mediterranean, the Caribbean, and South America and can only infect humans and rodents.
- *S. mekongi* is prevalent only in the Mekong river basin in Asia.

- *S. japonicum* is limited to China and the Philippines and can infect other mammals, in addition to humans, such as pigs, dogs, and water buffalos. As a result, it can be harder to control disease caused by this species.
- *S. intercalatum* is found in central Africa.
- *S. haematobium* occurs predominantly in Africa and the Eastern Mediterranean.

Intestinal schistosomiasis, caused by *Schistosoma japonicum*, *S. mekongi*, *S. mansoni*, and *S. intercalatum*, can lead to serious complications of the liver and spleen. Urinary schistosomiasis is caused by *S. haematobium*.

It is difficult to know how many individuals die of schistosomiasis each year because death certificates and patient records seldom identify schistosomiasis as the primary cause of death. Mortality estimates vary related to the type of schistosome infection but is generally low, for example, 2.4 of 100,000 die each year from infection with *S. mansoni*.

Causes and symptoms

All five species are contracted in the same way, through direct contact with fresh water infested with the free-living form of the parasite known as cercariae. The building of dams, irrigation systems, and reservoirs, and the movements of refugee groups introduce and spread schistosomiasis.

Eggs are excreted in human urine and feces and, in areas with poor sanitation, contaminate freshwater sources. The eggs break open to release a form of the parasite called miracidium. Freshwater snails become infested with the miracidium, which multiply inside the snail and mature into multiple cercariae that the snail ejects into the water. The cercariae, which survive outside a host for 48 hours, quickly penetrate unbroken skin, the lining of the mouth, or the gastrointestinal tract. Once inside the human body, the worms penetrate the wall of the nearest vein and travel to the liver where they grow and sexually mature. Mature male and female worms pair and migrate either to the intestines or the bladder where egg production occurs. One female worm may lay an average of 200 to 2,000 eggs per day for up to twenty years. Most eggs leave the blood stream and body through the intestines. Some of the eggs are not excreted, however, and can lodge in the tissues. It is the presence of these eggs, rather than the worms themselves, that causes the disease.

Early symptoms of infection

Many individuals do not experience symptoms. If present, it usually takes four to six weeks for symptoms

to appear. The first symptom of the disease may be a general ill feeling. Within twelve hours of infection, an individual may complain of a **tingling** sensation or light rash, commonly referred to as “swimmer’s itch,” due to irritation at the point of entrance. The rash that may develop can mimic **scabies** and other types of **rashes**. Other symptoms can occur two to ten weeks later and can include fever, aching, **cough**, **diarrhea**, or gland enlargement. These symptoms can also be related to avian schistosomiasis, which does not cause any further symptoms in humans.

Katayama fever

Another primary condition, called Katayama fever, may also develop from infection with these worms, and it can be very difficult to recognize. Symptoms include fever, lethargy, the eruption of pale temporary bumps associated with severe **itching** (urticarial) rash, liver and spleen enlargement, and bronchospasm.

Intestinal schistosomiasis

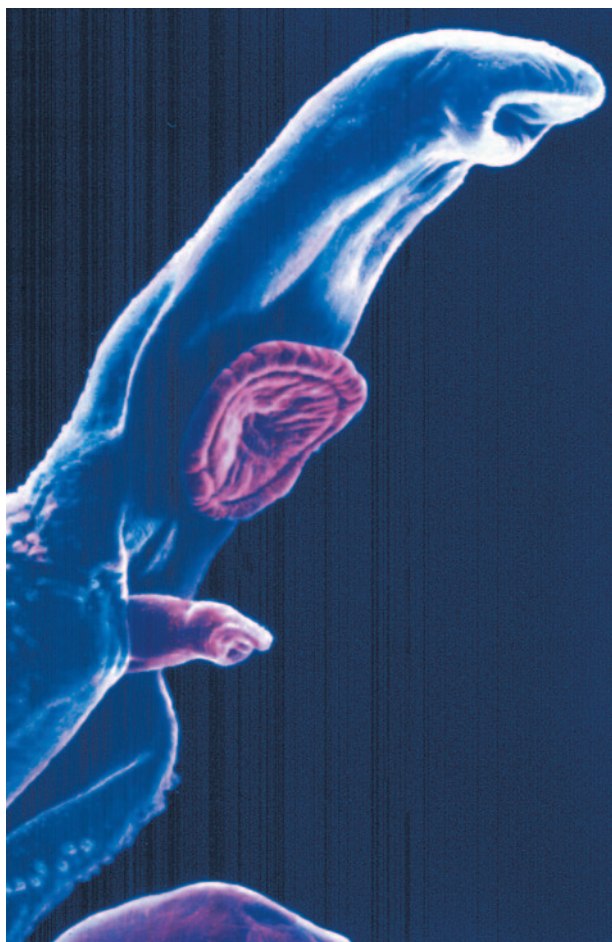
In intestinal schistosomiasis, eggs become lodged in the intestinal wall and cause an immune system reaction called a granulomatous reaction. This immune response can lead to obstruction of the colon and blood loss. The infected individual may have what appears to be a potbelly. Eggs can also become lodged in the liver, leading to high blood pressure through the liver, enlarged spleen, the build-up of fluid in the abdomen (**ascites**), and potentially life-threatening dilations or swollen areas in the esophagus or gastrointestinal tract that can tear and bleed profusely (esophageal varices). Rarely, the central nervous system may be affected. Individuals with chronic active schistosomiasis may not complain of typical symptoms.

Urinary tract schistosomiasis

Urinary tract schistosomiasis is characterized by blood in the urine, **pain** or difficulty urinating, and frequent urination and are associated with *S. haematobium*. The loss of blood can lead to **iron deficiency anemia**. A large percentage of persons, especially children, who are moderately to heavily infected experience urinary tract damage that can lead to blocking of the urinary tract and **bladder cancer**.

Diagnosis

Proper diagnosis and treatment may require a tropical disease specialist because the disease can be



A scanning electron microscopy (SEM) of the head region of the male and female adult flukes of *Schistosoma* sp. These worms cause schistosomiasis (bilharziasis) in humans. Flukes live in human blood vessels and their eggs contaminate freshwater. (Photo Researchers, Inc. Reproduced by permission.)

confused with malaria or typhoid in the early stages. The healthcare provider should do a thorough history of travel in endemic areas. The rash, if present, can mimic scabies or other rashes, and the gastrointestinal symptoms may be confused with those caused by bacterial illnesses or other intestinal parasites. These other conditions will need to be excluded before an accurate diagnosis can be made. As a result, clinical evidence of exposure to infected water along with physical findings, a negative test for malaria, and an increased number of one type of immune cell, called an eosinophil, are necessary to diagnose acute schistosomiasis.

Eggs may be detected in the feces or urine. Repeated stool tests may be required to concentrate and identify the eggs. Blood tests may be used to detect a particular antigen or particle associated with the

KEY TERMS

Ascites—The condition that occurs when the liver and kidneys are not functioning properly and a clear, straw-colored fluid is excreted by the membrane that lines the abdominal cavity (peritoneum).

Cercariae—The free-living form of the schistosome worm that has a tail, swims, and has suckers on its head for penetration into a host.

Miracidium—The form of the schistosome worm that infects freshwater snails.

schistosome that induces an immune response. Persons infected with schistosomiasis may not test positive for six months, and as a result, tests may need to be repeated to obtain an accurate diagnosis. Blood can be detected visually in the urine or with chemical strips that react to small amounts of blood.

Sophisticated imaging techniques, such as ultrasound, computed tomography scan (CT scan), and **magnetic resonance imaging** (MRI), can detect damage to the blood vessels in the liver and visualize polyps and ulcers of the urinary tract, for example, that occur in the more advanced stages. *S. haematobium* is difficult to diagnose with ultrasound in pregnant women.

Treatment

The use of medications against schistosomiasis, such as praziquantel (Biltricide), oxamniquine, and metrifonate, have been shown to be safe and effective. Praziquantel is effective against all forms of schistosomiasis and has few side effects. This drug is given in either two or three doses over the course of a single day. Oxamniquine is typically used in Africa and South America to treat intestinal schistosomiasis. Metrifonate has been found to be safe and effective in the treatment of urinary schistosomiasis. Patients are typically checked for the presence of living eggs at three and six months after treatment. If the number of eggs excreted has not significantly decreased, the patient may require another course of medication.

Prognosis

If treated early, prognosis is very good and complete recovery is expected. The illness is treatable, but people can die from the effects of untreated schistosomiasis. The severity of the disease depends on the number of worms, or worm load, in addition to how

long the person has been infected. With treatment, the number of worms can be substantially reduced, and the secondary conditions can be treated. The goal of the World Health Organization is to reduce the severity of the disease rather than to completely stop transmission of the disease. There is, however, little natural immunity to reinfection. Treated individuals do not usually require retreatment for two to five years in areas of low transmission. The World Health Organization has made research to develop a vaccine against the disease one of its priorities.

Prevention

Prevention of the disease involves several targets and requires long term community commitment. Infected patients require diagnosis, treatment, and education about how to avoid reinfecting themselves and others. Adequate healthcare facilities need to be available, water systems must be treated to kill the worms and control snail populations, and sanitation must be improved to prevent the spread of the disease.

To avoid schistosomiasis in endemic areas:

- contact the CDC for current health information on travel destinations.
- upon arrival, ask an informed local authority about the infestation of schistosomiasis before being exposed to freshwater in countries that are likely to have the disease.
- do not swim, stand, wade, or take baths in untreated water.
- treat all water used for drinking or bathing. Water can be treated by letting it stand for three days, heating it for five minutes to around 122°F (around 50°C), or filtering or treating water chemically, with chlorine or iodine, as with drinking water.
- Should accidental exposure occur, infection can be prevented by hastily drying off or applying rubbing alcohol to the exposed area.

Resources

ORGANIZATIONS

Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

Ruth E. Mawyer, RN

Schizencephaly see **Congenital brain defects**

Schizoaffective disorder

Definition

Schizoaffective disorder is a mental illness that shares the psychotic symptoms of schizophrenia and the mood disturbances of depression or bipolar disorder.

Description

The term schizoaffective disorder was first used in the 1930s to describe patients with acute psychotic symptoms such as **hallucinations** and **delusions** along with disturbed mood. These patients tended to function well before becoming psychotic; their psychotic symptoms lasted relatively briefly; and they tended to do well afterward. Over the years, however, the term schizoaffective disorder has been applied to a variety of patient groups. The current definition contained in the American Psychiatric Association's *Diagnostic and Statistical Manual of Mental Disorders IV (DSM-IV)* recognizes patients with schizoaffective disorder as those whose mood symptoms are sufficiently severe to warrant a diagnosis of depression or other full-blown mood disorder and whose mood symptoms overlap at some period with psychotic symptoms that satisfy the diagnosis of **schizophrenia** (e.g. hallucinations, delusions, or thought process disorder).

Causes and symptoms

The cause of schizoaffective disorder remains unknown and subject to continuing speculation. Some investigators believe schizoaffective disorder is associated with schizophrenia and may be caused by a similar biological predisposition. Others disagree, stressing the disorder's similarities to **mood disorders** such as depression and bipolar disorder (manic depression). They believe its more favorable course and less intense psychotic episodes are evidence that schizoaffective disorder and mood disorders share a similar cause.

Many researchers, however, believe schizoaffective disorder may owe its existence to both disorders. These researchers believe that some people have a biologic predisposition to symptoms of schizophrenia that varies along a continuum of severity. On one end of the continuum are people who are predisposed to psychotic symptoms but never display them. On the other end of the continuum are people who are destined to develop outright schizophrenia. In the middle are those who may at some time show symptoms of

schizophrenia, but require some other major trauma to set the progression of the disease into motion. It may be an early brain injury—either through a complicated delivery, prenatal exposure to the flu virus or illicit drugs; or it may be emotional, nutritional or other deprivation in early childhood. In this view, major life stresses, or a mood disorder like depression or **bipolar disorder**, may be sufficient to trigger the psychotic symptoms. In fact, patients with schizoaffective disorder frequently experience depressed mood or **mania** within days of the appearance of psychotic symptoms. Some clinicians believe that “schizomanic” patients are fundamentally different from “schizodepressed” types; the former are similar to bipolar patients, while the latter are a very heterogeneous group.

Symptoms of schizoaffective disorder vary considerably from patient to patient. Delusions, hallucinations, and evidence of disturbances in thinking—as observed in full-blown schizophrenia—may be seen. Similarly, mood fluctuations such as those observed in major depression or bipolar disorder may also be seen. These symptoms tend to appear in distinct episodes that impair the individual’s ability to function well in daily life. But between episodes, some patients with schizoaffective disorder remain chronically impaired while some may do quite well in day-to-day living.

Diagnosis

There are no accepted tissue or brain imaging tests or techniques to diagnose schizophrenia, mood disorders, or schizoaffective disorder. Instead, physicians look for the hallmark signs and symptoms of schizoaffective disorder described above, and they attempt to rule out other illnesses or conditions that may produce similar symptoms. These include:

- **Mania.** True manic patients can experience episodes of hallucinations and delusions similar to those seen in schizoaffective disorder; but these do not persist for long periods after the mania recedes, as they do in schizoaffective disorder.
- **Psychotic depression.** Patients with psychotic depression experience hallucinations and delusions similar to those seen in schizoaffective disorder; but these symptoms do not persist after the depressive symptoms recede, as they do in schizoaffective disorder.
- **Schizophrenia.** Depressed mood, mania, or other symptoms may be present in patients with schizophrenia, but patients with schizoaffective disorder will meet all the criteria set out for a full-blown mood disorder.
- **Medical and neurological disorders** that mimic psychotic/affective disorders.

Treatment

Antipsychotic medications used to treat schizophrenia and the **antidepressant drugs** and mood stabilizers used in depression and bipolar disorder are the primary treatments for schizoaffective disorder.

Unfortunately these treatments have not been well studied in controlled investigations. Studies suggest that traditional antipsychotics such as haloperidol are effective in treating psychotic symptoms. Newer generation antipsychotics, such as clozaril and risperidone, have not been as well studied, but also appear effective. For patients with symptoms of bipolar disorder, lithium is often the mood stabilizer of choice; and it is often augmented with an anticonvulsant such as valproate. For those with depressive symptoms, the evidence supporting the use of antidepressant medications in addition to antipsychotic medications is more mixed. **Electroconvulsive therapy** (electric shock) is frequently tried in patients who otherwise do not respond to antidepressant or mood stabilizing drugs.

While the mainstay of treatment for schizoaffective disorder is antipsychotic medications and mood stabilizers, certain forms of psychotherapy for both patients and family members can be useful. Therapy designed to provide structure and help augment patients’ ability to solve problems may aid in improving patients’ ability to function in the day-to-day world, reducing **stress** and the risk of recurrence. Vocational and other rehabilitative training can help patients to work on skills they need to develop. Whereas hospitalization may be necessary for acute psychotic episodes, half-way houses and day hospitals can provide needed treatment while serving as a bridge for patients to reenter the community.

Alternative treatment

While alternative therapies should never be considered a replacement for medication, these treatments can help support people with schizoaffective disorder and other mental illnesses. Dietary modifications that eliminate processed foods and emphasize whole foods, along with nutritional supplementation, may be helpful. Acupuncture, homeopathy, and botanical medicine can support many aspects of the person’s life and may help decrease the side effects of any medications prescribed.

Prognosis

In general, patients with schizoaffective disorder have a more favorable prognosis than do those with schizophrenia, but a less favorable course than those with a pure mood disorder. Medication and other

KEY TERMS

Bipolar disorder—Also referred to as manic depression, it is a mood disorder marked by alternating episodes of extremely low mood (depression) and exuberant highs (mania).

Mood disorder—A collection of disorders that includes major depression and bipolar disorder. They are all characterized by major disruptions in patients' moods and emotions.

Schizophrenia—A major mental illness marked by psychotic symptoms, including hallucinations, delusions, and severe disruptions in thinking.

interventions can help quell psychotic symptoms and stabilize mood in many patients, but there is great variability in outcome from patient to patient.

Prevention

There is no known way to prevent schizoaffective disorder. Treatment with antipsychotic and mood stabilizing drugs may prevent recurrences. Some researchers believe prompt treatment can prevent the development of full-blown schizophrenia, but this remains the subject of some disagreement.

Resources

ORGANIZATIONS

American Psychiatric Association. 1400 K Street NW, Washington DC 20005. (888) 357-7924. <<http://www.psych.org>>.

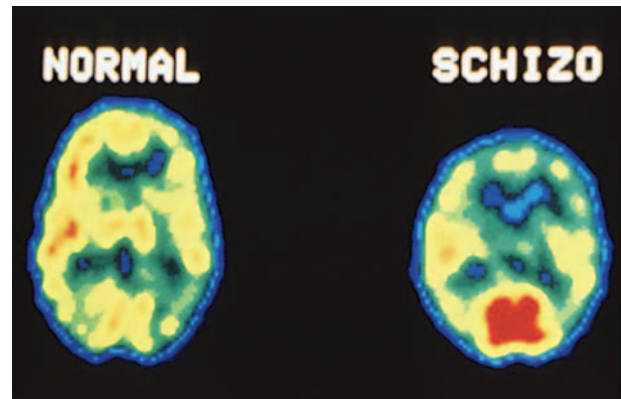
National Alliance for Research on Schizophrenia and Depression. 60 Cutter Mill Road, Suite 200, Great Neck, NY 11021. (516) 829-0091. <<http://www.mhsource.com>>.

Richard H. Camer

Schizophrenia

Definition

Schizophrenia is a psychotic disorder (or a group of disorders) marked by severely impaired thinking, emotions, and behaviors. Schizophrenic patients are typically unable to filter sensory stimuli and may have enhanced perceptions of sounds, colors, and other



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features of their environment. Most schizophrenics, if untreated, gradually withdraw from interactions with other people, and lose their ability to take care of personal needs and grooming.

The prevalence of schizophrenia is thought to be about 1% of the population around the world; it is thus more common than diabetes, **Alzheimer's disease**, or **multiple sclerosis**. In the United States and Canada, patients with schizophrenia fill about 25% of all hospital beds. The disorder is considered to be one of the top ten causes of long-term disability worldwide.

Description

The course of schizophrenia in adults can be divided into three phases or stages. In the acute phase, the patient has an overt loss of contact with reality (psychotic episode) that requires intervention and treatment. In the second or stabilization phase, the initial psychotic symptoms have been brought under control but the patient is at risk for relapse if treatment is interrupted. In the third or maintenance phase, the patient is relatively stable and can be kept indefinitely on antipsychotic medications. Even in the maintenance phase, however, relapses are not unusual and patients do not always return to full functioning.

The English term schizophrenia comes from two Greek words that mean "split mind." It was observed around 1908, by a Swiss doctor named Eugen Bleuler, to describe the splitting apart of mental functions that he regarded as the central characteristic of schizophrenia.

Recently, some psychotherapists have begun to use a classification of schizophrenia based on two main types. People with Type I, or positive schizophrenia, have a rapid (acute) onset of symptoms and

tend to respond well to drugs. They also tend to suffer more from the “positive” symptoms, such as **delusions** and **hallucinations**. People with Type II, or negative schizophrenia, are usually described as poorly adjusted before their schizophrenia slowly overtakes them. They have predominantly “negative” symptoms, such as withdrawal from others and a slowing of mental and physical reactions (psychomotor retardation).

There are five subtypes of schizophrenia:

Paranoid

The key feature of this subtype of schizophrenia is the combination of false beliefs (delusions) and hearing voices (auditory hallucinations), with more nearly normal emotions and cognitive functioning (cognitive functions include reasoning, judgment, and memory). The delusions of paranoid schizophrenics usually involve thoughts of being persecuted or harmed by others or exaggerated opinions of their own importance, but may also reflect feelings of jealousy or excessive religiosity. The delusions are typically organized into a coherent framework. Paranoid schizophrenics function at a higher level than other subtypes, but are at risk for suicidal or violent behavior under the influence of their delusions.

Disorganized

Disorganized schizophrenia (formerly called hebephrenic schizophrenia) is marked by disorganized speech, thinking, and behavior on the patient's part, coupled with flat or inappropriate emotional responses to a situation (affect). The patient may act silly or withdraw socially to an extreme extent. Most patients in this category have weak personality structures prior to their initial acute psychotic episode.

Catatonic

Catatonic schizophrenia is characterized by disturbances of movement that may include rigidity, stupor, agitation, bizarre posturing, and repetitive imitations of the movements or speech of other people. These patients are at risk for **malnutrition**, exhaustion, or self-injury. This subtype is presently uncommon in Europe and the United States. **Catatonia** as a symptom is most commonly associated with **mood disorders**.

Undifferentiated

Patients in this category have the characteristic positive and negative symptoms of schizophrenia but

do not meet the specific criteria for the paranoid, disorganized, or catatonic subtypes.

Residual

This category is used for patients who have had at least one acute schizophrenic episode but do not presently have strong positive psychotic symptoms, such as delusions and hallucinations. They may have negative symptoms, such as withdrawal from others, or mild forms of positive symptoms, which indicate that the disorder has not completely resolved.

The risk of schizophrenia among first-degree biological relatives is ten times greater than that observed in the general population. Furthermore the presence of the same disorder is higher in monozygotic twins (identical twins) than in dizygotic twins (nonidentical twins). The research concerning adoption studies and identical twins also supports the notion that environmental factors are important, because not all relatives who have the disorder express it. There are several chromosomes and loci (specific areas on chromosomes which contain mutated genes), which have been identified. Research is actively ongoing to elucidate the causes, types and variations of these mutations.

Most patients are diagnosed in their late teens or early twenties, but the symptoms of schizophrenia can emerge at any age in the life cycle. The male/female ratio in adults is about 1.2:1. Male patients typically have their first acute episode in their early twenties, while female patients are usually closer to age 30 when they are recognized with active symptoms.

Schizophrenia is rarely diagnosed in preadolescent children, although patients as young as five or six have been reported. Childhood schizophrenia is at the upper end of the spectrum of severity and shows a greater gender disparity. It affects one or two children in every 10,000; the male/female ratio is 2:1.

Causes and symptoms

Theories of causality

One of the reasons for the ongoing difficulty in classifying schizophrenic disorders is incomplete understanding of their causes. It is thought that these disorders are the end result of a combination of genetic, neurobiological, and environmental causes. A leading neurobiological hypothesis looks at the connection between the disease and excessive levels of dopamine, a chemical that transmits signals in the brain (neurotransmitter). The genetic factor in schizophrenia has been underscored by recent findings that

first-degree biological relatives of schizophrenics are ten times as likely to develop the disorder as are members of the general population.

Prior to recent findings of abnormalities in the brain structure of schizophrenic patients, several generations of psychotherapists advanced a number of psychoanalytic and sociological theories about the origins of schizophrenia. These theories ranged from hypotheses about the patient's problems with **anxiety** or aggression to theories about **stress** reactions or interactions with disturbed parents. Psychosocial factors are now thought to influence the expression or severity of schizophrenia rather than cause it directly.

As of 2004, migration is a social factor that is known to influence people's susceptibility to **psychosis**. Psychiatrists in Europe have noted the increasing rate of schizophrenia and other psychotic disorders among immigrants to almost all Western European countries. Black immigrants from Africa or the Caribbean appear to be especially vulnerable. The stresses involved in migration include family breakup, the need to adjust to living in large urban areas, and social inequalities in the new country.

Another hypothesis suggests that schizophrenia may be caused by a virus that attacks the hippocampus, a part of the brain that processes sense perceptions. Damage to the hippocampus would account for schizophrenic patients' vulnerability to sensory overload. As of 2004, researchers are focusing on the possible role of the herpes simplex virus (HSV) in schizophrenia, as well as human endogenous retroviruses (HERVs). The possibility that HERVs may be associated with schizophrenia has to do with the fact that antibodies to these retroviruses are found more frequently in the blood serum of patients with schizophrenia than in serum from control subjects.

Symptoms of schizophrenia

Patients with a possible diagnosis of schizophrenia are evaluated on the basis of a set or constellation of symptoms; there is no single symptom that is unique to schizophrenia. In 1959, the German psychiatrist Kurt Schneider proposed a list of so-called first-rank symptoms, which he regarded as diagnostic of the disorder.

These symptoms include:

- delusions
- somatic
- hallucinations
- hearing voices commenting on the patient's behavior
- thought insertion or thought withdrawal

Somatic hallucinations refer to sensations or perceptions concerning body organs that have no known medical cause or reason, such as the notion that one's brain is radioactive. Thought insertion and/or withdrawal refer to delusions that an outside force (for example, the FBI, the CIA, Martians, etc.) has the power to put thoughts into one's mind or remove them.

POSITIVE SYMPTOMS. The positive symptoms of schizophrenia are those that represent an excessive or distorted version of normal functions. Positive symptoms include Schneider's first-rank symptoms as well as disorganized thought processes (reflected mainly in speech) and disorganized or catatonic behavior. Disorganized thought processes are marked by such characteristics as looseness of associations, in which the patient rambles from topic to topic in a disconnected way; tangentially, which means that the patient gives unrelated answers to questions; and "word salad," in which the patient's speech is so incoherent that it makes no grammatical or linguistic sense. Disorganized behavior means that the patient has difficulty with any type of purposeful or goal-oriented behavior, including personal self-care or preparing meals. Other forms of disorganized behavior may include dressing in odd or inappropriate ways, sexual self-stimulation in public, or agitated shouting or cursing.

NEGATIVE SYMPTOMS. Schizophrenia includes three so-called negative symptoms. They are called negative because they represent the lack or absence of behaviors. The negative symptoms that are considered diagnostic of schizophrenia are a lack of emotional response (affective flattening), poverty of speech, and absence of volition or will. In general, the negative symptoms are more difficult for doctors to evaluate than the positive symptoms.

Diagnosis

A doctor must make a diagnosis of schizophrenia on the basis of a standardized list of outwardly observable symptoms, not on the basis of internal psychological processes. There are no specific laboratory tests that can be used to diagnose schizophrenia. Researchers have, however, discovered that patients with schizophrenia have certain abnormalities in the structure and functioning of the brain compared to normal test subjects. These discoveries have been made with the help of imaging techniques such as **computed tomography scans** (CT scans).

When a psychiatrist assesses a patient for schizophrenia, he or she will begin by excluding physical

conditions that can cause abnormal thinking and some other behaviors associated with schizophrenia. These conditions include organic brain disorders (including traumatic injuries of the brain), temporal lobe epilepsy, Wilson's disease, prion diseases, Huntington's chorea, and **encephalitis**. The doctor will also need to rule out **heavy metal poisoning** and **substance abuse** disorders, especially amphetamine use.

After ruling out organic disorders, the clinician will consider other psychiatric conditions that may include psychotic symptoms or symptoms resembling psychosis. These disorders include mood disorders with psychotic features; delusional disorder; dissociative disorder not otherwise specified (DDNOS) or **multiple personality disorder**; schizotypal, schizoid, or paranoid **personality disorders**; and atypical reactive disorders. In the past, many individuals were incorrectly diagnosed as schizophrenic. Some patients who were diagnosed prior to the changes in categorization should have their diagnoses, and treatment, reevaluated. In children, the doctor must distinguish between psychotic symptoms and a vivid fantasy life, and also identify learning problems or disorders. After other conditions have been ruled out, the patient must meet a set of criteria specified:

- the patient must have two (or more) of the following symptoms during a one-month period: delusions; hallucinations; disorganized speech; disorganized or catatonic behavior; negative symptoms
- decline in social, interpersonal, or occupational functioning, including self-care
- the disturbed behavior must last for at least six months
- mood disorders, substance **abuse** disorders, medical conditions, and developmental disorders have been ruled out

Treatment

The treatment of schizophrenia depends in part on the patient's stage or phase. Psychotic symptoms and behaviors are considered psychiatric emergencies, and persons showing signs of psychosis are frequently taken by family, friends, or the police to a hospital emergency room. A person diagnosed as psychotic can be legally hospitalized against his or her will, particularly if he or she is violent, threatening to commit **suicide**, or threatening to harm another person. A psychotic person may also be hospitalized if he or she has become malnourished or ill as a result of failure to feed, dress appropriately for the climate, or otherwise take care of him- or herself.

A patient having a first psychotic episode should be given a CT or MRI (**magnetic resonance imaging**) scan to rule out structural brain disease.

Antipsychotic medications

The primary form of treatment of schizophrenia is antipsychotic medication. **Antipsychotic drugs** help to control almost all the positive symptoms of the disorder. They have minimal effects on disorganized behavior and negative symptoms. Between 60–70% of schizophrenics will respond to antipsychotics. In the acute phase of the illness, patients are usually given medications by mouth or by intramuscular injection. After the patient has been stabilized, the antipsychotic drug may be given in a long-acting form called a depot dose. Depot medications last for two to four weeks; they have the advantage of protecting the patient against the consequences of forgetting or skipping daily doses. In addition, some patients who do not respond to oral neuroleptics have better results with depot form. Patients whose long-term treatment includes depot medications are introduced to the depot form gradually during their stabilization period. Most people with schizophrenia are kept indefinitely on antipsychotic medications during the maintenance phase of their disorder to minimize the possibility of relapse.

As of the early 2000s, the most frequently used antipsychotics fall into two classes: the older dopamine receptor antagonists, or DAs, and the newer serotonin dopamine antagonists, or SDAs. (Antagonists block the action of some other substance; for example, dopamine antagonists counteract the action of dopamine.) The exact mechanisms of action of these medications are not known, but it is thought that they lower the patient's sensitivity to sensory stimuli and so indirectly improve the patient's ability to interact with others.

DOPAMINE RECEPTOR ANTAGONIST. The dopamine antagonists include the older antipsychotic (also called neuroleptic) drugs, such as haloperidol (Haldol), chlorpromazine (Thorazine), and fluphenazine (Prolixin). These drugs have two major drawbacks: it is often difficult to find the best dosage level for the individual patient, and a dosage level high enough to control psychotic symptoms frequently produces extrapyramidal side effects, or EPS. EPSs include parkinsonism, in which the patient cannot walk normally and usually develops a tremor; dystonia, or painful **muscle spasms** of the head, tongue, or neck; and akathisia, or restlessness. A type of long-term EPS is called **tardive dyskinesia**, which features slow, rhythmic, automatic movements.

Schizophrenics with **AIDS** are especially vulnerable to developing EPS.

SEROTONIN DOPAMINE ANTAGONISTS. The serotonin dopamine antagonists, also called atypical antipsychotics, are newer medications that include clozapine (Clozaril), risperidone (Risperdal), and olanzapine (Zyprexa). The SDAs have a better effect on the negative symptoms of schizophrenia than do the older drugs and are less likely to produce EPS than the older compounds. The newer drugs are significantly more expensive in the short term, although the SDAs may reduce long-term costs by reducing the need for hospitalization. They are also presently unavailable in injectable forms. The SDAs are commonly used to treat patients who respond poorly to the DAs. However, many psychotherapists now regard the use of these atypical antipsychotics as the treatment of first choice; in particular, clozapine appears to be more effective than other antipsychotics in controlling persistent aggression in some patients.

NEWER DRUGS. Some newer antipsychotic drugs have been approved by the Food and Drug Administration (FDA) in the early 2000s. These drugs are sometimes called second-generation antipsychotics or SGAs. Aripiprazole (Abilify), which is classified as a partial dopaminergic agonist, received FDA approval in August 2003. Two drugs that are still under investigation, a neurokinin antagonist and a serotonin 2A/2C antagonist respectively, show promise in the treatment of schizophrenia and **schizoaffective disorder**.

Psychotherapy

Most schizophrenics can benefit from psychotherapy once their acute symptoms have been brought under control by antipsychotic medication. Psychoanalytic approaches are not recommended. Behavior therapy, however, is often helpful in assisting patients to acquire skills for daily living and social interaction. It can be combined with occupational therapy to prepare the patient for eventual employment.

Family therapy

Family therapy is often recommended for the families of schizophrenic patients, to relieve the feelings of guilt that they often have as well as to help them understand the patient's disorder. The family's attitude and behaviors toward the patient are key factors in minimizing relapses (for example, by reducing stress in the patient's life), and family therapy can often strengthen the family's ability to cope with the stresses caused by the schizophrenic's illness. Family therapy focused on communication skills and problem-solving

strategies is particularly helpful. In addition to formal treatment, many families benefit from support groups and similar mutual help organizations for relatives of schizophrenics.

Prognosis

One important prognostic sign is the patient's age at onset of psychotic symptoms. Patients with early onset of schizophrenia are more often male, have a lower level of functioning prior to onset, a higher rate of brain abnormalities, more noticeable negative symptoms, and worse outcomes. Patients with later onset are more likely to be female, with fewer brain abnormalities and thought impairment, and more hopeful prognoses.

The average course and outcome for schizophrenics are less favorable than those for most other mental disorders, although as many as 30% of patients diagnosed with schizophrenia recover completely and the majority experience some improvement. Two factors that influence outcomes are stressful life events and a hostile or emotionally intense family environment. Schizophrenics with a high number of stressful changes in their lives, or who have frequent contacts with critical or emotionally over-involved family members, are more likely to relapse. Overall, the most important component of long-term care of schizophrenic patients is complying with their regimen of antipsychotic medications.

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KEY TERMS

Affective flattening—A loss or lack of emotional expressiveness. It is sometimes called blunted or restricted affect.

Akathisia—Agitated or restless movement, usually affecting the legs and accompanied by a sense of discomfort. It is a common side effect of neuroleptic medications.

Catatonic behavior—Behavior characterized by muscular tightness or rigidity and lack of response to the environment. In some patients rigidity alternates with excited or hyperactive behavior.

Delusion—A fixed, false belief that is resistant to reason or factual disproof.

Depot dosage—A form of medication that can be stored in the patient's body tissues for several days or weeks, thus minimizing the risk of the patient forgetting daily doses. Haloperidol and fluphenazine can be given in depot form.

Dopamine receptor antagonists (DAs)—The older class of antipsychotic medications, also called neuroleptics. These primarily block the site on nerve cells that normally receive the brain chemical dopamine.

Dystonia—Painful involuntary muscle cramps or spasms.

Extrapyramidal symptoms (EPS)—A group of side effects associated with antipsychotic medications. EPS include parkinsonism, akathisia, dystonia, and tardive dyskinesia.

First-rank symptoms—A set of symptoms designated by Kurt Schneider in 1959 as the most important diagnostic indicators of schizophrenia. These symptoms include delusions, hallucinations, thought insertion or removal, and thought broadcasting. First-rank symptoms are sometimes referred to as Schneiderian symptoms.

Hallucination—A sensory experience of something that does not exist outside the mind. A person can experience a hallucination in any of the five senses. Auditory hallucinations are a common symptom of schizophrenia.

Huntington's chorea—A hereditary disease that typically appears in midlife, marked by gradual loss of brain function and voluntary movement. Some of its symptoms resemble those of schizophrenia.

Negative symptoms—Symptoms of schizophrenia characterized by the absence or elimination of certain behaviors. DSM-IV specifies three negative symptoms: affective flattening, poverty of speech, and loss of will or initiative.

Neuroleptic—Another name for the older type of antipsychotic medications given to schizophrenic patients.

Parkinsonism—A set of symptoms originally associated with Parkinson disease that can occur as side effects of neuroleptic medications. The symptoms include trembling of the fingers or hands, a shuffling gait, and tight or rigid muscles.

Positive symptoms—Symptoms of schizophrenia that are characterized by the production or presence of behaviors that are grossly abnormal or excessive, including hallucinations and thought-process disorder. DSM-IV subdivides positive symptoms into psychotic and disorganized.

Poverty of speech—A negative symptom of schizophrenia, characterized by brief and empty replies to questions. It should not be confused with shyness or reluctance to talk.

Psychotic disorder—A mental disorder characterized by delusions, hallucinations, or other symptoms of lack of contact with reality. The schizophrenias are psychotic disorders.

Serotonin dopamine antagonist (SDA)—The newer second-generation antipsychotic drugs, also called atypical antipsychotics. SDAs include clozapine (Clozaril), risperidone (Risperdal), and olanzapine (Zyprexa).

Wilson disease—A rare hereditary disease marked by high levels of copper deposits in the brain and liver. It can cause psychiatric symptoms resembling schizophrenia.

Word salad—Speech that is so disorganized that it makes no linguistic or grammatical sense.

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ORGANIZATIONS

- American Psychiatric Association. 1400 K Street NW, Washington DC 20005. (888) 357-7924.
< <http://www.psych.org> >.
- National Alliance for the Mentally Ill (NAMI). Colonial Place Three, 2107 Wilson Blvd., Suite 300 Arlington, VA 22201. (703) 524-7600 HelpLine: (800) 950-NAMI.
< <http://www.nami.org/> >.
- National Institute of Mental Health (NIMH). 6001 Executive Boulevard, Room 8184, MSC 9663, Bethesda, MD 20892-9663. (301) 443-4513.
< <http://www.nimh.nih.gov> >.
- Schizophrenics Anonymous. 15920 W. Twelve Mile, Southfield, MI 48076. (248) 477-1983.
- United States Food and Drug Administration (FDA). 5600 Fishers Lane, Rockville, MD 20857-0001. (888) INFO-FDA. < <http://www.fda.gov> >.

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Schwannoma see **Brain tumor**

Sciatic nerve pain see **Sciatica**

Sciatica

Definition

Sciatica refers to **pain** or discomfort associated with the sciatic nerve. This nerve runs from the lower part of the spinal cord, down the back of the leg, to the foot. Injury to or pressure on the sciatic nerve can cause the characteristic pain of sciatica: a sharp or burning pain that radiates from the lower back or hip, possibly following the path of the sciatic nerve to the foot.

Description

The sciatic nerve is the largest and longest nerve in the body. About the thickness of a person's thumb, it spans from the lower back to the foot. The nerve

originates in the lower part of the spinal cord, the so-called lumbar region. As it branches off from the spinal cord, it passes between the bony vertebrae (the component bones of the spine) and runs through the pelvic girdle, or hip bones. The nerve passes through the hip joint and continues down the back of the leg to the foot.

Sciatica is a fairly common disorder and approximately 40% of the population experiences it at some point in their lives. However, only about 1% have coexisting sensory or motor deficits. Sciatic pain has several root causes and treatment may hinge upon the underlying problem.

Of the identifiable causes of sciatic pain, lumbosacral radiculopathy and back strain are the most frequently suspected. The term lumbosacral refers to the lower part of the spine, and radiculopathy describes a problem with the spinal nerve roots that pass between the vertebrae and give rise to the sciatic nerve. This area between the vertebrae is cushioned with a disk of shock-absorbing tissue. If this disk shifts or is damaged through injury or disease, the spinal nerve root may be compressed by the shifted tissue or the vertebrae.

This compression of the nerve roots sends a pain signal to the brain. Although the actual injury is to the nerve roots, the pain may be perceived as coming from anywhere along the sciatic nerve.

The sciatic nerve can be compressed in other ways. Back strain may cause **muscle spasms** in the lower back, placing pressure on the sciatic nerve. In rare cases, infection, **cancer**, bone inflammation, or other diseases may be causing the pressure. More likely, but often overlooked, is the piriformis syndrome. As the sciatic nerve passes through the hip joint, it shares the space with several muscles. One of these muscles, the piriformis muscle, is closely associated with the sciatic nerve. In some people, the nerve actually runs through the muscle. If this muscle is injured or has a spasm, it places pressure on the sciatic nerve, in effect, compressing it.

In many sciatica cases, the specific cause is never identified. About half of affected individuals recover from an episode within a month. Some cases can linger a few weeks longer and may require aggressive treatment. In some cases, the pain may return or potentially become chronic.

Causes and symptoms

Individuals with sciatica may experience some lower back pain, but the most common symptom is

pain that radiates through one buttock and down the back of that leg. The most identified cause of the pain is compression or pressure on the sciatic nerve. The extent of the pain varies between individuals. Some people describe pain that centers in the area of the hip, and others perceive discomfort all the way to the foot. The quality of the pain also varies; it may be described as **tingling**, burning, prickly, aching, or stabbing.

Onset of sciatica can be sudden, but it can also develop gradually. The pain may be intermittent or continuous, and certain activities, such as bending, coughing, sneezing, or sitting, may make the pain worse.

Chronic pain may arise from more than just compression on the nerve. According to some pain researchers, physical damage to a nerve is only half of the equation. A developing theory proposes that some nerve injuries result in a release of neurotransmitters and immune system chemicals that enhance and sustain a pain message. Even after the injury has healed, or the damage has been repaired, the pain continues. Control of this abnormal type of pain is difficult.

Diagnosis

Before treating sciatic pain, as much information as possible is collected. The individual is asked to recount the location and nature of the pain, how long it has continued, and any accidents or unusual activities prior to its onset. This information provides clues that may point to back strain or injury to a specific location. Back pain from disk disease, piriformis syndrome, and back strain must be differentiated from more serious conditions such as cancer or infection. Lumbar stenosis, an overgrowth of the covering layers of the vertebrae that narrows the spinal canal, must also be considered. The possibility that a difference in leg lengths is causing the pain should be evaluated; the problem can be easily be treated with a foot orthotic or built-up shoe.

Often, a straight-leg-raising test is done, in which the person lies face upward and the health-care provider raises the affected leg to various heights. This test pinpoints the location of the pain and may reveal whether it is caused by a disk problem. Other tests, such as having the individual rotate the hip joint, assess the hip muscles. Any pain caused by these movements may provide information about involvement of the piriformis muscle, and piriformis weakness is tested with additional leg-strength maneuvers.

Further tests may be done depending on the results of the **physical examination** and initial pain treatment. Such tests might include **magnetic resonance imaging**

(MRI) and **computed tomography scans** (CT scans). Other tests examine the conduction of electricity through nerve tissues, and include studies of the electrical activity generated as muscles contract (**electromyography**), nerve conduction velocity, and evoked potential testing. A more invasive test involves injecting a contrast substance into the space between the vertebrae and making x-ray images of the spinal cord (**myelography**), but this procedure is usually done only if surgery is being considered. All of these tests can reveal problems with the vertebrae, the disk, or the nerve itself.

Treatment

Initial treatment for sciatica focuses on pain relief. For acute or very painful flare-ups, bed rest is advised for up to a week in conjunction with medication for the pain. Pain medication includes **acetaminophen**, **nonsteroidal anti-inflammatory drugs** (NSAIDs), such as **aspirin**, or **muscle relaxants**. If the pain is unremitting, opioids may be prescribed for short-term use or a local anesthetic will be injected directly into the lower back. Massage and heat application may be suggested as adjuncts.

If the pain is chronic, different pain relief medications are used to avoid long-term dosing of NSAIDs, muscle relaxants, and opioids. **Antidepressant drugs**, which have been shown to be effective in treating pain, may be prescribed alongside short-term use of muscle relaxants or NSAIDs. Local anesthetic injections or epidural steroids are used in selected cases.

As the pain allows, physical therapy is introduced into the treatment regime. Stretching exercises that focus on the lower back, buttock, and hamstring muscles are suggested. The exercises also include finding comfortable, pain-reducing positions. Corsets and braces may be useful in some cases, but evidence for their general effectiveness is lacking. However, they may be helpful to prevent exacerbations related to certain activities.

With less pain and the success of early therapy, the individual is encouraged to follow a long-term program to maintain a healthy back and prevent re-injury. A physical therapist may suggest exercises and regular activity, such as water **exercise** or walking. Patients are instructed in proper body mechanics to minimize symptoms during light lifting or other activities.

If the pain is chronic and conservative treatment fails, surgery to repair a **herniated disk** or cut out part or all of the piriformis muscle may be suggested, particularly if there is neurologic evidence of nerve or nerve-root damage.

KEY TERMS

Disk—Dense tissue between the vertebrae that acts as a shock absorber and prevents damage to nerves and blood vessels along the spine.

Electromyography—A medical test in which a nerve's ability to conduct an impulse is measured.

Lumbosacral—Referring to the lower part of the backbone or spine.

Myelography—A medical test in which a special dye is injected into a nerve to make it visible on an x ray.

Piriformis—A muscle in the pelvic girdle that is closely associated with the sciatic nerve.

Radiculopathy—A condition in which the spinal nerve root of a nerve has been injured or damaged.

Spasm—Involuntary contraction of a muscle.

Vertebrae—The component bones of the spine.

Alternative treatment

Massage is a recommended form of therapy, especially if the sciatic pain arises from muscle spasm. Symptoms may also be relieved by icing the painful area as soon as the pain occurs. Ice should be left on the area for 30-60 minutes several times a day. After 2-3 days, a hot water bottle or heating pad can replace the ice. **Chiropractic** or **osteopathy** may offer possible solutions for relieving pressure on the sciatic nerve and the accompanying pain. **Acupuncture** and **biofeedback** may also be useful as pain control methods. Body work, such as the **Alexander technique**, can assist an individual in improving posture and preventing further episodes of sciatic pain.

Prognosis

Most cases of sciatica are treatable with pain medication and physical therapy. After 4-6 weeks of treatment, an individual should be able to resume normal activities.

Prevention

Some sources of sciatica are not preventable, such as disk degeneration, back strain due to **pregnancy**, or accidental falls. Other sources of back strain, such as poor posture, overexertion, being overweight, or wearing high heels, can be corrected or avoided. Cigarette **smoking** may also predispose people to pain, and should be discontinued.

General suggestions for avoiding sciatica, or preventing a repeat episode, include sleeping on a firm mattress, using chairs with firm back support, and sitting with both feet flat on the floor. Habitually crossing the legs while sitting can place excess pressure on the sciatic nerve. Sitting a lot can also place pressure on the sciatic nerves, so it's a good idea to take short breaks and move around during the work day, long trips, or any other situation that requires sitting for an extended length of time. If lifting is required, the back should be kept straight and the legs should provide the lift. Regular exercise, such as swimming and walking, can strengthen back muscles and improve posture. Exercise can also help maintain a healthy weight and lessen the likelihood of back strain.

Resources

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Julia Barrett

SCID see **Severe combined immunodeficiency**

Scleral buckling see **Retinal detachment**

Scleroderma

Definition

Scleroderma is a progressive disease that affects the skin and connective tissue (including cartilage, bone, fat, and the tissue that supports the nerves and blood vessels throughout the body). There are two major forms of the disorder. The type known as localized scleroderma mainly affects the skin. Systemic scleroderma, which is also called systemic sclerosis, affects the smaller blood vessels and internal organs of the body.

Description

Scleroderma is an autoimmune disorder, which means that the body's immune system turns against itself. In scleroderma, there is an overproduction of abnormal collagen (a type of protein fiber present in connective tissue). This collagen accumulates throughout the body, causing hardening (sclerosis), scarring (fibrosis), and other damage. The damage may affect the appearance of the skin, or it may involve only the internal organs. The symptoms and severity of scleroderma vary from person to person.

Scleroderma occurs in all races of people all over the world, but it affects about four females for every male. Among children, localized scleroderma is more common, and systemic sclerosis is comparatively rare. Most patients with systemic sclerosis are diagnosed between ages 30 and 50. In the United States, about 300,000 people have scleroderma. Young African-American women and Native Americans of the Choctaw tribe have especially high rates of the disease. In 2003, researchers reported that they had identified 12 different genetic markers associated with scleroderma in the Choctaw population.

Causes and symptoms

The cause of scleroderma is still a puzzle. Although the accumulation of collagen appears to be a hallmark of the disease, researchers do not know why it occurs. Some theories suggest that damage to blood vessels may cause the tissues of the body to receive an inadequate amount of oxygen—a condition called **ischemia**. Some researchers believe that the resulting damage causes the immune system to overreact, producing an autoimmune disorder. According to this theory of scleroderma, the immune system gears up to fight an invader, but no invader is actually present. Cells in the immune system called antibodies react to the body's own tissues as if they were foreign. The antibodies turn against the already damaged blood vessels and the vessels' supporting tissues. These immune cells are designed to deliver potent chemicals in order to kill foreign invaders. Some of these cells dump these chemicals on the body's own tissues instead, causing inflammation, swelling, damage, and scarring.

Most cases of scleroderma have no recognizable triggering event. Some cases, however, have been traced to exposure to toxic (poisonous) substances. For example, coal miners and gold miners, who are exposed to high levels of silica dust, have above-average rates of scleroderma. Other chemicals associated with the disease include polyvinyl chloride, benzene, toluene, and epoxy resins. In 1981, 20,000 people in Spain were stricken with a syndrome similar to scleroderma when their cooking oil was accidentally contaminated. Certain medications, especially a drug used in **cancer** treatment called bleomycin (Blenoxane), may lead to scleroderma. Some claims of a scleroderma-like illness have been made by women with silicone **breast implants**, but a link has not been proven in numerous studies.

Symptoms of systemic scleroderma

A condition called Raynaud's phenomenon is the first symptom in about 95% of all patients with

systemic scleroderma. In Raynaud's phenomenon, the blood vessels of the fingers and/or toes (the digits) react to cold in an abnormal way. The vessels clamp down, preventing blood flow to the tip of the digit. Eventually, the flow is cut off to the entire finger or toe. Over time, oxygen deprivation may result in open ulcers on the skin surface. These ulcers can lead to tissue death (**gangrene**) and loss of the digit. When Raynaud's phenomenon is the first sign of scleroderma, the next symptoms usually appear within two years.

SKIN AND EXTREMITIES. Involvement of the skin leads to swelling underneath the skin of the hands, feet, legs, arms, and face. Swelling is followed by thickening and tightening of the skin, which becomes taut and shiny. Severe tightening may lead to abnormalities. For example, tightening of the skin on the hands may cause the fingers to become permanently curled (flexed). Structures within the skin are damaged (including those producing hair, oil, and sweat), and the skin becomes dry and scaly. Ulcers may form, with the danger of infection. Calcium deposits often appear under the skin.

In systemic scleroderma, the mouth and nose may become smaller as the skin on the face tightens. The small mouth may interfere with eating and dental hygiene. Blood vessels under the skin may become enlarged and show through the skin, appearing as purplish marks or red spots. This chronic dilation of the small blood vessels is called telangiectasis.

Muscle weakness, joint **pain** and stiffness, and **carpal tunnel syndrome** are common in scleroderma. Carpal tunnel syndrome involves scarring in the wrist, which puts pressure on the median nerve running through that area. Pressure on the nerve causes **numbness**, **tingling**, and weakness in some of the fingers.

DIGESTIVE TRACT. The tube leading from the mouth to the stomach (the esophagus) becomes stiff and scarred. Patients may have trouble swallowing food. The acid contents of the stomach may start to flow backward into the esophagus (esophageal reflux), causing a very uncomfortable condition known as **heartburn**. The esophagus may also become inflamed.

The intestine becomes sluggish in processing food, causing bloating and pain. Foods are not digested properly, resulting in **diarrhea**, weight loss, and anemia. Telangiectasis in the stomach or intestine may cause rupture and bleeding.

RESPIRATORY AND CIRCULATORY SYSTEMS. The lungs are affected in about 66% of all people with systemic scleroderma. Complications include **shortness of breath**, coughing, difficulty breathing due to tightening of the tissue around the chest, inflammation of the air sacs in the lungs (alveolitis), increased risk of



Scleroderma is a serious, progressive disease caused by the overproduction and accumulation of collagen throughout the body, resulting in hardening (sclerosis) and scarring (fibrosis) of the skin and connective tissue. (Photo Researchers, Inc. Reproduced by permission.)

pneumonia, and an increased risk of cancer. For these reasons, lung disease is the most likely cause of **death** associated with scleroderma.

The lining around the heart (pericardium) may become inflamed. The heart may have greater difficulty pumping blood effectively (heart failure). Irregular heart rhythms and enlargement of the heart also occur in scleroderma.

Kidney disease is another common complication. Damage to blood vessels in the kidneys often causes a major rise in the person's blood pressure. The blood pressure may be so high that there is swelling of the brain, causing severe headaches, damage to the retinas of the eyes, seizures, and failure of the heart to pump blood into the body's circulatory system. The kidneys may also stop filtering blood and go into failure. Treatments for high blood pressure have greatly improved these kidney complications. Before these treatments were available, kidney problems were the most common cause of death for people with scleroderma.

Other problems associated with scleroderma include painful dryness of the eyes and mouth, enlargement and destruction of the liver, and a low-functioning thyroid gland.

Diagnosis

Diagnosis of scleroderma is complicated by the fact that some of its symptoms can accompany other connective-tissue diseases. The most important symptom is thickened or hardened skin on the fingers, hands, forearms, or face. This symptom is found in 98% of people with scleroderma. It can be detected in the

course of a physical examination. The person's medical history may also contain important clues, such as exposure to toxic substances on the job. There are a number of nonspecific laboratory tests on blood samples that may indicate the presence of an inflammatory disorder (but not specifically scleroderma). The antinuclear antibody (ANA) test is positive in more than 95% of people with scleroderma.

Other tests can be performed to evaluate the extent of the disease. These include a test of the electrical system of the heart (an electrocardiogram), lung-function tests, and x-ray studies of the gastrointestinal tract. Various blood tests can be given to study kidney function.

Treatment

Mainstream treatments

As of early 2004 there is no cure for scleroderma. A drug called D-penicillamine has been used to interfere with the abnormal collagen. It is believed to help decrease the degree of skin thickening and tightening, and to slow the progress of the disease in other organs. Taking vitamin D and using ultraviolet light may be helpful for localized scleroderma. One group of British researchers reported in 2003 that long-wavelength ultraviolet A light is particularly effective in treating localized scleroderma. **Corticosteroids** have been used to treat joint pain, **muscle cramps**, and other symptoms of inflammation. Other drugs have been studied that reduce the activity of the immune system (immunosuppressants). Because these medications can have serious side effects, they are used in only the most severe cases of scleroderma.

The various complications of scleroderma are treated individually. Raynaud's phenomenon requires that people try to keep their hands and feet warm constantly. Nifedipine is a medication that is sometimes given to help control Raynaud's. Thick ointments and creams are used to treat dry skin. **Exercise** and massage may help joint involvement; they may also help people retain more movement despite skin tightening. An exercise regimen for stretching the mouth opening has been reported to be a helpful alternative to surgery in managing this condition. Skin ulcers need prompt attention and may require **antibiotics**. People with esophageal reflux will be advised to eat small amounts more often, rather than several large meals a day. They should also avoid spicy foods and items containing **caffeine**. Some patients with esophageal reflux have been successfully treated with surgery. Acid-reducing medications may be given

for heartburn. People must be monitored for the development of high blood pressure. If found, they should be promptly treated with appropriate medications, usually ACE inhibitors or other **vasodilators**. When fluid accumulates due to heart failure, diuretics can be given to get rid of the excess fluid.

Patients with scleroderma may also benefit from some form of counseling or psychotherapy, as they are at increased risk of depression. One study found that 46% of the patients in its sample met the criteria for a depressive disorder.

Alternative treatments

One alternative therapy that some naturopaths have used in treating patients with scleroderma is superoxide dismutase (SOD), an antioxidant enzyme used in its injectable form. More research, however, needs to be done on the benefits of this treatment.

Prognosis

The prognosis for people with scleroderma varies. Some have a very limited form of the disease called morphea, which affects only the skin. These individuals have a very good prognosis. Other people have a subtype of systemic scleroderma called limited scleroderma. For them, the prognosis is relatively good. Limited scleroderma is characterized by limited involvement of the patient's skin and a cluster of five symptoms called the CREST syndrome. CREST stands for:

- C = Calcinosis
- R = Raynaud's disease (phenomenon)
- E = Esophageal dysmotility (stiffness and malfunctioning of the esophagus)
- S = Sclerodactyly (thick, hard, rigid skin over the fingers)
- T = Telangiectasias

In general, people with very widespread skin involvement have the worst prognosis. This level of disease is usually accompanied by involvement of other organs and the most severe complications. Although women are more commonly stricken with scleroderma, men more often die of the disease. The two factors that negatively affect survival are male sex and older age at diagnosis. The most common causes of death include heart, kidney, and lung diseases. About 65% of all patients survive 11 years or more following a diagnosis of scleroderma.

As of early 2004 there are no known ways to prevent scleroderma. People can try to decrease occupational exposure to high-risk substances.

KEY TERMS

Autoimmune disorder—A disorder in which the body's immune cells mistake the body's own tissues as foreign invaders; the immune cells then work to destroy tissues in the body.

Collagen—The main supportive protein of cartilage, connective tissue, tendon, skin, and bone.

Connective tissue—A group of tissues responsible for support throughout the body; includes cartilage, bone, fat, tissue underlying skin, and tissues that support organs, blood vessels, and nerves throughout the body.

Fibrosis—The abnormal development of fibrous tissue; scarring.

Limited scleroderma—A subtype of systemic scleroderma with limited skin involvement. It is sometimes called the CREST form of scleroderma, after the initials of its five major symptoms.

Localized scleroderma—Thickening of the skin from overproduction of collagen.

Morphea—The most common form of localized scleroderma.

Raynaud phenomenon/Raynaud disease—A condition in which blood flow to the body's tissues is reduced by a malfunction of the nerves that regulate the constriction of blood vessels. When attacks of Raynaud's occur in the absence of other medical conditions, it is called Raynaud disease. When attacks occur as part of a disease (as in scleroderma), it is called Raynaud phenomenon.

Sclerosis—Hardening.

Systemic sclerosis—A rare disorder that causes thickening and scarring of multiple organ systems.

Telangiectasias—Very small arteriovenous malformations, or connections between the arteries and veins. The result is small red spots on the skin known as "spider veins."

Resources

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ORGANIZATIONS

- American College of Rheumatology. 60 Executive Park South, Suite 150, Atlanta, GA 30329. (404) 633-3777. <<http://www.rheumatology.org>>.
- National Organization for Rare Disorders, Inc. (NORD). 55 Kenosia Avenue, P. O. Box 1968, Danbury, CT 06813. (800) 999-6673 or (203) 744-0100. <<http://www.rarediseases.org>>.
- Scleroderma Foundation. 12 Kent Way, Suite 101, Byfield, MA 01922. (978) 463-5843 or (800) 722-HOPE. Fax: (978) 463-5809. <<http://www.scleroderma.org>>.

Rebecca J. Frey, PhD

Sclerotherapy for esophageal varices

Definition

Sclerotherapy for esophageal varices (also called endoscopic sclerotherapy) is a treatment for esophageal bleeding that involves the use of an endoscope and the injection of a sclerosing solution into veins.

Purpose

In most hospitals, sclerotherapy for esophageal varices is the treatment of choice to stop esophageal bleeding during acute episodes, and to prevent further incidences of bleeding. Emergency sclerotherapy is often followed by preventive treatments to eradicate distended esophageal veins.

Precautions

Sclerotherapy for esophageal varices cannot be performed on an uncooperative patient, since movement during the procedure could cause the vein to tear or the esophagus to perforate and bleed. It should not be performed on a patient with a perforated gastrointestinal tract.

Description

Esophageal varices are enlarged or swollen veins on the lining of the esophagus which are prone to bleeding. They are life-threatening, and can be fatal in up to 50% of patients. They usually appear in patients with severe **liver disease**. Sclerotherapy for esophageal varices involves injecting a strong and irritating solution (a sclerosant) into the veins and/or the area beside the distended vein. The sclerosant injected into the vein causes **blood clots** to form and stops the bleeding. The sclerosant injected into the area beside the distended vein stops the bleeding by thickening and swelling the vein to compress the blood vessel. Most physicians inject the sclerosant directly into the vein, although injections into the vein and the surrounding area are both effective. Once bleeding has been stopped, the treatment can be used to significantly reduce or destroy the varices.

Sclerotherapy for esophageal varices is performed by a physician in a hospital, with the patient awake but sedated. Hyoscine butylbromide (Buscopan) may be administered to freeze the esophagus, making injection of the sclerosant easier. During the procedure, an endoscope is passed through the patient's mouth to the esophagus to view the inside. The branches of the blood vessels at or just above where the stomach and esophagus come together, the usual site of variceal bleeding, are located. After the bleeding vein is identified, a long, flexible sclerotherapy needle is passed through the endoscope. When the tip of the needle's sheath is in place, the needle is advanced, and the sclerosant is injected into the vein or the surrounding area. The most commonly used sclerosants are ethanolamine and sodium tetradecyl sulfate. The needle is withdrawn. The procedure is repeated as many times as necessary to eradicate all distended veins.

Sclerotherapy for esophageal varices controls acute bleeding in about 90% of patients, but it may have to be repeated within the first 48 hours to achieve this success rate. During the initial hospitalization, sclerotherapy is usually performed two or three times. Preventive treatments are scheduled every few weeks or so, depending on the patient's risk level and healing rate. Several studies have shown that the risk of recurrent bleeding is much lower in patients treated with sclerotherapy: 30-50%, as opposed to 70-80% for patients not treated with sclerotherapy.

Preparation

Before sclerotherapy for esophageal varices, the patient's vital signs and other pertinent data are recorded, an intravenous line is inserted to administer fluid or blood, and a sedative is prescribed.

Aftercare

After sclerotherapy for esophageal varices, the patient will be observed for signs of blood loss, lung complications, **fever**, a perforated esophagus, or other complications. Vital signs are monitored, and the intravenous line maintained. **Pain** medication is usually prescribed. After leaving the hospital, the patient follows a diet prescribed by the physician, and, if appropriate, can take mild pain relievers.

Risks

Sclerotherapy for esophageal varices has a 20-40% incidence of complications, and a 1-2% percent mortality rate. Complications can arise from the sclerosant or the endoscopic procedure. Minor complications, which are uncomfortable but do not require active treatment or prolonged hospitalization, include transient chest pain, difficulty swallowing, and fever, which usually go away after a few days. Some people have allergic reactions to the solution. Infection occurs in up to 50% of cases. In 2-10% of patients, the esophagus tightens, but this can usually be treated with dilatation. More serious complications may occur in 10-15% of patients treated with sclerotherapy. These include perforation or bleeding of the esophagus and lung problems, such as aspiration **pneumonia**. Long-term sclerotherapy can damage the esophagus, and increase the patient's risk of developing **cancer**.

Patients with advanced liver disease complicated by bleeding are very poor risks for this procedure. The surgery, premedications, and anesthesia may be sufficient to tip the patient into protein intoxication and

KEY TERMS

Endoscope—An instrument used to examine the inside of a canal or hollow organ. Endoscopic surgery is less invasive than traditional surgery.

Esophagus—The part of the digestive canal located between the pharynx (part of the digestive tube) and the stomach.

Sclerosant—An irritating solution that stops bleeding by hardening the blood or vein it is injected into.

Varices—Swollen or enlarged veins, in this case on the lining of the esophagus.

hepatic **coma**. The blood in the bowels acts like a high protein meal; therefore, protein intoxication may be induced.

Resources

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Lori De Milto

Scoliosis

Definition

Scoliosis is a side-to-side curvature of the spine.

Description

When viewed from the rear, the spine usually appears perfectly straight. Scoliosis is a lateral (side-to-side) curve in the spine, usually combined with a rotation of the vertebrae. (The lateral curvature of scoliosis should not be confused with the normal set of front-to-back spinal curves visible from the side.) While a small degree of lateral curvature does not cause any medical problems, larger curves can cause postural imbalance and lead to muscle **fatigue** and **pain**. More severe scoliosis can interfere with breathing and lead to arthritis of the spine (spondylosis).

Approximately 10% of all adolescents have some degree of scoliosis, though fewer than 1% have curves which require medical attention beyond monitoring. Scoliosis is found in both boys and girls, but a girl's spinal curve is much more likely to progress than a boy's. Girls require scoliosis treatment about five times as often. The reason for these differences is not known.

Causes and symptoms

Four out of five cases of scoliosis are *idiopathic*, meaning the cause is unknown. While idiopathic scoliosis tends to run in families, no responsible genes had been identified as of 1997. Children with idiopathic scoliosis appear to be otherwise entirely healthy, and have not had any bone or joint disease early in life. Scoliosis is not caused by poor posture, diet, or carrying a heavy bookbag exclusively on one shoulder.

Idiopathic scoliosis is further classified according to age of onset:

- **Infantile.** Curvature appears before age three. This type is quite rare in the United States, but is more common in Europe.
- **Juvenile.** Curvature appears between ages 3 and 10. This type may be equivalent to the adolescent type, except for the age of onset.
- **Adolescent.** Curvature appears between ages of 10 and 13, near the beginning of **puberty**. This is the most common type of idiopathic scoliosis.
- **Adult.** Curvature begins after physical maturation is completed.

Causes are known for three other types of scoliosis:

- **Congenital scoliosis** is due to congenital abnormal formation of the bones of the spine, and is often associated with other organ defects.
- **Neuromuscular scoliosis** is due to loss of control of the nerves or muscles which support the spine. The most common causes of this type of scoliosis are **cerebral palsy** and **muscular dystrophy**.
- **Degenerative scoliosis** may be caused by degeneration of the discs which separate the vertebrae or arthritis in the joints that link them.

Scoliosis causes a noticeable asymmetry in the torso when viewed from the front or back. The first sign of scoliosis is often seen when a child is wearing a bathing suit or underwear. A child may appear to be standing with one shoulder higher than the other, or to have a tilt in the waistline. One shoulder blade may

appear more prominent than the other due to rotation. In girls, one breast may appear higher than the other, or larger if rotation pushes that side forward.

Curve progression is greatest near the adolescent growth spurt. Scoliosis that begins early on is more likely to progress significantly than scoliosis that begins later in puberty.

More than 30 states have screening programs in schools for adolescent scoliosis, usually conducted by trained school nurses or gym teachers.

Diagnosis

Diagnosis for scoliosis is done by an orthopedist. A complete medical history is taken, including questions about family history of scoliosis. The **physical examination** includes determination of pubertal development in adolescents, a neurological exam (which may reveal a neuromuscular cause), and measurements of trunk asymmetry. Examination of the trunk is done while the patient is standing, bending over, and lying down, and involves both visual inspection and use of a simple mechanical device called a scoliometer.

If a curve is detected, one or more x rays will usually be taken to define the curve or curves more precisely. An x ray is used to document spinal maturity, any pelvic tilt or hip asymmetry, and the location, extent, and degree of curvature. The curve is defined in terms of where it begins and ends, in which direction it bends, and by an angle measure known as the Cobb angle. The Cobb angle is found by projecting lines parallel to the vertebrae tops at the extremes of the curve; projecting perpendiculars from these lines; and measuring the angle of intersection. To properly track the progress of scoliosis, it is important to project from the same points of the spine each time.

Occasionally, **magnetic resonance imaging (MRI)** is used, primarily to look more closely at the condition of the spinal cord and nerve roots extending from it if neurological problems are suspected.

Treatment

Treatment decisions for scoliosis are based on the degree of curvature, the likelihood of significant progression, and the presence of pain, if any.

Curves less than 20 degrees are not usually treated, except by regular follow-up for children who are still growing. Watchful waiting is usually all that is required in adolescents with curves of 20–30 degrees, or adults with curves up to 40 degrees or slightly more, as long as there is no pain.



This woman suffers from scoliosis, or curvature of the spine.
(Custom Medical Stock Photo. Reproduced by permission.)

For children or adolescents whose curves progress to 30 degrees, and who have a year or more of growth left, bracing may be required. Bracing cannot correct curvature, but may be effective in halting or slowing progression. Bracing is rarely used in adults, except where pain is significant and surgery is not an option, as in some elderly patients.

Two general styles of braces are used for daytime wear. The Milwaukee brace consists of metal uprights attached to pads at the hips, rib cage, and neck. The underarm brace uses rigid plastic to encircle the lower rib cage, abdomen, and hips. Both these brace types hold the spine in a vertical position. Because it can be worn out of sight beneath clothing, the underarm brace is better tolerated and often leads to better compliance. A third style, the Charleston bending brace, is used at night to bend the spine in the opposite direction. Braces are often prescribed to be worn for 22–23 hours per day, though some clinicians allow or encourage removal of the brace for **exercise**.

Bracing may be appropriate for scoliosis due to some types of neuromuscular disease, including spinal muscular atrophy, before growth is finished. Duchenne muscular dystrophy is not treated by bracing, since surgery is likely to be required, and since later surgery is complicated by loss of respiratory capacity.

Surgery for idiopathic scoliosis is usually recommended if:

- the curve has progressed despite bracing
- the curve is greater than 40–50 degrees before growth has stopped in an adolescent
- the curve is greater than 50 degrees and continues to increase in an adult
- there is significant pain

Orthopedic surgery for neuromuscular scoliosis is often done earlier. The goals of surgery are to correct the deformity as much as possible, to prevent further deformity, and to eliminate pain as much as possible. Surgery can usually correct 40–50% of the curve, and sometimes as much as 80%. Surgery cannot always completely remove pain.

The surgical procedure for scoliosis is called *spinal fusion*, because the goal is to straighten the spine as much as possible, and then to fuse the vertebrae together to prevent further curvature. To achieve fusion, the involved vertebra are first exposed, and then scraped to promote regrowth. Bone chips are usually used to splint together the vertebrae to increase the likelihood of fusion. To maintain the proper spinal posture before fusion occurs, metal rods are inserted alongside the spine, and are attached to the vertebrae by hooks, screws, or wires. Fusion of the spine makes it rigid and resistant to further curvature. The metal rods are no longer needed once fusion is complete, but are rarely removed unless their presence leads to complications.

Spinal fusion leaves the involved portion of the spine permanently stiff and inflexible. While this leads to some loss of normal motion, most functional activities are not strongly affected, unless the very lowest portion of the spine (the lumbar region) is fused. Normal mobility, exercise, and even contact sports are usually all possible after spinal fusion. Full recovery takes approximately six months.

Alternative treatment

Numerous alternative therapies have been touted to provide relief and help for individuals with scoliosis, but none have been proven beneficial in clinical trials. These include massage, physical therapy, and electrical stimulation. In addition, alternatives such as

KEY TERMS

Cobb angle —A measure of the curvature of scoliosis, determined by measurements made on x rays.

Rolfing —A system of soft tissue manipulation and movement education to realign and reorient the body.

Scoliometer —A tool for measuring trunk asymmetry; it includes a bubble level and angle measure.

Spondylosis —Arthritis of the spine.

physical therapy, **rolfing**, or chiropractic manipulation may provide improved flexibility, stronger muscles, and pain relief, but cannot prevent or correct the curvature of the spine or its natural progression.

Although important for general health and strength, exercise has not been shown to prevent or slow the development of scoliosis. It may help relieve pain from scoliosis by helping to maintain range of motion. Aquatic exercise, in particular, can increase flexibility and improve posture, balance, coordination, and range of motion. Because it decreases joint compression, it can lessen the pain caused by scoliosis or surgery.

Good **nutrition** is also important for general health, but no specific dietary regimen has been shown to control scoliosis development. In particular, dietary calcium levels do not influence scoliosis progression.

Chiropractic treatment may relieve pain, but it cannot halt scoliosis development, and should not be a substitute for conventional treatment of progressing scoliosis. **Acupuncture** and **acupressure** may also help reduce pain and discomfort, but they cannot halt scoliosis development either.

Prognosis

The prognosis for a person with scoliosis depends on many factors, including the age at which scoliosis begins and the treatment received. More importantly, mostly unknown individual factors affect the likelihood of progression and the severity of the curve. Most cases of mild adolescent idiopathic scoliosis need no treatment and do not progress. Untreated severe scoliosis often leads to spondylosis, and may impair breathing. Degenerative arthritis of the spine, **sciatica**, and severe physical deformities can also result if severe scoliosis is left untreated. Finally, scoliosis can also poorly affect the individual's self-esteem and cause serious emotional problems.

Prevention

There is no known way to prevent the development of scoliosis. Progression of scoliosis may be prevented through bracing or surgery.

Exercise and physical fitness are of paramount importance for all individuals affected with scoliosis. They not only work to maintain flexibility and health, but decrease the likelihood of **osteoporosis**, which in these individuals, can be extremely debilitating.

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ORGANIZATIONS

American Physical Therapy Association. Scoliosis, P.O. Box 37257, Washington, DC 20013.

Center for Spinal Disorders, PC. 8515 Pearl Street, Suite 350, Thornton, CO 80229. (303) 287-2800; fax: (303) 287-7357. <<http://www.centrforspinaldisorders.com>> .

National Scoliosis Foundation. 72 Mount Auburn St., Watertown, MA 02172. (617) 926-0397.

The Scoliosis Association. PO Box 811705, Boca Raton, FL 33481-0669. (407) 368-8518.

OTHER

"Chiropractic Treatment of Scoliosis." Scoliosis World.

Liz Meszaros

Scrotal nuclear medicine scan

Definition

Scrotal nuclear medicine scan is a study of the blood circulation in the scrotum using radioactive contrast agent to highlight obstruction.

Purpose

This test is used almost exclusively to differentiate infection in the testis (testicle) from twisting and infarction. Infection is called **epididymitis** because it mostly involves a collection of tubules on top of the testicle called the epididymis. Twisting of the testis shuts off its blood supply and is called testicular

KEY TERMS

Radioisotope—An unstable form of an element that gives off radiation to become stable.

Scrotum—The bag of skin below the penis that contains the testes.

torsion. Both conditions cause a very painful, swollen testis on one side. Both occur most often in young men, although infection usually occurs at a slightly greater age. The infection increases the blood supply, and the torsion cuts off the blood supply. This is an ideal situation for a blood flow study.

The distinction is critically important, because **testicular torsion** must be untwisted immediately or the testis will die. On the other hand, epididymitis responds to **antibiotics**, and surgery might further injure it.

Description

A radioisotope, technetium-99, combined in a chemical (pertechnetate) is injected intravenously while the patient is under a special machine that detects radiation. This radiation detector, called a gamma camera, scans the scrotum at one minute intervals for about five minutes, then less often for another 10 or 15 minutes. It then creates pictures (either x ray or polaroid) that reveal where the isotope went in the scrotum. Since both sides are scanned, even greater accuracy is obtained by comparison.

Preparation

This procedure is usually done as an emergency to determine the need for immediate surgery.

Risks

The amount of radiation is so slight that even the sensitive testicular tissue is at minimum risk.

Normal results

Blood flow appears unobstructed.

Abnormal results

Three possible possible images appear. They are:

- Increased blood flow indicating infection
- No blood flow indicating testicular torsion

- Blood flow illuminated in a “donut” shaped pattern that indicates torsion that has resolved itself within the last few days.

Resources

BOOKS

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J. Ricker Polsdorfer, MD

Scrotal sonogram see **Scrotal ultrasound**

Scrotal ultrasound

Definition

Scrotal ultrasound is an imaging technique used for the diagnosis of suspected abnormalities of the scrotum. It uses harmless, high-frequency sound waves to form an image. The sound waves are reflected by scrotal tissue to form a picture of internal structures. It is not invasive and involves no radiation.

Purpose

Ultrasound of the scrotum is the primary imaging method used to evaluate disorders of the testicles and surrounding tissues. It is used when a patient has acute **pain** in the scrotum. Some of the problems for which the use of scrotal ultrasound is valuable include an absent or undescended testicle, an inflammation problem, testicular torsion, a fluid collection, abnormal blood vessels, or a mass (lump or tumor).

A sudden onset of pain in the scrotum is considered a serious problem, as delay in diagnosis and treatment can lead to loss of function. **Epididymitis** is the most common cause of this type of pain. Epididymitis is an inflammation of the epididymis, a tubular structure that transports sperm from the testes. It is most often caused by bacterial infection, but may occur after injury, or arise from an unknown cause. Epididymitis is treatable with antibiotics, which usually resolves pain quickly. Left untreated, this condition can lead to **abscess** formation or loss of blood supply to the testicle.

Testicular torsion is the twisting of the spermatic cord that contains the blood vessels which supply the testicles. It is caused by abnormally loose attachments of tissues that are formed during fetal development.

Torsion can be complete, incomplete, or intermittent. Spontaneous detorsion, or untwisting, can occur, making diagnosis difficult. Testicular torsion arises most commonly during adolescence, and is acutely painful. Scrotal ultrasound is used to distinguish this condition from inflammatory problems, such as epididymitis. Testicular torsion is a surgical emergency; it should be operated on as soon as possible to avoid permanent damage to the testes.

A scrotal sac with an absent testicle may be the result of a congenital anomaly (an abnormality present at birth), where a testicle fails to develop. More often, it is due to an undescended testicle. In the fetus, the testicles normally develop just outside the abdomen and descend into the scrotum during the seventh month. Approximately 3% of full-term baby boys have undescended testicles. It is important to distinguish between an undescended testicle and an absent testicle, as an undescended testicle has a very high probability of developing **cancer**.

Ultrasound can be used to locate and evaluate masses in the scrotum. Most masses within the testicle are malignant or cancerous, and most outside the testicle are benign. Primary cancer of the testicles is the most common malignancy in men between the ages of 15-35. Fluid collections and abnormalities of the blood vessels in the scrotum may appear to the physician as masses and need evaluation by ultrasound. A hydrocele, the most common cause of painless scrotal swelling, is a collection of fluid between two layers of tissue surrounding the testicle. An abnormal enlargement of the veins which drain the testicles is called a varicocele. It can cause discomfort and swelling, which can be examined by touch (palpated). Varicocele is a common cause of male infertility.

Precautions

Clear scrotal ultrasound images are difficult to obtain if a patient is unable to remain still.

Description

The patient lies on his back on an examining table. The technologist will usually take a history of the problem, then gently palpate the scrotum. A rolled towel is placed between the patient's legs to support the scrotum. The penis is lifted up onto the abdomen and covered. A gel that enhances sound transmission is put directly on the scrotum. The technologist then gently places a transducer (an electronic imaging device) against the skin. It is moved over the area creating images from reflected sound waves, which

KEY TERMS

Hydrocele— A collection of fluid between two layers of tissue surrounding the testicle; the most common cause of painless scrotal swelling.

Varicocele— An abnormal enlargement of the veins which drain the testicles.

appear on a monitor screen. There is no discomfort from the study itself. However, if the scrotum is very tender, even the slight pressure involved may be painful.

Normal results

A normal study would reveal testicles of normal size and shape, with no masses.

Abnormal results

An abnormal result of an ultrasound of the scrotum may reveal an absent or undescended testicle, an inflammation problem, testicular torsion, a fluid collection, abnormal blood vessels, or a mass.

Resources

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Scrub typhus

Definition

Scrub **typhus** is an infectious disease that is transmitted to humans from field mice and rats through the bite of mites that live on the animals. The main symptoms of the disease are **fever**, a wound at the site of the bite, a spotted rash on the trunk, and swelling of the lymph glands.

Description

Scrub typhus is caused by *Rickettsia tsutsugamushi*, a tiny parasite about the size of bacteria that

belongs to the family Rickettsiaceae. Under the microscope, rickettsiae are either rod-like (bacilli) or spherical (cocci) in shape. Because they are intracellular parasites, they can live only within the cells of other animals.

R. tsutsugamushi lives primarily in mites that belong to the species *Leptotrombidium* (*Trombicula*) *akamushi* and *Leptotrombidium deliense*. In Japan, some cases of scrub typhus have been reportedly transmitted by mites of the species *Leptotrombidium scutellare* and *Leptotrombidium pallidum*. The mites have four-stage life cycles: egg, larva, nymph, and adult. The larva is the only stage that can transmit the disease to humans and other vertebrates.

The tiny chiggers (mite larvae) attach themselves to the skin. During the process of obtaining a meal, they may either acquire the infection from the host or transmit the rickettsiae to other mammals or humans. In regions where scrub typhus is a constant threat, a natural cycle of *R. tsutsugamushi* transmission occurs between mite larvae and small mammals (e.g., field mice and rats). Humans enter a cycle of rickettsial infection only accidentally.

Scrub typhus is also known as *tsutsugamushi disease*. The name *tsutsugamushi* is derived from two Japanese words: *tsutsuga*, meaning something small and dangerous, and *mushi*, meaning creature. The infection is called scrub typhus because it generally occurs after exposure to areas with secondary (scrub) vegetation. It has recently been found, however, that the disease can also be prevalent in such areas as sandy beaches, mountain deserts, and equatorial rain forests. Therefore, it has been suggested that the names mite-borne typhus, or chigger-borne typhus, are more appropriate. Since the disease is limited to eastern and southeastern Asia, India, northern Australia and the adjacent islands, it is also commonly referred to as tropical typhus.

The seasonal occurrence of scrub typhus varies with the climate in different countries. It occurs more frequently during the rainy season. Certain areas such as forest clearings, riverbanks, and grassy regions provide optimal conditions for the infected mites to thrive. These small geographic regions are high-risk areas for humans and have been called scrub-typhus islands.

Causes and symptoms

The incubation period of scrub typhus is about 10 to 12 days after the initial bite. The illness begins rather suddenly with shaking chills, fever, severe

headache, infection of the mucous membrane lining the eyes (the conjunctiva), and swelling of the lymph nodes (lymphadenopathy). A wound (lesion) is often seen at the site of the chigger bite. Bite **wounds** are common in whites but rare in Asians.

The initial lesion, which is about 1 cm (0.4 in) in diameter and flat, eventually becomes elevated and filled with fluid. After it ruptures, it becomes covered with a black scab (eschar). The patient's fever rises during the first week, generally reaching 40–40.5°C (104–105°F). About the fifth day of fever, a red spotted rash develops on the trunk, often extending to the arms and legs. It may either fade away in a few days or may become spotted and elevated (maculopapular) and brightly colored. **Cough** is present during the first week of the fever. An infection of the lung (pneumonitis) may develop during the second week.

In severe cases, the patient's pulse rate increases and blood pressure drops. The patient may become delirious and lose consciousness. Muscular twitching may develop. Enlargement of the spleen is observed. Inflammation of the heart muscle (interstitial **myocarditis**) is more common in scrub typhus than in other rickettsial diseases. In untreated patients, high fever may last for more than two weeks. With specific therapy, however, the fever breaks within 36 hours. The patient's recovery is prompt and uneventful.

Diagnosis

Patient history and physical examination

Differentiating scrub typhus from other forms of typhus as well as from fever, typhoid and meningococcal infections is often difficult during the first several days before the initial rash appears. The geographical location of scrub typhus, the initial sore caused by the chigger bite, and the occurrence of specific proteins capable of destroying the organism (antibodies) in the blood, provide helpful clues and are useful in establishing the diagnosis.

Laboratory tests

Diagnostic procedures involving the actual isolation of rickettsiae from the blood or other body tissues are usually expensive, time-consuming, and hazardous to laboratory workers. As a result, several types of tests known as serological (immunological) tests are used widely to confirm the clinical diagnosis in the laboratory.

Specific antibodies develop in the body in response to an infection. The development of antibodies during the recovery period indicates that an

immune response is present. The formation of antibodies is the basic principle of a serological test. Three different tests are available to diagnose rickettsial infections. The most widely used is the Weil-Felix test. This test is based on the fact that some of the antibodies that are formed in the body during a rickettsial infection can react with certain strains (OX-2 and OX-19) of *Proteus* bacteria and cause them to clump (agglutinate). The clumping is easily seen under the microscope. The Weil-Felix test is easy and inexpensive to perform, with the result that it is widely used. The Weil-Felix test, however, is not very specific. In addition, the clumping is not detectable until the second week of the illness, which limits the test's usefulness in early diagnosis.

A second test known as a complement fixation (CF) test is based on the principle that if antibodies are formed in the body in response to the illness, then the antigen and the antibody will form complexes. These antigen-antibody complexes have the ability to inactivate, or fix, a protein that is found in blood serum (serum complement). The serum complement fixation can be measured using standardized biochemical tests and confirms the presence of antibodies. A third test known as the fluorescent antibody test uses fluorescent tags that are attached to antibodies for easy detection. This test has been developed using three strains of *Rickettsia tsutsugamushi* and has proven to be the most specific for diagnosis.

Treatment

Scrub typhus is treated with **antibiotics**. Chloramphenicol (Chloromycetin, Fenicol) and tetracycline (Achromycin, Tetracyclon) are the drugs of choice. They bring about prompt disappearance of the fever and dramatic clinical improvement. If the antibiotic treatment is discontinued too quickly, especially in patients treated within the first few days of the fever, relapses may occur. In patients treated in the second week of illness, the antibiotics may be stopped one to two days after the fever disappears.

Antibiotics are given intravenously to patients too sick to take them by mouth. Patients who are severely ill and whose treatment was delayed may be given **corticosteroids** in combination with antibiotics for three days.

Prognosis

Before the use of antibiotics, the mortality rate for scrub typhus varied from 1–60%, depending on the geographic area and the rickettsial strain. Recovery

KEY TERMS

Agglutinin—An antibody that causes particulate antigens such as bacteria or other cells to clump together.

Endemic area—A geographical region where a particular disease is prevalent.

Eschar—A hard crust or scab. In scrub typhus, an eschar forms over the initial sore from the chigger bite.

Intracellular parasite—An organism which can only feed and live within the cell of a different animal.

Maculopapular rash—A rash characterized by raised, spotted lesions.

Prophylactic dosage—Giving medications to prevent or protect against diseases.

Rickettsia—A rod-shaped infectious microorganism that can reproduce only inside a living cell. Scrub typhus is a rickettsial disease.

Serological tests—Tests of immune function that are performed using the clear yellow liquid part of blood.

also took a long time. With modern treatment methods, however, deaths are rare and the recovery period is short.

Prevention

General precautions

As of early 2004 there are no effective vaccines for scrub typhus. In endemic areas, precautions include wearing protective clothing. Insect repellents containing dibutyl phthalate, benzyl benzoate, diethyl toluamide, and other substances can be applied to the skin and clothing to prevent chigger bites. Clearing of vegetation and chemical treatment of the soil may help to break up the cycle of transmission from chiggers to humans to other chiggers.

Prophylactic antibiotic dosage

It has been shown that a single oral dose of chloramphenicol or tetracycline given every 5 days for a total of 35 days, with 5-day nontreatment intervals, actually produces active immunity to scrub typhus. This procedure is recommended under special circumstances in certain areas where the disease is endemic.

Resources

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ORGANIZATIONS

Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

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Rebecca J. Frey, PhD

Scurvy

Definition

Scurvy is a condition caused by a lack of vitamin C (ascorbic acid) in the diet. Signs of scurvy include tiredness, muscle weakness, joint and muscle aches, a rash on the legs, and bleeding gums. In the past, scurvy was common among sailors and other people deprived of fresh fruits and vegetables for long periods of time.

Description

Scurvy is very rare in countries where fresh fruits and vegetables are readily available and where processed foods have vitamin C added. Vitamin C is an important antioxidant vitamin involved in the development of connective tissues, lipid and vitamin metabolism, biosynthesis of neurotransmitters, immune function, and wound healing. It is found in fruits,

especially citrus fruits like oranges, lemons, and grapefruit, and in green leafy vegetables like broccoli and spinach. In adults, it may take several months of vitamin C deficiency before symptoms of scurvy develop.

Currently, the recommended dietary allowance (RDA) for vitamin C is 50–60 mg/day for adults; 35 mg/day for infants; 40–45 mg/day for children 1–14; 70 mg/day during **pregnancy**; and 90–95 mg/day during **lactation**. The body's need for vitamin C increases when a person is under **stress**, **smoking**, or taking certain medications.

Causes and symptoms

A lack of vitamin C in the diet is the primary cause of scurvy. This can occur in people on very restricted **diets**, who are under extreme physiological stress (for example, during an infection or after an injury), and in chronic alcoholics. Infants can develop scurvy if they are weaned from breast milk and switched to cow's milk without an additional supplement of vitamin C. Babies of mothers who took extremely high doses of vitamin C during pregnancy can develop infantile scurvy. In children, the deficiency can cause painful swelling of the legs along with **fever**, **diarrhea**, and **vomiting**. In adults, early signs of scurvy include feeling weak, tired, and achy. The appearance of tiny red blood-blisters to larger purplish blotches on the skin of the legs is a common symptom. Wound healing may be delayed and **scars** that had healed may start to break down. The gums swell and bleed easily, eventually leading to loosened teeth. Muscle and joint **pain** may also occur.

Diagnosis

Scurvy is often diagnosed based on the symptoms present. A dietary history showing little or no fresh fruits or vegetables are eaten may help to diagnose vitamin C deficiency. A blood test can also be used to check the level of ascorbic acid in the body.

Treatment

Adult treatment is usually 300–1,000 mg of ascorbic acid per day. Infants should be treated with 50 mg of ascorbic acid up to four times per day.

Prognosis

Treatment with vitamin C is usually successful, if the deficiency is recognized early enough. Left untreated, the condition can cause **death**.



An x-ray image of an infant suffering from scurvy. (Photograph by Lester V. Bergman, Corbis Images. Reproduced by permission.)

KEY TERMS

Ascorbic acid—Another term for vitamin C, a nutrient found in fresh fruits and vegetables. Good sources of vitamin C in the diet are citrus fruits like oranges, lemons, limes, and grapefruits, berries, tomatoes, green peppers, cabbage, broccoli, and spinach.

Recommended dietary allowance (RDA)—The daily amount of a vitamin the average person needs to maintain good health.

Prevention

Eating foods rich in vitamin C every day prevents scurvy. A supplement containing the RDA of vitamin C will also prevent a deficiency. Infants who are being weaned from breast milk to cow's milk need a supplement containing vitamin C.

Resources

BOOKS

Stein, Jay H., editor. "Ascorbic Acid (Vitamin C) Deficiency." In *Internal Medicine*. St. Louis: Mosby, 1998.

Altha Roberts Edgren

Seafood poisoning see **Fish and shellfish poisoning**

Seasonal affective disorder

Definition

Seasonal affective disorder (SAD) is a form of depression most often associated with the lack of daylight in extreme northern and southern latitudes from the late fall to the early spring.

Description

Although researchers are not certain what causes seasonal affective disorder, they suspect that it has something to do with the hormone melatonin. Melatonin is thought to play an active role in regulating the "internal body clock," which dictates when humans feel like going to bed at night and getting up in the morning. Although seasonal affective disorder is most common when light is low, it may occur in the spring, and it is then often called reverse SAD.

Causes and symptoms

The body produces more melatonin at night than during the day, and scientists believe it helps people feel sleepy at nighttime. There is also more melatonin in the body during winter, when the days are shorter. Some researchers believe that excessive melatonin release during winter in people with SAD may account for their feelings of drowsiness or depression. One variation on this idea is that, during winter, people's internal clocks may become out of sync with the light-dark cycle, leading to a long-term disruption in melatonin release.

Seasonal affective disorder, while not an official category of mental illness listed by the American Psychiatric Association, is estimated to affect 10 million Americans, most of whom are women. Another 25 million Americans may have a mild form of SAD, sometimes called the "winter blues" or "winter blahs." The risk of SAD increases the further from the equator a person lives.

The symptoms of SAD are similar to those of other forms of depression. People with SAD may feel sad, irritable, or tired, and may find themselves sleeping too much. They may also lose interest in normal or pleasurable activities (including sex), become withdrawn, crave carbohydrates, and gain weight.

Diagnosis

Doctors usually diagnose seasonal affective disorder based on the patient's description of symptoms, including the time of year they occur.

Treatment

The first-line treatment for seasonal affective disorder is light therapy, exposing the patient to bright artificial light to compensate for the gloominess of winter. Light therapy uses a device called a light box, which contains a set of fluorescent or incandescent lights in front of a reflector. Typically, the patient sits for 30 minutes next to a 10,000-lux box (which is about 50 times as bright as ordinary indoor light). Light therapy appears to be safe for most people. However, it may be harmful for those with eye diseases. The most common side effects are vision problems such as eye strain, headaches, irritability, and **insomnia**. In addition, hypomania (elevated or expansive mood, characterized by hyperactivity and inflated self esteem) may occasionally occur.

Recently, researchers have begun testing whether people who do not completely respond to light therapy can benefit from tiny doses of the hormone melatonin to reset the body's internal clock. Early results look promising, but the potential benefits must be confirmed in larger studies before this type of treatment becomes widely accepted.

Like other types of **mood disorders**, seasonal affective disorder may also respond to medication and psychotherapy. The four different classes of drugs used for mood disorders are:

- heterocyclic antidepressants (HCAs), such as amitriptyline (Elavil)
- selective serotonin reuptake inhibitors (SSRIs), such as fluoxetine (Prozac), paroxetine (Paxil), and sertraline (Zoloft)
- monoamine oxidase inhibitors (MAO inhibitors), such as phenelzine sulfate (Nardil) and tranylcypromine sulfate (Parnate)
- Lithium salts, such as lithium carbonate (Eskalith), often used in people with bipolar mood disorders, are often useful with SAD patients; many SAD patients also suffer from **bipolar disorder** (excessive mood swings; formerly known as manic depression)

A number of psychotherapy approaches are useful as well. Interpersonal psychotherapy helps patients recognize how their mood disorder and their interpersonal relationships interact. Cognitive-behavioral therapy explores how the patient's view of the world may be affecting mood and outlook.

Prognosis

Most patients with seasonal affective disorder respond to light therapy and/or **antidepressant drugs**.

KEY TERMS

Cognitive behavioral therapy—Psychotherapy aimed at helping people change their attitudes, perceptions, and patterns of thinking.

Melatonin—A naturally occurring hormone involved in regulating the body's "internal clock."

Serotonin—A chemical messenger in the brain thought to play a role in regulating mood.

Resources

ORGANIZATIONS

American Psychiatric Association. 1400 K Street NW, Washington DC 20005. (888) 357-7924. <<http://www.psych.org>>.

National Depressive and Manic Depressive Association (NDMDA). 730 N. Franklin St., Ste. 501, Chicago, IL 60610. (800) 826-3632. <<http://www.ndmda.org>>.

National Institute of Mental Health. Mental Health Public Inquiries, 5600 Fishers Lane, Room 15C-05, Rockville, MD 20857. (888) 826-9438. <<http://www.nimh.nih.gov>>.

Robert Scott Dinsmoor

Seasonal depression see **Seasonal affective disorder**

Seatworm infection see **Enterobiasis**

Seborrheic dermatitis

Definition

Seborrheic **dermatitis** is a common inflammatory disease of the skin characterized by scaly lesions usually on the scalp, hairline, and face.

Description

Seborrheic dermatitis appears as red, inflamed skin covered by greasy or dry scales that may be white, yellowish, or gray. It can affect the scalp, eyebrows, forehead, face, folds around the nose and ears, the chest, armpits (axilla), and groin. Dandruff and cradle cap are mild forms of seborrheic dermatitis, and appear as fine white scales without inflammation.



This young boy is afflicted with seborrheic dermatitis.
(Custom Medical Stock Photo. Reproduced by permission.)

Causes and symptoms

The cause of seborrheic dermatitis is unclear, though it is has been linked to genetic or environmental factors. *Pityrosporum ovale*, a species of yeast normally found in hair follicles, has been proposed as one possible causative factor. A high fat diet and alcohol ingestion are thought to play some role. Other possible risk factors include:

- stress and **fatigue**
- weather extremes (e. g. hot, humid weather or cold, dry weather)
- oily skin
- infrequent shampoos
- obesity
- Parkinson's disease
- AIDS
- use of drying lotions that contain alcohol
- other skin disorders (for example **acne**, **rosacea**, or psoriasis)

Mild forms of the disorder may be asymptomatic. Symptoms also disappear and reappear, and vary in

KEY TERMS

Acne—A chronic inflammation of the sebaceous glands that manifests as blackheads, whiteheads, and/or pustules on the face or trunk.

Psoriasis—A skin disorder of chronic, itchy scaling most commonly at sites of repeated minor trauma (e.g. elbows, knees, and skin folds). It affects up to 2% of the population in Western countries—males and females equally.

Rosacea—A chronic inflammation of the face, with associated scattered round nodules and increased reactivity of the facial capillaries to heat. It is most common in females, aged 30–50 years.

intensity over time. When scaling is present, it may be accompanied by **itching** that can lead to secondary infection.

Diagnosis

The diagnosis of seborrheic dermatitis is based on assessment of symptoms, accompanied by consideration of medical history.

Treatment

Treatment consists of vigorous shampoos with preparations that assist with softening and removing the scaly accumulations. For mild cases, a non-prescription shampoo with selenium sulfide or zinc pyrithione may be used. For more severe problems, the doctor may prescribe shampoos containing coal tar or scalp creams containing cortisone. The antiseborrheic shampoo should be left on the scalp for approximately five minutes before rinsing out. Hydrocortisone cream may also be ordered for application to the affected areas on the face and body. Application of the hydrocortisone should be discontinued when the condition clears and restarted with recurrence.

Prognosis

This chronic condition may be characterized by long periods of inactivity. Symptoms in the acute phase can be controlled with appropriate treatment.

Prevention

The condition cannot be prevented. The severity and frequency of flare-ups may be minimized with

frequent shampoos, thorough drying of skin folds after bathing, and wearing of loose, ventilating clothing. Foods that appear to worsen the condition should be avoided.

Resources

BOOKS

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Kathleen D. Wright, RN

Secobarbital see **Barbiturates**

Secondary erythrocytosis see **Secondary polycythemia**

Secondary polycythemia

Definition

Secondary polycythemia is an acquired form of a rare disorder characterized by an abnormal increase in the number of mature red cells in the blood.

Secondary polycythemia is also called secondary erythrocytosis.

Description

Polycythemia means too many red blood cells. The resulting excess of red cells thickens the blood and impedes its passage through small blood vessels.

Secondary polycythemia usually affects people between the ages of 40 and 60.

Types of secondary polycythemia

Known as spurious polycythemia, **stress** polycythemia, or Gaisbock's syndrome, relative polycythemia is characterized by normal numbers of red blood cells but decreased levels of plasma (the fluid part of the blood). Overweight, middle-aged white men who smoke, have high blood pressure, and are on diuretic medicines to remove excess water from their bodies may develop Gaisbock's syndrome.

In smoker's polycythemia, the number of red blood cells is elevated. Plasma levels are abnormally low.

Causes and symptoms

Smoking, which impairs red blood cells' ability to deliver oxygen to body tissues, can cause secondary polycythemia. So can the following conditions:

- carbon monoxide **poisoning**
- chronic heart or lung disease
- hormonal (endocrine) disorders
- exposure to high altitudes
- kidney cysts
- tumors of the brain, liver, or uterus.

Causes of spurious polycythemia include:

- burns
- diarrhea
- hemoconcentration (higher-than-normal concentration of cells and solids in the blood, usually due to becoming dehydrated or taking **diuretics**)
- stress

Weakness, headaches, and **fatigue** are usually the first symptoms of secondary polycythemia. Patients may feel lightheaded or experience **shortness of breath**.

Visual disturbances associated with this disorder include distorted vision, blind spots, and flashes of light. The gums and small cuts are likely to bleed, and the hands and feet may burn. Extensive **itching** often occurs after taking a bath or shower.

Pain in the chest or leg muscles is common. The face often becomes ruddy, then turns blue after **exercise** or other exertion. Confusion and ringing in the ears (**tinnitus**) may also occur.

Diagnosis

A very important part of diagnosing secondary polycythemia is differentiating it from primary polycythemia (also called polycythemia rubra vera or Vaquez' disease). Unlike secondary polycythemia, primary polycythemia cannot be traced to an underlying condition such as smoking, high altitude, or chronic lung disease.

Doctors diagnose polycythemia by measuring oxygen levels in blood drawn from an artery. A patient whose oxygen level is abnormally low probably has secondary polycythemia. Erythropoietin may also be measured. Levels of this hormone, which stimulates the bone marrow to produce red blood cells, may be normal or elevated in a patient with secondary polycythemia. Red blood cell mass is also frequently measured in diagnosing the disorder.

Imaging studies are sometimes performed to determine whether the spleen and liver are enlarged and to detect erythropoietin-producing kidney lesions. Other diagnostic procedures include chest x rays and an electrocardiogram (EKG).

Treatment

Secondary polycythemia is treated primarily by treating the underlying condition causing the disorder. For example, patients with Gaisbock's syndrome are often taken off diuretics and encouraged to lose weight. Lung disorders, such as chronic obstructive pulmonary disease (COPD), may cause secondary polycythemia; treating the lung disorder generally improves the polycythemia.

Some medications may also be taken to treat symptoms caused by polycythemia. For example, **anti-histamines** can alleviate itching, and **aspirin** can soothe burning sensations and bone pain.

Until the underlying condition is controlled, doctors use bloodletting (**phlebotomy**) to reduce the number of red blood cells in the patient's body. In most instances, a pint of blood is drained from the patient as needed and tolerated, until the hematocrit (the proportion of red cells in the blood) reaches an acceptable level. **Chemotherapy** is not used to treat secondary polycythemia; however, it may be used to treat the primary form.

Prognosis

Curing or removing the underlying cause of this disorder generally eliminates the symptoms.

Resources

OTHER

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Maureen Haggerty

SED rate see **Erythrocyte sedimentation rate**

Sedation

Definition

Sedation is the act of calming by administration of a sedative. A sedative is a medication that commonly induces the nervous system to calm.

Purpose

The process of sedation has two primary intentions. First, sedation is recommended to allow patients the ability to tolerate unpleasant diagnostic or surgical procedures and to relieve **anxiety** and discomfort. Second, sedation for uncooperative patients may expedite and simplify special procedures that require little or no movement. Additionally, sedation is often desirable to diminish fear associated with operative procedures. Sedation is typically used for common diagnostic tests that require prolonged **immobilization** such as **magnetic resonance imaging** (MRI) and computed axial tomography (CAT) scanning. Some cases that require sedation may also necessitate the use of **analgesics** to decrease **pain** associated with a procedure or test.

Precautions

Benzodiazepines (common sedative medication) have a cumulative effect. This means that if the patient has not had time to metabolize the previous dose and ingests more, then the sedative effect may increase. Because of these additive effects, these medications taken with other sedatives or alcohol (also a sedative hypnotic drug) may increase chances for accidental **death**. In general, most of the medications that induce sedation may alter breathing and cardiac stability. In patients with preexisting lung and/or heart disease, these medications should be monitored closely or not prescribed.

Description

The future of anesthetic care involves the simultaneous administration of several drugs including IV medications and inhaled anesthetics. An extensive survey of death in 100,000 cases published in 1988 revealed that death within seven days was 2.9 times greater when one or two anesthetic drugs were used than when using three or more medications. As of 2000 this study is accepted as standard practice and multiple IV anesthetics is the preferable recommendation for optimal patient care.

The procedure for sedation is usually explained to the patient by an attending clinician. An IV access line is set in place for fluid replacement and injection of medications. A history is usually taken to assess risk and choice of medication. The patient typically signs consent forms and the possible side effects are explained. The day before the test, the patient may be required to maintain specified dietary restriction.

For outpatient surgery there are two types of sedation, conscious and unconscious sedation. Patients receiving conscious sedation are capable of rational responses, and they are able to maintain their airway for ventilation. The hallmark of conscious sedation is that it does not alter respiratory, cardiac, or reflex functions (nerve reflexes from the brain) to the level that requires external support for these vital functions. Patients receiving conscious sedation are cooperative, have stable vital signs (pulse, respiratory rate, and temperature), shorter recovery room convalescence, and lower risk of developing drug-induced complications. Unconscious sedation is a controlled state of anesthesia, characterized by partial or complete loss of protective nerve reflexes, including the ability to independently breathe and respond to commands. The patient is unable to cooperate, has labile (fluctuating) vital signs, prolonged recovery room convalescence, and higher risk of anesthetic complications.

Preparation

Usually procedures for conscious sedation do not require preoperative or pre-testing orders. Clinical situations for unconscious sedation typically involve eating and drinking protocols starting the day before the procedure.

The age and physical status of the patient is useful in determining sensitivity. A detailed past history, especially prior experiences with sedatives and other anesthetics is an important part of preparatory assessment. It is important to determine if there were any untoward side effects associated with a previous medication. Patient positioning is important to prevent blood pressure changes or nerve damage associated with abnormal position.

Patients are also monitored for pulse rate, respiration, blood pressure, and temperature. Additionally, the heart is monitored using **electrocardiography** (ECG). Ventilation is assessed using a pulse oximeter. This machine is clipped with a special probe on one finger and can measure the levels of oxygen and carbon dioxide, which are reliable indicators of respiratory status.

Aftercare

The major goal for recovery room monitoring is assessment of residual drug effects. Recovery room monitoring primarily focuses on heart stability, respiratory adequacy and return to previous brain functioning.

KEY TERMS

Baseline— A return to an original state.

Diazepam— One of the most commonly used sedative-hypnotic medications.

Risks

The original forms of diazepam (Valium, a very common sedative) caused irritation of veins and phlebitis. Newer forms of diazepam (Dizac) are chemically improved to lower the possibility of vein irritation. Age and physical health are important risk factors. Preexisting medical conditions such as high blood pressure and heart and lung disease may increase the chance of developing undesirable side effects.

Normal results

Normal or uncomplicated results for sedation include alleviation of anxiety and discomfort. Coupled with analgesic, patients are usually pain-free. The normal progression post procedure or post operatively would be to return to baseline brain functioning, unassisted breathing, and normal heart rate and rhythm.

Abnormal results

Patients may have excessive **nausea and vomiting** associated with narcotic analgesia (if this is indicated). Excessive drowsiness can occur secondary to benzodiazepine-induced sedation. The patient can also develop hypoventilation (a decrease in ventilation), airway obstruction, high or low blood pressure, abnormal heart rhythms, **nausea, vomiting**, and shivering.

Resources

BOOKS

- Fleisher, Gary R., et al. *Textbook of Pediatric Emergency Medicine*. 4th ed. Lippincott Williams & Wilkins, 2000.
- Miller, Ronald D., et al, editors. *Anesthesia*. 5th ed. Churchill Livingstone, Inc., 2000.

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Sedative-hypnotic drugs see **Anti-insomnia drugs**

Sedimentation rate see **Erythrocyte sedimentation rate**

Seizure disorder

Definition

A seizure is a sudden disruption of the brain's normal electrical activity accompanied by altered consciousness and/or other neurological and behavioral manifestations. Epilepsy is a condition characterized by recurrent seizures that may include repetitive muscle jerking called convulsions.

Description

There are more than 20 different seizure disorders. One in ten Americans will have a seizure at some time, and at least 200,000 have at least one seizure a month.

Epilepsy affects 1–2% of the population of the United States. Although epilepsy is as common in adults over 60 as in children under 10, 25% of all cases develop before the age of five. One in every two cases develops before the age of 25. About 125,000 new cases of epilepsy are diagnosed each year, and a significant number of children and adults that have not been diagnosed or treated have epilepsy.

Most seizures are benign, but a seizure that lasts a long time can lead to status epilepticus, a life-threatening condition characterized by continuous seizures, sustained loss of consciousness, and respiratory distress. Non-convulsive epilepsy can impair physical coordination, vision, and other senses. Undiagnosed seizures can lead to conditions that are more serious and more difficult to manage.

Types of seizures

Generalized epileptic seizures occur when electrical abnormalities exist throughout the brain. A partial seizure does not involve the entire brain. A partial seizure begins in an area called an epileptic focus, but may spread to other parts of the brain and cause a generalized seizure. Some people who have epilepsy have more than one type of seizure.

Motor attacks cause parts of the body to jerk repeatedly. A motor attack usually lasts less than an hour and may last only a few minutes. Sensory seizures begin with **numbness** or **tingling** in one area. The sensation may move along one side of the body or the back before subsiding.

Visual seizures, which affect the area of the brain that controls sight, cause people to see things that are not there. Auditory seizures affect the part of the brain that controls hearing and cause the patient to imagine voices, music, and other sounds. Other types of

seizures can cause confusion, upset stomach, or emotional distress.

GENERALIZED SEIZURES. A generalized tonic-clonic (grand-mal) seizure begins with a loud cry before the person having the seizure loses consciousness and falls to the ground. The muscles become rigid for about 30 seconds during the tonic phase of the seizure and alternately contract and relax during the clonic phase, which lasts 30–60 seconds. The skin sometimes acquires a bluish tint and the person may bite his tongue, lose bowel or bladder control, or have trouble breathing.

A grand mal seizure lasts between two and five minutes, and the person may be confused or have trouble talking when he regains consciousness (post-ictal state). He may complain of head or muscle aches, or weakness in his arms or legs before falling into a deep sleep.

PRIMARY GENERALIZED SEIZURES. A primary generalized seizure occurs when electrical discharges begin in both halves (hemispheres) of the brain at the same time. Primary generalized seizures are more likely to be major motor attacks than to be absence seizures.

ABSENCE SEIZURES. Absence (petit mal) seizures generally begin at about the age of four and stop by the time the child becomes an adolescent.

Absence seizures usually begin with a brief loss of consciousness and last between one and 10 seconds. A person having a petit mal seizure becomes very quiet and may blink, stare blankly, roll his eyes, or move his lips. A petit mal seizure lasts 15–20 seconds. When it ends, the person who had the seizure resumes whatever he was doing before the seizure began. He will not remember the seizure and may not realize that anything unusual has happened. Untreated, petit mal seizures can recur as many as 100 times a day and may progress to grand mal seizures.

MYOCLONIC SEIZURES. Myoclonic seizures are characterized by brief, involuntary spasms of the tongue or muscles of the face, arms, or legs. Myoclonic seizures are most apt to occur when waking after a night's sleep.

A jacksonian seizure is a partial seizure characterized by tingling, stiffening, or jerking of an arm or leg. Loss of consciousness is rare. The seizure may progress in characteristic fashion along the limb.

Limp posture and a brief period of unconsciousness are features of akinetic seizures, which occur in young children. Akinetic seizures, which cause the child to fall, also are called drop attacks.

PARTIAL SEIZURES. Simple partial seizures do not spread from the focal area where they arise. Symptoms are determined by the part of the brain affected. The patient usually remains conscious during the seizure and can later describe it in detail. In 2003, it was reported that people who experience partial seizures are twice as likely to have sleep disturbances as people their same age and gender.

COMPLEX PARTIAL SEIZURES. A distinctive smell, taste, or other unusual sensation (aura) may signal the start of a complex partial seizure.

Complex partial seizures start as simple partial seizures, but move beyond the focal area and cause loss of consciousness. Complex partial seizures can become major motor seizures. Although a person having a complex partial seizure may not seem to be unconscious, he does not know what is happening and may behave inappropriately. He will not remember the seizure, but may seem confused or intoxicated for a few minutes after it ends.

Causes and symptoms

The origin of 50-70% of all cases of epilepsy is unknown. Epilepsy sometimes is the result of trauma at birth. Such causes include insufficient oxygen to the brain; **head injury**; heavy bleeding or incompatibility between a woman's blood and the blood of her newborn baby; and infection immediately before, after, or at the time of birth.

Other causes of epilepsy include:

- head trauma resulting from a car accident, gunshot wound, or other injury.
- alcoholism
- brain **abscess** or inflammation of membranes covering the brain or spinal cord
- phenylketonuria (PKU, a disease that is present at birth, often is characterized by seizures, and can result in **mental retardation**) and other inherited disorders
- infectious diseases like **measles**, **mumps**, and diphtheria
- degenerative disease
- lead **poisoning**, mercury poisoning, **carbon monoxide poisoning**, or ingestion of some other poisonous substance
- genetic factors

Status epilepticus, a condition in which a person suffers from continuous seizures and may have trouble breathing, can be caused by:



This patient's brain is exposed during surgery in order for surgeons to remove the mass responsible for his epilepsy. (Custom Medical Stock Photo. Reproduced by permission.)

- suddenly discontinuing anti-seizure medication
- hypoxic or metabolic encephalopathy (brain disease resulting from lack of oxygen or malfunctioning of other physical or chemical processes)
- acute head injury
- blood infection caused by inflammation of the brain or the membranes that cover it

Diagnosis

Personal and family medical history, description of seizure activity, and physical and neurological examinations help primary care physicians, neurologists, and epileptologists diagnose this disorder. Doctors rule out conditions that cause symptoms that resemble epilepsy, including small strokes (transient ischemic attacks, or TIAs), **fainting**, (syncope), pseudoseizures, and sleep attacks (narcolepsy.)

Neuropsychological testing uncovers learning or memory problems. Neuroimaging provides views of brain areas involved in seizure activity.

The electroencephalogram (EEG) is the main test used to diagnose epilepsy. EEGs use electrodes placed on or within the skull to record the brain's electrical activity and pinpoint the exact location of abnormal discharges.

The patient may be asked to remain motionless during a short-term EEG or to go about his normal activities during extended monitoring. Some patients are deprived of sleep or exposed to seizure triggers, such as rapid, deep breathing (hyperventilation) or flashing lights (photic stimulation). In some cases, people may be hospitalized for EEG monitorings that can last as long as two weeks. Video EEGs also document what the patient was doing when the seizure occurred and how the seizure changed his behavior.

Other techniques used to diagnose epilepsy include:

- Magnetic resonance imaging (MRI), which provides clear, detailed images of the brain. Functional MRI (fMRI), performed while the patient does various tasks, can measure shifts in electrical intensity and blood flow and indicate which brain region each activity affects.
- Positron emission tomography (PET) and single photon emission tomography (SPECT) monitor blood flow and chemical activity in the brain area being tested. PET and SPECT are very effective in locating the brain region where metabolic changes take place between seizures.

Treatment

The goal of epilepsy treatment is to eliminate seizures or make the symptoms less frequent and less severe. Long-term anticonvulsant drug therapy is the most common form of epilepsy treatment.

Medication

A combination of drugs may be needed to control some symptoms, but most patients who have epilepsy take one of the following medications:

- Dilantin (phenytoin)
- Tegretol (carbamazepine)
- Barbita (phenobarbital)
- Mysoline (primidone)
- Depakene (valproic acid, sodium valproate)
- Klonopin (clonazepam)
- Zarontin (ethosuximide).

Dilantin, Tegretol, Barbita, and Mysoline are used to manage or control generalized tonic-clonic and complex partial seizures. Depakene, Klonopin, and Zarontin are prescribed for patients who have absence seizures.

Neurontin (gabapentin), Lamictal (lamotrigine), and topiramate (Topamax) are among the medications more recently approved in the United States to treat adults who have partial seizures or partial and grand mal seizures. Another new medication called Levetiracetam (Keppra) has been approved and shows particularly good results in reducing partial seizures among elderly patients with few side effects. This is important, because elderly patients often have other conditions and must take other medications that might interact with seizure medications. In 2003, Keppra's manufacturer was working on a new anti-epileptic drug from the same chemical family as Keppra that should be more potent and effective. Available medications frequently change, and it the physician will determine the best treatment for an individual patient. A 2003 report found that monotherapy, or using just one medication rather than a combination, works better for most patients. The less complicated the treatment, the more likely the patient will comply and better manager the seizure disorder.

Even a patient whose seizures are well controlled should have regular blood tests to measure levels of anti-seizure medication in his system and to check to see if the medication is causing any changes in his blood or liver. A doctor should be notified if any signs of drug toxicity appear, including uncontrolled eye movements; sluggishness, **dizziness**, or hyperactivity; inability to see clearly or speak distinctly; **nausea** or **vomiting**; or sleep problems.

Status epilepticus requires emergency treatment, usually with Valium (Ativan), Dilantin, or Barbita. An intravenous dextrose (sugar) solution is given to patients whose condition is due to low blood sugar, and a vitamin B₁ preparation is administered intravenously when status epilepticus results from chronic alcohol withdrawal. Because dextrose and thiamine are essentially harmless and because delay in treatment can be disastrous, these medications are given routinely, as it is usually difficult to obtain an adequate history from a patient suffering from status epilepticus.

Intractable seizures are seizures that cannot be controlled with medication or without **sedation** or other unacceptable side effects. Surgery may be used to eliminate or control intractable seizures.

Surgery

Surgery can be used to treat patients whose intractable seizures stem from small focal lesions that can be removed without endangering the patient, changing the patient's personality, dulling the



This abstract artwork is based on a patient's description of what an epileptic seizure feels like. Epileptic seizures are caused by chaotic electrical activity in the brain. They can be triggered by a variety of factors, such as illness or stress, although the underlying causes is not completely understood. (Illustration by John Bavosi, Photo Researchers, Inc. Reproduced by permission.)

patient's senses, or reducing the patient's ability to function.

Each year, as many as 5,000 new patients may become suitable candidates for surgery, which most often is performed at a comprehensive epilepsy center. Potential surgical candidates include patients with:

- partial seizures and secondarily generalized seizures (attacks that begin in one area and spread to both sides of the brain)
- seizures and childhood **paralysis** on one side of the body (hemiplegia)
- complex partial seizures originating in the temporal lobe (the part of the brain associated with speech, hearing, and smell) or other focal seizures. (However, the risk of surgery involving the speech centers is that the patient will lose speech function.)
- Generalized myoclonic seizures or generalized seizures featuring temporary paralysis (akinetic) or loss of muscle tone (atonal)

A **physical examination** is conducted to verify that a patient's seizures are caused by epilepsy, and surgery is not used to treat patients with severe psychiatric disturbances or medical problems that raise risk factors to unacceptable levels.

Surgery is never indicated unless:

- the best available anti-seizure medications have failed to control the patient's symptoms satisfactorily
- the origin of the patient's seizures has been precisely located
- there is good reason to believe that surgery will significantly improve the patient's health and quality of life.

Every patient considering epilepsy surgery is carefully evaluated by one or more neurologists, neurosurgeons, neuropsychologists, and/or social workers. A psychiatrist, chaplain, or other spiritual advisor may help the patient and his family cope with the **stress** that occurs during and after the selection process.

TYPES OF SURGERY. Surgical techniques used to treat intractable epilepsy include:

- **Lesionectomy.** Removing the lesion (diseased brain tissue) and some surrounding brain tissue is very effective in controlling seizures. Lesionectomy is generally more successful than surgery performed on patients whose seizures are not caused by clearly defined lesions, but removing only part of the lesion lessens the effectiveness of the procedure.
- **Temporal resections.** Removing part of the temporal lobe and the part of the brain associated with feelings, memory, and emotions (the hippocampus) provides good or excellent seizure control in 75-80% of properly selected patients with appropriate types of temporal lobe epilepsy. Some patients experience post-operative speech and memory problems.
- **Extra-temporal resection.** This procedure involves removing some or all of the frontal lobe, the part of the brain directly behind the forehead. The frontal lobe helps regulate movement, planning, judgment, and personality, and special care must be taken to prevent post-operative problems with movement and speech. Extra-temporal resection is most successful in patients whose seizures are not widespread.
- **Hemispherectomy.** This method of removing brain tissue is restricted to patients with severe epilepsy and abnormal discharges that often extend from one side of the brain to the other. Hemispherectomies most often are performed on infants or young children who have had an extensive brain disease or disorder since birth or from a very young age.
- **Corpus callosotomy.** This procedure, an alternative to hemispherectomy in patients with congenital hemiplegia, removes some or all of the white matter that separates the two halves of the brain. Corpus callosotomy is performed almost exclusively on children who are frequently injured during falls caused

by seizures. If removing two-thirds of the corpus callosum doesn't produce lasting improvement in the patient's condition, the remaining one-third will be removed during another operation.

- **Multiple subpial transection.** This procedure is used to control the spread of seizures that originate in or affect the "eloquent" cortex, the area of the brain responsible for complex thought and reasoning.

Other forms of treatment

KETOGENIC DIET. A special high-fat, low-protein, low-carbohydrate diet sometimes is used to treat patients whose severe seizures have not responded to other treatment. Calculated according to age, height, and weight, the ketogenic diet induces mild **starvation** and **dehydration**. This forces the body to create an excessive supply of ketones, natural chemicals with seizure-suppressing properties.

The goal of this controversial approach is to maintain or improve seizure control while reducing medication. The ketogenic diet works best with children between the ages of one and 10. It is introduced over a period of several days, and most children are hospitalized during the early stages of treatment.

If a child following this diet remains seizure-free for at least six months, increased amounts of carbohydrates and protein gradually are added. If the child shows no improvement after three months, the diet is gradually discontinued. A 2003 study of the diet and its effect on growth noted that if used, clinicians should recommend adequate intake of energy and protein and a higher proportion of unsaturated to saturated dietary fats. The report also recommended use of vitamin and mineral supplements with the diet.

Introduced in the 1920s, the ketogenic diet has had limited, short-term success in controlling seizure activity. Its use exposes patients to such potentially harmful side effects as:

- staphylococcal infections
- atunted or delayed growth
- low blood sugar (**hypoglycemia**)
- excess fat in the blood (hyperlipidemia)
- disease resulting from calcium deposits in the urinary tract (urolithiasis)
- disease of the optic nerve (optic neuropathy)

VAGUS NERVE STIMULATION. The United States Food and Drug Administration (FDA) has approved the use of vagus nerve stimulation (VNS) in patients over the age of 16 who have intractable partial seizures.

This non-surgical procedure uses a pacemaker-like device implanted under the skin in the upper left chest, to provide intermittent stimulation to the vagus nerve. Stretching from the side of the neck into the brain, the vagus nerve affects swallowing, speech, breathing, and many other functions, and VNS may prevent or shorten some seizures. A 2003 report said that this treatment has reduced partial seizures by 50% or more in about one-third of patient with no adverse effects.

First aid for seizures

A person having a seizure should not be restrained, but sharp or dangerous objects should be moved out of reach. Anyone having a complex partial seizure can be warned away from danger by someone calling his/her name in a clear, calm voice.

A person having a grand mal seizure should be helped to lie down. Tight clothing should be loosened. A soft, flat object like a towel or the palm of a hand should be placed under the person's head. Forcing a hard object into the mouth of someone having a grand mal seizure could cause injuries or breathing problems. If the person's mouth is open, placing a folded cloth or other soft object between his teeth will protect his tongue. Turning his head to the side will help him breathe. After a grand mal seizure has ended, the person who had the seizure should be told what has happened and reminded of where he is.

Alternative treatment

Stress increases seizure activity in 30% of people who have epilepsy. Relaxation techniques can provide some sense of control over the disorder, but they should never be used instead of anti-seizure medication or used without the approval of the patient's doctor. **Yoga**, **meditation**, and favorite pastimes help some people relax and manage stress more successfully. **Biofeedback** can teach adults and older adolescents how to recognize an aura and what to do to stop its spread. Children under 14 are not usually able to understand and apply principles of biofeedback. **Acupuncture** treatments (acupuncture needles inserted for a few minutes or left in place for as long as 30 minutes) make some people feel pleasantly relaxed. **Acupressure** can have the same effect on children or on adults who dislike needles.

Aromatherapy involves mixing aromatic plant oils into water or other oils and massaging them into the skin or using a special burner to waft their fragrance throughout the room. Aromatherapy oils affect the body and the brain, and undiluted oils should never be applied directly to the skin. Ylang ylang, chamomile,

KEY TERMS

Acupressure—Needleless acupuncture.

Acupuncture—An ancient Chinese method of relieving pain or treating illness by piercing specific areas of the body with fine needles.

Biofeedback—A learning technique that helps individuals influence automatic body functions.

Epileptologist—A physician who specializes in the treatment of epilepsy.

or lavender can create a soothing mood. People who have epilepsy should not use rosemary, hyssop, sage or sweet fennel, which seem to make the brain more alert.

Dietary changes that emphasize whole foods and eliminate processed foods may be helpful. Homeopathic therapy also can work for people with seizures, especially constitutional homeopathic treatment that acts at the deepest levels to address the needs of the individual person.

Prognosis

People who have epilepsy have a higher-than-average rate of **suicide**; sudden, unexplained **death**; and drowning and other accidental fatalities.

Benign focal epilepsy of childhood and some absence seizures may disappear in time, but remission is unlikely if seizures occur several times a day, several times in a 48-hour period, or more frequently than in the past.

Seizures that occur repeatedly over time and always involve the same symptoms are called stereotypic seizures. The probability that stereotypic seizures will abate is poor.

About 85% of all seizure disorders can be partially or completely controlled if the patient takes anti-seizure medication according to directions; avoids seizure-inducing sights, sounds, and other triggers; gets enough sleep; and eats regular, balanced meals.

Anyone who has epilepsy should wear a bracelet or necklace identifying his seizure disorder and listing the medication he takes.

Prevention

Eating properly, getting enough sleep, and controlling stress and fevers can help prevent seizures. A person who has epilepsy should be careful not to

hyperventilate. A person who experiences an aura should find a safe place to lie down and stay there until the seizure passes. Anticonvulsant medications should not be stopped suddenly and, if other medications are prescribed or discontinued, the doctor treating the seizures should be notified. In some conditions, such as severe head injury, brain surgery, or **subarachnoid hemorrhage**, anticonvulsant medications may be given to the patient to prevent seizures.

Resources

PERIODICALS

Dilorio, Colleen, et al. "The Epilepsy Medication and Treatment Complexity Index: Reliability and Validity Testing." *Journal of Neuroscience Nursing* June 2003: 155–158.

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Finn, Robert. "Partial Seizures Double Risk of Sleep Disturbances (Consider in Diagnosis, Management)." *Clinical Psychiatry News* June 2003: 36–41.

Liu, Yeou-Mei Christiana, et al. "A Prospective Study: Growth and Nutritional Status of Children Treated With the Ketogenic Diet." *Journal of the American Dietetic Association* June 2003: 707.

"New Drug Candidate Shows Promise." *Clinical Trials Week* April 7, 2003: 26.

ORGANIZATIONS

American Epilepsy Society. 342 North Main Street, West Hartford, CT 06117-2507. (860) 586-7505. <<http://www.aesnet.org>>.

Epilepsy Concern International Service Group. 1282 Wynnewood Drive, West Palm Beach, FL 33417. (407) 683-0044.

Epilepsy Foundation of America. 4351 Garden City Drive, Landover, MD 20785. (800) 332-1000. <<http://www.efa.org>>.

Epilepsy Information Service. (800) 642-0500.

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Selective abortion see **Abortion, selective**

Selective mutism see **Mutism**

Selective serotonin reuptake inhibitors

Definition

Selective serotonin reuptake inhibitors are medicines that relieve symptoms of depression.

Purpose

Selective serotonin reuptake inhibitors are used to treat serious, continuing depression that interferes with a person's ability to function. Like other **anti-depressant drugs**, they help reduce the extreme sadness, hopelessness, and lack of interest in life that are typical in people with depression. Selective serotonin reuptake inhibitors also are used to treat **panic disorder**, obsessive compulsive disorder (OCD), and have shown promise for treating a variety of other conditions, such as **premenstrual syndrome**, eating disorders, **obesity**, **self-mutilation**, and **migraine headache**.

As of late 2003, SSRIs have been found to have other off-label applications, including treatment of **premature ejaculation** and **diabetic neuropathy**.

Description

Selective serotonin reuptake inhibitors, also known as SSRIs or serotonin boosters, are thought to work by correcting chemical imbalances in the brain. Normally, chemicals called neurotransmitters carry signals from one nerve cell to another. These chemicals are constantly being released and taken back up at the ends of nerve cells. Selective serotonin reuptake inhibitors act on one particular neurotransmitter, serotonin, reducing its re-entry into nerve cells and thus allowing serotonin to build up. Although scientists are not exactly sure how it works, serotonin is involved in the control of moods, as well as other functions such as sleep, body temperature, and appetite for sweets and other carbohydrates. Somehow, drugs that prevent the uptake of serotonin improve the moods of people with serious depression, OCD, and some types of **anxiety disorders**.

Selective serotonin reuptake inhibitors are available only with a doctor's prescription and are sold in tablet, capsule, and liquid forms. Commonly used selective serotonin reuptake inhibitors are fluoxetine (Prozac), paroxetine (Paxil), sertraline (Zoloft), and fluvoxamine (Luvox).

Recommended dosage

The recommended dosage depends on the type of **SSRI** and the type and severity of depression for which it is being taken. Dosages may be different for different people. It is important for people taking SSRIs to take the drug exactly as prescribed. Taking larger or more frequent doses or taking the drug for longer than directed, for example, can cause unwanted effects.

SSRIs are about as effective as other antidepressants. About 60–80% of people taking the drugs as directed will find that their conditions improve. However, it may take four weeks or more for the effects of this medicine to be felt. Therefore, when people begin SSRI therapy, it is important to continue taking the medication, even if an improvement in mood doesn't begin immediately.

People who take SSRIs should ask their doctors about how to stop taking the medication. Usually, doctors advise patients to taper down gradually to reduce the chance of withdrawal symptoms or SSRI discontinuation syndrome.

SSRIs may be taken with food to prevent stomach upset.

Precautions

There have been reports that some patients taking SSRIs have an increase in thoughts about **suicide**. It is not clear whether the medicine causes this effect because suicidal thoughts are very often a part of depression itself. While some patients may experience worsening of such thoughts early in the treatment of their depression, there is no credible evidence that SSRIs alone cause people to become suicidal or violent.

Serious and possibly life-threatening reactions may occur when SSRIs are used in combination with **monoamine oxidase inhibitors** (MAO inhibitors), such as Nardil and Parnate, which also are used to treat depression. These reactions also are possible when a person stops taking an SSRI and immediately begins taking an MAOI. SSRIs and MAO inhibitors should never be taken at the same time. When switching from an SSRI to an MAOI or vice versa, it may be necessary to allow two to five weeks or more between stopping one and starting the other. The physician prescribing the medications should tell the patient exactly how much time to allow before beginning the other medication.

People with a history of manic disorders should use any antidepressant, including an SSRI, with caution.

It is important to see a doctor regularly while taking SSRIs. The doctor will check to make sure the medicine is working as it should and will watch for unwanted side effects. The doctor may also need to adjust the dosage during this period.

Some people feel drowsy, dizzy, or lightheaded when using SSRIs. The drugs may also cause blurred vision in some people. Since SSRIs can sometimes cause drowsiness, driving or operating heavy

machinery should be undertaken cautiously, particularly when the person first begins taking the medication.

These medicines make some people feel light-headed, dizzy, or faint when they get up after sitting or lying down, a condition known as **orthostatic hypotension**. People may try to lessen the problem by getting up gradually and holding onto something for support if possible. If the problem is severe or doesn't improve, the patient should discuss it with his or her doctor.

Because SSRIs work on the central nervous system, they may add to the effects of alcohol and other drugs that slow down the central nervous system, such as **antihistamines**, cold medicine, allergy medicine, sleep aids, medicine for seizures, tranquilizers, some **pain** relievers, and **muscle relaxants**. They may also add to the effects of anesthetics, including those used for dental procedures. Anyone taking SSRIs should check with his or her doctor before taking any of the drugs mentioned above.

SSRIs may occasionally cause **dry mouth**, although this side effect is much more common with an older class of antidepressants known as tricyclics. To temporarily relieve the discomfort, doctors sometimes suggest chewing sugarless gum, sucking on sugarless candy or ice chips, or using saliva substitutes, which come in liquid and tablet forms and are available without a prescription. If the problem continues for more than two weeks, check with a doctor or dentist. Mouth dryness that continues over a long time may contribute to **tooth decay** and other dental problems.

Changes in sexual functioning are among the more common side effects with SSRIs. Depending on the particular SSRI prescribed, 8–15% of patients may report these side effects. The most common problem for men is delayed ejaculation. Women may be unable to have orgasms. A doctor should be contacted if any changes in sexual functioning occur.

Special conditions

People with certain medical conditions or who are taking certain other medicines can have problems if they take SSRIs. Before taking these drugs, a patient should let the doctor know about any of these conditions:

ALLERGIES. Anyone who has had unusual reactions to SSRIs in the past should let his or her doctor know before taking the drugs again. The doctor should also be told about any **allergies** to foods, dyes, preservatives, or other substances.

PREGNANCY. In studies of laboratory animals, some SSRIs have caused **miscarriage** and other

problems in pregnant females and their offspring. However, at least two studies in humans (by Pastuszak in 1993 and Kuhlman in 1998) have shown SSRIs to be safe during **pregnancy**, and newer studies done in 2003 have reported that SSRIs do not appear to increase the risks of **birth defects** in the offspring. Still, women who are pregnant or who may become pregnant should check with their doctors before using SSRIs.

BREASTFEEDING. SSRIs pass into breast milk and some may occasionally cause unwanted side effects in nursing babies whose mothers take the drugs. These effects include **vomiting**, watery stools, crying, and sleep problems. Women who are breastfeeding should talk to their doctors about the use of SSRIs. They may need to switch to a different medicine while breastfeeding. If SSRIs must be taken, it may be necessary to stop breastfeeding while being treated with these drugs. However, several studies in people (for example, Yoshida in 1998) have indicated that SSRIs in breast milk have no effect on infant development.

DIABETES. SSRIs may affect blood sugar levels. People with diabetes who notice changes in their blood or urine tests while taking this medicine should check with their doctors.

OTHER MEDICAL CONDITIONS. Before using SSRIs, people with any of these medical problems should make sure their doctors are aware of their conditions: diabetes, **kidney disease**, **liver disease**, seizure disorders, current or past drug **abuse** or dependence, or diseases or conditions that affect the metabolism or blood circulation.

Side effects

The most common side effects are **anxiety** and nervousness (reported by 5–13% of people taking various SSRIs), tremor (5–14%), trouble sleeping (2–8%), tiredness or weakness (4–15%), **nausea** (11–26%), **diarrhea** (11–26%), **constipation** (1–8%), loss of appetite (3–18%), weight loss (1–13%), dry mouth (10–22%), **headache** (1–5%), sweating (5–9%), trouble urinating (1–2%), and decreased sexual ability (8–15%). Many of these problems diminish or disappear as the body adjusts to the drug and do not require medical treatment unless they interfere with normal activities. Persistent problems, such as **sexual dysfunction**, should be discussed with the doctor.

More serious side effects are possible, but extremely rare. People taking SSRIs who notice unusual joint or muscle pain; breathing problems; chills or **fever**; excessive excitement, fast talking, or actions

that are out of control; or mood swings should contact their doctors. People who develop skin **rashes** or **hives** after taking an SSRI should stop taking the medication and contact their doctors as soon as possible. Other rare side effects may occur. Anyone who has unusual symptoms after taking an SSRI should get in touch with his or her doctor.

Side effects may continue for some time after treatment with this medicine ends. How long the effects continue depends on how long the drug was taken and how much of it was used. In most cases, doctors recommend that patients taper off SSRIs rather than abruptly stopping them, because of the risk of developing a condition known as SSRI discontinuation syndrome. This syndrome can mimic serious illness. People who experience agitation, confusion, or restlessness; **dizziness** or lightheadedness; vision problems; tremor; sleep problems; unusual tiredness or weakness; **nausea and vomiting** or diarrhea; headache; excessive sweating; runny nose; or muscle pain for more than a few days after stopping or tapering an SSRI should consult their doctors.

Interactions

SSRIs may interact with other medicines. When this happens, the effects of one or both of the drugs may change or the risk of side effects may be greater. Anyone who takes SSRIs should let the doctor know about all other medicines he or she is taking. Among the drugs that may interact with SSRIs are:

- such central nervous system (CNS) depressants as medicine for allergies, colds, hay fever, and **asthma**; sedatives; tranquilizers; prescription pain medicine; muscle relaxants; medicine for seizures; sleep aids; **barbiturates**; and anesthetics.
- blood thinners (anticoagulants)
- such monoamine oxidase inhibitors (MAOIs) as Nardil or Parnate, used to treat conditions including depression and Parkinson's disease
- the antiseizure drug phenytoin (Dilantin)
- the food supplement (and sleep aid) tryptophan, which has been withdrawn from the United States market, but may be found in some herbal preparations
- digitalis and other heart medicines
- **St. John's wort** (*Hypericum perforatum*). St. John's wort is a herb used in Europe and the United States to relieve mild-to-moderate symptoms of depression. Research indicates that it acts as an SSRI and not as an MAO inhibitor, as previously believed. People

KEY TERMS

Anesthetic—Medicine that causes a loss of feeling, especially of pain. Some anesthetics also cause a loss of consciousness.

Anxiety—Worry or tension in response to real or imagined stress, danger, or dreaded situations. Physical reactions, such as fast pulse, sweating, trembling, fatigue, and weakness may accompany anxiety.

Central nervous system—The brain and spinal cord.

Depression—A mental condition in which people feel extremely sad and lose interest in life. People with depression may also have sleep problems and loss of appetite and may have trouble concentrating and carrying out everyday activities.

Metabolism—All the physical and chemical changes that occur in cells to allow growth and maintain body functions. These include processes that break down substances to yield energy and processes that build up other substances necessary for life.

Obsessive-compulsive disorder—An anxiety disorder in which people cannot prevent themselves from dwelling on unwanted thoughts, acting on urges, or performing repetitious rituals, such as washing their hands or checking to make sure they turned off the lights.

Off-label application—The use of a prescription medication to treat conditions outside the indications approved by the Food and Drug Administration (FDA).

Premenstrual syndrome—(PMS) A set of symptoms that occur in some women 2–14 days before they begin menstruating each month. Symptoms include headache, fatigue, irritability, depression, abdominal bloating, and breast tenderness.

who are using St. John's wort to relieve depression should not take a prescription SSRI at the same time.

The list above does not include every drug that may interact with SSRIs. Patients should be sure to check with a doctor or pharmacist before combining SSRIs with any other prescription or nonprescription (over-the-counter) medicine, including herbal preparations.

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American Psychiatric Association (APA). 1400 K Street, NW, Washington, DC 20005. (888) 357-7924. <<http://www.psych.org>>.

National Center for Complementary and Alternative Medicine (NCCAM) Clearinghouse. P.O. Box 7923, Gaithersburg, MD 20898-7923. (888) 644-6226. <<http://nccam.nih.gov>>.

National Institute of Mental Health (NIMH). 6001 Executive Boulevard, Room 8184, MSC 9663, Bethesda, MD 20892-9663. (301) 443-4513. <www.nimh.nih.gov>.

U. S. Food and Drug Administration (FDA). 5600 Fishers Lane, Rockville, MD 20857. (888) 463-6332. <<http://www.fda.gov>>.

Nancy Ross-Flanigan
Rebecca J. Frey, PhD

Self-mutilation

Definition

Self-mutilation is a general term for a variety of forms of intentional self-harm without the wish to die. Cutting one's skin with razors or knives is the most common pattern of self-mutilation. Others include

biting, hitting, or bruising oneself; picking or pulling at skin or hair; burning oneself with lighted cigarettes, or amputating parts of the body.

Description

Self-mutilation has become a major public health concern as its incidence appears to have risen since the early 1990s. One source estimates that 0.75% of the general American population practices self-mutilation. The incidence of self-mutilation is highest among teenage females, patients diagnosed with borderline personality disorder, and patients diagnosed with one of the dissociative disorders. Over half of self-mutilators were sexually abused as children, and many also suffer from eating disorders.

Self-mutilation should not be confused with current fads for **tattoos** and body **piercing**. In some cases, however, it may be difficult to distinguish between an interest in these fads and the first indications of a disorder.

The relationship of self-mutilation to **suicide** is still debated even though statistics show that nearly 50% of individuals who injure themselves also attempt suicide at some point in their lives. Many researchers think that suicide attempts reflect feelings of rejection or hopelessness, while self-mutilation results from feelings of shame or a need to relieve tension.

Causes and symptoms

Several different theories have been proposed to explain self-mutilation:

- self-mutilation is an outlet for strong negative emotions, especially anger or shame, that the person is afraid to express in words or discuss with others.
- self-mutilation represents anger at someone else directed against the self.
- self-mutilation relieves unbearable tension or **anxiety**. Many self-mutilators do report feeling relief after an episode of self-cutting or other injury.
- self-mutilation is a technique for triggering the body's biochemical responses to **pain**. **Stress** and trauma release endorphins, which are the body's natural pain-killing substances
- self-mutilation is a way of stopping a dissociative episode. Dissociation is a process in which the mind splits off, or dissociates, certain memories and thoughts that are too painful to keep in conscious awareness. Some people report that they feel

“numb” or “dead” when they dissociate, and self-injury allows them to feel “alive.”

- self-mutilation is a symbolic acting-out of the larger culture’s mistreatment of women. This theory is sometimes offered to explain why the great majority (about 75%) of self-mutilators are girls and women

The symptoms of self-mutilation typically include wearing long-sleeved or baggy clothing, even in hot weather; and an unusual need for privacy. Self-mutilators are often hesitant to change their clothes or undress around others. In most cases the person has also shown signs of depression.

Diagnosis

Self-mutilation is usually diagnosed by a psychiatrist or psychotherapist. A family practitioner or nurse who notices **scars**, **bruises**, or other physical evidence of self-injury may refer the person to a specialist for evaluation.

Treatment

Persons who mutilate themselves should seek treatment from a therapist with some specialized training and experience with this behavior. Most self-mutilators are treated as outpatients, although there are some inpatient programs, such as S.A.F.E., for adolescent females. A number of different treatment approaches are used with self-mutilators, including psychodynamic psychotherapy, **group therapy**, journaling, and behavioral therapy.

Although there are no medications specifically for self-mutilation, antidepressants are often given, particularly if the patient meets the diagnostic criteria for a depressive disorder.

Alternative treatment

Mindfulness training, which is a form of **meditation**, has been used to teach self-mutilators to observe and identify their feelings in order to have some control over them.

Prognosis

The prognosis depends on the presence and severity of other emotional disorders, and a history of sexual **abuse** and/or suicide attempts. In general, teenagers without a history of abuse or other disorders have a good prognosis. Patients diagnosed with borderline personality disorder and/or a history of

KEY TERMS

Borderline personality disorder (BPD)—A pattern of behavior characterized by impulsive acts, intense but chaotic relationships with others, identity problems, and emotional instability.

Dissociation—The splitting off of certain mental processes from conscious awareness.

Dissociative disorders—A group of mental disorders in which dissociation is a prominent symptom. Patients with dissociative disorders have a high rate of self-mutilation.

Endorphins—Pain-killing substances produced in the human body and released by stress or trauma. Some researchers think that people who mutilate themselves are trying to trigger the release of endorphins.

attempted suicide are considered to have the worst prognosis.

Prevention

Some society-wide factors that influence self-mutilation, such as the high rate of sexual abuse of children and media stereotypes of women, are difficult to change. In general, however, young people who have learned to express themselves in words or through art and other creative activities are less likely to deal with painful feelings by injuring their bodies.

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American Psychiatric Association. 1400 K Street, NW. Washington, DC 20005. (202) 682-6220. < <http://www.psych.org> >.

Focus Adolescent Services. (877) 362-8727. < <http://www.focusas.com> >.

National Institute of Mental Health. 5600 Fishers Lane, Rockville, MD 20857. (301) 443-4513. Fax: (301) 443-4513. < <http://www.nimh.nih.gov> >.

Rebecca J. Frey, PhD

Semen analysis

Definition

Semen analysis evaluates a man's sperm and semen. It is done to discover cause for infertility and to confirm success of **vasectomy**.

Purpose

Semen analysis is an initial step in investigating why a couple has been unable to conceive a child. Abnormalities of sperm and semen can cause male **infertility**. Semen is the thick yellow-white male ejaculate containing sperm. Sperm are the male sex cells that fertilize the female egg (ovum). They contain the genetic information that the male will pass on to a child.

Vasectomy is an operation done to sterilize a man by stopping the release of sperm into semen. Success of vasectomy is confirmed by the absence of sperm in semen.

Description

The semen analysis test is usually done manually, though computerized test systems are available. Many laboratories base their procedures on standards published by the World Health Organization (WHO).

The volume of semen in the entire ejaculate is measured. The appearance, color, thickness, and pH is noted. A pH test looks at the range from a very acid solution to a very alkaline solution. Semen, like many other body fluids, has a standard pH range that would be considered optimal for fertilization of the egg to take place. The thick semen is then allowed to liquify; this usually takes 20-60 minutes.

Drops of semen are placed on a microscope slide and examined under the microscope. Motility, or movement, of 100 sperm are observed and graded in categories, such as rapid progressive or immotile.

The structure of sperm (sperm morphology) is assessed by carefully examining sperm for abnormalities in the size and shape in the head, tail, and neck regions. WHO standards define normal as a specimen with less than 30% abnormal forms. An alternative classification system (Kruger's) measures the dimensions of sperm parts. Normal specimens are allowed 14% or less abnormalities.

Sperm are counted by placing semen in a special counting chamber. The sperm within the chamber are counted under a microscope. White blood cells are

recorded; these may indicate a reproductive tract infection. Laboratories may test for other biochemicals such as fructose, zinc, and citric acid. These are believed to contribute to sperm health and fertility.

Results of semen analysis for infertility must be confirmed by a second analysis seven days to three months after the first. Sperm counts may vary from day to day.

Semen analysis to confirm success of vasectomy is concerned only with discovering if sperm are still present. Semen is collected six weeks after surgery. If sperm are seen, another specimen is collected 2 to 4 weeks later. The test is repeated until two consecutive specimens are free of sperm.

Preparation

A man should collect an entire ejaculate, by masturbation, into a container provided by his physician. To examine the best quality sperm, the specimen must be collected after two to three days of sexual abstinence, but not more than five to seven days. The specimen must not come into contact with any spermicidal agents used by a female partner for birth control purposes. The man should not have alcohol before the test.

A semen specimen to investigate infertility must be brought to the testing laboratory within one hour of obtaining it. Timing is not as critical for the postvasectomy test but the semen must be kept at body temperature. The most satisfactory sample is one obtained in the lab rather than at home.

Normal results

WHO standards have established these normal values:

- volume less than or equal to 2.0 mL
- sperm count greater than or equal to 20 million per mL
- motility (movement of the sperm) value is greater than or equal to 50% with forward progression, or greater than or equal to 25% with rapid progression within 60 minutes of ejaculation
- morphology greater than or equal to 30% with normal forms
- white blood cell count less than 1 million per mL.

If infertility continues, despite normal semen analysis and female studies, further tests are done to evaluate sperm function.

KEY TERMS

Infertility—The inability of a man and woman to conceive a child after 12 months of unprotected sexual intercourse.

Morphology—The size and shape of sperm.

Motility—The movement of sperm within the semen.

Abnormal results

Abnormalities of semen volume and liquidity, and sperm number and morphology decrease fertility. These abnormalities may be inherited or caused by a hormone imbalance, medications, or a recent infection. Further tests may be done to determine the cause of abnormalities.

Resources

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Nancy J. Nordenson

Senile tremor *see* **Tremors**

Seniors' health

Definition

Seniors' health refers to the physical and mental conditions of senior citizens, those who are in their 60s and older. The proportion of people age 65 years and older in the United States is on the rise and will continue to increase through 2030.

Purpose

For a senior, the **aging** process and a person's lifestyle will affect health. People who maintain a healthy weight, exercise regularly, eat nutritionally, and don't smoke reduce the risk for many health conditions. This wellness allows people to live longer and to remain independent for more years. **Smoking, obesity** (excess weight), and lack of exercise shorten life and increase the risk for many health conditions. According to a 2003 report, about 80% of people in

the United States age 65 and older have at least one chronic (long-lasting) condition and 50% have two.

Diet and exercise

Proper diet and regular **exercise** form the foundation of senior health. A nutritional diet and physical activity can help prevent diseases such as **cancer, stroke**, heart disease, and diabetes. A healthy diet also can help manage diabetes, high blood pressure, and heart disease.

As people age, there is more of a need to exercise on a regular basis. According to the American Heart Association, the inactive person loses from 3–5% of muscle fiber each decade after age 30. That loss would total 30% of lost muscle fiber at age 60. Exercise helps to boost muscle strength. It can help improve balance and coordination, and therefore help to prevent falls.

Organizations including the heart association advise that regular physical activity helps prevent bone loss (**osteoporosis**) and the risk of conditions such as heart disease, Type II diabetes, **colon cancer, stress**, and depression. In addition, exercise can help extend the lives of people with conditions such as diabetes, high blood pressure, and high cholesterol. Good health later in life helps to prevent serious illness or **death** from common infections as well. If a senior catches the flu, for instance, it can have more detrimental effects than in a healthier, younger person. When the **SARS** outbreak occurred in 2002 and 2003, clinicians expressed concern about the elderly Americans and again expressed the importance of diet and exercise. As people age, their immune system response weakens. Seniors need to be proactive in keeping their systems strong.

Osteoporosis

Osteoporosis is a condition in which bones become less dense (solid). Bones become brittle, thinner, and break easily. Although osteoporosis is associated with aging, it is only the risk of osteoporosis that increases as a person ages. It is linked to approximately 70% of bone fractures in people age 46 and older. According to the National Institutes of Health (NIH), one out of two women over age 50 will experience an osteoporosis-related fracture. So will one out of eight men over 50.

Osteoporosis is associated primarily with the changes that occur to women during menopause. During **menopause**, there is a decrease in the level of estrogen, the hormone that helps maintain bone mass. Other causes of osteoporosis include lack of exercise and a diet deficient in vitamin D.

Osteoporosis is largely preventable, however, research released in 2003 said that evidence is increasing to suggest that the condition starts as far back as in the womb. If this is true, it still is preventable, but by the behavior of the mother carrying a child. More research needs to be done, but it is clear that childhood growth rates are linked to hip **fractures** that occur decades later.

Osteoarthritis

Osteoarthritis is a joint disease in which cartilage wears out and bones rub against each other. This condition can occur gradually over time as activities performed throughout the years cause wear on joints. In addition, bones thin as a person ages.

Excess weight and injuries can aggravate this condition. About 16 million Americans experience some form of osteoarthritis. It generally affects the neck, fingers, lower back, knees, and toes. Symptoms include **pain**, stiffness, swelling, and creaking. The pain may disrupt sleep, and joint stiffness may make it difficult for a person to dress.

Falls

More than two million Americans each year fall and experience serious injuries, according to the American Academy of Otolaryngology-Head and Neck Surgery. For seniors, fall-related injuries can reduce mobility and hinder independence.

As people age, their reflexes slow down so it may be more difficult to prevent a fall. Deteriorating vision and hearing can affect balance, which can cause an accidental fall. Furthermore, conditions such as arthritis, **dizziness**, and sleeping disorders can increase the likelihood of a fall. In addition, a person may fall at the start of a condition such as a stroke or **heart attack**.

Falls can result in broken bones or fractures because bones are weakened by osteoporosis. In addition, healing takes longer. Head injuries could affect sight and hearing. Injuries sustained during falls could reduce an active person's mobility and independence.

Vision

Eyesight changes as people age. Generally, people are in their 40s when they experience presbyopia, a form of farsightedness. This is a progressive condition involving a decrease in the eye's ability to focus on close objects (near vision). By age 65, little near focusing ability remains.

Glaucoma is a condition caused by pressure from the build-up of a large amount of fluid in the eye. This progressive condition is often seen in people in their 50s. It starts with the gradual loss of peripheral vision. If not treated, it can lead to some vision loss.

People in their 60s may experience the first signs of age-related macular degeneration (AMD). It is a progressive condition affecting the retina. The macula in the retina distinguishes detail. Degeneration in the macula could cause scarring and a gradual reduction in vision. The person experiences a circle of blindness, an area of sightlessness that grows as the condition progresses.

More than half of people age 65 or older will be diagnosed with **cataracts**. Cataract refers to the loss of the transparency in the lens of the eye. As the loss progresses, the person is able to see less detail. This condition generally affects both eyes.

Hearing

Presbycusis, age-related **hearing loss**, is a progressive condition. It usually starts with a difficulty in hearing high-frequency sound such as people talking. A senior has less trouble with low-frequency tones. Background noise will make it even more difficult to hear. Presbycusis affects approximately 25% of people between the ages of 65 and 75 and half of those over 75. Many people diagnosed with this condition say they have lost hearing in both ears. They also report feelings of dizziness and that they experience a ringing in their ears.

Sleep disorders

Sleep patterns change when a person ages. Many people in their 60s and 70s experience less time in the stages of deep sleep known as delta sleep. Despite this change, many healthy older people don't experience **sleep disorders**. Overall health plays a role in whether a senior experiences trouble sleeping.

Obesity is linked to **snoring** and sleep apnea. Snoring can turn into apnea. A person with apnea stops breathing for up to one minute until the brain restarts the breathing process. This action could be repeated several hundred times each night.

Furthermore, a senior's sleep can be disrupted by conditions such as arthritis, osteoporosis, and **Alzheimer's disease**. Insomnia, or the inability to stay asleep, is a symptom of conditions including depression, **anxiety**, chronic pain, and restless leg syndrome (RLS).

RLS involves movement of legs when a person is at rest. The person moves legs in response to a **tingling** sensation in the upper leg, calf, or foot. In other cases, legs move involuntarily. Sensations that trigger movement can re-occur within seconds.

A person with RLS is likely to have PLMD (periodic limb movement disorder). A sleeping person with this condition will kick legs or move arms repeatedly. These involuntary movements can last from 20 seconds to an hour. Approximately 45% of the elderly have a mild form of PLMD, according to the National Sleep Foundation.

The cause of these disorders is not known. They are thought to be caused by a chemical reaction in the brain. In addition, the conditions may be hereditary.

Mental health

While age has little effect on the mind, social and emotional factors affect an older person's health. After a lifetime of work or raising a family, retirement brings several challenges. A person who has been identified for years by a profession may experience a sense of lost identity.

A senior may find that the thinking process has changed. Learning something new may take longer. However, older people have excellent recall of new information.

Memory loss may be a concern, particularly since this is a symptom of Alzheimer's disease.

Dementia

Alzheimer's disease is a form of **dementia**, a condition in which mental abilities decline. Symptoms of dementia include memory loss that goes beyond forgetting a word or where an item was placed. The person with dementia may never recognize family members or remember how to perform functions such as preparing a meal. Sometimes they experience a change in personality, with some uncharacteristic aggression or **paranoia**.

Alzheimer's disease is the most prevalent form of dementia. Although the cause of this condition is not known, the risk of Alzheimer's increases as a person ages. In 2000, the condition affected one in 15 people over the age of 65. The ratio rises to one in three people age 85 and older.

Alzheimer's is a progressive condition. In most cases, after five to eight years, a patient with this condition is unable to perform basic functions. There is no known cure for Alzheimer's. However, as of 2003, the U.S. Food and Drug Administration had approved four medications that could help delay the degenerative process.

Leading causes of death in persons 65 and older

Cause of death	Number of deaths	Death rate (per 100,000 population)	Percentage of all deaths in those >65 years old
All causes	1,542,493	4,963.2	100.0
Heart disease	594,858	1,914.0	38.6
Malignant neoplasms, including neoplasms of lymphatic and hematopoietic tissues	345,387	1,111.3	22.4
Cerebrovascular diseases	125,409	403.5	8.1
Chronic obstructive pulmonary disease and associated conditions	72,755	234.1	4.7
Pneumonia and influenza	70,485	226.8	4.6
Diabetes mellitus	35,523	114.3	2.3
Accidents and adverse effects	26,213	84.3	1.7
Motor vehicle accidents	7,210	23.2	0.5
All other accidents and adverse effects	19,003	61.1	1.2
Nephritis, nephrotic syndrome, and nephrosis	17,306	55.7	1.1
Atherosclerosis	17,158	55.2	1.1
Septicemia	15,351	49.4	1.0
All other causes, residual	222,048	2,045.9	14.4

Precautions

A health condition may result in a doctor recommending against some forms of exercise. However, even if a person can't jog, other forms of exercise include those designed for people in wheelchairs and those who are bedridden.

Treatments for menopause and osteoporosis include Raloxifene, a medication that may cause **blood clots**.

Description

The cost of treatment varies. Cost of medical treatment will be determined by the type of procedure and whether a person has medical insurance. Health plan and Medicare coverage and copayments impact an individual's cost for various preventions and treatments.

Nutrition

Nutrition plays an important role in senior health. Not only does a well-balanced diet keep a person from becoming obese, that same diet is a safeguard against health conditions that seniors face. Proper diet can help prevent a condition like diabetes or keep it from worsening.

The senior diet should consist of foods that are low in fat, particularly saturated fat and cholesterol. A person should choose foods that provide nutrients such as iron and calcium. Other healthy menu choices include:

- fish, skinless poultry, and lean meat.
- proteins such as dry beans (red beans, navy beans, and soybeans), lentils, chickpeas, and peanuts.
- low-fat dairy products
- vegetables, especially those that are dark green and leafy
- citrus fruits or juices, melons, and berries
- whole grains like wheat, rice, oats, corn, and barley
- whole grain breads and cereals

Exercise

Physical activity should be rhythmic, repetitive, and should challenge the circulatory system. It also should be enjoyable so that a senior gets in the habit of exercising regularly for 30 minutes each day. It may be necessary to check with a doctor to determine the type of exercise that can be done.

Walking is recommended for weight loss, stress release, and many other conditions. Brisk walking is said to produce the same benefits as jogging. Other forms of exercise can include gardening, bicycling, hiking, swimming, dancing, skating or ice-skating. If weather prohibits outdoor activities, a person can work out indoors with an exercise video.

Exercise also offers a chance to socialize. In some cities, groups of seniors meet for regular walks at shopping malls. Senior centers offer exercise classes ranging from line dancing to belly dancing.

Costs for exercise range from the price of walking shoes to the fees for joining a gym.

Osteoporosis

Prevention is the best method of treating osteoporosis. Methods of preventing osteoporosis include regular weight-bearing exercise such as walking, jogging, weight lifting, **yoga**, and stair climbing.

People should not smoke since smoking makes the body produce less estrogen. Care should be taken to avoid falling.

Diet should include from 1,000–1,300 mg. of calcium each day. Sources of calcium include:

- leafy, dark-green vegetables such as spinach, kale, mustard greens, and turnip greens
- low-fat dairy products such as milk, yogurt, and cheeses such as cheddar, Swiss, mozzarella, and parmesan; also helpful are foods made with milk such as pudding and soup
- canned fish such as salmon, sardine, and anchovies
- tortillas made from lime-processed corn
- tofu processed with calcium-sulfate
- calcium and vitamin D tablets

MEDICAL TREATMENT. An x ray will indicate bone loss when much of the density has decreased. A more effective way of detecting osteoporosis is the DEXA-scan (dual-energy x-ray absorptiometry). This whole-body scan will indicate whether a person is at risk for fractures. It could be useful for people at risk for osteoporosis as well as women near the age of menopause or older. People should ask their doctors about whether this test is needed.

During menopause, a woman loses estrogen. A pill or skin patch containing estrogen and progesterone eases symptoms of menopause has been used to treat osteoporosis. This treatment is known as **hormone replacement therapy (HRT)**. In 2002, the Women's Health Initiative found that HRT produced harmful effects in postmenopausal women, including increased incidence of **breast cancer**, heart disease and dementia. The effects were bad enough to stop the study. In 2003, researchers were looking for alternatives to HRT for women who had been using the hormones for osteoporosis. Until an alternative is identified, women and physicians have been advised to closely weigh the risks and benefits of hormone therapy. Several drugs are available to help reduce the risk of fractures in seniors with osteoporosis. In 2003, the FDA approved a new treatment option called Teriparatide. Some alternative treatments show promise in studies, including SAME, (S-adenosylmethionine). However, long-term safety and effectiveness of SAME have yet to be established.

Osteoarthritis

Treatments for osteoarthritis range from preventative measures such as walking to joint replacement surgery. Treatment costs vary from no cost for

soaking a joint in cold water, the price of over-the-counter remedies to fees for surgery.

Preventive and maintenance remedies include low-impact exercise such as swimming and walking, along with maintaining proper posture. Nutritional aids include foods rich in vitamin C such as citrus fruits and broccoli. Also recommended is daily consumption of 400 international units of Vitamin E. Cutting back on fats, sugar, salt, cholesterol, and alcohol helps relieve the symptoms of osteoarthritis.

HOME REMEDIES AND PHYSICAL THERAPY. The Arthritis Foundation recommends several remedies for easing pain. To treat inflammation, a person should use a cold treatment. Methods include soaking the affected area in cold water or applying an ice pack. To soothe aches and stimulate circulation, a person applies heat to the affected area for 20 minutes. This should be done three times a day.

Over-the-counter (OTC) remedies such as **aspirin** and ibuprofen and salves containing capsaicin can be helpful. Furthermore, a doctor may recommend anti-inflammatory medications.

SURGICAL TREATMENT. If osteoarthritis is suspected, a doctor's diagnosis will include an assessment of whether joint pain is part of a patient's medical history. The doctor may take an x ray to determine the presence of cartilage loss and how much degeneration occurred.

Acupuncture may be helpful in treating mild osteoarthritis. Generally, a person should have one to two treatments a week for several weeks. Afterward, one treatment is recommended. An assessment of results should be made after 10 treatments.

In cases of severe osteoarthritis, **joint replacement** surgery or joint **immobilization** may be required. Joints are replaced with metal, plastic, or ceramic material.

Fall prevention

Fall prevention starts with regular exercise such as walking. This improves balance and muscles. The walk route should be on level ground. Other methods for preventing falls include:

- when rising from a chair or bed, a senior should move slowly to avoid dizziness
- people who smoke should quit
- shoes with low heels and rubber soles are recommended
- medications should be monitored because of side effects that increase the probability of a fall
- vision and hearing should be checked periodically

- fall-proofing the home, including the installation of lighting, especially on stairways, clearing clutter and electrical cords that can cause falls, and installing handrails and strips in bathtubs and rails on stairs.

MEDICAL TREATMENT FOR FALLS. After a fall, a senior may need first aid treatment for cuts or fractures. The doctor may evaluate whether medications cause balance problems. If indicated, the doctor may examine the patient's central nervous system function, balance, and muscle/joint function. A hearing or vision test may be ordered.

Corrective measures could include adjusting prescriptions, vision surgery or having the patient use a cane or walker.

Vision

A person diagnosed with **presbyopia** may need bifocals or reading glasses to read print that appears too small. These lenses may need to be changed as vision changes over the years. Eventually, a person relies on glasses to focus on items that are near. Other seniors who never needed corrective lenses may need to wear eyeglasses. Publishers aware of this condition produce books with large print.

A senior should schedule periodic vision exams because early treatment helps prevent or lessen a risk of cataracts or glaucoma. Diet also plays a role in vision care. Dark green vegetables like broccoli are said to help prevent cataracts from progressing. Physical exercise is thought to reduce the pressure associated with glaucoma.

Glaucoma can be treated with eyedrops. Surgery can remove cataracts. The affected lens is removed and replaced with a permanent synthetic lens called an intraocular lens. There was no successful treatment for age-related **macular degeneration** as of 2001.

Hearing

An audiologist can administer tests to determine the amount of hearing loss. Although there is no cure for presbycusis, **hearing aids** can help a senior affected by age-related hearing loss. If this treatment is not effective, the person might need to learn to read lips.

Sleep disorders

Losing weight can help with conditions such as snoring and **sleep apnea**. A doctor may advise the senior to quit smoking, reduce alcohol consumption, or to sleep on his or her side. In some cases, a doctor may refer the senior to a sleep disorder clinic. The senior may be prescribed a continuous positive airway

pressure device. Known as a CPAP, the device is placed over the nose. It sends air into the nose.

PLMD and restless leg syndrome may be treated with the prescription drug Dopar. These disorders could be signs of kidney or circulation conditions. Treatment of those conditions should end these sleeping disorders.

Insomnia treatments include exercising and treating depression, stress, and other causes of sleeplessness.

Mental health

After retirement, a senior must find activities and interests to provide a sense of fulfillment. Otherwise, feelings of loneliness and isolation can lead to depression and susceptibility to poor health.

Activities that stimulate a person physically and intellectually contribute to good health. A senior can start an exercise program, take up hobbies, take classes, or volunteer. Senior centers offer numerous activities. Lunch programs provide nutritional meals and companionship. This is important because a senior living alone may not feel motivated to prepare healthy meals.

Dementia

Diagnosis of Alzheimer's disease starts with a thorough medical examination. The doctor should administer memory tests. Blood tests may be required, as well as a CT scan or MRI scan of the brain. If Alzheimer's is diagnosed, the doctor may prescribe medication to slow down progression of this form of dementia.

As of 2003, the FDA had approved four prescription medications for treatment of Alzheimer's. Tacrine, donepezil, riviastigmine, and galantamine are cholinesterase inhibitors that enhance memory. Modest improvement was reported in clinical trials on donepezil, riviastigmine, and galantamine. Tacrine's possible side effects include liver damage, so it is seldom prescribed.

Preparation

Before beginning a weight loss or exercise program, seniors should check with their doctors. The doctor will determine whether a patient is at a healthy weight, or needs to gain or lose weight. The medical professional should be informed about a health condition or a family history of a condition like heart disease. The doctor may order a physical exam or recommend a specific exercise program.

Exercise preparation

A senior should select a form of exercise enjoyable enough to become a regular routine. Suitable clothing or equipment such as walking shoes or a bicycle helmet should be purchased. If a person is active for more than a half-hour, the American Heart Association recommends drinking water every 15 minutes.

In addition to packing a water bottle, a person should pick an exercise buddy. Exercising with a friend or a group makes the activity more enjoyable. In addition, a person is more apt to stick with a routine if a buddy is involved.

Before exercising, a warmup with slow stretching exercises is recommended. This could take longer for a senior because muscular elasticity slows down as a person ages. The exercise session should end with a cool-down that includes slow stretches.

Aftercare

Some recovery time may be needed after surgery. However, a healthy person will heal more quickly. A senior needs to maintain a schedule of regular exercise in order to remain mobile. Otherwise, a minor illness could make them dependent on others for daily care, according to the American Heart Association.

If mobility becomes limited due to a condition such as osteoarthritis, equipment like a walker and devices that make it easier to open bottles and grip cutlery can be helpful.

Risks

Exercising too long or too strenuously can be physically harmful. The over-exertion could cause the person to lose interest in exercise and put off establishing a regular routine. Experts recommend starting out slowly and building up to more intense or longer sessions. This is particularly important for a sedentary person.

Osteoporosis

The long-term effects of hormone replacement therapy have ruled this treatment out for some women.

Normal results

Seniors who stay active and eat nutritionally will be at less risk for conditions such as diabetes. A senior also should seek mental stimulation and social interaction. These provide enjoyment, boost self-esteem, and help reduce feelings of isolation and depression.

Although eyesight and hearing will weaken, glasses and hearing aids help seniors keep the senses of sight and hearing.

When surgery is required for osteoarthritis, hip replacement surgery is extremely successful. In about 98% of surgeries, flexibility returns and pain is eased. Knee replacement surgery also is effective.

If a person maintains a healthy lifestyle, the ability to avoid falls and recover from them is increased.

After a fall, seniors need to build up physical strength and the confidence needed so they don't fear falling again. Care should be taken so that seniors don't feel isolated by their injuries. Isolation could lead to decreased mobility and loss of independence.

There is no cure for Alzheimer's disease. However, several medications have proved moderately effective in stopping memory loss. Since Alzheimer's is progressive, a person diagnosed with this condition should make arrangements for the future. Finances should be taken care of and plans should be made for future care. Family should be brought into the discussion.

After diagnosis, a person should stay active for as long as possible. Not only does this help with enjoying this stage of life, activities can help to fight depression. Alzheimer and other support groups can be helpful. In addition, modifications to environment can be effective.

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< <http://www.alz.org> >.
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- American Dietetic Association. 216 W. Jackson Blvd., Chicago, IL 60606-6995. (312) 899-0040. < <http://www.eatright.org> >.
- American Heart Association. 7272 Greenville Ave., Dallas, TX 75231. (800) AHA-USA1.
< <http://www.americanheart.org> >.
- National Institute on Aging. P.O. Box 8057, Gaithersburg, MD 20898-8057. (800) 222-2225.
< <http://www.nih.gov> >.
- National Osteoporosis Foundation. 1232 22nd St., NW, Washington, DC 20037. (800) 624-BONE.
< <http://www.osteoporosis.org> >.
- National Sleep Foundation. 1522 K St., NW, Suite 500 Washington, DC 20005. Fax: (202) 347-3472. < <http://www.sleepfoundation.org> >.

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Sensory hearing loss see **Hearing loss**

Sensory integration disorder

Definition

Sensory integration disorder or dysfunction (SID) is a neurological disorder that results from the brain's inability to integrate certain information received from the body's five basic sensory systems. These sensory systems are responsible for detecting sights, sounds, smell, tastes, temperatures, **pain**, and the position and movements of the body. The brain then forms a combined picture of this information in order for the body to make sense of its surroundings and react to them appropriately. The ongoing relationship between behavior and brain functioning is called sensory integration (SI), a theory that was first pioneered by A. Jean Ayres, Ph.D., OTR in the 1960s.

Description

Sensory experiences include touch, movement, body awareness, sight, sound, smell, taste, and the pull of gravity. Distinguishing between these is the

process of sensory integration (SI). While the process of SI occurs automatically and without effort for most, for some the process is inefficient. Extensive effort and attention are required in these individuals for SI to occur, without a guarantee of it being accomplished. When this happens, goals are not easily completed, resulting in sensory integration disorder (SID).

The normal process of SI begins before birth and continues throughout life, with the majority of SI development occurring before the early teenage years. The ability for SI to become more refined and effective coincides with the **aging** process as it determines how well motor and speech skills, and emotional stability develop. The beginnings of the SI theory by Ayres instigated ongoing research that looks at the crucial foundation it provides for complex learning and behavior throughout life.

Causes and symptoms

The presence of a sensory integration disorder is typically detected in young children. While most children develop SI during the course of ordinary childhood activities, which helps establish such things as the ability for motor planning and adapting to incoming sensations, others' SI ability does not develop as efficiently. When their process is disordered, a variety of problems in learning, development, or behavior become obvious.

Those who have sensory integration dysfunction may be unable to respond to certain sensory information by planning and organizing what needs to be done in an appropriate and automatic manner. This may cause a primitive survival technique called "fright, flight, and fight," or withdrawal response, which originates from the "primitive" brain. This response often appears extreme and inappropriate for the particular situation.

The neurological disorganization resulting in SID occurs in three different ways: the brain does not receive messages due to a disconnection in the neuron cells; sensory messages are received inconsistently; or sensory messages are received consistently, but do not connect properly with other sensory messages. When the brain poorly processes sensory messages, inefficient motor, language, or emotional output is the result.

According to Sensory Integration International (SII), a non-profit corporation concerned with the impact of sensory integrative problems on people's lives, the following are some signs of sensory integration disorder (SID):

- oversensitivity to touch, movement, sights, or sounds
- underreactivity to touch, movement, sights, or sounds
- tendency to be easily distracted
- social and/or emotional problems
- activity level that is unusually high or unusually low
- physical clumsiness or apparent carelessness
- impulsive, lacking in self-control
- difficulty in making transitions from one situation to another
- inability to unwind or calm self
- poor self concept
- delays in speech, language, or motor skills
- delays in academic achievement

While research indicates that sensory integrative problems are found in up to 70% of children who are considered learning disabled by schools, the problems of sensory integration are not confined to children with learning disabilities. SID transfers through all age groups, as well as intellectual levels and socioeconomic groups. Factors that contribute to SID include: premature birth; **autism** and other developmental disorders; learning disabilities; delinquency and **substance abuse** due to learning disabilities; stress-related disorders; and brain injury. Two of the biggest contributing conditions are autism and attention-deficit hyperactivity disorder (**ADHD**).

Diagnosis

In order to determine the presence of SID, an evaluation may be conducted by a qualified occupational or physical therapist. An evaluation normally consists of both standardized testing and structured observations of responses to sensory stimulation, posture, balance, coordination, and eye movements. These test results and assessment data, along with information from other professionals and parents, are carefully analyzed by the therapist who then makes recommendations about appropriate treatment.

Treatment

Occupational therapists play a key role in the conventional treatment of SID. By providing sensory integration therapy, occupational therapists are able to supply the vital sensory input and experiences that children with SID need to grow and learn. Also referred to as a "sensory diet," this type of therapy

involves a planned and scheduled activity program implemented by an occupational therapist, with each “diet” being designed and developed to meet the needs of the child’s nervous system. A sensory diet stimulates the “near” senses (tactile, vestibular, and proprioceptive) with a combination of alerting, organizing, and calming techniques.

Motor skills training methods that normally consist of adaptive physical education, movement education, and gymnastics are often used by occupational and physical therapists. While these are important skills to work on, the sensory integrative approach is vital to treating SID.

The sensory integrative approach is guided by one important aspect—the child’s motivation in selection of the activities. By allowing them to be actively involved, and explore activities that provide sensory experiences most beneficial to them, children become more mature and efficient at organizing sensory information.

Alternative treatment

Sensory integration disorder (SID) is treatable with occupational therapy, but some alternative methods are emerging to complement the conventional methods used for SID.

Therapeutic body brushing is often used on children (not infants) who overreact to tactile stimulation. A specific non-scratching surgical brush is used to make firm, brisk movements over most of the body, especially the arms, legs, hands, back and soles of the feet. A technique of deep joint compression follows the brushing. Usually begun by an occupational therapist, the technique is taught to parents who need to complete the process for three to five minutes, six to eight times a day. The time needed for brushing is reduced as the child begins to respond more normally to touch. In order for this therapy to be effective, the correct brush and technique must be used.

A report in 1998 indicated the use of cerebral electrical stimulation (CES) as being helpful to children with conditions such as moderate to severe autistic spectrum disorders, learning disabilities, and sensory integration dysfunction. CES is a modification of Transcutaneous Electrical Nerve Stimulation (TENS) technology that has been used to treat adults with various pain problems, including arthritis and carpal tunnel syndrome. TENS therapy uses a low voltage signal applied to the body through the skin with the goal of replacing painful impressions with a massage-like sensation. A much lower

KEY TERMS

Axon—A process of a neuron that conducts impulses away from the cell body. Axons are usually long and straight.

Cortical—Regarding the cortex, or the outer layer of the brain, as distinguished from the inner portion.

Neurotransmission—When a neurotransmitter, or chemical agent released by a particular brain cell, travels across the synapse to act on the target cell to either inhibit or excite it.

Proprioceptive—Pertaining to proprioception, or the awareness of posture, movement, and changes in equilibrium and the knowledge of position, weight, and resistance of objects as they relate to the body.

Tactile—The perception of touch.

Vestibular—Pertaining to the vestibule; regarding the vestibular nerve of the ear which is linked to the ability to hear sounds.

signal is used for CES than that used for traditional TENS, and the electrodes are placed on the scalp or ears. Occupational therapists who have studied the use of CES suggest that CES for children with SID can result in improved brain activity. The device is worn by children at home for 10 minutes at a time, twice per day.

Music therapy helps promote active listening. Hypnosis and biofeedback are sometimes used, along with psychotherapy, to help those with SID, particularly older patients.

Prognosis

By providing treatment at an early age, sensory integration disorder may be managed successfully. The ultimate goal is for the individual to be better able to interact with his or her environment in a more successful and adaptive way.

Resources

ORGANIZATIONS

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Sepsis

Definition

Sepsis refers to a bacterial infection in the bloodstream or body tissues. This is a very broad term covering the presence of many types of microscopic disease-causing organisms.

Description

Sepsis is also called **bacteremia**. Closely related terms include septicemia and septic syndrome. In the general population, the incidence of sepsis is two people in 10,000. The number of deaths from sepsis each year has almost doubled in the United States since 1980 because more patients are developing the condition. There are three major factors responsible for this increase: a rise in the number of organ transplants and other surgical procedures that require suppressing the patient's immune system; the greater number of elderly people in the population; and the overuse of **antibiotics** to treat infectious illnesses, resulting in the development of drug-resistant bacteria.

Causes and symptoms

Sepsis can originate anywhere bacteria can gain entry to the body; common sites include the genitourinary tract, the liver and its bile ducts, the gastrointestinal tract, and the lungs. Broken or ulcerated skin can also provide access to bacteria commonly present in the environment. Invasive medical procedures, including dental work, can introduce bacteria or permit them to accumulate in the body. Entry points and equipment left in place for any length of time present a particular risk. **Heart valve replacement**, catheters, ostomy sites, intravenous (IV) or arterial lines, surgical **wounds**, or surgical drains are examples. IV drug users are at high risk as well.

People with inefficient immune systems, HIV infection, spinal cord injuries, or blood disorders are at particular risk for sepsis and have a higher **death** rate (up to 60%); in people who have no underlying

KEY TERMS

Bacteremia—The medical term for sepsis.

Prophylactic—Referring to medications or other treatments given to prevent disease.

chronic disease, the death rate is far lower (about 5%). The growing problem of antibiotic resistance has increased the incidence of sepsis, partly because ordinary preventive measures (such as prophylactic antibiotics) are less effective.

Cancer patients are at an increased risk of developing sepsis because **chemotherapy** and other forms of treatment for cancer weaken their immune systems.

The most common symptom of sepsis is **fever**, often accompanied by chills or shaking, or other flu-like symptoms. A history of any recent invasive procedure or dental work should raise the suspicion of sepsis and medical help should be sought.

Diagnosis

The presence of sepsis is indicated by blood tests showing particularly high or low white blood cell counts. The causative agent is determined by blood culture.

In some cases the doctor may order imaging studies to rule out **pneumonia**, or to determine whether the sepsis has developed from a ruptured appendix or other leakage from the digestive tract into the abdomen.

Treatment

Identifying the specific causative agent ultimately determines how sepsis is treated. However, time is of the essence, so a broad-spectrum antibiotic or multiple antibiotics will be administered until blood cultures reveal the culprit and treatment can be made specific to the organism. Intravenous antibiotic therapy is usually necessary and is administered in the hospital. The patient's chances of survival are increased by rapid admission to an intensive care unit followed by aggressive treatment with antibiotics.

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- American Public Health Association (APHA). 800 I Street NW, Washington, DC 20001-3710. (202) 777-APHA. <<http://www.apha.org>>.
- Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

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Sepsis syndrome see **Septic shock**

Septal deviation see **Deviated septum**

Septic arthritis see **Infectious arthritis**

Septic shock

Definition

Septic **shock** is a potentially lethal drop in blood pressure due to the presence of bacteria in the blood.

Description

Septic shock is a possible consequence of **bacteremia**, or bacteria in the bloodstream. Bacterial toxins, and the immune system response to them, cause a dramatic drop in blood pressure, preventing the delivery of blood to the organs. Septic shock can lead to multiple organ failure including **respiratory failure**, and may cause rapid **death**. Toxic shock syndrome is one type of septic shock.

Causes and symptoms

During an infection, certain types of bacteria can produce and release complex molecules, called endotoxins, that may provoke a dramatic response by the body's immune system. Released in the bloodstream, endotoxins are particularly dangerous, because they become widely dispersed and affect the blood vessels themselves. Arteries and the smaller arterioles open wider, increasing the total volume of the circulatory system. At the same time, the walls of the blood vessels become leaky, allowing fluid to seep out into the tissues, lowering the amount of fluid left in circulation. This combination of increased system volume and decreased fluid causes a dramatic decrease in blood pressure and reduces the blood flow to the organs. Other changes brought on by immune response may cause coagulation of the blood in the extremities, which can further decrease circulation through the organs.

Septic shock is seen most often in patients with suppressed immune systems, and is usually due to bacteria acquired during treatment at the hospital. The immune system is suppressed by drugs used to treat **cancer**, autoimmune disorders, organ transplants, and diseases of immune deficiency such as **AIDS**. **Malnutrition**, chronic drug **abuse**, and long-term illness increase the likelihood of succumbing to bacterial infection. Bacteremia is more likely with preexisting infections such as urinary or gastrointestinal tract infections, or skin ulcers. Bacteria may be introduced to the blood stream by surgical procedures, catheters, or intravenous equipment.

Toxic shock syndrome most often occurs in menstruating women using highly absorbent tampons. Left in place longer than other types, these tampons provide the breeding ground for *Staphylococcus* bacteria, which may then enter the bloodstream through small tears in the vaginal lining. The incidence of toxic shock syndrome has declined markedly since this type of tampon was withdrawn from the market.

Symptoms

Septic shock is usually preceded by bacteremia, which is marked by **fever**, malaise, chills, and **nausea**. The first sign of shock is often confusion and decreased consciousness. In this beginning stage, the extremities are usually warm. Later, they become cool, pale, and bluish. Fever may give way to lower than normal temperatures later on in sepsis.

Other symptoms include:

- rapid heartbeat
- shallow, rapid breathing
- decreased urination.
- reddish patches in the skin

Septic shock may progress to cause “adult respiratory distress syndrome,” in which fluid collects in the lungs, and breathing becomes very shallow and labored. This condition may lead to ventilatory collapse, in which the patient can no longer breathe adequately without assistance.

Diagnosis

Diagnosis of septic shock is made by measuring blood pressure, heart rate, and respiration rate, as well as by a consideration of possible sources of infection. Blood pressure may be monitored with a catheter device inserted into the pulmonary artery supplying the lungs (Swan-Ganz catheter). Blood cultures are done to determine the type of bacteria responsible. The levels of oxygen, carbon dioxide, and acidity in the blood are also monitored to assess changes in respiratory function.

Treatment

Septic shock is treated initially with a combination of **antibiotics** and fluid replacement. The antibiotic is chosen based on the bacteria present, although two or more types of antibiotics may be used initially until the organism is identified. Intravenous fluids, either blood or protein solutions, replace the fluid lost by leakage. Coagulation and hemorrhage may be treated with transfusions of plasma or platelets. Dopamine may be given to increase blood pressure further if necessary.

Respiratory distress is treated with mechanical ventilation and supplemental oxygen, either using a nosepiece or a tube into the trachea through the throat.

Identification and treatment of the primary infection site is important to prevent ongoing proliferation of bacteria.

KEY TERMS

Bacteremia—Invasion of the bloodstream by bacteria.

Prognosis

Septic shock is most likely to develop in the hospital, since it follows infections which are likely to be the objects of treatment. Because of this, careful monitoring and early, aggressive therapy can minimize the likelihood of progression. Nonetheless, death occurs in at least 25% of all cases.

The likelihood of recovery from septic shock depends on many factors, including the degree of immunosuppression of the patient, underlying disease, promptness of treatment, and type of bacteria responsible. Mortality is highest in the very young and the elderly, those with persistent or recurrent infection, and those with compromised immune systems.

Prevention

The risk of developing septic shock can be minimized through treatment of underlying bacterial infections, and prompt attention to signs of bacteremia. In the hospital, scrupulous aseptic technique on the part of medical professionals lowers the risk of introducing bacteria into the bloodstream.

Resources

OTHER

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Richard Robinson

Septoplasty

Definition

Septoplasty is a surgical procedure to correct the shape of the septum of the nose. The nasal septum is the separation between the two nostrils. In adults, the septum is composed partly of cartilage and partly of bone.